Alexander Grothendieck

La Longue Marche à travers la théorie de Galois

1981

Reading text

Université de Montpellier, Institut Montpelliérain Alexander Grothendieck, archives Grothendieck
Cotes 140-1, 140-2, 140-3, 140-4
1 588 scans

13 November 2026

About this edition

This volume contains Alexander Grothendieck's research notes of 1981, held in the Montpellier archive: cote 140-1, a draft (37 scans); cote 140-2, manuscript pages 1–292 (575 scans); cote 140-3, pages 293–650 (696 scans); and cote 140-4, pages 650bis–787 (280 scans). All 116 packages and all 245 translation units are included in scan order. This edition works from the supplied transcription, translation and checking records; it proposes no new reading of the handwriting.

The corrected text follows the author's explicit final state. Cancelled words leave the running text; insertions are integrated and shown in teal. Every difference is recorded in the annotated master. The translation follows the supplied English package texts and glossary, with one documented correction from "structure with loops" to "loop structure". The author's letters, indices, arrows and forms such as Sl, Gl, Autext and the doubled Γ remain unchanged. Doubtful and illegible readings are retained.

This reading form omits the variant apparatus and certainly cancelled wording. Scan and manuscript-page numbers are confined to the margin. Uncertainty inside cancelled passages remains visible in linked fragments, indicated by a dagger at its original place and collected at the end of the scan. These uncertainty fragments preserve the transcription's doubts without restoring cancelled passages to the sentence. The archive's institutional footer is omitted; each independently transcribed manuscript number remains.

Every blue passage, red [illeg. n] marker and scan mark links to its scan in the Montpellier viewer. A marginal note stays in the margin when it fits the note column in at most three lines. Longer notes appear at their insertion point in the text block, in small type, with a rule and the label *Marginal note*. Wide formulae and diagrams have repairs at identified source locations. Diagrams needing more room appear on landscape pages.

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Sources and earlier work

Université de Montpellier, Institut Montpelliérain Alexander Grothendieck, archives Grothendieck. `https://grothendieck.umontpellier.fr/archives-grothendieck/`

Earlier transcriptions were consulted in the second pass as witnesses and are not reproduced here.

- Mateo Carmona's edition of the first part, transcribed from the typescript, and the typeset text deciphered and transcribed by Mateo Carmona with the collaboration of Matthias Künzer (agrothendieck.github.io).
- the draft edition of the Centre for Grothendieckian Studies (Grothendieck Institute, Mondovì), February 2026.
- the table of contents compiled by Grothendieck himself, as published by the Grothendieck Circle.

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Cote 140-1

[scan 1]

[scan 2]

"Long March"

$$\dagger \cdot | \,; \qquad \alpha = \underline{u} a^{-1}$$

$$-v|\nu d^2 + wcd + c^2$$

$$\underline{u} = \alpha \underline{a}$$

$$\det(d\rho + c) = d^2 \det \rho + cd \underline{\operatorname{Tr} \rho} + c^2$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\rho = \begin{pmatrix} 0 & v_1|\nu_1 \\ v|\nu & w \end{pmatrix}$$

$$a = \alpha^{-1} u$$

$$d_0\rho + c_0 = \begin{pmatrix} c_0 & d_0 v_1|\nu_1 \\ d_0 v|\nu & d_0 w + c_0 \end{pmatrix}$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} c & d \\ dv|\nu & dw + c \end{pmatrix} = \begin{pmatrix} ac + bdv|\nu & \\ & \end{pmatrix}$$

$$\underline{a} = \alpha^{-1} u = \frac{1}{\delta_\alpha} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \begin{pmatrix} \dagger\mu & \dagger\lambda \\ 0 & 1 \end{pmatrix}$$

$$= \frac{1}{\delta_\alpha} \begin{pmatrix} \dagger d\mu & d\lambda - b \\ -c\mu & -c\lambda + a \end{pmatrix} = d_0\rho + c_0|d_0$$

$$c_0 = d\mu/| \neq \delta_\alpha = \frac{-dv|\nu}{-v^2|\nu^2 + wcd + c^2}$$

$$\dagger$$

$$d_0 v|\nu = \frac{-c\mu}{\delta_\alpha}$$

$$d_0 = -$$

$$c_0 = \frac{d\mu}{\delta_\alpha}$$

$$d_0 = \frac{-c\mu}{\delta_\alpha v|\nu}$$

$$\dagger \qquad \dagger$$

$$\boxed{\underline{a} = \frac{\mu}{\delta_\alpha} \left(-| = \frac{c}{v|\nu} \rho + d \right)}$$

$$\det a \overset{?}{=} \delta_\alpha^{-1} \mu$$

$$\det \underline{a} = \left(\frac{\mu}{\delta_\alpha} \right)^2 \left[\frac{c^2}{v^2|\nu^2} (-vv_1|\nu\nu_1) - | + \frac{dc}{v|\nu} w + d^2 \right] \ldots$$

†

1

Uncertainty retained from cancelled passages.

140-1-s002-D001 $\Rightarrow | =$

140-1-s002-D002 $A|B$

140-1-s002-D003 B

140-1-s002-D004 $d|\delta$

140-1-s002-D005 $v|\nu \quad v_1|\nu_1$

140-1-s002-D006 $\cdot$

140-1-s002-D007 $u|\mu \quad \delta_\alpha|\delta$

140-1-s002-D008 $\underline{a}$

[scan 3]

[scan 4]

[p. 1]

$\widetilde{\Pi}_n^{\circ D}|\widetilde{\Pi}_n^D \simeq | = \dagger\{\tilde\rho, \tilde\sigma \mid \tilde\rho^n = \tilde\sigma^2\}$ $\quad \boxed{\tilde\rho^n = \tilde\sigma^2 = \tilde\omega_0}$ central element

$$1 \to \mathbb{Z} \to \widetilde{\Pi}_n^{\circ D}|\widetilde{\Pi}_n^D \text{ — } \Pi_n^D \text{ — } 1$$

central extension

Let us put

$$\boxed{\tilde\rho_0 = \tilde\sigma\tilde\rho\tilde\sigma^{-1}\tilde\rho} = \tilde\sigma(\tilde\rho)\tilde\rho \qquad \boxed{\tilde\rho_i = \tilde\rho^i(\tilde\rho_0)}$$
$$= (\tilde\sigma\tilde\rho)^2\tilde\omega_0^{-1}$$

† $\tilde\rho_i$ lifts ρ_i

We have

$$\tilde\rho_{n-1}\tilde\rho_{n-2}\cdots\tilde\rho_0 = \underbrace{\tilde\rho^{n-1}\tilde\rho_0\tilde\rho^{-(n-1)}}\,\underbrace{\tilde\rho^{n-2}\tilde\rho_0\tilde\rho^{-(n-2)}}\cdots(\tilde\rho\tilde\rho_0\tilde\rho^{-1})\tilde\rho_0$$
$$= \tilde\rho^{n-1}\tilde\rho_0\tilde\rho^{-1}\tilde\rho_0\tilde\rho^{-1}\cdots\tilde\rho^{-1}\dagger\tilde\rho_0\tilde\rho^{-1}\tilde\rho_0 \qquad (n \text{ factors } \tilde\rho_0$$
$$= \tilde\rho^n\underbrace{(\tilde\rho^{-1}\tilde\rho_0)\cdots(\tilde\rho^{-1}\tilde\rho_0)}_{n \text{ factors}} = \underbrace{\tilde\rho^n}_{\tilde\omega_0}(\tilde\rho^{-1}\tilde\rho_0)^n$$

now

$$\tilde\rho^{-1}\tilde\rho_0 = \tilde\rho^{-1}\tilde\sigma\tilde\rho\tilde\sigma^{-1}\tilde\rho = (\tilde\rho^{-1}\tilde\sigma)(\tilde\rho)$$

. therefore

$$(\tilde\rho^{-1}\tilde\rho_0)^n = (\tilde\rho^{-1}\tilde\sigma)(\tilde\rho^n) = (\tilde\rho^{-1}\tilde\sigma)(\tilde\omega_0) = \tilde\omega_0$$

whence

$$\boxed{\tilde\rho_{n-1}\tilde\rho_{n-2}\cdots\tilde\rho_0 = \tilde\omega_0^2}$$

Let $\widetilde{\Pi}_n$ be † the inverse image of $\Pi_n \subset \Pi_n^D$ in $\widetilde{\Pi}_n^D$ †

†

$$1 \text{ — } \mathbb{Z} \longrightarrow \widetilde{\Pi}_n \longrightarrow \Pi_n \text{ — } 1$$
$$\| \qquad\qquad \|$$
$$\left\{\tilde\rho_0, \ldots, \tilde\rho_{n-1}, \tilde\omega_0 \;\middle|\; \begin{array}{l}\tilde\omega_0 \text{ central,} \\ \tilde\rho_{n-1}\tilde\rho_{n-2}\cdots\tilde\rho_0 = \tilde\omega_0^2\end{array}\right\}$$

Let

$$\widetilde{\Pi}_n^\circ = \left\{\tilde\rho_0, \ldots, \tilde\rho_{n-1} \;\middle|\; \begin{array}{l}\tilde\rho_{n-1}\tilde\rho_{n-2}\cdots\tilde\rho_0 \text{ central} \quad \text{i.e.} \\ \tilde\rho_{n-1}\tilde\rho_{n-2}\cdots\tilde\rho_0 = \tilde\rho_{n-2}\tilde\rho_{n-3}\cdots\tilde\rho_0\tilde\rho_{n-1} = \cdots\end{array}\right\}$$

$\tilde\rho|\rho$

Then

$$1 \longrightarrow \mathbb{Z} \longrightarrow \widetilde{\Pi}_n^{\circ} \longrightarrow \overset{\dagger}{\Pi}_n \longrightarrow 1$$

generated by

$$\bar{\omega} = \tilde{\rho}_{n-1}\tilde{\rho}_{n-2}\cdots\tilde{\rho}_0$$

$$\widetilde{\Pi}_n^{\dagger} \simeq \,|\, \subset \widetilde{\Pi}_n^{\circ} \underset{\mathbb{Z}\bar{\omega}}{\wedge} \mathbb{Z}\omega$$

$$\bar{\omega}_0 \to \omega_0^2$$

3

Uncertainty retained from cancelled passages.

140-1-s004-D001 {
140-1-s004-D002 +
140-1-s004-D003 o
140-1-s004-D004 subgroup $\widetilde{\Pi}_n^D|\Pi_n^D$ generated
140-1-s004-D005 invariant
140-1-s004-D006 $\widetilde{\Pi}_n^D \neq$
140-1-s004-D007 can
140-1-s004-D008 ∘

[scan 5]

[scan 6]

[p. 2]

$\widetilde{\Pi}_n^{\circ}$ normal in $\widetilde{\Pi}_n$ (subgroup of index 2)

is it normal in $\widetilde{\Pi}_n^D$? Invariant|Invariance under $\tilde{\rho}$ it is clear — hence normal in $\widetilde{\Pi}_n^{D+}$. Invariant|Invariance under $\tilde{\sigma}$?

subgroup generated by the $\tilde{\rho}_i$

in Π_n^D

$$\sigma^{*|-1}(\rho_0) = \rho_1, \quad \sigma^{*|-1}(\rho_1) = \rho_0, \quad \sigma^{*|-1}(\rho_2) = \rho_0(\rho_{n-1}), \quad \cdots$$
$$\sigma^{*|-1}(\rho_i) = \rho_n \cdots \rho_{n+2-i}(\rho_{n+1-i})\cdots$$

in $\widetilde{\Pi}_n^D$

$$\tilde{\sigma}^{*|-1}(\tilde{\rho}_0) = \tilde{\rho}_1\tilde{z}_0, \quad \tilde{\sigma}^{*|-1}(\tilde{\rho}_1) = \tilde{\rho}_0\tilde{z}_1 \quad \cdots$$
$$\tilde{\sigma}^{*|-1}(\tilde{\rho}_i) = (\tilde{\rho}_n \cdots \tilde{\rho}_{n+2-i})(\tilde{\rho}_{n+1-i})\tilde{z}_i|\hat{z}_i$$
$$z_{n-1}z_{n-2}\cdots z_0 = 1$$

in $(\widetilde{\Pi}_n^D)_{\mathrm{ab}}$

$$\tilde{\sigma}^{*|-1}(\overset{\zeta}{\tilde{\rho}_0}) = \zeta_1|\gamma_1 + \dagger z_0,$$
$$\tilde{\sigma}^{*|-1}(\zeta_1|\gamma_1) = \zeta_0|\gamma_0 + z_1, \quad \cdots$$
$$\tilde{\sigma}^{*|-1}(\zeta_i|\rho_i) = \zeta_{n+1-i}|\gamma_{n+1-i} + z_i$$

$z_i = \tilde{\omega}_0^{\nu_i|\gamma_i}$ the condition for invariance of $\widetilde{\Pi}_n^{\circ}|\Pi_n^{\circ}$ is that the $\nu_i|\gamma_i \equiv 0$ (2). We are going to try to † calculate at least z_0, z_1.

$\tilde{\rho}_0 = \tilde{\sigma}\tilde{\rho}\tilde{\sigma}^{-1}\tilde{\rho}$

$$\tilde{\sigma}(\tilde{\rho}_0) = \tilde{\sigma}^{*|-1}(\tilde{\rho}_0) = \tilde{\rho}\tilde{\sigma}^{-1}\tilde{\rho}\tilde{\sigma}$$
$$\tilde{\rho}_1 = \tilde{\rho}\tilde{\sigma}\tilde{\rho}\tilde{\sigma}^{-1}$$
$$\tilde{\rho}_1^{-1}\tilde{\sigma}^{-1}(\tilde{\rho}_0) = \tilde{\sigma}\tilde{\rho}^{-1}\underbrace{\underbrace{\tilde{\sigma}^{-1}\underbrace{\tilde{\rho}^{-1}\tilde{\rho}}\tilde{\sigma}^{-1}}_{\tilde{\sigma}^{-2}=\tilde{\rho}^{-n}}\tilde{\rho}\tilde{\sigma}}_{\tilde{\rho}^{-n}}$$
$$= \tilde{\sigma}\underbrace{\tilde{\rho}^{-n}}_{\tilde{\omega}_0^{-1}|\tilde{\omega}_1^{-1}}\tilde{\sigma} = \dagger\tilde{\omega}_0^{-1}\underbrace{\tilde{\sigma}^2}_{\tilde{\omega}_0} = 1$$

†

$$\tilde{\sigma}^{*|-1}(\tilde{\rho}_1) = \tilde{\sigma}^{-1}(\tilde{\rho}_1) = \tilde{\sigma}^{-1}\tilde{\rho}\tilde{\sigma}\tilde{\rho}$$

$$\tilde{\rho}_0^{-1}\tilde{\sigma}(\tilde{\rho}_1) = \tilde{\rho}^{-1}\,\tilde{\sigma}^{*|-1}\,\tilde{\rho}^{-1}\,\underbrace{\tilde{\sigma}^{-1}\tilde{\sigma}^{-1}}_{\omega_0^{-2}}\,\tilde{\rho}\,\tilde{\sigma}\,\tilde{\rho} = 1$$

$$\underbrace{\qquad\qquad}_{\omega_0^{-2|-1}}$$

$$\underbrace{\qquad\qquad\qquad\qquad\qquad}_{1}$$

†

$$\tilde{\rho}_i = \tilde{\rho}^i\tilde{\sigma}\tilde{\rho}\tilde{\sigma}^{-1}\tilde{\rho}^{1-i}$$

$$\tilde{\sigma}(\tilde{\rho}_i) = \tilde{\sigma}\tilde{\rho}^i\tilde{\sigma}\tilde{\rho}\tilde{\sigma}^{-1}\tilde{\rho}^{1-i}\tilde{\sigma}^{-1}$$

$$\tilde{\rho}_n \cdots \tilde{\rho}_{n+2-i|n-2-i}\tilde{\rho}_{n+1-i}\tilde{\rho}_{n+2-i}^{-1}$$

5

Uncertainty retained from cancelled passages.

140-1-s006-D001 $\tilde{\rho}$

140-1-s006-D002 them

140-1-s006-D003 z_0

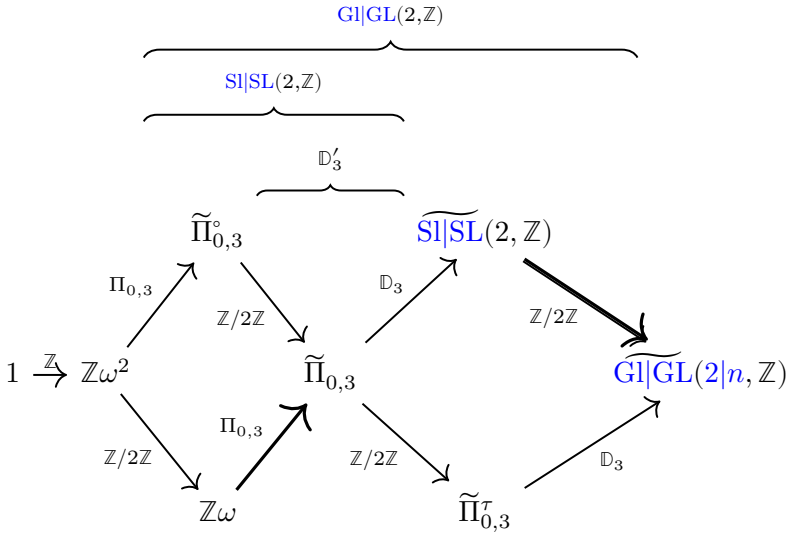
140-1-s006-D004 $\tilde{\rho}$

140-1-s006-D006 $\sigma|\rho$

[scan 7]

[scan 8]

[p. 3]



$$\mathbb{D}_3' = \left\{ \rho, \sigma \;\middle|\; \begin{array}{l} \rho^3 = \sigma^2,\ \sigma^4 = 1\ (= \rho^6), \\ \sigma(\rho) = \rho^{-1} \end{array} \right\},$$

m12

Marginal note. m12 N.B. this extension is not split (the 2 $\rho_{\mathbb{D}_3'}|\sigma_{\mathbb{D}_3'}$ which lift $\rho_{\mathbb{D}_3}|\sigma_{\mathbb{D}_3}$ are of order 6, not 3) unlike :

$$\mathbb{D}_6 \to \mathbb{D}_3$$

$$1 \to \mathbb{Z}/2\mathbb{Z} \text{ — } \mathbb{D}_3' \text{ — } \mathbb{D}_3 \to 1$$

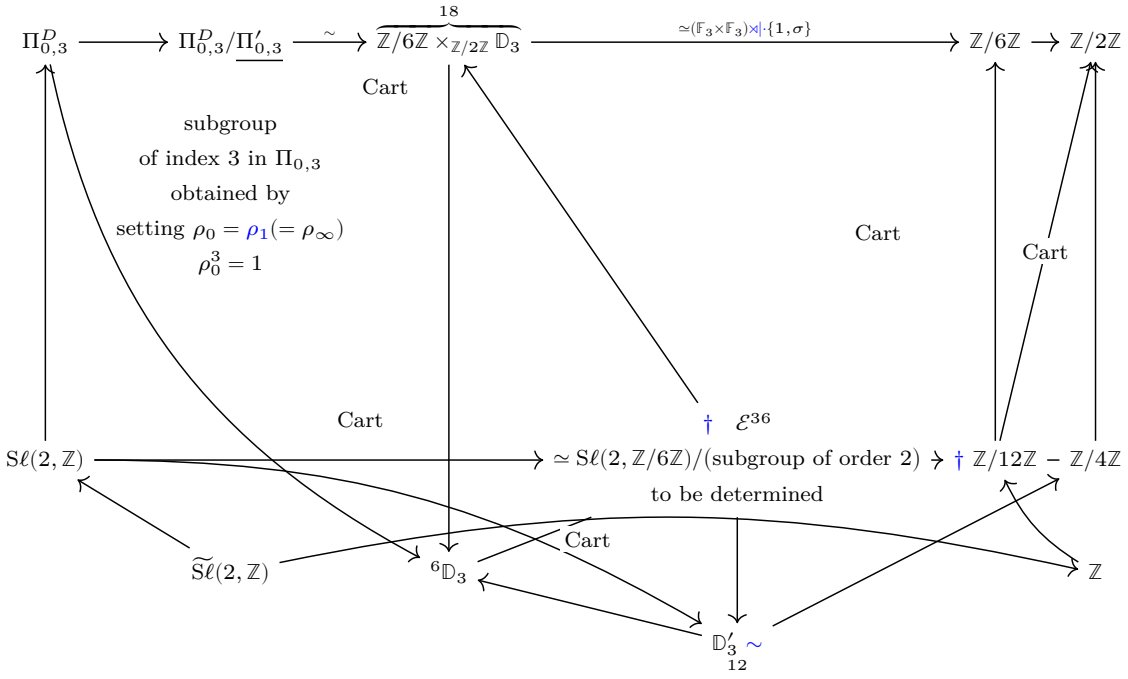
$$\begin{array}{ccccccccc}
1 & \to & \mathbb{Z}/2\mathbb{Z} \times \Pi_{0,3} & \to & \mathrm{Sl}|\mathrm{Gl}(2,\mathbb{Z}) & \longrightarrow & \mathbb{D}_3 & \longrightarrow & 1 \\
 & & \| & & \downarrow & & \downarrow & & \\
1 & \to & \mathbb{Z}/2\mathbb{Z} \times \Pi_{0,3} & - & \mathrm{Gl}|\mathrm{GL}(2,\mathbb{Z}) & - & \mathbb{D}_3 \times \{1,\tau\} & - & 1
\end{array}$$

$$\begin{array}{ccccccccc}
 & & & & \mathbb{Z}/2\mathbb{Z} & & & & \\
 & & & & \downarrow & & & & \\
1 & \to & \Pi_{0,3} & \longrightarrow & \mathrm{Gl}|\mathrm{GL}(2,\mathbb{Z}) & \longrightarrow & \mathbb{D}'_3 \times \{1,\tau\} & \longrightarrow & 1 \\
 & & \| & & \downarrow 2 & & \downarrow 2 & & \\
1 & \to & \Pi_{0,3} & \longrightarrow & \mathrm{Gl}|\mathrm{GL}(2,\mathbb{Z})' & \longrightarrow & \mathbb{D}_3 \times \{1,\tau\}|\{\pm 1\} & \to & 1
\end{array}$$

$$\wr$$

$$\Pi_1(U_{0,3}|M_{0,3}, \mathbb{D}_3 \times \{1,\tau|\gamma\}; P_+)$$

7
[scan 9]
[scan 10]



$$\Pi^D_{0,3} = \{\rho, \sigma \mid \rho^3 = \sigma^2 = 1\}$$

$$\mathrm{S}\ell(2,\mathbb{Z}) = \{\rho, \sigma \mid \underbrace{\rho^3 = \sigma^2}_{\omega},\ \rho^6 = 1\dagger\ (= \sigma^4)\}$$

$$\Pi^D_{0,3}/\Pi'_{0,3} \simeq \{\rho, \sigma \mid \rho^3 = \sigma^2 = (\sigma\rho)^6 = 1,\ (\sigma\rho)^2 = (\rho\sigma)^2\} \quad 18$$

$$\downarrow \rho \mapsto 4 \bmod 6,\ \sigma \mapsto 3 \bmod 6 \qquad \overset{\rho \mapsto \rho}{\sigma \mapsto \sigma} \searrow$$

$$\mathbb{Z}/6\mathbb{Z} \qquad \mathbb{D}_3 = \{\rho, \sigma \mid \rho^3 = \sigma^2 \dagger = 1,\ \sigma(\rho) = \rho^{-1}\}$$

i.e. $(\sigma\rho)^2 = 1$

$$\mathcal{E} = \left\{ \rho, \sigma \;\middle|\; \underbrace{\rho^3 = \sigma^2}_{\omega},\ \rho^6 = 1\ (=\sigma^4),\ \dagger\ \underbrace{(\sigma\rho)^2\omega}_{\rho_0 = \sigma\rho\sigma^{-1}\rho}\ \dagger \text{invariant under} \rho \text{ and of order} \mid 3 \right\} \qquad 36$$

$$\text{i.e. } (\sigma\rho)^6 = \omega, \qquad (\sigma\rho)^2 = (\rho\sigma)^2$$

$$\begin{cases} \rho & \mapsto & 10|14 \bmod 12 \\ \sigma & \mapsto & 3 \bmod 12 \\ \omega & \mapsto & 6 \bmod 12 \end{cases}$$

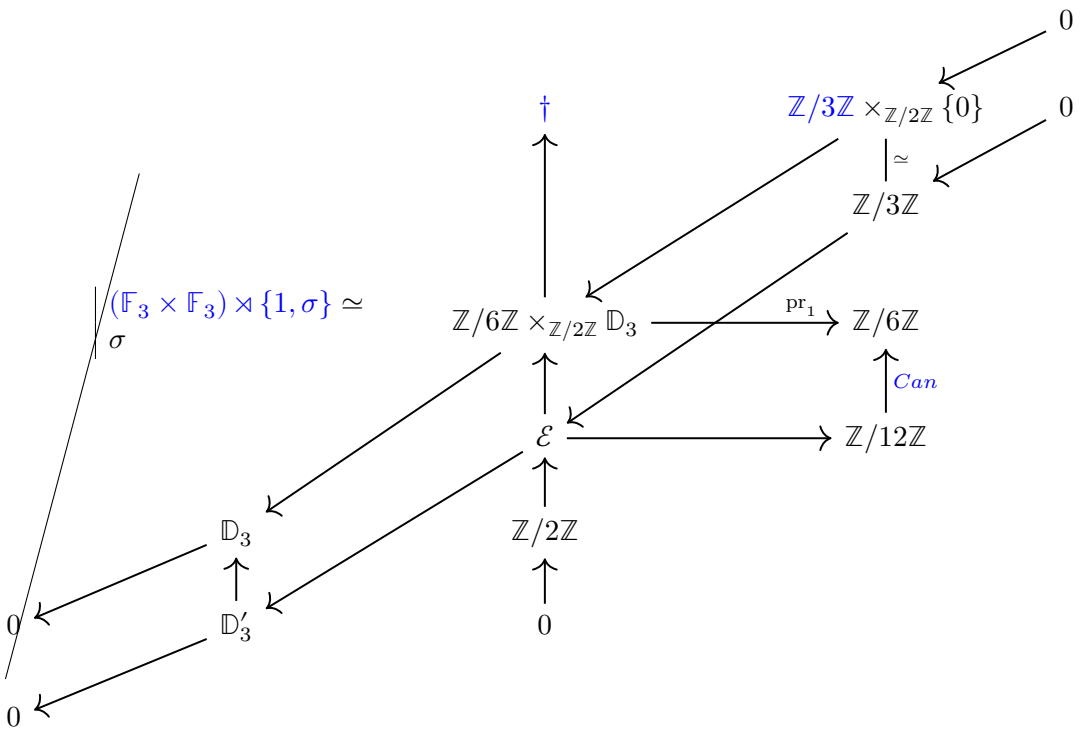
$$\downarrow$$

$$\leftrightarrows\ \mathbb{Z}/12\mathbb{Z} \simeq (\mathrm{S}\ell(2,\mathbb{Z}))_{\mathrm{ab}}$$

$$\sigma\rho = \varepsilon_0 \longmapsto 1 \bmod 12$$
$$\dagger$$
$$(\sigma\rho)^2\omega = \rho_0 \longmapsto 8 \bmod 12$$
$$\dagger$$

$$\mathbb{D}'_3 = \left\{ \rho, \sigma \mid \rho, \sigma \mid \underbrace{\rho^3 = \sigma^2}_{\omega} = (\sigma\rho)^2,\ \rho^6 = 1\ (=\sigma^4) \right\} \quad \dagger$$

$$\underset{\underset{(\sigma\rho)^2\omega.}{\|}}{\rho_0} \longmapsto (2 \bmod 6, 1)$$



9

Uncertainty retained from cancelled passages.

140-1-s010-D001 $\mathrm{S}\ell(2,\mathbb{Z})/\varepsilon$

140-1-s010-D002 $\mathbb{Z}/2\mathbb{Z}$

140-1-s010-D003 ω

140-1-s010-D005 2
140-1-s010-D006 ρ
140-1-s010-D007 ω
140-1-s010-D008 is
140-1-s010-D009 $(\sigma\rho)^2\omega = \rho_0$
140-1-s010-D010 $\mathbb{Z}/3\mathbb{Z}$
140-1-s010-D011 $(\sigma\rho)^2\omega$
140-1-s010-D012 $(\sigma\rho)^2\omega$

[scan 11]

m15

[scan 12]

[p. 4]

Marginal note. m15 (n odd)

Proposition There exist liftings $\tilde{\rho}, \tilde{\sigma}$ of ρ, σ such that we have $\tilde{\rho}^n = \tilde{\sigma}^2$ †, and the two sides are then of the form $\tilde{\omega}^\nu$, one of the †, ν well-defined quantity|† modulo $2n$ (independently of the choices of the liftings $\tilde{\rho}, \tilde{\sigma}$) and $\nu \equiv \dfrac{n^2+1}{2}$ †, or :

$$\nu \bmod 2n \ \in \left\{ \left(-\frac{n-1}{2}, \frac{n+1}{2}\right) \right\}$$

and ν is odd. Which implies that there exists a ν_0 unique in $\left\{-\dfrac{n-1}{2}, \dfrac{n+1}{2}\right\}$ that is $\equiv \nu\ (2n)$. For ν fixed, $\tilde{\rho}, \tilde{\sigma}$ are uniquely determined. Thus we find|This provides presentation

$$\widetilde{\Pi}_n^D = \{\tilde{\rho}, \tilde{\sigma}, \tilde{\omega} \mid \tilde{\omega} \text{ central}, \ \tilde{\rho}^n = \tilde{\sigma}^2 = \tilde{\omega}^{\nu_0|\nu}\}\,.$$

$$\begin{array}{ccccc}
1 & & & & 0 \\
\downarrow & & & & \downarrow \\
\mathbb{Z}\tilde{\omega} & & \tilde{\omega} = (1, 2n) & & \mathbb{Z} \\
\downarrow & & & & \downarrow 2n \\
\widetilde{\Pi}_n^D & & \longrightarrow & & \mathbb{Z} \\
\downarrow & & & & \downarrow \\
\Pi_n^D = \{\rho, \sigma \mid \rho^n = \sigma^2 = 1\} & & \Longrightarrow & & (\Pi_n^D)_{\mathrm{ab}} \simeq \mathbb{Z}/2n\mathbb{Z} \\
\downarrow & & & & \\
1 & & & & \rho \longmapsto (n+1 \bmod 2n) = (1-n \bmod 2n) \\
 & & & & \sigma \longmapsto n \bmod 2n
\end{array}$$

$$\begin{array}{ll}
\tilde{\rho}_0 = (\rho, n+1) & \tilde{\rho}_0^n = (\overbrace{\rho^n}^{1}, n(n+1)) = \dagger \\
\tilde{\sigma}_0 = (\sigma, n) & \tilde{\sigma}_0^2 = (\underbrace{\sigma^2}_{1}, 2n)
\end{array}$$

$$\tilde{\rho}_0^n / \tilde{\sigma}_0^2 = (1, n(n-1)) = \tilde{\omega}^\alpha$$

$$\alpha = \frac{n-1}{2} \qquad (\alpha = 1 \text{ if } n = 3!|?)$$

$$\left(\tilde{\rho}_0 \tilde{\omega}^\beta\right)^n = \left(\tilde{\sigma}_0 \tilde{\omega}^\gamma\right)^2 \qquad \text{i.e.} \quad \tilde{\rho}_0^n \tilde{\sigma}_0^{-2} = \tilde{\omega}^{\dagger 2\gamma - n\beta}$$

$$\alpha + 1$$

$$\dagger \qquad 2\gamma - n\beta = \alpha = \frac{n-1}{2}$$

$$\tilde{\rho} \overset{\text{definition}}{=} \tilde{\rho}_0 \tilde{\omega}^{\overset{\alpha}{\tilde{\beta}}} = \left(\rho, n+1+\underbrace{2n\overbrace{\frac{n-1}{2}}^{\beta=\alpha}}_{n(n-1)}\right) = (\rho, n^2+1)$$

$$\tilde{\sigma} \overset{\text{definition}}{=} \tilde{\sigma}_0\tilde{\omega}^{\gamma} = \left(\sigma, \dagger n + 2n \underbrace{\underbrace{\underbrace{\gamma}_{\frac{(n-1)(n+1)}{4}=\frac{n^2-1}{4}}}_{\frac{n^3-n}{2}}}_{\frac{n^3+n}{2}} \right) = \left(\sigma, \dagger \frac{n^3+n}{2}\right)$$

$$\left| \begin{array}{l} (\alpha+1).2 - 1.n = 1 \\ \underbrace{\alpha(\alpha+1)}_{\gamma}.2 - \underbrace{\alpha}_{\beta}.n = \alpha \end{array} \right. \qquad \gamma =$$

m16

Marginal note. m16

$$\left| \begin{array}{l} \tilde{\rho} = \tilde{\rho}_0\tilde{\omega}^{2x} \\ \tilde{\sigma} = \tilde{\sigma}_0\tilde{\omega}^{nx} \\ \\ x = \end{array} \right.$$

$$\tilde{\rho}_0^n = (1, n^3+n) = \tilde{\omega}^{\overbrace{\frac{n^2+1}{2}}^{\nu}}$$

$$\tilde{\sigma}_0^2 = (1, n^3+n) = \tilde{\omega}^{\frac{n^2+1}{2}}$$

11

Uncertainty retained from cancelled passages.

140-1-s012-D001 ρ
140-1-s012-D002 for all
140-1-s012-D003 well-defined quantity
140-1-s012-D004 $\mathbb{Z}$
140-1-s012-D005 $\tilde{\omega}^{n+1}$
140-1-s012-D006 $n\beta - 2\gamma$
140-1-s012-D007 n
140-1-s012-D009 4
140-1-s012-D010 $\frac{n^3-n}{2}$

[scan 13]

[scan 14]

[p. 4bis]

$$\tilde{\rho}^{*|n} = (\rho, n^2+1) \qquad \left(\tilde{\rho}\tilde{\omega}^{\beta}\right)^n = \left(\tilde{\sigma}\tilde{\omega}^{\gamma}\right)^2$$

$$\tilde{\sigma}^{2|*} = \left(\sigma, \frac{n^3+n}{2}\right)$$

$$\tilde{\rho}^n \simeq \tilde{\sigma}^{2|n} = \tilde{\omega}^{\frac{n^2+1}{2}} \qquad \Updownarrow$$

$$\tilde{\omega}^{\nu}\tilde{\omega}^{\beta n} = \tilde{\omega}^{\nu}\tilde{\omega}^{\gamma 2}$$

$$\beta n = 2\gamma$$

$$n^2 - n \qquad \frac{n-1}{2} \qquad \dagger$$

$$\mathbb{Z}^2 \longrightarrow \mathbb{Z}$$

$$(\beta, \gamma)$$

$$\begin{matrix}\beta = 2x \\ \gamma = nx\end{matrix} \qquad \beta n = 2\gamma = (2n)x$$

$$\nu + \underline{2nx}$$

$$\frac{n^2+1}{2} \bmod 2n \qquad \frac{n^2+1}{2}. \bmod 2n$$

$$13 \bmod 10 \;/\; 3 \qquad n = 3 \quad 5 \bmod 6 =$$

$$n = 2\alpha + 1$$

$$\frac{n^2+1}{2} = \frac{4\alpha^2+4\alpha+2}{2} = \underbrace{2\alpha^2+2\alpha+1}_{(4\alpha+2)\frac{\alpha}{2}+\alpha+1 \quad \text{if } \alpha \text{ even}} \quad \bmod \quad 4\alpha+2$$
$$\underbrace{(4\alpha+2)\frac{\alpha+1}{2}}_{2\alpha^2+3\alpha+1} - \alpha \quad \text{if } \alpha \text{ odd}$$

$$\frac{n^2+1}{2} \equiv \alpha + 1 = \frac{n+1}{2} \quad (2n) \qquad \text{if } \alpha \text{ even}$$

$$\frac{n^2+1}{2} \equiv -\alpha = \frac{1-n}{2} \quad (2n) \qquad \text{if } \alpha \text{ odd}$$

$$\rho^n = \tilde{\sigma}^2 = \tilde{\omega}^{\frac{n+1}{2}} \quad \text{if } \frac{n+1}{2} \text{ is odd i.e. } \alpha = \frac{n-1}{2} \text{ even}$$

$$\tilde{\rho}^n = \tilde{\sigma}^2 = \tilde{\omega}^{-\frac{n-1}{2}} \quad \text{if } \frac{n-1}{2} \text{ is odd}$$

so we find a

$$\tilde{\rho}^n = \tilde{\sigma}^2 = \tilde{\omega}^\nu$$

where

$$\nu \in \left\{-\frac{n-1}{2}, \dagger\frac{n+1}{2}\right\}, \quad \nu \text{ odd}$$

which suffices to characterize ν. 13

Uncertainty retained from cancelled passages.

140-1-s014-D001 $\beta =$

140-1-s014-D002 2

$\dagger$ [scan 15]

m19

Marginal note. m19

$$\sigma\rho\sigma\rho = 1$$

$$.2 \quad 4$$

$$\rho \longmapsto \begin{matrix}2 \bmod 6 \\ 4 \bmod 6\end{matrix}$$

$$\sigma \text{ —— } 3 \bmod 6$$

$$\rho_0 \qquad \rho_1 \qquad \rho_\infty \qquad 5|\varepsilon$$
$$\rho_0 = \dagger\sigma\rho\sigma^{-1}\rho = (\sigma\rho)^2\omega$$

14

principle and the procedures for these subdivisions are left to the initiative of the Committees. However, the proportions fixed in article 2 will have to be respected in a subcommittee.

ARTICLE 5.-

Under the chairmanship of the President of the U.S.T.L., a committee composed of all the Vice-Presidents and of the Vice-President of the Scientific Council, is in particular charged with determining the number of seats to be filled in each of the expanded Committees of specialists, and, in general,with giving its opinion on all the problems that might arise in connection with their establishment and their operation.

ARTICLE 6.-

All provisions of the internal regulations contrary to the present text are repealed.

Uncertainty retained from cancelled passages.

140-1-s015-D001 $g''|g'$ $\mathrm{sg}^+|\mathrm{sg}^t$ ext V $0|1$ $\mathbb{Z}/2\mathbb{Z}$ $\omega\rho_0$ -1 -1 -1

140-1-s015-D012 $\rho\sigma$

[scan 16]

[p. 5]

$M_{1,1} \simeq (U_{0,3}, \mathbb{D}'_3)$, where $\mathbb{D}'_3$ is defined as the fiber product

$$\begin{array}{ccc} \mathbb{D}_3 = \mathfrak{S}_3|\mathfrak{G}_3 & \xrightarrow{\text{can}} & \mathbb{Z}/2\mathbb{Z} \\ \uparrow & & \uparrow \text{can} \\ \mathbb{D}'_3 & \longrightarrow & \mathbb{Z}/4\mathbb{Z} \end{array}$$

to be checked

OK

so $\mathbb{D}'_3$ is an extension (nontrivial|quite trivial) of $\mathbb{D}_3$ by $\mathbb{Z}/2\mathbb{Z}$; $\dagger$ we let $\mathbb{D}'_3$ act on $U_{0,3}$ via $\mathbb{D}_3$, so with triviality of the element ω of order 2 of the kernel of $\mathbb{D}'_2|\mathbb{D}'_3 \to \mathbb{D}_2|\mathbb{D}_3$.

We thus have

$$\begin{array}{ccc} \mathcal{T}_{11} & \simeq & \Pi_1 \overbrace{(U_{0,3}, \mathbb{D}'_3; P_+)}^{\Pi_{0,3}^{\mathbb{D}'_3}} \\ \wr & & \\ \mathrm{S}\ell(2,\mathbb{Z}) & & \end{array}$$

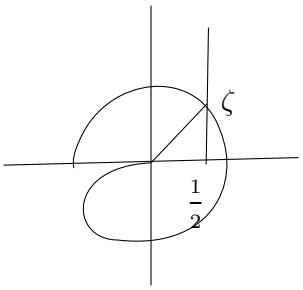
m22

Marginal note. m22 depends on the choice of a point of the Poincaré half-plane above P_+, we will take|or take $\zeta = \exp(2i\pi/6)$ a

and thus

m23

Marginal note. m23 $\Pi_\rho|\Pi_0$ $\Pi_{0,3}$



$$\begin{array}{ccc}
\Pi^{\mathbb{D}'}_{0,3} & \longrightarrow & \Pi^{\mathbb{D}}_{0,3} = \{\rho, \sigma \mid \rho^3 = \sigma^2 = 1\} \\
\downarrow & \text{cart} & \Downarrow \\
\dagger\,\mathbb{D}'_3 & \longrightarrow & \dagger\,\mathbb{D}_3 \\
\downarrow & \text{cart} & \downarrow \\
\mathbb{Z}/4\mathbb{Z} & \longrightarrow & \mathbb{Z}/2\mathbb{Z}
\end{array}$$

m25

$\rho \mapsto 0$
$\sigma \mapsto 1$

Marginal note. m25 We lift ρ, σ to $\dagger$ ρ', σ'
$\rho' = (\rho, 2), \quad \sigma' = (\sigma, \dagger 3)$
we have $\rho'^6 = \sigma'^4 = 1$,
$\rho'^3 = \sigma'^2 = (\underbrace{1, 2 \bmod 4}_{\omega})$.

m26

Marginal note. m26 We then easily verify that
$\mathrm{S}\ell(2, \mathbb{Z}) \simeq \{\rho', \sigma' \mid \rho'^6 = \sigma'^4 = 1,\ \rho'^3 = \sigma'^2\}$.

m27

Marginal note. m27



Thus $\Pi^{\mathbb{D}'}_{0,3} \simeq \mathrm{S}\ell(2, \mathbb{Z})$ appears as an extension of $\Pi^{\mathbb{D}}_{0,3}$ by $\mathbb{Z}/2\mathbb{Z}$, inverse image $\dagger$ the extension $\mathbb{D}'_3$ of $\mathbb{D}_3$ by $\mathbb{Z}/2\mathbb{Z}$, or of the extension $\mathbb{Z}/4\mathbb{Z}$ of $\mathbb{Z}/2\mathbb{Z}$ by $\mathbb{Z}/2\mathbb{Z}$, via $\Pi^{\mathbb{D}}_{0,3} \to \mathbb{D}_3$ resp. composite $\Pi^{\mathbb{D}}_{0,3} \to \dagger\mathbb{Z}/2\mathbb{Z}$. It is an extension of $\mathbb{D}'_3$ by a group $\simeq \Pi_{0,3}$, with basis $\rho_0, \rho_1, \rho_\infty$ — 15

Uncertainty retained from cancelled passages.
140-1-s016-D001 when
140-1-s016-D002 $\mathbb{Z}/4\mathbb{Z}$
140-1-s016-D003 $\mathbb{Z}/2\mathbb{Z}$

140-1-s016-D005 ρ, σ
140-1-s016-D006 2
140-1-s016-D007 of
140-1-s016-D008 $\mathbb{D}_3$

[scan 17]

†
†
m29

Marginal note. m29 Unoriented case

†
†
†
†

$$\mathrm{Pol}_{\tau|t} \qquad \mathrm{Pol}_\tau^{\mathrm{or}} \simeq \mathrm{Pol}_\tau \dagger_{/\Omega}$$

16

Uncertainty retained from cancelled passages.
140-1-s017-D001 $X_0|X_\bullet$
140-1-s017-D003 i.e.|among others $\hookrightarrow | \subset$ $G_0|H_S$
140-1-s017-D007 $A|R$
140-1-s017-D009 oriented edges $S|B$
140-1-s017-D010 A_3 $\simeq | =$ relation triples or flags|triples of flags $\leq | <$
140-1-s017-D016 $A_{2\Omega}$
140-1-s017-D017 $/\Omega$

[scan 18]
[p. 6]

$S\mathcal{T}_{1,1} \simeq$ inverse image of $\mathrm{S}\ell(2,\mathbb{Z})$ in $\dagger \mathrm{S}\ell(2,\mathbb{R})^\sim | \mathrm{G}\ell(2,\mathbb{R})^\sim$

$$\simeq \{\rho, \sigma \mid \rho^3 = \sigma^2 \ (= \omega)\}$$

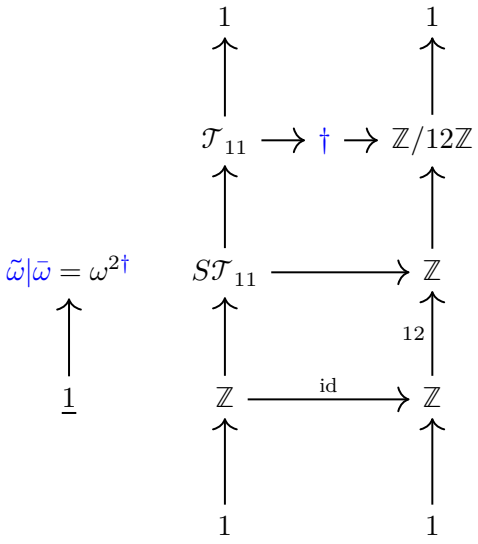
m31

Marginal note. m31 (central elem^{ent}|el^{ement}

$$\mathcal{T}_{11} = S\mathcal{T}_{11}/\langle \omega^2 \rangle$$

$$(\mathcal{T}_{11})_{\mathrm{ab}} \simeq \mathbb{Z} \quad (\mathcal{T}_{11})_{\mathrm{ab}} \simeq \mathbb{Z}/12\mathbb{Z}$$

$$\rho \longmapsto 2 \qquad \rho \longmapsto 2 \bmod 12$$
$$\sigma \longmapsto 3 \qquad \sigma \longmapsto 3 \bmod 12$$
$$\omega \longmapsto 6|8$$



m32

Marginal note. [m32] so $S\mathcal{T}_{11}$ is obtained from $\mathcal{T}_{11}$ by inverse image of the canonical extension of $\mathbb{Z}/12\mathbb{Z}$ by $\mathbb{Z}$, via $\mathcal{T}_{11} \to \mathbb{Z}/12\mathbb{Z}$.

†

† 17

Uncertainty retained from cancelled passages.

140-1-s018-D001 $G\ell$

140-1-s018-D003 $\mathcal{T}_{ab}$

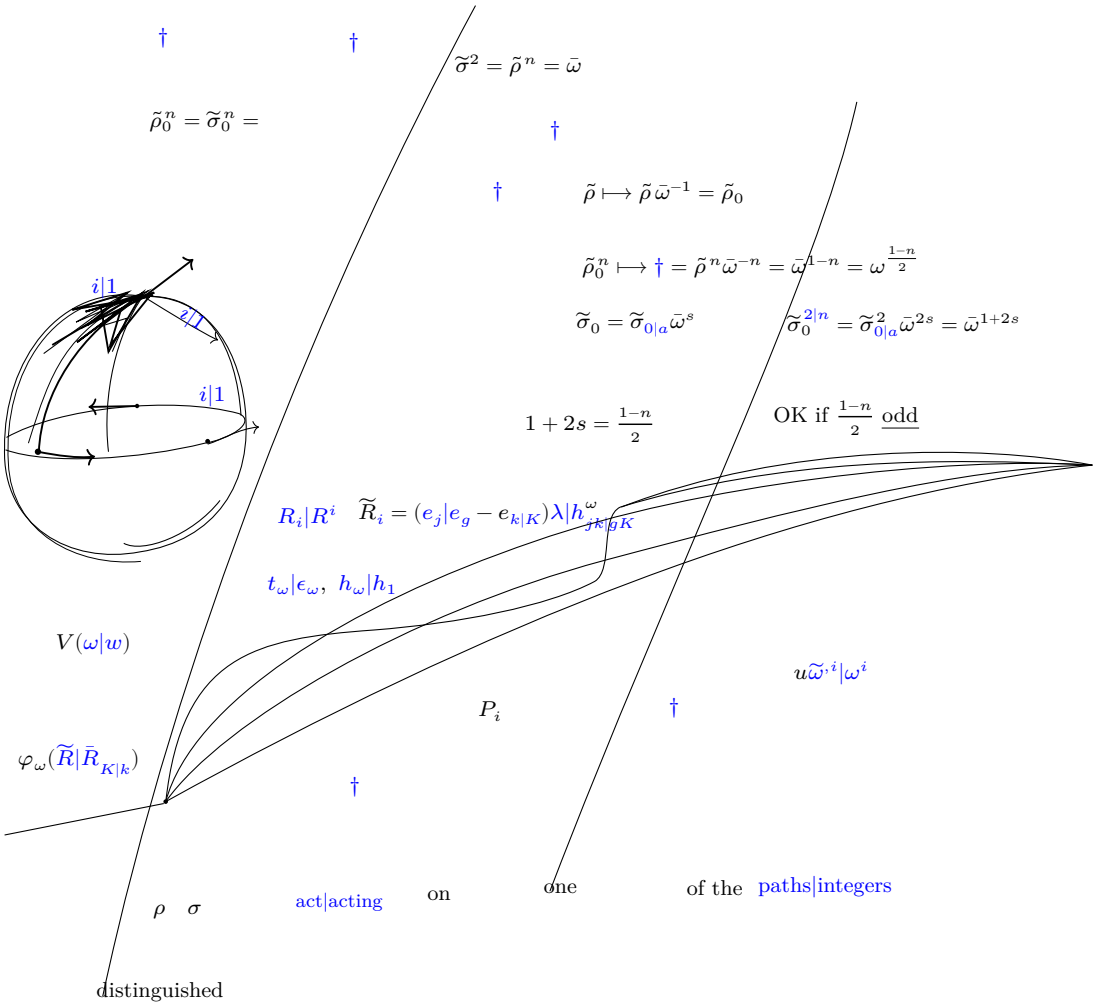
140-1-s018-D004 4

140-1-s018-D006 $K_0|N|k_0$ then special $K'_f N|K'_{fN}$ [illeg. 1] covering|i.e. has is|be morphism normalized it works not strict normalizations even|necessarily

140-1-s018-D008 8.|β. Fundamental|Fundamental groups|group shall show|show

[scan 19]

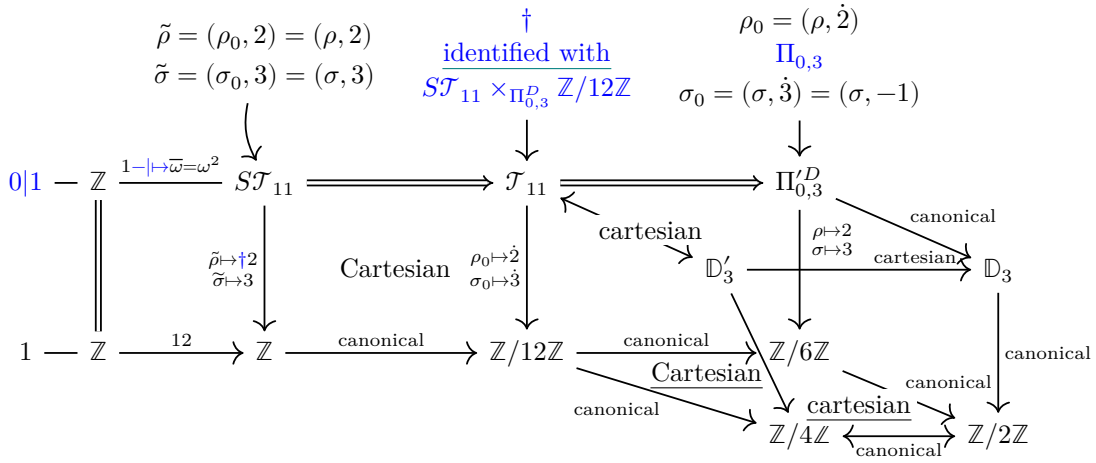
8v



Uncertainty retained from cancelled passages.

140-1-s019-D001 $\mathbb{Z}|\widetilde{\sigma}$
140-1-s019-D002 $\tilde{\rho}|\rho$
140-1-s019-D003 $\tilde{\rho}_0\bar{\omega}|\bar{\omega}$
140-1-s019-D004 $\tilde{\rho}|\widetilde{\sigma}$
140-1-s019-D005 $(\omega)|(\rho)$
140-1-s019-D006 $t_\omega|R_\omega$
140-1-s019-D007 $H_i|H_2$
140-1-s019-D008 $R_K R|R_\kappa R$

[scan 20]
⑦



m35

Marginal note. m35 for $\mathcal{T}_{11}$ identified with $\Pi^D_{0,3} \times_{\mathbb{Z}/2\mathbb{Z}} \mathbb{Z}/4\mathbb{Z}$

Have we indeed | $S\mathcal{T}_{11} \simeq S\Pi^D_{0,3}$??

it would rather seem, however

$$\dagger \quad PM_{11} \simeq \left(\underbrace{T^*U_{0,3}}_{\substack{\text{bundle} \\ \text{tangent}}}, \mathfrak{S}_3 \right)$$

(and not $PU_{0,3}$!!)

$$\text{thus} \quad \mathcal{T}_{11} \simeq \pi_1(T^*U_{0,3}, \mathfrak{S}_3) \ \ldots$$

19

Uncertainty retained from cancelled passages.

140-1-s020-D001 $\Pi^D_{0,3}$

140-1-s020-D002 4

140-1-s020-D003 $\mathcal{T}_{11}$

†

[scan 21]

20

Uncertainty retained from cancelled passages.

140-1-s021-D001 $g_i|q_i$ $g_i|q_i$ $g_i|q_i$ $\mathfrak{Z}\pi|\mathfrak{z}\pi$ $\widetilde{F}|\widehat{F}$ E $*|\cdot$ Operation compatible $\mathfrak{Z}(F)|\mathfrak{z}(F)$ central by outer automorphisms $\mathfrak{Z}(F)|\mathfrak{z}(F)$ compatible F $\pi_0|\pi_\infty$ 1 π_1

[scan 22]

21

[scan 23]

A) Lemma S † regular integral scheme such that $\mathrm{Pic}(S) = 0$, characteristic of the generic point $\neq 2$ Then †

$$H^1(U_{0,3}, \mathfrak{S}_3; \mathbb{G}_m] \simeq \mathbb{Z}/6\mathbb{Z}$$

(†generic point|if generic is 2, we find $\mathbb{Z}/3\mathbb{Z}$)

m39

Marginal note. m39 Proof we use exact sequence

$$0 \longrightarrow H^1(\mathfrak{S}_3, H^0(U_{0,3}, \mathbb{G}_m) \longrightarrow H^1(\text{mixed})$$

$$H^0(\mathfrak{S}_3, \underbrace{H^1(U_{0,3}; \mathbb{G}_m)}_{0})$$

and

$$0 \to H^0(S, \mathbb{G}_m) \to H^0(U_{0,3}, \mathbb{G}_m) \to h_1(V(J))|h^1(V(J)) \to 0$$

I say that if the scheme $\dagger\Sigma_J$ is defined as $\mathbb{P}(V_J)$, i.e.

$$0 \longrightarrow V_J \longrightarrow \mathcal{O}^J \longrightarrow \mathcal{O} \longrightarrow 0$$

then the principal bundle

$$P\Sigma_J = \check{V}(\mathcal{O}_{\Sigma_J}(1))^*$$

is a generator of the $H^1(\dagger\Sigma_J^*, \dagger\mathfrak{S}_J; \mathbb{G}_m)$.

I will first find a trivialization of

$$\underline{\mathcal{O}}(1)^{\otimes 6} = \mathcal{O}(6) \simeq \mathcal{T}^{\otimes 3}(\omega)|\mathcal{J}^{\otimes 3}(\omega),$$

in fact a section of $\mathcal{O}(6)$ over $\overline{\Sigma}$ in its entirety (which $\dagger$ is, invertible on Σ^*. We $\dagger$ have

$$\Gamma(\Sigma_J, \mathcal{O}(6)) \simeq \operatorname{Sym}^6 V_J$$

and we will take

$$\varphi|\Delta \in \Gamma(\Sigma_J, \mathcal{O}(6)), \qquad \varphi|\Delta = \dagger - \prod_{\substack{i,j\in J \\ i\neq j}} (e_i - e_j)$$

$$= \prod_{i\in J} \underbrace{(e_j - e_h)^2}_{J=\{i,j,h\}}.$$

Thus, over $\dagger\Sigma_J^*$, the structure|structure group $\dagger$ $\mathbb{G}_m$ of $P\Sigma_J^*$ is reduced to|: μ_6.

N.B. Unless I am mistaken, we have

$$\dagger \quad \operatorname{Sym}^*(V_J)^{\mathfrak{S}_3} \simeq \mathcal{O}(\gamma_2|\varphi_2, \gamma_3|\varphi_3)/(2\gamma_3^2 - \gamma_2^3)$$

$$\begin{cases} \gamma_2 = \sum_{i\in J} (e_j - e_h)^2 \\ \gamma_3 = \prod_{i\in J} (e_j - |+ e_h - 2e_i) \end{cases} \qquad \text{and} \quad \Delta = \gamma_2^3$$

22

Uncertainty retained from cancelled passages.

140-1-s023-D001 satisfying|strictly
140-1-s023-D002 H^1
140-1-s023-D003 residual
140-1-s023-D004 $U_{0,3}$
140-1-s023-D005 $U_{0,3}$
140-1-s023-D006 $\mathfrak{S}_3$
140-1-s023-D008 vanishes only
140-1-s023-D009 taking
140-1-s023-D010 $\sum_{i\in J}$
140-1-s023-D011 U_J
140-1-s023-D012 of
140-1-s023-D013 Sym

[scan 24]

23

[scan 25]

B)

†

†

†

Since γ_2 vanishes at † P_ω, γ_3 at the Q_i, it follows that there is no invariant section of an $\mathcal{O}(n)^*$ † over Σ_J^* † for $n \leq 6$, hence that the order of $\mathrm{cl}(\mathcal{O}(1))$ in $H^1(\Sigma_J, \mathfrak{S}_J; \mathbb{G}_m)$ is indeed 6, which proves that $\mathcal{O}(1)$ is a generator of the mixed H^1. 24

Uncertainty retained from cancelled passages.

140-1-s025-D001 If invertible will have therefore

140-1-s025-D002 $V_J|W_J$ $J|3$

140-1-s025-D003 Remark

140-1-s025-D004 the

140-1-s025-D005 for

140-1-s025-D006 in degree

[scan 26]

25

[scan 27]

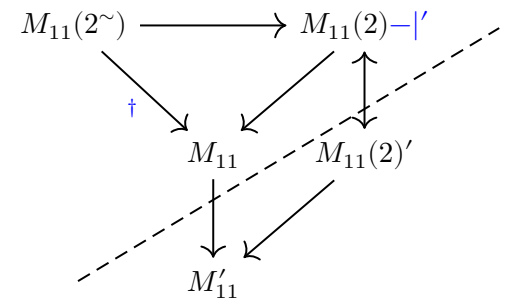
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MATHEMATICS 5

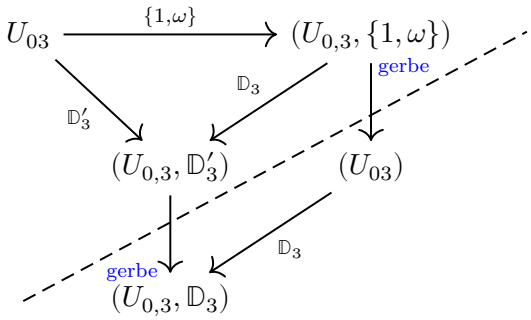
$$\mathbb{D}_3' \longrightarrow \mathbb{D}_3$$

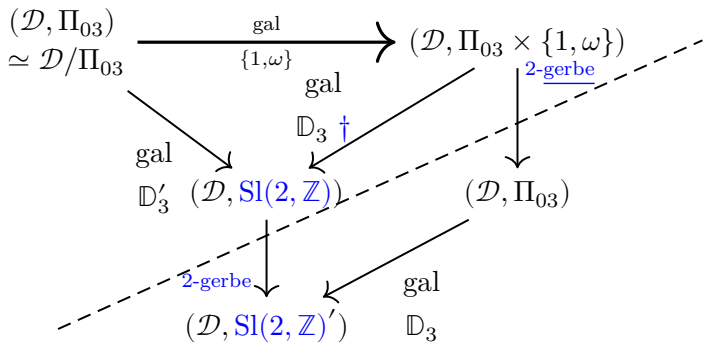
$$\begin{array}{ccc} PM_{11}(2^\sim) \overset{?}{\simeq} PU_{0,3} & \longrightarrow & PM_{11}(2) \simeq TU_{0,3} \\ \updownarrow & \text{cart} & \downarrow \\ U_{0,3} \simeq M_{11}(\tilde{2}^\sim|2^\sim) & \longrightarrow & (U_{0,3}, \{1,\omega\}) \\ & & \simeq M_{11}(2) \end{array}$$

†

$H^1(TU_{03}, \mathbb{D}_3; \mu_2)$







26

Uncertainty retained from cancelled passages.

140-1-s027-D001 $x \quad x^2 \wedge \omega_0$

140-1-s027-D002 $\mathcal{D}/\Pi_{0,3}|\mathbb{Z}/2\,\Pi_{0,3}$

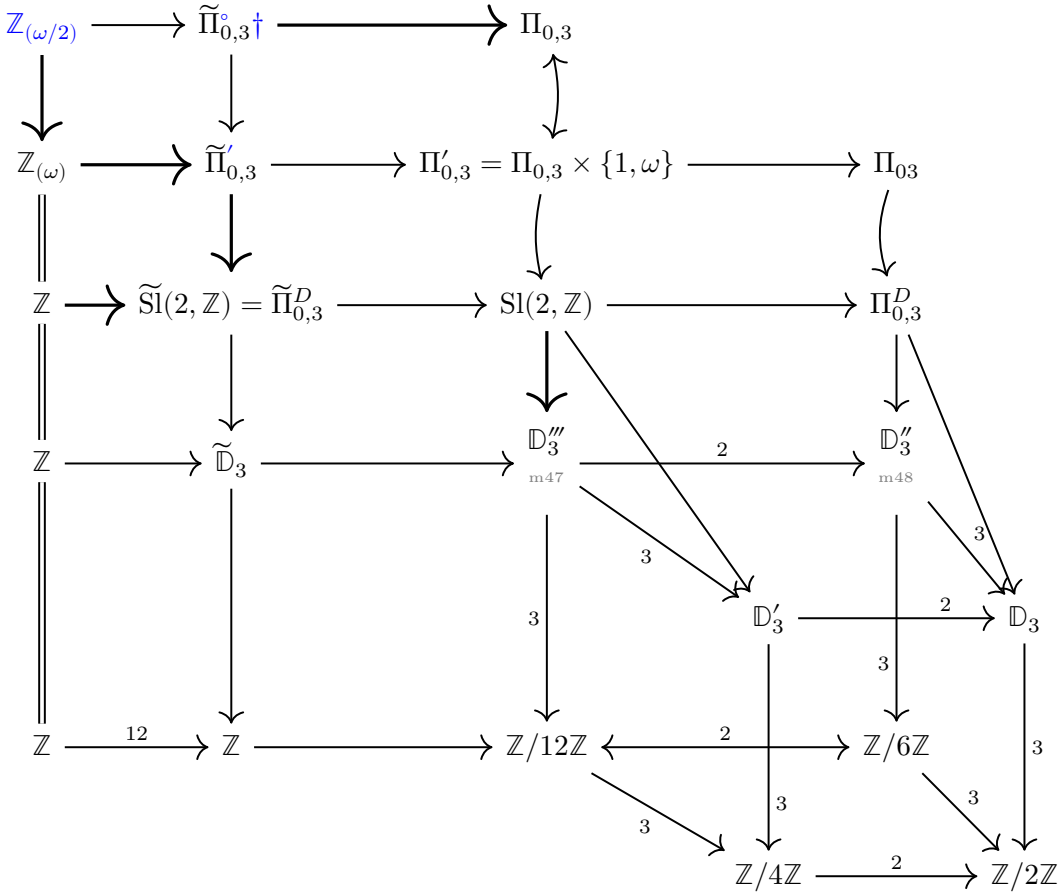
140-1-s027-D003 $\{1,\omega\}$

[scan 28]

27

[scan 29]

†



Marginal note. m47 3

Marginal note. m48 12|10

bottom|good

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \mathbb{Z} & \longrightarrow & \widetilde{\Pi}^{D}_{0,3} & \longrightarrow & \Pi^{D}_{0,3} & \longrightarrow & 1 \\ & & \downarrow{\scriptstyle 4} & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & \mathbb{Z} & \longrightarrow & \widetilde{\Pi}^{D}_{0,3}(4) & \longrightarrow & \Pi^{D}_{0,3} & \longrightarrow & 1 \end{array}$$

$$\begin{array}{ccccccccc} & & & & \widetilde{\Pi}^{D}_{0,3} & & \Pi^{D}_{0,3} & \longrightarrow & 1 \\ & & & & & & \uparrow & & \\ 1 & \longrightarrow & \overset{\omega}{\mathbb{Z}} & \longrightarrow & \widetilde{\Pi}^{D}_{0,3}(4) & \longrightarrow & \Pi^{D}_{0,3} \times \mathbb{Z}/4\mathbb{Z} & \longrightarrow & 1 \\ & & & & & & \uparrow & & \\ 1 & \longrightarrow & \mathbb{Z} & \longrightarrow & ? & \longrightarrow & \mathrm{Sl}(2,\mathbb{Z}) & \longrightarrow & 1 \end{array}$$

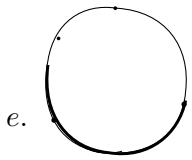
28

Uncertainty retained from cancelled passages.

140-1-s029-D001 $''|1 \quad \tilde{\varepsilon}|\widetilde{E} \quad 12 \quad 2 \quad \varepsilon|E \quad \widetilde{\mathbb{D}}^{*}_{3}|\widetilde{\mathbb{D}}'_{3}$

140-1-s029-D002 $\widetilde{\Pi}_{0,3}$

† [scan 30]



† $e_i - e_j =$

$\xi_i.$

$\underline{\mathcal{O}}_x(1)|\underline{\mathcal{O}}_a(1) \qquad \ni \xi_{ij}(x)$

$\cup$

$\lambda \qquad \lambda^6 = \prod_{i,j} \xi_{ij}(x)$

†

$(e_i - e_j)(\underline{Q_n|Q_k}) \qquad$ †

$\varphi \in V \qquad e_i - e_j($†

$e_i - e_j \bmod L_{Q_k}$

$\begin{array}{ccc} \frac{1}{2} & \frac{1}{2} & 1 \\ -\frac{1}{2} & -\frac{1}{2} & -1 \end{array}$

$$\left(\frac{1}{2}\right)^{4} \qquad \sqrt[3]{\frac{1}{2}}$$

†

m51

Marginal note. m51

$$\widetilde{Q}_m|\widetilde{R}_m - \widetilde{Q}_j|\widetilde{R}_j$$

$$\dagger \quad \gamma_h \; i\gamma_h$$

$$e_j - e_i \quad -(e_i - e_h)$$

29

Uncertainty retained from cancelled passages.

140-1-s030-D001 $\simeq$ $\Gamma(X, \underline{\mathcal{O}}(1))$ generated $\Pi(e_i - e_j)|P(e_i - e_j)$ how are sections permuted

140-1-s030-D005 $\mathcal{O}(6)|\mathcal{O}(1)$

140-1-s030-D006 03 $\sharp|*$

140-1-s030-D007 $e_i - e_j$

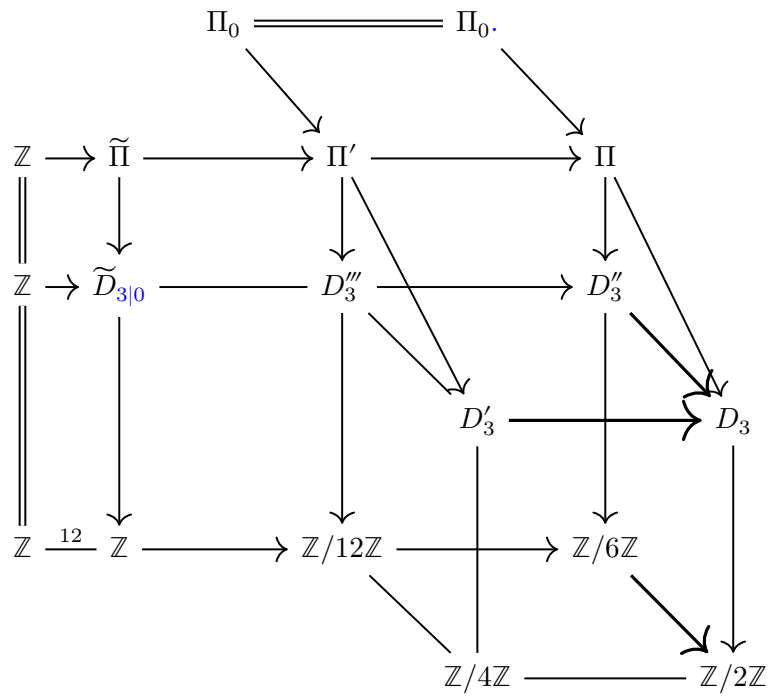
140-1-s030-D009 Q_k

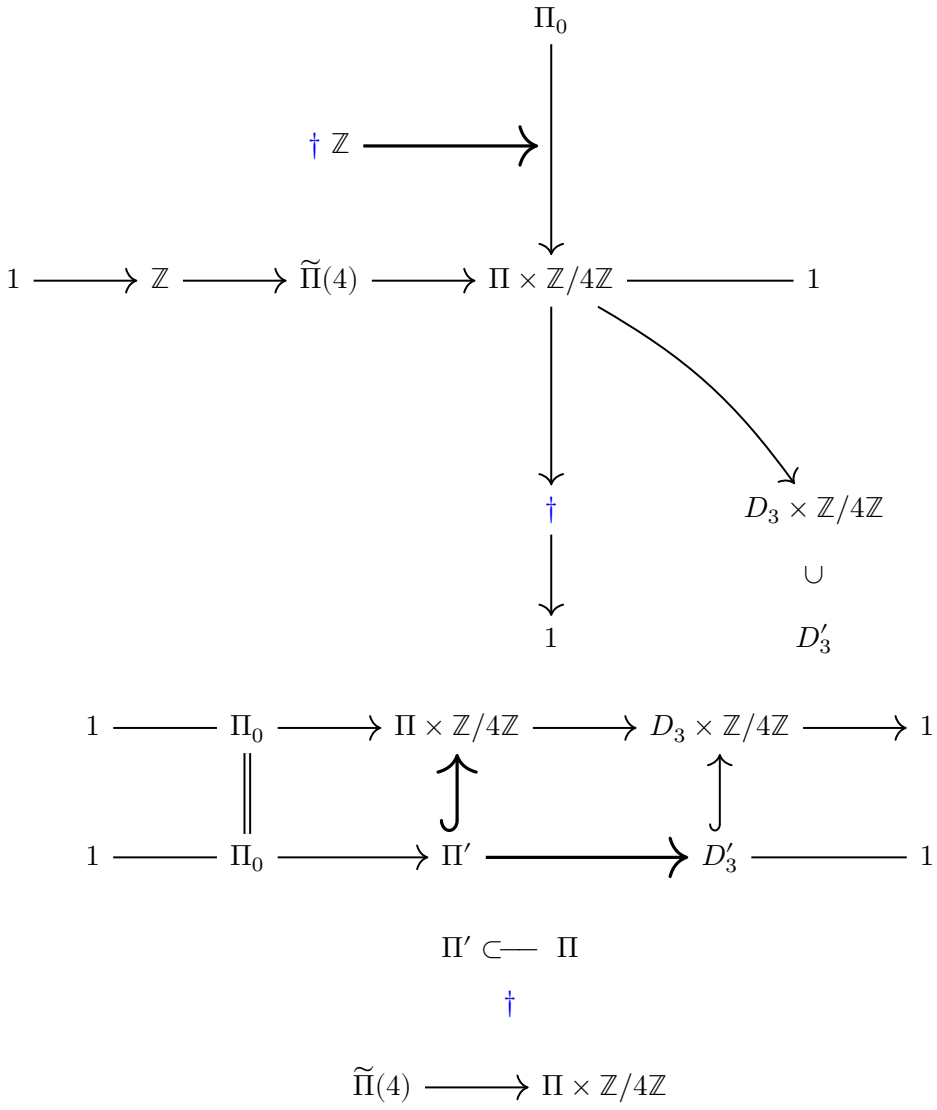
140-1-s030-D010 cellular|cellulation surface map gives to|: operators|operations satisfying what|what corresponding|corresponding operators|operations comes comes thus A pleasant

140-1-s030-D014 ξ

[scan 31]

†





30

Uncertainty retained from cancelled passages.

140-1-s031-D001 D_3''
140-1-s031-D003 $\widetilde{\Pi}$
140-1-s031-D004 D_3
140-1-s031-D005 $\widetilde{\Pi}$ $\mathbb{Z}/4|\mathbb{Z}/4\mathbb{Z}$

$\dagger$

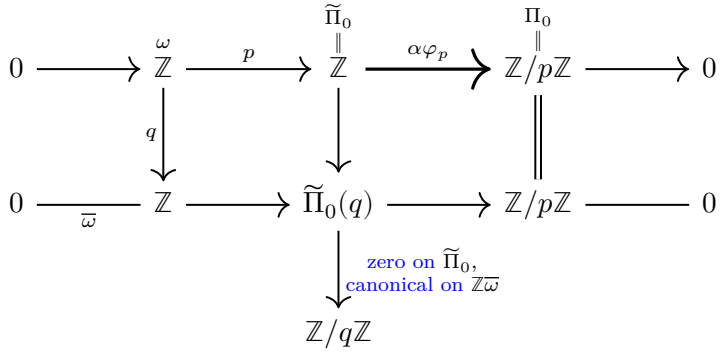
[scan 32]

31

Uncertainty retained from cancelled passages.

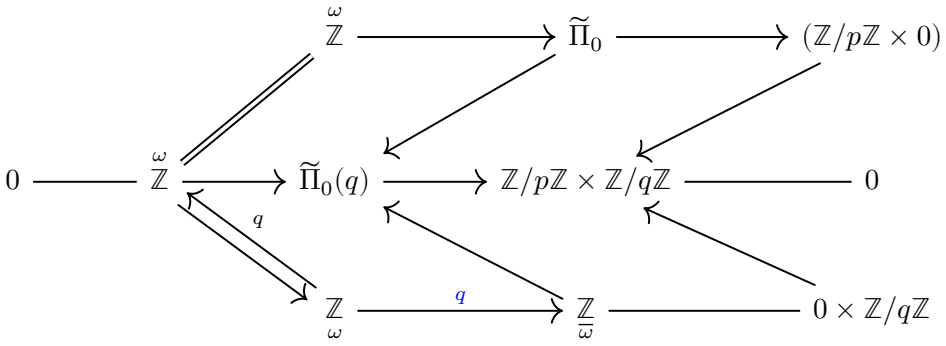
140-1-s032-D001 is not able to indefinitely set indefinitely whether we therefore have a finite cycle or an infinite define $\hat{a}|a$ then notion of circuit|notion of connectedness defined in a restricted sense defined extended all of $=$ opposite combinatorial automorphism $\overline{S}$ $\subset|=$ compactify partially $\overline{X}$ adding K° $\simeq \Pi_{0,3}$ contains A' and stable $\forall$ is not|this is not to|with in its entirety|is entire stability freeness of the action on passing from satisfying must be identified|are identified triples whatsoever recover

[scan 33]

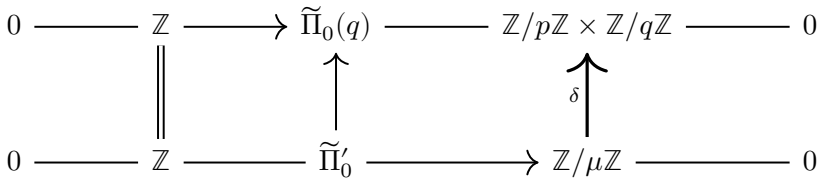


m55

Marginal note. m55 $\varphi_p : \mathbb{Z} \longrightarrow \mathbb{Z}/p\mathbb{Z}$ canonical homomorphism
$\alpha \in (\mathbb{Z}/p\mathbb{Z})^*$
†



$\alpha\eta_p$



†

$$\zeta'|\xi' \dagger = \mathrm{cl}(\widetilde{\Pi}'_0) \in \mathrm{Ext}^1_\dagger(\mathbb{Z}/\mu\mathbb{Z}, \mathbb{Z}) \simeq \mathbb{Z}/\mu\mathbb{Z}$$

$$\dagger \quad \zeta'|\xi' = \delta^* \Bigg(\underbrace{\mathrm{cl}(\widetilde{\Pi}_0(q)|q)}_{\zeta|\xi \in \mathrm{Ext}^1(\mathbb{Z}/p\mathbb{Z}\times\mathbb{Z}/q\mathbb{Z},\mathbb{Z})} \Bigg)'$$

$$\zeta|\xi = \mathrm{pr}_1^* \Bigg(\underbrace{\mathrm{cl}(\widetilde{\Pi}_0)}_{\alpha\eta_p} \Bigg) + \mathrm{pr}_2^* \Bigg(\underbrace{\mathrm{cl}(\mathbb{Z})}_{\eta_q} \Bigg)$$

$\delta_p : \mathbb{Z}/\mu\mathbb{Z} \overset{\mathrm{can}}{\frown} \mathbb{Z}/p\mathbb{Z} \quad \zeta'|\xi' =$
$\delta_q : \mathbb{Z}/\mu\mathbb{Z} \overset{\mathrm{can}}{\frown} \mathbb{Z}/q\mathbb{Z}$

$\delta_p^*(\eta_p) = q'\eta_\mu \qquad \delta_p^{*|'}(\alpha\eta_p) = q'\alpha\eta_\mu$

$\delta_q^{*|'}(\eta_q) = p'\eta_\mu$

32

Uncertainty retained from cancelled passages.

140-1-s033-D001 β

140-1-s033-D002 The extension is [illeg. 1]

140-1-s033-D003 $\zeta|\xi$

140-1-s033-D004 $\mathbb{Z}$

140-1-s033-D005 $=$

† [scan 34]

1)

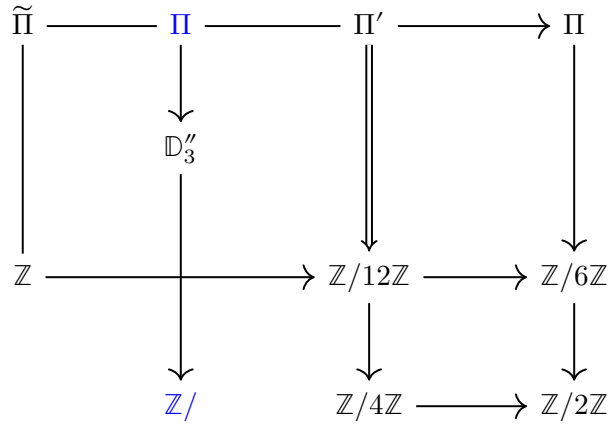
$$S\mathcal{T}_{11} \simeq \widetilde{\mathrm{Sl}}(2,\mathbb{Z}) \simeq \{\rho,\sigma \mid \underbrace{\rho^3=\sigma^2}_{\omega}\} \simeq \underset{\{\rho,\sigma|\rho^3=\sigma^2=1\}}{\Pi^D_{0,3}} \times_{\mathbb{Z}/6\mathbb{Z}} \mathbb{Z}$$

$$\simeq S\Pi^D_{0,3}$$

†

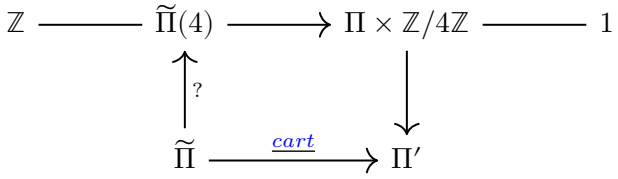
$$\rho \longmapsto 2$$

$$\sigma \longrightarrow 3$$



$\Pi_0 = \mathbb{Z}/6\mathbb{Z}$

$\Pi_0 \times 7|1$



27-11-81

Mr. Grothendieck
To be filled out urgently
Thank you
R. Boucher

33

Uncertainty retained from cancelled passages.

140-1-s034-D001 4) $\widetilde{\mathrm{Sl}}$ $\mathcal{T}_{1,1}$ SL Π

140-1-s034-D002 α

[scan 35]

$$(p' + q'\alpha)\eta_\mu$$

$$p = 6,\ q = 4 \qquad (\alpha = -1\ ?)$$

$$\dagger\ \mu = 12 \qquad p' = 3,\ q' = 2$$

$$\dagger\ (3-2)\eta_\mu\ !$$



.

$$\Pi^D_{0,3} \xrightarrow{-\alpha} \mathbb{Z}/6\mathbb{Z}$$

$$\rho \longmapsto -2 \bmod 6$$

$$\sigma \longmapsto 3 \bmod 6$$

$$\varepsilon_0 = \sigma\rho \longmapsto 1 \bmod 6$$

$$\rho_0 = \varepsilon_0^2 \longmapsto 2 \bmod 6$$

†
m63

Γ|√
ξ₀′|ξ₀
t′r|t′h

Marginal note. m63 defined directly by the action of $\Pi^D_{0,3}$ on the principal cover† of $(U_{0,3}, \mathfrak{S}_3)$ with group $\mathbb{Z}/6\mathbb{Z}$, defined by the canonical multisection|multiplication of $\dagger\mathcal{O}_{U_{0,3}}(1)^* = PU_{0,3}$

$$\widetilde{\Pi}^D_{0,3}$$

†

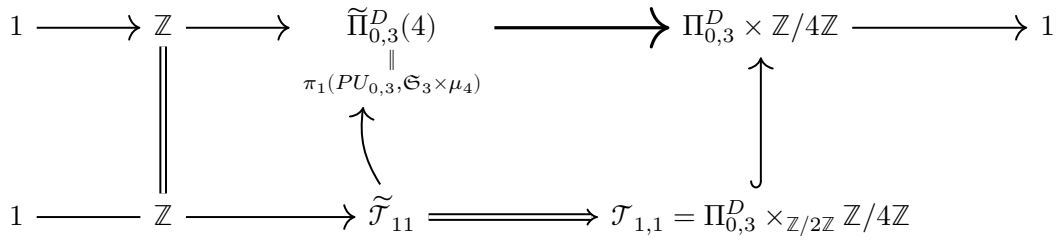
$$U'_{0,3} \subset PU_{0,3} \quad +$$

$$\mu_6\text{-rev}^{\mathrm{t}} \searrow \quad \swarrow$$

$$U_{0,3}$$

†

$$\begin{array}{ccccc}
 & & \Pi^D_{0,3} & \xrightarrow{\alpha'} & \mathbb{Z}/6\mathbb{Z} \\
 & & \uparrow & & \uparrow \\
\pi_1(PU_{0,3}, \mathfrak{S}_3) & = & \widetilde{\Pi}^D_{0,3} & \longrightarrow & \mathbb{Z} \\
 & & \uparrow & & \uparrow 6 \\
 & & \omega' \in \mathbb{Z} & \xrightarrow{=} & \mathbb{Z}
\end{array}$$



†

$$\widetilde{\Pi}^D_{0,3}\ni\tilde{\rho}^{\sharp|'}=(\rho,-2)\quad\tilde{\rho}^{\sharp 3}=\tilde{\sigma}^{\sharp 2}=\omega'$$
$$\tilde{\sigma}^{\sharp|'}=(\sigma,-3)$$

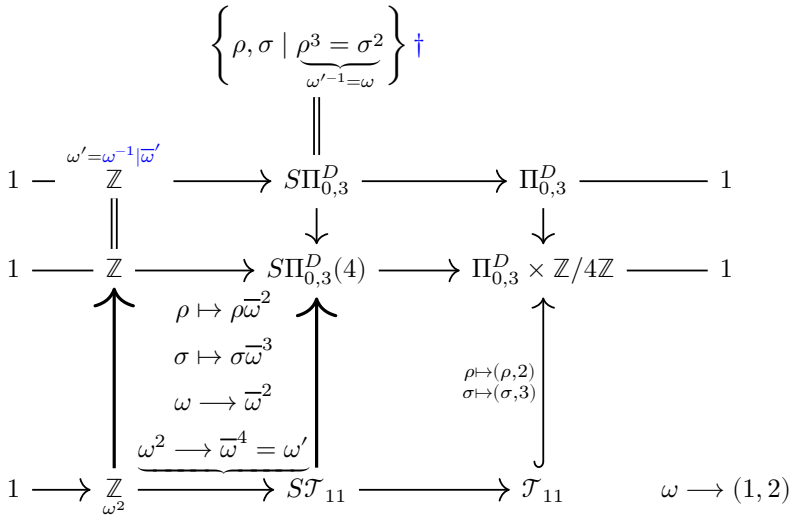
$$\tilde{\rho},\quad\tilde{\sigma}\quad\in\widetilde{\Pi}^D_{0,3}=\left\{\tilde{\rho},\tilde{\sigma}\mid\underbrace{\tilde{\rho}^3=\tilde{\sigma}^2}_{\omega'^{-1}|\overline{\omega}^{-1}=\omega|\infty}\ \dagger\right\}$$
$$\downarrow\quad\downarrow$$
$$\rho\quad\sigma\qquad\qquad\simeq\widetilde{\mathcal{T}}_{11}$$

34

Uncertainty retained from cancelled passages.

140-1-s035-D001 p
140-1-s035-D002 $p'+$
140-1-s035-D003 φ
140-1-s035-D004 ing
140-1-s035-D005 $\mathcal{O}(1)$
140-1-s035-D006 (1)
140-1-s035-D007 $\longrightarrow$
140-1-s035-D008 $\widetilde{\Pi}^D_{0,3}\quad\mathbb{Z}/4\mathbb{Z}$
140-1-s035-D009 $\omega\in\mathbb{Z}$

[scan 36]



m65

Marginal note. m65 $S\Pi^D_{0,3}(4)=\left\{\rho,\sigma,\overline{\omega}\mid\begin{matrix}\rho^3=\sigma^2=\overline{\omega}^{-4}\\ \overline{\omega}\text{ central}\end{matrix}\right\}$

N.B. $\dot{\overline{\omega}}|\overline{\omega}^4=\omega^{-1}$
$=\omega'$ †

$$\overbrace{S\Pi^{D}_{0,3}(4)}^{\pi_1(U_{0,3},\mathfrak{S}_3\times\mu_4)} \longrightarrow \Pi^{D}_{0,3}$$

35

Uncertainty retained from cancelled passages.

140-1-s036-D001 $\omega|\infty$

140-1-s036-D002 ω

[scan 37]

36

Cote 140-2

[scan 1]

[scan 2]

[scan 3]

The "long march" through Galois theory... I

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[Brace on the right joining entries 3 and 4.]

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Uncertainty retained from cancelled passages.

140-2-s003-D001 6.

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[To the left of the last two entries, brace extended upward by an arrow pointing to the entry "The injectivity of $\Gamma_{\mathbb{Q}}$".]

Uncertainty retained from cancelled passages.
140-2-s004-D001 [illeg. 1]
140-2-s004-D002 extensions

[scan 5]
814|714
752|252
832
[scan 6]
[p. 1]
W|VI

1. Multi-Galois topoi.

Proposition 1.1. Let $\mathcal{E}$ be a category. Equivalent conditions:

a) $\mathcal{E}$ is a topos, and every object of $\mathcal{E}$ is local^{ly} constant.

b) $\mathcal{E}$ is equivalent to a category $\widehat{C}$, where C is a groupoid.

m72

Marginal note. m72 N.B. We shall see (below) that we can choose C canonically

b') $\exists$ a family $(G_i)_{i\in I}$ of groups and an equivalence of categories

$$\mathcal{E} \cong \prod \mathrm{Ens}(G_i)$$

c) Ad-hoc exactness conditions, of the type of those given in SGA 1...

Proof. $b \Longleftrightarrow b' \Longrightarrow a$ immediate. For $a) \Longrightarrow b)$, I am less sure, perhaps one has to assume that $\mathcal{E}$ is locally connected, and that it has sufficiently many fiber functors i.e. sufficiently many points.

Definition 1.2. If the equivalent conditions b) b') above are satisfied, one says that C is a|a multi-Galois topos (or a multi-Galois category).

m73

Marginal note. [m73] Proposition 1.3 a) If $\mathcal{E}$ is multi-Galois, every induced topos $\mathcal{E}_{/S}$ too.
b) every sum of multi-Galois topoi (i.e. † product of multi-Galois categories) is so|is too.

Uncertainty retained from cancelled passages.
140-2-s006-D002 [illeg. 1]

Proposition 1.4 Let|Let $\mathcal{E}$ be a topos, C the category of points of $\mathcal{E}$ (opposite to the category of fiber functors on $\mathcal{E}$). The canonical functor [scan 7]

$$\mathcal{E} \times C^o \longrightarrow (\text{Ens})$$

induces a canonical functor

$$\mathcal{E} \longrightarrow \text{Hom}(C^o, (\text{Ens})) \overset{\text{déf}}{=} \widehat{C}$$

This being so: ($\mathcal{E}$ being multi-Galois)

a) C is a groupoid (called the "fundamental groupoid" of the multi-Galois topos $\mathcal{E}$, and often denoted $\Pi_1(\mathcal{E})$).

b) $\mathcal{E} \longrightarrow \widehat{C}$ is an equivalence of categories.

An object S of a topos $\mathcal{E}$ is said to be 0-connected if it is $\neq \emptyset_{\mathcal{E}}$ (it|i.e. is not an initial object) and if it is connected (i.e. $S \simeq S' \amalg S''$ implies S' or $S'' \simeq \emptyset_{\mathcal{E}}$) — this also means that the induced topos $\mathcal{E}_{/S}$ is 0-connected i.e. is not the initial topos ("empty topos", i.e. equivalent to the terminal category) and that it is connected, i.e. ... One says that S is 1-connected

[scan 8]

(or "simply connected") if it is 0-connected, and if every † object S' of $\mathcal{E}_{/S}$ † locally constant [p. 2]
is is †|constant — which depends again only on the induced topos $\mathcal{E}_{/S}$, which will then be said to be 1-connected.

Proposition 1.5. Let $\mathcal{E}$ be a multi-Galois topos, and let S be an object of $\mathcal{E}$. Equivalent conditions

a) S is 1-connected

b) S (is)|is 0-connected and projective

c) the covariant functor represented by S

$$T \longmapsto \text{Hom}_{\mathcal{E}}(S, T)$$

is a fiber functor, or again as it is already left exact) it commutes with arbitrary $\varinjlim$

N.B. (it suffices that it commute with sums and with passage to the quotient...)

d) $\mathcal{E}_{/S}$ is equivalent to the point topos

e) (If $\mathcal{E} = \widehat{C}$, C a groupoid) the functor S on C is representable

Uncertainty retained from cancelled passages.
140-2-s008-D001 covering of S is
140-2-s008-D002 [illeg. 1]
140-2-s008-D003 constant

Definition 1.6 One then says, sometimes, that S (or better, the topos $\mathcal{E}_{/S}$) is a universal covering of the multi-Galois topos $\mathcal{E}$. [scan 9]

Proposition 1.7 † Let $\mathcal{E}_0$ be the full subcategory of $\mathcal{E}$ formed by the 1-connected objects (of the multi-Galois category $\mathcal{E}$), † and $\Pi_1(\mathcal{E})$ the fundamental groupoid of $\mathcal{E}$. We have by 1.5 a canonical functor

$$\mathcal{E}_0 \longrightarrow \Pi_1(C)$$

(associating: $\forall\ S \in \mathrm{Ob}\,\mathcal{E}_0$ the fiber functor it represents, or rather the corresponding "point" of C), which is (not only fully faithful, but even) an equivalence of categories: every fiber functor on $\mathcal{E}$ is representable (by an essentially unique (1-connected) object, as is only right...).

Uncertainty retained from cancelled passages.

140-2-s009-D001 Taking [illeg. 1]

140-2-s009-D002 the functor

† [scan 10]

Corollary 1.8. Let P be a "point" of $\mathcal{E}$ (associated with a fiber functor F_P). There exists a [p. 3]
1-connected object S of $\mathcal{E}$ and a lifting $P \xrightarrow{\alpha} \mathcal{E}_{/S}$ (i.e. an $\alpha \in F(S)$),

[Under $P \xrightarrow{\alpha} \mathcal{E}_{/S}$, $\mathcal{E}$ is placed at bottom center. Two dotted arrows go respectively from P and from $\mathcal{E}_{/S}$ to $\mathcal{E}$. Between them, a rightward arrow with $\sim$ above it, double-lined|single-lined. Two side brackets enclose this lower part of the diagram.]

and this determines (S, α) up to unique isomorphism.

In fact, S is the unique object|† of $\mathcal{E}$ that represents F_P...

Definition 1.9. One says that S (or $\mathcal{E}_{/S}$) is the universal covering pointed over P determined by the point P.

Scholium 1.10 Giving oneself a "point" of the : multi-Galois topos $\mathcal{E}$, or giving oneself a universal covering, amounts essentially to the same: each determines the other...

Uncertainty retained from cancelled passages.

140-2-s010-D001 Definition 1.8 [illeg. 1] The object

140-2-s010-D002 object

[scan 11]

Proposition 1.11|1.10. Let $\mathcal{E}, \mathcal{E}'$ be two multi-Galois topoi, $\Pi_1(\mathcal{E}), \Pi_1(\mathcal{E}')$ their fundamental groupoids. The obvious functor

$$\underline{\mathrm{Hom}}_{\mathrm{top}}(\mathcal{E}, \mathcal{E}') \longrightarrow \underline{\mathrm{Hom}}(\underset{C}{\Pi_1(\mathcal{E})}, \underset{C'}{\Pi_1(\mathcal{E}')})$$

is an equivalence of categories, one finds a quasi-inverse equivalence by composing

$$\underline{\mathrm{Hom}}(\dagger C, C') \longrightarrow \underset{\cap}{\underline{\mathrm{Hom}}_{\mathrm{top}}(\widehat{C}, \widehat{C'})} \simeq \underline{\mathrm{Hom}}_{\mathrm{top}}(\mathcal{E}, \mathcal{E}').$$
$$\underline{\mathrm{Hom}}(\widehat{C'}, \widehat{C})$$

(or|for $\widehat{C} \simeq \mathcal{E}$, $\widehat{C'} \simeq \mathcal{E}'$)

† 1.12 Making explicit the case where $\mathcal{E}$ $\mathcal{E}'$ are 0-connected and pointed, hence given as $\mathcal{E} \simeq \mathrm{Ens}(G)$, $\mathcal{E}' \simeq \mathrm{Ens}(G')$

... 10

Uncertainty retained from cancelled passages.

140-2-s011-D001 [*illeg.* 1]

140-2-s011-D002 [illeg. 1]

[scan 12]

2. Application|Applications to coverings of topoi. [p. 4]

Theorem 2.1. Let $\mathcal{E}$ be a locally connected topos (i.e. every object of which is a sum of 0-connected objects) and locally simply connected (i.e. admitting a system of generators which are 1-connected). m75

Marginal note. [m75] perhaps one must suppose|supposed that $\mathcal{E}$ has "enough points" i.e. enough fiber functors…

†Then the category $\mathcal{E}_{lc}$ of local[ly] constant objects of $\mathcal{E}$ is a topos multi-Galois†, and the inclusion

$$(2.1.1.) \qquad \mathcal{E}_{lc} \hookrightarrow \mathcal{E}$$

commutes with finite $\varprojlim$ (N.B. in fact, without hypotheses on the topos $\mathcal{E}$, $\mathcal{E}_{lc}$ is stable under finite $\varprojlim$) and with arbitrary $\varinjlim$.

Definition 2.2 We denote this topos by $\mathcal{E}_{[1]}$, call it the multi-Galois envelope of $\mathcal{E}$, and the morphism of topoi transposed from the inclusion (2.1.1): 11

Uncertainty retained from cancelled passages.
140-2-s012-D001 [illeg. 6]
140-2-s012-D002 ne

[scan 13]

$$\mathcal{E} \longrightarrow \mathcal{E}_{[1]}$$

is given the name canonical morphism.

N.B. This is the equivalent in topos theory of the operation of "killing the Π_i for $i \geq 2$". One can also define an † $\mathcal{E}_{[0]}$ and a sequence

$$\mathcal{E} \longrightarrow |\!-\mathcal{E}_{[1]} \longrightarrow |\!-\mathcal{E}_{[0]}$$

[$\mathcal{E}_0$ is the discrete topos defined by $\Pi_0(\mathcal{E})$, which has a satisfactory meaning as soon as $\mathcal{E}$ locally connected…]. With suitable hypotheses on $\mathcal{E}$ (of the type "local contractibility"), one should be able to define the $\mathcal{E}_{[i]}$ for $\forall\ i \in \mathbb{N}$, and canonical morphisms

$$\mathcal{E} \longrightarrow \cdots \quad \mathcal{E}_{[i]} \longrightarrow |\!-\mathcal{E}_{[i-1]} \longrightarrow \cdots \quad \mathcal{E}_{[1]} \longrightarrow \mathcal{E}_0.$$

12

Uncertainty retained from cancelled passages.
140-2-s013-D001 $\mathcal{E}_\infty$

[scan 14]

2.3. The fundamental groupoid of $\mathcal{E}$ is defined as having for objects the points of $\mathcal{E}$ (which induce points of $\mathcal{E}_{[1]}$ thanks to $\mathcal{E}_{[1]} \longrightarrow \mathcal{E}$), and for morphisms the † morphisms of points of $\mathcal{E}_{[1]}$. We thus have canonical functors [p. 5]

$$\mathrm{Pt}(\mathcal{E}) \xrightarrow{\alpha} \Pi_1(\mathcal{E}) \overset{\beta}{\underset{\approx}{\longrightarrow}} \mathrm{Pt}(\mathcal{E}_{[1]}) \overset{\text{definition}}{=} \Pi_1(\mathcal{E}_{[1]})$$

where β is an equivalence (but not surjective on objects), and where of course α is not necessarily an equivalence, nor even fully faithful, or merely faithful. For example if $\mathcal{E}$ is 1-connected i.e. $\Pi_1(\mathcal{E})$ equivalent to the one-point category, it does not necessarily follow that the Hom in|of $\mathrm{Pt}(\mathcal{E})$ are all of cardinality ≤ 1! 13

Uncertainty retained from cancelled passages.
140-2-s014-D001 points

[scan 15]

Since a poin[t] P of $\mathcal{E}$ defines a poin[t] (still denoted P by abuse) of $\mathcal{E}_{[1]}$, one can therefore define the universal covering of $\mathcal{E}$ based at this point, as an object S 1-connected in $\mathcal{E}_{[1]}$ – it is characterized in $\mathcal{E}$ by being locally constant, 1-connected, and endowed with a lifting

$$P \longrightarrow \mathcal{E}_{/S}.$$

But since α is not an equivalence of categories (although it is † essentially surjective if one assumes that $\mathcal{E}$ has enough points…) one can not say that $\forall$ universal covering of $\mathcal{E}$ is defined by a poin^{t} of $\mathcal{E}$, defined up to unique isomorphism… 14

Uncertainty retained from cancelled passages.
140-2-s015-D001 [illeg. 1]

[scan 16]

3. "pro-multi-Galois" variants, resp. profinite (restricting ourselves, to simplify, to the case of locally connected topoi…). [p. 6]

4. Supplement-remorse on multi-Galois categories, which makes precise the intention that for a topos $\mathcal{E}$, the datum of an object $S \in \mathcal{E}$ defines an induced topos $\mathcal{E}_{/S} \longrightarrow \mathcal{E}$, and that S is reconstructed up to unique isomorphism from knowledge of the induced topos as a topos over $\mathcal{E}$. Here, † $\mathcal{E}$ being multi-Galois, $\mathcal{E}_{/S}$ too – and the question arises when a morphism of multi-Galois|multi-Galois topoi $\mathcal{E}' \longrightarrow \mathcal{E}$ can be regarded as 15

Uncertainty retained from cancelled passages.
140-2-s016-D001 [illeg. 1] is

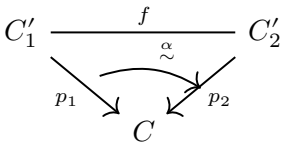
[scan 17]

an induction morphism. Now if $C = \Pi_1(\mathcal{E})$, $C' = \Pi_1(\mathcal{E}')$, the datum of $\mathcal{E}' \longrightarrow |-\mathcal{E}$† is equivalent to the datum of a functor $C' \longrightarrow C$. One finds that $\mathcal{E}' \longrightarrow |-\mathcal{E}$† is an induction morphism iff $C' \longrightarrow C$ is faithful. Thus, one finds an equivalence between the category $\mathcal{E}$ (of objects S of the multi-Galois category $\mathcal{E} \simeq \widehat{C}$, † where C is a groupoid, arbitrary if one insists) and the category whose objects are the "groupoids C' over C", with a faithful structural functor $C' \longrightarrow C$, the morphisms from C'_1 to C'_2 being the isomorphism classes of a|of a † pairs (f, α) of a functor $f : C'_1 \longrightarrow |-C'_2$ and an a bit hasty! 16

Uncertainty retained from cancelled passages.
140-2-s017-D001 ′
140-2-s017-D003 ′
140-2-s017-D005 [illeg. 1]
140-2-s017-D006 [illeg. 1]

[scan 18]

isomorphism of functors $\alpha : p_2 f \simeq p_1$ [p. 7]



m83

Marginal note. m83 Specify the isomorphisms between pairs (f, α) and † (g, β)…

In the case where for example C is the category reduced to a single object, an automorphism group G, this description of the category $\mathcal{E} = \mathrm{Ens}(G)$ is obviously a bit cumbersome, but it fits well into certain contexts below.

Thus, if k is a base field, the † category $\mathcal{E}$ of étale schemes over k is described, in terms of a separable closure k|the k_s of k and of the profinite group $\Gamma = \mathrm{Gal}(k_s/k)$, as the profinite groupoids over the profinite groupoid (pt, Γ)…We want to insert this description into a "Galois" "description" of the|of [certain] schemes [smooth quasi-projective of dimension ≤ 1] over k, at least if k a field of finite type over $\mathbb{Q}$. 17

Uncertainty retained from cancelled passages.
140-2-s018-D001 pairs
140-2-s018-D002 $\mathcal{E}$

[scan 19]

5. Let K be an extension, of finite type over $\mathbb{Q}$, and let us choose an algebraic closure $\overline{K}$ of K. 5.1 We consider pairs (X, S), where: † We set $\Gamma = \mathrm{Gal}(\overline{K}/K)$

a) X is a projective and smooth scheme over K, of dimension ≤ 1;

b) S is a finite reduced subscheme of X, (then [illeg. 1] étale over K), contained † (in) the union of the irreducible components of dimension 1 of X. (thus finite étale over $k|K$)

The morphisms $(X', S') \longrightarrow |-(X, S)$ will by definition be the morphisms of schemes

$$f : X' \longrightarrow X$$

such that

$$S' = f^{-1}(S)_{\mathrm{red}}$$

i.e. such that $\operatorname{supp} S' = f^{-1}(\operatorname{supp} S)$.

We are looking for a "Galois description" of this category, or at least of a full sub-category $\mathcal{V}_K$ which we are going to 18

Uncertainty retained from cancelled passages.
140-2-s019-D001 X
140-2-s019-D002 [illeg. 2]

[scan 20]

define now. [p. 8]

Lemma 5.2. Let Ω be an algebraically closed field, X a smooth projective algebraic curve, connected over Ω, S a finite subset of $X(\Omega)$, g the genus of X, $n = \operatorname{card} S$. †$U = X \setminus S$, Equivalent conditions

a) $\Pi_1(U)$ nonabelian

b) $\mathrm{Aut}(U)$ finite

c) For $\forall$ connected reduced scheme X of finite type over Ω, the set of nonconstant morphisms from X to U is finite

d) We are † in one of the following three cases: 1°) $g \geq 2$ 2°) $g = 1$, $n \geq 1$ 3°) $g = 0$, $n \geq 3$.

m88

Marginal note. m88 e) (if $\Omega \in | \subset \mathbb{C}$) The universal covering of $X(\mathbb{C}) \setminus S(\mathbb{C})$ is isomorphic to the half-plane of Poincaré.

m89

Marginal note. m89 ?!|?? f) (if $\Omega = \overline{\mathbb{Q}}$ $S \neq \emptyset$) The universal covering of $X \setminus S = U$ is isomorphic to that of $\mathbb{P}^1_\Omega \setminus \{0, 1, \infty\}$.

Definition 5.3. We then say that (X, S) is anabelian.

As this condition is (by d) for example) invariant under extension of the algebraically closed base field, we extend this definition to the case of a pair (X, S), with (X, S) as in (5.1) (N.B. we look separately at the connected components of $X_{\overline{K}}$...). 19

Uncertainty retained from cancelled passages.
140-2-s020-D002 X_0
140-2-s020-D004 X

[scan 21]

From now on, in (5.1) we are going to confine ourselves to the case of anabelian pairs (X, S).

5.4. To a pair (X, S) (not necessarily anabelian) — more generally to $\forall$ scheme X locally of finite type over $S|\xi$, we associate an object "of a Galois nature", namely the fundamental groupoid (profinite) $\Pi_1(X)$ of X (formed if we like of the universal coverings of X), equipped with the canonical † homomorphism

$$\begin{array}{ccc} \Pi_1(X) & \longrightarrow & \Pi_1(K) \\ & & \simeq\downarrow \\ & & [\mathrm{Tors}(\Gamma)] \end{array}$$

A morphism of K-schemes

$$X' \xrightarrow{f} X$$

defines a functor

$$\Pi_1(X') \xrightarrow{\Pi_1(f)} \Pi_1(X)$$

and a commutation isomorphism α 20

Uncertainty retained from cancelled passages.
140-2-s021-D002 functor

[scan 22]
[p. 9]

$$\begin{array}{ccc} \Pi_1(X') & \xrightarrow{\Pi_1(f)} & \Pi_1(X) \\ & \overset{\alpha}{\simeq} & \\ & \Pi_1(K) & \end{array}$$

We thus find a functor, from the category of schemes X locally of finite type over K, into the "category of profinite groupoids over $\Pi_1(K)$", defined as in n° 4.

When we pass to the category of connected K-schemes locally of finite type, † equipped with a geometric point, over $\overline{K}/K$ (i.e. with an $x \in X$, a separable closure $\overline{k(x)}$ of $k(x)$ and a K-morphism $\overline{K} \hookrightarrow \overline{k(x)}$), this corresponds to a functor from K-schemes locally of finite type and connected, pointed over $\overline{K}/K$ (in the preceding sense) to the category of † profinite groups † π equipped with a homomorphism (of profinite groups)

$$\pi \longrightarrow \Gamma$$

m92

Marginal note. m92 it is better to say: equipped with a universal covering 21

Uncertainty retained from cancelled passages.
140-2-s022-D001 and
140-2-s022-D003 extensions
140-2-s022-D004 [illeg. 1]

[scan 23]

(whose †image will moreover necessarily be open hence of finite index, for objects coming from X as above).

Conjecture 5.5. The restriction of the preceding functor $X \longmapsto (\Pi_1(X)$ over $\Pi_1(K))$ to smooth projective schemes of dimension ≤ 1 and anabelian (i.e. such that (X, S) is anabelian, where S is the union of the components of dimension 0) is fully faithful. This is a bit false cf. § 9

It comes to the same thing to say this:

Conjecture 5.5 bis The functor which, to any X as in 5.5. and moreover connected, (of dimension 0 or 1), equipped with a geometric point ξ over $\overline{K}$, associates † profinite group $\pi_1(X, \xi)$ over $\Gamma = \pi_1(K, \xi)$), is a ditto 22

Uncertainty retained from cancelled passages.
140-2-s023-D001 a
140-2-s023-D004 the profinite extension

[scan 24]

fully faithful functor. [p. 10]

We must all the same make explicit the morphisms $(X, \xi) \longrightarrow (X', \xi')$ in the source category: morphisms of K-schemes $X \xrightarrow{f} X'$, equipped with a morphism of $\Pi_1(X')$ (or|in class of paths) † $f(\xi) \simeq \xi'$.

These conjectures reduce to Galois theory, for X of dimension 0. For X of dimension 1, they concern only X such that the connected components of $X_{\overline{K}}$ are of genus ≥ 2 (or, what comes to the same thing, introducing the finite extension $K' = H^0(X, \underline{O}_X)$ of K, so that X is geometrically connected over K', such that X as an algebraic † curve over K' is of genus ≥ 2. We see easily (taking $X' = \mathrm{Spec}(K)$, X † $= \mathbf{P}^1_K | \mathbb{P}^1_K$ or an elliptic curve over K) that they become false otherwise — 23

Uncertainty retained from cancelled passages.
140-2-s024-D001 of
140-2-s024-D002 K'-scheme
140-2-s024-D003 [illeg. 1] geometric

[scan 25]

this is why it was necessary to introduce S, plus the anabelian hypothesis on (X, S), in order to associate to (X, S) a richer structure than $\Pi_1(X)$ over $\Pi_1(K)$. We shall find conjectures (for example) for X a geometrically connected curve over K of genus †1 (resp. †0), provided that S † is of degree ≥ 1 (resp. ≥ 3).

6. Local analysis of (X, S) at an $s \in S$.

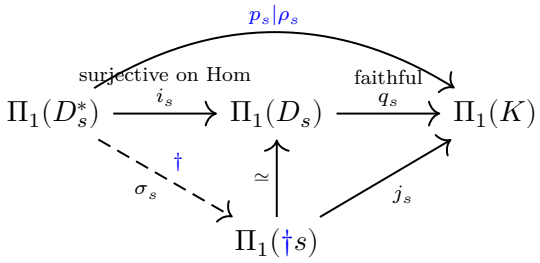
We are interested in the case where $\dim_s(X) = 1$, i.e. where s is not an isolated point of X.

Let O_s be the henselization (or the completion, if we insist) of $\underline{O}_{X,s}$, K_s its field of fractions, $D^*_s = \mathrm{Spec}(K_s)$, — we identify s with $\mathrm{Spec}\, k(s)$ ($k(s)$ is the residue field of the gauge ring O_s). Let us consider $D_s = \mathrm{Spec}(O_s)$, — ("arithmetic disk relative to|: $k(s)$"), thus $D^*_s = D_s \setminus \{s\} =$ ("punctured disk") $\longrightarrow D_s$, 24

Uncertainty retained from cancelled passages.
140-2-s025-D002 0
140-2-s025-D003 1
140-2-s025-D004 restriction of

[scan 26]
[p. 11]

we have
(6.1)



,
for the choice of a geometric point ξ_s of D_s^* over $\overline{K}/K$ (i.e. of an algebraic closure $\overline{K}_s$ of K_s and of a K-injection $\overline{K} \hookrightarrow \overline{K}_s | \overline{K} \subset \overline{K}_s$), this diagram of groupoids is reflected in a homomorphism of groups
(6.2)

$$\begin{array}{ccccc} \pi_1(D_s^*\dagger, \xi_s) & \xrightarrow{\text{surjective}} & \pi_1(k(s), \xi_s) & \xhookrightarrow{\text{injective}} & \Gamma\dagger \\ \wr & & \wr & & \\ \mathrm{Gal}(\overline{K}_s/K_s) & & \mathrm{Gal}(\overline{k(s)}/k(s)) & & \end{array}$$

m99

Marginal note. m99 N.B. the choice of $\overline{K}_s$ implies a choice of $\overline{k}_s$ — this is the dotted arrow in (6.1)

whose kernel, as we know from Kummer, is canonically isomorphic to $T_s(\overline{k_s})|T_\varepsilon(\overline{k_s}) \simeq T(\overline{K}_s)$ [$\simeq T(\overline{K})$]. We want to express the datum of this privileged isomorphism as an additional structure on (6.1) — † i.e.; on the groupoid $\Pi_1(D_s^*)$ over $\Pi_1(s)$ (i.e.|or over $\Pi_1(K)$) —.

We can say it as follows: † If to every † $\xi \in \Pi_1(D_s^*)$, we associate the 25

Uncertainty retained from cancelled passages.
140-2-s026-D001 $p_s|\rho_s$
140-2-s026-D002 [illeg. 1]
140-2-s026-D003 ξ
140-2-s026-D004 [illeg. 1]
140-2-s026-D005 groupoid
140-2-s026-D006 In
140-2-s026-D007 ξ

[scan 27]

kernel of

$$\mathrm{Aut}(\xi) \longrightarrow \mathrm{Aut}(i(\xi))$$

(which is also the kernel of the composite

$$\mathrm{Aut}(\xi) \longrightarrow \mathrm{Aut}(i(\xi)) \longrightarrow \mathrm{Aut}(p_s|\rho_s(\xi)\,\dagger = q_s(i_s(\xi)))$$

we find an abelian group, which depends (up to canonical isomorphism) only on $i(\xi) = \xi'$ [this, and the rest of the sentence, works whenever we have a functor † of connected groupoids, with abelian kernels and surjective on Hom], †, and for varying ξ' forms a local system on $\Pi_1(D_s)$, can call the † relative π_1 of the groupoid $\Pi_1(D_s^*)$ over the the groupoid $\Pi_1(D_s)$. That said, we have an isomorphism of local systems of groups [black right-angle mark, ⌝ or 7]

$$\pi_1(\Pi_1(D_s^*) \text{ over } \Pi_1(D_s)) \simeq q_s^*(\mathcal{T}_K)$$

where $\mathcal{T}_K$ is the Tate local system on K.

11|"
26

Uncertainty retained from cancelled passages.
140-2-s027-D001 [illeg. 1]
140-2-s027-D003 is surjective
140-2-s027-D004 where
140-2-s027-D006 π_1

[scan 28]
[p. 12]

Let us now set

$$D_S = \coprod_{s \in S} D_s \qquad (\text{"arithmetic multidisk at } S\text{"})$$

$$D_S^* = \coprod_{s \in S} D_s^* \qquad (\text{"arithmetic multiannulus at } S\text{"})$$

We have a homomorphism of groupoids

$$\Pi_1(D_S^*) \xrightarrow{\dagger \sigma_S} \Pi_1(S) \quad (\xrightarrow{j_S} \Pi_1(K))$$

and a canonical isomorphism

$$\pi_1(\Pi_1(D_S)/\Pi_1(S)) \simeq j_S^*(\mathcal{T}(K))$$

With this set up, we also have a morphism

$$D_S^* \xrightarrow{\rho_S} X \setminus S$$

inducing

$$\Pi_1(D_S^*) \xrightarrow{\rho_S} \Pi_1(X \setminus S).$$

7. Thus, to (X, S) as in 5.1, we associate:
m104

Marginal note. m104

$$\Pi_U \xrightarrow{\varphi} \Pi_0$$

m105

Marginal note. m105 This groupoid $\Pi_0|\Pi_1$ [illeg. 5] of the connected components which are those of $X \setminus S$, or again of X

27

Uncertainty retained from cancelled passages.
140-2-s028-D001 [illeg. 1]

[scan 29]

†
m107

Marginal note. m107

$$\Pi_D \xrightarrow{\psi} \Pi_0$$

†
†
†

28

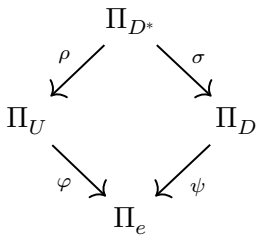
Uncertainty retained from cancelled passages.
140-2-s029-D001 corresponding
140-2-s029-D002 Π_D
140-2-s029-D005 Π_U
140-2-s029-D006 $\Pi_U|\Pi_D$ $\Pi_U|\Pi_D$ $\sigma|\rho$ [illeg. 1]

[scan 30]

†|6°) A [p. 13]

1°) Three (profinite) groupoids Π_U, Π_D, Π_{D^*} (in addition to $\Pi_e = \Pi_1(\mathrm{Spec}(K))$ $(\simeq \mathrm{Tors}(\Gamma))$)

2°) Four functors (of profinite groupoids)



3°) A commutation isomorphism

$$\alpha : \varphi\rho \simeq \psi\sigma$$

(which is even the identity in the case of the system coming from (X, S), but † it is better to forget that this is so).

These data satisfy the † preliminary conditions

a) σ induces an isomorphism on the π_0, and epimorphisms on Hom, and it has abelian kernel.

b) ψ is faithful

[c) ρ is faithful ...

d) φ is epimorphic modulo finite groups on Aut ...]

29

Uncertainty retained from cancelled passages.
140-2-s030-D001 6°) A
140-2-s030-D003 [illeg. 1]
140-2-s030-D004 [illeg. 1]

[scan 31]

The condition a) already allows us to define the relative π_1

$$\pi_1(\sigma) = \pi_1(\Pi_{D^*}/\Pi_D),$$

which is a local system of profinite|profinite abelian groups on Π_D i.e. a functor $(\Pi_D)^o \to$ Ens, and the last datum:

4°) A Kummerian isomorphism

$$\dagger\, k : \quad \pi_1(\sigma) \simeq \psi^*(\mathcal{T})$$

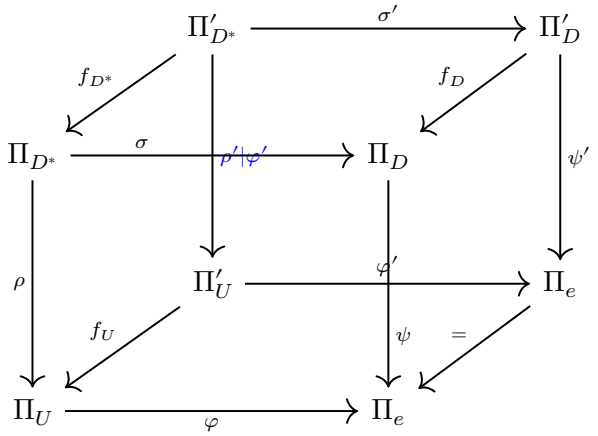
If we have two systems of this nature

$$\Pi = (\Pi_U, \Pi_D, \Pi_{D^*}, \varphi, \psi, \rho, \sigma, \alpha, k)$$

and $\Pi' = (\Pi'_U, \Pi'_D, ...)$, a morphism from Π' to Π is a system of 3 functors

$$f_U : \Pi'_U \to \Pi_U$$
$$f_D : \Pi'_D \to \Pi_D$$
$$\dagger f_{D^*} : \Pi'_{D^*} \to \Pi_{D^*}$$

and of four† commutation isomorphisms $\alpha_{D^*D^*}|\alpha_{D^*D}, \alpha_{D^*U}, \alpha_{Ue}, \alpha_{De}$ for the four faces of the prism



†

Uncertainty retained from cancelled passages.

140-2-s031-D002 [*illeg.* 1]

140-2-s031-D003 [*illeg.* 1]

140-2-s031-D004 [illeg. 1]

140-2-s031-D005 [illeg. 2]

[scan 32]

satisfying an equation of compatibility with α, α' which I do not write down — meaning that [p. 14] the two † isomorphisms u, v of functors

[Two parallel arrows from Π'_{D^*} to Π_e. The upper arrow is labeled $\varphi \circ \rho \circ f_{D^*}$; the lower arrow is labeled $\mathrm{id} \circ \psi' \circ \sigma'$. Between the two, two downward arrows, labeled u and v.]

obtained respectively, u by using successively $\alpha, \alpha_{D^*D}, \alpha_{De}$, † v by using successively $\alpha_{D^*U}, \alpha_{Ue}, \alpha'$, are equal. [This compatibility could be expressed by interpreting the datum of Π as that of a fibered category Π over the † "category square"

$$Q :$$

[Diamond with vertices D^* at the top, U on the left, D on the right, e at the bottom. The side $D^* \to U$ is labeled ρ_0; the other three sides are labeled σ_0 between D^* and D, φ_0 between U and e, ψ_0 between D and e. The arrowheads of these three lines are not clearly discernible.]

where $\varphi_0\rho_0 = \psi_0\sigma_0$.

with restriction to $\{e\}$ prescribed, and taking cartesian functors between fibered categories…]. Moreover, we require another compatibility, namely that

---

Uncertainty retained from cancelled passages.

140-2-s032-D001 [illeg. 1]

140-2-s032-D002 and

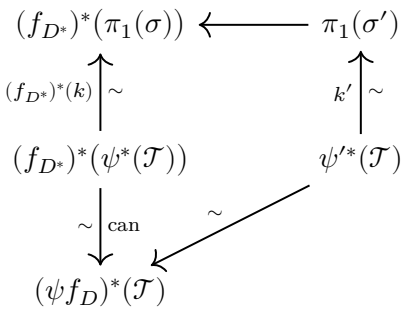
140-2-s032-D003 of diagram in the shape

[scan 33]

the homomorphism of local systems of abelian groups on Π'_D

$$(f_D)^*(\pi_1(\sigma)) \longleftarrow \pi_1(\sigma')$$

defined with the help of: $f_{D^*}, f_D, \alpha_{D^*D}$ make the following diagram of †isomorphisms of local systems on Π'_D commutative



deduced from $\alpha_{De} : \psi' \simeq \psi \circ f_D$ †

m110

Marginal note. m110 no, that|this only works for the cases of étale morphisms, otherwise we must bring in multiplication by "degrees of ramification" $d_{i'}$ ($i' \in \pi_0(\Pi'_{D^*})$).

For $\Pi, \Pi^*|\Pi'$ fixed, the systems

$$(f, \alpha) := (f_U, f_D, f_{D^*}, \alpha_{D^*D}, \alpha_{D^*U}, \alpha_{U,e}, \alpha_{D,e})$$

above form a category in a natural way — in fact a groupoid — by taking as morphisms μ from (f', α') to (f, α) the triples of morphisms (profinite functors)

$$f'_U \xrightarrow{\mu_U} f_U, \qquad f'_D \xrightarrow{\mu_D} f_D, \qquad f'_{D^*} \xrightarrow{\mu_{D^*}} f_D$$

Uncertainty retained from cancelled passages.

140-2-s033-D002 [illeg. 1]

140-2-s033-D004 [illeg. 1]

[scan 34]

satisfying four compatibility conditions with α_{D^*D} and α'_{D^*D}, with α_{D^*U} and α'_{D^*U}, with α_{Ue} and $\alpha'_{U,e}$, with $\alpha_{D,e}$ and $\alpha'_{D,e}$ respectively. (i.e. we are working in a full subcategory of the 2-category of the Hom between fibered categories over Q, with fiber at e fixed…). I have [p. 15]

the impression that this groupoid $\underline{\mathrm{Hom}}((f',\alpha'),(f,\alpha))$ is always rigid, i.e. $\mathrm{Aut}(f,\alpha)$ is always reduced to the trivial group — I'm too lazy to check — hence that if (f',α') and (f,α) are isomorphic, the isomorphism in question is unique. At any rate, we shall put

$\mathrm{Hom}((f',\alpha'),(f,\alpha)) = \pi_0\,\underline{\mathrm{Hom}}((f',\alpha'),(f,\alpha))$

Whence a category of † systems $(\Pi_U, \Pi_D, \Pi_{D^*}, \varphi, \psi, \rho, \sigma, \alpha, \dagger k)$

Uncertainty retained from cancelled passages.

140-2-s034-D002 (X,S)

140-2-s034-D003 [*illeg.* 1]

[scan 35]

We have a functor from pairs (X,S) (i.e.|where X a scheme locally of finite type over K, S a subscheme closed † † of X étale over K (such|such that $\forall s \in S$, X is smooth of relative dimension 1 at s) to this messy category $\mathcal{B}$.

m111

Marginal note. m111 Morphisms|The morphisms $(X',S') \to (X,S)$ are the morphisms $f : X' \to X$ such that $f^{-1}(S)_{\mathrm{reduced}} = S'$.

Messy conjecture 7.1. When we restrict ourselves to the (X,S) such that X smooth projective of dimension ≤ 1, and which moreover are anabelian, then the preceding functor is fully faithful.

N.B. Faithfulness is easy…

Description of the messy category in terms of group theory:

Let $I = \pi_0(\Pi_{D^*}) \simeq \pi_0(\Pi_D)$. Let us choose an element|element D_i^* in each component of Π_{D^*}, and let $D_i = \sigma(D_i^*)$. For every i, let us choose an isomorphism

$$\boxed{\psi(D_i) \xrightarrow[\simeq]{\partial_{D_i}|\lambda_{D_i}} \mathrm{Spec}(\overline{K})}.$$

† Up to replacing the object by an isomorphic "subobject",

Uncertainty retained from cancelled passages.

140-2-s035-D002 [illeg. 1]

140-2-s035-D003 discrete|discrete

140-2-s035-D004 whence [illeg. 3]

[scan 36]

[p. 16]

we may assume that Π_{D^*} is the category sum of the categories $[\mathcal{E}_i]$ defined by the $\mathcal{E}_i = \mathrm{Aut}(D_i^*)$, Π_D †the category sum of categories $[\Gamma_i]$ defined by the $\Gamma_i = \mathrm{Aut}(D_i)$, the functor σ being expressed by a system of homomorphisms

$$\sigma_i : \mathcal{E}_i \to \Gamma_i \qquad (i \in I)$$

which are surjective with abelian kernels, the functor ψ by a system of inclusions $\psi_i : \Gamma_i \subset \Gamma$ and the datum of k is in fact equivalent to isomorphisms

$$k_i : \mathrm{Ker}\,\sigma_i \simeq T(\overline{K})$$

compatible with the operations of Γ_i and Γ on the two sides|sides respectively and the inclusions ψ_i.

m113

Marginal note. m113 *Again N.B. Two different bases define different splittings!

We can say that the datum of (σ, ψ, k) is expressed the datum of the system $(\mathcal{E}_i)_{i\in I}$, of a system of exact sequences †

$$\boxed{1 \to T(\overline{K}) \xrightarrow{k_i} \mathcal{E}_i \xrightarrow{p_i} \Gamma}$$

such that the $\Sigma_i = \operatorname{Im} p_i$ are open, and that k_i is compatible with the operations of $\Sigma_i \simeq \operatorname{Coker} k_i$ (i.e. of $\mathcal{E}_i$ and Γ on the two terms respectively).

m114

Marginal note. m114 N.B. The extensions of the Γ_i by $T(\overline{K})$ obtained from geometric situations split — [illeg. 1] the choice of a uniformizer at s_i defines a splitting, and even [illeg. 1] the choice of a basis of the tangent space at s…

Uncertainty retained from cancelled passages.

140-2-s036-D002 [illeg. 1]

140-2-s036-D005 [illeg. 5] [illeg. 1]

[scan 37]

Assuming on the other hand (to simplify) $\dagger\Pi_U$ connected, † and choosing an element $\widetilde{U}$ of Π_U and an isomorphism

$$\varphi(\widetilde{U}) \xrightarrow{\partial_U|\lambda_U} \operatorname{Spec}(\overline{K})$$

thus (up to replacing Π_U by an equivalent groupoid) we may assume Π_U reduced to $\widetilde{U}$, and Π_U is then given by a group $\mathcal{E}$, and φ by a group homomorphism

$$\boxed{\mathcal{E} \xrightarrow{\dagger\, \varphi_{\widetilde{U},\partial_U}|\varphi_{\widetilde{U},\lambda_U}} \Gamma}$$

(or p) whose image $\Sigma \subset \Gamma$ is thus an open subgroup of Γ, and the kernel will be denoted $\pi|\Pi$

$$\boxed{1 \to \pi|\Pi \xrightarrow{\kappa|\iota} \mathcal{E} \xrightarrow{\varphi|p} \Sigma_0 \to 1}$$

[Below Σ_0, a vertical inclusion sign toward Γ.]

m115

Marginal note. m115 Well noted In geometric situations N.B. (This extension of group kernels splits, i.e. ∃ an open subgroup Σ_0 in Γ, † and a lifting $\Sigma_0 \to \mathcal{E}$…

[Having replaced the initial Π_U by a smaller full subcategory, we shall be obliged to modify ρ up to isomorphism, in practice by choosing for each $i \in I$ an isomorphism

$$\rho(D_i) \xrightarrow{\mu_i} \widetilde{U}].$$

†

Uncertainty retained from cancelled passages.

140-2-s037-D001 X

140-2-s037-D002 i.e.

140-2-s037-D004 [*illeg.* 1]

140-2-s037-D005 of index [illeg. 1]

140-2-s037-D006 [illeg. 1]

[scan 38]

[p. 17]

whereupon ρ is made explicit simply by group homomorphisms

$$\boxed{\rho_i : \mathcal{E}_i \longrightarrow \mathcal{E}}$$

It remains to make explicit the commutation isomorphism

$$\alpha : \varphi\rho \to \psi\sigma$$

which is defined by a system of elements

$$\boxed{\gamma_i \in \Gamma \qquad (i \in I)}$$

such that

$$(\alpha)|(*) \qquad \varphi \circ \rho_i = \mathrm{int}(\gamma_i) \circ p_i.$$

which incidentally implies that ρ_i † maps $\mathrm{Ker}\, p_i$ into $\mathrm{Ker}\, p$, i.e. induces a homomorphism of exact sequences

$$\begin{array}{ccccccccc} 1 & \longrightarrow & T(\overline{K}) & \longrightarrow & \mathcal{E}_i & \longrightarrow & \dagger\Sigma_i & \longrightarrow & 1 \\ & & \downarrow & & \downarrow{\scriptstyle \rho_i} & & \downarrow & & \\ 1 & \longrightarrow & \pi|\Pi & \longrightarrow & \mathcal{E} & \longrightarrow & \dagger\Sigma_0 & \longrightarrow & 1 \end{array}$$

[To the right of Σ_i, as a superscript, $\subset \Gamma$; below Σ_0, an inclusion sign and Γ.]

with an induced injective homomorphism $\Gamma_i \to \Gamma_0$ — and this being posited, the compatibility relation $(\alpha)|(*)$ becomes a relation on group monomorphisms

$$\begin{array}{ccc} \Gamma_i & \hookrightarrow & \Gamma \\ & \searrow & \\ & \Gamma_0 \hookrightarrow & \Gamma \end{array}$$

Uncertainty retained from cancelled passages.

140-2-s038-D001 [illeg. 1]

140-2-s038-D002 $\mathcal{E}_i$

140-2-s038-D003 $\mathcal{E}_0$

[scan 39]

†

Thus, we have a relatively simple description of the "reduced" messy systems (i.e. where in the groupoids Π_U, Π_{D^*}, Π_D, each connected component has exactly one object, and where moreover we force in some sense $\Pi_1(K)|\Pi_e(K)$ to have only the object $\mathrm{Spec}(\overline{K})$. But we are found again on turning for|by the determination of the morphisms of the systems

$$\mathcal{G} = (I, G_i, p_i, k_i, \mathcal{E}, \varphi, \rho_i, \gamma_i) \quad \text{and} \quad \mathcal{G}' = (I', G'_{i'}, ...),$$

(or p?) † say in the direction

$$f : \mathcal{G}' \to \mathcal{G}.$$

We therefore need (for f_{D^*}) a map

$$\boxed{\dagger\tau = \tau_f : I' \to I},$$

and $\forall i'|i$ a homomorphism

$$\boxed{G'_{i'} \xrightarrow{f_{i'}} G_{\tau i'}},$$

† (taking f_D into account) inducing by passage to the quotient homomorphisms

$$\Gamma'_{i'} \xrightarrow{g_{i'}} \Gamma_{\tau i'}$$

†

m116

Marginal note. m116 The "simplified messy objects" are thus the systems of homomorphisms of exact sequences

$$\begin{array}{ccccccc} 1 & \longrightarrow & T(\overline{K}) & \xrightarrow{k_i} & \mathcal{E}_i & \xrightarrow{p_i} & \Gamma \\ & & \downarrow{\scriptstyle \rho_i^0} & & \downarrow{\scriptstyle \rho_i} & & \downarrow{\scriptstyle \mathrm{int}(\gamma_i)} \\ 1 & \longrightarrow & \pi|\Pi & \xrightarrow{\kappa} & \mathcal{E} & \xrightarrow{p} & \Gamma \end{array}$$

Uncertainty retained from cancelled passages.

140-2-s039-D001 arising $\pi|\Pi$ subgroup

140-2-s039-D002 [illeg. 2]

140-2-s039-D003 [*illeg.* 1]

140-2-s039-D004 by [illeg. 1]

140-2-s039-D005 but

[scan 40]

of open subgroups of Γ, and the datum $f_{D^*}, f_D, \alpha_{D^*,D}, \alpha_{D,e}$ is then equivalent to the datum [p. 18]
of elements α_i $(i \in I|i' \in I')$ of Γ, $\boxed{\alpha_{i'} \in \Gamma}$ $(i' \in I')$ such that †

$$g_{i'}(\gamma') = \mathrm{int}(\alpha_{i'})(\gamma') \qquad \forall i' \in I', \quad \gamma' \in \Gamma'_{i'}.$$

The datum of f_U is equivalent to the datum of a group homomorphism

$$\boxed{f_{\mathcal{E}} : \mathcal{E}' \to \mathcal{E}}$$

that of $\alpha_{D,e}$ is equivalent to the datum of

$$\boxed{\alpha \in \Gamma}$$

such that

$$(\alpha)|(*) \qquad \varphi f_{\mathcal{E}} = \mathrm{int}(\alpha)\varphi$$

so that $f_{\mathcal{E}}$ induces a homomorphism of exact sequences

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \pi|\Pi' & \longrightarrow & \mathcal{E}' & \longrightarrow & \Gamma'_0 & \longrightarrow & 1 \\ & & \downarrow & & \downarrow{\scriptstyle f_{\mathcal{E}}} & & \downarrow{\scriptstyle f_{\Gamma_0}|f_{\Sigma_0}} & & \\ 1 & \longrightarrow & \pi|\Pi & \longrightarrow & \mathcal{E} & \longrightarrow & \Gamma_0 & \longrightarrow & 1 \end{array}$$

[Above Γ'_0, Γ with a vertical inclusion sign; below Γ_0, an inclusion sign and Γ.]

and with this, the condition called $(\alpha)|(*)$ becomes a condition on inclusions of subgroups of Γ:

$$f_{\Gamma_0}(\gamma') = \mathrm{int}(\alpha)\gamma' \quad \text{if} \quad \gamma' \in \Gamma'_0.$$

We must still make explicit (in addition to the $f_{D^*}, f_D, f_U, \alpha_{D^*D}, \alpha_{De}, \alpha_{Ue}$ already made explicit) the

Uncertainty retained from cancelled passages.
140-2-s040-D001 [illeg. 2]

[scan 41]

datum of the commutativities α_{D^*U}, and write the compatibility conditions with α, α' and k, k'. The datum of $\alpha_{D^*,U}$ is equivalent to that of systems of elements

$$\boxed{\beta_{i'} \in \mathcal{E}'} \qquad (i' \in I')$$

†such that one has, in the diagrams

$$\begin{array}{ccc} G'_{i'} & \xrightarrow{\rho'_{i'}} & \mathcal{E}' \\ {\scriptstyle f_{i'}}\downarrow & & \downarrow{\scriptstyle f_{\mathcal{E}}} \\ G_{\tau i'} & \xrightarrow[\rho_{\tau i'}]{} & \mathcal{E} \end{array}$$

the relation

$$f_{\mathcal{E}}\rho'_{i'} = \mathrm{int}(\beta_{i'})\rho_{\tau i'} \circ f_{i'}.$$

It remains to express the two compatibilities of $(f_{D^*}, f_D, f_U, \alpha_{D^*D}, \alpha_{De}, \alpha_{D^*U}, \alpha_{Ue})$ with itself and with K — the second compatibility is simply the compatibility of the $f_{i'}$ with the $k'_{i'}, k_i$, i.e.

$$f_{i'} \circ k'_{i'} = k_i \qquad (i = \tau i')$$

and the first unless I am mistaken is expressed by the "commutativity"

$$\dagger\alpha\gamma'_{i'} = \rho(\beta_{i'})|p(\beta_{i'})\gamma_i\alpha_{i'} \qquad \forall i' \in I'$$

$$\boxed{\alpha_{i'} = \gamma_i^{-1}\rho|p(\beta_{i'}^{-1})\alpha\gamma'_{i'}}$$

40

Uncertainty retained from cancelled passages.
140-2-s041-D001 [illeg. 1]
140-2-s041-D003 $\rho(\beta_{i'})|p(\beta_{i'})$

[scan 42]

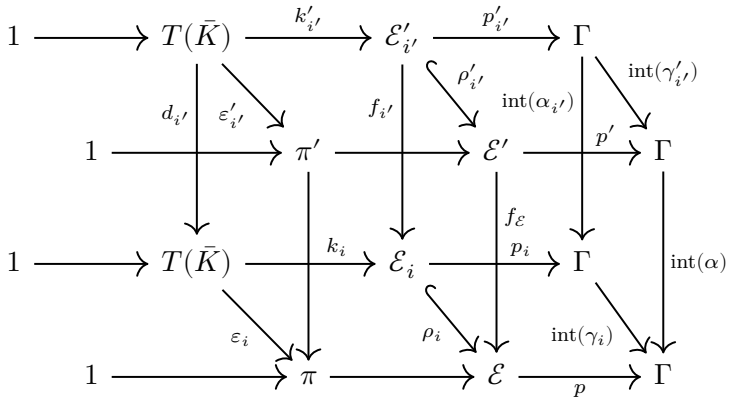
In summary, the homomorphisms, into a system of † commutative diagrams [p. 19]

$$\begin{array}{ccccccc} 1 & \to & T(\bar{K}) & \xrightarrow{k_i} & \mathcal{E}_i & \xrightarrow{p_i} & \Gamma \\ & & & \searrow & \downarrow{\scriptstyle \rho_i} & \searrow{\scriptstyle \mathrm{int}(\gamma_i)} & \\ & & 1 & \longrightarrow & \pi & \to \mathcal{E} \xrightarrow[p]{} & \Gamma \end{array} \qquad (i \in I)$$

from an analogous system, relative to a set of indices I', is given by a map

$$\tau : I' \longrightarrow I$$

and for every $i' \in I'$, setting $i = \tau(i')$, of a system of vertical arrows $f_{i'} : \mathcal{E}'_{i'} \longrightarrow \mathcal{E}_i$, and of a vertical arrow $f_{\mathcal{E}} : \mathcal{E}' \longrightarrow \mathcal{E}$, finally of a system of elements $\beta_{i'} \in \mathcal{E}$ and of an $\alpha \in \Gamma$, fitting into the system of diagrams



[Prism: the second and fourth rows are shifted to the right. The vertical arrow $d_{i'}$ between the two $T(\bar{K})$ is drawn with a double stroke; $\pi' \longrightarrow \pi$ without a label. Below the arrow p_i, near the tip of ρ_i, $\beta_{i'}$ overwritten; farther along the same arrow, near the tip toward Γ, an overwritten sign [illeg. 1].]

m118

Marginal note. m118 N.B. Since the $\mathcal{E}_i \hookrightarrow \mathcal{E}$ are injective, $f_{i'}$ is known when one knows $f_{\mathcal{E}}$ and $\beta_{i'}$ (the existence of|of the $f_{i'}$ is therefore a condition on the pairs $f_{\mathcal{E}}, \beta_{i'}$ [illeg. 1] namely that $\mathrm{int}(\beta_{i'})^{-1} f_{\mathcal{E}} \rho'_{i'}$, $\mathcal{E}'_{i'}$ into $\rho_i(\mathcal{E}_i)|p_i(\mathcal{E}_i)$)

m119

Marginal note. m119 N.B. We may assume the $\gamma'_{i'}, \gamma_i$ equal to 1 (by choosing the image of $\psi|\varphi\sigma(D_i^*)$ over $\mathrm{Spec}(\bar{K})$ via the isomorphism of $\varphi(\rho(D_i^*)) \simeq \varphi(\widetilde{U}|\widetilde{U}_i)$ with $\mathrm{Spec}\,\bar{K}$

where we have set

$$(*) \quad \alpha_{i'} = \gamma_i^{-1} p(\beta_{i'}^{-1}) \alpha \gamma'_{i'} \qquad \text{i.e.} \quad \dagger\ \alpha \gamma'_{i'} = p(\beta_{i'}) \gamma_i \alpha_{i'}$$

and i.e.|where the rear vertical face of the prisms is commutative (two conditions, on two squares), the front vertical face too (this is one condition, on the right-hand square, the other square 41

Uncertainty retained from cancelled passages.
140-2-s042-D001 [illeg. 1]
140-2-s042-D006 [*illeg.* 1]

[scan 43]

being commutative follows from this by definition of $\pi' \longrightarrow \pi$ as induced by $f_{\mathcal{E}}$ ...), the left vertical face of the right-hand cube being commutative modulo the commutativity isomorphism $\mathrm{int}(\beta_{i'})$, and the right vertical face being †commutative (not only, by the preceding condition, on

$$\Gamma'_{i'} = \mathrm{Im}(\mathcal{E}'_{i'} \xrightarrow{p'_{i'}} \Gamma),$$

but on the whole of Γ) by virtue of the more precise relation $(*)$.

Messy category made precise (corresponding to a Π_U that is connected):
These objects are the †homomorphisms of groups that are profinite †

$$\mathcal{E} \xrightarrow{p} \Gamma$$

(giving rise to an exact sequence:

$$1 \longrightarrow \pi \xrightarrow{k|\kappa} \mathcal{E} \xrightarrow{p} \Gamma$$

) and a finite set of subgroups (indexed by a set I of indices)

$$\mathcal{E}_i \overset{\rho_i}{\hookrightarrow} \mathcal{E},$$

and of isomorphisms

$$\mathcal{E}_i \cap \pi \overset{k_i}{\simeq} T(\bar{K})$$

42

Uncertainty retained from cancelled passages.
140-2-s043-D001 cont
140-2-s043-D002 systems
140-2-s043-D003 indexed

[scan 44]

A homomorphism from a system $(\mathcal{E}', p', (\mathcal{E}'_{i'})_{i' \in I'}, (k'_{i'})_{i' \in I'})$ into a system $(\mathcal{E}, p, (\mathcal{E}_i)_{i \in I}, (k_i)_{i \in I})$ [p. 20] is given by by a homomorphism

$$f : \mathcal{E}' \longrightarrow \mathcal{E}$$

$$\beta_{i'} \in \mathcal{E} \quad (i' \in I'), \qquad \alpha \in \Gamma,$$

such that one has the conditions:



m122

Marginal note. m122 The $\beta_{i'}$ not unique (if $\beta_{i'}$ works, also $\gamma\beta_{i'}$, with $\gamma|\gamma_i \in \operatorname{Im} k_i$). But α unique ?!|??

1°) $p \circ f_{\mathcal{E}} = \operatorname{int}(\alpha)\, p'$

2°) $\forall i' \in I', \quad \exists i \in I$ (unique!) such that

$$\operatorname{int}(\beta_{i'}^{-1}) f(\mathcal{E}'_{i'}) \subset \mathcal{E}_i$$

and an integer $d_{i'} \in \mathbb{N}^*$ such that

$$\operatorname{int}(\beta_{i'}^{-1}) f k'_{i'} = k_i \circ (d_{i'} \cdot \mathrm{id}_{T(\bar{K})})$$

m124

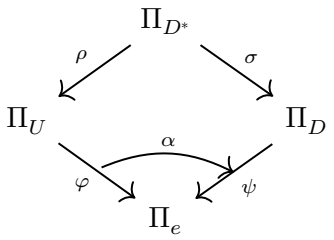
N.B.: $d_{i'}$ is also unique…

Marginal note. m124 [thus $\operatorname{int}(\beta_{i'}^{-1})\overbrace{\operatorname{Im}(f k'_{i'})|\operatorname{Im} k'_{i'}}^{L'_{i'}} \subset \underbrace{\operatorname{Im} k_i}_{L_i}$ and even † $\operatorname{int}(\beta_{i'}^{-1}) L'_{i'} = d_{i'} L_i$.]

I realize that it is better to replace the $\beta_{i'}$ by the $\beta_{i'}^{-1}$, i.e. to take the commutation isomorphism rather in the direction

$$f_{\mathcal{E}}\rho'_{i'} \xrightarrow{\beta_{i'}} \rho_i f_{i'}$$

than in the opposite direction. Moreover, conceptually the diagram



is too complicated, it suffices to give oneself

$$\boxed{\Pi_{D^*} \xrightarrow{\rho} \Pi_U \xrightarrow{\varphi} \Pi_e}$$

and to deduce Π_D by canonical factorization 43

Uncertainty retained from cancelled passages.
140-2-s044-D001 this

[scan 45]

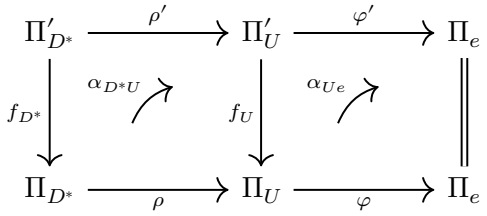
of the homomorphism of groupoids $\Pi_{D^*} \longrightarrow \Pi_e$ into a homomorphism which induces an isomorphism on π_0 and an epimorphism on the π_1, followed by a homomorphism which is (an epimorphism on π_0, and for good reason!, and which is) faithful. Then the 1-morphisms from

$$\Pi'_{D^*} \xrightarrow{\rho'} \Pi'_U \xrightarrow{\varphi'} \Pi_e$$

to

$$\Pi_{D^*} \xrightarrow{\rho} \Pi_U \xrightarrow{\varphi} \Pi_e$$

are the †quintuples of 2 functors and 3 isomorphisms of functors $(f_{D^*}, f_U, \alpha_{D^*U}, \alpha_{Ue}, K$ giving a diagram with commutation data



,
m126

Marginal note. m126 moreover, † $\pi_1(\varphi\rho) = \pi_1(\Pi_{D^*}/\Pi_e)$ can be defined as a local system of groups (commutative|abelian) on Π_{D^*}, and we can then define|write $\boxed{\bar{k}|k : (\varphi\rho)^* T_K|\mathcal{T}_K \simeq \pi_1(\varphi\rho)}$.

m127

Marginal note. m127 with one condition on the induced morphism $\pi_1(\varphi'\rho')\simeq | \longrightarrow f_{D^*}^*(\pi_1(\varphi\rho))$, which modulo the Kummer isomorphisms must be multiplication by a d ramification index (which is a map $\pi_0(\Pi'_{D^*}) \longrightarrow \mathbb{N}^*$). [N.B. multiplication by an integer that is positive [illeg. 5] constant…]

if we have two such 1-morphisms

$$f = (f_{D^*}, f_U, \alpha_{D^*U}, \alpha_{Ue}) \quad \text{and} \quad g = (g_{D^*}, g_U, \beta_{D^*U}, \beta_{Ue})$$

a 0-morphism|homomorphism from f to g is formed of a pair of μ_{D^*}, μ_U of isomorphisms of functors

$$\mu_{D^*} : f_{D^*} \simeq g_{D^*}, \qquad \mu_U : f_U \longrightarrow g_U$$

compatible with $\alpha_{D^*U}, \beta_{D^*U}$, and with α_{Ue}, β_{Ue}, 44

Uncertainty retained from cancelled passages.
140-2-s045-D002 pairs
140-2-s045-D003 [illeg. 1]

[scan 46]

(two compatibility conditions on the two squares). [p. 21]

If we restrict ourselves to connected Π_U, Π'_U, we find by choosing as above an object $\widetilde{U}$ of Π_U and an isomorphism

$$\boxed{\varphi(\widetilde{U}) \overset{\lambda}{\simeq} \Omega}$$

(where $\Omega = \mathrm{Spec}(\bar{K})$ is the reference object of Π_e ...), a group $\mathcal{E}$ and a homomorphism

$$\mathcal{E} \overset{\varphi}{\to} \Gamma$$

(which is changed by an inner automorphism if we change λ, and $\mathcal{E}$ itself replaced by an isomorphic group, with isomorphism defined modulo inner isomorphisms, if we change the reference object $\widetilde{U}$). Likewise, choosing a $\widetilde{U}_i$ in each component $i \in \pi_0(\Pi_{D^*})$, and isomorphisms

$$\lambda_i : \rho(\widetilde{U}_i) \simeq \widetilde{U},$$

we find groups $\mathcal{E}_i$ and homomorphisms

$$\mathcal{E}_i \overset{\rho_i}{\longrightarrow} \mathcal{E}.$$

If we change λ_i, ρ_i is changed by an inner automorphism of $\mathcal{E}$. If we change $\widetilde{U}_i$, $\mathcal{E}_i$ is replaced by an isomorphic group, the isomorphism defined up to an inner automorphism. 45

[scan 47]

Let us consider the composites

$$\mathcal{E}_i \longrightarrow \mathcal{E} \overset{p}{\longrightarrow} \Gamma, \qquad \mathcal{E}_i \overset{p_i}{\longrightarrow} \Gamma$$

whence kernel, † $K|k$ is defined by a system of isomorphisms

$$T(\bar{K}) \overset{k_i}{\simeq} \mathrm{Ker}\, p_i \qquad (i \in I)$$

fitting into a family of exact sequences

$$1 \longrightarrow T(\bar{K}) \overset{k_i}{\longrightarrow} \mathcal{E}_i \overset{p_i}{\longrightarrow} \Gamma$$

(N.B. the image of p_i is an open subgroup) k_i being compatible with the actions of $\mathcal{E}_i$, when we let $\mathcal{E}_i$ act on $T(\bar{K})$ through the action of Γ on $T(\bar{K})$, and on itself by inner automorphisms. Introducing also $\pi = \mathrm{Ker}\, p$, we thus find homomorphisms of exact sequences

$$(D)\qquad \begin{array}{ccccccc} 1 & \longrightarrow & T(\bar{K}) & \xrightarrow{k_i} & \mathcal{E}_i & \xrightarrow{p_i} & \Gamma \\ & & {\scriptstyle \varepsilon_i\,\dagger}\downarrow & & \downarrow{\scriptstyle \rho_i} & & \| \\ 1 & \longrightarrow & \pi & \xrightarrow[k|\kappa]{} & \mathcal{E} & \xrightarrow[p]{} & \Gamma \end{array} \qquad i \in I$$

We can say that Π_U defines an "outer group" $[\mathcal{E}]$, and $\Pi_U \longrightarrow \Pi_e$ a homomorphism of outer groups

$$[\mathcal{E}] \longrightarrow [\Gamma],$$

likewise Π_{D^*} defines a system of "outer groups" $[\mathcal{E}_i]$, and $\dagger\ \Pi_{D^*} \longrightarrow \Pi_U$ 46

Uncertainty retained from cancelled passages.

140-2-s047-D001 [illeg. 1]

140-2-s047-D002 [*illeg.* 1]

140-2-s047-D004 of the homomorphism

[scan 48]

a system of outer homomorphisms [p. 22]

$$[\mathcal{E}_i] \longrightarrow [\mathcal{E}],$$

but how in terms of outer groups to express the Kummer data k_i?

Returning to the system of diagram D (relative to an index set I.) and to an analogous D' (with index set I')

$$D'\qquad \begin{array}{ccccccc} 1 & \longrightarrow & T(\bar{K}) & \xrightarrow{k'_{i'}} & \mathcal{E}'_{i'} & \xrightarrow{p'_{i'}} & \Gamma \\ & & {\scriptstyle \varepsilon'_{i'}}\downarrow & & \downarrow{\scriptstyle \rho'_{i'}} & & \| \\ 1 & \longrightarrow & \pi|\pi' & \xrightarrow[k'|\kappa']{} & \mathcal{E}' & \xrightarrow[p']{} & \Gamma \end{array} \qquad i' \in I|I'$$

a 1-morphism f from D' to D is made explicit by

a) A homomorphism

$$\boxed{f_{\mathcal{E}} : \mathcal{E}' \longrightarrow \mathcal{E}}$$

[†is made explicit in terms of the choice of an isomorphism describes the † functor f_U

$$\nu : f(\widetilde{U}') \simeq \widetilde{U}$$

[another choice modifies $f_{\mathcal{E}}$ by an $\mathrm{int}(\beta)$, $\beta \in \mathcal{E}$)]

b) A map $\tau : I' \longrightarrow I$, and for every $i' \in I'$, putting $i = \tau(i')$, a group homomorphism

$$\boxed{\mathcal{E}'_{i'} \xrightarrow{f_{i'}} \mathcal{E}_i}$$

describes the functor f_{D^*}

47

Uncertainty retained from cancelled passages.

140-2-s048-D002 corresponding

140-2-s048-D003 depends

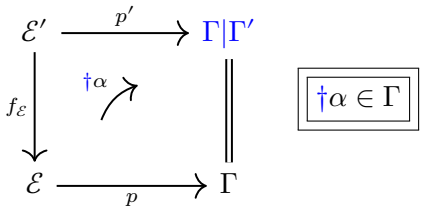
[scan 49]

[is made explicit in terms of the choice of an isomorphism

$$f_{D^*}(\widetilde{U}'_{i'}) \overset{\nu_{i'}}{\simeq} \widetilde{U}_i,$$

and modified by $\mathrm{int}(\beta_{i'})$, $\beta_{i'} \in \mathcal{E}_i$, if we change $\nu_{i'}$]

c) A commutation datum for

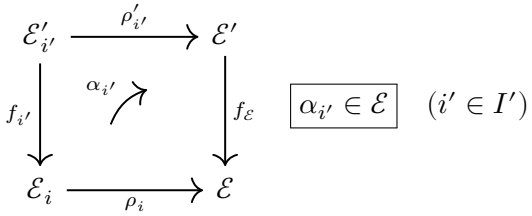


i.e.

α describes α_{Ue}

$$\dagger \qquad \boxed{pf_{\mathcal{E}} = \mathrm{int}(\alpha)p'}$$

d) $\forall i'[\in I|I']$ a commutation datum for

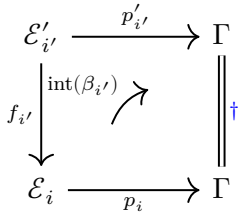


i.e.:

$(\alpha_{i'})$ describes α_{D^*U}

$$\boxed{\rho_i f_{i'} = \mathrm{int}(\alpha_{i'}) f_{\mathcal{E}} \rho'_{i'}}$$

Let us note that c, d together define a commutation datum



with $\beta_{i'} = p(\alpha_{i'})\alpha$ i.e.

$$p_i f_{i'} = \mathrm{int}(\beta_{i'}) p'_{i'}$$

48

Uncertainty retained from cancelled passages.

140-2-s049-D001 β

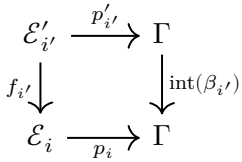
140-2-s049-D002 β

140-2-s049-D003 β

140-2-s049-D004 γ

[scan 50]

i.e. we have commutativity in [p. 23]



so $f_{i'}$ induces a homomorphism of exact sequences

$$\begin{array}{ccccccc} 1 & \longrightarrow & T(\bar{K}) & \xrightarrow{k'_{i'}} & \mathcal{E}'_{i'} & \xrightarrow{p'_{i'}|f'_{i'}} & \Gamma \\ & & \downarrow{\scriptstyle f^0_{i'}} & & \downarrow{\scriptstyle f_{i'}} & & \downarrow{\scriptstyle \mathrm{int}(\beta_{i'})} \\ 1 & \longrightarrow & T(\bar{K}) & \xrightarrow[k_i]{} & \mathcal{E}_i & \xrightarrow[p_i]{} & \Gamma \end{array}$$

795

and we must express the compatibility of $f^0_{i'}$ with the ramification index $d_{i'} \in \mathbb{N}$, we find

$$\boxed{f^0_{i'} = d_{i'}\chi(\beta_{i'})\,\mathrm{id}_{T(\bar{K})}}$$

where

$$\chi : \Gamma \longrightarrow \widehat{\mathbb{Z}}^*$$

is the canonical character.

This being so, let us determine, for a 1-morphism

$$f = (f_{\mathcal{E}}, \tau, (f_{i'}), \alpha, (\alpha_{i'}|\alpha_i))$$

and another

$$f' = (f'_{\mathcal{E}}, \tau', (f'_{i'}), \alpha', (\alpha'_{i'}|\alpha'_i)),$$

the 0-morphisms from one to the other.

49

[scan 51]

corresponding|correspond to systems of isomorphisms.

$$\mu_{\mathcal{D}^*} : f_{\mathcal{D}^*} \longrightarrow f'_{\mathcal{D}^*}, \qquad \mu_{U^*}\ f_U \longrightarrow g_U.$$

This corresponds conditions $\tau' = \tau$, $d'_{i'} = d_{i'}\ \forall i' \in I'$ and to the datum of this follows from the other conditions... m139

Marginal note. m139 let us express the existence of an isomorphism $\mu_{\mathcal{D}^*}$ describes μ_U

a) $\boxed{\mu \in \mathcal{E}}$, defining an isomorphism between

$$f_{\mathcal{E}} : \mathcal{E}' \longrightarrow \mathcal{E} \quad \text{and} \quad f'_{\mathcal{E}} : \mathcal{E}' \longrightarrow \mathcal{E}$$

i.e. † such that

$$\boxed{f'_{\mathcal{E}} = \mathrm{int}(\mu) f_{\mathcal{E}}}\ )$$

b) $\boxed{\mu_{i'} \in \mathcal{E}_i}$ $(i' \in I',\ i = \tau(i'))$ satisfying describes $\mu_{\mathcal{D}^*}$

$$\boxed{f'_{i'} = \mathrm{int}(\mu_{i'}) f_{i'}}),$$

the $\underset{\in \mathcal{E}}{\mu}$, $\underset{\in \mathcal{E}_i}{\mu_{i'}}$ being related to the $\underset{\in|\notin \mathcal{E}}{\alpha}, \alpha'$ †|in $(\in \mathcal{E})$ to the $\alpha_{i'}, \alpha'_{i'}$ $(\in \Gamma)$ in the following way this already implies $d_{i'} = d'_{i'}$

$$\boxed{\alpha' = p(\mu)\alpha}$$

$$\boxed{\alpha'_{i'} = \rho_i(\mu_{i'})\alpha_{i'}} \qquad \forall i' \in I'$$

or again

$$\boxed{\begin{cases} p(\mu) = \alpha'\alpha^{-1} \\ \rho_i(\mu_{i'}) = \alpha'_{i'}\alpha_{i'}^{-1} \end{cases}}$$

50

Uncertainty retained from cancelled passages.
140-2-s051-D001 making
140-2-s051-D002 in

[scan 52]

Note that, since ρ_i is injective, the second relations, giving the $\rho_i(\mu_{i'})$, determine the $\mu_{i'}$ in [p. 24]
a unique way, the existence of these $\mu_{i'}$ is equivalent to the relations

$$\boxed{\alpha'_{i'}\alpha_{i'}^{-1} \in \rho_i(\mathcal{E}_i)} \qquad i' \in I'.$$

As for the relation $p(\mu) = \alpha'\alpha^{-1}$, it determines μ modulo multiplication (on the right, say) by a † $\mu_0 \in \operatorname{Ker} p = \pi$, but which must be such that we still have

$$f'_{\mathcal{E}} = \operatorname{int}(\mu\mu_0) f_{\mathcal{E}}$$

(in addition to $f'_{\mathcal{E}} = \operatorname{int}(\mu) f_{\mathcal{E}}$) which means that

$$\operatorname{int}(\mu_0) f_{\mathcal{E}} = f_{\mathcal{E}} | f'_{\mathcal{E}}$$

i.e.

$$\boxed{\mu_0 \in \operatorname{Centr}_\pi f_{\mathcal{E}}(\mathcal{E}')}$$

So that is the exact indeterminacy in the choice of a 0-morphism the|of the 1-morphisms f and f' from $\mathcal{D}'$ to $\mathcal{D}$. If for example we are under the conditions where $\pi' \longrightarrow \pi$ induced by $f_{\mathcal{E}} : \mathcal{E}' \longrightarrow \mathcal{E}$ has an image of finite index (open image) — case of a "nonconstant morphism"! — then μ_0 must centralize an open subgroup of 51

Uncertainty retained from cancelled passages.
140-2-s052-D001 [illeg. 1]

[scan 53]

π — unless I am mistaken this too implies $\mu_0 = 1$, so it does seem that † (in the cases corresponding to dominant homomorphisms of algebraic curves) a 0-morphism from an f to an f' (if it exists) is unique, i.e. Hom $(\mathcal{D}', \mathcal{D}^*)$ is a discrete category.

We can interpret $\mathcal{D}$ as $\mathcal{E}$ equipped with subgroups $\mathcal{E}_i \subset \mathcal{E}$, and with $p : \mathcal{E} \longrightarrow \Gamma_*|\Gamma$ (kernel π) finally with isomorphisms

$$k_i : \mathrm{T}(\overline{K}) \xrightarrow{\sim} \mathcal{E}_i \cap \pi = L_i$$

(commuting with the action of $\mathcal{E}_i$). If we have a|a different system

$$(\mathcal{E}',\ (\mathcal{E}'_{i'})_{i' \in I'},\ p' : \mathcal{E}' \longrightarrow \Gamma',\ (k'_{i'})_{i' \in I'}),$$

$\mathcal{E}'_{i'} \subset \mathcal{E}'$ a morphism from the first to the second is defined by a

$$f : \mathcal{E}' \longrightarrow \mathcal{E}$$

satisfying the following conditions

b) pf is a conjugate of f' [let $\alpha \in \Gamma$ be such that $pf = \operatorname{int}(\alpha)p'$]

a) $\forall i' \in I', \quad \exists i \in I$ such that $f(\mathcal{E}'_{i'})$ is contained in a conjugate of $\mathcal{E}_i$ [let $\alpha_{i'} \in \mathcal{E}$ be such that $f(\mathcal{E}'_{i'}) \subset | = \operatorname{int}(\alpha_{i'}^{-1})(\mathcal{E}_i)$] i.e. $\operatorname{int}(\alpha_{i'}) f(\mathcal{E}'_{i'}) \subset \mathcal{E}_i$ 52

Uncertainty retained from cancelled passages.
140-2-s053-D001 [illeg. 1]

[scan 54]

N.B. The $i \in I$ corresponding to $i' \in I'$ is unique, whence $\tau : I' \longrightarrow I$. If the $\alpha_{i'}$ are chosen, [p. 25]
we deduce † induced homomorphisms $f_{i'} : \mathcal{E}'_{i'} \longrightarrow \mathcal{E}_i$,

$$f_{i'} = (\mathrm{int}(\alpha_{i'}) \circ f)|\mathcal{E}'_{i'},$$

which then induces, via the $k'_{i'}, k_i$, an endomorphism

$$f^0_{i'} : \mathrm{T}(\overline{K}) \longrightarrow \mathrm{T}(\overline{K}).$$

m146

Marginal note. m146 N.B. A priori $f^0_{i'}$ is multiplication by $\chi_{i'} \in \widehat{\mathbb{Z}}^*|\widehat{\mathbb{Z}}$. The centralizer in Γ of an open subgroup is reduced to the identity.

We require that $\alpha \in \Gamma$ (which conjugates $f_{\mathcal{E}}$ to $f'_{\mathcal{E}}$}, and is probably uniquely determined by this condition) and $\alpha_{i'}$ (which is probably determined modulo † composition on the left with an element of the normalizers|of the normalizer of $\mathcal{E}_i$ (at least of the transporters of the $\mathcal{E}_i$ of an open subgroup of $\mathcal{E}_i$ — it is the same if f is an isomorphism) can be chosen in such a way that, putting

$$\beta_{i'} = p(\alpha_{i'})\alpha,$$

we have

$$f^0_{i'} = (d_{i'}\chi(\beta_{i'}))\,\mathrm{id}_{\mathrm{T}(\overline{K})}$$

i.e. $\chi_{i'} = d_{i'}\chi(\beta_{i'})$ m147

Marginal note. m147 This condition on the systems of the $\alpha_{i'}$ does not change (α remaining fixed) if we change $\alpha_{i'}$ to $\mu_{i'}\alpha_{i'}$, where $\mu_{i'} \in \mathcal{E}_i$.

$d_{i'} \in \mathbb{N}^*$ is a natural integer obviously uniquely determined } when α and the $\alpha_{i'}$ are chosen). 53

Uncertainty retained from cancelled passages.
140-2-s054-D001 $\forall i'$
140-2-s054-D002 conjugation

[scan 55]

8, Taxonomic reflection.

The case $\dim X = 0$ translates in the paradigm of the system $(\mathrm{T}(\overline{K}) \overset{k_i}{\longrightarrow} \mathcal{E}_i \overset{\rho_i}{\hookrightarrow} \mathcal{E} \overset{p}{\to} \Gamma)$ † into the condition † p injective, † and thus $I = \emptyset$ — so this case is described † simply by giving an open subgroup $\mathcal{E} \subset \Gamma$, i.e. a finite subextension L of $\overline{K}/K$ (N.B. of course we will have $X' = \mathrm{Spec}\, L'|L$, and the choices made on X — i.e. on $\Pi_{\mathcal{D}^*} \longrightarrow \Pi_U \longrightarrow \Pi_e$ — leading|lead to this description — amount here to the choice of such a K-embedding of L† $= \mathrm{H}^0(X, \mathcal{O}_X)$ into $\overline{K}/K$)
$=\emptyset$

The case $\dim X = 1$ translates into the fact that $\mathcal{E} \longrightarrow \Gamma$ is not injective — as for I, it may be empty or not in this case. If it is empty, the description is thus simply in terms of a homomorphism of profinite groups $\mathcal{E} \overset{p}{\to} \Gamma$ (where Γ given in advance $= \mathrm{Gal}(\overline{K}/K)$ } $= \mathrm{Aut}_{\Pi_e}(\Omega)$).

Suppose now given both

$$\mathcal{D} = (\mathcal{E}, p, I, (\rho_i : \mathcal{E}_i \hookrightarrow \mathcal{E}), (k_i : \mathrm{T}(\overline{K}) \longrightarrow \mathcal{E}_i))$$

and

$$\mathcal{D}'=(\mathcal{E}', p', I', (\rho'_{i'} :), (k'_{i'} :)),$$

and let us return to the question of the description of the morphisms from $\mathcal{D}'$ to $\mathcal{D}$. We are going to distinguish four cases, according to the two values 54

Uncertainty retained from cancelled passages.
140-2-s055-D001 [illeg. 2]
140-2-s055-D002 $\pi|I$
140-2-s055-D003 is|of
140-2-s055-D004 [illeg. 1]
140-2-s055-D005 K'

[scan 56]

$0, 1$ possible for $n = \dim \mathcal{D}$ and $n' = \dim \mathcal{D}'$ respectively. [p. 26]

I) $n = 0, n' = 0$, † $I = I' = \emptyset$ and the data reduce to open subgroups $\mathcal{E} \overset{p}{\hookrightarrow} \Gamma$, $\mathcal{E}' \overset{p'}{\hookrightarrow} \Gamma$. In the cumbersome description above, the considerations relating to I, I' drop out and the morphism (a)
$\mathcal{D}' \longrightarrow \mathcal{D}$ amounts to a class of † pairs (f, α), where $f : \mathcal{E}' \longrightarrow \mathcal{E}$ and $\alpha \in \Gamma$, such that we have

$$pf = \mathrm{int}(\alpha) f$$

— which implies that f is already determined by α, being induced by $\mathrm{int}(\alpha)$ [α subject to $\alpha \in \mathrm{Trans}_\Gamma(\mathcal{E}', \mathcal{E}) = \mathrm{Trans}_\Gamma(L, L')$] †

$$\begin{array}{ccc} \mathcal{E}' & \xrightarrow{f} & \mathcal{E} \\ {\scriptstyle p'}\downarrow & & \downarrow{\scriptstyle p} \\ \Gamma & \xrightarrow[\sim]{\mathrm{int}(\alpha)} & \Gamma \end{array}$$

[which induces thus|indeed a homomorphism from $L^*|L \subset \overline{K}$ to $L' \subset \overline{K}$, hence a homomorphism $\mathrm{Spec}(L') \longrightarrow \mathrm{Spec}(L)$ in the opposite direction].

The condition that $\alpha, \alpha' \in \Gamma$ define the same morphism $\mathcal{D}' \longrightarrow \mathcal{D}$ (or again $\mathrm{Spec}\, L' \longrightarrow \mathrm{Spec}\, L$) is described by the existence of a $\mu \in \mathcal{E}$ such that $\alpha' = \mu\alpha$. The fact that we give a 1-1 correspondence between

$$\mathrm{Hom}_K(X', X) \simeq \mathrm{Hom}_K(L, L')$$

with $\mathrm{Hom}(\mathcal{D}', \mathcal{D})$ made explicit too, is clear. 55

Uncertainty retained from cancelled passages.
140-2-s056-D001 then|thus
140-2-s056-D003 homomorphisms
140-2-s056-D004 $\mathcal{E}'$

[scan 57]

II) $n' = 1, n = 0$. Here $n = 0$ already implies $I = \emptyset$. For $\mathrm{Hom}(X', X) \neq \emptyset$, we must have $I' = \emptyset$ [by the condition $f^{-1}(S) = S'$ which implies $S' = \emptyset$, if $f : (X', S') \longrightarrow (X, S)$] $f^{-1}(\emptyset) = \emptyset$ — in the description of the morphisms $\mathcal{D}' \longrightarrow \mathcal{D}$, this corresponds to the fact that we can have a map $\tau : I' \longrightarrow I (= \emptyset)$ only if $I' = \emptyset$. † Restricting ourselves therefore to the case $I' = \emptyset$ (otherwise $\mathrm{Hom}(X', X) = \mathrm{Hom}(\mathcal{D}', \mathcal{D}) = \emptyset$ and we are happy), we thus find that $\mathcal{D}$ amounts to giving $\mathcal{E} \overset{p}{\hookrightarrow} \Gamma$ an open subgroup, and $\mathcal{D}'$ to giving a homomorphism of profinite groups $p' : \mathcal{E}' \longrightarrow \Gamma$, whose image will be an open subgroup of Γ, say $\Gamma'_0 \subset \Gamma$, corresponding to a finite subextension L' of $\overline{K}/K$. [m153] this we cannot

Marginal note. [m153] Restricting ourselves therefore to the case

The choices made concerning X' imply precisely|that we have a fixed isomorphism $L' \simeq \mathrm{H}^0(X', \mathcal{O}_{X'})$. That said, a K-morphism from $(X', \emptyset)$ to $(X, \emptyset)$ i.e. from $X' \longrightarrow X$ amounts (since X affine) to giving a K-morphism $\operatorname{Spec} L' \longrightarrow X$ ($\simeq \operatorname{Spec} L$) from the affine envelope, i.e. a K-homomorphism $L \longrightarrow L'$. (which depends only on L, L', or again on the † subgroups $\mathcal{E} \subset \Gamma$ and $\Gamma_0 \subset \Gamma$ 56

Uncertainty retained from cancelled passages.

140-2-s057-D003 [illeg. 1]

140-2-s057-D004 [illeg. 1]

[scan 58]

of Γ). On the other hand the description in terms of the diagrams $\mathcal{D}, \mathcal{D}'$ amounts to saying [p. 27]
that a morphism is determined by a pair (f, α), $f : \mathcal{E}' \longrightarrow \mathcal{E}$ and $\alpha \in \Gamma$, such that we again have commutativity

$$\begin{array}{ccc} \mathcal{E}' & \xrightarrow{f} & \mathcal{E} \\ {\scriptstyle p'}\downarrow & & \downarrow{\scriptstyle p} \\ \Gamma & \xrightarrow{\operatorname{int}(\alpha)} & \Gamma \end{array}$$

$$pf = \operatorname{int}(\alpha) p'$$

which again implies that f is determined by α (namely, induced by $\operatorname{int}(\alpha)$), and α being subject to the sole condition

$$\alpha \in \operatorname{Transp}_\Gamma(\Gamma'_0, \mathcal{E})$$

— and α, α' describing|describe the $=$ homomorphism iff

$$\exists \mu \in \mathcal{E} \quad \text{such that} \qquad \Longleftarrow \mid \Longleftrightarrow$$

$$\alpha' = \mu\alpha$$

so we again have

$$\begin{aligned} \operatorname{Hom}(\mathcal{D}', \mathcal{D}) &\simeq \mathcal{E} \backslash \operatorname{Transp}_\Gamma(\Gamma'_0, \Gamma) \ \dagger \\ &\simeq \operatorname{Transp}_\Gamma(L, L') / \operatorname{Fix}_\Gamma(L) \\ &\simeq \operatorname{Hom}_{K\text{-ext}\mid K\text{-cat}}(L, L') \end{aligned}$$

so we again find in this case the|the full faithfulness.

In fact, we could have treated together 57

Uncertainty retained from cancelled passages.

140-2-s058-D001 [*illeg.* 1]

[scan 59]

the two cases $(n' = 0, n' = 1)$ where $n = 0$, and the description of the $\operatorname{Hom}(X, X')$ in terms of $\mathcal{D}, \mathcal{D}'$ is valid moreover, without making draconian hypotheses (projective, smooth, dim ≤ 1, anabelian) on X'. †

Indeed, since X is finite étale over K, its inverse image $X_{X'} =$ † $X' \times_K X$ over X' is finite étale over X', and

$$\operatorname{Hom}_K(X, X') \simeq \dagger \operatorname{Sections}(X_{X'}/X')$$

can be described in terms of the category $\widehat{\Pi_1(X')}$ ($= \Pi_U$) of finite local systems on X' i.e. of finite local systems on Π_U. Following this description, we arrive at the 58

Uncertainty retained from cancelled passages.
140-2-s059-D002 thus|in K-homomorphisms
140-2-s059-D003 [illeg. 1]
140-2-s059-D004 (

[scan 60]

common description given in I, II, without any other hypothesis on X than 0-connectedness [p. 28]
(to be able to describe $\Pi_U = \Pi_{X'}$ over Π_e in terms of a group homomorphism $p' : \mathcal{E}' \longrightarrow \Gamma$ ...).

When we are given (X', S'), with S' not necessarily empty, X' 0-connected (and X' smooth of dimension 1 at the points of S' (it would suffice = that S' not disconnect|disconnect locally (étale) X', ...), thus $X' - S'$ 0-connected then the datum $\Pi_1(X' - S') \longrightarrow \Pi_1(e_K)$ is described † by $\mathcal{E}' \xrightarrow{p'} \Gamma$ (independently of the consideration of the $\mathcal{E}_i$ etc) and hence $\mathrm{Hom}_K(X' - S', X)$ is described as we just said. But we have (with the normality hypotheses on X' at the points of S')

$$\mathrm{Hom}_K(X' - S', X) = \mathrm{Hom}_K(X', X).$$

This encourages modifying the definition of morphisms $(X', S') \longrightarrow (X, S)$ given at the|by beginning, via $f : X' \longrightarrow X$ with $f^{-1}(S)_{\text{réd}} = S'$, by requiring this not for f, but only for the restriction of f to the union of the irreducible components of X' 59

Uncertainty retained from cancelled passages.
140-2-s060-D004 [illeg. 1]

[scan 61]

† which are sent into irreducible components of X † that are non-discrete i.e. not reduced to a point. In the corresponding description $\mathcal{D}' \to \mathcal{D}$, we specified that one requires the datum of a map $I' \to I$ (and of the corresponding data $f_{i'}, \alpha_{i'}$) only if † $n \neq 0$ i.e. $p_{si}|p_X : \mathcal{E} \to \Gamma$ not injective, — in the contrary case (implying $I = \emptyset$) if by chance $I' \neq \emptyset$, one drops the knowledge of I' and of the corresponding structure elements.

III) $n' = 0$, $n = 1$. Here one thus has $I' = \emptyset$, $p' : \mathcal{E}' \hookrightarrow \Gamma$ i.e. $\mathcal{D}'$ reduces to the datum of an open subgroup $\mathcal{E}' \hookrightarrow \Gamma$ of Γ. On the side of $(X', S' = \emptyset)$ and (X, S), the condition $f^{-1}(S)_{\text{réd}} = S' = \emptyset$ on the K-morphism $f : X' \to X$ implies that f factors through $X' \Longrightarrow \mathrm{Spec}(L') \longrightarrow X \setminus S$, regardless of whether $S = \emptyset$ or not, i.e. whether $I = \emptyset$ or not. Thus it is a matter of describing

$\mathrm{Hom}_{K\text{-sch}}(X', X \setminus S)$, in terms of $\begin{matrix} \Pi_{U'} \\ \| \\ \Pi_1(X') \end{matrix} \to \Pi_e = \Pi_1(e_K)$ 60

Uncertainty retained from cancelled passages.
140-2-s061-D001 constant|constant
140-2-s061-D002 [illeg. 1]
140-2-s061-D003 $I \neq \emptyset$

[scan 62]
[p. 29]

and of $\begin{matrix} \Pi_U \\ \| \\ \Pi_1(X \setminus S) \end{matrix} \to \Pi_e$. Here the possible additional data relating to I do not serve explicitly in the description, in terms of diagrams (of groupoids). Unless I am mistaken, an easy reduction (Galois descent) brings us back to the case where $X' = \mathrm{Spec}\, K$ i.e. $\mathcal{E}' \Leftrightarrow \Gamma$. The description of the|in homomorphisms $\mathcal{D}' \to \mathcal{D}$ is then still in terms of pairs (f, α), where $f : \mathcal{E}' = \Gamma \to \mathcal{E}$ and where $\alpha \in \Gamma$, with the everlasting condition $pf = \mathrm{int}(\alpha)p'$, which becomes (as $p' = \mathrm{id}$)

$$pf = \mathrm{int}(\alpha).$$

† Here of course α no longer describes f (but the converse is true, here knowledge of f hence of $pf \in \mathrm{Aut}(\Gamma)$ implies that of α, i.e. $\mathrm{Cent}(\Gamma) = \{1\}$ …). The pairs (f, α) and (f', α') define the $=$ morphism of diagrams, iff $\exists \mu \in \mathcal{E}$ such that $f' = \mathrm{int}(\mu) \circ f$, $\alpha' = p(\mu)\alpha$. As (if $\exists (f, \alpha)$) p is surjective, 61

Uncertainty retained from cancelled passages.
140-2-s062-D001 α and α'

[scan 63]

one sees|wants that, on choosing μ over † α^{-1}, and replacing (f, α) by $(f', \alpha') = (\mathrm{int}(\mu)f, p(\mu)\alpha)$, one can always describe a morphism $\mathcal{D}' \to \mathcal{D}$ by (f, α) with $\alpha = 1$ hence by a section f of the homomorphism $\mathcal{E} \to \Gamma$, and two sections f, f' (corresponding to $(f, 1), (f', 1)$) define the $=$ homomorphism from $\mathcal{D}'$ to $\mathcal{D}$, iff $\exists \mu \in \mathcal{E}$ such that $p(\mu) = 1$ i.e. $\mu \in \pi = \mathrm{Ker}(p)$, and such that $f' = \mathrm{int}(\mu) \circ f$.

Thus, the morphisms from $\mathcal{D}'$ (corresponding to $X' = \mathrm{Spec}\, K$) to $\mathcal{D}$ correspond exactly to the classes of splittings of $\mathcal{E} \overset{p}{\to} \Gamma$, modulo inner automorphisms by $\mu \in \pi = \mathrm{Ker}\, p$. The "messy conjecture" in case Ш is thus equivalent to this: m160

Marginal note. m160



†

$$\Gamma((X,S)/K) \longrightarrow \begin{array}{c}\pi\text{-conjugacy classes of}\\ \text{sections of } \mathcal{E} \to \Gamma \text{ i.e.}\\ \text{of sections of } \Pi_1(X) \to \Pi_1(e_K)\end{array}$$

62

Uncertainty retained from cancelled passages.
140-2-s063-D001 [illeg. 1]
140-2-s063-D002 Hom(K

[scan 64]

is bijective (if (X, S) is an allowed "anabelian" pair). m162 [p. 30]

Marginal note. m162 false as it stands, cf. number 9 below — one needs additional conditions on the f … Perhaps true if $I = \emptyset$

I know in any case that this map is injective — this holds whenever one has a scheme U over K (here $X \setminus S$) which embeds in an algebraic group G extension of an abelian variety by a torus, and then follows from the "absolute" Mordell-Weil theorem: $G(K)$ is a finitely generated $\mathbb{Z}$-module. In fact, it suffices to know the splitting class of the extension $\mathcal{E}^\natural$ of Γ by

$$\pi_{\mathrm{ab}} = \pi/\overline{[\pi, \pi]} \simeq H_1(\mathrm{Alb}|U),$$

† corresponding to a point of U rational/K, to know this point. Thus the result of faithfulness is obtained, with less draconian hypotheses on X than the anabelian hypothesis (with the notation of lemma 5.2, this means that $g = 0 \Rightarrow n|\nu \geq 2$ i.e. that there is no irreducible component of $X_{\overline{K}}|U_{\overline{K}}$ which is isomorphic to $\mathbb{P}^1_{\overline{K}}$ or $\mathbb{E}^1_{\overline{K}}$ …). 63

Uncertainty retained from cancelled passages.
140-2-s064-D001 to deduce from it

[scan 65]

N.B. This suggests an approach to the Mordell conjecture, via a better knowledge of extensions of Γ by π: the fact that there is (for $g \geq 2$) only a finite number of π-conjugacy classes of splittings of such an extension .... Likewise for an approach to Fermat's theorem, via † a good knowledge of an extension of Γ by $\pi = \pi_1(\mathbb{P}^1_{\mathbb{Q}} \setminus \{0, 1, \infty\})$ ....

Case IV $n = 1, n' = 1$ i.e. X, X' of dimension 1.

a) If $I = I' = \emptyset$, I have nothing to add to the description of morphisms $f : X \to X'$ in terms of morphisms of diagrams of groupoids. Here † the condition $f^{-1}(S)_{\text{réd}} = S'$ is not a restriction on f — one therefore does not exclude constant maps. These (thanks to † (thanks to ...)

64

Uncertainty retained from cancelled passages.
140-2-s065-D001 [illeg. 1]
140-2-s065-D002 [illeg. 1]
140-2-s065-D004 $I' \neq \emptyset$ [illeg. 5] only i.e. $S' = \emptyset$ [illeg. 7] S' is empty therefore

[scan 66]

the study of the|of the case II) are moreover described in a "fully faithful" way by homomorphisms of diagrams — They correspond to the cases $f : \mathcal{E}' \to \mathcal{E}$ which are zero on π'. [p. 31]

b) The case $I' = \emptyset$, $I \neq \emptyset$ (i.e. $S' = \emptyset$, $S \neq \emptyset$) means, for the condition $f^{-1}(S)_{\text{réd}} = S'$ i.e. $f^{-1}(S) = \emptyset$, that f must map X' into $X \setminus S$, therefore that the f corresponding to a morphism from $(X', \emptyset)$ to (X, S) are constant. The messy conjecture in this case therefore expresses essentially that at the level of the homomorphisms $\mathcal{E}' \xrightarrow{p'} \Gamma$ and $\mathcal{E} \xrightarrow{p} \Gamma$, every group homomorphism $\mathcal{E}' \xrightarrow{f} \mathcal{E}$ such that $pf = \text{int}(\alpha)p'$ for $\alpha \in \Gamma$ [one is reduced to the case where $\alpha = 1$, p, p' surjective i.e. to homomorphisms of extensions of Γ by the|some groups $\pi = \pi_1(\overline{X \setminus S})$ and $\pi' = \pi_1(\overline{X}')$)] is trivial on π' [i.e. "is" a section]. One should try to verify this point directly, which (modulo case III) would establish the "messy conjecture" in that case. 65

[scan 67]

c) The case $I' \neq \emptyset$, $I = \emptyset$ implies that $\text{Hom}((X, S), (X', S')) = \emptyset$ (since $f^{-1}(S) = f^{-1}(\emptyset) = \emptyset \neq S'$!), which does indeed correspond to $\text{Hom}(\mathcal{D}', \mathcal{D}) = \emptyset$ since $\nexists \tau : I' \to I$.

d) In the case $I' \neq \emptyset$, $I \neq \emptyset$, the condition $f^{-1}(S)_{\text{réd}} = S'$ implies that f is not constant (otherwise one would have $f^{-1}(S)_{\text{réd}} = \emptyset$ or X'), hence f is dominant and finite, generically étale. [Thus, such f † can occur only if I and I' are either both $\emptyset$ (case a)), or both $\neq \emptyset$ (present case d)). The fact that in these cases, the diagrammatic Galois description is faithful (perhaps not fully) follows easily from the analogous result in case III. Full faithfulness on the other hand is a mysterious thing, = if the result corresponding † † to case III were established (and from it we 66

Uncertainty retained from cancelled passages.
140-2-s067-D002 are allowed [illeg. 1]
140-2-s067-D003 in
140-2-s067-D004 [illeg. 1]

[scan 68]

seem far, = if K finite over $\mathbb{Q}$ ...) [p. 32]

Elementary reductions To prove the "messy conjecture", one is reduced (by finite extensions of the fields K) to the case where $\mathcal{E} \to \dagger\Gamma$ (and, if one insists, also $\mathcal{E}' \to \Gamma$) is an epimorphism.

Unless I am mistaken, knowledge of case III for finitely generated extensions of K, implies the general case for K (with enough technical sweat …).

Returning however to the general case when $\mathcal{E} \to \Gamma$ is surjective, in every equivalence class of systems $(f : \mathcal{E}' \to \mathcal{E}, \alpha \in \Gamma, \dagger\ \tau : I' \to I, \dagger\ (\alpha_{i'} \in \mathcal{E}_{\tau(i')=i})_{i' \in I'})$, one can find a system with $\alpha = 1$, so that $f : \mathcal{E}' \to \mathcal{E}$ is compatible with the homomorphisms to Γ (i.e. the extension structures). When one restricts oneself to such systems, the|their equivalence by conjugation is effected by a system $(\mu, (\mu_{i'})_{i' \in I'})$ with $\mu \in \pi$ $(= \operatorname{Ker} p)$, and 67

Uncertainty retained from cancelled passages.

140-2-s068-D001 K

140-2-s068-D002 f_i,

140-2-s068-D003 $f_{i'} : \mathcal{E}'_{i'}$

[scan 69]

the $\mu_{i'} \in \mathcal{E}_{i=\tau(i')}$ as before. Thus, $f : \mathcal{E}' \to \mathcal{E}$ is a homomorphism of extensions, defined modulo inner automorphisms by an element of $\pi = \operatorname{Ker}(\mathcal{E} \to \Gamma)$, and satisfying conditions made explicit elsewhere. It could be (as we have already remarked) that knowledge of the † π-conjugacy class of f suffices to determine the "messy morphism" into which f fits (if $I, I' \neq \emptyset$).

[This is linked to the question of whether the † † $\forall d \in \mathbb{N}^*$, $\operatorname{Trans}_{\mathcal{E}}(L_i^d, L_i)$ is reduced to $\mathcal{E}_i$ (where $L_i = \operatorname{Ker}(p_i : \mathcal{E}_i \to \Gamma) = \operatorname{Im}(\kappa_i : T(\overline{K}) \to \mathcal{E}_i)$). In that case, $\mathcal{E}_i$ would be known in terms of $L_i \subset \pi$, and $=$ in terms of the rays|kernels of subgroups that it defines in π, and a fortiori in terms of the rays|kernels of homomorphisms defined by $\kappa_i : T(\overline{K}) \to \pi$ …. † 68

Uncertainty retained from cancelled passages.

140-2-s069-D002 [illeg. 1]

140-2-s069-D003 [illeg. 1]

140-2-s069-D004 tran

140-2-s069-D005 $\mathcal{E}$

[scan 70]

Case where the $\mathcal{E}_i \to \Gamma$ surjective (i.e. the $k(s_i) = K$ i.e. s_i rational over K). Then, if $(f,$ † $\tau,$ † $(\alpha_{i'}))$ is a homomorphism from $\mathcal{D}'$ to $\mathcal{D}$, † after † correcting by $\mu_i \in \mathcal{E}_i$ having the $=$ image as the $\alpha_{i'}$ in Γ, one may assume $\alpha_{i'} \in \pi$, in addition to $\alpha = 1$ (obtained by correcting by a suitable μ). Thus one describes the homomorphism $\mathcal{D}' \to \mathcal{D}$ by $(f, 1, \tau, (\alpha_{i'}))$, the $\alpha_{i'} \in \pi$. The † condition † here is that $f : \mathcal{E}' \to \mathcal{E}$ be a group homomorphism over Γ, that † $f(\mathcal{E}'_{i'}) \subset \operatorname{int}(\alpha_{i'}^{-1})(\mathcal{E}_{\tau(i')=i})$, and that the homomorphism induced $\mathcal{E}'_{i'} \to \mathcal{E}_i$ by † $\operatorname{int}(\alpha_{i'})f|\operatorname{int}(\alpha_{i'}^{-1})f$ † induces on $T(\overline{K})$ (via $\kappa_{i'}, \kappa_i$) the homomorphism of multiplication by d_i and nothing more — how beautiful! Two systems $(f, \tau, (\alpha_{i'}))$ and $(f', \tau', (\alpha'_{i'}))$ define the $=$ morphism, iff $\tau = \tau'$ and if there exists [p. 33]

$$\mu \in \pi, \quad \mu_{i'} \in \begin{matrix} L_{i=\tau(i')} \\ \| \\ \kappa_i(T(\overline{K})) \end{matrix}$$

such that

$$f' = \operatorname{int}(\mu) f$$

$$\alpha'_{i'} = \mu_{i'} \alpha_{i'}.$$

† Let us note that a priori, for (f, τ) fixed, the $\alpha_{i'}$ are determined modulo right multiplication by elements of $\operatorname{Transp}_{\pi}(\dagger(L_i^{d_{i'}}, L_i),$ 69

Uncertainty retained from cancelled passages.

140-2-s070-D002 α,
140-2-s070-D003 [illeg. 1]
140-2-s070-D004 for [illeg. 1]
140-2-s070-D005 modifying
140-2-s070-D006 systems
140-2-s070-D007 [illeg. 1]
140-2-s070-D008 int$(\alpha_{i'}^{-1})$
140-2-s070-D009 $f_i =$
140-2-s070-D010 of $\mathcal{E}_i$ of
140-2-s070-D011 [illeg. 1]
140-2-s070-D012 π

[scan 71]

which is probably $= L_i = \mathrm{Ker}(p_i : \mathcal{E}_i \to \Gamma) = 1$|the $\mathrm{Im}(\kappa_i : T(\overline{K}) \to \mathcal{E}_i \cap \pi)$. Thus we find that † the π-conjugacy class of groups over Γ of $f : \mathcal{E}' \to \mathcal{E}$ suffices to determine the homomorphism $\mathcal{D}' \to \mathcal{D}$ in the messy category. I presume that a little sweat would make it possible to prove that this still works in the general case (without assuming the $\Gamma_i = \mathrm{Im}(\mathcal{E}_i \to \Gamma)$ equal to Γ). This means (in the present case) that the essential structure which describes (X, S) is that of an † "outer extension" of Γ by a group π [surjective outer homomorphism from a group, with kernel a π], with π equipped with a loop structure i.e. of homomorphisms outer

$$\kappa_i : T\dagger|\mu\,(= T(\overline{K})\,\dagger \overset{\text{noncanonically}^t}{\simeq} \widehat{\mathbf{Z}}) \to \pi$$

(satisfying certain conditions), the 70

Uncertainty retained from cancelled passages.

140-2-s071-D001 f
140-2-s071-D003 [illeg. 1]
140-2-s071-D005 [illeg. 1]
140-2-s071-D006)

[scan 72]
[p. 34]

homomorphisms from $\mathcal{E}' \to \Gamma$ (kernel π') into $\mathcal{E} \to \Gamma$ (kernel π) being the π-conjugacy classes of homomorphisms $\mathcal{E}' \to \mathcal{E}$ of groups over Γ, inducing homomorphisms $\pi' \to \pi$ compatible with the loop structure.

9. Tangential structure at the $s \in S$. m173

Marginal note. m173 [The sections of extensions "of the second type"]

Returning to the "intrinsic" description of the (X, S) by $\Pi_{D^*} \to \Pi_U \to \Pi_e$, an $s_i \in S$ thus corresponds to a connected component $\Pi_{D_i^*}$ of Π_{D^*}, or a Π_{D_i} of its quotient Π_D — consider

$$\Pi_{D_i^*} \xrightarrow{\sigma_i = \Pi_{f_{D_i^* D_i}} | \sigma_i = \sigma_{D_i^* D_i}} \Pi_{D_i}.$$

We are going to describe certain sections (up to isomorphism) of this † morphism of groupoids, i.e. pairs (γ, λ) functors $\Pi_{D_i} \overset{\gamma}{\to} \Pi_{D_i^*}$ and an isomorphism $\sigma_i \circ \gamma \overset{\lambda}{\simeq} \mathrm{id}$. If we choose a $\widetilde{D}_i^* \in \mathrm{Ob}\,\Pi_{D_i^*}$ (universal covering of D_i^*), whence a surjective homomorphism

$$\mathcal{E}_i = \mathrm{Aut}(\widetilde{D}_i^*) \to \Gamma_i = \mathrm{Aut}(\sigma_i(\widetilde{D}_i^*) = \widetilde{D}_i),$$

the datum of $(\gamma, \lambda_i|\lambda)$, and of an isomorphism $\gamma(\widetilde{D}_i) \simeq \widetilde{D}_i^*$ giving the identity isomorphism on applying the σ_i, amounts to that of a homomorphism $\Gamma_i \to \mathcal{E}_i$. — and the isomorphism classes of the pairs (γ, λ) correspond to the L_i-conjugacy classes 71

Uncertainty retained from cancelled passages.
140-2-s072-D001 functor

[scan 73]

$(L_i = \mathrm{Ker}(\dagger\mathcal{E}_i \to \Gamma_i))$ of sections of the extension $\mathcal{E}_i$ of Γ_i by L_i.
This being so, we have

$$\Gamma_i = \mathrm{Gal}(\overline{K}_i/K_i).$$

where $K_i = k(s_i)$, $\overline{K}_i$ is the algebraic closure of K_i defined by $\widetilde{D}_i$ (a priori, it is not canonically isomorphic to $\overline{K}$ over K…), and the classes of † splittings of extensions form a torsor Σ_i (if there are any, and we know that there are…) under $H^1(\Gamma_i, L_i)$, or again, since $\kappa_i : L_i \xleftarrow{\sim} T(\overline{K}_i)$, under

$$H^1(\mathrm{Gal}(\overline{K}_i/K_i), T(\overline{K}_i)) \simeq H^1(K_i, T_e|T_\ell(G_m)) \simeq \varprojlim_n H^1(K_i, \mu_n).$$

Now the Kummer exact sequence gives

$$\begin{array}{ccccccccc} 0 & \to & H^0(K_i, \dagger G_m)_n & \to & H^1(K_i, \mu_n) & \to & {}_nH^1(K_i, G_m) & \to & 0 \\ & & \| & & & & \| & & \\ & & (K_i^*)_n & & & & 0 & & \end{array}$$

i.e. $H^1(K_i, \mu_n) \simeq (K_i^*)_n$, hence

$$H^1(K_i, T(G_m)) \simeq \varprojlim_n (K_i^*)_n$$

72

Uncertainty retained from cancelled passages.
140-2-s073-D001 [illeg. 1]
140-2-s073-D003 extensions
140-2-s073-D004 $\overline{K}$

[scan 74]

Note that we have a Kummer homomorphism [p. 35]

$$K_i^* \to \underbrace{H^1(K_i, T(G_m))}_{\mathcal{H}_i} \simeq \varprojlim (K_i^*{}_n)$$

whose kernel consists of the $x \in K_i^*$ such that for $\forall\ n \in \mathbf{N}$, we have $x \in K_i^{*n}$. Since K_i is of finite type over $\mathbf{Q}$, this homomorphism is injective, i.e. K_i^* is identified with a subgroup of $\mathcal{H}_i$.

This being so, we are going to define, in the torsor Σ_i under $\mathcal{H}_i$ of splittings of D_i^* over D_i, a sub-K_i^*-torsor (i.e. an element of Σ_i/K_i^*). For this, 73

[scan 75]

† m177

Marginal note. m177 make a general local argument, dropping the index i.

Let us consider more generally the case of a field k $(= K_i)$ of characteristic 0, and of a † k-algebra $\mathcal{O}$ which is a henselian gauge with residue field k. let L be the fraction field of $\mathcal{O}$ We are going, for every uniformizer t of $\mathcal{O}$, to define a section (γ, λ) of the|of the † † groupoid|groupoids of algebraic closures of L, to that of universal coverings of $\mathrm{Spec}(\mathcal{O})|\,\text{fét}(\mathcal{O})$ or, what amounts to the same thing, of algebraic closures of k. For this,

let

$$\mathcal{O}(n, t|s) = \mathcal{O}[T_n]/(T_n^n - s), \quad \text{and}$$

$$\mathcal{O}(\infty, t) = \varinjlim_n \mathcal{O}(n, t), \quad \text{by } T_{n\,t|s} \mapsto T^m_{mn\,t|s}.$$

† † $L(n,t) = \mathcal{O}(n,t) \otimes_{\mathcal{O}} K$ = fraction field of $\mathcal{O}(n,t)$, $L(\infty,t) = \varinjlim_n L(n,t)$ the fraction field of $\mathcal{O}(\infty,t)$. This is 74

Uncertainty retained from cancelled passages.

140-2-s075-D001 just [illeg. 1] $\overline{K}_i$ which is

140-2-s075-D004 [illeg. 1]

140-2-s075-D005 Π_K

140-2-s075-D006 the category

140-2-s075-D007 again a

140-2-s075-D008 Put

[scan 76]

an algebraic extension of L, and for ever^y † finite, or ind-finite extension k' of k, putting [p. 36] $\mathcal{O}' = \mathcal{O} \otimes_k k'$ (which is also a henselian gauge over k' with residue field k', with fraction field $L' \simeq L \otimes_k k'$), we have canonical isomorphisms †

$$\mathcal{O}(n|\infty, t) \otimes_{\mathcal{O}} \mathcal{O}' \simeq \mathcal{O}'(n|\infty, t)$$

$$\mathcal{O}(\infty, t) \otimes_{\mathcal{O}} \mathcal{O}' \simeq \mathcal{O}'(\infty, t)$$

whence

$$L(n|\infty, t) \otimes_k \overline{k}'|k' \simeq L'(n|\infty, t)$$

$$L(\infty, t) \otimes_k k' \simeq L'(\infty, t)$$

Now if $k'|\overline{k}'$ is algebraically closed, $\mathcal{O}'$ strictly henselian, we know that $L'(\infty,t)$ is then algebraically closed. Besides † since $\overline{k}'|k' \subset \mathcal{O}'(\infty,t) \subset L'(\infty,t)$, in this case $k\,k'$ † is identified with the algebraic closure $\overline{k}$ of k in $L'(\infty,t) = \overline{L}$. So we have indeed found a section of Π_{D^*} over Π_D.

But if we replace s by $s' = su^n$, with $u \in \mathcal{O}^*$, we find isomorphisms

$$\lambda(n,u): \quad \begin{array}{ccc} \mathcal{O}(n,s) & \longrightarrow & \mathcal{O}(n,s') \\ \| & & \| \\ \mathcal{O}[T_n]/(T_n^n - s) & & \mathcal{O}[T_n]/(T_n^n - s') \end{array}$$

by $T_n \to uT_n$ whence by passing to the limit

$$\lambda(\infty, u|n) : \mathcal{O}(\infty, s) \longrightarrow \mathcal{O}(\infty, s')$$

inducing $L(\infty,s) \xrightarrow{\sim} L(\infty,s')$ an isomorphism of extensions of L 75

Uncertainty retained from cancelled passages.

140-2-s076-D001 [illeg. 1]

140-2-s076-D002 [illeg. 1]

140-2-s076-D004 [illeg. 1]

140-2-s076-D005 [illeg. 1]

[scan 77]

whence an isomorphism between the corresponding sections of the Π_L over Π_k. This implies that if s, s' differ by a $u|v \in \mathcal{O}^*$ such that its image in k is 1 † i.e. $u|v \in 1 + m$ (which is a divisible subgroup of $\mathcal{O}^*$] then s, s' define isomorphic sections. Thus we find a canonical map

$$\begin{array}{ccc} & & \text{isomorphism classes of sections of} \\ (m/m^2)^* & \to & \Pi_L \text{ over } \Pi_k \quad \text{i.e. of} \\ & & \mathrm{Gal}(\overline{K}/L) \to \mathrm{Gal}(\overline{k}/k) \\ \dagger(\text{set of} & & \\ \text{bases of } m/m^2) & & (\text{which is a torsor under} \\ \text{which is a} & & H^1(\mathrm{Gal}(\overline{k}/k), T(\overline{k})) \\ \text{torsor under} & & \text{i.e. under } H^1(k, T(G_m)) \ldots) \\ k^* & & \end{array}$$

We see at once that this map is compatible with the homomorphism

$$\Theta : k^* \to H^1(k, T(G_m))$$

on the groups acting on these torsors. In the case where this homomorphism is injective (for example k † of finite type over $\mathbf{Q}$) we thus find an embedding of m/m^2 as a sub $\Theta(k^*)$-torsor of the $H^1(k, T(G_m))$-torsor 76

Uncertainty retained from cancelled passages.

140-2-s077-D001 $u|v$ [illeg. 2]

140-2-s077-D002 (and

140-2-s077-D003 embedding

[scan 78]

of splittings of Π_{D^*} ($\simeq$ groupoid of algebraic closures of L) over Π_D ($\simeq$ groupoid of algebraic [p. 37]
closures of k).

Returning to the case of the † $\Pi_{D^*} \to \Pi_U \to \Pi_e$ $\kappa|K \ldots$, and of the †

$$\mathcal{D} : T(\overline{K}^*) \xrightarrow{\kappa_i} \mathcal{E}_i \hookrightarrow \mathcal{E} \xrightarrow{p} \Gamma$$

which make them explicit, we thus find an additional structure on $\mathcal{E}$ or $\mathcal{D}$ via † the extensions $\mathcal{E}_i$ of $\Gamma_i \subset \Gamma$ by $T(\overline{K}^*)$, namely † $\forall i \in I$ sub-K_i^*-torsors in the torsor under $H^1(K_i, T(G_m)) \simeq \varprojlim (K_i^*)_n$ of classes of splittings of $\mathcal{E}_i$. We ought to make explicit the behavior of this additional structure with respect to morphisms (coming from geometric situations) and the relation with the norm of differential 1-forms — I'm too lazy to make it explicit at length and in detail!

The new thing I retain above all is that for $I \neq \emptyset$ and the $\mathcal{E}_i \to \Gamma$ surjective (to simplify), we find $\forall i$ a|some "transcendental|transcendental family|families" of splittings of $\mathcal{E} \to \Gamma$ (via splittings of $\mathcal{E}_i$ over Γ) — essential^{ly} indexed 77

Uncertainty retained from cancelled passages.

140-2-s078-D001 [illeg. 1]

140-2-s078-D002 [illeg. 1]

140-2-s078-D003 on

140-2-s078-D004 some

[scan 79]

by $H^1(K_i|K, T(G_m)) \simeq \varprojlim (K^*)_n$ (more precisely, by a torsor under the aforesaid)

— in this family, a subfamily of splittings which one can call "geometric", indexed by K^* (more precisely, a subtorsor…). We check that the classes of π-splittings obtained for|by distinct

indices i, i' are distinct, even on reducing to the corresponding π-splittings of the extension of Γ by π_{ab} deduced from the extension $\mathcal{E}$ of Γ by π (I am not going into the details of the verifications, via Mordell-Weil…) and distinct from the splittings associated with the rational points/K of $X\backslash S$.

m182

Marginal note. m182



We ought to correct the messy conjecture in the case ($\overline{\text{III}}$), by stating (with all reservations, again!) that there are (perhaps) no other splittings of the extension $\mathcal{E}$ of Γ by π than those 78

[scan 80]

10. Adjustment of the hypotheses (remorse). [p. 38]

I realize that in the definition of the morphisms $(X', S') \xrightarrow{f} (X, S)$, the hypothesis $f^{-1}(S)_{\mathrm{red}} = S'$ is too narrow — we must take arbitrary morphisms $f : X'\backslash S' \to X\backslash S$ (extending of course to $\hat{f} : X' \to X$). We will thus have $S' \supset \hat{f}^{-1}(S)_{\mathrm{red}}$, but S' may be strictly larger than $\hat{f}^{-1}(S)_{\mathrm{red}}$. We must then adjust the description of the "messy category" accordingly — the objects remain the diagrams of groupoids

$$\mathcal{D} : \Pi_{D^*} \to \Pi_U \to \Pi_e$$

(plus Kummer datum) but a morphism f from a $\mathcal{D}'$ into a $\mathcal{D}$ no longer necessarily defines a $\Pi'_{D^*} \to \Pi_{D^*}$ (in addition to $\Pi'_U \to \Pi_U$) one must specify a subset I'_f of $I' = \pi_0(\Pi'_{D^*})$, and specify only

$$\Pi'_{D^*, I'_f} \xrightarrow{f_{D^*}} \Pi_{D^*}$$

(with commutation datum α_{D^*U} relative to the square with f_U). Of course, in the description in terms of $\mathcal{E}', \mathcal{E}$ etc., we require that for $i' \in I'\backslash I'_f$, $f_{\mathcal{E}} : \mathcal{E}' \to \mathcal{E}$ is trivial on $\mathcal{E}'_{i'} \subset \mathcal{E}'$ — and τ is defined as $I'_f \to I$; some|the data relating to the $i' \in$ 79

[scan 81]

$I'_{f'}$ ($f_{i'} : \mathcal{E}_{i'} \longrightarrow \mathcal{E}_{i=\tau(i')}$, the $\alpha_{i'} \in \mathcal{E}$) are the same as in the case considered previously.

N.B. This means in fact that we begin to "plug the holes" (of X') corresponding to the $i' \in I' \setminus I'_{f'}$, by replacing $\mathcal{E}'$ by the †quotient group of $\mathcal{E}'$ by the invariant subgroup generated by the $L_{i'} = \kappa_{i'}(T(\overline{K}))$ ($i' \in I' \setminus I'_{f'}$), and $\mathcal{E}_{i'}$ ($i' \in I'_{f'}$) by their images in the said quotient group, and forgetting $I' \setminus I'_{f'}$, i.e. replacing I' by $I'_{f'}$.

[To do things properly, we would have to express these operations also at the level of diagrams of groupoids

$$\Pi'_{D^*} \longrightarrow \Pi'_U \longrightarrow \Pi'_e|\Pi_e$$

...].

This means then that in fact, we have brought ourselves back to the situation considered at the beginning, where $S' = \hat{f}^{-1}(S)$… So in the end the difference between the two points of view is not enormous, and †the one adopted at the beginning has the advantage of greater simplicity (everything is relative!) 80

†

Uncertainty retained from cancelled passages.

140-2-s081-D002 sub-

140-2-s081-D005 [illeg. 2] spell out [illeg. 2] some|the [illeg. 1]

140-2-s081-D006 [illeg. 1]

[scan 82]

11. Conditions on the systems of †groupoids obtained from geometric situations. [p. 39]

We have already mentioned some in passing, for example that the kernel groups of $\Pi_{D^*} - \Pi_e$ are abelian (with the Kummer isomorphism κ) and that the images are open subgroups, and the = for the image of the † π_1 in $\Pi_U - \Pi_e$. We are going to spell out again and make precise this in the framework of the interpretation in terms of

$$T(\overline{K}) \xrightarrow{\kappa_i} \mathcal{E}_i \xhookrightarrow{\rho_i} \mathcal{E} \xrightarrow{p} \Gamma$$

(the "dimension 1" case — we do not exclude the case $I = \emptyset$ — the "dimension 0" case being trivial…)

a) The image Σ of p is open, more precisely exists open subgroup $\Gamma' \subset \Gamma$ over which $\mathcal{E}$ has a section [and even an "admissible section" in a sense to be defined later — so as to exclude the "trivial" sections coming from the $\mathcal{E}_i$…].

b) The image Σ_i of $\mathcal{E}_i$ in Γ is open, and the extension $\mathcal{E}_i$ of Σ_i by L_i ($\simeq T(\overline{K})$) is trivial. [In fact, we have a privileged class of splittings, called "algebraic", forming a torsor under $K_i^* \hookrightarrow \varprojlim (K_i^{*n})_n | (K_i^*)_n$, cf. number 9]|? — but 81

Uncertainty retained from cancelled passages.

140-2-s082-D001 diagrams

140-2-s082-D002 [illeg. 1]

[scan 83]

we are not going to consider for the moment this additional piece of structure…)

The rest of the conditions essentially concerns the structure of the groups $\pi = \mathrm{Ker}(p : \mathcal{E} \to \Gamma)$ with its conjugacy classes of subgroups (or rather the outer homomorphisms κ_i : †T (= $T(\overline{K}^*)$) $\longrightarrow \pi$, and the way Σ operates by outer automorphisms on π. These conditions are expressed in a particularly simple way when the $\Sigma_i \subset \Gamma$ are $= \Gamma$ ("the points of S all rational over K") i.e. $\mathcal{E}_i \to \Gamma$ surjective, a fortiori $\Sigma = \Gamma$ i.e. $\mathcal{E} \to \Gamma$ surjective. We are going to restrict ourselves to this case. The general case follows by a finite extension of the base field [over which each points of S not rational over K i.e. each $\Sigma_i \neq \Gamma$ †gives|give rise to n_i points, † $n_i = [\Gamma : \Sigma_i]$ } — and $X_{K'}$ † split|splits into $n = (\Gamma : \Sigma)$ connected components which are geometrically connected] — but I am too lazy to spell out as I should, in the context of groupoids or of † homomorphisms of profinite groups, the operation of extension of base fields… It is understood that 82

Uncertainty retained from cancelled passages.

140-2-s083-D003 $\widehat{\mathbb{Z}}(1)$

140-2-s083-D005 split up [illeg. 1]

140-2-s083-D006 equal

140-2-s083-D007 [illeg. 1]

140-2-s083-D008 extensions

[scan 84]

the conditions I am going to describe will be invariant under extension of the|of the base [p. 40] field.

†

c) $\forall i$†, the outer homomorphism

$$\kappa_i : T \longrightarrow \pi$$

is compatible with the outer action of Γ (operating on T by the cyclotomic character χ, and on π thanks to the extension $\mathcal{E}$ of Γ by π). In other words, for $\forall g \in \mathcal{E}$, exists an $\underline{\alpha \in \pi}$ (N.B. not just $\alpha \in \mathcal{E}$!) such that we have,

$$\mathrm{int}(g)\kappa_i(\xi|\zeta) = \mathrm{int}(\alpha)\kappa_i(\chi(p(g))\xi|\zeta)$$

i.e.

$$\mathrm{int}(\alpha^{-1}g)\kappa_i(\xi|\zeta) = \kappa_i(\chi(p(g))\xi|\zeta)$$

This means then that 1°) $\alpha^{-1}g$ normalizes $L_i = \kappa_i(T)$. [and this means =, probably, that $\alpha^{-1}g \in \mathcal{E}_i$ — and that we can find such an $\alpha \in \pi$ such that $\alpha^{-1}g \in \mathcal{E}_i$ comes from the hypothesis $\mathcal{E}_i \to \Gamma$ surjective — we take a $\beta \in \mathcal{E}_i$ $=\alpha^{-1}g$ having the = image as g in Γ and we take $\alpha = g\beta^{-1}$] and that 2°) the inner action of $\beta = \alpha^{-1}g$ on T is nothing other than by multiplication by $\chi(g) = \chi(\beta)$ — which (for $\beta \in \mathcal{E}_i$) is nothing other than the 83

Uncertainty retained from cancelled passages.

140-2-s084-D001 $\xi|\zeta$ $T(\overline{K}^*)$

140-2-s084-D002 [illeg. 1]

[scan 85]

condition already spelled out that the group homomorphism $\kappa_i : T \to \mathcal{E}_i$ is compatible with the action of $\mathcal{E}_i$, operating on T via $\chi \circ p \mid \mathcal{E}_i$, and on itself by inner automorphisms.

So condition c) is not really new — I spell it out again in slightly different terms, because of its importance. It implies that the operation m190

Marginal note. m190



of Γ on π is very nontrivial (since $\chi : \Gamma \to \widehat{\mathbb{Z}}^*$ has an open image!) — it was not = obvious a priori (without deep arithmetic reasons!) — in view of the structure of π that we are going to give — that there exist such operations of Γ on a π!

This condition will be supplemented by a nontriviality condition in the style of Weil.

d) $\exists \eta \in T^*$. (a basis of T), and $\alpha_i \in \pi$ (in order to conjugate κ_i into $\kappa_i' = \mathrm{int}(\alpha_i) \circ \kappa_i$) finally an integer $g \geq 0$ and elements $x_i, y_i \in \pi$ $(1 \leq i \leq g)$, such that we have 84

[scan 86]

1°) the $\kappa_i'(\eta)$, † and the $x_i, y_i | x_j, y_j$ generate the profinite group π [p. 41]

2°) They satisfy the relation

$$\boxed{[x_1, y_1].[x_2, y_2] \ldots [x_g, y_g]\, \kappa_1'(\eta)\kappa_2'(\eta) \ldots \kappa_{\nu|n}'(\eta) = 1}$$

3°) This relation, with the generators considered describes π (as a profinite group) by generators and relations…

N.B. We know that under these conditions, g is uniquely determined, for example by the fact that $\pi_{ab}/\sum \kappa_i(T)$ is a free $\widehat{\mathbb{Z}}$-module of rank $2g$ — π_{ab} being free of rank $2g + \nu - 1$ where $\nu = \mathrm{card}(I)$.

For the choice of η, we know that if η works, then $\forall\ \chi(\alpha)\eta$ too (where $\alpha \in \Gamma$) — up to taking conjugates. So the η that work contain a subtorsorof T^* (under the open subgroup $\chi(\Gamma)$ of $\widehat{\mathbb{Z}}^*$.

† 85

Uncertainty retained from cancelled passages.

140-2-s086-D001 [illeg. 1]

140-2-s086-D004 =

[scan 87]

† In fact, since the structure of the group π is independent of K, taking $K = \mathbb{Q}$ (and admitting that there exists a smooth projective geometrically connected curve of genus g over $\mathbb{Q}$, having ν rational points over $\mathbb{Q}$!), we find that if the ℓ_i, † $(1 \leq i \leq \nu$†) fit into a privileged system of generators (with some x_α, y_α) then for $\forall \rho \in \widehat{\mathbb{Z}}^*$, we can find conjugates ℓ_i' of the ℓ_i^ρ which fit in the = way. Even for $\rho = -1$ this is not entirely trivial…

[p. 42] (therefore|in $\chi : \Gamma \to \widehat{\mathbb{Z}}^*$ epimorphism)

† Finally, we are going to state a draconian condition of nontriviality of the operation of Γ on π. Let $\mathcal{E}'$ be a subgroup open (hence of finite index) of $\mathcal{E}$, $\pi' = \pi \cap \mathcal{E}' = \mathrm{Ker}(\mathcal{E}' \to \Gamma')$ † 86

Uncertainty retained from cancelled passages.

140-2-s087-D001 Taking P $\nu - 1|\nu$ generators (

140-2-s087-D002 x_j, y_j

140-2-s087-D003 $1 \leq j \leq g$

140-2-s087-D005 [illeg. 2]

140-2-s087-D006 [illeg. 1] $\mathcal{E}|\Gamma$

[scan 88]

Γ' the image of $\mathcal{E}'$ in Γ. We thus get an extension of Γ' $(= \mathrm{Gal}(\overline{K}'/K'))$ over π', hence also by $(\pi'_{ab})_{(\ell)}$ (ℓ being a prime number), which is (we know this by d)) a free $\mathbb{Z}_\ell$-module of finite type, on which Γ' operates

[With a little work, we should be able to put on $\mathcal{E}' \to \Gamma$ a "loop structure" i.e. some $\mathcal{E}'_{i'}$, as for $\mathcal{E}$, and describe in $(\pi'_{ab})_{(\ell)}$ the sum of the images of the † $L'_{i'}$, on which Γ' operates therefore via the character χ.

m195

Marginal note. m195 This is done very elegantly in the setting

$$\Pi_{D^*} \longrightarrow \Pi_U \longrightarrow \Pi_e$$

We are interested in the quotient of $(\pi'_{ab})_{(\ell)}$ by this submodule|these submodules which is relatively trivial (the "abelian variety" part of the ℓ-adic module under consideration). This being said

ⓒ we require that the representation of Γ' on this be "pure of weight 1" — † and that the characteristic polynomials of the Frobenius elements (which have coefficients $\in \mathbb{Z}$, not just $\in \mathbb{Z}_\ell$) be independent of ℓ.

m196

Marginal note. m196



87

Uncertainty retained from cancelled passages.
140-2-s088-D001 [illeg. 1]
140-2-s088-D002 i.e.

[scan 89]

12. The topological analogy [p. 43]

Let X be a compact orientable (topological) surface, S a finite subset of X. If X is connected, we consider the set ω of the|its two orientations, we use it to twist the group $\mathbb{Z}$, whence a group T

$$T = \mathbb{Z} \underset{\pm 1}{\wedge} \{\omega\}|\omega.$$

isomorphic (noncanonically) to $\mathbb{Z}$. More generally, suppose given such a group T, i.e. an $\omega \in \mathrm{ObEns}_2$ — a T-orientation of X will by definition be an element of the set $\mathrm{Or}(X) \underset{\{\pm 1\}}{\wedge} \omega$.

[in the preceding case, X will thus be canonically T-oriented…]

Let us consider the fundamental groupoid of $X \setminus S = U$, say Π_U, and the fundamental groupoid Π_{D^*} of germs of spaces of X around S, deprived of S † groupoids ? (of germs of universal coverings of $X \setminus S$ in a neighborhood of S). We thus have a canonical functor

$$\Pi_{D^*} \longrightarrow \Pi_U$$

and on the other hand, † if X is T-oriented, we have an additional structure 88

Uncertainty retained from cancelled passages.
140-2-s089-D002 of the
140-2-s089-D003 [illeg. 2]

[scan 90]

of interest on the groupoid Π_{D^*}: the local system of its π_1 is canonically isomorphic to T.

Associating to every X which is T-oriented the system

$$\Pi_{D^*} \longrightarrow \Pi_{X|U}, \qquad \kappa : T_{(\Pi_{D^*})} \simeq \pi_1(\Pi_{D^*}/e),$$

we find a 2-equivalence between the topological 2 category of † pairs (X, S) of a (for homeomorphisms up to isotopy…) compact T-oriented manifold, and of the 2-category of the preceding systems (for equivalences), (which satisfy the conditions

a) $\pi_0(\Pi_{D^*})$, † $\pi_0(\Pi_X)$ finite

b) For every|∀ connected component of Π_U, say Π_{U_0}, and the part $\Pi_{D_0^*}$ above it, spelling out the groupoid situation by a|a group π and a family of homomorphisms $T \xrightarrow{\kappa_i} \pi$, we have (N.B. the whole thing depends|depending on choices, but π is intrinsic as an outer group, and the κ_i as homomorphisms that are outer — define|defining a "T-loop structure on S —) we have the following. 89

Uncertainty retained from cancelled passages.

140-2-s090-D001 manifolds

140-2-s090-D002 and

[scan 91]

∃ generator $g \in T$ (i.e. $g \in T^*$) and an order i_1–i_ν on the set I, † conjugates $l'_i|l'_{i_\alpha}$ of the $\kappa_{i_\alpha}(g)$, and $g \in \mathbb{N}$ and $x_\alpha, y_\alpha \in \pi$ $(1 \leq \alpha \leq g)$ such that we have [p. 44]

$$\left(\prod_1^g [x_\alpha, y_\alpha]\right) \prod_1^\nu l_i = 1$$

and that this be a defining relation of π.

There is however a grain of salt for $I = \emptyset$ (in which case T serves no purpose apparently (in the groupoid description of $(X, S) \simeq| = (X, \emptyset) = X$) one must then, † place of the "Kummerian" data κ, give oneself

$$H^2(\pi, \mathbb{Z}) \overset{\kappa}{\simeq} T$$

m200

Marginal note. m200 One must introduce therefore this as an additional datum, in the definition of groups with loops the exclusion of the cases $X_0 \simeq S^2$ (in the case of empty S) is then particularly convincing.

Finally, one must † (even|= with this grain of salt) exclude the case $X = S^2$, $S = \emptyset$ (as a connected component) — i.e. on the groups side, the case of a component of Π_U where $\pi = (1)$ † and $I = \emptyset$. [If I remember correctly, there is no reason to exclude (S^2, pt) (i.e. $X \setminus S = U \simeq \mathbb{R}^2 \simeq \mathbb{E}^1_{\mathbb{C}|e}$) 90

Uncertainty retained from cancelled passages.

140-2-s091-D001 [illeg. 2]

140-2-s091-D002 in

140-2-s091-D003 [illeg. 1]

140-2-s091-D004 [illeg. 2]

[scan 92]

where nevertheless we have $\pi = 0$. But except in these two cases, (corresponding to the case $g = 0$, $\nu = 1$) the homomorphism $\kappa_i : T \longrightarrow \pi$ is injective. If $T = \mathbb{Z}$ the datum of the κ_i is equivalent to that of elements $l_i \in \pi$, and we recover the usual definition of loop structures.

This is made explicit in the thesis of Yves Ladegaillerie — what is missing there, is (among other things) the consideration of arrows between (X, S) other than homeomorphisms (modulo isotopies); for example maps † $X' \xrightarrow{f} X$ such that $S' = f^{-1}(S)$ [and that X' be étale over $X \setminus S$, if one insists!], which reduces to the preceding case — the more general case where one assumes only that X' is a ramified covering of X (possibly ramified also outside S') and $S' \supset f^{-1}(S)$ (but S' possibly larger) would call for 91

Uncertainty retained from cancelled passages.
140-2-s092-D003 [illeg. 3]

[scan 93]

a careful study, with an ad-hoc notion of isotopy .... [p. 45]

Another important direction (notably for the study of the|of the unoriented, nonorientable case) is the introduction of finite groups of homeomorphisms, not necessarily respecting the orientation. To treat the case of a change of orientation, note that in the groupoid description $(\Pi_{D^*} \longrightarrow \Pi_U, \kappa)$ of a (X, S) that is T-oriented, the passage to the opposite orientation is expressed by keeping $\Pi_{D^*} \longrightarrow \Pi_U$ as is, and replacing κ by $\bar{\kappa} = \kappa^{-1}$

$$\bar{\kappa}(\zeta|\xi) = \kappa(-\zeta|\xi) = \kappa(\zeta|\xi)^{-1}$$

(If $T = \mathbb{Z}$, on the system $(\pi, (l_i))$, this amounts to replacing the l_i by the l_i^{-1} ...) m203

Marginal note. m203 and ditto (if $I = \emptyset$) for $\kappa : T \simeq H^2(\pi, \mathbb{Z})$ replaced by $-\kappa$.

Let then Γ be a group (a priori not necessarily finite) which operates on (X, S), hence on $U = X \setminus S$ and on Π_U, Π_{D^*}. We then find mixed fundamental groupoids † by the well-known construction 92

Uncertainty retained from cancelled passages.
140-2-s093-D001 [illeg. 1]

[scan 94]

$$\Pi_{D^*,\Gamma} \longrightarrow \Pi_{U,\Gamma} \longrightarrow \Pi_{e,\Gamma}$$

category of Γ-torsors i.e. of 1-connected objects in Γ-Sets
corresponding to the functors in the opposite direction

étale|étale Γ-coverings of D^* $\longleftarrow$ étale|étale Γ-coverings of U $\longleftarrow$ Γ-Sets

The connected components of $\Pi_{D^*,\Gamma}$ correspond to the orbits of Γ in S † $\simeq \pi_0(\Pi_{D^*})$, likewise for those of $\Pi_{U,\Gamma}$. The † connected $\Pi_{U,\Gamma}$ i.e. the connected topos $\widehat{\Pi}_{U,\Gamma}$, is the one where|with Γ transitive on $\pi_0(U) \simeq \pi_0(X)$ — at the cost of replacing X by a connected component X_0, and Γ by the subgroup|subgroup $\Sigma \subset \Gamma$ that stabilizes it, we would be reduced (for the study of the topoi $\widehat{\Pi}_{U,\Gamma}$ and of morphisms of topoi

$$\widehat{\Pi}_{D^*,\Gamma} \longrightarrow \widehat{\Pi}_{U,\Gamma} \longrightarrow (\widehat{\Pi}_{e,\Sigma} \longrightarrow)\widehat{\Pi}_{e,\Gamma}$$

to the case of (X_0, S_0, Σ). But the analogy|the analogue 93

Uncertainty retained from cancelled passages.
140-2-s094-D001 [illeg. 1]
140-2-s094-D002 topological

[scan 95]

that I have in mind with the "arithmetic" case makes this reduction inopportune in the [p. 46] general case (for in the arithmetic †, we do not restrict ourselves either to the case where $\Sigma = \Gamma$

i.e. $\mathcal{E} \longrightarrow \Gamma$ surjective, i.e. the connected component X geometric- ally connected over K). Note here that (X, S, Γ) is recovered from the $(X_0, S_0, \Gamma_0 = \Sigma)$ as an amalgamated sum

$$X = X_0 \wedge|\widehat{\times}_{\Sigma}\Gamma \quad ....$$

When we express (for $\Pi_{U|X,\Gamma}$ connected) the situation in terms of group theory, we therefore find † a mixed fundamental group

$$\mathcal{E}_{U|X,\Gamma} = \pi_1(U, \Gamma)$$

and a homomorphism

$$\mathcal{E}_{U,\Gamma} \longrightarrow \Gamma$$

— (Surjective iff † U connected (we have $\Gamma/\Sigma \simeq \pi_0(U)$)) finally a set of indices † I ($\simeq S/\Gamma \simeq \pi_0(\Pi_{D^*,\Gamma})$) and mixed fundamental groups 94

Uncertainty retained from cancelled passages.

140-2-s095-D001 case

140-2-s095-D002 [illeg. 1]

140-2-s095-D004 Γ/Σ

140-2-s095-D005 [illeg. 1]

[scan 96]

$$\mathcal{E}_i \xrightarrow{\sim} \pi_1|\Pi_1(D_i^*, \Gamma) \simeq \pi_1(D_{i_0}^*, \Sigma_{i_0}|\Gamma_{i_0})$$

where $D_{i_0}^*$ is a connected component of the punctured multidisk D_i^* (corresponding to the choice of an $s_i \in S$), and where $\Sigma_{i_0}|\Gamma_{i_0} \subset \Gamma$ is its stabilizer (i.e. the stabilizer of s_i in Γ), which is † (if Γ_i is finite and operates faithfully in a neighborhood of s_i) a cyclic or dihedral group ... We therefore find, if X is T oriented|oriented, an extension of Γ_{i_0} by T, the homomorphism $\mathcal{E}_i \longrightarrow \Sigma_{i_0}|\Gamma_{i_0} \subset \Gamma|\Gamma_i$ being induced of course via $\mathcal{E}_{U,\Gamma} \longrightarrow \Gamma$ and $\mathcal{E}_i \longrightarrow \mathcal{E}_{U,\Gamma}$.

One must also link the action of Γ on X to the orientation of X — for this we suppose given a character

$$\chi : \Gamma \longrightarrow \mathbb{Z}^* = \{\pm 1\}$$

and require that for $g \in \Gamma$, g_X preserve the orientation if $\chi(g) = 1$, reverse it if $\chi(g) = -1$. This implies that †we have an|the homomorphism or isomorphism

$$\pi_1(\Pi_{D^*,\Gamma}/\Pi_{e,\Gamma}) \overset{\kappa_\Gamma}{\underset{\sim}{\leftarrow}} T$$

m207

Marginal note. m207 local system of the kernels of the $\pi_1(\xi) \longrightarrow \pi_1(\varphi p(\xi))$ for $\xi \in \mathrm{Ob}\,\Pi_{D^*,\Gamma}$. 95

Uncertainty retained from cancelled passages.

140-2-s096-D001 a group

140-2-s096-D002 the

[scan 97]

T being regarded as a local system on $\Pi_{\Gamma,e}$ † i.e. as a Γ-group, thanks to the action of Γ via the character χ. It amounts to the same to say that $\forall i \in I$ the map [p. 47]

$$\kappa_i : T \xrightarrow{\sim} L_i = \mathrm{Ker}(\mathcal{E}_i \longrightarrow \Gamma)$$

as a homomorphism from T to $\mathcal{E}_i$ is compatible with the action of $\mathcal{E}_i$ (operating on T via χp_i ($p_i : \mathcal{E}_i \longrightarrow \Gamma$)), [illeg. 1] and on itself by inner automorphisms ...

If we exclude the case where (X_0, S_0) is $\simeq (S^2, \emptyset)$ or $(S^2, 1 \text{ pt})$ (i.e. the case $\pi = 0$), the $\kappa_i : T \longrightarrow \pi$ are injective, hence also the $\mathcal{E}_i \longrightarrow \mathcal{E}_{U,\Gamma}$, hence the $\mathcal{E}_i$ can be regarded as subgroups of $\mathcal{E}_{U,\Gamma}$.

§ Here it † would be particularly inadvisable (even|= if we assume $\Sigma = \Gamma$ i.e. $X = X_0$ i.e. X connected) to suppose that the Γ_i be|are $= \Gamma$ i.e. that the $s \in S$ are fixed by Γ ! 96

Uncertainty retained from cancelled passages.

140-2-s097-D001 [illeg. 1]

140-2-s097-D003 [illeg. 1]

[scan 98]

Since the center of π is †reduced to 1 (if we except † the case $(X_0, S_0) \simeq (S^2, \text{two points})$ i.e. $U_0 \simeq \mathbb{C}^*$ — case of the annulus —), the datum of an extension of Σ ($\subset \Gamma$) by π amounts (up to a unique isomorphism) to the datum of a homomorphism

$$\Sigma|\Gamma \longrightarrow \operatorname{Aut} \operatorname{ext}(\pi)$$

— $\mathcal{E}$ is reconstructed as the inverse image of the extension

$$1 \longrightarrow \pi \longrightarrow \operatorname{Aut}(\pi) \longrightarrow \operatorname{Autext} \pi \longrightarrow 1$$

.

But here the outer automorphisms relative to the $\alpha \in \Sigma$ respect the loop structure † of π, modulo the sign $\chi(\alpha)$ — i.e. for $g \in \mathcal{E}$, and $s \in S$ $\exists$ $\alpha \in \pi$ and $s' \in S$ (s' unique !) such that

$$\dagger \quad \operatorname{int}(g)\kappa_s(\zeta|\xi) = \operatorname{int}(\alpha)\kappa_{s'}(\chi(p(g))\zeta|\xi)$$

$\forall \zeta|\xi \in T|\mathbb{Z}$, i.e.

$$\operatorname{int}(\alpha^{-1}g)\kappa_s(\zeta|\xi) = \dagger\kappa_i|\kappa_{s'}(\chi(p(g))\zeta|\xi)$$

97

Uncertainty retained from cancelled passages.

140-2-s098-D001 [illeg. 1]

140-2-s098-D002 resp.

140-2-s098-D006 [illeg. 1]

140-2-s098-D007 $\varphi|g$

140-2-s098-D008 χ

[scan 99]

† Thus, the operation of Σ on S_0 (hence of Γ on $S = S_0 \wedge_\Sigma \Gamma$) is known, from the outer [p. 48] operation of $\Sigma \subset \Gamma$ on π endowed with its "T-loop" structure † (i.e. the outer homomorphisms $\kappa_s : T \longrightarrow \pi$) — hence also $I = S_0/\Sigma \simeq S/\Gamma$.

Can we reconstruct $\mathcal{E}_i$ ($i \in I$) from the † extension structure? We see, by virtue of the choices made, that if $i \in I$ (hence i an orbit of Σ in S_0) $\exists$ $s \in i$ such that $\operatorname{Ker}(\mathcal{E}_i \longrightarrow \Gamma)$ is none other than $L_s = \kappa_s(T)$, and we therefore have $\mathcal{E}_i \subset \operatorname{Norm}_{\mathcal{E}}(L_s)$, But we have

$$\operatorname{Norm}_{\mathcal{E}}(L_s) \cap \pi = \operatorname{Norm}_{\pi}(L_s) = L_s$$

$= | \neq \mathcal{E}_i \cap \pi$, and on the other hand the image of $\operatorname{Norm}_{\mathcal{E}}(L_s)$ in $\Sigma \subset \Gamma$ is not $\subset \Sigma_s$ = stabilizer s in Σ = Image of $\mathcal{E}_i$ in Σ, and|hence in summary

$$\mathcal{E}_i = \operatorname{Norm}_{\mathcal{E}}(L_s)$$

98

Uncertainty retained from cancelled passages.
140-2-s099-D001 [illeg. 2]
140-2-s099-D002 [illeg. 1]
140-2-s099-D003 if

[scan 100]

Conversely, the datum of $\mathcal{E} \longrightarrow \Gamma$ and of the $\mathcal{E}_i$, $\kappa_i : T \longrightarrow \mathcal{E}_i$ gives back the loop structure of π, by taking the κ_i and all their distinct outer conjugates by the $\alpha \in \Sigma$ (modified by $\chi(\alpha)$ ...). Thus the datum of the situation $\mathcal{E} \overset{p}{\to} \Gamma$, $\mathcal{E}_i \subset \mathcal{E}$ (family of subgroups, each defined modulo conjugation in $\mathcal{E}$) is equivalent to the datum of

a) $\pi = \operatorname{Ker} p$, with its T-loop structure (†finite set of outer homomorphisms from T to π)

b) A sub- group $\Sigma \subset \Gamma$, and a homomorphism

$$\Sigma \longrightarrow \text{extended Teichmüller of } \pi$$

(outer automorphisms of π, respecting the loop structure modulo sign compatible with the character $\chi|\Sigma$ and the "sign" character on extended Teichmüller. In fact, (X, S, Γ), or (U, Γ), defines π only as an outer group with T-loops on which Γ operates in a way compatible with χ. 99

Uncertainty retained from cancelled passages.
140-2-s100-D002 conjugacy classes

In summary, we have a †canonical functor Isotopic category of (X, S, Γ), X compact surface †, T oriented, S discrete subset, Γ operating by χ-automorphisms ($\sigma \in \Gamma$preserving|preserves the orientation if $\chi(\sigma) = +1$, reversing it if $\chi(g) = -1$), Γ transitive on $\Pi_0(X)$, † and if (X_0, S_0) is a connected component of (X, S), we want that †if $X_0 \simeq S^2$, we have card $S_0 \geq 3$ †. † m213 [scan 101] [p. 49]

Marginal note. m213 arrows: isotopy classes of homeomorphisms respecting all the structures.

↓

Category of outer groups with T-loops π, π noncommutative (i.e. $\pi \neq 0, \mathbb{Z}$); on which a subgroup $\Sigma \subset \Gamma$ of Γ operates (this description being equivalent: a description in terms of systems of groupoids

$$\Pi_{D^*,\Gamma} \longrightarrow \Pi_{U,\Gamma} \longrightarrow \Pi_{e,\Gamma} \quad \text{and } \kappa \dots,$$

more conceptual in certain contexts...). m214

Marginal note. m214 Beware of the case $S = \emptyset$ in the definition of the structures: the [illeg. 1] (datum of $H^2(\pi, \mathbb{Z}) \overset{\kappa}{\simeq} T$); we must of course take into account during the action of Σ on $H^2(\pi, \mathbb{Z})$ † (via its outer action on π) and on T (via χ) and require the compatibility.

† I presume that the proof of the fact that this functor is fully faithful presents no essential difficulties, m215

Marginal note. m215 It would really be worth doing it very carefully...

using what is known for $\Gamma = 1$. But the fact that, for|† Γ a given finite group, †it is essentially surjective 100

Uncertainty retained from cancelled passages.

140-2-s101-D001 [illeg. 1]
140-2-s101-D002 [illeg. 1]
140-2-s101-D003 [illeg. 1]
140-2-s101-D004 [illeg. 2]
140-2-s101-D005 [illeg. 3]
140-2-s101-D006 [illeg. 1] isomorphism
140-2-s101-D007 and on T
140-2-s101-D008 For definition
140-2-s101-D009 for
140-2-s101-D010 that it is an isomorphism

is an open problem on which people are stumped. Of course we may assume $\Sigma = \Gamma$, and [scan 102]
$\Gamma \subset$ extended Teichmüller and the question is whether every finite subgroup of the extended Teichmüller is realized as a group operating on (X_0, S_0) [illeg. 1] in an essentially unique way. More precisely, if † $A =$ group of †homeomorphisms (or diffeomorphisms, if X_0 is differentiable) of X_0, A^0 its connected component|of (N.B. A^0 is contractible, †in the anabelian cases) hence $A/A^0 = \mathcal{T}_{g,\nu}$ (Teichmüller group for genus g and ν holes), the question comes down to: this whether for every homomorphism from a finite group Σ into † $\mathcal{T}_{g,\nu}$, (we may assume $\Sigma \subset \mathcal{T}_{g,\nu}$), it lifts to a homomorphism into A, and whether two such liftings are conjugate by an $a \in A^0$ (isotopy in the strict sense of two liftings $\sim$).

Having arrived at such a simple reinterpretation|interpretation of the 1-category ("1-isotopic") deduced from the 2-category of systems $(\Pi_{D^*,\Gamma} \longrightarrow \Pi_{U,\Gamma} \longrightarrow \Pi_{e,\Gamma}, \kappa)$, in terms 101

Uncertainty retained from cancelled passages.

140-2-s102-D001 [illeg. 1]
140-2-s102-D002 diffe
140-2-s102-D003)
140-2-s102-D004 [illeg. 1]

of outer groups with loops (π[equipped with a set of outer homomorphisms $\kappa_i : T \longrightarrow \pi$, and [scan 103]
failing that with a $\kappa : T \longrightarrow H^2(\pi, \mathbb{Z})$) (to do it properly, one should write $T^{(\otimes -1)} \simeq H^2(\pi, \mathbb{Z})$, [p. 50]
but here we have a canonical isomorphism $T^{\otimes -1} \simeq T$ i.e. $T^{\otimes 2} \simeq \mathbb{Z}$ ...)], the question arises of how to recover, (up to an equivalence defined up to unique isomorphism) this diagram, in terms of κ; everything comes down to the description of the categories $\Pi_{D^*,\Gamma}$ and $\Pi_{U,\Gamma}$, and the two functors $\Pi_{D^*,\Gamma} \longrightarrow \Pi_{U,\Gamma}$, $\Pi_{U,\Gamma} \longrightarrow \Pi_{e,\Gamma}$; or what amounts to the =, of the corresponding (multi-Galois) topoi and morphisms of topoi.

a) Description of $\Pi_{U,\Gamma}$ and of $\Pi_{U,\Gamma} \longrightarrow \Pi_{e,\Gamma}$.

Let more generally π be an outer group, † whose center is trivial (this corresponds to the anabelian hypothesis!), let us show how we associate with it a classifying topos B_π, which (as a category of sheaves) will be Ens(π)) in a "functorial" way (for Isomorphisms|isomorphisms). Everything comes down to seeing how, to a conjugacy 102

Uncertainty retained from cancelled passages.

140-2-s103-D002 such that

class Θ of isomorphisms [scan 104]

$$\pi' \overset{u}{\to} \pi,$$

we associate an "inverse image" functor Ens(π) $\longrightarrow$ Ens(π'), defined up to unique isomorphism. Let us consider for every u in the class Θ, the functor "u-restriction of operations"

$$\mathrm{Ens}(\pi) \overset{u^*}{\longrightarrow} \mathrm{Ens}(\pi')$$

[which †defines, therefore, with $u_* = (u^{-1})^*$, an †equivalence of the Topoi

$$B_{\pi'} \overset{u_\bullet}{\tilde{\longrightarrow}} B_\pi$$

] We shall, between these equivalences, for $u \in \Theta$, define a transitive system of isomorphisms (which therefore lets us identify them with each other!).

Let|Let $u, u' \in \Theta$, whence u^*, u'^*; we have by hypothesis a $g \in \pi$ such that $u' = \mathrm{int}(g) \circ u$, furthermore g is determined modulo an element of $\mathrm{Centr}_\pi u(\pi') = \mathrm{Centr}(\pi) = 1$; so here g is unique. But g can be used to define a functorial isomorphism between

$$i_{u',u} : u^* \overset{\sim}{\longrightarrow} u'^*$$

by taking, for $E \in \mathrm{Ob}\,\mathrm{Ens}(\pi)$,

$$i_{u',u}(E) : \underset{\|}{u^*(E)} \longrightarrow \underset{\|}{u'^*(E)}, \qquad i_{u',u}(E) = g_E.$$
$$E \qquad\qquad E$$

103

Uncertainty retained from cancelled passages.
140-2-s104-D001 is
140-2-s104-D002 isomorphism

N.B. The argument works for every conjugacy class of homomorphisms of groups $\pi' \longrightarrow \pi$ (i.e. every outer homomorphism) whose centralizer in π is reduced to 1 (which for an epimorphism reduces to the hypothesis $\mathrm{Centr}(\pi) = 1$). m220 [scan 105] [p. 51]

Marginal note. m220 works for anabelian loops|anabelian groups with loops, and the homomorphisms of such whose image is of finite index… (for the centralizer in such a group π of a subgroup of finite index is still reduced to e|reduced to 1).

If now a group Σ operates on the outer group π, then by the preceding result we can say that it also operates on the topos B_π, whence a topos $B_{\pi,\Gamma}$, (we can also say that † Since $\mathrm{Centr}(\pi) = 1$) the operation of †Σ on π defines an extension $\mathcal{E}$ of Σ by π, whence a topos $B_{\pi,\Gamma} = B_{\mathcal{E}}$, † We recover of course also $B_{\pi,\Gamma} \longrightarrow B_\Gamma$. To reassure ourselves, we should make sure that if we have a homomorphism of outer groups $\pi' \overset{\Theta}{\longrightarrow} \pi$ commuting with the action of a Σ, with $\mathrm{Centr}_\pi(\Theta) = (1)$, then there exists a $\Theta_\Gamma : B_{\pi',\Gamma} \longrightarrow B_{\pi,\Gamma}$ defined up to canonical isomorphism. Now every $u \in \Theta$ defines a homomorphism of extensions u_Γ (of Σ by π' resp. π) $\mathcal{E}' \longrightarrow \mathcal{E}$ above|below u, and if we pass from u to $u' = \mathrm{int}(g) \circ u$, we will have 104

Uncertainty retained from cancelled passages.
140-2-s105-D001 [illeg. 1]
140-2-s105-D002 π
140-2-s105-D003 and

$u'_\Gamma = \mathrm{int}_{\mathcal{E}}(g) \circ u_\Gamma$, and we finish as above, with uniqueness of g. [scan 106]

Now take an injective outer homomorphism

$$\dagger\Lambda = [\kappa] : T \longrightarrow \pi \qquad (T \text{ commutative}),$$

m223 (initial of "loops")

Marginal note. m223 Injectivity in the case we are interested in follows from the anabelian hypothesis.

suppose that (for $\kappa \in [\kappa]$)

$$\mathrm{Centr}_\pi \kappa = \kappa(T)$$

(which does not depend on the|on the choice of κ in $[\kappa]$). I will then define a connected abelian groupoid Π_Λ, and an isomorphism of its π_1 with T (hence a topos, which plays the role of $\widehat{\Pi}_{D|D^*}$). An object will be a $\kappa \in [\kappa]$. A homomorphism from κ to κ' will be an element $g \in \mathrm{Transp}_\pi(\kappa, \kappa')$. The composition of homomorphisms obvious. We do indeed get a connected groupoid, in which $\mathrm{Aut}(\kappa) = \mathrm{Centr}_\pi(\kappa) \overset{\text{by hypothesis}}{=} \kappa(T) \underset{\sim}{\overset{\kappa}{\leftarrow}} T$.

OK. Of course, we have a homomorphism $\Pi_\Lambda \longrightarrow \dagger$ one-point groupoid, hence on the classifying|classifying topoi defined by π $B_\Lambda \longrightarrow B_\pi$. We have to see the behavior of this 105

Uncertainty retained from cancelled passages.

140-2-s106-D001 λ

140-2-s106-D005 π

construction under outer homomorphisms. So let [scan 107] [p. 52]

$$\Theta = f_\pi \dagger = [u] : \pi' \longrightarrow \pi$$

be an outer homomorphism such that $\mathrm{Centr}_\pi(\Theta) = 1$, and $\Lambda' : T \longrightarrow \pi'$ an injective outer homomorphism such that $\kappa' \in \Lambda' \Rightarrow \mathrm{Centr}_{\pi'}(\kappa') = \mathrm{Im}\,\kappa'$ and let us consider the composite

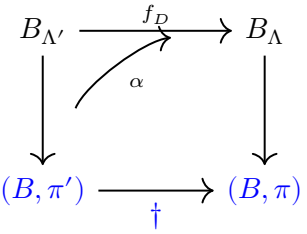
$$\dagger f_\pi \circ \Lambda' : T \longrightarrow \pi.$$

Let $d \in \mathbb{Z}$ and consider

$$\Lambda = f_\pi \circ \Lambda' \circ (d.\,\mathrm{id}_T)$$

suppose that $\kappa \in \Lambda \Rightarrow \mathrm{Centr}_\pi\,\kappa = \mathrm{Im}\,\kappa$ We are going to define a homomorphism from $\Pi_{\Lambda'}$ to Π_Λ, hence a morphism of Topoi $B_{\Lambda'} \longrightarrow B_\Lambda$, and a commutation datum α for the

(∗)



$B(f_\pi)$ defined above

$\dagger$ [illeg. 2] obviously [illeg. 5] $\dagger$ [illeg. 2].

$\kappa' : T \longrightarrow \pi'$ an object of $\Pi_{\Lambda'}$, we want to define (up to unique isomorphism) an object κ of Π_Λ. For every $u \in f_\pi = \Theta$, we consider $u \circ \kappa \circ (d.\,\mathrm{id}_T)$, there is between them a transitive system of isomorphisms for u varying in Θ, we can identify them with one another. 106

Uncertainty retained from cancelled passages.

140-2-s107-D001 Θ

140-2-s107-D002 $\Lambda =$

140-2-s107-D003 $B(f_\pi)$

140-2-s107-D004 of Topoi

140-2-s107-D005 [illeg. 2] to the [illeg. 1]

The functoriality of this object with respect to varying κ' is obvious: [we can say that $u \in \Theta$ defines a homomorphism of groupoids [scan 108]

$$u_\Lambda : \Pi_\Lambda \longrightarrow \Pi_{\Lambda'},$$

and between these there is a transitive system of isomorphisms }. In fact, for fixed u we have a commutative diagram of homomorphisms of groupoids

$$\begin{array}{ccc} \Pi_{\Lambda'} & \xrightarrow{u_\Lambda} & \Pi_{\Lambda|\Lambda e} \\ \downarrow & & \downarrow \\ (e,\pi') & \xrightarrow{u} & (e,\pi) \end{array}$$

and between these diagrams there is a transitive system of isomorphisms, hence the diagram $(*)$ and the commutation datum α.

When we have a family Λ' (or a set) † of outer homomorphisms $\Lambda'_{i'}$ $(i' \in I') : T \longrightarrow \pi'$, and a family $\Lambda{=}(\Lambda_i)_{i\in I}$ of outer homomorphisms $T \longrightarrow \pi$, and some $\tau : I' \longrightarrow I$, †and $(d_{i'})_{i'\in I'}$ the|some $d_{i'} \in \mathbb{Z}$, such that (for $\Theta : \pi' \longrightarrow \pi$ a given outer homomorphism) $\forall i' \in I'$, setting $i = \tau(i')$, we have

$$\Lambda_i = \Lambda_{i'} \circ \Theta \circ (d_{i'}.\,\mathrm{id}).$$

[N.B. if $I \longrightarrow \mathrm{Homext}(T,\pi)$ injective, these conditions show that τ is determined by Θ, and if 107

Uncertainty retained from cancelled passages.

140-2-s108-D004 [illeg. 1]

140-2-s108-D005 [illeg. 1]

in addition $\mathbb{Z}$ operates faithfully on T, for example if $T \simeq \mathbb{Z}$, then $(d_{i'})$ is also unique ...] then [scan 109] [p. 53]
we construct †

$$\Pi_{D^*,\Lambda'} = \text{groupoid sum of the } \Pi_{\Lambda'_{i'}}$$

and the $=$ for $\Pi_{D^*,\Lambda}$, and we get an essentially |isocommutative diagram of topoi

$$\begin{array}{ccc} B_{D^*,\Lambda'} & \longrightarrow & B_{D^*,\Lambda} \\ \downarrow & & \downarrow \\ B_{\pi'} & \longrightarrow & B_{\pi} \end{array}$$

Suppose now (whew!) that Σ operates on outer groups with loops ... I give up — it is obvious that everything works fine!

B. Return to the arithmetic case, if we want to describe in "Galoisian" terms pairs (X,S) † anabelian connected over a field K of finite type over $\mathbb{Q}$. [N.B. if we take a K of finite type over $\mathbb{F}_p$, we would have to restrict ourselves to the fundamental groups "prime to p", apart from that|that our developments could be carried out all the $=$ ...] 108

Uncertainty retained from cancelled passages.

140-2-s109-D001 B

140-2-s109-D003 connected

It has become clear that in terms of an algebraic closure $\overline{K}$ of K, and of the profinite Galois [scan 110]
group $\Gamma = \mathrm{Gal}(\overline{K}/K|k)$, the simplest description is in terms of outer groups with loops, and of outer actions of open subgroups $\Sigma \subset \Gamma$ on them. To be precise, we choose a connected component $\overline{X}_0$ of $\overline{X}$ (or what amounts to the $=$, $\overline{U}_0$ of $\overline{U} = (X \setminus S)_{\overline{K}}$), let Σ be its stabilizer in Γ (it is replaced by a conjugate, when we change $\overline{U}_0$). Then Σ operates on the scheme $\overline{U}_0|\overline{X}_0$, hence operates by outer automorphisms on its π_1 (considered as an outer group). Now on this there is a $T(\overline{K})$-loop structure, with as index set

$$I = S_0(\overline{K}) \subset \qquad S(\overline{K}) \qquad\qquad [\text{empty if|iff } S = \emptyset]$$

$$\wr|$$

$$S_0(\overline{K}) \wedge_\Sigma \Gamma$$

109
[scan 111]
[p. 54]

and the operation of Σ † on π is compatible with this loop structure, and the cyclotomic character $\chi : \Gamma \to \widehat{\mathbb{Z}}^*$ (rather, $\chi \mid \Sigma$). Thus the operation of Σ on π implies its action on I, whence $S(\overline{K}) \simeq I_0|\overline{S}_0 \wedge_\Sigma \Gamma$ as a Σ-set — we thus recover the étale K-scheme S.

But better, we recover the whole diagram

$$\Pi_{\overline{D}_0^*} \longrightarrow [\Pi_{\overline{U}_0} \to |-|\, \Pi_{\overline{U}} \longrightarrow \Pi_e$$

and the operation of Σ on it, whence the diagram of topoi of coverings† — or if we prefer, the diagram

$$\Pi_{D^*} \longrightarrow \Pi_U \longrightarrow \Pi_e$$

(with the notation from the beginning of these notes, which here would become $\Pi_{D^*,\Gamma} \to \Pi_{U,\Gamma} \to \Pi_{e,\Gamma}$), plus of course K…

The morphisms $(X', S') \xrightarrow{f} (X, S)$ [$S' \supset f^{-1}(S)$, f dominant] † are described † simply (via the choice of an $\overline{X}_0'$ over an $\overline{X}_0$) by outer homomorphisms (159)

110

Uncertainty retained from cancelled passages.

140-2-s111-D002 on it
140-2-s111-D003 classifying|étale [illeg. 1]
140-2-s111-D004 correspond
140-2-s111-D005 simply to the

[scan 112]

π'† $(= \pi_1(\overline{X}_0'))$ $\longrightarrow$ π $(= \pi_1(\overline{X}_0))$, compatible with the actions of Σ' $(\subset \Sigma)$ and of Σ, and with the anabelian loop structures (which is expressed with the aid of a map $\tau : I' \subset \overline{S}_0'$† $\longrightarrow I = \overline{S}_0$ compatible with Σ', and a system of natural numbers $(d_i)_{i \in I'}$). N.B. $I' = \overline{S}_0' \cap f^{-1}(\overline{S}_0)$

We must however complete, † for $I = \emptyset$ i.e. $S = \emptyset$, the definition of the loop structure, by the datum of an isomorphism

$$\kappa : T^{\otimes -1} \simeq H^2(\pi, \widehat{\mathbb{Z}})$$

[N.B., in characteristic $p \geq 0$, we must restrict ourselves to the l-adic components with $l \neq p$] i.e.

$$\widehat{\mathbb{Z}} \simeq H^2(\pi, T)$$

compatible with the action of Σ — and we must require, in the "Galoisian" interpretation of $f : (X', S') \to (X, S)$, when $S = S' = \emptyset$ that the homomorphism $\pi \to \pi'$ induces a commutative diagram

111

Uncertainty retained from cancelled passages.

140-2-s112-D001 [*illeg.* 1]
140-2-s112-D003 [*illeg.* 1]
140-2-s112-D004 [illeg. 1]

[scan 113]
[p. 55]

$$\begin{array}{ccc} \widehat{\mathbb{Z}} & \xrightarrow[\sim]{\kappa^*|\kappa} & H^2(\pi, T) \\ {\scriptstyle d}\downarrow & & \downarrow{\scriptstyle f^*} \\ \widehat{\mathbb{Z}} & \xrightarrow[\sim]{\kappa'} & H^2(\pi', T) \end{array}$$

, where $d(\in \mathbb{N})$ is the degree (defined † uniquely by this condition, as|or also but the order of Coker f^*). †

When $S^*|S = \emptyset$, $S'|S'_0 \neq \emptyset$, there should be still be a compatibility for the Kummerian data (of different natures|a different nature on π', where there really are "loops", and on π, where it is purely cohomological)|s. § The question is probably equivalent to that of describing a "cohomological" Kummerian structure on the group $\tilde{\pi}'$, obtained from a π' with loops (with $I' \neq \emptyset$) by dividing by the said loops. The question is the same, it seems, in the topological setting, or the arithmetic setting, I shall have to come back to it. It would be necessary that for every

m234

Marginal note. m234 paradigm of the passage from (X, S) to X : "plugging the holes"… 112

Uncertainty retained from cancelled passages.

140-2-s113-D001 in terms

140-2-s113-D003 [illeg. 1]

[scan 114]

homomorphism $\pi' \to \pi$ of groups with T-loops, the corresponding homomorphism $\hat{\pi}'|\tilde{\pi}' \to \tilde{\pi}$ also respect (for a suitable degree) the loop structure.

When we are interested anabelian systems $(\overline{X}, \overline{S})$ defined directly over $\overline{K}$, with $\overline{X}$ connected say, this is expressed by an outer group with T-loops, endowed with an action (not of Γ but of the) profinite group kernel defined by Γ. If we consider the † "Γ-automorphisms" of such an object (formed of a $\gamma \in \Gamma$ and an outer automorphism f of π respecting the loop structure, f and γ being compatible in an obvious sense…), we find a group (pro-finite ???) G and a homomorphism ("discrete" ???) $G \to \Gamma$ with image an open subgroup of Γ, and with kernel G_0 the group of Galois|outer loop automorphisms of π which commute with the action of the|of the "kernel"† † (which G_0 is conjecturally by the "messy conjecture", isomorphic to to the finite group $\mathrm{Aut}_{\overline{K}}(\overline{X}, \overline{S})$…). 113

Uncertainty retained from cancelled passages.

140-2-s114-D002 [illeg. 1]

140-2-s114-D004 [illeg. 1]

140-2-s114-D005 typical

[scan 115]

In terms of this exact sequence [p. 56]

$$1 \longrightarrow |-G_0 \longrightarrow G \longrightarrow |-\Gamma$$

the "restriction of $(\overline{X}, \overline{S})$ to the field K" is therefore expressed (we hope, in a fully faithful way, if the messy conjecture is valid) by a splitting $\Gamma \to G$ of $G \to \Gamma$…

13bis. Return to the notion of group with loops

Let π be a group, $([L_i])_{i \in I}$ a family of conjugacy classes of subgroups of π. We say that this defines a † "loop structure" on π [of type (g, ν)], if $\exists\, g \in \mathbb{N}$, †$|\forall i \in I$ an $L_i \in [L_i]$, a generator $l_i \in L_i$, elements $x_\alpha, y_\alpha \in \pi$ $(1 \leq \alpha \leq g)$ finally an order $i_1 \ldots i_\nu$ on I ($\nu =$ cardinality(I)), such that (putting $l_\alpha = l_{i_\alpha}$ for simplicity)

$$[x_1, y_1][x_2, y_2] \ldots [x_g, y_g] l_1 \ldots l_\nu = 0$$

is|are a presentation of the group π. We do not exclude a priori either the case $g = 0$, or the case $\nu = 0$, i.e. I empty.

We deduce from this:

a) If $\nu \neq 0$, π is free (on $2g+\nu-1$ generators) — hence free nonabelian unless $g=0$, $\nu \leq 2$ †. 114

Uncertainty retained from cancelled passages.
140-2-s115-D001 some
140-2-s115-D002 $\forall i \in I$
140-2-s115-D004 [illeg. 1]

[scan 116]

b) If $\nu = 0$, the only case where π abelian is the one $g \leq 1$. [Thus $\pi \simeq \pi_{g,\nu}$ is abelian iff $g=0$, $\nu \leq 3$ or $g=1$, $\nu \leq$ | $=0$†] m238

Marginal note. m238 To say that|Although † $\pi = 0$ †) † $g=0$, $\nu = 0, 1$, and outside these cases the L_i are $\simeq \mathbb{Z}$.

In any case, we have a canonical exact sequence of free $\mathbb{Z}$-modules of finite type

$$0 \longrightarrow |-T_\pi \longrightarrow \prod_{i \in I} L_i \overset{i}{\to} \pi_{\mathrm{ab}} \longrightarrow |-\tilde{\pi}_{\mathrm{ab}} \longrightarrow 0$$

m239

Marginal note. m239 N.B. Unless $\pi = 0$, i.e. $g=0$, $\nu = 0, 1$, the L_i are $\simeq \mathbb{Z}$.

where, for $I \neq \emptyset$, $\pi \neq 0$ T_π is $\simeq \mathbb{Z}$ (defined as $\operatorname{Ker} i$), and where the projections

$$T_\pi \longrightarrow L_i$$

are isomorphisms. In the case $I \neq \emptyset$, we call T_π the $\mathbb{Z}$-module of orientations of π (endowed with the family of (L_i)) — we define, † if $I = \emptyset$, ($\nu = 0$), but $g \neq 0$ (so $\pi \neq 1$ — the only case $\pi = 1$ is that where $g=0$, $\nu = 0$ or 1) by the condition

$$T_\pi = \underbrace{H^2(\pi, \mathbb{Z})}_{\substack{\mathbb{Z}\text{-module} \\ \text{free of rank 1}}}{}^{\otimes -1}$$

[we shall establish below a relation between the two definitions of T_π]. Thus π_{ab} is free of rank $2g+\nu-1$ if $\nu \neq 0$, $2g$ if $\nu = 0$. (so g is uniquely determined by π and $\nu = \operatorname{card} I$. 115

Uncertainty retained from cancelled passages.
140-2-s116-D001 or 1
140-2-s116-D002 [illeg. 1]
140-2-s116-D003 [illeg. 1]
140-2-s116-D004 [illeg. 2]
140-2-s116-D005 [illeg. 1] if [illeg. 2]

[scan 117]

Note that [if $\nu \neq 0$, and] except for the "abelian" cases $g=0$, $\nu = 1, 2$, the conjugacy classes [p. 57] of the L_i ($i \in I$) are distinct, so the loop structure of π can be described as the datum of a set of ν conjugacy classes of subgroups of π. Moreover, we see that $\forall\, g \in \pi$ which normalizes an L_i centralizes it (this is obvious in any case for $\nu \geq 2$, for then $L_i \to \pi_{\mathrm{ab}}$ is injective), which implies that the $\{L_i\}$ of one and the same class $[L_i]$ are canonically isomorphic to one another † which gives an intrinsic meaning to the term $\prod L_i$ in the exact sequence above, and to the canonical isomorphism $T_\pi \to L_i$…m241

Marginal note. m241 see separately the case $\nu = 1$: normalizer of the subgroup generated by $\prod [x_\alpha, y_\alpha]$, in the free group generated by the generators x_α, y_α…

If a group Σ operates on the loop structure $(\pi, ([L_i]))$ it operates on the invertible $\mathbb{Z}$-module T_π, whence a character

$$\chi : \Sigma \longrightarrow \mathbb{Z}^* = \{\pm 1\}$$

m242

Marginal note. m242 We assume from now on that we are in the anabelian case, or at least, we exclude $g = 0$, $\nu \leq 2$

116

Uncertainty retained from cancelled passages.
140-2-s117-D002 [illeg. 1] isomorphic

[scan 118]

conversely, if we have a character χ on Σ given in advance, we shall speak of an †action of Σ on $(\pi, ([L_i]))$ compatible with χ. If we are given an invertible $\mathbb{Z}$-module T, a T-loop structure on a group with loops π (with $\pi \neq 0$), endowed with an isomorphism $T \simeq T_\pi$, †

We have profinite variants (or prime-to-p profinite ones, if p is a given prime…) — but already now there are two points to note, to be checked in this case:

$$\begin{cases} \mathrm{Norm}_\pi(L_i) = \mathrm{Centr}_\pi(L_i) = L_i \text{ in the case } \nu = 1, \text{ otherwise no problem} \\ \mathrm{Centr}(\pi) = (1) \quad \text{(anabelian case)} \end{cases}$$

more generally, the centralizer of every open subgroup of π is reduced to 1…m244

Marginal note. m244 This is not proved for the moment in any profinite anabelian case!

117

Uncertainty retained from cancelled passages.
140-2-s118-D001 [illeg. 1]
140-2-s118-D002 [illeg. 3]

[scan 119]

14. Cohomological digression (on the|the "plugging the holes). [p. 58]

Let X be a locally noetherian regular scheme of dimension 1, S a closed subscheme reduced discrete such that $\forall s \in S$, $\dim_s S = 1$ (so S defined by a closed discrete subset of X), $U = X \setminus S$. m246

Marginal note. m246 It might perhaps be better to start with the purely topological case|the purely topological cases of a surface… and make the link with discrete groups.

We want to make explicit by Galoisian means the étale sheaves of sets constructible on X such that $F \mid U$ is locally constant. By the topos|carpet of Artin on the opens of the topos, they correspond to triples

$$(F_U, F_S, \varphi)$$

where F_U is a constructible locally constant sheaf on U, F_S a sheaf (necessarily|necessarily locally constant) and φ a morphism

$$F_S \xrightarrow{\varphi} j^* i_* (F_U)$$

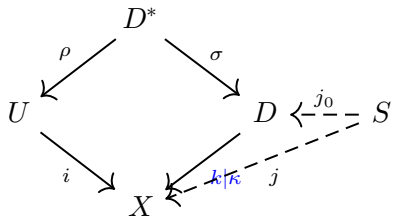
(where $i : U \hookrightarrow | \rightarrow X$ and $j : S \rightarrow X$ are the inclusions). Any constructibility condition being set aside, an étale sheaf F on X corresponds to such a triple — we restrict ourselves here to the F such that F_U † comes from the fundamental topos $B_{\Pi_1(U)|\pi_1(U)}$ of U †…(There is no need to assume F_U has finite fibers, 118

Uncertainty retained from cancelled passages.
140-2-s119-D001 comes
140-2-s119-D002 [illeg. 1]

[scan 120]

for what follows).

Let $\forall s \in S$ $\underline{O}_s$ be a henselization of $\underline{O}_{X,s}$, $D_s = \operatorname{Spec} \underline{O}_s$ ("disk at s"), $D_s^* = D_s - \{s\}$ $= \operatorname{Spec} K_s$ (K_s field of fractions of $\underline{O}_s$) ("pointed disk" at s) $D = \coprod_{s\in S} D_s$, $D_s^*|D^* = \coprod_{s\in S} D_s^*$. We have a commutative diagram



[The dotted branch $S \xrightarrow{j_0} D$ is placed in parentheses.]

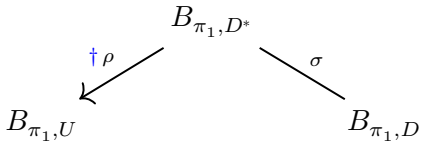
and we see (by Artin) that this diagram † allows us to express the sheaves étale on X † † as systems (F_U, F_D, φ) with F_U an étale sheaf on U, F_D an étale sheaf "essentially locally constant" on D [N.B. the inclusion $S \to D$ defines an equivalence between the category of these sheaves on D, and that of étale sheaves on S] and a morphism of sheaves $F_D \to \sigma_*\rho^* F_U$ or again 119

Uncertainty retained from cancelled passages.
140-2-s120-D002 makes of
140-2-s120-D004 as amalgamated
140-2-s120-D006 $D^*_{\text{ét}}$

[scan 121]
[p. 59]

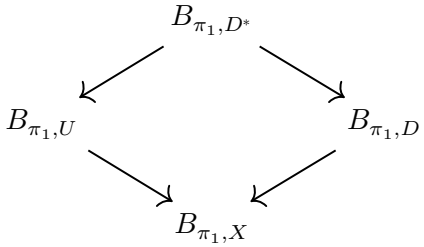
$$\boxed{\sigma^*(F_D) \xrightarrow{\varphi} \rho^*(F_U)}$$

Now, † if we restrict ourselves to the F_U such that F_U is itself essentially^t locally^t constant, then the data F_U, F_D, φ involve only essentially^t locally constant sheaves, i.e. fundamental topoi (multi-Galois) associated with the schemes under consideration, † thus are described entirely in terms of the diagrams



(*)

– moreover the case where φ is an isomorphism corresponds precisely to the case of locally essentially constant sheaves on X, i.e. of sheaves on $B_{\pi_1,X}$ – which fundamental topos thus appears as the amalgamated sum of the preceding diagram of topoi, fitting into the square



Uncertainty retained from cancelled passages.
140-2-s121-D001 this datum if F_U is a
140-2-s121-D002 namely
140-2-s121-D003 [*illeg.* 1]

Suppose for simplicity U connected, let us choose a universal covering $\widetilde{U}$ of U, and a universal covering $\widetilde{D}_i^*$ of each D_i^* – whence a universal covering $\widetilde{D}_i$ of D_i – and isomorphisms (= "classes of paths") between $\rho_!(\widetilde{D}_i^*)$ and $\widetilde{U}$ – thus the diagram of topoi (*), – or of fundamental groupoids – is made explicit in terms of a diagram [scan 122]

$$\begin{array}{ccc} & (\mathrm{Gal}(\overline{K}_i/K_i)) & \\ & \| & \\ & \mathcal{E}_i & \\ \swarrow^{\rho_i|\varphi_i} & & \searrow^{\sigma_i} \\ \mathcal{E} & & \Gamma_i \text{ [illeg. 1]} \\ \| & & \| \\ (\pi_1(U,\widetilde{U}) = \mathrm{Aut}_U(\widetilde{U})) & & (\mathrm{Gal}(\overline{k(s)}/k(s))) \end{array}$$

and the datum of an F as considered on X amounts to the datum of a system $(E_U, (E_i)_{i\in I}, (\varphi_i)_{i\in I})$, where E_U is an $\mathcal{E}$-set, E_i a Γ_i-set, $(\forall i \in I)$, and φ_i an $\mathcal{E}_i$-homomorphism from E_i to E_U (i.e. a a Γ_i-homomorphism

$$\varphi_i : \ E_i \longrightarrow E_U^{\Pi_i},$$

where † $\Pi_i = \mathrm{Ker}(\sigma_i)$ fits into the exact sequence [scan 123] [p. 60]

$$1 \longrightarrow \Pi_i \longrightarrow \mathcal{E}_i \longrightarrow \Gamma_i \longrightarrow 1$$

The case where the $E_i \longrightarrow E_U$ are isomorphisms, i.e. the E_i are identified|being identified † all with E_U, with trivial action of|of the Π_i on E_U, corresponds to the case where F is essentially locally constant|constant on X, whence at once

$$\pi_1(X) = \mathcal{E} = \pi_1(U) \ / \ {}^{\text{normal subgroup generated}}_{\text{by the } \rho_i(\Pi_i)} .$$

Our aim is that of a Galois-theoretic calculation (if possible) of the cohomology of X for the F under consideration. Let $B_{X,U}$ be the topos whose sheaves are triples (F_U, F_D, φ) as above, which thus maps to the topos $B_{\pi_1,X}$, corresponding to the pairs for which φ is an isomorphism, and receives the topos $X_{\text{ét}}$:

Uncertainty retained from cancelled passages.
140-2-s123-D001 Π_i is
140-2-s123-D002 [illeg. 1]

[scan 124]

$$X_{\text{ét}} \xrightarrow{f} B_{X,U} \xrightarrow{g} B_{\pi_1,X}$$

We ask ourselves the question

a) "Explicit" calculation, in terms of the cohomology of profinite groups, of cohomology in the abracadabra topos $B_{X,U}$. (N.B. † the cohomology † of $B_{\pi_1,X}$ is none other than the profinite Galois cohomology of $\pi_1(X) = \mathcal{E}/\ldots$ calculated above…).

b) Check whether for a torsion sheaf F on $B_{X,U}$ the canonical homomorphism

$$H^*(B_{X,U}, F) \longrightarrow H^*(X_{\text{ét}}, f^*F)$$

is an isomorphism.

N.B. Up to now, the hypothesis $\dim X = 1$ has not been used, nor the noetherian hypothesis – only the fact that S is a closed discrete subset, X connected.... For the calculations which

---

Uncertainty retained from cancelled passages.

140-2-s124-D001 b) the calculation of

140-2-s124-D002 [illeg. 1]

follow, corresponding to the † case where the $D_s|O_s$ are gauges with residue field $k(s)$ of characteristic 0, we are going to suppose that [scan 125] [p. 61]

$$\Pi_i \simeq \widehat{\mathbb{Z}} \qquad \text{(not canonically)}$$

– in fact, by Kummer theory we have a system of Tate local systems $T_{\mathcal{E}}$ (on $\mathcal{E}$), T_i (on Γ_i), and $\mathcal{E}_i$-isomorphisms $T_{\mathcal{E}} \simeq T_i$, † (i.e. a T_X on which $\widetilde{\mathcal{E}}|\overline{\mathcal{E}} = \mathcal{E}/\ldots$ operates) and we shall have a canonical Kummer isomorphism

$$T_i \xrightarrow[\sim]{\kappa_i|k_i} \Pi_i.$$

We would like to be able to paraphrase, on $B_{X,U}$, the well-known exact cohomology sequence

$$\longrightarrow H^i(X, F) \longrightarrow H^i(U, F) \longrightarrow H^{i+1}_S(X, F) \longrightarrow \cdots$$

relative to the open U of the topos X, and its closed "complement" S. Now, † just as X fits into a diagram of topoi

$$\begin{array}{ccccc} & & D^*_{\text{ét}} & & \\ & \swarrow & & \searrow & \\ U_{\text{ét}} & & & & D_{\text{ét}} \xleftarrow{\xi} S_{\text{ét}} \\ & \searrow & & \swarrow & \\ & & X_{\text{ét}} & & \end{array}$$

,

Uncertainty retained from cancelled passages.

140-2-s125-D001 [illeg. 1]

140-2-s125-D002 [illeg. 1]

140-2-s125-D003 [illeg. 2]

likewise $B_{X,U}$ fits into [scan 126]

$$\begin{array}{ccccc} & & B_{\pi_1,D^*} & & \\ & \swarrow & & \searrow & \\ B_{\pi_1,U} & & & & B_{\pi_1,D} \xleftarrow[\approx]{\xi_{\text{ét}}} B_{\pi_1,S} \\ & \searrow & & \swarrow & \\ & & B_{X,U} & & \end{array}$$

[N.B. the finite $\varinjlim$ and arbitrary $\varinjlim$ in $B_{X,U}$ i.e. for the (F_U, F_D, φ), are calculated "term by term"] I say that $B_{\pi_1,U} \longrightarrow B_{X,U}$ is identified with an induction morphism, relative to the object † (again denoted U by abuse of notation) of $B_{X,U}$ defined by $F_U =$ final sheaf e_U, $F_D =$ initial sheaf $\emptyset_D$ (and φ is|being then fixed !). Indeed, the objects of $B_{X,U}$ over U are identified with the (F_U, F_D, φ) with $F_D = \emptyset_D$ (which φ fixes !) so they form a category equivalent to that of the F_U, i.e. to $B_{\pi_1,U}$. We see that the residual topos (being identified with the category of F† $= (F_U, F_D, \varphi)$ such

Uncertainty retained from cancelled passages.

140-2-s126-D001 [illeg. 1]

140-2-s126-D003 U $E|e$ i

that $F_U = e_U$ final sheaf, is likewise identified with the category $B_{\pi_1,D}$ of the F_D † – the canonical functor "inverse image on $B_{\pi_1,D}$ of the direct image of F on $B_{\pi_1,U}$" being none other $\sigma_*\rho^*$, so that we recover the description † typical of Artin of a topos obtained by "gluing" an † open and the complementary closed. We thus find an exact sequence [scan 127] [p. 62]

$$\longrightarrow H^i(B_{X,U}, F) \longrightarrow H^i(B_{\pi_1,U}, F_U) \longrightarrow H^{i+1}_{S(\text{or } D)}(B_{X,U}, F) \longrightarrow \cdots$$

mapping into the analogous exact sequence relative to $X_{\text{ét}}, U_{\text{ét}}, S_{\text{ét}}$. † (We have still not used any special hypothesis on S…).

To check that we have isomorphisms at the level of the $H^i(B_{X,U}, F) \longrightarrow H^i(X_{\text{ét}}, F)$, it suffices by the five lemma to prove it at the level of the $H^i(B_{\pi_1,U}, F) \longrightarrow H^i(U_{\text{ét}}, F)$ [i.e. check|let us check that the cohomology of $U_{\text{ét}}$ "is calculated Galois-theoretically"] and at the level of the H^i_S. † Let us look,

Uncertainty retained from cancelled passages.

140-2-s127-D001 [illeg. 2]

140-2-s127-D002 [illeg. 1]

140-2-s127-D003 object

140-2-s127-D004 For [illeg. 2]

140-2-s127-D005 Now

first at the latter – we have $H^0_S(B_{X,U}, E) =$ set of sections of F [i.e. of pairs of a section of F_U and one of F_D, corresponding to each other under φ, i.e. of an element x of E_U invariant under $\mathcal{E}$, and of $x_i \in E_i$ invariant under the Γ_i, such that $x = \varphi_i(x_i) \quad \forall i$] such that $x = 0$ † [scan 128]

$$= \prod_i (\text{Ker}(E_i \longrightarrow E_U)$$

we observe that this functor factors through the functor H^0_S relative to the "inclusion" † analogous †

$$B \underbrace{{}_{\pi_1, D^*|D_*}}_{\text{replacing } \pi_1,U} \longrightarrow B_{D,D^*} \quad \cdots$$

and the H^i_S on $D_{X,U}$ are calculated like † those in the local situation $B_{D\dagger,D^*\dagger}$ – where we find the usual calculation in terms of the group homomorphism $\mathcal{E}_i \longrightarrow \Gamma_i$.

which I have forgotten a bit !

But in the present case, the hypothesis $\dim_s X = 1$ at the points of $s \in S$, † implies that the localized situation of topoi $D^*_{a|\text{ét}} \hookrightarrow D_{a|\text{ét}}$ is already entirely defined in terms of the fundamental topoi of the

Uncertainty retained from cancelled passages.

140-2-s128-D001 of the|in systems

140-2-s128-D002 $D^* \subset D$ or

140-2-s128-D003 [illeg. 1] [illeg. 2]

140-2-s128-D004 which is none other than
140-2-s128-D005 [*illeg.* 1]
140-2-s128-D006 [*illeg.* 1]
140-2-s128-D007 regular

† i.e. the topos B_{D,D^*} is equivalent to $D_{\text{ét}}$ – now the $H^i_S(X_{\text{ét}}, F)$ are indeed calculated on $D_{\text{ét}}$…We thus find [scan 129] [p. 63]

Proposition. The homomorphism of exact cohomology sequences under consideration is an isomorphism, provided we know that the homomorphism $H^i(\mathcal{E}, F_{\mathcal{E}}) \xrightarrow{\sim} H^i(U_{\text{ét}}, F_U)$ is an isomorphism for $\forall i$ (F_U an étale|étale local sheaf on $U_{\text{ét}}$ defined by a group $F_{\mathcal{E}}$ on which $\mathcal{E} = \pi_1(U)$ operates).

Example : OK If U is an affine curve over algebraically closed k F † prime to characteristic (since OK for $i = 0, 1$, and for $i = 2$, the two sides are zero (the Galois cohomology|cohomology, since the fundamental group prime to p is free…) – whence we deduce the analogous case for $U|X$ affine over arbitrary k, then even if U is not affine but simply quasi-projective ? † m251

Marginal note. m251 Corollary. In this case, the cohomology of X with coefficients in a locally constant F_X is also calculated Galois-theoretically ? No, because of genus 0 !!! In the case of smooth projective curves of genus $\neq 0$ a special argument is needed.

We are now interested in the case where X projective (connected) over algebraically closed k, we thus see|we thus know

Uncertainty retained from cancelled passages.
140-2-s129-D001 of the terms of
140-2-s129-D002 [illeg. 1]
140-2-s129-D004 [illeg. 1]

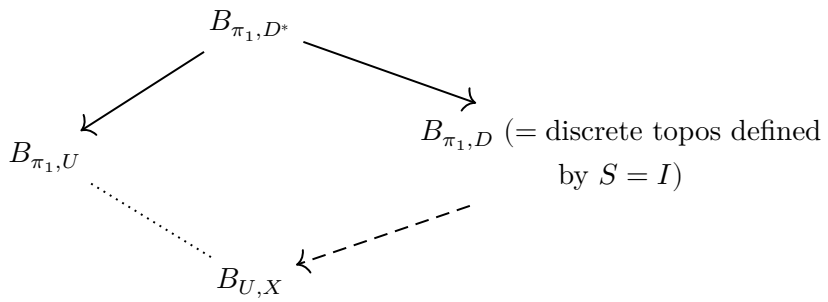
that : a) If $S = \emptyset$ i.e. $X = U$, the cohomology of U with (coefficients in local systems) is that of π. Thus…a canonical isomorphism [scan 130]

$$H^2(\pi, T) \simeq H^2(X, T) \quad (\simeq \widehat{\mathbb{Z}})$$

m252

Marginal note. m252 with grain|grains of salt in characteristic $p > 0$

b) If $S \neq \emptyset$, the $H^i_!(U, -)$ are determined|described and calculated by Galois-theoretic means, in terms of the † groups with loops, making it possible to reconstruct the situation



by postulating that the component B_{π_1,D^*_i} is none other than B_{L_i} (with the notations of the preceding number^o|case). We then find canonical isomorphisms (by local calculations)

$$H^2_{S_i|s_i}(B_{U,X}, L_i) \simeq \widehat{\mathbb{Z}} \quad \text{i.e.} \quad H^2_{S_i|s_i}(B_{U,X}, \widehat{\mathbb{Z}}) \simeq L_i^{\otimes -1}$$

(if $g \neq 0$)

$$H^2_{S_i|s_i}(B_{U,X}, \widehat{\mathbb{Z}}) \xrightarrow{\sim} H^2(X, \widehat{\mathbb{Z}}) \simeq H^2(\pi, \widehat{\mathbb{Z}})$$

whence in † putting together, canonical isomorphisms

$$\dagger \quad L_i \simeq H^2(\pi, \widehat{\mathbb{Z}})^{\otimes -1}.$$

m253

Marginal note. m253 But in all cases (even if $g = 0$, so long as we do not have $\nu = 0$) † we find $H^2_!(B_{\pi_1,U}, \widehat{\mathbb{Z}}) \simeq \widehat{T}_\pi^{\otimes -1} | T_\pi^{\otimes -1}$ canonically, whence a cohomological description of the orientation modules, common to the case without holes and with holes…

Uncertainty retained from cancelled passages.

140-2-s130-D003 systems
140-2-s130-D004 summary
140-2-s130-D005 H
140-2-s130-D006 [illeg. 1]

[scan 131]
[p. 64]

We find that the composites

$$T_\pi \xrightarrow{\sim} L_i \longrightarrow H^2(\pi, \widehat{\mathbb{Z}})^{\otimes -1}$$

do not depend on the choice of i, so that the orientation module T_π of the group profinite π with loops π is identified with the dual of $H^2(\pi, \widehat{\mathbb{Z}})$ (if $g \neq 0$).

Let us now consider a homomorphism of groups with loops

$$\pi' \xrightarrow{f} \pi$$
$$I'_0 \xrightarrow{\tau} I$$

associated with an isomorphism $T_{\pi'} \simeq T_\pi$ and a degree map $d : I'_0 \longrightarrow \mathbb{N}$ † (N.B. $i' \in I' \setminus I'_0 \longrightarrow f(L'_{i'}) = (1)$.

We would like to deduce from this a diagram of morphisms of topoi

$$\begin{array}{ccccc} B_{\Pi_1 U'} & \text{——} & B_{\dagger X',U'} & \text{——} & B_{\Pi_1 X'} \\ \downarrow & & \downarrow & & \downarrow \\ B_{\Pi_1 U} & \text{——} & B_{\dagger X,U} & \text{——} & B_{\Pi_1 X} \end{array}$$

and a homomorphism trace on the cohomology with proper supports of U', U (defined in terms of cohomology on $B_{X',U'}, B_{X,U}$ relatively) † inducing † an isomorphism

$$H^2_!(U', \widehat{\mathbb{Z}}) \longrightarrow H^2_!(U, \widehat{\mathbb{Z}})$$

Uncertainty retained from cancelled passages.

140-2-s131-D001 integers [illeg. 3]
140-2-s131-D002 Π_1
140-2-s131-D003 Π_1
140-2-s131-D004 which an isomorphism
140-2-s131-D005 for

[scan 132]

which is precisely (contragredient|contragredient of) (the isomorphism of orientation modules associated with f....

14. Where we return to mixed morphisms (corresponding, in the topological setting, to the case of $f : (X', S') \longrightarrow (X, S)$ such that we have $[f(X' \setminus S') \subset X \setminus S$, i.e. $S' \supset f^{-1}(S)$, but] † $(S' \neq f^{-1}(S)$, i.e. there are points of $S'|S$ that are sent into $U = X \setminus S$. X, X' T-oriented

In the topossic and groupoidal paradigm, (X, S) is described by a diagram of groupoids

$$\Pi_{D^*} \longrightarrow \Pi_U \qquad (\text{and } k : T_{\Pi_{D^*}} \simeq \text{ local system of the } \pi_1 \text{ on } \Pi_{D^*})$$

† or of topoi

$$B_{D^*} \longrightarrow B_U$$

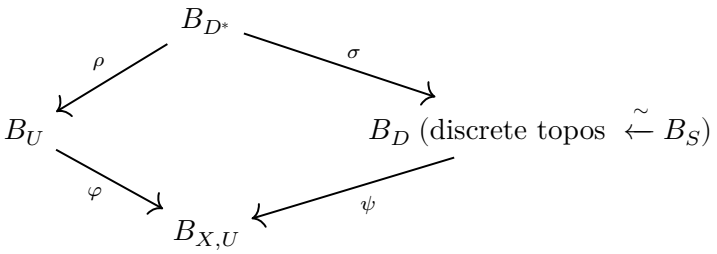
and † isomorphism k of the local system of the π_1 on B_{D^*}, into the locally constant system T, † which allows us to define B_D $(= B_{\pi_0 D_*})$ (the discrete topos) and the topos $B_{X,U}$, fitting into the diagram † of topoi

---

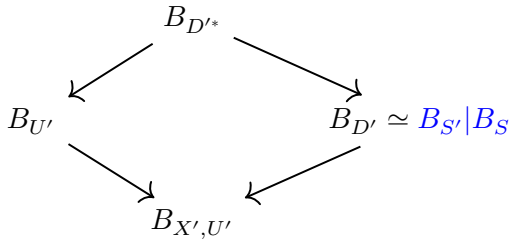
Uncertainty retained from cancelled passages.

140-2-s132-D001 [illeg. 1]
140-2-s132-D004 [illeg. 1]
140-2-s132-D005 an
140-2-s132-D006 [illeg. 4] X
140-2-s132-D008 $B_{\Pi_1 D^*}$

[scan 133]
[p. 65]



where φ, ψ are morphisms of inclusion|induction of subtopoi, φ open ψ closed, complementary to one another, and likewise for (X', S'), described by an analogous diagram



That said, we absolutely want, in a description of $f : (X', S') \longrightarrow (X, S)$, for it to allow us to recover not only $B_{U'} \longrightarrow B_U$ (which will be secured by giving an $f_\pi : \pi' \longrightarrow \pi$), but also

$$B_{X',U'} \longrightarrow B_{X,U} \qquad g|q.$$

We will have †

$$S' = \underset{\underset{f^{-1}(S)}{\|}}{S'_0} \amalg S'_1, \quad \text{hence} \quad D' = D'_0 \amalg D'_1,$$

Uncertainty retained from cancelled passages.
140-2-s133-D001 S

[scan 134]
[p. 66]

and in fact f induces $S'_1 \overset{f}{\to} U$ (and not $S'_1 \longrightarrow S$) which must be made explicit, at the level † of multi-Galois topoi, by a morphism

$$\begin{array}{ccc} B_{D'_1} & \longrightarrow & B_U \\ & \simeq & \\ B_{S'_1} & & \end{array}$$

i.e. by giving a family of universal coverings of U, parameterized by $S'_1 = I'_1$, or (if such a universal covering is chosen, U being connected, hence a $\pi = \mathrm{Aut}(\widetilde{U})$) by a family of torsors under † π, $(P_{i'})_{i' \in I'_1}$. This being † laid down we will be able to describe, in terms of

$$\begin{array}{ccc} B_{D'^*_0} & \longrightarrow & B_{U'} \\ \downarrow & \overset{\sim}{\nearrow} & \downarrow \\ B_{D^*} & \longrightarrow & B_U \end{array}$$

and of

$$B_{D'^*_1} \longrightarrow B_{D'_1} \longrightarrow B_U$$

† the morphism of topoi $B_{X',U'} \longrightarrow B_{X,U}$ — I pass over the details of the description [m255]

Marginal note. [m255] It seems that we are entitled to give the description of the morphisms of topoi $B_{X',U'} \longrightarrow B_{X,U}$ that induce $U' \longrightarrow U$ and that on D'^*_1 $(= D'^*|U$ †, factor through $D'^*_1 \longrightarrow D'_1$...

Uncertainty retained from cancelled passages.
140-2-s134-D001 [illeg. 2]
140-2-s134-D002 the
140-2-s134-D003 defined
140-2-s134-D004 a
140-2-s134-D005 [illeg. 2]

[scan 135]

When we give ourselves a group of operators Γ on the situation $(X', S') \longrightarrow (X, S)$ hence on the groupoidal or topossic situation, we must take this into account in the description above. Thus I'_0 and I'_1 are stable under Γ, and the morphism † $B_{D'_1} \longrightarrow B_U$ must be stable under Γ. † † Which also means, probably, that we have a morphism of topoi from $B_{D'_1,\Sigma} = B_{I'_1,\Sigma|\Gamma}$ into $B_{U,\Gamma}$. Choosing for every orbit of Γ in I'_1 a representative $s' \in I'_1$, and considering its stabilizer $\Gamma_{s'} \subset \Gamma$, we must therefore for every such orbit i.e. † every i' give ourselves a $\Gamma_{s'}$-object of the category of universal coverings of (U, Γ). Simultaneously|Equivalently Describing (U, Γ) in terms

Uncertainty retained from cancelled passages.
140-2-s135-D002 [illeg. 1]
140-2-s135-D003 nice $E_f = \bigcup E_i$
140-2-s135-D004 [illeg. 1] Hom $s'_{i'}$
140-2-s135-D007 every

[scan 136]

of an extension $\mathcal{E}$ of $\Sigma \subset \Gamma$ by π (by choosing a universal covering $\widetilde{U}$ of U), the datum of $B_{\Gamma_{s'}} \longrightarrow B_{X,U}$ compatible with everything amounts unless I am mistaken to the datum of a right torsor P under π (allowing us to twist the reference universal covering $\widetilde{U}$ of $U^{\Gamma}|U$, with P hence also to twist $\mathcal{E}$ by P), and of a splitting of $\mathcal{E}^P \longrightarrow \Gamma$ over $\Gamma_{s'}$. [The isomorphism class of such an object $\longrightarrow$ a π-conjugacy class of splittings of $\mathcal{E} \longrightarrow \Gamma$ over $\Gamma_{s'}$...). [p. 67]

An automorphism of such a pair $P,\ \Gamma_{s'} \overset{q}{\to} \mathcal{E}^P$ corresponds to an $\alpha \in \pi$ that centralizes $\Gamma_{s'}$, i.e. that is fixed by $\Gamma_{s'}$ operating on π. (If the $\Gamma_{s'}$ operate strongly enough on π for us to know that $\pi^{\Gamma_{s'}} = (1)$, the object $(P, q : \Gamma_{s'} \longrightarrow \mathcal{E}^P)$ is defined up to unique isomorphism by the isomorphism class † (class of splittings of $\mathcal{E}$ over $\Gamma_{s'}$). It seems probable to me that this is always the case in the arithmetic case, where $\Gamma_{s'}$ is an open subgroup of the Galois group $\Gamma_{k|K}$ (whose outer operation is draconian !). !!!|[three vertical strokes]

Uncertainty retained from cancelled passages.
140-2-s136-D003 of it

[scan 137]

This goes in the direction of the conjecture that a splitting class of $\mathcal{E}$ over Γ is "as good" as a rational point of U over K — and allows us to paradigm what such a rational point would allow us to obtain....

⑮ Return to the topological case Let Γ be a finite group operating on a group with loops π — let us suppose that this comes from a topological situation, Γ operating on (X, S). There are points to check (anabelian|anabelian cases) m257

Marginal note. m257 Structure of the critical Γ-orbits in terms of extension...

$$\text{a) Operation of } \Gamma \text{ trivial} \iff \Gamma \longrightarrow \underset{\text{trivial}}{\text{Autext}(\pi)}$$

in other words : an automorphism of finite order of (X, S) can be isotopic to the identity (or even merely homotopic, it comes to the same thing anyway) only if it is trivial.

This is even known in "abstract" Algebraic Geometry, once we admit that u preserves|preserves a † complex structure, † this is precisely

Uncertainty retained from cancelled passages.
140-2-s137-D001 complex
140-2-s137-D002 conformal

[scan 138]

so if we admit that there is no wildness.... Probably every action of a finite group Γ leaves conformal structure invariant, † hence its restriction to [p. 68]

$$\Gamma^0 = \text{Ker}(\Gamma \overset{\chi}{\to} \{\pm 1\})$$

leaves complex structure invariant.

Let us suppose from now on that Γ operates faithfully, $\Gamma \neq 1$, † The choice of a complex structure on $Y = X/\Gamma^0$ defines one on X; (N.B.: † Γ/Γ^0 still operates on X/Γ^0 in fact if $\Gamma \neq \Gamma^0$ i.e. $\Gamma/\Gamma^0 \simeq \{\pm 1\}$, X/Γ^0 is endowed with a "structure of a † real algebraic curve conformal structure"...) stable under Γ^0 — and (stable under Γ if we choose that of Y † left invariant under the element|the element nontrivial in Γ/Γ^0 (if there is one).). m258

Marginal note. m258 works|as every time we have a ramified covering of conformal surfaces

Every $x \in U^\Gamma$ defines a class † † of π-conjugacy of splittings of the extension $\mathcal{E}$ of Γ by π, as we see by taking x as base point. ! Note that U^Γ is a finite set if $\Gamma^0 \neq 1$, [i.e. Γ not reduced to an † anti-involution [illeg. 1]). We find a map

N.B. $U^\Gamma \neq \emptyset$ implies that Γ^0 is cyclic...

$$U^\Gamma \longrightarrow \text{set of classes of } \pi\text{-conjugacy of splittings of } \mathcal{E} \longrightarrow \Gamma,$$

Uncertainty retained from cancelled passages.

140-2-s138-D002 [illeg. 1]
140-2-s138-D003 [illeg. 1]
140-2-s138-D004 [illeg. 1]
140-2-s138-D006 [illeg. 2]
140-2-s138-D007 invariant
140-2-s138-D009 [illeg. 1]
140-2-s138-D010 class
140-2-s138-D011 [illeg. 1]

[scan 139]

i.e. set|to the of liftings of $\Gamma \longrightarrow \text{Autext}\,\pi$ to $\Gamma \longrightarrow \text{Aut}(\pi)$, modulo π-conjugation. [p. 69]

Question (If $\Gamma^0 \neq (1)$) Is this map bijective ??

[If $\Gamma^0 = \{1\}$, $\Gamma/\Gamma^0 \simeq \{\pm 1\}$, then X^Γ is the set of real points of a real algebraic curve and $U^\Gamma = X^\Gamma - S^\Gamma$ is the complement of a finite subset in it, we have

$$\pi_0(X^\Gamma - S^\Gamma) \longrightarrow \pi\text{-classes of splittings of } \mathcal{E} \text{ over } \Gamma$$

and the † analogous question of bijectivity arises, for extensions of ± 1 by π...]

For the injectivity of the map in the case $\Gamma^0 \neq (1)$, we may suppose $\Gamma = \Gamma_a^0|\Gamma^0 \simeq \mathbb{Z}/p\mathbb{Z}$, with p prime, i.e. Γ generated by a complex automorphism $\gamma|y$ of order p, which defines (if $x, y \in U^\Gamma$, $x \neq y$) an automorphism of order p of $\pi_1(U, x)$, † u_x, that is (hence π-conjugacy class of automorphisms of π of order p), and another u_y. We have to prove that u_x, u_y are not conjugate under π

Uncertainty retained from cancelled passages.

140-2-s139-D001 [illeg. 1]
140-2-s139-D002 say

[scan 140]

Theorem Let $U|X$ be † a connected topological space (†, by paths), Γ a finite group operating on U, $\widetilde{U}$ a universal covering of U, hence an extension group

$$1 \longrightarrow \pi \xrightarrow{i} \mathcal{E} \xrightarrow{p} \Gamma \longrightarrow 1$$

operating faithfully on $\widetilde{U}$ ($\pi = \text{Aut}_U(\widetilde{U}) \simeq \pi_1(U)$...). For every fixed point $x \in U^\Gamma$, Γ operates on the universal covering punctured|pointed over x, say † $R_x|\mathcal{R}_x$, leaving fixed the point marked $\tilde{x}$ of R_x over x, hence a splitting of the extension relative to R_x

$$1 \longrightarrow \pi_x \xrightarrow{i_x} \mathcal{E}_x \underset{\sigma_x}{\overset{p_x}{\rightleftarrows}} \Gamma \longrightarrow 1$$

and for every isomorphism ℓ ("path") $R_x \simeq \widetilde{U}$, inducing an isomorphism of extensions $\mathcal{E}_x \xrightarrow{\sim} \mathcal{E}$ (defined of|in modulo composition with an inner automorphism induced by an $\alpha \in \pi$...

) we find by transport of structure a splitting $\sigma_{x,\ell}$ of $\mathcal{E} \xrightarrow{p} \Gamma$, (which, for varying ℓ, is defined up to an inner automorphism by an $\alpha \in \pi$). We thus find a map

$$(*) \quad U^{\Gamma} \longrightarrow \begin{array}{c} \text{classes of } \pi\text{-conjugacy of splittings} \\ \text{of the extension } \mathcal{E} \text{ of } \Gamma \text{ by } \pi \\ = \text{classes of } \pi\text{-conjugacy of subgroups } \textit{sections } \Gamma' \text{ of } \mathcal{E} \end{array}$$

Uncertainty retained from cancelled passages.

140-2-s140-D001 a surface
140-2-s140-D002 [illeg. 2]
140-2-s140-D005 [illeg. 1]

[scan 141]

The image of this map thus consists of the π-conjugacy classes of the section subgroups $\Gamma' \subset \mathcal{E}$ such that $\widetilde{U}^{\Gamma'} \neq \emptyset$, and for such a Γ', the set of $x \in U^{\Gamma}$ which † give as image this conjugacy class is the image of $U^{\Gamma'}$ in U. Finally, if x is in this image, the set $\widetilde{U}_x^{\Gamma'}$ of the $\tilde{x} \in \widetilde{U}^{\Gamma'}$ over x († hypothesis on a|x) is a torsor under $\pi^{\Gamma'}$ i.e. if $\tilde{x} \in (\widetilde{U}^{\Gamma'})_x$, † and if $\alpha \in \pi$, then [p. 70]

$$\alpha.\tilde{x} \in \widetilde{U}^{\Gamma'} \quad \Longleftrightarrow \quad \alpha \in \pi^{\Gamma'}$$

(Trivial verification, as for that matter all the preceding assertions). Thus, the following conditions are equivalent:

m260

Marginal note. m260 N.B. $\widetilde{U}_x$ being identified with $\mathrm{Isom}_U(R_x, \widetilde{U})$, $\widetilde{U}_x^{\Gamma'}$ is identified with $\mathrm{Isom}_{U,\Gamma}(R_x, \widetilde{U}_{\Gamma'})$ of the two isomorphisms that commute|of covering isomorphisms commuting with the action of Γ and $\pi^{\Gamma} = (1)$ means that this isomorphism is unique i.e. † the conjugacy class of the sections [illeg. 2].

a) $(*)$ is a bijection, and $\forall$ subgroup Γ' of $\mathcal{E}$ a section of the extension, † we have $\pi^{\Gamma'} = \{1\}$.

b) $\forall\, \Gamma'$ as in a), Γ' operating on $\widetilde{U}$ has a fixed point and only one.

This being so, let us prove the

Fundamental lemma Let Γ be a finite finite group, operating faithfully on a space $D \simeq \mathbb{R}^2$, let Γ° be the subgroup of Γ [p. 140]

Uncertainty retained from cancelled passages.

140-2-s141-D001 to it
140-2-s141-D002 $\neq \emptyset$
140-2-s141-D003 of
140-2-s141-D004 [illeg. 3]
140-2-s141-D005 [illeg. 1]

[scan 142]

(of index 0 or 1) consisting of the $g \in \Gamma$ such that g_D preserves the orientation, and suppose $\Gamma^\circ \neq \{1\}$ (i.e. Γ is reduced neither to the trivial group, nor to the group † $\{1, \sigma\}$, where σ is an anti-involution of D. Then

a) Γ has a fixed point and only one in D i.e. $\mathrm{card}\, D^{\Gamma} = 1$

b) † Γ° is cyclic, and if $\Gamma \neq \Gamma^\circ$, Γ is a dihedral group.

More precisely, there exists a homeomorphism

$$D \simeq \mathbb{C}$$

such that on $\mathbb{C}$ by the group † of homeomorphisms of $\mathbb{C}$ transformed from † Γ is either the group of homotheties by $\mu_n(\mathbb{C})$ (if $\Gamma = \Gamma^\circ$ of order n), or the associated dihedral group $z \longrightarrow \zeta\tau^{\varepsilon}(z)$ ($\zeta \in \mu_n(\mathbb{C})$, $\varepsilon = \pm 1$) where σ † is complex conjugation.

Proof of the fundamental lemma.

† [p. 141]

Uncertainty retained from cancelled passages.

140-2-s142-D001 [illeg. 1]

140-2-s142-D002 And

140-2-s142-D004 [illeg. 1]

140-2-s142-D006 fixing

140-2-s142-D007 [illeg. 1]

140-2-s142-D008 without a poin^{t}

[scan 143]

† [p. 71]

a) Suppose first that one can find a C^2 structure on D invariant under Γ — then a standard argument shows that there exists a conformal structure invariant under Γ, now the fundamental theorem of conformal mapping shows that then

either $D \simeq$ interior Δ of the unit disk Δ or of the Poincaré half-plane
or $D \simeq \mathbb{C}$ (conformal isomorphisms).

In the first case, the group of conformal automorphisms is

$$\simeq \mathrm{Sl}(2,\mathbb{R})^{\sim} \mid \mathrm{Gl}(2,\mathbb{R})^{\sim} = \{u \in \mathrm{Gl}(2,\mathbb{R}) \mid \det u = \pm 1\}$$

($u = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ operating on † $\widehat{\mathbb{C}}$ by † $\Theta_u \circ \tau^{\det u}$, where τ is complex conjugation and $\Theta_u(z) = \dfrac{az+b}{cz+d}$, and leaving the Poincaré half-plane stable), and every compact subgroup is contained in a conjugate of the maximal compact subgroup, which (on passing back to the unit disk Δ) is identified with the group $O(2,\mathbb{R})$ of [p. 142]

Uncertainty retained from cancelled passages.

140-2-s143-D002 a) $D^{\Gamma} \neq \emptyset$

140-2-s143-D006 [illeg. 1]

140-2-s143-D007 $z \longmapsto \dfrac{az+b}{cz+d}$

[scan 144]

transformations of the unit disk of the form

$$z \longrightarrow \zeta \tau^{\varepsilon}(z) \qquad \begin{array}{c} \zeta \in \mathbb{U} = \{\zeta \in \mathbb{C} \mid |\zeta| = 1\} \\ \varepsilon \in \{\pm 1\} \\ \tau \text{ complex conjugation} \end{array}$$

Every finite subgroup of this group is of one of the types made explicit above — we win, by using a homeomorphism $[0,1[\longrightarrow [0,+\infty[$ to define a homeomorphism † $\Delta \simeq \mathbb{C}$ commuting with the action of $G =$ †$O(2,\mathbb{R})$.

In the case $D \simeq \mathbb{C}$, we see that † the group of conformal automorphisms of $\mathbb{C}$ is the group of transformations $az+b$ or $a\bar{z}+b$, in which a maximal compact subgroup is the same group †$O(2,\mathbb{R})$ as just now — and every compact subgroup (a fortiori, every finite subgroup) is contained in a conjugate of this one. We win again.

The rest of the work consists essential^{ly} in showing that the hypothesis of non-wildness [p. 143]

Uncertainty retained from cancelled passages.

140-2-s144-D001 [illeg. 1]

140-2-s144-D002 S

140-2-s144-D003 is

140-2-s144-D004 S

[scan 145]

is always satisfied, at least for Γ°. Suppose first $\Gamma = \Gamma^\circ$. We assume that all is proved for [p. 72]
the orders $<$ card(Γ).

b) Γ has a fixed poin^{t} $o|0$, i.e. $D^\Gamma \neq \emptyset$.

Otherwise, the stabilizer subgroups Γ_x of the $x \in D$ being of order $<$ † card Γ, by the induction hypothesis, the Γ_x have the structure stated in the theorem, hence $D^\Gamma = U|D/\Gamma = U$ is a topological surface and $D \longrightarrow U|D \longrightarrow D^\Gamma$ is a ramified covering, let us choose a conformal structure on U ; there is a unique conformal structure on D such that $D \longrightarrow U$ is † "conformal" (holomorphic or anti-holomorphic), this is invariant under Γ, and by a) Γ has a fixed poin^{t}, contradiction.

c) Γ has no fixed poin^{t} other than $o|0$. We let Γ operate faithful^{ly} on $D^* = D \setminus \{o\}|D^* = D \setminus \{0\}$, and we have to prove that $D^{*\Gamma} = \emptyset$. [p. 144]

Uncertainty retained from cancelled passages.

140-2-s145-D004 [illeg. 1]

140-2-s145-D007 conformal|holomorphic

[scan 146]

Let then $x \in D^{*\Gamma}$ — we are going to reach a contradiction. Consider the universal covering $\widetilde{D}^*$ of D^* pointed at x, so Γ operates on $\widetilde{D}^*$ with a fixed poin^{t} $\tilde{x}$ over x. Here $\pi = \pi_1(D^*) \simeq \mathbb{Z}$, and Γ operates on it trivial^{ly} (since $\Gamma = \Gamma^\circ$) so $\Gamma \times \mathbb{Z}$ operates on $\widetilde{D}^*$. We may assume $D = \mathbb{C}$, $o = 0|0 = 0$, $x = 1$, $D^* = \mathbb{C}^*$, $\widetilde{D}^* = \mathbb{C}$, $\tilde{x} = 0$, $\widetilde{D}^* \longrightarrow \widetilde{D}$ given by $\exp 2i\pi z$, † and $\mathbb{Z}$ operating on $\mathbb{C}$ by $\Theta_n z = z + n$ $(n \in \mathbb{Z})$.

It remains to prove that if a finite group Γ operates on $\mathbb{C}$ commuting † with the action of $\mathbb{Z}$ on $\mathbb{C}$, and leaving fixed the poin^{t} $o|0$, then Γ operates trivially (which contradicts the hypothesis of faithfulness of the operation).

We are reduced to the.

Lemma Let u be a homeomorphism of finite order of $\mathbb{C}$, [p. 145]

Uncertainty retained from cancelled passages.

140-2-s146-D002 so

140-2-s146-D004 with

[scan 147]

commuting with $z \longmapsto z + 1$, and leaving the origin invariant. Then $u = \mathrm{id}$ [p. 73]

We may assume $\exists$ prime number p such that $u^p = \mathrm{id}$, i.e. that u corresponds to an operation of $\Gamma = \mathbb{Z}/p\mathbb{Z}$ on $\mathbb{C}$. All the poin^{ts} of $\mathbb{Z} \subset \mathbb{C}$ are fixed by Γ. Passing to $\widehat{\mathbb{C}} \simeq S^2$, we find that ∞ is an accumulation poin^{t} of the fixed poin^{ts} under Γ — if Γ did not operate trivially, this would be decidedly very wild! We ought to be able to finish by the Adams spectral sequence... I am not going into these mazes...

d) The purely topological part being admitted|thus assumed|assumed proved|proved, we thus conclude from this, if $\Gamma \neq \Gamma^\circ$, $\Gamma^\circ \neq (1)$, as D^{Γ° is invariant under Γ, as D^{Γ° is reduced to a poin^{t}, this one is invariant under all of Γ, not just Γ°. On the other hand, We know enough now to know that if Γ finite group operates on a surface paracompact U, the Γ_x $(x \in U)$ respecting the orientation, then $U/\Gamma = V$ [p. 146]

[scan 148]

is a surface — and $U \longrightarrow V = U/\Gamma$ is a ramified covering and choosing a conformal structure on this one, we find by inverse image a conformal structure on U invariant under Γ, For the case $U = D$ we finish by a), for the complement to the fundamental lemma. [But to do things properly, one would have to prove that there always exists invariant conformal structure, † Γ finite group operates on surfaces † paracompact — so U/Γ is a surface with boundary.... Here what is missing is the analysis of the action of an anti-involution of a surface in a neighborhood of a fixed poin^{t}...]

m261

Marginal note. m261 One would have to prove that if τ is an involutive anti-automorphism of D, then $\exists$ isomorphism $D \simeq \mathbb{C}$ such that τ becomes $z \longmapsto \overline{z}$ (hence $D^{\tau} \simeq \mathbb{R}$!) which should make it possible to prove, if $\Gamma \simeq \mathbb{Z}/2\mathbb{Z}$ operates by anti-automorphisms on U (oriented|orientable $U \neq S^2$) that $\pi_0(U^{\Gamma}) \Longrightarrow \pi$-conjugacy classes of sections of $\mathcal{E}$ over Γ is bijective.

Consequence of the fundamental lemma:

Theorem. Let U be a paracompact topological surface 0-connected, Γ a finite group operating faithfully on U, we assume $\widetilde{U}$ † noncompact (i.e. U not homeomorphic to S^2 nor [p. 147]

Uncertainty retained from cancelled passages.

140-2-s148-D003 [illeg. 2]

140-2-s148-D004 [illeg. 1]

140-2-s148-D005 [illeg. 1]

[scan 149]

to the real projective plane), we assume that † that moreover if U is orientable, then † [p. 74] assume that the subgroup Γ° of Γ of the $g \in \Gamma$ which preserve an orientation is $\neq (1)$ [hence Γ not reduced either to 1, or to 1 and|Id an anti-involution], and if U nonorientable, that † $\Gamma \not\simeq | \neq (1), \{\pm 1\}$ i.e. $\operatorname{card}\Gamma \geq 3$. This being so, † let us consider the extension $\mathcal{E}$ of Γ by $\pi = \pi_1(U)$, and the map

$$U^{\Gamma} \longrightarrow \begin{array}{c}\text{Classes of } \pi\text{-conjugacy of} \\ \text{splittings of } \mathcal{E} \longrightarrow \Gamma\end{array}$$

we have this:

a) This map is bijective

b) For every section subgroup $\Gamma' \subset \mathcal{E}$, we have $\pi^{\Gamma'} = (1)$

c) If $U^{\Gamma} \neq \emptyset$ i.e. If there exists a splitting, then Γ is cyclic or dihedral (and in the case U oriented, Γ° is cyclic)

N.B. The hypotheses on U ensure that $\widetilde{U} \simeq D$, and those on Γ that Γ operating on D satisfies the hypotheses of the fundamental lemma. [p. 148]

Uncertainty retained from cancelled passages.

140-2-s149-D001 $\Gamma \neq (1)$ [illeg. 1]

140-2-s149-D003 $\Gamma \neq \{\pm 1\}$ [illeg. 1]

140-2-s149-D005 cardinality

140-2-s149-D006 the map

[scan 150]

†

Corollary Suppose U orientable, † Γ preserving the orientation. Let $U^{!}$ be the set of the $x \in U$ such that $\Gamma_x \neq (1)$. (which is therefore a discrete subset of U).

To every $x \in U^{!}$, let us associate the conjugacy class of the subgroups of $\mathcal{E}$ partial π sections of $\mathcal{E}$ over Γ_x which correspond|corresponding to this|this $x \in U^{\Gamma_x}$. Then

a) the partial sections thus obtained are maximal among those which are $\neq (1)$

b) The map from $U^!$ to the set|one of the π-conjugacy classes of maximal partial sections $\neq (1)$, is bijective

m262

Marginal note. m262 N.B. This map commutes with the natural actions of Γ!

c) For every such partial section, we have $\pi^{\Gamma'} = (1)$, i.e. the † automorphisms of the universal covering R_x of U which commute with the action of Γ_x are trivial

— there is a unique isomorphism commuting with the action of Γ_x between this torsor, and the torsor deduced from $\widetilde{U}$ by twisting by the π-torsor † P_x of the Γ'' in the class † $[\Gamma']$... [p. 149]

Uncertainty retained from cancelled passages.

140-2-s150-D001 inner automorphisms|functoriality $\mathcal{E}$-set which is a

140-2-s150-D006 [illeg. 1]

140-2-s150-D009 Γ_x-

140-2-s150-D010 of the

140-2-s150-D011 [illeg. 1]

[scan 151]

Let us prove a). Let $\Gamma' \subset \mathcal{E}$ be a partial section over Γ_x deduced from $x \in U^!$. Let $\Gamma'' \supset \Gamma'_x$ [p. 75] be another subgroup|subgroup let $\Gamma_1 \supset \Gamma_x$ be its image in|of Γ such that $\Gamma'' \cap \pi = (1)$ i.e. $\Gamma'' \hookrightarrow \Gamma$ [illeg. 1]. By the preceding theorem, it is defined by a unique $y \in U^{\Gamma_1} | y \in U^{\Gamma''_*}$, and it is clear that this y does not change if we replace the action of Γ_1 on U by the action of a smaller group $\neq (1)$ (and the induced section) such as Γ_x, hence Γ_1 † fixes x hence (by definition of Γ_x) $\Gamma_1 = \Gamma_x$ hence $\Gamma'' = \Gamma'$, †

†b) Let x, y give the same image $[\Gamma']$ $[\Gamma'']$, hence $\Gamma_x = \Gamma_y$, say Γ_1, and applying the theorem to Γ_1 operating on U, we find $x = y$. Let, on the other hand, Γ'_0 be a maximal partial section $\neq 1$, $\Gamma_0 \subset \Gamma$ its image, by the theorem applied to the action of Γ_0 on U, $\exists x \in U$ † such that $x \in U^{\Gamma'_0}$ i.e. $\Gamma'_0 \subset \Gamma_x$, and such that $[\Gamma'_0]$ is defined by x, but if $[\Gamma']$ is defined by x for the action of Γ_x in its entirety, we will have $[\Gamma'_0] \subset [\Gamma']$, hence by the maximal character of $[\Gamma'_0]$, we will have $[\Gamma'_0] = [\Gamma']$, which † proves b). On the other hand c) is clear. 150

Uncertainty retained from cancelled passages.

140-2-s151-D001 [illeg. 1]

140-2-s151-D002 q.e.d.

140-2-s151-D003 The injectivity

140-2-s151-D004 fixed

140-2-s151-D005 [illeg. 1]

[scan 152]

Let us now return to the case where $U = X \setminus S$, X a compact T-oriented surface and S a finite subset, anabelian case — so we have a "fully faithful" description of U by a π with loop structure, and we would like to convince ourselves that † the outer operation of Γ on π, when Γ preserves orientation (to simplify) also suffices to describe fully the object (U, Γ) in the appropriate isotopic category. m264

Marginal note. m264 we should return to the general case for what follows...

We already recover a description of $U^!$ in terms of the action of π, let J ($\simeq U^!$), the set of π-conjugacy classes of maximal partial sections $\neq 1$ of $\mathcal{E}$ over Γ, For every $j \in J$, j is a π-torsor as a π-conjugacy class of sections [that it is a π-torsor follows from $\pi^{\Gamma'} = (1)$, a priori † the stabilizer|the stabilizers of Γ' is precisely $\pi^{\Gamma'}$] In fact, 151

Uncertainty retained from cancelled passages.
140-2-s152-D001 [illeg. 2]
140-2-s152-D003 it is a $\pi/\pi^{\Gamma'}$-torsor

[scan 153]

the set of all these † maximal partial sections is naturally an $\mathcal{E}$-set (by conjugation) † on [p. 76] which π operates freely, and this $\mathcal{E}$-set is canonically identified with $\widetilde{U}|U^! = \widetilde{U}^!$, for the structure of $\mathcal{E}$-sets.

Let $S' = S \cup U^!$, hence $U' = X \setminus S'$, we must try to reconstruct (in terms of the extension $\mathcal{E}$ of Γ by the group with loops π) the outer group with loops corresponding to X', S' i.e. to U', and the outer action of Γ on it. † †

But first we should make sure of the intrinsic character of the definition of $J\ (\simeq U^!)|(\simeq \widetilde{U}^!)$ as a Γ-set, in terms of the outer group π, and of the outer action † of Γ on it. 152

Uncertainty retained from cancelled passages.
140-2-s153-D001 classes
140-2-s153-D002 and
140-2-s153-D004 We hope to find a [illeg. 5]
140-2-s153-D005 = [illeg. 5] obvious
140-2-s153-D008 Γ

[scan 154]

This is rather obvious moreover (in terms precisely of the π-conjugacy classes of partial liftings of $\Gamma \to \mathrm{Autext}(\pi)$ to $\mathrm{Aut}(\pi)$), we would however also like an intrinsic description of $\widetilde{U}|U^!$, in|or a paradigm for the map of Γ-spaces $U^! \longrightarrow U$ we must therefore describe a morphism of topoi with operation of Γ on it

$$B_J \xrightarrow{\rho} B_{\mathcal{E}}$$

which therefore corresponds to an † "inverse image" functor $\rho^*|?^*$ (compatible with action of Γ) †

$$\begin{array}{ccc} B_{\mathcal{E}} & \longrightarrow & B_J \\ \wr\| & & \wr\| \\ \pi\text{-Sets} & & \text{Ens. over } J \end{array}$$

This † is nothing other than the contracted product over π with the set of maximal partial sections $\neq 1$ of $\mathcal{E}$ over Γ 153

Uncertainty retained from cancelled passages.
140-2-s154-D001 morphism
140-2-s154-D002 [illeg. 1]
140-2-s154-D003 which

[scan 155]

Let us return to the extension $\mathcal{E}$ of Γ by π † arising from a geometric situation (which [p. 77] extension, in the anabelian case, is already defined in terms of an outer operation of Γ on π) we know that it satisfies additional draconian conditions! † (To formulate them, we will assume the action of Γ faithful): every subgroup $\Gamma' \subset \mathcal{E}$ such that $\Gamma' \cap \pi = (e)$ ("partial section") is cyclic (if $\Gamma' = \Gamma'^0$) or † dihedral, (if $\Gamma' \neq \Gamma'^0$) and the set of π-conjugacy classes of such subgroups is

finite. Every section subgroup $\Gamma' \neq 1$ is contained in a unique section subgroup that is maximal. m268

Marginal note. m268 good|so



(We have established it at least in the case $\Gamma = \Gamma^0$, we should return to the general case, I think it remains true as it stands, to be checked…) m269

Marginal note. m269

[A vertical stroke to the left of the two lines "I think … checked…)".]

Moreover, we saw an additional structure appear on the group $\mathcal{E}$ [which in the anabelian case is identified with a subgroup of $\mathrm{Aut}(\pi)$], namely a map

$$\mu_{\mathcal{E}} : \text{elements } u \text{ of finite order in } \mathcal{E} \longrightarrow T \otimes \mathbb{Q}/\mathbb{Z},$$

obtained by noting that if $u \in \mathcal{E}$ is of finite order n, whence $\mathbb{Z}/n\mathbb{Z} \underset{i}{\hookrightarrow} \mathcal{E}$, we have|i.e. $\mathrm{Im}\, i \cap \pi = (1)$ (π has no torsion elements), hence $\mathbb{Z}/n\mathbb{Z} \hookrightarrow \Gamma$, whence|of a subgroup $\Gamma_1 \subset \Gamma$ (i.e. the image of m270

Marginal note. m270 N.B. For a u of finite order in $\mathcal{E}$, we must know|have, if $u \notin \mathcal{E}^0$, that u is of order 2 exactly, [illeg. 5] to define $\mu(u)$! 154

Uncertainty retained from cancelled passages.
140-2-s155-D001 with faithful operation of Γ
140-2-s155-D002 to
140-2-s155-D003 [illeg. 1]

[scan 156]

u in Γ has the same order n), and the lifting $\Gamma_1 \hookrightarrow \mathcal{E}$ defines an $x \in U^{\Gamma_1}$, and u_1 operating on U while leaving x fixed therefore corresponds in a neighborhood of x to a "multiplier", which (for a chosen local orientation) is a primitive nth root of 1 i.e. an element of $\mathbb{Z}/n\mathbb{Z}$, and which for varying orientation is interpreted intrinsically as an element of $T \otimes \mathbb{Z}/n\mathbb{Z} \subset T \otimes \mathbb{Q}/\mathbb{Z}$.

The map μ satisfies the obvious conditions

$$\begin{cases} \mu(\alpha u \alpha^{-1}) = \mu(u) & \text{if } \alpha \in \pi \\ \mu(u^n) = n\mu(u) & \text{if } n \in \mathbb{Z} \end{cases}$$
$$\text{order } \mu(u) = \text{order of } u$$

I do not know whether this map μ can be defined intrinsically in terms of the extension $\mathcal{E}$, or whether on the contrary there can exist two extensions † $\mathcal{E}, \mathcal{E}'$ of Γ by π, defined by geometric situations, of Γ operating on U resp. U', and an isomorphism of extensions $\mathcal{E} \longrightarrow \mathcal{E}'$ which is not compatible with the functions μ, μ'. The simplest nontrivial case to look at is the 155

Uncertainty retained from cancelled passages.
140-2-s156-D003 isomorphic

[scan 157]

abelian case (where $\pi \simeq \mathbb{Z}$ † (card $I = 2$) or $\pi \simeq \mathbb{Z}^2$ († $I = \emptyset$)), In the first case $\pi \simeq \mathbb{Z}$, we|† [p. 78]
must have (to have a faithful action of $\Gamma = \Gamma^0$) † Γ^0 cyclic, $\mathcal{E} \simeq \mathbb{Z}$, there is no element of finite order in $\mathcal{E}$ except 1|e, so the question does not arise. The case $\pi \simeq \mathbb{Z}^2$ is more interesting, †

If π is a free $\mathbb{Z}$-module of †rank n, we have an exact sequence

$$0 \to \pi \to \pi \otimes_{\mathbb{Z}} \mathbb{R} \longrightarrow X_0(\pi) \longrightarrow 0$$

Torus of dimension n associated to π

and if Γ operates on π, the cohomology exact sequence (taking into account $H^i(\Gamma, \pi \otimes_{\mathbb{Z}} \mathbb{R}) = 0$ for $i > 0$) gives

$$H^2(\Gamma, \pi) \xleftarrow{\sim} H^1(\Gamma, X_0(\pi))$$

extension class|extension classes of Γ by π

classes of $X_0(\pi)$ torsors, with operation of Γ on them compatible with its action on $X_0(\pi)$

thus the datum of an extension $\mathcal{E}$ of Γ by π amounts essentially to that of a Γ-$X_0(\pi)$-torsor X (inhomogeneous n-torus on which Γ operates — hence it operates on its group of translations X_0, hence on $\pi = \pi_1(X_0, 0)$…). If it were true for $n = 2$ that every operation 156

Uncertainty retained from cancelled passages.
140-2-s157-D001 [illeg. 1]
140-2-s157-D002 cardinality
140-2-s157-D004 the
140-2-s157-D005 [illeg. 1]
140-2-s157-D006 and can be reformulated
140-2-s157-D007 finite type

[scan 158]

of Γ on a toroidal surface ($\simeq S^1 \times S^1$) is isomorphic to such a standard action, then the intrinsic character of the map μ in this case would be established — which would not for all that make the computation of μ uninteresting, which takes its values in $T \otimes \mathbb{Q}/\mathbb{Z}$, where here

$$T \simeq \bigwedge^2_{\mathbb{Z}} \pi \quad (\text{dual|dimension 2 of } H^2(\pi, \mathbb{Z}) \simeq \bigwedge^2 \underbrace{H^1(\pi, \mathbb{Z})}_{\downarrow\wr \simeq \ \pi^\vee}).$$

The question comes down to this: We have a split extension of a cyclic †group $\mathbb{Z}/n\mathbb{Z}$ with generator u by π — thus described entirely by an automorphism θ of π such that $\theta^n = \mathrm{id}_\pi$), describe in terms of this an element of $T \otimes \mathbb{Q}/\mathbb{Z}$.

Answer: the standard|standard of which geometric situation corresponds to $X = X_0(\pi)$, with 0 as a fixed point under Γ. If u reverses orientation, it is of order 1 or|i.e. 2, there is no problem, otherwise it is in $\pi \otimes_{\mathbb{Z}} \mathbb{R}$ a rotation around 0 (of order 2, 3, 4 or 6) which is readily specified by an element of † $T \otimes \mathbb{Z}/\nu\mathbb{Z}$ (if ν is of order|the order). It is also (if we identify $T \otimes \mathbb{Z}/\nu\mathbb{Z}$ with $\mu_\nu(\mathbb{C}^*)$, assuming $T \simeq \mathbb{Z}$ given i.e. π oriented i.e. $X_0(\pi) = X$ oriented) one of the two 157

Uncertainty retained from cancelled passages.
140-2-s158-D001 group
140-2-s158-D004 [illeg. 1]

[scan 159]

eigenvalues of $u \otimes_{\mathbb{Z}} \mathbb{C}$ (automorphism of the vector space over $\mathbb{C}$ of dimension 2 $\pi \otimes_{\mathbb{Z}} \mathbb{C}$). [p. 79]

This shows us, in the case of algebraic geometry over an algebraically closed field Ω, that if $X = X_0$ is an elliptic curve u an automorphism of finite order, we have as a paradigmatic description not only $\pi = \pi_1(X, 0)$ ($\widehat{\mathbb{Z}}$-module free of rank 2), (and the action of u on π), but as an additional structure one of the two solutions in Ω of the characteristic equation of u (with integer coefficients)

$$T^2 + aT + b = 0 \qquad (a = -\operatorname{Tr} u,\ b = \det u).$$

It is clear that this additional structure cannot be deduced from knowledge of the action of u on π alone. m275

Marginal note. m275



for $\Omega = \mathbb{C}$ [illeg. 2] comes down to [illeg. 1] orientation|orientations of [illeg. 1]

But there remains the question whether this † $\mu(u) \in \mu_n^*(\Omega)$ can be deduced from knowledge at least of the outer action of u on the group "with 1 loop" corresponding to the geometric situation — i.e. an outer automorphism † of order n of such a group (or whether here again one must decidedly regard it as an additional datum). m276

Marginal note. m276 yes, it can, thanks to the consideration of the "subgroups [illeg. 2] of $\mathcal{E}$" [illeg. 2] which define extensions

$$\begin{array}{ccccccccc} 1 & \longrightarrow & T & \longrightarrow & \mathcal{E}_i & \longrightarrow & \Gamma_i & \longrightarrow & 1 \\ & & & \searrow^{n_i \mathrm{id}_T} & \wr\vert & & & & \\ & & & & T & & & & \end{array}$$

whence $\Gamma_i \simeq T/n_i T$ [illeg. 2]

But if this were indeed so, it would imply on the other hand that the construction of $u \in$ group|these groups with 158

Uncertainty retained from cancelled passages.
140-2-s159-D001 latter
140-2-s159-D002 [illeg. 2]

[scan 160]

(1) loop(s), with the operation of Γ on it, cannot be carried out either with the help of the mere † knowledge of the action of Γ ($= \mathbb{Z}/n\mathbb{Z}$) on π. It is this question of "drilling holes" that we must therefore in the end, at the very end, tackle!

As for the question of whether †in † the anabelian case, the map

$$\mu : \underbrace{\infty}_{\substack{\text{set of}\\ \text{elements}\\ \text{of finite order in } \mathcal{E}}}\mathcal{E} \longrightarrow T \otimes \mathbb{Q}/\mathbb{Z})$$

is deducible from the action of Γ on π [if it is faithful, $\mathcal{E}$ is thus identified with a subgroup of $\mathrm{Autext}_{\ell|l}(\pi)$ and we may ask whether μ is not then definable on this in its entirety], or it is|whether it is an additional datum one must have in order to reconstruct the geometric situation — the question remains entirely open. [m278]

Marginal note. [m278] this does indeed seem so, taking into account [illeg. 1] that for Γ solvable, [illeg. 3], we know [illeg. 1] that every action † of Γ on π [illeg. 3] realize geometrically

16 Plugging and drilling holes: General topos-theoretic preliminaries

Let us consider a situation

$$B_{D^*} \longrightarrow B_U$$

(of multi-Galois topoi) [m279]

Marginal note. [m279] N.B. such a situation of topoi describes 2-faithfully the topological situations (X, S), or U, provided that no irreducible component of U is $\simeq S^2$ —

159

Uncertainty retained from cancelled passages.

140-2-s160-D001 [illeg. 2]
140-2-s160-D002 [illeg. 1]
140-2-s160-D003 [illeg. 1]
140-2-s160-D004 [illeg. 1]

[illeg. 2], and to the † fact that they remain very close to the language and|† to topological intuition, it is superior to the p^t of view "groups with loops", which corresponds to the computational approach rather.) [scan 161] [p. 80]

†

I realize that I need to return to the notation for the † various topoi associated with such a datum. But I am going to be guided by the situation of numbers° .– where we have a regular scheme X of dimension 1, a closed subscheme S of dimension 0, whence $U = X - S$ — in this case † $S_{\text{ét}}|X_{\text{ét}}$ can not be reconstructed from B_{D^*} as $B_{\pi_0 B_{D^*}}|B_I$ [p. 160]

Uncertainty retained from cancelled passages.

140-2-s161-D001 the
140-2-s161-D002 and
140-2-s161-D003 necessarily^t open definition B'_U B_U $\sim$
140-2-s161-D006 [illeg. 1]
140-2-s161-D007 [illeg. 2]

with $I = \pi_0(B_{D^*})$, we must take into account the residue fields $k(s)$ $(s \in S)$ i.e. the Galois groups $\mathrm{Gal}(\overline{k(s)}/k(s))$. So there is reason to return to a starting situation (which is adapted to the arithmetico-geometric case) of morphisms of multi-Galois topoi [scan 162]

$$(*) \qquad \begin{array}{ccccc} & & B_{D^*} & & \\ & \overset{\rho}{\swarrow} & & \overset{\sigma}{\searrow} & \\ B_U & & & & B_S \end{array}$$

(Careful, we write B_S, not B_D, which will have another meaning), where σ induces an isomorphism on the Π_0 [and is surjective on the Hom]. For the moment, we are not going to make any particular hypothesis on this situation, which for B_U connected, and in terms of the choices [† of universal coverings $\widetilde{D}_i^*$ of the connected components D_i^* of D^*, and of a universal covering $\widetilde{U}$ of B_U, and of the isomorphism between the $\rho_!(\widetilde{D}_i^*)$ and $\widetilde{U}$) is made explicit by the datum of a group $\mathcal{E}$ (or Π) $(= \mathrm{Aut}(\widetilde{U}))$ † $\mathcal{E}_i$ (or Π_i $(= \mathrm{Aut}(\widetilde{D}_i^*))$ † $(i \in I)$, Γ_i $(i \in I)$ [p. 161]

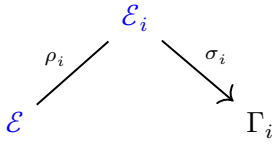
Uncertainty retained from cancelled passages.

140-2-s162-D001 D

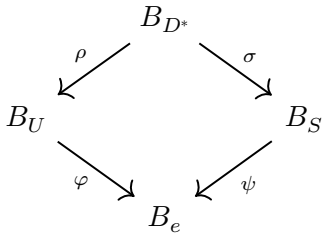
140-2-s162-D002 and

140-2-s162-D003 [illeg. 2]

and group homomorphisms [scan 163] [p. 81]

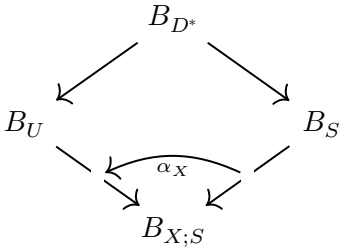


with the σ_i surjective. † [When there is a base field K for the geometric situation, then in the topos-theoretic description, putting $e = \mathrm{Spec}\, K$ and denoting by B_e the étale topos $e_{\text{ét}}$ of e, i.e. B_Γ where|if $\Gamma = \mathrm{Gal}(\overline{K}/K)$, the diagram $*$ fits into



, with a commutativity datum for the square considered... as considered at the beginning of these notes].

In terms of $*$, we construct by "gluing" of B_U and of B_S (via the gluing functor $\sigma_*\rho^*$) a mixed topos, which is no longer multi-Galois in general, previously denoted $D_{X,U}|B_{X,U}$, and which I prefer now † to denote|denoted $B_{X;S}$ † [to recall that these are sheaves on X, but having singularities only on S] †. It fits into a diagram of topoi



with an arrow [p. 162]

Uncertainty retained from cancelled passages.

140-2-s163-D001 and

140-2-s163-D002 [illeg. 1]

140-2-s163-D003 [illeg. 2]

140-2-s163-D004 [illeg. 1]

"of commutation" α which is such only by abuse of language — it is not an isomorphism † but a morphism of functors and nothing more [scan 164]

$$\sigma^*\psi^*\dagger \xrightarrow{\alpha_X} \rho^*\varphi^*.$$

In the same way, gluing D^* and S via σ_* (i.e. replacing B_U by B_{D^*} in the preceding construction of $D_{X;S}|B_{X;S}$), we find a topos, not multi-Galois in general, denoted $B_{D;S}$. In the † geometric model where we have X, S as above, B_X corresponds to sheaves on X which are essentially|set|in locally constant on U (and on S, where they have no merit) i.e. on U arise by inverse image under $U \longrightarrow B_{\Pi_1 U}$ from a sheaf on $B_{\Pi_1 U}$.

and in the same way $B_{D;S}$ corresponds to sheaves on $D = \coprod_i \mathrm{Spec}(\mathcal{O}_i =$ henselization of $\mathcal{O}_{X,i|s})$ which are locally constant on $D^* = D - S$ (and on S, without merit!) but with the dimension hypothesis made – — in|we fact

$$B_{D;S} \simeq D_{\text{ét}}$$

(D_i has only 2 points, $D_i^* = \{\eta_i\}$ —). Thus $B_{D;S}$ ($\simeq D_{\text{ét}}$) [p. 163]

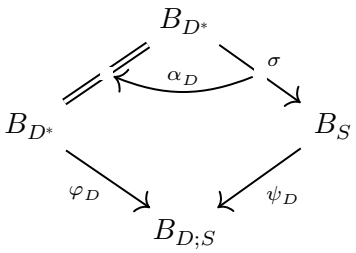
Uncertainty retained from cancelled passages.

140-2-s164-D001 functors

140-2-s164-D002 u

140-2-s164-D003 present

(when we start from X, S) fits into a triangle of morphisms of topoi [scan 165] [p. 82]



m281

Marginal note. m281 N.B. in the geometric case, † three topoi of this diagram are étale topoi ($D^*_{\text{ét}}, D_{\text{ét}}, S_{\text{ét}}$) and the arrows φ_D and ψ_D correspond to morphisms but not [illeg. 1] of topoi)

α_D, like α_X above, is just a plain homomorphism (not an isomorphism in general).

† When we speak of the canonical morphism from B_{D^*} to $B_{D;S}$ it is of φ_D (and not of $\psi_D\sigma$) that we are speaking — in the geometric case we have $B_{D^*} \simeq D^*_{\text{ét}}$, and φ_D corresponds to the inclusion of schemes

$$D^* \supset | = D \setminus S \longrightarrow D$$

and $\psi_D : B_S \longrightarrow B_{D;S}$ to the inclusion of schemes $S \longrightarrow D$ whereas neither σ nor $\psi_D\sigma$ correspond in general to morphisms of schemes.

We thus obtain a diagram of topoi † [p. 164]

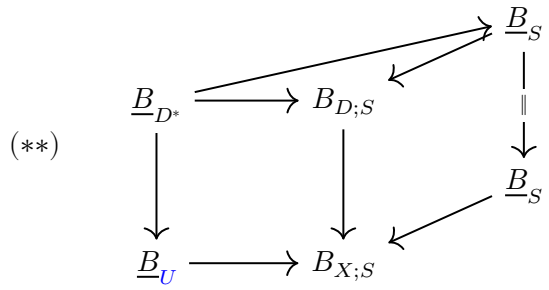
Uncertainty retained from cancelled passages.

140-2-s165-D001 [illeg. 1]

140-2-s165-D002 topos.

140-2-s165-D003 ρ_X B_{D^*} φ_D [*illeg.* 1] [*illeg.* 1]

[scan 166]

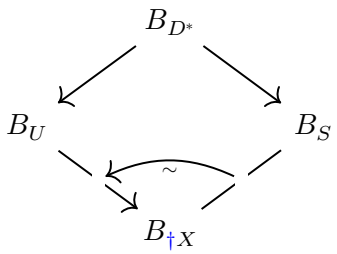


m282

Marginal note. m282 Careful, the triangle is not essentially commutative, but we have pseudo-commutativity…

where the three underlined topoi $\underline{B}_U$, $\underline{B}_S$ and $\underline{B}_{D^*}$ are multi-Galois, and $B_{D;S}$ and $B_{X;S}$ are composite (obtained by gluing two multi-Galois topoi). Unless I am mistaken the two squares are [not only essentially^t commutative but also] 2-cartesian (in the 2-category of topoi).

Passing to the B_{Π_1}? of the topoi considered (corresponding to the locally constant objects on these topoi) we find a topos $B_X = B_{\Pi_1 B_{X,S}} | B_{\Pi_1 B_{X;S}}$ as an amalgamated sum of topoi in the diagram

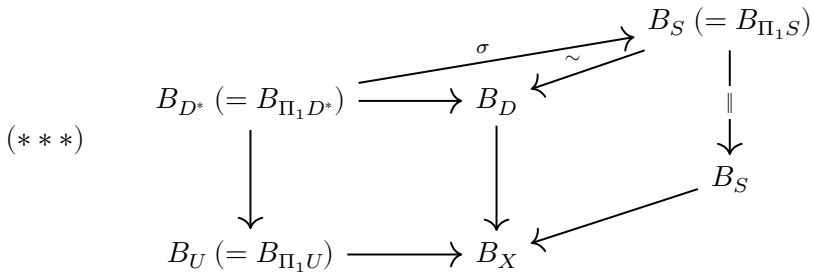


(this square is indeed essentially commutative) and in the same way (replacing B_U by B_{D^*}) a B_D, which is however (via $B_S \longrightarrow B_D$) isomorphic (rather: equivalent|equal) to B_S. The diagram (**) then becomes [p. 165]

Uncertainty retained from cancelled passages.

140-2-s166-D003 Π_1

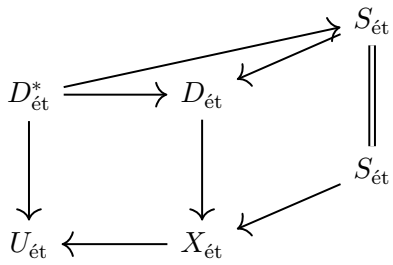
[scan 167]
[p. 83]



where this time the upper triangle is indeed essentially surjective (that is indeed how we defined σ, in the geometric case!) and we have a morphism from (**) to (***), which † for the arrows which are|are not essentially (semi-identities), $B_{D;S} \longrightarrow B_D$ and $B_{X;S} \longrightarrow B_X$, have a marked tendency to be "acyclic" i.e. to induce isomorphisms on cohomology (we should look at [illeg. 1] this point). Finally, in the geometric case, we have some analogous|canonical diagrams

of morphisms of étale topoi †

$(***)$

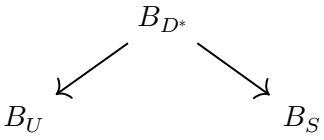


upper triangle not essentially commutative. [p. 166]

Uncertainty retained from cancelled passages.
140-2-s167-D003 is
140-2-s167-D006 [illeg. 1]

where all the arrows except $D^*_{ét} \longrightarrow S_{ét}$ are induced by morphisms of topoi. Here $(***)$ maps to the corresponding diagram $(**)$, inducing isomorphisms of topoi for D^*, D, S, and for $U_{ét} \longrightarrow B_U$ and $X_{ét} \longrightarrow X$ inducing morphisms which have less tendency to be acyclic, but which are † all the same in important cases, recalled in an earlier section… [scan 168]

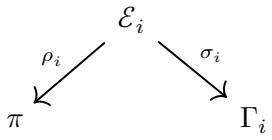
The constructions of $(**)$ and $(***)$ in terms of the starting diagram



are purely formal, and independent of every hypothesis. The construction of a multi-Galois B_X, as an amalgamated sum, can be interpreted as the translation (at the level of fundamental groupoids) of an operation of "plugging holes". In the computational context [p. 167]

Uncertainty retained from cancelled passages.
140-2-s168-D001 however

(with a choice of the $\widetilde{U}, \widetilde{U}_i$, $\rho_!(\widetilde{U}_i) \simeq \widetilde{U}$) we have [scan 169] [p. 84]



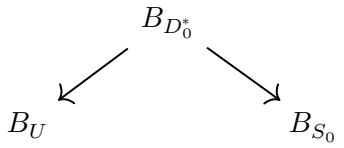
, putting $\widetilde{X}$ = image of $\widetilde{U}$ under $i_!$ $(i : B_U \longrightarrow B_{\Pi_1 X})$ B_X is described in terms of this $\widetilde{U}$ as the classifying topos B_{π_X}, where $\pi_X = \pi_1(X; \widetilde{X})$ is computed as the quotient of π by the invariant subgroup generated by the $\rho_i(L_i)$, where $L_i = \operatorname{Ker} \sigma_i \subset \mathcal{E}_i$.

If we are given a direct summand $B_{D^*_0}$ of B_{D^*} (corresponding to a subset I_0 of $I = \pi_0(B_{D^*})$) we find in the same way an amalgamated sum of B_U and of B_{S_0} under $B_{D^*_0}$, denoted † $B_{U \cup S_1} \longleftarrow B_U$, which is visualized as a partial plugging of holes, interpreted at the level of fundamental groupoids. In the geometric case, if we put $I = I_0 \amalg I_1$, i.e. $S = S_0 \amalg S_1$, we [p. 168]

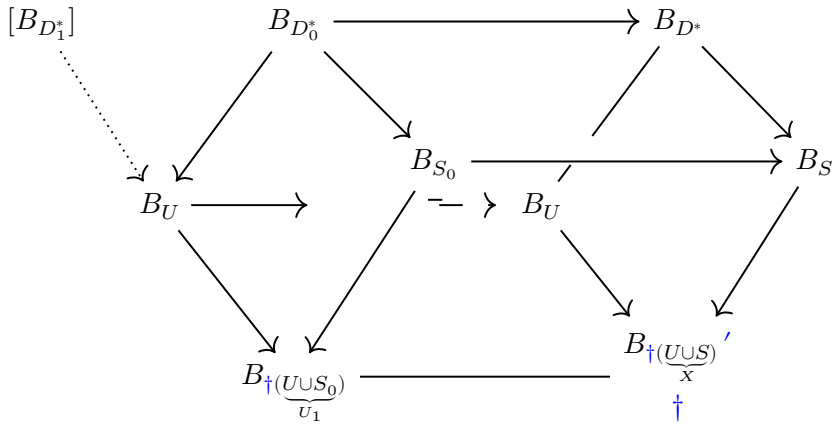
Uncertainty retained from cancelled passages.
140-2-s169-D004 $B_{\Pi_1(U \cup S_0)}$

can interpret this topos as $B_{\Pi_1 U_1}$, where $U_1 = X \setminus S_1$. [scan 170]

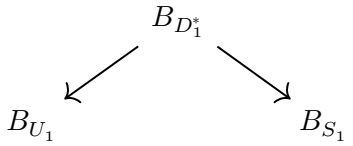
Of course, we have a homomorphism of cocartesian diagrams of topoi relative to



† the one relative to B_{D^*} † mapping into B_U, and † B_S



, and on the other hand we have a composite $B_{D_1^*} \longrightarrow B_U \longrightarrow B_{U_1}$ which with $B_{D_1^*} \longrightarrow B_{S_1}$ gives a diagram of topoi



of the same type as at the beginning, which we can use to construct another † amalgamated sum. And it is obvious that this one is canonically equivalent to B_X, the amalgamated sum corres- [p. 169]

Uncertainty retained from cancelled passages.

140-2-s170-D002 from below
140-2-s170-D003 the situation
140-2-s170-D004 B_{S_0}
140-2-s170-D005 Π_1
140-2-s170-D006 Π_1
140-2-s170-D007 [*illeg.* 1] Π_1
140-2-s170-D011 thus|in

ponding to this initial intuition... [scan 171]

† [p. 85]

All these operations are essentially trivial and without mystery, and independent of all|the special hypotheses of the "groups with loops" type. A particular interest attaches to the case where $B_S|B_D$ is a discrete topos i.e. identifies with B_I, † $I = \pi_0(B_{D^*})$, so 170

Uncertainty retained from cancelled passages.

140-2-s171-D001 letters graphic forms suggesting|being concerned with in this instance|the origin schemes classifying $\Pi_1|\pi_1$ [illeg. 1] finally set|essentially or better suggest $\Pi_1|\pi_1$ $\Pi_1|\pi_1$
140-2-s171-D005 where

that we start simply with a morphism of multi-Galois topoi [scan 172]

$$B_{D^*} \longrightarrow B_U$$

but where in addition we have a group Γ (discrete, say — or profinite in the arithmetic context) which operates on B_{D^*}, B_U ; the preceding morphism commuting with the action of Γ.

Let us note that the datum of an action of Γ on a topos B makes it possible to construct a topos (B, Γ), and a morphism $(B, \Gamma) \longrightarrow B_\Gamma$ (classifying topos of Γ i.e. a Γ-torsor of (B, Γ) — and

B identifies with the topos induced by $\mathcal{X} = (B/\Gamma)$ on this Γ torsor. Conversely, the datum of a topos $\mathcal{X}$ and of a Γ-torsor U of $\mathcal{X}$, and of an isomorphism of B with the induced topos $\mathcal{X}_{/U}$, † (identified with U, or with B) allows one to recover operations of Γ on B, via the operations on U. Thus, the datum 171

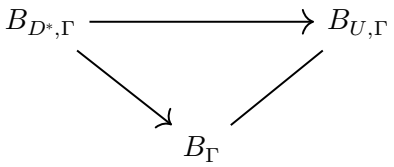
Uncertainty retained from cancelled passages.

140-2-s172-D001 by

of an operation of Γ on a topos B amounts to that of the datum of B as a Galois covering [scan 173] with group Γ of another topos (essentially unique, then denoted (B, Γ) ...). Thus, to have † Γ [p. 86] operate on $B_{D^*} \longrightarrow B_U$ is the same thing as inserting this † arrow into a diagram like

$$\begin{array}{ccc} B_{D^*}\dagger & \longrightarrow & B_U \\ \downarrow & & \downarrow \\ B_{D^*,\Gamma} & \longrightarrow & B_{U,\Gamma} \end{array}$$

where the vertical arrows are Γ-Galois and the square is 2-Cartesian, or again (independently of the prior datum of B_{D^*}, B_U) it is giving oneself an essentially commutative triangle of morphisms of topoi



172

Uncertainty retained from cancelled passages.

140-2-s173-D001 B_{D^*}

140-2-s173-D002 diagram

140-2-s173-D003 ,Γ

If Γ operates on a multi-Galois topos, we see that (B, Γ) is also multi-Galois, and the situation [scan 174] of a multi-Galois topos B_U and of an operation of Γ on it amounts to that of a multi-Galois topos $B_{U,\Gamma}$ and of a morphism $B_{U,\Gamma} \longrightarrow | \longleftrightarrow B_\Gamma$.

So the datum of a situation $B_{D^*} \longrightarrow B_U$ of multi-Galois topoi, and of an operation of Γ on it, amounts exactly to that of homomorphisms of multi-Galois topoi

$$B_{D^*,\Gamma} \longrightarrow B_{U,\Gamma} \longrightarrow B_\Gamma$$

†|The connectedness $B_{U,\Gamma}$ is 0-connected iff $\operatorname{card} \pi_0(B_U)/\Gamma = 1$, i.e. B_U nonempty and Γ transitive on $\pi_0(B_U)$. — Let us note that the situation 173

Uncertainty retained from cancelled passages.
140-2-s174-D001 The connectedness

envisaged at the beginning, with (X, S) over a field K, whence † $B_{D^*} \longrightarrow B_U \longrightarrow B_\Gamma$ "$e_{ét}$ [scan 175]
($\Gamma = \mathrm{Gal}(\overline{K}/K)$) can be interpreted as deduced from the "geometric" situation $(\overline{X}, \overline{S})$ over $\overline{K}$, [p. 87]
or $B_{\overline{D}^*} \longrightarrow B_{\overline{U}}$, taking into account the operations of Γ on it. It turns out that for many questions, it is this "geometric" interpretation (in the strict sense, i.e. over $\overline{K}$ algebraically closed) with operations of a Galois group Γ, that is the most convenient.

If we look at an operation of Γ on a topos (B_{D^*} say), it operates on the discrete topos $B_{I|\pi_0}$ ($I = \pi_0(B_{D^*})$), and $B_{D^*} \longrightarrow B_I$ is compatible with the actions of Γ. † But the topos $B_{(I,\Gamma)}$ is also that of Γ-sets over I (on which Γ operates) — an induced topos of B_Γ — † its components that are connected correspond 174

Uncertainty retained from cancelled passages.
140-2-s175-D001 B
140-2-s175-D002 In
140-2-s175-D004 fibers over of

to the orbits of Γ on I. If Γ is transitive on I nonempty (or if we look at one such orbit...), [scan 176]
choosing $i \in I$, the topos in question identifies with B_{Γ_i}, where Γ_i is the Stabilizer of † i in Γ.

N.B. Giving an operation of Γ on a discrete topos B ($= B_I$), when we interpret it as a morphism of a multi-Galois topos $B' = (B, \Gamma) \longrightarrow B_\Gamma$, is characterized by the fact that the morphism of groupoids describing it is injective on arrows, i.e. in terms of a system of homomorphisms of groups $\Gamma_j \longrightarrow \Gamma$ ($j \in J$ [$\simeq I/\Gamma$]), by the condition that these homomorphisms be injective. [and then I

is reconstructed as the direct sum of the Γ/Γ_j ...]. Thus a diagram

$$B_{D^*} \longrightarrow B_U$$

175

Uncertainty retained from cancelled passages.
140-2-s176-D001 [illeg. 1]

with action of Γ is equivalent to the datum of a diagram [scan 177]
[p. 88]

$$B_{D^*,\Gamma} \longrightarrow B_{U,\Gamma} \longrightarrow B_\Gamma$$

and this is completed (using the action of Γ on $\pi_0(B_{D^*}]$ — B_{D^*} being itself deduced from $B_{D^*,\Gamma} \longrightarrow B_\Gamma$ as the inverse image of the universal torsor) † to

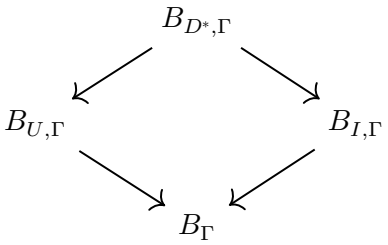
$$B_{D^*,\Gamma} \longrightarrow B_{I,\Gamma} \longrightarrow B_\Gamma$$

where $B_{D^*,\Gamma} \longrightarrow B_{I,\Gamma}$ can also be described directly in terms of $B_{D^*,\Gamma} \longrightarrow B_\Gamma$ (as in an earlier numbe^{r}), at the level of the fundamental groupoids, by a passage to the quotient, in|of the each connected component|connected components of $\Pi_1 B_{D^*,\Gamma}$, by the kernels of the maps on the homomorphisms deduced|by the homomorphisms deduced from $\Pi_1 B_{D^*,\Gamma} \longrightarrow \Pi_1 B_\Gamma$.

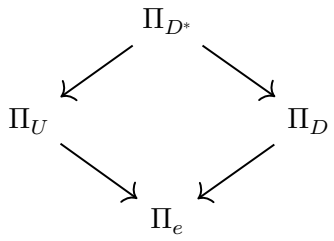
— i.e. this is the canonical factorization of a homomorphism of groupoids|groupoids into (bijective|bijective (for objects), epimorphic † (for homomorphisms)) followed by monomorphic (for homomorphisms). We thus find an 176

Uncertainty retained from cancelled passages.
140-2-s177-D001 as|with
140-2-s177-D002 [illeg. 1]

essentially commutative square [scan 178]



which † corresponds to the diagram of groupoids at the beginning of the notes † † † (§ 7)



I have the impression of having more or less understood the mechanism of the actions of groups on multi-Galois topoi, and how † the operation of passing from a topos B with operation of Γ to the "quotient" topos (B,Γ) = "B/Γ", commutes with operations of the type passing from

[$B_{D^*} \longrightarrow B_U$; below B_{D^*}, B_S, connected to B_{D^*} by a short vertical stroke thicker at its lower end : arrow $B_{D^*} \to B_S$|insertion pointer.]

to a $B_{X,S}$, via a B_X ("plugging the holes"). † The time 177

Uncertainty retained from cancelled passages.

140-2-s178-D001 is none other than the one that

140-2-s178-D002 that

140-2-s178-D003 Π

140-2-s178-D004 made

140-2-s178-D005 this commutes

140-2-s178-D006 I

seems therefore ripe at last to explain oneself about|with the hypothetical inverse operation of "drilling of|of the holes". [scan 179] [p. 89]

17. Supplement on the operations of finite groups on surfaces. (Supplement to § 7|§ 15)

Let U be a paracompact connected surface ! such that $\pi_1(U) \neq (1)$ i.e. $U \not\simeq S^2, \mathbb{R}^2$. We say that U is "anabelian" if π nonabelian (in which case Centre$(\pi) = 1$), i.e.|and if $U \not\simeq \mathbb{C}^*, S^1 \times S^1$. Let Γ be a finite group operating on U. Equivalent conditions :

b)|a) $\forall g \in \Gamma$, g_U is isotopic to the identity.

b') $\forall g \in \Gamma$, g_U is homotopic to the identity.

c) The outer operation of Γ on $\pi_1(U)$ is trivial.

d) (anabelian case) The extension $\mathcal{E}$ of Γ by π is isomorphic to a product extension, or (abelian case) † it is central. †

a)|d) (anabelian case) Γ operates trivially, (abelian|anabelian case) $\exists$ † structure of a topological group † on U (thus $U \simeq \mathbb{C}^*$or$S^1 \times S^1$), [illeg. 3] † and a homomorphism of groups $\Gamma \longrightarrow U$, operating by translations

m292

Marginal note. m292 That must be the only counterexample (in the case † reversal of an orientation ...) In any case a counterexample must be such that (if Γ faithful, $\Gamma \simeq (\pm 1)$), operates by anti involution...

m293

Marginal note. m293 It is prudent to assume that Γ operates preserving an orientation of U (assumed orientable), otherwise we have trouble, for example with $z \longrightarrow \overline{z}^{-1}$ from $\mathbb{C}^*$ into $\mathbb{C}^*$. It would be necessary to clarify|clear up the general situation (abelian case) in the theorem [illeg. 4] case of an anti-involution).

178

Uncertainty retained from cancelled passages.

140-2-s179-D002 [illeg. 5]
140-2-s179-D003 [illeg. 3]
140-2-s179-D004 [illeg. 3]
140-2-s179-D005 $U \simeq \mathbb{C}^*, S^1 \times S^1$
140-2-s179-D006 $\mathbb{C}$ [illeg. 4]
140-2-s179-D007 [illeg. 2]

Proof $a) \Longrightarrow b) \Longrightarrow b') \Longrightarrow c)$ trivial. [scan 180]

$c) \Longrightarrow d)$ In anabelian case, this comes from the fact that Centre$(\pi) = 1$ (and an extension of Γ by π is already defined by the outer action, † as the inverse image of the example $1 \longrightarrow \pi \longrightarrow \mathrm{Aut}(\pi) \longrightarrow \mathrm{Autext}\,\pi \longrightarrow 1$]. In the abelian case, this is trivial.

$d) \Longrightarrow a)$ Is the not very|not obvious part. † We may assume|Let us assume that Γ operates faithfully, Anabelian case † If we had|with $\Gamma \neq 1$, let us fix a splitting of the extension $\mathcal{E}$ of Γ by π, we must have by the theorem of numbe^{r} 15 $\pi^{\Gamma} = | \simeq 1$, now

the canonical splitting of $\pi \times | \rtimes \Gamma$ over Γ gives $\pi^{\Gamma} = \pi$, absurd.

Abelian case $U \simeq \mathbb{C}^*$ (more intrinsically, U a torsor under $U_0 = \pi \otimes_{\mathbb{Z}} \mathbb{C}/\pi$) or $U \simeq S^1 \times S^1$ (more intrinsically, U a torsor under $U_0 = \pi \otimes_{\mathbb{Z}} \mathbb{C}/\pi$).

I say that an action of Γ on U is here (up to † homeomorphism|conjugation) defined by an action of $\Gamma|\pi$ on the torsor — the class

not proved!
179

Uncertainty retained from cancelled passages.

140-2-s180-D001 taken
140-2-s180-D002 For the
140-2-s180-D005 [illeg. 1]
140-2-s180-D006 isotopy

[scan 181]

of isomorphism of such a torsor is moreover identified, by the cohomology exact sequence [p. 90] associated with the exact sequence

$$0 \longrightarrow \pi \longrightarrow \pi \otimes_{\mathbb{Z}} \mathbb{C}|\mathbb{R} \longrightarrow U_0 \longrightarrow 0$$

or $\pi \otimes_{\mathbb{Z}} \mathbb{R}$ with a class of extensions of Γ by π. But to say that the action of Γ on π is trivial, means that the action of Γ on the torsor U under U_0 is by translations.

Corollary-scholium N.B. The "abelian" case is not quite proved, for want of having established the † classification topological of the operations of a finite group on $\mathbb{C}^*$ or on $S^1 \times S^1$. However if in the abelian case we assume in advance that $U^{\Gamma} \neq \emptyset$, then † it is still true that the trivial operation of Γ on π is equivalent to the triviality of the action of Γ on U. For we are reduced to the case where Γ operates faithfully, and to proving in this case that if the operation of Γ on the π is trivial, we have $\Gamma = 1$. And we do as above in the anabelian case.

Uncertainty retained from cancelled passages.
140-2-s181-D001 topological
140-2-s181-D002 c)

[scan 182]
[p. 91]

† 18 Drilling holes. Application to outer actions of finite groups

Let Π' be a group with loops of type $(g, \nu+1)(\nu+1 \in \mathbb{N}^*)|(\nu \in \mathbb{N}^*)$, nonabelian i.e. †, if $g = 0$, we have $\nu \geq 2$ † Let T be its orientation module, and I' ($\operatorname{card} I' = \nu$) the set of its conjugacy classes of loop subgroups. Let us fix an $i \in I'$, and let $L'_i \subset \Pi'$ be in the class i.

When we are given only an outer group †with loops $[\Pi']$ (which † is equivalent to the datum of $B_{D'^*} \longrightarrow B_{U'}$), the datum of an † $i \in I'$ is equivalent to that of a connected component of $B_{D'^*}$, and that of a realization † of the categorical group (i.e. of a pair of a group $\Pi'_{s|\alpha}$, and a categorical isomorphism $\Pi' \longrightarrow [\Pi']_d$) is equivalent up to unique isomorphism †(universal covering $\widetilde{U}'$ of U', by $\Pi' = \operatorname{Aut}(\widetilde{U}')$. of $\Pi_1 B_{U'}$, Finally the datum of a pair (Π', L'_i) (L'_i in the conjugacy class i) is equivalent (up to unique isomorphism) to the datum of an object $\widetilde{D}^*_i$ of $\Pi_1 B_{D'^*_i}$ – we take the image $\widetilde{U}'$ of $\widetilde{D}^{*\prime}$ in $\Pi_1 B_{U'}$, and

$$\Pi' = \operatorname{Aut}(\widetilde{U}'), \qquad L'_i = \operatorname{Im}(\operatorname{Aut}(\widetilde{D}'^*_i) \longrightarrow \operatorname{Aut}(\widetilde{U}')).$$

m297

Marginal note. m297 I am going to temporarily drop the "primes"!

When we are given at the outset|whence a an object $\widetilde{U}'_0$ of $\Pi_1 B_{U'}$, whence $\Pi' = \operatorname{Aut}(\widetilde{U}'_0)$, then

Uncertainty retained from cancelled passages.
140-2-s182-D001 Supplement|Deleted
140-2-s182-D003 [illeg. 1]
140-2-s182-D004 $\nu \geq 2$ 2|3 $\geq$
140-2-s182-D007 [illeg. 1]
140-2-s182-D008 is done
140-2-s182-D009 [illeg. 1]
140-2-s182-D010 (L_i, Π)
140-2-s182-D011 [illeg. 3]

[scan 183]

the datum of an $L_i \subset \pi$ in the class i is equivalent to the datum of a pair $(\widetilde{D}_i, \lambda|\,\mathrm{id})$ of a $\widetilde{D}_i \in \operatorname{Ob} \Pi_1 B_{D_i}$ and an isomorphism from $\rho_!(\widetilde{D}_i) =: | = \widetilde{U}$ onto $\widetilde{U}_0$.

When the topos-theoretic situation is realized from a topological situation (X, S), and we define $\widetilde{U}_0$ by means of a base point $a \in U$, then the standard way of defining an $L_i \subset \pi = \operatorname{Aut}(\widetilde{U}_0) = \pi_1(U, a)$ is to choose a small disk Δ_i around s_i in X,



and a point b_i on the boundary, and a homotopy class of paths (in $\dagger U|X - (\mathring{\Delta}_i - \{s_i\}) = V_i$) from a to b_i, and to take the group L_i generated by either one of the two corresponding loops around s_i (which give † opposite elements of π). We see that we thus find a surjective map from

$$\operatorname{Isom}_{\Pi_1 V_i}(a, b_i) \simeq \operatorname{Isom}_{\Pi_1 U}(a, b_i)$$

onto the set of $L_i \subset \pi$ in the class $i|\Pi$, a map compatible with the

Uncertainty retained from cancelled passages.
140-2-s183-D002 X
140-2-s183-D003 two

[scan 184]

natural action of π (operating on $\mathrm{Isom}_{\Pi_1 U}(a, b_i)$ in the obvious way by composition, and on the set of L_i by inner automorphisms. † Here, the link with the "abstract" topos-theoretic description is established as follows: the choice of b_i can serve as a base|fixed point for defining a universal covering of $\Delta_i \smallsetminus \{s_i\} \overset{\sim}{\leftarrow} D_i^*$, whence an object $\rho_!(\widetilde{D}_i^*(b_i))$, and † the $L_i \subset \pi$ correspond (according to the abstract description) to the isomorphisms from $\widetilde{U}_0 = \widetilde{U}(a_i)$ onto $D_i^*(b_i)|\widetilde{D}_i^*(b_i)$ modulo composition with an automorphism of $D_i^*(b_i)|\widetilde{D}_i^*(b_i)$ coming from an automorphism of $D_i^*(b_i)|\widetilde{D}_i^*(b_i)$ – but the isomorphisms $\widetilde{U}(a_i) \overset{\sim}{\to} \rho_!(\widetilde{D}_i^*(b_i))$ correspond precisely to the classes of paths from a to b_i. [p. 92]

We return to the notation π', U' – ‖

Let us consider an object of $\Pi_1 B_{D_i^{*\prime}}$, i.e. a pair (π', L_i'). Let π be the quotient group of π' by the invariant subgroup of π' generated by L_i'. I say that up to unique isomorphism (actual isomorphism of groups, not merely outer!) it does not

Uncertainty retained from cancelled passages.
140-2-s184-D001 Then
140-2-s184-D003 this object

[scan 185]

depend on the choice of the object $\widetilde{D}_i^*$ of $\Pi_1 B_{D_i'^*}$. Indeed, $\pi = \pi(\widetilde{D}_i^*)$ depends functorially on $\widetilde{D}_i^*$, and everything comes down to seeing that this functor is constant, i.e. that the operation of

$$\mathrm{Aut}(\widetilde{D}_i^*) \simeq T \simeq L_i \quad \text{on} \quad \pi = \pi(\widetilde{D}_i^*)$$

is trivial. Now let $u \in L_i$, the|its automorphism of π which it defines is defined by the inner automorphism $\mathrm{int}(u)$ by passage to the quotient, hence (since u becomes 1 in π) it is trivial. Easy.

We thus find a functor

$$\begin{array}{c}\text{Outer groups with loops } \pi' \\ \text{of type } g, \nu+1 \text{ equipped with a} \\ \text{loop class } i \in I(\pi')\end{array} \longrightarrow \begin{array}{c}\text{groups} \\ \text{(realized)} \\ \text{with loops, of} \\ \text{type } g, \nu\dagger\end{array}$$

$$(\pi', i) \longmapsto \beta(\pi', i) \qquad [\text{"plugging the hole } i\text{"}]$$

which is expressed by a group homomorphism

$$\mathrm{Aut}\,\mathrm{ext}(\pi', i) \longrightarrow \mathrm{Aut}\,\pi \tag{*|5}$$

where $\pi = \beta(\pi', i)$

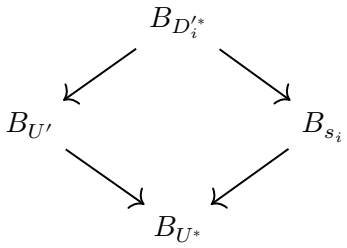
$$\left(\begin{array}{l} \overset{\text{definition}}{=} \text{ set of categorical automorphisms} \\ \text{of } \pi' \text{ respecting the loop} \\ \text{structure and fixing the loop} \\ \text{class } i \end{array}\right)$$

Uncertainty retained from cancelled passages.
140-2-s185-D001 +1

[scan 186]

From the geometric point of view, this fact (existence of this functor), merely expresses the fact that after "plugging the hole" i, we have a $U = U' \cup \{s_i\}$ equipped with a point s_i which we can use as a base point canonical to calculate $\pi_1(U)$. [p. 93]

We can also say that the choice of i makes it possible to construct the partial amalgamated sum (plugging holes partially) B_{U^*}



and $B_{s_i} \longrightarrow B_{U^*}$ supplies a geometric point of B_{U^*} which makes it possible to describe a canonical object of $\Pi_1 B_{U^*}$ (universal covering relative to this point) whence canonically a group π, which of course is a group with loops of type $g, \nu - 1$.

Theorem Suppose g, ν are such that not

[scan 187]

only the groups π'_s of type g, ν, but also the groups π (of type $g, \nu - 1$) are anabelian, i.e.

$$2g + \nu \dagger \geq \mid \leq 4 \quad [\text{thus } g \geq 2, \text{ or } g \geq \mid = 1, \nu \geq 2, \text{ or } g = 0, \nu \geq 4].$$

Then the preceding functor $(\pi', i) \longrightarrow \pi$ is an equivalence of categories, in other words (since these are 0-connected groupoids)

$$\operatorname{Aut}\operatorname{ext}(\pi', i) \longrightarrow \operatorname{Aut}(\pi)$$

is an isomorphism.

Let us note for this that we have the exact sequence

$$1 \longrightarrow \pi \longrightarrow \operatorname{Aut}(\pi) \longrightarrow \operatorname{Aut}\operatorname{ext}(\pi) \longrightarrow 1 :$$

(for $\operatorname{Centre}\pi = 1$ by the anabelian hypothesis on π), now we are going to define an exact sequence

$$1 \longrightarrow \pi \longrightarrow \operatorname{Aut}\operatorname{ext}(\pi', i) \longrightarrow \operatorname{Aut}\operatorname{ext}(\pi) \mid \operatorname{Aut}\operatorname{ext}(\pi') \longrightarrow 1$$

and a homomorphism of extensions from this one

Uncertainty retained from cancelled passages.
140-2-s187-D001 +1

[scan 188]

into the preceding one (which will necessarily be an isomorphism). [p. 94|99]

We may assume $T = \mathbb{Z}$, we consider the standard|standard groups with loops $\Pi_{g,\nu+1}$, $\Pi_{g,\nu}$. Let us put

$$\operatorname{Aut}\operatorname{ext}_{\text{lac}}(\Pi_{g,\nu}) = \mathcal{T} \mid \mathfrak{T}_{g,\nu}^{00\mid\infty}$$

inducing the identity on $I(\Pi_{g,\nu})$ and on $T(\Pi_{g,\nu})$ (Teichmüller group of index g, ν), we know that (for $2g + \nu \geq 3$) $\mathcal{T}_{g,\nu}$ is the fundamental group of the complex modular topos $M_{g,\nu}$ of complex

curves († projective nonsingular) connected, of genus g, with a system of ν given distinct points $s_1 - s_\nu$. [m300]

Marginal note. [m300] N.B. the universal covering of $M_{g,\nu}$ is contractible, Teichmüller.

Now the modular topos $M_{g,\nu+1}$ is none other than the "universal complex curve † of genus g with ν numbered holes" over $M_{g,\nu}$ [for giving ourselves a curve U' of genus g with $\nu+1$ holes $x_1 - x_{\nu+1}$ amounts to the same as giving ourselves one U with ν numbered holes $x_1 - x_\nu$, plus a point of U †), whence a homotopy exact sequence

$$\begin{array}{ccccccccc} \pi_2|\Pi_2(M_{g,\nu},\xi) & \longrightarrow & \pi_1(U) & \longrightarrow & \pi_1(M_{g,\nu+1}) & \to & \pi_1(M_{g,\nu}) & \to & \pi_0(U) \\ & & \| & & \| & & \wr & & \| \\ 0 & \longrightarrow & \pi = \Pi_{g,\nu} & \longrightarrow & \mathcal{T}^{00}_{g,\nu+1} & \longrightarrow & \mathcal{T}^{00}_{g,\nu} & \longrightarrow & 1 \end{array}$$

Uncertainty retained from cancelled passages.
140-2-s188-D001 [illeg. 1]
140-2-s188-D003 [illeg. 1]
140-2-s188-D005 pointed

[scan 189]

whence an extension structure

$$1 \longrightarrow \Pi_{g,\nu} \longrightarrow \mathcal{T}^{00}_{g,\nu+1} \longrightarrow \mathcal{T}^{00}_{g,\nu} \longrightarrow 1$$

which is (up to passage to a subgroup of index "2ν!) this announced extension structure. & the verifications of compatibility are left ….

We therefore have a quasi-inverse functor (defined up to unique isomorphism)

$$\begin{array}{ccc} \text{groups with loops} \\ \text{of type } g,\nu \\ (2g+\nu \geq 3) \end{array} \longrightarrow \begin{array}{c} \underline{\text{outer}} \text{ groups with} \\ \text{loops of type } (g,\nu+1), \\ \text{with a distinguished "loop} \\ \text{class"} \\ (\text{pairs } (\pi', i \in I(\pi'))) \end{array}$$

† At the topos-theoretic level, when we have a system $B_{D^*} \longrightarrow B_U$ of type g,ν $(2g+\nu \geq 3)$, and a "point" of B_U i.e. a $\widetilde{U} \in \mathrm{Ob}\,\Pi_1 B_U$, then we can in a canonical way find a $B_{D_i^*}$ (T-groupoid connected, where T is the orientation module), and a system

$$B_{D^*} \amalg B_{D_i^*} \longrightarrow B_{U'},$$

of type $g,\nu+1$,

Uncertainty retained from cancelled passages.
140-2-s189-D002 Now

[scan 190]

in such a way that $(B_{D^*}, B_U, \widetilde{U})$ is deduced from it by the operation of "plugging the hole D_i^*". [p. 95]

These constructions are so functorial, that they commute with actions of groups Γ. If a group Γ operates on a group with loops $\pi_\dagger$ of type g,ν, (not merely by outer automorphisms), then we deduce from it an outer operation of Γ on a π' with loops of type $g,\nu+1$, which fixes a privileged loop class $i \in I$. ∧

But then, in the corresponding extension

$$1 \longrightarrow \pi' \longrightarrow E' \longrightarrow \Gamma \longrightarrow 1,$$

Let us choose an $L'_i \subset \pi'$ and let $\mathcal{E}'_i|E'_i$ be the normalizer of L_i in $\mathcal{E}'|E'$ – we find

$$1 \longrightarrow \underset{\simeq T}{L'_i} \longrightarrow \mathcal{E}'_i|E'_i \longrightarrow \Gamma \longrightarrow 1$$

Uncertainty retained from cancelled passages.

140-2-s190-D001 g, ν

(this works without a finiteness hypothesis on Γ, the universal case being the one where Γ is the Teichmüller group of π' fixing i i.e. $\mathrm{Autext}_{\mathrm{lac}}(\pi'; i) \simeq \mathrm{Aut}_{\mathrm{lac}}(\pi)$). I say that if Γ operates faithfully on π i.e. if its outer operation on π' is faithful, and if Γ is finite, Then Γ is necessarily cyclic (case $\Gamma = \Gamma^0$) or dihedral ($\Gamma \neq \Gamma^0$), and that if in addition $\Gamma = \Gamma^0$, then † $\mathcal{E}_i \simeq \mathbb{Z}|\mathcal{E}'_i \simeq \mathbb{Z}$ i.e. ∃! isomorphism $T \xrightarrow{\sim} \mathcal{E}_i|T \xrightarrow{\sim} \mathcal{E}'_i$ such that $T \xrightarrow{k'_i} \mathcal{E}'_i$ identifies with $n\,\mathrm{id}_T$ (so that we find $\Gamma \simeq \mathcal{E}'_i/n\mathcal{E}'_i \simeq T \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}$, if $n = \mathrm{card}\,\Gamma$ ... [scan 191]

Changing notation, this amounts to the Theore[m] Let Γ be a finite group operating faithfully on an outer|outer[ly] group with loops π of type † $(g, \nu + 1)$ $(\nu \geq 0)$, leaving an $i \in I(\pi)$ fixed, Then Γ is cyclic (if $\Gamma = \Gamma^0$) or dihedral (if $\Gamma \neq \Gamma^0$), and in the extension [m302]

Marginal note. [m302] to the same there is equivalence, if in the theorem, we assume the type (g, ν) anabelian — otherwise the theorem is a little more general [p. 190]

Uncertainty retained from cancelled passages.

140-2-s191-D001 $\mathcal{E}'_i$ is $\simeq$

140-2-s191-D002 [illeg. 1]

corresponding † $\mathcal{E}$ of Γ by π, If $\mathcal{E}_i$ is the normalizer of an L_i in $\mathcal{E}$ [so that we have $\mathcal{E}_i \to \Gamma$ epimorphic i.e. an exact sequence $1 \to L_i \to \mathcal{E}_i \to \Gamma \to 1$]. the inverse image $\mathcal{E}^0_i$ of Γ^0 is $\simeq \mathbb{Z}$ so (if $n = \mathrm{card}\,\Gamma^0$)$= [\mathcal{E}^0_i : L_i]$ † $x \to x^n$ is an isomorphism $\mathcal{E}^0_i \simeq L_i$, which taking k_i into account gives an isomorphism $\mathcal{E}^0_i \simeq T$, whose composite with † $T \underset{\sim}{\xrightarrow{k_i}} L_i \subset \mathcal{E}^0_i$ is $n\,\mathrm{id}_T$, so that we have a canonical isomorphism [scan 192] [p. 96]

$$\Gamma^0 \simeq \mathcal{E}^0_i/L_i \simeq T/nT$$

(obviously independent of the choice of L_i ...)

Proof. Let us consider the extension $\mathcal{E}^0_i$ of Γ^0 by $L_i \simeq T$ ($\overset{\text{noncanonical}}{\simeq}$ $\mathbb{Z}$), since $\mathcal{E}^0_i|\Gamma^0$ operates trivial[ly] on T torsion-free, this extension (by the cohomology exact sequence associated with $0 \to T \to T \otimes_{\mathbb{Z}} \mathbb{Q} \to T \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z} \to 0$) is canonical[ly] isomorphic to † the extension defined by a homomorphism $\Gamma^0 \to T_n = T/nT$, as inverse image of the extension $0 \to T \xrightarrow{n} T \to T_n \to 0$ of T_n by T. I say that [p. 191]

Uncertainty retained from cancelled passages.

140-2-s192-D001 [illeg. 2]

140-2-s192-D003 n id

140-2-s192-D005 $\mathcal{E}_i$

140-2-s192-D006 an

this homomorphism is an isomorphism (from which would follow|will follow the other assertions), or what amounts to the same since Γ^0 and T_n are both of order n, that it is a monomorphism. Replacing Γ^0 by the kernel of $\Gamma^0 \to \mathbb{Z}/n\mathbb{Z}$, we may assume that the homomorphism in question [scan 193]

is zero, and we must prove that this implies that the action of Γ^0 is trivial (which, since by hypothesis the action of Γ is faithful, implies $\Gamma^0 = \{1\}$, OK.) So theorem is reduced to the

fundamental lemma. Every direct outer automorphism u of finite order n of a group with loops π, which fixes a loop class i and is such that the extension of $\mathbb{Z}/n\mathbb{Z}$ by $L_i \in i$ defined by u is trivial, is trivial.

The hypothesis means that u lifts to an automorphism u_0 of π, which normalizes L_i, and which is also of order n, (or of finite order, this amounts to the same|⇒ taking into account that † T has no elements of finite order) [p. 192]

Uncertainty retained from cancelled passages.
140-2-s193-D001 the elements of

then u_0 is trivial, i.e. this is equivalent to the Corollary. Every direct automorphism of finite order u_0 of a group with loops π, which is direct and which normalizes a loop subgroup L_i (i.e. which centralizes L_i), is reduced to the identity. m304 [scan 194] [p. 97]

In the lemma or its corollary

Marginal note. m304 N.B. (the case $g = 0$, $\nu + 1 = 1$ or $\nu + 1 = 2$ is trivial, the case $\nu + 1 = 3$ ($(g, \nu) = (0, 2)$ abelian!) is not trivial on the other hand, nor the case $g = 1$, $\nu + 1 = 1$ i.e. $(g, \nu) = (1, 0)$ abelian. Yet the result should be valid also in this|these cases.)

To prove it, we are reduced at once to the case where u_0 is such that $u_0^p = \mathrm{id}$, where p is a prime number — i.e. u_0 corres- ponds to an operation in the strict sense (not on^{ly} outer) of $\mathbb{Z}/p\mathbb{Z}$ † by automorphisms of π. But (whether or not we can find such a p) let us consider the case where we know that the outer operation of $\Gamma = \mathbb{Z}/n\mathbb{Z}$ [n = order of u] on π is realized geometrical^{ly}, by a (faithful) operation of $\Gamma = \mathbb{Z}/n\mathbb{Z}$ on a U of type $(g, \nu + 1)$. † m305

Marginal note. m305 $U = X \setminus S$, $S \simeq I$, X compact of genus g the puncture|the part s_i of S corresponding to a fixed point of Γ operating on U.

†

[p. 193]

Uncertainty retained from cancelled passages.
140-2-s194-D002 by
140-2-s194-D003 π-conjugacy classes [illeg. 1]
140-2-s194-D004 liftings automorphisms [illeg. 1] so a compatible with Γ

We express the hypothesis of the "genuine" operation of Γ on π, † centralizing an L_i, by saying that the outer operation gives a trivial extension of Γ by $L_i \simeq T$. But we know moreover in the geometric case and faithful operation that it is not trivial ! [scan 195]

It would therefore suffice, † to prove the fundamental lemma, to know that every outer action [faithful] of a cyclic group on a group with loops of type $(g, \nu+1)$ is realizable, and it even suffices to know this for a cyclic group of prime order. is Now unless I am mistaken this result is well known (even for solv- able groups ...). — so that the funda- mental lemma seems proved. I have only a doubt whether it is proved in the general case of a (g, ν), † or only for $g \geq 2$, $\nu = 0$. But † † m306

Marginal note. m306 as a theorem of existence of points fixed by a group [illeg. 1] operating of Teichmüller

†

[p. 194]

Uncertainty retained from cancelled passages.
140-2-s195-D001 [illeg. 1]
140-2-s195-D002 [illeg. 1]
140-2-s195-D003 [illeg. 1]

140-2-s195-D004 [illeg. 2]
140-2-s195-D005 [illeg. 2]
140-2-s195-D006 [illeg. 1]

† if this is indeed so, I think that (for $g \geq 2$ at the very least) we get by with dévissage for the case of arbitrary ν — and the case $g = 0, 1$ would require also a separate treatment. I will come back to this later, and prefer for the moment to admit the "fundamental lemma", and examine consequences and corollaries of it. [scan 196] [p. 98]

For an outer faithful action of a finite Γ on an anabelian group with loops π, we † corresponding to an extension $\mathcal{E}$ of Γ by π, we have therefore established a) that the partial splittings of it can only be made over † subgroups Γ' of Γ such that Γ'^0 is cyclic, and Γ' dihedral if $\Gamma' \neq \Gamma'^0$, and that b) for every class of [illeg. 2] of such partial splittings over $\Gamma' \subset \Gamma^0$, we have a corresponding canonical isomorphism $\Gamma' \simeq T_n$ $(= T/nT)$, where $n = \operatorname{card} \Gamma'$. m307

Marginal note. m307 i.e. we have established the existence of a canonical map [illeg. 1] from the elements of finite order of $\mathrm{Aut}_{\mathrm{lac}}(\pi)$ into $T \otimes \mathbb{Q}/T$, satisfying the conditions examined previously

These are results which we had [p. 195]

Uncertainty retained from cancelled passages.
140-2-s196-D001 [illeg. 1]
140-2-s196-D002 have established
140-2-s196-D003 cyclic

previously obtained for the case of a realizable operation. [There was moreover no reason to restrict ourselves to the anabelian case, as long as we assume $\pi \neq 0$ (essentially empty case!), which includes the abelian cases $g = 0$, $\nu = 2$ and $g = 1$, $\nu = 0$, for which a direct treatment is possible, and has already been given essentially, those cases † where|if one starts with an extension (not with an "outer" action i.e. here an action plain and simple of Γ on $\pi \simeq \mathbb{Z}$ or $\mathbb{Z}^2$), being always realizable. (N.B. † In this case, the hypothesis of a faithful action is replaced by the one that the extension is † a product extension over|under no subgroup $\Gamma' \subset \Gamma^0$ with $\Gamma' \neq 1$ ....). [scan 197]

There remain however other results of the realizable case, which would have to be examined in the general case :

c) If Γ is a † finite subgroup of $\mathrm{Aut}_{\mathrm{lac}}(\pi)$ i.e. a finite group operating faithfully on π, do we have $\pi^\Gamma = \{1\}$? m308

Marginal note. m308 We may assume $\Gamma = \Gamma^0$ hence Γ generated by a direct automorphism u or Γ generated by an anti-automorphism of order 2 [p. 196]

Uncertainty retained from cancelled passages.
140-2-s197-D001 being
140-2-s197-D002 the hypothesis
140-2-s197-D003 not splittable
140-2-s197-D004 discrete

d) If Γ', Γ'' are finite subgroups of $\mathcal{E}$ (extension of the finite group Γ by π, corresponding to an † outer (preserving the orientation) faithful operation), † such that $\Gamma' \cap \Gamma'' \neq (1)$, then ? are Γ' and Γ'' contained in the π-conjugate of a finite subgroup Γ''' of $\mathcal{E}$ [i.e. every section subgroup $\neq 1$ is contained in a unique maximal section subgroup, modulo π-conjugation m309 [scan 198] [p. 99]

Marginal note. m309 N.B. Would become false by † placing in the group $\mathrm{Aut}_{\mathrm{lac}}(\pi)$ [illeg. 2] one as a subgroup [illeg. 2] one of the two subgroups contained in [illeg. 4]

e) For every subgroup, [illeg. 2] † Γ' of Γ, is the set of classes † of π-conjugation of liftings of Γ' into $\mathcal{E}$ finite ? [We may assume $\Gamma' = \Gamma$, and Γ cyclic (if $\Gamma = \Gamma^0$) or dihedral (otherwise)] In case c)|e), we may assume Γ cyclic of prime order, and it is OK if it is established that an outer operation [illeg. 3] that [illeg. 4] $\pi_{g,\nu}$ † d). m310

Marginal note. m310 In the same way e) is established if we admit that the outer operations of cyclic groups on a $\pi \simeq \pi_{g,\nu}$ are realizable

†

†

Ultra strong!
If $g = 0$
[p. 197]

Uncertainty retained from cancelled passages.

140-2-s198-D001 respect
140-2-s198-D002 [illeg. 1]
140-2-s198-D004 [illeg. 1]
140-2-s198-D005 [illeg. 1]
140-2-s198-D006 of extensions
140-2-s198-D007 [illeg. 3]
140-2-s198-D009 [illeg. 2] juncture|junction $\Gamma = \Gamma' \cap \Gamma'' \neq (1)$
140-2-s198-D010 Theorem. Let Γ_0 [illeg. 1] be a finite subgroup of $A = \mathrm{Aut}_{\mathrm{lac}}(\pi)^0$, $\Gamma_0 \neq 1$, [illeg. 1] and let us consider Then the normalizer N of $\Gamma|\Gamma_0$ in A is equal to its centralizer $Z|\mathcal{Z}$

†

[scan 199]
[p. 198]

Uncertainty retained from cancelled passages.

140-2-s199-D001 and the set of elements of finite order of $Z|\mathcal{Z}$ is a cyclic group $F|\Gamma$, contained in which is also the largest finite subgroup of A containing Γ_0, and is contained in the center of $Z|\mathcal{Z}$.

The fact that the normalizer N centralizes Γ_0 comes from the fact that we have a canonical isomorphism (compatible with the transports of structures)

$$\Gamma_0 \xrightarrow{\sim} T_{n_0} \qquad (\text{if } n_0 = \operatorname{card} \Gamma_0)$$

(N.B. if $g \in \mathrm{Aut}_{\mathrm{lac}}(\pi)$ reversing the orientation, and normalizes|centralizes Γ_0, then g (int g) induces the symmetry $\sigma \to \sigma^{-1}$ on Γ_0, by the same argument) [illeg. 1] Let us note that if u is an element of finite order of $Z|\mathcal{Z}$, so that Γ_0, u generate a finite commutative subgroup of $Z|\mathcal{Z}$, we see that as we know that a finite subgroup of A is cyclic, it follows that u is contained in a cyclic subgroup of $Z|\mathcal{Z}$ contain^{ing} Γ_0, therefore say Γ'. let us now choose a geometric realization of the outer action of Γ_0 on [illeg. 1] π, say (X, S) with $U = X - S$;

†

[scan 200]
[p. 100]
[p. 199]

Uncertainty retained from cancelled passages.

140-2-s200-D001 we know that the lifting of the outer action of Γ_0 to an action is described by a unique point $a \in U$. Let I say that Γ' is realizable (or its outer action on $\pi \simeq \pi_1(U, a)$) is realizable in a unique way by an action of Γ' and inducing the already given action of $\Gamma|\Gamma_0$ on (X, S)), fixing a, the fact that it is indeed realizable in this way comes from the fact (which we admit) that every outer action of a cyclic group (here Γ') on a group with loops π of type (g, ν) is realizable, and from the uniqueness of such a realization (in the case of Γ_0). For uniqueness, this [illeg. 1] comes down to the

not true!

Lemma. Let u_0 be an automorphism of finite order $n_0 \neq 1$ of an orientable surface assumed $X^{u_0} \neq \emptyset$. $U = X - S$, X (compact, [illeg. 1], and leaving fixed a point $a \in U$ [illeg. 1] let $d \in \mathbb{N}^*$, (If there exists an automorphism u of [illeg. 1] (X, S) such that $u^d = u_0$, $u(a) = a$, then u is uniquely determined by its class of action on $\pi_1(U, a)$; and in any event the cyclic group of automorphisms of (X, S) generated by u is unique^{ly} determined (it is of order nd).

false!

Proof. †

[scan 201]

Let us prove d). We identify Γ', Γ'' with subgroups of Γ, and put $\Gamma_0 = \Gamma' \cap \Gamma''$. Let $\mathcal{E}^!$ be the subgroup of $\mathcal{E}$ consisting of the $g \in \mathcal{E}$ such that $\mathrm{int}(g).\Gamma_0$ is π-conjugate to Γ_0 and whose image in Γ † centralizes Γ_0. † We have $\mathcal{E}^! \supset \pi$, and $\mathcal{E}^!$ is therefore the inverse image of a subgroup $\Gamma^!$ of Γ, which centralizes Γ_0 †, and which contains Γ' and Γ'' (for $\mathcal{E}^!$ contains the centralizer of Γ_0

in $\mathcal{E}$, hence Γ' and Γ''). Replacing Γ by $\Gamma^!$, $\mathcal{E}$ by $\mathcal{E}^!$ if necessary, we may suppose $\mathcal{E} = \mathcal{E}^!$, $\Gamma = \Gamma^!$ i.e. that $\Gamma_0 \subset$ Centre of Γ, and that $\Gamma_0 \hookrightarrow \mathcal{E}$ invariant modulo π-conjugation under Γ.

†

Uncertainty retained from cancelled passages.

140-2-s201-D001 conformal [illeg. 1] invariant

140-2-s201-D005 [illeg. 1]

140-2-s201-D006 [illeg. 1]

140-2-s201-D007 [illeg. 1]

140-2-s201-D008 b

† [scan 202]

† [p. 101]

We are going to construct a section of $\mathcal{E}$ over Γ in all its entirety, as follows. Let $\tilde{\Gamma} \subset \mathcal{E}$ be the centralizer of Γ_0 in $\mathcal{E}$, we have|⇒ $\tilde{\Gamma} \cap \pi = (1)$ (for this means $\pi^{\Gamma_0} = (1)$) hence $\tilde{\Gamma} \longrightarrow \Gamma$ injective, I say that it is surjective. Indeed, let $\gamma \in \Gamma$, $g \in \mathcal{E}$ over γ, by hypothesis on Γ $\exists\ \alpha \in \pi$ such that $\mathrm{int}(g).\Gamma_0 = \mathrm{int}(\alpha).\Gamma_0$ i.e. we may suppose $\mathrm{int}(g)\Gamma_0 = \Gamma_0$, † i.e. g normalizes Γ_0, but since Γ_0 is central in Γ this means that g centralizes Γ_0 i.e. $g \in \tilde{\Gamma}$.

Uncertainty retained from cancelled passages.

140-2-s202-D001 (X, S, a) $\Gamma_0|\Gamma$ the same|= can of the generated subgroup generated subgroup by

140-2-s202-D005 $\Gamma/\Gamma_0|\Gamma'/\Gamma_0$ isotypically|isotopically

140-2-s202-D010 hence

Thus $\tilde{\Gamma} \simeq \Gamma$ is a finite subgroup of $\mathrm{Aut}(\pi)$, containing Γ' and Γ'', q.e.d. [scan 203]

N.B. If we had not assumed Γ finite at the outset, it would still be true that Γ', Γ'' generate a subgroup $\tilde{\Gamma} \subset \mathrm{Aut}(\pi)$ such that $\tilde{\Gamma} \cap \pi = (1)$, but that does us a fat lot of good †

Uncertainty retained from cancelled passages.

140-2-s203-D002 we can find some Autext(π, i) $i \in I(\pi)$ [illeg. 2] i [illeg. 2]

19. Teichmüller tower. [scan 204]

Let $g \in \mathbb{N}$, and X_g be a compact orientable surface of genus g, † [|] $(a_{g,i})_{i \in \mathbb{N}^*} = (a_{g,1}, a_{g,2}, \ldots)$]|[a sequence of distinct points of X_g. We put for $\forall \nu \in \mathbb{N}$ [p. 102]

$$(1) \qquad S_{g,\nu} = \{x_{g,i} | a_{g,i} \mid 0 \leq i \leq \nu - 1\} \qquad \text{thus } \nu = \mathrm{card}\, S_{g,\nu}$$

$$(2) \qquad U_{g,\nu} = X_g - S_{g,\nu}.$$

We thus have $S_{g,0} = \emptyset$, $U_{g,0} = X_g$; The $S_{g,\nu}$ form a strictly increasing sequence of finite subsets of X_g, the $U_{g,\nu}$ a strictly decreasing sequence of open subsets of X_g. In what follows we shall take $U_{g,\nu}$ as the standard surface (oriented, or orientable), of type (g, ν). N.B. we have $a_{g,\nu} \in U_{g,\nu}$

Let

$$(3) \qquad A_g = \mathrm{Aut}(X_g)$$

be the group of automorphisms of X_g, equipped with the topology of uniform convergence of u and of its inverse. We put

$$(4) \qquad A_{g,\nu} = \{u \in A_g \mid u(S_{g,\nu}) = S_{g,\nu}\} \subset A_g$$

thus $A_{g,\nu}$ is a subgroup of A_g. We have a canonical isomorphism $u \mapsto u|U_{g,\nu}$

$$A_{g,\nu} \xrightarrow{\sim} \mathrm{Aut}(U_{g,\nu}). \qquad (5)$$

m315

Marginal note. m315 N.B. This is probably|probably an isomorphism topologically, when $A_{g,\nu}$ is equipped with the topology induced by A_g, and $\mathrm{Aut}(U_{g,\nu})$ with the usual topology of compact [illeg. 1] convergence of u and its inverse.

† We have † canonical surjective homomorphism

$$(6) \qquad A_{g,\nu} \longrightarrow \mathrm{Aut}(S_{g,\nu})(\overset{(\text{canonical})}{\simeq} \mathfrak{S}_\nu),$$

Uncertainty retained from cancelled passages.
140-2-s204-D001 the set of orientations
140-2-s204-D002 Let
140-2-s204-D003 a

whose kernel is denoted by $A^!_{g,\nu}$ [scan 205]

$$(7) \qquad A^!_{g,\nu} = \{u \in A_g \mid u(a_{g,i}) = a_{g,i},\ \forall i \in [0, \dots, \nu-1]\}$$

hence exact sequence

$$(8) \qquad 1 \longrightarrow A^!_{g,\nu} \longrightarrow A_{g,\nu} \longrightarrow \mathfrak{S}_\nu \longrightarrow 1.$$

Let $A^\circ_{g,\nu}$ be the identity component of the group $A_{g,\nu}$, we thus have

$$(9) \qquad A^\circ_{g,\nu} (= (A^!_{g,\nu})^\circ) \subset A^!_{g,\nu}.$$

Let us put

$$(10) \qquad \Gamma_{g,\nu} = A_{g,\nu}/A^\circ_{g,\nu} = \pi_0(A_{g,\nu}) \quad (\text{Teichmüller group of type } g, \nu)$$

We also put $\Gamma_g = \Gamma_{g,0} (= \Gamma^!_{g,0})$

$$(11) \qquad \Gamma^!_{g,\nu} = A^!_{g,\nu}/A^\circ_{g,\nu} = \pi_0(A^!_{g,\nu}).$$

The exact sequence (8) thus gives an exact sequence

$$(12) \qquad 1 \longrightarrow \Gamma^!_{g,\nu} \longrightarrow \Gamma_{g,\nu} \longrightarrow \mathfrak{S}_\nu \longrightarrow 1.$$

We have canonical homeomorphisms

$$(13) \qquad A_g/A_{g,\nu} \simeq \begin{array}{c} \text{open subset } \mathrm{Sym}^\nu(X)^* \text{ of the † symmetric prod-} \\ \text{uct } \nu^{\text{th}}\, \mathrm{Sym}^\nu(X_g) \text{ of } X_g \\ \text{consisting of the finite subsets} \\ \text{of cardinality } \nu \\ = \mathcal{P}_\nu(X_g) \end{array}$$

m317

Marginal note. m317 N.B. the marked points of the homogeneous spaces on the left-hand sides in|of (13), (14) go to the systems of points $\{a_{g,0}, a_{g,1}, \dots, a_{g,\nu-1}\}$ or $(a_{g,0}, \dots, a_{g,\nu-1})$.

$$(14) \qquad A_g/A^!_{g,\nu} \simeq \begin{array}{c} \text{open subset } (X^\nu_g)^* \text{ of } X^\nu_g \text{ consisting of the} \\ \nu\text{-tuples of distinct points} \\ \simeq \mathrm{Mon}(\dagger I_\nu, X) \quad (\text{where } I_\nu \\ = \{0, \dots, \nu-1\}. \end{array}$$

Uncertainty retained from cancelled passages.
140-2-s205-D001 X_g
140-2-s205-D002 X_g

This homeomorphism respecting the natural actions of $\mathfrak{S}_\nu \simeq A_{g,\nu}/A'_{g,\nu}$. [scan 206]
The $A'_{g,\nu}$ for variable ν form a decreasing sequence of groups in A_g [p. 103]

$$(15) \qquad A_g = |\supset A'_{g,0} \supset A'_{g,1} \supset A'_{g,2} \supset \cdots \supset A'_{g,\nu} \supset \cdots$$

and the corresponding morphisms between homogeneous spaces of A_g fit into the commutative diagram of homeomorphisms [illeg. 1]

(16)

$$\begin{array}{ccc} A_g/A'_{g,\nu} & \xrightarrow{\sim} & \mathrm{Mon}(I_\nu, X) \\ \downarrow & & \downarrow \\ A_g/A'_{g,\nu'} & \xrightarrow{\sim} & \mathrm{Mon}(I_{\nu'}, X) \end{array}$$

$0 \leq \nu' \leq \nu$ where the second vertical arrow is that of restriction from I_ν to $I_{\nu'} \subset I_\nu$. [illeg. 1]
For $\nu = 1$ we have $A'_{g,1} = A_{g,1}$, and the isomorphism (13) (or if one prefers, (14)) is written

$$(17) \qquad A_g/A_{g,1} \simeq X_g$$

(homeomorphism compatible with the actions of A_g)

Moreover if to $\forall x \in X_g$ we associate its stabilizer $A_{g,x}$ in A_g, we find an obviously surjective map

[scan 207]

$$(18) \qquad X_g \longrightarrow \begin{array}{c} \text{set of conjugates of the sub-} \\ \text{group } A_{g,1} \text{ of } A_g \\ \simeq A_g/\operatorname{Norm}_{A_g}(A_{g,1}) \end{array}$$

which is identified via (17) with the canonical map on the homogeneous spaces

$$A_g/A_{g,1} \longrightarrow A_g/\operatorname{Norm}_{A_g}(A_{g,1})$$

deduced from the inclusion $A_{g,1} \subset \operatorname{Norm}_{A_g}(A_{g,1})$, and we see at once that (18) is bijective i.e. that †

$$(19) \qquad A_{g,1} = \operatorname{Norm}_{A_g}(A_{g,1}),$$

more generally for $\forall \nu$, we have

$$(20) \qquad A_{g,\nu} = \operatorname{Norm}_{A_g}(A_{g,\nu}) = \operatorname{Norm}_{A_g}(A'_{g,\nu})$$

which means that the canonical maps of homogeneous A_g-sets

$$\mathcal{P}_\nu(X_g) \longrightarrow \text{set of conjugates of } A_{g,\nu}$$

$$(21) \qquad S \longmapsto \text{stabilizer } A_{g,S} \text{ of } S$$

and

$$(22) \qquad | \quad S \longmapsto A'_{g,S}$$

are not only surjective, but even bijective. This comes from the fact that we recover S in terms of $A_{g,S}$ (or of $A'_{g,S}$ † as

Uncertainty retained from cancelled passages.
140-2-s207-D001 [illeg. 1]
140-2-s207-D005 (and of the group

[scan 208]
[p. 104]

$$(23)\qquad S = \{x \in X_g \mid ux = x \quad \forall u \in A^!_{g,S}\} = X_g^{(A^!_{g,S})}$$

$$(24)\qquad \begin{aligned} S &= \{x \in X_g \mid A_{g,S}.x \text{ finite}\} \\ &= \{x \in X_g \mid A_{g,S}.x \neq X_g\} \end{aligned}$$

$\forall S \in \mathcal{P}_\nu(X_g)$

We also have some|† canonical maps of homogeneous spaces under A_g

$$(25)\qquad S \longmapsto A^\circ_{g,S}$$

$$\mathcal{P}_\nu(X_g) \longrightarrow \begin{array}{l} \text{set of conjugates} \\ \text{of the } A^\circ_{g,\nu} \text{ in } A_g \\ \simeq A_g/\operatorname{Norm}(A^\circ_{g,\nu}) \end{array}$$

which is not only surjective, but also injective, for we have the following relation, which strengthens (23) and (24), $\forall S \in \mathcal{P}_\nu(X_g)$

$$(26)\qquad \begin{aligned} S &= \{x \in X_g \mid A^\circ_{g,S}.x = \{x\}\} = X_g^{A^\circ_{g,S}} \\ &= \{x \in X_g \mid A^\circ_{g,S}.x \text{ finite}\} \\ &= \{x \in X_g \mid A^\circ_{g,S}.x \neq X_g\}. \end{aligned}$$

† Thus we have for $\forall \nu$

$$(27)\qquad A_{g,\nu} = \operatorname{Norm}_{A_g}(A_{g,\nu}|A_{g,x}) = \operatorname{Norm}_{A_g}(A^!_{g,\nu}) = \operatorname{Norm}_{A_g}(A^\circ_{g,\nu}).$$

Let G be a topological group, equipped with a conjugacy class $\mathcal{E}|\mathcal{X}$ of subgroups, G_1 in this class. We say that

Uncertainty retained from cancelled passages.
140-2-s208-D001 some
140-2-s208-D002 [illeg. 2]

[scan 209]

$G, \mathcal{X}$ is a Teichmüller † pair of type g, if there exists an isomorphism of topological groups $G \simeq A_g$, taking $\mathcal{E}|\mathcal{X}$ to the conjugacy class of A_1. It amounts to the same thing to say that $\mathcal{X}$, † equipped with the topology of a homogeneous space under G, is a compact orientable surface of genus g, † say $\mathcal{X}|\mathfrak{X}$, and that † the map natural

$$(28)\qquad G \longrightarrow \operatorname{Aut}(\mathcal{X})$$

is a homeomorphism of topological groups. We then see that by $(G, \mathcal{X}) \longrightarrow \mathcal{X}$ from the category of Teichmüller pairs of type g, to that of compact oriented surfaces, is an equivalence of categories. It follows that † for an automorphism u of the topological group G, u is inner iff it

Uncertainty retained from cancelled passages.
140-2-s209-D001 datum
140-2-s209-D004 as
140-2-s209-D006 [illeg. 2]
140-2-s209-D007 [illeg. 3]
140-2-s209-D008 every automorphism

preserves the class $\mathcal{X}$ i.e. iff $u(G_1)$ is conjugate to G_1. Moreover the center of G is trivial, i.e. [scan 210]

We can give an analogous description for the category of oriented surfaces of genus g with ν [p. 105]
holes (i.e. homeomorphic to $U_{g,\nu}$) in terms of a topo- logical group $G_\nu(\simeq A_{g,\nu})$ and of a conjugacy class $U_\nu|\widetilde{U}_\nu$ of subgroups H_ν of it ($\simeq A_{g,\nu}\cap A_{g,\nu+1}$), with the conditions that $(G_\nu,\{H_\nu\})$ be isomorphic to $(A_{g,\nu}, A_{g,\nu}\cap A_{g,\nu+1})|\}$, or else that U_ν (space [illeg. 1] topology of a homogeneous space) be an oriented surface of type (g,ν), and that the canonical homomorphism

$$G \longrightarrow \mathrm{Aut}(U_\nu) \tag{29}$$

be a homeomorphism of topological groups. We then find an equivalence between the category of Teichmüller pairs $(G_\nu,\{H_\nu\})$ of type (g,ν), and that of orientable surfaces of type (g,ν). We find again

that the automorphisms of such a G_ν ($\simeq A_{g,\nu}$) which fix the class U_ν i.e. which transform|transport H_ν into a conjugate, are inner, and that the center of H_ν is $\{1\}$. [scan 211]

In fact, U_ν can also be interpreted as a homogeneous space under G_ν°, and not only under G_ν, more generally we have that G_ν° is transitive on U_ν, and even on $\mathcal{P}_{\nu'}(U_\nu)$ for every $\nu'\in\mathbb{N}^*$.

Returning to the model situation with $U_{g,\nu}$, we find

$$\begin{aligned} U_{g,\nu} &\simeq A_{g,\nu}^{\dagger|\bullet}/\underbrace{A_{g,(\nu,\nu+1)}}_{\substack{\| \text{ definition}\\ A_{g,\nu}\cap A_{g,\nu+1}}} \\ &\simeq A_{g,\nu}^{\circ}/\underbrace{B_{g,(\nu,\nu+1)}}_{\substack{\| \text{ (definition)}\\ A_{g,\dagger\{\nu+1\}}\cap A_{g,\nu}^{\circ}=\{u\in A_{g,\nu}^{\circ}|u.a_{g,\nu}=a_{g,\nu}\}}} . \end{aligned} \tag{30}$$

We thus have † † by inverse diagram of inclusions of subgroups of $A_{g,\nu}$: † 210

Uncertainty retained from cancelled passages.

140-2-s211-D001 ∘

140-2-s211-D002 (ν,

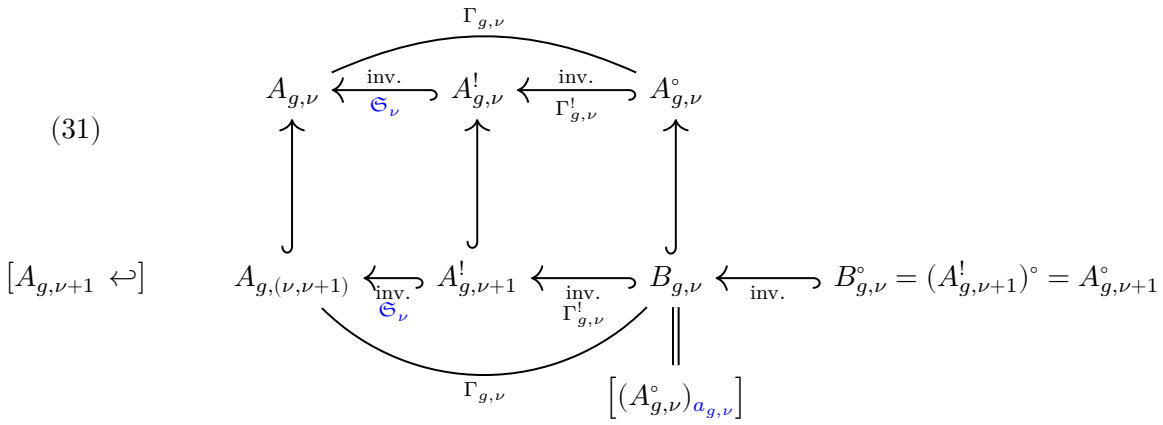
140-2-s211-D003 some

140-2-s211-D004 inclusions

140-2-s211-D005 $A_{g,[illeg.\ 1]}\ \supset\ A_{g,\nu}\ \supset$

[scan 212]
[p. 106]

(31)



m321

Marginal note. m321 N.B. $A_{g,(\nu,\nu+1)}$, stabilizer [illeg. 2] in ($A_{g,\nu}=\mathrm{Aut}(U_{g,\nu})$) of the marked point $a_{g,\nu}$ of [illeg. 1], deserves a simpler notation, for example $A^\bullet_{g,\nu}$ (the exponent • here to emphasize [illeg. 3] pointed|pointed in $U_{g,\nu}$).

The two squares are cartesian, and the horizontal inclusions (except the one in brackets in the 2nd row) are inclusions of invariant subgroups invariant even (in the largest group, $A_{g,\nu}$ resp. $A_{g,(\nu,\nu+1)}$).

[The equality $B^\circ_{g,\nu} = A^{!\circ}_{g,\nu+1}$ † (which refines the trivial inclusion $B^\circ_{g,\nu} \subset A^{!\circ}_{g,\nu+1}$ by the reverse inclusion $A^{!\circ}_{g,\nu+1} \subset B^\circ_{g,\nu}$, equivalent to $A^{!\circ}_{g,\nu+1} \subset B_{g,\nu}$) comes from the inclusion

$$A^{!\circ}_{g,\nu+1} = A^\circ_{g,\nu+1} \subset A^{!\circ}_{g,\nu} = A^\circ_{g,\nu} \quad (\text{whence : } A^\circ_{g,\nu+1} \subset A_{g,\nu+1} \cap A^\circ_{g,\nu} = B_{g,\nu}).$$

]

The three vertical inclusions define three homogeneous spaces, and † inclusion homomorphisms between them which are bijective

$$(U_{g,\nu} \simeq) \quad A_{g,\nu}/A_{g,(\nu,\nu+1)} \xleftarrow{\sim} A^!_{g,\nu}/A^!_{g,\nu+1} \xleftarrow{\sim} A^\circ_{g,\nu}/B_{g,\nu},$$

with (transitivity !) and † † the quotient groups denoted $\mathfrak{S}_\nu$ and $\Gamma^!_{g,\nu}$ defined by the horizontal inclusions of the first row, are isomorphic via the vertical inclusions to the corresponding quotients in the second row. Thus the extension of groups

$$(12) \qquad 1 \longrightarrow \Gamma^!_{g,\nu} \longrightarrow \Gamma_{g,\nu} \longrightarrow \mathfrak{S}_\nu \longrightarrow 1$$

can be deduced equally well from 211

Uncertainty retained from cancelled passages.
140-2-s212-D001 by
140-2-s212-D002 A_g
140-2-s212-D003 likewise by
140-2-s212-D004 arrows [illeg. 2]

the first row of (31), or from the first three terms of the second, in particular [scan 213]

$$\begin{cases} \mathfrak{S}_\nu \simeq A_{g,(\nu,\nu+1)}/A^!_{g,\nu+1} \\ \Gamma^{\bullet|}_{g,\nu} \simeq A_{g,(\nu,\nu+1)}/B_{g,\nu} \\ \Gamma^!_{g,\nu} \simeq A^!_{g,\nu+1}/B_{g,\nu}. \end{cases} \tag{32}$$

Thus the canonical torsor with group $A_{g,(\nu,\nu+1)} = (A_{g,\nu})_{a_{g,\nu}}$ over the homogeneous space $U_{g,\nu}$, and the one of the group $A^!_{g,\nu+1}$ $(\subset A_{g,(\nu,\nu+1)})$ which gives rise to it, is in either case deduced, by extension of the structure groups, from a torsor with group $B_{g,\nu}$ (whose fiber at $x \in U_{g,\nu}$ is the space of $u \in A^\circ_{g,\nu}$ such that $u(a_{g,\nu}) = x$). $= (A^\circ_{g,\nu})_{a_{g,\nu}}$

The associated Galois covering, with group $B_{g,\nu}/B^\circ_{g,\nu}$, is therefore also the homogeneous space $A^\circ_{g,\nu}/A^\circ_{g,\nu+1}$, which is obviously connected, and pointed over $a_{g,\nu} \in U_{g,\nu}$. Put † † 212

$$\widetilde{U}_{g,\nu} = A^\circ_{g,\nu}/A^\circ_{g,\nu+1}. \tag{33}$$

Theorem. $\widetilde{U}_{g,\nu}$ is simply connected (if $(g,\nu) \neq (1,0)$ and $(g,\nu) \neq (0,2)$) (and even contractible if $(g,\nu) \neq (0,0)$) and thus identifies with the universal covering pointed at $a_{g,\nu+1}$ of $U_{g,\nu}$.
m324

Marginal note. m324 This relation|This assertion plausible [illeg. 3] in the anabelian cases $2g + \nu \geq 3$, we shall see [illeg. 3] in abelian case [illeg. 4] reasonable [illeg. 3]. We must exclude the cases $(1,0)$ and $(0,2)$.

Corollary. We have a canonical isomorphism

$$B_{g,\nu}/B^\circ_{g,\nu} \simeq \pi_1(U_{g,\nu}; a_{g,\nu}) \overset{\text{definition}}{=} \Pi_{g,\nu}. \tag{34}$$

Uncertainty retained from cancelled passages.
140-2-s213-D002 Theorem
140-2-s213-D003 [illeg. 2]

Proof of the Theorem, in terms of "well-known" things. The † homotopy exact sequence for the fibration of $A^\circ_{g,\nu}$ over [scan 214] [p. 107]

$$A^\circ_{g,\nu+1}/A^\circ_{g,\nu} \simeq \widetilde{U}_{g,\nu}$$

†

$$\longrightarrow \pi_{i+1}(\widetilde{U}_{g,\nu}) \longrightarrow \pi_i(A^\circ_{g,\nu+1}) \longrightarrow \pi_i(A^\circ_{g,\nu\dagger}) \longrightarrow \pi_i(\widetilde{U}_{g,\nu}) \longrightarrow \pi_{i-1}(A^\circ_{g,\nu+1}) \tag{35}$$

and it shows that $\pi_1(\widetilde{U}_{g,\nu})$ is isomorphic to the cokernel of

$$\pi_1(A^\circ_{g,\nu}) \longrightarrow \pi_1(A^\circ_{g,\nu+1}),$$

whose kernel is a quotient of $\pi_2(\widetilde{U}_{g,\nu})$...

If $(g,\nu) \neq 0$, we know that $\pi_i(\widetilde{U}_{g,\nu}) = 0$ for $i \geq 2$ (for every covering of $U_{g,\nu}$) hence †

$$\left\{ \pi_i(A^\circ_{g,\nu+1}) \longrightarrow \pi_i(A^\circ_{g,\nu\dagger}) \text{ is } \begin{cases} \text{bijective if } i \geq 2 \\ \text{injective with invariant image if } i = 1 \end{cases} \right. \quad \begin{array}{c} \text{if } (g,\nu) \neq (0,0) \\ \\ [\text{the cokernel is } \simeq \pi_1(\widetilde{U}_{g,\nu})] \end{array} \tag{36}$$

and we want to prove that it is even bijective for $i = 1$. The thing to remember easy is this :

† 213

Uncertainty retained from cancelled passages.
140-2-s214-D001 [illeg. 1]
140-2-s214-D002 gives
140-2-s214-D003 +1
140-2-s214-D004 [illeg. 1]
140-2-s214-D005 +1
140-2-s214-D006 iff [illeg. 1] [illeg. 1] [illeg. 2] to

† [scan 215]

Well-known theorem. Equivalent conditions on the pair $(g,\nu) \in \mathbb{N} \times \mathbb{N}$:

a) $2g + \nu \geq 3$ i.e. we are not in one of the four exceptional cases $(1,0)$, $(0,0)$, $(0,1)$, $(0,2)$.
b) $\pi_1(U_{g,\nu})$ nonabelian
c) † $A^\circ_{g,\nu}$ is simply connected († $(g,\nu) \neq (0,1)$)
d) $A_{g,\nu}$ is contractible and $(g,\nu) \neq (0,1)$.

† abelian case

In any case, whether or not these conditions are satisfied, we have

$$\pi_i(A^\circ_{g,\nu}) = 0 \qquad i \geq 2, \quad (g,\nu) \neq 0$$

(i.e. the $A^\circ_{g,\nu}$ are classifying spaces of discrete groups $\pi|K$, with the sole exception of $A^\circ_{0,0} \simeq \mathrm{Aut}(S^2)^\circ$).

Taking (36) into account, this Theorem is equivalent to the relations

$$\begin{array}{lll} \pi_i(A^\circ_{g,0}) = 0 & \text{if } g \geq 2,\ i \geq 1 & \text{i.e. for } g \geq 2,\ A_{g,0} \text{ is contractible} \\ \pi_i(A^\circ_{1,1}) = 0 & \text{for } i \geq 1 & \text{i.e. } A^\circ_{1,1} \text{ is contractible} \\ \pi_i(A^\circ_{0,3}) = 0 & \text{for } i \geq 1 & \text{i.e. } A^\circ_{0,3} \text{ is contractible.} \end{array} \tag{37}$$

$$\left.\begin{array}{ll} \pi_i(A^\circ_{1,0}) = 0 & \text{for } i \geq 2 \\ \pi_i(A^\circ_{0,1}) = 0 & \text{for } i \geq 2 \end{array}\right\} \quad \text{i.e. } A^\circ_{1,0} \text{ and } A^\circ_{0,1} \text{ are } K(\pi,1). \tag{38}$$

†

and they have as a consequence that $\widetilde{U}_{g,\nu}$ is simply connected in the anabelian cases $2g+\nu \geq 3$. To know how things stand 214

Uncertainty retained from cancelled passages.

140-2-s215-D001 and the
140-2-s215-D003 π
140-2-s215-D004 and
140-2-s215-D005 In the case of simply [illeg. 3] contractible
140-2-s215-D007 also as a consequence

in the abelian cases, we must make precise the topological structure of $A^\circ_{g,\nu}$ in the four "abelian" [scan 216]
cases $2g + \nu \leq 2$. Now we have the [p. 108]

Corollary. † For $U = U_{g,\nu}$ † to have abelian π_1 (i.e. not satisfying the equivalent conditions of the theorem), it is necessary and sufficient that the topological space U † be able, either to be endowed with a topological group structure (which will be necessarily

$$\simeq \mathbb{U} \times \mathbb{U} \quad (\text{case } (1,0)), \qquad \simeq \mathbb{C}^* \quad (\text{case } (0,2)) \quad \text{or} \quad \simeq \mathbb{C} \quad (\text{case } (0,1))),$$

m328

Marginal note. m328 $\mathbb{U} \overset{\text{definition}}{=} \{z \in \mathbb{C}^* \mid |z| = 1\}$.

or with a structure of a homogeneous space under a topological group †. (we can take $\mathrm{SO}(3,\mathbb{R})$ or $\mathrm{GP}(1,\mathbb{C})|\,\mathrm{GL}(1,\mathbb{C})$ for the case of S^2), by a connected subgroup.

In all cases, the topological group in question can be described as follows: We choose a complex † structure on $X_g = \widehat{U}_{g,\nu} = \widehat{U}$ (the compactification pure of U), and we take the induced complex structure on U (i.e. we choose an algebraic complex curve structure on U…) and we take $G \simeq \mathrm{Aut}^\circ_{\mathbb{C}}(U)$, the component of the identity of the group of complex automorphisms of U, i.e. of the complex automorphisms of $\widehat{U}$ which leave invariant|fix $S_\nu = \widehat{U} \setminus U$.

This being so, the inclusion $G \subset \mathrm{Aut}^\circ(U) \simeq A^\circ_{g,\nu}$ 215

Uncertainty retained from cancelled passages.

140-2-s216-D001 The conditions
140-2-s216-D002 [illeg. 1]
140-2-s216-D003 isomorphic
140-2-s216-D004 [illeg. 1]
140-2-s216-D005 topology

is a homotopy equivalence. [scan 217]

The fact that if U is a homogeneous space of a group that is topological, by a connected subgroup, , then $\pi_1(U)$ is abelian, comes from the homotopy exact sequence, and from the fact that the π_1 of a topological group is commutative. The converse in the case of topological surfaces [illeg. 1] remarkable! The corollary gives us

m330

Marginal note. m330 We have a more precise result: If k is the smallest integer such that $(g, \nu+k)$ is an anabelian pair, then $\dim G = k$, and G is simply transitive on the set of k-tuples $(x_1, \dots, x_k)$ of distinct points of $U_{g,\nu}$, whence it follows at once that every element of $A_{g,\nu}$ can be written in a unique way as a product gu, with $g \in G$, $u \in A_{g,(\nu,\nu+k)}$, so † $A_{g,\nu}$ is homeomorphic to $A_{g,(\nu,\nu+k)} \times G$. As G is connected, we conclude from this

$$\Gamma_{g,\nu} \simeq \Gamma_{g,(\nu,\nu+k)} \subset \Gamma_{g,\nu+k},$$

and $A^\circ_{g,\nu} \simeq G \times A^\circ_{g,\dagger\nu+k}$, and as $A^\circ_{g,\nu+k}$ is contractible|∞-connected, we conclude from this [illeg. 2] $G \xrightarrow{\simeq} | \simeq A^\circ_{g,\nu}$.

$$\begin{aligned} &\pi_1(A^\circ_{1,0}) \simeq \mathbb{Z} \times \mathbb{Z} \quad \text{i.e. } A^\circ_{1,0} \simeq K(\mathbb{Z} \times \mathbb{Z}, 1) \\ &\pi_1(A^\circ_{0,2}) \simeq \mathbb{Z} \qquad \text{i.e. } A^\circ_{0,\nu|2} \simeq K(\mathbb{Z}, 1) \\ &\pi_1(A^\circ_{0,1}) \simeq \mathbb{Z} \qquad \text{i.e. } A^\circ_{0,1} \simeq \dagger K(\mathbb{Z}, 1). \end{aligned} \tag{39}$$

m333

$G \simeq \mathbb{U} \times \mathbb{U}|U$ (elliptic curve)

Marginal note. m333 $G \simeq \mathrm{Aff}(1, \mathbb{C})$ (homotopic to $\mathbb{C}^*$ by the inclusion)

$G \simeq \mathbb{C}^* = \mathrm{GL}(1, \mathbb{C})$

$G \simeq \mathrm{GP}(1, \mathbb{C})$

$A^\circ_{0,0} = \mathrm{Aut}(S^2) \simeq \mathrm{Aut}(\mathbb{P}^1_{\mathbb{C}})$ is homotopic by the inclusion to its subgroup $\mathrm{GP}(1, \mathbb{C})$ (hence to its maximal compact subgroup $\mathrm{SO}(3, \mathbb{R})/\{\pm 1\}$), which is not a $K(\pi, 1)$, and whose π_1 is $\simeq \mathbb{Z}/2\mathbb{Z}$.

We are going to use this corollary to determine the structure of $\widetilde{U}_{g,\nu}$ (as a Galois covering of $U_{g,\nu}$) for the "abelian" cases. Let us note for this that 216

Uncertainty retained from cancelled passages.
140-2-s217-D002 [illeg. 2]
140-2-s217-D003 [*illeg.* 1]
140-2-s217-D004 [*illeg.* 1]

† [scan 218]

the inclusion $G \subset A^\circ_{g,\nu}$ induces an inclusion $G_0 \hookrightarrow B_{g,\nu}$, where † [p. 109]

$$G_0 = G \cap B_{g,\nu} = \text{stabilizer of } a_{g,\nu} \text{ in } G. \tag{40}$$

and as

$$G/G_0 \xrightarrow{\sim} A^\circ_{g,\nu}/B_{g,\nu} \simeq U_{g,\nu}$$

(G being transitive on $U_{g,\nu} = U$) we find that the fact that $G \longrightarrow A^\circ_{g,\nu}$ is a homotopy equivalence, is equivalent to $G_0 \longrightarrow B_{g,\nu}$ being one. Now in all the cases considered, G_0 is already connected: it is reduced to 1 in the cases $(1,0)$ and $(0,2)$, [so in this case the corollary is equivalent to the contractibility of $B_{g,\nu}$], and it is $\simeq \mathbb{C}^*$ in the case $(0,1)$, finally it is $\mathrm{Aff}(1, \mathbb{C})$ in the case $(0,0)$. So we find 217

Uncertainty retained from cancelled passages.
140-2-s218-D001 $\mathbb{U} \times \mathbb{U} \quad B_{g,\nu} =$
140-2-s218-D003 [illeg. 1]

Corollary. In the abelian cases $2g + \nu < 3$ (and only in those, of course) $B_{g,\nu}$ is connected (i.e. $= A^\circ_{g,\nu+1}$), i.e. $A^\circ_{g,\nu}/A^\circ_{g,\nu+1} \longrightarrow U_{g,\nu}$ is a homeomorphism. So $A^\circ_{g,\nu}/A^\circ_{g,\nu+1} \longrightarrow U_{g,\nu}$ makes the left-hand side a universal covering of $U_{g,\nu}$ iff (g,ν) is distinct from the two pairs $(1,0)$ and $(0,2)$. [scan 219]

Let us therefore assume $(g,\nu) \neq (1,0), (0,2)$ (morally we are working in the anabelian cases, the † included abelian cases $(0,1)$ and $(0,0)$ being † of no interest for what is to follow, it seems to me). Let us take up diagram (31) again, where in the second row the quotient group $B_{g,\nu}/B^\circ_{g,\nu}$ is therefore identified with:

$$\Pi_{g,\nu} = \pi_1(U_{g,\nu}, a_{g,\nu}).$$

We therefore find

Theorem. Suppose $(g,\nu) \neq (1,0)$ and $(0,2)$, (the subgroup $A_{g,(\nu,\nu+1)}/A^{\circ}_{g,\nu}|A^{\circ}_{g,\nu+1}$ of the group $= \Gamma'_{g,\nu+1}$ of Teichmüller

$$\Gamma_{g,\nu+1} = A_{g,\nu+1}/A^{\circ}_{g,\nu+1}$$

formed of the elements that fix $a_{g,\nu}$ admits a composition series of length 3, whose successive factors are $\mathfrak{S}_\nu|\mathfrak{S}_{\nu+1}$ †, $\Gamma^{!}_{g,\nu}$ and $\Pi_{g,\nu}$ — deduced from a structure 218

Uncertainty retained from cancelled passages.
140-2-s219-D001 [illeg. 1]
140-2-s219-D002 [illeg. 1]
140-2-s219-D003 [illeg. 2] $\Gamma^{!}_{g,\nu+1}$

of extension on $\Gamma^{!}_{g,\nu+1}$ [scan 220]

$$1 \longrightarrow \Pi_{g,\nu} \longrightarrow \Gamma^{!}_{g,\nu+1} \longrightarrow \Gamma^{!}_{g,\nu} \longrightarrow 1. \tag{41}$$ [p. 110]

[The outer operations of $\Gamma^{!}_{g,\nu}$ on $\Pi_{g,\nu}$ are those deduced by passage to the quotient from those of $A^{!}_{g,\nu} \subset \mathrm{Aut}(U_{g,\nu})$ operating by outer automorphisms on $\Pi_{g,\nu} = \pi_1(U_{g,\nu}, a_{g,\nu})$.

Another way of saying things is this. Let us distinguish, in $\Gamma_{g,\nu+1}$ operating on $S_{\nu+1}$ † ($\simeq I_{\nu+1}$), the stabilizer of the last element $a_{g,\nu}$, i.e. of its complement, namely $\Gamma'_{g,\nu+1} \simeq A_{g,(\nu,\nu+1)}/A^{\circ}_{g,\nu+1}$, this|the subgroup of index $\nu+1$, inverse image of $\mathfrak{S}_\nu$ of $\mathfrak{S}_{\nu+1}$ under $\Gamma_{g,\nu+1} \longrightarrow \mathfrak{S}_{\nu+1}$.

This being so, we have an evident homomorphism of discrete groups, deduced from the inclusion † $A_{g,(\nu,\nu+1)} \hookrightarrow A_{g,\nu+1}$:

$$\Gamma'_{g,\nu+1} \longrightarrow \Gamma_{g,\nu} \tag{42}$$

m338

Marginal note. m338 N.B. This homomorphism still makes sense without conditions on (g,ν) and it is an isomorphism in the abelian case $2g+\nu \leq 2$.

and this homomorphism is surjective (without conditions on (g,ν)), finally (for $(g,\nu) \neq (1,0),(0,1)$) its kernel is canonically isomorphic to $\Pi_{g,\nu}$, so that we have an extension N.B. N.B. †

$$1 \longrightarrow \Pi_{g,\nu} \longrightarrow \Gamma'_{g,\nu+1} \longrightarrow \Gamma_{g,\nu} \longrightarrow 1, \tag{43}$$

219

Uncertainty retained from cancelled passages.
140-2-s220-D001 [illeg. 1]
140-2-s220-D002 [illeg. 1]
140-2-s220-D003 $(0,2)$ [illeg. 2] [illeg. 3] [illeg. 1] $\simeq$

[scan 221]

if|where the corresponding outer operations of $\Gamma_{g,\nu}$ on $\Pi_{g,\nu}$ are those deduced by passage to the quotient from those of † $A_{g,\nu} \simeq \mathrm{Aut}(U_{g,\nu})$ on $\pi_1(U_{g,\nu}, a_{g,\nu}) = \Pi_{g,\nu}$.

The extension structure (41) is deduced from (41)|(43) by inverse image under the inclusion

$$\Gamma^{!}_{g,\nu} \longrightarrow \Gamma_{g,\nu}.$$

Suppose we are in the anabelian case $2g+\nu \geq 3$. As for every extension structure by a kernel with trivial center, there is a canonical homomorphism into the canonical extension structure associated to $\Pi_{g,\nu}$

$$\begin{array}{ccccccccc} 1 & \to & \Pi_{g,\nu} & \hookrightarrow & \Gamma'_{g,\nu+1} & \longrightarrow & \Gamma_{g,\nu} & \longrightarrow & 1 \\ & & \| & & \downarrow & & \downarrow & & \\ 1 & \to & \Pi_{g,\nu} & \to & \mathrm{Aut}(\Pi_{g,\nu}) & \to & \mathrm{Autext}(\Pi_{g,\nu}) & \to & 1 \end{array} \tag{44}$$

† where the central vertical arrow † is obtained by associating to $g \in \Gamma'_{g,\nu+1}$ the restriction to $\Pi_{g,\nu} \subset \Gamma'_{g,\nu+1}$ of the inner automorphism int(g). [p. 220]

Theorem well knownIn the anabelian case* $(2g + \nu \geq 3)$

$$\Gamma_{g,\nu} \xrightarrow{\sim} \mathrm{Autext}_{\mathrm{lac}}(\Pi_{g,\nu}),$$

Which amounts to the same, by (44)

$$\Gamma'_{g,\nu+1} \xrightarrow{\sim} \mathrm{Aut}_{\mathrm{lac}}(\Pi_{g,\nu}).$$

the image of $\Gamma_{g,\nu} \longrightarrow \mathrm{Autext}\,\Pi_{g,\nu}$ consists of the outer automorphisms which respect the loop structure of $\Pi_{g,\nu}$ of type g, ν.

m342

Marginal note. m342 * N.B. In fact this remains true for the pair|case $(1,0)$ (cf. below) and even in the case $(0,2)$ if we define ad hoc the notion of group with loops $(0,2)$.

Uncertainty retained from cancelled passages.

140-2-s221-D002 inclusion

140-2-s221-D003 [illeg. 1]

140-2-s221-D004 Theorem

(vacuous condition if $\nu = 0$, for that matter…). [scan 222]

Corollary In the anabelian case $(2g + \nu \geq 3)$, the functor $X \longmapsto \pi_1(X)$ from the isotopic [p. 111] category of surfaces of type g, ν to the outer groups with loops of type (g, ν), is an equivalence of categories, as is the functor $(X, s) \longmapsto \pi_1(X, s)$ from the isotopic category of pointed|punctured surfaces of type (g, ν), to the groups with loops of type (g, ν).

m343

Marginal note. m343 N.B. The arrows in the|the isotopic category are the isotopy classes of homeomorphisms.

m344

Marginal note. m344 It is even an equivalence at the level of ∞-isotopic categories which expresses only the fact that † $A^0_{g,\nu}$ and $A^0_{g,\nu+1}$ are contractible.

It remains to examine to what extent we can adapt these results to the "abelian" case. Let us recall that in this case, we have

$$(45) \qquad \Gamma'_{g,\nu+1} \xrightarrow{\sim} \Gamma_{g,\nu\dagger} \qquad \begin{array}{l} \text{if } 2g + \nu \leq 2 \\ (\text{“abelian” cases}) \end{array}$$

thus $\Gamma_{g,\nu}$ identifies with the subgroup of $\Gamma_{g,\nu+1}$ formed by the elements which fix $a_{g,\nu}$.

1) Case $g = 1$, $\nu = 0$ thus $\Gamma_{1,0} \simeq \Gamma'_{1,1} = \Gamma_{1,1}$.

Let us consider the canonical homomorphism

$$(46) \qquad \Gamma_1 = \Gamma_{1,0} \longrightarrow \mathrm{Autext}(\Pi_{1,0}) = \mathrm{Aut}(\Pi_{1,0})$$

$$(\simeq \mathrm{Gl}(2, \mathbb{Z}) \text{ when we have chosen a basis of } \Pi_{1,0} \simeq \mathbb{Z}^2)$$

which corresponds to the homomorphism

$$(46 \text{ bis}) \qquad \Gamma_{1,1} \longrightarrow \mathrm{Aut}\,\pi_1(U_{1,0}, a_{1,0}) \qquad (\simeq \mathrm{Gl}(2, \mathbb{Z})),$$

deduced both of them by passage to the quotient from the † obvious operations of $A_{1\dagger} = A_{1,0}$= Aut(X_1), and of [p. 221]

Uncertainty retained from cancelled passages.
140-2-s222-D001 [illeg. 1]
140-2-s222-D003 +1
140-2-s222-D004 [illeg. 1]
140-2-s222-D005 0

$A_{1,1} = \mathrm{Aut}(X_{1,1}|U_{1,1}, a_{1,0})$. It is immediate that this homomorphism is surjective, but less obvious that it is an isomorphism, i.e. that every homeomorphism of X_1 which induces the identity the Π_1 is isotopic † to the identity — this is nevertheless a true (and known) result. [scan 223]

In the present case where † X_1 † identifies with the underlying topological space of a topological group H (which is the case in all the abelian cases except that of $g = 0$, $\nu = 0$ of the 2-sphere) †, by translations|translation we have a natural homomorphism

$$(47) \qquad H \longrightarrow A_{g,\nu} \simeq \mathrm{Aut}_{\mathrm{top}} H$$

allowing us to identify H with a topological subgroup of $A_{g,\nu}$, and we find that the map

$$(48) \qquad (g, u) \longmapsto gu \qquad H \times \underbrace{A_{g,\nu+1}}_{\substack{\text{Stabilizer}\\ \text{of } 1_H \text{ in}\\ \longleftarrow A_{g,\nu} \simeq \mathrm{Aut}_{\mathrm{top}}(H)}} \longrightarrow A_{g,\nu}$$

is a homeomorphism. † This gives (45) again (since H is connected) [p. 222]

Uncertainty retained from cancelled passages.
140-2-s223-D003 [illeg. 1]
140-2-s223-D004 [illeg. 1]
140-2-s223-D005 [illeg. 2]
140-2-s223-D006 [illeg. 1]
140-2-s223-D007 As

and makes it considerably more precise, by [scan 224] [p. 112]

$$(49) \qquad \pi_i(A_{g,\nu+1}) \simeq \pi_i(A_{g,\nu}) \times \pi_i(H)$$

† Non-spherical abelian cases, i.e. any|only of the three cases $(g = 1, \nu = 0)$, $(g = 0, \nu = 1)$, $(g = 0, \nu = 2)$.

In the "limiting" cases $(1, 0)$ and $(0, 2)$ (when (g, ν) is abelian, $(g, \nu + 1)$ anabelian), since $A^0_{g,\nu} \simeq H \times A^0_{g,\nu+1}$ with $A^0_{g,\nu+1}$ contractible, we find again that $H \longrightarrow A^0_{g,\nu}$ is a homotopism, †

(Let us note that in the cases envisaged$(1, 0)$, $(0, 2)$ and $(0, 1)$, if we choose a complex structure on the|the pure compactification $\widehat{X}$ of $X = U_{g,\nu}$ and choose $a_{g,\nu}$ as origin, there is a unique $\mathbb{C}$-algebraic group structure on $\widehat{X}|X$ admitting 1 as unit, and † the † identity component of the group of automorphisms of the algebraic variety X is precisely none other than the group of translations, in the limiting cases $(1, 0)$ and $(0, 2)$. [p. 223]

Uncertainty retained from cancelled passages.
140-2-s224-D001 [illeg. 1] [illeg. 1] trivial
140-2-s224-D002 [illeg. 2]
140-2-s224-D003 [illeg. 2]
140-2-s224-D004 [illeg. 3]

2) Case $g = 0$, $\nu = 2$, thus $H \simeq \mathbb{C}^*$, [scan 225]

$$A_{g,\nu|2} \simeq H \times A_{g,\nu+1} \quad \text{i.e.} \quad A_{g,2} \simeq \mathbb{C}^* \times \underline{A_{g,3}}$$

thus

$$(50)\qquad \begin{cases} \Gamma_{g,2} \simeq \Gamma'_{g,3} \\ A^0_{g,2} \overset{\simeq}{\leftarrow} \mathbb{C}^* \quad \text{(homotopy equivalence)} \end{cases}$$

† $\Pi_{①,2} \simeq \mathbb{Z}$†, let us consider (rather $\Pi_{0,2}$)

$$(51)\qquad \Gamma_{g,2} \overset{\rho}{\to} \mathrm{Aut}\left(\begin{array}{c} \Pi_{①,2} \\ \| \\ \pi_1(U_{0,2}) \end{array} \right) \dagger \simeq \{\pm 1\} \simeq \mathfrak{S}_2$$

identifying with

$$(52)\qquad \Gamma'_{g,3} \longrightarrow \mathrm{Aut}\left(\begin{array}{c} \Pi_{①,2} \\ \| \\ \pi_1(U_{0,2}, a_{0,2}) \end{array} \right) \simeq \{\pm\}$$

This time the homomorphism (51) is not bijective — it identifies with the canonical homomorphism

$$\dagger\ (51\text{ bis})\qquad \Gamma_{g,2} \overset{\rho}{\to} \mathfrak{S}_2$$

with kernel $\Gamma^!_{g,2}$ — and $\Gamma^!_{g,\nu|2}$ is not reduced to 1, for example (if we take $U_{0,2} = \mathbb{C}^*$)

$z \longrightarrow \overline{z}^{\dagger}$ defines an element of $\Gamma^!_{g,2}$ which is not equal to 1. Let us note † now the canonical homomorphism (defined without restriction on (g,ν)) [p. 224]

Uncertainty retained from cancelled passages.

140-2-s225-D001 As
140-2-s225-D002 [illeg. 1]
140-2-s225-D003 $X_{0,2}$
140-2-s225-D004 Γ_g
140-2-s225-D005 -1
140-2-s225-D006 [illeg. 1]

[scan 226]

$$(53)\qquad \Gamma_{g,\nu} \overset{\chi}{\to} \{\pm 1\}$$

[p. 113]

m346

Marginal note. m346 This homomorphism is always surjective. We shall denote its kernel by $\Gamma^+_{g,\nu}$ (and no longer $\Gamma^0_{g,\nu}$!).

via the action of $\Gamma_{g,\nu}$ on the orientation module $T_g = T$ of $U_{g,\nu}$ † (which is defined, for $(g,\nu) \neq (0,0),(0,1)$, $(0,2)$, in terms of the loop structure of $\Pi_{g,\nu}$). In the present case, putting together ρ and $\rho|\chi$, we find a homomorphism

$$(54)\qquad g \longmapsto (\rho(g), \chi(g)) : \quad \Gamma_{g,2} \longrightarrow \mathfrak{S}_2 \times \{\pm 1\} \simeq \dagger(\pm 1) \times (\pm 1)$$

which is obviously surjective (if g corresponds to $z \longmapsto z^{-1}$, its image is $(-1,-1)|(-1,1)$, if it corresponds to $z \longmapsto \overline{z}$, its image is $(1,-1)$).

m347

Marginal note. m347 N.B. This suggests that for an isotopic description of the category of surfaces X of type $(0,2)$, we must use the pair of $\pi_1(X)$ $(\simeq \mathbb{Z})$ and its orientation module T; likewise|ditto below concerning the cases of types|the case of type $(0,1)$, but then $\pi = 0$ and T suffices.

I say that it is bijective, † $\Gamma_{g,2} \xrightarrow[\rho,\chi]{\sim} \mathfrak{S}_2 \times \pm 1$ which amounts to the same, that the restriction of ρ (51 bis) to the kernel $\Gamma^+_{0,2}$ of $\chi : \Gamma_{g,2} \longrightarrow \{\pm 1\}$ is an isomorphism, or what amounts to the same

Theorem (well known!)

$$\Gamma^{!+}_{0,2} = 1, \tag{55}$$

or again: $\rho^+_{0,2} : \Gamma^+_{0,2} \longrightarrow \mathfrak{S}_2$ is an isomorphism, or $\chi^!_{0,2} : \Gamma^!_{0,2} \xrightarrow{\sim} \pm 1$.

[N.B. If we want a statement common to the two "limiting abelian" cases $(1,0)$ and $(0,2)$, we shall say that in these cases, $\Gamma^+_{g,\nu} \longrightarrow \mathrm{Aut}(\Pi)$ [p. 225]

Uncertainty retained from cancelled passages.

140-2-s226-D001 [illeg. 1]

140-2-s226-D002 $\mathbb{Z}_2$

140-2-s226-D003 [illeg. 3]

is injective, and has as image † the group of automorphisms of the free $\mathbb{Z}$-module Π (of rank 2 or 1) which are of determinant 1.] [scan 227]

This is equivalent, modulo the isomorphisms $\Gamma'_{g,3} \simeq \Gamma_{g,2}$, to the †

Corollary

$$\begin{cases} \Gamma^+_{g|0,3} \xrightarrow{\rho^+_{0,3}} \mathfrak{S}_3 & \text{is an isomorphism, and} \\ \Gamma_{0,3} \xrightarrow{(\rho_{0,3},\chi_{0,3})} \mathfrak{S}_3 \times (\pm 1) & \text{also} \end{cases} \tag{56}$$

†

3) Case $g = 0$, $\nu = 1$ thus $H \simeq \mathbb{C}$, thus

$$A_{g,1} \overset{\text{homeomorphism}}{\simeq} \mathbb{C} \times A_{g,2} \simeq \underbrace{\mathbb{C} \times \mathbb{C}^*}_{\mathrm{Aff}(1,\mathbb{C})} \times A_{g,3} \tag{57}$$

(which is simply transitive on pairs of distinct elements) †

m348

Marginal note. m348 N.B. Since $A^0_{g,3}$ is contractible, this indeed shows again|shows that the inclusion of $G = \mathrm{Aff}(1,\mathbb{C}) = \mathrm{Aut}_{\mathbb{C}}(X) = \mathrm{Aut}_{\mathbb{C}}(X)^0$ into $A_{g,1}$ is a homotopism.

taking on $U_{g,1}$ a complex algebraic group structure $\simeq \mathbb{C}$. [p. 226]

Uncertainty retained from cancelled passages.

140-2-s227-D001 SL

140-2-s227-D002 [illeg. 1]

140-2-s227-D003 [illeg. 1] to be determined|is determined product

140-2-s227-D006 whence

We have here $\Gamma_{g,1} \simeq \Gamma'_{g,2} = \Gamma^!_{g,1|2} \xrightarrow[\sim]{\chi} (\pm 1)$ [scan 228]

i.e. [p. 114]

Corollary We have an isomorphism via the sign

$$\Gamma_{0|g,1} \xrightarrow[\sim]{\chi_{0|g,1}} \{\pm 1\} \tag{58}$$

4) Case $g = 0$, $\nu = 0$. Here there is no H i.e. no topological group structure on $X \simeq \dagger X_0 = U_{0,0} \simeq S^2$, but taking a complex structure on X (whence $X \overset{\mathbb{C}\text{-isomorphism}}{\simeq} \mathbb{P}^1_{\mathbb{C}}$) we find a group G of † $\mathbb{C}$-automorphisms $G \simeq \mathrm{GP}(1,\mathbb{C})$, and

$$G \hookrightarrow A_0 = A_{00}, \tag{59}$$

let us prove again that it is an of homotopism. Since G is simply transitive on|† the triples of 3 distinct points of X_0, we again find a homeomorphism

$$(60) \qquad G \times A^!_{0,3} \longrightarrow A_0 \qquad (g,u) \longmapsto gu$$

whence

$$(61) \qquad \Gamma_0 \simeq \Gamma^!_{03} \xrightarrow[\sim]{\chi_{0,3}} \{\pm 1\}$$

[p. 227]

Uncertainty retained from cancelled passages.

140-2-s228-D001 S^2

140-2-s228-D002 homeomorphisms

140-2-s228-D004 on

and taking into account that the identity component $A^0_{0,3}$ of $A^!_{0,3}$ is contractible, we indeed find again that (59) is an of homotopism. [scan 229]

Common conclusion to all cases. It is appropriate to include in the notion of "group with loops" also the four abelian cases, and we do so as follows:

1) Type $(1,0)$ — we do not have to supplement the general definition, which here amounts to saying that Π is an abelian group, a free $\mathbb{Z}$-module of rank 1|2. Its orientation module T can indeed be defined as $H_2(\Pi,\mathbb{Z})$, or as $\Lambda^2_{\mathbb{Z}}\Pi$. The † sign of an automorphism of Π is given|given by its action on T, it is also its determinant.

2) Type $(0,2)$, $\Pi \overset{\text{not canonical}}{\simeq} \mathbb{Z}$. Here we must give ourselves, in addition to Π, an invertible $\mathbb{Z}$-module T, and the loop structure is defined by one of the two isomorphisms $T \simeq \mathbb{Z}$. An automorphism of the loop structure is therefore defined by (any pair † u_Π, u_T of an automorphism of Π and one of T. [p. 228]

Uncertainty retained from cancelled passages.

140-2-s229-D002 action

140-2-s229-D004 [illeg. 1]

3 and 4) Types $(0,1)$ and $(0,0)$, which are those where $\Pi = \pi_1(U_{g,\nu})$ is zero. By definition, a structure of group with loops of type (g,ν) is defined here by the datum of an "orientation module" and nothing more, which is an invertible $\mathbb{Z}$-module T, the automorphisms are those of T. m349 [scan 230] [p. 115]

Marginal note. m349 and we must give it|ourselves in addition the type, i.e. (with $g = 0$ understood) the $\nu \in \{0,1\}$.

With these definitions, and for fixed g, ν, we have the

Theorem Let $(g,\nu) \in \mathbb{N} \times \mathbb{N}$. The natural functor from the isotopic category of surfaces of type (g,ν) to the category of outer groups with loops of type (g,ν), is an equivalence of|of the categories. (N.B. as arrows we take the classes of isomorphisms, on one side as on the other).

But this theorem is of course not fully satisfactory in the abelian case|the abelian cases, for example the datum of an object of the isotopic category (made explicit|made explicit by its|an outer π_1 with loops), in the case of an action of a group, does not even allow us to recover [p. 229]

the extension of this group by the said Π_1 ! [scan 231]

What is wrong, one senses it well enough, is the fact that this equivalence of categories, is 1|0-isotopic(i.e. takes into account the Π_0 of the spaces of homeomorphisms), but neglects

the internal topological structure of the topological spaces † Homéo(X, X'), which one brutally replaces by their Π_0 — neglecting the homotopy structure of the connected components, which are (as torsors under topological groups $\simeq A^\circ_{g,\nu}$) homeomorphic to $A^\circ_{g,\nu}$. It is † in the anabelian case, and in that case only, that this group is ∞-connected. In the abelian case, experience proves that, the preceding description must be replaced by Ladegaillerie's in terms † $B_{D^*} \longrightarrow B_u$ or $\Pi_{D^*} \longrightarrow \Pi_u$, † it then becomes in all cases $\neq (0,0)$ ∞-faithful (i.e. takes into account the $\Pi_i(A^\circ_{g,\nu})$). The only case [m350]

Marginal note. [m350] N.B. There is no [illeg. 1] k here — [illeg. 2] $T|J$ is described intrinsically from the structure of topoi or of groupoids

230

Uncertainty retained from cancelled passages.

140-2-s231-D001 continuous maps [illeg. 1]
140-2-s231-D002 precisely
140-2-s231-D003 [illeg. 1]
140-2-s231-D004 [illeg. 2]

entirely refractory (and of importance, it must be said !) is that of type $0, 0$, i.e. of surfaces homeomorphic to the sphere S^2 — the homotopy type of † [scan 232] [p. 116]

$$A^\circ_0 = A^+_0\dagger \simeq \mathrm{GP}(1,\mathbb{C}) \simeq \underset{\text{(quaternions)}}{S^3/\pm 1}$$

and in particular its|its $\Pi_i|\Pi_1$, not being at all well known. To begin with one would have to † make explicit the group-category $\mathcal{G}$, with invariants $\Pi_0 \xrightarrow{\chi} \{\pm 1\}$ and $\Pi_1 \xrightarrow{\sim} \Pi_1(A^\circ_0) \simeq \pm 1$, deduced from $\dagger A_0$ by killing the higher homotopy groups Π_i $(i \geq 2)$ of A_0, or if one prefers, from the group of conformal automorphisms, of the Riemann sphere (split extension of $\mathbb{Z}_2|\mathbb{Z}/2$ by $\mathrm{GP}(1,\mathbb{C})$. The 2-isotopic 2-category of surfaces homeomorphic to † $\mathrm{GP}(1,\mathbb{C})$ is then described by the 2-groupoid of 1-torsors under the preceding group-category. [m352]

Marginal note. [m352] We may assume|Observe that $\mathcal{G}$ has only two objects corresponding to the identity and complex conjugation of the Riemann sphere $\widehat{\mathbb{C}}|\overline{\mathbb{C}}$, in fact it is splittable|split and even canonically split !…

231

Uncertainty retained from cancelled passages.

140-2-s232-D001 [illeg. 1]
140-2-s232-D002 A°_0
140-2-s232-D003 find the
140-2-s232-D004 $\mathrm{GP}(1,\mathbb{C})$
140-2-s232-D005 S^2

20 2-isotopic description of the category of topological spheres [scan 233]

A digression !

Here is a way to find a 1-isotopic description (and even 2-isotopic, as it happens) by using a canonical covering group $\widetilde{A}_0$ of A_0 †, fitting into the exact sequence

$$(62) \qquad 1 \longrightarrow \underset{\|}{\mu_2(\mathbb{C})}_{\{\pm 1\}} \longrightarrow \widetilde{A}_0 \longrightarrow A_0 \longrightarrow 1,$$

which contains the † corresponding exact sequence of subgroups

$$(63) \qquad 1 \longrightarrow \mu_2 \longrightarrow \widetilde{\mathrm{Sl}}(2,\mathbb{C}) \longrightarrow \widetilde{\mathrm{GP}}(1,\mathbb{C}) \longrightarrow 1$$

where $\widetilde{\mathrm{Sl}}(2,\mathbb{C})$ is made up of the $\frac{1}{2}$-linear automorphisms, of $\mathbb{C}^2$ (for an unspecified $\mathbb{R}$-automorphism ρ, id or complex conjugation τ), such that the corresponding ρ-linear automorphism of $\bigwedge^2 \mathbb{C}^2 \simeq \mathbb{C}$ is the identity if $\rho = \mathrm{id}$, and is $\tau : z \longrightarrow \bar{z}$ if $\rho = \tau$ (i.e., independently of

the|of the case, †one considers a unimodular vector space V over $\mathbb{R}$, whence $V_{\mathbb{C}}$ over $\mathbb{C}$, with a restriction of $\bigwedge^2 V_{\mathbb{C}}$ to $\mathbb{R}$, whence|thus a complex conjugation on $\bigwedge^2 V_{\mathbb{C}}$ — and a † basis of $\bigwedge^2 V_{\mathbb{C}}$ invariant under this — and one is interested in the $\rho|\mathbb{R}$-linear ($\rho \in \mathrm{Aut}_{\mathbb{R}}\mathbb{C}$) automorphisms of $V_{\mathbb{C}}$ which on $\bigwedge^2 V_{\mathbb{C}}$ are id or complex conjugation …) 232

Uncertainty retained from cancelled passages.
140-2-s233-D001 of
140-2-s233-D002 [illeg. 1]
140-2-s233-D003 [illeg. 1]
140-2-s233-D004 of the

$\widetilde{\mathrm{GP}}(1,\mathbb{C})$ identifies with the group of conformal automorphisms † of the Riemann sphere † $\mathbb{P}^1_{\mathbb{C}} \simeq \mathbb{P}^1(\mathbb{C}^2)$, or automorphisms of $\mathbb{P}^1_{\mathbb{C}}$ as an $\mathbb{R}$-scheme. We choose here|let act on $X_0 = \mathbb{P}^1_{\mathbb{C}}$, so that we have a canonical inclusion [scan 234] [p. 117]

$$\widetilde{\mathrm{GP}}(1,\mathbb{C}) \longrightarrow A_0 \qquad (64)$$

which is (by what precedes (59)) is a homotopy equivalence. Unless I am mistaken it follows (unless I am mistaken) that the classification of extensions of the topological group A_0 by a group discrete|say μ is equivalent, by the restriction functor, to the category of corresponding extensions for † $\mathrm{GP}(1,\mathbb{C})$, which makes it possible to construct $\widetilde{A}_0$, up to unique isomorphism. This being so, the datum of a surface compact oriented X of genus 0, which is equivalent to that of a torsor $[\mathrm{Isom}(\mathbb{P}^1_{\mathbb{C}}, X)|(\mathrm{Isom}(\mathbb{P}^1_{\mathbb{C}}, X))$ under A_0, defines 1°) the torsor under (± 1) deduced from it by $\chi : A_0 \longrightarrow \{\pm 1\}$ (which is surjective, with kernel A_0°) and φ 2°) of the [m356]

Marginal note. [m356] N.B. $\widetilde{\mathrm{GP}}(1,\mathbb{C})$ is also the group of $\mathbb{R}$-linear automorphisms of the $\mathbb{R}$-algebra $M_2(\mathbb{C})$ † (and the ∞-isotopic classification of 2-spheres is therefore equivalent to that of simple algebras of rank 4 over † a not specified quadratic extension of $\mathbb{C}$…) 233

Uncertainty retained from cancelled passages.
140-2-s234-D001 [illeg. 1] not here
140-2-s234-D002 [illeg. 1]
140-2-s234-D003 $\widetilde{A}_0$
140-2-s234-D004 [illeg. 3]
140-2-s234-D005 [illeg. 1]

groupoid of liftings of this †A_0 torsor to an $\widetilde{A}_0$-torsor, which is a connected groupoid (= gerbe) banded by the abelian band $\underbrace{\mu_2(\mathbb{C}) = (\pm 1)}_{\mu}$, and in which consequently $\mathrm{Tors}(\mu) \simeq \mathrm{Ens}_2$ operates [scan 235]

(it is a 1-torsor under the group-category $\mathrm{Tors}(\mu)$). Thus associating to every X the pair $(\omega, \mathcal{R})$ of the associated μ-torsor ω (i.e. the 2-element set of the two orientations of X, or what amounts to the same, the module of orientations of X), plus the μ-groupoid $\mathcal{R}$, one finds a 2-functor from the 2-isotopic 2-category of topological 2-spheres, to the 2-category of pairs $(\omega, \mathcal{R})$, and this is an equivalence of 2-categories (from this point of view one feels like saying that it is 3-faithful, but since † essential surjectivity on objects is trivial, the name 2-faithful is better, and even (since in describing it we took care to preserve, 234

Uncertainty retained from cancelled passages.
140-2-s235-D001 functor
140-2-s235-D002 [illeg. 1]

the Π_1 (in addition to the Π_0) of the connected components of the (A_0-torsors) $\mathrm{Isom}(X_0, X)$, which are A_0°-torsors thus with Π_1 Isomorphic|isomorphic to $\mu = \{\pm 1\}$, it is even better to call it 1-faithful — but in fact it is even 2-faithful (in the corresponding sense) thanks to the fact that [scan 236] [p. 118]

$$(65)\quad \Pi_2(A_0^\circ) \xleftarrow{\sim} \Pi_2(\mathrm{GP}(1,\mathbb{C})) \xleftarrow{\sim} \Pi_2(\dagger S^3/\pm 1) \qquad \Pi_2(S^3) = 0$$

On the other hand it is not 3-faithful, for

$$(66)\qquad \Pi_i(A_0^\circ) \xleftarrow{\sim} \Pi_i(S^3) \qquad \text{for } i \geq 2$$

and $\Pi_3(S^3) \simeq \mathbb{Z} \neq 0$ hence $\Pi_3(A_0^\circ) \simeq \mathbb{Z} \neq 0$,

We are looking for an ∞-isotopic description, which takes into account the homotopy groups of all orders (to be determined !) of A_0° i.e. of S^3 (or again, of $\mathrm{Sl}(2,\mathbb{C})$, or of $\mathrm{GP}(1,\mathbb{C})$. Up for competition ! 235

Uncertainty retained from cancelled passages.

140-2-s236-D001 $\mathrm{Sl}(2,\mathbb{C})$

But already for the modest description proposed, with pretensions to being 1 or 2-isotopic, for lack of having written things carefully, I am not too sure whether the description given is really correct — I am a little worried by the fact that I have not †imposed any relations between the μ-torsor ω, and the μ-groupoid $\mathscr{R}$. More generally let G be a topological group ($\widetilde{A}_0$, or $\widetilde{\mathrm{GP}}(1,\mathbb{C})$), [scan 237] [p. 119]

topological group G

such that $\dagger G^\circ$ is simply connected; †

$$(67)\qquad \Pi_1(G^\circ) = 0$$

let

$$(68)\qquad \mu = \mathrm{Centre}(G^\circ), \qquad \Gamma = G/G^\circ$$

m360

Marginal note. m360 we assume μ discrete (thus G° identifies with the universal covering group of $G^\circ/\mu = \mathscr{H}^\circ$ (if $\mathscr{H} = G/\mu$))

then the topological group extension G of Γ by G° defines

$$(69)\qquad \Gamma \longrightarrow \mathrm{Aut}\,\mathrm{ext}(G^\circ) \qquad (\longrightarrow \mathrm{Aut}(\mu).)$$

and the set of classes of extensions corresponding to a given outer operation of Γ on G° is in a natural way a torsor under $H^2(\Gamma,\mu)$ (if it is not empty, which is|we have excluded by the starting hypothesis, starting from G…) 236

Uncertainty retained from cancelled passages.

140-2-s237-D001 arranged

140-2-s237-D003 $\widetilde{A}_0^\circ$

140-2-s237-D005 set $\widetilde{A}_0^\circ$

This category of extensions is moreover equivalent † to that of splittings of a certain group-category defined by Sinh via (69), whose Π_0 and Π_1 are respectively †Γ and μ — which is thus here split by the datum of the extension G. (In the case that interests us, $\Gamma \simeq \mu \simeq \{\pm 1\}$, and this extension is even split, and corresponds to an operation of order 2 of Γ on G°, which I very [scan 238]

much doubt is an inner automorphism. In fact, I have the impression that in the two cases we are dealing with ($\widetilde{\mathrm{GP}}(1,\mathbb{C})$ and $\widetilde{A}_0$) that (69) is an isomorphism $\Gamma = \{\pm 1\} \xrightarrow{\sim} \mathrm{Autext}(G^\circ$

This being so, the datum of a torsor under $\mathcal{H} = G/\mu$ defines 1°) a torsor under Γ, $\mathcal{H} \longrightarrow \Gamma \simeq \mathcal{H}/\mathcal{H}^\circ$; 2°) the groupoid of liftings of liftings of this torsor to a torsor under G, which is a connected μ-groupoid (on which μ operates). m362

Marginal note. m362 We assume that Γ operates trivially on μ i.e. $\mu \subset \mathrm{Centre}(G)$

If we † take this up again in terms of bundles over a base space S we find 237

Uncertainty retained from cancelled passages.
140-2-s238-D001 corresponds
140-2-s238-D002 Π_0
140-2-s238-D003 exp

again on S (for every topological $\mathcal{H}_S$-torsor) a pair $(\omega, \mathcal{R}|R)$ of a Γ-torsor, and of a μ-gerbe on S, which are described (up to isomorphism [illeg. 1] equivalence up to equivalence|and equivalence) by $H^1(S,\Gamma)$ and $H^2(S,\mu)$ respectively. We would like to isolate conditions on G and on S for the map [scan 239] [p. 120]

$$(70) \qquad H^1(S,\mathcal{H}) \longrightarrow H^1(S,\Gamma) \times H^2(S,\mu)$$

to be bijective. m364

Marginal note. m364 N.B. ω does not depend on the choice of the extension $\widetilde{G}$ of $\mathcal{H}$ by $\mathfrak{z}|\zeta \simeq \Pi_1(\mathcal{H}^\circ)$; on the other hand $\mathcal{R}$ depends on it (and one would have to see how)

Injectivity : If the image of a $\zeta \in H^1(S,\mathcal{H})$, in $H^2(S,\mu)$ is zero, the element of $H^1(S,\mathcal{H})$ lifts to an element $\tilde{\zeta}$ of $H^1(S,\widetilde{\mathcal{H}} = G)$; whose image in $H^1(S,\Gamma)$ under $G \longrightarrow \Gamma$ is the same as that of ζ, so if it is trivial, one sees that $\tilde{\zeta}$ comes from a $\tilde{\zeta}_0 \in H^1(S,G^\circ)$. If we know that $H^1(S,G^\circ) = (1)$, we win. By the sledgehammers of homotopy, it works if $\exists n \in \mathbb{N}$ with

$$\Pi_i(G^\circ) = 0 \quad \text{if} \quad i \leq n$$

(thus $\Pi_i(B_{G^\circ}) = 0$ if $i \leq n+1$) S a CW-complex of dimension $\leq n+1$ m365

Marginal note. m365 in the case we are dealing with, one can take $n = 2$

For surjectivity, note that for|first an 238

element of $H^2(S,\mu)$, the obstruction to its lifting to an element ζ of $H^1(S,\mathcal{H})$ is in $H^2(S,G)$ [by the associated exact sequence of cohomology : $1 \longrightarrow \mu_S \longrightarrow G_S \longrightarrow \mathcal{H}_S \longrightarrow 1$] one must express that a certain † gerbe banded by G_S is neutral — brr!. But let us rather start from the element $\omega|\tilde{\zeta}$ of $H^1(S,\Gamma)$, if the extension of Γ by $\dagger\mathcal{H}^\circ$ is split, then one can find a $\zeta_0 \in H^1(S,\mathcal{H}_S)$ which gives rise to ω. It has a certain obstruction ρ_0 in $H^2(S,\mu)$, and it is a matter of correcting ζ_0 to ζ, in such a way that the obstruction becomes the given $\rho \in H^2(S,\mu)$. Using ζ_0 to twist $\mathcal{H}_S$ into $\mathcal{H}'_S$, and (via the operation of $\mathcal{H}_S = G_S/\mu_S$ on G_S, taking into account that $\mu_S \subset$ centre G_S) to twist G_S also into G'_S, whence [scan 240]

$$1 \longrightarrow \mu \longrightarrow G'_S \longrightarrow \mathcal{H}'_S \longrightarrow 1$$

one finds that the ζ having the same image in $H^1(S,\Gamma)$ 239

Uncertainty retained from cancelled passages.
140-2-s240-D001 [illeg. 1]
140-2-s240-D002 G°

that ξ_0 correspond $1-1$ to the $\xi' \in H^1(S, \mathcal{H}'_S)$. [scan 241]

For such a † ξ', let $\delta'(\xi') \in H^2(S,\mu)$ be the obstruction to lifting it to $H^1(S, \mathcal{G}'_S|\mathcal{G}_S)$, and [p. 121]
$\delta(\xi') = \delta(\xi)$ the obstruction to lifting it to $H^1(S, \mathcal{G}_S)$. Unless I am mistaken one has

$$\delta'(\xi') = \delta(\xi') - \rho_0$$

i.e.

$$\delta(\xi') = \delta(\xi) = \delta'(\xi') + \rho_0$$

and one wants $\delta(\xi) = \rho$ i.e.

$$\dagger \quad \delta'(\xi') = \rho - \rho_0$$

and the question again comes down to † the surjectivity of $H^1(S, \mathcal{H}'_S) \longrightarrow H^2(S,\mu)$ † — I am not getting anywhere. One ought to consult competent people, like Giraud or Larry Breen. One feels that one ought to work with a Gr-stack $\underline{\underline{\mathcal{H}}}$ of coefficients, with invariants $\underline{\pi}_0 \simeq \Gamma$ and $\underline{\pi}_1 \simeq \mu$, (but not necessarily a Picard stack!) and $H^1(S, \underline{\underline{\mathcal{H}}}) \simeq$ Classes of maps from S to $B\underline{\underline{\mathcal{H}}}$, which is a connected space with $\pi_1(B\underline{\underline{\mathcal{H}}}) \simeq \Gamma$, $\pi_2(B\underline{\underline{\mathcal{H}}}) = \mu$. Here the Postnikoff class in $H^3(\Gamma,\mu)$ is zero, 240

Uncertainty retained from cancelled passages.
140-2-s241-D001 $\mathcal{H}'_S$
140-2-s241-D002 $\delta(\xi')$
140-2-s241-D003 [illeg. 3]
140-2-s241-D004 [illeg. 1]

[but perhaps there is no reason to cling too much to this hypothesis, corresponding to the existence of an extension $\mathcal{G}$ of $\mathcal{H}$ by $\pi_1(\mathcal{H}^\circ)$ † which gives back the universal extension of $\mathcal{H}^\circ$ by $\pi_1(\mathcal{H}^\circ)$] one has a homomorphism $B\mathcal{H}$ † $\longrightarrow B\underline{\underline{\mathcal{H}}}$ which induces an isomorphism for the π_i for $i \leq 2$, and for $i \leq n$ (where $n \geq 2$ is given) iff $\pi_i(B\mathcal{H}) = 0$ for $2 < i \leq n$, i.e. $\pi_i(\mathcal{H}) = 0$ for $2 \leq i \leq n-1$ [in the case we are interested in $\mathcal{H} = A_0$, one can take $n = 3$). This implies that for every C-W-complex S of dimension $\leq n$, one has (Hot denoting the morphisms in the unpointed homotopy category) [scan 242]

$$\mathrm{Hot}(S, B\mathcal{H}) \xrightarrow{\sim} \mathrm{Hot}(S, B\underline{\underline{\mathcal{H}}})$$

i.e.|i.e.

$$H^1(S,\mathcal{H}) \xrightarrow{\simeq} H^1(S, \dagger\underline{\underline{\mathcal{H}}})$$

But if the Postnikoff-Sinh invariant $K|k \in H^3(\Gamma,\mu)$ is zero, † (as well as the action of Γ on μ) then unless I am mistaken $B\underline{\underline{\mathcal{H}}}$ is identified with a product $K(\Gamma,1) \times K(\pi,2)$, (this "identification" depending precisely on the 241

Uncertainty retained from cancelled passages.
140-2-s242-D001 [illeg. 1]
140-2-s242-D002 [illeg. 1]
140-2-s242-D003 B
140-2-s242-D004 [illeg. 1]

choice of the $H^2(\pi_1,\pi_2) \simeq H^2(\Gamma,\mu)$!), and hence [scan 243]
[p. 122]

$$H^1(S, \underline{\underline{\mathcal{H}}}) \xrightarrow{\simeq} H^1(S,\Gamma) \times H^2(S,\mu)$$

for every space S, hence for S a C-W-complex of dimension $\leq n$

$$(71) \qquad H^1(S,\mathcal{H}) \xrightarrow{\simeq} H^1(S,\Gamma) \times H^2(S,\mu)$$

So the classification of bundles of topological spheres over a topological space S, for a C-W complex S of dimension ≤ 3, really does work. Of course, the fact that one is obliged here to restrict oneself to † S of dimension ≤ 3 is due to the fact that we have found † (via $\underline{\underline{\mathcal{H}}}$) only a 2-isotopic (and not ∞-isotopic) description of the category of topological 2-spheres.

I would like to take up the classification again for an arbitrary CW-complex S, using the fact that the inclusion

$$\widetilde{\mathrm{GP}}(1,\mathbb{C}) \hookrightarrow A_0$$

242

Uncertainty retained from cancelled passages.
140-2-s243-D001 [illeg. 1]
140-2-s243-D002 only

is a homotopy equivalence, and therefore induces a bijection [scan 244]

$$H^1(S,\widetilde{\mathrm{GP}}(1,\mathbb{C})) \longrightarrow H^1(S,A_0)$$

and even an ∞-equivalence of the ∞-categories of torsors over S, with † groups $\widetilde{\mathrm{GP}}$ (corresponding to bundles of conformal spheres, or of projective lines over an $\mathbb{R}$-algebra (unspecified) isomorphic to $\mathbb{C}$) and with group A_0 — corresponding to sphere bundles over S.

The first invariant of a bundle with group $\widetilde{\mathrm{GP}}$ is a torsor $\eta|T$ under $(\mathbb{Z}/2\mathbb{Z})_S$ † defined (up to isomorphism not unique) by a χ ($\in H^1(S,\mathbb{Z}/2\mathbb{Z})$). [m371]

Marginal note. [m371] i.e. an étale covering|double covering of degree 2. The datum of $\eta|\chi$ amounts to the datum of a system T of twisted integers over S.

This torsor will serve to twist $\mathbb{C}$ (via the faithful operation of $\mathbb{Z}/2\mathbb{Z}$ on the $\mathbb{R}$-extension $\mathbb{C}$), whence a locally † constant bundle of quadratic extensions of $\mathbb{R}$ †, say C, and one will enrich S with the sheaf $\underline{C}_S$ (or $\underline{C}$) of continuous sections of C (which is a way of twisting $\mathrm{GP}(1,\mathbb{C})_S$ into $\mathrm{GP}(1,C)_S = \mathrm{GP}(1,\underline{C})$) and the point is to † describe in an understandable way — passing if need be, to the †n-isotopic ones, for $n = \ldots$ — the category 243

Uncertainty retained from cancelled passages.
140-2-s244-D002 [illeg. 1]
140-2-s244-D005 [illeg. 2]
140-2-s244-D007 [illeg. 1]
140-2-s244-D009 [illeg. 1]
140-2-s244-D012 therefore
140-2-s244-D014 ∞

— depending on the choice of χ via $\underline{C}$ — of Azumaya algebras of rank 4 over $\underline{C}$, or again of [scan 245] bundles of projective lines over $\underline{C}$, or of torsors under $\mathrm{GP}(1,\underline{C})$. [p. 123]

Now using the exact sequence

$$1 \longrightarrow \mu_{2S} \longrightarrow \mathrm{Sl}(2,\underline{C}_S) \longrightarrow \mathrm{GP}(1,\underline{C}_S) \longrightarrow 1$$

one associates to an object of the category — say a torsor under $\mathrm{GP}(1,\underline{C}_S)$ — the fibered (over variable open subsets of S) category of these liftings to $\mathrm{Sl}(2,\underline{C}_S)$, which is a gerbe $\mathcal{G}$ banded by μ_2. It is this gerbe that is the second complete invariant — finer than its class up to equivalence, which is identified with a $\xi \in H^2(S,\mu_2) = H^2(S,\{\pm 1\})$ (playing the role of a Brauer group)

An isomorphism of bundles, with invariants $(T,\mathcal{G})$, $(T',\mathcal{G}')$, will define an isomorphism $T \simeq T'$ (whence an isomorphism $\underline{C}_T \simeq \underline{C}_{T'}$ whence an isomorphism $\mathrm{GP}(1,\underline{C}_T) \simeq \mathrm{GP}(1,\underline{C}_{T'})$) — and an equivalence of gerbes — one should 244

spell out that for two isomorphisms f,g of the torsors, whence [scan 246]

$$f_T, g_T : T \overset{\simeq}{\underset{\sim}{\overset{\dagger}{\rightrightarrows}}} T' \qquad\qquad f_{\mathcal{G}}, g_{\mathcal{G}} : \mathcal{G} \overset{\simeq}{\underset{\simeq}{\rightrightarrows}} \mathcal{G}'$$

and for every homotopy h_t $(0 \leq t \leq 1)$ from f to g, one finds $f_T = g_T$ and an isomorphism $h_* : f_{\mathcal{G}} \overset{\sim}{\to} g_{\mathcal{G}}$ of equivalences of gerbes, which depends only on the homotopy class of this homotopy. One thus finds a 2-functor, from the 2-category of torsors over S with group $\widetilde{\mathrm{GP}}(1,\mathbb{C})$, to the 2-category formed by pairs $(T, \mathcal{G})$ of a system of twisted integers, ($\Longleftrightarrow$ of a torsor over S with group † $\mathbb{Z}/2$) and of a † $\mu_2(\mathbb{C})$-gerbe $\mathcal{G}$ over S. This 2-functor, unless I am mistaken, is 2-faithful under the conditions spelled out above († (dimension $S \leq 4$), and is 3-faithful (i.e. the injective map $H^1(S, \widetilde{\mathrm{GP}}) \longrightarrow |- H^1(S, \mathbb{Z}/2\mathbb{Z}) \times H^2(S, \mu_2)$ is surjective) 245

Uncertainty retained from cancelled passages.

140-2-s246-D002 $\overset{\sim}{\to}$

140-2-s246-D006 [illeg. 1]

140-2-s246-D007 $\{\pm 1\} =$

140-2-s246-D008 [illeg. 2]

if one even has dimension $S \leq 3$. One expects it to be 1-faithful as soon as dimension $S \leq 5$, 0-faithful as soon as dimension $S \leq 6$. [N.B. the assertion "0-faithful" means that if above one has two homotopies h, h' from f to g, such that $h_* = h'_*$, then h and h' are homotopic. The condition "1-faithful" means 0-faithful and that every isomorphism from $f_{\mathcal{G}}$ to $g_{\mathcal{G}}$ † is of the form h_* (with h well determined up to homotopy, by the preceding condition of 0-faithfulness). The 2-faithful condition means that in addition, for every † $\varphi : T \simeq T'$, $\psi_{\mathcal{G}} : \mathcal{G} \overset{\approx}{\to} \mathcal{G}'$, there exists an isomorphism of bundles f such that $\varphi = f_T$, † and an isomorphism $\psi \simeq f_{\mathcal{G}}$. Finally 3-faithful means that in addition, for every pair $T, \mathcal{G}$, $\exists\, T', \mathcal{G}'$ coming from a bundle, and an isomorphism $T \simeq T'$ and an equivalence $\mathcal{G} \approx \mathcal{G}'$. These categorical notions should simply interpret that, if one looks at $\mathcal{H} = \widetilde{\mathrm{GP}}(1,\mathbb{C}) \overset{\alpha}{\to} \underline{\underline{\mathcal{H}}}$ — where $\underline{\underline{\mathcal{H}}}$ is deduced from $\mathcal{H}$ by "killing the homotopy gr°ups π_i for $i \geq 2$", so that [scan 247] [p. 124] ?? 246

Uncertainty retained from cancelled passages.

140-2-s247-D001 comes from a

140-2-s247-D003 given

140-2-s247-D004 [illeg. 1]

α induces an isomorphism of the π_i for $i \leq 1$ (and even for $i \leq n$ if $\pi_i(\mathcal{H}) = 0$ for $1 < i \leq n$) whence [scan 248]

$$B_\alpha : B\mathcal{H} \longrightarrow B\underline{\underline{\mathcal{H}}}$$

m377

Marginal note. m377 N.B. $B\underline{\underline{\mathcal{H}}}$ is deduced from $B\mathcal{H}$ by killing the π_i for $i \geq 3$.

(inducing an isomorphism of the π_i for $i \leq n+1$), then taking for a given space S the corresponding map of spaces of continuous maps

$$\underline{\mathrm{Cont}}(S, B\mathcal{H}) \longrightarrow |- \underline{\mathrm{Cont}}(S, B\underline{\underline{\mathcal{H}}})$$

this induces an isomorphism for the π_i $(0 \leq i \leq 2)$. † It would seem that 0-faithfulness is a condition of injectivity for the π_2, 1-faithful bijectivity for the π_2 and injectivity for the π_1, 2-faithful bijectivity for π_2 and π_1 and injectivity for π_0, 3-faithful bijectivity for π_0, π_1, π_2.

Since $\pi_i(\mathrm{Cont}(S, B)) \simeq \pi_0(S \times S^i, B)$, it would seem that one is led, for 0-faithfulness, to make the draconian hypothesis that $S \times S^2$ of dimension ≤ 4, for 1-faithfulness that $S \times S^2$ of dimension ≤ 3 i.e. S of dimension ≤ 1 ? (This|That would then imply 3-faithfulness…) 247

Uncertainty retained from cancelled passages.
140-2-s248-D004 Since

Let us take the arithmetic analogue, of a description (more or less "faithful") † of a "profinite" nature of the projective lines (possibly twisted) defined over a finitely generated field K over $\mathbb{Q}$ (more generally over an arbitrary scheme S), these a priori form a category and nothing more — a groupoid (not necessarily connected) I do not know how to make a 2-category out of them reasonably, because for two isomorphisms f, g $(= f \circ u)$ of such torsors, to define the homotopisms from f to g, i.e. for a "form" G of $\mathrm{GP}(1)_S$ and a section u of G, to define the "homotopies" from u to the identity, † or rather a notion which replaces the notion of a homotopy class of such homotopies. This would be probably a lifting of u to a † section of the † simply connected covering $\widetilde{G}$ of G ! [scan 249] [p. 125] 248

Uncertainty retained from cancelled passages.
140-2-s249-D001 [illeg. 1]
140-2-s249-D003 [illeg. 1]
140-2-s249-D005 If [illeg. 1] [illeg. 1] every section [illeg. 1] if K is algebraic at least sections [illeg. 1]
140-2-s249-D006 $\widetilde{\mathrm{GP}}$

I have something rather amusing here, but which I am not going to pursue — in any case|any case it is clear that the "description" of the projective lines (over a finitely generated field, say) that one ends up with, has nothing faithful about it — not even 0-faithful ! Thus all the automorphisms of $\mathbb{P}^1_K$ coming from $\mathrm{Sl}(2, K)$ will be identified with the identity — that is a bit brutal ! But I realize that the conceptual work around the "Brauer" themes is not finished — that there are things to be understood for the study of the category (which should be a 2-category !) of Azumaya Algebras † fixed degree (here 4)… [scan 250] 249

Uncertainty retained from cancelled passages.
140-2-s250-D001 of

[scan 251] [p. 126]

21 Connection with Teichmüller spaces.

First a supplement connected with the diagram of inclusions (31) of number 19, in which $A_{g,(\nu,\nu+1)}$ deserves the simpler notation $A^{\bullet}_{g,\nu}$ (group of automorphisms of $U_{g,\nu}$ pointed at $a_{g,\nu}$). Consider the inclusion|the filtration of subgroups of A_g

$$A^{\circ}_{g,\nu} \supset B_{g,\nu} \supset A^{\circ}_{g,\nu+1} \tag{72}$$

†

giving, on dividing by $A_{g,\nu+1}$ and in the anabelian case $2g + \nu \geq 3$, the fibration (73)

$$\begin{array}{ccc} \widetilde{U}_{g,\nu} & \longrightarrow & U_{g,\nu} \\ & \simeq & \\ A^{\circ}_{g,\nu}/A^{\circ}_{g,\nu+1} & & \end{array}$$

We seek the largest subgroup of A_g that normalizes these|the triple (72), and which † operates therefore † the fibration $\widetilde{U}_{g,\nu} \to U_{g,\nu}$ (where $\widetilde{U}_{g,\nu}$ is the universal covering of $U_{g,\nu}$ based at $a_{g,\nu}$); since the normalizers of $A^{\circ}_{g,\nu}$ and $A^{\circ}_{g,\nu+1}$ respectively are $A_{g,\nu}$ and $A_{g,\nu+1}$, the group in question must be contained in their intersection $A^{\bullet}_{g,\nu}$, which normalizes † $B_{g,\nu} = A^{\bullet}_{g,\nu} \cap A^{\circ}_{g,\nu}$. It is therefore this group that should be made to operate on the fibration (73) – the subgroup formed † of the elements of $A_{g,\nu}$ whose operation

Uncertainty retained from cancelled passages.
140-2-s251-D001 $A^{\bullet}_{g,\nu}$
140-2-s251-D002 defines
140-2-s251-D003 on
140-2-s251-D004 also
140-2-s251-D005 [illeg. 7]

[scan 252]

on $U_{g,\nu}$ is isotopic to the identity being precisely $B_{g,\nu}$ – so it is the group Γ

$$A^{\bullet}_{g,\nu}/B_{g,\nu} \simeq \Gamma_{g,\nu}$$

that, as it should, operates up to isotopy. We would like to describe the group of all automorphisms of the topological fibration $\widetilde{U}_{g,\nu}$ over $U_{g,\nu}$ which will a priori be an extension of the group

$$\begin{aligned} A_{g,\nu} = \mathrm{Aut}(U_{g,\nu}) \quad \text{by} \quad \Pi_{g,\nu} &= \pi_1(U_{g,\nu}|\widetilde{U}_{g,\nu}, a_{g,\nu}) \\ &= \mathrm{Aut}_{U_{g,\nu}}(\widetilde{U}_{g,\nu}). \end{aligned}$$

† This extension is split canonical[ly] † over $A^{\bullet}_{g,\nu}$, stabilizer at the base point $a_{g,\nu}$ (and we thus recover the operation of $A^{\bullet}_{g,\nu}$ for the fibration (73). Now we already have an extension $\Gamma'_{g,\nu+1}$ of $\Gamma_{g,\nu}$ by $\Pi_{g,\nu}$, whence by inverse image under $A_{g,\nu} \to \Gamma_{g,\nu} = A_{g,\nu}/A^{\circ}_{g,\nu}$, an extension (which I am going to denote by $\widetilde{A}_{g,\nu}$) of $A_{g,\nu}$ by $\Pi_{g,\nu}$:

$$\widetilde{A}_{g,\nu} \simeq A_{g,\nu} \times_{\Gamma_{g,\nu}} \Gamma'_{g,\nu+1} \tag{74}$$

= quotient group of the subgroup † $\widetilde{A}^{\sharp|\sharp}_{g,\nu}$ of $A_{g,\nu} \times A^{\bullet}_{g,\nu}$ formed of the pairs (u, v), $u \equiv v$ † modulo $A^{\circ}_{g,\nu}$, by the subgroup † $1 \times A^{\circ}_{g,\nu+1}$.

We thus have two (and = three!) extension structures on $\widetilde{A}_{g,\nu}$
(75)

$$1 \to \Pi_{g,\nu} \to \widetilde{A}_{g,\nu} \xrightarrow{p} A_{g,\nu} \longrightarrow 1$$

$$1 \to A^{\circ}_{g,\nu} \to \widetilde{A}_{g,\nu} \to \Gamma'_{g,\nu+1} \to 1$$

$$1 \to (A^{\circ}_{g,\nu} \times | \rtimes \Pi_{g,\nu}) \to \widetilde{A}_{g,\nu} \to \Gamma_{g,\nu} \to 1$$

m381

Marginal note. m381 N.B. The second is meant to express that $\widetilde{A}^{\circ}_{g,\nu} = A^{\circ}_{g,\nu}$ and $\pi_0(\widetilde{A}_{g,\nu}) \simeq \Gamma'_{g,\nu+1}$ (the canonical extension of $\Gamma_{g,\nu}$ by $\Pi_{g,\nu}$).

†

Uncertainty retained from cancelled passages.
140-2-s252-D001 [illeg. 1]
140-2-s252-D002 split
140-2-s252-D003 H
140-2-s252-D004 $A^{\circ}_{g,\nu}$
140-2-s252-D005 $B^{\circ}_{g,\nu}$
140-2-s252-D006 Now

[scan 253]

I would like to make $\widetilde{A}_{g,\nu}$ operate on $\widetilde{U}_{g,\nu}$, i.e. make † $H = \widetilde{A}^{\natural|\sharp}_{g,\nu}$ operate with operation trivial on $A^{\circ}_{g,\nu+1}$, and this respecting the following conditions [p. 127]

m382

Marginal note. m382 trivial datum on $1 \times A^{\circ}_{g,\nu+1}$. Cf. generalization below in the margin.

a) † (compatibility with $\widetilde{A}_{g,\nu} \to A_{g,\nu}$) the pair $(u, v) \in H$ operates on $\widetilde{U}_{g,\nu}$ by an automorphism compatible with the automorphism u of $U_{g,\nu}$ (we shall say that it is a u-automorphism).

b) (compatibility with $\Pi_{g,\nu} \to \widetilde{A}_{g,\nu}$) If $u = 1$, [so $v \in A^{\circ}_{g,\nu}$, so (since $v \in A^{\bullet}_{g,\nu}$) $v \in B_{g,\nu}$] then the operation of $(u, v) = (1, v)$ on $\widetilde{U}_{g,\nu}$ is none other than that defined by $v \bmod B^{\circ}_{g,\nu} = A^{\circ}_{g,\nu+1}$, which is an element of $\Pi_{g,\nu} \simeq \mathrm{Aut}(\widetilde{U}_{g,\nu}/U_{g,\nu})$, or again, it is that defined by translation on the right by v^{-1}.

m383

Marginal note. m383 specification of the operation on $1 \times B_{g,\nu}$.

c) Compatibility with † the operation already † obtained above of $A^{\bullet}_{g,\nu}$ on $\widetilde{U}_{g,\nu}$: if $(u, v) \in H$ is such that $u = v$ (so $u = v \in A^{\bullet}_{g,\nu}$) then $(u, v) = (u, u)$ operates via the inner automorphism defined by u.

m384

Marginal note. m384 specification of the operation on $\delta(A^{\bullet}_{g,\nu})$ (δ: diagonal arrow).

d) Compatibility with $A^{\circ}_{g,\nu} \to \widetilde{A}_{g,\nu}$: The operation of $A^{\circ}_{g,\nu}$ on $\widetilde{U}_{g,\nu}$ is continuous (which, together with a), determines this operation uniquely † and the existence a priori of such an operation (follows from $\pi_1(A^{\circ}_{g,\nu}) = 0$)).

m385

Marginal note. m385 specification of the operation on $(A^{\circ}_{g,\nu}) \times 1$.

every element (u, v) of H can be written uniquely as a product of an element (v, v) of $\delta(A^{\bullet}_{g,\nu})$, with an element $(v^{-1}u, 1)$ of $A^{\circ}_{g,\nu} \times 1$ (the first group normalizing the second)

m386

Marginal note. m386 N.B. The operation of $A^{\circ}_{g,\nu} \times 1$ is described most simply by translations of $A^{\circ}_{g,\nu}$ on the homogeneous space $\widetilde{U}_{g,\nu}$ of $A^{\circ}_{g,\nu}$.

Uncertainty retained from cancelled passages.

140-2-s253-D001 [illeg. 1]
140-2-s253-D003 [illeg. 1]
140-2-s253-D004 $\widetilde{A}_{g,\nu} \to \Gamma'_{g,\nu+1}$
140-2-s253-D005 [illeg. 1]
140-2-s253-D006 [illeg. 2]

[scan 254]

and an operation of this $\frac{1}{2}$ direct product is given indeed by † specifying operations of the two factor groups, satisfying a compatibility condition, which is verified here by transport of structure. We have indeed defined an operation of $H = \widetilde{A}^{\natural|\sharp}_{g,\nu}$ on the fibration (73), † satisfying conditions c) and d) by the very construction, it remains to verify a) and b).

Now for a), it suffices to verify it separately for the elements of H in $\delta(A^{\bullet}_{g,\nu})$ and in $A^{\circ}_{g,\nu} \times 1$, where this is trivial by construction in both cases. It remains to verify b), i.e. to make explicit the operation of an element $(1, v)$, $v \in B_{g,\nu}$, which we write as

$$(\qquad (1, v) = \dagger(v, v)(v^{-1}, 1)$$

hence

$$\rho\dagger(1, v) = \rho(v, v)\rho(v^{-1}, 1).$$

Now $\rho(v, v)$ is induced by $\mathrm{int}(v)$, i.e. $x \mapsto vxv^{-1}$ and $\rho(v^{-1}, 1)$ by left translation: $x \mapsto v^{-1}x$, so the composite operates by $x \mapsto xv^{-1}$. Phew!

Uncertainty retained from cancelled passages.

140-2-s254-D001 [illeg. 1]
140-2-s254-D002 [illeg. 2]
140-2-s254-D003 v^{-1}
140-2-s254-D004 $\widetilde{A}$

[scan 255]

We now assume that X_g is equipped with a C^∞ structure, and we replace in the considerations above the groups of homeomorphisms by groups of diffeomorphisms. [p. 128]

m387

Marginal note. m387 N.B. A $C^{\nu|r}$ structure with $\nu|r \geq 2$ would suffice

$$\begin{aligned} E_g = {} & \text{set of conformal structures} \\ & \text{on } X_g \text{ compatible with its} \\ & \text{structure } C^\infty \end{aligned} \tag{76}$$

† We see at once that E_g is an ∞-connected topological space, as a quotient of the ∞-connected space (a convex set) of Riemannian structures, by the † ∞-connected group of C^∞ maps from X_g to $\mathbb{R}^{+*}$. On E_g the group A_g operates – but of course E_g is no longer a homogeneous space. Let us note at once

$$\begin{aligned} E_g/A_g \simeq {} & \text{set of isomorphism classes} \\ & \text{of surfaces } \dagger \\ & \text{conformal compact orientable} \\ & \text{of genus } g \end{aligned} \tag{77}$$

And if we choose one of the two orientations of X_g, so that E_g is identified with the set of complex structures on X_g compatible with its C^∞ structure and its orientation, then E_g/A_g is identified with the space of isomorphism classes of compact complex curves (nonsingular) connected of genus g, modulo passage to the complex conjugate. On the other hand

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Uncertainty retained from cancelled passages.

140-2-s255-D003 Then
140-2-s255-D004 group
140-2-s255-D006 compact

[scan 256]

$$\begin{aligned} E_g/A_g^+ \simeq {} & \text{set of isomorphism classes} \\ & \text{of curves } \mathbb{C}\text{-algebraic smooth projective} \\ & \text{connected of genus } g. \end{aligned} \tag{78}$$

More generally

$$E_g/A_{g,\nu} \simeq \text{set of isomorphism classes of compact surfaces conformal equipped with multipunctures|with multipuncturings of type } (g,\nu) \tag{79}$$

$$E_g/A^+_{g,\nu} \simeq \text{set of isomorphism classes of curves † } \mathbb{C} \text{ algebraic [smooth projective connected of genus } g] \text{ equipped with a multipuncturing, of type } (g,\nu) \tag{80}$$

if X_g oriented.

These are the "coarse moduli spaces" (Mumford's "coarse moduli"), which we may denote by $M_g^{\natural|\sharp}$ and $M_{g,\nu}^{\natural|\sharp}$, and which are indeed too coarse for the most important geometric uses. Much more interesting are the quotients $E_g/A^\circ_{g,\nu}$, [illeg. 1]

Uncertainty retained from cancelled passages.
140-2-s256-D003 complex

[scan 257]
[p. 129]

$$\widetilde{M}_{g,\nu} = E_g/A^\circ_{g,\nu} \qquad (\widetilde{M}_g = \widetilde{M}_{g,0} = E_g/A^\circ_g) \tag{81}$$

on which the group †

$$\Gamma_{g,\nu} = A_{g,\nu}/A^\circ_{g,\nu}$$

operates by passage to the quotient of the operation of $A_{g,\nu}$.

† The space $\widetilde{M}_{g,\nu}$ (with the operation of $\Gamma_{g,\nu}$) is called the Teichmüller space of type g,ν. Of course we recover $M_{g,\nu}^{\natural|\sharp}$ from $\widetilde{M}_{g,\nu}$ and the operation of $\Gamma_{g,\nu}$ on it, by

$$M_{g,\nu}^{\natural|\sharp} = \widetilde{M}_{g,\nu}/\Gamma_{g,\nu} \tag{82}$$

Theorem (Teichmüller) The Teichmüller space $\widetilde{M}_{g,\nu}$ is homeomorphic to $\mathbb{C}^\mu$, where $\mu = 3g-3+\nu$ (in the anabelian cases $2g+\nu \geq 3$), and $\mu = 3g-3+(\nu+\delta)$

$$\delta = \dim_{\mathbb{C}} G$$

in the general case, G being group of algebraic automorphisms of a complex $U_{g,\nu}$ (so $\delta = 1$ in the "limiting" cases $(1,0)$ and $(0,2)$, and more generally δ increases by 1 each time for g fixed, and $(g,\nu-1)$ abelian, one passes from ν to $\nu-1$), hence
m390

Marginal note. m390 We can also say that $\mu = 3g-3+\nu'$, where $\nu' = \nu$ in the anabelian case, and where ν' in the "general" case is the least integer $\geq \nu$ such that (g,ν') is "anabelian".

Uncertainty retained from cancelled passages.
140-2-s257-D001 [illeg. 1]
140-2-s257-D002 [illeg. 1]

[scan 258]

$$\begin{cases} \mu = 1 & \text{in the case } (1,0) \\ \mu = 0 & \text{in the cases } (0,2),(0,1),(0,0), \end{cases} \qquad (82_1|82')$$

i.e. $\widetilde{M}_{g,\nu}$ is in these cases reduced to a point, i.e. the action of $A^\circ_{g,\nu}$ on $E_g = E_0$ is transitive – (and these are with the anabelian case $(0,3)$ the only three cases where this is so, †).

m391

Marginal note. m391 i.e. on surfaces oriented spherical with $\nu \leq 3$, all complex structures are isotopic.

Starting from $\widetilde{M}_{0,3}$, or from $\widetilde{M}_{1,1}$ or from $\widetilde{M}_{g,0} = \widetilde{M}_g$ with $g \geq 1$, when ν increases the "complex dimensions" of $\widetilde{M}_{g,\nu}$ increase by the same amount (On the other hand $\widetilde{M}_{1,0} \longleftarrow \widetilde{M}_{1,1}$ is a homeomorphism †).

We can make Teichmüller's theorem more precise as follows

Corollary In the anabelian case, $A^\circ_{g,\nu}$ operates freely on E_g, so that E_g becomes a torsor over $\widetilde{M}_{g,\nu}$, with structure group $A^\circ_{g,\nu}$.

We deduce from this that $\widetilde{M}_{g,\nu}$ plays the role of a classifying space for $A^\circ_{g,\nu}$, and that

$$\pi_{i|\nu}(\widetilde{M}_{g,\nu}) \xrightarrow{\sim} \pi_{\nu-1|i-1}(A^\circ_{g,\nu}) \qquad (83)$$

and the fact that $\widetilde{M}_{g,\nu}$ is ∞-connected (contained in Teichmüller's theorem) is equivalent to

Uncertainty retained from cancelled passages.

140-2-s258-D002 [illeg. 4]

140-2-s258-D003 because $\simeq$ [illeg. 1]

[scan 259]

the fact that $A^\circ_{g,\nu}$ is so – which is the "well-known theorem" recalled in number 19. [p. 130]

I do not dwell here on the interpretation of the points of $\widetilde{M}_{g,\nu}$ as isomorphism classes of complex curves, equipped with a suitable "Teichmüller rigidification", and the rigidified moduli spaces point of view, which makes it possible to verify a priori that $\widetilde{M}_{g,\nu}$ is equipped with the structure of a nonsingular complex manifold – but already the fact that $\widetilde{M}_{g,\nu}$ is simply connected (which can express by interpreting $\widetilde{M}_{g,\nu}$ as the universal covering of the modular topos $M_{g,\nu}$) is quite deep, which does not seem to be able to fit into the framework of topology (or of differential topology (C^∞) alone...

m392

Marginal note. m392 N.B. To prove that the $\widetilde{M}_{g,\nu}$ are ∞-connected, we are easily reduced to the case of $M_{g,0}$ (if $g \geq 2$), or of $M_{1,1}$ (punctured elliptic case), by using the fibrations $A^\circ_{g,\nu}/A^\circ_{g,\nu+1} \simeq \widetilde{U}_{g,\nu}$ (which is ∞-connected...). The case $g = 1$ is moreover easy and well understood...

[scan 260]

Let us note that the subgroup inclusions

$$A^\circ_{g,\nu+1} \subset A^\circ_{g,\nu} \cdots \subset A^\circ_{g,0} = A^\circ_g$$

define a tower of continuous maps

$$\cdots \longrightarrow \widetilde{M}_{g,\nu+1} \longrightarrow \widetilde{M}_{g,\nu} \longrightarrow \cdots \longrightarrow \widetilde{M}_{g,0} = \widetilde{M}_g \qquad (84)$$

where

$$\widetilde{M}_{g,\nu+1} \longrightarrow \widetilde{M}_{g,\nu} \qquad (85)$$

is for (g,ν) anabelian a fibration with fiber $A^\circ_{g,\nu}/A^\circ_{g,\nu+1} \simeq \widetilde{U}_{g,\nu}$.

[it therefore has contractible fiber, and the ∞-connectedness of $\widetilde{M}_{g,\nu+1}$ is therefore equivalent to that of $\widetilde{M}_{g,\nu}$ – which brings us back to the case of $\widetilde{M}_g$ if $g \geq 2$, of $M_{1,1}$ and of $M_{0,3}$ if $g = | \neq | \geq 1$ or 0].

In the abelian case (g,ν), we find

$$\widetilde{M}_{g,\nu+1} \simeq \widetilde{M}_{g,\nu} \qquad \text{(if } (g,\nu) \text{ abelian)} \tag{85}$$

which † means that in these cases, if we have two complex structures α, β on X which are congruent by $u \in A^\circ_{g,\nu}$, (i.e. $\beta = u.\alpha$) they are = congruent by $A^\circ_{g,\nu+1}$.

Indeed, if G is the identity|neutral component of the group of complex automorphisms of $U_{g,\nu}$ for β, we can write $u = gv$ with $g \in G$, $v \in A^\circ_{g,\nu+1}$ hence $\beta = gv.\alpha$ hence (since $g^{-1}\beta = \beta$) $\beta = v\alpha$, q.e.d.

Uncertainty retained from cancelled passages.

140-2-s260-D002 follows precisely|stems, follows

Thus $\widetilde{M}_{1,1} \simeq \widetilde{M}_{1,0}$ – and it is immediate that the latter is $\simeq$ the Poincaré half-plane. Likewise [scan 261]
$\widetilde{M}_{3,0} \xrightarrow{\sim} \widetilde{M}_{2,0} \simeq \widetilde{M}_{1,0} \simeq \widetilde{M}_{00|0,0}$, and since A^0_0 is simply made up of the automorphisms of [p. 131]
$X_0 \simeq S^2$ which preserve orientation, we see that two complex structures on S^2 are isotopic (since they are isomorphic, and an isomorphism preserves orientation i.e. is $\in A^0_0$) this proves that the $M_{0,i}$ $(i \leq 3)$ are reduced to a point|points ! Thus the Teichmüller Theorem is fairly evident if $g = 0$ or 1, it is the case $g \geq 2$ that is deep…

The Teichmüller space (more generally, any space E ∞-connected on which A_g operates in such a way that $A^0_{g,\nu}$ operates freely) will allow us to interpret the canonical extension $\Gamma'_{g,\nu+1}$ of $\Gamma_{g,\nu}$ by $\Pi_{g,\nu}$ (anabelian cases (g,ν)) as the mixed fundamental group of a space (homotopic to $U_{g,\nu}$) on which $\Gamma_{g,\nu}$ operates. (We were bothered previously, because $\Gamma_{g,\nu}$ did not itself operate on $U_{g,\nu}$, but only the group $A_{g,\nu}$ of which $\Gamma_{g,\nu}$ is a quotient, we had the impression

that passage to the quotient by $A^0_{g,\nu}$ was nevertheless inessential, since $A^0_{g,\nu}$ is ∞-connected – [scan 262]
and moreover we had found an extension $\widetilde{A}_{g,\nu}$ of $\Gamma'_{g,\nu+1}$ by this same ∞-connected group $A^0_{g,\nu}$, which operates on the universal covering, † with a homomorphism surjective $\widetilde{A}_{g,\nu} \to A_{g,\nu}$ with kernel $\simeq \Pi_{g,\nu}$ compatible with this operation…

Now if $\widetilde{M} = E/A^0_{g,\nu}$, E is an $A^0_{g,\nu}$-torsor with base $\widetilde{M}$, which we can use to twist † $U_{g,\nu}$ on which $A^0_{g,\nu}$ operates continuously, we thus find a bundle $\mathcal{X}$ ($\simeq U_{g,\nu} \times E/A^0_{g,\nu}$ operating diagonally) over $\widetilde{M}$, with fiber $U_{g,\nu}$ (this is for $E = E_g$ the universal bundle of complex curves of type (g,ν) with Teichmüller rigidification…). Since E and $A^0_{g,\nu}$

Uncertainty retained from cancelled passages.

140-2-s262-D001 such a way

140-2-s262-D003 [illeg. 1]

are ∞-connected, so is $\widetilde{M}$, hence the inclusion of a fiber into the bundle $\mathcal{X}$ over $\widetilde{M}$ is a homotopy [scan 263]
equivalence. Now $\Gamma_{g,\nu}$ operates on $\mathcal{X}$ (in a way compatible with its operation on $\widetilde{M}$), whence [p. 132]
the construction of an extension of $\Gamma_{g,\nu}$ by $\pi_1(\mathcal{X}) \simeq \pi_1(U_{g,\nu}) \simeq \Pi_{g,\nu}$ – we have done all that was needed to prove that this is indeed essentially $\Gamma'_{g,\nu}$ – I dispense with giving the details…

23 Back to surfaces with group|groups of operators. [scan 264]

Our point of view will be, not that of outer groups with loops, but that of multi-Galois topoi [p. 133]
and morphisms between them, more flexible, as we have seen. Before bringing in operations of groups, † let us introduce the 2-category of orientable surfaces U (N.B. not oriented !) † † admissible (i.e. of the form $X \setminus S$, where X is compact, S finite, and every connected component

of X of genus 0 meets S), as some|in Ladegaillerie (taking as 2-morphisms between † homeomorphisms $f, g\colon U \rightrightarrows | \Rightarrow U'$ the homotopy classes of paths from f from|in g in the space $\underline{\mathrm{Isom}}(U, U')$ (which, with the hypotheses made, has as connected components † torsors under products of groups $A^0_{g,\nu}$ (with $(g,\nu) \neq 0,0|(g,\nu) \neq (0,0)$) – so these are $K(\Pi|\pi, 1)$.

We find a 2-functor from the

Uncertainty retained from cancelled passages.

140-2-s264-D001 let us note that a

140-2-s264-D002 [illeg. 1]

140-2-s264-D003 bounded

140-2-s264-D004 [illeg. 1]

140-2-s264-D005 pro

preceding 2 category to that of morphisms of multi-Galois topoi (or, if we prefer, to that of morphisms of groupoids), denoted [scan 265]

$$B_{D^*} \xrightarrow{\rho} B_U$$

, where for two objects of the 2-category $B_{D^*} \to B_U$ and $B_{D'^*} \to B_{U'}$, the category of morphisms from one to the other is made up of essentially commutative|in commutative diagrams of morphisms of topoi (or of functors of groupoids)

$$\begin{array}{ccc} B_{D'^*} & \xrightarrow{\rho'} & B_{U'} \\ {\scriptstyle f_{D^*}}\downarrow & \overset{\rho}{\swarrow}\alpha & \downarrow{\scriptstyle f_U} \\ B_{D^*} & \xrightarrow{\rho} & B_U \end{array}$$

where f_{D^*} and f_U are morphisms, and † $\alpha\colon \rho f_{D^*} \simeq f_U \rho'$ a commutativity datum, † the morphisms between an f and a g being defined ad hoc…

It may be useful to regard the

Uncertainty retained from cancelled passages.

140-2-s265-D002 α

140-2-s265-D003 [illeg. 1]

objects of the 2-category as cofibered topoi (or cofibered groupoids) over the "arrow" category [scan 266] [p. 134]

$$\overbrace{\underline{\underline{D^*}} \xrightarrow{\underline{\varrho}} \underline{\underline{U}}}^{\Delta_1}$$

having two objects D^* and U, and a single nonidentity arrow $D^* \to U$. In terms of a functor between groupoids, $\Pi_{D^*} \to \Pi_U$, the associated † cofibered category ("in groupoids") Π is † defined by

$$\mathrm{Ob}\,\Pi = \mathrm{Ob}\,\Pi_{D^*} \amalg \mathrm{Ob}\,\Pi_U$$

the arrows between two objects of Π_{D^*}, or two objects of Π_U, being those in Π_{D^*} resp. in Π_U, and the arrows from $\widetilde{D}^* \in \mathrm{Ob}\Pi_{D^*}$ to $\widetilde{U} \in \mathrm{Ob}\,\Pi_U$ being the † $\rho_!(\widetilde{D}^*) \xrightarrow{\sim} \widetilde{U}$ in $\Pi_{B_U}|\Pi_{B_{D^*}}$. We have a canonical functor $\Pi \to \Delta_1$ which is "cofibering" and † for which all arrows of Π cocartesian. [When we prefer to work

Uncertainty retained from cancelled passages.
140-2-s266-D001 category [illeg. 1]
140-2-s266-D003 [illeg. 2]
140-2-s266-D005 isomorphisms
140-2-s266-D007 [illeg. 1]

with topoi, and we identify the morphisms ρ of topoi by the inverse image functors ρ^* (N.B. we have a sequence of three adjoint functors $\rho_!\ \rho^*\ \rho_*$) then $B_U \xrightarrow{\rho^*} B_{D^*}$ is described by a category † fibered in topoi over Δ – i.e. a fibered category B such that the fiber categories are topoi, and the base change^t functors are left exact|exact and commute with arbitrary $\varinjlim$. [scan 267]

Let $\mathcal{S}$ be the 2-category of admissible surfaces, $\mathcal{M}$ that of morphisms of multi-Galois topoi (without conditions). We have a 2-functor of 2-categories

$$\mathcal{S} \longrightarrow \mathcal{M}$$

we know how to describe the 2-essential image by the condition "loop structure", which defines a 2-full sub-2-category of $\mathcal{M}$, say $\mathcal{M}_{\text{lac}}$ (admissible objects of $\mathcal{M}$). Moreover, [m393]

Marginal note. [m393] N.B. we exclude for example $B_{D^*} \simeq$ "empty topos", $B_U \simeq$ "point topos" i.e. $\Pi_{D^*} = \emptyset$, $\Pi_U \simeq$ one-point category

Uncertainty retained from cancelled passages.
140-2-s267-D001 co

we know that the functor is 2-faithful, hence induces an equivalence of 2-categories [scan 268] [p. 135]

$$\mathcal{S} \longrightarrow \mathcal{M}_{\text{lac}}$$

m394

Marginal note. [m394] There is no [illeg. 3] † † in the case B_{D^*} = "empty topos" over B_U – namely an isomorphism $T \simeq H^2(U_i, Z)$ for every connected component [illeg. 1], – for we do not assume that we are working with Automorphisms|structures [illeg. 1] orientation ! (In the arithmetic analog, it is|it will be no longer the same, of course…)

If now Γ is a group, and if we consider admissible surfaces with an action of Γ, they define topoi (or groupoids) B_{D^*}, B_U with an operation of Γ on them [|(or topoi cofibered over the groupoid $[\Gamma]$ defined by Γ] and morphisms between them commuting with Γ. We can regard such a datum as that of a multi-Galois topos (or groupoid) cofibered over $\Delta \times [\Gamma]$. † † But the datum of such a morphism of multi-Galois topoi with an operation of Γ, is equivalent to that of a morphism of multi-Galois topoi

$$B_{D^*,\Gamma} \longrightarrow B_{U,\Gamma} \longrightarrow B_\Gamma$$

Uncertainty retained from cancelled passages.
140-2-s268-D001 [illeg. 4]
140-2-s268-D002 structure [illeg. 2]
140-2-s268-D003 [illeg. 1]
140-2-s268-D004 [illeg. 1]

This seems a conceptually convenient point of view, especially when we vary Γ. We can replace Γ by a groupoid † Π (playing the role of a fundamental groupoid) and set out to describe [scan 269]

the 2-category $\mathcal{RS}$ of functors from Π to the category of admissible surfaces (or again the 2-category of "admissible fibered surfaces" over the|the multi-Galois|multi-Galois topos $\widehat{\Pi}^{\circ}|\widehat{\Pi}^{0}$ corresponding|corresponding to Π) – we describe it by † the 2-category $\mathcal{RM}$ of diagrams of multi-Galois topoi (or of groupoids)

$$B_{D^*,\Pi} \longrightarrow B_{U,\Pi} \longrightarrow B_{\Pi}$$

$(*)$ m395

Marginal note. m395 Rigidity of "representation|representations" (of groups)|(of groupoids)

finally we have a 2-functor of 2-categories, that of representations of groupoids in the category of admissible surfaces, and that of diagrams of multi-Galois topoi (or of groupoids)

$(*)$. Taking for every connected component B_{Π_i} of B_{Π} a universal covering $\widetilde{B}_{\Pi_i}$, and taking the fiber products, we recover

Uncertainty retained from cancelled passages.
140-2-s269-D001 [illeg. 1]
140-2-s269-D002 [illeg. 1]

as expected a diagram $(**)$ [scan 270] [p. 136]

$$B_{D_i^*}\,\dagger \xrightarrow{\rho_i|f_i} B_{U_i} \longrightarrow \widetilde{B}_{\Pi_i} \simeq \text{point topos}$$

and an action of $\Gamma_i = \mathrm{Aut}_{B_{\Pi_i}}\widetilde{B}_{\Pi_i}$ on it – and the|the † family of these for variable i allows us to recover the complete situation…We shall say that the diagram $(*)$ is "admissible", if the $\rho_i|f_i$ in $(**)$ are admissible, † whence a 2-category $\mathcal{RM}_{\mathrm{lac}}$, and a 2-functor

$$\mathcal{RS} \longrightarrow \mathcal{RM}_{\mathrm{lac}}$$

We look at the † full categories, obtained by restricting to Π whose fundamental groups are finite, whence an induced 2-functor

$$\mathcal{R}_f\mathcal{S} \longrightarrow R_f\widetilde{\mathcal{M}}_{\mathrm{lac}}$$

and we propose to study its faithfulness properties. I conjecture that it is a

Uncertainty retained from cancelled passages.
140-2-s270-D001 Π
140-2-s270-D002 set
140-2-s270-D003 [illeg. 1]
140-2-s270-D005 of the

[scan 271]

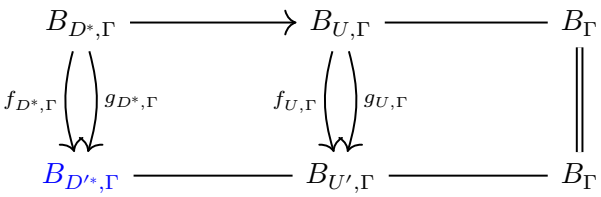
equivalence of 2-categories, i.e. that it is 3 faithful.

Of course we are reduced all the same to the case of groupoids "connected" Π_f defined by a group (finite if need be), and we are also reduced by essentially trivial arguments to looking at the case of diagrams $B_{D^*} \to B_U$ with B_U connected.

On the geometric side, the situation would be given by a $U = X \setminus S$ of type (g,ν) $((g,\nu) \neq (0,0))$, and an operation of Γ on it — we may suppose $U = U_{g,\nu}$, so we give $\Gamma \to A_{g,\nu}$. For the

question of i-faithfulness with $i \leq 2$, we may suppose that it is the same group Γ that operates; †

1) 0-faithfulness. † Let U, U' be given with an operation of Γ on them, $f, g : U \xrightarrow{\sim} U'$ commuting with Γ, and α, β two homotopies from f to g i.e. two paths from f to g in $\underline{\mathrm{Isom}}_\Gamma(U, U')$, we suppose that in the topossic description|the topossic descriptions



we have $\alpha_*^{D^*} = \beta_*^{D^*} : f_{D^*} \to g_{D^*}$ and $\alpha_*^U = \beta_*^U : f_U \to g_U$. To prove that α and β are homotopic. 270

Uncertainty retained from cancelled passages.
140-2-s271-D001 [illeg. 4]
140-2-s271-D002 [illeg. 1]

[scan 272]

m397

[p. 137]

Marginal note. m397 i.e. $f \in A_{g,\nu}$ commuting with the image of Γ under $\Gamma \to A_{g,\nu}$.

We may suppose $g = \mathrm{id}$, $U = U_{g,\nu}$, so $f \in A_{g,\nu}^\Gamma$, and we have two paths α, β from 1 to † f in the space $A_{g,\nu}^\Gamma$. We suppose that the two corresponding isomorphisms between the identity of Π_{D^*} and $f_{D^*,\Gamma} : \Pi_{\dagger D^*_{S_\nu},\Gamma} \to \Pi_{D^*_{S_\nu},\Gamma}$ on the one hand, between the identity of $\Pi_{U,\Gamma}$ and $f_{U,\Gamma}$ on the other hand, are the same, we want to prove that α and β are homotopic. Of course, the hypothesis † on α, β concerning the action of Γ is satisfied a fortiori upon restricting to a smaller group, for example † $\Gamma' = 1$, and in fact we see that it is equivalent for Γ, and for its subgroup 1. † the already known result (Ladegallerie|Ladegaillerie) for $\Gamma = 1$, shows that this means that if two paths in $A_\dagger^\Gamma$ from 1 to f are homotopic in A, they are so in A^Γ or again that

let us put $A = A_{g,\nu}$

$$\boxed{\pi_1(A^\Gamma) \to \pi_1(A) \text{ is injective.}}$$

In the case $\pi_1(A) = 0$ (case (g, ν) anabelian) this thus comes down to proving that $\pi_1(A^\Gamma) = 0$. 271

Uncertainty retained from cancelled passages.
140-2-s272-D001 [illeg. 1]
140-2-s272-D002 B
140-2-s272-D003 [illeg. 1]
140-2-s272-D004 Γ
140-2-s272-D005 [illeg. 1]
140-2-s272-D006 g, ν

[scan 273]

2) 1-faithfulness. This means (in addition to 0-faithfulness) that $\forall$ isomorphism between † B_f and B_g comes from a † path from f to g. With the preceding reductions|the preceding reduction, we may suppose $g = \mathrm{id}$, so we have $f \in A^\Gamma$, whence $f^* : (B_{D^*} \to B_U) \to (B_{D^*} \to B_U)$ (commuting † Γ respecting Γ), and we have an isomorphism with the identity — unless I am mistaken this condition 272

Uncertainty retained from cancelled passages.

140-2-s273-D001 f

140-2-s273-D002 path

140-2-s273-D003 with

[scan 274]

Let A be a topological group, whence two invariants [p. 138]

$$\pi_0 = \pi_0(A), \qquad \pi_1 = \pi_1(A)$$

the first a group not necessarily commutative, † the second a commutative group on which π_0 operates. We define a Gr-category $\underline{A}$ with invariants π_0, π_1 and corresponding to this operation of π_0 on π_1, by taking as underlying category the † fundamental † groupoid (naive) of A, and as composition functor the one deduced from $A \times A \to A$ (associativity in $\underline{A} = \Pi A$ is strict …)

This being so, every torsor under A defines a 1-torsor under the Gr-category $\underline{A}$. More generally, for ∀ topological space S (or ∀ topos which is locally a topological space) every torsor on S with group A_S defines an $\underline{A}_S$-stack on S which is a † stack of A_S-torsors. If $\pi_i(A) = 0$ for $i \geq 2$, then we find 273

Uncertainty retained from cancelled passages.

140-2-s274-D001 [illeg. 1]

140-2-s274-D002 [illeg. 3]

140-2-s274-D003 Hom

140-2-s274-D004 $\underline{A}_S$

[scan 275]

thus for $\forall n \in N,\ n \geq 2$ an n-equivalence between the n-category of torsors on S with group A_S, and the n-category deduced from the 2-category of gerbes $\underline{A}_S$-torsors by extending it in a discrete way …

But let us consider † a group Γ, and consider the classification of A_{B_Γ}-torsors on the classifying topos — i.e. that of A torsors, with an operation of Γ on them (commuting with the action of A). These objects form a 2-category, whose connected components correspond to the conjugacy classes of homomorphisms from Γ to A. If we have one homomorphism $\Gamma \to A$, it defines an action on the trivial torsor. 1_A If u, v are two homomorphisms, the isomorphisms from $(1_A, u)$ onto $(1_A, v)$ correspond to the elements of the set “

$$\mathrm{Transp}_A(u, v) = \{g \in A \mid v = \mathrm{int}(g).u\}$$

But we shall make a 0-isotopic category, 274

Uncertainty retained from cancelled passages.

140-2-s275-D001 [illeg. 1]

[scan 276]

by replacing $\mathrm{Transp}(u, v)$ by its π_0, and a 1-isotopic category by replacing it by its fundamental groupoid $\Pi_1 \mathrm{Transp}(u, v)$. Thus an isomorphism from u to v is still a point of $\mathrm{Transp}(u, v)$, but if we have two such isomorphisms f, g, the isomorphisms $f \simeq g$ are the classes of paths in $\mathrm{Transp}(u, v)$ from f to g. Thus, the category $\underline{\mathrm{Aut}}(u)$ is equivalent to $\Pi_1 \underbrace{\mathrm{Trans}(u, u)}_{A^u}$ [p. 139]

$$\underline{\mathrm{Aut}}(u) \simeq \Pi_1 \mathrm{Trans}(u)$$

and the † category $\mathrm{Isom}(u, v)$ is either empty (if u, v are not conjugate), or a torsor under $\underline{\mathrm{Aut}}(u)$.

But for every torsor P over A on which Γ operates, †the † 1-torsor $\Pi_1 P$ under $\Pi_1 A = \underline{A}$ is equipped with operations of Γ, whence a 275

Uncertainty retained from cancelled passages.
140-2-s276-D002 the
140-2-s276-D003 [illeg. 1]
140-2-s276-D004 associated a

[scan 277]

$(\Gamma, -\underline{A})$ torsor. We thus find a functor of 2-categories

2 category of $\Gamma - A$-torsors [or again, of homomorphisms $\Gamma \to A$] $\longrightarrow$ 2 category of 1-torsors under A, with operations of Γ on them.

In short, we have just repeated on the classifying topos B_Γ the construction made above for a topological space S. We would also like to express conditions for this 2-functor to be a 2-equivalence. For this, we should first have an understanding of the classification of the classes of equivalence|of equivalences of objects of the second category — which are, for their part, of a purely algebraic nature, in terms of the Gr-category $\underline{A}$ and of Γ. Even if we restrict ourselves to 276

[scan 278]

trivial torsors over $\underline{A}$ (which is legitimate) we would have to make explicit what it means that [p. 140] Γ operates on the trivial torsor. We observe that this means that we have a homomorphism of Gr-categories from the discrete Gr-category defined by Γ to $\underline{A}$. Thus the 2-category envisaged is the one whose objects are the homomorphisms of Gr-categories

$$\Gamma \xrightarrow[\dagger]{u} \underline{A}$$

and for two such morphisms u, v, we must define (not only a set $\mathrm{Hom}(u,v)$ but) a category $\underline{\mathrm{Hom}}(u,v)$, as the subcategory $\underline{\mathrm{Transp}}(u,v)$ of $\underline{A}$, to be defined ad-hoc.

Let us consider for simplicity the case where $\pi_1 = 0$, then $\underline{A}$ reduces to π_0, and the category of homomorphisms $\Gamma \to \underline{A}$ to that of homomorphisms $\Gamma \to \pi_0$. The 277

Uncertainty retained from cancelled passages.
140-2-s278-D001 $\underset{v}{\longrightarrow}$

[scan 279]

fact that the functor of 2-categories above † is an equivalence of 2|1-categories then translates step by step as follows :

1°) 0 faithfulness means that for $\forall$ homomorphism $u : \Gamma \to A$, we have $\pi_1(A^u) = 0$

N.B. A^u and A^{0u} have the same identity component|identity components, so the condition is written : $\boxed{\pi_1((A^0)^u) = 0}$.

2°) 1-faithfulness means that moreover, if u, v are homomorphisms $\Gamma \rightrightarrows A$, and † $\alpha, \beta \in \mathrm{Transp}_A(u,v)$ † have the same image in $\mathrm{Transp}_{\pi_0}(\underline{u}, \underline{v})$ † † then α, β are in the same † connected component of $\mathrm{Transp}_A(u,v)$. † † † To see this, we may suppose $u = v$, and we are reduced to 3°) expressing that the map $\pi_0(A^u) \to \pi_0$ is injective i.e. that † its kernel is reduced to 1, i.e. that the open subgroup $(A^u) \cap A^0 = (A^0)^u$ of A^u is connected, i.e.

$$\boxed{\pi_0((A^0)^u) = 0}$$

278

Uncertainty retained from cancelled passages.

140-2-s279-D001 that it
140-2-s279-D003 $\alpha, \beta \in$ Transp [illeg. 1]
140-2-s279-D004 [illeg. 1] [illeg. 2]
140-2-s279-D005 [illeg. 1] [illeg. 3] (u, v) [illeg. 1]
140-2-s279-D006 [illeg. 1]
140-2-s279-D007 [illeg. 1] identical
140-2-s279-D008 This
140-2-s279-D009 conjugate to one another
140-2-s279-D010 To
140-2-s279-D011 A^u

[scan 280]

3°) 2-faithfulness means that in addition to the preceding conditions, which ensure that for u, v fixed, the functor $\underline{\mathrm{Hom}}(u, v) \to \underline{\mathrm{Hom}}(\underline{u}, \underline{v})$ is fully faithful, that the latter is essentially surjective, i.e. that if u, v are such that $\underline{u}, \underline{v} : \Gamma \to \pi_0$ are conjugate, (equal if we wish) u, v are already conjugate (by an element^t of π_0). In other words : for $\forall$ homomorphism $\underline{u} : \Gamma \to \pi_0 = A/A^0$, if there exists a lifting of u to $\Gamma \to A$, it is unique up to conjugation. [p. 141]

m408

Marginal note. m408 We must say it [illeg. 1] in a stronger way : $\mathrm{Transp}(u, v) \to \mathrm{Transp}(\underline{u}, \underline{v})$ surjective — this is equivalent to saying that if $\underline{u} = \underline{v}$, then u and v are conjugate by an element of A^0.

4°) 3-faithfulness means that in addition, every homomorphism $\underline{u} : \Gamma \to \pi_0$ lifts to $u : \Gamma \to A$.

In summary, if $\pi_1(A^0) = 0$, 3-faithfulness means that for $\forall$ homomorphism $\underline{u} : \Gamma \to \pi_0 = A/A^0$, there exists lifting $u : \Gamma \to A$, unique up to conjugation, and that then $(A^0)^u$ is connected and simply connected, 279

[scan 281]

I presume that in the general case where we do not assume $\Pi_1(\mathcal{A}^0) = 0$, we must replace the condition $\Pi_1(\mathcal{A}^{0u}) = 0$ by

$$\Pi_1(\mathcal{A}^{0u}) \longrightarrow \Pi_1(\mathcal{A}^0)$$

an isomorphism.

m410

Marginal note. m410 Too brutal! The condition in question saying $\Pi_1(\mathcal{A}^{0\Gamma}) \xrightarrow{\sim} \Pi_1(\mathcal{A}^0)^\Gamma$ – but we must go over all the conditions carefully and modify them ad hoc… Cf.… intercalary|inserted sheet

(The nicest thing would be for it to be a homotopism – it is this probably that would express that we have an ∞-equivalence…)

I feel like proving first 1°, 2°, 3° in the case $\mathcal{A} = \mathcal{A}_{g,\nu}$, restricting myself if need be to the anabelian case (probably the hardest in fact! } but less conceptually tangled), and of course to the case where Γ is finite. The first task will of course be that|then of determining $\mathcal{A}^{0u}$ and its topological structure, to try to prove that $\mathcal{A}^{0u}$ is contractible in this case

[scan 282]

24 Attempt at Determination of|of the $\mathcal{A}^{0\Gamma}$. Link with the relation $\Pi^\Gamma_{g,(\nu,\nu+n-1)} = (1)$, and work program… [p. 142]

Let us consider an operation (faithful if we insist) of the finite group Γ on the surface $U \subset X = \widehat{U}$ of type (g, ν), $U = X - S$, let $\mathcal{A} = \mathrm{Aut}(U)$, so $\Gamma \subset \mathcal{A}$ and $\mathcal{A}^u = \mathcal{A}^\Gamma$ is the centralizer of Γ. Let us first assume $\Gamma = \Gamma^+$. Obviously $f \in \mathcal{A}^\Gamma$ implies that f † induces an automorphism g of $X/\Gamma \equiv | \simeq X'$ † X is a Galois covering with group Γ.

Let S' be the image of S in X' (so $S = X \mid S'$), and let $S^!$ be the set of ramification points of Γ in U, $S'^!$ its image in $U' = X' \backslash S'$ (N.B. $U = X \mid U'$). Thus g is an automorphism of (X', S'), taking $S'^!$ into itself, and † thus inducing an automorphism of $V' = U' \backslash S'^! = X' - (S' \cup S'^!)$, † The inverse † image $V = X \mid V'$ is étale over V', it is even a $\Gamma_{V'}$-torsor over V' and the inverse image of the latter under $g \mid V'$ is (via f) isomorphic to V. The datum of f is equivalent to the datum of an automorphism of the † Γ-covering V of V', inducing an

Uncertainty retained from cancelled passages.

140-2-s282-D001 sends U
140-2-s282-D002 being
140-2-s282-D003 such that
140-2-s282-D004 over which
140-2-s282-D005 [illeg. 2]
140-2-s282-D006 [illeg. 1]

[scan 283]

isomorphism of V' which, extended to X', sends S' into itself. This gives an interpretation of $\mathcal{A}^\Gamma$ in terms of constructions on X'.

Let us consider the fundamental group Π' of V', with the loop structure indexed by $S' \amalg S'^!$, so we have a surjective homomorphism

$$\Pi' \xrightarrow{\varphi} \Gamma$$

which on each of the subgroups with|of loops of index $s' \in S'^!$ is not trivial, and an automorphism g of V defines an outer automorphism $\mathfrak{g}|\overline{g}$ of Π' – the condition to put on it is that its composite with φ is conjugate to|of φ, and that $\mathfrak{g}|\overline{g}$ takes $S'^!$ into itself.

m411

Marginal note. m411 we choose a point $a \in V$ and its image $a' \in V'$ as base points, to make explicit [illeg. 1] $\pi|\Pi'$

This being so, † g is defined by $\mathfrak{g}|\overline{g}$ up to isotopy † (if we leave out degenerate cases (g', ν') †, namely $(0, \nu')$ with $\nu' = 0, 1, 2$ which will have to be examined separately…) and g being chosen, f is defined modulo multiplication by an element of the center of Γ, but we already know that $\mathcal{A}^0 \cap \Gamma = (1)$, so if we restrict ourselves to examining elements $f \in \mathcal{A}^{0\Gamma}$ then f will be uniquely determined by g

Uncertainty retained from cancelled passages.

140-2-s283-D001 g
140-2-s283-D002 or|and
140-2-s283-D003 [illeg. 2]

[scan 284]
[p. 143]

We are thus led to the following problem:

Let $V' = X' \setminus (S' \amalg S'^!)$ be an † admissible surface of type (g', ν') (X' compact, $\nu' = \operatorname{card} S' + \operatorname{card} S'^!$), equipped with a subset $S'^!$ of the set of its points at infinity and a base point x', whence $\Pi' = \Pi_1(X'|V', x')$. We are given a Galois covering connected|named V of V' with finite group Γ, pointed over x', hence defined by a surjective homomorphism

$$\Pi' \xrightarrow{\varphi} \Gamma$$

and we assume $V'|V$ ramified at each of the $s' \in S'^!$, i.e. (for a choice of the group with loops|group of loops † $L_{s'}$ corresponding to s') that $L_{s'} \to \Gamma$ is not trivial. We consider the compactified set|space † X of V, and the set S resp. $S^!$ of points of X over S' resp. $S'^!$. Let g be an automorphism of $V|V'$, defining an outer automorphism $g_\bullet$ of Π', we assume that

$$g^*(V'/V) \overset{V\text{-isomorphism}}{\simeq} V' \quad \text{i.e. that } \varphi \circ g_\bullet \text{ is conjugate to } \varphi$$

m412

Marginal note. m412 We may assume that g fixes x', so $g_\bullet$ is a well-defined automorphism of Π'.

(|{which expresses that g comes by passage to the quotient from an automorphism f of V, defined modulo multiplication by an element of $z \in \text{Center } \Gamma$). Let $U = V \cup S^! = X \setminus S$ (ramified covering over $U' = V' \cup S'^! = X' \setminus S'$), We want † to express that † among the f which lift g, there is one (necessarily|necessarily unique!) which is isotopic to 1 in $\underline{\text{Aut}}(U)$ — i.e. which induces the trivial outer automorphism of $\Pi_1(U)$. We would like

Uncertainty retained from cancelled passages.

140-2-s284-D001 orientable
140-2-s284-D002 [illeg. 1]
140-2-s284-D003 for|by
140-2-s284-D004 [illeg. 1]
140-2-s284-D005 g comes by an

[scan 285]

to prove that these are exactly those which † are isotopic to 1 in $\underline{\text{Aut}}(V')$ — or again that for such an f, f is necessarily isotopic to 1 in $\underline{\text{Aut}}(V)$ † i.e. is in $\underline{\text{Aut}}(V)^0$ ($\subset \underline{\text{Aut}}(V|U)^0$) and not merely in $\underline{\text{Aut}}(U)^0$. Let us note when $\underline{\text{Aut}}(V')^0$ is 1-connected that $\underline{\text{Aut}}(V')^0$ lifts (by virtue of general principles) to † †

$$\underline{\text{Aut}}(V')^0 \longrightarrow \underline{\text{Aut}}(V)^{0\Gamma} \subset \underline{\text{Aut}}(U)^{0\Gamma}.$$

Thus the stated condition on g of isotopy to 1 is certainly sufficient. Let us note moreover that the results already obtained imply that

$$\underline{\text{Aut}}(U)^{\dagger\Gamma} \cap \underline{\text{Aut}}(U)^0 = \underline{\text{Aut}}(U)^{0\Gamma}$$

induces the identity on $S^!$ ($= U^!$ with an earlier notation). Now let B † $\subset \underline{\text{Aut}}(U)^0 \cap \underline{\text{Aut}}(U|V, S^!)_{=|\epsilon}$ be the subgroup of points which fix the $s \in S^!$, so that

$$\underline{\text{Aut}}(U)^0/B \simeq \underline{\text{Mon}}(S^!, U),$$

and $B^0 = \underline{\text{Aut}}(V)^0$ — a known argument shows us that

$$B/B^0 \simeq \Pi_1(\underline{\text{Mon}}(S^!, U)).$$

Since $\underline{\text{Aut}}(U)^{0\Gamma} \subset B$ i.e. $\underline{\text{Aut}}(U)^{0\Gamma} = B^\Gamma$, we † :

$$\begin{array}{ccccccccc} 1 & \longrightarrow & B^{0\Gamma^0} & \longrightarrow & B^\Gamma & \longrightarrow & (B/B^0)^\Gamma & \longrightarrow & H^1(\Gamma, B^0) \\ & & \wr\| & & \wr\| & & & & \|? \\ & & \underline{\text{Aut}}(V)^0 & & \underline{\text{Aut}}(U)^{0\Gamma} & & & & 1 \end{array}$$

m413

Marginal note. m413 thus deduce from

$$1 \longrightarrow B^0 \longrightarrow B \longrightarrow B/B^0 \longrightarrow 1$$

Uncertainty retained from cancelled passages.

140-2-s285-D001 induce
140-2-s285-D002 or $\underline{\mathrm{Aut}}(U)$
140-2-s285-D003 $\underline{\mathrm{Aut}}$
140-2-s285-D004 $\underline{\mathrm{Aut}}$
140-2-s285-D005 0
140-2-s285-D007 $\subset \underline{\mathrm{Aut}}(U)^0$
140-2-s285-D008 thus is|being connected [illeg. 2]

[scan 286]

Thus we find that the identity component of $\underline{\mathrm{Aut}}(U)^{0\Gamma}$ is $\simeq \underline{\mathrm{Aut}}(V)^0$ (thus is ∞-connected if V anabelian), and its Π_0 is [p. 144]

$$\subset (B/B^0)^\Gamma \simeq \Pi_1(\mathrm{Mon}(S^!, U))^\Gamma.$$

We would like to prove that the subgroup of|of the this Π_1 of invariants under Γ is reduced to 1,

$$\underline{H^0(\Gamma, \Pi_1(\mathrm{Mon}(S^!, U)))} \quad ?$$

We would like this to be true even independently of realizability hypotheses, when we are given a homomorphism from Γ into the Teichmüller group of a V, and we make some suitable hypotheses....

Conjecture Let Π be an outer group with loops of type (g, ν), I the set of indices|of index of the classes of loops, Γ a finite group operating on Π by outer automorphisms. Then there exists an outer group with loops Π', with set|with an I' of loop classes, an outer operation|action of Γ on Π', a subset $I'^!$ of I' stable under|for Γ, an outer homomorphism for "plugging the holes of I'"

$$\Pi' \longrightarrow \Pi$$

compatible with the action of Γ, such that

[scan 287]

1°) The stabilizer in Γ of $\forall$ element of $I'^!$ is not reduced to 1

2°) The extension of Γ by Π' deduced from the outer action of Γ has no elements of finite order $\neq 1$ (i.e. for no elemen^t of $\Gamma \neq 1$, is the outer action on Π' realized as|by an automorphism of Π' of finite order)

Moreover, the outer Γ-group with loops Π', equipped with the morphism $\Pi' \longrightarrow \Pi$, is determined up to unique isomorphism.

† Comment. The existence of Π', $\Pi' \longrightarrow \Pi$ is obvious in the "realizable" case. Uniqueness – up to unique isomorphism, even in the realizable case is not obvious, nor even proved. The kernel of †

$$\begin{array}{ccc} \mathrm{Autext}_{\mathrm{lac}}(\Pi', \Pi) & \longrightarrow & \mathrm{Autext}(\Pi) \\ \| & & \\ \mathrm{Autext}_{\mathrm{lac}}(\Pi', I'^!) & & \end{array}$$

is precisely the group Π_1 from earlier [m415]

Marginal note. [m415] rather, an extension of $\mathfrak{S}_{I'}$ by this Π_1

and to show that the functor from the † pairs $(\Pi', I'^!)$ of an outer|category Γ-group Π', † and of a subset

Uncertainty retained from cancelled passages.
140-2-s287-D001 Comment.
140-2-s287-D003 [illeg. 1]
140-2-s287-D004 [illeg. 1]
140-2-s287-D005 [illeg. 1]

[scan 288]

$I'^!$ of $I(\Pi')$ stable under Γ, such that the corresponding extension of Γ by Π' is "torsion-free" and that the stabilizer of $\forall\ i' \in I'^!$ in Γ is nontrivial, to outer Γ-groups, is (not only essentially surjective, but) fully faithful, is already problematic. Faithfulness means precisely that $\Pi_1^\Gamma = 1$, full faithfulness is stronger|stronger than the fact that trivial [p. 145]

$$\mathrm{Autext}(\Pi', \Pi)^\Gamma \longrightarrow \mathrm{Autext}(\Pi)^\Gamma$$

is surjective (in the realizable case, this surjectivity would be a consequence of the fact that $\forall$ class of homeomorphisms commuting with Γ contains a homeomorphism commuting with Γ) – we must add to this that any|any other lifting of Γ as an outer operation|group of Π' [i.e. a $\Gamma \longrightarrow \mathrm{Autext}_{\mathrm{lac}}(\Pi', \Pi)$

$$\begin{array}{c} \mathrm{Autext}_{\mathrm{lac}}(\Pi', \Pi) \\ \wr \\ \mathrm{Autext}_{\mathrm{lac}}(\Pi', I) \end{array}$$

], † having the =|same action of Γ on $I'^!$, is conjugate by|to the [illeg. 1]

Uncertainty retained from cancelled passages.
140-2-s288-D001 such that

previous one|previous one by an element of the kernel Π_1 of [scan 289]

$$\mathrm{Autext}_{\mathrm{lac|loc}}(\Pi', I'^!) \longrightarrow \mathrm{Autext}_{\mathrm{lac|loc}}(\Pi) \times \mathfrak{S}_{I'^!} \quad \ldots$$

We are going to restate the preceding conjecture in a slightly more general form. We are going to introduce a notation. If Γ is a finite group operating by outer automorphisms in an orientation-preserving way on a group with loops Π anabelian, giving rise to an extension $\mathcal{E}|E$ of Π by Γ, we will denote by $\Phi = \Phi(\Pi, \Gamma)$ ("fixed points") the set of Π-conjugacy classes of maximal partial sections $\neq (1)$ of the extension. (The intrinsic character of $\Phi(\Pi, \Gamma)$, independently of the particular choices of $\mathcal{E}|E$, has already been noted). It is clear that Φ is a Γ-set, we also know that it is finite.

Let now (Π', Γ) be an outer Γ-group with loops, with Γ operating in an orientation-preserving way ($\Gamma = \Gamma^+$) and faithfully (to simplify) let us fix a subset $I'^!$ of $I' = I(\Pi')$, stable under Γ, such that $\forall i' \in I'^!$, we have $\Gamma_{i'} \neq (1)$, and consider the quotient outer group, defined by plugging $I'^!$, namely Π †, – it is clear that Γ still operates on it. We want first to define a bijection

Uncertainty retained from cancelled passages.
140-2-s289-D001 $\Pi'_{I'^!}$

[scan 290]

$$(*) \qquad \Phi(\Pi, \Gamma) \xleftarrow{\sim} \Phi(\Pi', \Gamma)\amalg|\sqcup I'^! \qquad \text{[p. 146]}$$

m416

Marginal note. m416 assuming Π, Γ also anabelian, to be more sure!

a) Map $I' \longrightarrow \Phi(\Pi, \Gamma)$

Let $L_{i'} \subset \Pi'$ be of class $i' \in I'$, $Z_{i'}$ its centralizer in the extension $\mathcal{E}|E'$ of Γ by Π' so that $L_{i'}$ is an extension of $\Gamma_{i'}$ by $L_{i'}$. The passage to the quotient $\mathcal{E}|E' \longrightarrow \mathcal{E}|E$ defines a partial section $\Gamma_{i'} \hookrightarrow \mathcal{E}|E$, contained in a unique maximal partial section (probably it is already maximal =|itself?…). m417

Marginal note. m417 Yes, because for [illeg. 2] Γ cyclic, [illeg. 5] realizable…! ?

We thus find a map $I' \longrightarrow \Phi(\Pi, \Gamma)$, obviously compatible with the actions of Γ.

b) Map $\Phi(\Pi', \Gamma) \longrightarrow \Phi(\Pi, \Gamma)$

Every partial section $\neq 1$ of $\mathcal{E}|E'$ over Γ obviously defines one of $\mathcal{E}|E$ over Γ by passing to the quotient, we know as above that a maximal section gives a maximal section, by reducing to the case Γ cyclic. m418

Marginal note. m418 not clear, since the hypothesis that the $\Gamma_{i'} \neq (1)$ risks|risk not being satisfied|satisfied

c) Bijectivity of (*). It is not † proved (except if the initial situation is realizable, or if $I'^!$ reduced to a single element …but then Γ cyclic and the situation is realizable…)

Uncertainty retained from cancelled passages.
140-2-s290-D002 reduced

[scan 291]

This † granted we have defined a functor

$$(\pi', \Gamma, I'^!) \longmapsto (\pi, \Gamma, I'^!)$$

going from the category of outer Γ-groups with loops π' equipped with a subset $I'^!$ of $I' = I(\pi')$ stable such that $s' \in I' \Rightarrow \Gamma_{s'} \neq (1)$, under Γ, to the category of outer Γ-groups with loops, equipped with a subset $I'^!$ of $\Phi(\pi, \Gamma)$. The conjecture is that the functor (itself somewhat conjectural, unless we restrict to realizable source situations, whence target situations also realizable) is an equivalence of categories, i.e. 1°) fully faithful and 2°) essentially surjective.

The point 2°) is clear, when we restrict on both sides to realizable situations. But even in this case, the point 1°) is not proved. † I think that † if I' is reduced to one element, then the † properties of the "drilling a hole" functor will allow us to establish it easily. We also see that to establish it, we are

Uncertainty retained from cancelled passages.
140-2-s291-D001 posited
140-2-s291-D004 It may be that
140-2-s291-D005 the properties [illeg. 2] on the [illeg. 2] faithfulness
140-2-s291-D006 $\pi)\Gamma|\pi, \Gamma \quad I'^!|I''$

[scan 292]

reduced to establishing the bijectivity of the $(*)$, and the theorem of equivalence in the case where Γ is transitive on $I'^!$. [p. 147]

To summarize, the Program of attack on the Problem of realizability of an outer Γ-group with loops would be the following

I) Establish the bijectivity of the $(*)$ in the general case

(II) Establish the full faithfulness of the preceding functor (by reducing if necessary to the case where $I'^!$ is one orbit (singular) of Γ in $I' = I(\pi')$.

(III) In the case where $\Phi(\pi, \Gamma) = \emptyset$ i.e. the extension $\mathcal{E}$ has no element of finite order $\neq 1$, prove that $\mathcal{E}$, equipped with the set of $\mathcal{E}$-conjugacy classes of the centralizers of the L_i $(i \in I(\pi))$ in $\mathcal{E}$, is a group with loops.

Moreover, to have the indispensable preliminary properties of the sets $\Phi(\pi, \Gamma)$, we would have to begin by carrying out this program for the

[scan 293]

cyclic groups (operating in a direct way), and proceed in this case by dévissage.

If the complete classification theorem (as an equivalence of the 2-categories –of operations that are topological and of isotopic|homological operations –) were true, points I, II, III of this program should be so too, and it would therefore be a good program of approach. Points I and II should be valid, as soon as the 2-functor between 2-categories were 2-faithful (not necessarily 3-faithful), at least for realizable situations. We would thus say that an outer action of Γ on a π', equipped with an $I'^{!}$, is admissible, if

$$\mathrm{Aut}\,\mathrm{ext}(\pi', \Gamma, I'^{!}) \xrightarrow{\sim} \mathrm{Aut}(\pi, \Gamma, I'^{!})$$

and if moreover every † other action of Γ on π', the same action on π and the same map $I'^{!} \longrightarrow |-\Phi(\pi, \Gamma)$, is conjugate to the original action

Uncertainty retained from cancelled passages.

140-2-s293-D004 good

[scan 294]

[p. 148]

This posited, the necessary and sufficient condition for realizability of an outer faithful action of Γ on a π would then be that this action lifts (by "drilling holes" † for $I'^{!} = \Phi(\pi, \Gamma)$) to an action admissible, of Γ on a $(\pi', I'^{!})$, † (by the very definition, this lifting would be † then essentially unique) and that moreover the corresponding extension $\mathcal{E}'$ of Γ by π' be a group with loops.

25. Special Teichmüller groups.

Let us return to the notation of number 19, We are going to define a subgroup $SA^{\dagger|!}_{g,\nu}$ of $A^{!}_{g,\nu}$

$$SA^{\dagger|!}_{g,\nu}\{ = \{u \in A_g \mid u \text{ induces the identity on a neighborhood of } S_{g,\nu}\}$$

(set of automorphisms of $U_{g,\nu}$ which) i.e. induce|induces the identity in|in the complement of a compact set.

Unlike the other subgroups of A_g considered so far, this one is not a closed subgroup. N.B. In the case where we work on compact surfaces

Uncertainty retained from cancelled passages.

140-2-s294-D001 for $I'^{!}$

140-2-s294-D002 [illeg. 1]

140-2-s294-D003 by some

140-2-s294-D004 !

140-2-s294-D005 !

[scan 295]

with boundary instead of "punctured|punctured" compact surfaces (without boundary), it would be appropriate to take the group of automorphisms which induce the identity on the boundary.

$SA_{g,\nu}$ is an invariant subgroup of $A_{g,\nu}$, the quotient $A_{g,\nu}/SA_{g,\nu}$ being isomorphic to the group of germs of automorphisms of X_g in a neighborhood of $S_{g,\nu}|SA_{g,\nu}$, or again the group of germs at infinity of automorphisms of $U_{\nu,g}|U_{g,\nu}$ (in the complements of compact sets…)

We will have obviously, since

$$SA_{g,\nu} \subset A_{g,\nu}$$

$$(SA_{g,\nu})^0 \subset SA^0_{g,\nu}|A^0_{g,\nu},$$

let us put

$$S\Gamma^!_{g,\nu} = SA_{g,\nu}/SA^0_{g,\nu},$$

we will have a canonical homomorphism

$$S\Gamma^!_{g,\nu} \longrightarrow \Gamma^{!+}_{g,\nu}.$$

which we are going to interpret algebraically, in terms of the interpretation of $\Gamma^{!+}_{g,\nu} = \Gamma^!$ as

[scan 296]

the group of outer automorphisms of the group with loops $\pi_{g,\nu}$, inducing the identity on $S_{g,\nu} = I(\pi_{g,\nu})$. [p. 149]

† For $i \in I = S_{g,\nu}$, let us choose an † L_i of class i in $\pi_{g,\nu}$ — which amounts to giving oneself a "point" of the|of $B_{D^*_i}$ (≃ a universal covering $\widetilde{D}^*_i$) † and an isomorphism between its image in $B_{U=U_{g,\nu}}$ with the † reference "point" $s = s_{g,\nu}$, which served to define $\pi_{g,\nu}$ as $\mathrm{Aut}_{B_U}(s_\dagger) \simeq \pi_1(B_U, s)$.

This said, if u is an automorphism of $\pi = \pi_{g,\nu}$, the fact that it respects (strictly, by inducing the identity on $S_{g,\nu}$) the loop structure, is expressed by the existence of a family of elements $g_i \in \pi$, such that

$$u(\ell) = \mathrm{int}\, g_i^{-1}(\dagger \ell^{\dagger\alpha}) \qquad (i \in I, \forall \ell \in L_i)$$

— which|which g_i are determined by u † modulo right multiplication by a $\lambda_i \in L_i$. If we likewise have $v, (h_i)$, then For $\ell \in L_i$ we have

where $\alpha = \chi(u)$

$$\begin{aligned}(vu)(\ell) = v(\underbrace{u(\ell)}_{\mathrm{int}(g_i^{-1})\ell^\alpha}) &= v(\mathrm{int}(g_i^{-1}))v(\ell)^\alpha \\ &= \mathrm{int}(v(g_i^{-1}))\,\mathrm{int}(h_i^{-1}) \cdot (\ell^{\alpha\beta}) \\ &= \mathrm{int}(v(g_i^{-1}) \cdot h_i^{-1}) \cdot \ell^{\alpha\beta} = \mathrm{int}((h_i \cdot v(g_i))^{-1})\ell^{\alpha\beta}\end{aligned}$$

thus vu is compatible with the system of † $h_i v(g_i)$ Let us put

N.B. $\alpha = \chi(u)$ $\beta = \chi(v)$

Uncertainty retained from cancelled passages.

140-2-s296-D001 [illeg. 1]
140-2-s296-D003 ℓ
140-2-s296-D004 on
140-2-s296-D005 element
140-2-s296-D006 g, ν
140-2-s296-D007 $\chi(u)$
140-2-s296-D008 $\chi(u)$
140-2-s296-D009 [illeg. 1]
140-2-s296-D011 $u|v$

[scan 297]

$$(v, (h_i)) \cdot (u, (g_i)) = (vu, h_i v(g_i)\dagger),$$

S

We then find a group structure on the set $S\mathcal{E}^!$ of $(u, (g_i))$, a subgroup of the semidirect product of $\mathcal{E}^! = \mathrm{Autext}_{\mathrm{lac}}(\pi, I)$ identity on I by π^I, on which $\mathcal{E}^!$ operates in the obvious way. The natural homomorphism

$$S\mathcal{E}^! \longrightarrow \mathcal{E}^!$$

is surjective, and its kernel is canonically isomorphic to $\prod_{i\in I} L_i \simeq T|\mathbb{T}^I$, where $T|\mathbb{T}$ is the orientation module. † whence an extension structure † m422

Marginal note. m422 where $\mathcal{E}$ operates on $T|\mathbb{T}^I$ via its action on $T|\mathbb{T}$

$$1 \longrightarrow T|\mathbb{T}^I \longrightarrow S\mathcal{E}^! \longrightarrow \mathcal{E}^! - 1$$

†

Uncertainty retained from cancelled passages.

140-2-s297-D001 h_i

140-2-s297-D002 homomorphism homomorphisms

140-2-s297-D003 central

140-2-s297-D004 $\longrightarrow$ $T|\mathbb{T}$ $\longrightarrow$ $\longrightarrow$ on $T|\mathbb{T}$ $\simeq$ $\mathbb{Z}$!|! !|!

[scan 298]

[p. 150]

†

I would like to interpret this extension as the inverse image of a canonical extension of $\Gamma^!$ by $T|\mathbb{T}^I$ (canonical in any case, once the $\widetilde{D}_i^*$ have been chosen). For this, note that if $\ddot{u}|\dot{u}$ is an autoequivalence|autoequivalence † of the situation $B_{D^*} \longrightarrow B_U$ inducing the identity on I we can consider the $\dot{u}(\widetilde{D}_i^*)$, and we can consider the $\tilde{\dot{u}}$ over $\dot{u}$, namely the systems $(\dot{u}, \gamma_i)$, where for every|every $i \in I$, $\gamma_i : \widetilde{D}_i^* \longrightarrow \dot{u}(\widetilde{D}_i^*)$ [determined modulo|by element|element of $T|\mathbb{T}$]. † The group of isomorphism classes of automorphisms|automorphisms of $(B_{D^*} \longrightarrow B_U, (\widetilde{D}_i^*))$, or if we prefer, of isomorphism classes of automorphisms|automorphisms of

$$(B_I \longrightarrow B_{D^*} \longrightarrow B_U)$$

Uncertainty retained from cancelled passages.

140-2-s298-D001 $\forall|\mathrm{tt}$ which we isomorphism $\widetilde{U}(i)|U(i)$ universal automorphism

140-2-s298-D002 in

140-2-s298-D003 The set of

[scan 299]

inducing the identity on I, is therefore an extension of $\Gamma^!$ by $T|\mathbb{T}^I$, say $S\Gamma^!$. We are going to define a homomorphism of exact sequences

$$\begin{array}{ccccccccc} 1 & \longrightarrow & T^I & \longrightarrow & S\mathcal{E}^! & \longrightarrow & \mathcal{E}^! & \longrightarrow & 1 \\ & & \| & & \downarrow & & \downarrow & & \\ \dagger & \longrightarrow & T^I & \longrightarrow & S\Gamma^! & \longrightarrow & \Gamma^! & \longrightarrow & 1 \end{array}$$

which proves that $S\mathcal{E}|\varepsilon^!$ is indeed the inverse image of † the extension $S\Gamma^!$ of $\Gamma^!$ by $T|\mathbb{T}^I$.

For this, we note that $\mathcal{E}|\varepsilon^!$ is the group of isomorphism classes of autoequivalences|of autoequivalences of

$$\begin{array}{ccc} B_{D^*} & & \\ \downarrow & & \\ B_U & \longleftarrow & \text{points} \end{array}$$

and $S\mathcal{E}|\varepsilon^!$ that of isomorphism classes of † autoequivalences of

$$\begin{array}{ccc} B_{D^*} & \longleftarrow & B_I \\ \downarrow & \nwarrow & \downarrow \\ B_U & \longleftarrow & \text{point } s \end{array} \qquad \begin{array}{l} \text{isomorphism} \\ \text{of} \\ \text{commutativity} \end{array}$$

which maps naturally to that of isomorphism classes of autoequivalences of the 2-diagrams

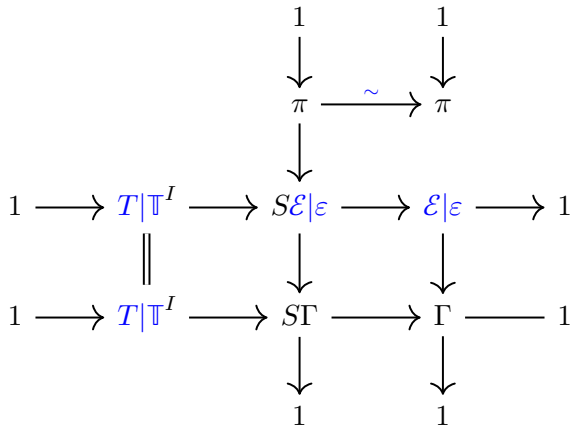
$$B_I \longrightarrow B_{D^*} \longrightarrow B_U$$

.

Uncertainty retained from cancelled passages.
140-2-s299-D002 [*illeg.* 1] Γ
140-2-s299-D003 [illeg. 1]
140-2-s299-D005 [illeg. 1]

[scan 300]
[p. 151]

In fact, in all this there was no reason at all to restrict ourselves to the † automorphisms of π (outer or total) inducing the identity on I. In any case we find extensions $S\mathcal{E}|\varepsilon$ of $\mathcal{E}|\varepsilon$ by $T|\mathbb{T}^I$, $S\Gamma$ of Γ by $T|\mathbb{T}^I$, and a homomorphism of extensions



so that we can consider $S\mathcal{E}|\varepsilon$ as an extension of Γ by $T|\mathbb{T}^I \times \pi$

$$1 \longrightarrow T|\mathbb{T}^I \times \pi \longrightarrow S\mathcal{E}|\varepsilon \longrightarrow \Gamma \longrightarrow 1$$

from which the extension $\mathcal{E}|\varepsilon$ resp. $S\Gamma$ is obtained by dividing by $T|\mathbb{T}^{I|S}$ resp. by π.

†|Returning now to the model case $U_{g,\nu} = X_{g,\nu} \setminus S_{g,\nu}$, we will use|use the|some notations †

$$\mathcal{E}|\varepsilon_{g,\nu} = \mathrm{Aut}\dagger_{\mathrm{lac}}(\pi|\Pi_{g,\nu})$$

($\overset{\mathrm{can}}{\simeq} \Gamma_{g,\nu+1}$ if $(g,\nu) \neq (0,0),(0,1)$ i.e. †) [illeg. 1]

and

$$S\Gamma_{g,\nu} = S\mathrm{Autext}_{\mathrm{lac}}(\pi|\Pi_{g,\nu})$$

(extension of $\Gamma_{g,\nu}$ by $T|\mathbb{T}^I$)

and $S\mathcal{E}|\varepsilon_{g,\nu}$ ($\simeq S\Gamma'_{g,\nu+1}/T|\mathbb{T}|\Gamma_{s_\nu|g,\nu}$ † where $S\Gamma'_{g,\nu+1}$ denotes the subgroup of $S\Gamma'_{g,\nu+1}$ of the elements which fix the last element s_ν …)

Uncertainty retained from cancelled passages.
140-2-s300-D001 outer
140-2-s300-D002 Returning
140-2-s300-D004 $S\mathcal{E}_{g,\nu}$
140-2-s300-D005 ext
140-2-s300-D006 $\pi_{g,\nu} \neq 0$

[scan 301]

We denote by $S\Gamma^{!}_{g,\nu}$ and $S\mathcal{E}|\varepsilon^{!}_{g,\nu}$ the subgroups|subgroups of the preceding $S\Gamma_{g,\nu}$, $S\mathcal{E}|\varepsilon_{g,\nu}$ which induce the identity on $I = S_{g,\nu}$.

And let us finally return to the relations with $SA|\mathcal{A}_{g,\nu}$, we are going to define

$$SA_{g,\nu} \longrightarrow S\Gamma^{!+}_{g,\nu}$$

by noting that if a $u \in A_{g,\nu}$ induces the identity in a neighborhood of $S_{g,\nu}$, then $u_\bullet|u_i(\widetilde{D}_i^*|\widetilde{D}_i) \simeq | = \widetilde{D}_i^*$ and we shall be able to define an element of $S\Gamma^{!}_{g,\nu}$ by taking as γ_i the identities. The fact that the homomorphism obtained is trivial on $(SA_{g,\nu})^0$ is probably trivial, hence

$$SA_{g,\nu}/(SA_{g,\nu})^0 \longrightarrow S\Gamma^{!+}_{g,\nu}.$$

To say that this is surjective means that $\forall$ automorphism $\in A^{!+}_{g,\nu}$ (i.e. all|the automorphisms of $U_{g,\nu}$ inducing the identity on $S_{g,\nu}$ and respecting the orientation) is isotopic (by an isotopy, if one wishes, which remains the identity outside a small neighborhood of $S_{g,\nu}$) to an automorphism which is the identity on a neighborhood, this is easy. Injectivity perhaps more delicate, it comes down essen

[p. 300]

[scan 302]

tially to: determining the kernel of

[p. 152]

$$SA|\mathcal{A}_{g,\nu}/(SA_{g,\nu})^0 \longrightarrow \Gamma^{!}_{g,\nu}, \quad \text{i.e.} \quad SA_{g,\nu} \cap A^0_{g,\nu}/(SA_{g,\nu})^0$$

as $T^I = T^{S_{g,\nu}}$. We can probably reduce to the context of compact surfaces with boundary (let us denote by a $'$ the corresponding topological groups), we have

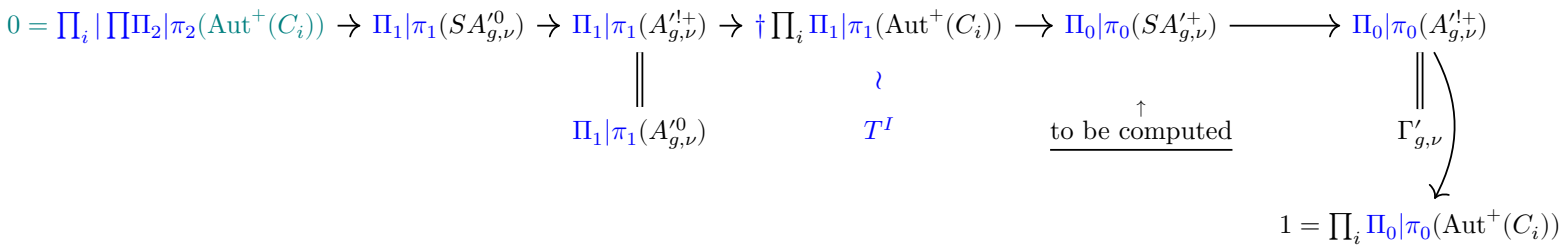
$$1 \longrightarrow SA'^{+}_{g,\nu} \longrightarrow A'^{!+}_{g,\nu} \longrightarrow \prod_{i \in S_{g,\nu}} \mathrm{Aut}^{+}(C_i) \longrightarrow 1$$

m424

Marginal note. m424 homotopic to the standard circle † group $\mathbb{U}$, twisted by $T|\Pi$

where the $+$ indicates the automorphisms preserving the orientation, and the C_i are the connected components of the boundary. We conclude from this the homotopy exact sequence

[Diagram continued on the landscape page; 140-2-302-D1]


$0 = \prod_i | \prod \Pi_2 | \pi_2(\mathrm{Aut}^+(C_i))$
$\Pi_1 | \pi_1(SA'^0_{g,\nu})$
$\Pi_1 | \pi_1(A'^{!+}_{g,\nu})$
$\dagger \prod_i \Pi_1 | \pi_1(\mathrm{Aut}^+(C_i))$
$\Pi_0 | \pi_0(SA'^+_{g,\nu})$
$\Pi_0 | \pi_0(A'^{!+}_{g,\nu})$
$\Pi_1 | \pi_1(A'^0_{g,\nu})$
$\wr$
T^I
to be computed
$\Gamma'_{g,\nu}$
$1 = \prod_i \Pi_0 | \pi_0(\mathrm{Aut}^+(C_i))$

$$\left[\text{and} \quad \Pi_i|\pi_i(SA'_{g,\nu}) \xrightarrow{\sim} \Pi_i|\pi_i(A'^0_{g,\nu}) \quad \text{if } i \geq 2\right]$$

i.e.

$$\begin{cases} 0 \longrightarrow \Pi_1|\pi_1(SA'^+_{g,\nu}) \longrightarrow \Pi_1|\pi_1(A'^0_{g,\nu}) \longrightarrow T^I \longrightarrow \Pi_0|\pi_0(SA'^{!+}_{g,\nu}) \longrightarrow \Gamma'^{!+}_{g,\nu} \longrightarrow 1. \\ \Pi_i|\pi_i(SA'^+_{g,\nu}) \xrightarrow{\sim} \dagger\Pi_i|\pi_i(A'^0_{g,\nu}) \quad \text{for } i \geq 2 \end{cases}$$

†

We have an obvious homomorphism: (by 'gluing disks') $A'^0_{g,\nu} \longrightarrow A_{g,\nu}$ inducing $SA'_{g,\nu} \longrightarrow SA_{g,\nu}$, and these are surely homotopy equivalences, so † the preceding exact sequence [p. 301]

Uncertainty retained from cancelled passages.

140-2-s302-D001 [illeg. 1]
140-2-s302-D002 Π_0
140-2-s302-D003 [*illeg.* 1]
140-2-s302-D004 [illeg. 1]
140-2-s302-D006 [illeg. 1]

[scan 303]

must be interpretable as an exact sequence

$$\begin{cases} 0 \longrightarrow \Pi_1|\pi_1((SA|\mathcal{A}_{g,\nu})^0) \longrightarrow \Pi_1|\pi_1(A^0_{g,\nu}) \longrightarrow T^I \longrightarrow \Pi_0|\pi_0(SA^{!+}_{g,\nu}) \longrightarrow \Gamma^{!+}_{g,\nu} \longrightarrow 1 \\ \Pi_i|\pi_i((SA_{g,\nu})^0) \xrightarrow{\sim} \Pi_i|\pi_i(A^0_{g,\nu}) \quad \text{for } i \geq 2. \end{cases}$$

In the anabelian case, we do indeed find, since $\Pi_1|\pi_1(A^0_{g,\nu}) = 0$, an extension structure

$$\boxed{1 \longrightarrow T^{S_{g,\nu}} \longrightarrow \Pi_0|\pi_0(SA_{g,\nu}) \longrightarrow \Gamma^{!+}_{g,\nu} \longrightarrow 1}$$

† and moreover

$$\Pi_i|\pi_i((SA_{g,\nu})^0) = 0 \quad \text{for } i \geq 2.$$

In the abelian case, we must make explicit †

$$\Pi_1|\pi_1(A^0_{g,\nu}) \longrightarrow T^{S_{g,\nu}}$$

and we are going to distinguish the two abelian cases (with the understanding that we have $I \neq \emptyset$!) — we have|i.e. always $g = 0$, $\nu = 1$ or 2. †

In any case, introducing a complex|complex analytic structure and the group G, identity component of the group of complex automorphisms, we have have

$$G \xrightarrow{\approx|\sim} A^0_{g,\nu}.$$

† :

a) Case $g = 0$, $\nu = 2$ † $G \simeq \mathbb{C}^*$, we see that if the loop structure of $\pi|\Pi$ is defined by the two isomorphisms $k_i|\kappa_i : T \longrightarrow \pi|\Pi$ ($i \in I = \{s_{g0}|s_{0,0}|s_\infty, s_{0g}|s_{0,1}|s_0\}$) [p. 302]

Uncertainty retained from cancelled passages.

140-2-s303-D001 from which we deduce $\Pi_0|\pi_0$ S
140-2-s303-D002 [illeg. 1]
140-2-s303-D004 As
140-2-s303-D005 thus
140-2-s303-D006 thus $= | \neq$

[scan 304]

then $\Pi_1|\pi_1(G) \longrightarrow T^I$ is identified with $T \longrightarrow T^I$ whose components are these|the two $k_i|\kappa_i$ [p. 153] (which are symmetric, so this is an injective homomorphism whose image is the kernel of the sum map $T^I \longrightarrow T$, so here the kernel of

$$\Pi_0|\pi_0(SA|\mathcal{A}_{g,\nu}) \to \Gamma^{!+}_{g,\nu}$$
$$\|$$
$$0\dagger$$

(of which|thus $\Pi_0|\pi_0(SA_{g,\nu})$ itself) is isomorphic, not to T^I, but to its quotient T. As for $(SA'_{g,\nu})^0$, all its $\Pi_i|\pi_i$ $(i \geq 1)$ are zero — it is again ∞-connected.

b) Case $g = 0$, $\nu = 1$. Then $G \simeq \mathrm{Aff}(1, \mathbb{C})$, $\Pi_1|\pi_1(G) \simeq T$ and $\Pi_1|\pi_1(G) \longrightarrow T^I = T$ is an isomorphism. Here, $\dagger$ the kernel of the homomorphism

$$\Pi_0|\pi_0(SA_{g,\nu}) \to \Gamma^{!+}_{g,\nu}$$
$$\wr$$
$$0$$

is zero $\dagger$

[V-shaped mark to the right of this 0, at the end of the first struck-out line.]

i.e. $\Pi_0|\pi_0(SA_{g,\nu}) = 0$. In this case too, we find that $\Pi_i|\pi_i(SA^0_{g,\nu}) = 0$ for every $i \geq 1$. We therefore find

Theorem Suppose $\nu > | \geq 0$. If (g, ν) is anabelian, then $\Pi_0|\pi_0(SA_{g,\nu})$ is canonically isomorphic to $S\Gamma^{!+}_{g,\nu}$. In all cases, $(SA_{g,\nu})^0$ is ∞-connected.

[p. 303]

Uncertainty retained from cancelled passages.

140-2-s304-D001 zero

140-2-s304-D002 again

140-2-s304-D003 formally|formally [illeg. 1]

We are going to make explicit a compatibility relation for $S\Gamma$ relative to a π with loops, fixing an $i \in I = \mathcal{I}(\pi)$, hence a stabilizer $\Gamma_i \subset \Gamma$ whose inverse image $(S\Gamma)_i$ in $S\Gamma$ is thus an extension of Γ_i by T^I. Using the projection $T^I \overset{\mathrm{pr}_i}{\longrightarrow} T$, we find an extension of Γ_i by T, which we are going to describe in another way. The extension $S\Gamma$ is defined intrinsically by the datum of I realizations $(\pi_i)_{i\in I}$ of the group outer π by groups, with in each π_i an $L_i \subset \pi_i$ of class i (that is it, the datum of a system of the $(\widetilde{D}^*_i)$!). For an outer automorphism $\dagger$ with loops $u \in \Gamma$ of π, $\dagger$ for $\forall i$ it $\dagger$ is possible to realize it by $u_i \in \mathrm{Aut}_{\mathrm{lac|loc}}(\pi_i)$, with $u_i(L_i) \subset L_i$ — this u_i is determined modulo right multiplication by an $\mathrm{int}(k_i|\kappa_i(\alpha))$, where $\alpha \in T$. Of course, $S\Gamma$ is the subgroup of the|of the [scan 305] [p. 304]

Uncertainty retained from cancelled passages.

140-2-s305-D001 u

140-2-s305-D002 [illeg. 1]

140-2-s305-D003 there are ways

$\prod_{i\in I} \mathrm{Aut}_{\mathrm{lac|loc}}(\pi_i)$, $\dagger$ the inverse image of the diagonal under the natural homomorphism from [scan 306] [p. 154]

this product to $\mathrm{Autext}_{\mathrm{lac|loc}}(\pi)^I$, † which is indeed extension of the said diagonal (identified with Γ) by T^I. If we look at the subextension obtained by restriction to Γ_i, we see that the projection $T^I \xrightarrow{\mathrm{pr}_i} T$ is stable under Γ_i; thus we deduce from it an extension of Γ_i by T, which is none other than the group of automorphisms with loops of π_i which normalize L_i — which is indeed an extension of Γ_i by L_i (for intersection with $\pi_i \subset \mathrm{Aut}_{\mathrm{lac|loc}}(\pi_i)$ is|being reduced to L_i).

This shows us that the extension of Γ by T^I has a marked tendency not to be trivial, for this is so (for $i \in I$ fixed) for the extension of Γ_i by T to which it gives rise. If for example we have a finite subgroup $G \subset \Gamma_i^+$, the induced extension is never trivial if $G \neq 1$, we have seen this. In fact, G must be cyclic and its inverse image in the extension in question is $\simeq T$… [p. 305]

Uncertainty retained from cancelled passages.
140-2-s306-D001 formed of [illeg. 1]
140-2-s306-D002 whence an

25bis. Case of the two groups. Return to the notations [scan 307]

We place ourselves first to fix ideas in the topological and discrete case, but the motivation is the case of an algebraic curve U over a finitely generated field K, where we have both the group $G_K = \mathrm{Aut}_K(U)$ and $\Gamma = \mathrm{Gal}(\overline{K}/K)$ which operate by outer automorphisms on $\pi_1(U_{\overline{K}})$. In this case, G and Γ commute, but we can look more generally at the case of the group (bigger than G_K) $G_{\overline{K}} = \mathrm{Aut}_{\overline{K}}(U)$, on which Γ operates (in a not necessarily trivial way — this operation describes a finite étale algebraic group over K. [p. 155]

Anabelian case so $G|G_K$ finite

So suppose we have a surface U (orientable, $U = X - S$, X compact connected, S finite) on which|which operate two groups G, Γ, the action of Γ normalizing that of G — so we have a group $\mathcal{G} = \Gamma.G$ (semidirect product, for a certain action of Γ on G) which operates on U. We suppose G finite, but not necessarily Γ finite. [p. 306]

[scan 308]

We suppose chosen a universal covering $\widetilde{U}$ of U, hence a group with loops $\pi = \mathrm{Aut}(\widetilde{U})$, on which $\mathcal{G}$ operates by outer automorphisms, hence extension

$$(1) \qquad 1 \longrightarrow \pi \longrightarrow \mathcal{E}|\varepsilon \longrightarrow \mathcal{G} \longrightarrow 1$$

m426

Marginal note. m426 We are going to suppose the action of $\mathcal{G}$ faithful

If the action of $\mathcal{G}$ on U modulo isotopy is faithful, then $\mathcal{G} \hookrightarrow \mathrm{Autext}(\pi)$, and the preceding extension is the inverse image of the Teichmüller extension of π

$$1 \longrightarrow \pi \longrightarrow \mathrm{Aut}_{\mathrm{lac}}(\pi) \longrightarrow \mathrm{Autext}_{\mathrm{lac}}(\pi) \longrightarrow 1.$$

† We shall have to look at other universal coverings than $\widetilde{U}$, and their isomorphisms with $\widetilde{U}$. When $\widetilde{U} = \widetilde{U}(P)$ is the universal covering based at a certain $P \in U$, then for universal coverings $U(Q)$ based at † a point†|points, the U-isomorphisms $\widetilde{U}(P) \xrightarrow{\sim} \overline{U}(Q)|\widetilde{U}(Q)$ correspond thus to the classes of paths from P to Q.

Let † $Q \in U$ be such that its isotropy group G_Q † in G is such that $G_Q^+ \neq 1$. (So G_Q^+ is cyclic). Choosing a [p. 307]

Uncertainty retained from cancelled passages.
140-2-s308-D001 Let
140-2-s308-D002 [illeg. 1]
140-2-s308-D003 s
140-2-s308-D004 [illeg. 1]

140-2-s308-D005 [illeg. 1]

[scan 309]

class of paths λ from P to Q, identifying $\overline{U}(P)|\widetilde{U}(P)$ with $\overline{U}(Q)|\widetilde{U}(Q)$, we find an operation [p. 156]
of G_Q on $\widetilde{U}(P)$ i.e. a lifting $G_Q \xrightarrow{r_{G_Q}} \mathcal{E}|\varepsilon$ in the extension (1) — the change of class of paths from λ to λ' will give a lifting r'_{G_Q} which † will be conjugate to r by a unique element of $\pi = \pi_1(U, P)$ (uniqueness comes from $\pi^{G_Q} = \{1\}$), namely the one which makes one pass from one path to another.

Let us consider the stabilizer Γ_Q of Q in Γ, † which of course operates on $\widetilde{U}(Q)$ just like G_Q (in fact it is $\mathcal{G}_Q$ which operates hence $r : \mathcal{G}_Q \to \mathcal{E}|\varepsilon$ …), so via λ we also have a lift[ing]|lifting $r_{\Gamma_Q} : \Gamma_Q \longrightarrow \mathcal{E}|\varepsilon$, which has even the property of normalizing r_{G_Q} (i.e. operation of † Γ_Q on it, which is the one coming from the given operation of Γ on G). m427

Marginal note. m427 N.B. As the set † of points Q is finite, and $\mathcal{G}$ operates on it, the orbit of Q under $\mathcal{G}$ is finite, i.e. $\mathcal{G}_Q$ of finite index in $\mathcal{G}$. and likewise Γ_Q in Γ.

To simplify, let us suppose all the same that Γ operates trivial[ly] on G i.e. $\mathcal{G} = \Gamma \times G$, then $r_{\Gamma_Q}(\Gamma_Q) \subset \mathcal{E}|\varepsilon$ is contained in the centralizer of r (or of $r_{G_Q}(G_Q)$), [p. 308]

Uncertainty retained from cancelled passages.
140-2-s309-D001 is
140-2-s309-D002 [illeg. 2]
140-2-s309-D003 [illeg. 1]
140-2-s309-D004 [illeg. 1]

[scan 310]

and since $\pi^{r(G_Q)} = 1$, so the homomorphism

$$\mathrm{Centr}(r_\dagger(G_Q)) \longrightarrow \mathcal{G}$$

is injective, the lifting in question r_{Γ_Q} is uniquely determined by the preceding condition. m428

Marginal note. m428 N.B. Independent[ly]|Independently of every|the hypothesis that Γ centralizes G, the normalizer of $r(G_Q)$ in π, equal to its "centralizer", is reduced to 1, so $\mathrm{Norm}(r(G_Q)) \longrightarrow \mathcal{G}$ is injective

In fact, the image of $\mathrm{Centr}(r(G_Q))$ in $\mathcal{G}$ contains $\mathcal{G}_Q$, and † even the lifting $\mathcal{G}_Q \xrightarrow{r_{\mathcal{G}_Q}} \mathcal{E}|\varepsilon$. The|The interesting thing is that the|the choice of a lifting† of the|of the (small) group G_Q, already imposes the choice of a lifting of the|of the (big) group Γ_Q, or $\mathcal{G}_Q$.

I say that the image in $\mathcal{G}$ of the centralizer (and even of the normalizer) of $r(G_Q)$ is $\mathcal{G}_Q$ itself (a priori it contains it). Returning|Let us return to $\widetilde{U}(Q)$ itself, this means that † if $g \in \mathcal{G}$ is such that there exists an automorphism $\tilde{g}$ of $\widetilde{U}(Q)$ which lifts g, while † normalizing † the action of G_Q, then $g \in \mathcal{G}_Q$ (and $\tilde{g}$ is the obvious lifting). Indeed, if $\tilde{g}$ † normalizes the action of G_Q, it leaves invariant|sends the set of fixed points of G_Q in $\widetilde{U}(Q)$, which is reduced † to the point $\widetilde{Q}|\overline{Q}$ [p. 309]

Uncertainty retained from cancelled passages.
140-2-s310-D001 G_Q
140-2-s310-D002 [illeg. 5]
140-2-s310-D004 s
140-2-s310-D006 [illeg. 1]
140-2-s310-D007 commuting
140-2-s310-D008 with
140-2-s310-D009 commutes

140-2-s310-D010 [illeg. 1]

[scan 311]

The set $\widetilde{U}|U^{!}$ of points $Q \in U$ such that $G_Q^+ \neq (1)$ is stable under the action of $\mathcal{G}$, and is [p. 157] identified (with this action) with the set of maximal liftings modulo π of (cyclic) subgroups $\neq 1$ of $G_\dagger^+$. When one knows, for a partial lifting r of a G_Q into $\mathcal{E}$ what its centralizer in $\mathcal{E}$ is i.e. the corresponding lifting of $\mathcal{G}_Q$, well|then in the $=$ way for the conjugates of r by any element g (not only of π, but $=$ of $\mathcal{E}$), by simple conjugation.

So the cases to be determined correspond in practice to the orbits of $\mathcal{G}$ in $U^!$.

One may be interested in describing $\mathcal{G}$ or Γ as a subgroup of $\operatorname{Aut}\operatorname{ext}_{\text{lac}}(\pi) = \Gamma'$ giving rise to the extension $\mathcal{E}'$ of Γ' by π (thus $\mathcal{E} \subset \mathcal{E}'$). But for every lifting r of a G_Q, let us consider $\operatorname{Centr}_{\mathcal{E}'}(r)$, $\dagger$. we shall have|we still have $\operatorname{Centr}_{\mathcal{E}'}(r) \cap \pi = (1)$ i.e. we find a section over the image of this centralizer in Γ', say $\Gamma'(Q)$. This image depends only on Q i.e. on the π-conjugacy class of r or of $r(G_Q)$, and is replaced by a G-conjugate when Q is replaced by a G-conjugate. This [p. 310]

Uncertainty retained from cancelled passages.

140-2-s311-D001 Q

140-2-s311-D002 [illeg. 1]

[scan 312]

said, the intersection $\Gamma'^\natural$ of the $\Gamma'(Q)$, for $Q \in U^!$, is a subgroup of Γ' which contains the intersection $\dagger$ $\mathcal{G}^\natural$ of the $\mathcal{G}_Q^\dagger$, and the centralizer of G in $\Gamma'^\natural$ contains in the $=$ way the intersection $\Gamma^\natural$ of the Γ_Q, which is an invariant subgroup of finite index in Γ. And one can then propose to oneself whether it is possible to characterize at least the|a finite subgroup $\Gamma^\natural$ of Γ as $\operatorname{Centr}_{\Gamma'^\natural}(G)$, and to recover from it perhaps Γ as the normalizer of $\Gamma^\natural$ in $\operatorname{Centr}_G(\Gamma)$.

I am more particularly interested in the profinite variant of this, in the case where $U = \mathbb{P}^1_{\mathbb{Q}} \setminus \{0, 1, \infty\}$, $G = \mathfrak{S}_3$, $\Gamma =$ Galois group of $\overline{\mathbb{Q}}/\mathbb{Q}$, $P|\rho = \exp(2i\pi/6)$. algebraic closure in $\mathbb{C}$

I have not checked that $\Gamma \longrightarrow \operatorname{Autext}_{\text{lac}}(\hat{\pi})$ is injective, this does not prevent doing calculations in $\operatorname{Aut}_{\text{lac}}(\hat{\pi})$, be they heuristic for the moment. [p. 311]

Uncertainty retained from cancelled passages.

140-2-s312-D001 of

140-2-s312-D002 $\natural$

[scan 313]

I am running into notational troubles – too many groups are denoted by the letter Γ (possibly [p. 158] with primes, subscripts, superscripts …). There are three types of groups that enter into my reflections:

a) the Teichmüller groups and its variants, which play the role of the|of "universal" groups operating (possibly modulo isotopy) on surfaces, or on outer groups $\dagger$ with loops. These groups tend to be infinite

b) Groups (most often finite) operating on topological surfaces, or on geometric algebraic curves (without, in that case, moving the base field).

m430

Marginal note. m430 Cases b) and c) sometimes mix (in a G groups $\mathcal{G}$ extension of a Galois group Γ by a finite group G) in the case of G Algebraic Geometry.

c) Profinite Galois groups (hence infinite), chiefly of fields of finite type over $\mathbb{Q}$, operating "arithmetically" on surfaces and their geometric $\hat{\pi}_1$.

It is because of the deep analogies between cases b) and c), and their close relations with case a), that I had been led to adopt common notations, but [p. 312]

Uncertainty retained from cancelled passages.
140-2-s313-D001 with loops

[scan 314]

which in the long run end up leading to collisions. There is thus reason to review the notations. I will reserve the letter G and variants for geometric actions (case b)) of groups, most often finite, the letter Γ and variants for Galois groups, the letter $\mathcal{G}$ for mixed groups.

As for the "universal" groups of Teichmüller type, like those denoted $\Gamma_{g,\nu}$ previously, I will instead denote them $\mathcal{T}$, $\mathcal{T}_{g,\nu}$ (initial of "Teichmüller", whereas Γ, G are the initial of Galois).

The Galois group over $\mathbb{Q}$ of the algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$ in $\mathbb{C}$ † deserves a special letter, I will denote it $\mathbf{\Gamma}$. The quotient † $\mathrm{Norm}_{\widehat{\widehat{\mathcal{T}}}_{g,\nu}}(\widehat{\mathcal{T}}_{g,\nu})/\widehat{\mathcal{T}}_{g,\nu}$, which is more akin to a Galois group than to a Teichmüller group, will be denoted $\mathbf{\Gamma}_{g,\nu}$ (bold letter!). In the [p. 313]

Uncertainty retained from cancelled passages.
140-2-s314-D001 of $\overline{\mathbb{Q}}$ over
140-2-s314-D002 by

[scan 315]

case $(g,\nu) = (0,3)$ which occupies me more particularly, $\mathbf{\Gamma}_{0,3}$ is identified with the † centralizer [p. 159] of $G = \mathfrak{S}_3 = \mathcal{T}^+_{0,3}$ in $\widehat{\widehat{\mathcal{T}}}_{0,3}$ – it is contained in $\widehat{\widehat{\mathcal{T}}}^!_{03}$, and perhaps equal. We have a canonical homomorphism [m431]

Marginal note. [m431] $\widehat{\widehat{\mathcal{T}}}$ is here the $\frac{1}{2}$ direct product of $\mathfrak{S}_3 = \mathcal{T}^+$ by $\mathcal{T}^!$

$$\mathbf{\Gamma} \longrightarrow \mathbf{\Gamma}_{03} \qquad (\text{more generally } \mathbf{\Gamma} \longrightarrow \mathbf{\Gamma}_{g,\nu})$$

for which I do not know for the moment whether it is injective, and still less whether it is surjective. The preceding reflections suggest conditions on the image, which are especially interesting if one admits the relations

$$\hat{\pi}^{\rho} = \hat{\pi}^{\sigma_0} = (1)$$

We have denoted† by $\mathcal{E}_{g\nu}$ the canonical extension of $\mathcal{T}_{g,\nu}$ by $\pi_{g,\nu}$ (which for (g,ν) anabelian is identified with $\mathcal{T}'_{g,\nu+1}$).

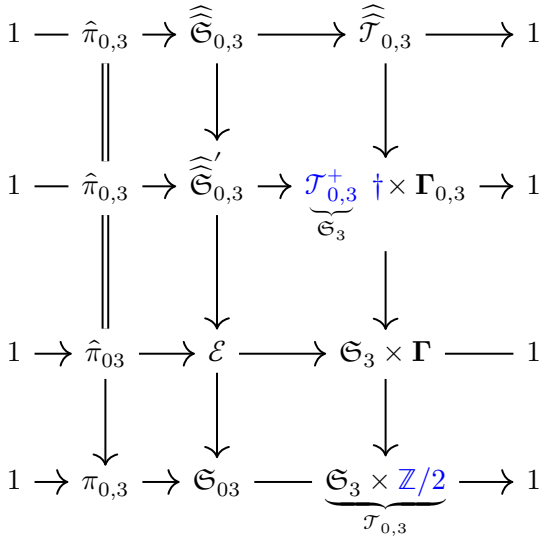
The outer operation of a group G, Γ or $\mathcal{G}$ also defines an extension by π (or by $\hat{\pi}$), which we have also denoted|denoted by the letter $\mathcal{E}$ (initial of extension) – there is again a collision of notations. [p. 314]

Uncertainty retained from cancelled passages.
140-2-s315-D001 normalizer
140-2-s315-D002 will

[scan 316]

I will take the letter $\mathfrak{S}$ (which is a counterpart to $\mathcal{T}$) for these extensions in the universal Teichmüller-style cases, (writing $\mathfrak{S}_{g,\nu}$ instead of the $\Gamma'_{g,\nu+1}$, since the viewpoint is different…), and keeping the letter $\mathcal{E}$ in the|the preceding|preceding cases. So $\mathcal{E}$ tends to be a subextension of an $\mathfrak{S}$.

Admitting that $\mathbf{\Gamma} \longrightarrow \mathbf{\Gamma}_{0,3}$ is injective, we would thus have



m432

Marginal note. m432 N.B. We denote by $\widehat{\widehat{\mathfrak{S}}}'_{g,\nu}$ the normalizer of $\widehat{\mathfrak{S}}^{+}_{g,\nu}$ in $\widehat{\widehat{\mathfrak{S}}}_{g,\nu}$, an extension of $\mathbf{\Gamma}_{g,\nu}$ by $\widehat{\mathfrak{S}}^{+}_{g,\nu}$

If|Let $\gamma \in \mathbf{\Gamma}_{0,3}$, †for it to be in the image of $\mathbf{\Gamma}$ †it is necessary that it|that they admit|admit a lifting u which commutes with ρ, and a lifting which commutes with σ_0 (which|Which, as soon|of the as $\gamma \in \widehat{\widehat{\mathcal{T}}}_{0,3}$, implies [p. 315]

Uncertainty retained from cancelled passages.

140-2-s316-D001 $\mathbf{\Gamma}$

140-2-s316-D002 which is

140-2-s316-D003 have|is

[scan 317]

already that γ in $\widehat{\widehat{\mathcal{T}}}_{0,3}$ commutes with $\mathcal{T}^{+}_{0,3} = \mathfrak{S}_3$, i.e. that they is in $\mathbf{\Gamma}_{0,3}$). It could be that [p. 160] every element of $\mathbf{\Gamma}_{0,3}$ already has this property, hence that this condition does not impose|make any genuine restriction on the image of $\mathbf{\Gamma}$ in $\mathbf{\Gamma}_{0,3}$. [p. 316]

[scan 318]

26 Profinite Teichmüller groups. [p. 161]

m433

Marginal note. m433 Discretifications and prediscretifications.

Let π be a profinite group with loops of type g, ν, T the invertible $\widehat{\mathbb{Z}}$-module of its orientations. We assume we are in the anabelian case, and we shall admit that then

$$\mathrm{Centre}(\pi) = \{1\} \tag{1}$$

more generally that the centralizer in π of every open subgroup of π is reduced to $\{1\}$. We shall thus again have a canonical exact sequence of profinite groups

$$1 \longrightarrow |-\pi \longrightarrow \mathrm{Aut}_{\mathrm{lac}}(\pi) \longrightarrow \mathrm{Autext}_{\mathrm{lac}}(\pi) \longrightarrow 1 \tag{2}$$

We shall also put †

$$\mathrm{Autext}_{\mathrm{lac}}(\pi) = \widehat{\widehat{\mathcal{T}}}(\pi) \tag{3}$$

we|and will call it the extended Teichmüller group of π. We shall also put $\widehat{\widehat{\mathfrak{S}}}(\pi) = \mathrm{Aut}_{\mathrm{lac}}(\pi)$, so that we can write (2) as

$$1 \longrightarrow \pi \longrightarrow \widehat{\widehat{\mathfrak{S}}}(\pi) \longrightarrow \widehat{\widehat{\mathcal{T}}}(\pi) \longrightarrow 1 \tag{2'}$$

Let us call a "basis" of π a set of elements of π $(x_i, y_i)_{1\leq i\leq g}$, $(\ell_j)_{1\leq j\leq \nu}$, the ℓ_j generating[t] the different groups with loops satisfying

$$\dagger \quad \ell_\nu \ell_{\nu-1} \cdots \ell_1 [x_g, y_g] \cdots [x_1, y_1] = 1 \tag{4}$$

and such that this is a generating relation. [p. 317]

Uncertainty retained from cancelled passages.
140-2-s318-D001 Autext
140-2-s318-D002 $\prod_i \quad [x_i, y_i] \quad \cdots [x_g, y_g]\ell_1$

[scan 319]

m434

Marginal note. [m434] or again, one defines $\Pi_{g,\nu}$ as the discrete group with|with the generators the x_i, y_i, ℓ_j and with defining relation (4)

If one chooses in $\Pi_{g,\nu}$ a (discrete) basis (corresponding definition|Definition), whence a basis of $\widehat{\Pi}_{g,\nu}$, one will have an obvious bijection

$$\mathrm{Bases}(\pi) \overset{\sim}{\leftarrow} \mathrm{Isom}_{\mathrm{lac}}(\widehat{\Pi}_{g,\nu}, \pi) \tag{5}$$

The set of bases of π is a homogeneous set in|under $\dagger$ $\widehat{\widehat{\mathfrak{S}}}(\pi) = \mathrm{Aut}_{\mathrm{lac}}(\pi)$. Let|If the $\dagger$ basis corresponds to an isomorphism $u : \widehat{\Pi}_{g,\nu} \longrightarrow \pi$, the group Π_0 generated by the x_i, y_i, ℓ_j is none other than $\dagger u(\Pi_{g,\nu})$. the subgroups of π which can be obtained in this way are called the discretifications of π. These form a homogeneous set under $\widehat{\widehat{\mathfrak{S}}} = \mathrm{Aut}_{\mathrm{lac}}(\pi)$, canonically isomorphic to the quotient of the right $\widehat{\widehat{\mathfrak{S}}}_{g,\nu}$-$\dagger$set $\mathrm{Isom}_{\mathrm{lac}}(\widehat{\Pi}_{g,\nu}, \pi)$ by $\mathfrak{S}_{g,\nu}$:

$$\mathrm{Discrt}(\pi) \overset{\sim}{\leftarrow} \mathrm{Isom}_{\mathrm{lac}}(\widehat{\Pi}_{g,\nu}, \pi)/\mathfrak{S}_{g,\nu} \tag{6}$$

The automorphisms (profinite with loops) of π fixing a discretification Π_0 are identified with the automorphisms of the discrete group with loops Π_0.

$$\mathrm{Aut}_{\mathrm{lac}}(\pi, \Pi_0) \simeq \mathrm{Aut}_{\mathrm{lac}}(\Pi_0) \tag{7}$$

[p. 318]

Uncertainty retained from cancelled passages.
140-2-s319-D001 [illeg. 1]
140-2-s319-D002 Π_0 d
140-2-s319-D003 Π_0
140-2-s319-D005 torsor

[scan 320]
[p. 162]

The bijection (5) puts on the set of Bases of π a bitorsor structure under $\widehat{\widehat{\mathfrak{S}}}(\pi), \widehat{\widehat{\mathfrak{S}}}_{g,\nu}$, and the|the corresponding|corresponding topologies|topology makes it a profinite set. The set of discretifications of π, which is a quotient of the preceding set, inherits a quotient topology, which is none other than the quotient topology of the right-hand side of (6). $\dagger$ This space is not Hausdorff (N.B. $\mathfrak{S}_{g,\nu}$ is always infinite), for the group $\mathfrak{S}_{g,\nu} \subset \widehat{\widehat{\mathfrak{S}}}_{g,\nu}$ is not closed. We denote by $\widetilde{\mathrm{Discrt}}' \,|\, \mathrm{Discrt}'(\pi)$ the associated Hausdorff topological space, identified with

$\mathrm{Isom}_{\mathrm{lac}}(\widehat{\Pi}_{g,\nu},\pi)/\overline{\mathfrak{S}}_{g,\nu}$, where $\overline{\mathfrak{S}}_{g,\nu}$ denotes the closure of $\mathfrak{S}_{g,\nu}$ in $\widehat{\overline{\mathfrak{S}}}_{g,\nu}$. We moreover have an obvious surjective homomorphism $\widehat{\mathfrak{S}}_{g,\nu} \longrightarrow \widehat{\overline{\mathfrak{S}}}|\overline{\mathfrak{S}}_{g,\nu}$ whose image is $\overline{\mathfrak{S}}_{g,\nu}$, we shall admit that it is injective, and shall identify $\overline{\mathfrak{S}}_{g,\nu}$ with $\widehat{\mathfrak{S}}_{g,\nu}$. Thus

$$\mathrm{Discrt}'(\pi) \simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\Pi}_{g,\nu},\pi)/\widehat{\mathfrak{S}}_{g,\nu} \tag{8}$$

[p. 319]

Uncertainty retained from cancelled passages.
140-2-s320-D001 exceptional $(g,\nu) = (0,3)$ where

[scan 321]

An element of Discrét$'(\pi)$ is called a class of discretifications of πor prediscretification of π. If π_0 is a discretification, we denote its class by $\pi_0^\natural$. The set of classes of discretifications on π is a homogeneous space under $\widehat{\widehat{\mathfrak{S}|\mathcal{G}}}(\pi)$, the stabilizer of $\pi_0^\natural$ is identified with|being identified with : the closure of $\mathfrak{S}|\mathcal{G}(\pi_0) = \mathrm{Aut}_{\mathrm{lac}}(\pi_0)$ in $\widehat{\widehat{\mathfrak{S}|\mathcal{G}}}(\pi)$, or again : $\widehat{\mathfrak{S}|\mathcal{G}}(\pi_0)$. The latter always contains π.

$$\mathrm{Aut}_{\mathrm{lac}}(\pi,\pi_0^\natural) \simeq \widehat{\mathfrak{S}|\mathcal{G}}(\pi_0)$$

for every discretification $\pi_0^\natural$, the subgroup of $\widehat{\widehat{\mathfrak{S}|\mathcal{G}}}(\pi)$ of loop automorphisms which fix $\pi_0^\natural$ is denoted $\widehat{\mathfrak{S}|\mathcal{G}}(\pi_0^\natural)$, it is therefore the inverse image of a subgroup of $\widehat{\widehat{\mathcal{T}}}(\pi)$, denoted $\widehat{\mathcal{T}}(\pi_0^\natural)$. We thus have an inclusion of extension structures

(10)

$$\begin{array}{ccccccccc}
1 & \longrightarrow & \pi & \longrightarrow & \widehat{\mathfrak{S}|\mathcal{G}}(\pi_0^\natural) & \longrightarrow & \widehat{\mathcal{T}}(\pi_0^\natural) & \longrightarrow & 1 \\
 & & & & \downarrow & & \downarrow & & \\
1 & \longrightarrow & \pi & \longrightarrow & \widehat{\widehat{\mathfrak{S}|\mathcal{G}}}(\pi) & \longrightarrow & \widehat{\widehat{\mathcal{T}}}(\pi) & \longrightarrow & 1
\end{array}$$

m435

Marginal note. m435 N.B. The set of classes of discretifications of π is described [illeg. 2] of the outer group π, as $\mathrm{Isom\,ext}(\widehat{\Pi}_{g,\nu},\pi)^\dagger$ divided by $\widehat{\mathcal{T}}_{g,\nu}$. We moreover have an arrow of degree 2

$$\begin{array}{c}
\mathrm{Isom\,ext}(\widehat{\Pi}_{g,\nu},\pi)/\widehat{\mathcal{T}}^{+}_{g,\nu} \\
\downarrow \\
\mathrm{Isom\,ext}(\widehat{\Pi}_{g,\nu},\pi)/\widehat{\mathcal{T}}_{g,\nu}
\end{array}$$

which makes it possible for every|the class of discretifications, to define its two orientations, and speak of oriented discretifications.

[p. 320]

Uncertainty retained from cancelled passages.
140-2-s321-D002 [illeg. 1]

[scan 322]

(One shows by arithmetico-geometric means that the inclusion $\widehat{\mathcal{T}} \longrightarrow \widehat{\widehat{\mathcal{T}}}$ is never an isomorphism.) We thus find a map [p. 163]

$$\text{Discrét}'(\pi) \longrightarrow \text{Closed subgroups of } \widehat{\widehat{\mathcal{T}}}(\pi) \tag{11}$$

obviously compatible with the action of † $\widehat{\widehat{\mathfrak{S}|\mathcal{G}}}(\pi)$ (transport of structure) – operating on the right via $\widehat{\widehat{\mathcal{T}}}(\pi)$ and its inner automorphisms on itself.

We thus find a conjugacy class, well determined of|of the closed subgroups of $\widehat{\widehat{\mathcal{T}}}(\pi)$, which we call its "geometric" Teichmüller subgroups. † It could moreover be that $\widehat{\widehat{\mathcal{T}}}_{g,\nu}$ is its own normalizer in $\widehat{\widehat{\mathcal{T}}}_{g,\nu}$ – equivalent to the assertion that (11) is injective|bijective|injection : The datum of a class of discretificationsof π would be equivalent to that of a geometric Teichmüller subgroup in † its extended Teichmüller group $\widehat{\widehat{\mathcal{T}}}$.

m436

Marginal note. m436 this is completely screwy and ultra-false – a moment of aberration ! This becomes clear in what follows…The sensible question (with which I must have at the time confused) is whether $\widehat{\mathcal{T}}$ is invariant in $\widehat{\widehat{\mathcal{T}}}$, i.e. whether the normalizer is $\widehat{\widehat{\mathcal{T}}}$ all|all of it !

[p. 321]

Uncertainty retained from cancelled passages.

140-2-s322-D001 $\widehat{\widehat{\mathcal{T}}}$

140-2-s322-D002 $\natural$ [illeg. 1]

140-2-s322-D003 $\widehat{\widehat{\mathcal{T}}}$

[scan 323]

Note that we have the canonical homomorphisms

$$\widehat{\widehat{\mathcal{T}}}\dagger \overset{\chi}{\to} \widehat{\mathbb{Z}}^* \tag{12}$$

$$\widehat{\widehat{\mathcal{T}}} \longrightarrow \mathfrak{S}_I|\mathcal{G}_I \tag{13}$$

m437

Marginal note. m437 where $I = I(\pi)$ is the set|the set of conjugacy classes of loop subgroups of π

inducing on $\widehat{\mathcal{T}}(\pi_0^\natural)$ † corresponding homomorphisms – whence the definition of the subgroups $\widehat{\widehat{\mathcal{T}}}^!$, $\widehat{\widehat{\mathcal{T}}}^+$, $\widehat{\widehat{\mathcal{T}}}^{!+}$ and likewise for $\widehat{\mathcal{T}}$. Note that χ takes on $\widehat{\mathcal{T}}$ only the values ± 1.

The subgroup $\widehat{\mathcal{T}}^+$ of outer automorphisms of the profinite groups|groups with loops π, †fixing a class of discretifications $\pi_0^\natural$ and with multiplier $+1$, plays a very special role. In contrast to that which one can conjecture about $\widehat{\mathcal{T}}$ (of which $\widehat{\mathcal{T}}^+$ is an open|invariant subgroup of index 2), one would rather suppose that $\widehat{\mathcal{T}}^+$ is invariant in $\widehat{\widehat{\mathcal{T}}}$. In any case, the subgroups thus defined in $\widehat{\widehat{\mathcal{T}}}$ via classes of discretifications of π † (perhaps

[p. 322]

Uncertainty retained from cancelled passages.

140-2-s323-D001 $(\pi_0^\natural)$

140-2-s323-D002 continuous

140-2-s323-D003 [illeg. 2]

140-2-s323-D005 [illeg. 1]

[scan 324]

there is only a single and unique one!) will be called the strict geometric Teichmüller subgroups. The choice of such a subgroup of $\widehat{\widehat{\mathcal{T}}}$ is, in any case, an element of structure distinctly weaker than that of a class of discretifications, and even than that of a geometric Teichmüller subgroup (not strict).

[p. 164]

To specify the relations between these two notions, let us first recall that $\widehat{\mathcal{T}}^+ = \Sigma$ is deduced from $\widehat{\mathcal{T}}$ as the kernel of $\chi \mid \widehat{\mathcal{T}} : \widehat{\mathcal{T}} \longrightarrow \{\pm 1\}$. On the other hand, (for a chosen strict geometric Teichmüller subgroup $\widehat{\mathcal{T}}^+$) let us consider

$$(14) \qquad \mathcal{N} | N_\Sigma = \mathcal{N} = \mathrm{Norm}_{\widehat{\widehat{\mathcal{T}}}}(\Sigma) \qquad (\Sigma = \widehat{\mathcal{T}}^+)$$

(perhaps always equal to all of $\widehat{\widehat{\mathcal{T}}}$)

obviously (if Σ comes from a $\widehat{\mathcal{T}}$)

$$(15) \qquad \underset{=\widehat{\mathcal{T}}^+}{\Sigma} \subset \widehat{\mathcal{T}} \subset \mathcal{N}_\Sigma.$$

The profinite group $\mathcal{N}/\Sigma$ associated with † Σ is denoted $\mathbf{\Gamma}_{\dagger\Sigma}$ † † (if Σ is unique, we write $\mathbf{\Gamma}_\pi$); [p. 323]

Uncertainty retained from cancelled passages.

140-2-s324-D008 structure

140-2-s324-D010 $\widehat{\mathcal{T}}^+$

140-2-s324-D011 [illeg. 1]

140-2-s324-D012 [illeg. 2]

[scan 325]

and the $\widehat{\mathcal{T}}$† giving rise to the same Σ correspond † to a conjugacy class of elements of order 2 of $\mathbf{\Gamma}_{\dagger\Sigma}$. It is clear † in any case that we have an injective map

$$(16) \qquad \begin{array}{c}\text{set of geometric}\\ \text{Teichmüller subgroups}\\ \widehat{\mathcal{T}} \text{ of } \widehat{\widehat{\mathcal{T}}} \text{ such that}\\ \widehat{\mathcal{T}}^+ = \Sigma\end{array} \longrightarrow \begin{array}{c}\text{set of elements|elements}\\ \text{of order 2 of } \mathbf{\Gamma}_\Sigma\end{array} \quad :$$

and that its image is stable under conjugation. Let us show that two|the elements of the image are conjugate under $\mathbf{\Gamma}_\Sigma$. Indeed, let

$$\widehat{\mathcal{T}},\ \widehat{\mathcal{T}}' \supset \Sigma \quad \text{such that} \quad \Sigma = \widehat{\mathcal{T}}^+ = \widehat{\mathcal{T}}'^+.$$

† $\exists\, g \in \widehat{\widehat{\mathcal{T}}}$ such that $\widehat{\mathcal{T}}' = \mathrm{int}(g)\mathcal{T}$, and we will have|have therefore $\widehat{\mathcal{T}}'^+ = \mathrm{int}_g \widehat{\mathcal{T}}^+$ i.e. $g \in \mathcal{N}$, OK.

The elements of order 2 thus obtained in $\mathbf{\Gamma}_\Sigma$ are called the canonical involutions. We will give a remarkable conjectural interpretation of them below. The set $\widehat{\widehat{\mathcal{T}}}/\mathcal{N}$ is identified with the set of subgroups of [p. 324]

Uncertainty retained from cancelled passages.

140-2-s325-D001 $\mathcal{G}$

140-2-s325-D003 to the classes

140-2-s325-D004 $\widehat{\mathcal{T}}^+$

140-2-s325-D005 that we|that

140-2-s325-D006 [illeg. 1] obviously

[scan 326]

strict geometric Teichmüller (once the "origin" element Σ has been chosen). More intrinsically, we will have: [p. 165]

$$(17) \qquad \begin{array}{c}\text{Set of strict geometric}\\ \text{Teichmüller subgroups}\\ \text{of } \widehat{\widehat{\mathcal{T}}}(\pi)\end{array} \overset{\sim}{\leftarrow} \mid \simeq \mathrm{Isomext}_{\mathrm{lac}}(\widehat{\Pi}_{g,\nu}, \pi)/\mathcal{N}_{g,\nu}$$

where we set, as is only right

$$(18) \qquad \mathcal{N}_{g,\nu} = \mathrm{Norm}_{\widehat{\widehat{\mathcal{T}}}_{g,\nu}}(\widehat{\mathcal{T}}^{+}_{g,\nu})$$

(perhaps equal to all of $\widehat{\widehat{\mathcal{T}}}_{g,\nu}$!)

Let us now consider the modular topos over $\mathbb{Q}$ of algebraic curves of type (g,ν), denoted $M_{g,\nu\mathbb{Q}}$, or simply $M_{g,\nu}$. If we choose a universal covering $\widetilde{M}_{g,\nu}$ of $M_{g,\nu\mathbb{Q}}$ (hence a universal covering of Spec$\mathbb{Q}$ i.e. an algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$) we can specify the $\pi_1(M_{g,\nu\mathbb{Q}})$ as the group of $M_{g,\nu\mathbb{Q}}$-automorphisms of this covering|covering; † call it the arithmetic Teichmüller group of type (g,ν) (relative to the choice of $\widetilde{M}_{g,\nu\mathbb{Q}}$). We will thus have an exact sequence [p. 325]

Uncertainty retained from cancelled passages.
140-2-s326-D001 and of it make [illeg. 1]

[scan 327]

$$(19) \qquad 1 \longrightarrow \pi_1(M_{g,\nu\overline{Q}}) \longrightarrow \pi_1(M_{g,\nu}) \longrightarrow \mathrm{Gal}(\overline{Q}/Q) \longrightarrow 1$$

†

(where $\overline{Q}$ and the base points are made explicit as has been said) Let us note that by the transcendental theory of Teichmüller, we have an isomorphism (defined modulo operation induced by $\pi_1(M_{g,\nu})$ on $\pi_1(M_{g,\nu,\overline{Q}})$)

$$(20) \qquad \pi_1(M_{g,\nu,\overline{Q}}) \simeq \widehat{\mathcal{T}}^{+}_{g,\nu} \qquad \text{(profinite compactification of } \mathcal{T}^{+}_{g,\nu})$$

Let us on the other hand consider the scheme. $U|\mathcal{U}_{g,\nu}$ over $M_{g,\nu}$, "universal" curve of type (g,ν)

We thus have, on choosing a universal covering $\widetilde{U|\mathcal{U}}_{g,\nu}$ of it over the chosen universal covering $\widetilde{M}^{\natural}_{g,\nu}$ of $M_{g,\nu}$, an exact sequence

(21) and (22) which maps to:

$$\begin{array}{ccccccccc}
(21) & 1 & \longrightarrow & \pi & \longrightarrow & \pi_1(U|\mathcal{U}^{\dagger}_{g,\nu}) & \longrightarrow & \pi_1(M_{g,\nu}) & \longrightarrow & 1 \\
 & & & \| & & \downarrow & & \downarrow & & \\
(22) & 1 & \longrightarrow & \pi & \longrightarrow & \widehat{\widehat{\mathfrak{S}|\mathcal{G}}}(\pi) & \longrightarrow & \widehat{\widehat{\mathcal{T}}}(\pi) & \longrightarrow & 1
\end{array}$$

and the same for $U|\mathcal{U}_{g,\nu\overline{Q}} \longrightarrow M_{g,\nu\overline{Q}|Q}$ [p. 326]

Uncertainty retained from cancelled passages.
140-2-s327-D001 $\widehat{\mathcal{T}}_{g,\nu}$ Γ
140-2-s327-D002 ∼

[scan 328]
[p. 166|168]

$$(\overline{21})|(\overline{21}') \qquad 1 \longrightarrow \pi \longrightarrow \pi_1(U|\mathcal{U}_{g,\nu\overline{Q}}) \longrightarrow \pi_1(M_{g,\nu\overline{Q}}) \longrightarrow 1$$

†

We of course have an isomorphism of groups with loops|ends

$$\pi \simeq \widehat{\Pi}_{g,\nu} \tag{23}$$

whose indeterminacy we would like to determine precisely

Let $M_{g,\nu C}$ denote the complex modular Teichmüller †topos, defined using the C^∞ surface $U|\mathcal{U}_{g,\nu}$, it is equipped with the canonical universal covering $\widetilde{M}_{g,\nu C}$ (§ 21) [m438]

Marginal note. [m438] $\widetilde{M}_{g,\nu C} = E_g/A^0_{g,\nu}|A^\circ_{g,\nu}$

with group of automorphisms $\mathcal{T}^+_{g,\nu}$ precisely –|... $U|\mathcal{U}_{g,\nu C}$ being itself also equipped with a universal covering over the preceding one, $\widetilde{U|\mathcal{U}}_{g,\nu C} = E_g/A^0_{g,\nu+1}|A^\circ_{g,\nu+1}$. This gives rise to the exact sequence of discrete groups (in the general topology context)

$$\begin{array}{lccccccc} (24) & 1 \to \Pi_{g,\nu} & \longrightarrow & \mathfrak{S}|\mathcal{G}^+_{g,\nu} & \longrightarrow & \mathcal{T}^+_{g,\nu} & \longrightarrow & 1 \\ & \downarrow & & \| & & \| & & \\ & & & \pi_1(U|\mathcal{U}_{g,\nu C}) & & \pi_1(M_{g,\nu C}) & & \\ & \downarrow & & \downarrow & & \downarrow & & \\ \text{(mapping to} & 1 \to \Pi_{g,\nu} & \to & \mathrm{Aut}_{\mathrm{lac}}\,\dagger(\Pi_{g,\nu}) & \to & \mathrm{Aut}\,\mathrm{ext}_{\mathrm{lac}}(\Pi_{g,\nu}) & \to & 1 \\ & & & \| & & \| & & \\ & & & \mathfrak{S}|\mathcal{G}_{g,\nu} & & \mathcal{T}_{g,\nu} & & \end{array}$$

by an isomorphism|homomorphism of extension structures), which by passage to the profinite completions gives the analogous exact sequence for the [p. 327]

Uncertainty retained from cancelled passages.

140-2-s328-D001 ($\overline{22}$)

140-2-s328-D002 [illeg. 1]

140-2-s328-D004 ext

[scan 329]

algebraic multiplicities over $\mathbb{C}$

$$\begin{array}{lccccccc} (25) & 1 \to \widehat{\Pi}_{g,\nu} & \longrightarrow & \dagger\,\widehat{\mathcal{G}}_{g,\nu} & \longrightarrow & \widehat{\mathcal{T}}^+_{g,\nu} & \longrightarrow & 1 \\ & \| & & \downarrow & & \updownarrow & & \\ & \widehat{\Pi}_{g,\nu} & \longrightarrow & \mathrm{Aut}\,\dagger_{\mathrm{lac}}(\widehat{\Pi}_{g,\nu}) & \longrightarrow & \mathrm{Autext}_{\mathrm{lac}}(\widehat{\Pi}_{g,\nu}) & & \\ & \backslash\backslash\, \widehat{\widehat{\mathcal{G}}}_{g,\nu} & & & & \backslash\backslash\, \widehat{\widehat{\mathcal{T}}}_{g,\nu} & & \end{array}$$

This situation moreover arises canonically from an analogous situation over $\overline{\mathbb{Q}}_0$ = algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$, the choice of $\widetilde{U|\mathcal{U}}_{g,\nu\mathbb{C}}$ defining a $\widetilde{U|\mathcal{U}}_{g,\nu\overline{\mathbb{Q}}_0}$, hence a $\widetilde{M}_{g,\nu\overline{\mathbb{Q}}_0}$.

To define an isomorphism between this sequence (25) and the exact sequence ($\overline{21}$) (equipped with 2 homomorphisms|its homomorphism to (22)) one must

1°) choose an isomorphism $\overline{\mathbb{Q}} \simeq \overline{\mathbb{Q}}_0$ — the indeterminacy is in $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$; which makes it possible to identify $\overline{\mathbb{Q}}$ and $\overline{\mathbb{Q}}_0$;

2°) Choose an isomorphism († $\overline{\mathbb{Q}} = \overline{\mathbb{Q}}_0$) of $(\widetilde{U|\mathcal{U}}_{g,\nu})_0$ (constructed by transcendental|transcendental means over $\mathbb{C}$ and descended to $\overline{\mathbb{Q}}_0$) with $\widetilde{U|\mathcal{U}}_{g,\nu\overline{\mathbb{Q}}}$ — the indeterminacy is in $\pi_1(M_{g,\nu\overline{\mathbb{Q}}})$.

In summary, we have a class of isomorphisms of (24)|(21)2|? and (25), defined modulo automorphisms † [illeg. 1] once we have chosen $\overline{\mathbb{Q}} \simeq \overline{\mathbb{Q}}_0$ inner in the profinite group $\pi_1(U|\mathcal{U}_{g,\nu})$. (These|The corresponding isomorphisms transform the Standard|standard oriented class of discretification of $\widehat{\Pi}_{g,\nu}$ into † exactly one class of oriented discretifications of π. [p. 328]

Uncertainty retained from cancelled passages.
140-2-s329-D001 $\widehat{\Pi}$
140-2-s329-D002 ext
140-2-s329-D004 between
140-2-s329-D005 [illeg. 1]
140-2-s329-D007 a class

[scan 330]

† [p. 167]

Changing the choice of the isomorphism to its complex conjugate, we find the opposite oriented quasi- discretification.

We see on the other hand that the image of $\pi_1(M_{g,\nu\overline{\mathbb{Q}}}) \subset \pi_1(M_{g,\nu})$ in $\widehat{\widehat{\mathcal{T}}}(\pi)$ is none other than the $\widehat{\mathcal{T}}^+$ corresponding|corresponding to an arbitrary one of the chosen classes of discrete quasi-rigidifications — the subgroup does not depend on the choice of this class — which merely expresses that the image of $\pi_1(M_{g,\nu\overline{\mathbb{Q}}})$ in $\widehat{\widehat{\mathcal{T}}}(\pi)$, considered|considered as a subgroup of the image of the larger group $\pi_1(M_{g,\nu})$, is normal|normal in it — which follows from the fact that $\pi_1(M_{g,\nu\overline{\mathbb{Q}}})$ is normal in $\pi_1(M_{g,\nu\mathbb{Q}|\overline{\mathbb{Q}}})$. Thus π [p. 329]

Uncertainty retained from cancelled passages.
140-2-s330-D001 canonically class of quasi $\pi_0^\natural$ $U|\mathcal{U}$ $\mathfrak{S}|\mathcal{G}$ $\pi_0^\natural$ $\pi_0^\natural$ $\longrightarrow$

[scan 331]

is canonically endowed (not with a discrete oriented quasi-rigidification, which depends on the choice of an isomorphism $\overline{\mathbb{Q}} \simeq \overline{\mathbb{Q}}_0$ i.e. of an embedding $\overline{\mathbb{Q}} \subset \mathbb{C}$) but at least with a strict geometric Teichmüller group $\Sigma \subset \widehat{\widehat{\mathcal{T}}}(\pi)$, namely the image of $\pi_1(M_{g,\nu\overline{\mathbb{Q}}})$, isomorphic to This being so, We have a canonical homomorphism of group extensions

(26)
$$\begin{array}{ccccccccc} 1 & \to & \pi_1(M_{g,\nu\overline{\mathbb{Q}}}) & \longrightarrow & \pi_1(M_{g,\nu}) & \longrightarrow & \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) & \longrightarrow & 1 \\ & & \downarrow{\scriptstyle |2} & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & \Sigma & \longrightarrow & \mathcal{N}|N_\Sigma & \longrightarrow & \Gamma_\Sigma & \longrightarrow & 1 \\ & & & & \| & & & \searrow & \\ & & & & \mathrm{Norm}_{\widehat{\widehat{\mathcal{T}}}(\pi)}(\Sigma) & & & & \end{array}$$

m439

Marginal note. m439 group of †outer "arithmetic" automorphisms of π (endowed with Σ)

For the choice of an isomorphism $\overline{\mathbb{Q}} \simeq \overline{\mathbb{Q}}_0$ i.e. of an embedding $\overline{\mathbb{Q}} \subset \mathbb{C}$, the image of complex conjugation in $\mathrm{Gal}(\overline{\mathbb{Q}}_0/\mathbb{Q})$ is none other than the element of order 2 of Γ_Σ associated with the geometric Teichmüller subgroup, associated with the canonical quasi-discretification (oriented, but no matter) on π (defined by $\overline{\mathbb{Q}} \simeq | \xrightarrow{\sim} \overline{\mathbb{Q}}_0$).

Uncertainty retained from cancelled passages.
140-2-s331-D001 "arithmetic

[scan 332]
[p. 168]

The natural conjecture here is that

$$(27) \qquad \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \Gamma_\Sigma$$

from the Galois group to the arithmetic outer automorphisms of (π, Σ) is an isomorphism. We can supplement it by the conjecture, still bolder, that moreover Σ is invariant in $\widehat{\widehat{\mathcal{T}}}(\pi)$ ($\Longleftrightarrow$ $\Sigma_{g,\nu}$ invariant in $\widehat{\widehat{\mathcal{T}}}_{g,\nu}$), which would thus mean that $\mathcal{N}|N_\Sigma = \widehat{\widehat{\mathcal{T}}}$, or again that † Conjectural!!

$$(28) \qquad \pi_1(M_{g,\nu}) \xrightarrow{\sim} \widehat{\widehat{\mathcal{T}}}(\pi) = \mathrm{Aut}\,\mathrm{ext}_{\mathrm{lac}}(\pi).$$

and $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ would be identified with the "arithmetic" quotient of the group of outer loop automorphisms of π.

Let now U be an algebraic curve of type g, ν over $\overline{\mathbb{Q}}$. It is defined by a point of $M_{g,\nu\overline{\mathbb{Q}}}$ over $\mathrm{Spec}\,\overline{\mathbb{Q}}$, and as the fiber of $\mathcal{U}_{g,\nu}$ at this point. Let us choose a universal covering of U, whence a group $\pi(U)$, canonically isomorphic to the group π a-

Uncertainty retained from cancelled passages.
140-2-s332-D001 $\pi_1(M_{g,\nu})$

[scan 333]

bove, once an isomorphism has been chosen between $\widetilde{\mathcal{U}}_{g,\nu}$ and the universal covering of $\mathcal{U}_{g,\nu}$ deduced from $\widetilde{U}$, which gives an indeterminacy in $\pi_1(\mathcal{U}_{g,\nu\overline{\mathbb{Q}}}) \simeq \hat{\mathcal{S}}^+$.

Thus, once an isomorphism $\overline{\mathbb{Q}} \simeq \overline{\mathbb{Q}}_0$ has been chosen, one finds on $\pi_1(U)$ an oriented quasi-discretification (obvious moreover to define directly, by transcendental means) changed into its opposite by changing the isomorphism $\overline{\mathbb{Q}}_2|\overline{\mathbb{Q}} \simeq \overline{\mathbb{Q}}_0$ by complex conjugation. On the other hand, the subgroup $\widehat{\mathcal{T}}^+$ of $\widehat{\widehat{\mathcal{T}}}(\pi)$ corresponding to this quasi-discretification (oriented, little do we care) does not depend on the choice of the isomorphism in question $\overline{\mathbb{Q}} \simeq \overline{\mathbb{Q}}_0$, it is canonically associated with π as the fundamental group of a U over † an algebraically closed field

[N.B. one could take any algebraically closed field of characteristic 0, no need for it to be a|over $\overline{\mathbb{Q}}$, cf. below].

Uncertainty retained from cancelled passages.
140-2-s333-D001 $\overline{\mathbb{Q}}$

[scan 334]
[p. 169]

†

If the conjecture on Γ holds, it would follow that the datum of a † profinite group with loops π of type (g, ν), endowed with a strict geometric Teichmüller group $\Sigma \subset \widehat{\widehat{\mathcal{T}}}(\pi)$ (which perhaps is not an additional structure at all — Σ would be uniquely determined by π !) : would be equivalent to the datum 1°) of an algebraically closed extension $\overline{\mathbb{Q}}$ of $\mathbb{Q}$ (whose Galois group would be $N_{\widehat{\mathcal{T}}}(\Sigma)/\Sigma = \Gamma_\Sigma$, and the Γ_0-|Π_0-torsor $\mathrm{Isom}(\overline{\mathbb{Q}}_0, \overline{\mathbb{Q}}_1)$ would be identified

Uncertainty retained from cancelled passages.
140-2-s334-D001 homomorphism
140-2-s334-D002 outer

[scan 335]

with the set ($\subset \mathrm{Isom}(\widehat{\Pi}_{g,\nu}, \pi)/\widehat{\mathcal{T}}^{0}_{g,\nu}$ of † oriented quasi-rigidifications giving rise to $\pi_{g,\nu}|\Pi_{g,\nu}$) and of a universal covering of $\mathcal{U}_{g,\nu\overline{\mathbb{Q}}}$ — or simply, of a universal covering $\widetilde{\mathcal{U}}_{g,\nu}$ of $\mathcal{U}_{g,\nu} = \mathcal{U}_{g,\nu\mathbb{Q}}$. m441

Marginal note. m441 If one gives oneself only π as an outer group, with Σ, this amounts to the datum of a universal covering of $M_{g,\nu}$.

The datum of an isomorphism $\overline{\mathbb{Q}} \simeq \overline{\mathbb{Q}}_0$ thus amounts to that of an oriented quasi-rigidification on π compatible with Σ (condition perhaps automatically satisfied…). m442

Marginal note. m442 N.B. If one gives up choosing $\widetilde{U}$, this is equivalent|amounts to working with outer groups with loops.

That said, the datum of an algebraic curve of type g, ν over † an algebraically closed field $\overline{\mathbb{Q}}$, and of a universal covering of it would amount to the datum of a preceding datum (namely † a universal covering of an $M_{g,\nu}$), plus a point of $M_{g,\nu,\overline{\mathbb{Q}}}$ over $\overline{\mathbb{Q}}$ † defining this universal covering — and, this last datum would be identified with a † germ of splitting of the extension

$$1 \longrightarrow \pi_1(M_{g,\nu\overline{\mathbb{Q}}}) \longrightarrow \pi_1(M_{g,\nu}) \longrightarrow \pi_1(\mathbb{Q}) \longrightarrow 1$$

The profinite interpretation (in terms of the outer groups with loops π) is then † a (π, Σ), and a germ of splitting of

$$1 \longrightarrow \Sigma \longrightarrow N_\Sigma \longrightarrow \Gamma_\Sigma \longrightarrow 1.$$

If one wants a curve over a finite subextension K' of $\overline{\mathbb{Q}}/\mathbb{Q}$, corresponding to a subgroup of finite index $\Gamma' \subset \Gamma_\Sigma$, it will be a matter of a partial † splitting $\Gamma' \longrightarrow N_\Sigma$.

Uncertainty retained from cancelled passages.

140-2-s335-D001 [illeg. 1]
140-2-s335-D003 ([illeg. 1] would correspond
140-2-s335-D005 profinite outer group
140-2-s335-D006 this
140-2-s335-D007 point
140-2-s335-D008 $\longrightarrow N_\Sigma \longrightarrow \Gamma_\Sigma \longrightarrow 1$
140-2-s335-D009 [illeg. 1]

[scan 336]
[p. 170]

But surely, if this † description of anabelian algebraic curves is fully faithful, it is not 2-faithful, i.e. conditions on a splitting are needed, which will have to be examined later.

The description of an † algebraic curve of type g, ν over an arbitrary fixed field of finite type , characteristic 0|over $\mathbb{Q}$ K, endowed with an algebraically closed extension $\overline{K}$ (thus $\Gamma = \mathrm{Gal}\,\overline{K}/K' \longrightarrow \Gamma_0 = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, where $\overline{\mathbb{Q}}$ is the algebraic closure of $\mathbb{Q}$ in $\overline{K}$), in terms of $\Gamma \longrightarrow \Gamma_0$, would be the following: datum of an outer group with loops π of type g, ν, of a Teichmüller subgroup Σ in $\widehat{\mathcal{T}}(\pi)$, of an isomorphism $\overline{\mathbb{Q}} \simeq$ field defined by this situation (having as Galois group N_Σ/Σ, so that we find an isomorphism $\Gamma_0 \simeq N_\Sigma/\Sigma$) finally of a lifting of $\Gamma \longrightarrow \Gamma_0$ † (which describes an arithmetically outer action of Γ on (π, Σ)) into an outer action $\Gamma \longrightarrow N_\Sigma$ of Γ on (π, Σ) ; with conditions which one will have to try

Uncertainty retained from cancelled passages.

140-2-s336-D001 representation
140-2-s336-D002 group
140-2-s336-D004 [illeg. 2]

to bring out (necessary and sufficient conjecturally) on this lifting. [scan 337]

Let us note that every embedding $\overline{K} \subset \mathbb{C}$ defines canonically|canonically on the outer group $\pi = \pi_1(U_{\overline{K}})$ an oriented quasi-discretification, replaced by the opposite when the embedding is replaced by its complex conjugate. † I say that these quasi-discretifications all define Σ as the group of automorphisms, and that the quasi-discretification defined by $\overline{K} \hookrightarrow \mathbb{C}$ depends only on its restriction to $\overline{\mathbb{Q}} \subset \mathbb{C}$ — in a more precise way, it is the one one defines directly with the help of $\overline{\mathbb{Q}} \subset \mathbb{C}$ i.e. of $\overline{\mathbb{Q}} \simeq \overline{\mathbb{Q}}_0$ — But there is reason also to examine the way to obtain discretifications. Thus, if one has a|a U is defined over $\overline{\mathbb{Q}}_0$ (i.e. an outer $\pi|\Pi$ of type (g, ν), endowed with an oriented quasi-discretification

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Uncertainty retained from cancelled passages.

140-2-s337-D003 that|that

$\pi_0^{\natural+}|\pi_0^{\zeta+}$, and with a germ of †splitting of [scan 338] [p. 171]

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \Sigma_\dagger & \longrightarrow & N_\Sigma & \longrightarrow & \Gamma_\Sigma & \longrightarrow & 1 \\ & & \| & & & & & & \\ & & \operatorname{Aut}\operatorname{ext}_{\mathrm{lac}}(\pi, \pi_0^{\natural+}|\pi_0^{\zeta+}) & & & & & & \end{array}$$

)|, which makes the kernel of|of the group defined by Γ_Σ operate outerly on π, the rational points/$\overline{\mathbb{Q}}_0$ corresponding|correspond to the π-conjugacy classes of the germs of liftings of this operation, outer|outer into a genuine operation. Let us therefore suppose such a point given, i.e. † one has an open subgroup $\Gamma^{\pi+}|\Gamma' \subset \Gamma_\Sigma$ which does indeed operate on π, the operation defined modulo inner automorphisms (itself unique since $\pi^\Gamma|\overline{\pi}^\Gamma = (1)$!) In this situation, it would be necessary to define in π $(\simeq \pi_1(U_{P|p}, P))$ a discretification oriented $\pi_0 \subset \pi$, and not just an oriented quasi-discretification ! (Of course, it must be in the class which we have for ourselves given in advance of oriented discretifications, which we had precisely denoted $\pi_0^{\natural+}|\pi_0^{\zeta+}$. We shall return to this — and to the analogous situation over an arbitrary base field K of finite type....|...

Uncertainty retained from cancelled passages.

140-2-s338-D001 liftings

140-2-s338-D002 $\pi_0^{\natural+}$

140-2-s338-D003 Γ_Σ operates

[scan 339]

I refer to the beginning of the following § (§ 27) for the change in terminology, "quasi-discretification" becoming a "prediscretification". But there is reason also to introduce a finer notion, corresponding — to the [illeg. 1] π_1 (profinite) of an algebraic variety X defined over an algebraically closed field $\overline{K}$, when one embeds $\overline{K}$ into $\mathbb{C}$: there is a discretification modulo inner automorphisms|automorphisms — we shall call it a strict prediscretification. In the case of a π with loops, the homogeneous in $\widehat{\widehat{\mathcal{T}}}(\pi)$ of this is identified with

$$\operatorname{Isom}_{\mathrm{lac}}(\widehat{\Pi}_{g,\nu}, \pi)/\widehat{\Pi}_{g,\nu} \cdot \mathcal{T}^{\dagger}_{g,\nu} \simeq \operatorname{Isom}\operatorname{ext}_{\mathrm{lac}}(\widehat{\Pi}_{g,\nu}, \pi)/\mathcal{T}_{g,\nu}$$

— this notion also|also depends only on the outer profinite group with loops defined by $\Pi_{g,\nu}$), and † is equivalent to that of an "outer basis" in π modulo the action of $\mathcal{T}_{g,\nu}$, i.e. of an isomorphism of π with the "standard" $\widehat{\Pi}_{g,\nu}$; modulo the action of $\mathcal{T}_{g,\nu}$. The map

$$\text{Outer bases of } \pi \longrightarrow \text{Prediscretifications of } (\pi)$$

makes therefore of the left-hand side a relative torsor on the right, over the one on the right, under the group $\mathcal{T}_{g,\nu}$. †

A prediscretification of (π, Σ) corresponds to a choice of an isomorphism of the algebraic closure $\overline{\mathbb{Q}}_{(\pi,\Sigma)} = \overline{\mathbb{Q}}$ associated with (π, Σ), with $\overline{\mathbb{Q}}_0$. When one has given (π, Σ) as corresponding to an algebraic curve over $\overline{\mathbb{Q}}$, i.e. that one gives oneself|has given oneself a germ of liftings of Γ_Σ into $N_\Sigma \subset \widehat{\widehat{\mathcal{T}}}(\pi)$ i.e. a germ|group of outer action of Γ_Σ on π, every|whose prediscretification must give rise to a strict discretification — and a † discretification is equivalent on lifting back a germ of action ("admissible") of Γ_Σ on π ...

Uncertainty retained from cancelled passages.

140-2-s339-D003 $\widehat{\wedge}$

140-2-s339-D004 [illeg. 1]

140-2-s339-D006 [illeg. 1]

140-2-s339-D007 [illeg. 1]

[scan 340]

27. Change of type (g, ν) : a) plugging holes. [p. 172]

I am going to change terminology, calling prediscretification what I had called (a little pejoratively !) quasi-discretification. Indeed, we foresee that a prediscretification — compatible with Σ —) (which amounts morally|morally to the embedding (modulo complex conjugation) into $\mathbb{C}$ of the algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$ canonically associated with|$*$ $(\pi, \Sigma(\subset \widehat{\widehat{\mathcal{T}}}(\pi)))$, gives rise as soon as the outer-arithmetic action of $\Gamma_\Sigma = N_\Sigma/\Sigma$ on π is lifted to a germ of action on π, to a genuine discretification of π. Ditto|The same for oriented prediscretifications. The group N_Σ/Σ deserves [m443]

Marginal note. [m443] morally|normally, $\Sigma = \widehat{\mathcal{T}}$ (but there is no $\mathcal{T}$!)

a name as well — I am going to call it the arithmetic Teichmüller group associated with (π, Σ), and its elements will be called the arithmetically outer automorphisms of π — deduced from an outer automorphism (normalizing Σ) : by disregarding the "geometric" outer automorphisms (i.e. precisely those $\in \Sigma$) — † as the ordinary outer automorphisms were described by disregarding the inner automorphisms of π. We † shall note that the "multiplier" character

$$\chi : \widehat{\widehat{\mathcal{T}}}(\pi) \longrightarrow \widehat{\mathbb{Z}}^*$$

is trivial on Σ, by construction of Σ, and passes to Γ_Σ

$$\chi_\Sigma : \Gamma_\Sigma \longrightarrow \widehat{\mathbb{Z}}^*$$

Uncertainty retained from cancelled passages.

140-2-s340-D001 of

140-2-s340-D002 [illeg. 1]

[scan 341]

Of course, this character will again be called, "multiplier", and|or "cyclotomic character". † If our fundamental conjecture is true, this character identifies $(\Gamma_\Sigma)_{\mathrm{ab}}$ with $\widehat{\mathbb{Z}}^*$.

On the other hand, if $I = I(\pi)$, the canonical homomorphism

$$\widehat{\widehat{\mathcal{T}}} \longrightarrow \mathfrak{S}_I | \mathcal{S}_I$$

is † not trivial on Σ, but induces a surjective homomorphism

$$\Sigma \longrightarrow \mathfrak{S}_I^\dagger$$

whence a subgroup $\Sigma^!$ such that

$$\Sigma/\Sigma^! \xrightarrow{\sim} \mathfrak{S}_I.$$

We see at once that $\forall\ \gamma|g \in \widehat{\widehat{\mathcal{T}}}$ which normalizes Σ normalizes $\Sigma^! = \Sigma \cap \widehat{\widehat{\mathcal{T}}}^!$, whence|i.e. $\Sigma^!$ is also invariant in $\mathcal{N}_\Sigma$. † Let $\mathcal{N}^!_\Sigma = \mathcal{N}_\Sigma \cap \widehat{\widehat{\mathcal{T}}}^!$, we have a Cartesian diagram of subgroups of $\widehat{\widehat{\mathcal{T}}}$

morally, $\Sigma^! \simeq \widehat{\mathcal{T}}^{!+}$

$$\begin{array}{ccc} \mathcal{N}^! & \overset{\mathfrak{S}_I}{\hookrightarrow} & \mathcal{N} \\ {\scriptstyle \Gamma_\Sigma}\uparrow & & \uparrow{\scriptstyle \Gamma_\Sigma} \\ \Sigma^! & \underset{\mathfrak{S}_I}{\hookrightarrow} & \Sigma \end{array}$$

giving an injective homomorphism

$$\mathcal{N}/\Sigma^! \longrightarrow \underbrace{\mathcal{N}/\mathcal{N}^!}_{\mathfrak{S}_I} \times \underbrace{\mathcal{N}/\Sigma}_{\Gamma_\Sigma}$$

of which|when we see at once that it is bijective m445

Marginal note. m445 and|for $\Sigma/\Sigma^! \xrightarrow{\sim} \mathcal{N}/\mathcal{N}^!$ $(\simeq \mathfrak{S}_I)$

$$\mathcal{N}/\Sigma^! \xrightarrow{\sim} \mathcal{N}/\mathcal{N}^! \times \mathcal{N}/\Sigma \simeq \mathfrak{S}_I \times \Gamma_\Sigma$$

and consequently we also have

$$\mathcal{N}^!/\Sigma^! \xrightarrow{\sim} \mathcal{N}/\Sigma = \Gamma_\Sigma,$$

i.e. the group Γ_Σ of arithmetically outer automorphisms of (π, Σ) can also be described via the automorphisms [p. 340]

Uncertainty retained from cancelled passages.

140-2-s341-D001 It

140-2-s341-D002 [illeg. 1]

140-2-s341-D003 !

140-2-s341-D004 [illeg. 3]

[scan 342]

that are outer and induce the identity on I. [p. 173]

In the case $(g, \nu) = (0, 3)$, we have $\Sigma^! = (1)$, $\Rightarrow$ hence $\Sigma \xrightarrow{\sim} \mathfrak{S}_3$. $\mathcal{N}$ is the normalizer of $\mathfrak{S}_3$ in $\widehat{\widehat{\mathcal{T}}}$, $\mathcal{N}^! \xrightarrow{\sim} \Gamma_\Sigma$ is its centralizer, and $\mathcal{N}$ is identified with the direct product of the two.

†

Now let $I|I' \subset I$, $\nu' = \operatorname{card} I'$, and let us consider the outer group π' obtained from π by "plugging the holes at $I \setminus I'$", whence a homomorphism

$$\pi^* \longrightarrow \pi'.$$

† Let us consider the bases of π, for which † the ν' first l_i, are in † groups with loops † $L_{i'}$ $(i' \in I')$ — we call them adapted to I'; they form a torsor $\subset \operatorname{Isom}_{\text{lac}}(\hat{\pi}_{g,\nu}, \pi)$ under the subgroup of † $\widehat{\widehat{\mathcal{G}}}_{g;(\nu,\nu'\dagger)}$ of $\widehat{\widehat{\mathcal{G}}}_{g,\nu} = \operatorname{Aut}_{\text{lac}}(\hat{\pi}_{g,\nu})$ consisting of the $u|h \in \widehat{\widehat{\mathcal{G}}}_{g,\nu}$ whose image in $\mathfrak{S}_I$ leaves invariant † the set of the ν'† first elements (or, of the $\nu - \nu'$ last its|their complementary set,).

m446

Marginal note. [m446] Whence also: ∀ basis, ∀ discretification, oriented discretification of π defines one of π', whence ditto for classes of π, π' conjugation, i.e. for|by the strict prediscretifications (possibly oriented), and also for|by the closures of the classes, i.e. the † prediscretifications and oriented prediscretifications. It is only the case of arithmetizations|arithmetizations that † would require a finer analysis…

For such a basis, we find a corresponding basis of π'. The datum of (π, I') is equivalent to that of a torsor under $\widehat{\widehat{\mathcal{G}}}_{g,(\nu,\nu'\dagger)}$, that of a π' to † the datum of a torsor under $\widehat{\widehat{\mathcal{G}}}_{g,\nu'}$, and the passage from π to π' is described by the change of group of operators

$$\widehat{\widehat{\mathcal{G}}}_{g,(\nu,\nu')} \longrightarrow \widehat{\widehat{\mathcal{G}}}_{g,\nu'}$$

this homomorphism sends $\widehat{\mathcal{G}}_{g,(\nu,\nu')} = \Sigma_{g,(\nu,\nu')}$ into $\widehat{\mathcal{G}}_{g,\nu'} = \Sigma_{g,\nu'}$, [p. 341]

$\Sigma_{g,\nu}^{\cap}$

Uncertainty retained from cancelled passages.

140-2-s342-D002 [illeg. 2]
140-2-s342-D003 For a map
140-2-s342-D005 [illeg. 2]
140-2-s342-D007 bases
140-2-s342-D008 in
140-2-s342-D009 [illeg. 1]
140-2-s342-D010 +2
140-2-s342-D012 also
140-2-s342-D013 +2
140-2-s342-D014 [illeg. 1]
140-2-s342-D015 requires
140-2-s342-D017 +2
140-2-s342-D018 [illeg. 1]

[scan 343]

and even $\mathcal{G}_{g,(\nu,\nu')}$ into $\mathcal{G}_{g,\nu'}$, also π into π'. It follows that every prediscretification of π defines one of π', the same for strict prediscretifications, discretifications, outer bases "adapted to I'" †. Finally, if we have a "prearithmetization" of π, i.e. a $\Sigma \subset \widehat{\widehat{\mathcal{T}}}(\pi)$, do we deduce from it a prearithmetization of π' — i.e. (for example) if two prediscretifications of π are compatible i.e. † have the same group $\Sigma \subset \widehat{\widehat{\mathcal{T}}}(\pi)$, of outer automorphisms, is the same true of their images?

†

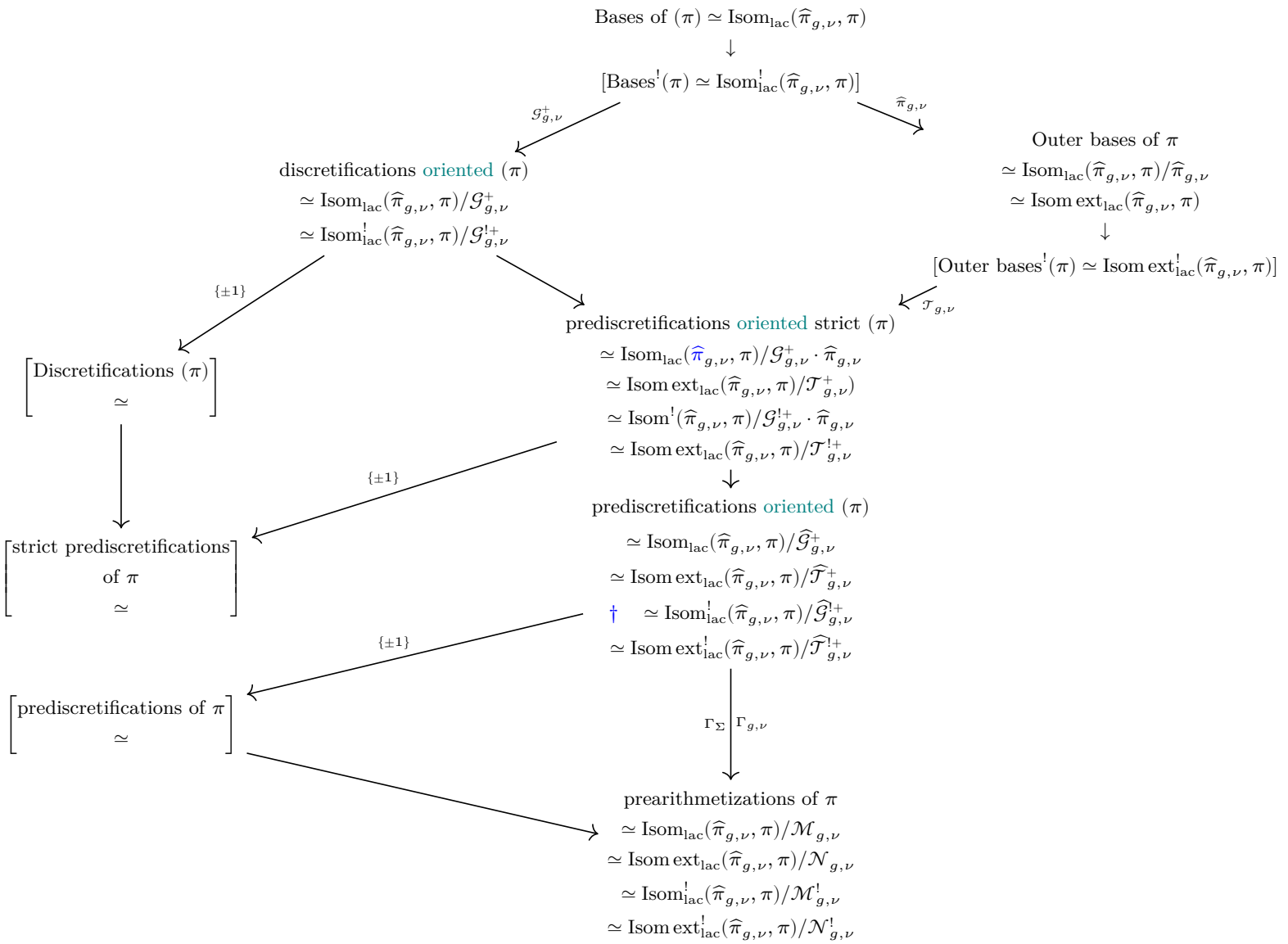
To see more clearly, we are going to write in diagram form the remarkable sets associated with π, and this in two ways, one without using any particular structure|particular structures on $I = I(\pi)$, the other using a total order, which makes it possible, in the torsor $\mathrm{Isom}_{\mathrm{lac}}(\hat{\pi}_{g,\nu}, \pi)$ under $\widehat{\widehat{\mathcal{G}}}_{g,\nu}$, to define the subtorsor under $\widehat{\widehat{\mathcal{G}}}^{!}_{g,\nu}$ which we may denote † $\mathrm{Isom}^{!}_{\mathrm{lac}}(\hat{\pi}_{g,\nu}, \pi)$. [p. 342]

Uncertainty retained from cancelled passages.

140-2-s343-D002 (i.e.) for which
140-2-s343-D003 define
140-2-s343-D004 [ ] [which sends Σ
140-2-s343-D005 [illeg. 1]

[scan 344]

[Diagram continued on the landscape page; 140-2-344-D1] [p. 174]


Bases of $(\pi) \simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)$
$[\mathrm{Bases}^{!}(\pi) \simeq \mathrm{Isom}^{!}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)]$
$\mathcal{G}^{+}_{g,\nu}$
$\widehat{\pi}_{g,\nu}$
discretifications oriented (π)
$\simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{G}^{+}_{g,\nu}$
$\simeq \mathrm{Isom}^{!}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{G}^{!+}_{g,\nu}$
Outer bases of π
$\simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\widehat{\pi}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)$
$[\mathrm{Outer\ bases}^{!}(\pi) \simeq \mathrm{Isom}\,\mathrm{ext}^{!}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)]$
$\{\pm 1\}$
$\mathcal{T}_{g,\nu}$
prediscretifications oriented strict (π)
$\simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{G}^{+}_{g,\nu} \cdot \widehat{\pi}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{T}^{+}_{g,\nu})$
$\simeq \mathrm{Isom}^{!}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{G}^{!+}_{g,\nu} \cdot \widehat{\pi}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{T}^{!+}_{g,\nu}$
Discretifications (π)
$\simeq$
$\{\pm 1\}$
prediscretifications oriented (π)
$\simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\widehat{\mathcal{G}}^{+}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\widehat{\mathcal{T}}^{+}_{g,\nu}$
$\dagger \quad \simeq \mathrm{Isom}^{!}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\widehat{\mathcal{G}}^{!+}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}^{!}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\widehat{\mathcal{T}}^{!+}_{g,\nu}$
strict prediscretifications of π
$\simeq$
$\{\pm 1\}$
Γ_{Σ}
$\Gamma_{g,\nu}$
prediscretifications of π
$\simeq$
prearithmetizations of π
$\simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{M}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{N}_{g,\nu}$
$\simeq \mathrm{Isom}^{!}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{M}^{!}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}^{!}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{N}^{!}_{g,\nu}$

m450

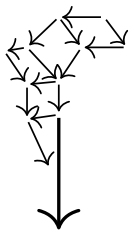
not separated
not separated

Marginal note. m450



m451

Marginal note. m451



Please turn over
[p. 343]

Uncertainty retained from cancelled passages.
140-2-s344-D002 ≃

[scan 345]

N.B.
We set †

$$\Gamma_{g,\nu} = \mathcal{N}_{g,\nu}/\Sigma_{g,\nu} = \mathcal{N}^{!}_{g,\nu}/\Sigma^{!}_{g,\nu}$$

(where $\Sigma_{g,\nu} = \widehat{\mathcal{T}}^{+}_{g,\nu}$, $\Sigma^{!}_{g,\nu} = \widehat{\mathcal{T}}^{!+}_{g,\nu}$). We can also describe it as the group Γ_Σ associated with a profinite outer group with loops π equipped with an oriented prediscretification α (nothing more). If we have two such pairs $(\pi, \alpha), (\pi', \alpha')$, whence Γ_Σ and $\Gamma_{\Sigma'}$ ($\Sigma = \mathrm{Aut}\,\mathrm{ext}_{\mathrm{lac}}(\pi, \alpha)$, $\Sigma' = \mathrm{Aut}\,\mathrm{ext}_{\mathrm{lac}}(\pi', \alpha')$), we indeed find a canonical isomorphism

$$\Gamma_\Sigma \xrightarrow{\sim} \Gamma_{\Sigma'}$$

by taking any outer isomorphism with loops u of π with π', transforming α into α' — the associated isomorphism $\Gamma_\Sigma \longrightarrow \Gamma_{\Sigma'}$ obviously does not depend on the choice of u.

This being said, the pairs (π, Σ) of a profinite|profinite group|groups with loops of type (g, ν), equipped with a prearithmetization, with as morphisms the arithmetically outer isomorphisms (relative to Σ and Σ' ...), form a connected groupoid $\Pi_{g,\nu}$, having an "origin" object (defined up to unique isomorphism as being $(\hat{\pi}_{g,\nu}, \Sigma_{g,\nu})$, or at will any (π, Σ_α) coming from a (π, α) (α an oriented prediscretification of π), and whose group of automorphisms is precisely $\Gamma_{g,\nu}$. [p. 344]

Uncertainty retained from cancelled passages.
140-2-s345-D001 [illeg. 1]

[scan 346]
[p. 175]

We observe that, with the exception of the two homogeneous spaces (under $\widehat{\widehat{\mathcal{G}}}(\pi)$, but not necessarily|necessarily under $\widehat{\widehat{\mathcal{G}}}(\pi, I')$, or even under $\widehat{\widehat{\mathcal{G}}}^{!}(\pi)$) Bases (π) and Outer bases (π), the other four are in fact homogeneous spaces under $\widehat{\widehat{\mathcal{G}}}^{!}$, and are expressed as quotients of the $\widehat{\widehat{\mathcal{G}}}^{!}_{g,\nu}$-torsor $\mathrm{Isom}^{!}(\hat{\pi}_{g,\nu}, \pi)$, by the four subgroups

$$\mathcal{G}^{!}_{g,\nu}, \quad \mathcal{G}^{!}_{g,\nu} \cdot \hat{\pi}_{g,\nu}, \quad \widehat{\mathcal{G}}^{!}_{g,\nu}, \quad \mathcal{M}^{!}_{g,\nu}$$

(the last defined as the inverse image of $\mathcal{N}^{!}_{g,\nu}$ in $\widehat{\widehat{\mathcal{G}}}_{g,\nu}$, just as $\mathcal{M}_{g,\nu}$ is defined as the inverse image of $\mathcal{N}_{g,\nu}$). We find on the other hand, if|for $I' = \{i_0, \dots, i_{\nu'-1}\}$, a|some homomorphism

$$\widehat{\widehat{\mathcal{G}}}^{!}_{g,\nu} \longrightarrow \widehat{\widehat{\mathcal{G}}}^{!}_{g,\nu'},$$

sending $\mathcal{G}^{!}_{g,\nu}$ into $\mathcal{G}^{!}_{g,\nu'}$, † by sending $\pi_{g,\nu}$ into $\pi_{g,\nu'}$, thus $\hat{\pi}_{g,\nu}$ into $\hat{\pi}_{g,\nu'}$, $\widehat{\mathcal{G}}^{!}_{g,\nu} = \Sigma^{!}_{g,\nu}$ into $\widehat{\mathcal{G}}^{!}_{g,\nu'} = \Sigma^{!}_{g,\nu'}$, and the subgroup $\mathcal{G}^{!}_{g,\nu} \cdot \hat{\pi}_{g,\nu}$ of † $\widehat{\mathcal{G}}^{!}_{g,\nu}$ into the subgroup $\mathcal{G}^{!}_{g,\nu'} \cdot \hat{\pi}_{g,\nu'}$ of $\widehat{\mathcal{G}}^{!}_{g,\nu'}$. Finally, since $\Sigma^{!}_{g,\nu}$ maps into $\Sigma^{!}_{g,\nu'}$, † $\mathcal{N}^{!}_{g,\nu}$ maps into the normalizer of $\Sigma^{!}_{g,\nu'}$ (in $\widehat{\widehat{\mathcal{T}}}^{!}_{g,\nu'}$), I say that this is none other than $\mathcal{N}^{!}_{g,\nu'}$. Changing notation, this amounts to the

Lemma The $\mathcal{N}^{!}_{\Sigma} = \mathrm{Norm}_{\widehat{\widehat{\mathcal{T}}}^{!}}(\Sigma^{!})$ † $(= \mathrm{Norm}_{\widehat{\widehat{\mathcal{T}}}}(\Sigma^{!}) \cap \widehat{\widehat{\mathcal{T}}}^{!})$. not proved †

Obviously we have $\mathcal{N}^{!}_{\Sigma} \subset \mathrm{Norm}_{\widehat{\widehat{\mathcal{T}}}}(\Sigma^{!}) \cap \widehat{\widehat{\mathcal{T}}}^{!}$, conversely let $g \in \widehat{\widehat{\mathcal{T}}}^{!}$ which normalizes $\Sigma^{!}$, let us show that it normalizes Σ i.e. that it is $\in \mathcal{N}$ (thus $\in \mathcal{N}^{!} = \mathcal{N} \cap \widehat{\widehat{\mathcal{T}}}^{!}$). [p. 345]

Uncertainty retained from cancelled passages.

140-2-s346-D001 [illeg. 1] $\widehat{\mathcal{G}}$ $\widehat{\pi}_{g,\nu}$

140-2-s346-D002 [illeg. 1]

140-2-s346-D003 [illeg. 1]

140-2-s346-D005 [illeg. 1]

140-2-s346-D006 [illeg. 4] $\mathrm{Norm}_{\widehat{\mathcal{T}}}(\Sigma^{!})/\Sigma^{!}$

[scan 347]

It's not clear. I nevertheless feel like proving|pronouncing † the conjecture The canonical map

$$\text{Prediscretifications|Prediscretifications}^{+}(\pi) \xrightarrow{\sim} \text{Prediscretifications|Prediscretifications}^{+}(\pi')$$

is bijective. In order that two oriented prediscretifications α, β of π † have the same group of outer automorphisms $\Sigma_{\alpha} = \Sigma_{\beta}$, it is necessary and sufficient † that it be so. for their images $\Sigma_{\alpha'}$ and $\Sigma_{\beta'}$, i.e. that $\Sigma_{\alpha'} = \Sigma_{\beta'}$; so that the preceding bijection induces a bijection

$$\text{Prearithmetizations}(\pi) \xrightarrow{\sim} \text{Prearithmetizations}(\pi').$$

Finally, the preceding bijections are compatible with the homomorphism of the groups of operators $\widehat{\widehat{\mathcal{T}}}(\pi, I') \longrightarrow \widehat{\widehat{\mathcal{T}}}(\pi')$, † a fortiori a $\widehat{\widehat{\mathcal{T}}}^{!}(\pi) \longrightarrow \widehat{\widehat{\mathcal{T}}}^{!}(\pi')$, which implies that it thereby induces|that it induces them (for a given prearithmetization $\Sigma \subset \widehat{\widehat{\mathcal{T}}}(\pi)$ of π, giving $\Sigma' \subset \widehat{\widehat{\mathcal{T}}}(\pi')$ in π'; with $\Sigma^{!} \longrightarrow \Sigma'^{!}$) a homomorphism $\mathcal{N}^{!}_{\Sigma}\dagger \longrightarrow \mathcal{N}^{!}_{\Sigma'}\dagger$, † and the corresponding homomorphism

$$\Gamma_{\Sigma} = \mathcal{N}^{!}_{\Sigma}/\Sigma^{!} \longrightarrow \dagger\Gamma_{\Sigma'} = \mathcal{N}^{!}_{\Sigma'}/\Sigma'^{!}$$

is an isomorphism.

To convince oneself of this it suffices to look at † (choosing $\pi = \pi_{g,\nu}|\widehat{\pi}_{g,\nu}$, I' † reduced to the|being a last element|element of I_ν) the case where $I' = \{i'|i\}$, We then have a homomorphism

$$\widehat{\widehat{\mathcal{T}}}(\pi, i) \longrightarrow \widehat{\widehat{\mathcal{G}}}(\pi').$$

(the outer loop automorphisms of π fixing i define well-determined automorphisms of π' — not just outer ones) [p. 346]

Uncertainty retained from cancelled passages.

140-2-s347-D001 [illeg. 1]
140-2-s347-D002 be
140-2-s347-D004 for their images
140-2-s347-D006 which induces
140-2-s347-D008 (Σ)
140-2-s347-D009 (Σ')
140-2-s347-D010 which is likewise
140-2-s347-D012 $\mathcal{N}^{!}_{\Sigma'}/\Sigma'^{!}$
140-2-s347-D013 $\pi = \widehat{\pi}_{g,\nu}$
140-2-s347-D014 [illeg. 1]

[scan 348]

i.e. $\widehat{\widehat{\mathcal{G}}}(\pi, i) \longrightarrow \widehat{\widehat{\mathcal{G}}}(\pi')$ is trivial on $\widehat{\pi}$, and thus passes to the quotient as a $\widehat{\widehat{\mathcal{T}}}(\pi, i) \longrightarrow \widehat{\widehat{\mathcal{G}}}(\pi, i)$. [p. 176] I assume, by analogy with the discrete case, that this homomorphism is an isomorphism, so that we have an exact sequence m454

Marginal note. m454



m455

Marginal note. m455 recall that we assume π anabelian|nonabelian $(2g + \nu \geq 3)$

$$1 \longrightarrow \pi' \longrightarrow \widehat{\widehat{\mathcal{T}}}(\pi, i) \longrightarrow \widehat{\widehat{\mathcal{G}}}(\pi', i) \longrightarrow 1$$

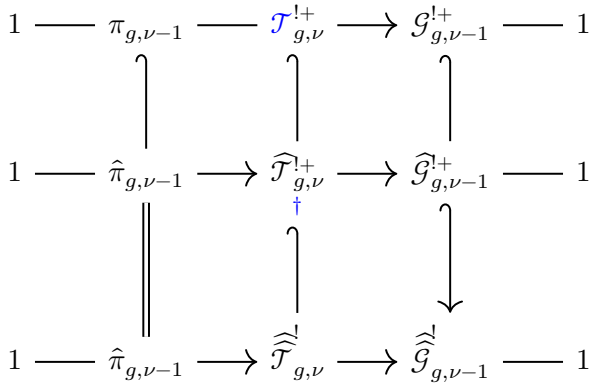
or again

$$1 \longrightarrow \widehat{\pi}_{g,\nu-1} \longrightarrow \widehat{\widehat{\mathcal{T}}}_{g,(\nu,\nu-1)} \longrightarrow \widehat{\widehat{\mathcal{G}}}_{g,\nu-1} \longrightarrow 1$$

which contains the exact sequence

$$1 \longrightarrow \pi^{*|+}_{g,\nu-1} \longrightarrow \mathcal{T}_{g,(\nu,\nu-1)} \longrightarrow \mathcal{G}_{g,\nu-1} \longrightarrow 1$$

as|known sub†, sequence|is exact. It is convenient to work instead with



The map

[↪ 140-2-348-M1]

Prediscretifications|Prediscretifications$^{+}(\pi)$

$\longrightarrow$ Prediscretifications|Prediscretifications$^{+}(\pi')$

is thus identified with $\widehat{\widehat{\mathcal{T}}}^{!}_{g,\nu}/\widehat{\mathcal{T}}^{!}_{g,\nu}$, we read off the diagram that this is $\simeq \widehat{\widehat{\mathcal{G}}}^{!\dagger}_{g,\nu-1}/\widehat{\mathcal{G}}^{!+}_{g,\nu-1}$, whence bijectivity on the sets of oriented prediscretifications.

In the group $\widehat{\widehat{\mathcal{T}}}^{!}_{g,\nu} = \mathcal{T}$, we have an invariant subgroup $\hat{\pi}_{g,\nu-1} = \pi'$, and a subgroup Σ $(= | \longleftarrow \widehat{\mathcal{T}}^{!+}_{g,\nu})$ [p. 347]

Uncertainty retained from cancelled passages.

140-2-s348-D002 particular|dense part

140-2-s348-D003 [*illeg.* 1]

140-2-s348-D004 +

[scan 349]

between π' and $\mathcal{T}$, † giving a subgroup $\Sigma' \simeq \mathcal{T}(\pi')(= \widehat{\mathcal{G}}^{!+}_{g,\nu-1})$ in $\mathcal{T}' = \mathcal{T}/\pi'$ $(\simeq \widehat{\widehat{\mathcal{G}}}^{!}_{g,\nu-1})$, and we know well that then

$$\mathcal{T}/\Sigma \simeq \mathcal{T}'/\Sigma',$$

and † that the normalizer $\mathcal{N}$, of Σ in $\mathcal{T}$ is the inverse image of the normalizer $\mathcal{N}'$ of Σ' in $\mathcal{T}'$, so that $\mathcal{N}/\Sigma \simeq \mathcal{N}'/\Sigma'$, which proves what we wanted.

Corollary We have canonical isomorphisms

$$\Gamma_{g,\nu} \xrightarrow{\sim} \Gamma_{g,\nu-1} \xrightarrow{\sim} \cdots \quad \dagger$$

m456

Marginal note. m456 We also see that $\Sigma|\Sigma^{!}$ invariant in $\mathcal{T}|\widehat{\mathcal{T}}$ is equivalent to Σ' invariant in $\widehat{\widehat{\mathcal{T}}}'$

so that if $g \geq 2$, we have canonically $\Gamma_{g,\nu} \simeq \Gamma_{g,0}$, on the other hand $\Gamma_{1,\nu} \simeq \Gamma_{1,1}$ $(g = 1, \nu \geq 1)$, $\Gamma_{0,\nu} \simeq \Gamma_{0,3}$ $(\nu \geq 3)$.

N.B. The "fact" † assumed is far from obvious — even that $\widehat{\widehat{\mathcal{T}}}^{!}_{g,(\nu,\nu-1)} \longrightarrow \widehat{\widehat{\mathcal{T}}}_{g,\nu-1}|\widehat{\widehat{\mathcal{M}}}_{g,\nu-1}$ is surjective, or that $\widehat{\widehat{\mathcal{T}}}_{g,(\nu,\nu-1)} \longrightarrow \widehat{\widehat{\mathcal{G}}}_{g,\nu-1}$ is injective, is far from obvious, and is perhaps altogether false !

The assumed fact amounts to a statement of the existence of a functor in the opposite direction "drilling holes", which|what in any case remains not understood. [p. 348]

Uncertainty retained from cancelled passages.
140-2-s349-D001 giving by
140-2-s349-D002 $N(\Sigma)$
140-2-s349-D003 $\Gamma_{g,0}$
140-2-s349-D004 [illeg. 1]

[scan 350]

28. Change of type (g, ν) (continued) : passage to a finite covering. [p. 177]

Let π be a profinite group with loops of type (g, ν), anabelian as always, π' a subgroup of finite index. We know (or we|we thereby verify, by passing to the discrete case) that † equipped with the traces on|under π of the loop subgroups of π, π' is a group with loops.

†

The loop subgroups are also|thus the † subgroups L' such that

$$\text{a)} \quad L' = \mathrm{Centr}_\pi(L') \cap \pi'.$$

$$\text{b)} \quad L = | : \mathrm{Centr}_\pi(L') \text{ is a loop subgroup in } \pi.$$

We put $d(L') = [L : L']$, and we thus have a function

$$d : I' = I(\pi') \longrightarrow \mathbb{N}^*,$$

and a map $I' \xrightarrow{\varphi} I$, such that $\forall i \in I$, we have

$$(1) \qquad \sum_{\substack{i' \in I' \\ \text{over } i}} d(i') = n \qquad (\text{where|here } n = [\pi : \pi'])$$

We will have the Hurwitz formula

$$(2) \qquad \dagger\ 2g' - 2 = n(2g-2) + \overbrace{\sum_{i' \in I'} (d(i') - 1)}^{n \,\mathrm{card}\, I - \mathrm{card}\, I'}$$

[p. 349]

Uncertainty retained from cancelled passages.
140-2-s350-D001 [illeg. 1] discretified
140-2-s350-D002 n int(g) } n counted|to count int(g) $d_\alpha|d_{i\alpha}$ $L'_{i\alpha}|N'_{i\alpha}$ $\cap\pi'$ for conjugacy
140-2-s350-D006 the
140-2-s350-D007 $2g|2\alpha$

i.e. [scan 351]

$$(3) \qquad 2g' - 2 + \mathrm{card}\, I' = n(2g - 2 + \mathrm{card}\, I)$$

(N.B. † $(2g - 2 + \mathrm{card}\, I)$ is the negative of the Euler–Poincaré characteristic with compact supports of the curve of which $\Pi|\pi$ is the fundamental group…) ✓
m457

Marginal note. m457 N.B. saying that we are in the anabelian case means that $-\mathrm{EP}_!$ is ≥ 1 — this condition|relation is therefore preserved by passage to a covering|to coverings. The integer $-\mathrm{EP}_!$ measures strict positivity [illeg. 1] degree of anabelianity, in some sense…

Obviously, every discretification of $\Pi|\pi$ defines one of $\Pi|\pi'$ — and the same holds for oriented discretifications, modulo a bit of cohomology [which amounts to saying, in the discrete case for example|for example) that

$$\begin{array}{ccc} H^2_!(U|\mathcal{U}, \mathbb{Z}|\mathbf{Z}) & \longrightarrow & H^2_!(U'|\mathcal{U}', \mathbb{Z}|\mathbf{Z}) \\ \wr & & \wr \\ T & & T' \end{array}$$

can be written in the form $n\theta^{-1}$, where θ is a well-determined isomorphism (the trace isomorphism) $\theta|\text{of } T' \overset{\sim}{\to} T$.

It is not clear to me at first sight whether the map

$$\text{Discret}(\Pi|\pi) - | \longrightarrow \text{Discret}(\Pi|\pi')$$

is surjective, or injective. Injectivity (for every $\Pi|\pi'$ of finite index in $\Pi|\pi$) would mean that: two discretifications $\Pi|\pi_0, \Pi|\pi_0'$ of $\Pi|\pi$ which are commensurable, i.e. such that $\Pi|\pi_0 \cap \Pi|\pi_0'$ is of finite index in $\Pi|\pi_0$ and in $\Pi|\pi_0'$, † equal

---

Uncertainty retained from cancelled passages.

140-2-s351-D001 n

140-2-s351-D002 that it be so

Suppose that $\Pi|\pi'$ is an invariant subgroup in $\Pi|\pi$, put $G = \Pi|\pi'/\Pi|\pi$ [scan 352]

Conjecture The map $\text{Discret}(\Pi|\pi) \longrightarrow \text{Discret}(\Pi|\pi')$ is injective, and its image consists of the discretifications $\Pi|\pi_0' \subset \Pi|\pi'$ such that $G \longrightarrow \text{Aut}\,\text{ext}_{\text{lac}}(\Pi|\pi') \simeq \text{Aut}\,\text{ext}_{\text{lac}}(\widehat{\Pi|\pi_0}') \simeq \widehat{\widehat{\mathcal{T}}}(\widehat{\Pi|\pi_0}')$ factors through $\mathcal{T}(\Pi|\pi_0') = \text{Autext}_{\text{lac}}(\Pi|\pi_0')$ (and in fact, even through $\mathcal{T}(\Pi|\pi_0')^+$). [p. 178]

m458

Marginal note. m458 In other words, $\forall g \in \Pi|\pi$, $\exists h \in \Pi|\pi'$ such that $\text{int}(hg)(\Pi|\pi_0') \subset \Pi|\pi_0'$ i.e. $\Pi|\pi'|\Pi|\pi_0' \cdot \Pi|\pi_0 = \Pi|\pi$, where $\Pi|\pi_0$ is the normalizer of $\Pi|\pi_0'$ in $\Pi|\pi$.

The condition for a † $\Pi|\pi_0'$ to be in the image is obviously necessary. Let us show that it is sufficient, and that the $\Pi|\pi_0^*$ giving rise to $\Pi|\pi_0'$ by $\Pi|\pi_0 \cap \Pi|\pi'$ is unique. We have an outer action of G on $\Pi|\pi_0'$, which defines the outer action on $\widehat{\Pi|\pi_0}' \simeq \Pi|\pi'$, now $\Pi|\pi$ is recovered canonically from the latter (since $\text{Centre}(\widehat{\Pi|\pi}') = 1$), and we see that it is the profinite completion of the extension $\Pi|\pi_0$ of G by $\Pi|\pi_0'$, defined by the outer action of G on $\Pi|\pi_0'$. We do indeed have a generalized discretification $\Pi|\pi_0$ of $\Pi|\pi$, in the sense of just the structure of a † profinite group, and it induces $\Pi|\pi_0'$ — but we must see that it is compatible with the loop structure of $\Pi|\pi$.

Uncertainty retained from cancelled passages.

140-2-s352-D001 [illeg. 1]

140-2-s352-D003 group

But this is made explicit, starting from the loop structure of $\Pi|\pi'$, we take the loop subgroups L' of $\Pi|\pi'$, and their Centralizers in $\Pi|\pi$. Let us do the same for $\Pi|\pi_0 \supset | > \Pi|\pi_0'$, we should find a loop structure on $\Pi|\pi_0$, giving rise to the sames d_i. I have the feeling that this should not be too nasty to prove — it would be as if we already knew that every "torsion-free" extension of a finite group by a group with loops is (in the canonical way) a group with loops… [scan 353]

Corollary In any case, the map $\text{Discret}(\Pi|\pi) \longrightarrow \text{Discret}(\Pi|\pi')$ is injective.

N.B. It should not always be bijective, for there must be an outer|outer action of a G on a $\widehat{\Pi|\pi_0}'$ i.e. homomorphisms $G \longrightarrow \widehat{\widehat{\mathcal{F}}}(\widehat{\Pi|\pi_0}')$, which do not factor|factor through $\mathcal{T}(\Pi|\pi_0')$, and which nevertheless is as good|good from the profinite group point of view as that|those coming from a $\Pi|\pi$ and a subgroup

[scan 354]

invariant $\Pi|\pi'$. In fact, let us start from such a discrete situation $\Pi|\pi_0 \supset | > \Pi|\pi_0'$, whence [p. 179] $G = \Pi|\pi_0/\Pi|\pi_0'$ and $G \longrightarrow \mathrm{Autext}_{\mathrm{lac}}(\Pi|\pi_0')\dagger = \mathcal{T}(\Pi|\pi_0') \subset \widehat{\mathcal{F}}|\widehat{\widehat{\mathcal{F}}}(\widehat{\Pi|\pi_0}')$, and conjugate G by an element γ of $\widehat{\widehat{\mathcal{F}}}(\Pi|\pi_0')$, so as to find $G \longrightarrow \widehat{\mathcal{F}}(\widehat{\Pi|\pi_0}')$ which does $\dagger$ not factor through $\mathcal{T}(\Pi|\pi_0')$. We find an extension $\widehat{\Pi|\pi}|\Pi|\pi^{\natural|\sharp}$ of G by $\widehat{\Pi|\pi_0}'$, isomorphic to $\widehat{\Pi|\pi_0}$ (as an extension of G by $\widehat{\Pi|\pi_0}'$), hence the loop structure of $\widehat{\Pi|\pi_0}'$ defines one on $\Pi|\pi^{\natural|\sharp}$ (so as to be induced by the latter), but nevertheless the chosen discretification $\Pi|\pi_0$ of $\widehat{\Pi|\pi_0}$ is not induced by one of $\Pi|\pi$. To

$*$ Carry out the construction, it suffices to find an element $\gamma \in \widehat{\widehat{\mathcal{F}}}(\Pi|\pi_0')\ \dagger$ such that $\mathrm{int}(\gamma).G \not\subset | \not\in \mathcal{T}, \dagger$

[A large part of the canceled passage is enclosed by a frame (left, bottom and right sides) and crossed out with four long slanting strokes. After the inclusion $\gamma^{-1}.\Pi|\pi_0' \subset \widehat{\Pi|\pi_0}'$ on the last line, stroke curved into a hook, directed toward the following inclusion.]

i.e. such that $G\ \dagger$ does not stabilize $\gamma^{-1}(\Pi|\pi_0') \subset \widehat{\Pi|\pi_0}'$

Such a γ always exists....

Uncertainty retained from cancelled passages.

140-2-s354-D001 (π_0')

140-2-s354-D002 [illeg. 2]

140-2-s354-D004 finite

140-2-s354-D005 u of order [illeg. 1] coming in $\Pi|\pi$ or lifting $\mathcal{G}|\mathcal{C}$ $\Pi|\pi$ $\Pi|\pi$ conjugation a $\Pi|\pi$ $\Pi|\pi$ fixes $\Pi|\pi$ $\Pi|\pi$

140-2-s354-D009 does not fix

[scan 355]

Of course, there is no reason, for two discretifications $\Pi|\pi_0'|\Pi|\pi_0, \Pi|\pi_1$ of $\Pi|\pi$, that if $\Pi|\pi_0, \Pi|\pi_1$ are $\Pi|\pi$-conjugate (i.e. define the same strict prediscretification|prediscretification of $\Pi|\pi$), the same should hold for $\Pi|\pi_0'$ and $\Pi|\pi_1'$ under the inner action of $\Pi|\pi'$. (It would be necessary for $\Pi|\pi_0, \Pi|\pi_1$ to be conjugate by $\Pi|\pi'$, and not just by $\Pi|\pi$). On the other hand, I presume that if $\Pi|\pi_0, \Pi|\pi_1$ are "adherent" to one another in the space of discretifications|discretifications of $\Pi|\pi$, i.e. if they define the same prediscretification, the same holds for $\Pi|\pi_0', \Pi|\pi_1'$, and that the map

$$\mathrm{Prédiscrt}^+(\Pi|\pi\dagger) \longrightarrow \mathrm{Prédiscrt}^+(\Pi|\pi'\dagger) \tag{4}$$

which we obtain in this way, is again bijective, and that it in turn passes to the quotient, to define a bijective map

$$\text{Préarith} \,|\, \text{Préarithm}(\Pi|\pi\dagger) \longrightarrow \text{Préarith} \,|\, \text{Préarithm}(\Pi|\pi'\dagger), \tag{5}$$

and that if Σ, Σ' correspond to one another under the latter, that we have a canonical isomorphism

$$\mathbb{\Gamma}|\Gamma_\Sigma \longrightarrow \mathbb{\Gamma}|\Gamma_{\Sigma'}, \tag{6}$$

compatible,

Uncertainty retained from cancelled passages.

140-2-s355-D004 o

140-2-s355-D005 o

140-2-s355-D006 o

140-2-s355-D007 o

[scan 356]

with the actions of these groups on the set P_Σ of prediscretifications of $\Pi|\pi$ compatible with Σ, and the set $P_{\Sigma'}$ of prediscretifications of $\Pi|\pi'$ compatible with Σ' (which are resp. left torsors under $\mathbb{\Gamma}|\Gamma_\Sigma, \mathbb{\Gamma}|\Gamma_{\Sigma'}$). At the same time, we will find therefore that Σ is unique for $\Pi|\pi$ i.e. invariant under $\widehat{\widehat{\mathcal{T}}}(\Pi|\pi)$, iff Σ' is unique in $\Pi|\pi'$ i.e. invariant under $\widehat{\widehat{\mathcal{T}}}(\Pi|\pi')$. [p. 180]

In any case, the fundamental conjecture which interprets the $\mathbb{\Gamma}|\Gamma_{g,\nu|v}$ as $\mathbb{\Gamma}|\Gamma_0 = \mathrm{Gal}(\overline{\mathbb{Q}}_0/\mathbb{Q})$ would have|will have at least as a consequence that the map $\mathrm{Discr}^+(\Pi|\pi)\text{–}| \longrightarrow \mathrm{Discr}^+(\Pi|\pi')$ passes to the quotient in two ways, to give a commutative diagram

$$\begin{array}{ccc}
\mathrm{Discr}^+(\Pi|\pi) & \longrightarrow & \mathrm{Discr}^+(\Pi|\pi') \\
\downarrow & & \downarrow \\
P|p = \mathrm{Prédiscr}^+(\Pi|\pi) & \longrightarrow & P|p' = \mathrm{Prédiscr}^+(\Pi|\pi') \\
\downarrow & & \downarrow \\
A = \mathrm{Arithm}(\Pi|\pi) & \longrightarrow & A' = \mathrm{Arithm}(\Pi|\pi')
\end{array}$$

and that the † lower square is cartesian (without prejudging the injectivity or the bijectivity of the map $A \longrightarrow A'$). So it would be necessary

Uncertainty retained from cancelled passages.

140-2-s356-D001 diagram|diagram

[scan 357]

that † if $\Pi|\pi_0, \Pi|\pi_1$ are two discretifications (oriented) of $\Pi|\pi$, which define the same prediscretification (oriented), then the same holds for their traces on $\Pi|\pi'$, $\Pi|\pi'_0$ and $\Pi|\pi'_1$, with respect to $\Pi|\pi'$. But already if $\Pi|\pi_0$ and $\Pi|\pi_1$ are $\Pi|\pi$-conjugate (i.e. define same discretification (oriented, strict)), this is not so clear — † but yes, for we will have $\Pi|\pi = \Pi|\pi' \cdot \Pi|\pi_0$ hence if $\Pi|\pi_1 = \mathrm{int}(u)\Pi|\pi_0$ writing $u = u'u_0$ with $u' \in \Pi|\pi'$, $u_0 \in \Pi|\pi_0$, we will have

$$\Pi|\pi_1 = \mathrm{int}(u')\,\mathrm{int}(u_0)\Pi|\pi_0 = \mathrm{int}(u')\Pi|\pi_0,$$

hence $\Pi|\pi'_1 = \mathrm{int}(u')\Pi|\pi'_0$ hence $\Pi|\pi'_0$ and $\Pi|\pi'_1$ are conjugate.

Let us try to proceed in the same way in the general case of a $u \in \mathcal{C}|\mathcal{G}(\widehat{\Pi|\pi_0})^+$, such that $\Pi|\pi_1 = u(\Pi|\pi_0)$, writing if possible $u = u'u_0$, with $u' \in \mathcal{C}|\mathcal{G}(\widehat{\Pi|\pi_0, \Pi|\pi'_0})^+$, and $u_0 \in \mathcal{C}|\mathcal{G}(\Pi|\pi_0)^+$, where we define $\mathcal{C}|\mathcal{G}(\Pi|\pi_0, \Pi|\pi'_0)^+$ as the group of i-automorphisms|l-automorphisms direct|discrete of $\Pi|\pi_0$ which stabilize $\Pi|\pi'_0$, and $\widehat{\mathcal{C}|\mathcal{G}}(\Pi|\pi_0, \Pi|\pi'_0)^+$ is its profinite compactification. We would then have $\Pi|\pi_1 = u'(u_0(\Pi|\pi_0)) = \dot{u}'(\Pi|\pi_0)$, where † $\dot{u}'$ denotes the image of u' in $\mathcal{T}|\widehat{\mathcal{C}(\Pi|\pi'_0)}$. In any case, there is no problem if $\Pi|\pi'$† is a † characteristic subgroup of $\Pi|\pi$, for then we have a homomorphism

Uncertainty retained from cancelled passages.

140-2-s357-D001 already
140-2-s357-D002 unless
140-2-s357-D003 [illeg. 1]
140-2-s357-D004 o
140-2-s357-D005 closed

discrete $\mathcal{C}|\mathcal{G}(\Pi|\pi_0) \longrightarrow \mathcal{C}|\mathcal{G}(\Pi|\pi_0')$, defining the|the $\widehat{\mathcal{C}|\mathcal{G}}(\widehat{\Pi|\pi_0}) \longrightarrow \widehat{\mathcal{C}|\mathcal{G}}(\widehat{\Pi|\pi_0}')$. Since the open characteristic subgroups of $\Pi|\pi$ are cofinal, we are reduced (for the questions of factorizability of $D(\Pi|\pi) \longrightarrow D(\Pi|\pi')$ into $P(\Pi|\pi) \longrightarrow P(\Pi|\pi')$ and $A(\Pi|\pi) \longrightarrow A(\Pi|\pi')$ and of bijectivity of the maps $P(\Pi|\pi) \longrightarrow P(\Pi|\pi')$ and $A(\Pi|\pi) \longrightarrow A(\Pi|\pi')$) we are reduced to the case where $\Pi|\pi'$ is a characteristic subgroup. We have indeed|seen then factorizability of $D(\Pi|\pi) \longrightarrow D(\Pi|\pi')$ into $P(\Pi|\pi) \longrightarrow P(\Pi|\pi')$, we want to show|let us show that this map is injective ... This † ⓐ means that the inverse image in $\widehat{\widehat{\mathcal{C}|\mathcal{G}}}(\widehat{\Pi|\pi_0})$ of $\widehat{\mathcal{C}|\mathcal{G}}(\widehat{\Pi|\pi_0}')^+$ is $\widehat{\mathcal{C}|\mathcal{G}}(\widehat{\Pi|\pi_0})^+$, the surjectivity ⓑ means that every element of $\widehat{\widehat{\mathcal{C}|\mathcal{G}}}(\widehat{\Pi|\pi_0}')$ is congruent modulo $\mathcal{C}|\widehat{\mathcal{G}(\Pi|\pi_0}')^+$ to an element of the image of $\widehat{\widehat{\mathcal{C}|\mathcal{G}}}(\Pi|\pi_0)$. The factorizability into $A(\Pi|\pi) \longrightarrow A(\Pi|\pi')$ means that by ⓒ the preceding map † $\widehat{\widehat{\mathcal{C}|\mathcal{G}}}(\Pi|\pi_0) \hookrightarrow \widehat{\widehat{\mathcal{C}|\mathcal{G}}}(\Pi|\pi_0')$, (N.B. it is immediate that it is injective) † sends

[scan 358] [p. 181]

$$M(\Pi|\pi_0) = \mathrm{Norm}_{\widehat{\widehat{\mathcal{C}}}|\widehat{\mathcal{C}}(\Pi|\pi_0)}(\widehat{\mathcal{C}|\mathcal{G}}^+(\Pi|\pi_0))$$

into $M(\Pi|\pi_0')$, the injectivity of $A(\Pi|\pi) \longrightarrow A(\Pi|\pi')$ that

Uncertainty retained from cancelled passages.

140-2-s358-D002 this means
140-2-s358-D003 of
140-2-s358-D004 is

[scan 359]

ⓓ we even have that $M(\Pi|\pi)$ is exactly|exact^{ly} the inverse image of $M(\Pi|\pi')$, the surjectivity of $A(\Pi|\pi) \twoheadrightarrow | \longrightarrow A(\Pi|\pi')$ that ⓔ every element of $\widehat{\widehat{\mathcal{C}|\mathcal{G}}}(\Pi|\pi_0')$ is congruent modulo $M(\Pi|\pi_0')$ to an element of $\widehat{\widehat{\mathcal{C}|\mathcal{G}}}(\Pi|\pi_0)$ — and these conditions together imply that the canonical homomorphism

weaker than (b)

$$\begin{array}{ccc} \mathbb{\Gamma}_{\dagger\Pi|\pi_0} & \longrightarrow & \mathbb{\Gamma}_{\Pi|\pi_0'} \\ \| & & \| \\ M(\Pi|\pi_0)/\widehat{\mathcal{C}|\mathcal{G}}(\Pi|\pi_0) & & M^{\sharp|*}(\Pi|\pi_0')/\widehat{\mathcal{C}|\mathcal{G}}(\Pi|\pi_0') \end{array}$$

is an isomorphism – it even suffices for this to have (c)|(e) (to be able to define this map) and (a) (for its injectivity), and (b) and the strengthening (d) of (c)|(e) (for its bijectivity|surjectivity. So we see that everything hangs on the properties of the homomorphism of group inclusions

$$(7) \qquad \begin{array}{ccccc} \widehat{\mathcal{G}}^+(\pi_0) & \subset & M(\pi_0) & \subset & \widehat{\widehat{\mathcal{G}}}(\pi_0) \\ \Big\uparrow & & \Big[\Big\downarrow ? \Big] & & \Big\uparrow \\ \widehat{\mathcal{G}}^+(\pi_0') & \subset & M(\pi_0') & \subset & \widehat{\widehat{\mathcal{G}}}(\pi_0') \end{array}$$

with the task of proving a) b) (which concern only the composite square) and c)|e).
†

Uncertainty retained from cancelled passages.

140-2-s359-D001 π

140-2-s359-D002 will be immediate c)|e) $\mathcal{C}|\mathcal{G}$ $\Pi|\pi$ $\mathcal{C}|\mathcal{G}$

[scan 360]

Perhaps the properties a) b) c) are completely false – yet a) (which corresponds to the injectivity of $P(\Pi|\pi) \longrightarrow P(\Pi|\pi')$) seems fairly plausible. Property b) (which would express the surjectivity of $P(\Pi|\pi) \longrightarrow P(\Pi|\pi')$) is much more problematic, even doubtful. The trouble is that the group $\widehat{\widehat{\mathcal{C}|\mathcal{G}}}(\Pi|\pi_0)$ can be "much smaller" than $\widehat{\widehat{\mathcal{C}|\mathcal{G}}}(\Pi|\pi_0')$, we realize this by passing to the quotient by the subgroup $\widetilde{\Pi|\pi_0'} = \Pi|\pi'$ (contained in the smallest of the groups in (7), and normal in the largest), we find on the second line the groups $\widehat{\mathcal{T}}^+(\Pi|\pi_0')$, † $\mathcal{N}|N(\Pi|\pi_0')$ and $\widehat{\widehat{\mathcal{T}}}(\Pi|\pi_0')$, on the first extensions of the corresponding|corresponding groups for $\Pi|\pi_0$ ($\widehat{\mathcal{T}}^+(\Pi|\pi_0)$, $\mathcal{N}|N(\Pi|\pi_0)$ and $\widehat{\widehat{\mathcal{T}}}(\Pi|\pi_0)$) by the finite group $G = \Pi|\pi_0/\Pi|\pi_0' \simeq \Pi|\pi/\Pi|\pi'$, † denoted|denoted by $\sim$'s, so that .|–

[p. 182]

(8)

$$\begin{array}{ccccccc} G & \subset & \widetilde{\widehat{\mathcal{T}}}^{+:}(\pi_0) & \subset & \widetilde{\mathcal{N}}(\pi_0) & \subset & \widetilde{\widehat{\widehat{\mathcal{T}}}}(\pi_0) \\ & \searrow & \uparrow & & \left[\uparrow ? \downarrow\right] & & \downarrow \\ & & \widehat{\mathcal{T}}^+(\pi_0') & \subset & \mathcal{N}(\pi_0') & \subset & \widehat{\widehat{\mathcal{T}}}(\pi_0') \end{array}$$

, which shows that the groups on the first line

Uncertainty retained from cancelled passages.

140-2-s360-D001 [illeg. 1]

140-2-s360-D002 let them be|let it be $\widetilde{\widehat{\mathcal{T}}}^+|\widetilde{\mathcal{T}}^+$

normalize the subgroup $G \subset \widehat{\mathcal{T}}^+(\pi_0')$ (N.B. in fact we have $G \subset \mathcal{T}^+(\pi_0')$; † and π_0 is reconstructed from π_0' and from this subgroup of π_0') — †we should be able to show without too much trouble that † in (8), $\widetilde{\widehat{\mathcal{T}}}^+(\pi_0)$ and $\widetilde{\widehat{\widehat{\mathcal{T}}}}(\pi_0)$ are precisely the normalizers of G in $\widehat{\mathcal{T}}^+(\pi_0')$ and in $\widehat{\widehat{\mathcal{T}}}(\pi_0')$ (which would indeed imply a), besides) — but I see no plausible reason why (b) should be true — which would mean, essentially, that there is no more "arithmetic transcendence" defined by the ("very big" !) $\widehat{\widehat{\mathcal{T}}}(\pi_0')$ (modulo $\widehat{\mathcal{T}}(\pi_0')^+$), than that defined by the ("much smaller") group $\widetilde{\widehat{\widehat{\mathcal{T}}}}(\pi_0) = \mathrm{Norm}_{\widehat{\widehat{\mathcal{T}}}(\pi_0|\pi_0')}(G)$ … Besides these conditions a) b) are not imperative for the fundamental conjecture linking the $\pi_{g,\nu}|\Pi_{g,\nu}$ to Γ_0 to be consistent —

[scan 361]

[p. 360]

Uncertainty retained from cancelled passages.

140-2-s361-D001 [illeg. 1]

140-2-s361-D002 I think

140-2-s361-D003 [illeg. 1]

on the other hand we would absolutely need c) in order at least to define $A(\pi) \longrightarrow |-A(\pi')$ and (if Σ, Σ' correspond) $\Gamma_\Sigma \longrightarrow |-\Gamma_{\Sigma'}$, in the case $\pi_a'|\pi_0'$ †characteristic in π (and in all cases if in addition we admit a), which seems rather plausible, and gives the injectivities required. But one does wonder why an element of † $\widehat{\widehat{\mathcal{T}}}(\pi_0')$ simply because it †is in the †normalizer R of G †in $\widehat{\widehat{\mathcal{T}}}(\pi_0')$, hence also R, and it normalizes $R \cap \widehat{\mathcal{T}}^+(\pi_0')$, should also normalize $\widehat{\mathcal{T}}^+(\pi_0')$ itself ! It is

[scan 362]

[p. 183]

true that we made draconian hypotheses at the outset (starting from a characteristic subgroup π_0' of π_0) which must surely be reflected in special properties of $G \subset \mathcal{T}^+(\pi_0')$. Perhaps in the end the trick of reducing to characteristic subgroups to examine [p. 361]

Uncertainty retained from cancelled passages.
140-2-s362-D001 invariant
140-2-s362-D002 [illeg. 1]
140-2-s362-D003 normalizes
140-2-s362-D004 normalizes
140-2-s362-D005 the same

the situation is itself not so clever. To see more clearly, we could already try to understand the [scan 363]
case where (without assuming π' characteristic) we assume π' of index 2 in π, hence invariant and $G \simeq \mathbb{Z}/2$, or else we take † the kernel of the canonical homomorphism $\pi \longrightarrow (\pi_{\mathrm{ab}})_2$ ($\simeq (\mathbb{Z}/2\mathbb{Z})^{2g}$ if $\nu = 0$, $\simeq (\mathbb{Z}/2\mathbb{Z})^{2g+|-\nu-1}$ if $\nu \geq 1$), taking for example $\pi = \widehat{\Pi}_{0,3}$ — situation of the Fermat curve $x^2 + y^2 + z^2 = 0$ (octahedral covering of $\mathbb{P}^1_{\mathbb{Q}} \setminus \{0, 1, \infty\}$ …)

But let us provisionally admit the hypotheses of injectivity (relatively innocuous), plus the condition c), not innocuous at all — whence, if Σ and Σ' correspond, an injective homomorphism

$$(9) \qquad \Gamma_\Sigma \hookrightarrow \Gamma_{\Sigma'},$$

and let us see what we could get from it. even admitting, provisionally (to see) that (9) is an isomorphism — † that is to say the full force of the main conjecture $\Gamma_0 \simeq \Gamma_{g,\nu}$. [p. 362]

Uncertainty retained from cancelled passages.
140-2-s363-D001 [illeg. 2]
140-2-s363-D004 Which|What

[scan 364]

Let|Let π be a profinite group with loops of type (g, ν), π' of type (g', ν'). Let us call a [p. 184]
"correspondence" between π and π' a pair consisting of a profinite $G|\pi''$ with loops , and of homomorphisms with loops $G|\pi'' \overset{p}{\to} \pi$, $G|\pi'' \overset{q}{\to} \pi'$ which are therefore each|each composed|composed of a "plugging holes" morphism, and of an isomorphism onto a subgroup|subgroup of finite index. Let|Let $J_p|I_p''$, $J_q|I_q'' \in J|I'' = I(\pi'')$ be the subsets of $J|I''$ which correspond to the "holes plugged by p" resp. by q, we assume if necessary $J_p|I_p'' \cap J_q|I_q'' = \emptyset$ (otherwise one could factor through the same quotient $\tilde{\pi}''$ of π''). In other words if $\overline{J}_p|\overline{I}_p''$, $\overline{J}_q|\overline{I}_q''$ are the complements of $J_p|I_p''$, $J_q|I_q''$ in $J|I''$ (so that we have : $\overline{J}_p|\overline{I}_p'' \to I = I(\pi)$, $\overline{J}_q|\overline{I}_q'' \to I' = I(\pi')$) we have

$$\overline{J}_p|\overline{I}_p'' \cup \overline{J}_q|\overline{I}_q'' = J|I''.$$

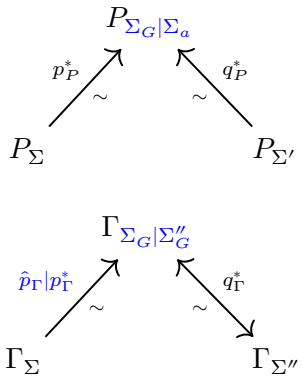
†

We are now going to suppose π equipped with a $\Sigma \in A(\pi)$, π' equipped with a $\Sigma' \in A(\pi')$, we shall call an "arithmetic correspondence" between π and π' a quadruple $(G|\pi'', p, q, \Sigma_G|\Sigma'')$, where $(G|\pi'', p, q)$ are as above, and $\Sigma_G^a|\Sigma_G \in A(G)$, such that we have $\Sigma_G|\Sigma_a = p^*(\Sigma) = q^*(\Sigma')$. † If P_Σ, [p. 363]

Uncertainty retained from cancelled passages.
140-2-s364-D004 Using|Let us use paragraphs above
140-2-s364-D005 We suppose

[scan 365]

$P_{\Sigma'}$, $P_{\Sigma_G|\Sigma_a}$ are resp. the torsors under Γ_Σ, $\Gamma_{\Sigma'}$, $\Gamma_{\Sigma''}$ that we know, we have two diagrams|diagram of isomorphisms of torsors



, whence by composition an isomorphism|homomorphism of †

$$(P_\Sigma, \Gamma_\Sigma) \xrightarrow{\sim} (P_{\Sigma'}, \Gamma_{\Sigma'})$$

We call "arithmetically outer isomorphisms" of (π, Σ) with (π', Σ'), any isomorphism of torsors that we can obtain in this way. Two correspondences are said to be equivalent † from the point of view † arithmetically outer, if they define the same|= arithmetically outer isomorphism. The composition is defined by composition of the actions on $(P_\Sigma, \Gamma_\Sigma)$ — we see that this comes from a correspondence. We find therefore a groupoid, which we check without difficulty is connected. I say that The automorphisms of (π, Σ) "are" the automorphisms of $(P_\Sigma, \Gamma_\Sigma)$ defined by elements of Γ_Σ. † This is easy…

m460

Marginal note. m460 We shall call it the groupoid of "outer arithmetic curves". [p. 364]

Uncertainty retained from cancelled passages.

140-2-s365-D001 $P_\Sigma \to$

140-2-s365-D002 from the point of

140-2-s365-D003 arithmetic|arithmetic outer|outer

140-2-s365-D004 obvious not

[scan 366]

In this groupoid, there is a transitive system of isomorphisms between the † $(\hat{\pi}_0, \overline{\Sigma}_{\pi_0})$ (where π_0 is a discrete group with loops) more generally between the (π, Σ_α), where $\alpha \in P(\pi)$ is a prediscretification of π — let us therefore define an arithmetically outer isomorphism of $\Pi_{g,\nu}$ π … The|Their group of these|its automorphisms is denoted Γ_0. The groupoid of virtual outer arithmetic curves is therefore identified with the category of torsors under Γ_0, A prediscretification (we could also call it an arithmetically outer rigidification) is therefore nothing other than an arithmetically outer isomorphism between this reference element, and π. [p. 365]

[p. 185]

Uncertainty retained from cancelled passages.

140-2-s366-D001 [illeg. 1]

[scan 367]

29 Critique of the preceding approach. [p. 186]

The approach of the|of the preceding § seems frankly|in the end very brutal. I even have doubts whether the|the conjecture in|on the properties of the "plugging holes" functor is true as it stands. It is true that for a profinite group † with loops π, if we fix $i \in I(\pi)$, we have a canonical homomorphism

$$\widehat{\mathcal{F}}(\pi, i) = \underset{\substack{\text{stabilizer} \\ \text{of } i \text{ in } \widehat{\mathcal{F}}(\pi)}}{\underline{\widehat{\mathcal{F}}(\pi)_i}} \longrightarrow \widehat{\mathcal{G}}(\pi'), \tag{1}$$

but it is problematic whether this is an isomorphism — and even whether it is injective, or whether it is surjective. But, choosing a discretification $\pi_0 \subset \pi$, whence ditto π'_0 for π', the preceding homomorphism does indeed induce an isomorphism

$$\mathcal{T}(\pi_0)_i \simeq \mathcal{G}(\pi'_0) \tag{1}$$

whence

$$\widehat{\mathcal{T}}(\pi_0) \xrightarrow{\sim} \widehat{\mathcal{G}}(\pi'_0) \tag{3}$$

and consequently, the exact sequence $\dagger\hat{\pi}'_0 \longrightarrow \widehat{\mathcal{G}}(\pi'_0) \longrightarrow \widehat{\mathcal{T}}(\pi'_0) \longrightarrow 1$ gives

$$1 \longrightarrow \hat{\pi}'_0 \longrightarrow \widehat{\mathcal{T}}(\pi_0) \longrightarrow \widehat{\mathcal{T}}(\pi'_0) \longrightarrow 1 \tag{4}$$

and consequently we find an injective homomorphism

$$\begin{array}{ccc} \hat{\pi}'_0 & & \widehat{\widehat{\mathcal{T}}}(\pi_0)_i \\ \| \hookrightarrow | \leftrightarrow & \widehat{\mathcal{T}}(\pi_0)_i \hookrightarrow & \| \\ \pi' & & \widehat{\widehat{\mathcal{T}}}(\pi) \end{array}$$

hence a homomorphism

$$i_{\pi_0} : \quad \pi' \longrightarrow \widehat{\widehat{\mathcal{T}}}(\pi)_i \tag{5}$$

for which the|the composite with (1) is the canonical injection $\pi' \hookrightarrow \widehat{\widehat{\mathcal{G}}}(\pi')$. Replacing the discretification π_0 by another, π_1, we find a priori another homomorphism $i_{\pi_1} : \pi' \longrightarrow \widehat{\widehat{\mathcal{T}}}(\pi)_i$. We may assume $\pi_1 = u(\pi_0)$, $u \in \widehat{\widehat{\mathcal{T}}}(\pi)_i$; [p. 366]

Uncertainty retained from cancelled passages.

140-2-s367-D002 [illeg. 1]

140-2-s367-D003 $\widehat{\pi}_0$

[scan 368]

and by transport of structure we find

$$i_{\pi_1} = i_{u(\pi_0)} = \mathrm{int}(u) \circ i_{\pi_0} \circ u_{\pi'}^{-1}$$

where $u_{\pi'}$ is the automorphism of π' defined by u via (1). To say that i_{π_0} is independent of the choice of the discretification π_0, also amounts to saying that its image in $\widehat{\widehat{\mathcal{T}}}(\pi)_i$ is an invariant subgroup — and then the action of $\widehat{\widehat{\mathcal{T}}}(\pi)_i$ on $\dagger$ the invariant subgroup $i_{\pi_0}(\pi) = i(\pi)$ $\dagger$ via inner automorphisms, is none other than that defined by (1). In this case, we find by passage to the quotient a homomorphism

$$\widehat{\widehat{\mathcal{T}}}(\pi)_i/\pi' \longrightarrow \widehat{\widehat{\mathcal{T}}}(\pi') \tag{6}$$

whose injectivity resp. surjectivity would be equivalent to that of (1). But it is not at all obvious that it is always so.

If it were not so, it would be necessary to look at the subgroup of $\widehat{\widehat{\mathcal{T}}}(\pi)_i$ formed of the $\gamma \in \widehat{\widehat{\mathcal{T}}}(\pi)_i$ which normalize the $\dagger$ subgroup $i_{\pi_0}(\pi')$, and such that the action induced on $i_{\pi_0}(\pi')$ corresponds to that given by the action (1) of $\widehat{\widehat{\mathcal{T}}}(\pi)$ on π'. This subgroup $H_{\pi_0,i}$ (which contains $\widehat{\mathcal{T}}(\pi_0)_i$) $\dagger$ therefore depends a priori on the chosen discretification. Replacing π_0 by $u(\pi_0)$ (where $\dagger$ $u \in \widehat{\widehat{\mathcal{T}}}(\pi_0|\pi)_i$) replaces it by $\mathrm{int}(u).H$ — hence H [p. 367]

Uncertainty retained from cancelled passages.

140-2-s368-D001 π

140-2-s368-D002 [illeg. 1]

140-2-s368-D005 injecting|injected

140-2-s368-D006 [illeg. 1]

140-2-s368-D007 u

[scan 369]

at least does not change, if we vary π_0 within their|a class of prediscretification. [p. 187]

We can do more symmetric things, by considering for every $i \in I$ the corresponding quotient π'_i of π, whence, for a chosen discretification π_0, homomorphisms

$$i_{\pi_0,i} : \pi'_i \longrightarrow \widehat{\widehat{\mathcal{T}}}(\pi)_i \subset \widehat{\widehat{\mathcal{T}}}(\pi)$$

and we shall define $H_{\pi_0} \subset \widehat{\widehat{\mathcal{T}}}(\pi)$ as the subgroup of the $\gamma \in \widehat{\widehat{\mathcal{T}}}(\pi)$ which "permute the $\dagger i_{\pi_0,i}$ among themselves" in an evident sense. Thus we have

$$H_{\pi_0} \cap \widehat{\widehat{\mathcal{T}}}(\pi)_i = H_{\pi_0,i} \tag{7}$$

and

$$H \supset \widehat{\mathcal{T}}(\pi_0) \tag{8}$$

The point I have in view, is that the subgroup of $\widehat{\widehat{\mathcal{T}}}(\pi_0)$ image of $\pi_1(M_{g,\nu})$, must not only normalize $\widehat{\mathcal{T}}(\pi_0)^+$, but moreover be contained in this H. This is a condition I have encountered by the back door, †thanks to the hypothesis (may be very hasty) that (1) is an isomorphism, which implied (tacitly) that we had $H = \widehat{\widehat{\mathcal{T}}}(\pi_0)$ in its entirety. It is possible that $\widehat{\widehat{\mathcal{T}}}(\pi_0)$ is a group so enormous|unreasonable and pathological, that there will never be any question of saying reasonable (and true) things about † this whole group (such as the bijectivity of (1), for example [p. 368]

Uncertainty retained from cancelled passages.

140-2-s369-D001 [illeg. 1]

140-2-s369-D002 under

140-2-s369-D003 the image of

[scan 370]

and that we are obliged to work with smaller subgroups, which remain close to the discrete (with all the same some additional "arithmetic" aspects, due to the $\Gamma_0 = \mathrm{Gal}(\overline{\mathbb{Q}}_0/\mathbb{Q})$!). In fact, there is (for $I = I(\pi) \neq \emptyset$ i.e. $\nu \neq 0$) in the group $\widehat{\mathcal{T}}(\pi_0)$ a simplicial structure of successive extensions, which is going to be respected by the outer action of the Galois group, and which we should take into account. It is part of the "structure at|: ∞" in the π_1 of the † modular multiplicities $M_{g,\nu}$, which even for $\nu = 0$ is probably nontrivial, and it may be necessary to take it into account, to manage to put our finger on $\Gamma_{\mathbb{Q}}$.

Without trying to give from the outset an a priori description of $\Gamma_{\mathbb{Q}}$ "in the $\widehat{\widehat{\mathcal{T}}}_{g,\nu}$", we are going to proceed in a more inductive way, starting from the presence of $\Gamma_{\mathbb{Q}}$ (for arithmetico-geometric reasons), and trying to bring out properties of this presence, perhaps strong enough to end up giving a purely algebraic characterization. [p. 369]

Uncertainty retained from cancelled passages.

140-2-s370-D001 spaces

[scan 371]

[p. 188]

Let us recall that, via the choice of $U_{g,\nu}$ (differentiable), we had been able to construct by transcendental means a $\widetilde{M}_{g,\nu\mathbb{C}}$and a $\widetilde{\mathcal{U}}_{g,\nu\mathbb{C}}|\mathcal{U}_{g,\nu\mathbb{C}} = \widetilde{M}_{g,\nu+1\,\mathbb{C}}|M_{g,\nu+1\,\mathbb{C}}$, hence a $\widetilde{M}_{g,\nu\mathbb{Q}}$, †and a $\widetilde{\mathcal{U}}_{g,\nu\mathbb{Q}}$, hence a $\pi_1(\mathcal{U}_{g,\nu,\mathbb{Q}})$, with a filtration in three steps, whose factors are respectively canonically isomorphic to

$$\hat{\pi}_{g,\nu}, \quad \widehat{\mathcal{T}(\pi_{g,\nu})}^{+}, \quad \Gamma_{\mathbb{Q}} = \mathrm{Gal}(\overline{\mathbb{Q}}_0/\mathbb{Q}) \tag{9}$$

the group maps (we presume injectively) into $\widehat{\widehat{\mathcal{G}}}(\pi_{g,\nu})$, † inducing an isomorphism between its subgroup $\pi_1(\mathcal{U}_{g,\nu\overline{\mathbb{Q}}_0})$ with $\widehat{\mathcal{G}}(\pi_{g,\nu})$, let us denote its image by $M_{g,\nu}$. So it is a subgroup closed

$$\widehat{\mathcal{G}}_{g,\nu} \subset M_{g,\nu} \subset \widehat{\widehat{\mathcal{G}}}_{g,\nu} \tag{10}$$

and $\widehat{\mathcal{G}}^{+}_{g,\nu}$ is normal in $M_{g,\nu}$(but not $\widehat{\mathcal{G}}_{g,\nu}$!). The datum †of such an $M_{g,\nu}$ is equivalent to that of an $N_{g,\nu}$

$$\widehat{\mathcal{T}}_{g,\nu} \subset N_{g,\nu} \subset \widehat{\widehat{\mathcal{T}}}_{g,\nu}, \tag{11}$$

with $\widehat{\mathcal{T}}^{+}_{g,\nu}$ normal in $N_{g,\nu}$.

We set

$$\Gamma_{g,\nu} = N_{g,\nu}/\widehat{\mathcal{T}}^{+}_{g,\nu} \simeq M_{g,\nu}/\widehat{\mathcal{G}}^{+}_{g,\nu}. \tag{12}$$

m461

Marginal note. m461 N.B. In $\Gamma_{g,\nu}$ we have a canonical element of order 2 $\tau_{g,\nu}$, corresponding to the elements of $\dagger\widehat{\mathcal{T}}_{g,\nu} - |\!\searrow \widehat{\mathcal{T}}^{+}_{g,\nu} = \widehat{\mathcal{T}}^{-}_{g,\nu}$.

We have a surjective homomorphism

$$\Gamma_{\mathbb{Q}} \longrightarrow \Gamma_{g,\nu} \tag{13}$$

which we presume is bijective — i.e. that the $\Gamma_{g,\nu}$ are canonically isomorphic to one another. 370

Uncertainty retained from cancelled passages.

140-2-s371-D001 a|a
140-2-s371-D002 by
140-2-s371-D003 of
140-2-s371-D004 *A*

[scan 372]

Let π_0 now be a discrete group with loops of type g, ν, then we define subgroups $M_{\pi_0}\dagger$, $N_{\pi_0}\dagger$

$$\widehat{\mathcal{G}}(\pi_0) \subset M_{\pi_0}\dagger \subset \widehat{\widehat{\mathcal{G}}}(\pi_0) \tag{14}$$

$$\widehat{\widehat{\mathcal{T}}}(\pi_0) \subset N_{\pi_0}\dagger = M(\pi_0)/\hat{\pi}_0 \subset \widehat{\mathcal{T}}(\pi_0) \tag{15}$$

$\Gamma_{\pi_0} = M_{\pi_0}/\widehat{\mathcal{G}}(\pi_0)$

$\hookleftarrow ?\ \widehat{\mathcal{T}}$

using an isomorphism $\pi_0 \xrightarrow{\sim} \pi_{g,\nu}$ and proceeding by transport of structure — the result does not depend on it. More generally, let us start with a pair (π, α) of a profinite group π, equipped with a prediscretification α, which we can therefore interpret as a class of isomorphisms $\hat{\pi}_{g,\nu} \longrightarrow \pi$, defined|defined modulo composition on the right by a $u \in \widehat{\mathcal{G}}_{g,\nu}$. Then by transport of structure we deduce from it groups M_α, N_α

$$\widehat{\mathcal{G}}_\alpha\dagger \subset M_\alpha \subset \widehat{\widehat{\mathcal{G}}}(\pi) \tag{16}$$

$$\widehat{\mathcal{T}}_\alpha\dagger \subset \underset{\underset{M_\alpha/\pi}{\|}}{N_\alpha} \subset \widehat{\mathcal{T}(\pi)} \tag{17}$$

with $\widehat{\mathcal{G}}(\pi)^+$ normal in M_α i.e. $\widehat{\mathcal{T}}^+(\pi)$ normal in N_α. We set $\widehat{\widehat{\mathcal{T}}}$?

$$\Gamma_\alpha = M_\alpha/\widehat{\mathcal{G}}^+_\alpha\dagger \simeq N_\alpha/\widehat{\mathcal{T}}^+_\alpha \tag{18}$$

We have a canonical element †

$$\tau_\alpha \in \Gamma_\alpha, \qquad \tau_\alpha^2 = 1, \quad \tau_\alpha \neq 1 \tag{19}$$

371

Uncertainty retained from cancelled passages.

140-2-s372-D001 π_0
140-2-s372-D002 π_0
140-2-s372-D003 π_0
140-2-s372-D004 π_0
140-2-s372-D005 π
140-2-s372-D006 π
140-2-s372-D007 π
140-2-s372-D008 whose

[scan 373]

Let π now be a profinite group with loops of type (g,ν). We call an arithmetization of π the [p. 189] datum of an element of

$$\begin{aligned} A(\pi) &= \underline{\mathrm{Isom}}_{\mathrm{lac}}(\hat\pi_{g,\nu}, \pi)/M_{g,\nu} \\ &\simeq \underline{\mathrm{Isomext}}_{\mathrm{lac}}(\hat\pi_{g,\nu}, \pi)/N_{g,\nu} \end{aligned} \tag{20}$$

Let $P(\pi)$ be the set of prediscretifications oriented of π

$$\begin{aligned} P(\pi) &= \underline{\mathrm{Isom}}_{\mathrm{lac}}(\hat\pi_{g,\nu}, \pi)/\widehat{\mathcal{G}}^+_{g,\nu} \\ &\simeq \mathrm{Isomext}_{\mathrm{lac}}(\hat\pi_{g,\nu}, \pi)/\widehat{\mathcal{T}}^+_{g,\nu} \end{aligned} \tag{21}$$

then $\Gamma_{g,\nu}$ operates freely on the right on $P(\pi)$, and

$$A(\pi) \simeq P(\pi)/\Gamma_{g,\nu} \tag{22}$$

— $P(\pi)$ is a relative torsor over $A(\pi)$, with group $\Gamma_{g,\nu}$. For $a \in A(\pi)$, let M_a be the subgroup of $\widehat{\widehat{\mathcal{G}}}(\pi)$ of automorphisms of (π, a), it operates on the $\Gamma_{g,\nu}$-torsor P_a of oriented discretifications of π over a. For † $\alpha \in P_a$, let $\widehat{\mathcal{G}}^{*|\dagger}_\alpha \subset \widehat{\widehat{\mathcal{G}}}(\pi)$, then $\widehat{\mathcal{G}}^+_\alpha$ † does not depend on α (we shall denote it|in view of the notation ([illeg. 1])![?! $\widehat{\mathcal{G}}_a$), it is also the kernel of the operation of M_a on P_a. We have [m467]

Marginal note. [m467] but. Careful $\widehat{\mathcal{G}}^{*|\dagger}_\alpha$ depends on α i.e. $\tau_\alpha \in \Gamma_a$ depends on α

$$\widehat{\mathcal{G}}^+_a \subset M_a \subset \widehat{\widehat{\mathcal{G}}}(\pi) \tag{23}$$

hence also

372

Uncertainty retained from cancelled passages.

140-2-s373-D001 every
140-2-s373-D002 +
140-2-s373-D003 trivi
140-2-s373-D004 +

[scan 374]

$$\widehat{\mathcal{T}}_a^+ \subset N_a \subset \widehat{\widehat{\mathcal{T}}}(\pi) \tag{24}$$

setting

$$N_a = M_a/\pi, \qquad \widehat{\mathcal{T}}_a^+ = \widehat{\mathcal{G}}_a^+/\pi. \tag{25}$$

We set

$$\Gamma_a = M_a/\widehat{\mathcal{G}}_a^+ \simeq N_a/\widehat{\mathcal{T}}_a^+ \tag{26}$$

then Γ_a identifies with the commutant of $\Gamma_{g,\nu}$ operating on P_a, i.e. with $\Gamma_{g,\nu}$ "twisted" by the torsor P_a. The choice of an $\alpha \in P_a$ amounts therefore to the choice of such an isomorphism †

$$\Gamma_a \overset{\sim}{\longleftrightarrow} \Gamma_{g,\nu} \tag{27}$$

which is changed by an inner automorphism if we change $\alpha|\alpha$...

We have the category of † arithmetized outer groups with loops of type g, ν — it is a connected groupoid, with a fixed origin $(\hat{\pi}_{g,\nu}, \underbrace{a_{g,\nu}}_{\text{arithmetization "canonical"}})$, † whose group of automorphisms is $N_{g,\nu}$, an extension of $\Gamma_{g,\nu}$ by $\widehat{\mathcal{T}}_{g,\nu}$, $\simeq \pi_1(M_{g,\nu\mathbb{Q}|\overline{\mathbb{Q}}})$ (computed with respect to the canonical universal covering of $M_{g,\nu\mathbb{Q}}$...) if $\Gamma_{\mathbb{Q}} \longrightarrow \Gamma_{g,\nu}$ injective!

373

Uncertainty retained from cancelled passages.

140-2-s374-D001 $\simeq$

140-2-s374-D002 arith|ext

140-2-s374-D003 corresponding in|in

[scan 375]

To give oneself an object of this category is essentially the = thing (up to unique isomorphism) as to give oneself a universal covering $\widetilde{M}_{g,\nu\mathbb{Q}}$ of $M_{g,\nu}\overline{\mathbb{Q}}$ — or a torsor under $N_{g,\nu}$; we then consider the family of curves of type g, ν over $\widetilde{M}_{g,\nu\mathbb{Q}}|M_{g,\nu\mathbb{Q}}$ [p. 190]

$$\mathcal{U}_{g,\nu} \times_{M_{g,\nu}} \widetilde{M}_{g,\nu\mathbb{Q}} = \mathcal{U}_{g,\nu}(\widetilde{M}_{g,\nu\mathbb{Q}}) \tag{28}$$

as being "the" algebraic curve whose π_1 outer (which is therefore the π_1 of this scheme...) is the given outer group with loops.

If we took the arithmetized groups with loops, not outer, we would have again a connected groupoid with "origin" $(\hat{\pi}_{g,\nu}, a_{g,\nu})$, with a group of automorphisms which is $M_{g,\nu} \simeq \pi_1(\mathcal{U}_{g,\nu\mathbb{Q}})$. The datum of an object of this category amounts to the datum of a universal covering (not only of $M_{g,\nu\mathbb{Q}}$, but) of $\mathcal{U}_{g,\nu\mathbb{Q}}$, i.e. $\widetilde{\mathcal{U}}_{g,\nu\mathbb{Q}}$, which indeed makes it possible to serve as a reference universal covering 374

[scan 376]

for the relative curve (28), allowing one then to specify its fundamental group as a genuine group with loops (not just an outer group).

We can finally also look at the category of profinite groups with loops arithmetized, where we take as morphisms the arithmetically outer isomorphisms.

We again find a connected groupoid with marked origin $(\pi_{g,\nu}, a_{g,\nu})$, whose group of automorphisms is now $\Gamma_{g,\nu}$. Now the groups $\hat{\pi}_0$ (π_0 a discrete group with loops of type g, ν) are canonically isomorphic to one another. The category is (conjecturally, assuming $\Gamma_{\mathbb{Q}} \to \Gamma_{g,\nu}$ an isomorphism) equivalent to that of universal coverings of Spec $\mathbb{Q}$, i.e. to that of algebraic closures of $\mathbb{Q}$, the origin element corresponding to $\overline{\mathbb{Q}}_0$. 375

[scan 377]

[p. 191]

When we give ourselves a π with arithmetization a, then there † corresponds to it therefore canonically an algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$. The $\Gamma_{\mathbb{Q}}$ torsor of isomorphisms $\mathrm{Isom}(\overline{\mathbb{Q}}_0, \overline{\mathbb{Q}})$, i.e. of embeddings $\overline{\mathbb{Q}} \hookrightarrow | \subset \mathbb{C}$, is in $1-1$ correspondence with the set P_a of oriented prediscretifications of π giving rise to a.

Let us note that every α defines an element $\tau_\alpha \in \Gamma_a$, $\tau_\alpha^2 = 1$ ($\tau_\alpha \neq 1$), hence a canonical map

$$P_a \longrightarrow {}_2\Gamma_a \tag{29}$$

We see that this map is compatible with the action of the group ± 1 on P_a

$$\tau_\alpha = \tau_{-\alpha}$$

If it is true that $\Gamma_{\mathbb{Q}} \xrightarrow{\sim} \Gamma_{g,\nu}$, then † (29) induces by passage to the quotient a bijective map

$$\begin{array}{ccc} P_a/\pm 1 & \xrightarrow{\sim} & \text{set of elements} \\ & & \text{of order 2 of } \Gamma_a \\ \| & & \\ P_a^\natural = \text{set of prediscretifications} & & \\ \text{(not oriented) over } a) & & \end{array} \tag{30}$$

376

Uncertainty retained from cancelled passages.

140-2-s377-D001 gives|given

140-2-s377-D002 of

[scan 378]

This comes from the known fact that in $\Gamma_{\mathbb{Q}}$, the only elements of order 2 are the conjugates of τ, and that the centralizer of τ in $\Gamma_{\mathbb{Q}}$ is reduced to $\{1, \tau\}$.

We can say that the datum of a discretification (not oriented) $\alpha^\natural$ over the arithmetization a, amounts to the datum of an Archimedean valuation on the algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$ defined by (π, a) (i.e. of an isomorphism $\overline{\mathbb{Q}}_1 \simeq \overline{\mathbb{Q}}_0$ modulo complex conjugation)

†

377

Uncertainty retained from cancelled passages.

140-2-s378-D002 30

[scan 379]

30 Properties of the $N_{g,\nu}$, $\Gamma_{g,\nu}$. m475

[p. 192]

Marginal note. m475 a) Properties related to finite subgroups|subgroups.

We would like to bring out properties, inspired by the geometrico-arithmetic context, but which can|† be formulated|formulated in a purely algebraic way – and which are strong enough perhaps to end up characterizing the $N_{g,\nu}$, †

Let us return to a (π, a), an arithmetized group with loops, heuristically, it corresponds to the datum of an algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$, and (if π is given as an outer group) of an algebraic family, parametrized by a universal covering $\widetilde{M}_{g,\nu\overline{\mathbb{Q}}}$ of $M_{g,\nu\overline{\mathbb{Q}}}$, of algebraic curves of type (g,ν). In this perspective, to reduce this family to one algebraic curve – i.e. to give oneself an algebraic curve of type (g,ν) over $\overline{\mathbb{Q}}$ – must amount to giving oneself a kernel of lifting of the surjective homomorphism

$$N_a \longrightarrow \Gamma_a \qquad (\text{kernel } \widehat{\mathcal{T}}_a^+) \tag{1}$$

378

Uncertainty retained from cancelled passages.
140-2-s379-D001 can
140-2-s379-D002 [illeg. 1]

[scan 380]

The datum of such a lifting over an open subgroup Γ', amounts to the datum of an algebraic curve of type g, ν, defined over hum-|hom- the extension finite $K' \subset \overline{\mathbb{Q}}$ defined by Γ'. ?

It is a matter here not of|of the arbitrary continuous liftings, but of liftings having particular properties (some of which very deep, of the "Weil" kind… but perhaps consequences of much simpler properties). The properties † to be brought out|† should in any case be † stable under passage to a smaller open subgroup. Among these properties, there would be that these|a germs|germ of liftings could|could in their turn be lifted to $\widehat{\widehat{\mathcal{G}}}(\pi)$ itself – in an equally "admissible" way in a sense to be specified – and =|even that there would be "many" (classes of π-conjugacy) of such 379

Uncertainty retained from cancelled passages.
140-2-s380-D002 thus
140-2-s380-D003 thus brought out
140-2-s380-D004 invariant

[scan 381]

germs of|germ of liftings, for a|an germ of outer action on π already given — which corresponds [p. 193] to the naive fact that the algebraic curve $U_\dagger$ over $\overline{\mathbb{Q}}$ defined by this outer action has "many point^s" rational/$\overline{\mathbb{Q}}$. In fact, it would suffice, in the limit, to speak of the † properties of these more complete liftings to a germ of a genuine action of † $\mathbb{\Gamma}_a$ on π (not only outer) — from which the outer actions are|will to be|be deduced by composition with

$$M_a \longrightarrow N_a \simeq M_a/\pi.$$

† We thus suppose given a closed subgroup

$$\Gamma \subset \dagger M_a \subset \widehat{\widehat{\mathcal{G}}}(\pi)$$

such that

$$\text{ⓐ} \qquad \Gamma \longrightarrow \mathbb{\Gamma}_a = M_a/\pi \quad \text{is|is injective,}$$

and has|has as its image an open subgroup of $\mathbb{\Gamma}_a$.

— it being understood that two subgroups conjugate under π — or even, sometimes, under M_a — are not considered essentially distinct. We consider, at the same time as Γ, subgroups

Uncertainty retained from cancelled passages.
140-2-s381-D001 g, ν
140-2-s381-D002 [illeg. 1]
140-2-s381-D003 [illeg. 1]
140-2-s381-D004 [illeg. 1]
140-2-s381-D005 $\mathcal{G}$

[scan 382]

Γ' that are open. We surely want, for such open subgroups

$$\text{ⓑ} \qquad \pi^{\Gamma'} = (1)$$

If we denote by $\tilde{\Gamma}$ the image of Γ in N_a, so that we have an extension (1)

$$1 \longrightarrow \pi \longrightarrow \mathcal{E} \longrightarrow \tilde{\Gamma} \longrightarrow 1, \qquad \Gamma \uparrow \sim \tilde{\Gamma}$$

we can regard Γ as a section of this extension, Γ' as a partial section. The hypothesis $\pi^{\Gamma'} = (1)$ ensures the rigidity of the category of points of U with values in $\overline{\mathbb{Q}}$, paradigmed by that of germs of splittings|splitting of the extension (1)'.

!|?

We can also look at the sets

$$\widehat{\mathcal{G}}_a^{+\Gamma'} = \dagger\, \mathrm{Centr}_{M_a}(\Gamma') \cap \widehat{\mathcal{G}}_a^{+} \quad \text{and}$$

$$\widehat{\mathcal{T}}_a^{+\Gamma'} = \mathrm{Centr}_{N_a}(\tilde{\Gamma}') \cap \widehat{\mathcal{T}}_a^{+},$$

we thus have an exact sequence

$$(2) \qquad 1 \longrightarrow \pi^{\Gamma'} \longrightarrow (\widehat{\mathcal{G}}_a^{+})^{\Gamma'} \longrightarrow (\widehat{\mathcal{T}}_a^{+})^{\Gamma'}$$

which, taking into account the hypothesis $\pi^{\Gamma'} = (1)$,

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Uncertainty retained from cancelled passages.

140-2-s382-D001 $N_a^{\Gamma'}|M_a^{\Gamma'}$

[scan 383]

[p. 194]

gives

$$(3) \qquad (\widehat{\mathcal{G}}_a^{+})^{\Gamma'} \hookrightarrow (\widehat{\mathcal{T}}_a^{+})^{\Gamma'}.$$

The right-hand side of (3), by our hypothetical dictionary, should be canonically the|isomorphic group of automorphisms of the curve U over $K'_a|K'$ hence be a finite group, and even † the inductive limit (for Γ' decreasing) — which is the group of automorphisms of U over $\overline{\mathbb{Q}}$ — is finite. Denoting by a superscript $\natural$ the corresponding germ of a group, we thus want

$$\text{©} \qquad Z = (\widehat{\mathcal{T}}_a^{+})^{(\Gamma^{\natural})} \left(= \mathrm{Centr}_{N_a|M_a}(\tilde{\Gamma}^{\natural}) \cap \widehat{\mathcal{T}}_a^{+}\right)$$

is a finite group

— which is a counterpart to (b), and expresses that the automorphism groups of the points of $M_{g,\nu\overline{\mathbb{Q}}}$ over $\overline{\mathbb{Q}}$ are finite.

In fact, we would like Z to be conjugate in $\dagger M_a$ to a subgroup of to $\mathcal{T}^+$ (if we suppose that we have a discretification † π_0 of π, making it possible to define $\mathcal{T}$ — otherwise, we can express this

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Uncertainty retained from cancelled passages.

140-2-s383-D001 by

140-2-s383-D002 $\widehat{\mathcal{T}}$

140-2-s383-D004 [illeg. 1]

[scan 384]

property by saying that there exists a discretification π_0 of π, compatible with a, such that $Z \subset \mathcal{T}(\pi_0)\ldots)$

Let us consider now an arbitrary finite subgroup $G \subset \widehat{\mathcal{T}}_a^{+}$. [N.B. If we take only $G \subset M_a$, so that the image of G in $\Gamma_a = M_a/\widehat{\mathcal{T}}_a^{+}$ is a finite subgroup, then if it is true that $\Gamma_a \simeq \mathcal{G}|\mathfrak{G}$, this image must be of order 1 or 2, and in the second case must define a $\tau \in \Gamma_a$ corresponding

$\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow$?

to a well-determined prediscretification (not oriented) of π. This case should be reused again in what follows…]. Let us suppose, or admit, that G is contained in $\mathcal{T}^+$, [m482]

Marginal note. [m482] this is probably not automatic (if we do not|N' suppose $N_a = \mathrm{Norm}_{\widehat{\mathcal{F}}}(\widehat{\mathcal{T}}_a|\widehat{\mathcal{T}}_a^+)$)

for a suitable discretification in the class a, and that the outer action of G thus obtained on a discrete π_0 of type (g,ν) is|are always realizable. It is therefore also realizable by|for an action of G

[scan 385]

on a complex, hence algebraic, structure, which means that we can find a † germ|germs of admissible lifting [p. 195]

$$\Gamma_a \longrightarrow N_a$$

which centralizes G. Let us consider on the other hand

$$(4) \qquad \mathrm{Centr}_{\widehat{N}_a}(G) \longrightarrow \Gamma_a$$

whose kernel is $\mathrm{Centr}_{\widehat{\mathcal{T}}_a^+}(G) = \widehat{\mathcal{T}}_a^{+G}$. If we placed|placing ourselves in the discrete context † (with a discretification $\pi_0 \subset \pi$ invariant|invariant under G) we would have, by the standard topological conjectures, that $\widehat{\mathcal{T}}_a^{+G}|\widehat{\widehat{\mathcal{T}}}_a^{+G}$ is the group of automorphisms, in the isotopic category, of the action of G on a surface U of type (g,ν), described by the given outer operation of G on π_0. This group has no reason to be finite — if it were, the homomorphism (4) would have finite kernel, hence would be an isomorphism of the|of kernels of groups, and there would be a

Uncertainty retained from cancelled passages.
140-2-s385-D001 [illeg. 1]
140-2-s385-D002 for

[scan 386]

unique germ of lifting $\Gamma_a^\natural \longrightarrow M_a$ which would centralize G — which would mean that there is only one way to realize the topological operation of G on U by a complex analytic operation, hence algebraic — but this is surely not the case, the set of fixed points of G operating on Teichmüller space $\widetilde{M}_{g,\nu}$ is not reduced to a point — it is (in the topological context) a schematic|arithmetic multiplicity which can be † of arbitrary $\mathbb{R}|\mathbb{C}$ dimension — It is of dimension > 0 in any case, if the quotient † U/G is not of genus 0.

To keep in mind in any case, as a plausible property :

(d) For $\forall$ finite subgroup G of $\widehat{\mathcal{T}}_a^+$ (or at least if $G \subset \mathcal{T}^+$, when we have a discretification $\pi_0 \subset \pi$ in a), looking at $\mathrm{Centr}_{\widehat{\widehat{\mathcal{T}}}|\widehat{\mathcal{T}}}(G) = ;Z(G)$, $N_a \cap Z(G) \longrightarrow \Gamma_a$ has an open image (and there are even some

Uncertainty retained from cancelled passages.
140-2-s386-D003 [illeg. 2]
140-2-s386-D004 [illeg. 1]

[scan 387]

germs of admissible liftings $\Gamma_a \to Z(G)$) [p. 196]

This implies a nontrivial property for the $g \in N_a|M_a$ [illeg. 1] as elements of $\widehat{\widehat{\mathcal{T}}}(\pi)$, or better of $N' = \mathrm{Norm}_{\widehat{\mathcal{T}}}(\widehat{\mathcal{T}})$: † [m483]

Marginal note. [m483] namely : $\exists n \in \mathbb{N}^*$ such that g^n is congruent modulo $\widehat{\mathcal{T}}_a^+$ to an element of $Z(G)$…

Let us indeed consider the image $Z'_G|Z'(G)$ of $Z(G) \cap N'$ in $\Gamma' = N'/\widehat{\mathcal{T}}$, † where obviously $\mathbb{\Gamma}_a \subset \Gamma'$, and † the image of $Z(G) \cap N_a$ in $\mathbb{\Gamma}_a$ is none other than † $Z'_G \cap \mathbb{\Gamma}_a$. Saying that it is open|open i.e. of index finite|finite, therefore implies that $\forall g \in \mathbb{\Gamma}_a$, † $\exists n \in \mathbb{N}^*$ such that $g^n \in Z'_G$. †

Let us note that in $\mathcal{T}(\pi_{g,\nu})$ there are only a finite number of conjugacy classes of finite subgroups, so if we look at their centralizers Z_G in $\widehat{\widehat{\mathcal{T}}}(\pi_{g,\nu})$, and their images in $\Gamma'_{g,\nu} = \mathrm{Norm}_{\widehat{\widehat{\mathcal{T}}}_{g,\nu}}(\widehat{\mathcal{T}}^+_{g,\nu})/\widehat{\mathcal{T}}^+_{g,\nu}$, we find only a finite number of subgroups of $\Gamma'_{g,\nu}$, let $Z'_{g,\nu}$ be their intersection. We thus see that this|the subgroup $\mathbb{\Gamma}_{g,\nu}$ of $\Gamma'_{g,\nu}$ is such that $\mathbb{\Gamma}_{g,\nu} \cap Z'_{g,\nu}$ is open in $\mathbb{\Gamma}_{g,\nu}$

Uncertainty retained from cancelled passages.

140-2-s387-D001 ,|;

140-2-s387-D002 then is|then let

140-2-s387-D003 the property

140-2-s387-D004 [illeg. 1]

140-2-s387-D005 [illeg. 1]

140-2-s387-D006 To tell the truth

[scan 388]

Let us now pass to finite subgroups of $\widehat{\widehat{\mathcal{G}}}(\pi)$ itself — or at least of M_a — or, what comes to almost the same thing, of $\widehat{\mathcal{G}}^+_a$. We shall, as earlier, restrict ourselves to $G \subset \mathcal{G}(\pi_0)$, for a suitable discretification $\in a$, for otherwise there would be hardly anything to say. We know that then G is cyclic — it|and corresponds to an action of G on a topological U, with fixed point. We can realize it in a complex way, which we will express|express by saying that there exists an admissible lifting $\mathbb{\Gamma}^\natural_a \longrightarrow \dagger M_a$, which centralizes G. We know moreover, if $G \neq 1$, that †

$$(5) \qquad \pi^G = (1)$$

(or rather, we know it in the discrete case — we admit it in the profinite context) —

Uncertainty retained from cancelled passages.

140-2-s388-D001 $\mathcal{G}$

140-2-s388-D002 π

[scan 389]

[p. 197]

which we can again interpret as a rigidity property — thus

$$\mathrm{Centr}_{\widehat{\widehat{\mathcal{G}}}}(G) = \widehat{\widehat{\mathcal{G}}}^G \longrightarrow \widehat{\widehat{\mathcal{T}}}^G$$

is injective, so if we have a $\Gamma''|\tilde{\Gamma} \subset N_a|M_a$ which is in the image, then there exists a unique lifting $\Gamma''|\tilde{\Gamma} \longrightarrow \widehat{\widehat{\mathcal{G}}}$ which centralizes G i.e. such that it is $\Gamma \longrightarrow \widehat{\widehat{\mathcal{G}}}^G$.

But in fact what we shall know, for a given $\Gamma \subset M_a$ (defining an algebraic curve via its outer action on a π_1) is that $\Gamma \subset M_a^G$ (i.e. the outer action commutes with a given outer action of G i.e. G operates on the algebraic curve) — and we would nevertheless like to conclude from this that when we have lifted $G \longrightarrow \widehat{\mathcal{T}}^+$ to|or $G \longrightarrow \widehat{\mathcal{G}}^+$ (i.e. when we have given ourselves a fixed point of the action of G on U): then|therefore the action of $\Gamma^\natural$ lifts (i.e. $\Gamma^\natural \longrightarrow M_a^G$) automatically|automatic, to $\Gamma^\natural \longrightarrow M_a$ in such a way as to commute|commuting|commuting,.... Is this a property of the choice of the|of the choices of the M_a, or of that of the lifting $\Gamma^\natural \longrightarrow | \hookrightarrow N_a$, or of G??

[scan 390]

In the discrete setting, unless I am mistaken, if we have a faithful action direct|discrete, of finite G on π_0, then

$$\mathcal{G}(\pi_0)^G \longrightarrow \mathcal{T}(\pi_0)^G$$

(which is injective, by $\pi_0^G = 1$) is also surjective. No, there is a mistake — that would mean as it stands that if G operates faithfully on a topological surface U of type g, ν, with a fixed point P, then the automorphisms which commute with G fix P (instead of merely permuting the fixed points among themselves…). So we must not expect $\forall$ element of $N_a^G | M_a^G$, nor even of $\widehat{\mathcal{T}}_a^G$, to lift to M_a^G — but rather this: for $G \in | \subset \mathcal{T}_a(\pi_0)$ a given finite subgroup, the π-conjugacy classes of liftings to $\widehat{\mathcal{G}}$, † which are interpreted as fixed points of an action of G on some U, form a finite set, on which $\widehat{\widehat{\mathcal{T}}}(\pi)^G$ operates in a natural way, and the|the stabilizers|stabilizer of a point P i.e. of a conjugacy class of lifting^{s}|liftings

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Uncertainty retained from cancelled passages.

140-2-s390-D002 to

lifts uniquely to $\widehat{\widehat{\mathcal{G}}}^G$. This should be reexamined more carefully later on. But it seems to me that we find no new properties here of the $\Gamma \subset \mathcal{M}_a$ or of M_a itself, i.e. of Γ_a as a profinite subgroup of $\Gamma' = \text{Norm}_{\widehat{\widehat{\mathcal{T}}}}(\widehat{\mathcal{T}})/\widehat{\mathcal{T}}$. [scan 391] [p. 198]

31 Digression on the liftings of an outer action of†a finite group G on a profinite group with loops π.

We assume the outer action faithful, i.e. $G \subset \widehat{\widehat{\mathcal{T}}}(\pi) = \widehat{\widehat{\mathcal{T}}}$, and that $G = G^+$. We assume moreover that there exists an invariant discretification π_0, i.e. such that $G \subset \mathcal{T}(\pi_0)$.

If we assume G cyclic, then unless I am mistaken it is proved that the situation is realizable topologically (thus: algebraically in a complex analytic way ...).

Uncertainty retained from cancelled passages.

140-2-s391-D001 e

In the discrete case, the meaning of the π_0-conjugacy classes of lifting $G \longrightarrow \mathcal{G}(\pi_0)$ is well understood; likewise for the partial liftings We find a † canonical realization of the fundamental groupoid associated to the outer group π_0, by a finite groupoid $[U^!]$ on which G operates in the strict sense, whose points correspond to the π_0-conjugacy classes of maximal partial sections $\neq 1$, the group $\dagger\mathcal{T}(\pi_0)^G$ operates in an equally canonical way on this groupoid. If $P \in U^!$ corresponds to a lifting $G \longrightarrow \mathcal{G}(\pi_0)$, then an element $\dagger\dot{u} \in \mathcal{T}(\pi_0)^G$ is in the image of $\mathcal{G}(\pi_0)^G$, i.e. can be lifted to $u \in \mathcal{G}(\pi_0)$ commuting with G, iff $\dot{u}(P) = P$ [i.e. iff (if $\dot{u}$ is lifted in an arbitrary way to v) $v(G)$ is π_0-conjugate to G i.e. iff $\exists g \in \pi_0$ such that $v(G) = \text{int}(g)(G)$ i.e. † $u \overset{\text{definition}}{=} \text{int}(g^{-1})v$ fixes G. † (in which case of course it centralizes, (by the hypothesis $\dot{u} \in \mathcal{T}^G$)) Thus our brilliant assertion, is a tautology] [scan 392] if $U^! \neq \emptyset | \phi$

Uncertainty retained from cancelled passages.

140-2-s392-D002 category

140-2-s392-D003 [illeg. 1]

140-2-s392-D004 g

140-2-s392-D005 which

140-2-s392-D006 (Thus

and hence the lifting is unique. As $U^!_{*|\varphi}$ is finite, hence the stabilizer $\mathcal{T}(\pi_0)^G_P$ of P in $\mathcal{T}(\pi_0)^G$ has finite index, and so it is this subgroup of finite index that lifts merrily and canonically. [scan 393] [p. 199]

What can one say in the profinite context? Of course, we have a canonical map

$$\begin{array}{c} \dagger \text{ classes of } \pi_0\text{-conjugacy} \\ \text{of liftings of} \\ G \text{ into } G \longrightarrow \mathcal{G}(\pi_0) \\ \text{i.e. of splittings of} \\ \text{the extension } 1 \longrightarrow \pi_0 \longrightarrow E \longrightarrow G \longrightarrow 1 \end{array} \longrightarrow \begin{array}{c} \text{classes of } \pi_0\text{-} \\ \text{conjugacy of liftings of} \\ G \text{ into } G \longrightarrow \widehat{\widehat{\mathcal{G}}}(\pi_0) \\ \text{i.e. of splittings} \\ \text{of the extension} \\ 1 \longrightarrow \hat{\pi}_0 \longrightarrow \widehat{E} \longrightarrow G \longrightarrow 1 \\ \| \\ \pi \end{array}$$

We should first examine

a) The question of the bijectivity of this map.

b) The question whether $\pi^G = 1$ (rigidity|rigidity), for $G \neq (1)$. – (for a given lifting, in E for simplicity)

I feel like conjecturing shamelessly that this is so — which would imply for example that for every such lifting G, there is

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Uncertainty retained from cancelled passages.

140-2-s393-D001 Set

[scan 394]

an open subgroup $(\widehat{\widehat{\mathcal{T}}}|\widehat{\mathcal{T}}^G)_P$ of the group (perhaps really immense a priori!) $\widehat{\widehat{\mathcal{T}}}^G$, which lifts canonically so as to commute with $\widetilde{G}|G$....

We should clearly develop, at the same time as a theory of outer operations of finite groups on discrete groups with loops (which for the moment is extremely conjectural|conjectural) an analogous theory in the profinite case — I †will have probably to come back to it later on.

Uncertainty retained from cancelled passages.

140-2-s394-D001 [illeg. 1]

[scan 395]

32 Return to the arithmetic aspects of plugging holes: relations between $\Gamma_{g,\nu}$ and $\Gamma_{g,\nu-1}$. [p. 200]

Of course, we † have that

$$\mathcal{N}_{g,\nu} \longrightarrow \mathfrak{S}_\nu$$

is surjective (since $\mathcal{T}_{g,\nu} \longrightarrow \mathfrak{S}_\nu$ is), hence we will have

$$(1) \qquad 1 \longrightarrow |-\mathcal{N}^!_{g,\nu} \longrightarrow |-\mathcal{N}_{g,\nu} \longrightarrow |-\mathfrak{S}_\nu \longrightarrow |-1$$

and the Cartesian diagram of subgroups of $\widehat{\widehat{\mathcal{T}}}_{g,\nu}$

$$(2) \qquad \begin{array}{ccc} \widehat{\mathcal{T}}^+_{g,\nu} & \hookrightarrow & \mathcal{N}_{g,\nu} \\ \uparrow & & \uparrow \\ \widehat{\mathcal{T}}^{!+}_{g,\nu} & \hookrightarrow & \mathcal{N}^!_{g,\nu} \end{array}$$

m485

Marginal note. m485 † $\mathcal{N}_{g,\nu} = \mathcal{N}^!_{g,\nu} \cdot \widehat{\mathcal{T}}^+_{g,\nu}$ i.e. $\mathcal{N}_{g,\nu}$ known from $\mathcal{N}^!_{g,\nu}$

will give an isomorphism

$$(3) \qquad \mathcal{N}^!_{g,\nu}/\widehat{\mathcal{T}}^{!+}_{g,\nu} \xrightarrow{\sim} \mathcal{N}_{g,\nu}/\widehat{\mathcal{T}}^+_{g,\nu} = \mathbb{\Gamma}_{g,\nu}$$

In the case of a (π, a), (π an outer group with loops, a an arithmetization), we will likewise have

$$(4) \qquad \begin{cases} \mathcal{N}^!_a/\widehat{\mathcal{T}}^+_a \simeq \mathcal{N}_a/\widehat{\mathcal{T}}^+_a = \mathbb{\Gamma}_a. \\ \mathcal{N}_a = \mathcal{N}^!_a \widehat{\mathcal{T}}^+_a \end{cases}$$

Let us now consider † for $\forall i \in I$ the stabilizer † $\mathcal{N}_i$ of i in $\mathcal{N}$, and the group with loops (not outer!) π'_i of type $g, \nu - 1$, deduced from π by "plugging hole i", m486

Marginal note. m486 We write $\mathcal{N}$ etc. instead of $\mathcal{N}_a$. We assume [illeg. 2] $(g, \nu - 1)$, i.e. $2g - 2 + \nu$ [illeg. 1] ≥ 1, i.e. $2g + \nu \geq 4$

Uncertainty retained from cancelled passages.
140-2-s395-D001 see that
140-2-s395-D002 [illeg. 1]
140-2-s395-D003 subset $i \in I = I(\pi)$ [illeg. 3] (g, ν) and a [illeg. 2] $I = I(\pi)$ [illeg. 1]
140-2-s395-D004 [illeg. 1]

[scan 396]

we then have, for a chosen discretification, † α compatible with the arithmetization, an injective homomorphism|homomorphism

$$(5) \qquad \pi'_i \xrightarrow{\varphi_i} \widehat{\mathcal{T}}^{+!}_{i|a} \subset \mathcal{N}^!_{i|a} \subset \widehat{\widehat{\mathcal{T}}}_i$$

† † † such that the composite

$$\begin{array}{ccccc} \pi'_i & \longrightarrow & \widehat{\widehat{\mathcal{T}}}_i & \longrightarrow & \widehat{\widehat{\mathcal{G}}}(\pi'_i) \\ & & & & \uparrow \\ & & & & \pi'_i \end{array}$$

is the canonical inclusion. This being so, we want the image of (5) (which may not be invariant in $\widehat{\widehat{\mathcal{T}}}_i$) to be invariant in $\mathcal{N}_i$, or what amounts to the same, in $\mathcal{N}^!$ (for $\mathcal{N}_i = \mathcal{N}^! \cdot \widehat{\mathcal{T}}^+_i$ and $\widehat{\mathcal{T}}^+_i$ leaves this image invariant). † Returning to the universal situation, this means|means m488

rather: it is true that!

Marginal note. m488 This also means that φ_i does not depend on the choice of the discretification with datum α...

etc $\mathcal{N}^!_{g,\nu} \subset \widehat{\widehat{\mathcal{T}}}_{g,\nu}$ is contained in the normalizer of the $\pi'_i \hookrightarrow \widehat{\mathcal{T}}^{!+}_{g,\nu}$ $(0 \leq i \leq \nu - 1)$ which are therefore invariant in $\mathcal{N}^!_{g,\nu}$ †.

Moreover $\widehat{\widehat{\mathcal{T}}}^!_{g,\nu} \xrightarrow{\psi_i} \widehat{\widehat{\mathcal{G}}}^!_{g,\nu-1}$ induces an † isomorphism|isomorphism

$$(6) \qquad \mathcal{N}^!_{g,\nu} \xrightarrow{\sim} \mathcal{M}^!_{g,\nu-1}$$

m489

Marginal note. m489 N.B. It suffices it seems to me to prove it for one i to deduce it for the others †

N.B. [illeg. 4]

Uncertainty retained from cancelled passages.

140-2-s396-D001 a

140-2-s396-D002 cannot|can only [illeg. 2]

140-2-s396-D004 which lifts the composite

140-2-s396-D005 e|c

140-2-s396-D007 [illeg. 1]

140-2-s396-D008 if is also anabelian

140-2-s396-D009 [illeg. 1]

140-2-s396-D010 [illeg. 1]

[scan 397]

This assertion breaks up into two: first then|of all that ψ_i does indeed map $\mathcal{N}^!_{g,\nu}$ into $\mathcal{M}^!_{g,\nu-1}$ – and this follows from the arithmetico-geometric definition of these groups – this implies on the other hand, since ψ_i also induces an isomorphism † [p. 201]

$$\widehat{\mathcal{T}}^+_{g,\nu} \xrightarrow{\sim} \widehat{\mathcal{G}}^+_{g,\nu-1},$$

that it induces a homomorphism

$$(7) \qquad \begin{array}{ccc} & \overset{\lambda_i}{\underset{\dagger}{\longrightarrow}} & \\ \mathbb{\Gamma}_{g,\nu} & & \mathbb{\Gamma}_{g,\nu-1} \\ \wr\vert & & \wr\vert \\ \mathcal{N}_{g,\nu}/\widehat{\mathcal{T}}^+_{g,\nu} & & \mathcal{M}_{g,\nu-1}/\widehat{\mathcal{G}} \end{array}$$

† and the injectivity (resp. surjectivity) of (6) is equivalent to that of (7). Moreover (7) fits, by construction, into a commutative diagram

(8)

$$\begin{array}{ccc} & \mathbb{\Gamma}_{\mathbb{Q}} & \\ {\scriptstyle \Theta_{g,\nu}}\swarrow & & \searrow{\scriptstyle \Theta_{g,\nu-1}} \\ \mathbb{\Gamma}_{g,\nu} & \longrightarrow & \mathbb{\Gamma}_{g,\nu-1} \end{array}$$

with $\Theta_{g,\nu}$ and $\Theta_{g,\nu-1}$ surjective, so it is indeed quite obvious that (6) is surjective. Bijectivity means|means that $\Theta_{g,\nu}$ and $\Theta_{g,\nu-1}$ have the same kernel (whereas a priori it could be that $\Theta_{g,\nu}$ has a smaller kernel i.e. corresponds to a representation of $\mathbb{\Gamma}_{\mathbb{Q}}$ "less unfaithful" than $\Theta_{g,\nu-1}$. Moreover,

Uncertainty retained from cancelled passages.

140-2-s397-D001 [illeg. 1]

140-2-s397-D002 ψ_i

140-2-s397-D003 arithmetic

[scan 398]

it is obvious (even, a priori, without using the definition elsewhere explicit|made explicit of the $\mathcal{N}_{g,\nu}$) that the homomorphism †λ_i (7) does not depend on the choice of $0 \leq i \leq \nu - 1$ – for example since we pass from one to the other by applying operations of $\widehat{\mathcal{T}}^+$ (since $\widehat{\mathcal{T}}^+$ operates transitively on I) and $\widehat{\mathcal{T}}^+$ operates trivially|trivially on $\mathbb{\Gamma}_{g,\nu}$ ...

We can say that (6) describes $\mathcal{M}^!_{g,\nu-1}$ (hence $\mathcal{N}^!_{g,\nu-1}$) in terms of $\mathcal{N}_{g,\nu}$ – but the reverse is less clear, for lack of knowing whether

$$\widehat{\widehat{\mathcal{T}}}^!_{g,\nu} \xrightarrow{\psi_i} \widehat{\widehat{\mathcal{G}}}_{g,\nu-1}$$

is injective; if we knew this, we could describe $\mathcal{N}^!_{g,\nu}$ as the inverse image of $\mathcal{M}^!_{g,\nu-1}$ ... † It would not be impossible moreover that (f) should be false, i.e. that the $\Theta_{g,\nu}$ should be unfaithful, but less and less so as we increase ν – in passing to the projective limit $\Theta_{g,\infty}$, only, would we have

---

Uncertainty retained from cancelled passages.

140-2-s398-D002 ψ_i

140-2-s398-D003 Then

[scan 399]

(perhaps!) a faithful representation of $\mathbb{\Gamma}_{\mathbb{Q}}$. But until an indication to the contrary, I prefer [p. 202] to work hypothetically with (f), which is expressed by the fundamental exact|exact sequences

$$(9) \qquad 1 \longrightarrow \hat{\pi}_{g,\nu-1} \xrightarrow{\varphi_i} \mathcal{N}^!_{g,\nu}|\mathcal{M}^!_{g,\nu} \overset{\psi'_i}{\longleftrightarrow} | \xrightarrow{\psi'_i} \mathcal{N}^!_{g,\nu-1}|\mathcal{M}^!_{g,\nu-1} \longrightarrow 1$$

m491

Marginal note. m491 & we could include it in a slightly larger exact sequence, with $(\mathcal{N}_{g,\nu})_i$ and $\mathcal{N}_{g,\nu-1}$ ...

which extends the exact|exact sequence

$$(10) \qquad 1 \longrightarrow \hat{\pi}_{g,\nu-1} \xrightarrow{\varphi_i} \widehat{\mathcal{T}}^!_{g,\nu} \xrightarrow{\psi'_i} \widehat{\mathcal{T}}^!_{g,\nu-1} \longrightarrow 1$$

(and takes the place of the perhaps failing exact|exact sequence

$$?? \qquad 1 \longrightarrow \hat{\pi}_{g,\nu-1} \longrightarrow \widehat{\widehat{\mathcal{T}}}^!_{g,\nu} \longrightarrow \widehat{\widehat{\mathcal{T}}}^!_{g,\nu-1} \longrightarrow 1 \qquad ??$$

)

When (π, a) is an outer group with loops equipped with an arithmetization – what one could call a "virtual algebraic curve" – then what precedes allows us to define, on each of the π'_i of type $(g, \nu-1)$ associated to the $i \in I(\pi)$, an arithmetization canonically associated to a, say a',

[scan 400]

and we find then|thus an exact|exact sequence

$$1 \longrightarrow \pi'_i \longrightarrow (\mathcal{N}_a)_i \xrightarrow{\psi_i} \mathcal{N}_{a'_i} \longrightarrow 1 \qquad (11)$$

such that we have

$$(\widehat{\mathcal{T}}^+_a)_i = \psi_i^{-1}(\widehat{\mathcal{T}}^+_{a'_i}), \qquad (12)$$

$$\downarrow$$

inducing

$$(11') \qquad \begin{array}{ccccccccc} 1 & \longrightarrow & \pi'_i & \longrightarrow & \mathcal{N}^!_a & \xrightarrow{\psi^!_i} & \mathcal{N}^!_{a'_i} & \longrightarrow & 1 \\ & & & & \cup & & \cup & & \\ & & & & \widehat{\mathcal{T}}^{+!}_a & & \widehat{\mathcal{T}}^{+!}_{a'_i} & & \\ & & & & \| & & & & \\ & & & & \psi_i'^{-1}(\widehat{\mathcal{T}}^{+!}_{a_i} & & & & \end{array}$$

and inducing by passage to the quotients

$$\mathbb{\Gamma}_a \xrightarrow{\sim} \mathbb{\Gamma}_{a_i} \tag{13}$$

Moreover, the map which is canonical

$$\text{Discrt}|\text{Discr}^+(\pi) \longrightarrow \text{Discrt}|\text{Discr}^+(\pi'_i)$$

defines by passage to the quotients

$$\mathbb{P}_a \xrightarrow{\sim} \mathbb{P}_{a_i} \tag{14}$$

compatible with the actions of $\mathbb{\Gamma}_a$, $\mathbb{\Gamma}_{a'_i}$ and (13).

I make here no assertion about a so-called|so-called bijectivity between set of arithmetizations of π, and set of

[scan 401]
[p. 203]

arithmetizations|arithmetization of π'_i — which would amount to the bijectivity of

$$\widehat{\widehat{\mathcal{F}}}/\mathcal{N}_a \longrightarrow \widehat{\widehat{\mathcal{G}}}'/M^{\natural}_{a'}|M^{\sharp}_{a'} \simeq \widehat{\widehat{\mathcal{F}}}'/\mathcal{N}_{a'},$$

which would hardly have any reason to be except if one admitted $\widehat{\widehat{\mathcal{F}}}(\pi) \xrightarrow{\sim} \widehat{\widehat{\mathcal{G}}}|\widehat{\mathcal{G}}(\pi'_i)$, which seems quite problematic to me.

m492

Marginal note. m492



However, one finds, by the "plugging the holes" functor, an equivalence between the category of † outer arithmetized profinite groups with loops, of type (g, ν), and of arithmetized profinite groups with loops (not outer!), of type $(g, \nu - 1)$.

One can propose to try to specify † the type of properties which will characterize $\mathcal{N}_{g,\nu}$ in $\mathcal{N}'_{g,\nu} = \text{Norm}_{\widehat{\widehat{\mathcal{F}}}_{g,\nu}}(\widehat{\mathcal{T}}^+_{g,\nu})$, † †or again $\mathbb{\Gamma}_{g,\nu}$ † in $\mathbb{\Gamma}'_{g,\nu} = \mathcal{N}'_{g,\nu}/\widehat{\mathcal{T}}^+_{g,\nu}$. We have natural homomorphisms 400

Uncertainty retained from cancelled passages.

140-2-s401-D001 algebraic curves
140-2-s401-D002 by
140-2-s401-D003 think
140-2-s401-D004 We shall have
140-2-s401-D005 for [illeg. 1]

[scan 402]

$$\begin{vmatrix} \mathbb{\Gamma}'_{g,\nu} \longrightarrow \mathrm{Autext}(\widehat{\mathcal{T}}^{+}_{g,\nu}) \\ \mathbb{\Gamma}'_{g,\nu} \longrightarrow \mathrm{Aut\,ext}(\widehat{\mathcal{T}}^{+!}_{g,\nu}) \end{vmatrix} \tag{15}$$

and I presume that $\mathbb{\Gamma}_{g,\nu}$ will be describable as the inverse image of a suitable closed subgroup of one or the other of the right-hand sides, i.e. that one can describe it in terms of the|of properties of outer operations on $\widehat{\mathcal{T}}^{+}_{g,\nu}$ or on $\widehat{\mathcal{T}}_{g,\nu}$. The conditions ⓔ in any case are indeed of this type (property|properties of normalizing invariant|inscribed subgroups † π'_i of $\mathcal{T}^{+!}_{g,\nu}$ ...). Of course, one could impose conditions on outer automorphisms (of $\widehat{\mathcal{T}}'^{+!}_{g,\nu}$, say), which are stable under successive passage to $\widehat{\mathcal{T}}'^{+!}_{g,\nu-1}$, $\widehat{\mathcal{T}}'^{+!}_{g,\nu-2}$ etc ... and which are generated by the conditions ⓔ. But it is not at all certain that this 401

Uncertainty retained from cancelled passages.
140-2-s402-D002 $\mathcal{T}'_i$

[scan 403]

will suffice|suffices to describe the † $\mathbb{\Gamma}_{g,\nu} \subset \Gamma'_{g,\nu}$, if only because the condition becomes vacuous [p. 204] for † the limiting case $\nu = 0$ (if $g \geq 2$; or for the cases $g = 1$, $\nu = 1$, or $g = 0$, $\nu = 3$).

It is possible that one will have to bring in properties of the π_1 of the $M_{g,\nu\overline{\mathbb{Q}}_a}$, linked to the compactification. It is practically only in the case of $(g, \nu) = (0, 3)$ that one should not expect conditions of this kind at all.

Let us call "potential algebraic curve|potential algebraic curves" (understood, over an unspecified algebraic closure of $\mathbb{Q}$,) the datum of a|a profinite outer group π with loops|loop of type (g, ν), equipped with an arithmetization a, and with a germ of "admissible" liftings †

$$\mathbb{\Gamma}^{\natural}_{a|\bullet} \longrightarrow \mathcal{N}_a. \tag{16}$$

They form a category (for isomorphisms, for the moment) If one gives oneself a torsor P under $\mathbb{\Gamma}_{g,\nu}$ (which trivially|morally amounts to giving oneself an algebraic 402

Uncertainty retained from cancelled passages.
140-2-s403-D001 $\mathcal{N}_{g,\nu}$
140-2-s403-D002 $\nu = 0$
140-2-s403-D005 [illeg. 1]

closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$ ...) the potential curves of type (g, ν) relative|relative to this torsor [scan 404] (morally|morally, corresponding|correspond to algebraic curves over $\overline{\mathbb{Q}}$ ...) are those equipped in addition with an isomorphism ("inner")

$$(17) \qquad \mathbb{P}_a \xrightarrow{\sim} P$$

(defining an isomorphism

$$(18) \qquad \mathbb{\Gamma}_a \longrightarrow \Gamma_P = \mathrm{Aut}_{\mathbb{\Gamma}_{g,\nu}}(P).$$

m497

Marginal note. m497 The case where P is the trivial torsor, is that of the † outer π (?)|(18) equipped with an oriented prediscretification α (whence an † exact sequence $1 \longrightarrow \widehat{\mathcal{T}}_\alpha \longrightarrow \mathcal{N}_\alpha \longrightarrow \mathbb{\Gamma}_\alpha \longrightarrow 1$)

This time, one expects to find finite sets of isomorphisms, instead of torsors under considerable profinite groups.

We consider the † Points of π, as the π-conjugacy classes of "admissible" liftings of (16), into|or

$$(19) \qquad \mathbb{\Gamma}^{\natural}_{a|\bullet} \longrightarrow \mathcal{M}_a \subset \widehat{\widehat{\mathcal{F}}}(\pi)$$

i.e. a germ of genuine action of $\mathbb{\Gamma}_\bullet$ on π, with (20)

$$\boxed{\pi^{\mathbb{\Gamma}^{\natural}_\bullet} = 1}$$

(this for † 403

Uncertainty retained from cancelled passages.

140-2-s404-D001 curves

140-2-s404-D002 [illeg. 1]

140-2-s404-D003 P

140-2-s404-D004 the ad

admissibility. [scan 405]

Preliminary question Knowledge of π, of $\widehat{\mathcal{F}}^{+}_{a} \subset \widehat{\widehat{\mathcal{F}}}(\pi)$, and of a subgroup $\Gamma' \subset \widehat{\widehat{\mathcal{F}}}(\pi)$ (normalizing Σ_a, such that $\Gamma' \cap \Sigma_a|\Sigma_\bullet = (1)$) would it make it possible to recover the arithmetization|arithmetization of π — and, to begin with, to recover $\mathcal{N}_a$ (in which $\Gamma' \cdot \Sigma$ is of finite index)? It would suffice|will suffice, to be able to answer in the affirmative, to know that every outer isomorphism $\pi \longrightarrow \pi_{g,\nu}$, which sends Σ onto $\widehat{\mathcal{F}}_{g,\nu} = \Sigma_{g,\nu}$, and every|the subgroup of finite index $\Gamma' \cdot \Sigma$ in $\mathcal{N}_a$ into $\mathcal{N}_{g,\nu}$, is compatible with the arithmetizations|arithmetizations. Now this amounts exactly|exactly to: [p. 205]

Ⓖ (optional, all the same!) For every open subgroup $\mathcal{N}'$ of $\mathcal{N}_{g,\nu}$, the elements of $\widehat{\widehat{\mathcal{F}}}_{g,\nu}$ normalizing $\widehat{\mathcal{F}}_{g,\nu}$ and which carry $\mathcal{N}'$ into $\mathcal{N}_{g,\nu}$, are in $\mathcal{N}_{g,\nu}$ (a fortiori, $\mathcal{N}_{g,\nu}$ would be its own normalizer in $\mathrm{Norm}_{\widehat{\widehat{\mathcal{F}}}_{g,\nu}}(\widehat{\mathcal{F}}_{g,\nu})$. 404

With the "points of U" (defined as conjugacy classes of "admissible" liftings (19) i.e. satisfying (20), one makes a groupoid, having π as its outer fundamental group, and on which $\Gamma^{\natural|\sharp} \subset \mathcal{N}_a$ operates strictly (N.B. to take a rest, one now gives oneself $\Gamma^{\natural}|\Gamma$ itself, not only|only a germ — morally|morally, this means that one has a curve defined over a finite subextension K of $\overline{\mathbb{Q}}/\mathbb{Q}$ …}|). The fixed points of Γ correspond to the rational points/K, the fixed points under a closed subgroup Γ' to the points rational over the subextension K' of $\overline{\mathbb{Q}}/\mathbb{Q}$ associated with Γ' …) 405 [scan 406]

[scan 407]

Now let [p. 206]

$$\boxed{I' \subset I = I(\pi)}$$

be a subset of I, stable under Γ. One finds by "plugging the holes at I'" an outer group with loops π', and an outer homomorphism

$$(20) \qquad \pi^{\natural|\sharp} \longrightarrow \pi'$$

Besides, π' inherits an arithmetization a' from π – and we shall have

$$(21) \qquad \begin{cases} \mathbb{\Gamma}_a \xrightarrow{\sim} \mathbb{\Gamma}_{a'} \\ \mathbb{P}_a \xrightarrow{\sim} \mathbb{P}_{a'} \\ \wr \\ P \end{cases} \rightsquigarrow$$

so π' is defined over the|the same P as $\mathbb{\Gamma}_a$ (i.e. by making holes in U to find U', we have not disturbed the algebraically closed|absolute base field $\overline{\mathbb{Q}}$...). Besides we shall have a canonical homomorphism

$$\widehat{\mathcal{T}}(\pi) \longrightarrow \widehat{\mathcal{T}}(\pi')$$

inducing

(22)

$$\begin{array}{ccc} \mathcal{N}_a & \longrightarrow & \mathcal{N}_{a'} \\ \cup & & \cup \\ \widehat{\mathcal{T}}_a & \text{———} & \widehat{\mathcal{T}}_{a'} \end{array}$$

406
[scan 408]

(inducing precisely $\mathbb{\Gamma}_a \longrightarrow \mathbb{\Gamma}_{a'}$ on passing to the quotients), and one finds, on composing

$$\Gamma \subset | \hookrightarrow \mathcal{N}_a \longrightarrow \mathcal{N}_{a'}$$

a homomorphism which is also|also injective

(23) $\qquad \Gamma \hookrightarrow \mathcal{N}_{a'}$

which is a quasi-section|quasi section of $\mathcal{N}_{a'}$ over $\mathbb{\Gamma}_{a'}$, So †under the proviso of admissibility, one finds on † $\pi'|\pi$ a structure of potential algebraic curve, relative to the same P.

†

† I am really feeling cramped, for lack of a definition in due form of "admissible" – 407

Uncertainty retained from cancelled passages.

140-2-s408-D001 that|by
140-2-s408-D002 [illeg. 1]
140-2-s408-D003 perhaps by every|∀
140-2-s408-D004 [illeg. 1] I

[scan 409]

I am going to come back to this very soon — but for the moment, I feel like noting that, [p. 207]
† if $I' \neq \emptyset$, the outer group π' can be described by a genuine groupoid, having I' as its set of objects, and on which Γ operates in the strict sense (this is pure algebra, independently of the arithmetization business...) The fact that Γ operates trivially on the $i' \in I'$ implies that for every|$\forall$ $i' \in I'$, we have a † canonical homomorphism

(24) $\qquad \Gamma \longrightarrow \pi'(i')$

which lifts its outer|outer action, whence a unique π'-conjugacy class of liftings of the outer action of Γ on π' to a genuine action of Γ on π' — one hopes that it satisfies ${\pi'}^{\Gamma} = 1$ — and one thus has a canonical map

(25) $\qquad I' \longrightarrow \mathrm{Pts}(\pi', \alpha', \Gamma)$

408

Uncertainty retained from cancelled passages.

140-2-s409-D001 operated trivially
140-2-s409-D002 lifting of

[scan 410]

I say that this map is injective

This is "obvious" when one interprets "admissible outer action" as "realizable by a genuine algebraic curve", and "admissible liftings" as "realizable by genuine rational points / $K^+|K$, field of the fixed elements|invariants of Γ". But one would of course like reasons internal to the datum of the $(\mathcal{N}_{g,\nu})$, and) properties of these data! †

m505

Marginal note. m505 We are reduced|are going to come back to the case Cardinality $I' = 2$...

This point being admitted (putting in parentheses the two possible|essential definitions of admissibility, to which we shall come back below) one finds a "plugging holes" functor

Algebraic curves potential over P, †|relative to an open subgroup † Γ of Π_P, and equipped with a set I' of points at ∞	$\longrightarrow$	Algebraic curves potential over P relative to Γ, equipped with a subset of the set of "Points rational|invariant/Γ"

m506

Marginal note. m506 for this one needs the usual anabelian restrictions

409

Uncertainty retained from cancelled passages.
140-2-s410-D001 And
140-2-s410-D002 relative
140-2-s410-D003 of a set

[scan 411]
[p. 208]

and unless I am mistaken, it has become obvious that this functor is an equivalence of categories (and as a consequence, there is the functor in the opposite direction : drill holes at "points" invariant under $\Gamma_{a|0}$ (or at arbitrary points, at the cost of passing to a smaller Γ').

for the isomorphisms

But I realize that † this is not at all obvious – I am going to try to elucidate the situation axiomatically. We have fixed a genus g, we suppose given, for varying ν (such that (g,ν) anabelian) closed subgroups $N_{g,\nu} \subset \widehat{\widehat{\mathcal{T}}}_{g,\nu}$, normalizing $\widehat{\mathcal{T}}_{g,\nu}$, and satisfying the essential condition that the $\psi_i : \widehat{\widehat{\mathcal{T}}}_{g,\nu} \longrightarrow \widehat{\widehat{\mathcal{G}|\mathcal{S}}}_{g,\nu-1}$ induce

$$M_{g,\nu} \xrightarrow{\sim} M_{g,\nu-1}$$

Which makes it possible to define the notion of arithmetization of a profinite π of anabelian type (g,ν), and the theory of

Uncertainty retained from cancelled passages.
140-2-s411-D001 [illeg. 1]

[scan 412]

plugging an arbitrary number of holes, and of drilling just 1 hole, in the category of groups of type g,ν, arithmetized.

We suppose on the other hand given|given a full subcategory of the category of profinite m509

Marginal note. m509 we suppose it : stable under taking open subgroups.

groups (which could reduce to the open subgroups of Γ_g, common value of the $\Gamma_{g,\nu}$ or of open subgroups of the group Γ_a, a an arithmetization), and a notion of outer operations "admissible" of such Γ on the arithmetized π. We suppose

① If Γ operates admissibly on π, every open subgroup too († and conversely ?)

② If moreover Γ leaves invariant a subset I' of $I = I(\pi)$, then by "plugging I'", the operation of Γ on π' is admissible.

m510

Marginal note. m510 It suffices to postulate it|prove it for card $I' = 1$, i.e. plugging a|a single hole – at least if we know that the admissibility condition depends only on the action of the kernels of the groups.

† † When we have an actual action, (not outer)|(non-outer) of Γ on a (π, a), then the functor of "drilling holes" gives an outer action on a (π^0, a^0) of type $g, \nu+1$, and we say that the starting action is

Uncertainty retained from cancelled passages.

140-2-s412-D002 and

140-2-s412-D003 [illeg. 1]

140-2-s412-D004 We propose to define a notion of admissible action [illeg. 2]

[scan 413]

admissible, if the outer action deduced is. [p. 209]

We thus find an equivalence between the category of † systems $(\pi, a, \Gamma, \psi, i)$ of a (π, a) of type $(g, \nu+1)$, with an admissible outer operation ψ of a Γ on it, and an $i \in I(\pi)$ stable under Γ, with the category of $(\pi', a', \Gamma, \psi')$ of (π', a') of type (g, ν), with a genuine admissible action ψ' of Γ on it.

† Condition (1) ensures that † if Γ operates † † actually, in a way admissible, then the outer action deduced is admissible (but the converse will not be true – there will be "pathological" liftings of a given admissible categorical|outer action…). We suppose moreover

m511

Marginal note. m511



③ If Γ operates actually in an admissible way on π, then $\pi^\Gamma = (1)$.

† This implies that the category of Γ-Points of π, or if you like of sections of $B_{\pi,\Gamma}$ ($\simeq B_u$) over B_Γ, is rigid.

Uncertainty retained from cancelled passages.

140-2-s413-D001 categories

140-2-s413-D002 We must [illeg. 3]

140-2-s413-D003 If a genuine action

140-2-s413-D004 genuinely

140-2-s413-D005 admissible|admissibly

140-2-s413-D006 Then

[scan 414]

But † let us consider the inductive limit category of the † categories of sections of $B_{\pi,\Gamma} \simeq B_u$ over $B_{\Gamma'}$ (Γ' an open subgroup), it is discrete and corresponds to the set

$$\mathrm{Pt}_{\mathrm{ad}}(B_{\pi,\Gamma^\natural}) \simeq \varinjlim_{\substack{\Gamma' \text{ ouvert} \\ \text{de } \Gamma}} \mathrm{Pt}_{\mathrm{ad}}(B_{\pi,\Gamma'}).$$

m512

Marginal note. [m512] N.B. The groupoid with strict operators $\underline{\mathrm{Pt}}(B_{\pi,\Gamma})$ depends functorially on (π,Γ), [illeg. 1] for outer morphisms † more [illeg. 3] of Γ, if †.

† If $\mathrm{Pt}(B_{\pi,\Gamma}) \neq \emptyset$, then (from the mere fact that $\pi^{\Gamma'} = (1)$ for $\forall$ open subgroup of $\Gamma'|\Gamma$) the categorical group π can be described canonically|commutatively by a groupoid profinite $\underline{\mathrm{Pt}}(B_{\pi,\Gamma^\natural})$ having $\mathrm{Pt}(B_{\pi,\Gamma})$ as object set, † linked|linked by the categorical group π, on which Γ operates in the strict sense, with the stabilizer Γ_P of $\forall$ point being open. and $\Gamma_P \to \mathrm{Aut}(P)$ ($\simeq \pi$ mod inner automorphisms)

†

being continuous. We have here if to $\forall\ P \in \mathrm{Pt}(B_{\pi,\Gamma})$ we associate the isomorphism class of †

Uncertainty retained from cancelled passages.

140-2-s414-D001 let us call|let us suppose
140-2-s414-D002 [illeg. 1]
140-2-s414-D003 in
140-2-s414-D004 [illeg. 4]
140-2-s414-D005 For [illeg. 2] [illeg. 1]
140-2-s414-D006 and|then
140-2-s414-D007 independently of formed of a profinite group satisfying
140-2-s414-D008 groups of liftings

[scan 415]

† of sections of $B_{\pi,\Gamma}$ over B_Γ, we find a bijection – which characterizes the Γ-groupoid in question up to unique isomorphism. We suppose [p. 210]

④ If Γ operates admissibly on (π, a), then there exists an open subgroup Γ' of Γ † whose categorical operation lifts to an admissible actual operation – i.e. $\mathrm{Pt}_{\mathrm{ad}}(B_{\Gamma,\pi}) \neq \emptyset$.

†

Let us note that in the situation envisaged, of plugging a hole $i \in I(\pi)$ of a π, in addition to the homomorphism

$$\widehat{\widehat{\mathcal{T}}}(\pi)_i \xrightarrow{\psi_i} \widehat{\widehat{\mathcal{G}|\mathcal{S}}}(\pi'_i)$$

associated, giving by composition

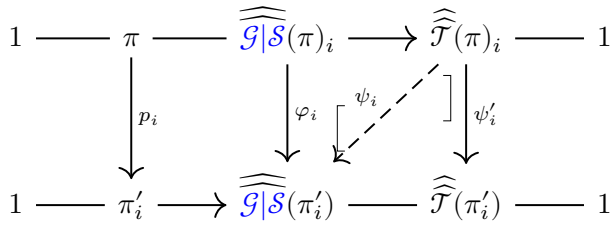
$$\widehat{\widehat{\mathcal{T}}}(\pi)_i \xrightarrow{\psi'_i} \widehat{\widehat{\mathcal{T}}}(\pi'_i)$$

Uncertainty retained from cancelled passages.

140-2-s415-D001 groups
140-2-s415-D002 operate in a way
140-2-s415-D003 [illeg. 1] admissible lifting [illeg. 2]

[scan 416]

whence an outer action of $\widehat{\widehat{\mathcal{T}}}(\pi)$ on π'_i, in such a way that the $\pi \xrightarrow{p_i} \pi'_i$ commute|commuting with the outer actions of $\widehat{\widehat{\mathcal{T}}}(\pi)$, † we have a canonical homomorphism of extensions, coming from this outer action and the homomorphism $p_i : \pi \longrightarrow \pi'_i$,



which is not the composite|composite

$$\widehat{\widehat{\mathcal{G}|\mathcal{S}}}(\pi)_i \longrightarrow \widehat{\mathcal{T}}(\pi)_i \overset{\psi_i}{\longrightarrow} \widehat{\widehat{\mathcal{G}|\mathcal{S}}}(\pi'_i).$$

(it can be defined whenever we have a group Γ, here $\widehat{\mathcal{T}}(\pi)$, which † commuting|commutes as outer actions on two outer groups π, π', and we have a homomorphism $\beta|p$ † $\pi \longrightarrow \pi'$ which commutes with the action of Γ, and such that the center of π and the|the centralizer|centralizers of its image|$\beta(\pi)$ in π' are trivial…) This is also here the "transport of structure" homomorphism, which for every|† loop automorphism u of π, associates the corresponding|canonical automorphism of π'_i.

We have†|† forgotten to specify:

69 The canonical homomorphism $(\widehat{\widehat{\mathcal{G}|\mathcal{S}}}_{g,\nu})_i \longrightarrow \widehat{\widehat{\mathcal{G}|\mathcal{S}}}_{g,\nu-1}$ associated with $i \in [0, \nu-1]$, sends $(M_{g,\nu})_i$ into $M_{g,\nu-1}$ so it fits into a homomorphism|diagram of exact sequences

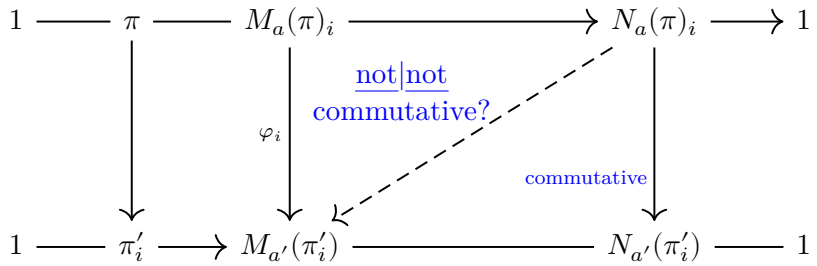
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Uncertainty retained from cancelled passages.

140-2-s416-D001 [illeg. 1]
140-2-s416-D003 with trivial center
140-2-s416-D004 [illeg. 1]
140-2-s416-D005 every
140-2-s416-D006 vait
140-2-s416-D007 had

[scan 417]
[p. 211]

(26)



[The horizontal segments issuing from the two 1 on the left have no visible arrowhead. The right ends of the segments π–$M_a(\pi)_i$ at the top, $M_{a'}(\pi'_i)$–$N_{a'}(\pi'_i)$ and $N_{a'}(\pi'_i)$–1 at the bottom are thickened, with no arrowhead legible with certainty. An ink blot (overwriting) follows the parenthesis of $M_a(\pi)_i$. The word commutative is written in the right-hand triangle, between the dotted arrow and the right-hand vertical arrow; it is not a legible label of the dotted arrow.]

.

This being so, if we have an admissible outer operation $\Gamma \longrightarrow N_a(\pi)$ of Γ on π, fixing i, hence also through $N_a(\pi) \longrightarrow N_{a'}(\pi'_i)$ on π'_i, we can consider that † lifting $\Gamma \longrightarrow M_a(\pi)$ for π defines by composition a lifting $\Gamma \longrightarrow M_{a'}(\pi'_i)$ for π'_i. We easily conclude from ② that if the first is admissible, the second is. We thus find

$$(27) \qquad \mathrm{Pt}_{\mathrm{ad}}(B_{\pi,\Gamma}) \longrightarrow \mathrm{Pt}_{\mathrm{ad}}(B_{\pi'_i,\Gamma})$$

(and passing to the limit, a map of Γ-sets

$$(28) \qquad \mathrm{Pt}_{\mathrm{ad}}(B_{\pi,\Gamma^\natural}) \longrightarrow \mathrm{Pt}_{\mathrm{ad}}(B_{\pi'_i,\Gamma^\natural})$$

– which moreover corresponds to a homomorphism of Γ groupoids

$$(29) \qquad \underline{\mathrm{Pt}}_{\mathrm{ad}}(B_{\pi,\Gamma^\natural}) \longrightarrow \underline{\mathrm{Pt}}_{\mathrm{ad}}(B_{\pi'_i,\Gamma^\natural}).$$

This being so, let $P \in \mathrm{Pt}_{\mathrm{ad}}(B_{\pi'_i,\Gamma})$ be the "canonical point" of $\mathrm{Pt}_{\mathrm{ad}}B_{\pi'_i,\Gamma} = \left(\mathrm{Pt}_{\mathrm{ad}}(B_{\pi'_i,\Gamma^\natural})\right)^\Gamma$.

Uncertainty retained from cancelled passages.

140-2-s417-D001 every

[scan 418]

We ask:

⑤ The map (27) induces a bijection.



$$\mathrm{Pt}_{\mathrm{ad}}(B_{\pi,\Gamma}) \longrightarrow \mathrm{Pt}_{\mathrm{ad}}(B_{\pi'_i,\Gamma}) \setminus \{P\}$$

or again (passing to the limit) a bijection of Γ-sets

$$\mathrm{Pt}_{\mathrm{ad}}(B_{\pi,\Gamma^\natural}) \simeq \mathrm{Pt}_{\mathrm{ad}}(B_{\pi'_i,\Gamma^\natural}) \setminus \{P\}$$

This means three things:

a) The canonical homomorphism $\Gamma \longrightarrow M_{a'}(\pi'_i)$, the composite of $\Gamma \longrightarrow N_a(\pi)$ and $N_a(\pi) \simeq M_{a'}(\pi'_i)$, is not π'_i-conjugate to a composite homomorphism $\Gamma \overset{\lambda}{\to} M_a(\pi) \overset{\varphi_i}{\longrightarrow} M_{a'}(\pi'_i)$, where $\Gamma \overset{\lambda}{\to} M_a(\pi)$ is an admissible lifting.

b) If $\lambda, \mu : \Gamma \rightrightarrows M_a(\pi)$ are two admissible liftings of $\Gamma \longrightarrow N_a(\pi)$ † such that $\varphi_i \circ \lambda$ and $\varphi_i \circ \mu : \Gamma \rightrightarrows M_a(\pi)$ are π'_i-conjugate, then λ and μ are already π-conjugate.

c) Every admissible lifting of $\Gamma \longrightarrow N_{a'}(\pi'_i)$ to $\Gamma \longrightarrow M_{a'}(\pi'_i)$, comes by composition from an admissible lifting $\Gamma \longrightarrow M_a(\pi)$.

Uncertainty retained from cancelled passages.

140-2-s418-D001 that

[scan 419]

Thanks to this, the equivalence of categories between systems $(\pi, a, i, \Gamma, \theta : \Gamma \underset{\text{adm}}{\longrightarrow} N_a)$, and [p. 212] the systems $(\pi', a', \Gamma, \theta' : \Gamma \longrightarrow N_{a'}, P)$, where $P \in \mathrm{Pt}(B_{\pi',\Gamma^\natural})^\Gamma$, becomes perfectly precise at the level of points: the Γ-rational Points of $B_{\pi,\Gamma^\natural|\dagger}$ are identified with one|the set † of the Γ-rational points of $B_{\pi'}$, distinct from P.

This then allows us (which was not feasible in the discrete context!) to extend the equivalence of categories to an equivalence

(30)

category of systems $(\pi, a\|\alpha, I', \Gamma, \theta : \Gamma \longrightarrow \cong \| \simeq$ $N_a)$ of an arithmetized π (by $\alpha\|a$) of type (g, ν), of $I' \subset I(\pi)$, of an admissible outer action of Γ on π such that Γ † fixes each point of I'	category of systems $(\pi', a'\|\alpha', I', \Gamma^*\|\Gamma, \theta' :$ $\Gamma \longrightarrow N_{a'})$ of an arithmetized π' (by $a'\|\alpha'$) of type (g, ν') $[\nu' = \nu - \mathrm{card}(I')]$ with an admissible outer operation of Γ on it, and a subset $I' \subset \mathrm{Pt}_{ad}(B_{\pi',\Gamma})$ (hence $I' \subset \mathrm{Pt}_{ad}(B_{\pi',\Gamma})^{\Gamma}$) i.e. I' formed of points invariant|fixed under Γ, i.e. "Γ-rational"

m513

Marginal note. m513 with $2g + (\underbrace{\nu - \mathrm{card}\, I'}_{\nu'}) \geq 3$

We thus have a quasi-inverse functor: drilling holes at I' !

Uncertainty retained from cancelled passages.

140-2-s419-D001 , $\Gamma^{\natural}$

140-2-s419-D002 rational

140-2-s419-D003 operates

[scan 420]

It is to be noted that in this approach, we have had to restrict ourselves to the case † of a set of points † $I' \subset I(\pi)$ not only stable under Γ, but † $\subset I^{\Gamma}$ or again to a subset $I' \subset \mathrm{Pt}(B_{\pi',\Gamma^{\natural}})$ not only stable under Γ, but even\|= $\subset \mathrm{Pt}(B_{\pi',\Gamma^{\natural}})^{\Gamma}$. To show that without this restriction, the natural functor † that corresponds is nevertheless an equivalence, we are reduced to this

⑥ Let $(\pi, a, I' \subset I(\pi))$ be with (π, a) an arithmetized categorical profinite\|profinite group with loops of type g, ν, whence (π', a') — let Γ be, with Γ' an open subgroup operating on (π, a) in an admissible way $(\Gamma \longrightarrow N_a(\pi))$, and leaving I' stable, hence it operates in an admissible way on (π', a'), Suppose given a lifting of this action of Γ' to an admissible action of Γ on $N_{a'}$, which leaves invariant

$$\begin{array}{ccc} I' & \subset & \mathrm{Pt}(B_{\pi',\Gamma^{\natural}}). \\ & & \wr \| \\ & & \mathrm{Pt}(B_{\pi',\Gamma'^{\natural}}) \end{array}$$

$$\begin{array}{ccc} \dagger\ (N_a)_{I'} & \longleftarrow & \Gamma' \\ \downarrow & \nearrow & \downarrow \\ N_{a'} & \longleftarrow & \Gamma \end{array}$$

Then there exists a unique admissible action of $\Gamma\|\Gamma'$ on (π, a), which extends that of Γ' and which gives rise to the given one on π'.

Uncertainty retained from cancelled passages.

140-2-s420-D001 of the\|of

140-2-s420-D002 [illeg. 1]

140-2-s420-D003 formed

140-2-s420-D005 is

140-2-s420-D006 M_a

Is uniqueness at all|all events clear? † Let us consider the extension $\mathcal{E}$ of Γ by $L = \mathrm{Ker}((N_a)_{I'} \to N_a|N_0)$, inverse image of the extension $(N_a)_{I'}$ (of $N_{a'}|N_{0'}$ by L) via $\Gamma \to N_{a'}|N_{0'}$, we have a partial splitting of this extension over the subgroup Γ' of Γ, and the assertion is that this splitting extends, in a way that is unique, to Γ. We may suppose Γ' normal in Γ, and we identify Γ' with a subgroup of $\mathcal{E}$. All † the sections of $\mathcal{E}$ over Γ are identified with section subgroups $\tilde{\Gamma}$ of $\mathcal{E}$. For such a $\tilde{\Gamma}$, we have of course $\tilde{\Gamma} \subset \mathrm{Norm}_{\mathcal{E}}(\Gamma')$ moreover † we obviously have [scan 421] [p. 213]

$$\mathrm{Norm}_{\mathcal{E}}(\Gamma') \cap L = L^{\Gamma'} \tag{31}$$

and I presume we must have $L^{\Gamma'} = 1$ (which generalizes condition ③ above…) Now this condition implies uniqueness – namely $\tilde{\Gamma} = \mathrm{Norm}_{\mathcal{E}}(\Gamma')$ – and existence means that $\mathrm{Norm}_{\mathcal{E}}(\Gamma') \to \Gamma$ is an epimorphism (hence an isomorphism…).

But we will have to try to make precise, in certain contexts (at least that of

Uncertainty retained from cancelled passages.

140-2-s421-D002 Choosing a

140-2-s421-D003 sections

140-2-s421-D004 if we [illeg. 2]

arithmetically faithful actions, i.e. $\Gamma \to \mathbf{\Gamma}_a$ injective – which corresponds normally|presumably to the case of algebraic curves defined over algebraic extensions of $\mathbb{Q}$ – the notion of an "admissible" action. For this, we must return to the relation between (potential) curves and finite coverings, which also gives a way of varying g. [scan 422]

But I feel like first taking up again under|on another aspect (perhaps more general) the preceding formalism, which can perhaps also be applied to the case of discrete groups with loops (I am thinking of the formalism of the $U^!|\mathcal{U}^!$, linked to outer actions of finite groups on such profinite groups with loops).

Let $\mathcal{X}$ be a set $\neq \emptyset$ (morally, a set of the "points" of an algebraic curve, or of a topological surface…),

We suppose given, for every finite subset I of $\mathcal{X}$, with complement $U_I = \mathcal{X} \setminus I$, a groupoid Π_{U_I}, having U_I as its set of objects. Moreover, for $J \supset I$ i.e. $U_J \subset U_I$, we suppose given a homomorphism of groupoids [scan 423] [p. 214]

$$\Pi_{U_J} \to \Pi_{U_I}$$

which on objects is the inclusion $U_J \subset | \hookrightarrow U_J|U_I$. We suppose transitivity (strict) i.e. † a covariant functor from the category of subsets U of $\mathcal{X}$ complementary to finite subsets (with the inclusions), to the category of groupoids, which on the set of objects coincides with the obvious functor…

m514

Marginal note. m514 If $\mathcal{X}$ is finite and $I = \mathcal{X}$, we suppose all the same|however Π_{U_I} $(= \Pi_\emptyset)$ is not the empty groupoid, but a groupoid (which maps into the preceding ones…) but we will not be able to require the strict transitivity. We will need|would need perhaps [illeg. 1] anabelian hypotheses on Π_{U_I} in this case…

We suppose moreover that for every $i \in \mathcal{X}$, we give ourselves a † Π_i (or $\Pi_{D_i^*}$), and † for $i \in I \in \mathcal{P}_f(\mathcal{X})$, a homomorphism of groupoids

$$\Pi_{D_i^*} \to \Pi_{U_I}$$

compatible with the transition morphisms $\Pi_{U_I} \to \Pi_{U_J}$ $(I \supset J)$. We suppose that for fixed I, we find on Π_{U_I} a structure of groupoid with loops by

$$\Pi_{D_I^*} = \coprod_{i \in I} \Pi_{D_i^*} \to \Pi_{U_I}.$$

Uncertainty retained from cancelled passages.
140-2-s423-D002 which
140-2-s423-D005 groupoid [illeg. 1]
140-2-s423-D006 for every

without moreover excluding the case where $I = \emptyset$, and where ($\Pi_{\mathcal{X}}$ being connected, say) its|the genus is 0 — † m515 [scan 424]

Marginal note. m515 We suppose [illeg. 4] that [illeg. 1] $\mathcal{X}$ [illeg. 3] † †

We suppose moreover that if $J \supset I$ i.e. $U_J \subset U_I$, and for $i \in J \setminus I$, the composite homomorphism

$$\Pi_{D_i^*} \longrightarrow \Pi_{U_J} \longrightarrow \Pi_{U_I}$$

is "constant", in such a way that Π_{U_I} is obtained from Π_{U_J} by "plugging the holes" at $J \setminus I$.

Up to now, all|all this could be visualized for example by starting from an orientable topological surface † X, and a subset [illeg. 1] $\mathcal{X}$ of X †meeting †every connected component; by taking for Π_I the restriction to $\mathcal{X} \setminus I$ of the fundamental groupoid of $X \setminus I$ (if $I \neq \mathcal{X}$; if $\mathcal{X} = I$, which requires $\mathcal{X}$ finite, we will take the fundamental groupoid of $X \setminus I = X \setminus \{\mathcal{X}\}$; which we will have trouble mapping into the others with strict transitivity, no matter!

But, we are mainly interested in the case $\mathcal{X}$ infinite. Where we take X an algebraic curve over

Uncertainty retained from cancelled passages.
140-2-s424-D001 thus
140-2-s424-D002 for [illeg. 2]
140-2-s424-D003 two points
140-2-s424-D004 $X \setminus S$ [illeg. 4]
140-2-s424-D005 [illeg. 1]
140-2-s424-D006 every connected component

an algebraically closed field, $\mathcal{X}$ a subset of X meeting every connected component, and we define the Π_{U_I} as before. [scan 425] [p. 215]

Let now Γ be a group which operates on the situation (strictly, be it understood) thus on the set $\mathcal{X}$, the groupoids Π_{U_I}, the $\Pi_{D_i^*}$, (N.B. which it permutes among themselves), commuting with the homomorphisms d|of

$$\Pi_{U_I} \longrightarrow \Pi_{U_J}$$

and the $\Pi_{D_i^*} \longrightarrow \Pi_{U_I}$.

(We will have difficulties, if $\mathcal{X}$ is finite, for Π_{U_I} when $I = \mathcal{X}$, for the strict commutation † $\Pi_{U_{\mathcal{X}}} \longrightarrow \Pi_{U_J}$ .. but let us pass on...) We † suppose defined a notion of admissible subgroup of Γ, (they are in any case $\neq 1$) such that † if Γ' is admissible then every subgroup $\Gamma'' \supset \Gamma'$ also. † We † †

†

†

†

Uncertainty retained from cancelled passages.
140-2-s425-D002 permutes among themselves
140-2-s425-D003 of the
140-2-s425-D004 the hypothesis|the hypotheses
140-2-s425-D005 [illeg. 1]
140-2-s425-D006 following

140-2-s425-D007 suppose
140-2-s425-D010 U_I stable under Γ_P i.e. I stable
140-2-s425-D011 ② [illeg. 1] [illeg. 1] find $\pi_1(U_I, P)|\Pi_1(U_I, P)$
140-2-s425-D014 $\pi_1(U_I, P)|\Pi_1(U_I, P)$ call say that a [illeg. 1] meets every

† We assume [scan 426]

① $\forall P \in \mathcal{X}$, $\Gamma_P \neq \{1\}$, and Γ_P of finite index in Γ (i.e. the orbit of P finite)

We want conditions which ensure that $\mathcal{X}$ and the Π_{U_I} † etc, can|may be reconstructed up to † canonical isomorphism with the help of the purely groupoid-theoretic "bare" or topossic (with operators Γ) corresponding|corresponding data. † We can say it in learned topossic language, but we are going to express it in terms of the theory of groups with loops. Let I be a subset of $\mathcal{X}$ stable under $\Gamma^*|\Gamma'$, we would like to lay down conditions which † allow|will allow us to express (for such I) the situation of the groupoids Π_V associated with the $V \supset U_I$ (i.e. the U_J with $J \subset I$), the $\Pi_{D_i^*}$ ($i \in I$), and the operation of Γ on them, in terms of the † sole operation of Γ on the outer group with loops $\pi_1(\Pi_{U_I})$ associated with I —

Uncertainty retained from cancelled passages.
140-2-s426-D001 simply contains
140-2-s426-D002 for
140-2-s426-D003 the help
140-2-s426-D004 [illeg. 1]
140-2-s426-D005 we take subgroup
140-2-s426-D006 operation

or rather (for this is immediate, by the operation of drilling holes), in the reverse direction, [scan 427] show how, † † † subject to anabelianity $\zeta|\xi$ say of $\Pi_{\mathcal{X}}|B_{\Pi_{\mathcal{X}}}$ (to fix ideas), we can ± reconstruct [p. 216] Π_{U_I} and the action of Γ on it ....|..., starting from $\Pi_{\mathcal{X}}$ (or rather, from the topos $B_{\Pi_{\mathcal{X}}}$) and from ⇝|⟶ the action of Γ on it. We want at least an intrinsic description of the elements of $\mathcal{X}$, and of the action of Γ on it, via this "soft" action — which, for X connected, again amounts to an outer action of Γ on a group with loops (without loops!) π...

A first condition, in weak form, it is|is that (assuming $\mathcal{X}$ connected, and calling : $\pi(I)$ the outer group $\pi_1(\Pi_{U_I})$, for every finite subset I of $\mathcal{X}$) that for $\pi(I)$ anabelian (which will be the case for I "large enough",

Uncertainty retained from cancelled passages.
140-2-s427-D001 for a invariant
140-2-s427-D002 [illeg. 5]
140-2-s427-D003 reconstruct [illeg. 2]

at least if $\mathcal{X}$ is infinite so|of that we can take I of arbitrarily large cardinality...) the obvious [scan 428] map

$$2^\circ) \quad \dagger U_I \longrightarrow \mathrm{Pt}(B_{\pi(I),\Gamma\natural})|\mathrm{Pt}(B_{\pi(I)},\Gamma\natural) \simeq \begin{array}{c}\dagger\\ \text{classes of } \pi(I)\text{-conjugation of|of the}\\ \text{germs of}\\ \text{splittings of}\\ \text{the extension } \mathcal{E}(I)\\ \text{of } \dagger\Gamma \text{ by } \pi(I)\end{array}$$

(compatible|compatible with the actions of Γ) be injective. But to go further, we would have to be able to give a characterization of the image. Let us note (placing ourselves henceforth in the profinite context) that if we give ourselves on the $\Pi(I)|\pi(I)$ some arithmetizations, compatible with the outer homomorphisms of plugging holes $\pi(I) \longrightarrow \pi(J)$, and with the action of Γ on the $\pi(I)$, we therefore have a canonical homomorphism from Γ into the common group $\mathbf{\Gamma}$ of

arithmetically (outer automorphisms of these arithmetizations. (and even $\Gamma \longrightarrow M_{\pi(\emptyset)=\pi}$, — which is already a markedly stronger datum). We can therefore

Uncertainty retained from cancelled passages.
140-2-s428-D002 $\mathrm{Pt}(B_{\pi(I),\Gamma\natural})|\mathrm{Pt}(B_{\pi(I)},\Gamma\natural)$
140-2-s428-D003 $\mathcal{X}$
140-2-s428-D005 $\mathcal{E}$

suppose given a notion of admissible lifting of such an outer arithmetic action, in such a way that the preceding morphism is a bijection [scan 429] [p. 217]

$$U_I = \mathcal{X}|\mathfrak{X} \setminus I \xrightarrow{\sim} \mathrm{Pt}_{\mathrm{ad}}(B_{\pi(I)},\Gamma\natural)$$

This means moreover, in practice, that the notion of admissibility satisfies conditions $1^\circ) : 6^\circ)$ seen above.

As for knowing what should be called operations † "admissible|admissible" of Γ on an outer (profinite)|, group with loops, this remains conjectural for the moment.

We could by way of experiment conjecture † that the following notion would work Let us call|Let us call "admissible" every such action

$$\Gamma \longrightarrow \widehat{\widehat{\mathcal{F}}}(\pi)$$

which respects an arithmetization a, i.e. which factors through the corresponding N_a, and such that the image of $\Gamma \longrightarrow N_a/\widehat{\mathcal{F}}_a = \mathbf{\Gamma}_a$ is open|open, †, and that

Uncertainty retained from cancelled passages.
140-2-s429-D001 outer
140-2-s429-D002 every [illeg. 2]
140-2-s429-D004 the conditions

† [scan 430]

$\forall$ homomorphism of plugging holes at $i \in I(\pi)$, $\pi \longrightarrow \pi'_i$, $N(\pi)_i \longrightarrow M(\pi'_i)$, the corresponding homomorphism Γ_i, nor of any open subgroup Γ' of Γ_i, does not normalize an L'_j $(j \in I(\pi'_i) = I \setminus \{i\})$.

perhaps we would even have to impose $\pi^{(\Gamma'^+)|\langle\Gamma'^+\rangle} = (1)$, where Γ'^+ is the kernel of $\Gamma' \xrightarrow{\chi} \widehat{\mathbb{Z}}^*$ — in any case we expect — that we then have $\pi^{\Gamma'} = (1)$, and if this were not the case, we would have to introduce it into the definition — for each operation of plugging holes relative to $I' \subset I(\pi)$, and to the choice of an $i \in I'$, allowing us to define $\Gamma_{(I',i)} \longrightarrow M(\pi(I'))$. †
m519

Marginal note. m519 More generally, for $i \in I' \subset I = I(\pi)$, † we impose that the true action of $\Gamma_{I',i}$ on $\pi_i(I')|\Pi_i(I')$ corresponding to i normalize no subgroup L_j $(j \in I \setminus I')$ — we are reduced for this to the case where $I' = I \setminus \{j\}$, hence where $\pi(I')|\Pi(I')$ has (only a single class loops… (if|and $g \geq 1$)

Let us note that a true action of Γ on π (not only|only outer) which lifts a given+ admissible action, is it too admissible (in the sense that it defines an admissible action on a π' of type $(g, \nu+1)$), iff this action (rather its germ) normalizes no L_i $(i \in I(\pi))$, and that

Uncertainty retained from cancelled passages.
140-2-s430-D001 considered $1^\circ : 6^\circ$ are satisfied|satisfied
140-2-s430-D002 [illeg. 2] premature every pair|all pairs corresponding [illeg. 1] K formed $\pi \longrightarrow \pi'_i$ too small for i.e. $\mathbf{\Gamma}|\Pi$ $\widehat{\mathcal{F}}(\pi)|\widehat{\mathcal{G}}(\pi)$ is not
140-2-s430-D005 it is necessary

[scan 431] [p. 218]

it is likewise for the action induced on each $\pi(I')$ ($I' \subset I = I(\pi)$) [it suffices to take the $\pi(I')$ for I' of the form $I \setminus \{j\}$, $j \in I$] — and that finally the action of $\Gamma^\natural$ † on $\pi'(I)$ (deduced from π' by plugging the holes at I, so that $\pi'(I)$ has a single class of loops; $\pi'(I)$ is also deduced from the actual action of Γ on $\pi(I)$ — with 0 classes of loops — by making "a hole" corresponding) relative to a point $i \in I$, normalizes no loop subgroups… But I presume that the first condition (action of $\Gamma^\natural$ normalizing none of the L_i) implies the others. at least if $g \neq 0$

But one must in the definition of admissibility also take into account condition (4°), †which implies the existence of enough liftings… .

I come (painfully!) to a

Provisional profinite conjecture For Γ a given profinite group, an outer action on an arithmetized anabelian (π, a) of type (g, ν) is called admissible, if Hoaxed cf. below…

a) † The homomorphism $\Gamma \longrightarrow \mathbf{\Gamma}_a$ has open image

b) The actual actions deduced by pluggings of|of the holes ($i \in I' \subset I = I(\pi)$) normalize no loop subgroups.

Uncertainty retained from cancelled passages.

140-2-s431-D001 an

140-2-s431-D002 [illeg. 1]

140-2-s431-D003 [illeg. 1]

[scan 432]

c) There exist infinitely many π-conjugacy classes of the germs † splittings of the extension of Γ by π, which normalize no loop subgroup of π.

m522

Marginal note. m522 This condition c) probably excludes cases like $\Gamma = \mathcal{N}_{g,\nu}|N_{g,\nu}$ with $(g, \nu) \neq (0, 3)$, for the extension $\mathcal{M}_{g,\nu}|M_{g,\nu}$ of $\mathcal{N}_{g,\nu}|N_{g,\nu}$ by $\widehat{\pi}_{g,\nu}|\widehat{\Pi}_{g,\nu}$ probably admits no germs of [illeg. 1] ($\overset{?}{\Longleftrightarrow}$ $\mathcal{U}_{g,\nu}|U_{g,\nu}$ over $\mathcal{M}_{g,\nu}|\mathcal{N}_{g,\nu}$ has|admits probably no étale multisection|étale multisections…)

An actual action of Γ on π is called actual, if the outer action is actual, and if it normalizes no loop subgroup.

This being so:

a) If Γ operates actually in an admissible way, we have

$$\pi^\Gamma = (1)$$

(perhaps even $\pi^{\Gamma^+|\Gamma^\dagger} = (1)$) † †

Uncertainty retained from cancelled passages.

140-2-s432-D001 [illeg. 1]

140-2-s432-D003 sub-

140-2-s432-D006 [illeg. 1]

[scan 433]

b) If $i \in I$ is stable under Γ operating by outer automorphisms in an admissible way, then the actual action on $\pi' = \pi(i)$ normalizes no L'_j ($j \in I \setminus \{i\}$). [p. 219]

c) If Γ operates actually on π in an admissible way, while leaving $i \in I$ fixed, then the corresponding actual action (by †passage to the quotient) on $\pi'^\natural = \pi(i)|R(i)$ normalizes no L'_j ($j \in I \setminus \{i\}$).

d) If Γ operates actually on π in an admissible way, then the map

$$\beta \qquad \mathrm{Pt}_{\mathrm{ad}}(B_{\pi,\Gamma^\natural}) \longrightarrow \mathrm{Pt}_{\mathrm{ad}}(B_{\pi',\Gamma^\natural})$$

defined by c) is injective, and the complement of its image is equal to $\{P\}$, where P is defined by the "hole" i.

But this last part of the assertion, going beyond mere injectivity — and characterizing the image, now seems to me quite doubtful —

Uncertainty retained from cancelled passages.
140-2-s433-D003 [illeg. 3]

[scan 434]

indeed, it suffices to look at a parameter scheme S, of finite type over $Q|\mathbb{Q}$ (a $K(\pi,1)$ preferably, for example an algebraic curve) and to take for Γ (not the fundamental group of its generic point, i.e. of a field, but) the fundamental group of S itself. The datum of an admissible outer action of Γ on a π corresponds morally to that of a family of curves U of type g,ν parameterized by S. Condition c) is satisfied for example if U comes from an algebraic curve over the base field itself, so no problem – and it is plain that surjectivity goes haywire. The difficulty comes from the fact that two distinct sections of U over S may not be disjoint !
m524

Marginal note. m524 The splittings of extensions correspond to the sections of U over S

[scan 435]
[p. 220]

One must be able to speak of sections "strictly|strictly distinct" i.e. distinct at every point of S – which, in profinite translation, amounts to comparing two splittings of the extension of Γ by π, with †the germs of splittings over $\mathbf{\Gamma}'_{\cdot} \subset \mathbf{\Gamma}|\mathbf{\Gamma}' \subset \mathbf{\Gamma}$ of Γ (as an extension of $\mathbf{\Gamma}_Q$ by a geometric part) which correspond to the points of S – i.e. precisely the admissible (germs of splittings). One has the impression of going around in a vicious circle or almost – it would seem that one should not try to swallow too much at once – deal from the outset with all profinite groups Γ at once – but rather restrict oneself first to those which, morally, correspond to $\pi_1|\Pi_1$ of schemes of finite type over $Q|\mathbb{Q}$ – some $K(\pi,1)$ let us say, or even varieties

Uncertainty retained from cancelled passages.
140-2-s435-D001 two

[scan 436]

elementary in Artin's sense – and in the definitions resp. conjectures, pull oneself up by one's shoelaces, by inducting on the dimension.

At the first notch then (dimension 0) one would restrict oneself to actions of groups Γ which are (modulo at least passage to an open subgroup) "arithmetically faithful" for example|for example $\Gamma \to \mathbf{\Gamma}_a$ injective.

In|In that case, the conjecture as stated just now seems reasonable, or at least not necessarily crazy. The decisive test,|,: it is true, would be the possibility of obtaining the "curves" thus defined as coverings of $\mathbb{P}^1 \setminus \{0,1,\infty\}$ …

[scan 437]
[p. 221]

33 Topological digression : Anti-involutions of oriented algebroid surfaces
m525

Marginal note. m525 In the end I look only at compact surfaces – Yet the noncompact case would also be very interesting to look at – to try to recover by an "algebraic" route, from the outer automorphism of order 2 in π_1, the arrangement of the curve's "points at infinity" on the connected components of † X^σ $(X = \widehat{U})$

We call an algebroid surface a surface U of the form $\widehat{X} \setminus S$, X a compact orientable surface, S finite. Then X is determined up to unique homeomorphism as the "pure compactification"

of U, denote it|we denote it $\widehat{U}$. † The homeomorphisms of U onto itself are identified with the morphisms of X which map S into itself. The orientations of X are in 1-1 correspondence with those of U. The anti-involutions of U (assumed oriented) are in 1-1 correspondence with those of X which preserve S.

We then find an equivalence between the category of oriented algebroid surfaces U, equipped with an anti-involution σ, and of compact surfaces with boundary Y, equipped with a finite subset T, to (U,σ) corresponds $(\widehat{U}/\sigma, (\widehat{U}\setminus U)/\sigma)$ – and to (Y,T) corresponds $\tilde{Y}-\tilde{T}$, where $\tilde{Y}$ is the "oriented double" of Y (which is a compact oriented surface), and $\tilde{T}$ the inverse image of T in $\tilde{Y}$.

We are now interested in the

Uncertainty retained from cancelled passages.

140-2-s437-D001 [illeg. 1]

140-2-s437-D002 The

[scan 438]

case where U is connected of type (g,ν), first examining the case $\nu=0$, i.e. $U=\widehat{U}$, $S=\emptyset$ (to which the general case will reduce). The condition $\nu=0$ corresponds to the case where $T=\emptyset$ – thus the $U=X$ considered are identified with the oriented doublings of certain compact manifolds with boundary Y. Which ones ? Obviously Y must be connected, and nonorientable.

m526

Marginal note. m526 it is not true that Y is necessarily nonorientable – only in the case where $X^\sigma=\emptyset$

So Y is characterized, up to homeomorphism, by its genus γ and the number μ of connected components of the boundary – it can be obtained from the projective plane Y_0 over the reals by cutting out γ disjoint disks in it, and gluing Möbius strips back in, then removing again μ open disks, – which gives

$$\chi(Y) = \underset{1}{\chi(Y_0)} - \dagger\gamma\cdot\underbrace{\chi_!(\text{open disk})}_{+1} + \gamma\chi_!(\text{open Möbius strip}) \\ +|-\mu\underbrace{\chi_!(\text{open disk})}_{+1}$$

now the open Möbius strip (obtained from Y_0 by removing a closed disk) has a|a $\chi_!$ equal to :

$$\underset{1}{\chi(Y_0)} - \underset{1}{\chi(\text{closed disk})}, \quad \text{i.e. } 0, \quad \text{hence}$$

$$(1) \qquad \chi(Y) = 1-(\gamma+\mu)$$

m527

Marginal note. m527 Y compact connected nonorientable surface without boundary of genus γ

On the other hand, we will have|have

$$\chi(X) = \chi(X\dagger\setminus X^\sigma)$$

for X^σ is a union of circles so its χ is zero,

Uncertainty retained from cancelled passages.

140-2-s438-D004 χ

140-2-s438-D005)

[scan 439]

[p. 222]

Now, since $X \setminus X^\sigma$ is an étale covering of order 2 of Int Y, we have

$$\chi(X \setminus X^\sigma) = 2\chi(\mathrm{Int}(Y))$$

finally

$$\chi(Y) = \chi(\underline{Y \setminus \partial Y}) = \chi(\mathrm{Int}\, Y)$$

for the same reason as before ($\chi(\partial Y) = 0$), hence finally

$$(2) \qquad \chi(X) = 2\chi(Y) = 2(1 - (\gamma + \mu))$$

† or again, since $\chi(X) = 2 - 2g$

$$(3) \qquad g = \gamma + \mu$$

Thus we find [illeg. 1]

Proposition The category of oriented connected compact surfaces of genus g equipped with an anti-involution σ is equivalent to that of nonorientable compact manifolds with boundary, of type (γ, μ) (γ the genus, μ the number of holes), with $\gamma + \mu = g$.

m529

Marginal note. m529 no, we find only a full subcategory of that of all the (X, σ). On the other hand there are also the oriented doublings $Y \amalg_{\partial Y} Y$ of compact orientable surfaces with nonempty boundary – if Y is of type (γ, μ), X is of genus g with $g = 2\gamma + \mu - 1$ (where $\gamma \geq 0$, $\mu \geq 1$)

Corollary There are exactly $g + 1$ isomorphism classes of systems (X, σ), classified by $0 \leq \mu \leq g$, where $\mu = \mathrm{card}\, \pi_0 | \Pi_0(X^\sigma)$.

m530

Marginal note. m530 This only works by restricting oneself to the (X, σ) such that (X/σ) nonorientable – cf.|cf. opposite † the † case X/σ orientable

$X = X_g$ is a compact oriented surface of

†

Uncertainty retained from cancelled passages.

140-2-s439-D001 [illeg. 1]

140-2-s439-D002 for

140-2-s439-D003 general

140-2-s439-D004 oriented [illeg. 1] algebroid

[scan 440]

genus g, this also means that in the group $A_g = \mathrm{Aut}(X_g)$, there are exactly $g + 1$ conjugacy classes of elements σ satisfying

$$(4) \qquad \sigma^2 = 1, \qquad \mathrm{sg}(\sigma) = +1.$$

(i.e. which are anti-involutions). They give $g + 1$ conjugacy classes of elements of $A_g / A_g^\circ = \mathcal{T}_g$, satisfying the same|same relations. We will show (we hope!) that these classes are distinct, and that every $\sigma \in \mathcal{T}_g$ satisfying relations (4) is in one of these classes.

We will admit the

Lemma Every anti-involution of the unit disk (case|or of $\mathbb{C}$) is conjugate to $z \mapsto \bar{z}$, and consequently the set of its fixed points is homeomorphic to $\mathbb{R}$, and in particular is connected.

Theorem Let U be an oriented surface (Hausdorff connected paracompact), not isomorphic to S^2, σ an anti-involution of U, $\widetilde{U}$ a universal covering of U, whence $\pi = \mathrm{Aut}(\widetilde{U}/U)$ and an extension $E|\mathcal{E}$ of $\mathbb{Z}/2$ by π, formed

[scan 441]

of topological automorphisms of $\widetilde{U}$ compatible with the equivalence relation defined by [p. 223]
$\widetilde{U} \longrightarrow |-U$, and inducing on U the automorphism id_U or σ.

Then

a) (For the record) U^σ is a closed submanifold of U of dimension 1.

b) For $\forall\ x \in U^\sigma$, one finds a π-conjugacy class of †splittings of the extension $\mathcal{E}$, in the usual way, whence a map †

$$U^\sigma \longrightarrow \mathrm{Sc}(\mathcal{E}, \mathbb{Z}/2) = \text{set of classes under } \pi\text{-conjugation of splittings of } \mathcal{E} \text{ over } \mathbb{Z}/2, \text{ or again of elements of order 2 of } \mathcal{E} \text{ which are not in } \pi. \tag{5}$$

This map factors through a bijection

$$\pi_0(U^\sigma) \simeq \mathrm{Sc}(\mathcal{E}, \mathbb{Z}/2) \tag{6}$$

c) Let Z_i be a connected component of U^σ, corresponding to a $\sigma_i \in \mathcal{E}^-$ of order 2. Then †

$$\pi^{\sigma_i} = \{1\} \quad \text{sss|iff} \quad Z_i \simeq \mathbb{R}, \quad \dagger \tag{7}$$

d) If $Z_i \not\simeq \mathbb{R}$ i.e. † $Z_i \simeq S^1|\mathbb{S}^1$, then, for a chosen orientation of Z_i, denot-

Uncertainty retained from cancelled passages.

140-2-s441-D002 of
140-2-s441-D003 4|5
140-2-s441-D006 [illeg. 1] [illeg. 1]
140-2-s441-D007 and
140-2-s441-D008 [illeg. 4]

ing by g_i the element of $\pi = \pi_1(U)$ that it defines (defined up to conjugation), and by $\varphi_i : \mathbb{Z} \longrightarrow \pi$ [scan 442]
the homomorphism $\varphi_i(n) = g_i^n$ from $\mathbb{Z}$ to π, one has:

1°) φ_i is injective

2°) after possibly replacing φ_i by a conjugate, one has

$$(9)|(8) \qquad \pi^{\sigma_i} = \mathrm{Im}\,\varphi_i$$

Proof One finds the map (5) by †, for $\forall\ x \in U^\sigma$, the canonical operation of $\mathbb{Z}/2$ on the universal covering $\widetilde{U}(x)$ pointed at x, and by taking the corresponding operations on $\widetilde{U}$, deduced via the isomorphisms $\widetilde{U} \simeq \widetilde{U}(x)$. One sees that the † operations σ'† obtained on $\widetilde{U}$ (for fixed x), are those for which † $\widetilde{U}^{\sigma'}$ has a point over x — hence the set of x which give rise to †the class of a σ', are the elements of the image of $\widetilde{U}^{\sigma'}$ under the projection $\widetilde{U} \longrightarrow U$. Since by the lemma $\widetilde{U}^{\sigma'}$ is nonempty and connected, it follows that

Uncertainty retained from cancelled passages.

140-2-s442-D001 taking
140-2-s442-D003 only
140-2-s442-D004 [illeg. 1]
140-2-s442-D005 [illeg. 1]
140-2-s442-D006 a

its image in U^σ is so too — we are going to see that its image is exactly a connected component [scan 443]
[p. 224]

of U^σ — which will prove both the factorizability of (5) through † $\pi_0(U^\sigma)$ (rather trivial in any case) and the fact that the resulting map (6) is bijective.

Let $\widetilde{Z}_i$ be a universal cover[ing]|covering of Z_i, and let us take the corresponding universal covering $\widetilde{U}_i$ of U, so that one has (8)|(9)

$$\begin{array}{ccc} \widetilde{Z}_i & \longrightarrow & \widetilde{U}_i \\ \downarrow & & \downarrow \\ Z_i & \longrightarrow & U \end{array}$$

and by functoriality σ also operates on this diagram (operating trivial[ly] on $Z_i, \widetilde{Z}_i$), let σ_i be its operation on $\widetilde{U}_i$. One thus sees that $\widetilde{Z}_i$ maps into $\widetilde{U}_i^{\sigma_i}$, which therefore maps onto Z_i — which already finishes proving b).

† Let us consider † the image † of $\widetilde{Z}_i$ in $\widetilde{U}_i|\widetilde{U} \mid Z_i$, this is an open and closed subset (as the image of a morphism of étale cover[ings]|coverings over the same component Z_i), so it is

Uncertainty retained from cancelled passages.

140-2-s443-D001 the

140-2-s443-D003 simpl[y]|simply a

140-2-s443-D004 the image

140-2-s443-D005 induced

fortiori an open and closed subset of $\widetilde{U}_i^{\sigma_i}$, and since this space is connected, it is equal to it. [scan 444]
Moreover $\widetilde{Z}_i \longrightarrow \widetilde{U}^{\sigma_i}$ makes †of $\widetilde{Z}_i$ an étale covering of its image $\widetilde{U}_i^{\sigma_i}$ in $\widetilde{U}_i \mid Z_i$, $*$ and since $\widetilde{U}_i^{\sigma_i}$ is simpl[y]|simply connected, one finds final[ly]|finally

$$(10) \qquad \widetilde{Z}_i \xrightarrow{\sim} \widetilde{U}_i^{\sigma_i}$$

When Z_i is simpl[y]|simply connected i.e. $\widetilde{Z}_i \simeq Z_i$, this also means that $\widetilde{U}_i^{\sigma_i} \longrightarrow Z_i$ is a homeomorphism, (thus $\widetilde{Z}_i = Z_i \longrightarrow \widetilde{U}_i^{\sigma_i}$ is the inverse homeomorphism). Since, for $x \in Z_i$, † and $\tilde{x} \in (\widetilde{U}_i^{\sigma_i})_x$, the $\tilde{x}'$ of $\widetilde{U}_i^{\sigma_i}$ over x are † the elements of the form $\tilde{x} \cdot \gamma$, with $\gamma \in \pi_i^{\sigma_i}$ ($\pi_i = \mathrm{Aut}(\widetilde{U}_i/U)$) it follows that one indeed has

$$(11) \qquad \pi_i^{\sigma_i} = 1$$

which is essentially formula (7).

In the case † Z_i not simpl[y]|simply connected, setting

$$(12) \qquad T_i = \mathrm{Aut}(\widetilde{Z}_i/Z_i) \underset{f}{\simeq} \pi_1(Z_i; \widetilde{Z}_i)$$

one finds a canonical homomorphism

$$(13) \qquad \varphi_i : T_i \longrightarrow \pi_i$$

and injectivity in (10) is equivalent to (13) being injective. Of course,

Uncertainty retained from cancelled passages.

140-2-s444-D003 [illeg. 1]

140-2-s444-D004 the elements

140-2-s444-D005 [illeg. 1]

140-2-s444-D006 denoting

the choice of a generator g_i° of T_i amounts to the choice of an orientation of Z_i, and $g_i =$ [scan 445]
[p. 225]

$\varphi_i(g_i^\circ) \in \pi_i$ is then the corresponding element of $\pi_1(U)$ in question in the statement (less precise, since it only speaks of a conjugacy class). It is clear that σ_i (operating trivial[ly] on T_i, and on π_i by transport of structure) commutes with the injective homomorphism φ_i, hence φ_i induces $T_i \longrightarrow \pi_i^{\sigma_i}$. I say that this is in fact an isomorphism

$$(14) \qquad \varphi_i : T_i \xrightarrow{\sim} \pi_i^{\sigma_i}$$

– it remains to prove surjectivity. But if $\gamma \in \pi_i^{\sigma_i}$, then $\tilde{x}\gamma \in \widetilde{U}_i^{\sigma_i}$, hence $\tilde{x}\gamma \in \varphi_i(\widetilde{Z}_i)$ which means that $\tilde{x}\gamma = \tilde{x}\varphi_i(g)$ † for $g \in T_i$ hence $\gamma \in \operatorname{Im}\varphi_i$, q.e.d.

Corollary 1 Let $\sigma_i \in \mathcal{T}_g$ $(0 \leq i \leq g)$, corresponding to an anti-involution σ_i of X_g such that $\operatorname{card}\pi_0(X_g^{\sigma_i}) = i$. Then a) There are exactly i Π_g-conju- and X_g/σ nonori-entable....

Uncertainty retained from cancelled passages.
140-2-s445-D003 hence

gacy classes of actual automorphisms of order 2 of Π_g, in the class σ_i (which proves that if $i \neq j$, σ_i and σ_j are not conjugate in $\mathcal{T}_g$ † ...) [scan 446]

b) If $u_i \in \mathcal{G}_g = \operatorname{Aut}_{\text{lac}}(\Pi_g)$ is of order 2, in the class σ_i, then †

$$(14) \qquad \Pi_g^{u_i} \simeq \mathbb{Z}$$

c) If $u_i, u_i' \in \mathcal{G}_g$, † are of order 2 of class σ_i, then $\exists\, h \in \mathcal{G}_g^+$ such that

$$(15) \qquad u_i' = h u_i h^{-1} = \operatorname{int}(h) u_i$$

(N.B. Necessarily, the image of h in $\mathcal{T}_g^+$ will be in $(\mathcal{T}_g^+)^{\sigma_i} = \operatorname{Centr}_{\mathcal{T}_g}(\sigma_i)^+$.)

Proof a) and b) are special cases of the theorem, applied to an anti-involution σ_i of X_g, with $\pi_0(X_g^{\sigma_i})$ of cardinality i. Let $Y_{g,i} = X_g^{\sigma_i}$, a compact surface with boundary nonorientable connected of type $(g-i, i)$, it is well known, and immediate, that the group of auto-

Uncertainty retained from cancelled passages.
140-2-s446-D001 [illeg. 1]
140-2-s446-D002 Π_g
140-2-s446-D003 satisfy both

morphisms of the|of† such a manifold is transitive on $\pi_0(\partial Y)$, hence on lifting to (X_g, σ_i), that the group of automorphisms of (X_g, σ_i) is transitive on $\pi_0(X_g^{\sigma_i})$, † which one can interpret also as the set of Π_g-conjugacy classes of u_i as in b), c). This therefore implies that there exists $\dot{h} \in \mathcal{T}_g^+|\mathcal{G}_g^+$, commuting with σ_i, such that – denoting by h a lifting|lifting in $\mathcal{G}_g^+$ – one has u_i' $\Pi|\pi$-conjugate to $\operatorname{int}(h).u_i$, which also means that one can after possibly modifying h), choose it so that one has (15) [scan 447] [p. 226]

Corollary 2 Under the conditions of the preceding corollary, setting $\Pi_g^{u_i} = T$ $(\simeq \mathbb{Z})$, one has a diagram of exact sequences

$$(16) \qquad \begin{array}{ccccccc} 1 & \longrightarrow & T & \longrightarrow & \dagger\mathcal{G}_g^{+u_i} & \longrightarrow & \mathcal{T}_g^{+\sigma_i} \quad \dagger \\ & & \| & & \downarrow & & \downarrow \\ 1 & \longrightarrow & T & \longrightarrow & \mathcal{G}_g^{\dagger u_i} & \longrightarrow & \mathcal{T}_g^{\dagger\sigma_i} \quad \dagger \end{array}$$

and the index of $\mathcal{G}_g^{+u_i}$ in $\mathcal{T}_g^{+\sigma_i}$ (and of $\mathcal{G}_g^{u_i}$ in $\mathcal{T}_g^{\sigma_i}$) is i. Since the last two vertical arrows are inclusions of subgroups of index 2, it suffices to treat the† assertion concerning $\mathcal{G}^+$, $\mathcal{T}^+$, †

Uncertainty retained from cancelled passages.

140-2-s447-D001 Y
140-2-s447-D002 [illeg. 2]
140-2-s447-D003 $\mathrm{Centr}_{\mathcal{G}_g}$
140-2-s447-D004 $\longrightarrow 1$
140-2-s447-D005 $+$
140-2-s447-D006 $+$
140-2-s447-D007 $\longrightarrow 1$
140-2-s447-D008 es
140-2-s447-D009 [illeg. 2] [illeg. 1] [illeg. 1]

† [scan 448]

Now $\mathcal{T}_g^{+\sigma_i}$ operates transitively on the set of Π-conjugacy classes of the u_i', by c), on the other hand the stabilizer in $\mathcal{T}_g^{+\sigma_i}$, for this action, of u_i, consists of the $\dot\gamma \in \mathcal{T}_g^{+\sigma_i}$ such that † $\mathrm{int}(\gamma)u_i$ is Π-conjugate to u_i, i.e. such that there exists $\alpha \in \Pi$, with $\mathrm{int}(\gamma)u_i = \mathrm{int}(\alpha)u_i$ i.e. $\alpha^{-1}\gamma \in \mathcal{G}_g^{+u_i}$, which means that $\dot\gamma$ is in the image of $\mathcal{G}_g^{+u_i}$, q.e.d.

† †

Uncertainty retained from cancelled passages.

140-2-s448-D001 $\dot\gamma$ u_i' [illeg. 1] $h\gamma$ $h\gamma$
140-2-s448-D004 there exists
140-2-s448-D005 as
140-2-s448-D006 [illeg. 1] 0-connected homeomorphic [illeg. 1] is necessary and sufficient least [illeg. 1] when [illeg. 2] orientable as correspond connected

I feel like constructing a geometric figure where one can bring out simultaneously topological [scan 449]
anti-involutions which give rise to the $\sigma_i \in \mathcal{T}_g^\natural | \mathcal{T}_g^\sharp$ (up to conjugation..), and to the various u_i [p. 227]
$\in \mathcal{G}_g$ associated with a σ_i (which will then be easy, by the general recipe). We know that σ_i $\in \mathcal{T}_g$ is obtained by regarding an X_g as the oriented double of a $Y = Y_{g-\dagger i,i}$, thus starting from the † real projective plane $Y_{\dagger 0}$, by making g holes in it, one † fills $g-i$ back in with Möbius bands, leaving the i other holes as they are. Let D_j $(1 \leq j \leq g)$ be the disjoint closed disks corresponding to the "holes, thus † †

$$V_{0,j} = Y_0 \setminus \bigcup_j \overset{\circ}{D}_j \tag{17}$$

is a manifold with boundary (nonorientable), contained both in Y_0, and in $Y = Y_{g-i,i}$, and even coinciding with Y at the points of

$$\partial Y = \bigcup_{g-i+1 \leq j \leq g} \partial D_j .$$

Let X_0 be the oriented sphere which covers Y_0,

Uncertainty retained from cancelled passages.

140-2-s449-D002 2
140-2-s449-D003 [illeg. 1]
140-2-s449-D004 g
140-2-s449-D005 D_0
140-2-s449-D007 [illeg. 1]
140-2-s449-D008 Then

and U_0 † $= X_0|V_0$, its restriction over $\bigcup D_j$, it is thus the complement of a union of $2g$ disks [scan 450]

$$(18) \quad \dagger \quad U_0\dagger = X_0 \setminus \bigcup_{1\leq j\leq g} \Delta_j,$$

where $\Delta_j \longrightarrow D_j$ is a trivial 2†-sheeted covering of D_j ($\Delta_j = \Delta'_j \amalg \Delta''_j \simeq D_j \times \varepsilon_j|\mathcal{E}_j$, where $\varepsilon_j|\mathcal{E}_j$ is a set with 2 elements, which is identified with the set of the two orientations of D_j...). † Let on the other hand † M_j $(1 \leq j \leq g-i)$ be the Möbius band whose boundary has|. been glued back to V_0 by a homeomorphism

$$(19) \qquad \partial M_j \simeq \partial D_j$$

let|over

$$(20) \qquad M = \coprod_{1\leq j\leq g-i} M_i$$

so that

$$(21) \qquad Y_{g-|+i,i} = V_0 \amalg_{\partial M} M.$$

†

Let then $X = X_g$ be the oriented double|double of $Y_{g-i,i}$ defined by (21), let $\widetilde{M} = \coprod_{1\leq j\leq g-i} \widetilde{M}_j$ be the inverse image of M in it, † an étale covering of degree 2 of M, †

$$(22|21) \qquad \dagger \quad X_{g,i} = X_g \setminus \mathrm{Int}(\widetilde{M}),$$

Uncertainty retained from cancelled passages.

140-2-s450-D001 [illeg. 1]
140-2-s450-D002 U_0
140-2-s450-D003 U_g^i
140-2-s450-D005 g
140-2-s450-D006 constructs [illeg. 1]
140-2-s450-D007 [illeg. 1]
140-2-s450-D008 $X_{g-i,i}|X_{g,i}$
140-2-s450-D009 X_g [illeg. 4]
140-2-s450-D010 [illeg. 3]
140-2-s450-D011 [*illeg.* 1]

[scan 451]
[p. 228]

and we will have

$$(23) \qquad X_g = X_{g,i} \amalg_{\partial\widetilde{M}} \widetilde{M}$$

by a well-determined homeomorphism of $\partial\widetilde{M}$ with an open and closed subset of $\partial X_{g,i}$. Moreover, we will have a canonical continuous map

$$(24) \qquad U_0 \longrightarrow X_{g,i}$$

which defines a homeomorphism of $X_{g,i}$ with the|the † surface obtained|obtained by contracting the $\underset{\|}{\partial\Delta_j}$ $\subset U_0$ $(g-i+1 \leq j \leq g)$ using the projections $\partial\Delta_j \simeq (\partial D_j) \times \varepsilon_j \longrightarrow D_j$.
$(\partial D_j)\times\varepsilon_j$

On the other hand, each $\widetilde{M}_j$ $(1 \leq j \leq g-i)$, the covering of the orientations of the Möbius band, is isomorphic to the cylinder $S^1 \times I$, and its canonical anti-involution has no fixed point, and is identified with:

$$(z,t) \longrightarrow (-z^\dagger, -t)$$

(where S^1 is identified with the complex numbers of modulus 1, and $I : [-1;+1]$).

Moreover, X_0 oriented|oriented, with its anti-involution of the cover^{ing}|covering of Y_0 is identified with the ordinary sphere †, in a Euclidean vector space of dimension 3, with the anti-involution $x \longmapsto -x$.

Uncertainty retained from cancelled passages.
140-2-s451-D001 manifold
140-2-s451-D003 [*illeg.* 1]
140-2-s451-D004 of the

[scan 452]

In summary:

Proposition We can obtain (for $0 \leq i \leq g$ fixed) the pairs (X_g, σ_i), up to homeomorphisms, by taking the † oriented Euclidean sphere X_0, with its standard antipodism τ, by taking a set of $2g$ disks D'_j $(j \in J)$ mutually disjoint, stable under τ, and a subset $I' \subset J/\tau = I$ (I of cardinality g) with card $I' = i$, and by proceeding as follows: for every $j \in I$, let $\Delta_j = D'_j \cup D''_j$ be the union of the two corresponding disks, and $S_j = \partial\Delta_j/\tau$, so that S_j is a circle, let

introduce it later

$$T_j = S_j \times \mathbb{I}^{\xi_j}$$

(where $\mathbf{I}|\mathbb{I} = [-1, +1]$), $\varepsilon_j = \{j' \in J|\mathbf{I} \mid j' \text{ over } i\} \simeq$ set of orientations of S_j, $\mathbf{I}^{\varepsilon_j}|\mathbb{I}^{\varepsilon_j}$ twisted by ε_j, so that we have a canonical homeomorphism

$$\partial T_j \simeq \partial\Delta_j. \tag{25}$$

i.e. $\partial T \simeq \partial V_0$ † reverses it
m536

Marginal note. m536 so that $X_g = V_0|U_0 \amalg_{\partial\Delta} T$ (where $\Delta = \coprod_1^g \Delta_j$, $T = \coprod_1^g T_j$) is oriented|oriented
m537

Marginal note. m537 N.B. T_j is canonically oriented and the isomorphism (25) respects as it should|should the orientation

This being understood, the antipodism τ induces on each $\partial T_i|\partial T_j \simeq S_j \times \varepsilon_j$, †the canonical anti-involution † coming from this expression of ∂T_i, which interchanges the two connected components.

For every|$\forall$ $j \in I|I'$, we extend $\dot{\tau}|\partial T_j$ to T_j as two anti-involutions $\dot{\tau}_j$, $\dot{\tau}'_j$ of T_j, † in such a way that

Uncertainty retained from cancelled passages.
140-2-s452-D001 oriented
140-2-s452-D002 id
140-2-s452-D003 [illeg. 1]
140-2-s452-D004 which
140-2-s452-D006 [illeg. 1]

[scan 453]
[p. 229]

a) , $\dot{\tau}_j$ has no fixed points

b) , $\dot{\tau}'_j$ has a set of fixed points homeomorphic to S_j by the projection $T_j^{\dot{\tau}_j|\dot{\tau}'_j} \longrightarrow S_j$ (cf. below } for particular and explicit choices }. — we see|will see that we can even|= suppose $T_j^{\dot{\tau}_j|\dot{\tau}'_j} = S_j \times \{0\}$).

Let, for every subset $I' \subset I$, $\tau_{I'}$ be the anti-involution of $X_g = V_0 \amalg_{\partial T} T$ (where $V_0 = X_0 \setminus \bigcup_{j\in J} D'_j$) which coincides with τ on V_0, with τ'_j on T_j for $j \in I'$, with $\tau_j^0|\dot{\tau}_j$ for $j \in I \setminus I'$.

Then we have

$$X_g^{(\tau_{I'})} = \bigcup_{j\in I'} (S_j \times \{0\}) \tag{26}$$

so if $\operatorname{Card} I' = i$ $(0 \leq i \leq g)$, then $\tau_{I'}$ is a σ_i.

Choice of the $\dot\tau_j$, $\dot\tau'_j$. We choose an isomorphism $S_j \simeq \mathbb{U}|\mathbf{U} \overset{\text{df}}{=} \{z \in \mathbb{C} \mid |z| = 1\}$, whence an orientation of S_j, and a bijection $\varepsilon_j \simeq \{\pm 1\}$, whence $\mathbf{I}^{\varepsilon_j}|\mathbb{I}^{\varepsilon_j} \simeq I|\mathbf{I} = [-1, +1]$ and we take;

$$\begin{cases} \dot\tau_{j|g}(z,t) = \dagger\ (-z\exp \pi i t, -t) \\ \dot\tau'_{j|g}(z,t) = (z\dagger, -t) \end{cases} \tag{27}$$

m538

Marginal note. m538 N.B. For the definition of|of the $\dot\tau'_j$, we do not need the choice of an isomorphism $S_j \simeq \mathbb{U}|\mathbf{U}$ (27)

We do have $\dot\tau^2 = \mathrm{id}$, $\dot\tau'^2 = \mathrm{id}$

$$\dot\tau|\partial T = \dot\tau'|\partial T = \tau|\partial T,$$

Moreover

Uncertainty retained from cancelled passages.

140-2-s453-D003 $-z|z$

140-2-s453-D004 $\exp \pi i t$

[scan 454]

We note that

$$(\dot\tau'\dot\tau)^2 = \dot\tau'\dot\tau\dot\tau'\dot\tau = ((z,t) \longmapsto (z\exp 2i\pi t, t))$$

this is not the identity map — the set of its fixed points is equal to: † the set of (z,t) such that $t \in \{-1, 0, +1\}$ — i.e.

$$T_j^{(\dot\tau'_j\dot\tau_j)^2} = S_j \times [\partial\mathbf{I}|\mathbb{I}^{\varepsilon_j} \cup \{0\}] \tag{28}$$

†

†

Let $\rho_j =$.

$$\dot\rho_j = \dot\tau'_j\dot\tau_j^{\dagger} \qquad [(z,t) \longmapsto (-z\exp \pi i t, t)] \tag{29}$$

— this is an automorphism of T_j which is the identity on ∂T_j. For every subset I' of $I = J/\tau$, let

(30) $\rho_{I'}$ be the automorphism of X_g which is the identity on $V_0 = X_g - \bigcup_{j\in I-I'} T_j$, and which is ρ_j on T_j for $j \in I'$.

m539

Marginal note. m539 Setting $\rho_j = \rho_{\{j\}}$, we will simply have $\rho_J = \prod_{j\in J} \rho_j$. Unless I am mistaken, ρ_j generates the group $S\Gamma^{!+}(T_j)|S\Gamma^{!*}(T_j) \simeq \mathbb{Z}$, so the ρ_j generate a group $\simeq \mathbb{Z}^I$. (oriented on[illeg. 1])

N.B. We have $\rho_j = \tau_{\dagger\{j\}}\tau_{\emptyset}$.

We will therefore have

$$\begin{cases} \tau_{I'}\tau_{I''} = \rho^{\dagger}_{I'_0}\rho^{-1\,\dagger}_{I''_0} \\ \dagger \\ \dagger \end{cases} \tag{32}$$

† where † $I'_0 = I' \setminus I' \cap I''$, $I''_0 = I'' \setminus I' \cap I''$.

on the other hand we will obviously have

$$[\rho_J, \rho_K] = 1 \qquad \text{for } J, K \subset I. \tag{33}$$

Remark. Instead of an isomorphism $S_j \simeq \mathbb{U}$, suppose rather that we are given on

Uncertainty retained from cancelled passages.

140-2-s454-D001 S_j
140-2-s454-D002 We have
140-2-s454-D003 be able to
140-2-s454-D004 [*illeg.* 1]
140-2-s454-D006 $\emptyset$
140-2-s454-D007 [*illeg.* 1]
140-2-s454-D008 [*illeg.* 1]
140-2-s454-D009 $\tau'_{I'}\tau'_{I''}$ $\rho_{I'_0}\rho^{-1}_{I''_0}$
140-2-s454-D010 $(\tau_{I'}\tau_{I''})$ [*illeg.* 1] I'_0 [*illeg.* 1] I''_0 [*illeg.* 1]
140-2-s454-D011 and the
140-2-s454-D012 J $I' \cap I''$

[scan 455]

S_j a torsor structure under $\mathbb{U}^{\varepsilon_j}$ ($\mathbb{U}$ twisted by ε_j, thanks|taken to the automorphism of order 2 $z \longmapsto z^{-1} = \bar{z}$ of $\mathbb{U}$) We can therefore define [p. 230]

$$\exp : \mathbb{R}^{\varepsilon_j}_{\mathbf{I}\|^{\varepsilon_j}} \longrightarrow \mathbb{U}^{\varepsilon_j} \tag{36}$$

in the obvious way, whence anti-involutions $\dot{\tau}_j, \dot{\tau}'_j : T_j \xrightarrow{\sim} T_j$ by the formulas (27).

If for example we choose disks D'_j such that their boundaries are Euclidean circles, then there is on each † $\partial D'_j$ a torsor structure under a group $\mathbb{U}^{\varepsilon_j}$, invariant|invariant under antipodism, and which therefore passes to the quotient. Hence every automorphism of $(X_0, (D'_j))$ which respects this † additional structure of torsors — and in particular every automorphism which respects the metric structure, † operates on X_g commuting with the system of τ_j, τ'_j in an obvious sense.

Equal^{ly}, if we retain on X the|the conformal|conformal structure sole^{ly}, and if we

Uncertainty retained from cancelled passages.

140-2-s455-D002 S_j
140-2-s455-D003 shows that the
140-2-s455-D004 respects

[scan 456]

chooses in each D'_j a "center" a'_j, in a way compatible with the involution, then the choice of the $a'_j \in \mathrm{Int}(D'_j)$ defines a torsor structure on $\partial D'^*_j$, and these structures are invariant under conformal transformations which respect the set of points a'_j.

m541

Marginal note. m541 N.B. We no longer assume necessarily that D_j is a circle, only that ∂D_j not too † wild. If ∂D_j is a circle, this definition coincides with the preceding one iff $a_j|a'_j$ is the center of the|of this circle…

Re-N.B. To define τ_j on T_j, considerably less data suffice than a topological torsor structure on T_j. Writing for $(z,t) \in S_j \times \mathbf{I}|\mathbb{I}^{\varepsilon_j}$

$$\tau_j(z,t) = (u_t(z), -t) \tag{37}$$

where $u_t : S_j \xrightarrow{\sim} S_j$ is a homeomorphism depending continuously on t, writing that $\tau_j|\partial T_j = \tau|\partial T_j$, we find the condition

$$\text{a)} \quad u_t = \dagger\mathrm{id} \quad \text{if } t \in \partial\mathbf{I}|\mathbb{I}^{\varepsilon_j} \simeq \varepsilon_j$$

m542

Marginal note. m542 this is written, if S_j is oriented, $u_1 = u_{-1} = 0$.

writing that $\tau_j^2 = \mathrm{id}$ we find the condition

$$\text{b)} \quad u_{-t} = u_t^{-1} \qquad (t \in \mathbf{I}|\mathbb{I}^{\varepsilon_j})$$

and writing that $T_j^{\tau_j} = \emptyset$ we find

$$\text{c)} \quad u_0 \text{ without fixed points.}$$

If an orientation is chosen, the $t \longmapsto u_t$ satisfying a) b) c) correspond to the maps $[0,1] \xrightarrow{t \mapsto u_t} \dagger\mathrm{Aut}(S_j|S_1)$ such that $u_1 = \mathrm{id}$, u_0 without fixed point †and of order 2. (The case considered above is the one where $t|* \longmapsto u_{1-t}$ comes from a continuous representation $\mathbb{R} \longrightarrow \mathrm{Aut}(S_j|S_1)$…)

Uncertainty retained from cancelled passages.

140-2-s456-D001 [illeg. 1]

140-2-s456-D002 $u_1 =$

140-2-s456-D003 Homéom

140-2-s456-D004 [illeg. 2]

[scan 457]

[p. 231]

Remarks We may set ourselves the task of determining the structure of all anti-involutions τ on an oriented cylinder $T \simeq S \times \mathbf{I}^{\varepsilon}|\mathbb{I}^{\varepsilon}$ ($\varepsilon = \mathrm{Or}(S)$) which † have no fixed point on the boundary — which already implies that τ permutes|permute the two connected components of the boundary. More generally, the anti-involutions τ of an oriented †surface with boundary X, having no fixed point on ∂X, correspond to manifolds with boundary Y equipped with a subset both open and closed $(\partial Y)'$ of ∂Y — by associating to such a $(Y, (\partial Y)')$ its oriented "doubling" relative to $(\partial Y)'$ — to X, σ corresponding $(X/\sigma, \mathrm{Im}\, X^{\sigma} \longrightarrow X/\sigma)$. If X is connected, compact, Y is compact, nonorientable, suppose that its type (γ, j), and † let $0 \leq i \leq j$, and, $i|i' = \mathrm{card}(\pi_0((\partial Y)'))$= $\mathrm{card}\,\pi_0(X^{\sigma})$. Thus $\chi(Y) = 1 - (\gamma + j)$, and we know|see at once that (for the χ of a circle is zero)

$$\chi(X) = \chi_!\left(\underbrace{X \setminus (X \mid \partial Y)}_{X|\mathrm{Int}(Y)}\right)$$
$$= 2\chi_!(\mathrm{Int}\, Y) = 2\chi(Y) = 2(1 - (\gamma + j))$$

thus if X is of type (g, ν), we will have $\chi(X) = 2 - 2g-|+\nu$

$$\dagger \quad 2 - 2g-|+\nu = 2(1 - (\gamma + j))$$

i.e.

$$(38) \qquad g + \frac{\nu}{2} = \gamma + j$$

m543

Marginal note. m543 not true! If Y orientable of type (γ, j) (same|with $0 \leq i \leq j$) we will have

$$(39) \qquad \begin{cases} g + \dfrac{\nu}{2} = 2\gamma + j - 1 \\ \nu = 2(j - i) \end{cases}$$

i.e. $i = j - \dfrac{\nu}{2}$ (as in the case opposite)

Uncertainty retained from cancelled passages.
140-2-s457-D001 induce
140-2-s457-D003 manifold
140-2-s457-D005 cardinality
140-2-s457-D008 $g + \nu$

[scan 458]

(this requires that ν be even, which was obvious a priori, since σ must permute the elements of $\pi_0(\partial X)$ among themselves without there being a fixed point…) † But I forgot to note that ν is determined as a function of $(\gamma, j, i = \mathrm{card}((\partial Y)'))$ by † † †

$$\nu = 2(j - i) \tag{39}$$

i.e. $i = j - \frac{\nu}{2}$.

Thus here (for $g = 0, \nu = 2$) we must look for (γ, j, i), with $\gamma \in \mathbb{N}$, $0 \leq i \leq j \in \mathbb{N}$, such that we have $\nu = 2(j - i)$ i.e. $j - i = 1$ i.e. $i = j - 1$, and $\gamma + j = 1$, which gives the only possibility (since i must be ≥ 0 hence $j \geq 1$)

a) $\gamma = 0,\ j = 1,\ i = 0$:|.

†

Case a) corresponds to the case where † $T^\sigma = \emptyset$, † is obtained from the real projective plane by making one hole|holes in it, with boundary (hence Möbius strip) and by taking the † oriented doubling.

The case where $T^\sigma \neq \emptyset$ will necessarily give a quotient Y that is oriented, we must have for its type (γ, j) (since $j \geq 1$), the sole case:

Uncertainty retained from cancelled passages.
140-2-s458-D001 find|will find
140-2-s458-D005 $i^\varepsilon|i$
140-2-s458-D006 thus
140-2-s458-D007 must have
140-2-s458-D008 $\gamma = 1|\gamma = 0 \quad j = 0|j = 1$
140-2-s458-D009 [illeg. 1]
140-2-s458-D010 [illeg. 2] necessarily
140-2-s458-D013 oriented covering

[scan 459]
[p. 232]

b) $\gamma = 0,\ j = 2,\ i = 1$ (oriented quotient)

obtained from the sphere with two holes i.e. from the cylinder, by taking the oriented doubling with respect to one of the holes.

These are indeed the two cases obtained resp.[t] by $\dot\tau$ and $\dot\tau'$ in formulas (27).

I would now like to construct a situation (under the conditions of the proposition) bringing out a maximum of symmetries — it is true that we could work with † the automorphisms of X_0 commuting with the antipodism τ, preserving the set of the D'_j, and respecting (say) topological torsors|torsor structures on the set of the $\partial D'_j$ — or what naturally amounts to the same (up to an indeterminacy of multiplication by $\tau|2$) the automorphisms of Y_0 which † preserve|preserving $D = \bigcup_{i \in I} D_i$, and respect|respecting torsor|torsors structures on the connected components ∂D_i of ∂D. We expect that (working modulo isotopy) we will have|obtain a group
m544

Marginal note. m544 no, without indeterminacy, by lifting defined|to the sphere so as to respect the orientations|the orientation…

Uncertainty retained from cancelled passages.
140-2-s459-D002 the [illeg. 1] of the
140-2-s459-D003 [illeg. 1]

[scan 460]

which will be close to a braid group, and probably computable without much trouble — and by putting it together with the group generated by the preceding operators, we will perhaps find a start for generating for example $\mathcal{T}_g|\Pi_g$ by generators (anti-involutive?) and relations.

Let us consider the †group G of $A_g = \operatorname{Aut} X_g$ generated by the $\dagger\tau_{I'}$ $(I' \in I)$, Let $\mathcal{H}$ be the subgroup generated by the ρ_j $(j \in I)$. We have

$$\begin{aligned}\tau_{I'}\rho_j\tau_{I'} &= \dagger\tau_{I'}(\tau_{\{j\}}\tau_\emptyset)\tau_{I'} \\ &= (\tau_{I'}\tau_{\{j\}})\underbrace{(\tau_\emptyset\tau_{I'})}_{\dagger\rho_{I'}} \in \mathcal{H}\end{aligned}$$

by formula (31)|(32), so $\mathcal{H}$ is an invariant subgroup.

Formulas (32) show that $G/\mathcal{H}$ is a commutative group, †and even that it is isomorphic to ± 1 by the orientation character (all the $\tau_{I'}$ are equal modulo $\mathcal{H}$).

Uncertainty retained from cancelled passages.
140-2-s460-D001 sub
140-2-s460-D002 [illeg. 1]
140-2-s460-D005 $(\tau_{I'}(\tau_{\{j\}}\tau_\emptyset)\tau_{I'})$
140-2-s460-D006 $\rho_{I'_0}$
140-2-s460-D007 isomorphic to

[scan 461]
[p. 233]

Let us consider as reference element of $\mathcal{H}^-$ the element

$$(40) \qquad \tau = \tau_\emptyset \qquad \text{(anti-involution without fixed points of } X_g\text{)}$$

we then find by (32) that for $I' \subset I$

$$(41) \qquad \tau_{I'} = \rho_{I'}\tau = \left(\prod_{j\in I'}\rho_j\right)\cdot\tau \quad \dagger$$

m545

Marginal note. m545 we see at once, since $\tau^2 = 1$, $\tau_{I'}^2 = 1$, that $\tau_{I'} = \tau_{I'}^{-1} = \tau\rho_{I'}^{-1}$ hence $\tau\rho_{I'}\tau = \rho_{I'}^{-1}$ which is (42)|which ⇒ (42)

moreover we will have, for $j|i \in I$

$$\tau\rho_i\tau = \tau_\emptyset\left(\tau_{\{j|i\}}\tau_{\{\emptyset\}}\right)\tau_{\{0\}} = \tau_\emptyset\tau_{\{j\}} = \rho_j^{-1}$$

$$(42) \qquad \tau\rho_i\tau^{-1} = \rho_i^{-1}.$$

Thus $\mathcal{G}$ appears as the semidirect product of $\mathcal{H} \simeq \mathbb{Z}^I$, and of † $\{\pm 1\} \simeq \{1, \tau\}$ operating on it by the symmetry $\rho \longmapsto \rho^{-1}$. So for every $\rho \in \mathcal{H}$, † we have $(\tau\rho)^2 = 1$, i.e. for every $\sigma \in \mathcal{H}^{-1}$, σ † is an anti-involution ! Since σ agrees with τ on $V_0 = X_g - \bigcup_{j|i\in I}\operatorname{Int}(T_j)$, we see that the set of fixed points of σ is contained in $\bigcup_{i\in I}\operatorname{Int}(T_i)$, and to calculate the index of σ, i.e. $\operatorname{card}\pi_0(X_g^\sigma)$, it suffices to take the sum of the

460

Uncertainty retained from cancelled passages.
140-2-s461-D001 $= \rho_h$
140-2-s461-D002 [illeg. 2]
140-2-s461-D003 $\tau\rho$
140-2-s461-D004 [illeg. 1]

[scan 462]

indices in the T_j. Now in T_j, we have (29) $\rho_j(z,t) = (-z\exp\pi it, t)$, and by induction

$$(43) \qquad \rho_j^{n_j}(z,t) = \big((-1)^{n_j} z \exp n_j \pi i t, t\big)$$

hence

$$(44) \qquad \tau\rho_j^{n_j}(z,t) = \big((-1)^{n_j+1} z \exp(n_j+1)\pi i t, -t\big)$$

and † $(z,t) \in T_j$ is a fixed point of $z|\tau\rho_j^{n_j}$ iff|without $t = 0$, † and n_j is odd, hence

$$(45) \qquad T_j^{\tau\rho_j^{n_j}} = \begin{cases} \emptyset & \text{if } n_j \text{ even} \\ S_j \times \{0\} & \text{if } n_j \text{ odd} \end{cases}$$

hence

$$(46) \qquad \text{Index of } u = \tau \prod_j \rho_j^{n_j} \text{ is equal to the cardinality of the set } I'_u \text{ of the } j \in I \text{ such that } n_j \text{ is|are odd}$$

It follows for general reasons that u is conjugate in $\mathcal{A}_g = \mathrm{Aut}(X_g)$ (by an element of $A_g^+|\widetilde{A}_g^+$) to $\tau_{I'} = \rho_{I'}\tau = \tau\rho_{I'}^{-1}$ or also $\rho_{I'}^{-1}\tau = \tau\rho_{I'}$ — †

†
461

Uncertainty retained from cancelled passages.
140-2-s462-D001 [illeg. 1]
140-2-s462-D002 if|and
140-2-s462-D003 but [illeg. 4] [illeg. 2] in the group too small [illeg. 4] [illeg. 2] the questions $(\tau\rho)^{-1}$ $\tau\rho\rho_{I'}$ enlarging it completing configuration
140-2-s462-D005 $(\tau\rho)(\tau\rho')(\tau\rho)^{-1}$ [illeg. 1] i.e.|i.e. [illeg. 3] conjugate [illeg. 1] $\tau\rho''$

[scan 463]
[p. 234]

But if $\tau\rho', \tau\rho'' \in \mathcal{G}^-$, $(\rho', \rho'' \in \mathcal{H})$, and if $\rho \in \mathcal{H}$, the relation

$$\rho(\tau\rho')\rho^{-1} = \tau\rho''$$

is equivalent to

$$\underbrace{\tau\rho\tau}_{\rho^{-1}}\rho'\rho^{-1} = \rho'' \qquad \text{i.e.} \qquad \rho^2 = \rho'\rho''^{-1}$$

hence $\tau\rho'$ and $\tau\rho''$ are conjugate by an element of $\mathcal{H}$ iff|without $\rho'\rho''^{-1} \in \mathcal{H}^2$ – which makes the preceding observation more precise…

The way richest in simple symmetries for arranging the $2g$ antipodal holes D_j seems to me to be the following. We † consider the Euclidean sphere, with the action of the dihedral group $\mathbb{D}_{2g}|\mathbf{D}_{2g}$ – for example, when it is the Riemann sphere that is considered as Riemannian, by the choice of the antipodism as being

$$(47) \qquad \tau z = -\frac{1}{\bar{z}}$$

we take the standard action of the dihedral group (with the points $0, \infty$ as poles, and the unit circle $\mathbb{U} = \{z \in \mathbb{C} : |z| = 1\}$ as equator), by writing $\mathbb{D}_n | \mathbf{D}_n$ as $\subset \mathrm{O}(\dagger 2, \mathbb{R})$, as a $\frac{1}{2}$ direct product of $\dagger\{\pm 1\}$ by $\mu_n(\mathbb{C}) = \mu_n$, the pair (ζ, α) $(\zeta \in \mu_n, \alpha \in \dagger\{\pm 1\})$ operating by

$$u_{\zeta,\alpha}(z) = \zeta z^{\alpha} \tag{48}$$

462

Uncertainty retained from cancelled passages.
140-2-s463-D001 [illeg. 1]
140-2-s463-D003 3
140-2-s463-D004 $\mathbb{Z}/2\mathbb{Z}$
140-2-s463-D005 $\mathbb{Z}/2\mathbb{Z}$

[scan 464]

We can moreover enlarge it to a group $\mathbb{D}_n^{\sim} = \mathbb{D}_n \times \mathbb{Z}_{/2}$, where the second factor $\mathbb{Z}_{/2}$ is generated by the antipodism (47) (which commutes with $\mathbb{D}_n$ — likewise moreover does the anti-involution $z \longmapsto \frac{1}{\bar{z}}$, which has as its set of fixed points $\mathbb{U}$ and is nothing other than the symmetry with respect to the equator, their composite $z \longmapsto -z$ being the symmetry with respect to the axis of the poles …) hence to consider the transformations

$$u_{\zeta,\alpha,\beta} = \zeta u_{\zeta,\alpha} \tau^{\beta} \qquad (\zeta \in \mu_n,\ \alpha \in \{\pm 1\},\ \beta \in \mathbb{Z}_{/2})$$

hence

$$u_{\zeta,\alpha,\beta} z = \begin{cases} u_{\zeta,\alpha}(z) = \zeta z^{\alpha} & \text{si } \beta = 0 \\ u_{\zeta,\alpha}(-\frac{1}{\bar{z}}) = -\zeta \bar{z}^{-\alpha} & \text{si } \beta = 1 \end{cases},$$

but we will remember that τ reverses the orientation, hence †is to be †handled with caution as regards "transport of structure" in the present situation.

Here $n = 2g$, $\mathbb{D}_{2g|n}$ is of order $4g$, and $\mathbb{D}_n^{\sim}$ of order $8g$. We take on the equator an orbit $\mathcal{J}|\mathbb{J}$, of $\mu_n = \mu_{2g}$, for example the|here precisely † (μ_{2g} the set of $2g$th roots of unity itself, viewed† as a subset of the sphere) as the set of "centers" of the disks D'_i. We choose a disk D'_0 around the point|point P_0 (small enough for what is to follow), and we take the transforms|transforms of D'_0 by the $\zeta \in \mu_n$.
m550

Marginal note. m550 it is simply necessary that D'_0 not meet $\zeta_0|\zeta D'_0$, where $\zeta = \exp \frac{2i\pi}{2g}$

463

Uncertainty retained from cancelled passages.
140-2-s464-D002 [illeg. 1]
140-2-s464-D003 more
140-2-s464-D004 the
140-2-s464-D005 [illeg. 1]

[scan 465]

These choices having been made, the surface X_g is determined without ambiguity|ambiguity, [p. 235] and the group $\mathbb{D}_{2g}$ operates on it by transport of structure, permuting the g cylinders T_i among themselves, corresponding|corresponding to the elements of $\mathcal{J}|\mathbb{J}/\tau = \mathcal{J}|\mathbb{J}/\{\pm 1\}$ i.e. to the pairs of antipodal elements of $S|\mathcal{J}$, i.e. to the "diagonals" of the polygon with 2μ sides which they †determine on the equator. The group $\mathbb{D}_{2g}$ normalizes † the group $\mathcal{G}$, to be precise we will have, for $u \in \mathbb{D}_{2g}$

$$u \tau_{I'} u^{-1} = \tau_{u(I')} \tag{49}$$

for $I' \subset I = \mathcal{J}|\mathbb{J}/\tau$, taking into account the operation of $\mathbb{D}_{2g}$ on I. We will thus have in particular, denoting by $\tau_g = \tau_\emptyset$ the extension of the antipodism of the sphere X_0 (or rather of $\tau|V_0$) to an antipodism without fixed points of X_g, denoted τ previously:

$$u\tau_g u^{-1} = \tau_g \qquad \text{i.e. } \tau_g \text{ commutes with the action of } \mathbb{D}_n \tag{50}$$

$$\dagger \qquad u\rho_j u^{-1} = \rho_{u(j)}$$

We can therefore say that $\mathbb{D}_{2g}$ operates on $\mathcal{G}$, whence a $\frac{1}{2}$-direct product $\mathbb{D}_{2g}.\mathcal{G}$, which therefore operates on X_g. As for the question of extending likewise the action on X_0, of $\mathbb{D}_{2g}^{\sim}$ as a whole to an action on X_g, 464

Uncertainty retained from cancelled passages.
140-2-s465-D001 form
140-2-s465-D002 [illeg. 1]
140-2-s465-D004 (51)

[scan 466]

this has been done without warning, to τ_{X_0} corresponding naturally $\tau_{X_g} = \tau_g$, which indeed has the good taste to commute with the action of $\mathbb{D}_{2g}$. [It might seem more natural, it is true, in a spirit of "transport of structure (in spite of and against everything?)", to associate to τ_{X_0} the operation $\tau_I$given|given by (27), which on each T_j would be equal to τ'_j (and on V_0, of course, coincides with τ_{X_0}), $\tau'_j(z,t) = (z,-t)$, the set of fixed points of τ_I^2 being formed of the g middle circles $S_j \times \{0\}$ of the g tubes T_j. We will have by (41) $\tau_I = \rho_I \tau_g$, hence τ_I commutes also with $\mathbb{D}_n$, since τ and $\rho_I = \prod_{i\in I} \rho_i$ commute with it. But it does|to me does not seem important for the moment which convention we adopt]

We can therefore also say that the group $\mathbb{D}_{2g}^{\sim}$ operates on $\mathcal{H}$ — this operation extending that of $\tau_g\dagger$, now identified with an element of $\mathbb{D}_{2g}^{\sim}$, i.e. with the antipodism in $\mathbb{D}_{2g}^{\sim}$ — and the $\frac{1}{2}$ direct product

$$\mathbb{D}_{2g}^{\sim}.\mathcal{H} \supset \mathcal{G} = \dagger\{1, \tau_g\}.\mathcal{H} \tag{51}$$

465

Uncertainty retained from cancelled passages.
140-2-s466-D002 by
140-2-s466-D003 $\mathcal{H}$

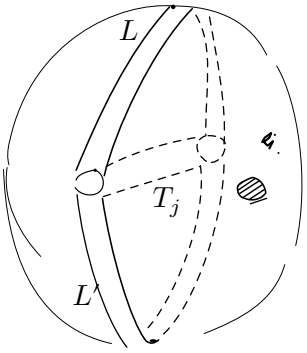
[scan 467]

operates on X_g. [p. 236]

One should now, along these lines:

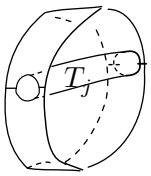
1) Make explicit the outer action of this $\frac{1}{2}$ direct product on the fundamental group π_g, with very special attention to the action of the group $(\mathbb{Z}/2\mathbb{Z})^3 \subset \mathbb{D}_{2g}^{\sim}$ which stabilizes one of the cylinders T_j — which in Euclidean space of dimension 3 is interpreted as the group of changes of sign relative to the|a corresponding orthonormal system of axes.

2) Extend the action on π_g of the (more or less) "special" Teichmüller group of each of the T_j, to the action of an analogous group of a larger set obtained by adding to it a "strip" L (whence a torus with 1 hole, and its special Teichmüller group, which come in naturally*) or even|see the still larger set obtained by also putting in the antipodal strip L' (this set presents itself as a cylinder $(L \cup L')$, or "culvert", in which one would have put a tube (T_j) crosswise, and has the topological structure of a torus with two holes.



m555

Marginal note. m555 * This is essentially an $\mathrm{SL}(2,\mathbb{Z})$ — rather a remarkable central extension of $\mathrm{SL}(2,\mathbb{Z})$ by $\mathbb{Z}$, which recalls that of $\mathrm{SL}(2,\mathbb{R})$ by $\mathbb{Z}$… (universal covering of $\mathrm{SL}(2,\mathbb{R})$)



466

[scan 468]

It is possible that it will be necessary to look closely at this last one, to study the relations among the elements of $\mathcal{T}_g$ coming from the preceding sets… An amusing job will be to manage to write down the generators of the group π_g, and above all the famous relation, in a relative geometric arrangement of the "handles", which is a "mixed" arrangement — and no longer nicely in single file!

†

While waiting to get to the heart of the structure of the Teichmüller group in this way, I will already try to describe generators and relations for the interesting subgroup we have just written down, $\mathbb{D}_{2g}^{\sim} \cdot H$. I will take the everlasting generators $\varepsilon_0, \varepsilon_1$ of $\mathbb{D}_{2g}$

$$\varepsilon_0^2 = \varepsilon_1^2 = 1, \qquad (\varepsilon_0\varepsilon_1)^{2g} = 1$$

and adjoin to them $\tau = \tau_g$, and † $\rho_0\tau = \tau'$ ($\rho_0 = \rho_{j_0}$, j_0 marked point of $I = J/\tau$) satisfying

$$\tau^2 = \tau'^2 = 1, \qquad \dagger$$

$$(\tau\varepsilon_0)^2 = (\tau\varepsilon_1)^2 = 1 \quad \dagger$$

$$(\varepsilon_1\tau')^2 = 1 \qquad \varepsilon_1 \text{ commutes with } \tau'$$

m557

Marginal note. m557 N.B. ε_0 moves the vertex of the frame, ε_1 the edge and not the vertex, hence $\varepsilon_1(j_0) = j_0$

m558

Marginal note. m558 express the commutation of τ † with $\varepsilon_0, \varepsilon_1$

467

Uncertainty retained from cancelled passages.
140-2-s468-D001 appearing
140-2-s468-D002 [illeg. 1]
140-2-s468-D003 $[\varepsilon_1, \tau]|[\tau, \varepsilon_1]$
140-2-s468-D004 $(\tau'\varepsilon_0)^2 = (\tau'\varepsilon_1)^2$
140-2-s468-D005 [illeg. 1]

[scan 469]
[p. 237]

$$((\varepsilon_0\varepsilon_1)^g\tau')^2 = 1$$

m560

Marginal note. m560 (the involution $(\varepsilon_0\varepsilon_1)^g$ commutes with τ')

unless I am mistaken, that gives a set of generators and relations for $\mathbb{D}^{\sim}_{2g} \cdot \mathcal{H}$ — in summary $\varepsilon_0, \varepsilon_1, \tau, \tau'$

$$(52) \quad \begin{cases} \varepsilon_0^2 = \varepsilon_1^2 = \tau^2 = \tau'^2 = 1 \\ (\tau\varepsilon_0)^2 = (\tau\varepsilon_1)^2 = (\tau'\varepsilon_1)^2 = 1 \\ (\varepsilon_0\varepsilon_1)^{2g} = 1 \\ (\tau'(\varepsilon_0\varepsilon_1)^g)^2 = 1 \end{cases}$$

This is perhaps not a very clever choice of generators, in the sense that $\varepsilon_0, \varepsilon_1$, are not at all on the same footing as τ, τ' — they are not anti-involutions. It would be better to take $\varepsilon'_0 = \tau\varepsilon_0$, $\varepsilon'_1 = \tau\varepsilon_1$ so as to obtain

$$(52\text{bis}) \quad \begin{cases} \varepsilon_0'^2 = \varepsilon_1'^2 = \tau^2 = \tau'^2 = 1 \\ (\tau\varepsilon'_0)^2 = (\tau\varepsilon'_1)^2 = (\tau'(\tau\varepsilon'_1))^2 = 1 \\ (\varepsilon'_0\varepsilon'_1)^{2g} = 1 \\ (\tau'(\varepsilon'_0\varepsilon'_1)^g)^2 = 1 \end{cases}$$

these are essentially "the same" relations except for the last one in the second line.

Perhaps this little game with the tiny group $\mathbb{D}^{\sim}_n$ operating on $\mathcal{H}$ is a bit of a diversion — the group $\mathcal{H} \simeq \mathbb{Z}^g$ in $\mathcal{T}_g$ strongly suggests the situation 468

[scan 470]

of a maximal torus in a $\frac{1}{2}$ simple group, one really wants to have|see an intrinsic characterization of it in $\mathcal{T}_g$ — or in $\mathcal{S}_g|\mathcal{G}_g|\mathcal{C}_g$, which is possible since † as a group of actual|actual topological transformations, it leaves fixed the points of $V_0 \subset X_g$ — for example the two poles, which one can take as base|base points to construct universal coverings and fundamental|fundamental groups. These are perhaps the maximal free-abelian|abelian subgroups.

One would like to study the normalizer — is it exactly what comes from the automorphisms of $X_g - \bigcup \mathring{T}_i \simeq V_0$?

do the $\mathrm{SL}(2, \mathbb{Z}_j)$ associated with the "strips" through the tube T_j play a role analogous to that of the groups $\mathrm{SL}(2, K)$ or $\mathrm{GP}(1, k)|\mathrm{GL}(1, k)$ "of rank 1" in the theory of reductive algebraic groups? If Jean does not|me "bite" 469

Uncertainty retained from cancelled passages.
140-2-s470-D001 [illeg. 1]

[scan 471]
[p. 238]

not to the situation, perhaps submit it to Carlos, who is at ease both with discrete groups and their generators and relations, and with the theory of $\frac{1}{2}$ simples – and topology ....

[scan 472]
[p. 239]

33 Study of finite coverings – relation between the $\mathcal{N}_{g,\nu}, \Gamma_{g,\nu}$ for variable g.

Let $(\pi\dagger)$ be an arithmetized profinite outer group with loops of type (g,ν), π' an open subgroup|subgroup of π, whence an injective map

$$\text{Discrét}(\pi) \hookrightarrow \text{Discrét}(\pi'),$$

we would like to prove that it is compatible with the equivalence relation †of arithmetization|of arithmetizability, in † such a way that every arithmetization of π gives one of π', and likewise for the † prediscretifications.

We would like to establish at the same time that the maps

$$\begin{array}{rcl} & P^+(\pi) \longrightarrow P^+(\pi') & \text{on the oriented} \\ & & \text{prediscretifications} \\ \text{and} & A(\pi) \longrightarrow |-A(\pi') & \text{on the arithmetizations} \end{array}$$

are injective. To prove these points, we are reduced to the case where π' is characteristically invariant. (Cf. § 28, diagram 7, ff) Choosing

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Uncertainty retained from cancelled passages.

140-2-s472-D001 *a*

140-2-s472-D002 of the [illeg. 1]

140-2-s472-D003 prove

140-2-s472-D004 discre

[scan 473]

a discretification π_0 of π, hence π'_0 of π', we then find

$$\widehat{\widehat{\mathcal{G}}}(\pi_0) \longrightarrow \widehat{\widehat{\mathcal{G}}}(\pi'_0)$$

and we have

① The homomorphism $\widehat{\widehat{\mathcal{G}}}(\pi_0) \longrightarrow \widehat{\widehat{\mathcal{G}}}(\pi'_0)$ sends $\mathcal{M}(\pi_0)$ into $\mathcal{M}(\pi'_0)$

we will therefore have a diagram (a variant|relative of the one in § 28)

(1)
$$\begin{array}{ccccc} \widehat{\mathcal{G}}^+(\pi_0) & \hookleftarrow & \mathcal{M}(\pi_0) & \hookleftarrow & \widehat{\widehat{\mathcal{G}}}(\pi_0) \\ \downarrow & & \downarrow & & \downarrow \\ \widehat{\mathcal{G}}^+(\pi'_0) & \hookleftarrow & \mathcal{M}^{*|\sharp}(\pi'_0) & \hookleftarrow & \widehat{\widehat{\mathcal{G}}}(\pi'_0) \end{array}$$

which implies that a discretification π_1 on π which gives the same oriented prediscretification as π_0 (resp. the same arithmetization) defines a discretification π'_1 of π' which gives the same oriented prediscretification (resp. the same arithmetization) as π'_0. Thus we do have $P^+(\pi) \longrightarrow P^+(\pi')$, $A(\pi) \longrightarrow A(\pi')$, we still have to express injectivity, which amounts to the relations

$$(2) \quad \widehat{\mathcal{G}}^+(\pi_0) = \widehat{\mathcal{G}}^+(\pi'_0) \cap \widehat{\widehat{\mathcal{G}}}(\pi_0), \qquad \mathcal{M}(\pi_0) = \mathcal{M}(\pi'_0) \cap \widehat{\widehat{\mathcal{G}}}(\pi_0)$$

[scan 474]

i.e. a loop automorphism of π^+ which, on π', belongs to $\widehat{\mathcal{G}}(\pi'_0)$ † resp. to $\mathcal{M}(\pi'_0)$, is already [p. 240] in $\widehat{\mathcal{G}}(\pi_0)$ resp. in $\mathcal{M}(\pi_0)$.

The † marked assertion being admitted, the unmarked assertion also means that the canonical homomorphism

$$(3) \qquad \mathbb{\Gamma}_a = \mathcal{M}_a(\pi)/\widehat{\mathcal{G}}^+_a(\pi) \longrightarrow \mathbb{\Gamma}_{a'} = \mathcal{M}_{a'}(\pi')/\widehat{\mathcal{G}}^+_{a'}(\pi')$$

is injective. By construction (via $\mathbb{\Gamma}_{\mathbb{Q}}$) it is also surjective, so we would find:

② the canonical homomorphism (3) is bijective

and we would also find a bijection

$$(4) \qquad P_a(\pi) \longrightarrow P_{a'}(\pi')$$

between arithmetizations of π of type a, and arithmetizations of π' of type a', compatible with the isomorphism (3).

Finally, a result in the case π' characteristically invariant would extend at once to the case of an arbitrary open $\pi' \subset \pi$.

Proceeding for example as in §28 by means of "correspondences" outer arithmetic, between

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Uncertainty retained from cancelled passages.

140-2-s474-D001 [illeg. 2]

140-2-s474-D002 [illeg. 1]

[scan 475]

"potential curves", we find a connected groupoid (pointed by the $\hat{\pi}_{g,\nu}$, for example, † which are canonically isomorphic there), whose outer π_1 is the common value of the $\mathbb{\Gamma}_{g*} = \mathbb{\Gamma}_{g,\nu}$.

To find an the isomorphism canonical between $\mathbb{\Gamma}_{g,\nu}$ and $\mathbb{\Gamma}_{g',\nu'}$, we choose a † correspondence between a $\pi_{g,\nu}$ and $\pi_{g',\nu'}$, for example by regarding † both of them (passing if need be: from g,ν to $g,\tilde{\nu}$ with $\tilde{\nu} \geq \nu$, and likewise for g',ν' to $(g',\tilde{\nu}')$ with $\tilde{\nu}' \geq \nu'$) as finite subgroups of the same $\pi_{0,3}$ and whence

$$\mathbb{\Gamma}_{0,3} \overset{\sim}{\to} \Gamma_{g,\tilde{\nu}}, \qquad \Gamma_{0,3} \overset{\sim}{\to} \Gamma_{g',\tilde{\nu}'}.$$

We would now like to see what, in the preceding yoga, is independent of any conjecture. For example, the fact that the surjective homomorphisms canonical $\mathbb{\Gamma}_{g,\nu} \longrightarrow \mathbb{\Gamma}_{g,\nu'}$ (g the same, ν' constant) is not established.

However, if we knew that in the situation of (π,π') with π' characteristically invariant we had

$$\widehat{\mathcal{F}}(\pi'_0) \cap \widehat{\mathcal{G}}^{+}(\pi_0) = \widehat{\mathcal{G}}^{+}(\pi_0)$$

Uncertainty retained from cancelled passages.

140-2-s475-D001)

140-2-s475-D002 common multiple

140-2-s475-D003 N of $2g-2+\nu$ and $2g'-2+\nu'$

[scan 476]

[p. 241]

(which is a statement of a relatively innocuous nature about profinite groups with loops) we would conclude the bijectivity of $\mathbb{\Gamma}_{\pi_0} \longrightarrow \mathbb{\Gamma}_{\pi'_0}$ in this situation, and we would conclude from this that the kernel of $\mathbb{\Gamma}_{\mathbb{Q}} \longrightarrow \mathbb{\Gamma}_{0,3}$ † maps trivially into the $\mathbb{\Gamma}_{g,\nu}$ for $\forall \nu$ – i.e. that it maps into the $\mathbb{\Gamma}_{g,\nu}$ through the intermediary of $\mathbb{\Gamma}_{0,3}$ – and that moreover $\mathbb{\Gamma}_{0,3} \longrightarrow \Gamma_{g,\nu}$ is † an isomorphism for $\forall(g,\nu)$ with ν sufficiently large – it suffices that the surface of type (g,ν) † (or of type (g,ν') with a $\nu' \leq \nu$) can be obtained as an étale covering of $X_{0,3}$, i.e. that $\pi_{g,\nu'}$ is realized (with its structure of loops) as a subgroup|subgroup of finite index † of $\pi_{0,3}$. Then, † considering a π'' $\simeq |\sim\pi_{g'',\nu''} \subset | \in \pi_{g,\nu'}$ of finite index and characteristic in $\pi_{0,3}$, we will have

!−A

Uncertainty retained from cancelled passages.

140-2-s476-D001 maps|maps it

140-2-s476-D002 lack|for lack I am going to show kernels|kernel $\mathbb{Q}$ in finite|finite

140-2-s476-D003 can

140-2-s476-D005 of in

140-2-s476-D006 taking

[scan 477]

$$\begin{array}{ccc} \Gamma_{0,3} & \xrightarrow{\sim} & \Gamma_{g'',\nu''} \\ \uparrow & & \uparrow{\scriptstyle |2} \\ \Gamma_{\mathbb{Q}} & \longrightarrow & \Gamma_{g,\nu'} \end{array}$$

commutative diagram.

whence our assertion $\Gamma_{0,3} \xrightarrow{\sim} \Gamma_{g,\nu'}$, and a fortiori the surjective homomorphism $\Gamma_{0,3} \longrightarrow \Gamma_{g,\nu}$ which factors it $(\nu \geq \nu')$ is bijective. Let us note for example that the $\Gamma_{0,\nu} \longrightarrow \Gamma_{0,3}$ $(\nu \geq 3)$ are then bijective. But †† note that in any case, we obtain a result for a g, ν only if $\nu \geq 3$ – so this says nothing about the possible faithfulness for example of $\Gamma_{0,3} \longrightarrow |-\Gamma_{g,0} = \Gamma_g$ $(g \geq 2)$ or $\Gamma_{0,3} \longrightarrow \Gamma_{1,1} = \Gamma_1$.

m564

Marginal note. m564 We no longer make any conjectural hypothesis (such as injectivity in (3)

Let an algebraic curve U of type g, ν be defined over a subfield K finite over $\mathbb{Q}$ of $\overline{\mathbb{Q}}_0$, so it defines an outer action of the open subgroup $\Gamma = \Gamma_K \subset \Gamma_{\mathbb{Q}}$ on $\pi_1(U_{\overline{\mathbb{Q}}_0})$. Enlarging K if need be, † and enlarging

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Uncertainty retained from cancelled passages.

140-2-s477-D001 [illeg. 3]

140-2-s477-D002 [illeg. 1]

140-2-s477-D004 to we may suppose that

[scan 478]

ν to ν' (i.e. making holes to find U') we may suppose that U' is an étale covering of † $(U_{0,3})_K$. [p. 242]

m565

Marginal note. m565 we put $\pi' = \pi_1(U'_{\overline{\mathbb{Q}}_0})$ $\pi = \pi_1(U_{\overline{\mathbb{Q}}})$

† † Then let $\Theta|\Theta'$ be the kernel of

$$\begin{array}{ccc} \Gamma'|\Gamma & \longrightarrow & \Gamma_{0,3} \\ \cap & & \\ \Gamma_{\mathbb{Q}} & & \end{array}$$

, it therefore operates inner automorphisms|automorphisms on $\hat{\pi}_{03|0,3}$, so †the image of $\Theta' = \Gamma' \cap |\wedge \Theta$ in the group of outer automorphisms|automorphisms of $\pi' \subset \hat{\pi}_{0,3}$ is a finite subgroup – a fortiori its image in $\Gamma_{\pi'}$, and in $\mathcal{N}(\pi)$ and in Γ_{π}. The same holds (choosing a base|good point rational over K in U') for the actual operation in|of $\pi_1(U,P)|\pi_1(U',P)$ – for it suffices to apply the preceding result to $U\backslash|-\{P\}$. We also conclude from this that † the kernels of the homomorphisms $\Gamma_{0,\nu} \longrightarrow \Gamma_{0,3}$ $(\nu \geq 3)$ are finite. We easily see that for every abelian variety A defined over K, the operation of Θ' on the $\prod_\ell T_\ell(A)$ takes place through a finite quotient group.

It becomes very difficult to imagine, how it † could happen that $\Theta'|\Theta$ does not operate, in fact, trivially ! I have

Uncertainty retained from cancelled passages.

140-2-s478-D001 $\mathbb{P}^1_K$

140-2-s478-D002 Let us choose

140-2-s478-D003 Let us consid-er U relative Γ' actually

140-2-s478-D005 its

140-2-s478-D006 such

140-2-s478-D007 Θ

[scan 479]

even the impression that I can show thanks to the result of the Russian which Deligne pointed out to me, † that $\mathbb{\Gamma}_{\mathbb{Q}} \longrightarrow \mathbb{\Gamma}_{0,3}$ is an isomorphism ! That|Tha would mean that the outer actions of $\mathbb{\Gamma}_{\mathbb{Q}}$ on π_1's $\simeq \overline{\pi}_{g,\nu}$ can be interpreted (at least for ν large enough, g being fixed) as splittings of the extension $\mathcal{N}_{g,\nu}$ of $\mathbb{\Gamma}_{g,\nu}$ by $\widehat{\mathcal{T}}_{g,\nu}$ — or of the extension $\mathcal{M}_{g,\nu}$ of $\mathbb{\Gamma}_{g,\nu}$ by $\widehat{\mathcal{G}}_{g,\nu}$, when it is a matter of actual actions.

Of course, the essential question which then arises (admitting $\mathbb{\Gamma}_{\mathbb{Q}} \simeq \mathbb{\Gamma}_{0,3}$) is to characterize $\mathbb{\Gamma}_{0,3}$ algebraically, as well as the $\mathcal{N}_{g,\nu}$ — and to give an algebraic description also of the "admissible" splittings — i.e. corresponding really and truly to algebraic curves over algebraic number fields.

Uncertainty retained from cancelled passages.

140-2-s479-D001 I think

[scan 480]

that these are exactly the actions lifting the given|given outer action, which normalize no loop subgroup|subgroups, or again such that $\hat{\pi}_{0,3}^{\Gamma''+}$ † (1) (But it is conceivable that more subtle conditions may have to be added to it, involving the Frobenius ...) We may propose to find a description of the outer "admissible" actions, † without having to pass through admissible actual actions — and to endea-|endevor to give a more or less plausible definition of them. The difficulty of course|course comes from the fact that if Γ'' operates by outer automorphisms on † $\hat{\pi}_{0,3}$, it does not thereby operate by outer automorphisms on a given open subgroup π' of $\hat{\pi}_{0,3}$. But we are going to suppose precisely that such an action exists, † in such a way that † $\pi' \longrightarrow \hat{\pi}_{0,3}$ is compatible with the outer actions, and that the action of Γ'' on π is deduced (modulo isomorphism) from [p. 243]

Uncertainty retained from cancelled passages.

140-2-s480-D002 [illeg. 1]

140-2-s480-D003 in terms

140-2-s480-D004 [illeg. 1]

140-2-s480-D005 compatible

140-2-s480-D006 with

[scan 481]

For this second point, we have an immediate answer (assuming the $\mathcal{N}_{g,\nu}|N_{g,\nu}$ already known). Let a|a (π, a) be of type g, ν, with † a partial lifting of $\mathbf{\Gamma}_a$ to|a $\Gamma'_{a|\cdot} \hookrightarrow \mathcal{N}_a|N_a$. It will be admissible iff we can find a (π', a') of type (g, ν'), and an embedding of π' as a subgroup|subgroup of finite index in $\hat{\pi}_{03}$ compatible with a', and an open subgroup Γ Γ'' of $\mathbf{\Gamma}_{03}$, and an admissible actual action of Γ'' † on $\hat{\pi}_{03}$ which lifts the given outer action, and which leaves π' invariant, so that up to isomorphism, (π, a) and the germ of action of Γ' on it is deduced from (π', a') by the action of Γ'' on it, by "plugging" a batch of holes (and forgetting an actual action in favor of the outer action). One needs in this approach simply to know how to specify algebraically what is meant by an "admissible" action of $\Gamma'' \subset \mathbf{\Gamma}_{0,3}$ on $\hat{\pi}_{0,3}$ — it being understood that this must correspond to the points of $U_{0,3}$ rational over † the number field defined by Γ''. One can conjecture 480

Uncertainty retained from cancelled passages.
140-2-s481-D001 when
140-2-s481-D002 admissible
140-2-s481-D003 $\mathbb{Q}$

[scan 482]

this one, by "plugging the holes". [p. 244]

It seems that on the same principle one can give a description of the $\mathcal{N}_{g,\nu}|N_{g,\nu}$ (hence of the $\mathbf{\Gamma}_{g,\nu}$) in terms of $\mathbf{\Gamma}_{0,3}$. Using the transition homomorphisms $\widehat{\widehat{\mathcal{T}}}_{g,\nu} \longrightarrow \widehat{\widehat{\mathcal{T}}}_{g,\nu'}$, it suffices to do so, when g is fixed, for large ν — large enough that there exists an algebraic curve of type g, ν over $\mathbb{Q}$ which is an étale covering of $U_{0,3}$ over $\mathbb{Q}$.

We therefore consider a corresponding embedding of $\pi_{g,\nu}$ into $\pi_{0,3}$, and decree that if † we can make $\mathrm{ext}^t\ \mathbf{\Gamma}_{0,3}$ operate on $\pi_{g,\nu}$, so that $\pi_{g,\nu} \longrightarrow \pi_{0,3}$ commutes with these outer actions, then the subgroup of $\widehat{\widehat{\mathcal{T}}}_{g,\nu}$ generated by $\widehat{\mathcal{T}}_{g,\nu}$ and $\mathbf{\Gamma}_{0,3}$ is † $\mathcal{N}_{g,\nu}|N_{g,\nu}$. 481

Uncertainty retained from cancelled passages.
140-2-s482-D001 $\mathbf{\Gamma}_{03}|\mathcal{T}_{03}$
140-2-s482-D002 $\widehat{\mathcal{T}}$

34. Heuristic profinite description of the category of algebraic curves defined over † finite subextensions K of $\overline{\mathbb{Q}}_0/\mathbb{Q}$ (i.e. of $\mathbb{C}/\mathbb{Q}$). [scan 483] [p. 245]

We restrict ourselves to geometrically connected curves (for convenience), anabelian (out of provisional necessity !, cf. below on the way to get rid of this restriction). The datum of a universal covering of $U_{\overline{\mathbb{Q}}_0}$ defines an extension structure

$$(1) \qquad 1 \longrightarrow \pi \longrightarrow \mathcal{E} \longrightarrow \Gamma^+ \longrightarrow 1 \qquad \text{or}$$

(2) or again we have a homomorphism $\mathcal{E} \longrightarrow \mathbf{\Gamma}$ (image Γ^+ open in $\mathbf{\Gamma}$, kernel called π), where $\Gamma^+ \subset \mathbf{\Gamma}_{\mathbb{Q}} \simeq \mathbf{\Gamma}_{0,3}$ is the open subgroup|subgroup corresponding to K. Giving oneself such an extension (assuming Centre$(\pi) = 1$) amounts to giving oneself an outer action of Γ' on π. A first question: must one put the loop structure of π in the † data $\mathcal{E}$ of the object (1) (or (2)|(1')) supposed to describe $U\mathcal{E}/K$?

Conjecture It is not worth the trouble – the loop structure of π is the only loop structure invariant under the outer action of Γ^+ (or of $\Gamma^\natural$). The loop subgroups L_i are the maximal subgroupsin π, such that $\mathrm{Norm}_{\mathcal{E}}(L_i) \longrightarrow \Gamma^+$ has as image an open subgroup of $\Gamma^+|\Gamma$. m568

Marginal note. m568 and every subgroup $\neq 1$ of π which has this property is contained in a unique L_i ...

I also presume that every homomorphism †between extensions $\mathcal{E}$ of Γ^+ by a π, $\mathcal{E}'$ of Γ by a π' ($\mathcal{E}, \mathcal{E}'$ coming from algebraic curves U, U') and such that the image of $\mathcal{E}$ in $\mathcal{E}'$ is open, respects 482

Uncertainty retained from cancelled passages.
140-2-s483-D001 $\mathbb{Q}$
140-2-s483-D003 [illeg. 1]
140-2-s483-D004 of

necessarily the loop structure, and in fact comes from a morphism (unique, as we know) of algebraic curves. [scan 484]

At any rate, if one admits the description of the loop subgroups, it will be clear that the image under u of an L_i will be, either (1), or contained in a unique L'_j ...

This would mean that the functor from algebraic curves over K(morphisms the dominant morphisms), to profinite outer groups on which Γ operates, would be fully faithful. The functor "extension of the base field" from K to K' corresponds to the functor: restriction of an outer|outer group (or of an extension structure) from Γ to Γ' (open subgroup). The finite étale coverings of U correspond to $\mathcal{E}$-sets: [$\widetilde{U}$ being chosen]…

To describe the essential image of this functor, we are not reduced to conjectures: we start from the canonical extension $\mathcal{E}_{03}$ of $\mathbf{\Gamma}_{03} = \mathbf{\Gamma}$ by $\hat{\pi}_{03}$, we take an open subgroup|subgroup $\mathcal{E}'$ of $\mathcal{E}_{03}$, with image $\Gamma \subset \mathbf{\Gamma}_{03}$, kernel $\pi' \subset \hat{\pi}_{0,3}$, and in the canonical loop structure of $\pi'$of genus g (induced by that of $\hat{\pi}_{03}$) we take an $I|\overline{I} \subset I(\pi')$such that $2g + \mathrm{Card}(I) \geq 3$, stable under Γ, and we "plug the holes" at $I(\pi') \setminus I$. Likewise, to describe when 483

an actual action, lifting an admissible outer action, is itself admissible – † i.e. can be obtained [scan 485]
from an algebraic †curve U, pointed by a point rational over K. This does not mean for all that, [p. 246|245]
that if U is given by an outer action of Γ on a π, that the conjugacy classes of admissible liftings of this action indeed all come from points of U rational/K. But we see at once that this will be the case, as soon as one admits full faithfulness for isomorphisms

N.B. Even for the automorphisms of $U_{0,3} = \mathbb{P}^1_{\mathbb{Q}} \setminus \{0, 1, \infty\}$, this full faithfulness is not at all clear. One would have to prove that † the commutant of $\mathbf{\Gamma}_{0,3}$ in $\widehat{\widehat{\mathcal{T}}}_{0,3}$ is reduced to $\mathfrak{S}_3|\mathcal{S}_3$. The situation is less hopeless, when we give ourselves, along with the structure of a profinite outer group with operators Γ, an arithmetization of π (which supposes that we have made explicit a loop structure) invariant under † Γ – thus in $\widehat{\widehat{\mathcal{T}}}(\pi)$ we have at our disposal an $N(\pi)|\mathcal{N}(\pi)$, and $\Gamma \hookrightarrow N(\pi)|\mathcal{N}(\pi)$. From this point of view, for a homomorphism $\pi' \longrightarrow \pi$ of outer Γ-groups (arithmetized), it is understood that it is compatible with the arithmetizations, i.e. if π'' is the image of π' in π, 484

Uncertainty retained from cancelled passages.

140-2-s485-D001 can

140-2-s485-D002 [illeg. 1]

140-2-s485-D003 [illeg. 1]

140-2-s485-D004 [illeg. 1]

[scan 486]

we see that the arithmetization of π'' deduced from that of π' by "passage to the quotient", be the one induced by π. † In particular, for the automorphisms(outer) of π it is understood that † not only do they commute with Γ, but also that they are in $N(\pi)$. (which implies that they are in $\widehat{\mathcal{T}}(\pi)$, for the centralizer in $\mathbf{\Gamma}_\pi$ of every open subgroup of $\mathbf{\Gamma}_\pi$ is (1) !) In the case of $U_{0,3}$, we have $N_{0,3} \simeq \mathfrak{S}_3 \times \mathbf{\Gamma}_{0,3}$, and we know that the center of $\mathbf{\Gamma}_{03} \simeq \mathbf{\Gamma}$ is trivial – thus we do indeed find that the group of automorphisms of this structure is reduced to|: $\mathfrak{S}_3$!

This point of view is nevertheless probably superfluous – for we presume that for the given outer action of Γ, there is a unique arithmetization invariant under Γ, and that the "admissible" homomorphisms of the "admissible" outer Γ-groups, as they have been defined previously, automatically respect this arithmetization. If this were not so, one would of course have to introduce the arithmetizations into the structure.

† 485

Uncertainty retained from cancelled passages.

140-2-s486-D001 The interest

140-2-s486-D002 [illeg. 1]

140-2-s486-D003 Here as $\mathbf{\Gamma}_\pi|\Gamma_\pi = \mathbf{\Gamma}_{\mathbb{Q}}$ [illeg. 1] is described from the fact that [illeg. 2] we are going to|we for ourselves oriented [illeg. 3] base point

[scan 487]
[p. 247]

The question arises of finding a more † simplistic description of the outer actions of a Γ on a π which are "admissible". † Here, we shall start from a π for which we already fix a loop structure and an arithmetization a (invariant|invariant under Γ), so † that $\Gamma \subset N(\pi)$, $\Gamma \longrightarrow \mathbf{\Gamma}_\pi$ injective with open image, † i.e. Γ corresponds to a partial splitting (or germ of splitting) of †

$$1 \longrightarrow \widehat{\mathcal{T}}(\pi) \longrightarrow N(\pi) \longrightarrow \mathbf{\Gamma}_\pi \longrightarrow 1$$

Let (g, ν) be the type of π; † if $g \geq 1$ the simplest criterion, it is that the "$\Gamma^\natural$-points" of|of the π' deduced from π by plugging all the holes, be distinct. † (N.B. if $g = 1$, † † by plugging all the holes we land in an abelian case – but that does not matter). If $g = 0$, there is no condition for $\alpha|\nu = 3$, and if $\nu > 3$, let us set|setting aside a † subset $I' \subset I(\pi)$ with card $I' = 3$, the condition is that on plugging the holes of|in $I \setminus I'$, the $\Gamma^\natural$-points of the|of π deduced from the points of $I \setminus I'$ are distinct.

One will note that in this conjectural perspective, in the limiting cases $(g, 0)$ $(g \geq 2)$, $(1, 1)$, $(0, 3)$, we impose no condition a priori on the liftings. 486

Uncertainty retained from cancelled passages.

140-2-s487-D001 simple and
140-2-s487-D002 [illeg. 1]
140-2-s487-D003 [illeg. 2]
140-2-s487-D004 [illeg. 1]
140-2-s487-D005 [illeg. 1]
140-2-s487-D006 [illeg. 2]
140-2-s487-D007 Except
140-2-s487-D008 this
140-2-s487-D009 plugging
140-2-s487-D011 a

[scan 488]

This is perhaps very brutal – and it could be that we find liftings which do not correspond to an algebraic curve – i.e. which in fact are not admissible, even if they appear to be.

To understand something of this, it seems to me that we must return to an interpretation of the germs of splittings called "admissible" of an extension such as

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \widehat{\mathcal{T}}_{g,\nu} & \longrightarrow & N_{g,\nu} & \longrightarrow & \mathbf{\Gamma}_{g,\nu} & \longrightarrow & 1 \\ & & & & & & \uparrow \wr & & \\ & & & & & & \mathbf{\Gamma}_{\mathbb{Q}} \simeq \mathbf{\Gamma}_{0,3} & & \end{array}$$

as corresponding to the algebraic points of a modular variety (rather here, a multiplicity) $M_{g,\nu\mathbb{Q}}|M_{g,\nu,\mathbb{Q}}$, of which $N_{g,\nu}$ is the arithmetic fundamental group, and $\widehat{\mathcal{T}}_{g,\nu}$ the geometric fundamental group.

I would like to examine this situation more closely, later on. For the moment, it seems prudent not to make any|any hasty conjectures for † a direct description (i.e. not via U_{03}) of "admissible" outer operations. We shall therefore work for the moment with|on this notion in the "constructive" form (via U_{03}). 487

Uncertainty retained from cancelled passages.

140-2-s488-D001 [illeg. 1]

[scan 489]

If we have an outer π of type $(g\nu)|(g, \nu)$ with an admissible outer operation of $\Gamma \subset \mathbf{\Gamma}$, we [p. 248] expect that there will be a canonical oriented prediscretification on π_x (not invariant|invariant

under Γ, of course !|) – but such that the corresponding arithmetization is). We shall even, for an admissible actual action of $\Gamma^\natural$ on π, (lifting the given outer action), define a canonical corresponding oriented discretification $\pi_0 \subset \pi$ (replaced by a conjugate, when we conjugate † the lifting $\Gamma^\natural \to \mathcal{E}$ by a $g \in \pi$) – all these oriented discretifications (corresponding to the different "points" of $B_{\pi,\Gamma^\natural}$) define one and the same oriented pre-discretification – and even one and the same strict oriented prediscretification.

The actual action of $\Gamma^\natural$ on π can be obtained (up to isomorphism) as follows: we take|shall take an open subgroup $\mathcal{E}' \subset \mathcal{E}_{03}$, whence extension

$$1 \to \pi' \to \mathcal{E}' \to \Gamma' \to 1$$

with $\pi' \subset \hat{\pi}_{03}$, hence $\pi' = \hat{\pi}'_0$ ($\pi'_0 = \pi' \cap \pi|\Pi_{03}$). We have an operation of Γ' on $I(\pi') = I(\pi'_0)$, we take it trivial (at the cost of passing to an open subgroup of Γ'), we choose $i \in I' \subset I(\pi')$, we plug the holes at I', whence π, an oriented discretification π_0, and an outer action of Γ' on it, which is lifted to an actual action thanks to i. We do indeed find in this way a 488

Uncertainty retained from cancelled passages.
140-2-s489-D001 the action

[scan 490]

"lattice" – an oriented discretification in π – I say that it does not depend on the choices that have been made – in particular, that the automorphisms|automorphisms of $(\pi, a, \Gamma^\natural)$ carry π_0 into itself. Moreover, if we have two points x, y of $B_{\pi,\Gamma^\natural}$, whence $\pi(x)$, $\pi(y)$, among the classes of paths from x to y (which make|form a bitorsor under $\pi(y), \pi(x)$) there are some for which † $\pi(x) \to \pi(y)$ sends $\pi_0(x)$ into $\pi_0(y)$, – when we restrict ourselves to these, we find a subgroupoid of the fundamental groupoid $\Pi_{1|\cdot}(\pi, \Gamma^\natural)$, which this time is a connected loop groupoid. We shall take note that Γ does not operate on this groupoid, although it operates on the set of its objects. 489

Uncertainty retained from cancelled passages.
140-2-s490-D001 [illeg. 2]

[scan 491]

35 The injectivity of $\Gamma_{\mathbb{Q}} \longrightarrow \mathrm{Autext}_{\mathrm{lac}}(\widehat{\Pi}_{0,3})$. [p. 249]

Theorem 1 Let $\Gamma = \mathrm{Gal}(\overline{\mathbb{Q}}_0/\mathbb{Q})$, where $\overline{\mathbb{Q}}_0$ is the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$, let us consider the canonical homomorphism proved modulo lemma 2 below —

$$(1) \qquad \Gamma \overset{\Theta}{\longrightarrow} \widehat{\mathcal{T}}_{0,3} = \mathrm{Autext}_{\mathrm{lac}}(\widehat{\Pi}_{0,3}).$$

This homomorphism is injective.

Proof.

Lemma 1 Let $x \in \mathbb{Q}$, $x \neq 0, 1$ i.e. $x \in U_{0,3}(\mathbb{Q})$, let us choose a path on $\mathbb{P}^1(\mathbb{C})$ from $P = -\overline{j}|\overline{J}$ (base point for defining $\Pi_{03} = \pi_1(U_{0,3}(\mathbb{C}), P)$ — N.B. $U_{03} = \mathbb{P}^1_{\mathbb{Q}} \setminus \{0, 1, \infty\}$) hence an isomorphism $\pi_1(U_{0,3\overline{\mathbb{Q}}_0}, x) \simeq \widehat{\Pi}_{0,3}$, and by transport of structure an actual action of Γ on $\widehat{\Pi}_{0,3}$ (not just outer). Then K=Ker Θ operates trivially on $\widehat{\Pi}_{0,3}$.

Proof. It suffices to see that the outer operation of K on $\pi_1(\overline{V})$, where $V = U_{0,3} \setminus \{x\}$, is trivial. By the result of Белый, there exists a morphism (defined over $\mathbb{Q}$, this is essential)

$$\mathbb{P}^1_{\mathbb{Q}} \longrightarrow \mathbb{P}^1_{\mathbb{Q}},$$

†étale over $U_{0,3} = \mathbb{P}^1_{\mathbb{Q}} \setminus \{0, 1, \infty\}$, and such that $x \longmapsto 0$. Let U' be the induced étale covering of $U_{03} = U$, † $\pi' = \pi_1(\overline{U'})$ its geo- 490

Uncertainty retained from cancelled passages.
140-2-s491-D002 [illeg. 1]
140-2-s491-D003 [illeg. 2]

[scan 492]

metric fundamental group, on which Γ hence $K \subset \Gamma$ operates by outer automorphisms. Then $\pi(\overline{V})$ is a quotient of π', respecting the outer operations of Γ, and it suffices to prove that K operates trivially on π'. But π' is identified with an open subgroup of $\widehat{\Pi}_{0,3} = \pi$ (respecting the outer operations of K, on which the outer operation of K is trivial. It follows that the outer operation of K on π' factors through a finite quotient group.
m579

Marginal note. m579

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \pi' & \longrightarrow & \mathcal{E}'_K & \longrightarrow & K & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \| & & \\ 1 & \longrightarrow & \pi & \longrightarrow & \mathcal{E}_K & \longrightarrow & K & \longrightarrow & 1 \end{array}$$

[Indeed, the outer action of K factors through the actual action of the extension group $\mathcal{E}_K$, which by the construction of K is given by a homomorphism $\mathcal{E}_K \longrightarrow \pi/Z|\mathcal{Z}$ ($Z = \mathrm{Centre}(\pi)$), † the induced|induced action of $\mathcal{E}'_K$ is given by the composite $\mathcal{E}'_K \longrightarrow \mathcal{E}_K \longrightarrow \pi/Z$, whose image lies in N/Z, where N is the normalizer of π' in π. Thus the image of K in $\mathrm{Autext}(\pi')$ is contained in that of $N/\pi' \longrightarrow \mathrm{Autext}(\pi')$, and N/π' is finite] 491

Uncertainty retained from cancelled passages.
140-2-s492-D001 [illeg. 1]

[scan 493]

Returning to $\pi_1(\overline{V})$, we thus see that the image of K in $\mathrm{Autext}\,\pi_1(\overline{V})$ is finite, hence the image of K in $\mathrm{Aut}\,\widehat{\Pi}_{03}$ is finite. It moreover consists of inner automorphisms, so Lemma 1 will be a consequence of [p. 250]

Lemma 2 Every inner automorphism of finite order of $\widehat{\Pi}_{0,3}$ (free profinite group on two generators) is trivial. More precisely: $\widehat{\Pi}_{0,3}$ has a center reduced to $\{1\}$, and every element of $\widehat{\Pi}_{0,3}$ of finite order is reduced to 1.

This is a statement of pure profinite algebra, which I put off until later, to be done with the "geometric" part of the proof of the theorem.

Lemma 3 For every|every nonempty open $V \subset \dagger\mathbb{P}^1_{\mathbb{Q}}$, considering the outer action of Γ on $\pi_1(\overline{V})$, its restriction to K is trivial.

Passing if need be to a smaller 492

Uncertainty retained from cancelled passages.
140-2-s493-D002 $U_{0,3}$

[scan 494]

open, we may suppose by Белый that V is an étale covering of U_{03} — and the preceding argument then shows that the outer action of K factors through a finite quotient group — but this is not sufficient for our purpose, and does not by itself imply that this action is trivial.

(Besides, at the point I have reached, we could have replaced U by any quasi-projective smooth geometrically connected algebraic curve — to find that K operates on the outer group

$\pi_1(\overline{V})$ via a finite group. But here the hypothesis $V \subset \mathbb{P}^1_{\mathbb{Q}}$ implies that there exists $y \in V(\mathbb{Q})$, let $x \in U(\mathbb{Q})$ be its image in $U = U_{0,3}$.

Taking † y, x as base points for the fundamental groups, we now find an actual homomorphism of fundamental groups 493

Uncertainty retained from cancelled passages.
140-2-s494-D005 the

[scan 495]
[p. 251]

$$\pi' = \pi_1(\overline{V}, y) \hookrightarrow \pi\ (\simeq \widehat{\Pi}_{0,3}) = \pi_1(\overline{U}, x),$$

compatible with an actual action of Γ on these groups. By lemma 1 the induced action of K on $\pi = \pi_1(\overline{U}, x)$ is trivial, hence also its action on the subgroup π'. A fortiori the outer action is trivial, q.e.d.

We can now prove

Lemma 4 $K = (1)$ (i.e. the theorem!)

Indeed, under the conditions of lemma 2, the action of †K on the set $I = S(\overline{\mathbb{Q}}_0)$, where $S = \mathbb{P}^1_{\mathbb{Q}} \setminus V$, is deduced from the outer action of †K on $\pi_1(\overline{V})$, if card $I \geq 2$, as the action on the conjugacy classes of "loop subgroups|subgroups". As K operates trivially on the outer group, it operates trivially on I. This being true for every V, we see that the action of K on $\mathbb{P}^1(\overline{\mathbb{Q}}_0) = \overline{\mathbb{Q}}_0 \cup \{\infty\}$ is trivial, i.e. its action on $\overline{\mathbb{Q}}_0$ is trivial, hence $K = (1)$, q.e.d. Whew! 494

Uncertainty retained from cancelled passages.
140-2-s495-D002 Γ
140-2-s495-D003 Γ

[scan 496]

It remains to set down the proof of lemma 2, which I am going to reformulate in a more general form:

Theorem 2 Let π be a free profinite group on a set finite I of cardinality ≥ 2.

Then:

a) The center of π is equal to (1).

b) There is not in π any element of finite order other than 1.

In the end I am stuck on a) – just as I do not see why the center of Γ (and of every open subgroup) must be reduced to (1). Consult Deligne about this!

For b), here is an expedient.

Let K be an algebraically closed field of characteristic 0, and let us consider the function field $L = K(x)$ of † $U = \mathbb{P}^1_K \setminus$† $(n+1$ points) † $(n = \text{card}\, I)$. Then $\pi \simeq \pi_1(U)$, and it is a quotient of $\mathcal{E}_L = \text{Gal}(\overline{L}/L)$. Since π is free, $\mathcal{E}_L \longrightarrow \pi$ lifts to $\pi \longrightarrow \mathcal{E}_L$, and it suffices to see 495

Uncertainty retained from cancelled passages.
140-2-s496-D002 $U_{0,3\,K}|(U_{0,3})_K$
140-2-s496-D003 $I|\Sigma$
140-2-s496-D004 $(I$ [illeg. 1]

[scan 497]
[p. 252]

that $\mathcal{E}_L$ has no element of finite order $\neq 1$.

But according to Artin, there is no automorphism τ of finite order $\neq 1$ of an algebraically closed field $\overline{L}$, except in the case where $\overline{L}$ is a quadratic extension of a † maximal †ordered field R, † which is the field of invariants of τ (hence τ of order 2 exactly).

† But in this case|case we would have $R \supset L$, and $L \supset K$, now in K the element -1 is a square, so R cannot be ordered; absurd!

Corollary 1 of Theorem 1 Let $X|U$ be an algebraic curve (smooth, geometric[ally] connected, quasi-projective) over k a finite extension of $\mathbb{Q}$, † (I presume that the outer action of $\mathcal{E}_k^{\bar{k}}$ on $\pi_1(X|U_{\bar{k}})$ is faithful if U anabelian – failing to be able to prove it, I state:) Then there exists a nonempty open subset V of $X|U$ such that the outer action of $\mathcal{E}_k^{\bar{k}}$ on $\pi_1(V_{\bar{k}})$ is faithful. 496

Uncertainty retained from cancelled passages.

140-2-s497-D001 [illeg. 1]
140-2-s497-D002 real
140-2-s497-D003 fixed|finite
140-2-s497-D004 [illeg. 1]
140-2-s497-D005 there|A

[scan 498]

Proof. By Белый, we know that we can find $V \subset X$, which is an étale covering of $U_\dagger = (U_{03})_k$, I say that this V will do. So let K be the kernel of the outer operation of $\Gamma = \mathcal{E}_k^{\bar{k}}$ on $\pi' = \pi_1(V_{\bar{k}})$. Let y be a closed point of V, rational over the finite extension k' of k, corresponding to an open subgroup $\Gamma' = \mathcal{E}_{k'}^{\bar{k}}$ of Γ, let $K' = \Gamma' \cap K$. Let x be the image of y in U, we have

$$\pi' = \pi_1(V_{\bar{k}}, y) \hookrightarrow \pi = \pi_1(U_{\bar{k}}, x),$$

and by construction the outer action of K' on π' is trivial, hence K' operates on π' by inner automorphisms. Now we have

m587

Marginal note. [m587] proved only modulo verification of (2), (4) below – for the corollary (2) is moreover sufficient...

Lemma 5 Let π be a profinite group with loops, π' an open subgroup, then every automorphism u of π which leaves π' stable and induces on π' an inner automorphism (resp. the identity) is inner (resp. the identity). 497

Uncertainty retained from cancelled passages.

140-2-s498-D001 03

[scan 499]

Let us admit this lemma for the moment – it follows that the elements of K', which operate [p. 253] on π inducing inner automorphisms on π', induce|induce themselves inner automorphisms on π itself, i.e. that the outer action of K' on π is trivial. By theorem 1, we know on the other hand that the outer action of Γ' on π is faithful, hence $K' = (1)$. It follows that K is finite. But by Artin we know that the only finite subgroups $\neq (1)$ of Γ are those generated by a "complex conjugation " τ, corresponding|corresponding to a|to the ordered|ordered maximal|maximal subfield between k and $\bar{k}$. But for such a τ we have $\chi(\tau) = -1$, hence the outer action of τ cannot be trivial – we win. In fact, the proof has shown this:

Corollary 2 Let $k|K$ be a field of characteristic 0, 498

[scan 500]

†U a curve algebraic (smooth etc.) over k, such|such that $\mathcal{E}_k^{\bar{k}} = \Gamma$ operates faithfully on $\pi_1(U_{\bar{k}})$ (outer operation).

Then for every geometric[ally] connected étale covering V of U, the outer operation of † Γ on $\pi_1(V_{\bar{k}})$ is also faithful.

Remains to prove lemma 5. It suffices to prove it in the resp. case – the other follows from it at once. If π has at least one class of loops (case of a curve algebraic affine i.e. nonproper),

then π is free – and lemma 5 †is valid precisely for such groups. If $(\ell_i|h_i)_{1\leq i\leq \nu}$ is a system of generators, let L_i be the closed subgroup generated by $\ell_i|h_i$, we shall admit that $\forall n \in \mathbb{N}^*$ not proved!

$$(2) \qquad L_i = \mathrm{Centr}_\pi(L_i^n|\ell_i^n)$$

This granted, if the automorphism u of π induces the identity on π', $\exists n \in \mathbb{N}^*$ such that $\forall 1 \leq i \leq \nu$ $u(\ell_i|{h_i}^n) = \ell_i|{h_i}^n{}_r$, hence $u(\ell_i|h_i)$ centralizes $\ell_i|{h_i}^n = u(\ell_i|h_i)^n$, 499

Uncertainty retained from cancelled passages.

140-2-s500-D001 type

140-2-s500-D002 π

140-2-s500-D006 is found|is proved

[scan 501]

thus by $(*)$ we have $u(l_i) \in L_i$ i.e. $u(L_i) \subset L_i$, but we have $\mathrm{Aut}(L_i) \simeq \widehat{\mathbb{Z}}^*$, and an endomorphism|automorphism of the|of a † $\widehat{\mathbb{Z}}$-module|modules free|free of rank 1 L_i is known when one knows it on an $nL_i \overset{\sim}{\leftarrow} L_i$. Thus $\forall i$ we have $u(l_i) = l_i$, thus $u = \mathrm{id}$, q.e.d. [p. 254]

In the case where π is not free, take a basis $(x_i, y_i)_{1\leq i\leq g}$ – so that π is defined by the generating relation

$$\prod_1^g [x_i, y_i] = 1, \qquad (3)$$

let Λ_i, Λ_i' be the closed subgroups generated by x_i, y_i. (They are moreover all conjugate under $\mathrm{Aut}(\pi)$…) I admit that these groups are $\simeq \widehat{\mathbb{Z}}$ (i.e. that the surjective homomorphisms $\widehat{\mathbb{Z}} \longrightarrow \Lambda_i$, $\widehat{\mathbb{Z}} \longrightarrow \Lambda_i'$ sending 1 to the x_i, y_i are injective – this † this is moreover trivial on passing to π_{ab}) and that we have, by analogy with (2), for every $n \in \mathbb{N}^*$

$$\mathrm{Centr}_\pi(\Lambda_i^n) = \Lambda_i, \qquad \mathrm{Centr}_\pi(\Lambda_i'^n) = \Lambda_i', \qquad (4)$$

Uncertainty retained from cancelled passages.

140-2-s501-D001 L_i

140-2-s501-D004 should not be hard

[scan 502]

and we finish as above.

Remark The strongest faithfulness result in the direction of the present §, concerning actions of $\mathbb{\Gamma}$ and of its open subgroups, would be the following: if V is an anabelian algebraic curve over the finite extension k of $\mathbb{Q}$, with the algebraic closure $\bar{k}$, then not only should the outer action of $\mathcal{E}_k^{\bar{k}} = \Gamma$ on $\pi_1(V_{\bar{k}})$ be faithful, i.e.

$$\Gamma \longrightarrow \widehat{\widehat{\mathcal{T}}}(\pi)$$

m592

Marginal note. m592 this homomorphism[m] is automatically factored through the subgroup $\mathcal{N}(\pi)$ which interests us [illeg. 1]|normalizes $\widehat{\mathcal{T}}(\pi)$

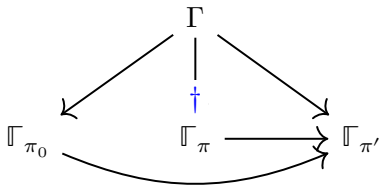
injective, but even the composite homomorphism

$$\Gamma \longrightarrow \mathcal{N}_4(\pi)|\mathcal{N}_\Delta(\pi) \longrightarrow \mathcal{N}(\pi)/\widehat{\mathcal{T}}^+(\pi) \simeq \mathbb{\Gamma}_\pi$$

should be injective (in which case it will even be an isomorphism, since it is surjective by the very construction of $\mathcal{N}(\pi), \mathbb{\Gamma}_\pi$…) Some heuris-

[scan 503]

tic arguments made previously (probably less convincing than those of the present § !) seem to indicate that this would be the case at least when V is an étale covering of $U = (U_{03})_k$, in which case indeed the homomorphism $\Gamma \longrightarrow \mathbb{\Gamma}_\pi$ fits into a commutative diagram [p. 255]



where $\pi_0 = \pi(U_{\bar{k}})$, so that π is an open subgroup of π_0, and $\pi''|\pi'$ is a suitable open subgroup, characteristic in π_0. If we could show that $\mathbb{r}_{\pi_0} \longrightarrow \mathbb{r}_{\pi'}$ is injective (which is $\pm$ a matter of profinite algebra), this would likewise be true of the composite $\Gamma \hookrightarrow | \subset \mathbb{r}_{\pi_0} \longrightarrow \mathbb{r}_{\pi'}$, thus also of $\Gamma \longrightarrow \mathbb{r}_{\pi}$. Thus modulo this hypothesis on profinite groups, and using Belyi, we find that

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Uncertainty retained from cancelled passages.
140-2-s503-D001 $\longrightarrow$

for every algebraic curve V over k, $\exists\, V' \subset V$ nonempty open, such that [scan 504]

$$\Gamma \longrightarrow \mathbb{r}_{\pi'} \qquad (\text{where } \pi' = \pi_1(V'_{\bar{k}}))$$

is injective. As for knowing whether $\Gamma \longrightarrow \mathbb{r}_{\pi}$ is itself already injective – or † what amounts to the same thing, whether $\Gamma \longrightarrow \mathbb{r}_{g}$ for $g \geq 2$, and $\Gamma \longrightarrow \mathbb{r}_{1,1}$ are injective, I have no plausible heuristic reason to convince me of it at present – perhaps|may be this is entirely false ? The more or less convincing argument recalled previously (assuming we manage to justify it) would show only that if $\Gamma \longrightarrow \mathbb{r}_{\pi}$ is injective for one π (which depends only on its type (g, ν)) then it is so for the π' open in π. † We know it at

---

Uncertainty retained from cancelled passages.
140-2-s504-D001 [illeg. 1]
140-2-s504-D002 it|-it

present (modulo very little, at any rate) for the types $(0, \nu)$ $(\nu \geq 3)$ exactly – no more, no less – and we could perhaps deduce it for types (g, ν) which are deduced from them by "finite covering". But it is already clear that we do not obtain the types $(g, 0)$ in this way $(g \geq 2)$, nor even any (g, ν) with $g \geq 1$, $\nu \in \{0, 1, 2\}$.|- That is to say we are far from the mark…The first really interesting case would thus be the case (1,1), where $\widehat{\mathcal{T}}^{+}_{1,1} \simeq$ Sl|SL$(2, \mathbb{Z})^{\wedge}$ Operating|operating by outer automorphisms on $\widehat{\Pi}_{1,1}$ (a free group on two generators, again – as if by chance) – is the "arithmetic" action of $\mathbb{r}_{\mathbb{Q}}$ on $\widehat{\Pi}_{1,1}$ (i.e. the outer action modulo $\widehat{\mathcal{T}}^{+}_{1,1} \simeq$ Sl|SL$(2, \mathbb{Z})^{\wedge}$) faithful ? [scan 505] [p. 256] torus with one hole !

36 The isomorphism $\mathbb{r}_{\mathbb{Q}} \xrightarrow{\sim} \mathbb{r}_{1,1}$, and the injectivity of $\mathbb{r}_{\mathbb{Q}} \longrightarrow \mathrm{Aut}\,\mathrm{ext}_{\mathrm{lac}}(\widehat{\mathcal{T}}^{+}_{11}) \simeq \mathrm{Aut}\,\mathrm{ext}_{\mathrm{lac}}$(Sl|SL$(2, \mathbb{Z})^{\wedge}$) m594 [scan 506] [p. 257]

Marginal note. m594 The reflections|reflections of the present §, a bit halting, will be taken up again in a way|way less ponderous in the § that follows…

Let Π'_{03} be the quotient group of Π_{03} defined by the relations

$$(1) \qquad l_1^2 = l_\infty^3 = 1$$

– this is therefore the "oriented cartographic group for triangulated structures",|; the finite $\Pi'_{0,3}$-sets correspond|corresponding to oriented finite maps (possibly having flattened loops) whose faces are triangles or mono-angles. The outer operation of $\mathbb{r}_{\mathbb{Q}}$ (but not that of $\mathfrak{S}_3|\mathcal{S}_3$!) on $\widehat{\Pi}_{03}$ passes to the quotient as an action on $\widehat{\Pi}'_{0,3}$. We can improve theorem 1 of the preceding § by the|the

Proposition The outer action of $\mathbb{r}_{\mathbb{Q}}$ on $\widehat{\Pi}'_{0,3}$ is faithful.

To show this, we shall embed $\Pi_{0,3}$ as a subgroup of finite index in $\Pi'_{0,3}$, whence an analogous embedding †

$$(2) \qquad \widehat{\Pi}_{0,3} \hookrightarrow \widehat{\Pi}'_{0,3}$$

which will be compatible with the outer action of Γ.

Uncertainty retained from cancelled passages.
140-2-s506-D003 [illeg. 1]

[scan 507]

For this, consider the quotient scheme

$$Y = \underbrace{\mathbb{P}^1_{\mathbb{Q}}}_{X} / \mathfrak{S}_3$$

with the three points a_0, a_1, a_∞ of Y, rational over $\mathbb{Q}$, a_0 corresponding to the orbit $\{0, 1, \infty\}$, a_1 to the orbit $\{-1, \frac{1}{2}, 2\}$, and a_∞ to the orbit $\{-j, -\bar{j}\}$ of $\mathfrak{S}_3$ (which exhausts the set of geometric "singular" orbits). We know that Y is a projective line (over $\overline{\mathbb{Q}}$ a priori), which we pin by a_0, a_1, a_∞ as points $0, 1, \infty$, (thus $Y \simeq \mathbb{P}^1_{\mathbb{Q}}$ – the † morphism $X \longrightarrow Y$ is thus identified with a well-determined morphism $j = \exp \frac{2i\pi}{3}$

$$(3) \qquad f : \begin{array}{ccc} \mathbb{P}^1_{\mathbb{Q}} & \longrightarrow & \mathbb{P}^1_{\mathbb{Q}} \\ \| & & \| \\ X & & Y \end{array}$$

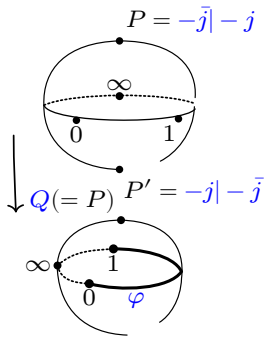
which makes X a Galois covering|covering of Y, with group $\mathfrak{S}_3$, étale over|over $\mathbb{P}^1_{\mathbb{Q}} \setminus \{0, 1, \infty\}$, with ramification indices at these points $2, 2, 3$. From the point of view of the geometry of maps (placing oneself|immersing oneself over the base field $\mathbb{C}$), we consider the

Uncertainty retained from cancelled passages.
140-2-s507-D002 [illeg. 1]

[scan 508]

map determined on X by the spherical triangles $(0, 1, \infty)$ (on the real axis as equator) – which is thus the universal weighted triangular map – as the inverse image of the universal map (having a single vertex 0, a single edge (folded) $0 \longrightarrow 1$, a single face, with center ∞). m596 [p. 258 (259)]

Marginal note. m596



Every topological † étale covering of $U_{03}(\mathbb{C}) \subset X$ thus gives a topological étale covering of $U_{0,3}(\mathbb{C}) \subset Y$, having † over 1 ramification 2, over ∞ ramification 3 exactly|exactly, which corresponds to the "forgetful" functor (of the weighting), where a weighted triangular map is considered as just a map. From the point of view of fundamental groups we thus find an † equivalence of categories

†

$$(\dagger 4) \qquad \left(\text{Étale coverings of } X \setminus \{0,1,\infty\} = U_{0,3}\right) \simeq \left[\begin{array}{c}\text{Coverings of } Y \setminus \{0,1,\infty\} = U_{0,3} \\ \text{with ramification subordinate to} \\ \text{the signature } 2\{1\} + 3\{\infty\}\end{array}\right]$$

$$\wr \qquad\qquad \swarrow \text{the object } (X \setminus \{0,1,\infty\}) \text{ of the said category}$$

$$\wr$$

$$\Pi_{0,3}\text{-sets} \qquad\qquad (\Pi'_{0,3}\text{-Sets})/\dagger E$$

where E is the $\Pi'_{0,3}$-set corresponding to the object $X \setminus \{0,1,\infty\}$ of the category of étale coverings of $Y \setminus \{0,1,\infty\}$ with ramification †subordin-ate to $2\{1\} + 3\{\infty\}$

†

Uncertainty retained from cancelled passages.

140-2-s508-D001 of
140-2-s508-D003 [illeg. 1]
140-2-s508-D004 homomorphism
140-2-s508-D005 $X(\mathbb{C})$
140-2-s508-D008 [*illeg.* 1]
140-2-s508-D009 E
140-2-s508-D010 imposed
140-2-s508-D011 $\simeq \quad \Pi'^{\circ}_{0,3}$

[scan 509]

Here, † † the base point of $Y - \{0,1,\infty\}$ chosen;|, to describe the coverings † (with ramification subordinate to …) by a fundamental group with ramification, being again $P = -\bar{j}$, we shall have [p. 260|250]

$$(5) \qquad E = f^{-1}_{\mathbf{C}|\mathbb{C}}(P)$$

and we can thus make explicit the category $(\Pi'_{03}\text{-Sets})/E$;

Where E is a homogeneous space, isomorphic to the quotient of Π'_{03} by the subgroup|subgroup Π'°_{03} kernel of the suitable surjective homomorphism

$$(6) \qquad \left\{\begin{array}{ccc}\Pi'_{0,3} & \longrightarrow & \mathfrak{S}_3|\mathcal{S}_3 \\ \ell_0 & \longmapsto & \text{element of order 2} \\ \ell_1 & \longmapsto & \text{element of order 2} \\ \ell_\infty & \longmapsto & \text{element of order 3}\end{array}\right\} \quad \text{to be computed !}$$

We choose a $Q_0 \in E$ as "origin" of E – so as to be able to identify E as a $\Pi'_{0,3}$-set with $\Pi'_{03}/\Pi'^{\circ}_{03} \simeq \mathfrak{S}_3$ – then $(\mathrm{Ens}\,\Pi'_{03})/E$ is identified, by the functor $F \longmapsto F_{Q_0}$, with $\mathrm{Ens}(\Pi'^{\circ}_{03})$.

We thus find in summary an equivalence of categories (depending on the choice of Q_0)

$$\Pi_{03}\text{-sets} \quad \simeq \quad \Pi'^{\circ}_{03}\text{-sets}$$

which is itself described by a bitorsor under $(\Pi'^{\circ}_{0,3}, \Pi_{03})$ – and, up to isomorphism (not unique !), by an isomorphism

$$\Pi_{03} \simeq \Pi'^{\circ}_{0,3} \quad -$$

the isomorphisms thus obtained (for

Uncertainty retained from cancelled passages.
140-2-s509-D001 [illeg. 3]
140-2-s509-D002 [illeg. 1]
140-2-s509-D003 f^{-1} (

[scan 510]

varying origins of the bitorsor) form|forming exactly a conjugacy class under Π'°_{03} – i.e. an outer isomorphism $\Pi_{03} \simeq \Pi'^{\circ}_{03}$. When in addition the choice of Q_0 varies, we find a composite|composites [p. 249]

$$\Pi_{0,3} \longrightarrow \Pi'^{\circ}_{03} \longrightarrow \Pi'_{03}$$

† a Π'_{03}-conjugacy class, i.e. an outer homomorphism.

I have slaved away a lot for not much – for lack of having written down the necessary|elementary general functorialities for "fundamental groups with ramification". For example the mysterious bifunctor I from earlier is made up of classes of paths from $P \in X(\mathbb{C})|X_{\mathbb{C}} \setminus \{0,1,\infty\}$ (base point for computing $\Pi_{0,3}$) to Q_0 (playing the role of a new base point, having the merit of mapping to the one – P – which serves to compute Π'_{03}, fundamental group with ramification on $Y(\mathbb{C}) \setminus \{0,1,\infty\}$ ...) It would perhaps be more convenient to define direct^{ly}|directly a functor in the opposite direction:|.

$$(7) \qquad \begin{array}{c}\text{Coverings of } Y(\mathbb{C})\setminus(0,1,\infty) \text{ subordinate}\\ \text{to the signature } 2\{1\}+3\{\infty\}\end{array} \longrightarrow \begin{array}{c}\text{Étale|Étale coverings of}\\ X(\mathbb{C}) \setminus \{0,1,\infty\}\end{array}$$

by inverse image, followed|followed by a normalization

Uncertainty retained from cancelled passages.
140-2-s510-D002 exactly

[scan 511]

– Here one uses the fact that the object $X(\mathbb{C}) \setminus \{0,1,\infty\}$ of the category of covering^{s} with maximum ramification imposed|imposed, attains precisely a maximum, at the points $1, \infty$ where the condition comes in. One finds that the corresponding functor [p. 261]

$$\Pi'_{03}\text{-Ens} \longrightarrow (\Pi_{0,3}\text{-Ens}) \qquad (7 \text{ bis})$$

has the exactness properties of a functor associated to a morphism of Galois topoi

$$\overset{B}{B|D}_{\Pi_{03}} \longrightarrow B_{\Pi'_{0,3}}$$

(commutes with $\varinjlim$ of any kind, left exact) and is therefore defined by a $(\Pi^{\dagger}_{03}, \Pi'_{03})$ set which is a (right) torsor for $\overrightarrow{\Pi}'_{03}$. The choice of an origin for this torsor then also defines a corresponding homomorphism $\Pi_{0,3} \longrightarrow \Pi'_{03}$.
m597

Marginal note. m597 This bitorsor is also the set of $(Y(\mathbb{C})\setminus\dagger\{0\})$ isomorphisms|morphisms of the "universal covering with ramification imposed" of this space, onto the † † ordinary universal covering of $X(\mathbb{C}) \setminus \{0,1,\infty\}$ ...

Let us return to the arithmetic situation over $\mathbb{Q}$, where one has not only the geometric profinite fundamental groups $\widehat{\Pi}_{03}$, $\widehat{\Pi}'_{03}$, but also extensions

$$\begin{cases} 1 \longrightarrow \widehat{\Pi}_{03} \longrightarrow \mathcal{E}^{\dagger}_{0,3} \longrightarrow \Gamma_{\dagger} \longrightarrow 1 \\ 1 \longrightarrow \widehat{\Pi}'_{03} \longrightarrow \mathcal{E}'^{\dagger}_{0,3} \longrightarrow \Gamma \longrightarrow 1 \end{cases}, \qquad (8)$$

interpreted as fundamental groups of U_{03}, resp. of U_{03} with imposed ramification $2\{1\}+3\{\infty\}$. The arguments

Uncertainty retained from cancelled passages.
140-2-s511-D001 ∧
140-2-s511-D002 [illeg. 2]
140-2-s511-D003 with ramification
140-2-s511-D004 ordinary
140-2-s511-D005 ∧
140-2-s511-D006 $\mathbb{Q}$
140-2-s511-D007 ∧

[scan 512]

above extend to this setting, and thus provide a homomorphism of extensions

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \widehat{\Pi}_{03} & \longrightarrow & \mathcal{E}_{03} & \longrightarrow & \Gamma & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \| & & \\ 1 & \longrightarrow & \widehat{\Pi}'_{03} & \longrightarrow & \mathcal{E}'_{03} & \longrightarrow & \Gamma & \longrightarrow & 1 \end{array} \tag{9}$$

, modulo conjugation by an inner automorphism of Π'_{03}.
m598

Marginal note. m598 (9 bis) $\widehat{\Pi}_{0,3} \longrightarrow \widehat{\Pi}'_{0,3}$ fits into an exact sequence

$$\boxed{1 \longrightarrow \widehat{\Pi}_{0,3} \longrightarrow \widehat{\Pi}'_{0,3} \longrightarrow \mathfrak{S}_3 \longrightarrow 1}$$

and ditto for the discrete groups

$$\boxed{1 \longrightarrow \Pi_{0,3} \longrightarrow \Pi'_{0,3} \longrightarrow \mathfrak{S}_3 \longrightarrow 1}.$$

That said, let K be the kernel of the outer operation of Γ on Π'_{03}. One then sees that the outer operation of K on the open subgroup $\widehat{\Pi}_{03}$ takes place through a finite group (in fact, through the intermediary of $\widehat{\Pi}'_{03}/\widehat{\Pi}_{03} \simeq \mathfrak{S}_3$ ...) – which implies, by theorem 1 of the § above, that the group K itself is finite. Thus by Artin one has $K = 1$ or $K = (1, \tau)$, τ complex conjugation. But since $\chi(\tau) = -1$ it is obvious that the outer operation of τ on Π'_{03} (nor even on $(\Pi'_{03})^*_{\mathrm{ab}}$ †) is not trivial... † This proves the proposition.

$* \simeq \mathbb{Z}/2\mathbb{Z} + \mathbb{Z}/3\mathbb{Z}$

Uncertainty retained from cancelled passages.
140-2-s512-D003 [illeg. 2]
140-2-s512-D004 Which

[scan 513]
[p. 262]

I now feel like interpreting Π'_{03} as $\mathcal{T}^+_{1,1}$/center (isomorphic to $\{\pm 1\}$),|, and ditto for $\widehat{\Pi}'_{03} \simeq \widehat{\mathcal{T}}_{11}/\{\pm 1\}$, isomorphism|isomorphisms which is compatible with the outer action of Γ, One has (for the record)

$$\mathcal{T}_{1,1} \xrightarrow{\sim} \mathrm{Gl}(2, \mathbb{Z}), \tag{10}$$

the isomorphism being obtained as follows:

$$\mathcal{T}_{1,1} \xrightarrow[\substack{\text{plugging}\\ \text{the hole}}]{\text{iso}} \mathcal{T}_{1,0} \xrightarrow{\text{iso}} \mathrm{Autext}\left(\underbrace{\Pi_{10}}_{\mathbb{Z}^2}\right) \simeq \mathrm{Gl}(2, \mathbb{Z}) \tag{11}$$

One thus has

$$\mathcal{T}^+_{11} \simeq \mathrm{Sl}(2, \mathbb{Z}) \tag{12}$$

and

$$\mathrm{Cent}(\mathcal{T}_{11}) \simeq (\pm 1) \tag{13}$$

(identifies with the subgroup of homotheties of $\mathrm{Sl}(2,\mathbb{Z})$). It is known that one has in $\mathrm{Sl}(2,\mathbb{Z})/(\pm 1)$ two generators λ_3, λ_2, satisfying respectively

$$\lambda_3^3 = 1, \quad \lambda_2^2 = 1 \tag{14}$$

and such that these relations are a presentation of $\mathrm{Sl}(2,\mathbb{Z})/(\pm 1)$. Unless I am mistaken, these elements|elements come from elements|elements u_6, u_4 of $\mathrm{Sl}(2,\mathbb{Z})$ itself, satisfying

$$u_6^6 = 1, \quad u_4^4 = 1 \tag{15}$$

[scan 514]

and which correspond to the only elements of order 6 and of order 4 (up to conjugation and taking inverses). In fact, every element of finite order of $\mathrm{Sl}(2,\mathbb{Z})$ is conjugate to a power of|of u_4, or u_6. The subgroups generated resp. by u_4 and u_6, cyclic of order 4 and 6, are the groups of automorphisms of the two exceptional elliptic curves (called "anharmonic") (in characteristic 0).

I have no formula at hand for "placing" † u_4 and u_6, so that they generate the group $\mathrm{Sl}(2,\mathbb{Z})$† modulo its center (which probably implies that they generate $\mathrm{Sl}(2,\mathbb{Z})$ — but one will pay attention|well. Pay attention to the additional relation. †

$$(16) \qquad u_4^2 = u_6^3 \qquad \varepsilon$$

(this is † precisely the element -1 of the center of $\mathrm{Sl}(2,\mathbb{Z})$). Perhaps I do not need it (the explicit formula for u_4, u_6). One then will define|defines

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Uncertainty retained from cancelled passages.

140-2-s514-D003 u_4
140-2-s514-D004 $/\pm 1$
140-2-s514-D005 to of the
140-2-s514-D006 [illeg. 2]

[scan 515]
[p. 263]

$$(17) \qquad \Pi'_{03} \xrightarrow{\sim} \mathrm{Sl}(2,\mathbb{Z})/(\pm 1)$$

by

$$(18) \qquad \begin{array}{ccc} \ell'_1 & \longmapsto & \lambda_2 \\ \ell'_\infty & \longmapsto & \lambda_3 \end{array}$$

$$[\text{thus} \qquad \ell'_0 \longmapsto (\lambda_3\lambda_2)^{-1} = \lambda_2\lambda_3^{-1} = \lambda_2\lambda_3^2],$$

where $\ell'_0, \ell'_1, \ell'_\infty$ are the generators of Π'_{03} corresponding to those $\ell_0, \ell_1, \ell_\infty$ of Π_{03}, with the defining relations

$$(19) \qquad \ell'_\infty \ell'_1 \ell'_0 = 1 \qquad \ell_1'^2 = 1, \quad \ell_\infty'^3 = 1$$

so that Π'_{03} is indeed the group with generators ℓ'_1, ℓ'_∞ satisfying: $\ell_1'^2 = 1, \ell_\infty'^3 = 1$.

I now want to convince myself that the homomorphism corresponding to (17)

$$(20) \qquad \widehat{\Pi}'_{03} \longrightarrow \mathrm{Sl}(2,\mathbb{Z})^\wedge/\pm 1 \simeq \widehat{\mathcal{T}}^+_{1,1}/(\pm 1)$$

is compatible with the outer operation of $\Gamma = \Pi_{\mathbb{Q}}$. For this, I feel like defining directly the composite with $\widehat{\Pi}_{0,3}|\Pi_{0,3} \longrightarrow \widehat{\Pi}'_{03}$

$$\dagger \qquad \widehat{\Pi}_{03} \longrightarrow \widehat{\mathcal{T}}^+_{1,1}/\pm 1$$

and even lifting it to a homomorphism

$$(21) \qquad \widehat{\Pi}_{03} \longrightarrow \widehat{\mathcal{T}}^+_{11}\ .$$

Uncertainty retained from cancelled passages.
140-2-s515-D002 (21)

[scan 516]

At the level of discrete groups, one does define

$$(22) \qquad \Pi_{03} \longrightarrow \mathrm{Sl}(2,\mathbb{Z})$$

$$\begin{cases} \ell_1 & \longmapsto & u_4 \\ \ell_\infty & \longmapsto & u_6 \\ \ell_0 & \longmapsto & (u_6u_4)^{-1} = u_4^{-1}u_6^{-1} \end{cases}$$

and $\mathrm{Sl}(2,\mathbb{Z}) \simeq \mathcal{T}^{+}_{11}$ identifies with the quotient of $\Pi_{0,3}$ by the invariant subgroup generated by the elements

$$\ell_1^4,\ \ell_\infty^6,\ (\ell_1^2)(\ell_\infty^3)^{-1}\ .$$

Besides, Π_{03} can be interpreted as a Teichmüller group $\mathcal{T}^{!+}_{0,4}$, and (22) as a homomorphism
m600

Marginal note. m600 indeed, $\mathcal{T}^{!}_{g,\nu}$ is an extension of $\mathcal{T}^{!}_{g,\nu-1}$ by $\Pi_{g,\nu-1}$ (if $(g,\nu-1)$ anabelian) and ditto for the $\mathcal{T}^{+}$, $\mathcal{T}^{!+}_{0,3} = (1)$!

† (23) $\qquad \mathcal{T}^{!+}_{0,4} \longrightarrow \mathcal{T}^{+}_{11}$

which I suspect extends to a homomorphism

$$(24) \qquad \begin{array}{ccc} \mathcal{T}^{!}_{0,4} & \longrightarrow & \mathcal{T}_{11} \\ \wr & & \\ \text{extension of } \{1,\tau\} \text{ by } \Pi_{0,3}, \text{ which is none other than the cartographic group triangulated weighted unoriented} & & \wr \\ & & \mathrm{Gl}(2,\mathbb{Z}) \end{array}$$

not yet defined

I interpret the first side of (23) as the Π_1 "transcendental geometric" of the|of the modular|modular topos

Uncertainty retained from cancelled passages.
140-2-s516-D001 (24)

[scan 517]
[p. 264]

$\mathbb{M}^{!}_{0,4}$, classifying projective lines with four distinct points numbered from 1 to 4; and the second side|right-hand side of (23) is the transcendental π_1 of the modular topos $M_{1,1}$, classifying elliptic curves (with a fixed origin). One should therefore be able to define, at the level of modular multiplicities over $\mathbb{Q}$

$$(25) \qquad (\mathbb{M}^{!}_{0,4})_{\mathbb{Q}} \longrightarrow (\mathbb{M}^{\dagger}_{1,1})_{\mathbb{Q}}$$

which would give rise to (23). †

But one sees at once that one has a canonical isomorphism

$$(26) \qquad \mathbb{M}^{!}_{0,4} \simeq U_{0,3}$$

(I drop the subscripts $\mathbb{Q}$), and giving (25) thus also amounts to giving a family of elliptic curves over $U_{0,3}$.

But if λ is the "variable" $\in U_{0,3}$", taking "the" quadratic covering E_λ of † $\mathbb{P}^1$ ramified at $0, 1, \infty, \lambda$,

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Uncertainty retained from cancelled passages.

140-2-s517-D003 !

140-2-s517-D004 also means

140-2-s517-D005 [illeg. 1]

[scan 518]

one finds an elliptic curve, with four marked points above $0, 1, \infty, \lambda$ — of which one can take the point above 0 as origin — then the other points are the three points of order 2, indexed resp. by $1, \infty, \lambda$.

More precisely: over any base scheme S, except that 2 should be invertible there (residual characteristics all $\neq 2$), one has a functor which goes from the category of systems $(E, \alpha, \beta\dagger)$ of a relative|relative elliptic|elliptic curves|curve (homogeneous|homogeneous) † over S, and † α, β† two sections of E which form a basis of ${}_2E$ as a locally constant scheme in $\mathbb{F}_2$-Modules (a local system of $\mathbb{F}_2$-modules) on S [giving rise to: $\gamma = \alpha + \beta$, as a third rogue — so that (β, γ),|. † etc are in fact also bases) — to the category of projective-line bundles P over S, with four sections $u_0, u_1, u_\infty, \lambda$ marked distinct at every point, by associating to E the quotient $E/\pm 1$, equipped with the sections $u_0, u_1, u_\infty, \lambda$ which are the images resp. of $0, \alpha, \beta, \gamma = \alpha + \beta$

m602

Marginal note. m602 it amounts to the same thing to say that one has|the three sections α, β, γ of ${}_2E$ over S, which are $\forall s \in S$, † distinct and hence|everywhere $\neq 1$.

Uncertainty retained from cancelled passages.

140-2-s518-D004 γ

140-2-s518-D005 with [illeg. 1]

140-2-s518-D006 whose

140-2-s518-D007 γ

140-2-s518-D009 (γ, α)

140-2-s518-D010 everywhere

[scan 519]

m603

[p. 265]

Marginal note. m603 The pages which follow are needlessly complicated, with the introduction of the $M|\mathbb{M}'_{1,1}$, $\mathcal{T}'_{1,1}$ etc. – it is enough to go at it brutally|simply with the elliptic curve $y^2 = \sqrt{x(x-1)(x-\lambda)}$, over $U_{0,3}$, to [illeg. 1] have $U_{0,3} \longrightarrow M|\mathbb{M}_{1,1}$ cf. below…

and this functor is almost an equivalence of categories. What it lacks in order to be one, is that for a relative elliptic curve E, the symmetry $x \mapsto -x$ of E – which is a nontrivial automorphism – operates trivially on the corresponding object $E/\pm 1$. So one must take the|the stack over (Sch), obtained from that of elliptic curves by starting by taking $\mathrm{Isom}(E, F)' = \mathrm{Isom}(E, F)/\pm 1$, then one will take the stack associated to a prestack (the objects over S are the "relative elliptic curves up to symmetry over S"). This|This|The stack is representable by a modular multiplicity $M|\mathbb{M}'_{1,1}$, † having as its geometric fundamental group $\mathrm{Sl}(2, \mathbb{Z})\hat{}/\pm 1$ precisely, obtained for example from the modular varieties with Jacobi rigidification of level n, $M|\mathbb{M}_{11}[n]$ – with group $\mathrm{Sl}(2, \mathbb{Z}/n\mathbb{Z})$ operating on it, so that

$$M|\mathbb{M}_{11} \simeq (M|\mathbb{M}_{1,1}[n], \mathrm{Sl}(2, \mathbb{Z}/n\mathbb{Z})) \tag{27}$$

moreover|one notes that the central subgroup $(\pm 1)|\{\pm 1\}$ of $\mathrm{Sl}(2,\mathbb{Z}/n\mathbb{Z})$ operates trivially on $M|\mathbb{M}_{11}[n]$, so one can let the quotient group $\mathrm{Sl}(2,\mathbb{Z}/n\mathbb{Z})'\,\dagger = \mathrm{Sl}(2,\mathbb{Z}/n\mathbb{Z})/\pm 1$ operate, and set

Uncertainty retained from cancelled passages.
140-2-s519-D002 obtained from [illeg. 1]
140-2-s519-D004 [*illeg.* 1]

[scan 520]

$$M|\mathbb{M}'_{11} = (M|\mathbb{M}_{11}[n], \mathrm{Sl}(2,\mathbb{Z}/n\mathbb{Z})') \tag{28}$$

(which|Which manifestly^{t}|manifestly does not depend on the choice of n, $n \geq 2$). One thus finds a morphism

$$\begin{array}{ccc} M|\mathbb{M}^{!}_{0,4} & \longrightarrow & M|\mathbb{M}'_{11} \\ \| & & \\ U_{0,3} & & \end{array} \tag{29}$$

which makes $M|\mathbb{M}^{!}_{0,4}$ a Galois covering with group $\mathfrak{S}_3$ of $M|\mathbb{M}'_{11}$, and more precisely

$$M|\mathbb{M}^{!}_{0,4} \xrightarrow{\sim} \underset{\text{Galois covering with group } \mathrm{Sl}(2,\mathbb{F}_2)\simeq\mathfrak{S}_3}{M|\mathbb{M}'^{\,\dagger}_{1,1}[2]} \longrightarrow M|\mathbb{M}'_{11}\,\dagger \tag{30}$$

the geometric fundamental group of $M|\mathbb{M}_{11}[2]$ being moreover isomorphic to the kernel of the homomorphism $\mathrm{Sl}(2,\mathbb{Z})'^{\wedge} \longrightarrow \mathrm{Sl}(2,\mathbb{Z}/2\mathbb{Z})$ which factors the canonical homomorphism $\mathrm{Sl}(2,\mathbb{Z})^{\wedge}_{\dagger} \longrightarrow \mathrm{Sl}(2,\mathbb{Z}/2\mathbb{Z})$.

Passing to the fundamental groups for $M|\mathbb{M}^{!}_{0,4} = U_{0,3}$ and $M|\mathbb{M}'_{11}$, one finds a homomorphism|homomorphism of exact sequences

$\dagger$

$$\begin{array}{ccccccccc} & & & & \mathcal{T}^{!+}_{0,4}|\widehat{\mathcal{T}}^{!+}_{0,4} & & N_{0,4}|\mathcal{N}_{0,4} & & \\ & & & & \| & & \| & & \\ 1 & \longrightarrow & \widehat{\Pi}_{0,3} & \longrightarrow & \widehat{\Pi}_{0,3} & \longrightarrow & \mathcal{E}_{0,3} & \longrightarrow \Gamma \longrightarrow & 1 \\ & & & & \downarrow & & \downarrow & \Downarrow & \\ 1 & \longrightarrow & \widehat{\mathcal{T}}'^{+}_{1,1} & \longrightarrow & \widehat{\mathcal{T}}'^{+}_{1,1} & \longrightarrow & \mathcal{E}'_{11} & \longrightarrow \Gamma \longrightarrow & 1 \\ & & & & \wr & & & & \\ & & & & \widehat{\mathcal{T}}^{+}_{11}/\{\pm 1\}|(\pm 1) \simeq \mathrm{Sl}(2,\mathbb{Z})^{\wedge}/\pm 1 & & & & \end{array} \tag{31}$$

Uncertainty retained from cancelled passages.
140-2-s520-D001 [*illeg.* 1]
140-2-s520-D002 [*illeg.* 1]
140-2-s520-D003 [*illeg.* 1]
140-2-s520-D004 (31)

[scan 521]

which identifies $\mathcal{E}_{0,3}$ with a subgroup open[illeg. 1] of index 6 of $\mathcal{E}'_{1,1} = \mathcal{E}_{1,1}/(\pm 1)$, and ditto for $\widehat{\Pi}_{0,3}$ in $\widehat{\mathcal{T}}'^{+}_{1,1}$. We thus find an iso$\dagger$morphism [p. 266]

$$\widehat{\Pi}_{0,3} \xrightarrow{\sim} \begin{array}{c} \mathrm{Ker}(\widehat{\mathcal{T}}'^{+}_{1,1} \longrightarrow \mathfrak{S}_3) \\ \wr \\ \left(\mathrm{Ker}\left(\begin{array}{ccc} \widehat{\mathcal{T}}^{+}_{11} & & \mathfrak{S}_3 \\ \wr & \longrightarrow & \wr \\ \mathrm{Sl}(2,\mathbb{Z})^{\wedge} & & \mathrm{Sl}(2,\mathbb{F}_2) \end{array}\right)\right)/\pm 1 \end{array} \tag{32}$$

compatible with the outer actions of Γ. We can also say that we have an extension structure

$$(33)\qquad \begin{array}{ccccccccc} 1 & - & \widehat{\Pi}_{0,3} & \longrightarrow & \widehat{\mathcal{T}}'^{+}_{1,1} & \longrightarrow & \mathfrak{S}_3 & \longrightarrow & 1 \\ & & & & \wr & & \wr & & \\ & & & & \mathrm{Sl}(2,\mathbb{Z})^{\wedge}/\pm 1 & \to & \mathrm{Sl}(2,\mathbb{Z}_{/2}) & & \end{array}$$

compatible with the actions of Γ (Γ operating trivially on $\mathfrak{S}_3$).

I finally realize that what we get here is far from 22 — this homomorphism (22) is not at all injective, on the other hand it is surjective — and its values are not in $\mathrm{Sl}(2,\mathbb{Z})/\pm 1$, but in $\mathrm{Sl}(2,\mathbb{Z})$ itself! So I shall have to return again to a way of giving

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Uncertainty retained from cancelled passages.

140-2-s521-D001 [illeg. 1]

a remarkable arithmetico-geometric meaning to (22), and its extension to the corresponding profinite groups. For the moment, I confine myself to exploiting (33) a little. First of all I note that the arguments made in the scheme-theoretic setting, work also in the transcendental, complex analytic case, and then yield [scan 522]

$$(34)\qquad \Pi_{0,3} \overset{\sim}{\to} \mathrm{Ker}\left(\begin{array}{ccc} \mathcal{T}'^{+}_{11} & & \mathfrak{S}_3 \\ \| & \longrightarrow & \| \\ \mathrm{Sl}(2,\mathbb{Z}/2\mathbb{Z})/\pm 1 & & \mathrm{Sl}(2,\mathbb{Z}/2\mathbb{Z}) \end{array} \right)$$

compatible with (32), or again an exact sequence

$$(35)\qquad \begin{array}{ccccccccc} 1 & - & \Pi_{0,3} & \longrightarrow & \mathcal{T}'^{+}_{11} & \longrightarrow & \mathfrak{S}_3 & \longrightarrow & 1 \\ & & & & \wr & & \wr & & \\ & & & & \mathrm{Sl}(2,\mathbb{Z})/\pm 1 & & \mathrm{Sl}(2,\mathbb{Z}/2\mathbb{Z}) & & \end{array}$$

,)|.,

† We would like to interpret (35) and (33) as corresponding to the analogous exact sequences linked to diagram (9), relating $\Pi_{0,3}$ and $\Pi'_{0,3}$ — identifying $\Pi'_{0,3}$ with $\mathcal{T}'^{+}_{11}$, $\widehat{\Pi}'_{0,3}$ with $\widehat{\mathcal{T}}'^{+}_{11}$. I shall return to this later.

Remark One may ask whether the homomorphism

$$\widehat{\Pi}_{0,3} \longrightarrow \mathrm{Sl}(2,\mathbb{Z})^{\wedge}/\pm 1 \simeq \widehat{\mathcal{T}}^{+}_{11}/(\pm 1)$$

lifts (in a more or less natural way)

Uncertainty retained from cancelled passages.

140-2-s522-D001 which

to a homomorphism [scan 523]

$$\widehat{\Pi}_{0,3} \longrightarrow \mathrm{Sl}(2,\mathbb{Z})^{\wedge} \simeq \dagger\widehat{\mathcal{T}}^{+}_{1,1}$$ [p. 267]

and ditto for the discrete groups

$$\Pi_{0,3} \longrightarrow \mathrm{Sl}(2,\mathbb{Z}) \simeq \mathcal{T}^{+}_{11}.$$

m604

Marginal note. m604 Yes, one can lift, and it is more or less trivial... cf. below...

Of course, since $\Pi_{0,3}$ and $\widehat{\Pi}_{03}$ are free (with two generators) one can always lift — †in exactly four ways, incidentally. (which necessarily, in the profinite setting, respect the discrete lattices). But can one do this while respecting the operations of Γ? For this it would suffice to be able to lift $U_{0,3} \longrightarrow M'_{1,1}$ to $U_{0,3} \longrightarrow M_{11}^{t|\ell}$ (N.B. M_{11} is a † $\mathbb{Z}/2$-gerbe over M'_{11}) — and I suspect, [m605]

Marginal note. [m605] not quite cf. the next page for a more reasonable formulation...

according to the "anabelian" yoga that I am trying to develop, that the converse must be true — that every lifting of $\widehat{\Pi}_{03} \longrightarrow \widehat{\mathcal{T}}'^{+}_{11}$† commuting with the operations of Γ is defined by such a lifting $U_{0,3} \longrightarrow M_{1,1}$. Now the existence of such a lifting $U_{0,3} \longrightarrow M_{1,1}$ would mean exactly the existence of a family of elliptic curves over $U_{0,3}$, with

Uncertainty retained from cancelled passages.
140-2-s523-D001 Sl
140-2-s523-D002 the indeterminacy up to
140-2-s523-D004 [illeg. 1]
140-2-s523-D006 Π

Jacobi rigidification of level 2, which corresponds to the tautological invariant λ. I doubt that there is one, as I doubt that the lifting in terms of profinite groups with operations can be made. [scan 524]

To tell the truth, since $\widehat{\mathcal{T}}^{+}_{11} = \mathrm{Sl}(2,\mathbb{Z})^{\wedge}$ no longer has trivial center, it is no longer reasonable in fact to want to "express everything" by the outer operations of Γ on $\widehat{\mathcal{T}}^{+}_{11}$, we must rather return to the extension $\mathcal{E}_{11} = \mathcal{E}_{M_{1,1}}$ of Γ by $\widehat{\mathcal{T}}^{+}_{1,1}$, and the question is whether the homomorphism

$$\mathcal{E}_{0,3} \longrightarrow \mathcal{E}'_{1,1} = \mathcal{E}_{1,1}/\pm 1$$

lifts to

$$\mathcal{E}_{0,3} \overset{?}{\to} \mathcal{E}_{11},$$

†which is more stringent than finding a lifting $\widehat{\Pi}_{0,3} \longrightarrow \mathcal{T}^{+}_{11}$ compatible with the outer operations of Γ. To appreciate this difference, I note that Γ operates on the set E of the four liftings of $\widehat{\Pi}_{0,3} \longrightarrow \widehat{\mathcal{T}}^{+}_{1,1}$, (interpreted as outer homomorphisms), and the

Uncertainty retained from cancelled passages.
140-2-s524-D002 choosing a

question posed in the weak form is whether there exists an element of E invariant under Γ. If there were none, there would in any case be an open subgroup of Γ, of index 2 or 4 † which would leave an element invariant — so at the cost of passing to the † corresponding finite extension of $\mathbb{Q}$, we find the lifting — and at the cost of passing to a slightly larger extension, the four possible liftings commute with the outer action of Γ. On the other hand, nothing proves that one can find a "germ of liftings" † of $\mathcal{E}_{0,3} \longrightarrow \mathcal{E}'_{1,1}$ to $\mathcal{E}^{G}_{0,3} \longrightarrow \mathcal{E}^{G}_{1,1}$ (germs taken with respect to the subgroups open|open of $\mathcal{E}_{0,3}$ † containing $\widehat{\Pi}_{0,3}$, or even of all the open subgroups) The obstruction to †lifting over $\mathcal{E}_{0,3}$ †itself, i.e. to splitting a certain extension of $\mathcal{E}_{0,3}$ by † ± 1), is [scan 525] [p. 268]

$$\alpha \in H^2(\mathcal{E}_{0,3}, \mathbb{Z}_{/2}) \quad . \tag{36}$$

— and nothing says that this cohomology class can be killed, by passing to an open subgroup of $\mathcal{E}_{0,3}$.



Uncertainty retained from cancelled passages.

140-2-s525-D002 [illeg. 1]
140-2-s525-D004 [illeg. 1]
140-2-s525-D005 $\mathcal{E}_{0,3} \longrightarrow \mathcal{E}_{11}$
140-2-s525-D006 [illeg. 1]
140-2-s525-D007 [illeg. 3]
140-2-s525-D008 [illeg. 3]
140-2-s525-D009 $\mathbb{Z}$

[scan 526]

But in geometric terms, † the relative elliptic curves sought over $U_{0,2}|U_{0,3}$ form †|the sections of a stack over $(U_{0,3})_{\text{et}}$, which is a $\dagger\mathbb{Z}_{/2}$-gerbe — the obstruction therefore lies in a Brauer group

$$\beta \in H^2(U_{0,3}, \mathbb{Z}_{/2}), \tag{37}$$

now since $U = U_{0,3}$ is an algebraic † affine curve over a field (it would suffice that it not be of type $0, 0$) it follows that the canonical homomorphism (with arbitrary torsion coefficients)

$$H^*(\mathcal{E}_U, -) \longrightarrow H^*(U, -)$$

is an isomorphism. Besides, it must be $\pm$ trivial that β is the image of α — which confirms the intuition|intention that † if the lifting is possible at the level of profinite fundamental groups (arithmetico-geometric|arithmetico-geometric) it is also possible at the level of the modular multiplicities themselves, so that we have an existence of a relative elliptic curve over $U_{0,3}$ which…

Uncertainty retained from cancelled passages.

140-2-s526-D001 soon
140-2-s526-D002 the
140-2-s526-D003 $\{\pm 1\}$
140-2-s526-D004 [illeg. 1]
140-2-s526-D005 the

[scan 527]
[p. 269]

It is true that it is well known that a cohomology class (37) with torsion coefficients is killed, on passing to a finite extension of the † base field ($\mathbb{Q}$ in this case). In fact, we have a spectral sequence

$$H^*(U_{0,3}, \mathbb{Z}_{/2}) \Longleftarrow E_2^{pq} = H^p(\dagger\mathbb{Q}, H^q(\overline{U}_{0,3}, \mathbb{Z}_{/2})|)) \tag{38}$$

and here $H^q(\overline{U}_{0,3}) = 0$ for $q \geq 2$, so we find an exact sequence

$$\begin{array}{ccccccccccc} & & & & & & \dagger & & & & \\ \cdots & & E_2^{0,1} & \longrightarrow & E_2^{2,0} & \longrightarrow & \dagger H^2(U_{0,3}) & \longrightarrow & E_2^{1,1} & \longrightarrow & E_2^{3,0} & \cdots \\ & & \| & & \| & & & & \| & & \| & \\ & & H^0(\mathbb{Q}, H^1(\overline{U}) & & H^2(\mathbb{Q}, \mathbb{Z}_{/2}) & & & & H^1(\mathbb{Q}, H^1(\overline{U})) & & H^3(\mathbb{Q}, \mathbb{Z}_{/2}|\mathbb{F}_2) & \\ & & & & & & & & \wr & & & \\ & & & & & & & & \mathbb{F}_2^2 & & & \end{array}$$

I realize at last that the existence of a † lifting, – i.e. of the hypothetical elliptic curve over $U_{0,3}$ – is completely trivial, it suffices to define it|it by the equation

$$y = \sqrt{x|\dot{x}(x-1)(x-\lambda)}$$

i.e.

$$y^2 - x(x-1)(x-\lambda) = F(x, y; \lambda] = 0$$

which defines an affine plane curve,

Uncertainty retained from cancelled passages.

140-2-s527-D001 $\mathbb{Q}$

140-2-s527-D002 $\overline{U}_{0,3}$

140-2-s527-D003 $H^1(\mathbb{Q})$ $H^1(U_{0,3}, \mathbb{Z}/2)$ $H^1(\overline{U}_{0,3}, -)$

140-2-s527-D004 E

140-2-s527-D005 [illeg. 1]

we pass|or pass to projective coordinates [scan 528]

$$F(x, y, z; \lambda) = y^2 z - x^3 + (1 + \lambda)x^2 z - \lambda x z^2 = 0$$

† I am not going to go further into the question here. to what extent this choice is "natural".

† It gives in any case a homomorphis-† as precise as can be

$$(39) \qquad M^!_{0,4} = U_{0,3} \longrightarrow M_{1,1}$$

hence a homomorphism|homomorphism of extensions

$$(40) \qquad \begin{array}{ccccccccc} & & & & \widehat{\mathcal{T}}^{!+}_{0,4} & & N^!_{0,4}|\mathcal{N}^!_{0,4} & & \\ & & & & \wr & & \wr & & \\ 1 & \longrightarrow & \widehat{\Pi}_{0,3} & \longrightarrow & \mathcal{E}_{0,3} & \longrightarrow & \Gamma & \to & 1 \\ & & \downarrow & & \downarrow & & \| & & \\ 1 & \longrightarrow & \widehat{\mathcal{T}}^{+}_{11} & \longrightarrow & \mathcal{E}_{11} & \longrightarrow & \Gamma & \to & 1 \\ & & \wr & & \| & & & & \\ & & \mathrm{Sl}(2,\mathbb{Z})^\wedge & & N_{1,1}|\mathcal{N}_{1,1}\dagger & & & & \end{array}$$

defined modulo automorphism|automorphism inner by an element of $\widehat{\mathcal{T}}^+_{11} \simeq \mathrm{Sl}(2,\mathbb{Z})^\wedge$, and at the level of discrete|discrete groups

$$\Pi_{0,3} \longrightarrow \mathcal{T}^+_{1,1}$$

† and even

$$(42) \qquad \Pi_{0,3} \hookrightarrow \overbrace{\mathrm{Ker}\left(\begin{array}{ccc} \mathcal{T}^+_{1,1} & \longrightarrow & \mathfrak{S}_3 \\ \| & & \| \\ \mathrm{Sl}(2,\mathbb{Z}) & & \mathrm{Sl}(2,\mathbb{Z}/2) \end{array}\right)}^{\Gamma_2}$$

(and ditto for the profinite groups). Here $\Pi_{0,3}$ is such that the congruence subgroup on the second side|level † i.e. Γ_2, is|is the direct product of this|the subgroup|subgroups $\Pi_{0,3}$ and of

Uncertainty retained from cancelled passages.

140-2-s528-D001 for|then

140-2-s528-D002 Once

140-2-s528-D003 m outer

140-2-s528-D004 $\simeq \mathrm{Sl}(2,\mathbb{Z})^\wedge$

140-2-s528-D007 but

140-2-s528-D008 Γ

its center ± 1. (But in fact, there are exactly four subgroups of Γ_2 which realize this decomposition. So as not to make anyone|anyone jealous, one could look at the intersection of the four, which is a subgroup Γ_2 of index a divisor of $2^4 = 16$ it is the intersection of the kernels of|of the four homomorphisms $\Gamma_2 \longrightarrow$ Centre$(\Gamma_2) = \{\pm 1\}$, [m606] [scan 529] [p. 270]

Marginal note. m606 N.B. That must be the kernel of $\mathrm{Sl}(2,\mathbb{Z})^{\wedge|*} \longrightarrow \mathrm{Sl}(2,\mathbb{Z}/4\mathbb{Z})$

As a subgroup of $\Pi_{0,3}$, its index is a divisor of 8 – It wouldn't surprise me if it were just the commutator group of $\Pi_{0,3}$,) which is of index 4 – there is a whole situation there to elucidate… .
m608

it is true and it is $\simeq$|even trivial…

Marginal note. m608 It is not clear is invariant as a subgroup of † $\mathcal{T}^{+}_{1,1} \simeq \mathrm{Sl}(2,\mathbb{Z})$, i.e. stable under the outer action of $\mathfrak{S}_3$ on $\mathrm{Sl}(2,\mathbb{Z})$ – I presume so, since $\mathfrak{S}_3$ also operates by outer automorphisms on $\Pi_{0,3}$ a priori… $\mathcal{T}^{+}_{11}/\Pi_{0,3}$ would be an interesting central extension of $\mathfrak{S}_3$ by $\{\pm 1\}$…

Of course, one should also make the homomorphism (42) explicit (defined modulo inner automorphisms) by its values on the generators l_0, l_1, l_2 – I shall perhaps return to this later.

Uncertainty retained from cancelled passages.
140-2-s529-D003 $\mathfrak{S}$

[scan 530]

Theorem The natural outer action of Γ_Q on $\widehat{\mathcal{T}}^{+}_{1,1}$, and even on $\widehat{\mathcal{T}}^{+\prime}_{1,1} = \widehat{\mathcal{T}}_{1,1}/(\pm 1)$, is faithful.

We use the fact that we have an injective outer homomorphism

$$\widehat{\Pi}_{0,3} \hookrightarrow \widehat{\mathcal{T}}^{+}_{1,1}$$

compatible with the action of Γ, proceeding as for the proposition at the beginning (where $\widehat{\Pi}'_{0,3}$ played the role of $\widehat{\mathcal{T}}^{\prime +}_{1,1}$).

Corollary The canonical homomorphisms (surjective)

$$\Gamma_Q \longrightarrow \Gamma_{1,\nu} \qquad (\nu \geq 1) \tag{43}$$

are injective, hence isomorphisms.

It suffices to prove this for $\Gamma_Q \longrightarrow \Gamma_{1,1}$ (since $\Gamma_{1,1}$ is a quotient of $\Gamma_{1,\nu}$, $\nu \geq 1$ …), now we have a homomorphism † $\Gamma_{1,1} \longrightarrow \mathrm{Aut\,ext}\,\widehat{\mathcal{T}}^{+}_{1,1}$ and we can consider the composite

Uncertainty retained from cancelled passages.
140-2-s530-D001 [illeg. 1]

[scan 531]
[p. 271]

$$\Gamma_{\mathbb{Q}} \text{–|} \longrightarrow \Gamma_{1,1} \longrightarrow \mathrm{Aut\,ext}\,\widehat{\mathcal{T}}^{+}_{1,1} \longrightarrow \mathrm{Aut\,ext}\,\widehat{\mathcal{T}}^{+\prime}_{1,1},$$

by the preceding theorem this composite is injective, hence also $\Gamma_{\mathbb{Q}} \longrightarrow \Gamma_{1,1}$, q.e.d.

Remark We really have the impression, with the proposition at the beginning, and the preceding result christened "theorem", of having proved the same thing twice, and with the same proof at that! So it is time, after this detour, to make sure that this really is so, i.e. that the homomorphism $\widehat{\Pi}_{0,3} \longrightarrow \dagger\widehat{\mathcal{T}}^{+\prime}_{1,1}$ which we have just used does indeed identify with the homomorphism $\widehat{\Pi}_{0,3} \longrightarrow \widehat{\Pi}'_{0,3}$, by means of a suitable isomorphism (which remains to be described), commuting with the outer actions of Γ, which we had set out to construct (cf. (10) to (20)) — and we lost it along the way, in

Uncertainty retained from cancelled passages.
140-2-s531-D001 Π

[scan 532]

trying to †find $\Pi'_{0,3} \xrightarrow{\sim} \mathcal{T}^{+\prime}_{1,1}$ by factorization in the composite homomorphism

$$\Pi_{0,3} \overset{\text{can.}}{\underset{\text{surj}}{\longrightarrow}} \Pi'_{0,3} \longrightarrow \mathcal{T}^{+\prime}_{1,1}$$

— and in a sense we went astray, for the "natural" homomorphism $\varphi|\rho\,\Pi_{0,3} \longrightarrow \mathcal{T}^{+\prime}_{1,1}$ which we stumbled upon was not at all the one we had in mind: I was way off base. So I must return to the charge!

Uncertainty retained from cancelled passages.
140-2-s532-D001 prove

[scan 533]
[p. 272]

37 Theory of Moduli of elliptic curves via Legendre (rigidification of level 2).)

I realize that I have made some nevertheless quite simple things rather complicated, in the preceding §First a pang of conscience about surface topology:

Proposition Let|Let† X be a paracompact topological* surface, G a discrete group operating properly on X, we assume that for every $x \in X$, the stabilizer G_x operates faithfully on a neighborhood of x (so that G operates faithfully — in fact the two must be equivalent) preserving the local orientation (hence G_x is a cyclic group), so that $Y = X/G$ is a paracompact topological surface. Consider the set

$$X^! = \{x \in X \mid G_x \neq (1)\}$$

— which is a discrete subset of X stable under G — and the image $Y^! \subset \mid \in Y$ of $X^!$ — a discrete subset of Y — and the "signature" on $(Y, Y^!)$, for which the coefficient d_y $(y \in Y^!)$ is the ramification index $\operatorname{ord} G_x$ at the $x \in X$ over y. For every G-covering X' of X, let us then consider $X'/G = Y'$ as a space over Y. Then:

m609

Marginal note. m609 Variant $\mathcal{G}$ operates † †on X, with|its invariant subgroup G in $\mathcal{G}$ satisfying the conditions opposite. We then have, putting † $Y = X/G$ with an operation of $\mathcal{G}/G = \mathcal{H}$ on it:

$$\dagger\underline{\text{Rev}}.(X, \mathcal{G}) \cong \mid \simeq \underline{\text{Rev}}((Y, \underline{d}), \mathcal{H}),$$

hence

$$\Pi_1(X, \mathcal{G}, x) \xrightarrow{\sim} \Pi_1((Y, \underline{d}, \mathcal{H}), y).$$

Application

$$\mathfrak{S}_{0,3} \simeq \mathcal{T}'_{1,1} (\simeq \text{Gl}|\text{Sl}(2, \mathbb{Z})/\pm 1)$$

extension of $\mathcal{T}_{0,3} = \mathfrak{S}_3 \times \mathbb{Z}/2$ by Π_{03}

Uncertainty retained from cancelled passages.
140-2-s533-D001 [illeg. 1]
140-2-s533-D002 [illeg. 2]
140-2-s533-D003 operates
140-2-s533-D004 [illeg. 1]
140-2-s533-D005 [*illeg.* 1]

[scan 534]

a) Y' is a topological surface, a ramified covering of Y, subordinate to the signature $\underline{d}\,\dagger = \underline{d}(X/Y)$.

b) The functor $X' \longmapsto Y' = X'/G$ is an equivalence of categories

$$(1) \qquad \mathrm{Rev}(X, G) \xrightarrow{\approx} \mathrm{Rev}_{\underline{d}}(Y)$$

N.B. We can make explicit a quasi-inverse functor, by associating to every ramified covering Y' of Y compatible with $\underline{d}$, † the "normalized" (X, G)-covering (in a topological sense easy to make explicit) of $X \times_Y Y'$. We have an analogous statement for regular locally noetherian schemes of dimension 1, where the quasi-inverse functor is obtained by normalizing $X \times_Y Y'$.

Corollary Let $x \in X - X^!$, y its image in Y. We assume Y connected (i.e. that G operates transitively on the set of connected components of X). Then we have a canonical isomorphism

$$(2) \qquad \Pi_1(X, G; a|x) \xrightarrow{\sim} \Pi_1((Y, \underline{d}), y)$$

Uncertainty retained from cancelled passages.

140-2-s534-D002 [*illeg.* 1]

140-2-s534-D003 [illeg. 2]

[scan 535]

[p. 273]

whence, for X connected, a canonical exact sequence

$$(3) \qquad 1 \longrightarrow \dagger\Pi_1(X, x) \longrightarrow \Pi_1((Y, \underline{d}), y) \longrightarrow G \longrightarrow 1$$

N.B. We have ? an analogous corollary in the scheme-theoretic context, X and Y being regular schemes of dimension 1. — or, as the case may be, smooth relative schemes of relative dimension 1 over an S, but in this case one would have to (if I need it) take a little care to specify the statement|the statements in a reasonable form.

Example Let us take $X = \mathbb{P}^1(\mathbb{C}) \setminus \{0, 1, \infty\} = U_{0,3}(\mathbb{C})$, $G = \mathfrak{S}_3$ operating in the usual way, thus (with the † conventions of the preceding §), † identifying $\mathbb{P}^1(\mathbb{C})/(G = \mathfrak{S}_3)$ with $\mathbb{P}^1_{\mathbb{C}}$, with

$$(4) \qquad \begin{array}{rcll} (0, 1, \infty) & \longmapsto & 0 & d_0 = 2 \\ (2, -1, \frac{1}{2}) & \longmapsto & 1 & d_1 = 2 \\ (j, \bar{j}) & \longmapsto & \infty & d_\infty = 3 \end{array}$$

we find $Y = \mathbb{P}^1(\mathbb{C}) \setminus \{0\}$, and $\underline{d} = 2\{1\} + 3\{\infty\}$.

We thus find a canonical isomorphism

$$(5) \qquad \Pi^1((U_{0,3}(\mathbb{C}), \mathfrak{S}_3), x) \simeq \Pi_1((\mathbb{P}^1(\mathbb{C}) \setminus \{0\}, \underline{d}), y)$$

Uncertainty retained from cancelled passages.

140-2-s535-D001 $G|1$

140-2-s535-D002 notations

140-2-s535-D003 $\mathbb{P}^1(\mathbb{C})$

[scan 536]

now|or unfortunately we cannot take $x = P = j$, we have to take $x \in \mathbb{P}^1(\mathbb{C}) \setminus \{0, 1, \infty, 2, -1, \frac{1}{2}, j, \bar{j}\}$ — the most convenient thing will be to take x such that $y = f(x) \in | \leq | \leftarrow$ is the typical base-p^t $Q = j$ for calculating the π_1 with ramification — for such an x we thus find a canonical isomorphism

$$(5\,\text{bis}) \qquad \Pi_1((U_{0,3}(\mathbb{C}), \mathfrak{S}_3), x) \simeq \Pi'_{0,3}$$

and by choosing an $\mathfrak{S}_3$-class of paths from the initial base point $P = j$ on $X_0|X = U_{0,3}(\mathbb{C})$ to x, whence a corresponding isomorphism of the first side of (5bis) † with $\Pi_{0,3}$, we find a canonical isomorphism

$$(6) \qquad \Pi_1((U_{0,3}(\mathbb{C}), \mathfrak{S}_3), P) \simeq \Pi'_{0,3}$$

(which changes by an inner automorphism when one modifies the choice of an $\mathfrak{S}_3$-class of paths), whence the exact sequence

$$(7) \qquad 1 \longrightarrow \Pi_{0,3} \longrightarrow \Pi'_{0,3} \longrightarrow \mathfrak{S}_3 \longrightarrow 1,$$

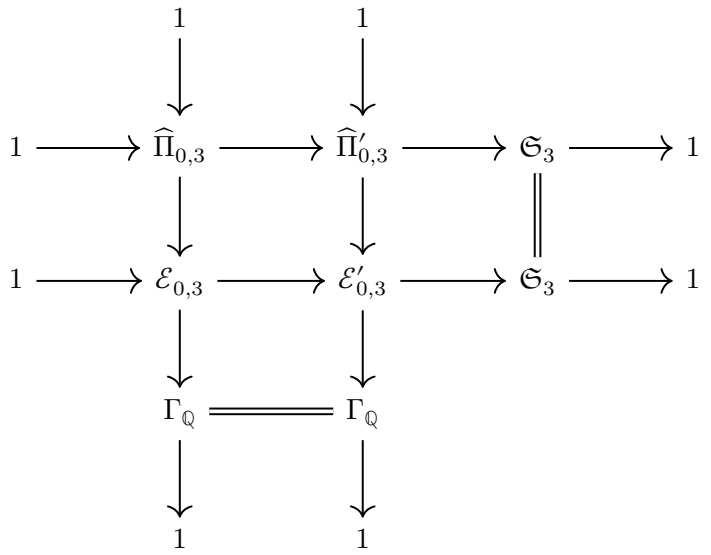
which is none other than (9bis) of the preceding § but interpreted here much more transparently as

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Uncertainty retained from cancelled passages.
140-2-s536-D001 [illeg. 1]

[scan 537]
[p. 274]

the homotopy exact sequence for $\mathfrak{S}_3$ operating on $U_{0,3}(\mathbb{C})$. And we find likewise, in the arithmetico-geometric setting, working on the scheme $U_{0,3} = U_{0,3\mathbb{Q}}$ and its quotient $U_{0,3}/\mathfrak{S}_3 \simeq \mathbb{P}^1_{\mathbb{Q}} \setminus \{0\}$, with the signature $2\{1\} + 3\{\infty\}$ on it, the diagram of exact sequences



[(8) to the left of the second horizontal sequence. To the left and a little below $\widehat{\Pi}_{0,3}$: $\widehat{\mathcal{T}}^+_{0,4}$, followed by two short parallel oblique strokes (an = sign equating it with $\widehat{\Pi}_{0,3}$). To the left and a little below $\mathcal{E}_{0,3}$: $N^!_{0,4}$ (a short horizontal stroke against the $^!$), followed by two short parallel oblique strokes (an = sign toward $\mathcal{E}_{0,3}$). To the left and a little below the first $\Gamma_{\mathbb{Q}}$: $\Gamma_{0,4}$, followed by two short parallel oblique strokes (an = sign toward $\Gamma_{\mathbb{Q}}$).]

We now want to relate $U_{0,3}$, with the action of $\mathfrak{S}_3$ on it, to the modular multiplicity M_{11} for elliptic curves (subscripts $\mathbb{Q}$ understood). It is a pity, by the way, to work only over $\mathbb{Q}$ — I am going to work instead, over $\mathbb{Z}[1/2]$ — i.e. in residue|always characteristic $\neq 2$.

[scan 538]

In the preceding § I identified, somewhat murkily, $U_{0,3} = M^!_{0,4}$, with $M_{1,1}[2]'$, (over which $M_{1,1}[2]$ †is a gerbe banded by $\mathbb{Z}/2$). The technical troubles (rather, the conceptual complications) are due to the fact that for elliptic curves (more generally, for abelian varieties of arbitrary dimension), if n is an integer ≥ 3, the "Jacobi rigidification of level n" really is a rigidification (i.e. every automorphism of a relative elliptic curve E which induces the identity on ${}_nE$ — kernel subscheme of $n.\mathrm{id}_E$ — is the identity — but it is no longer the same for $n = 2$, where $u|n$

is the identityid_E up above an open and closed subset of S, and $-\mathrm{id}_E$ on the complement. This suggests, for a better understanding, covering $M_{1,1}[2]$ by an $M_{1,1}(n)$, where n is a multiple of 2 — then $M_{1,1}(n)$ becomes an ordinary scheme of finite type over $\operatorname{Spec}\mathbb{Z}$ (its image in $\operatorname{Spec}\mathbb{Z}$ will be|will be the open set $\operatorname{Spec}\mathbb{Z}[1/n]$) and over $\operatorname{Spec}[1/n]$, the †modular topos

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Uncertainty retained from cancelled passages.

140-2-s538-D002 should be|was to be [illeg. 1]

140-2-s538-D003 scheme|scheme

[scan 539]

[p. 275]

$M_{1,1}$ is recovered as

$$M_{1,1} \simeq \mid \cong (M_{1,1}[n], \overbrace{\mathrm{Gl}(2, \mathbb{Z}/n\mathbb{Z})}^{\Gamma_n}) \tag{9}$$

(over $\operatorname{Spec}\mathbb{Z}[1/n]$)

m610

Marginal note. m610 careful, it really is $\mathrm{Gl}(2, \mathbb{Z}/n\mathbb{Z})$, and not $\mathrm{Sl}(2, \mathbb{Z}/n\mathbb{Z})$ that one must take here

Here $M_{1,1}[n]$ is the scheme representing the functor on $(\mathrm{Sch})_{\mathbb{Z}|S}$

$$S \longmapsto \begin{array}{c}\text{set of relative elliptic curves}\\ \text{over } S, \text{ equipped with an isomorphism}\\ (\mathbb{Z}/n\mathbb{Z})^2_S \longrightarrow {}_nE\end{array}$$

m611

Marginal note. m611 this functor is|→ representable iff $n \geq 3$ (so it is not for $n = 1, 2$)

on which the group $\Gamma_n = \mathrm{Gl}(2, \mathbb{Z}/n\mathbb{Z}) = \mathrm{Aut}((\mathbb{Z}/n\mathbb{Z})^2)$ operates in the obvious way. The modular scheme ("coarse", as opposed to the modular multiplicity or topos) is described over $\mathbb{Z}/n\mathbb{Z}$ by

$$\widetilde{M}_{1,1} \simeq M_{1,1}[n]/\Gamma_n \tag{10}$$

(over $\mathbb{Z}[1/n]$)

N.B. The scheme $\widetilde{M}_{1,1}$ over $\operatorname{Spec}\mathbb{Z}$ can be described as "the representable envelope" of the nonrepresentable functor

$$S \longmapsto \begin{array}{c}\text{isomorphism|Isomorphism classes of relative}\\ \text{elliptic curves over } S \ldots\end{array}$$

– ditto over a base scheme (for example $\mathbb{Q}$, or $\operatorname{Spec}\mathbb{Z}[1/n]$, arbitrary …

[scan 540]

One must be careful that as a scheme over $\mathbb{Z}[1/n]$ (or over $\mathbb{Q}$, passing to the generic|generic fiber) $M_{1,1}[n]$ is not relatively connected (i.e. does not have geometrically connected fibers). In fact, an isomorphism

$$(\mathbb{Z}/n\mathbb{Z})^2_S \simeq {}_nE$$

implies, by passing to the $2^{\text{ième}|\text{ème}}$ exterior power, an isomorphism

$$(\mathbb{Z}/n\mathbb{Z})_S \xrightarrow{\sim} \Lambda^2_{\mathbb{Z}/n\mathbb{Z}}\, {}_nE \simeq \mu_n^{\otimes -1}(S)$$

whence an isomorphism

$$\mathbb{Z}/n\mathbb{Z} \simeq \mu_n(S)$$

which identifies (taking † the image of 1 mod n) with a section of $\mu_n^*(S)$, where μ_n^* is the relative $(\mathbb{Z}/n\mathbb{Z})^*$-torsor over $\operatorname{Spec}\mathbb{Z}[1/n]$ of "primitive nth|n-th roots of 1". We thus have a canonical morphism

$$M_{1,1}[n] \longrightarrow \mu_n^* \tag{11}$$

and it is this morphism that has geometrically connected fibers in characteristic 0. Passing to the limit over variable n, we find over $\mathbb{Q}$

$$M_{1,1}[\infty] = \varprojlim_n M_{1,1}[n] \longrightarrow \mu_\infty^* \overset{\text{iso.can}}{\simeq} \operatorname{Spec}\Sigma \tag{12}$$

Uncertainty retained from cancelled passages.

140-2-s540-D001 and

[scan 541]

(12) where $\Sigma \subset \mathbb{C}$ is the subextension that is maximal cyclotomic in $\overline{\mathbb{Q}}_0$. [p. 276]

† We recover the $M_{1,1}[n']$, for $n' \mid n$, $n' \neq 1, 2$, from $M_{1,1}[n]$ with the action of Γ_n on it, by

$$M_{1,1}[n'] \simeq M_{1,1}[n] \wedge_{\Gamma_n} \Gamma_{n'} \simeq M_{1,1}[n]/K_{n,n'} \tag{13}$$

$$\text{over } \operatorname{Spec}\mathbb{Z}[1/n]$$

where

$$K_{nn'} = \operatorname{Ker}(\Gamma_n \longrightarrow \Gamma_{n'}). \tag{14}$$

m612

Marginal note. m612 N.B. $K_{n,n'}$ operates freely on the scheme $M_{1,1}[n]$ (if $n' \mid n$, $n' \neq 1, 2$) — but of course not if $n' = 1$ or 2.

If however we apply this formula in the inadmissible case|the inadmissible cases $n' = 1$ or 2, we find for the case $n' = 1$, not $M_{1,1}$ but $\widetilde{M}_{1,1}$ (coarse modular scheme), and for $n' = 2$, not $M_{1,1}[2]$ (which is not a scheme either) but what we had called in the preceding § $M_{1,1}[2]'$.

We can † make explicit $M_{1,1}$ and $\widetilde{M}_{1,1}$ over all of $\operatorname{Spec}\mathbb{Z}$, in terms of the $M_{1,1}[n]$, by describing them over the two open sets $\operatorname{Spec}(\mathbb{Z}[1/n])$, $\operatorname{Spec}(\mathbb{Z}[1/n'])$ which cover $\operatorname{Spec}(\mathbb{Z})$ (i.e. for $(n, n') = 1$) by (9) and (10), (taking first n, then n' in them) and by "gluing" over $\operatorname{Spec}(\mathbb{Z}[\frac{1}{nn'}])$ by these same formulas, applied to nn'…

Uncertainty retained from cancelled passages.

140-2-s541-D001 The

140-2-s541-D002 determine|define

[scan 542]

We have moreover (for the record)

$$\underline{\text{Theorem}} \qquad \widetilde{M}_{1,1} \simeq \mathbb{E}^1_{\mathbb{Z}} \qquad (\text{over } \operatorname{Spec}\mathbb{Z})$$

We can pin down such an isomorphism (defined|defines a priori modulo the integral affine group $X \longmapsto \pm X + n$, $n \in \mathbb{Z}$) by noting that there are in $\widetilde{M}_{1,1}$ two distinguished sections over $\mathbb{Z}$, whose values at the generic point correspond to the two isomorphism classes of elliptic curves in characteristic 0 (over $\mathbb{C}$, for example) which have automorphisms $\neq \mathrm{id}, -\mathrm{id}$ — the two groups of automorphisms that we obtain are moreover $\mathbb{Z}/4\mathbb{Z}$ and $\mathbb{Z}/6\mathbb{Z}$ (very easy for example by the transcendental|transcendental route). These are what are called unless I am mistaken the "anharmonic" elliptic curves. On an $M_{1,1}[n]$, the group of rational automorphisms $/k$ of an

elliptic curve described by a point x of $M_{1,1}[n](k)$ is canonically identified with the stabilizer of x in Γ_n:

$$(15) \qquad \mathrm{Aut}(E_x) \simeq (\Gamma_n)_x$$

[scan 543]

— of course, we have suitable statements over an arbitrary base scheme, not necessarily a field. This already shows that the automorphisms of elliptic curves are linked in an essential way to the ramification of Γ_n operating on $M_{1,1}[n]|M_{1,n}[n]$. We thus find: [p. 277]

Proposition (Over Spec $\mathbb{Q}$) There are exactly two orbits of Γ_n operating on $M_{1,1}[n]$ which are "critical", and which correspond to stabilizers isomorphic resp. to $\mathbb{Z}/4\mathbb{Z}$ and $\mathbb{Z}/6\mathbb{Z}$, which are the groups of automorphisms of the corresponding ("anharmonic") elliptic curves. $n \neq 1, 2$

We will note|note that Γ_n does not operate faithfully|faithfully on $M_{1,1}[n]$ — the kernel of this operation is the central subgroup $\{\pm 1\}$, image of the corresponding group of $\mathrm{Gl}(2, \mathbb{Z})$ — it corresponds to the group of "universal" automorphisms $\pm \mathrm{id}_E$ of elliptic curves E. The quotient group

$$(16) \qquad \Gamma'_n = \Gamma_n / \{\pm 1\}$$

† operates faithfully|faithfully, and we can look at the "mixed topos" $(M_{1,1}[n], \Gamma'_n)$ — we find

Uncertainty retained from cancelled passages.

140-2-s543-D002 (on|on

at once that it does not depend (up to canonical equivalence|equivalence) on the choice of n [except that the open set of Spec $\mathbb{Z}$ over which "it makes sense" depends on n...] – and. in fact, we have: [scan 544]

m614

Marginal note. m614 thus on condition of working over a base scheme (such as Spec $\mathbb{Z}[\frac{1}{nn'}]$) where n, n' are both invertible...

$$(17) \qquad (M_{1,1}[n], \Gamma'_n) \simeq M'_{1,1}$$
$$(\text{over } \mathrm{Spec}\, \mathbb{Z}[1/n])$$

where the modular schematic multiplicity $M'_{1,1}$ is described|described, in terms of elliptic curves "defined modulo symmetry", as in the § preceding.

Remark. The preceding proposition seems to depend on n – but we see a priori (thanks to the fact that the $\Gamma_{n,n'}$ $(n' \mid n,\ n' \neq 1, 2)$ (operating|operate freely) that if it is valid for one n, it is valid for all.

† Let us note also

Corollary of the proposition. (characteristic 0) The stabilizers in Γ'_n (the group which operates faithfully on $M_{1,1}[n]$) of the points of the two critical orbits are resp. $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/3\mathbb{Z}$. † valid for $n \neq | = | \geq 2$, cf.|cf. below

Uncertainty retained from cancelled passages.

140-2-s544-D003 Corollary

140-2-s544-D005 [illeg. 1]

We have assumed here that $n \neq 1, 2$ – but nothing prevents us from also considering on [scan 545] [p. 278]

$$(18) \qquad M_{1,1}[2]' \overset{\text{dfn}}{=} M_{1,1}[2m]/\Gamma_{2m,2} = M_{1,1}[2m]/\Gamma_{2m,2}|\Gamma'_{2m,2}$$

(where m is any integer, ≥ 2) the action of $\Gamma_2 \simeq \mathrm{Gl}(2, \mathbb{Z}/2\mathbb{Z}) \simeq \mathrm{Sl}(2, \mathbb{Z}/2\mathbb{Z}) \simeq \mathfrak{S}_3|\mathcal{S}_3 \simeq \Gamma_{2m}/\Gamma_{2m,2}$. † It is no longer true, to be sure, that $\Gamma_{2m,2}$ operates freely on $M_{1,1}[2m]$ – for $\Gamma_{2m,2}$

contains the element $-1 \in \Gamma_{2m}$ corresponding to the universal symmetry $-\mathrm{id}_E$ – i.e. to the element † -1 of $\mathrm{Gl}(2,\mathbb{Z})$. But the general rigidity result recalled at the beginning implies that

$$(19) \qquad \Gamma'_{2m,2} \overset{\text{dfn}}{=} \Gamma'_{2m,2}/\pm 1 \simeq \mathrm{Ker}(\Gamma'_{2m} \longrightarrow \Gamma'_2 \simeq \mathfrak{S}_3|\mathcal{S}_3)$$

operates † still freely on $M_{1,1}[2m]$ – and this implies that the corollary of the proposition is still valid for the action of Γ'_2 ($\overset{\sim}{\leftarrow} \Gamma_2 \simeq \mathfrak{S}_3|\mathcal{S}_3$) on $M_{1,1}[2]'$.

Terminological digression. $M_{1,1} = M_{1,1}[1]$ and the

Uncertainty retained from cancelled passages.

140-2-s545-D001 Whereas|Whereas

140-2-s545-D002 of

140-2-s545-D004 fi

[scan 546]

[p. 278]

278 → 278′

[p. 278']

$M_{1,1}[n]$ ($n \in \mathbb{N}^*$) are defined a|a priori as modular schematic multiplicities, i.e. a) as locally ringed topoi locally isomorphic to the étale topos of a scheme (in this case, of finite type over $\mathrm{Spec}\,\mathbb{Z}$, or over $\mathbb{Q}$ if we work in characteristic 0) and b) defined, up to equivalence itself defined up to unique isomorphism|isomorphism, as "representing" the † corresponding "functors"

$$S \longmapsto \begin{array}{c}\text{category of elliptic curves}\\ \text{relative over } S\text{, with rigidification}\\ \text{of Jacobi of level } n \text{ i.e.}\\ \text{isomorphism} \quad (\mathbb{Z}/n\mathbb{Z})_S \simeq {}_nE\\ = \mathcal{E}ll_n(S)\end{array}$$

thus by construction we have for every scheme S an equivalence

$$\mathcal{E}ll_n(S)\simeq \,|\cong \underline{\mathrm{Hom}}_{\substack{\text{topoi locally ringed}\\ \text{i.e. morphisms of multiplicities}}}(S, M_{1,1}[n])$$

For the categories $\mathcal{E}ll_n(S)$ to be discrete, † or again for $M_{1,1}[n]$ to be a scheme, or again for the automorphisms of an

Uncertainty retained from cancelled passages.

140-2-s546-D001 [illeg. 1]

140-2-s546-D002 [illeg. 1]

[scan 547]

[p. 279]

elliptic curve respecting|respect a Jacobi rigidification of level n to be the identity, it is necessary and sufficient that $n \geq 3$ i.e. $n \neq 1, 2$. Thus $M_{1,1}[2]$ is not a scheme – but $M'_{1,1}[2]$ is one, and $M_{1,1}[2] \to M_{1,1}[2]'$ is a † local isomorphism (described entirely by a $\mathbb{Z}/2\mathbb{Z}$-gerbe over $M_{1,1}[2]'$). † It is necessary all the same to take the trouble to define

$$\begin{array}{ccc} M_{1,1}[2] & \longrightarrow & M_{1,1}[2]' \\ \wr \text{ “theorem”} & & \wr \text{ “definition”} \\ (M_{1,1}[2m], \Gamma_{2m,2}) & & M_{1,1}[2m]/\Gamma'_{2m,2} \\ & & \| \\ & & M_{1,1}(2m)|[2m]/\Gamma_{2m,2} \end{array} \tag{20}$$

as the canonical homomorphism of “passage to the coarse quotient” – which makes sense whenever a discrete group (say)§ G operates on a scheme (say) X, as a morphism

$$(X, G) \longrightarrow X/G = Y \tag{21}$$

whose inverse image functor (sheaves on $Y = X/G \longrightarrow G$-sheaves on X) is obvious. Here we have a normal subgroup $\mathfrak{z}|z$ of G which operates trivially|trivially on X, such that $G/\mathfrak{z}|z$ operates freely, which means that the morphism

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Uncertainty retained from cancelled passages.

140-2-s547-D001 [illeg. 1]

140-2-s547-D002 We consider of course

[scan 548]

(21) factors as

$$(X, G) \longrightarrow \left(X, \underbrace{G/\mathfrak{z}|z}_{G'}\right) \xrightarrow{\sim} Y = X/G = X/G', \tag{22}$$

so that the morphism of multiplicities (21) is also identified, simply, with

$$\begin{array}{ccc} (M_{1,1}[2m], \Gamma_{2m,2}) & \longrightarrow & (M_{1,1}[2m], \Gamma'_{2m,2}) \\ & & \wr \\ & & M_{1,1}[2]' \end{array} \tag{23}$$

On the other hand, if we want to go down as far as $M_{1,1}[1]' = M'_{1,1}$, by passing to the quotient (in the “coarse” sense) of $M_{1,1}[n]$ by Γ'_n, the morphism

$$\begin{array}{ccc} (M_{1,1}[n], \Gamma'_n) & \longrightarrow & M_{1,1}[n]/\Gamma'_n \\ \| & & \| \\ M'_{1,1} & & \widetilde{M}_{1,1} \end{array} \tag{24}$$

is no longer an “isomorphism” i.e. is no longer an equivalence, for Γ'_n does not operate freely – it has (even in characteristic 0) ramification of degree $2, 3$.

[If however we exclude the anharmonic elliptic curves, its|it would work – it is "only in the neighborhood" of these elliptic curves that (24) is not an isomorphism…

[scan 549]

In fact, the arguments of the preceding § establish an isomorphism (valid over the Spec $\mathbb{Z}[1/n]$ [p. 280] of $\mathbb{Z}[1/2]$)

$$(25) \qquad M_{1,1}[2]' \simeq M^!_{0,4} \simeq U_{0,3}$$

(valid over Spec $\mathbb{Z}[1/2]$)

m616

Marginal note. m616 N.B. the image of $M_{1,1}[2]'$ in Spec $\mathbb{Z}$ is Spec $\mathbb{Z}[1/n|2]$, while that of $M^!_{0,4} \simeq U_{0,3}$ is Spec $\mathbb{Z}$ in all its entirety

and this isomorphism is compatible with the operations of †

$$(26) \qquad \Gamma'_2 \simeq \Gamma_2 = \mathrm{Gl}(2, \mathbb{Z}/2\mathbb{Z}) = \mathrm{Sl}(2, \mathbb{Z}/2\mathbb{Z})$$

on $M_{1,1}[2]'$ on the one hand, of $\mathfrak{S}_3|\mathcal{S}_3$ on $U_{0,3}$ on the other hand|hands, when we consider the isomorphism

$$(27) \qquad \begin{array}{ccc} \Gamma_2 & \xrightarrow{\sim} & \mathfrak{S}_3|\mathcal{S}_3 \\ \| & & \\ \mathrm{Gl}(2, \mathbb{F}_2) & & \end{array}$$

which associates, to every automorphism of $\mathbb{F}_2^2$, its action on the three nonzero elements, $\underline{\alpha} = (1,0)$, $\underline{\beta} = (0,1)$, and $\underline{\gamma} = (1,1) = \underline{\alpha} + \underline{\beta}$ (so that we have moreover) †

$$(28) \qquad \gamma = \alpha + | - \beta, \quad \alpha|\beta = \beta + | - \gamma, \quad \beta = \gamma + \alpha.$$

It would be necessary to make this compatibility explicit, by considering the natural action of $\mathfrak{S}_4|\mathcal{S}_4$ on $M^!_{0,4}$, $M^!_{0,n}$ being the modular multiplicity defined by

Uncertainty retained from cancelled passages.

140-2-s549-D001 $\Gamma'_2 \simeq \Gamma$

140-2-s549-D002 [illeg. 2]

[scan 550]

$$(29) \qquad \underline{\mathrm{Hom}}_{\text{multiplicities}}(S, M^!_{0,n}) \simeq \begin{array}{c} \text{category of} \\ \text{relative curves|lines} \\ \text{over } S, \text{ locally isomorphic to } \mathbb{P}^1_S \\ \text{(for étale topology) and equipped} \\ \text{with } n \text{ sections mutually} \\ \text{disjoint numbered} \\ \dagger \quad s_1|s_0, s_2, \dots s_n \end{array}$$

m617

Marginal note. m617 If we want representability for $n = 0, 1, 2$, it no longer suffices to work with the étale topology|the étale topologies — we need fppf topologies for example…

N.B. this really is representable by a multiplicity iff $n \geq 3$, i.e. if the stack in|of groupoids which we want to represent is "rigid" — and then the multiplicity is even a scheme, isomorphic moreover canonical|canonically to the sub-scheme

$$\underline{\mathrm{Mon}}(I_{n-3}, U_{0,3}) \subset U_{0,3}^{I_{n-3}},$$

where here

$$I_{n-3} = \{i \in \mathbb{N} \quad 3 < i \leq n\dagger\}$$

— thus †

$$M^!_{0,3} \xrightarrow{\sim} \operatorname{Spec} \mathbb{Z} \quad \text{(terminal scheme)}, \quad \text{and}$$

$$M^!_{0,4} \xrightarrow{\sim} U_{0,3} = \mathbb{P}^1_{\mathbb{Z}} \setminus \{\text{sections } \dagger\ 0, 1, \infty\} = \operatorname{Spec} \mathbb{Z}[T]\Big[\frac{1}{T(T-1)}$$

But we will keep in mind that there is a natural operation of $\mathfrak{S}_n|\mathcal{S}_n$ on $M^!_{0,n}$, hence on $\underline{\mathrm{Mon}}(I_{n-3}, U_{0,3})$ — † whereas we see a priori only the action of $\mathfrak{S}_3|\mathcal{S}_3 \times \mathfrak{S}_{n-3}|\mathcal{S}_{n-3}$ on the latter: the canonical isomorphism

$$(30) \qquad M^!_{0,n} \simeq \mathrm{Mon}(I_{n-3}, U_{0,3}) \subset U_{0,3}^{I_{n-3}}$$

$$(\text{where } I_{n-3} = \{i \in \mathbb{N} \mid \dagger 3 < i \leq n\})$$

takes account only of the natural operations of the|of the

Uncertainty retained from cancelled passages.

140-2-s550-D001 [*illeg.* 1]
140-2-s550-D002 −3
140-2-s550-D003 [illeg. 1]
140-2-s550-D004 [illeg. 1]
140-2-s550-D005 of which|therefore
140-2-s550-D006 2

[scan 551]
[p. 281]

canonical bijection

$$\tilde{I} \xrightarrow{\sim} V(I)^* \overset{\mathrm{df}}{=} V(I) \setminus \{0\} \tag{33}$$

by associating to a partition $I = I' \cup I''$, with I', I'' of cardinality 2, the only permutation (even)|† of I which leaves invariant I', I'' and induces|induces on each its nontrivial permutation. (The elements of $\mathfrak{S}_I$ obtained in this way are also the even permutations of order 2, † which together with the identity thus form the subgroup $V(I)$).

We find, if $i \in I$, whence $I \setminus \{i\}$ of cardinality 3, † a canonical map

$$I \setminus \{i\} \longrightarrow \tilde{I} \tag{34}$$

by associating to $j \in I \setminus \{i\}$ the partition of I into $\{i, j\}$ and its complement. This bijection being compatible with the transport of structures|structure, we conclude that the corresponding isomorphism

$$\mathfrak{S}_{I \setminus \{i\}} \simeq \mathfrak{S}_{\tilde{I}} \tag{35}$$

† is † also the composite

$$\underset{\substack{\text{stabilizer of} \\ i \text{ in } \mathfrak{S}_I}}{\mathfrak{S}_{I \setminus \{i\}}} \hookrightarrow \mathfrak{S}_I \longrightarrow \mathfrak{S}_{\tilde{I}}, \tag{36}$$

hence the four symmetric subgroups of index 3 $\mathfrak{S}_{I \setminus \{i\}}$ of $\mathfrak{S}_I$ $(i \in I)$.|,

Uncertainty retained from cancelled passages.

140-2-s551-D001 (even)
140-2-s551-D003)
140-2-s551-D004 that
140-2-s551-D005 which
140-2-s551-D006 compat

are canonically isomorphic to $\mathfrak{S}_{\tilde{I}}$ — hence to one another, † by isomorphisms which † are † [scan 552]
moreover † deduced from the canonical bijections †

$$(\dagger 37) \qquad I \setminus \{i\} \xrightarrow{\sim} I \setminus \{j\} \qquad (i \neq j,\ i,j \in I)$$

canonical, equal to the transposition (not to the identity) on $I \setminus \{i,j\}$. This bijection is in fact the composite †

$$I \setminus \{i\} \xrightarrow{\sim} \tilde{I} \xrightarrow{\sim} I \setminus \{j\} \qquad —$$

† and for variable i, j, these isomorphisms form a transitive system of isomorphisms.

Returning to the action of $\mathfrak{S}_4$ on $M'_{0,4}$, the key to understanding is given by this:

Proposition Consider the subgroup $V(\dagger I)$ of $\mathfrak{S}_{\dagger I}$ formed by 1 and the even involutions. Then the action of $V(\dagger I)$ on $M'_{0,\dagger I}$ is trivial, &|If therefore $\mathfrak{S}_I$ acts on $M'_{0,I}$ through $\mathfrak{S}_I/V(I) = \mathfrak{S}_{\tilde{I}}$.

m618

Marginal note. m618 N.B. $M'_{0,I}$ is defined in terms of I, for every set. I of cardinality. ≥ 3, as $M'_{0,n}$

Corollary Let $i \in I$, then the action of $\mathfrak{S}_{I\setminus\{i\}} \subset \mathfrak{S}_I$ on $M'_{0,I}$ is deduced from that of $\mathfrak{S}_{\tilde{I}}$ via the canonical isomorphism $\mathfrak{S}_{I\setminus\{i\}} \xrightarrow{\sim} \mathfrak{S}_{\tilde{I}}$

Uncertainty retained from cancelled passages.

140-2-s552-D001 but
140-2-s552-D002 not
140-2-s552-D003 not
140-2-s552-D004 [illeg. 1]
140-2-s552-D005 equal
140-2-s552-D006 39
140-2-s552-D008 do not form a
140-2-s552-D009 system
140-2-s552-D010 4
140-2-s552-D011 4
140-2-s552-D012 4
140-2-s552-D013 4

† of the subgroups [scan 553]
[p. 280|1bis]

$$(31) \qquad \mathfrak{S}_3 \times \mathfrak{S}_{n-3} \subset \mathfrak{S}_n$$

of $\mathfrak{S}_n$ on $M'_{0,n}$, and not of that|those of the whole of $\mathfrak{S}_n$. In the case $n = 4$, the operation of $\Gamma'_2 = \mathrm{Sl}(2, \mathbb{F}_2) \simeq \mathfrak{S}_3$ on $M'_{0,4}$ which interests us, is the one which corresponds to the subgroup $\mathfrak{S}_1 \times \mathfrak{S}_3 \subset \mathfrak{S}_4$, which fixes the first element (corresponding to the zero section of an elliptic curve — † and which permutes the three following ones corresponding to the three elements of order 2 of a geometric elliptic curve in characteristic $\neq 2$, which are the three other poin^{t}s fixed by $x \mapsto -x$, whereas the naive|naive operation of $\mathfrak{S}_3$ on $U_{0,3}$ corresponds to the subgroup $\mathfrak{S}_3 \times \mathfrak{S}_1$ of $\mathfrak{S}_4$, which fixes the first three elements. A little question of scheme-theoretic gymnastics (independent|independent now of the geometry of elliptic curves, but a matter of projective

geometry of dimension 1): how to pass from one of these †actions of $\mathfrak{S}_3$ to the other? We feel like proving that they are the same!

Uncertainty retained from cancelled passages.

140-2-s553-D001 of

140-2-s553-D002 which

140-2-s553-D003 representations

Let us note that if I is a set with 4 elements, then associated to it is a three-element set of "partitions of type $(2,2)$" $P_{2,2}(I) = \tilde{I}$, and the homomorphism [scan 554]

$$\begin{array}{ccc} \mathfrak{S}_I & \longrightarrow & \mathfrak{S}_{\tilde{I}} \\ \wr\vert & & \wr\vert \\ \mathfrak{S}_4 & & \mathfrak{S}_3 \end{array} \tag{32}$$

is surjective — its kernel is a subgroup isomorphic (not canonical[ly]) to $\mathbb{F}_2 \times \mathbb{F}_2$ — hence a commutative group $V(I)$ killed by 2, i.e. a vector space over $\mathbb{F}_2$, and as such of dimension 2.

In fact, $V(I)$ operating on I makes I a $V(I)$ torsor — so I is canonically equipped with a structure of|of an affine †plane over $\mathbb{F}_2$, and $V(I) \subset \mathfrak{S}_I$ is its group of translations, $\mathfrak{S}_I$ being interpreted|is interpreted as the group of affine automorphisms of I (every permutation of I is affine — there † is on the set I 1 and only 1 affine plane structure over $\mathbb{F}_2$), and $\mathfrak{S}_{\tilde{I}}$ as the group $\mathrm{Aut}(V(I))$ — which suggests interpreting $\tilde{I}$ as the set of the three nonzero elements of $V(I)$, and indeed, we observe that † we have a

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Uncertainty retained from cancelled passages.

140-2-s554-D001 space

140-2-s554-D002 not

140-2-s554-D003 the

[scan 555]

coming from $I \setminus \{i\} \xrightarrow{\sim} \tilde{I}$. [p. 282]

In particular, for $I = [1,2,3,4]$, to compare the action of $\mathfrak{S}_{\{1,2,3\}}$ and of $\mathfrak{S}_{\{2,3,4\}}$ on $M'_{0,4} \simeq U_{0,3}$ — where the first is the standard action of $\mathfrak{S}_3$ on $U_{0,3}$, the second that of $\mathfrak{S}_3$ on $M_{1,1}[2]'$, we use the isomorphism

$$\mathfrak{S}_{\{1,2,3\}} \simeq \mathfrak{S}_{\{2,3,4\}} \tag{38}$$

made explicit in (37), corresponding to the bijection

$$1 \mapsto 4, \quad 2 \mapsto 3, \quad 3 \mapsto 2$$

of the index sets, which therefore corresponds to the automorphism

$$\mathrm{int}(\sigma_1) : \mathfrak{S}_3 \longrightarrow \mathfrak{S}_3 \tag{39}$$

now identifying the two index sets $[1,2,3]$, $[2,3,4]$ with $\{0,1,\infty\}$, and denoting by $\sigma_i \in \mathfrak{S}_{\{0,1,\infty\}}$ (for $i \in \{0,1,\infty\}$) the transposition of the elements $\neq i$. Thus

Corollary The canonical isomorphism

$$U_{0,3} \xrightarrow{\wp} M_{1,1}[2]' \qquad (\text{over } \mathbb{Z}[1/2]),$$

when we let $\mathfrak{S}_3 = \mathfrak{S}_{\{0,1,\infty\}}$ operate on both sides, is compatible with its operations, via the automorphism $\mathrm{int}(\sigma_1|\sigma_i)$, i.e. [scan 556]

$$\varphi(u.x) = (\sigma_1|\sigma_i\, u\, \sigma_1^{-1}|\sigma_i^{-1})\varphi(x) \tag{40}$$

We must all the same prove proposition 4, which is equivalent of course to this

Corollary Let X be a relative projective line over scheme S, I set with four elements, $(\dagger s_i)_{i\in I}$ a family of mutually|mutually disjoint sections, and σ an even †involution of I (corresponding to a partition $I = \{i_1, i_2\}\cup\{i_3, i_4\}$ by $\sigma i_1 = i_2$, $\sigma i_2 = i_1$, $\sigma i_3 = i_4$, $\sigma i_4 = i_3$). Then $\exists$ automorphism (obviously unique) u of X, such that $u \circ s_i = s_{\sigma i}$.

Proof We may suppose $X = \mathbb{P}^1_S$, $s_1|s_{i_1} = 0$, $s_{i_2} = \infty$, hence $s_{i_2}|s_{i_4}, s_{i_3}$ identify with sections of $\mathcal{O}^*_S$, we will take † u defined by

$$u(z) = \lambda/z|Z \qquad \lambda \in \Gamma(S, \mathcal{O}^*_S)$$

which exchanges 0 and ∞, and the condition for exchanging s_2, s_3 will be|is written $\lambda/s_3 = s_4$, $\lambda/s_4 = s_3$ i.e. $\lambda = s_3 s_4$ (N.B. it sufficed|would suffice that (s_1, s_2, s_3) and $(s_1, s_2|s_1, s_4)$ are|be mutually disjoint — no need for s_3, s_4 to be mutually disjoint ...)

Uncertainty retained from cancelled passages.

140-2-s556-D001 σ_i

140-2-s556-D002 transposition

140-2-s556-D003 [illeg. 1]

[scan 557]

Let us again work over $\mathbb{Q}$ (to fix ideas), and consider the fundamental groups of the modular multiplicities $M_{1,1}$, $M'_{1,1}$. We start from [p. 283]

$$\left\{\begin{array}{ccccc} M_{1,1} & \simeq & (M_{1,1}[n], \Gamma_n) & & (n \geq 3) \\ \downarrow & & \downarrow & & \\ M'_{1,1} & \simeq & (M_{1,1}[n], \underset{\Gamma_n/\pm 1}{\Gamma'_n}) & & (n \geq 2),\ \text{replacing for } n=2 \quad M_{1,1}[2] \text{ by } M_{1,1}[2]' \end{array}\right. \tag{41}$$

which gives|give isomorphisms of outer groups

$$\left\{\begin{array}{ccccc} \dagger\pi_1(M_{1,1}) & \simeq & \pi_1(M_{1,1}[n], \Gamma_n) & & (n \geq 3) \\ \downarrow & & \downarrow & & \\ \pi_1(M'_{1,1}) & \simeq & \pi_1(M_{1,1}[n], \Gamma'_n) & & (n \geq 2),\ \text{with same grain of salt for } n=2) \end{array}\right. \tag{42}$$

since $M_{1,1}[n]$ is connected (even if it is not geometrically connected), we find [m619]

Marginal note. [m619] this is Kronecker's theorem on the irreducibility of the cyclotomic equation

$$\left\{\begin{array}{ccccccccc} 1 & — & \pi_1(M_{1,1}[n]) & \to & \pi_1(M_{1,1}\dagger) & — & \Gamma_n & \to & 1 \\ & & \downarrow \sim & & \downarrow & & \downarrow & & \\ 1 & — & \pi_1(M_{1,1}[n]) & \to & \pi_1(M_{1,1}\dagger') & — & \Gamma'_n & — & 1 \end{array}\right. \tag{43}$$

which allows us to reconstruct|reconstruct $\pi_1(M_{1,1})$, as an extension of Γ_n, when we know|know $\pi_1(M'_{1,1})$ as an extension of Γ'_n, by inverse image via $\Gamma_n \to \Gamma'_n$. We find †as is only right

Proposition $\pi_1(M_{1,1})$ is a central extension of $\pi_1(M'_{1,1})$ by $\{\pm 1\}$.

On the other hand, we have the exact sequences, † of homotopy over $\operatorname{Spec}\mathbb{Q}$

Uncertainty retained from cancelled passages.

140-2-s557-D001 {

140-2-s557-D002 [n]

140-2-s557-D003 [n]

140-2-s557-D004 [illeg. 2]

140-2-s557-D005 [illeg. 5]

[scan 558]

$$(44)\qquad \begin{array}{ccccccccc} 1 & \longrightarrow & \pi_1(\overline{M}_{1,1}) = \widehat{\mathcal{T}}^{+}_{1,1} & \longrightarrow & \pi_1(M_{1,1}) = \mathcal{E}_{1,1} = N_{1,1} & \longrightarrow & \Gamma_{\mathbb{Q}} & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & \pi_1(\overline{M'_{1,1}}) & \longrightarrow & \pi_1(M'_{1,1}) & \longrightarrow & \Gamma_{\mathbb{Q}} & \longrightarrow & 1 \end{array}$$

and the transcendental theory provides canonical outer isomorphisms

$$(45)\qquad \begin{cases} \pi_1(\overline{M}_{1,1}) & \simeq & \mathrm{Sl}(2,\mathbb{Z})^{\wedge} \\ \downarrow & & \downarrow \\ \pi_1(\overline{M'_{1,1}}) & \simeq & \mathrm{Sl}(2,\mathbb{Z})'^{\wedge} = \mathrm{Sl}(2,\mathbb{Z})^{\wedge}/\pm 1, \end{cases}$$

m620

Marginal note. m620 Careful not to confuse $\mathrm{Sl}(2,\mathbb{Z})^{\wedge}$ with $\mathrm{Sl}(2,\widehat{\mathbb{Z}})$!!!

†

We have the following essential compatibility between (43) (44) (45): the commutativity|compatibility of

$$(46)\qquad \begin{array}{ccccccccc} 1 & \longrightarrow & \pi_1(\overline{M}_{1,1}) & \longrightarrow & \pi_1(M_{1,1}) & \longrightarrow & \Gamma_{\mathbb{Q}} & \longrightarrow & 1 \\ & & \wr\downarrow & & \downarrow & \dagger & \downarrow & & \\ & & \mathrm{Sl}(2,\mathbb{Z})^{\wedge} & & & & \chi_n \text{ cyclotomic character} & & \\ & & \downarrow \text{surjective} & & & & & & \\ 1 & \longrightarrow & \mathrm{Sl}(2,\mathbb{Z}/n\mathbb{Z}) & \hookrightarrow & \Gamma_n = \mathrm{Gl}(2,\mathbb{Z}/n\mathbb{Z}) & \xrightarrow{\det} & (\mathbb{Z}/n\mathbb{Z})^* & \longrightarrow & 1 \end{array}$$

Passing to the limit over n, this gives a commutative diagram

$$(47)\qquad \begin{array}{ccccccccc} 1 & \longrightarrow & (\pi_1(\overline{M}_{1,1}) \simeq \mathrm{Sl}(2,\mathbb{Z})^{\wedge}) & \longrightarrow & \pi_1(M_{1,1}) & \longrightarrow & \Gamma_{\mathbb{Q}} & \longrightarrow & 1 \\ & & \text{surjective}\downarrow & & \downarrow & & \downarrow\chi & & \\ 1 & \longrightarrow & \mathrm{Sl}(2,\widehat{\mathbb{Z}}) & \hookrightarrow & \mathrm{Gl}(2,\widehat{\mathbb{Z}}) & \longrightarrow & \widehat{\mathbb{Z}}^* & \longrightarrow & 1 \end{array}$$

– diagram to which we will return.

Uncertainty retained from cancelled passages.
140-2-s558-D001 whence [illeg. 1]
140-2-s558-D002 $\longrightarrow \Gamma_n = \mathrm{Gl}(2, \mathbb{Z}_n)$

[scan 559]

For now I am going to exploit the second isomorphism (42), $n = 2$: [p. 284]

$$(48) \qquad \begin{aligned} \pi_1(M'_{1,1}) &\simeq \dagger N'_{1,1} \simeq \pi_1(M_{1,1}[2]', \Gamma_2 = \mathfrak{S}_3) \\ &\simeq \pi_1(U_{0,3}, \mathfrak{S}_3), \end{aligned}$$

where the last isomorphism corresponds to the isomorphism $\mathrm{int}(\sigma_1) : \mathfrak{S}_3 \simeq \mathfrak{S}_3$ made explicit in (40).

We do indeed find, as foreseen in the preceding § – and in an entirely conceptual way, the isomorphism of extensions

$\dagger$

$$(49) \qquad \begin{array}{ccccccccc} 1 & \to & \pi_1(\overline{M'_{1,1}}) = \widehat{\mathcal{T}}^{+\prime}_{1,1} \simeq \mathrm{Sl}(2,\mathbb{Z})^\wedge & \to & \pi_1(M'_{1,1}) \simeq N'_{1,1} & \to & \mathbb{\Gamma}_{\mathbb{Q}} & — & 1 \\ & & \sim\downarrow & & \sim\downarrow & & \| & & \\ 1 & \longrightarrow & \pi_1(\overline{U}_{0,3}; \mathfrak{S}_3) & \longrightarrow & \pi_1(U_{0,3}, \mathfrak{S}_3) & — & \mathbb{\Gamma}_{\mathbb{Q}} & — & 1 \\ & & \simeq & & \simeq & & & & \\ & & \widehat{\Pi}'_{0,3} & & \mathcal{E}'_{0,3} & & & & \end{array}$$

Besides, taking up these reflections again in the transcendental context, we see that this isomorphism

$$(50) \qquad \begin{array}{ccc} \widehat{\Pi}'_{0,3} & \xrightarrow{\sim} & \pi_1(\overline{M'_{1,1}}) \\ \| & & \simeq \\ \widehat{\Pi}_{0,3}/\{\ell_1^2, \ell_\infty^3\} & & \mathrm{Sl}(2,\mathbb{Z})'^\wedge = \mathrm{Sl}(2,\mathbb{Z})^\wedge/\pm 1 \end{array}$$

is associated with an isomorphism

$$(51) \qquad \boxed{\begin{array}{ccc} \Pi'_{0,3} & \xrightarrow{\sim} & \mathrm{Sl}(2,\mathbb{Z})' \simeq \mathrm{Sl}(2,\mathbb{Z})/\pm 1 \\ \| & & \\ \Pi_{0,3}/\{\ell_1^2, \ell_\infty^3\} & & \end{array}}$$

Uncertainty retained from cancelled passages.
140-2-s559-D001 $\mathcal{T}^+_{1,1}$
140-2-s559-D004 49 $\widehat{\mathcal{T}}^{+\prime}_{1,1}$

[scan 560]

deduced from the isomorphism of complex analytic multiplicities

$$(52) \qquad M^{\mathrm{an}\prime}_{1,1} \simeq (U^{\mathrm{an}}_{0,3}\dagger, \mathfrak{S}_3)$$

We should all the same make the homomorphism (51) explicit – which is defined only modulo inner automorphisms, a priori, by explicit formulas – whereas its definition here comes out in a purely conceptual, "geometric" way (in the sense of relative algebraic geometry, of rational

and elliptic curves over arbitrary analytic spaces). This is a key calculation, surely instructive, which should not present any particular difficulty…

I have somewhat dropped along the way in all this the "coarse" modular scheme $\widetilde{M}_{1,1}$, after having asserted that it is $\simeq \mathbb{E}^1_{\mathbb{Z}}$, and that there are two marked sections, whose values in characteristic 0 correspond † two isomorphism classes of anharmonic elliptic curves. In terms of the isomorphism

Uncertainty retained from cancelled passages.
140-2-s560-D001 (ℂ)
140-2-s560-D003 to the

[scan 561]
[p. 285]

$$(53) \qquad \widetilde{M}_{1,1} \simeq M_{1,1}[2]'/\Gamma'_2 \simeq U_{0,3}/\mathfrak{S}_3$$

one|now (valid over $\operatorname{Spec}\mathbb{Z}[1/2]$)

one does indeed find, over $S = \operatorname{Spec}\mathbb{Z}[1/2]$, that $\widetilde{M}_{1,1}$ is obtained from a relative scheme Y, which is † locally (étale) isomorphic to $\mathbb{P}^1_S$, by removing a section s (whose existence already implies that Y is globally|globally isomorphic to $\mathbb{P}^1_S$ - hence $Y - s(S)$ isomorphic to $\mathbb{E}^1_S$. In this approach one thus feels like denoting by ∞ (not by 0, as in the theory of maps!) this section, which corresponds to the "points at infinity" $(0, 1, \infty)$ of $U_{0,3}$, or again to the "point at infinity" of $\widetilde{M}_{1,1}$. It is moreover easy to check a priori that (by the Deligne-Mumford compactification of $M_{1,1}$ and of $\widetilde{M}_{1,1}$) that $\widetilde{M}_{1,1}$ over the whole of $\operatorname{Spec}\mathbb{Z}$ is of the form $Y - \operatorname{Im} s$, where Y is smooth proper over $\operatorname{Spec}\mathbb{Z}$, and s a section - hence what one knows for example about the generic fiber, and the fact that $\mathbb{Z}$ is principal, imply that $Y \simeq \mathbb{P}^1_{\mathbb{Z}}$, and

Uncertainty retained from cancelled passages.
140-2-s561-D001 [illeg. 1]

[scan 562]

that one therefore has $\widetilde{M}_{1,1} \simeq \mathbb{E}^1_{\mathbb{Z}}$. To choose this isomorphism, one uses the two sections of $\mathbb{E}^1_{\mathbb{Z}}$, corresponding to the anharmonic curves. These two sections are not disjoint, they meet in characteristic 2 and characteristic 3. (As for what happens in characteristic $\neq 2$ i.e. over $\operatorname{Spec}\mathbb{Z}[1/2]$, one sees it clearly from the isomorphism 53 - for in characteristic 3 the two orbits

$$(2, -1, 1/2) \quad \text{and} \quad (j, \bar{j})$$

coincide and collapse into one and the same point). But taking one of these sections as zero section (unless I am mistaken it is the one corresponding to the elliptic curve richest in symmetries, namely|i.e. of|the group $\mathbb{Z}/6\mathbb{Z}$, that one takes - it is a matter of convention, of course - it is therefore also the orbit $\{j, \bar{j}\}$, corresponding to a stabilizer $\simeq \mathbb{Z}/3\mathbb{Z}$) - this determines the isomorphism $\widetilde{M}_{1,1} \simeq \mathbb{E}^1_{\mathbb{Z}}$ up to sign. The other section then becomes an integer of the form $\pm 2^a 3^b$ ($a, b \in \mathbb{N}^*$ - integers well determined whose value I|I have do not know by heart), and one finishes

[scan 563]
[p. 286]

normalizing by requiring that the sign be plus. Thus

Theorem (for the record - this is well known) There is a unique isomorphism

$$(54) \qquad \widetilde{M}_{1,1} \simeq \mathbb{E}^1_{\mathbb{Z}},$$

which in characteristic 0 gives to the anharmonic curve with automorphism group $\mathbb{Z}/6\mathbb{Z}$ the †invariant 0, and to the one with group $\mathbb{Z}/4\mathbb{Z}$ an invariant † > 0 (which will necessarily be $2^a 3^b$, with $a, b \in \mathbb{N}^*$ well determined, but forgotten by me…). $b = 3$, cf. below

In terms of the isomorphism

$$\dagger \qquad \widetilde{M}_{1,1} \simeq U_{0,3}/\mathfrak{S}_3 \subset \mathbb{P}^1/\mathfrak{S}_3$$

(valid over $\operatorname{Spec}\mathbb{Z}[1/2]$) this corresponds to an isomorphism between $\mathbb{P}^1/\mathfrak{S}_3$ and $\mathbb{P}^1$ which to the orbit $(0,1,\infty)$ assigns ∞ (not 0) and to the orbit $(j,\bar{j})$ assigns 0 (not ∞), to the orbit $(2,-1,1/2)$ the famous 2^a3^b (and not 1). The function

modular invariant

$$(55) \qquad j(\lambda) \in \mathbb{Q}(\lambda)$$

which realizes this isomorphism, i.e. the|the † morphism

$$(56) \qquad \mathbb{P}^1_{\mathbb{Z}} \xrightarrow{j} \mathbb{P}^1_{\mathbb{Z}}$$

corresponding (of degree 6, with

$$j \circ u = j \qquad (u \in \mathfrak{S}_3)$$

)

Uncertainty retained from cancelled passages.
140-2-s563-D002 [illeg. 1]
140-2-s563-D003 [illeg. 1]
140-2-s563-D004 (55)
140-2-s563-D006 morphism

[scan 564]

is therefore related|related to the one inspired by the theory of maps, i.e. $f(\lambda)$, by

$$(57) \qquad j(\lambda) = 2^a3^b f(\lambda)^{-1}.$$

Moreover $f(\lambda)$ (or $j(\lambda)$) is readily computed from knowledge of its zeros and its poles, with their multiplicities, † and the "normalization" by the value of f (or of j) on one of the sections $2,-1,1/2$; one finds immediately:

$$(58) \qquad f(z) = \frac{3^3}{2^2}\frac{z^2(z-1)^2}{(z^2-z+1)^3}$$

hence

$$(59) \qquad j(\lambda) = 2^{a+2}3^{b-3}\frac{(z^2-z+1)^3}{z^2(z-1)^2}.$$

Since $j(\lambda)$ must still make sense in characteristic 3 and not be constant, this shows moreover $b = 3$, so (59) can also be written

$$\dagger \qquad j(\lambda) = 2^{a+2}\frac{(z^2-z+1)^3}{z^2(z-1)^2}$$

- but one will recall|recalls that this formula is not relevant - does not allow one to compute †the invariant of an elliptic curve - except in characteristic $\neq 2$, fortunately! To †

Uncertainty retained from cancelled passages.
140-2-s564-D001 [illeg. 2]
140-2-s564-D002 (60)
140-2-s564-D003 discriminant
140-2-s564-D004 [illeg. 1]

[scan 565]

describe [p. 287]

$$(60) \qquad M_{1,1} \longrightarrow \mathbb{E}^1_{\mathbb{Z}}$$

m623

Marginal note. m623 and in particular to determine the other critical modular invariant $2^a 3^3$, i.e. determine a

also in a neighborhood of 2 - say in characteristic $\neq 3$ - one needs a study of elliptic curves which no longer goes by way of the representation (of Legendre, unless I am mistaken) $y^2 = \sqrt{x(x-1)(x-\lambda)}$, or again, by way of the Jacobi rigidification of level 2, but on the other hand by the one of level 3.

Remarks One sees at once that the quotient $\mathbb{P}^1_{\mathbb{Z}}/\mathfrak{S}_3$ identifies with $\mathbb{P}^1_{\mathbb{Z}}$, uniquely by taking as images of the orbits $(0, 1, \infty)$ and $(j, \bar{j})$ the sections 0 and ∞ (for example, following the convention corresponding to the theory of maps), and by taking as image of the orbit $(2, -1, 1/2)$ a rational which is > 0. This is possible a priori, for one notes that the orbits $(0, 1, \infty)$ and $(j, \bar{j})$ do not coincide in any characteristic, hence correspond to disjoint sections of $Y = \mathbb{P}^1_{\mathbb{Z}}/\mathfrak{S}_3$, whereas the remaining|constant orbit

[scan 566]

$(2, -1, 1/2)$ coincides with the first in characteristic 2, with the second in characteristic 3 (and these are the only coincidences that can occur). One therefore sees a priori that the third section will be of the form $2^\alpha/3^\beta$, with $\alpha, \beta \in \mathbb{N}^*$. The explicit calculation is moreover obvious, the composite morphism

$$(61) \qquad \mathbb{P}^1_{\mathbb{Z}} \xrightarrow{f_1^{*|\circ}} \mathbb{P}^1_{\mathbb{Z}}/\mathfrak{S}_3 \simeq \mathbb{P}^1_{\mathbb{Z}}$$

must be of the form $f_1^{*|\circ} = c.f^{*|\circ}$, with the constant $c \in \mathbb{Q}$ chosen in such a way that $cf^{*|\circ}$ reduces properly in all characteristics (which determines c modulo|modulo the sign) and that $f_1(-1) =^\dagger cf^{*|\circ}(-1) = c > 0$ (which removes the indeterminacy of the sign). One then finds

$$(62) \qquad f_1(\lambda) = \frac{\lambda^2(\lambda-1)^2}{(\lambda^2-\lambda+1)^2} = \frac{2^2}{3^3} f(\lambda)$$

hence

$$(63) \qquad f_1(-1) = \frac{2^2}{3^3} \qquad (\text{so } \alpha = 2,\ \beta = 3).$$

But one will take note that in a neighborhood of characteristic 2, these calculations no longer concern $\widetilde{M}_{1,1}$ and $\widetilde{\widetilde{M}}|\widetilde{\widetilde{M}}_{1,1}$, where the configuration of the three sections $\dagger$ is not the same as here...

Uncertainty retained from cancelled passages.

140-2-s566-D002 [illeg. 1]

140-2-s566-D004 [illeg. 1] $\Delta_\infty, \Delta_0|\Lambda_\infty, \Lambda_0$

[scan 567]

Following the momentum of these reflections, it would be natural to look more closely at the [p. 288] scheme

$$(64) \qquad M_{1,1}[4], \quad \text{on which acts } \Gamma_4 = \mathrm{Gl}(2, \mathbb{Z}/4)$$
$$\text{via} \quad \Gamma'_4 = \Gamma_4/\pm 1$$

and the homomorphism $\otimes$

$$(65) \qquad M_{1,1}[4] \longrightarrow M_{1,1}[2]' \simeq U_{0,3}$$

Compatible †

$$\Gamma'_4 = \mathrm{Gl}(2,\mathbb{Z}/4)/\pm 1 \longrightarrow \Gamma'_2 = \Gamma_2 \simeq \mathrm{Gl}(2,\mathbb{Z}/2) \simeq \mathfrak{S}_3$$

m624

Marginal note. m624

[Above the arrow, a crossed-out inscription: †. To the left of (65), a reference line connects the sign ⊗ to the marginal note.]

m625

Marginal note. m625 in fact the kernel of the homomorphism $\mathrm{Gl}(k,A)$†$\longrightarrow \mathrm{Gl}(k,A/J)$ is always a commutative group ⊛ if J is a square-zero ideal of a ring A - so here $\mathrm{Ker}(\mathrm{Gl}(2,\mathbb{Z}/4\mathbb{Z}) \longrightarrow \mathrm{Gl}(2,\mathbb{Z}/2\mathbb{Z}))$ is a commutative group, and even a vector space over $\mathbb{F}_2$ - it is isomorphic to $\mathbb{F}_2^4$

⊛ and even an A/J $(A/J)^4$

An immediate calculation, taking into account that

$$(\mathbb{Z}/4\mathbb{Z})^* \simeq \{\pm 1\}$$

, gives

$$\mathrm{card}\,\Gamma_4 = 2\underbrace{\mathrm{card}\,\mathrm{Sl}(2,\mathbb{Z}/4\mathbb{Z})}_{8\,\mathrm{card}\,\mathrm{Sl}(2,\mathbb{Z}/2\mathbb{Z})} = 16.\underbrace{\mathrm{card}\,\Gamma_2}_{6}$$

hence

$$(66) \qquad \mathrm{card}(\Gamma'_{4,2} = \mathrm{Ker}(\Gamma'_4 \longrightarrow \Gamma'_2)) = 8$$

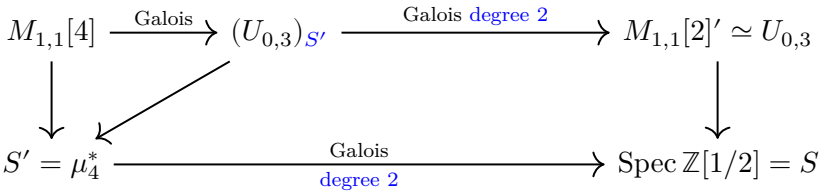
i.e. $M_{1,1}[4]$ is a Galois covering of $M_{1,1}[2]'$, of order 8. Let us note that it is not geometrically connected over $\mathbb{Q}$, or over $\mathrm{Spec}\,\mathbb{Z}[1/2]$, - since $M_{1,1}[4]$ lies over the quadratic extension μ_4^* †, of $\mathrm{Spec}\,\underbrace{\mathbb{Z}[1/2]}_{\Lambda}$ say, which is none other than $\Lambda[i]$. One has †a † factorization of $M_{1,1}[4] \longrightarrow M_{1,1}[2]'$ †

Uncertainty retained from cancelled passages.

140-2-s567-D001 [illeg. 1]
140-2-s567-D002 [illeg. 1]
140-2-s567-D003 [illeg. 1]
140-2-s567-D005 ≃ Spec
140-2-s567-D006 [illeg. 2]
140-2-s567-D007 Let us choose the point [illeg. 1]
140-2-s567-D008 over connected component

[scan 568]

(67)



where now

$$M_{1,1}[4] \longrightarrow M_{1,1}[2]'_{S'} = (U_{0,3})_{S'}$$

is a Galois covering with group the kernel of $\Gamma'_{4,2} \overset{\det}{\longrightarrow} \{\pm 1\}$, †or again that of $\mathrm{Sl}(2,\mathbb{Z}/4\mathbb{Z})' \longrightarrow \mathrm{Sl}(2,\mathbb{Z}/2\mathbb{Z})'$, which we shall denote by $S\Gamma'_{4,2}$, which is of order 4.

Proposition The group

$$(68) \qquad S\Gamma_{4,2} = \mathrm{Ker}(\mathrm{Sl}(2,\mathbb{Z}/4\mathbb{Z}) \longrightarrow \mathrm{Sl}(2,\mathbb{Z}/2\mathbb{Z}))$$

is isomorphic to † $\mathfrak{sl}(2,\mathbb{F}_2)$ (group of matrices of trace zero with coefficients in $\mathbb{F}_2$), with the obvious action|the obvious actions of $\mathrm{Sl}(2,\mathbb{F}_2)$ on it, and the subgroup ± 1 corresponds to scalar matrices - the quotient †$S\Gamma'_{4,2}$ of the two isomorphic to † $\mathbb{F}_2^2$ (in a silly way and not canonical at all!). m627

Marginal note. m627 N.B. It is the 4 complements of this subspace $\mathbb{F}_2 \hookrightarrow | \subset \mathfrak{sl}(2,\mathbb{F}_2)$ † which form a torsor under|over the dual of $V = \mathfrak{sl}/\mathbb{F}_2|\mathfrak{sl}(\mathbb{F}_2)$ which should correspond to the four † liftings of $\pi_{0,3} \hookrightarrow T^{+\prime}_{1,1} = \mathrm{Sl}(2,\mathbb{Z})'$, i.e.|to $\pi_{0,3} \hookrightarrow \mathrm{Sl}(2,\mathbb{Z})$ cf. pages 259, 260 above on these questions

† Viewing $M_{1,1}[4]$ as a Galois covering of $M_{1,1}[2]'_{S'} = (U_{0,3})_{S'}$, one finds geometrically "the" universal abelian covering of $U_{0,3}$ with group killed by 2 (and for good reason), which has as Galois group $(\mathbb{F}_2)^{\{0,1,\infty\}}$/Diagonale.

Uncertainty retained from cancelled passages.
140-2-s568-D001 [illeg. 1]
140-2-s568-D002 [illeg. 2]
140-2-s568-D003 is
140-2-s568-D004 [illeg. 1]
140-2-s568-D005 commutative
140-2-s568-D006 [illeg. 1]
140-2-s568-D007 finally

[scan 569]
[p. 289]

Finally, everything is essentially tautological - but an amusing calculation to do will be to "redescribe the action of $\mathrm{Sl}(2,\mathbb{F}_2)$ on $\mathfrak{sl}(2,\mathbb{F}_2)$/scalar matrices" in such a way as to identify it with the action of $\mathfrak{S}_3$ on $\mathbb{F}_2^{\{0,1,\infty\}}$/diagonal.

As a covering of †

$$(69) \qquad \widetilde{M}_{1,1\,S'} \simeq M_{1,1}[4]/S\Gamma'_4 \simeq M_{1,1}[2]'_{S'}/\Gamma'_2,$$

$M_{1,1}[4]$ is a Galois covering of order 24 (= 4.6), with group $\mathrm{Sl}(2,\mathbb{Z}/4\mathbb{Z})'$. It would be quite rare|welcome if one did not have an isomorphism

$$(70) \qquad \mathrm{Sl}(2,\mathbb{Z}/4\mathbb{Z})' \simeq \mathfrak{S}_4 \qquad \text{(automorphisms of the octahedron)}$$

and if $M_{1,1}[4]_{S'}$ were its covering † † of $\mathbb{P}^1_{S'} \setminus \{0,1,\infty\}$, with ramification compatible with $2\{1\}$ † $+3\{\infty\}$, which corresponds to the octahedron & in the theory of maps - one should have over $S' = \mathrm{Spec}\,\mathbb{Z}[1/2][i]$ a canonical isomorphism †

$$(71) \qquad M_{1,1}[4]_{S'} \simeq \text{Fermat curve} \quad x^2+y^2+z^2=0 \quad (\text{over } S')$$

m628

Marginal note. m628 it would be better to write the Fermat equation here $x_0^2 + x_1^2 + x_\infty^2 = 0$

(whose automorphism group is precisely an extension of $\mathfrak{S}_3$ by $\mu_2^{\{0,1,\infty\}}$/diag!).

Uncertainty retained from cancelled passages.
140-2-s569-D001 $\widetilde{M}$
140-2-s569-D003 étale
140-2-s569-D004 [illeg. 1]
140-2-s569-D005 [illeg. 1]

[scan 570]

To do things properly, one would have to make explicit the relation between elliptic curves E/S (over a scheme S over $\mathbb{Z}[1/2][i]$) equipped with a Jacobi rigidification of level 4 e_1, e_2 with $e_1 \wedge e_2 = [i]^{-1}$, and "nontrivial" solutions of the equation

$$(72) \qquad x^2 + y^2 + z^2 = 0$$

over S (i.e. the systems of sections $x, y, z \in \Gamma(S, \mathcal{O}_S^*)$ satisfying (72), modulo multiplication by a "scalar" $\alpha \in \Gamma(S, \mathcal{O}_S^*)$ - or again, breaking the symmetry by putting $X = x/z$, $Y = y/z$, † the solutions of the equation

$$(73) \qquad X^2 + Y^2 = -1 \qquad (= i^2)$$

$(X, Y \in \Gamma(S, \mathcal{O}_S^*))$...

One has thus interpreted $M_{1,1}[2]'$, and (modulo some checks and a somewhat more thorough study) $M_{1,1}[4]$, in relation with regular weighted triangular maps - the "dihedral|dihedral" map of index 3|of order 3 or universal weighted triangular map (three vertices $0, 1, \infty$,

[An almost circular outline, with a horizontal oval. Three filled dots are marked on the oval: 0 on the left, ∞ on the upper branch on the right, 1 on the lower branch on the right. The three portions of the oval join these three points.]

three edges, two faces which are triangles, type $(p, q) = (3, 3)$), and the octagonal map, which is a covering of order 4 of this one (†étale away from the vertices $\{0, 1, \infty\}$),

There would be reason also to look at

Uncertainty retained from cancelled passages.

140-2-s570-D001 [illeg. 1]

140-2-s570-D003 [illeg. 2]

[scan 571]

more closely the congruential modular varieties $M_{1,1}[3]$ and $M_{1,1}[5]$, corresponding to "geo- [p. 290]
metric" modular groups:

$$\left(\begin{cases} S\Gamma_3' = \underbrace{\mathrm{Sl}(2, \mathbb{F}_3)}_{\substack{\text{order } 4.3.2 \\ =24}} / \pm 1 \quad \text{of order } 12 \ {}_{\dagger}\subset \underbrace{\mathrm{GP}(1, \mathbb{F}_3)}_{\text{order } 24} \simeq \mathfrak{S}_4 \\ \text{in fact, we must have } S\Gamma_4' \simeq A_4 \\ S\Gamma_5' = \underbrace{\mathrm{Sl}(2, \mathbb{F}_5)}_{\substack{\text{order } 6.5.4 \\ =120}} / \pm 1 \quad \text{of order } 60 \subset \underbrace{\mathrm{GP}(1, \mathbb{F}_5)}_{\text{order } 120} \simeq \text{group of the biicosahedron} \end{cases} \right. \tag{74}$$

Unless I am mistaken, these groups are|† resp. the groups A_4 of the oriented tetrahedron, and the one A_5 of the oriented icosahedron. If I remember rightly, the cases $n = 2, 3, 4, 5$ exhaust the cases of congruential modular †curves $M_{1,1}[n]$ which are rational – it is a remarkable thing that through them one finds all the regular finite polyhedra with triangles for faces [m629]

Marginal note. [m629] except one, with "folded-back edges" – the one in figure (76), cf. below...

– it was clear a priori|a-priori, from the isomorphism|isomorphism between $\mathcal{T}_{1,1}'$ and the "triangula- oriented cartographic group" $\pi_{0,3}'$, that we should find again all the finite regular polyhedra with|† triangles for faces in this way, but not that this should be with congruence|congruences subgroups, very precisely. The table obtained is then the following

Uncertainty retained from cancelled passages.

140-2-s571-D001 ∞

140-2-s571-D002 are

140-2-s571-D003 varieties

140-2-s571-D004 tri

[scan 572]

group $G = S\Gamma'_n$	modular curve	type of regular oriented polyhedron
$D_3 \simeq \mathfrak{S}_3$	$M_{1,1}[2]'$	spherical bitriangle
A_4	$M_{1,1}[3]$	tetrahedron
$\mathfrak{S}_4$	$M_{1,1}[4]$	octahedron
A_5	$M_{1,1}[5]$	icosahedron

(75)

As other rational modular curves (not congruential) Galois over $\widetilde{M}_{1,1}$, there would still be the quotients of the preceding ones by invariant subgroups of the $S\Gamma'_n$, distinct from the whole group and from 1. One finds as possible quotients

a) Case $\mathfrak{S}_3$; the quotient ± 1 (via the sign)

b) Case A_4: the quotient $\mathbb{Z}/3\mathbb{Z} = C_3$ (via $\mathfrak{S}_4 \to \mathfrak{S}_3$, inducing $A_4 \to C_3$)

c) Case $\mathfrak{S}_4$: the quotient $\mathfrak{S}_3$, and the quotient $\{\pm 1\}$ of the latter.

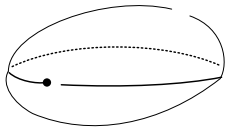
(For d), there is nothing, A_5 being a simple group)

†

The rational modular curves obtained are reduced to the two cases a) and b) (which are found again|found again in c)). In case a), one finds the quadratic covering of $Y = \mathbb{P}^1(\mathbb{C}) - \{0, 1, \infty\}$, which is unramified at ∞ (for the ramification index must divide 2 and 3), which therefore corresponds to the covering

$$\dagger \qquad y = \sqrt{x(x-1)}$$

and to the regular spherical map of type $(2, 1)$ (vertices of index 2, faces of index 1)



formed|formed of one vertex, on an edge (equator) bounding two faces which are monogons.

Case b) is the cyclic covering of order 3: ramified only at 0 and at ∞ (at 1 the ramification index must be a divisor of 2 and of 3, hence is 1) hence regular spherical map of

Uncertainty retained from cancelled passages.

140-2-s572-D001 The varieties

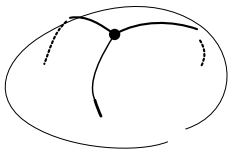
140-2-s572-D002 $\sqrt{x(x+1)}$

[scan 573]

[p. 291]

type $(3, 3)$, with edges which are loops †folded back

(76)



: 1 single vertex, three folded-back edges coming out of it, one triangular face. [These modular

varieties are too close to $\widetilde{M}_{1,1}$ – with an automorphism group (quotient of $\Gamma'_\infty = \mathrm{Sl}(2, Z|\mathbb{Z})$) too small, to be able to be rigidifying (the Galois-Poincaré group of the covering would have to be a multiple of 12(multiple of 4 and of 6 ?!), for † the modular curve to have a chance of being rigidifying. These cases thus seem distinctly less interesting.

There would be reason moreover, in each of the three remaining cases of (75) (thus setting aside the first case, which corresponds to $M_{1,1}[2]'$ and is more or less understood), to make explicit the relation between the points of the corresponding rational curves, and between elliptic curves with Jacobi rigidifications, of level $3, 4, 5$ respectively. And there are surely precise things in Klein, but one would like to do things a little finer, which are valid simultaneously in all characteristics|every characteristic i.e. over general base schemes. This concern seems intimately linked to the question of the operations of $\Gamma_{\mathbb{Q}}$ on the isomorphism classes of maps, and

Uncertainty retained from cancelled passages.
140-2-s573-D001 [illeg. 1]
140-2-s573-D002 [illeg. 1]

[scan 574]

more particularly of regular spherical maps. It is quite extraordinary that Jean Malgoire, in the three years that the question has been there, has not yet touched it – God knows it is juicy!

Supplement on the cross-ratio.

Let x_1, x_2, x_3, x_4 be †sections everywhere distinct of $\mathbb{P}^1_S$, we want to construct the corresponding section λ of $U|\mathbb{U}_{0,3} = \mathbb{P}^1_S - \{0, 1, \infty\}_S$. Let a, b, c, d be sections (local) of $\mathcal{O}_S$, with † $ad - bc$ invertible, such that

$$u = u_{a,b,c,d} : \quad z \longmapsto \frac{az + b}{cz + d} \tag{77}$$

transforms (x_1, x_2, x_3) into $(0, 1, \infty)$ – we shall therefore have

$$\lambda = u(x_4). \tag{78}$$

The conditions on $\xi|\dagger$ (a, b, c, d) for $ux_1 = 0$ etc. are

$$\begin{cases} ax_1 + b = 0 \\ ax_2 + b = cx_2 + d \\ cx_3 + d = 0 \end{cases}$$

which give, if for example a is invertible (otherwise, we can assume c invertible…) on solving for b, c, d

$$\begin{cases} b = -ax_1 \\ cx_2 + d = a(x_2 - x_1) \\ cx_3 + d = 0 \end{cases} \qquad \begin{vmatrix} c = a\dfrac{x_2 - x_1}{x_2 - x_3} \\ d = -a\dfrac{x_2 - x_1}{x_2 - x_3}x_3 \end{vmatrix}$$

whence

$$\lambda = \frac{ax_4 + b}{cx_4 + d} = \frac{a(x_4 - x_1)}{a\frac{x_2 - x_1}{x_2 - x_3}x_4 - a\frac{x_2 - x_1}{x_2 - x_3}x_3} = \frac{(x_4 - x_1)(x_2 - x_3)}{(x_2 - x_1)(x_4 - x_3)}$$

Uncertainty retained from cancelled passages.
140-2-s574-D001 at|or
140-2-s574-D002 [illeg. 1]
140-2-s574-D003 [illeg. 1]

[scan 575]
[p. 292]

$$\lambda(x_1, x_2, x_3, x_4) = \frac{(x_4 - x_1)(x_2 - x_3)}{(x_2 - x_1)(x_4 - x_3)} \tag{79}$$

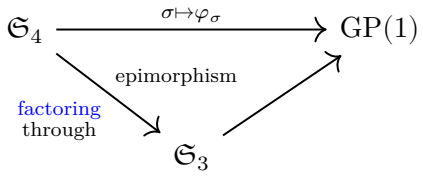
As a rational function in x_1, x_2, x_3, x_4, (with coefficients in a fixed integral domain) it is characterized by the two properties: a) invariance with respect to a (same) homographic transformation on the variables, x_1, x_2, x_3, x_4,

$$f(ux_1, ux_2, ux_3, ux_4) = f(x_1, x_2, x_3, x_4),$$

and b)

$$f(0, 1, \infty, x_4) = x_4$$

(with a grain of salt to give a meaning to the first side. We have moreover c) a remarkable symmetry property, which we † make explicit by defining a homomorphism from $\mathfrak{S}_4$ into the homographic group (morally|normally, GP(1)) †



in such a way that we have

$$\lambda(x_{\sigma^{-1}1}, x_{\sigma^{-1}2}, x_{\sigma^{-1}3}, x_{\sigma^{-1}4}) = \varphi_\sigma(\lambda(x_1, x_2, x_3, x_4)) \tag{80}$$

...

Uncertainty retained from cancelled passages.

140-2-s575-D002 [illeg. 1]

140-2-s575-D003 [illeg. 1]

Cote 140-3

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40 Topologico-isotopic paraphrase of Legendre's theory 307′

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42 Recapitulation on the group $\mathfrak{S}_{0,3} \simeq$ ... and on the loop automorphisms of $\widehat{\mathfrak{S}}^+_{0,3}$ and of $\widehat{\Pi}_{0,3}$: 331

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VIII Calculations in $\mathrm{Sl}(2,\widehat{\mathbb{Z}})'$: cocycles under σ_∞ 346

m631

Marginal note. m631 N.B. the calculations in these sections VI, VII, VIII partly replaced by those of §46, with more flexible|simple notations…

43 Digression on the 1-cocycles of cyclic groups 350

44 Return to notations(the $\rho|\beta|g, h, f, \alpha, \beta$ ...). Putting into equations in $\mathrm{Sl}(2,\widehat{\mathbb{Z}})$. 356

45 The homomorphism $\widetilde{M}_{0,3} \longrightarrow \mathrm{Gl}(2,\widehat{\mathbb{Z}})'$, and construction of the extension $\widetilde{N}_{1,1}$ of $\tilde{\Gamma}$ by $\widehat{\mathrm{Sl}(2,\mathbb{Z})}$. 384

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Uncertainty retained from cancelled passages.

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[scan 5]

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m636

Marginal note. m636 rectification of the "relation of conjugation" [illeg. 1] related †

9°) Recapitulation on the universal case. Structure of the $N^*_\sigma, N^*_\rho|N^\circ_\sigma, N^*_\rho$. The canonical homomorphism $M \xrightarrow{\sim} M^{\rightleftarrows|\leftrightarrows}_{\rho,\sigma}/H_\rho \hookrightarrow M_{\rho,\sigma}/H_\rho$. † 614

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m637

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4

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[scan 6]
[p. 291bis]

38 Explicit determination of the homomorphism $\Pi'_{0,3} \overset{\sim}{\to} \mathfrak{S}^+_{0,3}$ (and of $\Pi'^{\sim}_{0,3}$ (unoriented triangulated cartographic group) $\overset{\sim}{\to} \mathfrak{S}_{0,3}$).

I feel like coming back to the proposition of the preceding § (page 262) with connected surface X, discrete group G operating properly, the stabilizers G_x of the $x \in X$ † preserving the orientation and operating faithfully in a neighborhood of x … taking $Y = X/G$, equipped|equipped with the "signature" $\underline{d} = d(X/Y)$ defined by the ramification — whence an Isomorphism|isomorphism

$$(1) \qquad \Pi = \Pi_1((X,G),x) \overset{\sim}{\to} \Pi_1((Y,\underline{d}),y) = \Pi'$$

($x \in X$, y image of x in Y) which I would like to study more closely.

We have in $\Pi_1((Y,\underline{d}),y)$, for every $b \in \mathrm{Supp}\underline{d} = Y^!$, "truncated-loop subgroups", more precisely a conjugacy class of|of the subgroups quotients of $\mathbb{Z}/d_b\mathbb{Z}$, and more precisely still, a canonical outer homomorphism

$$(2) \qquad T_b|T_y \otimes_{\mathbb{Z}} \mathbb{Z}/d_b\mathbb{Z} \longrightarrow \Pi'$$

obtained by passage to the quotient from the outer homomorphism

$$(3) \qquad T_b|T_y \xrightarrow{k_y|k_b} \Pi_1(Y \setminus Y^!) \qquad (\longrightarrow \Pi_1((Y,\underline{d})))$$

(where T_b is the free $\mathbb{Z}$-module of rank 1 of "twisted integers" at b on Y). On the other hand, if $a \in X$ lies above b, the stabilizer G_a is † a cyclic group of finite order $d_a = d_b$ — We have = an obvious canonical isomorphism

$$(4) \qquad G_a \simeq T_a \otimes_{\mathbb{Z}} \mathbb{Z}/d_a\mathbb{Z} \simeq T_b \otimes \mathbb{Z}/d_b\mathbb{Z}.$$

(N.B. $T_a \simeq T_b$ canonically)

5

Uncertainty retained from cancelled passages.

140-3-s006-D001 being

140-3-s006-D002 thus

[scan 7]

On the other hand, to a is associated|associated a conjugacy class of splittings of the extension

$$(5) \qquad 1 \longrightarrow \Pi_1(X,x) \longrightarrow \Pi_1((X,G),x) \longrightarrow G \longrightarrow 1,$$

(cf. § 15)

$$(6) \qquad k_a : G_a \longrightarrow \Pi_1((X,G),x)$$

these classes of $\Pi_1(X,x|e)$-conjugacy of splittings (if $X \neq S^2$) correspond one-to-one to the fixed points of G_a operating on X. If we take the outer homomorphism defined|defined by (6) (thus conjugating|conjugating not only by the $\in \Pi_1(X)$, but by those of $\Pi_1(X,G)$ in its entirety) we find a 1-1 correspondence between the set of outer homomorphisms (6) which lift the inclusion

$$G_a \hookrightarrow G$$

(considered as an outer homomorphism), and the set of orbits of G in $X^!$, i.e. the points of $Y^!$. In summary

Proposition We have a bijection between $Y^!$ and the set of "maximal" outer homomorphisms $\varphi = \varphi_b : \Gamma \longrightarrow \Pi|\pi$ of subgroups $\Gamma \subset G$, † $\Gamma \neq 1$, into $\Pi_1(X,G)$ which lift the inclusion $\Gamma \subset G$ (the said Γ are necessarily cyclic, the Γ

6

Uncertainty retained from cancelled passages.
140-3-s007-D002 (cyclic|(finite

[scan 8]

in question corresponding|corresponding to a $b \in Y^!$ being the stabilizers G_a in G of the † [p. 292bis]
$a \in X_b = X^!_b$. Moreover, the composite of the homomorphisms

$$(7) \qquad T_b \otimes \mathbb{Z}/n\mathbb{Z} \overset{(4)}{\simeq} G_a \xrightarrow{\varphi_a = \varphi_b} \Pi_1(X, G) \overset{(1)}{\sim} \to \Pi_1((Y, \underline{d}))$$

is none other (we might have guessed!) than "the "truncated loops" homomorphism (2).

Let us now suppose that X is of the form $\widehat{X} \setminus S$, $\widehat{X}$ a compact surface, S a finite subset of $\widehat{X}$ †

$$(8) \qquad \begin{cases} X = \widehat{X} \setminus S \quad \text{thus} \quad Y = \hat{Y} \setminus T \quad \neq \emptyset \quad \text{with} \\ \hat{Y} = \widehat{X}/G \end{cases}$$

Let us consider, then, for every $a|b \in S$, the outer homomorphism

$$(9) \qquad k_a : T_a \longrightarrow \Pi_1(X) \hookrightarrow \Pi_1(X, G)$$

which I am going to compare with the composite

$$(10) \qquad \dagger \quad T_b \xrightarrow{k_b} \Pi_1(Y) \longrightarrow \Pi_1(Y, \underline{d})$$

(for $b = \hat{f}(a)$), by means of the canonical isomorphism

$$\dagger\ (11) \qquad T_a \simeq T_b$$

If d_a again denotes the ramification index at a, we shall have a commutative compatibility diagram: 7

Uncertainty retained from cancelled passages.
140-3-s008-D001 $a \in (X^!)_a | a \in (X^!)_b$
140-3-s008-D002 thus
140-3-s008-D003 k_b :
140-3-s008-D004 (12)

[scan 9]

$$(12) \qquad \begin{array}{ccccc} T_a & \xrightarrow{k_a} & \Pi_1(X) & \longrightarrow & \Pi_1(X, G) \\ \downarrow \wr \text{ can} & & & & \\ T_b & & & & \downarrow \wr \\ \downarrow d_a \cdot \mathrm{id} & & & & \\ T_b & & \xrightarrow{k_b} & & \Pi_1(Y, \underline{d}) \end{array}$$

We are going to apply this to the situation of $X = U^{\mathrm{an}}_{0,3}$, $G = \mathfrak{S}_3$, $\widehat{X} = \mathbb{P}^{1\mathrm{an}}$, $\hat{Y} \simeq \mathbb{P}^{1\mathrm{an}}$, with the map of passage to the quotient given|given by (page 286' (58))

$$(13) \qquad f(z) = \frac{3^3}{2^2} \frac{z^2(z-1)^2}{(z^2 - z + 1)^3}$$

$$f : \left\{ \begin{array}{l} 0 \\ 1 \\ \infty \end{array} \right. \longrightarrow 0 \qquad \left\{ \begin{array}{l} 2 \\ -1 \\ 1/2 \end{array} \right. \longrightarrow 1 \qquad \left\{ \begin{array}{l} -\bar{j} \\ -j \end{array} \right. \longrightarrow \infty$$

Thus we see, with the notations of , that the generators l'_0, l'_1, l'_∞ of $\Pi'_{0,3}$ (satisfying

$$(14) \qquad l'_\infty l'_1 l'_0 = 1 \qquad {l'_1}^2 = {l'_\infty}^3 = \cdot 1 \qquad)$$

correspond, respectively, under the isomorphism

$$(15) \qquad \begin{array}{c} \mathfrak{S}^+_{0,3} = \Pi_1(U^{\mathrm{an}}_{0,3}, \mathfrak{S}_3) \\ \text{cartographic group} \\ \text{triangulated weighted oriented} \\ \underline{\text{mixed}} \end{array} \quad \overset{\sim}{\to} \quad \begin{array}{c} \Pi'_{0,3} = \mathcal{T}^{+\prime}_{1,1} \\ \text{cartographic group triangulated|triangul-} \\ \text{oriented} \qquad \simeq \mathrm{Sl}(2,\mathbb{Z})' \end{array}$$

to the elements denoted †

$$(16) \qquad \left\{ \begin{array}{ll} \varepsilon_0, \varepsilon_1, \varepsilon_\infty & \longmapsto \quad l'_0 \\ (\text{such that } \varepsilon_i^2 = l_i) & \\ \sigma_0, \sigma_1, \sigma_\infty & \longmapsto \quad l'_1 \\ \rho & \longmapsto \quad l'_\infty \end{array} \right| \quad \begin{array}{l} \\ \\ \\ \\ \text{mod} \\ \text{conjugacy} \end{array}$$

or ρ^{-1} ?

8

Uncertainty retained from cancelled passages.

140-3-s009-D001 $\rho, \varepsilon_0, \varepsilon_1$ [illeg. 2]

[scan 10]

[p. 293]

†

This shows that we can find in $\Pi = \mathfrak{S}^+_{0,3}$ conjugates ε'_0 of ε_0, σ'_0 of σ_0, such that we have

$$(17) \qquad \rho\sigma'_0\varepsilon'_0 = 1, \quad \text{in addition to} \quad \rho^3 = {\sigma'_0}^2 = 1$$

and in such a way that these relations provide a presentation of Π (thus in particular, that $\varepsilon'_0, \sigma'_0, \rho$ generate } Π). Drawing inspiration, moreover, from the isomorphism which extends (15) and is obtained in like manner

$$(16) \qquad \mathfrak{S}_{0,3} \simeq \Pi_1 \left(U^{\mathrm{an}}_{0,3}, \underbrace{\mathfrak{S}_3 \times \mathbb{Z}/2|\mathbb{Z}_2}_{\mathcal{T}_{0,3}} \right) \longrightarrow \begin{array}{c} \mathcal{T}'_{1,1} \\ \text{cartographic group} \\ \text{triangulated non-} \\ \text{oriented} \\ \simeq \mathrm{Gl}(2,\mathbb{Z}) \end{array}$$

we should find on the first side three generators|generators $\tau'_0, \tau'_1, \tau'_\infty$ (corresponding to the analogous generators $\sigma_0, \sigma_1, \sigma_\infty$ of the corresponding|corresponding cartographic groups) satisfying|satisfying

$$(17) \qquad {\tau'_0}^2 = {\tau'_1}^2 = {\tau'_\infty}^2 = 1, \quad \left(\begin{array}{c} \underbrace{\tau'_{0|\infty}\tau'_\infty} \\ \text{corresponding to} \\ l'_1 \end{array} \right)^2 = \left(\begin{array}{c} \underbrace{\tau'_1\tau'_0} \\ \text{corresponding to} \\ l'_\infty \end{array} \right)^3 = 1$$

in such a way that these are † some defining relations for $\mathfrak{S}_{0,3}$. It would be necessary that we have

$$(18) \qquad \begin{array}{ccccccc} \tau'_\infty\tau'_1 & = & \varepsilon'_0; & \tau'_0\tau'_\infty & = & \sigma'_0, & \tau'_1\tau'_0 & = & \rho \\ & & \updownarrow & & & \updownarrow & & & \updownarrow \\ & & l'_0 & & & l'_1 & & & l'_\infty \end{array}$$

9

Uncertainty retained from cancelled passages.
140-3-s010-D001 The isomorphism
140-3-s010-D003 some

[scan 11]

X topological surface
$\mathcal{G}$ discrete group operating on X
G invariant subgroup of $\mathcal{G}$, operating properly on X, so that the stabilizers G_x $(x \in X)$ operate faithfully in a neighborhood of x, while preserving the local orientation at x †
$Y = X/G$ (topological surface), $f\colon X \to Y$
$X^!$ discrete subset|subset of X, stable under $\mathcal{G}$,
d_X "signature" on X with support $X^!$, invariant under $\mathcal{G}$

$$i \mapsto d_X(i) \qquad X^! \longmapsto \mathbb{N}$$

$Y^! \simeq X^!/G \subset Y$ image of $X^!$ in Y
We assume $x \in X$, $G_x \neq \{1\} \Rightarrow x \in X^!$

$$d_Y\colon Y^! \to \mathbb{N}, \qquad d_Y(y) = \nu_x d_X(x)$$
$$\text{if } y = f(x), \quad \nu_x = [G_x]$$

$\mathrm{Rev}(X, d_X, \mathcal{G})$ category of ramified $\mathcal{G}$-coverings of X, subordinate to d_X
$\mathrm{Rev}(Y, d_Y, \mathcal{H})$ category of ramified $\mathcal{H}$-coverings of Y, subordinate to d_Y
For $X' \in \mathrm{Ob}\,\mathrm{Rev}(X, d_X, \mathcal{G})$, let us consider X'/G as a surface on which $\mathcal{H} = \mathcal{G}/G$ operates, over $Y = X'/G|X/G$. Then 10

Uncertainty retained from cancelled passages.
140-3-s011-D002 [illeg. 1]

[scan 12]
[p. 294]

a) $X'/G \in \mathrm{Ob}\,\mathrm{Rev}(Y, d_Y, \mathcal{H})$
b) We thus find|give a functor

$$\mathrm{Rev}(X, d_X, \mathcal{G}) \longrightarrow \mathrm{Rev}(Y, d_Y, \mathcal{H}) \tag{19}$$

c) This functor is an equivalence of categories.
d) Description of a quasi-inverse functor by "normalized" inverse image.
Remarks. We may assume d_X with values > 0 i.e. d_Y with values > 0 (otherwise, if $S = \{x \in X^! \mid d_X(x) = 0\}$, let us replace X by $X \setminus S$), Y by $Y \setminus T$ $(T = S/G)$.
We may assume $d_X(x) = 1 \Rightarrow [G_x] \neq 1$ (otherwise, replace $X^!$ by $X^! \setminus S$ $S = \{x \in X^! \mid d_X(x) = 1,\ [G_x] = 1\}$)
The most interesting case for us is the one where $\forall x \in X^!$, we have $d_X(x) = 1$ and $G_x \neq \{1\}$ (thus $X^! = \{x \in X \mid G_x \neq \{1\}\}$), thus|in the case where

$$\mathrm{Rev}(X, d_X, \mathcal{G}) = \mathrm{Rev}(X, \mathcal{G}) \tag{20}$$

and where $\mathrm{Rev}(\dagger Y, d_Y, \mathcal{H})$ can therefore be interpreted as a $\mathrm{Rev}(X, \mathcal{G})$ (without ramification data) 11

Uncertainty retained from cancelled passages.
140-3-s012-D002 [*illeg.* 1]

[scan 13]

Corollary Let $x \in X \setminus X^!$, y its image in $Y \setminus Y^!$, we have a canonical isomorphism

$$\Pi_1((X, d_X, \mathcal{G}), |; x) \xrightarrow{\sim} \Pi_1((Y, d_Y, \mathcal{H}), |; y) \tag{21}$$

We are going to make explicit this isomorphism in terms of calculations of paths.
An elementα of $\Pi_1((X, d_X, \mathcal{G}), |; x)$ comes from a pair (γ, ℓ), with

$$\gamma \in \mathcal{G}, \qquad \ell \text{ class of paths in } X \setminus X^! \text{ from } x \text{ to } \gamma^{-1}x$$

(with|for (γ, ℓ) and (γ', ℓ') define the same element, iff|sss $\gamma' = \gamma$, and the element $\ell'^{-1}\ell$ of $\Pi_1(X \setminus X^!, x)$ is in the invariant subgroup kernel of

$$\Pi_1(X \setminus X^!, x) \longrightarrow \Pi_1((X, d_X), x)$$

– i.e. the invariant subgroup generated by the † $\lambda_a^{d_X(a)}$, where for every $a \in X^!$, λ_a denotes the class in $\Pi_1(X \setminus X^!, x|a)$ of a "loop from x around a").

Then let $\dot{\gamma}$ be the image of $\mathcal{G}$ in $\mathcal{H}$ – † let $f_U \colon X \setminus X^! \overset{\text{df|def}}{=} U \longrightarrow Y \setminus Y^! \overset{\text{df|def}}{=} V \simeq U/G$ be the map 12

Uncertainty retained from cancelled passages.
140-3-s013-D002 $k|\ell$
140-3-s013-D003 $f_0|f$

[scan 14]
[p. 295]

which is canonical, then the image β of $\alpha \in \Pi_1(X, d_X, \mathcal{G}; x)$ defined by (γ, ℓ) in $\Pi_1(Y, d_Y, \mathcal{H}; y)$ is $(\dot{\gamma}, f_U(\ell))$
(N.B. $f_U(\ell)$ is a class of paths in $V = Y \setminus Y^!$ from $y = f_U(x)$ to $\dot{\gamma}^{-1}(y) = f_U(\gamma^{-1}x)$).
N.B. We see at once that if (γ, ℓ) and (γ', ℓ') define the same α, then they also define the same β, which comes from the fact that

$$\Pi_1(f_U, a) : \Pi_1(X \setminus X^!, x) \longrightarrow \Pi_1(Y \setminus Y^!, y)$$

maps a λ_a into a $\lambda_b^{\nu(a)}$ ($b = f(a)$, $\nu(a) = [G_a]$ = degree of ramification of f at a) hence $\lambda_a^{d_X(a)}$ into $\lambda_b^{d_X(a)\nu(a) = d_Y(b)}$.
Let us return to the situation of (13)

$$X = U_{0,3}^{\text{an}}, \quad \mathcal{G} = \mathcal{T}_{0,3} = \mathfrak{S}_3 \times \mathbb{Z}/2, \quad G = \mathcal{T}_{0,3}^{+} = \mathfrak{S}_3$$

m650

Marginal note. m650 $X^! = \{\dagger 2, -1, \frac{1}{2}; -\bar{j}, -j\}$

$$Y \simeq \mathbb{P}^{1\,\text{an}} \setminus \{0\}, \quad Y^! = \{1, \infty\},$$

$\mathcal{H} = \mathbb{Z}/2\mathbb{Z}$, map of passage to the quotient from $\widehat{X} = \mathbb{P}^{1\,\text{an}}$ to $\widehat{Y} = \mathbb{P}^{1\,\text{an}}$ given by (13)

$$f(z) = \frac{3^3}{2^2} \frac{z^2(z-1)^2}{(z^2 - z + 1)^3}$$

13

Uncertainty retained from cancelled passages.
140-3-s014-D002 [*illeg.* 1]

[scan 15]

To calculate $\Pi_1(Y, \underline{d}_Y, \mathbb{Z}/2\mathbb{Z}) = \widetilde{\Pi}'_{0,3}$ (carto-graphic triangulated unoriented group), we took the base point $y = -\bar{j}$, so we are going to choose an $x \in U^{\mathrm{an}}_{0,3}$ over y, such that

$$f(x) = y = -\bar{j}$$

— there is † a choice in a set $f^{-1}(\{y\})$ with six elements, of which three in the northern hemisphere (the other three in the southern hemisphere), these are permuted transitively by ρ. They lie respectively in the spherical triangles $(-\bar{j}, 0, \frac{1}{2})$, $(-\bar{j}, 1, 2)$, $(-\bar{j}, \infty, -1)$ (mapped by f homeomorphically onto the upper spherical triangle $(0, 1, \infty)$, in the upper hemisphere) with center $-\bar{j}$). The choice of $x|a$ thus amounts to the choice of one of the three spherical triangles, or again of the 3 spherical segments $(0, \frac{1}{2})$, $(1, 2)$, $(\infty, -1)$ — I feel like taking $(0, \frac{1}{2})$, corresponding to the triangle $(-\bar{j}, 0, \frac{1}{2})$.

The three involutive generators † $\tau'_0, \tau'_1, \tau'_\infty$ of $\Pi_1(Y, d_Y, \mathbb{Z}/2\mathbb{Z}|\mathbb{Z}/2)$ over|over $\tau \in \mathbb{Z}/2\mathbb{Z}$ are obtained by taking paths $\ell'_0, \ell'_1, \ell'_\infty$ from north pole $y = -\bar{j}$ to the south pole $\tau y = -j$, which are the|some great circle arcs crossing the equator resp. at $2, -1, \frac{1}{2}$. 14

Uncertainty retained from cancelled passages.
140-3-s015-D001 places|points
140-3-s015-D002 [illeg. 1]

[scan 16]
[p. 296]

These paths are the images under f of paths on X, let them be $\lambda_0 : \overrightarrow{x, z_0}$, $\lambda_1 : \overrightarrow{x, z_1}$, $\lambda_\infty : \overrightarrow{x, z_\infty}$, which each cross the barycentric subdivision of the double spherical triangle $(0, 1, \infty)$ in exactly one point, situated respectively on the side $(\frac{1}{2}, -\bar{j})$, $(-\bar{j}, 0)$ and $(0, \frac{1}{2})$ of the spherical triangle $(0, \frac{1}{2}, -\bar{j})$.
m653

Marginal note. m653 N.B. The points z_0, z_1, z_∞ are characterized resp.^{t} as the only points of X over $-j = f(\dot{\tau}|\tau x)$, lying in|in the spherical triangles $(-\bar{j}, \frac{1}{2}, 1)$, $(0, -\bar{j}, -1)$, $(\frac{1}{2}, 0, -j)$.

The points † z_0, z_1, z_∞ over τy are respectively of the form $g_0 x = \dot{\sigma}_\infty \dot{\tau} x$, $g_1 x = \dot{\sigma}_0 \dot{\tau} x$, $g_\infty x = \dagger \dot{\tau} x$ (i.e. $g_0 = \dot{\sigma}_\infty \dot{\tau}$, $g_1 = \dot{\sigma}_0 \dot{\tau}$, $g_\infty = \dot{\tau}$). Thus the involutive generators $\tau'_0, \tau'_1, \tau'_\infty$ of $\widetilde{\Pi}'_{0,3} = \Pi_1(Y, d_Y, \mathbb{Z}/2\mathbb{Z}; y)$ are the images of the elements †, (denoted by the same sign)

$$\tau'_0 = (\dot{\sigma}_0 \dot{\tau}, \lambda_0); \quad \tau'_1 = (\dot{\sigma}_1 \dot{\tau}, \lambda_1), \quad \tau'_\infty = (\dot{\sigma}_\infty \dot{\tau}, \lambda_\infty)$$

of $\Pi_1(X, \mathcal{G}; x)$. We are going to use the great circle arc ℓ from $-\bar{j} = P$ (base point usual|usual for the calculation of $\Pi_1(X, \mathcal{G})$) to x, to identify $\Pi_1(X, \mathcal{G}; x)$ with $\Pi_1(X, \mathcal{G}; P)$.

We can then take †

$$\tau'_0 = (\dot{\sigma}_0 \dot{\tau}, \lambda'_0), \quad \tau'_1 = (\dot{\sigma}_1 \dot{\tau}, \lambda'_1), \quad \tau'_\infty = (\dot{\sigma}_\infty \dot{\tau}, \lambda'_\infty) \tag{21}$$

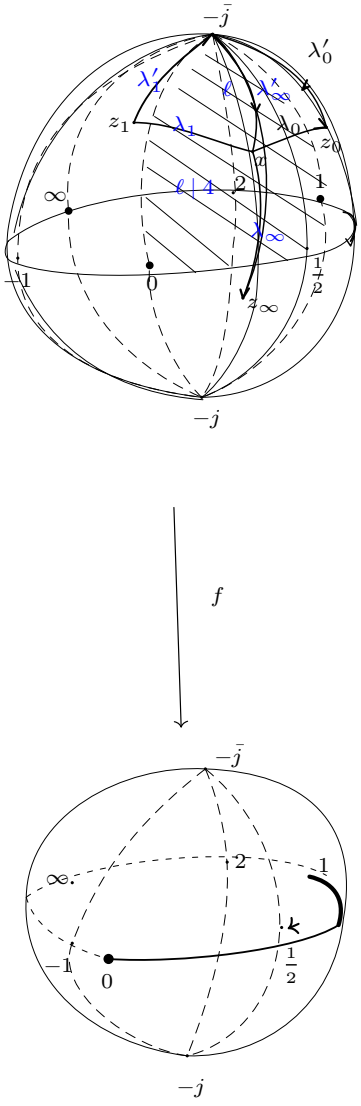
where $\lambda'_0, \lambda'_1, \lambda'_\infty$ are the † paths from † from $P = -\bar{j}$ to † $g_i P | \dot{g}_i P$ † † 15

Uncertainty retained from cancelled passages.
140-3-s016-D001 the
140-3-s016-D002 $\dot{\sigma}_\infty$

140-3-s016-D003 0
140-3-s016-D006 as
140-3-s016-D008 loops
140-3-s016-D009 loops|ends
140-3-s016-D010 [illeg. 1]
140-3-s016-D011 [illeg. 1]
140-3-s016-D012 respectively

[scan 17]

†
†



given by

$$\lambda'_i = (\dagger\dot{g}_i(\ell^{-1}))(\lambda_i)\ell.$$

We therefore have

$$\lambda'_0 = \underbrace{(\dot{\sigma}_\infty \dot{\tau})}_{\substack{\text{symmetry} \\ \text{with respect to the} \\ \text{plane } (\infty, -\bar{j}, \frac{1}{2}, -j)}} (\ell^{-1}) \circ \lambda_0 \circ \ell$$

all|is homotopic to the trivial loop in $U_{0,3}$

$$\lambda'_1 = \underbrace{\dot\sigma_0\dot\tau}_{\substack{\text{symmetry}\\ \text{with respect to the equator}\\ (0,-\bar j,2,-j)}} \qquad (\ell^{-1})\circ\lambda_1\circ\ell$$

homotopic to the trivial loop in $U_{0,3}$

$$\lambda'_\infty = (\underbrace{\tau}_{}(\ell^{-1}))\circ\lambda_\infty\circ\ell$$

homotopic to the arc of the great circle $(-\bar j, \frac{1}{2}, -j)$ from $P = -\bar j|\bar j$ toward $\dot\tau P = -j$ 16

Uncertainty retained from cancelled passages.

140-3-s017-D001 only^{t} other situated

140-3-s017-D003 characterized respectively lie $-\bar j$

140-3-s017-D004 $\tau_i \mid \ell_i$

[scan 18]

In summary, with the notations of [C] page 7 and following, we shall therefore have [p. 297]

$$\begin{cases} \tau'_0 = \tilde\sigma_\infty & (\text{over } \dot\sigma_\infty\dot\tau)\\ \tau'_1 = \tilde\sigma_0 & (\text{over } \dot\sigma_0\dot\tau)\\ \tau'_\infty = \tau_\infty & (\text{over } \dot\tau)\end{cases} \tag{22}$$

N.B. $\tau_\infty = \sigma_\infty\tilde\sigma_\infty = \tilde\sigma_\infty\sigma_\infty$

We must check the essential relations

$$\tau_0'^2 = \tau_1'^2 = \tau_\infty'^2 = (\underbrace{\tau'_0\tau'_\infty}_{\sigma_\infty})^2 = (\underbrace{\tau'_1\tau'_0}_{\rho^{-1}})^3 = 1 \tag{23}$$

the first four are clear, the last reads

$$(\tilde\sigma_0\tilde\sigma_\infty)^3 = 1$$

now we have

$$\tilde\sigma_0\tilde\sigma_\infty = \rho^{-1} \qquad ([C], \text{ formula 13,})$$

whence what we want.

Let us compute (in accordance with (18)) $\tau'_\infty\tau'_1$, $\tau'_0\tau'_\infty$, $\underbrace{\tau'_1\tau'_0}_{\rho^{-1}}$ — which will have to correspond to $\ell'_0, \ell'_1, \ell'_\infty$ —

$$\ell'_0 = \tau'_\infty\tau'_1 = \tau_\infty\tilde\sigma_0 = \sigma_\infty\underbrace{\tilde\sigma_\infty\tilde\sigma_0}_{\rho} = \sigma_\infty\rho$$
$$\ell'_1 = \tau'_0\tau'_\infty = \sigma_\infty$$
$$\ell'_\infty = \tau'_1\tau'_0 = \rho^{-1}$$

i.e.

$$\ell'_0 = \sigma_\infty\rho, \quad \ell'_1 = \sigma_\infty, \quad \ell'_\infty = \rho^{-1} \tag{24}$$

with the characteristic relations

$$\ell'_\infty\ell'_1\ell'_0 = 1, \qquad \ell'^2_{1|\infty} = 1, \qquad \ell'^3_\infty = 1 \tag{25}$$

Everything is OK!

m657

Marginal note. m657 We express the distinguished elements $\rho, \sigma_0, \sigma_1, \sigma_\infty, \ell_0, \ell_1, \ell_\infty, \tau_0, \tau_1, \tau_\infty$ of $\Pi_1(U^{\mathrm{an}}_{0,3}; \mathfrak{S}_3 \times \mathbb{Z}/2)$ in terms of the $\tau'_0, \tau'_1, \tau'_\infty$ as follows:

$$\rho = \tau'_0\tau'_1$$
$$\tau_\infty = \tau'_\infty, \quad \tau_0 = \rho\tau_\infty\rho^{-1} = \tau'_0\tau'_1\tau'_\infty\tau'_1\tau'_0,$$
$$\tau_1 = \rho^{-1}\tau_\infty\rho = \ldots = \tau'_1\tau'_\infty\tau'_1$$

$= \tau'_\infty\tau'_0$

$$\sigma_\infty = \tau'_0\tau'_\infty, \quad \sigma_0 = \rho\sigma_\infty\rho^{-1} = \tau'_0\tau'_1\tau'_0\tau'_\infty\tau'_1\tau'_0,$$
$$\sigma_1 = \rho^{-1}\sigma_\infty\rho = \tau'_1\tau'_\infty\tau'_0\tau'_1$$
$$\ell_0 = \tau_\infty\tau_1 = \tau'_\infty\tau'_1\tau'_\infty\tau'_1 = (\tau'_\infty\tau'_1)^2$$
$$\ell_1 = \tau_0\tau_\infty = \tau'_0\tau'_1\tau'_\infty\tau'_1\tau'_0\tau'_\infty$$
$$\ell_\infty = \tau_1\tau_0 = (\underbrace{\tau'_1\tau'_\infty\tau'_1}\,\tau'_0)^2$$
$$\varepsilon_0 = \ell'_0 = \dagger\tau'_\infty\tau'_1$$
$$\varepsilon_1 = \tau'_0\tau'_1\tau'_\infty\tau'_0$$
$$\varepsilon_\infty = \tau'_1\tau'_\infty\tau'_1\tau'_0$$

17

Uncertainty retained from cancelled passages.
140-3-s018-D002 $\tau'_1\tau'_\infty$

[scan 19]

I would also like to make explicit the relation of $\widetilde{\Pi}'_{0,3}$ unoriented carto-graphic group $\simeq \mathfrak{S}_{0,3}$ $= \mathrm{Aut}_{\mathrm{loc|ext}}(\Pi_{0,3})$ with $\mathrm{GP}(1,\mathbb{Z}) \simeq \mathrm{Gl}(2,\mathbb{Z})' \overset{\mathrm{def}}{=} \mathrm{Gl}(2,\mathbb{Z})/\{\pm 1\}$. Let us first recall the operation of

$$\mathrm{GP}(1,\mathbb{R}) \simeq \mathrm{Gl}(2,\mathbb{R})/\mathbb{R}^* \simeq \mathrm{Gl}(2,\mathbb{R})^{\pm 1}/\{\pm 1\} \quad \dagger \tag{26}$$

on the Poincaré half-plane D

$$D = \{z \in \mathbb{C} \mid \Im(z) > 0\} = \{x + iy \mid x \in \mathbb{R}, y \in \mathbb{R}^{*+}\}. \tag{27}$$

We make the matrix $u = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ operate by

$$f_u(z) = \begin{cases} \dfrac{az+b}{cz+d} & \text{if } \det u > 0 \\ \dfrac{a\bar{z}+b}{c\bar{z}+d} & \text{if } \det u < 0 \end{cases} \tag{28}$$

In this way, $\mathrm{GP}(1,\mathbb{R})$ is isomorphic to the group of conformal automorphisms of D.
m659

Marginal note. m659 N.B. $\mathrm{GP}(1,\mathbb{R})$ has two connected components. Its identity component $\mathrm{GP}(1,\mathbb{R})^+$ is the image of $\mathrm{Sl}(2,\mathbb{R})$ (or of|of $\mathrm{Gl}(2,\mathbb{R})^+$) and is identified with $\mathrm{Sl}(2,\mathbb{R})/\pm 1$.

Moreover, the datum of an elliptic curve E over $\mathbb{C}$ is equivalent to that of a space V of dimension 1 over $\mathbb{C}$, and of a "lattice" $\pi \subset V$ (discrete subgroup of rank 2). The canonical orientation of V corresponds to a generator $\eta|\gamma$ of $\bigwedge^2_{\mathbb{Z}} \pi$ — thus π is "oriented". The choice of a rigidification

equivalence of categories [illeg. 1].
18

Uncertainty retained from cancelled passages.
140-3-s019-D001 GP(|acts

[scan 20]

of unimodular Jacobi type for π, is equivalent to that of a basis (e_1, e_2) of π (as a $\mathbb{Z}$-module) such that $e_1 \wedge e_2 = \eta|\gamma$. The element e_1 (as a basis of V over $\mathbb{C}$) moreover allows us to identify [p. 298]

V with $\mathbb{C}$, e_1 becoming 1. Thus the datum of E with its unimodular Jacobi rigidification, is equivalent to that of an element $e_2 = \tau$ of $\mathbb{C}|E$, such that

1°) $(1, \tau)$ generate|generates a discrete subgroup of rank 1, i.e. $\tau \notin \mathbb{R}$ i.e. $\Im\tau \neq 0$

2°) the orientation of the said subgroup defined by the basis $(1, \tau)$ is "positive", i.e. $\Im\tau > 0$, i.e. $\tau \in D$.

When we look at the operation of $\mathrm{Sl}(2, \mathbb{Z})$ on the set of isomorphism classes of rigidified elliptic curves, set identified with D, we find that $\begin{pmatrix} a, b \\ c, d \end{pmatrix}$ operates by $\tau \longmapsto \dfrac{a\tau + b}{c\tau + d}$, i.e. it is the natural operation of $\mathrm{Sl}(2, \mathbb{Z})$ on D, through $\mathrm{Sl}(2, \mathbb{Z})' \simeq \mathrm{GP}(1, \mathbb{Z})^+$.

This group operates properly, It has exactly two critical orbits, which corres- 19

[scan 21]

pond respectively to the τ which with 1 generate the one lattice of algebraic integers of $\mathbb{Q}(i)$ (i.e. $\mathbb{Z} + i\mathbb{Z}$), and the lattice of integers of $\mathbb{Q}(j)$, made up of numbers of the form $aj + b\bar{j}$ ($j = \dfrac{-1 + i\sqrt{3}}{2} = \exp\dfrac{2i\pi}{3}$); the corresponding ramification is 2 and 3.

We have a canonical (analytic) isomorphism between

$$D/\mathrm{GP}(1, \mathbb{Z})^+ \simeq \mathbb{P}^{1\,\mathrm{an}} \setminus \{0\} = Y, \tag{29}$$

characterized by the condition that the critical orbit of ramification 3 correspond to the point ∞, and that of ramification 2 to the point 1. We thus find an isomorphism (putting $\mathcal{G} = \mathrm{GP}(1, \mathbb{Z})$, $G = \mathrm{GP}(1, \mathbb{Z})^+ = \mathrm{Sl}(2, \mathbb{Z})'$)

(30)

$$\begin{array}{ccc} \pi_1(D, \mathcal{G}, x) & \xrightarrow{\sim} & \pi_1^{\pm|*}(Y, \underline{d}; y) \\ \| & & \\ \mathcal{G} & & \end{array}$$

which I would like to make explicit, by identifying $\tau'_0, \tau'_1, \tau'_\infty$ as elements of $\mathrm{Gl}^{\pm'|\pm}(2, \mathbb{Z})'$. ($x \in D \setminus D^{\mathrm{I}}|D'$) 0 20 [scan 22] [p. 299]

39 †The congruence conditions for the outer operation of $\Gamma_{\mathbb{Q}}$ on $\widehat{\Pi}_{0,3}$.

Let us consider the diagram

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \widehat{\mathcal{T}}^+_{1,1} & \longrightarrow & N_{1,1} & \longrightarrow & \Gamma_{\mathbb{Q}} & \longrightarrow & 1 \\ & & \wr & & & & & & \\ & & \mathrm{Sl}(2, \mathbb{Z})^\wedge & & \downarrow & & \downarrow \chi & & \\ & & \downarrow & & & & & & \\ 1 & \longrightarrow & \mathrm{Sl}(2, \widehat{\mathbb{Z}}) & \longrightarrow & \mathrm{Gl}(2, \widehat{\mathbb{Z}}) & \xrightarrow{\det} & \widehat{\mathbb{Z}}^* & \longrightarrow & 1 \end{array} \tag{1}$$

It shows that the outer operation of $\Gamma_{\mathbb{Q}}$ on $\widehat{\mathcal{T}}^+_{1,1} \simeq \mathrm{Sl}(2, \mathbb{Z})^\wedge$ passes to the|to the quotient as an outer action on $\mathrm{Sl}(2, \widehat{\mathbb{Z}})$, and this outer action, moreover, can be made entirely explicit with the help of the cyclotomic character

$$\chi : \Gamma_{\mathbb{Q}} \longrightarrow \widehat{\mathbb{Z}}^*. \tag{2}$$

To make this last point precise, let us note that the second row of (1) gives a homomorphism

$$\widehat{\mathbb{Z}}^* \longrightarrow \mathrm{Autext}(\mathrm{Sl}(2, \widehat{\mathbb{Z}})), \tag{3}$$

and the canonical homomorphism

$$\Gamma_{\mathbb{Q}} \longrightarrow \dagger\mathrm{Autext}(\mathrm{Sl}(2, \widehat{\mathbb{Z}})) \tag{4}$$

21

Uncertainty retained from cancelled passages.
140-3-s022-D001 Properties
140-3-s022-D002 Sl(2,

[scan 23]

is none other than the composite of (2) and (3). Moreover, the homomorphism (4) factors as

$$\widehat{\mathbb{Z}}^*/\widehat{\mathbb{Z}}^{*2} \longrightarrow \mathrm{Autext}(\mathrm{Sl}(2,\widehat{\mathbb{Z}})) \tag{5}$$

(comes from the fact that the elements|elements of $\mathrm{Gl}(2,\widehat{\mathbb{Z}})$ which are in the center $\widehat{\mathbb{Z}}^*$ operate trivially †on $\mathrm{Sl}(2,\widehat{\mathbb{Z}})$, and that the composite

$$\widehat{\mathbb{Z}}^* \xrightarrow{\text{(center)}} \mathrm{Gl}(2,\widehat{\mathbb{Z}}) \xrightarrow{\det} \widehat{\mathbb{Z}}|\mathbb{Z}^{*|2}$$

is none other than $\lambda \longmapsto \lambda^2$). More precisely, we see that for the automorphism of $\mathrm{Sl}(2,\widehat{\mathbb{Z}})$ induced by a $u \in \mathrm{Gl}(1|2,\widehat{\mathbb{Z}})$ to be inner, i.d.|i.e. of the form $\mathrm{int}\,(u_0)|u_0$ with $u_0 \in \mathrm{Sl}(2,\widehat{\mathbb{Z}})$, it is necessary and sufficient that there exist $u_0 \in \mathrm{Sl}(2,\widehat{\mathbb{Z}})$ such that $u^{\dagger|\circ}u_0^{-1}$ commutes with $\mathrm{Sl}(2,\widehat{\mathbb{Z}})$, and this means that it is a scalar – i.e. $u \in \mathrm{Sl}(2,\widehat{\mathbb{Z}}) \cdot \widehat{\mathbb{Z}}^*,|.$ In other words the homomorphism (5) is injective, and consequently, 22

Uncertainty retained from cancelled passages.
140-3-s023-D001 from the point
140-3-s023-D002 0

[scan 24]
[p. 300]

the kernel of (4) is none other than the kernel of the composite

$$\Gamma_{\mathbb{Q}} \xrightarrow{\chi} \widehat{\mathbb{Z}}^* \longrightarrow \widehat{\mathbb{Z}}^*/\widehat{\mathbb{Z}}^{*2} \tag{6}$$

i.e. (by class field theory for $\mathbb{Q}$) the outer operation of $\Gamma_{\mathbb{Q}}$ on $\mathrm{Sl}(1|2,\widehat{\mathbb{Z}})$ factors through an outer operation of $(\Gamma_{\mathbb{Q}\,\mathrm{ab}})_2$ – the "multiquadratic" quotient of $\Gamma_{\mathbb{Q}}$ – and the latter is faithful.
m668

Marginal note. m668 N.B. The outer operation of $\Gamma_{\mathbb{Q}}$ on $\mathrm{Sl}(2,\mathbb{Z}/n|m\mathbb{Z})$ takes place through the intermediary of $((\mathbb{Z}/n|m\mathbb{Z})^*)_2|(\mathbb{Z}/n|m\mathbb{Z})_2^*$

Thus, if we identify $\Gamma_{\mathbb{Q}}$ † with a subgroup of $\mathrm{Autext}\,\mathrm{Sl}(2,\widehat{\mathbb{Z}})^\wedge$, this subgroup Γ is subject to fairly strong conditions : † leave invariant the normal subgroup kernel of

$$\mathrm{Sl}(2,\mathbb{Z})^\wedge \longrightarrow \mathrm{Sl}(2,\widehat{\mathbb{Z}}) \tag{7}$$

and induce on $\mathrm{Sl}(2,\widehat{\mathbb{Z}})$ †an outer operation coming from a homomorphism $\Gamma \longrightarrow \widehat{\mathbb{Z}}^*/\widehat{\mathbb{Z}}^{*2}$. We can moreover make this homomorphism precise as follows. Let us consider in $\mathrm{Sl}(2,\widehat{\mathbb{Z}})$ the "loop-subgroup" coming from the point at 23

Uncertainty retained from cancelled passages.
140-3-s024-D001 by (4)
140-3-s024-D002 [illeg. 1]
140-3-s024-D003 the

[scan 25]

infinity of $M_{1,1\,\mathbb{Q}}$, it corresponds to a class of outer homomorphisms

$$(8) \qquad T = T_\infty(\overline{\mathbb{Q}}_0) \longrightarrow \mathrm{Sl}(2,\overset{\dagger|\wedge}{\mathbb{Z}})^\wedge$$

whose image in $\mathrm{Sl}(2,\widehat{\mathbb{Z}})$ unless I am mistaken, is (up to conjugation) a "unipotent" subgroup

$$(9) \qquad \begin{pmatrix} 1 & 0 \\ n & 1 \end{pmatrix}, \qquad n \in \widehat{\mathbb{Z}}.$$

This outer homomorphism is respected by the operation of Γ, whence a well-determined operation of Γ on † T i.e. a homomorphism †

$$(10) \qquad \Gamma \xrightarrow{\chi} \mathrm{Aut}(T) \simeq \widehat{\mathbb{Z}}^*,$$

which will be precisely the cyclotomic character. More precisely, for an|An outer automorphism u of $\mathrm{Sl}(2,\mathbb{Z})^\wedge$ (or only of $\mathrm{Sl}(2,\widehat{\mathbb{Z}})$) which leaves invariant the|the † conjugacy|conjugacy class of the image of (8) [or only|only of the composite $T \longrightarrow \mathrm{Sl}(2,\widehat{\mathbb{Z}})$], we check that there exists a unique $\chi(u) \in \mathrm{Aut}(T) \simeq \widehat{\mathbb{Z}}^*$, such that we have commutativity in

do the verification
24

Uncertainty retained from cancelled passages.

140-3-s025-D001 ∧

140-3-s025-D002 Sl(2,

140-3-s025-D004 [illeg. 1] χ

140-3-s025-D005 image

[scan 26]
[p. 301]

$$(11) \qquad \begin{array}{ccc} T & \xrightarrow{\kappa|k} & \mathrm{Sl}(2,\mathbb{Z})^\wedge \\ \downarrow{\scriptstyle \chi(u)} & & \downarrow{\scriptstyle u} \\ T & \longrightarrow & \mathrm{Sl}(2,\mathbb{Z})^\wedge \end{array}$$

— or even, only in

$$(12) \qquad \begin{array}{ccc} T & \xrightarrow{\kappa|k} & \mathrm{Sl}(2,\widehat{\mathbb{Z}}) \\ \downarrow{\scriptstyle \chi(u)} & & \downarrow{\scriptstyle \dot{u}} \\ T & \longrightarrow & \mathrm{Sl}(2,\widehat{\mathbb{Z}}) \end{array} \qquad - - -$$

whence a character χ (10) on the group of all outer automorphisms u of $\mathrm{Sl}(2,\widehat{\mathbb{Z}})$ which have the property in question.

That said, Γ is contained in this subgroup Γ, and moreover in the subgroup of outer automorphisms which moreover preserve the kernel of (7), and which operate (outerly!) in the quotient $\mathrm{Sl}(2,\widehat{\mathbb{Z}})$ according to the composite

$$\Gamma : \xrightarrow{\chi} \widehat{\mathbb{Z}}^* \longrightarrow \widehat{\mathbb{Z}}^*/\widehat{\mathbb{Z}}^{*2} \hookrightarrow \mathrm{Autext}_{\mathrm{lac!}}(\mathrm{Sl}(2,\widehat{\mathbb{Z}})).$$

Besides, let us note that the center $\{\pm 1\}$ of $\mathrm{Sl}(2,\mathbb{Z})^\wedge$ not being trivial, the outer operation 25

[scan 27]

of $\Gamma_{\mathbb{Q}}$ on $\widehat{\mathcal{T}}^+_{1,1}$ does not allow us a priori to reconstruct (up to unique isomorphism) the extension $N_{1,1}$ of $\Gamma_{\mathbb{Q}}$ by $\widehat{\mathcal{T}}^+_{1,1}$ of (1). It gives only the extension † $N'_{1,1}$ of $\Gamma_{\mathbb{Q}}$ by

$$\widehat{\mathcal{T}}^{+\prime}_{1,1} \simeq \mathrm{Sl}(2,\mathbb{Z})^\wedge/\{\pm 1\} \overset{\text{dfn|déf}}{=} \mathrm{Sl}(2,\widehat{\mathbb{Z}})'$$

(by inverse image of the extension

$$(13) \quad 1 \to \mathrm{Sl}(2,\widehat{\mathbb{Z}})' \to \mathrm{Aut}\,\dagger_{\mathrm{lac}}(\mathrm{Sl}(2,\mathbb{Z})^\wedge) \to \mathrm{Autext}_{\mathrm{lac}}\ \mathrm{Sl}(2,\mathbb{Z})^\wedge$$
$$\downarrow$$
$$1$$

This extension $N'_{1,1}$ has also been interpreted as the extension $M_{0,3} \simeq N'_{0,4}$ of $\Gamma_{\mathbb{Q}}$ by $\widehat{\mathfrak{S}}_{0,3} \simeq \widehat{\mathcal{T}}'_{0,4}$ (group which is itself an extension of $\mathfrak{S}_3 \simeq \mathrm{Sl}(2, \mathbb{Z}/2\mathbb{Z})$ by $\widehat{\Pi}_{0,3}$). We already know that the outer operation of $\Gamma_{\mathbb{Q}}$ on the kernel $\widehat{\Pi}_{0,3}$ of $\mathrm{Sl}(2, \overset{\dagger|\wedge}{\mathbb{Z}})^{\wedge\prime} \longrightarrow \mathrm{Sl}(2, \mathbb{Z}/2|\mathbb{Z}_2)$ is faithful, and commutes with the outer operation of $\mathfrak{S}_3 \simeq \mathrm{Sl}(2, \mathbb{Z}/2\mathbb{Z}) \simeq \mathrm{Sl}(2, \mathbb{Z})^{\wedge\prime}/\widehat{\Pi}_{0,3}$. But this commutation can at pres- 26

Uncertainty retained from cancelled passages.
140-3-s027-D001 [illeg. 1]
140-3-s027-D002 ext
140-3-s027-D003 $\wedge$

ent be interpreted as a consequence of the fact that the outer operation of $\Gamma_{\mathbb{Q}}$ on $\mathfrak{S}_3 \simeq \mathrm{Sl}(2, \mathbb{Z}/2\mathbb{Z})$ is trivial, itself (from the present viewpoint) a consequence of the fact that it takes place through the intermediary of a (faithful) operation of $(\mathbb{Z}/n\mathbb{Z})^*_{/2|2}$, now for $n = 2$ $(\mathbb{Z}/n\mathbb{Z})^*_2 = 1$ (a relation which is on the other hand false for † other|other values, $n \neq \dagger 1, 2$ i.e. $n \geq 3$). This commutation relation, of a fairly trivial nature, thus appears at present as a very special case of the markedly subtler congruential relations. We can try to express them † directly in terms of the outer operation of $\Gamma_{\mathbb{Q}}$ on $\widehat{\Pi}_{0,3}$, by introducing the kernels of the homomorphisms $\widehat{\mathcal{T}}^{+\prime|+/n}_{1,1} \longrightarrow \mathrm{Sl}(2, 2m)$, which are open subgroups : [scan 28] [p. 302]

$$(14) \qquad \Pi(2m) \subset \widehat{\Pi}_{0,3}$$

normal and invariant under $\mathfrak{S}_3$, kernels 27

Uncertainty retained from cancelled passages.
140-3-s028-D001 the
140-3-s028-D002 2,
140-3-s028-D003 in

of homomorphisms [scan 29]

$$(15) \qquad \varphi_{2m} : \widehat{\Pi}_{0,3} \longrightarrow \mathrm{Sl}(2, \mathbb{Z}/2m)'$$

which it should of them|not be very difficult to make explicit, † by "reduction modulo $2m$" of a homomorphism

$$(16) \qquad \varphi_{\infty} : \widehat{\Pi}_{0,3} \longrightarrow \mathrm{Sl}(2, \widehat{\mathbb{Z}})'$$

which has already been mentioned previously – homomorphism whose image is the kernel subgroup of $\mathrm{Sl}(2, \widehat{\mathbb{Z}})' \longrightarrow \mathrm{Sl}(2, \mathbb{Z}/2\mathbb{Z})$, i.e. we have an exact sequence

$$(17) \qquad \dagger 1 \longrightarrow \Pi(\infty) \longrightarrow \widehat{\Pi}_{0,3} \longrightarrow \mathrm{Sl}(2, \widehat{\mathbb{Z}})' \longrightarrow \mathrm{Sl}(2, \mathbb{Z}/2\mathbb{Z}) \longrightarrow 1$$

$$(18) \qquad 1 \longrightarrow \Pi(2m) \longrightarrow \widehat{\Pi}_{0,3} \longrightarrow \mathrm{Sl}(2, \mathbb{Z}/2m\mathbb{Z})' \longrightarrow \mathrm{Sl}(2, \mathbb{Z}/2\mathbb{Z}) \longrightarrow 1.$$

This being laid down, these|the exact sequences allow us in a natural way to define an outer operation of $\widehat{\mathbb{Z}}^*/\widehat{\mathbb{Z}}^{*2}$ on $\widehat{\Pi}_{0,3}/\Pi(\infty)$, or again of the $(\mathbb{Z}/2m\mathbb{Z})^*_2$ on the $\widehat{\Pi}_{0,3}/\Pi(2m)$, and the "condition of congruence|congruences" is expressed by saying that the cyclotomic character χ of $\Gamma_{\mathbb{Q}}$ (which we can regard as deduced from the outer operation of 28

Uncertainty retained from cancelled passages.
140-3-s029-D001 in terms
140-3-s029-D002 $2m$

$\Gamma_{\mathbb{Q}}$ on the|the group|groups with loops $\widehat{\Pi}_{0,3}$), which gives a|in composite [scan 30] [p. 303]

$$(19) \qquad \Gamma_{\mathbb{Q}} \overset{\chi}{\to} \widehat{\mathbb{Z}}^* \overset{\text{can.}}{\longrightarrow} \widehat{\mathbb{Z}}^*/\widehat{\mathbb{Z}}^{*2},$$

of|hence an outer operation of $\Gamma_{\mathbb{Q}}$ on $\widehat{\Pi}_{0,3}/\Pi(\infty)$, is compatible with the outer operation of $\Gamma_{\mathbb{Q}}$ on $\widehat{\Pi}_{0,3}$ itself.

For each integer m, this condition is made explicit by the fact that $\Pi(2m)$ is invariant under the outer operation of $\Gamma_{\mathbb{Q}}$, and that the induced outer operation of|on $\widehat{\Pi}_{0,3}/\Pi(2m)$ is the one deduced from the character obtained by composing (19) with

$$\widehat{\mathbb{Z}}^*/\widehat{\mathbb{Z}}^{*2} \longrightarrow (\mathbb{Z}/2m|m\mathbb{Z})_2^* :$$

$$(20) \qquad \Gamma_{\mathbb{Q}} \longrightarrow (\mathbb{Z}/m\mathbb{Z})_2^*.$$

We should look whether this formulation is really equivalent to that of the conditions of congruences|congruence in $\widehat{\mathcal{T}}_{1,1}^{+\prime}$ in its full entirety, † and not only a consequence; exercise|exercises ! On the other hand, the|the 29

Uncertainty retained from cancelled passages.
140-3-s030-D001 even if it means there [illeg. 1]

[scan 31]

question whether the conditions do indeed constitute a restriction, an effective one, on an outer automorphism of the group with loops $\widehat{\Pi}_{0,3}$, commuting with the action of $\mathfrak{S}_3$, may very well be out of reach, for lack of knowing how actually to construct such outer automorphisms (one|or [illeg. 1] some nontrivial elements of $\widehat{\mathcal{T}}|\widehat{\mathcal{T}}_{0,3}^{!*}$) in any other way than precisely by arithmetic means, with the help of the operation of $\Gamma_{\mathbb{Q}}$ on $\widehat{\Pi}_{0,3}$! (One should all the same make another effort in this direction sometime …)

I also ask myself the question to|of what extent one can reconstruct, in terms of the theory of profinite groups with loops and of the † outer automorphisms of $\widehat{\Pi}_{0,3}$, or of $\dagger\widehat{\mathfrak{S}}_{0,3}^{+} \simeq \widehat{\mathcal{T}}_{1,1}^{+\prime}$, 30

Uncertainty retained from cancelled passages.
140-3-s031-D001 their
140-3-s031-D002 [illeg. 1]

[scan 32]

a) the finer extension $N_{1,1}$ of $\Gamma_{\mathbb{Q}}$ by $\widehat{\mathcal{T}}_{1,1}^{+}$, which is thus an extension of $N'_{1,1}$ (supposed to be known) by the group $\{\pm 1\}$ [p. 304]

b) the homomorphism

$$\boxed{N_{1,1} \longrightarrow \mathrm{Gl}(2,\widehat{\mathbb{Z}})} \qquad (21)$$

of (1), and failing that, the homomorphism

$$N'_{1,1} \simeq \dagger M_{0,3}^{\dagger} \longrightarrow \mathrm{Gl}(2,\widehat{\mathbb{Z}})/\{\pm 1\}. \qquad (22)$$

I will begin|begin with question b).

If one has the extension $N_{1,1}$ of $\Gamma_{\mathbb{Q}}$ by $\widehat{\mathcal{T}}_{1,1}^{+} \simeq \mathrm{Sl}(2,\mathbb{Z})^{\wedge}$, one has a homomorphism

$$N_{1,1} \longrightarrow \mathrm{Aut}_{\mathrm{lac}}\left(\underbrace{\widehat{\mathcal{T}}_{1,1}^{+}}_{\mathrm{Sl}(2,\mathbb{Z})^{\wedge}}\right) \qquad (23)$$

which passes to the quotient as a homomorphism

$$N_{1,1} \longrightarrow \mathrm{Aut}_{\mathrm{lac}}\left(\mathrm{Sl}(2,\widehat{\mathbb{Z}})\right) \qquad (24)$$

now one has an obvious homomorphism

$$\dagger \quad \mathrm{Gl}(2,\widehat{\mathbb{Z}}) \longrightarrow \mathrm{Aut}_{\mathrm{lac}}\left(\mathrm{Sl}(2,\widehat{\mathbb{Z}})\right)$$

which factors as

$$\mathrm{Gl}(2,\widehat{\mathbb{Z}}) \longrightarrow \mathrm{Gl}(2,\widehat{\mathbb{Z}})/\widehat{\mathbb{Z}}^* \simeq \mathrm{GP}(1,\widehat{\mathbb{Z}}) \hookrightarrow \mathrm{Aut}_{\mathrm{lac}}\left(\mathrm{Sl}(2,\widehat{\mathbb{Z}})\right) \tag{25}$$

and it turns out that (24) factors as

$$N_{1,1} \longrightarrow \underset{\simeq \mathrm{Gl}(2|1,\widehat{\mathbb{Z}})/\widehat{\mathbb{Z}}^*}{\mathrm{GP}(1,\widehat{\mathbb{Z}})} \hookrightarrow \mathrm{Aut}_{\mathrm{lac}}\left(\mathrm{Sl}(2,\widehat{\mathbb{Z}})\right) \tag{26}$$

31

Uncertainty retained from cancelled passages.
140-3-s032-D002 $\mathfrak{S}$
140-3-s032-D003 $+$
140-3-s032-D004 [*illeg.* 1]

[scan 33]

which thus makes it possible to reconstruct, from the extension $N_{1,1}$, a homomorphism

$$N_{1,1} \overset{\varphi}{\to} \mathrm{GP}(1,\widehat{\mathbb{Z}}) \simeq \mathrm{Gl}(2,\widehat{\mathbb{Z}})/\widehat{\mathbb{Z}}^*, \tag{27}$$

a little weaker than the homomorphism (21) we are trying to construct. But let us note that the homomorphism

$$\begin{array}{ccc} \mathrm{Gl}(2,\widehat{\mathbb{Z}}) & \xrightarrow{u\mapsto(\dot{u},\det(u))} & \mathrm{GP}(1,\widehat{\mathbb{Z}}) \quad \times \widehat{\mathbb{Z}}^* \\ & & \downarrow \\ & & \widehat{\mathbb{Z}}^*/\widehat{\mathbb{Z}}^{*2} \end{array} \tag{28}$$

has as its kernel the set of $\lambda \in \widehat{\mathbb{Z}}^*$ such that $\lambda^2 = 1$ – i.e. the group

$${}_2\widehat{\mathbb{Z}}^* \simeq (\mathbb{Z}/2\mathbb{Z})^P \tag{29}$$

P= | : set of prime numbers

Thus, one finds that the homomorphism

$$\dagger N_{1,1} \xrightarrow{u\mapsto(\varphi(u),\chi(u))} \mathrm{GP}(1,\widehat{\mathbb{Z}}) \times \widehat{\mathbb{Z}}^* \tag{30}$$

(deduced from (27) and from χ) § m680

Marginal note. m680 this is yet another condition on the extension $N_{1,1}$ of $\Gamma_{\mathbb{Q}}$ by $\widehat{\mathcal{T}}^+_{1,1}$, i.e. [illeg. 1] of the outer operation of $\Gamma_{\mathbb{Q}}$ on $\widehat{\mathcal{T}}^+_{1,1}$

factors through the image of (28), and thus defines a homomorphism

$$\dagger N_{1,1} \overset{\psi}{\to} \mathrm{Gl}(2,\widehat{\mathbb{Z}})/{}_2\widehat{\mathbb{Z}}^* \tag{31}$$

which is still a weakened version of (21). 32

Uncertainty retained from cancelled passages.
140-3-s033-D001 $\Gamma_{\mathbb{Q}}$
140-3-s033-D002 $\Gamma_{\mathbb{Q}}$

[scan 34]
[p. 305]

In fact, the homomorphism (31) factors as

$$[N_{1,1} \longrightarrow]\ \underset{\|}{N'_{1,1}} \overset{\psi'}{\longrightarrow} \mathrm{Gl}(2,\widehat{\mathbb{Z}})/{}_2\widehat{\mathbb{Z}}^* \tag{31}$$
$$N_{1,1}/\pm 1$$

and the homomorphism ψ' can surely be constructed directly from the extension $N'_{1,1}$ of $\Gamma_{\mathbb{Q}}$ by $\widehat{\mathcal{T}}'_{1,1} \simeq \widehat{\mathfrak{S}}^{+}_{0,3} \simeq \mathrm{Sl}(2,\widehat{\mathbb{Z}})'$, just as ψ a moment ago: it suffices to check that the canonical homomorphism

$$\mathrm{GP}(1,\widehat{\mathbb{Z}}) \longrightarrow \mathrm{Aut}_{\mathrm{lac}}(\mathrm{Sl}(2,\widehat{\mathbb{Z}})') \tag{33}$$
$$\wr$$
$$\mathrm{Gl}(2,\widehat{\mathbb{Z}})/\widehat{\mathbb{Z}}^*$$

is again injective. The question here (in terms of the theory of profinite groups…) how to †make explicit the lifting of ψ' of (31) to the canonical homomorphism

$$\dagger \quad N'_{1,1} \longrightarrow \mathrm{Gl}(2,\widehat{\mathbb{Z}})' \overset{\mathrm{dfn}}{=} \mathrm{Gl}(2,\widehat{\mathbb{Z}})/\pm 1 \tag{34}$$

— once this homomorphism has been obtained, one deduces from it the extension $N_{1,1}$ by inverse image of the extension structure

$$1 \longrightarrow \{\pm 1\} \longrightarrow \mathrm{Gl}(2,\widehat{\mathbb{Z}}) \longrightarrow \mathrm{Gl}(2,\widehat{\mathbb{Z}})' \longrightarrow 1, \tag{35}$$

as well as the homomorphism (21) $N_{1,1} \longrightarrow \mathrm{Gl}(2,\widehat{\mathbb{Z}})$. 33

Uncertainty retained from cancelled passages.
140-3-s034-D001 [illeg. 1]
140-3-s034-D002 $\Gamma_{\mathbb{Q}}$

[scan 35]

So in summary, the essential question here seems to be the determination of the lifting (34) of (32).

Let us take up our arguments again. We take a subgroup Γ of $\mathrm{Autext}_{\mathrm{lac}}(\widehat{\Pi}_{0,3})$, commuting with $\mathfrak{S}_3$, whence an extension † (denoted $M_{0,3}$ or $N'_{1,1}$) of Γ by $\widehat{\mathfrak{S}}^{+}_{0,3}$, or again of $\mathfrak{S}_3 \times \Gamma$ by $\widehat{\Pi}_{0,3}$, † whence a homomorphism

$$M_{0,3}(N'_{1,1})|(=N'_{1,1}) \longrightarrow \mathrm{Aut}_{\mathrm{lac}}\left(\underbrace{\widehat{\mathfrak{S}}^{+}_{0,3}}_{\mathrm{Sl}(2,\widehat{\mathbb{Z}}|\mathbb{Z})'^{\wedge}}\right) \tag{36}$$

inducing on $\widehat{\mathfrak{S}}^{+}_{0,3}$ †the operation of this group on itself that is inner. One has, moreover, an injective homomorphism

$$\mathrm{GP}(1,\widehat{\mathbb{Z}}) \hookrightarrow \mathrm{Aut}(\mathrm{Sl}(2,\widehat{\mathbb{Z}})') \tag{37}$$

and $\Gamma \subset \mathrm{Autext}_{\mathrm{lac}}(\widehat{\Pi}_{0,3})$ is subject (unless this is already a consequence of the hypothesis of commutation with $\mathfrak{S}_3$) to the condition that the outer† homomorphism

$$\widehat{\mathfrak{S}}^{+}_{0,3} \longrightarrow \mathrm{Sl}(2,\widehat{\mathbb{Z}})' \tag{38}$$
$$\wr$$
$$\mathrm{Sl}(2,\mathbb{Z})'^{\wedge}$$

34

Uncertainty retained from cancelled passages.
140-3-s035-D001 [illeg. 1]
140-3-s035-D002 Using
140-3-s035-D003 [illeg. 2]
140-3-s035-D004 [illeg. 1] [illeg. 1]

[scan 36]

be compatible with the outer actions of Γ, on $\widehat{\mathfrak{S}}^{+}_{0,3}$ by tautology (since Γ consists of elements of $\mathrm{Autext}_{\mathrm{lac}}(\widehat{\Pi}_{0,3})$ which commute with $\mathfrak{S}_3$), on $\mathrm{Sl}(2,\widehat{\mathbb{Z}})'$ via $\widehat{\mathbb{Z}}^*/\widehat{\mathbb{Z}}^{*2}$ and the "cyclotomic" character [p. 306]

$$\chi : \Gamma \longrightarrow \widehat{\mathbb{Z}}^* \tag{39}$$

deduced from the tautological outer operation of Γ on $\widehat{\Pi}_{0,3}$ (respecting the loop structure).

This "congruence condition" † † also|thus that (36) defines $M_{0,3}|N'_{1,1} \longrightarrow \operatorname{Aut} \operatorname{Sl}(2,\widehat{\mathbb{Z}})'$, which (factors through (37)), as a well-determined homomorphism

$$\varphi' : M_{0,3}^{\dagger} \longrightarrow \operatorname{GP}(1,\widehat{\mathbb{Z}}), \tag{40}$$

whence, using in addition the said cyclotomic character, a homomorphism

$$\dagger \quad M_{0,3} \xrightarrow{u\mapsto(\varphi'(u),\chi(u))} \operatorname{GP}(1,\widehat{\mathbb{Z}}) \times \widehat{\mathbb{Z}}^* \tag{41}$$

and the congruential condition in fact assures us, more precisely, that the latter factors as a homomorphism

$$M_{0,3} \xrightarrow{\psi'} \operatorname{Gl}(2,\widehat{\mathbb{Z}})/{}_2\widehat{\mathbb{Z}}^*, \tag{42}$$

in view of the injection 35

Uncertainty retained from cancelled passages.

140-3-s036-D001 means

140-3-s036-D002 implies

140-3-s036-D003 +

140-3-s036-D004 M

[scan 37]

$$\begin{aligned} (43) \qquad & \operatorname{Gl}(2,\widehat{\mathbb{Z}})/{}_2\widehat{\mathbb{Z}}^* \hookrightarrow \operatorname{GP}(1,\widehat{\mathbb{Z}}) \times \widehat{\mathbb{Z}}^* \\ & u \bmod {}_2\widehat{\mathbb{Z}}^* \longmapsto (u \bmod \widehat{\mathbb{Z}}^*, \det u). \end{aligned}$$

Let us consider the canonical homomorphism (surjective)

$$(44) \qquad \underset{\substack{\text{dfn}\,\|\\ \operatorname{Gl}(2,\widehat{\mathbb{Z}})/\pm 1}}{\operatorname{Gl}(2,\widehat{\mathbb{Z}})'} \longrightarrow \operatorname{Gl}(2,\widehat{\mathbb{Z}})/{}_2\widehat{\mathbb{Z}}^*,$$

whose kernel is isomorphic:

$$(45) \qquad \nu|V = (\mathbb{Z}/2\mathbb{Z})^P/\text{diag} \qquad (P = \text{set of prime numbers}).$$

The canonical homomorphism ψ' (42), deduced from the congruence conditions, defines by inverse image a central extension

$$(46) \qquad 1 \longrightarrow \nu|V \longrightarrow \widetilde{M}|\overline{M}_{0,3} \longrightarrow M_{0,3} \longrightarrow 1,$$

and the datum of a homomorphism (34)

$$(47) \qquad M_{0,3} \longrightarrow \operatorname{Gl}(2,\widehat{\mathbb{Z}})'$$

which would lift (42), is equivalent to that of a splitting of this extension (46). The question to elucidate now: is it possible to make explicit such a canonical lifting of (42), or splitting of (46), in terms of the 36

sole datum of $\Gamma \subset \operatorname{Autext}_{\text{lac}}(\widehat{\Pi}_{0,3})$, satisfying the congruential conditions stated above? [scan 38]

Let us note that on the subgroup $\widehat{\mathcal{T}}^{+}_{0,3} \simeq \operatorname{Sl}(2,\widehat{\mathbb{Z}}|\mathbb{Z})'^{\wedge\dagger}$ of $M_{0,3}$, the †lifting sought (47) is already known, as being [p. 307]

$$\operatorname{Sl}(2,\mathbb{Z})'^{\wedge} \longrightarrow \operatorname{Sl}(2,\widehat{\mathbb{Z}})' \hookrightarrow \operatorname{Gl}(2,\widehat{\mathbb{Z}})'.$$

In other words, the sought-for splitting of (46) is already known over the subgroup $\widehat{\mathcal{T}}^{+}_{0,3}$ of $M_{0,3}$.

The indeterminacy of the liftings or splittings being a priori in †

$$\mathrm{Hom}(M_{0,3}, \underline{\nu|V}),$$

taking account of the preceding condition, it will be in

$$(48) \qquad \mathrm{Ker}(\mathrm{Hom}(M_{0,3}, \underline{\nu|V}) \longrightarrow \mathrm{Hom}(\widehat{\mathcal{T}}^{+}_{0,3}, \underline{\nu|V})) \simeq \mathrm{Hom}(\Gamma, \nu|V) -$$

if one assumes that $\Gamma/\Gamma_{\mathrm{ab}} \overset{\chi}{\to} \widehat{\mathbb{Z}}^*$ is an isomorphism, the indeterminacy would still be in $\mathrm{Hom}((\widehat{\mathbb{Z}}^*)_2, \nu|V)$ — that's big ! It 37

Uncertainty retained from cancelled passages.

140-3-s038-D001 +

140-3-s038-D002 homomorphism

140-3-s038-D003 [illeg. 1]

takes a somewhat clever element to justify this lifting — one guesses that it is linked precisely [scan 39]
to the interpretation of an $M_{0,3}$ as an $N'_{1,1}$ — i.e. to the introduction, into a "profinite theory of|of the rational curves", of the "profinite theory of elliptic curves".

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40 Topologico-isotopic|Topologico-isotopic paraphrase (in terms of fundamental groups) of the Theory of Legendre

To feel out the terrain, I feel like starting first from the situation of discrete groups, which we have better in hand, one can well say so !

Let $C_{0,3}$ be the category of (discrete) groups with loops of type $0, 3$ (with isomorphisms as arrows), whose standard representative is $\Pi_{0,3}$ — this category is therefore canonically equivalent to that of torsors under the group

$$(49) \qquad \mathrm{Aut}_{\mathrm{lac}}(\Pi_{0,3}) \simeq \mathfrak{S}_{0,3} \simeq \mathrm{Gl}(2, \mathbb{Z})^{\wedge}/\pm 1 \simeq \mathrm{GP}(1, \mathbb{Z})$$

Let $C^{\mathrm{ext}}_{1,1}$ be the category of outer groups with loops of type $(1, 1)$, †whose standard representative is $\Pi_{1,1} = \{x, y, l \mid [x, y]l = 1\}$; this category is 38

Uncertainty retained from cancelled passages.

140-3-s039-D001 it is

equivalent to that of torsors under the group $\mathrm{Autext}_{\mathrm{lac}}(\Pi_{1,1}) \overset{\sim}{\to} \mathcal{T}_{1,1} \simeq \mathrm{Gl}(2, \mathbb{Z})$ — the isomor- [scan 40] [p. 308]
phism

$$(50) \qquad \mathrm{Autext}_{\mathrm{lac}}(\Pi_{1,1}) \longrightarrow \mathrm{Gl}(2, \mathbb{Z})$$

.|: it is obtained by associating to every outer automorphism of $\Pi_{1,1}$ its action on

$$(51) \qquad (\Pi_{1,1})_{\mathrm{ab}} \simeq \mathbb{Z}^2$$

One thus has a canonical functor

$$(52) \qquad L : \dagger\, C^{\mathrm{ext}}_{1,1} \longrightarrow C_{0,3}$$

defined (up to canonical isomorphism) by the condition of making the diagram commutative L, first letter of "Legendre"

(53)

$$\begin{array}{ccc} C_{1,1}^{\mathrm{ext}} & \longrightarrow & C_{0,3} \\ \downarrow\wr & & \downarrow \\ \mathrm{Tors}(\mathrm{Gl}(2,\mathbb{Z})) & \longrightarrow & \mathrm{Tors}(\mathrm{GP}(1,\mathbb{Z})) \end{array}$$

where the|1'

$$\mathrm{Tors}(\mathrm{Gl}(2,\mathbb{Z})) \longrightarrow \mathrm{Tors}(\mathrm{GP}(1,\mathbb{Z}))$$

is the functor of change of group of operators|operations relative to:

$$(54) \qquad \mathrm{Gl}(2,\mathbb{Z}) \overset{\mathrm{can}}{\longrightarrow} \mathrm{GP}(1,\mathbb{Z}) \simeq \mathrm{Gl}(1|2,\mathbb{Z})/\pm 1$$

39

Uncertainty retained from cancelled passages.
140-3-s040-D001 *A*

One can also introduce the groupoid $C_{1,1}^{\mathrm{cat}\,\prime}$, obtained from the groupoid $C_{1,1}^{\mathrm{ext}}$ by passing to the|taking a quotient by the universal automorphism $-\mathrm{id}$ — corresponding to the center $\{\pm 1\}$ of the fundamental|fundamental groups of this groupoid. — one obtains a connected groupoid equivalent canonical[ly] to that of torsors under $\mathrm{Gl}(2,\mathbb{Z})' = \mathrm{Gl}(2,\mathbb{Z})/(\pm 1)$, from which|whence one now obtains an equivalence of categories [scan 41]

$$(55) \qquad C_{1,1}^{\mathrm{ext}\,\prime} \overset{\approx}{\to} C_{0,3}$$

making commutative

(56)

$$\begin{array}{ccc} C_{1,0|1}^{\mathrm{ext}\,\prime} & \overset{\approx}{\longrightarrow} & C_{0,3} \\ \downarrow & & \downarrow \\ \mathrm{Tors}(\mathrm{Gl}(2,\mathbb{Z})') & \overset{\approx}{\longrightarrow} & \mathrm{Tors}(\mathrm{GP}(1,\mathbb{Z})) \end{array}$$

,

where $\mathrm{Tors}(\mathrm{Gl}(2,\mathbb{Z})') \overset{\approx}{\to} \mathrm{Tors}(\mathrm{GP}(1,\mathbb{Z}))$ is the tautological functor associated to the isomorphism

$$(57) \qquad \mathrm{Gl}(2,\mathbb{Z})' \overset{\approx}{\to} \mathrm{GP}(1,\mathbb{Z}).$$

In short, I am in the process of making a topological isotopic paraphrase, in terms of the fundamental groups with loops, of Legendre's theory of the relations between elliptic curves and rational curves. I would like now to make explicit the Legendre equivalence (55), which had been obtained thanks to the knowledge of the appropriate Teichmüller groups relative to the two sides and of an ad-hoc isomorphism between these two groups, directly in terms of the fundamental groups of type|types $\Pi_{0,3}$ or|and $\Pi_{1,1}$. [scan 42] [p. 309]

If π' is a group with loops of type 1.1|1, 1, it is appropriate first to note that there is in $\mathrm{Autext}_{\mathrm{lac}}(\pi')$ a canonical element of order 2 — in fact the unique nontrivial element of the center — denoted as is only right $(-1)_{\pi'}$. [m690]

Marginal note. [m690] N.B. $(-1)_{\pi'}|\mathrm{Aut}_{\pi'}$ preserves the orientation i.e. has multiplier|multiplicity $+1$

Thus, one can consider π' as endowed canonical[ly] with an † outer operation of $G = \mathbb{Z}/2\mathbb{Z}$ on [scan 43]

it, whence a canonical extension

$$(58) \qquad 1 \longrightarrow \pi' \text{ — } \mathcal{E} \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 1.$$

The topological model teaches us that there are exactly three π'- conjugacy classes of splittings of this extension, corresponding to the three points of order 2 of a topological group $E \simeq \mathbf{U} \times \mathbf{U}$ — the three fixed points of $x \mapsto -x$ in $E \setminus \{0\}$. The theory of "drilling holes simultaneously", relative to a situation of a finite group of outer operators of a|of a group with loops (theory which even in the discrete case has not really been written — for the moment there are conjectural statements) then makes it possible m691

Marginal note. m691 but here, one is in a case [illeg. 2] can get by with drillings [illeg. 1] of one hole at a time, repeated successively...

Uncertainty retained from cancelled passages.
140-3-s043-D001 action|other

[scan 44]

to construct canonically an outer group with loops $\tilde{\pi}'$ of type $\Pi_{1,4}$, with an outer operation of $G = \mathbb{Z}/2\mathbb{Z}$ on this group, and a subset $J = I \setminus \{i_0\}$ of cardinality 3 of $I = I(\tilde{\pi}')$ (which is of cardinality 4), J stable under G i.e. i_0 fixed by G, and an isomorphism of the|of group $\tilde{\pi}'(J)$ of type $\Pi_{1,1}$ obtained by "plugging the holes at J" with π' [p. 310]

$$(59) \qquad \tilde{\pi}'(J) \simeq \pi',$$

isomorphism which is compatible with the operations of G. In the topological model, $\tilde{\pi}'$ is the categorical fundamental group of the space $E \setminus {}_2E$, space on which G operates freely.
m692

Marginal note. m692 in fact, G operates trivially on $I = I(\tilde{\pi}')$

The extension $\tilde{\mathcal{E}}$ of G by $\tilde{\pi}'$ corresponding|corresponding to the action of † G on $\tilde{\pi}'$ (60)

$$1 \text{ —— } \tilde{\pi}' \text{ —— } \tilde{\mathcal{E}} \text{ —— } \mathbb{Z}/2\mathbb{Z} \longrightarrow 1$$

, which is identified thus with the mixed fundamental group of $(E \setminus {}_2E, G)$, is thus identified also-

Uncertainty retained from cancelled passages.
140-3-s044-D001 $\tilde{\mathcal{E}}$

[scan 45]

with the fundamental group of

$$(61) \qquad V = (E \setminus {}_2E)/\{\pm 1\}$$

which is a topological surface of type $(0, 4)$.

In other words, $\tilde{\mathcal{E}}$ can be considered as a group with loops of type $(0, 4)$, for a loop structure easy to describe in terms of that of $\tilde{\pi}'$, and for which one has a canonical isomorphism

$$(62) \qquad I(\tilde{\mathcal{E}}) \simeq I(\tilde{\pi}')/G \simeq I(\tilde{\pi}').$$

Thus one has an $i_0 \in I(\tilde{\mathcal{E}})$, which allows one to "plug the hole at i_0" to obtain an actual group with loops of type $(0, 3)$, i.e. †

$$(63) \qquad \pi = \mathcal{L}(\pi')$$

Taking up in the reverse direction the relation between π and π', one can proceed as follows: to an actual group with loops of type $(0, 3)$, one associates by

Uncertainty retained from cancelled passages.
140-3-s045-D001 $\pi =$

[scan 46]

the "drilling holes" functor an outer group with loops $\tilde{\pi}(= \tilde{\mathcal{E}}| \in \tilde{\mathcal{E}})$ of type $(0,4)$, endowed with an element $i_0 \in I(\tilde{\pi})$ such that † [p. 311]

$$(64) \qquad I(\tilde{\pi}) \simeq \{i_0\} \amalg \underbrace{I(\pi)}_{\text{cardinality } 3}$$

One then has (T being the orientation module of π)

$$(65) \qquad \tilde{\pi}_{\mathrm{ab}} \simeq T^{I(\tilde{\pi})} \qquad (\tilde{\pi}_{\mathrm{ab}})_2 \simeq (\mathbb{Z}/2\mathbb{Z})^{I(\tilde{\pi})}$$

and one has an (outer) homomorphism

$$(66) \qquad \tilde{\pi} \longrightarrow \mathbb{Z}/2\mathbb{Z}$$

canonical, defined by the condition of being "ramified" at all the $i \in I(\tilde{\pi})$ — i.e. corresponding to the homomorphism

$$(67) \qquad (\tilde{\pi}_{\mathrm{ab}})_2 \xrightarrow{\sim} (\mathbb{Z}/2\mathbb{Z})^{I(\tilde{\pi})} \longrightarrow \mathbb{Z}/2\mathbb{Z}$$

sum of the coefficients. The kernel $\tilde{\pi}'$ of (66) is a group with loops of type $(1,4)$ — but it is not quite an outer group — it is not defined "modulo|up to inner automorphisms", only modulo (in addition) the outer operation of symmetry of $\mathbb{Z}/2\mathbb{Z}$

Uncertainty retained from cancelled passages.
140-3-s046-D001 [illeg. 1]

[scan 47]

on it. If one chooses the group $\tilde{\pi}$ (in its "outer class") – one will in any case have an exact sequence

$$(68) \qquad 1 \longrightarrow \tilde{\pi}' \longrightarrow \tilde{\pi} \text{ — } \mathbb{Z}/2\mathbb{Z} \longrightarrow 1$$

and $\tilde{\pi}'$ is a group with loops, with

$$(69) \qquad I(\tilde{\pi}') \simeq I(\tilde{\pi})$$

By plugging the holes corresponding to $I(\pi) \subset I(\tilde{\pi}) \simeq | = I(\tilde{\pi}')$, which is of cardinality 3, one obtains a group of type $(1,1)$, with operation of $\mathbb{Z}/2\mathbb{Z}$ on it which is none other than the canonical operation. But depending on the indeterminacies of the choices made, it is a little less than an outer group of type $(1,1)$ i.e. an object of $C^{\mathrm{ext}}_{1,1}$ – it is such an element, "modulo symmetry" – i.e. an object of $C^{\mathrm{ext}\,\prime}_{1,1}$. To have an object of $C^{\mathrm{ext}}_{1,1}$, one must have "chosen a kernel" $\tilde{\pi}'$ for the homomorphism which is outer $\tilde{\pi} \longrightarrow | \text{ —} \mathbb{Z}/2\mathbb{Z}$, as an outer group – this is the choice of an object

[scan 48]

in a $(\mathbb{Z}/2\mathbb{Z})$-gerbe, i.e. in a connected groupoid admitting $\mathbb{Z}/2\mathbb{Z}$ as fundamental group. [p. 312]

Modulo a more careful write-up, it seems to me that these conceptual reflections do indeed prove and without a flaw|flaws that one has a canonical isomorphism (at least modulo inner|inner automorphisms) 3

$$(70) \qquad \mathcal{T}_{1,1}/\{\pm 1\} \xrightarrow{\sim} \mathfrak{S}_{0,3}$$

m694

Marginal note. m694 in fact, one can make it|it precise modulo inner automorphisms arising from|from the $\mathfrak{S}_{0,3}^{1+[i,+} \simeq \Pi_{0,3}$

– the indeterminacy modulo inner automorphisms arising quite certainly from the choice of an isomorphism between $\mathcal{L}(\Pi_{1,1})$ and $\Pi_{0,3}$.

This fact is quite independent, it seems to me, of the fact that the homomorphism which is natural

$$\mathcal{T}_{1,1} \xrightarrow{\sim} \mathrm{Gl}(2,\mathbb{Z})$$

is an isomorphism. I see no objection to identifying $\mathcal{T}_{1,1}$ purely and simply with $\mathrm{Gl}(2|1,\mathbb{Z})$ on occasion,|; on the other hand it is more questionable to identify $\mathfrak{S}_{0,3}$ with $\mathcal{T}_{1,1}/\pm 1$ or with $\mathrm{Gl}(2,\mathbb{Z})'$

[scan 49]

except only as an outer group, because of the indeterminacy by inner automorphisms in (70).

From the point of view of analytic modular multiplicities, one has a canonical morphism

$$(71) \qquad \mathbb{M}_{1,1}^{\mathrm{an}} \longrightarrow \underbrace{\mathbb{U}|\mathbb{M}_{0,3}^{\mathrm{an}}}_{\substack{\text{complex|complete analytic curve}\\ \text{of type } (0,3) \text{ “universal”}}} \simeq \mathbb{M}_{0,4}'^{\mathrm{an}} \simeq (U_{0,3}^{\mathrm{an}}, \mathfrak{S}_3)$$

m695

Marginal note. m695 with a little more care, one could include in the reflections also complex conjugation, so as|in order not to have in|in (73) to restrict oneself to the subgroups $\mathcal{T}^+$ and $\mathfrak{S}^+$ in question

which induces an isomorphism

$$(72) \qquad \mathbb{M}_{1,1}^{\mathrm{an}\prime} \xrightarrow{\sim} M_{0,3}^{\mathrm{an}},$$

and the effect of (71), on the outer fundamental groups is none other than the categorical|outer homomorphism

$$(73) \qquad \mathcal{T}_{1,1}^{+} \longrightarrow \mathfrak{S}_{0,3}^{+}$$

induced by (70) – is it possible to make it precise in a useful way, as an actual homomorphism, not merely outer? There is a canonical universal covering|covering $\widetilde{\mathbb{M}}_{1,1}^{\mathrm{an}}$ of $\mathbb{M}_{1,1}^{\mathrm{an}}$, provided by the space of|of the

[scan 50]

Teichmüller (the Poincaré half-plane D of § 38) – relative to this universal covering one does [p. 313] indeed have a canonical isomorphism

$$(74) \qquad \pi_1(M|\mathbb{M}_{1,1}^{\mathrm{an}}) \xrightarrow{\sim} \mathrm{Sl}(2,\mathbb{Z})$$

(not merely outer), which is moreover none other, essentially, than (50) – (restricted to the subgroup of index 2 $\mathcal{T}_{1,1}^{+}$ of $\mathcal{T}_{1,1}$). One does indeed have a canonical morphism.

$$(75) \qquad D \longrightarrow U_{0,3}^{\mathrm{an}} \qquad (\simeq M|\mathbb{M}_{1,1}[2]')$$

m696

Marginal note. m696 For the definition in form|terms of this morphism, cf. §

compatible with a homomorphism on the groups of operators

$$(76) \qquad \mathrm{Sl}(2,\mathbb{Z})' \longrightarrow \underbrace{\mathrm{Sl}(2,\mathbb{Z}/2\mathbb{Z}) \xrightarrow{\sim} \mathfrak{S}_3}_{\substack{\text{this isomorphism|the isomorphism which is appropriate}\\ \text{here is made explicit in}\\ \S\ \ \text{, page}}}$$

so that D appears as a universal covering of $U^{\mathrm{an}}_{0,3|3}$, hence also of $(U^{\mathrm{an}}_{0,3}, \mathfrak{S}_3)$, and $\mathrm{Sl}(2,\mathbb{Z})'$ as the mixed π_1 relative to this particular universal covering. To be able to identify it with $\pi_1((U^{\mathrm{an}}_{0,3}, \mathfrak{S}_3); P = -\bar{j}|-j)$, it

must therefore choose a point of D above $-\bar{j}$ (which will then make it possible to identify D with the universal covering punctured|pointed at P of $U_{0,3}$) – which introduces precisely the indeterminacy by an element of the kernel of $\mathrm{Sl}(2|1,\mathbb{Z})' \longrightarrow \mathrm{Sl}(2,\mathbb{Z}/2\mathbb{Z}) \simeq \mathfrak{S}_3$, itself isomorphic to $\Pi_{0,3}$. But one can take for example [scan 51]

$$j = \exp\frac{2i\pi}{3} = \frac{-1+i\sqrt{3}}{2} \tag{77}$$

as a point above $P = -\bar{j}$ – by twisting some arms so that it is indeed above P, and not $\bar{P} = -j$, otherwise it is $-\bar{j} = \dfrac{1+i\sqrt{3}}{2} = \exp\dfrac{2i\pi}{6}$ that one would choose. [N.B. One finds that the three elements of $\mathrm{Sl}(2,\mathbb{Z})' \simeq \mathrm{GP}(1,\mathbb{Z})^+$ which transform j into $-\bar{j}$ are $\begin{pmatrix}1&1\\0&1\end{pmatrix}$, $\begin{pmatrix}1&0\\1&1\end{pmatrix}$, $\begin{pmatrix}0&-1\\1&0\end{pmatrix}$, none of which is $\equiv \begin{pmatrix}1&0\\0&1\end{pmatrix}$ mod 2, which shows that j and $-\bar{j}$ † are not congruent

Uncertainty retained from cancelled passages.

140-3-s051-D002 in

mod the kernel $\Pi(2)$ of $\mathrm{Sl}(2|1,\mathbb{Z}) \longrightarrow \mathrm{Sl}(2,\mathbb{Z}/2\mathbb{Z})$, i.e. that they do not have the same image in $U^{\mathrm{an}}_{0,3} \simeq D/\Pi(2)$, so their images are $P = -\bar{j}$ and $\bar{P} = -j$ (or the other way around)] The choice in question is characterized by the fact that $\tau \in D$ above P is † a root of unity, which is after all quite natural. One can therefore say that one has "normalized" the categorical homomorphism (73) into an actual homomorphism – and by the same token also (70). [scan 52] [p. 314]

One would like to take up again the preceding constructions in the context of profinite groups – unfortunately I lack the necessary foundations, and above all a theory of "hole-making" or "drilling holes" at the level of profinite groups with loops – even some|in a † fashion conjec-

Uncertainty retained from cancelled passages.

140-3-s052-D002 thus

140-3-s052-D003 [illeg. 2]

tural. Yet one feels that it is indeed in this direction that there is a "group-theoretic" understanding of the homomorphism † (40) [scan 53]

$$\widehat{M}_{0,3} \longrightarrow \mathrm{Gl}(2,\widehat{\mathbb{Z}})/\pm 1 \simeq \mathrm{GP}(1,\widehat{\mathbb{Z}}),$$

which is the main motivation for the reflections of the present §, and of the preceding ones. This should be an incentive for a renewed reflection on the question of hole-making in the profinite context…But I realize already now that † the question whether there is an extension of the constructions of the present §, to the profinite context, independently of the introduction of elements of "arithmetic" structure (such as the datum|given of a subgroup $\Gamma \subset \mathrm{Autext}_{\mathrm{loc}}(\widehat{\Pi}_{0,3})$, satisfying a certain number of $\pm$ subtle conditions, of the congruential relations kind) is subordinate to that,

Uncertainty retained from cancelled passages.
140-3-s053-D001 $\mathfrak{S}$
140-3-s053-D002 the

[scan 54]

whether these congruential relations really are relations of an arithmetic nature, specific to [p. 315] outer automorphisms coming from Γ_Q, or on the contrary $\pm$ automatic consequences of a purely "algebraic" formalism of profinite groups, which would be modeled on the discrete case. Without prejudging in any way the answer and the degree of plausibility of such a formalism, I could as an experiment try to bring one out, to test|try where it will lead me.

§41 Condition for $\widehat{\Pi}'_{0,3} \xrightarrow{\sim} \widehat{\mathfrak{S}}^+_{0,3}$ to commute with the operations of Γ_Q ; Calculation of outer automorphisms of $\widehat{\Pi}_{0,3}$ and $\widehat{\mathfrak{S}}^+_{0,3}$.

Before embarking on this path, I would still like to elucidate a question, concerning other possible necessary conditions (in addition to the congruential conditions of §38|39) for a categorical automorphism of $\widehat{\Pi}_{0,3}$ to come

[scan 55]

from Γ_Q. For this, let us take up again the composite outer homomorphism

$$\Pi_{0,3} \xrightarrow{\text{epimorphism}} \Pi'_{0,3} \xrightarrow{\sim} \mathfrak{S}^+_{0,3} \simeq \pi_1((U^{\mathrm{an}}_{0,3}, \mathfrak{S}_3), P) \quad (\varphi)$$

(1)
$$= \Pi_{0,3}/\{\ell_1^2, \ell_\infty^3\}$$
$$\simeq \text{cartographic group}$$
$$\text{oriented triangulated}$$

defined by

$$(2) \qquad \begin{cases} \varphi(\ell_0) = \ell'_0 = \sigma_\infty \rho \\ \varphi(\ell_1) = \ell'_1 = \sigma_\infty \\ \varphi(\ell_\infty) = \ell'_\infty = \rho^{-1} \end{cases}$$

(with the notation of [C, …] …), which by passing to the completions gives an outer homomorphism

$$\widehat{\varphi} : \widehat{\Pi}_{0,3} \xrightarrow{\text{epimorphism}} \widehat{\mathfrak{S}}^+_{0,3} \simeq \pi_1((U_{0,3\,\overline{Q}_0}, \mathfrak{S}_3), P).$$

(3)
$$\cap$$
$$\widehat{\widehat{\mathcal{T}}}_{0,3} = \mathrm{Aut}^{\dagger}_{\mathrm{loc}}(\widehat{\Pi}_{0,3})$$

This homomorphism is defined here †by the "brutal formulas" (2), without reference to a particular subgroup Γ of $\mathrm{Aut}\,\mathrm{ext}'_{\mathrm{loc}}(\widehat{\Pi}_{0,3})$. If one is given a † subgroup of this kind, commuting with $\mathfrak{S}_3$, it automatically operates on the outer group $\widehat{\mathfrak{S}}^+_{0,3}$, since it

Uncertainty retained from cancelled passages.
140-3-s055-D001 int|ext
140-3-s055-D002 [illeg. 1]
140-3-s055-D003 [illeg. 1]

[scan 56]

fits into an exact|exact sequence [p. 316]

$$(4)\qquad 1 \longrightarrow \widehat{\mathfrak{S}}^{+}_{0,3} \longrightarrow M_{0,3} \longrightarrow \Gamma \longrightarrow 1.$$

containing the exact|exact sequence

(5)

$$\begin{array}{ccccccc} 1 & \longrightarrow & \widehat{\Pi}_{0,3} & \longrightarrow & \mathcal{E}_{0,3} & \longrightarrow & \Gamma & \longrightarrow & 1 \\ & & \downarrow & & \uparrow & & \| & & \\ 1 & \longrightarrow & \widehat{\mathfrak{S}}^{+}_{0,3} & \longrightarrow & M_{0,3} & \longrightarrow & \Gamma & \longrightarrow & 1. \end{array}$$

The fact that the homomorphism (3) commutes with the action of $\Gamma_Q = \Gamma$ seems to me † a priori a nontrivial condition on Γ – more precisely, on the elements of Γ separately, which one should really manage to make explicit (likewise for the congruential conditions), in terms of the ± explicit calculations of [C]. This condition is equivalent to the condition that the homomorphism (3) fits into a homomorphism of extensions

---

Uncertainty retained from cancelled passages.

140-3-s056-D002 to be

[scan 57]

m697

Marginal note. m697 N.B. $\mathcal{E}_{0,3}$ is the group denoted by $M^{!}_{0,3}$ in § 46

$$(6)\qquad \begin{array}{ccccccccc} 1 & \longrightarrow & \widehat{\Pi}_{0,3} & \longrightarrow & \mathcal{E}_{0,3} & \longrightarrow & \Gamma & \longrightarrow & 1 \\ & & \downarrow{\scriptstyle \widehat{\varphi}} & & \downarrow{\scriptstyle \Phi} & & \downarrow{\scriptstyle 1} & & \\ 1 & \longrightarrow & \widehat{\mathfrak{S}}^{+}_{0,3} & \longrightarrow & M_{0,3} & \longrightarrow & \Gamma & \longrightarrow & 1 \end{array}$$

,

by means of which $M_{0,3}$ (interpreted as the fundamental group "with signature" of $(\mathbb{P}^1_{\mathbb{Q}} - \{0\}, 2\{1\} + 3\{\infty\})$) is identified with the quotient of $\widehat{\Pi}_{0,3}$ by the invariant closed subgroup generated by ℓ_1^2, ℓ_∞^3.

I am beginning to suspect that there will be a whole litany of conditions of the same nature on Γ, which one should really manage to bring out in a more general way.

[scan 58]

I|i am going to take up again the calculations made in [C], concerning the outer automorphisms|automorphisms of $\widehat{\Pi}_{0,3}$ which commute with $\mathfrak{S}_3$, † They can be realized uniquely as actual automorphisms u which commuting|commute with σ_∞; (and which moreover commute|commuting with ρ as outer automorphisms). They correspond bijectively to the pairs [p. 317]

A.

$$(7)\qquad (g_0, p)\qquad \begin{cases} g_0 \in \widehat{\Pi}_{0,3} \bmod L_0 \\ p \in \widehat{\mathbb{Z}}^* \quad \text{(multiplier)} \end{cases}$$

satisfying the following conditions (where † $\gamma \in \widehat{\mathbb{Z}}$ † is uniquely determined by u (or again, by the corresponding outer automorphism)), where we have set

$$h_0 = \sigma_\infty(g_0)^{-1} g_0 = [\sigma_\infty, g_0^{-1}] :$$

m698

Marginal note. m698 N.B. Admitting that $\widehat{\Pi}_{0,3}^{\sigma_\infty} = \{1\}$, one sees that g_0 is (?) known when one knows h_0…

m699

Marginal note. m699 N.B. in fact, we will see that (9) is superfluous|superfluous, and $\gamma = \frac{p-1}{2}$.

$$\boxed{\begin{aligned} &(9) \quad \ell_{1|0}^p \cdot \mathrm{int}(h_0)(\ell_{0|\infty}^p)\,\mathrm{int}(\ell_1^\gamma \rho(h_0)^{-1})(\ell_\infty^p) = 1 \\ &(10) \quad h_0 \ell_0^{-\gamma} \rho^{-1}(h_0) = \ell_1^\gamma \rho(h_0)^{-1} \ell_\infty^\gamma \end{aligned}}$$

† The homomorphism u is given by

m700

Marginal note. m700 N.B. (10) is also written †

$$(10\text{ bis}) \qquad (\ell_0^\gamma h_0^{-1}) \cdot \rho(\ell_0^\gamma h_0^{-1}) \cdot \rho^2(\ell_0^\gamma h_0^{-1}) = 1$$

Uncertainty retained from cancelled passages.
140-3-s058-D001 which|that
140-3-s058-D003 $\beta_0|\rho_0$
140-3-s058-D004 has to be|have to be
140-3-s058-D005 [illeg. 2]
140-3-s058-D006 (11)

[scan 59]
[p. 318]

$$(11) \qquad \begin{cases} u(\ell_0) = \mathrm{int}(g_0)(\ell_0^p) \\ u(\ell_1) = \mathrm{int}(\underbrace{\sigma_\infty(g_0)}_{g_1})(\ell_1^p) \\ u(\ell_\infty) = \mathrm{int}(g_0 \ell_0^{-\gamma|\sigma} \rho^{-1}(h_0))(\ell_\infty^p) \end{cases},$$

and one must in all rigor, in addition to conditions (9),(10), require that the two elements $u(\ell_0), u(\ell_1)$ given by (11) form generators of the profinite group $\widehat{\Pi}_{0,3}$. (11 bis)

I am going to calculate the action of u on $\widehat{\mathfrak{S}}_{0,3}^+$, considering this group as generated by the elements

$$(12) \qquad \ell_1' = \sigma_\infty, \qquad \ell_\infty' = \rho^{-1}$$

so one simply has to calculate $\mathrm{int}(u).\sigma_\infty = u(\sigma_\infty) = u\sigma_\infty u^{-1}$ and $u(\rho) = \mathrm{int}(u).\rho = u\rho u^{-1}$. One has by construction (u commuting with σ_∞)

$$(13) \qquad u \circ \sigma_\infty u^{-1} = \sigma_\infty,$$

and on the other hand one verifies that

$$(14)\qquad \begin{cases} \rho^{-1}u\rho = \mathrm{int}(h)u \\ \text{with} \quad h = (\rho^{-1}\sigma_\infty)(g_0)\ell_0^\gamma g_0^{-1} \end{cases}$$

m702

Marginal note. m702 i.e. $u\rho u^{-1} = \rho\,\mathrm{int}(h)$.
$= \mathrm{int}(\rho(h))\rho$
i.e. $\rho(h) = \sigma_\infty(g_0)\ell_1^\gamma\rho(g_0)^{-1}$

so one finds, in terms of an operation on the group $\widehat{\mathfrak{S}}^+_{0,3}$ generated by $\ell'_1 = \sigma_\infty$ and $\ell'_\infty = \rho^{-1}$ (or ρ)

$$(15)\qquad \begin{cases} u(\sigma_\infty) = \sigma_\infty \\ u(\rho) = g.\rho \end{cases} \quad \text{with} \quad g = \sigma_\infty(g_0)\ell_1^{\gamma|\tau}\rho(g_0)^{-1}$$

[scan 60]

N.B. The relation $u(\sigma_\infty)^2 = 1$ is trivial, the relation $u(\rho)^3 = 1$ reduces to

†

$$(16)\qquad g\rho(g)\rho^2(g) = 1$$

which is a consequence of the explicit expression (15) for g, and of formula (10).

Let us note that one † should have, & taking up again

$$\ell'_0 = \sigma_\infty\rho = \varepsilon_0|e_0 \qquad (\text{whence} \quad \ell'^2_0 = \ell_0)$$

$$(17)\qquad u(\ell'_0) = \mathrm{int}(g_0)(\ell'^p_0)$$



?

— this relation when|when squared is none other than the first relation (11)

$$u(\ell_0) = \mathrm{int}(g_0)(\ell_0^p).$$

[The relation (17) is a consequence of it, if one admits that the centralizer of $L_0 = \ell_0^{\widehat{\mathbb{Z}}}$ in $\widehat{\Pi}_{0,3}$ is reduced to L_0…]

One should also have the relation

$$(18)\qquad u(\rho)\dagger\text{conjugate of}\rho^p \quad \dagger$$

† to which we are going to return – it is not clear to me that it is a consequence of the conditions considered on u up to now.

†

Uncertainty retained from cancelled passages.
140-3-s060-D002 =
140-3-s060-D003 will have
140-3-s060-D004 ≡ | =
140-3-s060-D005 mod|modulo
140-3-s060-D006 [illeg. 1]
140-3-s060-D007 [illeg. 2]

[scan 61]
[p. 319]

Let us now consider the homomorphism (3)

$$\begin{cases} \widehat{\varphi}: & \widehat{\Pi}_{0,3} \longrightarrow \widehat{\mathfrak{S}}^{+}_{0,3} \\ & \ell_0 \longmapsto \ell'_0 = \sigma_\infty \rho = \varepsilon_0 \\ & \ell_1 \longrightarrow \ell'_1 = \sigma_\infty \\ & \ell_\infty \longmapsto \ell'_\infty = \rho^{-1} \end{cases} \tag{18}$$

and let us write that it commutes, as an outer homomorphism, with u. This therefore means that there exists a

$$\lambda \in \widehat{\mathfrak{S}}^{+}_{0,3} \tag{19}$$

such that we have

$$\widehat{\varphi}(ux) = \lambda\, u\widehat{\varphi}(x)\, \lambda^{-1} \qquad \forall x \in \widehat{\Pi}_{0,3}, \tag{20}$$

and it suffices to express this condition for x equal to one of the two generators ℓ_0, ℓ_1 of $\widehat{\Pi}_{0,3}$

For $x = \ell_0$ relation (20) is written

$$u(\ell'_0) = \mathrm{int}(\lambda^{-1}\widehat{\varphi}(g_0))(\ell_0'^{p}) \tag{21}$$

which, compared with (17), gives

$$\mathrm{int}(g_0^{-1}\lambda^{-1}\widehat{\varphi}(g_0))(\ell_0'^{p}) = \ell_0'^{p} \quad \text{hence}$$

$$g_0^{-1}\lambda^{-1}\widehat{\varphi}(g_0) = \ell_0'^{\beta} \qquad (\beta \in \widehat{\mathbb{Z}})$$

or again

$$\boxed{\lambda = \widehat{\varphi}(g_0)\ell_0'^{-\beta}g_0^{-1}} = \widehat{\varphi}(g_0 h_0^{-1}) = \widehat{\varphi}(\sigma_\infty(g_0)) \tag{22}$$

↘ cf. below (24 bis)

For $x = \ell_1$ relation (20) is written

[scan 62]

$$u(\ell'_1) = \mathrm{int}(\lambda^{-1}\widehat{\varphi}(g_1))(\ell_1'^{p}) \tag{23}$$

or again

$$\sigma_\infty = \mathrm{int}(\lambda^{-1}\widehat{\varphi}(g_1))(\sigma_\infty)$$

where ($g_1 = \sigma_\infty(g_0) = \sigma_\infty g_0 \sigma_\infty$)

m707

Marginal note. m707 For $\sigma_\infty^p = \sigma_\infty$ since $\sigma_\infty^2 = 1$ and the image of p in $(\mathbb{Z}/2\mathbb{Z})^*$ is 1!

or again σ_∞ commutes with

$$\lambda^{-1}\widehat{\varphi}(g_1) \overset{\text{cf. (22)}}{=} g_0\ell_0'^{\beta}\widehat{\varphi}(\underbrace{g_0^{-1}\sigma_\infty(g_0)}_{h_0^{-1}}) = g_0\ell_0'^{\beta}\widehat{\varphi}(h_0^{-1})$$

or again with its inverse $\widehat{\varphi}(h_0)\ell_0'^{-\beta}g_0^{-1}$ which is also written

$$\underbrace{\sigma_\infty(\widehat{\varphi}(h_0).\ell_0'^{-\beta}.g_0^{-1})} = \widehat{\varphi}(h_0).\ell_0'^{-\beta}.g_0^{-1}$$

$$\sigma_\infty(\widehat{\varphi}(h_0)).\underbrace{\varepsilon_1^{-\beta}}_{(\rho\sigma_\infty)^{-\beta}}.\sigma_\infty(g_0)^{-1} = \widehat{\varphi}(h_0)\ell_0'^{-\beta}g_0^{-1}$$

$$\sigma_\infty(\widehat{\varphi}(h_0))(\underbrace{\rho\sigma_\infty}_{\varepsilon_1})^{-\beta}h_0 = \widehat{\varphi}(h_0).\varepsilon_0^{-\beta}$$

$$\boxed{\overbrace{\varepsilon_1^{-\beta} h_0 \varepsilon_0^{\beta}}^{\sigma_\infty(g_0\varepsilon_0^\beta)^{-1}(g_0\varepsilon_0^\beta)} = \sigma_\infty(\widehat{\varphi}(h_0))^{-1}\widehat{\varphi}(h_0)} \tag{24}$$

m709

Correct? ↘

Marginal note. m709 N.B. If g_0 is replaced by $g_0' = g_0\ell_0^\lambda$ ($\lambda \in \widehat{\mathbb{Z}}$), h_0 is replaced by $h_0' = \ell_1^{-\lambda} h_0 | g_0 \ell_0^\lambda$, and we find that β is " replaced by $\beta' = \beta - 3\lambda$. This shows ? that we can in a unique way [illeg. 1] normalize the choice of g_0, in such a way that we have $\beta \in \{0, 1, 2\}$. (It would be necessary to elucidate whether β actually takes all three values modulo 3…)

m710

Marginal note. m710 N.B. This equation is equivalent to the simpler equation

(24 bis) $$\boxed{\widehat{\varphi}(h_0) = \sigma_\infty^{\varepsilon} g_0 \varepsilon_0^\beta}$$

(25)
$$\begin{cases} h_0 \text{ given by (8) in terms of } g_0 & \\ \varepsilon_0 = \ell_0' = \sigma_\infty \rho & (\varepsilon_0^2 = \ell_0) \\ \varepsilon_1 = \rho \sigma_\infty = \sigma_0 \rho & (\varepsilon_1^2 = \ell_1) \end{cases}$$

Setting, for † $\mu \in \widehat{\mathfrak{S}}_{0,3}^+$,

$$\tilde{\mu} = \sigma_\infty(\mu)^{-1}\mu = \sigma_\infty \mu^{-1} \sigma_\infty \mu = [\sigma_\infty, \mu] \tag{26}$$

(so that $h_0 = \tilde{g}_0$) the formula (24)|(25) takes the form

$$\varepsilon_1^{-\beta} \tilde{g}_0 \varepsilon_0^\beta = (\widehat{\varphi}(\tilde{g}_0))^{\sim}. \tag{27}$$

Uncertainty retained from cancelled passages.
140-3-s062-D001 *g*

[scan 63]
[p. 320]

In summary:

Proposition Let us consider an outer automorphism u of the group with loops $\widehat{\Pi}_{0,3}$, realized by $g_0 \in \widehat{\Pi}_{0,3}$ modulo L_0, and $p \in \widehat{\mathbb{Z}}^*$, (and $\gamma \in \widehat{\mathbb{Z}}^*$ uniquely determined by terms|in terms of g_0 and p) by formulas (11), by means of an actual automorphism (still denoted u) which commutes with σ_∞ — the data g_0, p (and γ) being therefore required to satisfy conditions (9) and (10) above, plus the generation condition (11 bis). (We extend u to an automorphism of $\widehat{\mathfrak{S}}_{0,3}^+$ by int(u), cf. (15)) For the outer homomorphism $\widehat{\varphi} : \widehat{\Pi}_{0,3} \longrightarrow \widehat{\mathfrak{S}}_{0,3}^+$ defined by (18) to commute with the outer action of u, it is necessary and sufficient that we can find a $\beta \in \widehat{\mathbb{Z}}$, such that we have (24), or what amounts to the same, (27),

better: 24 bis

This condition (for β fixed) is also equivalent to formula (20) for $x = \ell_\infty$ (λ being always given by (22)), † i.e. to the formula

$$u(\ell_\infty') = \mathrm{int}(\lambda^{-1}\widehat{\varphi}(g_\infty))(\ell_\infty'^{\,p}), \qquad \text{where} \tag{28}$$

$$g_\infty = g_0 \ell_0^{-\gamma} \rho^{-1}(h_0) \qquad (\text{and } \lambda \text{ given by (22)}). \tag{29}$$

Uncertainty retained from cancelled passages.
140-3-s063-D001 (2

[scan 64]

formula which thus makes precise the presumptive formula (18). It is also written, more explicitly

$$\rho^{-1}g^{-1} = \mathrm{int}(g_0\varepsilon_0^{\beta}\widehat{\varphi}(\ell_0^{\gamma}\rho^{-1}(h_0)))(\rho^{-p})$$

$$g_0\varepsilon_0^{\beta+\gamma}\widehat{\varphi}(\rho^{-1}(h_0))$$

i.e. (taking inverses, and making g explicit)

$$(30) \qquad \boxed{\dagger h_0^{-1}\ell_1^{\gamma}\rho = \mathrm{int}(\dagger\varepsilon_0^{\dagger\alpha}\widehat{\varphi}(\rho^{-1}(h_0)))(\rho^{p\dagger})}$$

m712

Marginal note. m712 N.B. We put $\alpha = \beta + \gamma$. The existence of $\beta \in \widehat{\mathbb{Z}}$ satisfying (24), is equivalent to that of $\alpha \in \widehat{\mathbb{Z}}$ satisfying (30).

The equivalence of (24) and (30) may seem a little strange, owing to the fact for example that in (24) the exponent p does not enter explicitly, but only h_0, whereas formula (30) brings in the value of $p \in \widehat{\mathbb{Z}}^*$ modulo 3 (which gives $\rho^p = \rho$ or $\rho^p = \rho^{-1}$). But in any event, whether it be in the form (24) where|or in the form (30), it does seem that the condition in question is highly non-tautological!

To do things properly, it would be necessary to first make explicit the congruential conditions of § 39 in terms of g_0, p.

Uncertainty retained from cancelled passages.
140-3-s064-D001 [*illeg.* 1]
140-3-s064-D002 g_0
140-3-s064-D003 $\beta + \gamma$
140-3-s064-D004 $-p$

[scan 65]
[p. 321]

I would like to compare the outer automorphisms of $\widehat{\Pi}_{0,3}$, and those of $\widehat{\mathfrak{S}}^+_{0,3}$, taking account of the exact sequence

$$(31) \qquad \begin{array}{ccccccccc} 1 & \longrightarrow & \widehat{\Pi}_{0,3} & \longrightarrow & \widehat{\mathfrak{S}}^+_{0,3} & \longrightarrow & \mathfrak{S}_3 & \longrightarrow & 1 \\ & & & & \wr & & & & \\ & & & & \widehat{\Pi'}_{0,3} & & & & \end{array}$$

Let us note that every outer automorphism u of $\widehat{\mathfrak{S}}^+_{0,3}$ which leaves invariant the normal subgroup $\widehat{\Pi}_{0,3}$, and induces on the quotient $\mathfrak{S}_3$ a trivial outer automorphism, can be written as obtained from an automorphism u_0 of $\widehat{\mathfrak{S}}^+_{0,3}$, stabilizing $\widehat{\Pi}_{0,3}$, and inducing the identity on the quotient $\mathfrak{S}_3$ — and u_0 is then determined modulo composition with an element $\mathrm{int}(g)$, where

$g \in \widehat{\mathfrak{S}}^+_{0,3}$ is such that its image in $\mathfrak{S}_3$ is central, i.e. $= 1$, i.e. $g \in \widehat{\Pi}_{0,3}$. Thus

$$(32) \qquad \begin{array}{c}\text{group of outer automorphisms}\\ \text{of } \widehat{\mathfrak{S}}^+_{0,3} \text{ which}\\ \text{induce on } \mathfrak{S}_3 \text{ a}\\ \text{trivial outer automorphism (hence leave invariant}\\ \widehat{\Pi}_{0,3})\end{array} \simeq \begin{array}{c}\text{group of}\\ \text{automorphisms of}\\ \widehat{\mathfrak{S}}^+_{0,3} \text{ which}\\ \text{induce}\\ \text{the identity on}\\ \text{its quotient } \mathfrak{S}_3,\\ \underline{\text{modulo}} \text{ the sub-}\\ \text{group of}\\ \text{inner automorphisms}\\ \text{by } g \in \widehat{\Pi}_{0,3}\end{array}$$

[scan 66]

Is such an automorphism u_0 determined by its restriction to $\widehat{\Pi}_{0,3}$? In other words, an automorphism of $\widehat{\mathfrak{S}}^+_{0,3}$ $(\simeq \mathrm{Sl}(2|1,\mathbb{Z})'^{\wedge})$ which induces the identity on $\widehat{\Pi}_{0,3}$ and on $\mathfrak{S}_3$, is it trivial? It suffices to prove that its restriction to the inverse image $\widehat{\Pi}_{0,\infty|0,3} \cdot \{1, \sigma_\infty\}$ of $\{1, \dot\sigma_\infty\} \subset \mathfrak{S}_3$ is trivial|trivial (by symmetry this will also be the case on replacing $\dot\sigma_\infty$ by $\dot\sigma_0, \dot\sigma_1$, now these three subgroups generate $\widehat{\mathfrak{S}}^+_{0,3}$. We will have $f(\sigma_\infty) = \sigma_\infty g$, with $g \in \widehat{\Pi}_{0,3}$ [and we must have $\sigma_\infty g \sigma_\infty g = 1$ i.e.

$$g \cdot \sigma_\infty(g) = 1$$

] moreover † we will have to have that

$$\mathrm{int}(\sigma_\infty g)|\widehat{\Pi}_{0,3} = \dagger\, \mathrm{int}(\sigma_\infty)|\widehat{\Pi}_{0,3}$$

whence

$$\mathrm{int}(g)|\widehat{\Pi}_{0,3} = \mathrm{id} \quad \text{i.e.} \quad g \in \mathrm{Centre}(\widehat{\Pi}_{0,3}) \quad \text{i.e.} \quad g = 1,$$

OK.

But under what condition does an automorphism of $\widehat{\Pi}_{0,3}$ extend to an automorphism of $\widehat{\mathfrak{S}}^+_{0,3}$ which induces the identity on $\mathfrak{S}_3$? We see at once that it is necessary for this that the outer automorphism of $\widehat{\Pi}_{0,3}$ which it induces commute with the action of $\mathfrak{S}_3$.

Uncertainty retained from cancelled passages.

140-3-s066-D002 we have

140-3-s066-D003 $f(x\sigma_\infty)$

[scan 67]

This is sufficient, for let $\Gamma \subset \mathrm{Aut\,ext}(\widehat{\Pi}_{0,3})$ be the group of outer automorphisms of $\widehat{\Pi}_{0,3}$ which have this property (we are not concerned for the moment with the loop structure of $\widehat{\Pi}_{0,3}$), we have $\Gamma \cap \mathfrak{S}_3 = (1)$ (identifying $\mathfrak{S}_3$ with a subgroup of $\mathrm{Aut\,ext}\,\widehat{\Pi}_{0,3}$) for † the center of $\mathfrak{S}_3$ is trivial, hence [p. 322|332]

this is evident|evidently

$$\dagger\ \mathfrak{S}_3 \times \Gamma \hookrightarrow \mathrm{Autext}(\widehat{\Pi}_{0,3}),$$

and we thus have an extension structure

$$1 \longrightarrow \widehat{\Pi}_{0,3} \longrightarrow \mathcal{E} \longrightarrow \mathfrak{S}_3 \times \Gamma \longrightarrow 1 \qquad (33)$$

(for the center of $\widehat{\Pi}_{0,3}$ is trivial), or again an extension structure

$$1 \longrightarrow \widehat{\mathfrak{S}}^+_{0,3} \longrightarrow \mathcal{E} \longrightarrow \Gamma \longrightarrow 1 \qquad (34)$$

which defines an outer action of Γ on $\widehat{\mathfrak{S}}^{+}_{0,3}$ — which is the outer action sought. Thus

Proposition the group of outer automorphisms of $\widehat{\Pi}_{0,3}$ which commute with $\mathfrak{S}_3$ is canonically isomorphic to the group of outer automorphisms of $\widehat{\mathfrak{S}}^{+}_{0,3}$ which induce on $\mathfrak{S}_3$ the trivial outer automorphism.

But every automorphism of $\mathfrak{S}_3$ is inner!

Uncertainty retained from cancelled passages.

140-3-s067-D001 $\mathfrak{S}$

140-3-s067-D002 $\Gamma\times$

[scan 68]

Corollary the group of automorphisms of $\widehat{\Pi}_{0,3}$ which commute as outer actions with $\mathfrak{S}_3$, is isomorphic to the group of automorphisms of $\widehat{\mathfrak{S}}^{+}_{0,3}$ which induce on $\mathfrak{S}_3$ the trivial automorphism (by restriction of the latter)

Corollary

$$\begin{array}{ccccccccc} 1 & \text{—} & \widehat{\Pi}_{0,3} & \longrightarrow & \mathrm{Aut}\left(\widehat{\Pi}_{0,3} \,\middle|\, \begin{array}{l}\text{comm.}\\ \text{ext.}^{t}\\ \text{with } \mathfrak{S}_3\end{array}\right) & \longrightarrow & \mathrm{Autext}\left(\widehat{\Pi}_{0,3} \,\middle|\, \begin{array}{l}\text{comm.}\\ \text{with } \mathfrak{S}_3\end{array}\right) & \longrightarrow & 1 \\ & & & & \uparrow \sim \text{ restriction to } \widehat{\Pi}_{0,3} & & \uparrow \sim & & \\ 1 & \text{—} & \widehat{\Pi}_{0,3} & \text{—} & \underbrace{\mathrm{Aut}\left(\widehat{\mathfrak{S}}^{+}_{0,3} \,\middle|\, \begin{array}{l}\text{identity on the}\\ \text{quotient } \mathfrak{S}_3\end{array}\right)}_{\mathcal{E}^{!}} & \to & \mathrm{Autext}\left(\widehat{\mathfrak{S}}^{+}_{0,3} \,\middle|\, \begin{array}{l}\dagger\text{trivial}\\ \text{on the}\\ \text{quotient } \mathfrak{S}_3\end{array}\right) & \to & 1 \end{array} \tag{35}$$

†

This extension $\mathcal{E}^{!}$ of Γ by $\widehat{\Pi}_{0,3}$ is splitnoncanonically, the splitting|splitting subgroup consists of the automorphisms of $\widehat{\Pi}_{0,3}$ which commute with σ_∞, or again the automorphisms of $\widehat{\mathfrak{S}}^{+}_{0,3}$ which †fix with σ_∞ (and which fix ρ mod $\widehat{\Pi}_{0,3}$) m715

Marginal note. m715 comes from the fact that there † exists † a [illeg. 2] $\widehat{\Pi}_{0,3}$-conjugation [illeg. 1] of order 2 above σ_∞ …

$$\begin{cases} u(\sigma_\infty) = \sigma_\infty \\ u(\rho) = g.\rho \end{cases} \qquad (g \in \widehat{\Pi}_{0,3}) \tag{36}$$

Uncertainty retained from cancelled passages.

140-3-s068-D001 finite

140-3-s068-D002 inducing Let [illeg. 1] u_0 $=|:=$ commuting [illeg. 2] well [illeg. 1]

140-3-s068-D004 commute

140-3-s068-D005 [illeg. 2]

140-3-s068-D006 [illeg. 1]

[scan 69]

writing that u is an endomorphism (for we recall that $\widehat{\mathfrak{S}}^{+}_{0,3}$ is defined by the generators σ_∞, ρ with the relations [p. 323]

$$\sigma_\infty^2 = \rho^3 = 1 \tag{37}$$

we find the condition

$$(g\rho)^3 = 1 \qquad \text{i.e.} \qquad g\,\rho(g)\,\rho^2(g) = 1. \tag{38}$$

m716

Marginal note. m716 moreover it is necessary in order that σ_∞ and $g\rho$ generate $\widehat{\Pi}_{0,3}$, with only relations $\sigma_\infty^2 = (g\rho)^3 = 1$.

Corollary. Let $u \in \mathcal{E}^!$ be an automorphism of $\widehat{\mathfrak{S}}^+_{0,3}$ which induces the identity on $\mathfrak{S}_3$. Equivalent conditions:

a) $\exists\ p \in \widehat{\mathbb{Z}}^*$ such that $u(\underbrace{\ell'_0}_{\varepsilon_0})$ conjugate in $\widehat{\mathfrak{S}}^+_{0,3}$ to $\ell_0'^p$ (this p is then unique, and we can conjugate by an element of $\widehat{\Pi}_{0,3}$).

b) $\exists\ p \in \widehat{\mathbb{Z}}^*$ such that $u(\ell_0)$ is conjugate in $\widehat{\Pi}_{0,3}$ to ℓ_0^p (this p is then unique).

c) $u_0 \overset{\text{dfn}}{=} u \mid \widehat{\Pi}_{0,3}$ respects the loop structure of $\widehat{\Pi}_{0,3}$.

Moreover, the p in a) and b) is the same.

Proof. If $u(\ell'_0) = \mathrm{int}(g_0)(\ell_0'^p)$, with $g_0 \in \widehat{\mathfrak{S}}^+_{0,3}$, squaring we will have $u(\ell_0) = \mathrm{int}(g_0)(\ell_0^p)$, this shows that the image $\dot{g}_0$ of g_0 in $\mathfrak{S}_3$ leaves invariant 0 i.e. is † 1 or $\dot{\sigma}_0$, if it is 1 we do have b), otherwise we put|† $g_0 = h\sigma_0$ ($h \in \widehat{\Pi}_{0,3}$) †

$$u(\ell_0) = \mathrm{int}(h)\ \underbrace{\sigma_0(\ell_0^p)}_{(\sigma_0(\ell_0))^p}$$

now we know that $\sigma_0(\ell_0) = \mathrm{int}(a)\ell_0$ (with $a \in \widehat{\Pi}_{0,3}$), whence $\sigma_0(\ell_0)^p = \mathrm{int}(a)\ell_0^p$, thus $u(\ell_0) = \mathrm{int}(ha)(\ell_0^p)$, whence b) with the same p.

Uncertainty retained from cancelled passages.

140-3-s069-D001 [illeg. 1]

140-3-s069-D003 know q

140-3-s069-D004 whence

[scan 70]

Since we know the uniqueness of p in b), in a) too. If we have

$$u_0(\ell_0) = \mathrm{int}(g_0)(\ell_0^p) \qquad (g_0 \in \widehat{\Pi}_{0,3}) \tag{39}$$

we deduce from this, writing

$$\dagger \qquad \rho u_0 \rho^{-1} = \mathrm{int}(g^{-1})u_0 \qquad (a|g \in \widehat{\Pi}_{0,3})$$

$$u_0(\ell_1) = \mathrm{int}(a|g)\,\rho\,u_0\,\underbrace{\underbrace{\underbrace{\rho^{-1}(\ell_1)}_{\ell_0}}_{\mathrm{int}(g_0)(\ell_0^p)}}_{\mathrm{int}(\rho(g_0))(\ell_1^p)} = \mathrm{int}(g\rho(g_0))(\ell_1^p)$$

i.e.

$$u_0(\ell_1) = \mathrm{int}(g_1)(\ell_1^p) \qquad \text{with} \quad g_1 = g\rho(g_0) \tag{40}$$

and likewise

$$u_0|u(\ell_\infty) = \mathrm{int}(g_\infty)(\ell_\infty^p) \qquad \text{with} \quad \begin{aligned} g_\infty &= g\rho(g_1) \\ &= g\rho(g)\rho^2(g_0) \end{aligned} \tag{41}$$

thus b) ⇒ c). Finally let us prove c) ⇒ a).

We have by hypothesis

$$\begin{array}{ccc} u_0(\ell_0) & = & \mathrm{int}(g)\ell_0^p \\ \text{"} & & \text{"} \\ u(\varepsilon_0)^2 & & \mathrm{int}(g)(\ell|\varepsilon_0^{2p|p})^2 = (\mathrm{int}\, g(\ell_0^p|\varepsilon_0^p))^2 \end{array}$$

I say that we then have

$$u(\varepsilon_0) = \mathrm{int}(g)(\varepsilon_0^p).$$

It suffices to recall that †the only element of $\widehat{\mathfrak{S}}^+_{0,3}$ with square equal to ℓ_0 is ε_0… OK check

---

Uncertainty retained from cancelled passages.

140-3-s070-D001 =

140-3-s070-D004 every

[scan 71]

[p. 324]

†

Henceforth we restrict ourselves to the automorphisms (actual or outer) of $\widehat{\Pi}_{0,3}$ which respect the loop structure, and commute with $\mathfrak{S}_3$ (as outer actions) — which correspond to the automorphisms (actual or outer) of $\widehat{\mathfrak{S}}^+_{0,3}$ which induce the identity on $\mathfrak{S}_3$, and satisfy a) above. †

The outer automorphisms of this nature correspond therefore to the actual automorphisms u of $\widehat{\Pi}_{0,3}$ which commute with σ_∞, i.e. u of $\widehat{\mathfrak{S}}^+_{0,3}$ which fix σ_∞, and are determined by a $g \in \widehat{\Pi}_{0,3}$ satisfying (38), and such that in addition there exist $p \in \widehat{\mathbb{Z}}^*$ and $g_0 \in \widehat{\Pi}_{0,3}$ satisfying

$$(39) \qquad u(\ell_0') = \mathrm{int}(g_0)(\ell_0'^{\,p}) \qquad \text{i.e.} \qquad u(\ell_0) = \mathrm{int}(g_0)(\ell_0^p)$$

(where|i.e. g_0 is determined modulo $L_0 = \ell_0^{\widehat{\mathbb{Z}}}$ …). Calculations made elsewhere show then that we must have

$$(40) \qquad g = \sigma_\infty(g_0)\, \ell_1^\gamma\, \rho(g_0)^{-1}$$

m718

Marginal note. m718 with $\gamma \in \widehat{\mathbb{Z}}$ well determined (independent of the choice of g_0)

i.e.

$$(41) \qquad \sigma_\infty(g_0)^{-1}\, \underbrace{g\rho(g_0)}_{g_1} \in L_1 = \mathcal{N}|N|\mathrm{Norm}_{\widehat{\Pi}_{0,3}}(L_1)$$

Uncertainty retained from cancelled passages.

140-3-s071-D001 have

140-3-s071-D002 [illeg. 1]

[scan 72]

Indeed, we have by hypothesis on g_0 (cf. (39))

$$(42) \qquad u(L_0) = \mathrm{int}(g_0)(L_0)$$

†|hence

$$\begin{array}{l} u\sigma_\infty(L_0) = u(L_1) \overset{40}{=} \mathrm{int}(g_1)(L_1) \\ \| \quad (\text{since } u\sigma_\infty = \sigma_\infty u) \\ \sigma_\infty u(L_0) \underset{(42)}{=} \sigma_\infty(\mathrm{int}(g_0)(L_0)) = \mathrm{int}(\sigma_\infty(g_0))(\sigma_\infty(L_0)) = \mathrm{int}(\sigma_\infty(g_0))(L_1) \end{array}$$

hence

$$\mathrm{int}(\sigma_\infty(g_0)^{-1} g_1)(L_1) \subset L_1$$

therefore

$$\sigma_\infty(g_0)^{-1}\, \underbrace{g_1}_{g\rho(g_0)} \in \mathcal{N}_{\widehat{\Pi}_{0,3}}(L_1) = L_1$$

hence

$$\sigma_\infty(g_0)^{-1} g_1 | g \rho(g_0) = \ell_1^\gamma \qquad (\gamma \in \widehat{\mathbb{Z}})$$

hence (40). We will note, to return to it, that the element g defined in terms of a g_0 and of γ indeed depends only on g_0 modulo L_0. † ✓

m720

Marginal note. m720



Thus; the datum of u_0 † a loop outer automorphism of $\widehat{\Pi}_{0,3}$ commuting with $\mathfrak{S}_3$, or again an$|u$ outer automorphism of $\widehat{\mathfrak{S}}^+_{0,3}$ inducing the identity in $\mathfrak{S}_3$ and † leaving invariant the conjugacy class of the loop subgroup $L'_0 = \varepsilon_0^{\widehat{\mathbb{Z}}} = \ell_0'^{\widehat{\mathbb{Z}}} = \widehat{\varphi}(L_0)$, or again that of a loop automorphism u_0 of $\widehat{\Pi}_{0,3}$ commuting with σ_∞, and in the outer automorphism quotient with ρ, or again of an automorphism u of $\widehat{\mathfrak{S}}^+_{0,3}$ which fixes σ_∞, and which fixes ρ modulo $\widehat{\Pi}_{0,3}$ — is equivalent to the datum of $g_0 \in \widehat{\Pi}_{0,3}$ modulo L_0, and of $\gamma \in \widehat{\mathbb{Z}}$, satisfying the relation

Uncertainty retained from cancelled passages.

140-3-s072-D001 hence

140-3-s072-D002 [illeg. 1]

140-3-s072-D003 satisfying

140-3-s072-D004 transforming

[scan 73]

m721

[p. 325]

Marginal note. m721

$$(38)\qquad \underline{g\rho(g)\rho^2(g)=1},$$

where g is given by (40), and where u is expressed in terms of g hence of g_0,γ by

$$(36)\qquad \begin{cases} u(\sigma_\infty)=\sigma_\infty \\ u(\rho)=g\rho \qquad \left(=\underbrace{\sigma_\infty(g_0)\ell_1^\gamma\rho(g_0)^{-1}}_{g}\,\rho\right). \end{cases}$$

†

†

†

†

†

†

†

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†

The relation (42), or the apparently stronger relation

$$(43)\qquad u(L_0')\subset \mathrm{int}(g_0)(L_0')$$

reads

$$\mathrm{int}(g_0^{-1})\qquad u(\underbrace{\sigma_\infty\rho}_{\underbrace{\ell_0'=\varepsilon_0}_{\underbrace{\sigma_\infty g\rho=g_0\sigma_\infty\ell_1^\gamma\rho g_0^{-1}}_{\sigma_\infty\ell_1^\gamma\rho=\underbrace{\sigma_\infty\ell_1^\gamma\sigma_\infty}_{\sigma_\infty(\ell_1)^\gamma}\underbrace{\sigma_\infty\rho}_{\dagger\varepsilon_0}}}})\qquad \in L_0'$$

$$\underbrace{\ell_0^\gamma=\varepsilon_0^{2\gamma}}_{\varepsilon_0^{2\gamma+1}}$$

m724

Marginal note. m724 N.B. we have $\mathrm{int}(g_0^{-1})u(\underbrace{\ell_0'}_{\varepsilon_0})=\varepsilon_0^{2\gamma+1}$

is satisfied automatically, we will have even

Uncertainty retained from cancelled passages.

140-3-s073-D002 cf. = above

140-3-s073-D003 σ_∞

140-3-s073-D005 2|4 $L_0'|L_0$ now

140-3-s073-D006 g_0^{-1}

140-3-s073-D010 i.e.

140-3-s073-D011 [*illeg.* 1]

140-3-s073-D014 i.e.

140-3-s073-D015 $\ell_0^\gamma \quad \varepsilon_0^{2\gamma}$
140-3-s073-D016 ...
140-3-s073-D017 $\mathrm{int}(g_0)$
140-3-s073-D018 $\sigma_\infty\rho$

[scan 74]

$$(44) \qquad \underset{\varepsilon_0}{u(\ell_0')} = \mathrm{int}(g_0)(\underset{\varepsilon_0}{\ell_0'^p}) \qquad \text{with} \quad \underline{\underline{p = 2\gamma + 1}}$$

hence (39), on raising to the square:

$$u(\ell_0) = \mathrm{int}(g_0)(\ell_0^p).$$

† N.B. We will note that if $p \in \widehat{\mathbb{Z}}^*$, then $p-1 \in 2\widehat{\mathbb{Z}}$ i.e. $\frac{p-1}{2} \in \widehat{\mathbb{Z}}$, since it suffices to check the analogous relation for a $p \in \mathbb{Z}_\ell^*$, for|that $p-1 \in 2\mathbb{Z}_\ell$, now if $\ell \neq 2$ we have $2\mathbb{Z}_\ell = \mathbb{Z}_\ell$ OK, there remains the case $\ell = 2$ $p \in \mathbb{Z}_2^*$ so the image of p in $\mathbb{Z}/4\mathbb{Z}$ is 1 or 3 so $p-1 \equiv 0$ or 2 mod 4 † so $\frac{p-1}{2} \in \mathbb{Z}_2$, OK.

†

It therefore remains to express condition (38), and we write it †

$$\underbrace{\sigma_\infty(g_0)\ell_1^\gamma\rho(g_0^{-1})}\ \underbrace{\rho\sigma_\infty(g_0)\ell_\infty^\gamma\rho^2(g_0^{-1})}\ \rho^2\sigma_\infty(g_0)\ell_0^\gamma g_0^{-1} = 1$$

i.e., on applying $\mathrm{int}(g_0^{-1})$ and †setting again (cf. (8))

$$(45) \qquad g_0^{-1}\sigma_\infty(g_0) = h_0^{-1} \qquad \text{i.e.} \quad h_0 = \sigma_\infty(g_0)^{-1}g_0 = [\sigma_\infty, g_0^{-1}]$$

$$h_0^{-1}\ell_1^\gamma\ \underbrace{\rho(h_0^{-1})\ell_\infty^\gamma}\ \rho^2(h_0^{-1})\ell_0^\gamma = 1$$

or again, on applying $\mathrm{int}(\ell_0^\gamma)$

$$(46) \qquad \boxed{\underbrace{(\ell_0^\gamma h_0^{-1})}\cdot\rho(\ell_0^\gamma h_0^{-1})\cdot\rho^2(\ell_0^\gamma h_0^{-1}) = 1} \qquad \left(\text{where } \gamma = \frac{p-1}{2}\right)$$

which is none other than (10). As for the condition (9)|(7), we see that it is superfluous, i.e., automatically satisfied.

Uncertainty retained from cancelled passages.
140-3-s074-D002 We
140-3-s074-D003 [illeg. 3]
140-3-s074-D004 It therefore remains to expri
140-3-s074-D005 in the form
140-3-s074-D006 [illeg. 2]

[scan 75]
[p. 326]

Let us sum up:

Theorem An outer automorphism with loops|with loops of $\widehat{\Pi}_{0,3}$ commuting with $\mathfrak{S}_3$, lifts uniquely as an automorphism of the group with loops|with loops $\widehat{\Pi}_{0,3}$, commuting with σ_∞ (and commuting with ρ modulo inner|inner automorphisms), or again as the restriction of an automorphism u of $\widehat{\mathfrak{S}}_{0,3}^+$ which fixes σ_∞, leaves invariant ρ modulo $\widehat{\Pi}_{0,3}$ (i.e. fixes $\dot{\rho}$ in $\mathfrak{S}_3$), and which leaves invariant the conjugacy class of the closed subgroup $L_0' = \widehat{\varphi}(L_0)$ † generated by $\ell_0' = \varepsilon_0 = \sigma_\infty\rho$. (or again of the closed subgroup $L_0 = (L_0')^2$ generated by ℓ_0). The datum of such a u is equivalent to that of its multiplier $p \in \widehat{\mathbb{Z}}^*$, and of a $g_0 \in \widehat{\Pi}_{0,3}/L_0$ satisfying the following conditions

a) Setting

$$(47)\qquad \begin{cases} h_0 = \sigma_\infty(g_0)^{-1} g_0 = [\sigma_\infty, g_0^{-1}] \quad (\in \widehat{\Pi}_{0,3}) \\ \gamma = \dfrac{p-1}{2} \in \widehat{\mathbb{Z}} \end{cases}$$

we have relation (46) above (invariant under replacement of g_0 by $g_0 \ell_0^\lambda$, $\lambda \in \widehat{\mathbb{Z}}$)

$$(48)\qquad (\underbrace{\ell_0^\gamma h_0^{-1}}_{\xi_0}) \rho(\ell_0^\gamma h_0^{-1}) \rho^2(\ell_0^\gamma h_0^{-1}) = 1$$

m725

Marginal note. m725 i.e. $\xi_0 \rho(\xi_0) \rho^2(\xi_0) = 1$ with $\xi_0 = \ell_0^\gamma h_0^{-1}$

b) σ_∞ and $g\rho$ generate topologically $\widehat{\mathfrak{S}}^+_{0,3}$, where g is given in terms of g_0 by 40

$$[(40)]\qquad g = \sigma_\infty(g_0) \ell_1^\gamma \rho(g_0)^{-1} = \sigma_\infty(g_0 \ell_0^\gamma) \rho(g_0^{-1})$$

m726

Marginal note. m726
$= g_0(h_0^{-1} \ell_1^\gamma) \rho(g_0)^{-1}$
$= (g_0 \ell_0^{-\gamma})(\ell_0^\gamma h_0^{-1})(\rho(g_0 \ell_0^{-\gamma}))^{-1}$

† (N.B. relation (46) is equivalent to: $(g\rho)^3 = 1$ i.e. to

$$(38)\qquad g\rho(g)\rho^2(g) = 1,$$

expressed directly in terms of g_0, $\gamma = \frac{p-1}{2}$), with the generating relations $\sigma_\infty^2 = (g\rho)^3 = 1$. The (36) automorphism u is given in terms of g_0 and p by

Uncertainty retained from cancelled passages.
140-3-s075-D001 [illeg. 3]
140-3-s075-D002 [illeg. 3]

[scan 76]

$$[(36)]\qquad \begin{cases} u(\sigma_\infty) = \sigma_\infty \\ u(\rho) = g\rho \end{cases} \qquad \text{where } g \text{ given by } [(40)])$$

We will then have

$$(49)\qquad \begin{cases} u(\ell_0) = \mathrm{int}(g_0) \ell_0^p \\ u(\ell_1) = \mathrm{int}(g_1) \ell_1^p = \mathrm{int}(\sigma_\infty(g_0))(\ell_1^p) \\ u(\ell_\infty) = \mathrm{int}(g_\infty)(\ell_\infty^p) \end{cases}$$

m728

Marginal note. m728
$g_1 = g\rho(g_0) = \sigma_\infty(g_0)\ell_1^\gamma$
$= \sigma_\infty(g_0 \ell_0^\gamma)$
$g_\infty = g\rho(g_1) = \dagger g\rho(g)\rho^2(g_0)$
$= \rho^2(g^{-1} g_0) = \rho(g_0 \ell_0^\gamma) h_0$

†

cf. proposition above β resp. α are uniquely determined

Uncertainty retained from cancelled passages.

140-3-s076-D001 g

140-3-s076-D002 surjective $0,3|3$ def find $30|38$ h_0^{-1} $\rho^{-1}(h_0)|\rho^{-1}h_0)$ $\rho^{-1}(h_0)|\rho^{-1}h_0)$

[scan 77]

Scholium In the theorem, the $g \in \widehat{\Pi}_{0,3}$ (which enters directly in writing u) is expressed by means of the $g_0 \in \widehat{\Pi}_{0,3}$ and $\gamma = \frac{p-1}{2}$, † (g_0 defined modulo L_0) — condition (46) being in fact a condition which is expressed directly in terms of g_0, or better still, in terms of [p. 327]

$$(50) \qquad \xi_{\dagger|0} = \ell_0^{\gamma} h_0^{-1} \in \widehat{\Pi}_{0,3} \quad (\text{whence } h_0 = \xi_{\dagger|0}^{-1} \ell_0^{\gamma})$$

as

$$(51) \qquad \xi\rho(\xi)\rho^2(\xi) = 1,$$

which is equivalent to

$$(52) \qquad (\xi\rho)^3 = 1.$$

The solutions of equation (51), or what amounts to the same thing of (52), correspond to the splittings of the|the sub-extension of $\widehat{\mathfrak{S}}_{0,3}^{+}$ of $\mathbb{Z}/3\mathbb{Z}$ by $\widehat{\Pi}_{0,3}$, generated by $\rho \in \mathfrak{S}_3$, † i.e. to the elements of order 3 of $\widehat{\mathfrak{S}}_{0,3}^{+}$ above $\dot{\rho}$. On the set of these splittings resp. of these elements $\xi\rho$, $\widehat{\Pi}_{0,3}$ operates by inner automorphisms:

$$(53) \qquad \mathrm{int}(\lambda)(\xi\rho) = \lambda\xi\rho\lambda^{-1} = \lambda\xi\rho\lambda^{-1}\rho^{-1}\rho = (\lambda\xi\rho(\lambda)^{-1})\rho$$

i.e. it operates on the set of solutions of (51), by

$$(54) \qquad {}^{\lambda}\xi = \lambda\xi\rho(\lambda)^{-1}$$

N.B. $\lambda \mapsto {}^{\lambda}\xi$ is injective

Uncertainty retained from cancelled passages.

140-3-s077-D001 [illeg. 2]

140-3-s077-D002 0

140-3-s077-D003 0

140-3-s077-D004 [illeg. 1]

[scan 78]

Let us admit (leaving it to be checked later) that there are exactly two conjugacy classes: that of $\xi_0 = 1$, corresponding to the tautological splitting, †in such a way that for such ξ we will have

$$(55) \qquad \xi = \lambda\rho(\lambda)^{-1} \qquad (\lambda \in \widehat{\Pi}_{0,3}),$$

and on the other hand the conjugacy class of the lift^{t} of order 3 $\sigma_\infty\rho^{-1}\sigma_\infty^{-1} = \sigma_\infty\rho^{-1}\sigma_\infty$ of $\dot{\rho}$, corresponding to

$$\sigma_\infty\rho^{-1}\sigma_\infty = \xi_0\rho, \quad \text{with}$$

$$\begin{cases} \xi_0 = \sigma_\infty\rho^{-1}\sigma_\infty\rho^{-1} = \sigma_\infty \underbrace{(\rho^{-1}\sigma_\infty)}_{\varepsilon_0^{-1}}(\rho^{-1}\sigma_\infty)\sigma_\infty \\ \quad = \sigma_\infty\varepsilon_0^{-2}\dagger\sigma_\infty = \sigma_\infty(\ell_0^{-1}) = \ell_1^{-1} \end{cases}$$

(and we do indeed have

$$\ell_1^{-1}\rho(\ell_1^{-1})\rho^2(\ell_1^{-1}) = \ell_1^{-1}\ell_\infty^{-1}\ell_0^{-1} = (\ell_0\ell_\infty\ell_1)^{-1} = 1$$

), This orbit corresponds to solutions of the form †

$$(56) \qquad \xi = \lambda\ell_0^{-1}\rho(\lambda)^{-1} \qquad (\lambda \in \widehat{\Pi}_{0,3})$$

[we will note that ℓ_0^{-1} and ℓ_1^{-1} are in the same orbit

$$\ell_1^{-1} = \dagger \ell_0(\ell_0^{-1})\rho(\ell_0)^{-1}$$

]

i.e. †

†

A) Case † where ξ is of the form (55), hence

$$(56|57) \qquad h_0 = \rho(\lambda)\lambda^{-1}\ell_0^\gamma$$

Uncertainty retained from cancelled passages.

140-3-s078-D001 [illeg. 1]
140-3-s078-D002 =
140-3-s078-D003 [illeg. 1]
140-3-s078-D004 $\ell_0\ell_1$
140-3-s078-D005 [illeg. 1] 0
140-3-s078-D007 a)
140-3-s078-D008 A

[scan 79]
[p. 328]

It remains then to solve the equation in g_0 (47)

$$(58) \qquad \sigma_\infty(g_0)^{-1}g_0 = \underbrace{\rho(\lambda)\lambda^{-1}\ell_0^\gamma}_{h_0}$$

For ∀|a fixed h_0, there is at most a single g_0 which satisfies this condition, and such a g_0 exists iff , which is written

using that $\widehat{\Pi}_{0,3}^{\sigma_\infty} = \{1\}$

$$(59) \qquad \sigma_\infty(h_0)\cdot h_0 = 1 \qquad \text{or again } h_0\sigma_\infty(h_0) = 1$$

which is also written, taking up again the expression $h_0 = \xi^{-1}\ell_0^\gamma$

$$\dagger \qquad \sigma_\infty(\xi)^{-1}\ell_1^\gamma\xi^{-1}\ell_0^\gamma = 1 \qquad \text{i.e.}$$

$$(60) \qquad \sigma_\infty(\xi) = \ell_1^\gamma\cdot\xi^{-1}\cdot\ell_0^\gamma$$

In summary

Corollary The u which do not change the $\widehat{\Pi}_{0,3}$-conjugacy class of ρ correspond 1-1 to the systems $(\lambda, p) \in L_0\backslash\widehat{\Pi}_{0,3}\times\widehat{\mathbb{Z}}^*$, such that — setting $\xi = \lambda\rho(\lambda)^{-1}$, $\gamma = \frac{p-1}{2}$, we have relation (60). We will note that for λ hence ξ fixed, there exists at most one $\gamma \in \widehat{\mathbb{Z}}$ such that relation (60) holds, hence p is determined by λ, subject to the additional condition that the element $2\gamma+1$ of $\widehat{\mathbb{Z}}$ be in $\widehat{\mathbb{Z}}^*$ (i.e. $\exists\gamma' \in \widehat{\mathbb{Z}}^*$ with $2\gamma\gamma' + \gamma + \gamma' = 0$)

Uncertainty retained from cancelled passages.

140-3-s079-D003 60

[scan 80]

N.B. If λ is replaced by $\lambda' = \ell_0^\alpha\lambda$, $\xi = \lambda\rho(\lambda)^{-1}$ is replaced by $\xi' = \ell_0^\alpha\cdot\xi\rho(\ell_0)^{-\alpha}$, and $h_0 = \xi^{-1}\ell_0^\gamma$ is replaced by $h_0' = \rho(\ell_0)^\alpha h_0\ell_0^{-\alpha}$, finally g_0 is replaced by $g_0' = g_0\ell_0^{-\alpha}$, OK. ! (Equation (60) for ξ or for ξ' is equivalent)

Finally, whether u changes or does not change the $\widehat{\Pi}_{0,3}$-conjugacy class of ρ, the clearest approach for determining the u in question seems to me to be this:

1°) We choose a $\xi \in \widehat{\Pi}_{0,3}$ satisfying the equation

$$(A) \qquad \xi\rho(\xi)\rho^2(\xi) = 1$$

(we will take it either in the form (55), or in the form (56)† :

$$(B) \qquad \xi = \lambda\rho(\lambda)^{-1} \quad \text{or} \quad \xi = \lambda\ell_0^{-1}\rho(\lambda)^{-1} \quad \dagger, \quad \text{with } \lambda \in \widehat{\Pi}_{0,3}$$

)

2°) We choose $\gamma \in \widehat{\mathbb{Z}}$ such that

$$(C) \qquad \sigma_\infty(\xi) = \ell_1^\gamma \xi^{-1} \ell_0^\gamma$$

m733

Marginal note. m733 here, ξ is thus given in terms of λ by by|by (B)

which determines γ uniquely. We will in 1°)|2°) have to restrict ourselves to the ξ (i.e. to the λ) for which such a γ exists, and such moreover that $p = 2\gamma + 1$ be $\in \widehat{\mathbb{Z}}^*$.

Uncertainty retained from cancelled passages.

140-3-s080-D001)

140-3-s080-D002 }|ξ

[scan 81]

[p. 329]

3°) We then construct

$$h_0 = \xi^{-1}\ell_0^\gamma \in \widehat{\Pi}_{0,3}$$

The relation in 2°) ensures that $h_0\sigma_\infty(h_0) = 1$ i.e. that h_0 can be put in the form

$$h_0 = \sigma_\infty(g_0)^{-1} g_0$$

with $g_0 \in \widehat{\Pi}_{0,3}$.

m734

Marginal note. m734 N.B. We should make sure on the other hand that knowledge of g_0 does not determine γ i.e. p i.e. that † the γ such that $\xi = \ell_0^\gamma h_0^{-1}$ satisfies (A) is not † unique. To be clarified!

4°) We can now put

$$g = \sigma_\infty(g_0)\ell_1^\gamma\rho(g_0)^{-1} = (g_0\ell_0^{-\gamma})\xi\rho(g_0\ell_0^{-\gamma})^{-1}$$

and we † have what is needed to define u, via

$$\begin{cases} u(\sigma_\infty) = \sigma_\infty \\ u(\rho) = g\rho \end{cases}$$

Here, the only delicate equation, which the existence (not the uniqueness, in the good case) of a solution, is problematic, is equation C (in $\gamma \in \widehat{\mathbb{Z}}$) — plus the further condition that $2\gamma+1 \in \widehat{\mathbb{Z}}^*$. This is therefore a †condition on $\lambda \in \widehat{\Pi}_{0,3}$ — or rather two types of condition, according as ξ is defined in terms of λ by one or the other formula (B).

I expect that we are in one or

80

Uncertainty retained from cancelled passages.
140-3-s081-D001 [illeg. 2]
140-3-s081-D002 [illeg. 1]
140-3-s081-D003 define
140-3-s081-D005 question

[scan 82]

the other case of formulas (B), according as p mod 3 is congruent to 1, or to -1 (N.B. it must not be congruent to 0, since p mod $3 \in (\mathbb{Z}/3\mathbb{Z})^*$) i.e. according as $2\gamma \equiv 0$ or -2 (3) i.e. $\gamma \equiv 0$ or -1 (3) (N.B. γ must not be $\equiv 1$ (3), otherwise $2\gamma + 1 = p$ would be congruent to 0 mod (3)). This is in any case what will be the case if $u \in \Gamma_{\mathbb{Q}_0}$.

To test this, let us reduce the †group $\widehat{\mathfrak{S}}^+_{0,3}$ modulo the' equation|equations $\ell_0^3 = 1$ i.e. $(\sigma_\infty \rho)^6 = 1$

so that $\widehat{\mathfrak{S}}^+_{0,3}$ is replaced by an extension of $\mathfrak{S}_3$ by the group defined by the generators $\ell_0, \ell_1, \ell_\infty$ satisfying

$$\ell_0{}^3 = \ell_1^3 = \ell_\infty^3 = (\ell_\infty \ell_1 \ell_0) = 1$$

— or better, let us use

$$\widehat{\mathfrak{S}}^+_{0,3} \simeq \mathrm{Sl}(2, \mathbb{Z})'^\wedge$$

and reduce directly in

$$\mathrm{Sl}(2, \mathbb{Z}/6\mathbb{Z})' \simeq \mathrm{Sl}(2, \mathbb{Z}/2\mathbb{Z}) \times \mathrm{Sl}(2, \mathbb{Z}/3\mathbb{Z})'.$$

81

Uncertainty retained from cancelled passages.
140-3-s082-D001 [illeg. 1]

[scan 83]

But to do the calculations, we must be able to identify σ_∞ and ρ as elements of $\mathrm{Sl}(2, \widehat{\mathbb{Z}}|\mathbb{Z})$ — [p. 330] an explicitation which we shall carry out presently.

Returning to (24 bis), and putting $h_0|h_\infty = \xi^{-1}\ell_0^\gamma$ into it, we get

$$g_0 = \widehat{\varphi}(\xi)\varepsilon_0^{\gamma-\beta} \tag{61}$$

We thus think that u corresponds to $(\xi, g_0, \gamma, \beta)$, with $\xi, g_0 \in \widehat{\Pi}_{0,3}$ and $\gamma, \beta \in \widehat{\mathbb{Z}}$ satisfying [illeg. 1]

$$\begin{cases} \xi\rho(\xi)\rho^2(\xi) = 1 \\ \xi \underbrace{\sigma_\infty(g_0)^{-1} g_0}_{h} = \ell_0^\gamma \\ \dagger g_0 = \widehat{\varphi}(\xi)\varepsilon_0^{\gamma-\beta} \\ 2\gamma + 1 \in \widehat{\mathbb{Z}}^* \end{cases} \tag{62}$$

from which we can eliminate g_0, and we find equations in (ξ, r, s) $(r = \gamma - \beta,\ s = \gamma + \beta)$

$$\begin{cases} \xi\rho(\xi)\rho^2(\xi) = 1 \\ \sigma_\infty\big(\widehat{\varphi}(\xi)\big)^{-1}\widehat{\varphi}(\xi) = \varepsilon_1^r \xi^{-1} \varepsilon_0^s \\ \underbrace{r + s + 1}_{p} \in \widehat{\mathbb{Z}}^* \end{cases} \tag{63}$$

m738

Marginal note. m738 (thus $\xi = \lambda\rho(\lambda)^{-1}$ or $\xi = \lambda\ell_0^{-1}\rho(\lambda)^{-1}$

Here, λ is determined modulo transformation into $\lambda' = \dagger\ell_0^{\alpha}\lambda$ thus ξ becomes $\xi' = \ell_0^{\alpha}\xi\ell_1^{-\alpha}$ 82

Uncertainty retained from cancelled passages.
140-3-s083-D001 $\widehat{\varphi}(\xi)$
140-3-s083-D004 $\ell_0^{\alpha}\xi$

[scan 84]

42 Recapitulation on the group [p. 331]

$$(\mathfrak{S}_{0,3}) \simeq (\Pi_1(U_{0,3}^{\mathrm{an}}, \mathfrak{S}_3; -\bar{j})) \simeq (\widetilde{\Pi}'_{0,3} \simeq \Pi_1((\mathbb{P}^{1\mathrm{an}}, 0\{0\} + 2\{1\} + 3\{\infty\}), \{1, \tau\}; -\bar{j}))$$

$$\simeq (\mathrm{Gl}(2,\mathbb{Z})' = \mathrm{Gl}(2,\mathbb{Z})/\{\pm 1\} \simeq \mathrm{GP}(1,\mathbb{Z})) \simeq (\mathcal{T}'_{1,1} = \mathcal{T}_{1,1}/\{\pm 1\})$$

and on the "loop" automorphisms of $\widehat{\mathfrak{S}}^{+}_{0,3}$ and of $\widehat{\Pi}_{0,3}$ (Fundamental formulary).
m740

Marginal note. m740 † $\mathfrak{S}_{0,3}$ defined as $\mathrm{Aut}_{\mathrm{lac}}(\Pi_{0,3})$ — we give five different views of it ! (In fact, there are some ten of them, cf. (V) below…)

I) $\mathcal{T}_{0,3}$
The group $\mathfrak{S}_{0,3}$ is presented as an extension

$$(1) \qquad \begin{array}{ccccccccc} 1 & \longrightarrow & \Pi_{0,3} & \longrightarrow & \mathfrak{S}_{0,3} & \longrightarrow & \mathcal{T}_{0,3} & \longrightarrow & 1 \\ & & \| & & & & \wr & & \\ & & \Pi_1(U_{0,3}^{\mathrm{an}}; -\bar{j}) & & & & \mathfrak{S}_3 \times \mathbb{Z}/2\mathbb{Z} & & \end{array}$$

where $\mathfrak{S}_3 \times \mathbb{Z}/2\mathbb{Z}$ is regarded as a group operating on $U_{0,3}^{\mathrm{an}} = \mathbb{P}^{1\mathrm{an}} \setminus \{0,1,\infty\} = \mathbb{C} \setminus \{0,1\}$, † — it is the group of conformal automorphisms of the|of $\mathbb{P}^{1\mathrm{an}}$ which stabilize $\{0,1,\infty\}$. The group $\dagger\mathfrak{S}_3 = \mathfrak{S}_{\{0,1,\infty\}}$ is the subgroup of †conformal automorphisms, it is identified with the group of permutations of $\{0,1,\infty\}$, its six elements are id;|1; $\dot{\sigma}_0, \dot{\sigma}_1, \dot{\sigma}_\infty$ (the three transpositions), $\dot{\rho}, \dagger\dot{\rho}^{-1}$ (cyclic permutations) given by the explicit formulas (2)

$$(2) \qquad \begin{cases} \dot{\sigma}_0(z) = \dfrac{z}{z-1} = 1 + \dfrac{1}{1-z} \mid = 1 + \dfrac{1}{z-1}, \quad \dot{\sigma}_1(z) = \dfrac{1}{z}, \quad \dot{\sigma}_\infty(z) = 1 - z, \\ \dot{\rho}(z) = \dfrac{1}{1-z}, \quad \dot{\rho}^{-1}(z) = 1 - \dfrac{1}{z} = \dfrac{z-1}{z}, \end{cases}$$

with

$$(3) \qquad \begin{cases} \dot{\sigma}_0\dot{\sigma}_1 = \dot{\sigma}_1\dot{\sigma}_\infty = \dot{\sigma}_\infty\dot{\sigma}_0 = \dot{\rho}, \\ \dot{\rho}\dot{\sigma}_0\dot{\rho}^{-1} = \dot{\sigma}_1, \quad \dot{\rho}\dot{\sigma}_1\dot{\rho}^{-1} = \dot{\sigma}_\infty, \quad \dot{\rho}\dot{\sigma}_\infty\dot{\rho}^{-1} = \dot{\sigma}_0 \end{cases}$$

$$(4) \qquad \dot{\sigma}_0^2 = \dot{\sigma}_1^2 = \dot{\sigma}_\infty^2 = \dot{\rho}^3 = 1.$$

The group $\mathcal{T}'_{0,3} = \mathbb{Z}/2\mathbb{Z} \subset \mathcal{T}_{0,3}$ appears as the center, 83

Uncertainty retained from cancelled passages.
140-3-s084-D001 [illeg. 1]
140-3-s084-D002 automorphisms
140-3-s084-D004 $\mathfrak{S}_3$
140-3-s084-D005 elements
140-3-s084-D006 $\dot{\rho}$

it is generated by complex conjugation τ

$$(5) \qquad , \quad \tau(z) = \bar{z},$$

the five other elements of $\mathcal{T}_{0,3}$ are

$$\dagger \quad (6) \qquad \tau\dot{\sigma}_0 = \dot{\tau}_0, \quad \tau\dot{\sigma}_1 = \dot{\tau}_1, \quad \tau\dot{\sigma}_\infty = \dot{\tau}_\infty, \quad \tau\rho, \quad \tau\rho^{-1} = (\tau\rho)^{-1},$$

the anti-involutions|anti-involutions τ'_i $(i = 0, 1, \infty)$ are particularly interesting, given by

$$(7) \qquad \dot{\tau}'_0(z) = \frac{\bar{z}}{\bar{z} - 1}, \quad \dot{\tau}'_1(z) = \frac{1}{\bar{z}}, \quad \dot{\tau}'_\infty(z) = 1 - \bar{z}.$$

The fixed points † of the [illeg. 1] conformal automorphisms of † $(\mathbb{P}^{1\text{an}}, \{0, 1, \infty\})$ are the following

$$(8) \qquad \begin{cases} (\mathbb{P}^1_{\mathbb{C}})^{\dot{\sigma}_0} = \{0, 2\}, \\ (\mathbb{P}^1_{\mathbb{C}})^{\dot{\sigma}_1} = \{1, -1\}, \\ (\mathbb{P}^1_{\mathbb{C}})^{\dot{\sigma}_\infty} = \{\infty, \frac{1}{2}\} \\ \mathbb{P}^{1\,\dot{\rho}}_{\mathbb{C}} = \mathbb{P}^{1\,\dot{\rho}^{-1}} = \{-\bar{j}, -j\} \end{cases}$$

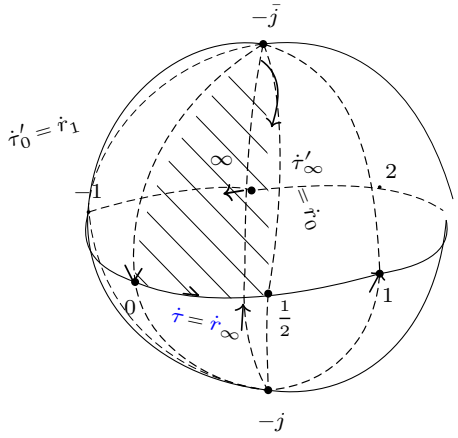
where $j = \exp 2i\pi/3 = \dfrac{-1 + i\sqrt{3}}{2}$ primitive root $3^{\text{ième|ème}}$ of 1

$-\bar{j} = \exp 2i\pi/6 = \dfrac{1 + i\sqrt{3}}{2}$ primitive root $6^{\text{ième|ème}}$ of 1

N.B. $\mathbb{P}^{1\,\dot{\rho}\tau}_{\mathbb{C}} = \mathbb{P}^{1\,\dot{\rho}^{-1}\tau}_{\mathbb{C}} = \emptyset$

$$(9) \qquad \begin{cases} \mathbb{P}^{1\,\tau}_{\mathbb{C}} = \mathbb{P}^1_{\mathbb{R}} \\ \mathbb{P}^{1\,\dot{\tau}'_0}_{\mathbb{C}} = \{x + iy \mid (x-1)^2 + y^2 = 1\} \\ \quad = \{z \mid z + \bar{z} = z\bar{z}\} = \frac{1}{z} \mid = \frac{1}{2} = \frac{1}{2} - \mathbb{U}, \\ \mathbb{P}^{1\,\dot{\tau}'_1}_{\mathbb{C}} = \{z \mid z\bar{z} = 1\} = \{x + iy \mid x^2 + y^2 = 1\} = \mathbb{U} \\ \mathbb{P}^{1\,\dot{\tau}'_\infty}_{\mathbb{C}} = \{z \mid z + \bar{z} = 1\} = \{x + iy \mid x = \frac{1}{2}\} = \frac{1}{2} + i\mathbb{R}; \end{cases}$$

The most suggestive picture is probably that of the Riemann sphere, where $\tau, \dot{\tau}'_0, \dot{\tau}'_1, \dot{\tau}'_\infty$ appear as the †reflections with respect to the corresponding "gr^{eat}|great circles", which for τ is "the equator" and for the others are the great circles "meridians", passing through the poles $-\bar{j}, -j$, which effect the cutting up of the equator into six circular segments, by the subdivision points $0, \frac{1}{2}, 1, 2, \infty, -1$.



Let us also note the formulas

$$(10) \qquad \begin{cases} \dot{\rho}\dot{\tau}'_0\dot{\rho}^{-1} = \dot{\tau}'_1, \quad \dot{\rho}\dot{\tau}'_1\dot{\rho}^{-1} = \dot{\tau}'_\infty, \quad \dot{\rho}\dot{\tau}'_\infty\dot{\rho}^{-1} = \dot{\tau}'_0 \\ \dot{\tau}'_0\dot{\tau}'_1 = \dot{\tau}'_1\dot{\tau}'_\infty = \dot{\tau}'_\infty\dot{\tau}'_0 = \dot{\rho} \end{cases}$$

84

Uncertainty retained from cancelled passages.
140-3-s085-D001 $\tau\sigma$
140-3-s085-D002 in $U_{0,3}$
140-3-s085-D003 [illeg. 2]
140-3-s085-D004 symmetries

[scan 86]

The four great circles (9) cut the sphere into 12 spherical triangles, which form fundamental [p. 332] domains for the group $\mathcal{T}_{0,3} \simeq \mathfrak{S}_3 \times \mathbb{Z}/2$ of order 12, and are permuted among themselves in a simply transitive way. †(This is also|thus the barycentric subdivision of the double spherical triangles determined by the equator $\mathbb{P}^1_{\mathbb{R}}$, and its three subdivision points $0, 1, \infty$.) Let us take one of these spherical triangles, say $(-\bar{j}, 0, \frac{1}{2})$, the reflections with respect to its three sides, i.e. $\tau, \dot{\tau}'_\infty, \dot{\tau}'_0$, form a system of generators of $\mathcal{T}_{0,3}$, with the relations

$$(11) \qquad \tau^2 = \dot{\tau}'^2_\infty = \dot{\tau}'^2_0 = \left(\underbrace{\tau\dot{\tau}'_\infty}_{\dot{\sigma}_\infty}\tau\right)^2 = \left(\underbrace{\dot{\tau}'_0\dot{\tau}'_\infty}_{\dot{\rho}^{-1}}\right)^3 = \left(\underbrace{\tau\dot{\tau}'_0}_{\dot{\sigma}_0}\dagger\right)^2 = 1$$

as for $\mathfrak{S}_3 = \mathcal{T}^+_{0,3}$, it is generated by $\tau\dot{\tau}'_\infty = \dot{\sigma}_\infty$, $\dot{\tau}'_0\dot{\tau}'_\infty = \dot{\rho}^{-1|1}$ and $\tau\dot{\tau}'_0 = \dot{\sigma}_0$, with the relations

$$(12) \qquad \dot{\sigma}_0\dot{\rho}\dot{\sigma}_\infty = \dot{\sigma}^2 = \dot{\rho}^3 = \dot{\sigma}_0^2 = 1.$$

II) The group $\Pi_{0,3}$ is generated by $\ell_0, \ell_1, \ell_\infty$, linked by the generating relation

$$(13) \qquad \ell_\infty\ell_1\ell_0 = 1$$

The subgroup of index 2 $\mathfrak{S}'_3 = (\mathcal{T}_{0,3})_{-\bar{j}} = \mathcal{T}_{0,3}$, stabilizer of $-\bar{j}$, consisting of the $(g, \tau^\varepsilon) \in \mathfrak{S}_3 \times \mathbb{Z}/2\mathbb{Z}$ such that $\mathrm{sg}(g) = (-1)^\varepsilon$, operates on $\Pi_{0,3}$. This subgroup consists of $1, \dot{\tau}_0, \dot{\tau}_1, \dot{\tau}_\infty, \dot{\rho}, \dot{\rho}^{-1}$, and their operations on $\Pi_{0,3}$ are denoted respectively $1, \tau'_0, \tau'_1, \tau'_\infty, \rho, \rho^{-1}$.
m745

Marginal note. m745 N.B. $\tau'_0, \tau'_1, \tau'_\infty$ were previously denoted (in [C]) $\sigma_0, \sigma_1, \sigma_\infty|\bar{\sigma}_0, \bar{\sigma}_1, \bar{\sigma}_\infty$. 85

Uncertainty retained from cancelled passages.
140-3-s086-D001 Taking|Let us take
140-3-s086-D002 τ

[scan 87]

We identify these elements with elements of the group $\mathrm{Aut}_{\mathrm{lac}}(\Pi_{0,3}) \simeq \mathfrak{S}_{0,3}$, † hence † for $g \in \mathfrak{S}'_3$, $x \in \Pi_{0,3}$, we set $g(x) = gxg^{-1}$; and we find:

$$\begin{cases} \rho(\ell_0) = \rho\ell_0\rho^{-1} = \ell_1, & \rho(\ell_1) = \rho\ell_1\rho^{-1} = \ell_\infty, & \rho(\ell_\infty) = \rho\ell_\infty\rho^{-1} = \ell_0, \\ \dot{\tau}'_0|\tau'_0(\ell_0) = \ell_0^{-1}, & \dot{\tau}'_0|\tau'_0(\ell_1) = \ell_\infty^{-1}, & \dot{\tau}'_0|\tau'_0(\ell_\infty) = \ell_1^{-1}, \\ \tau'_1(\ell_1) = \ell_1^{-1}, & \tau'_1(\ell_\infty) = \ell_0^{-1}, & \tau'_1(\ell_0) = \ell_\infty^{-1}, \\ \tau'_\infty(\ell_\infty) = \ell_\infty^{-1}, & \tau'_\infty(\ell_0) = \ell_1^{-1}, & \tau'_\infty(\ell_0|\ell_1|\ell_\infty) = \ell_1^{-1} \end{cases} \tag{14}$$

$$\therefore \quad \tau'_0\tau'_1 = \tau'_1\tau'_\infty = \tau'_\infty\tau'_0 = \rho \qquad \text{(and of course we have } \tau'^2_0 = \tau'^2_1 = \tau'^2_\infty|\tau^2_\infty = \rho^3 = 1). \tag{15}$$

In The group $\mathfrak{S}_{0,3}$ we define elements $\sigma_0, \sigma_1, \sigma_\infty$, over $\dot{\sigma}_0, \dot{\sigma}_1, \dot{\sigma}_\infty$, by

$$\begin{cases} \sigma_0(\ell_1) = \ell_\infty, & \sigma_0(\ell_\infty) = \ell_1, & \sigma_0(\ell_0) = (\ell_1\ell_\infty)^{-1} \\ \sigma_1(\ell_\infty) = \ell_0, & \sigma_1(\ell_0) = \ell_\infty, & \sigma_1(\ell_1) = (\ell_\infty\ell_0)^{-1} \\ \sigma_\infty(\ell_0) = \ell_1, & \sigma_\infty(\ell_1) = \ell_0, & \sigma_\infty(\ell_\infty) = (\ell_0\ell_1)^{-1} \end{cases} \tag{16}$$

We have

$$\begin{cases} \rho\sigma_0\rho^{-1} = \sigma_1, & \rho\sigma_1\rho^{-1} = \sigma_\infty, & \rho\sigma_\infty\rho^{-1} = \sigma_0 \\ \rho\tau_0\rho^{-1} = \tau_1, & \rho\tau_1\rho^{-1} = \tau_\infty, & \rho\tau_\infty\rho^{-1} = \tau_0 \end{cases}, \tag{17}$$

$$\sigma_0\sigma_1 = \rho\ell_0 = \dagger\ell_1\rho, \quad \sigma_1\sigma_\infty = \rho\ell_1 = \ell_\infty\rho, \quad \sigma_\infty\sigma_0 = \rho\ell_\infty = \ell_0\rho^{-1}. \tag{18}$$

m747

Marginal note. m747 N.B. $\sigma_0, \sigma_1, \sigma_\infty \in \mathfrak{S}^+_{0,3}$, $\tau'_0, \tau'_1, \tau'_\infty \in \mathfrak{S}^-_{0,3}$, $\tau_0, \tau_1, \tau_\infty \in \mathfrak{S}^-_{0,3}$.

We thus have elements of $\mathfrak{S}_{0,3}$ on top of|of the $1, \dot\sigma_0, \dot\sigma_1, \dot\sigma_\infty, \dot\rho, \dot\rho^{-1}, \dot\tau'_0, \dot\tau'_1, \dot\tau'_\infty$, but not yet over τ (which makes it possible to have some over $\tau\dot\rho, \tau\dot\rho^{-1}$), we set $\tau_i = \underbrace{\sigma_i}_{\text{over } \dot\sigma_i} \underbrace{\tau'_i}_{\text{over } \dot\tau'_i} = \tau'_i\sigma_i$ (which is over τ, † τ_i is characterized † by the condition of being over τ, of square 1, and of commuting with σ_i (or again, with τ'_i) m748

Marginal note. m748 recall why we have this characterization.

$$[\tau_0, \sigma_0] = [\tau_1, \sigma_1] = [\tau_\infty, \sigma_\infty] = 1 \tag{19}$$

m749

Marginal note. m749 $\tau_0 = \sigma_0\tau'_0 = \tau'_0\sigma_0$ $\quad \tau_1 = \sigma_1\tau'_1 = \tau'_1\sigma_1$ $\quad \tau_\infty = \sigma_\infty\tau'_\infty = \tau'_\infty\sigma_\infty$

$$\tau_0^2 = \tau_1^2 = \tau_\infty^2 = 1 \qquad \text{and } \tau_0, \tau_1, \tau_\infty \text{ over } \tau. \tag{20}$$

We find

$$\tau_\infty\tau_1 = \ell_0, \quad \tau_0\tau_\infty = \ell_1, \quad \tau_1\tau_0 = \ell_\infty. \tag{21}$$

To do things properly, we should make explicit the topological description (given in [C]) of $\sigma_0, \sigma_1, \sigma_\infty, \rho, \tau'_0, \tau'_1, \tau'_\infty, \tau_0, \tau_1, \tau_\infty$. 86

Uncertainty retained from cancelled passages.

140-3-s087-D004 [illeg. 3] [illeg. 1]
140-3-s087-D005 [illeg. 1]
140-3-s087-D006 ρ
140-3-s087-D008 [illeg. 1]
140-3-s087-D009 [illeg. 1]

[scan 88]
[p. 333]

We have

$$\begin{cases} (\sigma_\infty\rho)^2 = (\rho\sigma_1)^2 = \ell_0 = \varepsilon_0^2, & \varepsilon_0 \overset{\text{definition|definition}}{=} \sigma_\infty\rho = \rho\sigma_1\dagger \\ & (\text{NB } \sigma_\infty = \rho\sigma_1\rho^{-1}) \\ (\sigma_0\rho)^2 = (\rho\sigma_\infty)^2 = \ell_1 = \varepsilon_1^2, & \varepsilon_1 = \sigma_0\rho = \rho\sigma_\infty \\ (\sigma_1\rho)^2 = (\rho\sigma_\infty|\sigma_0)^2 = \ell_\infty = \varepsilon_\infty^2, & \varepsilon_\infty = \sigma_1\rho = \rho\sigma_0 \end{cases} \tag{22}$$

The element $\varepsilon_i \in \mathfrak{S}_{0,3}$ (in fact $\varepsilon_i \in \mathfrak{S}^+_{0,3}$) is over $\dot\sigma_i$, it is characterized by the condition

$$\varepsilon_i^2 = \ell_i.$$

In the group $\mathcal{T}_{0,3}$ of order 12, there are seven involutive elements $\dot\sigma_0, \dot\sigma_1, \dot\sigma_\infty, \tau, \dot\tau'_0, \dot\tau'_1, \dot\tau'_\infty$, † two elements of order 3 $\dot\rho, \dot\rho^{-1}$, and two of order 6 $\dot\rho\tau, \dot\rho^{-1}\tau$ (in addition to the identity).

Except for the last two (for which this is not possible*) We have lifted each of these elements to elements of the same order in $\mathfrak{S}_{0,3}$. We know that the possibilities, mod $\Pi_{0,3}$-conjugation, of|of the such liftings, correspond to the connected components of the fixed points in|of $U = U^{\mathrm{an}}_{0,3}$ of the corresponding topological automorphisms. — for $\dot\sigma_0, \dot\sigma_1, \dot\sigma_\infty, \dot\tau'_0, \dot\tau'_1, \dot\tau'_\infty$ there is only one possibility mod conjugation (we have chosen $\sigma_0, \sigma_1, \sigma_\infty, \tau_0, \tau_1, \tau_\infty$ in the way that seemed most convenient for the calculations), for τ there are three — these are the three choices $\tau_0, \tau_1, \tau_\infty$ — for $\dot\rho, \dot\rho^{-1}$ there are two, corresponding|corresponding to the two poles $-\bar j, -j$ — we have chosen the one that corresponds to the pole $-\bar j$. The liftings corresponding to $-j$ can be deduced from them by conjugating by a τ_i: 87

* indeed $\mathbb{P}^{1\dot\rho\tau} = \mathbb{P}^{1\dot\rho^{-1}\tau} = \emptyset$

Uncertainty retained from cancelled passages.
140-3-s088-D001 dfn =
140-3-s088-D002 four

[scan 89]

$$\rho_0 = \tau_0\rho\tau_0^{-1} = \ell_\infty^{-1}\rho = \rho\ell_1^{-1}, \qquad \rho_1 = \tau_1\rho\tau_1^{-1} = \ell_0^{-1}\rho = \rho\ell_\infty^{-1}, \qquad \rho_\infty = \tau_\infty\rho\tau_\infty^{-1} = \ell_1^{-1}\rho = \rho\ell_0^{-1} \tag{23}$$

are of order 3, over $\dot\rho$, and $\Pi_{0,3}$-conjugate to one another*, but not $\Pi_{0,3}$-conjugate to ρ.
m753

Marginal note. m753 * N.B.

$$\rho\rho_0\rho^{-1} = \rho_1$$
$$\rho\rho_1\rho^{-1} = \rho_\infty$$
$$\rho\rho_\infty\rho^{-1} = \rho_0$$

Let us also note the corollary formulas of the general theory – and of the fact that the connected components of the sets of fixed points which appear here are 1-connected –

$$\Pi^{\sigma_i}_{0,3} = \Pi^{\rho}_{0,3} = \Pi^{\tau_i}_{0,3} = \Pi^{\tau'_i}_{0,3} = \{1\} \tag{24}$$

If we look at the critical points of $U^{\mathrm{an}}_{0,3}$, relative to the operation of $\mathcal{T}_{0,3}$, we find the †two orbits $(2, -1, \frac{1}{2})$ and $(-\bar j, -j)$, the stabilizer of a point of these orbits is of order 4 resp. 6. For the point 2 (opposite 0), this stabilizer is the group $\mathrm{D}_2 \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, generated by $\dot\sigma_0, \dot\tau_0$ (or σ_0, τ'_0) – which has indeed been lifted multiplicatively to $\mathfrak{S}_{0,3}$. Likewise|=, for the point $-\bar j$, it is $\mathfrak{S}'_3 \simeq \mathrm{D}_3$, which has also been lifted. By the general theory we know that the $\Pi_{0,3}$-conjugacy classes of such a lifting correspond to the fixed points of the group of operators under consideration on $U_{0,3}$, for $\mathrm{D}_2^{(0)}$ there is only one fixed point, namely 2, for $\mathfrak{S}'_3 = \mathrm{D}_3$ there are two, $-\bar j$ and $-j$, we had chosen $-\bar j$. We find|will find the other conjugacy class 88

Uncertainty retained from cancelled passages.
140-3-s089-D002 points [illeg. 1]

[scan 90]

by conjugating by any one of the τ_i $(\tau_0, \tau_1, \tau_\infty)$. [p. 334]

III Relation with the unoriented triangul- cartographic group,

$$\begin{aligned}\widetilde{\Pi}'_{0,3} &= \Pi_1(\mathbb{P}^{1\circ}_{\mathbb{C}} | \mathbb{P}^{1\bullet}_{\mathbb{C}},\ 0\{0\} + 2\{1\} + 3\{\infty\}; -\bar j) \\ &= (\Pi_{0,3} \cdot \{1, \tau\})/(\ell_1^2, \ell_\infty^3),\end{aligned}$$

and with $\mathrm{Sl}(2, \mathbb{Z})$.

We have † involutive generators $\sigma_0, \sigma_1, \sigma_2$ of $\widetilde{\Pi}'_{0,3}$, (σ_i in a map frame †changes the †of dimension i of the frame, so for a biarc σ_0 and σ_2 are resp. the †change^{s} of orientation tangential|tangential resp. transverse, σ_1 the passage to the "adjacent" biarc



) with the generating relations

$$(25) \qquad \begin{cases} (\sigma'_0)^2 = (\sigma'_1)^2 = (\sigma'_2)^2 = (\sigma'_2\sigma'_0)^2 = 1 \\ \underbrace{(\sigma'_1\sigma'_0)}_{\rho_f}{}^3 = 1 \end{cases}$$

m755

Marginal note. m755 we could also write σ'_∞ instead of σ'_2. We would have done better to denote them here outright r_0, r_1, r_∞ (instead of $\sigma_0, \sigma_1, \sigma_2$).

The subgroup $\Pi'_{0,3}$ of index 2 †(oriented triangul' cartographic group) is generated by

$$\sigma'_1\sigma'_0 = \rho_f = \ell'_\infty, \qquad \sigma'_2\sigma'_1 = \rho_s = \ell'_0, \qquad \sigma'_0\sigma'_2 = \sigma'|\sigma = \ell'_1,$$

with the generating relations

$$(26) \qquad \rho_f\sigma\rho_s\dagger = 1 \qquad \sigma^2 = \rho_f^3 = 1$$

This is therefore the quotient of $\Pi_{0,3}$ by ℓ_1^2, ℓ_∞^3

$$(27|17) \qquad \begin{cases} \Pi_{0,3} & \longrightarrow & \Pi'_{0,3} \\ \ell_0 & \longmapsto & \ell'_0 = \rho_s \\ \ell_1 & \longmapsto & \ell'_1 = \sigma \\ \ell_\infty & \longmapsto & \ell'_\infty = \rho_f \end{cases},$$

† This homomorphism extends to

† 89

Uncertainty retained from cancelled passages.

140-3-s090-D001 [illeg. 1]
140-3-s090-D002 interchanges
140-3-s090-D003 [illeg. 2]
140-3-s090-D004 passages [illeg. 2]
140-3-s090-D005 [illeg. 1]
140-3-s090-D006 ρ_s
140-3-s090-D007 [illeg. 1]
140-3-s090-D008 (28)|(27)

[scan 91]

$$(28) \qquad \begin{cases} \widetilde{\Pi}_{0,3} = \Pi_{0,3} \cdot \{1, \tau\} \longrightarrow \widetilde{\Pi}'_{0,3} \\ \qquad\qquad \tau_0, \tau_1, \tau_\infty \longmapsto \sigma'_0, \sigma'_1, \sigma'_\infty \end{cases}$$

This being so, we define an isomorphism †

$$\widetilde{\Pi}'_{0,3} \xrightarrow{\sim} \mathfrak{S}_{0,3}$$

$$(29)\qquad (\sigma'_0, \sigma'_1, \sigma'_\infty) \longrightarrow \left(\begin{array}{ccc} \dagger\ \tau'_\infty & , \quad \tau'_0 & , \ \tau_\infty \\ \text{over } \dot\tau'_\infty & \text{over } \dot\tau'_0 & \text{over } \tau \\ \| & \| & \\ \dot\sigma_\infty \tau & \dot\sigma_0 \tau & \end{array} \right)$$

also denoted r_0, r_1, r_∞ (fundamental reflections)

m757

Marginal note. m757 This isomorphism is associated with a canonical homomorphism $\varphi : \widetilde{\Pi}_{0,3} = \Pi_{0,3} \cdot \{1, \tau\} \xrightarrow{\text{epimorphism}} \mathfrak{S}_{0,3}$ inducing a canonical homomorphism $\varphi : \Pi_{0,3} \xrightarrow{\text{epimorphism}} \mathfrak{S}^+_{0,3}$ allowing us to identify $\mathfrak{S}_{0,3}$ and $\mathfrak{S}^+_{0,3}$ with the quotients of the left-hand sides|sides by the relations $\ell_1^2 = \ell_\infty^3 = 1$ …

which induces an isomorphism

$$(30)\qquad \begin{cases} \Pi'_{0,3} \xrightarrow{\sim} \dagger \mathfrak{S}^+_{0,3} \\ \rho_s = \ell'_0 \longmapsto \varepsilon_0 = \sigma_\infty \rho \\ \sigma = \ell'_1 \longmapsto \sigma_\infty \\ \rho_f = \ell'_\infty \longmapsto \rho^{-1} \end{cases}$$

m758

Marginal note. m758 N.B. $\begin{cases} \tau_\infty \tau'_0 = \varepsilon_0 = \ell'_0 & r_\infty r_1 = \ell'_0\ (= \varepsilon_0) \\ \tau'_\infty \tau_\infty = \sigma_\infty = \ell'_1 & \text{i.e. } r_0 r_\infty = \ell'_1\ (= \sigma_\infty) \\ \tau'_0 \tau'_\infty = \rho^{-1} = \ell'_\infty & r_1 r_0 = \ell'_\infty\ (= \rho^{-1}) \end{cases}$

We thus find that $\mathfrak{S}_{0,3}$ has † generating relations

$$(31)\qquad {\tau'_\infty}^2 = {\tau'_0}^2 = \tau_\infty^2 = (\underbrace{\tau'_\infty \tau_\infty}_{\sigma_\infty})^2 = (\underbrace{\tau'_0 \tau'_\infty}_{\rho^{-1}})^3 = 1,$$

and $\mathfrak{S}^+_{0,3}$ has generating relations

$$(32)\qquad \ell'_\infty \ell'_1 \ell'_0 = {\ell'_1}^2 = {\ell'_\infty}^3 = 1$$

or also simply, eliminating ℓ'_0 and setting $\ell'_1 = \sigma_\infty$ and ${\ell'_\infty}^{-1} = \rho$

$$(33)\qquad \boxed{\sigma_\infty^2 = \rho^3 = 1}$$

For the record, we are going to express the notable elements of $\mathfrak{S}^+_{0,3}$ in terms of σ_∞, ρ:

Uncertainty retained from cancelled passages.

140-3-s091-D001 [illeg. 1]

140-3-s091-D002 σ'_1

140-3-s091-D003 Π

140-3-s091-D004 gen

[scan 92]

[p. 335]

$$(34)\qquad \begin{cases} \varepsilon_0 = \sigma_\infty \rho\ \dagger & \ell_0 = \sigma_\infty \rho \sigma_\infty \rho = (\sigma_\infty \rho)^2 \\ \varepsilon_1 = \rho \sigma_\infty & \ell_1 = \rho \sigma_\infty \rho \sigma_\infty = (\rho \sigma_\infty)^2 \\ \varepsilon_\infty = \rho^{-1} \sigma_\infty \rho^{-1} & \ell_\infty = (\rho^{-1} \sigma_\infty \rho^{-1})^2 = \rho^{-1} \sigma_\infty \rho^{-1} \sigma_\infty \rho^{-1} \end{cases},$$

$$(35)\qquad \begin{cases} \sigma_0 = \rho^{\dagger}\sigma_\infty \rho^{\dagger -1} & (= \dagger\varepsilon_1\rho^{-1} = \rho\varepsilon_1^{-1}) \\ \sigma_1 = \rho^{-1}\sigma_\infty\rho^{\dagger} & (= \rho^{-1}\varepsilon_0\dagger = \varepsilon_\infty\rho^{-1}) \end{cases}$$

and in $\mathfrak{S}_{0,3}^{-}$ we have, in terms of $\tau'_\infty, \tau'_0, \tau_\infty$, the expressions for † $\tau_0, \tau_1, (\tau_\infty, \tau'_0), \tau'_1, (\tau'_\infty)$

$$(36)\qquad \begin{cases} \tau_0 = \dagger\rho\tau_\infty\rho^{-1} = \tau'_\infty\tau'_0\tau_\infty\tau'_0\tau'_\infty & (= r_0r_1r_\infty r_1r_0) \\ \tau_1 = \rho^{-1}\tau_\infty\rho = \tau'_0\tau'_\infty\tau_\infty\tau'_\infty\tau'_0 & (= r_1r_0r_\infty r_0r_1) \\ \tau'_1 = \rho\tau'_0\rho^{-1} = \tau'_\infty\dagger\tau'_0\tau'_\infty & (= r_0r_1r_0) \end{cases}$$

and

$$\begin{cases} \tau'_\infty = r_0 \\ \tau'_0 = r_1 \\ \tau_\infty = r_\infty. \end{cases}$$

IV) Relation with $\mathrm{Gl}(2,\mathbb{Z})' = \mathrm{Gl}(2,\mathbb{Z})/\pm 1 \simeq \mathrm{GP}(1,\mathbb{Z})$.
This group operates on the Poincaré half-plane

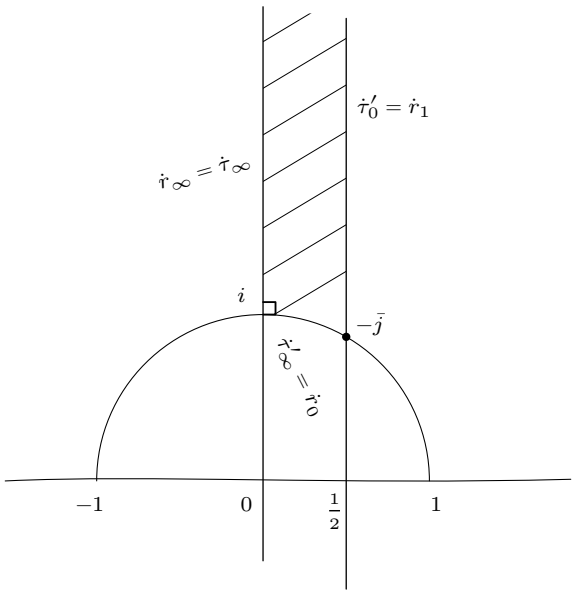
$$D = \{z \in \mathbb{C} \mid \Im z > 0\}$$

by

$$(37)\qquad \begin{cases} u_g(z) = \dfrac{az+b}{cz+d} & \text{if } ad-bc = +1 \\ u_g(z) = \dfrac{a\bar{z}+b}{c\bar{z}+d} & \text{if } ad-bc = -1 \end{cases} \qquad \begin{matrix} g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \\ a,b,c,d \in \mathbb{Z} \\ ad-bc \in \{\pm 1\} \end{matrix}$$

We have in D a hatched fundamental domain, which corresponds to the spherical triangle $(-\bar{j}, 0, \frac{1}{2})$ by the homomorphism|homeomorphism of the natural covering, i corresponding to $\frac{1}{2}$, $-\bar{j}$ to $-\bar{j}$, and the point at infinity to 0. †The reflections with respect to the three sides of this triangle are given by

$$(38)\qquad \dagger r_0|\tau_0(z) = \frac{1}{\bar{z}}, \qquad r_1|\tau_1(z) = 1-\bar{z}, \qquad r_\infty|\tau_\infty(z) = -\bar{z}$$



† and the pairwise products of these reflections

Uncertainty retained from cancelled passages.

140-3-s092-D001 [*illeg.* 1]
140-3-s092-D002 2
140-3-s092-D003 2
140-3-s092-D004 ρ
140-3-s092-D005 2
140-3-s092-D006 ρ
140-3-s092-D007 $\tau_0, \tau_1, \tau_\infty, \tau'_0, \tau'_1, \tau'_\infty$
140-3-s092-D008 $\tau_0\tau'_0\tau_\infty\tau'_\infty$
140-3-s092-D009 τ'_0
140-3-s092-D010 we find
140-3-s092-D011 τ_0
140-3-s092-D012 They determine

[scan 93]

are

$$(39)\quad \begin{cases} r_\infty r_1 | \tau_\infty\tau_1(z) = \ell'_0(z) = z - 1 \ \dagger \\ r_0 r_\infty | \tau_0\tau_\infty(z) = \ell'_1(z) = -\dfrac{1}{z} \\ r_1 r_0 | \tau_1\tau_0(z) = \ell'_\infty(z) = 1 - \dfrac{1}{z} = \dfrac{z-1}{z} \end{cases}$$

We thus find an isomorphism

$$(40)\quad \begin{cases} \mathfrak{S}_{0,3} \xrightarrow{\sim} \mathrm{Gl}(2,\mathbb{Z})' \simeq \mathrm{Gl}(2,\mathbb{Z})/\pm 1 \simeq \mathrm{GP}(1,\mathbb{Z}) \\ r_0 = \tau'_\infty \longmapsto r_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ r_1 = \tau'_0 \longmapsto r_1 = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix} \\ r_\infty = \tau_\infty \longmapsto r_\infty = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \end{cases}$$

m759

Marginal note. m759 it is better to change some|of signs cf. Formulary|cf. formulary

and a presentation of $\mathrm{GP}(1,\mathbb{Z})$ by

$$(41)\quad r_0^2 = r_1^2 = r_\infty^2 = (r_0 r_\infty)^2 = (r_1 r_0)^3 = 1.$$

This isomorphism induces an isomorphism

$$(42)\quad \begin{cases} \mathfrak{S}^+_{0,3} \xrightarrow{\sim} \mathrm{Sl}(2,\mathbb{Z})' = \mathrm{Sl}(2,\mathbb{Z})/\pm 1 \simeq \mathrm{GP}(1,\mathbb{Z})^+ \\ \ell'_1 = \sigma_\infty \longmapsto \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \\ \ell'_\infty = \rho^{-1} \longmapsto \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix} \end{cases}$$

$$(\text{hence} \quad \rho \longmapsto \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}).$$

We henceforth identify the elements of $\mathfrak{S}_{0,3}$ with their images in Gl|Sl$(2,\mathbb{Z})' \simeq \mathrm{GP}(1,\mathbb{Z})$. Let us note the following formulary (where we use that for a matrix $u = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{Sl}(2,\mathbb{Z})$, † †, we have

Uncertainty retained from cancelled passages.
140-3-s093-D001 ;
140-3-s093-D002 [illeg. 1]
140-3-s093-D003 for

[scan 94]
[p. 336]

$$(43)\quad u^{-1} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \quad) :$$

m760

Marginal note. m760 N.B. 2. † $u \in \mathrm{Gl}(2, A)$, A an arbitrary ring, we have $\det(u) \cdot u^{-1} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ †.

$$(44)\quad \begin{cases} \varepsilon_0 = \sigma_\infty \rho = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, & \varepsilon_0^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \\ \varepsilon_1 = \rho \sigma_\infty = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, & \varepsilon_1^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \\ \varepsilon_\infty = \rho^{-1}\sigma_\infty\rho^{-1} = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}, & \varepsilon_\infty^{-1} = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix} \\ \ell_0 = \varepsilon_0^2 = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} & \sigma_0 = \begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix} \\ \ell_1 = \varepsilon_1^2 = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} & \sigma_1 = \begin{pmatrix} 1 & -2 \\ 1 & -1 \end{pmatrix} \\ \ell_\infty = \varepsilon_\infty^2 = \begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix} & \sigma_\infty = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \end{cases}$$

m761

Marginal note. m761 Cf. "formulary" for a more complete list of relations and expressions.

$$(45)\quad \rho = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \qquad \rho^{-1}|\rho^{-1\prime} = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix},$$

$$(46)\quad \ell_\infty \ell_1 \ell_0 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

the nontrivial element of the center of $\mathrm{Sl}(2, \mathbb{Z})$ which (fortunately!) gives 1 in $\mathrm{Sl}(2, \mathbb{Z})'$

The group $\Pi_{0,3}$ is recovered as the kernel of the homomorphism

$$\begin{array}{ccc} \mathrm{Sl}(2, \mathbb{Z})' & \longrightarrow & \mathrm{Sl}(2, \mathbb{Z}/2\mathbb{Z}) \\ & & \wr\vert \\ & & \mathfrak{S}_3 \end{array}$$

the exact sequence

$$(47)\quad \begin{array}{ccccccccc} 1 & \longrightarrow & \Pi_{0,3} & \text{——} & \mathfrak{S}^+_{0,3} & \text{——} & \mathcal{T}^{+\prime}_{0,3} & \text{——} & 1 \\ & & & & & & \wr\vert & & \\ & & & & & & \mathfrak{S}_3 & & \end{array}$$

is recovered as

$$\begin{array}{ccccccccc} & & 1 & \longrightarrow & \Pi(2)' & \longrightarrow & \mathrm{Sl}(2,\mathbb{Z})' & \longrightarrow & \mathrm{Sl}(2,\mathbb{Z}/2\mathbb{Z}) & \longrightarrow & 1 \\ (48) & & & & \wr & & & & \wr & & \\ & & & & \Pi_{0,3}^{\beta} & & & & \mathfrak{S}_3 & & \end{array}$$

Uncertainty retained from cancelled passages.

140-3-s094-D001 If det

140-3-s094-D002 And

[scan 95]

It makes our hands|our hand itch to †work in $\mathrm{Gl}(2,\mathbb{Z})$ itself, a central extension of $\mathfrak{S}_{0,3}$ — in fact, this extension is interpreted as $\mathcal{T}_{1,1}$ — but we shall come back to it, later…

V) Let us sum up the avatars of the same group, which we have encountered|encountered so far

1) $\mathfrak{S}_{0,3} \overset{\text{dfn}}{=} \mathrm{Aut}_{\mathrm{lac}}(\Pi_{0,3})$ with subgroup $\mathfrak{S}_{0,3}^{+}$ (preserving orientation)

2) $\mathcal{T}'_{0,4} \simeq \mathrm{Autext}_{\mathrm{lac}}(\Pi_{0,4})'$ outer automorphisms of $\Pi_{0,4}$ respecting the loop structure and fixing the loop class $L_4|L_\lambda$ (and permuting $L_0, L_1, L_\infty|L_3$)

† † with $\mathcal{T}'^{+}_{0,4}$ (preserving orientation)

3) $A'_{0,4}/A'^{0}_{0,4}$ group of connected components of

$$A'_{0,4} \subset \mathrm{Auttop}\left(\begin{array}{c} \dagger \quad U_{0,4}^{\mathrm{top}} \\ \| \\ \mathbb{P}^1_{\mathbb{C}}\setminus\{0,1,\infty;\frac{1}{2}\} \end{array}\right)$$

subgroup of automorphisms that fix the point at infinity $a_{0,4} = \frac{1}{2}$

with $A'^{+}_{0,4}/A'^{0}_{0,4}$ (preserving orientation

4) $\Pi_1((U^{\mathrm{an}}_{0,3}, \overbrace{\mathfrak{S}_3 \times \mathbb{Z}_2|\mathbb{Z}/2}^{\mathcal{T}_{0,3}}), -\bar{j})$ (described in the manner of J. Malgoire) with the subgroup $\Pi_1((U^{\mathrm{an}}_{0,3}, \mathfrak{S}_3), -\bar{j})$ ($\mathfrak{S}_3 = \mathcal{T}^{+}_{0,3}$)

5) $\Pi_1((\mathbb{P}^{1\,\mathrm{an}}_{\mathbb{C}}, 0\cdot\{0\} + 2\{1\} + 3\{\infty\}), -\bar{j})$ (in the manner of J. Malgoire) with the subgroup $\Pi_1(\mathbb{P}^{1\,\mathrm{an}}_{\mathbb{C}}, 0\{0\} + 2\{1\} + 3\{\infty\})$

6) Triangulated cartographic group † $\Gamma_{\infty,3}$

$$\Gamma_{\infty,3}\} = \{r_0, r_1, r_\infty \mid r_0^2 = r_1^2 = r_\infty^2 = (r_0 r_\infty)^2 = (r_1 r_0)^3 = 1\}$$

(with oriented triangulated cartographic subgroup

$$\Gamma^{\circ}_{\infty,3} = \{\underset{\| \; r_\infty r_1}{\rho_s}\ ,\ \underset{\| \; r_0 r_\infty}{\sigma}\ ,\ \underset{\| \; r_1 r_0}{\rho_f} \mid \rho_f\sigma\rho_s = \sigma^2 = \rho_f^3 = 1\}.$$

)

m762

Marginal note. m762 this is also the group $\widetilde{\Pi}_{0,3}/(\ell_1^2, \ell_\infty^3)$, where $\widetilde{\Pi}_{0,3} = \Pi_{0,3}\cdot\{1,\tau\}$

7) $\Pi_1(\mathcal{U}^{\mathrm{conf}}_{0,3}, \mathrm{pt}(U_{0,3}, -\bar{j}))$ universal conformal surface of type $(0,3)$ with subgroup $\Pi_1(\mathcal{U}^{\mathrm{an}}_{0,3}, \mathrm{pt}(U_{0,3}, -\bar{j}))$ universal holomorphic surface of type $(0,3)$

m763

Marginal note. m763 N.B. $\mathcal{U}_{0,3}^{\mathrm{conf}} \simeq (U_{0,3}, \mathcal{T}_{0,3})$ which makes it possible to identify 7) with 4)

8) $\mathcal{T}'_{1,1} = \mathcal{T}_{1,1}/\underbrace{\{\pm 1\}}_{\text{center}}$, with $\mathcal{T}_{1,1} \simeq \mathrm{Autext}_{\mathrm{lac}}(\Pi_{1,1}) \simeq A_{1,1}/A^0_{1,1}$

($A_{1,1} \simeq$ topological automorphisms of $U^\circ_{1,1}|U_{1,1}$) with subgroup $\mathcal{T}'^{+}_{1,1} = \mathcal{T}^+_{1,1}/\{\pm 1\}$

Uncertainty retained from cancelled passages.

140-3-s095-D001 fool around
140-3-s095-D002 $\mathcal{T}'^{+}_{0,4}$
140-3-s095-D003 $A'_{0,4}/A'^{0}_{0,4}$ Auttop 1
140-3-s095-D004 {
140-3-s095-D005 [illeg. 1]

[scan 96]

9) $\Pi_1(M^{\mathrm{conf}}_{1,1}, \mathrm{pt})' \simeq \Pi_1()/\mathrm{centre}\{\pm 1\} \simeq \Pi_1({M'_{1,1}}^{\mathrm{conf}}, \mathrm{pt})$ [p. 337]

where $M^{\mathrm{an}}_{1,1}$ is the conformal|conformal modular multiplicity of conformal elliptic curves (i.e. conformal curves of type $1,1$), ${M'_{1,1}}^{\mathrm{an}}$ that of conformal elliptic curves "modulo symmetry"; with subgroup

$$\Pi_1(M^{\mathrm{an}}_{1,1}, \mathrm{pt})' \simeq \Pi_1()/\mathrm{centre}\{\pm 1\} \simeq \Pi_1({M'_{1,1}}^{\mathrm{an}}, \mathrm{pt})$$

where the base point is the conformal elliptic resp. analytic curve, quotient of $\mathbb{C}$ by $\mathbb{Z} + \mathbb{Z}(-\bar{j})$. N.B. The modular multiplicities are $(D, \mathrm{Gl}(2,\mathbb{Z}))$, $(D, \mathrm{Sl}(2,\mathbb{Z}))$ in the conformal resp. analytic setting for $M_{1,1}$, and $(D, \mathrm{Gl}(2,\mathbb{Z})' = \mathrm{GP}(1,\mathbb{Z}))$ resp. $(D, \mathrm{Sl}(1,\mathbb{Z})'|\mathrm{Sl}(2,\mathbb{Z})' = \mathrm{GP}(1,\mathbb{Z})^+)$ in the conformal resp. analytic setting for $M'_{1,1}$.

10) $\mathrm{Gl}(2, \mathbb{Z}^*|\mathbb{Z})' \simeq \mathrm{GP}(1,\mathbb{Z})$, with the subgroup $\mathrm{Sl}(2,\mathbb{Z})' \simeq \mathrm{GP}(1,\mathbb{Z})^+$.

[scan 97]

(VI) The loop automorphisms of $\widehat{\Pi}_{0,3}$ and of $\widehat{\mathfrak{S}}^+_{0,3}$

We have a canonical commutative diagram:

(49)

$$\begin{array}{ccc}
\begin{array}{l}\mathrm{Autext}(\widehat{\mathfrak{S}}^+_{0,3})' \\ = \text{group of outer automorphisms} \\ \text{inducing} \\ \text{the identity in the} \\ \text{quotient } \mathfrak{S}_3 \simeq \mathcal{T}^+_{0,3}\end{array} & \xrightarrow[\sim]{\text{restriction}} & \begin{array}{l}\mathrm{Autext}^{\rho}(\widehat{\Pi}_{0,3})' \\ = \text{group of outer automorphisms} \\ \text{that} \\ \text{commute with} \\ \text{the outer action of } \mathfrak{S}_3\end{array} \\
\uparrow \sim & & \uparrow \sim \\
\begin{array}{l}\mathrm{Aut}(\widehat{\mathfrak{S}}^+_{0,3}, \sigma_\infty)' \\ = \text{group of automorphisms} \\ \text{of } \widehat{\mathfrak{S}}^+_{0,3} \text{ that } \dagger\text{fix } \sigma_\infty, \\ \text{and that fix } \rho \bmod \widehat{\Pi}_{0,3}\end{array} & \xrightarrow[\sim]{\text{restriction}} & \begin{array}{l}\mathrm{Aut}(\widehat{\Pi}_{0,3}, \sigma_\infty) \\ = \text{group of automorphisms} \\ \text{that commute} \\ \text{with } \sigma_\infty, \text{ and that} \\ \text{commute} \\ \mathrm{ext}^{\mathrm{t}}. \text{ with } \rho\end{array}
\end{array}$$

The $\in \dagger$ $\mathrm{Aut}(\widehat{\mathfrak{S}}^+_{0,3}, \sigma_\infty)'$ are of the form $\dagger$

$$(50) \qquad \begin{cases} u(\sigma_\infty) = \sigma_\infty \\ u(\rho) = g\rho \end{cases} \qquad (\text{i.e. } u\rho u^{-1}\rho^{-1} = g)$$

m765

Marginal note. m765 We write the formulas in $\mathrm{Aut}(\widehat{\mathfrak{S}}^+_{0,3})' = G$, group of automorphisms of $\widehat{\mathfrak{S}}^+_{0,3}$ that induce the identity in $\widehat{\mathfrak{S}}^+_{0,3}/\Pi_{0,3} \simeq \mathfrak{S}_3$, is identified with the group $\mathrm{Aut}(\Pi_{0,3})' =$ gr. of automorphisms of $\Pi_{0,3}$ that commute outer with $\mathfrak{S}_3$; – we shall have $\Pi_{0,3} \subset \widehat{\mathfrak{S}}^+_{0,3} \subset \widehat{\mathfrak{S}}_{0,3} \subset \mathfrak{g}.|g.\, G$. N.B. We assume here that the center of $\widehat{\mathfrak{S}}^+_{0,3}$ is trivial.

$g \in \widehat{\Pi}_{0,3}$ † is subject to the cocycle condition

$$(51) \qquad \boxed{g\rho(g)\rho^2(g) = 1}$$

which expresses the relation

$$(52) \qquad (g\rho)^3 = 1\ .$$

We thus find a $1-1$ correspondence between the elements of $\mathrm{Aut}(\widehat{\mathfrak{S}}^+_{0,3}, \sigma_\infty)'$ and the $g \in \widehat{\Pi}_{0,3}$ satisfying (51), and such moreover that

(53) $\sigma_\infty, g\rho$ with the relations $\sigma_\infty^2 = (g\rho)^3 = 1$ present $\widehat{\mathfrak{S}}^+_{0,3}$

Uncertainty retained from cancelled passages.

140-3-s097-D001 fix

140-3-s097-D002 [illeg. 1]

140-3-s097-D003 elements

140-3-s097-D004 [illeg. 1]

[scan 98]

(which I am going to call the "preliminary condition"). [p. 338]

†

For a $u \in G = \mathrm{Aut}(\widehat{\mathfrak{S}}^+_{0,3})'$! (not necessarily fixing σ_∞), we have equivalent conditions

$$(54) \quad \begin{cases} \text{(i)} & u(\overset{\varepsilon_0}{\ell'_0}) \in \text{conjugate of } \dagger\varepsilon_0^{\widehat{\mathbb{Z}}} = L'_0 \text{ (in } \widehat{\mathfrak{S}}^+_{0,3}) \\ \text{(ii)} & u(\ell_0) \in \text{conjugate of } \ell_0^{\widehat{\mathbb{Z}}} = L_0 \text{ (in } \widehat{\Pi}_{0,3}) \\ \text{(iii)} & u(L_i) \text{ conjugate of } L_i \quad \text{for } i = 0, 1, \infty \\ \text{(iv)} & \exists\ p \in \widehat{\mathbb{Z}}^*,\ g_0 \in \widehat{\Pi}_{0,3} \quad \boxed{u(\varepsilon_0) = \mathrm{int}(g_0)\varepsilon_0^p} \\ \text{(v)} & \exists\ p \in \widehat{\mathbb{Z}}^*,\ g_0 \in \widehat{\Pi}_{0,3} \quad \boxed{u(\ell_0) = \mathrm{int}(g_0)(\ell_0^p)} \end{cases}$$

We shall then also have

$$(55) \qquad \dagger \quad \begin{array}{ll} u(\ell_1) = \mathrm{int}(g_1)(\ell_1^p) & g_1 = g\rho(g_0) \\ u(\ell_\infty) = \mathrm{int}(g_\infty)(\ell_\infty^p) & g_\infty = g\rho(g_1) = g\rho(g)\rho^2(g_0)\ . \end{array}$$

The element †

$$(55) \qquad p \in \widehat{\mathbb{Z}}^* \qquad p = \chi(u)$$

is well determined by u, hence by g(if u given by (50)), we shall call it the multiplier of u. The element †

$$(56) \qquad g_0 \in \widehat{\Pi}_{0,3}$$

is determined a priori modulo $L_0 = \ell_0^{\widehat{\mathbb{Z}}}$ (change

$$(57) \qquad g_0 \longmapsto g_0\ell_0^\alpha$$

).

m766

Marginal note. m766 Notations $\mathrm{Autext}_{\mathrm{lac}}(\widehat{\mathfrak{S}}^{+}_{0,3})$, $\mathrm{Aut}_{\mathrm{lac}}(\widehat{\mathfrak{S}}^{+}_{0,3})$, $\mathrm{Aut}_{\mathrm{lac}}(\widehat{\mathfrak{S}}^{+}_{0,3})'$, $\mathrm{Aut}_{\mathrm{lac}}(\widehat{\mathfrak{S}}^{+}_{0,3}, \sigma_\infty)'$, $\mathrm{Aut}_{\mathrm{lac}}(\widehat{\Pi}_{0,3}, \sigma_\infty)$ ...

If the equivalent conditions (54|55) are satisfied, we say that u respects the loop structure of $\widehat{\mathfrak{S}}^{+}_{0,3}$ (or of $\Pi_{0,3}$) – it being understood that $\widehat{\mathfrak{S}}^{+}_{0,3}$ is equipped with the "loop-group" $\varepsilon_0^{\widehat{\mathbb{Z}}} \simeq L'_0$. We shall restrict|restrict ourselves from now on to such automorphisms. N.B. These conditions depend only on the outer automorphism defined by u, in $\widehat{\Pi}_{0,3}$ or in $\widehat{\mathfrak{S}}^{+}_{0,3}$. We

Uncertainty retained from cancelled passages.

140-3-s098-D001 u thus defined by a

140-3-s098-D002 ℓ

140-3-s098-D003 u

140-3-s098-D004 is

140-3-s098-D005 g_0

[scan 99]

are going to take an interest in the canonical liftings of such outer automorphisms given by (49), (50), to actual automorphisms.

The defining formula (50) for u involves only $g \in \widehat{\Pi}_{0,3}$, in principle g_0 mod L_0 in $\widehat{\Pi}_{0,3}/L_0$ is deduced from g. But in practice g_0 plays as important a role as g – we shall see that to a large extent it determines g (formula (62)).

If there were no need to introduce g_0, to express the conditions (54), we could easily parametrize the g satisfying (51) – hence corresponding to †liftings of $\dot\rho \in \mathfrak{S}_3$ to elements of order 3 of $\widehat{\mathfrak{S}}^{+}_{0,3}$ – by noting that there are exactly two conjugacy classes of such liftings, (corresponding to the fixed points $-\bar j, -j$ of $\dot\rho$ operating on $U^{\mathrm{an}}_{0,3}$), which are parametrized by

$$(58)\qquad \begin{cases} g = \lambda_0^{\sharp|*}\rho(\lambda_0^{\sharp|*})^{-1} = [\lambda_0^{\sharp|*}, \rho] & \text{(principal class, corresponding|corresponding to the "north pole" } -\bar j) \\ g = \lambda_0^{\sharp|*\prime}\ell_1^{-1}\rho(\lambda_0^{\sharp|*\prime})^{-1} & \text{(adjoint class, corresponding to the "south pole" } -j) \end{cases}$$

m767

Marginal note. m767 These|The parametric representations are unambiguous|unique.

where $\lambda_0, \lambda_0^{\sharp|*\prime} \in \widehat{\Pi}_{0,3}$. We shall see below that these cases correspond respectively to the cases

$$(59)\qquad p \equiv 1\ (3) \qquad \text{and} \qquad p \equiv -1\ (3)\ .$$

Uncertainty retained from cancelled passages.

140-3-s099-D001 splittings of

[scan 100]

But here the situation gets considerably more involved through the need to take account of the † conditions (54) (for example in the form (iv)) – hence that of introducing g_0, p jointly. It is then appropriate to introduce the quantities [p. 339]

$$(60)\qquad \gamma = \frac{p-1}{2} \in \widehat{\mathbb{Z}}$$

(N.B. since $p \in \widehat{\mathbb{Z}}^*$, its class in $\mathbb{Z}/2\mathbb{Z} = \widehat{\mathbb{Z}}/2\widehat{\mathbb{Z}}$ is 1, hence $p-1 \in 2\widehat{\mathbb{Z}}$, whence the possibility of finding γ),

$$(61)\qquad h_0 = g_0^{-1}\sigma_\infty(g_0)$$

m768

Marginal note. m768 h_0, relative to the automorphism σ_∞ of order 2, plays a role symmetric to ξ relative to the automorphism ρ of order 3 – this does not show up either in the notation.

† We check indeed|thus that g is expressed by means of g_0 and of γ (hence of p) by means of the formula (equivalent to (54) (iv) or (v))

$$(62)\qquad \boxed{g = \sigma_\infty(g_0)\ell_1^\gamma \rho(g_0)^{-1}} \left(= (g_0\ell_0^{-\gamma}) \left(\underbrace{\ell_0^\gamma h_0}_{\xi} \right) \rho(g_0\ell_0^{-\gamma})^{-1} \right)$$

m769

Marginal note. m769 N.B. This expression for g is not changed if $g_0 \longmapsto g_0\ell_0^\alpha$.

Thus the u that interest us are described by the pairs (g_0, γ) $(g_0 \in \widehat{\Pi}_{0,3}/L_0,\ \gamma \in \widehat{\mathbb{Z}})$ satisfying the conditions

$$(63)\qquad 2\gamma + 1 \in \widehat{\mathbb{Z}}^* \qquad (\text{NB}\quad 2\gamma + 1 = p = \chi(u))$$

and condition (51), where g is expressed as a function of g_0 and of γ by (62). This relation (51) is in turn equivalent to the relation

$$(64)\qquad \boxed{\xi\rho(\xi)\rho^2(\xi) = 1}$$

where

$$(65)\quad \xi = \ell_0^\gamma h_0 \qquad \dagger|\text{with (66)} \quad \begin{aligned} h_0 &\overset{\text{dfn}}{=} g_0^{-1}\sigma_\infty(g_0) \\ &= [g_0^{-1}, \sigma_\infty] \end{aligned}$$

m770

Marginal note. m770 N.B. We call h_0 here what we previously denoted by h_0^{-1}.

Uncertainty retained from cancelled passages.
140-3-s100-D001 the|the
140-3-s100-D002 with respect|p.v.
140-3-s100-D004 with

[scan 101]

Thus, the u are parametrized by the pairs $(g_0 \in \widehat{\Pi}_{0,3}/L_0,\ \gamma \in \widehat{\mathbb{Z}})$, satisfying conditions (63) and (64), which (by means of (65)) is written in explicit form

$$\boxed{(\ell_0^\gamma \underbrace{g_0^{-1}\sigma_\infty(g_0)}_{h_0})\rho(\ell_0^\gamma \underbrace{g_0^{-1}\sigma_\infty(g_0)}_{h_0})\rho^2(\ell_0^\gamma \underbrace{g_0^{-1}\sigma_\infty(g_0)}_{h_0}) = 1} \qquad (65\text{bis})$$

m771

Marginal note. m771 it would be better to regard this equation as an "equation in h_0" playing a role symmetric to "the equation in ξ" (68)

("equation in g_0" — as opposed to the "equation in ξ" below). (Moreover, we must add the preliminary condition, for the record...)

We can also consider that u (via g) is defined by $(\xi \in \widehat{\Pi}_{0,3},\ \gamma \in \widehat{\mathbb{Z}})$, which indeed allow us to recover

$$h_0 = \ell_0^{-\gamma}\xi$$

by (65), then g_0 by (66)

$$g_0^{-1}\sigma_\infty(g_0) = h_0$$

which indeed determines g_0 uniquely, thanks to the fact (which we admit provisionally) that $\widehat{\Pi}_{0,3}^{\sigma_\infty} = \{1\}$. The possibility of finding g_0 satisfying this condition is equivalent (we shall admit this here too) to the condition

$$\sigma_\infty(h_0)h_0 = 1 \tag{67}$$

which, in terms of ξ, γ, is also written in the form

[scan 102]
[p. 340]

$$\boxed{\sigma_\infty(\xi) = \ell_1^\gamma \xi^{-1} \ell_0^\gamma}. \tag{68}$$

Equations (64), (68) are called the "equations in ξ" — as opposed to the "equation in g_0" (66), which at first sight has the advantage of being a single one, and the disadvantage of being more complicated. In summary, the u that interest us ($u \in \mathrm{Aut}_{\mathrm{lac}}(\widehat{\mathfrak{S}}_{0,3}^+, \sigma_\infty)'$) are given|described by the pairs †($\xi \in \widehat{\Pi}_{0,3}^+,\ \gamma \in \widehat{\mathbb{Z}}$), satisfying the conditions

rather "the equation in h_0"

$$\begin{aligned} &\text{a)} \quad 2\gamma + 1 \in \widehat{\mathbb{Z}}^* && (63)\\ &\text{b)} \quad \xi\rho(\xi)\rho^2(\xi) = 1 && (64)\\ &\text{c)} \quad \sigma_\infty(\xi) = \ell_1^\gamma \xi^{-1} \ell_0^\gamma && (68) \end{aligned}$$

d) Preliminary condition for the record (53) (which apparently are not expressed directly in a convenient way in terms of ξ, γ).

For two pairs (ξ, γ), (ξ', γ') to define the same u, it is necessary and sufficient that we have

$$\gamma = \gamma', \qquad \xi' = \ell_0^{-\alpha} \xi \ell_1^\alpha \qquad (\alpha \in \widehat{\mathbb{Z}}). \tag{69}$$

Condition (64) can also be replaced by a parametric representation

$$\begin{cases} \xi = \lambda\rho(\lambda)^{-1} \\ \xi = \lambda\ell_0^{-1}\rho(\lambda)^{-1} \end{cases} \quad \text{or} \tag{70}$$

with $\lambda \in \widehat{\Pi}_{0,3}$ — in which case equation (68)

Uncertainty retained from cancelled passages.
140-3-s102-D001 (ξ, γ)

[scan 103]

can be interpreted as an "equation in λ", which according to the two cases takes the form

$$\begin{cases} \sigma_\infty(\lambda\rho(\lambda)^{-1}) = \ell_1^\gamma(\rho(\lambda)\lambda^{-1})\ell_0^\gamma \\ \qquad\qquad \text{or} \\ \sigma_\infty(\lambda\ell_0^{-1}\rho(\lambda)^{-1}) = \ell_1^\gamma(\rho(\lambda)\ell_0\lambda^{-1})\ell_0^\gamma, \end{cases} \tag{71}$$

Here for (λ, γ) and (λ', γ') to define the same u, it is necessary and sufficient that we have †

$$\lambda' = \ell_0^{-\alpha}\lambda, \qquad \gamma = \gamma'. \tag{72}$$

Remarks 1) Up to the "harmless" factor ℓ_0^γ, which|that distinguishes h_0 and ξ (65), we can say that the search for the $u \in \mathrm{Aut}_{\mathrm{lac}}(\widehat{\mathfrak{S}}_{0,3}^+, \sigma_\infty)$ is equivalent to that for the † simultaneous solutions of the two cocycle equations

$$\begin{cases} \xi\rho(\xi)\rho^2(\xi) = 1 \\ h_0\sigma_\infty(h_0) = 1 \end{cases} \tag{73}$$

relative to the groups $\mathbb{Z}/3\mathbb{Z}$, operating on $\widehat{\Pi}_{0,3}$ via ρ, and $\mathbb{Z}/2\mathbb{Z}$, operating via σ_∞ — whose solutions in parametric forms are respectively

$$\begin{cases} \xi = \lambda\rho(\lambda)^{-1} \quad \text{or} \quad \xi = \lambda\ell_0^{-1}\rho(\lambda)^{-1} \\ \quad \text{(according as } \gamma \equiv 0 \ (3) \text{ or } \gamma \equiv 1|-1 \ (3) \\ \quad\quad \text{i.e. } p \equiv 1 \ (3) \text{ or } p \equiv -1 \ (3)) \\ h_0 = |\ \equiv g_0^{-1}\sigma_\infty(g_0) \end{cases} \tag{74}$$

m773

Marginal note. m773 N.B. expression (62) for g brings out equally well this double intervention of σ_∞ and of ρ in the determination of the u!!!

m774

Marginal note. m774 (N.B. we could put everywhere g_0 for g_0^{-1}...)

Uncertainty retained from cancelled passages.
140-3-s103-D001 [illeg. 1]
140-3-s103-D002 equations

[scan 104]

3) The relation between the λ_0 in the parametric representation of g (58), and the λ in that of ξ (74 a) is implicit in the last expression (62) for g, namely [p. 341]

$$\lambda_0 = g_0\ell_0^{-\gamma}\lambda. \tag{75}$$

4) Let us note here also, that the quantities

$$\begin{array}{cccccccc} p, & \gamma, & g, & g_0, & h_0, & \xi, & \lambda_0, & \lambda \\ \cap & \cap & \cap & \cap & \cap & \cap & \cap & \cap \\ \widehat{\mathbb{Z}}^* & \widehat{\mathbb{Z}} & \widehat{\Pi}_{0,3} & \widehat{\Pi}_{0,3}/L_0 & & & \widehat{\Pi}_{0,3} & \end{array}$$

resp. λ_0', λ' (with for u constant the indeterminacies specific to g_0, h_0, ξ, λ) can be considered for u variable, as functions on the group $\mathrm{Autext}_{\mathrm{lac}}(\widehat{\mathfrak{S}}_{0,3}^+)' \simeq \mathrm{Autext}_{\mathrm{lac}}(\widehat{\Pi}_{0,3})'$ (subgroup of the outer automorphisms of the group with loops $\widehat{\Pi}_{0,3}$ that commute with $\mathfrak{S}_3$), with values in the sets $\widehat{\mathbb{Z}}^*, \widehat{\mathbb{Z}}, \widehat{\Pi}_{0,3}, \widehat{\Pi}_{0,3}/L_0, \ , \ , \widehat{\Pi}_{0,3}, \ .$ In particular, we thus find 8 functions continuous on $\Gamma_{\mathbb{Q}}$, with values in the appropriate spaces — starting with the cyclotomic character. m775

Marginal note. m775 We shall see ([illeg. 2] less!) that we can normalize g_0, and thereby also $h_0, \xi, \lambda, \lambda_0$, so as to remove all ambiguity in its|these definition|definitions...

[scan 105]

Let us note in this connection the equivalent conditions

$$\begin{cases} \text{(i)} \quad p = 1 \\ \text{(ii)} \quad \gamma = 0 \\ \text{(iii)} \quad g = \sigma_\infty(g_0)\rho(g_0)^{-1} \\ \text{(iv)} \quad h_0 = \xi \\ \text{(v)} \quad \sigma_\infty(\xi) = \xi^{-1} \qquad (\text{“equation in } \xi\text{” simplified!}) \\ \text{(vi)} \quad h_0\rho(h_0)\rho^2(h_0) = 1 \end{cases} \tag{75bis}$$

all meaning that $\chi(u) = 1$. Thus the search for the $u \in \mathrm{Aut}_{\mathrm{lac}}(\widehat{\mathfrak{S}}^{+}_{0,3}, \sigma_\infty)'$ with multiplier 1 ($u \in \mathrm{Aut}_{\mathrm{lac}}(\widehat{\mathfrak{S}}^{+}_{0,3}, \sigma_\infty)'^{0}$) is thus indeed equivalent to that for the simultaneous solutions of the equations of cocycles (73), or of the parametric equations ("coboundary") (74).

2) Returning to the general case (arbitrary multiplier), the only case that I know how to write down explicitly (except for $u = 1$) is the one that corresponds to $\tau_\infty \in \mathfrak{S}^{*}_{0,3}$, which corresponds to the characteristics

$$\begin{cases} u = \tau_\infty, \quad \chi(u) = p = -1, \quad \gamma = -1 \\ g = \ell_1^{-1} \\ g_0 = 1, \quad h_0 = 1, \quad \xi = \ell_0^{-1} \\ \lambda'_0 = 1, \quad \lambda' = \ell_0^{\gamma} g_0^{-1} \lambda_0 = \ell_0^{-1} \\ [\, \lambda_0 = \tau_\infty, \quad \lambda = \ell_0^{-1} \tau_\infty \,] \end{cases} \tag{75ter}$$

← cf. below

It would certainly be no luxury to manage to write down a bunch of other u, to see in particular whether the crowds|crowd of necessary conditions that we are led to derive for the elements coming from $\Gamma_{\mathbb{Q}}$, are not automatically satisfied!

[scan 106]
[p. 342]

(VII) Calculations in $\mathrm{Sl}(2, \widehat{\mathbb{Z}})' = \mathrm{Sl}(2, \widehat{\mathbb{Z}})/\pm 1$ (cocycles under ρ).

m777

Marginal note. m777 Calculations "for conscience's sake", which will not be of use perhaps any more!

This group is a (very small!) quotient of $\widehat{\mathrm{Sl}(2, \mathbb{Z})'}$, which has the advantage of lending itself to convenient explicit algebraic calculations, in terms of matrices. Let us denote, by abuse of language, by the same letters $g, g_0, h_0, \xi, \lambda_0, \lambda$ elements of † $\widehat{\Pi}_{0,3}$, (or of $\widehat{\mathfrak{S}}^{+}_{0,3}$ more generally) and their images in $\Pi(2)' \subset \mathrm{Sl}(2, \widehat{\mathbb{Z}})'$ † (or in $\mathrm{Gl}(2, \widehat{\mathbb{Z}})' \simeq \mathrm{Gl}(2, \widehat{\mathbb{Z}})/\{\pm 1\}$). † in general, let us consider a matrix

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{Gl}(2, \widehat{\mathbb{Z}}) \tag{76}$$

we then have

$$\begin{cases} \dagger\rho(g) = \rho g \rho^{-1} = \begin{pmatrix} c+d & -c \\ c+d-a-b & a-c \end{pmatrix}, \quad \det(g)\rho(g^{-1}) = \begin{pmatrix} a-c & c \\ a+b-c-d & c+d \end{pmatrix} \\ \rho^2(g) = \rho^{-1} g \rho = \dagger \begin{pmatrix} d-b & a+b-c-d \\ -b & a+b \end{pmatrix}, \quad \det(g)\rho^2(g^{-1}) = \begin{pmatrix} a+b & c+d-a-b \\ \dagger b & d-b \end{pmatrix} \end{cases} \tag{77}$$

$$\begin{cases} [g, \rho] = g\rho g^{-1}\rho^{-1} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \cdot \dagger \\ \delta A = (a^2 + b^2 + ab) - (ac + bd + bc) \qquad \delta B = ac + bd + bc \\ \delta C = -(c^2 + d^2 + cd) + (ac + bd + ad) \quad \delta D = c^2 + d^2 + cd \end{cases} \tag{78}$$

m778

Marginal note. m778 N.B. $C + D - B = 1$† $\delta = \det g = ad - bc$ †

We shall write the equation

$$\dagger \qquad g\rho(g)\rho^2(g) = 1$$

in the equivalent form

$$g\rho(g) = \rho^{-1}(g^{-1})(= \rho^2(g^{-1})) \qquad \text{in } \mathrm{Gl}(2, \widehat{\mathbb{Z}})' \tag{79}$$

where

$$g\rho(g) = \begin{pmatrix} P & Q \\ R & S \end{pmatrix} \tag{80}$$

Uncertainty retained from cancelled passages.

140-3-s106-D001 [illeg. 1]
140-3-s106-D002 [illeg. 1]
140-3-s106-D004 shall set
140-3-s106-D005 ρ
140-3-s106-D006 (
140-3-s106-D007 c
140-3-s106-D008 $\frac{1}{\det(g)}$
140-3-s106-D009 $\det g = \delta$
140-3-s106-D010 [illeg. 1]
140-3-s106-D011 79

[scan 107]

$$(81)\qquad \begin{cases} P = (a+b)(c+d-b) & Q = -a(c-b) - bc = \dagger\delta - a(c+d-b) \\ R = (c+d)^2 - d(a+b) & S = ad - c(c+d) = \underset{\text{if } \det g=1}{\delta} - c(c+d-b) \\ \quad \underset{\text{if } \det g=+1}{=} (c+d)(c+d-b) - \delta & \end{cases}$$

where we have set $\delta = ad - bc = \det g$.

Comparing (81) and the expression (77) for $\rho^2(g^{-1})$, equation (79) – which is also written

$$g\rho(g) = \pm\rho^2(g^{-1})$$

gives the following solutions.

m779

Marginal note. m779 Up to replacing g by $-g$, † we are reduced to determining the solutions of the equation

$$(82)\qquad g|\rho\rho(g)\rho^2(g) = +|\pm 1 \qquad \text{i.e.} \quad g|\rho\rho(g) = +|\pm\rho^{-1}(g^{-1}) \qquad \underline{\text{in } \mathrm{Gl}(2,\widehat{\mathbb{Z}})}$$

† We distinguish the solutions according to the factors of the product

$$\mathrm{Sl}(2,\widehat{\mathbb{Z}})' \simeq \prod_{\ell \text{ premier}} \mathrm{Sl}(2,\mathbb{Z}_\ell)'.$$

m780

Marginal note. m780 N.B. We shall use|use this study of equation (82) only in the case $\det g = 1$, i.e. $\delta = 1$, unless I am mistaken

The relation

$$\frac{1}{\delta}(a+b) = P = (a+b)(c+d-b) \qquad \text{in } \mathbb{Z}_\ell$$

gives the two solutions $a+b=0$ or $c+d-b=1$ (since $\mathbb{Z}_\ell$ is an integral domain).

†

m781

Marginal note. m781 The necessity of this formula (84) was obvious a priori.

†

Uncertainty retained from cancelled passages.
140-3-s107-D002 1
140-3-s107-D003 [illeg. 5]
140-3-s107-D004 (82) [illeg. 2] $g\rho(g) = +\rho^{-1}(g^{-1})$
140-3-s107-D005 at once [illeg. 1] $R =$ a $\frac{1}{\delta^2}|\frac{1}{\delta}$ [illeg. 5] [illeg. 1]
140-3-s107-D007) ad [*illeg.* 1] $c+d-b$ [*illeg.* 1]

[scan 108]
[p. 343]

1°) $a+b \neq 0$ (in $\mathbb{Z}_\ell$) whence

$$(83) \qquad \boxed{c+d-b = \frac{1}{\delta}}.$$

Given (83), we verify that the formulas (82), – where it remains to check

$$Q = \frac{c+d-(a+b)}{\delta} \leftarrow | = \left(= \frac{1}{\delta^2} - \frac{a}{\delta}\right), \quad R = \frac{b}{\delta}$$
$$S = \frac{d-b}{\delta}. \left(= \frac{1}{\delta^2} - \frac{c}{\delta}\right)$$

are equivalent to:

$$(84) \qquad \boxed{\delta^3 = 1}$$

(whose necessity was obvious a priori).
m782

Marginal note. m782 N.B. The conditions $c+d-b = \frac{1}{\delta}$, $\delta^3 = 1$ are sufficient [illeg. 1] for equations (82), ! (independently of the question whether $a+b=0$ or $a+b \neq 0$) – they are necessary if we assume that we are not in case 2° below

2°) $a+b=0$ (in $\mathbb{Z}_\ell$) whence

$$(85) \qquad a = -b \qquad \text{i.e.} \quad g = \begin{pmatrix} -b & b \\ c & d \end{pmatrix}$$

$$(86) \quad \begin{cases} \det g = -b(c+d) = \delta \quad \text{whence} \\ \dagger\ \delta + b(c+d) = 0 \qquad\qquad d = -\frac{\delta}{b} - c \end{cases}$$

i.e.

$$g = \begin{pmatrix} -b & b \\ c & -\frac{\delta}{b} - c \end{pmatrix},$$

and

$$\begin{cases} P = 0, & Q = \delta + b(c+d-b) = -b^2 \\ R = (c+d)^2 = \frac{\delta^2}{b^2}, & S = \delta + c\left(\frac{\delta}{b} + b\right); \end{cases}$$

†

$$\rho^2(g^{-1}) = \begin{pmatrix} 0 & -1/b \\ b/\delta & -\frac{1}{\delta}\left(\frac{\delta}{b} + b + c\right) \end{pmatrix},$$

and equation (82) is then written

$$(87) \quad \begin{cases} \text{a)} & b^3 = \delta^3 = 1. \\ \text{b)} & c((b\delta)^2 + (b\delta) + 1) + (\delta^2 + b^2\delta + b) = 0. \end{cases}$$

Uncertainty retained from cancelled passages.
140-3-s108-D001 $b+$
140-3-s108-D002 here

[scan 109]

We again distinguish two cases:
a) $b\delta = 1$ i.e. $b = \frac{1}{\delta}$, then equation 87 b) becomes

$$3c + 3\frac{1}{\delta} = 0 \qquad \text{i.e.}$$

$$(88) \qquad c = -\frac{1}{\delta} = -\delta^2 \ \dagger \qquad \dagger$$

corresponds to

$$(90) \qquad g = \dagger \quad \begin{pmatrix} -\delta'\dagger & \delta'\dagger \\ -\delta'\dagger & 0 \end{pmatrix}$$

where $\delta' = 1/\delta = \delta^2$, and where, I recall, δ is subject to the sole condition

$$\delta^3 = 1.$$

b) $b\delta \neq 1$ thus (since $(b\delta)^3 = 1$) $(b\delta)^2 + b\delta + 1 = 0$) and equation (87 b) becomes

$$(\alpha) \qquad \delta^2 + b^2\delta + b = 0. \qquad \dagger$$

We shall have

$$\begin{cases} b = \dagger u\frac{1}{\delta} = u\delta^2 \quad \text{with} \quad u^3 = 1,\ u \neq 1 \qquad \text{i.e.} \\ u^2 + u + 1 = 0 \end{cases}$$

and equation α), reduces to

$$u^2|\delta^2 + u\dagger + 1 = 0.$$

$\dagger$
$\dagger$

Uncertainty retained from cancelled passages.
140-3-s109-D002 $3c + (2\delta^2 + 1) = 0$
140-3-s109-D003 $3c + 3\delta^2 = 0$
140-3-s109-D005 $-\frac{1}{3}(2\delta^2 + 1)$ $4|1$
140-3-s109-D006 ^{2}
140-3-s109-D007 ^{2}
140-3-s109-D008 ^{2}
140-3-s109-D009 i.e. a
140-3-s109-D010 δ
140-3-s109-D013 ^{2}
140-3-s109-D014 [illeg. 1]
140-3-s109-D015 [illeg. 1] obviously [illeg. 3] means also that $\frac{b}{\delta} \neq \delta$

[scan 110]
[p. 344]

Proposition Let $g \in \mathrm{Gl}(2, \widehat{\mathbb{Z}}_\ell)$, $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $a, b, c, d \in \widehat{\mathbb{Z}}_\ell|\mathbb{Z}_\ell$, $ad - bc = \delta \in \widehat{\mathbb{Z}}^*_\ell|\mathbb{Z}^*_\ell$. For us to have

$$g\rho(g)\rho^2(g) = 1 \qquad (82|81)$$

it is necessary and sufficient that we have $\delta^3 = 1$, and that we be in one of the following two cases, which are mutually exclusive:

$$1°) \quad c + d - b = 1$$

$$2°) \quad g = -\delta' \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} = -\delta' \rho^{-1} \qquad \left(\text{where } \delta' = \frac{1}{\delta} = \delta^2\right)$$

†

Corollary The g for which we have 1°) and $a + b \neq 0$ are those of the form

†

$$\begin{cases} g = \begin{pmatrix} a & b \\ c & b - c + \frac{1}{\delta} \end{pmatrix} \qquad † \\ \text{with} \quad c = \dfrac{1}{a+b}(a(b + \delta') - \delta), \end{cases}$$

those for which we have 1°) and $a + b = 0$ are the matrices of the form

$$\begin{cases} g = \begin{pmatrix} -u\delta' & u\delta' \\ c & -\delta' u' - c \end{pmatrix} \\ \text{where } \delta' = 1/\delta = \delta^2, \ u' = \dfrac{1}{u} = u^2, \quad u \text{ being a solution} \\ \text{of the cyclotomic equation } u^2 + u + 1 = 0 \quad \text{i.e.} \\ \text{root|roots primitive|primitive } 3^{\text{e}} \text{ of unity} \end{cases}$$

Uncertainty retained from cancelled passages.

140-3-s110-D002 iff|sss when of

140-3-s110-D004 u $c|0$

140-3-s110-D005 for c $(a(b + \delta^2))|(ab + a\delta')$

[scan 111]
[p. 345]

$$= \begin{pmatrix} 0 & -1| + 1 \\ 1 & -1 \end{pmatrix}$$

Remarks † 1°) Obviously $-\rho$ is also a solution of (87) (just like ρ^{-1}) † (just like $-\rho^{-1}$), but it falls respectively under cases † 1°) † †

2°) To †determine the solutions in $\mathrm{Gl}(2, \widehat{\mathbb{Z}})$ of $g\rho(g)\rho^2(g) = 1$, it suffices to choose for $\forall\, \ell$ such a solution in $\mathrm{Gl}(2, \widehat{\mathbb{Z}})$, using the result|results (86); we shall note that the "cases" obtained 1°) to 3°) can vary from one ℓ to another. † We should keep in mind that cases 1°), 2°), 3°) correspond resp.^{t} to families of solutions depending on 2, 1 and 0 parameters.

3°) I do not know whether a $\xi \in \mathrm{Sl}(2, \widehat{\mathbb{Z}})'|^*$, $\xi = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$, which satisfies $g\rho(g)\rho^2(g) = 1$ admits a parametric representation of the form (58) by $\lambda_0 \rho(\lambda_0)^{-1}$ or $\lambda_0'(-l_1^{-1})\rho(\lambda_0')^{-1} = \lambda_0(-l_1^{-1}|l_0^{-1})\rho(\lambda_0)^{-1}$ with $\lambda_0 = \lambda_0' \tau_\infty$ (it suffices to look separately in the $\mathrm{Sl}(2, \mathbb{Z}_\ell)'$). I do not know even if we assume $g \equiv \mathrm{id}\ (2)$, (which is a condition on the component in $\mathrm{Sl}(2, \mathbb{Z}_2)'$ only). We see in

m783

Marginal note. m783 N.B. $g = l_0^{-1}$ is not a solution of this the equation|equation, on the other hand $g = -l_0^{-1}$ is…

Uncertainty retained from cancelled passages.
140-3-s111-D002 N.B.
140-3-s111-D003 and ρ thus
140-3-s111-D004 [illeg. 1]
140-3-s111-D005 and under case
140-3-s111-D006 A, resp.
140-3-s111-D007 [illeg. 1]
140-3-s111-D008 $-1|1$

[scan 112]

any|any case from formulas (78) that if ξ is of the form $\lambda_0\rho(\lambda_0)^{-1}$ (if $\xi\rho(\xi)\rho^2(\xi)=1$, of course), † † we will have $C+D-B=1$ — i.e. †we are in the "general" case with two parameters. † OK, cf. (99) below.

(Without nevertheless being able to exclude the case where we should at the same time $A+B=0$, i.e. the case with one parameter

$$\xi=\begin{pmatrix}-u & u\\ C & -u'-C\end{pmatrix}$$

with u a primitive third root of 1, $u'=u^{-1}=u^2$ the other primitive root.

Uncertainty retained from cancelled passages.
140-3-s112-D002 then indeed has case A)
140-3-s112-D003 [illeg. 1]
140-3-s112-D004 [illeg. 1]
140-3-s112-D005 dispense with

[scan 113]
[p. 346]

(VIII) — Calculations in $\mathrm{Sl}(2,\widehat{\mathbb{Z}})$: cocycles under σ_∞.

Let us take again a $g\in\begin{pmatrix}a & b\\ c & d\end{pmatrix}\in\mathrm{Sl}(2,\widehat{\mathbb{Z}})$, and compute $\sigma_\infty g\sigma_\infty^{-1}$, we find, thanks to

$$\sigma_\infty=\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}\qquad(48)$$

† the formulas

$$(90)\quad\begin{cases}\sigma_\infty g\sigma_\infty^{-1}=\begin{pmatrix}d & -c\\ -b & a\end{pmatrix}={}^t(g^{-1}) & \text{the contragredient of } g\ !\\ \text{i.e.}\\ \sigma_\infty g^{-1}\sigma_\infty^{-1}={}^t g & \text{thus}\\ [\sigma_\infty,g^{-1}]=({}^t g)(g)=\begin{pmatrix}a^2+b^2 & ac+bd\\ ac+bd & c^2+d^2|c^2-d^2\end{pmatrix}\end{cases}$$

$=\sigma_\infty g^{-1}\sigma_\infty|\sigma_\infty^{-1}g$
m785

Marginal note. m785 N.B. we have $\sigma_\infty^{-1}=\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}=-\sigma_\infty$ and † $\sigma_\infty^2\neq 1$ in $\mathrm{Sl}(2,\widehat{\mathbb{Z}})$!
m786

Marginal note. m786 N.B. If $g\in\mathrm{Gl}(2,A)$, A any ring (for example $A=\mathbb{Z}_\ell$), we have $\sigma_\infty g\sigma_\infty^{-1}=\begin{pmatrix}d & -c\\ -b & a\end{pmatrix}=(\det g)^t(g^{-1})$

thus, taking the inverse of both sides in the last formula

$$(91)\qquad \underbrace{g^{-1}\sigma_\infty(g)}_{({}^t g\cdot g)^{-1}} = \begin{pmatrix} c^2+d^2 & -(ac+bd) \\ -(ac+bd) & a^2+b^2 \end{pmatrix}.$$

On the other hand, let † $h_0 \in \mathrm{Sl}(2,\widehat{\mathbb{Z}})$ we have

$$h_0\sigma_\infty(h_0) = h_0({}^t h_0^{-1})$$

thus

$$(*)\qquad h_0\sigma_\infty(h_0) = 1 \iff h_0 = {}^t h_0.$$

Finally, let

$$(91)\qquad \xi = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

and consider the equation (68)

$$(**)\qquad \sigma_\infty(\xi) = l_1^\gamma \xi^{-1} l_0^\gamma$$

which is also equivalent to (*), with h_0 given by (65|66) †, l_0, l_1 by (44).

Uncertainty retained from cancelled passages.

140-3-s113-D001 [illeg. 1]

140-3-s113-D002 [illeg. 1]

140-3-s113-D003 r

140-3-s113-D004 $\gamma|\delta$ [illeg. 4]

[scan 114]

$$(92)\qquad \begin{aligned} h_0 = l_0^{-\gamma}\xi &= \begin{pmatrix} 1 & +2\gamma \\ 0 & 1 \end{pmatrix}\begin{pmatrix} A & B \\ C & D \end{pmatrix} \\ &= \begin{pmatrix} A+2\gamma C & B+2\gamma D \\ C & D \end{pmatrix} \end{aligned}$$

thus "the equation in ξ" (**) (written in the quotient $\mathrm{Sl}(2,\widehat{\mathbb{Z}})'$ of $\widehat{\mathfrak{S}}^+_{0,3}$, is written $B+2\gamma D = C$ i.e.

$$(93)\qquad \underbrace{C-B}_{1-D} = 2\gamma D \qquad \text{i.e.} \qquad \boxed{pD=1} \text{ in case|one case}$$

†

on recalling that $2\gamma+1 = p$

(We should keep in mind that the equation (**) in $\mathrm{Sl}(2,\widehat{\mathbb{Z}})'$ means $h_0\sigma_\infty(h_0) = \pm 1$ i.e. $h_0 = \pm {}^t h_0$, but the sign $-$ would imply that we have $D = -D$ i.e. $2D = 0$ so $D = 0$, which is impossible if we assume that $\xi = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ comes from $\widehat{\Pi}_{0,3}$ so is in $\Pi'(2)$ i.e. $\equiv$ †1 (2) (so that $D \equiv 1$ (2)). Thus we must indeed have (93).

Let us now take the parametric representation (70) of ξ (taking account of the equation $\xi\rho(\xi)\rho^2(\xi) = 1$ † (64))

$$(94)\qquad \begin{aligned} &\xi = \lambda\rho(\lambda)^{-1} && (\lambda \in \mathrm{Sl}(2,\widehat{\mathbb{Z}})) \\ \text{or}\quad &\xi = \lambda'(-l_1^{-1})\rho(\lambda')^{-1} = \lambda\rho(\lambda)^{-1} && \begin{pmatrix} \lambda = \lambda'\tau_\infty \\ \lambda' \in \mathrm{Sl}(2,\widehat{\mathbb{Z}}) \end{pmatrix} \end{aligned}$$

† with $\lambda \in \mathrm{Gl}(2,\widehat{\mathbb{Z}})^{*|\circ}$ † $\det\lambda = \pm 1$, (in fact $\lambda \equiv$ †1 (2) in the case that interests us, but no matter…)

Uncertainty retained from cancelled passages.
140-3-s114-D001 [illeg. 2]
140-3-s114-D003 ±
140-3-s114-D004 ξ
140-3-s114-D005 λ, λ′ ∈ Sl
140-3-s114-D006 case I
140-3-s114-D007 ±

[scan 115]

then formula (78) gives A, B, C, D as quadratic functions in a, b, c, d, (paying Attention to the sign $\delta = -1$ in case 2°) † thus (93) is written † [p. 347]

$$(96) \qquad \boxed{(2\gamma+1)D = 1 \qquad \text{i.e.} \quad pD = 1} \qquad D = \delta(c^2 + d^2 + cd),$$

i.e. the multiplier p is obtained as a function of $\xi \in \widehat{\Pi}_{0,3}$ by taking the image of ξ in $\mathrm{Sl}(2, \widehat{\mathbb{Z}})'$, in the form $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ (there is a sign ambiguity, removed by the choice of λ such that $\xi = \lambda\rho(\lambda)^{-1}$, the image of λ in $\mathrm{Gl}(2, \widehat{\mathbb{Z}})'$ not|in determining a matrix (still denoted λ) of $\mathrm{Gl}(2, \widehat{\mathbb{Z}})$ except up to a sign, but $\lambda\rho(\lambda)^{-1}$ no longer depends on the choice of this sign…), by the formula

$$(97) \qquad p = \frac{1}{D} \qquad (D = \delta(c^2 + d^2 + cd))$$

Let us now note that on reduction modulo 3, the quadratic form $c^2 + d^2 + cd = (c+d)^2 - cd$ takes only the values 0 and 1 (clear if $c = 0$, if $d = 0$, — so we find a perfect square in these cases, otherwise $c, d = \pm 1$, and $f(c, d) = 1 + 1 \pm 1 \equiv 1$ or 0 † (3). Now here we must find a unit so the value 0 excluded, so

Uncertainty retained from cancelled passages.
140-3-s115-D002 $B - C|B + C \quad \delta(c^2 + d^2 + cd - \delta) \quad D =$
140-3-s115-D003 or
140-3-s115-D006 [illeg. 1]

[scan 116]

m788

Marginal note. m788 N.B. We have $D \equiv p \equiv \delta$ (3) (98) knowledge of D (or p, or δ) modulo 3 † tells us — which case we have in the parametric representation of ξ. Useless now to treat these cases separately — since we have $\xi = \lambda\rho(\lambda)^{-1}$, $\lambda = \lambda'\tau_\infty$, $\det\lambda = -1$.

$$p \equiv \delta \quad (3)$$

(where $\delta = \det\lambda \in \{\pm 1\}$, $\xi = \lambda\rho(\lambda)^{-1}$) † i.e.

$$\begin{cases} \text{Case } \xi = \lambda\rho(\lambda)^{-1} \quad (\lambda \in \mathrm{Sl}(2, \widehat{\mathbb{Z}})) \iff p \equiv 1 \ (3) \quad \dagger \iff \gamma \equiv 0 \ (3) \\ \text{Case } \xi = \lambda' l_1^{-1}\rho(\lambda)^{-1}|\rho(\lambda')^{-1} \quad (\lambda' \in \mathrm{Sl}(2, \widehat{\mathbb{Z}})) \iff p \equiv -1 \ (3) \iff \gamma \equiv -1 \ (3). \end{cases}$$

B) Case $\xi = \lambda' l_0^{-1}\rho(\lambda')^{-1} = (\lambda' l_0^{-1})(\rho\lambda'^{-1}\rho^{-1})$.
Using the second formula (77) we find

$$\xi = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} a-c & c \\ a+b-c-d & c+d \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

with

$$(99)\qquad \begin{cases} A = 3a^2 + b^2 + 3ab - 3ac - 2ad - bc - bd & B = 3ac + 2ad + bc + bd \\ C = -(3c^2 + 3cd + \dagger d^2) + 3ac + ad + 2bc + \dagger bd & D = 3c^2 + 3cd + d^2 \end{cases},$$

N.B. $C + D - B = -1 | -\delta$.

and formula (93) (which expresses "the equation in ξ" (**), or the one in h_0 (*)) is written, † taking account of

$$B - C = -1 + D,$$

in the form †

$$2\gamma D = -1 - D \qquad \text{i.e.}$$

$$(100)\qquad \boxed{(2\gamma + 1)D = -1 \qquad \text{i.e.} \quad pD = -1},$$

m790

Marginal note. m790 as before we cannot have $h_0 = -{}^t h_0$, for we cannot have $D = 0$ if we assume that $\xi \equiv 1$ (2), hence $D \equiv 1$ (2).

so here the multiplier p is expressed as a function of the image of ξ in $\mathrm{Sl}(2, \widehat{\mathbb{Z}})'$, put in the form $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ where $A, B, C, D \in \widehat{\mathbb{Z}}$ — and on removing the sign ambiguity as before — by the formula

$$(101)\qquad p = -\frac{1}{D} \qquad (D = 3c^2 + 3cd + d^2 | 3(c^2 + 3cd + d^2)).$$

Uncertainty retained from cancelled passages.

140-3-s116-D001 already|thus
140-3-s116-D002 [illeg. 4] [illeg. 3]
140-3-s116-D003 $g = \lambda_0 \rho(\lambda_0)^{-1}$
140-3-s116-D004 2
140-3-s116-D005 2
140-3-s116-D006 thus again in the form
140-3-s116-D007 2γ

[scan 117]
[p. 348]

Here it is clear that the only values modulo 3 represented by the quadratic form $3c^2 + 3cd + d^2 \equiv d^2$ (3) are 0 and 1, whence

$$(102)\qquad p \equiv -1 \ (3) \quad (\text{i.e. } \gamma \equiv -1 \ (3)) \quad \text{if } \xi = \lambda l_0^{-1} \rho(\lambda)^{-1}$$

$$(\Longleftrightarrow g = \lambda_0 l_0^{-1} \rho(\lambda_0)^{-1})$$

This study thus does show that we are in the case $\xi = \lambda\rho(\lambda)^{-1}$, resp. $\xi = \lambda l_0^{-1}\rho(\lambda)^{-1}$, according as $p = \chi(u) \equiv 1$ or $\equiv -1$ (3), i.e. according as $\gamma \equiv 0$ or $\gamma \equiv -1$ (3).

Remarks. 1°) Placing ourselves in $\mathrm{Sl}(2, \widehat{\mathbb{Z}})'$ (or rather, in its † subgroup $\Pi(2)' = \mathrm{Ker}(\mathrm{Sl}(2, \widehat{\mathbb{Z}})' \to \mathrm{Sl}(2, \mathbb{Z}/2\mathbb{Z}))$ in place of $\widehat{\Pi}_{0,3}$, to † determine the solutions of "the equation in ξ" (**) (cf. (68)), when ξ is put in parametric form (94), we see that as a condition on $\lambda = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, this equation is relatively innocuous and very simple — it simply says that $pD = 1$, where $D = c^2 + d^2 + cd$ in the first case, $D = -(c^2 + d^2 + cd)$ in the second. If we do not fix $p \in \widehat{\mathbb{Z}}^*$ beforehand, this condition is simply that the element

Uncertainty retained from cancelled passages.
140-3-s117-D002 $\Gamma|F$
140-3-s117-D003 solve the equation

[scan 118]

D of $\widehat{\mathbb{Z}}$ be a unit, i.e.

$$(103) \quad D \in \widehat{\mathbb{Z}}^*$$

— and we then determine p uniquely in terms of D , hence of c and d and the sign δ by $D = \delta(c^2 + d^2 + cd)$, ($a$ and b are of no use) — hence also in terms of $\lambda \in \widehat{\Pi}_{0,3} \cdot (1, \tau)|\{1, \tau\}$, itself uniquely determined by ξ.

N.B. We will have p

Thus knowledge of $\xi \in \widehat{\Pi}_{0,3}$ suffices to determine the multiplier $p = 2\gamma + 1 \in \widehat{\mathbb{Z}}^*$, hence also $\gamma \in \widehat{\mathbb{Z}}$, and $h_0 = l_0^{-\gamma}\xi$, hence also g_0 (by formula (66), $g_0^{-1}\sigma_\infty(g_0) = h_0$), and consequently g by formula (62), $g = \sigma_\infty(g_0)l_1^\gamma\rho(g_0)^{-1}$, hence u by (50). It is interesting to make sure that $D|$it † does not change when we make the change (69)

$$\xi \longmapsto \xi' = l_0^{-\alpha}\xi l_1^\alpha,$$

in matrix terms, we have indeed

$$(104) \quad l_0^{-\alpha}\xi l_1^\alpha = \begin{pmatrix} 1 & 2\alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 2\alpha & 1 \end{pmatrix} = \begin{pmatrix} A + 2\alpha(B + C) + 4\alpha^2 D & B + 2\alpha D \\ C + 2\alpha D & D \end{pmatrix}.$$

2/ But we should take care that in full rigor mere knowledge of the image of ξ in $\mathrm{Sl}(2, \widehat{\mathbb{Z}})'$ allows us to determine the multiplier p only up to sign — on the other hand knowledge of the image of λ in $\mathrm{Gl}(2, \widehat{\mathbb{Z}})'$ — hence of a matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{Sl}|\mathrm{Gl}(2, \widehat{\mathbb{Z}})$ defined up to sign — allows us to determine

Uncertainty retained from cancelled passages.
140-3-s118-D002 [illeg. 1]

[scan 119]

p as the inverse of D, given by $D = \delta(c^2 + cd + d^2)$ †. where $\delta \in \{\pm 1\}$ is determined by the condition $D \equiv \delta$ (3). [p. 349]

3/ We know by arithmetic means (since $\Pi_{\mathbb{Q}} \xrightarrow{\chi} \widehat{\mathbb{Z}}^*$ is surjective) that we must find any $p \in \widehat{\mathbb{Z}}^*$ in the form $1/D$ (or again, in the form D), where $D = c^2 + cd + d^2$ if $p \equiv 1$ (3), and $D = -3(c^2 + cd)-|+d^2$ if $p \equiv -1$ (3), with $c, d \in \widehat{\mathbb{Z}}$, and even $c \equiv 0$ (2), $d \equiv 1$ (2). If we want the direct verification, (We are reduced to verifying it separately for the various $\mathbb{Z}_\ell$, i.e. to proving the

Corollary. 1°)|°) If ℓ † prime $\neq 3$, then $\forall \beta|p \in \mathbb{Z}_\ell^*$ can be put equally well in the forms $c^2 + cd + d^2$, or $-(3(c^2 + cd) + d^2)$ (with $c, d \in \widehat{\mathbb{Z}}_\ell|\mathbb{Z}_\ell$). If $\ell = 2$, we can even suppose $c \equiv 0$, $d \equiv 1$ (2). For $p \in \mathbb{Z}_\ell^*$, $\ell = 3$, it can be put in the form $c^2 + cd + d^2$ (resp. $-3(c^2 + cd)-|+d^2$) iff $p \equiv 1$ (3) (resp. $p \equiv -1$ (3)), cases which are mutually exclusive. m792

Marginal note. m792 we don't care about the|about $-(3(c^2 + cd) + d^2)$!

4/ Returning to the "innocuous" reduction in $\mathrm{Sl}(2, \widehat{\mathbb{Z}})'$ of "the equation in λ" as meaning simply that $c^2 + d^2 + cd \in \widehat{\mathbb{Z}}^*$, we might hope

Uncertainty retained from cancelled passages.
140-3-s119-D002 or
140-3-s119-D003 [illeg. 1]

[scan 120]

to have a method for constructing even in $\widehat{\Pi}_{0,3}$ big batches of solutions of the equation in question. But I have absolutely not managed it — not even to construct a single one explicitly, apart from those which correspond to the case $u = 1$, and $u = \tau_\infty$.

A question: Let $\lambda_0 \in \mathrm{Sl}(2, \widehat{\mathbb{Z}})'^{|*}$, $\lambda_0 \equiv 1\ (2)$, be such that $\xi_0 = \lambda_0\rho(\lambda_0)^{-1}$ (say, or $\xi = \lambda(-l_0^{-1})\rho(\lambda_0)^{-1}$) satisfies in $\mathrm{Sl}(2, \widehat{\mathbb{Z}})'$ an "equation in ξ" for a suitable $\gamma = \frac{p-1}{2}$ $(p \in \widehat{\mathbb{Z}}^*)$ — i.e. $c^2 + d^2 + cd \in \widehat{\mathbb{Z}}^*$ (resp. $3(c^2 + cd) + d^2 \in \widehat{\mathbb{Z}}^*$). Does there then exist a $\lambda|\xi \in \widehat{\Pi}_{0,3}$ above λ_0, such that $\xi = \lambda\rho(\lambda)^{-1}$ (resp.…) satisfies an equation in ξ as well (necessarily with the same γ) ?

[scan 121]
[p. 350]

43 – Digression on the 1-cocycles of cyclic groups.

Let|Let Π be a group, ρ an automorphism of Π such that

$$\rho^n = \mathrm{id} \tag{1}$$

where $n \in \mathbb{N}^*$. We thus have an operation of the cyclic group $\mathbb{Z}/n\mathbb{Z}$ on Π, whence a semidirect product

$$\widetilde{\Pi} = \Pi.\,\mathbb{Z}/n\mathbb{Z}, \tag{1|2}$$

and an extension structure

(3)

$$1 \longrightarrow \Pi \longrightarrow \widetilde{\Pi} \underset{\curvearrowleft}{\longrightarrow} \mathbb{Z}/n\mathbb{Z} \longrightarrow 1.$$

We denote by $\dot\rho$ †the generator of $\mathbb{Z}/n\mathbb{Z}$, and we|to we identify ρ with† the image of the latter in $\widetilde{\Pi}$, by the canonical splitting †

$$\mathbb{Z}/n\mathbb{Z} \longrightarrow \widetilde{\Pi} \qquad 1 \bmod n \longmapsto \rho. \tag{4}$$

We thus have

$$\rho(g) = \rho g \rho^{-1} \qquad (g \in \Pi). \tag{5}$$

The splittings of the extension (3) correspond bijectively to the elements ρ' of $\widetilde{\Pi}$ over $\dot\rho$, such that

$$\rho'^n = 1. \tag{6}$$

To say that ρ' is over $\rho|\dot\rho$ means that we have

$$\rho' = \rho'_g \overset{\text{dfn}}{=} g\rho \qquad \text{with } g \in \Pi, \tag{7}$$

and in terms of such a $g \in \Pi$ equation (6) Is|is rewritten as follows

$$\boxed{g\rho(g)\rho^2(g)\cdots\rho^{n-1}(g) = 1} \tag{8}$$

Uncertainty retained from cancelled passages.
140-3-s121-D003 with
140-3-s121-D004 [illeg. 1]
140-3-s121-D005 [illeg. 1]

[scan 122]

†
†
†

The group Π operates by conjugation on the set of splittings of the extension (3), or what amounts to the same thing, on the set of ρ' above $\rho|\dot{\rho}$, satisfying (6), We will moreover have, for $\alpha \in \Pi$

$$\begin{aligned} \alpha\rho'_g\alpha^{-1} &= \dagger\alpha g\rho\alpha^{-1} = \alpha g\rho\alpha^{-1}\rho^{-1}\rho \\ &= (\alpha g\rho(\alpha)^{-1})\rho = \rho'_{\alpha g\rho(\alpha)^{-1}} \end{aligned} \tag{9}$$

i.e. the operation by conjugation of Π on the set of splittings of (3), is interpreted as the operation $\Theta = \Theta_\rho$ of Π on the set of solutions of (8) given, for $\alpha \in \Pi$, by

$$\Theta_\rho(\alpha) : g \longmapsto \alpha g\rho(\alpha)^{-1} \tag{10}$$

(We will say that $\alpha g\rho(\alpha)^{-1}$ is the ρ-conjugate of g by the element $\alpha \in \Pi$). The Π-conjugacy classes of|of the splittings of (3) thus correspond to the Π-ρ-conjugacy classes of solutions of (8).

Among these classes, the "trivial" class †of the splitting canonical for (3), i.e. the class of $g = 1$, is given parametrically as the set of

$$g = \alpha\rho(\alpha)^{-1} \qquad (\alpha \in \Pi). \tag{11}$$

Let g, g' be two congruent|conjugate solutions of

Uncertainty retained from cancelled passages.
140-3-s122-D003 $\alpha\rho\alpha^{-1} =$
140-3-s122-D004 [illeg. 2]
140-3-s122-D005 [illeg. 1]
140-3-s122-D007 $(\alpha g\alpha^{-1})(\alpha\rho\alpha^{-1}) = \alpha$
140-3-s122-D008 of the extension

[scan 123]
[p. 351]

the equation (8), so that we have

$$g' = \alpha g\rho(\alpha)^{-1} \qquad \text{i.e. } g' = \Theta(\alpha).g \tag{12}$$

for a suitable $\alpha \in \Pi$. When do we have uniqueness of the|of α – more generally, what can we say, for fixed g, g', of the set of $\alpha \in \Pi$ such that we have (12) (which we can call the ρ-transporter of g into g', in the group Π). It is obviously a right coset modulo|module the subgroup $\Pi(\rho, g)$ of Π which stabilizes g

$$\begin{aligned} \Pi(\rho, g) &= \{\beta \in \Pi \mid \Theta_\rho(\beta).g = g\} \\ &= \{\beta \in \Pi \mid \beta g\rho(\beta)^{-1} = g\} \\ &= \{\beta \in \Pi \mid \{\mathrm{int}(\beta)\}(g\rho) = g\rho\} \\ &= \Pi^{\rho'_g} = \{\beta \in \Pi \mid \mathrm{int}(\underbrace{g\rho}_{\rho'_g})\beta = \beta\} \\ &= \text{Centralizer of } \rho'_g \text{ in } \Pi. \end{aligned} \tag{13}$$

In other words, if α satisfies (12), the set of α' which satisfy the same relation is none other than the set $\alpha\Pi(\rho, g)$:

$$\alpha' g \rho(\alpha')^{-1} = \alpha g \rho(\alpha)^{-1} \Longleftrightarrow \alpha' \in \alpha\Pi(\rho, g) = \alpha\Pi^{\rho'_g} \quad \text{i.e. } \alpha^{-1}\alpha' \in \Pi^{\rho'_g} \tag{14}$$

Thus we have uniqueness of the α satisfying (12) † iff

$$\Pi^{\rho'_g} = 1, \tag{15}$$

a condition which depends only on the conjugacy class of the splitting † of (3) defined by g i.e. by ρ'_g.

Uncertainty retained from cancelled passages.

140-3-s123-D006 iff

140-3-s123-D007 under consideration

[scan 124]

In fact, not only the group Π, but the group $\widetilde{\Pi}$ itself operates by conjugation on the set of splittings of the extension (3), This is in particular so for ρ, and we have

$$\rho(\rho'_g) \overset{\text{dfn}}{=} \rho\rho'_g\rho^{-1} = \rho(g\rho)\rho^{-1} = (\rho g \rho^{-1}).\rho = g'_{\rho(g)}|\rho'_{\rho(g)}, \tag{16}$$

i.e. the operation of ρ on the set of splittings in question, is identified with the operation induced by|ρ the permutation ρ of Π on the set of solutions of (8) ($g \in \Pi$). Let us note moreover that g and $\rho(g)$ † are Π-conjugate, for

$$\rho(g) = \alpha g \rho(\alpha)^{-1} \qquad \text{with } \alpha = g^{-1}. \tag{17}$$

Let us note that †the operation of $\widetilde{\Pi}$†, on the set of splittings of (3), or again on the set of solutions of (8), is always|thus given by

$$\Theta_\alpha : g \longmapsto \alpha g \rho(\alpha)^{-1} \tag{10}$$

where now we assume $\alpha \in \widetilde{\Pi}$ (and not necessarily $\alpha \in \Pi$). The $\widetilde{\Pi}$-conjugacy classes are reduced|identical to the Π-conjugacy classes, (as it follows follows from (17)) † In particular, if $\alpha \in \widetilde{\Pi}$, the element

$$\alpha\rho(\alpha)^{-1} \qquad (\alpha \in \widetilde{\Pi}) \tag{18}$$

of $\widetilde{\Pi}$ is in Π, and it † satisfies (8), and

Uncertainty retained from cancelled passages.

140-3-s124-D003 [illeg. 1]

140-3-s124-D005 [illeg. 4]

140-3-s124-D006 ρ

140-3-s124-D007 has just seen it.

140-3-s124-D008 [illeg. 2]

[scan 125]

[p. 352]

is in the principal class, i.e. we will have †

$$\alpha\rho(\alpha)^{-1} = \alpha_0\rho(\alpha_0)^{-1} \qquad \text{for } \alpha_0 \in \Pi \text{ suitable}$$

In fact, we can write $\alpha = \alpha_0\rho^i$, with $\alpha_0 \in \Pi$, $i \in \mathbb{Z}$, whence $\rho(\alpha)^{-1} = \rho^{-i}\rho(\alpha_0)^{-1}$ whence (19). More generally, we will have, †

$$\alpha = \alpha_0\rho^i \qquad (\dagger\alpha_0 \in \Pi,\ i \in \mathbb{Z})$$

the relation †

$$\alpha g\rho(\alpha^{-1}) = \alpha_0 \underbrace{\rho^i g \rho^{-i}}_{\dagger} \rho(\alpha_0^{-1})|\rho(\alpha_0)^{-1}$$
$$= \dagger\alpha_0\rho^i(g)\rho(\alpha_0)^{-1}$$

with|for

$$\rho^i(g) = \beta g\rho(\beta)^{-1} \qquad \beta = (g\rho(g)\cdots\rho^{i-1}(g))^{-1}$$

thus

$$\begin{cases} \alpha g\rho(\alpha^{-1}) = (\alpha_0\beta)g\rho(\alpha_0\beta)^{-1} \\ \beta = (g\rho(g)\cdots\rho^{i-1}(g))^{-1} \end{cases} \tag{19}$$

(for $\alpha = \alpha_0\rho^i$, $i \in \mathbb{N}|\mathbb{N}^*$).

Of course, the representation (18) of a solution of equation (8), † when we no longer assume $\alpha \in \Pi$ but only $\alpha \in \widetilde{\Pi}$, is no longer one-to-one, since for $\alpha = \rho$ for example we find † $g = \rho\rho^{-1} = 1$.

†Let us suppose more generally that we have a group †

$$\mathcal{E}|\varepsilon \supset \widetilde{\Pi}$$

(as in the † case where $\Pi = \Pi_{0,3}$, $\widetilde{\Pi}$ inverse image of $\{1, \dot\rho, \dot\rho^2\} \subset \mathfrak{S}_3$ in $\mathfrak{S}_{0,3}$ extension of $\mathcal{T}_{0,3}$ by $\Pi_{0,3}$), in which Π is normal, so that †we have an extension

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Uncertainty retained from cancelled passages.

140-3-s125-D001 (19)
140-3-s125-D002 if α, α_0
140-3-s125-D003 $\alpha \in \widetilde{\Pi}$
140-3-s125-D004 (19)
140-3-s125-D005 $\rho^i(g)$
140-3-s125-D006 $\alpha_0\rho^i g\rho(\alpha_0\rho^i)^{-1}$
140-3-s125-D007 is not
140-3-s125-D008 the element 1
140-3-s125-D009 in
140-3-s125-D010 (20)
140-3-s125-D011 group
140-3-s125-D012 $\mathcal{E}|\varepsilon$

[scan 126]

†

$$1 \longrightarrow \Pi \longrightarrow \mathcal{E}|\varepsilon \longrightarrow G \longrightarrow 1 \tag{20}$$

m793

Marginal note. m793 N.B. the extension (3) is the inverse image of (20) under the inclusion $\mathbb{Z}/n\mathbb{Z} \hookrightarrow G$

with

$$\mathbb{Z}/n\mathbb{Z} \hookrightarrow G, \qquad 1 \bmod n \longmapsto \dot\rho \in G \tag{21}$$

(element of order n) $\widetilde{\Pi}$ being identified with the inverse image of $\mathbb{Z}/n\mathbb{Z}$ in $\mathcal{E}|\varepsilon$.

We cannot say in general, if ρ' is an element of $\mathcal{E}|\varepsilon$ above $\dot\rho$ (thus $\rho' \in \widetilde{\mathcal{E}|\varepsilon}$). (even if it satisfies $\rho'^n = 1$), that for $\forall\ \alpha \in \mathcal{E}|\varepsilon$, $\mathrm{int}(\alpha)(\rho')$ is still above $\dot\rho$: for this to be so, it is necessary and sufficient that $\dot\alpha \in G$ centralize $\dot\rho$. So let $\mathcal{E}|\varepsilon_\rho$

$$\mathcal{E}|\varepsilon_\rho = \{\alpha \in \mathcal{E}|\varepsilon \mid [\dot\alpha, \dot\rho] = 1\} \tag{22}$$

it is a subgroup of $\mathcal{E}|\varepsilon$ containing $\widetilde{\Pi}$, which †itself operates on the set of † ρ' above $\dot{\rho}$, in particular on † the set of [illeg. 1] which satisfying (8), by the same operation

$$(23) \qquad \alpha\rho'_g\alpha^{-1} = \rho'_{\alpha g\rho(\alpha)^{-1}}$$

so that we can also consider this operation as given by

$$(24) \qquad \begin{cases} \Theta_{\rho,\mathcal{E}|\varepsilon_\rho}(\alpha) = \Theta(\alpha) : g \longmapsto \alpha g\rho(\alpha)^{-1} \\ (\alpha \in \mathcal{E}|\varepsilon_\rho). \end{cases}$$

This time, an operation $\Theta(\alpha)$ may permute the Π-conjugacy classes of the † solutions, of (8) in Π. We|It † will have on $\mathcal{E}|\varepsilon_\rho$

Uncertainty retained from cancelled passages.

140-3-s126-D001 set such|such
140-3-s126-D002 for ∀
140-3-s126-D003 ρ'
140-3-s126-D004 [illeg. 1]
140-3-s126-D006 various
140-3-s126-D007 [illeg. 1]

[scan 127]
[p. 353]

an extension structure

$$(25) \qquad \begin{array}{ccccccccc} 1 & \longrightarrow & \Pi & \longrightarrow & \mathcal{E}|\varepsilon_\rho & \longrightarrow & G^\rho & \longrightarrow & 1 \\ & & & & & & \| & & \\ & & & & & & \mathrm{Centr}_G(\rho) = \{g \in G \mid \rho g g|\rho^{-1} = g\} & & \\ & & & & & & = \{g \in G \mid [\rho, g] = 1\} & & \end{array}$$

and $\mathcal{E}|\varepsilon_\rho$ operates on the set of Π-conjugacy classes of splittings of (3), (or the Π-ρ-conjugacy classes of solutions of (8)) via an operation of G^ρ on it.

Let $\dagger\alpha \in \mathcal{E}|\varepsilon$, let us consider the element

$$\dagger\alpha\rho(\dagger\alpha)^{-1} = \alpha(\rho\alpha\rho^{-1})^{-1} = \alpha\rho\alpha^{-1}\rho^{-1} = [\alpha, \rho]$$

It is in Π iff we have †

$$[\dot{\alpha}, \dot{\rho}] = 1 \qquad \text{i.e.} \quad \dot{\alpha} \in G^\rho \qquad \text{i.e.} \quad \alpha \in \mathcal{E}|\varepsilon_\rho.$$

For a solution g of equation (8), it may happen that we cannot put it|it in the standard form $\alpha\rho(\alpha)^{-1}$ with $\alpha \in \Pi$, but that we succeed with $\alpha \in \mathcal{E}|\varepsilon_\rho$. For such an α to exist, it is necessary and sufficient that the $\Pi\rho$-conjugacy class $[g]_\rho$ of g (among the solutions of (8)) be in the orbit of the trivial conjugacy class $[1]_\rho$, for the action under consideration of G^ρ on this set of conjugacy classes; then the $\alpha \in \mathcal{E}|\varepsilon_\rho$ which satisfy

$$(26) \qquad g = \alpha\rho(\alpha)^{-1}$$

Uncertainty retained from cancelled passages.

140-3-s127-D001 g
140-3-s127-D002 g
140-3-s127-D003 g
140-3-s127-D004 that

[scan 128]

† satisfy the relation

$$(27) \qquad \dot\alpha[1]_\rho = [g]_\rho,$$

and for $\forall\ \dot\alpha \in G^\rho$ satisfying this relation (27), there exists at least one representative $\alpha \in \mathcal{E}|\varepsilon_\rho$ satisfying (6) (which is pure tautology).

Under what condition do we have uniqueness of the|of the $\alpha \in \mathcal{E}|\varepsilon_\rho$ satisfying (26)? The set of these α is a right coset under the subgroup of $\mathcal{E}|\varepsilon_\rho$ formed of the α such that †

$$\alpha\rho(\alpha)^{-1} = 1 \qquad \text{i.e.} \quad \alpha = \rho(\alpha) \qquad \text{i.e.} \quad [\alpha, \rho] = 1,$$

this group is none other than †

$$(28) \qquad \mathcal{E}|\varepsilon^\rho = \mathrm{Centr}_{\mathcal{E}|\varepsilon}(\rho) \subset \mathcal{E}|\varepsilon_\rho.$$

which group contains ρ hence the sub-group $\simeq \mathbb{Z}/n\mathbb{Z}$ of $\mathcal{E}|\varepsilon$ generated by ρ – so it is never trivial – i.e. we never have uniqueness (except in the trivial case where † $n = 1$ i.e. $\rho = 1$, † but then $\mathcal{E}|\varepsilon^\rho = \mathcal{E}|\varepsilon$ and we would need $\mathcal{E}|\varepsilon = 1$ …!)

But if we assume $\Pi^\rho = \{1\}$ i.e.

$$(29) \qquad \mathcal{E}|\varepsilon^\rho \cap \Pi = \{1\}$$

†

$$\mathcal{E}|\varepsilon^\rho_\dagger \underset{\text{injective}}{\overset{\text{can}}{\hookrightarrow}} G^\rho \subset G,$$

then the $\alpha \in \mathcal{E}|\varepsilon^\rho$ solutions of (26) are in one-to-one correspon-

---

Uncertainty retained from cancelled passages.

140-3-s128-D003 are some [illeg. 1]
140-3-s128-D004 [illeg. 1]
140-3-s128-D005 $\mathcal{E}|\varepsilon$
140-3-s128-D006 ρ
140-3-s128-D007 [illeg. 1]
140-3-s128-D008 then
140-3-s128-D009 ρ

[scan 129]
[p. 354]

dence with their images $\dot\alpha$ in G, i.e. with the transporter in G from $[1]_\rho$ into|to $[g]_\rho$. If we choose an element of this transporter, we have chosen by that very fact an α, solution of (26).

In the case of $\quad \Pi = \Pi_{0,3} = \Pi_1(U_{0,3}), \qquad \mathcal{E}|\varepsilon = \mathfrak{S}_{0,3} = \Pi_1(U_{0,3}, \mathcal{T}_{0,3}), \quad \dagger$
$$G = \mathcal{T}_{0,3} \simeq \mathfrak{S}_3 \times \mathbb{Z}/2\mathbb{Z}$$

with $\rho, \dot\rho$ as usual, the set of conjugacy classes of the †splittings of (3) i.e. of the solutions of (8), is identified with $U^{\dot\rho}_{0,3}$, which has two elements, the two "poles" $-\bar j, -j$ – one of which, chosen as base point, corresponds to the trivial class.

† Here

$$G^\rho = \mathcal{T}|\mathcal{G}^\rho_{0,3} \simeq \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$$

is the group generated by ρ and τ, and ρ operates trivially|trivially on the set of the two classes, while τ permutes them (i.e. $\mathbb{Z}/2|n\mathbb{Z}$ generated by τ operates on it in a simply transitive way). Thus if we set

$$(30) \qquad \mathcal{E}|\varepsilon' = \widetilde\Pi|\Pi_{0,3} = \Pi \cdot \{1, \tau_\infty\}$$
$$= \text{inverse image of } \{1, \tau_\infty\} \subset G$$
$$\text{in } \mathcal{E}|\varepsilon = \mathfrak{S}_{0,3}$$

m794

Marginal note. m794 († unoriented weighted triangulated cartographic group …

we see that

$$\mathcal{E}|\varepsilon'^{\rho} = \{1\}, \tag{31}$$

and the set of solutions in $\underline{\Pi_{0,3}}$ of the equation

$$g\rho(g)\rho^2(g) = 1 \tag{32}$$

Uncertainty retained from cancelled passages.
140-3-s129-D001 $G = \mathcal{T}_0$
140-3-s129-D002 solutions
140-3-s129-D003 Here the [illeg. 1] $\mathbb{Z}/2\mathbb{Z}$
140-3-s129-D004 [illeg. 1]

[scan 130]

is in 1-1 correspondence with the set of elements of $\widetilde{\Pi}_{0,3} = \mathcal{E}|\varepsilon'$ (the unoriented weighted triangulated cartographic group, having as generators the elements $\tau_0, \tau_1, \tau_\infty$ related by the relations $\tau_0^2 = \tau_1^2 = \tau_\infty^2 = 1$ and no more, with $l_0 = \tau_\infty\tau_1$, $l_1 = \tau_0\tau_\infty$, $l_\infty = \tau_1\tau_0$ …) It should still be likewise when we place ourselves in the groups $\Pi^{\wedge}_{0,3}$ and $\widetilde{\Pi}^{\wedge}_{0,3} = \widetilde{\Pi}_{0,3} \cdot \{1, \tau_\infty\}$, if we admit (something to be checked later)

$$(\widetilde{\Pi}^{\wedge}_{0,3})^{\rho} = \{1\} \tag{33}$$

and that the elements of finite order of $\widehat{\mathfrak{S}}_{0,3}$ are conjugates of elements of finite order of $\mathfrak{S}_{0,3}$ …

Let us note that here we have

$$\tau_\infty\rho(\tau_\infty)^{-1} = \tau_\infty\rho(\tau_\infty) = l_1^{-1}, \qquad \dagger \tag{34}$$

in virtue of the formulas (23) of § 42

$$\rho_\infty \overset{\text{dfn|definition}}{=} \tau_\infty\rho\tau_\infty^{-1} = l_1^{-1}\rho = \rho l_0^{-1}.$$

The situation is even simpler for the element|element σ_∞ of $\mathfrak{S}_{0,3}$, operating on Π by an automorphism of order $n = 2$, for here every solution of the corresponding equation

$$\dagger \quad \dagger h\,\sigma_\infty(\dagger h) = 1 \qquad (h \in \Pi) \tag{35}$$

can be put in the standard form

$$h = \beta\sigma_\infty(\beta)^{-1} \qquad (\beta \in \Pi) \tag{36}$$

and this in a unique way. This follows indeed from the general theory …, taking into account that

$$U_{0,3}^{\dot{\sigma}_\infty} = \{\tfrac{1}{2}\} \tag{37}$$

Uncertainty retained from cancelled passages.
140-3-s130-D001 in
140-3-s130-D002 (36)
140-3-s130-D003 g
140-3-s130-D004 g

[scan 131]
[p. 355]

the set of fixed points of $U_{0,3}$ under $\dot\sigma_\infty$ is reduced to a single element, which also implies that †

$$\Pi_{0,3}^{\sigma_\infty} = \{1\}.$$

We shall also assume that these facts extend to the elements of $\widehat\Pi_{0,3}$ — which will be the case if (in addition to what has been assumed about it with regard to ρ) we assume that we still have

$$\widehat\Pi_{0,3}. \tag{38}$$

Note that here we have

$$G^{\sigma_\infty} = \mathcal{T}_{0,3}^{\sigma_\infty} \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}, \tag{39}$$

subgroup generated by $\dot\sigma_\infty, \dot\tau|\tau$, whose nonzero elements are $\dot\sigma_\infty, \tau, \dot\tau'_\infty$ — which lift to $\sigma_\infty, \tau_\infty, \tau'_\infty$; and we have

$$\mathcal{E}^{\sigma_\infty} = \mathcal{G}_{0,3}^{\sigma_\infty} \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}, \tag{40}$$

group generated by $\sigma_\infty, \tau_\infty$ (the other nontrivial element being τ'_∞). If then we write a solution $h \in \pi$ (of (35) in the form (36)), with $\beta \in \mathcal{G}_{0,3}$ only (not only $\beta \in \pi$), then the indeterminacy is in the group (40) of order (4) ; More precisely, if β_0 is the solution in π, then the other solutions are of the form

$$\beta = \beta_0 \sigma_\infty^{\varepsilon} \tau_\infty^{\varepsilon'} \qquad \varepsilon, \varepsilon' \in \mathbb{Z}/2\mathbb{Z} \tag{41}$$

130

Uncertainty retained from cancelled passages.
140-3-s131-D001 [illeg. 1]

[scan 132]

(with $\varepsilon, \varepsilon'$ uniquely determined). In order to have $\beta \in \mathcal{G}_{0,3}^+$, it is necessary and sufficient that $\varepsilon' = 0$! 131

[scan 133]

m797

[p. 356]

Marginal note. m797 These notations will be revised once again in § 46

44 Return to the notation. Setting up equations in $\mathrm{Sl}(2, \widehat{\mathbb{Z}})$.

Let u be an automorphism of $\widehat\Pi_{0,3}$ commuting up to inner automorphisms : $\mathcal{G}_3$† — or what amounts to the same, an automorphism of $\widehat{\mathcal{G}}_{0,3}^+$ which induces the identity in $\mathcal{T}_{0,3}^+ = \mathcal{G}_3$. Then u is known when we know its actions on the two generators σ_∞, ρ of $\widehat{\mathcal{G}}_{0,3}^+$, which will resp. be elements of $\widehat{\mathcal{G}}_{0,3}^+$ over $\dot\sigma_\infty$ and $\dot\rho$,

namely

$$\boxed{u(\sigma_\infty) = h\sigma_\infty, \qquad u(\rho) = g\rho} \qquad (h, g \in \widehat\Pi_{0,3}) \tag{1}$$

where h, g are subject to the conditions

$$\boxed{h\sigma_\infty(h) = 1 \qquad g\rho(g)\rho^2(g) = 1}, \tag{2}$$

expressing respectively the relations

$$\boxed{(h\sigma_\infty)^2 = (g\rho)^3 = 1}. \tag{3}$$

(There is in addition the preliminary condition to check — namely that (3) are defining relations for $\widehat{\mathcal{G}}_{0,3}^+$ …).

†For the moment, there is no need to impose any relation between g and h — except the relations (2) on each of these elements — since $\widehat{\mathcal{G}}_{0,3}^+$ is the group with defining relations

$$\sigma_\infty^2 = \rho^3 = 1.$$

132

Uncertainty retained from cancelled passages.
140-3-s133-D001 [illeg. 1]
140-3-s133-D002 [illeg. 1]

[scan 134]

The elements g and h, thus taken as automorphisms of $\widehat{\Pi}_{0,3}$, express the commutation relations of u with ρ and σ_∞ respectively :

$$[u,\rho] = \underbrace{u\rho u^{-1}}_{u(\rho)}\rho^{-1} = \mathrm{int}(g), \qquad [u,\sigma_\infty] = u\sigma_\infty u^{-1}\sigma_\infty = \mathrm{int}(h) \tag{4}$$

— it is a remarkable thing that knowledge of the elements g, h of π satisfying (4), already implies|imply knowledge of u (this|which to the fact that σ_∞, ρ generate $\mathcal{G}^+_{0,3}$ ...).

Let us now write the compatibility of u with the loop structure, †We had it †expressed|expressed previously by a relation of the form

$$u(\varepsilon_0) = \mathrm{int}(g_0)(\varepsilon_0^p) \quad \text{or} \quad u(\ell_0) = \mathrm{int}(g_0)\ell_0^p,$$

for

$$p \in \widehat{\mathbb{Z}}^*, \qquad g_0 \in \widehat{\Pi}_{0,3}$$

suitably chosen. The notation g_0 is not all that happy, since g and g_0 play quite dissimilar roles. The subscript 0 in g_0 was intended to allow us to write, more generally

$$\dagger \quad u(\ell_i) = \mathrm{int}(g_i)\ell_i^p \qquad i \in \{0, 1, \infty\}.$$

133

Uncertainty retained from cancelled passages.
140-3-s134-D003 [illeg. 3]
140-3-s134-D004 formulated
140-3-s134-D005 g_i

[scan 135]
[p. 357]

I prefer to use the letter f_0 — with the option of dropping the subscript if no confusion is possible. The letter p for a multiplier is not a choice that is very happy, for later on † we shall surely need this letter to denote a characteristic — I prefer the letter μ (Greek initial of "multiplier"), and the loop relation would therefore take the form

$$\dagger \qquad u(\varepsilon_0) = \mathrm{int}(f_0)\varepsilon_0^\mu. \qquad \dagger$$

In fact, I prefer to replace f_0 in it by f_0^{-1}, to obtain later on formulas in which the symmetry of the roles played by ρ and by σ_∞ appears more clearly. — in particular I would like the formula defining h_0 in terms of g_0 (§ 43, formula (65,)

$$h_0 = g_0^{-1}\sigma_\infty(g_0)$$

to become instead

$$h_0 = g_0\sigma_\infty(g_0)^{-1},$$

in harmony with

$$\xi = \lambda\rho(\lambda)^{-1} \quad \ldots$$

Thus the "loop|loop relation" takes the form

$$u(\varepsilon_0) = \mathrm{int}(f_0^{-1})\varepsilon_0^\mu \qquad f_0 \in \widehat{\Pi}_{0,3},\ \mu \in \widehat{\mathbb{Z}}^*, \tag{5}$$

†

$$u(\ell_0) = \mathrm{int}(f_0^{-1})\ell_0^\mu, \tag{6}$$

from which we immediately deduce

134

Uncertainty retained from cancelled passages.
140-3-s135-D001 p
140-3-s135-D002 (5)
140-3-s135-D003 or
140-3-s135-D004 rather we

[scan 136]

$$\begin{cases} u(\varepsilon_1) = \mathrm{int}(f_1^{-1})(\varepsilon_1^{\mu}), & u(\ell_1) = \mathrm{int}(f_1^{-1})(\ell_1^{\mu}), \\ u(\varepsilon_\infty) = \mathrm{int}(f_\infty^{-1})(\varepsilon_\infty^{\mu}), & u(\ell_\infty) = \mathrm{int}(f_\infty^{-1})(\ell_\infty^{\mu}), \end{cases} \tag{7}$$

where|or

$$f_1 = \rho(f_0)g^{-1}, \qquad f_\infty = \rho(f_1)g^{-1} = \rho^2(f_0)\rho(g)^{-1}g^{-1}. \tag{8}$$

In what follows, we shall simply write f instead of f_0, and we are going to transform equation (5) † into

$$\mathrm{int}(f)u(\varepsilon_0) = \varepsilon_0^{\mu}, \tag{8bis}$$

by making the left-hand side explicit

$$\begin{aligned} \mathrm{int}(f)u(\varepsilon_0) = f\,u(\sigma_\infty\rho)f^{-1} = f\underbrace{u(\sigma_\infty)}_{h\sigma_\infty}\underbrace{u(\rho)}_{g\rho}\,f^{-1} = fh\sigma_\infty g\rho f^{-1} \\ = fh\sigma_\infty g\sigma_\infty^{-1}\underbrace{\sigma_\infty\rho}_{\varepsilon_0}\,f^{-1} = fh\sigma_\infty(g)\,\underbrace{\varepsilon_0 f^{-1}\varepsilon_0^{-1}}\,\varepsilon_0 \end{aligned}$$

i.e.

$$\mathrm{int}(f)u(\varepsilon_0) = [fh\sigma_\infty(g)(\sigma_\infty\rho)(f)^{-1}]\varepsilon_0 = [f, h\sigma_\infty(g\rho(f)^{-1}]\varepsilon_0 \tag{9}$$

where the four|three factors $f, h, \sigma_\infty(g\rho(f)^{-1}$ in [ ] are $\in \widehat{\Pi}_{0,3}$. Thus (5)|(9) or (8), takes the form

$$fh\sigma_\infty(g\rho(f)^{-1}) = \varepsilon_0^{\mu-1}\ \dagger$$

which gives, setting

$$\mu - 1 = 2\nu \qquad (\nu \in \widehat{\mathbb{Z}}) \tag{10}$$

(so we write ν for what was denoted by γ previously) we find the relation

$$fh\sigma_\infty(g\rho(f)^{-1}) = \ell_0^{\nu}$$

or again, transforming by σ_∞

$$\sigma_\infty(f)\sigma_\infty(h)g\rho(f)^{-1} = \ell_1^{\nu}$$

; that is 135

Uncertainty retained from cancelled passages.
140-3-s136-D001 [illeg. 1]
140-3-s136-D002 [*illeg.* 1]

[scan 137]
[p. 358]

$$\boxed{\sigma_\infty(h).g = \sigma_\infty(f)^{-1}\ell_1^{\nu}\rho(f)}. \tag{11}$$

This relation, which relates h and g via f, can rightly be called the "loop relation". We see, from this relation, that † not only does knowledge of u i.e. of the|of h and g determine f and μ i.e. f and ν, but also conversely, once f and ν are known, g and h determine each other via (11).
m802

Marginal note. m802 f determined modulo L_0 only $f \mapsto \ell_0^n f$ $(n \in \widehat{\mathbb{Z}})$.

Note that for u corresponding to : †, g, h, f, if $\gamma \in \widehat{\Pi}_{0,3}$, the automorphism $u' = \mathrm{int}(\gamma) \circ u$ corresponds to

$$g' = \gamma g \rho(\gamma^{-1}), \quad h' = \gamma h \sigma_\infty(\gamma)^{-1}, \quad f' = f\gamma^{-1}. \qquad (12)$$

m803

Marginal note. m803 This holds more generally if $\gamma \in \widetilde{\mathcal{G}}_{0,3} | \mathcal{G}_{0,3}$, and even for $\gamma \in \widehat{\widetilde{\mathcal{G}}}_{0,3} | \widehat{\mathcal{G}}_{0,3}$.

The equations (2) "in h" and "in g" are solved parametrically in a unique way by

$$g = \alpha\rho(\alpha)^{-1}, \quad h = \beta\sigma_\infty(\beta)^{-1} \qquad (13)$$

where

$$\begin{cases} \alpha \in \widehat{\widetilde{\Pi}}_{0,3} = \widehat{\Pi}_{0,3}.\{1, \tau_\infty\} \\ \beta \in \widehat{\Pi}_{0,3} \end{cases} \qquad (14)$$

to be precise, giving g, h satisfying (2) is equivalent resp. to giving α, β in $\widehat{\widetilde{\Pi}}_{0,3}$ and $\widehat{\Pi}_{0,3}$, via the formulas (13). 136

Uncertainty retained from cancelled passages.
140-3-s137-D001 [illeg. 1]
140-3-s137-D002 $f|\xi$

[scan 138]

The elements g and h being thus made explicit in terms of α and β, the loop relation (11) can be considered as a relation between α, β, f, ν

$$\sigma_\infty(\beta\sigma_\infty(\beta)^{-1})(\alpha\rho(\alpha)^{-1}) = \sigma_\infty(f)^{-1}\ell_1^\nu \rho(f), \qquad (15)$$

where † ν is uniquely determined by α, β, and f is determined in $L_0 \backslash \widehat{\Pi}_{0,3}$. (We shall note that the right-hand side of (15) does not change, on changing f to $\ell_0^n f$, with $n \in \widehat{\mathbb{Z}}$).

The main asymmetry, it seems to me, between σ_∞ and ρ in this theory, is that † we always have $\beta \in \widehat{\Pi}_{0,3}$, whereas α is not necessarily in $\widehat{\Pi}_{0,3}$ – it is, as we have seen, iff $\nu \equiv 0$ (3) i.e. $\mu \equiv 1$ (3).

† The automorphism

$$u' = \mathrm{int}(\beta^{-1})u$$

will therefore be congruent to u modulo inner automorphisms, and will correspond to

$$\begin{cases} g' = \beta^{-1} g \rho(\beta) = (\beta^{-1}\alpha)\rho(\beta^{-1}\alpha)^{-1} \\ h' = \beta^{-1} h \sigma_\infty(\beta) = 1 \\ f' = f\beta \end{cases} \qquad (16)$$

– thus u is congruent modulo inner automorphisms 137

Uncertainty retained from cancelled passages.
140-3-s138-D001 are
140-3-s138-D002 [illeg. 1]
140-3-s138-D003 [illeg. 1] [illeg. 1]

[scan 139]
[p. 359]

to a unique u' such that $u'(\sigma_\infty) = \sigma_\infty$ i.e. a u' which commutes with σ_∞. But in the symmetric notation|the symmetric notations we shall take

$$u'' = \mathrm{int}(\alpha^{-1})u$$

corresponding to

$$\begin{cases} g'' = \alpha^{-1} g \rho(\alpha) = 1 \\ h'' = \alpha^{-1} h \sigma_\infty(\alpha) = (\alpha^{-1}\beta)\sigma_\infty(\alpha^{-1}\beta)^{-1} \\ f'' = f\alpha \end{cases} \qquad (17)$$

but u'' will be only $\widehat{\widetilde{\Pi}}_{0,3}$-conjugate † of $u''|u$ – it will be $\widehat{\Pi}_{0,3}$-conjugate to an automorphism which commutes with ρ iff g is $\Pi_{0,3}$-ρ-conjugate to 1 i.e. iff $\alpha \in \widehat{\Pi}_{0,3}$ i.e. iff $\mu \equiv 1$ (3) or again $\nu \equiv 0$ (3).

The calculations of the preceding §§ concerned the case where $h = 1$, i.e. u commutes with σ_∞ – so [illeg. 1] there remain the quantities g and f, plus the multiplier μ † – once μ is known i.e. ν known, g and f determine one another by (11) (f remaining an element of $L_0\backslash\widehat{\Pi}_{0,3}$).

m806

Marginal note. m806 We shall see that μ is already determined by f, at least modulo sign, cf. below.

For fixed f, the equation in g can be interpreted as an equation in f, or better, introducing

$$\varphi = f\sigma_\infty(f)^{-1} \qquad (12)$$

m807

Marginal note. m807 This notation is useful|valid only if $\beta = 1$ [illeg. 2]

m808

Marginal note. m808 it is essentially the element denoted h_0 previously

138

Uncertainty retained from cancelled passages.
140-3-s139-D001 by
140-3-s139-D005 or for

[scan 140]

an "equation in φ" –

$$(\ell_0^\nu\varphi)\rho(\ell_0^\nu\varphi)\rho^2(\ell_0^\nu\varphi) = 1, \qquad (13)$$

or again, putting

$$\boxed{\xi = \ell_0^\nu\varphi \quad \text{i.e. } \varphi = \ell_0^{-\nu}\xi} \qquad (14)$$

we can interpret it as "equation in ξ"

$$\xi\rho(\xi)\rho^2(\xi) = 1. \qquad (15)$$

Thus, the loop automorphisms of $\widehat{\Pi}_{0,3}$ which commute with σ_∞, with multiplier $\mu = 2\nu + 1$, correspond to the pairs of solutions (ξ, φ) of the equations

$$\boxed{\varphi\sigma_\infty(\varphi) = 1 \qquad \xi\rho(\xi)\rho^2(\xi) = 1} \qquad (16)$$

m810

Marginal note. m810 We shall find|find that both ξ and φ already determine μ, hence ν.

satisfying in addition the compatibility condition (14) (particularly simple † if $\mu = 1$ i.e. $\nu = 0$, since then $\varphi = \xi$!). We recover g (hence u) as a function † of (φ, ξ) by the loop formula (11), which here takes the form

$$g = \sigma_\infty(f)^{-1}\ell_1^\nu\rho(f) = (\ell_0^\nu f)^{-1}(\underbrace{\ell_0^\nu\varphi}_{\xi})\rho(\ell_0^\nu f) \qquad (17)$$

$f \in \widehat{\Pi}_{0,3}$ being uniquely determined in terms of φ by formula (12).

Symmetrically, we are going to look at the loop automorphisms u of $\Pi_{0,3}$ which com- 139

Uncertainty retained from cancelled passages.
140-3-s140-D001 (17)
140-3-s140-D004 [illeg. 3]

[scan 141]
[p. 360]

mute with ρ, i.e. those for which we have $g = 1$, hence described in terms of h, f,

$$u(\sigma_\infty) = h\sigma_\infty, \quad u(\rho) = 1_\dagger \qquad (18)$$

$$u(\varepsilon_0) = \mathrm{int}(f^{-1})(\varepsilon_0^\mu) \qquad (19)$$

this last relation taking the form (11)

$$\sigma_\infty(h) = \sigma_\infty(f)^{-1}\ell_1^\nu\rho(f), \qquad (20)$$

which can also be written, applying σ_∞

$$h = f^{-1}.\ell_0^\nu.(\sigma_\infty\rho)(f) = | = \qquad (21)$$

"The equation in h" (2) the

$$h\sigma_\infty(h) = 1 \qquad (*)$$

is thus written as an "equation in f", with

$$h\sigma_\infty(h) = f^{-1}\ell_0^\nu \underbrace{(\sigma_\infty\rho)(f)\sigma_\infty(f)^{-1}}_{\sigma_\infty(\underbrace{\rho(f)f^{-1}}_{\psi^{-1}})}\ell_1^\nu\rho(f)\dagger$$

so this is written, in terms of $\psi = f\rho(f)^{-1}$

$$\underbrace{\ell_0^\nu\sigma_\infty(\psi^{-1})}\,\ell_1^\nu\,\underbrace{\rho(f)f^{-1}}_{\psi^{-1}} = 1$$

i.e. (passing to the inverse)

$$\psi\ell_1^{-\nu}\underbrace{\sigma_\infty(\psi)\ell_0^{-\nu}}_{\sigma_\infty(\psi\ell_1^{-\nu})} = 1 \quad \text{i.e.} \quad \underbrace{\ell_0^{-\nu}\psi}_{\eta}\,\underbrace{\ell_1^{-\nu}\sigma_\infty(\psi)}_{\sigma_\infty(\underbrace{\ell_0^{-\nu}\psi}_{\eta})} = 1.$$

Thus, putting

$$\psi \overset{\text{dfn}}{=} f\rho(f)^{-1} \qquad (22)$$

$$\boxed{\eta = \ell_0^{-\nu}\psi \quad \text{i.e. } \psi = \ell_0^\nu\eta} \qquad (23)$$

, we find that the loop automorphisms u of $\widehat{\Pi}_{0,3}$ which commute with ρ correspond to the triples (η, ψ, ν), $\eta, \psi \in \widehat{\Pi}_{0,3}$, $\nu \in \widehat{\mathbb{Z}}$ (with $2\nu + 1 \in \widehat{\mathbb{Z}}^*$), satisfying 140

Uncertainty retained from cancelled passages.
140-3-s141-D001 ρ
140-3-s141-D003 [*illeg.* 1]

[scan 142]

$$\boxed{\psi\rho(\psi)\rho^2(\psi)=1, \qquad \eta\sigma_\infty(\eta)=1}, \qquad (24)$$

m813

Marginal note. m813 We shall find [illeg. 1] again, by both ψ and η

and in addition the compatibility relation (23) – particularly simple if $\mu=1$ i.e. if $\nu=0$, for then it means $\psi=\eta$! But in addition to these relations, we must assume that the solution ψ of equation (24) "in ψ" belongs|belong to the principal family, i.e. that, when we write ψ in parametric form

$$\psi = f\rho(f)^{-1} \qquad (22)$$

with $f\in\widehat{\widetilde{\Pi}}_{0,3}=\widehat{\Pi}_{0,3}.\{1,\tau_\infty\}$, we indeed have $f\in\widehat{\Pi}_{0,3}$ (hence that the image of f in $\widehat{\widetilde{\Pi}}_{0,3}/\widehat{\Pi}_{0,3}\simeq\{1,\tau_\infty\}\simeq\mathbb{Z}/2$ is zero…).

Having thus determined f as a function of ψ, determine|determines h by (21), whence u †.

Remarks 1) If this additional condition on ψ is not satisfied, the † formula (21) still makes sense and defines $h\in\widehat{\Pi}_{0,3}$, satisfying "the equation in h" $h\sigma_\infty(h)=1$ hence defining an automorphism u of $\widehat{\Pi}_{0,3}$, commuting with ρ.

We shall then have

$$u(\ell_0)=\mathrm{int}(f^{-1})(\ell_0^\mu)$$

or again, putting

$$f=\tau_\infty f_0 \qquad \text{with } f_0\in\widehat{\Pi}_{0,3}$$
$$u(\ell_0)=\mathrm{int}(f_0^{-1})(\tau_\infty(\ell_0^\mu))=\mathrm{int}(f_0^{-1})(\ell_0^{-\mu})$$

141

Uncertainty retained from cancelled passages.
140-3-s142-D001 [illeg. 3]
140-3-s142-D002 determination

[scan 143]

i.e. u is a loop automorphism, with multiplier $-\mu=-(2\nu+1)$. [p. 361]

2) Let us note that the indeterminacy of f (for fixed u, i.e. fixed g,h) lies in

$$f\longmapsto f'=\ell_0^n f\,\dagger \qquad (n\in\widehat{\mathbb{Z}}) \qquad (25)$$

which gives

$$\begin{cases}\varphi\longmapsto\varphi'=\ell_0^n\varphi\ell_1^{-n}, & \psi\longmapsto\psi'=\ell_0^n\psi\ell_1^{-n},\\ \xi\longmapsto\xi'=\ell_0^n\xi\ell_1^{-n}, & \eta\longmapsto\eta'=\ell_0^n\eta\ell_1^{-n},\end{cases} \qquad (26)$$

m815

Marginal note. m815 N.B. We similarly have $\alpha_0\longmapsto\alpha_0'=\ell_0^n\alpha_0$ $\beta_0\longmapsto\beta_0'=\ell_0^n\beta_0$.

(N.B. we have $\ell_1=\sigma_\infty(\ell_0)=\rho(\ell_0)$, so ξ' and ψ' are the ρ-transforms of ξ,ψ by $\ell_0^{-n}|\ell_0^n$, and φ',η' the σ_∞-transforms of φ,η by $\ell_0^{-n}|\ell_0^n$ — in particular the equations (16) (in φ and in ξ) and (24) (in ψ and in η) are indeed invariant under these transformations, as are moreover the compatibility relations (14) resp. (23).

3) We have the following formulas (rewriting of (11))

$$\sigma_\infty(h)g = (\ell_0^\nu f)^{-1}\xi\rho(\ell_0^\nu f) = \sigma_\infty(f^{-1}\ell_0^\nu)\eta(f^{-1}\ell_0^\nu)^{-1} \tag{27}$$

— expressing $\sigma_\infty(h)g$ as the ρ-conjugate of ξ by $(\ell_0^\nu f)^{-1}$, and symmetrically as the σ_∞-conjugate of η by $\sigma_\infty(f^{-1}\ell_0^\nu)$. If $h=1$, it is the first expression that is interesting (implying the equation in ξ), for $g=1$, it is the second, which we can also rewrite (applying σ_∞ to the 2 sides)

$$h = (f^{-1}\ell_0^\nu)(\eta^{-1})\sigma_\infty(f^{-1}\ell_0^\nu)^{-1} \tag{28}$$

142

Uncertainty retained from cancelled passages.
140-3-s143-D001 f

[scan 144]

expressing h as σ_∞-transform of η^{-1} by $f^{-1}\ell_0^\nu$. If we want to express it as σ_∞-transform of η (instead of $\sigma_\infty(\eta) = \eta^{-1}$), we write

†

$$\eta^{-1} = \sigma_\infty(\eta) = \eta^{-1}.\eta.\sigma_\infty(\eta)$$

whence

$$h = (f^{-1}\ell_0^\nu\eta^{-1})\eta\sigma_\infty(f^{-1}\ell_0^\nu\eta^{-1})^{-1}. \tag{28bis}$$

These formulas allow us to relate, in the case $h=1$ the solutions of the † equations

$$g = \alpha\rho(\alpha)^{-1} \qquad \text{and} \qquad \xi = \alpha_0\rho(\alpha_0)^{-1} \tag{29}$$

by

$$\alpha = (\ell_0^\nu f)^{-1}\alpha_0 = f^{-1}\ell_0^{-\nu}\alpha_0, \tag{30}$$

and in the case where $g=1$, the solutions of the two equations

$$h = \beta\sigma_\infty(\beta)^{-1}, \qquad \eta = \beta_0\sigma_\infty(\beta_0)^{-1} \tag{31}$$

by

$$\beta = (f^{-1}\ell_0^\nu\eta^{-1})\beta_0. \tag{32}$$

The asymmetry between these two formulas ((32) being a little more complicated than (30) by the extra factor η^{-1} in the coefficient of β_0) comes from the fact that f was defined via the loop formula|formulas (6), first form, where ℓ_0 is defined as the square of $\sigma_\infty\rho = \varepsilon_0$, and not of $\rho\sigma_\infty = \varepsilon_1$ 143

Uncertainty retained from cancelled passages.
140-3-s144-D001 $\eta^{-1} = \eta^{-1}\eta\sigma_\infty(\eta^{-1})$
140-3-s144-D002 two

[scan 145]

If we now identify $\widehat{\mathcal{G}}_{0,3}$ with $\mathrm{Gl}(2,\mathbb{Z})'^\wedge$, and look at the images of the elements g, h, f and so on in the quotient [p. 362]

$$\mathrm{Gl}(2,\widehat{\mathbb{Z}})^{\pm 1\prime} = \mathrm{Gl}(2,\widehat{\mathbb{Z}})^{\pm 1}/\pm 1$$

— which we denote by the same letters, by abuse of language — we are led to calculate with elements of $\mathrm{Gl}(2,\widehat{\mathbb{Z}})$, rather than of $\mathrm{Gl}(2,\widehat{\mathbb{Z}})'$. We will choose α, β in $\widehat{\Pi}_{0,3}$ and $\widetilde{\widehat{\Pi}}_{0,3}$ in accordance with (13), hence

$$\alpha, \beta \in \mathrm{Gl}(2,\widehat{\mathbb{Z}})^{\pm 1} \quad \underline{\text{modulo sign}}, \tag{33}$$

but the corresponding elements

$$g = \alpha\rho(\alpha)^{-1}, \quad h = \beta\sigma_\infty(\beta)^{-1} \quad \in \mathrm{Sl}(2,\widehat{\mathbb{Z}})^\dagger \tag{34}$$

are determined in $\mathrm{Gl}(2,\widehat{\mathbb{Z}})^\dagger$ without sign ambiguity. Likewise

$$\overset{(\text{case } h=1)}{\alpha_0} \text{ and}|, \quad \overset{(\text{case } g=1)}{\beta_0} \in \mathrm{Gl}(2,\widehat{\mathbb{Z}})^{\pm 1} \quad \text{modulo sign,} \tag{35}$$

and then

$$\xi = \alpha_0\rho(\alpha_0)^{-1} \quad (\text{case } h=1), \quad \eta = \beta_0\sigma_\infty(\beta_0)^{-1} \quad (\text{case } g=1) \in \mathrm{Sl}(2,\widehat{\mathbb{Z}}) \tag{36}$$

are determined without sign ambiguity. We have †

$$\dagger f \in \mathrm{Sl}(2,\widehat{\mathbb{Z}}) \quad \text{modulo sign} \quad (\text{for } f \in \widehat{\Pi}_{0,3} \text{ fixed}) \tag{37}$$

and we deduce from this

$$\varphi = f\sigma_\infty(f)^{-1}, \quad \psi = f\rho(f)^{-1} \in \mathrm{Sl}(2,\widehat{\mathbb{Z}}) \tag{38}$$

without sign ambiguity.

In the choice of the † representatives

†

144

Uncertainty retained from cancelled passages.

140-3-s145-D002 ± 1

140-3-s145-D003 ± 1

140-3-s145-D004 This done|This fixed (34)|(37)

140-3-s145-D006 $\varphi =$

140-3-s145-D008 repr

140-3-s145-D009 (39) [illeg. 1] $\varphi|\xi$

[scan 146]

of $\varepsilon_0 \in \mathrm{Sl}(2,\widehat{\mathbb{Z}})'$ in $\mathrm{Sl}(2,\widehat{\mathbb{Z}})$ there is sign ambiguity

$$\varepsilon_0 = \pm \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix},$$

m819

Marginal note. m819 though the sign − would hardly be a reasonable choice — one likes to see a unipotent element † generating a 1-parameter subgroup…

but for

$$\ell_0 = \varepsilon_0^2 = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} \tag{39}$$

there is no ambiguity of sign. This being so, the relations

$$\xi = \ell_0^\nu \varphi \qquad (\text{case } h=1) \tag{40}$$

and

$$\eta = \ell_0^{-\nu} \psi \qquad (\text{case } g=1) \tag{41}$$

are they true without a correcting sign ??|?!

The case $h = 1$ corresponds to the data (in $\widehat{\widehat{\Pi}}_{0,3}$) of ξ and †f, and ν, satisfying

$$\xi\rho(\xi)\rho^2(\xi) = 1, \qquad \xi = \ell_0^\nu(\underbrace{f\sigma_\infty(f)^{-1}}_{\varphi}),$$

† or again to the data

$$\alpha_0 \in \widehat{\widehat{\Pi}}_{0,3}, \quad f \in \widehat{\Pi}_{0,3}, \quad \nu \in \widehat{\mathbb{Z}}$$

satisfying the relations

$$\underbrace{\alpha_0\rho(\alpha_0)}_{\xi} = \ell_0^\nu \underbrace{f\sigma_\infty(f)^{-1}}_{\varphi} \tag{42}$$

in terms of these data we define

$$\begin{cases} \xi = \alpha_0\rho(\alpha_0) \\ \varphi = f\sigma_\infty(f)^{-1} \\ g = \sigma_\infty(f)^{-1}\ell_1^\nu\rho(f) = (\ell_0^\nu f)^{-1}(\ell_0^\nu\varphi)\rho(\ell_0^\nu f) \end{cases} \tag{43}$$

This recalled, we lift α_0, f to $\mathrm{Gl}(2,\widehat{\mathbb{Z}})^{\pm 1}$, 145

Uncertainty retained from cancelled passages.

140-3-s146-D001 [illeg. 1]

140-3-s146-D002 φ

140-3-s146-D003 is

[scan 147]

with sign ambiguity, † and †ξ and φ are thus determined in Sl|Gl$(2,\widehat{\mathbb{Z}})$, independently of the [p. 363]
choices of signs, † and likewise g.

We will have|have therefore in $\mathrm{Sl}(2,\widehat{\mathbb{Z}})$

$$\xi = \varepsilon\ell_0^\nu\varphi \qquad \text{i.e.} \qquad \overset{\dagger}{\alpha_0\rho(\alpha_0)} = \varepsilon\ell_0^\nu f\sigma_\infty(f)^{-1}, \tag{44}$$

in Sl$(2,\widehat{\mathbb{Z}})$, with a sign $\varepsilon \in \{\pm 1\}$ that is well determined, and the question that arises is whether we always|always have $\varepsilon = +1$.

† Let us note that the formula

$$\rho(\ell_0) = \ell_1$$

is true in $\mathrm{Sl}(2,\widehat{\mathbb{Z}})$ (without a correcting sign) so the last expression for g in (43) is valid in $\mathrm{Sl}(2,\widehat{\mathbb{Z}})$ (without a sign), which shows that $g \in \mathrm{Sl}(2,\widehat{\mathbb{Z}})$, † is a ρ-transform of $\ell_0^\nu\varphi$, † the latter is a ρ-transform of g. † We have

$$g\rho(g)\rho^2(g) = \varepsilon_0.1 \qquad \varepsilon \in \{\pm 1\},$$

and we deduce from this, if $g' = \alpha g\rho(\alpha)^{-1}$ † ($\alpha \in \mathrm{Gl}(2,\widehat{\mathbb{Z}})$), that we have

$$g'\rho(g')\rho^2(g') = \varepsilon_0.1$$

hence in particular for $g' = \ell_0^\nu\varphi = \varepsilon\xi$ (44) , with $\varepsilon \in \{\pm 1\}$ † now ξ given by (42) satisfies

$$\xi\rho(\xi)\rho^2(\xi) = 1$$

(N.B. we do not have $\rho^3 = 1$ in $\mathrm{Sl}(2,\widehat{\mathbb{Z}})$, but $\rho^3 = -1$, which implies that the inner automorphism † defined|defines by ρ is of order 3…), hence

$$\varepsilon_0 = \varepsilon,$$

146

Uncertainty retained from cancelled passages.

140-3-s147-D001 but we
140-3-s147-D002 [illeg. 2]
140-3-s147-D003 [illeg. 2]
140-3-s147-D004 $\varepsilon\ell_0^\nu$
140-3-s147-D005 Defining $\rho_0|\rho$
140-3-s147-D006 is not
140-3-s147-D007 therefore
140-3-s147-D008 [illeg. 1] [illeg. 3]
140-3-s147-D011 $\rho^2 \in$
140-3-s147-D013 [illeg. 2]
140-3-s147-D014 that it

[scan 148]

i.e. the question of the validity of the formula

$$\xi = \ell_0^\nu \varphi$$

in $\mathrm{Sl}(2, \widehat{\mathbb{Z}})$ is equivalent to that of the validity of the formula

$$g\rho(g)\rho^2(g) = 1,$$

which we leave open for the moment. Let us note for reference the following formulas in $\mathrm{Gl}|\mathrm{Sl}(2, \widehat{\mathbb{Z}})$

$\det \alpha_0 \in \pm 1,\ \det f = 1,$

$$\text{(Case } h = 1\text{)}$$

$$\begin{cases} \alpha_0, f \in \mathrm{Gl}(2, \widehat{\mathbb{Z}})^{\pm 1} \quad \dagger \text{ (arbitrary signs, we choose them)} \\ \left.\begin{aligned} \xi &\overset{\text{dfn}}{=} \alpha_0 \rho(\alpha_0)^{-1} \\ \varphi &\overset{\text{dfn}}{=} f\sigma_\infty(f)^{-1} \end{aligned}\right| \in \mathrm{Sl}(2, \widehat{\mathbb{Z}}) \\ [g \overset{\text{dfn}}{=} \sigma_\infty(f)^{-1}\ell_1^\nu \rho(f) = (\ell_0^\nu f)^{-1}(\ell_0^\nu \varphi)\rho(\ell_0^\nu f)] \\ \text{hence|therefore} \quad \xi = \dagger \cdot \varepsilon \ell_0^\nu \varphi \qquad \varepsilon \in \{\pm 1\} \\ \left[\begin{aligned} &g\rho(g)\rho^2(g) = \varepsilon.1 \\ &g = \varepsilon.\alpha\rho(\alpha)^{-1} \qquad \alpha = (\ell_0^\nu f)^{-1}\alpha_0 = f^{-1}\ell_0^{-\nu}\alpha_0 \end{aligned}\right] \end{cases} \tag{45}$$

m822

Marginal note. m822 It now seems more practical to me to define g by $g = \alpha\rho(\alpha)^{-1}$, so as thus to have $g\rho(g)\rho^2(g) = 1$, and to place the sign $\varepsilon = \varepsilon_2$ in the formula $g = \varepsilon\sigma_\infty(f^{-1})\ell_1^\nu\rho(f)$ — the sign [illeg. 1] does not occur|only occurs [illeg. 3] in $\xi = \varepsilon\ell_0^\nu\varphi$.

N.B. we write [illeg. 1] $\varepsilon = \varepsilon_2$

Symmetrically

$$\text{(Case } g = 1\text{)}$$

$$\begin{cases} \beta_0, f \in \mathrm{Sl}(2, \widehat{\mathbb{Z}}) \qquad \text{(modulo signs, we choose them)} \\ \eta \overset{\text{dfn}}{=} \beta_0\sigma_\infty(\beta_0)^{-1} \\ \psi \overset{\text{dfn}}{=} \psi|f\rho(f)^{-1} \\ h \overset{\text{dfn}}{=} f^{-1}\ell_0^\nu\sigma_\infty(\rho(f)) = \end{cases} \tag{46}$$

147

Uncertainty retained from cancelled passages.
140-3-s148-D001 with
140-3-s148-D002 ℓ_0^{ν}

[scan 149]
[p. 364]

I am going to look at the case $h = 1$ (cf. (45)), setting

$$\begin{cases} \alpha_0 = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \\ f = \begin{pmatrix} r & s \\ t & u \end{pmatrix} \end{cases} \tag{47}$$

m825

Marginal note. m825 change of signs in $\alpha_0, -f|f$ is inessential

m826

Marginal note. m826 N.B. In the "arithmetic" case (i.e. f comes|coming from $\widehat{\Pi}_{0,3}$, α_0 from $\widehat{\widetilde{\Pi}}_{0,3} = \widehat{\Pi}_{0,3}.\{1, \tau_\infty\}$, we will have $\alpha_0, f \equiv 1$ (2) i.e. $a, d, r, u \equiv 1$ (2), $b, c, s, t \equiv 0$ (2)

if|where † $a, b, c, d, r, s, t, u \in \widehat{\mathbb{Z}}$, with

$$\boxed{ab - cd = \delta \in \{\pm 1\}} \tag{48}$$

$$\boxed{ru - st = 1}. \tag{49}$$

This being set

$$\xi = \alpha_0 \rho(\alpha_0)^{-1} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \tag{50}$$

$$\varphi = f \underbrace{\sigma_\infty(f)^{-1}}_{{}^{\iota}f} = \begin{pmatrix} R & S \\ T & U|u \end{pmatrix} = ? \begin{pmatrix} r^2 + s^2 & rt + su \\ rt + su & t^2 + u^2 \end{pmatrix} \tag{51}$$

so that we will have (cf. § 42, formula (78)

$$\begin{cases} \delta A = (a^2 + b^2 + ab) - (ac + bd + bc) & \delta B = ac + bd + bc \\ \delta C = ac + bd + ad - (c^2 + d^2 + cd) & \delta D = c^2 + d^2 + cd \end{cases}, \tag{52}$$

and (§ 42, (90) and 91)

N.B. we have $\delta = \varepsilon_3$

$$R = r^2 + s^2, \qquad S = T = rt + su, \qquad U = t^2 + u^2. \tag{53}$$

† †We will have the formulas

$$C + D+|-B = 1 \tag{54}$$

$$S = T. \tag{55}$$

The essential relation linking α_0 and f, i.e. linking the "variables" a, b, c, d with u, r, s, t, is the relation

$$\boxed{\xi = \varepsilon \ell_0^{\nu} \varphi, \qquad \varepsilon \in \{\pm 1\}} \tag{56}$$

which is also written $\ell_0^{-\nu} \xi = \varepsilon \varphi$, i.e.

"loop relation" 148

Uncertainty retained from cancelled passages.
140-3-s149-D001 [illeg. 1]
140-3-s149-D002 [illeg. 1]
140-3-s149-D003 [illeg. 2]

[scan 150]

$$\begin{pmatrix} A+2\nu C & B+2\nu D \\ C & D \end{pmatrix} = \varepsilon \begin{pmatrix} R & S \\ T(=S) & U \end{pmatrix}. \tag{57}$$

But this condition implies first that the matrix on the left-hand side is symmetric (since $S = T$) i.e.

$$2\nu D = C - B$$

a condition which is also equivalent, by (54), to

$$2\nu D = 1 - D \quad \text{i.e.} \quad \boxed{\mu D = 1} \quad \dagger \quad \text{i.e.} \quad D = 1/\mu, \quad \mu = 1/D \tag{58}$$

where $\mu = 2\nu + 1$, which already implies μ and D invertible.

Symmetrically, if we write (57) in the form

$$\varepsilon \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \varepsilon \begin{pmatrix} R - 2\nu T & S - 2\nu U \\ T & U \end{pmatrix},$$

the condition $C + D - B = 1$ becomes, for the entries of the matrix φ

$$T + U - (S - 2\nu U) = \varepsilon\ ;$$

i.e.

$$(2\nu + 1)U = \varepsilon \quad \text{i.e.}$$

$$\mu U = \varepsilon \quad \text{i.e.} \quad U = \varepsilon/\mu, \quad \mu = \varepsilon/U \tag{59}$$

which is also equivalent, taking (58) into account, to the relation

$$D = \varepsilon U \quad \text{or} \quad U = \varepsilon D \tag{60}$$

which is none other than the equality of the entries in the bottom rows of the matrices (57). 149

Uncertainty retained from cancelled passages.
140-3-s150-D001 1 [*illeg.* 1] μ 1

[scan 151] [p. 365]

We can write out the relations (57), as † four relations on the 9 variables $a, b, c, d, r, s, t, u, \mu$ in addition to the relations (48), (49);

$$\begin{cases} \boxed{\begin{aligned} D &= \varepsilon U \\ \mu D &= 1 \\ C &= \varepsilon T \end{aligned}} \quad \dagger i.e. \quad \dagger \quad \mu U = \varepsilon \\ A = \varepsilon(R - \dagger \underbrace{(\mu - 1)}_{2\nu} \underbrace{T}_{C}) \end{cases} \tag{61}$$

m831

Marginal note. m831 N.B. this last one is a consequence of the others, taking $C + D - B = 1$ and $AD - BC = 1$ into account

i.e. – again

$$\begin{cases} \boxed{\begin{aligned} \mu(c^2+d^2+cd) &= \delta \\ c^2+d^2+cd &= \varepsilon\delta(t^2+u^2) \\ \underline{ac}+\underline{bd}+\underline{ad}-(c^2+d^2+cd) &= \delta\varepsilon(rt+su) \end{aligned}} \\ a^2+b^2+ab-(ac+bd+bc) = \delta\varepsilon(r^2+s^2-(\mu-1)(rt+su)) \end{cases} \tag{62}$$

consequence of the preceding ones

When the signs ε, δ are fixed, we find with (48), (49) a system of 6 equations in 9 variables $a, b, c, d, r, s, t, u, \mu$ – we should be able to check that the scheme (over $\operatorname{Spec}\mathbb{Z}$) of solutions of these equations is of relative dimension 3, and of relative dimension 1 over the "factors" (a, b, c, d) resp. (r, s, t, u). We thus have an adelic point of this absolute scheme – a point with coefficients in $\widehat{\mathbb{Z}}$…

no, it is of total relative dimension 4

Theorem The multiplier μ is determined as a function of $\xi|\zeta = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ (considered as an element of $\mathrm{Sl}(2, \widehat{\mathbb{Z}})$), by the relation

$$\mu D = 1 \qquad (58)$$

(+) and it is determined up to sign by $\varphi = \begin{pmatrix} R & S \\ T & U \end{pmatrix}$ by the relation

$$\mu U = \varepsilon \qquad (59) \qquad \text{where } \varepsilon \in \{\pm 1\},$$

150

Uncertainty retained from cancelled passages.

140-3-s151-D001 [illeg. 1]

140-3-s151-D002 $U \quad \varepsilon$

140-3-s151-D003 $D = 1/\mu$

140-3-s151-D004 2ν

and where ε is the sign which occurs in the formulas [scan 152]

$$\begin{cases} \xi|\zeta = \varepsilon l_0^{\nu}\varphi \qquad (56) \\ g\rho(g)\rho^2(g) = \varepsilon \qquad (45). \\ g = \varepsilon\alpha\rho(\alpha)^{-1} \end{cases}$$

We have $\varepsilon = 1$ iff $\mu \equiv 1$ (4), $\varepsilon = -1$ iff $\mu \equiv -1$ (4). i.e. $\nu \equiv 0$ (2) i.e. $\nu \equiv 1$ (2)

m835

Marginal note. m835 in other words, we have $\mu \equiv \varepsilon$ (4) since $U = t^2 + u^2 \equiv 1$ (4) hence $1/U \equiv (1)$ (4) hence $\mu = \varepsilon/U \equiv \varepsilon$ (4). This would be the time to include also the|here the relation $\mu \equiv \delta$ (3) cf. with the appropriate proof…

To prove this, we recall that

$$U = t^2 + u^2$$

note that the only values modulo 4 taken by t^2 and u^2 are 0 and 1, hence those taken by $t^2 + u^2$ are $0, 1$ and 2, and the only unit among these is 1. Thus † $1/\mu$ can be put in the form $\varepsilon(t^2 + u^2)$ modulo 4 iff $1/\mu \equiv \varepsilon$ (4) or again $\mu \equiv \varepsilon$ (4) q.e.d.

We had seen in a quite analogous way, thanks to the formula $\mu D = 1 \quad (58)$ and

$$D = \delta(c^2 + d^2 + cd)$$

that $\mu \equiv \delta$ (3) i.e. $\delta = +1$ (resp. -1) iff $\mu \equiv 1$ (3) (resp. $\mu \equiv -1 \bmod 3$). We † shall denote, for all|the prime numbers p, by $\varepsilon_p|\beta_p$ the value at μ of the "quadratic character" relative to p, equal to 1 iff $\mu_p \in (\mathbb{Z}_p^*)^2$. We shall therefore have

$$\begin{cases} \delta = ad - bc = \varepsilon_3|\beta_3(\mu) = \varepsilon_3 \\ \varepsilon = D/U = \varepsilon_2|\beta_2(\mu) = \varepsilon_2, \end{cases} \tag{63}$$

and knowing ε_3 (resp. ε_2, resp. both) is equivalent to knowing μ modulo 3|2 (resp. modulo 4, resp. modulo 12). m836

Marginal note. m836 $(\mathbb{Z}/12\mathbb{Z})^* \simeq (\mathbb{Z}/4\mathbb{Z})^* \times (\mathbb{Z}/3\mathbb{Z})^* \simeq \{\pm 1\} \times \{\pm 1\}$

151

Uncertainty retained from cancelled passages.

140-3-s152-D001 μ

140-3-s152-D003 could thus

We are also going to compute † (cf. formulas (45)) [scan 153] [p. 366]

$$\alpha = (l_0^\nu f)^{-1}\alpha_0 = f^{-1}l_0^{-\nu}\alpha_0$$

and

$$g = \varepsilon\alpha\rho(\alpha)^{-1} = \underbrace{\sigma_\infty(f)^{-1}}_{{}^t f}\, l_1^\nu \rho(f) = (l_0^\nu f)^{-1}\xi|\zeta\,\rho(l_0^\nu f).$$

We find

$$\begin{aligned} \alpha &= \underbrace{\begin{pmatrix} u & -s \\ -t & r \end{pmatrix}}_{f^{-1}} \underbrace{\begin{pmatrix} 1 & 2\nu \\ 0 & 1 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix}}_{\begin{pmatrix} a+2\nu c & b+2\nu d \\ c & d \end{pmatrix}} \\ &= \begin{pmatrix} u(a+2\nu c) - cs & u(b+2\nu d) - ds \\ -t(a+2\nu c) + cr & -t(b+2\nu d) + dr \end{pmatrix}, \end{aligned} \tag{64}$$

m838

Marginal note. m838 N.B. Unlike α_0 and f, and $\xi|\zeta, \varphi$ which are deduced from them, α † shares with g the advantage and the interest of being determined without ambiguity (up to sign) by an automorphism u of $\widehat{\Pi}_{0,3}$, whereas f etc. are [illeg. 1] choices of ambiguities $f \mapsto l_0^n f$ etc.…

$$\begin{aligned} g &= \underbrace{\begin{pmatrix} r & t \\ s & u \end{pmatrix}}_{\sigma_\infty(f^{-1}) = {}^t f} \underbrace{\begin{pmatrix} 1 & 0 \\ 2\nu & 1 \end{pmatrix}}_{l_1^\nu | l_0^\nu} \underbrace{\begin{pmatrix} t+u & -t \\ t+u-r-s & r-t \end{pmatrix}}_{\rho(f)\ \text{(cf. §43 formulas (77))}} \\ &= \begin{pmatrix} r & t \\ s & u \end{pmatrix}\begin{pmatrix} t+u & -t \\ \mu(t+u) - r - s & -\mu t + r \end{pmatrix} \\ &= \begin{pmatrix} 1+\mu t(t+u) & -\mu t^2 \\ -1+\mu u(t+u) & 1-\mu ut \end{pmatrix}, \end{aligned}$$

i.e.

$$g = \begin{pmatrix} 1+\mu t(t+u) & -\mu t^2 \\ -1+\mu u(t+u) & 1-\mu ut \end{pmatrix}. \tag{65}$$

It appears that g is expressed simply in terms of f i.e. of r, s, t, u directly, and not in terms of α_0 or $\xi|\zeta$ i.e. via the a, b, c, d or A, B, C, D. On the other hand, to express α (64), we must bring in both the r, s, t, u and the a, b, c, d.

It was natural, in connection with the canonical homomorphism

$$\mathcal{N}_{1,1} \longrightarrow \mathrm{Gl}(2, \widehat{\mathbb{Z}})$$

(cf. §39, (11)) to look for a matrix †, which by abuse of language I am going to denote by the same letter, u, as the automorphism under consideration of $\widehat{\Pi}_{0,3}$ with which we had started (just as I denote by the same letter the elements g, f etc. of $\widehat{\Pi}_{0,3}$ or of $\widehat{\mathfrak{S}}_{0,3}|\hat{\mathcal{S}}_{0,3}$, and their images in $\mathrm{Gl}(2, \widehat{\mathbb{Z}})'$, or even corresponding elements in $\mathrm{Gl}(2, \widehat{\mathbb{Z}})$ itself), 152

Uncertainty retained from cancelled passages.

140-3-s153-D001 [illeg. 1]

140-3-s153-D002 [illeg. 1]

140-3-s153-D003 [illeg. 1]

[scan 154]

$$u \in \mathrm{Gl}(2, \widehat{\mathbb{Z}})$$

† with determinant equal to μ, and inducing on $\mathrm{Sl}(2, \widehat{\mathbb{Z}})'$, by an inner automorphism, the same automorphism as the u at the outset (by passage to the quotient), which is made explicit by the formulas

$$\det u = \mu \tag{66}$$

$$u\sigma_\infty u^{-1} = \varepsilon'\sigma_\infty \qquad \varepsilon' \in \{\pm 1\} \tag{67}$$

$$u\rho u^{-1} = \varepsilon'' g\rho \qquad \varepsilon'' \in \{\pm 1\} \tag{68}$$

In fact, (68) † is written

$$[u, \rho] = u\rho(u)^{-1} = \varepsilon'' g$$

and applying to both sides the operation $x \mapsto x\rho(x)\rho^2(x)$, and taking into account that $g\rho(g)\rho^2(g) = \varepsilon_2$, and $x\rho(x)\rho^2(x) = 1$ for $x = u\rho(u)^{-1}$, we find

$$\varepsilon'' = \varepsilon_2 \tag{68 bis}$$

and (68) is thus equivalent to

$$u\rho(u)^{-1} = \varepsilon_2 g = \alpha\rho(\alpha)^{-1}$$

† i.e. to

$$u = \alpha u_0 \quad \text{with } [u_0, \rho] = 1 \quad (\text{i.e. } u_0\rho(u_0)^{-1} = 1). \tag{69}$$

We have already remarked that (67), (68) determine u only modulo multiplication by a scalar $\lambda \in \widehat{\mathbb{Z}}^*$, and taking (66) into account, u is determined only modulo λ satisfying $\lambda^2 = 1$ i.e. modulo ${}_2\widehat{\mathbb{Z}}^* \simeq \{\pm 1\}^{\mathbb{P}}$ (where $\mathbb{P}$ is the set of prime numbers), but we nevertheless hope to hit upon a canonical choice.

In my first calculations, I hadn't yet brought out|dug up (69), where the condition u_0 commutes with ρ was expressed in the form

$$u_0 = p\,\mathrm{id} + q\rho = \begin{pmatrix} p & q \\ -q & p+q \end{pmatrix} \quad (p, q \in \widehat{\mathbb{Z}}), \tag{70}$$

153

Uncertainty retained from cancelled passages.

140-3-s154-D001 satisf

140-3-s154-D003 should have

140-3-s154-D005 whence

[scan 155]

[p. 367]

so that

$$\det u_0 = p^2 + pq + q^2 \tag{71}$$

and that (66) is written (taking (69) into account, where $\det \alpha = \delta = \varepsilon_3$)

$$p^2 + pq + q^2 = \mu\varepsilon_3 \tag{72}$$

which, in view of the relation

$$c^2 + cd + d^2 = \varepsilon_3/\mu \qquad (52 \text{ and } 58)$$

suggested relations of the form

$$p = \mu c,\ q = \mu d, \quad \text{or} \quad p = \mu d,\ q = \mu c.$$

In fact, expressing (67) as an equation in p, q (added to|: 72), I found (up to sign of course which here is inessential)

$$p = \mu d,\ q = \mu c \qquad \text{i.e.} \qquad u_0 = \mu(c\rho + d) \tag{73}$$

but working in the case $\varepsilon' = +1$ exclusively. (I had thought I could conclude, a little too hastily, that the sign ε' was always $+1$, before realizing that this was hardly compatible with class field theory, and suspecting that we should have $\varepsilon' = +1$ iff $p \equiv 1$ (4).) By making this rather cumbersome detour via the (a, b, c, d), I nevertheless arrived, after many twists and turns (calculation errors etc.) at the formula where there were no longer any a, b, c, d at all, and of a disconcerting simplicity:

$$\underline{\underline{u}} = \mu(t\sigma_\infty + u) = \mu \begin{pmatrix} u & t \\ -t & u \end{pmatrix}, \tag{74}$$

which manifestly corresponds to the case $\varepsilon_2'|\varepsilon' = 1$, i.e. the one where $\underline{u}$ commutes with σ_∞. As (⟺ probably, $\varepsilon_2 = 1$)
m842

Marginal note. m842 I underline the letter u which denotes a matrix, to distinguish it from the u which denotes a scalar.

[scan 156]

$$\det(t\sigma_\infty + u) = t^2 + u^2 = \varepsilon_2/\mu \quad (= 1/\mu \text{ if } \varepsilon_2 = +1),$$

we shall indeed have, in the case $\varepsilon_2 = 1$, †

$$\det \underline{u} = \mu^2 \det(t\sigma_\infty + u) = \mu^2\, 1/\mu = \mu$$

i.e., (66). On the other hand, we shall have

$$\begin{aligned}
[u, \rho] &= [t\sigma_\infty + u, \rho] \\
&= \underbrace{\begin{pmatrix} u & t \\ -t & u \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}}_{\begin{pmatrix} -t & u+t \\ -u & u-t \end{pmatrix}} \mu \underbrace{\begin{pmatrix} u & -t \\ t & u \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}}_{\begin{pmatrix} u-t & -u \\ u+t & -t \end{pmatrix}}. \\
&= \mu \begin{pmatrix} (u+t)^2 - ut + t^2 & -t^2 \\ ut - t^2 & u^2 + t^2 - ut \end{pmatrix} = \mu \begin{pmatrix} \frac{1}{\mu} + t(u+t) & -t^2 \\ t(u-t) & \frac{1}{\mu} - ut \end{pmatrix} \\
&= \begin{pmatrix} 1 + \mu t(u+t) & -\mu t^2 \\ \mu t(u\dagger - t^\dagger) & 1 - \mu ut \end{pmatrix},
\end{aligned}$$

m844

Marginal note. m844 We could also apply the formulas §43 (78) —

and comparing with (65) we do indeed find

$$[u, \rho] = g$$

taking into account that

$$\mu(u^2 + t^2) = \varepsilon_2 = 1$$

thus

$$-1 + \mu u(t + u) = \underbrace{-1 + \mu u^2}_{-\mu t^2} + \mu u t = \mu t(u - t) \text{ –!|!}$$

I would now like, without launching into new calculations, to find "by guesswork" an expression for u, in the † where $\varepsilon_2 = -1$, and where we expect to find in (67) the sign $\varepsilon' = -1$ — i.e. that u be an antisimilitude[illeg. 1]

$$\underline{u}\,{}^t\underline{u} = -\det \underline{u}$$

But † the matrix

$$\tau_\infty = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

is the typical case of such an antisimilitude, and I expect that the matrix

$$\underline{u} = \mu\tau_\infty(t\sigma_\infty + u) = \mu \begin{pmatrix} 1| -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} u & t \\ -t & u \end{pmatrix} = \mu \begin{pmatrix} u| -u & t| -t \\ -t & -u \end{pmatrix}$$

(it is = an orthogonal reflection)

155

Uncertainty retained from cancelled passages.

140-3-s156-D001 therefore
140-3-s156-D002 t
140-3-s156-D003 2
140-3-s156-D004 case
140-3-s156-D005 and

[scan 157]
[p. 368]

will do. We shall have, of course

$$u\,{}^t u = \mu^2 \begin{pmatrix} u & t \\ t & -u \end{pmatrix} \begin{pmatrix} u & t \\ t & -u \end{pmatrix}$$
$$= \mu^2 \begin{pmatrix} u^2 + t^2 & 0 \\ 0 & u^2 + t^2 \end{pmatrix} = \mu^2 \begin{pmatrix} -1/\mu & 0 \\ 0 & -1/\mu \end{pmatrix}$$

i.e.

$$u\,{}^t u = -\mu.\mathrm{Id}$$

and

$$\det u = \mu^2(-u^2 - t^2) = \mu^2(1/\mu) = \mu$$

taking into account that now

$$u^2 + t^2 = \varepsilon_2/\mu = -\frac{1}{\mu} \qquad \text{since } \varepsilon_2 = -1.$$

Finally to check (68) let us compute

$$[u,\rho] = \left[\begin{pmatrix} u & t \\ t & -u \end{pmatrix}, \rho\right]$$
$$= \underbrace{\begin{pmatrix} u & t \\ t & -u \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}}_{\begin{pmatrix} -t & u+t \\ u & -u+t \end{pmatrix}} \frac{1}{\mu}|\mu \underbrace{\begin{pmatrix} -u & -t \\ -t & u \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}}_{\begin{pmatrix} -u-t & u|-u \\ u-t & t \end{pmatrix}}$$
$$= \mu \begin{pmatrix} u(u+t) & t^2 \\ -(u^2+t^2)-u(u-t) & u^2+t^2-ut \end{pmatrix} = \begin{pmatrix} \mu u(u+t) & \mu t^2 \\ 1-\mu u(u-t) & -1-\mu ut \end{pmatrix}$$

(taking into account that $\mu(u^2+t^2) = -1$), † do we indeed have

$$[u,\rho] = -g$$

taking into account formula (65) which gives g explicitly in terms of r,s,t,u, we find that this expected|precise formula is equivalent to

$$2\mu ut = 0 \qquad \text{i.e.} \qquad ut = 0$$

really, this is screwy! But I see that if we replace in the definition of u t by $-t$, i.e. if we define

$$\underline{u} = \mu(t\sigma_\infty + u)\tau_\infty = \mu \begin{pmatrix} u & t \\ -t & u \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \mu \begin{pmatrix} u & -t \\ -t & -u \end{pmatrix}$$

(instead of $\underline{u} = \mu \begin{pmatrix} u & t \\ t & -u \end{pmatrix}$), then $\underline{u}$ is always an antisimilitude of determinant μ, and moreover now

$$[u,\rho] = \begin{pmatrix} \mu u(u-t) & \mu t^2 \\ 1-\mu u(u+t) & -1+\mu ut \end{pmatrix} = -g$$

156

Uncertainty retained from cancelled passages.

140-3-s157-D001 whence|therefore

[scan 158]

which does indeed reduce to checking

$$\mu u(u-t) = -(1+\mu t(t+u))$$

i.e.

$$\mu[u^2 - +t^2+] = -1 \qquad \text{O.K.}$$

In summary, we find

Theorem Let $\alpha_2 \in \mathbb{Z}/2\mathbb{Z}$ be defined by $(-1)^{\alpha_2} = \varepsilon_2$ (thus $\alpha_2 \equiv 0\ (2)$ iff $\mu \equiv 1\ (4)$, $\alpha_2 \equiv 1\ (2)$ iff $\mu \equiv -1\ (4)$). Put

$$\underline{u} = \mu(t\sigma_\infty + u)\tau_\infty^{\alpha_2} = \mu \begin{pmatrix} u & t \\ -t & u \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & \varepsilon_2 \end{pmatrix}^{\alpha_2}, \tag{75}$$

We then have the formulas

$$\det \underline{u} = \mu \tag{76}$$

$$[\underline{u}, \sigma_\infty] = -|\cdot\varepsilon_2.\mathrm{id} \qquad \text{i.e.} \quad \underline{u}(\sigma_\infty) = \varepsilon_2.\sigma_\infty \tag{77}$$

$$[\underline{u}, \rho] = \varepsilon_2.g \qquad \text{i.e.} \quad \underline{u}(\rho) = (\varepsilon_2 g).\rho \tag{78}$$

$= \begin{pmatrix} u & \varepsilon_2 t \\ -t & \varepsilon_2 u \end{pmatrix}$

Corollary Let u be an automorphism of the group with loops $\widehat{\Pi}_{0,3}$, commuting as outer actions with $\mathfrak{S}_3|\mathcal{S}_3$ (i.e. with σ_∞ and with ρ), μ its multiplier, let us also denote by u the canonical extension of u to $\widehat{\mathfrak{S}}^+_{0,3}|\widehat{\mathcal{S}}^+_{0,3} \simeq \mathrm{Sl}(2,\mathbb{Z})^{\wedge\prime}$, inducing the identity on $\mathfrak{S}_3|\mathcal{S}_3 \simeq \widehat{\mathfrak{S}}^+_{0,3}|\widehat{\mathcal{S}}^+_{0,3}/\widehat{\Pi}_{0,3}$, and consider the quotient $\mathrm{Sl}(2,\widehat{\mathbb{Z}})'$ of $\mathrm{Sl}(2,\mathbb{Z})^{\wedge\prime}$. Then the action of u passes to an action on the quotient $\mathrm{Sl}(2,\widehat{\mathbb{Z}})'$, and the latter is induced by an †"inner|inner" automorphism of $\mathrm{Gl}(2,\widehat{\mathbb{Z}})$, defined by a matrix $\underline{u} \in \mathrm{Gl}(2,\widehat{\mathbb{Z}})$ such that $\det u = \mu$ 157

Uncertainty retained from cancelled passages.
140-3-s158-D004 inner

[scan 159]

(which describes the outer action of u on $\mathrm{Sl}(2,\widehat{\mathbb{Z}})'$ in terms of the multiplier μ). [p. 369]
m851

Marginal note. m851 N.B. To check the corollary, we are reduced to the case where u commutes with σ_∞, if necessary conjugating by an inner automorphism of[illeg. 1]…
m852

Marginal note. m852 We have done this, for the moment, for u commuting with σ_∞.

In fact, we have much better still, since we are going to define canonically modulo sign, in terms of u, the matrix $\underline{u}$ by formula (75), where $t, u \in \widehat{\mathbb{Z}}$ are defined as follows: † the datum of the automorphism u defines $f \in \widehat{\Pi}_{0,3}$, modulo the operation $f \mapsto \ell_0^n f$ (for $n \in \widehat{\mathbb{Z}}$), and we again denote by f the image of f in $\mathrm{Sl}(2,\widehat{\mathbb{Z}})/\pm 1$, which is thus a matrix $\begin{pmatrix} r & s \\ t & u \end{pmatrix}$, defined modulo sign, We shall note that in $\mathrm{Sl}(2,\widehat{\mathbb{Z}})$ we have

$$\ell_0^n \begin{pmatrix} r & s \\ t & u \end{pmatrix} = \begin{pmatrix} 1 & -2n \\ 0 & 1 \end{pmatrix} \begin{pmatrix} r & s \\ t & u \end{pmatrix} = \begin{pmatrix} r-2nt & s-2nu \\ t & u \end{pmatrix} \tag{79}$$

($n \in \widehat{\mathbb{Z}}$), so (t,u) does not change (modulo sign) when we replace f by $\ell_0^n f$ — so (t,u) is determined modulo sign by the automorphism u of $\widehat{\Pi}_{0,3}$. † The analogous observation applies moreover to $\alpha_0 = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, which is determined modulo sign and modulo left multiplication by an ℓ_0^n — the † vector (c,d) is determined modulo sign, hence the element

$$c\rho + d$$

158

Uncertainty retained from cancelled passages.
140-3-s159-D002 We have
140-3-s159-D003 [illeg. 1]
140-3-s159-D004 [illeg. 1]

[scan 160]

is determined modulo sign, and †unless I am mistaken we must have a relation

$$\underline{u} \overset{?}{=} \mu\alpha(c\rho+d) = \mu f^{-1}\ell_0^{-\nu}\alpha_0(c\rho+d) \tag{80}$$

(cf. (69) (73)), at least if $\varepsilon_2 = 1$ — otherwise we may perhaps have to correct by a factor τ_∞ somewhere. †
m854

Marginal note. [m854] N.B. this formula is valid with the corrective factor $\delta = \varepsilon_3$, cf. (82) below…

$$\ell_0^{-\nu}\alpha_0(c\rho+d) = \begin{pmatrix} 1 & 2\nu=\mu-1 \\ 0 & 1 \end{pmatrix} \underbrace{\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} d & c^\dagger \\ -c & c+d \end{pmatrix}}_{\underbrace{\begin{pmatrix} ad-bc=\delta & ac+bc+bd=\delta B \\ 0 & c^2+cd+d^2=\delta D \end{pmatrix}}_{\delta\begin{pmatrix} 1 & B \\ 0 & D \end{pmatrix}}}$$

$$\ell_0^{-\nu}\alpha_0(c\rho+d) = \delta\begin{pmatrix} 1 & B+2\nu D=C \\ 0 & D \end{pmatrix}$$

i.e.

$$\ell_0^{-\nu}\alpha_0(c\rho+d) = \delta\begin{pmatrix} 1 & C \\ 0 & D \end{pmatrix} \tag{81}$$

whence

$$\begin{aligned}\mu\alpha(c\rho+d) = (\mu f^{-1})(\ell_0^{-\nu}\alpha_0(c\rho+d)) &= \mu\delta\begin{pmatrix} u & -s \\ -t & r \end{pmatrix}\begin{pmatrix} 1 & C \\ 0 & D \end{pmatrix} \\ &= \mu\delta\begin{pmatrix} u & uC-sD \\ -t & -tC+rD \end{pmatrix} \\ &= \delta\begin{pmatrix} \mu u & -s+\mu uC \\ -\mu t & r-\mu tC \end{pmatrix}\end{aligned}$$

Now (61)

$$\begin{aligned} C = \varepsilon_2 T = \varepsilon_2(rt+su) \qquad & \text{whence} \\ -s+\mu uC = -s+\varepsilon_2\mu urt|\mu turt+\varepsilon_2\mu u^2 s & \\ \overset{\text{if } \varepsilon_2=+1}{=} s(\underbrace{-1+\varepsilon_2 u^2|\varepsilon_2\mu u^2}_{-\varepsilon_2\mu t^2}) + \varepsilon_2\mu urt & \\ = \varepsilon_2\mu t(\overbrace{-st+ur|st+ur}^{+1}) & \\ = \varepsilon_2\mu t & \end{aligned}$$

159

Uncertainty retained from cancelled passages.

140-3-s160-D001 [illeg. 1]

140-3-s160-D002 desire my|my calculations|calculation

140-3-s160-D003 $+d$

[scan 161]

[p. 370]

$$\begin{aligned} r-\mu tC \overset{\text{if } \varepsilon_2=1}{=} & \underbrace{r-\varepsilon_2\mu rt^2}_{r(1-\varepsilon_2\mu t^2)=\varepsilon_2 r\mu u^2} - \varepsilon_2\mu stu \\ & = \varepsilon_2\mu u(ru-st) = \varepsilon_2\mu u, \end{aligned}$$

thus

$$\mu\alpha(c\rho+d) = \delta\mu\begin{pmatrix} u & \varepsilon_2 t \\ -t & \varepsilon_2 u \end{pmatrix} \overset{\dagger}{=} \delta\underline{u},$$

i.e.

$$\underline{u} = (\delta\mu)\alpha(c\rho+d) = (\delta\mu)(f^{-1}\ell_0^{-\nu}\alpha_0)(c\rho+d), \tag{82}$$

† where $\delta = ad-bc \in \{\pm 1\}$, $\delta = \varepsilon_3$ (thus $\delta \equiv \mu$ (3)).

Formula which can also be written with account † taken of (75)

$$\begin{cases} \dagger(t\sigma_\infty + u)\tau_\infty^{\alpha_2} = \delta\alpha(c\rho + d) \qquad \dagger \\ \qquad \text{where} \quad \alpha = f^{-1}\ell_0^{-\nu}\alpha_0, \quad \dagger \quad \delta = \varepsilon_3 \in \{\pm 1\}, \quad \delta = ad - bc, \quad \delta \equiv \mu \ (3) \end{cases} \tag{83}$$

or again

$$f(t\sigma_\infty + u)\tau_\infty^{\alpha_2} = \delta\ell_0^{-\nu}\alpha_0(c\rho + d) \tag{84}$$

$$\underbrace{\begin{pmatrix} r & s \\ t & u \end{pmatrix}\begin{pmatrix} u & t \\ -t & u \end{pmatrix}}_{\begin{pmatrix} 1 & rt+su \\ 0 & t^2+u^2 \end{pmatrix}}\begin{pmatrix} 1 & 0 \\ 0 & \varepsilon_2 \end{pmatrix}\begin{pmatrix} 1 & rt+su \\ 0 & t^2+u^2 \end{pmatrix}\dagger \qquad \underbrace{\begin{pmatrix} 1 & 2\nu \\ 0 & 1 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} d & c \\ -c & c+d \end{pmatrix}}_{\begin{pmatrix} 1 & C \\ 0 & D \end{pmatrix}} \quad \text{cf. (81)}$$

$$\overset{\dagger}{=} \begin{pmatrix} 1 & \varepsilon_2 T \\ 0 & \varepsilon_2 U \end{pmatrix}$$

and in this form the formula was in fact known, as the essential content of the loop relation in matrix form (56) or (57)

$$: \quad \varepsilon_2 U = D, \qquad \varepsilon_2 T = C \qquad !$$

Remark In fact, under a false appearance of arithmetic, we have done by|for the back door some scales in linear algebra, valid over 160

Uncertainty retained from cancelled passages.

140-3-s161-D001 if $\varepsilon_2 = 1$
140-3-s161-D002 if $\varepsilon_2 = 1$ (i.e. $\mu \equiv 1$ (4))
140-3-s161-D004 [illeg. 1]
140-3-s161-D005 f
140-3-s161-D006 if $\varepsilon_2 = 1$
140-3-s161-D007 [*illeg.* 1]
140-3-s161-D008 $u \quad t \quad t \quad u$
140-3-s161-D010 $=$

[scan 162]

arbitrary rings and formulable|formulable probably under much more general and more "comprehensive" conditions — concerning equations of the form

$$[\alpha_0, \rho] = \xi$$

where $\rho \in \mathrm{Gl}(2, A)$ is given, †as well as ξ, and $\alpha_0 \in \mathrm{Gl}(2, A)$ is the unknown (possibly subject to conditions of the type $\det \alpha_0 = \delta$, δ fixed), as well as simultaneous equations in α_0 and f

$$[\alpha_0, \rho] = \varepsilon.\mathcal{E}_0|\varepsilon_0(\mu - 1)[f, \sigma] \qquad \text{cf. (56)} \tag{*}$$

where $\rho, \sigma \in \mathrm{Gl}(2, A)$ are given, as well as a scalar $\varepsilon \in A^*$, and $\lambda \longmapsto \mathcal{E}_0|\varepsilon_0(\lambda)$ is a "one-parameter subgroup", for example we can assume

$$\mathcal{E}_0|\varepsilon_0(\lambda) = \begin{pmatrix} 1 & -\lambda \\ 0 & 1 \end{pmatrix} = \varepsilon_0^\lambda,$$

$$\varepsilon_0 = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$

— assuming if necessary a relation of the form $\sigma\rho = \varepsilon_0$.

Modulo some pleasant adjustments, all the formulas of the present § should be able to be developed in this more general context (as well as substantial complements) (— but I do not know whether I am going to spell out 161

Uncertainty retained from cancelled passages.
140-3-s162-D001 and [illeg. 1]

[scan 163]

here these delightful scales of linear algebra, or even of scheme-theoretic interpretations of [p. 380]
the kind that the scheme of solutions of (*) (for f, α_0, μ variable) is a smooth relative scheme of relative dimension 4, which etc.

I will only spell out, in the present adelic context, manifestly too special, the

Proposition Let $\dagger c, d \quad \dagger t, u \in \widehat{\mathbb{Z}}$, for them to correspond to data $\alpha_0 = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $f = \begin{pmatrix} r & s \\ t & u \end{pmatrix}$ satisfying

$$\underbrace{ad - bc}_{\substack{\text{definition} \\ = \varepsilon_3}} \in \{\pm 1\}, \quad ru - st = 1, \quad \alpha_0 \rho (\alpha_0)^{-1} = \varepsilon_2 \ell_0^{\nu} (f \sigma_\infty (f)^{-1})$$

with $\varepsilon_2 \in \{\pm 1\}$ and $\nu \in \widehat{\mathbb{Z}}$ suitable|suitable (cf. (48) (49) (53)), it f|is necessary and † it suffices that one have

$$c^2 + cd + d^2 = \pm(t^2 + u^2) \quad \in \widehat{\mathbb{Z}}^* \tag{85}$$

Then † $(\varepsilon_2, \varepsilon_3, 2\nu + 1) \in \{\pm 1\} \times \{\pm 1\} \times \widehat{\mathbb{Z}}^*$ is determined modulo sign (i.e. modulo passing to: $(-\varepsilon_2, -\varepsilon_3, -\nu)$) by a, b, c, d, by the relation (where $\mu = 2\nu + 1$) †

†

$$\frac{1}{\mu} = \varepsilon_2 (t^2 + u^2) = \varepsilon_3 (c^2 + cd + u^2) \tag{86}$$

by

†

† (which determines ε_3, μ when one has chosen ε_2) and one has

$$\begin{cases} \varepsilon_2 \equiv \mu \quad (4) \\ \varepsilon_3 \equiv \mu \quad (3) \\ \nu = (\mu - 1)/2 \end{cases} \tag{87}$$

162

Uncertainty retained from cancelled passages.
140-3-s163-D001 a, b
140-3-s163-D002 r, s
140-3-s163-D003 [illeg. 1]
140-3-s163-D004 signs ν
140-3-s163-D007 [illeg. 1]
140-3-s163-D008 $\equiv | =$ $(c^2 + cd + d^2)$ $\in \widehat{\mathbb{Z}}^*$
140-3-s163-D009 $\mu | \frac{1}{\mu}$ ε_3
140-3-s163-D010 [illeg. 3]

[scan 164]

The solutions are then obtained as follows: one chooses $\varepsilon_2, \varepsilon_3, \mu | \mu_2$ as above (two possibilities), and arbitrarily $r, s \in \widehat{\mathbb{Z}}$ satisfying

$$ur - ts = 1$$

(there are always some (for $r^2+s^2 \in \widehat{\mathbb{Z}}^*$ — hence $\forall p \in \mathbb{P}$, r_p and s_p are not both in $p\mathbb{Z}_p$) and if r_0, s_0 is a solution, the others are obtained in parametric form in λ

$$(r,s) = (r_0,s_0) +|- \lambda(t|u, u|t) = (r_0 +|- \lambda t, s_0 +|- \lambda u). \tag{87}$$

Having chosen r, s hence $f = \begin{pmatrix} r & s \\ t & u \end{pmatrix} \in \mathrm{Sl}(2, \widehat{\mathbb{Z}})$,

† 163

Uncertainty retained from cancelled passages.
140-3-s164-D002 r U t^2 [illeg. 1] t^2+ to

[scan 165]

it remains to choose a, b so as to satisfy the two relations $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \varepsilon_3$ and $C = \varepsilon_2 T$ (cf. [p. 381] (61) (62)) i.e.

$$\begin{cases} d\underline{a} - c\underline{b} = \varepsilon_3 \\ (c+d)\underline{a} + d\underline{b} = \varepsilon_2\varepsilon_3 \underbrace{(\underline{r}t + \underline{s}u)}_{S} + \underbrace{(c^2+cd+d^2)}_{1/\mu\varepsilon_3} \end{cases} \tag{88}$$

which is a system of two linear equations in $\underline{a}, \underline{b}$, whose determinant is as if by chance $c^2+cd+d^2 = 1/\mu\varepsilon_3 \in \widehat{\mathbb{Z}}^*$ — which ensures a unique solution!

It seems more natural perhaps to start from the multiplier μ, and to look for the systems $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} r & s \\ t & u \end{pmatrix}$ corresponding to it, but putting the emphasis on (c,d) (t,u) in the first place. Using μ one determines $\varepsilon_2, \varepsilon_3$ by

$$\mu \equiv \varepsilon_2 \ (2), \qquad \mu \equiv \varepsilon_3 \ (3),$$

† which allows one to † represent $1/\mu$ in the form

$$1/\mu = \varepsilon_2(t^2+u^2) = \varepsilon_3(c^2+cd+d^2)$$

(with $t, u, c, d \in \widehat{\mathbb{Z}}$, the systems (t,u) as well as (c,d) "depend on 1 parameter", each in a very concrete sense that one could probably make precise if one cared to, †with explicit parameters † σ, τ… Once c, d, t, u have been chosen, one chooses $r, s|t$ as has 164

Uncertainty retained from cancelled passages.
140-3-s165-D001 then
140-3-s165-D002 define
140-3-s165-D003 a wide choice
140-3-s165-D004 [illeg. 1]

[scan 166]

been said (with a parameter λ entering linearly), which then determines r, s uniquely, in a way that involves λ in a linear-affine way.

In fact, for c, d, t, u and $\varepsilon_2, \varepsilon_3$ fixed, so as to satisfy: $\varepsilon_2(t^2+u^2) = \varepsilon_3(c^2+cd+d^2)$, the three equations relating a, b, r, s are linear-affine

$$\begin{cases} d\underline{a} - c\underline{b} = \varepsilon_3 \\ u\underline{r} - t\underline{s} = 1 \\ (c+d)\underline{a} + d\underline{b} - \varepsilon_2\varepsilon_3(u\underline{s} + t\underline{r}) = c^2+cd+d^2 \end{cases} \tag{89}$$

m862

Marginal note. m862 we have underlined the "unknown coefficients" $\underline{a}, \underline{b}, \underline{r}, \underline{s}$ to be determined

and the system is "of rank 3" † — the set of its solutions is an "affine space of dimension 1" over $\widehat{\mathbb{Z}}$…

Remark We have not for the moment taken account of the congruence conditions

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv 1 \ (2) \qquad \begin{pmatrix} r & s \\ t & u \end{pmatrix} \equiv 1 \ (2)$$

which express that these matrices arise modulo signs by "reduction" from elements of $\widehat{\widetilde{\Pi}}_{0,3}$, $\widehat{\Pi}_{0,3}$. For the quantities (c, d) † one finds the consequences

$$\begin{cases} d \equiv \pm 1 \ (4) & \text{hence } d^2 \equiv 1 \ (4) \\ c \equiv 0, 2 \ (4) & \text{hence } c^2 \equiv 0 \ (4) \end{cases} \Longrightarrow \begin{array}{r} c^2 + d^2 + cd \equiv 1 + cd \ (4) \\ \equiv 1 + c \ (4) \end{array}$$

hence

$$c^2 + d^2 + cd \equiv 1 + c \ \dagger(4) \quad \dagger \quad \text{hence}$$

hence

$$\mu = \varepsilon_3(c^2 + cd + d^2) \equiv \dagger\varepsilon_3(1 + c) \ (4)$$

165

Uncertainty retained from cancelled passages.

140-3-s166-D004 det
140-3-s166-D005 fixed
140-3-s166-D006 (4)
140-3-s166-D007 $\Longleftrightarrow \quad cd$
140-3-s166-D008 $\Longleftrightarrow c \equiv 0 \ (4)$

[scan 167]

We thus see that knowledge of c † modulo 4, and that of the sign $ad - bc = \varepsilon_3$, allows one to find the sign ε_2 † by [p. 382]

$$\overset{\dagger}{\varepsilon_2 \equiv \varepsilon_3(1 + c) \ (4)} \quad \text{i.e.} \quad \varepsilon_2\varepsilon_3 \equiv 1 + c \ (4) \tag{90}$$

If one sets out, for c, d, t, u fixed, satisfying

$$\begin{cases} c \equiv t \equiv 0 \ (2) \qquad d \equiv u \equiv 1 \ (2) \\ (c^2 + cd + d^2) \equiv | = \dagger\varepsilon(t^2 + u^2) \in \widehat{\mathbb{Z}}^* \end{cases} \tag{91}$$

where ε $\dagger\in \{\pm 1\}$ is the sign defined by

$$\varepsilon \equiv 1 + c \ (4)$$

($\varepsilon = 1$ if $c \equiv 0 \ (4)$, $\varepsilon = -1$ if $c \equiv 2 \ (4)$), to find the corresponding a, b, r, s, satisfying conditions (89), and moreover the congruence conditions

$$b \equiv s \equiv 0 \ (2) \qquad a \equiv r \equiv 1 \ (2), \tag{92}$$

one is led to lift an already given solution modulo 2 of this system of equations to a solution. In terms of the $\widehat{\mathbb{Z}}$- torsor $\mathcal{T}$ of solutions, this means that one gives oneself an element of the corresponding quotient $\mathbb{Z}/2\mathbb{Z}$-torsor — the set of its liftings is identified with a torsor under $2\widehat{\mathbb{Z}} \simeq \widehat{\mathbb{Z}}$… No mystery!

166

Uncertainty retained from cancelled passages.

140-3-s167-D001 [illeg. 1]
140-3-s167-D002 by
140-3-s167-D003 ε_2 c
140-3-s167-D004 $(1+c$
140-3-s167-D005 [illeg. 1]

[scan 168]

Remark The additive group $\widehat{\mathbb{Z}}$ operates on the set of solutions $(\alpha_0, f) = \left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}, \begin{pmatrix} r & s \\ t & u \end{pmatrix}\right)$ of the equations under consideration (48)(49)(56)

$$\det \alpha_0 \in \{\pm 1\}, \quad \det f = 1, \quad \alpha_0 \rho(\alpha_0)^{-1} = \varepsilon \ell_0^{\nu}(f \sigma_\infty(f)^{-1}), \quad \varepsilon \in \{\pm 1\}, \tag{93}$$

by

$$\begin{aligned}(\alpha_0, f) \longmapsto (\alpha_0', f') &= (\ell_0^n \alpha_0, \ell_0^n f) \quad \dagger \\ &= \left(\begin{pmatrix} a-2nc & b-2nd \\ c & d \end{pmatrix}, \begin{pmatrix} r-2nt & s-2nu \\ t & u \end{pmatrix}\right)\end{aligned} \tag{93|94}$$

($n \in \widehat{\mathbb{Z}}$). Up to the factor 2 in $2n$, this is precisely the indeterminacy in † the set of solutions of (93), when c, d, t, u are fixed (cf. (87) and the proposition that goes with it). (the factor 2 is ensured only when one considers|in considering solutions satisfying $\alpha_0' \equiv f_0' \equiv \mathrm{Id}$ (2)) One can † say that what is determined intrinsically in $\mathrm{Gl}(2, \widehat{\mathbb{Z}})^{\pm 1\prime}|\mathrm{Gl}(2, \widehat{\mathbb{Z}})^{\pm 1}$ in terms of the loop automorphism $\underline{u}$ of $\widehat{\Pi}_{0,3}$ commuting with σ_∞, is neither more nor less than †

$$\text{a)} \quad \text{the multiplier } \mu \in \widehat{\mathbb{Z}}^* \tag{95}$$

† (hence|whence $\varepsilon_2 = \varepsilon_2|\xi_2(\mu)$, $\varepsilon_3 = \varepsilon_3|\xi_3(\mu)$)

b) the quantities † the $c, d, t, u \in \widehat{\mathbb{Z}}$ satisfying the everlasting relation (86)

$$\varepsilon_2(t^2 + u^2) = \varepsilon_2(c^2 + cd + d^2) = 1/\mu \tag{96}$$

and the congruence relations

$$c \equiv t \equiv 0 \ (2) \qquad d \equiv u \equiv 1 \ (2) \tag{97}$$

It is true that we have seen that there is a natural way in $\widehat{\Pi}_{0,3}$ to normalize the choice of the f, so as to remove the ambiguity|ambiguity of the $\ell_0^n f$ (cf. § 41 (24bis) where this was touched upon — I will come back to it again later…) but it seems to me not very interesting 167

Uncertainty retained from cancelled passages.

140-3-s168-D001 a
140-3-s168-D002 [illeg. 1]
140-3-s168-D003 thus
140-3-s168-D004 values
140-3-s168-D005 which gives $\varepsilon_2|\xi_2$ $\varepsilon_3|\xi_3$
140-3-s168-D006 δ ε_3 satisfying

[scan 169]

for the moment to ignore this possibility of normalization – which brings out more clearly, indeed, the crucial character, in all these l-adic calculations, of the role played by the four quantities $c, d, t, u \in \widehat{\mathbb{Z}}$. [p. 383]

[scan 170]
[p. 384]

45 The homomorphism $\widetilde{M}_{0,3} \longrightarrow \mathrm{Gl}(2,\widehat{\mathbb{Z}})'$, construction of the extension † $\widetilde{N}_{1,1}$ of $\tilde{\Gamma}$ by $\mathrm{Sl}(2,\mathbb{Z})^\wedge$.

We denote by $\widetilde{M}_{0,3}$ the group of loop automorphisms of $\widehat{\mathcal{G}}^+_{0,3}$ which † induce up to conjugacy † on the quotient $\mathfrak{S}_3|\mathcal{G}_3 \simeq \widehat{\mathcal{G}}^+_{0,3}/\widehat{\Pi}_{0,3}$ the trivial action – we have thus a structure of an extension

$$(1) \qquad 1 \longrightarrow \widehat{\mathcal{G}}^+_{0,3} \longrightarrow \widetilde{M}_{0,3} \longrightarrow \widetilde{N}_{0,3} \longrightarrow 1,$$

m866

Marginal note. m866 (We shall denote by $\widetilde{M}^!_{0,3}$ the subgroup consisting of the automorphisms which induce strict^{ly} the ident^{ity} in $\mathfrak{S}_3|\mathcal{G}_3$ – it corresponds to $\widetilde{N}^!_{0,3}$ – automorphisms that are outer loop automorphisms fixing the loop classes (and †|induce the identity on $\mathfrak{S}_3|\mathcal{G}_3$ up to conjug^{acy}))

$\widetilde{N}_{0,3}$ denotes the group of outer automorphisms of $\widehat{\mathcal{G}}^+_{0,3}$ which † induce the identity in $\mathfrak{S}_3|\mathcal{G}_3$. We note that

$$\Gamma = \Gamma_{\mathbb{Q}} \hookrightarrow \tilde{\Gamma},$$

and that the inverse image of $\Gamma_{\mathbb{Q}}$ in the extension (1) is none other than $M_{0,3}$ – but for the moment I have no very plausible characterization to propose of $\Gamma_{\mathbb{Q}}$ as a subgroup of $\tilde{\Gamma}$. We have a canonical splitting of the extension (1), via the subgroup $\tilde{\Gamma}_\sigma$ of M consisting of the automorphisms $u \in \widetilde{M}_{0,3}$ which commute with σ_∞,

Uncertainty retained from cancelled passages.

140-3-s170-D001 [illeg. 1]
140-3-s170-D003 commute
140-3-s170-D004 $\mathfrak{S}_3|\mathcal{G}_3$ which is
140-3-s170-D005 induce
140-3-s170-D006 [illeg. 3]

[scan 171]

so that $\widetilde{M}_{0,3}$ presents itself as a $\frac{1}{2}$ direct product

$$(3) \qquad \widetilde{M}_{0,3} \simeq \underbrace{\widehat{\mathcal{G}}^+_{0,3}}_{\mathrm{Sl}(2,\widehat{\mathbb{Z}})'^\wedge} \cdot \tilde{\Gamma}_\sigma.$$

Let us consider the canonical homomorphism

$$(4) \qquad \widehat{\mathcal{G}}^+_{0,3} \simeq \mathrm{Sl}(2,\mathbb{Z})'^\wedge \xrightarrow{\varphi'_0} \mathrm{Sl}(2,\widehat{\mathbb{Z}})' \subset \mathrm{Gl}(2,\widehat{\mathbb{Z}})'$$

and the map

$$(5) \qquad \begin{aligned} \tilde{\Gamma}_\sigma &\xrightarrow{\varphi'_\sigma} \mathrm{Gl}(2,\widehat{\mathbb{Z}})' \\ \underline{u} &\longmapsto \mu(t\sigma_\infty + u)\tau_\infty^\varepsilon = \mu \begin{pmatrix} u & \varepsilon t \\ -t & \varepsilon u \end{pmatrix}, \end{aligned}$$

where ††

$$t, u \in \widehat{\mathbb{Z}}, \quad \mu \in \widehat{\mathbb{Z}}^*, \quad \varepsilon = \varepsilon_2 \in \{\pm 1\}, \quad \alpha = \alpha_2 \in \mathbb{Z}/2\mathbb{Z}$$

are the quantities (depending on $\underline{u} \in \tilde{\Gamma}_\sigma$) considered in the § above, with

$$\varepsilon = (-1)^\alpha.$$

If need be, we shall write $t(\underline{u})$, $\underline{u}(u)$, $\alpha(\underline{u})$, $\varepsilon(\underline{u})$ to indicate the dependence with respect to $\underline{u}$.

Let us consider the (continuous) map

$$(6) \qquad \varphi' : \widetilde{M}_{0,3} \longrightarrow \mathrm{Gl}(2,\widehat{\mathbb{Z}})'$$

defined by

$$(7) \qquad \varphi'(a.\underline{u}) = \varphi'_0(a)\varphi'_\sigma(\underline{u}) \qquad \begin{cases} a \in \widehat{\mathcal{G}}^+_{0,3} \\ \underline{u} \in \tilde{\Gamma}_\sigma \end{cases}$$

Uncertainty retained from cancelled passages.

140-3-s171-D002 $t, u, \alpha = \alpha_2$ [illeg. 2]

140-3-s171-D003 $\varepsilon = \varepsilon_2$

[scan 172]

[p. 385]

Theorem The map φ' is a group homomorphism.

This means, taking into account that φ_0 is already a group homomorphism, the theorem means that

a) φ'_σ is a group homomorphism, i.e.

$$(8) \qquad \varphi'_\sigma(\underline{u}'\underline{u}) = \varphi'_\sigma(\underline{u}')\,\varphi'_\sigma(\underline{u}) \qquad (\underline{u}, \underline{u}' \in \tilde{\Gamma}_\sigma)$$

b) we have, for $a \in \mathcal{G}^+_{0,3} | \widehat{\mathcal{G}}^+_{0,3}$, $\underline{u} \in \tilde{\Gamma}_\sigma$

$$(9) \qquad \varphi'_0(\underbrace{\underline{u}a\underline{u}^{-1}}_{\underline{u}(a)}) = \varphi'_\sigma(\underline{u})\varphi'_0(a)\varphi'_\sigma(\underline{u}^\dagger)^{-1}.$$

But the formula (9) is a fact already established, † since the † very construction of the matrix $\varphi'_\sigma(\underline{u}) = \mu(t\sigma_\infty + u)\tau^\varepsilon_\infty$ in the preceding paragraphs was made in a way such that this formula should be verified (cf. (67)(68), and the corollary to theorem (77)…). To verify (8), we shall first make explicit the quantities g, h, f for a product of two operators $u', u'' | u', u \in \widetilde{M}_{0,3}$, in terms of their values for u', u themselves. An immediate calculation gives, in this sense, some "cocycle relations"

$$(10) \qquad \begin{cases} g(\underline{u}'\underline{u}) = \underline{u}'(g(\underline{u})).g(\underline{u}') \\ h(\underline{u}'\underline{u}) = \underline{u}'(h(\underline{u})).h(\underline{u}') \\ f(\underline{u}'\underline{u}) = f(\underline{u}').\underline{u}'(f(\underline{u})) \end{cases}$$

Uncertainty retained from cancelled passages.

140-3-s172-D001 -1

140-3-s172-D003 consequence of (8) [illeg. 1]

140-3-s172-D004 [illeg. 1]

[scan 173]

Remarks 1) We shall moreover have likewise, with the notations of the § above

$$(11) \qquad \begin{cases} \alpha(u'u) = u'(\alpha(u)).\alpha(u') \\ \beta(u'u) = u'(\beta(u)).\beta(u') \end{cases}$$

† 2) The formulas (10) are a little more friendly|friendly for f, than for g and h, α and β (– one would feel like changing notation yet again, and also replacing g and h by g^{-1}, h^{-1} (as previously a g_0 had become f^{-1}…).

Let us again denote by the same letters $g(u), h(u)$ etc. the images of these quantities in $\mathrm{Sl}(2, \widehat{\mathbb{Z}})'$, † the analogous formulas in $\mathrm{Sl}(2, \widehat{\mathbb{Z}})'$ (or $(\mathrm{Gl}(2, \widehat{\mathbb{Z}})^{\pm 1})'$ for † the α of (11)) which are written

$$(12) \qquad \begin{cases} g(\underline{u}'\underline{u}) = \mathrm{int}(\varphi'_\sigma(\underline{u}'))(g(\underline{u})).g(\underline{u}') \\ h(\underline{u}'\underline{u}) = \mathrm{int}(\varphi'_\sigma(\underline{u}'))(h(\underline{u})).h(\underline{u}') \\ f(\underline{u}'\underline{u}) = \dagger f(\underline{u}').(\mathrm{int}(\varphi_\sigma(\underline{u}'))(f(\underline{u}))) \\ \alpha(\underline{u}'\underline{u}) = \mathrm{int}\,\varphi'_\sigma(\underline{u}')(\alpha(\underline{u})).\alpha(\underline{u}') \\ \beta(\underline{u}'\underline{u}) = \mathrm{int}\,\varphi'_\sigma(\underline{u}')(\beta(\underline{u})).\beta(\underline{u}') \end{cases}$$

in $\mathrm{Gl}(2,\widehat{\mathbb{Z}})'$.

These formulas in $\mathrm{Gl}(2,\widehat{\mathbb{Z}})'$ do indeed have an entirely explicit character from the point of view of effective computation, unlike

Uncertainty retained from cancelled passages.
140-3-s173-D001 (N.B.
140-3-s173-D002 (10) give
140-3-s173-D003 (11)
140-3-s173-D004 $\mathrm{int}(\varphi'_\sigma(\underline{u}'))$

[scan 174]
[p. 386]

the case of the initial formulas in $\widehat{\Pi}_{0,3}$ or $\widehat{\widetilde{\Pi}}_{0,3}$, where quantities like $\underline{u}'(\underbrace{g(\underline{u})}_{g})$ cannot be made explicit † by simple algebraic formulas in terms of the data g, h, f, g', h', f' † $= g(\underline{u})$ etc. which determine $\underline{u}$ and $\underline{u}'$.

These formulas are supplemented, for the record, by

$$\mu(\underline{u}'\underline{u}) = \mu(\underline{u}')\mu(\underline{u}) \tag{13}$$

implying in particular

$$\varepsilon_2(\underline{u}'\underline{u}) = \varepsilon_2(\underline{u}')\varepsilon_2(\underline{u}). \tag{14}$$

To carry out the calculation of $\varphi_\sigma(\underline{u}' \,.\, \underline{u})$ and verify that it is equal to $\varphi_\sigma(\underline{u}')\varphi_\sigma(\underline{u})$, we † shall introduce the coefficients r, s, t, u of $f = f(\underline{u})$, r', s', t', u' of $f' = f(\underline{u}')$, and r'', s'', t'', u'' of $f'' = f(\underline{u}'\underline{u})$, likewise μ, μ', μ'', $\varepsilon, \varepsilon', \varepsilon''$, whence

$$\dagger \qquad \mu'' = \mu\mu', \quad \varepsilon'' = \varepsilon\varepsilon' \tag{15}$$

and the formula in f in (12) is therefore written

$$f'' = f'\underline{u}' f \underline{u}'^{-1} \tag{16}$$

where we have written, as in the § above, $\underline{u}'$ instead of $\varphi_\sigma(\underline{u}')$, † In explicit terms

$$\begin{pmatrix} r'' & s'' \\ t'' & u'' \end{pmatrix} = \begin{pmatrix} r' & s' \\ t' & u' \end{pmatrix} \mu' \begin{pmatrix} u' & \varepsilon' t' \\ -t' & \varepsilon' u' \end{pmatrix} \begin{pmatrix} r & s \\ t & u \end{pmatrix} \dagger | \frac{\mu'}{\mu'} \begin{pmatrix} \varepsilon' u' & -\varepsilon' t' \\ t' & u' \end{pmatrix}$$

N.B. $\det \underline{u}' = \mu'$

Uncertainty retained from cancelled passages.
140-3-s174-D001 [illeg. 1]
140-3-s174-D002 resp.
140-3-s174-D003 shall have
140-3-s174-D004 relations
140-3-s174-D005 In
140-3-s174-D006 $\frac{\mu'}{\mu'}$

[scan 175]

The product of the last three terms on the right-hand side is not encouraging:

$$\underline{u}' f \underline{u}'^{-1} = \mu' \dagger \begin{pmatrix} \varepsilon' r u'^2 + s t' u' + t t' u' + \varepsilon' u t'^2 & -\varepsilon' r t' u' + s u'^2 - t t'^2 + \varepsilon' u u' t' \\ -\varepsilon' r u' t' - s t'^2 + t u'^2 + \varepsilon' u u' t' & -\varepsilon' r t'^2 - s u' t' - t u' t' + \varepsilon' u u'^2 \end{pmatrix},$$

but the product with f' simplifies miraculously, thanks to the relation

$$t'^2 + u'^2 = \varepsilon'/\mu'$$

and we find

$$\dagger \qquad f'' = \begin{pmatrix} r'' & s'' \\ t'' & u'' \end{pmatrix} = \dagger \begin{pmatrix} \varepsilon|* \text{ mess} & \text{mess} \\ \varepsilon' tu' + ut' & -\varepsilon' tt' + uu' \end{pmatrix}$$

whence

$$t'' = \frac{1}{\mu'}(\varepsilon' tu' + ut') \qquad u'' = \frac{1}{\mu'}(-tt' + uu') \tag{18}$$

and hence (taking account of $\mu'' = \mu\mu'$, $\varepsilon'' = \varepsilon\varepsilon'$)

$$\dagger \qquad \underline{u}'' = \mu\mu' \dagger \begin{pmatrix} -\varepsilon' tt' + uu' & \varepsilon tu' + \varepsilon\varepsilon' ut' \\ -\varepsilon' tu' - ut' & -\varepsilon\varepsilon' tt' + \varepsilon\varepsilon' uu' \end{pmatrix}$$

and the formula to be checked u,

$$\underline{u}'' = \underline{u}'\underline{u}$$

reduces to

$$\mu\mu' \begin{pmatrix} -tt' + uu' & \varepsilon tu' + \varepsilon\varepsilon' ut' \\ -\varepsilon' tu' \dagger - ut' & -\varepsilon \dagger tt' + \varepsilon\varepsilon' uu' \end{pmatrix} = \mu' \begin{pmatrix} u' & \varepsilon' t' \\ -t' & \varepsilon' u' \end{pmatrix} \mu \begin{pmatrix} u & \varepsilon t \\ -t & \varepsilon u \end{pmatrix}$$

which is indeed true, OK !.

Here again, one feels that these calculations ought to fit into a carpet of linear algebra that I am too lazy to unroll in extenso. One would have to paraphrase the group Or rather $\widetilde{M}_{0,3}$ – described by these systems (g, h, f) – in terms (α, β, f) of the reductions of g, h, f

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Uncertainty retained from cancelled passages.

140-3-s175-D001 $\frac{1}{\mu'}$
140-3-s175-D002 (17)
140-3-s175-D003 $\frac{1}{\mu'}$
140-3-s175-D004 (19)
140-3-s175-D005 μ'
140-3-s175-D006 ε'
140-3-s175-D007 ε'

[scan 176]
[p. 387]

in $\mathrm{Gl}(2, \widehat{\mathbb{Z}})^{\pm 1\prime}_{\mathfrak{g}}$, then replacing $\widehat{\mathbb{Z}}$ by an arbitrary ring, and of course $\mathrm{Gl}(2, \widehat{\mathbb{Z}})'$ by $\mathrm{Gl}(2, A|\widehat{\mathbb{Z}})$ itself. Perhaps I will come back to this eventually....

Having thus constructed

$$\varphi : \widetilde{M}_{0,3} \longrightarrow \mathrm{Gl}(2, \widehat{\mathbb{Z}})' = \mathrm{Gl}(2, \widehat{\mathbb{Z}})/\pm 1$$

we deduce from it, by inverse image of the central extension $\mathrm{Gl}(2, \widehat{\mathbb{Z}})$ of $\mathrm{Gl}(2, \widehat{\mathbb{Z}})'$ by $\dagger\, \{\pm 1\}$, a central extension of $\widetilde{M}_{0,3}$ by $\{\pm 1\}$, which I denote by $\widetilde{N}_{1,1}$, and we have a homomorphism of extensions

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \{\pm 1\} & \longrightarrow & \widetilde{N}_{1,1} & \longrightarrow & \dagger \widetilde{M}_{0,3} & \longrightarrow & 1 \\ & & \| & & \downarrow \varphi & & \downarrow \varphi' & & \\ 1 & \longrightarrow & \{\pm 1\} & \longrightarrow & \mathrm{Gl}(2, \widehat{\mathbb{Z}}) & \longrightarrow & \mathrm{Gl}(2, \widehat{\mathbb{Z}})' & \longrightarrow & 1 \end{array} \tag{19}$$

$\dagger$

Uncertainty retained from cancelled passages.
140-3-s176-D001 $\mathbb{Z}$
140-3-s176-D002 $\mathcal{T}$
140-3-s176-D003 $\widetilde{N}_{1,1}$ $\widetilde{\mathcal{T}}^{+}_{1,1}|\widehat{\mathcal{T}}^{+}_{1,1}$ $\widetilde{N}_{1,1}$ $\widehat{\mathrm{Sl}(2,\mathbb{Z})}'$ $\mathcal{T}$ $\widehat{\mathcal{T}}^{+}_{1,1}|\widetilde{\mathcal{T}}^{+}_{1,1}$

[scan 177]

One should check that the homomorphism φ' we have just constructed coincides on $M_{0,3} \subset \widetilde{M}_{0,3}$ with the well-known "modular" homomorphism of § 39 (1) – it suffices to check it on the subgroup $\Gamma_{\mathbb{Q},\sigma}$ of $\tilde{\Gamma}_\sigma$ corresponding to $\Gamma_{\mathbb{Q}}$ – this should not be very difficult, and I will assume it here. This † assumed, we find that the inverse image of $M_{0,3}$ in † $\widetilde{N}_{1,1}$ is none other (up to canonical isomorphism) than $N_{1,1}$ itself – in any case the inverse image of $\widehat{\mathcal{G}}^{+}_{0,3}$ † is none other than $\widehat{\mathcal{T}}^{+}_{1,1} \simeq \mathrm{Sl}(2,\mathbb{Z})^\wedge$. We will then have (with this identification)

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \widehat{\mathcal{T}}^{+}_{1,1} \\ & & \| \ \mathrm{Sl}(2,\mathbb{Z})^\wedge & \longrightarrow & \dagger\widetilde{N}_{1,1} & \longrightarrow & \tilde{\Gamma} & \longrightarrow & 1 \\ & & \downarrow & & \downarrow{\scriptstyle\varphi} & & \downarrow{\scriptstyle\chi} \\ 1 & \longrightarrow & \mathrm{Sl}(2,\widehat{\mathbb{Z}}) & \longrightarrow & \mathrm{Gl}(2,\widehat{\mathbb{Z}}) & \xrightarrow{\det} & \widehat{\mathbb{Z}}^* & \longrightarrow & 1 \end{array} \tag{20}$$

Uncertainty retained from cancelled passages.
140-3-s177-D001 being
140-3-s177-D002 $\mathcal{T}$
140-3-s177-D003 on
140-3-s177-D004 $\mathcal{G}$

[scan 178]

so we have reconstructed|reconstructed in terms of loop automorphisms of $\widehat{\Pi}_{0,3}$ the diagram [p. 388]
(1) of § 39, or rather a diagram a larger, into which that of § 39 embeds – it would still remain to characterize $\Gamma_{\mathbb{Q}}$ as a subgroup of $\tilde{\Gamma}$.

The diagram deduced directly from the preceding theorem

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \widehat{\mathcal{G}}^{+}_{0,3} & \longrightarrow & \widetilde{M}_{0,3} & \longrightarrow & \tilde{\Gamma} & \longrightarrow & 1 \\ & & \downarrow & & \downarrow{\scriptstyle\varphi'} & & \downarrow{\scriptstyle\chi} \\ 1 & \longrightarrow & \mathrm{Sl}(2,\widehat{\mathbb{Z}})' & \longrightarrow & \mathrm{Gl}(2,\widehat{\mathbb{Z}})' & \xrightarrow[\det]{} & \widehat{\mathbb{Z}}^* & \longrightarrow & 1 \end{array} \tag{21}$$

can be regarded as being deduced from the preceding one by reduction modulo the central subgroups $\{\pm 1\}$ in the two rows.

What becomes in (20) of the splitting (3) of $\widetilde{M}_{0,3}$? Is there a privileged way, for every element of $\widetilde{N}_{1,1}$ – i.e. a pair of a $u = (g,h,f) \in \widetilde{M}_{0,3}$, and a $\underline{u}_0 \in \mathrm{Gl}(2,\widehat{\mathbb{Z}})$

[scan 179]

which lifts the element $\underline{u} = \varphi'_\sigma(u)$ of $\mathrm{Gl}(2,\widehat{\mathbb{Z}})'$, to "normalize" it modulo † composition with an element of $\mathrm{Sl}(2,\mathbb{Z})^\wedge$, so that it has (say) a commutation property with σ_∞, of the type $u(\sigma_\infty) = \sigma_\infty$, † $u_0|u(\sigma_\infty) = \pm\sigma_\infty$?

Another way of posing the question, in terms of the subgroup $\tilde{\Gamma}_\sigma \subset \widetilde{M}_{0,3}$, is the following: is there a canonical lifting $\tilde{\Gamma}_\sigma \longrightarrow \widetilde{N}_{1,1}$, thus assigning to every $\underline{u} = (g,f) \in \tilde{\Gamma}_\sigma$ (That is a $\underline{u} = \mu(t\sigma_\infty + u)\tau^\alpha_\infty \in \mathrm{Gl}(2,\widehat{\mathbb{Z}})',|\}$ † i.e. a $\underline{u}$ defined modulo sign) a representative $\underline{u}_0 \in \mathrm{Gl}(2,\widehat{\mathbb{Z}})$ – i.e. a way of removing this ambiguity of sign? In fact, if we replace $\tilde{\Gamma}_\sigma$ by $\Gamma_{\mathbb{Q},\sigma}$, the Standard †yogas suggest that the liftings in

Uncertainty retained from cancelled passages.
140-3-s179-D001 [illeg. 1]
140-3-s179-D004 [illeg. 1]
140-3-s179-D005 modulo
140-3-s179-D006 [illeg. 1]

[scan 180]

question should|must correspond to elliptic curves over $\mathbb{Q}$, equipped with a Jacobi rigidification of level 2, corresponding (under the isomorphism $\dagger\mathcal{M}'_{1,1\,\mathbb{Q}} \simeq U_{0,3\,\mathbb{Q}}$) to the point 1/2 of $U_{0,3}(\mathbb{Q})$. The indeterminacy in the choice of such an elliptic curve is precisely in $H^1(\mathbb{Q}, \{\pm 1\})$ (it is a datum of a quadratic extension of|to $\mathbb{Q}$) which is identified with $\mathrm{Hom}(\Gamma_{\mathbb{Q}}, \dagger\{\pm 1\})$ – so this is indeed also the indeterminacy for a lifting. This proves essentially that the idea suggested by the general yoga is indeed correct|correct – and this shows at the same time that there is no (even on $\Gamma_{\mathbb{Q},\sigma}$) really privileged choice. That there are such liftings (on $\Gamma_{\mathbb{Q},\sigma}$) comes from the fact that there are indeed elliptic [p. 389]

Uncertainty retained from cancelled passages.
140-3-s180-D001 $\mathcal{T}$
140-3-s180-D002 μ

[scan 181]

curves answering the question, the simplest being given by the equation

$$(22) \qquad y^2 - x(x-1)(x-1/2) = 0$$

– and there is the choice made ! It would remain to try to make it explicit in a computational formulary, if this is not hopeless. I should examine this question more closely, sometime.

[scan 182]

46 Definitive notations (?) – new recapitulation [p. 390]
(for the loop automorphisms of $\widehat{\Pi}_{0,3}$ and of $\widehat{\mathcal{G}}^{+}_{0,3} \simeq \mathrm{Sl}(2,\mathbb{Z})'^{\wedge}$).
Ⓘ The groups $\widetilde{M}_{0,3}$ and $\widetilde{N}^{!}_{0,3}$.
We set
(1) $\widetilde{M}_{0,3}$ = group of automorphisms of $\widehat{\mathcal{G}}^{+}_{0,3} \simeq \mathrm{Sl}(2,\mathbb{Z})^{\wedge\prime}$ which leave $\widehat{\Pi}_{0,3}$ invariant, and which respect the "loop structure" i.e. satisfy $u(\varepsilon_i|\varepsilon_0) = \mathrm{int}(f^{-1})(\varepsilon_i^{\mu})$, with $f \in \widehat{\mathcal{G}}^{+}_{0,3}$, $i \in \{0,1,\infty\}$, $\mu \in \widehat{\mathbb{Z}}^*$ (OBSERVATION $f \in \widehat{\Pi}_{0,3}$, and $f|f_0$ is then determined in $L_0\backslash\widehat{\Pi}_{0,3}$)
$\overset{\text{restr.}}{\underset{\sim}{\longrightarrow}}$ group of loop automorphisms of $\widehat{\Pi}_{0,3}$ which commute as outer actions with the natural outer action of $\mathfrak{S}_3|\mathcal{G}_3$.
m868

Marginal note. m868 N.B. $\widetilde{M}_{0,3}$ is also the normalizer of $\widehat{\mathcal{G}}^{+}_{0,3}$ in $\widehat{\widehat{\mathcal{G}}}_{0,3}$, and likewise $\widetilde{N}_{0,3}$ the normalizer|normalizer of $\widehat{\mathcal{T}}_{0,3} = \mathcal{T}_{0,3}$ in $\widehat{\widehat{\mathcal{T}}}_{0,3}$

We have an extension structure

$$(2) \qquad 1 \longrightarrow \widehat{\Pi}_{0,3} \longrightarrow \widetilde{M}_{0,3} \longrightarrow \widetilde{N}_{0,3} \longrightarrow 1,$$

where †

$$(3) \qquad \widetilde{N}_{0,3} \overset{\text{dfn}}{=} \widetilde{M}_{0,3}/\widehat{\Pi}_{0,3}$$

$\simeq$ group of outer loop automorphisms of $\widehat{\Pi}_{0,3}$, commuting with $\mathfrak{S}_3|\mathcal{G}_3$.

The natural homomorphism: $\mathfrak{S}_3|\mathcal{G}_3 \longrightarrow \mathrm{Aut}(\mathfrak{S}_3|\mathcal{G}_3)$ is an isomorphism

$$(4) \qquad \mathfrak{S}_3|\mathcal{G}_3 \overset{x\mapsto \mathrm{int}(x)}{\overset{\sim}{\longrightarrow}} \mathrm{Aut}(\mathfrak{S}_3|\mathcal{G}_3),$$

and the natural homomorphism

$$\widetilde{M}_{0,3} \longrightarrow \mathrm{Aut}(\mathfrak{S}_3|\mathcal{G}_3 = \widehat{\mathcal{G}}^{+}_{0,3}/\widehat{\Pi}_{0,3})$$

then defines a homomorphism

$$(5) \qquad \widetilde{M}_{0,3} \longrightarrow \mathfrak{S}_3|\mathcal{G}_3$$

which can also be interpreted as the composite

$$(6) \qquad \widetilde{M}_{0,3} \hookrightarrow \underset{\substack{\text{all the loop automorphisms}\\ \text{of } \widehat{\Pi}_{0,3}}}{\widehat{\widehat{\mathcal{G}}}_{0,3}} \xrightarrow[\text{loop classes}]{\text{action on the}} \mathfrak{S}_3|\mathcal{G}_3,$$

Uncertainty retained from cancelled passages.

140-3-s182-D002 N_0

[scan 183]

and which is trivial on $\widehat{\Pi}_{0,3}$, hence factors as

$$(7) \qquad \widetilde{N}_{0,3} \longrightarrow \mathfrak{S}_3|\mathcal{G}_3.$$

The kernels of (5), (7), are denoted $\widetilde{M}^{!}_{0,3}$, $\widetilde{N}^{!}_{0,3}$ respectively, so that we have a diagram of exact sequences

$$\begin{array}{ccccccccc} & & & & 1 & & & & \\ & & & & \downarrow & & & & \\ & & & & \Pi_{0,3} & & & & \\ & & & & \downarrow & & & & \\ (8) & & 1 & \longrightarrow & \widetilde{M}^{!}_{0,3} & \longrightarrow & \widetilde{M}_{0,3} & \longrightarrow & \mathfrak{S}_3|\mathcal{G}_3 & \longrightarrow & 1 \\ & & & & \downarrow & & \downarrow & & \| & & \\ (9) & & 1 & \longrightarrow & \widetilde{N}^{!}_{0,3} & \longrightarrow & \widetilde{N}_{0,3} & \longrightarrow & \mathfrak{S}_3|\mathcal{G}_3 & \longrightarrow & 1, \\ & & & & \downarrow & & & & & & \\ & & & & 1 & & & & & & \end{array}$$

(9) being interpretable as the quotient of (8) by $\widehat{\Pi}_{0,3}$.

Let us note also the corresponding extension structure

$$(9\text{bis}) \qquad 1 \longrightarrow \widehat{\Pi}_{0,3} \longrightarrow \widetilde{M}^{!}_{0,3} \longrightarrow \widetilde{N}^{!}_{0,3} \longrightarrow 1$$

We also have a canonical homomorphism

$$(10) \qquad \mathfrak{S}_3|\mathcal{G}_3 \simeq \widehat{\mathcal{G}}^{+}_{0,3}/\widehat{\Pi}_{0,3} \hookrightarrow \widetilde{N}_{0,3},$$

whose composite with 7 is the identity, whence a canonical isomorphism

$$(11) \qquad \widetilde{N}_{0,3} \simeq \mathfrak{S}_3|\mathcal{G}_3 \times \widetilde{N}^{!}_{0,3}$$

On the other hand, we also have an inclusion

$$(12) \qquad \begin{array}{c} \mathcal{T}_{0,3} \\ \wr\downarrow\simeq \\ \mathfrak{S}_3|\mathcal{G}_3\times\mathbb{Z}/2\mathbb{Z} \end{array} \hookrightarrow \widetilde{N}_{0,3}$$

which on the factor $\mathfrak{S}_3|\mathcal{G}_3 = \mathcal{T}^+_{0,3}$ coincides with (10), and which maps the nontrivial element τ of $\mathbb{Z}/2\mathbb{Z} = \mathcal{T}^!_{0,3}$ to the element, also denoted τ, of the second factor $\widetilde{N}^!_{0,3}$ of (11).

[scan 184]
[p. 391]

The composite homomorphism

$$(13) \qquad \widetilde{M}_{0,3} \hookrightarrow | \subset \widehat{\widehat{\mathcal{G}}}_{0,3} \xrightarrow[\chi]{\text{multiplier}} \widehat{\mathbb{Z}}^*$$

† a homomorphism

$$(14) \qquad \widetilde{M}_{0,3} \overset{\chi}{\to} \widehat{\mathbb{Z}}^*$$

surjective, factoring through $\widetilde{N}_{0,3}$ and even through $\widetilde{N}_{0,3}/\mathfrak{S}_3|\mathcal{G}_3$

$$(15) \qquad \begin{array}{ccc} \widetilde{N}_{0,3} & \xrightarrow{\quad\chi\quad} & \widehat{\mathbb{Z}}^* \\ & \searrow \quad \nearrow & \\ & \widetilde{N}_{0,3}/\mathfrak{S}_3|\mathcal{G}_3 & \\ & \simeq \widetilde{N}^!_{0,3} & \end{array} ,$$

and giving rise to extension structures

$$(16) \qquad \begin{array}{ccccccccc} 1 & \longrightarrow & \widetilde{M}^+_{0,3} & \longrightarrow & \widetilde{M}_{0,3} & \xrightarrow{\chi} & \widehat{\mathbb{Z}}^* & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \| & & \\ 1 & \longrightarrow & \widetilde{N}^+_{0,3} & \longrightarrow & \widetilde{N}_{0,3} & \xrightarrow{\chi} & \widehat{\mathbb{Z}}^* & \longrightarrow & 1 \end{array} ,$$

where we have set

$$(17) \qquad \begin{cases} \widetilde{M}^+_{0,3} = \mathrm{Ker}(\widetilde{M}_{0,3} \to \widehat{\mathbb{Z}}^*) \\ \widetilde{N}^+_{0,3} = \mathrm{Ker}(\widetilde{N}_{0,3} \to \widehat{\mathbb{Z}}^*) \end{cases}$$

– the second row of (16) may be interpreted as the quotient of the first row by $\widehat{\Pi}_{0,3}$.

We have a canonical embedding

$$(18) \qquad \Gamma_{\mathbb{Q}} \overset{\text{dfn}}{=} \mathrm{Gal}\left(\underbrace{\overline{\mathbb{Q}}_0}_{\substack{\text{algebraic} \\ \text{closure of } \mathbb{Q} \\ \text{in } \mathbb{C}}} / \mathbb{Q}_0 \right) \hookrightarrow | \subset \widetilde{N}^!_{0,3}$$

sending the automorphism of complex

Uncertainty retained from cancelled passages.
140-3-s184-D001 gives|defines

[scan 185]

conjugation τ to the element already denoted (τ) (cf. (12)). The inverse image of this|the subgroup $\Gamma_{\mathbb{Q}}$ †of $\widetilde{N}^{!}_{0,3}$ under $\widetilde{M}_{0,3} \to \widetilde{N}^{!}_{0,3} \simeq \widetilde{N}_{0,3}/\mathfrak{S}_3|\mathcal{G}_3$, i.e. the inverse image of †

$$(19) \qquad N_{0,3} \overset{\text{dfn}}{=} \mathfrak{S}_3|\mathcal{G}_3 \times \Gamma_{\mathbb{Q}} \subset \widetilde{N}_{0,3} = \mathfrak{S}_3|\mathcal{G}_3 \times \widetilde{N}^{!}_{0,3},$$

is the extension † denoted $M_{0,3}$ elsewhere:
m869

Marginal note. m869 [We have a †sub-extension of the extension $\widetilde{M}_{0,3}$ of (2)]

$$(20) \qquad \begin{array}{ccccccccc} 1 & \longrightarrow & \widehat{\Pi}_{0,3} & \longrightarrow & M_{0,3} & \longrightarrow & \overbrace{\mathfrak{S}_3|\mathcal{G}_3 \times \Gamma_{\mathbb{Q}}}^{N_{0,3}} & \longrightarrow & 1 \\ & & \downarrow{\scriptstyle \iota} & & \downarrow\downarrow & & \downarrow{\scriptstyle (\alpha,\beta)\to\beta} & & \\ 1 & \longrightarrow & \widehat{\mathcal{G}}^{+}_{0,3} & \longrightarrow & M_{0,3} & \longrightarrow & \Gamma_{\mathbb{Q}} & \longrightarrow & 1 \end{array} .$$

Of course, it is in $M_{0,3}$ and $N_{0,3}$ rather than in $\widetilde{M}_{0,3}$ and $\widetilde{N}_{0,3}$ that we are interested – hence in the remarkable subgroup $\Gamma_{\mathbb{Q}}$ of $\widetilde{N}^{!}_{0,3}$. We are going to "determine" by computation the group $\widetilde{N}^{!}_{0,3}$ – and it will be a matter in the rest of the work of finding a maximum of necessary conditions on elements of $\widetilde{N}^{!}_{0,3}$ for them to come from $\Gamma_{\mathbb{Q}}$ – until finding, if possible, a characterization (at least plausible, if not proved!) of the elements of $\Gamma_{\mathbb{Q}}$ – i.e. of $\Gamma_{\mathbb{Q}}$ itself.

Uncertainty retained from cancelled passages.
140-3-s185-D001 M
140-3-s185-D002 $\mathfrak{S}_3|\mathcal{G}_3$
140-3-s185-D003 of
140-3-s185-D004 [illeg. 1]

[scan 186]
[p. 392]

II Computational determination of the elements of $\widetilde{N}^{!}_{0,3}$.
1) Determination by h, g, f.
†
Recall that

$$(21) \qquad \begin{array}{l} \widetilde{N}^{!}_{0,3}|\widetilde{M}^{!}_{0,3} = \text{automorphisms of } \widehat{\mathcal{G}}^{+}_{0,3} \simeq \mathrm{Sl}(2,\mathbb{Z})^{\wedge\prime}|\widehat{\mathrm{Sl}(2,\mathbb{Z})^{\prime}} \text{ that stabilize} \\ \widehat{\Pi}_{0,3}, \quad \text{induce the identity on } \mathfrak{S}_3|\mathcal{G}_3 = \widehat{\mathcal{G}}^{+}_{0,3}/\widehat{\Pi}_{0,3}, \\ \text{and preserve the loop structure (defined} \\ \text{by the class of } \varepsilon_0) \end{array}$$

A $u \in \widetilde{M}^{!}_{0,3}$ is determined by a system

$$(22) \qquad (h, g, f) \quad \dagger \quad (h, g, f \in \widehat{\Pi}_{0,3})$$

†via the formulas

f determined modulo $f \mapsto \ell_0^n f$, $n \in \widehat{\mathbb{Z}}$

$$(23)\qquad \begin{cases} u(\sigma_\infty) = h\sigma_\infty \\ u(\rho) = g\rho \end{cases}$$

†

the condition that these formulas define an automorphism (or at least an endomorphism) of $\widehat{\mathcal{G}}^+_{0,3}$ (which will necessarily induce the identity on $\mathfrak{S}_3|\mathcal{G}_3$) is expressed by the formulas

$$(24)\qquad (h\sigma_\infty)^2 = (g\rho)^3 = 1,$$

which can also be written

$$(25)\qquad \boxed{\begin{cases} h\sigma_\infty(h) = 1 \\ g\rho(g)\rho^2(g) = 1 \end{cases}},$$

and the condition that u preserves the loop structure is expressed by

$$(26)\qquad u(\varepsilon_0) = \mathrm{int}(f^{-1})(\varepsilon_0^\mu) \qquad \mu \in \widehat{\mathbb{Z}}^* \text{ suitable},$$

equivalent|equivalent to the condition

loop relation in $\widehat{\mathcal{G}}^+_{0,3}$ (raw form)

Uncertainty retained from cancelled passages.

140-3-s186-D001 specification $\widetilde{M}^!_{0,3}|\widetilde{N}^!_{0,3}$

140-3-s186-D002 $\in (\widehat{\Pi}_{0,3})^3 \quad \times \quad \widehat{\Pi}_{0,3}$

140-3-s186-D003 satisfying

140-3-s186-D004 $u(\varepsilon_0)$ int (

[scan 187]

loop relation in $\widehat{\Pi}_{0,3}$ (raw form)

$$(27)\qquad u(\ell_0) = \mathrm{int}(f^{-1})(\ell_0^\mu),$$

it is easily made explicit in terms of h, g, f (preferably in form (26)) taking into account that $\varepsilon_0 = \sigma_\infty\rho$

$$\underbrace{(h\sigma_\infty)(g\rho)}_{h\sigma_\infty(g)\underbrace{\sigma_\infty\rho}_{\varepsilon_0}} = \underbrace{\mathrm{int}(f^{-1})\varepsilon_0^\mu}_{f^{-1}\varepsilon_0^\mu f}, \quad \text{i.e}$$

$$fh\sigma_\infty(g)\qquad \underbrace{\varepsilon_0 f^{-1}}_{\varepsilon_0(f^{-1})\varepsilon_0 = \sigma_\infty(\rho(f)^{-1})\varepsilon_0} = \varepsilon_0^\mu \qquad \text{i.e.}$$

$$fh\sigma_\infty(g\rho(f)^{-1}) = \varepsilon_0^{\mu-1} \qquad \text{i.e., setting}$$

$$(28)\qquad \nu \overset{\text{dfn}}{=} \frac{\mu-1}{2} \in \widehat{\mathbb{Z}}$$

the relation (taking into account that $\varepsilon_0^2 = \ell_0$)

possible, since $\mu \in \widehat{\mathbb{Z}}^*$ hence $\mu \equiv 1$ (2)

$$fh\sigma_\infty(g\rho(f)^{-1}) = \ell_0^\nu,$$

or again, applying σ_∞, since $\sigma_\infty(\ell_0) = \ell_1$

$$\sigma_\infty(f)\sigma_\infty(h)g\rho(f)^{-1} = \ell_1^\nu$$

that is, finally

$$(29)\qquad \boxed{\sigma_\infty(h)g = \sigma_\infty(f)^{-1}\ell_1^\nu\rho(f)}$$

(loop relation in h, g, f).

m874

Marginal note. m874 N.B. $\nu \in \widehat{\mathbb{Z}}$ is uniquely determined by g, h, and we have also $2\nu + 1 \in \widehat{\mathbb{Z}}^*$ $2\nu + 1$ is the multiplier…

N.B. One must in addition require, for h, g, f to really define an automorphism, that (24) be a presentation-generation of $\widehat{\mathcal{G}}^+_{0,3}$ – in what follows, we shall always tacitly take this "preliminary condition" as understood.

[scan 188]
[p. 393]

2) Determination by α_0, β_0, f.

There is a one-to-one correspondence between the pairs (g, h) satisfying (25), and the pairs

$$(\alpha_0, \beta_0) \qquad (\alpha_0 \in \Pi^{\wedge\sim}_{0,3} = \Pi^{\wedge}_{0,3} \cdot \{1, \tau_\infty\}, \quad \beta_0 \in \Pi^{\wedge}_{0,3}) \tag{31}$$

† via the relations

$$\begin{cases} g = \alpha_0 \rho(\alpha_0)^{-1} = [\alpha_0, \rho] \\ h = \beta_0 \sigma_\infty (\beta_0)^{-1} = [\beta_0, \sigma_\infty] \end{cases} \tag{32}$$

Thus, the $u \in \mathcal{M}^{!\sim}_{0,3} | \mathcal{N}^{!\sim}_{0,3}$ correspond to the systems

$$(\alpha_0, \beta_0, f) \qquad \alpha_0 \in \Pi^{\sim\wedge}_{0,3}, \quad \beta_0, f \in \Pi^{\wedge}_{0,3}, \quad f \text{ determined mod } f \mapsto \ell_0^n f \ (n \in \widehat{\mathbb{Z}}) \tag{33}$$

satisfying the condition on loops (29), considered as a condition in α_0, β_0, f, † substituting in it for g, h the values given by (32). The relation is written

$$\sigma_\infty(\beta_0)\beta_0^{-1}\alpha_0\rho(\alpha_0)^{-1} = \sigma_\infty(f^{-1})\ell_1^\nu \rho(f)$$

or again

$$\sigma_\infty(f\beta_0) \underbrace{\beta_0^{-1}\alpha_0}_{(f\beta_0)^{-1}(f\alpha_0)} \rho(f\alpha_0\dagger)^{-1} = \ell_1^\nu.$$

or again

$$\boxed{(f\alpha_0)\rho(f\alpha_0)^{-1} = ((f\beta_0)\sigma_\infty(f\beta_0)^{-1})\ell_1^\nu} \tag{34}$$

(loop relation in α_0, β_0, f). As with (29), $\nu \in \widehat{\mathbb{Z}}$ is uniquely determined in terms of|of the α_0, β_0, by the possibility of finding f satisfying relation (34).

The advantage of introducing α_0, β_0 instead

Uncertainty retained from cancelled passages.
140-3-s188-D001 satisf
140-3-s188-D002 by
140-3-s188-D003 *f*

[scan 189]

of g, h is to make unnecessary (since automatic) the relations (25), – only relation (34) remains to be checked. The relations (23) describing u become here:

$$u(\sigma_\infty) = \mathrm{int}(\beta_0)(\sigma_\infty), \qquad u(\rho) = \mathrm{int}(\alpha_0)(\rho) \tag{35}$$

3) Determination by (α, β, f).

The particular form of (34), where the triple $(\alpha_0\ \beta_0\ f)$ enters only through the corresponding pair $(f\alpha_0, f\beta_0)$, leads to setting

$$\alpha = f\alpha_0\dagger, \qquad \beta = f\beta_0, \tag{35}$$

we find that the $u \in \mathcal{M}^{!\sim}_{0,3}$ are determined by the triples

$$(\alpha,\beta,f) \qquad \begin{array}{c} \alpha \in \Pi^{\widetilde{\wedge}}_{0,3}, \quad \beta, f \in \Pi^{\wedge}_{0,3}, \quad \alpha,\beta,f \text{ modulo} \\ \alpha \mapsto \ell_0^n \alpha, \quad \beta \mapsto \ell_0^n \beta, \quad f \mapsto \ell_0^n f \ (n \in \widehat{\mathbb{Z}}) \\ (\text{the same } n \text{ for } \alpha,\beta,f) \end{array} \tag{36}$$

m875

Marginal note. m875 A disadvantage is the indeterminacy of the pair (α,β), under $(\alpha,\beta) \longrightarrow (\ell_0^n\alpha, \ell_0^n\beta)$ $(n \in \widehat{\mathbb{Z}})$, but this is a minor drawback, which will be removed below.

satisfying the condition

$$\boxed{(\alpha\rho(\alpha)^{-1}) = (\beta\sigma_\infty(\beta)^{-1}) \cdot \ell_1^\nu} \tag{37}$$

(loop relation in α,β). The advantage of this relation is that f no longer appears in it.

We recover α_0, β_0, g, h in terms of α,β,f by the formulas

$$\alpha_0 = f^{-1}\alpha, \qquad \beta_0 = f^{-1}\beta \tag{38}$$

$$g = \alpha_0\rho(\alpha_0)^{-1} = f^{-1}(\underbrace{\alpha\rho(\alpha)^{-1}}_{\dagger})\rho(f) = f^{-1}\xi\rho(f) \tag{39}$$

$$h = \beta_0\sigma_\infty(\beta_0)^{-1} = f^{-1}(\beta\sigma_\infty(\beta)^{-1}) = f^{-1}\eta\sigma_\infty(f), \tag{40}$$

where we have set

$$\xi = \alpha\rho(\alpha)^{-1}, \qquad \eta = \beta\sigma_\infty(\beta)^{-1} \qquad (\xi,\eta \in \Pi^{\wedge}_{0,3}) \tag{41}$$

Uncertainty retained from cancelled passages.

140-3-s189-D001 =

140-3-s189-D002 [*illeg.* 1]

[scan 190]

The formulas (23) – (35) determining u in terms of the computational data, become here [p. 394]

$$u(\sigma_\infty) = \mathrm{int}(f^{-1}\beta)(\sigma_\infty) \qquad u(\rho) = \mathrm{int}(f^{-1}\alpha)(\rho) \tag{42}$$

i.e.

$$\dagger\ u = \mathrm{int}(f^{-1})u_{\alpha,\beta}, \tag{42bis}$$

where $u_{\alpha,\beta}$ depends only on α,β:

$$\boxed{u_{\alpha,\beta}(\sigma_\infty) = \mathrm{int}(\beta)(\sigma_\infty) = \eta\sigma_\infty} \quad \boxed{u_{\alpha,\beta}(\rho) = \mathrm{int}(\alpha)(\rho) = \xi\rho} \tag{42ter}$$

Formula (37) can be interpreted as a relation in ξ,η

$$\xi = \eta\ell_1^\nu. \tag{43}$$

4) Determination by (ξ,η,f).

The datum of the pair $(\alpha,\beta) \in \Pi^{\widehat{\sim}\wedge}_{0,3} \times \Pi^{\wedge}_{0,3}$, is equivalent to that of the pair (ξ,η) deduced from it by (41), subject to the equations

$$\boxed{\begin{cases} \xi\rho(\xi)\rho^2(\xi) = 1 \\ \eta\sigma_\infty(\eta) = 1 \end{cases}} \tag{44}$$

Thus, the datum of $u \in \mathcal{M}^{!\sim}_{0,3}$ is equivalent to that of a triple

$$(\xi, \eta, f) \qquad \begin{matrix} \xi, \eta, f \in \Pi^{\wedge}_{0,3} \quad \text{modulo} \\ \xi \mapsto \ell_0^n \xi \ell_1^n |\ell_1^{-n}, \quad \eta \mapsto \ell_0^n \eta \ell_1^n |\ell_1^{-n}, \quad f \mapsto \ell_0^n f \end{matrix} \tag{45}$$

satisfying the relations (44), and the relation

$$\boxed{\xi = \eta \ell_1^{\nu}}, \tag{46}=(43)$$

which plays the role of "loop relation in ξ, η". The advantage of this presentation is

---

Uncertainty retained from cancelled passages.

140-3-s190-D001 i.e.

[scan 191]

the maximal simplicity of the equations (44),(46) – and the fact that f does not enter into them: † the first two equations are equations in ξ alone and in η alone, and the third relates ξ and η in a particularly simple way – in fact if $\mu = 1$, i.e. $\nu = 0$, it is the relation $\xi = \eta$!

The drawback is that there are three equations instead of one, and that we have no explicit expression for $u(\sigma_\infty)$ and $u(\rho)$ in terms of ξ, η, f: we must first, for this, solve the equations m876

Marginal note. m876 but yes, we do have a nice explicit expression (42ter):

$$u(\sigma_\infty) = \mathrm{int}(f^{-1})(\eta \sigma_\infty)$$
$$u(\rho) = \mathrm{int}(f^{-1})(\xi \rho)$$

!!

$$\begin{cases} \alpha \rho(\alpha)^{-1} = \xi & (\alpha \in \Pi^{\sim\wedge}_{0,3}) \\ \beta \sigma_\infty(\beta)^{-1} = \eta & (\beta \in \Pi^{\wedge}_{0,3}) \end{cases} \tag{47}$$

in α, β, to be able then to apply (42).

Another [!] drawback, of importance, is that when we work with the images of the elements under consideration in $\mathrm{Gl}(2, \widehat{\mathbb{Z}})^{\pm\prime|\pm 1}$, the knowledge in this group of ξ, η (even in $\mathrm{Sl}(2, \widehat{\mathbb{Z}})$ itself) is † a rather weak datum, which in no way allows us to recover α, β as elements of $\mathrm{Gl}(2, \widehat{\mathbb{Z}})^{\pm\prime|\pm 1}$, cf. below.

Uncertainty retained from cancelled passages.

140-3-s191-D001 these are

140-3-s191-D002 too

[scan 192]

5) Composition with the|with automorphisms that are inner. Cocycle relations for a composite $u'' = u'u$. [p. 395]

Let $\gamma \in \Pi^{\wedge}_{0,3}$, and let us consider the composite

$$u' = \mathrm{int}(\gamma) \circ u \tag{48}$$

it corresponds to invariants

$$f', g', h'|\beta', \alpha'_0, \beta'_0, \alpha', \beta', \xi', \eta'$$

given by

$$\begin{cases} f' = f\gamma^{-1}, \quad \dagger \\ g' = \gamma g \rho(\gamma)^{-1}, \quad h' = \gamma h \sigma_\infty(\gamma)^{-1} \\ \alpha'_0 = \gamma \alpha_0, \quad \beta'_0 = \gamma \beta_0 \\ \alpha' = \alpha, \quad \beta' = \beta \\ \xi' = \xi, \quad \eta' = \eta \end{cases} \tag{49}$$

Thus α and β do not change when we replace $u|f$ by $\mathrm{int}(\gamma)u$, they depend only on the outer loop automorphism defined by u, i.e. on the image of u in $\mathcal{N}^{!\sim}_{0,3}$. This is a substantial advantage of the use of (α, β, f) or of (ξ, η, f), in preference to (α_0, β_0, f) or of (g, h, f).

Now let

$$u, u' \in \mathcal{M}^{!\sim}_{0,3}, \quad \text{let } u'' = u'u \tag{50}$$

let

$$\begin{cases} f, g, h, \alpha_0, \beta_0, \alpha, \beta, \xi, \eta \\ f', g', h', \alpha'_0, \beta'_0, \alpha', \beta', \xi', \eta' \qquad \text{and} \\ f'', g'', h'', \alpha''_0, \beta''_0, \alpha'', \beta'', \xi'', \eta'' \end{cases} \tag{51}$$

Uncertainty retained from cancelled passages.

140-3-s192-D001 $g' = \gamma g$

[scan 193]

be the invariants relative to u, u', u'' respectively, – we want to express (as far as possible) those of u'' in terms of those of u, u', and we find

$$\begin{cases} g'' = u'(g)g' & \alpha''_0 = u'(\alpha_0)\alpha'_0 * & \alpha'' = (f'u')(\alpha).\alpha' * \\ h'' = u'(h)h' & \beta''_0 = u'(\beta_0)\beta'_0 & \beta'' = (f'u')(\beta).\beta' \end{cases} \tag{52}$$

but $f'' = f'u'(f)\dagger$.

m877

Marginal note. m877 $* \; u'(\alpha_0)$ and $(f'u')(\alpha)$ make sense a priori only if $\alpha_0, \alpha \in \widehat{\Pi}_{0,3}$ i.e. $\mu \equiv 1$ (3), otherwise (for example) we can still define $u'(\alpha_0) = u'\alpha_0 u'^{-1}$, but it will no longer be † [illeg. 2] in $\widehat{\Pi}_{0,3}$, and the formula $\alpha''_0 = u'(\alpha_0)\alpha'_0$ will make sense, but will be false !

We thus see that if we consider (for u varying in $\mathcal{M}^{!\sim}_{0,3}$) $f, g, h, \alpha_0, \beta_0, \alpha, \beta$ as functions

$$\mathcal{M}^{!\sim}_{0,3} \longrightarrow \widehat{\Pi}_{0,3}, \text{ or } \widetilde{\Pi}^{\wedge}_{0,3}$$

then $f, g^{-1}, h^{-1}, \alpha_0^{-1}, \beta_0^{-1}$ satisfy the relation for cocycles, and α^{-1}, β^{-1} do too when we restrict ourselves to the† subgroup for which $f = 1$, i.e. the automorphisms $u \in \mathcal{M}^{!\sim}_{0,3}$ which normalize † $L_0 = \{\ell_0^n \mid n \in \widehat{\mathbb{Z}}\}$. To tell the truth, this statement † for f, α, β is a little lame, because of the indeterminacies for f, entailing a corresponding indeterminacy for α, β, $f \mapsto \ell_0^n f$, $\alpha \mapsto \ell_0^n \alpha$, $\beta \mapsto \ell_0^n \beta$. To construct canonical cocycles corresponding to $f, \alpha^{-1}, \beta^{-1}$, it would be necessary to introduce a group extension of $\mathcal{M}^{!\sim}_{0,3}$ by $\widehat{\mathbb{Z}}$, on the model of the special Teichmüller groups, on which f, α, β would be defined without ambiguity. There is no need here to go into detail…

Uncertainty retained from cancelled passages.

140-3-s193-D001 ξ

140-3-s193-D002 in

140-3-s193-D003 [illeg. 1]

140-3-s193-D004 the

140-3-s193-D005 [illeg. 1]

[scan 194]

[p. 396]

III Calculations in $\mathrm{Gl}(2, \widehat{\mathbb{Z}})'$ and in $\mathrm{Gl}(2, \widehat{\mathbb{Z}})$.

For a given $u \in \mathcal{M}^{!\sim}_{0,3}$, whence the quantities

$$f, g, h, \alpha_0, \beta_0, \alpha, \beta, \xi, \eta : \tag{53}$$

(all in $\widehat{\Pi}_{0,3}$, except α_0, α which are only in $\widetilde{\Pi}^{\wedge}_{0,3} \simeq \widehat{\Pi}_{0,3}.\{1, \tau_\infty\}$) we denote by the same letters the|their images in

$$\mathrm{Gl}(2,\widehat{\mathbb{Z}})' \overset{\text{dfn}}{=} \mathrm{Gl}(2,\widehat{\mathbb{Z}})/\pm 1,$$

they are in fact in

$$\mathrm{Sl}(2,\widehat{\mathbb{Z}})' = \mathrm{Sl}(2,\widehat{\mathbb{Z}})/\pm 1,$$

except α_0, α which are only in $\mathrm{Gl}(2,\widehat{\mathbb{Z}})^{\pm'|\pm 1}$

$$\det \alpha_0 = \det \alpha = \varepsilon_3 \in \{\pm 1\} \tag{54}$$

where ε_3 is defined by

$$\varepsilon_3 \equiv \mu \quad (3), \tag{55}$$

[illeg. 1] otherwise.

$$\begin{gathered} f, g, h, \beta_0, \beta, \xi, \eta \in \mathrm{Sl}(2,\widehat{\mathbb{Z}})' \quad \text{i.e.} \\ \det f = \det g = \det h = \det \beta_0 = \det \beta = \det \xi = \det \eta = 1 \end{gathered} \tag{56}$$

m878

Marginal note. m878 N.B. f is determined modulo transformation into $\ell_0^n f$, and thereby α, β are subject to the same indeterminacy – more precisely we have an indeterminacy $(\alpha, \beta, f) \mapsto (\ell_0^n \alpha, \ell_0^n \beta, \ell_0^n f)$, (in addition to that of the signs). In what follows, we shall assume therefore † f, α, β chosen in $\widehat{\Pi}_{0,3}$, and that the images in $\mathrm{Gl}(2,\widehat{\mathbb{Z}})'$ have been lifted to $\mathrm{Gl}(2,\widehat{\mathbb{Z}})$…

m879

Marginal note. m879 cf. page 398|348 or better|in a better form formulary below

(N.B. We † $\det(-1) = 1$, hence det factors

$$\mathrm{Gl}(2,\widehat{\mathbb{Z}})' \overset{\det}{\longrightarrow} \widehat{\mathbb{Z}}^\times,$$

to give an exact sequence

$$1 \longrightarrow \mathrm{Sl}(2,\widehat{\mathbb{Z}})' \longrightarrow \mathrm{Gl}(2,\widehat{\mathbb{Z}})' \longrightarrow \widehat{\mathbb{Z}}^\times \longrightarrow 1 \tag{57}$$

).

We shall choose representatives

$$f, \alpha|\alpha_0, \beta_0|\beta \in \mathrm{Gl}(2,\widehat{\mathbb{Z}}) \tag{58}$$

which is equivalent to the choice of three signs,

Uncertainty retained from cancelled passages.

140-3-s194-D002 [illeg. 1]

140-3-s194-D003 function

[scan 195]

and in terms of these choices we define the other quantities by

†

$$\begin{cases} \xi = \alpha\rho(\alpha)^{-1}, & \eta = \beta\dagger\sigma_\infty(\beta)^{-1} \\ \alpha_0 = f^{-1}\alpha, & \beta_0 = f^{-1}\beta \\ g = \alpha_0\rho(\alpha_0)^{-1} = f^{-1}\xi\rho(f), & h = \beta_0\rho(\beta_0)^{-1} = f^{-1}\eta\rho(f). \end{cases} \tag{59}$$

We shall note that ξ and η, † g, h are independent of the choice of signs that have been made. The essential relation linking $(f, \alpha_0|\alpha, \beta_0|\beta)$, or rather α, β (in addition to the relations (59), which play the role of defining relations) is the "loop relation", in matrix form, which takes the form (corresponding to (46) or to (43))

$$\boxed{\xi = \varepsilon\eta\ell_1^\nu} \qquad \varepsilon \in \{\pm 1\} \tag{60}$$

where ε† is a sign which no longer depends on any choice – we have seen|we have in † (cf. also formulary below)

$$\varepsilon = \varepsilon_2 \quad \text{defined in terms i.e. of } \mu \text{ by} \quad \mu \equiv \varepsilon_2 \quad (4), \tag{61}$$

Uncertainty retained from cancelled passages.

140-3-s195-D001 $\alpha_0|\alpha$ $\beta_0|\beta$

140-3-s195-D002 ρ

140-3-s195-D003 or

140-3-s195-D004 ε_2

140-3-s195-D005 [illeg. 1]

[scan 196]
[p. 397]

and where ℓ_1 is defined in matrix form by

$$\dagger \qquad \ell_1 = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$$

whence

$$\ell_1^\nu = \begin{pmatrix} 1 & 0 \\ 2\nu & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \mu - 1 & 1 \end{pmatrix}. \tag{62}$$

Remark

(Relation with the notations of § 44 (page 356 ff, in particular formulary after page 364). In the groups $\widehat{\Pi}_{0,3}$ and $\widetilde{\Pi}^{\wedge}_{0,3}$, the letters f, g, h of loc. cit. have|are the same meaning as here, but not the letters α, β etc. I shall distinguish by underlines, in the case of deviation between the two notations, the letters with the † meaning used|studied in loc. cit, we find

$$\underline{\alpha} = \alpha_0, \qquad \underline{\beta} = \beta_0 \tag{i}$$

from page 357 on we practically always assume $\underline{\beta} = \beta_0 = 1$, (i.e. $u(\sigma_\infty) = \sigma_\infty$) so that

$$\boxed{\beta = f\beta_0 = f} \tag{ii}$$

and loc. cit (17) gives

$$\underline{\varphi} = f\sigma_\infty(f)^{-1} = \beta\sigma_\infty(\beta)^{-1} = \eta \tag{iii}$$

whereas (loc. cit (14))

$$\underline{\xi} = \ell_0^\nu \underline{\varphi} = \ell_0^\nu \eta,$$

comparing with

$$\xi = \eta\ell_1^\nu \qquad (46)$$

we find

$$\dagger \quad \underline{\xi} = \ell_0^\nu \xi \ell_1^{-\nu} = \ell_0^\nu \xi \rho(\ell_0^\nu)^{-1}. \tag{iv}$$

i.e. $\underline{\xi}$ is the ρ-transform of ℓ_0^ν. The relation (loc. cit (19|29))

$$\underline{\xi} = \underline{\alpha}_0 \rho(\underline{\alpha}_0)^{-1}$$

compared with $\xi = \alpha\rho(\alpha)^{-1}$ and with (iv)†, gives

Uncertainty retained from cancelled passages.
140-3-s196-D001 (62)
140-3-s196-D002 meaning
140-3-s196-D004 (*iv*)
140-3-s196-D005 [illeg. 1]

[scan 197]

$$\dagger \qquad \underline{\alpha}_0 = \ell_0^{\nu}\alpha. \tag{v}$$

The formulas † (i) to (v) make it possible to translate the formulary of loc. cit. in terms of the formulary that I am going to make explicit.

We make α and β explicit by their matrices

$$\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \beta = \begin{pmatrix} r & s \\ t & u \end{pmatrix}, \tag{63}$$

where $a, b, c, d, r, s, t, u \in \widehat{\mathbb{Z}}$, with

$$ad - bc = \dagger\delta = \varepsilon_3, \qquad ru - st = 1. \tag{64}$$

we shall have

$$\xi = \alpha\rho(\alpha)^{-1} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \tag{65}$$

with (cf. page 342, (78)|(18))

$$\begin{cases} \delta A = (a^2 + ab + b^2) - (ac + bd + bc), & \delta B = ac + bd + bc \\ \delta C = -(c^2 + cd + d^2) + (ac + bd + ad), & \delta D = c^2 + cd + d^2 \end{cases}, \tag{66}$$

$$\eta = \beta\sigma_\infty(\beta)^{-1} = \begin{pmatrix} R & S \\ T & U \end{pmatrix}, \quad \text{with} \tag{67}$$

$$R = r^2 + s^2, \quad S = T = rt + su, \quad U = t^2 + u^2. \tag{68}$$

The loop relation (60) takes the form

$$\begin{aligned} \begin{pmatrix} A & B \\ C & D \end{pmatrix} &= \varepsilon \begin{pmatrix} R & S \\ T & \dagger U \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 2\nu & 1 \end{pmatrix} \\ &= \varepsilon \begin{pmatrix} R + 2\nu S & S \\ T + 2\nu U & U \end{pmatrix}, \end{aligned} \tag{69}$$

Uncertainty retained from cancelled passages.
140-3-s197-D001 [*illeg.* 1]
140-3-s197-D002 [illeg. 2]
140-3-s197-D003 [*illeg.* 1]
140-3-s197-D004 *T*

[scan 198]
[p. 398]

Let us note the relations (consequences of (66), (67)):

$$(70) \qquad AD - BC = 1 \qquad RU - ST = 1$$

$$(71) \qquad C + D - B = 1 \qquad T = |-S = 1 \quad \text{i.e. } T = S,$$

This †loop relation (69) is equivalent to the system of relations

$$(72)\qquad \boxed{\begin{matrix} D=\varepsilon U \\ \mu D=1 \\ B=\varepsilon S \end{matrix}} \qquad (\text{where } \mu=2\nu+1),$$

i.e. to the system of equations

$$(73)\qquad \begin{cases} c^2+cd+d^2=\varepsilon(t^2+u^2)=\delta/\mu \\ ac+bd+bc=\varepsilon(rt+su) \end{cases}$$

It is from the|from †first two relations (73) that we derive at once

$$(74)\qquad \delta\equiv\mu\ (3), \qquad \varepsilon\equiv\mu\ (4)$$

which characterize the signs δ,ε as

$$(75)\qquad \delta=\varepsilon_3(\mu)=\varepsilon_3, \qquad \varepsilon=\varepsilon_2(\mu)=\varepsilon_2,$$

cf. Theorem page 365.

The formula (83) on page 370 takes here the very simple form (valid without conditions other than those specified so far on α,β,f — f moreover not occurring † for the moment in the formulary):

Uncertainty retained from cancelled passages.

140-3-s198-D001 cocycle

140-3-s198-D004 the comparison of the first

140-3-s198-D005 in

[scan 199]

$$(76)\qquad \underset{\varepsilon_3\in\{\pm1\}}{(\delta)}\quad \alpha(c\rho+d)=\beta(t\sigma_\infty+u)\tau_\infty^{\alpha_2} \qquad (\alpha=\alpha_2)$$

where $\alpha_2=\alpha$ is defined by the condition

$$(-1)^{\alpha_2}=\varepsilon \quad (=\varepsilon_2).$$

Making (76) explicit by matrix calculation, we have

$$\delta\alpha(c\rho+d)=\delta\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} d & c \\ -c & c+d \end{pmatrix}=\dagger\begin{pmatrix} 1 & B \\ 0 & D \end{pmatrix}$$

$$\beta(t\sigma_\infty+u)\tau_\infty^{\alpha}=\begin{pmatrix} r & s|-s \\ t|-t & u \end{pmatrix}\begin{pmatrix} u & \varepsilon_2 t|\varepsilon t \\ -t & \varepsilon u \end{pmatrix}=\begin{pmatrix} 1 & \varepsilon S \\ 0 & \varepsilon U \end{pmatrix}$$

so that (76) reduces to the equations

$$D=\varepsilon U, \quad B=\varepsilon S,$$

thus the loop relation $\xi=\varepsilon\eta\ell_1^\nu$ (60|68, 69) is equivalent to the conjunction of (76) and of the relation

$$(77)\qquad \mu D=1.$$

This being laid down, we shall set (for (α, β) fixed so as to satisfy the loop formula)

$$(78) \qquad u_{\alpha,\beta} \overset{\text{dfn}}{=} (\mu\delta).\alpha(c\rho + d) = \mu\beta(t\sigma_\infty + u)\tau_\infty^\alpha$$
$$= \mu \begin{pmatrix} 1 & B \\ 0 & D \end{pmatrix} = \begin{pmatrix} \mu & \mu B \\ 0 & 1 \end{pmatrix}$$

so that $u_{\alpha,\beta}$ in fact depends only on α, or only on β and the sign α_2 (which, for its part, is not determined by β, but by α as the value of $\delta(c^2 + cd + d^2) \bmod 4$…); and we shall set, for a triple (α, β, f) of elements of $\mathrm{Gl}(2, \widehat{\mathbb{Z}})$, $\det f = 1$, α and $\beta|\det\beta$ related by the loop formula

m880

Marginal note. m880 in fact, $u_{\alpha\beta}$ depends only on $\xi = \alpha\rho(\alpha)^{-1}$, and even only on the image $\overline{\xi}|\xi$ of ξ in $\mathrm{Sl}(2, \widehat{\mathbb{Z}})|\mathrm{Sl}(2, \widehat{\mathbb{Z}})'$, the sign being characterized by the condition $\xi\rho(\xi)\rho^2(\xi) = +1$) i.e. depends only on $B, D(= 1/\mu \ldots)$. It should be denoted u_ξ, cf. page 410.

Uncertainty retained from cancelled passages.
140-3-s199-D001 δ

[scan 200]
[p. 399]

$$(79) \qquad u_{\alpha,\beta,f} \overset{\text{dfn}}{=} f^{-1}u_{\alpha,\beta} = (\mu\delta)\,\overbrace{f^{-1}\alpha}^{\alpha_0}(c\rho + d) =$$
$$= \mu\,\underbrace{f^{-1}\beta}_{\beta_0}(t\sigma_\infty + u)\tau_\infty^{\alpha_2}$$

and one finds, as one should,

$$(80) \qquad \begin{cases} u_{\alpha,\beta,f}(\sigma_\infty) = \mathrm{int}(\beta_0)(\sigma_\infty) \\ u_{\alpha,\beta,f}(\rho) = \mathrm{int}(\alpha_0|\beta_0)(\rho) \end{cases},$$

m881

Marginal note. m881



m882

Marginal note. m882 N.B. if|If u, σ are elements of a group G, one of course sets $u(\sigma) = u\sigma u^{-1}$.

m883

Marginal note. m883 this implies that, † $\mathrm{int}(u_{\alpha,\beta,f})$ has the same effect, on|in $\mathrm{Sl}(2, \widehat{\mathbb{Z}})'$ as the automorphism u of $\widehat{\mathcal{G}}^+_{0,3}|\widetilde{\mathcal{G}}^{\rightarrow}_{0,3}$ defined (when there is one), by passage to the quotient in $\mathrm{Sl}(2, \widehat{\mathbb{Z}})'$.

formulas which reduce to formulas in which f has again disappeared

$$(81) \qquad \begin{cases} u_{\alpha,\beta}(\sigma_\infty) = \mathrm{int}(\beta)(\sigma_\infty) \\ u_{\alpha,\beta}(\rho) = \mathrm{int}(\alpha|\beta)(\rho) \end{cases},$$

which follow trivially from the two expressions (78) for $u_{\alpha,\beta}$!

The formulas (80) are naturally completed by

$$(82) \qquad u_{\alpha,\beta,f}(\varepsilon_0) = \dagger\varepsilon\,\mathrm{int}(f^{-1})(\varepsilon_0^\mu) \qquad \dagger$$

$$(\text{where } \varepsilon = \varepsilon_2 \in \{\pm 1\})$$

which in turn reduces to

$$(83) \qquad u_{\alpha,\beta}(\varepsilon_0) = \varepsilon.\varepsilon_0^\mu$$

where f no longer appears. Since $\varepsilon_0 = \sigma_\infty\rho$, this formula reduces to:

$$u_{\alpha,\beta}(\sigma_\infty)u_{\alpha,\beta}(\rho) = \varepsilon.\varepsilon_0^\mu,$$

which in view of (81) reduces to:

$$\mathrm{int}(\beta)(\sigma_\infty)\,\mathrm{int}(\alpha|\beta)(\rho) = \varepsilon.\varepsilon_0^\mu$$

†

†

Uncertainty retained from cancelled passages.

140-3-s200-D001 by
140-3-s200-D004 $f\varepsilon_0$
140-3-s200-D005 $(\varepsilon = \varepsilon_2)$
140-3-s200-D006 then
140-3-s200-D007 $\alpha\sigma_\infty\alpha^{-1}\beta\rho\beta^{-1}$

[scan 201]

now the first side is written

†

†

$$\beta\sigma_\infty\beta^{-1}\alpha\rho\alpha^{-1} = \underbrace{(\beta\sigma_\infty\beta^{-1}\sigma_\infty)}_{\beta\sigma_\infty(\beta)^{-1}}\underbrace{(\sigma_\infty\rho)}_{\varepsilon_0}\underbrace{(\rho^{-1}\alpha\rho\alpha^{-1})}_{\rho^{-1}(\alpha)\alpha^{-1}=\rho^{-1}(\alpha\rho(\alpha)^{-1})}$$
$$= \eta.\varepsilon_0.\rho^{-1}(\xi) = \eta.\underbrace{(\varepsilon_0\rho^{-1})}_{\sigma_\infty}(\xi).\varepsilon_0 = \eta\sigma_\infty(\xi)\varepsilon_0$$

so relation (83) is also written

$$\eta\sigma_\infty(\xi) = \varepsilon\varepsilon_0^{\mu-1} \quad (= \varepsilon\ell_0^\nu)$$

now one has $\xi = \varepsilon\eta\ell_1^\nu$ (loop relation (60))

so

$$\eta\sigma_\infty(\xi) = \varepsilon\eta\sigma_\infty(\eta)\ell_0^\nu = \varepsilon\ell_0^\nu, \quad \text{OK.}$$

Remark. All these formulas are valid in $\mathrm{Gl}(2,\widehat{\mathbb{Z}})$, without ambiguity of signs, once α, β, f are chosen. If one changes the signs of α or|and of β or both, the loop relation is not modified — and $u_{\alpha,\beta}$ is not changed either, nor $u_{\alpha,\beta,f}$ (if one does not touch f). On the other hand, a change of sign on f implies a change of sign on $u_{\alpha,\beta,f}$. But such a change of sign of course changes nothing as to the validity of formulas (80), (82). When one makes

Uncertainty retained from cancelled passages.
140-3-s201-D001 $\beta\sigma_\infty\beta^{-1}\alpha\rho\alpha^{-1} = (\beta\sigma_\infty\beta^{-1}\sigma_\infty)(\sigma_\infty\rho)(\rho^{-1}\alpha\rho\alpha^{-1})$
140-3-s201-D002 $= (\beta\sigma_\infty(\beta)^{-1})\varepsilon_0(\rho^{-1}(\alpha)\beta^{-1})$

[scan 202]
[p. 400]

on (α, β, f) the changes the

$$(84) \qquad \alpha' = \ell_0^n \alpha, \qquad \beta' = \ell_0^n \beta, \qquad f' = \ell_0^n f \qquad (n \in \mathbb{Z})$$

then †

$$(85) \qquad \begin{cases} c' = c, \quad d' = d \qquad t' = t, \quad u' = u \\ \underbrace{f'^{-1}\alpha'}_{\alpha'_0} = \underbrace{f^{-1}\alpha}_{\alpha_0} \qquad \underbrace{f'^{-1}\beta'}_{\beta'_0} = \underbrace{f^{-1}\beta}_{\beta_0} \qquad \dagger \end{cases}$$

$$\dagger \qquad \dagger$$

finally

$$\delta'(= \det \alpha') = \delta(= \det \alpha) \quad (\text{since } \det \ell_0^n = 1)$$

$$(86) \qquad \begin{cases} \text{hence} \quad \dagger \quad \mu'(= \delta'/(c'^2 + c'd' + d'^2)) = \mu(= \delta/(c^2 + cd + d^2)) \\ \qquad\qquad\qquad\qquad \text{hence} \quad \varepsilon'_2 = \varepsilon_2, \quad \alpha'_2 = \alpha_2. \end{cases}$$

this is twice what is needed to check with (79) that

$$(87) \qquad u_{\alpha',\beta',f'} = u_{\alpha,\beta,f}.$$

Let us also recall (cf. proposition page 384)

Proposition. Let $\mu \in \widehat{\mathbb{Z}}^*$, hence the signs $\varepsilon_2 = \varepsilon_2(\mu)$ and $\varepsilon_3 = \varepsilon_3(\mu)$ (defined by $\mu \equiv \varepsilon_2$ (4), $\mu \equiv \varepsilon_3$ (3)), and let $c, d, t, u \in \widehat{\mathbb{Z}}$. For there to exist $a, b, r, s \in \widehat{\mathbb{Z}}$

$$\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \beta = \begin{pmatrix} r & s \\ t & u \end{pmatrix} \quad \text{satisfying (64)}$$

$$ad - bc = \varepsilon_3 \qquad ru - st = 1$$

and the loop relation

$$\alpha\rho(\alpha)^{-1} = \varepsilon_2(\beta\sigma_\infty(\beta^{-1}))\ell_1^\nu \qquad \text{where } \nu = (\mu - 1)/2$$

(i.e. relations (73)) it is necessary and sufficient that one have

$$(88) \qquad \varepsilon_3(c^2 + cd + d^2) = \varepsilon_2(t^2 + u^2) = 1/\mu,$$

and the set|the set of pairs (α, β) satisfying these conditions is a torsor under $\widehat{\mathbb{Z}}$, operating
m885

Marginal note. m885 N.B. For fixed μ, one can always find c, d, t, u satisfying relation (88) — fortunately!

Uncertainty retained from cancelled passages.
140-3-s202-D001 [illeg. 1]
140-3-s202-D002 $\mu' = \mu | u' = u$
140-3-s202-D004 $\delta' = \delta$
140-3-s202-D005 $u_{\alpha',\beta'}$
140-3-s202-D006 f

[scan 203]

by

$$(\alpha, \beta) \longmapsto (\varepsilon_0^n \alpha, \varepsilon_0^n \beta).$$

†

Corollary For one to be able to find α, β such that one has in addition $\alpha \equiv \beta \equiv 1$ (2), it is necessary and sufficient that one have

$$c \equiv t \equiv 0 \ (2) \qquad d \equiv u \equiv 1 \ (2), \tag{89}$$

and then the set of solutions (α, β) satisfying these additional conditions is a torsor under $\widehat{\mathbb{Z}}$, operating by

$$(\alpha, \beta) \longmapsto (\ell_0^n \alpha, \ell_0^n \beta).$$

Scholium One can thus say that for an element u of $\mathcal{N}_{0,3}^{!\sim}$, which defines thus|two α, β in $\Pi_{0,3}^{\sim\wedge}, \Pi_{0,3}^{\wedge}$ (modulo the transformations $(\alpha, \beta) \mapsto (\ell_0^n \alpha, \ell_0^n \beta)$), (satisfying the loop relation), there are attached to it the "adelic invariants" †

$$\Theta = (\mu, c, d, t, u) \tag{90}$$

(c, d, t, u determined up to ambiguities|ambiguities of sign, and no more) satisfying (88) — and this is the best one can do in the present context: the a, b, r, s have|is not an invariant character. When one has an element $\underline{u}$ of $\mathcal{M}_{0,3}^{!\sim}|\mathcal{N}_{0,3}^{!\sim}$ — i.e. an actual automorphism, not just outer — one has in addition an $\underline{f} \in \Pi_{0,3}^{\wedge}$, giving an $f \in \mathrm{Sl}(2, \widehat{\mathbb{Z}})'$, whence

$$f = \begin{pmatrix} j & k \\ \ell & m \end{pmatrix} \qquad \frac{\text{defined modulo signs}}{\text{and modulo } f \longmapsto \ell_0^n f} \tag{91}$$

Uncertainty retained from cancelled passages.

140-3-s203-D001 [illeg. 1]

140-3-s203-D002 [illeg. 3]

[scan 204]

[p. 401]

†

But this additional datum is a little too weak to reconstruct the element

$$\underline{\underline{u}} = \varphi(\underline{u})\dagger = u_{\alpha,\beta,f}$$

of $\mathrm{Gl}(2, \widehat{\mathbb{Z}})'$ associated to u.

It is better to say that for fixed μ, c, d, t, u one has a $\widehat{\mathbb{Z}}$-torsor of $(\alpha, \beta) \equiv (1, 1)$ (2) satisfying the conditions stated in|of the proposition §|, and its corollary, let $T = T_\Theta$, one then makes $\widehat{\mathbb{Z}}$ operate diagonally on $T \times \mathrm{Sl}(2, \widehat{\mathbb{Z}})$ by

$$(\alpha, \beta, f) \longmapsto (\ell_0^n \alpha, \ell_0^n \beta, \ell_0^n f)$$

and one looks at the quotient — this is thus the twist by $\check{T}$ (torsor opposite to T) of $\mathrm{Sl}(2, \widehat{\mathbb{Z}})$, on which $\widehat{\mathbb{Z}}$ operates by $f \longmapsto \ell_0^n f$. it is the datum of an element f of

$$\check{T}_\Theta \wedge_{\widehat{\mathbb{Z}}} \mathrm{Sl}(2, \widehat{\mathbb{Z}})\dagger$$

m886

Marginal note. m886 it is an $\mathrm{Sl}(2, \widehat{\mathbb{Z}})$-torsor on the right

which makes it possible to define an element

$$\underline{\underline{u}} = u_{\alpha,\beta,f},$$

which changes sign if f is replaced by $(f(-1))$.

Decidedly, these formulations are not exciting, and it is becoming clear that it will be necessary to introduce explicit-

Uncertainty retained from cancelled passages.
140-3-s204-D001 thus
140-3-s204-D004 déf
140-3-s204-D006 $= \mathcal{G}_\Theta$

[scan 205]

ly and study with a minimum of care the adelic group formed by the systems (α, β, f) satisfying the loop relation, and † the canonical homomorphism from $\mathcal{M}^{!\sim}_{0,3}$ into the said group.

But I would like before that to finish the notational scales.

†

Uncertainty retained from cancelled passages.
140-3-s205-D002 consider
140-3-s205-D003 [illeg. 1]

[scan 206]
[p. 402]

IV The three fundamental special cases.

We return to the elements g, h, f etc. in $\Pi^{\wedge}_{0,3}$ and $\Pi^{\sim\wedge}_{0,3}$ which correspond to u, which can be described|described in four different ways, by one or another of the four systems

$$(g, h, f), \quad (\alpha_0, \beta_0, f), \quad (\alpha, \beta, f), \quad (\xi, \eta, f). \tag{92}$$

1°|1er Case: †

$$h = 1, \quad \text{i.e. } \beta_0 = 1, \quad \text{i.e. } \beta = f, \quad \text{i.e. } \eta = f\sigma_\infty(f)^{-1}, \tag{93}$$

i.e.

$$[u, \sigma_\infty] = 1. \tag{94}$$

For a u whatsoever (i.e. an arbitrary system (g, h, f) etc., there is one and only one inner automorphism $\mathrm{int}(\gamma)$ such that $u' = \mathrm{int}(\gamma)u$ satisfies the preceding property i.e. commutes with σ_∞, † this will be for

$$\gamma = \beta_0^{-1} \qquad \text{(cf. formulas (49))}.$$

†

The u satisfying (93), expressed in terms of triples (ξ, η, f) satisfying the three relations (44) (46) (not involving f), † thus correspond (by the relation $\eta = f\sigma_\infty(f)^{-1}$ of (93)) to the † pairs (ξ, f) of elements of $\Pi^{\wedge}_{0,3}$, satisfying

$$\xi\, \rho(\xi)\, \rho^2(\xi) = 1 \tag{95}$$

$$\xi = (f\sigma_\infty(f)^{-1})\ell_1^\nu \tag{96}$$

with $\nu \in \widehat{\mathbb{Z}}$, as a variable, such that $2\nu + 1 \in \widehat{\mathbb{Z}}^*$
m887

Marginal note. m887 N.B. one will have $\xi = fg\rho(f)^{-1}$. in place of f it is permissible|it is legitimate to write here β — but then [illeg. 2] β determined by [illeg. 2] ℓ_0^n [illeg. 2] $f \longmapsto \ell_0^n f$ [illeg. 2]

Uncertainty retained from cancelled passages.
140-3-s206-D001 [illeg. 1] [illeg. 1]
140-3-s206-D002 [illeg. 1]
140-3-s206-D003 [illeg. 1]
140-3-s206-D004 [illeg. 1]

[scan 207]

or to the pairs (α, β) of elements of $\widehat{\Pi}_{0,3}$ satisfying the loop relation (37)

$$(97) = (37) \qquad (\alpha\rho(\alpha)^{-1}) = (\beta\sigma_\infty(\beta)^{-1})\ell_1^\nu$$

(which we will then complete by taking $f = \beta$).

m888

Marginal note. m888 α, β determined modulo $(\alpha, \beta) \longmapsto (\ell_0^n\alpha, \ell_0^n\beta)$.

Up to changing α to $\ell_0^\nu\alpha = \underline{\alpha}_0$, these were the conventions of § 44|§ 4 h (where the formula thus takes the form

$$\underbrace{\underline{\alpha}_0\rho(\underline{\alpha}_0)^{-1}}_{\xi} = \ell_0^\nu \underbrace{(f\sigma_\infty(f)^{-1})}_{\varphi=\eta}$$

).

Taking (79) into account, the conditions

†

(93) (94) also take the form

$$(98) \qquad u_{\alpha,\beta,f} = \mu(t\sigma_\infty + u)\tau_\infty^{\alpha_2},$$

and imply that $\underline{u} = u_{\alpha,\beta,f}$ † normalizes the subalgebra of the matrix algebra $M_2(\widehat{\mathbb{Z}})$ generated by σ_∞ (taking into account the relation

$$(99) \qquad \tau_\infty(\sigma_\infty) = -\sigma_\infty \qquad \text{in } \mathrm{Gl}(2, \widehat{\mathbb{Z}}).$$

If we look at the subgroup of $\mathcal{M}_{0,3}^{!\sim}$ consisting of the u satisfying the equivalent conditions (93) (94) (98)

$$(100) \qquad \mathcal{M}_{0,3}^{!\sim}\left(\frac{1}{2}\right) = \{u \in \mathcal{M}_{0,3}^{!\sim} \mid [u, \sigma_\infty] = 1\} \xrightarrow{\sim} \mathcal{N}_{0,3}^{!\sim}$$

we find that the composite homomorphism

Uncertainty retained from cancelled passages.

140-3-s207-D001 (98) α_0

140-3-s207-D002 [illeg. 1]

[scan 208]

[p. 403]

(101)

$$\begin{array}{ccccc} \mathcal{N}_{0,3}^{!\sim} & \xrightarrow{\sim} & \mathcal{M}_{0,3}^{!}(\frac{1}{2})^{\sim} & \longrightarrow & \mathcal{M}_{0,3}^{!\sim} \xrightarrow[u\longmapsto \underline{u}=(u_{\alpha,\beta,f})]{} \mathrm{Gl}(2, \widehat{\mathbb{Z}})' \\ & \searrow & \downarrow k_{1/2} & \nearrow & \\ & & \mathcal{N}_{0,3}^{!\sim}(\frac{1}{2}) & & \end{array}$$

m889

Marginal note. m889 N.B. $\mathcal{M}_{0,3}^{!}(\frac{1}{2})^{\sim}$ is stable under $x \longmapsto \sigma_\infty x\sigma_\infty^{-1}$ and under $x \longmapsto \tau_\infty x\tau_\infty^{-1}$ (since σ_∞ and τ_∞ commute with σ_∞).

factors through the said normalizer — if we prefer to view it in terms of the group $\mathrm{Gl}(2,\widehat{\mathbb{Z}}) \simeq M_2(\widehat{\mathbb{Z}})^*$, rather than of the algebra of matrices, it factors through the normalizer of the maximal torus †

$$(102) \qquad T_\alpha | T_\sigma = \{ t\sigma_\infty + u \quad (t,u \in \widehat{\mathbb{Z}} | \mathbb{Z},\ \underbrace{t^2+u^2}_{=\det(t\sigma_\infty+u)} \in \widehat{\mathbb{Z}}^*).$$

I suspect that there is only the splitting $k_{1/2}$ of (101) of the extension $\mathcal{M}^{!\sim}_{0,3}$ (of $\mathcal{N}^{!\sim}_{0,3}$ by $\widehat{\Pi}_{0,3}$, † or of $\mathcal{N}^{!\sim}_{0,3}$ by $\widehat{\mathcal{G}}^{+}_{0,3}$) that factors through the normalizer.

m890

Marginal note. m890 Surely bogus: there are also all the curves isogenous to the elliptic curve with modular invariant $j|J=0$ (corresponding the orbit $\{\frac{1}{2}, 2, -1\}$ in $U_{0,3}$ under $\mathfrak{S}_3$).

2° Case This is the case

$$(103) \qquad g=1 \quad \text{i.e. } \alpha_0 = 1 \quad \text{i.e. } \alpha = f \quad \text{i.e. } \xi = f\rho(f)^{-1}, \text{ i.e.}$$

$$(104) \qquad [u,\rho] = 1.$$

For an arbitrary u, there is a $\gamma \in \widetilde{\widehat{\Pi}}_{0,3} = \widehat{\Pi}_{0,3} \cdot \{1, \tau_\infty\}$ † such that $u' = \mathrm{int}(\gamma) \circ u = \gamma u$ commutes|commuting thus with ρ, namely $\gamma = \alpha_0^{-1}$, †But we will pay Attention to the fact that γ is not always in $\widehat{\Pi}_{0,3}$, in fact

$$(105) \qquad \gamma = \alpha_0^{-1} \in \widehat{\Pi}_{0,3} \iff \varepsilon_3(u) = 1 \quad \text{i.e. } \mu(u) = 1 \ (3)$$

Uncertainty retained from cancelled passages.

140-3-s208-D001 formed by

140-3-s208-D002 I think

140-3-s208-D004 unique

140-3-s208-D005 We thus have

[scan 209]

If we denote by $\mathcal{M}^{!\sim}_{0,3}(-\bar{j})$ the subgroup of $\mathcal{M}^{!\sim}_{0,3}$ consisting of the u satisfying (103), (104), its image in $\mathcal{N}^{!\sim}_{0,3}$ is therefore the subgroup of index 2 kernel of

$$u \longmapsto \underbrace{\varepsilon_3(\mu(u))}_{\sim} = \varepsilon_3(u)$$

i.e.

$$(106) \qquad \mathcal{N}^{!\sim}_{0,3}(-\bar{j}) = \{ u \in \mathcal{N}^{!\sim}_{0,3} \mid \varepsilon_3(u) = 1 \quad \text{i.e. } \mu(u) \equiv 1 \ (3) \}$$

m891

Marginal note. m891 N.B. $\mathcal{M}^{!\sim}_{0,3}(-\bar{j})$ is stable under $x \longmapsto \rho x \rho^{-1}$.

and we have

$$(107) \qquad \mathcal{M}^{!\sim}_{0,3}[-\bar{j}] \xrightarrow{\sim} \mathcal{N}^{!\sim}_{0,3}[-\bar{j}].$$

Inverting it we thus find a splitting

(108)

$$\mathcal{N}^{!\sim}_{0,3}(-\bar{j}) \xrightarrow{\sim} \mathcal{M}^{!\sim}_{0,3}(-\bar{j}) \hookrightarrow \mathcal{M}^{!\sim}_{0,3}, \qquad k(-\bar{j})$$

, whose composite with $\mathcal{M}^{!\sim}_{0,3}$ is a homomorphism which †factors through the maximal torus T_ρ of $\mathrm{Gl}(2,\widehat{\mathbb{Z}})'$, formed by the †

$$(109) \qquad \{ c\rho + d \mid c,d \in \widehat{\mathbb{Z}},\ c^2 + cd + d^2 \in \widehat{\mathbb{Z}}^* \}.$$

I suspect that this is the only splitting of $\mathcal{M}^{!\sim}_{0,3}$ (or even of $\mathcal{M}^{!}_{0,3}$) which has this property. This follows from the equivalent form of conditions (103) (104), provided by (79):

bogus, for same reasons as just now…

Uncertainty retained from cancelled passages.
140-3-s209-D002 factors
140-3-s209-D003 $c\rho + d$

[scan 210]
[p. 404]

$$(110) \qquad \underline{u} = u_{\alpha,\beta,f} = \mu(c\rho + d)$$

(N.B. $\delta = \varepsilon_3(\mu|u)$ is equal to 1 in this case).

The u in question correspond to the pairs (η, f) satisfying

$$(111) \qquad \eta\sigma_\infty(\eta) = 1$$

$$(112) \qquad (f\rho(f)^{-1}) = \eta\ell_1^\nu$$

m893

Marginal note. m893 which we will complete|complete by $\xi = f\rho(f)^{-1}$.
m894

Marginal note. m894 We can also write β instead of f, $f(=\beta)$ determined modulo $f \mapsto \ell_0^n f$, id. for α, β.

or again to the pairs (α, β) satisfying the loop relation (97) = (37), which we then complete by $f = \alpha$.

3° Case It is the one where

$$(113) \qquad f \equiv 1 \ (L_0) \quad \text{i.e. } \alpha \equiv \alpha_0 \ (L_0 \text{ on the left}) \quad \text{i.e. } \beta \equiv \beta_0 \ (L_0 \text{ on the left})$$

or again $u(\varepsilon_0) = \varepsilon_0^\mu$ $(\mu \in \widehat{\mathbb{Z}}^*)$ i.e.

$$(114) \qquad [u, \varepsilon_0] = \ell_0^\nu \qquad (\nu \in \widehat{\mathbb{Z}} \ - \ \nu = \tfrac{\mu-1}{2}),$$

i.e. u normalizes $L_0 = \{\ell_0^n \mid n \in \widehat{\mathbb{Z}}\}$ — or again (enlarging $\widehat{\Pi}_{0,3}$) u normalizes $L_0' = \{\varepsilon_0^n \mid n \in \widehat{\mathbb{Z}}\}$. N.B. $L_0 = L_0' \cap \widehat{\Pi}_{0,3}$.

In this case, we remove|can remove the indeterminacy $(\alpha, \beta, f) \longmapsto (\ell_0^n\alpha, \ell_0^n\beta, \ell_0^n f)$ by requiring

$$(115) \qquad f = 1,$$

which determines α, β uniquely.

group $\mathcal{M}^{!\sim}_{0,3}[0]$ of the $u \in \mathcal{M}^{!\sim}_{0,3}$ which normalize L_0 is in one-to-one correspondence with the set of pairs (α, β) $(\alpha \in \widetilde{\widehat{\Pi}}_{0,3} = \widehat{\Pi}_{0,3} \cdot \{1, \tau_\infty\},\ \beta \in \widehat{\Pi}_{0,3})$ satisfying
m896

Marginal note. m896 We do not lift|lift Therefore [illeg. 2] by $\mathcal{M}^{!\sim}_{0,3}(0)$, [illeg. 2] $\mathcal{M}_{0,3}$ and $\mathcal{N}_{0,3}$ since it does not cover|covers only [illeg. 2] section of the extension [illeg. 2] by $\widehat{\Pi}_{0,3}$ — we [illeg. 2] (Proposition [illeg. 1]) section $\mathcal{M}^{!\sim}_{0,3}(0)$ further page 412[1]).

[scan 211]

the everlasting loop relation

$$(115) \qquad (\alpha\rho(\alpha)^{-1}) = (\beta\sigma_\infty(\beta)^{-1})\ell_1^\nu \qquad (\text{for } \nu \in \widehat{\mathbb{Z}} \text{ suitable})$$

(by $(\alpha,\beta)\longmapsto g|\xi=\alpha\rho(\alpha)^{-1}; h|\eta=\beta\sigma_\infty(\beta)^{-1}$) or again to the pairs (g,h) (or (ξ,η)) of elements of $\widehat{\Pi}_{0,3}$ satisfying

$$(116)\quad (=(44)+(46))$$
$$g\rho(g)\rho^2(g)=1,\qquad h\sigma_\infty(h)=1,\qquad g=h\ell_1^\nu\quad(\nu\in\widehat{\mathbb{Z}}\text{ suitable}),$$

m897

Marginal note. m897 In all this, I drop the "preliminary condition" on the generating relations…

to the pair † (g,h) corresponds the automorphism u defined by

$$(117)=(23)\qquad u(\sigma_\infty)=h\sigma_\infty,\qquad u(\rho)=g\rho$$

or again by

$$(118)\qquad u(\sigma_\infty)=\mathrm{int}(\alpha)\sigma_\infty,\qquad u(\rho)=\mathrm{int}(\beta)\rho.$$

In the group $\mathcal{M}^{!\sim}_{0,3}[0]$, there is the subgroup

$$(119)\qquad \mathcal{M}^{!\sim}_{0,3}[0]\cap\widehat{\Pi}_{0,3}\simeq L_0$$

the inclusion

$$(120)\qquad \widehat{\mathbb{Z}}\xrightarrow[\sim]{\ell_0} L_0\hookrightarrow\mathcal{M}^{!\sim}_{0,3}$$

being described by

$$(121)\qquad \lambda=\ell_0^n\longmapsto(\alpha=\lambda|\ell_0^n,\beta=\lambda|\ell_0^n)$$

†

or again by

$$(122)\qquad \lambda=\ell_0^n\longmapsto(f=\ell_0^n\ell_1^{-n},g=\ell_0^n\ell_1^{-n})$$

i.e. $f=g=\lambda\sigma_\infty(\lambda^{-1})=\lambda\rho(\lambda)^{-1}$, multiplication on the left by the $\lambda=\ell_0^n$ in $\mathcal{M}^{!\sim}_{0,3}[0]$ being made explicit|made explicit by

$$(123)\qquad (\alpha,\beta)\longmapsto(\ell_0^n\alpha,\ell_0^n\beta)$$

$$(124)\qquad (\xi,\eta)\longmapsto\ell_0^n\xi\ell_1^{-n},\ \ell_0^n\eta\ell_1^{-n}).$$

Uncertainty retained from cancelled passages.
140-3-s211-D001 as
140-3-s211-D002 i.e. [illeg. 2]

[scan 212]
[p. 405]

Since we have an isomorphism

$$\mathcal{M}^{!\sim}_{0,3}[0]/L_0\xrightarrow{\sim}\mathcal{N}^{!\sim}_{0,3},$$

we find the

Corollary The group $\mathcal{N}^{!\sim}_{0,3}$ is in canonical bijection with the set of pairs $(\alpha,\beta)\in\Pi^{\sim\wedge}_{0,3}\times\Pi^{\wedge}_{0,3}$ satisfying the loop relation (97)

$$\alpha\rho(\alpha)^{-1}=(\beta\sigma_\infty(\beta)^{-1})\ell_1^\nu\qquad(\text{for }\nu\in\widehat{\mathbb{Z}}\text{ suitable}),$$

modulo the transformations $(\alpha,\beta)\mapsto(\ell_0^{n|m}\alpha,\ell_0^{n|m}\beta)$.

The group law (of $\mathcal{M}^{!\sim}_{0,3}(0)$ or of its quotient $\mathcal{N}^{!\sim}_{0,3}$) is made explicit in terms of the (g,h), or of the (α,β), by the "cocycle formulas" (52): if u', u and $u'' = u' \cdot u$ are given by $(g',h'), (g,h), (g'',h'')$, or by $(\alpha',\beta'), (\alpha,\beta)$ and (α'',β''), we will have

$$\begin{cases} g'' = u'(g)g' & h'' = u'(h)h' \\ \alpha'' = u'(\alpha)\alpha' & \beta'' = u'(\beta)\beta'. \end{cases} \tag{125}$$

Remark on the notation $\mathcal{M}^{!\sim}_{0,3}(q)$, for $q = \frac{1}{2}, -\bar{j}, 0$. In fact for every algebraic point

$$q \in \mathcal{U}_{0,3}(\mathbb{C}) = \mathbb{C} \setminus \{0,1\} = \mathbb{P}^1_{\mathbb{C}|\mathbb{Q}}(\mathbb{C}) \setminus \{0,1,\infty\} \tag{126}$$

we expect † a partial splitting

$$k_q : \mathcal{N}|\mathcal{M}^{!\sim}_{0,3}(q) \longrightarrow \mathcal{M}^{!\sim}_{0,3}, \qquad \operatorname{Im} k_q \overset{\text{definition}}{=} \mathcal{M}^{!\sim}_{0,3}(q) \tag{127}$$

of the extension $\mathcal{M}^{!\sim}_{0,3}$ of $\mathcal{N}^{!\sim}_{0,3}$ by $\Pi^{\wedge}_{0,3}$, or
m898

Marginal note. m898 In fact, in a way [illeg. 1] more precise|precise, k_q will only be determined up to an inner automorphism...

Uncertainty retained from cancelled passages.
140-3-s212-D001 to find

[scan 213]

at least we will have a partial splitting over $\Gamma_{\mathbb{Q}}(q)$, which we will try to determine ; $\mathcal{N}^{!\sim}_{0,3}(q)$ denotes a hypothetical subgroup of $\mathcal{N}^{!\sim}_{0,3}$ (which we would have to manage to determine, if indeed [illeg. 1] not too [illeg. 1]...), inducing on $\Gamma_{\mathbb{Q}}$ the open subgroup $\Gamma_{\mathbb{Q}}(q)$ of $\Gamma_{\mathbb{Q}}$ stabilizer of q.
m899

Marginal note. m899 The definition is forced for example when q is an element of a cyclotomic extension of $\mathbb{Q}$, in such a way that $\mathbb{Q}(q)$ corresponds to a finite discrete quotient group of $\widehat{\mathbb{Z}}^*$ and $\Gamma_{\mathbb{Q}}(q)$ is the kernel of

$$\Gamma_{\mathbb{Q}} \xrightarrow{\chi} \widehat{\mathbb{Z}}^* \longrightarrow G$$

...

The analysis of cases 1° and 2° above in fact amounted to the determination of such splittings, for $q = \frac{1}{2}$ and for $q = -\bar{j}$ — in the first case $\Gamma_{\mathbb{Q}}(q)$ is the whole of $\Gamma_{\mathbb{Q}}$, in the second it is the subgroup of index 2 kernel of $\varepsilon_2 \circ \chi : \Gamma_{\mathbb{Q}} \longrightarrow \{\pm 1\}$, and idem for $\mathcal{N}|\mathcal{M}^{!\sim}_{0,3}(q)$. By transforming by ρ and ρ^2, we †would deduce from the splittings|from the splitting $k_{1/2}$ the splittings k_2 and k_{-1} — by transforming by any one of the $\sigma_{i|\rho}$, †, we would deduce from the splitting $k_{-\bar{j}}$ a splitting of type $k_{-j|-1}$.

As for $\mathcal{M}^{!\sim}_{0,3}[0]$, it corresponds in principle to the point

---

Uncertainty retained from cancelled passages.
140-3-s213-D001 would lead
140-3-s213-D002 $\tau_i|\sigma_i$

[scan 214]

0 of $\widehat{\mathcal{U}}_{0,3} = \mathbb{P}^1_{\mathbb{Q}}$, there †two other analogous subgroups $\mathcal{M}^{!\sim}_{0,3}[q]$, with $q = 1$, $q = \infty$, corresponding to the two other points "at infinity" of $\mathcal{U}_{0,3}$, and which are deduced from the preceding [p. 406]

one by applying ρ, ρ^2. Here, we have groups which do not correspond strictly to even partial sections of $\mathcal{M}^{!\sim}_{0,3}$ over $\mathcal{N}^{!\sim}_{0,3}$, since the intersection with $\Pi^{\wedge}_{0,3}$ is not reduced to 1, but it is one of the three loop groups of type L_0, L_1, L_∞.

But we know that the choice of a uniformizer|uniformizer of the local ring of $\widehat{\mathcal{U}}_{0,3}$ at 0 (say) determines a splitting of $\mathcal{M}^{!\sim}_{0,3}(0)|[0]$, regarded as an extension of $\mathcal{N}^{!}_{0,3}$ by L_0 — and later on we will have to try to grasp †such a splitting (associated with a uniformizer) from the viewpoint of our calculation, as a privileged way †, for every datum mod L_0 of (α, β) satisfying a loop relation, of "normalizing" the choice of the (α, β) mod L_0. I will come back to this later, surely.

Uncertainty retained from cancelled passages.

140-3-s214-D001 correspond to it by

140-3-s214-D002 this

140-3-s214-D003 normalizing

[scan 215]

[p. 407]

V Recapitulation in terms of $\mathcal{M}^{!\sim}_{0,3}[0]$

It finally appears that the simplest way to study the loop automorphisms of $\Pi^{\wedge}_{0,3}$ (commuting outer^{ly} with $\mathfrak{S}_3$, i.e. the elements of † $\mathcal{N}^{!\sim}_{0,3}$) is to begin with the study of the subgroup $\mathcal{M}^{!\sim}_{0,3}[0]$, normalizer of L_0 in $\mathcal{M}^{!\sim}_{0,3}$. Setting for $u \in \mathcal{M}^{!\sim}_{0,3}[0]$

$$u(\sigma_\infty) = \eta\sigma_\infty, \qquad u(\rho) = \xi\rho \qquad (\eta, \xi \in \Pi^{\wedge}_{0,3}) \tag{128}$$

i.e.

$$[u, \sigma_\infty] = \eta \qquad [u, \rho] = \xi \tag{129}$$

which express the outer commutation of u with $\mathfrak{S}_3$ (indeed generated by σ_∞ and ρ), the condition of normalizing L_0 is expressed by a relation of the form

$$u(\varepsilon_0) = \varepsilon_0^\mu \quad \text{or} \quad u(\ell_0) = \ell_0^\mu \qquad (\mu \in \widehat{\mathbb{Z}}^*). \tag{130}$$

m900

Marginal note. m900 (130) is also equivalent, given $[u, \rho] = \xi$, to

$$\begin{cases} u(\varepsilon_1) = \mathrm{int}(\xi)(\varepsilon_1^\mu) \\ \qquad \text{and to} \\ u(\varepsilon_\infty) = \mathrm{int}(\rho^{-1}(\xi^{-1})\,[\textit{illeg. 1}])(\varepsilon_\infty^\mu) \end{cases} \tag{130bis}$$

Now we have, since $\varepsilon_0 = \sigma_\infty\rho$ † †

$$u(\varepsilon_0) = u(\sigma_\infty)u(\rho) = \eta\sigma_\infty\xi\rho = \eta\sigma_\infty(\xi)\underbrace{\sigma_\infty\rho}_{\varepsilon_0} = \eta\sigma_\infty(\xi)\varepsilon_0$$

so (130) can be written

$$\begin{gathered} \eta\sigma_\infty(\xi) = \varepsilon_0^{\mu-1} \qquad \text{i.e., setting } \mu - 1 = 2\nu \quad (\nu \in \widehat{\mathbb{Z}}) \\ \eta\sigma_\infty(\xi) = \ell_0^\nu \qquad \text{i.e., applying } \sigma_\infty \\ \sigma_\infty(\eta)\xi = \ell_1^\nu. \end{gathered} \tag{131}$$

Now the relations

$$(\eta\sigma_\infty)^2 = 1, \qquad (\xi\rho)^3 = 1 \tag{132}$$

can also be written

$$\boxed{\eta\sigma_\infty(\eta) = 1, \qquad \xi\rho(\xi)\rho^2(\xi) = 1}, \tag{133}$$

whence $\sigma_\infty(\eta) = \eta^{-1}$, so that (131) can also be written in a more convenient way

$$\boxed{\xi = \eta\ell_1^\nu} \qquad \text{(loop relation)} \tag{134}$$

Uncertainty retained from cancelled passages.
140-3-s215-D001 [illeg. 1]
140-3-s215-D003 is written
140-3-s215-D005 $\eta\sigma_\infty$

[scan 216]

We thus find the

Theorem The elements u of $\mathcal{M}|\mathcal{N}^{!\sim}_{0,3}[0]$ — i.e. the outer automorphisms of $\Pi^\wedge_{0,3}$ which commute outer^{ly} with $\mathfrak{S}_3$ and which normalize L_0 — are in one-to-one correspondence (via formulas (128) or 129) with the pairs (ξ,η) $(\xi,\eta\in\Pi^\wedge_{0,3})$, satisfying relations (133)(134), and such that moreover ("preliminary condition") relations (132) (equivalent : (133)) are †generating for $\widehat{\mathcal{G}}^+_{0,3}$. († In (134) ν is a suitable element of $\widehat{\mathbb{Z}}$, obviously uniquely determined, and subject to the additional condition $2\nu+1\in\widehat{\mathbb{Z}}^*$) (which is implied by the preliminary condition, and perhaps implies it...)
m901

Marginal note. m901 this condition $2\nu+1\in\widehat{\mathbb{Z}}^*$ is in fact a consequence of equations (133), (134), cf. below, theorem page 409!

And the obvious inclusion

$$\begin{aligned} L_0 &\hookrightarrow \mathcal{M}^{!\sim}_{0,3}(0)|(\omega)\\ \ell_0^n=\lambda &\longmapsto \mathrm{int}(\lambda)\end{aligned} \tag{135}$$

is given by

$$\ell_0^n=\lambda\longmapsto(\xi=\lambda\rho(\lambda)^{-1}=\ell_0^n\ell_1^{-n},\quad \eta=\lambda\sigma_\infty(\lambda)^{-1}=\ell_0^n\ell_1^{-n}) \tag{136}$$

and more generally the left action of L_0 on $\mathcal{M}^{!\sim}_{0,3}(0)|(\omega)$ is given, for $\lambda=\ell_0^n\in L_0$, by

$$(\xi,\eta)\longmapsto(\xi'=\ell_0^n\xi\ell_1^{-n},\quad \eta'=\ell_0^n\eta\ell_1^{-n}) \tag{137}$$

so

$$\mathcal{M}^{!\sim}_{0,3}[0]/L_0\simeq\mathcal{N}^{!\sim}_{0,3}$$

is isomorphic to the quotient of the set of pairs (ξ,η) satisfying (133)(134) [and the preliminary condition], by the operations (137) — taking into account a † natural group structure : to put on this set of pairs (ξ,η), given by

Uncertainty retained from cancelled passages.
140-3-s216-D001 primitive
140-3-s216-D002 [illeg. 1]
140-3-s216-D003 operation

[scan 217]
[p. 408]

$$\underbrace{(\xi',\eta')}_{\dagger}(\xi,\eta)=(\xi'',\eta'') \tag{138}$$

with

$$\xi''=u'(\xi)\xi',\qquad \eta''=u'(\eta)\eta' \tag{139}$$

u' being the automorphism defined by (ξ',η').

Corollary Every $u\in\mathcal{M}^{!\sim}_{0,3}$ can be written under|in the form

$$u=f^{-1}u_{\xi,\eta}=\mathrm{int}(f^{-1})\circ u_{\xi,\eta}, \tag{140}$$

with $f \in \widehat{\Pi}_{0,3}$ and (ξ, η) satisfying the conditions of the preceding theorem, so that we will have

$$u(\sigma_\infty) = h\sigma_\infty, \qquad u(\rho) = g\rho, \quad \text{with} \tag{141}$$

$$h = f^{-1}\eta\sigma_\infty(f), \qquad g = f^{-1}\xi\rho(f). \tag{143}$$

For us to have

$$f^{-1}u_{\xi,\eta} = f'^{-1}u_{\xi',\eta'}$$

it is necessary and sufficient that there exist $n \in \mathbb{Z}$ such that

$$\xi' = \ell_0^n \xi \ell_1^{-n}, \qquad \eta' = \ell_0^n \eta \ell_1^{-n}, \qquad f' = \ell_0^n f. \tag{144}$$

N.B. Thanks to the loop formula, once the multiplier μ i.e. ν is fixed (and in fact μ is already uniquely determined by the knowledge of ξ alone, as is shown by the adelic calculations in $\mathrm{Gl}(2, \widehat{\mathbb{Z}})'$ ...), ξ and η determine one another by (134), so we can eliminate one or the other from the system of equations (133), (134), which becomes either

$$\xi\rho(\xi)\rho^2(\xi) = 1, \qquad (\dagger\xi\ell_1^{-\nu})\sigma_\infty(\xi\ell_1^{-\nu}) = 1 \quad \text{i.e.} \quad \sigma_\infty(\xi) = \ell_1^\nu \xi^{-1} \ell_0^\nu \tag{145}$$

or

$$\eta\sigma_\infty(\eta) = 1, \qquad (\eta\ell_1^\nu)\rho(\eta\ell_1^\nu)\rho^2(\eta\ell_1^{\nu|-\nu}) = 1. \tag{146}$$

Uncertainty retained from cancelled passages.

140-3-s217-D001 u'

140-3-s217-D002 η

[scan 218]

The adelic calculation. We again denote by ξ, η the images of ξ, η in $\mathrm{Sl}(2, \widehat{\mathbb{Z}})'$, which lift to elements, again denoted ξ, η, of $\mathrm{Sl}(2, \widehat{\mathbb{Z}})$.

There is a sign ambiguity, which we remove for ξ by imposing

$$\xi\rho(\xi)\rho^2(\xi) = \underline{\underline{+}}1 \qquad (\text{in } \mathrm{Sl}(2, \widehat{\mathbb{Z}})) \tag{147}$$

for η there is no simple way to remove the ambiguity (if we wish to admit provisionally the possibility of writing in the form $\beta\sigma_\infty(\beta)^{-1}$) of which|thus we will have a priori only

$$\eta\sigma_\infty(\eta) = \dagger \pm 1, \tag{148}$$

with no possibility of changing the sign in the 2° case by the change $\eta \mapsto -\eta$! Let us set

m902

Marginal note. m902 but yes, we can make the choice of η by requiring $\xi = +\eta\ell_1^\nu$ (but this convention is not compatible with a convention made elsewhere $\eta = \beta\sigma_\infty(\beta)^{-1}$).

$\in \widehat{\Pi}_{0,3}$

$$\xi = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \qquad \eta = \begin{pmatrix} R & S \\ T & U \end{pmatrix}. \tag{148}$$

We will have

$$AD - BC = 1 \qquad RU - ST = 1 \tag{149}$$

and relation (147) is then equivalent to :

$$C + D - B = 1, \tag{150}$$

† (at least in each ℓ-adic component we will have to have either this relation, or

$$\xi_\ell = \begin{pmatrix} A_\ell & B_\ell \\ C_\ell & D_\ell \end{pmatrix} = -\rho^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 0 \end{pmatrix},$$

† proposition on page 394 – but this solution is incompatible with the condition $\boxed{\xi \equiv 1\ (2)}$, whence $A, D \equiv 1\ (2)$ and $B, C \equiv 0\ (2)$ ... so (150) is true in $\mathbb{Z}_\ell$ for $\forall \ell$, hence in $\widehat{\mathbb{Z}}$ ...).

Uncertainty retained from cancelled passages.

140-3-s218-D001 ε

140-3-s218-D002 whereas

140-3-s218-D003 cf.

[scan 219]
[p. 409]

Relation (148) can also be written

$$\eta = \pm\sigma_\infty(\eta^{-1})$$

now $\sigma_\infty(\eta^{-1}) = {}^t\eta$, so (148) can be written

$$\eta = \pm{}^t\eta,$$

but if we had the sign $-$, we would have $U = -U$ so $2U = 0$ so $U = 0$, which is incompatible with

$$\eta \equiv \pm 1 | 1 \quad (2)$$

which implies $R, U \equiv 1\ (2)$, $S, T \equiv 0\ (2)$. We will therefore have $\eta = {}^t\eta$ i.e.

$$\eta\sigma_\infty(\eta) = +1, \qquad \text{i.e.} \tag{151}$$

$$S = T. \tag{152}$$

The loop relation (134) can be written

$$\begin{aligned} \begin{pmatrix} A & B \\ C & D \end{pmatrix} &= \varepsilon \begin{pmatrix} R & S \\ T & U \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 2\nu & 1 \end{pmatrix} \\ &= \varepsilon \begin{pmatrix} R + 2\nu S & S \\ T + 2\nu U & U \end{pmatrix} \qquad \text{with } \varepsilon \in \{\pm 1\} \end{aligned} \tag{153}$$

m904

Marginal note. m904 Observation here $\varepsilon = 1$, by choosing the sign of η suitably (153) (the sign convention does not coincide with a previous convention centered on the α, β, but from the present viewpoint it seems more friendly).

and it is equivalent to the relations

$$B = \varepsilon S, \qquad D = \varepsilon U \qquad \text{and} \tag{154}$$

†

$$\mu D = 1 \qquad \text{where } \mu = 2\nu + 1 \tag{155}$$

equivalent|being equivalent also ; given (154), to

$$\mu U = \varepsilon, \tag{155bis}$$

and the latter merely expresses (taking (152) into account) the relation

$$\varepsilon \overbrace{(T + 2\nu U)}^{A'} + \overbrace{\varepsilon U}^{B'} - \overbrace{\varepsilon S}^{C'} = 1$$

paraphrasing

$$A + B - C = 1,$$

it

Uncertainty retained from cancelled passages.
140-3-s219-D002 [illeg. 1]

[scan 220]

it merely expresses (in view of (154) and (150) the relation

$$C = \overbrace{\varepsilon(T + 2\nu U)}^{C'}.$$

Thus the relations (154) (155) express the fact that the two sides of (153) † three coefficients equal † pairwise, $B = B'$, $C = C'$, $D = D'$, and the equality for the last

$$A = \overbrace{\varepsilon(R + 2\nu S)}^{A'}$$

then follows from this thanks to the relations ,

$$AD - BC = 1, \qquad A'D' - B'C' = 1,$$

which give

$$A = \frac{1 + BC}{D}, \qquad A' = \frac{1 + B'C'}{D'}$$

(N.B. we know that $D = D'$ is invertible, by (155)) hence $A = A'$.

We thus find the

Theorem For a solution (ξ, η) in $\widehat{\Pi}_{0,3} \times \widehat{\Pi}_{0,3}$ of the system of equations (133) (134), again denoting by ξ, η elements of $\mathrm{Sl}(2, \widehat{\mathbb{Z}})$ corresponding to them, the sign ambiguity for ξ being removed by relation (147), we will have the relations (154) and (155), which imply the following consequences:

a) $\mu = 2\nu + 1$ is invertible, and it is entirely determined by ξ – it is determined by η only up to sign.

Uncertainty retained from cancelled passages.
140-3-s220-D001 all their terms
140-3-s220-D002 [illeg. 1]

[scan 221]
[p. 410]

Remark

In view of the relations (149) (150)

$$AD - BC = 1, \qquad A|C + D - B = 1,$$

and of the fact that D is invertible, the matrices ξ are entirely determined by the coefficients B, D — the pairs (ξ, η) of matrices $\in \mathrm{Sl}(2, \widehat{\mathbb{Z}})$, satisfying

$$\begin{cases} \xi\rho(\xi)\rho^2(\xi) = 1, \quad \eta\sigma_\infty(\eta) = 1, \quad \xi = \eta\ell_1^\nu \qquad (\nu \in \widehat{\mathbb{Z}} \text{ suitable}) \\ \xi \equiv 1 \quad (2), \qquad \eta \equiv 1 \quad (2) \end{cases} \tag{156}$$

m905

Marginal note. m905 Of course, the introduction of the congruences here is entirely incidental — the algebraic statement is independent of such conditions...

are in one-to-one correspondence with the pairs (B, D), $B \in \widehat{\mathbb{Z}}$, $D \in \widehat{\mathbb{Z}}^*$, $B \equiv 0\ (2)$, from which we deduce

$$\xi = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \qquad \eta = \begin{pmatrix} R & S \\ T & U \end{pmatrix}, \qquad \text{and } \nu \in \widehat{\mathbb{Z}} \text{ by}$$

$$C = B - D + 1, \qquad A = \frac{1 + BC}{D} = \frac{1}{D}(1 + B^2 - BD + B) \tag{157}$$
$$= \mu(1 + B + B^2) - B$$

$$\mu = \frac{1}{D}, \qquad \nu = \frac{\mu - 1}{2}, \tag{158}$$

†

$$\eta = \xi \ell_1^{-\nu} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -2\nu & 1 \end{pmatrix} = \begin{pmatrix} A - 2\nu B & B \\ C - 2\nu D & D \end{pmatrix} \qquad \text{i.e.}$$

m906

Marginal note. m906 we have normalized the sign of η as said in (156)

$$\text{i.e.} \quad U = D, \quad S = T = B, \quad R = A - 2\nu B = \mu(1 + B^2) = \frac{1 + B^2}{D} \tag{159}$$

Put (compare page 398' (78))

$$\underline{u}_{\xi\dagger} = \mu \begin{pmatrix} 1 & B \\ 0 & D \end{pmatrix} = \begin{pmatrix} \mu & \mu B \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{D} & B/D \\ 0 & 1 \end{pmatrix} \in \mathrm{Gl}(2, \widehat{\mathbb{Z}}) \tag{160}$$

so that knowledge of $\underline{u}_\xi$ is equivalent to that of B and D, i.e. to that of $\xi \in \mathrm{Sl}(2, \widehat{\mathbb{Z}})$, or again to that of the pairs (ξ, η) (η normalized by $\eta = +\xi\ell_1^{-\nu}$). (In loc. cit. this element was denoted $u_{\alpha,\beta}$... but it depends only on $\xi \in \widehat{\Pi}_{0,3}$, and even only on $\underline{\xi} \in \mathrm{Sl}(2, \widehat{\mathbb{Z}})$ — and knowledge of it is equivalent to that of $\underline{\xi} \in \mathrm{Sl}(2, \widehat{\mathbb{Z}})$ —) †

m908

Marginal note. m908 N.B. We deduce ξ, η from knowledge of $\underline{u}$ by the formulas †

$$\begin{cases} \underline{u} \cdot {}^t\underline{u} = \mu\eta \\ \mu = \det \underline{u} \end{cases} \tag{161}$$

whence

$$\begin{cases} \eta = \mu^{-1}\underline{u} \cdot {}^t\underline{u} = \mu^{-1}, \\ \xi = \eta\ell_1^{-\nu} = \mu^{-1}(\underline{u} \cdot {}^t\underline{u})\ell_1^{-\nu} \end{cases} \tag{162}$$

Uncertainty retained from cancelled passages.

140-3-s221-D001 (159)

140-3-s221-D002 , η

140-3-s221-D003 have $\widetilde{\widehat{\Pi}}_{0,3}$ which is determined|ξ is determined by or in|or of

140-3-s221-D004 [illeg. 1]

[scan 222]

† We will have † †

$$(163) \qquad \underline{u}_\xi(\sigma_\infty) = \eta\sigma_\infty = (\xi\ell_1^{-\nu})\sigma_\infty, \qquad \underline{u}_\xi(\rho) = \xi\rho$$

$$(164) \qquad \underline{u}_\xi(\varepsilon_0) = \varepsilon_0^\mu \qquad \text{and} \qquad u_\xi(\ell_0] = \ell_0^\mu$$

m909

Marginal note. m909 The formulas (163) can also be written

$$(\underline{u}, \sigma_\infty)|[\underline{u}, \sigma_\infty] = \eta, \quad [\underline{u}, \rho] = \xi \qquad (163 \text{ bis}),$$

they give the way to reconstruct (ξ, η) using $\underline{u}_\xi = \underline{u}$.

If we consider an arbitrary invertible matrix

$$\underline{u} = \begin{pmatrix} L & M \\ N & P \end{pmatrix} \in \mathrm{Gl}(2, \widehat{\mathbb{Z}})$$

the relation

$$\dagger \qquad \underline{u}(\varepsilon_0) = \varepsilon_0^\mu \quad \text{i.e.} \quad u\varepsilon_0 = \varepsilon_0^\mu \qquad (\text{where } \mu \in \widehat{\mathbb{Z}}^*)$$

†is written

$$\begin{pmatrix} L & -L+M \\ N & -N+P \end{pmatrix} = \begin{pmatrix} L-\mu N & M-\mu P \\ N & P \end{pmatrix}$$

and is equivalent to: †

$$N = 0, \qquad L = \mu P$$

i.e.

$$\underline{u} = \begin{pmatrix} \mu P & M \\ 0 & P \end{pmatrix}, \qquad \text{whence} \quad \det u = \mu P^2$$

the additional condition

$$(165) \qquad \det \underline{u} = \mu$$

gives the condition

$$\dagger \qquad P^2 = \underline{1},$$

the most obvious solution being $P = 1$, which gives just the matrices

$$\underline{u} = \begin{pmatrix} \mu & M \\ 0 & 1 \end{pmatrix} = \underline{u}_\xi$$

Uncertainty retained from cancelled passages.

140-3-s222-D001 the map
140-3-s222-D002 $\mathcal{M}^{!\sim}_{0,3}(0)$
140-3-s222-D003 i.e. $\eta = (\underline{u}, \sigma_\infty)$
140-3-s222-D004 [*illeg.* 1]
140-3-s222-D005 is equivalent to
140-3-s222-D006 N
140-3-s222-D007 [*illeg.* 1]

[scan 223]
[p. 411]

where $\xi = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ † is given|given in terms of B, D by $D = \dfrac{1}{\mu}$, $B = DM = M/\mu$.

Theorem The map

$$(165) \qquad \begin{array}{ccc} \mathcal{M}^{!\sim}_{0,3}[0] & \longrightarrow & \mathrm{Gl}(2, \widehat{\mathbb{Z}})^\dagger \\ u_{\xi,\eta} & \longmapsto & \underline{u}_\xi \\ u & & \end{array}, \qquad \text{or again } u \longmapsto \underline{u}$$

$\underline{u}_\xi$ is given by (160)) is a group homomorphism.

m910

Marginal note. m910 N.B. Since we have restricted ourselves to $\mathcal{M}^{!\sim}_{0,3}(0)$, we get a homomorphism into $\mathrm{Gl}(2,\widehat{\mathbb{Z}})$, not just into $\mathrm{Gl}(2,\widehat{\mathbb{Z}})/\pm 1$.

m911

Marginal note. m911 N.B. This map factors through the triangular subgroup|subgroup $\mathcal{B}$ below (169)

Proof Let us consider $u = u_{\xi,\eta}$, † $\underline{u}$ is its image in $\mathrm{Gl}(2,\widehat{\mathbb{Z}})$, and if $x \in \widehat{\Pi}_{0,3}$, $\underline{x}$ its image in $\mathrm{Sl}(2,\widehat{\mathbb{Z}})$: †

$$\underline{u(x)} = \underline{u}(\underline{x}) \left(\overset{\text{definition}}{=} \underline{u}\,\underline{x}\,\underline{u}^{-1}\right) \tag{166}$$

for it suffices to see it for $x = \sigma_\infty$ and for $x = \rho$, where this formula reduces to (161). This noted, if $u = u_{\xi,\eta} = u_\xi$, $u' = u_{\xi',\eta'} = u_{\xi'}$ are two elements of $\mathcal{M}^{!\sim}_{0,3}(0)$, $u'' = u_{\xi'',\eta''} = u_{\xi''}$ their product, we will have (139)

$$\dagger \qquad \xi'' = u'(\xi)\xi'$$

whence

$$\underline{\xi}'' = \underline{u'(\xi)}\,\underline{\xi'} = \underline{u}'(\underline{\xi})\,\underline{\xi}', \quad \text{i.e.}$$

$$\dagger \quad (167) \qquad \underline{\xi}'' = \mathrm{int}(\underline{u}_{\underline{\xi}'})(\underline{\xi}) \,.\, \underline{\xi}'.$$

and it suffices to verify that if three solutions †

$$\xi = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad \xi' = \begin{pmatrix} A' & B' \\ C' & D' \end{pmatrix}, \quad \xi'' = \begin{pmatrix} A'' & B'' \\ C'' & D'' \end{pmatrix}$$

of the equations $AD - BC = 1$, $C + D - B = 1$, $D \in \widehat{\mathbb{Z}}^*$, are related by (167), then we have

$$\underline{u}_{\underline{\xi}''} = \underline{u}_{\underline{\xi}'} \,.\, \underline{u}_{\underline{\xi}} \tag{168}$$

Uncertainty retained from cancelled passages.

140-3-s223-D001 satisfies
140-3-s223-D002 $'|*$
140-3-s223-D003 let
140-3-s223-D005 therefore in
140-3-s223-D006 (167)
140-3-s223-D007 (167)
140-3-s223-D008 [illeg. 1]

[scan 224]

Since ξ and $\underline{u} = u_\xi$ determine each other (163bis) it † amounts to the same thing to say that if, on the group

$$\begin{array}{l} \mathcal{B} \subset \mathrm{Gl}(2,\widehat{\mathbb{Z}}) \\ \| \\ \left\{ \begin{pmatrix} \mu & M \\ 0 & 1 \end{pmatrix} \;\middle|\; \mu, M \in \widehat{\mathbb{Z}},\ \mu \text{ invertible} \right\} \end{array} \tag{169}$$

we define (cf. (162))

$$\begin{cases} F_\xi : \mathcal{B} \longrightarrow \mathrm{Sl}(2,\widehat{\mathbb{Z}}) \\ F_\xi(\underline{u}) = (\underbrace{\det \underline{u}}_{\mu})^{-1} \underbrace{\dagger}_{[\underline{u},\rho]} \dagger \end{cases} \tag{170}$$

then we have the 1-cocycle relation

$$F_\xi(\underline{u}'\underline{u}) = \mathrm{int}(\underline{u}')(F_\xi(\underline{u})) \,.\, F_\xi(\underline{u}'). \tag{171}$$

By multiplicativity of det, we are reduced † to seeing that

$$\underline{u} \longmapsto [\underline{u}, \rho] = \underline{u}\,\rho(\underline{u})^{-1}$$

satisfies the cocycle relation, which is trivial.

Corollary The map

$$\begin{array}{rcl} \mathcal{M}^{!\sim}_{0,3} = \mathcal{M}^{!\sim}_{0,3}[0] \,.\, \widehat{\mathcal{G}}^{+}_{0,3} & \longrightarrow & \mathrm{Gl}(2, \widehat{\mathbb{Z}})' \\ u_\xi \,.\, a & \longrightarrow & (\underline{u}_\xi)' \,.\, \underline{a} \end{array} \tag{172}$$

where $(\underline{u}_\xi)'$ is the reduction of $\underline{u}_\xi$ modulo ± 1, and where $\underline{a}$ denotes the image of a in $\mathrm{Sl}(2, \widehat{\mathbb{Z}})'$) is a group homomorphism.

Follows at once from the Theorem and the formulas (163).

m912

Marginal note. m912 careful, this is not a semidirect product, since $\mathcal{M}^{!\sim}_{0,3}(0) \cap \widehat{\mathcal{G}}^{+}_{0,3} = L_0 \neq \{1\}$, we must therefore show that the defining formula does not depend on the way of writing a u in the form $u_\xi a$ — this amounts to seeing that for $\lambda \in L_0$, these two images [illeg. 2] as an element of one or the other factor, are the same.

Uncertainty retained from cancelled passages.

140-3-s224-D001 [illeg. 1]

140-3-s224-D002 $\underline{u}\,{}^t\underline{u}$

140-3-s224-D003 $\mu - 1 = 2\nu$

140-3-s224-D004 $\varepsilon_1^{-(\det \underline{u} - 1)} | \ell_1^{-(\det \underline{u} - 1)}$

140-3-s224-D005 $(\underline{u}, \underline{u}' \in \mathcal{B})$

[scan 225]
[p. 412]

From now on I am going to specify|specify the pair $(\underline{\xi}, \underline{\eta})$ from the adelic point of view (in $\mathrm{Gl}(2, \widehat{\mathbb{Z}})$) using $\underline{u}_\xi$, which is therefore of the form

$$\underline{u} = \begin{pmatrix} \mu & M \\ 0 & 1 \end{pmatrix} \tag{173}$$

with $M \equiv 0$ (2) if $\underline{u}$ comes from $\mathcal{M}^{!\sim}_{0,3}(0)$ from which we get ξ, η by the formulas (163 bis)|(167 bis),

$$\underline{\xi} = [\underline{u}, \underline{\rho}], \qquad \underline{\eta} = [\underline{u}, \underline{\sigma}_\infty]. \tag{174}$$

For $\underline{u} \in \mathcal{M}^{!\sim}_{0,3}(0)$, defined by (ξ, η), as long as we do not bring in the possibility of writing ξ and η uniquely as

$$\xi = \alpha\rho(\alpha)^{-1}, \quad \eta = \beta\sigma_\infty(\beta)^{-1}$$

with $\alpha \in \widetilde{\widehat{\Pi}}_{0,3} | \widehat{\widetilde{\Pi}}_{0,3}$, $\beta \in \widehat{\Pi}_{0,3}$, the only adelic invariants that we can associate to it directly in a natural way is the pair $(\underline{\xi}, \underline{\eta})$, or again (what amounts to the same thing, but is prettier) $\underline{u}$. The elements †

$$\lambda = \ell_0^n \in L_0 \hookrightarrow \mathcal{M}^{!\sim}_{0,3}[0]$$

correspond to

$$\begin{aligned} \xi = \eta = \ell_0^n \ell_1^{-n} &= \begin{pmatrix} 1 & -2n \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -2n & 1 \end{pmatrix} \\ &= \begin{pmatrix} A = 1 + 4n^2 & B = -2n \\ C = -2n & D = 1 \end{pmatrix} \end{aligned} \tag{175}$$

Marginal note. m913 please do the forgotten check which is referred to in the note in the margin just before…

$$\underline{\lambda} = \underline{\underline{\ell_0}}^{\,n} | \underline{\ell_0}^{\,n} = \begin{pmatrix} 1 & -2n \\ 0 & 1 \end{pmatrix} \tag{176}$$

moreover we have

$$\underbrace{\begin{pmatrix} 1 & -2n \\ 0 & 1 \end{pmatrix}}_{\underline{\ell_0}^{\,n}} \begin{pmatrix} \mu & M \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \mu & M-2n \\ 0 & 1 \end{pmatrix} \tag{177}$$

which shows that † the quotients of the subgroups $\mathcal{B}'_1$ of $\mathcal{B}$ (169) consisting of the

Uncertainty retained from cancelled passages.
140-3-s225-D001 [illeg. 1]
140-3-s225-D002 [illeg. 1]

matrices $\{\ \underline{u} \in B$ such that $u \equiv 1\ (2)$ i.e. $M \equiv 0\ (2)\}$, by the closed subgroup $\underline{L}_0$ generated by $\ell_0 = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$, is identified with the group of matrices $\begin{pmatrix} \mu & 0 \\ 0 & 1 \end{pmatrix}$ — the maximal torus T_0 of B, more precisely we have a $\frac{1}{2}$ direct product structure [scan 226]

$$\begin{array}{ccccc} \underline{B}_1 & \simeq & T_0 & \cdot & \underline{L}_0 \\ \| & & \| & & \| \\ \left\{ \begin{pmatrix} \mu & M \\ 0 & 1 \end{pmatrix} \middle| \begin{array}{l} \mu \in \widehat{\mathbb{Z}}^* \\ M \in 2\widehat{\mathbb{Z}} \end{array} \right\} & & \left\{ \begin{pmatrix} \mu & 0 \\ 0 & 1 \end{pmatrix} \middle| \mu \in \widehat{\mathbb{Z}}^* \right\} & & \left\{ \begin{array}{c} \ell_0^n \\ \| \\ \begin{pmatrix} 1 & -2n \\ 0 & 1 \end{pmatrix} \end{array} \middle| n \in \widehat{\mathbb{Z}} \right\} \end{array} \tag{178}$$

This shows that if we are interested in the quotient $\mathcal{M}^{!\sim}_{0,3}(0)/L_0 \xrightarrow{\sim} \mathcal{N}^{!\sim}_{0,3}$, then the only abelian invariant we know how to associate with it in the present context (when we are not working with α, β) is the multiplier μ. But better still, we find thanks to the decomposition (178) in $\mathrm{GL}(2, \widehat{\mathbb{Z}})$ a canonical splitting of the extension $\mathcal{M}^{!\sim}_{0,3}(0)$ of $\mathcal{N}^{!\sim}_{0,3}$ by L_0 :

Proposition For $\forall\, \dot{u} \in \mathcal{N}^{!\sim}_{0,3}$, there is a unique representative $u = u_{\dot{u}} | u = u_\xi \in \mathcal{M}^{!\sim}_{0,3}[0]$ such that we have $\underline{u} \in T_0$ i.e. $\underline{u}$ of the form $\begin{pmatrix} \mu & 0 \\ 0 & 1 \end{pmatrix}$ [i.e. $\xi = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ of the form $\begin{pmatrix} A & 0 \\ C & D \end{pmatrix} = \begin{pmatrix} *|\frac{1}{D} & 0 \\ 1-D & D \end{pmatrix}$], — In other words, the subgroup 225

†

[scan 227] [p. 413]

$$\mathcal{M}^{!\sim}_{0,3}(0) \overset{\mathrm{df}}{=} \left\{ u \in \mathcal{M}^{!\sim}_{0,3}(0) \;\middle|\; \underline{u} \in T_0 \quad \text{i.e. } \underline{u} = \begin{pmatrix} \mu & 0 \\ 0 & 1 \end{pmatrix} \right\} \tag{179}$$

is a section subgroup of the extension $\mathcal{M}^{!\sim}_{0,3}[0]$ of $\mathcal{N}^{!\sim}_{0,3}$ by L_0, hence also of the extension $\mathcal{M}^{!\sim}_{0,3}$ of $\mathcal{N}^{!\sim}_{0,3}$ by $\widehat{\Pi}_{0,3}$, or of the extension $\mathcal{M}^{\sim}_{0,3}$ of $\mathcal{N}^{!\sim}_{0,3}$ by $\widehat{\mathcal{G}}^{+}_{0,3}$.

This|That comes from the fact that if $u \in \mathcal{M}^{!\sim}_{0,3}[0]$ defines $\underline{u} = \begin{pmatrix} \mu & M \\ 0 & 1 \end{pmatrix}$, thus with $M \equiv 0\ (2)$, i.e. $M \in 2\widehat{\mathbb{Z}}$, there is a unique $n \in \widehat{\mathbb{Z}}$ such that $\underline{\ell_0^n u} = \begin{pmatrix} \mu & M-2n \\ 0 & 1 \end{pmatrix}$ is in T_0 i.e. such that $\ell_0^n u \in \mathcal{M}^{!\sim}_{0,3}(0)$, namely $n = \dfrac{M}{2}$.

If $(\xi, \dagger\eta)$ satisfy conditions (133)(134) i.e. define a $u = u_{\xi,\eta}$ (cf. Theorem page 407'), then we shall say that (ξ, η) is normalized if we have $u_{\xi,\eta} \in \mathcal{M}^{!\sim}_{0,3}(0)$. Likewise, if an arbitrary element of $\mathcal{M}^{!\sim}_{0,3}$ is written in the form

$$u = \cdot f^{-1} \cdot u_{\xi,\eta} = f^{-1} u_{\xi},$$

we say that this representation (tainted with the indeterminacy (136)) is normalized, if $u_{\xi,\eta} = u_{\xi}$ is normalized, [i.e. if for $\xi \simeq \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ we have $B = 0$]. Thus, u can be written uniquely in the form

$$u_{f,\xi,\eta} = f^{-1} u_{\xi}, \tag{179}$$

226

Uncertainty retained from cancelled passages.

140-3-s227-D001 $\mathcal{M}^{!\sim}_{0,3}(0)$

140-3-s227-D002 ε

with $f|\xi$ normalized. In fact, this corresponds to the $\frac{1}{2}$-direct product decomposition [scan 228]

$$\mathcal{M}^{!\sim}_{0,3} \simeq \mathcal{M}^{!\sim}_{0,3}(0) \cdot \widehat{\Pi}_{0,3}, \tag{180}$$

moreover we likewise have

$$\mathcal{M}^{\sim}_{0,3} \simeq \mathcal{M}^{!\sim}_{0,3}(0) \cdot \widehat{\mathcal{G}}^{+}_{0,3}. \tag{181}$$

Remark We should at some point try to relate these canonical splittings to the choice of a uniformizer at 0 of $\mathbb{P}^1_{\mathbb{Q}} = \widehat{U}_{0,3\mathbb{Q}}$. Of course, the same question also arises at|for the other two "points at infinity" $1, \infty$ of $U_{0,3\mathbb{Q}}$, relative to the section subgroups

$$\mathcal{M}^{!\sim}_{0,3}(q) \qquad q \in \{0, 1, \infty\} \tag{182}$$

corresponding to these points at infinity — the construction of a section-group $\mathcal{M}^{!\sim}_{0,3}(0)$ in the normalizer of L_0, obviously applies without change for L_1, L_∞. We shall moreover have

$$\rho(\mathcal{M}^{!\sim}_{0,3}(q)) = \mathcal{M}^{!\sim}_{0,3}(\dot{\rho}|\rho(q)) \qquad \text{for } q \in \{0, 1, \infty\}, \tag{183}$$

so the three $\mathcal{M}^{!\sim}_{0,3}(q)$ are obtained from any one of them by applying ρ, ρ^2. 227

47. Return to the homomorphism $\widehat{\Pi}_{0,3} \xrightarrow{\widehat{\varphi}} \widehat{\mathcal{G}}^{+}_{0,3}$ and the condition of commutation with it up to inner automorphisms. The groups $\mathcal{M}^{!\approx}_{0,3}, \mathcal{M}^{\approx}_{0,3}$ and the homomorphism $\Phi : \mathcal{M}^{!\approx}_{0,3} \longrightarrow \mathcal{M}^{\approx}_{0,3}$. [scan 229] [p. 415]

Recall (§41) that $\widehat{\varphi}$ is described by

$$\widehat{\varphi}(\ell_0) = \varepsilon_0 = \sigma_\infty \rho, \quad \widehat{\varphi}(\ell_1) = \sigma_\infty, \quad \widehat{\varphi}(\ell_\infty) = \rho^{-1}. \tag{1}$$

Let

$$u = u_{\alpha,\beta,f} \quad \in \mathcal{M}^{!\sim}_{0,3}$$

be a loop automorphism of $\widehat{\Pi}_{0,3}$, commuting with $\mathcal{G}_3$ up to inner automorphisms, defined by

$$u(\sigma_\infty) = h\sigma_\infty = \mathrm{int}(\beta_0)\sigma_\infty \quad \begin{pmatrix} h = \beta_0 \sigma_\infty(\beta_0)^{-1}, \\ \beta_0 = f^{-1}\beta \end{pmatrix} \tag{2}$$

$$u(\rho) = g\rho = \mathrm{int}(\alpha_0)\rho \quad \begin{pmatrix} g = \alpha_0 \rho(\alpha_0)^{-1}, \ \dagger \\ \alpha_0 = f^{-1}\alpha \end{pmatrix} \tag{3}$$

and let us write that u commutes with φ up to inner automorphisms, i.e. that there exists $\dagger$

$$\lambda \in \widehat{\Pi}_{0,3} \tag{4}$$

such that we have

$$\widehat{\varphi}u(x) = \mathrm{int}(\lambda)u\widehat{\varphi}(x) \qquad \forall x \in \widehat{\Pi}_{0,3}. \tag{5}$$

It suffices to check (5) for $x = \ell_0$ and $x = \ell_1$.

1) Case $x = \ell_0$ †, whence

$$u(x) = \dagger\mathrm{int}(f^{-1})(\ell_0^\mu)$$
$$\widehat{\varphi}(u(x)) = \mathrm{int}(\widehat{\varphi}(f)^{-1})\widehat{\varphi}(\ell_0^\mu) = \mathrm{int}(\widehat{\varphi}(f)^{-1})\varepsilon_0^\mu$$
$$u\widehat{\varphi}(x) = u(\varepsilon_0) = \mathrm{int}(f^{-1})(\varepsilon_0^\mu)$$

so the relation (5) reads

$$\mathrm{int}(\dagger\widehat{\varphi}(f)\lambda f^{-1})(\varepsilon_0^\mu) = \varepsilon_0^\mu$$

228

Uncertainty retained from cancelled passages.

140-3-s229-D001 $=$

140-3-s229-D002 $\lambda \in$

140-3-s229-D003 $= (\sigma_\infty\rho)^2$

140-3-s229-D005 $= h\sigma_\infty g\rho = h(\sigma_\infty g\sigma_\infty^{-1})\sigma_\infty\rho$

140-3-s229-D006 $f\lambda^{-1}$

is equivalent to [scan 230]

$$\dagger\widehat{\varphi}(f)\lambda f^{-1} \in L_0^{\sim} \quad \left(= \mathrm{Centr}_{\widehat{\Pi}_{0,3}}(\varepsilon_0^\mu)\right)$$

m918

Marginal note. m918 N.B. we do not necessarily have $\widehat{\varphi}(f) \in \widehat{\Pi}_{0,3}$, so neither [illeg. 1] $\widehat{\varphi}(f)\lambda f^{-1} \in \widehat{\Pi}_{0,3}$ ...

i.e. to the existence of $r \in \widehat{\mathbb{Z}}$ such that we have

$$\dagger\widehat{\varphi}(f)\lambda f^{-1} = \varepsilon_0^r,$$

i.e.

$$\lambda = \widehat{\varphi}(f)^{-1}\varepsilon_0^r f. \tag{6}$$

This thus makes it possible to eliminate $\exists\lambda$ from the second relation :

2) Case $x = \ell_1$, whence

$$u(x) = \mathrm{int}(f_1^{-1})(\ell_1^\mu) \qquad (f_1^{-1} = \dagger g\rho(f^{-1}))$$
$$\widehat{\varphi}(u(x)) = \mathrm{int}(\widehat{\varphi}(f_1)^{-1})\sigma_\infty^\mu = \mathrm{int}(\widehat{\varphi}(f_1))\sigma_\infty$$

(taking into account that $\sigma_\infty^\mu = \sigma_\infty$ since $\mu \equiv 1$ (2)). †

$$u\widehat{\varphi}(x) = u(\sigma_\infty) = \mathrm{int}(\beta_0)(\sigma_\infty),$$

So the relation (5) reads

$$\mathrm{int}(\widehat{\varphi}(f_1)^{+1}\lambda\beta_0)\sigma_\infty = \sigma_\infty,$$

which also means that there exists $i|\alpha \in \mathbb{Z}/2\mathbb{Z}$ such that we have †

$$\widehat{\varphi}(f_1)^{+1}\lambda\beta_0 = \sigma_\infty^i \quad \text{i.e.} \quad \widehat{\varphi}(f_1^{-1})^{+1} = \lambda\beta_0\sigma_\infty^i,$$

and replacing λ by its value (6), and f_1^{-1} by its value $g\rho(f^{-1})$†, we find the condition

†

$$\widehat{\varphi}(g\rho(f)^{-1}) = \widehat{\varphi}(f)^{-1}\varepsilon_0^r \underbrace{f\beta_0}_{\beta}\sigma_\infty^\alpha \quad \text{i.e.}$$

$$\widehat{\varphi}(\underbrace{fg\rho(f)^{-1}}_{\xi=\alpha\rho(\alpha)^{-1}}) = \varepsilon_0^r\beta\sigma_\infty^\alpha$$

229

Uncertainty retained from cancelled passages.

140-3-s230-D001 $f\lambda^{-1}$

140-3-s230-D002 $f\lambda^{-1}$

140-3-s230-D003 $\rho(f)$

140-3-s230-D004 $u\widehat{\varphi}(x) = \mathrm{int}(\widehat{\varphi}(f_1))$

140-3-s230-D005 [illeg. 1]

140-3-s230-D006 f

140-3-s230-D007 $\widehat{\varphi}(\rho(f)^{-1})\widehat{\varphi}(g)^{-1} = \widehat{\varphi}(f)^{-1}\ell_0^r f\beta_0\sigma_\infty^\alpha$

[scan 231]
[p. 416]

Thus :

Theorem For

$$\widehat{\varphi} : \widehat{\Pi}_{0,3} \longrightarrow \widehat{\mathcal{G}}^+_{0,3}$$

to commute up to inner automorphisms with the element $u \in \mathcal{M}^{!\sim}_{0,3}$, corresponding to the invariants $g, h, f, \alpha_0, \beta_0, \alpha, \beta, \xi, \eta$ (cf. §46), it is necessary and sufficient that we have

$$\widehat{\varphi}(\xi) = \varepsilon_0^r \beta \sigma_\infty^{i|\alpha} \qquad \text{for } r \in \widehat{\mathbb{Z}}, \quad \alpha \in \mathbb{Z}/2\mathbb{Z} \text{ suitable.} \tag{7}$$

m920

Marginal note. m920 N.B. r, α are well determined when α and β (or again ξ and β) are chosen.

We shall then have (5), with λ given by (6) ; $\lambda = \widehat{\varphi}(f)^{-1}\ell_0^r f$.

Note that u defines the triple (α, β, f) only modulo a transformation of the form †

$$(\alpha, \beta, f) \longmapsto (\alpha' = \ell_0^n \alpha,\ \beta' = \ell_0^n \beta,\ f' = \ell_0^n f) \tag{8}$$

where $n \in \widehat{\mathbb{Z}}$. We shall then have (putting $\xi' = \alpha'\rho(\alpha')^{-1} = \ell_0^n \xi \ell_1^{-n}$)

$$\widehat{\varphi}(\xi') = \varepsilon_0^{r'} \beta' \sigma_\infty^{i'|\alpha'} \tag{*}$$

with $r' \in \widehat{\mathbb{Z}}$, $i'|\alpha' \in \mathbb{Z}/2\mathbb{Z}$ also well determined. Now we shall have

$$\begin{aligned} \widehat{\varphi}(\xi') &= \widehat{\varphi}(\ell_0^n \xi \ell_1^{-n}) \\ &= \widehat{\varphi}(\ell_0^n)\widehat{\varphi}(\xi)\widehat{\varphi}(\ell_1^{-n}) \\ &= \varepsilon_0^n \widehat{\varphi}(\xi)\sigma_\infty^{-n} \end{aligned}$$

†† and the relation (*) reads

$$\widehat{\varphi}(\xi) = \varepsilon_0^{r'-n|r'-2n} \beta' \sigma_\infty^{i'-n|\alpha'-n}$$

or again, since $\beta' = \ell_0^n \beta = \varepsilon_0^{2n}\beta$

$$\widehat{\varphi}(\xi) = \varepsilon_0^{r'+n} \beta \sigma_\infty^{\alpha'-n|i'-n},$$

which, compared with (7), gives

$$r' = r - n, \qquad i'|\alpha' = i|\alpha + n \bmod 2. \tag{9}$$

230

Uncertainty retained from cancelled passages.
140-3-s231-D003 $\mapsto$ $\alpha' = \ell_0^n \alpha,\ \beta$
140-3-s231-D004 [illeg. 1]
140-3-s231-D005 [illeg. 1]

[scan 232]

This implies the

Corollary 1 Let us suppose that u commutes with $\widehat{\varphi} : \widehat{\Pi}_{0,3} \longrightarrow \widehat{\mathcal{G}}^{+}_{0,3}$ up to inner automorphisms. Then it is possible in a unique way to describe $u|\varphi$ by a triple (α, β, f) (a priori determined modulo a transformation of type (8), in such a way that we have (putting $\xi = \alpha\rho(\alpha)^{-1}$, as usual) a relation

$$\boxed{\widehat{\varphi}(\xi) = \beta\sigma_\infty^i} \qquad \text{for } i \in \mathbb{Z}/2\mathbb{Z} \text{ suitable.} \tag{10}$$

m922

Marginal note. m922 We should examine whether this definition coincides with that on the page 410', 413|417, formulated without involving $\widehat{\varphi}$.

Such a choice will be called normalized.

Thus the $u \in \mathcal{M}^{!\sim}_{0,3}$ which commute with $\widehat{\varphi}$ up to inner automorphisms correspond one-to-one to the triples

$$(\alpha, \beta, f), \qquad \alpha \in \widetilde{\widehat{\Pi}}_{0,3}, \quad \beta, f \in \widehat{\Pi}_{0,3}$$

satisfying, on the one hand the loop relation

$$\xi \doteq | = \eta\ell_1^\nu \qquad \begin{array}{c} \text{where } \xi = \alpha\rho(\alpha)^{-1}, \quad \eta = \beta\sigma_\infty(\beta)^{-1}, \\ \text{and where } \nu \in \widehat{\mathbb{Z}} \text{ suitable} \\ (\text{such that } 2\nu + 1 \in \widehat{\mathbb{Z}}^*) \end{array} \tag{11}$$

on the other hand relation (10) (for suitable $i \in \mathbb{Z}/2\mathbb{Z}$) — neither relation moreover involving f. 231

[scan 233]

Let us note that $i \in \mathbb{Z}/2\mathbb{Z}$ is defined in terms of ξ (taking into account $\beta \in \widehat{\Pi}_{0,3} = \mathrm{Ker}(\widehat{\mathcal{G}}^{+}_{0,3} \to \mathcal{G}_3|\mathfrak{S}_3)$) by reducing (10) modulo $\widehat{\Pi}_{0,3}$. Denoting by [p. 417]

$$\widehat{\varphi}^\bullet : \widehat{\Pi}_{0,3} \longrightarrow \mathcal{G}_3|\mathfrak{S}_3 \tag{12}$$

the composite

$$\widehat{\Pi}_{0,3} \xrightarrow{\widehat{\varphi}} \widehat{\mathcal{G}}^{+}_{0,3} \xrightarrow{\text{can}} \mathcal{G}_3|\mathfrak{S}_3,$$

formula (10) also gives

$$\widehat{\varphi}^{\bullet|\circ}(\xi) = \sigma_\infty^i \tag{13}$$

which does determine $i \in \mathbb{Z}/2\mathbb{Z}$ in terms of $\xi \in \widehat{\Pi}_{0,3}$, and implies an intrinsic condition on the element ξ of $\widehat{\Pi}_{0,3}$, namely that its image in $\mathcal{G}_3|\mathfrak{S}_3$ under $\widehat{\varphi}^{\bullet|\circ}$ should indeed be in the subgroup $\{1, \sigma_\infty\}$ of $\mathcal{G}_3|\mathfrak{S}_3$. In fact, the kernel of $\widehat{\varphi}^{\bullet|\circ}$ is the inverse subgroup of $\widehat{\Pi}_{0,3}$ generated by $\ell_0^2, \ell_1^2, \ell_\infty^3$, and the condition being considered therefore means that modulo this subgroup, ξ should be of the form ℓ_1^i.

By formula (10), we can eliminate β from the datum (α, β, f), and we can replace the datum α by the datum ξ. Thus: 232

[scan 234]

Corollary 2 The elements of the ^{s}/group of $\mathcal{M}^{!\sim}_{0,3}$ formed by the u which commute with $\widehat{\varphi}$ up to inner automorphisms, is † in one-to-one correspondence with the set of pairs (ξ, f) † $(\xi, f \in \widehat{\Pi}_{0,3})$ satisfying the conditions (not|by involving anything but ξ)

$$\boxed{\xi\rho(\xi)\rho^2(\xi) = 1} \tag{14}$$

$$\boxed{\xi = (\widehat{\varphi}(\xi)\,\sigma_\infty(\widehat{\varphi}(\xi))^{-1})\ell_1^\nu} \tag{15}$$

$(\nu \in \widehat{\mathbb{Z}}$ suitable such that $2\nu + 1 \in \widehat{\mathbb{Z}}^*)$.

m925

Marginal note. m925 we also need the "preliminary" "condition"…

For such a ξ, $\widehat{\varphi}^{\bullet|\circ}(\xi)$ in $\mathcal{G}_3|\mathfrak{S}_3$ is of the form σ_∞^i, with $i \in \mathbb{Z}/2\mathbb{Z}$, and putting

$$\begin{cases} \beta = \widehat{\varphi}(\xi)\sigma_\infty^i \quad (\in \widehat{\Pi}_{0,3}) \\ \alpha \in \widehat{\Pi}_{0,3} \quad \text{the unique element of } \widetilde{\widehat{\Pi}}_{0,3} = \widehat{\Pi}_{0,3} \cdot \{1, \tau_\infty\} \\ \qquad\qquad \text{such that } \alpha\rho(\alpha)^{-1} = \xi \end{cases} \tag{16}$$

u is obtained from ξ, f by the formulas

in fact, we do not really need α, β

$$\begin{cases} u(\sigma_\infty) = \mathrm{int}(f^{-1}\beta)(\sigma_\infty) \\ \qquad\quad = \mathrm{int}(f^{-1})(\mathrm{int}(\beta)(\sigma_\infty)) \\ \qquad\quad = \mathrm{int}(f^{-1})(\xi\ell_1^{-\nu}\sigma_\infty) \\ u(\rho) = \mathrm{int}(f^{-1}\alpha)(\rho) \\ \qquad\quad = \mathrm{int}(f^{-1})(\mathrm{int}(\alpha)(\rho)) \\ \qquad\quad = \mathrm{int}(f^{-1})(\xi\rho) \end{cases} \tag{17}$$

We shall note that in these formulas, f enters only through composition with $\mathrm{int}(f)$. If we denote, for $\xi \in \widehat{\Pi}_{0,3}$ 233

Uncertainty retained from cancelled passages.

140-3-s234-D001 [illeg. 1]

140-3-s234-D002 ∈

[scan 235]

m928

[p. 418]

Marginal note. m928 This notation is already in use for the general considerations of the preceding §$u_{\xi,\eta}$ depends in any case only on ξ…

satisfying (14)(15), by u_ξ the automorphism corresponding, in terms of Corollary 2, to the pair $(\xi, 1)$, so that

$$\begin{cases} u_\xi(\sigma_\infty) = \dagger\, \overbrace{(\xi\ell_1^{-\nu})}^{\eta|\nu}\, \sigma_\infty\,, \\ u_\xi(\rho) = \xi\rho \end{cases} \tag{18}$$

formulas (17) can also be written, simply

$$u_{\xi,f} = \mathrm{int}(f^{-1})\, u_\xi. \tag{19}$$

On the other hand formulas (5), (6) give here

$$\widehat{\varphi} \circ u_\xi = u_\xi \circ \widehat{\varphi} \tag{20}$$

where, of course, on the first side|side u_ξ denotes the restriction of u_ξ to $\widehat{\Pi}_{0,3}$.

Let us note

Corollary 3 The loop automorphisms of $\widehat{\mathcal{G}}^+_{0,3}$ of the form u_ξ (ξ normalized|normalized*) are exactly those for which we have the formulas (20) (i.e. which † commute with $\widehat{\varphi}$) and which moreover normalize L_0.

* in the sense of page 416'



234

Uncertainty retained from cancelled passages.
140-3-s235-D002 int(α)
140-3-s235-D004 [illeg. 1]

[scan 236]

Indeed, in terms of the triples (α, β, f) "special" these properties are expressed by the simple conditions f of the form ℓ_0^n (which expresses the condition of normalizing L_0) and that λ given by (6) case|for $r = 0$, i.e.

$$\lambda = \widehat{\varphi}(f)^{-1} f \tag{21}$$

be equal to 1, i.e. the condition

$$\widehat{\varphi}(f) = f, \tag{22}$$

which in this case reads

$$\varepsilon_0^n = \underbrace{\ell_0^n}_{\varepsilon_0^{2n}} \qquad \text{i.e.} \quad \varepsilon_0^n = 1$$

which implies $f|f_0 = \ell_0^n = 1$, hence u of the form $u_{(\alpha,\beta,1)} = u_\xi$, where $\xi = \alpha\rho(\alpha)^{-1}$.

Corollary 4 The u_ξ normalized form a subgroup of $\mathrm{Norm}_{\mathcal{M}^{!\sim}_{0,3}}(L_0)$, i.e. $\mathcal{M}^{!\approx}_{0,3}(0)$, whose †intersection with $\widehat{\Pi}_{0,3}$ is reduced to $\{1\}$, and whose image in $\mathcal{N}^{!\sim}_{0,3}$ is the subgroup $\mathcal{N}^{!\approx}_{0,3}$ formed by the outer loop automorphisms of $\widehat{\mathcal{G}}^+_{0,3}$, stabilizing $\widehat{\Pi}_{0,3}$ (i.e. $u \in \mathcal{M}^{+\hat{}}_{0,3}$) such that $\widehat{\varphi} : \widehat{\Pi}_{0,3} \to \widehat{\mathcal{G}}^+_{0,3}$ commutes with u up to inner automorphisms. We therefore have an isomorphism

$$\mathcal{M}^{!\approx}_{0,3}(0) = \{u_\xi \mid \xi \in \widehat{\Pi}_{0,3} \text{ solution of (14)(15)}\} \xrightarrow{\sim} \mathcal{N}^{!\approx}_{0,3}. \tag{23}$$

235

Uncertainty retained from cancelled passages.
140-3-s236-D002 image

[scan 237]

It only remains to check that a u_ξ can be an inner automorphism only if $u_\xi = 1$ i.e. $\xi = 1$ (hence $\nu = 0$). Now u_ξ is a $u_{\alpha,\beta} = u_{\alpha,\beta,1}$ (α, β defined via ξ by $\alpha\rho(\alpha)^{-1} = \xi$, $\beta\sigma_\infty(\beta)^{-1} = \xi\ell_1^{-\nu}$), which is an inner automorphism iff we have $\alpha = \ell_0^n$, $\beta = \ell_0^n$ (cf. proposition on page 404), i.e. $\xi = \ell_0^n\ell_1^{-n}$, $\eta = \beta\sigma_\infty(\beta)^{-1} = \ell_0^n\ell_1^{-n}$, and † $\xi = \eta\ell_1^\nu$ reads $\ell_1^\nu = 1$ i.e. $\nu = 0$. Relation (15) therefore becomes (since [p. 419]

$$\widehat{\varphi}(\xi) = \widehat{\varphi}(\ell_0^n\ell_1^{-n}) = \varepsilon_0^n\sigma_\infty^n$$
$$\sigma_\infty(\widehat{\varphi}(\xi)) = \varepsilon_1^n\sigma_\infty^n$$

) † the relation

$$\ell_0^n \ell_1^{-n} = \varepsilon_0^n \sigma_\infty^{-n} \varepsilon_1^n \sigma_\infty^n,$$

i.e. (since $\ell_0 = \varepsilon_0^2$, $\ell_1 = \varepsilon_1^2$) if n even

$$\varepsilon_0^n = \dagger \varepsilon_1^{3n} \qquad \text{whence } n = 0,$$

and if n odd

$$\varepsilon_0^n = \dagger \underbrace{\sigma_\infty \varepsilon_1^n \sigma_\infty}_{\sigma_\infty (\varepsilon_1)^n = \varepsilon_0^n} \varepsilon_1^{2n}$$

i.e. $\varepsilon_1^{2n} = 0$, i.e. again $n = 0$, q.e.d.

m932

Marginal note. m932 N.B. [illeg. 1] $\mathcal{M}^{!\sim}_{0,3}[0]$ — a [illeg. 2] $\mathcal{N}^{!\sim}_{0,3}$ in its entirety, but it remains to be seen whether the other… [illeg. 1]

Scholium We have thus found a canonical partial section of $\mathcal{M}^{!\sim}_{0,3}$ over the subgroup $\mathcal{N}^{!\approx}_{0,3}$ of $\mathcal{N}^{!\sim}_{0,3}$. Since 236

Uncertainty retained from cancelled passages.

140-3-s237-D003 for

140-3-s237-D004 gives

140-3-s237-D005 σ_∞

140-3-s237-D006 ε_0

[scan 238]

this subgroup contains $\Gamma_{\mathbb{Q}}$, we therefore find a section of the extension $\mathcal{M}^!_{0,3}$ of $\Gamma_{\mathbb{Q}}$ by $\widehat{\Pi}_{0,3}$, which is in|in fact a section of the subgroup of $\mathcal{M}^!_{0,3}$ which is the normalizer of L_0. We should check that this canonical section is associated with the choice of a uniformizer at the point 0 of $\widehat{U}_{0,3\mathbb{Q}}$, † whose image in $\mathfrak{m}_0/\mathfrak{m}_0^2$ should be well determined, where $\mathfrak{m}_0$ denotes the maximal ideal of the local ring at 0 in $\widehat{U}_{0,3\mathbb{Q}} = \mathbb{P}^1_{\mathbb{Q}}$) — we should determine this element of $\mathfrak{m}_0/\mathfrak{m}_0^2$ — We feel like taking an element which is, not only a basis over $\mathbb{Q}$, but even (in the situation over the integers) a basis over $\mathbb{Z}$ — † condition which indeed determines an element of $\mathfrak{m}_0/\mathfrak{m}_0^2$ up to sign — this would be in fact the choice corresponding to the uniformizer z, or $-z$. But is it really the right one? We should come back to this! 237

Uncertainty retained from cancelled passages.

140-3-s238-D002 and which will be determined

140-3-s238-D003 this

[scan 239]

Let us work henceforth in the subgroup $\mathcal{M}^{!\approx}_{0,3}$ of $\mathcal{M}^{!\sim}_{0,3}$, consisting of the u which commute as outer actions with $\widehat{\varphi}$, i.e. inverse image of $\mathcal{N}^{!\approx}_{0,3}$, so that we have [p. 420]

$$1 \longrightarrow \widehat{\Pi}_{0,3} \longrightarrow \mathcal{M}^{!\approx}_{0,3} \longrightarrow \mathcal{N}^{!\approx}_{0,3} \longrightarrow 1 \qquad (24)$$
$$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \cup \Gamma_{\mathbb{Q}}$$

We have therefore found a section subgroup

$$\mathcal{M}^{!\approx}_{0,3}(0) \xrightarrow{\sim} \mathcal{N}^{!\approx}_{0,3} \qquad (25)$$

†

$$\{u_\xi \mid \xi \in \widehat{\Pi}_{0,3} \text{ solution of (14) and (15)}\} \subset \mathcal{M}^{!\approx}_{0,3}.$$

In terms of this we can therefore write $\mathcal{M}^{!\approx}_{0,3}$ as a $\frac{1}{2}$ direct product

$$\mathcal{M}^{!\approx}_{0,3} \simeq \mathcal{M}^{!\approx}_{0,3}(0) \cdot \widehat{\Pi}_{0,3}, \tag{26}$$

by writing every element of $\mathcal{M}^{!\approx}_{0,3}$ uniquely as

$$\begin{gathered} \dagger\, \mathrm{int}(f^{-1}) \circ u_0 | u_\xi \\ (u_0|u_\xi \in \mathcal{M}^{!\approx}_{0,3}(0) \text{ i.e. } u_0 = u_\xi \text{ with } \xi \text{ solution of (14)(15)}) \\ f \in \widehat{\Pi}_{0,3} \text{ well determined} \end{gathered} \tag{27}$$

These are precisely the $u_{\xi,f}$ of (19), i.e., [illeg. 1]

$$u_{\xi,f} = u_{\alpha,\beta,f} \tag{28}$$

with α, β determined in terms of ξ by

$$\alpha\rho(\alpha)^{-1} = \xi, \quad \dagger\beta\sigma_\infty(\beta)^{-1} = \xi\ell_1^{-\nu}, \tag{29}$$

238

Uncertainty retained from cancelled passages.

140-3-s239-D001 ξ [illeg. 2]

140-3-s239-D002 $u_{\xi,f}$

140-3-s239-D004 ε

[scan 240]

this is how we can interpret the normalized triples (α, β, f) of corollary 1.

Rescholium (Recapitulation) The group $\mathcal{N}^{!\approx}_{0,3} \simeq \mathcal{M}^{!\approx}_{0,3}(0)$ is in 1-1 correspondence with the set of $\xi \in \widehat{\Pi}_{0,3}$ satisfying the two relations

$$\begin{cases} \xi\rho(\xi)\rho^2(\xi) = 1 & \text{for} \\ \xi = (\widehat{\varphi}(\xi)\sigma_\infty(\widehat{\varphi}(\xi))^{-1})\ell_1^\nu & \nu \in \widehat{\mathbb{Z}} \text{ suitable} \end{cases} \tag{30}$$

or again (putting $\xi = \alpha\rho(\alpha)^{-1}$, $\alpha \in \widehat{\widetilde{\Pi}}_{0,3}$) with the set of $\alpha \in \widehat{\widetilde{\Pi}}_{0,3}$ satisfying the second relation (30), in which we replace ξ by $\alpha\rho(\alpha)^{-1}$. The element u_ξ of $\mathcal{M}^{!\approx}_{0,3}(0)$ defined by ξ is defined by the formulas (18), i.e. by

$$u_\xi(\sigma_\infty) = (\xi\ell_1^{-\nu})\sigma_\infty, \qquad u_\xi(\rho) = \xi\rho. \tag{31}$$

Finally, if we have u_ξ and $u_{\xi'}$, then

$$u_{\xi'}u_\xi = u_{\xi''}, \tag{32}$$

where ξ'' is given by the formulas on page 395' (52) — noting that here $\xi = g$, $\xi' = g'$, $\xi'' = g''$: †

†

239

Uncertainty retained from cancelled passages.

140-3-s240-D001 whence$|f'' = 1$

140-3-s240-D002 $\alpha''\rho$

[scan 241]

[p. 421]

† $$\xi'' = u'(\xi) \cdot \xi'. \tag{33}$$

Let us now set

$$\mathcal{N}^{\approx}_{0,3} = \mathcal{G}_3 \times \mathcal{N}^{!\approx}_{0,3} \subset \mathcal{N}^{\sim}_{0,3} = \mathcal{G}_3 \times \mathcal{N}^{!\sim}_{0,3} \tag{34}$$

and let $\mathcal{M}^{\approx}_{0,3} \subset \dagger\mathcal{M}^{\sim}_{0,3}$ be the inverse image in $\mathcal{M}^{\sim}_{0,3}$ of $\mathcal{N}^{\approx}_{0,3} \subset \mathcal{N}^{\sim}_{0,3}$, so that we have an extension structure

$$1 \longrightarrow \dagger\widehat{\Pi}_{0,3} \longrightarrow \mathcal{M}^{\approx}_{0,3} \twoheadrightarrow | \longrightarrow \underset{\simeq \mathcal{G}_3 \times \mathcal{N}^{!\approx}_{0,3}}{\mathcal{N}^{\approx}_{0,3}} \longrightarrow 1 \tag{35}$$

containing the subextension (24) as the inverse image of $\dagger$ $\dagger$

$$\mathcal{N}^{!\approx}_{0,3} = \mathrm{Ker}(\mathcal{N}^{\approx}_{0,3} \longrightarrow \mathcal{G}_3).$$

$\dagger$ We have $\dagger$ a further extension structure on $\mathcal{M}^{\approx}_{0,3}$

$$1 \longrightarrow \widehat{\mathcal{G}}^{+}_{0,3} \longrightarrow \mathcal{M}^{\approx}_{0,3} \longrightarrow \mathcal{N}^{!\approx}_{0,3} \longrightarrow 1, \tag{36}$$

containing (24) as a subextension of the same quotient by a smaller group.

The subgroup $\mathcal{M}^{!\approx}_{0,3}(0)$, which is a section of (24), is therefore also a section of (36), hence a canonical isomorphism

$$\mathcal{M}^{\approx}_{0,3} \simeq \underbrace{\mathcal{N}|\mathcal{M}^{!\approx}_{0,3}(0)}_{\simeq \mathcal{N}^{!\approx}_{0,3}} \cdot \widehat{\mathcal{G}}^{+}_{0,3} \tag{37}$$

($\frac{1}{2}$-direct product). 240

Uncertainty retained from cancelled passages.

140-3-s241-D001 [illeg. 1]
140-3-s241-D002 $\mathcal{G}$
140-3-s241-D003 $\widehat{\mathcal{G}}^{+}_{0,3}$
140-3-s241-D005 the
140-3-s241-D006 [illeg. 7]
140-3-s241-D007 (36)
140-3-s241-D008 also

[scan 242]

Let us now consider the map

$$\begin{array}{ccc} & \mathcal{M}^{!\approx}_{0,3} & \longrightarrow \mathcal{M}^{\approx}_{0,3} \\ \Phi : & \wr & \wr \\ & \mathcal{M}^{!\approx}_{0,3}(0) \cdot \widehat{\Pi}_{0,3} & \mathcal{M}^{!\approx}_{0,3}(0) \cdot \widehat{\mathcal{G}}^{+}_{0,3} \end{array} \tag{38}$$

which is given|given by

$$\Phi(u \cdot a) = u\widehat{\varphi}(a) \tag{39}$$

($u \in \mathcal{M}^{!\approx}_{0,3}(0)$, $a \in \widehat{\Pi}_{0,3}$). It follows from the characterization of $\mathcal{M}^{!\approx}_{0,3}(0)$ (corollary 3) — $\dagger$ more particularly (from the fact that the elements of $\mathcal{M}^{!\approx}_{0,3}(0)$ actually commute (not only as outer actions) with $\widehat{\varphi}$, that Φ is a group homomorphism. This is of course none other than the homomorphism Φ considered in § 41 (page 316') (up to restriction to the subgroups $\mathcal{M}^{!}_{0,3} = \mathcal{E}_{0,3}$ and $\mathcal{M}_{0,3}$) — in loc. cit. this homomorphism could moreover be defined in an effective way — not only up to $\widehat{\Pi}_{0,3}$-conjugation or up to $\widehat{\mathcal{G}}^{+}_{0,3}$-conjugation — and with a little care we should be able to verify the equality of these homomorphisms Φ from § 41 and from here, in an effective way. 241

Uncertainty retained from cancelled passages.
140-3-s242-D001 i.e.

Calculation with|of $x = \ell_\infty$. [scan 243]

I am going to return to the determination of the $u_\xi \in \mathcal{M}^{!\sim}_{0,3}[0] = \mathrm{Norm}_{\mathcal{M}^{!\sim}_{0,3}}(L_0)$ which are normalized i.e. which commute with $\widehat{\varphi}$ i.e. satisfy (5) with $\lambda = 1$. For $x = \ell_0$ this formula, by virtue of the particular form of u_ξ, is tautological [p. 422]

$$(\widehat{\varphi} u \ell_0 = \widehat{\varphi}(\ell_0^\mu) = \varepsilon_0^\mu, \quad u\widehat{\varphi}(\ell_0) = u(\varepsilon_0) = \varepsilon_0^\mu),$$

so we want|see that it should come|comes to the same thing to impose it for $x = \ell_1$, or for $x = \ell_\infty$. For $x = \ell_1$, the condition takes the form (10) — remarkable in that it gives a calculation of the|of "delicate" †parameter β (whose existence au priori|a priori, in terms of ξ, rests on foundations which have not yet been established…), in terms of the "coarse" parameter ξ — taking into account the fact that $i \in \mathbb{Z}/2\mathbb{Z}$ itself is determined just as easily in terms of ξ.

We are going to give this condition a form rather different in appearance, by writing the commutation relation (5) for $x = \ell_\infty$, instead of $x = \ell_1$. Let us first note that we have the relation

$$\tau'_\infty(\rho) = \rho^{-1} \tag{40}$$

For

$$\tau'_\infty(\rho) = \sigma_\infty(\underbrace{\tau_\infty(\rho)}_{\ell_1^{-1}\rho}) = \dagger\ell_0^{-1}\ \underbrace{\sigma_\infty(\rho)}_{\underbrace{\sigma_\infty\rho\sigma_\infty^{-1}}_{\varepsilon_0}} = \varepsilon_0^{-1}\sigma_\infty^{-1} = \rho^{-1},$$

242

Uncertainty retained from cancelled passages.
140-3-s243-D001 variable
140-3-s243-D002 $(\sigma_\infty \ell_1^{-1} \sigma_\infty^{-1})$ ℓ_0^{-1}

[scan 244]

it follows that if $u \in \mathcal{M}^{!\sim}_{0,3}$ satisfies

$$u(\rho) = \rho^{\mu|j} \qquad \mu|j \in \mathbb{Z}/3\mathbb{Z} \text{ suitable} \tag{41}$$

then we necessarily have $j \equiv -1$ or $+1$ (3) and if †

$$\alpha_3(j) \in \mathbb{Z}/2\mathbb{Z} \quad \text{characterized by} \quad j \equiv (-1)^{\alpha_3(j)}\ (3) \tag{42}$$

we will therefore have $u(\rho) = \rho$ or $u(\rho) = \rho^{-1}$ according as $\alpha_3(j) = 0$ or $\alpha_3(j) = 1$, and in any case we will have

$$u(\rho) = {\tau'_\infty}^{\alpha_3}(\rho). \tag{43}$$

This being so, let us put $x = \ell_\infty$ in (5):

$$\widehat{\varphi} u(\ell_\infty) = \widehat{\varphi}(\mathrm{int}(f_\infty^{-1})\ell_\infty^\mu) = \mathrm{int}(\widehat{\varphi}(f_\infty^{-1}))\rho^{-\mu}$$

(since $\widehat{\varphi}(\ell_\infty) = \rho^{-1}$), and

$$u(\widehat{\varphi}(\ell_\infty)) = u(\rho^{-1}) = u(\rho)^{-1} = \dagger\mathrm{int}(\alpha)(\rho^{-1})$$

so formula (5) for $\lambda = 1$ becomes

$$\mathrm{int}(\widehat{\varphi}(f_\infty)\alpha)(\rho^{-1}) = \rho^{-\mu} \qquad \text{i.e.}$$

$$\mathrm{int}(\widehat{\varphi}(f_\infty)\alpha)\rho = \rho^\mu$$

and by virtue of (43) †we have †

$$\rho^{\mu} = {\tau'_{\infty}}^{\alpha_3=\alpha_3(\mu)}(\rho) \overset{\text{df|déf}}{=} \operatorname{int}({\tau'_{\infty}}^{\alpha_3})(\rho)$$

so the condition becomes

$$\operatorname{int}({\tau'_{\infty}}^{\alpha_3}\widehat{\varphi}(f_{\infty})\alpha)\rho = \rho \tag{44}$$

i.e. that the element ${\tau'_{\infty}}^{\alpha_3}\widehat{\varphi}(f_{\infty})\alpha$ of $\widehat{\mathcal{G}}_{0,3} = \widehat{\mathcal{G}}^{+}_{0,3} \cdot \{1, \tau_{\infty}\}$ commutes with ρ. 243

Uncertainty retained from cancelled passages.

140-3-s244-D001 α is [illeg. 1] [illeg. 1]

140-3-s244-D002 $(\xi\rho)^{-1}$ $(\rho^{-1}\xi^{-1}\rho)\rho^{-1}$

140-3-s244-D003 this is equivalent $\rho^{\mu} = {\tau'_{\infty}}^{\alpha_3}(\rho)$

140-3-s244-D004 that

[scan 245]

m942

[p. 423]

Marginal note. m942 one should give its formulas in a precise way, take [illeg. 2]!!!

Now according to the general principles (not yet established, to tell the truth, in the profinite setting) the commutant of ρ in the group in question is reduced to the group generated by ρ, $\{1, \rho, \rho^2\}$, so that the relation (*) takes the form

$$\tau'^{\alpha_3}_{\infty}\widehat{\varphi}(f_{\infty})\alpha = \rho^{j} \qquad \text{for } j \in \mathbb{Z}/3\mathbb{Z} \text{ suitable} \quad \text{(well determined)}.$$

We make it explicit by using the value of f_{∞}

$$f_{\infty}^{-1} = g\rho(g)\rho^2(f_0^{-1}) = g\rho(g) = \rho^2(g)^{-1} = \rho^{-1}(\xi^{-1}),$$

N.B. here $\begin{cases} f_0 = f = 1 \\ g = \xi = \alpha\rho(\alpha)^{-1} \end{cases}$ thus the relation † becomes

$$\tau'^{\alpha_3}_{\infty}\widehat{\varphi}(\rho^{-1}(\xi))\alpha = \rho^{j}$$

i.e., †

$$\widehat{\varphi}(\rho^{-1}(\xi)) = \underbrace{\tau'^{\alpha_3}_{\infty}\rho^{j}}_{\tau'^{\alpha_3}_{\infty}(\rho^{j})\,\tau'^{\alpha_3}_{\infty}} \alpha^{-1} \qquad \dagger$$

or also † since $\tau'_{\infty}(\rho) = \rho^{-1}$ thus $\tau'^{\alpha_3}_{\infty}(\rho^{j}) = \rho^{(-1)^{\alpha_3}j} = \rho^{j'}$

$$\widehat{\varphi}(\rho^{-1}(\xi)) = \rho^{j'}\tau'^{\alpha_3}_{\infty}\alpha^{-1} \tag{44}$$

where $\alpha_3 = \alpha_3(\mu) \in \mathbb{Z}/2\mathbb{Z}$ defined by $(-1)^{\alpha_3} \equiv \mu$ (3) and $j' \in \mathbb{Z}/3\mathbb{Z}$ (well determined).

We shall transform this relation a little further, by writing

$$\tau'^{\alpha_3}_{\infty} = (\sigma_{\infty}\tau_{\infty})^{\alpha_3} = \sigma^{\alpha_3}_{\infty}\tau^{\alpha_3}_{\infty} \qquad \text{whence}$$

$$\tau'^{\alpha_3}_{\infty}\alpha^{-1} = \sigma^{\alpha_3}_{\infty}\underbrace{(\alpha\tau^{\alpha_3}_{\infty})}_{\widetilde{\alpha}}{}^{-1} \tag{*}$$

[p. 244]

Uncertainty retained from cancelled passages.
140-3-s245-D001 [illeg. 2]
140-3-s245-D002 to the
140-3-s245-D003 44|43
140-3-s245-D004 *44bis*|*43bis*

[scan 246]

where †

$$\tilde{\alpha} = \alpha\tau_\infty^{\alpha_3}\dagger \tag{45}$$

has the advantage over α itself of being in $\widehat{\Pi}_{0,3}$ in all cases (N.B. we have $\alpha \in \widehat{\Pi}_{0,3} \cdot \{1, \tau_\infty\}$, and $\alpha \in \widehat{\Pi}_{0,3}$ iff $\alpha_3 = 0$). It is equal to α iff $\alpha_3 = 0$ and in the case $\alpha_3 = 1$ † is directly related to ξ by the formula

$$\xi = \tilde{\alpha}\ell_1^{-1}\rho(\tilde{\alpha})^{-1} \tag{46}$$

(deduced from (45), from $\xi = \alpha\rho(\alpha)^{-1}$, and from

$$\tau_\infty\rho(\tau_\infty)^{-1} = \ell_1^{-1} \tag{**}$$

). Thus (44) takes the definitive form

$$\boxed{\widehat{\varphi}(\rho^{-1}(\xi)) = \rho^{j'}\sigma_\infty^{\alpha_3}\{\tilde{\alpha}^{-1}} \qquad (\tilde{\alpha} = \alpha\tau_\infty^{\alpha_3}) \tag{47}$$

m943

Marginal note. m943 all the factors of this formula are defined in $\widehat{\mathcal{G}}_{0,3}^{+}|\widetilde{\mathcal{G}}_{0,3}^{+}$, i.e. of multiplier +1

which can also|thus be written|describe, on passing to inverses

$$\widehat{\varphi}(\rho^{-1}(\xi^{-1})) = \tilde{\alpha}\sigma_\infty^{\alpha_3}\rho^{-j'} \tag{47 bis}$$

or again, solving for $\tilde{\alpha}$

$$\underset{\alpha\tau_\infty^{\alpha_3}}{\tilde{\alpha}} = \widehat{\varphi}(\rho^{-1}(\xi^{-1}))\rho_\dagger^{j'}\sigma_\infty^{\alpha_3}. \tag{48}$$

We shall compare this formula with formula (10), which is solved for β in the form

$$\widehat{\varphi}(\xi) = \beta\sigma_\infty^{i} \tag{49}=(10)$$

$$\beta = \widehat{\varphi}(\xi)\sigma_\infty^{i} \tag{50}$$

[p. 245]

Uncertainty retained from cancelled passages.
140-3-s246-D001 $\alpha\tau_\infty|\alpha_0$
140-3-s246-D002 (α_3)
140-3-s246-D003 [illeg. 1]
140-3-s246-D004 ∞

[scan 247]
[p. 424]

The fact that formula (47) is indeed equivalent to (10), or again to (49), when $\alpha \in \widehat{\widetilde{\Pi}}_{0,3}|\widetilde{\widetilde{\Pi}}_{0,3}$ and $\beta \in \widehat{\Pi}_{0,3}$ are related by $\xi = \eta\ell_1^{\nu}$ (where $\xi = \alpha\rho(\alpha)^{-1}$, $\eta = \beta\sigma_\infty(\eta)^{-1}$), would deserve a direct verification. The interest of (47), or of its form solved for α (48), just as of (10)= (49) or (50), consists in the fact that these formulas give the "delicate" parametersα, β, in terms of the

"coarse parameter" ξ. Both are formulas in $\widehat{\mathcal{G}}^{+}_{0,3}$ — on reducing modulo $\widehat{\Pi}_{0,3}$, they give formulas in $\mathfrak{S}_3|\mathcal{S}_3$, namely (taking into account that $\tilde{\alpha}, \beta \in \widehat{\Pi}_{0,3}$): †

$$\begin{cases} \widehat{\varphi}\cdot(\rho^{-1}(\xi)) = \dot{\rho}^{j'}\dot{\sigma}_{\infty}^{\alpha_3} \\ \widehat{\varphi}\cdot(\xi) = \dot{\sigma}_{\infty}^{i} \end{cases} \tag{51}$$

which gives back|gives exactly †, in terms of the elements $\widehat{\varphi}\cdot(\xi)$ and $\widehat{\varphi}\cdot(\rho^{-1}(\xi))$ of $\mathfrak{S}_3|\mathcal{S}_3$, the values of $i, \alpha_3 \in \mathbb{Z}/2\mathbb{Z}$ and of $j' \in \mathbb{Z}/3\mathbb{Z}$. (We should see whether i, j' can, like α_3, be made explicit in terms of the multiplier μ, or even in terms of its [m944]

Marginal note. [m944] and also see whether (i, α_3, j') can take all sorts of values in $\mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/3$ [p. 246]

Uncertainty retained from cancelled passages.
140-3-s247-D002 i.e.
140-3-s247-D003 the

[scan 248]

quadratic residues $\varepsilon_\ell(\mu)$) We can supplement them with a formula deduced from one and the other, taking into account that

$$\xi\rho(\xi)\underbrace{\rho^2(\xi)}_{\rho^{-1}(\xi)} = 1 \qquad \text{i.e.} \qquad \rho^{-1}(\xi)\xi = \rho(\xi)^{-1}$$

we therefore find, multiplying (47) and (10)

$$\widehat{\varphi}(\underbrace{\rho^{-1}(\xi)\cdot\xi}_{\rho(\xi)^{-1}}) = \rho^{j'}\sigma_{\infty}^{\alpha_3}\tilde{\alpha}^{-1}\beta\sigma_{\infty}^{i}$$

i.e. again

$$\widehat{\varphi}(\rho(\xi)) = \sigma_{\infty}^{i}(\beta^{-1}\tilde{\alpha})\sigma_{\infty}^{\alpha_3}\rho^{-j'} \tag{52}$$

i.e., in the form solved for $\beta^{-1}\tilde{\alpha}$

$$\underbrace{\beta^{-1}\tilde{\alpha}}_{(\beta^{-1}\alpha)\tau_{\infty}^{\alpha_3}} = \sigma_{\infty}^{i}\widehat{\varphi}(\rho(\xi))\sigma_{\infty}^{\alpha_3}\rho^{j'}. \tag{53}$$

We can consider that the three main "delicate parameters" are the quantities β, α and $\beta^{-1}\alpha$, which are expressed directly, in a remarkably simple way, in terms of the values of $\widehat{\varphi}$ for ξ, $\rho^{-1}(\xi)$ and $\rho^{-1}|\rho^{-2}(\xi) = \rho(\xi)$, which would dispense † probably (by [p. 247]

Uncertainty retained from cancelled passages.
140-3-s248-D003 [illeg. 1]

[scan 249]

carrying out the calculations in terms of ξ, η exclusively, which would require a little more [p. 425] effort...) with appealing to delicate results not yet proved, of the type $\xi = \alpha\rho(\alpha)^{-1}$, $\eta = \beta\sigma_{\infty}(\beta)^{-1}$, to justify the introduction of these parameters α, β. (recapitulation)

$$\begin{cases} \widehat{\varphi}(\xi) = \beta\sigma_{\infty}^{i} \\ \widehat{\varphi}(\rho^{-1}(\xi)) = \rho^{j'}\sigma_{\infty}^{\alpha_3}\tilde{\alpha}^{-1} \qquad (\tilde{\alpha} = \alpha\tau_{\infty}^{\alpha_3}) \\ \widehat{\varphi}(\underbrace{\rho^{-2|-1}(\xi)}_{\rho(\xi)}) = \underbrace{\sigma_{\infty}^{i}(\beta^{-1}\tilde{\alpha})\sigma_{\infty}^{\alpha_3}}_{\sigma_{\infty}^{i}(\beta^{-1}\tilde{\alpha})\sigma_{\infty}^{\alpha_3+i}\rho^{-j'}}\ \rho^{-j'} \end{cases} \tag{54}$$

Suppose for simplicity

$$\alpha_3 = 0 \quad \text{i.e.} \quad \alpha = \tilde{\alpha} \quad \text{i.e.} \quad \alpha \in \widehat{\Pi}_{0,3} \tag{55}$$

Then we obtain from the second formula (54), taking inverses

$$\widehat{\varphi}(\rho^{-1}(\xi^{-1})) = \alpha\rho^{-j'}$$

and the $x\rho(x)^{-1}$ of both sides

$$\xi = \widehat{\varphi}(\rho^{-1}(\xi^{-1}))\rho\Big(\widehat{\varphi}(\rho^{-1}(\xi^{-1}))\Big)^{-1} \tag{56}$$

which already implies the first formula (14) in the fundamental formulas (14) (15). [p. 248]

[scan 250]

We thus find in this case ($\mu \equiv 1$ (3) i.e. $\nu \equiv 0$ (3)) that the two relations (14) (15) are also [p. 425] equivalent to the set|one of the two relations [m946]

Marginal note. [m946] Valid only if $\nu \equiv 0$ (3) i.e. $\mu \equiv 1$ (3) i.e. $\varepsilon_3(\mu) = 1$.

$$\boxed{\begin{aligned} &(57) \quad \xi = \widehat{\varphi}(\rho^{-1}(\xi^{-1}))\rho(\widehat{\varphi}(\rho^{-1}(\xi^{-1})))^{-1} \\ &(58) \quad \xi = \widehat{\varphi}(\xi)\,\sigma_\infty(\widehat{\varphi}(\xi))^{-1}\ell_1^\nu \end{aligned}}$$

with $\nu \in \widehat{\mathbb{Z}}$, $\nu \equiv 0$ (3) which therefore characterize the $u_0|u_\cdot = u_\xi \in \mathcal{M}^{!\sim}_{0,3}|\mathcal{M}^{!\approx}_{0,3}(0)$, whose multiplier μ is $\equiv 1$ (3).

Remark The only element of $\mathcal{M}^{!\approx}_{0,3}(0) \simeq \mathcal{N}^{!\approx}_{0,3}$ that I know how to make explicit, apart from the unit element (and one even in $\mathcal{N}^{!\sim}_{0,3} \supset \mathcal{N}^{!\approx}_{4,3}|\mathcal{N}^{!\approx}_{g,3}$) is †the unique element $\tau \in \mathcal{T}^!_{0,3} \subset \mathcal{N}^{!\approx}_{0,3}$, which in $\mathcal{M}^{!\approx}_{0,3}(0)$ corresponds to the element τ_∞, † $= u_\xi = u_{\xi,\eta}$

$$(59) \qquad \begin{aligned} &h = \beta_0 = 1, \quad f = 1, \quad \alpha = \alpha_0 = \tau_\infty, \quad \xi = g = \ell_1^{-1}, \quad \eta = 1 \\ &\nu = -1, \quad \mu = -1, \end{aligned}$$

m947

Marginal note. [m947] Careful, here $\alpha_3 = 1$ so (57) is not valid|applicable.

and ξ, ν do satisfy † $\xi\rho(\xi)\rho^2(\xi) = 1$ i.e. $\ell_0^{-1}\ell_1^{-1}\ell_\infty^{-1} = 1$ † and the relations (54) page above [p. 249]

Uncertainty retained from cancelled passages.
140-3-s250-D001 [illeg. 1]
140-3-s250-D002 [illeg. 1]
140-3-s250-D005 (57) (58)
140-3-s250-D006 ρ^{-1} ε_0 $\varepsilon_0\varepsilon_1^{-1}$

[scan 251]

with [p. 426]

$$(60) \qquad i = 1, \quad \alpha_3 = 1 \quad \text{i.e.} \quad \varepsilon_3 = -1, \quad j' = -1$$

Remark We can ask the question, in general, of determining the elements $u_{\alpha,\beta} = u_{\xi,\eta}$ of $\mathcal{M}^{!\sim}_{0,3}[0]$ (not necessarily normalized in one sense or the other) such that we have (of course by hypothesis, $f = 1$; but also) $\beta = 1$ i.e. $\eta = 1$, † (N.B. We already know that if $\alpha = 1$ i.e. $\xi = 1$, then $u = 1$) We shall have to have $\xi = \eta\ell_1^\nu = \ell_1^\nu$, and it remains to check †

$$\xi\rho(\xi)\rho^2(\xi) = 1 \quad \text{with} \quad \xi = \ell_1^\nu,$$

$\mathrm{Sl}(2,\widehat{\mathbb{Z}})'$, so that

$$\underline{\xi} = \begin{pmatrix} 1 & 0 \\ 2\nu & 1 \end{pmatrix}$$

this means (page 344|342) that we have either

$$\underline{\xi} = \pm\underline{\rho}^{-1}. \quad = \pm \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix}$$

– which is clearly not the case – or

$$C + D - B = \pm 1 \quad \text{i.e.} \quad 2\nu + 1 = \mu = \pm 1 \quad \text{i.e.}$$

$\nu = 0$ or -1 – these are the two trivial cases $u_\xi = \mathrm{id}$ and $u_\xi = \tau_\infty$.
m949

Marginal note. m949 We are needlessly repeating arguments already made, we know very well that the multiplier is (up to sign when the normalization of ξ has not been done) equal to one of the "coefficients [illeg. 1]" of ξ, hence equal to ± 1.

[p. 250]

Uncertainty retained from cancelled passages.
140-3-s251-D003 [illeg. 3]
140-3-s251-D004 [illeg. 2]

[scan 252]

Let us note the|the

Proposition. The commutant of τ (the nontrivial element of $\mathcal{T}^!_{0,3} \simeq \mathbb{Z}/2\mathbb{Z} \subset \mathcal{N}^{!\sim}_{0,3}|\mathcal{M}^{!\sim}_{0,3}$) in $\mathcal{N}^{!\sim}_{0,3}$ is reduced to $\{1,\tau\}$.
m950

Marginal note. m950 N.B. This proposition [illeg. 3] in § 46 VI|II, [illeg. 4] of [illeg. 2] of $\widehat{\Gamma}$.

We shall do the calculation in the isomorphic group $\mathcal{M}^{!\sim}_{0,3}(0)$, where τ corresponds to τ_∞, in fact it turns out that the centralizer of τ in the whole of $\mathcal{M}^{!\sim}_{0,3}[0]$ (the big normalizer of L_0 in $\mathcal{M}^{!\sim}_{0,3}$, formed of the u_ξ without normalization conditions) is reduced to $\{1,\tau\}$. Indeed, we must express the commutation of a $u = u_\xi$ with $u_{\xi'} = \tau_\infty$, where $\xi' = \ell_1^{-1}$, we have

$$\begin{cases} u_{\xi'} u_\xi = u_{\xi''} & \text{with } \xi'' = \dot{u}_{\xi'}(\xi)\xi' \\ u_\xi u_{\xi'} = u_{\xi'''} & \text{with } \xi''' = \dot{u}|u_\xi(\xi')\xi \end{cases}$$

and we must express $\xi'' = \xi'''$, i.e.

$$(61) \qquad \tau_\infty(\xi)\ell_1^{-1} = u_\xi(\ell_1^{-1})\xi.$$

Now we have

$$\dot{u}|u_\xi(\varepsilon_1) = \mathrm{int}(\xi)(\varepsilon_1^\mu) \qquad \text{whence}$$
$$\dot{u}|u_\xi(\underbrace{\ell_1^{-1}}_{\varepsilon_1^{-2}}) = \mathrm{int}(\xi)(\ell_1^{-\mu})$$

and (61) is written

$$\tau_\infty(\xi)\ell_1^{-1} = \xi\ell_1^{-\mu} \qquad \text{i.e.}$$
$$(62) \qquad \xi^{-1}\tau_\infty(\xi) = \ell_1^{1-\mu} = \ell_1^{-2\nu}.$$

[p. 251]

[scan 253]

This equation has the solution [p. 427]

$$(63) \qquad \xi = \ell_1^{\nu}$$

in $\widehat{\Pi}_{0,3}$ (for $\tau_\infty(\ell_1) = \ell_1^{-1}$ hence $\tau_\infty(\xi) = \ell_1^{-\nu}$) and this solution is unique if|once we admit that the discrete results extend to the profinite case, so as to have

$$(64) \qquad (\widehat{\Pi}_{0,3})^{\tau_\infty} = \{1\}.$$

Now ξ must moreover satisfy the relation

$$\xi\rho(\xi)\rho^2(\xi) = 1$$

and we have just seen that this implies $\nu = \pm 1$, i.e. $u|u \in \{\mathrm{id}|1, \tau_\infty\}$. [p. 252]

[scan 254]

48. Construction of a group scheme. [p. 428]

I) Study of the map $\alpha \longmapsto \alpha\rho(\alpha)^{-1} = [\alpha, \rho]$.

Let|Let G be a group scheme over a ring A (for example $A = \mathbb{Z}$, $G = \mathrm{Sl}(2)_{\mathbb{Z}}$† or $\mathrm{Gl}(2)_{\mathbb{Z}}$), ρ a section of G, and let us consider the morphism

$$(1) \qquad F_\rho : \ \alpha \longmapsto \alpha\rho(\alpha)^{-1} = [\alpha, \rho] : \qquad G \longrightarrow G.$$

Let

$$\dagger \quad T_\rho = \mathrm{Centr}_G(\rho) = G^\rho \subset G$$

then

$$(2) \qquad F_\rho(\alpha') = F_\rho(\alpha) \quad \Longleftrightarrow \quad \dagger\alpha' \text{ section of } \alpha T_\rho$$

thus we have a canonical monomorphism induced by F_ρ

$$(3) \qquad G/T_\rho \hookrightarrow G,$$

whose image is to be characterized.

In the case that concerns us, the quotient group G_{ab} is representable (it is the trivial group for $G = \mathrm{Sl}(2)_{\mathbb{Z}}$, and it is $\mathbb{G}_m$ if $G = \mathrm{Gl}(2)_{\mathbb{Z}}$), In any case, a first condition for a section $\xi \in G(S)$ over an AScheme S, for it to be in the image, is that we have

$$(4) \qquad \delta(\xi) = 1$$

[p. 253]

Uncertainty retained from cancelled passages.

140-3-s254-D001 , $\mathbb{Z}$)

140-3-s254-D002 Z_ρ

140-3-s254-D003 α

[scan 255]

$$(5) \qquad \delta : G \longrightarrow G_{\mathrm{ab}}$$

is the canonical homomorphism. A second condition is that $\xi\rho$, which must be (locally) of the form $\alpha\rho\alpha^{-1}$, be conjugate locally to ρ – and this condition is moreover necessary and sufficient, tautologically, for ξ to belong to the image of F_ρ, and thus implies condition (4). But the latter may sometimes serve as an ingredient, in a [illeg. 1] of conditions of a "computational" nature, which are necessary and sufficient for us to have $\xi\rho \sim \rho$.

In the case of a reductive group G, the condition of †conjugation locally of sections is moreover fairly well understood, since in particular the work of Steinberg. We have a "scheme of invariants" I_G, and a canonical morphism

$$(6) \qquad G \longrightarrow I_G$$

(which is universal (over A, and also after every base change) for morphisms from G into schemes, which are invariant under inner automorphisms – i.e. I_G is the representable envelope of $G/\mathrm{int}(G)$ sheaf of conjugacy [p. 254]

Uncertainty retained from cancelled passages.
140-3-s255-D002 equivalence

classes. [scan 256]

$$(7) \qquad G/\mathrm{int}(G) \longrightarrow I_G. \qquad \text{[p. 429]}$$

To tell the truth, † this morphism is an epimorphism (for the|the fppf topology), but not an isomorphism i.e. not a monomorphism – in other words, for two sections (in this case ρ, $\xi\rho$) of G, it is indeed necessary, but not always sufficient, that their images in I_G be equal, for them to be conjugate locally^t fppf. But in the case where the two sections are -regular in the sense of Steinberg (which is a condition on each fiber) the condition is sufficient. It is also so if the two sections are $\frac{1}{2}$ simple (which is also a condition on each fiber).

In the case of a group locally isomorphic to Sl(2), to GP(1), or Gl(2), the regular sections in the sense of Steinberg are moreover those which are central at no point – which is indeed the case for the ρ or the $\sigma_\infty = \sigma$ of the preceding §On the other hand, I_G is then well known, [p. 255]

Uncertainty retained from cancelled passages.
140-3-s256-D001 [illeg. 1]

E locally free of rank 2, it is canonically isomorphic to $\mathbb{E}^1$ or to $\mathbb{E}^1 \times \mathbb{G}_m$ (depending on whether the rank is 1 or 2) in the case of †a form of Sl(2) †, it is given by the Trace [scan 257]

$$(7) \qquad \begin{array}{ccc} G & \overset{\dagger}{\longrightarrow} & I_G \simeq \mathbb{E}^1, \\ x & \longmapsto & \mathrm{Tr}\, x \end{array}$$

in the case of a form of Gl(2) it is given by

$$(8) \qquad \begin{array}{ccc} G & \longrightarrow & I_G = \mathbb{E}^{1*} \times \mathbb{E}^1. \\ x & \longmapsto & (\det x, \mathrm{tr}\, x) \end{array}$$

We thus find (for the record)

Proposition Let G be a form of Sl(2) or Gl(2) over a scheme S, let ρ be a section of G which is central in no fiber. For a section ξ of G to be (locally) in the image of F_ρ ((1) or (3)), it is necessary and sufficient that it satisfy the following conditions

a) †$\xi\rho$ is central in no fiber, i.e. in no fiber do we have $\xi = \lambda\rho^{-1}$, where λ † a central element

b) $\mathrm{tr}\, \xi\rho = \mathrm{tr}\, \rho$

c) (if G a form of Gl(2)) $\det \xi = 1$ i.e. ξ a section of Sl(2). is equivalent to: $\det(\xi\rho) = \det \rho$ [p. 256]

Uncertainty retained from cancelled passages.
140-3-s257-D002 Sl(2)|Gl(2)
140-3-s257-D003 or GP(1)
140-3-s257-D004 det
140-3-s257-D005 ρ
140-3-s257-D006 is|being

Corollary When these conditions are satisfied, the scheme of solutions in α of the equation [scan 258] [p. 430]

$$[\alpha, \rho] = \xi$$

is a (right) torsor under T_ρ, which is a smooth commutative group scheme over S, of relative dimension 1 (if G a form of $\mathrm{Sl}(2)$) or 2 (if G a form of $\mathrm{Gl}(2)|\mathrm{Sl}(2)$).

Examples † $G = \mathrm{Gl}(2)_{\mathbb{Z}}$, $\rho = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}$ is the element of $\mathrm{Sl}(2, \mathbb{Z})$ already so denoted (starting from page 335). It is indeed regular, † likewise the element $\sigma = \sigma_\infty = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

In general, the datum of a form of $\mathrm{Gl}(2)$ is equivalent to that of a form of $\mathrm{Sl}(2)$, or of $\mathrm{GP}(1)$, or again to that of an Azumaya Algebra of rank 4 over S, let M be one – i.e. an Algebra locally (fppf) isomorphic to $M_2(\mathcal{O}_S)$ – by

$$(9) \qquad \begin{cases} \mathbb{G}|G(M) &= \text{scheme of units of } M \\ S\mathbb{G}|G(M) &= \text{——————————} \quad \text{of determinant 1} \\ P\mathbb{G}|G(M) &= S\mathbb{G}|G(M)/\text{Centre} \simeq \mathbb{G}|G(M)/\mathbb{G}_m|G_m \end{cases}$$

Thus|In the datum corresponds to that of an invertible section of M, and [p. 257]

Uncertainty retained from cancelled passages.
140-3-s258-D001 1
140-3-s258-D002 and

the regularity condition means that the subalgebra generated on each fiber is of rank 2 (and not reduced to the scalars) – which implies that ρ generates †a|unique sub-algebra of degree two [scan 259]

$$(10) \qquad \mathcal{O}_S + \mathcal{O}_S\rho = A_\rho$$

which is locally a direct summand in M as a module, i.e. with formation commuting|commuting with base change … And we shall have

$$(11) \qquad \begin{array}{c} T_\rho = \text{scheme of units of } A_\rho \\ (\text{case of } G = \mathbb{G}|G(M)) \end{array}$$

or

$$(11\text{bis}) \qquad \begin{array}{c} T_\rho = \text{scheme of units of } A_\rho \\ \text{of norm equal to 1} \\ (\text{case of } G = S\mathbb{G}|G(M)) \end{array}$$

We shall note that T_ρ is a torus (necessarily maximal) iff ρ is $\frac{1}{2}$ simple, i.e. $\frac{1}{2}$ simple in each fiber – which is equivalent, since it is regular, to the fact that in each fiber its two eigenvalues are distinct. For

$$(12) \qquad \rho = \begin{pmatrix} 0 & 1 \\ -1 & 0|1 \end{pmatrix} \qquad (\rho^6 = 1,\ \rho^3 = -1)$$

this is † the case in characteristic $\neq 3$ – thus in the|in the strict sense T_ρ (as a group scheme over $\mathrm{Spec}\,\mathbb{Z}$) is not a torus – it [p. 258]

Uncertainty retained from cancelled passages.
140-3-s259-D001 [illeg. 2]
140-3-s259-D002 is

[scan 260]

becomes † a product of a unipotent with the center $\mathbb{G}_m|G_m$ in characteristic 3. Likewise, for [p. 431]

$$\sigma = \sigma_\infty = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_\infty^4 = 1, \quad \sigma^2 = -1, \qquad (13)$$

σ is $\frac{1}{2}$ simple in all characteristics except in characteristic 2, where it is unipotent – and T_σ is a torus except in characteristic 2, where it is a product of the center $\mathbb{G}_m|G_m$ with a unipotent.

Let us return to the general case

When we take the centralizer in $S\mathbb{G}|G(M)$, we find

$$ST_\rho \subset T_\rho, \qquad ST_\rho = \mathrm{Ker}\left(T_\rho \xrightarrow[\text{is induced|induced by the determinant}]{\text{norm}} \mathbb{G}_m|G_m\right) \tag{13}$$

The advantage of the T_ρ over ST_ρ is that every torsor under T_ρ = scheme of units of A_ρ is locally trivial in the Zariski sense, whereas this is no longer necessarily so for a torsor under ST_ρ. Thus, when we set out to solve the equation in α

$$[\alpha, \rho] = \xi,$$

for given ξ satisfying the conditions of the proposition, this is possible (Zariski locally) in $\mathbb{G}|G(M)$, but perhaps not in $S\mathbb{G}|G(M)$. [p. 259]

Uncertainty retained from cancelled passages.

140-3-s260-D001 unipotent in characteristic 3

[scan 261]

Let us return to the special case

$$\rho = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \qquad A = \mathbb{Z}$$

then

$$\begin{gathered} A_\rho = \text{ring of integers of the field} \\ \mathbb{Q}(\sqrt[3]{-1}), \quad \rho \text{ corresponds to} \\ \exp(2i\pi/6) = \frac{1+i\sqrt{3}}{2} \end{gathered} \tag{14}$$

When we give ourselves ξ,

$$\xi = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad \text{whence } \xi\rho = \begin{pmatrix} -B & A+B \\ -D & C+D \end{pmatrix} \qquad \dagger$$

the conditions † b), c) of the proposition

$$\det \xi = 1, \quad \mathrm{Tr}(\xi\rho) = \mathrm{Tr}\,\rho$$

become

$$AD - BC = 1 \qquad C + D - B = 1 \tag{16}$$

– quite familiar conditions! They imply the possibility † (over an arbitrary scheme) of finding Zariski locally an $\alpha \in \Gamma\mathrm{Gl}(2)$ such that †

$$\alpha\rho(\alpha)^{-1} = [\alpha, \rho] = \xi, \tag{*}$$

m953

Marginal note. m953 at least if we suppose that the condition of regularity is not violated on any fiber…

and over $\mathbb{Z}$ it is even possible globally, for the obstruction is an † invertible A_ρ-Module, now A_ρ is principal unless I am mistaken i.e. †

$$\mathrm{Pic}(A_\rho) = 1 \tag{17}$$

[p. 260]

Uncertainty retained from cancelled passages.

140-3-s261-D002 15
140-3-s261-D003 of
140-3-s261-D006 finding
140-3-s261-D007 satisfies
140-3-s261-D008 torsor under
140-3-s261-D010 Pic of

[scan 262]

But if α is a solution, we shall have a priori [p. 432]

$$\det \alpha \in \mathbb{Z}^* = \{\pm 1\} \tag{18}$$

and other solution will be of the form $\alpha' = \alpha f$, with $f \in A_\rho^*$, whence

$$\det \alpha' = \det \alpha \cdot \det f = \det \alpha$$

for

$$\det f = f\bar{f} > 0 \quad \text{hence} \quad \det f = +1,$$

thus the sign in (18) † does not depend on the choice of α – if it is -1, one cannot find a solution of equation (*) in $\mathrm{Sl}(2, \mathbb{Z})$. The sign depends only on ξ i.e. only on A, B, C, D – and in fact † only on B, D (which already determine ξ, thanks to formulas (16) at least if D invertible) – the computations made elsewhere suggest that this sign is also † equal to the remainder of D mod $3 - 1$ if $D \equiv 1$ (3), -1 if $D \equiv -1$ (3).

Returning to the general case, if we have found a relation

$$[\alpha, \rho] = \alpha\rho(\alpha)^{-1} = \xi \qquad \alpha \in \Gamma\mathbb{G}|G(M), \tag{19}$$

in order † to find a solution α' which is $\in \Gamma S\mathbb{G}|G(M)$ i.e. $\det \alpha' = 1$, we know that the solutions $\alpha' \in \Gamma\mathbb{G}|G(M)$ † are given by

$$\alpha' = \alpha f \qquad f \in \Gamma\underline{\mathcal{O}}_S^*,$$

[p. 261]

Uncertainty retained from cancelled passages.

140-3-s262-D002 is
140-3-s262-D003 if D
140-3-s262-D004 [illeg. 1]
140-3-s262-D005 that there exist
140-3-s262-D006 sts

[scan 263]

and the condition becomes

$$\det \alpha' = 1 \quad \text{i.e.} \quad \det(\alpha f) = 1 \quad \text{i.e.}$$

$$\det(f^{-1}) = \det \alpha$$

thus the necessary and sufficient condition for the existence of a solution is that the section $\det \alpha$ of $\mathcal{O}_S^*$ be in the image of the norm homomorphism

$$N|\mathcal{N}_{A_\rho/A} : A_\rho^* \longrightarrow A^*, \tag{20}$$

i.e. the obstruction is in the group

N.B. $S = \operatorname{Spec} A$

$$A^*/N|\mathcal{N}_{A_\rho/A} A_\rho^* \tag{21}$$

which is a particularly well-known invariant in class field theory!

N.B. $N(c\rho+d) = c^2+cd+d^2$

Unless I am mistaken, it follows from class field theory that for ρ given by (12), and for $A = \underbrace{\mathbb{Z}_\ell}_{\text{integers } \ell\text{-adic}}$ (ℓ a prime number), this quotient group is zero in all cases, except for † $\ell = 3$, where it is isomorphic to $\mathbb{Z}/2\mathbb{Z}$. †

arbitrary base|Ring

Let us note that (still in the special case (12)) if

$$\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \tag{22}$$

immediate computations already made show that [p. 262]

Uncertainty retained from cancelled passages.

140-3-s263-D001 $\rho|\ell$

140-3-s263-D002 [illeg. 1]

[scan 264]

[p. 433]

$$(ad-bc)D = (c^2+cd+d^2) \tag{23}$$

thus D is invertible iff $c^2+cd|d+d^2 = \mathcal{N}_{A_\rho/A}(c\rho+d)$ is i.e. iff $c\rho+d$ is invertible, and then

$$ad-bc = (c^2+cd+d^2)/D$$

which shows that the sign of $\det\alpha = ad-bc$ is then equal to the value of D mod 3.

for $c^2+cd+d^2 \equiv 1$ (3)

(But can D be equal to 0 mod 3? then mod 3 the matrix ξ would depend only on the parameters A, B

$$\xi = \begin{pmatrix} A & B \\ 1+B & 0 \end{pmatrix}$$

with $-B(1+B) = 1$ i.e. $B^2+B+1 = 0$ thus $B^3 = 1$ thus (since 3) $B = 1$ thus we will have $\xi = \begin{pmatrix} A & 1 \\ -1 & 0 \end{pmatrix}$, †

$$\xi\rho = \begin{pmatrix} -1 & A+1 \\ 0 & -1 \end{pmatrix},$$

this matrix is regular iff $A \neq -1$, OK.)|} Thus a priori we cannot exclude the case where D can take a zero value mod 3…)

When we take the section σ of $\mathrm{Sl}(2)_{\mathbb{Z}}|\mathrm{Gl}(2)_{\mathbb{Z}}$, given by

$$\sigma = \sigma_\infty = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \qquad \sigma_\infty^4 = 1, \quad \sigma_\infty^2 = -1 \tag{24}$$

we again have (unless I am mistaken)

case|for $A = \mathbb{Z}$.

$$\mathrm{Pic}(A_\sigma) = 0|1. \tag{25}$$

The numerical condition on

$$\eta = \begin{pmatrix} R & S \\ T & U \end{pmatrix} \tag{26}$$

to be locally in the image

Uncertainty retained from cancelled passages.
140-3-s264-D003 [illeg. 4]

[scan 265]

of F_σ (in addition to the regularity condition on $\eta\sigma$)

$$\eta = \beta\sigma(\beta)^{-1} = [\beta, \sigma] \tag{27}$$

is that we have

$$\det \eta = 1, \quad \mathrm{Tr}(\eta\sigma) = \mathrm{Tr}\sigma \tag{28}$$

now

$$\eta\sigma = \begin{pmatrix} R & S \\ T & U \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} S & -R \\ U & -T \end{pmatrix}$$

and (28) becomes

$$RU - ST = 1, \quad S - T = 0 \quad \text{i.e. } S = T \tag{29}$$

i.e. η must be symmetric. † If this condition is satisfied, it follows from (25) — when the base is $\mathbb{Z}$ — that we can find a solution $\beta \in \mathrm{Gl}(2, \mathbb{Z})$ of (27), the obstruction to finding a solution in $\mathrm{Sl}(2, \mathbb{Z})$ is again a sign, namely an element in

$$\mathbb{Z}^*/\mathcal{N}(A_\sigma^*) \simeq \pm 1$$

— as we have, for

$$\beta = \begin{pmatrix} r & s \\ t & u \end{pmatrix}. \tag{30}$$

† satisfying (27), the relation

ε

$$(rt - su)U = t^2 + u^2 = \mathcal{N}(t\sigma + u) \tag{31}$$

Uncertainty retained from cancelled passages.
140-3-s265-D001 [illeg. 1]
140-3-s265-D002 the relation

[scan 266]
[p. 434]

we see as before that U is invertible iff $t\sigma + u$ is i.e. $t^2 + u^2$ is, and that in this case, the obstruction † (base $\mathbb{Z}$, or $\mathbb{Z}_\ell$) is the residue of U mod 4 (which is equal to ± 1)…

(arbitrary base)

Let us summarize for reference

Proposition. Let $A = \mathbb{Z}$, or $\mathbb{Z}_\ell$, or $\widehat{\mathbb{Z}}$, and let us consider the two equations for $t^2 + u^2 \equiv 1$ (4)

$$\begin{cases} [\alpha, \rho] \overset{\text{déf}}{=} \alpha\rho(\alpha)^{-1} = \xi \\ [\beta, \sigma] = \beta\sigma(\beta)^{-1} = \eta \end{cases}$$

where $\rho = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}$, $\sigma = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, ξ and $\eta \in \mathrm{Gl}(2, A)$ given, with unknowns α resp. β in $\mathrm{Gl}(2, A)$. We assume that

$$\xi = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \quad \text{resp. } \eta = \begin{pmatrix} R & S \\ T & U \end{pmatrix}$$

satisfy the necessary numerical conditions for the existence of a solution

$$AD - BC = 1 \qquad B + D - C = 1 \tag{16}$$

$$RU - ST = 1 \qquad S - T = 0 \quad \text{i.e. } S = T. \tag{29}$$

Then:

1) For there to be a solution α (resp. β) in $\mathrm{Gl}(2, A)$, it is necessary|necessary and sufficient that $\xi\rho = \begin{pmatrix} -B & A+B \\ -D & C+D \end{pmatrix}$ be regular, i.e. nonscalar on each fiber (resp. that $\eta\sigma = \begin{pmatrix} S & -R \\ U & -T = -S \end{pmatrix}$ be so) —

Uncertainty retained from cancelled passages.

140-3-s266-D001 of

[scan 267]

† i.e. that on no fiber, ξ (resp. η) be a scalar multiple of ρ^{-1} (resp. σ^{-1}) — and for this it suffices that D (resp. U) be invertible (since the † corresponding in|of

$$\rho^{-1} = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{or} \quad \sigma^{-1} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

is zero).

2) When D (resp. U) is invertible, there is always a solution α resp. β, with $\det\alpha \in \pm 1$ (resp. $\det\beta \in \{\pm 1\}$).

3) For there to be a solution in $\mathrm{Sl}(2, A)$, it is necessary and sufficient that we have $D \equiv 1$ (3) (resp. $U \equiv 1$ (4)) — a condition which is automatic if $A = \mathbb{Z}_\ell$ with $\ell \neq 3$ (resp. with $\ell \neq 2$).

II) Study in Gl(2) † of the equation

$$[\alpha, \rho] = [\beta, \sigma]\varepsilon_1(\dagger\mu - 1). \tag{32}$$

Study first of the equation in ξ, η.

We always assume that G is a form of $\mathrm{Gl}(2)$, associated with an Azumaya algebra $\mathcal{M}$ over the base $S = \mathrm{Spec}\, A$, and that we have two regular sections ρ, σ of G i.e. two invertible sections of $\mathcal{M}$. We are also given a "1-parameter subgroup" (additive) of G, i.e. a homomorphism

$$\varepsilon_1 : \mathbf{G}_a \longrightarrow SG, \qquad \lambda \longmapsto \varepsilon_1(\lambda). \tag{33}$$

Uncertainty retained from cancelled passages.

140-3-s267-D001 this it suffices

140-3-s267-D002 coefficient

140-3-s267-D003 $\mathbb{Z}$

140-3-s267-D004 [*illeg.* 1]

[scan 268]

[p. 435]

The case which interests us more particularly is that where $G = \mathrm{Gl}(2)_{\mathbb{Z}}$, ρ and σ as above, and

$$\varepsilon_1(\lambda) = \varepsilon_1^\lambda = \begin{pmatrix} 1 & 0 \\ \lambda & 1 \end{pmatrix}. \tag{34}$$

We will note that here

$$\varepsilon_1 = \varepsilon_1(1) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \dagger\rho\sigma \tag{35}$$

We regard (32) as equations relating the quantities α, β, μ — α, β variable local sections of G, μ a section of $\mathbb{E}^1|\mathbf{E}^1$ — the hypotheses we are going to make imply|will imply moreover that μ will in fact be a section of $\mathbf{G}_m$ (if, together with α, β, it satisfies relation (32)). †

We will begin by studying the systems $(\xi, \eta, \mu|\beta)$ satisfying

$$\begin{cases} \xi = \eta\varepsilon_1(\mu - 1) & (33) \\ \det\xi = \det\eta = 1 & (34) \\ \mathrm{Tr}\xi\rho = \underbrace{\mathrm{Tr}\rho}_{\|\,i} \text{ and } \mathrm{Tr}\eta\sigma = \underbrace{\mathrm{Tr}\sigma}_{\|\,j} & (35) \end{cases}$$

m963

Marginal note. m963 N.B. We will have $\det\varepsilon_1(\mu-1) = 1$ thus $\det\xi = \det\eta$, by the first relation.

We can eliminate η from these equations, thanks to (33), we find

Uncertainty retained from cancelled passages.

140-3-s268-D001 ℓ_1

140-3-s268-D002 [illeg. 1]

[scan 269]

$$(36) \qquad \det\xi = 1, \qquad \mathrm{Tr}(\xi{-}1)\rho = \dagger 0, \qquad \mathrm{Tr}(\xi\varepsilon_1(1-\mu)\dagger{-}1)\sigma\dagger = 0.$$

†

The third relation (36), on expanding with respect to the coefficients of ξ, can be written

$$(37) \qquad P(\xi)\mu{-}|{+}Q(\xi) = 0$$

where the coefficients P, Q are affine linear functions with respect to those of ξ, give rise to the identity (in ξ, μ)

$$(38) \qquad (\mathrm{Tr}(\xi\varepsilon_1(1-\mu)\dagger{-}1)\sigma\dagger) = P(\xi)\mu - Q(\xi)\dagger$$

†

The constant term $-Q(\xi)$ being † identically equal to the value of the left-hand side of (38) for $\mu = 0$, i.e.

$$(39) \qquad \begin{aligned} -Q(\xi) &= \mathrm{Tr}((\xi\varepsilon_1 - 1)\sigma). \\ &= \mathrm{Tr}\,\xi(\varepsilon_1\sigma) - \mathrm{Tr}\,\sigma \end{aligned}$$

We assume that we have

$$(40) \qquad \varepsilon_1\sigma = -\lambda_0\rho \qquad \begin{matrix} \lambda \text{ central section of } G \\ \text{i.e. } \lambda \text{ invertible scalar} \end{matrix}$$

so that

$$(40 \text{ bis}) \qquad \boxed{\varepsilon_1 = -\lambda_0\rho\sigma^{-1}}$$

m964

Marginal note. m964 N.B. If $\det\rho = \det\sigma = 1$, then we must have $\lambda_0^2 = 1$ thus if A integral, $\lambda_0 \in \{\pm 1\}$.

and (39) becomes, by virtue †

$$\dagger \qquad \mathrm{Tr}\,\xi(\varepsilon_1\sigma) = -\lambda_0|\lambda_0\,\mathrm{Tr}\,\xi\rho \overset{\text{by (36)}}{=} -\lambda_0\,\mathrm{Tr}\,\rho, \quad \text{thus}$$

Uncertainty retained from cancelled passages.

140-3-s269-D001 i
140-3-s269-D002 σ
140-3-s269-D003 j
140-3-s269-D004 set simplify
140-3-s269-D007 σ
140-3-s269-D008 $-j$
140-3-s269-D009 $= 0$
140-3-s269-D010 $\mathrm{Tr}(\sigma\xi\varepsilon_1(1-\mu))$
140-3-s269-D011 equal to
140-3-s269-D012 the equation
140-3-s269-D013 $-$

[scan 270]
[p. 436]

we find

$$\dagger\ (41) \qquad Q(\xi) = \underbrace{\lambda_0 \operatorname{Tr}\rho + \operatorname{Tr}\sigma}_{c} \qquad \text{(constant, independent of } \xi)$$

and we assume

$$(42) \qquad c = \lambda_0 \operatorname{Tr}\rho + \operatorname{Tr}\sigma \quad \dagger\text{section invertible of } \mathcal{O}_S.$$

(In the case which interests us, we will have

$$\sigma^{-1} = -\sigma, \quad \varepsilon_1 = \rho\sigma = -\rho\sigma^{-1}, \quad \text{i.e. } \lambda_0 = 1$$

$$\dagger \quad c = \lambda_0 \underbrace{\operatorname{Tr}\rho}_{1} + \underbrace{\operatorname{Tr}\sigma}_{0} = 1 \qquad \text{OK.}$$

)

This being so, the third equation (36) becomes (taking account of the first two)

$$(43) \qquad P(\xi)\mu = c$$

which also means that $P(\xi)$ is invertible, and

$$(44) \qquad \mu = c/P(\xi).$$

Thus, μ is uniquely determined by ξ, by equation (44) — and μ will necessarily be invertible.

In summary:

Proposition If we assume that we have an expression (40 bis) for the unipotent element $\varepsilon_1 = \varepsilon_1(1)$, and that we have (42) ($\lambda_0 \operatorname{Tr}\rho + \operatorname{Tr}\sigma$ invertible), then

Uncertainty retained from cancelled passages.

140-3-s270-D001 42|37
140-3-s270-D002 integer|section element
140-3-s270-D003 Tr

[scan 271]

the set of solutions $\dagger$ (ξ, η, μ) of the simultaneous equations (33)(34)(35) is in 1-1 correspondence with the set of ξ sections|section of $G = G(\mathcal{M})$ satisfying the conditions

$$(45) \qquad \begin{cases} \det\xi = 1, \quad \operatorname{Tr}\xi\rho = \operatorname{Tr}\rho, \quad \text{and} \\ P(\xi) \text{ invertible section of } \mathcal{O}_S. \end{cases}$$

In terms of ξ, the corresponding solution of the system (33 : 35) is given by

$$(\xi,\ \eta = \xi\varepsilon_1(1-\mu),\ \mu = c/P(\xi))$$

— and μ (the "multiplier") is necessarily invertible.

N.B. If we had solved for η, we would have found a condition of the form

$$P'(\eta)\mu - Q'(\eta) = 0$$

with

$$Q'(\eta) = \operatorname{Tr} \eta\varepsilon_0^{-1}\rho - \operatorname{Tr} \rho$$

and the natural condition would be

$$\varepsilon_0^{-1}\rho = -\lambda_0'\sigma \qquad (\lambda_0' \text{ invertible scalar}),$$

which is equivalent to (40 bis) with $\lambda_0' = 1/\lambda_0$, and implies that

$$Q'(\eta) = c^{\mathrm{te}} = \lambda_0' \operatorname{Tr} \sigma - \operatorname{Tr} \rho = \dagger c',$$

Uncertainty retained from cancelled passages.

140-3-s271-D001 of the|in

140-3-s271-D002 λ_0'

[scan 272]
[p. 437]

we see at once that

$$c' = \lambda_0' c$$

thus the condition c invertible is equivalent to c' invertible, and we will then have the system of equations in η

$$\det \eta = 1, \quad \operatorname{Tr} \eta\sigma = \operatorname{Tr} \sigma, \quad P'(\eta)\mu = c' \quad \text{i.e.} \quad \begin{cases} P'(\eta) \text{ invertible} \\ \mu = c'/P'(\eta) \end{cases}.$$

We see therefore that the hypotheses made are in fact symmetric in ξ, η, and allow a symmetric treatment of the solution, either via ξ, or via η.

Determination of $P(\xi)$. We will have

$$\varepsilon_1(\lambda) = 1 + N_1\lambda = (1 + N_1)^\lambda$$

N_1 a square-zero matrix

$$N_1^2 = 1,$$

and

$$\operatorname{Tr}(\xi\varepsilon_1(1-\mu)\sigma) - \operatorname{Tr} \sigma = \operatorname{Tr} \xi\sigma - \operatorname{Tr} \sigma + \lambda(1-\mu) \operatorname{Tr} \xi N_1\sigma$$

thus

$$(47) \qquad P(\xi) = \dagger \operatorname{Tr} \xi \underbrace{(-N_1\sigma)}_{\| \; N_1\rho=\rho+\sigma}$$

Moreover we will have

$$(48) \qquad \varepsilon_1 = 1 + N_1$$

and formula (40 bis) reads

$$(49) \qquad N_1 = -(\lambda_0\rho\sigma^{-1} + 1)$$

i.e.

or again

Uncertainty retained from cancelled passages.
140-3-s272-D001 —

[scan 273]

$$N_1\sigma = -(\lambda_0\rho + \sigma)$$

thus we will have

$$(50) \qquad P(\xi) = \operatorname{Tr} \dagger\xi(\lambda_0\rho + \sigma)$$

Corollary to the Proposition The scheme of simultaneous solutions of the equations (33) : (35) in ξ, η, μ is a smooth relative scheme, of relative dimension 2 — for the equation $\operatorname{Tr}(\xi\rho) = c^{\text{te}}$ always defines a smooth subscheme of dimension 1 of † $SG(\mathcal{M})$ (as soon as ρ is a section that is zero on no fiber), — of which we must then take the open subset defined by "$P(\xi)$ invertible".

In the case which interests us,

$$\rho = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \qquad \sigma = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

†

setting|for

$$\xi = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \qquad \eta = \begin{pmatrix} R & S \\ T & U \end{pmatrix}$$

we will have

$$P(\xi) = D, \qquad P'(\eta) = U$$
$$(51) \qquad \eta\varepsilon_1(\mu - 1) = \begin{pmatrix} R + (\mu - 1)S, & S \\ T + (\mu - 1)U & U \end{pmatrix}$$

Uncertainty retained from cancelled passages.
140-3-s273-D001 ξ
140-3-s273-D002 Sl(2)|SG
140-3-s273-D003 [illeg. 1]

[scan 274]
[p. 438]

and the equation

$$\xi = \eta\varepsilon_1(\mu - 1)$$

† implies

$$(52) \qquad B = S, \quad D = U,$$

and in fact it is equivalent to one of these two equations and to

$$(53) \qquad \mu D = 1 \qquad (\Longleftrightarrow \mid \Longrightarrow \mu U = 1)$$

when we already assume that ξ, η satisfy the equation of the determinants and the traces

$$(54) \quad \begin{cases} AD - BC = 1 & RU - ST = 1 \\ C|A + D - B = 1 & T - S = 1 \end{cases}$$

The † scheme of solutions of the simultaneous equations (33) to|: (35) is then that of the (D, B) with D invertible, or again of the (S, U) with S invertible. We can identify it with the scheme of matrices

$$(55) \qquad \begin{pmatrix} \mu & M \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1/D & B/D \\ 0 & 1 \end{pmatrix}$$

(cf. page 410 [illeg. 1]), which normalize the 1-parameter subgroup

$$\dagger \quad \varepsilon_0(\lambda) = \dagger \begin{pmatrix} 1 & -\lambda \\ 0 & 1 \end{pmatrix} = \rho^{-1}\varepsilon_1(\lambda)\rho.$$

Uncertainty retained from cancelled passages.
140-3-s274-D001 is equivalent to
140-3-s274-D003 [illeg. 1]
140-3-s274-D004 $\rho \quad \varepsilon$
140-3-s274-D005 [*illeg.* 1]

[scan 275]

Let us return to the general case, where we nevertheless assume that ε_1 given by (40 bis), † let us put

$$(56) \qquad \varepsilon_0 = \rho^{-1}\varepsilon_1\rho = -\lambda_0\sigma^{-1}\rho = 1 + N_0, \quad \text{with}$$

$$(57) \qquad N_0 = \rho^{-1}N_1\rho = -(\lambda_0\sigma^{-1}\rho + 1),$$

so that $N_0^2 = 0$, and

$$(58) \qquad \varepsilon_0 \quad \lambda \longmapsto \varepsilon_0(\lambda) \overset{\text{dfn}}{=} 1 + \lambda N_0$$

is a 1-parameter subgroup of $SG(\mathcal{M})$, say L_0, whose normalizer in $G(\mathcal{M})$ will be denoted by B_0. We have two homomorphisms

(59)

$$B_0 \overset{\chi(\underline{u})=\det(\underline{u})}{\underset{\chi'(\underline{u}) \text{ given}}{\rightrightarrows}} \mathbf{G}_m|G_m$$

by

$$\underline{u}(\varepsilon_0(\lambda)) = \varepsilon_0(\chi'(\underline{u})\lambda)$$

or again (in linear terms in $\mathcal{M}$)

$$(60) \qquad \underline{u}N_0\underline{u}^{-1} = \chi'(\underline{u})N_0$$

m966

Marginal note. m966 I underline $\underline{u}$, so as not to confuse it with a coefficient of $\eta|\beta = \begin{pmatrix} r & s \\ t & u \end{pmatrix}$!

And we consider the subgroup

$$(61) \qquad \begin{cases} \widetilde{B}_0^* \subset B_0 \quad \text{defined by } \chi(\underline{u}) = \chi'(\underline{u}) \\ \text{i.e. by } \dagger \\ \underline{u}N_0\underline{u}^{-1} = \det(\underline{u})N_0 \end{cases}$$

so we have

$$L_0 \subset \widetilde{B}_0^*.$$

and we even have an exact sequence of group schemes

$$1 \longrightarrow L_0 \longrightarrow \widetilde{B}_0^* \xrightarrow{\det} \mathbf{G}_m|G_m \longrightarrow 1.$$

m967

Marginal note. m967 we would need to check [illeg. 4] † This is false, cf. below.

Uncertainty retained from cancelled passages.
140-3-s275-D001 then
140-3-s275-D002 det
140-3-s275-D003 [illeg. 1]

[scan 276]
[p. 439]

Let $\underline{u}$ be a section of $\widetilde{B}_0^*$, I feel like proving that if we put

$$(63) \qquad \xi = [\underline{u}, \rho], \quad \eta = [\underline{u}, \sigma],$$

we will automatically have the relation

$$(64) \qquad \xi = \eta\varepsilon_1(\mu - 1) \qquad \text{where } \mu = \det \underline{u}.$$

To reassure ourselves, let us first take $\underline{u}$ in L_0, i.e.

$$\underline{u} = \varepsilon_0(\lambda)$$

whence

$$[\underline{u}, \rho] = \varepsilon_0(\lambda)\rho\varepsilon_0(\lambda)^{-1}\rho^{-1} = \varepsilon_0(\lambda)\varepsilon_1(-\lambda)$$
$$[\underline{u}, \sigma] = \varepsilon_0(\lambda)\sigma\varepsilon_0(-\lambda)\sigma^{-1}$$

Now

$$\sigma \underbrace{\varepsilon_0(-\lambda)}_{1+N_0\lambda} \sigma^{-1} = 1 + (\sigma \underline{N_0} \sigma^{-1})\lambda \ \dagger$$

now † by virtue of (57) we have

$$\sigma N_0 \sigma^{-1} = -(\lambda_0 \rho \sigma^{-1} + 1) \overset{(49)}{=} N_1$$

so we have

$$(65) \qquad \begin{cases} \sigma\varepsilon_0(\lambda)\sigma^{-1} = \varepsilon_1(\lambda) & \text{and similarly} \\ \rho\varepsilon_0(\lambda)\rho^{-1} = \varepsilon_1(\lambda) \end{cases}$$

so we do indeed find

$$[\underline{u}, \rho] = [\underline{u}, \sigma] = \varepsilon_0(\lambda)\varepsilon_1(-\lambda)$$

i.e. (64), since here $\mu = \det u = 1$, so $\mu - 1 = 0$.

Uncertainty retained from cancelled passages.
140-3-s276-D001 =
140-3-s276-D002 [illeg. 1]

[scan 277]

We have seen that ρ and σ both transform N_0 into N_1, hence the additive 1-parameter subgroup

$$L_0 = \left\{ \underset{=1+\lambda N_0}{\varepsilon_0(\lambda)} \;\middle|\; \lambda \text{ in } \mathbf{G}_a | G_a \right\}$$

into the analogous subgroup

$$L_1 = \left\{ \underset{=1+\lambda N_1}{\varepsilon_1(\lambda)} \;\middle|\; \lambda \text{ in } \mathbf{G}_a | G_a \right\},$$

and hence $\widetilde{B}_0$ into $\widetilde{B}_1$ (defined in terms of L_1, or of N_1, as $\widetilde{B}_0^*$ in terms of L_0). The formula (64|69) can also be written

$$\rho \underline{u}^{-1} \rho^{-1} = \sigma \underline{u}^{-1} \sigma^{-1} \varepsilon_1(\mu - 1)$$

or also, replacing $\underline{u}$ by $\underline{u}^{-1}$

$$(*|**) \qquad \rho\underline{u}\rho^{-1} = \sigma\underline{u}\sigma^{-1}\varepsilon_1\left(\frac{1}{\mu} - 1\right).$$

Now we † have

$$\rho\underline{u}\rho^{-1} = \dagger\,\mathrm{int}(\lambda)(\sigma\underline{u}\sigma^{-1})$$

where $\underline{\lambda = \rho\sigma^{-1}}$, and the formula $(*)|(**)$ can also be written

$$(**): \qquad \mathrm{int}(\lambda)(v) = v\cdot\varepsilon_1\left(\frac{1}{\mu} - 1\right) \qquad \begin{array}{l}\text{for } v \text{ in } \underline{B_1'|\widetilde{B}_1} \\ \text{and } \mu = \det v;\end{array}$$

and by virtue of (40 bis) $\mathrm{int}\,\lambda = \mathrm{int}(\varepsilon_1)$, so the formula reduces to

$$\mathrm{int}(\varepsilon_1)(v) = v\cdot\varepsilon_1\left(\frac{1}{\mu} - 1\right)$$

Uncertainty retained from cancelled passages.

140-3-s277-D003 [illeg. 1]

140-3-s277-D004 λ

[scan 278]
[p. 440]

or again

$$\dagger$$

$$\dagger$$

$$\varepsilon_1 v\varepsilon_1^{\dagger}(-1) = v\varepsilon_1\left(\frac{1}{\mu} - 1\right) \quad \text{i.e.}$$

$$\underbrace{\varepsilon_1}_{1+N_1}\ v = v\underbrace{\varepsilon_1\left(\frac{1}{\mu}\right)}_{1+\frac{1}{\mu}N_1} \qquad \text{i.e. } v + N_1 v = v + \frac{1}{\mu}vN_1$$

that is, finally

$$vN_1v^{-1} = \mu N_1 \quad (= \det(v).N_1)$$

which is nothing other than the definition of $\widetilde{B}_1^*$!

We thus find a †morphism

$$(66) \qquad \left\{\begin{array}{rcl}\widetilde{B}_0 & \longrightarrow & \mathcal{X}_{\rho,\sigma} \\ \underline{u} & \longmapsto & ([\underline{u},\rho],[\underline{u},\sigma],\det\underline{u})\end{array}\right.$$

from $\widetilde{B}_0^*$ into the set of solutions of the system (33) to (35), is it an isomorphism?

Note that the equality $[\underline{u},\rho] = [\underline{u}',\rho]$ means that $u^{-1}u' \in \Gamma T_\rho$ so † saying that (66) is a monomorphism simply means that

$$\dagger \qquad T_\rho \cap \widetilde{B}_0 = \{1\}$$

in fact the analysis carried out must show

Uncertainty retained from cancelled passages.

140-3-s278-D001 $v_0|v$
140-3-s278-D003 $+|-$
140-3-s278-D004 -1
140-3-s278-D005 mono
140-3-s278-D006 [illeg. 3]
140-3-s278-D007 67

[scan 279]

that this intersection is also equal to $T_\sigma \cap \widetilde{B}_0^{*|\prime}$, hence contained in $T_\sigma \cap T_\rho$. I am going to assume

$$T_\sigma \cap T_\rho = \mathbf{G}_m \ (= \text{Cent}\, G(M)) \tag{68}$$

(is $\supset$ | $\supseteq$, it [illeg. 2] $\subset$ | $=$) — it will suffice|suffices for this to assume that in each fiber, $1, \sigma, \rho$ are linearly independent — this is of course the case in the case we are interested in. But Thus

$$T_\sigma \cap T_\rho \cap \widetilde{B}_0^{*|\prime} \subset \mathbf{G}_m \cap \widetilde{B}_0^*,$$

but

$$\mathbf{G}_m \cap \widetilde{B}_0^* = \mu_2$$

For if λ † is a local section of $\mathbf{G}_m$, then

$$\lambda N_0 \lambda^{-1} = N_0 = 1. N_0|N_3$$

but $\det \lambda = \lambda^2$, and the condition that λ † (considered as a section of B_0) be in $B_0^{*|\prime}$, is indeed $\lambda^2 = 1$. †

We find † in this way

$$T_\sigma \cap \widetilde{B}_0^{*|\prime} = T_\rho \cap \widetilde{B}_0^{*|\prime} = \mu_2 \tag{69}$$

and the† morphism (66) is not a monomorphism, but factors

Uncertainty retained from cancelled passages.

140-3-s279-D002 [illeg. 1]
140-3-s279-D003 be
140-3-s279-D004 B_0
140-3-s279-D005 [illeg. 1]
140-3-s279-D006 [illeg. 1]

[scan 280]
[p. 441]

as a monomorphism

$$\widetilde{B}_0/\mu_2 \hookrightarrow \mathcal{X}_{\rho,\sigma}. \tag{70}$$

To tell the truth, we see that $\widetilde{B}_0$ is not the "right" group to introduce here — in the special case $\varepsilon_0 = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$, † we have

$$B_0 = \left\{ \begin{pmatrix} p & \ell|l \\ 0 & q \end{pmatrix} \mid pq \text{ inv.} \right\}$$

and the relation † $\chi'(\underline{u}) = \underbrace{\chi(\underline{u})}_{\det \underline{u} = pq}$ can be written, by an immediate computation

$$p/q = pq \qquad \text{i.e.} \quad q^2 = 1 \tag{71}$$

whereas the "right" group we are interested in is defined by

$$B'_0 = \left\{ \begin{pmatrix} p & \ell|e \\ 0 & 1 \end{pmatrix} \mid p \text{ inv} \right\},$$

thus with $q = 1$. To give the description in the general case, we note that the determinant function on B_0^* decomposes intrinsically into two factors

$$\det \underline{u} = p(\underline{u})q(\underline{u}), \tag{72}$$

which we can distinguish from each other precisely by the condition that

$$\underline{u} N_0 \underline{u}^{-1} = \frac{p(\underline{u})}{q(\underline{u})} N_0, \tag{73}$$

and the condition which is going to describe the "right" group for us is not (71), but rather

$$q = q(\underline{u}) = 1, \tag{74}$$

Uncertainty retained from cancelled passages.
140-3-s280-D001 B_0
140-3-s280-D002 for

[scan 281]

which thus defines a group that is smooth (even in characteristic 2 !) and this time fits into a canonical exact sequence

$$1 \longrightarrow L_0^*|L_0 \longrightarrow B'_0 \overset{\det}{\longrightarrow} \mathbf{G}_m \longrightarrow 1 \tag{74}$$

We now have

$$B'_0 \cap \mathbf{G}_m = \{1\}, \tag{75}$$

which implies that the morphism induced by (66)

$$B'_0 \hookrightarrow \mathcal{X}_{\rho,\sigma} \tag{76}$$

is a monomorphism.

To show that it is an isomorphism, it now suffices, by virtue of general principles, to show that it is surjective on the points with values in an algebraically closed field (if that can|for simplify life for us). I drop the details of the|of verifications.

†

Remark|Remark We can consider, by condition (40 bis), that we start from a unipotent section $\varepsilon_1 = 1 + N_1 \quad (N_1^2 = 0)$ † of $SG(M)$, and from a section ρ, and that σ is deduced from it by †the

Uncertainty retained from cancelled passages.
140-3-s281-D001 Remark
140-3-s281-D002 ε_1
140-3-s281-D003 a

[scan 282]
[p. 442]

formula

$$\sigma = -\varepsilon_1^{-1}\rho \ (= -(1 - N_1)\rho \quad \text{i.e.} \quad \sigma = -\rho + N_1\rho, \quad \text{i.e.} \quad \rho + \sigma = N_1\rho) \tag{77}$$

m969

Marginal note. m969 I take the constant which occurs in|in the case that interests us most, where $-\sigma^{-1} = \sigma$, so that (40 bis) becomes $\varepsilon_1 = \rho\sigma$.

(I drop the constant pure|for λ_0, for † the situation is not changed when we replace ρ, σ by scalar multiples, for F_ρ and F_σ is not changed …) Condition (42) then becomes

$$c = \operatorname{Tr}\rho + \operatorname{Tr}\sigma = \operatorname{Tr}(1 - \underbrace{\varepsilon_1^{-1}}_{1-N_1})\rho = \operatorname{Tr} N_1\rho \in \Gamma\mathcal{O}_S^*$$

m970

Marginal note. m970 N.B. we have

$$\begin{cases} \rho + \sigma = N_1\rho = \rho N_0 \\ \qquad = -N_1\sigma = -\sigma N_0 \end{cases} \tag{78 bis}$$

let $\gamma_0|\varepsilon_0$ [illeg. 4]

(invertible section of $\mathcal{O}_S$). We assume that $1, \rho, \sigma$ are linearly independent in each fiber, it amounts to the same to say that $1, \rho$, and $N_1\rho$ are, or again that $\rho^{-1}, 1$ and N_1 are, or that $1, \sigma, N_1\sigma$ are, or $\sigma^{-1}, 1, N_1$. In summary

m971

Marginal note. m971 N.B. (After possibly multiplying the|the ρ by an invertible scalar) we $\operatorname{Tr}\gamma_0|\delta_0 = 1$ ($\gamma_0|\delta_0 = N_1\rho$) ! (which normalizes entirely, with the relation $\varepsilon_1 = -\rho\sigma^{-1}$, the arbitrary|intrinsic scalar factors of ρ, σ)

Proposition. Let $M|\mathcal{M}$ be an Azumaya algebra over the base scheme S, N_1 a section of square zero † i.e. $\varepsilon_1 = 1 + N_1$ a unipotent section, ρ, σ † some sections related by

$$\varepsilon_1 = -\rho\sigma^{-1} \quad \text{i.e.} \quad \sigma = -\varepsilon_1^{-1}\rho \quad \text{i.e.} \quad \rho + \sigma = N_1\rho \quad \text{i.e.} \quad \sigma + \rho|\rho + \sigma = -N_1\sigma \qquad (40|78)$$

† (which thus determines each of the three sections †|ε_1 ρ, σ and ε_1 (or N_1) in terms of the other two, and we can choose arbitrarily ρ and † ε_1 i.e. ρ and N_1 of square zero) — we

Uncertainty retained from cancelled passages.

140-3-s282-D001 [illeg. 1]
140-3-s282-D002 which
140-3-s282-D003 a
140-3-s282-D004 [illeg. 1]
140-3-s282-D005 ε_1
140-3-s282-D006 N_1

[scan 283]

suppose moreover †(2°) † $1, \rho, \sigma$ are linearly independent in each fiber i.e. $1, \rho^{-1}, N_1$ linearly independent in each fiber i.e. $1, \sigma^{-1}, N_1$ linearly independent in each fiber, †and (3°) $\operatorname{Tr} N_1\rho$ invertible.

m972

Marginal note. m972 [illeg. 2] $1, \rho^{-1}, \varepsilon_1(= 1 + N_1)$ [illeg. 1]. [illeg. 2] $1, \sigma^{-1}, \varepsilon_1(= 1 + N_1)$ [illeg. 1].

m973

Marginal note. m973 N.B. 2°) is a consequence of 3°), which implies primo, [illeg. 3] suitable basis*

$$\varepsilon_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad \rho = \begin{pmatrix} 0 & 1 \\ a & b \end{pmatrix} \quad (a \text{ invertible}),$$

$$\dagger = N_1\rho = \cdots = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \text{thus } \sigma = \begin{pmatrix} 0 & -1 \\ -a & 1-b \end{pmatrix}.$$

* and on normalizing so as to have $\operatorname{Tr} N_1\rho = 1$.

(N.B. we have $N_1\rho = -N_1\sigma = \rho + \sigma$). This being so, we will have

$$\operatorname{int}(\rho^{-1})\varepsilon_1 = \operatorname{int}(\sigma^{-1})\varepsilon_1, \quad \text{i.e. } \varepsilon_0.$$

(thus $\operatorname{int}(\rho)\varepsilon_0 = \operatorname{int}(\sigma)\varepsilon_0 = \varepsilon_1$)

†

$$\varepsilon_0 = 1 + N_0.$$

† set

$$\varepsilon_0(\lambda) = 1 + \lambda N_0, \qquad \varepsilon_1(\lambda) = 1 + \lambda N_1,$$

(additive 1-parameter subgroups) and let B_0 be the (Borel) normalizer in $G(\mathcal{M})$ of the

$$L_0 = \text{subgroup scheme } \varepsilon_0(\mathrm{G}_a)$$
$$L_1 = \text{_____} \quad \varepsilon_1(\mathrm{G}_a),$$

(thus $\rho(L_0) = \sigma(L_0) = L_1$), let B_0' be the subgroup scheme of B_0 defined in terms of ε_0, by (74), so that B_0 fits into an exact sequence

$$1 \longrightarrow L_0 \hookrightarrow B_0 \overset{\det}{\longrightarrow} \mathrm{G}_m \longrightarrow 1,$$

and that $\underline{u} \in \Gamma B_0$ implies

$$\underline{u}(N_0) = \det(\underline{u})N_0, \qquad \text{i.e.}$$
$$\underline{u}(\varepsilon_0(\lambda)) = \varepsilon_0(\det(\underline{u})\lambda).$$

This being so, consider the morphism

$$B_0' \longrightarrow \dagger G(\mathcal{M}) \times G(\mathcal{M}) \times \mathbb{E}^1|\mathbb{E}'$$
$$\underline{u} \longmapsto ([\underline{u}, \rho], [\underline{u}, \sigma], \det(\underline{u})).$$

[p. 282]

Uncertainty retained from cancelled passages.

140-3-s283-D001 that [illeg. 4]
140-3-s283-D002 [illeg. 1]
140-3-s283-D003 [illeg. 4]
140-3-s283-D004 [*illeg.* 1]
140-3-s283-D005 [illeg. 1] subgroup
140-3-s283-D006 set
140-3-s283-D007 $\mathcal{X}$

[scan 284]

This takes its values in $\mathcal{X}_{\rho,\sigma}$, the scheme of solutions † simultaneous equations (33) to (35), [p. 443] and induces an isomorphism

$$B_0' \simeq | \overset{\sim}{\to} \mathcal{X}_{\rho,\sigma}. \tag{79}$$

† Let us recall also|thus for the record

Corollary The projection

$$\mathcal{X}_\rho \longrightarrow G(\mathcal{M})$$
$$(\zeta, \eta, \mu) \longmapsto \zeta$$

is a monomorphism, and the image consists of the ζ satisfying

$$\det\zeta = 1, \quad \operatorname{Tr}\zeta\rho = \operatorname{Tr}\rho, \quad \operatorname{Tr}\zeta \underbrace{(\rho+\sigma)}_{\| \; -N_1\sigma = +N_1\rho = \rho N_0 = -\sigma N_0} \text{ invertible}$$

Likewise the projection

$$\begin{gathered}\mathcal{X}_\rho \longrightarrow G(\mathcal{M}) \\ (\zeta,\eta,\mu) \longmapsto \eta\end{gathered}$$

is a monomorphism, and the image consists of the η satisfying

$$\det\eta = 1, \quad \operatorname{Tr}\eta\sigma = \operatorname{Tr}\sigma, \quad \operatorname{Tr}\eta(\rho+\sigma) \text{ invertible}$$

Finally, we will have †

$$\dagger\operatorname{Tr}\zeta(\rho+\sigma) = \operatorname{Tr}\eta(\rho+\sigma) = 1/\mu. \tag{80}$$

Scholium By transport of structure, we find a group|groups structure on $\mathcal{X}_{\rho,\sigma}$. [p. 283]

Uncertainty retained from cancelled passages.

140-3-s284-D001 of the

140-3-s284-D002 Corollary.

140-3-s284-D003 [illeg. 1]

140-3-s284-D004 ζ

[scan 285]

Making it explicit, we find

$$\begin{array}{rcl} \text{if } \underline{u} & \text{corresponds to} & \zeta,\eta,\mu \\ \underline{u}' & — & \zeta',\eta',\mu'|\mu'' \\ \underline{u}'' = \underline{u}'\underline{u} & — & \zeta'',\eta'',\mu'' \end{array}$$

we will have

$$\mu'' = \mu\mu', \quad \zeta'' = u'(\zeta)\zeta', \quad \eta'' = u'(\eta)\eta' \tag{81|21}$$

where we set, as usual

$$u'(x|X) = u'x|Xu'^{-1},$$

† for two elements x, u' of a group.

III) Study of the equation (Continued)

$$\begin{gathered}\alpha\rho(\alpha)^{-1} = \beta\sigma(\beta)^{-1}\varepsilon_1(\mu-1) \\ (\alpha,\beta\in\Gamma G(\mathcal{M}), \quad \mu\in\Gamma\mathbb{E}^1_S|\mathcal{O}^*_S)\end{gathered} \tag{82}$$

i.e.

$$[\alpha,\rho] = [\beta,\sigma]\varepsilon_1(\mu-1). \tag{82bis}$$

Let $\mathcal{G}_{\rho,\sigma}$ be the scheme of solutions, we thus have a morphism

$$\begin{gathered}\mathcal{G}_{\rho,\sigma} \longrightarrow \mathcal{X}_{\rho,\sigma} \\ (\alpha,\beta,\mu) \longmapsto (\alpha\rho(\alpha)^{-1}, \beta\dagger\sigma(\beta)^{-1}, \mu)\end{gathered} \tag{83}$$

and it follows from the preceding theorem that this is † an epimorphism, more precisely, [p. 284]

Uncertainty retained from cancelled passages.
140-3-s285-D001 for
140-3-s285-D002 ρ
140-3-s285-D003 [illeg. 1]

[scan 286]
[p. 444]

We have a canonical section

$$\begin{array}{ccc} \mathcal{G}_{\rho,\sigma} & \longrightarrow & \mathcal{X}_{\rho,\sigma} \\ & \nwarrow & \uparrow \\ & & B'_0 \end{array} \qquad \begin{array}{c} ([\underline{u},\rho],[\underline{u},\sigma],\det \underline{u}) \\ \uparrow \\ \underline{u} \end{array} \tag{84}$$

$$(\underline{u},\underline{u},\det \underline{u}) \longleftarrow\!\shortmid \underline{u}$$

This section takes its values in the "diagonal", † which corresponds to the pairs $(\underline{u},\mu)$ ($\underline{u} \in \Gamma\mathcal{M}^*,\ \mu \in \Gamma\mathbb{E}^1|\mathbb{E}'$) such that

$$[\underline{u},\rho] = [\underline{u},\sigma]\varepsilon_1(\mu - 1), \tag{85}$$

which we will try|try to determine as well.

Combining the results of (I, and II) we see that the general solution of (82) will be formed|provided by the,

$$(\underline{ua},\underline{ub},\mu = \det \underline{u})$$

where $\underline{u}$ is a section of B'_0, $\underline{a}$ a section of T_ρ, $\underline{b}$ a section of T_σ.

To be precise

Theorem. Consider

$$\begin{aligned} B'_0 \times T_\rho \times T_\sigma &\longrightarrow G \times G \times \mathbb{E}^1|\mathbb{E}' \\ (\underline{u},\underline{a},\underline{b}) &\longmapsto (\underline{ua}^{-1},\underline{ub}^{-1},\det \underline{u}) \end{aligned} \tag{86}$$

[p. 285]

Uncertainty retained from cancelled passages.
140-3-s286-D001 form

[scan 287]

It takes its values in $\mathcal{G}_{\rho,\sigma}$, and is a monomorphism — hence induces an isomorphism

$$B'_0 \times T_\rho \times T_\sigma \overset{\sim}{\to} \mathcal{G}_{\rho,\sigma} \tag{87}$$

More precisely, let † (α,β,μ) be a section of $\mathcal{G}_{\rho,\sigma}$, then there exist then some unique sections $\underline{a}$ of T_ρ, $\underline{b}$ of T_σ, such that we have

$$\alpha\underline{a},\beta\underline{b} \in \Gamma B'_0 \tag{88}$$

† and we will then necessarily have

$$\alpha\underline{a} = \beta\underline{b}\ (\overset{\text{df}}{=} \underline{u})$$

whence of course

$$([\alpha,\rho],[\beta,\rho]) = ([\underline{u},\rho],[\underline{u},\dagger\sigma])$$

and the unique μ satisfying (82) is therefore $\det \underline{u}$ † ; thus

$$(\alpha,\beta,\mu) = (\underline{ua}^{-1},\underline{ub}^{-1},\det \underline{u}) \quad \text{(uniquely)} \tag{89}$$

†

Proof. It remains only to make explicit the uniqueness of α,β satisfying (88), which is a consequence of

$$T_\rho \cap B'_0 = T_\sigma \cap B'_0 = \{1\} \tag{90}$$

which we had forgotten to make explicit in (II).

[p. 286]

Uncertainty retained from cancelled passages.
140-3-s287-D001 locally
140-3-s287-D004 [illeg. 2]
140-3-s287-D005 ρ
140-3-s287-D006 [illeg. 1]
140-3-s287-D007 Proof. It

[scan 288]

In terms of the sole datum of α, we will therefore have an $\underline{a}$ uniquely determined by the [p. 445]
condition $\alpha\underline{a} \in \Gamma B_0'$, and we will have

$$(91) \qquad \underline{a} = c_0\rho + d_0 \qquad \begin{matrix} c_0, d_0 \in \Gamma(\mathcal{O}_S) \\ \text{determined by } \alpha \end{matrix}$$

likewise we will have

$$(92) \qquad \underline{b} = t_0\sigma + u_0 \qquad \begin{matrix} t_0, u_0 \in \Gamma(\mathcal{O}_S) \\ \text{determined by } \beta \end{matrix}$$

in such a way therefore that we will have

$$(93) \qquad \alpha\underline{a} = \beta\underline{b} \qquad (\overset{\text{dfn}}{=} \underline{u})$$

In the numerical case

$$\rho = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \qquad \sigma = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

setting

$$\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \beta = \begin{pmatrix} r & s \\ t & u \end{pmatrix}$$

we will have (cf. page 398′ (76)(78)) unless I am mistaken

$$(94) \qquad \underline{a} = \frac{1}{\delta D}(c\rho + d)$$

$$\left(= \frac{1}{c^2 + cd + d^2}(c\rho + d)\right)$$

where $\delta D = c^2 + cd + d^2$, — D being the corner coefficient of

$$\zeta = \alpha\rho(\alpha)^{-1} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

$$(95) \qquad \underline{b} = \frac{1}{\delta' U}(t\sigma + u)$$

$$= \frac{1}{t^2 + u^2}(t\sigma + u)$$

where $\delta' U = t^2 + u^2$, — U the corner coefficient of

$$\eta = \beta\sigma(\beta)^{-1} = \begin{pmatrix} R & S \\ T & U|u \end{pmatrix},$$

m974

Marginal note. m974 $\delta = \det\alpha = ad - bc$

$\delta' = \det \beta = ru - st$

m975

Marginal note. m975 We will have to come back to these formulas in the general case

the only trouble, in making the link with the "brutal" computations of loc. cit, is in the factor τ_∞^α on page 398′ (78), where $\alpha = \alpha_2(\mu) \in \mathbb{Z}/2.\mathbb{Z}$, in the case where $\alpha = 1$ — [p. 287]

[scan 289]

at first sight there seems to be an incompatibility between the (simpler) theory developed here, and this factor τ_∞^α — which I must think has very good reasons for being there ! And there it is — in loc. cit. we wrote the equation of the loops with the possibility of a sign

$$\zeta = \varepsilon\eta \qquad \underbrace{\ell_1^\nu}_{\varepsilon_1(\mu-1)\text{ with the current notation}} , \qquad \varepsilon \in \{\pm 1\},$$

m976

Marginal note. m976 here ζ and η are † defined as $\alpha\rho(\alpha)^{-1}$, $\beta\sigma(\beta)^{-1}$

this is thus a generalization relative to the equation set down at the beginning, where we assumed $\varepsilon = +1$. We will therefore have to later take up the preceding theory again, in the more general context of an equation

$$(95|96) \qquad \underset{\|}{\zeta}_{\alpha\rho(\alpha)^{-1}} = \varepsilon . \underset{\|}{\eta}_{\beta\sigma(\beta)^{-1}} \varepsilon_1(\mu-1) \qquad \varepsilon \text{ section of } \mathcal{O}_S$$

where we will have to have (taking the determinants of the two sides) †

$$\dagger \qquad \varepsilon^2 = 1$$

— but there is probably no reason to suppose that we have $\varepsilon \in \{\pm 1\}$. But I will first continue the calculations in the present case $\varepsilon = +1$. [p. 288]

Uncertainty retained from cancelled passages.

140-3-s289-D001 [illeg. 1]

140-3-s289-D002 [illeg. 1]

140-3-s289-D003 det

[scan 290]

IV A canonical pinning [p. 446]

Let us recall that giving (over a base scheme S) a form G of $\mathrm{Gl}(2)_S$, or SG of $\mathrm{Sl}(2)_S$, or PG of $\mathrm{GP}(1)_S$, or again an Azumaya|Azumaya algebra $\mathcal{M}$ (form of $M_2(\mathcal{O}_S)$), amounts to giving a bundle of projective lines $\mathcal{P}|P$ over S. $\mathcal{P}|P$ can be interpreted as the scheme of Borel subgroups B of G, or SB of SG, or PB of PG, or as the scheme of "one-parameter additive "Borelian" subgroups" (i.e. identical with the unipotent part of a suitable Borel) †L of G, or of SG, or of PG (N.B. The canonical morphisms $SG \longrightarrow G \longrightarrow PG$ induce bijections between the sets of Borel subgroups of the three groups in question, and also between the sets of "Borelian" 1-parameter subgroups of these groups). Likewise, $\mathcal{P}|P$ can be interpreted in terms of the sections N † of $\mathcal{M}$ which are nonzero on every fiber and such that ,

$$(1) \qquad \det N = 0, \quad \mathrm{Tr}\, N = 0 \qquad \text{hence } N^2 = 0$$

(the "strictly nilpotent" and regular (nonzero on each fiber) sections), to every such section will correspond|corresponds a 1-parameter Borelian subgroup of SG, as the image of $\mathbf{G}_a|\mathrm{G}_a$ under

$$(2) \qquad \lambda \longmapsto e_N(\lambda) = \exp(\lambda N) = 1 + \lambda N.$$

But it is obvious that if N and N' are two such [p. 289]

Uncertainty retained from cancelled passages.
140-3-s290-D001 [illeg. 1]
140-3-s290-D002 [illeg. 1]

[scan 291]

sections which are homothetic, i.e. $N' = aN$ with $a \in \Gamma(S, \mathcal{O}_S)$, (in fact, we will have $a \in \Gamma(S, \mathcal{O}_S^*)$) then they define the same 1-parameter subgroup, — so the latter in fact correspond to sections of the projective bundle (of relative dimension 3) $\check{P}(\mathcal{M})$, or if we take $\operatorname{Tr} N = 0$ into account, † to sections of the projective bundle (of relative dimension 2 — thus a bundle of projective planes) $\check{P}({}_1\mathcal{M}|1\mathcal{M})$ (where ${}_1\mathcal{M}|1\mathcal{M} \subset \mathcal{M}$ is defined precisely by the condition $\operatorname{Tr} N = 0$) — namely the sections of the † "conic" defined by the equation

$$\det N = 0.$$

The N's moreover also correspond to the operators

$$e = e_N = \dagger e_N(1) = 1 + N$$

satisfying the conditions

$$(3) \qquad \det e = 1, \quad \operatorname{Tr} e = 2$$

which is indeed equivalent, on setting $e = 1 + N$, i.e. on defining N by

$$N = e - 1$$

to the conditions (1) [for Tr and det of two|of the two operators N and e related by $e = \lambda + N$ mutually determine † by

$$\begin{cases} \operatorname{Tr} e = 2 \operatorname{Tr} \lambda + \operatorname{Tr} N, & \det e = \lambda^2 + \lambda \operatorname{Tr} N + \det N \\ \operatorname{Tr} N = -2 \operatorname{Tr} \lambda + \operatorname{Tr} e, & \det N = \lambda^2 - \lambda \operatorname{Tr} e + \det e \end{cases}.$$

[p. 290]

Uncertainty retained from cancelled passages.
140-3-s291-D002 [illeg. 1]
140-3-s291-D003 quadric
140-3-s291-D004 $1 + N$
140-3-s291-D005 one another

[scan 292]

The sections of G (in fact of SG) satisfying (3), and which are not $= 1$ on any fiber, are called the † regular unipotent sections — two such sections e, e' are considered "equivalent" if we have [p. 447]

Steinberg's terminology

$$(4) \qquad e' = e^{\lambda} \dagger \overset{\text{def|dfn}}{=} e(\lambda) \qquad (\lambda \in \Gamma\mathbf{G}_a|\mathrm{G}_a = \Gamma\mathcal{O}_S)$$

$$\overset{\text{def}}{=} (1 + \lambda N) \qquad \dagger$$

we will have necessarily^{t} $\lambda \in \Gamma\mathbf{G}_m|\mathrm{G}_m$

$$= 1 + \lambda(e - 1) = (1 - \lambda) + \lambda e.$$

We thus see that the regular (strictly) nilpotent sections of $\mathcal{M}$ correspond 1-1 to the regular unipotent sections, and the equivalence relation of homothety becomes that of equivalence by $e \longmapsto e^{\lambda}$ — and the equivalence classes of either correspond to the Borelian additive subgroups of G (or SG, or PG via $SG \longrightarrow PG$) — or again $\longrightarrow$ (taking the normalizers of these in G, SG or PG) to the Borels of these groups, i.e. to the sections of $\mathcal{P}|P$ over S.

Suppose such a section x_∞ of $\mathcal{P}|P$ is given, this thus allows us to consider $\mathcal{P}|P$ as obtained from an affine line bundle †

$$(5) \qquad E \quad (= \mathcal{P}|P \setminus \{x_\infty\}),$$

† [p. 291]

Uncertainty retained from cancelled passages.
140-3-s292-D002 strict|strictly
140-3-s292-D003 $= e$
140-3-s292-D004 always
140-3-s292-D006 [illeg. 3]
140-3-s292-D007 A vector bundle [illeg. 4] [illeg. 3] invertible

[scan 293]

and x_∞ is also given by a Borelian 1-parameter subgroup

$$L \subset SG \subset G \tag{6}$$

(which, under $SG \longrightarrow PG$, again gives a Borelian additive subgroup $L \hookrightarrow PG$ in PG ...). Locally, L will be generated by a regular unipotent section

$$e = 1 + N \qquad (N \text{ regular nilpotent}) \tag{7}$$

as the subgroup of the

$$e^\lambda = \underline{e}(\lambda)|e_\lambda(\lambda) = e_N(\lambda) = 1 + \lambda N, \tag{8}$$

but as we have said, e, N are not determined intrinsically in terms of $L \subset SG$. More precisely, L is canonically endowed with a vector bundle structure (preserved by its normalizer in G), † to|: wit precisely that obtained by transport of structure from that of G_a under

$$\underline{e} : G_a \xrightarrow{\sim} L \qquad \underline{e}(\lambda) = e^\lambda = 1 + \lambda N,$$

and L is thus identified with the vector bundle canonically|canonical associated with the line bundle $\mathcal{N} \subset {}_1M$ † defined by the N's

$$L \simeq \check{V}(\mathcal{N}), \tag{9}$$

[p. 292]

Uncertainty retained from cancelled passages.
140-3-s293-D001 [illeg. 1]
140-3-s293-D003 sub

[scan 294]

the choice of e, or of N, corresponds simply to that of a basis N of $\mathcal{N}$, or of a section e of L^* [p. 448]
† i.e. of a section e of L which is $\neq 1$ on every fiber.
m978

Marginal note. m978 Let us note that L is also canonically isomorphic to the vector bundle of translations of the affine line bundle $\mathbb{P}|P \setminus \{x|x_\infty\}$.

Let us return to equation (32), and note that ρ and σ enter into it only modulo transformations $\rho \mapsto \lambda\rho$, $\sigma \mapsto \lambda'\sigma$ $(\lambda, \lambda' \in \Gamma G_m)$ (which do not change the commutators $[\alpha, \rho]$ and $[\beta, \sigma]$); † We can express this by saying that ρ, σ enter only through their images in $\Gamma(PG)$, and moreover we have

$$\begin{aligned} PG \xrightarrow{\sim} \underline{\mathrm{Aut}}_{Sgr|S\text{-gr}}(G) \simeq \underline{\mathrm{Aut}}_{S\text{-gr}}(SG) \simeq \underline{\mathrm{Aut}}_{Sgr|Sg}(PG) \\ \simeq \underline{\mathrm{Aut}}_{S\text{-Alg}}(M) \simeq \underline{\mathrm{Aut}}_{S}(\mathbb{P}|P) \end{aligned} \tag{10|40}$$

and our remark amounts to saying that ρ and σ enter only through their action on G (or on SG), which is also clear from the formulas

$$[\alpha, \rho] = \alpha\rho(\alpha)^{-1}, \qquad [\beta, \sigma] = \beta\sigma(\beta)^{-1}.$$

On the other hand, the factor $\varepsilon_1(\mu - 1)$ of (32) can be regarded as a section of a † additive subgroup L_1 of G (in fact, of SG), which we suppose of [p. 293]

Uncertainty retained from cancelled passages.
140-3-s294-D001 everywhere $\neq$
140-3-s294-D002 the data
140-3-s294-D004 false bundle

[scan 295]

course Borelian — equation (32) is then written in the form (where μ no longer enters, only α and β)

$$[\beta,\sigma]^{-1}[\alpha,\rho]\in\Gamma L_1 \tag{11}$$

or

$$[\dagger\alpha,\rho]=[\beta,\sigma]\cdot\gamma \qquad \text{with } \gamma\in\Gamma L_1. \tag{12}$$

In the form (12), we can regard it as an equation relating α,β,γ.

Let us consider the image †

$$\widetilde{L}_1\subset PG\dagger \tag{13}$$

of L_1 under $G\longrightarrow PG$, the relation (40 bis) is then written in the form

$$\tilde{\varepsilon}_1\overset{\text{dfn}}{=}\tilde{\rho}\tilde{\sigma}^{-1}\in\Gamma\widetilde{L}_1 \qquad (\underset{\text{can}}{\overset{\sim}{\longleftarrow}}\Gamma L_1). \tag{14}$$

† This section therefore comes from a section well determined † in terms of $\tilde{\rho},\tilde{\sigma}$

$$\varepsilon_1\in\Gamma L_1, \tag{15}$$

described in terms of ρ,σ directly by the condition that we have a relation of the form

$$\varepsilon_1=-\lambda_0\rho\sigma^{-1} \qquad (\lambda_0\in\Gamma G_m) \qquad (16)=(40\text{ bis})$$

for convenience — (because of the application to the special case that interests us most …).
m979

Marginal note. m979 N.B. λ_0 is uniquely determined by ρ,σ if 2 invertible on S; more generally, if 2 regular †element of $\Gamma(S,\mathcal{O}_S)$. (But in characteristic 2 λ_0 is not determined except modulo μ_2 !!!). [p. 294]

Uncertainty retained from cancelled passages.
140-3-s295-D001 β
140-3-s295-D002 $\widetilde{L}_1|\widetilde{L}_0$
140-3-s295-D003 ℓ
140-3-s295-D004 [illeg. 1]
140-3-s295-D005 and
140-3-s295-D006 section

[scan 296]

Let us return to the general situation of a form G of $\mathrm{Gl}(2)_S|\mathrm{GL}(2)_S$, and suppose that it can [p. 449] be written in the form

$$G=\underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-Mod}}(F), \tag{17}$$

where F is an $\mathcal{O}_S$-Module locally free of rank 2,

This is the case as soon as we have a section x of $\mathbb{P}|P$ over S, i.e. a Borel additive subgroup L of G, or of SG, (in fact we can then take for F the extension of $\mathcal{O}_S$ by $\mathcal{N}$ which determines the affine-line bundle $\mathbb{P}|P\setminus\{x\}$). But the|the $\mathbb{P}|P$ is realized as †

$$\mathbb{P}|P=\mathbf{P}|P(F)\simeq\check{\mathbf{P}}|\check{P}(F) \tag{18}$$

the projective bundle associated with F, † i.e. the sections of $\mathbb{P}|P$ are identified with the "lines" $D \subset F$.

We will have

$$M = \underline{\mathrm{End}}_{\mathcal{O}_S\text{-mod}}(F)\dagger \simeq \check{F} \otimes_{\mathcal{O}_S} F \tag{19}$$

and the sub-Module $\mathcal{N}$ associated with the section x_D of $\mathbb{P}|P$ † is none other than †

$$\mathcal{N} = D^\circ \otimes D \qquad (D^\circ \simeq (F/D)^\vee \subset \check{F}, \text{ the orthogonal to } D \text{ in } \check{F}) \tag{20}$$

its sections are the $u : F \longrightarrow F$ such that

$$u(F) \subset D, \quad u(D) = \{0\}. \tag{21}$$

[p. 295]

Uncertainty retained from cancelled passages.

140-3-s296-D001 [illeg. 1]

140-3-s296-D002 Thus

140-3-s296-D003 $\simeq$

140-3-s296-D006 is

140-3-s296-D007 [illeg. 1]

[scan 297]

Proposition Let G be a form of $\mathrm{Gl}(2)_S|\mathrm{GL}(2)_S$, ρ, σ two sections of G such that $\tilde{\rho}\tilde{\sigma}^{-1} = \tilde{\varepsilon}_1$ is a [illeg. 1] unipotent-† section of PG, linked by the relation (16). Then the following conditions are equivalent

(i) ε_1 is † a regular section (i.e. † $\neq 1$ on each fiber), i.e. $\varepsilon_1 = 1 + N_1$, with N_1 a basis of $\mathcal{N}_1$) and $\tilde{\rho}^{-1}(x_1)$ is a section of $\mathbb{P}|P$ $\neq x_1$ in each fiber.

m980

Marginal note. m980 N.B. We have denoted by x_1 the section of $\mathbb{P}|P$ over S corresponding to L_1.

(ii) Ditto|Item, with $\tilde{\sigma}^{-1}$ in place of $\tilde{\rho}$.

m981

Marginal note. m981 (i bis) (ii bis) | Ditto|Item, with $\tilde{\sigma}, \tilde{\rho}$ in place of $\tilde{\sigma}^{-1}, \tilde{\rho}^{-1}$.

(iii) We have (42), i.e. $c = \lambda_0 \mathrm{Tr}\,\rho + \mathrm{Tr}\,\sigma$ is invertible.

Proof. We have, thanks to the fact that $\tilde{\rho}\tilde{\sigma}^{-1} \in \Gamma\widetilde{L}_1$, normalizes $\widetilde{L}_1$, or what amounts to the same, L_1, the relation

$$\tilde{\rho}^{-1}(L_1) = \tilde{\sigma}^{-1}(L_1)$$

or what amounts to the same, we have

$$\tilde{\rho}^{-1}(x_1) = \dagger\tilde{\sigma}^{-1}(x_1), \tag{22|24}$$

which proves the equivalence of (i) and (ii).

Let us show that $(i) \Longleftrightarrow (iii)$. We may assume that S is the spectrum of an algebraically closed field, if it suits us, and we may assume $\lambda_0 = 1$ (at the cost † of replacing ρ by $\lambda_0\rho$). If we had $\varepsilon_1 = 1$ i.e. $1 = -\rho\sigma^{-1}$ i.e. $\rho + \sigma = 0$, we would have † $\mathrm{Tr}(\rho + \sigma) = \mathrm{Tr}\,\rho + \mathrm{Tr}\,\sigma = 0$, so (iii) implies the regularity of ε_1 i.e. of N_1, † we may assume therefore that ε_1 is regular. We have to show that if we have such a regular unipotent † endomorphism $\varepsilon_1 = 1 + N_1$, of F, associated with a $D \subset F$, [p. 296]

Uncertainty retained from cancelled passages.
140-3-s297-D002 regular
140-3-s297-D003 [illeg. 2]
140-3-s297-D004 [illeg. 1]
140-3-s297-D006 $\tilde{\rho}^{-1}$
140-3-s297-D007 [illeg. 1]
140-3-s297-D009 Tr
140-3-s297-D010 [illeg. 1]
140-3-s297-D011 operator

[scan 298]

and an † endomorphism σ of F, whence $\rho = -\varepsilon_1\sigma$, then † $c = \operatorname{Tr}\sigma + \operatorname{Tr}\rho = \operatorname{Tr}\sigma - \operatorname{Tr}\varepsilon_1\sigma =$ [p. 450]
$-\operatorname{Tr} N_1\sigma$ is invertible iff $D + \sigma^{-1}.D = F$ i.e. $\sigma D + D = F$, † Let us take a basis e_0, e_1 of F such that e_1 generates D, and N_1 is written as the matrix $\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, let $\sigma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ we have $N_1\sigma = \begin{pmatrix} 0 & 0 \\ a & b \end{pmatrix}$, $\operatorname{Tr} N_1\sigma = b$, $\sigma e_1 = ae_0 + be_1$, and $D + \sigma D = D + b\mathcal{O}_S e_{2|0}$ is equal to $F = \mathcal{O}_S^2$ iff b invertible, q.e.d.

Let us suppose these equivalent conditions satisfied, † which are also expressed by saying that L_1 and

$$L_0 = \tilde{\rho}^{-1}(L_1) = \tilde{\sigma}^{-1}(L_1) \tag{23}$$

are distinct on each fiber, i.e. define Borels $B_1, B_0 \subset G$ such that

$$T = B_1 \cap B_0 \tag{24}$$

is a torus (maximal) of G, † The latter is also the † intersection of the stabilizers B_0, B_1 in G of the sections x_0, x_1 of $\mathcal{P}|P$ which correspond to L_0, L_1 (satisfying

$$x_1 = \tilde{\rho}x_0 = \tilde{\sigma}x_0 \tag{25}$$

), and is the inverse image of the subgroup|of the subgroups (privileged|privileged toric|toric [p. 297]

Uncertainty retained from cancelled passages.
140-3-s298-D001 operator [illeg. 1]
140-3-s298-D002 [illeg. 2]
140-3-s298-D003 iff
140-3-s298-D004 in P
140-3-s298-D005 which
140-3-s298-D006 [illeg. 1]

[scan 299]

called|whose $\tilde{T}$ in $PG = \underline{\mathrm{Aut}}_S(\mathcal{P}|P)$. We can also say that the datum of two distinct sections † x_0, x_1 of the projective bundle $\mathcal{P}|P$ † amounts to that of the Borel B_1 corresponding to the choice of x_1), and of the maximal torus T of B_1 (which is then written

$$B_1 = T.L_1 \qquad (\textit{produit } \tfrac{1}{2} \text{ direct}) \tag{26}$$

m982

Marginal note. m982 the pair (B_1, T) is called a Killing pair.

which allows us then to recover x_0 as one corresponding to the unique Borel B_0 such that $T = B_0 \cap B_1$. (The Borels B_0 such that $B_0 \neq B_1$ on each fiber, i.e. such that $B_0 \cap B_1$ is a torus, correspond one-to-one to the maximal tori T of B_1, by the preceding formula $T = B_0 \cap B_1$.)

This also amounts to: the datum, in addition to B_1, i.e. to x_1, of a section x_0, of the affine-line bundle $U_1 = \mathcal{P}|P \smallsetminus \{x_1\}$, † i.e. of an isomorphism from U_1 onto the vector bundle L_1 of its translations

$$U_1 \overset{\text{dfn}}{=} \mathcal{P}|P \smallsetminus \{x_1\} \overset{\sim}{\to} L_1. \tag{27}$$

To finish "pinning" the situation (i.e. [p. 298]

Uncertainty retained from cancelled passages.

140-3-s299-D001 points

140-3-s299-D002 makes it possible to

140-3-s299-D003 then makes it possible

[scan 300]

reducing $\mathcal{P}|P$ to the standard projective bundle $\mathbb{P}^1_S$) we still have to give ourselves a basis of [p. 451]
L_1 i.e. a section x_∞ of

$$L_1^* \simeq U_1 \smallsetminus \{x_0\} \simeq \mathcal{P}|P \smallsetminus \{x_1, x_0\}. \tag{28}$$

Now the section ε_1 of L_1 is precisely a section of L_1^* – so we have a complete pinning of the situation. In terms of $(F, D = D_1)$ corresponding to L_1,

† we will have

$$\begin{cases} \rho^{-1}(D_1) = \sigma^{-1}(D_1) \overset{\text{dfn}}{=} D_0 \\ F = D_0 \oplus D_1 \end{cases} \tag{29}$$

†

$$N_1|D_0 : \quad D_0 \overset{\sim}{\to} D_1 \tag{30}$$

and (at the cost of tensoring with $D_1^{\otimes -1}$) we may assume $D_1 \simeq \mathcal{O}_S$, whence

$$\begin{cases} F \simeq \mathcal{O}_S^2 = \mathcal{O}_S e_0 + \mathcal{O}_S e_1 \\ D_0 = \mathcal{O}_S e_0, \quad D_1 = \mathcal{O}_S e_1, \\ N_1 e_1 = 0, \quad N_1 e_0 = e_1, \end{cases} \tag{31}$$

[p. 299]

Uncertainty retained from cancelled passages.

140-3-s300-D001 we may assume we have [illeg. 1] x OPS 29 1

140-3-s300-D002 30

[scan 301]

i.e.

$$N_1 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad \varepsilon_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}. \tag{32}$$

On the other hand †

$$\rho D_0 = | \neq \sigma D_0 = D_1$$

whence

$$\rho\sigma^{-1}(D_1) = D_1$$

We choose ρ, σ sections in Gl(2)) over $\tilde{\rho}, \tilde{\sigma}$ (sections of $\mathrm{GP}(1) \simeq \mathrm{Gl}(2)/\mathrm{G}_m$ satisfying $\tilde{\rho}\tilde{\sigma}^{-1} = \tilde{\varepsilon}_1$, in such a way as to have

$$\boxed{\rho\sigma^{-1} = -\varepsilon_1} \tag{33}$$

which determines each of the two sections ρ, σ as a function of the other (=|itself determined modulo a scalar, in terms of $\tilde{\sigma}$ resp. of $\tilde{\rho}$). As

$$\rho D_0 = D_1,$$

we will have

$$\rho = \begin{pmatrix} 0 & u|U \\ v|V & w|W \end{pmatrix}, \qquad \det \rho = -uv|UV \text{ invertible} \tag{*}$$

and we can remove the scalar ambiguity|ambiguity in ρ by † normalizing by

$$\rho = \begin{pmatrix} 0 & 1 \\ v|V & w|W \end{pmatrix} \qquad v|V \in \Gamma \mathrm{G}_m. \tag{34}$$

[p. 300]

Uncertainty retained from cancelled passages.

140-3-s301-D001 we will have D

140-3-s301-D005 writing

[scan 302]

Let us note that (33), can be written $\rho = -(1 + N_1)\sigma$ i.e. † [p. 452]

$$\rho + \sigma = -N_1\sigma \tag{35}$$

multiplying by N_1 on the left we get

$$N_1\rho + N_1\sigma = 0$$

(since $N_1^2 = 0$), moreover

$$N_1\rho = \rho N_0, \qquad N_1\sigma = \sigma N_0$$

(for $N_1 = \rho N_0 \rho^{-1} = \sigma N_0 \sigma^{-1}$), so (35) can be written

$$\rho + \sigma = -N_1\sigma = +N_1\rho = -\sigma N_0 = +\rho N_0 \overset{\text{definition}}{=} N. \tag{35|36}$$

moreover when ρ is in the general form $\rho = \begin{pmatrix} 0 & u \\ v & w \end{pmatrix}$ we find

$$N\dagger = \begin{pmatrix} 0 & 0 \\ 0 & u \end{pmatrix}, \qquad N^2 = \begin{pmatrix} 0 & 0 \\ 0 & u^2 \end{pmatrix}$$

so the normalization (34) $u = 1$ also means

$$\rho + \sigma = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \dagger, \tag{37}$$

or again (since u is invertible)

$$N^2 = N \qquad (\text{where } N = \rho + \sigma) \tag{37 bis}$$

or again that σ is given in terms of ρ by

$$\sigma = \begin{pmatrix} 0 & -1 \\ -v & 1 - w \end{pmatrix}. \tag{38}$$

moreover, $N_0 = \rho N_1 \rho^{-1}$ will be of the form $\begin{pmatrix} 0 & \alpha \\ 0 & 0 \end{pmatrix}$, an immediate calculation shows that α is given by the formula $u = \alpha v$, i.e. (if we normalize by $u = 1$)

m983

Marginal note. m983 N.B. We see at once that the conditions considered so far imply that $1, \rho, \sigma$ are linearly independent in each fiber, i.e. $T_\rho \cap T_\sigma = \mathbb{G}_m$! i.e. $\widetilde{T}_\rho \cap \widetilde{T}_\sigma = \{1\}$.

Uncertainty retained from cancelled passages.
140-3-s302-D001 [illeg. 1]
140-3-s302-D002 [*illeg.* 1]
140-3-s302-D003 *by*

[scan 303]

$$N_0 = \begin{pmatrix} 0 & 1/v \\ 0 & 0 \end{pmatrix}, \qquad \varepsilon_0 = \begin{pmatrix} 1 & 1/v \\ 0 & 1 \end{pmatrix}. \tag{39}$$

m984

Marginal note. m984 N.B. We will have

$$\boxed{\sigma^{-1}\rho = -\varepsilon_0}\,\dagger \tag{39 bis}$$

For $\sigma^{-1}\rho = \sigma^{-1}(\rho\sigma^{-1})\sigma = \sigma^{-1}(-\varepsilon_1)\sigma = -\sigma^{-1}(\varepsilon_1) = -\varepsilon_0$.

These canonical normalizations having been made, let us return to the proposition on page 436 ; since ε_1 has been chosen (in as canonical a way as can be) there is no longer any drawback to †leaving equation (32) (put into the apparently more intrinsic form (11) or (12)) in the † initial form, where a priori we were thinking of introducing the parameter γ with

$$[\alpha, \rho] = [\beta, \rho]\varepsilon_1(\gamma)\,; \quad -$$

but it appears that it is $\gamma + 1 = \mu$ that has a particularly remarkable significance in the theory — so we keep|will keep its form (32), with the factor $\varepsilon_1(\mu - 1)$.

The expression (97) for $P(\zeta)$, thanks to (37), can be written

$$P(\zeta) = D \qquad \text{for } \zeta = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \tag{40}$$

Uncertainty retained from cancelled passages.
140-3-s303-D001 N_0
140-3-s303-D003 [illeg. 1]
140-3-s303-D004 [illeg. 1]

[scan 304]
[p. 453]

Thus the proposition tells us that the scheme $\mathcal{X}_{\rho,\sigma}$ of simultaneous solutions in (ζ, η, μ) of

$$\det\zeta = \det\eta = 1, \qquad \zeta = \eta\varepsilon_1(\mu - 1), \qquad \operatorname{Tr}(\zeta\rho) = \operatorname{Tr}\rho, \qquad \operatorname{Tr}\eta\dagger\sigma = \operatorname{Tr}\sigma|\alpha \tag{41}$$

is isomorphic, via $(\zeta, \eta, \mu) \longrightarrow \zeta$, to that of the $\zeta = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ satisfying the conditions

$$\underbrace{\det\zeta}_{AD-BC} = 1, \qquad \underbrace{\operatorname{Tr}\zeta\rho}_{(C+wD)+vB} = \underbrace{\operatorname{Tr}\rho}_{w}, \qquad \text{and } D \text{ an invertible section of } \mathcal{O}_S \tag{42}$$

and we recover μ in terms of ζ by

$$\mu = 1/D \tag{43}$$

(here $c = \operatorname{Tr}(\rho + \sigma) = \operatorname{Tr}\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = 1$), whence η by $\eta = \zeta\varepsilon_1(1 - \mu)$.

The subgroup scheme B_0' of page 442′ is none other than the scheme of matrices

$$B_0' = \left\{ \begin{pmatrix} \mu & \lambda \\ 0 & 1 \end{pmatrix} \,\middle|\, \mu \in \Gamma\mathbb{G}_m,\ \lambda \in \Gamma\mathbb{G}_a \right\} \tag{44}$$

We are going to make explicit the isomorphism (79)

$$\underline{u} \longmapsto ([\underline{u}, \rho], [\underline{u}, \sigma], \det \underline{u}) \tag{45}$$

in terms of the coefficients μ, λ of B_0', with the help of the parameters v, w in $\rho = \begin{pmatrix} 0 & 1 \\ v & w \end{pmatrix}$.

Uncertainty retained from cancelled passages.

140-3-s304-D001 $\zeta|\rho$

[scan 305]

We find †

$$\begin{aligned}
[\underline{u}, \rho] &= \begin{pmatrix} \mu & \lambda \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ v & w \end{pmatrix} \frac{1}{\mu} \begin{pmatrix} 1 & -\lambda \\ 0 & \mu \end{pmatrix} \begin{pmatrix} -\frac{1}{v} \end{pmatrix} \begin{pmatrix} w & -1 \\ -v & 0 \end{pmatrix} \\
&= -\frac{1}{\mu v} \begin{pmatrix} \lambda v & \mu + \lambda w \\ v & w \end{pmatrix} \begin{pmatrix} w + \lambda v & -1 \\ -\mu v & 0 \end{pmatrix} \\
&= -\frac{1}{\mu v} \begin{pmatrix} \dagger v^2\lambda^2 - v\mu^2 - vw\lambda\mu + vw\lambda & -\lambda v \\ v^2\lambda - vw\mu + vw & -v \end{pmatrix} \\
&= \frac{1}{\mu} \begin{pmatrix} -v\lambda^2 + w\lambda\mu + \mu^2 - w\lambda & \lambda \\ -v\lambda + w\mu - w & 1 \end{pmatrix}
\end{aligned}$$

m985

Marginal note. m985 now|and for $\sigma = \begin{pmatrix} 0 & -1 \\ v' & w' \end{pmatrix}$ with $v' = -v$, $w' = 1 - w$

$$\begin{aligned}
[\underline{u}, \sigma] &= \begin{pmatrix} \mu & \lambda \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ v' & w' \end{pmatrix} \frac{1}{\mu} \begin{pmatrix} 1 & -\lambda \\ 0 & \mu \end{pmatrix} \begin{pmatrix} \frac{1}{v'} \end{pmatrix} \begin{pmatrix} w' & 1 \\ -v' & 0 \end{pmatrix} \\
&= \frac{1}{\mu v'} \begin{pmatrix} \lambda v' & -\mu + \lambda w' \\ v' & w' \end{pmatrix} \begin{pmatrix} w' + \lambda v' & 1 \\ -\mu w' & 0 \end{pmatrix} \\
&= \frac{1}{\mu v'} \begin{pmatrix} v'^2\lambda^2 + v'\mu^2 - v'w'\lambda\mu + v'w'\lambda & \lambda v' \\ v'^2\lambda - v'w'\mu + v'w' & v' \end{pmatrix} \\
&= \frac{1}{\mu} \begin{pmatrix} v'\lambda^2 - \dagger w'\lambda\mu + \mu^2 + w'\lambda & \lambda \\ v'\lambda - w'\mu + w' & 1 \end{pmatrix} \\
&= \frac{1}{\mu} \begin{pmatrix} -v\lambda^2 + (w-1)\lambda\mu + \mu^2 - (w-1)\lambda & \lambda \\ -v\lambda + (w-1)\mu - (w-1)\dagger & 1 \end{pmatrix}
\end{aligned}$$

In summary

$$\begin{cases}
\zeta = [\underline{u}, \rho] = \dfrac{1}{\mu} \begin{pmatrix} \mu^2 + (-v\lambda + w(\mu - 1))\lambda & \lambda \\ -\dagger v\lambda + w(\mu - 1) & 1 \end{pmatrix} \\
\eta = [\underline{u}, \sigma] = \dfrac{1}{\mu} \begin{pmatrix} \mu^2 + (-v\lambda + (w-1)(\mu-1))\lambda & \lambda \\ -v\lambda + (w-1)(\mu-1) & 1 \end{pmatrix} \quad \text{and, of course} \\
\mu = \det \underline{u} \ \dagger
\end{cases} \tag{46}$$

so we have, if $\zeta = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$

$$B = \lambda/\mu, \quad D = 1/\mu \qquad \text{i.e.} \tag{47}$$

$$\mu = 1/D, \quad \lambda = B/D, \qquad \text{i.e.|of which } \underline{u} = \begin{pmatrix} 1/D & B/D \\ 0 & 1 \end{pmatrix} \tag{48}$$

Uncertainty retained from cancelled passages.

140-3-s305-D001 $\underline{u}(\rho) =$

140-3-s305-D002 [*illeg.* 2] $-\mu)vw + \lambda$

140-3-s305-D003 μ

140-3-s305-D004 λ

140-3-s305-D005 (

140-3-s305-D006 $= \mu$

[scan 306]
[p. 453bis]

Likewise, if $\eta = \begin{pmatrix} R & S \\ T & U \end{pmatrix}$, we will have

$$\begin{cases} S = B = \lambda/\mu, & U = D = 1/\mu \qquad \text{whence} \\ \dagger\ \mu = 1/U, & \lambda = S/U \end{cases} \tag{47 bis}$$

so

$$\underline{u} = \begin{pmatrix} 1/U\dagger & S/U \\ 0 & 1 \end{pmatrix}. \tag{48 bis}$$

Let us now consider the equation

$$[\alpha, \rho] = [\beta, \sigma]\varepsilon_1(\mu - 1), \qquad \text{with} \tag{49}$$

$$\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \beta = \begin{pmatrix} r & s \\ t & u \end{pmatrix}, \tag{50}$$

and set, for more elegant calculations

$$\rho = \begin{pmatrix} 0 & v_1 \\ v & w \end{pmatrix}, \quad \sigma = \begin{pmatrix} 0 & v_1' \\ v' & w' \end{pmatrix} \tag{51}$$

(we have written v_1, v_1' instead of u, u', so as not to interfere with the notation u for one of the coefficients of β), with

$$v' = -v, \quad v_1' = -v_1|v', \quad w + w' = v_1, \qquad \text{let us set} \tag{52}$$

$$\delta_\alpha = \det\alpha = ad - bc, \quad \delta_\beta = \det\beta = ru - st, \tag{53}$$

m986

Marginal note. m986 We do not set $v_1 = 1$, so as not to have to redo the same calculation for $[\beta, \sigma]$ as for $[\alpha, \rho]$!

we then find

$$\begin{aligned} \zeta = [\alpha, \rho] &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 & v_1 \\ v & w \end{pmatrix} \frac{1}{\delta_\alpha} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \dagger \frac{1}{-vv_1} \begin{pmatrix} w & -v_1 \\ -v & 0 \end{pmatrix} \\ &= -\frac{1}{\delta_\alpha vv_1} \begin{pmatrix} bv & av_1 + bw \\ dv & cv_1 + dw \end{pmatrix} \begin{pmatrix} dw + bv & -dv_1 \\ -cw - av & cv_1 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \end{aligned}$$

with

$$\begin{cases} (-\delta_\alpha vv_1)A = \ldots & \text{(a bore).}, \quad (-\delta_\alpha vv_1)B = -vv_1bd + v_1^2ac + v_1wbc \\ (-\delta_\alpha vv_1)C = \ldots & \text{(a bore)}, \quad (-\delta_\alpha vv_1)D = -vv_1d^2 + v_1wcd + v_1^2c^2 \end{cases} \tag{54}$$

m987

Marginal note. m987 N.B. the formulas for B, D simplify by v_1

Uncertainty retained from cancelled passages.
140-3-s306-D003 *and*
140-3-s306-D004 $(=\mu)$
140-3-s306-D005 (

[scan 307]

and we likewise find

$$\eta = (\beta,\sigma)|[\beta,\sigma] = \begin{pmatrix} R & S \\ T & U \end{pmatrix}$$

with

$$\begin{array}{ll} (-vv_1\delta_\beta)R = \ldots & (-vv_1\delta_\beta)S = (v_1v)su + v_1^2tr - v_1w'st \\ (-vv_1\delta_\beta)T = \ldots & (-vv_1\delta_\beta)U = -(v_1v)u^2 - v_1w'tu + v_1^2t^2. \end{array} \tag{55}$$

N.B. we can divide by v_1.

We deduce from this in particular †

$$\begin{cases} D = \dfrac{-v\dagger d^2 + \dagger wcd + v_1c^2}{-v\dagger(ad-bc)} \\ \mu = 1/D = \dfrac{-v\dagger(ad-bc)}{-v\dagger d^2 + \dagger wcd + v_1c^2} \\ \lambda = \dagger B/D = \dfrac{-v\dagger bd + v_1ac + \dagger wbc}{-v\dagger d^2 + \dagger wcd + v_1c^2}. \end{cases} \tag{56}$$

which thus makes it possible to make explicit

$$\underline{u} = \begin{pmatrix} 1/D = \mu & B/D = \lambda \\ 0 & 1 \end{pmatrix}$$

in terms of a, b, c, d (and the parameters u_1, v, w which determine ρ).

For $\underline{v_1 = 1}$, the formula for D in (56), can be written

$$D = \det(d\rho + c)/ - v\delta_\alpha = \det(d\rho + c)/\det\rho\det\alpha \tag{57}$$

†

$$D(\alpha) \text{ invertible} \quad \Longleftrightarrow \quad d\rho + c \text{ invertible.} \tag{58}$$

but I realize that this is a flat form,

Uncertainty retained from cancelled passages.
140-3-s307-D001 [illeg. 1]
140-3-s307-D002 v_1
140-3-s307-D003 v_1
140-3-s307-D005 v_1
140-3-s307-D006 v_1
140-3-s307-D007 v_1
140-3-s307-D008 v_1
140-3-s307-D011 C/D
140-3-s307-D012 v_1
140-3-s307-D014 v_1
140-3-s307-D015 v_1
140-3-s307-D016 v_1
140-3-s307-D018 to [illeg. 1]

[scan 308]
[p. 454]

V Study(of the equation) in $GP(1)$, and the condition $\operatorname{Tr}\sigma = 0$.

Up to now we have studied equation (32)

$$[\alpha, \rho] = [\beta, \sigma]\, \varepsilon_1(\mu - 1) \tag{*}$$

in $G\dagger$ ($\simeq \mathrm{Gl}(2)$), or what amounts to the same, as a relation in $\mathrm{Sl}(2)$. Here α, β are †variable sections of $G\dagger \simeq \mathrm{Gl}(2)$, but $[\alpha, \rho]$ and $[\beta, \sigma]$ depend only on the corresponding sections $\tilde\alpha, \tilde\beta$ of

$$PG\ (\simeq GP(1)) \simeq G/\underbrace{G_m}_{\text{Center of } G} \simeq \mathrm{Aut}_{1|l}(G)$$

(likewise they depend on ρ, σ only via $\tilde\rho, \tilde\sigma$). We can therefore consider that it is an equation in $SG \simeq \mathrm{Sl}(2)$, relating $\tilde\alpha, \tilde\beta, \mu$.

We can propose to study the corresponding equation in $PG \simeq GP(1)$,

$$[\tilde\alpha, \tilde\rho] = [\tilde\beta, \tilde\sigma] \cdot \tilde\varepsilon_1(\mu - 1) \tag{1}$$

which can also be written in $\mathrm{Gl}(2)$, in the equivalent form

$$[\alpha, \rho] = \varepsilon[\beta, \sigma]\, \varepsilon_1^{\dagger}(\mu - 1), \tag{2}$$

where ε is a section of G_m, of course uniquely determined|determined by α, β, μ, (and even by $\tilde\alpha, \tilde\beta, \mu$)). We will necessarily have

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Uncertainty retained from cancelled passages.

140-3-s308-D001 ℓ_2

140-3-s308-D003 [illeg. 1]

140-3-s308-D004 ℓ_2

140-3-s308-D006 N_0

[scan 309]

$$\varepsilon^2 = 1 \tag{3}$$

(take the determinant of the two sides). We regard (2) as an equation relating the variables $\alpha, \beta, \mu, \varepsilon$ (α, β in $\mathrm{Gl}(2)$, μ, ε in $\mathbb{E}^1|\mathbb{E}'$ – and in fact necessarily in $\mathbb{G}_m|G_m$, and ε already in $\mu_2 = {}_2\mathbb{G}_m|G_m$), or also $\tilde\alpha, \tilde\beta, \mu, \varepsilon$. Let us put, by analogy with (II),

$$\zeta = [\alpha, \rho], \qquad \eta = \dagger\varepsilon[\underline{\underline{\beta}}, \sigma]\dagger, \tag{4}$$

so that (2) takes the form

$$\zeta = \eta\varepsilon_1(\mu - 1) \tag{5}$$

as in (II). To be able to apply (II) to this situation, we will have to assume that $\eta = \varepsilon[\beta, \sigma]$ satisfies the same preliminary conditions as

$$\eta_0 = [\beta, \sigma]$$

itself, namely $\det \eta = 1$ ($\iff \varepsilon^2 = 1$) and †

$$\underbrace{\mathrm{Tr}\,\eta}_{\varepsilon \mathrm{Tr}\,\eta_0 = \varepsilon \mathrm{Tr}\,\sigma} = \mathrm{Tr}\,\sigma \qquad \dagger$$

i.e.

$$\varepsilon \mathrm{Tr}\,\sigma = \mathrm{Tr}\,\sigma.$$

Uncertainty retained from cancelled passages.

140-3-s309-D002 [*illeg.* 1]
140-3-s309-D003 $\varepsilon_1 t$|and
140-3-s309-D004 that
140-3-s309-D005 *for*

[scan 310]

Lemma. Let σ be a section of G (a form of Gl(2), associated|associated with an Azumaya [p. 455] algebra $\mathcal{M}|M$) – i.e. a section of $\mathcal{M}|M$ such that $\det\sigma$ invertible. Let A_σ be the scheme of sections η † of G satisfying the conditions

$$\det\eta = 1 \quad (\text{i.e. } \eta \text{ in } SG), \qquad \operatorname{Tr}\eta\circ\sigma = \operatorname{Tr}\sigma.$$

Equivalent conditions a) b) c) :

a) A_σ is stable under translations by $\mu_2 \subset \mathbb{G}_m|G_m$:

$$\eta \longmapsto \varepsilon\eta \qquad (\varepsilon^2 = 1)$$

b) $\operatorname{Tr}\sigma = 0$

†

d)|c) (Under the conditions of the preceding number IV (page 452), with the normalizations (34) (38) for ρ, σ) We have

$$\sigma = \begin{pmatrix} 0 & -1 \\ v' & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ -v & 0 \end{pmatrix} \tag{6}$$

i.e. (with the notation of loc. cit.) we have $w = 1$,

$$\text{i.e.} \qquad \rho = \begin{pmatrix} 0 & 1 \\ v & 1 \end{pmatrix}. \tag{7}$$

d) We have $\sigma^2 \in \Gamma\mathbb{G}_m|G_m$, i.e. $\tilde{\sigma}^2 = 1$

Finally, these conditions imply the following d), and are equivalent to it|them if σ † is not a homothety on any fiber (thus i.e. σ is regular in the sense of Steinberg).

The equivalence $a) \iff b)$ means also|thus that $\mathbb{E}^1|\mathbb{E}'^{\mu_2} = (0)$, which is well known and immediate to verify. The equivalence of b) and d) is immediate, remains to prove $b) \iff c)$. † Now we have the equation of|of Hamilt-Cayley|Hamilton-Cayley

$$\sigma^2 - (\operatorname{Tr}\sigma)\sigma + \det\sigma = 0$$

Uncertainty retained from cancelled passages.

140-3-s310-D002 [illeg. 1]
140-3-s310-D003 $\mathbb{G}_m|G_m$
140-3-s310-D004 when σ
140-3-s310-D006 Let us assume|Assuming satisfied

[scan 311]

which shows that $\operatorname{Tr}\sigma = 0 \Longrightarrow \sigma = \det\sigma\cdot 1$, and conversely if σ^2 is a scalar, we will have that $(\operatorname{Tr}\sigma)\sigma = \sigma^2 - \det\sigma\cdot 1$ is one, which is possible (if σ is † a scalar on no fiber) only if $\operatorname{Tr}\sigma = 0$.

Let us assume these conditions satisfied for what follows, † Let

$$\mathcal{G}^*_{\rho,\sigma} \subset G \times G \times \mathbb{E}^1|\mathbb{E}' \times \mathbb{E}^1|\mathbb{E}' \tag{8}$$

be the scheme of solutions in $(\alpha, \beta, \mu, \varepsilon)$ of equation (2), which is also the inverse image in $G \times G \times \mathbb{E}^1|\mathbb{E}' \times \mathbb{E}^1|\mathbb{E}'$ of a subscheme †

$$\widetilde{\mathcal{G}}^*_{\rho,\sigma} \subset PG \times PG \times \mathbb{E}^1|\mathbb{E}' \dagger \tag{9}$$

consisting of the solutions (α, β, μ) of (1). The projection onto the second|last factor $\mathbb{E}^1$ gives a morphism

$$\mathcal{G}^*_{\rho,\sigma} \overset{\underline{\varepsilon}}{\to} \mu_2, \tag{10}$$

Uncertainty retained from cancelled passages.

140-3-s311-D001 not

140-3-s311-D002 find [illeg. 1]

140-3-s311-D003 σ [illeg. 1]

140-3-s311-D004 $\mathbb{E}^1|\mathbb{E}'$

which moreover factors through $\widetilde{\mathcal{G}}^*_{\rho,\sigma}$: [scan 312] [p. 456]

$$\tilde{\varepsilon} : \widetilde{\mathcal{G}}^*_{\rho,\sigma} \longrightarrow \mu_2. \tag{11}$$

Let us note that we have

$$\dagger \qquad \mathcal{G}_{\rho,\sigma} \subset \mathcal{G}^*_{\rho,\sigma}$$

more precisely

$$\mathcal{G}_{\rho,\sigma} = \underline{\varepsilon}^{-1}(1) \tag{12}$$

which leads us to introduce also

$$\underset{\substack{\| \text{ definition} \\ \underline{\tilde{\varepsilon}}^{-1}(1)}}{\widetilde{\mathcal{G}}_{\rho,\sigma}} = \widetilde{\mathcal{G}}^*_{\rho,\sigma}, \tag{13}$$

so that $\mathcal{G}_{\rho,\sigma}$ is the inverse image in $G \times G \times \mathbb{E}^1|\mathbb{E}'$ of $\widetilde{\mathcal{G}}_{\rho,\sigma}$.

We have a morphism

$$\begin{cases} (\alpha, \beta, \mu, \varepsilon) \longrightarrow \left(\underbrace{(\zeta = \alpha\rho(\alpha)^{-1},\ \eta = \varepsilon\beta\sigma(\beta)^{-1},\ \mu)}_{\in \Gamma \mathcal{X}_{\rho,\sigma}},\ \underset{\in \Gamma\mu_2}{\underset{\smile}{\varepsilon}} \right) \\ \text{from} \quad \mathcal{G}^*_{\rho,\sigma} \longrightarrow \mathcal{X}_{\rho,\sigma} \times \mu_2 \end{cases} \tag{14}$$

and the inverse image of a section $((\zeta, \eta, \mu), \varepsilon)$ of the right-hand side is the scheme of solutions in α, β of the equations

$$\underset{\alpha}{{}^{(a)}\alpha\rho(\alpha)^{-1}} = \zeta, \qquad {}^{(b)}\beta\sigma(\beta)^{-1} = \varepsilon\eta \tag{15}$$

Uncertainty retained from cancelled passages.

140-3-s312-D001 (11)

Now|We we have, via the isomorphism [scan 313]

$$B'_0 \overset{\sim}{\to} \mathcal{X}_{\rho,\sigma}$$

a privileged solution $\alpha = \underline{u}$ of the first equation (15), which is characterized as the unique solution which is † section of B'_0 (uniqueness resulting from

$$T_\alpha \cap B'_0 = \{1\}$$

). But we will have, for this u

$$\underbrace{u\sigma(u)^{-1}}_{[u,\sigma]} = \eta \qquad \text{(and not } \varepsilon\eta \text{ !)}$$

to find in terms of u a solution of the equation (15 b) $\beta\sigma(\beta)^{-1} = \varepsilon\eta$, we are led to try in the form

$$\beta = u\tau \quad \text{whence} \quad \beta\dagger\sigma(\beta)^{-1} = \dagger u\tau\sigma(\tau)^{-1}\sigma(\dagger u)^{-1}, \tag{16}$$

trying to determine τ so that we have

$$\underbrace{\tau\sigma(\tau)^{-1}}_{[\tau,\sigma]} = \varepsilon, \tag{17}$$

which indeed implies

$$\beta\sigma(\beta)^{-1} = \dagger u\varepsilon\sigma(\dagger u^{-1}) = \varepsilon\dagger u\sigma(\dagger u)^{-1} = \varepsilon\eta$$

i.e. (15 b). †

Uncertainty retained from cancelled passages.

140-3-s313-D001 a
140-3-s313-D002 ρ
140-3-s313-D003 β
140-3-s313-D004 β
140-3-s313-D005 β
140-3-s313-D006 β
140-3-s313-D007 β
140-3-s313-D008 β
140-3-s313-D009 Lemma

The relation (17) can also be written [scan 314] [p. 457]

(take the inverse)

$$\tau\sigma\tau^{-1}\sigma^{-1} = \varepsilon \quad \text{or} \quad \sigma\tau\sigma^{-1}\tau^{-1} = \varepsilon \quad \text{or} \tag{18}$$

$$\sigma = \varepsilon\tau(\sigma) \quad \text{or} \quad \tau(\sigma) = \varepsilon\sigma \tag{19}$$

which implies that τ normalizes T_σ

$$\tau(T_\sigma) = T_\sigma \quad \text{i.e.} \quad \tau \in \Gamma N_\sigma \tag{20}$$

where

$$N_\sigma \overset{\text{definition}}{=} \mathrm{Norm}_G(T_\sigma). \tag{21}$$

Corollary to the lemma (page 455) the equivalent conditions of the lemma † are implied by the following ; and imply it if σ is regular|regular :

(e) $\forall$ fppf local section ε of μ_2, exists local^{ly}|locally section † τ of G such that we have (17) (or again (19))

(f) (If 2 invertible on S) idem, with $\varepsilon = -1$.

If we have (e), then we see that for every † ε, we must have $\mathrm{Tr}\,\varepsilon\sigma = \mathrm{Tr}\,\sigma$ now $\mathrm{Tr}\,\varepsilon\sigma = \varepsilon\,\mathrm{Tr}\,\sigma$, hence (varying ε) $\mathrm{Tr}\,\sigma \in \Gamma(\mathbb{E}^1|\mathbb{E}'^{\mu_2} = 0)$ hence $\mathrm{Tr}\,\sigma = 0$. If we assume it only for $\varepsilon = -1$, we will have $\mathrm{Tr}\,\sigma = -\mathrm{Tr}\,\sigma$ i.e. $2\,\mathrm{Tr}\,\sigma = 0$, i.e. $\mathrm{Tr}\,\sigma = 0$ if 2 invertible. Conversely, let us assume $\mathrm{Tr}\,\sigma = 0$, then $\mathrm{Tr}\,\varepsilon\sigma = 0$, moreover $\det(\varepsilon\dagger) = \varepsilon^2 = 1$, hence ε satisfies the "numerical conditions" to be of the form $[\tau,\sigma]$ for suitable τ. For

Uncertainty retained from cancelled passages.
140-3-s314-D002 are equivalent
140-3-s314-D004 [illeg. 1]
140-3-s314-D005 [illeg. 1]
140-3-s314-D006 σ

ensure the local existence of τ, it remains to make sure that $\varepsilon\sigma$ is a regular section in the sense of Steinberg i.e. is nowhere a scalar, which follows from the fact that this is so for σ. [scan 315]

Thus, under the present conditions, let $N'_\sigma \subset N_\sigma$ be the subscheme of G (or of $N_\sigma|N$) defined by

$$N'_\sigma = \{\tau \mid [\tau, \sigma] \in \Gamma\mu_2\} \tag{22}$$

then N'_σ is stable under right translation by T_σ — better, it is a subgroup scheme of N_σ (containing T_σ), for if τ, τ' are such that

$$\tau(\sigma) = \varepsilon\sigma, \qquad \tau'\sigma = \varepsilon'\sigma,$$

we will have

$$\tau'\tau(\sigma) = \varepsilon'\varepsilon \cdot \sigma,$$

at the same time we see that

$$\tau \longmapsto \varepsilon$$

is a group homomorphism $N'_\sigma \longrightarrow \mu_2$, and we will have an exact sequence of group schemes

$$1 \longrightarrow T_\sigma \hookrightarrow N'_\sigma \xrightarrow{\varepsilon_\sigma} \mu_2 \longrightarrow 1. \tag{23}$$

We will note that if $\tau \in \Gamma G$ normalizes T_σ, we will have

$$\tau(\sigma) = a\sigma + b \qquad a, b \in \Gamma\mathcal{O}_S,$$

and taking the traces of the two sides we find (taking into account that $\operatorname{Tr}\tau(\sigma) = \operatorname{Tr}\sigma = 0$) [scan 316]
$\operatorname{Tr} b.1 = 0$ i.e. $2b = 0$, so if b is invertible $b = 0$, hence $\tau(\sigma) = a\sigma$ and we necessarily have $a^2 = 1$ [p. 458|488]
i.e. a is an ε. Thus

$$N'_\sigma = N_\sigma \quad \text{in residual characteristic } \neq 2. \tag{24}$$

† More generally, over a base S such that 2 be † a regular section (i.e. not a zero divisor) in $\Gamma(S, \mathcal{O}_S)$, we see that every section of $N'_\sigma|N_\sigma$ is in fact a section of N_σ. But this is no longer so in characteristic 2, for

$$\sigma' = a\sigma + b \qquad (a \in \Gamma G_m,\ b \in \Gamma\mathcal{O}_S)$$

as soon as it satisfies

$$\operatorname{Tr}\sigma' = \sigma|0, \quad \det\sigma' = \det\sigma$$

is conjugate local[ly] to σ — and † the conditions are written (since $\det(a\sigma + b) = a^2 \det\sigma + ab\underbrace{\dagger\operatorname{Tr}\sigma}_{0} + b^2$

$$\begin{cases} -va^2 + b^2 = -v \\ 2b = 0 \qquad \text{(automatic in characteristic 2)} \end{cases}$$

which in no way imply $b = 0$ ....

† Let us now consider the product group

$$B'_0 \times A_\rho \times N'_\sigma,$$

Uncertainty retained from cancelled passages.
140-3-s316-D001 But in characteristic
140-3-s316-D003 a zero divisor
140-3-s316-D004 it suffices for
140-3-s316-D005 det
140-3-s316-D006 [illeg. 1]

[scan 317]

and the morphism of schemes

$$\begin{cases} B'_0 \times A_\rho \times N'_\sigma \longrightarrow G \times G \times \mathbb{E}^1|\mathbb{E}' \times \mathbb{E}^1|\mathbb{E}' \\ \qquad (\underline{u}, \underline{a}, \underline{b}) \longmapsto (\alpha = \underline{u}\underline{a}^{-1}, \beta = \underline{u}\underline{b}^{-1}, \mu = \det \underline{u}, \varepsilon = \varepsilon_\sigma(\underline{b})), \end{cases} \tag{25}$$

I say that the image lies in $\mathcal{G}^*_{\rho,\sigma}$, i.e. that we will indeed have

$$\alpha\rho(\alpha)^{-1} = \varepsilon(\beta\sigma(\beta)^{-1})\varepsilon_1(\mu - 1). \tag{$*$}$$

Indeed

$$\begin{aligned} \alpha\rho(\alpha)^{-1} &= \underline{u}\rho(\underline{u})^{-1} \\ \beta\rho(\beta)^{-1} &= \underline{u}\underbrace{\underline{b}^{-1}\sigma(\underline{b}^{-1})^{-1}}_{\varepsilon_\sigma(\underline{b})^{-1} = \varepsilon_\sigma(\underline{b})}\sigma(\underline{u})^{-1} = \varepsilon\underline{u}\sigma(\underline{u})^{-1} \end{aligned}$$

and we know (by II) that

$$\underline{u}\rho(\underline{u})^{-1} = \underline{u}\sigma(\underline{u})^{-1}\varepsilon_1(\mu - 1)$$

hence $(*)$, since $\varepsilon^2 = 1$. But moreover the morphism (25) is a monomorphism, for in terms of $(\alpha, \beta, \mu, \varepsilon)$, u is determined (in terms of α already), (as the unique section of B'_0 which is of the form $\alpha\underline{a}^{-1}$, with $\underline{a} \in \Gamma T_\rho$ (for $T_\rho \cap B'_0 = \{1\}$)), and $\underline{b}$ is then determined by the relation $\beta = \underline{u}\underline{b}^{-1}$. We thus find that (25) is an isomorphism

$$\begin{cases} B'_0 \times A_\rho \times N'_\sigma \xrightarrow{\sim} \mathcal{G}^*_{\rho,\sigma} \\ \qquad (\underline{u}, \underline{a}, \underline{b}) \longmapsto (\alpha = \underline{u}\underline{a}^{-1}, \beta = \underline{u}\underline{b}^{-1}, \mu = \det \underline{u}, \varepsilon = \varepsilon_\sigma(\underline{b})) \end{cases} \tag{26}$$

[scan 318]
[p. 459]

If we have two sections $(\underline{u}, \underline{a}, \underline{b})$ and $(\underline{u}', \underline{a}', \underline{b}')$ of $B'_0 \times T_\rho \times N'_\sigma$, under what conditions do we have

$$\tilde{\alpha} = \tilde{\alpha}', \quad \tilde{\beta} = \tilde{\beta}' \ ? \tag{27}$$

(27)

It clearly suffices to have

$$\underline{a}' = \lambda_1\underline{a}, \qquad \underline{b}' = \lambda_2\underline{b} \qquad \lambda_1, \lambda_2 \in \Gamma G_{\mathrm{m}} \tag{28}$$

and I say that these conditions are also necessary. This comes from the fact that $B'_0 \hookrightarrow PG \simeq GP(1)$ is a monomorphism (let $\widetilde{B}'_0$ be the image of B'_0), that † $A_\rho \to PG$ and $N'_\sigma \longrightarrow PG$ have kernel G_{m} (let|let $\tilde{T}_\rho$ and $\widetilde{N}'_\sigma$ be their images), and that we have in PG

$$\tilde{T}_\rho \cap \widetilde{B}'_0 = \{1\}, \tag{29}$$

so that † $\underline{\tilde{u}} \in \Gamma\widetilde{B}'_0$ and $\underline{\tilde{a}} \in \Gamma\widetilde{A}_\rho$ are defined uniquely in terms of $\tilde{\alpha}$ by the relation

$$\dagger \qquad \tilde{\alpha} = \underline{\tilde{u}}\underline{\tilde{a}}^{-1}$$

– and that then $\underline{\tilde{b}}$ is defined in terms of $\tilde{\alpha}, \tilde{\beta}$ by the relation

$$\tilde{\beta} = \underline{\tilde{u}}\underline{\tilde{b}}^{-1}.$$

This shows that † we have an isomorphism

$$\begin{cases} \widetilde{B}'_0 \times \widetilde{A}_\rho \times \widetilde{N}'_\sigma \xrightarrow{\sim} \widetilde{\mathcal{G}}^*_{\rho,\sigma} \subset PG \times PG \times \mathbb{E}^1|\mathbb{E}' \\ (\underline{\tilde{u}}, \underline{\tilde{a}}, \underline{\tilde{b}}) \longmapsto (\underline{\tilde{u}}\underline{\tilde{a}}^{-1}, \underline{\tilde{u}}\underline{\tilde{b}}^{-1}, \delta(\underline{\tilde{u}})) \end{cases} \tag{30}$$

Uncertainty retained from cancelled passages.
140-3-s318-D001 the
140-3-s318-D002 [illeg. 1]
140-3-s318-D003 [*illeg.* 1]
140-3-s318-D004 $\mathcal{G}$

[scan 319]

where

$$\delta(\underline{\tilde{u}}) = \det(\underline{u}) \tag{31}$$

when $\underline{\tilde{u}}$ comes from $\underline{u} \in \Gamma B'_0$ – or, what amounts to the same (cf. page 441) $\delta(\underline{\tilde{u}})$ is defined by

$$\underline{\tilde{u}}(N_0) = \delta(\underline{\tilde{u}}) N_0. \tag{32}$$

m990

Marginal note. m990 N.B. The centralizer of $\tilde{\sigma}$ in $\widetilde{G} = PG$ is not $\widetilde{T}_\sigma$, but $\widetilde{N}'_\sigma$ †.

Remark. Let us note that a section $(\tilde{\alpha}, \tilde{\beta}, \mu)$ † of $\widetilde{\mathcal{G}}^*_{\rho,\sigma}$ remains † determined, modulo the action of $\widetilde{N}'_\sigma$ on β, to be precise, in terms of $\tilde{\alpha}$ – condition|conditions to be satisfied by $\tilde{\alpha}$, so that there exist $\tilde{\beta}, \mu$ † which complete it to a section of $\widetilde{\mathcal{G}}^*_{\rho,\sigma}$, being the following: the coefficient D of $\zeta = \alpha\rho(\alpha)^{-1} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ is invertible (which makes it possible to determine μ by $\mu = 1/D$), then we determine $\tilde{\eta}$ † by $\tilde{\eta} = \tilde{\zeta}\tilde{\varepsilon}_1(\mu - 1)$, and finally we find the $\tilde{\beta}$ that correspond by the condition

$$\tilde{\beta}\tilde{\sigma}(\tilde{\beta})^{-1} = \tilde{\eta}.$$

m991

Marginal note. m991 N.B. $\tilde{\alpha}$ depends on three parameters, while $\widetilde{\mathcal{G}}^*_{\rho,\sigma}$ is of|admits relative dimension $4 = 2+1+1$ (for $\widetilde{\mathcal{G}}^*_{\rho,\sigma} \simeq \widetilde{B}'_0 \times \widetilde{T}_\rho \times \widetilde{N}'_\sigma$).

m992

Marginal note. m992 N.B. We have lifted locally^t $\tilde{\alpha}$ to $\alpha \in \Gamma G$, ζ does not depend on the choice of lifting.

Likewise, a section $(\alpha, \beta, \mu, \varepsilon)$ of $\mathcal{G}^*_{\rho,\sigma}$ is determined by † † α alone modulo † N'_σ – we first find $\underline{u}, \underline{a}$ in terms

Uncertainty retained from cancelled passages.
140-3-s319-D001 [illeg. 2]
140-3-s319-D002 $\in \Gamma$
140-3-s319-D003 entirely
140-3-s319-D004 [illeg. 1]
140-3-s319-D005 [illeg. 1]
140-3-s319-D006 α alone
140-3-s319-D007 [illeg. 2]
140-3-s319-D008 [illeg. 2]

[scan 320]
[p. 460]

of α as above, hence $\mu = \det \underline{u}$, hence and it † remains to determine β as $\beta = \underline{ub}^{-1}$, with $\underline{b} \in \Gamma N'_\sigma$ arbitrary, ε being defined then in terms of the choice of $\underline{b}$ as being $\varepsilon_\sigma(\underline{b})$. The condition for fixed $\alpha_0|\alpha$ to be completable by $(\beta, \mu, \varepsilon)$ to $(\alpha, \beta, \mu, \varepsilon) \in \Gamma \mathcal{G}^{\alpha|*}_{\rho,\sigma}$, is always the condition that the coefficient D †

$$\zeta = \alpha\rho(\alpha)^{-1} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \tag{33}$$

be invertible, which makes it possible to will determine|determine μ by

$$\mu = 1/D. \tag{34}$$

Let us put

$$\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \tag{35}$$

and make $\alpha\rho(\alpha)^{-1}$ explicit; with $\rho = \begin{pmatrix} 0 & 1 \\ v & 1 \end{pmatrix}$ $\delta = \det \alpha = ad - bc$

$$\begin{aligned} \alpha\rho(\alpha)^{-1} &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 & 1 \\ v & \dagger 1 \end{pmatrix} \frac{1}{\delta} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \begin{pmatrix} -\frac{1}{v} \end{pmatrix} \begin{pmatrix} \dagger 1 & -1 \\ -v & 0 \end{pmatrix} \\ &= -\frac{1}{\delta v} \begin{pmatrix} bv & a + b\dagger \\ dv & c + d\dagger \end{pmatrix} \begin{pmatrix} d\dagger + bv & -d \\ -c\dagger - av & c \end{pmatrix}, \end{aligned}$$

hence

$$\dagger$$

$$\begin{cases} (-\delta v)A = vb(d + vb|bv) - (a + b)(c + va) & (-\delta v)B = -vbd + c(a + b) \\ (-\delta v)C = vd(d + vb) - (c + d)(c + va) & (-\delta v)D = -vd^2 + cd + c^2, \end{cases} \tag{36}$$

the most essential quantity thus being

$$\begin{cases} (-\delta v)D = -vd^2 + cd + c^2 = \det(d\rho + c), \\ \dagger \text{ where} \\ \qquad \begin{cases} \dagger\ \delta = \det \alpha = ad - bc, \\ \quad v \text{ the parameter|coefficient which defines} \\ \quad \rho = \begin{pmatrix} 0 & 1 \\ v & 1 \end{pmatrix} \end{cases} \end{cases} \tag{37}$$

Uncertainty retained from cancelled passages.

140-3-s320-D002 $\alpha|\eta$
140-3-s320-D003 $+$
140-3-s320-D004 w
140-3-s320-D005 w
140-3-s320-D006 w
140-3-s320-D007 w
140-3-s320-D008 w
140-3-s320-D009 w
140-3-s320-D010 $= \frac{1}{\delta v}$
140-3-s320-D011 and after that|and after calculation
140-3-s320-D012 α

and we have [scan 321]

$$(38) \qquad (-\delta v|\nu)B = ac + bc - v|\nu bd$$

which gives $\underline{u}$ again in terms of a, b, c, d, thanks to the formula $\overline{\text{IV}}(48)$ (page 453′)

$$(39) \qquad \underline{u} = \begin{pmatrix} 1 & B/D\dagger \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \dfrac{-v|\nu(ad-bc)}{-v|\nu d^2+cd+d^2} & \dfrac{-ac+bc-v|\nu bd}{-v|\nu d|a^2+cd+d^2} \\ 0 & 1 \end{pmatrix}.$$

Remark. On the other hand, the datum of β alone (without the "sign" ε) does not allow one to recover μ. But the datum of (β, ε) does, for from it one deduces the knowledge of

$$(40) \qquad \eta = \varepsilon\beta\sigma(\beta)^{-1} \overset{\text{dfn}}{=} \begin{pmatrix} R & S \\ T & U \end{pmatrix},$$

which will be related to $\zeta = \alpha\rho(\alpha)^{-1}$ by

$$\zeta = \eta\varepsilon_1(\mu - 1)$$

and by the theory of the|of (II) we will have

$$(41) \qquad \mu = 1/U$$

(but in terms of $\beta\sigma(\beta)^{-1} \overset{\text{dfn}}{=} \begin{pmatrix} R_0 & S_0 \\ T_0 & U_0 \end{pmatrix}$, we will have †)

$$(42) \qquad \mu = \frac{\varepsilon}{U_0} \qquad)$$

i.e. it is a priori not known from U_0 alone, but only via U_0 and ε. †

To make this point precise, we are going to look at when and how, for fixed (β, ε), one can complete with α, μ so as to have a [p. 320]

Uncertainty retained from cancelled passages.

140-3-s321-D001 $\underline{u} =$

140-3-s321-D002 [illeg. 1]

140-3-s321-D003 R

section $(\alpha, \beta, \mu, \varepsilon)$ of $\mathcal{G}^*_{\rho,\sigma}$. A necessary condition is that U_0 be invertible (it depends only on β, not on ε) — and then one can choose ε arbitrarily in $\Gamma\mu_2$, $\varepsilon \in \Gamma\mu_2$ $\eta = \varepsilon\beta\sigma(\beta)^{-1}$ will then be a section of the subscheme A_σ † of G of solutions of $\det\eta = 1$, $\operatorname{Tr}\eta\sigma = \operatorname{Tr}\sigma$ (cf. lemma page 455), hence by the theory of (II), for † $\mu = 1/U = \varepsilon/U_0$, † and defining [scan 322] [p. 461]

$$\zeta = \eta\varepsilon_1(\mu - 1),$$

ζ will indeed be expressible locally in the form $\alpha\rho(\alpha)^{-1}$, the indeterminacy being moreover in T_ρ via $\alpha \longmapsto \alpha' = \alpha\underline{b}$, $\underline{b} \in \Gamma T_\rho$.

This does show that μ can thereof|not take, for a fixed β, all the values ε/U_0, where $\beta\sigma(\beta)^{-1} = \begin{pmatrix} R_0 & S_0|C_0 \\ T_0 & U_0 \end{pmatrix}$ ... [p. 321]

Uncertainty retained from cancelled passages.

140-3-s322-D002 [illeg. 1]

140-3-s322-D003 completing taking

140-3-s322-D004 [illeg. 2]

[scan 323]

[p. 462]

(VI) Characterization of the|of the modular triple $(\varepsilon_1, \rho, \sigma)$.

1°) Let G, SG, PG be associated forms of $\mathrm{Gl}(2), \mathrm{Sl}(2), \mathrm{GP}(1)$ over a base S — thus corresponding: a torsor under $\mathrm{GP}(1)_S$, or again: a bundle of projective lines P over S, or an Azumaya Algebra M of rank 1|4, and let us consider two sections [illeg. 1] $\tilde\rho, \tilde\sigma$ of PG, such that

$$(1) \qquad a) \quad \tilde\varepsilon_1 = \tilde\rho\tilde\sigma^{-1}$$

be † a regular unipotent section, and that the section

$$(2) \qquad \tilde\varepsilon_0 \overset{\text{dfn}}{=} \tilde\rho^{-1}(\tilde\varepsilon_1) = \tilde\sigma^{-1}(\tilde\varepsilon_1)$$

which is deduced from it be † in each fiber inequivalent to $\tilde\varepsilon_0$ (i.e. that the sections x_0, x_1 of P which correspond to $\varepsilon_0, \varepsilon_1$ be † distinct in each fiber).

m994

Marginal note. m994 N.B. If $\varepsilon_0, \varepsilon_1$ are the canonical liftings of $\tilde\varepsilon_0, \tilde\varepsilon_1$ into $G|P$, regular, with $\widetilde{\sigma}(\varepsilon_0) = \tilde\rho(\varepsilon_0) = \varepsilon_1|\widetilde{\sigma}(\tilde\varepsilon_0) = \tilde\rho(\tilde\varepsilon_0) = \tilde\varepsilon_1$ one can lift $\tilde\rho, \widetilde{\sigma}$ in a unique way to ρ, σ so as to satisfy $-\sigma^{-1}\rho = \varepsilon_0 \quad (\Longleftrightarrow -\rho\sigma^{-1} = \varepsilon_1)$, $(\rho+\sigma)^2 = (\rho+\sigma)$.

Then there exists a unique* pinning

$$(3) \qquad \dagger \quad \mathrm{Gl}(2)_S \xrightarrow[\sim]{\varphi} G$$

and sections $v|\nu, w|\omega, v|\nu', w|\omega'$ of $\mathcal{O}_S$, satisfying

$$(4) \qquad v|\nu + v|\nu' = 0, \qquad w|\omega + w|\omega' = 1, \qquad v|\nu \in \Gamma\mathcal{O}_S^*$$

such that $\tilde\rho, \tilde\sigma$ are the images, under the composite $\mathrm{Gl}(2)_S \xrightarrow[\sim]{\varphi} G \longrightarrow PG$, of the sections

$$(5) \qquad \rho = \begin{pmatrix} 0 & 1 \\ v|\nu & w|\omega \end{pmatrix}, \qquad \sigma = \begin{pmatrix} 0 & -1 \\ v|\nu' & w|\omega' \end{pmatrix}$$

m995

Marginal note. m995 N.B.* Uniqueness of $v|\nu, w|\omega$ (i.e. of ρ_0, σ_0) is not true without some restrictions on S, it suffices that every section τ of $\mathcal{O}_S$ such that $2\tau = 0,\ \tau^2 = 0$ be zero. [p. 322]

Uncertainty retained from cancelled passages.

140-3-s323-D001 [illeg. 1]

140-3-s323-D002 everywhere

140-3-s323-D003 everywhere

140-3-s323-D004 $G_0 =$

of $\mathrm{Gl}(2)_S$ — and conversely, for every pinning φ and every system $\dagger v|\nu, w|\omega, v|\nu', w|\omega'$ satisfying (4) — i.e. [scan 324]

$$(4\text{bis}) \qquad v|\nu' = -v|\nu, \quad w|\omega' = 1 - w|\omega, \quad \begin{matrix} v|\nu \in \Gamma(S, \mathcal{O}_S^*) \\ w|\omega \in \Gamma(S, \mathcal{O}_S) \end{matrix},$$

the images $\tilde\rho, \tilde\sigma$ under the composite $\mathrm{Gl}(2)_S \longrightarrow PG \simeq G/\mathbf{G}_m$ of the sections ρ, σ of $\mathrm{Gl}(2)_S$ given by (5) satisfy the conditions above ($\tilde\rho\tilde\sigma^{-1} \overset{\text{dfn}}{=} \tilde\varepsilon_1$ is regular unipotent, and everywhere inequivalent to $\tilde\sigma^{-1}(\tilde\varepsilon_1) = \tilde\rho^{-1}(\tilde\varepsilon_1) \overset{\text{dfn}}{=} \tilde\varepsilon_0$).

Setting then

$$(6) \qquad \varepsilon_{1|0} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad \varepsilon_0 = \begin{pmatrix} 1 & 1/v|\nu \\ 0 & 1 \end{pmatrix}$$

we will have

$$(7) \qquad \rho\sigma^{-1} = -\varepsilon_1, \qquad \sigma^{-1}\rho = -\varepsilon_0$$

$$(7\text{bis}) \qquad \varepsilon_0 = \rho^{-1}(\varepsilon_1) = \sigma^{-1}(\varepsilon_1), \qquad \varepsilon_1 = \rho(\varepsilon_0) = \sigma(\varepsilon_0)$$

whence the images under $\mathrm{Gl}(2)_S \longrightarrow PG$ of $\varepsilon_1, \varepsilon_0$ are $\tilde{\varepsilon}_1, \tilde{\varepsilon}_0$. We have the formulas

$$(8) \qquad \mathrm{Tr}\,\sigma = w|\omega', \quad \mathrm{Tr}\,\rho = w|\omega, \quad \mathrm{Tr}(\sigma+\rho) = 1$$

$$(9) \qquad \sigma + \rho = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \overset{\text{dfn}}{=} N\dagger$$

$$(10) \qquad \mathrm{Tr}\,\zeta N = D \quad \text{if} \quad \zeta = \begin{pmatrix} A & B \\ C & \underline{D} \end{pmatrix},$$

$$(11) \qquad \det\sigma = \det\rho = -v|\nu = -v'|v'.$$

[p. 323]

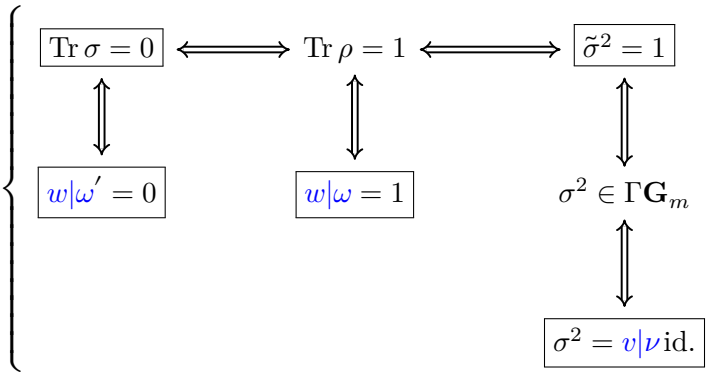
Uncertainty retained from cancelled passages.

140-3-s324-D001 [illeg. 4]

140-3-s324-D004 0

2°) The following conditions are equivalent [scan 325] [p. 463]

(12)



They are also equivalent to this: denoting by N'_σ the subgroup scheme of $\mathrm{Gl}(2)$ defined by the condition

$$(13) \qquad \tau(\tilde{\sigma}) = \tilde{\sigma} \quad \text{i.e.} \quad \tau(\sigma) = \varepsilon\sigma, \quad \varepsilon \in \Gamma\mathbf{G}_m$$

(we will necessarily have $\varepsilon^2 = 1$ i.e. $\varepsilon \in \Gamma\mu_2$) the morphism

$$(14) \qquad \begin{array}{ccc} N'_\sigma & \longrightarrow & \mu_2 \\ \tau & \longmapsto & \varepsilon \end{array}$$

is an epimorphism — so that we will have an exact sequence

$$(15) \qquad 1 \longrightarrow T_\sigma \longrightarrow N'_\sigma \longrightarrow \mu_2 \longrightarrow 1,$$

where †

$$(16) \qquad T_\sigma = \mathrm{Centr}_{\mathrm{Gl}(2)_S}(\sigma) = \{t\sigma + u \mid \det(t\sigma + u) \text{ invertible}\}$$

$$(17) \qquad \text{We have} \quad N'_\sigma \subset N_\sigma \overset{\text{dfn}}{=} \mathrm{Norm}_{\mathrm{Gl}(2)}(T_\sigma).$$

If 2 is a regular section of $\mathcal{O}_S$, then every section of N_σ over S is a section of N'_σ — thus if|in the residue characteristics are|is $\neq 2$ i.e. 2 invertible, we will have

$$(18) \qquad N'_\sigma = N_\sigma \qquad (\text{residue characteristics } \neq 2).$$

The condition under consideration means that σ, ρ take the form [p. 324]

Uncertainty retained from cancelled passages.
140-3-s325-D002 Let us set

[scan 326]

$$(19)\qquad \sigma = \begin{pmatrix} 0 & -1 \\ v|\nu' & 0 \\ \| & \\ -v|\nu & \end{pmatrix}, \qquad \rho = \begin{pmatrix} 0 & 1 \\ v|\nu & 1 \end{pmatrix}$$

with $v|\nu + v|\nu' = 0$.

These results of 1°), 2°) are only a recapitulation of part of the results of the preceding sections IV, V. †, we have just studied conditions which amount to the determination the value of|the values of $w|\omega$ (or again of $w|\omega'$) in a explicit way. We are going to turn to conditions which are to|will serve to fix $v|\nu$ (or again $v|\nu'$) in an explicit way.

3°) Equivalent conditions

$$(20)\qquad \begin{cases} \det\sigma = 1 \iff \det\rho = 1 \iff v|\nu = -1 \iff v|\nu' = 1 \\ \qquad\qquad \iff \varepsilon_0 = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}. \end{cases}$$

These equivalences follow at once from (6) and (11).|! The conditions also mean that ρ and σ have the form

$$(21)\qquad \sigma = \begin{pmatrix} 0 & -1 \\ 1 & w|\omega' \end{pmatrix}, \qquad \rho = \begin{pmatrix} 0 & 1 \\ -1 & w|\omega \end{pmatrix}$$
$$(w|\omega + w|\omega' = 1)$$

4°) Conjunction of 2°) and 3°).

Equivalent conditions: [p. 325]

Uncertainty retained from cancelled passages.
140-3-s326-D001 Here

[scan 327]
[p. 464]

$$(22)\qquad \begin{cases} \text{Conjunction of the conditions of 2° and of the conditions of 3°} & \iff & \boxed{\sigma^2 = -1} \text{ i.e. } \sigma^2 + 1 = 0 & \iff & \boxed{\tilde\sigma^2 = \tilde\rho^3 = 1} \\ \Updownarrow & & & & \\ v|\nu = -1, w|\omega = 1 & \iff & \rho = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} & \iff & \operatorname{Tr}\rho = \det\rho = 1 \\ \Updownarrow & & & & \\ v|\nu' = 1, w|\omega' = 0 & \iff & \sigma = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} & \iff & (\operatorname{Tr}\sigma = 0, \det\sigma = 1) \end{cases}$$

m996

Marginal note. m996

$\boxed{\tilde\sigma^2 = \tilde\rho^3 = 1} \Longleftrightarrow$

$\boxed{\rho^2 - \rho + 1 = [illeg.\ 1]}$

cf. corollary 2 below

Proof. The fact that the conjunction of the conditions of 2°) and 3°) is equivalent to each of the conditions of the 2^e and 3^e rows, is trivial. The equivalence with $\sigma^2 = -1$ comes from the fact that † the condition 2° means that σ^2 be|is scalar, i.e. $\sigma^2 = v|\nu\,\mathrm{id}$, † whereas 3° means $v|\nu = -1$, so $\sigma^2 = -1$ is indeed equivalent to the conjunction.

m997

Marginal note. m997 for another proof of the equivalence of $2°) + 3°)$ with $\tilde\sigma^2 = \tilde\rho^3 = 1$, cf. lemma below

Let us note that 2°) is also written $\tilde\sigma^2 = 1$, and is equivalent to: $w|\omega' = 0$, i.e. $w|\omega = 1$, and we then find, writing †

$$\rho = \begin{pmatrix} 0 & 1 \\ v|\nu & 1 \end{pmatrix},$$

$$\rho^3 = \begin{pmatrix} v|\nu & v|\nu + 1 \\ v|\nu(v|\nu + 1) & 2v|\nu + 1 \end{pmatrix} \tag{23}$$

thus ρ^3 can be a scalar i.e. $\tilde\rho^3 = 1$ only if the two diagonal terms are equal, now their difference is $(2v|\nu + 1) - v|\nu = v|\nu + 1$, [p. 326]

Uncertainty retained from cancelled passages.

140-3-s327-D002 [illeg. 1]

140-3-s327-D003 [illeg. 2]

140-3-s327-D004 $\rho =$

[scan 328]

so the condition reads $v|\nu = -1$, i.e. this is condition 3°, but then the expression (23) for ρ^3 shows us that $\rho^3 = -1$.

Remarks|Remark The preceding conditions imply the formulas

$$\varepsilon_0 = \sigma\rho \tag{24}$$

$$\varepsilon_1 = \rho\sigma \tag{25}$$

$$\rho\varepsilon_1 = \varepsilon_0^{-1} \tag{26}$$

$$\rho^2\sigma = \varepsilon_0^{-1} \tag{27}$$

each of which except the last (27) implies them in turn — for example we have $\varepsilon_0 = -\sigma^{-1}\rho$ (7), so (24) is equivalent to $\sigma = -\sigma^{-1}$ i.e. $\sigma^2 = -1$, and (25) is equivalent to (24) by applying $\mathrm{int}(\rho)$ to both sides.

We have moreover

$$\rho\varepsilon_1 = \begin{pmatrix} 0 & 1 \\ v|\nu & w|\omega \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ v|\nu + w|\omega & w|\omega'|w|\omega \end{pmatrix}, \qquad \varepsilon_0^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

which shows that $\rho\varepsilon_1 = \varepsilon_0^{-1}$ is equivalent to:

$$w|\omega = 1, \quad v|\nu + w|\omega = 0 \qquad \text{i.e.} \quad w|\omega = 1, v|\nu = -1 \quad \text{i.e.} \quad \rho = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix},$$

OK. Composing (25) (in the form $\rho\varepsilon_1 = \rho^2\sigma$) and (26) we find (27) as a consequence of the conditions $2°) + 3°)$; †Let us try to prove converse implication †by a brute-force calculation

$$\rho^2\sigma = \begin{pmatrix} -v|\nu w|\omega & w|\omega - (v|\nu + w|\omega^2) \\ -v|\nu(v|\nu + w|\omega^2) & -w|\omega(w|\omega^2 + 2v|\nu) + v|\nu - w|\omega^2 \\ & |\,-w|\omega(w|\omega^2 + 2v|\nu) + v|\nu + w|\omega^2 \end{pmatrix},$$

$$\varepsilon_0^{-1} = \begin{pmatrix} 1 & -1/v|\nu \\ 0 & 1 \end{pmatrix}$$

† we can have the equality $\rho^2\sigma = -\varepsilon_0^{-1}|\varepsilon_0^{-1}$ only if $-v|\nu(v|\nu + w|\omega^2) = 0$, this implies|which implies ($v|\nu$ invertible!) $v|\nu + w|\omega^2 = 0$, †i.e. $v|\nu = -w|\omega^2$, [p. 327]

Uncertainty retained from cancelled passages.

140-3-s328-D001 the

140-3-s328-D002 follows

140-3-s328-D005 $v|\nu$

140-3-s328-D006 whence

[scan 329]

[p. 465]

$$\rho^2\sigma = \begin{pmatrix} w|\omega^3 & w|\omega \\ 0 & \dagger w|\omega^3 \end{pmatrix},$$

This shows that † m998

Marginal note. m998 † $v|\nu = -w|\omega^2$ † i.e. $v|\nu = -1, w|\omega = 1$

(27bis) $\quad \rho^2\sigma = \varepsilon_0^{-1} \iff \{v|\nu + w|\omega^2 = 0,\ w|\omega^3 = 1$

$(\Rightarrow v|\nu = -w|\omega^2 = -1/w|\omega$ i.e. $w|\omega = -1/v|\nu,$ OK$)$

But (27), on multiplying by ρ, is also equivalent to

$$\dagger\rho^3\sigma = \dagger\rho\varepsilon_0^{-1}$$

now we know that $\varepsilon_0 = -\sigma^{-1}\rho$ (7) i.e. $\varepsilon_0^{-1} = -\dagger\rho^{-1}\sigma|\sigma_0$ i.e. $\rho\varepsilon_0^{-1} = -\sigma$, so the preceding relation also reads $\rho^3\sigma = -\sigma$ or again

$$(28) \qquad \boxed{\rho^3 = -1}.$$

We can see † the equivalence of (28) with the conditions (27bis) by computing ρ^3 in terms of $v|\nu, w|\omega$, we find

$$(29) \qquad \rho^3 = \begin{pmatrix} v|\nu w|\omega & v|\nu + w|\omega^2 \\ v|\nu(v|\nu + w|\omega^2) & w|\omega(2v|\nu + w|\omega^2) \end{pmatrix}$$

And we can have $\tilde{\rho}^3 = 1$ i.e. ρ^3 a scalar only if †

$$v|\nu + w|\omega^2 = 0 \qquad \text{i.e.} \quad v|\nu = -w|\omega^2$$

which implies

$$\rho^3 = \begin{pmatrix} -w|\omega^3 & 0 \\ 0 & -w|\omega^3 \end{pmatrix}$$

i.e. we find

Lemma We have $\tilde{\rho}^3 = 1$ iff $v|\nu + w|\omega^2 = 0$ i.e. $v|\nu = -w|\omega^2$ i.e. we have

$$(30) \qquad \rho = \begin{pmatrix} 0 & 1 \\ -w|\omega^2 & w|\omega \end{pmatrix},$$

and then we have

$$(31) \qquad \rho^3 = -w|\omega^3 \cdot 1\ ,$$

so we have

$$(32) \qquad \rho^3 = -1 \iff \dagger \quad \begin{matrix} w|\omega^3 = 1 \text{ i.e. } w|\omega \in \Gamma\mu_3 \\ \underline{\text{and}} \quad v|\nu = -w|\omega^2\ (= -1/w|\omega) \end{matrix}$$

[p. 328]

Uncertainty retained from cancelled passages.

140-3-s329-D001 $-w(1+2w)$
140-3-s329-D002 iff $w|\omega$
140-3-s329-D003 We have in addition to [illeg. 4]
140-3-s329-D004 [illeg. 2]
140-3-s329-D005 (27) [*illeg.* 1]
140-3-s329-D006 u
140-3-s329-D007 σ
140-3-s329-D008 this
140-3-s329-D010 $v|\nu$ $w|\omega$
140-3-s329-D011 $v|\nu = -w|\omega^2$

[scan 330]

Corollary 1 The equivalent conditions (22) are also equivalent to

$$(33) \qquad \rho^3 = -1 \quad \text{and} \quad \det\rho = 1$$

or again to

$$(34) \qquad \rho^3 = -1 \quad \text{and} \quad \dagger\mathrm{Tr}\rho = 1$$

in other words, considering the three relations

$$(35) \qquad \det\rho = 1, \quad \mathrm{Tr}\,\rho = 1, \quad \rho^3 = -1$$

the conjunction of two of these imply|implies the third, and is equivalent to the equivalent conditions of (22). false

† [p. 329]

Uncertainty retained from cancelled passages.

140-3-s330-D001 det
140-3-s330-D002 36|30 $w|\omega$ $v|\nu$ $w|\omega$ $v|\nu$ $=|\neq$ det

[scan 331]

Another way of finding certain equivalences is via the Hamilton–Cayley equations for σ and ρ [p. 466]

$$(36) \qquad \dagger\sigma^2 - \underbrace{(\mathrm{Tr}\,\sigma)}_{w|\omega'=1-w|\omega}\sigma + \underbrace{\det\sigma}_{v|\nu'=-v|\nu} = 0$$

$$(37) \qquad \rho^2 - \underbrace{(\mathrm{Tr}\,\rho)}_{w|\omega}\rho + \underbrace{\det\rho}_{-v|\nu} = 0.$$

Equation (36) gives again that the conjunction of $\mathrm{Tr}\,\sigma = 0$, $\det\sigma = 1$ implies $\sigma^2 + 1 = 0$ i.e. $\sigma^2 = -1$, and conversely, taking into account that 1 and σ are sections linear[ly] independent in|of $\mathrm{M}_2(\underline{\mathcal{O}}_S|\mathcal{O}_S)$. Likewise|= (37) shows

$$(38) \qquad (\mathrm{Tr}\,\rho = 1 \text{ and } \det\rho = 1) \iff (\rho^2 - \rho + 1 = 0).$$

and we have the implication

$$(39) \qquad \rho^2 - \rho + 1 = 0 \quad \Longrightarrow \quad \rho^3 + 1 = 0 \text{ i.e. } \rho^3 = -1$$

by virtue of the algebraic identity

$$\rho^3 + 1 = (\rho + 1)(\rho^2 - \rho + 1).$$

So

Corollary 2. The equivalent conditions of (22) are also equivalent to the relation

$$(40) \qquad \rho^2 + \rho + 1 = 0.$$

[p. 330]

Uncertainty retained from cancelled passages.
140-3-s331-D001 ρ

[scan 332]

†
5°) A return to 1°), concerning the situation

$$(41)\qquad \begin{cases} \tilde\rho,\tilde\sigma \text{ with } 1^\circ) \quad \tilde\varepsilon_0 = \tilde\sigma^{-1}\tilde\rho \quad \underline{\text{regular}} \text{ unipotent } \dagger \quad (\text{i.e.} \\ \tilde\varepsilon_1 = \tilde\rho\tilde\sigma^{-1} = \tilde\rho(\tilde\varepsilon_0) = \tilde\sigma(\tilde\varepsilon_0) \quad \underline{\text{regular}} \text{ unipotent}) \text{ and} \\ 2^\circ) \quad \tilde\varepsilon_0 \text{ and } \tilde\varepsilon_1 \text{ inequivalent in each fiber.} \end{cases}$$

Let $\varepsilon_0,\varepsilon_1$ be the regular unipotent liftings in G — identified with sections of $M|\mathcal{M}$

$$(42)\qquad \varepsilon_0 = 1+N_0, \quad \varepsilon_1 = 1+N_1,$$

m1000

Marginal note. m1000 N_0 and N_1 strictly nilpotent i.e. their determinant and their trace † zero

with

$$(43)\qquad \tilde\sigma(\varepsilon_0) = \tilde\rho(\varepsilon_0) = \varepsilon_1, \quad \text{i.e.,} \quad \tilde\sigma(N_0) = \tilde\rho(N_0) = N_1.$$

We can then lift $\tilde\sigma$ and $\tilde\rho$ canonically to sections σ,ρ of G (or of $M|\mathcal{M}$) satisfying the following conditions

$$(44)\qquad \begin{cases} 1^\circ) \ -\sigma^{-1}\rho = \varepsilon_0 & [\Longleftrightarrow -\rho\sigma^{-1} = \varepsilon_1] \\ \dagger & \left[\begin{array}{l} \Longleftrightarrow \rho+\sigma = -N_1\sigma \ (= -\sigma N_0) \\ \Longleftrightarrow \rho+\sigma = N_1\rho \ (= \rho N_0) \end{array}\right] \\ & \dagger \\ 2^\circ) \ (\rho+\sigma)^2 = \rho+\sigma & \Longrightarrow (\rho+\sigma)^2 = u(\rho+\sigma) \quad \text{with} \\ & \qquad u \in \Gamma G_m \end{cases}$$

Moreover, there exists a † unique triple $(\varphi,\rho_0,\sigma_0)$

$$(45)\qquad \begin{cases} G_0 = \mathrm{Gl}(2)_S \xrightarrow{\varphi} G \quad (\text{pinning}) \\ \rho_0,\sigma_0 \in \Gamma(G_0), \quad \rho_0 = \begin{pmatrix} 0 & 1 \\ v|\nu & w|\omega \end{pmatrix}, \quad \sigma_0 = \begin{pmatrix} 0 & -1 \\ v|\nu' & w|\omega' \end{pmatrix} \\ (v|\nu + v|\nu' = 0, w|\omega + w|\omega' = 1), \quad v|\nu \in \Gamma G_m \\ \text{i.e. } v|\nu' = -v|\nu, w|\omega' = 1 - w|\omega \end{cases}$$

such that we have

$$(46|48)\qquad \varphi(\rho_0) = \rho, \quad \varphi(\sigma_0) = \sigma,$$

which implies

$$(47)\qquad \varphi((\varepsilon_0)_0) = \varepsilon_0, \quad \varphi((\varepsilon_1)_0) = \varepsilon_1$$

[p. 331]

Uncertainty retained from cancelled passages.
140-3-s332-D001 over scheme ( )
140-3-s332-D003 i.e.
140-3-s332-D004 [illeg. 1]
140-3-s332-D005 $2^\circ)(\rho+\sigma)$
140-3-s332-D006 $\Longleftrightarrow \rho+\sigma =$
140-3-s332-D007 isomorphism

[scan 333]

where † [p. 467]

$$(48)\qquad (\varepsilon_0)_0 = -\sigma_0^{-1}\rho_0 = \begin{pmatrix} 1 & \frac{1}{v|\nu} \\ 0 & 1 \end{pmatrix}, \qquad \dagger(\varepsilon_1)_0 = -\rho_0\sigma_0^{-1} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

— and conversely, for ρ_0, σ_0 as above, and any isomorphism $\varphi : G_0 = \mathrm{Gl}(2)_S \xrightarrow{\sim} G$, the images $\tilde\rho, \tilde\sigma$ of ρ, σ in PG satisfy (41), and the images ρ, σ of ρ_0, σ_0 under φ are liftings of $\tilde\rho, \tilde\sigma$ satisfying (44).

Modular pinning theorem. Let G, SG, PG be forms (over scheme S) of $\mathrm{Gl}(2), \mathrm{Sl}(2), \mathrm{GP}(1)$, associated with an Azumaya algebra M of rank 4.

1.) Let us call a modular †triple in PG a triple of sections $(\tilde\rho, \tilde\sigma, \tilde\varepsilon_0)$ of PG, satisfying conditions (49)(50)(51).

†

$$(49)\qquad \tilde\varepsilon_0 = \tilde\sigma^{-1}\tilde\rho \quad \text{i.e.} \quad \tilde\rho^{-1}\tilde\sigma\tilde\varepsilon_0 = 1$$

[so that

$$(49\text{ bis})\qquad \tilde\varepsilon_1 \overset{\text{dfn}}{=} \tilde\rho\tilde\sigma^{-1} = \tilde\rho(\tilde\varepsilon_0) = \tilde\sigma(\tilde\varepsilon_0)$$

];

(50) $\tilde\rho^3 = \tilde\sigma^2 = 1$, $\tilde\varepsilon_0$ is regular unipotent ($\iff \tilde\varepsilon_1$ is regular unipotent);

(51) $\tilde\varepsilon_0$ and $\tilde\varepsilon_1$ † inequivalent † in each fiber (which means, introducing the liftings

$$\varepsilon_0 = 1 + N_0, \qquad \varepsilon_1 = 1 + N_1$$

m1001

Marginal note. m1001 (N_0 and N_1 strictly nilpotent,... cf. (42))

to G, that N_0 and N_1 not proportional in every fiber).

2) Let us call a modular triple of|in G a [p. 332]

Uncertainty retained from cancelled passages.

140-3-s333-D001 [illeg. 1]
140-3-s333-D002 $(\varepsilon_1)_0$
140-3-s333-D003 [illeg. 1]
140-3-s333-D004 49 regular
140-3-s333-D005 everywhere
140-3-s333-D006 on the

[scan 334]

triple of sections $(\rho, \sigma, \varepsilon_0)$ satisfying the † following conditions

$$(52)\qquad -\sigma^{-1}\rho = \varepsilon_0 \quad (\iff \rho^{-1}\sigma\varepsilon_0 = -1.)$$
$$\left[\begin{array}{ll} \iff & -\rho\sigma^{-1} = \varepsilon_1 \quad \text{where } \varepsilon_1 = \rho\varepsilon_0\rho^{-1} \\ \iff & -\rho\sigma^{-1} = \varepsilon_1 \quad \text{where } \varepsilon_1 = \sigma\varepsilon_0\sigma^{-1} \end{array}\right]$$

$$(53)\qquad (\rho+\sigma)^2 = \rho + \sigma$$

(54) ε_0 † is regular unipotent i.e. $\varepsilon_0 = 1 + N_0$ with N_0 † regular nilpotent ($\mathrm{Tr}\, N_0 = \det N_0 = 0$, N_0 nonzero in each fiber).

m1002

Marginal note. $^{\text{m1002}} \Longleftrightarrow \begin{cases} \det \varepsilon_0 = 1 \\ \operatorname{Tr} \varepsilon_0 = 2 \\ \varepsilon_0 \neq 1 \text{ in every fiber} \end{cases}$

($\Longleftrightarrow$ † $\varepsilon_1 \overset{\text{dfn}}{=} \rho\varepsilon_0\rho^{-1} = \sigma\varepsilon_0\sigma^{-1}$ is regular unipotent)

(55) · ε_0 and ε_1 are inequivalent in each fiber ($\Longleftrightarrow$ N_0 and N_1 not proportional in every fiber).

(56) We have one of the four equivalent conditions

$$\boxed{\sigma^2+1=0} \qquad \boxed{\rho^2+|-\rho+1=0}$$
$$\boxed{\operatorname{Tr}\sigma=0, \det\sigma=1} \qquad \boxed{\operatorname{Tr}\rho=\det\rho=1} \quad .$$

† 3) Let us call the standard modular triple in $\mathrm{Gl}(2)_S$ (resp. in $\mathrm{GP}(1)_S$) the triple

$$(57) \quad \left(\rho_0 = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \quad \sigma_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \varepsilon_0 = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}\right)$$

(resp. its image $(\tilde\rho_0, \tilde\sigma_0, \tilde\varepsilon_0)$ in $\Gamma\mathrm{GP}(1)_S$) — which is indeed a modular triple.

This being so, we have the commutative diagram of canonical maps [p. 333]

Uncertainty retained from cancelled passages.

140-3-s334-D001 five
140-3-s334-D002 are
140-3-s334-D003 unipotent
140-3-s334-D004 ε_0
140-3-s334-D006 3

[scan 335]
[p. 468]

$$\varphi \longmapsto (\varphi(\rho_0), \varphi(\sigma_0), \varphi(\varepsilon_0))$$

(58)

$$\begin{array}{ccc} \mathrm{Isom}(\mathrm{Gl}(2), G) & \longrightarrow & \mathrm{TripMod}(G) \\ \wr & & \downarrow \\ \mathrm{Isom}(\mathrm{GP}(1), PG) & \longrightarrow & \mathrm{Tripmod}(PG) \end{array} \qquad \begin{array}{c} (\rho, \sigma, \varepsilon_0) \\ \downarrow \\ (\tilde\rho, \tilde\sigma, \tilde\varepsilon_0) \end{array}$$

$$\psi \longmapsto (\psi(\tilde\rho_0), \psi(\tilde\sigma_0), \psi(\tilde\varepsilon_0))$$

and we have the

Pinning theorem. All the arrows of (58) are bijections.

This theorem only summarizes some †of the main results of the reflections of the|of the present § VI. Let us note that the injectivity of the horizontal arrows means also that an † $\mathrm{Gl}(2)_S = G_0 \longrightarrow G$ (resp. $\mathrm{GP}(1)_S = PG_0 \longrightarrow G|PG$ is known when we know its effect on σ and ρ (thus also on ε_0] ← resp. when we know its effect on $\tilde\sigma$ and $\tilde\rho$) — which also means the same|= thing for the automorphisms of $\mathrm{Gl}(2)_S$ (resp. of $\mathrm{GP}(1)_S$) which also means that

$$(59) \qquad T_\rho \cap T_\sigma = \dagger\ G_m$$

(the intersection of the centralizers of ρ, σ in $\mathrm{Gl}(2)_S$ is reduced|reduced to the center G_m) — resp. the stronger relation

$$(60) \qquad N'_\rho \cap N'_\sigma = G_m,$$

[p. 334]

Uncertainty retained from cancelled passages.
140-3-s335-D002 reflections
140-3-s335-D003 isomorphism
140-3-s335-D004 $\mu_2|\{1\}$

[scan 336]

where N'_ρ, N'_σ are the group subschemes of $\mathrm{Gl}(2)_S$ defined respectively as stabilizers of $\tilde{\rho}$, $\tilde{\sigma}$ in $\mathrm{Gl}(2)_S$, i.e.

$$(61) \qquad \begin{cases} N'_\rho = \{g \mid g(\rho) = \varepsilon\rho,\ \varepsilon \text{ scalar}\}, \\ N'_\sigma = \{g \mid g(\sigma) = \varepsilon\sigma,\ \varepsilon \text{ scalar}\} \end{cases}$$

m1003

Marginal note. m1003 N.B. we will have necessarily $\varepsilon^2 = 1$

Remark It is plausible, † In view of these results, that $\mathrm{GP}(1)_{\mathbb{Z}}$ — as an "absolute" sheaf of groups (over $\operatorname{Spec}\mathbb{Z}$) for fppf, is generated by the homomorphisms

$$\mathbb{Z}/2\mathbb{Z} \xrightarrow{\alpha \mapsto \tilde{\sigma}_0^\alpha} \mathrm{GP}(1)_{\mathbb{Z}}$$
$$\mathbb{Z}/3\mathbb{Z} \xrightarrow{\beta \mapsto \tilde{\rho}_0^\beta} \mathrm{GP}(1)_{\mathbb{Z}}$$
$$\mathbb{G}_a \xrightarrow{\lambda \mapsto \tilde{\varepsilon}_0(\lambda)} \mathrm{GP}(1)_{\mathbb{Z}}$$

satisfying the|the sole relation (R)

$$(62) \qquad \tilde{\rho}_0^{-1}\tilde{\sigma}_0\tilde{\varepsilon}_0(1) = 1 \qquad \dagger$$

— it would be well worth the trouble to clear up this question, as well as the analogous question for $\mathrm{Gl}(2)_{\mathbb{Z}}$, where the groups $\mathbb{Z}/2\mathbb{Z}$, $\mathbb{Z}/3\mathbb{Z}$ must be replaced by $\mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}/6\mathbb{Z}$, and the relation (62), by

$$(63) \qquad \rho_0^{-1}\sigma_0\tilde{\varepsilon}_0(1) = \rho_0^3 = \sigma_0^2.$$

[p. 335]

Uncertainty retained from cancelled passages.
140-3-s336-D002 [illeg. 5]
140-3-s336-D003 (?)

[scan 337]

Remark I have the distinct impression that the package of conditions (51) to (56) is redundant, [p. 469] and wonder for example whether the set|one of the three simple algebraic conditions

$$(64) \qquad \sigma^2 + 1 = \rho^2 - \rho + 1 = 0, \qquad (\rho + \sigma)^2 = \rho + \sigma$$

does not imply the other conditions (defining ε_0 by (51|52)). The third condition (64) means that

$$(65) \qquad N = \rho + \sigma$$

is idempotent

$$(66) \qquad N^2 = N,$$

and in terms of N we will have

$$\sigma^{-1}N = \sigma^{-1}\rho + 1 \quad \text{i.e.}$$

$$(67)\qquad \begin{cases} \varepsilon_0 \overset{\text{dfn}}{=} -\sigma^{-1}\rho = 1-\sigma^{-1}N = 1+N_0 & N_0 = -\sigma^{-1}N \\ & \quad = \sigma N \\ \varepsilon_1 \overset{\text{dfn}}{=} -\rho\sigma^{-1} = 1-N\sigma^{-1} = 1+N_1 & N_1 = -N\sigma^{-1} = N\sigma \end{cases}$$

m1004

Marginal note. m1004 N.B. We must † specify † [illeg. 4] first relations (64) [illeg. 3] $\operatorname{Tr}\sigma = 0$, $\det\sigma = 1$, $\operatorname{Tr}\rho = 0|1$, $\det\rho = 1$ must require that [illeg. 2] are regular i.e. not homotheties in any fiber [illeg. 1].

and the condition ε_0 † regular unipotent is expressed by the conditions

$$(68)\qquad \det\sigma N = \operatorname{Tr}\sigma N = 0, \qquad \begin{matrix} \sigma N \text{ nonzero in every fiber.} \\ \text{i.e. } N \text{ nonzero in every fiber} \end{matrix}$$

Now the third of these (68) follows from the others, for if we had $N = 0$ on a fiber, we could suppose S the spectrum of a field, $N = 0$, i.e. $\rho = -\sigma$, so † we would have both

$$\sigma^2 + 1 = 0 \qquad \sigma^2 + \sigma + 1 = 0 \qquad \text{whence } \sigma = 0, \text{ absurd.}$$

I also say that in no fiber do we have $N = 1$, otherwise we would have in this fiber $\sigma = 1 - \rho$ and substituting in $\sigma^2 + 1 = 0$ we find †

$$(1-\rho)^2 + 1 = 0 \quad \text{i.e.} \quad \rho^2 - 2\rho + 2 = 0,$$

[p. 336]

Uncertainty retained from cancelled passages.

140-3-s337-D001 perhaps

140-3-s337-D002 that in

140-3-s337-D003 strict^{ly} unipotent

140-3-s337-D005 [illeg. 1]

140-3-s337-D007 [illeg. 1]

[scan 338]

comparing with $\rho^2 - \rho + 1 = 0$, we find † by subtraction $\rho - 1 = 0$ i.e. $\rho = 1$, but the relation $\rho^2 - \rho + 1 = 0$ becomes $1 = 0$, absurd.

Thus the idempotent operator N is of rank 1 in each fiber, so [illeg. 1] (unless I am mistaken) † that N is (in terms of $\mathcal{M} \simeq \check{F} \otimes F$) local^{ly} decomposable in the form $x' \otimes x$, with $\langle x, x' \rangle = 1$, and

$$\operatorname{Tr} N = 1, \quad \det N = 0$$

so the relation $\det(\sigma N) = 0$ in (68) is also automatic, since $\det\sigma N = \det\sigma \det N = 0$. There remains only the relation

$$\operatorname{Tr} \underbrace{\sigma N}_{\substack{\sigma^2+\sigma\rho \\ =\sigma\rho-1}} = 0$$

(which is also written

$$(69)\qquad \operatorname{Tr}\sigma\rho = 2 \quad (\Longleftrightarrow \operatorname{Tr}\sigma^{-1}\rho = -2)$$

for which we would have to see whether or not it follows from the other relations.

Let us take

$$(70)\qquad \begin{cases} \rho = \rho_0 = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} \\ \sigma = \underline{u}\sigma_0\underline{u}^{-1} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \\ \text{where } \underline{u} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma\mathrm{Sl}(2), \text{ i.e. } ad - bc = 1 \end{cases}$$

We find [p. 337]

Uncertainty retained from cancelled passages.
140-3-s338-D001 $\rho - 1$
140-3-s338-D003 says

[scan 339]
[p. 470]

$$(70\text{bis}) \qquad \sigma = \begin{pmatrix} bd+ac & -(a^2+b^2) \\ c^2+d^2 & -(bd+ac) \end{pmatrix}$$

whence

$$N = \rho + \sigma = \begin{pmatrix} (bd+ac) & 1-(a^2+b^2) \\ -1+(c^2+d^2) & 1-(bd+ac) \end{pmatrix}$$

$$N-1 = \begin{pmatrix} (bd+ac)-1 & 1-(a^2+b^2) \\ -1+(c^2+d^2) & -(bd+ac) \end{pmatrix} \quad = -\det(N)N^{-1} \text{ if } \det N \text{ invertible}$$

and we find

$$(71) \qquad N^2 - N = N(N-1) = -\det(N).1$$

so

$$(72) \qquad N^2 = N\dagger \iff \det N = 0\ ; \quad (\Longrightarrow \operatorname{Tr} N = 1)$$

Now we have

$$\begin{aligned} \det N &= bd+ac-(bd{+}ac)^2\dagger \\ &\quad -|\ = [-1-(a^2+b^2)(c^2+d^2)+a^2+b^2+c^2+d^2] \\ &= bd+ac---2a\underline{bc}d+1++\underline{a^2d^2}+\underline{b^2c^2}+\dagger \\ &\qquad\qquad -a^2-b^2-c^2-d^2 \\ &= (bd+ac)-(a^2+b^2+c^2+d^2)+\underbrace{(ad-bc)^2}_{1}+1 \\ &= 2+(bd+ac)-(a^2+b^2+c^2+d^2) \end{aligned}$$

so the condition (72)|(71) is equivalent to

$$(73) \qquad a^2+b^2+c^2+d^2 = 2+(bd+ac),$$

on the other hand we have by 70, (70bis)

$$\rho\sigma = \begin{pmatrix} c^2+d^2 & -(bd+ac) \\ c^2+d^2-(bd+ac) & a^2+b^2-(bd+ac) \end{pmatrix}$$

so

$$(74) \qquad \operatorname{Tr}\sigma\rho = \operatorname{Tr}\rho\sigma = a^2+b^2+c^2+d^2-(bd+ac) = 2$$

[p. 338]

Uncertainty retained from cancelled passages.
140-3-s339-D002 $\iff$
140-3-s339-D003 $+1$
140-3-s339-D007 b^2d^2

This calculation strongly suggests that the relations † (64) already imply (68)… it is even very nearly a proof… If we admit this point, it only remains to verify that the preceding conditions (64) (i.e. essentially already (51) to (56) except (55)) also imply (55), i.e. the independence of the $\varepsilon_0, \varepsilon_1$ in every fiber, i.e. that N_0, N_1 are non-homothetic in every fiber. Now (67) [scan 340]

$$(*) \qquad N_1 = \tilde{\sigma}(N_0),\ \dagger = \tilde{\rho}(N_1|N_0)\ \dagger :$$

If we had on a fiber

$$(\dagger) \qquad N_1 = \lambda N_0$$

we would conclude from this, by applying using $\tilde{\sigma}^2 = 1$, and $\tilde{\rho}^3 = 1$, the relations

†

†

$$\lambda^2 = 1, \quad \lambda^3 = 1 \qquad \text{whence } \lambda = 1$$

i.e. $N_0 = N_1$ i.e. N_0 commutes with σ, ρ thus also $N = -\sigma N_0 = \rho^{-1} N_0$ commutes with σ, ρ, and since $N = \sigma + \rho$, σ and ρ commute. But then [p. 339]

Uncertainty retained from cancelled passages.

140-3-s340-D001 pa|by
140-3-s340-D002 and
140-3-s340-D003 $\widetilde{\sigma}(N_1)$
140-3-s340-D004 74
140-3-s340-D007 (**) $= \lambda^2$
140-3-s340-D011 –

[scan 341]

$$(\sigma+\rho)^2 = \underbrace{\sigma^2}_{-1} + \underbrace{\rho^2}_{\rho^{-1}} + 2\sigma\rho = \rho - 2 + 2\sigma\rho = \qquad \text{[p. 471]}$$

thus the relation $(\sigma+\rho)^2 = \sigma + \rho$ is written

$$-2 + 2\sigma\rho = \sigma + \qquad \text{i.e.}$$

$$\dagger \qquad 2\sigma\rho = \sigma + 2 \quad \text{that is} \quad \underbrace{2\sigma^2\rho}_{-2\rho} = \underbrace{\sigma^2}_{-1} + 2\sigma \quad \text{i.e.} \quad 2(\sigma+\rho) = 1$$

† ‖

†

† i.e.

$$(**) \qquad 2N = 1,$$

whence, taking the determinants

$$4 \underbrace{\det N}_{0} = 1 \qquad \text{i.e.} \quad 0 = 1, \quad \underline{\text{absurd}}.$$

Remark I would not be surprised if we also had conditions equivalent to (51) to (56) in the form

$$(75) \qquad \sigma^2 + 1 = \rho^{3|2} - | + \rho + 1 = 0, \quad \varepsilon_0 \overset{\text{dfn}}{=} -\sigma^{-1}\rho \text{ is regular unipotent}$$

[p. 340]

Uncertainty retained from cancelled passages.

140-3-s341-D003 (**)
140-3-s341-D004 (**) det
140-3-s341-D006 2
140-3-s341-D008 it

requiring in addition, if need be, that σ, ρ also be regular i.e. be scalars on no fiber. One would have to convince oneself that this implies that [scan 342]

$$N \overset{\text{dfn}}{=} \rho + \sigma$$

does satisfy the idempotence relation

$$(75) \qquad N^2 - N = 0$$

(which is equivalent, if N regular, to

$$(77) \qquad \operatorname{Tr} N = 1, \quad \det N = 0 \qquad (?)$$

But N is regular, for if on a fiber it were a scalar we would have that (restricting to this fiber) ρ and σ commute, we would have

$$-\sigma^{-1}\rho = \sigma\rho = \varepsilon_0 = 1 + N_0$$
$$-''\rho\sigma^{-1} = \rho\sigma = \varepsilon_1 = 1 + N_1,$$

†

$$(1-\sigma\rho)^2 = 0 \quad \text{i.e.} \quad 1 - 2\sigma\rho + \underbrace{\sigma^2}_{-1}\underbrace{\rho^2}_{\rho-1} = 0 \quad \text{i.e.}$$

$$1 - 2\sigma\rho + 1 - \rho = 0 \quad \text{i.e.} \quad 2 = \rho + 2\sigma\rho$$

that is, multiplying by σ

$$2\sigma = \rho\sigma + 2\underbrace{\sigma^2}_{-1}\rho \quad \text{i.e.} \quad 2\underbrace{(\sigma+\rho)}_{N} = \rho\sigma$$

But by hypothesis we have $N = \lambda.1$, λ a scalar, whence $\rho\sigma = 2\lambda$, and taking the determinants

$$\det \underbrace{(\rho\sigma)}_{1} = 4\lambda^2$$

i.e. † 2 invertible and $\lambda = \dfrac{\varepsilon}{2}$, with $\varepsilon^2 = 1$,

$$\text{i.e.} \qquad \rho\sigma = \varepsilon \qquad \varepsilon^2 = 1,$$
$$\rho^2\underbrace{\sigma^2}_{-1} = 1 \qquad \underbrace{\rho^2}_{\rho-1} = -1, \quad \text{i.e.} \quad \rho = 0, \quad \underline{\text{absurd}}$$

[p. 341]

Uncertainty retained from cancelled passages.

140-3-s342-D001 Taking $\varepsilon_0 = \det \rho$

140-3-s342-D003 4

† Thus N is regular, it remains to prove (77). [scan 343]

But $\operatorname{Tr} N = \underbrace{\operatorname{Tr}\sigma}_{0} + \underbrace{\operatorname{Tr}\rho}_{1} = 1$, OK, remains to prove $\det(\rho+\sigma) = 0$ but [m1006] [p. 472]

Marginal note. [m1006]



$$\det(\rho+\sigma) =$$

$''$ $\det(\dagger\sigma(1+\sigma^{-1}\rho) = \det\sigma \quad \underbrace{\det(1+\sigma^{-1}\rho)}_{\underbrace{\det\sigma^{-1}\rho}_{(\det\sigma)^{-1}\det\rho} + \mathrm{Tr}(\underbrace{\sigma^{-1}\rho}_{-\sigma\rho})+1}$

$$= \overbrace{\det\sigma}^{1} + \overbrace{\det\rho}^{1} + \underbrace{\det\sigma}_{1}\,\mathrm{Tr}(\sigma^{-1}\rho)$$

m1007

Marginal note. m1007 calculation already done See page – ? (cf. formula (69))

$$\text{i.e.} \qquad \det(\sigma+\rho) = 2+\mathrm{Tr}(\sigma^{-1}\rho) = 2-\mathrm{Tr}\,\sigma\rho, \quad \text{thus}$$

the condition $\det N = 0$ is equivalent to (69)

$$\mathrm{Tr}\,\sigma\rho = 2$$

To † prove this relation (or versions|neighboring ones tested|such), we can think of proceeding as earlier, except that I have started from (68) instead of (75). This is not entirely elucidated – to return to the attack! [p. 342]

Uncertainty retained from cancelled passages.

140-3-s343-D001 by

140-3-s343-D004 ρ

140-3-s343-D005 it

VII Recapitulation on pinnings. † We denote by G, SG, PG forms of $\mathrm{G}\ell(2)_S$, $\mathrm{S}\ell(2)_S$, $\mathrm{GP}(1)_S$ associated to the same Azumaya Algebra $\mathcal{M}$, i.e. to a bundle of projective lines $\mathbb{P}|\mathcal{P}$, over S. 1) [scan 344] [p. 473]
Regular sections of G. Let ρ be a section of $\mathcal{M}$ (not necessarily invertible), put

$$(1) \qquad v|\nu = \det\rho, \quad w = \mathrm{Tr}\,\rho$$

so that we have the Hamilton-Cayley equation

$$(2) \qquad \rho^2 - w\rho + v|\nu = 0 \qquad \text{i.e.} \quad \rho^2 = w\rho - v|\nu$$

Proposition Equivalent conditions: a) ρ is not a scalar on any fiber. a') $\mathcal{A}_\rho = \underline{\mathcal{O}}_S + \underline{\mathcal{O}}_S\rho$ is a submodule locally|locally a free direct summand of rank 2 of $\mathcal{M}$. m1008

Marginal note. m1008 N.B. Because of (2), $\mathcal{A}_\rho$ is a sub-Algebra of $\mathcal{M}$.

b) (If $\mathcal{M} = \underline{\mathrm{End}}_{\mathcal{O}_S}(F)$) There exists locally (Zariski) a section $e_0|E_0$ of F such that $(e_0, \dagger e_1 \overset{\text{dfn}}{=} \rho(e_0))$ form a basis of F b') (If $\mathcal{M} = \underline{\mathrm{End}}_{\mathcal{O}_S}(F)$) There exists locally Zariski a "line subbundle" $D_0 \subset F$, such that

$$F = D_0 + D_1, \quad \text{where } D_1 = \rho(D_0)$$

b") (If $\mathcal{M} = \underline{\mathrm{End}}_{\mathcal{O}_S}(F)$.) The $\mathcal{O}_S[T]$-module F_ρ (: T is|being an indeterminate, which operates on F by ρ) is † † cyclic locally (Zariski). c) (If $\mathcal{M} = \underline{\mathrm{End}}_{\mathcal{O}_S}(F)$) The $\underline{\mathcal{O}}_S[T]$-module F_ρ is|→ isomorphic to

$$\underline{\mathcal{O}}_S[T]/(T^2 - wT + v|\nu)$$

c') (case $\mathcal{M} = \underline{\mathrm{End}}(F)$) ∃ locally|locally Zariski a basis e_0, e_1 of F, such that the matrix of ρ with respect|p.v. to this basis [p. 343]

Uncertainty retained from cancelled passages.
140-3-s344-D001 1)
140-3-s344-D003 ρ.
140-3-s344-D004 isomorphic
140-3-s344-D005 locally

[scan 345]

is

$$(3) \qquad \rho_{e_0,e_1} = \begin{pmatrix} 0 & -v|\nu \\ 1 & w \end{pmatrix}$$

m1009

Marginal note. m1009 (i.e. $\rho e_0 = e_1$ whence $\rho e_1 = -v\rho e_0 - we_0$

c") (Case $\mathcal{M} = \underline{\mathrm{End}}_{\underline{\mathcal{O}}_S}(F)$) $\exists$ locally Zariski a basis (e_0, e_1) of F such that the matrix of ρ with respect to it is

$$(4) \qquad \rho_{e_0,e_1} = \begin{pmatrix} 0 & 1 \\ -v & w \end{pmatrix}$$

d) The commutant of ρ in $\mathcal{M}$ is reduced to $\mathcal{A}_\rho = \underline{\mathcal{O}}_S + \underline{\mathcal{O}}_S\rho$, and it remains so after every base change $S' \to S$.

d') Item|Item in each fiber.

Proof $a \iff a'$ trivial (linear algebra of the Modules $\mathcal{M}$ and of the submodules $\underline{\mathcal{O}}_S 1$ locally direct summands)

$b \iff b' \iff b''$ trivial, $b) \Longrightarrow a)$ as well.

$a \Longrightarrow b$ it suffices to choose e_0 on a †fiber so as not to be an eigenvector under ρ (possible over a field, for a nonscalar operator) then to lift in a neighborhood.

$b \Longleftrightarrow c) \Longleftrightarrow c')$ trivial ; †the bases for which ρ_{e_0,e_1} is of the form $\begin{pmatrix} 0 & -v \\ 1 & w \end{pmatrix}$ are precisely the bases of the form $(e_0, e_1 = \rho e_0)$.

$c') \Longleftrightarrow c'')$ means that if ρ satisfies the conditions under consideration, ${}^t\rho$ does too. († and conversely it follows from this that the condition for ${}^t\rho$ implies it for $\rho = {}^t({}^t\rho)$†, i.e. $c'') \longrightarrow c')$) — but this is immediate in the form a).

Thus the conditions a) to c") are equivalent, moreover $d) \Longrightarrow d') \Longrightarrow a)$ is trivial, it remains to prove that b (for example) $\Rightarrow d)$. Since b stable under base change, it suffices to see that [p. 344]

Uncertainty retained from cancelled passages.
140-3-s345-D003 field
140-3-s345-D004 for the
140-3-s345-D005 therefore
140-3-s345-D006 }

[scan 346]

N.B. In fact, $G = \underline{\mathcal{M}}^*$ operates on $\underline{\mathcal{M}}$ (by inner automorphisms) or on G, or SG, via $PG = \underline{\mathcal{M}}^*/\mathbb{G}_m$, and PG is an fppf quotient of SG (and even|= for the topology (étale), if 2 invertible). [p. 474]

So we see that in the preceding conjugation criterion, in the formula

$$(6) \qquad \rho' = \mathrm{int}(\alpha)\rho = \alpha\rho\alpha^{-1},$$

we may assume $\alpha \in \Gamma \mathrm{S}\ell(2)|SG$.

†Corollary 2 Let $\rho, \zeta|\xi$ be two sections of G, ρ being regular. For us to be able to find locally Zariski an $\alpha \in \Gamma G$ such that

$$(7) \qquad [\alpha, \rho] = \zeta \qquad \text{i.e.} \quad \alpha\rho\alpha^{-1} = \zeta\rho$$

it is necessary and sufficient that we have

a) $\zeta\rho$ is regular, i.e. ζ † non-homothetic to ρ^{-1} on any fiber

b) $\det\zeta = 1$, $\operatorname{Tr}\zeta\rho = \operatorname{Tr}\rho$

(We can then take α in SG…)

This is|These are only the conditions of corollary 1 † for conjugation of $\zeta\rho$ to ρ, the condition $\det\zeta\rho = \det\rho$ also being written as $\det\zeta\det\rho = \det\rho$ i.e. $\det\zeta = 1$.

Corollary 3 The indeterminacy of α in the solution of (7) is of the form $\alpha \mapsto \alpha\underline{t}$, with $\underline{t}$ a section of $\underline{\mathcal{A}^*_\rho}$, i.e. $t = c\rho + d$, with $c, d \in \Gamma\underline{\mathcal{O}_S}$, $\det(c\rho + d)$ invertible. i.e.

⑧ $c^2v|\nu + wcd + d^2$ invertible (where, $v = \det\rho$, $w = \operatorname{Tr}\rho$, cf. (1)). [p. 345]

Uncertainty retained from cancelled passages.

140-3-s346-D002 ε

140-3-s346-D003 $\neq$

140-3-s346-D004 of

[scan 347]

if u is a section of $\mathcal{M}$ which commutes with ρ, then $u = a + b\rho$ $(a, b \in \Gamma\underline{\mathcal{O}_S})$. By a) a and b are unique, and this shows that the question is fppf local. We may assume therefore $\mathcal{M} = \underline{\operatorname{End}}(F)$, F equipped with a basis $(e_0, e_1 = \rho(e_0))$. † We will have, if

$$u(e_0) = ae_0 + be_1 = u'(e_0) \qquad \text{where} \quad u' = a + b\rho,$$

and if u commutes with ρ, we will have

$$u(e_1) = u\rho(e_0) = \rho u(e_0)$$

likewise

$$u'(e_1) = u'\rho(e_0) = \rho u'(e_0) \qquad \text{for } u' \text{ commutes with } \rho$$

therefore

$$u(e_0) = | \neq u(e'_0)|u'(e_0), \quad u(e_1) = u'(e_1) \qquad \text{therefore } u = u',$$

q.e.d.

Definition : A section of $\mathcal{M}$ which satisfies the equivalent conditions of the proposition is called regular. Item|Item for sections of G, SG, interpreted as sections of $\mathcal{M}$ — such a section is regular iff at no point is it a scalar, i.e. is not in the center. This remains|is stable under multiplication by an invertible scalar — whence by passage to the quotient we deduce the notion of a regular section of PG — these are the sections which are not $= 1$ on any fiber.

Corollary 1 Let ρ, ρ' be two sections of $\mathcal{M}$, ρ being regular. For them to be conjugate (locally Zariski) via $\mathcal{M}^*$, it is necessary and sufficient that a) ρ' be regular (Condition of regularity †) and b) we have

$$(5) \qquad \det\rho = \det\rho', \quad \operatorname{Tr}\rho = \operatorname{Tr}\rho'$$

("Numerical condition" [p. 346]

Uncertainty retained from cancelled passages.
140-3-s347-D002 and
140-3-s347-D011 [illeg. 1]

[scan 348]

This follows from the equivalence $a) \iff d)$ of the proposition. [p. 475]

Proposition Let

$$(9) \qquad \varepsilon = 1 + N$$

be a section of $\mathcal{M}$. Equivalent conditions

$$\text{a)} \quad (10) \quad \begin{matrix} \det \varepsilon = \det 1 \quad (=1) \\ \operatorname{Tr} \varepsilon = \operatorname{Tr} 1 \quad (=2) \end{matrix} \qquad \text{and } \varepsilon \text{ regular}$$

m1010

Marginal note. m1010 N.B. $\det(1+N) = 1 + \operatorname{Tr} N + \det N \quad \operatorname{Tr}(1+N) = 2 + \operatorname{Tr} N.$

$$\text{b)} \quad (11) \quad \det N = \operatorname{Tr} N = 0 \qquad \text{and } N \text{ regular}$$

c) (If $\mathcal{M} = \underline{\operatorname{End}}_{\underline{\mathcal{O}}_S}(F)$) $\exists$ line bundle $D \subset F$ such that we have

$$N(F) = D, \quad N(D) = 0$$

(Then D is uniquely determined by ε i.e. by N, D determines N modulo homothety)

c') (If $\mathcal{M} = \underline{\operatorname{End}}_{\underline{\mathcal{O}}_S}(F)$). $\exists$ locally basis|Zariski e_0, e_1 such that $\varepsilon = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ i.e. $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

d) $\exists$ Borel $B \subset G$, such that ε is a section everywhere $\neq 1$ of B_u.

d') †$\exists$ Borel $SB \subset SG$, such that ε is a section everywhere $\neq 1$ of $(SB)_u$.

† Moreover, D, B, SB are well determined by ε i.e. by N, and determine N up to homothety.

Definition : we that ε is (strictly) unipotent iff we have $\det \varepsilon = 1$, $\operatorname{Tr} \varepsilon = 2$, that N is strictly nilpotent iff|if $\det N = \operatorname{tr} N = 0$ (whence $N^2 = 0$ by Hamilton-Cayley). We say that ε is regular unipotent resp. regular nilpotent (dropping the qualifier "strictly") if in addition it is regular. [p. 347]

Uncertainty retained from cancelled passages.
140-3-s348-D003 If
140-3-s348-D004 Proof

[scan 349]

Proposition. Let P be a section of $\mathcal{M}$. Equivalent conditions

$$\begin{matrix} \text{a)} & P \text{ is regular, and } P^2 = P \\ \text{b)} & \det P = 0, \quad \operatorname{Tr} P = 1 \end{matrix}$$

c) (If $\mathcal{M} = \underline{\operatorname{End}}_{\underline{\mathcal{O}}_S}(F)$) $\exists$ basis e_0, e_1 of F such that the matrix of P is

$$P_{e_0, e_1} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \qquad \text{i.e.}$$

P is the projection of $\underline{\mathcal{O}}_S e_0 \oplus \underline{\mathcal{O}}_S e_1$ onto the factor $\underline{\mathcal{O}}_S e_1$

d) $\exists$ (Zariski locally) isomorphism $\mathcal{M} \underset{\sim}{\overset{\varphi|\psi}{\leftarrow}} \mathrm{M}_2(\underline{\mathcal{O}}_S)$, taking $\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ to P.

c') (If $\mathcal{M} = \underline{\mathrm{End}}_{\underline{\mathcal{O}}_S}(F)$) P is a projector whose image D_1 and kernel D_0 are of rank 1.

Remark † For the proof, we will note that the numerical condition b) already implies that P is regular, for if P were a scalar λ (on a fiber) we would have

$$\det P = \lambda^2 = 0, \quad \mathrm{Tr}\, P = 2\lambda = 1$$

hence λ would be invertible and would have square zero, absurd. Note that this argument shows, more generally, that if ρ is a section of $\mathcal{M}$ such that

$$(12) \qquad \delta(\rho) = (\mathrm{Tr}\,\rho)^2 - 4\det\rho = w^2 - 4v|\nu$$

m1011

Marginal note. m1011 (= discriminant of the Hamilton-Cayley polynomial of ρ)

is an invertible section of $\underline{\mathcal{O}}_S$, then ρ is a regular section of $\mathcal{M}$.

The P † satisfying the conditions of the preceding proposition are called the elementary projectors in $\mathcal{M}$. Proof|their.

Corollary We have a one-to-one correspondence between the elementary projectors in $\mathcal{M}$, and the Killing pairs (B, T) in G (or SB, ST in SG, or PB, PT in PG), by [p. 348]

Uncertainty retained from cancelled passages.
140-3-s349-D001 Corollary
140-3-s349-D003 in

[scan 350]

associating to † P the † torus $T = \underline{A}^*_P$, and the Borel B_0 formed of the α which stabilize † [p. 476] D_0 † † hence with respect|p.v. † to e_0, e_1 we will have

$$(13) \qquad \begin{cases} T = \left\{ \begin{pmatrix} \mu_0 & 0 \\ 0 & \mu_1 \end{pmatrix} \;\middle|\; \mu_0, \mu_1 \in \Gamma G_m \right\}, \\ B_0 = \left\{ \begin{pmatrix} \mu_0 & \lambda \\ 0 & \mu_1 \end{pmatrix} \;\middle|\; \mu_0, \mu_1 \in \Gamma G_m,\ \lambda \in \Gamma G_a \right\}. \end{cases}$$

The Killing pair opposite to (T, B_0) is (T, B_1), with

$$(14) \qquad B_1 = \left\{ \begin{pmatrix} \mu_0 & 0 \\ \lambda & \mu_1 \end{pmatrix} \;\middle|\; \mu_0, \mu_1 \in \Gamma G_m,\ \lambda \in \Gamma G_a \right\},$$

in a slightly more intrinsic way, B_1 † is the stabilizer of D_1. The elementary projector associated to the Killing pair opposite (T, B_1) is $1 - N$.

Remark. The scheme of the ε_1 regular unipotents, or again of the regular nilpotent N is a bundle of punctured lines (i.e. a G_m-torsor) over the scheme $\mathcal{P}$ of Borels (of G, of SG or of PG, it amounts to the same|=), it is smooth of relative dimension 2 — the scheme of elementary projectors N is isomorphic to that of Killing pairs, smooth-fibered of relative dimension 1 over the scheme of Borels. The fiber product of these two relative schemes over $\mathcal{P}$ = scheme of Borels, is isomorphic to the scheme of pinnings of G (or of SG, or of PG) i.e. of isomorphisms $\mathrm{G}\ell(2)_S \xrightarrow{\sim} G$ (resp. $\mathrm{S}\ell(2)_S \xrightarrow{\sim} SG$, [p. 349]

Uncertainty retained from cancelled passages.
140-3-s350-D001 N
140-3-s350-D002 sub
140-3-s350-D005 the
140-3-s350-D006 induce
140-3-s350-D007 the pair of the
140-3-s350-D010 commuting
140-3-s350-D011 will have

[scan 351]

resp. $\mathrm{GP}(1)_S \xrightarrow{\sim} PG$), which is a torsor under $\mathrm{GP}(1)_S \simeq \underline{\mathrm{Aut}}_{S-\mathrm{gr}} \mathrm{G}\ell(2)_S \simeq \underline{\mathrm{Aut}}_{S-\mathrm{gr}} \mathrm{S}\ell(2)_S \simeq \underline{\mathrm{Aut}}_{S-\mathrm{gr}}(\mathrm{GP}(1)_S)$. We say that ε and $\dagger P$ are associated, (or $N = \varepsilon - 1$ and P are associated), if they define the same|= Borel, i.e. †come from a pinning. We check that this amounts to the following relations

$$PN = 0, \qquad NP = N \tag{15}$$

or again, in terms of P and $\varepsilon = 1 + N$

$$P\varepsilon = P, \qquad \varepsilon P = P + \underbrace{\varepsilon - 1}_{N} \tag{16}$$

This is really pretty!

We now consider the question of †the conjugation of the sections $\tilde{\rho}, \tilde{\rho}'$ of PG, defined by sections ρ, ρ' of G. We assume $\tilde{\rho}$ regular i.e. $\neq 1$ in each fiber, i.e. ρ regular — † † the conjugation of $\tilde{\rho}, \tilde{\rho}'$ by a section u of G (say) means that we have

$$\rho' = \lambda u \rho u^{-1}, \qquad \lambda \in \Gamma G_m \tag{17}$$

whence we deduce

$$\det \rho' = \lambda^2 \det \rho \qquad \mathrm{Tr}\, \rho' = \lambda \,\mathrm{Tr}\, \rho$$

and hence

$$\frac{\mathrm{Tr}^2 \rho'}{\det \rho'} = \frac{\mathrm{Tr}^2 \rho}{\det \rho}. \tag{18}$$

Let us put, for a section of G

$$\dagger\Delta(\rho) = \frac{\mathrm{Tr}^2 \rho}{\det \rho}, \tag{19}$$

[p. 350]

Uncertainty retained from cancelled passages.
140-3-s351-D001 N
140-3-s351-D003 come
140-3-s351-D005 the equivalence
140-3-s351-D007 A
140-3-s351-D008 a
140-3-s351-D009 $\delta|\Delta$

[scan 352]

we have done what was needed for us to find a $G \longrightarrow \mathbb{E}^1$ which passes to the quotient as [p. 477]

$$\begin{array}{c} PG \xrightarrow{\Delta} \mathbb{E}^1 \\ \Delta(\tilde{\rho}) = \Delta(\rho) = \dfrac{(\mathrm{Tr}\, \rho)^2}{\det \rho}, \end{array} \tag{20}$$

and which is|remains invariant under inner automorphisms.

Proposition. In order for two sections $\tilde{\rho}, \tilde{\rho}'$ of PG to be † conjugate fppf locally, $\tilde{\rho}$ being regular, it is necessary † that $\tilde{\rho}'$ be regular, and that we have

$$(21) \qquad \Delta(\tilde{\rho}') = \Delta(\tilde{\rho})$$

i.e. (in terms of ρ, ρ' lifting $\tilde{\rho}, \tilde{\rho}'$) that ρ' be regular and satisfy (18). These conditions are sufficient when $\Delta(\tilde{\rho})$ invertible i.e. $\operatorname{Tr}\rho$ invertible. It remains to prove sufficiency. Let us note that fppf locally, exists $\lambda \in \Gamma G_m$ such that

$$\det(\rho') = \det(\lambda\rho) \qquad \text{i.e.} \quad \lambda^2 = \frac{\det\rho'}{\det\rho}.$$

Thus we may assume (at the cost of replacing ρ by $\lambda\rho$) that $\det\rho = \det\rho'$. Then (18) amounts to

$$(22) \qquad \operatorname{Tr}^2\rho' = \operatorname{Tr}^2\rho.$$

the question amounts to seeing that $\exists\varepsilon \in \Gamma\mu_2$, such that

$$\operatorname{Tr}\rho' = \varepsilon\operatorname{Tr}\rho \quad (= \operatorname{Tr}\varepsilon\rho),$$

we will also have

$$\det\rho' = \det\varepsilon\rho \quad (= \underbrace{\varepsilon^2}_{1}\det\rho = \det\rho),$$

whence local conjugation of ρ' and $\varepsilon\rho$, we will be done. But if τ', τ are two sections of $\mathcal{O}_S$, with τ invertible, such that $\tau'^2 = \tau^2$, we will indeed have $\tau' = \varepsilon\tau$ ($\varepsilon = \tau'/\tau$), and $\varepsilon^2 = 1$, OK. [p. 351]

Uncertainty retained from cancelled passages.

140-3-s352-D001 equi

140-3-s352-D002 and sufficient

[scan 353]

Remark. If $\Delta(\rho)$ not invertible there are obvious counterexamples, for example with $(\operatorname{Tr}\rho)^2 = 0$, $(\operatorname{Tr}\rho')^2 = 0$, $\det\rho = \det\rho' = 1$, $\operatorname{Tr}\rho$ and $\operatorname{Tr}\rho'$ not being homothetic. We can take

$$\rho = \begin{pmatrix} 0 & 1 \\ -1 & w \end{pmatrix}, \qquad \rho' = \begin{pmatrix} 0 & 1 \\ -1 & w' \end{pmatrix}$$

$w^2 = w'^2 = 0$, w, w' not homothetic, for example base ring $A = A_0[w, w']/(w^2, ww', w'^2)$ (dual numbers|numbers on the free module of rank 2 generated by w, w').

2) Relative position of two sections ρ, σ of G, or $\tilde{\rho}, \tilde{\sigma}$ the two sections of PG.

In PG we are interested in

$$(23) \qquad \tilde{\varepsilon}_0 = \tilde{\sigma}^{-1}\tilde{\rho} \quad \text{and} \quad \tilde{\varepsilon}_1 = \tilde{\rho}\tilde{\sigma}^{-1} = \tilde{\rho}(\tilde{\varepsilon}_0) = \tilde{\sigma}(\tilde{\varepsilon}_1)$$

and in the condition that ① $\tilde{\varepsilon}_0$, † (hence $\tilde{\varepsilon}_1$) be|are regular unipotent — hence $\tilde{\varepsilon}_0, \tilde{\varepsilon}_1$ come canonically from the regular unipotents $\varepsilon_0, \varepsilon_1$ of SG, satisfying

$$(24) \qquad \varepsilon_1 = \tilde{\rho}(\varepsilon_0) = \tilde{\sigma}(\varepsilon_0)$$

$$(25) \qquad \begin{cases} \varepsilon_0 = (\text{Invertible scalar})\,\sigma^{-1}\rho \\ \varepsilon_1 = (- \;\; - \;\; - \;\; -)\,\rho\sigma^{-1} \end{cases}$$

m1015

Marginal note. m1015 for a more convenient|suitable lifting of σ and ρ

and ② $\tilde{\varepsilon}_0, \tilde{\varepsilon}_1$ be in each fiber "non-equivalent", i.e. , let us put

$$(26) \qquad \varepsilon_0 = 1 + N_0, \qquad \varepsilon_1 = 1 + N_1$$

i.e. $\varepsilon_0 = -\dagger\sigma^{-1}\rho$, $\varepsilon_1 = -\dagger\rho\sigma^{-1}$

[p. 352]

Uncertainty retained from cancelled passages.
140-3-s353-D001 $\tilde{\varepsilon}_1$, both
140-3-s353-D002 λ_0
140-3-s353-D003 λ_0

[scan 354]

that N_0 and N_1 be everywhere non-homothetic|nonhomothetic. [p. 478]

Then $\tilde{\varepsilon}_0(\mathbb{G}_a)$, $\tilde{\varepsilon}_1(\mathbb{G}_a)$, $\tilde{\varepsilon}_0 \in \Gamma\tilde{\varepsilon}_0(\mathbb{G}_a)$ define a canonical † pinning of PG hence of G …We will then have, for this pinning

$$(26) \qquad \rho = \begin{pmatrix} 0 & u \\ v & w \end{pmatrix} \qquad \sigma = \begin{pmatrix} 0 & u' \\ v' & w' \end{pmatrix}$$

with $u, v, u', v' \in \Gamma(\mathbb{G}_m)$, $w, w' \in \Gamma\mathbb{G}_a$, †

and we will have

$$(27) \quad \begin{cases} -\sigma^{-1}\rho = \begin{pmatrix} -\dfrac{v}{v'} & \dfrac{uw' - wu'}{u'v'} \\ 0 & -\dfrac{u}{u'} \end{pmatrix} = \varepsilon_0 = \begin{pmatrix} 1 & ? \\ 0 & 1 \end{pmatrix} \\ -\rho\sigma^{-1} = \begin{pmatrix} -\dfrac{u}{w'|u'} & 0 \\ \dfrac{vw' - | + wv'}{w'|u'v'} & -\dfrac{v}{v'} \end{pmatrix} = \varepsilon_1 \overset{\text{déf.}}{=} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \end{cases}$$

by definition of our pinning

so we will need

$$(28) \qquad u/u' = -1, \quad v/v' = -1, \qquad \text{i.e.} \quad u' = -u, \; v' = -v$$

† so

$$-\rho\sigma^{-1} = \begin{pmatrix} 1 & 0 \\ \dfrac{1}{u}(w' + w) & 1 \end{pmatrix}$$

and we need

$$(29) \qquad w' + w = u \qquad \text{i.e.} \qquad w' = w - u$$

in other words combining (26)(27)

$$(30) \qquad \rho + \sigma = uP \quad \text{where} \quad P = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

is the elementary projector associated with the pinning under consideration. m1018

Marginal note. m1018 which determines σ in terms of ρ, the coefficient u of ρ, and P

And we will have

$$(31) \qquad -\sigma^{-1}\rho = \varepsilon_0 = \begin{pmatrix} 1 & u/v \\ 0 & 1 \end{pmatrix}, \qquad N_0 = \begin{pmatrix} 0 & u/v \\ 0 & 0|1 \end{pmatrix}$$

[p. 353]

Uncertainty retained from cancelled passages.
140-3-s354-D001 splitting
140-3-s354-D003 ρ
140-3-s354-D004 $+|-$

[scan 355]

We would like to give intrinsic "numerical criteria" for ρ, σ to have the preceding properties…

Proposition Let ρ, σ be two † sections of G,

Put

ε $\qquad \varepsilon_0 = -\sigma^{-1}\rho, \qquad \varepsilon_1 = -\rho\sigma^{-1}$

so that we thus have

$$\varepsilon_1 = \sigma(\varepsilon_0) = \rho(\varepsilon_0).$$

Equivalent conditions

a) 1°) ε_0 † is regular|regular unipotent|unipotent † ($\Leftrightarrow \varepsilon_1$ regular unipotent) and 2°) ε_0 and ε_1 † independent in each fiber, — i.e. putting

$$\varepsilon_0 = 1 + N_0, \qquad \varepsilon_1 = 1 + N_1$$

N_0 ($\Leftrightarrow N_1$) is regular nilpotent, and N_0 and N_1 are everywhere nonhomothetic.

b) We have

$$\overset{\dagger}{(32)} \quad \left\{ \begin{array}{ll} a) & \det\sigma = \det\rho \\ b) & \operatorname{Tr}\sigma^{-1}\rho = -2 \\ c) & u \overset{\text{dfn}}{=} \operatorname{Tr}(\sigma+\rho) \text{ is invertible} \end{array} \right\}$$

† † m1019

Marginal note. m1019 these two relations express $\det(\varepsilon_0) = 1$, $\operatorname{Tr}\varepsilon_0 = 2$ i.e. ε_0 is unipotent

m1020

Marginal note. m1020 this expresses that ε_0 is regular and independent of ε_1, cf. loc. cit.

cf. proposition page 449′

c) We have

$$(33) \quad \left\{ \begin{array}{ll} a) & \det\sigma = \det\rho \\ b) & \det(\sigma+\rho) = 0 \\ c) & \operatorname{Tr}(\sigma+\rho) \text{ invertible} \end{array} \right.$$

d) We have $\det\sigma = \det\rho$, and there exist an elementary projector P and an invertible scalar $u \in \Gamma\mathbb{G}_{\mathrm{m}}$, such that we have

$$(34) \qquad \sigma + \rho = uP$$

m1022

Marginal note. m1022 N.B. P and u are uniquely determined, $u = \operatorname{Tr}(\sigma+\rho)$, $P = \frac{1}{u}(\sigma+\rho)$

[p. 354]

Uncertainty retained from cancelled passages.

140-3-s355-D001 regular
140-3-s355-D004 are
140-3-s355-D005 everywhere
140-3-s355-D007 not equivalent, i.e.
140-3-s355-D008 (32)
140-3-s355-D009 (33)
140-3-s355-D010 (34)

[scan 356]
[p. 479]

e) $\exists$ isomorphism $\mathrm{G}\ell(2) \overset{\varphi}{\xrightarrow{\sim}} G$, such that we have

$$(35) \quad \left\{ \begin{array}{l} \varphi^{-1}(\varepsilon_1) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \\ \varphi^{-1}(\varepsilon_0) = \begin{pmatrix} 1 & \dagger? \\ 0 & 1 \end{pmatrix} \end{array} \right.$$

N.B. this isomorphism is unique

f) $\exists$ isomorphism $\mathrm{G}\ell(2) \simeq G$ such that we have

$$(36)\qquad \begin{cases} \varphi^{-1}(\rho) = \begin{pmatrix} 0 & u \\ v & w \end{pmatrix} \qquad \varphi^{-1}(\sigma) = \begin{pmatrix} 0 & w'|u' \\ v' & w' \end{pmatrix} \\ \text{with} \qquad u + u' = v + v' = 0, \qquad w + w' = u \end{cases}$$

m1024

Marginal note. m1024 N.B. We will have $P = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$

m1025

Marginal note. m1025 this φ is again unique; We will have

$$(37)\qquad \begin{cases} u = \mathrm{Tr}(\sigma + \rho)\ (= -u') \\ w = \mathrm{Tr}\,\rho \quad (w' = \mathrm{Tr}\,\sigma) \\ -uv = \det\rho = \det\sigma \end{cases}$$

which describes u, w, v in terms of ρ, hence also u', v', w' ...

†

We will have in (35), with

$$(38)\qquad \varphi^{-1}(\varepsilon_0) = \begin{pmatrix} 1 & u/v \\ 0 & 1 \end{pmatrix},$$

and we will have

$$(39)\qquad \sigma + \rho = uP, \quad \text{with } \varphi^{-1}(P) = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Easy proof. The equivalence $a) \Leftrightarrow b)$ seen|seen in proposition page 449′. For $b) \Leftrightarrow c)$, we write

$$\det(\sigma+\rho) = \det\{\sigma(1+\sigma^{-1}\rho)\} = \det\sigma \underbrace{\det(1+\sigma^{-1}\rho)}_{1+\mathrm{Tr}\,\sigma^{-1}\rho+\underbrace{\det\sigma^{-1}\rho}_{(\det\sigma)^{-1}\det\rho}}$$

$$= \det\sigma + \det\rho + (\det\sigma)\,\mathrm{Tr}(\sigma^{-1}\rho)$$

so by virtue of the hypothesis

$$\det\sigma = \det\rho$$

common to $b)$ and $c)$, we will have

$$(*|6)\qquad \det(\sigma+\rho) = \det\sigma\,(2 + \mathrm{Tr}(\sigma^{-1}\rho))$$

so the first side|side is zero iff $\mathrm{Tr}\,\sigma^{-1}\rho = -2$, which was to be proved. [p. 355]

Uncertainty retained from cancelled passages.

140-3-s356-D003 1

140-3-s356-D004 (37)

[scan 357]

Putting $Q = \sigma + \rho$, the equivalence $b \Leftrightarrow c$ comes from the equivalence (valid for every section Q of M)

$$(40)\quad (\det Q = 0, \mathrm{Tr}\,Q \text{ invertible}) \Longleftrightarrow \begin{pmatrix} \exists \text{ elementary projector } P, \text{ and } u \in \Gamma\mathbb{G}_{\mathrm{m}}, \\ \text{such that } Q = uP \end{pmatrix}$$

which follows easily from the results of the preceding n° (it is a corollary of the proposition page 475′, which I forgot to make explicit.

The equivalence with $e)$, $f)$ is also immediate, † and merely summarizes certain preceding calculations.

m1026

Marginal note. m1026 Corollary Under these conditions $1, N_0, N_1, P$ form a basis of M, and ρ and σ are independent linear combinations of † N_0, N_1, P

$$\begin{cases} \rho = v(N_0 + N_1) + wP \\ \sigma = -v(N_0 + N_1) + (u - w)P \end{cases}$$

ρ, σ form an obviously|obvious basis of the sub-Module|submodule of M generated by $(N_0 + N_1)$ and P, hence

$$\rho, \sigma, N_0, 1 \qquad \text{and} \qquad \rho, \sigma, N_1, 1$$

also form bases of M, and in them we can replace either ρ, or σ by P; i.e. in $N_0, N_1, P, 1$ we can replace N_0, or N_1 by ρ or σ.



Our point of view here is that we start from $\tilde{\rho}, \tilde{\sigma}_1$, which we want to describe — the pinning is indeed already defined in terms of $\tilde{\rho}, \tilde{\sigma}$, but there remains the question of normalizing the liftings of $\tilde{\rho}, \tilde{\sigma}$ to ρ, σ. We have already made a first restriction by the formulas

$$\varepsilon_1 = -\sigma^{-1}\rho$$

by eliminating the scalar factor λ_0 — we can normalize|simplify further by multiplying ρ, σ by the factor u^{-1}, so as to obtain the simple expression.

$$(40) \qquad \operatorname{Tr}(\rho + \sigma) = 1 \quad \text{i.e.} \quad \operatorname{Tr}\rho + \operatorname{Tr}\sigma = 1 \quad \text{i.e.} \quad u = 1,$$

which thus means that ρ, σ are of the [p. 356]

Uncertainty retained from cancelled passages.

140-3-s357-D001 it is in|it is a

140-3-s357-D002 N_0

[scan 358]

[p. 480]

form

$$(41) \qquad \rho = \begin{pmatrix} 0 & 1 \\ v & w \end{pmatrix}, \qquad \sigma = \begin{pmatrix} 0 & -1 \\ v' & w' \end{pmatrix}$$

$$v + v' = 0, \quad w + w' = 1 \qquad \text{i.e. } v' = -v, \quad w' = 1 - w,$$

so

$$(42) \qquad \varepsilon_0 = \begin{pmatrix} 1 & 1/v \\ 0 & 1 \end{pmatrix},$$

where we have identified, by φ, $\mathrm{G}\ell(2)_S$ and G.

† Is. there uniqueness of ρ, σ in terms of $\tilde{\rho}, \tilde{\sigma}$, † by the conditions $\varepsilon_1 = -\rho\sigma^{-1}$ (i.e. $\varepsilon_0 = -\sigma\rho^{-1}$) and (40)? I would first like to check carefully an assertion made in passing a certain number of times, namely that the morphism

$$G \longrightarrow PG \simeq G/\mathbb{G}_{\mathrm{m}}$$

induces a bijection

(43)

$$\begin{array}{c}\text{unipotents } \dagger \\ \text{regular of} \\ G\end{array} \longrightarrow \begin{array}{c}\text{unipotents} \\ \text{regular} \\ \text{of } PG\end{array}$$

,

(at least locally † fppf †)

the surjectivity of this map could here serve as a definition for the regular unipotents of PG. So it is a matter of testing injectivity, which also means this: if $\varepsilon, \varepsilon'$ are two regular unipotent sections of M, such that $\tilde{\varepsilon} = \tilde{\varepsilon}'$ i.e. $\varepsilon' = \lambda\varepsilon$ ($\lambda \in \Gamma\mathbb{G}_m$), then $\varepsilon = \varepsilon'$, i.e. $\lambda = 1$. Taking the determinants of the two sides, we will have

$$\lambda^2 = 1, \tag{44}$$

m1027

Marginal note. m1027 So, we will say that a section of PG is regular unipotent if it is locally fppf the image of a regular unipotent section of G.

[p. 357]

Uncertainty retained from cancelled passages.

140-3-s358-D001 will note entirely determined

140-3-s358-D002 by

140-3-s358-D004 *strict*

140-3-s358-D005 fppf

140-3-s358-D006 Zariski

taking traces we find

[scan 359]

$$2\lambda = 2 \tag{45}$$

— these are the necessary and sufficient conditions on λ for † (ε being regular unipotent) $\lambda\varepsilon$ to be still strict^{ly} unipoten^{t} (hence necessari^{ly} regular unipotent). If 2 is invertible on S, more generally if 2 is a regular section of $\mathcal{O}_S$, † these conditions imply $\lambda = 1$, OK, but †outside of this hypothesis (for example in characteristic 2 † over a nonreduced scheme) this is no longer necessarily so. Let us write

$$\lambda = 1 + \nu$$

the conditions (44) (45) read

$$2\nu = 0, \quad \nu^2 = 0, \tag{46}$$

thus for the map (43) to be an injection on S, it is necessary and sufficient that the only sections of S which satisfy (46) are the zero section.

In general, denoting by μ_2' the subgroup scheme of μ_2 defined by (45) (it is in fact an absolute infinitesimal group scheme, † finite over $\mathbb{Z}$, not flat over $\mathbb{Z}$, isomorphic to the spectrum of

$$\mathbb{Z}[T]/(T^2 - 1, 2T - 2) \simeq \mathbb{Z}[S]/(S^2, 2S),$$

also isomorphic to the scheme of dual numbers over $\mathbb{Z}$, relative to the module $\mathbb{Z}/2\mathbb{Z}$ over $\mathbb{Z}$, †

$$\mu_2' \simeq \mathrm{Spec}(\mathbb{Z}\dagger \oplus \underbrace{\mathbb{Z}/2\mathbb{Z}}_{\text{square-zero ideal}}) \quad \ldots$$

[p. 358]

Uncertainty retained from cancelled passages.

140-3-s359-D001 λ

140-3-s359-D002 is reduced

140-3-s359-D003 in the case

140-3-s359-D004 even if

140-3-s359-D005 over

140-3-s359-D006 $\mu_2' \simeq$

140-3-s359-D007 ε

It turns out that μ'_{2S} operates freely by translations on the scheme of regular unipotents (or not) of G, and we have [scan 360] [p. 481]

$$\underline{\text{Unip.rég}}(G)/\mu'_{2S} \xrightarrow{\sim} \underline{\text{Unip.rég}}(PG) \quad . \tag{47}$$

This seems at first sight to be in contradiction with the sequence of Borels

$$\begin{array}{ccccc} \dagger\ B & & \underline{\text{Borels}}(G) & \xrightarrow[B\mapsto B/G_m]{\sim} & \text{Borels}(PG) \qquad \dagger\widetilde{B} \\ \Big\downarrow & & \Big\downarrow\wr & & \Big\downarrow\wr \qquad \Big\downarrow \\ B_u & & \text{“Subgroups} & & \text{“Subgroups} \qquad \widetilde{B}_u \\ & & \dagger\text{unipotent with 1} & \xrightarrow{\sim} & \text{unipotent with 1 parameter”} \\ & & \text{parameter” of } G & & \text{of } PG \end{array} \tag{48}$$

$$\text{that is} \quad L \longmapsto p(L)$$

(where $p : G \to PG$ is the canonic[al] morphism)

Could the difficulty come from the fact that the vertical arrows (surjective, by definition of the notion of “1-parameter unipotent subgroups” of G or of PG) might not be injective, outside the hypothesis that 2 is invertible (i.e. residue characteristic $\neq 2$)? I feel like saying that

$$\underline{B} = \text{Norm}(B_u), \quad \widetilde{B} = \text{Norm}(\widetilde{B}_u) \tag{49}$$

are formulas which allow us to recover $B, \widetilde{B}$ with the help of $B_u, \widetilde{B}_u$,|” — and I have † indeed already checked the first of these formulas, by dint of linear algebra in M (in the form: the 1-parameter unipotent subgroups of G correspond 1-1 to the sections of $\mathbb{P}$ over S, i.e. to the line subbundles D of F…) — [p. 359]

Uncertainty retained from cancelled passages.

140-3-s360-D001 1

140-3-s360-D002 B

140-3-s360-D003 U

140-3-s360-D004 done

but this same point in PG, it is not very trivial. [scan 361]

The difficulty must rather come from the fact that a|that no regular unipotent section $\tilde{\varepsilon}$ of PG does not define (without restriction on S) a 1-parameter unipotent subgroup — whereas for a section ε of regular unipotents|regular unipotent of G

$$\varepsilon = 1 + N,$$

we define one by the formula

$$\varepsilon(t) = 1 + tN\,.$$

We will note that if we replace ε by $\varepsilon' = \lambda\varepsilon$ (with λ satisfying (44) (45) $\lambda^2 = 1$, $2\lambda = 2$), We have

$$\varepsilon' = (1+\nu)(1+N) = 1 + N' \qquad N' = (1+\nu)N + \nu$$

and

$$\varepsilon'(t) = 1 + tN'$$

I say that if $\lambda \neq 1$ i.e. $\nu \neq 0$, † the composite homomorphisms

$$\underline{\tilde{\varepsilon}}, \underline{\tilde{\varepsilon}}' : \mathbb{G}_{a|0} \longrightarrow G \longrightarrow PG$$

† associated to $\varepsilon, \varepsilon'$ will be distinct, although they have the same value for $t = 1$. Indeed, † the derivatives at the origin of $\underline{\tilde{\varepsilon}}, \underline{\tilde{\varepsilon}}'$ are identified with the composites ‖

$$\underline{\mathcal{O}}_S \longrightarrow M \longrightarrow M/\underline{\mathcal{O}}_S = \mid \simeq P\mathfrak{g}$$

[p. 360]

Uncertainty retained from cancelled passages.

140-3-s361-D003 we then have

140-3-s361-D004 will be

140-3-s361-D005 we will have

which send 1 to $\widetilde{N}$ resp. to $\widetilde{N}'$, (where the $\sim$ here denotes passage to the additive quotient) we have [scan 362] [p. 482]

$$\widetilde{N}' = (1+\nu)\widetilde{N}$$

and $\widetilde{N}$ is a section of $P\mathfrak{g}$ everywhere nonzero — thus we can have

$$\widetilde{N}' = \widetilde{N} \quad \text{i.e.} \quad (1+\nu)\widetilde{N} = \widetilde{N} \quad \text{i.e.} \quad \nu\widetilde{N} = 0$$

only if $\nu = 0$, i.e. $\lambda = 1 + \nu = 1$, q.e.d.

In conclusion, we must consider that we give ourselves in PG, not only $\tilde{\rho}$ and $\tilde{\sigma}$ (whence $\tilde{\varepsilon}_0 = \tilde{\sigma}^{-1}\tilde{\rho}$, $\tilde{\varepsilon}_1 = \tilde{\rho}\tilde{\sigma}^{-1}$), but also "1-parameter subgroups" (parametrized)

$$\dagger \xrightarrow{\dagger} \quad ,$$

$$\begin{cases} \underline{\tilde{\varepsilon}}_0 : \mathbb{G}_{a|0} \longrightarrow PG \\ \\ \underline{\tilde{\varepsilon}}_1 : \mathbb{G}_{a|0} \longrightarrow PG \end{cases} . \tag{49}$$

related of course by

$$\begin{cases} \underline{\tilde{\varepsilon}}_1 = \rho(\underline{\tilde{\varepsilon}}_0)(= \sigma(\underline{\tilde{\varepsilon}}_0)) \quad \text{i.e.} \\ \underline{\tilde{\varepsilon}}_1(t) = \rho(\underline{\tilde{\varepsilon}}_0(t)) = \sigma(\underline{\tilde{\varepsilon}}_0(t|u)) \end{cases} \tag{50}$$

which means precisely that we give ourselves, in addition to $\tilde{\rho}, \tilde{\sigma}$, a lifting† regular unipoten^{t} ε_0 of $\tilde{\varepsilon}_0$ († the choice is therefore not a priori unique), and we deduce ε_1 from it by

$$\dagger \quad \varepsilon_1 = \tilde{\rho}(\varepsilon_0) = \tilde{\sigma}(\varepsilon_0)$$

m1029

Marginal note. m1029 we will necessarily have a relation $\sigma^{-1}\rho = -\lambda_0\varepsilon_0$ hence † $\sigma(\varepsilon_0)|\sigma^{-1}(\varepsilon_0) = \rho(\varepsilon_0) =$ [p. 361]

Uncertainty retained from cancelled passages.

140-3-s362-D002 $a|0$

140-3-s362-D003 $\underline{\tilde{\varepsilon}}_0$

140-3-s362-D005 [illeg. 1]

140-3-s362-D006 which|that

140-3-s362-D007 ρ

140-3-s362-D008 int(ε_0)|i.e. $\sigma(\varepsilon_0)$ $\quad \varepsilon_0$

† Thus, we have adjusted the † description of the data, so as to † be able to give precise normalization statements — namely [scan 363] [p. 4]

Proposition Let $\tilde{\rho}, \tilde{\sigma}$ be sections of PG, and $\underline{\tilde{\varepsilon}}_0$ a "strict" additive subgroup with 1 parameter m1030

Marginal note. m1030 (i.e. a monomorphism and whose image is a $\widetilde{B}_u$)

$$\underline{\tilde{\varepsilon}}_0 : \mathbb{G}_{a|0} \longrightarrow |\!-\!- PG \quad .$$

such|such that we have $\times|*$

$$\tilde{\sigma}^{-1}\tilde{\rho} = \underline{\tilde{\varepsilon}}_0(1) \overset{\text{dfn}}{=} \tilde{\varepsilon}_0 \tag{51}$$

whence

$$\begin{cases} \tilde{\sigma}(\underline{\tilde{\varepsilon}}_0) = \tilde{\rho}(\underline{\tilde{\varepsilon}}_0) \overset{\text{dfn}}{=} \underline{\tilde{\varepsilon}}_1, \quad \text{and} \\ \tilde{\rho}\tilde{\sigma}^{-1} = \underline{\tilde{\varepsilon}}_1(1) \overset{\text{dfn}}{=} \tilde{\varepsilon}_1 = \rho(\tilde{\varepsilon}_0) = \sigma(\tilde{\varepsilon}_0) \end{cases} \tag{52}$$

Then:

1) ("For the record") $\exists$ unique $\varepsilon_0, \varepsilon_1$ † regular unipotent sections of G such that (setting $\varepsilon_i = \dagger 1 + N_i$, $\underline{\varepsilon}_i(t) = 1 + tN_i$) we have

$$\underline{\tilde{\varepsilon}}_i(t) = \varepsilon_i(t)^{\sim} \ . \tag{53}$$

2) We will then have

$$\dot{\varepsilon}_1|\varepsilon_1 = \tilde{\sigma}(\varepsilon_0) = \tilde{\rho}(\varepsilon_0) \tag{54}$$

†

†

3) Let|Being ρ, σ be liftings of $\tilde{\rho}, \tilde{\sigma}$, we will then have

$$\dagger \qquad \varepsilon_0 = -\lambda_0\sigma^{-1}\rho, \quad \varepsilon_1 = -\lambda_0\rho\sigma^{-1} \tag{55}$$

for $\lambda_0 \in \Gamma G_m$ uniquely determined. For a suitable choice of ρ (when σ is already chosen) we can assume $\lambda_0 = 1$ (which determines ρ uniquely in † terms of σ, and conversely) [p. 362]

Uncertainty retained from cancelled passages.

140-3-s363-D001 And

140-3-s363-D002 precision|position

140-3-s363-D003 adjust

140-3-s363-D004 strict^{ly}

140-3-s363-D006 Id+

140-3-s363-D008 ρ

140-3-s363-D010 in

140-3-s363-D011 $\rho\sigma^{-1}\rho$

140-3-s363-D012 function

[scan 364]

4) so that we have [p. 483]

$$\varepsilon_0 = -\sigma^{-1}\rho \qquad (\text{hence } \varepsilon_1 = -\rho\sigma^{-1}) \tag{56}$$

4) ρ and σ being chosen so as to satisfy (56), for $\varepsilon_0, \varepsilon_1$ to be independent (in each fiber), i.e. for us to have N_0 and N_1 nonhomothetic in every fiber, it is necessary and sufficient that we have

$$\mathrm{Tr}(\rho+\sigma) \quad \text{invertible} \tag{57}$$

(cf. also proposition page 478' for other equivalent conditions).

5) For $\varepsilon_0, \varepsilon_1$ to be independent (in each fiber), it is necessary and sufficient that we can find (locally fppf) ρ, σ above $\tilde\rho, \tilde\sigma$ so as to satisfy (56) and

$$\mathrm{Tr}(\rho+\sigma) = 1 \qquad \text{i.e.} \quad \mathrm{Tr}\,\rho + \mathrm{Tr}\,\sigma = 1 \tag{57'}$$

and this determines ρ, σ uniquely (in terms of $\tilde\rho, \tilde\sigma, \tilde\varepsilon_0$).

$\rho+\sigma$ an elementary projector $\Longleftrightarrow$

Remarks 1) The conditions on (ρ, σ) considered in the proposition on page 478', and the normalization $u|v_1 = 1$ i.e. $\mathrm{Tr}(\sigma+\rho) = 1$ i.e. $\sigma+\rho$ elementary projector, are symmetric in (ρ, σ): they are satisfied for (ρ, σ) iff they are for (σ, ρ) — † to be more precise let us put

$$\rho' = \sigma, \ \sigma' = \rho \qquad \varepsilon_0' = \underbrace{-\sigma'^{-1}\rho'}_{=-\rho^{-1}\sigma}, \quad \varepsilon_1' = \underbrace{-\rho'\sigma'^{-1}}_{-\sigma\rho^{-1}} \tag{58}$$

then we will have

$$\varepsilon_0' = \varepsilon_0^{-1}, \qquad \varepsilon_1' = \varepsilon_1^{-1}. \tag{59}$$

Likewise, (ρ^{-1}, σ^{-1}) satisfies the same conditions on page 478', and putting|taking [p. 363]

Uncertainty retained from cancelled passages.

140-3-s364-D003 the interchange of ρ and σ|the interchange is made

[scan 365]

$$\rho'' = \rho^{-1}, \ \sigma'' = \sigma^{-1}, \qquad \varepsilon_0'' = \underbrace{-\sigma''^{-1}\rho''}_{-\sigma\rho^{-1}}, \quad \varepsilon_1'' = \underbrace{-\rho''\sigma''^{-1}}_{-\rho^{-1}\sigma} \tag{60}$$

we will have

$$\varepsilon_0'' = \varepsilon_{1|0}^{-1}, \qquad \varepsilon_1'' = \varepsilon_0^{-1}, \tag{61}$$

but we will take note that in general we will no longer have $\mathrm{Tr}(\rho''+\sigma'') = 1$, but (by an immediate matrix calculation)

$$\mathrm{Tr}(\rho''+\sigma'') = \mathrm{Tr}(\rho^{-1}+\sigma^{-1}) = -\frac{1}{v|\nu} = (\det\rho)^{-1} = (\det\sigma)^{-1} \tag{62}$$

(assuming of course already $v_1|u_1 = 1$ i.e. $\mathrm{Tr}(\rho+\sigma) = 1$) — this is equal to 1 only if

$$\det\rho \ (= \det\sigma) = 1 \tag{63}$$

— this is the condition of the section|relation 3°) of $(A1)_2|(41)_2$ (page 463'), which therefore also means that the normalization $v_1|u_1 = 1$ is stable under the transformation $\rho \mapsto \rho^{-1}$, $\sigma \mapsto \sigma^{-1}$.

On the other hand, the choice of the pinning associated with (ρ, σ) (i.e. $\tilde\rho, \tilde\sigma$) [in which the emphasis has been put on $\varepsilon_1 = -\rho\sigma^{-1}$, which receives the simple form

$$\varepsilon_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix},$$

whereas $\varepsilon_0 = -\sigma^{-1}\rho$ has the more complicated form

$$\varepsilon_0 = \begin{pmatrix} 1 & v_1/v|u_1/v \\ 0 & 1 \end{pmatrix} \qquad \left(= \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \text{ in the modular case}\right)$$

] implicitly brings in the order in which we give ourselves ρ, σ — if we change it, we change the pinning by the change from ε_1 to $\varepsilon_1' = \varepsilon_1^{-1}$.

In fact, the situation offered the choice between four "natural" pinnings, associated with the pair of [p. 364]

[scan 366]

one-parameter subgroups $\{L_0, L_1\}$, corresponding to the four sections $\varepsilon_0, \varepsilon_0^{-1}, \varepsilon_1, \varepsilon_1^{-1}$ of the [p. 484] one † and of the other.

We preferred to normalize with the help of L_1, because of the most convenient form of the loop equation (which I finally arrived at moreover after some rather heavy peregrinations, where it is L_0 that I obstinately kept favoring in the formulas — cf. §44, the definitive notation appears only in §45 —. The preference given to † $-\rho\sigma^{-1}$ (denoted ε_1) over its inverse $-\sigma\rho^{-1}$ (denoted ε_1^{-1}) comes from the preference (justified by the facts) given to ρ over $\sigma = \sigma_\infty$, in the loop formula written in the form

$$[\alpha, \rho] = [\beta, \sigma]\varepsilon_1(\nu)$$

instead of writing it

$$[\beta, \sigma] = [\alpha, \rho]\varepsilon_1'(\nu) \qquad (\varepsilon_1' = \varepsilon_1^{-1})$$

So it is ε_1 that then deserved to have the simplest possible matrix form — we took

$$\varepsilon_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \ \ldots$$

2) In terms of the systems

$$\begin{array}{ccc} \tilde{\ell}'_\infty, & \tilde{\ell}'_1, & \tilde{\ell}'_0 \\ \| & \| & \| \\ \tilde{\rho}^{-1} & \tilde{\sigma} & \tilde{\varepsilon}_0 \end{array} \quad \text{satisfying} \quad \tilde{\ell}'_\infty \cdot \tilde{\ell}'_1 \cdot \tilde{\ell}'_0 = 1 \tag{64}$$

(i.e. $\tilde{\rho}^{-1} \cdot \tilde{\sigma} \cdot \tilde{\varepsilon}_0 = 1$ i.e. $\tilde{\varepsilon}_0 = \tilde{\sigma}^{-1}\tilde{\rho}$), the group $\mathfrak{S}_3'|\mathcal{G}_3'$ operates on them (cf. §42 I) by cyclic permutations, and by the change into $(\tilde{\varepsilon}_0^{-1}, \tilde{\sigma}^{-1}, (\tilde{\rho}^{-1})^{-1} = \tilde{\rho})$, which in its turn by cyclic permutation gives $(\tilde{\sigma}^{-1}, \tilde{\rho}, \tilde{\varepsilon}_0^{-1})$ — [p. 365]

Uncertainty retained from cancelled passages.

140-3-s366-D002 or

140-3-s366-D003 ε

[scan 367]

the passage from $(\tilde{\rho}^{-1}, \tilde{\sigma}, \tilde{\varepsilon}_0)$ to $(\tilde{\sigma}^{-1}, \tilde{\rho}, \tilde{\varepsilon}_0^{-1})$ then corresponds to the interchange $(\rho, \sigma) \mapsto (\sigma, \rho)$. How † to interpret the transformation $(\rho, \sigma) \mapsto (\rho^{-1}, \sigma^{-1})$, i.e. $(\tilde{\rho}^{-1}, \tilde{\sigma}, \tilde{\varepsilon}_0) \mapsto (\tilde{\rho}, \tilde{\sigma}^{-1}, \tilde{\varepsilon}_1)$?

Let us interpret the triples as homomorphisms.

$$\Pi_{0,3} \xrightarrow{\varphi|\psi} \Gamma PG$$

$$\ell_0 \longmapsto \ell'_0$$

$$\ell_1 \longmapsto \ell'_1$$

$$\ell_\infty \longmapsto \ell'_\infty$$

we see that the change $(\rho, \sigma) \mapsto (\sigma, \rho)$ corresponds to the composition of φ with the automorphism of $\Pi_{0,3}$ such that

$$\ell_0 \mapsto \ell_1^{-1} \qquad \ell_1 \mapsto \ell_0^{-1} \qquad \text{thus } \ell_\infty \mapsto \ell_\infty^{-1}$$

i.e. with $\underline{\tau'_\infty}$. Likewise the transformation $(\rho, \sigma) \mapsto (\rho^{-1}, \sigma^{-1})$ corresponds to composition with the automorphism

$$\ell_0 \mapsto \ell_0^{-1} \qquad \text{hence } \ell_\infty \mapsto \ell_0 \ell_1$$

$$\ell_1 \mapsto \ell_1^{-1}$$

, which is none other than $\underline{\tau_\infty} = \sigma_\infty \tau'_\infty$.

Thus the group of four transformations † on the set of pairs (ρ, σ) that we considered in $\underline{1^\circ)}$ is interpreted † as the subgroup $\simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ generated by $\tau'_\infty, \tau_\infty$ in $\mathrm{Aut}_{\mathrm{lac|ext}}(\Pi_{0,3}) = \mathfrak{S}'_{0,3} | \mathcal{G}_{0,3}$, which stabilizes $L_0 \subset \Pi_{0,3}$ (thus its action on the|this third factor $\tilde{\varepsilon}_0 = \ell'_0$ will be to preserve it or replace it by its inverse, † which will have the virtue of † preserving its unipotent character, which we care about !). [p. 366]

Uncertainty retained from cancelled passages.

140-3-s367-D001 to apply some to|by applying to

140-3-s367-D002 which operates|that we have

140-3-s367-D003 as

140-3-s367-D006).

140-3-s367-D007 it|it

[scan 368]

$\overline{\text{VIII}}$ Recapitulation on the equation [p. 485]

$$[\alpha, \rho] = [\beta, \sigma] \varepsilon_1(\nu). \tag{1}$$

The groups †, $S\mathcal{G}^*_{\rho,\sigma}$ and its variants.

The essential result we have arrived at in ($\overline{\text{III}}$), is the possibility of parametrizing the solutions of such an equation in α, β, ν (under suitable hypotheses on ρ, σ) by a group $B'_0 \times T_\rho \times T_\sigma$, where B'_0 is a "pre-Borel|pri-Borel" in G (taking, for a suitable basis, the form

$$B'_0 = \left\{ \begin{pmatrix} \mu & \lambda \\ 0 & 1 \end{pmatrix} \;\middle|\; \begin{array}{l} \mu \in \Gamma \mathbb{G}_m \\ \lambda \in \Gamma \mathbb{G}_a \end{array} \right.$$

and T_ρ, T_σ are the centralizers|stabilizers of the regular sections ρ, σ of G, by the morphism

$$\left\{ \begin{array}{ccc} B'_0 \times T_\rho \times T_\sigma & \longrightarrow & \mathcal{G}_{\rho,\sigma} \\ (\underline{u}, \underline{a}, \underline{b}) & \longmapsto & (\underline{u}\underline{a}^{-1}, \underline{u}\underline{b}^{-1}, \nu | \delta(\underline{u})) \end{array} \right. \tag{2}$$

(where the fact that $\nu(\underline{u}) = \det(\underline{u}) - 1$ is not of capital importance for the moment).

m1032

Marginal note. m1032 We now start from $\rho, \sigma, B'_0, \varepsilon_1$ (hence L_1) and we ask the question under what conditions, one can have a parametrization (2).

To say that this morphism from the first|from the first ones into $G \times G \times E'$ maps indeed into the solutions of (1), means also that we have identically

$$\underline{u}\rho(\underline{u})^{-1} = \underline{u}\sigma(u)^{-1}\varepsilon_1(\underline{\nu}(u))$$

which can also be written, canceling $\underline{u}|$ = and taking inverses

$$\rho(\underline{u}) = \varepsilon_1(-\underline{\nu}(u))\sigma(\underline{u}). \tag{3}$$

The possibility of finding $\underline{\nu} : B'_0 \longrightarrow \mathbb{G}_a$ (morphism of schemes) will give|would give, via (2), a morphism $B'_0 \times T_\rho \times T_\sigma \longrightarrow \mathcal{G}_{\rho,\sigma} \overset{\text{dfn}}{=}$ scheme [p. 367]

Uncertainty retained from cancelled passages.

140-3-s368-D001 $\mathcal{G}^{*0}_{\rho,\sigma}$

[scan 369]

of solutions of (1), is therefore equivalent to that † σ and ρ † the same action on B'_0, modulo multiplication on the left by † the subgroup † unipoten^{t} with one parameter L_1 which enters into the very writing of the equation (1) we are studying. This is hard to conceive unless one admits that $L_1 \subset \rho(B'_0) \overset{\text{df}}{=} B'_1$, which moreover implies $\sigma(B'_0) \subset B'_1$ (and by a known argument $\sigma(B'_0) = B'_1$) and $L_1 = (B'_1)_u$. Conversely, if we have $\rho(B'_0) = \sigma(B'_0)$, † i.e. σ^{-1}†ρ normalizes B'_0, i.e. $\sigma^{-1}\rho$ a section of B_0, then $\sigma(u) = \sigma u \sigma^{-1}$ and $\rho(u) = \rho u \rho^{-1}$ are two sections of B'_1 which have the same determinant – hence abelianizatio^{n}, hence which do coincide modulo $(B'_1)_u$.

Thus we see that the essential property for obtaining solutions of (1) via (2), is to have

$$(4) \qquad (\sigma^{-1}\rho)(B'_0) = B'_0 \qquad \text{i.e.} \quad \sigma^{-1}\rho \in \Gamma B_0 \,..$$

m1033

Marginal note. m1033 N.B. This condition (if $\sigma^{-1}\rho$ regular, i.e. ρ and σ are not homothetic in any fiber), practically determines B_0 – † over a field, there is an ambiguity possible between|of order two (the two Borels containing the element in question).

For uniqueness of the representation of a triple (α, β, ν) in the form $(\underline{ua}^{-1}, \underline{ub}^{-1}, \underline{\nu}(\underline{u}))$, it is necessary and it suffices that we have

$$(5) \qquad T_\sigma \cap T_\rho \cap B'_0 = \{1\},$$

m1034

Marginal note. m1034 It suffices for example that we have $T_\rho \cap T_\sigma = G_m$ and $G_m \cap B'_0 = \{1\}$.

† the indeterminacy is [p. 368]

Uncertainty retained from cancelled passages.

140-3-s369-D001 that
140-3-s369-D003 have
140-3-s369-D004 L_1
140-3-s369-D005 L_1
140-3-s369-D006 then for every
140-3-s369-D007 $\rho(-)$
140-3-s369-D009 determines
140-3-s369-D011 our having ρ =

[scan 370]

in the form [p. 486]

$$(\underline{u},\underline{a},\underline{b}) \longmapsto (\underline{u}l,\underline{a}l,\underline{b}l)$$

, with l a section of $T_\rho \cap T_\sigma \cap B'_0$. There remains the question of the surjectivity of (2) – can every solution of (1) indeed be written in the form $(ua^{-1},\underline{u}b^{-1},\nu(\underline{u}))$? It amounts to the same (if ρ,σ are regular) to say that † we have surjectivity in

$$B'_0 \longrightarrow \mathcal{X}_{\rho,\sigma}$$

(6)

$$\underline{u} \longmapsto ([\underline{u},\rho],[\underline{u},\sigma],\underline{\nu}(\underline{u}))$$

where $\mathcal{X}_{\rho,\sigma}$ is the scheme of simultaneous solutions in † ζ,η,ν of

$$(7) \qquad \det\zeta = \det\eta = 1, \quad \operatorname{Tr}\zeta\rho = \operatorname{Tr}\rho, \quad \operatorname{Tr}\eta\sigma = \operatorname{Tr}\sigma, \quad \zeta = \eta\varepsilon_1(\nu).$$

It was essentially to obtain this surjectivity that we had been led in section (II) to make the draconian hypotheses (pages 435',436) which are expressed, at the cost of an étale, by the conditions †

$$(8) \qquad \begin{cases} \text{a)} & \sigma^{-1}\rho \text{ a section of } (B_0)_u = L_0 \quad \text{(and not only of } B_0\text{)} \\ \text{b)} & B_0 \text{ and } B_1 = \rho(B_0) = \dagger\sigma(B_0) \text{ in "general position",} \\ & \text{i.e. } L_0 \text{ and } L_1 = \rho(L_0) = \sigma(L_0) \text{ independent;} \\ & \text{(in each fiber)} \end{cases}$$

but it is not clear whether these conditions (suggested by earlier calculations, and indeed giving rise to some|some very pretty calculations) are really indispensable for the result of parametrizati- [p. 369]

Uncertainty retained from cancelled passages.

140-3-s370-D001 it

140-3-s370-D003 $\underline{\nu}$,

140-3-s370-D004 that|for

140-3-s370-D005 ρ

[scan 371]

In our context it is quite natural to assume $\varepsilon_0 = \sigma^{-1}\rho| - \sigma^{-1}\rho$ unipotent (or at least quasi-unipotent, i.e. † an iterate ε_0^n unipotent) but condition b) of (8), more subtle, appears less imperative. The sequel will show whether this restriction is judicious.

In the case where one considers an equation

$$(9) \qquad \underbrace{[\alpha,\rho]}_{\zeta} = \varepsilon\,\underbrace{[\beta,\sigma]}_{\eta}\,\varepsilon_1(\nu) \qquad \dagger$$

with a parameter ε (which can in particular take the important value -1) $\in \Gamma\mu_2$, one|where we have for the first time in these calculations destroyed the symmetry between ρ and σ, by introducing the condition $\operatorname{Tr}\sigma = 0$ as a condition of stability of the scheme $\mathcal{X}_\sigma$ of the η satisfying

$$\det\eta = 1, \qquad \operatorname{Tr}\eta\sigma = \operatorname{Tr}\sigma$$

under μ_2. The essential result we were after, was the isomorphism

$$B'_0 \times T_\rho \times N'_\sigma \xrightarrow{\ \sim\ } \mathcal{G}^*_{\rho,\sigma}$$

(10)

$$(\underline{u},\underline{a},\underline{b}) \longmapsto (\alpha = \underline{u}\underline{a}^{-1}, \beta = \underline{u}\underline{b}^{-1}, \mu = \det\underline{u}, \varepsilon = \varepsilon(\underline{b}))$$

where $N'_\sigma \subset N_\sigma$ is now the normalizer of T_σ – defined by the condition

$$(11). \qquad \underline{b}\,\sigma(\underline{b})^{-1} = \varepsilon(\underline{b})\dagger \qquad \text{i.e.} \quad \underline{b}(\sigma) = \varepsilon(\underline{b})\sigma$$

[p. 370]

Uncertainty retained from cancelled passages.
140-3-s371-D001 a par
140-3-s371-D002 ε
140-3-s371-D003 σ

[scan 372]

That the morphism (10) does map into $\mathcal{G}^*_{\rho,\sigma}$, is trivial by definition of N'_σ and via the analogous fact for (2) – without a hypothesis of the kind $\operatorname{Tr}\sigma = 0$. But the possibility of obtaining (local[ly]) any ε via a $\underline{b} \in \Gamma N'_\sigma$ – or simply $\varepsilon = -1$, if 2 invertible) requires precisely $\operatorname{Tr}\sigma = 0$. [p. 487]

Of course, we are going to consider $\mathcal{G}^*_{\rho,\sigma}$ as a group scheme, by transport of structure from the product group structure of $B'_0 \times T_\rho \times N'_\sigma$. We single out in it the subgroup

$$(12) \qquad \mathcal{G}^{\flat|b*}_{\rho,\sigma}(0) \quad \text{corresponding to :} \quad T'_0 \times T_\rho \times N'_\sigma$$

where

$$(13) \qquad T_0 = B'_0 \cap B'_1$$

is the distinguished maximal torus of B'_0, which the canonical pinning is described as

$$(14) \qquad T_0 = \left\{ \begin{pmatrix} \mu & 0 \\ 0 & 1 \end{pmatrix} \right\} \Big| \mu \in \Gamma G_m \Big\}.$$

We likewise set

$$(15) \qquad \mathcal{G}^{\flat|b}_{\rho,\sigma}(0) = \mathcal{G}_{\rho,\sigma} \cap \mathcal{G}^{*\flat|b}_{\rho,\sigma}(0), \quad \text{corresponding to :} \ T'_0 \times T_\rho \times T_\sigma$$

[p. 371]
[scan 373]

We shall thus have semidirect product structures

$$(16) \qquad \mathcal{G}_{\rho,\sigma} \simeq \mathcal{G}^{\flat|b}_{\rho,\sigma}(0) \cdot \Lambda_0, \qquad \mathcal{G}^*_{\rho,\sigma} \simeq \mathcal{G}^{*\flat|b}_{\rho,\sigma}(0) \cdot \Lambda_0,$$

where

$$(17) \qquad \begin{cases} \Lambda_0 \subset \mathcal{G}_{\rho,\sigma}, \quad \Lambda_0 = \left\{ \left(\alpha = \underset{\varepsilon_0^{n|u}}{\lambda}, \beta = \underset{\varepsilon_0^{n|u}}{\lambda}, \mu = 1 \right) \Big| \lambda \in \Gamma L_0 \right\} \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad (n|u \in \Gamma G_a) \\ \text{corresponding to:} \quad L_0 \times \{1\} \times \{1\} \subset B'_0 \times T_\rho \times T_\sigma. \end{cases}$$

We shall finally consider another subgroup scheme of $\mathcal{G}^*_{\rho,\sigma}$, by the conditions

$$(18) \qquad \begin{cases} \det\alpha \in \Gamma\mu_2 \quad \text{i.e. } (\det\alpha)^2 = 1 \\ \det\beta = 1 \end{cases} \Bigg| \quad \text{whence } S\mathcal{G}^{*|\flat}_{\rho,\sigma} \subset \mathcal{G}^*_{\rho,\sigma}$$

Remark † $S\mathcal{G}^*_{\rho,\sigma}$ can be interpreted as the scheme of systems

$$(\alpha, \beta, \mu, \varepsilon, \varepsilon')$$

satisfying

$$(19) \qquad [\alpha, \rho] = \varepsilon[\beta, \sigma]\varepsilon_1(\mu - 1), \quad \det\alpha = \varepsilon', \quad \det\beta = 1, \quad \underline{\varepsilon^2 = \varepsilon'^2 = 1}$$

m1035

Marginal note. m1035 N.B. $\varepsilon^2 = 1$ already follows from the first relation.

In terms of the systems

$$(\underline{u}, \underline{a}, \underline{b}) \in \Gamma B'_0 \times T_\rho \times N'_\sigma$$

the equations (19) (of which the first|first is automatic), reduce to:

$$(20) \qquad \det\underline{u} = \det\underline{b} = \varepsilon' \det\underline{a}, \qquad \varepsilon'^2 = 1,$$

[p. 372]

Uncertainty retained from cancelled passages.
140-3-s373-D002 $\varepsilon' \longleftarrow (\alpha,\beta), |\varepsilon'(\alpha,\beta),$

which show that $S\mathcal{G}^*_{\rho,\sigma}$ is indeed a subgroup of $\mathcal{G}^*_{\rho,\sigma}$. This subgroup contains Λ_0 (corresponding: $L_0 \times \{1\} \times \{1\} | \{(1,1)\} \subset B'_0 \times T_\rho \times N'_\sigma$) — which corresponds to the case where † [scan 374] [p. 488]

$$\det \underline{u} = \det \underline{b} = \det \underline{a} = 1 \qquad (\varepsilon' = 1), \tag{21}$$

and we have a $\frac{1}{2}$ direct product structure

$$S\mathcal{G}^*_{\rho,\sigma} \simeq S\mathcal{G}^{*\flat|b}_{\rho,\sigma}(0) \,.\, \Lambda_0 \tag{22}$$

where

$$S\mathcal{G}^*_{\rho,\sigma}(0) = S\mathcal{G}^*_{\rho,\sigma} \cap \mathcal{G}^{*\flat|b}_{\rho,\sigma}(0) \tag{23}$$

corresponds to the subgroup|subgroup $\dagger T_0 | T'_0 \times T_\rho \times N'_\sigma$ defined by the same equations (20). Of course, we shall also have

$$S\mathcal{G}_{\rho,\sigma}, \quad S\mathcal{G}^{\flat|b}_{\rho,\sigma}(0) \quad (\text{without } *) \tag{24}$$

intersections of $S\mathcal{G}^*_{\rho,\sigma}$, $S\mathcal{G}^{\flat|b*}_{\rho,\sigma}(0)$ with $S\mathcal{G}_{\rho\sigma}$ — deduced from the expression of these in terms of $(B'_0, T_0|T'_0), T_\rho, N'_\sigma$ by replacing N'_σ with T_σ.

To make explicit the structure of the $S\mathcal{G}^*_{\rho,\sigma}$, we are reduced via (22) to studying the group

$$(25) \quad S\mathcal{G}^{*\flat|b}_{\rho,\sigma}(0) \hookrightarrow T_0|T'_0 \times T_\rho \times N'_\sigma$$ [p. 373]

Uncertainty retained from cancelled passages.
140-3-s374-D001 we have
140-3-s374-D003 B'_0

described by equations (20). But we have [scan 375]

$$T_0 \xrightarrow{\sim} \mathbb{G}_m$$

$$\underline{u} \longmapsto \det \underline{u}$$

, which implies that the projection

$$S\mathcal{G}^{*\flat|b}_{\rho,\sigma}(0) \longrightarrow T_\rho \times N'_\sigma$$

$$(\underline{u}\underline{a}^{-1}, \underline{u}\underline{b}^{-1}, \det \underline{u}, \varepsilon = \varepsilon(\underline{b})) \longrightarrow (\underline{a}, \underline{b})$$

$$\wr$$

$$(\underline{u}, \underline{a}, \underline{b})$$

is a monomorphism, and induces an isomorphism of $S\mathcal{G}^*_{\rho,\sigma}(0)$ onto the subgroup|subgroup of $T_\rho \times N'_\sigma$ defined|defined by the equation

$$\det \underline{b} = \varepsilon' \det \underline{a}, \qquad \varepsilon'^2 = 1 \tag{21}$$

we shall have

$$(\det \underline{b})^2 = (\det \underline{a})^2.$$

In terms of the homomorphisms (determinants, in short)

$$T_\rho \xrightarrow{\delta_\rho} \mathbb{G}_m, \qquad N'_\sigma \xrightarrow{\delta_\sigma} \mathbb{G}_m$$

this amounts to taking the inverse image of the diagonal of $\mathbb{G}_m \times \mathbb{G}_m$ under $\delta_\rho^2, \delta_\sigma^2$. We see at once (since $\delta_\rho, \delta_\sigma$ are epimorphisms) that the homomorphism

$$(22) \quad S\mathcal{G}^{*\flat|b}_{\rho,\sigma}(0) \longrightarrow \mu_2 \times \mu_2$$

$$(\alpha, \beta, \mu, \varepsilon, \varepsilon') \longmapsto (\varepsilon, \varepsilon')$$

$$\text{i.e.} \quad (\underline{u}, \underline{a}, \underline{b}) \longmapsto \left(\underline{\varepsilon}(\underline{b}), \underbrace{\det \underline{b} / \det \underline{a}}_{\det(\underline{b}\underline{a}^{-1})} \right)$$

is an epimorphism, and thus gives rise to an exact sequence [p. 374]

[scan 376]

[p. 489]

$$(23) \quad 1 \to S\mathcal{G}^{*\circ\flat|b}_{\rho,\sigma}(0) \to S\mathcal{G}^{*\flat|b}_{\rho,\sigma}(0) \to \mu_2 \times \mu_2 \to 1$$

where $S\mathcal{G}^{\circ\flat|\circ}_{\rho,\sigma}(0)$ is isomorphic to the subscheme of $T_\rho \times T_\sigma$ defined by the equation

$$\det \underline{a} = \det \underline{b} \tag{24}$$

and thus in turn breaks down into

$$(25) \quad 1 \to ST^{\flat|b}_\rho \times ST^{\flat|b}_\sigma \to S\mathcal{G}^{\circ\flat|b}_{\rho,\sigma}(0) \longrightarrow \mathbb{G}_m \longrightarrow 1$$

$$\dagger(\underline{a}, \underline{b}) \longmapsto \det \underline{a} = \det$$

where we have set

$$\begin{cases} ST^{\flat|b}_\rho = \operatorname{Ker}\left(T_\rho \xrightarrow{\delta_\rho} \mathbb{G}_m\right) \\ ST^{\flat|b}_\sigma = \operatorname{Ker}\left(T_\sigma \xrightarrow{\delta_\sigma} \mathbb{G}_m\right) \end{cases} \tag{26}$$

If for example ρ (non-s|non-g) is $\frac{1}{2}$ simple, hence T_ρ a maximal torus, then $ST^{\flat|b}_\rho$ is itself a torus (of dimension 1) — ditto for σ. Thus, $\dagger$ in the "modular case"

$$\rho = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \qquad \sigma = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

$ST^{\flat|b}_\rho$ is a torus except in characteristic 3, and $ST^{\flat|b}_\sigma$ a torus except in characteristic 2. Thus, except in characteristic 2,3, $S\mathcal{G}^{\circ\flat|b}_{\rho,\sigma}(0)$ is a torus of dimension 3.

To finish determining the structure of $S\mathcal{G}^*_{\rho,\sigma}$ in terms of $\mathcal{G}^{*\circ}_{\rho,\sigma}(0)$ and of the decomposition (22) as a semidirect product, we must make explicit the operation of [p. 375]

Uncertainty retained from cancelled passages.
140-3-s376-D001 ($\underline{u}$, $|\underline{u}$
140-3-s376-D003 [illeg. 1]

$S\mathcal{G}^{*\circ}_{\rho,\sigma}(0)$ on Λ_0, by interpreting $S\mathcal{G}^{*\circ}_{\rho,\sigma}(0)$ as a subgroup of $T_\rho \times N'_\sigma$, and Λ_0 as $L_0 \simeq \mathbb{G}_{a|0}$ via $\lambda \mapsto \varepsilon_0(\lambda)$. [scan 377]

Now the relations

$$\underset{\dagger}{\underline{u}}(\varepsilon_0(\lambda))\underline{u}^{-1} = \varepsilon_0(\det(\underline{u})\lambda), \qquad \det(\underline{u}) = \det(\underline{b}) = \delta_\sigma(\underline{b})$$

show that the operation takes place via the composite character

$$(27) \quad S\mathcal{G}^{*\circ|\flat}_{\rho,\sigma}(0) \longrightarrow N'_\sigma \xrightarrow{\delta_\sigma} \mathbb{G}_m$$

, OK.

For what follows, we shall need a canonical homomorphism

$$(28) \quad \begin{array}{ccc} \mathcal{G}^*_{\rho,\sigma} & \xrightarrow{\theta|\Theta} & G \\ & \underset{\dagger}{\swarrow} & \nearrow \\ & B_0 \times T_\rho \times N'_\sigma|T'_\sigma & \end{array}$$

which identifies with the first projection

$$(\underline{u}, \underline{a}, \underline{b}) \longmapsto \underline{u}$$

of the product, but considered|considered as a morphism into G thanks to the inclusion

$$B'_0 \hookrightarrow | \subset G.$$

We shall also often set (

$$\underline{u} = \theta|\Theta(\alpha, \beta, \mu, \varepsilon),$$

thus|whence we shall have [p. 376]

Uncertainty retained from cancelled passages.
140-3-s377-D001 $\varepsilon_0(\lambda)$
140-3-s377-D002 α ε $\|| =$ $\|| =$

[scan 378]
[p. 490]

$$(29) \quad \begin{cases} \underline{u}(\rho) = \underset{\|\,\text{dfn.}}{\underline{u}\rho\underline{u}^{-1}} = \underset{\|}{\xi\rho}\ \underset{\text{int}(\alpha)\rho}{} , & \underline{u}(\sigma) = \underset{\|}{\eta\sigma}\ \underset{\varepsilon\,\text{int}(\beta)(\sigma)}{} , \\ \det \underline{u} = \mu \end{cases} \qquad \text{with } \xi = \alpha\rho(\alpha)^{-1}, \quad \eta = \varepsilon\beta\sigma(\beta)^{-1}$$

watch the factor ε!

Proposition The morphism of schemes Θ is characterized by formulas (29), modulo a morphism $\mathcal{G}^*_{\rho,\sigma} \xrightarrow{\lambda} \mu_2$.

Indeed, the first two formulas, which describe the action of $\underline{u}$ on ρ resp. σ, determine $\underline{u}$ modulo T_ρ resp. modulo T_σ, hence (together) modulo $T_\rho \cap T_\sigma = \mathbb{G}_{\mathrm{m}}$, the third determines it modulo SG, – hence with the preceding ones, modulo μ_2 – i.e. another morphism Θ' having the same properties will be of the form

$$\Theta'(g) = \lambda(g)\Theta(g) \qquad \text{where } \lambda : \mathcal{G}^*_{\rho,\sigma} \longrightarrow \mathcal{E}_2|\varepsilon_2.$$

q.e.d.

Remark Let us note moreover that † Θ is normalized by

$$\Theta(\underbrace{1,1,1,1}_{1}) = 1$$

which imposes † †corresponding normalization $\lambda(1) = 1$; on λ – but unless I am mistaken this already implies (at least if ρ, σ are $\frac{1}{2}$ simple hence T_ρ, T_σ tori…) that λ is a homomorphism of group schemes, which here is necessarily trivial, on the component $B'_0 \times T_\rho \times T_\sigma$ (at least in characteristic $\neq 2$) so we find essentially as the only possible indeterminacy the multiplication of $\underline{u}$ by ε † (corresponding to the canonical homomorphism (nontrivial) from N'_σ/T_σ onto μ_2)… †

m1037

Marginal note. m1037 If we worked with $S\mathcal{G}_{\rho,\sigma}$ instead of $\mathcal{G}^*_{\rho,\sigma}$, the indeterminacy would be given by multiplication by$|\mapsto \varepsilon^i\varepsilon'^j$ $(i,j \in \mathbb{Z}/2\mathbb{Z})$.

377

Uncertainty retained from cancelled passages.

140-3-s378-D001 every
140-3-s378-D002 an
140-3-s378-D003 additional
140-3-s378-D005 all
140-3-s378-D006 Error

[scan 379]

(IX) The groups $\mathcal{M}_{\rho,\sigma}$ and $S\mathcal{M}_{\rho,\sigma}$. [p. 490]

Let ρ, σ be as always two sections of G, satisfying the general conditions (cf. for example proposition page 478'|470'), plus the condition $\operatorname{Tr}\sigma = 0$ (cf. V – for example lemma page 455 and its corollary page 457) – we could carry out their construction also without this additional condition (contenting ourselves with working with $\mathcal{G}_{\rho,\sigma}$ instead of $\mathcal{G}^*_{\rho,\sigma}$) – but it matters little probably for what follows. We can if need be suppose ρ, σ normalized so that we have $\operatorname{Tr}(\rho+\sigma) = 1$.

We denote by $S\mathcal{M}_{\rho,\sigma}$ the product scheme

$$(1) \qquad S\mathcal{M}_{\rho,\sigma} = S\mathcal{G}^*_{\rho,\sigma} \times_S SG$$

m1039

Marginal note. m1039 In|For what follows, the "officialization" [illeg. 1] of the group schemes|structures $S_0\mathcal{M}_{\rho,\sigma}$, cf. page 559 ff.

Its sections are denoted

$$(\alpha, \beta, \mu, \varepsilon, f)\ ;$$

f thus being an arbitrary section of SG $(\simeq \mathrm{Sl}(2)_S)$.

In the case where $S = \operatorname{Spec}(A)$, $A = \widehat{\mathbb{Z}}$,

$$G = \mathrm{Gl}(2)_S, \quad \rho = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \quad \sigma = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

and where we thus have a canonical homomorphism

$$(2) \qquad \widehat{\mathcal{G}|\mathcal{C}}_{0,3} \longrightarrow G(\widehat{\mathbb{Z}})'^{\pm 1} \simeq \mathrm{Gl}|\mathrm{Sl}(2,\widehat{\mathbb{Z}})^{\pm 1}/\{\pm 1\}$$

with values in the subgroup of elements|elements of determinant ± 1, 378

[scan 380]

sending $\widehat{\mathcal{G}|\mathcal{C}}^{+}_{0,3}$ into $SG(\widehat{\mathbb{Z}})' = \mathrm{Sl}(2,\widehat{\mathbb{Z}})' = \mathrm{Sl}(2,\widehat{\mathbb{Z}})/\{\pm 1\}|\pm 1$, we are going to construct a variant of the group $\mathcal{M}^{\sim}_{0,3}$, (cf. page 390), which maps canonically into the product (1), which will lead us to define a group structure on $S\mathcal{M}_{\rho,\sigma}$, so as to make the said morphism multiplicative.

Let us first consider the "special" variant $S\mathcal{M}^{\sim}_{0,3}$ of $\mathcal{M}^{\sim}_{0,3}$, an extension of $\mathcal{M}^{\sim}_{0,3}$ by $\widehat{\mathbb{Z}}^3$, deduced from an extension of

$$\mathcal{A}|\mathcal{H}^{\sim}_{0,3} = \mathcal{M}^{\sim}_{0,3}/\widehat{\Pi}_{0,3} \subset \widehat{\widehat{\mathcal{T}}}_{0,3}|\widehat{\mathcal{T}}_{0,3},$$

† and even from an extension of $\widehat{\widehat{\mathcal{T}}}_{0,3}|\widehat{\mathcal{T}}_{0,3}$ (via $\mathcal{M}^{\sim}_{0,3} \longrightarrow \widehat{\widehat{\mathcal{T}}}_{0,3}|\widehat{\mathcal{T}}_{0,3}$), consisting of the

$$(3) \qquad (u, f_0|g_0, f_1|g_1, f_\infty|g_\infty) \qquad \left(u \in \widehat{\widehat{\mathcal{T}}}_{0,3}|\widehat{\mathcal{T}}_{0,3}, \quad g_0, g_1, g_\infty \in \widehat{\Pi}_{0,3}\right)$$

satisfying

$$(4) \qquad u(\ell_i) = \mathrm{int}(f|g^{-1}_{\dot{u}(i)})(\ell^{\mu}_{\dot{u}(i)})$$

where μ is the multiplier of u, and $\dot{u}$ the action of u on the set $\{0,1,\infty\}$ of loop classes. The product is defined by

$$(5) \qquad \begin{aligned}&(u', f'_0, f'_1, f'_\infty)(\dagger, f_0, f_1, f_\infty)\\ &= (u'u,\ f'_0u'(f_0),\ f'_1u'(f_1),\ f'_\infty u'(f_\infty))\end{aligned}$$

restricting ourselves (deflation!) to the case $u, u' \in \mathcal{M}^{!\sim}_{0,3}$ i.e. $\dot{u}, \dot{u}' = 1$. 379

Uncertainty retained from cancelled passages.

140-3-s380-D001 p

140-3-s380-D002 *u*

[scan 381]

We have a surjective homomorphism $S\mathcal{M}^{\sim}_{0,3} \longrightarrow \mathcal{M}^{\sim}_{0,3}$, whose kernel consists of the [p. 491]

$$(6) \qquad (1, f_0, f_1, f_\infty) \qquad f_i \in L_i$$

thus we have

$$(7) \qquad 1 \longrightarrow \widehat{\mathbb{Z}}^3 \longrightarrow S\mathcal{M}^{\sim}_{0,3} \longrightarrow \mathcal{M}^{\sim}_{0,3} \longrightarrow 1$$

We are going to define an interesting subgroup

$$(8) \qquad S_0\mathcal{M}^{\sim}_{0,3} \subset \mathcal{M}^{\sim}_{0,3},$$

now using the definition of $\mathcal{M}^{\sim}_{0,3}$, namely via the property of the $u \in \mathcal{M}^{\sim}_{0,3}$ of commuting with ρ as outer actions, so that we will have

$$(9) \qquad u\rho u^{-1} = g\rho \qquad \text{i.e. } [u,\rho] = g, \quad g \in \widehat{\Pi}_{0,3}$$

This allows us, when we have already chosen $f_i \in \widehat{\Pi}_{0,3}$ satisfying

$$(10) \qquad u(\ell_i) = \mathrm{int}(f_i^{-1})(\ell_i^{\mu})$$

(supposing for simplicity $u \in \mathcal{M}^{!\sim}_{0,3}$), to deduce from it the † f_{i+1}, f_{i+2} via the formula

$$(11) \qquad f_{i+1}^{-1} = g\rho(f_i^{-1})\,, \qquad \left[\begin{array}{l}\text{and } f_{i+2}^{-1} = g\rho(f_{i+1}^{-1}) \\ \qquad\qquad = g\rho(g)\rho^2(f_i^{-1})\end{array}\right]$$
$$[\text{where } g = [u,\rho] \in \widehat{\Pi}_{0,3})$$

(we have set $\dot\rho(i) = i+1$ by abuse of language, i.e. $0+1=1$, $1+1=\infty$, $\infty+1=0$!) – we then recover|will recover

$$f_i^{-1} = f|g^{-1}_{(i+2)+1} = g\rho(f_{i+2}^{-1}) = \underbrace{g\rho(g)\rho^2(g)}_{1}\ \underbrace{\rho^3(f_i^{-1})}_{f_i^{-1}}\,.$$

380

Uncertainty retained from cancelled passages.
140-3-s381-D001 forms|formulas

[scan 382]

The $(u, f_0, f_1, f_\infty) \in S\mathcal{M}^{!\sim}_{0,3}$ which satisfy (11) for $\forall\ i \in \{0,1,\infty\}$ (it suffices that it satisfy it|them for two indices, for this to be satisfied for all three) form a subgroup of $S\mathcal{M}^{!\sim}_{0,3}$, which I denote $S_0\mathcal{M}^{!\sim}_{0,3}$ – it also identifies with the group of pairs (u, f_0) $(u \in \mathcal{M},\ f_0 \in \widehat{\Pi}_{0,3})$ satisfying [p. 492]

$$(12) \qquad u(\ell_0) = \mathrm{int}(f_0^{-1})(\ell_0^\mu) \qquad (\text{where } \mu = \chi(u)).$$

We have an extension structure

$$(13) \qquad 1 \longrightarrow \widehat{\mathbb{Z}} \longrightarrow S_0\mathcal{M}^{!\sim}_{0,3} \longrightarrow \mathcal{M}^{!\sim}_{0,3} \longrightarrow 1\ ,$$

and a homomorphism of extensions

$$(14) \qquad \left\{\begin{array}{ccccccccc} 1 & \longrightarrow & \widehat{\mathbb{Z}} & \longrightarrow & S_0\mathcal{M}^{!\sim}_{0,3} & \longrightarrow & \mathcal{M}^{!\sim}_{0,3} & \longrightarrow & 1 \\ & & \downarrow{\scriptstyle \text{diagonal}} & & \downarrow & & \Downarrow & & \\ 1 & \longrightarrow & \widehat{\mathbb{Z}}^3 & \longrightarrow & S\mathcal{M}^{!\sim}_{0,3} & \longrightarrow & \mathcal{M}^{!\sim}_{0,3} & \longrightarrow & 1 \end{array}\right.$$

(where $S\mathcal{M}^{!\sim}_{0,3}$ is defined as the reverse|inverse image of $\mathcal{M}^{!\sim}_{0,3} \subset\ |\ \in \mathcal{M}^{\sim}_{0,3}$ in $S\mathcal{M}^{\sim}_{0,3}$). We can therefore say that the extension $S\mathcal{M}^{!\sim}_{0,3}$ (of $\mathcal{M}^{!\sim}_{0,3}$ by $\widehat{\mathbb{Z}}^3$) is deduced, by the diagonal homomorphism

$$(15) \qquad \begin{array}{ccc} \widehat{\mathbb{Z}} & \longrightarrow & \widehat{\mathbb{Z}}^3 \\ \underline{n} & \longmapsto & (\underline{n}_0 = \underline{n},\ \underline{n}_1 = \underline{n},\ \underline{n}_\infty = \underline{n}) \end{array}\ ,$$

from a "finer" extension $S_0\mathcal{M}^{!\sim}_{0,3}$ of $\mathcal{M}^{!\sim}_{0,3}$ by $\widehat{\mathbb{Z}}$. 381 [scan 383]

The calculations of §46 V (page 407 and following) make it possible on the other hand to make explicit the pairs $(u|\alpha, f_0 = f)$ in terms of systems

$$(16) \qquad (\alpha, \beta, f) \ \Big|\ \begin{array}{c} \alpha \in \widehat{\widetilde{\Pi}}_{0,3} = \widehat{\Pi}_{0,3} \cdot \{1, \tau_\infty\} = \text{inverse image|inverse image of } \{1,\tau\} \\ \text{under } \widehat{\mathcal{G}|\mathcal{C}}_{0,3} \longrightarrow \mathcal{T}_{0,3} \\ \beta, f \in \widehat{\Pi}_{0,3} \end{array}$$

Marginal note. m1044 to (u, f) we associate (fu, f) by putting $fu \in \mathcal{M}^{!\sim}_{0,3}[0]$ in the form $u_{\alpha,\beta}$

satisfying relations (not involving f)

$$(17) \qquad \alpha\rho(\alpha)^{-1} = \beta\sigma(\beta)^{-1} \underbrace{\varepsilon_1(\mu-1)}_{=\ell_1|h^\nu \text{ if } \mu-1=2\nu}$$

where $\mu \in \widehat{\mathbb{Z}}^*$ is defined † ambiguity|unambiguously by α — and even from knowledge alone of $\xi = \underline{\alpha}\rho(\underline{\alpha})^{-1}$, where $\underline{\alpha}$ is the image of α in $\mathrm{Gl}(2, \widehat{\mathbb{Z}})^{\pm 1}$.

The case $f = 1$ corresponds to a subgroup of $S_0\mathcal{M}^{!\sim}_{0,3}$, which is identified with the normalizer of L_0 in $\mathcal{M}^{!\sim}_{0,3}$, which we had denoted $\mathcal{M}^{!\sim}_{0,3}[0]$ (§46 V ...) :

$$(18) \qquad \underset{\substack{\| \text{ dfn} \\ \mathrm{Norm}_{\mathcal{M}^{!\sim}_{0,3}}(L_0)}}{\mathcal{M}^{!\sim}_{0,3}[0]} \simeq \left\{ (\alpha, \beta, f) \,\middle|\, \begin{array}{l} f = 1,\ \alpha, \beta \\ \text{satisfying (17)} \end{array} \right\} \subset S_0\mathcal{M}^{!\sim}_{0,3}.$$

If we denote by $u_{\xi,\eta}|u_{\alpha,\beta}$ † or simply u_ξ the automorphism defined in terms of

$$(19) \qquad \xi = \alpha\rho(\alpha)^{-1}, \qquad \eta = \beta\sigma(\beta)^{-1}$$

382

Uncertainty retained from cancelled passages.

140-3-s383-D001 without any

140-3-s383-D003 or u_ξ

by [scan 384]

$$(20) \qquad u(\rho) = \xi\rho, \qquad u(\sigma) = \eta\sigma \qquad \text{[p. 493]}$$

(N.B. η is defined in terms of ξ thanks to (17), where $\mu = \mu(\xi)$ is determined by ξ), we will have by construction (loc. cit.)

$$(21) \qquad u_{\alpha,\beta,f} = \mathrm{int}(f^{-1})u_{\alpha,\beta}$$

and to the triple (α, β, f) corresponds the element $(u_{\alpha,\beta,f}, f)$ of $S_0\mathcal{M}^{!\sim}_{0,3}$:

$$(22) \qquad (\alpha, \beta, f) \longmapsto (u_{\alpha,\beta}, f) \in S_0\mathcal{M}^{!\sim}_{0,3}.$$

Having thus made the elements of $S_0\mathcal{M}^{!\sim}_{0,3}$ explicit, we define a map

$$S_0\mathcal{M}^{!\sim}_{0,3} \longrightarrow S\mathcal{M}_{\rho,\sigma}(\widehat{\mathbb{Z}})' = S\mathcal{G}^*_{\rho,\sigma}(\widehat{\mathbb{Z}})' \times \mathrm{Sl}(2, \widehat{\mathbb{Z}})'$$

$$(23) \qquad (\alpha, \beta, f) \longmapsto \left(\underline{\alpha}, \underline{\beta}, \mu = \mu(\underline{\xi} = \underline{\alpha}\rho(\underline{\alpha})^{-1}), \underset{\uparrow \varepsilon}{\varepsilon_2(\mu)}, \underline{f} \right)$$

where the underlined $\underline{\alpha}, \underline{\beta}, \underline{f}$ & denote the images of α, β, f in $\mathrm{Sl}(2, \widehat{\mathbb{Z}})'$, and the "accent" $'$ in $S\mathcal{M}_{\rho,\sigma}(\widehat{\mathbb{Z}})'$ denotes the set of systems $(\underline{\alpha}, \underline{\beta}, \mu, \varepsilon, \underline{f})$ †

$$\in \mathrm{Gl}(2, \widehat{\mathbb{Z}})'^{\pm 1} \times \mathrm{Sl}(2, \widehat{\mathbb{Z}})' \times \widehat{\mathbb{Z}}^* \times \{\pm 1\} \times \mathrm{Sl}(2, \widehat{\mathbb{Z}})'$$

satisfying in $\mathrm{Sl}(2, \widehat{\mathbb{Z}})$ (not in $\mathrm{Sl}(2, \widehat{\mathbb{Z}})'$!), the loop relation †

$$(24) \qquad \underline{\alpha}\rho(\underline{\alpha})^{-1} = \varepsilon\underline{\beta}\sigma(\underline{\beta})^{-1}\varepsilon_1(\mu - 1)$$

(which does not depend on the choices of signs ...) 383

Uncertainty retained from cancelled passages.
140-3-s384-D002 defined of elements of
140-3-s384-D003 taken|by

We have "done what was needed" so that the restriction of (23) to $\dagger\mathcal{M}^{!\sim}_{0,3}[0]$ ((18), identified with the (α,β,f) for which $f=1$) is a homomorphism into the group $S\mathcal{G}_{\rho,\sigma}$ — by defining precisely in a clever way ad hoc the composition law in $S\mathcal{G}^*_{\rho,\sigma}$. I am going to recheck it all the same to reassure myself, by considering $u=u_{\alpha,\beta}$, $u'=u_{\alpha',\beta'}$, $u''=uu'$ in the form $u_{\alpha'',\beta''}$, and recalling the "cocycle relation" which gives, up to one of the two signs, (α'',β'') in terms of $(\alpha',\beta'),(\alpha,\beta)$ (§46 II 5°)) [scan 385]

$$(25)\qquad \alpha''=u'_{\infty|\omega}(\alpha)\alpha',\qquad \beta''=u'(\beta)\beta',\qquad \text{in addition to } \mu''=\mu\mu'$$

Let then|therefore $\underline{\alpha},\underline{\beta},\underline{\alpha}',\underline{\beta}'$ be elements of $\mathrm{Gl}(2,\widehat{\mathbb{Z}})^{\pm1}$ (determined up to sign) which lift the images of $\alpha,\alpha',\beta,\beta'$ in $\mathrm{Gl}(2,\widehat{\mathbb{Z}})'$, we will therefore have $(\underline{\alpha},\underline{\beta},\mu,\varepsilon_2(\mu))$ and $(\underline{\alpha}',\underline{\beta}',\mu',\varepsilon_2(\mu'))$ in $S\mathcal{G}^*_{\rho,\sigma}(\widehat{\mathbb{Z}})$, and applying Θ we find $\underline{u},\underline{u}'\in\mathrm{Gl}(2,\widehat{\mathbb{Z}})$ (in fact, $\in B'_0(\widehat{\mathbb{Z}})$) such that

$$(26)\qquad \begin{cases}\underline{u}'(\rho)=\mathrm{int}(\underline{\alpha}')\rho & \underline{u}'(\sigma)=\varepsilon_2(\mu')\,\mathrm{int}(\underline{\beta}')(\sigma)\\ \det\underline{u}'=\mu'\end{cases}$$

(and relations item|idem for u) and we will then have, for every $x\in\dagger\widehat{\mathcal{G}|\mathcal{C}}_{0,3}$

$$(27)\qquad \underline{u'(x)}=\underline{u}'(\underline{x})$$

(cf. ...) Thus, for suitable liftings $\underline{\alpha}'',\underline{\beta}^{''|'}$ of the images of $\alpha^{''|'},\beta^{''|'}$ in $\mathrm{Gl}(2,\widehat{\mathbb{Z}})^{\pm1}$, we will have

$$(28)\qquad \underline{\alpha}''=\underline{u}'(\underline{\alpha})\underline{\alpha}',\qquad \underline{\beta}''=\underline{u}'(\underline{\beta})\underline{\beta}',$$
$$\text{in addition to } \mu''=\mu\mu'\quad \text{hence}\quad \varepsilon''_2=\varepsilon_2\varepsilon'_2.$$

384

Uncertainty retained from cancelled passages.
140-3-s385-D001 S_0
140-3-s385-D003 Π

and it remains to check that the element [scan 386] [p. 494]

$$g''=(\underline{\alpha}'',\underline{\beta}'',\mu''=\mu\mu',\varepsilon''_2=\varepsilon'_2\varepsilon_2)\in\dagger$$
$$\Gamma(\widehat{\mathbb{Z}};\mathrm{Gl}(2)\times\mathrm{Sl}(2)\times\mathbf{E}'\times\mu_2)$$

is indeed the product of the elements

$$g'=(\underline{\alpha}',\underline{\beta}',\mu',\varepsilon')\quad\text{and}\quad g=(\underline{\alpha},\underline{\beta},\mu,\varepsilon).$$

That it is OK for μ'' and ε'' is trivial, it remains to check that in $\mathcal{G}^*_{\rho,\sigma}$, the product is indeed (as regards the components in α and in β) given by formulas (28) — which will be valid, of course, over any base, and under the general conditions of V (or even of III, if we are interested in $\mathcal{G}_{\rho,\sigma}$, instead of $\mathcal{G}^*_{\rho,\sigma}$). We will have

$$(29)\qquad \begin{cases}\underline{\alpha}=\underline{u}\,\underline{a}^{-1}, & \underline{\beta}=\underline{u}\,\underline{b}^{-1} & \underline{\alpha},\underline{\alpha}'\in T_\rho(\widehat{\mathbb{Z}})\\ \underline{\alpha}'=\underline{u}'\,\underline{a}'^{-1}, & \underline{\beta}'=\underline{u}'\,\underline{b}'^{-1} & \underline{\beta},\underline{\beta}'\in N'_\sigma(\widehat{\mathbb{Z}})\end{cases}$$

and by definition of the products in $\mathcal{G}^*_{\rho,\sigma}$, the product of † $g'g$ will therefore have components $\underline{\alpha}''_0$ and $\underline{\beta}''_0$

$$\underline{\alpha}''_0 = (\underline{u}'\underline{u})(\underline{a}'\underline{a})^{-1}, \qquad \underline{\beta}''_0 = (\underline{u}'\underline{u})(\underline{b}'\underline{b})^{-1}$$

and it remains to prove

$$\underline{\alpha}''_0 = \underline{\alpha}'', \qquad \underline{\beta}''_0 = \underline{\beta}'', \quad \text{i.e. the relations}$$

$$(30) \qquad \underline{u}'(\underline{\alpha})\cdot\underline{\alpha}' = (\underline{u}'\underline{u})(\underline{a}'\underline{a})^{-1} \qquad \underline{u}'(\underline{\beta})\underline{\beta}' = (\underline{u}'\underline{u})(\underline{b}'\underline{b})^{-1}$$

where † $\underline{\alpha}, \underline{\beta}, \underline{\alpha}', \underline{\beta}'$ are given in terms of † $\underline{u}, \underline{a}, \underline{b}, \underline{u}', \underline{a}', \underline{b}'$ by (29). We therefore have

$$\begin{aligned}\underline{u}'(\underline{\alpha})\underline{\alpha}' &= \underline{u}'(\underline{u}\,\underline{a}^{-1})\underline{u}'^{-1}\,\underline{u}'\underline{a}'^{-1} \\ &= (\underline{u}'\underline{u})(\underline{a}^{-1}\underline{a}'^{-1}) = (\underline{u}'\underline{u})(\underline{a}'\underline{a})^{-1}, \qquad \dagger\end{aligned}$$

† and likewise for the second relation 30, which proves our assertion. ✓|/

385

Uncertainty retained from cancelled passages.

140-3-s386-D001 Γ
140-3-s386-D002 computed at
140-3-s386-D003 $\underline{u}$
140-3-s386-D004 $\underline{\alpha}, \underline{\beta}, \underline{u}, \underline{a}$
140-3-s386-D005 OK
140-3-s386-D006 $\underline{u}'(\underline{\beta})\underline{\beta}'$

Let us note on this occasion that the systems [scan 387]

$$(31) \qquad (\alpha, \beta, \mu = 1, \varepsilon = 1) \quad \text{with} \quad \alpha \in \Gamma\dagger\mathbb{G}_m, \qquad \beta \in \Gamma\dagger\mathbb{G}_m$$

are in $\Gamma|S\mathcal{G}^{\circ|0}_{\rho,\sigma}$. [which are indeed stable under multiplication, † $(\alpha, \beta, \mu, \varepsilon)$, of α and β each by a section of $\dagger\mathbb{G}_m$, which does not change $\tilde{\alpha}, \tilde{\beta}$ ...] they correspond, by the isomorphism $\mathcal{G}_{\rho,\sigma} \simeq B'_0 \times T_\rho \times T_\sigma$, to the systems $(\underline{u}, \underline{a}, \underline{b})$ with $\underline{u} = 1$, $\underline{a} \in \Gamma\mathbb{G}_m$, $\underline{b} \in \Gamma\mathbb{G}_m$) and form a subgroup (with ordinary multiplication † in α β as group law) — if α, β are sections of μ_2, we find a subgroup "by nature|in action trivial" $\mu_2 \times \mu_2$ of $S\mathcal{G}_{\rho,\sigma}$. The action by translation of this subgroup $\simeq \mathbb{G}_m \times \mathbb{G}_m$ on $\mathcal{G}^*_{\rho,\sigma}$ is given by multiplication of the components in question. This shows that the fact of working with $\underline{\alpha}, \underline{\beta}$ defined only "up to signs" simply amounts to working in the quotient groups of $\underline{\mathcal{G}^*_{\rho,\sigma}(\widehat{\mathbb{Z}})}$ (or of $S\mathcal{G}^*_{\rho,\sigma}(\widehat{\mathbb{Z}})$) by the subgroup

$$\boxed{\begin{array}{c}\Sigma_0\ . \\ \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}\end{array}}$$

of the elements $(\alpha, \beta, 1, 1)$ with $\alpha, \beta \in \{\pm 1\}$. This 386

Uncertainty retained from cancelled passages.

140-3-s387-D001 $\{\pm 1\}$
140-3-s387-D002 $\{\pm 1\}$
140-3-s387-D004 for
140-3-s387-D005 μ_2
140-3-s387-D006 on the|on the

[scan 388]

makes precise at the same time the meaning of the expression: we have a homomorphism of [p. 495] groups ....

Let us now return to the map (23) with arbitrary f, and make precise the composition law of $S_0\mathcal{M}^{!\sim}_{0,3}$ in terms of (α,β,f) i.e. in terms of the $u_{\alpha,\beta,f}$. Let us consider the subgroup formed of the $(1,1,f)$, which corresponds to the pairs (u,f) with $u=\mathrm{int}(f^{-1})$, i.e. to the pairs $(\mathrm{int}(f^{-1}),f)$ — they form a "group isomorphic to $\widehat{\Pi}_{0,3}$, and we have a $\frac{1}{2}$ direct product structure:

$$(31)\qquad S_0\mathcal{M}^{!\sim}_{0,3}\simeq \mathcal{M}^{!\sim}_{0,3}[0]\cdot\widehat{\Pi}_{0,3}$$

where

$$(32)\qquad \begin{cases} \widehat{\Pi}_{0,3} \longrightarrow S_0\mathcal{M}^{!\sim}_{0,3} \\ f\longmapsto (\dagger\,\mathrm{int}(f),f)\longleftrightarrow (1,1,f^{-1}), \end{cases}$$

the operation of $\mathcal{M}^{!\sim}_{0,3}[0]$ on $\widehat{\Pi}_{0,3}$ † the tautological operation (deduced from the inclusion $\mathcal{M}^{!\sim}_{0,3}[0]\subset\mathcal{M}^{!\sim}_{0,3}\subset \underline{\mathrm{Aut}}_{\mathrm{lac}}(\widehat{\Pi}_{0,3})$).

The invariant subgroup $\widehat{\mathbb{Z}}$ of $S_0\mathcal{M}^{!\sim}_{0,3}$ is given by the pairs $(u,f)=(1,\ell_0^n)$ $(n\in\widehat{\mathbb{Z}})$ which we also put in the form (α,β,f) where α,β are such that $u_{\alpha,\beta}=fu=\mathrm{int}(\ell_0^n)$, hence $\alpha=\beta=\ell_0^n$, i.e.

$$(33)\qquad \widehat{\mathbb{Z}}\longrightarrow S_0\mathcal{M}^{!\sim}_{0,3}\quad\text{is identified with}\quad n\longmapsto(\dagger\ell_0^n,\beta=\ell_0^n,f=\ell_0^n)$$

387

Uncertainty retained from cancelled passages.

140-3-s388-D003 f,

140-3-s388-D004 being

140-3-s388-D005 $\alpha=$

[scan 389]

We will have

$$(f'^{-1}u_{\alpha',\beta'})(f^{-1}u_{\alpha,\beta})=\dagger\underbrace{f'^{-1}u_{\alpha',\beta'}(f^{-1})}_{f''^{-1}}(u_{\alpha',\beta'}u_{\alpha,\beta})$$

i.e. the multiplication in $S_0\mathcal{M}^{!\sim}_{0,3}$ is given, in terms of the components $(\alpha',\beta',f')(\alpha,\beta,f)$, for|by $\dagger(\alpha'',\beta'',f'')$, where

$$(34)\qquad \begin{cases} u_{\alpha'',\beta''}=u_{\alpha',\beta'}u_{\alpha,\beta} \\ f''=u_{\alpha',\beta'}(f)f' \end{cases}\quad\text{i.e.}\quad \begin{cases}\alpha''=u_{\alpha',\beta'}(\alpha)\alpha'\\ \beta''=u_{\alpha',\beta'}(\beta)\beta'\end{cases},\quad\text{and}$$

Returning now to the definition of $S\mathcal{M}_{\rho\sigma}$ (1), we are going to introduce there the corresponding multiplication, which amounts to defining $S\mathcal{M}_{\rho\sigma}$ as a $\frac{1}{2}$ direct product

$$(35)\qquad S\mathcal{M}_{\rho,\sigma}\overset{\text{dfn}}{=}S\mathcal{G}^*_{\rho,\sigma}\cdot SG\qquad(\text{product }\tfrac{1}{2}\text{ direct})$$

by the operation of $S\mathcal{G}^*_{\rho,\sigma}$ on G via the homomorphism Θ (page 489'|488'), and by defining|will define the|the morphism

$$(36)\qquad \begin{cases} S\mathcal{M}_{\rho,\sigma}\longrightarrow S\mathcal{G}^*_{\rho,\sigma}\times_S G\\ \dagger U\cdot f\longmapsto (U,f^{-1})\end{cases}$$

m1052

Marginal note. m1052 Watch out for the passage from f to f^{-1}!

where $U = (\alpha, \beta, \mu, \varepsilon)$ is in $\Gamma S\mathcal{G}_{\rho,\sigma}$. 388

Uncertainty retained from cancelled passages.
140-3-s389-D001 $f'^{-1}f^{-1}(fu_{\alpha',\beta'}f^{-1})$
140-3-s389-D002 g"=
140-3-s389-D004 $(\alpha, \beta, \mu, \varepsilon)|(\alpha, \beta)$

[scan 390]

We have thus done what was needed, in the case $S = \mathrm{Spec}(\widehat{\mathbb{Z}})$ etc, to have a homomorphism of groups [p. 496]

$$(37) \qquad S_0\mathcal{M}^{!\sim}_{0,3} \longrightarrow S\mathcal{M}_{\rho,\sigma}(\widehat{\mathbb{Z}})/\widetilde{\Sigma}\dagger$$

where $\Sigma_0 \simeq \mathbb{Z}_{/2} \times \mathbb{Z}_{/2}$ is the subgroup $\dagger$ of $S\mathcal{G}^*_{\rho,\sigma}$ introduced on page 494', which operates trivially on G hence is invariant in $S\mathcal{M}_{\rho,\sigma}$, hence in $S\mathcal{M}_{\rho,\sigma}(\widehat{\mathbb{Z}})$, and where

$$(38) \quad \widetilde{\Sigma} = \Sigma_0 \times \underbrace{\{\pm 1\}}_{\substack{\text{this factor identified with the center of } SG(\widehat{\mathbb{Z}}) \\ =\mathrm{Sl}(2,\widehat{\mathbb{Z}}),\ \text{contained in}\ \mathrm{Gl}(2,\widehat{\mathbb{Z}}) \\ =G(\widehat{\mathbb{Z}})}} \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$$

is the corresponding normal group in the $\frac{1}{2}$ direct product — the fact of passing to the quotient by Σ expresses the three ambiguities|ambiguities of signs, on α, on β and on f.

Let us now consider the homomorphism

$$(39) \quad \left| \begin{array}{l} \mathbb{G}_a \longrightarrow S_0|S\mathcal{M}^{\dagger}_{\rho,\sigma} \\ \lambda \longmapsto (\alpha{=}\varepsilon_0(\lambda|-\lambda), \beta{=}\varepsilon_0(-\lambda), \mu = 1, \varepsilon = 1, f = \varepsilon_0(\lambda)) \end{array} \right.$$

it is a homomorphism of groups, and even makes $\mathbb{G}_a$ a normal subgroup. To verify this the most convenient thing is to identify $S_0|S\mathcal{M}^{!\sim}_{\rho,\sigma}$ with the $\frac{1}{2}$-direct product (35), 389

Uncertainty retained from cancelled passages.
140-3-s390-D001 $(\widehat{\mathbb{Z}})$
140-3-s390-D002 "powerless"
140-3-s390-D003 ! ∼

[scan 391]

where we identify $S\mathcal{G}^*_{\rho,\sigma}$ with a subgroup of $B'_0 \times T_\rho \times N'_\sigma$, operating on G by inner automorphisms via B'_0.

The preceding morphism thus corresponds to

$$(40) \qquad \begin{array}{ccccc} \lambda & \overset{i}{\longmapsto} & (\lambda, 1, 1, & \lambda^{-1} &) = \dagger \\ \cup & & & \cup & \\ \Gamma B'_0 & & & \Gamma G & \end{array}$$

m1055

Marginal note. m1055 $\dagger$ the canonical homomorphism of groups

$\dagger$

†

We identify $S\mathcal{G}^*_{\rho,\sigma}$ with a subgroup of $\mathcal{G}^*_{\rho,\sigma}$, and likewise the $\frac{1}{2}$ direct product $S\mathcal{M}^*_{\rho,\sigma}$ with a subgroup of the $\frac{1}{2}$ direct product

$$(41) \qquad {}^{\dagger}\tilde{S}\mathcal{M}_{\rho,\sigma} = \mathcal{G}^*_{\rho,\sigma} \cdot G \simeq (B'_0 \times T_\rho \times N'_\sigma) \cdot G,$$

and we are going to see that the image of the monomorphism i (40) in $\mathcal{G}^*_{\rho,\sigma} \cdot G$ is even normal in this larger group. So as not to get lost in the notation, let us denote by

$$(42) \qquad i_B,\ i_\rho,\ i_\sigma,\ i_G$$

the morphisms of inclusions|of inclusion

$$\left.\begin{array}{l} B'_0 \longrightarrow \dagger \\ T_\rho \longrightarrow \\ N'_\sigma | N''_\sigma \longrightarrow \\ G \longrightarrow \end{array}\right\} \; i' \; {}^{\dagger}\tilde{S}\mathcal{M}_{\rho,\sigma} \cdot G$$

390

Uncertainty retained from cancelled passages.

140-3-s391-D001 j
140-3-s391-D002 j
140-3-s391-D003 where U_0
140-3-s391-D004 $a|\rho \quad b|\beta \quad \varepsilon$
140-3-s391-D005 $S\mathcal{M}^*_{\rho,\sigma}$
140-3-s391-D006 $\mathcal{G}^*_{\rho,\sigma}$
140-3-s391-D007 $\mathcal{G}^*_{\rho,\sigma}|\mathcal{M}_{\rho,\sigma}$

[scan 392]

it being understood, of course, that we let $\mathcal{G}^*_{\rho,\sigma}$ operate on G by inner automorphisms via the factor B'_0. We must check that [p. 497]

$$i_\dagger(L_0) = \{i_B(\lambda) i_G(\lambda^{-1}) \mid \lambda \in \Gamma L_0\}$$

is invariant under † the $i_\rho(x)$, $i_\sigma(x)$, $i_B(x)$ and $i_G(x)$. — For the first two, this is trivial — with trivial action on L_0.

If $x \in \Gamma B'_0$, we will have

$$\begin{aligned} i_B(x)(i(\lambda)) &= i_B(x)(i_B(\lambda) i_G(\lambda^{-1})) = \\ &= \underbrace{i_B(x)(i_B(\lambda))}_{i_B({}^{\dagger}x(\lambda))} \cdot \underbrace{i_B(x)(i_G(\lambda^{-1}))}_{\substack{i_G(x)(i_G(\lambda^{-1})) \\ = i_G(x(\lambda)^{-1}]}} \\ &= i(x(\lambda)) \end{aligned}$$

m1058

Marginal note. m1058 We see at once that $i(\lambda)$ commutes with G, hence is invariant under G !

which shows that †

$$L_0 \xrightarrow{\ i\ } \mathcal{G}^*_{\rho,\sigma} \cdot G = \tilde{S}\mathcal{G}^*_{\rho,\sigma}$$

commutes with the natural action of B'_0 on both sides (on the second via π_B,

$$i_B : B'_0 \longrightarrow \tilde{S}\mathcal{M}^{\flat|*|\natural}_{\dagger\ \rho,\sigma}$$

). Finally

$$i_G(\dagger x)(i(\lambda)) = i_G(x) \cdot (i_B(\lambda) i_G(\lambda^{-1})) \cdot i_G(x)^{-1}$$

now by definition, $i_B(\lambda)$ and $i_G(\lambda)$ operate in the same way on G, hence $i(\lambda) = i_B(\lambda) i_G(\lambda^{-1})$ operates trivially on it, hence the last factor $i_G(x)^{-1}$ can jump over|under the factor $i(\lambda)$, we find

$$i_G(x) i_G(x)^{-1} i(\lambda) = i(\lambda)$$

O.K.
391

Uncertainty retained from cancelled passages.

140-3-s392-D001 B
140-3-s392-D002 $\Gamma T_\rho \quad \Gamma T_\sigma \quad i_T$
140-3-s392-D003 $\lambda|x$
140-3-s392-D004 $\mathbb{G}_a|L_0$
140-3-s392-D005 0
140-3-s392-D006 f

[scan 393]

Remark $\tilde{S}\mathcal{M}_{\rho,\sigma}$ is a flat group of relative dimension $\underset{B'_0}{2} + \underset{T_\rho}{2} + \underset{N'_\sigma}{2} + \underset{G}{4} = 10$, $S\mathcal{M}_{\rho,\sigma}$ is of relative dimension $\underset{S\mathcal{G}^*_{\rho,\sigma}}{4} + \underset{SG}{3} = 7$, on passing to the quotient by $\underset{\simeq \mathbb{G}_a}{i(L_0)}$ gives|to give groups $\dagger$ denoted $\widetilde{\mathcal{M}}_{\rho,\sigma}$, $\mathcal{M}_{\rho,\sigma}$, of relative dimension 9 resp. 6, giving rise to a diagram

(43)

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \mathbb{G}_a & \xrightarrow{\lambda \mapsto \varepsilon_0(\lambda)} & S\mathcal{M}_{\rho,\sigma} & \longrightarrow & \mathcal{M}_{\rho,\sigma} & \longrightarrow & 1 \\ & & \| & & \downarrow \text{can.} & & \downarrow & & \\ 1 & \longrightarrow & \mathbb{G}_a & \longrightarrow & \tilde{S}\mathcal{M}_{\rho,\sigma} & \longrightarrow & \mathcal{M}_{\rho,\sigma}|\widetilde{\mathcal{M}}_{\rho,\sigma} & \longrightarrow & 1 \end{array}$$

The passage to the quotient presents no difficulties — the quotients are in fact representable †as 1/2 direct products

$$(44) \qquad \begin{cases} \dagger\mathcal{M}_{\rho,\sigma} \simeq \dagger\, S\mathcal{G}^*_{\rho,\sigma}(0) \cdot SG \\ \widetilde{\mathcal{M}}^{\dagger}_{\rho,\sigma} \simeq \mathcal{G}^{*\dagger}_{\rho,\sigma}(0|\omega) \cdot \dagger G \end{cases}$$

where the subgroup schemes $S\mathcal{G}^{*\flat|\natural|\circ}_{\rho,\sigma}(0|\omega)$ and $\mathcal{G}^{*\flat|\natural|\circ}_{\rho,\sigma}(0|\omega)$ of $S\mathcal{G}^*_{\rho,\sigma}$, $\mathcal{G}^*_{\rho,\sigma}$ are defined in VIII, $\mathcal{G}^{*\flat|\natural|\circ}_{\rho,\sigma}(0|\omega)$ being identified with $T_0 \times T_\rho \times N'_\sigma$, and $S\mathcal{G}^{*\flat|\natural|\circ}_{\rho,\sigma}(0)$ with its intersection with $S\mathcal{G}^*_{\rho,\sigma}$ (defined by the supplementary conditions (20) on page 487′). 392

Uncertainty retained from cancelled passages.

140-3-s393-D001 this
140-3-s393-D002 by the|by
140-3-s393-D003 S
140-3-s393-D004 T_0
140-3-s393-D005 $\sim$
140-3-s393-D006 $\flat|\natural|\circ$
140-3-s393-D007 S

[scan 394]
[p. 498]

Returning now to the arithmetic situation, we find by passage to the quotient via (37) a canonical homomorphism in view of (13)

$$(45)\qquad \begin{array}{ccc} \dagger\mathcal{M}^{!\sim}_{0,3} & \longrightarrow & \underbrace{\mathcal{M}_{\rho,\sigma}(\widehat{\mathbb{Z}})}_{\simeq} \qquad /\Sigma \\ & & \underbrace{\dagger T_0|\hat{T}_0 \times T_\rho(\widehat{\mathbb{Z}}) \times T_\sigma(\widehat{\mathbb{Z}})}_{\text{operating on } \mathrm{Sl}(2,\widehat{\mathbb{Z}}) \text{ via the factor } T_0} \cdot \mathrm{Sl}(2,\widehat{\mathbb{Z}}) \end{array}$$

Σ being the subgroup $\simeq \{\pm 1\}^3$, product of three factors $\mathbb{Z}/2\mathbb{Z}$ included resp.[t] in $T_\rho(\widehat{\mathbb{Z}})$, $T_\sigma(\widehat{\mathbb{Z}})$ and in $\mathrm{Sl}(2,\widehat{\mathbb{Z}})$.

Remark This homomorphism was obtained here, essentially, with the help of the homomorphism from $\mathcal{M}^{!\sim}_{0,3}(0)$

$$(46)\qquad \begin{array}{ccc} \mathcal{M}^{!\sim}_{0,3}(0) & & = \mathrm{Norm}_{\mathcal{M}^{!\sim}_{0,3}}(L_0) \\ \text{subgroup} & & \text{(cf. §46, V)} \\ \text{of the automorphisms} & & \\ \text{“normalized”} & & \\ & \searrow \qquad \swarrow & \\ & S\mathcal{G}^{*\flat|\natural|\circ}_{\rho,\sigma}(0)(\widehat{\mathbb{Z}})/\Sigma_0 \hookleftarrow S\mathcal{G}^{*}_{\rho,\sigma}(\widehat{\mathbb{Z}})/\Sigma_0 & \end{array}$$

and of the composite

$$\begin{array}{ccccc} \dagger\widehat{\Pi}_{0,3} & \hookleftarrow & \widehat{\mathcal{G}|\mathcal{C}}^{+}_{0,3} & \longrightarrow & \mathrm{Sl}(2,\widehat{\mathbb{Z}})' \\ & & \cup|\wr & & \\ & & \mathrm{Sl}(2,\mathbb{Z})'^{\wedge} & & \end{array}$$

by writing

$$(47)\qquad \mathcal{M}^{!\sim}_{0,3} \simeq \mathcal{M}^{!\sim}_{0,3} \cdot \widehat{\Pi}_{0,3} \quad !|:$$

393

Uncertainty retained from cancelled passages.

140-3-s394-D001 S_0

140-3-s394-D002 B'_0

140-3-s394-D003 $\mathcal{G}$

[scan 395]

But we also have a semidirect product

$$(48)\qquad \mathcal{M}^{\sim}_{0,3} \simeq \mathcal{M}^{!\sim}_{0,3}(0|\omega) \cdot \mathcal{G}|\mathcal{C}^{+}_{0,3}$$

which, combined directly with

$$\widehat{\mathcal{G}|\mathcal{C}}^{+}_{0,3} \longrightarrow \mathrm{Sl}(2,\widehat{\mathbb{Z}})'$$

gives a homomorphism

$$(49)\qquad \mathcal{M}^{\sim}_{0,3} \longrightarrow (\mathcal{M}_{\rho,\sigma}(\widehat{\mathbb{Z}}) \simeq S\mathcal{G}^{*}_{\rho,\sigma}(\widehat{\mathbb{Z}}) \cdot \mathrm{Sl}(2,\widehat{\mathbb{Z}}))/\Sigma$$

which extends the homomorphism (45), defined first on its normal subgroup $\mathcal{M}^{!\sim}_{0,3}$ of index 6. 394

[scan 396]

$\underline{\overline{\mathrm{X}}}$ Study of the group $\mathcal{M}_{\rho,\sigma}$ and of $\mathcal{M}_{\rho,\sigma} \to \mathrm{Gl}(2)$ [p. 499]

By construction of the semidirect product

$$\mathcal{M}^{\sim}_{\rho,\sigma} \;=\; \mathcal{G}^{*\flat|\natural|\circ}_{\rho,\sigma}(\omega|0)\cdot G \qquad \simeq \qquad \underbrace{(T_0 \times T_\rho \times N'_\sigma)}_{\substack{\text{operating on } G \\ \text{via } T_0 \hookrightarrow G}} \cdot G$$

$$\cup$$

(1)

$$\mathcal{M}_{\rho,\sigma} \;=\; S\mathcal{G}^{*\flat|\natural|\circ}_{\rho,\sigma}(\omega|0)\cdot SG \hookrightarrow \qquad \underbrace{(T_\rho \times N'_\sigma)}_{\substack{\text{operating on } G \\ \text{via } N'_\sigma \xrightarrow{\det} \mathbb{G}_m \xleftarrow[\sim]{\det} T_0 \hookrightarrow G}} \cdot G$$

we have a group homomorphism

$$(2)\qquad \begin{cases} \mathcal{M}^{\sim}_{\rho,\sigma} \xrightarrow{\Theta} G \\ \Theta(i_B(x)i_{\dagger\rho}(y)i_\sigma|i'_\sigma(z)i_G(f)) = x.f \end{cases}$$

$$(x \in \Gamma T_0, \quad y \in \Gamma T_\rho, \quad z \in \Gamma N'_\sigma, \quad f \in \Gamma G).$$

The composite homomorphism

$$(3)\qquad \begin{array}{ccccc} S\mathcal{M}^{\sim}_{\rho,\sigma} & \longrightarrow & \mathcal{M}^{\sim}_{\rho,\sigma} & \xrightarrow{\Theta} & G \\ \wr & & & & \\ \underbrace{{}^{\dagger}\mathcal{G}^{*\flat|\natural|\circ}_{\rho,\sigma}}_{B'_0\times T_\rho \times N'_\sigma} \cdot G & & & & \end{array}$$

described by (composite still denoted Θ)

$$\Theta(i_B(x)i_\rho(y)i_\sigma(z)i_G(f|\varphi)) = x.f$$

and Θ will indeed be trivial on the

$$i(\lambda) = i_B(\lambda)i_G(\lambda^{-1}) \qquad (\lambda \in \Gamma L_0)$$

and therefore passes to the quotient as $\Theta : \mathcal{M}^{\sim}_{\rho,\sigma} \to G$. 395

Uncertainty retained from cancelled passages.

140-3-s396-D001 $\rho|B$

140-3-s396-D002 $\mathcal{M}$

I am interested above all in the induced homomorphism [scan 397]

$$(4)\qquad \dagger\mathcal{M}_{\rho,\sigma} \xrightarrow{\Theta} G \simeq \mathrm{Gl}(2)_S$$

it is obvious|obviously an epimorphism, for its image contains trivially SG, so it suffices to see that the composite with

$$G/SG \xrightarrow[\sim]{\det} \mathbb{G}_m$$

is an epimorphism, i.e. that the composite with det is one, now this composite is also the homomorphism (again denoted χ)

$$(5)\qquad \begin{array}{ccccc} \mathcal{M}_{\rho,\sigma} & \xrightarrow{\substack{\text{canonical}\\ \text{projection}}} & S\mathcal{G}^{*\flat|\sharp}_{\rho,\sigma}(0)\dagger & \xrightarrow{\text{multiplier}} & \mathbb{G}_m \\ \wr & & & & \\ S\mathcal{G}^{*\flat|\sharp}_{\rho,\sigma}(0)\,.\,SG & & (\alpha,\beta,\mu,\varepsilon) & \longmapsto & \mu \end{array}$$

which is an epimorphism — in terms of

$$(6)\qquad S\mathcal{G}^{*\flat|\sharp}_{\rho,\sigma}(0) \hookrightarrow \dagger T_0 \times T_\rho \times N'_\sigma$$

this will be

$$(\underline{u},\underline{a},\underline{b}) \longmapsto \det \underline{u}\ ,$$

in terms of

$$(7)\qquad S\mathcal{G}^{*\flat|\sharp}_{\rho,\sigma}(0) \hookrightarrow T_\rho \times N'_\sigma$$

this will be

$$(\underline{a},\underline{b}) \longrightarrow \det \underline{b}$$

(cf. VIII, page 489 in particular (25)).

Let us define

$$(8)\qquad \mathcal{M}^+_{\rho,\sigma} \overset{\text{definition}}{=} \mathrm{Ker}\left(\ \mathcal{M}_{\rho,\sigma} \xrightarrow{\chi} \mathbb{G}_m\ \right),$$

we shall thus have a homomorphism of exact sequences 396

Uncertainty retained from cancelled passages.

140-3-s397-D001 $\mathcal{A}$

140-3-s397-D002 ∞

140-3-s397-D003 $\mathbb{G}_m$

[scan 398]
[p. 500]

$$(9)\qquad \begin{array}{ccccccccc} 1 & \longrightarrow & \mathcal{M}^+_{\rho,\sigma} & \longrightarrow & \mathcal{M}_{\rho,\sigma} & \longrightarrow & \mathbb{G}_m & \longrightarrow & 1 \\ & & \downarrow & & \downarrow{\scriptstyle\Theta} & & \downarrow{\scriptstyle\dagger} & & \\ 1 & \longrightarrow & \dagger SG & \hookrightarrow & G & \xrightarrow{\det} & \mathbb{G}_m & \longrightarrow & 1 \end{array}$$

which we shall compare with §39 page 299 (1)

As $\mathcal{M}_{\rho,\sigma}$ is flat of relative dimension 6, and G idem|idem of relative dimension 4, the kernel of the epimorphism $\Theta\colon \mathcal{M}_{\rho,\sigma} \longrightarrow G$ is necessarily flat of relative dimension 2. Let us denote it by $I_{\rho,\sigma}$, it is contained in the kernel $J_{\rho,\sigma}$ (of relative dimension 3) of the composite

$$\mathcal{M}_{\rho,\sigma} \longrightarrow G \longrightarrow PG\ ,$$

so that we have

$$(10)\qquad 1 \longrightarrow I_{\rho,\sigma} \longrightarrow J_{\rho,\sigma} \longrightarrow \mathbb{G}_m \longrightarrow 1 \longrightarrow 1$$

and a composition series of $\mathcal{M}_{\rho,\sigma}$

$$(11)\qquad \begin{array}{ccccccc} 1 & \subset & I_{\rho,\sigma} & \subset & J_{\rho,\sigma} & \subset & \mathcal{M}_{\rho,\sigma} \\ & & \llcorner\!\!\!

Uncertainty retained from cancelled passages.
140-3-s399-D001 x

the group [scan 400] [p. 501]

$$\mathcal{M}_{\rho,\sigma}/\dot{\mathcal{M}}^{\circ}_{\rho,\sigma} \quad \underset{\text{component of the identity}}{\simeq} \quad \mathcal{G}^{*}_{\rho,\sigma}(0)/\mathcal{G}^{*}_{\rho,\sigma}(0)^{\circ} \simeq \mu_2 \times \mu_2 \tag{15}$$

(cf. page 489 (23)), we shall thus have

$$\begin{array}{l} \mathcal{M}^{\circ}_{\rho,\sigma} = \mathcal{G}^{*}_{\rho,\sigma}(0)^{\circ} \,.\, SG \;\text{—}\; (T_0 \times T_\rho \times T_\sigma) \,.\, SG \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \wr\!\mid \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad T_\rho \times T_\sigma \times \underbrace{(T_0 \,.\, SG)}_{\underset{G}{\simeq}} \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \wr\!\mid \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad T_\rho \times T_{\sigma|\rho} \times G \end{array}$$

$$\begin{array}{l} \mathcal{M}^{\circ}_{\rho,\sigma} \hookrightarrow T_\rho \times T_\sigma \times G \\ \qquad\qquad \simeq \left\{ \begin{pmatrix} x & , & y & , & f \\ \cup\!\!\!\cap & & \cup\!\!\!\cap & & \cup\!\!\!\cap \\ \Gamma T_\rho & & \Gamma T_\sigma & & \Gamma G \end{pmatrix} \;\middle|\; \det x = \det y = \det f \right\} \end{array} \tag{16}$$

Thus the kernel

$$\left\{ \begin{array}{l} J^{\circ}_{\rho,\sigma} \overset{\text{definition}}{=} \operatorname{Ker} \begin{pmatrix} \mathcal{M}^{\circ}_{\rho,\sigma} & \longrightarrow & PG \\ & & \wr\!\mid \\ & & G/\mathbb{G}_m \end{pmatrix} \\ \qquad \simeq \left\{ \begin{pmatrix} x & , & y & , & \lambda \\ \cup\!\!\!\cap & & \cup\!\!\!\cap & & \cup\!\!\!\cap \\ \Gamma T_\rho & & \Gamma T_\sigma & & \Gamma \mathbb{G}_m \end{pmatrix} \;\middle|\; \det x = \det y = \lambda^2 \right\} \dagger \\ \qquad \simeq T_\rho \times_{\mathbb{G}_m} T_\sigma = \{(x,y) \mid \det x = \det y\} \\ \qquad \simeq S\mathcal{G}^{*}_{\rho,\sigma}(0)^{\circ} \end{array} \right. \tag{17}$$

Finally, we deduce from this that

$$\begin{array}{l} I^{\circ}_{\rho,\sigma} \overset{\text{definition}}{=} \operatorname{Ker} \begin{pmatrix} \mathcal{M}^{\circ}_{\rho,\sigma} & \longrightarrow & G \end{pmatrix} \\ \qquad \simeq ST_\rho \times ST_\sigma = \left\{ \begin{pmatrix} x & , & y \\ \cup\!\!\!\cap & & \cup\!\!\!\cap \\ \Gamma T_\rho & & \Gamma T_\sigma \end{pmatrix} \;\middle|\; \det x = \det y = 1 \right\}. \end{array} \tag{18}$$

Let us recapitulate: 399

Uncertainty retained from cancelled passages.
140-3-s400-D001 $\simeq T_\rho \times_{\mathbb{G}_m} T_\sigma$

Theorem The group [scan 401]

$$\begin{array}{l} \mathcal{M}_{\rho,\sigma} \overset{\text{definition}}{\simeq} \mathcal{G}^{*}_{\rho,\sigma}(0) \,.\, SG \longleftarrow \text{cf. definition page 487' (20)} \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \cup \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad T_0 \times T_\rho \times N'_\sigma \end{array} \tag{19}$$

is an extension of $\mu_2 \times \mu_2$ by $\mathcal{M}^\circ_{\rho,\sigma}$

$$(20) \qquad 1 \longrightarrow \mathcal{M}^\circ_{\rho,\sigma} \longrightarrow \mathcal{M}_{\rho,\sigma} \longrightarrow \mu_2 \times \mu_2 \longrightarrow 1 \ .$$

deduced from

$$\mathcal{G}^*_{\rho,\sigma}\dagger \longrightarrow \mu_2 \times \mu_2$$

$$(\alpha, \beta, \mu, \varepsilon) \longrightarrow (\det \alpha, \varepsilon(= \det \underline{b}))$$

In $\mathcal{M}^\circ_{\rho,\sigma}$ we †the subgroup central (in the whole of $\mathcal{M}_{\rho,\sigma}$)

$$(21) \qquad ST_\rho \times ST_\sigma \hookrightarrow \mathcal{G}^*_{\rho,\sigma}(0) \hookrightarrow \mathcal{M}_{\rho,\sigma} \ ,$$

and we have an exact sequence

† (21)|(22)

$$\boxed{1 \longrightarrow ST_\rho \times ST_\sigma \longrightarrow \mathcal{M}^\circ_{\rho,\sigma} \xrightarrow{\Theta} G \longrightarrow 1}$$
$$\uparrow$$
$$SG$$

We have † in $\mathcal{M}^\circ_{\rho,\sigma}$ a structure of an extension

read †

NB
$J^\circ_{\rho,\sigma}$ is also
central in $\mathcal{M}_{\rho,\sigma}$
and $J^\circ_{\rho,\sigma} \cap SG \simeq$

$$x \longmapsto \chi(x), \varepsilon(x), \varepsilon'(x)$$

$$(23) \qquad \boxed{1 \longrightarrow ST_\rho \times ST_\sigma \times SG \longrightarrow \mathcal{M}_{\rho,\sigma} \longrightarrow \mathbb{G}_m \times \mu_2 \times \mu_2 \longrightarrow 1}$$

i.e.

$$(24) \qquad \mathcal{M}^{\circ+}_{\rho,\sigma} \simeq ST_\rho \times ST_\sigma \times SG, |\}$$

† nonexistent

†

We have a canonical homomorphism

$$(25) \qquad \begin{array}{ccc} \mu_2 \times \mu_2 \times \mu_2 \hookrightarrow \mathcal{M}^{\circ+}_{\rho,\sigma} & \overset{\text{definition}}{=} & \mathrm{Ker}(\mathcal{M}^\circ_{\rho,\sigma} \xrightarrow{\chi} \mathbb{G}_m) \\ & & \wr\, (24) \\ & & ST_\rho \times ST_\sigma \times SG \end{array}$$

400

Uncertainty retained from cancelled passages.
140-3-s401-D001 (0)
140-3-s401-D003 have two
140-3-s401-D005 (22)
140-3-s401-D006 therefore
140-3-s401-D008 $\times|$.
140-3-s401-D009 using
140-3-s401-D011 $\longrightarrow$

[scan 402]

thanks to the canonical inclusions [p. 502]

$$\mu_2 \hookrightarrow ST_\rho, \qquad \mu_2 \hookrightarrow ST_\sigma, \qquad \mu_2 \hookrightarrow SG \tag{26}$$

giving rise to canonical homomorphisms already considered elsewhere

$$\Sigma = (\mathbb{Z}/2\mathbb{Z})^3 \longrightarrow \Gamma\mathcal{M}^{\circ+}_{\rho,\sigma}\dagger \subset \Gamma\mathcal{M}^{+|*}_{\rho,\sigma}. \tag{27}$$

I am going to introduce for the group $\mathcal{M}_{\rho,\sigma}$ notations which paraphrase the notations $\mathcal{M}_{0,3}$ etc., in the context of group schemes. The invariant subgroup SG of $\mathcal{M}_{\rho,\sigma}$ will also be denoted $\widetilde{G}^+_{\rho,\sigma}$ (the analogue of $\widetilde{\widetilde{G}}^+_{0,3}$), the quotient

$$\dagger\mathcal{M}_{\rho,\sigma}/\widetilde{G}^+_{\rho,\sigma} \simeq \tau^!_{\rho,\sigma} \tag{28}$$

paraphrases $\tau^!_{0,3}$, we have moreover

$$\tau^!_{\rho,\sigma} \simeq S\mathcal{G}^*_{\rho,\sigma}(0) \hookrightarrow T_0 \times T_\rho \times N'_\sigma \tag{29}$$

m1072

Marginal note. m1072 Seen hanging around a bit the sign !, (which has fallen from the sky — even in the section following the X) it is $\Gamma_{\rho,\sigma}$ (and not $\tau_{\rho,\sigma}$ without the exponent !) that one ought to denote this group by !

— in what follows the simpler and more suggestive notation $\tau_{\rho,\sigma}$ will be used in preference to $S\mathcal{G}^*_{\rho,\sigma}(0)$. We thus have

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \widetilde{G}^+_{\rho,\sigma} & \longrightarrow & \mathcal{M}_{\rho,\sigma} & \to & \tau^!_{\rho,\sigma} & \to & 1 \\ & & \wr| & & & & & & \\ & & SG \simeq \mathrm{Sl}(2)_S & & & & & & \end{array} \tag{30}$$

and a canonical splitting as a semidirect product

$$\mathcal{M}_{\rho,\sigma} \simeq \mathcal{M}_{\rho,\sigma}(0) \,.\, \widetilde{G}^+_{\rho,\sigma} \tag{31}$$

401

Uncertainty retained from cancelled passages.
140-3-s402-D001 $+$
140-3-s402-D002 $\mathcal{M}$

[scan 403]

where

$$\begin{array}{ccc} \mathcal{M}_{\rho,\sigma}(0) & \hookrightarrow & \mathcal{M}_{\rho,\sigma} \\ & \searrow^{\sim} & \downarrow{\scriptstyle \mathrm{can}} \\ & & \tau^{!}_{\rho,\sigma} \end{array} \tag{32}$$

is a subgroup isomorphic to $\tau^{!}_{\rho,\sigma}$ — which corresponds to the subgroup $\mathcal{M}^{!\sim}_{0,3}(0)$ in $\mathcal{M}^{\sim}_{0,3}$ (§46, V).

Remark. One would like to introduce a $\tau_{\rho,\sigma}$ extension (or product) of $\tau^{!}_{\rho,\sigma}$ by a group such as $\widetilde{G}_3$ (which would be denoted $\tau^{+}_{\rho,\sigma}$), in relation to a subgroup $\pi_{\rho,\sigma}$ of $\widetilde{G}^{+}_{\rho,\sigma}$ such that $\widetilde{G}^{+}_{\rho,\sigma}/\pi_{\rho,\sigma} \simeq \widetilde{G}_3$ — but this is not possible in the context of group schemes (even in the modular special case), since $\widetilde{G}^{+}_{\rho,\sigma} \simeq \mathrm{Sl}(2)_S$ is † relatively connected over S. On the other hand one can, in the "modular case" $\rho = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}$, $\sigma = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ introduce a canonical section

$$\dot{\tau}|\dot{t} = \dot{\tau}_\infty|\dot{t}_\infty \in \Gamma(\tau^{!}_{\rho,\sigma}) \tag{33}$$

whose representative τ_∞ in $\Gamma\mathcal{M}^{!}_{\rho,\sigma}(0)$ is explicitly given as

$$\tau_\infty = \begin{pmatrix} -1 & 0 \\ 0 & +1 \end{pmatrix}. \tag{34}$$

(It would be interesting to see whether, outside the 402

Uncertainty retained from cancelled passages.

140-3-s403-D002 [illeg. 1]

[scan 404]

modular case, one can introduce "anti-involutions" which generalize the geometric reflections [p. 503]

$$\begin{cases} \dagger r_0 = \tau'_\infty = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ r_1 = -\tau'_0 = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix} \qquad \text{(cf. p. 335′, §42, IV)} \\ r_\infty = \tau_\infty = \begin{pmatrix} -1 & 0 \\ 0 & +1 \end{pmatrix} \end{cases} \tag{35}$$

I am convinced that one can. Besides, had we not brought to light in $\mathcal{M}_{\rho,\sigma}$ a subgroup $\mu_2 \times \mu_2 \times \mu_2$, † giving the three remarkable sections (corresponding respectively to the sections -1 of the three factors μ_2)? But I am getting things mixed up here — these sections were sections of $\mathcal{M}^{\circ}_{\rho,\sigma}$ — whereas the sections † $\tau_\infty, \tau'_\infty, \tau'_0$ are of multiplier † $Z_{\rho,\sigma} \overset{\mathrm{df}}{=} (\mathrm{Centr}\,\mathcal{M}^{\circ}_{\rho,\sigma})$. -1, thus correspond to $\varepsilon = \varepsilon' = -1$, thus correspond to the "least trivial" section $(-1,-1)$ of $\mathcal{M}_{\rho,\sigma}/\mathcal{M}^{\circ}_{\rho,\sigma} = \mu_2 \times \mu_2$.

↪

⟶

†

⟶

⟶

†

403

Uncertainty retained from cancelled passages.

140-3-s404-D001 τ

140-3-s404-D002 corresponding in

140-3-s404-D004 modular

140-3-s404-D005 isomorphic to

140-3-s404-D017 "injective"

140-3-s404-D024 composite side

† † [scan 405]

$\hookrightarrow$ †

Let us do the calculations to determine the invariants α β ... corresponding to r_0, r_1, r_∞.

i) $r_\infty = \tau_\infty = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \in \mathrm{Gl}(2,\mathbb{Z}) \subset \mathrm{Gl}(2,\widehat{\mathbb{Z}})^{\pm 1}$

We even have: $\tau_\infty \in T_0'|B_0'(\mathbb{Z})$, it is the only element that is nontrivial in $T_0'(\mathbb{Z}) \simeq \mathbb{G}_m(\mathbb{Z}) \simeq \{\pm 1\}$. †

The inner automorphism it defines in $\Pi_{0,3}$ or in $\mathrm{Sl}(2,\mathbb{Z})$ — or their profinite completions — stabilizes the subgroup $\widetilde{L}_0$ generated by ε_0 (i.e. L_0 generated by $\varepsilon_0^2 = \ell_0$)

$$\tau_\infty(\varepsilon_0) = \varepsilon_0^{-1}, \qquad \tau_\infty(\ell_0) = \ell_0^{-1}$$

(∗)

thus

$$\tau_\infty \in \mathcal{M}_{0,3}^{!\sim}[0]$$

† † it is of the form $u_{\alpha,\beta}$, † the calculations already done give

$$\alpha = \tau_\infty, \qquad \zeta|\xi = g = \alpha\rho(\alpha)^{-1} = -\ell_1^{-1} = -\begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$$

(watch the sign − when doing the calculations in $\mathrm{Sl}(2,\mathbb{Z})$ and not in $\mathrm{Sl}(2,\mathbb{Z})'$!)

$$\beta = \underline{1}, \qquad \eta = h = 1,$$
$$\mu = -1, \qquad \varepsilon = \varepsilon_2(\mu) = -1, \qquad \varepsilon' = \varepsilon_3(\mu)|\varepsilon_3^b(\mu) = -1.$$

we have here $B = 0, D = -1$, thus

$$u = \begin{pmatrix} \dagger\frac{1}{D} = \mu & \frac{B}{D} = \mu D|\mu B \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \tau_\infty$$

$$\underline{a} = \alpha^{-1}u = \tau_\infty^{-1}\tau_\infty = \underline{1}, \qquad \underline{b} = \beta^{-1}u = \tau_\infty.$$

thus

$$r_\infty = \tau_\infty = \dagger(\underline{u}_\infty = \tau_\infty, \underline{a}_\infty = 1, \underline{b}_\infty = \tau_\infty) \in T_0 \times T_\rho \times N_\sigma'. \tag{36}$$

the nontrivial element of ${}_2T_0(\mathbb{Z}) \simeq {}_2T_0'(\mathbb{Z})$

N.B. Let more generally

$$\underline{u} = \begin{pmatrix} \mu & \lambda \\ 0 & 1 \end{pmatrix} \in \dagger\, B_0'(\widehat{\mathbb{Z}}) \subset \mathrm{Gl}(2,\mathbb{Z})$$

and let us consider the automorphism of $\mathrm{Sl}(2,\mathbb{Z})'\simeq$ which it induces, hence an automorphism ($\in \mathcal{M}_{0,3}^{!\sim}$) of $\widehat{\widetilde{G}}_{0,3}^{+}$, what are these invariants α, β etc.? We shall be able obvious- 404

Uncertainty retained from cancelled passages.

140-3-s405-D001 trivial centrality
140-3-s405-D002 $\mathbb{G}_m\times$
140-3-s405-D004 T_ρ
140-3-s405-D007 The
140-3-s405-D008 etc.
140-3-s405-D009 we find
140-3-s405-D010 we
140-3-s405-D011 μ
140-3-s405-D012 $(\underline{u}_\infty=-1,\underline{a}_\infty=1,\underline{b}_\infty=\tau_\infty)$
140-3-s405-D013 Gl

ly take $f=1$, and we will have [scan 406] [p. 504]

$$\alpha=\underline{u},\qquad \text{whence}\qquad \underline{a}=1,$$

on the other hand, let us note that $\mu\in\{\pm1\}$i.e. $\mu=(-1)^k$ $(k\in\mathbb{Z}/2\mathbb{Z})$, we will have $\varepsilon=\varepsilon'=\mu$, and if $\mu=1$ (i.e. $u\in \mathrm{Sl}(2,\mathbb{Z})$) we will have $\beta=\underline{u}$ (hence $\underline{b}=1$, otherwise $\beta=\dagger u\tau_\infty$ hence $\underline{b}=\tau_\infty$), in all cases we will have

$$\beta=u\tau_\infty^k,\qquad \underline{b}=\tau_\infty^k,$$

in summary

$$\underline{u}=\left(\begin{array}{ccc}\underline{u}=u, & \underline{a}=1, & \underline{b}=\tau_\infty^k\\ \cup & \cup & \cup\\ \Gamma B_0' & \Gamma T_\rho & \Gamma N_\sigma'\end{array}\right)\in \dagger\, S\mathcal{G}|\mathcal{G}^*_{\rho,\sigma}(0)(\widehat{\mathbb{Z}})\subset\left(B_0'\times T_\rho\times N_\sigma'\right)(\widehat{\mathbb{Z}}).\qquad(37)$$

$\mathcal{M}_{\rho,\sigma}[0](\widehat{\mathbb{Z}})$

$\dagger$

But this is not an element in $\dagger$ normalized form — $\dagger$ i.e. it is not in $\Gamma\mathcal{G}^*_{\rho,\sigma}(0)$, unless $\lambda=0$. $\dagger$

2°) $r_1=-\tau_0'=\begin{pmatrix}-1 & 1\\ 0 & 1\end{pmatrix}$, this is precisely a case that falls under the preceding remarks, we will have

$$r_1|\tau_1=-\tau_0'=\left(\underline{u}_1=-\tau_0'=\begin{pmatrix}-1 & 1\\ 0 & 1\end{pmatrix},\ \underline{a}=1,\ \underline{b}=\tau_\infty\right)\in\mathcal{M}_{\rho,\sigma}[0](\mathbb{Z})\qquad(38)$$

m1077

Marginal note. m1077 N.B. reducing modulo $L_0\subset\widetilde{G}^+_{\rho,\sigma}$, τ_∞ and r_1 have the same image in $\tau_{\rho,\sigma}$, namely $(-1,1,\tau_\infty)$.

3°) $r_0=\tau_\infty'=\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$, this time we have an automorphism of $\widetilde{G}^+_{0,3}$ or $\widetilde{G}^{+\wedge}_{0,3}$ which does not normalize L_0 — that is better, if we want r_0,r_1,r_∞ to generate $\mathrm{Gl}(2,\mathbb{Z})$ (or even "only" $\mathrm{Gl}(2,\mathbb{Z})'$) !

We have $\tau_\infty'=\tau_\infty\sigma_\infty$, so its image in $\mathcal{M}_{\rho,\sigma}(\mathbb{Z})\subset\mathcal{M}_{\rho,\sigma}(\widehat{\mathbb{Z}})$ is $\dagger$ the product of the images of τ_∞ and σ_∞ — the first was computed in 1°, the second is tautological since $SG\simeq\widetilde{G}^+_{\rho,\sigma}$ and $\sigma_\infty\in SG(\mathbb{Z})$. We will note that $\dagger$ $r_0=\tau_\infty'$ thus has the same image 405

Uncertainty retained from cancelled passages.

140-3-s406-D002 τ_∞
140-3-s406-D003 $\mathcal{M}$
140-3-s406-D004 apply the

140-3-s406-D005 preceding paragraph|preceding argument
140-3-s406-D006 we
140-3-s406-D007 We will have
140-3-s406-D008 like
140-3-s406-D009 τ'_∞

[scan 407]

in $\tau_{\rho,\sigma}(\mathbb{Z})$ as $r_\infty = \tau_\infty$ and $r_1 = -\tau'_0$, it is always the same element † [p. 504]

$$(39) \qquad (-1, 1, \tau_\infty) \in \left(\underset{\wr \mathbb{G}_m}{T_0} \times T_\rho \times N'_\sigma \right)(\mathbb{Z})$$

Let us moreover compute

$$T_0 \times T_\rho \times N'_\sigma(\mathbb{Z}) = T_0(\mathbb{Z}) \times T_\rho(\mathbb{Z}) \times N'_\sigma(\mathbb{Z})$$

We have

$$(40) \quad \begin{cases} T_0 \simeq \mathbb{G}_m, \quad T_0(\mathbb{Z}) \simeq \{\pm 1\} & \\ T_\rho(\mathbb{Z}) \simeq (\mathbb{Z} + \mathbb{Z}.\rho)^* \simeq \mathbb{Z}/6\mathbb{Z} & \text{group generated by } \rho \\ N'_\sigma(\mathbb{Z}) \simeq \{1, \tau_\infty\}.T_\sigma(\mathbb{Z}) & \text{(semidirect product)} \\ T_\sigma(\mathbb{Z}) \simeq (\mathbb{Z} + \mathbb{Z}\sigma)^* \simeq \mathbb{Z}/4\mathbb{Z} & \text{group generated by } \sigma \\ N'_\sigma(\mathbb{Z}) \simeq D_4 & \text{dihedral group of order 4 (“symmetries of the square”)} \end{cases}$$

whence

$$(41) \quad (T_0 \times T_\rho \times N'_\sigma)(\mathbb{Z}) \simeq \underbrace{(\mathbb{Z}/2\mathbb{Z})}_{\substack{\text{group generated} \\ \text{by } \tau_\infty}} . (\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z})$$

τ_∞ operates trivially on the first two factors and by symmetry on the last

$$\simeq \underbrace{\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z} \times \underbrace{(\mathbb{Z}/2\mathbb{Z}.\mathbb{Z}/4\mathbb{Z})}_{D_4}}_{\downarrow}$$

On the other hand m1079

Marginal note. m1079 N.B. the condition $\det \underline{a} = \varepsilon' \det \underline{u}$, $\varepsilon'^2 = 1$ is automatic, since $\mathbb{G}_m(\mathbb{Z}) = \{\pm 1\}$.

$$(42) \quad S\mathcal{G}^*_{\rho,\sigma}(0)(\mathbb{Z}) = \tau_{\rho,\sigma}\dagger(\mathbb{Z}) = \left\{ \underbrace{(u, a, b)}_{\in T_0 \times T_\rho \times N'_\sigma(\mathbb{Z})} \mid \det u = \det b \right\}$$
$$\simeq (T_\rho \times N'_\sigma)(\mathbb{Z}) \simeq \mathbb{Z}/6\mathbb{Z} \times D_4$$

† † 406

Uncertainty retained from cancelled passages.
140-3-s407-D001 [illeg. 1]
140-3-s407-D002 [0]
140-3-s407-D004 arrow specified by
140-3-s407-D005 $\mathcal{M}$

[scan 408]

the remarkable element, common image of $r_0 = \tau'_\infty$, $r_1 = -\tau'_0$, $r_\infty = \tau_\infty$ corresponds to the element $(1, \tau_\infty)$ of $\mathbb{Z}/6\mathbb{Z} \times (D_4 \simeq \{1, \tau_\infty\}.\mathbb{Z}/4\mathbb{Z})$. [p. 505]

This also gives us an expression for †

$$(43) \qquad \begin{cases} \mathcal{M}_{\rho,\sigma}(\mathbb{Z}) \simeq \underbrace{\tau_{\rho,\sigma}(\mathbb{Z})}_{\mathcal{M}_{\rho,\sigma}(0)(\mathbb{Z})} . \mathrm{Sl}(2,\mathbb{Z}) \\ \qquad\qquad (\text{semidirect product}) \\ \qquad\qquad \simeq (\mathbb{Z}/4\mathbb{Z} \times D_4).\mathrm{Sl}(2,\mathbb{Z}) \end{cases}$$

where $\mathbb{Z}/4\mathbb{Z} \times D_4$ operates on $\mathrm{Sl}(2,\mathbb{Z})$ via the composite canonical homomorphism

(44)

$$\mathbb{Z}/4\mathbb{Z} \times D_4 \xrightarrow[\text{projection}]{\text{canonical}} D_4 \xrightarrow[\text{canonical}]{\text{homomorphism}} \{1, \tau_\infty\}$$

τ_∞ operating on $\mathrm{Sl}(2,\mathbb{Z})$ by inner automorphism with the matrix

$$\tau_\infty = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

!

We note that the canonical epimorphism

$$\mathcal{M}_{\rho,\sigma} \longrightarrow \mu_2 \times \mu_2$$

† through homomorphism $\tau_{\rho,\sigma}$, and induces homomorphisms on the integral points

(45)

$$\mathcal{M}_{\rho,\sigma}(\mathbb{Z}) \to \tau_{\rho,\sigma}(\mathbb{Z}) \simeq \mathbb{Z}/\dagger 6\mathbb{Z} \times D_4 \to \begin{matrix} (\mu_2 \times \mu_2)(\mathbb{Z}) \\ \simeq \{\pm 1\} \times \{\pm 1\} \end{matrix}$$

This homomorphism is zero on $\mathbb{Z}/6\mathbb{Z} \times \underbrace{\mathbb{Z}/4\mathbb{Z}}_{D_4^+}$ † (corresponds to determinants equal to 1), and on τ_∞ takes the value $(-1,-1)$. We will note therefore that this homomorphism on the integral points is 407

Uncertainty retained from cancelled passages.

140-3-s408-D001 hence

140-3-s408-D002 factors

140-3-s408-D003 4

140-3-s408-D004 det a

[scan 409]

not surjective.

Center of $\mathcal{M}_{\rho,\sigma}$= $\mathcal{M}_{\rho,\sigma}(0).SG$. It is the direct product of the the subgroup of the center of $\mathcal{M}_{\rho,\sigma}(0)$ which operates trivially on SG, by the subgroup of the points †invariant under $\mathcal{M}_{\rho,\sigma}$ center of SG. The latter center is well known, it is μ_2, † and $\mathcal{M}_{\rho,\sigma}$ operates on it trivially.

m1083

Marginal note. m1083 false, an element $g.f$ of a 1/2 direct product can be central without g operating trivially on the invariant group G.

On the other hand the subgroup of

$$\mathcal{M}_{\rho,\sigma}(0) \simeq \tau_{\rho,\sigma} \subset B'_0 \times T_\rho \times N'_\sigma$$

which operates trivially on SG, is the intersection of $\tau_{\rho,\sigma}$ with $(\dagger 1 \times T_\rho \times N'_\sigma)$, for B'_0 operates trivially † (since

$$B'_0 \cap \underbrace{\mu_2}_{\substack{\text{center of} \\ SG}} = \{1\} \tag{46}$$

. It is therefore the subgroup of the

$$\left\{ \left(1, \underbrace{\underline{a}}_{\in \Gamma T_\rho}, \underbrace{\underline{b}}_{\in \Gamma N'_\sigma} \right) \mid (\det \underline{a})^2 = 1,\ \det \underline{b} = 1 \right\} \tag{47}$$

Remains : determine the center of $\tau_{\rho,\sigma}$, extension of $\mu_2 \times \mu_2$ by $\tau^0_{\rho,\sigma} \simeq \mathcal{G}^*_{\rho,\sigma}(0)^0 \subset T^0_0 \times T_\rho \times \dagger T_\sigma$. no!
Since $\tau^0_{\rho,\sigma}$ is commutative, it suffices to determine the action of $\mu_2 \times \mu_2$ on $\tau^0_{\rho,\sigma}$, and to determine its subgroup of invariants. m1085

Marginal note. m1085 This will give at least the intersection of the center with $\tau^0_{\rho,\sigma}$. 408

Uncertainty retained from cancelled passages.
140-3-s409-D002 fixed
140-3-s409-D003 is the same
140-3-s409-D004 μ_2
140-3-s409-D005 $\subset$
140-3-s409-D006 N'_σ

[scan 410]
[p. 506]

Let us note that we have

$$\tau_{\rho,\sigma} = S\mathcal{G}^*_{\rho,\sigma}(0) \subset \mathcal{G}^*_{\rho,\sigma}(0) \simeq T_0 \times T_\rho \times N'_\sigma$$

and we have a homomorphism of exact sequences
(48)

$$\begin{array}{ccccccccc}
1 & \longrightarrow & \tau^0_{\rho,\sigma} & \longrightarrow & \tau^0_{\rho,\sigma} & \xrightarrow{(\underline{u},\underline{a},\underline{b}) \mapsto (\varepsilon(\underline{b}), \det \underline{a}/\det(\underline{u}))} & \mu_2 \times \mu_2 & \longrightarrow & 1 \\
 & & \downarrow & & \downarrow & & \downarrow{\scriptstyle \text{first projection}} & & \\
1 & \longrightarrow & T_0 \times T_\rho \times \dagger T_\sigma & \longrightarrow & T_0 \times T_\rho \times N'_\sigma & \longrightarrow & \mu_2 & \longrightarrow & 1
\end{array}$$

,

which shows that the action of $\mu_2 \times \mu_2$ on † $\tau^0_{\rho,\sigma}$ takes place through the projection of $\mu_2 \times \mu_2$ onto its first factor, and this by the operation of the latter on $T_0 \times T_\rho \times T_\sigma$ which operates trivially on $T_0 \times T_\rho$, and which on T_σ is the canonical operation of μ_2 on T_σ, $\sigma \mapsto \varepsilon\sigma$, whose invariants are reduced to the scalars $\mathbb{C}_m$. This reasoning shows, more generally, that the trace of the center of $\tau_{\rho,\sigma}$ on the subgroup $\tau'_{\rho,\sigma} \subset \tau_{\rho,\sigma}$ kernel of

$$\tau_{\rho,\sigma} \longrightarrow \mu_2 \times \mu_2 \overset{1^{\text{st}}\text{ projection}}{\longrightarrow} \mu_2$$

$$(\underline{u},\underline{a},\underline{b}) \longmapsto \varepsilon(\underline{b})$$

is also the trace, on this subgroup, of $T_0 \times T_\rho \times \mathbb{G}_m$. Moreover, since μ_2 operates faithfully on T_σ via $\sigma \mapsto \varepsilon\sigma$, we see at once that it also operates faithfully on $\tau'_{\rho,\sigma}$ (which indeed maps epimorphically onto T_σ !) so we have

$$\underline{\text{Centre}}(\tau_{\rho,\sigma}) \subset \tau'_{\rho,\sigma} = \text{Ker}(\tau_{\rho,\sigma} \overset{(\underline{u},\underline{a},\underline{b})\mapsto\varepsilon(\underline{b})}{\longrightarrow} \mu_2)$$

409

Uncertainty retained from cancelled passages.

140-3-s410-D001 N'_σ

140-3-s410-D002 τ

[scan 411]

Thus in summary

$$(49)\quad \begin{cases} \underline{\text{Centre}}(\mathcal{G}^*_{\rho,\sigma}(0)) = T_0 \times T_\rho \times \mathbb{G}_m \\ \underline{\text{Centre}}\left(\underset{S\mathcal{G}^*_{\rho,\sigma}(0)}{\underline{\tau_{\rho,\sigma}}}\right) = \dagger(T_0 \times T_\rho \times \mathbb{G}_m) \cap \tau_{\rho,\sigma} \\ \qquad = \left\{ \left(\underset{\substack{\in\Gamma T_0 \\ \simeq \Gamma\mathbb{G}_m}}{\underline{u}}, \underset{\in\Gamma T_\rho}{\underline{a}}, \underset{\in\Gamma\mathbb{G}_m}{\lambda} \right) \mid u = \lambda^2,\ \dagger\det\underline{a} = \varepsilon'\lambda^2,\ \varepsilon'^2 = 1 \right\} \end{cases}$$

The subgroup of the center which operates trivially on SG is thus, by (47), given by the additional restriction $u = 1$, it thus identifies with

$$(50)\quad \begin{cases} 1 \times ST'_\rho \times \mu_2 \subset \tau^{a|0}_{\rho,\sigma} \subset T_0 \times T_\rho \times N'_\sigma \\ \text{where } ST'_\rho \subset T_\rho \\ \qquad\qquad \| \{\underline{a} \mid (\det\underline{a}]^2 = 1\} \end{cases}$$

Thus

$$(51)\quad \dagger$$

false the center is larger,

The group of integral points of this group, in the modular case, is reduced:

$$(52)\quad \begin{cases} \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \\ \simeq \mathbb{Z}/3\mathbb{Z} \times (\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}), \end{cases}$$

† m1089

Marginal note. m1089 N.B. This group is in fact contained in $\mathcal{M}^0_{\rho,\sigma}(\mathbb{Z})$.

product of a cyclic group of order 3 (naturally generated by $-\rho$) with a group $(\mathbb{Z}/2\mathbb{Z})^3$, which is none other than the group Σ already in (IX).

More interesting would probably be the 410

Uncertainty retained from cancelled passages.
140-3-s411-D001 Centre
140-3-s411-D002 $\underline{a}$
140-3-s411-D004 ST'_ρ
140-3-s411-D006 Group of

[scan 412]

center of $\mathcal{M}^\circ_{\rho,\sigma}$, which will thus be the direct product of the center μ_2 of SG (on which the commutative group $\mathcal{M}^\circ_{\rho,\sigma}(0) \simeq \tau^\circ_{\rho,\sigma}$ operates trivially) with the subgroup of $\tau^\circ_{\rho,\sigma} \subset T_0 \times T_\rho \times T_\sigma$ which operates trivially on SG, namely † [p. 507]

false

$$\tau^\circ_{\rho,\sigma} \cap 1 \times T_\rho \times T_\sigma = \{(1, \underline{a}, \underline{b}) \mid \det \underline{b} = 1,\ \det \underline{a}\dagger = 1\}$$
$$\simeq ST_\rho \times ST.$$

m1092

Marginal note. m1092 (and not just $(\det a)^2 = 1$, since we are in $\tau^\circ_{\rho,\sigma}$

Thus
We
thus in the modular case we have
†
†
†

false ! Center is of dimension 3

411

Uncertainty retained from cancelled passages.
140-3-s412-D001 $\tau_{\rho,\sigma}$
140-3-s412-D003 2
140-3-s412-D005 54|14
140-3-s412-D006 $\mathcal{M}_{\rho,\sigma}$ although the latter
140-3-s412-D009 dévissage|discussion

[scan 413]

XI Recapitulation on the group $\mathcal{M}_{\rho,\sigma}$. [p. 508]

Preliminaries

Ⓣ Given, over a scheme S, a form G of Gl(2), hence forms SG, PG of Sl(2), GP(1)|PGl(1), and two sections $\tilde\rho, \tilde\sigma$ of PG, and a 1-parameter † additive bordered|based subgroup

$$\underline{\tilde\varepsilon}_0 : G_a \hookrightarrow PG$$

$\varepsilon_0|G_a$ (which amounts to the same thing, a † regular unipotent section of ε_0 of G, or a regular nilpotent section N_0 in the form $\mathcal{M}$ of $\mathrm{M}_2(\mathcal{O}_S)$ associated to G) — satisfying the conditions

(1) $\tilde\sigma^{-1}\tilde\rho = \underline{\tilde\varepsilon}_0(1)$, $\underline{\tilde\varepsilon}_0$ and $\underline{\tilde\varepsilon}_1 = \rho(\underline{\tilde\varepsilon}_0)(= \sigma(\underline{\tilde\varepsilon}_0))$ are independent on each fiber

(2) $\operatorname{Tr}\sigma = 0$ if σ is a local lifting of $\tilde\sigma$,

We define canonical liftings ρ, σ of $\tilde\rho, \tilde\sigma$ to G, † by the conditions

(3) † $\begin{cases} \varepsilon_0 = -\sigma^{-1}\rho & \text{i.e. } \varepsilon_1 = -\rho\sigma^{-1} \\ \operatorname{Tr}(\rho+\sigma) = 1 & \text{i.e. } \operatorname{Tr}\rho = 1, \end{cases}$

and the constructions which follow make sense.

We construct a pinning

$$\dagger\ (5)\quad \dagger\ \mathrm{Gl}(2) \xrightarrow{\sim} G$$

by the condition that ε_1 take the form

$$(6)\qquad \varepsilon_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

whence

$$(7)\quad \begin{cases} \rho = \begin{pmatrix} 0 & 1 \\ v\vert\upsilon & 1 \end{pmatrix}, \qquad \sigma = \begin{pmatrix} 0 & -1 \\ -v\vert\upsilon & 0 \end{pmatrix} \qquad (v\vert\upsilon \in \Gamma G_m \\ \det\rho = \det\sigma = -v\vert\upsilon \end{cases}$$

$$(8)\qquad \varepsilon_0 = \begin{pmatrix} 1 & 1/v\vert\upsilon \\ 0 & 1 \end{pmatrix},$$

412

Uncertainty retained from cancelled passages.

140-3-s413-D001 unipotent|multiplicative
140-3-s413-D002 of
140-3-s413-D003 satisfy
140-3-s413-D004 $\sigma\rho$
140-3-s413-D005 4
140-3-s413-D006 $\tau_{\rho,\sigma}$

[scan 414]

we consider the subgroups

$$(9)\quad \begin{cases} L_0 = \{\underline{\varepsilon}_0(\lambda)\} = \left\{ \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix} \right\} & (\lambda \in \Gamma G_a) \\ L_1 = \sigma(L_0) = \rho(L_0) = \{\underline{\varepsilon}_1(\lambda)\} = \left\{ \begin{pmatrix} 1 & 0 \\ \lambda & 1 \end{pmatrix} \right\} & \setminus \mathrm{id} \\ B_0' = \left\{ \begin{pmatrix} \mu & \lambda \\ 0 & 1 \end{pmatrix} \mid \mu \in \Gamma G_m,\ \lambda \in \Gamma G_a \right\} \\ T_0 = \left\{ \begin{pmatrix} \mu & 0 \\ 0 & 1 \end{pmatrix} \mid \mu \in \Gamma G_m \right\} \end{cases}$$

whence

$$(10)\qquad B_0' = T_0.L_0 \qquad (\tfrac{1}{2}\text{-direct product})$$

and we have isomorphisms

$$(11)\quad \begin{cases} \underline{\varepsilon}_0 : G_a \xrightarrow{\sim} L_0 \qquad \lambda \longrightarrow \varepsilon_0(\lambda) = \begin{pmatrix} 1 & \lambda/v|v \\ 0 & 1 \end{pmatrix} \\ \underline{\varepsilon}_1 : G_a \xrightarrow{\sim} L_1 \qquad \lambda \longrightarrow \varepsilon_1(\lambda) = \begin{pmatrix} 1 & 0 \\ \lambda & 1 \end{pmatrix} \\ \text{thus } \underline{\varepsilon}_1(\lambda) = \rho(\,|\{\,\varepsilon_0(\lambda)) = \sigma(\varepsilon_0(\lambda)). \end{cases}$$

We consider

$$(12)\quad \dagger \begin{cases} T_\rho = \text{scheme of units of } \mathcal{A}_\rho = \underline{\mathcal{O}}_S + \underline{\mathcal{O}}_S\rho \\ \qquad\qquad\qquad\qquad \simeq \underline{\mathcal{O}}_S[X]/(X^2 - X - v|v) \\ \quad \dagger \\ T_\sigma = \text{scheme of units of } \mathcal{A}_\sigma = \underline{\mathcal{O}}_S + \underline{\mathcal{O}}_S\sigma \\ \qquad\qquad\qquad\qquad \simeq \underline{\mathcal{O}}_S[Y]/(Y^2 - v|v) \end{cases}$$

$$(13)\quad \begin{array}{c} N'_\sigma \\ \cap \\ G \end{array} = \{g \in \Gamma G \mid g(\sigma) = \varepsilon\sigma,\ \varepsilon^2 = 1\}$$

whence

$$(14)\quad 1 \longrightarrow T_\sigma \longrightarrow N'_\sigma \xrightarrow{\varepsilon_\sigma} \mu_2 \longrightarrow 1.$$

We set

$$(15)\quad \begin{cases} \mathcal{G}_{\rho,\sigma} = B'_0 \times T_\rho \times T_\sigma \\ \mathcal{G}^*_{\rho,\sigma} = B'_0 \times T_\rho \times N'_\sigma. \end{cases}$$

413

Uncertainty retained from cancelled passages.

140-3-s414-D002 $\mathcal{G}_{\rho,\sigma}$

140-3-s414-D003 of

[scan 415]

m1099

[p. 509]

Marginal note. m1099 $\mathcal{M}_{\rho,\sigma}[0],\ \mathcal{M}_{\rho,\sigma}(0)\,,$ $\tau_{\rho,\sigma}|\mathcal{G}_{\rho,\sigma},\ \tau^\circ_{\rho,\sigma}\,,$ $\mathcal{M}_{\rho,\sigma},\ \widetilde{\mathcal{G}}^+_{\rho,\sigma}\quad \dagger$ $\tau^{\circ+}_{\rho,\sigma}$

② We introduce

$$(16)\quad \mathcal{M}_{\rho,\sigma}[0] \subset \mathcal{B}^*_{\rho,\sigma}|\mathcal{G}^*_{\rho,\sigma} = B'_0 \times T_\rho \times N'_\sigma \overset{\text{dfn}}{=} \{(\underline{u}, \underline{a}, \underline{b}) \mid \det \underline{u} = \det \underline{b} = \varepsilon' \det \underline{a},\ \varepsilon'^2 = 1\}$$

$$(17)\quad \mathcal{M}_{\rho,\sigma}(0) \subset \mathcal{M}_{\rho,\sigma}[0] \qquad \text{defined by } \underline{u} = 1 \quad \text{i.e.}$$
$$\mathcal{M}_{\rho,\sigma}(0) = \mathcal{M}_{\rho,\sigma}[0] \cap (T_0 \times T_\rho \times N'_\sigma) \qquad \text{in } \mathcal{G}^*_{\rho,\sigma}.$$

$$(18)\quad L_0 \hookrightarrow \mathcal{M}_{\rho,\sigma}[0]$$

$$\lambda \longmapsto (\lambda,1,1)$$

thus we have

$$(19)\qquad \mathcal{M}_{\rho,\sigma}[0] \simeq \mathcal{M}_{\rho,\sigma}[0)|(0)\,.\,L_0 \qquad \text{product } \tfrac{1}{2}\text{-direct}$$

We have a homomorphism

$$(20)\quad \dot{\mathcal{M}}|\mathcal{M}_{\rho,\sigma}[0] \longrightarrow G_m \times \mu_2 \times \mu_2$$

denoted $\{g \mapsto \chi(g), \varepsilon_2(g), \varepsilon_3(g)$

for $g = (\underline{u}, \underline{a}, \underline{b})$ we have

$$\chi(g) = \det \underline{u} = \det \underline{b}$$
$$\varepsilon_2(g) = \varepsilon_\sigma(b) \qquad \text{(cf. (13)(14)}$$
$$\varepsilon_3(g) = \det(\underline{a})/\det \underline{b}$$
$$\text{i.e. } \begin{cases} \det a = \varepsilon_3 \mu \\ \text{where } \mu = \chi(g) \end{cases}$$

trivial on L_0 (thus factoring through the quotient). We also set

$$(21)\qquad \tau_{\rho,\sigma} \simeq \mathcal{M}_{\rho,\sigma}[0]/L_0 \qquad \text{where}$$

m1100

Marginal note. m1100 This notation is not at all adequate (wrongly inspired by $\tau_{g,\nu|r}$) the sequel shows (and what precedes already showed) that the notation $\Gamma_{\rho,\sigma}$ would be suitable, inspired by $\Gamma_{g,\nu}$ and $\Gamma_{0,3}$.

and the composite

$$(22)\quad \mathcal{M}_{\rho,\sigma}(0) \hookrightarrow \mathcal{M}_{\rho,\sigma}[0] \longrightarrow \tau_{\rho,\sigma}$$

is an isomorphism, whence we have

$$(23)\quad \mathcal{M}|\mathcal{G}_{\rho,\sigma}(0) \simeq \tau_{\rho,\sigma}|\tau'_{\rho,\sigma} \hookrightarrow T_0 \times T_\rho \times N'_\sigma$$

414

Uncertainty retained from cancelled passages.
140-3-s415-D001 $\mathcal{M}^\circ_{\rho,\sigma}$

[scan 416]

Composition with the canonical projection to $T_\rho \times N'_\sigma$ gives a monomorphism
m1102

Marginal note. m1102 for we shall have

$$T_0 \xrightarrow{\det}_{\sim} \mathbb{G}_m$$

$$\begin{array}{ccc} \mathcal{T}_{\rho,\sigma} & \hookrightarrow & T_\rho \times N'_\sigma \\ \wr\downarrow & & \\ \{(\underline{a},\underline{b}) \mid (\det \underline{a}/\det \underline{b})^2 = 1\} & & \text{i.e. } \det \underline{a} = \varepsilon' \det \underline{b} \text{ with } \varepsilon'^2 = 1 \end{array} \tag{24}$$

and we shall therefore have

$$\begin{cases} \chi(\underline{a},\underline{b}) = \det \underline{b} \\ \varepsilon_2(\underline{a},\underline{b}) = \varepsilon_\sigma(\underline{b}) \\ \varepsilon_3(\underline{a},\underline{b}) = \det(\underline{a})/\det(\underline{b}) = \det \underline{b}/\det \underline{a} \end{cases} \tag{25}$$

We define

$$\begin{array}{ll} \mathcal{T}^\circ_{\rho,\sigma} & = \operatorname{Ker} \mathcal{T}_{\rho,\sigma} \xrightarrow{(\varepsilon_2,\varepsilon_3)} \mu_2 \times \mu_2 \\ \wr\downarrow & \\ T_\rho \times_{\mathbb{G}_m} T_\sigma & \end{array} \tag{26}$$

so we have

$$1 \longrightarrow \mathcal{T}^\circ_{\rho,\sigma} \longrightarrow \mathcal{T}_{\rho,\sigma} \longrightarrow \mu_2 \times \mu_2 \longrightarrow 1 \tag{27}$$

and a canonical exact sequence

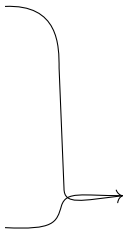
$$1 \longrightarrow \overbrace{ST_\rho \times ST_\sigma}^{\mathcal{T}^{\circ+}_{\rho,\sigma}} \longrightarrow \mathcal{T}^\circ_{\rho,\sigma} \xrightarrow{\chi} \mathbb{G}_m \longrightarrow 1, \tag{28}$$

where we define

$$\begin{cases} ST_\rho = \{g \in \Gamma T_\rho \mid \det g = 1\} \\ ST_\sigma = \{g \in \Gamma T_\sigma \mid \det g = 1\} \end{cases} . \tag{29}$$

m1103

Marginal note. m1103



The operation of $\mathcal{T}_{\rho,\sigma} \simeq \mathcal{M}_{\rho,\sigma}(0)$ on L_0 is given by

$$g(\varepsilon_0(\lambda)) = \varepsilon_0(\mu\lambda), \quad \text{where } \mu = \chi(g) \tag{30}$$

We let $\mathcal{M}_{\rho,\sigma}(0)(\simeq \mathcal{T}_{\rho,\sigma})$ operate on SG, by

$$\begin{array}{ccccccc} \mathcal{M}_{\rho,\sigma}(0) & \xrightarrow{\text{projection}} & T_0 & \xrightarrow[\text{(monomorphism)}]{\text{canonical}} & PG & \xrightarrow{\sim} & \underline{\operatorname{Aut}}_{\mathrm{gr}}(SG) \\ \cap & & \cap & & \wr\downarrow & & \\ T_0 \times T_\rho \times N'_\sigma & & G & & G/\mathbb{G}_m \simeq \underline{\operatorname{Aut}}(G) & & \end{array} \tag{31}$$

415

[scan 417]
[p. 510]

and we define

$$(32) \qquad \mathcal{M}_{\rho,\sigma} \simeq \mathcal{M}_{\rho,\sigma}(0) \,.\, SG$$

$\frac{1}{2}$ direct product

We thus have a canonical homomorphism

$$(33) \qquad SG \overset{j}{\hookrightarrow} \mathcal{M}_{\rho,\sigma}$$

whose image is denoted $\widetilde{\mathcal{G}}^{+}_{\rho,\sigma}$, so

$$(34) \qquad 1 \longrightarrow \widetilde{\mathcal{G}}^{+}_{\rho,\sigma} \longrightarrow \mathcal{M}_{\rho,\sigma} \longrightarrow \mathcal{T}_{\rho,\sigma} \longrightarrow 1$$

$$(35) \qquad \begin{array}{ccc} \mathcal{M}_{\rho,\sigma} \simeq & \mathcal{M}_{\rho,\sigma}(0) & .\,\widetilde{\mathcal{G}}^{+}_{\rho,\sigma} \\ & \wr\vert & \\ & \mathcal{T}_{\rho,\sigma} & \end{array}$$

$\frac{1}{2}$ direct product.

N.B. We use the notation $\mathcal{T}_{\rho,\sigma}$ for the quotient group

$$(36) \qquad \mathcal{T}_{\rho,\sigma} \simeq \mathcal{M}_{\rho,\sigma}/\widetilde{\mathcal{G}}^{+}_{\rho,\sigma},$$

the notation $\mathcal{M}_{\rho,\sigma}[0)$ for the subgroup of $\mathcal{M}_{\rho,\sigma}$.

We define a canonical embedding

$$(37) \qquad \mathcal{M}_{\rho,\sigma}[0] \overset{i}{\hookrightarrow} \mathcal{M}_{\rho,\sigma}$$

which extends the inclusions of the subgroups|of the subgroup $\mathcal{M}_{\rho,\sigma}(0)$ common to $\mathcal{M}_{\rho,\sigma}[0]$ and to $\mathcal{M}_{\rho,\sigma}$, and which on L_0 is given by

$$(38) \qquad i(\lambda) = j(\lambda) \qquad \text{(cf. (33))}$$

(which is indeed compatible with the operations of $\dagger\mathcal{M}_{\rho,\sigma}(0)$ on L_0 on the one hand, on $SG \simeq \widetilde{\mathcal{G}}^{+}_{\rho,\sigma}$ on the other).

416

Uncertainty retained from cancelled passages.

140-3-s417-D001 $\mathcal{T}|\mathcal{M}$

[scan 418]

Another way of saying it is to observe that $\mathcal{M}_{\rho,\sigma}(0)$ operating on $SG \simeq \widetilde{\mathcal{G}}^{+}_{\rho,\sigma}$ normalizes the subgroup $j(L_0)$ – which we henceforth identify with L_0), so that we shall have a subgroup

$$\mathcal{M}_{\rho,\sigma}(0) \,.\, L_0 \subset \mathcal{M}_{\rho,\sigma}$$

which is|will be canonically isomorphic to $\mathcal{M}_{\rho,\sigma}[0]$ defined above, thanks to the expression (19|49) of the latter as a $\frac{1}{2}$-direct product.

Remark. We have

$$\dagger \qquad \mathcal{M}_{\rho,\sigma}[0] \subset \underline{\mathrm{Norm}}_{\mathcal{M}_{\rho,\sigma}}(L_0)$$

By construction, $\dagger$ the inclusion in the reverse direction would amount to saying that L_0 is its own normalizer in $SG \simeq \widetilde{\mathcal{G}}^{+}_{\rho,\sigma}$, which of course is not true at all – even though the analogous fact is true in the profinite group $\widehat{\widetilde{\mathcal{G}}^{+}_{0,3}}$, for the subgroup generated by ε_0, or in $\Pi_{0,\hat{3}}$ for the one generated by l_0 (subject to verification ! besides which I still haven't done).

417

Uncertainty retained from cancelled passages.
140-3-s418-D002 39
140-3-s418-D003 we have for $\supset$

[scan 419]
[p. 511]

m1104

Marginal note. m1104 $\mathcal{M}^{\circ}_{\rho,\sigma}, \mathcal{M}^{+}_{\rho,\sigma}, \mathcal{M}^{+\circ}_{\rho,\sigma}, \mathcal{T}^{\circ}_{\rho,\sigma}, \mathcal{T}^{+}_{\rho,\sigma}, \mathcal{T}^{\circ+}_{\rho,\sigma}$ ... some central subgroups and computations of integral points...

3°

We denote by $\mathcal{M}^{\circ}_{\rho,\sigma}$, $\mathcal{M}_{\rho,\sigma}[0]^{\circ}$, $\dagger\mathcal{M}_{\rho,\sigma}[0)^{\circ}$ identity components, we shall have for example

$$(39)\qquad \begin{array}{ccccccccc} 1 & \longrightarrow & \mathcal{M}^{\circ}_{\rho,\sigma} & \longrightarrow & \mathcal{M}_{\rho,\sigma} & \to & \mu_2 \times \mu_2 & \to & 1 \\ & & \wr| & & & & & & \\ & & \mathcal{M}^{\circ}_{\rho,\sigma}(0)\,.\,\widetilde{\mathcal{G}}^{+}_{\rho,\sigma} & & & & & & \end{array}$$

$\frac{1}{2}$ direct product

$$(40)\qquad \mathcal{M}_{\rho,\sigma}(0)^{\circ} \xrightarrow{\sim} \mathcal{T}^{\circ}_{\rho,\sigma} \simeq T_\rho \times_{\mathbb{G}_m} T_\sigma$$

$$(41)\qquad \begin{array}{l} \mathcal{M}^{\circ}_{\rho,\sigma} \text{ is the inverse image of } \mathcal{T}^{\circ}_{\rho,\sigma} \\ \text{under } \mathcal{M}_{\rho,\sigma} \overset{\text{canonical}}{\longrightarrow} \mathcal{T}_{\rho,\sigma}\ , \end{array}$$

we have a canonical composite homomorphism

$$(42)\qquad \mathcal{M}_{\rho,\sigma} \longrightarrow \mathcal{T}_{\rho,\sigma} \xrightarrow{\chi} \mathbb{G}_m$$

also denoted χ, its kernel is denoted $\mathcal{M}^{+}_{\rho,\sigma}$, and we define in the $=$ way $\mathcal{M}^{\circ+}_{\rho,\sigma}$, so

$$(43)\qquad \begin{array}{ccccccccc} 1 & \longrightarrow & \mathcal{M}^{+}_{\rho,\sigma} & \longrightarrow & \mathcal{M}_{\rho,\sigma} & \xrightarrow{\chi} & \mathbb{G}_m & \longrightarrow & 1 \\ & & \cup| \downarrow & & \cup| \downarrow & & \cup| \downarrow & & \\ 1 & \longrightarrow & \mathcal{M}^{\circ+}_{\rho,\sigma} & \longrightarrow & \mathcal{M}^{\circ}_{\rho,\sigma} & \xrightarrow{\chi} & \mathbb{G}_m & \longrightarrow & 1 \end{array}$$

$$(44)\qquad \begin{cases} \mathcal{M}^{+}_{\rho,\sigma} \simeq \underbrace{\mathcal{M}^{+}_{\rho,\sigma}(0)}_{\substack{\wr| \\ \mathcal{T}^{+}_{\rho,\sigma}}} .\widetilde{\mathcal{G}}^{+}_{\rho,\sigma} \\ \mathcal{M}^{\circ+}_{\rho,\sigma} \simeq \underbrace{\mathcal{M}^{\circ+}_{\rho,\sigma}(0)}_{\substack{\wr| \\ \mathcal{T}^{\circ+}_{\rho,\sigma} \simeq ST_\rho \times ST_\sigma \\ \text{cf. (28)}}} .\widetilde{\mathcal{G}}^{+}_{\rho,\sigma} \end{cases}$$

Thus we have

$$(45)\qquad \begin{array}{l} \mathcal{M}^{\circ+}_{\rho,\sigma}(0) \overset{\text{dfn}}{=} \operatorname{Ker}\left(\mathcal{M}^{\circ}_{\rho,\sigma}(0) \xrightarrow{\chi,\varepsilon_2,\varepsilon_3} \mathbb{G}_m \times \mu_2 \times \mu_2 \right) \\ \simeq ST_\rho \times ST_\sigma \end{array}$$

and this subgroup is central in $\mathcal{M}^{\circ}_{\rho,\sigma}$ (not in $\mathcal{M}_{\rho,\sigma}$), more precisely
418

Uncertainty retained from cancelled passages.
140-3-s419-D001 $\mathcal{M}$

[scan 420]

we have

$$(46)\qquad \mathcal{M}^{\circ+}_{\rho,\sigma} \overset{\text{dfn}}{=} \mathrm{Ker}(\mathcal{M}_{\rho,\sigma} \xrightarrow{(\chi,\varepsilon_2,\varepsilon_3)} \mathbb{G}_m \times \mu_2 \times \mu_2)\dagger$$
$$\simeq \underbrace{ST_\rho \times ST_\sigma}_{\subset \mathcal{M}_{\rho,\sigma}(0)} \times \widetilde{\mathcal{G}}^+_{\rho,\sigma}$$

m1105

Marginal note. m1105 The whole end of the number should come after (or before) the number 6° about the "radicalities"

m1106

Marginal note. m1106 ∘ Careful, this inclusion is not an equality, cf. below…

$$(47)\qquad \underline{\mathrm{Centr}}(\mathcal{M}^\circ_{\rho,\sigma}) \supset \underbrace{1 \times ST_\rho \times ST_\sigma}_{\cap\ \mathcal{M}_{\rho,\sigma}(0)^\circ} \times \underset{\cap\ \widetilde{\mathcal{G}}^+_{\rho,\sigma} \simeq SG}{\mu_2}\ .$$

Hence a subgroup

$$(48)\qquad \mu_2 \times \mu_2 \times \mu_2 \subset 1 \times \underbrace{ST_\rho \times ST_\sigma \times \mu_2}_{\cap\ \mathrm{Centr}(\mathcal{M}^\circ_{\rho,\sigma})}$$

which is in fact contained in the Center of $\mathcal{M}_{\rho,\sigma}$ as a whole, for

$$(49)\qquad \underline{\mathrm{Centr}}(\mathcal{M}_{\rho,\sigma}) \supset \underbrace{1 \times S'T_\rho \times \mu_2}_{\cap\ \mathcal{M}_{\rho,\sigma}(0)} \times \underset{\cap\ \widetilde{\mathcal{G}}^+_{\rho,\sigma} \simeq SG}{\mu_2}$$

with

$$(50)\qquad S'T_\rho \overset{\text{dfn}}{=} \{g \in \Gamma T_\rho \mid (\det g)^2 = 1\}$$

id.

so that we have

$$(51)\quad 1 \longrightarrow ST_\rho \longrightarrow S'T_\rho \xrightarrow{\varepsilon_3} \mu_2 \longrightarrow 1$$

We thus have

$$(52)\qquad \mathrm{Centr}(\mathcal{M}_{\rho,\sigma}) \cap \mathcal{M}^\circ_{\rho,\sigma} \supset \underset{\cap\ T_0}{1} \times \underset{\cap\ T_\rho}{ST_\rho} \times \underset{\cap\ T_\sigma}{\mu_2} \times \underset{\cap\ \widetilde{\mathcal{G}}^+_{\rho,\sigma}}{\mu_2}\ .$$

id.

The "modular case" is the one where

$$(53)\qquad \begin{cases} \det\rho = 1 \quad \text{i.e.} \quad \det\sigma = 1 \quad \text{i.e.} \quad \nu = -1 \\ \text{i.e.} \quad \rho = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} \quad \text{i.e.} \quad \sigma = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \end{cases}$$

(cf. VI for other characterizations, such as $\sigma^2 = -1$, or $\rho^2 - \rho + 1 = 0$, or $\tilde\sigma^2 = \tilde\rho^3 = 1$ ...), this also means that ρ (as an element of T_ρ) is in ST_ρ hence in the center of $\mathcal{M}^\circ_{\rho,\sigma}$,
419

Uncertainty retained from cancelled passages.
140-3-s420-D001 ≃

[scan 421]
[p. 512]

or that σ as an element of T_σ is in ST_σ i.e. in the center of $\mathcal{M}^\circ_{\rho,\sigma}$.
So in this case we find a homomorphism

$$(54)\qquad \begin{cases} \mathbb{Z}/6\mathbb{Z}\times\mathbb{Z}/4\mathbb{Z} \longrightarrow \underline{\mathrm{Centr}}(\mathcal{M}^\circ_{\rho,\sigma}) \\ \\ (i,j) \longrightarrow i_\rho(\rho^i)i_\sigma(\sigma^j) \end{cases}$$

(where we define i_ρ, i_σ among the †three canonical morphisms

†

$$(55)\qquad \begin{cases} i_\rho : S'T_\rho \longrightarrow \mathcal{M}_{\rho,\sigma}(0) \subset \mathcal{M}_{\rho,\sigma} \\ i_\sigma : SN'_\sigma \longrightarrow \mathcal{M}_{\rho,\sigma}(0) \subset \mathcal{M}_{\rho,\sigma} \\ i_G : SG \simeq \widetilde{\mathcal{G}}^+_{\rho,\sigma} \longrightarrow \mathcal{M}_{\rho,\sigma}, \end{cases} \quad)$$

whose image, for 2 invertible on S, contains contains the image of $\mu_2\times\mu_2\times 1$ in (48). For 2 invertible on S, the homomorphism (53) is a monomorphism (but not in characteristic 2, of course). Over the absolute base $\mathbb{Z}$, we have moreover, in the modular case

$$(56)\qquad \begin{cases} \mathcal{T}_{\rho,\sigma}(\mathbb{Z}) \simeq T_\rho(\mathbb{Z})\times N'_\sigma(\mathbb{Z}) \simeq \mathbb{Z}/6\mathbb{Z}\times\mathbb{D}_4 \\ \mathcal{M}_{\rho,\sigma}(\mathbb{Z}) \simeq \mathcal{T}_{\rho,\sigma}(\mathbb{Z})\,.\,\mathrm{Sl}(2,\mathbb{Z}) \end{cases}$$

$\frac{1}{2}$ direct product, $\mathbb{D}|\mathcal{B}|\mathcal{T}_{\rho,\sigma} \simeq \mathbb{Z}/6\mathbb{Z}\times\mathbb{D}_4$ operating via the character

$$\mathbb{D}_4 \xrightarrow{\varepsilon_2} \mathbb{Z}/2\mathbb{Z} \simeq (1,\tau_\infty)|\{1,\tau_\infty\}$$

τ_∞ operating on $\mathrm{Sl}(2,\mathbb{Z})$ by automorphism induced by the matrix

$$\tau_\infty = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

m1109

Marginal note. m1109



Uncertainty retained from cancelled passages.
140-3-s421-D001 four
140-3-s421-D002 i_B

[scan 422]
the homomorphisms $\Theta, \widetilde{\Theta}$ $\Theta^\circ, \widetilde{\Theta}^\circ$

④ We define a homomorphism

$$(57)\quad \mathcal{M}_{\rho,\sigma} \xrightarrow{\ \Theta\ } G$$

$$\Theta \mid \widetilde{\mathcal{G}}^{+}_{\rho,\sigma} \simeq SG \quad \text{is the identity}$$
$$\Theta \mid \mathcal{M}_{\rho,\sigma}(0) \subset T_0 \times T_\rho \times N'_\sigma \quad \text{given by} \quad \Theta(\underline{u},\underline{a},\underline{b}) = \underline{u}$$

hence

$$(58)\qquad \Theta \mid \mathcal{M}_{\rho,\sigma}[0] \subset B'_0 \times T_\rho \times N'_\sigma \quad \text{given by} \quad \Theta(\underline{u},\underline{a},\underline{b}) = \underline{u}$$

We have

$$\begin{array}{ccccc} (59) & & \mathcal{M}_{\rho,\sigma}(0) & \subset & T_0 \times T_\rho \times N'_\sigma \\ & & \cup & \text{Cartesian} & \cup \\ \dagger & & \operatorname{Ker}\Theta & \subset & 1 \times T_\rho \times N'_\sigma \end{array}$$

$$\wr \quad \left\{ \left(1,\ \underset{\cup}{\underset{\Gamma T_\rho}{\underline{a}}}\ ,\ \underset{\cup}{\underset{\Gamma N'_\sigma}{\underline{b}}} \right) \middle| \det \underline{b} = 1, (\det \underline{a})^2 = 1 \right\}$$
$$\wr \quad S'T_\rho \times SN'_\sigma$$

where $SN'_\sigma = \operatorname{Ker}(N'_\sigma \xrightarrow{\text{determinant}} \mathbb{G}_m)$

Let

$$(60)\qquad \Theta^\circ = \Theta \mid \mathcal{M}^\circ_{\rho,\sigma} \qquad \dagger$$

we find

$$(61)\quad 1 \longrightarrow \underset{\wr}{\operatorname{Ker}\Theta^\circ} \longrightarrow \operatorname{Ker}\Theta \xrightarrow{(\varepsilon_3,\varepsilon_2)} \mu_2 \times \mu_2 \longrightarrow 1.$$
$$ST_\rho \times ST_\sigma$$

Put

$$(62)\qquad S\mathcal{R}^\circ_{\rho,\sigma} = 1 \times ST_\rho \times ST_\sigma = \operatorname{Ker}\Theta^\circ = (\operatorname{Ker}\Theta) \cap \mathcal{M}^\circ_{\rho,\sigma}$$

which is a central subgroup of $\mathcal{M}^\circ_{\rho,\sigma}$ (it is

Uncertainty retained from cancelled passages.
140-3-s422-D003 $\mathcal{R}|SG$
140-3-s422-D004 that we|we have

[scan 423]
[p. 513]

†normal but not central in $\mathcal{M}_{\rho,\sigma}$ as a whole)
the group

$$(63) \qquad \mathcal{M}_{\rho,\sigma}/S\mathcal{R}^{\circ}_{\rho,\sigma} = \mathcal{M}^{\natural}_{\rho,\sigma}$$

is therefore an extension of $\mu_2 \times \mu_2$ by

$$\mathcal{M}^{\circ}_{\rho,\sigma}/S\mathcal{R}^{\circ}_{\rho,\sigma} \overset{\text{dfn}}{=} \mathcal{M}^{\circ\natural}_{\rho,\sigma},$$

isomorphic to G via Θ° (since $S\mathcal{R}^{\circ}_{\rho,\sigma} = \operatorname{Ker}\Theta^{\circ}$), hence

$$(64) \quad 1 \longrightarrow G \longrightarrow \mathcal{M}^{\natural}_{\rho,\sigma} \longrightarrow \mu_2 \times \mu_2 \longrightarrow 1,$$

moreover the homomorphism

$$\Theta : \mathcal{M}_{\rho,\sigma} \longrightarrow G$$

being trivial on $\mathcal{R}_{\rho,\sigma}$ passes to the quotient as

$$(65) \quad \Theta^{\natural} : \mathcal{M}^{\natural}_{\rho,\sigma} \longrightarrow G,$$

whose restriction to G (via (64)) is the identity, so we find, via (64),

$$(66) \qquad \mathcal{M}^{\natural}_{\rho,\sigma} \simeq G \times \mu_2 \times \mu_2.$$

We thus find

$$(67) \quad \begin{array}{ccccccccc} 1 & \longrightarrow & S\mathcal{R}^{\circ}_{\rho,\sigma} & \longrightarrow & \mathcal{M}_{\rho,\sigma} & \xrightarrow{(\Theta,\varepsilon_3,\varepsilon_2)} & \mathcal{M}^{\natural}_{\rho,\sigma} & \longrightarrow & 1. \\ & & \wr| & & & & \wr| & & \\ & & ST_\rho \times ST_\sigma & & & & G \times \mu_2 \times \mu_2 & & \end{array}$$

To see how $\mathcal{M}^{\natural}_{\rho,\sigma} \simeq G \times \mu_2 \times \mu_2$ operates on $ST_\rho \times ST_\sigma$, we note that $\mathcal{M}^{\circ}_{\rho,\sigma}$ is the inverse image of $G \simeq \mathcal{M}^{\natural\circ}_{\rho,\sigma}$ in the extension (67), where $S\mathcal{R}_{\rho,\sigma}$ is central in $\mathcal{M}_{\rho,\sigma}$, so G operates trivially on it.
422

Uncertainty retained from cancelled passages.
140-3-s423-D001 invariant

[scan 424]

On the other hand we have

$$\begin{array}{ccccc} S\mathcal{R}^{\circ}_{\rho,\sigma} & \subset & \mathcal{M}_{\rho,\sigma}(0) & \xrightarrow[(\varepsilon_3,\varepsilon_2)]{\text{epimorphism}} & \mu_2 \times \mu_2 \\ \wr| & & \cap & & \\ ST_\rho \times ST_\sigma & & T_0 \times T_\rho \times N'_\sigma & & \end{array}$$

and we find that $\mu_2 \times \mu_2$ operates on $S\mathcal{R}_{\rho,\sigma}$ via its second factor (corresponding to $\varepsilon_2(\underline{u}, \underline{a}, \underline{b}) = \varepsilon_\sigma(\underline{b})$), an ε of the latter ($\varepsilon^2 = 1$) operates on $S\mathcal{R}_{\rho,\sigma} \simeq ST_\rho \times ST_\sigma$ leaving the first factor untouched, and operating on T_σ hence on ST_σ via

$$(68) \qquad \sigma \longrightarrow \varepsilon\sigma$$

m1111

Marginal note. m1111 (compatible with the Hamilton–Cayley equation $\sigma^2 - \nu = 0$ of σ)

†

⑤ Let

$$(69) \qquad \begin{array}{l} \mathcal{R}^\circ_{\rho,\sigma} \quad \text{inverse image of } \mathbb{G}_m \subset G \\ \text{in } \mathcal{M}^\circ_{\rho,\sigma} \text{ under } \Theta : \mathcal{M}^\circ_{\rho,\sigma} \longrightarrow G \end{array}$$

m1112

Marginal note. m1112 It would be worthwhile to begin right away with the introduction of $\mathcal{R}_{\rho,\sigma}$ and the realization (102) below of $\mathcal{M}_{\rho,\sigma}$, in terms of amalgamated sums of the products... cf. ⑥.

We thus have an extension structure on $\mathcal{R}^\circ_{\rho,\sigma}$

$$(70) \quad 1 \longrightarrow S\mathcal{R}^\circ_{\rho,\sigma} \longrightarrow \mathcal{R}^\circ_{\rho,\sigma} \longrightarrow \mathbb{G}_m \longrightarrow 1.$$

We will have

$$\mathcal{R}^\circ_{\rho,\sigma} = \left\{ \begin{pmatrix} \underset{\Gamma T_0}{\underset{\cup}{\underline{u}}} , & \underset{\Gamma T_\rho}{\underset{\cup}{\underline{a}}} , & \underset{\Gamma T_\sigma}{\underset{\cup}{\underline{b}}} \end{pmatrix} \cdot \underset{\Gamma SG}{\underset{\cup}{f}} \;\middle|\; \det u = \det \underline{a} = \det \underline{b}, \quad \underline{u} f \in \Gamma \mathbb{G}_m \right\}$$

i.e. $f = \lambda \underline{u}^{-1}$ where $\lambda \in \Gamma\mathbb{G}_m$, $\lambda^2 = \det u$

423

Uncertainty retained from cancelled passages.

140-3-s424-D001 is none other than ∘

[scan 425]
[p. 514]

$$\mathcal{R}^\circ_{\rho,\sigma} \simeq \left\{ \begin{pmatrix} j_{T_0}(\mu^2), & \underset{\Gamma T_\rho}{\underset{\cup}{\underline{a}}} , & \underset{\Gamma T_\sigma}{\underset{\cup}{\underline{b}}} \end{pmatrix} \cdot \underbrace{\mu j_G(\mu^{-2})}_{\substack{\text{N.B. the} \\ \text{determinant is 1} \\ \text{since } \det j_G(\mu^{-2}) = \mu^{-2}}} \;\middle|\; \begin{array}{c} \mu \in \Gamma\mathbb{G}_m, \\ \det \underline{a} = \det \underline{b}\dagger \\ \underline{a} \in \Gamma T_\rho, \qquad = \mu^2 \\ \underline{b} \in \Gamma T_\sigma, \end{array} \right\} \qquad (71)$$

where $j^+ | j_G$

$$\begin{cases} j(\mu) = \begin{pmatrix} \mu & 1 \\ 0 & 1 \end{pmatrix} \in \Gamma SG \qquad (\mu \in \Gamma\mathbb{G}_m) \\ j_{T_0}\dagger : \mathbb{G}_m \longrightarrow T_0 \quad \text{given by } j \\ j_G : \mathbb{G}_m \longrightarrow \dagger G \quad \text{given by } j \text{ above.} \end{cases} \qquad (72)$$

We thus find that the composite

$$\left\{\begin{array}{c} \mathcal{R}^{\circ}_{\rho,\sigma} \xhookrightarrow[\text{inclusion}]{} \mathcal{M}^{\circ}_{\rho,\sigma} \xrightarrow[\text{canonical}]{} \begin{array}{c}\mathcal{T}^{\circ}_{\rho,\sigma} \simeq T_\rho \times_{\mathbb{G}_m} T_\sigma \\ \| \\ \mathcal{M}^{\circ}_{\rho,\sigma}/\widetilde{\mathcal{G}}^{+}_{\rho,\sigma}\end{array} \\ \text{makes } \mathcal{R}^{\circ}_{\rho,\sigma} \text{ an algebraic covering} \\ \text{of } \mathcal{T}^{\circ}_{\rho,\sigma} \end{array}\right. \tag{73}$$

and more precisely that we have a cartesian diagram

$$\begin{array}{ccc} \mathcal{R}^{\circ}_{\rho,\sigma} & \xrightarrow{(73)} & T_\rho \times_{\mathbb{G}_m} T_\sigma \simeq \mathcal{T}^{\circ}_{\rho,\sigma} \\ \Big\downarrow{\scriptstyle \text{the canonical homomorphism of (70)}} & \text{cartesian} & \Big\downarrow{\scriptstyle \text{canonical}} \\ \mathbb{G}_m & \xrightarrow[\mu\longmapsto\mu^2]{} & \mathbb{G}_m \end{array} \tag{74}$$

(75) Proposition : $\mathcal{R}^{\circ}_{\rho,\sigma} = \underline{Centr}(\mathcal{M}^{\circ}_{\rho,\sigma})$

Obviously the center of $\mathcal{M}^{\circ}_{\rho,\sigma}$ is contained in $\mathcal{R}^{\circ}_{\rho,\sigma}$, precisely the inverse image of the center

424

Uncertainty retained from cancelled passages.

140-3-s425-D001 $= \lambda^2$

140-3-s425-D002 μ

140-3-s425-D003 S

[scan 426]

of $\mathcal{M}^{\circ\natural\sharp}_{\rho,\sigma} = \mathcal{M}^{\circ}_{\rho,\sigma}/S\mathcal{R}_{\rho,\sigma} \simeq G$, it remains to prove that $\mathcal{R}_{\rho,\sigma}$ is central in $\mathcal{M}^{\circ}_{\rho,\sigma}$. But $\mathcal{M}^{\circ}_{\rho,\sigma}$ is generated † by $\mathcal{R}^{\circ}_{\rho,\sigma}$ and by $\widetilde{\mathcal{G}}^{+}_{\rho,\sigma} \simeq SG$, since

$$\mathcal{R}^{\circ}_{\rho,\sigma} \longrightarrow \mathcal{T}^{\circ}_{\rho,\sigma} \simeq \mathcal{M}^{\circ}_{\rho,\sigma}/\widetilde{\mathcal{G}}^{+}_{\rho,\sigma}$$

is an epimorphism (73)(74), and since $\mathcal{R}_{\rho,\sigma}$ is commutative (74), it remains to show that it commutes with $\widetilde{\mathcal{G}}^{+}_{\rho,\sigma}$ i.e. operates trivially on $\widetilde{\mathcal{G}}^{+}_{\rho,\sigma}$. Now a section of $\mathcal{R}^{\circ}_{\rho,\sigma}$ is of the form

$$(j_{T_0}(\mu^2), \underline{a}, \underline{b}) \cdot (\lambda j_{T_0}(\mu^{-2}))$$

the first factor operating as

$$i. \qquad \operatorname{int} j_G(\mu^2)$$

the second as

$$\operatorname{int} j_G(\mu^{-2})$$

so their product operates trivially, q.e.d.

† Let us now note that

$$\mathcal{R}^{\circ}_{\rho,\sigma} \cap \underbrace{\widetilde{\mathcal{G}}^{+}_{\rho,\sigma}}_{\| SG} = \operatorname{Ker}(\mathcal{R}^{\circ}_{\rho,\sigma} \longrightarrow \mathcal{T}^{\circ}_{\rho,\sigma}) \underset{\text{cf. (74)}}{=} \mu_2 \tag{75}$$

(we can also say more directly, that

425

Uncertainty retained from cancelled passages.
140-3-s426-D002)
140-3-s426-D004 in

[scan 427]

$\mathcal{R}^\circ_{\rho,\sigma} \cap \widetilde{\mathcal{G}}^+_{\rho,\sigma}$, being contained in the center of $\widetilde{\mathcal{G}}^+_{\rho,\sigma} \simeq SG$ which is μ_2, is $\subset \mu_2$ — but this does not show equality. [p. 515]

Remark This gives a simpler description of $\mathcal{M}^\circ_{\rho,\sigma}$ than as a $\frac{1}{2}$ direct product, as follows. We define $\mathcal{R}^\circ_{\rho,\sigma}$ by diagram (74), whence an exact sequence

$$1 \longrightarrow \mu_2 \xrightarrow{j'} \mathcal{R}^\circ_{\rho,\sigma} \longrightarrow T_\rho \times_{\mathbb{G}_m} T_\sigma \longrightarrow 1 \tag{77}$$

and we define $\mathcal{M}^\circ_{\rho,\sigma}$ as a central amalgamated sum

$$\begin{array}{ccccccccc} & & & & \dagger & & & & \\ & & & & \dagger\widetilde{\mathcal{G}}^+_{\rho,\sigma} & & & & \\ 1 & \longrightarrow & \mu_2 & \longrightarrow & \mathcal{R}^\circ_{\rho,\sigma} \times & || & \longrightarrow & \mathcal{M}^\circ_{\rho,\sigma} & \longrightarrow 1 \\ & & & & & SG & & & \end{array} \tag{78}$$

$$\varepsilon \longmapsto (j'(\varepsilon), j''(\varepsilon^{-1}))$$

(where j' is given in (77) and $j'' : \mu_2 \longrightarrow SG = \widetilde{\mathcal{G}}^+_{\rho,\sigma}$ the canonical morphism).

Let us note that we have in the $=$ way a well-known †description of G in terms of SG

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \mu_2 & \longrightarrow & \mathbb{G}_m \times SG & \longrightarrow & G & \longrightarrow & 1 \\ & & & & \varepsilon & \longmapsto & (\varepsilon, j''(\varepsilon^{-1})) & & \end{array} \tag{79}$$

and we will have a homomorphism from (78) to (79), thanks to the canonical homomorphism

$$\mathcal{R}^\circ_{\rho,\sigma} \longrightarrow \mathbb{G}_m \qquad \text{of (74)}, \tag{80}$$

Uncertainty retained from cancelled passages.
140-3-s427-D001 $\mathcal{M}^\circ_{\rho,\sigma}$
140-3-s427-D002 $\mathcal{M}$
140-3-s427-D003 well-known

[scan 428]

which by passage to the quotient in|of the

$$\mathcal{R}^\circ_{\rho,\sigma} \times SG \xrightarrow{\widetilde{\Theta}^\circ} \mathbb{G}_m \times SG$$

makes it possible to reconstruct

$$\Theta^\circ : \quad \mathcal{M}^\circ_{\rho,\sigma} \longrightarrow G \ ,$$

just like the group $\mathcal{M}^\circ_{\rho,\sigma}$ it-=, in terms of the group schemes ($\pm$ toric) "in the air" T_ρ, T_σ and the epimorphisms

$$T_\rho \xrightarrow{\delta_\rho} \mathbb{G}_m \ , \qquad T_\sigma \xrightarrow{\delta_\sigma} \mathbb{G}_m \ ,$$

and the form G of $\mathrm{Gl}(2)_S$ of course (without having to use = embeddings $T_\rho \longrightarrow G, T_\sigma \longrightarrow G$, and still less sections ρ, σ of G ...). We find $\widetilde{\mathcal{G}}^+_{\rho,\sigma}$ as the image of SG, in $\mathcal{M}^\circ_{\rho,\sigma}$, †

$$\mathcal{T}^\circ_{\rho,\sigma} \overset{\text{dfn}}{=} \mathcal{M}^\circ_{\rho,\sigma}/\widetilde{\mathcal{G}}^+_{\rho,\sigma} \simeq \begin{array}{c} \mathcal{R}^\circ_{\rho,\sigma}/\mu_2 \\ \wr \\ T_\rho \times_{\mathbb{G}_m} T_\sigma \end{array}$$

But for our purpose, the splitting "askew" of the extension thus constructed of $\mathcal{T}^\circ_{\rho,\sigma}$ by $\widetilde{\mathcal{G}}^+_{\rho,\sigma} \simeq SG$, corresponding to the section subgroup $\mathcal{M}^\circ_{\rho,\sigma}(0)$, will play an essential role — by composing this section

427

Uncertainty retained from cancelled passages.

140-3-s428-D002 [illeg. 1] however prism|product

[scan 429]
[p. 516]

$$\mathcal{T}^\circ_{\rho,\sigma} \simeq T_\rho \times_{\mathbb{G}_m} T_\sigma \longrightarrow \mathcal{M}^\circ_{\rho,\sigma}$$

with

$$\mathcal{M}^\circ_{\rho,\sigma} \longrightarrow G$$

we find a homomorphism which factors in turn as

$$\mathcal{T}^\circ_{\rho,\sigma} \simeq T_\rho \times_{\mathbb{G}_m} T_\sigma \overset{\substack{\text{epimorphism} \\ \text{canonical}}}{\longrightarrow} \mathbb{G}_m \overset{\substack{\text{monomorphism} \\ j_G}}{\hookrightarrow} G$$

whose image is the torus T_0 ... .

Remark We can think of constructing $\mathcal{M}_{\rho,\sigma}$ in an analogous way, by introducing the inverse image — let us denote it by $\mathcal{R}^{\sim|\infty}_{\rho,\sigma}$ — of $\mathbb{G}_m \times \mu_2 \times \mu_2$ under

$$\mathcal{M}_{\rho,\sigma} \longrightarrow G \times \mu_2 \times \mu_2$$

(67) — whose intersection with $\widetilde{\mathcal{G}}^+_{\rho,\sigma} \simeq SG$ obviously equal to μ_2, since

m1113

Marginal note. m1113 N.B. $\mathcal{R}_{\rho,\sigma}$ is also, more simply, the kernel of the composite homomorphism

$$\widetilde{\Theta}: \quad \mathcal{M}_{\rho,\sigma} \overset{\Theta}{\longrightarrow} G \overset{\text{canonical}}{\longrightarrow} PG$$

so we have exact sequence

† (81) $$1 \longrightarrow \mathcal{R}_{\rho,\sigma} \longrightarrow \mathcal{M}_{\rho,\sigma} \longrightarrow PG \longrightarrow 1$$

$$\mathcal{R}^\circ_{\rho,\sigma} = \mathcal{R}^{\sim|\infty}_{\rho,\sigma} \cap \mathcal{M}^\circ_{\rho,\sigma} = (\mathcal{R}^{\sim|\infty}_{\rho,\sigma})^\circ \quad \text{(identity component)}$$

I am not going to spell out here the construction of $\mathcal{R}^{\sim|\infty}_{\rho,\sigma}$ in terms of T_ρ, N'_σ etc., not being sure that it will be needed later on.

428

Uncertainty retained from cancelled passages.
140-3-s429-D001 [*illeg.* 1]

[scan 430]

In the modular case, over the absolute base $\mathbb{Z}$, we will have

$$(81)\qquad \left(\begin{array}{ccc} S\mathcal{R}^{\circ}_{\rho,\sigma}(\mathbb{Z}) & \twoheadrightarrow \mathcal{T}^{\circ}_{\rho,\sigma}(\mathbb{Z}) \twoheadrightarrow & T_\rho(\mathbb{Z})\times T_\sigma(\mathbb{Z}) \simeq \mathbb{Z}/6\mathbb{Z}\times\mathbb{Z}/4\mathbb{Z} \\ \| & & \text{since } T_\rho(\mathbb{Z}) = ST_\rho(\mathbb{Z}) \\ ST_\rho(\mathbb{Z})\times ST_\sigma(\mathbb{Z}) & & T_\sigma(\mathbb{Z}) = ST_\sigma(\mathbb{Z}) \end{array}\right.$$

whence, via (74)

$$(82)\qquad \mathcal{R}^{\circ}_{\rho,\sigma}(\mathbb{Z}) \simeq (\mathbb{Z}/6\mathbb{Z}\times\mathbb{Z}/4\mathbb{Z})\times\{\pm 1\}$$

the projections onto the two factors of the right-hand side correspond precisely to the homomorphisms

$$\mathcal{R}^{\circ}_{\rho,\sigma} \longrightarrow \mathcal{T}^{\circ}_{\rho,\sigma}\ , \qquad \mathcal{R}^{\circ}_{\rho,\sigma} \longrightarrow \mathbb{G}_m$$

of (74), so that the inclusion

$$(83)\qquad \mu_2(\mathbb{Z}) = \{\pm 1\} \twoheadrightarrow \mathcal{R}^{\circ}_{\rho,\sigma}(\mathbb{Z}) \simeq (\mathbb{Z}/6\mathbb{Z}\times\mathbb{Z}/4\mathbb{Z})\times\{\pm 1\}$$

is the obvious inclusion $\varepsilon \longmapsto (0)\times\varepsilon$.

A more telling way of saying this is that the canonical inclusion

$$(84)\qquad \begin{array}{ccccc} S\mathcal{R}^{\circ}_{\rho,\sigma} & \times & \mu_2 & \hookrightarrow & \mathcal{R}^{\circ}_{\rho,\sigma} \\ \| & & \dagger & & \\ ST_\rho\times ST_\sigma & & \| & & \\ \cap & & \underline{\mathrm{Centr}}(SG) & & \\ 1\,\mathcal{M}^{\circ}_{\rho,\sigma}(0) & & & & \end{array}$$

induces an isomorphism

$$(85)\qquad \underbrace{S\mathcal{R}^{\circ}_{\rho,\sigma}(\mathbb{Z})}_{\mathbb{Z}/6\mathbb{Z}\times\mathbb{Z}/4\mathbb{Z}} \times \underbrace{\mu_2(\mathbb{Z})}_{\{\pm 1\}} \xrightarrow{\ \sim\ } \mathcal{R}^{\circ}_{\rho,\sigma}(\mathbb{Z})$$

429

Uncertainty retained from cancelled passages.
140-3-s430-D002 $\mathbb{G}_m$

[scan 431]
[p. 517]

†
m1114

Marginal note. m1114 Mixed groups of the radical
$(\mathcal{R}_{\rho,\sigma}, S\mathcal{R}_{\rho,\sigma}, \mathcal{R}^{\circ}_{\rho,\sigma}, S\mathcal{R}^{\circ}_{\rho,\sigma}\ldots)$.

Ⓒ I am going to return to

$$(86)\qquad \mathcal{R}_{\rho,\sigma} \overset{\mathrm{dfn}}{=} \mathrm{Ker}\left(\ \mathcal{M}_{\rho,\sigma} \xrightarrow{\widetilde{\Theta}} PG\ \right),$$

giving rise to the exact sequence

$$(87)\qquad 1 \longrightarrow \mathcal{R}_{\rho,\sigma} \longrightarrow \mathcal{M}_{\rho,\sigma} \longrightarrow PG \longrightarrow 1.$$

We will have

$$(88)\qquad \underset{\text{defined in 69}}{\mathcal{R}^{\circ}_{\rho,\sigma}} = \mathcal{R}_{\rho,\sigma} \cap \mathcal{M}^{\circ}_{\rho,\sigma} = \begin{array}{c}\text{identity component}\\ \text{of } \mathcal{R}_{\rho,\sigma}\end{array}.$$

Besides we also have

$$(89)\qquad \mathcal{R}_{\rho,\sigma} = \Theta^{-1}(\mathbb{G}_m) \qquad \left(\Theta:\ \mathcal{M}_{\rho,\sigma} \longrightarrow G\right)$$

whence an exact sequence

$$(90)\qquad \begin{array}{ccccccccc} 1 & \to & S\mathcal{R}_{\rho,\sigma} & \to & \mathcal{R}_{\rho,\sigma} & \overset{\kappa}{\to} & \mathbb{G}_m & \to & 1 \\ & & \| \text{ dfn} & & & & & & \\ & & \left\{\begin{array}{l}\text{Ker } \dagger(\mathcal{R}_{\rho,\sigma} \overset{\kappa}{\to} \mathbb{G}_m)\\ \|\\ \text{Ker}(\mathcal{M}_{\rho,\sigma} \overset{\Theta}{\to} G)\end{array}\right. & & & & & & \end{array}$$

where κ is defined by the condition that

$$(91)\qquad \begin{array}{ccccccc} \mathcal{R}_{\rho,\sigma} & \overset{\kappa}{\to} & \mathbb{G}_m & \overset{\sim}{\to} & \underline{\text{Center}}(G) & \to & G \\ & \searrow & & & & \nearrow & \\ & & & \overset{\Theta}{} & & & \end{array}$$

be commutative.

Let us note in passing that the homomorphism χ of (42) $\mathcal{M}_{\rho,\sigma} \longrightarrow \mathbb{G}_m$ can also be deduced from Θ (57) by composition

$$(92)\qquad \begin{array}{ccc} \mathcal{M}_{\rho,\sigma} & \xrightarrow{\Theta} & G \\ & {\scriptstyle\chi}\searrow \quad \swarrow{\scriptstyle\det} & \\ & \mathbb{G}_m & \end{array} \qquad)|,$$

and we will therefore have

$$(93)\qquad \chi(g) = \kappa(g)^2 \quad \text{if } g \in \Gamma\mathcal{R}_{\rho,\sigma}$$

i.e.

$$\begin{array}{ccc} \mathcal{R}_{\rho,\sigma} & \xrightarrow{\kappa} & \mathbb{G}_m \\ & {\scriptstyle\chi|\mathcal{R}_{\rho,\sigma}}\searrow \quad \swarrow{\scriptstyle\lambda\mapsto\lambda^2} & \\ & \mathbb{G}_m\,\dagger & \end{array} \qquad \text{commutative.}$$

Uncertainty retained from cancelled passages.

140-3-s431-D001 subgroup $\frac{1}{2}$

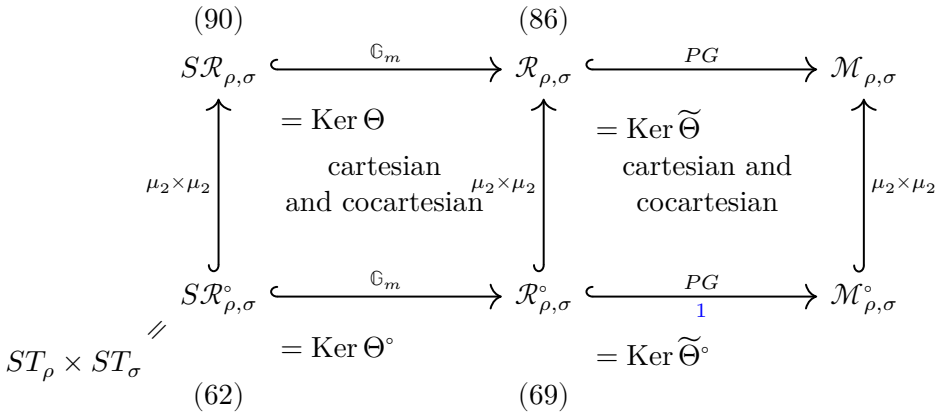
140-3-s431-D002 $\kappa|\Theta$

140-3-s431-D003 $\mathbb{G}_m$

[scan 432]

We have the diagram of inclusions

(94)



m1115

Marginal note. m1115 N.B.
There is a check to be made, which reduces to seeing that

$$S\mathcal{R}_{\rho,\sigma} \to \mu_2 \times \mu_2 \quad \text{is an epimorphism}$$

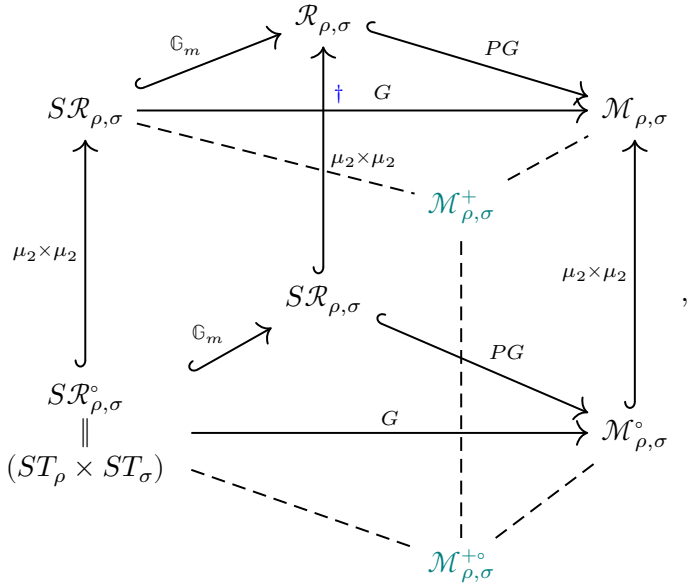
(its kernel is $S\mathcal{R}^\circ_{\rho,\sigma}$ by definition), hence a fortiori

$$\mathcal{R}_{\rho,\sigma} \text{ — } \mu_2 \times \mu_2 \quad \text{an epimorphism}$$

(its kernel is $\mathcal{R}^\circ_{\rho,\sigma}$ by definition) — but the check is trivial.

the second row is formed of the identity components of the first, which we can rewrite more prettily still

(95)



,

in the form of a prism whose three lateral squares are both cartesian and cocartesian — thus giving rise to quotients (on parallel sides), isomorphic — the quotients that occur are

$$\mu_2 \times \mu_2, \quad \mathbb{G}_m, \quad PG \quad \text{and the extension } G \text{ of } PG \text{ by } \mathbb{G}_m.$$

We read off from this diagram, for example, the exact sequence

(96)

$$1 \to \begin{array}{c} S\mathcal{R}^\circ_{\rho,\sigma} \\ \simeq \\ ST_\rho \times ST_\sigma \end{array} \to \mathcal{R}_{\rho,\sigma} \xrightarrow{(\kappa,\varepsilon_3|\mathcal{R}_{\rho,\sigma},\varepsilon_2|\mathcal{R}_{\rho,\sigma})} \mathbb{G}_m \times \mu_2 \times \mu_2 \to 1 \quad ,$$

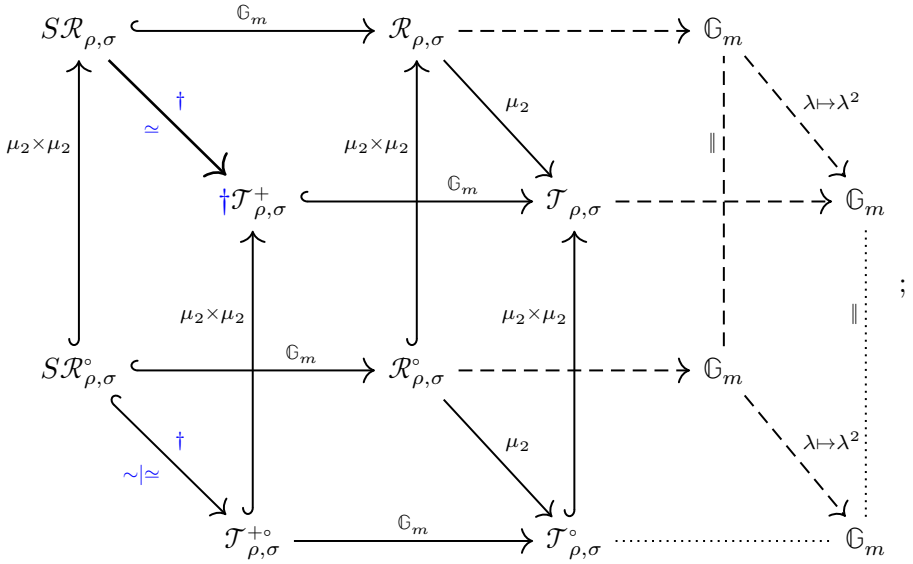
Uncertainty retained from cancelled passages.
140-3-s432-D001 G

[scan 433]
[p. 518]

which makes explicit the structure of $\mathcal{R}_{\rho,\sigma}$ in terms of the very simple groups $ST_\rho, ST_\sigma, \mathbb{G}_m, \mu_2$.

We have a cube, built on the left rear vertical face of (95) (which becomes the vertical rear face)

(95)



;

whose vertical rear (also contained in (95)) and front faces are cartesian and cocartesian, and the 4 horizontal arrows that go from the former to the latter are morphisms of passage to the quotient, all having kernels isomorphic either to μ_2 (marked above these arrows), or to 1 († †we make them into arrows † which are isomorphisms $\xrightarrow{\sim}$), which † expresses|makes explicit|implies the relations,

$$(97)\qquad \begin{cases} \mathcal{R}_{\rho,\sigma}\cap SG = \dagger\mathcal{R}^\circ_{\rho,\sigma}\cap SG = \dagger\mu_2\dagger \\ S\mathcal{R}_{\rho,\sigma}\cap SG = S\mathcal{R}^\circ_{\rho,\sigma}\cap SG = \{1\}. \quad \dagger \end{cases}$$

†

Uncertainty retained from cancelled passages.
140-3-s433-D002 μ_2
140-3-s433-D003 S
140-3-s433-D004 μ_2
140-3-s433-D005 marked by
140-3-s433-D006 formerly
140-3-s433-D007 of inclusion|of isomorphism
140-3-s433-D008 which implies the relations
140-3-s433-D010 $S\mathcal{R}^\circ_{\rho,\sigma}\cap SG$
140-3-s433-D011 $S\mathcal{R}^\circ_{\rho,\sigma}\cap SG$
140-3-s433-D012 μ_2
140-3-s433-D013 $\neq\mu_2$ center
140-3-s433-D014 it suffices to see that the|we must see that the equal|isomorphic obviously $\operatorname{Ker}\Theta$

[scan 434]
[p. †519]

we note

$$\underbrace{\mathcal{R}^{\dagger}_{\rho,\sigma}}_{\operatorname{Ker}\widetilde{\Theta}}\cap SG=\operatorname{Ker}(\widetilde{\Theta}|SG),\qquad \underbrace{\mathcal{R}^{\circ}_{\rho,\sigma}}_{\operatorname{Ker}\widetilde{\Theta}^{\circ}}\cap SG=\operatorname{Ker}(\widetilde{\Theta}^{\circ}|SG),$$

now

$$\widetilde{\Theta}|SG=\widetilde{\Theta}^{\circ}|SG=\text{canonical morphism}\quad SG\longrightarrow PG$$

whose kernel is indeed μ_2.

On the other hand we have

$$\underbrace{S\mathcal{R}_{\rho,\sigma}}_{\operatorname{Ker}\Theta}\cap SG=\operatorname{Ker}(\Theta|SG)=\{1\}$$

since $\Theta|SG$ is the canonical inclusion $SG\subset G$, a fortiori $S\mathcal{R}^{\circ}_{\rho,\sigma}\cap SG=\{1\}$. The fact that we have isomorphisms

$$(98)\qquad \begin{cases} S\mathcal{R}_{\rho,\sigma}\overset{\sim}{\to}\mathcal{T}^{+}_{\rho,\sigma}\simeq S'T_{\rho}\times SN'_{\sigma} \\ S\mathcal{R}^{\circ}_{\rho,\sigma}\overset{\sim}{\to}\mathcal{T}^{+\circ}_{\rho,\sigma}\simeq ST_{\rho}\times ST_{\sigma} \end{cases}$$

follows, † for the second, from the comparison of (62) and (28), for the first this is deduced from it by the five lemma, since

$$S\mathcal{R}_{\rho,\sigma}/S\mathcal{R}^{\circ}_{\rho,\sigma}\longrightarrow\mathcal{T}^{+}_{\rho,\sigma}/\mathcal{T}^{+\circ}_{\rho,\sigma}\simeq\mu_2\times\mu_2.$$

The fact that † we have isomorphisms (98) implies that in

$$\underset{\operatorname{Ker}|\operatorname{Ker}'(\chi:\mathcal{M}_{\rho,\sigma}\to\mathbb{G}_m)}{\mathcal{M}^{+}_{\rho,\sigma}}\simeq\underbrace{\mathcal{M}^{+}_{\rho,\sigma}(0)}_{\simeq\mathcal{T}^{+}_{\rho,\sigma}}.\underbrace{\widetilde{\mathcal{G}}^{+}_{\rho,\sigma}}_{SG}\qquad\begin{array}{c}(\text{product }\frac{1}{2}\text{ direct}\\ \text{cf. (44)})\end{array}$$

regarded as an extension of $\mathcal{T}^{+}_{\rho,\sigma}$ by SG, there is, alongside the splitting by $\mathcal{M}^{+}_{\rho,\sigma}(0)\subset\mathcal{M}^{+}_{\rho,\sigma}$, induced by $\mathcal{M}_{\rho,\sigma}(0)$, another, splitting,

433

Uncertainty retained from cancelled passages.

140-3-s434-D001 8

140-3-s434-D002 ∘

140-3-s434-D003 from

140-3-s434-D004 $S\mathcal{R}_{\rho}$

[scan 435]

linked in a much more natural way to the very structure of the group $\mathcal{M}_{\rho,\sigma}$ and to its homomorphisms $\overset{\Theta}{\longrightarrow}G$ and $\chi=\det\circ\Theta$ into G_m, namely the integral subgroup $SR_{\rho,\sigma}$, allowing us to write even $\mathcal{M}^{+}_{\rho,\sigma}$ as a direct product

$$\begin{array}{ccccc}\mathcal{M}^{+}_{\rho,\sigma} & \simeq & SR_{\rho,\sigma} & \times & \mathcal{G}|\mathcal{C}^{+}_{\rho,\sigma}\\ & & \wr\downarrow & & \wr\downarrow\,\|\\ & & S'T_{\rho}\times SN'_{\sigma} & & SG\end{array}\quad. \tag{99}$$

inducing an analogous decomposition

$$\begin{cases}\mathcal{M}^{+\circ}_{\rho,\sigma}\simeq\underbrace{SR^{\dagger\circ}_{\rho,\sigma}}_{ST_{\rho}\times ST_{\sigma}}\times\underbrace{\dagger\mathcal{G}|\mathcal{C}^{+}_{\rho,\sigma}}_{SG}\\ \qquad\simeq ST_{\rho}\times ST_{\sigma}\times SG,\end{cases} \tag{100}$$

m1116

Marginal note. m1116 this is (46), which I had forgotten along the way…

who can do better!. On the other hand, the fact that $R_{\rho,\sigma} \longrightarrow \mathcal{T}_{\rho,\sigma}$ is an isogeny mod μ_2 shows that we have, up to this isogeny, a "section" of|even † $\mathcal{M}_{\rho,\sigma}$ itself as an extension of $\mathcal{T}_{\rho,\sigma}$ by $\mathcal{G}|\mathcal{C}^{+}_{\rho,\sigma} \simeq SG$. In fact, we have

$$\begin{cases} R_{\rho,\sigma} \cap SG = \mu_2 \\ \dagger \\ \dagger \\ R_{\rho,\sigma} \text{ commutes with } SG \end{cases} \tag{101}$$

(trivial by definition of $R_{\rho,\sigma}$ as $\operatorname{Ker}\widetilde{\Theta}$

so we have an exact sequence [p. 434]

Uncertainty retained from cancelled passages.

140-3-s435-D001 +
140-3-s435-D002 S
140-3-s435-D003 $\mathcal{T}_{\rho}$
140-3-s435-D004 but no longer commutes
140-3-s435-D005 so we have an exact sequence

[scan 436]

(102) [p. 520]

$$1 \longrightarrow \mu_2 \longrightarrow R_{\rho,\sigma} \times SG \longrightarrow \mathcal{M}_{\rho,\sigma} \longrightarrow 1$$

$$\varepsilon \longmapsto \left(\varepsilon, \underset{\varepsilon^{-1}}{\varepsilon^{\dagger}}\right)$$

and likewise

(103)

$$1 \longrightarrow \mu_2 \longrightarrow R^{\circ}_{\rho,\sigma} \times SG \longrightarrow \mathcal{M}^{\circ}_{\rho,\sigma} \longrightarrow 1$$

already considered (a little timidly) in (78). It is decidedly time to make fully explicit the

description of $R_{\rho,\sigma}$ in terms of T_ρ, N'_σ ... — we obtain it, in the context of page 510, as

$$
\begin{aligned}
R_{\rho,\sigma} &= \left\{ \begin{pmatrix} \underset{\Gamma T_0}{\underline{u}} , \underset{\Gamma T_\rho}{\underline{a}} , \underset{\Gamma N'_\sigma}{\underline{b}} \end{pmatrix} \cdot \underset{SG}{f} \;\middle|\; \begin{array}{c} \det \underline{u} = \det \underline{b} = \varepsilon' \det \underline{a}, \quad \varepsilon'^2 = 1 \\ \underline{u} f = \lambda \qquad (\varepsilon', \lambda \in \Gamma G_m) \\ (\text{hence } \lambda^2 = \det \underline{u}) \end{array} \right\} \\
&\simeq \left\{ \begin{pmatrix} \underset{T_0}{\underline{u}}, \underset{T_\rho}{\underline{a}}, \underset{N'_\sigma}{\underline{b}}\,\dagger, \underset{G_m}{\lambda} \end{pmatrix} \;\middle|\; \det \underline{u} = \det \underline{b} = \varepsilon' \det \underline{a} = \lambda^2 \right\} \\
&\simeq \left\{ \begin{pmatrix} \underset{T_\rho}{\underline{a}}, \underset{N'_\sigma}{\underline{b}}, \underset{G_m}{\lambda}\,\dagger, \underset{\mu_2}{\varepsilon'} \end{pmatrix} \;\middle|\; \begin{array}{c} \det \underline{b} = \varepsilon' \det \underline{a} = \lambda^2 \\ \dagger \end{array} \right\} \\
&\simeq \left\{ \begin{pmatrix} \underset{T_\rho}{\underline{a}}, \underset{N'_\sigma}{\underline{b}}, \underset{G_m}{\lambda} \end{pmatrix} \;\middle|\; \det \underline{b} = \lambda^2, \quad \det \underline{a}^2 = \det \underline{b}^2 \right\} \\
&\simeq \left(\dagger(T_\rho, \delta_\rho^2) \times_{G_m} (N'_\sigma, \delta_\sigma^2) \right)_{(\delta_\sigma)} \times_{G_m} G_{m(\delta^2)}
\end{aligned}
\tag{104}
$$

or more simply, in terms of $\mathcal{T}_{\rho,\sigma}$:

†

(105)

$$
\begin{array}{ccc}
R_{\rho,\sigma} & \longrightarrow & \mathcal{T}_{\rho,\sigma} \\
{\scriptstyle\kappa}\downarrow & & \downarrow{\scriptstyle\chi} \\
G_m & \xrightarrow[\delta^2]{} & G_m
\end{array}
\qquad \text{diagram}
$$

$$
\begin{array}{ccc}
\lambda & \longmapsto & \lambda^2
\end{array}
\qquad \text{cartesian}
$$

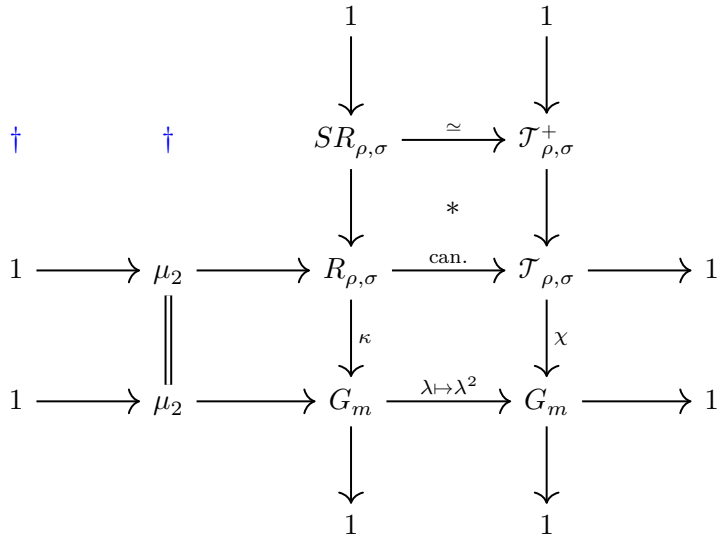
, [p. 435]

Uncertainty retained from cancelled passages.

140-3-s436-D001 +
140-3-s436-D003)
140-3-s436-D004)
140-3-s436-D005 $\det \underline{b} / \det \underline{a} \in \Gamma\mu_2$
140-3-s436-D006 {
140-3-s436-D008 $\mathcal{G}|\mathcal{C}$

[scan 437]

fitting into a diagram of exact sequences
(106)



which is nothing but an explicit description of the upper face (∗) of the cubical diagrams (96) — † we shall take note that this one is not cartesian — nor the lower face. Thus, we construct †
(107) group $\mathcal{T}_{\rho,\sigma}$, and the homomorphisms

$$\mathcal{T}_{\rho,\sigma} \xrightarrow{\chi} G_m$$

,

$$\mathcal{T}_{\rho,\sigma} \xrightarrow{(\varepsilon_3,\varepsilon_2)} \mu_2 \times \mu_2$$

[and thereby thanks to (105).–(106)) $R_{\rho,\sigma}$, $SR_{\rho,\sigma} \simeq \mathcal{T}^{+}_{\rho,\sigma}$, and also (by †replacing $\mathcal{T}_{\rho,\sigma}$ by $\mathcal{T}^{\circ}_{\rho,\sigma} = \mathrm{Ker}(\mathcal{T}_{\rho,\sigma} \longrightarrow \mu_2 \times \mu_2)$) $R^{\circ}_{\rho,\sigma}$, $SR^{\circ}_{\rho,\sigma} \simeq \mathcal{T}^{+\circ}_{\rho,\sigma}$ ⇒] in terms of the groups
(108) T_ρ, N'_σ and of

$$T_\rho \xrightarrow{\delta_\rho} G_m$$

,

$$N'_\sigma \xrightarrow{\delta_\sigma} G_m$$

$$N'_\sigma \xrightarrow{\varepsilon_\sigma} \mu_2$$

, [p. 436]

Uncertainty retained from cancelled passages.
140-3-s437-D001 1
140-3-s437-D003 μ_2
140-3-s437-D007 so
140-3-s437-D008 $\mathcal{T}_{\rho,\sigma}$ $\mathcal{T}_{\rho,\sigma}|R_{\rho,\sigma}$
140-3-s437-D009 simplifying

[scan 438]
[p. 521]

as a subgroup
(109)

$$\begin{array}{ccc} \mathcal{T}_{\rho,\sigma} \hookrightarrow T_0 & & \times T_\rho \times N'_\sigma \\ & \simeq \big\downarrow \delta_0 & \\ & G_m & \end{array}$$

$$\begin{array}{c} \| \\ \left\{ \left(\underset{T_0}{\underline{u}}, \underset{T_\rho}{\underline{a}}, \underset{N'_\sigma}{\underline{b}} \right) \;\middle|\; \delta_0(\underline{u}) = \delta_\sigma(\underline{b}) = \varepsilon' \delta_\rho(\underline{a}), \quad \varepsilon'^2 = 1 \right\} \\ \simeq \left\{ \left(\underset{T_\rho}{\underline{a}}, \underset{N'_\sigma}{\underline{b}} \right) \;\middle|\; \delta_\rho(\underline{a}) = \varepsilon' \delta_\sigma(\underline{b}), \quad \varepsilon'^2 = 1 \right\} \end{array}$$

with

$$\begin{cases} \chi(\underline{a}, \underline{b}) = \delta_\sigma | \delta_0(\underline{b}) \\ \varepsilon_3(\underline{a}, \underline{b}) = \delta_\rho(\underline{a}) / \delta_\sigma(\underline{b}) \; . \\ \varepsilon_2(\underline{a}, \underline{b}) = \varepsilon_\sigma(\underline{b}) \end{cases} \tag{110}$$

With the help of this, we can construct $\mathcal{M}_{\rho,\sigma}$ by (102), and of course the homomorphism
(111) Θ :

$$\mathcal{M}_{\rho,\sigma} \longrightarrow G$$

is deduced by passage to the quotient from the obvious homomorphisms
(112)

$$SG \hookrightarrow G$$

,

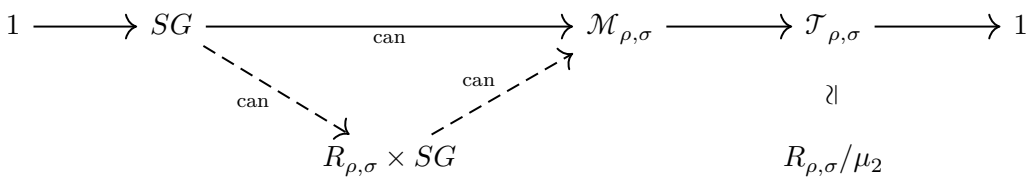
$$\begin{array}{ccc} R_{\rho,\sigma} & \longrightarrow & G \\ & \searrow^{\kappa} & \uparrow \\ & G_m \xrightarrow[\simeq]{} & \underline{\mathrm{Centr}}(G) \end{array}$$

whence we deduce the composites
(113)

$$\begin{cases} \mathcal{M}_{\rho,\sigma} \xrightarrow{\Theta} G \longrightarrow PG \\ \qquad\qquad \widetilde{\Theta} \\ \mathcal{M}_{\rho,\sigma} \xrightarrow{\Theta} G \xrightarrow{\det} G_m \\ \qquad\qquad \chi \end{cases}$$

We also deduce the exact sequence

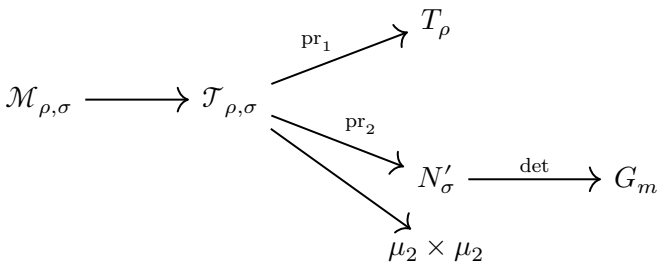
(114)



[p. 437]
[scan 439]

from the description (102), whence composites

(115)

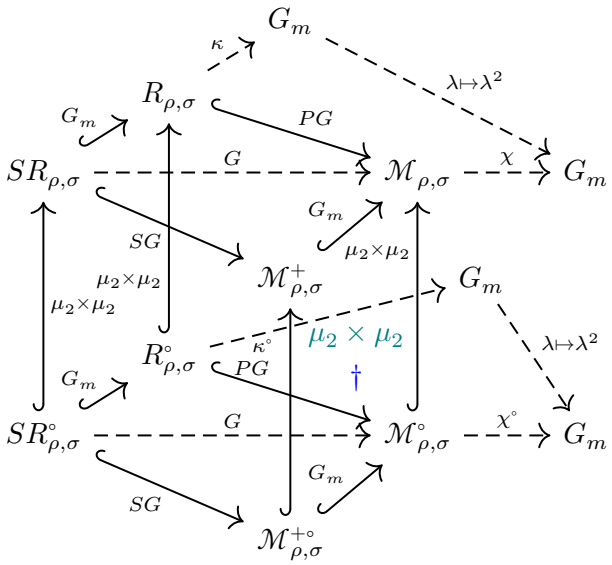


In all this, we have not needed the datum of|of the sections ρ, σ of G, nor to know anything at all about the way in which T_ρ, N'_σ have been obtained from such sections. But more delicate additional structures of the $\mathcal{M}_{\rho,\sigma}$ — such as the datum of the integral subgroups $\mathcal{M}_{\rho,\sigma}(w)|(\omega)$ — depend on them, and will perhaps allow us to recover them.

Remark|Remark The cube (96) is the "composite" of two cubical diagrams, which I am going to make explicit, the first of which completes (95) — which can be regarded as a partial diagram of this one:

[p. 438]
[scan 440]
[p. 522]

(116)



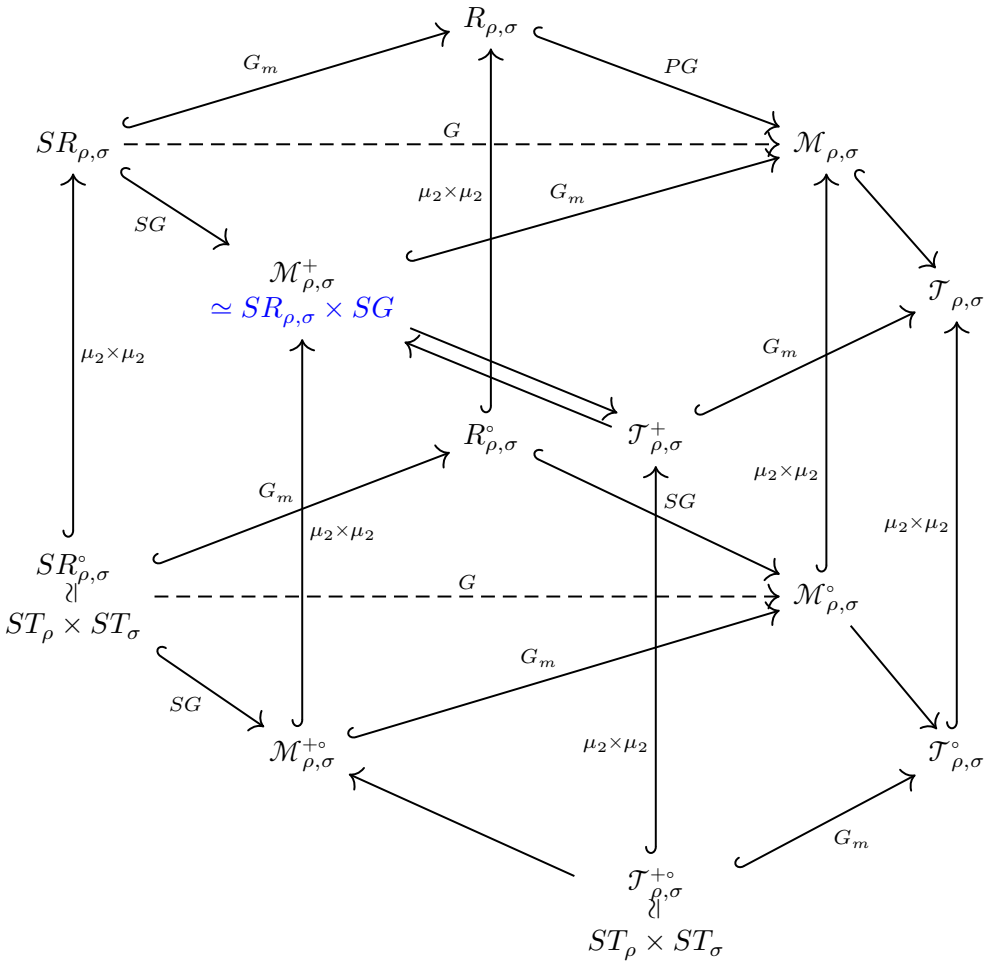
As in (95)(96), the lower face is deduced from the upper one by passing to the identity components, the four quotient groups are isomorphic to one another by the canonical arrows,

they are isomorphic to $\mu_2 \times \mu_2$ — itm|item in the cube attached below (117). The lateral faces of the cube are cartesian and. cocartesian, but not the upper faces nor the lower face. [p. 439]

Uncertainty retained from cancelled passages.

140-3-s440-D001 $\mu_2 \times \mu_2$

[scan 441]
[p. 523]



(117)

The composite front left face of the cube is of a particularly simple nature, since the middle terms $\mathcal{M}^+_{\rho,\sigma}$, $\mathcal{M}^{+\circ}_{\rho,\sigma}$ are identified simply with the direct products of the outer terms

$$\begin{array}{ll} SR_{\rho,\sigma} \xrightarrow{\sim} \mathcal{T}^+_{\rho,\sigma} & \text{and} \quad SR^\circ_{\rho,\sigma} \xrightarrow{\sim} \mathcal{T}^{+\circ}_{\rho,\sigma} \\ \simeq S'T_\rho \times SN'_\sigma & \simeq ST_\rho \times ST_\sigma \end{array}$$

with the group SG. This is almost the case too for the parallel face (back right face), up to the group μ_2 – this becomes true on on condition of replacing the middle terms $\mathcal{M}_{\rho,\sigma}$, $\mathcal{M}^\circ_{\rho,\sigma}$ by the μ_2-coverings

$$\mathcal{M}^\sim_{\rho,\sigma} \simeq R_{\rho,\sigma} \times SG, \qquad \mathcal{M}^{\circ\sim}_{\rho,\sigma} \simeq R^\circ_{\rho,\sigma} \times SG \tag{118}$$

(cf. (107)|(102), (103)), which will have also the virtue [p. 440]

[scan 442]

delectable of making the four squares of the upper and lower faces of the two attached cubes cartesian and cocartesian, which they were not before, regrettably.

To finish these scales on the radicals I am going to introduce a canonical homomorphism

$$G_m \overset{i_R}{\hookrightarrow} R^\circ_{\rho,\sigma} \tag{118}$$

via a canonical homomorphism

$$\begin{aligned} G_m &\overset{i_{\mathcal{T}}}{\hookrightarrow} \mathcal{T}^\circ_{\rho,\sigma} \simeq T_\rho \times_{G_m} T_\sigma \\ \lambda &\longmapsto (\lambda, \lambda) \end{aligned} \tag{119}$$

of which the composite|the composite with

$$\chi : \mathcal{T}^\circ_{\rho,\sigma} \longrightarrow G_m$$

is indeed given by $\lambda \mapsto \lambda^2$ (104), so that thanks to the description (105) of $R_{\rho,\sigma}$ in terms of $\mathcal{T}_{\rho,\sigma}$ as a fiber product (paraphrased as an expression of $R^\circ_{\rho,\sigma}$ in terms of $\mathcal{T}^\circ_{\rho,\sigma}$) we have a unique homomorphism (118), making the diagram commutative [p. 441]

[scan 443]

[p. 524]

$$\begin{array}{ccc} G_m & \overset{i_{\mathcal{T}}}{\longrightarrow} & \mathcal{T}^\circ_{\rho,\sigma} \\ & \underset{i_R}{\searrow} & \uparrow \text{can} \\ & & R^\circ_{\rho,\sigma} \end{array} \tag{120}$$

or, more prettily, the diagram of exact sequences

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \mu_2 & \longrightarrow & G_m & \overset{\lambda\mapsto\lambda^2}{\longrightarrow} & G_m & \longrightarrow & 1 \\ & & & & \| & & & & \end{array} \quad . \tag{121}$$

By construction, we find at once

$$G_m \overset{i_R}{\longrightarrow} R^\circ_{\rho,\sigma} \overset{\kappa}{\longrightarrow} G_m \qquad \text{(id : } G_m \to G_m\text{)} \tag{122}$$

whence direct product decompositions

$$\begin{cases} R_{\rho,\sigma} \simeq SR_{\rho,\sigma} \times G_m \simeq S'T_\rho \times SN'_\sigma \times G_m \\ R^\circ_{\rho,\sigma} \simeq SR^\circ_{\rho,\sigma} \times G_m \\ \qquad\quad \simeq ST_\rho \times ST_\sigma \times G_m \end{cases} , \tag{123}$$

which considerably sharpens (96) with which I was nevertheless so pleased!

In terms of this decomposition, we are going to make explicit the †homomorphism

$$\mu_2 \hookrightarrow R^\circ_{\rho,\sigma} \subset R_{\rho,\sigma} \qquad \left(\text{and} \quad R_{\rho,\sigma} \twoheadrightarrow \mathcal{T}_{\rho,\sigma} \ , \quad R^\circ_{\rho,\sigma} \twoheadrightarrow \mathcal{T}^\circ_{\rho,\sigma}\right)$$

which occurs in (102), (103), $\mu_2 = R_{\rho,\sigma} \cap SG = R^\circ_{\rho,\sigma} \cap SG$,

† [p. 442]

Uncertainty retained from cancelled passages.
140-3-s443-D001 h
140-3-s443-D002 We find that [illeg. 2]

[scan 444]

For the homomorphism

$$R_{\rho,\sigma} \longrightarrow \underset{T_{\rho,\sigma}}{\{(\underline{a},\underline{b}) \mid (\det \underline{a}/\det \underline{b})^2 = 1\}} \hookrightarrow T_\rho \times N'_\sigma$$

it suffices to make explicit the composites

$$\left\{\begin{array}{ccc} & & T_\rho \\ R_{\rho,\sigma} & \longrightarrow & T_{\sigma\sigma}|T_{\rho,\sigma} \\ \wr| & & \searrow \\ S'T_\rho \times SN'_\sigma \times G_m & & N'_\sigma \\ & & \\ (\underline{a},\underline{b},\lambda) & \longrightarrow & (\lambda\underline{a},\lambda\underline{b}) \end{array}\right. \tag{124}$$

m1117

Marginal note. m1117 we have as well

$$\underbrace{\det(\lambda\underline{a})^2}_{\substack{\lambda^2 \\ \text{since } (\det \underline{a})^2=1}} = \underbrace{\det(\lambda\underline{b})^2}_{\substack{\lambda^2 \text{ since} \\ \det \underline{b}=1}}$$

The kernel of this homomorphism therefore consists of the $(\lambda^{-1},\lambda^{-1},\lambda)$, with|where $\lambda \in \Gamma G_m$ such that $\lambda^{-1} \in \Gamma S'T_\rho$, $\lambda^{-1} \in \Gamma SN'_\sigma$, † i.e. $(\lambda^2)^2 = 1$ and $\lambda^2 = 1$, i.e. $\lambda \in \Gamma\mu_2$. Thus

$$\left\{\begin{array}{ccccc} \overset{\mu_2}{\|} & \xrightarrow{\text{can}} & \underbrace{R^\circ_{\rho,\sigma}}_{ST_\rho\times ST_\sigma\times G_m} & \subset & \underbrace{R_{\rho,\sigma}}_{S'T_\rho\times SN'_\sigma\times G_m} \\ R_{\rho,\sigma}\cap SG & & & & \\ & & \lambda & \longmapsto & (\lambda,\lambda,\lambda) \end{array}\right. . \tag{125}$$

We thus find the particularly simple explicit description

$$\mathcal{T}_{\rho,\sigma} \simeq (S'T_\rho \times SN'_\sigma \times G_m)/\begin{array}{c} \mu_2 \\ \uparrow \\ \mu_2 \text{ sent "diagonally" into the} \\ \text{triple product} \end{array} \tag{126}$$

We will note that we have canonical homomorphisms

$$\mu_2 \to ST_\rho \subset S'T_\rho \ , \quad \mu_2 \to ST_\sigma \subset SN'_\sigma \ , \quad \mu_2 \text{ — } G_m$$

(which are used in (125)) whence

$$\left\{\begin{array}{ccc} \mu_2\times\mu_2\times\mu_2 & \longrightarrow & \underbrace{ST_\rho \times ST_\sigma \times G_m}_{R^\circ_{\rho,\sigma}} \subset \underbrace{S'T_\rho \times SN'_\sigma \times G_m}_{R_{\rho,\sigma}} \\ (\lambda',\lambda'',\lambda) & \longmapsto & (\lambda',\lambda'',\lambda) \end{array}\right. \tag{127}$$

which is none other than (48)) [p. 443]

Uncertainty retained from cancelled passages.
140-3-s444-D002 that is|and

[scan 445]

The Center of $\mathcal{M}_{\rho,\sigma}$. It maps under $\widetilde{\Theta}$ into the center of PG which is trivial, hence [p. 525]

$$\underline{\text{Cent}}(\mathcal{M}_{\rho,\sigma}) \subset \operatorname{Ker}\widetilde{\Theta} = R_{\rho,\sigma} \simeq S'T_\rho \times SN'_\sigma \times G_m$$

whence obviously. Since $R_{\rho,\sigma}$ commutes with SG, and $R_{\rho,\sigma}\,\dagger \cdot SG = \mathcal{M}_{\rho,\sigma}$, we conclude

$$\underline{\text{Cent}}(\mathcal{M}_{\rho,\sigma}) = \underline{\text{Cent}}(R_{\rho,\sigma}) = S'T_\rho \times \underset{?}{\underline{\text{Cent}}}(SN'_\sigma) \times G_m \tag{128}$$

let us therefore determine $\text{Cent}(SN'_\sigma)$. Let us note that

$$\left\{\begin{array}{l}\mu_2 \simeq N'_\sigma/T_\sigma \quad \text{operates faithfully on } T_\sigma \\ \text{and } T_\sigma = G_m.ST_\sigma\end{array}\right. \qquad \left|\begin{array}{l}\text{hence} \quad \underset{\wr|}{\mu_2} \quad \text{operates faithfully} \\ \qquad\quad SN'_\sigma/T_\sigma \\ \text{on } ST_\sigma\end{array}\right.$$

whence

$$\underline{\text{Cent}}(SN'_\sigma) \subset ST_\sigma \quad \text{whence} \quad \underline{\text{Cent}}(SN'_\sigma) \simeq (ST_\sigma)^{\mu_2}. \tag{129}$$

Now we find that

$$T_\sigma^{\mu_2} \simeq G_m \tag{130}$$

† whence

$$\underline{\text{Cent}}(SN'_\sigma) = {ST_\sigma}^{\mu_2} = ST_\sigma \cap G_m = \mu_2 \tag{131|129}$$

†

[p. 444]

Uncertainty retained from cancelled passages.
140-3-s445-D001 ×
140-3-s445-D002 the section c g|that $c^2 = 0$ $d^2 = 1$ c hence for|hence if
140-3-s445-D008 regular

[scan 446]

†

whence finally|finally, namely|i.e.:

$$\underset{=}{\underline{\text{Cent}}}(\mathcal{M}_{\rho,\sigma}) \simeq S'T_\rho \times \mu_2 \times G_m \subset R_{\rho,\sigma} \simeq S'T_\rho \times SN'_\sigma \times G_m \tag{128}$$

whence

$$\underline{\text{Cent}}(\mathcal{M}_{\rho,\sigma}) \cap \mathcal{M}^\circ_{\rho,\sigma} = ST_\rho \times \mu_2 \times G_m \subset R^\circ_{\rho,\sigma} \simeq ST_\rho \times ST_\sigma \times G_m \tag{129}$$

Let us return to the integral sections in the absolute modular case

$$R_{\rho,\sigma}(\mathbb{Z}) \simeq \underbrace{S'T_\rho(\mathbb{Z})}_{T_\rho(\mathbb{Z})\simeq\mathbb{Z}/6\mathbb{Z}} \times \underbrace{SN'_\sigma(\mathbb{Z})}_{T_\sigma(\mathbb{Z})=\mathbb{Z}/4\mathbb{Z}} \times \underbrace{G_m(\mathbb{Z})}_{\mu_2(\mathbb{Z})\simeq\pm1} \tag{130}$$

$$\begin{aligned}\underline{\text{Cent}}(\mathcal{M}_{\rho,\sigma})[\mathbb{Z}]|(\mathbb{Z}) &\simeq S'(T_\rho[\mathbb{Z}]|(\mathbb{Z})) \times \mu_2(\mathbb{Z}) \times G_m(\mathbb{Z}) \\ &\simeq \mathbb{Z}/6\mathbb{Z} \times \{\pm1\} \times \{\pm1\}\end{aligned} \tag{131}$$

and for the record (cf. (75), p. 514)

$$\begin{aligned}\underbrace{\underline{\text{Cent}}(\mathcal{M}^\circ_{\rho,\sigma})}_{\substack{R^\circ_{\rho,\sigma} \\ \simeq ST_\rho \times ST_\sigma \times G_m}}(\mathbb{Z}) &\simeq ST_\rho(\mathbb{Z}) \times ST_\sigma(\mathbb{Z}) \times G_m(\mathbb{Z}) \\ &\simeq \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z} \times \{\pm1\}.\end{aligned} \tag{132}$$

In summary, we have the diagram of strict inclusions of group schemes

$$\begin{array}{lcc} \text{dimension 3} & \begin{array}{c}\underline{\mathrm{Cent}}(\mathcal{M}^\circ_{\rho,\sigma}) = R^\circ_{\rho,\sigma} \\ \simeq ST_\rho \times ST_\sigma \times G_m\end{array} \xhookrightarrow[\substack{\text{(quotient}\\ \mu_2\times\mu_2)}]{} \begin{array}{c} R_{\rho,\sigma} \\ \simeq S'T_\rho \times SN'_\sigma \times G_m\end{array} & \text{dimension 3} \\ & \uparrow \qquad\qquad\qquad \uparrow & \\ \text{dimension 2} & \begin{array}{c}\underline{\mathrm{Cent}}(\mathcal{M}_{\rho,\sigma}) \cap \mathcal{M}^\circ_{\rho,\sigma} \\ = ST_\rho \times \mu_2 \times G_m\end{array} \xhookrightarrow[\substack{\text{(quotient}\\ \mu_2)}]{} \begin{array}{c}\underline{\mathrm{Cent}}(\mathcal{M}_{\rho,\sigma}) \\ \simeq S'T_\rho \times \dagger\mu_2 \times G_m)\end{array} & \text{dimension 2} \end{array} \tag{133}$$

[p. 445]

Uncertainty retained from cancelled passages.

140-3-s446-D001 μ'_2 Cent this|a

140-3-s446-D004 SN'_σ

[scan 447]

valid in the general case – and which, in the absolute modular case, induces on the integral points isomorphisms for the two horizontal arrows, corresponding resp. to the two groups [p. 526]

$$\begin{cases} \underline{\mathrm{Cent}}(\mathcal{M}^\circ_{\rho,\sigma})(\mathbb{Z}) \simeq \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z} \times \{\pm 1\} & \text{and} \\ \underline{\mathrm{Cent}}(\mathcal{M}_{\rho,\sigma})(\mathbb{Z}) \simeq \mathbb{Z}/6\mathbb{Z} \times \{\pm 1\} \times \{\pm 1\} \end{cases} \tag{134}$$

and

which is a subgroup of it of index 2.

Finally, for the record let us point out once again the group

$$\Sigma = \{\pm 1\} \times \{\pm 1\} \times \{\pm 1\} \tag{135}$$

†

contained in $\underline{\mathrm{Cent}}(\mathcal{M}_{\rho,\sigma})(\mathbb{Z})$ (of order 8, of index 3), which plays a special role in the arithmetic application, since we will have a canonical homomorphism

$$\boxed{\mathcal{M}^\sim_{0,3} \longrightarrow \mathcal{M}_{\rho,\sigma}(\widehat{\mathbb{Z}})/\Sigma}\,. \tag{136}$$

[p. 446]

Uncertainty retained from cancelled passages.

140-3-s447-D001 $\pm 1\times$

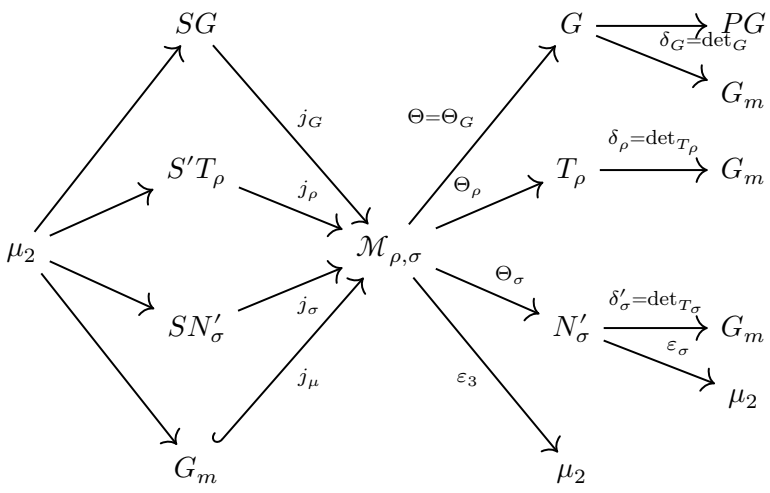
[scan 448]

m1118

Marginal note. m1118 Where we define $\mathcal{M}_{\rho,\sigma}$

Ⓣ Let us make a list|little recapitulation|recapitulatory of the main group homomorphisms considered in the course of section 6°, involving the group $\mathcal{M}_{\rho,\sigma}$

(134)



Let us note that this diagram can be reconstructed (with the essential properties which we are going to pass in review) from the data

$$(135)\quad \begin{cases} G_m \xrightarrow{i_\rho} T_\rho \xrightarrow{\delta_\rho} G_m & (\delta_\rho i_\rho(\lambda) = \lambda^2) \\ G_m \xrightarrow{i_\sigma} N'_\sigma \xrightarrow{\delta_\sigma} G_m & (\delta_\sigma i_\sigma(\lambda) = \lambda^2) \\ N'_\sigma \xrightarrow{\varepsilon_\sigma} \mu_2 & \varepsilon_\sigma i_\sigma(\lambda) = 1 \end{cases}$$

in addition to the familiar datum

$$(136)\quad \begin{array}{ccccccccc} & & 1 & & 1 & & & & \\ & & \uparrow & & \uparrow & & & & \\ & & G_m & = & G_m & & & & \\ & & {\scriptstyle \lambda\mapsto\lambda^2}\uparrow & & \uparrow{\scriptstyle \delta_G} & & & & \\ 1 & \to & G_m & \xrightarrow{j} & G & \to & PG & \to & 1 \\ & & \uparrow & & \uparrow & & \| & & \\ 1 & \to & \mu_2 & \to & SG & \to & PG & \to & 1 \\ & & \uparrow & & \uparrow & & & & \\ & & 1 & & 0|1 & & & & \end{array}$$

[p. 447]
[scan 449]
[p. 527]

Let us recall that we define, in terms of (135)

$$(137)\quad \begin{cases} ST_\rho = \operatorname{Ker}\delta_\rho, \quad S'T_\rho = \operatorname{Ker}\delta_\rho^2 \\ T_\sigma = \operatorname{Ker}\varepsilon_\sigma, \quad SN'_\sigma = \operatorname{Ker}\delta'_\sigma, \quad ST_\sigma = SN'_\sigma \cap T_\sigma = \operatorname{Ker}\delta_\sigma \\ \qquad\qquad (\delta_\sigma = \delta'_\sigma | T_\sigma) \\ \mu_2 \to ST_\rho \subset S'T_\rho \qquad \text{as composite} \qquad \mu_2 \subset G_m \xrightarrow{i_\rho} T_\sigma \\ \mu_2 \to ST_\sigma \subset SN'_\sigma \qquad \text{———————} \qquad \mu_2 \subset G_m \xrightarrow{i_\sigma} T_\sigma \end{cases},$$

which determines the left-hand part of diagram (134), where $\mathcal{M}_{\rho,\sigma}$ can be considered (and defined) as quotient of the sum of the four factors $SG, S'T_\rho, SN'_\sigma, G_m$, by μ_2 embedding itself in it diagonally.

This makes it possible in principle to reduce the description of the four fundamental morphisms from $\mathcal{M}_{\rho,\sigma}$ in (134) – $\Theta_G, \Theta_\rho, \Theta_\sigma$ and $\varepsilon_{3|2}$ – to their composites with the four homomorphisms $j_G, j_\rho, j_\sigma, j_\mu$ – subject to checking that the† of the four composites †commute, and that

their product of †their values on the images of an † $\varepsilon \in \Gamma\mu_2$ is trivial. We will have

$$(138)\quad \begin{cases} \Theta_G j_G : & SG \hookrightarrow G, \quad \text{canonical inclusion} \quad (136) \\ & \quad\dagger \\ & \Theta_G j_\rho,\ \Theta_G j_\sigma \quad \text{trivial} \\ \Theta_G j_\mu : & G_m \longrightarrow G_m \quad \text{canonical inclusion} \end{cases}$$

(conditions of compatibility|compatibilities satisfied !) [p. 448]

Uncertainty retained from cancelled passages.

140-3-s449-D002 images
140-3-s449-D003 coincide
140-3-s449-D004 that the|on the
140-3-s449-D005 $\varepsilon| \equiv$
140-3-s449-D007 $\Theta_G j_\rho \quad S'T_\rho \quad T_\rho \quad j$

[scan 450]

$$(139|135)\quad \begin{cases} & \Theta_\rho j_G\ , \quad \Theta_\rho j_\sigma \quad \text{trivial} \\ \Theta_\rho j_\rho : & S'T_\sigma \hookrightarrow T_\rho \quad \text{inclusion} \\ \Theta_\rho j_\mu : & G_m \xrightarrow{i_\rho} T_\rho \end{cases}$$

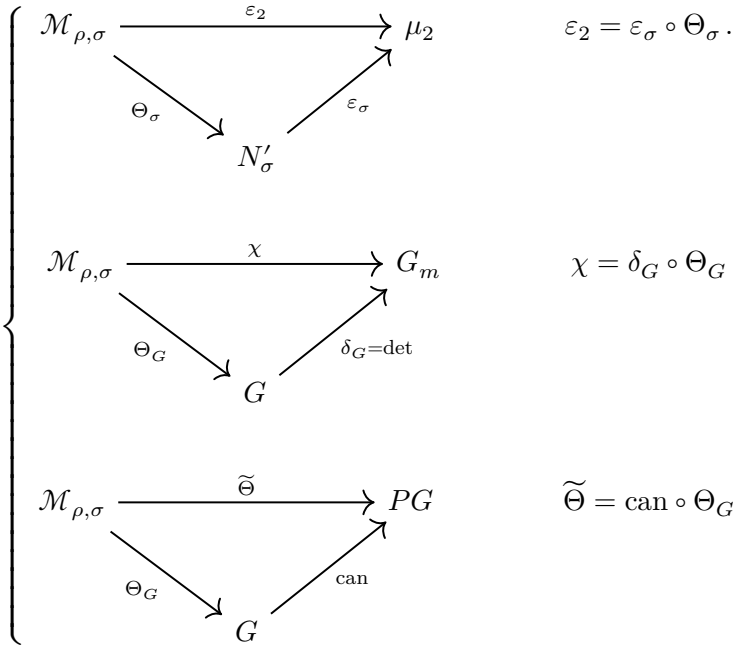
trivial compatibilities,

$$(140)\quad \begin{cases} & \Theta_\sigma j_G, \quad \Theta_\sigma j_\rho \quad \text{trivial} \\ \Theta_\sigma j_\sigma : & SN'_\sigma \hookrightarrow N'_\sigma \quad \text{inclusion} \\ \Theta_\sigma j_\mu : & G_m \xrightarrow{i_\sigma} N'_\sigma \end{cases}$$

trivial compatibilities,

$$(141)\quad \begin{cases} & \dagger\varepsilon_3 j_G, \quad \varepsilon_3 j_\sigma, \quad \varepsilon_3 j_\mu \quad \text{trivial} \\ \varepsilon_3 j_\rho : & S'T_\rho \xrightarrow{\delta_\rho} \mu_2 \\ & \quad\dagger \end{cases} .$$

We therefore define, in terms of (134)

(142)


we shall therefore have

$$(143) \qquad \mathcal{M}_{\rho,\sigma} \xrightarrow{(\chi,\varepsilon_3,\varepsilon_2)} G_m \times \mu_2 \times \mu_2 \ ,$$

epimorphism [p. 449]

Uncertainty retained from cancelled passages.

140-3-s450-D002 Θ

140-3-s450-D004 $\varepsilon_3 j_\sigma$

[scan 451]

and the other two composites $\delta_\rho\Theta_\rho$ and $\delta'_\sigma\Theta_\sigma$ of (134) are made explicit therefore|then by [p. 528] the formulas

$$(144) \qquad \begin{cases} \delta'_\sigma\Theta_\sigma = \chi \quad (\overset{\text{dfn}}{=} \delta_G\Theta_G) \\ \delta_\rho\Theta_\rho(g) = \varepsilon_3(g)\chi(g) \end{cases}$$

In terms of (143) we define

$$(145) \qquad \begin{cases} \mathcal{M}^{+}_{\rho,\sigma} = \mathrm{Ker}\left(\mathcal{M}_{\rho,\sigma} \xrightarrow{\chi} G_m \right) \\ \mathcal{M}^{\circ}_{\rho,\sigma} = \mathrm{Ker}\left(\mathcal{M}_{\rho,\sigma} \xrightarrow{(\varepsilon_3,\varepsilon_2)} \mu_2 \times \mu_2 \right) \\ \mathcal{M}^{+\circ}_{\rho,\sigma} = \mathcal{M}^{+}_{\rho,\sigma} \cap \mathcal{M}^{\circ}_{\rho,\sigma} = \mathrm{Ker}\left(\mathcal{M}_{\rho,\sigma} \xrightarrow{(\chi,\varepsilon_3,\varepsilon_2)} G_m \times \mu_2 \times \mu_2 \right) \end{cases} ,$$

we define

$$(146)\quad \begin{cases} R_{\rho,\sigma} = \mathrm{Ker}\left(\mathcal{M}_{\rho,\sigma} \xrightarrow{\widetilde{\Theta}} PG \right) \\ R^{\circ}_{\rho,\sigma} = \mathcal{M}_{\rho,\sigma} \cap R_{\rho,\sigma} = \mathrm{Ker}\left(\mathcal{M}^{\circ}_{\rho,\sigma} \xrightarrow{\widetilde{\Theta}^{\circ}} PG \right) \\ SR_{\rho,\sigma} = \mathrm{Ker}\left(\mathcal{M}_{\rho,\sigma} \xrightarrow{\Theta} G \right) \\ SR^{\circ+}_{\rho,\sigma} = \mathrm{Ker}\left(\mathcal{M}_{\rho,\sigma} \xrightarrow{(\Theta,\varepsilon_3,\varepsilon_2)} G \times \mu_2 \times \mu_2 \right) \\ \qquad = SR_{\rho,\sigma} \cap R^{\circ}_{\rho,\sigma} \end{cases}$$

whence the cartesian and cocartesian squares (94), and more richly, the prismatic – cubical diagrams (95|05) (96) (116) (117), and the dévissages

$$(147)\quad \begin{cases} R_{\rho,\sigma} \simeq S'T_\rho \times SN'_\sigma \times G_m \\ R^{\circ}_{\rho,\sigma} \simeq ST_\rho \times ST_\sigma \times G_m \\ SR_{\rho,\sigma} \simeq S'T_\rho \times SN'_\sigma \times 1 \\ SR^{\circ}_{\rho,\sigma} \simeq ST_\rho \times ST_\sigma \times 1 \end{cases},$$

m1119

Marginal note. m1119 whose meaning is evident from diagram (134)

[p. 450]
[scan 452]

$$\begin{cases} 1 \longrightarrow \mu_2 \longrightarrow R_{\rho,\sigma} \times SG \longrightarrow \mathcal{M}_{\rho,\sigma} \longrightarrow 1 \\ \qquad\quad \| \qquad\quad \downarrow{\scriptstyle \mu_2\times\mu_2} \qquad\quad \downarrow{\scriptstyle \mu_2\times\mu_2} \\ 1 \longrightarrow \mu_2 \longrightarrow R^{\circ}_{\rho,\sigma} \times SG \longrightarrow \mathcal{M}^{\circ}_{\rho,\sigma} \longrightarrow 1 \end{cases}, \qquad (148)$$

this is here the definition of $\mathcal{M}_{\rho,\sigma}$

$$\begin{cases} \mathcal{M}^{+}_{\rho,\sigma} \simeq SR_{\rho,\sigma} \times SG \\ \mathcal{M}^{+\circ}_{\rho,\sigma} \simeq SR^{\circ}_{\rho,\sigma} \times SG \end{cases}, \qquad (149)$$

Finally, let us define

$$\begin{cases} \mathcal{T}_{\rho,\sigma} \simeq \mathcal{M}_{\rho,\sigma} / \underset{\substack{\| \text{déf} \\ \mathcal{G}|\mathcal{C}^{+}_{\rho,\sigma}}}{\underline{SG}}, \\ \mathcal{T}^{\circ}_{\rho,\sigma} = \mathcal{M}^{\circ}_{\rho,\sigma}/SG = \mathrm{Ker}(\mathcal{T}_{\rho,\sigma} \longrightarrow \mu_2 \times \mu_2) \\ \mathcal{T}^{+}_{\rho,\sigma} = \mathcal{M}^{+}_{\rho,\sigma}/SG = \mathrm{Ker}(\mathcal{T}_{\rho,\sigma} \longrightarrow G_m) \\ \mathcal{T}^{+\circ}_{\rho,\sigma} = \mathcal{M}^{+\circ}_{\rho,\sigma}/SG = \mathrm{Ker}(\mathcal{T}_{\rho,\sigma} \longrightarrow \underline{G_m} \times \mu_2 \times \mu_2) \\ \qquad = \mathcal{T}^{\circ}_{\rho,\sigma} \cap \mathcal{T}^{+}_{\rho,\sigma} \quad \text{in } \mathcal{T}_{\rho,\sigma}, \end{cases} \qquad (150).$$

whence extension structures

$$1 \longrightarrow SG \longrightarrow \mathcal{M}_{\rho,\sigma} \longrightarrow \mathcal{T}_{\rho,\sigma} \longrightarrow 1$$

etc (itm|item with $\begin{matrix}\mathcal{M}^{\circ}_{\rho,\sigma},\mathcal{T}^{\circ}_{\rho,\sigma}\\ \mathcal{M}^{+}_{\rho,\sigma},\mathcal{T}^{+}_{\rho,\sigma}\\ \mathcal{M}^{+\circ}_{\rho,\sigma},\mathcal{T}^{+\circ}_{\rho,\sigma}\end{matrix}$)

and exact sequences

$$1 \longrightarrow \mathcal{T}^{\circ}_{\rho,\sigma} \longrightarrow \mathcal{T}_{\rho,\sigma} \longrightarrow \mu_2 \times \mu_2 \longrightarrow 1$$

etc (itm|item for $\begin{matrix}\mathcal{T}^{+}_{\rho,\sigma} \subset \mathcal{T}_{\rho,\sigma}\\ \mathcal{T}^{+\circ}_{\rho,\sigma} \subset \mathcal{T}_{\rho,\sigma}\\ \mathcal{T}^{+\circ}_{\rho,\sigma} \subset \mathcal{T}^{+}_{\rho,\sigma}\\ \mathcal{T}^{+\circ}_{\rho,\sigma} \subset \mathcal{T}^{\circ}_{\rho,\sigma}\end{matrix}$ [p. 451]

[scan 453]

finally isomorphisms [p. 529]

$$\begin{cases} R_{\rho,\sigma}/\mu_2 \xrightarrow{\sim} \mathcal{T}_{\rho,\sigma} \\ R^{\circ}_{\rho,\sigma}/\mu_2 \xrightarrow{\sim} \mathcal{T}^{\circ}_{\rho,\sigma} \\ SR_{\rho,\sigma} \simeq \mathcal{T}^{+}_{\rho,\sigma} \\ SR^{\circ}_{\rho,\sigma} \simeq \mathcal{T}^{+\circ}_{\rho,\sigma} \end{cases} . \tag{151}$$

8°

If U is an element of $\mathcal{M}_{\rho,\sigma}$, we associate to it its images under $\Theta = \Theta_G$, Θ_ρ, Θ_σ, and under † χ, ε_3, ε_2, we denote them respectively

$$\begin{cases} \underline{U} \in \Gamma G \\ \underline{a} \in \Gamma T_\rho \\ \underline{b} \in \Gamma N'_\sigma \\ \quad (\varepsilon, \varepsilon' \in \Gamma\mu_2 \\ \mu \in \Gamma G_m \end{cases} \tag{152}$$

m1121

Marginal note. m1121 satisfying

$$\begin{cases} \mu = \delta_G(\underline{U}) = \delta'_\sigma | \delta_\sigma(\underline{b}) = \varepsilon' \delta_\rho(a) \\ \varepsilon' | \varepsilon = \varepsilon_\sigma(\underline{b}) \end{cases} \tag{152'}$$

thus these quantities are determined by the mere knowledge of $\underline{U}, \underline{a}, \underline{b}$

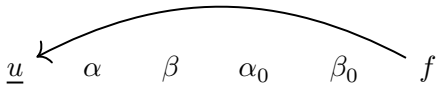
m1122

Marginal note. m1122

It would be necessary to make explicit that the equalities|relations (152)' characterize $\mathcal{M}_{\rho,\sigma}$ as a subgroup of $G \times T_\rho \times N'_\sigma$ – it would be necessary to carry out the verification, to see whether we have not forgotten, in (135) some hyp^{ses} to be made…

N.B. $G^{\pm 1} \overset{\text{déf}}{=} \operatorname{Ker} \delta^2_G$

with notations from § 46 and following. missing however, in|for the perspective of the integral points, are the elements previously denoted



, with

$$\begin{cases} u \in \Gamma T_0 \subset \Gamma G \\ \alpha, \alpha_0 \in \Gamma G^{\pm 1} \subset \Gamma G \\ \beta, \beta_0, f \in \Gamma SG. \end{cases}$$

linked to one another and to the preceding ones, by relations

$$\begin{cases} \alpha = \underline{u}\,\underline{a}^{-1} \\ \beta = \underline{u}\,\underline{b}^{-1} \\ \underline{U} = f^{-1}\underline{u} \end{cases} \quad \text{i.e.} \quad \underline{u} = \alpha\overline{a} = \beta b.$$

$$\alpha_0 = f^{-1}\alpha \qquad \text{i.e.} \quad \alpha = f\alpha_0$$

$$\beta_0 = f^{-1}\beta \qquad \text{i.e.} \quad \beta = f\beta_0$$

formulas in G

m1125

Marginal note. m1125 N.B. of these 6 quantities (153), $\underline{u}$ is distinguished by the fact of being multiplicative (like the quantities (152))

m1126

Marginal note. m1126 We could introduce

$$\begin{cases} \xi = \alpha\rho(\alpha)^{-1} \\ \eta = \varepsilon\,\beta\sigma(\beta)^{-1} \end{cases} \quad (153)$$

$$g = \alpha_0\rho(\alpha_0)^{-1}$$
$$h = \varepsilon\,\beta_0\sigma(\beta_0)^{-1}$$

but little do we care here.

[p. 452]

Uncertainty retained from cancelled passages.

140-3-s453-D001 $\varepsilon_3, \varepsilon_2$

[scan 454]

formulas which have a meaning only if we suppose given embeddings (or at least morphisms)

$$(155) \quad \begin{cases} T_\rho \hookrightarrow G \\ T_\sigma \hookrightarrow G \end{cases} \qquad \text{and a subgroup } T_0 \hookrightarrow G\,.$$

When we have such embeddings, the formulas (154) have a meaning, and define † (with the quantities (152) known) the quantities (153) in terms of one †of them, say $\underline{u}$, or † f, or α, or β (but not in terms of α_0 or of β_0).

In fact, we have a subgroup

(156)

$$\dagger \Big\downarrow \text{ such that } \begin{cases} T_0 \subset G \\ T_0 \xrightarrow{\delta_G|T_0}_{\sim} G_m \\ \text{i.e.} \quad G \simeq T_0.SG \\ \qquad\qquad \text{product 1/2 direct,} \end{cases}$$

so that for $U \in \Gamma G$, there exists a unique pair $(f, u) \in \Gamma(SG \times T_0)$ with

$$U = f^{-1}\underline{u}.$$

hence unique $\alpha, \beta, \alpha_0, \beta_0 \in \Gamma G$ satisfying (with the preceding f, u) (154) :

$$(157) \quad \alpha = \underline{u}\,\underline{a}^{-1}, \quad \beta = \underline{u}\,\underline{b}^{-1}, \quad \alpha_0 = f^{-1}\alpha = f^{-1}\underline{u}\,\underline{a}^{-1}, \quad \beta_0 = f^{-1}\dagger\beta = f^{-1}\underline{u}\,\underline{b}^{-1}.$$

[p. 453]

Uncertainty retained from cancelled passages.

140-3-s454-D001 all the

140-3-s454-D002 of them

140-3-s454-D003 [illeg. 1]

140-3-s454-D004 such

140-3-s454-D007 α

[scan 455]

[p. 530]

We will moreover have

$$\begin{aligned} \dagger \det \underline{u} &= \det \underline{U} = \mu \\ \det \alpha &= \underline{\det}(\underline{u}) \det(\underline{a})^{-1} = \varepsilon' \quad \text{by (152')} \\ \det \beta &= \det(\underline{u}) \det(\underline{b})^{-1} = 1 \\ \text{and} \quad \det \alpha_0 &= \det \alpha = \varepsilon' \\ \det \beta_0 &= \det \beta = 1 \end{aligned}$$

we will thus indeed have

$$\begin{aligned} \alpha, \alpha_0 \in \Gamma G^{\pm 1} &\overset{\text{definition}}{=} \{g \mid (\det g)^2 = 1\} \\ \beta, \beta_0 \in \Gamma SG & \end{aligned}$$

— but we have assumed, of course, that the homomorphisms $\delta_\rho, \delta_\sigma$ of (135) are induced, via the embeddings (155), by the determinant δ_G.

To arrive at the relation or of equation

$$(157) \qquad \underbrace{[\alpha, \rho]}_{\xi} = \varepsilon\, \underbrace{[\beta, \sigma]}_{\eta}\, \varepsilon_1(\mu - 1),$$

we must give ourselves sections ρ, σ of G († more precisely probably, resp.^{t} of maximal T_ρ and T_σ), and a strict unipotent $\varepsilon_1 \in \Gamma SG$, for the relation (157) to make sense. Let us replace α, β

by some|the values $\underline{u}\,\underline{a}^{-1}, \underline{u}\,\underline{b}^{-1}$, the relation becomes (using the commutativity of T_ρ, $\dagger$ which we had not really brought into play until now)

$$(*) \qquad [u,\rho] = \varepsilon \cdot \underline{u}(\underline{b}^{-1}\sigma(\underline{b})^{\dagger})^{\dagger}\sigma(\underline{u})^{-1}\varepsilon_1(\mu - 1).$$

[p. 454]

Uncertainty retained from cancelled passages.

140-3-s455-D001 $\det \underline{U} =$

140-3-s455-D002 or

140-3-s455-D003 and

140-3-s455-D004 -1

140-3-s455-D005 -1

[scan 456]

and will follow from the two conditions together

$$(158) \quad \begin{cases} \text{a)} & b^{-1}\sigma(b) = \varepsilon_\sigma(b) & \underline{\text{identically}} \text{ if } b \in \Gamma N'_\sigma \\ & \text{i.e.} \quad b(\sigma) = \varepsilon_\sigma(b).\sigma & \text{identically} \\ \text{b)} & [u,\rho] = [u,\sigma]\varepsilon_1(\mu - 1) & \text{identically in } u \in \Gamma T_0 \\ & & \text{for } \mu = \delta_{T_0}(u) \overset{\text{dfn}}{=} \det(u) \\ & \text{i.e.} \quad \rho(u) = \varepsilon_1(1-\mu)\sigma(u) & \text{identically in } u \end{cases}$$

which can also be written (applying σ^{-1} to the two sides|sides)

$$\sigma^{-1}\rho(u) = \varepsilon_0(1-\mu).u \qquad \text{where} \quad \varepsilon_0 = \sigma^{-1}(\varepsilon_1)$$

and is hardly conceivable unless $\underline{T_0 \text{ normalizes } L_0}$, and unless $\sigma^{-1}\rho \in \Gamma T_0.L_0$ is of the form

$$(159) \quad \begin{cases} \text{a)} & \widetilde{\sigma^{-1}\rho} = \widetilde{\varepsilon_0 u_0} \qquad u_0 \in \Gamma T_0 \\ \text{b)} & T_0 \text{ operates on } L_0 \text{ by the coaction|action induced} \\ & \qquad\qquad \text{by the determinant} \end{cases}$$

(There is all the same still some leeway, in relation|by to the draconian conditions envisaged in VII !)

But here equation (157) came a bit like hair in soup — there must be more convincing and demanding ways of bringing it in namely in connection with formulas

$$(160) \quad \begin{cases} \underline{U}(\rho) = \operatorname{int}(\alpha_0).\rho & (= \alpha_0\rho(\alpha_0)^{-1}\rho) \\ \underline{U}(\dagger\sigma) = \varepsilon \operatorname{int}(\beta_0)\sigma & (= \beta_0\sigma(\beta_0)^{-1}) \\ \underline{U}(\varepsilon_0) = \operatorname{int}(f^{-1})(\varepsilon_0^{\dagger}(\mu)) \end{cases} \qquad \text{i.e.} \quad \begin{aligned} [\underline{U},\rho] &= \alpha\rho(\alpha)^{-1} \\ [\underline{U},\sigma] &= \varepsilon\beta\sigma(\beta)^{-1} \end{aligned}$$

watch the sign ε

The third relation follows at once from $\underline{U} = f^{-1}\underline{u}$, and from the fact that by hypothesis (159 b) $\underline{u} \in \Gamma T_0$ operates on L_0 (identically) with the multiplier $\mu = \det(\underline{u})$. [p. 455]

Uncertainty retained from cancelled passages.

140-3-s456-D003 ρ

140-3-s456-D004 μ

[scan 457]

The other two formulas are made explicit successively, by writing $\alpha = \underline{u}\,\underline{a}^{-1}$, $\beta = \underline{u}\,\underline{b}^{-1}$, [p. 531] $\underline{U} = f^{-1}\underline{u}$

$$\alpha_0 = f^{-1}\alpha, \quad \beta_0 = f^{-1}\beta$$

$$(161)\qquad \begin{cases} \underline{u}(\rho) = \mathrm{int}(\alpha)(\rho) \\ \underline{u}(\sigma) = \varepsilon\,\mathrm{int}(\beta)(\sigma) \end{cases} \qquad \text{where } f \text{ no longer appears}$$

i.e. $\underline{u}(\rho) = \mathrm{int}(\underline{u})(\rho)$ since $\mathrm{int}(\underline{a}^{-1})\rho = \rho$ (as T_ρ commutative)

OK !

$$\underline{u}(\sigma) = \dagger\varepsilon\,\mathrm{int}(u)\underbrace{(\underline{b}^{-1}(\sigma))}_{\substack{=\varepsilon.\sigma \\ \text{cf. (158)(a)}}} = \mathrm{int}(u)\sigma \quad \text{OK !}$$

To prove (160) we have used only (158 a) and (159) (b). Let us now try to deduce from them the "loop relation" (157) — it must be a formal calculation, which I must already have done a dozen times, in contexts which become differentiated as we go along ! We eliminate f by writing the relations (160) in the equivalent form (161) and

$$(161\text{ bis})\quad \underline{u}(\varepsilon_0) = \varepsilon_0(\mu) \qquad \text{i.e.} \quad \underbrace{[\underline{u}, \varepsilon_0]}_{u\varepsilon_0(u)^{-1}} = \varepsilon_0(\mu - 1) \qquad \dagger$$

now
$$[\underline{u}, \varepsilon_0] = u\ \underbrace{\varepsilon_0(u)^{-1}}_{\text{i.e. } \sigma^{-1}\rho(u)^\dagger}$$
i.e. $\sigma([u, \varepsilon_0]) = \sigma(u)\rho(u)^{-1}$

now
$$u\rho u^{-1}\rho^{-1} = \xi$$
$$\rho(u^{-1}) = u^{-1}\xi$$
$$\rho(u) = \xi^{-1}u$$
of = |likewise $\sigma(u) = \dagger\eta^{-1}u$

$\xi \overset{\text{dfn}}{=} \alpha\rho(\alpha)^{-1}$

$\eta \overset{\text{dfn}}{=} \varepsilon\beta\sigma(\beta)^{-1}$

hence $\sigma([u, \varepsilon_0]) = \dagger\eta^{-1}\xi$

and (*)|(α) becomes, on applying σ

$$\eta^{-1}\xi = \varepsilon_1(\mu - 1) \qquad \text{i.e.} \quad \xi = \eta\varepsilon_1(\mu - 1) \qquad \text{i.e. (157)}$$

OK ! [p. 456]

Uncertainty retained from cancelled passages.

140-3-s457-D001 [*illeg.* 1]

140-3-s457-D002 int

140-3-s457-D004 -1

140-3-s457-D006 ε

140-3-s457-D007 $\xi^{-1}u\eta^{-1}u$

[scan 458]

In summary :

Proposition Let us suppose given, in addition to the data of the preceding section 7° (page 526' and following }|), in particular (135)(136)), † the following

$$(162)\quad \begin{cases} \text{a)} & \text{section } \rho \in \Gamma T_\rho, \quad \sigma \in \Gamma T_\sigma \quad \underline{\text{central}} \text{ sections} \\ \text{b)} & \text{Embeddings} \quad T_\rho \hookrightarrow G\ , \quad N'_\sigma \longrightarrow G \\ \text{c)} & \text{torus} \quad T_0 \hookrightarrow G \quad \text{such that } \delta_G : \quad T_0 \xrightarrow{\sim} G_m \\ & \text{i.e.} \quad G \simeq T_0.SG \quad \text{(semidirect)} \\ \text{d)} & \varepsilon_0 \in \Gamma G \quad \text{regular unipotent, defining} \\ & \text{thus} \quad \lambda \longmapsto \varepsilon_0(\lambda) \ : \quad G_a \hookrightarrow SG \end{cases}$$

and let us suppose

$$\text{a)}\quad \delta_G|T_\rho = \delta_\rho, \quad \delta_G|N'_\sigma = \delta_\sigma \qquad \dagger$$

$$\text{b)}\quad \mathrm{int}(\underline{b})(\sigma) = \varepsilon_\sigma(\underline{b})\sigma \quad \text{in } G \text{ identically for } \underline{b} \in \Gamma N'_\sigma|N^2_\sigma$$

(163)

$$\text{c)}\quad T_0 \text{ normalizes } L_0, \text{ and we have } u(\varepsilon_0(\lambda)) = \varepsilon_0(\dagger\mu\lambda) \quad \text{where } \mu = \delta_G(u)$$

$$\text{d)}\quad \widetilde{\sigma|\rho}^{-1}\tilde{\rho} \in \dagger\Gamma\tilde{\varepsilon}_0\widetilde{L}_0 \quad \text{i.e. } \exists \text{locally}^{\text{t}} \text{ section } \lambda_0 \in \Gamma G_m \text{ with } \lambda_0\tilde{\sigma}^{-1}\tilde{\rho} = \varepsilon_0 u_0, \quad \text{with } u_0 \in \Gamma T_0$$

m1129

links L_0, T_0

Marginal note. m1129 links L_0, T_0 | ρ, σ

Under these conditions, we have for every system (152) satisfying the relations (152') (i.e. in fact for every † section of $\mathcal{M}_{\rho,\sigma}$!) a unique system (153), satisfying the relations (154) — namely $\underline{U}$ can be written uniquely in the form

$$(164)\quad \underline{U} = f^{-1}\underline{u}, \qquad f \in \Gamma SG, \quad \underline{u} \in \Gamma T_0$$

and we put

$$(165)\quad \alpha = \underline{u}\,\underline{a}^{-1}, \quad \beta = \underline{u}\,\underline{b}^{-1}, \quad \alpha_0 = f^{-1}\alpha|\alpha_\rho = \dagger f^{-1}\underline{u}\,\underline{a}^{-1}, \quad \beta_0 = f^{-1}\beta = f^{-1}\underline{u}\,\underline{b}^{-1}.$$

[p. 457]

Uncertainty retained from cancelled passages.

140-3-s458-D001 $T_{\rho,\sigma}|T_\rho, \sigma$
140-3-s458-D004 normalizes L_0
140-3-s458-D005 $\delta_G(u)$
140-3-s458-D006 $\widetilde{T}_0$
140-3-s458-D007 syst
140-3-s458-D008 β

[scan 459]

All this being laid down, we then have the formulas (160), (161), (161 bis), and the "loop" formula (157). [p. 532]

We now ask ourselves the question under what conditions knowledge of (α, β, f) (for example) among the quantities (153), allows us to recover the quantities (152) — i.e. a section of $\mathcal{M}_{\rho,\sigma}$ — and what conditions must be imposed on (α, β, f). It would suffice (for uniqueness) to be able to construct $\underline{u}$, whence $\underline{U}, \underline{a}, \underline{b}$ by the first three formulas (154). † We must therefore find $\underline{a} \in \Gamma T_\rho$ such that $\alpha\underline{a}$ is a section of T_0, — indeterminacy in $\underline{a}$ is by $\Gamma T_0 \cap T_\rho$, and $\underline{b} \in \Gamma N'_\sigma$ such that $\beta\underline{b}$ is a section of T_0 — indeterminacy in † $\Gamma T_0 \cap N'_\sigma$. The indeterminacy in $\underline{u}$ is right multiplication by a section of $T_0 \cap T_\rho \cap N'_\sigma$, so it will suffice that we have

$$(164)\qquad T_0 \cap T_\rho \cap N'_\sigma = \{1\}.$$

m1130

Marginal note. m1130 in fact, we will even have $T_0 \cap T_\rho = \{1\}$ in the cases considered in II, III and following|sequel...

Assuming this relation satisfied, what conditions to put on (α, β, f) to come from a system (152) — i.e. from a section of $\mathcal{M}_{\rho,\sigma}$? [p. 458]

Uncertainty retained from cancelled passages.

140-3-s459-D001 If, on the other hand we have

140-3-s459-D002 $\Gamma N'_\sigma \cap$

[scan 460]

We see that if a section † φ_0 of $\mathcal{M}_{\rho,\sigma}$, † corresponding to α_0, β_0, f_0, is replaced by $\varphi_1 \overset{\text{definition}}{=} f^{-1}.\varphi_0$, corresponding to α_1, β_1, f_1, then

$$(165) \qquad \alpha_1 = \alpha_0, \quad \beta_1 = \beta_0, \quad f_1 = f_0 f,$$

for we see that if φ_0 corresponds to $(\underline{U}_0, \underline{a}_0, \underline{b}_0)$, then φ_1 to $(U_1, \underline{a}_1, \underline{b}_1)$, then we have

$$(166) \qquad U_1 = f^{-1} U_0, \quad \underline{a}_1 = \underline{a}_0, \quad \underline{b}_1 = \underline{b}_0$$

(for $\Theta_G \mid SG =$ the canonical inclusion $SG \hookrightarrow G$, and $\Theta_\rho, \Theta_\sigma$ trivial on SG). Thus † we will have

$$u_1 = u_0 \qquad (\text{for } \det U_1 = \det U_0)$$

whence (165). This shows that the conditions sought on (α, β, f) depend only on α, β, not on f. We would like them to be expressed by the validity of the loop formula(157), for a parameter ε, μ not specified ($\varepsilon^2 = 1$) (which will be then uniquely determined — and will allow the secondary definition of local notion|local action…).

m1131

Marginal note. m1131 to tell the truth, this condition is not sufficient even in the cases considered — we will find therefore|in a $\underline{u} \in B'_0$, and we must require that $\underline{u} \in T_0$ (already for (α, β) to be "normalized") i.e. $B = 0$, where $B = ac + bd + bc$, if $\underline{u} = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ (modular case) cf. proposition page 412'… To have it R ; itself being additional condition, it|which is necessary to replace|is necessary to consider $\mathcal{M}_{\rho,\sigma}$ by the product $\frac{1}{2}$ direct by L_0, on which we set $f_\Delta = 0$, i.e. $S\mathcal{M}_{\rho,\sigma}$.

The sufficiency of this condition will be the subject of the reflections in (II) (III), under conditions (more draconian than those of the preceding proposition (page 531') which we have imposed on the triple $(\varepsilon_0, \rho, \sigma)$. It [p. 459]

Uncertainty retained from cancelled passages.

140-3-s460-D001 φ

140-3-s460-D003 replaced

140-3-s460-D006 it

[scan 461]

is possible that the conditions at which we there arrived were more restrictive than necessary, [p. 533] and that I shall be obliged to come back to this later, in a context more general than that of forms of Gl(2), GP(1), Sl(2), but for the moment the detailed study made by taking our inspiration from the hypotheses brought out in † II, III will have served to clear the ground for us, to clarify our ideas on an important particular case.

Remark. Let us recall that we have seen that in fact μ is already defined in terms of α alone, and even in terms of $\alpha\rho(\alpha)^{-1} = \xi$; † But it is determined in terms of β — in fact, of $\beta\sigma(\beta)^{-1}$ — only modulo knowledge of ε (in practice, modulo sign…). But if we impose (for example) $\varepsilon = 1$, the datum of α determines β only modulo † $\beta \longrightarrow \beta T_\sigma$… [p. 460]

Uncertainty retained from cancelled passages.

140-3-s461-D001 the

140-3-s461-D002 indeed|therefore

140-3-s461-D003 β

[scan 462]

The structure of $\mathcal{M}_{\rho,\sigma}$ being well understood, let us return to the group $S\mathcal{M}_{\rho,\sigma}$(cf. page 495'), considered as the semidirect product of $\mathcal{M}_{\rho,\sigma}$ by the subgroup L_0 of SG, on which $\mathcal{M}_{\rho,\sigma}$ in fact operates via χ. Denoting by
(165)

$$\dagger\, \mathcal{E}|\varepsilon_{L_0} : G_a \xrightarrow{\sim} \mathrm{Ker}(S\mathcal{M}_{\rho,\sigma} \longrightarrow \mathcal{M}_{\rho,\sigma})$$

corresponding to ε_0, we will have by † construction

$$(166) \qquad U(\mathcal{E}|\varepsilon_{L_0}(\lambda)) = \mathcal{E}|\varepsilon_{L_0}(\mu\lambda) \qquad \text{where } \mu = \chi(U).$$

[p. 461]

Uncertainty retained from cancelled passages.
140-3-s462-D001 ε_0 U
140-3-s462-D002 definition

[scan 463]

† ℘ The subgroups $\mathcal{M}_{\rho,\sigma}(0)$, † $\mathcal{M}_{\rho,\sigma}[0]$, and the group $S\mathcal{M}_{\rho,\sigma}$, the isomorphism [p. 534]

$$S\mathcal{M}_{\rho,\sigma} \xrightarrow{\sim} \Sigma^{+}_{\rho,\sigma}$$

The group $\mathcal{M}_{\rho,\sigma}$ is an extension of $\mathcal{T}_{\rho,\sigma}$ by $\widetilde{G}|\widetilde{\mathcal{C}}^{+}_{\rho,\sigma} \simeq SG$, and the three subgroups which we are going to make explicit will be subgroups|subgroups, with the grain of salt that for the third, it will be a partial section only, over the subgroup $\mathcal{T}'_{\rho,\sigma}$ of index 2 in $\mathcal{T}_{\rho,\sigma}$, kernel of ε_3.

⟦Let us determine in a general way the sections of $\mathcal{M}_{\rho,\sigma}$ over $\mathcal{T}_{\rho,\sigma}$. Since we have
(1)

$$\begin{array}{ccc} R_{\rho,\sigma} & \longrightarrow & \mathcal{T}_{\rho,\sigma} \quad \text{epimorphism with kernel } \mu_2 \\ \wr| \simeq & & \\ S'T_\rho \times SN'_\sigma \times G_m & & \end{array}$$

it comes to the same to give oneself a section, or: a homomorphism
(2)

in the end this won't be of much use…

$$R_{\rho,\sigma} \xrightarrow{\dagger\psi} \mathcal{M}_{\rho,\sigma}$$

which is

$$(3) \qquad \begin{cases} \text{a)} & \text{trivial on } \mu_2 \\ \text{b)} & \dagger \quad \text{“lifts” (1)} \end{cases}$$

The homomorphisms $\Xi|\Sigma$ satisfying (3) correspond to the splittings of the trivial extension of $R_{\rho,\sigma}$ by SG (product extension), hence simply to the homomorphisms
(4)

$$R_{\rho,\sigma} \xrightarrow{\dagger\psi^0} SG$$

, [p. 462]

Uncertainty retained from cancelled passages.

140-3-s463-D001 XI
140-3-s463-D002 $\overline{J}|\overline{\jmath}$
140-3-s463-D005 $\Xi|\zeta$
140-3-s463-D007 $\Xi|\Sigma \quad \equiv | = \quad \widetilde{G}|\widetilde{\mathcal{C}}$
140-3-s463-D008 $\Xi|\zeta \quad 0$

[scan 464]

by associating to ψ^0† the homomorphism

$$(5) \qquad x \longmapsto \dagger\psi(x) = x\dagger\psi^0(x) = \dagger\psi^0(x).x$$

Thus condition a takes the form

$$(6) \qquad \dagger\psi_0^0(x) = x \quad \text{on } \mu_2 = R_{\rho,\sigma} \cap SG.$$

†

The ψ, ψ^0 corresponding to the † sections $\mathcal{M}^{(0)}_{\rho,\sigma}$, $\mathcal{M}^{(\frac{1}{2})}_{\rho,\sigma}$, $\mathcal{M}^{(-\overline{J})}_{\rho,\sigma}$ will be denoted resp.^{ly}

$$(\psi_0, \psi_0^0),\ (\psi_{1/2}, \psi_{1/2}^0),\ (\psi_{-\overline{J}}, \psi^0_{-\overline{J}}). \qquad]]$$

1°) Case of $\mathcal{M}^{(0)}_{\rho,\sigma}$. With the notations of page 529, it will correspond † $\underline{u}$ of $\mathcal{M}_{\rho,\sigma}$ satisfying one of the equivalent conditions

$$(7) \qquad f = 1, \quad \underline{U} = \underline{u}, \quad \underline{U} \in \Gamma T_0, \quad \alpha_0 = \alpha, \quad \beta_0 = \beta$$

so we have

$$(8) \qquad \mathcal{M}^{(0)}_{\rho,\sigma} = \Theta^{-1}(T_0).$$

(in terms of the description of $\mathcal{M}_{\rho,\sigma}$ on page 526' and following)

Let us check that this is a subgroup-section. We have

$$\overbrace{\mathcal{M}^{(0)}_{\rho,\sigma} \cap \widetilde{G|\mathcal{C}}^{+}_{\rho,\sigma}}^{(\Theta|SG)^{-1}(T_0)} = \{1\}$$

for $\Theta|(\widetilde{G|\mathcal{C}}^{+}_{\rho,\sigma} = SG)$ is the canonical inclusion $SG \hookrightarrow G$, now $SG \cap T_0 = \{1\}$. It remains to check that and

$$\mathcal{M}^{(0)}_{\rho,\sigma} \longrightarrow \mathcal{T}_{\rho,\sigma}$$

(which is a monomorphism by $(a)|(\alpha)$) is an epimorphism. But in [p. 463]

Uncertainty retained from cancelled passages.

140-3-s464-D001 ξ_0
140-3-s464-D002 ξ
140-3-s464-D003 ξ
140-3-s464-D004 $\xi(x)x$
140-3-s464-D005 ξ
140-3-s464-D006 ρ
140-3-s464-D007 "three"
140-3-s464-D008 $\underline{U}$ and the

[scan 465]
[p. 535]

terms of the canonical realization

$$(9)\qquad \mathcal{M}_{\rho,\sigma} \simeq \underbrace{R_{\rho,\sigma} \times SG}_{\widetilde{\mathcal{M}}_{\rho,\sigma}} / \underset{\text{diagonal}}{\mu_2}$$

and of the homomorphism m1135

Marginal note. m1135

$$\kappa : R_{\rho,\sigma} \longrightarrow \mathbf{G}_m$$

the canonical homomorphism making it possible to express $\Theta|R_{\rho,\sigma}$

$$(10)\qquad i_{T_0} : \mathbf{G}_m \xrightarrow{\sim} T_0 \hookrightarrow G$$

isomorphism|homomorphism inverse to the isomorphism

$$\delta_{T_0} = \det|T_0 : \quad T_0 \xrightarrow{\sim} \mathbf{G}_m,$$

we have, for an element

$$\begin{array}{cc} (\underline{r}, & U_0) \\ \wedge & \wedge \\ R_{\rho,\sigma} & SG \end{array}$$

of $\Gamma\widetilde{\mathcal{M}}_{\rho,\sigma}$

$$\Theta(\underline{r}, U_0) = \kappa(\underline{r})U_0, \quad \text{determinant } \kappa(r)^2$$

to say that this is $\in U$ therefore means we have

$$\kappa(\underline{r})U_0 = i_{T_0}(\kappa(\underline{r})^2) \quad \text{i.e. } \dagger$$

$$(11)\qquad U_0 = \kappa(\underline{r})^{-1} i_{T_0}(\kappa(\underline{r})^2)$$

so the inverse image $\widetilde{\mathcal{M}}^{(0)}_{\rho,\sigma}$ of $\mathcal{M}^{(0)}_{\rho,\sigma}$ in $\widetilde{\mathcal{M}}_{\rho,\sigma}$ consists of the

$$(12)\qquad (\underline{r}, \kappa(\underline{r})^{-1} i_{T_0}(\kappa(\underline{r}))^2) \in \Gamma\widetilde{\mathcal{M}}_{\rho,\sigma}$$

in other words this corresponds to a group section, with

$$(13)\qquad \psi^0_0(\underline{r}) = \kappa(\underline{r})^{-1} i_{T_0}(\kappa(\underline{r})^2).$$

As it should be, we do indeed have here a homomorphism

$$\psi^0_0 : R_{\rho,\sigma} \longrightarrow SG$$

and its value on μ_2 is the identity, for $\kappa(\underline{r})^2 = 1$ and we have

$$\psi^0_0(r) = \kappa(\underline{r})^{-1} = \begin{array}{c} \kappa(\underline{r}) \\ \| \\ \underline{r} \end{array}$$

Remark † To define this section, it sufficed [p. 464]

Uncertainty retained from cancelled passages.
140-3-s465-D001 $U_0 =$
140-3-s465-D002 qui

[scan 466]

(from the point of view of pure calculation) to give oneself an arbitrary homomorphism, denoted $i'_{T_0}|i_{T_0}$, from $\mathbf{G}_m$ to SG, and to set (13).

Before leaving $\mathcal{M}^{(0)}_{\rho,\sigma}$, I am going to introduce the slightly larger group (of dimension 4, instead of 3)

$$(14) \qquad \mathcal{M}^{[0]}_{\rho,\sigma} \overset{\text{dfn}}{=} \Theta^{-1}(B'_0)$$

where, let us recall, B'_0 is defined as the $\frac{1}{2}$ direct product

$$(15) \qquad B'_0 = T_0.L_0$$

This time, $U = U_0\kappa(r) \in \Gamma B'_0$ is not determined by knowledge alone of its determinant2 $\underline{\kappa}(r)^2$, which determines it only modulo L_0. This corresponds to the fact that $\mathcal{M}^{[0]}_{\rho,\sigma} \cap \overbrace{\widetilde{G|\mathcal{C}}^{+}}^{SG}_{\rho,\sigma}$ is no longer reduced to (1), but to L_0, which therefore appears as an invariant subgroup of $\mathcal{M}^{[0]}_{\rho,\sigma}$, and we obviously will have|have (taking into account that $\mathcal{M}^{(0)}_{\rho,\sigma} \subset \mathcal{M}^{[0]}_{\rho,\sigma}$ and

$$\mathcal{M}^{(0)}_{\rho,\sigma} \xrightarrow{\sim} \mathcal{T}_{\rho,\sigma}$$

) by|that

$$(16) \qquad \mathcal{M}^{[0]}_{\rho,\sigma} = \mathcal{M}^{(0)}_{\rho,\sigma}.L_0 \quad \text{product } \frac{1}{2} \text{ direct}$$

This is again a construction which makes sense as soon as we have a T_0 as above, † and a subgroup † L_0 of SG, [p. 465]

Uncertainty retained from cancelled passages.
140-3-s466-D001 [illeg. 2]
140-3-s466-D002 affine

[scan 467]

invariant under T_0. † [p. 536]

The operation of $\mathcal{M}^{[0]}_{\rho,\sigma}$ on SG takes place through Θ, only recently regarded as an epimorphism

$$\mathcal{M}^{[0]}_{\rho,\sigma} \xrightarrow{\Theta|\mathcal{M}^{[0]}_{\rho,\sigma}} B'_0$$

; this operation being specified, we can introduce m1136

Marginal note. m1136 for a more transparent definition, cf. pages 538', 539

$$(17) \qquad \dagger S\mathcal{M}_{\rho,\sigma} \overset{\text{definition}}{=} \mathcal{M}^{[0]}_{\rho,\sigma}.SG \quad \text{product } \frac{1}{2} \text{ direct}$$

in which SG is invariant, and $\mathcal{M}^{[0]}_{\rho,\sigma}$ is identified with a subgroup of $S\mathcal{M}_{\rho,\sigma}$. Since $\mathcal{M}^{(0)}_{\rho,\sigma} \subset \mathcal{M}^{[0]}_{\rho,\sigma}$, we have a canonical inclusion

$$(18) \qquad \underbrace{\mathcal{M}^{(0)}_{\rho,\sigma} : SG}_{\simeq \mathcal{M}_{\rho,\sigma}} \subset \underbrace{\mathcal{M}^{[0]}_{\rho,\sigma}.SG}_{= S\mathcal{M}_{\rho,\sigma}}$$

whose image is † identified with $\mathcal{M}_{\rho,\sigma}$. †

† I am beginning again to get lost in it, I am going to hang on — for an understanding of the "geometric" meaning of $S\mathcal{M}_{\rho,\sigma}$, $\mathcal{M}^{[0]}_{\rho,\sigma}$, to the notations of page 529, which deserves to be made explicit|made explicit by a [p. 466]

Uncertainty retained from cancelled passages.

140-3-s467-D001 We

140-3-s467-D002 $\mathcal{S}$

140-3-s467-D003 denoted

140-3-s467-D004 operates

140-3-s467-D005 Let us return|Let us resume

[scan 468]

Proposition Let us consider the homomorphism

$$(19) \quad \mathcal{M}_{\rho,\sigma} \longrightarrow G \times T_\rho \times N'_\sigma \times \mathbf{G}_m \times \mu_2^2|\mu_2\dagger$$

$$x \longmapsto (\Theta_G(x), \Theta_\rho(x), \Theta_\sigma(x), \chi(x), \varepsilon_2(x), \varepsilon_3(x))$$

This is a monomorphism, whose image consists of the systems $(\underline{U}, \underline{a}, \underline{b}, \mu, \varepsilon, \varepsilon')$ satisfying

†

$$(20) \qquad \begin{cases} \mu = \delta_G(\underline{U}) = \delta_\sigma(\underline{b}) = \varepsilon' \delta_\rho(\underline{a}) \\ \varepsilon = \varepsilon_\sigma(\underline{b}) \end{cases}$$

Corollary. The morphism.

$$(21) \quad \left| \begin{array}{ccc} \mathcal{M}_{\rho,\sigma} & \longrightarrow & G \times T_\rho \times N'_\sigma \\ x & \longmapsto & \Theta_G(x), \Theta_\rho(x), \Theta_\sigma(x) \end{array} \right.$$

is a monomorphism, the image consists of the triples $(\underline{U}, \underline{a}, \underline{b})$ satisfying m1137

Marginal note. m1137 N.B. we have $\begin{cases} \varepsilon = \varepsilon_\sigma(\underline{b}) \\ \varepsilon' = \delta_\rho(\underline{a})/\delta_\sigma(\underline{b}) \\ \mu = \delta_\sigma(\underline{b}) = \delta_G(\underline{U}) \end{cases}$

$$(22) \qquad \begin{cases} \delta_G(\underline{U}) = \delta_\sigma(\underline{b}), \\ [\delta_\rho(\underline{a})/\delta_\sigma(\underline{b})]^2 = 1. \end{cases}$$

Proof of the proposition. We have †

$$\mathrm{Ker}(\Theta_G, \dagger, \varepsilon_2, \varepsilon_3) = SR^{\circ|0}_{\rho,\sigma} \simeq ST_\rho \times ST_\sigma$$

and $\Theta_\rho, \Theta_\sigma$ † on $SR^{\circ|0}_{\rho,\sigma}$ are identified with the two projections, hence †

$$\mathrm{Ker}((\Theta_\rho, \Theta_\sigma)|SR^{\circ|0}_{\rho,\sigma}) = \{1\}, \quad \text{hence}$$

$$\mathrm{Ker}(\Theta_G, \chi, \varepsilon_2, \varepsilon_3, \Theta_\rho, \Theta_\sigma) = (1),$$

[p. 467]

Uncertainty retained from cancelled passages.

140-3-s468-D001 $(\mathbf{G}_m^3 \times \mathbf{G}_m \times \mu_2)$
140-3-s468-D003 $\mu = |\text{for} \quad \det_G |\delta_G$
140-3-s468-D004 $\mathrm{Ker}(\Theta_G)|\mu_G$
140-3-s468-D005 $\chi|\xi_2$
140-3-s468-D006 thus|of the
140-3-s468-D007 Θ

[scan 469]

hence a monomorphism. For the epimorphism, we are reduced to the equivalent form of the Corollary. — We may assume that we have a λ such that [p. 537]

$$\lambda^2 = \mu \overset{\text{dfn}}{=} \det \underline{U}$$

we want to write

$$\underline{U} = \Theta_{\widetilde{\mathcal{M}}} \begin{pmatrix} r & , & U_0 \\ \cup & & \cup \\ R_{\rho\sigma} & & SG \end{pmatrix} = \kappa(r) U_0$$

m1138

Marginal note. m1138 N.B. $\Theta_{\widetilde{\mathcal{M}}}$ the canonical homomorphism

$$\widetilde{\mathcal{M}}_{\rho,\sigma} \overset{\text{dfn}}{=} R_{\rho,\sigma} \times SG \longrightarrow G$$

which I had|wrote|write Θ_G.

with $U_0 \in \Gamma(SG)$, i.e. $\kappa(r)^2 = \det \underline{U}$ i.e. $\kappa(r)^2 = \mu$, we shall take r with †

$$\kappa(r) = \lambda,$$

† we therefore put

$$\underline{U}_0 = \lambda^{-1} U, \quad \underline{a}_0 = \lambda^{-1} \underline{a}, \quad \underline{b}_0 = \lambda^{-1} b$$

† and we are reduced to seeing that $\underline{U}_0, \underline{a}_0, \underline{b}_0$ are in the image of $\mathcal{M}_{\rho,\sigma}$ (in being|removing "corrected ones" $(\underline{U}, \underline{a}, \underline{b})$ by the image † $G \times T_\rho \times T_\sigma$ of the section λ of $G_m \hookrightarrow R_{\rho,\sigma}$). We are reduced to seeing that $(\underline{a}_0, \underline{b}_0)$ is in the image of

$$R_{\rho,\sigma} \longrightarrow \dagger T_\rho \times N'_\sigma.$$

But now $\underline{a}_0$ is in $S'T_\rho$, $\underline{b}_0$ in SN'_σ, and precisely $R_{\rho,\sigma} \simeq S'T_\rho \times SN'_\sigma$, we have won...

Remark. We cannot replace G by PG, i.e.

$$\mathcal{M}_{\rho,\sigma} \longrightarrow PG \times T_\rho \times N'_\sigma$$

(where we can still add $G_m \times \mu_2 \times \mu_2$ if we want!) is not a monomorphism — its kernel is precisely the subgroup μ_2 of $\widetilde{G|\mathcal{C}}^+_{\rho,\sigma} = SG$. [p. 468]

Uncertainty retained from cancelled passages.

140-3-s469-D001 κ
140-3-s469-D002 then
140-3-s469-D003 and we are
140-3-s469-D004 in the
140-3-s469-D005 T

[scan 470]

Let us consider, now that the quantities $\underline{U}, \underline{a}, \underline{b}$ (*and* $\mu, \varepsilon, \varepsilon'$) of page 529 (152) are understood, the quantities (153) of loc. cit. $\underline{u}, f, \alpha, \beta, \alpha_0, \beta_0$ (g, h ξ, η do not appear there, besides…) They are determined in terms of the $\underline{U}, \underline{a}, \underline{b}$, as soon as we know either $\underline{u}$ or f, by the formulas (154), $\underline{u}$ and f being related by the single relation

$$\underline{U} = f^{-1}\underline{u} \tag{23}$$

with in addition on f the condition

$$f \in \Gamma \underbrace{\widetilde{SG}}_{\widetilde{G|\mathcal{C}}^+_{\rho,\sigma}}, \tag{24}$$

plus a condition on u which, in loc. cit., was written

$$\underline{u} \in \Gamma T_0, \tag{25}$$

and was suggested by the calculations of §46 $\overline{\text{V}}$ (taken up again later in §48 $\overline{\text{II}}$, $\overline{\text{III}}$ etc). This condition ensures the uniqueness of $f, \underline{u}$ satisfying|satisfying (23) — that is, from the p^{t} of view of calculation, its main virtue! — but in certain contexts it is preferable to replace it by the more broad condition

$$\underline{u} \in \Gamma B'_0 \qquad \text{where} \quad \begin{matrix} B'_0 = T_0.L_0. \\ (\text{product } \tfrac{1}{2} \text{ direct}) \end{matrix} \tag{26}$$

[p. 469]

We no longer have † uniqueness for writing $\underline{U}$ in the form (23) — on the other hand we have an existence theorem in a simple form: [scan 471] [p. 538]

If we start with (α, β, f)†, sections of G such that

$$(\det \alpha)^2 = \det \beta = \det f = 1, \tag{27}$$

for this triple and as a result|consequence the quantities deduced from it:

$$\begin{cases} \alpha_0 = f^{-1}\alpha, & \beta_0 = f^{-1}\varepsilon\beta \\ \xi = \alpha\rho(\alpha)^{-1}, & \eta' = \beta\sigma(\beta)^{-1} \\ g = \alpha_0\rho(\alpha_0)^{-1}, & h = \beta_0\sigma(\beta_0)^{-1} \end{cases} \tag{28}$$

to come from a system $(x, \dagger\underline{u})$, with $x \in \Gamma\mathcal{M}_{\rho,\sigma}$, $\dagger\underline{u} \in \Gamma\dagger B'_0$ such that

$$f \overset{\text{dfn}}{=} \underline{u}\,\underline{U}^{-1}\dagger \in \Gamma\dagger SG, \quad \text{i.e.} \quad \det \underline{u} = \det \underline{U}, \tag{29}$$

by the formulas

$$\alpha = \underline{u}\,\underline{a}^{-1}, \quad \beta = \underline{u}\,\underline{b}^{-1}, \quad f = \underline{u}\,\underline{U}^{-1}, \tag{30}$$

it is necessary and sufficient that we have the relation (which does not involve f, only α, β):

$$\alpha\rho(\alpha)^{-1} = \varepsilon\beta\sigma(\beta)^{-1}\varepsilon_1(\mu - 1) \qquad \begin{matrix} \text{for : } \varepsilon \in \Gamma\mu_2 \\ \mu \in \Gamma G_m \\ \text{suitable.} \end{matrix} \tag{31}$$

(N.B. $\varepsilon|\varepsilon_1, \mu$ are uniquely determined (thus in terms of α β). Then the pair $(x, \underline{u})$ is unique. For $f = 1$, (i.e. $\underline{u} = \underline{U}$), We can say also that we have a 1-1 correspondence between the the set † sections of the|of the subgroups|subgroup $\mathcal{M}_{\rho,\sigma}[0]$ (not $\mathcal{M}_{\rho,\sigma}(0)$!) of $\mathcal{M}_{\rho,\sigma}$, and the set [p. 470]

Uncertainty retained from cancelled passages.

140-3-s471-D001 result|results
140-3-s471-D002 $\in$
140-3-s471-D006 f
140-3-s471-D007 f
140-3-s471-D008 $SG|T_0$
140-3-s471-D009 $\underline{u}|f$
140-3-s471-D010 B_0'
140-3-s471-D012 [illeg. 1]

[scan 472]

of pairs α, β satisfying the two relations

$$(\det\alpha)^2 = \det\beta = 1 \qquad \text{of (27)}$$

and the loop relation. For this, we have † no use for the group $S\mathcal{M}_{\rho,\sigma}$.

But returning to the case of arbitrary f, to interpret the algebraic sheaf on the triples (α, β, f), satisfying (27)(31), with the structure of a strange group on it via cocycle relation, (to which I shall have to return in more geometric terms…) and because the three quantities α, β, f mix intimately — there is reason to introduce $S\mathcal{M}_{\rho,\sigma}$ — which will be isomorphic to the|to scheme (in groups) of the common solutions of (27)(31).

From this point of view, $S\mathcal{M}_{\rho,\sigma}$ is derived from ρ as † scheme †

$$S\mathcal{M}_{\rho,\sigma} = \text{scheme of triples } (x, \underline{u}, f) \quad \text{with} \quad \begin{cases} x \text{ † in } \mathcal{M}_{\rho,\sigma}, \\ \text{†}\underline{u} \text{ in } B_0' \\ f \text{ in } SG \end{cases} \quad \begin{cases} \underline{U} = f^{-1}\underline{u} \\ \| \text{ definition} \\ \Theta_G(x) \end{cases} \tag{32}$$

Thus

$$(33) \quad S\mathcal{M}_{\rho,\sigma} \hookrightarrow \mathcal{M}_{\rho,\sigma} \times B_0' \quad \leftarrow$$

$$\wr|$$

$$\left\{ x, \underline{u} \;\middle|\; \chi|\gamma(x) = \underbrace{\delta_G(\underline{u})}_{\begin{matrix}\| \text{ dfn} \\ \det \underline{u}\end{matrix}} \right\}$$

or again, more simply [p. 471]

Uncertainty retained from cancelled passages.

140-3-s472-D001 besides

140-3-s472-D002 described by triples

140-3-s472-D003 of

140-3-s472-D004 $\in$

140-3-s472-D005 f

[scan 473]

[p. 539]

$$\left\{\begin{array}{l} S\mathcal{M}_{\rho,\sigma} \xrightarrow{\sim} \mathcal{M}_{\rho,\sigma} \underset{\mathbb{G}_m}{\times} B'_0 \\ \\ \text{where} \quad \left\{\begin{array}{l} \mathcal{M}_{\rho,\sigma} \xrightarrow{\chi} \mathbb{G}_m \\ \\ B'_0 \xrightarrow{\delta_{B'_0}=\delta_G|B'_0} \mathbb{G}_m \end{array}\right. \end{array}\right. \quad , \quad \text{Whew !} \tag{34}$$

This formula will serve us as the definition of $S\mathcal{M}_{\rho,\sigma}$ from now on – it defines at once a group law on $S\mathcal{M}_{\rho,\sigma}$, and we find on $S\mathcal{M}_{\rho,\sigma}$ two extension structures

$$\begin{array}{ccccccccc} 1 & \longrightarrow & L_0 & \longrightarrow & S\mathcal{M}_{\rho,\sigma} & \longrightarrow & \mathcal{M}_{\rho,\sigma} & \to & 1\,, \\ & & \| & & & & & & \\ & & \mathrm{Ker}(B'_0 \xrightarrow{\delta_{B'_0}} \mathbb{G}_m) & & \tilde{x} = (x, \underline{u}|y) & \longmapsto & x & & \end{array} \tag{35}$$

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \mathcal{M}^+_{\rho,\sigma} & \longrightarrow & S\mathcal{M}_{\rho,\sigma} & \to & B'_0 & \to & 1 \\ & & \| & & & & & & \\ & & \mathrm{Ker}(\mathcal{M}_{\rho,\sigma} \xrightarrow{\chi} \mathbb{G}_m) & & \begin{array}{c}\tilde{x}\\ \|\\ (x,\underline{u})\end{array} & \longmapsto & \underline{u} & & \end{array} \tag{36}$$

The subgroup $S\mathcal{M}_{\rho,\sigma}[0]$ of $S\mathcal{M}_{\rho,\sigma}$ corresponds to the condition $f = 1$ i.e. $\begin{array}{c}\underline{U}\\ \|\\ \Theta_G(x)\end{array} = \underline{u}$ (which already implies $\begin{array}{c}\det \underline{U}\\ \|\\ \chi(x)\end{array} = \det \underline{u}$ – it is also the subgroup of the (x, u) in $\mathcal{M}_{\rho,\sigma} \times B'_0$ such that we have $u = \Theta_G(x)$. Thus $S\mathcal{M}_{\rho,\sigma}[0]$ is the subgroup of $S\mathcal{M}_{\rho,\sigma}$ which is the isomorphic image of $\mathcal{M}_{\rho,\sigma}[0]$ † (cf. (14|19) page 536') under $x \longmapsto (x, \Theta_G(x))$:

†

$$S\mathcal{M}_{\rho,\sigma}[0] = \mathrm{Im}\left(\begin{array}{l} \mathcal{M}_{\rho,\sigma}[0] \hookrightarrow S\mathcal{M}_{\rho,\sigma} \\ \\ \quad x \longmapsto (x, \Theta_G(x)) \end{array}\right) \tag{37}$$

$$(\text{where} \quad \mathcal{M}_{\rho,\sigma}[0] = \Theta_G^{-1}(B'_0) \quad)$$

[p. 472]

Uncertainty retained from cancelled passages.
140-3-s473-D001 under
140-3-s473-D002 (

[scan 474]

Giving a splitting of (35) amounts to a splitting of

$$1 \longrightarrow L_0 \longrightarrow B'_0 \longrightarrow \mathbb{G}_m \longrightarrow 1\,,$$

now there is one ready-made via the subgroup T_0. So we deduce from this an embedding of $\mathcal{M}_{\rho,\sigma}$ into $S\mathcal{M}_{\rho,\sigma}$, which thereby does indeed become a 1/2-direct product.

$$S\mathcal{M}_{\rho,\sigma} \simeq \mathcal{M}_{\rho,\sigma} \cdot L_0 \tag{38}$$

$\mathcal{M}_{\rho,\sigma}$ operating moreover on L_0 via the character $\chi : \mathcal{M}_{\rho,\sigma} \longrightarrow \mathbb{G}_m$ – but this description now seems to me much less "telling" than (34), for it does not bring out the canonical homomorphism

$$S\mathcal{M}_{\rho,\sigma} \longrightarrow B'_0$$

which is nevertheless the essential thing for the introduction of $S\mathcal{M}_{\rho,\sigma}$. Once this is in place, we find ("for the record") the

Theorem We have a canonical isomorphism

$$S\mathcal{M}_{\rho,\sigma} \longrightarrow S\dagger\Sigma^*_{\rho,\sigma} \qquad \begin{array}{c}\text{group}\\ \text{scheme of the}\\ \text{systems } (\alpha,\beta,f)\\ \text{satisfying (17)}\\ \text{and (31)}\end{array} \tag{39}$$

†

[p. 473]

Uncertainty retained from cancelled passages.
140-3-s474-D001 $\mathcal{M}$
140-3-s474-D002 $\underline{a}|\Theta_\rho$

[scan 475]
[p. 540]

given by :

$$(x,\underline{u}) \longmapsto (\alpha,\beta,f) \quad \text{with} \quad \alpha = \underline{ua}^{-1}, \quad \beta = \underline{ub}^{-1}, \quad f = \underline{uU}^{-1}$$
$$\text{(how beautiful it is !)} \tag{40}$$
$$\text{where} \quad \underline{a} = \Theta_\rho(x), \quad \underline{b} = \Theta_\sigma(x), \quad \underline{U} = \Theta_G(x)$$

†The operation of L_0 on $\Sigma_{\rho,\sigma}$ by

$$(\alpha,\beta,f) \longrightarrow {}''|(\ell\alpha,\ell\beta,\ell f \tag{41}$$

corresponds to the operation of left translation by the subgroup L_0 of $S\mathcal{M}_{\rho,\sigma}$ (cf. (35)), consisting of the $(1,\ell)$ with ℓ in L_0, so we recover $\mathcal{M}_{\rho,\sigma} \simeq S\mathcal{M}_{\rho,\sigma}/L_0$ as the quotient (group) scheme of $S\Sigma^*_{\rho,\sigma}$ by the action (41) of L_0.

Remark In this description of the (α, β, f) the new quantities $\underline{u}$, $\underline{a}$, $\underline{b}$, $\underline{U}$ occur, which leads to considering the morphism

$$\begin{cases} S\mathcal{M}_{\rho,\sigma} \longrightarrow B'_0 \times T_\rho \times N'_\sigma \times G \\ \\ (x, \underline{u}) \longmapsto (\underline{u}, \underline{a}, \underline{b}, \underline{U}) \end{cases} \quad \text{with} \quad \begin{array}{l} \underline{a} = \Theta_\rho(x) \\ \underline{b} = \Theta_\sigma(x) \\ \underline{U} = \Theta_G(x) \end{array} \tag{42}$$

We have in this connection :

Proposition The homomorphism (42) is a monomorphism, its image consists of the systems satisfying

$$\det \underline{u} = \det \underline{b} = \det \underline{U} \quad \text{i.e. } \mu, \quad \text{and} \quad \det \underline{a}/\mu \in \Gamma\mu_2 \tag{43}$$

This is an immediate consequence of the corollary to the proposition on page 536', and of the definition of $S\mathcal{M}_{\rho,\sigma}$ as a fiber product. [p. 474]

Uncertainty retained from cancelled passages.

140-3-s475-D001 The

[scan 476]

10°) The section-subgroups $\mathcal{M}_{\rho,\sigma}(\sigma)$, $\mathcal{M}_{\rho,\sigma}(\rho)$.

We define $\mathcal{M}_{\rho,\sigma}(\sigma)$ by the condition

$$\beta_0 = 1 \quad \text{i.e.} \quad \beta = f \quad \text{i.e. cf. (40))}$$

$$\underline{U} = \underline{b} \tag{43}$$

It is evident from the proposition on page 536' that the subgroup

$$\begin{array}{c} \mathcal{M}_{\rho,\sigma}(\sigma) \subset \mathcal{M}_{\rho,\sigma}) \\ \| \text{ df} \\ \{(\underline{a}, \underline{b}, \underline{U}) \in \Gamma\mathcal{M}_{\rho,\sigma} \mid U = \underline{b}\} \end{array} \tag{44}$$

provides a section – when $\underline{a}, \underline{b}$ is given in $\mathcal{T}_{\rho,\sigma}$, i.e. satisfying the condition

$$\delta_\rho(\underline{a})/\delta_\sigma(\underline{b}) \in \Gamma\mu_2,$$

and the simplest way of lifting it to an element $(\underline{a}, \underline{b}, \underline{U})$ of $\mathcal{M}_{\rho,\sigma}$ isn't it by $(\underline{a}, \underline{b}, \underline{U} \text{df} = \underline{b})$! (And we will have indeed $\delta_G(\underline{U}) = \delta_\sigma(\underline{b})$.). We likewise define $\mathcal{M}_{\rho,\sigma}(\rho)$ by $\alpha_0 = 1$ i.e. $\alpha = f$ i.e.

$$\begin{array}{c} \mathcal{M}_{\rho,\sigma}(\rho) \subset \mathcal{M}_{\rho,\sigma} \\ \| \\ \{(\underline{a}, \underline{b}, \underline{U}) \text{ in } \mathcal{M}_{\rho,\sigma} \mid \underline{U} = \underline{a}\} \end{array} \tag{45}$$

which gives a partial section over the subgroup $\mathcal{T}'_{\rho,\sigma}$ of $\mathcal{T}_{\rho,\sigma}$ defined as

$$\begin{array}{c} \mathcal{T}'_{\rho,\sigma} \in \mid \subset \mathcal{T}_{\rho,\sigma} \\ \| \,\mathrm{df} \\ \big((\underline{a},\underline{b}) \in T_\rho \times N'_\sigma \mid \delta_\rho(\underline{a}) = \delta_{\sigma|\rho}(\underline{b})\big) \\ \| \\ \mathrm{Ker}(\mathcal{T}_{\rho,\sigma} \xrightarrow{\epsilon_3} \mu_2) \end{array} \tag{46}$$

so

$$\mathcal{T}'_{\rho,\sigma} = T_\rho \times_{\mathbb{G}_m} N'_\sigma, \quad \text{and we have} \tag{47}$$

[p. 475]
[scan 477]
[p. 541]

so

$$\mathcal{T}'_{\rho,\sigma}/\mathcal{T}^0_{\rho,\sigma} \xrightarrow{\epsilon_2} \mu_2 \ . \tag{48}$$

Remark These subgroups can be defined, and provide a section resp. a section over $\mathcal{T}'_{\rho,\sigma}$, as soon as we have (in the general situation with 7°|ɬ, page 526' ff) embeddings

$$T_\rho \hookrightarrow G$$

,

$$N'_\sigma \hookrightarrow G$$

, such that $\delta_\rho, \delta_\sigma$ are induced by δ_G.
In the modular case, we also write

$$\mathcal{M}_{\rho,\sigma}(1/2) = \mathcal{M}_{\rho,\sigma}(\sigma), \quad \mathcal{M}_{\rho,\sigma}(-\bar{j}) = \mathcal{M}_{\rho,\sigma}(\rho), \tag{49}$$

and we define more general^{ly}|generally

$$\mathcal{M}_{\rho,\sigma}(q), \qquad q \in \{0, 1, \infty\}; 1/2, 2|2, -1\} \tag{50}$$

starting from $\mathcal{M}_{\rho,\sigma}(0)$ and $\mathcal{M}_{\rho,\sigma}(1/2)$ by

$$\mathrm{int}(\rho_G)\mathcal{M}_{\rho,\sigma}(q) = \mathcal{M}_{\rho,\sigma}(\dot{\rho}(q)) \tag{51}$$

m1139

Marginal note. m1139 ρ_G is ρ as an element of the subgroup $\mathcal{G}^+_{\rho,\sigma} \simeq SG$ of $\underset{\sim}{\mathcal{M}}_{\rho,\sigma}$

$\dot{\rho}$ is the homographic transformation

$$\dot{\rho} : \qquad q \longmapsto \frac{1}{1-q} \tag{52}$$

(whose third iterate is the identity, just like the third iterate of $\tilde{\rho}$).

I would like : to define $\mathcal{M}_{\rho,\sigma}(-j)$, and † we have a choice within a well-determined Sl|SL$(2,\mathbb{Z})^\wedge$-conjugacy class, a choice more [p. 476.]

Uncertainty retained from cancelled passages.
140-3-s477-D003 as

restricted would be between [scan 478]

$$\mathrm{int}(\sigma_i)(\mathcal{M}_{\rho,\sigma}(0)) \qquad i \in \{0,1,\infty\} \tag{53}$$

[p. 477]

[scan 479]

XII Retrospective [p. 542]

After many unnecessarily complicated detours, I ended up quite simply constructing two group schemes over $\mathbb{Z}$

$$\mathcal{M}_{\rho,\sigma} \subset \dagger T_\rho \times N'_\sigma \times G \tag{1}$$

$$S\mathcal{M}_{\rho,\sigma} \subset B'_0 \times T_\rho \times N'_\sigma \times \dagger G \tag{2}$$

by

$$\mathcal{M}_{\rho,\sigma} = \left\{ \overbrace{(\dagger\underline{a}, \dagger\underline{b}, \dagger\underline{U})}^{x} \;\middle|\; \overbrace{\det \underline{U} = \det \underline{b}}^{\chi(x)} = \overset{\varepsilon_3(x)}{\underset{|}{\varepsilon'}} \det \underline{a}, \varepsilon'^2 = 1 \right\} \tag{3}$$

$$S\mathcal{M}_{\rho,\sigma} = \left\{ (\underline{u}, \underline{a}, \underline{b}, \underline{U}) \;\middle|\; \det \underline{U} = \det \underline{u} = \det \underline{b} = \varepsilon' \det \underline{a},\ \varepsilon'^2 = 1 \right\} \tag{4}$$

with

$$\mathcal{M}_{\rho,\sigma} \simeq S\mathcal{M}_{\rho,\sigma}/L_0 \tag{5}$$

thus $S\mathcal{M}_{\rho,\sigma} = B'_0 \times_{\mathbb{G}_m} \mathcal{M}_{\rho,\sigma}$

where

$$L_0 \hookrightarrow S\mathcal{M}_{\rho,\sigma}$$

$$\lambda \longmapsto (1,1,1,\lambda)$$

and

$$S\mathcal{M}_{\rho,\sigma} \simeq \underset{\frac{1}{2}\text{-direct product}}{\mathcal{M}_{\rho,\sigma} \,.\, L_0} \tag{6}$$

where

$$\begin{cases} \mathcal{M}_{\rho,\sigma} \hookrightarrow S\mathcal{M}_{\rho,\sigma} \\ \\ x = (\underline{U}, \underline{a}, \underline{b}) \longmapsto (\underline{U}, \underline{a}, \underline{b}, i_{T_0}(\dagger\mu)\dagger) \end{cases} \qquad \text{where } \mu = \chi(x). \tag{7}$$

Here (for reference) in the modular case

$$\rho = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} \qquad \sigma = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \tag{8}$$

$$\begin{cases} T_\rho = \left\{ (c\rho + d) \;\middle|\; \underbrace{\det(c\rho+d)}_{c^2+cd+d^2} \text{ inv.} \right\} \\ T'_\sigma = \left\{ (t\sigma + u) \;\middle|\; \underbrace{\det(t\sigma+u)}_{t^2+u^2} \text{ inv.} \right\}. \end{cases} \tag{9}$$

m1141

Marginal note. m1141
$$\begin{cases} B'_0 = \left\{ \begin{pmatrix} \mu & \lambda \\ 0 & 1 \end{pmatrix} \middle| \mu \text{ inv.} \right\} \\ T_0 = \left\{ \begin{pmatrix} \mu & 0 \\ 0 & 1 \end{pmatrix} \middle| \mu \text{ inv.} \right\} \\ i_{T_0}(\mu) = \begin{pmatrix} \mu & 0 \\ 0 & 1 \end{pmatrix} \\ L_0 = \left\{ \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix} \right\} \end{cases}$$

†

$$N'_\sigma = \left\{ \underline{b} \text{ in } G \middle| \underline{b}(\sigma) = \underset{\varepsilon_\sigma(\underline{b})}{\varepsilon} \sigma,\ \varepsilon^2 = 1 \right\}, \tag{10}$$

whence exact sequence

$$1 \longrightarrow T_\sigma \longrightarrow N'_\sigma \xrightarrow{\varepsilon_\sigma} \mu_2 \longrightarrow 1\ , \tag{11}$$

[p. 478]

Uncertainty retained from cancelled passages.
140-3-s479-D001 Gl(2)
140-3-s479-D002 B_0
140-3-s479-D003 $\underline{U}$
140-3-s479-D004 $\underline{a}$
140-3-s479-D005 $\underline{b}$
140-3-s479-D006 det
140-3-s479-D007 $(\underline{U})$
140-3-s479-D008 10

and the three homomorphisms [scan 480]

$$\mathcal{M}_{\rho,\sigma} \xrightarrow{\Theta_G} G\ , \quad \mathcal{M}_{\rho,\sigma} \xrightarrow{\Theta_\rho} T_\rho\ , \quad \mathcal{M}_{\rho,\sigma} \xrightarrow{\Theta_\sigma} N'_\sigma \tag{12}$$

given in the definition (1)(3) are completed by

$$\mathcal{M}_{\rho,\sigma} \xrightarrow{\chi=\det\circ\Theta_G} \mathbb{G}_m\ , \quad \mathcal{M}_{\rho,\sigma} \xrightarrow{\varepsilon_2=\varepsilon_\sigma\circ\Theta_\sigma} \mu_2\ , \quad \mathcal{M}_{\rho,\sigma} \xrightarrow{\varepsilon_3(x)=\frac{\det \underline{a}}{\det \underline{b}}} \mu_2 \tag{12 bis}$$

so that we find

$$\begin{gathered} \mathcal{M}_{\rho,\sigma} \xhookrightarrow{(\Theta_G,\Theta_\rho,\Theta_\sigma,\chi,\varepsilon_2,\varepsilon_3)} G \times T_\rho \times N'_\sigma \times \mathbb{G}_m \times \mu_2 \times \mu_2 \\ x \longmapsto (\underline{U}, \underline{a}, \underline{b}, \mu, \varepsilon, \varepsilon') \end{gathered} \tag{13}$$

whose image consists of the systems satisfying

$$\begin{cases} \det \underline{U} = \det \underline{b} = \varepsilon' \det \underline{a} = \mu \\ \qquad\qquad\qquad\qquad \underline{b}(\sigma) = \varepsilon\sigma \quad \text{i.e.} \\ \varepsilon = \varepsilon_\sigma(\underline{b}) \quad \text{i.e} \quad \underbrace{\sigma(\underline{b})}_{{}^t\underline{b}^{-1}} = \varepsilon \underline{b} \quad \text{i.e} \quad \underline{b}\,{}^t\underline{b} = \varepsilon \end{cases} \tag{14}$$

Likewise we have a monomorphism

$$\begin{aligned} S\mathcal{M}_{\rho,\sigma} &\hookrightarrow B'_0 \times T_\sigma \times N'_\sigma \times \underline{\dagger G} \times \mathbb{G}_m \times \mu_2 \times \mu_2 \\ x &\longmapsto (\underline{u}, \underline{a}, \underline{b}, \underline{U}, \mu, \varepsilon, \varepsilon') \end{aligned} \tag{15}$$

whose image consists of the systems satisfying

$$\begin{cases} \mu = \det \underline{u} = \det \underline{U} = \det \underline{b} = \varepsilon' \det \underline{a} \\ \varepsilon = \varepsilon_\sigma(\underline{b}) \quad \text{i.e} \qquad \underline{b}(\sigma) = \varepsilon(\sigma) \end{cases} \tag{16}$$

Let $\dagger$ on the other hand

$$\begin{array}{c} S\Sigma^*_{\rho,\sigma} \hookrightarrow G \times \dagger G \times G \times \mathbb{G}_a \times \mu_2 \\ \| \\ \left\{ \underbrace{(\alpha, \beta, f, \mu, \varepsilon)}_{y} \;\middle|\; \begin{array}{c} (\det \alpha)^2 = \det \beta = \det f = 1 \\ \underbrace{\alpha \rho(\alpha)^{-1}}_{\xi} = \underbrace{\varepsilon \beta \sigma(\beta)^{-1}}_{\eta} \varepsilon_1(\mu - 1) \end{array} \right\} \end{array} \tag{17}$$

(where we define

$$\varepsilon_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad \varepsilon_1(\nu) = \varepsilon_1^\nu \overset{\text{dfn}}{=} \begin{pmatrix} 1 & 0 \\ \nu & 1 \end{pmatrix}.) \tag{18}$$

[p. 479]

Uncertainty retained from cancelled passages.

140-3-s480-D001 B_0

140-3-s480-D002 [illeg. 1]

140-3-s480-D003 G

in fact the projection [scan 481] [p. 543]

$$\begin{cases} S\Sigma^*_{\rho,\sigma} \hookrightarrow G \times G \times G \\ \\ y \longmapsto (\alpha, \beta, f) \end{cases} \tag{19}$$

is already a monomorphism. We let act freely

$$L_0 = \operatorname{Ker}\left(B'_0 \xrightarrow{\det} \mathbb{G}_m \right) = \dagger \left\{ \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix} \right\} \quad \text{on } S\Sigma^*_{\rho,\sigma}$$

by

$$(\alpha, \beta, f) \longmapsto (l\alpha, l\beta, lf)\ , \tag{20}$$

and we set

$$\Sigma^*_{\rho,\sigma} \overset{\text{dfn}}{=} L_0 \backslash S\Sigma^*_{\rho,\sigma}. \tag{21}$$

This being so, we have an isomorphism

$$\begin{aligned} S\mathcal{M}_{\rho,\sigma} &\xrightarrow{\sim} S\Sigma^*_{\rho,\sigma} \\ (\underline{u}, \underline{a}, \underline{b}, \underline{U}) &\longmapsto \begin{cases} (\alpha, \beta, f), \\ \alpha = \underline{u}\,\underline{a}^{-1}, \quad \beta = \underline{u}\,\underline{b}^{-1}, \quad f = \underline{u}\,\underline{U}^{-1} \end{cases} \end{aligned} \tag{22}$$

transforming the action of L_0 on $S\dagger\mathcal{M}_{\rho,\sigma}$ by translation on the left into the action of L_0 described in (20), and inducing an isomorphism

$$\mathcal{M}_{\rho,\sigma} \xrightarrow{\sim} \Sigma^*_{\rho,\sigma} \tag{23}$$

by passage to the quotient.

We have canonical homomorphisms

$$S_0\mathcal{M}^{\sim}_{0,3} \longrightarrow S\mathcal{M}_{\rho,\sigma}(\widehat{\mathbb{Z}})/\Sigma \tag{24}$$

$$\mathcal{M}^{\sim}_{0,3} \longrightarrow \mathcal{M}_{\rho,\sigma}(\widehat{\mathbb{Z}})/\Sigma \tag{25}$$

the second deduced from the first by passage to the quotient, and coinciding on $\mathcal{G}^{+\wedge}_{0,3}$ with $\dagger$ [p. 480]

Uncertainty retained from cancelled passages.

140-3-s481-D002 {

140-3-s481-D004 Σ^*

140-3-s481-D005 and|the [illeg. 2] in groups

the composite homomorphism [scan 482]

$$\mathcal{G}^{+\wedge}_{0,3} \overset{\S42\ \overline{\mathrm{IV}}}{\sim} \underset{SG(\widehat{\mathbb{Z}})/SG(\widehat{\mathbb{Z}})\cap\Sigma}{\mathrm{Sl}(2,\widehat{\mathbb{Z}})'} \hookrightarrow \dagger\mathcal{M}_{\rho,\sigma}(\widehat{\mathbb{Z}})/\Sigma \tag{26}$$

thanks to the canonical homomorphism

$$\begin{array}{ccc} SG & \hookrightarrow & \mathcal{M}_{\rho,\sigma} \\ \| & & \\ \mathrm{Sl}(2)\dagger & = & \left\{ (\underline{a},\underline{b},\underline{U}) \,\middle|\, \underline{a} \overset{\text{in } \mathcal{M}_{\rho,\sigma}}{=} \underline{b} = 1 \right\} \end{array} \tag{27}$$

We have defined$\dagger$

$$\Sigma = \underset{(\mu_2)^3(\mathbb{Z})}{(\mathbb{Z}/2\mathbb{Z})^3} \hookrightarrow \mathcal{M}_{\rho,\sigma}(\mathbb{Z}) \quad \subset \quad \mathcal{M}_{\rho,\sigma}(\widehat{\mathbb{Z}}) \tag{28}$$

via the canonical homomorphism (central)

$$\mu_2^3 \longrightarrow \dagger\mathcal{M}_{\rho,\sigma} \quad \subset \quad T_\rho \times N'_\sigma \times G \tag{29}$$

defined via the canonical composites

$$\mu_2 \hookrightarrow \mathbb{G}_m \begin{cases} \to T_\rho \\ \hookrightarrow T_\sigma \hookrightarrow N'_\sigma \\ \to G \end{cases} . \tag{30}$$

By passage to the quotient, the homomorphism (25) induces

$$\Gamma^{\sim}_{\dagger 0,3} \longrightarrow \mathcal{T}_{\rho,\sigma}(\widehat{\mathbb{Z}})/\underbrace{\{\pm 1\}\times\{\pm 1\}}_{\Sigma_0} , \tag{31|34}$$

where

$$\begin{array}{ccc} \mathcal{T}_{\rho,\sigma} & \hookrightarrow & T_\rho \times N'_\sigma \\ & & \| \\ & & \{(\underline{a},\underline{b}) \,\big|\, \det \underline{a}/\det \underline{b} \in \Gamma\mu_2\} \end{array} \tag{32}$$

The giving — rather the construction made by us — of a group homomorphism (24) [p. 481]

Uncertainty retained from cancelled passages.

140-3-s482-D001 $\mathrm{Sl}(2,\widehat{\mathbb{Z}})'$

140-3-s482-D002 $\mathbb{Z}$

140-3-s482-D003 t

140-3-s482-D004 $\mathcal{T}$

140-3-s482-D005 0

[scan 483]

amounts to the construction of the four group homomorphisms [p. 544]

$$S_0\mathcal{M}^{\sim}_{0,3} \xrightarrow[\Theta^{\pi}_{B'_0}]{\text{giving } \underline{u}} B'_0(\widehat{\mathbb{Z}}) = \widehat{\mathbb{Z}}^* \,.\, \widehat{\mathbb{Z}} \qquad (\tfrac{1}{2}\text{-direct product}) \tag{33}$$

$$\mathcal{M}^{\sim}_{0,3} \xrightarrow[\Theta^{\pi}_{G}]{\text{giving } \underline{U} \text{ (mod sign)}} G(\widehat{\mathbb{Z}})/\pm 1 = \mathrm{Gl}(2,\widehat{\mathbb{Z}})/\pm 1 \tag{34}$$

$$\Gamma^{\sim}_{0,3} \xrightarrow[\Theta^{\pi}_{\rho}]{\text{giving } \underline{a} \text{ (mod sign)}} T_\rho(\widehat{\mathbb{Z}})/\pm 1 \qquad \begin{array}{c}(\text{adelic units} \\ (\text{mod } \pm 1) \text{ of the} \\ \text{field } \mathbb{Q}(\dagger j))\end{array} \tag{35}$$

$$\Gamma^{\sim}_{0,3} \xrightarrow[\Theta^{\pi}_{\sigma}]{\text{giving } \underline{b} \text{ (mod sign)}} T_\sigma(\widehat{\mathbb{Z}})/\pm 1 \qquad \begin{array}{c}(\text{adelic units} \\ (\text{mod } \pm 1) \text{ of the} \\ \text{field } \mathbb{Q}(i))\end{array} \tag{36}$$

linked|unlinked by the relations (4), which we are going to identify. We use the subgroup

$$\begin{array}{l}\mathcal{M}^{!\sim}_{0,3}[0]^- = \text{subgroup of } \mathcal{M}^{!\sim}_{0,3} \text{ which} \\ \qquad (\text{leaves invariant the classes of the loop subgroups} \\ \qquad \text{of } \Pi^{\wedge}_{0,3}, \text{ and which}) \\ \qquad \text{normalizes } L_{0\pi} \text{ (closed subgroup} \\ \qquad \text{generated by } l_0)\end{array} \tag{37}$$

and we note that

$$1 \longrightarrow L_{0\pi} \longrightarrow \mathcal{M}^{!\sim}_{0,3}[0] \longrightarrow \Gamma^{\sim}_{0,3} \longrightarrow 1 \tag{38}$$

and that

$$\begin{cases} \mathcal{M}^{\sim}_{0,3} = \mathcal{M}^{!\sim}_{0,3}[0] \,.\, \underbrace{\mathcal{G}^{+\wedge}_{0,3}}_{\substack{\text{invariant subgroup} \\ \text{of } \mathcal{M}^{\sim}_{0,3}}} \\ \text{with} \qquad \mathcal{M}^{!\sim}_{0,3}[0] \cap \mathcal{G}^{+\wedge}_{0,3} = L_{0\pi} \end{cases} \tag{39}$$

— of course (38) is a consequence of (39).

finally

$$S_0\mathcal{M}^{\sim}_{0,3} \simeq \underbrace{\mathcal{M}^{!\sim}_{0,3}[0] \,.\, \mathcal{G}^{+\wedge}_{0,3}}_{(\frac{1}{2}\text{-direct product})} \tag{40}$$

1°) The homomorphism (33) from $S_0\mathcal{M}^{\sim}_{0,3}$ † is trivial on the factor $\mathcal{G}^{+\wedge}_{0,3}$ [p. 482]

Uncertainty retained from cancelled passages.

140-3-s483-D001 $\sqrt[3]{1}$

140-3-s483-D002 it

[scan 484]

m1142

Marginal note. m1142 so the description of (33) amounts to the description of the sole homomorphism|first homomorphism

$$\mathcal{M}^{\sim}_{0,3}[0] \longrightarrow B'_0(\widehat{\mathbb{Z}})$$

Its † value depends only on the pair (α, β) † of the triple (α, β, f) which describes an element of $\mathcal{M}^{\sim}_{0,3}$), and the † definition of $\underline{u}$ in terms of α, β — and even only in terms of the image of $\xi = \alpha\rho(\alpha)^{-1}$ in $\mathrm{Sl}(2, \widehat{\mathbb{Z}})$ † and even only in $\mathrm{Sl}(2, \widehat{\mathbb{Z}})'$! — m1143

Marginal note. m1143 For one normalizes by the condition $\xi\rho(\xi)\rho^2(\xi) = \pm 1| + 1$!

to|has been made|made essentially in §45 or at least in §46 III, — we have

$$\underline{u} := \begin{pmatrix} 1/D & B/D \\ 0 & 1 \end{pmatrix} \qquad \text{where } \xi = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \text{ (in } \mathrm{Sl}(2, \widehat{\mathbb{Z}})) \tag{41}$$

or again, in terms of

$$\underline{\alpha} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad \text{with } ad - bc = | \in \varepsilon' \in \{\pm 1\}$$

$$\begin{cases} \varepsilon' D = (c^2 + cd + d^2)\dagger \\ \varepsilon' B = ac + bd + bc \end{cases} \qquad \dagger \tag{42}$$

This does not depend on the foundations not laid for profinite groups — the "calculations in ξ, η" are in fact sufficient. It is thanks to the description of this homomorphism that one can define

$$\begin{array}{c} \mathcal{M}^{\sim}_{0,3}(0) \subset \mathcal{M}^{\sim}_{0,3}[0] \\ \| \\ \{\underline{u} \mid \underline{u} \in T_0(\widehat{\mathbb{Z}}) \quad \text{i.e. } B = 0\} \\ \text{i.e. } ac + bd + bc = 0 \end{array} \tag{43}$$

(normalization condition), [p. 483]

Uncertainty retained from cancelled passages.

140-3-s484-D002 not

140-3-s484-D003 part|taken

140-3-s484-D004 posi

140-3-s484-D005 modulo ± 1 or

140-3-s484-D006 $/\varepsilon$

140-3-s484-D007 ε'

[scan 485]

and obtain canonical decompositions into $\frac{1}{2}$-direct|direct products [p. 545]

$$\mathcal{M}^{\sim}_{0,3}[0] \simeq \mathcal{M}^{\sim}_{0,3}(0) \,.\, L_{0\pi} \tag{44}$$

$$\mathcal{M}^{\sim}_{0,3} \simeq \mathcal{M}^{\sim}_{0,3}(0)\,.\,\mathcal{G}^{+\wedge}_{0,3}. \tag{45}$$

2°) To describe (34), since it is already known on $\mathcal{G}^{+\wedge}_{0,3}$, it suffices to describe it on $\mathcal{M}^{\sim}_{0,3}[0]$ or even only on $\mathcal{M}^{\sim}_{0,3}(0)$ (subject of course to compatibilities…). And on $\mathcal{M}^{\sim}_{0,3}[0]$ it coincides with (33) — whereas on $\mathcal{M}^{\sim}_{0,3}(0)$ it is practically trivial and reduces to the multiplier μ

$$\Theta^{\pi}_{G} = \Theta^{\pi}_{B'_0} \quad \text{on } \mathcal{M}^{\sim}_{\rho|0,3}[0] \tag{46}$$

$$\begin{cases} \Theta^{\pi}_{G}(\underline{\underline{u}}) = i'_{T_0}(\mu) & \text{where } \mu \text{ is the} \\ & \text{multiplier of } \underline{\underline{u}} \\ \text{if|where} \quad \underline{\underline{u}} \in \mathcal{M}^{\sim}_{0,3}(0) & \end{cases} \tag{47}$$

(this is even a possible definition of $\mathcal{M}^{\sim}_{0,3}(0)$).

Remark As Θ^{π}_{G} was constructed "by sheer strength of wrist|wrists" by way of $\Theta^{\pi}_{B'_0}$, i.e. $\underline{U}$ was constructed from $\underline{u}$, there was for a long time a sort of [p. 484]

[scan 486]

confusion in my mind between $\underline{u}$ and $\underline{U}$, all the more so since most of the time I restricted myself to working in $\mathcal{M}^{\sim}_{0|\rho,3}[0]$ (after having fumbled about quite a bit with more varied points of view, before realizing that technically this was the most elegant way to proceed — from §46 on). This confusion was cleared up for good only yesterday and today, by dint of putting my feet in my own dishes again and again…

3°) The homomorphisms (35)(36) were obtained only by working in depth on the parameters α, β for $\underline{u}$ — here again, it suffices to work in $\mathcal{M}^{\sim}_{0,3}[0]$, and to define

$$\begin{array}{ccc} \underline{\underline{u}} & \longmapsto & \underline{a} \\ \mathcal{M}^{\sim}_{0,3}[0] & \longrightarrow & T_{\rho}(\widehat{\mathbb{Z}})/\pm 1 \\ & \longrightarrow & N'_{\sigma}(\widehat{\mathbb{Z}})/\pm 1 \\ \underline{\underline{u}} & \longmapsto & \underline{b} \end{array}$$

[p. 485]

so that it is trivial on $L_0^{\dagger\pi}$. [scan 487]

[p. 546]

The quantities $\underline{a}, \underline{b}$ appeared in the calculations gradually starting with §44 (page 366′ and following), light began to be shed on their role only starting with §48 $\overline{\text{III}}, \overline{\text{IV}}, \overline{\text{V}}$ — and a complete understanding of their nature, from the point of view of the group schemes that enter into the theory, likewise came about only in these very last days. This understanding moreover remains for the moment rather formal, and one should go deeper in several directions,

a) Make explicit the meaning from the point of view of the class fields of $\mathbb{Q}(j)$ and of $\mathbb{Q}(i)$.

b) Make explicit the meaning for the theory of "congruential|congruential" maps (i.e. subordinate|subordinate to † representations of the quotient $\mathrm{Sl}(2,\widehat{\mathbb{Z}})'$ of $\mathrm{Sl}(2,\mathbb{Z})^{\wedge\prime}$) of the results obtained, [p. 486]

Uncertainty retained from cancelled passages.

140-3-s487-D001 $\xi|\wedge$

140-3-s487-D003 continuous

[scan 488]

c) Introduce on $\mathcal{M}_{\rho,\sigma}$ the elements of structure characteristic of arithmetic Galois groups – and make the connection with the typical constructions of this sort in class field theory.

d) Examine to what extent in this context one can bring in the seemingly draconian compatibility conditions on α, β of §47 linked to compatibility with

$$\varphi : \widehat{\Pi}'_{0,3} \xrightarrow{\sim} \mathcal{G}^{+\wedge}_{0,3}$$

– I foresee. moreover that the framework of the preceding calculations is too narrow to manage to bring these additional conditions into it.

e) What interests me most for the moment is a systematization of the preceding calculations in a framework much broader than that of the group SG alone and of the standard homomorphism [p. 487]

[scan 489]
[p. 547]

$$\mathcal{G}^{+\wedge}_{0,3} \longrightarrow \dagger G(\widehat{\mathbb{Z}})/\pm 1$$

– I have the feeling of having put my finger on a systematic procedure for constructing "motivic" representations of $\Gamma_{\mathbb{Q}}$, via representations of the "geometric" part $\mathcal{G}^{+\wedge}_{0,3}$ of

$$\mathcal{M}^{+}_{0,3} \simeq \Pi_1(M_{0,3\,\mathbb{Q}}, \mathfrak{S}_3) \simeq \Pi_1(M'_{1,1\,\mathbb{Q}})$$

– or better, via representations of the discrete group $\mathcal{G}^{+}_{0,3} \simeq \mathrm{Sl}(2,\mathbb{Z})'$ (or the "variants such as $\mathrm{Sl}(2,\mathbb{Z})$, $\Pi_{0,3}$ ...) with values in the integral points of group schemes (generically reductive ??) over $\mathbb{Z}$.

f) Yet there is another direction of investigation that attracts me a great deal, not necessarily linked to the introduction of group schemes, and which could be pursued in vitro|in addition within the framework of just the "big" profinite fundamental groups – [p. 488]

Uncertainty retained from cancelled passages.
140-3-s489-D001 *S*

[scan 490]

namely the study of certain special $\mathbb{Q}$-coverings of $U_{0,3\,\mathbb{Q}} = \mathbb{P}_{0,3} - \{0,1,\infty\}$, and information one should draw from them about additional properties of the "arithmetic" outer actions of $\Pi^{\wedge}_{0,3}$ (of the type of the results obtained in §47), and † calculations of splittings of the extension $\mathcal{M}_{0,3}$ of $\Gamma_{0,3}$ by $\mathcal{G}^{+\wedge}_{0,3}$ at points other than the "critical" points $0, 1, \infty, \frac{1}{2}, 2, -1, -\bar{j}, -j$.

(Not counting the question of establishing|of securing the foundations, via the method suggested by Deligne – and of studying the loop equation in the protilp.|pro. tilp. compactification $L_A|L_\Delta$ – and of continuing to investigate in the direction of the fundamental conjectures, notably on the arithmetization of complex points – etc etc). [p. 489]

Uncertainty retained from cancelled passages.
140-3-s490-D001 on

[scan 491]

Let us return for the moment to the question of identifying the four homomorphisms obtained [p. 548]
(33) to (36), in terms of classical things. One can consider the homomorphism (33) $\Theta^{\pi}_{B'_0}$ as being of an essentially technical nature, or at least as being subordinate to the homomorphism Θ^{π}_{G} of (34). Moreover, (35) and (36) are deduced from it, by composing them with the two homomorphisms

$$\Gamma^{\sim}_{0,3} \longrightarrow \mathcal{M}^{\sim}_{0,3}$$

(morally at least, for the moment !) associated with the † points $\frac{1}{2}$ and $-\bar{j}$ of $U_{0,3\,\mathbb{Q}}(\mathbb{C})$, and which correspond, at the level of group schemes, to the two splittings ψ_ρ, ψ_σ of $\overset{(3)}{\mathcal{M}}_{\rho,\sigma}$ over $\overset{(31)}{\mathcal{T}}_{\rho,\sigma}$, given respectively by

$$\begin{cases} \psi_\rho : (\underline{a}, \underline{b}) \longmapsto (\underline{a}, \underline{b}, \underline{a}) \\ \psi_\sigma : (\underline{a}, \underline{b}) \longmapsto (\underline{a}, \underline{b}, \underline{b}) \end{cases} \tag{48}$$

m1144

Marginal note. m1144

$$\begin{aligned} \dagger\, \mathcal{T}'_{\rho,\sigma} &\longrightarrow \mathcal{M}_{\rho,\sigma} \\ \mathcal{T}_{\rho,\sigma} &\longrightarrow \mathcal{M}_{\rho,\sigma}, \end{aligned}$$

with this grain of salt that the first is defined only on the subgroup $\mathcal{T}'_{\rho,\sigma}$ defined by

$$\begin{cases} \mathcal{T}'_{\rho,\sigma} \subset \mathcal{T}_{\rho,\sigma} \\ \quad \| \quad \left\{ \left(\begin{matrix} \underline{a} \\ \cap \\ T_\rho \end{matrix}, \begin{matrix} \underline{b} \\ N'_\sigma \end{matrix} \right) \middle| \ \det \underline{a} = \det \underline{b} \right\} \\ \quad \| \quad T_\rho \underset{\mathbb{G}_m}{\times} N'_\sigma. \end{cases} \tag{49}$$

[p. 490]

Uncertainty retained from cancelled passages.

140-3-s491-D001 the

140-3-s491-D003 $\mathcal{T}_{\rho,\sigma}|\psi_\rho$

and likewise † m1145 [scan 492]

Marginal note. m1145 $\begin{cases} \Psi^{\pi}_{-\bar{\mathrm{j}}|-\mathrm{j}} : \widetilde{\Gamma}'_{\mathbb{Q}} \simeq \widetilde{\mathcal{T}}'_{0,3} \longrightarrow T_\rho(\widehat{\mathbb{Z}})/\pm 1 \\ \widetilde{\Gamma}'_{\mathbb{Q}} = \{\gamma \in \widetilde{\Gamma}_{\mathbb{Q}} \mid \varepsilon_3(\mu) = 1, \text{ where } \mu = \chi(\gamma)\} \end{cases}$

(50

We thus see that the indeterminacies of the signs for $\Theta^{\pi}_{\rho}, \Theta^{\pi}_{\sigma}$ correspond precisely to the indeterminacies for Θ_G itself — everything hangs together! The latter is removed † tautologically when we replace $\widetilde{M}_{0,3}$ by the † ad-hoc extension $\widetilde{\mathcal{T}}_{1,1}$ by ± 1, so that we find [m1147]

Marginal note. [m1147] We will note however that Θ^{π}_{ρ}, unlike the splitting $\Psi^{\pi}_{\rho} = \Psi^{\pi}_{1/2}$, is defined on $\widetilde{\mathcal{T}}_{0,3}$ in its entirety.



$$(50) \qquad \widetilde{\mathcal{T}}_{1,1} \longrightarrow \mathrm{GL}(2, \widehat{\mathbb{Z}})$$

and one should † interpret this homomorphism as the fundamental "modular" homomorphism of the theory of elliptic curves (that the inverse image of † $\Gamma_{\mathbb{Q}} \subset \tilde{\Gamma}_{\mathbb{Q}}$ in $\widetilde{\mathcal{T}}_{1,1}$ can be interpreted as $\mathcal{T}_{1,1} \simeq \pi_1(M_{1,1\,\mathbb{Q}})$ must be more or less tautological, by construction.)

It is clear in any case that the two coincide for every $\gamma \in \widetilde{M}_{0,3}$ viewed as|when elements of

$$(51) \qquad \mu_2(\widehat{\mathbb{Z}}) = \{\pm 1\}^{\mathbb{P}} \qquad (\mathbb{P} = \text{set of prime numbers})$$

[p. 491]

Uncertainty retained from cancelled passages.

140-3-s492-D001 $\Psi^{\pi}_{-\mathrm{j}}$

140-3-s492-D002 by [illeg. 2]

140-3-s492-D003 covering|profinite with|of 2 sheets

140-3-s492-D004 be able to

140-3-s492-D005 Γ

i.e. they coincide modulo a homomorphism [scan 493]

[p. 549]

$$\dagger\tilde{\Gamma}_{\mathbb{Q}} \longrightarrow \mu_2(\widehat{\mathbb{Z}}) \simeq \{\pm 1\}^{\mathbb{P}}$$

i.e. a homomorphism

$$G_{\mathrm{m}}(\widehat{\mathbb{Z}})_2 = (\widehat{\mathbb{Z}}^*)_2 \simeq \{\pm 1\}^{\mathbb{P}} \longrightarrow \dagger_2 G_{\mathrm{m}}(\widehat{\mathbb{Z}}) = \mu_2(\widehat{\mathbb{Z}}) \simeq \{\pm 1\}^{\mathbb{P}}$$

— so it is not a certain indeterminacy that is at issue! It is morally certain that the two coincide — I have not tried to prove it.

As for (35) (36), they must surely correspond to the action of $\Gamma_{\mathbb{Q}}$ on the Tate modules of elliptic curves with exceptional isomorphisms, corresponding precisely to the points $(1/2, 2, -1)$ and $(-\bar{\mathrm{j}}, -\mathrm{j})$ of $M_{0,3\,\mathbb{Q}} = M_{1,1}[2]'_{\mathbb{Q}}$. The indeterminacy of sign must|of which correspond|corresponds to the indeterminacy in lifting a point of $M_{1,1}[2]'_{\mathbb{Q}}$ to a point of $M_{1,1}[2]_{\mathbb{Q}}$, i.e. in choosing an elliptic curve (with "rigidification" of level 2) corresponding to the Legendre data $(0, 1, \infty, q|\rho)$ with $\rho|q = 1/2$ or $\rho|q = -\bar{\mathrm{j}}$. [p. 492]

Uncertainty retained from cancelled passages.

140-3-s493-D001 $M_{0,3}$

140-3-s493-D002 μ

Even once the ambiguity of sign for Θ^{π}_{G} is removed by passing to the two-sheeted extension $\widetilde{\mathcal{T}}_{1,1}$ of $\widetilde{M}_{0,3}$, the indeterminacy of the signs for $\Theta^{\pi}_{\rho}, \Theta^{\pi}_{\sigma}$ is not removed for all that, for the splittings † $\Psi^{\pi}_{1/2}, \Psi^{\pi}_{-\bar{\mathrm{j}}}$) are not for all that splittings (even partial!) of † the extension $\widetilde{\mathcal{T}}_{1,1}$, but only of $\widetilde{M}_{0,3}$ [scan 494]

— and lifting them to splittings of $\widetilde{\mathcal{T}}_{1,1}$ over $\mathcal{T}_{1,1}$, amounts, very precisely, to choosing precisely elliptic curves with Legendre rigidification over the points $1/2, -\bar{\jmath}$. Thus, calculations made in terms of automorphisms of the groups with loops $\widehat{\Pi}_{0,3}$ or of $\mathcal{C}^{+\wedge}_{0,3}$, give back (without having looked for them!) results that fit very closely with particularly crucial arithmetico-geometric truths. [p. 493]

Uncertainty retained from cancelled passages.
140-3-s494-D001 $\Psi^{\pi}_{\rho}, \Psi^{\pi}_{\sigma}$
140-3-s494-D002 this

49. Homomorphisms of $M_{0,3}$, the general groups $M_{\rho,\sigma}$. [scan 495] [p. 550]
m1149

Where we get our feet wet…

Marginal note. m1149 All of this section I seems entirely useless for what follows, it is a preliminary groping to lead into section II page 553.

I) Consider a closed subgroup G of $M_{0,3}$, containing $\mathcal{C}^{+\wedge}_{0,3}$, hence the inverse image of a closed subgroup of $\tilde{\Gamma}_{0,3}$. We assume that the latter is "big enough" — say that it contains a subgroup of finite index in $\Gamma_{\mathbb{Q}} \subset \tilde{\Gamma}_{0,3}$. For the moment, we assume for simplicity that $G \subset \operatorname{Ker} \varepsilon_3$. In summary

$$(1) \qquad \mathcal{C}^{+\wedge}_{0,3} \subset G \subset \widetilde{M}_{0,3}, \qquad u \in G \Longrightarrow \mu(u) \equiv 1 \ (3).$$

This last hypothesis thus means that for every $u \in G$, we have $\alpha_0(u) \in \mathcal{C}^{\wedge+}_{0,3}$, so we have

$$(2) \qquad G \underset{\underline{\beta}_0}{\overset{\underline{\alpha}_0}{\rightrightarrows}} \mathcal{C}^{+}_{0,3} \ ,$$

characterized by

$$(2') \qquad \begin{cases} u(\rho) = \operatorname{int}(\alpha_0)\rho & \alpha_0 = \underline{\alpha}_0(u) \\ u(\sigma) = \operatorname{int}(\beta_0)\sigma & \beta_0 = \underline{\beta}_0(u). \end{cases}$$

We consider

$$(3) \qquad \begin{array}{c} G(0) = G \cap \widetilde{M}_{0,3}(0) \\ \cap \\ G[0] = G \cap \widetilde{M}_{0,3}[0] = \operatorname{Norm}_G(L^{\mathcal{C}}_0) \end{array}$$

so that

$$(4) \qquad G = G(0).\mathcal{C}^{+\wedge}_{0,3} \qquad \text{(semidirect product)}$$
$$(5) \qquad G[0] = G(0).L^{\mathcal{C}}_0 \qquad \text{id}$$
$$(6) \qquad G[0] \cap \mathcal{C}^{+\wedge}_{0,3} = L^{\mathcal{C}}_0.$$

We have set

$$(7) \qquad L^{\mathcal{C}}_0 = \text{closed subgroup of } \mathcal{C}^{+\wedge}_{0,3} \text{ generated by } \varepsilon_0$$

and we define likewise $L^{\mathcal{C}}_1, L^{\mathcal{C}}_{\infty}$. [p. 494]

Consider a profinite group $\dagger M$, and a (continuous) surjective homomorphism [scan 496]

$$(8) \qquad \varphi : G \longrightarrow \dagger M.$$

We set

$$(9) \qquad \begin{cases} \varphi(\mathcal{C}^{+\wedge}_{0,3}) = \mathcal{C} & \text{normal subgroup of } M \\ \varphi(G(0)) = \dagger M(0) \\ \varphi(G[0]) = \dagger M[0]. \\ \varphi(L^{\mathcal{C}}_0) = L^{\dagger}_0 \end{cases}$$

†

Thus these are closed subgroups of M, satisfying

$$(10) \qquad M(0) \subset M[0] \subset \mathrm{Norm}_M(L_0)$$
$$(11) \qquad L_0^{\dagger} \subset \mathcal{C} \cap M[0].$$

By abuse of language, we still denote by $\rho, \sigma, \varepsilon_0, \varepsilon_1$ … the images of these elements in $\mathcal{C}$, which therefore satisfy there the relations known in $\mathcal{C}_{0,3}^{+\wedge}$, in particular

$$(12) \qquad \sigma^2 = \rho^3 = 1, \qquad \sigma\rho = \varepsilon_0, \quad \rho\sigma = \varepsilon_1, \qquad \varepsilon_1 = \sigma(\varepsilon_0) = \rho(\varepsilon_0).$$

The restrictions of the maps α_0, β_0 of (2,2') to $M_{\rho,\sigma}[0]$ will rather be denoted α, β, where α, β are in fact maps

$$(13) \qquad \begin{array}{c} SG \overset{\text{dfn}}{=} G \\ \text{by } L_0^{\mathcal{C}},\ \text{inverse} \\ \text{image of } G \\ \text{in } S\widetilde{M}_{0,3} \\ = \{(u, f) \in G \times \mathcal{C}_{0,3}^{+\wedge} \mid fu \in G[0]\} \end{array} \qquad \overset{\alpha}{\underset{\beta}{\rightrightarrows}} \ \mathcal{C}_{0,3}^{+\wedge}$$

of which we look at the restrictions to the subgroup ($\simeq M_{\rho,\sigma}[0]$) of SG defined by the relation $f = 1$.

Composing $\underline{\alpha}, \underline{\beta}$:

$$\dagger G[0] \rightrightarrows \mathcal{C}_{0,3}^{+\wedge}$$

with $\varphi \mid \mathcal{C}_{0,3}^{+\wedge}$, we find maps, again denoted $\underline{\alpha}, \underline{\beta}$ (where $\underline{\alpha}^M, \underline{\beta}^M$ in case of possible confusion)

$$(14) \qquad G[0] \overset{\underline{\alpha}}{\underset{\underline{\beta}}{\rightrightarrows}} \mathcal{C}$$

thus satisfying

$$(15) \qquad \begin{cases} u(\rho) = \mathrm{int}(\alpha)\rho \\ u(\sigma) = \mathrm{int}(\beta)\sigma \end{cases} \qquad \begin{array}{c} \text{for } u \in M[0] \text{ of the form } \varphi(u_0), \\ \text{and } \alpha = \underline{\alpha}(u_0),\ \beta = \underline{\beta}(u_0). \end{array}$$

[p. 495]

Uncertainty retained from cancelled passages.

140-3-s496-D001 A
140-3-s496-D002 A
140-3-s496-D003 A
140-3-s496-D004 A
140-3-s496-D005 $\mathcal{C}$
140-3-s496-D006 Thus
140-3-s496-D007 $\mathcal{C}$
140-3-s496-D009 $M[0]$

[scan 497]

A priori, α and β depend on the choice of u_0 from which u comes not only on u — but [p. 551]

$$(16) \qquad \begin{cases} \alpha\rho(\alpha)^{-1} = [u, \rho] \\ \beta\sigma(\beta)^{-1} = [u, \sigma] \end{cases}$$

do not depend on it. If we put

$$(17) \qquad T_\rho = \underline{\mathrm{Centr}}_M(\rho) \qquad T_\sigma = \underline{\mathrm{Centr}}_M(\sigma)$$

we thus see that, modulo T_ρ resp. T_σ, α, β depend only on u.

Hypothesis $\alpha = \alpha(u_0), \beta = \beta(u_0)$ depend only on $u = \varphi(u_0)$ (for $u_0 \in \dagger G[0]$).

We shall express this hypothesis in a slightly different way, by introducing

$$(18) \qquad J = \mathrm{Ker}(G[0] \longrightarrow M[0]),$$

we thus want that for $\dagger$

$$u_0' = \delta_0 u_0 \quad \text{with } u_0 \in G[0],\ \delta_0 \in J, \quad \text{we have}$$

$$\alpha(u_0') = \alpha(u_0), \qquad \beta(u_0') = \beta(u_0)$$

Write

$$\alpha_0 = \underline{\alpha}(u_0), \quad \beta_0 = \underline{\beta}(u_0), \quad \alpha_0' = \underline{\alpha}(u_0'), \quad \beta_0' = \underline{\beta}(u_0'), \quad \overline{\alpha}_0 = \underline{\alpha}(\delta_0), \quad \overline{\beta}_0 = \underline{\beta}(\delta_1|\delta_0),$$

in G, so that we shall have, by the cocycle formulas of G

$$\alpha_0' = \delta_0(\alpha_0)\overline{\alpha}_0, \qquad \beta_0' = \delta_0(\beta_0)\overline{\beta}_0,$$

whence, applying φ and noting that

$$\varphi(\delta_0(\alpha_0)) = \underbrace{\varphi(\delta_0)}_{1}(\varphi(\alpha_0)) = \varphi(\alpha_0) \overset{\text{dfn}}{=} \alpha \quad \text{and likewise}$$

$$\varphi(\delta_0(\beta_0)) = \varphi(\beta_0) \overset{\text{dfn}}{=} \beta$$

$$(19) \qquad \alpha' = \dagger\alpha\overline{\alpha}, \qquad \beta' = \beta\overline{\beta} \qquad \text{in } \mathcal{C}$$

where $\alpha, \beta, \alpha', \beta', \overline{\alpha}, \overline{\beta}$ are the images under φ of the quantities α_0 etc in $\mathcal{C}^{+\wedge}_{0,3}$. $\dagger$ Thus the hypothesis means this

$$(20) \qquad \forall\, \delta_0 \in J \overset{\text{dfn}}{=} \mathrm{Ker}(G[0] \longrightarrow M[0]), \quad \text{we have} \quad \alpha^{\Pi}(\delta_0), \beta^{\Pi}(\delta_0) \in \mathrm{Ker}\varphi$$

$\dagger$ When this hypothesis is satisfied, we thus have maps [p. 496]

Uncertainty retained from cancelled passages.

140-3-s497-D001 M

140-3-s497-D002 α

140-3-s497-D004 φ

140-3-s497-D005 with

140-3-s497-D006 when

[scan 498]

$$(21) \qquad M[0] \underset{\underline{\beta}}{\overset{\underline{\alpha}}{\rightrightarrows}} \mathcal{C}$$

factoring (14), and thus satisfying

$$(22) \qquad \begin{cases} u(\rho) = \mathrm{int}(\alpha)\rho \\ u(\sigma) = \mathrm{int}(\beta)\sigma \end{cases} \qquad \text{for } u \in M[0], \quad \alpha = \underline{\alpha}(u), \quad \beta = \underline{\beta}(u),$$

which † paraphrase (2').

Moreover, let us put

$$(23) \qquad \zeta|\xi = \alpha\rho(\alpha)^{-1}, \qquad \eta = \beta\sigma(\beta)^{-1}$$

the relations (22) can be written thus

$$(24) \qquad u(\rho) = \zeta|\xi\rho, \qquad u(\sigma) = \eta\sigma$$

and we shall have

$$(25) \qquad \zeta|\xi = [u, \rho], \qquad \eta = [u, \sigma]$$

and moreover

$$(26) \qquad \zeta|\xi = \eta\varepsilon_1|\ell_1{}^{\nu} \qquad \text{where } \nu = \frac{\mu - 1}{2} \in \widehat{\mathbb{Z}} \quad (\mu \text{ the multiplier of } u).$$

which comes from the single relation in $\mathcal{C}^{+}_{0,3}$.

(We assume to fix ideas that the homomorphism

$$\dagger \qquad G \xrightarrow{\chi^G} \widehat{\mathbb{Z}}^* \qquad (\text{"multiplier"})$$

factors through

$$(27) \qquad G \longrightarrow M \xrightarrow{\chi^M} \widehat{\mathbb{Z}}^*,$$

so that we have a notion of multiplier ($\in \widehat{\mathbb{Z}}^*$) for the $u \in M$.)

Consider the natural homomorphism

$$(28) \qquad \begin{aligned} M[0] &\longrightarrow \mathrm{Aut}(\mathcal{C}) : \\ u &\longmapsto \mathrm{int}(u)|\mathcal{C} \end{aligned}$$

we see from (22) even|or even (24) that it is known when we know [p. 497]

Uncertainty retained from cancelled passages.

140-3-s498-D002 generalize|generalizing

140-3-s498-D003 (27)

[scan 499]

[p. 552]

the functions $\underline{\alpha}, \underline{\beta} : M[0] \longrightarrow \mathcal{C}$, and even when we know only $\zeta|\xi, \eta : M[0] \longrightarrow \mathcal{C}$. If moreover we put

for † ρ, σ generate topologically $\mathcal{C}$

$$(29) \qquad M'[0] = \mathrm{Im}(M[0] \longrightarrow \mathrm{Aut}(\mathcal{C}))$$

these latter functions $\zeta|\xi, \eta$ are already defined on $M'[0]$ — whereas it is not clear that the same holds for α, β, when $M[0]$ does not act|does not operate faithfully on $\mathcal{C}$. We thus have a map $(\zeta|\xi, \eta)$ from a certain subgroup $M'[0] \subset \mathrm{Norm}_{\mathrm{Aut}(\mathcal{C})}(L_0)$ to the set $\mathcal{R}$ of solutions of the equation in $\zeta|\xi, \eta$ (26) which we write|shall write simply

$$(30) \qquad \eta^{-1}\zeta|\xi \in L_1$$

a relation which implies (26) (with the precise form of the exponent ν) if we assume (for example) $\mathcal{C}|M$ big enough that $\mathcal{C}^{+\wedge}_{0,3} \xrightarrow[2]{} \mathrm{SL}|\mathrm{PSL}(2, \widehat{\mathbb{Z}})$ factors through $\mathcal{C}$ — which we shall assume, to make life easier for ourselves.

† More precisely, let †

$$(32)\qquad \left\{ \begin{array}{l} \dagger = \Sigma_{\mathcal{C},\rho,\sigma} \subset \mathcal{C}\times\mathcal{C} \\ \qquad\qquad \| \\ \left\{ (\zeta|\xi,\eta) \;\middle|\; \begin{array}{ll} \text{a)} & \zeta|\xi\rho \text{ conjugate to } \rho \quad \dagger\text{in } \mathcal{C} \\ \text{b)} & \eta\sigma \quad \text{———————} \quad \sigma \quad \text{———————} \\ \text{c)} & \eta^{-1}\zeta|\xi \in L_1 \quad \text{i.e.} \quad \exists \nu' \in \widehat{\mathbb{Z}} \text{ such that } \zeta|\xi = \eta\varepsilon_1^{\nu'} \end{array} \right. \end{array} \right\}$$

m1151

Marginal note. m1151

$$\underbrace{[\alpha,\rho] = \alpha\rho(\alpha)^{-1}}$$
$$\left\{ \begin{array}{l} \text{i.e. } \zeta|\xi \text{ of the form } [\alpha,\rho] \\ \text{resp. } \eta \text{ of the form } [\beta,\sigma] = \beta\sigma(\beta)^{-1} \end{array} \right.$$

† [p. 498]

Uncertainty retained from cancelled passages.
140-3-s499-D001 [illeg. 1]
140-3-s499-D002 $\mathcal{C}$
140-3-s499-D003 H)|II)
140-3-s499-D004 $\Sigma|\mathcal{R}$
140-3-s499-D005 $\mathcal{C}$
140-3-s499-D006 every $\mathcal{C}$ fixing must such|such $\zeta|\xi$ $\zeta|\xi$ satisfy|satisfies and

[scan 500]

and let $u \in \mathrm{Aut}(\mathcal{C})$ be an automorphism of $\mathcal{C}$, and let us put

$$(34)\qquad \left\{ \begin{array}{ll} \zeta|\xi = [u,\rho] = u\rho u^{-1}\rho^{-1} = u(\rho)\rho^{-1} & \text{i.e. } u(\rho) = \zeta|\xi\rho \\ \eta = [u,\sigma] = u(\sigma)\sigma^{-1} & \text{i.e. } u(\sigma) = \eta\sigma \end{array} \right.$$

then to say that $\zeta|\xi\rho$ $(= u(\rho))$ is conjugate to ρ means that u fixes the conjugacy class of ρ, to say that $\eta\sigma$ $(= u(\sigma))$ is conjugate of|to σ means that u fixes the conjugacy class of σ. Finally the condition $\eta^{-1}\zeta|\xi \in L_1$ can also be written, since cf. (24),(25)

$$\eta^{-1}\zeta|\xi = (\sigma u(\sigma^{-1}))(u(\rho)\rho^{-1}) = \sigma u(\sigma^{-1}\rho)\rho^{-1}$$
$$= \sigma\left(u(\underbrace{\sigma^{-1}\rho}_{\varepsilon_0})\dagger \underbrace{\rho^{-1}\sigma}_{\varepsilon_0^{-1}} \right)$$

† N.B. $\sigma^{-1} = \sigma$

$$= \sigma(u(\varepsilon_0)\varepsilon_0^{-1})$$

in the form

$$\sigma(u(\varepsilon_0)\varepsilon_0^{-1}) \in L_1 \qquad \text{i.e.} \qquad u(\varepsilon_0)\varepsilon_0^{-1} \in L_0$$

(since $L_0 = \sigma^{-1}(L_1)$) or again (since $\varepsilon_0 \in L_0$)

$$u(\varepsilon_0) \in L_0, \quad \text{i.e. (since } \varepsilon_0 \text{ generates } L_0)$$

m1154

Marginal note. m1154 In this argument, we have used only the relation $\sigma^{-1}\rho = \varepsilon_0$ and the fact that ε_0 generates L_0, and not that $\sigma^2 = 1$ or $\rho^3 = 1$ for example.

$$u(L_0) = L_0$$

i.e. that u normalizes L_0.

Thus let

$$(35)\qquad \left\{ \begin{array}{c} \widetilde{B}'_0 \subset \mathrm{Aut}(\mathcal{C}) \\ \| \text{ dfn} \\ \left\{ u \in \mathrm{Aut}(\mathcal{C}) \;\middle|\; \begin{array}{l} u \text{ fixes conjugacy class of } \rho \text{ and that of } \sigma \\ u \text{ normalizes } L_0 \end{array} \right\} \end{array} \right.,$$

so that we have a map (injective because ρ, σ generate $\mathcal{C}$)

$$(36)\qquad \begin{array}{ccc} \widetilde{B}'_0 & \hookrightarrow & \Sigma \\ u & \longmapsto & ([u,\rho] = u(\rho)\rho^{-1},\ [u,\sigma] = u(\sigma)\sigma^{-1}). \end{array}$$

[p. 499]

Uncertainty retained from cancelled passages.

140-3-s500-D001 σ

140-3-s500-D002 whence

[scan 501]

[p. 553]

$$(37)\qquad \left\| \begin{array}{l} \text{Fundamental hypothesis on } \mathcal{C},\ \text{equipped with } \rho, \sigma,\ \text{whence } \varepsilon_0 = \sigma^{-1}\rho, \\ \varepsilon_1 = \rho\sigma^{-1} = \sigma(\varepsilon_0) = \rho(\varepsilon_0) : \\ \text{a) } \mathcal{C} \text{ is generated topologically by } \rho, \sigma \\ \text{b) The map (36) } \dagger \text{ (injective by a)) is} \\ \quad \text{bijective.} \end{array} \right.$$

$\dagger$

Condition b) thus means that for $(\alpha, \beta) \in \mathcal{C} \times \mathcal{C}$, satisfying a relation

$$\alpha\rho(\alpha)^{-1} = \beta\sigma(\beta)^{-1}\varepsilon_1^{\nu'}, \qquad \nu''|\nu' \text{ suitable},$$

there exists $u \in \mathrm{Aut}(\mathcal{C})$ such that we have

$$\begin{array}{lll} [u,\rho] = \alpha\rho(\alpha)^{-1} & \text{i.e. } u(\rho) = \mathrm{int}(\alpha)(\rho) & \text{i.e. } u(\rho)|u\rho) = \zeta|\xi\rho\dagger \\ [u,\sigma] = \beta\sigma(\beta)^{-1} & \text{i.e. } u(\sigma) = \mathrm{int}(\beta)\sigma & \text{i.e. } u(\sigma) = \eta\sigma \end{array}$$

(we shall thus necessarily have $u \in B'_0$ }, and u will be unique if we have a)).

Thus, on the set of the|of $R|\mathcal{R}_{\rho,\sigma}$ of the $(\zeta|\xi, \eta)$ satisfying a) b) c), we shall have a group structure, +|and deduced|described by transport of structure from that of $\widetilde{B}'_{\dagger 0}$

etc.

[p. 500]

Uncertainty retained from cancelled passages.

140-3-s501-D002 is|by

140-3-s501-D003 $\zeta|\xi$ is|u

140-3-s501-D005 $\mathrm{int}(\zeta|\xi)\rho$

140-3-s501-D006 ρ

[scan 502]

The groups $M_{\rho,\sigma}$ etc

II Let us return to the situation above, forgetting (at least for the moment) G, φ etc.
m1157

Marginal note. m1157 For a systematic rectification of the notations that follow|following cf. page 564, remark.

Let $\mathcal{C}$ be a group — we place ourselves, for what follows, either in the category of discrete groups, or in that of profinite groups, as we choose. We shall (to fix ideas) take that of profinite groups. We suppose given

$$(1) \qquad \rho, \sigma \in \mathcal{C}$$

m1158

Marginal note. m1158 [i.e. we have between $\varepsilon_0 = \ell'_0$, $\sigma = \ell'_1$ and $\rho^{-1} = \ell'_\infty$ with relation (2) $\ell'_\infty \ell'_1 \ell'_0 = 1$]

we put

$$\varepsilon_0 = \sigma^{-1}\rho, \qquad \varepsilon_1 = \rho\sigma^{-1} = \sigma(\varepsilon_0) = \rho(\varepsilon_0)$$

$$(3) \qquad \begin{aligned} L_0 &= \text{closed subgroup generated by } \varepsilon_0 \\ L_1 &= \underline{\qquad\qquad\qquad\qquad\qquad\qquad} \ \varepsilon_1 \end{aligned} \quad , \text{ i.}$$

thus $L_1 = \rho(L_0) = \sigma(L_0)$.

We consider

$$\mathrm{Aut}(\mathcal{C}) \xrightarrow[\dagger]{\Phi=(\underline{\zeta|\xi},\underline{\eta})} \dagger\,\mathcal{C} \times \mathcal{C}$$

(4) defined by

$$(5) \qquad \begin{cases} \underline{\zeta|\xi}(u) = [u,\rho] = u\rho u^{-1}\rho^{-1} = u(\rho)\rho^{-1} & \text{thus } u(\rho) = \zeta|\xi\rho \quad (\zeta|\xi = \underline{\zeta|\xi}(u)) \\ \qquad\quad = u\rho(u)^{-1} & \\ \underline{\eta}(u) = [u,\sigma] = u\sigma u^{-1}\sigma^{-1} = u(\sigma)\sigma^{-1} & \text{thus } u(\sigma) = \eta\sigma \quad (\eta = \underline{\eta}(u)) \\ \qquad\quad = u\sigma(u)^{-1}. & \end{cases}$$

Thus for $u \in \mathrm{Aut}(\mathcal{C})$, and $\zeta|\xi = \underline{\zeta|\xi}(u)$, $\eta = \underline{\eta}(u)$, we have

$$(6)|(5) \qquad \begin{cases} u \text{ fixes the } \mathcal{C}\text{-conjugation class of } \rho & \iff \zeta|\xi\rho \text{ conjugate to } \rho \\ & \iff \zeta|\xi = \alpha\rho(\alpha)^{-1} \quad (\alpha \in \mathcal{C}) \\ u \text{ fixes the } \mathcal{C}\text{-conjugation class of } \sigma & \iff \eta\sigma \text{ conjugate to } \sigma \\ & \iff \eta = \beta\sigma(\beta)^{-1} \quad (\beta \in \mathcal{C}) \\ u \text{ normalizes } L_0 & \iff \eta^{-1}\zeta|\xi \in L_1 \iff \zeta|\xi = \eta\varepsilon_1^{\nu} \text{ with } \nu \in \widehat{\mathbb{Z}} \text{ suitable.} \end{cases}$$

[p. 501]

Uncertainty retained from cancelled passages.

140-3-s502-D002 φ

140-3-s502-D003 $\mathcal{C}$

Let|Let †

[scan 503]
[p. 554]

$$(7) \qquad \begin{aligned} &B_{\rho,\sigma} \subset \mathrm{Aut}(\mathcal{C}) \\ &\{u \mid \text{satisfying the three conditions } (6|5)\} \end{aligned}$$

$$\Sigma_{\rho,\sigma} \subset \mathcal{C} \times \mathcal{C}$$
$$\| \tag{8}$$
$$\left\{ (\zeta|\xi, \eta) \;\middle|\; \begin{array}{c} \zeta|\xi = \alpha\rho(\alpha)^{-1}, \quad \eta = \beta\sigma(\beta)^{-1} \quad (\alpha, \beta \in \mathcal{C}) \\ \text{i.e. } \zeta|\xi\rho \text{ conjugate to } \rho, \qquad \text{i.e. } \eta\sigma \text{ conjugate to } \sigma \\ \eta^{-1}\zeta|\xi \in L_1 \quad \text{i.e. } \zeta|\xi = \eta\varepsilon_1^{\nu} \quad (\nu \in \widehat{\mathbb{Z}}) \end{array} \right\},$$

so that $B_{\rho,\sigma}$ is the inverse image of $\Sigma_{\rho,\sigma}$ under (4), and we have an induced map

$$B_{\rho,\sigma} \xrightarrow{\varphi=\varphi_{\rho,\sigma}} \Sigma_{\rho,\sigma}. \tag{9}$$

We suppose

$$\boxed{\begin{cases} \text{a)} & \rho, \sigma \text{ generate } \mathcal{C} \text{ topologically} \\ \text{b)} & \text{the map (9) (injective by a))} \\ & \text{is surjective (thus bijective).} \end{cases}} \tag{10}$$

m1159

Marginal note. m1159 Careful, this condition is † satisfied for $\mathcal{C} = \mathrm{SL}(2, \widehat{\mathbb{Z}})'$! (it imposes † $\varepsilon = \varepsilon_3(\mu)$ on the group γ on writing $\zeta|\xi, \eta$...)

We then transport by φ the group structure of $B_{\rho,\sigma}$ to $\Sigma_{\rho,\sigma}$. We put †

$$u_{\zeta|\xi,\eta} = \varphi^{-1}(\zeta|\xi, \eta) \in B_{\rho,\sigma} \qquad ((\zeta|\xi, \eta) \in \Sigma_{\rho,\sigma}), \tag{11}$$

so that † $u_{\zeta|\xi,\eta}$ is characterized by

$$\begin{cases} u_{\zeta|\xi,\eta}(\rho) = \zeta|\xi\rho \\ u_{\zeta|\xi,\eta}(\sigma) = \eta\sigma \end{cases} \tag{12}$$

Hypothesis b) therefore means that, as soon as $(\zeta|\xi, \eta) \in \Sigma_{\rho,\sigma}$ i.e. $\zeta|\xi\rho$ conjugate to ρ, $\eta\sigma$ conjugate to σ and $\eta^{-1}\zeta|\xi \in L_1$, there exists [p. 502]

Uncertainty retained from cancelled passages.
140-3-s503-D001 $B_{\rho,\sigma}$
140-3-s503-D002 not
140-3-s503-D003 a condition of possibility
140-3-s503-D005 $\zeta|\xi$ $\zeta|\xi$
140-3-s503-D006 we shall have

an automorphism $u_{\zeta|\xi,\eta}$ of $\mathcal{C}$ satisfying (12) (and which will be unique by (10) a)). The group law on $\Sigma_{\rho,\sigma}$ is made explicit as follows: [scan 504]

$$(\zeta|\xi', \eta')(\zeta|\xi, \eta) = (\zeta|\xi'', \eta'') \quad \text{with} \quad \begin{cases} \zeta|\xi'' = u_{\zeta|\xi',\eta'}(\zeta|\xi)\zeta|\xi' \\ \eta'' = u_{\zeta|\xi',\eta'}(\eta)\eta'. \end{cases} \tag{13}$$

Let us consider the homomorphism †

$$\mathcal{C} \longrightarrow \mathrm{Aut}(\mathcal{C})$$

$$g \longmapsto \mathrm{int}(g)$$

and the inverse image of $B_{\rho,\sigma}$ is none other than the normalizer of L_0 in $\mathcal{C}$, i.e. N_0. We then want to construct a group $G_{\rho,\sigma}$, containing $\mathcal{C}$ as an invariant subgroup and into which † $B_{\rho,\sigma}$ maps, so that the tautological action of $B_{\rho,\sigma}$ on $\mathcal{C}$ is deduced from this homomorphism, and $G_{\rho,\sigma}$ is generated by $\mathcal{C}$ and the image of $B_{\rho,\sigma}$, denoted $B'_{\rho,\sigma}$, which will contain † N_0.

†If the center of $\mathcal{C}$ is trivial i.e.

$$\mathcal{C} \hookrightarrow \mathrm{Aut}(\mathcal{C})$$

($\mathcal{C}$ normal subgroup), †

$$\longrightarrow \dagger$$

$$\longmapsto$$

it suffices to take the subgroup $G_{\rho,\sigma}$ of $\mathrm{Aut}(\mathcal{C})$ generated by $\mathcal{C}$ and $B_{\rho,\sigma}$.

† But the hypothesis that the center of $\mathcal{C}$ be|is trivial is not a happy one — for example it is not satisfied if $\mathcal{C} = \mathrm{SL}(2,\widehat{\mathbb{Z}})/\pm 1$. It seems to me that the introduction of N_0 (over which in practice we have no control) [p. 503]

Uncertainty retained from cancelled passages.

140-3-s504-D002 Let us introduce
140-3-s504-D004 $B_{\rho,\sigma}$
140-3-s504-D005 thus
140-3-s504-D006 To|To
140-3-s504-D007 $\mathcal{C}$
140-3-s504-D009 $\mathcal{C}$
140-3-s504-D012 is|has commutative but

was not judicious either, and the condition $B'_{\rho,\sigma} \supset N_0$ — it suffices to have $B'_{\rho,\sigma} \supset L_0$. I shall therefore consider [scan 505] [p. 555]

$$(14) \quad \Big\{ S_0 G_{\rho,\sigma} \overset{\mathrm{dfn}}{=} B_{\rho,\sigma} \cdot \mathcal{C} \qquad \text{the product } \tfrac{1}{2} \text{ direct, and}$$

$$L_0 \overset{i}{\hookrightarrow} \dagger\, S_0 G_{\rho,\sigma}$$

$$g \longmapsto \mathrm{int}(g^{\mathcal{C}}) \cdot g^{-1}$$

the image of this homomorphism is a normal subgroup, for a) it commutes with $\mathcal{C}$ (trivial) and b) it is invariant under $B_{\rho,\sigma}$, for we shall have for $u \in B_{\rho,\sigma}$

$$\mathrm{int}(u)(i(g)) = \mathrm{int}(u)(\mathrm{int}(g) \cdot g^{-1}) = \mathrm{int}(u(g)) \cdot \underbrace{\mathrm{int}(u)(g^{-1})}_{u(g^{-1}) = u(g)^{-1}} = i(u(g))$$

(i.e. i commutes with the actions of $B_{\rho,\sigma}$, operating on L_0 by restriction to L_0 of the tautological|total operation of $B_{\rho,\sigma}$ on $\mathcal{C}$, and on $\dagger S_0 G_{\rho,\sigma} \supset B_{\rho,\sigma}$ by inner automorphisms).

We can therefore consider

$$(15) \qquad G_{\rho,\sigma} \overset{\text{dfn}}{=} S_0 G_{\rho,\sigma}/i(L_0)$$

then the inclusion homomorphisms

$$B_{\rho,\sigma} \hookrightarrow S_0 G_{\rho,\sigma}$$

,

$$\mathcal{C} \longrightarrow S_0 G_{\rho,\sigma}$$

give, by composition with

$$S_0 G_{\rho,\sigma} \longrightarrow G_{\rho,\sigma}$$

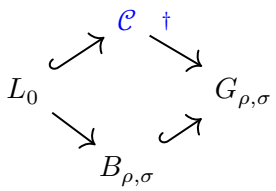
, homomorphisms

$$\dagger\mathcal{C} \overset{\dagger}{\longrightarrow} G_{\rho,\sigma}$$

,

$$B_{\rho,\sigma} \hookrightarrow G_{\rho,\sigma}$$

(16) making the diagram commutative



(17) We note

$$(18) \qquad \mathrm{Ker}(\mathcal{C} \longrightarrow G_{\rho,\sigma}) = L_0 \cap \mathrm{Centr}(\mathcal{C}) = \mathrm{Ker}(L_0 \longrightarrow B_{\rho,\sigma})$$

[p. 504]

Uncertainty retained from cancelled passages.

140-3-s505-D002 $\mathcal{C}$

140-3-s505-D003 $\mathcal{C}$

140-3-s505-D004 $B_{\rho,\sigma}$

140-3-s505-D005 ‹

140-3-s505-D006 ‹

In the case $\mathcal{C} = \mathrm{SL}(2, \widehat{\mathbb{Z}})'$, (with ρ, σ as usual) we shall have $L_0 \cap \mathrm{Centr}(\mathcal{C}) = \{1\}$. Thus it does not seem a priori that there are any major drawbacks to supposing that [scan 506]

$$(18) \qquad L_0 \cap \mathrm{Centr}(\mathcal{C}) = \{1\} \quad \text{i.e.}$$

$$\mathcal{C} \hookrightarrow G_{\rho,\sigma} \quad \text{injective}$$

Let us note that † the image of

$$L_0 \longrightarrow B_{\rho,\sigma}$$

$$\ell \longmapsto \mathrm{int}(\ell)$$

(composite of (14) and the projection $S_0 G_{\rho,\sigma} \longrightarrow B_{\rho,\sigma}$) is obviously invariant|invariant, and putting

$$(19) \qquad \Gamma^*_{\rho,\sigma}|\gamma_{\rho,\sigma} = B_{\rho,\sigma}/\operatorname{Im} L_0$$

we find

$$(20) \qquad G_{\rho,\sigma}/\mathcal{C}_{\rho,\sigma} \simeq B_{\rho,\sigma}/L_0 \simeq \Gamma|\gamma_{\rho,\sigma}.$$

m1160

Marginal note. m1160 i.e. homomorphism of exact sequences

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \dagger L_0 & \longrightarrow & B_{\rho,\sigma} & \to & \Gamma|\gamma_{\rho,\sigma} & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \| & & \\ 1 & \longrightarrow & \mathcal{C} & \longrightarrow & G_{\rho,\sigma} & \to & \Gamma|\gamma_{\rho,\sigma} & \longrightarrow & 1 \\ & & \| & & & & & & \\ & & SG_{\rho,\sigma} & & & & & & \end{array}$$

(20bis)

(N.B. In the case $\mathcal{C} = \mathrm{SL}(2,\widehat{\mathbb{Z}})'$; † etc.|and, $B_{\rho,\sigma} \subset \mathrm{GP}(\widehat{\mathbb{Z}})$ is the group previously denoted B'_0 (or rather its isomorphic image $\widetilde{B}'_0$), and the homomorphism dét | $B'_0 = B_{\rho,\sigma}$ (trivial on L_0) induces by passage to the quotient an isomorphism

$$\Gamma|\gamma_{\rho,\sigma} \xrightarrow{\sim} G_{\mathrm{m}}$$

(21) m1161

Marginal note. m1161 In the case where we start from $\mathrm{SL}(2,\widehat{\mathbb{Z}}), \rho, \sigma$, we end up with $G = \mathrm{GL}(2,\widehat{\mathbb{Z}})$…

and we see, using the isomorphism

$$B'_0 = \tilde{T}_0|\tilde{\Gamma}_0 \cdot L_0 \quad (\tfrac{1}{2} \text{ direct})$$

where $\tilde{T}_0|\overline{T}_0 \xrightarrow{\sim} B'_0/L_0 \simeq G_{\mathrm{m}}$

$$\text{i.e.} \qquad B_{\rho,\sigma} \simeq \Gamma|\gamma_{\rho,\sigma} \cdot L_0,$$

that in this case we have [p. 505]

$$(22) \qquad G_{\rho,\sigma} = \dagger \mathrm{GL}(2,\widehat{\mathbb{Z}})'.$$

[p. 505]

Uncertainty retained from cancelled passages.

140-3-s506-D001 is going to be|is its

140-3-s506-D002 $\mathcal{C}_{\rho,\sigma}$

140-3-s506-D003 $B_{\rho,\sigma}$

140-3-s506-D006 GL

[scan 507]

[p. 556]

But in the more general cases to which we aspire, one must not at all expect $\Gamma_{\rho,\sigma}$ to be so small — for example that it be commutative. For example in the "universal" case, $\mathcal{C} = \mathcal{C}^{+\wedge}_{0,3}$, we will have

$$B_{\rho,\sigma} = \widetilde{M}_{0,3}[0]' = \mathrm{Ker}(\widetilde{M}_{0,3}[0] \xrightarrow{\varepsilon_3} \{\pm 1\}) \tag{23}$$

hence

$$\Gamma_{\rho,\sigma} \simeq \tilde{\Gamma}'_{0,3} \quad (\supset \Gamma'_{\mathbb{Q}}\ !). \tag{24}$$

On the other hand, we should be able without inconvenience to postulate the existence of a canonical splitting of the extension $B_{\rho,\sigma}$ of $\Gamma_{\rho,\sigma}$ by L_0. In fact, if we assume, † † in the general case, where nevertheless we assume $\sigma^2 = \rho^3 = 1$, i.e. $\mathcal{C}$ a quotient of $\mathcal{C}^{+\wedge}_{0,3}$, that the quotient $\mathcal{C}$ of $\mathcal{C}^{+\wedge}_{0,3}$ is large enough to map to GP|CP($\widehat{\mathbb{Z}}$), then † $B_{\rho,\sigma} \simeq \Sigma_{\rho,\sigma}$ maps to $\widetilde{B}'_{0,3}|\widetilde{B}'_0$, and the inverse image of $\tilde{T}_0$ is a subgroup (which we could denote $T_{\rho,\sigma}$)

$$T_{\rho,\sigma} \subset B_{\rho,\sigma} \qquad T_{\rho,\sigma} \xrightarrow{\sim} \Gamma_{\rho,\sigma} \tag{25}$$

which is a section subgroup, for $B_{\rho,\sigma} \to \Gamma_{\rho,\sigma}$, and also (as a subgroup of $G_{\rho,\sigma}$) for $G_{\rho,\sigma} \to \Gamma_{\rho,\sigma}$, so that we will have

$$B_{\rho,\sigma} \simeq T_{\rho,\sigma}.L_0, \quad G_{\rho,\sigma} \simeq T_{\rho,\sigma}.\mathcal{C} \qquad (\tfrac{1}{2}\text{-direct products}) \tag{26}$$

[p. 506]

Uncertainty retained from cancelled passages.

140-3-s507-D001 $\mathcal{C}$ arbitrary

140-3-s507-D002 that [illeg. 1] is

140-3-s507-D003 that the

[scan 508]

Now let

$$\left\{\begin{array}{c} S_0 g^{\dagger}_{\rho,\sigma}\dagger \subset \mathcal{C}\times\mathcal{C} \\ \| \text{ dfn} \\ \left\{(\alpha,\beta) \;\middle|\; \underbrace{\alpha\rho(\alpha)^{-1}}_{\zeta|\xi} = \underbrace{\beta\sigma(\beta)^{-1}}_{\eta}\varepsilon_1^{\nu},\ \text{for } \nu \in \widehat{\mathbb{Z}} \text{ suitable}\right\} \\ \text{i.e.} \qquad \eta^{-1}\zeta|\xi \in L_1 \end{array}\right. \tag{27}$$

We have a canonical map

$$\left\{\begin{array}{ccc} S_0 g_{\rho,\sigma}\dagger & \longrightarrow & \Sigma_{\rho,\sigma} \\ (\alpha,\beta) & \longmapsto & (\zeta|\xi,\eta), \quad \begin{array}{ll} \zeta|\xi & = \alpha\rho(\alpha)^{-1} \\ \eta & = \beta\sigma(\beta)^{-1} \end{array}, \end{array}\right. \tag{28}$$

which is a surjection by definition of $\Sigma_{\rho,\sigma}$, and in fact by hypothesis (10) we have almost a canonical section

$$\begin{array}{ccc} \Sigma_{\rho,\sigma} & \longrightarrow & S_0 g_{\rho,\sigma}\dagger \\ \dagger & & \\ (\zeta|\xi,\eta) & \longmapsto & u_{\zeta|\xi,\eta} \end{array}$$

except that $u_{\zeta|\xi,\eta}$ is not in $\mathcal{C}$, but only in $G_{\rho,\sigma}$. We are going to consider the sets|the set of larger size

$$\begin{array}{c} g_{\rho,\sigma} \subset G_{\rho,\sigma} \times G_{\rho,\sigma} \\ \| \text{ dfn} \\ \left\{ (\alpha\beta) \;\middle|\; \underbrace{\alpha\rho(\alpha)^{-1}}_{\zeta|\xi} = \underbrace{\beta\sigma(\beta)^{-1}}_{\eta} \varepsilon_1^{\nu} \quad \text{for } \nu \in \widehat{\mathbb{Z}} \text{ suitable} \right\} \\ \dagger\zeta|\xi \end{array} \tag{29}$$

m1162

Marginal note. m1162 N.B. We will also have $\zeta|\xi = \alpha(\rho)\rho^{-1}$, $\eta = \beta(\sigma)\sigma^{-1}$ hence $\zeta|\xi, \eta \in \mathcal{C}$ (since $\mathcal{C}$ invariant in $G_{\rho,\sigma}$)

so that we will have

$$Sg_{\rho,\sigma} = g_{\rho,\sigma} \cap (\mathcal{C} \times \mathcal{C}). \tag{30}$$

(assuming (18) to simplify). [p. 507]

Uncertainty retained from cancelled passages.

140-3-s508-D001 $\wedge$
140-3-s508-D002 $\mathcal{C}$
140-3-s508-D003 $\mathcal{C}$
140-3-s508-D005 (29)
140-3-s508-D006 $\mathcal{C}$
140-3-s508-D007 η^{-1}

[scan 509]
[p. 508]
[p. 557]

We find again

$$\begin{cases} g_{\rho,\sigma} \longrightarrow \Sigma_{\rho,\sigma} \\ (\alpha,\beta) \longmapsto (\zeta|\xi, \eta), \quad \begin{array}{ll} \zeta|\xi & = \alpha\rho(\alpha)^{-1} = [\alpha,\rho] = \alpha(\rho)\rho^{+1|-1} \\ \eta & = \beta\sigma(\beta)^{-1} = [\beta,\sigma] = \beta(\sigma)\sigma^{-1} \end{array} \end{cases} \tag{31}$$

(by definition of $B_{\rho,\sigma}$ we do indeed find that the $(\zeta|\xi, \eta)$ associated with the (α, β) are in $\Sigma_{\rho,\sigma}$). This time hypothesis (10) implies that we have a canonical section

$$\begin{array}{c} \Sigma_{\rho,\sigma} \longrightarrow g_{\rho,\sigma} \\ (\zeta|\xi, \eta) \longrightarrow (u_{\zeta|\xi,\eta}, u_{\zeta|\xi,\eta}) \end{array} \tag{32}$$

and we find the same|the statements of § 48 III).

Theorem Let us consider subgroups of $\dagger G_{\rho,\sigma}$

$$\dagger \quad \begin{cases} Z_\rho = \underline{\text{Centr}}_{G_{\rho,\sigma}}(\rho) \\ Z_\sigma = \underline{\text{Centr}}_{G_{\rho,\sigma}}(\sigma) \end{cases}, \tag{33}$$

m1163

Marginal note. m1163 (N.B. in the case $\mathcal{C} = \text{SL}(2, \widehat{\mathbb{Z}})'$ etc., we will have

$$Z_\rho = T_\rho, \qquad Z_\sigma = N'_\sigma$$

and the product group

$$B_{\rho,\sigma} \times T_\rho \times T_\sigma \dagger$$

then we have a † bijective map

$$\begin{aligned} B_{\rho,\sigma} \times T_\rho | Z_\rho \times T_\sigma | Z_\sigma \dagger &\xrightarrow{\sim} g_{\rho,\sigma} \\ (u,a,b) &\longmapsto (ua^{-1}, ub^{-1}) \end{aligned} \tag{34}$$

m1164

Marginal note. m1164 by definition of $B_{\rho,\sigma}$ (7) and we see that $Z_\rho \to B_{\rho,\sigma}$ and $Z_\sigma \to B_{\rho,\sigma}$ are epimorphisms, hence also their composites $\delta_\rho, \delta_\sigma$ with $B_{\rho,\sigma} \to \Gamma_{\rho,\sigma}$. hence the maps (35) are epimorphisms.

Corollary. Let us consider the group homomorphisms $\delta_B : B_{\rho,\sigma} \to \Gamma_{\rho,\sigma}$,

$$\{\delta_G : G_{\rho,\sigma} \longrightarrow \Gamma_{\rho,\sigma}, \quad \delta_\rho : Z_\rho \longrightarrow \Gamma_{\rho,\sigma}, \quad \delta_\sigma : Z_\sigma \longrightarrow \Gamma_{\rho,\sigma}, \tag{35}$$

$\delta_B : B_{\rho,\sigma} \to \Gamma_{\rho,\sigma}$ δ_G is the canonical homomorphism of passage to the quotient $\mathcal{C}_{\rho,\sigma} = \mathcal{C}$, and $\delta_\rho, \delta_\sigma, \delta_B$ are the composites of δ_G with the inclusions $Z_\rho \subset G_{\rho,\sigma}$, $Z_\sigma \hookrightarrow G_{\rho,\sigma}$, $B_{\rho,\sigma} \subset G_{\rho,\sigma}$

Uncertainty retained from cancelled passages.

140-3-s509-D003 g
140-3-s509-D004 $Z_{\rho,\sigma}$
140-3-s509-D005 $= S\widetilde{\Gamma}_{\rho,\sigma}$
140-3-s509-D006 the
140-3-s509-D007 $S\widetilde{\Gamma}_{\rho,\sigma}$

m1165 [scan 510]

Marginal note. m1165 denoted $S_0\Gamma^b_{\rho,\sigma}$ in the § 48, IX ff

Let

$$\begin{aligned} \dagger S_0\Gamma^{b_0}_{\rho,\sigma} \subset B_{\rho,\sigma} \times Z_\rho \times Z_\sigma \\ \| \text{ dfn} \\ \{(u,a,b) \mid \delta_G(u) = \delta_\rho(a) = \delta_\sigma(b)\}. \end{aligned} \tag{36}$$

† i.e.

$$S\Gamma'_{\rho,\sigma} = B_{\rho,\sigma} \times_{\Gamma_{\rho,\sigma}} Z_\rho \times_{\Gamma_{\rho,\sigma}} Z_\sigma. \tag{37}$$

a bijection †

$$\begin{aligned} S_0\Gamma^{b_0}_{\rho,\sigma} &\xrightarrow{\sim} S_0 g_{\rho,\sigma} \quad \text{(defined in (27))} \\ (u,a,b) &\longmapsto \dagger(\zeta|\xi, \eta), \quad \text{with } \zeta|\xi = ua^{-1},\ \eta = ub^{-1}. \end{aligned} \tag{38}$$

m1166

Marginal note. m1166 N.B. The letter S here is inspired by the usage of Mister Ahlfors for the "special Teichmüller groups", and does not at all have the same meaning as in SZ_ρ, SZ_σ, where it indicates that we take the kernels of $\delta_\rho, \delta_\sigma$ (playing the role of a "determinant"…)

Scholium By the bijections (34) (38), we transport onto $g_{\rho,\sigma}$, $S_0 g_{\rho,\sigma}$ the natural group structures of $\tilde{\Gamma}_{\rho,\sigma}$, $\Gamma^{b_0}_{\rho,\sigma}$. Making this group structure explicit we find in particular for $Sg_{\rho,\sigma}$

$$\begin{cases} (\alpha',\beta')(\alpha,\beta) = \alpha''\beta'', \quad \text{with } \alpha'' = u'(\alpha)\alpha', \quad \beta'' = u'(\beta)\beta'. \\ u' = u_{\zeta|\xi',\eta'} : \quad \mathcal{C} \longrightarrow \mathcal{C}\ , \quad \text{defined by } \begin{cases} u(\rho) = \zeta|\xi\rho = \mathrm{int}(\alpha)\rho \\ u(\sigma) = \eta\sigma = \mathrm{int}(\beta)\sigma \end{cases} \end{cases} \tag{39|37}$$

(with $\zeta|\xi = \alpha\rho(\alpha)^{-1}$, $\eta = \beta\sigma(\beta)^{-1}$).

In $S_0\Gamma^{b_0}_{\rho,\sigma}$ we have a subgroup isomorphic to L_0, i.e. m1167

Marginal note. m1167

$$\begin{aligned} L_0 &\longrightarrow S\Gamma'_{\rho,\sigma} \\ \ell &\longmapsto (\ell,1,1) \end{aligned} \tag{39}$$

and we have exact sequence

$$1 \to L_0 \to S\Gamma'_{\rho,\sigma} \to \Gamma_{\rho,\sigma} \to 1 \tag{40}$$

† [p. 509]

Uncertainty retained from cancelled passages.

140-3-s510-D001 $S\widetilde{\Gamma}^0_{\rho,\sigma} = \widetilde{M}^0_{\rho,\sigma} \subset S_0\Gamma_{\rho,\sigma}$

140-3-s510-D002 $S\Gamma'_{\rho,\sigma}$

140-3-s510-D003 here

140-3-s510-D004 (ua^{-1}, ub^{-1})

140-3-s510-D005 Let us now put $\mathcal{C}$ a)|1°) *which* $\mathcal{C}$ $\mathcal{C}$ such that b)|2°) $\mathcal{C}$ $\mathcal{C}$ $\mathcal{T}|\Gamma$ $\mathcal{T}|\Gamma$ $\delta_{\mathcal{T}}$

[scan 511]

[p. 558]

$$\begin{aligned} L_0 &\longrightarrow S\dagger\Gamma^0_{\rho,\sigma} \subset B_{\rho,\sigma} \times Z_\rho \times Z_\sigma \\ \ell &\longmapsto (\ell,1,1) \end{aligned} \tag{40}$$

and the action of L_0 by left translation on $S\mathcal{T}'_{\rho,\sigma}|S\Gamma'_{\rho,\sigma}$ becomes, by (39), the action on $S\mathcal{G}|\mathcal{G}_{\rho,\sigma}$ given by

$$(\alpha,\beta) \longmapsto (\ell\alpha, \ell\beta). \tag{41}$$

Let us therefore set m1168

Marginal note. m1168 denoted $\Gamma|\mathcal{T}^0_{\rho,\sigma}$ in § 48, IX #.

$$\dagger\Gamma^0_{\rho,\sigma} \overset{\text{dfn}}{=} S\dagger\Gamma^0_{\rho,\sigma}/\operatorname{Im} L_0 \simeq Z_\rho \times_{\gamma_{\rho,\sigma}} Z_\sigma, \tag{42}$$

we find that (39) induces a canonical bijection

$$\dagger\Gamma^0_{\rho,\sigma} \xrightarrow{\sim} L_0\backslash S\mathcal{G}|\mathcal{G}_{\rho,\sigma}\ \dagger \tag{43}$$

Setting

$$\begin{cases} SZ_\rho \overset{\text{dfn}}{=} \operatorname{Ker}(Z_\rho \longrightarrow \Gamma_{\rho,\sigma}) = \operatorname{Centr}_{\mathcal{G}}(\rho) \\ SZ_\rho = \operatorname{Ker}(Z_\rho - \Gamma_{\rho,2|\sigma}) = \operatorname{Centr}_{\mathcal{G}}(\sigma) \end{cases} \tag{44}$$

Marginal note. m1169 also introduce $S\Gamma_{\rho,\sigma} = \mathrm{Ker}(\Gamma_{\rho,\sigma} \longrightarrow \gamma_{\rho,\sigma})$, $SG_{\rho,\sigma} = \mathrm{Ker}(G_{\rho,\sigma} \longrightarrow \gamma_{\rho,\sigma}) = \mathcal{G}_{\rho,\sigma} = \mathcal{G}$.

we thus have exact sequences

$$\left\{ \begin{array}{c} 1 \text{ — } SZ_\rho \to Z_\rho \text{ — } \Gamma_{\rho\sigma|\rho,\sigma} \text{ — } 1 \\ 1 \text{ — } SZ_\sigma \text{ — } Z_\sigma \text{ — } \Gamma_{\rho,\sigma} \text{ — } 1 \end{array} \right. , \tag{45}$$

and the expressions (37), (42) for $S\dagger\Gamma^0_{\rho,\sigma}$, $\dagger\Gamma^0_{\rho,\sigma}$ give

$$1 \to L_0 \times SZ_\rho \times SZ_\sigma \text{ — } S_0\dagger\Gamma^0_{\rho,\sigma} \text{ — } \gamma_{\rho,\sigma} \text{ — } 1 \tag{46}$$

$$1 \to SZ_\rho \times SZ_\sigma \to \dagger\dagger\Gamma^0_{\rho,\sigma} \text{ — } \gamma_{\rho,\sigma} \to 1 \ . \tag{47}$$

N.B. In the case $\mathcal{G} = \mathrm{Sl}(2, \widehat{\mathbb{Z}})$, ρ and σ "modular", we recover $\dagger$ the $S_0\dagger\Gamma^{04|\#}_{\rho,\sigma}$ and $\dagger\Gamma^0_{\rho,\sigma}$ of § 48 IX. [p. 510]

Uncertainty retained from cancelled passages.

140-3-s511-D001 $\mathcal{T}$
140-3-s511-D002 $\mathcal{T}$
140-3-s511-D003 $\mathcal{T}$
140-3-s511-D004 $\mathcal{T}$
140-3-s511-D005 $\simeq Z_\rho \times Z_\sigma$
140-3-s511-D009 $\mathcal{T}$
140-3-s511-D010 $\mathcal{T}$
140-3-s511-D011 $\mathcal{T}$
140-3-s511-D012 S
140-3-s511-D013 $\mathcal{T}$
140-3-s511-D014 [illeg. 1]
140-3-s511-D015 $\mathcal{T}$
140-3-s511-D016 $\mathcal{T}$

[scan 512]
[p. 559]

Now let

$$S_0M^{0*}_{\rho,\sigma} \subset S\dagger\Gamma^0_{\rho,\sigma} \times G_{\rho,\sigma} \quad (\subset B_{\rho,\sigma} \times Z_\rho \times Z_\sigma \times G_{\rho,\sigma})$$
$$\overset{\text{dfn}}{=}$$
$$\left\{ (u,a,b,U) \,\middle|\, \underbrace{\delta_B(u) = \delta_\rho(a) = \delta_\sigma(b) = \delta_G(U)}_{\text{relation in } \gamma_{\rho,\sigma} \text{ cf. (17)}} \right\} \tag{49}$$

$$M^0_{\rho,\sigma} \subset T_\rho \times T_\sigma \times G_{\rho,\sigma}$$
$$\overset{\text{dfn}}{=}$$
$$\left\{ (a,b,U) \,\middle|\, \underbrace{\delta_\rho(a) = \delta_\sigma(b) = \delta_G(U)}_{\text{relations in } \gamma_{\rho,\sigma}} \right\} \tag{50}$$

We thus have a canonical homomorphism

$$SM^0_{\rho,\sigma} \longrightarrow M^0_{\rho,\sigma}$$
$$(u,a,b,U) \longrightarrow \dagger(a,b,U) \tag{51}$$

whose kernel consists of the $(u,1,1,1)$ where $u \in B_{\rho,\sigma}$ is such that $\delta_B(u) = 1$ i.e. $u \in L_0$, whence an exact sequence

$$1 \longrightarrow L_0 \to SM^0_{\rho,\sigma} \to M^0_{\rho,\sigma} \longrightarrow 1 \tag{52}$$

and canonical homomorphisms

$$M^0_{\rho,\sigma} \xrightarrow{\Theta_G} G_{\rho,\sigma}, \quad M^0_{\rho,\sigma} \xrightarrow{\Theta_\rho} Z_\rho, \quad M^0_{\rho,\sigma} \xrightarrow{\Theta_\sigma} Z_\sigma \tag{53}$$

$$\left(\text{resp.} \quad (a,b,U) \mapsto \begin{matrix} U \\ a \\ b \end{matrix} \right),$$

giving back the analogous homomorphisms on $SM^0_{\rho,\sigma}$ by composition with (41), and moreover we have a canonical homomorphism

$$\begin{matrix} SM^0_{\rho,\sigma} & \xrightarrow{\Theta_B} & B_{\rho,\sigma} \\ (u,a,b,U) & \longmapsto & u \end{matrix} . \tag{54}$$

[p. 511]

Uncertainty retained from cancelled passages.

140-3-s512-D001 $\mathcal{T}$

140-3-s512-D002 $(u$

[scan 513]

We also have † the interpretation

$$S_0 M^0_{\rho,\sigma} \simeq M^0_{\rho,\sigma} \times_{\gamma_{\rho,\sigma}} B_{\rho,\sigma}; \tag{55}$$

($S_0 M^0_{\rho,\sigma}$ is expressed in terms of $M^0_{\rho,\sigma}$ and $B_{\rho,\sigma}$, and we see that the extension of $M_{\rho,\sigma}$ by L_0 is the inverse image of the extension $B_{\rho,\sigma}$ of $\gamma_{\rho,\sigma}$ by L_0. And we have

$$M^0_{\rho,\sigma} \simeq \underbrace{Z_\rho \times_{\gamma_{\rho,\sigma}} Z_\sigma}_{\dagger \Gamma^0_{\rho,\sigma}} \times_{\gamma_{\rho,\sigma}} G_{\rho,\sigma}. \tag{56}$$

whence we deduce (since $\delta_\rho, \delta_\sigma, \delta_G$ are epimorphisms)

$$1 \to \underbrace{\dagger\, SZ_\rho \times SZ_\sigma \times \mathcal{G}}_{\overset{\text{dfn}}{=} M^{0+}_{\rho,\sigma}} \to M'_{\rho,\sigma} | M^0_{\rho,\sigma} \xrightarrow{\delta_M^{\#|*}} \gamma_{\rho,\sigma} \to 1 \tag{57}$$

where we have † (44)

$$\dagger \quad \begin{cases} SZ_\rho = Z_\rho \cap \mathcal{G} = \underset{\dagger}{\text{Centr}}_{\mathcal{G}} \ (\rho) \\ SZ_\sigma = Z_\sigma \cap \mathcal{G} = \underset{\dagger}{\text{Centr}}_{\mathcal{G}} \ (\sigma) \end{cases}$$

Similarly we have

$$S_0 M^0_{\rho,\sigma} \simeq \underbrace{B_{\rho,\sigma} \times_{\gamma_{\rho,\sigma}} Z_\rho \times_{\gamma_{\rho,\sigma}} Z_\sigma}_{S_0 \dagger \Gamma^0_{\rho,\sigma}} \times_{\gamma_{\rho,\sigma}} G_{\rho,\sigma} \tag{49}$$

whence we have a canonical exact sequence

$$1 \to L_0 \times SZ_\rho \times SZ_\sigma \times \mathcal{G} \to S_0 M^0_{\rho,\sigma} \to \Gamma_{\rho,\sigma} \text{ — } 1 \tag{50}$$

Let us consider the canonical homomorphism

$$\begin{array}{ccc} \mathcal{G} & \xrightarrow{i_{\mathcal{G}}} & M^0_{\rho,\sigma} \\ & , & \\ U & \longmapsto & (1,1,U) \end{array} \tag{51}$$

[p. 512]

Uncertainty retained from cancelled passages.

140-3-s513-D001 of
140-3-s513-D002 $\mathcal{T}$
140-3-s513-D003 $\Gamma_\rho|T_\rho$
140-3-s513-D004 set
140-3-s513-D005 48|58
140-3-s513-D006 aut
140-3-s513-D007 aut
140-3-s513-D008 $\mathcal{T}$

[scan 514]

its image is the third factor in the kernel of (57), and we have a canonical exact sequence [p. 560] (deduced from (56)–(57))

$$1 \to \underset{\substack{\text{sometimes denoted} \\ \mathcal{G}_{\rho,\sigma}?}}{\mathcal{G}} \to M^0_{\rho,\sigma} \to \dagger\Gamma^0_{\rho,\sigma} \text{ — } 1 \tag{52}$$

m1170

Marginal note. m1170 (53′)

$$1 \to SZ_\rho \times SZ_\sigma \to M^0_{\rho,\sigma} \to G|G_1 \text{ — } 1$$

and similarly we have (via (49) or 50))

$$\begin{cases} (53) \quad 1 \to \mathcal{G} \to S_0 M^0_{\rho,\sigma} \to S_0 \dagger\Gamma^0_{\rho,\sigma} \to 1 \ . \\ \text{Let us consider the map} \\ \qquad S_0 M^0_{\rho,\sigma} \longrightarrow \mathcal{G} \times \mathcal{G} \times \mathcal{G} \\ (54) \\ \qquad \{u,a,b,U\} \longmapsto (ua^{-1}, ub^{-1}, uU^{-1}) = (\alpha,\beta,f) \end{cases}$$

we see † by the corollary on page 557 (cf. (37)) that this induces a bijection

$$S_0 M^0_{\rho,\sigma} \simeq (S\mathcal{G}|\mathcal{G}_{\rho,\sigma}) \times \mathcal{G}. \tag{55}$$

We thus have, for an $x \in S_0 M^0_{\rho,\sigma}$, on the one hand the quantities

$$\begin{array}{cccc} u, & a, & b, & U \\ \cup & \cup & \cup & \cup \\ B_{\rho,\sigma} & Z_\rho & Z_\sigma & G_{\rho,\sigma} \end{array} , \tag{56}$$

associated, on the other hand the elements

$$\alpha, \beta, f, \quad \zeta|\xi, \eta, \quad \alpha_0\ \beta_0, g, h \qquad \text{all } \in \mathcal{G} \tag{57}$$

where (α, β, f) defined in terms of (56) in (54), and where we have set

$$\begin{cases} \zeta = \alpha\rho(\alpha)^{-1}, \quad \eta = \beta\sigma(\beta)^{-1}, \quad \alpha_0 = f^{-1}\alpha\#|\zeta, \quad \beta_0 = f^{-1}\beta, \\ g = \alpha_0\rho(\alpha_0)^{-1} = f^{-1}\zeta\rho(f), \quad h = \beta_0\sigma(\beta_0)^{-1} = f^{-1}\eta\sigma(f) \end{cases} \tag{58}$$

[p. 513]

Uncertainty retained from cancelled passages.

140-3-s514-D001 $\mathcal{T}$

140-3-s514-D002 $\mathcal{T}$

140-3-s514-D003 [illeg. 1]

[scan 515]

We will have, in terms of α, β, f, the following expressions for the action of U on $\mathcal{G}$:

$$\begin{cases} U(\rho) = \mathrm{int}(f^{-1}\alpha)\dagger\rho = \mathrm{int}(f^{-1})(\zeta\rho) \\ U(\sigma) = \mathrm{int}(f^{-1}\beta)\dagger(\sigma) = \mathrm{int}(f^{-1})(\eta\sigma). \end{cases} \tag{59}$$

†

Remark|Remark Finally, in all these developments, we have not really used the so-called "fundamental hypothesis" (20)†, † except part a). In the absence of b), one should introduce

$$\begin{array}{c} \Sigma^{\mathrm{eff}}_{\rho,\sigma} \subset \Sigma_{\rho,\sigma} \subset \mathcal{G} \times \mathcal{G} \\ \| \\ \left\{(\zeta, \eta) \in \mathcal{G}|\Sigma_\rho \times \mathcal{G} \mid \exists u \in B_{\rho,\sigma},\ u(\rho) = \zeta\rho,\ u(\sigma) = \eta\sigma\right\}, \end{array} \tag{60}$$

which is thus nothing other than the image of the map (9): effective pairs (ζ, η) satisfying the conditions defining $\dagger\Sigma_{\rho,\sigma}$ (cf. (8)))

†

let us write

$$\begin{array}{c} S\mathcal{G}|\mathcal{G}^{\mathrm{eff}}_{\rho,\sigma} = \text{Inverse image of } \Sigma^{\mathrm{eff}}_{\rho,\sigma} \\ \text{under } \ S\mathcal{G}|\mathcal{G}_{\rho,\sigma} \longrightarrow \Sigma_{\rho,\sigma} \quad (28), \end{array} \tag{61}$$

then|by bijection|bis

and it is on $S\mathcal{G}|\mathcal{G}^{\mathrm{eff}}_{\rho,\sigma}$ that we will have a structure

[p. 514]

Uncertainty retained from cancelled passages.

140-3-s515-D001 $\mathrm{int}(\alpha)$

140-3-s515-D002 $\mathrm{int}(\beta)$

140-3-s515-D003 and $\zeta|\xi \quad \eta$.

140-3-s515-D004 b

140-3-s515-D005 and

140-3-s515-D006 the (5)

140-3-s515-D007 $\mathcal{G}|\mathcal{G}$ is|will be

[scan 516]

of a group isomorphic to|: $S_0\mathcal{T}^0_{\rho,\sigma}$, by the isomorphism (which replaces (38))

[p. 561]

$$\begin{array}{ccc} S_0\dagger\Gamma^0_{\rho,\sigma} & \simeq & S_0\mathcal{G}|\mathcal{G}^{\mathrm{eff}}_{\rho,\sigma}. \end{array} \tag{62}$$

$$(\underline{u}, \underline{a}, \underline{b}) \longrightarrow (\underbrace{\underline{u}\underline{a}^{-1}}_{\zeta}, \underbrace{\underline{u}\underline{b}^{-1}}_{\eta})$$

This being so, let us return to $M^{\sim}_{\rho,\sigma}$, and the exact sequence

$$1 \to \mathcal{G}^{+\wedge}_{0,3} \to M^{\sim}_{0,3|\rho,\sigma} \to \underbrace{\Gamma^{\sim}_{0,3}}_{\Gamma^{\sim}_{\mathbb{Q}}} \to 1 \qquad (63).$$

Consider the subgroup

$$(64) \qquad \begin{aligned} \Gamma'^{\sim}_{\mathbb{Q}} &= \mathrm{Ker}(\Gamma^{\sim}_{0,3} \xrightarrow{(\varepsilon_3,\dagger)} \{\pm 1\}\dagger) \\ &= \{\gamma \in \Gamma^{\sim}_{0,3} \mid \mu(\gamma) \equiv 1 \ (3\dagger)\}. \end{aligned}$$

† †

and let †

$$(65) \qquad M'^{\sim}_{0,3} = \text{Inverse image of } \Gamma'^{\sim}_{0,3} \text{ in } M^{\sim}_{0,3}.$$

Let $\mathcal{G}$ be a profinite group, and

$$(66) \qquad \psi : \mathcal{G}^{+\wedge}_{0,3} \longrightarrow \mathcal{G},$$

a continuous surjective homomorphism — which amounts to the same thing as giving two elements

$$(67) \qquad \rho_{\mathcal{G}}, \sigma_{\mathcal{G}} \in \mathcal{G}$$

(the images of ρ, σ under ψ) satisfying

$$(68) \qquad \rho^3_{\mathcal{G}} = \sigma^2_{\mathcal{G}} = 1.$$

[p. 515]

Uncertainty retained from cancelled passages.

140-3-s516-D001 $\mathcal{T}$
140-3-s516-D002 ε_2
140-3-s516-D003 $\times\{\pm 1\}$
140-3-s516-D004 12
140-3-s516-D005 [illeg. 3]
140-3-s516-D006 two factors
140-3-s516-D007 $M'^{\sim}_{0,3}$

[scan 517]

m1172

Marginal note. m1172 introduce it by equivalent conditions:
a)

$$M'^{\sim}_{0,3}[0] \longrightarrow \Sigma^{\mathrm{eff}}_{\rho,\sigma}$$

all
b) $\forall u \in M'^{\sim}_{0,3}$, $\exists u_{\mathcal{G}} \in \mathrm{Aut}\,\mathcal{G}$ such that $u_{M|\mathcal{G}} \circ \psi = \psi \circ u$
b') Idem with $u \in M'^{\sim}_{0,3}[0]$
c) $\exists \psi_M$ which extends ψ
c') $\exists \psi_G$ which extends ψ
We will therefore say that † automatic|automorphism
ψ is effective if (20) b) satisfied

Definition We will say that ψ is effective if for every $u \in M'^{\sim}_{0,3}[0]$, putting $u(\rho) = \zeta\rho$, $u(\sigma) = \eta\sigma$,

$$(\zeta_{\mathcal{G}}, \eta_{\mathcal{G}}) = (u(\zeta), u(\eta)) \in \Sigma_{\rho,\sigma}$$

effective (i.e. $\in \Sigma^{\mathrm{eff}}_{\rho,\sigma}$, i.e. $\exists u_{\mathcal{G}}$ automorphism $\mathcal{G}$ such that we have †

$$(69) \qquad u_{\mathcal{G}}(\rho_{\mathcal{G}}) = \zeta_{\mathcal{G}}\rho_{\mathcal{G}}, \qquad u_{\mathcal{G}}(\sigma_{\mathcal{G}}) = \eta_{\mathcal{G}}\sigma_{\mathcal{G}}$$

or, what amounts to the same thing, that we have commutativity in

$$\begin{array}{ccc} \mathcal{G}_{0,3}^{+\wedge} & \xrightarrow{u} & \mathcal{G}_{0,3}^{+\wedge} \\ \downarrow{\scriptstyle \psi} & & \downarrow{\scriptstyle \psi} \\ \mathcal{G} & \xrightarrow[u_{\mathcal{G}}]{} & \mathcal{G} \end{array} \qquad (70).$$

(Thus to say that this is the case for every $u \in M_{0,3}'^{\sim}$ means that these u transform $\operatorname{Ker}\psi$ into itself — or also, that $M_{0,3}'^{\sim}\dagger$ $(= M_{0,3}'^{\sim}[0].\mathcal{G}_{0,3}^{+\wedge})$ in its entirety stabilizes $\operatorname{Ker}\psi$. This condition is therefore automatically fulfilled when $\mathcal{G}, \rho_{\mathcal{G}}, \sigma_{\mathcal{G}}$ satisfy the condition|the conditions b) of (20).

We have done what was needed to have the [p. 516]

Uncertainty retained from cancelled passages.

140-3-s517-D001 effective

140-3-s517-D002 $u_{\mathcal{G}}\rho$

140-3-s517-D004 [0]

[scan 518]

Theorem Let $\psi = \psi_{\mathcal{G}} : \mathcal{G}_{0,3}^{+\wedge} \longrightarrow \mathcal{G}$ be an effective surjective homomorphism (cf. above). Then [p. 562] there exists a unique homomorphism

$$(71) \qquad \psi_M : \dagger M_{0,3}'^{\sim\dagger} \longrightarrow \dagger M^0_{\rho_{\mathcal{G}},\sigma_{\mathcal{G}}}$$

†

m1173

Marginal note. m1173 In fact, we have of course in the first place a homomorphism

$$S_0 M_{0,3}'^{\sim} \longrightarrow S_0 M^0_{\psi}.$$

having the following property: if † $u=$ | $\in M_{0,3}'^{\sim}$ is of the form $u_{\alpha,\beta,f}$ $(\alpha, \beta, f \in \mathcal{G}_{0,3}^{+\wedge}$, satisfying the loop equation $\alpha\rho(\alpha)^{-1} = \beta\sigma(\beta)^{-1}\ell_1^{\nu})$, then $\psi_M(u)$ is the element of † M^0_{ψ} defined by the invariants $\psi(\alpha), \psi(\beta); \psi(f)$.

Corollary We then have homomorphisms of exact sequences

$$\begin{array}{ccccccccc} 1 & \to & \mathcal{G}_{0,3}^{+\wedge} & \to & M_{0,3}'^{\sim} & \to & \Gamma_{\mathbb{Q}}'^{\sim} & \to & 1 \\ & & \downarrow{\scriptstyle \psi_{\mathcal{G}}} & & \downarrow{\scriptstyle \psi_M} & & \downarrow{\scriptstyle \psi_{\mathcal{T}}|\psi_\Gamma} & & \\ 1 & \longrightarrow & \mathcal{G} & \longrightarrow & M^0_{\psi} & \to & \dagger\Gamma^0_{\psi} & \to & 1 \end{array} \qquad (72).$$

(The homomorphism ψ_M is made explicit by means of three homomorphisms [p. 517]

Uncertainty retained from cancelled passages.

140-3-s518-D001 S_0

140-3-s518-D002 $*$

140-3-s518-D003 S_0

140-3-s518-D005 $\psi|\varphi \quad \mathcal{G} \quad \mathcal{G}$

140-3-s518-D008 $u \in$

140-3-s518-D009 $M^0_{\rho_{\mathcal{G}},\sigma_{\mathcal{G}}}$

140-3-s518-D010 $\mathcal{T}$

[scan 519]

$$\begin{cases} \psi_{T_\rho} : \Gamma'^{\sim}_{\mathbb{Q}} \longrightarrow Z_\rho | T_\rho \\ \psi_{T_\sigma} : \Gamma^{\sim}_{\mathbb{Q}} \longrightarrow Z_\sigma | T_\sigma \end{cases} \qquad (73)$$

$$\psi_G : M'^{\sim}_{0,3} \longrightarrow G_{\rho,\sigma} \qquad (74)$$

m1174

Marginal note. m1174 N.B. for simplicity we write ρ, σ instead of $\rho_{\mathcal{G}}, \sigma_{\mathcal{G}}$ as subscripts

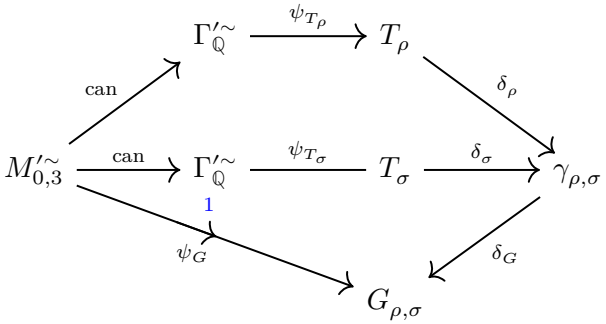
(For † $x \in M'^{\sim}_{0,3}$, $\dot{x}$ its image in $\Gamma'^{\sim}_{\mathbb{Q}}$, †

$$\dagger \quad a = \psi_{T_\rho | \Gamma_\rho}(\dot{x}), \quad b = \psi_{T_\sigma | \Gamma_\sigma}(\dot{x}), \quad U = \psi_G(x), \qquad (75)$$

we have

$$\delta_\rho(a) = \delta_\sigma(b) = \delta_G(U) \quad \text{in } \gamma_{\rho,\sigma} \qquad (76)$$

i.e. the three composite homomorphisms



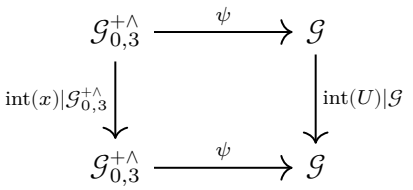
(77)

are equal. Moreover, we will have by (71),

$$\psi_G | \mathcal{G}^{+\wedge}_{0,3} = \psi_{\mathcal{G}}, \qquad (78)$$

and finally this:

(79) If $x \in \dagger M'^{\sim}_{0,3}$, $U = \psi_G(x)$, we have commutativity in



(80)

. . . . . [p. 518]

Uncertainty retained from cancelled passages.

140-3-s519-D002 u

140-3-s519-D003 $T_\rho|\Gamma_\rho$ $T_\rho|\Gamma_\rho$ $y|\dot{x}$ $\psi u|\psi(x)$

140-3-s519-D004 $\delta_G($

140-3-s519-D005 M

[scan 520]

[p. 563]

. We can regard all this as a statement of functoriality of the constructions made, with respect to a surjective homomorphism

$$\mathcal{G} \xrightarrow{\varphi=\varphi_{\mathcal{G}}} \mathcal{G}' \qquad (81)$$

m1175

Marginal note. m1175 It would have been better to take then

$$\mathcal{G}' \xrightarrow{\varphi} \mathcal{G}$$

, in order to be able to specialize to $\mathcal{G}'$

transforming the pair (ρ, σ) of topological generators of $\mathcal{G}$ into a pair (ρ', σ'). There is then a unique homomorphism

$$S_0 M^0_{\mathcal{G}} \xrightarrow{\varphi_{M^0}} S_0 M^0_{\mathcal{G}'} \qquad (82)$$

satisfying the following conditions:

m1176

Marginal note. m1176 this shows that φ_{SM} is determined when we know the four homomorphisms $\varphi_B, \varphi_\rho, \varphi_\sigma, \varphi_G$

a) $\varphi_M|\varphi_{SM}$ passes to quotients as|to homomorphisms

$$\begin{cases} \varphi_B : B_{\rho,\sigma} \longrightarrow Z_{\rho,\sigma}|\Sigma_{\rho,\sigma} \\ \varphi_\rho : Z_\rho \longrightarrow Z_{\rho'} \\ \varphi_\sigma : Z_\sigma \text{ — } Z_{\sigma'} \\ \varphi_G : G_{\rho,\sigma} \text{ — } G_{\rho',\sigma'} \end{cases} ; \qquad (83)$$

b) †

$$(84) \quad \begin{cases} \dagger \quad \varphi_G \qquad \text{(which shows that knowledge} \\ \text{[illeg. 1] of } \varphi_{SM} \text{ } b|se \text{ amounts to that of } \varphi_G); \end{cases}$$

c) We have commutativity in

$$\begin{array}{ccc} \mathcal{G} & \longrightarrow & G_{\rho,\sigma} \\ {\scriptstyle \varphi=\varphi_{\mathcal{G}}}\downarrow & & \downarrow{\scriptstyle \varphi_G} \\ \mathcal{G}' & \longrightarrow & G_{\rho',\sigma'} \end{array} \qquad (85)$$

(which shows, since $G_{\rho,\sigma} = B_{\rho,\sigma} \cdot \mathcal{G}$, that φ_G is known when we know φ_B;

d) For $u \in B_{\rho,\sigma}$, † putting $u' = \varphi_B(u)$, we have commutativity in

$$\begin{array}{ccc} \mathcal{G} & \xrightarrow{u} & \mathcal{G} \\ {\scriptstyle \varphi_{\mathcal{G}}}\downarrow & & \downarrow{\scriptstyle \varphi_{\mathcal{G}}} \\ \mathcal{G}' & \xrightarrow{u'} & \mathcal{G}' \end{array} \qquad (86)$$

[p. 519]

Uncertainty retained from cancelled passages.
140-3-s520-D002 are induced by
140-3-s520-D003 only
140-3-s520-D004 denoting

[scan 521]

– (which determines φ_B in terms of $\varphi = \varphi_{\mathcal{G}}$, thus this|it determines $\varphi_{S_0 M}$, and by passage to the quotient, also

$$(87) \qquad \varphi_M : M_{\rho,\sigma} \longrightarrow M_{\rho',\sigma'}.$$

Another way, more direct, of describing $\varphi_{S_0 M^0}$ or φ_M, in terms of the "parameters" (α, β, f) instead of the (u, a, b, U), is to say that one † transforms a $u_{\alpha,\beta,f}$ into the $u_{\alpha',\beta',f'}$, where α', β', f' are the images of α, β, f under φ:

$$(88) \qquad \varphi_{\dagger S_0 M_0 | M^0}\left(\underbrace{u_{\alpha,\beta,f}}_{\in S_0 M^0_{\rho,\sigma}}\right) = u_{\alpha',\beta',f'}, \qquad \begin{cases} \alpha' = \varphi(\alpha), \\ \beta' = \varphi(\beta), \\ f' = \varphi(f). \end{cases}$$

We thus recover, in a slightly more precise form, the preceding theorem, by † applying it to the case where $\mathcal{G} = \mathcal{G}^{+\wedge}_{0,3}$ (the "universal case")*, $\mathcal{G}'$ arbitrary — it is better in the notation (81) to interchange the role of $\mathcal{G}$ and $\mathcal{G}'$. m1177

Marginal note. m1177 * at least "universal" with respect to the additional hypothesis $\rho^3 = 1$, $\sigma^2 = 1$ in $\mathcal{G}$.

It is good to †make explicit the essential invariants (in the "universal" case:

$$(90) \qquad \mathcal{G}' = \mathcal{G}^{+\wedge}_{0,3}, \qquad B^{\dagger\,\prime}_{\rho',\sigma'} = M'^{\sim}_{0,3}[0], \qquad \gamma'^{|0}_{\rho',\sigma'} = \Gamma'^{\sim}_{\mathbb{Q}},$$
$$G'^{|0}_{\rho',\sigma'} = M'^{\dagger\sim}_{0,3}, \qquad Z'^{\dagger}_{\rho} = M'^{\sim}_{0,3}\left(-\frac{1}{3}\right) \mid \left(\frac{1}{3}\right), \qquad Z'^{\dagger}_{\sigma} = M'^{\sim}_{0,3}\left(\frac{1}{2}\right) \mid \left(-\frac{1}{2}\right),$$
$$\Gamma'_{\rho',\sigma'} = \Gamma'^{\sim}_{\mathbb{Q}} \quad \dagger.$$

hence [p. 520]

Uncertainty retained from cancelled passages.
140-3-s521-D002 $\varphi_{S_0 M^0}$ transforms
140-3-s521-D003 $\mathcal{G}$
140-3-s521-D004 putting
140-3-s521-D005 make explicit
140-3-s521-D007 0
140-3-s521-D008 0
140-3-s521-D009 0
140-3-s521-D010 0
140-3-s521-D011 $\prime|0$

[scan 522]
[p. 564]

$$\Gamma'_{\rho',\sigma'} \overset{\sim}{\to} \gamma'_{\rho',\sigma'}, \quad Z'_{\rho'} \overset{\sim}{\to} \gamma'_{\rho',\sigma'}, \quad Z'_{\sigma'} \overset{\sim}{\to} \gamma'_{\rho',\sigma'}, \quad \text{thus}$$
$$(91)|(92) \qquad S_0 M'_{\rho',\sigma'} \simeq \overset{S_0 G'_{\rho',\sigma'} \simeq |=}{S_0 M'^{\sim}_{0,3}}, \quad \dagger M'_{\rho',\sigma'} \overset{\sim}{\to} \overset{G'_{\rho',\sigma'} \overset{\sim}{\to}}{\dagger M'^{\sim}_{0,3}}.$$

m1178

Marginal note. m1178 hence

$$(92) \qquad \begin{cases} SZ'_{\rho'} = SZ'_{\sigma'} \\ \qquad = S\Gamma|\mathcal{T}'_{\rho',\sigma'} = 1 \end{cases} \quad (cf.) \text{ of } (92)$$

Remark In writing the relations (86), it became evident that the notations used in this section II (from page 0 553') were inadequate, and that they should be modified as follows: m1179

Marginal note. m1179 This had been suggested to me in the case $\mathcal{G} = \mathrm{Sl}(2, \widehat{\mathbb{Z}})$, modular pair ρ, σ which does not come from $\mathcal{G}^{+\wedge}_{0,3}$, because of the $\{\pm 1\}$! but in this case [illeg. 1] universal, should have served me as a guiding thread for my notations.

$$(93) \qquad \begin{matrix} \Sigma_{\rho,\sigma},\ B_{\rho,\sigma},\ \overset{S_0 G_{\rho,\sigma},}{G_{\rho,\sigma}}\ ,\ \gamma_{\rho,\sigma},\ \dagger,\ Z_\rho,\ Z_\sigma,\ \mathcal{G}_{\rho,\sigma},\ S\mathcal{G}_{\rho,\sigma} \\ Z_\rho,\ Z_\sigma,\ S_0\Gamma^{\dagger 0}_{\rho,\sigma},\ \Gamma^0_{\rho,\sigma},\ SZ_\rho,\ SZ_\sigma,\ S_0M^0_{\rho,\sigma},\ M^0_{\rho,\sigma}. \end{matrix}$$

We put a prime $'$ everywhere to make them $\Sigma'_{\rho,\sigma}$, $B'_{\rho,\sigma}$ etc, and we remove the exponent ^{0} from $S_0\Gamma^0_{\rho,\sigma}$, $\Gamma^0_{\rho,\sigma}$, $S_0M^0_{\rho,\sigma}$, $M^0_{\rho,\sigma}$ (which is thus replaced by a $'$).

Other important special cases to make explicit:

2°) $\mathcal{G} = \mathrm{Sl}(2, \widehat{\mathbb{Z}})/\pm 1$, (ρ, σ) modular pair.

$$(94) \quad \begin{cases} B'_{\rho,\sigma} \simeq \left\{ \begin{pmatrix} \mu & \lambda \\ 0 & 1 \end{pmatrix} \mid \mu \in \widehat{\mathbb{Z}}^*,\ \mu \equiv 1\ (3),\ \lambda \in \widehat{\mathbb{Z}} \right\} \\ \quad L_0 = \left\{ \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix} \mid \lambda \in \widehat{\mathbb{Z}} \right\} \\ \quad \dagger \\ G'_{\rho,\sigma} \simeq \mathrm{Gl}'(2, \widehat{\mathbb{Z}})/\pm 1, \quad \text{where} \quad \mathrm{Gl}'(2, \widehat{\mathbb{Z}}) \overset{\text{dfn}}{=} \{u \in \mathrm{Gl}(2, \widehat{\mathbb{Z}}) \mid \det u \equiv 1\ (3)\} \\ \quad \dagger \\ Z'_\rho = \{c\rho + d \mid c^2 + cd + d^2 \in \widehat{\mathbb{Z}}^*\}/\pm 1 \\ \quad \simeq T_\rho | \mathcal{T}_\rho(\widehat{\mathbb{Z}})/\mu_2(\mathbb{Z}) \\ \quad \dagger \\ Z'_\sigma = \dagger \\ \quad N'_\sigma(\widehat{\mathbb{Z}})/\mu_2(\mathbb{Z}) \quad \text{with notations of §48} \\ S_0M'_{\rho,\sigma} \simeq \dagger\ SM''_{\rho,\sigma} | SM'_{\rho,\sigma}(\widehat{\mathbb{Z}})/\Sigma \quad \text{with notations of §48} \\ M'_{\rho,\sigma} \simeq \dagger\ M'_{\rho,\sigma}(\widehat{\mathbb{Z}})/\Sigma \quad \text{id} \\ \Gamma'_{\rho,\sigma} \simeq \mathcal{T}'_{\rho,\sigma}(\widehat{\mathbb{Z}})/\Sigma_0 \quad \text{with notations of §48} \end{cases}$$

m1180

Marginal note. m1180 this is $G(\widehat{\mathbb{Z}})/\mu_2(\mathbb{Z})$ with notations of §48

[p. 521]

Uncertainty retained from cancelled passages.

140-3-s522-D001 $G_{\rho',\sigma'} =$
140-3-s522-D002 G
140-3-s522-D003 $\Gamma_{\rho,\sigma}$
140-3-s522-D004 $*$
140-3-s522-D005 $\mathrm{Gl}(\widehat{\mathbb{Z}})/\mu_2(\mathbb{Z})$
140-3-s522-D006 $G\,\mathbb{Z}\,B/\mu_2$
140-3-s522-D007 T_ρ with notations of
140-3-s522-D008 $\{t\sigma + u \mid t^2 + u^2 \in \widehat{\mathbb{Z}}^*\}/\pm 1$ subgroup
140-3-s522-D009 is the
140-3-s522-D010 $SM'_{\rho,\sigma}(\widehat{\mathbb{Z}})$

[scan 523]

3°) Let us take $\mathcal{G} = \mathrm{Sl}(2, \widehat{\mathbb{Z}})$, with modular pair.

$$(95)\quad \begin{cases} B'_{\rho,\sigma} = \left\{ \begin{pmatrix} \mu & \lambda \\ 0 & 1 \end{pmatrix} \mid \mu \in \widehat{\mathbb{Z}}^*,\ \mu \equiv 1\ (3),\ \lambda \in \widehat{\mathbb{Z}} \right\} \\ G'_{\rho,\sigma} = \mathrm{Gl}'(2, \widehat{\mathbb{Z}}) = \{u \in \mathrm{Gl}(2, \widehat{\mathbb{Z}}) \mid \det u \equiv 1\ (3)\} \\ Z'_\rho = \{c\rho + d \mid c^2 + cd + d^2 \in \widehat{\mathbb{Z}}^*\} \quad {}^{c,d \in \widehat{\mathbb{Z}},\ \text{such that}} \\ Z'_\sigma = T'_\sigma \cdot \{1, \tau_\infty\}, \quad \text{where} \quad \begin{cases} T'_\sigma \overset{\text{dfn}}{=} \{t\sigma + u \mid t, u \in \widehat{\mathbb{Z}},\ t^2 + u^2 \in \widehat{\mathbb{Z}}^*\} \\ \tau_\infty = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \end{cases} \\ S_0 M'_{\rho,\sigma} = S_0 M'_{\rho,\sigma}(\widehat{\mathbb{Z}})/\Sigma_0 \quad \text{with notations §48} \\ M'_{\rho,\sigma} = M'_{\rho,\sigma}(\widehat{\mathbb{Z}})/\Sigma_0 \quad \text{———————} \\ S_0\Gamma'_{\rho,\sigma} = S_0\mathcal{T}'_{\rho,\sigma}|S\mathcal{T}'_{\rho,\sigma}(\widehat{\mathbb{Z}})/\Sigma_0 \quad \text{———————} \\ \Gamma_{\rho,\sigma} = \mathcal{T}'_{\rho,\sigma}(\widehat{\mathbb{Z}})/\Sigma_0 \quad \text{———————} \end{cases}$$

† [p. 522]

Uncertainty retained from cancelled passages.

140-3-s523-D001 $M_{\rho,\sigma}|M_\rho$ with image $L_0^{\mathcal{G}}$ the the image factors through a $*$ i.e. factors homomorphism

[scan 524]

[p. 565]

III

Let us consider the extension $\mathrm{Gl}(2, \mathbb{Z})^\wedge$ of $\mathcal{G}_{0,3}^{\dagger\wedge}$ by $\{\pm 1\}$:

(1)

$$\begin{array}{ccccccccc} & & & & 1 & & 1 & & \\ & & & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & \{\pm 1\} & \longrightarrow & \mathrm{Sl}(2, \widehat{\mathbb{Z}}) & \longrightarrow & \mathcal{G}_{0,3}^{+\wedge} & \longrightarrow & 1 \\ & & \| & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & \{\pm 1\} & \longrightarrow & \mathrm{Gl}(2, \widehat{\mathbb{Z}}) & \longrightarrow & \mathcal{G}_{0,3}^{\wedge} & \longrightarrow & 1 \\ & & & & \downarrow & & \downarrow & & \\ & & & & \{\pm 1\} & \longrightarrow & \{\pm 1\} & & \\ & & & & \downarrow & & \downarrow & & \\ & & & & 1 & & 1 & & \end{array}$$

We are interested in the group $\dagger N_{1,1}^{\sim}$, defined extension of $M_{0,3}^{\sim}$ by $\{\pm 1\}$, as the inverse image of the extension $\mathrm{Gl}(2, \widehat{\mathbb{Z}})$ of $\mathrm{Gl}(2, \widehat{\mathbb{Z}})/\pm 1$ by $\{\pm 1\}$, via the canonical homomorphism

$$(2)\qquad M_{0,3}^{\sim} \longrightarrow \mathrm{Gl}(2, \widehat{\mathbb{Z}})' = \mathrm{Gl}(2, \widehat{\mathbb{Z}})/\pm 1$$

(cf. §45 page…), so that we find a homomorphism of exact sequences,
†
(3)

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \dagger\{\pm 1\} & \text{—} & N^{\sim}_{1,1}|N^{\sim}_{0,3} & \text{——} & M^{\sim}_{0,3} & \longrightarrow & 1 \\ & & \| & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & \{\pm 1\} & \longrightarrow & \mathrm{Gl}(2,\widehat{\mathbb{Z}}) & \longrightarrow & \mathrm{Gl}(2,\widehat{\mathbb{Z}})' & \text{—} & 1 \\ & & & & \dagger & & \dagger & & \\ & & & & \dagger & & \dagger & & \end{array}$$

We find, † a as the inverse image of $\mathcal{G}^{+\wedge}_{0,3} \subset M^{\sim}_{0,3}$ in $N_{1,1}$, a $\mathrm{Gl}(2,\mathbb{Z})^{\wedge}$, deduc- [m1181]

Marginal note. [m1181] N.B. $\mathcal{T}_{1,1} \simeq \mathrm{Gl}(2,\mathbb{Z})_{\wedge}|\mathrm{Gl}(2,\mathbb{Z})$. [p. 523]

Uncertainty retained from cancelled passages.

140-3-s524-D002 +
140-3-s524-D003 M
140-3-s524-D006 $\tau_{\infty}(\epsilon_0) = \epsilon_0^{-1}$
140-3-s524-D007 $M_{0,3}$
140-3-s524-D008 ↓
140-3-s524-D009 ↓
140-3-s524-D010 1
140-3-s524-D011 1
140-3-s524-D012 $\mathcal{T}^{\wedge}_{1,1}$

[scan 525]

ing a homomorphism

$$(4) \qquad \mathrm{Sl}(2,\mathbb{Z})^{\wedge}|\mathrm{Gl}(2,\mathbb{Z})^{\wedge} \longrightarrow \dagger N^{\sim}_{1,1}$$

whose restriction to $\mathrm{Sl}(2,\mathbb{Z})^{\wedge}$ fits into a canonical exact sequence
(5)

$$1 \text{ —— } \underbrace{\mathrm{Sl}(2,\mathbb{Z})^{\wedge}}_{\mathcal{T}^{+\wedge}_{1,1}} \longrightarrow N^{\sim}_{1,1} \longrightarrow \underset{\Gamma^{\sim}_{0,3}}{\Gamma^{\sim}_{\mathbb{Q}}} \text{ —— } 1$$

which in turn fits into the diagram
(6)

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \mathrm{Sl}(2,\mathbb{Z})^{\wedge} & \longrightarrow & N_{1,1}^{\sim} & \longrightarrow & \Gamma_{\mathbb{Q}}^{\sim} & \longrightarrow & 1 \\ & & \downarrow \text{epimorphism} & & \downarrow \text{epimorphism} & & \chi \downarrow \text{epimorphism} & & \\ 1 & \longrightarrow & \mathrm{Sl}(2,\widehat{\mathbb{Z}}) & \longrightarrow & \mathrm{Gl}(2,\widehat{\mathbb{Z}}) & \longrightarrow & \widehat{\mathbb{Z}}^{*} & \longrightarrow & 1 \end{array}$$

We can write

$$(7) \qquad M_{0,3}^{\sim} \simeq \overbrace{M_{0,3}^{\sim}(0)}^{\simeq\Gamma_{\mathbb{Q}}^{\sim}} \cdot \overbrace{\mathcal{G}_{0,3}^{+\wedge}}^{\simeq \mathrm{Sl}(2,\mathbb{Z})^{\wedge\prime} \simeq \mathrm{Sl}(2,\mathbb{Z})^{\wedge}/\pm 1}$$

(semidirect)

in terms of this decomposition, and taking into account that we have a canonical homomorphism

$$(8) \qquad M_{0,3}^{\sim}(0) \longrightarrow \mathrm{Gl}(2,\widehat{\mathbb{Z}})^{\sharp|\prime}$$

that|which taking the|there † restriction of (2) to $M_{0,3}^{\sim}(0)$, we find a canonical isomorphism [p. 524]

Uncertainty retained from cancelled passages.
140-3-s525-D001 M
140-3-s525-D003 homomorphism composed of

[scan 526]
[p. 566]

$$(9) \qquad N_{1,1}^{\sim} \simeq \underbrace{N_{1,1}^{\sim}(0)}_{M_{0,3}^{\sim}(0) \simeq \Gamma_{\mathbb{Q}}^{\sim}} . \mathrm{Sl}(2,\mathbb{Z})^{\wedge}.$$

Thus we have an operation of $N_{1,1}^{\sim}(0) \simeq \Gamma_{\mathbb{Q}}^{\sim}$ on $\mathrm{Sl}(2,\mathbb{Z})^{\wedge}$ which I would like to make explicit, let us use|using the † cartesian diagram

(10)

$$\begin{array}{ccccc} \mathrm{Sl}(2,\mathbb{Z}|\widehat{\mathbb{Z}})^{\wedge} & \longrightarrow & \mathcal{G}_{0,3}^{+\wedge} & \simeq & \mathrm{Sl}(2,\mathbb{Z})^{\wedge}/\pm 1 \\ \downarrow & \text{cartesian} & \text{induced by (2)} & \searrow & \downarrow q|q' \text{ canonical} \\ \mathrm{Sl}(2,\widehat{\mathbb{Z}}) & & \longrightarrow & & \mathrm{Sl}(2,\widehat{\mathbb{Z}})/\pm 1 \end{array}$$

An element $\tilde{\nu}|\tilde{\gamma}$ of $\Gamma_{\mathbb{Q}}^{\sim}$ corresponds to a pair †

$$(11) \qquad (\alpha,\beta) \in \dagger \quad \underbrace{\widehat{\Pi}_{0,3}^{\dagger\prime}}_{\{1,\tau_{\infty}\}.\widehat{\Pi}_{0,3}} \times \widehat{\Pi}_{0,3}$$

satisfying † the loop equation

$$(12) \qquad \alpha\rho(\alpha)^{-1} = \beta\sigma(\beta)^{-1}\ell_1^{\nu} \qquad (\nu \in \widehat{\mathbb{Z}})$$

which operates on $\mathcal{G}_{0,3}^{+\wedge} \simeq \mathrm{Sl}(2,\mathbb{Z})^{\wedge\prime}$ by

$$(13) \qquad \begin{cases} u(\rho) = \mathrm{int}(\alpha)\rho = \zeta\rho, & \zeta = \alpha\rho(\alpha)^{-1}, \\ u(\sigma) = \mathrm{int}(\beta)\sigma = \eta\sigma, & \eta = \beta\sigma(\beta)^{-1} \end{cases}$$

which is compatible with the operation † denoted $\underline{u}|\underline{\alpha}$ on $\mathrm{Sl}(2,\widehat{\mathbb{Z}})$ induced by $\mathrm{int}(\underline{u})$, where $\underline{u} \in \mathrm{Gl}(2,\widehat{\mathbb{Z}})$ is the element [p. 525]

Uncertainty retained from cancelled passages.
140-3-s526-D002 isomorphism
140-3-s526-D003 of elements
140-3-s526-D004 $\mathcal{G}_{0,3}$
140-3-s526-D005 $\wedge|'$
140-3-s526-D006 of|to
140-3-s526-D008 side [illeg. 2]

[scan 527]

defined as the image of u under (8), more precisely

$$(14)\qquad \tilde{u}|\underline{u} = i_{T_0}|i_{\Gamma_0}(\mu), \qquad \text{where } \mu = \chi(u) \text{ is the multiplier, and}$$

$$(15)\qquad \begin{cases} i_{T_0}|i_{\Gamma_0} : G_m \longrightarrow \mathrm{Gl}(2,\widehat{\mathbb{Z}}) \\ \qquad \mu \longmapsto \begin{pmatrix} \mu & 0 \\ 0 & 1 \end{pmatrix}. \end{cases}$$

We will have, more precisely

$$(16)\qquad \begin{cases} \underline{u}(\underline{\rho}) = \underline{\zeta}(\rho) & \text{where } \underline{\zeta} = \underline{\alpha}\rho(\underline{\alpha})^{-1} \\ \underline{u}(\underline{\sigma}) = \underline{\eta}(\sigma) & \text{where } \underline{\eta} = \varepsilon_{\varphi|\rho}\underline{\beta}\sigma(\underline{\beta})^{-1} \end{cases}$$

m1182

Marginal note. m1182 the underlined ones denote the elements of $\mathrm{Gl}(2,\widehat{\mathbb{Z}})$ — $\underline{\alpha}$ and $\underline{\beta}$ are not determined except up to|except up to a sign, but $\underline{\zeta}, \underline{\eta}$ are unambiguous|uniquely determined.

with, as a reminder

$$(17)\qquad \varepsilon = \varepsilon_2(\mu) \in \pm 1 \qquad \begin{cases} \varepsilon = +1 & \text{iff } \mu \equiv 1 \ (4) \\ \varepsilon = -1 & \text{iff } \mu \equiv -1 \ (4) \end{cases}$$

Of course, $\mathrm{Sl}(2,\mathbb{Z})^\wedge$ is generated topologically by ρ and σ, and we identify ± 1 with the central|center subgroup of $\mathrm{Sl}(2,\mathbb{Z})$ hence of $\mathrm{Sl}(2,\mathbb{Z})^\wedge$, we will also write $-g$ for the product of $g \in \mathrm{Sl}(2,\mathbb{Z})^\wedge$ by -1. We will therefore have in $\mathrm{Sl}(2,\mathbb{Z})^\wedge$

$$(18)\qquad \begin{cases} u(\rho) = \dagger\, \mathrm{int}(\tilde{\alpha})(\rho) = \zeta\rho \\ u(\sigma|\rho) = \varepsilon\, \mathrm{int}(\tilde{\beta})(\sigma) = \eta\sigma \end{cases}$$

and

with

$$(19)\qquad \begin{cases} \zeta = \tilde{\alpha}\rho(\tilde{\alpha})^{-1}, \qquad \eta = \varepsilon\tilde{\beta}\sigma(\tilde{\beta})^{-1} \\ \varepsilon = \varepsilon_2(\mu) \in \{\pm 1\} \subset \mathrm{Sl}(2,\mathbb{Z})^\wedge, \end{cases}$$

[p. 526]

Uncertainty retained from cancelled passages.
140-3-s527-D003 ζ

[scan 528]

where now $\tilde{\alpha}$ and $\tilde{\beta}$ denote arbitrary liftings of α and β to $\mathrm{Sl}(2,\mathbb{Z})^\wedge$ — they are determined only up to sign, but $\mathrm{int}(\tilde{\alpha}), \mathrm{int}(\tilde{\beta})$, ζ and η are unambiguously determined by u. † Let us also note: [p. 567]

$$(20)\qquad u(\varepsilon_0) = \varepsilon_0^\mu \quad \dagger$$

If now we have a surjective homomorphism

$$(20) \qquad N_{1,1}^{\sim} \xrightarrow{\Psi} G$$

inducing a homomorphism

$$(21) \qquad \mathrm{Sl}(2,\mathbb{Z})^{\wedge} \xrightarrow{\text{epimorphism}} \underset{\mathrm{Im}\,\Psi}{\mathcal{G}} \subset G,$$

the images of ρ, σ in $\mathcal{G}$ (denoted by the same letters) satisfying|satisfy

$$(22) \qquad \rho^6 = \sigma^4 = 1, \qquad \rho^3 = \sigma^2 \quad \text{and hence}$$

†

$$[\rho^3, \sigma] = [\sigma^2, \rho] = 1 \quad \text{i.e. } \rho^3, \sigma^2$$

† are central
m1183

Marginal note. m1183 thus in $\mathcal{G}$ there is a canonical involutive central element, denoted -1 . .

and we can, for $u \in N_{1,1}^{\sim}(0)$, corresponding to elements $\tilde{\alpha}, \tilde{\beta}$ of $\mathrm{Sl}(2,\mathbb{Z})^{\wedge}$ (determined modulo sign), hence ζ, η, try to find an endomorphism|automorphism $u_{\mathcal{G}}$ of $\mathcal{G}$, satisfying the appropriate relations

$$(23) \qquad u_{\mathcal{G}}(\rho_{\mathcal{G}}) = \zeta_{\mathcal{G}}\rho_{\mathcal{G}}, \qquad u_{\mathcal{G}}(\sigma_{\mathcal{G}}) = \eta_{\mathcal{G}}\sigma_{\mathcal{G}},$$

and bring into play our|to the $\tilde{\alpha}$ and $\tilde{\beta}$. But we are [p. 527]

Uncertainty retained from cancelled passages.
140-3-s528-D001 now
140-3-s528-D002 for
140-3-s528-D003 which suffices
140-3-s528-D004 imply

[scan 529]

obliged, since † we do not necessarily $\tilde{\alpha} \in \mathrm{Sl}(2,\mathbb{Z})^{\wedge}$, only $\tilde{\alpha} \in \mathrm{Gl}(2,\mathbb{Z})^{\wedge}$, to bring in the element τ_∞ of $\mathrm{Gl}(2,\mathbb{Z})^{\wedge}$, and the action of its image τ in G on $\mathcal{G}$. We will have, now taking

$$(24) \qquad \begin{aligned} \varepsilon_{0\mathcal{G}}^{\dagger} &= \sigma_{\mathcal{G}}^{\dagger}\rho_{\mathcal{G}}^{\dagger}, & \varepsilon_{1\mathcal{G}} &= \rho_{\mathcal{G}}\sigma_{\mathcal{G}}\dagger = \sigma_{\mathcal{G}}(\varepsilon_{0\mathcal{G}}) = \rho_{\mathcal{G}}(\varepsilon_{0\mathcal{G}}) \\ &= -\sigma_{\mathcal{G}}^{-1}\rho_{\mathcal{G}} & &= -\rho_{\mathcal{G}}\sigma_{\mathcal{G}}^{-1} \end{aligned}$$

m1184

Marginal note. m1184 which is not the usual convention up to now...

the relations

$$(25) \qquad \tau(\sigma) = \sigma^{-1}, \qquad \tau(\rho) = \sigma(\rho^{-1})$$

I drop the subscripts $\mathcal{G}$

which imply, via (24) and (22)

$$(26) \qquad \tau(\varepsilon_0) = \varepsilon_0^{-1}, \qquad \tau(\varepsilon_1) = \varepsilon_1^{-1},$$

and we introduce the group

$$(27) \qquad \dagger\mathcal{G}^{\bullet|\prime} \overset{\text{dfn}}{=} \underset{\frac{1}{2}\text{ direct}}{\{1,\tau\}.\mathcal{G}},$$

where $\{1,\tau\}$ operates on $\mathcal{G}$ by the formulas (25).

This said, we are in a position to paraphrase constructions already made elsewhere. [p. 528]

Uncertainty retained from cancelled passages.

140-3-s529-D001 in general
140-3-s529-D002 $\mathcal{G}|\wedge$
140-3-s529-D003 $\mathcal{G}|\wedge$
140-3-s529-D004 $\mathcal{G}|\wedge$
140-3-s529-D005 $=$
140-3-s529-D006 $\{1,\tau\}$

IV [scan 530]

So let $\mathcal{G}$ be a profinite group, equipped with the|with generators $\rho,\sigma\in\mathcal{G}$ satisfying the relations [p. 568]

$$(1)\qquad \rho^6=\sigma^4=1,\qquad \rho^3=\sigma^2\quad (\overset{\text{dfn}}{=}\text{“}-1\text{”}).$$

m1186

Marginal note. m1186 This simply means that we give ourselves a surjective homomorphism

$$\mathrm{SL}(2,\mathbb{Z})^\wedge\longrightarrow\mathcal{G}$$

A particularly interesting case is the one where $\rho^3=\sigma^2=1$, to which nevertheless I would not wish to limit myself. Let τ be an automorphism of $\mathcal{G}$ satisfying

$$(2)\qquad \tau(\dagger\sigma)=\begin{matrix}\sigma^{-1}\\ =-\sigma\end{matrix},\qquad \tau(\rho)=\sigma(\rho^{-1})$$

m1187

Marginal note. m1187 Let us introduce $\dagger$ right away the equivalent element $\tau'=\sigma^{-1}\tau$ satisfying $\tau'(\rho)=\rho^{-1}$, $\tau'(\sigma)=\sigma^{-1}$ $(\tau'(\varepsilon_0)=\varepsilon_1^{-1},\tau'(\varepsilon_1)=\varepsilon_0^{-1})$.

(which means that τ_∞ “passes to the quotient” as an automorphism τ of $\mathcal{G}$), which implies

$$(3)\qquad \tau^2=1$$

$$(4)\qquad \tau(\varepsilon_0)=\varepsilon_0^{-1},\qquad \tau(\varepsilon_1)=\varepsilon_1^{-1},\qquad \text{where}$$

$$(5)\qquad \begin{matrix}\varepsilon_0=\dagger\sigma\rho\\ =-\sigma^{-1}\rho\end{matrix},\qquad \varepsilon_1=\rho\sigma=\sigma(\varepsilon_0)=\sigma^{-1}(\varepsilon_0)=\rho(\varepsilon_0).$$

m1188

Marginal note. m1188 of the formula forgotten a sign $-$, whereas we could from the outset|ask simply give|take $\varepsilon_0=\sigma_\infty|\sigma^{-1}\rho\,\ldots$

Put

$$(5)\qquad \mathcal{G}'=\{1,\tau\}\,.\,\mathcal{G},$$

so we have

$$(6)\qquad 1\longrightarrow\mathcal{G}\longrightarrow\mathcal{G}'\longrightarrow\{1,\tau\}\longrightarrow 1.$$

Let $\dagger$

$$B_{\rho,\sigma}\subset\mathrm{Aut}(\mathcal{G})\times\widehat{\mathbb{Z}}^*$$

$$(7)\qquad =\left\{(u,\mu)\;\middle|\;\begin{array}{l}u(\varepsilon_0)=\varepsilon_0^\mu\\ u(\sigma)\text{ conjugate in }\mathcal{G}\text{ to }\sigma^\mu\\ u(\rho)\qquad\qquad\qquad\qquad\mathcal{G}^\dagger\text{ to }\rho^\mu\end{array}\right\}.$$

which is obviously a subgroup. The third condition can be made explicit moreover, 529

Uncertainty retained from cancelled passages.

140-3-s530-D002 ρ
140-3-s530-D003 the
140-3-s530-D004 α
140-3-s530-D005 (7)
140-3-s530-D007 ′

since $\rho^6 = 1$ i.e. ρ^μ depends only on the image of μ in [scan 531]

$$(\mathbb{Z}/6\mathbb{Z})^* \simeq (\mathbb{Z}/2\mathbb{Z})^* \times (\mathbb{Z}/3\mathbb{Z})^* \simeq (\mathbb{Z}/3\mathbb{Z})^* \simeq \pm 1,$$

† so

$$(8) \qquad \rho^\mu = \rho^{\varepsilon_3(\mu)}$$

that for $\varepsilon_3(\mu) = 1$ $u(\rho)$ is conjugate to ρ in $\mathcal{G}$, and for $\varepsilon_3(\mu) = -1$ it is conjugate to ρ^{-1} in $\mathcal{G}$, or again to ρ by an element of $\mathcal{G}'$ – i.e. an element of the form $\alpha_1 \tau_\infty$ with $\alpha_1 \in \mathcal{G}$ (since $\tau_\infty(\rho) = \sigma(\rho^{-1})$ precisely…). We see likewise that we have

$$(9) \qquad \sigma^\mu = \sigma^{\varepsilon_2(\mu)} \quad \begin{cases} = \sigma & \text{if } \mu \equiv 1 \quad (4) \\ \dagger\sigma^{-1} = -\sigma & \text{if } \mu \equiv -1 \quad (4) \end{cases}$$

$$\overset{\dagger}{=} \varepsilon_2(\mu)\sigma$$

†

Thus:

Proposition For a|one $(u, \mu) \in \mathrm{Aut}(\dagger\mathcal{G}) \times \widehat{\mathbb{Z}}^*$ to be in $B_{\rho,\sigma}$, it is necessary and sufficient that there exist $\alpha \in \mathcal{G}'^{\dagger}_{\dagger}$, $\beta \in \mathcal{G}$ such that we have 530

Uncertainty retained from cancelled passages.

140-3-s531-D001 only [illeg. 1]
140-3-s531-D007 =
140-3-s531-D008 $\varepsilon\sigma|\sigma$
140-3-s531-D009 even|= we write such $\alpha \in$
140-3-s531-D012 $B_{\rho,\sigma}$
140-3-s531-D013 ρ, σ
140-3-s531-D014 ε_3

[scan 532]

$$(10) \qquad u(\rho) = \mathrm{int}(\alpha)(\rho), \qquad u(\sigma) = \underbrace{\varepsilon_2(\mu)}_{\varepsilon}\, \mathrm{int}(\beta)(\sigma)$$ [p. 569]

and finally

$$(11) \qquad u(\varepsilon_0) = \varepsilon_0^\mu,$$

where in addition we have † i.e. $\alpha \equiv \tau$.

We shall note that (u, μ) is uniquely determined by μ and the knowledge of α and of β, and even only by μ and

$$(13) \qquad \zeta = \alpha\rho(\alpha)^{-1}, \qquad \eta = \varepsilon_2(\mu)\beta\sigma(\beta)^{-1}$$

since we shall then have u by

$$(14) \qquad u(\rho) = \zeta\rho, \qquad u(\sigma) = \eta\sigma.$$

Moreover, in the case where

$$(15) \qquad \begin{array}{ccc} \widehat{\mathbb{Z}} & \longrightarrow & \mathcal{G} \\ n & \longmapsto & \varepsilon_0^n \end{array}$$

is injective, μ is determined as a function of u as multiplier of the action of $\dagger u$ on $L_0^{\mathcal{G}} \subset \mathcal{G}$, image of (15), so (u, μ) is determined by u so by $(\alpha, \beta, \varepsilon_4 \Leftarrow | = \varepsilon_2(\mu))$. Moreover, if $\mathcal{G}$ is a group that maps into $\mathrm{SL}(2, \widehat{\mathbb{Z}})'$ (so as to send ρ, σ to the usual elements), we have seen that α all by itself 531

Uncertainty retained from cancelled passages.

140-3-s532-D001 mod ν_3 $\varepsilon_3(\mu)$ $\Longleftrightarrow$

140-3-s532-D005 u

already determines μ, so ε, so $\dagger$ the element of $B_{\rho,\sigma}$ is determined with the help of α, β alone. [scan 533]

But what conditions must (α, β, μ) satisfy to determine an element of $B_{\rho,\sigma}$? First of all, it is necessary that there indeed exist an automorphism u (necessarily[t] unique) satisfying (14) — this is a condition on ζ, η alone.

We must next express (11). $\dagger\,\dagger$

Let us introduce

$$\varepsilon_0' = \sigma^{-1}\rho = -\varepsilon_0. \tag{*}$$

whence

$$\varepsilon_0'^{\mu} = (-1)^{\mu}\varepsilon_0^{\mu} = -\varepsilon_0^{\mu} \qquad (\text{since } \mu \equiv 1\ (2))$$

$$u(\varepsilon_0') = -u(\varepsilon_0)$$

since $u(-1) = -1$ cf. below.

so the relation (11) is also equivalent to

$$u(\varepsilon_0') = \varepsilon_0'^{\mu}, \tag{**|*}$$

which, by the same old calculation, is written thus 532

Uncertainty retained from cancelled passages.

140-3-s533-D002 determines|all

140-3-s533-D004 we

140-3-s533-D005 so $\sigma(\eta^{-1}\zeta)\,\underbrace{\sigma\rho}_{\varepsilon_0}$ $\sigma(\varepsilon_0)|\sigma\rho$ ζ We shall get from this|We also have [illeg. 1] u σ $\varepsilon(\beta\sigma(\beta)^{-1})$ ρ u

[scan 534]

$$\eta^{-1}\zeta = \varepsilon_1'^{2\nu}\dagger, \quad \text{where } \nu = (\mu-1)/2, \quad \varepsilon_1' = -\varepsilon_1 = \rho\sigma^{-1} \qquad \text{[p. 570]}$$

we have

$$\varepsilon_1'^{2\nu} = (-1)^{2\nu}\varepsilon_1^{2\nu} = \varepsilon_1^{2\nu} = l_1^{\dagger\nu},$$

so the relation at the top takes the form

$$\boxed{\zeta = \eta l_1^{\nu}}, \quad \text{where } \nu = (\mu-1)/2. \tag{16}$$

We still have to prove the

Lemma Let u be an automorphism of $\mathcal{G}$ such that $u(\sigma)$ is conjugate in $\mathcal{G}$ to σ or σ^{-1}, then

$$\dagger \qquad \boxed{u(-1) = -1} \tag{17}$$

We will indeed have $u(\sigma) = \mathrm{int}(\alpha)(\sigma')$, where $\sigma' = \sigma$ or σ^{-1}, hence

$$u(\sigma^2) = \mathrm{int}(\alpha)(\sigma'^2) = \mathrm{int}(\alpha)(-1) = -1, \quad \text{OK.}$$

It remains to|: express finally that $u(\rho) = \mathrm{int}(\alpha)\rho$ is conjugate to ρ^μ, and $u(\sigma)$ to σ^μ, and this is † automatic for σ, since

$$u(\sigma) = \mathrm{int}(\beta)(\varepsilon_2(\mu)\sigma) = \mathrm{int}(\beta)(\sigma^\mu) \quad \text{(by (9))},$$

and for ρ this means that we have

$$\dagger \qquad \alpha \equiv \tau^{\nu_3(\mu)} \ (\mathcal{G}) \tag{17}$$

which we also write $\tau^{\nu_3(\mu)}\alpha \in \mathcal{G}$ where $\nu_3(\mu) \in \mathbb{Z}/2$ is the expression additive form|form of $\varepsilon_3(\mu)$

—

So we have found

Corollary The map

$$\begin{aligned} B_{\rho,\sigma} &\longrightarrow \mathcal{G} \times \mathcal{G} \times \widehat{\mathbb{Z}}^* \\ (u, \mu) &\longmapsto ([u,\rho], [u,\sigma], \mu) \end{aligned} \tag{18}$$

is injective, and it has as its image the set $\Sigma^{\mathrm{eff}}_{\rho,\sigma}$ of triples (ζ, η, μ) satisfying the following conditions:

1°) $\zeta\rho$ conjugate to ρ^μ " $\rho^{\varepsilon_3(\mu)}$ in $\mathcal{G}$

2°) $\eta|\zeta\sigma$ ——————— $\sigma^\mu = \sigma^{\varepsilon_2(\mu)}$ in $\mathcal{G}$

533

Uncertainty retained from cancelled passages.

140-3-s534-D001 $= l_1^\nu$

140-3-s534-D002 2

140-3-s534-D003 $u(-1) = -1 | u(\sigma)^2 = -1$

140-3-s534-D005 done by the conditions

140-3-s534-D006 $\alpha \equiv \tau^{\varepsilon_3(\mu)}$

3°) We have the loop relation (16). [scan 535]

4°) ∃ automorphism u of $\mathcal{G}$, satisfying $u(\rho) = \zeta\rho$, $u(\sigma) = \eta\sigma$.

†

The|The conjunction of the conditions 1°), 2°), 3°) are equivalent|is equivalent to this: † this one:

$$\begin{aligned} (A) \qquad & \exists (\alpha, \beta) \in \mathcal{G}' \times \mathcal{G}, \quad \text{with } \alpha \in \mathcal{G} \text{ sss|iff } \mu \equiv 1 \ (3) \\ & \text{i.e. } \dagger\ \alpha \bmod \mathcal{G} = \tau^{\nu_3(\mu)}, \quad \text{and such that we have} \\ & \zeta = \alpha\rho(\alpha)^{-1}, \qquad \eta = \varepsilon_2(\mu). \end{aligned}$$

Let us consider the homomorphism of|of the groups

$$(19)\qquad \begin{array}{ccc} L_0 & \xrightarrow{i_{L_0,B}} & B_{\rho,\sigma} \\ l & \longmapsto & (\mathrm{int}(l),1) \end{array}$$

It is injective if we always assume

$$(20)\qquad L_0 \cap \underline{\mathrm{Centr}}(\mathcal{G}) = \{1\},$$

its image is normal|invariant for we have

$$(21)\qquad u(i_{L_0,B}(l)) = i_{L_0,B}(u(l)) = i_{L_0,B}(l^{\mu}),$$

where μ is the multiplier. We set

$$(22)\qquad \gamma_{\rho,\sigma} = B_{\rho,\sigma}/\operatorname{Im} L_0,$$

whence canonical exact sequence

$$(23)\qquad 1 \longrightarrow L_0 \longrightarrow B_{\rho,\sigma} \longrightarrow \gamma_{\rho,\sigma} \longrightarrow 1$$

and we have a homomorphism (deduced from $(u,\mu) \mapsto \mu$ by factoring through $\gamma_{\rho,\sigma}$)

$$(24)\qquad \gamma_{\rho,\sigma} \xrightarrow{\chi_\gamma} \widehat{\mathbb{Z}}^*$$

whose kernel is denoted $\gamma^+_{\rho,\sigma}$: †

$$(25)\qquad 1 \longrightarrow \gamma^+_{\rho,\sigma} \longrightarrow \gamma_{\rho,\sigma} \longrightarrow \widehat{\mathbb{Z}}^* \longrightarrow 1$$

534

Uncertainty retained from cancelled passages.

140-3-s535-D001 is necessary and sufficient

140-3-s535-D002 this one completes them

140-3-s535-D003 α

140-3-s535-D005 of

[scan 536]
[p. 571]

Let us consider|Let us study the composite homomorphisms

$$(26)\qquad \begin{array}{ccccc} \gamma_{\rho,\sigma} & \xrightarrow{\chi_\gamma} & \widehat{\mathbb{Z}}^* & \xrightarrow{\varepsilon_2} & \pm 1 \\ \gamma_{\rho,\sigma} & \xrightarrow{\chi_\gamma} & \widehat{\mathbb{Z}}^* & \xrightarrow{\varepsilon_3} & \pm 1 \end{array}$$

and let

$$(27)\qquad \begin{cases} \gamma'_{\rho,\sigma}, \gamma''_{\rho,\sigma} \subset \gamma_{\rho,\sigma} & \text{subgroups of index 2,} \\ & \text{defined by} \\ \gamma'_{\rho,\sigma} = \operatorname{Ker}(\varepsilon_3 \circ \chi_\gamma), & \gamma''_{\rho,\sigma} = \operatorname{Ker} \varepsilon_2 \circ \chi_\gamma \end{cases}$$

Now let

$$(28)\qquad G_{\rho,\sigma} = \left(\underbrace{B_{\rho,\sigma}\cdot\mathcal{G}}_{\substack{(\frac{1}{2}\text{ direct})\\ \simeq S_0 G_{\rho,\sigma}}} \right) / L_0,$$

where

$$(29)\qquad \begin{array}{ccc} L_0 & \xrightarrow{i_{L_0,B}} & \underbrace{B_{\rho,\sigma}\cdot G_{\rho,\sigma}|\mathcal{G}} \\ l & \longmapsto & \mathrm{int}(l)\cdot l^{-1}, \end{array}$$

whose image is invariant. We have a commutative diagram

$$(30)\qquad \begin{array}{ccccc} & & \mathcal{G} & & \\ & \nearrow & & \searrow & \\ L_0 & & & & G_{\rho,\sigma} \\ & \searrow & & \nearrow & \\ & & B_{\rho,\sigma} & & \end{array}$$

— $\mathrm{Ker}(\mathcal{G} \longrightarrow G_{\rho,\sigma}) \simeq \mathrm{Ker}(L_0 \longrightarrow B_{\rho,\sigma}] \simeq L_0 \cap \mathrm{Centr}(\mathcal{G})$ is trivial according to|according to (20), which we are going to assume in what follows. We have a homomorphism of exact sequences

$$(31)\qquad \begin{array}{ccccccccc} 1 & \to & L_0 & \to & B_{\rho,\sigma} & \xrightarrow{\delta_B} & \gamma_{\rho,\sigma} & \to & 1 \\ & & \cup & & \downarrow & & \| & & \\ 1 & \to & \mathcal{G} & \to & G_{\rho,\sigma} & \xrightarrow{\delta_G} & \gamma_{\rho,\sigma} & & \\ & & & & & \searrow^{\chi_G} & \downarrow{\scriptstyle \chi_\gamma} & & \\ & & & & [& & \widehat{\mathbb{Z}}^* & &] \end{array}$$

535

Let us set [scan 537]

$$(32)\qquad \begin{cases} Z_\rho = \mathrm{Centr}_{G_{\rho,\sigma}}(\cdot\rho) \\ Z_\sigma = \mathrm{Centr}_{G_{\rho,\sigma}}(\sigma) \end{cases}$$



then the homomorphisms induced by δ_G

$$(33)\qquad Z_\rho \xrightarrow{\delta_\rho} \gamma_{\rho,\sigma}\ , \qquad Z_\sigma \xrightarrow{\delta_\sigma} \gamma_{\rho,\sigma}$$

have some images which contain $\gamma'_{\rho,\sigma}$ and $\gamma''_{\rho,\sigma}$ respectively, †

†

†

536

Uncertainty retained from cancelled passages.

140-3-s537-D002 sss|iff to

140-3-s537-D003 or equal sss|iff for this reduced|reduced sss|iff the satisfied but that it is satisfied if SL theorem can it happen case

140-3-s537-D014 for Z_σ $\widehat{\gamma}_{\rho,\sigma}|\gamma_{\rho,\sigma}$ surjections|surjective Z_ρ $\mathcal{G}^{+\wedge}_{0,3}$ reduced

[scan 538]

[p. 572]

Proposition 1°) The image $\delta_\rho(Z_\rho)$ contains $\gamma'_{\rho,\sigma}$, and is † reduced to $\gamma'_{\rho,\sigma}$ iff|iff ρ and ρ^{-1} not conjugate in $\mathcal{G}$ — therefore it is equal to $\gamma_{\rho,\sigma}$ i.e. $\delta_\rho : Z_\rho \longrightarrow \gamma_{\rho,\sigma}$ epimorphism iff|iff ρ and ρ^{-1} are conjugate in $\mathcal{G}$.

2°) The image $\delta_\sigma(Z_\sigma)$ contains $\gamma''_{\rho,\sigma}$, and it is reduced to $\gamma''_{\rho,\sigma}$ iff|iff σ and $\sigma^{-1} = -\sigma$ not conjugate in $\mathcal{G}$ — therefore|otherwise it is equal to $\gamma_{\rho,\sigma}$ i.e. $\delta_\sigma : Z_\sigma \longrightarrow \gamma_{\rho,\sigma}$ epimorphism iff|iff σ and $-\sigma^{-1}|\sigma^{-1} = -\sigma$ are conjugate in $\mathcal{G}$.

Corollary 1. If $\mathcal{G}$ contains as a quotient

$$SL(2,\mathbb{Z}/3\mathbb{Z})', \quad \text{with} \quad \rho = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \qquad \sigma = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

(mod ± 1), then the image of δ_ρ is $\gamma'_{\rho,\sigma}$.

Indeed, we see that ρ and $\pm\rho^{-1}$ are not conjugate in $\mathcal{G}$, for they are not in $SL(2,\mathbb{Z}/3\mathbb{Z})'$ — † Let us check this: ρ and $-\rho^{-1}$ already have distinct traces 1 and -1, $\in(\mathbb{Z}/3\mathbb{Z})^*$, they are not conjugate.

We have

$$(\underbrace{\tau\sigma}_{\tau'_\infty})(\rho) = \rho^{-1},$$

therefore the elements of $GL(2,\mathbb{Z}/3\mathbb{Z})$ which conjugate ρ to ρ^{-1} are of the form $f\tau'_\infty$, where $f = c|b\rho+d$, we want $\det(f\tau'_\infty) = 1$ i.e. $\det(f) = -1$ (since $\det(\tau'_\infty) = -1$), i.e. $c^2+cd+d^2 = -1$, impossible. 537

Uncertainty retained from cancelled passages.

140-3-s538-D001 eq

140-3-s538-D002 for example for

[scan 539]

Corollary 2 If $\sigma^2 = 1$, then $Z_\sigma \longrightarrow \gamma_{\rho,\sigma}$ is surjective. If On the other hand $\mathcal{G}$ contains as a quotient $SL(2,\mathbb{Z}/4\mathbb{Z})$ (without ′ !), with ρ,σ as usual, then

$$\mathrm{Im}(\delta_\sigma) = \gamma''_{\rho,\sigma}.$$

We must see that σ and $\sigma^{-1} = -\sigma$ not conjugate in $SL(2,\mathbb{Z}/4\mathbb{Z})$, using that $\tau_\infty(\sigma) = \sigma^{-1}$, therefore the elements|elements of $GL(2,\mathbb{Z}/4\mathbb{Z})$ which conjugate σ to σ^{-1} are the $g = f\tau_\infty$ with $f = t\sigma + u|n$, † and therefore that $\det(g) = 1$ means $\det f = -1$ i.e. $t^2 + u|n^2 = -1$, impossible.

† 538

Uncertainty retained from cancelled passages.

140-3-s539-D003 with

140-3-s539-D005 $\mathcal{M}|M$ $Z_\rho|N_\rho$ $G_{\rho,\sigma}|\gamma_{\rho,\sigma}$ }|)

[scan 540]

Let us note that by (2), (4), we have [p. 573]

$$(35) \qquad \tau_B \overset{\text{dfn}}{=} (\tau,-1) \in B_{\rho,\sigma}$$

And obviously $\tau_B \notin L_0 = B_{\rho,\sigma} \cap \mathcal{G}$. So we have an embedding

$$(36) \qquad \begin{cases} \mathcal{G}' = \{1,\tau\}.\mathcal{G} \hookrightarrow G_{\rho,\sigma} \\ \tau \longmapsto \tau_B \\ g \longmapsto g \quad \text{if } g \in \mathcal{G} \end{cases},$$

Marginal note. m1201 N.B. $\mathcal{G}'$ is a quotient of $\mathrm{GL}(2,\mathbf{Z})^\wedge$, and τ corresponds there to the image of the element $\tau_\infty = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ of $\mathrm{GL}(2,\mathbf{Z})$.

therefore the automorphism τ of $\mathcal{G}$ will be considered as induced by † $\mathrm{int}(\tau_B)$, where τ_B is itself considered as an element of $G_{\rho,\sigma}$.

† In connection with the equality

$$\tau_B(\sigma) = \sigma^{-1} \quad (= -\sigma)$$

which shows that τ_B normalizes the subgroup L_σ generated by σ (isomorphic to a quotient of $\mathbf{Z}/4\mathbf{Z}$), † therefore normalizes $Z_\sigma = \mathrm{Centr}_{G_{\rho,\sigma}}(L_\sigma)$, † let us introduce

$$(37) \qquad N_\sigma = \underset{(1/2 \text{ direct})}{\{1, \tau_B\}.Z_\sigma},$$

and the canonical homomorphism

$$(38) \qquad \begin{cases} N_\sigma \xrightarrow{i_\sigma} G_{\rho,\sigma} \\ \tau_B \longmapsto \tau_B, \qquad\qquad x \longmapsto x \quad \text{if } x \in Z_\sigma \end{cases} .$$

The image of this homomorphism is none other than the normalizer of σ in $G_{\rho,\sigma}$, for a † $g \in G_{\rho,\sigma}$ which normalizes L_σ must transform σ into a generator of L_σ, i.e. into σ or σ^{-1}, in the first case we will have $g \in Z_\sigma$, in the second $g(\sigma) = \tau_B(\sigma)$ i.e. $\tau_B^{-1} g \in Z_\sigma$ i.e. $g \in \tau_B Z_\sigma$. The homomorphism (38) is injective 539

Uncertainty retained from cancelled passages.

140-3-s540-D001 the ele

140-3-s540-D002 [illeg. 2]

140-3-s540-D003 group

140-3-s540-D004 then let us introduce

140-3-s540-D006 automorphism

iff $\tau_B \notin Z_\sigma$ i.e. $\sigma \neq \sigma^{-1}$ i.e. $\sigma^2 \neq 1$ i.e. "-1" $\neq 1$. [scan 541]

While we are at it, let us take advantage of the formula

$$\tau_B(\rho) = \sigma(\rho^{-1})$$

to introduce

$$(39) \qquad \tau' \underset{=}{=} \sigma^{-1}\tau \in \mathcal{G}' \subset G_{\rho,\sigma}$$

(which corresponds to the element

$$(40) \qquad \tau'_\infty = \sigma_\infty^{-1}\tau_\infty = -\sigma_\infty\tau_\infty = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \in \underbrace{\mathrm{GL}(2,\mathbf{Z})}_{\| \; \mathcal{T}_{1,1}} \Big),$$

† we will therefore have

$$(40) \qquad \tau'^2 = 1, \qquad \tau'(\rho) = \rho^{-1}.$$

Thus τ' normalizes the subgroup L_ρ (isomorphic to a quotient of $\mathbf{Z}/6\mathbf{Z}$) generated by ρ, therefore it normalizes the centralizer Z_ρ of L_ρ, and we can construct m1203

Marginal note. m1203 N.B. We have at the same time $\tau'(\sigma) = \sigma^{-1}$, therefore we can consider τ' as an element of N_σ, and † we have $N_\sigma = \{1, \tau'\}.Z_\sigma$ (which is 1/2 direct)

$$N_\rho \overset{\text{dfn}}{=} \underset{\text{(semidirect)}}{\{1, \tau'\}.Z_\rho} \tag{41}$$

and † † canonical homomorphism

$$\begin{cases} N_\rho \xrightarrow{\ i_\rho\ } G_{\rho,\sigma} \\ \tau' \longmapsto \tau', \qquad\qquad x \longmapsto x \quad \text{for } x \in Z_\rho \end{cases} . \tag{42}$$

The image of this homomorphism is the normalizer of L_ρ in $G_{\rho,\sigma}$ — we see this as a moment ago for i_σ, noting that ρ, ρ^{-1} are the only generators of L_ρ. The homomorphism i_ρ is injective iff $\tau' \notin Z_\rho$ i.e. † $\rho \neq \rho^{-1}$ i.e. $\rho^2 \neq 1$ i.e. $\rho \neq -1$ († N.B. $\rho = -1$ † † † that ρ commutes with σ, and that $\mathcal{G}$ is a quotient of $\mathbf{Z}/12\mathbf{Z}$, with ρ corresponding to 2 and σ to 3, but here $2\rho = 0$ therefore $\mathcal{G}$ is a quotient of $\mathbf{Z}/4\mathbf{Z}$ with ρ corresponding to 2, σ to $3 = -1$ ...).

m1204

Marginal note. m1204

$$\begin{cases} N_\rho \xrightarrow{\ \delta_\rho = \delta_G \circ i_\rho\ } \gamma_{\rho,\sigma} \\ N_\sigma \xrightarrow{\ \delta_\sigma = \delta_G \circ i_\sigma\ } \gamma_{\rho,\sigma} \\ \qquad \text{epimorphisms} \end{cases} \tag{43}$$

m1205

Marginal note. m1205 "We have done what was needed" to have epimorphisms φ.

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Uncertainty retained from cancelled passages.

140-3-s541-D002 which is
140-3-s541-D005 [illeg. 1]
140-3-s541-D006 a
140-3-s541-D007 of the
140-3-s541-D009 iff
140-3-s541-D010 in this case
140-3-s541-D011 implies
140-3-s541-D012 equivalent
140-3-s541-D013 in fact

† Let us situate|Let †

[scan 542]
[p. 574]

$$\begin{gathered} S\mathfrak{g}|g^*_{\rho,\sigma} \subset \mathcal{G}' \times \mathcal{G} \times \widehat{\mathbb{Z}}^* \\ \| \\ \left\{ (\alpha, \beta, \mu) \;\middle|\; \begin{array}{l} \alpha \equiv \tau^{\nu_3(\mu)} \bmod \mathcal{G} \quad \text{i.e. } \chi(\alpha) = \varepsilon_3(\mu) \\ \underbrace{\alpha\rho(\alpha)^{-1}}_{\zeta} = \underbrace{\varepsilon_2(\mu)\beta\sigma(\beta)^{-1}}_{\eta} \ell_1^\nu, \quad \text{where} \end{array} \right\} \end{gathered} \tag{44}$$

$$\nu = (\mu-1)/2, \qquad \underline{(\ell_1 = \varepsilon_1^2)}$$

†
whence

(45)

$$\left\{\begin{array}{l} \qquad\qquad \Sigma_{\rho,\sigma} \subset \mathcal{G}\times\mathcal{G}\times\widehat{\mathbb{Z}}^* \\ \qquad\qquad\qquad \| \quad \text{dfn} \\ S\mathfrak{g}|g^*_{\rho,\sigma} \longrightarrow \left\{(\zeta,\eta,\mu) \;\middle|\; \begin{array}{l} \zeta\rho \text{ conjugation in } \mathcal{G} : \rho^\mu \\ \eta|\zeta\sigma \qquad\qquad\qquad \sigma^\mu \\ \zeta = \dagger\eta\ell_1^\nu \end{array}\right. \\ \\ (\alpha,\beta,\mu) \longmapsto \left(\overbrace{\alpha\rho(\alpha)^{-1}}^{[\alpha,\rho]}, \varepsilon_2(\mu)\overbrace{\beta\sigma(\beta)^{-1}}^{[\beta,\sigma]}, \mu\right). \end{array}\right.$$

(where $\nu = (\mu-1)/2$ We denote by $\mathfrak{g}|g^{*\,\text{eff}}_{\rho,\sigma}$ the inverse image of $\Sigma^{\text{eff}}_{\rho,\sigma}$ (cf. corollary page 570). $\simeq B_{\rho,\sigma}$. By construction we thus find a surjective map [m1207]

Marginal note. [m1207] (46bis)

$$S\mathfrak{g}|g^{\text{eff}}_{\rho,\sigma} \xrightarrow{\text{epimorphism}} \Sigma^{\text{eff}}_{\rho,\sigma}$$

We regard $S\mathfrak{g}|g^*_{\rho,\sigma}$ as contained in a set that is larger

(47)
$$\begin{array}{c} \mathfrak{g}|g^*_{\rho,\sigma} \subset G_{\rho,\sigma}\times G_{\rho,\sigma}\times\widehat{\mathbb{Z}}^* \\ \| \\ \left\{(\alpha,\beta,\mu) \;\middle|\; \begin{array}{l} \chi(\alpha) \equiv \varepsilon_3(\dagger\mu)\chi(\beta); \\ \underbrace{\alpha\rho(\alpha)^{-1}}_{=\alpha(\rho)\rho^{-1}=\zeta} = \underbrace{\varepsilon_2(\mu)\beta\sigma(\beta)^{-1}}_{=\varepsilon_2(\mu)\beta(\sigma)\sigma^{-1}=\eta}\ell_1^\nu \end{array}\right\} \end{array}$$

so that we will have

(48)
$$\begin{aligned} S\mathfrak{g}|g^*_{\rho,\sigma} &= \mathfrak{g}|g^*_{\rho,\sigma} \cap (\mathcal{G}'\times\mathcal{G}\times\widehat{\mathbb{Z}}^*), \\ &= \left\{(\alpha,\beta,\mu) \in S\mathfrak{g}|Sg^*_{\rho,\sigma} \;\middle|\; \begin{array}{l} \delta_\sigma(\beta) = 1\dagger \\ \delta_\rho(\alpha\tau^{\nu_3(\mu)}) = 1 \end{array}\right\} \end{aligned}$$

541

Uncertainty retained from cancelled passages.
140-3-s542-D001 (43)
140-3-s542-D002 $\mathfrak{g}|g$
140-3-s542-D003 36 $\mathfrak{g}|g$ $*$ $\mathfrak{g}|g$ $\exists$
140-3-s542-D004 $\beta\sigma(\beta)^{-1}$
140-3-s542-D006 $\chi(\mu)|\chi(\beta)$
140-3-s542-D007 }

and we again have a canonical map [scan 543]

(49)

$$\begin{cases} \mathfrak{g}|g^{*}_{\rho,\sigma} \longrightarrow \Sigma^{*}_{\rho,\sigma} \\ (\alpha,\beta,\mu) \longmapsto ([\alpha,\rho],\varepsilon_2(\mu)[\beta,\sigma],\mu). \end{cases}$$

† We denote by $\mathfrak{g}|g^{\mathrm{eff}}_{\rho,\sigma}$ the inverse image of $\Sigma^{\mathrm{eff}}_{\rho,\sigma}$, and we find

(50)
$$\begin{aligned} S\mathfrak{g}|g^{\mathrm{eff}}_{\rho,\sigma} &= \mathfrak{g}|g^{\mathrm{eff}}_{\rho,\sigma} \cap (\mathcal{G}' \times \mathcal{G} \times \widehat{\mathbb{Z}}^{*}) \\ &= \left\{ (\alpha,\beta,\mu) \in \mathfrak{g}|g^{\mathrm{eff}}_{\rho,\sigma} \;\middle|\; \begin{array}{l} \delta_\sigma(\beta) = 1 \\ \delta_\rho|\delta_B(\alpha\tau'^{\nu_3(\mu)}) = 1 \end{array} \right\} \end{aligned}$$

We have a section of

(51)

$$\begin{array}{ccc} \mathfrak{g}|g^{\mathrm{eff}}_{\rho,\sigma} & \longrightarrow & \Sigma^{\mathrm{eff}}_{\rho,\sigma} \\ & \underline{\text{section}} \nwarrow & \uparrow \sim \\ & & B_{\rho,\sigma} \end{array} \qquad \begin{array}{c} \zeta = [u,\rho],[u,\sigma],\chi_B(u) \\ \overset{\eta=}{} \quad \overset{\mu=}{} \\ \uparrow \\ u\dagger \end{array}$$

given by

(52)

$$u\dagger \longmapsto (\dagger u,\ u\tau'^{\nu_2(\mu)},\ \mu = \chi_B(u))$$

since we will have (setting $\varepsilon = \varepsilon_2(\mu) \overset{\mathrm{dfn}}{=} | = (-1)^{\nu_2}$.)

$$[u,\rho] = u\rho(u)^{-1} = \zeta$$

$$\begin{aligned} \varepsilon\left[u\tau'^{\nu_2(\mu)=\nu_2},\sigma\right] &= \varepsilon u\tau'^{\nu_2}\sigma(u\tau|\tau'^{\nu_2})^{-1} \\ &= \varepsilon u\Big(\underbrace{\tau|\tau'^{\nu_2}\sigma(\tau|\tau'^{\nu_2})^{-1}}_{=\varepsilon}\Big)\sigma(u)^{-1} = \varepsilon u|u\sigma(u)^{-1} = \eta \end{aligned}$$

OK. So † the set † $\mathfrak{g}_{\rho,\sigma}(u,\mu)|g^{(u,\mu)}_{\rho,\sigma}$ of the $(\alpha,\beta,\mu)-$ giving the image

$$(\zeta,\eta,\mu)\dagger[|(\,[u,\rho],[u,\sigma],\overset{\mu=}{\chi_B}(u))$$

will be exactly 542

Uncertainty retained from cancelled passages.

140-3-s543-D001 whose i
140-3-s543-D002 $\varphi|*$
140-3-s543-D003 $\varphi|*$
140-3-s543-D004 $\alpha =$
140-3-s543-D005 the set|the
140-3-s543-D006 of the

$\dagger\ \mathfrak{g}_{\rho,\sigma}(u,\mu)|g^{(u,\mu)}_{\rho,\sigma} =$

$$(53)\qquad \left\{ \left(\overbrace{ua_0^{-1}}^{\alpha}, \overbrace{u\tau'^{\nu_2}b_0^{-1}}^{\beta|b}, \mu \right) \;\middle|\; a_0 \in Z_\rho,\ b_0 \in Z_\sigma,\ \nu_2 = \nu_2(\mu) \right\}$$

$$(= (ua_0^{-1}, ub^{-1}, \mu)\} \quad \text{where} \quad b = b_0\tau'^{\nu_2}$$

which we also write

$$(54)\qquad \mathfrak{g}_{\rho,\sigma}(u,\mu)|g^{(u,\mu)}_{\rho,\sigma} = \left\{ \left(\underbrace{ua_0^{-1}}_{\alpha}, \underbrace{u\, i_\sigma(b)}_{\beta}, \mu \right) \;\middle|\; a_0 \in Z_\rho,\quad \underset{\text{(cf. (37)(38))}}{b \in N_\sigma}\ ,\ \boxed{\varepsilon_\sigma(b) = \varepsilon_2(\mu)} \right\}$$

where $\varepsilon_\sigma : N_\sigma \longrightarrow \{\pm 1\}$ is defined in (37) as the canonical homomorphism fitting into the exact sequence

$$(55)$$

$$1 \longrightarrow Z_\sigma \longrightarrow N_\sigma \xrightarrow{\varepsilon_\sigma} \underbrace{\{\pm 1\}}_{\simeq\{1,\tau'\}} \longrightarrow 1.$$

If we look for the set of elements of $S\mathfrak{g}|g^{\text{eff}}_{\rho,\sigma}$ over (ζ,η,μ), we must in addition impose the conditions

$$(56)\qquad \dagger ua_0^{-1}\tau'^{\nu_3(\mu)} \in \mathcal{G},\quad u\, i_\sigma(b) \in \mathcal{G}$$

i.e.

$$\delta_G(ua_0^{-1}\tau'^{\nu_3}) = 1,\quad \delta_G(u\, i_\sigma(b)) = 1$$

that is, again

$$(57)\qquad \delta_G(u) = \underbrace{\delta_\sigma(b)}_{\text{cf. (43)}} = \underbrace{\delta_\rho}_{\text{cf. (43)}}(\tau'^{\nu_3}.a_0).$$

Let us therefore set

$$a = \tau'^{\nu_3}.a_0 \quad \text{in|in } N_\rho$$

(thus $a_0 = \tau'^{\nu_3}a$) we see that

$$(54)\qquad \mathfrak{g}_{\rho,\sigma}(u,\mu)|g^{(u,\mu)}_{\rho,\sigma} = \left\{ (ua^{-1}\tau'^{\nu_3}, ub^{-1}, \mu) \;\middle|\; \begin{array}{l} a \in N_\rho,\ b \in N_\sigma \text{ such that} \\ \varepsilon_\rho(a) = \varepsilon_3(\mu),\ \varepsilon_\sigma(b) = \varepsilon_2(\mu) \\ \delta_\rho(a) = \delta_\sigma(b) = \delta_G(u) \end{array} \right\}$$

Let us therefore introduce the group

$$(55)\qquad \dagger\mathcal{M}|M_{\rho,\sigma}[0] = B_{\rho,\sigma} \times_{\gamma_{\rho,\sigma}} N_\rho \times_{\gamma_{\rho,\sigma}} N_\sigma.$$

and the composite homomorphisms

$$(56)$$

$$\begin{cases} \dagger\mathcal{M}|M_{\rho,\sigma}[0] \longrightarrow N_\rho \xrightarrow{\varepsilon_\rho} \{\pm 1\} \\ \dagger\mathcal{M}|M_{\rho,\sigma}[0] \longrightarrow N_\sigma \xrightarrow{\varepsilon_\sigma} \{\pm 1\} \end{cases}$$

(again denoted $\varepsilon_\rho, \varepsilon_\sigma$ by abuse of notation)

Uncertainty retained from cancelled passages.
140-3-s544-D001 the
140-3-s544-D002 $\alpha =$
140-3-s544-D003 S_0
140-3-s544-D004 S_0
140-3-s544-D005 S_0
140-3-s544-D006 denoted

and [scan 545]
(57)

$$\dagger M_{\rho,\sigma}[0] \xrightarrow{\delta_{SM}} \gamma_{\rho,\sigma} \xrightarrow{\chi_\gamma} \widehat{\mathbb{Z}}^*$$

i.e. χ_{SM}, (or simply|simply χ)
consider
(58)

$$\dagger M_{\rho,\sigma}[0] \xrightarrow{\varepsilon_{\rho,\sigma}=(\varepsilon_\sigma,\varepsilon_\rho)} \{\pm 1\} \times \{\pm 1\}$$

and
(59)

$$\dagger M_{\rho,\sigma}[0] \xrightarrow{\varepsilon_{2,3}: x \longmapsto (\varepsilon_2(\chi(x)),\varepsilon_3(\chi(x)))} \{\pm 1\} \times \{\pm 1\}$$

and consider the subgroup of coincidences of the two preceding homomorphisms

$$(60)\quad \dagger M^{\mathrm{eff}}_{\rho,\sigma}[0] \overset{\mathrm{dfn}}{=} \{x \in S_0\{M_{\rho,\sigma}\} \mid \varepsilon_{\rho,\sigma}(x) = \varepsilon_{2,3}(x)\}$$
$$= \left\{ \begin{pmatrix} u & , & a & , & b \\ \cup & & \cup & & \cup \\ B_{\rho,\sigma} & & N_\rho & & Z_\sigma \end{pmatrix} \;\middle|\; \begin{array}{r} \delta_B(u) = \delta_\rho(a) = \delta_\sigma(b) \quad \text{i.e. } \Delta \in \gamma_{\rho,\sigma} \\ \varepsilon_\rho(a) = \dagger\varepsilon_3(\chi_\gamma(\Delta)), \quad \varepsilon_\sigma(b) = \varepsilon_2(\chi_\gamma(\Delta)) \end{array} \right.$$

We have a natural map
(61)

$$M^{\mathrm{eff}}_{\rho,\sigma}[0] \longrightarrow \mathcal{G}' \times \mathcal{G} \times \widehat{\mathbb{Z}}^*$$

$$(u,a,b) \longmapsto (\alpha,\beta,\mu)$$

$$\alpha = ua^{-1}\tau'^{\nu_\rho(a)}$$
, with $$\beta = ub^{-1}$$
$$\mu = \chi_B(u) = \chi_\rho(a) = \chi_\sigma(b)$$
m1212

Marginal note. m1212 N.B. we have identified $a \in N_\rho$, $b \in N_\sigma$ with their images $i_\rho(a)$, $i_\sigma(b)$ in $G_{\rho,\sigma}$.
m1213

Marginal note. m1213 N.B. We have set † $\nu_\rho(a) \in \mathbb{Z}/2$ such that $(-1)^{\nu_\rho(a)} = \varepsilon_\rho(a)$

and we have done what was needed for it to be injective and have as image $S\mathfrak{g}|g^{\mathrm{eff}}_{\rho,\sigma}$, hence induce a bijection
(62)

$$M^{\mathrm{eff}}_{\rho,\sigma}[0] \xrightarrow{\sim} S\mathfrak{g}|g^{\mathrm{eff}}_{\rho,\sigma}.$$

Remark. In fact the formulas for (61) define even a map
(62)

$$M_{\rho,\sigma}[0] \longrightarrow \mathcal{G}' \times \mathcal{G} \times \widehat{\mathbb{Z}}^*$$

and for a $(u,a,b) \in M_{\rho,\sigma}[0]$, its image (α,β,μ) is related to $u \in B_{\rho,\sigma}$ by the formulas 544

Uncertainty retained from cancelled passages.

140-3-s545-D001 S_0
140-3-s545-D002 S_0
140-3-s545-D003 S_0
140-3-s545-D004 S_0
140-3-s545-D005 $\varepsilon_3(\chi_B(u))$
140-3-s545-D006 (61)

[scan 546]
[p. 576]

$$(63)\quad \begin{cases} u(\rho) = \mathrm{int}(\alpha)(\rho) & (\mathcal{G}\text{—conjugate to } \rho^{\varepsilon'},\ \text{where } \varepsilon' = \varepsilon_\rho(a)) \\ u(\sigma) = \varepsilon\,\mathrm{int}(\beta)(\sigma) & (\mathcal{G}\text{-conjugate to } \sigma^{\varepsilon},\ \text{where } \varepsilon = \varepsilon_\sigma(b)) \\ \quad \dagger \end{cases}$$

(but without necessarily having $\varepsilon' = \varepsilon_3(\mu)$, $\varepsilon = \varepsilon_2(\mu)$).

This suggests a definition of a group a little bigger than $B_{\rho,\sigma}$, in which $B_{\rho,\sigma}$ will be included as a normal|$\triangleleft$ subgroup of index|of index $1, 2$ or 4 — and which will have the advantage, in the "good cases", of being the group of adelic points of a suitable group scheme over $\mathbb{Z}$.

On the other hand, to † eliminate the sign ε in (63), (which is present even if $(u,a,b) \in M_{\rho,\sigma}[0]^{\mathrm{eff}}$), I feel like replacing β by

†

$$(64)\quad \beta' = \tau'^{\nu}\beta \qquad (\text{where } \nu \in \mathbb{Z}/2\mathbb{Z} \text{ related by } (-1)^\nu = \varepsilon)$$

So that we will have

$$(65)\quad \dagger u(\sigma) = \mathrm{int}(\beta')(\sigma) \quad \text{i.e.} \quad [u,\sigma] = \beta'\sigma(\beta')^{-1}.$$

This is related to the convention that a|which a $u \in \Gamma^{\sim}_{\mathbb{Q}}$, which we were specifying by the pair $(\alpha,\beta) \in \widehat{\Pi}'_{0,3} \times \widehat{\Pi}_{0,3}$, will henceforth be specified by the pair $(\alpha,\beta') \in \widehat{\Pi}_{0,3} \times \widehat{\Pi}'_{0,3}$, with

$$(66)\quad \chi(\alpha) = \varepsilon_3(\mu), \quad \chi(\beta') = \varepsilon_2(\mu) \qquad (\text{where } \mu = \chi(u) \text{ is the multiplier})$$

which for the operation of u on †

$$\mathcal{G}^{+\wedge}_{0,3} = \mathrm{SL}(2,\mathbb{Z})^{\wedge\textcircled{1}} = \mathrm{SL}(2,\mathbb{Z})^{\wedge}/\pm 1$$

does not in fact change the particular formulas:

$$(67)\quad \begin{cases} u(\rho) = \mathrm{int}(\alpha).\rho, \quad u(\sigma) = \mathrm{int}(\beta)\sigma \quad (= \mathrm{int}(\beta').\sigma) \\ \alpha\rho(\alpha)^{-1} = \beta\underset{\dagger}{\sigma}(\beta)^{-1}\ell_1^\nu \quad (= \beta'\sigma(\beta')^{-1}\ell_1^\nu) \end{cases} \qquad \nu = \frac{\mu-1}{2}$$

545

Uncertainty retained from cancelled passages.

140-3-s546-D001 α

140-3-s546-D002 kill

140-3-s546-D003 β' is related

140-3-s546-D004 $\varepsilon|\chi$

140-3-s546-D006 $\mathcal{G}_{0,3}^{+\wedge}$

140-3-s546-D007 ε

[scan 547]

but for the operation in †
$\mathcal{T}_{1,1}^{+\wedge} \simeq \mathrm{SL}(2,\mathbb{Z})^\wedge$ itself gives simpler formulas, namely, instead of

$$(68)\quad \begin{cases} u(\rho) = \mathrm{int}(\alpha)\rho \qquad u(\sigma) = \varepsilon.\,\mathrm{int}(\beta)\sigma \\ \alpha\rho(\alpha)^{-1} = \varepsilon\beta\sigma(\beta)^{-1}\ell_1^\nu \end{cases},$$

the nicer formulas

$$(69)\quad \begin{cases} u(\rho) = \mathrm{int}(\alpha)\rho, \qquad u(\sigma) = \mathrm{int}(\beta')(\sigma) \\ \alpha\rho(\alpha)^{-1} = \beta'\sigma(\beta')^{-1}\ell_1^\nu \end{cases},$$

where the sign $\varepsilon = \varepsilon_2(\mu)$ has disappeared. It appears in the formulas

$$(70)\quad \chi(\alpha) = \varepsilon_3(\mu), \quad \chi(\beta') = \varepsilon_2(\mu).$$

It seems to me I have enough to be able to take up again "freehand" the preceding constructions and complete them with more adequate notations.

The group $\mathcal{G}'$ quotient of $\mathcal{T}_{1,1}^\wedge \simeq \mathrm{GL}(2,\mathbb{Z})^\wedge$ is moreover taking on increasing importance in our calculations, it is the one that deserves † the simple notation $\mathcal{G}$, the closed subgroup generated by ρ, σ becoming $\mathcal{G}^+$ — it corresponds to $\mathcal{T}_{1,1}^{+\wedge} \simeq \mathrm{SL}(2, \widehat{\mathbb{Z}}|\mathbb{Z})^\wedge$.

We will simply assume that $\tau \notin \mathcal{G}^+$ (τ being the image of τ_∞), or what amounts to the same, that

$$(71)\quad \tau' \notin \mathcal{G}^+$$

546

Uncertainty retained from cancelled passages.

140-3-s547-D001 $\mathcal{G} = \mathcal{T}_{1,1}$

140-3-s547-D002 also

[scan 548]
[p. 577]

It is ultimately this element, which normalizes both L_ρ and L_σ † via the remarkable formulas

$$(*) \quad \tau'(\rho) = \rho^{-1}, \qquad \tau'(\sigma) = \sigma^{-1}$$

that takes precedence over τ, which satisfies

$$(**) \quad \tau(\varepsilon_0) = \varepsilon_0^{-1}, \qquad \tau(\varepsilon_1) = \varepsilon_1^{-1} \quad \text{and even } \tau(\sigma) = \sigma^{-1}$$

(which is obviously very virtuous !), but whose action on ρ

$$\tau(\rho) = \sigma(\rho^{-1})$$

is less convenient.
m1217

Marginal note. m1217 Nevertheless τ' has over τ the serious disadvantage of not being an element of the group $\widehat{\Pi}'_{0,3} = \{1, \tau\}.\widehat{\Pi}_{0,3}$, where we have the rigidity relation $\widehat{\Pi}'^{\sigma}_{0,3} = \widehat{\Pi}'^{\rho}_{0,3} = \{1\}$ — whereas in $\{1, \tau'\}.\widehat{\Pi}_{0,3}$, the centralizer of σ is not reduced to 1, but is equal to $\{1, \tau'\}$ Moreover, τ' has the disadvantage of not being in $\mathcal{M}^{!\prime}_{0,3}$ ...

$$\overline{\text{V}}$$

Preliminary data

1°) So let $\mathcal{G}$ be a profinite group, equipped with a surjective homomorphism

$$= \mid \rightarrow$$

(1)

$$\mathcal{T}^{*\mid+\wedge}_{1,1} \simeq \mathrm{GL}(2, \mathbb{Z})^\wedge \xrightarrow{\text{epimorphism}} \mathcal{G}$$

,

which we will express by giving the three topological generators $\rho, \sigma, \tau' \in \mathcal{G}$, satisfying

$$(2) \quad \rho^6 = \sigma^4 = \tau'^2 = 1, \quad \begin{matrix} \rho^3 = \sigma^2 \\ (\text{denoted } -1) \end{matrix}, \quad \tau'\rho\tau'^{-1} = \rho^{-1}, \quad \tau'\sigma\tau'^{-1} = \sigma^{-1}.$$

We denote by $\mathcal{G}^+$ the closed subgroup generated by ρ, σ, which is thus the image of

$$\mathcal{T}^{+\wedge}_{1,1} \simeq \mathrm{SL}(2, \mathbb{Z})^\wedge,$$

547

Uncertainty retained from cancelled passages.
140-3-s548-D001 by

[scan 549]

and we assume that $\mathcal{G}^+ \neq \mathcal{G}$, hence

$$\begin{matrix} \mathcal{G} \simeq \{1, \tau'\} \,.\, \mathcal{G}^+ \\ (\tfrac{1}{2} \text{ direct}) \end{matrix} \tag{3}$$

and we have a canonical exact sequence

$$1 \longrightarrow \mathcal{G}^+ \longrightarrow \mathcal{G} \xrightarrow{\varepsilon_{\mathcal{G}}} \{\pm 1\} \longrightarrow 1 \tag{4}$$

We also denote by $\nu_{\mathcal{G}}$ the composite

$$\begin{array}{ccccc} & & \overset{\nu_{\mathcal{G}}}{\curvearrowright} & & \\ \mathcal{G} & \xrightarrow{\varepsilon_{\mathcal{G}}} & \{\pm 1\} & \xleftarrow{\sim} & \mathbb{Z}/2\mathbb{Z} \\ & & (-1)^{\nu} & \longleftarrow\!\shortmid & \nu \end{array} \quad . \tag{5}$$

In general, we will denote (by abuse of language) by the same letter elements of $\mathrm{Gl}(2,\mathbb{Z})$ — such as $\varepsilon_0, \varepsilon_1, \tau = \tau_\infty, \tau' = \tau'_\infty, \sigma = \sigma_\infty, \rho = \rho_\infty$ etc — and their images in $\mathcal{G}$.

The groups $Z_{\rho,\sigma}$, $\gamma_{\rho,\sigma}$

2°) Let

$$\begin{array}{c} Z_{\rho,\sigma} \ \subset \ \mathrm{Aut}(\mathcal{G}^+) \times \widehat{\mathbb{Z}}^* \times \{\pm 1\} \times \{\pm 1\} \\ \| \ \text{dfn} \\ \left\{ (u,\mu,\varepsilon,\varepsilon') \ \middle| \ \begin{array}{l} u(\varepsilon_0) = \varepsilon_0^{\mu} \\ u(\rho) \text{ conjugate in } \mathcal{G}^+ \text{ to } \rho^{\varepsilon'} \\ u(\sigma) \ \text{_____} \text{ to } \sigma^{\varepsilon} \end{array} \right\} \end{array} \tag{6}$$

This is a subgroup of the product group.

Let us put

$$\begin{cases} L_\rho = \text{subgroup of } \mathcal{G} \text{ generated by } \rho \\ L_\sigma = \text{_____} \ \sigma \end{cases} \tag{7}$$

so that we have epimorphisms

$$\begin{cases} \mathbb{Z}/6\mathbb{Z} \xrightarrow{\text{epimorphism}} L_\rho & n \longmapsto \rho^n \\ \mathbb{Z}/4\mathbb{Z} \longrightarrow L_\sigma & n \longmapsto \sigma^n \end{cases} , \tag{8}$$

[p. 548]

[scan 550]

and a commutative diagram of homomorphisms [p. 578]

$$\begin{array}{ccccc} & & L_\rho & & \\ & \nearrow & & \searrow & \\ \{\pm 1\} & & & & \mathcal{G}^+ \\ & \searrow & & \nearrow & \\ & & L_\sigma & & \end{array} \tag{9}$$

— If $\mathcal{G}$ is "big enough", we will moreover have that $\{\pm 1\} \hookrightarrow \mathcal{G}^+$ i.e. "-1" $\neq 1$ in $\mathcal{G}^+$, and even

$$L_\rho \cap L_\sigma = \{\pm 1\} \ .$$

The conditions (6) therefore imply that

$$\begin{cases} u(L_\rho) \ \mathcal{G}^+\text{-conjugate to } L_\rho \\ u(L_\sigma) \ \mathcal{G}^+\text{-conjugate to } L_\sigma \end{cases} \tag{10}$$

If $\mathcal{G}^+$ is "big enough" so that ρ and ρ^{-1}, σ and σ^{-1} are not conjugate in it, then an element $(u,\mu,\varepsilon,\varepsilon')$ of $Z_{\rho,\sigma}$ is known from the sole knowledge of (u,μ). If on the other hand the homomorphism

$$\begin{array}{ccc} \widehat{\mathbb{Z}} & \longrightarrow & \mathcal{G}^+ \\ n & \longmapsto & \varepsilon_0^n \end{array} \tag{11}$$

is injective, then μ is known when we know u. Thus if $\mathcal{G}^+$ is big enough to satisfy the three conditions considered (for example so that we have a "modular homomorphism"

$$\mathcal{G} \longrightarrow \mathrm{Gl}(2, \widehat{\mathbb{Z}}) \tag{12}$$

(sending ρ, σ, τ' to the elements with the same names), then an element of $Z_{\rho,\sigma}$ is known from the sole [p. 549]

[scan 551]

knowledge of $u \in \mathrm{Aut}(\mathcal{G}^+)$.

We have a homomorphism

$$\begin{aligned} L_0 &\xrightarrow{i_{L_0,Z}} Z_{\rho,\sigma} \\ \ell &\longmapsto (\mathrm{int}(\ell), 1, 1, 1) \end{aligned} \tag{13}$$

whose image is normal, the kernel is none other than $L_0 \cap \mathrm{Cent}(\mathcal{G}^+)$. Let us note that we have

$$L_0 \cap \mathrm{Cent}(\mathcal{G}^+) \subset L_0 \cap \mathrm{Cent}_{\mathcal{G}^+}(\rho) \subset L_0 \cap L_1 \tag{14}$$

for if $\ell \in L_0$ and ρ commute, i.e. if $\ell = \rho(\ell)$, since $\rho(L_0) = L_1$, we will have $\ell \in L_1$. Thus if $L_0 \cap L_1 = \{1\}$ (which is the case if (12) exists), and even if there exists only

$$\mathcal{G} \longrightarrow \mathrm{Gl}(2, \widehat{\mathbb{Z}})' \qquad \text{modular}, \tag{12'}$$

then we will have

$$L_0 \cap \mathrm{Cent}(\mathcal{G}^+) = \{1\}, \tag{15}$$

and (13) will be injective. We are going to assume this, to fix ideas.

We put

$$\gamma_{\rho,\sigma} = Z_{\rho,\sigma} / \mathrm{Im}\, L_0, \tag{16}$$

so that we will have a canonical exact sequence

$$1 \longrightarrow L_0 \xrightarrow{i_{L_0,Z}} Z_{\rho,\sigma} \xrightarrow{\delta_Z} \gamma_{\rho,\sigma} \longrightarrow 1 \quad . \tag{17}$$

The canonical projection homomorphisms $(u, \mu, \varepsilon, \varepsilon') \mapsto \mu$, $\varepsilon, \varepsilon'$ factor through $\gamma_{\rho,\sigma}$ and give canonical homomorphisms [p. 550]

[scan 552]

[p. 579]

$$\begin{cases} \gamma_{\rho,\sigma} \xrightarrow{\chi_\gamma} \widehat{\mathbb{Z}}^* \\ \gamma_{\rho,\sigma} \xrightarrow{\varepsilon_\gamma} \{\pm 1\} \qquad \text{i.e.} \qquad \gamma_{\rho,\sigma} \xrightarrow{(\chi_\gamma, \varepsilon_\gamma, \varepsilon'_\gamma)} \widehat{\mathbb{Z}}^* \times \{\pm 1\} \times \{\pm 1\} \\ \gamma_{\rho,\sigma} \xrightarrow{\varepsilon'_\gamma} \{\pm 1\} \end{cases} \tag{18}$$

the composites of these with $Z_{\rho,\sigma} \to \gamma_{\rho,\sigma}$ are denoted

$$\chi_Z = \chi_\gamma \circ \delta_Z, \quad \varepsilon_Z = \varepsilon_\gamma \circ \delta_Z, \quad \varepsilon'_Z = \varepsilon'_\gamma \circ \delta_Z, \tag{19}$$

so that

$$\chi_Z(u, \mu, \varepsilon, \varepsilon') = \mu, \quad \varepsilon_Z(u, \mu, \varepsilon, \varepsilon') = \varepsilon, \quad \varepsilon'_Z(u, \mu, \varepsilon, \varepsilon') = \varepsilon' \tag{20}$$

We would feel like denoting by $\gamma^{0|\circ}_{\rho,\sigma}$ the kernel of $(\varepsilon_\gamma, \varepsilon'_\gamma)$, by $\gamma^{+}_{\rho,\sigma}$ that of χ_γ, by $\gamma^{0|\circ+}_{\rho,\sigma}$ their intersection, and introducing under the notations $Z^{0|\circ}_{\rho,\sigma}$, $Z^{+}_{\rho,\sigma}$, $Z^{0|\circ+}_{\rho,\sigma}$ their inverse images in $Z_{\rho,\sigma}$, i.e. the kernels resp.^{t}|resp. of $(\varepsilon_Z, \varepsilon'_Z)$, of χ_Z and of $(\chi_Z, \varepsilon_Z, \varepsilon'_Z)$.

The groups $S_0G_{\rho,\sigma}$, $G_{\rho,\sigma}$

3°) † We have a tautological homomorphism

$$\begin{array}{ccc} Z_{\rho,\sigma} & \longrightarrow & \mathrm{Aut}(\mathcal{G}^+) \\ (u,\mu,\varepsilon,\varepsilon') & \longmapsto & u \end{array} \quad , \tag{21}$$

thus $Z_{\rho,\sigma}$ acts|operates on $\mathcal{G}^+$, and we can therefore consider the $\frac{1}{2}$ direct product

$$S_0G_{\rho,\sigma} = Z_{\rho,\sigma} \,.\, \mathcal{G}^+ \ . \quad (\tfrac{1}{2}\text{-direct}) \tag{22}$$

[p. 551]

Uncertainty retained from cancelled passages.

140-3-s552-D001 operates

[scan 553]

We have a homomorphism

$$\begin{array}{ccc} L_0 & \xrightarrow{i_{L_0,SG}} & S_0G_{\rho,\sigma} = Z_{\rho,\sigma} \,.\, \mathcal{G}^+ \\ \ell & \longmapsto & i_{L_0,Z}(\ell) \,.\, \ell^{-1} \end{array} \tag{23}$$

whose image is again normal, † We † put

$$G_{\rho,\sigma} = \underbrace{S_0G_{\rho,\sigma}}_{Z_{\rho,\sigma}.\mathcal{G}^+} / \operatorname{Im} L_0 \ \dagger, \tag{24}$$

so that we will have an exact sequence

$$1 \longrightarrow L_0 \xrightarrow{i_{L_0,SG}} S_0G_{\rho,\sigma} \longrightarrow G_{\rho,\sigma} \longrightarrow 1 \quad , \tag{25}$$

and a diagram commutative of homomorphisms

$$\begin{array}{ccccc} & & Z_{\rho,\sigma} & & \\ & \overset{i_{L_0,Z}}{\nearrow} & & \overset{i_Z}{\searrow} & \\ L_0 & & & & G_{\rho,\sigma} \\ & \underset{i_{L_0,\mathcal{G}}}{\searrow} & & \underset{i_{\mathcal{G}^+}}{\nearrow} & \\ & & \mathcal{G}^+ & & \end{array} \tag{26}$$

(a priori $\mathrm{Ker}(\mathcal{G}^+ \to G_{\rho,\sigma}) \simeq \mathrm{Ker}(L_0 \to Z_{\rho,\sigma}) \simeq L_0 \cap \mathrm{Cent}(\mathcal{G}^+)$, but by hypothesis (15) it is reduced to $\{1\}$), fitting into a commutative diagram of exact sequences

$$\begin{array}{ccccccccc} 1 & \longrightarrow & L_0 & \xrightarrow{i_{L_0,Z}} & Z_{\rho,\sigma} & \xrightarrow{\delta_Z} & \gamma_{\rho,\sigma} & \longrightarrow & 1 \\ & & \downarrow{\scriptstyle i_{L_0,\mathcal{G}}} & & \downarrow & & \| & & \\ 1 & \longrightarrow & \mathcal{G}^+ & \xrightarrow{i_{\mathcal{G}^+}} & G_{\rho,\sigma} & \xrightarrow{\delta_G} & \gamma_{\rho,\sigma} & \longrightarrow & 1 \end{array} \ . \tag{27}$$

We will put

$$\chi_G = \chi_\gamma \circ \delta_G, \quad \varepsilon_G = \varepsilon_\gamma \circ \delta_G, \quad \varepsilon'_\gamma \circ \delta_G \tag{28}$$

[p. 552]

Uncertainty retained from cancelled passages.
140-3-s553-D001 [illeg. 1]
140-3-s553-D002 is stable
140-3-s553-D003 $Z_\rho|\mathcal{G}$

[scan 554]
[p. 580]

hence a homomorphism

$$G_{\rho,\sigma} \xrightarrow{(\chi_G,\varepsilon_G,\varepsilon'_G)} \widehat{\mathbb{Z}}^* \times \{\pm 1\} \times \{\pm 1\} \ , \tag{29}$$

leading us to introduce $G^+_{\rho,\sigma}$, $G^0_{\rho,\sigma}$, $G^{+0}_{\rho,\sigma}$ as usual.

The injection $i_{\mathcal{G}^+} : \mathcal{G}^+ \hookrightarrow G_{\rho,\sigma}$ can be extended to an embedding † (denoted $i_{\mathcal{G}}$)

$$i_{\mathcal{G}} : \mathcal{G} \hookrightarrow G_{\rho,\sigma} \ , \tag{30}$$

which I define by its value on † $\tau = \sigma\tau'$, by

$$i_{\mathcal{G}}(\tau) = \dagger\, i_Z(\tau_Z) \tag{31}$$

m1222

Marginal note. m1222 If τ_γ denotes the image of $\tau = \tau_G \in G_{\rho,\sigma}$ in $\gamma_{\rho,\sigma} = G_{\rho,\sigma}/\mathcal{G}^+$, $\mathcal{G}$ is identified with the inverse image in $G_{\rho,\sigma}$ of the subgroup $\{1, \tau_\gamma\}$ of $\gamma_{\rho,\sigma}$.

$$\begin{cases} \tau_Z \in Z_{\rho,\sigma} \quad \text{defined as} \\ \tau_Z = \left(\underbrace{\mathrm{int}(\tau)|\mathcal{G}^+}_{\in \mathrm{Aut}(\mathcal{G}^+)}, \underset{\widehat{\mathbb{Z}}^*}{-1}, \underset{\{\pm 1\}}{-1}, \underset{\{\pm 1\}}{-1} \right). \end{cases} \tag{32}$$

From now on we identify $\mathcal{G}$ with a subgroup of $G_{\rho,\sigma}$, and we denote by the same letter elements of $\mathcal{G}$ (such as $\tau, \tau', \rho, \sigma, \varepsilon_0, \varepsilon_1$, etc) and their images in $G_{\rho,\sigma}$. m1223

Marginal note. m1223 Note that $\tau'_G \notin Z_{\rho,\sigma}$ on the other hand we have † $\tau_G \in Z_{\rho,\sigma}$ †.

We will moreover have

$$\chi_G|\mathcal{G} = \varepsilon_G|\mathcal{G} = \varepsilon'_G|\mathcal{G} = \varepsilon_{\mathcal{G}} : \begin{cases} \text{trivial on } \mathcal{G}^+ \\ -1 \text{ on } \mathcal{G}^- = \mathcal{G} \setminus \mathcal{G}^+. \end{cases} \tag{32bis}$$

m1224

Marginal note. m1224 The groups $Z_\rho, Z_\sigma, N_\rho, N_\sigma, SN_\rho, SN_\sigma$...

4°) We note that the element τ' of † $\mathcal{G} \subset G_{\rho,\sigma}$ normalizes L_ρ and L_σ, therefore also normalizes

$$\begin{cases} Z_\rho \overset{\text{dfn}}{=} \mathrm{Cent}_{G_{\rho,\sigma}}(\rho) = \mathrm{Cent}_{G_{\rho,\sigma}}(L_\rho), \\ Z_\sigma \overset{\text{dfn}}{=} \mathrm{Cent}_{G_{\rho,\sigma}}(\sigma) = \mathrm{Cent}_{G_{\rho,\sigma}}(L_\sigma) \end{cases} \tag{33}$$

[p. 553]

Uncertainty retained from cancelled passages.

140-3-s554-D001 [illeg. 1]
140-3-s554-D002 τ'
140-3-s554-D004 [*illeg.* 1]
140-3-s554-D005 τ'_G
140-3-s554-D006 [illeg. 1]
140-3-s554-D008 $G_{\rho,\sigma}$

[scan 555]

thus $\{1,\tau'\}$ acts on these groups, and we can consider

$$\begin{cases} N_\rho = \{1,\tau'\}\,.\,Z_\rho, & \dagger \\ N_\sigma = \{1,\tau'\}\,.\,Z_\sigma \end{cases} \qquad \text{(products } \tfrac{1}{2} \text{ direct).} \tag{34}$$

We have canonical exact sequences

$$\begin{array}{c} 1 \longrightarrow Z_\rho | L_\rho \longrightarrow N_\rho \xrightarrow{\varepsilon_\rho} \{\pm 1\} \longrightarrow 1 \\ \\ 1 \longrightarrow Z_\sigma \longrightarrow N_\sigma \xrightarrow{\varepsilon_\sigma} \{\pm 1\} \longrightarrow 1 \end{array} \quad , \tag{35}$$

and a commutative diagram of canonical homomorphisms

$$\begin{array}{ccccc} & & \mathbb{D}_\rho & \longrightarrow & N_\rho \\ \{1,\tau'\}\times\{\pm 1\} & \nearrow & & & \searrow^{i_\rho} \\ & \searrow & & & G_{\rho,\sigma} \\ & & \mathbb{D}_\sigma & \longrightarrow & N_\sigma \\ & & & & \searrow_{i_\sigma} \end{array} \tag{36}$$

where

$$\begin{cases} \mathbb{D}_\rho \simeq \mathbb{D}_6 \subset \mathrm{Gl}(2,\mathbb{Z}) \quad \text{generated by } \rho,\tau' \\ \mathbb{D}_\sigma \simeq \mathbb{D}_4 \subset \mathrm{Gl}(2,\mathbb{Z}) \quad \text{_____} \ \sigma,\tau' \\ \{1,\tau'\}\times\{\pm 1\} = \mathbb{D}_\rho \cap \mathbb{D}_\sigma \quad \text{in } \mathrm{Gl}(2,\mathbb{Z}), \end{cases} \tag{37}$$

†

$$\begin{cases} \mathbb{D}_\rho = \{1,\tau'\}\,.\,L_\rho \quad (\text{at least if } \rho \text{ of order 6 in } \mathcal{G}) \\ \mathbb{D}_\sigma = \{1,\tau'\}\,.\,L_\sigma \quad (\text{_____} \ \sigma \text{ of order 4 } \text{_____}), \end{cases} \tag{38}$$

and the homomorphisms

$$\mathbb{D}_\rho \longrightarrow N_\rho, \qquad \mathbb{D}_\sigma \longrightarrow N_\sigma$$

are obvious. The homomorphisms i_ρ, i_σ too, of course.

Let us recall that

$$\begin{cases} \dagger i_\rho \text{ injective} \iff \tau' \notin Z_\rho \iff \rho^2 \neq 1 \iff \rho \neq -1 \\ \qquad\qquad \iff \mathcal{G}^+ \text{ is } \dagger|\text{isomorphic to a quotient of } \mathbb{Z}/4\mathbb{Z}. \\ i_\sigma \text{ injective} \iff \tau' \notin Z_\sigma \iff \sigma^2 \neq 1 \quad \dagger \end{cases} \tag{39}$$

[p. 554]

Uncertainty retained from cancelled passages.

140-3-s555-D001 N

140-3-s555-D002 hence the homomorphisms

140-3-s555-D003 i'_ρ

140-3-s555-D004 isomorphic

140-3-s555-D005 $\Longleftrightarrow \sigma \neq \pm 1$

[scan 556]

[p. 581]

Letting $G_{\rho,\sigma}$ act on $\mathcal{G}^+$, a normal subgroup, via inner automorphisms, we see that every element of $G_{\rho,\sigma}$ transforms $L_\rho \subset \mathcal{G}^+$ into a $\mathcal{G}^+$-conjugate subgroup, thus is congruent modulo † $\mathcal{G}^+$ to an element of $\mathrm{Norm}_{G|\mathcal{G}}(L_\rho) = \mathrm{Im} i_\rho(N_\rho)$, and likewise for L_σ. In other words, setting

$$N^*_\rho \overset{\text{dfn}}{=} \{x \in N_\rho \mid \varepsilon_\rho(x) = \varepsilon'_G(i_\rho(x))\}, \qquad N^*_\sigma \overset{\text{dfn}}{=} \{x \in N_\sigma \mid \varepsilon_\sigma(x) = \varepsilon_G(i_\sigma(x))\} \tag{40}$$

the composite homomorphisms

$$\begin{cases} N^*_\rho \xrightarrow{i_\rho} G_{\rho,\sigma} \xrightarrow{\delta_G} \gamma_{\rho,\sigma}\ , \quad \delta_\rho \overset{\text{dfn}}{=} \delta_G \circ i_\rho \\ N^*_\sigma \xrightarrow{i_\sigma} G_{\rho,\sigma} \xrightarrow{\delta_G} \gamma_{\rho,\sigma}\ , \quad \delta_\sigma \overset{\text{dfn}}{=} \delta_G \circ i_\sigma \end{cases} \tag{40}$$

epimorphisms. † We thus have †

m1225

Marginal note. m1225 If $g \in G_{\rho,\sigma}$, and $\varepsilon = \varepsilon_G(g)$, $\varepsilon' = \varepsilon'_G(g)$, then

$$\mathrm{int}(g)(\rho) \underset{\mathcal{G}^+}{\sim} \rho^{\varepsilon'} = \tau^{\nu'}(\rho),$$
$$\mathrm{int}(g)(\sigma) \underset{\mathcal{G}^+}{\sim} \sigma^{\varepsilon} = \tau^{\nu}(\rho)|\tau^{\nu}(\sigma).$$

† (where $(-1)^\nu = \varepsilon$, $(-1)^{\nu'} = \varepsilon'$), thus † $\exists a = \tau^{\nu'} \,.\, a_0|ag \in N_\rho$ (where $a_0 = \tau^\nu g$) such that
a) $g \equiv a|a_0$ $(\mathcal{G}^+)$ and
b) $\varepsilon_\rho(a) = \varepsilon'|\varepsilon'_G = \varepsilon'_G(g)|\varepsilon_G(g)$ —
(and idem for N_σ…)
thus

$$\mathrm{Im}(N_\rho^{\dagger(+1)} \overset{\dagger}{\to} \gamma_{\rho,\sigma}) \supset \gamma'_{\rho,\sigma}\dagger,$$
$$\mathrm{Im}(N_\rho^{(-1)} \longrightarrow \gamma_{\rho,\sigma}) \supset \gamma_{\rho,\sigma} \setminus \gamma'_{\rho,\sigma}.$$

But this does not prove that we have commutativity! hence probably

† †

†

commutative diagram of † exact sequences

$$\begin{array}{ccccccccc} 1 & \longrightarrow & Z_\rho & \longrightarrow & N^*_\rho & \xrightarrow{\varepsilon_\rho} & \{\pm 1\} & \longrightarrow & 1 \\ & & \downarrow \text{epimorphism} & & \downarrow \text{epimorphism} & & \| & & \\ 1 & \longrightarrow & \gamma'_{\rho,\sigma} & \longrightarrow & \gamma_{\rho,\sigma}|Z_{\rho,\sigma} & \xrightarrow{\varepsilon'_\gamma} & \{\pm 1\} & \longrightarrow & 1 \end{array} \tag{41}$$

where we have set

$$\gamma'_{\rho,\sigma} = \mathrm{Ker}(\ \gamma_{\rho,\sigma} \xrightarrow{\varepsilon'_\gamma} \{\pm 1\}\ ); \tag{42}$$

Likewise

,

thus|hence this commutative diagram of exact sequences

$$\begin{array}{ccccccccc} 1 & \longrightarrow & Z_\sigma & \longrightarrow & N_\sigma^* & \xrightarrow{\varepsilon_\sigma} & \{\pm 1\} & \longrightarrow & 1 \\ & & \downarrow \text{epimorphism} & & \downarrow \text{epimorphism} & & \| & & \\ 1 & \longrightarrow & \gamma''_{\rho,\sigma} & \longrightarrow & \gamma_{\rho,\sigma} & \xrightarrow{\varepsilon_\gamma} & \{\pm 1\} & \longrightarrow & 1 \end{array} \quad , \qquad \text{where} \tag{43}$$

$$\gamma''_{\rho,\sigma} = \mathrm{Ker}(\ \gamma_{\rho,\sigma} \xrightarrow{\varepsilon_\gamma} \{\pm 1\}\ ). \tag{44}$$

[p. 555]

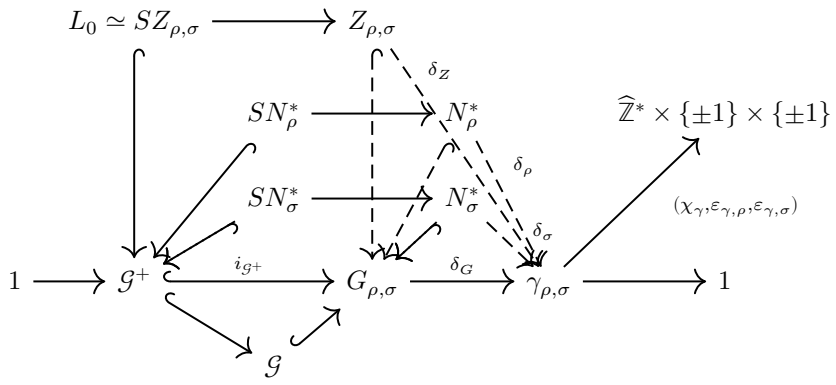
Uncertainty retained from cancelled passages.

140-3-s556-D001 $\mathrm{Norm}_{\mathcal{G}^+}(L_\rho)$
140-3-s556-D002 see [illeg. 3]
140-3-s556-D003 [illeg. 1]
140-3-s556-D004 thus
140-3-s556-D005 $\mathcal{G}^+$
140-3-s556-D006 $*$
140-3-s556-D007 δ_ρ
140-3-s556-D008 ε'
140-3-s556-D009 They are not [illeg. 2]
140-3-s556-D010 41
140-3-s556-D011 ε_ρ $\{\pm 1\}$ $\{\pm 1\}$
140-3-s556-D013 $\{\pm 1\}$

[scan 557]

N.B. The notations $\varepsilon_\gamma, \varepsilon'_\gamma : \gamma_{\rho,\sigma} \longrightarrow \{\pm 1\}$ can lead to confusion, the first referring|pertaining to the action of a u on L_σ (not on L_ρ). A slightly heavier notation, but not leading to confusion, would be $\varepsilon_{\gamma,\rho}, \varepsilon_{\gamma,\sigma}$ — or even just $\varepsilon_\rho, \varepsilon_\sigma$, the confusion with the homomorphisms of this name $N'_\rho|N^*_\rho \longrightarrow \{\pm 1\}$, $N_\sigma \longrightarrow \{\pm 1\}$ being harmless when $\rho^2 \neq 1, \sigma^2 \neq 1$, because of the compatibilities (41) (43).

We have thus constructed, with the aid of|of the (1) i.e. of $\mathcal{G}$ equipped with ρ, σ, τ', groups $\mathcal{G}^+$, $G_{\rho,\sigma}, \gamma_{\rho,\sigma}, Z_{\rho,\sigma}, N'_\rho, N'_\sigma$, fitting into a commutative diagram of groups



(45)

where we have set

$$\begin{cases} SZ_{\rho,\sigma} = i_{L_0,Z}(L_0) = \mathrm{Ker}(\delta_Z : Z_{\rho,\sigma} \longrightarrow \gamma_{\rho,\sigma}) \\ SN'_\rho = \mathrm{Ker}(\delta_\rho : N^*_\rho \longrightarrow \gamma_{\rho,\sigma}) = \mathrm{Cent}_{\mathcal{G}^+}(\rho) \\ SN'_\sigma = \mathrm{Ker}(\delta_\sigma : N^*_\sigma \longrightarrow \gamma_{\rho,\sigma}) = \mathrm{Cent}_{\mathcal{G}^+}(\sigma) \end{cases} \tag{46}$$

m1226

Marginal note. m1226 N.B. We can thus also denote them by SZ_ρ, SZ_σ instead of the SN^*_ρ, SN^*_σ.

[p. 556]
[scan 558]
[p. 582]

in such a way that we have exact sequences

$$\begin{cases} 1 \longrightarrow S\dagger N^*_\rho \longrightarrow N^*_\rho \xrightarrow{\delta_\rho} \gamma_{\rho,\sigma} \longrightarrow 1 \\ 1 \longrightarrow SN^*_\sigma \longrightarrow N^*_\sigma \xrightarrow{\delta_\sigma} \gamma_{\rho,\sigma} \longrightarrow 1 \end{cases} \tag{47}$$

m1227

Marginal note. m1227 We have

$$\begin{cases} \mathrm{Im}(SN_\rho \longrightarrow \mathcal{G}^+) = \mathrm{Norm}_{\mathcal{G}^+}(L_\rho) \\ \mathrm{Im}(SN_\sigma \longrightarrow \mathcal{G}^+) = \mathrm{Norm}_{\mathcal{G}^+}(L_\sigma) \end{cases} \tag{47bis}$$

(or again, if $\rho^2 \neq 1$ resp. $\sigma^2 \neq 1$, $N_\rho \cap \mathcal{G}$ and $N_\sigma \cap \mathcal{G}$)

mapping to the exact sequence

$$1 \longrightarrow \mathcal{G}^+ \longrightarrow G_{\rho,\sigma} \longrightarrow \gamma_{\rho,\sigma} \longrightarrow 1 \ ,$$

in a way analogous to (27). Moreover, the homomorphisms $i_\rho : N^*_\rho \longrightarrow G_{\rho,\sigma}$ and $i_\sigma : N^*_\sigma \longrightarrow G_{\rho,\sigma}$ have a marked tendency to be injective (cf. (39)), which we have marked on diagram (45) by the sign $\hookrightarrow$ (dotted inclusion sign).

We also introduce

$$\begin{cases} \chi_\rho = \chi_\gamma \circ \delta_\rho : \ \dagger N^*_\rho \longrightarrow \widehat{\mathbb{Z}}^* \\ \chi_\sigma = \chi_\gamma \circ \delta_{\dagger\sigma} : \ \dagger N^*_\sigma \longrightarrow \widehat{\mathbb{Z}}^* \end{cases} , \tag{48}$$

whose kernels (which contain $S\dagger N_\rho\dagger$ resp. SN_σ, and which we shall take care not to confuse with them) are denoted as is only right by $N^{*+}_\rho, N^{*+}_\sigma$.

N.B. We should still examine the composites

$$\begin{cases} N^*_\rho \longrightarrow \gamma_{\rho,\sigma} \xrightarrow{\varepsilon_{\gamma,\sigma}} \{\pm 1\} \\ N^*_\sigma \longrightarrow \gamma_{\rho,\sigma} \xrightarrow{\varepsilon_{\gamma,\rho}} \{\pm 1\} \end{cases} , \tag{49}$$

we shall see that in the "good cases", the restrictions of these homomorphisms to Z_ρ, Z_σ respectively are epimorphisms.

[p. 557]

Uncertainty retained from cancelled passages.

140-3-s558-D001 Z
140-3-s558-D002 Z
140-3-s558-D003 ρ
140-3-s558-D004 Z
140-3-s558-D005 Z
140-3-s558-D006 [*illeg.* 1]

[scan 559]
[p. 583]

†

(15)

†

Canceled
†

⑤ m1229

Marginal note. m1229 The groups $S_0\mathcal{M}_{\rho,\sigma}, \mathcal{M}_{\rho,\sigma}, S_0\Gamma_{\rho,\sigma}, \Gamma_{\rho,\sigma}$.

We set

$$\begin{cases} S_0\mathcal{M}_{\rho,\sigma} = \dagger Z_{\rho,\sigma} \underset{\gamma_{\rho,\sigma}}{\times} N^*_\rho \underset{\gamma_{\rho,\sigma}}{\times} N^*_\sigma \underset{\gamma_{\rho,\sigma}}{\times} G_{\rho,\sigma}. \\ \mathcal{M}_{\rho,\sigma} = N^*_\rho \underset{\gamma_{\rho,\sigma}}{\times} N^*_\sigma \underset{\gamma_{\rho,\sigma}}{\times} G_{\rho,\sigma} \end{cases} \tag{50}$$

$$\dagger \qquad \begin{cases} S_0\Gamma_{\rho,\sigma} = Z_{\rho,\sigma} \underset{\gamma_{\rho,\sigma}}{\times} N^*_\rho \underset{\gamma_{\rho,\sigma}}{\times} N^*_\sigma \\ \Gamma_{\rho,\sigma} = N^*_\rho \underset{\gamma_{\rho,\sigma}}{\times} N^*_\sigma, \end{cases} \tag{51}$$

whence †diagram of exact sequences

(52)
$$\begin{array}{ccccccccc} & & & & 1 & & 1 & & \\ & & & & \downarrow & & \downarrow & & \\ & & & & L_0 & = & L_0 & & \\ & & & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & \mathcal{G}^+ & \longrightarrow & S_0\mathcal{M}_{\rho,\sigma} & \longrightarrow & S_0\Gamma_{\rho,\sigma} & \longrightarrow & 1 \\ & & \| & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & \mathcal{G}^+ & \longrightarrow & \mathcal{M}_{\rho,\sigma} & \longrightarrow & \Gamma_{\rho,\sigma} & \longrightarrow & 1, \\ & & & & \downarrow & & \downarrow & & \\ & & & & 1 & & 1 & & \end{array}$$

and exact sequences

(53)
$$1 \longrightarrow \underset{\simeq L_0}{\underline{SZ_{\rho,\sigma}}} \times SN^*_\rho \times SN^*_\sigma \times \mathcal{G}^+ \longrightarrow S_0\mathcal{M}_{\rho,\sigma} \longrightarrow \gamma_{\rho,\sigma} \longrightarrow 1$$

(54)
$$1 \longrightarrow SN^*_\rho \times SN^*_\sigma \times \mathcal{G}^+ \longrightarrow \mathcal{M}_{\rho,\sigma} \longrightarrow \gamma_{\rho,\sigma} \longrightarrow 1$$

(55)
$$1 \longrightarrow \underset{\simeq L_0}{\underline{SZ_{\rho,\sigma}}} \times SN^*_\rho \times SN^*_\sigma \longrightarrow S_0\Gamma_{\rho,\sigma} \longrightarrow \gamma_{\rho,\sigma} \longrightarrow 1$$

(56)
$$1 \longrightarrow SN^*_\rho \times SN^*_\sigma \longrightarrow \Gamma_{\rho,\sigma} \longrightarrow \gamma_{\rho,\sigma} \longrightarrow 1.$$

[p. 558]

Uncertainty retained from cancelled passages.

140-3-s559-D002 $Z_\rho | N_\rho$ $Z_\rho | N_\rho$

140-3-s559-D003 B B Norm | Cent $L_0 | L_\rho$ $L_0 | L_\sigma$

140-3-s559-D004 It seems that in the case that we want to consider $\neq$ | = Cent

140-3-s559-D005 B

140-3-s559-D006 42
140-3-s559-D007 homomorphism

[scan 560]

We also have

$$S_0\mathcal{M}_{\rho,\sigma} \simeq \mathcal{M}_{\rho,\sigma} \underset{\gamma_{\rho,\sigma}}{\times} Z_{\rho,\sigma} \tag{57}$$

$$S_0\Gamma_{\rho,\sigma} \simeq \Gamma_{\rho,\sigma} \underset{\gamma_{\rho,\sigma}}{\times} Z_{\rho,\sigma} \tag{58}$$

i.e. the extensions $S_0\mathcal{M}_{\rho,\sigma}, S_0\Gamma_{\rho,\sigma}$ of $\mathcal{M}_{\rho,\sigma}, \Gamma_{\rho,\sigma}$ by L_0 are deduced from the extension $Z_{\rho,\sigma}$ of $\gamma_{\rho,\sigma}$ by L_0 by inverse image under $\mathcal{M}_{\rho,\sigma} \to \gamma_{\rho,\sigma}, \Gamma_{\rho,\sigma} \to \gamma_{\rho,\sigma}$ — and we also have †

$$S_0\mathcal{M}_{\rho,\sigma} \simeq \mathcal{M}_{\rho,\sigma} \underset{\Gamma_{\rho,\sigma}}{\times} S_0\Gamma_{\rho,\sigma} \tag{59}$$

i.e. the extension $S_0\mathcal{M}_{\rho,\sigma}$ of $\mathcal{M}_{\rho,\sigma}$ by L_0 is deduced from the extension $S_0\Gamma_{\rho,\sigma}$ of $\Gamma_{\rho,\sigma}$ by L_0, by inverse image under $\mathcal{M}_{\rho,\sigma} \to \Gamma_{\rho,\sigma}$.

We introduce an element

$$\dagger, \qquad \tau'_{\dagger\mathcal{M}} \in \mathcal{M}_{\rho,\sigma} \quad \dagger \quad \tau'_{\dagger\mathcal{M}} = \dagger(\overset{a}{\|}\tau'_\rho, \overset{b}{\|}\tau'_\sigma, \overset{u}{\|}\tau'_G) \tag{60}$$

† where $\dagger, \tau'_\rho, \tau'_\sigma, \tau'_G$ are the "avatars" of τ', taken resp. in $\dagger N^*_\rho, N^*_\sigma, G_{\rho,\sigma}$. These do indeed all have the same image in $G_{\rho,\sigma}$ and a fortiori in $\gamma_{\rho,\sigma}$, hence define an element of $\dagger\mathcal{M}_{\rho,\sigma}$, whose operation on $\mathcal{G}^+$ is that of τ. Whence an embedding

$$(61) \qquad \mathcal{G} \overset{i_{\mathcal{G},\mathcal{M}}}{\hookrightarrow} \mathcal{M}_{\rho,\sigma} \quad : \text{identity on } \mathcal{G}^+ \qquad \tau'_{\mathcal{G}} \longmapsto \tau'_{\mathcal{M}}.$$

[p. 559]

Uncertainty retained from cancelled passages.

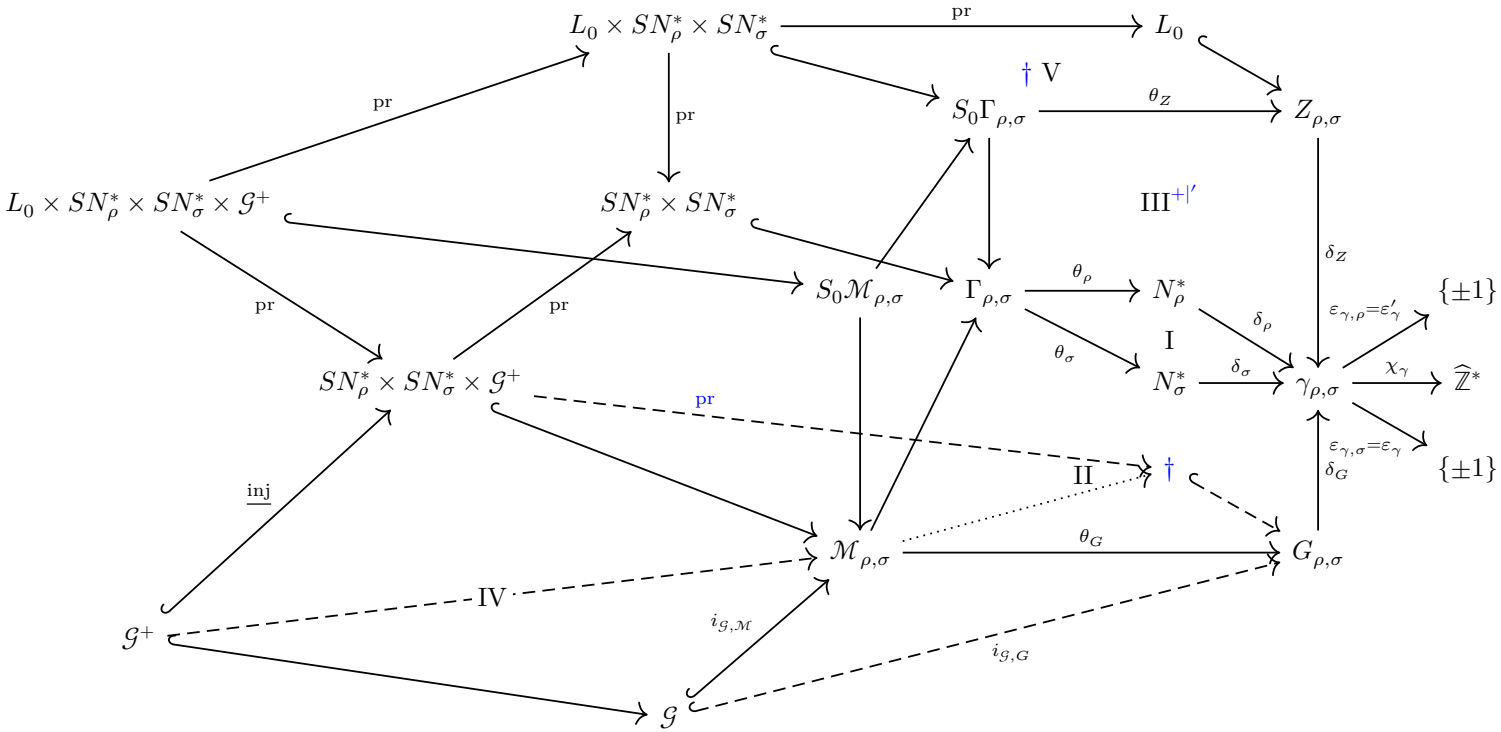
140-3-s560-D001 Γ
140-3-s560-D002 $\tau'_{\mathcal{M}}$
140-3-s560-D003 S_0
140-3-s560-D005 the
140-3-s560-D006 S_0
140-3-s560-D007 τ'_Z
140-3-s560-D008 of homomorphisms
140-3-s560-D010 τ'_Z
140-3-s560-D011 $Z_{\rho,\sigma}$
140-3-s560-D012 G

[scan 561]

[Diagram continued on the landscape page; 140-3-561-D1]

[p. 584]

(62)



$\|$

$\{1, \tau'\} \cdot \mathcal{G}^+$

We have summarized †some of the relations between the groups introduced so far in the commutative diagram (62), which I comment on briefly.

a) All the arrows are either injections ($\hookrightarrow$) or surjections ($\longrightarrow$), with the exception perhaps of the three arrows issuing from $\gamma_{\rho,\sigma}$.

b) The ten squares (six faces of a cube, and three|the squares marked $\mathrm{I}, \mathrm{II}, \mathrm{III}^{+|'}, \mathrm{IV}, \mathrm{V}$) are cartesian.

c) The five arrows marked pr are projections, canonical from product groups onto partial products. The arrow marked inj

$$\mathcal{G}^+ \longrightarrow SN^*_\rho \times SN^*_\sigma \times \mathcal{G}^+$$

is the canonical injection.

d) In order not to destroy the commutativity of the diagram, we have omitted to show in it the arrows (of inclusion, in particular)

$$Z_{\rho,\sigma} \longrightarrow G_{\rho,\sigma}, \qquad N^*_\rho \longrightarrow G_{\rho,\sigma}, \qquad N^*_\sigma \longrightarrow G_{\rho,\sigma}$$

which appeared in (45). [p. 560]

Uncertainty retained from cancelled passages.

140-3-s561-D001 $\mathcal{G}^+|N^*_\sigma$

140-3-s561-D002 IV

140-3-s561-D003 the main

[scan 562]

e) In order not to overload the diagram, I have not introduced in the diagram notations for certain useful composite arrows, such as

$$(63) \qquad \begin{cases} \delta_\Gamma : \Gamma_{\rho,\sigma} \longrightarrow \gamma_{\rho,\sigma}, & \dagger \\ \delta_{S_0\Gamma} : S_0\Gamma_{\rho,\sigma} \longrightarrow \gamma_{\rho,\sigma} \\ \delta_{\mathcal{M}} : \mathcal{M}_{\rho,\sigma} \longrightarrow \gamma_{\rho,\sigma} \\ \delta_{S_0\mathcal{M}} : S_0\mathcal{M}_{\rho,\sigma} \longrightarrow \gamma_{\rho,\sigma} \end{cases}$$

and

$$(64) \qquad \begin{cases} \chi_\Gamma = \chi_\gamma \circ \delta_\Gamma : \Gamma_{\rho,\sigma} \longrightarrow \widehat{\mathbb{Z}}^* \\ \chi_{S_0\Gamma} = \chi_\gamma \circ \delta_{S_0\Gamma} : S_0\Gamma_{\rho,\sigma} \longrightarrow \widehat{\mathbb{Z}}^* \\ \chi_{\mathcal{M}} = \chi_\gamma \circ \delta_{\mathcal{M}} : \mathcal{M}_{\rho,\sigma} \longrightarrow \widehat{\mathbb{Z}}^* \\ \chi_{S_0\mathcal{M}} = \chi_\gamma \circ \delta_{S_0\mathcal{M}} : S_0\mathcal{M}_{\rho,\sigma} \longrightarrow \widehat{\mathbb{Z}}^* \end{cases}$$

and the composites $\varepsilon_{?,\rho}, \varepsilon_{?,\sigma} : ? \longrightarrow \{\pm 1\}$, where ? can take † value Z, G (already introduced) N^*_ρ, N^*_σ (we will have $\varepsilon_{N_\rho,\rho} = \varepsilon_\rho\dagger, \quad \varepsilon_{N_\sigma,\sigma} = \varepsilon_\sigma\dagger$), $\Gamma, S_0\Gamma, \mathcal{M}, S_0\mathcal{M}$. The kernel of $\chi_?$ is denoted $?^+$, that of $\varepsilon_{?,\rho}$ is denoted $?'$, that of $\varepsilon_{?,\sigma}$ is denoted $?''$, we write

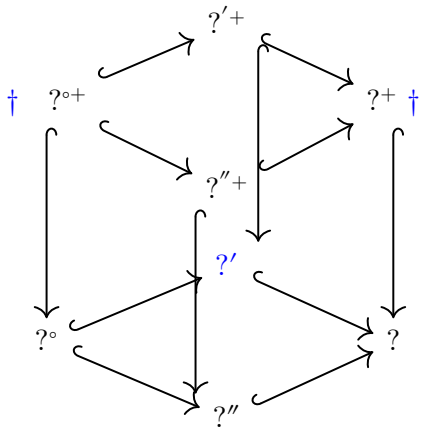
$$?^\circ \doteq ?' \cap ?'', \qquad ?^{+'} = ?^+ \cap ?', \quad ?^{+''} = ?^+ \cap ?'',$$

$$?^{+,\circ} = ?^+ \cap ?^\circ, \quad S? = \operatorname{Ker} \delta_?,$$

hence †

†

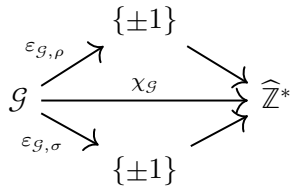
(65)



"cartesian" cube.

f) Let us also recall that we have commutativity in

(66)



, i.e. $\varepsilon_{\mathcal{G}} : \mathcal{G} \longrightarrow \{\pm 1\}$, [p. 561]

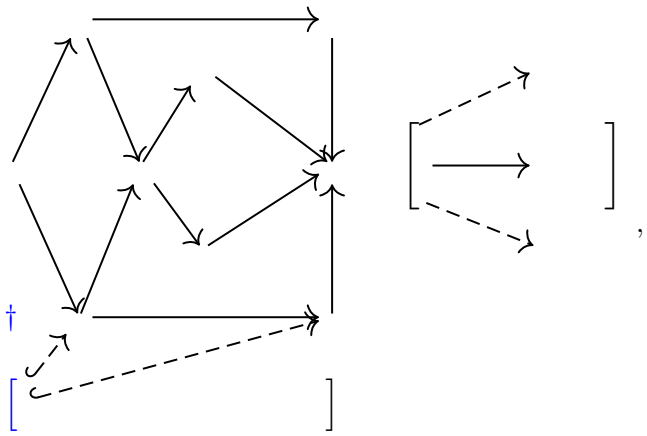
Uncertainty retained from cancelled passages.

140-3-s562-D001 $\delta_{S_0\Gamma}$ $S_0\Gamma_{\rho,\sigma}$ $\gamma_{\rho,\sigma}$
140-3-s562-D002 any
140-3-s562-D003 simply
140-3-s562-D004 simply
140-3-s562-D006 $S? \subset$
140-3-s562-D007 sense
140-3-s562-D008 $S? \subset$
140-3-s562-D009 we

[scan 563]

The kernel of $\varepsilon_{\mathcal{G}}$ is none other than $\mathcal{G}^+$. In its capacity as a subgroup of $\mathcal{M}_{\rho,\sigma}$ (resp. of $G_{\rho,\sigma}$) [p. 585]
$\mathcal{G}$ is identified with the inverse image in this group of the subgroup $\{1, \tau'_\Gamma\}$ of $\Gamma_{\rho,\sigma}$ (resp. of the subgroup $\{1, \tau'_\gamma\}$ of $\gamma_{\rho,\sigma}$), where $\tau'_\Gamma, \tau'_\gamma$ denote the images of $\tau' \in \mathcal{G}$ in $\Gamma_{\rho,\sigma}$ and $\gamma_{\rho,\sigma}$ respectively.

g) The diagram (62) has as its main part

(67) 

,

all the groups marked on this last diagram, except for $\widehat{\mathbb{Z}}^*, \{\pm 1\}, \{\pm 1\}$ (put in brackets [ ]) map to $\gamma_{\rho,\sigma}$, which is the terminal term of this diagram, linking 9 groups (not counting $\mathcal{G}$ † inserted † as †outside †of two dotted arrows) including $\gamma_{\rho,\sigma}$, obtained from four of them $Z_{\rho,\sigma}, N^*_\rho, N^*_\sigma, G_{\rho,\sigma}$ (which map to $\gamma_{\rho,\sigma}$ by epimorphisms) by taking fiber products over $\gamma_{\rho,\sigma}$ of certain of them — † the initial term being $S_0\mathcal{M}_{\rho,\sigma}$, which is the fiber product of the four preceding factors. Taking the kernels of the homomorphisms from these nine groups to $\gamma_{\rho,\sigma}$, we find a diagram of the same combinatorial type, linking [p. 562]

Uncertainty retained from cancelled passages.

140-3-s563-D001 $[\mathcal{G}^+]$
140-3-s563-D002 $\mathcal{G}^+$
140-3-s563-D003 by
140-3-s563-D004 terms
140-3-s563-D005 initial
140-3-s563-D006 the

[scan 564]

nine groups, formed from $L_0 = SZ_{\rho,\sigma}, SN^*_\rho, SN^*_\sigma$ and $\mathcal{G}^+ = SG_{\rho,\sigma}$ and from certain products of these groups (including the product over the empty index set, namely $\{1\} = \mathrm{Ker}(\gamma_{\rho,\sigma} \overset{\mathrm{id}}{\longrightarrow} \gamma_{\rho,\sigma})$), with as † canonical projections. We have inserted in (62) only of these product groups, with the canonical injections into the fiber products from which they arise, dispensing with including in it also|also $SN^*_\rho, SN^*_\sigma, SG_{\rho,\sigma} = \mathcal{G}^+$, and the canonical inclusions $SN^*_\rho \hookrightarrow N^*_\rho$, $SN^*_\sigma \hookrightarrow N^*_\sigma$, $\mathcal{G}^+ = SG_{\rho,\sigma} \hookrightarrow G_{\rho,\sigma}$, and the † complete diagram of the projections of the partial products of these groups and of L_0, which therefore lacks the projections onto these three groups, and the homomorphisms from these groups and from L_0 to the unit group. [p. 563]

Uncertainty retained from cancelled passages.

140-3-s564-D002 arrowisms|arrows
140-3-s564-D005 p

[scan 565]
[p. 586]

6°) The set $\Sigma_{\rho,\sigma}, S\mathcal{G}|g_{\rho,\sigma}$.

We put

$$\Sigma_{\rho,\sigma} \subset \mathcal{G}^+ \times \mathcal{G}^+ \times \widehat{\mathbb{Z}}^* \times \{\pm 1\} \times \{\pm 1\}$$

$$\| \tag{68}$$

$$\left\{ (\xi, \eta, \mu, \varepsilon', \varepsilon) \;\middle|\; \begin{array}{l} \xi\rho \ \mathcal{G}^+\text{-conjugate to } \rho^{\varepsilon'} \\ \eta\sigma \ \mathcal{G}^+\text{-conjugate to } \sigma^{\varepsilon} \\ \xi = \eta \underbrace{\varepsilon_1^{\mu-1}}_{\rho_1^{\nu} \text{ where } \nu = \frac{\mu-1}{2}} \quad \dagger \end{array} \right\}$$

N.B. If ρ not conjugate to ρ^{-1} in $\mathcal{G}^+$, ε' known when we know ξ, η — likewise if σ not conjugate to σ^{-1} (hence $\sigma^2 \neq 1$), ε' is determined by ξ, η. If this homomorphism $\widehat{\mathbb{Z}} \longrightarrow L_0$ $(n \longmapsto \varepsilon_0^n)$ is injective, then μ is determined by ξ, η, and the relation $\xi = \eta\varepsilon_1^{\mu-1}$ can also be written (without bringing in μ)

$$\eta^{-1}\xi \in L_1. \tag{69}$$

Let on the other hand

$$S\mathcal{G}_{\rho,\sigma} \subset \mathcal{G} \times \mathcal{G} \times \widehat{\mathbb{Z}}^*$$

$$\| \tag{70}$$

$$\left\{ (\alpha, \beta, \mu) \;\middle|\; \underbrace{[\alpha, \rho]}_{\xi} = [\beta, \sigma]\varepsilon_1^{\mu-1} \right\}$$

We have a map

$$\left\{ \begin{array}{ccc} S\mathcal{G}_{\rho,\sigma} & \longrightarrow & \Sigma_{\rho,\sigma} \\ (\alpha, \beta, \mu) & \longmapsto & (\xi, \eta, \mu, \varepsilon', \varepsilon) \end{array} \right. \tag{71}$$

with

$$\left\{ \begin{array}{l} \xi = [\alpha, \rho] \\ \eta = [\beta, \sigma] \\ \mu \text{ the same} \\ \varepsilon' = \varepsilon_{\mathcal{G}}(\alpha) \\ \varepsilon = \varepsilon_{\mathcal{G}}(\beta) \end{array} \right.$$

m1231

Marginal note. m1231 N.B. $\xi = \alpha\rho(\alpha)^{-1} \in \mathcal{G}^+$, since we have $\varepsilon_{\mathcal{G}}(\xi) = \dot{\alpha}|\bar{\alpha}\rho(\dot{\alpha}|\bar{\alpha}) = 1$ since ρ operates trivially on $\mathcal{G}/\mathcal{G}^+ \simeq \{\pm 1\}$, for the same reason $\eta = \beta\sigma(\beta)^{-1} \in \mathcal{G}^+$.

which is obviously surjective.

We define on the other hand [p. 564]

Uncertainty retained from cancelled passages.

140-3-s565-D001 (with

[scan 566]

$$\Sigma^{\text{eff}}_{\rho,\sigma} \subset \Sigma_{\rho,\sigma}$$

$$\| \tag{72}$$

$$\left\{ (\xi, \eta, \mu, \varepsilon', \varepsilon) \in \Sigma_{\rho,\sigma} \;\middle|\; \begin{array}{l} \exists u \in \mathrm{Aut}(\mathcal{G}^+) \text{ with} \\ u(\rho) = \xi\rho, \quad u(\sigma) = \eta\sigma \end{array} \right\}$$

(this u will then be unique, and conversely, knowledge of u determines ξ, η.) There is a 1-1 correspondence between the elements of $\Sigma^{\text{eff}}_{\rho,\sigma}$, and the systems $(u, \mu, \varepsilon', \varepsilon) \in \mathrm{Aut}(\mathcal{G}^+) \times \widehat{\mathbb{Z}}^* \times \{\pm 1\} \times \{\pm 1\}$, such that we have † i.e. $[u, \rho] = \xi$, $[u, \sigma] = \eta$

$$\left\{ \begin{array}{lll} u(\rho) \text{ is} & \mathcal{G}^+\text{-conjugate to} & \rho^{\varepsilon'} \\ u(\sigma) & \text{———————} & \sigma^{\varepsilon} \\ u(\varepsilon_0) = \varepsilon_0^{\mu} & \text{(which expresses } \xi = \eta\rho_1^{\nu}) & \end{array} \right.$$

— now this set is none other than the subgroup $Z_{\rho,\sigma}|\mathbb{Z}_{\rho,\sigma}$ of $\mathrm{Aut}(\mathcal{G}^+)\times\widehat{\mathbb{Z}}^*\times\{\pm1\}\times\{\pm1\}$. (cf. (6) page 577')

We thus find a bijection

$$(73)\qquad \begin{array}{ccc} Z_{\rho,\sigma} & \xrightarrow{\ \sim\ } & \Sigma^{\mathrm{eff}}_{\rho,\sigma} \\ \dagger\ u & \longmapsto & (\xi,\eta,\mu,\varepsilon',\varepsilon) \end{array} \qquad \left\{\begin{array}{l} \xi=[u,\rho] \\ \eta=[u,\sigma] \\ \mu=\chi(u) \\ \varepsilon'=\varepsilon'_Z(u) \\ \varepsilon=\varepsilon_Z(u) \end{array}\right. .$$

We denote by $S\mathcal{G}^{\mathrm{eff}}_{\rho,\sigma}$ the inverse image of $\Sigma^{\mathrm{eff}}_{\rho,\sigma}$ in $S\mathcal{G}_{\rho,\sigma}$ under (71), so we will have

$$S\mathcal{G}^{\mathrm{eff}}_{\rho,\sigma}\subset S\mathcal{G}_{\rho,\sigma}$$
$$\| \qquad (74)$$
$$\left\{ \{\alpha,\beta,\mu\}\in\mathcal{G}\times\mathcal{G}\times\widehat{\mathbb{Z}}^* \;\middle|\; \begin{array}{l} [\alpha,\rho]=[\beta,\sigma]\varepsilon_1^{\mu-1} \\ \exists u\in\mathrm{Aut}(\mathcal{G}^+)\text{ such that we have} \\ \left\{\begin{array}{ll} u(\rho)=\alpha(\rho) & (=\xi\rho,\ \text{where } \xi=[\alpha,\rho]) \\ u(\sigma)=\beta(\sigma) & (=\eta\sigma,\ \text{where } \eta=[\beta,\sigma]) \end{array}\right. \end{array} \right\}$$

so that we have a surjective homomorphism induced by (71)

$$(75)\qquad S\mathcal{G}^{\mathrm{eff}}_{\rho,\sigma}\longrightarrow\Sigma^{\mathrm{eff}}_{\rho,\sigma}\quad .$$

[p. 565]

Uncertainty retained from cancelled passages.

140-3-s566-D001 .

140-3-s566-D002 $(u,\mu,\varepsilon,\varepsilon')$

[scan 567]

[p. 587]

We have a canonical map

$$(76)\qquad \left\{\begin{array}{l} S_0\Gamma_{\rho,\sigma}=Z|\mathbb{Z}_{\rho,\sigma}\times_{\gamma_{\rho,\sigma}}N^*_\rho\times_{\gamma_{\rho,\sigma}}N^*_\sigma\longrightarrow S\mathcal{G}^{\mathrm{eff}}_{\rho,\sigma} \\ \qquad x=\{u,a,b\}|(u,a,b)\longmapsto(\alpha,\beta,\mu) \end{array}\right. \quad\text{with}\quad \left\{\begin{array}{l} \alpha=u\,\overbrace{a^{-1}\tau'^{\nu'(x)}}^{\in Z_\rho} \\ \beta=u\,\underbrace{b^{-1}\tau'^{\nu(x)}}_{\in Z_\sigma} \\ \mu=\chi_{S\Gamma}(x) \end{array}\right.$$

m1233

Marginal note. m1233 to be more correct, one should write

$$\left\{\begin{array}{l} \alpha=u\,i_\rho(a)^{-1}\tau'^{\nu'(x)} \\ \beta=u\,i_\sigma(b)^{-1}\tau'^{\nu(x)} \end{array}\right.$$

where we have defined $\nu'(x),\nu(x)|\nu'(x)\in\mathbb{Z}/2\mathbb{Z}$ by

$$\dagger\qquad (-1)^{\nu'(x)}=\varepsilon'(x),\qquad (-1)^{\nu(x)}=\varepsilon(x).$$

By definition of $S_0\Gamma_{\rho,\sigma}$ as a fiber product, we will indeed have $ua^{-1},ub^{-1}\in\mathcal{G}^+$, hence $\alpha,\beta\in\mathcal{G}$, we will moreover have

$$(77)\qquad \left\{\begin{array}{l} \alpha(\rho)=u\left(\underbrace{a^{-1}\left(\underbrace{\tau'^{\nu'(x)}(\rho)}_{\rho^{\varepsilon'(x)}}\right)}_{(\rho^{\varepsilon'(x)})^{\varepsilon'(x)}=\rho}\right)=u(\rho),\quad\text{and likewise} \\ \beta(\sigma)=u(\sigma), \end{array}\right. \qquad \text{i.e.}\quad \begin{array}{l}[\alpha,\rho]=[u,\rho]\\ [\beta,\sigma]=[u,\sigma]\end{array}$$

Marginal note. m1234 N.B. We have, for every $a \in N_\rho^*$, if $\varepsilon_\rho(a) = (-1)^{\nu_\rho(a)}$, $a\tau'^{\nu_\rho(a)} \in Z_\rho$, hence $[ua\tau'^{\nu_\rho(a)}, \rho] = [u, \rho]$. Likewise for σ ...

moreover the condition $\underbrace{[\alpha, \rho]}_{[u,\rho]} = \underbrace{[\beta, \sigma]}_{[u,\sigma]} \varepsilon_1^{\mu-1}$ † merely expresses the condition that $u(\varepsilon_0) = \varepsilon_0^\mu$ which enters into the very definition of $Z_{\rho,\sigma}$.

I say that the map in question (76) is injective, It suffices to see that when (α, β, μ) are known, we know u — indeed we † in terms of (α, β, μ)

$$(78) \qquad \begin{cases} \nu'(x) = \nu|\varepsilon_{\mathcal{G}}(\alpha), \quad \nu(x) = \nu|\varepsilon_{\mathcal{G}}(\beta), \quad \dagger\ \chi_{S\Gamma}(x) = \chi_Z(u) = \mu \\ \quad \text{whence} \quad a = \tau'^{\nu|\varepsilon_{\mathcal{G}}(\alpha)}\alpha^{-1}u, \quad b = \tau'^{\nu|\varepsilon_{\mathcal{G}}(\beta)}\beta^{-1}u \end{cases}$$

† so if u is known, we also know a and b hence x. On the other hand to determine in terms of α, β, μ, $u \in Z_{\rho,\sigma} \subset \mathrm{Aut}(\mathcal{G}^+) \times \widehat{\mathbb{Z}}^* \times \{\pm 1\} \times \{\pm 1\}$ it suffices, in addition to (78), to determine † † the element of $\mathrm{Aut}(\mathcal{G}^+)$ which it defines, which is [p. 566]

Uncertainty retained from cancelled passages.

140-3-s567-D001 (
140-3-s567-D003 here
140-3-s567-D006 then
140-3-s567-D007 $\mu =$
140-3-s567-D009 we
140-3-s567-D010 it
140-3-s567-D011 as

immediate from (77). We thus have † [scan 568]

$$(79) \qquad S_0\Gamma_{\rho,\sigma} \hookrightarrow S\mathfrak{G}^{\mathrm{eff}}_{\rho,\sigma}$$

it remains to see that this map is surjective — we have done what was needed for. Let us first note that the composite

$$S_0\Gamma_{\rho,\sigma} \hookrightarrow S\mathfrak{G}^{\mathrm{eff}}_{\rho,\sigma} \xrightarrow{\text{epimorphism}} \Sigma^{\mathrm{eff}}_{\rho,\sigma}(\xrightarrow{\sim} Z_{\rho,\sigma})$$

is none other, by (77) and (71) than †

$$(80) \qquad x = (u, a, b) \longmapsto (\xi, \eta, \mu, \varepsilon', \varepsilon) \qquad \begin{cases} \xi = [u, \rho] & \text{thus } u(\rho) = \xi\rho \\ \eta = [u, \sigma] & \text{thus } u(\sigma) = \eta\sigma \\ \mu = \chi_{S\Gamma}(x) & \\ \varepsilon' = \varepsilon'(x) & \\ \varepsilon = \varepsilon(x) & \end{cases}$$

thus its composite with $\Sigma^{\mathrm{eff}}_{\rho,\sigma} \xrightarrow{\sim} Z_{\rho,\sigma}$ is none other than the canonical projection

$$S_0\Gamma_{\rho,\sigma} \longrightarrow Z_{\rho,\sigma}$$

$$(u, a, b) \longmapsto u$$

which is surjective, with kernel $SN_\rho^* \times SN_\sigma^* = SZ_\rho \times SZ_\sigma$.

Let us consider on the other hand †

$$(81)\quad \begin{cases} \text{operation of } SN_\rho \times SN_\sigma & (a_0, b_0) \in S\dagger Z_\rho^* \times S\dagger Z_\sigma^* \\ \text{on } \mathfrak{G}_{\rho,\sigma} \subset \mathcal{G} \times \mathcal{G} \times \widehat{\mathbb{Z}}^* & (= SZ_\rho \times SZ_\sigma) \\ \qquad\qquad \| & \text{operates by} \\ \{(\alpha, \beta, \mu) \mid [\alpha, \rho] = [\beta, \sigma]\varepsilon_1^{\mu-1}\} & (\alpha, \beta, \mu) \longmapsto (\alpha a_0, \beta b_0, \mu) \end{cases},$$

It is clear that it operates freely, and that the quotient of $\mathfrak{G}_{\rho,\sigma}$ by this action is identified with $\Sigma_{\rho,\sigma}$, thus we have the result idem for $\mathfrak{G}^*_{\rho,\sigma}$ and $\Sigma^*_{\rho,\sigma}$ †. Now, for † [p. 567]

Uncertainty retained from cancelled passages.

140-3-s568-D002 d

140-3-s568-D004 α β μ $[\alpha, \rho]$ $[\beta, \sigma]$

140-3-s568-D006 the operation

140-3-s568-D009 N

140-3-s568-D010 N

140-3-s568-D012 [illeg. 1]

140-3-s568-D013 (u, a, b) $\varepsilon_\rho^{\tau'}(a_0^{-1})$ $\varepsilon_\sigma^{\tau'}(b_0^{-1})$

†

[scan 569]

$$(82)\qquad \varepsilon = (-1)^\nu, \quad \varepsilon' = (-1)^{\nu'}$$

†

[p. 588]

fixed, the preceding operation of (a_0, b_0) corresponds, under the map (76), to the operation

$$(83)\qquad (u, a, b) \longrightarrow (u, \tau^{\nu'}(a_0^{-1})a, \tau^\nu(b_0^{-1})b)\ ,$$

i.e. to left translation by $(\tau^{\nu'}, \tau^\nu)\underbrace{(a_0^{-1}, b_0^{-1})}_{(a_0, b_0)^{-1}}$. Whence surjectivity, and the desired isomorphism

$$(84)\qquad S_0\Gamma_{\rho,\sigma} \xrightarrow{\sim} S\mathfrak{G}^{\text{eff}}_{\rho,\sigma} \qquad (\text{cf. } 76)$$

The left translation by $\ell \in L_0$ in $S_0\Gamma_{\rho,\sigma}$ corresponds, under the preceding bijection, to the operation

$$(85)\qquad (\alpha, \beta, \mu) \longmapsto (\ell\alpha, \ell\beta, \mu)$$

which it was clear a priori stabilizes $S\mathfrak{G}^{\text{eff}}_{\rho,\sigma}$. Passing to the quotient, we thus find

$$(86)\qquad \Gamma_{\rho,\sigma} \xrightarrow{\sim} L_0 \backslash S\mathfrak{G}^{\text{eff}}_{\rho,\sigma}\ .$$

Finally, let us consider

$$(87)\quad \begin{cases} S_0 M_{\rho,\sigma} \longrightarrow S\mathfrak{G}^{\text{eff}}_{\rho,\sigma} \times \mathcal{G}^+ = \underbrace{\{(\alpha, \beta, \mu, f) \mid}_{([\alpha,\rho]=[\beta,\sigma]\varepsilon_1^{\mu-1})} \overbrace{\quad}^{(\in \mathcal{G}\times\mathcal{G}\times\widehat{\mathbb{Z}}^*\times\mathcal{G}^+} \\ x = (u, a, b, U) \longmapsto \dagger((\alpha, \beta, \mu), f) \\ \begin{cases} \alpha = ua^{-1}\tau'^{\dagger\nu'_\rho(a)} \\ \beta = ub^{-1}\tau'^{\nu_\sigma(b)} \\ \mu = \chi_{SM}(x) \quad (= \chi_Z(u) = \chi_\rho(a) = \chi_\sigma(b) = \chi_G(U)) \\ f = uU^{-1} \end{cases} \end{cases}$$

This map is bijective (trivial by (84)). [p. 568]

Uncertainty retained from cancelled passages.
140-3-s569-D001 i.e. $\varepsilon_\rho^{\tau'}$ (a)
140-3-s569-D002 it is thus [illeg. 2] the set
140-3-s569-D004 $ua^{-1}\tau'$ $ub^{-1}\tau'$ uU^{-1}
140-3-s569-D005 ν'

(7) Functorialities [scan 570]
Let us consider [p. 588]

$$(88) \qquad \mathcal{G} \xrightarrow{\varphi} \mathcal{G}' \qquad \text{surjective homomorphism of profinite groups}$$

let † $\rho_{\mathcal{G}'}, \sigma_{\mathcal{G}'}, \tau'_{\mathcal{G}'}, \ldots$ be the images of $\rho = \rho_{\mathcal{G}}, \sigma = \sigma_{\mathcal{G}}, \tau' = \tau'_{\mathcal{G}}, \ldots$ in $\mathcal{G}'$.

Proposition Equivalent conditions on φ :

a) The map induced by φ

$$(89) \qquad \Sigma^{\mathcal{G}}_{\rho,\sigma} \longrightarrow \Sigma^{\mathcal{G}'}_{\rho,\sigma}$$

†maps $\Sigma^{\mathcal{G}\,\mathrm{eff}}_{\rho,\sigma}$ into $\Sigma^{\mathcal{G}'\,\mathrm{eff}}_{\rho,\sigma}$.

b) For every $u \in \mathrm{Aut}\,\mathcal{G}^+$, which|that normalizes L_0, and such|such that $u(L_\rho) \underset{\mathcal{G}^+}{\sim} L_\rho$, $u(L_\sigma) \underset{\mathcal{G}^+}{\sim} L_\sigma$, there exists $u' \in \mathrm{Aut}(\mathcal{G}'^+)$ which makes commutative

$$(90) \qquad \begin{array}{ccc} \mathcal{G}^+ & \xrightarrow{\varphi} & \mathcal{G}'^+ \\ {\scriptstyle u}\downarrow & & -\downarrow{\scriptstyle u'} \\ \mathcal{G}^+ & \xrightarrow{\varphi} & \mathcal{G}'^+ \end{array},$$

in other words, u † † respects $\mathrm{Ker}\,\varphi$.

b') idem, assuming only, in place of u normalizing L_0, that $u(L_0) \underset{\mathcal{G}^+}{\sim} L_0$ (plus the other conditions)

c|b) ∃ group homomorphism

$$(91) \qquad \varphi_Z : \quad Z^{\mathcal{G}}_{\rho,\sigma} \xrightarrow{\dagger} Z^{\mathcal{G}'}_{\rho,\sigma}$$

† compatible with $\chi, \varepsilon_Z, \varepsilon'_Z$ and such that $\forall u \in Z^{\mathcal{G}}_{\rho,\sigma}$, if u' is its image we have commutativity in (90). (N.B. φ_Z will be † unique by|for ζ|?, and is compatible with $i_{L_0,Z}$) m1236

Marginal note. m1236 thus it induces $\gamma^{\mathcal{G}}_{\rho,\sigma} \longrightarrow \gamma^{\mathcal{G}'}_{\rho,\sigma}$

c') ∃ group homomorphism

$$(92) \qquad \varphi_G : \quad G^{\mathcal{G}}_{\rho,\sigma} \longrightarrow G^{\mathcal{G}'}_{\rho,\sigma}$$

compatible with $\chi_G, \varepsilon_G, \varepsilon'_G$ and satisfying the compatibility condition (90). (N.B. this φ_G [p. 569]

Uncertainty retained from cancelled passages.
140-3-s570-D001 ρ, σ, τ'
140-3-s570-D002 transforms
140-3-s570-D004 maps
140-3-s570-D005 stabilizes
140-3-s570-D006 φ_Z
140-3-s570-D007 with u'
140-3-s570-D008 [illeg. 1]

is then unique, and compatible with $i_{\mathcal{G},G}$ $i_{Z,G}$; †maps $\dagger Z_\rho^{\mathcal{G}}$ into $\dagger Z_\rho^{\mathcal{G}'}$, $\dagger Z_\sigma^{\mathcal{G}}$ into $\dagger Z_\sigma^{\mathcal{G}'}$, which extends to $\varphi_\rho :\ N_\rho^{\mathcal{G}} \longrightarrow N_\rho^{\mathcal{G}'}$ inducing $N_\rho^{*\mathcal{G}} \longrightarrow N_\rho^{*\mathcal{G}'}$, and extends to $\varphi_\sigma :\ N_\sigma^{\mathcal{G}} \longrightarrow N_\sigma^{\mathcal{G}'}$ inducing $N_\sigma^{*\mathcal{G}} \longrightarrow N_\sigma^{*\mathcal{G}'}$ ...) [scan 571] [p. 589]

†

We shall then say that φ is "extendable".

Corollary We can then extend φ to

$$(93) \qquad \varphi_M :\ S_0 G_{\rho,\sigma}^{\mathcal{G}} \longrightarrow S_0 G_{\rho,\sigma}^{\mathcal{G}'} \ ,$$

compatible with $\Theta_Z, \Theta_\rho, \Theta_\sigma, \Theta_G$ (taking into account $\varphi_Z, \varphi_\rho, \varphi_\sigma, \varphi_G$), and this in a way "compatible with everything", in particular with diagrams (45) (62).

Example Suppose $\Sigma_{\rho,\sigma}^{\mathcal{G}'} = \Sigma_{\rho,\sigma}^{\mathcal{G}'\,\mathrm{eff}}$, then the conditions of the proposition are automatically satisfied (whatever the surjective homomorphism $\mathcal{G} \longrightarrow \mathcal{G}'$).

Remark It may be that we are given a subgroup $Z_{\rho,\sigma}^{\natural|\sharp} \subset Z_{\rho,\sigma}^{\mathcal{G}}$, and that we consider † condition a) of the proposition only for $u \in Z_{\rho,\sigma}^{\natural|\sharp}$. But with the help of $Z_{\rho,\sigma}^{\natural|\sharp}$, we construct [m1237]

Marginal note. [m1237] We must suppose in addition that $\tau'_Z \in Z^*_{\rho,\sigma}$

$$(94) \quad G_{\rho,\sigma}^{\natural|\sharp} = Z_{\rho,\sigma}^{\natural|\sharp} \cdot \mathcal{G}^+, \quad Z_\rho^{\natural|\sharp} = Z_\rho \cap G_{\rho,\sigma}^{\natural|\sharp}, \quad Z_\sigma^{\natural|\sharp} = Z_\sigma \cap G_{\rho,\sigma}^{\natural|\sharp}$$

whence $N_\rho^{\natural|\sharp}$† and $N_\rho^{\natural|\sharp *}$†, $N_\sigma^{\natural|\sharp}$ and $N_\sigma^{\natural|\sharp *}$, $\gamma_{\rho,\sigma}^{\natural|\sharp}$, then $S_0 M_{\rho,\sigma}^{\natural|\sharp}$ and $M_{\rho,\sigma}^{\natural|\sharp}$. † This being so, we have an obvious variant of the [p. 570]

Uncertainty retained from cancelled passages.
140-3-s571-D001 transforms
140-3-s571-D002 N
140-3-s571-D003 N
140-3-s571-D004 N
140-3-s571-D005 N
140-3-s571-D006 it [illeg. 4]
140-3-s571-D008 a)
140-3-s571-D009 ρ, σ
140-3-s571-D010 ρ, σ
140-3-s571-D011 the

proposition, giving conditions for extension|extension of φ to $S_0 M_{\rho,\sigma}^* \longrightarrow S_0 M_{\rho,\sigma}^{\mathcal{G}'}$. [scan 572]

† [p. 571]

Uncertainty retained from cancelled passages.
140-3-s572-D001 { 1 ε

[scan 573]

8°) The universal case $\mathcal{G} = \mathrm{Gl}(2,\mathbb{Z})^\wedge$. [p. 590]

. Let us consider the exact sequence

$$(1) \quad 1 \longrightarrow \{\pm 1\} \longrightarrow \mathcal{N}_{1,1}^{\sim} \longrightarrow \mathcal{M}_{0,3}^{\sim} \longrightarrow 1$$
$$\qquad\qquad\qquad\qquad\qquad\qquad \| \qquad\qquad$$
$$\qquad\qquad\qquad\qquad\qquad\qquad \mathcal{M}$$

We have

$$(2) \qquad \{\pm 1\} = \text{Centr|Centre}(\mathcal{M}) = \text{Centr|Centre}(\widehat{\mathcal{G}}^{+}),$$

m1238

Marginal note. m1238 we have taken|we have set

$$\begin{aligned} \mathcal{G} &= \mathcal{T}_{1,1}|\mathcal{G}_{1,1} \simeq \mathrm{Gl}(2,\mathbb{Z}), \\ \widehat{\mathcal{G}} &= \widehat{\mathcal{T}}_{1,1} \simeq \mathrm{Gl}(2,\mathbb{Z})^{\wedge}, \\ \widehat{\mathcal{G}}^{+}|\mathcal{G}^{+} &= \mathcal{T}^{+}_{1,1} = \mathrm{Sl}(2,\mathbb{Z}), \\ \widehat{\mathcal{G}}^{+} &= \mathrm{Sl}(2,\mathbb{Z})^{\wedge}. \end{aligned}$$

whence by (1) a map

$$(3) \quad \mathcal{M}^{\sim}_{0,3} \longrightarrow \mathrm{Aut}(\mathcal{M}) \xrightarrow[\mathcal{G}^{+\wedge}=\mathrm{Sl}(2,\mathbb{Z})^{\wedge}]{\text{restriction to}} \mathrm{Aut}(\widehat{\mathcal{G}}^{+\dagger})$$

Besides, every automorphism of $\widehat{\mathcal{G}}^{+}$ preserves its center, hence passes to the quotient, whence

$$(4) \quad \mathrm{Aut}(\widehat{\mathcal{G}}^{+}) \longrightarrow \mathrm{Aut}(\widehat{\mathcal{G}}^{+}/\pm 1 \simeq \dagger\mathcal{G}^{+\wedge}_{0,3})$$

and the composite|the composite of (3) with (4) is the canonical homomorphism:

$$(5) \quad \begin{array}{ccc} \mathcal{M}^{\sim}_{0,3} & \longrightarrow & \mathrm{Aut}(\widehat{\mathcal{G}}^{+}) \\ & \searrow \quad \swarrow & \\ & \mathrm{Aut}(\widehat{\mathcal{G}}^{+}_{0,3} = \widehat{\mathcal{G}}^{+}/\{\pm 1\}) & \end{array}$$

This shows that (3) is injective, since $\mathcal{M}^{\sim}_{0,3} \longrightarrow \mathrm{Aut}(\widehat{\mathcal{G}}^{+}_{0,3})$ is.

Remark The map $\mathrm{Aut}(\widehat{\mathcal{G}}^{+}) \longrightarrow \mathrm{Aut}(\mathcal{G}^{+\wedge}_{0,3})$ is not injective on the other hand, † the elements of the kernel correspond to the homomorphisms $\widehat{\mathcal{G}}^{+} \longrightarrow \text{Centr|Centre}(\widehat{\mathcal{G}}^{+}) = \{\pm 1\}$, [p. 572]

Uncertainty retained from cancelled passages.

140-3-s573-D002 $\wedge$

140-3-s573-D004 $\mathcal{M}$

140-3-s573-D006 the

[scan 574]

or again to the homomorphisms from

$$(6) \quad \mathcal{G}^{+\wedge}_{\mathrm{ab}} \longrightarrow \{\pm 1\} \qquad \text{or again} \quad (\mathcal{G}^{+}_{\mathrm{ab}})_2 \longrightarrow \{\pm 1\}$$

Now we see that

$$\mathcal{G}^{+\wedge}_{\mathrm{ab}} = \left\{ \rho, \sigma \;\middle|\; \begin{array}{c} \rho^6 = \sigma^4 = [\rho,\sigma] = 1 \\ \rho^3 = \sigma^2 \end{array} \right\} = \mathbb{Z}/12\mathbb{Z}$$

m1239

Marginal note. m1239

$$\rho \longmapsto 2 \bmod 12$$

$$\sigma \longmapsto 3 \bmod 12$$

hence

$$(\mathcal{G}^{+}_{\mathrm{ab}})_2 \xrightarrow{\sim} \mathbb{Z}/2\mathbb{Z}$$

$$(7)\quad \begin{cases} \rho \longmapsto 0 \\ \sigma \longmapsto 1 \end{cases}$$

This isomorphism corresponds to the character $\mathcal{G}^{+\wedge} \longrightarrow \{\pm 1\}$ which is also obtained as the composite

$$(8)\quad \mathcal{G}^{+\wedge} \longrightarrow \underset{\substack{\text{symmetric group}\\ \text{on three letters}}}{\mathcal{G}_3} = \mathcal{G}^{+\wedge}/\underset{\substack{\text{inverse}\\ \text{image}\\ \text{of } \Pi^{\wedge}_{0,3}\\ \text{in } \mathcal{G}^{+\wedge}}}{\widehat{\Pi}} \xrightarrow[\text{of a permutation}]{\text{sign}} \{\pm 1\}$$

(sg|sgn)

Thus the only element Θ † from 1 of the kernel of (4) is given by

$$(9)\quad \Theta : x \longmapsto x \cdot \mathrm{sg}(x)$$

We shall note, † on setting

$$(10)\qquad L_0^{\mathcal{G}} = \varepsilon_0^{\widehat{\mathbb{Z}}}, \qquad L_0^{\Pi} = L_0^{\mathcal{G}} \cap \Pi = l_0^{\widehat{\mathbb{Z}}},$$

that the subgroup $\Theta(L_0^{\mathcal{G}}) \subset \mathcal{G}^{+\wedge}$ is not $\mathcal{G}$-conjugate to L_0, † for it is generated by † :

$$(11)\qquad \Theta(\varepsilon_0) = -\varepsilon_0 \qquad \begin{pmatrix} \text{for } \varepsilon_0 = \sigma\rho \\ \mathrm{sg}(\varepsilon_0) = -1 \end{pmatrix}$$

and we check that ε_0 is not conjugate to $(-\varepsilon_0)^{\mu} = \dagger - \varepsilon_0^{\mu}$ for any $\mu \in \widehat{\mathbb{Z}}^*$ (as we see [p. 573]

Uncertainty retained from cancelled passages.

140-3-s574-D002 different

140-3-s574-D005 Θ and

140-3-s574-D006 or

140-3-s574-D008 $\Theta\varepsilon$

140-3-s574-D010 ε_0

[scan 575]

at once, on †passing: $\mathcal{G}^{\wedge}_{0,3}/\Pi^{\wedge}_{0,3\mathrm{ab}}$ and noting that $l_0 = \varepsilon_0^2$ would be conjugate to: $(-\varepsilon_0^{\mu})^2 = l_0^{\mu}$, [p. 591] that we must have|which must have from it $\mu = 1$, now ε_0 and $-\varepsilon_0$ are not conjugate even in $\mathrm{Sl}(2, \widehat{\mathbb{Z}})$'|! ...). Now we shall be interested precisely in the automorphisms of|φ $\mathcal{G}^{+\wedge}$ which transform $L_0^{\mathcal{G}}$ into a conjugate subgroup. We shall note|note that on the other hand

$$(12).\qquad \Theta(l_0) = l_0 \qquad \text{hence} \qquad \Theta(L_0^{\Pi}) = L_0^{\Pi}$$

Proposition The homomorphism (3)

$$\mathcal{M}^{\sim}_{0,3} \longrightarrow \mathrm{Aut}(\widehat{\mathcal{G}}^{+})$$

is injective, and its image consists of the automorphisms$u$ of $\widehat{\mathcal{G}}^{+}$ which satisfy the following two conditions

a) u stabilizes $\widehat{\Pi} = \mathcal{G}^{+\wedge}[2] = \mathrm{Ker}(\mathcal{G}^{+\wedge} \longrightarrow \mathrm{Sl}(2, \mathbb{Z}/2\mathbb{Z}))$

b) u transforms $L_0^{\mathcal{G}}$ into a subgroup of $\mathcal{G}^{+\wedge}$ $\mathcal{G}^{+\wedge}$-conjugate to L_0; i.e. $\exists \mu \in \widehat{\mathbb{Z}}^*$ and $f_0 \in \mathcal{G}^{+\wedge}$ such that we have

$$(13) \qquad u(\varepsilon_0) = \mathrm{int}(f_0^{-1})(\varepsilon_0^{\mu}).$$

Proof. Injectivity has already been seen. †Let us show that conditions a)b) are necessary for u to †come from a $u_0 \in \mathcal{M}^{\sim}_{0,3}$, a) this is trivial, since u_0 stabilizes $\Pi^{\wedge|\sim}_{0,3}$, hence its [p. 574]

Uncertainty retained from cancelled passages.

140-3-s575-D001 *a*

140-3-s575-D002 Let us prove|Let us recall

140-3-s575-D003 belong

[scan 576]

inverse image in $\mathcal{G}^{+\wedge}$. For condition b), we are reduced to the case where $u_0 \in \mathcal{M}^{\sim}_{0,3}[0]$ (conjugating if necessary by an element of $\mathcal{G}^{+\wedge}_{0,3} \simeq \mathcal{G}^{+\wedge}/\pm 1$), in this case we have by construction of $\mathcal{N}^{\sim}_{1,1}$ from $\mathcal{M}^{\sim}_{0,3}$ (cf. III, pages|pages 665...)

$$(14) \qquad u_0(\varepsilon_0) = \varepsilon_0^{\mu}$$

where $\mu \in \widehat{\mathbb{Z}}^*$ is the multiplier, OK.

Conversely, let u be an automorphism of $\mathcal{G}^{+\wedge}$ which satisfies a),b), consider the automorphism $\tilde{u}$ of $\mathcal{G}^{+\wedge}/\pm 1 \simeq \mathcal{G}^{+\wedge}_{0,3}$ which it induces. We thus see that $\tilde{u}(\Pi^{\wedge}_{0,3}) = \Pi^{\wedge}_{0,3}$, and that $\tilde{u}(\varepsilon_0)$ is $\mathcal{G}^{\wedge}_{0,3}$-conjugate to an ε_0^{μ} ($\mu \in \widehat{\mathbb{Z}}^*|\widehat{\mathbb{Z}}$). Thus by definition, we have $\tilde{u} \in \mathcal{M}^{\sim}_{0,3}$ (cf. pages|pages 390 ff). Let u' be the automorphism of $\mathcal{G}^{+\wedge}$ induced by $\tilde{u}$ (via (3)) – then u' and u induce the same automorphism of $\mathcal{G}^{+\wedge}_{0,3} = \mathcal{G}^{+\wedge}/\{\pm 1\}$, thus we will have $u' = u \circ \Theta$, where $\Theta = u^{-1}u'$ is an automorphism which

1°) induces the identity on $\mathcal{G}^{+\wedge}/\{\pm 1\}$, and

2°) takes $L_0^{\mathcal{G}}$ to a conjugate subgroup

– we have seen that this implies $\Theta = 1$, q.e.d.

Let us now consider the exact †sequence

$$(15) \quad 1 \longrightarrow \pm 1 \longrightarrow \widehat{\Pi} \longrightarrow \Pi^{\wedge}_{0,3} \longrightarrow 0$$

$$\widehat{\Pi} \;\|\; \mathcal{G}^{+\wedge}[2]$$

[p. 575]

Uncertainty retained from cancelled passages.

140-3-s576-D003 lemma

[scan 577]

[p. 592]

and consider the splitting given by

$$(16)\quad \left\{ \qquad \Pi_{0,3} \longrightarrow \Pi = \mathcal{G}^+[2] = \mathrm{Ker}(\mathrm{Sl}(2,\mathbb{Z}) \longrightarrow \mathrm{Sl}(2,\mathbb{Z}/2\mathbb{Z}))$$

$$l_i \longmapsto \dagger\,\lambda_i \overset{\text{dfn}}{=} -l_i$$

There $\dagger$ is such a homomorphism, by virtue of

$$(17)\quad \begin{cases} (-l_\infty)(-l_1)(-l_0) = 1 \quad \text{in } \mathrm{Sl}(2,\mathbb{Z}) \quad \text{i.e.} \\ l_\infty l_1 l_0 = -1 \end{cases}$$

)

and it is obvious that this is a splitting. We also likewise have a splitting for $\widehat{\Pi}$ over $\widehat{\Pi}_{0,3}$. We denote by Π_0 the image of 16), i.e.

$$(18)\quad \begin{cases} \Pi_0 = \text{subgroup of } \Pi = \mathcal{G}^+[2] \text{ generated by} \\ \quad \lambda_0, \lambda_1, \lambda_\infty \qquad (\text{where } \lambda_i = -l_i) \end{cases}$$

and we thus have a subgroup $\widehat{\Pi}_0$ of $\widehat{\Pi}$.

Proposition a) The subgroup Π_0 of Π is of index 2, and it is normal in $\mathcal{G}^{\wedge|*}$.

b) The subgroup $\widehat{\Pi}_0$ of $\widehat{\Pi}$ is $\dagger$of index 2, and it is normal in $\mathcal{M}$ – or, what amounts to the same, stable under the action of $\mathcal{M}/\pm 1 \simeq \mathcal{M}^{\sim}_{0,3}$.

Proof Since

$$(19)\quad \begin{cases} \rho(\lambda_i) = \lambda_{i+1} \\ \sigma(\lambda_0) = \lambda_1^{\dagger}, \quad \sigma(\lambda_1) = \lambda_0^{\dagger} \\ \tau(\lambda_0) = \lambda_0^{-1}, \quad \tau(\lambda_1) = \lambda_1^{-1} \end{cases}$$

m1241

Marginal note. m1241

$$\sigma(\lambda_\infty) = \sigma(\lambda_1\lambda_0)^{-1} = \lambda_1^{-1}\lambda_0^{-1}$$
$$\dagger$$
$$\tau(\lambda_\infty) = \tau(\lambda_1\lambda_0)^{-1} = \lambda_0\lambda_1$$

we see that Π_0 is $\dagger$ normalized by ρ, σ, τ, hence by $\mathcal{G}$. That it is of index 2 is obvious. Thus $\widehat{\Pi}_0$ is of index 2 in $\widehat{\Pi}$ and normalized by $\widehat{\mathcal{G}}$. To see that it is [p. 576]

Uncertainty retained from cancelled passages.

140-3-s577-D001 $\lambda_i | l_i$ dfn
140-3-s577-D002 is
140-3-s577-D004 and
140-3-s577-D005 -1
140-3-s577-D006 -1
140-3-s577-D007 τ'
140-3-s577-D009 stable

[scan 578]

normalized by $\mathcal{M}$ in its full entirety, since

$$\mathcal{M} = \mathcal{M}(0)\,.\,\mathcal{G}^{+\wedge},$$

we are reduced to verifying that it is normalized by $\mathcal{M}(0) \simeq \mathcal{M}^{\sim}_{0,3}(0)$. Now we have, for $u \in \mathcal{M}^{\sim}_{0,3}(0)$

$$(20) \qquad \begin{cases} u(\rho) = \mathrm{int}(\alpha)\rho \\ u(\sigma) = \varepsilon_2\,\mathrm{int}(\beta)\sigma \\ u(\varepsilon_0) = \varepsilon_0^{\mu} \end{cases} \qquad \begin{cases} \varepsilon_2 = \varepsilon_2(\mu) \\ \alpha, \dagger \in \widehat{\Pi}\,.\,\{1,\tau\} \\ \beta \in \widehat{\Pi} \end{cases}$$

whence

$$\dagger \quad (21) \qquad \begin{cases} u(\ell_0) = \ell_0^{\mu} \\ u(-\ell_0) = (-\ell_0)^{\mu} \quad \text{i.e.} \quad u(\lambda_0) = \lambda_0^{\mu} \quad (\text{since } (-1)^{\mu} = -1) \end{cases}$$

)

† † thus $u(\lambda_0) \in \widehat{\Pi}_0$, it remains to prove that $u(\lambda_1) \in \widehat{\Pi}_0$, now

$$u(\lambda_1) = u(\rho(\lambda_0)) = \underbrace{(u\rho)}_{\dagger(\mathrm{int}(\alpha)\rho)u}(\lambda_0) = \mathrm{int}(\alpha)(\rho)\left(\underbrace{u(\lambda_0)}_{\lambda_0^{\mu}}\right) = \dagger.$$

† and since $\widehat{\Pi}_0$ is normalized by $\mathcal{G}^{\wedge}$ and in particular by $\dagger\alpha\rho\alpha^{-1}$, we indeed find $u(\lambda_1) \in \Pi_0$.

Corollary We have

$$(22) \qquad \begin{cases} \Pi \simeq \Pi_0 \times \mu \\ \widehat{\Pi} \simeq \widehat{\Pi}_0 \times \mu \end{cases}$$

where $\mu = \{\pm 1\} = \mathrm{Centr}(\mathcal{G}^{+\wedge})$.

† This is trivial, since we have a splitting of the central extension (15). [p. 577]

Uncertainty retained from cancelled passages.

140-3-s578-D001 β
140-3-s578-D003 (20)
140-3-s578-D004 remains see
140-3-s578-D005 Indeed|Finally
140-3-s578-D006 u
140-3-s578-D007 $\mathrm{int}(\alpha)(\lambda_1^{\mu})$
140-3-s578-D008 seen
140-3-s578-D009 ρ
140-3-s578-D010 note -1 of is not

[scan 579]

Let|Let [p. 593]

$$(23) \qquad \begin{cases} \mu_\tau = \{1,\tau\} \subset \mathcal{G} & D_\tau = \mu_\tau \times \mu \subset \mathcal{G} \\ \mu_\sigma = \{1,\sigma|\tau'\} \subset \mathcal{G} & D_\sigma = \mu_\sigma \times \mu \subset \mathcal{G} \end{cases}$$

so that we have

$$(24) \qquad \begin{array}{ll} D_\tau \cap \mathcal{G}^+ = D_\tau \cap \Pi = \mu & D_\tau \cap \Pi_0 = \{1\} \\ D_\sigma \cap \mathcal{G}^+ = D_\sigma \cap \Pi = \mu & D_\sigma \cap \Pi_0 = \{1\}, \end{array}$$

We set

$$(25) \qquad \begin{cases} \Pi_\tau = \mu_\tau\,.\,\Pi = D_\tau\,.\,\Pi_0, & \Pi_{0\tau} = \mu_\tau\,.\,\Pi_0 \\ \Pi_\sigma = \mu_\sigma\,.\,\Pi = D_\sigma\,.\,\Pi_0, & \Pi_{0,\sigma} = \mu_\sigma\,.\,\Pi_0 \end{cases}$$

idem on replacing $\mathcal{G}, \Pi$ by $\mathcal{G}^{\wedge}, \widehat{\Pi}$

so that we have

$$(26) \qquad \begin{cases} \Pi_\tau = \Pi_{0\tau} \times \mu \\ \Pi_\sigma = \Pi_{0,\sigma} \times \mu \end{cases},$$

and idem on putting in $\wedge$'s.

Normalization of α, β The relations 20, determine α, β only "modulo sign". We remove this ambiguity by taking

$$(27) \qquad \alpha \in \Pi_{0,\tau}, \qquad \beta \in \Pi_0.$$

The group $\mathcal{G}/\Pi_0$. It is described by generators ρ, σ, τ' and relations by

$$(28) \qquad \begin{cases} \rho^6 = \sigma^4 = \tau^2 = 1, \quad \rho^3 = \sigma^2 \\ \tau(\rho) = \sigma(\rho^{-1}), \quad \tau(\sigma) = \sigma^{-1} \\ \tau(\rho) = \rho \end{cases}$$

For we have

$$(29) \qquad \tau(\rho) = (-\ell_1^{-1})\rho$$

thus the last relation 28 reads $-\ell_1^{-1} = 1$ i.e. $\lambda_1 = 1$ (and implies $\dagger\lambda_\infty = \lambda_0 = 1$ on applying ρ). [p. 578]

Uncertainty retained from cancelled passages.

140-3-s579-D002 $-\ell_\infty = -\ell_0$

[scan 580]

In terms of ρ, σ, τ' we find likewise

$$(30) \qquad \begin{cases} \rho^6 = \sigma^4 = \tau'^2 = 1, \quad \rho^3 = \sigma^2 \\ \tau'(\rho) = \rho^{-1}, \quad \tau'(\sigma) = \sigma^{-1} \\ \sigma(\rho) = \rho^{-1} \end{cases}$$

The subgroup $\mathcal{G}^+/\Pi_0$ has the presentation in ρ, σ

$$(31) \qquad \left\{ \begin{cases} \rho^6 = \sigma^4 = 1, \quad \rho^3 = \sigma^2 \\ \sigma(\rho) = \rho^{-1} \end{cases} \right\}$$

What is the operation of $\mathcal{M}^{\sim}_{0,3} \simeq \mathcal{M}/\pm 1$ on $\mathcal{G}^+/\Pi_0$? † We have a structure of extension

$$(32) \qquad \begin{array}{ccccccccc} 1 & \longrightarrow & \mu & \longrightarrow & \mathcal{G}^+/\Pi_0 & \longrightarrow & \mathcal{G}^+/\Pi & \longrightarrow & 1 \\ & & \| & & & & \| & & \\ & & \pm 1 & & & & \mathcal{G}_3 & & \end{array}$$

and the automorphisms in question preserve this extension structure. The elements|elements of $\mathcal{M}^{!\sim}_{0,3} \simeq \mathcal{M}^!/\{\pm 1\}$ operate by inducing the identity in the quotient $\mathcal{G}_3$, and of course also in μ — now such an automorphism of the extension corresponds to a homomorphism

$$(33) \qquad \begin{array}{ccc} \mathcal{G}_3 & \longrightarrow & \mu \\ & & \| \\ & & \pm 1 \end{array} \qquad \text{i.e.} \qquad \mathcal{G}_{3\,\mathrm{ab}} \longrightarrow \mu$$

now

$$\mathcal{G}_{3\,\mathrm{ab}} \xrightarrow{\underline{\mathrm{sg}}} \{\pm 1\} = \mu$$

thus the group of automorphisms of $\mathcal{G}^+/\Pi_0$ inducing the identity in $\mathcal{G}_3$ is $\simeq \mathbb{Z}/2\mathbb{Z}$, a generator Θ is given by

$$(34) \qquad \Theta(x) = \mathrm{sg}(x)\,.\,x \quad \text{thus} \quad \begin{cases} \Theta(\rho) = \rho \\ \Theta(\sigma) = -\sigma \end{cases}$$

[p. 579]

Uncertainty retained from cancelled passages.
140-3-s580-D001 it is an

[scan 581]
[p. 594]

Comparing with the formulas (20), we find

$$(35)\qquad {}^{\dagger}u|_{\mathcal{G}^{+\wedge}/\widehat{\Pi}_0} = \Theta^{\varepsilon_2(u)} : \begin{matrix} \dot\rho \longmapsto \dot\rho \\ \dot\sigma \longmapsto \varepsilon_2(u)\dot\sigma \end{matrix}$$

i.e.

$$(35\text{ bis})\qquad \begin{matrix}[\dot u, \dot\sigma] \\ \| \\ \dot u(\dot\sigma)\dot\sigma^{-1}\end{matrix} = \varepsilon_2(u)^{\cdot} \quad \text{in } \mathcal{G}/\Pi_0.$$

m1244

Marginal note. m1244 This can also be written $[\dot u, \dot x] = \mathrm{sg}(x)^{\nu_2(u)}$ in $\mathcal{G}^{+\wedge}/\Pi_0$ ($u \in \mathcal{M}^!, x \in \mathcal{G}^{+\wedge}$) (careful, this goes wrong|does if we assume there for $u \in \mathcal{M}^!$ only $u \in \mathcal{M}$ ...)

Let us take up again the definition

$$(36)\qquad Z = Z_{\rho,\sigma} \overset{\text{dfn}}{=} \left\{ u \in \mathrm{Aut}(\mathcal{G}^{+\wedge}) \;\middle|\; \begin{matrix} u(L_\rho)\ \mathcal{G}^{+\wedge}\text{-conjugate to } {}^{\dagger}L_\rho \\ u(L_\sigma)\ \mathcal{G}^{+\wedge}\text{-conjugate : } L_\sigma \\ u(L_0) = L_0 \end{matrix} \right\}$$

so that we shall have, for $u \in Z$

$$(37)\qquad \begin{cases} u(\rho) & \mathcal{G}\text{-conjugate to } \rho^{\varepsilon'} \\ u(\sigma) & \mathcal{G}\text{-conjugate to } \sigma^{\varepsilon} \quad (= \varepsilon\sigma) \\ u(\varepsilon_0) = \varepsilon_0^{\mu} \end{cases}$$

with $\mu \in \widehat{\mathbb{Z}}^*$ and $\varepsilon, \varepsilon' \in \{\pm 1\}$ well determined, (for

$$\widehat{\mathbb{Z}} \longrightarrow \mathcal{G}$$

$$n \longmapsto \varepsilon_0^n$$

is injective, ρ and ρ^{-1} are not $\mathcal{G}^{+\wedge}$-conjugate, nor σ and σ^{-1}). I say that such a u comes from $\mathcal{M}^{\sim}_{0,3}$ — it remains to check, by virtue of the proposition..., that u stabilizes $\widehat{\Pi}$, which is generated by $\ell_0, \ell_1, \ell_\infty$. Now we shall have

$$u(\ell_0) = \ell_0^{\mu} \in \widehat{\Pi}$$

and

$$u(\ell_1) = \begin{matrix} \underbrace{u\rho} \\ \dagger \\ \dagger \\ \mathrm{int}(f)(\rho^{\varepsilon'})u \\ f \in \mathcal{G}^{+\wedge} \end{matrix} (\ell_0) = \mathrm{int}(f)(\rho^{\varepsilon'})(u(\ell_0)) = \mathrm{int}(f)(\rho)^{\varepsilon'}(\ell_0^{\mu})$$

now $\mathcal{G}^\wedge$ and a fortiori $\mathrm{int}(f)(\rho)^{\varepsilon'}$ normalizes †$\widehat{\Pi}$, thus $u(\ell_1) \in \widehat{\Pi}$ and since $u(-1) = -1$, whence:

$$u(\ell_\infty) = -u(\ell_1\ell_0)^{-1} = -u(\ell_0^{-1})u(\ell_1)^{-1}$$

we also have $u(\ell_\infty) \in \Pi$, OK.

If we identify u with an element of $\mathcal{M}^{\sim}_{0,3}$, we see that † in fact † [p. 580]

Uncertainty retained from cancelled passages.

140-3-s581-D001 u
140-3-s581-D003 ℓ_ρ
140-3-s581-D004 $\mathrm{int}(f)\rho u$
140-3-s581-D005 $\mathrm{int}(f)\rho^{\varepsilon'} u$
140-3-s581-D006 $\mathcal{G}$
140-3-s581-D008 it is
140-3-s581-D009 $Z|\mathcal{M}$

[scan 582]

$$u \in \mathcal{M}^{\sim}_{0,3}[0] \simeq \mathcal{M}^{!}[0] \subset \mathcal{M}$$

m1245

Marginal note. m1245 Careful we had not previously introduced $\mathcal{M}^{\sim}_{0,3}|\mathcal{M}_{0,3}[0]$, but only $\mathcal{M}^{!\sim}_{0,3}[0] \subset \mathcal{M}^{!\sim}_{0,3}$. Now it is not true that we shall have here $u \in \mathcal{M}^{!\sim}_{0,3}$, for example we may have $u = \varepsilon_0$. The image of u in $\mathcal{G}_3$ is in † $\{1, \dot\sigma_0\}$, thus by correcting by ε_0, we shall have an element of $\mathcal{M}^{!\sim}_{0,3}[0]$. Cf. more detailed correction on page 609 ff. $Z^!_{\rho,\sigma} = \mathcal{M}^!{}[0] = Z_{\rho,\sigma} \cap \mathcal{M}^!$.

† thus (20), more precisely

$$(38) \qquad \begin{cases} u(\rho) = \mathrm{int}(\alpha)\rho \\ u(\sigma) = \varepsilon_2\,\mathrm{int}(\beta)\sigma \\ u(\varepsilon_0) = \varepsilon_0^\mu \end{cases} \qquad \text{with †} \quad \alpha \in \widehat{\Pi}_{0\tau},\ \beta \in \widehat{\Pi}_0 \qquad \dagger \quad \begin{cases} \mu \in \widehat{\mathbb{Z}}^* \\ \end{cases}$$

in their turn imply|implying (37). In fact the formulas (38) for $u(\rho), u(\sigma), u(\varepsilon_0)$ uniquely determine $\mu \in \widehat{\mathbb{Z}}^*$, $\varepsilon_2 \in \{\pm 1\}$, $\alpha \in \widehat{\Pi}_{0,\tau}$, $\beta \in \widehat{\Pi}_0$, and we shall then necessarily have

$$(39) \qquad \varepsilon_2 = \varepsilon_2(\mu), \qquad \varepsilon_\tau(\alpha) = \varepsilon_3(\mu)$$

where ε_τ is the homomorphism that occurs in the exact sequence

$$(40) \qquad \dagger \quad 1 \longrightarrow \widehat{\Pi}_0 \longrightarrow \widehat{\Pi}_{0,\tau} \xrightarrow{\varepsilon_\tau} \{\pm 1\} \longrightarrow 1 \quad .$$

Thus we have proved

Proposition In the universal case ($\mathcal{G} = \mathcal{G}^{+\wedge}$) we have

$$Z^!_{\rho,\sigma} \xleftarrow{\sim} \mathcal{M}^!{}[0] \overset{\text{dfn}}{=} \mathcal{M}^{!\sim}_{0,3}[0] \quad ,$$

the components $u, \mu, \varepsilon', \varepsilon$ of an element of $Z^!_{\rho,\sigma}$ correspond resp. (in terms of $u_0 \in \mathcal{M}[0]$): u_0 identified with an element of $\mathrm{Aut}(\mathcal{G}^{+\wedge})$, $\mu = \chi(u_0)$ (the multiplier of u_0), $\varepsilon = \varepsilon_3(\mu)$, $\varepsilon' = \varepsilon_2(\mu)$.

Corollary We have

$$G_{\rho,\sigma} \simeq \mathcal{M} = \mathcal{N}^{\sim}_{1,1},$$
$$\gamma_{\rho,\sigma} \simeq \mathcal{M}/\mathcal{G}^{+\wedge} \simeq \mathcal{N}^{\sim}_{1,1}/\mathcal{G}^{+\wedge} \simeq \mathcal{M}^{\sim}_{0,3}/\mathcal{G}^{+\wedge}_{0,3} \simeq \Gamma^{\sim}_{\mathbb{Q}}.$$

[p. 581]

Uncertainty retained from cancelled passages.

140-3-s582-D001 sg

140-3-s582-D003 we have

140-3-s582-D004 α

140-3-s582-D005 ε_2

140-3-s582-D007 ε_0

[scan 583]

Remark It is tempting to write the first two relations (38) in the form [p. 595]

$$(41)\qquad \begin{cases} (\alpha^{-1}u)(\rho) = \rho & \text{i.e.} \quad \alpha^{-1}u \in Z_\rho^\sharp \\ (\beta^{-1}u)(\sigma) = \varepsilon_2\sigma & \text{whence|i.e.} \quad \beta^{-1}u \in N'_\sigma \\ \dagger \end{cases}$$

and then to introduce

$$(42)\qquad \begin{cases} a = \alpha^{-1}u \in Z_\rho^\sharp, & \text{thus } u = \alpha a, \quad \alpha = ua^{-1} \\ b = \beta^{-1}u \in N'_\sigma, & \text{thus } u = \beta b, \quad \beta = ub^{-1}, \end{cases}$$

and the developments of § 48 (notably XI) may lead one to expect that the quantities a, b, for $u \in \mathcal{M}[0]$ variable, depend multiplicatively on u. We are going to see however that this is not so.

Let indeed

$$\delta : \dagger\, G_{\rho,\sigma} \xrightarrow{\sim} \mathcal{M} \longrightarrow \gamma_{\rho,\sigma} \simeq \Gamma_{\mathbb{Q}}$$

be the canonical homomorphism, so that we will have

$$(43)\qquad \delta(u) = \delta(b) = \tau_\gamma^{\nu_3}\delta(a),$$

putting

$$\tau_\gamma = \delta(\tau), \qquad \varepsilon_3 = (-1)^{\nu_3} \quad (\nu_3 \in \mathbb{Z}/2\mathbb{Z})$$

If u' corresponds to b', a', ν'_3, we will likewise have

$$(43\,\text{bis})\qquad \delta(u') = \delta(b') = \tau_\gamma^{\nu'_3}\delta(a')$$

and if $u'' = u'u$ corresponds to a'', b'', ν''_3,

$$(43\,\text{ter})\qquad \delta(u'u) = \delta(b'') = \tau_\gamma^{\nu''_3}\delta(a''),$$

In fact, we will have

$$(44)\qquad \begin{array}{l} b'' = b'b, \quad \nu''_2 = \nu'_2 + \nu_2 \quad \text{and.|and} \quad \nu''_3 = \nu'_3 + \nu_3 \\ (\text{i.e. } \varepsilon''_2 = \varepsilon'_2\varepsilon_2 \quad \text{and} \quad \varepsilon''_3 = \varepsilon'_3\varepsilon_3) \end{array}$$

[p. 582]

Uncertainty retained from cancelled passages.

140-3-s583-D001 $u(\varepsilon_0)$

140-3-s583-D002 $\mathcal{M}$

[scan 584]

We have seen

$$\dagger u(\sigma) = \varepsilon_2\,\mathrm{int}(\beta)\sigma$$
$$u'(\sigma) = \varepsilon'_2\,\mathrm{int}(\beta')\sigma$$

whence

$$(u'u)(\sigma) = u'(u(\sigma)) = \varepsilon_2 \operatorname{int}(u'(\beta))u'(\sigma)$$
$$= \varepsilon_2\varepsilon'_2 \operatorname{int}(\dagger u'(\beta)\beta')\sigma$$

with $u'(\beta)\beta' \in \widehat{\Pi}_0$, which by uniqueness proves

$$(45) \qquad \beta'' = u'(\beta)\beta', \qquad \varepsilon''_2 = \varepsilon'_2\varepsilon_2$$

whence we indeed conclude

$$b'' \qquad = \quad \underset{\beta'^{-1}u'}{\underset{\smile}{b'}} \quad \underset{\beta^{-1}u}{\underset{\smile}{b}}$$
$$\|$$
$$\beta''^{-1}u'' = \beta''^{-1}(u'u)$$

i.e. $\beta''^{-1}u'u = \beta'^{-1}u'\beta^{-1}u$ i.e. $\beta''^{-1} = \beta'^{-1}u'(\beta^{-1})$
i.e. $\beta'' = u'(\beta)\beta'$ which is none other than (45).

The relation $\varepsilon''_3 = \varepsilon'_3\varepsilon_3$ is also evident, for example from the relation (which contains ε_3, and likewise $\varepsilon'_3, \varepsilon''_3$

$$(46) \qquad u(\rho)\ \mathcal{G}^{+\wedge}\text{-conjugate to } \rho^{\varepsilon_3}.$$

†

It remains therefore to examine the hypothetical relation

$$a'' = a'a$$

which † implies

$$(*) \qquad \delta(a'') = \delta(a')\delta(a)$$

but comparing (43) (43 bis) (43 ter) we find the relation

$$\dagger\delta(u')\delta(u) = \tau_\gamma^{\nu'_3}\delta(a')\tau_\gamma^{\nu_3}\delta(a)$$

$$\delta(u'u''|u) = \tau_\gamma^{\nu''_3}\delta(a'')$$

i.e. [p. 583]

Uncertainty retained from cancelled passages.

140-3-s584-D001 ε_2|and
140-3-s584-D002 β'
140-3-s584-D004 when we
140-3-s584-D005 is written|would be
140-3-s584-D006 δ

† [scan 585]
[p. 596]

$$\tau_\gamma^{\nu'_3+\nu_3}\delta(a'') = \tau_\gamma^{\nu'_3}\delta(a')\tau_\gamma^{\nu_3}\delta(a)$$

or again

$$\dagger\delta(a'') = \dagger\tau_\gamma^{\nu_3}\delta(a')\tau_\gamma^{\nu_3}\delta(a)\dagger$$

i.e.

$$(47) \qquad \delta(a'') = \dagger\tau_\gamma^{\nu_3}(\delta(a')) \cdot \delta(a).$$

Thus the relation (*) is equivalent to

$$(48) \qquad \delta(a') = \tau_\gamma^{\nu_3}(\delta(a')) \quad \text{i.e.} \quad [\delta(a'), \tau_\gamma^{\nu_3}] = 1 \quad \dagger$$
$$\text{i.e.} \quad [\delta(u'), \tau_\gamma^{\nu_3}] = 1 \quad \dagger$$

since $\delta(u) = \tau_\gamma^{\nu_3}\delta(a)$

This shows that if $\nu_3 = 1$ i.e. $\varepsilon_3 = -1$, then the relation (*) is equivalent to the commutation † of $\delta(u')$ † with τ_γ, †. It is a draconian condition on an element of $\Gamma^{\sim}_{\mathbb{Q}}$ that|which, for an element of $\Gamma_{\mathbb{Q}}$ itself in its version, implies that $u' \in \{1, \tau_\gamma\}$ — and by heuristic arguments, we have seen that it should even be so for every element of $\Gamma^{\sim}_{\mathbb{Q}}$. It appears that the success of the presentation of § 48 XI is due to the fact that in the situations|the situation considered|considered in the|the cases, the group $\gamma_{\rho,\sigma} \simeq \widehat{\mathbb{Z}}^*$ was commutative, or τ_γ central. Outside this case, the quantity

$$a = \alpha^{-1}u \in Z^\sharp_\rho \hookrightarrow G_{\rho,\sigma}$$

depends multiplicatively on the element $u \in \dagger\mathcal{M}^!{}[0]$, only on condition of restricting oneself to $u \in \mathcal{M}^!{}[0]'$.

m1247

Marginal note. m1247 †

$\xrightarrow[\dagger]{}$

[p. 584]

Uncertainty retained from cancelled passages.

140-3-s585-D001 $\tau_\gamma^{\nu_3''}$
140-3-s585-D003 ν_3
140-3-s585-D004 τ_γ
140-3-s585-D005 $\tau_\gamma^{\nu_3}$
140-3-s585-D006 int
140-3-s585-D007 or
140-3-s585-D008 (for
140-3-s585-D009 as
140-3-s585-D010 $\delta(a')$
140-3-s585-D011 commutation i.e. to
140-3-s585-D012 $Z_{\rho,\sigma}$
140-3-s585-D013 centralized|central
140-3-s585-D015 ε_3

[scan 586]

We are going to try to salvage things, by finding a canonical homomorphism

$$\begin{array}{ccc} \mathcal{M}^!{}[0] & \longrightarrow & \underbrace{Z^!_{\rho,\sigma} \times_{\gamma_{\rho,\sigma}} \dagger N^*_\rho \times_{\gamma_{\rho,\sigma}} N^*_\sigma}_{S_0\Gamma^!_{\rho,\sigma} \underset{\text{index } 2}{\subset} S_0\Gamma_{\rho,\sigma}} \\ \| & & \\ Z^!_{\rho,\sigma} & & \end{array}$$

m1248

Marginal note. m1248 N.B. we have

$$S_0\Gamma^!_{\rho,\sigma}/L^\Pi_0 \xrightarrow{\sim} S_0\Gamma_{\rho,\sigma}/L^{\mathcal{G}}_0$$

then

$$\gamma_{\rho,\sigma} \xrightarrow{\sim} \Gamma^{\sim}_{\mathbb{Q}}.$$

Let us note that the second side|set is formed of the triples

$$(49)\quad \{(u, \underset{\cup}{\dagger\mathcal{M}[0]}\ a, \underset{\cup}{N'_\rho}\ \dagger b, \underset{\cup}{N'_\sigma}) \mid u = fb = f'a, \text{ with } f, f' \in \mathcal{G}^{+\wedge} \text{ suitable}\}$$

$$\text{which implies} \quad \begin{matrix}\varepsilon_2(u) = \varepsilon_\sigma(b)\\ \varepsilon_3(u) = \varepsilon_\rho(a)\end{matrix}$$

and for such a triple, we will have

$$(49\text{ bis})\quad \begin{cases} u(\dagger\sigma) = \mathrm{int}(fb)\dagger(\sigma) = \mathrm{int}(f)\ \underbrace{\mathrm{int}(b)(\sigma)}_{\varepsilon_2(b)(\sigma)} = \dagger\varepsilon_2(u)\,\mathrm{int}(f)(\sigma) \\ u(\rho) = \mathrm{int}(f')(\rho^{\varepsilon_3(u)}) = \mathrm{int}(f')\varepsilon_3(\tau')(\rho) = \mathrm{int}(f'\varepsilon_3(\tau'))(\rho) \end{cases}$$

where we put

$$\varepsilon_3(\tau') \overset{\text{dfn}}{=} \begin{cases} 1 & \text{if } \varepsilon_3 = +1 \quad \text{i.e. } \varepsilon_3(\tau') = \sigma^{\nu_3} \\ \tau' & \text{if } \varepsilon_3 = -1 \quad \text{if } \varepsilon_3 = (-1)^{\nu_3} \end{cases}$$

If u corresponds to α, β (cf. (38)), the formulas (49 bis) are written also|thus respectively

$$\begin{cases} \mathrm{int}(f)(\sigma) = \mathrm{int}(\beta)(\sigma) & \text{i.e.} \quad f = \beta\sigma^i \quad (i \in \mathbb{Z}/4\mathbb{Z}) \\ \dagger \\ \mathrm{int}(f'\varepsilon_3(\tau'))(\rho) = \mathrm{int}(\alpha)(\rho) & \text{i.e.} \quad \underbrace{f'\varepsilon_3(\tau') = \alpha\rho^j}_{j\in\mathbb{Z}/6\mathbb{Z}} \end{cases}$$

Thus we find

Proposition For a fixed $u \in Z^!_{\rho,\sigma} = \mathcal{M}[0] \simeq \widetilde{\mathcal{M}}_{0,3}[0]$, the $(u, a, b) \in S_0\Gamma_{\rho,\sigma}$ over u are exactly the elements of

$$(50)\quad \left\{(u,a,b) \,\middle|\, \begin{matrix} a = f'^{-1}u & \text{with} & f' = \alpha\rho^j\varepsilon_3(\tau') & (j \in \mathbb{Z}/6\mathbb{Z}) \\ b = f^{-1}u & \text{with } f = \beta\sigma^i & & (i \in \mathbb{Z}/4\mathbb{Z}) \end{matrix} \right\}$$

m1249

Marginal note. m1249 where $\alpha, \beta, \varepsilon_2, \varepsilon_3$ are defined for u via (38)

Let us consider then the element (u, a, b) over u having the simplest expression in terms of α, β, by taking $i = 0|j$, $j = 0$, thus corresponding to the choices: [p. 585]

Uncertainty retained from cancelled passages.

140-3-s586-D001 Z
140-3-s586-D002 b
140-3-s586-D003 Z
140-3-s586-D004 ρ
140-3-s586-D005 $\varepsilon_2(b)|\,\mathrm{int}(b)$
140-3-s586-D006 ε_2
140-3-s586-D007 ε_3

[scan 587]
[p. 597]

$$(51)\qquad \begin{cases} a = \varepsilon_3(\tau')\alpha^{-1}u & \text{i.e.} \quad \alpha = ua^{-1}\varepsilon_3(\tau') \\ b = \beta^{-1}u & \text{i.e.} \quad \beta = ub^{-1} \end{cases},$$

I wonder whether this set-theoretic section of $S_0\Gamma^!_{\rho,\sigma}$ over $Z^!_{\rho,\sigma} = M^!{}[0]$ is multiplicative, or, which amounts to the same thing, whether a and b defined by (51) depend multiplicatively on † u.

Proposition. Let $(u,a,b) \in S_0\Gamma^!_{\rho,\sigma}$ (cf. (50)). For us to have $i=0$, $j=0$, i.e. for us to have (51) — i.e. for (u,a,b) † to belong to the preceding set-theoretic section — it is necessary and sufficient that we have the relations

$$(52)\qquad u \equiv b \equiv \varepsilon_3(\sigma)a\,\dagger \quad \mod \widehat{\Pi}_0.$$

m1250

Marginal note. m1250

m1251

Marginal note. m1251 where we have put $\varepsilon_3(\sigma) = \begin{cases} 1 & \text{if } \varepsilon_3 = +1 \\ \sigma & \text{if } \varepsilon_3 = -1 \end{cases}$

Proof. Since $\beta\dagger = ub^{-1}$ and $\dagger\ \dagger\alpha\varepsilon_3(\tau) = \dagger ua^{-1}\varepsilon_3(\tau'\tau) = ua^{-1}\varepsilon_3(\sigma^{-1})$ are both in $\widehat{\Pi}_0$, the condition is necessary. For sufficiency, we note that (52) reads, with the notation of (50), where $f = ub^{-1}$, $f' = ua^{-1}$ i.e. $f'\varepsilon_3(\sigma^{-1}) = ua^{-1}\varepsilon_3(\sigma^{-1})\ \dagger\ \dagger$,

N.B. $\tau'\tau = \sigma^{-1}$ $\tau'\sigma^{-1} = \tau$.

$$\dagger \qquad (53)\quad f, f'\varepsilon_3(\sigma^{-1}) \in \Pi_0$$

† and reducing modulo Π_0 in $\widehat{G}/\Pi_0$ this reads $\dot\beta\dot\sigma^i = 1$ and †, $\dot\alpha\dot\rho^j\varepsilon_3(\tau'\sigma^{-1})^{\cdot} = 1$, this last relation also reads $\dot\alpha\dot\rho^j\varepsilon_3(\tau)^{\cdot} = 1$ i.e. $\varepsilon_3(\dot\tau)\dot\alpha\dot\rho^j = 1$, $\varepsilon_3(\dot\tau)\dot\alpha\dot\rho^j = 1$ now $\varepsilon_3(\dot\tau)\dot\alpha$=1, the condition is

$$\dagger \qquad \dot\sigma^i = \dot\rho^j = 1 \qquad \text{i.e.} \quad i=0,\ j=0$$

q.e.d.. [p. 586]

Uncertainty retained from cancelled passages.

140-3-s587-D001 these
140-3-s587-D002 to be
140-3-s587-D003 $(\widehat{\Pi}_0)$
140-3-s587-D004 $\in$
140-3-s587-D005 $\alpha\varepsilon_3(\tau') = ua^{-1}$
140-3-s587-D006 α
140-3-s587-D007 ua
140-3-s587-D008 i.e. σ
140-3-s587-D009 f'
140-3-s587-D010 $f \in \Pi_0$
140-3-s587-D011 1
140-3-s587-D012 $\dot\alpha\dot\rho^j\varepsilon_3(\dot\tau) = 1$
140-3-s587-D016 (53)

† [scan 588]

Let|Let now u, corresponding to a, b, ε_3 and u', corresponding to a', b', ε'_3, let $u'' = u'u$, corresponding to a'', $b'' = b'b$ and $\varepsilon''_3 = \varepsilon'_3\varepsilon_3$, so that we shall have

$$\begin{array}{rcll} u & \equiv & \varepsilon_3(\sigma)a & (\widehat{\Pi}_0) \\ u' & \equiv & \varepsilon'_3(\sigma)a' & (\widehat{\Pi}_0) \\ u'' & \equiv & \underbrace{\varepsilon''_3(\sigma)}\, a'' & (\widehat{\Pi}_0) \\ \| & & & \\ u'u & \equiv & \dagger\varepsilon.\varepsilon'_3(\sigma).\varepsilon_3(\sigma).a'' & \text{where } \varepsilon \in \{\pm 1\},\ \varepsilon = -1 \text{ iff} \\ & & & \varepsilon_3 = -1 \quad \underline{\underline{\text{and}}} \quad \varepsilon'_3 = -1 \\ \| & & & \\ (\varepsilon'_3(\sigma)a')(\varepsilon_3(\sigma)a) & & & \end{array}$$

=
!

which gives

$$a'\varepsilon_3(\sigma)a \equiv \varepsilon\varepsilon_3(\sigma)a'' \quad (\widehat{\Pi}_0) \qquad \text{i.e.}$$

$$(53) \qquad a'' \equiv \varepsilon\varepsilon_3(\sigma|\sigma^{-1})(a').a$$

m1255

Marginal note. m1255 N.B. $\varepsilon_3(\sigma)$ and $\varepsilon_3(\sigma^{-1})$ operate in the same way.

Thus the expected relation

$$a'' = a'a \qquad \text{equivalent to} \quad a'' \equiv a'a \quad (\widehat{\Pi}_0)$$

is true if and only if † † † † †

m1256

Marginal note. m1256

⟶
⟶

[p. 587]

Uncertainty retained from cancelled passages.

140-3-s588-D001 $\gamma_{\rho,\sigma}$ $\Gamma_{\rho,\sigma}\rtimes$ G Γ_ρ is a of which covering|group
140-3-s588-D005 $\varepsilon''_3(\sigma)$
140-3-s588-D006 $a'' = \varepsilon$
140-3-s588-D008 $(\widehat{\Pi}_0\tau)$ $\varepsilon_3(a') = \varepsilon a'$
140-3-s588-D011 for
140-3-s588-D012 condition which
140-3-s588-D013 the|these τ' τ' M

[scan 589]
[p. 598]

$$(54) \qquad \varepsilon_3(\sigma)(a') \equiv \varepsilon a' \quad (\Pi_0|\widehat{\Pi}_0), \quad \text{where} \begin{cases} \varepsilon \in \{\pm 1\}, & \varepsilon \overset{\text{dfn}}{=} -1 \text{ iff } \varepsilon_3 = \varepsilon'_3 = -1 \\ \varepsilon_3(\sigma) \overset{\text{dfn}}{=} \begin{cases} 1 & \text{if } \varepsilon_3 = +1 \\ \sigma & \text{if } \varepsilon_3 = -1 \end{cases} \end{cases}$$

This condition (54) can be spelled out according to the cases:

$$(55) \qquad \begin{cases} a) & \text{if } \varepsilon_3 = 1, \text{ then } \varepsilon = 1 \text{ and (54) always satisfied} \\ b) & \text{if } \varepsilon_3 = -1, \text{ then (54) reads } \dagger\ \sigma(a') \equiv \varepsilon'_3.a' \quad (\widehat{\Pi}_0) \quad \text{i.e.} \\ & [\dot\sigma, \dot a'] = \dot\varepsilon'_3 \end{cases}$$

m1257

Marginal note. m1257 where the · denote the classes in $M/\widehat{\Pi}_0$

Since we have

$$a' \equiv \varepsilon'_3(\sigma)^{-1}u' = \begin{cases} u' & \text{if } \varepsilon'_3 = 1 \\ \sigma^{-1}u' & \text{if } \varepsilon'_3 = -1 \end{cases},$$

we find that condition (55, b) is also equivalent to:

$$(56) \quad \dagger \begin{cases} \dot\sigma(\dot u') = \dot\varepsilon'_3|\varepsilon'_3\dot u' & \text{in } M/\widehat{\Pi}_0, \quad \text{i.e.} \\ [\dot\sigma, \dot u'] = \dot\varepsilon'_3|\varepsilon'_3 \end{cases}$$

(equivalence Valid both for $\varepsilon'_3 = 1$ and $\varepsilon'_3 = -1$).

Now we have seen (35) that we have in $\widehat{G}/\widehat{\Pi}_0$)

$$\dot u'(\dot\sigma) = \varepsilon_2(u')\dot\sigma$$

i.e.

$$[\dot u', \dot\sigma] = \varepsilon_2(u') \quad \text{or again}$$

$$(57) \qquad [\dot\sigma, \dot u'] = \varepsilon_2(\dot u'|\hat u')^{\cdot}.$$

Comparing (56) and (57), we see that (56) takes the equivalent form

$$(58) \qquad \varepsilon_2(u') = \varepsilon_3(u').$$

We have thus proved:

Proposition. Let u, u', and $u'' = u'u \in M^!{}[0] = Z^!_{\rho,\sigma}$, corresponding to $\varepsilon_3, | \neq \varepsilon'_3$, $\varepsilon''_3 = \varepsilon'_3\varepsilon_3 \in \{\pm 1\}$, and:

$$a = \varepsilon_3(\tau')\alpha^{-1}u, \quad a' = \varepsilon'_3(\tau')\alpha'^{-1}u', \quad a'' = \underbrace{\varepsilon''_3(\tau')}_{\varepsilon'_3(\tau')\varepsilon_3(\tau')}\ \alpha''^{-1}\ \underbrace{u''}_{u'u},$$

with $a, a', a'' \in N'_\rho|N^*_\rho$, $a \equiv \varepsilon_3(\sigma)^{-1}u$, $a' \equiv ...$, $a'' \equiv ...$ (Π_0) The following conditions are equivalent:

m1258

Marginal note. m1258 N.B. $\begin{cases} \varepsilon_3 = \varepsilon_3(u) = \varepsilon_3(a) \\ \varepsilon'_3 = \varepsilon_3(u') = \varepsilon_3(a') \\ \varepsilon''_3 = \varepsilon_3(u'') = \varepsilon_3(a''|a') \end{cases}$

[p. 588]

Uncertainty retained from cancelled passages.

140-3-s589-D001 ε_3

140-3-s589-D002 $[\dot\sigma, \dot u'] = \varepsilon'_3$

[scan 590]

a) We have $a'' = a'a$

a′) We have $a'' \equiv a'a \quad (\widehat{\Pi}_0)$

b) We have $\varepsilon_3\dagger = 1$ i.e. $a(\rho) = \rho$ or $\sigma(a') \equiv \varepsilon'_3 a' \quad (\widehat{\Pi}_0)$

b′) We have $\varepsilon_3 = 1$ i.e. $a(\rho) = \rho$ or $\dot a'|a'(\sigma) \equiv \varepsilon'_3\sigma \quad (\widehat{\Pi}_0|\overline{\Pi}_0)$

c) We have $\varepsilon_3 = 1$ i.e. $a(\rho) = \rho$ or $\dagger\ \sigma(u') \equiv \varepsilon'_3 u' \quad (\widehat{\Pi}_0)$

c′) We have $\varepsilon_3 = 1$ i.e. $a(\rho) = \rho$, or $u'|\dot u'(\sigma) \equiv \varepsilon'_3\sigma \quad (\widehat{\Pi}_0)$

d) We have $\boxed{\varepsilon_3 = 1 \quad \text{or} \quad \varepsilon'_2 = \varepsilon'_3 \quad \text{i.e.} \mid \varepsilon'_2\varepsilon'_3 = 1}$

Corollary Over the two subgroups

$$(59)\qquad \begin{cases} Z'^{!\prime}_{\rho,\sigma} = M^{!\prime}[0] \overset{\text{dfn}}{=} \{u \in Z^!_{\rho,\sigma} \mid \varepsilon_3(u) = 1\} \\ Z'^{!\prime\prime}_{\rho,\sigma} = M^{!\prime\prime}[0] \overset{\text{dfn}}{=} \{u \in Z^!_{\rho,\sigma} \mid \varepsilon_2(u) = \varepsilon_3(u)\} \end{cases}$$

the set-theoretic section map

(60) A :

$$\begin{array}{ccc} Z^!_{\rho,\sigma} & \longrightarrow & S_0\Gamma^!_{\rho,\sigma} \\ \| & & \\ M^![0] & & \ni \end{array}$$

$$\cdot u \longmapsto (u,\ a = \varepsilon_3(\tau')\alpha^{-1}u,\ b = \beta^{-1}u)$$

is a multiplicative section. But it is not multiplicative over $M^![0]$ as a whole.

cf. proposition page 597

[p. 589]

Uncertainty retained from cancelled passages.

140-3-s590-D001 a

140-3-s590-D002 $\varepsilon_3 = 1$

[scan 591]

[p. 599]

Remark The elements of the inverse image|image $S_0\Gamma^{\prime!|!\prime}_{\rho,\sigma}$ of $Z'^{!\prime}_{\rho,\sigma}$ in $S_0\Gamma^!_{\rho,\sigma}$ which belong to the section under consideration are characterized (by virtue of the proposition on page 598) by the relation

$$(61)\qquad u \equiv b \equiv a \mod \widehat{\Pi}_0$$

m1260

Marginal note. m1260 (in addition to $\varepsilon_3(u) = 1$ which characterizes the elements of $S_0\Gamma'_{\rho,\sigma}$)

which makes it evident that this is a subgroup. I have not found a nice analogous expression for the membership of an element of $S_0\Gamma^{!\prime\prime}_{\rho,\sigma}$ (characterized by the relation $\varepsilon_2(u)\varepsilon_3(u) = 1$) in the corresponding section.

Proposition Let $S_0^{\Pi}\Gamma^!_{\rho,\sigma} \subset S_0\Gamma^!_{\rho,\sigma}$ be the subset of $S_0\Gamma^!_{\rho,\sigma}$ †

$$\begin{array}{ccc} S_0^{\Pi}\Gamma^!_{\rho,\sigma} & \subset & S_0\Gamma^!_{\rho,\sigma} \\ & & \prime\prime \end{array}$$

$$(62)\qquad \left\{ (u,a,b) \in \begin{array}{c} Z_{\rho,\sigma} \times N^*_\rho \times N^*_\sigma \\ \dagger \\ M[0] \end{array} \;\middle|\; \begin{array}{ll} u \equiv b & (\widehat{\Pi}_0) \\ u \equiv \dagger\varepsilon_3(\sigma)a & \mod \widehat{\Pi} = \widehat{\Pi}_0 \times \{\pm 1\} \end{array} \right\}$$

Then

a) $S_0^{\Pi}\Gamma^!_{\rho,\sigma}$ is a subgroup of $S_0\Gamma^!_{\rho,\sigma}$

b) The hom^{omorphism} †

$$S_0^{\Pi}\Gamma^!_{\rho,\sigma} \longrightarrow Z^!_{\rho,\sigma} = M^![0],$$

$$(u,a,b) \longmapsto u$$

is surjective, and its kernel is the subgroup, isomorphic to $\mu = \{\pm 1\}$, of elements $(1,\varepsilon,1)$ with $\varepsilon \in \mu = {}^{\dagger}N'_\rho \cap \widehat{\Pi} = L_\rho \cap \widehat{\Pi}$ [p. 590]

Uncertainty retained from cancelled passages.

140-3-s591-D001 of

140-3-s591-D002 $S_0\Gamma^!_{\rho,\sigma}$

140-3-s591-D003 $b \equiv$

140-3-s591-D004 is

140-3-s591-D005 $S_0\Gamma_\rho$

[scan 592]

Proof The fact that we have a subgroup follows from the calculation already made on pages 597′, 598, the condition that the product of two elements $(u',a',b')(u,a,b)$ of $S_0^{\Pi}\Gamma_{\rho,\sigma}$ lie in it still being expressed by the formula

$$[\dot\sigma, \dot u'] = \dot\varepsilon'_3 \qquad \underline{\text{in}} \quad M/\widehat{\Pi} \quad \text{(instead of } M/\widehat{\Pi}_0) \tag{56}$$

†or again

$$\begin{gathered}[\dot u', \dot\sigma] = \dot\varepsilon'_3 \qquad \text{i.e.} \quad \dot\varepsilon'_3 = \dot\varepsilon'_2 \\ \text{"} \\ \dot u'(\dot\sigma)\dot\sigma^{-1} = \dot\varepsilon'_2\end{gathered} \tag{56 bis}$$

Now in $\widetilde{\mathcal{G}}|\widetilde{\mathcal{C}}/\Pi_0$, the images of ε'_2 and ε'_3 both belong to the central subgroup Π/Π_0, hence they are trivial in $\widetilde{\mathcal{G}}|\widetilde{\mathcal{C}}/\Pi$, hence relation (56 bis) is trivially|trivially satisfied.

The surjectivity of the map $S_0^{\Pi}\Gamma^!_{\rho,\sigma} \longrightarrow Z^!_{\rho,\sigma}$ comes from the fact that $S_0^{\Pi}\Gamma^!_{\rho,\sigma}$ contains the set-theoretic section $A(Z^!_{\rho,\sigma})$ of the proposition on page 597. The kernel consists of the † $(1,a,b)$ with $a \in N^*_\rho \cap \Pi$, $b \in N^*_\sigma \cap \Pi_0$.

Now $N^*_\rho \cap \widetilde{\mathcal{G}}|\widetilde{\mathcal{C}} = {}^{\dagger}L_\rho$, $N^*_\sigma \cap \widetilde{\mathcal{G}}|\widetilde{\mathcal{C}} = L_\sigma$, and

$$\dagger$$

$$L_\rho \cap \widehat{\Pi} = L_\sigma \cap \widehat{\Pi} = \mu, \qquad L_\rho \cap \widehat{\Pi}_0 = L_\sigma \cap \widehat{\Pi}_0 = \{1\} \tag{57}$$

[p. 591]

Uncertainty retained from cancelled passages.

140-3-s592-D001 We have

140-3-s592-D002 u

140-3-s592-D003 Z_ρ

140-3-s592-D004 $\Pi_0|\Pi \quad \mu \quad \Pi$

[scan 593]

whence the characterization of the kernel. [p. 600]

Corollary Let us consider † $S_0\Gamma^!_{\rho,\sigma}$ as an extension of $Z^!_{\rho,\sigma} = M^![0]$ by

$$\dagger \quad L_\rho \times L_\sigma \simeq \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}, \qquad \text{hence}$$

(writing $L_\rho \simeq \mu \times L_{-\rho}\dagger \simeq \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$) an extension of

$$\underbrace{\mathbb{Z}/2\mathbb{Z}}_{\mu} \times \underbrace{\mathbb{Z}/3\mathbb{Z}}_{L_{-\rho}} \times \underbrace{\mathbb{Z}/4\mathbb{Z}}_{L_\sigma}.$$

†

This extension corresponds therefore canonically|canonically to three extensions, by $\mu, L_{-\rho}, L_\sigma$ respectively. The|The extensions|extension by $L_{-\rho} \times L_\sigma$ equipped two|with a canonical sections.

There remains an extension (this is precisely $S_0^{\Pi}\Gamma^!_{\rho,\sigma}$) of $Z^!_{\rho,\sigma}$ by μ.

This extension corresponds just to the ambiguity of the sign in choosing an $\alpha = \widehat{\Pi}$ over $\tilde{\alpha} \in \widehat{\Pi}/\pm 1$ defined by $u \in Z^!_{\rho,\sigma} = M^![0]$, which is reflected in the ambiguity in choosing the sign for $a \in N'_\rho|N^*_\rho$, linked to α by (51)

$$a = \varepsilon_3(\tau')\alpha^{-1}u, \qquad \alpha = ua^{-1}\varepsilon_3(\tau'), \tag{58}$$

We were able to remove this ambiguity by the requirement $\alpha \in \widehat{\Pi}_{0\tau|0,\tau} = \dagger\{1,\tau\}.\widehat{\Pi}_0$, and we [p. 592]

Uncertainty retained from cancelled passages.

140-3-s593-D001 Z

140-3-s593-D002 $\varepsilon|Z$

140-3-s593-D003 μ

140-3-s593-D004 co

140-3-s593-D007 Π_0

[scan 594]

have just observed that this choice, canonical though it is, is not multiplicative. At the same time, we find the complete explicitation of the α'' (normalized canonically|canonically by the condition $\alpha'' \in \widehat{\Pi}_{0\tau}$) associated with $u'' = u'u$, in terms of α' (associated with u') and α (associated with u) also normalized, in terms of a', a Indeed, the proposition on page 598 tells us in which cases † we have

$$a'' = a'a \tag{59}$$

namely iff we have

$$\varepsilon_3(u) = 1 \quad \text{or} \quad \varepsilon_{\dagger 2}(u')\varepsilon_3(u') = 1. \tag{59 bis}$$

But we now know that in any case we must have

$$a'' = \varepsilon a'a \qquad \varepsilon \in \{\pm 1\} \tag{60}$$

and we now find that $\varepsilon = +1$ iff we are in case (59 bis), -1 in the contrary case i.e. if $\varepsilon_3(u) = -1$ and $\varepsilon_2(u')\varepsilon_3(u') = +1$. In summary

Corollary Let $u, u' \in M^![0]$, whence $u'' = u'u$, let $\alpha, \alpha', \alpha'' \in \widehat{\Pi}_{0\tau}$ be associated (defined by $u(\rho) = \mathrm{int}(\alpha)\rho$ etc) and put [p. 593]

Uncertainty retained from cancelled passages.

140-3-s594-D001 exact|exactly

140-3-s594-D002 3|2

[scan 595]

[p. 601]

$$a = \varepsilon_3(\tau')\alpha^{-1}u, \qquad a' = \varepsilon'_3(\tau')\alpha'^{-1}u', \qquad a'' = \varepsilon''_3(\tau')\alpha''^{-1}u'' \tag{61}$$

where

$$\varepsilon_3(\tau') = \begin{cases} 1 & \text{if } \varepsilon_3 = 1 \\ \tau' & \text{if } \varepsilon_3 = -1 \end{cases} \qquad \text{and the } \overline{\text{m}} \text{ for } \varepsilon'_3(\tau'),\ \varepsilon''_3(\tau'),$$

and where $\varepsilon_3 = \varepsilon_3(u) = \varepsilon_3(\chi|\gamma(u))$, and likewise for $\varepsilon'_3, \varepsilon''_3$ — hence $\varepsilon''_3 = \varepsilon_3\varepsilon'_3$. We then have

$$a'' = \varepsilon a' a, \quad \text{with} \quad \varepsilon \in \{\pm 1\} \tag{62}$$

and

$$\begin{cases} \varepsilon = +1 & \text{iff} \quad \varepsilon_3 = 1 \text{ or } \varepsilon'_2\varepsilon'_3 = 1 \qquad \text{hence} \\ \varepsilon = -1 & \text{iff} \quad \varepsilon_3 = -1 \text{ and } \varepsilon'_2\varepsilon'_3 = -1. \end{cases} \tag{63}$$

† therefore

We will have|have therefore

$$\begin{cases} \alpha'' = u''a''^{-1}\varepsilon''_3(\tau') \\ \qquad = \varepsilon u' u\, a^{-1}\, \underbrace{a'^{-1}\varepsilon_3^{\dagger|''}(\tau')}\, \varepsilon'_3(\tau') \end{cases} \tag{*}$$

I † presume that we will have

not true in general

$$\alpha'' = \varepsilon u'(\alpha)\alpha' \qquad \text{where } \varepsilon \text{ given by (63).} \tag{64}$$

Indeed

$$\begin{cases} \alpha = u\, a^{-1}\varepsilon_3(\tau') \\ \alpha' = u'\, a'^{-1}\varepsilon'_3(\tau') \end{cases} \qquad \text{whence}$$

$$u'(\alpha) = u' u\, a^{-1}\varepsilon_3(\tau')u'^{-1}$$

$$u'(\alpha)\alpha' = u' u\, a^{-1}\, \underbrace{\varepsilon_3(\tau')a'^{-1}}\, \varepsilon'_3(\tau')$$

Formula (64), by virtue of (*), thus takes the form

$$a'^{-1}\varepsilon_3(\tau') = \varepsilon_3(\tau')a'^{-1}$$

i.e.

$$[\varepsilon_3(\tau'), a'^{-1}] = 1$$

it is not always satisfied if $\varepsilon_3\dagger = -1$ i.e. $\varepsilon_3(\tau') = \tau'$, where it represents again a draconian restriction on a'. The formula that is always correct is †

$$\alpha'' = (\varepsilon u'(\alpha)\alpha')\,.\,\underbrace{\varepsilon'_3(\tau')([a', \varepsilon_3(\tau')])}_{[\varepsilon'_3(\tau')(a'),\, \varepsilon_3(\tau')]}$$

[p. 594]

Uncertainty retained from cancelled passages.

140-3-s595-D001 and

140-3-s595-D003 ″

140-3-s595-D004 say|would say

140-3-s595-D007 u

140-3-s595-D008 (64)|(65)

[scan 596]

as $\varepsilon'_3(\tau')a' = \alpha'^{-1}u'$ (61), the formula a little convoluted: m1263

Marginal note. m1263 Sic! Here is the monster that gave me cold sweats for a morning or two!

$$\alpha'' = (\varepsilon u'(\alpha)\alpha')\,.\,\begin{bmatrix} \\ \underbrace{\alpha'^{-1}u'}_{\varepsilon'_3(\tau')a'}, \ \varepsilon_3(\tau') \\ \cap \\ \dagger \end{bmatrix}. \tag{65}$$

(where $\varepsilon \in \{\pm 1\}$ defined in|definition (63))

It|I remains to examine the question whether the extension $S_0^{\Pi}\Gamma_{\rho,\sigma}$ of $Z^!_{\rho,\sigma} \simeq M^!$[0] by $\mu = \{\pm 1\}$ is trivial or not, taking into account the fact that its restrictions to $Z^{!\prime}_{\rho,\sigma}$ and to $Z^{!\prime\prime\prime}_{\rho,\sigma}$ are trivial.

Let $S_0^{\Pi}\Gamma^{!\prime}_{\rho,\sigma}$, $S_0^{\Pi}\Gamma|W^{!\prime\prime\prime}_{\rho,\sigma})|]$ be the inverse images of $Z^{!\prime}_{\rho,\sigma}$, $Z^{!\prime\prime\prime}_{\rho,\sigma}$ in $S_0^{\Pi}\Gamma^!_{\rho,\sigma}$, hence

$$\begin{cases} S_0^{\Pi}|\Sigma^{\Pi}_{\rho k}\Gamma^{!\prime}_{\rho,\sigma} = S_0^{\Pi}\Gamma^!_{\rho,\sigma} \cap \Gamma^{!\prime}_{\rho,\sigma} \\ S_0^{\Pi}\Gamma^{!\prime\prime\prime}_{\rho,\sigma} = S_0^{\Pi}\Gamma^!_{\rho,\sigma} \cap \Gamma^{!\prime\prime\prime}_{\rho,\sigma} \end{cases} \tag{66}$$

and we have exact sequences of central extensions

$$\begin{cases} 1 \longrightarrow \mu \longrightarrow S_0^{\Pi}\Gamma^{!\prime}_{\rho,\sigma} \longrightarrow Z^{!\prime}_{\rho,\sigma} \longrightarrow 1 \\ \\ 1 \longrightarrow \mu \longrightarrow S_0^{\Pi}\Gamma^{!\prime\prime\prime}_{\rho,\sigma} \longrightarrow Z^{!\prime\prime\prime}_{\rho,\sigma} \longrightarrow 1 \end{cases}, \tag{67}$$

which are split by subgroups denoted respectively $S_0^{\Pi_0}\Gamma^{!\prime}_{\rho,\sigma}$, $S_0^{\Pi_0}\Gamma^{!\prime\prime|!\prime\prime\prime}_{\rho,\sigma}$, so that we have

$$\begin{cases} S_0^{\Pi}\Gamma^{!\prime}_{\rho,\sigma} \simeq S_0^{\Pi_0}\Gamma^{!\prime}_{\rho,\sigma} \times \mu \\ S_0^{\Pi}\Gamma^{!\prime\prime\prime}_{\rho,\sigma} \simeq S_0^{\Pi_0}\Gamma^{!\prime\prime\prime}_{\rho,\sigma} \times \mu. \end{cases} \tag{68}$$

[p. 595]

Uncertainty retained from cancelled passages.

140-3-s596-D001 $Z_\rho|N_\rho$

[scan 597]

[p. 602]

I ask myself whether these subgroups

$$\begin{cases} S_0^{\Pi_0}\Gamma^{!\prime}_{\rho,\sigma} \subset S_0^{\Pi}\Gamma^{!\prime}_{\rho,\sigma} \subset S_0^{\Pi}\Gamma^!_{\rho,\sigma} \\ S_0^{\Pi_0}\Gamma^{!\prime\prime\prime}_{\rho,\sigma} \subset S_0^{\Pi}\Gamma^{!\prime\prime\prime}_{\rho,\sigma} \subset S_0^{\Pi}\Gamma^!_{\rho,\sigma} \end{cases} \tag{69}$$

m1264

Marginal note. m1264 Subgroups of indices|of index 4 in $S_0^{\Pi}\Gamma_{\rho,\sigma}$

are invariant in $S_0^{\Pi}\Gamma^!_{\rho,\sigma}$, which would also mean (passing then to the quotient by these subgroups) that the extension $S_0^{\Pi}\Gamma^!_{\rho,\sigma}$ of $Z^!_{\rho,\sigma}$ by μ, split over $Z^{!\prime}_{\rho,\sigma}$ and over $Z^{!\prime\prime\prime}_{\rho,\sigma}$, in fact comes (with the said splittings) from extensions of $Z^!_{\rho,\sigma}/Z^{!\prime}_{\rho,\sigma}$ ($\simeq \pm 1$) and of $Z^!_{\rho,\sigma}/Z^{!\prime\prime\prime}_{\rho,\sigma}$ ($\simeq \pm 1$) by μ, which would then have to be characterized (as trivial or nontrivial).

Since the subgroups in question are normal in $S_0^{\Pi}\Gamma^{!\prime}$ resp. $S_0^{\Pi}\Gamma^{!\prime\prime\prime}$, if x is an element of $S_0^{\Pi}\Gamma^!$ which is not in $S_0^{\Pi}\Gamma^{!\prime}$ resp. $S_0^{\Pi}\Gamma^{!\prime\prime\prime}$, to say that $S_0^{\Pi_0}\Gamma^{!\prime}$ (resp. $S_0^{\Pi_0}\Gamma^{!\prime\prime\prime}$) is normal in $S_0^{\Pi}\Gamma^!$ as a whole simply means that it is normalized by x.

1°) Case of $S_0^{\Pi_0}\Gamma^{!\prime}$† in $S_0^{\Pi}\Gamma^!$. We shall take $\underline{x} = (u_0, a_0, b_0)$ with $u_0 = \tau$, hence $\varepsilon_3(u_0) = \varepsilon_2(u_0) = \chi(u_0) = -1$,

$$\begin{cases} \alpha_0 = \tau, \quad \beta_0 = 1, \quad a_0 = \varepsilon^0_{3,0}(\tau')\alpha_0^{-1}u_0 = \tau', \quad b_0 = \beta_0^{-1}u_0 = \tau) \\ \text{thus} \qquad \underline{x} = (u_0 = \tau,\ a_0 = \tau',\ b_0 = \tau) \end{cases} \tag{70}$$

[p. 596]

Uncertainty retained from cancelled passages.
140-3-s597-D002 ″

[scan 598]

thus we have

$$x \in S_0^{\Pi_0}\Gamma^{!'''} \underset{2}{\subset} S_0^{\Pi}\Gamma^{!}\dagger \quad \text{and} \quad x \notin S_0^{\Pi}\Gamma^{!'}$$

and we must see whether x normalizes $S_0^{\Pi_0}\Gamma^{!'}$.

Let †

$$(71) \qquad \underline{u}\dagger = \begin{pmatrix} u, & a, & b \\ \cup & \cup & \cup \\ Z & N'_\rho & N'_\sigma \end{pmatrix} \in S_0^{\Pi_0}\Gamma^{!'} \quad \text{i.e.} \quad \begin{cases} u \equiv a \equiv b \quad (\Pi_0) \\ \varepsilon_3(u) = 1 \end{cases}$$

and let us compute

$$\underline{x}\,\underline{u}\,\underline{x}^{-1} = \underline{x}\,\underline{u}\,x = (\tau(u), \tau'(a), \tau(b))$$

is it in $S_0^{\Pi_0}\Gamma'$ (it is [illeg. 1] only|certainly in $S_0^{\Pi}\Gamma'$) i.e. do we have

$$\dagger \quad (72) \qquad \tau(u) \equiv \tau'(a) \equiv \tau^{\dagger}(b) \quad (\Pi_0) \qquad ?$$

The relation $\tau(u) \equiv \tau(b) \equiv \tau(a)$ is an immediate consequence of (71), so (72) is equivalent to

$$\tau'(a) \equiv \tau(a) \quad (\Pi_0) \qquad (\text{i.e. } \sigma^{-1}(\tau(a)) = \tau(a)$$

i.e.

$$(73) \qquad — \qquad [\dot{\sigma}, \tau(a)^{\cdot}] = 1 \quad \text{in } \widetilde{\mathcal{G}}^*/\Pi_0$$

Or again, †, by virtue of (35 bis) N.B.

$$\varepsilon_2(\tau(a)) = 1 \qquad \text{i.e.} \quad \varepsilon_2(a) = 1$$

or finally (since $a \equiv u$ (Π_0))

$$(74) \qquad \varepsilon_2(u) = 1\,.$$

This condition is not a consequence of the condition $\varepsilon_3(u) = 1$, which shows that $S_0^{\Pi_0}\Gamma^{!'}$ is not invariant in $S_0^{\Pi}\Gamma^{!}$.

Is $S_0^{\Pi_0}\Gamma^{!'''}$ invariant in $S_0^{\Pi}\Gamma^{!}$?

Let

$$(75) \qquad x = (u_0, a_0, b_0) \in S_0^{\Pi_0}\Gamma^{!'} \setminus S_0^{\Pi}\Gamma^{!'''} \quad \text{thus} \quad \begin{array}{ll} \text{a)} & \begin{cases} \varepsilon_3(u_0) = 1 \\ \varepsilon_2(u_0) = -1 \end{cases} \\ & \dagger \\ \text{b)} & u_0 \equiv b_0 \equiv a_0 \quad (\Pi_0) \end{array}$$

[p. 597]

Uncertainty retained from cancelled passages.
140-3-s598-D001 $S_0^{\Pi}\Gamma^{!}$
140-3-s598-D002 $u \in$
140-3-s598-D003 $\in Z$
140-3-s598-D004 (70)
140-3-s598-D005 ′
140-3-s598-D006 obviously|equivalent
140-3-s598-D009 $-1|1$

[scan 599]
[p. 603]

and let us study its action on

$$(76)\qquad \underline{u} = (u,a,b) \in S_0^{\Pi_0}\Gamma^{!'''} \qquad \begin{cases} \text{a)} & \underbrace{\varepsilon_2(u)}_{\varepsilon_2}\,\underbrace{\varepsilon_3(u)}_{\varepsilon_3} = 1 \\ \text{b)} & u \equiv b \equiv \varepsilon_3(\sigma)a \quad (\widehat{\Pi}_0) \end{cases}$$

We have

$$(77)\qquad \underline{x}\,\underline{u}\,\underline{x}^{-1} = (u_0(u), a_0(a), b_0(b))$$

and the relation $\underline{x}\,\underline{u}\,\underline{x}^{-1} \in S_0^{\Pi_0}\Gamma^{!'''}$ is written

$$(78)\qquad u_0(u) \equiv b_0(b) \equiv \varepsilon_3'(\sigma)\,.\,a_0(a)\,.$$

Now the relations (75) b) and 76 b) immediately imply

N.B. $\varepsilon_3(u_0(u)) = \varepsilon_3(u)$

$$u_0(u) \equiv b_0(b) \equiv \underbrace{a_0(\varepsilon_3(\sigma)\,.\,a)}_{\varepsilon_3[a_0(\sigma)].a_0(a)}$$

thus the relation (78) is also equivalent to

$$(79)\qquad \varepsilon_3[\sigma] \equiv \varepsilon_3[a_0(\sigma)]$$

which is automatically satisfied if $\varepsilon_3 = 1$, and for $\varepsilon_3 = -1$ amounts to

$$(80)\qquad \sigma \equiv a_0(\sigma) \quad (\widehat{\Pi}_0)$$

$$\text{i.e.}\quad \varepsilon_2(a_0) = 1 \qquad \text{i.e.}\quad \varepsilon_2(u_0) = 1,$$

a relation which is false by hypothesis 75 a).

Thus we again find $S_0^{\Pi_0}\Gamma^{!'''}$ non-invariant in $S_0^{\Pi}\Gamma^{!}$. On the other hand, †let us set

$$(81)\qquad \begin{cases} S_0\Gamma^0 = \mathrm{Ker}(S_0\Gamma^0 \xrightarrow{(\varepsilon_2,\varepsilon_3)} \mu\times\mu) = S_0\Gamma^{!'} \cap S_0\Gamma^{!'''} \\ \qquad\qquad\qquad\qquad\qquad\qquad = S_0\Gamma^{!'} \cap S_0\Gamma^{!''} \\ \qquad\qquad\qquad\qquad\qquad\qquad = S_0\Gamma^{!''} \cap S_0\Gamma^{!'''} \\ S_0^{\Pi}\Gamma^0 = S_0\Gamma^{!0} \cap S_0^{\Pi}\Gamma^{!}, \\ S_0^{\Pi_0}\Gamma^0 = S_0\Gamma^{!0} \cap S_0^{\Pi_0}\Gamma^{!}, \end{cases}$$

so that we have an exact sequence

†

$$(82)\qquad 1 \longrightarrow \mu \longrightarrow S_0^{\Pi|\Pi_0}\Gamma^{!0} \longrightarrow S_0\Gamma^{!0} \longrightarrow 1\ ,$$

[p. 598]

Uncertainty retained from cancelled passages.
140-3-s599-D001 let us denote|let us define
140-3-s599-D003 b)

[scan 600]

canonically split by $S_0^{\Pi_0}\Gamma_0$ — we therefore find

$$(82)\qquad S_0^{\Pi}\Gamma^{!0} = S_0^{\Pi_0}\Gamma^{!0} \times \mu\,.$$

This being so, the preceding calculations show that $S_0^{\Pi_0}\Gamma^{!0}$, which is of index 8 in $S_0^{\Pi}\Gamma^{!}$, is an invariant subgroup. In summary:

Proposition Consider the subgroups (of indices 4, 4, 8) $S_0^{\Pi_0}\Gamma^{!\prime}$, $S_0^{\Pi_0}\Gamma^{!\prime\prime\prime}$, $S_0^{\Pi_0}\Gamma^{!0}$ of $S_0^{\Pi}\Gamma^!$ which are the images of the|of their homomorphic restrictions $Z^{!\prime}, Z^{!\prime\prime\prime}, Z^{!0}$ of the set-theoretic section: A defined by (51) (52) of $S_0^{\Pi}\Gamma^!$ over $Z = M[0]$ — which is multiplicative on these subgroups, and therefore defines splittings of the inverse images $S_0^{\Pi}\Gamma^{\prime|!\prime}, S_0^{\Pi}\Gamma^{\prime\prime\prime|!\prime\prime\prime}, S_0^{\Pi}\Gamma^{0|!0}$ of † $Z^{!\prime}, Z^{!\prime\prime\prime}, Z^{!0}$ in $S_0^{\Pi}\Gamma^!$:

$$(83) \qquad S_0^{\Pi}\Gamma^{!\prime} \simeq \dagger S_0^{\Pi_0}\Gamma^{!\prime} \times \mu, \qquad S_0^{\Pi}\Gamma^{!\prime\prime\prime} = S_0^{\Pi_0}\Gamma^{!\prime\prime\prime} \times \mu, \qquad S_0^{\Pi}\Gamma^{!0} = S_0^{\Pi_0}\Gamma^{!0} \times \mu.$$

m1268

Marginal note. m1268

> Which implies that there is no section †(multiplicative) of the extension $S_0^{\Pi}\Gamma$ of $Z^!$ by μ, which coincides with A on $Z^{!\prime}$ or on $Z^{!\prime\prime\prime}$.

We then have this:

a) $S_0^{\Pi_0}\Gamma^{!\prime}, S_0^{\Pi_0}\Gamma^{!\prime\prime\prime}$ are not normal in $S_0^{\Pi}\Gamma^!$.

b) $S_0^{\Pi_0}\Gamma^{!0}$ is normal in $S_0^{\Pi}\Gamma^!$, so that the extension $S_0^{\Pi}\Gamma^!$ of $Z^! \simeq M^![0]$ by μ is canonically isomorphic to the inverse image of an extension|category central and finite $S_0^{\Pi}\Gamma^!/(S_0^{\Pi_0}\Gamma^{!0})$ of $Z^!/Z^{!0} \simeq \mu \times \mu$ by μ. [p. 599]

Uncertainty retained from cancelled passages.

140-3-s600-D002 S_0^{Π}

140-3-s600-D003 μ

140-3-s600-D004 $A|_{Z^!}$ multiplicative

[scan 601]

Let us examine this central extension [p. 604]

$$(84) \qquad \begin{array}{ccccccccc} 1 & \longrightarrow & \mu & \longrightarrow & \mathcal{E} & \xrightarrow{(\varepsilon_3,\varepsilon_2)} & \mu \times \mu & \longrightarrow & 1 \\ & & & & \wr\vert & & & & \\ & & & & S_0^{\Pi}\Gamma_{\rho,\sigma}/S_0^{\Pi_0}\Gamma^0_{\rho,\sigma} & & & & \end{array}$$

fitting into the diagram of extensions,

$$(85) \qquad \begin{array}{ccccccccc} 1 & \longrightarrow & \mu & \longrightarrow & \mathcal{E} & \longrightarrow & \mu \times \mu & \longrightarrow & 1 \\ & & \uparrow{\scriptstyle 1} & & \uparrow & & \uparrow{\scriptstyle (\varepsilon_3,\varepsilon_2)} & & \\ 1 & \longrightarrow & \mu & \longrightarrow & S_0^{\Pi}\Gamma^!_{\rho,\sigma} & \longrightarrow & Z^!_{\rho,\sigma} & \simeq M[0] & \\ & & \downarrow & & \downarrow & & \downarrow{\scriptstyle 1} & & \\ 1 & \longrightarrow & L_\rho \times L_\sigma & \longrightarrow & S_0\Gamma^!_{\rho,\sigma} & \longrightarrow & Z^!_{\rho,\sigma} & \longrightarrow & 1 \end{array}$$

The interest of (84) is that it allows one to recover the extension $S_0\Gamma_{\rho,\sigma}$ of $Z^!_{\rho,\sigma}$ by $L_\rho \times L_\sigma$, by the covariant-contravariant law in the functor "extensions", from

$$(86) \qquad Z^!_{\rho,\sigma} \xrightarrow{(\varepsilon_3,\varepsilon_2)} \mu \times \mu \ , \qquad \begin{array}{ccc} \mu & \hookrightarrow & L_\rho \times L_\sigma \\ -1 & \longmapsto & (\rho^3, \sigma^2) \end{array} \ ,$$

("diagonal homomorphism")

as well as the subextension $S_0^{\Pi}\Gamma^!_{\rho,\sigma}$ of the latter (of $Z^!_{\rho,\sigma}$ by μ).

Let us first study the order of the elements of $\mathcal{E}$ — they are obviously divisors of 4, are they all of order 2 (or 1)? Let

$$\underline{\bar{u}|u} = \begin{pmatrix} u, & a, & b \\ \cup & \cup & \cup \\ Z & N_\rho^* & N_\sigma^* \end{pmatrix} \in S_0^{\Pi}\Gamma^! \qquad \text{i.e.} \quad \begin{cases} u \equiv b \quad (\widehat{\Pi}_0) \\ u \equiv \varepsilon_3[\sigma]a \quad (\widehat{\Pi}) \end{cases} \tag{87}$$

m1269

Marginal note. m1269 N.B. To prevent|introduce confusions, I write $\varepsilon_3[\sigma], \varepsilon_3[\tau']$ in preference to|of $\varepsilon_3(\sigma), \varepsilon_3(\tau')$ for $\sigma^{\nu_3}, \tau'^{\nu_3}$.

I † want to examine whether $\underline{u}^2 \in S_0^{\Pi_0}\Gamma^{!0}$. Obviously

$$\underline{u} = (u^2, a^2, b^2) \in S_0^{\Pi}\Gamma^{!0}$$

[p. 600]

Uncertainty retained from cancelled passages.

140-3-s601-D001 want

[scan 602]

it remains to see whether $\underline{u}^2 \in S_0^{\Pi_0}\Gamma^!$, i.e. whether we have † only $u^2 \equiv b^2$ (Π_0) and $\dagger u^2 \equiv a^2$ $(\widehat{\Pi})$ — which is clear and simply expresses $\underline{u}^2 \in S_0^{\Pi}\Gamma|\Gamma^!$ — but even

$$u^2 \equiv a^2 \mod \underline{\underline{\widehat{\Pi}_0}} \qquad (?) \tag{88}$$

The relation

$$u \equiv \varepsilon_3[\sigma].a \mod \Pi$$

can also be written

$$u \equiv \pm\varepsilon_3[\sigma].a \mod \Pi_0$$

and implies

$$u^2 \equiv \varepsilon_3[\sigma]a\ \varepsilon_3[\sigma]a \mod \Pi_0$$

which we can write

$$u^2 \underset{\text{mod } \widehat{\Pi}_0}{\equiv} \varepsilon_3\, \varepsilon_3[\sigma](a).a = \begin{cases} a^2 & \text{if } \varepsilon_3 = 1. \\ -\sigma(a).a & \text{if } \varepsilon_3 = -1 \end{cases} \tag{89}$$

Thus (88) is true if $\varepsilon_3 = 1$, whereas for $\varepsilon_3 = -1$, it is equivalent to

$$\sigma(a) \equiv -a \qquad (\widehat{\Pi}_0)$$

i.e.

$$[\sigma, a]^{\cdot} = -1 \quad \text{in } \widetilde{\mathcal{G}}|\widetilde{\mathcal{C}}/\Pi_0$$

i.e.

$$\underbrace{\varepsilon_2(a)}_{\| \ \varepsilon_2(u)} = -1 \qquad \text{i.e. } \varepsilon_2(u)\varepsilon_3(u) = 1.$$

Thus we find:

Proposition Let $x \in \mathcal{E}$ (cf. (84)). For $x^2 = 1$ to hold, it is necessary and sufficient that we have $\varepsilon_3(x) = 1$ or $\varepsilon_2(x)\varepsilon_3(x) = 1$, i.e. that † we have

$$x \in \mathcal{E}' \cup \mathcal{E}''', \tag{90}$$

[p. 601]

Uncertainty retained from cancelled passages.

140-3-s602-D001 not

140-3-s602-D002 a^2

140-3-s602-D003 belong to the subgroup or

[scan 603]

where [p. 605]

$$\mathcal{E}' = \{x|n \in \mathcal{E} \mid \varepsilon_3(x) = 1\}, \qquad \mathcal{E}''' = \{x|n \in \mathcal{E} \mid \varepsilon_2(x)\varepsilon_3(x) = 1\} \tag{91}$$

Corollary 1 $\mathcal{E}$ has exactly two elements of order 4, which are the two elements of $\mathcal{E}$ above the element $(-1, 1)$ of $\mu \times \mu$. These two elements are inverses of one another, these|and are the two generators of the only cyclic subgroup of order 4 of $\mathcal{E}$, which is none other than

$$\mathcal{E}'' = \{x|n \in \mathcal{E} \mid \varepsilon_2(x) = 1\} \tag{92}$$

Corollary 2 The extension $\mathcal{E}$ is not split.

For being central, it † would be isomorphic to $(\mu \times \mu) \times \mu$, so its elements would have order dividing 2.

We can now regard $\mathcal{E}$ as an extension of $\mathcal{E}/\mathcal{E}'' \simeq \{\pm 1\} = \mu$ by $\mathcal{E}''$ ($\simeq \mathbb{Z}/4\mathbb{Z}$). It would be necessary first to make explicit the operation of $\mathcal{E}/\mathcal{E}'' = \mu$ on $\mathcal{E}'|\mathcal{E}''$ — the element -1 of order 2 of μ induces an involutive automorphism of † $\mathcal{E}''$ moreover in $\mathrm{Aut}(\mathcal{E}'|\mathcal{E}'') \simeq (\mathbb{Z}/4\mathbb{Z})^* \simeq \{\pm 1\}$, so the automorphism in question † is multiplication by a sign $\varepsilon \ \pm 1$, which has to be determined. We are going to take as representative of $-1 \in \mathcal{E}/\mathcal{E}''$ the element of $\mathcal{E}$ coming from

$$\dagger \quad \underline{x} = (u_0 = \tau,\ a_0 = \tau',\ b_0 = \tau) \in S_0^{\Pi_0}\Gamma^{!'''} \qquad (\text{cf.}(70)) \tag{92}$$

which we are going to have operate on the image of a

$$\underline{u} = (u, a, b) \in S_0^{\Pi}\Gamma^!, \quad \dagger \text{ with } \varepsilon_3(u) = -1,\ \varepsilon_2(u) = 1 \tag{93}$$

$$\begin{pmatrix} u \equiv b\ \dagger & \mathrm{mod}\widehat{\Pi}_0 \\ u \equiv \sigma a & \mathrm{mod}\widehat{\Pi} \end{pmatrix}$$

(which is a generator of $\mathcal{E}''$). [p. 602]

Uncertainty retained from cancelled passages.

140-3-s603-D001 itself

140-3-s603-D003 $\mathbb{Z}/4\mathbb{Z}$

140-3-s603-D004 its

140-3-s603-D006 τ, a', τ

140-3-s603-D007 where

140-3-s603-D008 $\equiv a$

[scan 604]

We shall thus have

$$\underline{x}(\underline{u}) = (\tau(u), \tau'(a), \tau(b)) \equiv \dagger \underbrace{(1, \varepsilon, \varepsilon)(u, a, b)}_{(u, \varepsilon a, \varepsilon b)}$$

modulo $S_0^{\Pi_0}\Gamma^{!0|0}$, with a sign $\varepsilon \in \{\pm 1\}$ to be determined. The congruence condition can be written

$$(u\tau(u)^{-1},\ \varepsilon a\tau'(a)^{-1},\ \varepsilon b\tau'(b)|\tau(b)^{-1}) \in S_0^{\Pi_0}\Gamma^{!0|0}$$

that we are in $S_0^{\Pi}\Gamma^{!0}$ is obvious, † † valid|valid whatever the choice of the sign ε, † †it is the|the condition|conditions of congruence † which distinguishes|distinguish $S_0^{\Pi_0}\Gamma^!$ from $S_0^{\Pi}\Gamma^!$ which is essential, namely

$$u\tau(u)^{-1} \equiv \varepsilon a\tau'(a)^{-1} \qquad (\widehat{\Pi}_0) \qquad \text{i.e.} \tag{94}$$

†

The difficulty with this approach comes from the fact that the only elements of M that we know how to make explicit are those coming from $\widetilde{\mathcal{G}}|\widetilde{\mathcal{C}} \simeq \mathrm{Gl}(2,\mathbb{Z})$ itself; now for these we have $\varepsilon_2(u)\varepsilon_3(u) = 1$, whereas the u-here † above must satisfy $\varepsilon_2(u) = 1$, $\varepsilon_3(u) = -1$ so $\varepsilon_2(u)\varepsilon_3(u) = -1$. We can on the other hand work in the † quotient $\mathrm{Gl}(2,\widehat{\mathbb{Z}})$ of † M, and take in it

$$u = \begin{pmatrix} \mu & 0 \\ 0 & 1 \end{pmatrix} \quad \text{with} \quad \mu \in \widehat{\mathbb{Z}}^*, \quad \begin{cases} \mu \equiv -1 & (3) \\ \mu \equiv 1 & (4) \end{cases} \tag{95}$$

i.e. $\mu \equiv 5$ (12)

whereas τ corresponds to the matrix

$$\tau = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

∧
[p. 603]

Uncertainty retained from cancelled passages.

140-3-s604-D001 $(u, \varepsilon a, \varepsilon b)|(\varepsilon u, \varepsilon a, \varepsilon b)$
140-3-s604-D002 the choice
140-3-s604-D003 immaterial|independent
140-3-s604-D004 it is
140-3-s604-D005 it is|and
140-3-s604-D007 are essential
140-3-s604-D008 $u(\tau)\tau^{-1} \equiv \varepsilon a(\tau')\tau'^{-1}$
140-3-s604-D009 test
140-3-s604-D010 group $\widetilde{\mathcal{G}}|\widetilde{\mathcal{C}}$
140-3-s604-D011 $\widetilde{\mathcal{G}}|\widetilde{\mathcal{C}}$

[scan 605]
[p. 606]

we shall in any case have

$$u\tau(u)^{-1} = 1$$

let, on the other hand, in $\mathrm{Gl}(2,\widehat{\mathbb{Z}})$

$$\begin{cases} a = \tau' a_0 & a_0 = c\rho + d \qquad c^2 + cd + d^2 \in \widehat{\mathbb{Z}}^* \\ & \tau' = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \rho = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} \end{cases} \tag{96}$$

thus|whence

$$a = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} d & c \\ -c & c+d \end{pmatrix} = \begin{pmatrix} -c & c+d \\ d & c \end{pmatrix} \tag{97}$$

and

$$\mu = -(c^2 + cd + d^2) \tag{98}$$

We † work with the closed sub-group $\widehat{\Pi}_0$ in $\mathrm{Sl}(2,\widehat{\mathbb{Z}})$ generated by

a little brief...
cf. below...

$$-\ell_0 = -\begin{pmatrix} 1 & -2 \\ \dagger 0 & 1 \end{pmatrix}, \qquad -\ell_1 = -\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \qquad -\ell_\infty = -\begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix}$$

and in the quotient group of $\mathrm{Gl}(2,\widehat{\mathbb{Z}})$ by $\widehat{\Pi}_0$, which is † the extension of $\widehat{\mathbb{Z}}^*$ by

$$\mathrm{Sl}(2,\widehat{\mathbb{Z}})/\widehat{\Pi}_0 \simeq \mathrm{Sl}(2,\mathbb{Z})/\Pi_0 \simeq \left\{ (\rho,\sigma) \;\middle|\; \begin{array}{l} \rho^6=\sigma^4=1 \\ \rho^3=\sigma^2 \\ \sigma(\rho)=\rho^{-1} \end{array} \right\},$$

† and this|in a $\frac{1}{2}$ direct product thanks to

$$T_0 = \left\{ \begin{pmatrix} \dagger\zeta|\xi & 0 \\ 0 & 1 \end{pmatrix} \;\middle|\; \zeta|\xi \in \widehat{\mathbb{Z}}^* \right\} \overset{\det}{\simeq} \widehat{\mathbb{Z}}^*,$$

operating through $\varepsilon_2(\zeta|\xi)$ — cf. pages 593′, 594 — formulas (34)(35)(35 bis)]. We must suppose

$$\sigma a \equiv u \qquad (\widehat{\Pi}_0) \qquad \text{where } \sigma = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \tag{99}$$

thus

$$\sigma a = \begin{pmatrix} -d & -c \\ -c & c+d \end{pmatrix} \tag{100}$$

[p. 604]

Uncertainty retained from cancelled passages.

140-3-s605-D001 are going to
140-3-s605-D002 $-2|2$
140-3-s605-D003 the
140-3-s605-D004 with
140-3-s605-D005 ξ

[scan 606]

† The relation (99), † is written

$$\begin{cases} \sigma a \equiv u \quad (\mathrm{SL}(2,\widehat{\mathbb{Z}})) \quad \text{i.e.} \quad \det \sigma a = \det u \quad \text{i.e. (98)} \\ \sigma a \equiv u \quad (2) \end{cases} \tag{102}$$

† and this last congruence is written

$$\begin{gathered} c \equiv 0 \quad (2) \qquad d \equiv 1 \quad (2) \\ (-d \equiv \mu \quad (\dagger 2) \text{ is a consequence of it for } \mu \equiv 1 \equiv -1 \quad (2)) \,. \end{gathered} \tag{103}$$

N.B. It is not possible to have the † congruences mod 4 ($\sigma a \equiv u$ (4), for we would have $\mu \equiv -d \equiv -1$ (4), which is contrary to the hypothesis (95) $\mu \equiv 1$ (4).

†

Returning to the relation (99), we find this:

Lemma. Let $c, d \in \widehat{\mathbb{Z}}^\dagger$ satisfy

$$\mu \overset{\text{dfn}}{=} -(c^2+cd+d^2) \in \widehat{\mathbb{Z}}^* \tag{104}$$

m1272

Marginal note. m1272 N.B. We will necessarily have $\mu \equiv -1$ (3)

$$\mu \equiv 1 \quad (4) \qquad (\text{i.e. } c^2+cd+d^2 \equiv -1 \quad (4)). \tag{105}$$

$$c \equiv 0 \quad (2), \qquad d \equiv 1 \quad (2) \tag{106}$$

Let † d

$$a = \tau'(c\rho + d) = \begin{pmatrix} -c & c+d \\ d & c \end{pmatrix} \in \mathrm{GL}(2, \widehat{\mathbb{Z}}) \tag{107}$$

(matrix with determinant μ). Then we have a congruence

$$a\tau'(a)^{-1} \equiv \varepsilon \qquad \bmod \widehat{\Pi}_0 \subset \mathrm{GL}(2, \widehat{\mathbb{Z}}) \tag{108}$$

† the sign $\varepsilon \in \{\pm 1\}$ does not depend on the choices made for c, d.

Uncertainty retained from cancelled passages.

140-3-s606-D001 To have
140-3-s606-D002 suffices have
140-3-s606-D005 i.e.
140-3-s606-D008 4
140-3-s606-D009 [illeg. 3]
140-3-s606-D010 too strong This
140-3-s606-D012 *
140-3-s606-D013 a c|a=
140-3-s606-D014 and

[scan 607]
[p. 607]

Now is the time to calculate it! Taking up again

$$a_0 = c\rho + d \quad \in Z_\rho$$

we will have

$$a = \tau' a_0, \quad \tau'(a) = a_0 \tau', \quad a\tau'(a)^{-1} = \tau' a_0 \tau'^{-1} a_0^{-1} = \tau'(a_0) a_0^{-1}$$

where

$$\begin{cases} \tau'(a_0) = c\rho^{-1} + d = {}^t a_0 \\ a_0^{-1} = {}^t a_0 / \det a_0 = -\dfrac{1}{\mu}(c\rho^{-1} + d) \end{cases}$$

m1273

Marginal note. m1273 We set for $u \in \mathrm{M}_2(\widehat{\mathbb{Z}})$ ${}^t u = (\mathrm{Tr}\, u) - u$ and we have ${}^t\rho = \rho^{-1}$ (more generally ${}^t u = u^{-1}$ if $\det u = 1$)

thus

$$a\tau'(a)^{-1} = \tau'(a_0) a_0^{-1} = -\frac{1}{\mu}(c\rho^{-1} + d)^2 = \frac{1}{c^2 + cd + d^2}(c\rho^{-1} + d)^2$$

Now we have

$$(c\rho^{-1} + d)^2 = c^2 \underbrace{\rho^{-2}}_{-\rho} + 2cd \underbrace{\rho^{-1}}_{1-\rho} + d^2 = -(c^2 + 2cd)\rho + (2cd + d^2).$$

Now

$$\begin{aligned} c \equiv 0 \quad (2) &\Longrightarrow c^2, 2cd \equiv 0 \quad (4) \\ d \equiv 1 \quad (2) &\Longrightarrow d^2 \equiv 1 \quad (4) \qquad \text{thus} \end{aligned}$$

†

$$(*) \qquad (c\rho^{-1} + d)^2 \equiv 1 \quad (4)$$

And since by hypothesis

$$c^2 + cd + d^2 \equiv -1 \quad (4)$$

we will have

$$a\tau'(a)^{-1} = \frac{1}{c^2+cd+d^2}(c\rho^{-1}+d)^2 \equiv -1 \quad (4) \tag{109}$$

So we have proved, in fact

Proposition Let $c, d \in \mathbb{Z}|\widehat{\mathbb{Z}}$ be such that $\xi \overset{\text{dfn}}{=} c^2+cd+d^2 \in \widehat{\mathbb{Z}}^*$, let

$$a = \tau'(c\rho+d) = \begin{pmatrix} -c & c+d \\ d & c \end{pmatrix} \in \mathrm{GL}(2, \widehat{\mathbb{Z}}),$$

†

† $\quad \sigma a \equiv \begin{pmatrix} \xi & 0 \\ 0 & 1 \end{pmatrix} \quad (2) \qquad$ i.e.



$$c \equiv 0 \quad (2), \qquad d \equiv 1 \quad (2).$$

N.B. $\sigma = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

Uncertainty retained from cancelled passages.
140-3-s607-D003 ± a fortiori a
140-3-s607-D006 let us prove|let us suppose
140-3-s607-D007 110

[scan 608]

Then we have (for $\tau' = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$)

$$a\tau'(a)^{-1} \equiv \varepsilon_2(\xi) \quad (4)$$

(a fortiori we have the congruence mod $\widehat{\Pi}_0$).

Corollary The extension $\mathcal{E}$ of $\mu \times \mu$ by μ (84|89), or of μ by $\mathcal{E}'' \simeq \mathbb{Z}/4\mathbb{Z}$, is noncommutative. |

But here is a way of seeing it without calculation. If the extension were commutative, its isomorphism class would be given by an $\mathrm{Ext}_{\mathbb{Z}}|\mathrm{Ext}^1_{\mathbb{Z}}$ (or an $\mathrm{Ext}^{1|?}_{\mathbb{Z}/4\mathbb{Z}}$). Now the functor $\mathrm{Ext}|\mathrm{Ext}^1(-,\mu)$ is additive. Since the restriction of the extension to the subgroups $\mathcal{E}'/\mu$ and $\mathcal{E}'''/\mu$ of $\mu \times \mu$ is trivial, and $\mu \times \mu$ is the direct sum of these subgroups (which are resp. $1 \times \mu$ and the diagonal subgroup), it would follow that the extension would be trivial, but we have seen that this is not the case.

† The structure of the group $\mathcal{E}$ is readily made explicit, for example in its capacity as an extension of $\mathcal{E}/\mathcal{E}'' \simeq \mu$ by $\mathcal{E}'' \simeq \mathbb{Z}/4\mathbb{Z}$. Let us choose any element of $\mathcal{E} - \mathcal{E}''$, it is of order 2 (the only elements of order 4 being in $\mathcal{E}''$) so that we have

Uncertainty retained from cancelled passages.
140-3-s608-D002 In|And

[scan 609]
[p. 608]

$$\mathcal{E} \simeq \{1, x\} \,.\, \mathcal{E}'' \qquad \left(\tfrac{1}{2} \text{ direct}\right) \tag{110}$$

Let us note that if x operated trivially on $\mathcal{E}''$, if $\mathcal{E}$ were isomorphic to the product, then for an element u of $\mathcal{E}''$ of order 4, $x.u$ would be of order 4|8, which is absurd (third proof of the

fact that $\mathcal{E}/\mathcal{E}'' \simeq \mu$ operates nontrivially on $\mathcal{E}''$). Now from the mere fact that the operation of $\{1, x\}$ on $\mathcal{E}''$ is nontrivial., it is known, and the structure of $\mathcal{E}$ is determined – in fact we have

$$\mathcal{E} \simeq \mathbb{D}_4 \tag{111|110}$$

There is a preferred choice of x, by taking for x the element

$$\begin{cases} \tau_{\mathcal{E}} \overset{\text{dfn}}{=} \tau_{S_0\Gamma} \bmod S_0^{\Pi}\Gamma^0, & \text{where of course} \\ \tau_{S_0\Gamma} = (\tau, \tau', \tau), \end{cases} \tag{112}$$

so that we have a canonical representation

$$\mathcal{E} \simeq \{1, \tau_{\mathcal{E}}\} \,.\, \mathcal{E}'' \qquad \text{product } \tfrac{1}{2} \text{ direct} \tag{113}$$

$\mathcal{E}'' \simeq \mathbb{Z}/4\mathbb{Z}$
$\tau_{\mathcal{E}}(u) = u^{-1}$
for $u \in \mathcal{E}''$.

The homomorphism $\mathcal{E} \to \mu \times \mu$ is recovered from its restrictions to the two factors

$$\begin{aligned} &\mu \simeq \{1, \tau_{\mathcal{E}}\} \longrightarrow \mu \times \mu \qquad \tau_{\mathcal{E}} \longmapsto (-1, -1) \quad \text{(diagonal homomorphism)} \\ &\mathcal{E}'' \longrightarrow \mu \times \{1\} \hookrightarrow \mu \times \mu \end{aligned} \tag{114}$$

the unique nontrivial homomorphism $\mathcal{E}'' \to \mu$ deduced from the isomorphism

$$\mathcal{E}_2'' \overset{\text{dfn}}{=} \mathcal{E}''/2\mathcal{E}'' | \mathcal{E}''/{}_2\mathcal{E}'' \xrightarrow{\sim} \mu$$

[scan 610]

The fact that $\mathcal{E}$ is not commutative is also expressed by the fact that the canonical homomorphism

$$\mathcal{E}_{\text{ab}} \xrightarrow{\sim} \mu \times \mu \tag{115}$$

is an isomorphism, i.e. that the kernel subgroup μ of $\mathcal{E}_{\text{ab}}|\mathcal{E} \to \mu \times \mu$ is also the commutator subgroup.

We have seen that the extension $\mathcal{E}$ of $\mu \times \mu$ by μ is not split, but what about its inverse image, the extension $S_0^{\Pi}\Gamma^!$ of $Z^! \simeq \mathcal{M}[0]$ by μ? A priori, we only know that there is no splitting of this extension which extends the canonical splitting A we have over

$$Z^0 = \operatorname{Ker}\left(Z^! \xrightarrow{(\varepsilon_3, \varepsilon_2)} \mu \times \mu \right).$$

I presume that even $S_0^{\Pi}\Gamma''$, an extension of $Z^{!\prime\prime}$ by μ, is nontrivial, and even the extension it induces on the subgroup

$$\Gamma'_{\mathbb{Q}} = \Gamma_{\mathbb{Q}} \cap \underbrace{\Gamma_{\mathbb{Q}}'^{\sim}}_{\mathcal{M}(0)'} \subset \mathcal{M}^{\xi|*}(0) \subset \mathcal{M}^!{}[0] = Z^!.$$

It would be interesting to identify, more precisely, the extension over $\Gamma_{\mathbb{Q}}$, which it induces, which corresponds to a canonical element

$$c \in H^2(\operatorname{Spec} \mathbb{Q}, \mu_2) \subset \operatorname{Br}(\mathbb{Q}) \tag{115}$$

[scan 611]
[p. 609]

Unless I am mistaken we will have

$$c = \underset{\text{(cup-product)}}{\xi'\xi''}, \quad \text{where} \quad \xi', \xi'' \in H^2(\mathbb{Q}, \mu_2) \tag{115}$$

are the elements which describe respectively the extensions $\mathbb{Q}(\sqrt[3]{1}) = \mathbb{Q}(\sqrt{-3})$, and $\mathbb{Q}(\sqrt{3})$ — both contained, as well as $\mathbb{Q}(\sqrt{-1}) = \mathbb{Q}(i)$, in the biquadratic extension $\mathbb{Q}(\underset{\|}{i}, \underset{\|}{j}) = \mathbb{Q}(\sqrt[12]{1})$, $\sqrt{-1}$ $\sqrt[3]{1}$

the field of 12th roots of unity (whose Galois group is $(\mathbb{Z}/12\mathbb{Z})^* \simeq (\mathbb{Z}/3\mathbb{Z})^* \times (\mathbb{Z}/4\mathbb{Z})^* \simeq \mu \times \mu$).

It must be trivial — when one knows a little of the foundations — that c thus made explicit is $\neq 0$, and is not split either on passing to $\mathbb{Q}(i)$, the third rogue.

Let us now consider the central μ-grouping

N.B. $Z \simeq \mathcal{M}[0]$, $\gamma \simeq \widetilde{\Gamma}_{\mathbb{Q}}$

$$\begin{array}{ccc} \mu & \xhookrightarrow{i_{\rho,\mu}} & S_0\Gamma^!_{\rho,\sigma} = Z^!_{\rho,\sigma} \times_\gamma N^*_\rho \times_\gamma N^*_\sigma \\ \varepsilon & \longmapsto & (1, \varepsilon, 1) \end{array} \tag{116}$$

with values in

$$\mathrm{Ker}(S_0\Gamma^!_{\rho,\sigma} \longrightarrow Z^!_{\rho,\sigma}) \simeq L_\rho \times L_\sigma,$$

we thus have a canonical lifting homomorphism

$$\mathcal{M}^![0] \simeq Z^!_{\rho,\sigma} \simeq S_0^\Pi\Gamma^!_{\rho,\sigma}/\mu \longrightarrow S_0\Gamma^!_{\rho,\sigma}/i_{\rho,\mu}(\mu), \tag{117}$$

[scan 612]

which is made explicit in terms †|of the two fundamental canonical homomorphisms [m1278]

Marginal note. [m1278] This deserves to be made explicit in a summarizing theorem ...

$$\begin{array}{ccccc} Z^!_{\rho,\sigma} & \xrightarrow{\underline{a}} & N^*_\rho/\mu, & \qquad & Z^!_{\rho,\sigma} \xrightarrow{\underline{b}} N^*_\sigma \\ \wr & & & & \\ \mathcal{M}[0] & & & & \\ \wr & & & & \\ \widetilde{\Gamma}_{\mathbb{Q}} \,.\, L_0^\Pi & & & & \end{array} \tag{118}$$

where, I recall

$$\left\{ \begin{array}{l} N^*_\rho \subset \mathcal{M} = \widetilde{N}_{1,1} \simeq \widetilde{\Gamma}_{\mathbb{Q}} \,.\, \underbrace{\mathcal{G}^+}_{\mathrm{SL}(\widehat{2,\mathbb{Z}})^\wedge} \\ \| \\ \{u \in \mathcal{M} \mid u(\rho) = \dagger\rho^{\varepsilon_3(u)}\} \\ \dagger N^*_\sigma \subset \mathcal{M} \\ \| \\ \{u \in \mathcal{M} \mid u(\sigma) = \sigma^{\varepsilon_2(u)} \ (= \varepsilon_2(u)\sigma)\}. \end{array} \right. \tag{119}$$

The maps $\underline{a}, \underline{b}$ are given by the formulas (cf. 51)

$$\left\{ \begin{array}{ll} \underline{a}(u) = a = \varepsilon_3[\tau']\alpha^{-1}u, & \underline{b}(u) = b = \beta^{-1}u \quad \text{i.e.} \\ \alpha = ua^{-1}\varepsilon_3[\tau'], & \beta = ub^{-1} \end{array} \right. \tag{120}$$

where $\alpha \in \widehat{\Pi}_{0\tau}$, $\beta \in \widehat{\Pi}_0$ are characterized by

$$u(\rho) = \mathrm{int}(\alpha)(\rho), \quad u(\sigma) = \varepsilon_2(\sigma)\,\mathrm{int}(\beta)(\sigma). \tag{121}$$

The quantities † $a = \underline{a}(u) \in N_\rho^*/\mu$, $b = \underline{b}(u) \in N_\sigma^*$ are characterized in terms of $u \in Z^!$ (only up to a sign) by the congruences

$$u \equiv b \quad (\widehat{\Pi}_0), \qquad u \equiv \varepsilon_3[\sigma]a \quad (\widehat{\Pi}), \tag{122}$$

which imply a fortiori (reducing mod $\mathcal{G}^{+\wedge}$), that we have commutativity of the|of,

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Uncertainty retained from cancelled passages.

140-3-s612-D001 of the

140-3-s612-D002 fixes the group

140-3-s612-D003 {

140-3-s612-D004 a,

[scan 613]

[p. 619|610]

†

$$\begin{array}{ccccc} & & N_\rho^*/\{\pm 1\} & & \\ & \nearrow & & \searrow^{\delta_\rho} & \\ Z^!_{\rho,\sigma} & \overset{\delta_Z}{\dashrightarrow} & & & \gamma_{\rho,\sigma} \simeq \tilde{\Gamma}_{\mathbb{Q}} \\ & \searrow & & \nearrow_{\delta_\sigma} & \\ & & N_\sigma^* & & \end{array} \tag{123}$$

$\longrightarrow$ †

It would be time to rectify the confusion (page 594') between $Z = Z_{\rho,\sigma} \not\subset \mathcal{M}_{0,3}^{\circ|\bullet\sim}$, and $Z^! = Z^!_{\rho,\sigma} = \dagger Z \cap \mathcal{M}^{!\sim}_{0,3}$. We have in fact

$$\begin{array}{ccc} Z^\dagger_{\rho,\sigma} \simeq \mathcal{M}^{\xi}(0) \,.\, L_0^{\mathcal{G}} \subset \mathcal{M} & & \\ {\scriptstyle \frac{1}{2}\text{-direct}} & & \\ & \searrow^{\sim} & \\ & & \mathcal{M}^{\xi\sim}_{0,3}(0) \,.\, L_0^{\mathcal{G}_{0,3}} \subset \widetilde{\mathcal{M}}_{0,3} \end{array} \tag{124}$$

$$\begin{array}{ccc} Z^!_{\rho,\sigma} \simeq Z^!_{\rho,\sigma} \cap \mathcal{M}^! = \mathcal{M}(0) \,.\, L_0^{\Pi} \subset \mathcal{M} & & \\ {\scriptstyle \frac{1}{2}\text{-direct}} & & \\ & \searrow^{\sim} & \\ & & \widetilde{\mathcal{M}}_{0,3}(0) \,.\, L_0^{\Pi_{0,3}} \dagger (\subset \widetilde{\mathcal{M}}_{0,3}) \end{array} \tag{125}$$

The link between the two is given by the diagram

$$\begin{array}{ccccccccc} 1 & \to & L_0^{\Pi} & \to & Z^!_{\rho,\sigma} & \longrightarrow & \gamma_{\rho,\sigma} & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \| & & \\ 1 & \to & L_0^{\mathcal{G}} & \to & Z_{\rho,\sigma} & \longrightarrow & \gamma_{\rho,\sigma} & \longrightarrow & 1 \end{array} \tag{126}$$

$$\wr$$

$$\mathcal{M}(0) \simeq \tilde{\Gamma}_{\mathbb{Q}}$$

where $L_0^{\Pi} \longrightarrow L_0^{\mathcal{G}}$ is the inclusion of index 2 of

$$\begin{array}{ccc} L_0^{\Pi} = \ell_0^{\widehat{\mathbb{Z}}} & \hookrightarrow & \dagger L_0^{\mathcal{G}} = \varepsilon_0^{\widehat{\mathbb{Z}}} \\ \| & & \\ \varepsilon_0^{2\widehat{\mathbb{Z}}} & & \end{array} \tag{127}$$

Of course, it is indeed on $Z^!_{\rho,\sigma}$, not on $Z_{\rho,\sigma}$, that we defined (117), (118) whence (123).

Uncertainty retained from cancelled passages.

140-3-s613-D001 123|122

140-3-s613-D005 $\mathcal{M}|\Gamma$

140-3-s613-D006 $\mathcal{M}$

140-3-s613-D007 !

140-3-s613-D008 $\subset \widetilde{\mathcal{M}}_{0,3}$

140-3-s613-D009 $\ell_0^{\widehat{\mathbb{Z}}}|\varepsilon_0^{\widehat{\mathbb{Z}}}$

[scan 614]

I am interested in the restriction of the homomorphisms (123) to the subgroup L_0^{Π} of $Z^!_{\rho,\sigma}$ the characterization (122) shows that the induced homomorphism

$$L_0^{\Pi} \longrightarrow N_\rho^*/\{\pm 1\}$$

is trivial (for a in terms of u depends only on u modulo $\widehat{\Pi}$), whereas the induced homomorphism

$$L_0^{\Pi} \longrightarrow N_\sigma^*$$

factors through $L_0^{\Pi}/L_0^{\Pi_0} \simeq \mathbb{Z}/2\mathbb{Z}$ (for b in terms of u depends only on u modulo $\widehat{\Pi}_0$). Its value b on ℓ_0 is characterized|characterized by the conditions

$$\{ \qquad b \in N_\sigma^*, \qquad b \equiv \ell_0 \quad (\Pi_0)$$

which are obviously satisfied by $b = -1$. Thus There is reason to introduce also, whether we like it or not, the canonical central μ-grouping

$$\begin{array}{ccc} \mu & \hookrightarrow & N_\sigma^* \\ -1 & \longmapsto & -1 \end{array} \tag{128}$$

and we have commutativity of

$$\begin{array}{ccc} L_0^{\Pi} & \hookrightarrow & Z^!_{\rho,\sigma} \\ \downarrow & & \downarrow \\ L_0^{\Pi}/L_0^{\Pi_0} \simeq \mu & \hookrightarrow & N_\sigma^*. \end{array} \tag{129}$$

Thus we find, from the two homomorphisms (123),

[scan 615]
[p. 611]

of $Z^!_{\rho,\sigma}$, by passage to the quotient two canonical homomorphisms

$$\begin{cases} \gamma_{\rho,\sigma} \longrightarrow N^*_\rho/\{\pm 1\} \\ \| \\ \tilde{\Gamma}_{\mathbb{Q}} \\ \| \\ \gamma_{\rho,\sigma} \longrightarrow N^*_\sigma/\{\pm 1\}, \end{cases} \tag{130}$$

which are in fact sections of the

$$\begin{array}{ccc} N^*_\rho/\{\pm 1\} & \longrightarrow & \gamma_{\rho,\sigma} \\ & & \| \\ & & \tilde{\Gamma}_{\mathbb{Q}} \xleftarrow{\sim} \mathcal{M}(0) \\ & & \| \\ N^*_\sigma/\{\pm 1\} & \longrightarrow & \gamma_{\rho,\sigma}, \end{array}$$

so that we find decompositions,

$$\begin{cases} N^*_\rho/\pm 1 \simeq \gamma_{\rho,\sigma}\dagger \,.\, \underbrace{SZ_\rho/\{\pm 1\}}_{=L_\rho/\{\pm 1\}\simeq \mathbb{Z}/3\mathbb{Z}} \\ N^*_\sigma/\pm 1 \simeq \gamma_{\rho,\sigma} \,.\, \underbrace{SZ_\sigma/\pm 1}_{=L_\sigma/\pm \,\simeq \mathbb{Z}/2\mathbb{Z}} \end{cases} \tag{131}$$

($\frac{1}{2}$-direct products). The operation of $\gamma_{\rho,\sigma}$ on $L_\rho/\{\pm 1\} \simeq \mathbb{Z}/3\mathbb{Z}$ is given by the quadratic character ε_3, the one on $L_\sigma/\{\pm 1\} \simeq \mathbb{Z}/2\mathbb{Z}$ is given by the quadratic character ε_2.

The symmetry of presentation of the two homomorphisms (130) is a bit phony — for example the homomorphism $\tilde{\Gamma}_{\mathbb{Q}} = \gamma_{\rho,\sigma} \longrightarrow N^*_\sigma/\{\pm 1\}$ lifts in an obvious way to

$$\begin{array}{ccccc} \tilde{\Gamma}_{\mathbb{Q}} = \gamma_{\rho,\sigma} & & \longrightarrow & & N^*_\sigma, \\ & \nwarrow^{\sim} & & \nwarrow & \\ & \mathcal{M}(0) & \subset & \mathcal{M}[0] = Z^! & \end{array} \tag{132}$$

Uncertainty retained from cancelled passages.

140-3-s615-D001 ⋉|×

[scan 616]

Another way to see this is to note that

$$(132) \qquad \gamma_{\rho,\sigma} \simeq \mathcal{M}^!_0[0]/L_0^{\Pi_0}$$

where

$$\mathcal{M}_0^{\dagger !}[0] \overset{\text{dfn}}{=} \mathcal{M}_{\delta|0}(0).L_0^{\Pi_0} \underset{\substack{\text{subgroup} \\ \text{of index 2}}}{\subset} \mathcal{M}^!{}[0] = \mathcal{M}(0).L_0^{\Pi|\tau}$$

Now on $L_0^{\Pi_0}$, the homomorphism induced by

$$\mathcal{M}^!{}[0] = Z^!_{\rho,\sigma} \longrightarrow N^*_\sigma$$

is trivial. Thus we even have a † splitting of

$$\begin{array}{ccccccccc} 1 & \longrightarrow & L_\sigma & \longrightarrow & N^*_\sigma & \longrightarrow & \gamma_{\rho,\sigma} & \longrightarrow & 1 \\ & & & & & & \wr\vert & & \\ & & & & & & \tilde{\Gamma}_{\mathbb{Q}} & & \end{array} \tag{133}$$

as a $\frac{1}{2}$-direct product

$$\dagger \qquad N^*_\sigma \simeq \gamma_{\rho,\sigma}.L_\sigma$$

which it is better to write by concretizing the section subgroup as a subgroup of $G_{\rho,\sigma} \simeq \mathcal{M}$, i.e. then

$$\begin{aligned} N^{!*}_{0\sigma} &\overset{\text{dfn}}{=} \mathrm{Im}(\mathcal{M}_0[0] \longrightarrow N^*_\rho) \\ &= \mathrm{Im}(\mathcal{M}[0] \hookrightarrow N^*_\rho) \\ &= \{\beta^{-1}u \mid u \in \mathcal{M}(0),\ \beta \in \widehat{\Pi}_0 \text{ such that } u(\sigma) = \varepsilon_2(u)\,\mathrm{int}(\beta)(\sigma)\} \\ &= \{v \in N^*_\sigma \mid \exists\, \beta \in \widehat{\Pi}_0 \text{ such that } \beta v \in \mathcal{M}[0] = \mathrm{Norm}_{\mathcal{M}}(l_0) \cap \mathcal{M}^!\} \end{aligned} \tag{134}$$

m1279

Marginal note. m1279 N.B. We define $\mathcal{M}^!$ as the inverse image of $\mathcal{M}^{!\sim}_{0,3}$, i.e. the kernel of the canonical representation

$$\mathcal{M} \longrightarrow \underbrace{\mathcal{G}^{+\wedge}/\Pi^\wedge}_{\mathcal{G}_3} \xrightarrow{\sim} \mathrm{Aut}(\mathcal{G}^{+\wedge}/\Pi^\wedge)$$

Setting

$$\mathcal{M}^!_0 \overset{\text{dfn}}{=} \mathcal{M}(0).\Pi_0 \underset{\substack{\text{index}\\2}}{\subset} \mathcal{M}^! \overset{\text{dfn}}{=} \mathcal{M}(0).\Pi, \tag{135}$$

Uncertainty retained from cancelled passages.

140-3-s616-D001 *

140-3-s616-D002 isomorphism

140-3-s616-D003 (134)

[scan 617]

[p. 612]

so that we have

$$\mathcal{M}^! \simeq \mathcal{M}^!_0 \times \mu, \tag{136}$$

m1280

Marginal note. m1280 N.B. We have

$$\mathcal{M}^!_0 \xrightarrow{\sim} \mathcal{M}^{!\sim}_{0,3}$$

we thus have

$$(137)\qquad \begin{cases} N_{\sigma 0}^{\dagger *!} = N_\sigma^* \cap \mathcal{M}_0^! \\ N_\sigma^{*!} \overset{\text{dfn}}{=} N_\sigma^* \cap \mathcal{M}^! = N_{\sigma 0}^{*!} \times \mu, \end{cases}$$

and we find that

$$(138)\qquad N_{\sigma 0}^{*!|d!} \xrightarrow{\sim} \gamma_{\rho,\sigma} = \tilde{\Gamma}_{\mathbb{Q}}$$

† Let us set for short

$$\dagger$$

$$(139)\qquad \mathcal{M}(1/2) = N_{\sigma 0}^{*!} \qquad \text{cf. (137)}$$

we will thus have that $\mathcal{M}(1/2)$ is a|a group-section for $N_\sigma^{*|\wedge}$ extension of $\gamma_{\rho,\sigma}$ by $L_{\sigma|\rho}$, and a fortiori for $\mathcal{M}$ extension of $\gamma_{\rho,\sigma}$ by $\mathcal{G}^{+\wedge}$, and one|we even|$\simeq$ in addition|simple for $\mathcal{M}_0^!, \mathcal{M}^!$

$$(140)\qquad \begin{cases} \mathcal{M}_0^! \simeq \mathcal{M}(1/2).\widehat{\Pi}_0 \\ \mathcal{M}^! \simeq \mathcal{M}(1/2).\widehat{\Pi} \\ \mathcal{M} \simeq \mathcal{M}(1/2).\mathcal{G}^{+\wedge} \\ N_\sigma^* \simeq \mathcal{M}(1/2).L_\sigma \end{cases}$$

m1281

Marginal note. m1281 ⟵ watch out, $\mathcal{M}^!$ is inverse in $\mathcal{M}$ and $\mathcal{M}_0^!$ is inverse in $\mathcal{M}^!$, but not in $\mathcal{M}$ cf. below

[If we want to paraphrase by introducing

$$(141)\qquad N_{\rho 0}^{*!} \overset{\text{dfn}}{=} N_\rho^* \cap \mathcal{M}_0^! = \{v \in N_\rho^* \mid \exists\, \alpha \in \widehat{\Pi}_0 \text{ such that } \underbrace{\alpha v}_{u} \in \mathcal{M}(0)\}$$

it is thus one of the v that we can associate to the $u \in \mathcal{M}(0) \subset \mathcal{M}^![0] = Z_{\rho,\sigma}^!$, using the $\alpha \in \Pi_0$ such that $\alpha^{-1}u \in N_\rho^{*|\wedge}$ i.e.

$$u(\rho) = \mathrm{int}(\alpha)(\rho^{\varepsilon_3}) \qquad \text{where} \quad \varepsilon_3 = \varepsilon_3(u) = \varepsilon_3(v).$$

But since $u(\rho)$ must have the same image in $\mathcal{G}_3 = \mathcal{G}^{+\wedge}/\Pi$ as ρ, we must have $\varepsilon_3 = 1$, and then α is the element $\underline{\alpha}(u)$ associated to u and $v = \alpha^{-1}u$ is the canonical image of u in $N_\rho^{!*}$. Thus we

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Uncertainty retained from cancelled passages.

140-3-s617-D001 *d*

140-3-s617-D002 whence

140-3-s617-D003 *d*

[scan 618]

find

Proposition The homomorphism $N_{\rho 0}^{*!} \longrightarrow \gamma_{\rho,\sigma} = \tilde{\Gamma}_{\mathbb{Q}}$ is injective, and has image $\gamma'_{\rho,\sigma} = \{\gamma \in \gamma_{\rho,\sigma} \mid \varepsilon_3(\gamma) = 1\}$.

Corollary $N_{\rho,0}^{*!} \subset Z_\rho^{\xi|*!}$ i.e. $N_{\rho 0}^{*!} = \dagger Z_{0\rho|\rho 0}^{\xi|*!} \overset{\text{dfn}}{=} Z_\rho^{\xi|*} \cap \mathcal{M}_0^!$.

It is natural to set

$$(142)\qquad N_{\rho 0}^{*!} = Z_{\rho 0}^! = \mathcal{M}(-\bar{j})\,..|.$$

Since $N_\rho^{*!} \supset Z_\rho^! \supset \mu$, we see that we have

$$(142)\qquad \begin{cases} N_{\rho\dagger}^{*!} &= N_{\rho 0}^{*!} \times \mu \quad (\text{thus } N_\rho^{*!} = Z_\rho^!) \\ Z_\rho^{!\|} &= Z_{\rho 0}^! \times \mu \end{cases}$$

†

The subgroup $Z^!_{0,\rho} = \mathcal{M}(-\bar{j})$ is a section of $\mathcal{M}'$ over $\gamma'_{\rho,\sigma} \simeq \Gamma'^{\sim}_{\mathbb{Q}}$, and † $\mathcal{M}'$ contained|contained in $\mathcal{M}'^!_0, \mathcal{M}'^!, \dagger, Z^!_\rho, Z_\rho$, we will have decompositions into $\frac{1}{2}$ direct products

$$(143) \qquad \begin{cases} \mathcal{M}'^!_0 \simeq \mathcal{M}(-\bar{j}).\Pi_0 \\ \mathcal{M}'^! \simeq \mathcal{M}(-\bar{j}).\Pi \\ \mathcal{M}'^{\xi|*} \simeq \mathcal{M}(-\bar{j}).\mathcal{G} \\ Z^!_\rho \simeq \mathcal{M}^{\xi|*}(-\bar{j}) \times \mu \\ Z^{\xi|*}_\rho \simeq \mathcal{M}^{\xi|*}(-\bar{j}) \times L_\rho \qquad .... \end{cases}$$

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Uncertainty retained from cancelled passages.

140-3-s618-D001 N
140-3-s618-D002 0
140-3-s618-D003 N_ρ
140-3-s618-D004 is
140-3-s618-D006 $Z_{0\rho}$

[scan 619]
[p. 613]

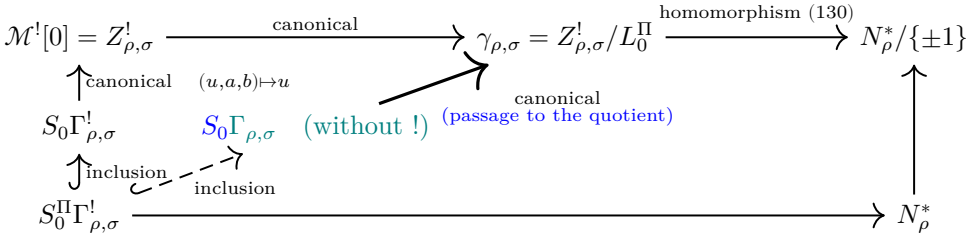
Let us return to diagram (130), I say that the homomorphism

$$\begin{array}{ccc} \gamma_{\rho,\sigma} & \longrightarrow & N^*_\rho/\{\pm 1\} \\ \| & & \\ \Gamma^{\sim}_{\mathbb{Q}} & & \end{array}$$

does not lift to a homomorphism from $\gamma_{\rho,\sigma}$ to N^*_ρ — in other words, that the extension of $\gamma_{\rho,\sigma}$ by $\{\pm 1\} = \mu$ which it defines is not trivial.

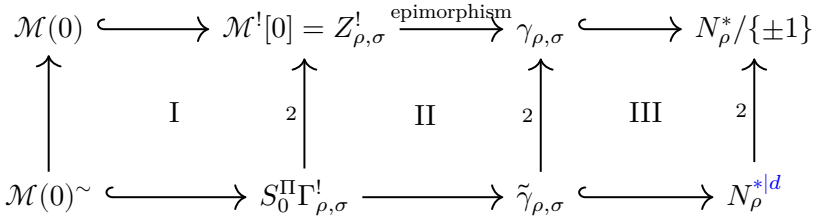
To identify this extension, let us consider the diagram N.B.

(144)



which shows that we have a diagram of cartesian squares

(145) †



where $\mathcal{M}(0)^\sim$ is defined|defined as the extension of $\mathcal{M}(0)$ by μ induced by the extension $S_0^\Pi\Gamma^!_{\rho,\sigma}$ of $\mathcal{M}^![0]$, which is thus identified also with the inverse|inverse image of

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Uncertainty retained from cancelled passages.

140-3-s619-D001 (146)|(144)

[scan 620]

the extension $\tilde{\gamma}_{\rho,\sigma}$ of $\gamma_{\rho,\sigma}$ by μ, under the composite isomorphism

$$\mathcal{M}(0) \xrightarrow{\sim} \gamma_{\rho,\sigma}.$$

So the extension $\tilde{\gamma}_{\rho,\sigma}$ of $\gamma_{\rho,\sigma}$ by μ is identified naturally|naturally with the extension $\mathcal{M}(0)^\sim$ of $\mathcal{M}(0)$ by μ which we have more or less convinced ourselves was not trivial.

Here is another way of obtaining the canonical homomorphism

$$\gamma_{\rho,\sigma} \longrightarrow N^*_\rho/\pm 1,$$

and the extension $\tilde{\gamma}_{\rho,\sigma}$ of $\gamma_{\rho,\sigma}$. Let us consider

$$(146) \qquad N^{*!}_\rho = N^*_\rho \cap \mathcal{M}^!$$

We then have

$$(147) \qquad N^{*!}_\rho \cap \mathcal{G}^+ = \mathrm{Ker}(N^{*!}_\rho \longrightarrow \underbrace{\gamma_{\rho,\sigma}}_{\Gamma^\sim_{\mathbb{Q}}}) = L_\rho \cap \Pi = \mu$$

on the other hand the homomorphism

$$(148) \qquad N^{*!}_\rho \longrightarrow \gamma_{\rho,\sigma} \quad \text{is an epimorphism}$$

For it suffices to see that for $\forall\, u \in \mathcal{M}(0)$, there exists an element of $N^{*!}_\rho$ which is congruent to it, modulo $\mathcal{G}$ Now we have

$$u(\rho) = \mathrm{int}(\alpha)(\rho)\dagger$$

with $\alpha \in \dagger\widehat{\Pi}_\tau = \{1,\tau\}.\widehat{\Pi}$, if $\varepsilon_3(u) = 1$ i.e. $\alpha \in \widehat{\Pi}$, $\alpha^{-1}u \in \dagger Z_\rho \subset N_\rho^*$ does the job. If $\dagger\ \varepsilon_3 = | \neq -1$, we have

$$\alpha = \alpha_0\tau, \qquad \alpha_0 \in \widehat{\Pi}.$$

and we have

$$u(\rho) = \alpha(\rho) = \alpha_0(\underbrace{\tau(\rho)}_{\sigma(\rho^{-1})}) = \alpha_0(\sigma(\rho^{-1}))$$

so

$$(\sigma^{-1}\alpha_0^{-1}u)(\rho) = \rho^{-1} \quad \text{so} \quad \sigma^{-1}\alpha_0^{-1}u \in N_\rho^*.$$

OK.

Uncertainty retained from cancelled passages.

140-3-s620-D002 $\mathrm{int}(\alpha_0\varepsilon_3[\tau])(\rho)$

140-3-s620-D003 $\widehat{\Pi}_\tau$

140-3-s620-D004 Z

140-3-s620-D005 $\varepsilon_3(u) = -1$

[scan 621]
[p. 614]

So we have an exact sequence
(149)

$$\begin{array}{ccccccccc} 1 & \to & \mu & \longrightarrow & Z_\rho^{*!} & \longrightarrow & \gamma_{\rho,\sigma} & \to & 1 \\ & & & & & & \wr\| & & \\ & & & & (\simeq \tilde{\gamma}_{\rho,\sigma}) & & \tilde{\Gamma}_{\mathbb{Q}} & & \end{array}$$

which makes $Z_\rho^{*!}$ a μ-extension of $\gamma_{\rho,\sigma}$, hence a canonical homomorphism

$$\gamma_{\rho,\sigma} \simeq Z_\rho^{*!}/\mu \longrightarrow N_\rho^*/\mu$$

which, of course, is none other than the first homomorphism (130).

9°) Recapitulation on the universal case.

Finally, I have the impression that I have again made long detours and contortions, to finally get lost in the sands, and lose the thread a little. So I am going to recapitulate what seems essential to me.

ⓐ

$$(1) \qquad Z_{\rho,\sigma} \simeq \mathcal{M}(0).L_0^{\mathcal{G}} = \mathcal{M}[0]\ \dagger$$

(operating on $\widehat{\mathcal{G}}^+$ by operation induced by the adjoint)

we distinguish inside it a subgroup of index 2

$$(2) \qquad Z'_{\rho,\sigma} \simeq \mathcal{M}(0).L_0^{\Pi}.$$

Uncertainty retained from cancelled passages.

140-3-s621-D001 [*illeg.* 1]

[scan 622]

We have

$$(3) \qquad G_{\rho,\sigma} \simeq \mathcal{M}$$

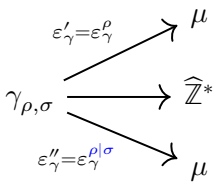
and, more precisely, the diagram
(4)

$$\begin{array}{ccccccccc} 1 & \longrightarrow & L_0 & \longrightarrow & Z_{\rho,\sigma} & \longrightarrow & \gamma_{\rho,\sigma} & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \| & & \\ 1 & \to & \mathcal{G}^{+\wedge}_{\rho,\sigma} & \to & G_{\rho,\sigma} & \to & \gamma_{\rho,\sigma} & \to & 1 \end{array}$$

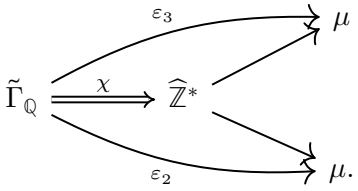
is identified with the diagram
(4 bis)

$$\begin{array}{ccccccccc} & & & & \overset{\mathcal{M}(0).L_0^{\mathcal{G}}}{\widehat{\mathcal{M}[0]}} & & & & \\ 1 & \to & L_0^{\mathcal{G}} & \to & & \to & \tilde{\Gamma}_{\mathbb{Q}} & \to & 1 \\ & & \downarrow & & \downarrow & & \wr & & \\ 1 & \to & \widehat{\mathcal{G}}^{+} & \longrightarrow & \mathcal{M} & \longrightarrow & \tilde{\Gamma}_{\mathbb{Q}} & \to & 1. \end{array}$$

Moreover, the homomorphisms
(5)



are identified:
(5 bis)



ⓑ We have /

$$(6) \qquad N_\sigma^* = \left\{ u \in \mathcal{M} \;\middle|\; u(\sigma) = \dagger\sigma^{\varepsilon_2(u)} \begin{array}{r} (= \varepsilon_2(u).\sigma) \\ = \varepsilon_2[\tau](\sigma) \\ = \varepsilon_2[\tau'](\sigma) \end{array} \right\}$$

† and

$$(7) \qquad \begin{cases} N_\sigma^* \cap \widehat{\mathcal{G}} = \mathbb{D}_\sigma = \{1,\tau\}.L_\sigma = \{1,\tau'\}.L_\sigma \\ N_\sigma^* \cap \mathcal{G}^{+\wedge} = L_\sigma \qquad (\simeq \mathbb{Z}/4\mathbb{Z}) \\ N_\sigma^* \cap \Pi = \mu \\ N_\sigma^* \cap \Pi_0 = \{1\} \end{cases} .$$

Uncertainty retained from cancelled passages.
140-3-s622-D002 $\varepsilon_2(u)$
140-3-s622-D003 Let us put

[scan 623]
[p. 615]

Let us put

$$(8)\qquad \begin{cases} & \mathcal{M}_0^! = \mathcal{M}(0).\Pi_0 \subset \mathcal{M}^! = \mathcal{M}(0).\Pi = \mathrm{Ker}(\mathcal{M} \to \mathcal{G}_3) \\ \text{so that} & \mathcal{M}^! \simeq \mathcal{M}_0^! \times \mu \end{cases}$$

and

$$(9)\qquad \begin{cases} \mathcal{M}(1/2) \overset{\text{dfn}}{=} \dagger N_{\sigma\dagger 0}^{*!} \overset{\text{dfn}}{=} \dagger N_\sigma^* \cap \mathcal{M}_0^! \\ N_\sigma^{*!} = N_\sigma^* \cap \mathcal{M}^! = \dagger, \end{cases}$$

then the homomorphism $\dagger\ \mathcal{M} \to \gamma_{\rho,\sigma}$ induces an isomorphism
(10)

$$\mathcal{M}(1/2) = N_{\sigma 0}^{*!} \xrightarrow{\sim} \gamma_{\rho,\sigma}$$

so we have splittings into 1/2 direct products

$$(11)\qquad \begin{cases} . \ \mathcal{M}^\dagger \simeq \mathcal{M}(1/2).\mathcal{G}^{+\wedge} \\ \mathcal{M}^! \simeq \mathcal{M}(1/2).\Pi \\ \mathcal{M}_0^! \simeq \mathcal{M}(1/2).\Pi_0 \\ N_\sigma^{*!} \simeq \mathcal{M}(1/2) \times \mu \\ N_\sigma^* \simeq \mathcal{M}(1/2).L_\sigma \end{cases} , \quad \text{and}$$

(12)

$$\gamma_{\rho,\sigma} \xrightarrow{\sim} \mathcal{M}(1/2) = N_{\sigma 0}^{*!} \hookrightarrow N_\sigma^*.$$

© We have

$$(13)\qquad N_\rho^* = \left\{u \in \mathcal{M} \mid u(\rho) = \rho^{\varepsilon_3(u)}\right\} \quad (= \varepsilon_3[\tau'](\rho))$$

and

$$(14)\qquad \begin{cases} N_\rho^* \cap \widehat{\mathcal{G}} = \mathbb{D}_\rho = \{1, \tau'\}.L_\rho \\ N_\rho^* \cap \mathcal{G} = L_\rho \qquad (\simeq \mathbb{Z}/6\mathbb{Z}) \\ N_\rho^* \cap \Pi = \mu \\ N_\rho^* \cap \Pi_0 = \{1\} \end{cases} .$$

Uncertainty retained from cancelled passages.
140-3-s623-D001 A
140-3-s623-D002 ρ
140-3-s623-D003 $Z_\sigma \mathcal{M}_0$
140-3-s623-D004 $\mathcal{M}(1/2)|\mathcal{M}(0)$
140-3-s623-D005 from
140-3-s623-D006 !

[scan 624]

Let

$$N^{*!}_{\rho 0} \overset{\text{dfn}}{=} N^*_\rho \cap \mathcal{M}^!_0, \qquad N^{*!}_\rho = N^*_\rho \cap \mathcal{M}^! . \tag{15}$$

we find

$$N^{*!}_{\rho 0} \subset Z_\rho = \mathrm{Cent}_{\mathcal{M}(0)|\mathcal{M}}(\rho) = \mathrm{Cent}_{\mathcal{M}}(L_\rho) \tag{16}$$

and we also note

$$N^{*!}_{\rho,0} = Z^!_{\rho 0} = \mathcal{M}'(-\bar{\jmath}). \tag{17}$$

We see that the homomorphism $\mathcal{M} \to \gamma_{\rho,\sigma} = \tilde{\Gamma}_{\mathbb{Q}}$ induces an isomorphism (18)

$$\begin{array}{ccc} \mathcal{M}'(-\bar{\jmath}) = Z^!_{\rho,0} & \xrightarrow{\quad\sim\quad} & \gamma'_{\rho,\sigma} = \tilde{\Gamma}_{\mathbb{Q}} \\ & & \| \text{ dfn} \\ & & \{\gamma \in \gamma_{\rho,\sigma} \mid \varepsilon_3(\gamma) = 1\} \end{array}$$

from which we deduce splittings into semidirect products (or even direct products)

$$\text{(19)} \qquad \begin{cases} \mathcal{M}' \simeq \mathcal{M}'(-\bar{\jmath}).\mathcal{G}^{+\wedge} \\ \mathcal{M}'^! \simeq \mathcal{M}'(-\bar{\jmath}).\Pi \\ \mathcal{M}'^!_0 \simeq \mathcal{M}'(-\bar{\jmath}).\Pi_0 \\ Z^!_\rho = | \simeq N^{*!}_\rho \simeq \mathcal{M}'(-j) \times \mu \\ \dagger Z_\rho \simeq \mathcal{M}'(-j) \times L_\rho \end{cases} .$$

$\dagger \longrightarrow$

$\dagger \longrightarrow$

Uncertainty retained from cancelled passages.

140-3-s624-D001 N^*_ρ

140-3-s624-D003 $\mathcal{M}'^!_0[0]|\mathcal{M}'^![0]$

140-3-s624-D014 $\mathcal{M}'^!_0[0]|\mathcal{M}'^![0]$

[scan 625]

[p. 616]

Now let

$$\mu_{\tau'} = \{1, \tau'\} \subset \mathcal{G}^\wedge \tag{20}$$

then $\mu_{\tau'}$ normalizes L_ρ, $\dagger$ for $\tau'(\rho) = \rho^{-1}$ whence $\tau'(L_\rho) = L_\rho$ hence it normalizes $Z_\rho \equiv | = \mathrm{Cent}_{\mathcal{M}}(L_\rho)$, hence also $Z^!_\rho = Z_\rho \cap \mathcal{M}^!$ (since $\mathcal{M}^!$ is invariant in $\mathcal{M}$) $= Z^!_{\rho,0} \times \mu = \mathcal{M}'(-\bar{\jmath}) \times \mu$. Moreover $\mu_{\tau'} \cap Z^!_\rho \subset \dagger\mu_{\tau'} \cap \mathcal{M}^! = \{1\}$. Let

$$\mathcal{N}^?_\rho = \mu_{\tau'} \cdot Z^!_\rho = \mu_{\tau'} \cdot \underbrace{(Z^!_{\rho,0} \times \mu)}_{\substack{\mathcal{M}(-\bar{\jmath}) \times \mu \\ \simeq \Gamma'^{\sim}_{\mathbb{Q}} \times \mu}} \quad \dagger. \tag{21}$$

($\frac{1}{2}$ direct product)

One should be aware that contrary to what this notation might suggest, $\mu_{\tau'}$ does not normalize $Z^!_{\rho,0}$ — otherwise one would have a decomposition|arrangement into a product

$$\mathcal{N}^?_\rho = \mathcal{N}^?_{\rho,0} \times \mu, \quad \text{where } \mathcal{N}^?_{\rho,0} = \dagger\mu_{\tau'} \cdot Z^?_{\rho,0}|Z^!_{\rho,0},$$

but this is not the case. We have an exact sequence

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \mu & \longrightarrow & \mathcal{N}^?_\rho & \longrightarrow & \gamma_{\rho,\sigma} & \longrightarrow & 1 \\ & & & & & & \| & & \\ & & & & & & \Gamma^\sim_{\mathbb{Q}} & & \end{array} \tag{22}$$

so that $\mathcal{N}^?_\rho$ is a quasi-section of multiplicity 2 of the extension $\mathcal{N}^*_\rho \longrightarrow \gamma_{\rho,\sigma}$, — of $\mathcal{M} \longrightarrow \gamma_{\rho,\sigma} = \Gamma^\sim_{\mathbb{Q}}$. Thus we find 624

Uncertainty retained from cancelled passages.

140-3-s625-D001 this

140-3-s625-D003 {1}

140-3-s625-D004 $\simeq \mathcal{M}_{\mathbb{Q}} \times \mu| \simeq \mathcal{M}_0 \times \mu$

140-3-s625-D005 $Z^!_{\rho,0} \times \mu$

[scan 626]

$$\begin{cases} \mathcal{M} = \mathcal{N}^?_\rho \wedge_{\mu_2} \mathcal{G}^{+\wedge} \\ \mathcal{N}^*_\rho = \mathcal{N}^?_\rho \wedge_{\mu_2} \dagger L_\rho \simeq \mathcal{N}^?_\rho \times \underbrace{L_{-\rho}}_{\simeq \mathbb{Z}/3\mathbb{Z}} \end{cases} \tag{23}$$

The extension (22) of $\gamma_{\rho,\sigma}$ by μ is the extension denoted by $\tilde{\gamma}_{\rho\sigma}$ in the preceding section, it is surely not split. It gives rise to a canonical homomorphism

$$\gamma_{\rho,\sigma} \xrightarrow{\sim} \mathcal{N}^?_\rho/\mu \longrightarrow \mathcal{N}^*_\rho/\mu. \tag{24}$$

Remark To do it properly, one would have to make explicit the operation of τ' on $Z^!_\rho \simeq Z^!_{\rho,0} \times \mu \xrightarrow{\sim} \Gamma'^\sim_{\mathbb{Q}} \times \mu$, in terms of the operation of the image $\tau_{\Gamma\dagger}$ of τ' on $\Gamma'^\sim_{\mathbb{Q}}$ by inner automorphism (N.B. τ_Γ is none other than the element of $\Gamma_{\mathbb{Q}} \subset \Gamma^\sim_{\mathbb{Q}}$, "complex conjugation") and of a homomorphism from $\Gamma'^\sim_{\mathbb{Q}}$ to μ i.e. of a "quadratic character" of $\Gamma'^\sim_{\mathbb{Q}}$ — it is a good bet that this is precisely ε_2 — so that $\Gamma^{0\sim}_{\mathbb{Q}} \simeq \Gamma'^\sim_{\mathbb{Q}} \cap \Gamma''^\sim_{\mathbb{Q}}$ appears as invariant under τ' …In short, it would be well to take up again the study of the extension made in the preceding section, by starting there from the point of view of the associated α, β, 625

Uncertainty retained from cancelled passages.

140-3-s626-D001 $\mathbb{Z}$

140-3-s626-D004 $\mathbb{Q}$

[scan 627]

in the present perspective, probably simpler. It would be well at the same time to make explicit another partial section, which would be over $\dagger\Gamma'''^\sim_{\mathbb{Q}}$, [p. 617]

$$\Gamma'''^\sim_{\mathbb{Q}} = \{x \in \Gamma^\sim_{\mathbb{Q}} \mid \varepsilon_2(x)\varepsilon_3(x) = 1\},$$

and which would perhaps deserve the notation $\mathcal{M}(\sqrt{3})$ or at least $\mathcal{M}(q)$, for a suitable generator $q|\rho$ of $\mathbb{Q}(\sqrt{3})$??? m1286

Marginal note. m1286 ? Ponder these coincidences. Attentive.|attentively

I realize that I am talking crap — having somewhat lost touch with the geometric content of my calculations. One must keep in mind that

$$\mathcal{M}_0^! \xrightarrow{\sim} \mathcal{M}_{0,3}^{!\sim} \tag{25}$$

corresponds up to $\sim$, to the π_1 of $U_{0,3\mathbb{Q}}$, the sections of $\mathcal{M}_0^! \longrightarrow \Gamma_{\mathbb{Q}}^{\sim} \simeq \gamma_{\rho,\sigma}$ thus correspond (up to $\sim$) to those of $\pi_1(U_{0,3\mathbb{Q}})$ over $\Gamma_{\mathbb{Q}}$, thus ("invariably|inevitably|unavowably") to the rational/$\mathbb{Q}$ points of $U_{0,3\mathbb{Q}}$ — when they are partial sections, to the points of $U_{0,3\mathbb{Q}}$ with values in finite extensions of $\mathbb{Q}$. Thus, the section †considered in b) does indeed correspond to the pt $\frac{1}{2}$ of $U_{0,3\mathbb{Q}}$, the †section partial $\mathcal{M}(-\bar{\jmath})$ considered here in c) corresponds to the point $-\bar{\jmath}$ of $U_{0,3\mathbb{Q}}$, which is in $U_{0,3\dagger}(\mathbb{Q}[\sqrt[3]{1}])$, so the notation $\mathcal{M}(-\bar{\jmath})$ is reasonable. As for $\mathcal{N}_{\rho}^{\sim}$, it is not $\subset \mathcal{M}_0^!$ — even on dividing by μ; one finds after division by μ a section 626

Uncertainty retained from cancelled passages.

140-3-s627-D003 Γ

140-3-s627-D005 privileged

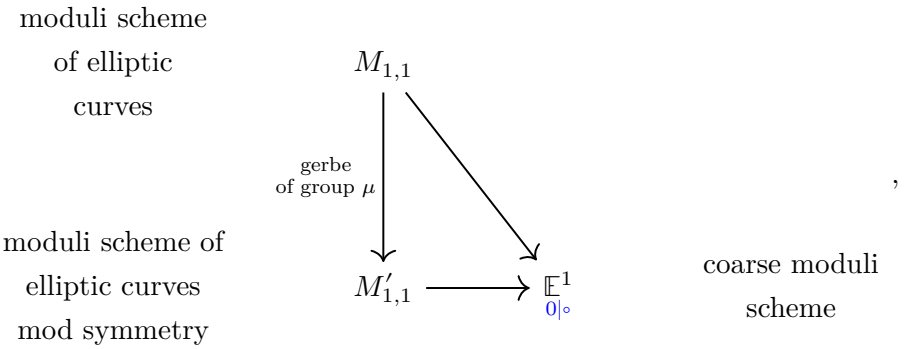
140-3-s627-D006 quasi-

140-3-s627-D007 $\mathbb{Q}$

[scan 628]

of $\mathcal{M}_{0,3}^{\sim}$ over $\Gamma_{\mathbb{Q}}^{\sim}$, but this is not a section of $\mathcal{M}_{0,3}^{!\sim}$. Now up to $\sim$, $\mathcal{M}_{0,3}^{\sim}$ is the π_1 of $M'_{1,1\mathbb{Q}}$, moduli scheme over $\mathbb{Q}$ of elliptic curves mod symmetry. Thus giving this section corresponds to giving such an "elliptic curve mod symmetry" over $\mathbb{Q}$. Passing to the subgroup $\Gamma_{\mathbb{Q}}'^{\sim}$ — which corresponds to the extension of the base field from $\mathbb{Q}$ to $\mathbb{Q}(j) = \mathbb{Q}(\sqrt[3]{1})$ one recovers the image in $\mathcal{M}_{0,3}^{!\sim}$ of the preceding section partial — to $\mathcal{M}(-\bar{\jmath})$ — this thus shows that it is still the "same" elliptic curve. I thus have the impression however that over the whole of $\Gamma_{\mathbb{Q}}^{\sim}$, this section of $\mathcal{M}_{0,3}^{\sim}$ cannot be lifted to a section of $\mathcal{M}_{\mu}|\mathcal{M}$ over $\Gamma_{\mathbb{Q}}^{\sim}$ — i.e. precisely that the extension $\mathcal{N}_{\rho}^{?|*}$ of $\Gamma_{\mathbb{Q}}^{\sim}$ by μ is nontrivial. Would this mean that there is no elliptic curve defined/$\mathbb{Q}$ with invariant 0? (Something that seems absurd — one must be able over a field to give any invariant…). ?

To be precise, we have the diagram of moduli schemes over $\mathbb{Q}$



, (26)

the fact that a point of $M'_{1,1}(\mathbb{Q})$ does not li- 627

[scan 629]

ft to a point of $M_{1,1}$, does not mean that its image in $\mathbb{E}^1$ does not lift to $M_{1,1}$. To be precise, [p. 618] the †diagram identifies with:

$$\begin{array}{ccc} M_{1,1} \simeq (U_{0,3}, \widetilde{\mathcal{G}}_3) & & \\ \downarrow & \searrow & \\ M'_{1,1} \simeq (U_{0,3}, \mathcal{G}_3) & \longrightarrow & U_{0,3}/\mathcal{G}_3 \simeq U_{0,3}/\widetilde{\mathcal{G}}_3 \simeq \mathbb{E}^1 \end{array} \quad , \tag{27}$$

where $\widetilde{\mathcal{G}}_3$ is the extension of $\mathcal{G}_3 = \mathrm{SL}(2,\mathbb{Z}/2\mathbb{Z})$ by μ, defined by

$$\widetilde{\mathcal{G}}_3 = \mathcal{G}^+/\pi_0 \simeq \mathrm{SL}(2,\mathbb{Z})/\begin{array}{c}\text{invariant subgroup}\\ \text{generated by}\\ -\ell_0\\ \text{invariant}\\ \text{subgroup}\\ \text{(of order 4) generated}\\ \text{by } -\ell_0, \text{ i.e. subgroup|subgroup}\\ \text{generated by}\\ -\ell_0, -\ell_1.\end{array} \simeq \mathrm{SL}(2,\mathbb{Z}/4\mathbb{Z})/ \tag{28}$$

by letting $\widetilde{\mathcal{G}}_3$ operate on $U_{0,3}$ through $\mathcal{G}_3$. A point of $M'_{1,1}$rational/$\mathbb{Q}$ identifies with $\mathcal{G}_3$-torsor T over $\mathbb{Q}$, equipped with a map

$$T \longrightarrow U_{0,3} \tag{29}$$

m1290

Marginal note. m1290 T is the torsor under $\mathrm{SL}(2,\mathbb{Z}/2\mathbb{Z})$ of rigidifications|privilegiations of the points of order 2 and $\widetilde{T}$ that of "pri-rigidifications|privilegiations|rigidifications of the points of order ≤ 4|4 made precise" (the first is defined for an elliptic curve mod symmetry, and the 2nd only for a genuine elliptic curve)

compatible with the actions of $\mathcal{G}_3$, a point of $M_{1,1}$ identifies with a $\widetilde{\mathcal{G}}_3$-torsor $\tilde{T}$ over $\mathbb{Q}$, equipped with a map

$$\tilde{T} \text{ —— } U_{0,3} \tag{30}$$

compatible with the operation of $\widetilde{\mathcal{G}}_3$. Denoting by $\dagger$

$$T = \tilde{T}/\mu$$

the corresponding $\mathcal{G}_3$-torsor, giving (30) is equivalent to giving (29). Thus the 628

Uncertainty retained from cancelled passages.
140-3-s629-D001 [illeg. 1]
140-3-s629-D003 T

[scan 630]

liftings to $M_{1,1}$ of a point of $M'_{1,1}$ rational/$\mathbb{Q}$ identify with the $\widetilde{\mathcal{G}}_3$-torsors $\tilde{T}$ which lift the $\mathcal{G}_3$-torsor T, i.e. (up to isomorphism) they correspond to the homomorphisms $\Gamma_{\mathbb{Q}} \longrightarrow \widetilde{\mathcal{G}}_3$ which lift the homomorphism $\Gamma_{\mathbb{Q}} \longrightarrow \mathcal{G}_3$ which defines T: m1292

Marginal note. m1292 we suppose now chosen a $\dagger$ point of $T(\overline{\mathbb{Q}}_0)$

$$\begin{array}{ccc} & & \widetilde{\mathcal{G}}_3 \\ & \nearrow & \downarrow \\ \Gamma_{\mathbb{Q}} & \longrightarrow & \mathcal{G}_3 \end{array} \quad -$$

the obstruction is, as it should be, an extension of $\Gamma_{\mathbb{Q}}$ by μ_2, i.e. an element of $H^2(\mathbb{Q},\mu_2) \simeq {}_2\mathrm{Br}(\mathbb{Q})$. In the case at hand, the image of T in $U_{0,3}$ is made up of $\{-\bar{\jmath}, -j|-\bar{\jmath}\}$ which is connected over $\mathbb{Q}$, hence $\Gamma_{\mathbb{Q}} \longrightarrow \mathcal{G}_3$ is surjective. m1293

Marginal note. m1293 corrected|erroneous, cf. correction below

In general, † in the Galois-theoretic translation in terms of splittings of $\mathcal{M}_{0,3}$ over $\Gamma_{\mathbb{Q}}$, the homomorphism $\Gamma_{\mathbb{Q}} \longrightarrow \mathcal{G}_3$ is the composite

$$\Gamma_{\mathbb{Q}} \xrightarrow{k|\kappa} \mathcal{M}_{0,3} \longrightarrow \mathcal{G}_3 \ ,$$

which is trivial iff † $k|\kappa$ is a section of

$$\mathcal{M}^{!}_{0,3} = \mathrm{Ker}(\mathcal{M}_{0,3} \longrightarrow \mathcal{G}_3)$$

over $\Gamma_{\mathbb{Q}}$. Here we have a section of $\mathcal{M}^{!}_{0,3}$ above the subgroup $\Gamma'_{\mathbb{Q}}$ of index 2, $k|\kappa \mid \Gamma'_{\mathbb{Q}} = 1$, hence $k|\kappa(\Gamma_{\mathbb{Q}}) \subset \mathcal{G}_3$ is of order 2, we see that 629

Uncertainty retained from cancelled passages.
140-3-s630-D001 geometric
140-3-s630-D002 one must
140-3-s630-D003 $\Gamma_{\mathbb{Q}}$

[scan 631]

it is the subgroup $\{1, \dot{\sigma}_\infty\}$ of $\mathcal{G}_3$. (for the image of $\tau' = \sigma\tau \in \mathcal{M}_{0,3}$ in $\mathcal{G}_3$ † (since that of τ is 1) is equal to that of $\sigma = \sigma_\infty$, i.e. $\dot{\sigma}_\infty$. This means that † we have [p. 619]

$$T \simeq T_0 \wedge_{\{1,\dot{\sigma}_\infty\}} \mathcal{G}_3$$

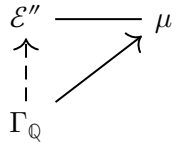
i.e. T is obtained from a μ-torsor (i.e. from a quadratic extension of $\mathbb{Q}$) — namely precisely $\mathbb{Q}(\sqrt[3]{1}) = \mathbb{Q}(j)$) by extension of the groups of operators $\mu \longrightarrow \mathcal{G}_3 \quad (-1 \longmapsto \dot{\sigma}_\infty)$. Thus, taking the inverse image of $\mu = \{1, \dot{\sigma}_\infty\}$ in $\widetilde{\mathcal{G}}_3$, we find an extension of μ by μ — it can hardly be anything but $\mathbb{Z}/4\mathbb{Z}$! — which I feel like denoting $\mathcal{E}''$

$$1 \to \mu \longrightarrow \overset{\mathbb{Z}/4\mathbb{Z}}{\overset{\cdot\cdot}{\mathcal{E}''}} \longrightarrow \mu \to 1$$

m1295

Marginal note. m1295 yes, this is obvious since $\sigma = \sigma_\infty$ in $\widetilde{\mathcal{G}}_3 = \mathrm{SL}(2,\mathbb{Z})/\Pi_0|\Pi$ is indeed of order 4, so $\mathcal{E}'' \simeq \mathbb{Z}/4\mathbb{Z}$ canonically…

and the problem of obstruction is to find a lifting



, (which † surely does not exist) — so the obstruction is identified as the extension of $\Gamma_{\mathbb{Q}}$ by μ, inverse image of the extension $\mathcal{E}''$. m1296

Marginal note. m1296 This is obvious in effect|fact, that we|since : $(\Gamma_{\mathbb{Q}})_{\mathrm{ab}} \simeq \widehat{\mathbb{Z}}^* \supset \mathbb{Z}_3^*$, now $(\mathbb{Z}_3^*)_4 \simeq \mathbb{Z}/2\mathbb{Z}$ (so already on $\mathbb{Z}_3^*$ it does not lift…

(One should look at why, in 630

Uncertainty retained from cancelled passages.
140-3-s631-D001 is
140-3-s631-D002 T
140-3-s631-D003 we

[scan 632]

classical terms, this obstruction is split, when one passes to the subgroup $\Gamma'''_{\mathbb{Q}}$ corresponding to the field $\mathbb{Q}(\sqrt{3})$)

Thus, the $\mathbb{Q}$-rational point under consideration of $M'_{1,1}$ does not lift to a $\mathbb{Q}$-rational point of $M_{1,1}$ — On the other hand, its image 0 in $\mathbb{E}^1 \simeq U_{0,3}/\mathcal{G}_3$ must indeed lift — i.e. there must indeed exist a $\widetilde{\mathcal{G}}_3$-torsor $\tilde{P}$, and a $\widetilde{\mathcal{G}}_3$-morphism †

$$\tilde{P} \longrightarrow U_{0,3} \qquad \text{i.e.} \qquad \begin{array}{ccc} P & \longrightarrow & U_{0,3} \\ \| & & \\ \tilde{P}/\mu & & \end{array}$$

.

Let us consider indeed the sub-scheme

$$\dagger \qquad Q = U_{0,3} \times_{\mathbb{E}^1} \{0\}$$

the inverse image of 0, i.e.

$$P_0 = Q_{\mathrm{red}} \ \dagger$$

it is a subscheme which, over a field $K \supset \mathbb{Q}(j)$, is a sum of two points, and which over $\mathbb{Q}$ is connected — it is a scheme canonical[ly] isomorphic to $\operatorname{Spec} \mathbb{Q}(-\bar{j})$. It is stable under $\mathcal{G}_3$, which operates on it via

$$\mathcal{G}_3 \xrightarrow{\text{sg}} \mu$$

and the Galois operation of μ on the $\mu_{\mathbb{Q}}$-torsor P_0. 631

Uncertainty retained from cancelled passages.
140-3-s632-D001 P
140-3-s632-D003 $Q = P$
140-3-s632-D004 $\simeq$

[scan 633]

Thus we have the diagram [p. 620]

$$\begin{array}{ccccc} \widetilde{\mathcal{G}}_3 & \xrightarrow{\text{can}} & \mathcal{G}_3 & \xrightarrow{\text{sg}} & \mu \\ & & & \nearrow & \\ & & \Gamma_{\mathbb{Q}} & & \end{array} \qquad (31)$$

m1299

Marginal note. m1299 N.B. the kernel of $\widetilde{\mathcal{G}}_3 \longrightarrow \mu$ is a central extension of $\mathbb{Z}/3\mathbb{Z}$ by $\mu = \mathbb{Z}/2\mathbb{Z}$, it is therefore trivial and isomorphic to $\mathbb{Z}/6\mathbb{Z}$ (kernel generated by ρ).

We thus see that the category of liftings of the $\mathbb{Q}$-rational point 0 of $\mathbb{E}^1$ to a $\mathbb{Q}$-rational point of $M'_{1,1}$ resp. $M_{1,1}$, is equivalent to that of $\mathcal{G}_3$-torsors P resp. of $\widetilde{\mathcal{G}}_3$-torsors $\tilde{P}$ over $\mathbb{Q}$, which lift the μ-torsor P_0.

We had chosen a $P = T$ in a particularly trivial|trivial way, using the splitting

$$\mu \xrightarrow{\sim} \{1, \dot\sigma_\infty\} \hookrightarrow \mathcal{G}_3$$

of $\mathcal{G}_3 \longrightarrow \mu$ — but the torsor T does not lift to a $\tilde T$. The isomorphism classes of points of $M'_{1,1}$ resp. of $M_{1,1}$ over the point 0 of $\mathbb{E}^1$, correspond likewise to the conjugacy classes (modulo kernels $\mathcal{G}_3^+, \widetilde{\mathcal{G}}_3^+$ of $\mathcal{G}_3 \longrightarrow \mu$ and $\widetilde{\mathcal{G}}_3 \longrightarrow \mu$) of liftings of $\Gamma_{\mathbb{Q}} \longrightarrow \mu$ to $\Gamma_{\mathbb{Q}} \longrightarrow \mathcal{G}_3$ resp. to $\Gamma_{\mathbb{Q}} \longrightarrow \widetilde{\mathcal{G}}_3$.

† Let us consider a lifting to

$$\Gamma_{\mathbb{Q}} \longrightarrow \mathcal{G}_3$$

, its image is a subgroup of $\mathcal{G}_3$, which is either 632

Uncertainty retained from cancelled passages.

140-3-s633-D001 From there

[scan 634]

$\{1, \dot\sigma_\infty\}$ or one of its conjugates, $\{1, \dot\sigma_0\}$, $\{1, \bar\sigma_1|\dot\sigma_1\}$, or it is the whole of $\mathcal{G}_3$. In the first case, we have seen that it does not lift to $\widetilde{\mathcal{G}}_3$. So to be able to lift, one must take an epimorphism $\Gamma_{\mathbb{Q}} \longrightarrow \mathcal{G}_3$ — so we leave the abelian setting over $\mathbb{Q}$, to enter the domain of abelian extensions of $\mathbb{Q}(j)$.

To examine in terms of Galois groups whether there exists such a lifting $\Gamma_{\mathbb{Q}} \longrightarrow \widetilde{\mathcal{G}}_3$, and † how to obtain them, appears as a pleasant and delectable exercise in class field theory, which I am not going to pursue for the moment, so as not to make this digression last forever. Returning later to this point, I would need at the same time to make explicit the relation between the choices|points of a lifting of the homomorphism from $\Gamma_{\mathbb{Q}}$, and splittings of the extension $\mathcal{M}_{0,3}$ of $\Gamma_{\mathbb{Q}}$ by $\mathcal{G}_{0,3}^{+\wedge}|\widetilde{\mathcal{G}}_{0,3}^{+\wedge}$, or $\mathcal{M}$ of $\Gamma_{\mathbb{Q}}$ by $\mathcal{G}^{+\wedge} \simeq \mathrm{SL}(2,\mathbb{Z})^\wedge$. 633

Uncertainty retained from cancelled passages.

140-3-s634-D002 which

[scan 635]

Let us return to the subgroups † [p. 621]

$$\begin{array}{ccccc} \mathcal{N}_\rho^? & \overset{3}{\subset} & \mathcal{N}_\rho^* & \subset & \mathcal{M} \\ 4\cup & & & & \\ Z^!_{\rho,0} = \mathcal{M}'(-\bar j) & & & & \end{array}$$

containing the|the partial subgroup|subgroups $\mathcal{M}(-\bar j) \simeq \tilde\Gamma'_{\mathbb{Q}}$.

Let

$$\begin{aligned} \mathcal{M}^0(-\bar j) &\overset{2}{\subset} \mathcal{M}'(-\bar j) \\ &\underset{\text{dfn|def}}{=} \mathrm{Ker}(\mathcal{M}'(-\bar j) \xrightarrow{\varepsilon_2} \mu) = \dagger\, Z_\rho \cap \dagger\, \mathcal{M}_0''^! \\ &= \mathcal{N}_\rho^* \cap \mathcal{M}_0^{0!} \end{aligned} \tag{32}$$

which is therefore a partial section up|up above

$$\begin{aligned} \tilde\Gamma^0_{\mathbb{Q}} &= \mathrm{Ker}(\tilde\Gamma_{\mathbb{Q}} \xrightarrow{(\varepsilon_2,\varepsilon_3)} \mu \times \mu)\dagger \\ &= \mathrm{Ker}(\tilde\Gamma_{\mathbb{Q}} \longrightarrow \widehat{\mathbb{Z}}^* \longrightarrow (\mathbb{Z}/12\mathbb{Z})^*) \end{aligned} \tag{33}$$

Modulo a verification which I am going to make below, τ' normalizes $\mathcal{M}^0(-\bar{j})$, let us consider the|the

$$\mathcal{M}'''(-\bar{j})\dagger \overset{\text{dfn|def}}{=} \underset{\text{(semidirect)}}{\mu_{\tau'} \,.\, \mathcal{M}^0(-\bar{j})} \qquad \text{where } \mu_{\tau'} = \{1, \tau'\} \tag{34}$$

it is a subgroup of index 4 of $\mathcal{N}_\rho^?$, and we now have

$$\dagger \quad \mathcal{M}'''(-\bar{j}) \xrightarrow{\sim} \tilde{\Gamma}'''_{\mathbb{Q}}, \tag{35}$$

it is a partial section up|up above $\tilde{\Gamma}'''_{\mathbb{Q}}$, whence isomorphisms 634

Uncertainty retained from cancelled passages.

140-3-s635-D001 $\mathcal{N}_\rho^?$
140-3-s635-D003 $\mathcal{N}_\rho^? | \mathcal{M}$
140-3-s635-D004 $\operatorname{Ker} \varepsilon_2$
140-3-s635-D005 $\neq | =$
140-3-s635-D008 $\mathcal{N}_{\rho,0}^?$
140-3-s635-D009 $\mathcal{N}_{\rho,0}^?$

[scan 636]

$\dagger$ 635

Uncertainty retained from cancelled passages.

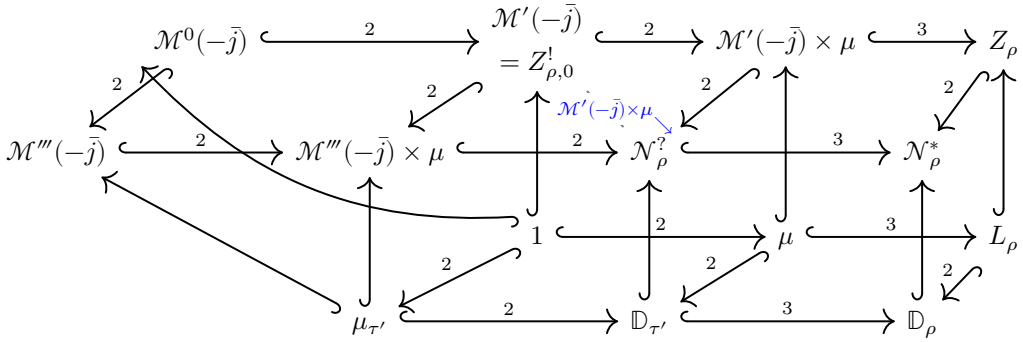
140-3-s636-D001 $\mathcal{M}'''$ $\mathbb{D}_6 | \mathbb{D}_\sigma$

[scan 637]
[p. 622]

$$\begin{cases} \mathcal{M}''' \simeq \mathcal{M}'''(-\bar{j}) \,.\, \mathcal{G}^{+\wedge} \\ \mathcal{N}_\rho^* \simeq \mathcal{M}'''(-\bar{j}) \,.\, L_\rho. \end{cases} \tag{36}$$

Summary diagram of the subgroups of $\mathcal{N}_\rho^*$ obtained so far:

(37)



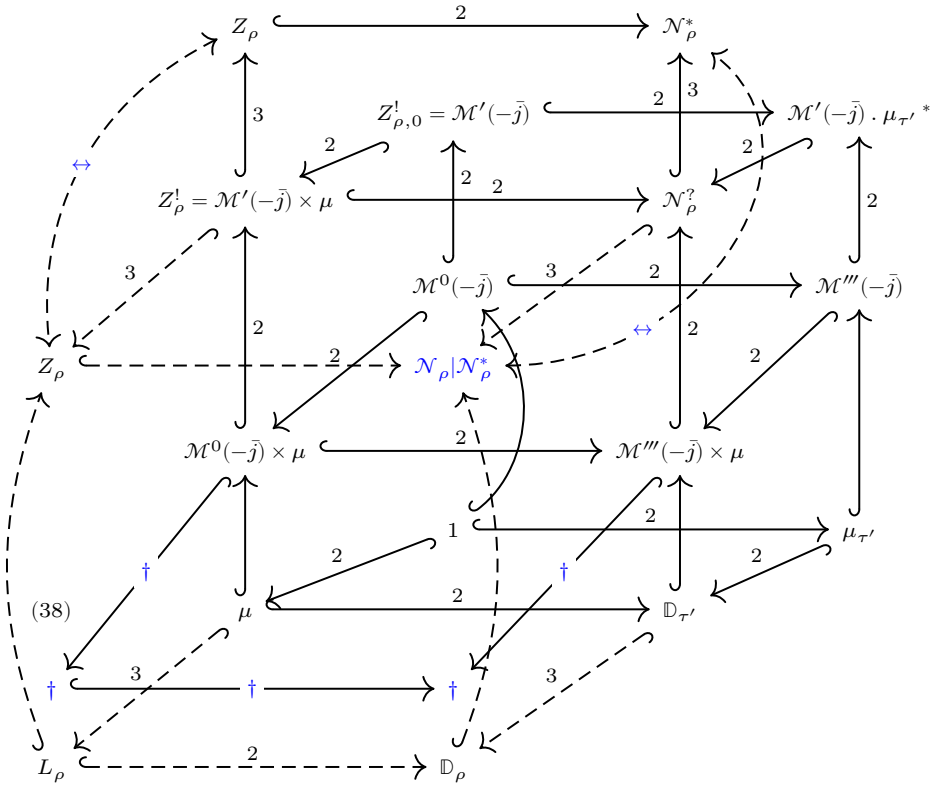
,

where we are here interested above all in the "upper face", containing|concerning subgroups of finite index in $\mathcal{N}_\rho^*$, intermediate between it and $\mathcal{M}_0(-\bar{j}) | \mathcal{M}^0(-\bar{j})$ (of index 24 in $\mathcal{N}_\rho^*$). I am going to examine the question of invariance of these subgroups in one Another, and the structure of the quotient groups.

I am going to redo a diagram which really|better shows the $\dagger$ relations between the groups in question. 636

Uncertainty retained from cancelled passages.
140-3-s637-D001 groups|relationships

[scan 638]



m1305

Marginal note. m1305 $*$ – Caution, $\mathcal{M}'(-\bar{j}) \cdot \mu_{\tau'}$ is not a subgroup, for $\mu_{\tau'}$ does not normalize $\mathcal{M}'(-\bar{j})$.

a) Subgroups of finite index in Z_ρ – in the diagram there are four of them

$$\begin{array}{ccc} Z^!_\rho = \mathcal{M}'(-\bar{j}) \times \mu & \xleftarrow{2} & \mathcal{M}'(-\bar{j}) = Z^!_{\rho,0} = \mathcal{N}^{*!}_{\rho,0} \\ \uparrow 2 & & \uparrow 2 \\ \mathcal{M}^0(-\bar{j}) \times \mu & \xleftarrow[2]{} & \mathcal{M}^0(-\bar{j}) = Z^{0!}_{\rho,0} \end{array} \tag{39}$$

are they invariant in Z_ρ? This is clear for the subgroups †

$$\begin{cases} \mathcal{M}'(-\bar{j}) \times \mu = Z_\rho \cap \mathcal{M}^! = Z^!_\rho, & \text{and for} \\ \quad\quad\quad\quad \dagger & \\ \mathcal{M}^0(-\bar{j}) \times \mu = Z_\rho \cap \mathcal{M}^! \cap \mathrm{Ker}(\varepsilon_2 : \mathcal{M} \to \mu) \end{cases} \tag{40}$$

– this shows that these groups are 637

Uncertainty retained from cancelled passages.
140-3-s638-D001 ↪
140-3-s638-D002 ↪
140-3-s638-D003 L_ρ
140-3-s638-D004 ↪

140-3-s638-D005 $\mathbb{D}_\rho$
140-3-s638-D006 $\mathcal{M}'(-\bar{j})$
140-3-s638-D007 $\mathcal{M}^0(-\bar{j}) \times \mu =$

[scan 639]

even invariant in $\mathcal{N}_\rho^*$, since †. $Z_\rho, \mathcal{M}^!, \operatorname{Ker}\varepsilon_2$ are stabilized by $\mathcal{N}_\rho^*$. Besides, $Z^!_{\rho,0} = \mathcal{M}'(-\bar{j})$ [p. 623] is invariant †in $Z^!_\rho = \mathcal{M}'(-\bar{j}) \times \mu$, to show that it is so in Z_ρ, it suffices to prove that it is normalized by ρ (since $Z_\rho = L_\rho \,.\, Z^!_\rho$ – this is $\pm$ trivial), now let us examine in general for † $x \in \mathcal{M}$ whether x normalizes †

$$\mathcal{M}^!_0 \overset{\text{dfn}}{=} \mathcal{M}(0) \,.\, \Pi_0,$$

Since it normalizes Π_0, it suffices to see whether for $u \in \mathcal{M}(0)$, we have

$$xux^{-1} \in \mathcal{M}^!_0$$

which is also equivalent to

$$xux^{-1} \equiv u \qquad (\Pi_0)$$

i.e.

$$[x, u] \in \Pi_0$$

now we know that for $x \in \mathcal{G}$, $u \in \mathcal{M}^!$, we have

$$\underbrace{[x,u]}_{x\,u(x)^{-1}}{}^{c} \equiv (\operatorname{sg}(x))^{\xi_2(u)|\varepsilon_2(u)\dagger} \mod \Pi_0 \qquad (*)$$

where

$$\operatorname{sg}: \qquad \mathcal{G} \xrightarrow{\text{can}} \mathcal{G}_3 \xrightarrow{\text{sg}} \mu.$$

This shows † that

Lemma

$$\begin{cases} \text{If } x \in \mathcal{G}^+,\ u \in \mathcal{M}^!_0, & \text{then equivalent conditions} \\ a) \quad xux^{-1} \in \mathcal{M}^!_0 \\ b) \quad xux^{-1} \equiv u \quad (\Pi_0) & \text{i.e.} \quad [x,u] \in \Pi_0 \\ c) \quad .\operatorname{sg}(x)^{\xi_2(u)|\varepsilon_2(u)\dagger} = 1 & \text{i.e.} \quad \operatorname{sg}(x) = 1 \quad \underline{\text{or}} \quad \varepsilon_2(u) = 1 \end{cases}$$

638

Uncertainty retained from cancelled passages.
140-3-s639-D001 $\mathcal{M}^!$
140-3-s639-D004 in
140-3-s639-D005 $x \in$
140-3-s639-D006 $\mathcal{M}^!_0$
140-3-s639-D007 $\xi_2(u)|\varepsilon_2(u)$
140-3-s639-D008 that we
140-3-s639-D010 $\xi_2(u)|\varepsilon_2(u)$

[scan 640]

†

Corollary 1 Let $x \in \mathcal{G}^{+\wedge}$. For x to normalize $\mathcal{M}^!_0$, it is necessary and sufficient that we have $\operatorname{sg}(x) = +1$.

Corollary 2. The subgroup $\mathcal{M}''^!_0 = \mathcal{M}''(0) \,.\, \Pi_0$ is normalized by $\mathcal{G}^{+\wedge}$.

This shows in particular that ρ normalizes $\mathcal{M}^!_0$, hence also $\mathcal{M}'(-\bar{j}) = Z^!_{\rho,0} = Z_\rho \cap \mathcal{M}^!_0$, which is therefore invariant in Z_ρ. The same therefore holds for its intersection with $\mathcal{M}^0(-\bar{j}) \times \mu$, i.e.

$\mathcal{M}^0(-\bar{j})$. (Or again; this group is equal to $Z_\rho \cap \mathcal{M}_0''^!$, and $\mathcal{M}_0''^!$ is normalized by ρ ...). Thus the four groups (39) are normal in Z_ρ.)

To see whether they are normal in $\mathcal{N}_\rho^* = \mu_{\tau'} . Z_\rho$, it suffices to see whether they are normalized by τ'. This is clear for $Z_\rho^! = Z_\rho \cap \mathcal{M}^!$ – whence the fact that $\mathcal{N}_\rho^? = \mu_{\tau'} . Z_\rho^!$ is a subgroup of $\mathcal{N}_\rho^*$. To see † which of the three other subgroups considered last are normalized, it would therefore be necessary to make explicit the action of τ' on $Z_\rho^! = \mathcal{M}'(-\bar{j}) \times \mu$. 639

Uncertainty retained from cancelled passages.

140-3-s640-D001 the|the condition first group is

140-3-s640-D005 whether the

[scan 641]

Let us note that $\tau \in \mathcal{M}(0) \subset \mathcal{M}_0^!$ normalizes $\mathcal{M}_0^!$, so †it operates † on $\mathcal{M}^! = \underbrace{\mathcal{M}_0^!}_{\simeq \widetilde{\Gamma}_{\mathbb{Q}} . \Pi_0} \times \mu$ by [p. 623]

the product operation of conjugation by τ in $\mathcal{M}_0^!$, and the identity in μ. On the other hand we have $\tau' = \tau\sigma$, we have seen that

$$(40) \qquad \sigma(x) \equiv \varepsilon_2(x)x \qquad (\Pi_0) \qquad x \in \mathcal{M}$$

hence

$$(41) \qquad \tau'(x) \equiv \varepsilon_2(x)\tau(x) \qquad (\Pi_0) \qquad x \in \mathcal{M}$$

so it follows

$$(42) \quad \begin{cases} \text{if } x \in \mathcal{M}_0^!, \quad \text{then } \tau'(x) \equiv x \quad (\Pi_0) \Longleftrightarrow \tau'(x) \in \mathcal{M}_0^! \Longleftrightarrow \varepsilon_2(x) = 1 \\ \text{in particular, } \mathcal{M}_0''^! \text{ is normalized by } \tau \end{cases}$$

Thus applying this to the elements of $Z_{\rho,0}^! = \mathcal{M}'(-\bar{j})$ in $Z_\rho^! = \mathcal{M}'(-\bar{j}) \times \mu$, we find

$$(43) \quad \begin{cases} \text{Let } x \in Z_{\rho,0}^! = \mathcal{M}'(-\bar{j}), \text{ then } \dagger \\ \tau'(x) \in Z_{\rho,0}^! \Longleftrightarrow \varepsilon_2(x) = 1 \end{cases}$$

Corollary $Z_{\rho,0}^! = \mathcal{M}'(-\bar{j})$ is not normalized by τ', hence is not invariant in $\mathcal{N}_\rho^*$, but $\mathcal{M}^0(-\bar{j}) = Z_\rho \cap \mathcal{M}_0''^!$ is normalized by τ', hence also its product with μ.

Thus the only one of the four subgroups (39) of Z_ρ which is not normal in $\mathcal{N}_\rho$ is $\mathcal{M}'(-\bar{j}) = Z_{\rho,0}^!$. The others give 640

Uncertainty retained from cancelled passages.

140-3-s641-D001 one|therefore

140-3-s641-D002 Z

140-3-s641-D004 τ

[scan 642]

rise, by $\frac{1}{2}$ direct product with $\mu_{\tau'}$, to subgroups|subgroups of $\mathcal{N}_\rho$, invariant in $\mathcal{N}_\rho$. The smallest of all these subgroups is $\mathcal{M}^0(-\bar{j})$, which is of index 24 in $\mathcal{N}_\rho^*$. The quotient group †

$$\dagger \qquad \mathcal{N}_\rho^* / \mathcal{M}^0(-\bar{j})$$

is therefore a group of order 24, which moreover receives the group $\mathbb{D}_\rho = \mu_{\tau'} . L_\rho \subset \mathcal{N}_\rho^*$, normalizer of L_ρ in $\mathrm{GL}(2, \mathbb{Z})$, a group isomorphic to D_6, and of order 12. The kernel of

$$(44) \quad \mathbb{D}_\rho \hookrightarrow \mathcal{N}_\rho^* / \mathcal{M}^0(\bar{j})$$

is obviously contained in $L_\rho = \mathbb{D}_\rho \cap Z_\rho$ and it is reduced to 1 for since we even have

$$L_\rho \cap Z^!_{\rho,0} \subset L_\rho \cap \mathcal{M}^!_0 = L_\rho \cap \Pi_0 = \{1\}.$$

Thus $\mathbb{D}_\rho$(of order 12) appears as a subgroup of index 2 of $\mathcal{N}^*_\rho/\mathcal{M}^0(-\bar{j})$, just as it is a subgroup of index 2 of $\mathcal{G}/\Pi_0$. It would be tempting 641

Uncertainty retained from cancelled passages.

140-3-s642-D001 is a

140-3-s642-D002 group

[scan 643]

then to establish an isomorphism [p. 624]

$$\mathcal{N}^*_\rho/\mathcal{M}^0(-\bar{j}) \overset{?}{\simeq} \mathcal{G}^\wedge/\Pi_0$$

compatible with these embeddings of $\mathbb{D}_\rho$. Let us try it this way|also:

Proposition Let us consider the composite homomorphism

$$(45)\quad \mathcal{M} \xrightarrow{\Theta_{\mathcal{M}}} \mathrm{GL}(2,\widehat{\mathbb{Z}}) \longrightarrow \mathrm{GL}(2,\mathbb{Z}/4\mathbb{Z}) \xrightarrow{\text{epimorphism of degree 4}} \mathcal{G}^{*|\wedge}/\Pi_0 \simeq \mathcal{G}^\wedge/\Pi_0^\wedge$$

$$\mathrm{GL}(2,\mathbb{Z})/\mathcal{G}[4] \quad (\mathcal{G}),\ \text{with } \uparrow \sim \text{ to } \mathrm{GL}(2,\mathbb{Z}/4\mathbb{Z})$$

which induces on $\mathcal{G}^\wedge \subset \mathcal{M}$ the canonical homomorphism $\mathcal{G}^\wedge \longrightarrow \mathcal{G}^\wedge/\Pi_0^\wedge$. Then the homomorphism induced

$$\mathcal{N}^*_\rho \longrightarrow \mathcal{G}/\Pi_0$$

has as kernel $\mathcal{M}^0(-\bar{j})$, and induces by passage to the quotient an isomorphism

$$(46)\quad \mathcal{N}^*_\rho/\mathcal{M}^0(-\bar{j}) \xrightarrow{\sim} \mathcal{G}/\Pi_0$$

Proof. The elements of $\mathcal{M}^0(-\bar{j})$ are the elements of the form

$$\left\{ v = \alpha^{-1}u \;\middle|\; \begin{array}{ll} u \in \mathcal{M}(0), & \alpha \in \Pi_0 \\ u(\rho) = \alpha(\rho) & (\text{thus } \varepsilon_3(u) = 1) \\ \varepsilon_2(u) = 1 & \end{array} \right\}$$

they are in 1-1 correspondence with the elements of

$$\mathcal{M}^0(0) = \{u \in \mathcal{M}(0) \mid \varepsilon_2(u) = \varepsilon_3(u) = 1\}.$$

The image of|of the 642

$v|u$ in $\mathrm{Gl}(2,\widehat{\mathbb{Z}})/\operatorname{Im}\widehat{\Pi}^\circ_0$, factored through $\mathcal{G}/\Pi_0$, is equal to that of u. Thus the assertion that [scan 644] $M^\circ(-\bar{j})$ is in the kernel of $M \longrightarrow \mathcal{G}/\Pi_0$, is equivalent to the assertion that $M^\circ(0)$ is in this kernel, hence that we have a commutative diagram

$$\begin{array}{ccc} M(0) & \longrightarrow & M \\ \downarrow{\scriptstyle (\varepsilon_2,\varepsilon_3)} & & \downarrow{\scriptstyle (45)} \\ \mu \times \mu & \xrightarrow{?} & \mathcal{G}/\Pi_0 \end{array}$$

Marginal note. m1312 † N.B. † the correction|condition ε_3 does not enter into the definition (47) of $M \longrightarrow \mathcal{G}/\Pi_0$, cf. further on.

Note that the composite

$$\begin{array}{ccccc} M & \longrightarrow & \mathcal{G}/\Pi_0 & \longrightarrow & (\dagger\mathbb{Z}/4\mathbb{Z})^* \simeq \{\pm 1\} \\ & \searrow & & \nearrow & \uparrow \\ & & \mathrm{Gl}(2,\mathbb{Z}/4\mathbb{Z})^* & \longleftarrow & \dagger \end{array}$$

is none other than ε_2. Note also that by the compatibility

$$\begin{array}{ccc} \mathcal{G} & \hookrightarrow & M \\ & \searrow^{\text{can.}} & \downarrow (45) \\ & & \mathcal{G}/\Pi_0 \end{array}$$

the operation of M on $\mathcal{G}/\Pi_0$ deduced from the homomorphism (45) is none other than the operation †of a topological group|of quotients, of automorphisms on $\mathcal{G}$ on passing to the quotient by Π_0, an operation which, for $u \in M^!$ and in particular on $M(\omega|0) \subset M^!$, is given by $\varepsilon_2(u)$ (cf. page 594|554). Thus [p. 643]

Uncertainty retained from cancelled passages.

140-3-s644-D002 [illeg. 2]
140-3-s644-D003 commutative
140-3-s644-D004 GL
140-3-s644-D005 $\mathrm{GL}(2,\mathbb{Z}), \mathbb{Z}$
140-3-s644-D006 regular

We see that $M'^{!}_{\dagger}$ maps into the center of $\mathcal{G}/\Pi_0|\mathcal{G}[\Pi_0]$, which is contained in $\mathcal{G}^+/\Pi_0$, and in fact equal to the image μ of Π/Π_0 in $\mathcal{G}^+/\Pi_0$ (since the center of $\mathcal{G}^+/\Pi \simeq \mathcal{G}_3$ is trivial). Thus we have a factorization [scan 645] [p. 625]

$$M'^{!}_{\dagger} \longrightarrow \mu \underset{\text{centre}}{\hookrightarrow} \mathcal{G}/\Pi_0$$

and we would have to verify that this homomorphism $M'^{!} \longrightarrow \mu$ is none other than ε_3, which is therefore trivial on $M^{\circ!}$ and a fortiori on $M^{\circ!}(0)$. In fact, we recognize that the kernel is not $\mathrm{Ker}(\varepsilon_3 \mid M'^{!})$ (which in fact|fact contains μ;) but $M'^{!}_0$ — we will return to this below. † [p. 644]

Uncertainty retained from cancelled passages.

140-3-s645-D001 $0|a$
140-3-s645-D003 $0|a$
140-3-s645-D005 fact|point find|let us find M

† Admitting this point, we therefore find [scan 646]

$$\mathrm{Ker}(N^*_\rho \longrightarrow \mathcal{G}/\Pi_0) = N^*_\rho \cap M_0^{\dagger\prime\prime!}\dagger$$

and since

$$N^*_\rho \cap M^!_0 = Z^!_{\rho 0} = M'(-\bar{\jmath}),$$

we find that the kernel sought is equal to:

$$M'(-\bar{\jmath}) \cap M'' = M^\circ(-\bar{\jmath}), \quad \text{cqfd.}$$

Thus we do have an injective homomorphism (46), and since the groups [p. 645]

Uncertainty retained from cancelled passages.

140-3-s646-D001 $\wedge|*\quad\wedge$ some

140-3-s646-D002 $\circ$

140-3-s646-D003 a

in question have the same order 24, this homomorphism is indeed an isomorphism, q.e.d. d) [scan 647]
Application to the structure of $\mathcal{M}_{\rho,\sigma}$ and to the definition of a homomorphism † $M \longrightarrow \mathcal{M}_{\rho,\sigma}/\mu_\rho$. [p. 626]
Recall that

$$\mathcal{M}_{\rho,\sigma} \overset{\text{déf}}{=} N^*_\rho \times_{\gamma_{\rho,\sigma}} N^*_\sigma \times_{\gamma_{\rho,\sigma}} G_{\rho,\sigma} \tag{48}$$

where †

$$\begin{cases} G_{\rho,\sigma} \simeq M = \underbrace{M(0)}_{\simeq M^{\sim}_{0,3}(0)} \cdot \mathcal{G}^{+\wedge} \\ \gamma_{\rho,\sigma} \overset{\text{dfn}}{=} G_{\rho,\sigma}/\underbrace{\mathcal{G}^{+\wedge}}_{\mathcal{G}_{\rho,\sigma}} \simeq \tilde{\Gamma}_{\mathbb{Q}}\ (\overset{\sim}{\leftarrow} M(0)), \\ N^*_\rho = \mathrm{Norm}_M(L_\rho), \qquad N^*_\sigma = \mathrm{Norm}_M(L_\sigma) \end{cases} \tag{49}$$

On the other hand, we have introduced the distinguished subgroups

$$\begin{cases} Z^!_{\sigma 0} = M(\frac{1}{2}) \subset N^*_\sigma \qquad \text{(p. 615 (9))} \\ \dagger N^?_\rho = \dagger Z^!_\rho \cdot \mu_{\tau'} \subset N^*_\sigma \qquad \text{(p. 616 (21))}. \end{cases} \tag{50}$$

The first of these groups does not contain μ, the second contains it

$$\begin{cases} Z^!_{\sigma 0} \cap \mu = \{1\} \\ N^?_\rho \dagger \supset \mu \end{cases} \tag{51}$$

and

$$(52) \quad \begin{cases} Z^!_{\sigma 0} \overset{\sim}{\longrightarrow} \gamma_{\rho,\sigma} = \tilde{\Gamma}_{\mathbb{Q}} \\ \dagger \\ 1 \to \mu \to N^?_\rho \to \gamma_{\rho,\sigma} \to 1 \end{cases} \qquad \begin{matrix} \tilde{\Gamma}_{\mathbb{Q}} \\ \| \\ \end{matrix} \qquad \text{exact} \downarrow$$

[p. 646]

Uncertainty retained from cancelled passages.

140-3-s647-D001 $\mathcal{M}$

140-3-s647-D002 $\mathcal{G}$

140-3-s647-D003 A

140-3-s647-D004 ρ

140-3-s647-D005 0

140-3-s647-D006 Ker $= \mu| \longrightarrow \mu$

[scan 648]

Now let

$$(53) \qquad \begin{matrix} \mathcal{M}^{\natural|\sharp}_{\rho,\sigma} & \subset & \mathcal{M}_{\rho,\sigma} \\ \| \text{dfn} & & \\ N^?_\rho \times_{\gamma_{\rho,\sigma}} Z^!_{\sigma 0} \times_{\gamma_{\rho,\sigma}} \underbrace{G_{\rho,\sigma}}_{M} & & \end{matrix}$$

We have a canonical †central embedding

$$(54)\qquad \mu\times\mu\times\mu \hookrightarrow \mathcal{M}_{\rho,\sigma}$$

$$(\epsilon,\epsilon',\epsilon'') \longmapsto (\epsilon,\epsilon',\epsilon'')$$

coming from the central embeddings †

$$\mu \hookrightarrow N_\rho^*,\qquad \mu \hookrightarrow N_\sigma^*,\qquad \mu \hookrightarrow G_{\rho,\sigma}$$

$$\mathcal{G}^+_{\rho,\sigma} = \mathcal{G}^+)$$

and we have

$$\mathcal{M}^{\natural|\sharp}_{\rho,\sigma}\cap(\mu\times\mu\times\mu) = (1\times\mu\times\mu)$$

hence

$$(55)\qquad \mu\times\mu \hookrightarrow \mathcal{M}^{\natural|\sharp}_{\rho,\sigma}.$$

We denote by μ_ρ the central subgroup μ of N_ρ or of $N_\rho^?$, or $\mu\times 1$ in $\mathcal{M}^{\natural|\sharp}_{\rho,\sigma}$. It is then clear from (52), that we have an exact sequence

$$(55)\qquad 1 \longrightarrow \mu_\rho \hookrightarrow \mathcal{M}^{\natural|\sharp}_{\rho,\sigma} \longrightarrow G_{\rho,\sigma} \longrightarrow 1$$

$$\wr$$

$$M$$

whence canonical homomorphisms|homomorphisms [p. 647]

Uncertainty retained from cancelled passages.

140-3-s648-D001 homomorphism

140-3-s648-D003 the

[scan 649]

[p. 627]

$$(56)\qquad M \xrightarrow{\sim} \mathcal{M}^{\natural|\sharp}_{\rho,\sigma}/\mu_\rho \hookrightarrow \mathcal{M}_{\rho,\sigma}/\mu_\rho.$$

m1313

Marginal note. m1313 We therefore set $\Gamma^{\natural|\sharp}_{\rho,\sigma} \overset{\text{dfn}}{=} \mathcal{M}^{\natural|\sharp}_{\rho,\sigma}/\mathcal{G}^+_{\rho,\sigma} \subset \Gamma_{\rho,\sigma} = \mathcal{M}_{\rho,\sigma}/\mathcal{G}^+_{\rho,\sigma}$.

Obviously the subgroup $\mathcal{M}^{\natural|\sharp}_{\rho,\sigma}$ of $\mathcal{M}_{\rho,\sigma}$ contains $\mathcal{G}_{\rho,\sigma} = SG_{\rho,\sigma}\simeq\mathcal{G}^{+\wedge}$, in fact we have a diagram of exact sequences

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \mu_\rho\times\mathcal{G}^\dagger_{\rho,\sigma} & \longrightarrow & \mathcal{M}^{\natural|\sharp}_{\rho,\sigma} & \longrightarrow & \gamma_{\rho,\sigma} & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \| & & \\ (57)\quad 1 & \to & \underbrace{SZ_\rho}_{L_\rho}\times\underbrace{SZ_\sigma}_{L_\sigma}\times\mathcal{G}^+_{\rho,\sigma} & \to & \mathcal{M}_{\rho,\sigma} & \to & \gamma_{\rho,\sigma}|\Gamma_{\rho,\sigma} & \to & 1 \end{array}$$

In the isomorphism (56), the subgroup $\mathcal{G}^{+\wedge}$ of M corresponds to the subgroup $\mu_\rho\times\mathcal{G}^+_{\rho,\sigma}/\mu_{\sigma|\rho}\simeq\mathcal{G}_{\rho,\sigma}$, it maps isomorphically onto the subgroup $\mathcal{G}_{\rho,\sigma}$ of $\mathcal{M}_{\rho,\sigma}/\mu_\rho$.

We thus find

$$
\begin{array}{ccccccccc}
 & & & & \mathcal{G}^{+}_{\rho,\sigma}|G^{+}_{\rho,\sigma} & & \gamma_{\rho,\sigma} & & \\
 & & & & \wr & & \wr & & \\
1 & \longrightarrow & \mathcal{G}^{+\wedge} & \longrightarrow & M & \longrightarrow & \tilde{\Gamma}_{\mathbb{Q}} & \longrightarrow & 1 \\
 & & \downarrow{\scriptstyle \sim} & & \downarrow{\scriptstyle \sim} & & \wr & & \\
(58)\quad 1 & \longrightarrow & \mathcal{G}^{+}_{\rho,\sigma} & \xrightarrow{\sim} & \mathcal{M}^{\natural|\sharp}_{\rho,\sigma}/\mu_\rho & \longrightarrow & \Gamma^{\natural|\sharp}_{\rho,\sigma}/\mu_\rho & \longrightarrow & 1 \\
 & & \downarrow{\scriptstyle 1} & & \downarrow & & \downarrow & & \\
1 & \longrightarrow & \mathcal{G}^{+}_{\rho,\sigma} & \longrightarrow & \mathcal{M}_{\rho,\sigma}/\mu_\rho & \longrightarrow & \Gamma_{\rho,\sigma}/\mu_\rho & \longrightarrow & 1
\end{array}
$$

Thus, we find in passing that the canonical homomorphism

$$
\begin{array}{ccccccccc}
(59)\quad 1 & \longrightarrow & \underbrace{SZ_\rho \times SZ_\sigma}_{L_\rho \times L_\sigma} & \longrightarrow & \Gamma_{\rho,\sigma} & \overset{\text{épi}}{\longrightarrow} & \gamma_{\rho,\sigma} & \longrightarrow & 1 \\
 & & & & & & \uparrow{\scriptstyle \sim} & & \\
 & & & & & & \dagger & & \Gamma^{\natural|\sharp}_{\rho,\sigma}
\end{array}
$$

admits a "bisection" (whose intersection with $SZ_\rho \times SZ_\sigma$ is $\mu_\rho \times 1$) [p. 648]

Uncertainty retained from cancelled passages.

140-3-s649-D001 $+$

140-3-s649-D003 $\Gamma^{\natural}_{\rho,\sigma}/\mu_\rho$

[scan 650]

$$
(60)\qquad \begin{cases} \Gamma_{\rho,\sigma} \simeq \Gamma^{\natural|\sharp}_{\rho,\sigma} \cdot \underbrace{S\Gamma_{\rho,\sigma}}_{SZ_\rho \times SZ_\sigma = L_\rho \times L_\sigma} \quad ; \quad \Gamma^{\natural|\sharp}_{\rho,\sigma} \cap S\Gamma_{\rho,\sigma} = \mu_\rho \\ \Gamma_{\rho,\sigma}/\mu_\rho \simeq \left(\underbrace{\Gamma^{\natural|\sharp}_{\rho,\sigma}/\mu_\rho}_{\gamma_{\rho,\sigma} \simeq \tilde{\Gamma}_{\mathbb{Q}}} \right) \cdot \left(L_\rho/\mu_\rho \times L_\sigma \right). \end{cases}
$$

To say what becomes of the remarkable subgroups

$$
M(0),\quad \begin{array}{c} M[0], \\ \| \\ M^{\mathcal{G}}[0] \\ \| \\ M(\omega|0).L^{\mathcal{G}}_0 \end{array} \quad \begin{array}{c} M^!_a[0], \\ \| \\ M^{\Pi}[0] \\ \| \\ M(\omega|0).L^{\Pi}_0 \end{array} \quad \begin{array}{c} M^!_0[0] \\ \| \\ M^{\Pi_0}[0] \\ \| \\ \dagger \\ M(\omega|0)L^{\Pi_0}_0 \end{array}
$$

$$
\begin{array}{c} \{M(\frac{1}{2})\}|M(\frac{1}{2}) \\ \| \\ Z^!_{\sigma 0} \end{array}, \quad M'(-\bar{\jmath}), \quad M'''(-\bar{\jmath}), \quad N^?_\rho
$$

of M under the homomorphism

$$M \xrightarrow{\sim} \mathcal{M}^{\natural|\sharp}_{\rho,\sigma}/\mu_\rho \hookrightarrow \mathcal{M}_{\rho,\sigma},$$

we would first have to introduce the appropriate groups in the general $\mathcal{M}_{\rho,\sigma}$. We are therefore going to return to these groups now. [p. 649]

Uncertainty retained from cancelled passages.

140-3-s650-D001 $M(\omega)L_0|M(0)L_0$

[scan 651]

VI The general groups $\mathcal{M}_{\rho,\sigma}$: recapitulation, refinement. [p. 628]

1°) Data. These are those of V 1°) (page 577). In addition to $\mathcal{G}$ (profinite) and to

$$\mathrm{Gl}(2,\mathbb{Z})^\wedge \longrightarrow \mathcal{G}$$

a continuous surjective homomorphism, i.e. generators ρ, σ, τ (or ρ, σ, τ') of $\mathcal{G}$, we also give ourselves an open subgroup

$$(1) \qquad \mathcal{G}^{\natural|\sharp}_{0|a} \subset \mathcal{G}^+, \quad \text{invariant in } \mathcal{G}^{*|+}.$$

In the universal case $\mathcal{G} = \mathrm{Gl}(2,\mathbb{Z})^\wedge$, the datum of $\mathcal{G}^{\natural|\sharp}_{0|a}$ may correspond to the point of view where we look at the subgroups $M^{\natural|\sharp}$ of M formed|formed of the u which normalize† $\mathcal{G}^{\natural|\sharp}$ and which operate trivially in $\mathcal{G}^+/\mathcal{G}^{\natural|\sharp}$ — this is † an open subgroup, which defines in $\tilde{\Gamma}_{\mathbb{Q}} = M/\mathcal{G}^+$ an open image subgroup $\tilde{\Gamma}^{\natural|\sharp}_{\mathbb{Q}}$, whose representations in suitable profinite groups we may propose to study. An interesting case will be the one where

$$\mathcal{G}^{\natural|\sharp} = \Pi_0, \quad \text{thus } M^{\natural|\sharp} = M''^{!},$$

or

$$\mathcal{G}^{\natural|\sharp} = \Pi, \quad \text{thus } M^{\natural|\sharp} = M^{!}.$$

If

$$\mathcal{G}^{\natural|\sharp} = \mathcal{G}, \quad \text{thus } M^{\natural|\sharp} = M,$$

we recover the preceding constructions. † [p. 650]

Uncertainty retained from cancelled passages.

140-3-s651-D002 ateur of|ent the

140-3-s651-D003 necessarily

140-3-s651-D004 We can

[scan 652]

An interesting case also is the one where $\mathcal{G}$ is the group of adelic integral points of a group scheme over $\mathbb{Z}$ (in this case, it is true that ρ and σ, τ generate perhaps|not then $\mathcal{G}^{+|*}$ only modulo a finite group i.e. they generate perhaps only an open subgroup. In this case, we should rather paraphrase the constructions that follow in the scheme-theoretic case I shall come back to this later on. An interesting choice of a $\mathcal{G}^{\natural|\sharp}$ is then the identity component of $\mathcal{G}$.

I hesitate to introduce notations $Z^{\natural|\sharp}_{\rho,\sigma}, \mathcal{M}^{\natural|\sharp}_{\rho,\sigma}$ etc — so as not to overload notations that are already heavy. So I prefer to use the simpler notations $Z_{\rho,\sigma}$, etc — it being understood that in these constructions, the group $\mathcal{G}^{\natural|\sharp}$ enters — with the option of reintroducing the notation $\natural|\sharp$ if need be. We put

$$(3) \qquad \mathcal{G}^{\natural|\sharp}_\tau = \overbrace{\{1,\tau\}}^{\mu_\tau} . \mathcal{G}^{\natural|\sharp} \subset \mathcal{G}.$$

[p. 651]
[scan 653]
[p. 629]

2°) $Z_{\rho,\sigma},\ \gamma_{\rho,\sigma}$.

Let

$$Z_{\rho,\sigma} \subset \mathrm{Aut}(\mathcal{G}^+) \times \widehat{\mathbb{Z}}^* \times \mu \times \mu$$
$$\| \text{ dfn}$$

$$(4) \qquad \left\{ (u,\mu,\varepsilon_2,\varepsilon_3) \;\middle|\; \begin{array}{l} \dagger \\ a)\ \ \exists \alpha \in \mathcal{G}_\tau^{\natural|\sharp},\ \text{with } \varepsilon_{\mathcal{G}}(\alpha)=\varepsilon_3, \quad u(\rho)=\alpha(\rho) \\ b)\ \ \exists \beta \in \mathcal{G}_\tau^{\natural|\sharp},\ \text{with } \varepsilon_{\mathcal{G}}(\beta)=\varepsilon_2, \quad u(\sigma)=\beta(\sigma) \\ c)\ \ u(\varepsilon_0)=\varepsilon_0^{\mu} \\ d)\ \ u \text{ normalizes } \mathcal{G}^{\natural|\sharp} \text{ and induces the identity on } (\mathcal{G}^+/\mathcal{G}^{\natural|\sharp})/\operatorname{Im}\Pi \end{array} \right\}.$$

m1314

Marginal note. m1314 cf. page 635 continuation † this condition.

†.

Proposition. $Z_{\rho,\sigma}$ is a subgroup of the product group.

Proof. The conditions c), d) define obviously each subgroup. Given that

$$\tau(\sigma) = -\sigma|\varepsilon,$$

condition b) also means that

$$(5^*) \qquad b') \quad u(\sigma) \text{ is } \mathcal{G}^{\natural|\sharp}\text{-conjugate to : } \ \dagger\sigma^{\varepsilon_2}(\dagger = \varepsilon_2\sigma)$$

and in this form it is clear that condition c') defines a subgroup. I do not think that condition a) by itself defines a subgroup. But if $\mathfrak{g}_{(b)}|G_{(b)}$ † denotes the subgroup of the product defined by these conditions †b), we have to show that $\mathfrak{g}|G_{\dagger(a,b)}$ is a subgroup of the $\mathfrak{g}|G_{\dagger(b)}$ — which will prove the proposition. Let $\underline{u},\underline{u}' \in \mathfrak{g}_{(b)}|G_{(b)}$, corresponding to $(u,\mu,\varepsilon_2,\varepsilon_3)$ and $(u',\mu',\varepsilon_2',\varepsilon_3')$, and to $\alpha,\alpha' \in \mathcal{G}_\tau^{\natural|\sharp}$, so that we have [p. 652]

Uncertainty retained from cancelled passages.

140-3-s653-D001 u normalizes $\mathcal{G}^{\natural}$ and induces the identity in $\mathcal{G}^+/\mathcal{G}^{\natural}$

140-3-s653-D002 a weaker condition

140-3-s653-D003 added in principle consequence in which case $\natural|\sharp$ induced by where

140-3-s653-D006 ε_2

140-3-s653-D007 $\varepsilon_2\sigma$

140-3-s653-D008 $\mathfrak{g}_{(a,b)}$

140-3-s653-D009 a)

140-3-s653-D010 (a,b,c)

140-3-s653-D012 (a,b)

[scan 654]

$$(6) \qquad u(\rho)=\alpha(\rho), \quad u'(\rho)=\alpha'(\rho), \quad \text{with} \quad \begin{array}{l} \varepsilon_{\mathcal{G}}(\alpha)=\varepsilon_3, \\ \alpha_{\mathcal{G}}(\alpha')=\varepsilon_3'. \end{array}$$

Let us prove † that $u'u \in \mathfrak{g}_{(a)(b)}|G_{(a,b)}$ — i.e. that it also satisfies (b)|(a). If $\varepsilon_3 = 1$ i.e. $\alpha \in \mathcal{G}^{\natural|\sharp} \subset \mathcal{G}^+$, $u'(\alpha)$ is defined and we have

$$\begin{aligned} (u'u)(\rho) &= u'(u(\rho)) = u'(\alpha(\rho)) = u'(\alpha)(u'(\rho)) = u'(\alpha)(\alpha'(\rho)) \\ &= (u'(\alpha)\alpha')(\rho) = \alpha''(\rho), \end{aligned}$$

with

$$(7) \qquad \left. \begin{array}{l} \alpha'' = u'(\alpha)\alpha' \in \mathcal{G}_\tau^{\natural|\sharp} \\ \varepsilon_{\mathcal{G}}(\alpha'') = \varepsilon_{\mathcal{G}}(\alpha') = \varepsilon_3' = \varepsilon_3\varepsilon_3' \end{array} \right| \text{Case } \varepsilon_3 = 1$$

(since $u'(\alpha) \in \mathcal{G}^{\natural|\sharp}$ i.e. $\varepsilon_{\mathcal{G}}(u'(\alpha)) = 1$, and $\varepsilon_3 = 1$.

In this case we win. If $\varepsilon_3 = -1$, we write

$$\alpha = \alpha_0 \tau \qquad \alpha_0 \in \mathcal{G}^{\natural|\sharp},$$

and we have

$$u(\rho) = (\alpha_0\tau)(\rho) = \alpha_0(\tau(\rho)) = \alpha_0(\sigma^{-1}(\rho^{-1})) = (\alpha_0\sigma)(\rho^{-1}),$$

where now $\alpha_0\sigma \in \mathcal{G}^+$, hence $u'(\alpha_0\sigma)$ is defined, and we will have

$$(u'u)(\rho) = u'(u(\rho)) = u'(\alpha_0\sigma)(\underbrace{u'(\rho^{-1})}_{\alpha'(\rho^{-1})}) = (u'(\alpha_0\sigma)\alpha')(\rho^{-1})$$

and since

$$\dagger \qquad \rho^{-1} = \sigma\tau(\rho)$$

we find

$$(8) \qquad (u'u)(\rho) = \dagger\alpha''(\rho)$$

with

$$\begin{cases} \alpha'' = \varepsilon \overbrace{u'(\alpha_0)}^{\in \mathcal{G}^{\natural|\sharp}} \overbrace{u'(\sigma)}^{\in \mathcal{G}^{+}} \overbrace{\alpha'}^{\in \mathcal{G}_\tau^{\natural|\sharp}} \underset{\dagger}{\sigma\tau} & \text{where} \\ \varepsilon \in \mu = \{\pm 1\} \hookrightarrow \,|\, \Rightarrow \text{Centre}\, \mathcal{G}^+ \end{cases}$$

$\dagger$ is chosen so that α'' (if possible) is in $\mathcal{G}_\tau^{\natural|\sharp}$. Let us note that all the factors of this product are in $\mathcal{G}_\tau^{\natural}\mathcal{G}^+|\mathcal{G}_\tau^{\natural}/\mathcal{G}^+$, in any case $\alpha'' \in \mathcal{G}$. [p. 653]

Uncertainty retained from cancelled passages.

140-3-s654-D001 ε_3
140-3-s654-D002 ρ^{-1}
140-3-s654-D003 $u'(\alpha_0\sigma)\alpha'\sigma\tau$
140-3-s654-D004 τ'
140-3-s654-D005 is

[scan 655]
[p. 630]

Let us reduce modulo $\mathcal{G}^{\natural|\sharp}$, the relation $\alpha'' \in \mathcal{G}_\tau^{\natural|\sharp}$ reads

$$\dot\alpha'' \in \{1, \dot\tau\} \quad \text{i.e.} \quad \dot\varepsilon\, u'(\sigma)^{\cdot}\dot\alpha'\dot\sigma\dot\tau \in \{1, \dot\tau\} \quad \text{i.e.} \quad \dot\varepsilon\, u'(\sigma)^{\cdot}\dot\alpha'\dot\sigma \in \{1, \dot\tau\}.$$

$\dagger$

We distinguish again two cases

1°) $\varepsilon_3' = 1$ i.e. $\dot\alpha' = 1$ $\dagger$ the relation reads

$$(*) \qquad \dot\varepsilon\, u'(\sigma)^{\cdot}\dot\sigma = 1$$

2°) $\varepsilon_3' = -1$ i.e. $\dot\alpha' = \dot\tau$, since $\tau\sigma = -\sigma\tau$, the relation reads

$$(**) \qquad -\dot\varepsilon\, u'(\sigma)^{\cdot}\dot\sigma = 1,$$

thus in all cases, the relation reads

$$(9) \qquad \begin{cases} u'(\sigma)^{\cdot} = (\varepsilon\varepsilon_3')^{\cdot}\dot\sigma^{-1} & \dagger \text{ i.e. again} \\ u'(\sigma)^{\cdot} = (-\varepsilon\varepsilon_3')^{\cdot}\dot\sigma \end{cases}$$

Now suppose now that we have b) for u', which $\dagger$implies

$$u'(\sigma)^{\cdot} = \varepsilon_2' \dagger\dot\sigma,$$

we see therefore that it suffices to choose ε so that $-\varepsilon\varepsilon'_3 = \varepsilon'_2$ i.e.

$$\varepsilon = -\varepsilon'_2\varepsilon'_3$$

thus we will have

$$\dagger \qquad \alpha'' = -\varepsilon'_2\varepsilon'_3\, u'(\underbrace{\alpha_0 \dagger \sigma}_{\alpha\tau\sigma=\alpha\tau'})\alpha' \underbrace{\sigma\tau}_{-\tau'}$$

i.e.

$$(10) \qquad \alpha'' = \varepsilon'_2\varepsilon'_3\, u'(\alpha\tau')(\alpha'\tau') \quad \Big| \quad \text{Case } \varepsilon_3 = -1$$

a formula which makes sense, since $\alpha\tau' \in \mathcal{G}^+$ hence $u'(\alpha\tau')$ is defined, and the preceding argument proves that the element in question †satisfies

$$(11) \qquad \alpha'' \in \mathcal{G}_\tau^{\natural|\sharp}, \quad \varepsilon_{\mathcal{G}}(\alpha'') = \underbrace{\varepsilon_{\mathcal{G}}(\alpha)}_{-1}\varepsilon_{\mathcal{G}}(\alpha')$$

by distinguishing cases (*) and (**) of (9). Same conclusion in the first case where $\varepsilon_{\mathcal{G}}(\alpha) = 1$ (7). This completes the proof. [p. 654]

Uncertainty retained from cancelled passages.

140-3-s655-D001 we would have
140-3-s655-D002 $\dot\varepsilon\, u'(\sigma)\dot\alpha'\dot\tau$
140-3-s655-D003 or
140-3-s655-D005 means
140-3-s655-D006 [illeg. 1]
140-3-s655-D007 (10)
140-3-s655-D008 τ
140-3-s655-D009 is in

[scan 656]

Remarks †

If $\mathcal{G}$ is "big enough", the projection homomorphism is injective

(12)

$$Z_{\rho,\sigma} \hookrightarrow \mathrm{Aut}(\mathcal{G}^+)$$

(cf. page 578)

Examples in the universal case

a) $\mathcal{G}^{\natural|\sharp} = \mathcal{G}^+ = \dagger\mathrm{Sl}(2,\mathbb{Z})^\wedge$. Condition|condition (4) d) is empty, conditions a), b) are equivalent respectively to the fact that $u(\rho)$ conjugate to ρ^{ε_3}, $u(\sigma)$ conjugate to σ^{ε_2}, i.e. comes back to definition (6) page 578, which gives

$$\dagger$$

$$(13) \qquad Z_{\rho,\sigma} = M(0).L_0^{\mathcal{G}} = M^{\mathcal{G}}[0] = M[0]$$

m1316

Marginal note. m1316 N.B. A priori we rather have $Z_{\rho,\sigma} \subset \widetilde{M}_{0,3}|M_{0,3}$, but

$$\begin{array}{ccc} M[0] & \xrightarrow{\sim} & \widetilde{M}_{0,3}[0] \\ \| & & \| \\ M(\omega|0).L_0^{\mathcal{G}} & \xrightarrow{\sim} & M_{0,3}(0).L_0^{\mathcal{G}_{0,3}} \end{array}$$

b) $\mathcal{G}^{\natural|\sharp} = \Pi = \Pi_0 \times \mu$. We now find

$$(14) \qquad Z_{\rho,\sigma} = M(0).L_0^{\Pi} = M^{\Pi}[0] = M^{!}[0]|M'^{!}[0]$$

(denoted $Z'_{\rho,\sigma}$ in V number° 8).

m1317

Marginal note. m1317 In this case, it is immediate that $Z_{\rho,\sigma}$ operates trivial[ly] even on $\mathcal{G}^{+}/\mathcal{G}_{\tau}^{\natural|\sharp}$, for this is so for the †.

m1318

Marginal note. m1318 $M^{!}[0]|M'^{!}[0] = M^{\Pi}[0]$.

m1319

Marginal note. m1319 make explicit again why (i.e. why) that $(\varepsilon_0, 1, 1, 1) \notin Z_{\rho,\sigma}$

c) $\mathcal{G}^{\natural|\sharp} = \Pi_0$. Conditions a) and b) of (4) have not changed with respect to|with respect to the †preceding case, since $\Pi = \Pi_0 \times \mu$ and μ is central, on the other hand the [p. 655]

Uncertainty retained from cancelled passages.

140-3-s656-D001 multiplicative all $\natural|\sharp$ $\natural|\sharp$

140-3-s656-D003 $\mathcal{G}$

140-3-s656-D005 $\omega|0$

140-3-s656-D006 N.B. all the subgroups contained by $\mathcal{G}^{\natural} = M(0)_2.L_0^{\Pi}|\mathcal{G}^{\natural} = M(\omega)_2.L_0^{\Pi}$

140-3-s656-D007 before

[scan 657]

condition d) becomes distinctly stricter, we find for example that $Z_{\rho,\sigma} \cap \mathcal{G}^{+} = L_0^{\Pi_0}$ (of index 2 in the corresponding group L_0^{Π} of the preceding case), and since the operation of $M(0) \subset M^{!}$ on $\mathcal{G}^{+}/\Pi_0$ is through the character $\varepsilon_2^{*|\#}$, we see that we now have [p. 631]

$$(15) \qquad Z_{\rho,\sigma} = M''(0).L_0^{\Pi_0} = M''[0]_0$$

$$\begin{cases} M''(0) = \mathrm{Ker}(M(0) \xrightarrow{\varepsilon_2} \mu) \\ M''[0]_0 = \mathrm{Ker}(M[0]_0 \xrightarrow{\varepsilon_2} \mu) \\ M[0]_0 = M(\omega|0).L_0^{\Pi_0} \end{cases}$$

m1322 †

†

Marginal note. m1322 This remark about a $\mathcal{G}_0^{\natural|\sharp}$ now becomes useless.

These two latter examples suggest the advisability, in the general case where we would have a subgroup † $\mathcal{G}_0^{\natural|\sharp}$ of $\mathcal{G}^{\natural|\sharp}$, invariant in $\mathcal{G}$, such that we have

$$(16) \qquad \mathcal{G}^{\natural|\sharp} = \mathcal{G}_0^{\natural|\sharp} \times \mu \quad \dagger$$

(here μ is the universal central subgroup $\mu = \{1, \rho^3\} = \{1, \sigma^2\}$) of defining $Z_{\rho,\sigma}$ in terms of $\mathcal{G}^{\natural|\sharp}$, and of choosing the α, β of definition (4) in †

$$(17) \qquad \mathcal{G}_{0\tau}^{\natural|\sharp} \overset{\mathrm{dfn}}{=} \mu_{\tau}.\mathcal{G}_0^{\natural|\sharp}.$$

Remark We are tempted to replace definition (4) (for a) and b)) by the apparently simpler definition of page 578, which makes it evident that we have a subgroup. But they † do not give, even if $\mathcal{G}^{\natural|\sharp} \subset \Pi$ (universal case) † a trivial action of † Z on $\mathcal{G}^{+}/\Pi$, as we † see immediately, † [p. 656]

Uncertainty retained from cancelled passages.

140-3-s657-D001 induce|produce $\natural|\sharp$ In the new version $\natural|\sharp$ $\natural|\sharp$ $M[0]_0^!|M[0]_0$ $\omega|0$ too a little too (old version)

140-3-s657-D002 introduced $M^!{}'[0]_0|M'^!{}[0]_0$ $0|\omega$ $L_0^{\Pi_0}|\Pi_0$

140-3-s657-D003 Z

140-3-s657-D004 $\subset\,|\,\Leftarrow$

140-3-s657-D005 $\mathcal{G}_{0\tau}^{\natural}$

140-3-s657-D006 are compatible

140-3-s657-D007 if we suppose

140-3-s657-D008 $\natural|\sharp$

140-3-s657-D009 see

140-3-s657-D010 ρ, σ

[scan 658]

except by restricting ourselves to some, $(u, \mu, \varepsilon_2, \varepsilon_3)$ with $\varepsilon_3 = 1$, since $\dot\rho^{-1}$ is not equal to $\dot\rho$! It is true (in the universal case for example) that $u(\rho)$ is conjugate in $\mathcal{G}^+$ to ρ^{ε_3}, but in general it will not be so by an element (say) of Π.

We set

$$(18)\qquad L_0^{\natural|\sharp} = \dagger\{l \in L_0^{\mathcal{G}} \mid (\mathrm{int}(l), 1, 1, 1) \in Z_{\rho,\sigma}\}$$

$$\big(\text{where } L_0^{\mathcal{G}} \subset \mathcal{G}^+ \text{ is the set of the } \varepsilon_0^n,\ n \in \widehat{\mathbb{Z}}\big)$$

m1323

Marginal note. m1323 N.B. We have $L_0^{\natural|\sharp} \supset L_0 \cap \mathcal{G}^{\natural|\sharp}$, and the inclusion can be strict, cf. example c) above…where $L_0^{\natural|\sharp} = L_0^{\Pi}$, $L_0 \cap \mathcal{G}^{\natural|\sharp} = L_0^{\Pi_0}$. which is surprising.

and we have a canonical homomorphism

$$(19)\qquad L_0^{\natural|\sharp} \xrightarrow{\ i\ } Z_{\rho,\sigma},$$

$$l \longmapsto (\mathrm{int}_{\mathcal{G}^+}(l), 1, 1, 1)$$

injective iff we have

$$(20)\qquad L_0^{\natural|\sharp} \cap \mathrm{Centr}(\mathcal{G}^+) = \{1\}$$

the image invariant. We set

$$(21)\qquad \gamma_{\rho,\sigma} = Z_{\rho,\sigma}/L_0^{\natural|\sharp},$$

whence

$$(22)\quad 1 \longrightarrow L_0^{\natural|\sharp} \longrightarrow Z_{\rho,\sigma} \xrightarrow{\delta_Z} \gamma_{\rho,\sigma} \longrightarrow 1,$$

$$(23)\quad Z_{\rho,\sigma} \longrightarrow \gamma_{\rho,\sigma} \begin{cases} \xrightarrow{\varepsilon_{3\gamma}} \mu \\ \xrightarrow{\chi_\gamma} \widehat{\mathbb{Z}}^* \\ \xrightarrow{\varepsilon_{2\gamma}} \mu \end{cases}$$

) deduced from the

$$(u,\mu,\varepsilon_2,\varepsilon_3) \begin{cases} \mapsto \varepsilon_3 \\ \mapsto \mu \\ \mapsto \varepsilon_2. \end{cases}$$

† [p. 657]

Uncertainty retained from cancelled passages.

140-3-s658-D002 L_0 $\natural|\sharp$

140-3-s658-D003 $\varepsilon_{2,\sigma}|\varepsilon_{2\gamma}$)

[scan 659]

We set [p. 632]

$$(24) \qquad \begin{cases} \gamma'_{\rho,\sigma} = \operatorname{Ker} \varepsilon_{3\gamma}, \\ \gamma''_{\rho,\sigma} = \operatorname{Ker} \varepsilon_{2\gamma}, \\ \gamma'''_{\rho,\sigma} = \operatorname{Ker} \varepsilon_{2\gamma}\varepsilon_{3\gamma},\ \gamma^\circ_{\rho,\sigma} = \gamma' \cap \gamma'' = \gamma'' \cap \gamma''' = \gamma''' \cap \gamma' \\ Z'_{\rho,\sigma}, Z''_{\rho,\sigma}, Z'''_{\rho,\sigma}, Z^\circ_{\rho,\sigma} \text{ are the inverse images of the } \dagger\gamma'_{\rho,\sigma} \text{ etc} \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \text{in } Z_{\rho,\sigma} \end{cases}$$

These are thus subgroups of index 1, or 2 or four in $\gamma_{\rho,\sigma}$ resp. $Z_{\rho,\sigma}$ — the index 4 being possible only for $\gamma^\circ_{\rho,\sigma}, Z^\circ_{\rho,\sigma}$.

Remarks 1) We thus have

$$(25) \qquad Z^\circ_{\rho,\sigma} \simeq \left\{ (u,\mu) \in \operatorname{Aut}(\mathcal{G}^+) \times \widehat{\mathbb{Z}}^* \;\middle|\; \begin{array}{ll} \text{a)} & u(\rho) \text{ is } \mathcal{G}^{\natural|\sharp}\text{-conjugate to } \rho \\ \text{b)} & u(\sigma) \quad \underline{\qquad\qquad} \quad \sigma \\ \text{c)} & u(\varepsilon_0) = \varepsilon_0^\mu \end{array} \right\},$$

m1324

Marginal note. m1324 More generally,

$$Z'''_{\rho,\sigma} \simeq \left\{ (u,\mu,\varepsilon) \in \operatorname{Aut}(\mathcal{G}^+) \times \widehat{\mathbb{Z}}^* \times \mu \;\middle|\; \begin{array}{ll} \text{a)} & u(\rho)\ \dagger\mathcal{G}^{\natural|\sharp}\text{-conjugate to } \varepsilon[\tau](\rho) \\ \text{b)} & u(\sigma) \text{ is } \mathcal{G}^{\natural|\sharp}\text{-conjugate to } \varepsilon[\tau](\sigma) \\ \text{c)} & u(\varepsilon_0) = \varepsilon_0^\mu \end{array} \right\},$$

(isomorphism of groups). The condition d) of (4) — stability of $\mathcal{G}^{\natural|\sharp}$ under u — is a consequence here of a) b) c), for we will have

$$u(\rho) \equiv \rho \quad (\mathcal{G}^{\natural|\sharp})$$
$$u(\sigma) \equiv \sigma \quad (\mathcal{G}^{\natural|\sharp})$$

which implies

$$(25\text{bis}) \qquad u(x) \equiv x \quad (\mathcal{G}^{\natural|\sharp}) \qquad \forall x \in \mathcal{G}^+,$$

i.e. the stability of $\mathcal{G}^{\natural|\sharp}$ and a trivial action of u on $\mathcal{G}/\mathcal{G}^{\natural|\sharp}$.

2) We also have the remarkable element

$$(26) \qquad \tau_Z \in Z_{\rho,\sigma}, \qquad \tau_Z = (\tau,-1,-1,-1)$$

m1325

Marginal note. m1325 That is one more reason not to require in (4) d) that u operate trivially on $\mathcal{G}^+/\mathcal{G}^{\natural|\sharp}$, which would make me|we would make lose τ !

3°) $S_0G_{\rho,\sigma}, G_{\rho,\sigma}, S_0G^{\natural|\sharp}_{\rho,\sigma}, G^{\natural|\sharp}_{\rho,\sigma}$.
Proceeding as on page 577, we find

$$Z_{\rho,\sigma} \longrightarrow \mathrm{Aut}(\mathcal{G}^+)$$

hence 1/2 direct products

$$(27) \qquad S_0G_{\rho,\sigma} \overset{\text{dfn}}{=} Z_{\rho,\sigma} \,.\, \mathcal{G}^+, \supset S_0G^{\natural|\sharp}_{\rho,\sigma} \overset{\text{dfn}}{=} Z_{\rho,\sigma} \,.\, \mathcal{G}^{\natural|\sharp}$$

and a homomorphism

$$(28) \quad L_0^{\natural|\sharp} \overset{i_{L_0,S_0G}|i_{L_0,G}}{\hookrightarrow} S_0G^{\natural|\sharp}_{\rho,\sigma} \qquad \subset \qquad S_0G_{\rho,\sigma}$$

$$l \longmapsto \dagger\, i_{L_0,Z}(l) \,.\, l^{-1}$$

[p. 658]

Uncertainty retained from cancelled passages.
140-3-s659-D001 Z_{ρ}
140-3-s659-D002 is $\varepsilon[\tau](\rho)$
140-3-s659-D004 $\iota(l)|i(l)$

[scan 660]

with invariant image, which allows us to set

$$(29) \qquad G_{\rho,\sigma} \overset{\text{dfn}}{=} \underbrace{S_0G_{\rho,\sigma}}_{Z_{\rho,\sigma}.\mathcal{G}^+} / \operatorname{Im} L_0^{\natural|\sharp} \,, \supset \; G^{\natural|\sharp}_{\rho,\sigma} = S_0G^{\natural|\sharp}_{\rho,\sigma} / \operatorname{Im} L_0^{\natural}|L_0^{\sharp}$$

hence exact sequences

$$(30) \quad \begin{array}{ccccccccc} 1 & \text{——} & L_0^{\natural|\sharp} & \longrightarrow & S_0G^{\natural|\sharp}_{\rho,\sigma} & \longrightarrow & G^{\natural|\sharp}_{\rho,\sigma} & \text{——} & 1, \\ & & \| & & \downarrow & & \downarrow & & \\ 1 & \text{——} & L_0^{\natural|\sharp} & \text{——} & S_0G_{\rho,\sigma} & \longrightarrow & G_{\rho,\sigma} & \text{——} & 1 \end{array}$$

and a commutative diagram

$$(31) \quad \begin{array}{ccccccc} & & & & Z_{\rho,\sigma} & & \\ & & & i_{L_0,Z} \nearrow & & \searrow i_Z & \\ & & L_0^{\natural|\sharp} & & \dagger & & G^{\natural|\sharp}_{\rho,\sigma} \\ & & & i_{\mathcal{G}^{\natural|\sharp}} & & \dagger & \\ & i_{L_0,\mathcal{G}} \searrow & & \dagger \nearrow & & & G_{\rho,\sigma} \\ & & \mathcal{G}^{\natural|\sharp} & & & & \\ & & & \searrow & & \nearrow i_{\mathcal{G}} & \\ & & & & \mathcal{G}^+ & & \end{array}$$

fitting into a commutative diagram of exact sequences

$$(32)\quad \begin{array}{ccccccccc} 1 & \longrightarrow & L_0^{\natural|\sharp} & \longrightarrow & Z_{\rho,\sigma} & \xrightarrow{\delta_Z} & \gamma_{\rho,\sigma} & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \Vert & & \\ 1 & \longrightarrow & \mathcal{G}^{\natural|\sharp} & \longrightarrow & G_{\rho,\sigma}^{\natural|\sharp} & \xrightarrow{\delta_G} & \gamma_{\rho,\sigma} & \longrightarrow & 1. \\ & & \downarrow & & \downarrow & & \Vert & & \\ 1 & \longrightarrow & \mathcal{G}^{+} & \longrightarrow & G_{\rho,\sigma} & \longrightarrow & \gamma_{\rho,\sigma} & \longrightarrow & 1 \end{array}$$

We consider the composites

$$(33)\quad G_{\rho,\sigma} \xrightarrow{\delta_G} \gamma_{\rho,\sigma} \begin{cases} \xrightarrow{\varepsilon_{2\gamma}} \mu \\ \xrightarrow{\chi_\gamma} \widehat{\mathbb{Z}}^* \\ \xrightarrow{\varepsilon_{3\gamma}} \mu \end{cases}$$

denoted

$$\chi_G = \chi_\gamma \circ \delta_G, \qquad \varepsilon_{2G} = \dagger\varepsilon_{2\gamma}\delta_G, \qquad \varepsilon_{3G} = \delta_G\varepsilon_{3\gamma}.$$

We now have

$$(34)\quad \begin{cases} G'_{\rho,\sigma}, G''_{\rho,\sigma}, G'''_{\rho,\sigma}, G^\circ_{\rho,\sigma} \quad \text{as in (24)} \\ G^+_{\rho,\sigma} = \operatorname{Ker}\chi_G, \quad G'^+_{\rho,\sigma}, G''^+_{\rho,\sigma}, G'''^+_{\rho,\sigma}, G^{\circ+}_{\rho,\sigma} \\ (\gamma^+_{\rho,\sigma} = \operatorname{Ker}\chi_\gamma \quad \text{denoted} \ \ldots\ldots\ ) \end{cases}$$

If

$$(35)\quad \tau_G \in G^{\natural|\sharp}_{\rho,\sigma} \subset G_{\rho,\sigma}, \quad \tau_\gamma \in \gamma_{\rho,\sigma}$$

idem with $G'^{\natural|\sharp}_{\rho,\sigma}$ ...

[p. 659]

Uncertainty retained from cancelled passages.

140-3-s660-D001 $\longrightarrow$
140-3-s660-D002 $G_{\rho,\sigma}$
140-3-s660-D003 $\longrightarrow$
140-3-s660-D004 $\varepsilon_G = \varepsilon_\gamma\delta_G$
140-3-s660-D005 χ

denote the images of τ_Z (26) in|in $G_{\rho,\sigma}, \gamma_{\rho,\sigma}$, we can make the identification

[scan 661]
[p. 633]

$$(36)\quad \begin{cases} \mathcal{G} \simeq \text{inverse image of } \{1, \tau_\gamma\} \subset \gamma_{\rho,\sigma} \\ \quad G_{\rho,\sigma}, \quad \text{under } G_{\rho,\sigma} \longrightarrow \gamma_{\rho,\sigma}, \\ \mathcal{G}^{\natural|\sharp} = \text{inverse image of } \{1, \tau_\gamma\} \subset \gamma_{\rho,\sigma} \text{ in } G^{\natural|\sharp}_{\rho,\sigma}, \end{cases}$$

and

$$(37)\quad \varepsilon_{\mathcal{G}} = \chi_G|_{\mathcal{G}} = \varepsilon_{2G}|_{\mathcal{G}} = \varepsilon_{3G}|_{\mathcal{G}} : \begin{cases} \text{trivial on } \mathcal{G}^+ \\ -1 \text{ on } \tau. \end{cases}$$

Let us consider the canonical map

$$(38)\quad \begin{array}{ccc} G^{\natural|\sharp}_{\rho,\sigma} \hookrightarrow G^{\dagger}_{\rho,\sigma} & \longrightarrow & \operatorname{Aut}(\mathcal{G}^+) \times \widehat{\mathbb{Z}}^* \times \mu \times \mu \\ \underline{u} & \longrightarrow & (\operatorname{int}(\underline{u})|_{\mathcal{G}^+}, \chi_G(u), \varepsilon_{2G}(u), \varepsilon_{3G}(u)) \end{array}$$

Proposition This is a homomorphism of groups: † the image of $G^{\natural|\sharp}_{\rho,\sigma}$ consists of the †quadruplets|quadruples

$$(u,\mu,\varepsilon_2,\varepsilon_3)\in \mathrm{Aut}(\mathcal{G}^+)\times\widehat{\mathbb{Z}}^*\times\mu\times\mu$$

satisfying the conditions

$$(39)\qquad \begin{cases} \text{a)} & u(\rho) \text{ is } \mathcal{G}^{\natural|\sharp}_{\tau}\text{-conjugate to } \varepsilon_3[\tau]\rho \\ \text{b)} & u(\sigma) \quad \text{---------------} \quad \varepsilon_2|\varepsilon_\tau[\tau]\sigma \\ \text{c)} & u(\varepsilon_0) \text{ is } \mathcal{G}^{\natural|\sharp}\text{-conjugate to } \varepsilon_0^{\mu} \\ \text{d)} & u \text{ stabilizes } \mathcal{G}^{\natural|\sharp} \quad \text{(automatic in view of a), b) if } \varepsilon_2=\varepsilon_3) \end{cases}$$

Its kernel is

$$(40)\qquad \dagger\left\{ f^{-1}\underline{u}\in G^{\natural|\sharp}_{\rho,\sigma} \;\middle|\; \begin{array}{l} \dagger\underline{u}=(u,1,1,1)\in Z^{\circ+}_{\rho,\sigma} \\ f\in\mathcal{G}^{\natural|\sharp} \end{array} \qquad \left\{ u=\mathrm{int}(f). \right\}.\right.$$

†

Corollary Let

$$(41)\qquad Z_0=\mathrm{Centr}_{\mathcal{G}^+}(L_0),\qquad Z_0^{\natural|\sharp}=Z_0\cap\mathcal{G}^{\natural|\sharp}$$
$$\left(=\{f\in\mathcal{G}^{\natural|\sharp}\mid(\mathrm{int}(f),1,1,1)\in Z^{\circ+}_{\rho,\sigma}|Z_{\rho,\sigma}\}\right)$$

so that $L_0^{\natural|\sharp}$ is identified with a subgroup [p. 660]

Uncertainty retained from cancelled passages.

140-3-s661-D002 $\natural|\sharp$
140-3-s661-D003 $G_{\rho,\sigma}$
140-3-s661-D005 pairs
140-3-s661-D008 $f^{-1}\underline{u}$
140-3-s661-D009 f
140-3-s661-D010 $E|\mathcal{E}$ Ker Z_0 $\natural|\sharp$ $\natural|\sharp$ } $\natural|\sharp$ $\mathcal{G}^{\natural}|\mathcal{G}$)

central in $Z_0^{\natural|\sharp}$. Let us consider the map [scan 662]

$$Z_0^{\natural|\sharp}\longrightarrow S_0G^{\natural|\sharp}_{\rho,\sigma}=Z_{\rho,\sigma}\cdot\mathcal{G}^{\natural|\sharp}$$

$$(42)\qquad \begin{array}{ccccc} f\mapsto f^{-1} & \cdot\underline{u}(f) & = \underline{u}(f)\cdot f^{-1} & \text{where } \underline{u}(f)=(\mathrm{int}(f),1,1,1)\in Z_{\rho,\sigma} \\ \cap & \cap & & \cap \\ \mathcal{G}^{\natural|\sharp} & Z_{\rho,\sigma} & & G^{\natural|\sharp}_{\rho,\sigma} \end{array}$$

This map is a homomorphism of groups, †it † † induces on $L_0^{\natural|\sharp}\subset Z_0^{\natural|\sharp}$ the canonical homomorphism $i_{L_0,SG}|i_{L_0,S_0G}$ (28), and $L_0^{\natural|\sharp}\subset Z_0^{\natural|\sharp}$ is the inverse image of

$$i_{L_0,SG}|i_{L_0,S_0G}(L_0^{\natural|\sharp})=\mathrm{Ker}(S_0G^{\natural|\sharp}_{\rho,\sigma}\longrightarrow G^{\natural|\sharp}_{\rho,\sigma}).$$

We thus find an injective homomorphism

$$(43)\qquad Z_0^{\natural|\sharp}/L_0^{\natural|\sharp}\hookrightarrow G^{\natural|\sharp}_{\rho,\sigma}$$

whose image is the kernel of (38), i.e. we have a canonical exact sequence

$$(44)\qquad \begin{array}{ccccccccc} 1 & \to & L_0^{\natural|\sharp} & \to & Z_0^{\natural|\sharp} & \to & G^{\natural|\sharp}_{\rho,\sigma} & \longrightarrow & \widetilde{G}^{\natural|\sharp}_{\rho,\sigma} & \longrightarrow & 1 \\ & & & & & & & & \cap & & \\ & & & & & & & & \mathrm{Aut}(\mathcal{G}^+)\times\widehat{\mathbb{Z}}^*\times\mu\times\mu & & \end{array}$$

where $\widetilde{G}^{\natural|\sharp}_{\rho,\sigma}$ consists of the $(u,\mu,\varepsilon_2,\varepsilon_3)$ satisfying the conditions (39).

Remark. This proposition brings out to what extent the † construction made here of $G^{\natural|\sharp}_{\rho,\sigma}$ is finer than the one where we would define|defined it simply as a group of automorphisms of $\mathcal{G}^+$ (we even have|of them need of the $\mu,\varepsilon,\varepsilon'$ into the bargain…). [p. 661]

Uncertainty retained from cancelled passages.

140-3-s662-D001 whose|definition

140-3-s662-D002 $Z_0|L_0$ L_0

140-3-s662-D004 induces|injective by projection|passage to the quotient $\natural|\sharp$

140-3-s662-D006 [illeg. 1]

[scan 663]

We are also going to consider the homomorphism (in fact actually brought into play by (38), [p. 634] even in cases not at all degenerate such as $\mathcal{G}^+ = \mathrm{SL}(2,\widehat{\mathbb{Z}})$)

$$(45)\qquad \begin{array}{ccc} G^{\natural|\sharp}_{\rho,\sigma} & \longrightarrow & \mathrm{Aut}(\mathcal{G}^+)\times\mu\times\mu \\ \underline{u} & \longmapsto & (\mathrm{int}(\underline{u})|\mathcal{G}^+, \varepsilon_3(\underline{u}), \varepsilon_2(\underline{u})) \end{array}$$

induced by (38), its image consists of the systems $(u,\varepsilon_3,\varepsilon_2)$ such that

$$(46)\qquad \left\{\begin{array}{ll} \text{a)} & u(\rho) \text{ is } \mathcal{G}^{\dagger\natural|\sharp}\text{-conjugate to } \varepsilon_3[\tau](\rho) \\ \text{b)} & u(\sigma) \quad \text{---------------} \quad \varepsilon_2[\tau](\sigma) \\ \text{c)} & u(L_0^{\mathcal{G}}) \text{ is } \mathcal{G}^{\natural|\sharp}\text{-conjugate to } L_0 \\ \text{d)} & u \text{ stabilizes } \mathcal{G}^{\natural|\sharp} \quad (\text{automatic if } \varepsilon_3=\varepsilon_2, \text{ in view of a) b)|in view of c) b)}) \end{array}\right.$$

Its kernel is

$$(47)\qquad \left\{ f^{-1}\underline{u} \in G^{\natural|\sharp}_{\rho,\sigma} \;\middle|\; \begin{array}{l} \underline{u}=(u,\mu,1,1)\in Z^{\circ}_{\rho,\sigma} \\ f\in\mathcal{G}^{\natural|\sharp} \end{array} \quad \text{such that} \quad u=\mathrm{int}(f) \right\}.$$

To appreciate the structure of the kernel, let

$$(48)\qquad \begin{array}{l} SN_0^{\dagger} = \mathrm{Norm}_{\mathcal{G}^+}(L_0), \qquad SN_0^{\dagger\natural|\sharp} = SN_0\cap\mathcal{G}^{\natural|\sharp} \\ \qquad\qquad = \left\{ f\in\mathcal{G}^{\natural|\sharp} \;\middle|\; \exists\mu\in\widehat{\mathbb{Z}}^* \text{ such that } \underbrace{(\mathrm{int}\, f,\mu,1,1)}_{\in Z_{\rho,\sigma}} \right\} \end{array}$$

and let

$$(49)\qquad \begin{array}{c} S\widetilde{N}_0\subset SN_0\times\widehat{\mathbb{Z}}^*, \qquad S\widetilde{N}_0^{\natural|\sharp}|\widetilde{N}_0^{\natural|\sharp} = S\widetilde{N}_0\cap(N_0^{\natural|\sharp}\times\widehat{\mathbb{Z}}^*) \\ \| \\ \{(u,\mu)\mid u(\varepsilon_0)=\varepsilon_0^{\mu}\} \end{array}$$

— thus we have $\widetilde{N}_0^{\natural|\sharp}$

$$(50)\qquad \begin{array}{ccc} S\widetilde{N}_0^{\natural|\sharp} & \xrightarrow{\sim} & SN_0^{\natural|\sharp} \\ (u,\mu) & \longmapsto & u \end{array} \qquad \text{if} \qquad \begin{array}{ccc} \widehat{\mathbb{Z}} & \longrightarrow & L_0^{\mathcal{G}} \\ n & \longmapsto & \varepsilon_0^n \end{array} \quad \text{is injective.}$$

idem for $S\widetilde{N}_0$.

We have a canonical injective homomorphism

$$(51)\qquad \begin{aligned} S\widetilde{N}_0^{\natural|\sharp} &\longrightarrow S_0\widetilde{G}_{\rho,\sigma}^{\natural|\sharp} = Z_{\rho,\sigma}\cdot\mathcal{G}^+ \\ (f,\mu) &\longmapsto f^{-1}\cdot\underbrace{(\mathrm{int}(f),\mu,1,1)}_{u} = u\cdot f^{-1} \end{aligned}$$

662

Uncertainty retained from cancelled passages.

140-3-s663-D001 +

140-3-s663-D003 $\mathcal{G}$

140-3-s663-D004 $\mathcal{G}$

[scan 664]

which induces on $L_0^{\natural|\sharp}$ the canonical homomorphism (28) (and which extends (42)) — it is still true that the inverse image † of † $i_{L_0,G}(L_0^{\natural|\sharp})$ is $L_0^{\natural|\sharp} \subset S\widetilde{N}_0^{\natural|\sharp}$, and we thus find a canonical exact sequence.

$$(52|51)\qquad 1 \longrightarrow L_0^{\natural|\sharp} \to S\widetilde{N}_0^{\natural|\sharp} \to G_{\rho,\sigma}^{\natural|\sharp} \longrightarrow \widetilde{\widetilde{G}}_{\rho,\sigma} \longrightarrow 1$$
$$\Big\uparrow$$
$$\mathrm{Aut}(\mathcal{G}^+)\times\mu\times\mu$$

where † $\widetilde{\widetilde{G}}_{\rho,\sigma}$ consists of the $(u,\varepsilon_2,\varepsilon_3)$ satisfying (†46) — i.e. we have

$$(52\text{ bis})\qquad S\widetilde{N}_0^{\natural|\sharp}/L_0^{\natural|\sharp} \simeq \mathrm{Ker}(G_{\rho,\sigma}^{\natural|\sharp} \longrightarrow \mathrm{Aut}(\mathcal{G}^+)\times\mu\times\mu;).$$

663

Uncertainty retained from cancelled passages.

140-3-s664-D001 of

140-3-s664-D002 $L_0^{\natural}$

140-3-s664-D003 with

140-3-s664-D004 4

[scan 665]

4°) $N_\rho, N_\sigma, Z_\rho, Z_\sigma, N_\rho^{\natural|\sharp}, N_\sigma^{\natural|\sharp}, Z_\rho^{\natural|\sharp}, Z_\sigma^{\natural|\sharp}$. [p. 635]

We put

$$(53)\qquad \begin{cases} N_\rho = \{a\in G_{\rho,\sigma} \mid a(\rho) = \rho^{\varepsilon_3},\ \text{where } \varepsilon_3 = \varepsilon_3(a)\} \\ N_\sigma = \{b\in G_{\rho,\sigma} \mid b(\rho) = \rho^{\varepsilon_2},\ \text{where}|\div\ \varepsilon_2 = \varepsilon_2(b)\} \end{cases}$$

N.B. There is no need after all to drag along|drag along $*$'s by giving a definition of the N_ρ, N_σ which would give bigger groups. (On page 580, 580' I got myself tangled up† on this subject...)

We put

$$(54)\qquad \begin{cases} Z_\rho = \mathrm{Cent}_{G_{\rho,\sigma}}(\rho) = \mathrm{Cent}_{G_{\rho,\sigma}}(L_\rho) = \mathrm{Ker}(N_\rho \xrightarrow{\varepsilon_3} \mu). \\ Z_\sigma = \mathrm{Cent}_{G_{\rho,\sigma}}(\sigma) = \mathrm{Cent}_{G_{\rho,\sigma}}(L_\sigma) = \mathrm{Ker}(N_\sigma \xrightarrow{\varepsilon_2} \mu) \end{cases}$$

$$(55)\qquad \begin{cases} SZ_\rho = Z_\rho\cap\mathcal{G}^+ = \mathrm{Ker}(Z_\rho \xrightarrow{\delta_Z} \gamma_{\rho,\sigma}) \\ \qquad = \mathrm{Cent}_{\mathcal{G}^+}(\rho) = \mathrm{Cent}_{\mathcal{G}^+}(L_\rho) \\ SZ_\sigma = Z_\sigma\cap\mathcal{G}^+ = \mathrm{Ker}(Z_\sigma \xrightarrow{\delta_{Z|z}} \gamma_{\rho,\sigma}) \\ \qquad = \mathrm{Cent}_{\mathcal{G}^+}(\sigma) = \mathrm{Cent}_{\mathcal{G}^+}(L_\sigma) \end{cases},$$

m1331

Marginal note. m1331 we define in the same way SN_ρ, SN_σ, but $SN_\rho = SZ_\rho$, $SN_\sigma = SZ_\sigma$.

whence

$$(56)\qquad \begin{array}{ccccccccc} 1 & \longrightarrow & Z_\rho & \longrightarrow & N_\rho & \xrightarrow{\varepsilon_{3\rho}} & \mu & \longrightarrow & \dagger \\ 1 & \text{——} & Z_\sigma & \text{——} & N_\sigma & \xrightarrow{\varepsilon_{2\sigma}} & \mu & \text{——} & 1 \end{array}$$

m1332

Marginal note. m1332 Exact sequences semidirect by $\mu_{\tau'}$ — the last also semidirect by μ_τ.

(The fact that we can put 1's at the end comes from the fact that $\tau' \in N_\rho \cap N_\sigma$ satisfies

$$(57)\qquad \varepsilon_2(\tau') = \varepsilon_3(\tau') = \dagger\chi(\tau') = -1.$$

We have a marvelous diagram (page 580'). 664

Uncertainty retained from cancelled passages.

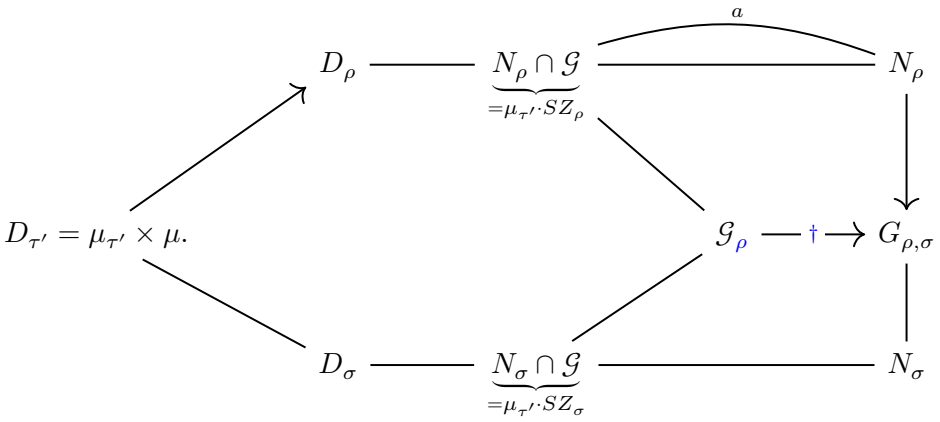
140-3-s665-D001 $\natural$

140-3-s665-D002 split|surjective

140-3-s665-D003 κ

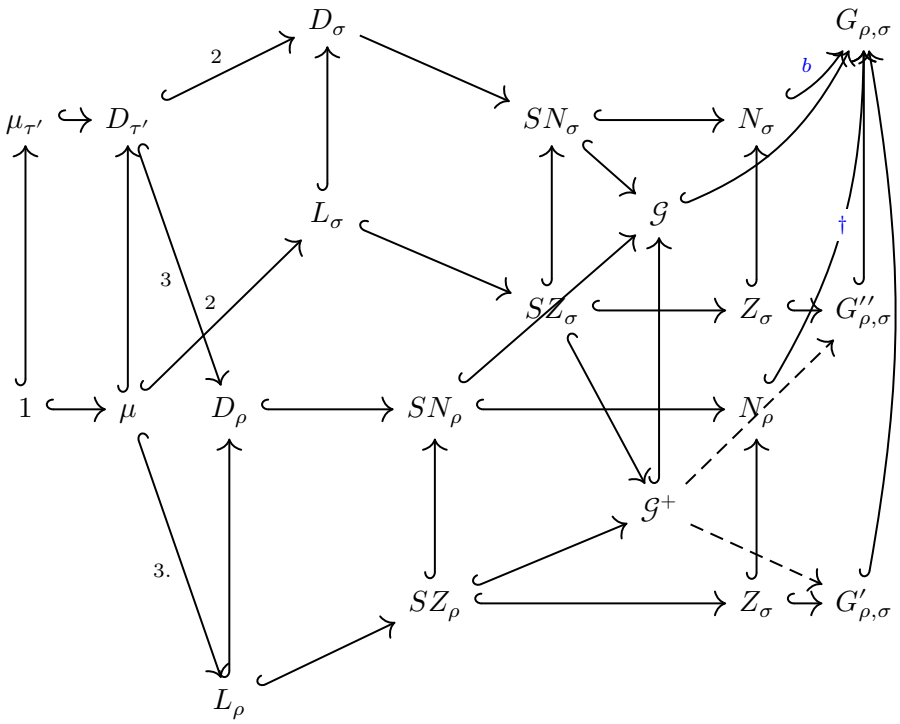
[scan 666]

(58)



or better

(59)



We define:

$$(60)\quad N_\rho^{\natural|\sharp}\dagger \quad N_\sigma^{\natural|\sharp}, Z_\rho^{\natural|\sharp}, Z_\sigma^{\natural|\sharp}, \dagger, SZ_\rho^{\natural|\sharp}, SZ_\sigma^{\natural|\sharp}$$

intersections of the N_ρ, N_σ etc. with $G_{\rho,\sigma}^{\natural|\sharp}$.

We have exact sequences

$$(61)\quad \begin{cases} 1 \longrightarrow SZ|N_\rho^{\natural|\sharp} \longrightarrow N_\rho^{\natural|\sharp} \xrightarrow{\delta_\rho^{\natural|\sharp}} \gamma_{\rho,\sigma} \longrightarrow 1 \\ 1 \longrightarrow SZ_\sigma^{\natural|\sharp} \longrightarrow N_\sigma^{\natural|\sharp} \xrightarrow{\delta_\sigma^{\natural|\sharp}} \gamma_{\rho,\sigma} \longrightarrow 1 \end{cases}$$

and homomorphisms of exact sequences 665

Uncertainty retained from cancelled passages.

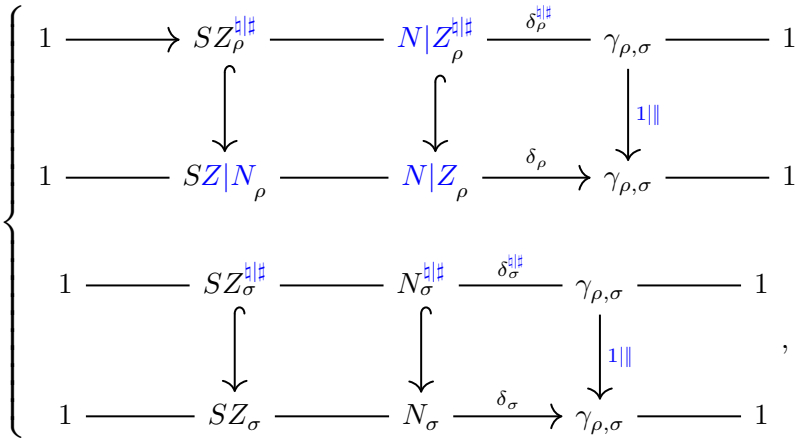
140-3-s666-D001 ↪

140-3-s666-D002 $a_{\rho,\sigma}$

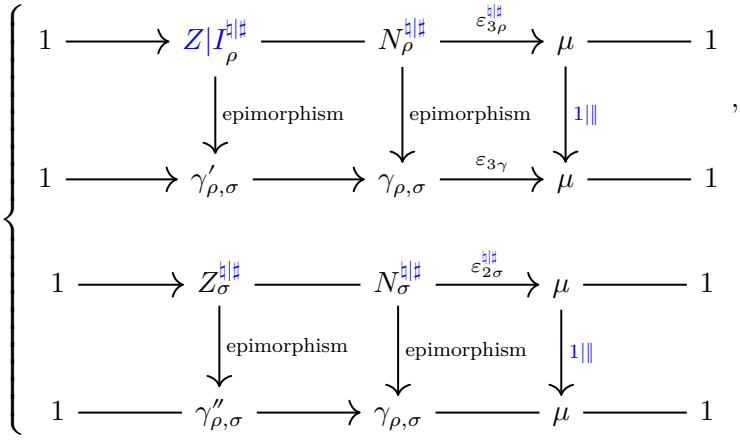
140-3-s666-D003 $= \quad N_\rho \cap G_{\rho,\sigma}^\natural$

140-3-s666-D004 $SN_\rho^\natural \quad SN_\sigma^\natural$

[scan 667]
[p. 636]



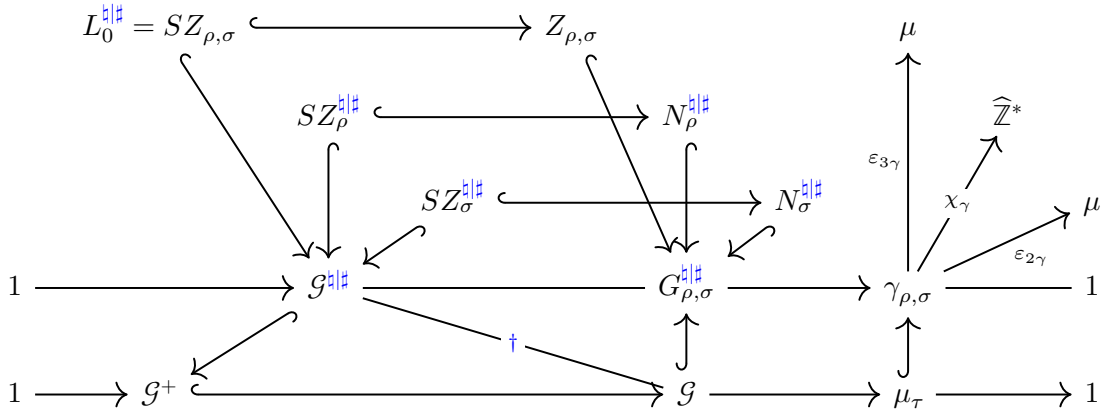
(62)



(63)

(and analogous diagrams † without $\natural|\sharp$).

† We thus have a diagram



(64)

(cf. page 581' (45)), and analogous diagrams without $\natural|\sharp$. 666

Uncertainty retained from cancelled passages.

140-3-s667-D001 with (63)

140-3-s667-D002 hence

140-3-s667-D003 ↪

[scan 668]

†

I would now like to define the analogue of the subgroups $M|\mathcal{M}(\frac{1}{2})$, $M|\mathcal{M}'(-\bar{j})$, $M|\mathcal{M}'''(-\bar{j})|M|\mathcal{M}''(-\bar{j})$, $N_\rho{}^?$ on page 615 ff, which implies that we have a group $\subset G^{\natural|\sharp}_{\rho,\sigma}$ playing the role of $M|\mathcal{M}^!_0$ — the ideal would be for it to be $G^{\natural|\sharp}_{\rho,\sigma}$ itself. But in the universal case $\mathcal{G}^+ = \mathrm{SL}|\mathrm{GL}(2,\mathbb{Z})^\wedge$, $\mathcal{G}^{\natural|\sharp} = \Pi_0$, hence $Z_{\rho,\sigma} = M|\mathcal{M}[0]^! = M|\mathcal{M}(0|\infty).L^\pi_0$, $L^{\natural|\sharp}_0 = L^\pi_0$ (we find $G_{\rho,\sigma} \simeq M|\mathcal{M}$, † $G^{\natural|\sharp}_{\rho,\sigma}$ = subgroup of $G_{\rho,\sigma} = M|\mathcal{M}$ generated by $\mathcal{G}^{\natural|\sharp} = \Pi_0$ and by $Z_{\rho,\sigma} = M|\mathcal{M}(0|\infty)L^\pi_0 = M|\mathcal{M}(0|\infty).\Pi = M|\mathcal{M}^!$, and not $M|\mathcal{M}(0|\infty).\Pi_0 = M|\mathcal{M}^!_0$†

The trouble comes ultimately from the fact that it is $Z_{\rho,\sigma}$ that is too big, we would like to define in $Z_{\rho,\sigma}$ a subgroup playing the role of $M|\mathcal{M}(0|\infty).\Pi_0$. There is no difficulty if we can define in † $Z_{\rho,\sigma}$ a subgroup which corresponds :|to $M|\mathcal{M}(0|\infty)$

$$Z_{\rho,\sigma}(0) \subset Z_{\rho,\sigma} \tag{65}$$

which gives a splitting

$$Z_{\rho,\sigma} \simeq Z_{\rho,\sigma}(0|\infty).L^{\natural|\sharp}_0 \qquad (\tfrac{1}{2}\ \text{direct}) \tag{66}$$

and we †will have then

$$\dagger G_{\rho,\sigma} \simeq Z_{\rho,\sigma}(0).\mathcal{G}^+, \qquad \dagger \qquad (\tfrac{1}{2}\ \text{direct}) \tag{67}$$

667

Uncertainty retained from cancelled passages.

140-3-s668-D001 of the case GL|SL $\natural|\sharp$

140-3-s668-D002 group

140-3-s668-D003 !

140-3-s668-D004 the

140-3-s668-D005 will possess

140-3-s668-D006 $M|\mathcal{M}$

140-3-s668-D007 $G^{\natural}_{\rho,\sigma}$ $\natural|\sharp$

[scan 669]
[p. 637]

and we † would set

$$\begin{cases} (G^{\natural|\sharp}_{\rho,\sigma})_0 \overset{\text{dfn|definition}}{=} Z_{\rho,\sigma}(0|\infty).\mathcal{G}^{\natural|\sharp} \subset G^{\natural|\sharp}_{\rho,\sigma} \subset \begin{matrix} G_{\rho,\sigma} \\ \| \\ Z_{\rho,\sigma}(0|\infty).\mathcal{G}^+ \end{matrix} \\ Z^{\natural|\sharp}_{\rho,\sigma} = \dagger Z_{\rho,\sigma} \\ \| \\ G^{\natural|\sharp}_{\rho,\sigma\,0} \cap Z_{\rho,\sigma} = Z_{\rho,\sigma}(0|\infty).(L_0 \cap \mathcal{G}^{\natural|\sharp}) \end{cases} \tag{68}$$

Let us recall a standard way of obtaining a canonical splitting of $Z_{\rho,\sigma}$ over $\gamma_{\rho,\sigma}$, when we have a (ρ,σ)-homomorphism †

†.

$$\mathcal{G}^+ \longrightarrow \mathrm{SL}(2,\widehat{\mathbb{Z}})' = \mathrm{SL}(2,\widehat{\mathbb{Z}})/\{\pm 1\} \tag{69}$$

commuting with the action of τ, which induces a homomorphism

$$Z_{\rho,\sigma} \longrightarrow B'_0 = \left\{ \begin{pmatrix} \mu & \lambda \\ 0 & 1 \end{pmatrix} \,\middle|\, \mu \in \widehat{\mathbb{Z}}^*;\ \lambda \in \widehat{\mathbb{Z}} \right\} \tag{70}$$

(see: the study of § 48)

inducing

$$\begin{cases} L_0^{\natural|\sharp} \longrightarrow L_0' = \left\{ \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix} \,\middle|\, \lambda \in \widehat{\mathbb{Z}} \right\} \\ \\ \varepsilon_0^n \longmapsto \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \end{cases} \tag{71}$$

†more precisely we have a diagram of exact sequences

$$\begin{array}{ccccccccc} 1 & \longrightarrow & L_0^{\natural|\sharp} & \longrightarrow & Z_{\rho,\sigma} & \longrightarrow & \gamma_{\rho,\sigma} & \longrightarrow & 1 \\ & & \downarrow{\scriptstyle \varepsilon_0^n \mapsto n} & & \downarrow & & \downarrow{\scriptstyle \chi_\gamma} & & \\ 1 & \longrightarrow & \widehat{\mathbb{Z}} & \xrightarrow[\lambda \mapsto \left(\begin{smallmatrix} 1 & \lambda \\ 0 & 1 \end{smallmatrix}\right)]{} & B_0'' & \xrightarrow[\left(\begin{smallmatrix} \mu & \lambda \\ 0 & 1 \end{smallmatrix}\right) \mapsto \mu]{\det} & \widehat{\mathbb{Z}}^* & \longrightarrow & 1 \end{array} \tag{72}$$

If the homomorphism (71) is an isomorphism, then every 668

Uncertainty retained from cancelled passages.

140-3-s669-D001 will define|would define

140-3-s669-D002 G

140-3-s669-D003 67

140-3-s669-D004 φ

140-3-s669-D006 thus|hence

[scan 670]

splitting of the second|second exact sequence defines by inverse image one of the first, † But in the typical cases which us, $L_0^{\natural|\sharp}$ (which is not small enough to give "the right" † $G_{\rho,\sigma}^{\natural|\sharp}$!) is not big enough on the other hand to give an isomorphism in (71) ! But † we can also make an additional hypothesis — namely that the inverse image under (70) of †"right|right" section

$$T_0' = \left\{ \begin{pmatrix} \mu & 0 \\ 0 & 1 \end{pmatrix} \,\middle|\, \mu \in \widehat{\mathbb{Z}}^* \right\} \subset B_0' \quad — \tag{73}$$

which is a subgroup

$$\begin{array}{l} \dagger Z_{\rho,\sigma}(0|\infty) \subset Z_{\rho,\sigma} \\ \quad \| \\ \left\{ u \in Z_{\rho,\sigma} \,\middle|\, \text{its image } \begin{pmatrix} \mu & \lambda \\ 0 & 1 \end{pmatrix} \text{ is in } T_0', \text{ i.e. } \lambda = 0 \right\} \end{array} \tag{74}$$

satisfying

$$Z_{\rho,\sigma}(0|\infty) \cap L_0^{\mu|\natural} = \{1\} \quad .) \tag{75}$$

be in fact a section subgroup, i.e. that $Z_{\rho,\sigma}(0|\infty) \hookrightarrow \gamma_{\rho,\sigma}$ be an epimorphism i.e. an isomorphism, i.e. that we do have

$$Z_{\rho,\sigma} = Z_{\rho,\sigma}(0|\infty).L_0^{\natural|\sharp}. \tag{76}$$

I have good reasons to think that (as soon as (69) exists) this condition is satisfied in all really useful cases. This|The choice of a 669

Uncertainty retained from cancelled passages.
140-3-s670-D001 and the
140-3-s670-D002 $M|\mathcal{M}$
140-3-s670-D003 Remark|Then
140-3-s670-D004 the
140-3-s670-D005 $M|\mathcal{M}$

[scan 671]

$Z_{\rho,\sigma}(0|\infty)$ has moreover the good property of being "functorial" in an obvious sense (cf. below the questions of functoriality), which will be essential for having a good formalism. This thus allows us to define a subgroup $(G^{\natural|\sharp}_{\rho,\sigma})_0 \subset G^{\natural|\sharp}_{\rho,\sigma}$ by (68). If its definition were not thus a little too convoluted, it is † group "finer" that would be better|better, and would in fact deserve the simpler notation $G^{\natural|\sharp}_{\rho,\sigma}$. [p. 638]

Another way of proceeding ! We look for a system of|of the adequate restrictions † the definition of $Z_{\rho,\sigma}$, which † imply that $L_0^{\natural|\sharp} = L_0 \cap \mathcal{G}^{\natural|\sharp}$, i.e. that for $l \in L_0$ a, we have

$$(\mathrm{int}(l), 1, 1, 1) \in Z_{\rho,\sigma} \Longleftrightarrow l \in \mathcal{G}^{\natural|\sharp}$$

But in the universal case, with $\mathcal{G}^{\natural|\sharp} = \Pi_0$, taking $l = l_0$, its image in $\mathcal{G}^{+|*}/\mathcal{G}^{\natural|\sharp} = \mathcal{G}/\Pi_0$ is the nontrivial central element of $\mathcal{G}/\Pi_0$, the inner automorphism it defines in $\mathcal{G}^+/\Pi_0$ (and even|= in $\mathcal{G}/\Pi_0$) is trivial — so it is not through † properties of this action that we will manage to exorcise it !

But perhaps it is premature to try already to "cut out" these groups †$G^{\natural|\sharp}_{\rho,\sigma}$, $M|\mathcal{M}^{\natural|\sharp}_{\rho,\sigma}$ 670

Uncertainty retained from cancelled passages.
140-3-s671-D001 this G
140-3-s671-D004 in
140-3-s671-D005 must
140-3-s671-D006 this
140-3-s671-D007 $M|\mathcal{M}$ $\natural|\sharp$

[scan 672]

"as fine as possible" — perhaps there are still some rather gross errors of orientation which will be rectified by the examination of particular cases. So I am going to give up for the time being continuing to invest † some energy in the|in cutting up, and shall therefore leave it at that in this direction, even if it means coming back to it later and refining the present constructions, which will however have to be compared when the occasion arises with those of § 48 XI. 671

Uncertainty retained from cancelled passages.
140-3-s672-D001 some

[scan 673]

5°) $S_0\mathcal{M}^{\natural|\sharp}_{\rho,\sigma}$, $\mathcal{M}^{\natural|\sharp}_{\rho,\sigma}$, $S_0\Gamma^{\natural|\sharp}_{\rho,\sigma}$, $\Gamma_{\rho,\sigma}$ ... . [p. 639]

We shall put as on page 583

$$\begin{cases} S_0\mathcal{M}_{\rho,\sigma} = Z_{\rho,\sigma} \times_{\gamma_{\rho,\sigma}} N_\rho^{\natural|\sharp} \times_{\gamma_{\rho,\sigma}} N_\sigma^{\natural|\sharp} \times_{\gamma_{\rho,\sigma}} G_{\rho,\sigma} \\ S_0\mathcal{M}^{\natural|\sharp}_{\rho,\sigma} = Z_{\rho,\sigma} \times_{\gamma} N_\rho^{\natural|\sharp} \times_{\gamma} N_\sigma^{\natural|\sharp} \times_{\gamma} G^{\natural|\sharp}_{\rho,\sigma} \end{cases} \tag{77}$$

$$(78)\qquad \begin{cases} S_0\Gamma_{\rho,\sigma} = Z_{\rho,\sigma} \times_\gamma N_\rho^{\natural|\sharp} \times N_\sigma^{\natural|\sharp} \\ \Gamma_{\rho,\sigma} = \dagger\, N_\rho^{\natural|\sharp} \times_\gamma N_\sigma^{\natural|\sharp} \end{cases}$$

$$(78)\qquad \begin{cases} \mathcal{M}_{\rho,\sigma} \simeq N_\rho^{\natural|\sharp} \times N_\sigma^{\natural|\sharp} \times G_{\rho,\sigma} \\ \mathcal{M}_{\rho,\sigma}^{\natural|\sharp} \simeq N_\rho^{\natural|\sharp} \times N_\sigma^{\natural|\sharp} \times G_{\rho,\sigma}^{\natural|\sharp}. \end{cases}$$

whence diagram of exact sequences

(79)

$$\begin{array}{ccccccccc}
\dagger\,1 & \to & \mathcal{G}^+ & \hookrightarrow & S_0\mathcal{M}_{\rho,\sigma} & \hookrightarrow & S_0\Gamma_{\rho,\sigma} & \to & 1 \\
 & & \uparrow & \searrow & \uparrow & \searrow & \uparrow & \searrow & \\
1 & \to & \mathcal{G}^+ & \hookrightarrow & \mathcal{M}_{\rho,\sigma} & \to & \Gamma_{\rho,\sigma} & \to & 1 \\
 & & \uparrow & & \uparrow & & \uparrow & & \\
1 & \to & \mathcal{G}^{\natural|\sharp} & \hookrightarrow & S_0\mathcal{M}_{\rho,\sigma_\dagger}^{\natural|\sharp} & \to & S_0\Gamma_{\rho,\sigma}^{\natural|\sharp} & & \\
 & & \sim\downarrow & \searrow & \uparrow & \searrow & \uparrow & \searrow & \\
1 & \to & \mathcal{G}^{\natural|\sharp} & \hookrightarrow & \dagger\mathcal{M}_{\rho,\sigma}^{\natural|\sharp} & \to & \Gamma_{\rho,\sigma}^{\natural|\sharp} & \to & 1
\end{array}$$

and homomorphisms of|homomorphisms of exact sequences

(80)|(89)

$$\begin{array}{ccccccccc}
1 & \to & \overbrace{SZ_{\rho,\sigma}}^{L_0^{\natural|\sharp}} \times SN_\rho^{\natural|\sharp} \times SN_\sigma^{\natural|\sharp} \times \mathcal{G}^{\natural|\sharp} & \to & S_0\mathcal{M}_{\rho,\sigma}^{\natural|\sharp} & \to & \gamma_{\rho,\sigma} & \to & 1 \\
 & & \downarrow & & \downarrow & & \| & & \\
1 & \to & \underbrace{SZ_{\rho,\sigma}}_{L_0^{\natural|\sharp}} \times SN_\rho^{\natural|\sharp} \times SN_\sigma^{\natural|\sharp} \times \mathcal{G}^+ & \to & S_0\mathcal{M}_{\rho,\sigma} & \to & \gamma_{\rho,\sigma} & \to & 1
\end{array}$$

(82)

$$\begin{array}{ccccccccc}
1 & \to & SN_\rho^{\natural|\sharp} \times SN_\sigma^{\natural|\sharp} \times \mathcal{G}^{\natural|\sharp} & \to & \mathcal{M}_{\rho,\sigma}^{\natural|\sharp} & \to & \gamma_{\rho,\sigma} & \to & 1 \\
 & & \downarrow & & \downarrow & & \| & & \\
1 & \to & SN_\rho^{\natural|\sharp} \times SN_\sigma^{\natural|\sharp} \times \mathcal{G} & \to & \mathcal{M}_{\rho,\sigma} & \to & \gamma_{\rho,\sigma} & \to & 1
\end{array}$$

$$(83)\qquad 1 \to L_0^{\natural|\sharp} \times SN_\rho^{\natural|\sharp} \times SN_\sigma^{\natural|\sharp} \to S_0\Gamma_{\rho,\sigma}^{\natural|\sharp} \to \gamma_{\rho,\sigma} \to 1$$

$$(84)\qquad 1 \to \dagger SN_\rho^{\natural|\sharp} \times SN_\sigma^{\natural|\sharp} \to \Gamma_{\rho,\sigma} \to \gamma_{\rho,\sigma} \to 1$$

672

Uncertainty retained from cancelled passages.

140-3-s673-D001 $Z_{\rho,\sigma}$

140-3-s673-D002 1

140-3-s673-D003 ↖
140-3-s673-D004 S_0
140-3-s673-D005 $L_0^{\natural|\sharp}\times$

[scan 674]

$$\begin{cases} S_0\mathcal{M}_{\rho,\sigma} \simeq \mathcal{M}_{\rho,\sigma} \times_{\gamma_{\rho,\sigma}} Z_{\rho,\sigma} \\ S_0\mathcal{M}^{\natural|\sharp}_{\rho,\sigma} \simeq \mathcal{M}^{\natural|\sharp}_{\rho,\sigma} \times_{\gamma_{\rho,\sigma}} Z_{\rho,\sigma} \end{cases} \tag{85}$$

$$\dagger \quad S_0\Gamma_{\rho,\sigma} \simeq \Gamma_{\rho,\sigma} \times_{\gamma_{\rho,\sigma}} Z_{\rho,\sigma} \tag{86}$$

$$\begin{cases} S_0\mathcal{M}_{\rho,\sigma} \simeq \mathcal{M}_{\rho,\sigma} \times_{\Gamma_{\rho,\sigma}} S_0\Gamma_{\rho,\sigma} \\ S_0\mathcal{M}^{\natural|\sharp}_{\rho,\sigma} \simeq \mathcal{M}^{\natural|\sharp}_{\rho|2,\sigma} \times_{\Gamma_{\rho,\sigma}} S_0\Gamma_{\rho,\sigma}. \end{cases} \tag{87}$$

Remark|Remark This deluge of groups makes one a little seasick. I am not sure that the variants S_0 ($S_0\mathcal{M}$, $S_0\mathcal{M}^{\natural|\sharp}$, $S_0\Gamma$) are very useful.

There remains the competition|comparison between invariants with or without $\natural|\sharp$, especially $\dagger$ in $\mathcal{M}_{\rho,\sigma}$ and $\mathcal{M}^{\natural|\sharp}_{\rho,\sigma}$ — whose "difference" is "the $\underset{=}{\mathrm{m}}|\overline{\mathrm{m}}$" as between $\mathcal{G}^+$ and $\mathcal{G}^{\natural|\sharp}$, or $\dagger$ and between $G_{\rho,\sigma}$ and $G^{\natural|\sharp}_{\rho,\sigma}$, to be precise $\mathcal{M}_{\rho,\sigma}$ is normal in $\mathcal{M}^{\natural|\sharp}_{\rho,\sigma}$, and $G^{\natural|\sharp}_{\rho,\sigma}$ in $G_{\rho,\sigma}$, we have isomorphisms

$$(88) \qquad \mathcal{G}^+/\mathcal{G}^{\natural|\sharp} \xrightarrow{\sim} G_{\rho,\sigma}/G^{\natural|\sharp}_{\rho,\sigma} \xrightarrow{\sim} \mathcal{M}_{\rho,\sigma}/\mathcal{M}^{\natural|\sharp}_{\rho,\sigma}.$$

The groups $G_{\rho,\sigma}, \mathcal{M}_{\rho,\sigma}$ play with respect to|vis-à-vis $G^{\natural|\sharp}_{\rho,\sigma}, \mathcal{M}^{\natural|\sharp}_{\rho,\sigma}$ the role of catchalls (in particular to allow us to consider the automorphisms ρ, σ of $G^{\natural|\sharp}_{\rho,\sigma}, \mathcal{M}^{\natural|\sharp}_{\rho,\sigma}$ as induced by inner automorphisms of the 673

Uncertainty retained from cancelled passages.
140-3-s674-D001 { S_0
140-3-s674-D002 the
140-3-s674-D003 pre

[scan 675]
[p. 640]

surrounding groups $G_{\rho,\sigma}, \mathcal{M}_{\rho,\sigma}$.

We have an element

$$(89) \qquad \begin{array}{ccccc} & & \tau'_{M|\mu} \in \mathcal{M}^{\natural|\sharp}_{\rho,\sigma} & & \dagger \text{ over } \tau'_\gamma \in \gamma_{\rho,\sigma} \\ & & \| & & \\ & (\tau'_\rho, & \tau'_\sigma, & \tau'_G) & \\ & \cup & \cup & \cup & \\ & N^{\natural|\sharp}_\rho & N^{\natural|\sharp}_{\rho|\sigma} & \overset{\dagger}{G}{}^{\natural|\sharp}_{\rho,\sigma} & \end{array}$$

whence $\dagger$ canonical embeddings

$$(90) \qquad \begin{array}{ccc} \mathcal{G}^{\natural|\sharp}_\tau & \hookrightarrow & \mathcal{M}^{\natural|\sharp}_{\rho,\sigma} \\ \downarrow & & \downarrow \\ \mathcal{G} & \hookrightarrow & \mathcal{M}_{\rho,\sigma} \end{array}$$

$\mathcal{G}^{\natural|\sharp}$ = inverse image in $\mathcal{M}_{\rho,\sigma}$ of $\{1, \tau_\gamma|\tau'_\gamma\} \subset \gamma_{\rho,\sigma}$

$\mathcal{G}$ = inverse image in $\mathcal{M}_{\rho,\sigma}$ of $\{1, \tau_\gamma\} \subset \gamma_{\rho,\sigma}$

We can reproduce the messy diagrams of page 584, with ♮|♯'s in the key signature…

I am going to see if I can in fact dispense with redeveloping the links with $\overline{Z}_{\rho,\sigma}$, $SG_{\rho,\sigma}$ etc — even if it means coming back to it later if need be. 674

Uncertainty retained from cancelled passages.
140-3-s675-D001 with image|whence
140-3-s675-D002 $\mathcal{G}$
140-3-s675-D003 some

[scan 676]

6°) Functorialities.
m1345

Marginal note. m1345 N.B. Up to now (in 1° to 5°) we had not used the hypothesis that ρ, σ generate $\mathcal{G}^+$

Cf. p. 588. In addition to the surjective homomorphism

$$(91) \qquad \mathcal{G} \xrightarrow{\varphi} \mathcal{G}'$$

m1346

Marginal note. m1346 It is better to write

$$\mathcal{G}_0 \to \mathcal{G}$$

cf. p. 643′ ff…

we assume that φ maps $\mathcal{G}^{\natural\sharp}$ into $\mathcal{G}'^{\natural\sharp}$.

Here we denote by $Z_{\rho,\sigma}, \mathcal{M}_{\rho,\sigma}$ etc. the invariants associated with $\mathcal{G}$, $Z_{\rho',\sigma'}$ etc. those of $\mathcal{G}'$. We have equivalent conditions as in loc. cit. — where we are going to drop the consideration of $\overline{Z}^{\mathcal{G}}_{\rho,\sigma}$… — which mean that for $\forall$

$$\underline{u} = (u, \mu, \varepsilon_2, \varepsilon_3) \in Z_{\rho,\sigma}$$

there exists an automorphism u' of $\mathcal{G}'$ making commutative

$$(92) \qquad \begin{array}{ccc} \mathcal{G}^+ & \xrightarrow{\varphi} & \mathcal{G}'^+ \\ \downarrow{\scriptstyle u} & & \downarrow{\scriptstyle u'} \\ \mathcal{G}^{++} & \xrightarrow{\varphi'} & \mathcal{G}'^+ \end{array}$$

which u' will be unique. It amounts to the =|same to say that the $u \in \mathrm{Aut}(\mathcal{G}^+)$ which come from $Z_{\rho,\sigma}$, i.e. which normalize L_0, and which transform $\xi_\rho|\ell_\rho, \xi_\sigma|\ell_\sigma$ into †† † elements $\mathcal{G}^{\natural\sharp}_\tau$-conjugate to ρ, σ, stabilize the kernel of φ (91).

We then find 675

Uncertainty retained from cancelled passages.
140-3-s676-D002 elements
140-3-s676-D003 groups
140-3-s676-D004 ♮|♯

[scan 677]
[p. 641]

$$(93)\qquad Z_{\rho,\sigma} \xrightarrow{\varphi_{Z|z}} Z_{\rho',\sigma'}$$

compatible with the actions of $Z_{\rho,\sigma}, Z_{\rho',\sigma'}$ on $\mathcal{G}^+$ resp. $\mathcal{G}'^+$, and with the homomorphisms

$$L_0^{\natural|\sharp} \subset Z_{\rho,\sigma} \curvearrowright \mathcal{G}^+, \qquad L_0'^{\natural|\sharp} \subset Z_{\rho',\sigma'} \curvearrowright \mathcal{G}'^+,$$

whence by the standard|Amalgamation construction and a homomorphism

$$(94)\qquad G_{\rho,\sigma} \xrightarrow{\varphi_G} G^{\dagger|\varepsilon}_{\rho',\sigma'}$$

inducing

$$(95)\qquad G^{\natural|\sharp}_{\rho,\sigma} \xrightarrow{\varphi_G^{\natural|\sharp}} G^{\natural|\sharp}_{\rho',\sigma'}$$

whence a homomorphism from the diagrams (64) relative to (ρ,σ), into the analogous diagram relative to (ρ',σ'), with or without $\natural|\sharp$.

No more is needed for all the constructions and diagrams of the preceding n°s, relative|relative to ρ,σ, to be mapped into those relative to ρ',σ'.

Remark Suppose that $\mathcal{G}'$, $\mathcal{G}'^{\natural|\sharp}$ satisfies the following condition: For every pair

$$(96)\quad \begin{cases} \text{of elements } \alpha',\beta' \in \mathcal{G}'^{\natural|\sharp}_{\tau}, \quad \text{such that we have} \\ \qquad [\alpha',\rho] = [\beta',\sigma]\,\varepsilon_1|\xi_1{}^{\mu-1} \qquad \text{for } \mu \in \widehat{\mathbb{Z}} \text{ suitable} \\ \text{there exists a } u' \in \mathrm{Aut}(\mathcal{G}'^+) \quad \text{such that} \\ (97)\qquad u'(\rho) = \alpha'(\rho), \quad u'(\sigma) = \beta'(\sigma) \end{cases}$$

i.e.

$$(98)\qquad \alpha'^{-1}u' \in \mathrm{Cent}_{\mathrm{Aut}}(\rho), \quad \beta'^{-1}u' \in \mathrm{Cent}_{\mathrm{Aut}}(\sigma)$$

676

Uncertainty retained from cancelled passages.

140-3-s677-D001 $\natural|\sharp$

[scan 678]

the relation of the Cent (96) being expressed in terms of u by the condition

$$u'(L_0^{\mathcal{G}'}) = L_0^{\mathcal{G}'} \qquad \text{i.e. } u' \text{ normalizes } \mathcal{G}'. \tag{99}$$

Then every (ρ,σ)-homomorphism $(\mathcal{G},\mathcal{G}^{\natural|\sharp}) \longrightarrow |-(\mathcal{G}',\mathcal{G}'^{\natural|\sharp})$ is necessarily admissible.

We are going to apply this to the case where $\mathcal{G}$ is universal — but to be able to say about it † what we feel like, I will first speak of the subgroups

$$\mathcal{M}(0|\infty),\ \mathcal{M}(\rho),\ \mathcal{M}(\sigma) \subset \mathcal{M}_{\rho,\sigma}.$$

677

Uncertainty retained from cancelled passages.
140-3-s678-D001 all

†
m1350

[scan 679]
[p. 642]

Marginal note. m1350 7°) The subgroups $\mathcal{M}^{\dagger}_{\rho,\sigma}[0|\rho]$, $\mathcal{M}^{\dagger}_{\rho,\sigma}(\rho)$, $\mathcal{M}^{\dagger}_{\rho,\sigma}(\sigma)$.

In the group

$$\begin{aligned} S_0\mathcal{M}^{\natural|\sharp}_{\rho,\sigma} &= Z_{\rho,\sigma} \times_{\gamma_{\rho,\sigma}} N^{\natural|\sharp}_{\rho} \times_{\gamma_{\rho,\sigma}} N^{\natural|\sharp}_{\sigma} \times_{\gamma_{\rho,\sigma}} G^{\natural|\sharp}_{\rho,\sigma} \\ &= \left\{ x = (u,a,b,U) \in Z_{\rho,\sigma} \times N_\rho \times N_\sigma \times G_{\rho,\sigma} \;\middle|\; \delta_Z(u) = \delta_\rho(a) = \delta_\sigma(b) = \delta_G(U) \right\} \end{aligned} \tag{100}$$

we introduce three subgroups

$$\left\{ \begin{array}{lll} S_0\mathcal{M}_{\rho,\sigma}[0] \subset \mid = S_0\mathcal{M}^{\natural|\sharp}_{\rho,\sigma} & \text{by the condition} & U = i_Z(u) \\ S_0\mathcal{M}_{\rho,\sigma}[\rho] \subset S_0\mathcal{M}^{\natural|\sharp}_{\rho,\sigma} & \text{———} & U = i_\rho(a) \\ S_0\mathcal{M}_{\rho,\sigma}[\sigma] \subset S_0\mathcal{M}^{\natural|\sharp}_{\rho,\sigma} & \text{———} & U = i_\sigma(b) \end{array} \right. \tag{101}$$

now using the three canonical homomorphisms

$$\left\{ \begin{array}{l} i_Z : Z_{\rho,\sigma} \hookrightarrow G_{\rho,\sigma} \\ i_\rho : N_\rho \hookrightarrow G_{\rho,\sigma} \\ i_\sigma : N_\sigma \hookrightarrow G_{\rho,\sigma} \end{array} \right. . \tag{102}$$

These three groups are manifestly all three isomorphic: $Z_{\rho,\sigma} \times_{\gamma_{\rho,\sigma}} {}^{\dagger}N^{\natural|\sharp}_{\rho} \times_{\gamma_{\rho,\sigma}} N^{\natural|\sharp}_{\sigma} = S_0\Gamma_{\rho,\sigma}$, to be precise, in the exact sequence (52)

$$1 \longrightarrow \mathcal{G}^{+} \longrightarrow S_0\mathcal{M}_{\rho,\sigma} \longrightarrow S_0\Gamma_{\rho,\sigma} \longrightarrow 1$$

the canonical homomorphism $S_0\mathcal{M}_{\rho,\sigma} \longrightarrow S_0\Gamma_{\rho,\sigma}$ induces isomorphisms between the groups (68) and $S_0\Gamma_{\rho,\sigma}$, the three groups are therefore section subgroups.

We define $\mathcal{M}_{\rho,\sigma}[0]$, $\mathcal{M}_{\rho,\sigma}(\rho)$ (or $\mathcal{M}_{\rho,\sigma}(\frac{1}{2})$) and $\mathcal{M}_{\rho,\sigma}(\sigma)$ (or $\mathcal{M}_{\rho,\sigma}(-\bar{j})$) as the images of $S_0\mathcal{M}_{\rho,\sigma}[0]$, $S_0\mathcal{M}_{\rho,\sigma}[\rho]$ and $S_0\mathcal{M}_{\rho,\sigma}[\sigma]$ under the canonical homomorphism $S_0\mathcal{M}_{\rho,\sigma} \longrightarrow \mathcal{M}_{\rho,\sigma}$, which

†
678

Uncertainty retained from cancelled passages.
140-3-s679-D001 646|686
140-3-s679-D002 $\natural|\sharp$
140-3-s679-D003 $\natural|\sharp$
140-3-s679-D004 $\natural|\sharp$
140-3-s679-D006 Z
140-3-s679-D007 69

[scan 680]

amounts to setting

$$\begin{cases} \mathcal{M}_{\rho,\sigma}[0] = \{(a,b,U) \in \mathcal{M}^{\natural|\sharp}_{\rho,\sigma} \mid U \in Z_{\rho,\sigma}\} \\ \mathcal{M}_{\rho,\sigma}(\rho) = \{(a,b,U) \in \mathcal{M}^{\natural|\sharp}_{\rho,\sigma} \mid U = a\} \\ \mathcal{M}_{\rho,\sigma}(\sigma) = \{(a,b,U) \in \mathcal{M}^{\natural|\sharp}_{\rho,\sigma} \mid U = b\}. \end{cases} \tag{103}$$

Thus the subgroups|subgroups $\mathcal{M}_{\rho,\sigma}(\rho), \mathcal{M}_{\rho,\sigma}(\sigma)$ of $\mathcal{M}^{\natural|\sharp}_{\rho,\sigma}$ are section groups over $\Gamma^{\natural|\sharp}_{\rho,\sigma} \simeq N^{\natural|\sharp}_{\rho} \times_{\gamma_{\rho,\sigma}} N^{\natural|\sharp}_{\sigma}$, hence

$$\begin{cases} \mathcal{M}^{\natural|\sharp}_{\rho,\sigma} \simeq \mathcal{M}_{\rho,\sigma}(\rho) \,.\, \mathcal{G}^{\natural|\sharp} \\ \mathcal{M}^{\natural|\sharp}_{\rho,\sigma} \simeq \mathcal{M}_{\rho,\sigma}(\sigma) \,.\, \mathcal{G}^{\natural|\sharp} \end{cases} \quad (\tfrac{1}{2} \text{ direct}), \tag{104}$$

and likewise|=

$$\begin{cases} \mathcal{M}_{\rho,\sigma} \simeq \mathcal{M}_{\rho,\sigma}(\rho) \,.\, \mathcal{G}^{+} \\ \mathcal{M}_{\rho,\sigma} \simeq \mathcal{M}_{\rho,\sigma}(\sigma) \,.\, \mathcal{G}^{+} \end{cases} \quad (\tfrac{1}{2} \text{ direct}). \tag{105}$$

On the other hand, we will have

$$\begin{array}{ccccccccc} 1 & \longrightarrow |- & L_0^{\natural|\sharp} & - & \mathcal{M}^{\dagger}_{\rho,\sigma}[0] & \longrightarrow & \Gamma_{\rho,\sigma} & - & 1 \\ & & \Big\downarrow{\scriptstyle 1} & & \Big\uparrow{\scriptstyle \sim} & & \Big\uparrow{\scriptstyle 1} & & \\ & & & & S_0\mathcal{M}_{\rho,\sigma}[0] & & & & , \\ & & \Big\downarrow{\scriptstyle 1} & & \Big\downarrow{\scriptstyle \sim} & & \Big\uparrow{\scriptstyle 1} & & \\ 1 & - & L_0^{\natural|\sharp} & \longrightarrow & S_0\Gamma_{\rho,\sigma} & \longrightarrow |- & \Gamma_{\rho,\sigma} & - & 1 \end{array} \tag{106}$$

so the extension $\mathcal{M}_{\rho,\sigma}[0]$ of $\Gamma_{\rho,\sigma}$ by $L_0^{\natural|\sharp}$ is identified with the extension $S_0\Gamma_{\rho,\sigma}$ of $\Gamma_{\rho,\sigma}$ by $L_0^{\natural|\sharp}$, i.e. with the inverse image of the extension $Z_{\rho,\sigma}$ of $\gamma_{\rho,\sigma}$ by $L_0^{\natural|\sharp}$, via the canonical homomorphism 679

Uncertainty retained from cancelled passages.

140-3-s680-D001 $\natural|\sharp$

[scan 681]
[p. 643]

$$\Gamma_{\rho,\sigma} = N^{\natural|\sharp}_{\rho} \times_{\gamma_{\rho,\sigma}} N^{\natural|\sharp}_{\sigma} \longrightarrow \gamma_{\rho,\sigma}. \tag{107}$$

Thus giving a section of $\dagger Z_{\rho,\sigma}$ over $\gamma_{\rho,\sigma}$ (cf. on this subject p. 636′ ff). defines one of $S_0\Gamma_{\rho,\sigma}$ over $\Gamma_{\rho,\sigma}$, hence of $\mathcal{M}_{\rho,\sigma}[0]$ over $\Gamma_{\rho,\sigma}$. When such a section is fixed, we denote this section group by

$$\mathcal{M}_{\rho,\sigma}(0|\infty) \subset \mathcal{M}^{\natural|\sharp}_{\rho,\sigma},$$

so we will have

$$\begin{cases} \mathcal{M}_{\rho,\sigma}[0] \simeq \mathcal{M}_{\rho,\sigma}(0|\infty) \,.\, L_0^{\natural|\sharp} \\ \mathcal{M}^{\natural|\sharp}_{\rho,\sigma}. \simeq \mathcal{M}_{\rho,\sigma}(0|\infty) \,.\, \mathcal{G}^{\natural|\sharp} \\ \mathcal{M}_{\rho,\sigma} \simeq \mathcal{M}_{\rho,\sigma}(0) \,.\, \mathcal{G} \end{cases} . \tag{108}$$

680

Uncertainty retained from cancelled passages.
140-3-s681-D001 Γ

[scan 682]

8°) Return to the universal case, and the homomorphism

$$M^{\natural|\sharp} \longrightarrow M^{\natural|\sharp}_{\rho,\sigma}/\mu_\rho$$

We now suppose that $\mathcal{G}$ corresponds to the universal case $\mathrm{GL}(2,\mathbb{Z})^\wedge$, with an arbitrary $\mathcal{G}^{\natural|\sharp}$. We shall denote the universal case by a subscript 0: $\mathcal{G}_0$, $\mathcal{G}_0^+$, $\mathcal{G}_0^{\natural|\sharp}$ etc., and $\mathcal{G}'$, $\mathcal{G}'^{\natural|\sharp}$ etc. the|the former $\mathcal{G}$, $\mathcal{G}^{\natural|\sharp}$ etc.

↙ Z_{ρ_0,σ_0} etc ... We shall write $\mathcal{G}_0^+$ instead of $\mathcal{G}_0^{+\wedge}$ ...

I set

$$(107) \qquad M^{\natural|\sharp} \subset M$$

$$\left\{ u \in M \;\middle|\; \begin{array}{ll} \text{a) } u(\rho) \text{ is } \mathcal{G}^{\natural|\sharp}\text{-conjugate to} & \varepsilon_3[\tau](\rho) \\ \text{b) } u(\sigma) \quad \text{---------------} & \varepsilon_2[\tau](\sigma) \\ \text{c) } u(\varepsilon_0) \quad \text{---------------} & \varepsilon_0^\mu \\ \text{d) } u \text{ stabilizes } M^{\natural|\sharp} & \end{array} \right\}$$

N.B. $M^{\natural|\sharp} \supset \mid \ni \mu$

where of course

$$\mu = \chi(u), \quad \varepsilon_2 = \varepsilon_2(u) = \varepsilon_2(\mu), \quad \varepsilon_3 = \varepsilon_3(u) = \varepsilon_3(\mu).$$

I say that $M^{\natural|\sharp}$ is a subgroup of M. We can see this by going back over the calculations on page 629 and following. I prefer to proceed as follows. 681

[scan 683]
[p. 644]

We have seen that we have

$$(108) \qquad \underset{\dagger}{Z_{\rho_0,\sigma_0}} \hookrightarrow M[0] = M(0) \cdot L_0^{\mathcal{G}} \simeq \widetilde{M}_{0,3}[0] = \widetilde{M}_{0,3}(0) \cdot L_0^{\mathcal{G}_{0,3}}$$

an inclusion which is an equality if $\mathcal{G}_0^{\natural|\sharp} = \mathcal{G}_0^+$, and which identifies $Z_{\rho,\sigma}$ with

$$M^!{}[0] \simeq M(0) \cdot L_0^\pi \quad \text{if} \quad \mathcal{G}_0^{\natural|\sharp} = \pi := \Pi_0.$$

[— in fact, if $\mathcal{G}_0^{\natural|\sharp} \subset \Pi_0$, then we find

$$(109) \quad Z_{\rho,\sigma} = \left\{ u = \underset{\cap\, M(0)}{u_0} \cdot \underset{\cap\, \dagger L_0^\pi}{\ell} \;\middle|\; \begin{array}{l} \text{a) } u(\rho) = \alpha(\rho) \quad \text{with } \alpha \in \mathcal{G}_{\bar\tau}^{\natural|\sharp} \\ \text{b) } u(\sigma) = \beta(\sigma) \quad \text{with } \beta \in \mathcal{G}_{\bar\tau}^{\natural|\sharp} \\ \text{d) } u\varepsilon| \in \text{stabilizes } \mathcal{G}_0^{\natural|\sharp} \end{array} \right\}$$

m1359

Marginal note. m1359 this explicit description does not seem to be of any use.

and we see that α, β in a) b) are determined by $u = u_0\ell$ uniquely (even if we|we suppose only $u \in M^!{}[0]$ and no more, and that we require † $\alpha, \beta \in \Pi_{0\bar\tau}$, † We know that this is a subgroup|subgroup of $\mathrm{Aut}(\mathcal{G}_0^+) \times \widehat{\mathbb{Z}}^* \times \mu \times \mu$ — and in fact, of $\mathrm{Aut}(\mathcal{G}_0^+)$ (proposition page 629). †

As this group operates on $\mathcal{G}$ via $\widetilde{M}_{0,3}$, = via $\widetilde{M}^!_{0,3}[0]$ it maps into $\widetilde{M}^!_{0,3}[0]$ by a homomorphism of groups, hence into † $M[0]^! \simeq \widetilde{M}_{0,3}[0]$ by a homomorphism of groups.] 682

Uncertainty retained from cancelled passages.

140-3-s683-D001 ♯

140-3-s683-D002 $\mathcal{G}$

140-3-s683-D004 $\alpha \in$|with $\mathcal{G}$ ♮

140-3-s683-D005 can|In short

140-3-s683-D006 Let us consider

140-3-s683-D008 $\widetilde{M}_{0,3}[0]$

[scan 684]

† Let us consider the system

$$(110)\quad (\mathcal{G}_0^{+\wedge}, Z_{\rho_0,\sigma_0},\quad L_0^{\natural|\sharp} \subset Z_{\rho_0,\sigma_0},\quad Z_{\rho_0,\sigma_0} \longrightarrow \mathrm{Aut}(\mathcal{G}_0^{+\wedge}),\quad L_0^{\natural|\sharp} \hookrightarrow \mathcal{G}_0^{+\wedge})$$

(cf. "digression" below 9°) which gives rise to

$$(110')\qquad G_{\rho_0,\sigma_0},\quad \mathcal{G}_0^{+\wedge} \subset G_{\rho_0,\sigma_0},\quad Z_{\rho_0,\sigma_0} \hookrightarrow G_{\rho_0,\sigma_0},$$

and let us consider the system analogous to (110*)

$$(111^{*|\sharp})\quad (\mathcal{G}_0^{+}, \underset{\substack{\| \\ M(0)\cdot L_0^{\pi}}}{M[0]^{!|\natural}},\quad \underset{\text{(generated by } \varepsilon_0)}{L_0^{\mathcal{G}}} \subset M[0]^{!|\natural},\quad \dagger^{!|\natural} \longrightarrow \mathrm{Aut}(\mathcal{G}_0^{+}),\quad L_0^{\mathcal{G}} \hookrightarrow \mathcal{G}_0^{+})$$

which gives rise (loc. cit (9°)) to the situation

$$(111')\qquad (M,\quad \mathcal{G}_0^{+} \subset M,\quad M[0]' \hookrightarrow \mathcal{G}_0^{+}).$$

Now thanks to (108), we have a homomorphism from (110|109|119) into (111), corresponding to the identity on $\mathcal{G}_0^{+}$, (108) on Z_{ρ_0,σ_0}, the canonical inclusion : † $L_0^{\natural|\sharp} \subset L_0^{\mathcal{G}}$. This yields a homomorphism from (110′) into (111′), in particular a canonical homomorphism of exact sequences 683

Uncertainty retained from cancelled passages.

140-3-s684-D001 The homomorphism

140-3-s684-D002 $M[0]$

140-3-s684-D003 on

[scan 685]
[p. 645]

(112)

$$\begin{array}{ccccccccc}
1 & \longrightarrow & \mathcal{G}_0^{+} & \longrightarrow & G_{\rho_0,\sigma_0} & \longrightarrow & \gamma_{\rho_0,\sigma_0} & \longrightarrow & 1 \\
 & & \downarrow{\scriptstyle 1} & & \downarrow & & \downarrow & & \\
1 & \longrightarrow & \mathcal{G}_0^{+} & \longrightarrow & M & \longrightarrow & \tilde{\Gamma}_{\mathbb{Q}} & \longrightarrow & 1
\end{array}$$

where

$$\begin{array}{ccc}
\gamma_{\rho_0,\sigma_0} & \longrightarrow & \tilde{\Gamma}_{\mathbb{Q}} \\
\wr & & \wr \\
Z_{\rho_0,\sigma_0}/L_0^{\mu} & & M[0]/L_0^{\mathcal{G}}
\end{array}$$

is the one induced by (108), and is identified with:

$$(113) \qquad Z_{\rho_0,\sigma_0}/L_0^{\natural|\sharp} \longrightarrow M[0]/L_0^{\mathcal{G}}$$

By the very definition of $L_0^{\natural|\sharp}$ as being $Z_{\rho_0,\sigma_0} \cap \mathcal{G}_0^+$ †, we see that (113) is injective hence $G_{\rho_0,\sigma_0} \longrightarrow M$ is always injective (it is an isomorphism if $\mathcal{G}_0^{\natural|\sharp} \supset \Pi_0$ ...). We deduce by comparison from (112) a homomorphism of exact sequences

(114)

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \mathcal{G}_0^{\natural|\sharp} & \longrightarrow & G_{\rho_0,\sigma_0}^{\natural|\sharp} & \longrightarrow & \gamma_{\rho_0,\sigma_0} & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & \mathcal{G}_0^{+} & \longrightarrow & M & \longrightarrow & \dagger\widetilde{\Gamma}_{\mathbb{Q}} & \longrightarrow & 1. \end{array}$$

We then see immediately that the image of $G_{\rho_0,\sigma_0}^{\natural|\sharp}$ in M is none other than $M^{\natural|\sharp}$ (107). We therefore have 684

Uncertainty retained from cancelled passages.

140-3-s685-D001 and|this

140-3-s685-D002 γ

[scan 686]

$$(115) \qquad \begin{cases} M^{\natural|\sharp} \underset{(107)}{\text{defined in}} \simeq G_{\rho_0,\sigma_0}^{\natural|\sharp}, \\ \gamma_{\rho_0,\sigma_0} \hookrightarrow \widetilde{\Gamma}_{\mathbb{Q}}\dagger \end{cases}$$

m1363

Marginal note. m1363

to be checked

↓

N.B. We have

$$(116) \qquad \left\{ \begin{array}{l} \gamma_{\rho_0,\sigma_0}\dagger = \Gamma_0^{\natural}|\Gamma_0^{\sharp} = \left\{ u \in \widetilde{\Gamma}_{\mathbb{Q}} \;\middle|\; \begin{array}{l} \text{the lifting } u_0 \\ \text{of } u \text{ to } u_0 \in M(0) \\ \text{satisfies the} \\ \text{conditions a) b) d)} \\ \text{of (107)} \end{array} \right\} \\ \\ (\text{notations} \\ \gamma_0 = \Gamma_0^{\natural|\sharp}, \text{ where } \Gamma_0 \text{ denotes } \widetilde{\Gamma}_{\mathbb{Q}}) \end{array} \right.$$

which shows at the same time in the process that $M^{\natural|\sharp}$ is indeed a subgroup of M.

We now have

$$(117) \qquad M_{\rho_0,\sigma_0}^{\natural|\sharp} \overset{\text{dfn|déf}}{\simeq} N|M_{\rho_0}^{\natural|\sharp} \times_{\gamma_{\rho_0,\sigma_0}} N_{\sigma_0}^{\natural|\sharp} \times_{\gamma_{\rho_0,\sigma_0}} \underbrace{G_{\rho_0,\sigma_0}^{\natural|\sharp}}_{\simeq M^{\natural|\sharp}},$$

where $G_{\rho_0,\sigma_0}^{\natural|\sharp} \simeq M^{\natural|\sharp}$ therefore|gives

$$N_{\rho_0}^{\natural|\sharp} \simeq | = \left\{ u \in M^{\natural|\sharp} \mid u(\rho) = \rho^{\varepsilon_3(u)} \right\}$$
$$N_{\sigma_0}^{\natural|\sharp} \simeq | = \left\{ u \in M^{\natural|\sharp} \mid u(\sigma) \equiv | = \sigma^{\varepsilon_2(u)} \right\}$$

and finally we have

$$(118)\qquad \begin{cases} M^{\natural|\sharp}_{\rho_0,\sigma_0} \simeq N_{M^{\natural|\sharp}}(\rho) \times_{\dagger\Gamma_0^{\natural|\sharp}} N_{M^{\natural|\sharp}}(\sigma) \times_{\dagger\Gamma_0^{\natural|\sharp}} M^{\natural|\sharp}. \\ \text{where } \Gamma_0^{\natural|\sharp} \neq | = \gamma_0 \simeq \gamma_{\rho_0,\sigma_0} \subset \tilde{\Gamma}_{\mathbb{Q}} \quad \text{made explicit in (116)}, \end{cases}$$

whence an exact|exact sequences|sequence

$$(119)$$

$$1 \to \overbrace{SZ_{\rho_0} \times SZ_{\sigma_0}}^{L_\rho \times L_\sigma} \to M^{\natural|\sharp}_{\rho_0,\sigma_0} \to \dagger M^{\natural|\sharp} \to 1$$
$$\downarrow{\scriptstyle \natural|\sharp}$$

where the kernel $SZ_{\rho_0} \times SZ_{\sigma_0}$ is a very small group, namely 685

Uncertainty retained from cancelled passages.

140-3-s686-D001 $\Leftarrow$
140-3-s686-D002 $= \Gamma_0$
140-3-s686-D003 γ_{ρ_0,σ_0}
140-3-s686-D004 γ_{ρ_0,σ_0}
140-3-s686-D006 γ

[scan 687]
[p. 646]

$$(120)\qquad \mathrm{Ker}\left(M^{\natural|\sharp}_{\rho_0,\sigma_0} \overset{\text{epimorphism}}{\longrightarrow} M^{\natural|\sharp}\right) \simeq SZ_{\rho_0} \times SZ_{\sigma_0} \simeq L_\rho \times L_\sigma \simeq \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}.$$

We are going to improve further|further the †situation, by cutting out of $M^{\natural|\sharp}_{\rho_0,\sigma_0}$ a group even|even smaller, which is an extension of $M^{\natural|\sharp}$ by μ.

†

(

Let us note that

$$(121)\qquad \begin{array}{ccc} M^{\natural|\sharp}_{\rho_0,\sigma_0} = N^{\natural|\sharp}_{\rho_0} \times_{\Gamma_0^{\natural|\sharp}} N^{\natural|\sharp}_{\sigma_0} \times_{\Gamma_0^{\natural|\sharp}} \dagger \underbrace{G|M^{\natural|\sharp}_{\rho_0,\sigma_0}}_{M^{\natural|\sharp}} \\ \cap \\ N_{\rho_0} \times_{\Gamma_0} N_{\sigma_0} \times_{\Gamma_0} M. \end{array}$$

and where this last fiber product is none other than the quantity $M^{(\natural|\sharp\,\prime)}_{\rho_0,\sigma_0}$ which would correspond to the|in the case † of $\mathcal{G}_0^{\natural|\sharp\,\prime} = \mathcal{G}_0^+$ in all|all its entirety (denoted $M_{\rho,\sigma}$ on page|page 626 and following). In this group we defined the subgroup, which was denoted $M^{\natural|\sharp}_{\rho,\sigma}$ †on page 626′, and which I am now going to denote instead by $M^{?|!}_{\rho_0,\sigma_0}$, defined as 686

where $\Gamma_0 \overset{\text{dfn|déf}}{=} \widetilde{\Gamma}_{\mathbb{Q}}$, and

Uncertainty retained from cancelled passages.

140-3-s687-D001 [illeg. 2]
140-3-s687-D003 $\sigma_0|\rho_0 \quad \natural|\sharp \quad \natural|\sharp \quad = |\cdot \quad \natural|\sharp \quad \cap$
140-3-s687-D004 $G^{\natural}_{\rho_0,\sigma_0}|G^{\sharp}_{\rho_0,\sigma_0}$
140-3-s687-D005 M_{ρ_0,σ_0}
140-3-s687-D006 of

† †

$$(122)\qquad \dagger M^{?} = N^{?}_{\rho|\rho_0} \times_{\Gamma_0} Z^{!}_{\sigma_0} \times_{\Gamma_0} M$$

where, we recall that

$$N_{\rho_0} = \mathrm{Norm}_M(L_{\rho_0}), \quad N_{\sigma|\sigma_0} = \mathrm{Norm}_M(L_{\sigma_0})$$

$$(123)\quad \left\{ \begin{array}{l} \dagger \qquad Z_{\sigma|\sigma_0} = \mathrm{Centr}_M(L_{\sigma_0}) = \mathrm{Centr}_M(\sigma_0) \\ \\ Z^{!}_{\sigma_0} = Z_{\sigma_0} \cap M^{!}_0 \,(= N_{\sigma_0} \cap M^{!}_0) \subset N_{\sigma_0} \\ \\ N^{?}_{\rho_0} = Z^{!}_{\rho_0} \cdot \mu_{\tau_1|\tau'} \subset N_{\rho_0})|, \quad \text{where } Z^{!}_{\rho_0} = Z_{\rho_0} \cap M^{!} \ ) \end{array} \right.$$

so that we do indeed have

$$M^{?} \subset N_{\rho_0} \times_{\gamma_0} N_{\sigma_0} \times_{\Gamma_0} M.$$

m1367

Marginal note. m1367 independent|independently of the choice of a $\mathcal{G}^{\natural|\sharp}$

This recalled, we are going to set

$$(124)$$

$$\begin{array}{ccc} M^{\natural|\sharp\dagger?}_{\rho_0,\sigma_0} & \subset & M^{\natural|\sharp}_{\rho_0,\sigma_0} \\ \| & & \\ M^{\natural|\sharp}_{\rho_0,\sigma_0} \cap M^{?} & & \text{(intersection} \\ \| & & \text{in } N_{\rho_0} \times_{\Gamma_0} N_{\sigma_0} \times_{\Gamma_0} M\text{)} \\ \dagger & & \end{array}$$

and let us consider the homomorphism|the homomorphism induced by (119)

$$(125)$$

$$\begin{array}{ccc} M^{\natural|\sharp?}_{\rho_0,\sigma_0} & \longrightarrow & M^{\natural|\sharp} \\ \downarrow & & \downarrow \\ M^{?} & \longrightarrow & M \end{array}$$

. 687

Uncertainty retained from cancelled passages.

140-3-s688-D001 $\rho_0|\rho_0, \sigma_0 \quad \natural|\sharp$

140-3-s688-D002 $Z^{!}_{\rho_0}$

140-3-s688-D003 $\rho_0|\rho_0, \sigma_0$

140-3-s688-D004 Z_{ρ_0}

140-3-s688-D006 $!|\natural$

140-3-s688-D007 $N|M$!|? $\natural|\sharp$

[scan 689]

We have seen (page 626′, (55)) that $M^? \longrightarrow M$ makes $M^?$ an extension of M with kernel [p. 647]
μ_ρ. Thus the kernel of $M^?_{\rho_0,\sigma_0} \longrightarrow M^{\natural|\sharp}$ is equal to $\mu_\rho \cap M^?_{\rho_0,\sigma_0}$ now it is clear that we have $\mu_\rho \subset M^{\natural|\sharp}_{\rho_0,\sigma_0}$ cf. (119|114), hence $\mu_\rho \subset M^{\natural|\sharp?}_{\rho_0,\sigma_0}$.

We thus find a †homomorphism of exact sequences

(126)

$$\begin{array}{ccccccccc} & & & & M^{\natural|\sharp}_{\rho_0,\sigma_0} & & & & \\ & & & & \cup & & & & \\ 1 & \to & \mu_\rho & \to & M^{\natural|\sharp?}_{\rho_0,\sigma_0} & \to & M^{\natural|\sharp} & & \\ & & \| & & \curlyvee & & \curlywedge & & \\ 1 & \to & \mu_\rho & \longrightarrow & M^? & \longrightarrow & M & \longrightarrow & 1 \end{array}$$

and it remains to see whether the †homomorphism

$$(\infty) \qquad M^?_{\rho_0,\sigma_0} \twoheadrightarrow M^{\natural|\sharp}$$

is an epimorphism, and if not, characterize its image. † Let us note that the definition, (124) also gives † taking into account the expressions (118) for † $M^{\natural|\sharp}_{\rho_0,\sigma_0}$ and (122) for $M^?$:

$$(127) \qquad M^{\natural|\sharp?}_{\rho_0,\sigma_0} \simeq N^{\natural|\sharp?}_{\rho_0} \times_{\dagger\Gamma_0^{\natural|\sharp}} Z^{\natural|\sharp!}_{\sigma_0 0} \times_{\dagger\Gamma_0^{\natural|\sharp}} M^{\natural|\sharp}$$

m1369

Marginal note. m1369

$$\begin{array}{ccc} \gamma_0 & \subset & \widetilde{\Gamma}_{\mathbb{Q}} \\ \| & & \\ \mathrm{Im}(M^{\natural|\sharp}) & & \end{array}$$

where we have set

$$(128) \qquad \begin{cases} N^{\natural|\sharp?}_{\rho_0} = N^?_{\rho_0} \cap N^{\natural|\sharp}_{\rho_0} = N^{?|!}_{\rho_0} \cap M^{\natural|\sharp} \\ Z^{\natural|\sharp!}_{\sigma_0 0} = Z^!_{\sigma_0 0} \cap N^{\natural|\sharp}_{\sigma_0} = Z^!_{\sigma_0 0_\dagger} \cap M^{\natural|\sharp} \end{cases}$$

688

Uncertainty retained from cancelled passages.

140-3-s689-D001 exact sequence
140-3-s689-D002 arrow
140-3-s689-D003 Consid|Let us consider
140-3-s689-D004 this homomorphism fits diagram
140-3-s689-D005 of|some
140-3-s689-D006 γ_{ρ_0,σ_0}
140-3-s689-D008 γ_{ρ_0,σ_0}
140-3-s689-D011 ?|$\natural$

[scan 690]

and the homomorphism $(\infty|\alpha)$ is none other than the third projection of the fiber product. Thus the image of $(\infty|\alpha)$ is none other than the inverse image, in $M^{\natural|\sharp}$ under $M^{\natural|\sharp} \longrightarrow \gamma_0 = \gamma_{\dagger\rho_0,\sigma_0}$ of the image of

$$(\infty\infty|\alpha\alpha) \qquad N^{\dagger\natural|\sharp?}_{\rho_0} \underset{\dagger\Gamma_0^{\natural|\sharp}}{\times} N^{\natural|\sharp!}_{\sigma_0 0} \longrightarrow \dagger\Gamma_0^{\natural|\sharp} (\subset \Gamma_0)$$

It thus remains to show that this homomorphism is surjective, or again that the homomorphisms

$$\begin{cases} N_{\rho_0}^{\natural|\sharp ?} \longrightarrow \dagger\Gamma_0^{\natural|\sharp} \\ N_{\sigma_0 0}^{\natural|\sharp !} \longrightarrow \Gamma_0^{\natural|\sharp} \end{cases}$$

are surjective (the first will then be an extension of $\Gamma_0^{\natural|\sharp}$ by μ_ρ, the second an isomorphism). So let $u \in \Gamma_0^{\natural|\sharp}$, coming from $u_0 \in M^{\natural|\sharp}\dagger \simeq G_{\rho,\sigma}^{\natural|\sharp}$ thus satisfying conditions a) b) c) d) of (107). We thus see that there exists $\beta_0 \in \mathcal{G}_0^{\natural|\sharp} \subset M^{\natural|\sharp}$ such that $\beta^{-1}u_0 \in N(\sigma_0)$.

m1371

Marginal note. m1371 Here I have avoided the verifications, thinking I would return to them in the following pages, it turns out that this is not urgent – it was more important to develop the special cases "Fermat tower|towers" to clarify the ideas of the general formulations of the $M_{\rho,\sigma}$….

689

Uncertainty retained from cancelled passages.

140-3-s690-D001 $\natural|\sharp$
140-3-s690-D002 $\natural|\sharp$
140-3-s690-D003 γ_{ρ_0,σ_0}
140-3-s690-D004 γ_{ρ_0,σ_0}
140-3-s690-D005 γ_{ρ_0,σ_0}
140-3-s690-D007 ρ,σ

[scan 691]
[p. 648]

9°) Digression on group formalism.

ⓐ

Let

$$(1) \qquad G, \quad \mathcal{G} \subset G, \quad N \overset{\varphi_N}{\hookrightarrow} G, \quad \text{hence} \quad SN = \dagger\varphi^{-1}(\mathcal{G}) \subset N$$

be the situation of a group G, invariant subgroup $\mathcal{G}$, and subgroup N such that

$$(\alpha) \qquad N \xrightarrow{\text{epimorphism}} G/\mathcal{G}\dagger, \qquad \text{i.e.} \quad N/SN \xrightarrow{\sim} G/\mathcal{G}$$

† We are going to reconstruct this situation from the following one

$$(2) \qquad \mathcal{G}, \quad N, \quad SN \subset N\dagger, \quad N \xrightarrow{\dagger\Theta} \mathrm{Aut}(\mathcal{G}), \quad SN \xrightarrow[\varphi=\varphi_{SN}]{\dagger} \mathcal{G}$$

where $\mathcal{G}, N$ are groups, SN a normal subgroup of N, $N \longrightarrow \mathrm{Aut}(\mathcal{G})$ a homomorphism of groups, i.e. an action of N on $\mathcal{G}$; finally $SN \longrightarrow \mathcal{G}$ a homomorphism of groups. We assume that †

$$(3) \quad \begin{cases} \text{a) the action of } SN \text{ on } \mathcal{G} \text{ induced|induced by that of } \mathcal{G}|G, \\ \text{is also that deduced from } SN \longrightarrow \mathcal{G}, \\ \text{via the adjoint operation of } \mathcal{G}, \\ \text{b) } SN \longrightarrow \mathcal{G} \text{ is compatible with the actions of } N. \end{cases}$$

and † † clear how (1) defines (2), let us show how a system (2) allows one to reconstruct a system (1).

690

Uncertainty retained from cancelled passages.

140-3-s691-D001 $N \cap \mathcal{G}$
140-3-s691-D002 $\overset{\text{dfn}}{=} \gamma$
140-3-s691-D003 is an epimorphism
140-3-s691-D004 Aut
140-3-s691-D005 φ
140-3-s691-D006 φ
140-3-s691-D008 $\mathcal{G}$
140-3-s691-D009 It is|I see
140-3-s691-D010 It is|I see

[scan 692]

We thus start from (2), we set

$$(4) \qquad S_0G = N.\mathcal{G} \qquad \text{(Ch direct|direct product)},$$

we define a homomorphism

$$(5) \quad \begin{cases} SN \overset{i}{\hookrightarrow} S_0G \\ u \longmapsto u.\varphi(u)^{-1} = \varphi(u)^{-1}.u \end{cases}$$

m1374

Marginal note. m1374 N.B. we have (by hypothesis (a)) for $u \in SN$, $x \in \mathcal{G}$

$$(*) \quad \begin{cases} u(x) = \underbrace{uxu^{-1}}_{\substack{\text{in } S_0G \\ =N.\mathcal{G}}} = \varphi(u)x\varphi(u)^{-1} \end{cases}$$

so if $x = \varphi(u)$, we have

$$u(x) = uxu^{-1} = x$$

i.e. $x = \varphi(u)$ and u commute.

Let us check that it is a homomorphism †

$$i(u'u) = (u'u)\varphi(u'u)^{-1}$$
$$i(u')i(u) = u'\varphi(u')^{-1}u\varphi(u)^{-1}$$
$$= u'u \underbrace{(u^{-1}\varphi(u'^{-1})u)}_{\substack{=\varphi(u^{-1})\varphi(u'^{-1})\varphi(u) \\ \text{by a)}}} \varphi(u^{-1})^{\dagger}$$
$$= u'u\varphi(u^{-1}u'^{-1}) = (u'u)\varphi((u'u)^{-1}) = (u'u)\varphi(u'u)^{-1} \qquad \text{ok.}$$

We have not yet used the fact that is normal in N, and hypothesis (b) this implies that the image of (4)|(5) is normal in S_0G, for we have, if $a \in N$, $u \in SN$

$$\dagger \qquad \dagger \quad a(u.\varphi(u)^{-1})a^{-1} = \underbrace{aua^{-1}}_{\in SN} . \underbrace{a\varphi(u^{-1})a^{-1}}_{=\varphi(au^{-1}a^{-1})=\varphi(au^{-1}a^{-1})^{-1}}$$

i.e. the homomorphism (5) † commutes with the actions of N on SN and on $S_0G \supset N$ – thus the image of SN is normalized by $N \subset S_0G$. We still have to check that it is also normalized by $\mathcal{G}$, †but we have better:

The image of $SN \overset{i}{\to} S_0G$ commutes with $\mathcal{G}$, i.e. 691

Uncertainty retained from cancelled passages.

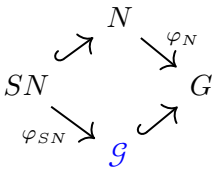
140-3-s692-D001 we
140-3-s692-D003 -1
140-3-s692-D006 (6)(6)
140-3-s692-D007 $(*)$
140-3-s692-D008 is
140-3-s692-D009 $\mathcal{G}$

[scan 693]

$\forall u \in SN$, $u\varphi(u)^{-1}$ induces the trivial operation on $\mathcal{G}$ — which is indeed trivial by (3) a). [p. 649]
We can therefore set

$$(6) \qquad G \overset{\text{déf|dfn}}{=} S_0 G/i(SN),$$

we then find a commutative diagram
(7)



,
we see that the image of $\mathcal{G}$ in G is an invariant subgroup, and that we have an isomorphism on the quotients:

$$(8) \qquad N/SN \xrightarrow{\sim} G/\mathcal{G},$$

which implies that we also have an isomorphism

$$(9) \qquad \mathrm{Ker}(\varphi_{SN} : SN \longrightarrow \mathcal{G}) \xrightarrow{\sim} \mathrm{Ker}(N \xrightarrow{\varphi_N} G);$$

(10)

$$\begin{array}{l} \varphi_N : N \longrightarrow G \text{ is an embedding} \\ \Updownarrow \\ \varphi_{SN} : SN \longrightarrow \mathcal{G} \text{ is an embedding.} \end{array}$$

We thus find equivalences of categories, inverse to one another, between the categories of 692
[scan 694]
situations (1), and that of situations (2).
ⓑ Let us now consider a situation

$$(11) \qquad \begin{cases} G, \quad \mathcal{G} \subset G, \quad (\varphi_i : N_i \longrightarrow G)_{i \in I} \\ \text{hence } SN_i = \varphi_i^{-1}(\mathcal{G}) \end{cases}$$

under the condition

$$(12) \qquad \forall i \in I, \quad \varphi_i : N_i \longrightarrow G/\mathcal{G} = \gamma \quad \text{epimorphism}$$

can we reconstruct it from

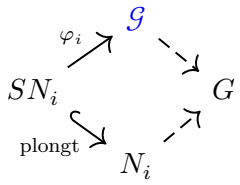
$$(13) \qquad \mathcal{G}, \ (N_i, SN_i, \varphi_i, \Theta_i)_{i \in I}$$

where $\forall i \in I$, $(N_i, SN_i, \varphi_i, \Theta_i)$ satisfies the conditions (3) a) and b) — where|and of course $\forall i \in I$

$$(14) \qquad \dagger\Theta_i : N_i \longrightarrow \mathrm{Aut}(\mathcal{G}), \qquad \varphi_i : SN_i \longrightarrow \mathcal{G}.$$

It is tempting to reconstruct G with the help of|of the $\mathcal{G}$ and of the N_i, as a sort of "amalgamated sum" relative to:

†



693

Uncertainty retained from cancelled passages.

140-3-s694-D001 φ_i

140-3-s694-D002 (15)

[scan 695]

†

[p. 650]

But it is more plausible to say, that for every i, we reconstruct via|here $N_i, SN_i, \varphi_i, \Theta_i$ the situation (for i fixed)

†(15)

$$\begin{array}{ccc} \dagger N_i & \longrightarrow & G_i(\simeq G) \\ \uparrow & & \uparrow \\ SN_i & \longrightarrow & \mathcal{G} \end{array}$$

and that we reconstruct (11) itself by giving ourselves a transitive system of isomorphisms between the G_i ($i \in I$), inducing the identity on $\mathcal{G}$. Such a datum implies in the first place the datum of a transitive system of isomorphisms between the

$$(16) \qquad \gamma_i \overset{\text{déf|dfn}}{=} N_i/SN_i$$

— which we will identify therefore with one and the same group

$$(17) \qquad \begin{array}{c} \gamma, \text{ common quotient group} \\ \text{of the } N_i \text{ (by the } SN_i). \end{array}$$

Let us now identify the $G/\mathcal{G}_i = \gamma_i$ with γ

$$(18) \qquad G_i/\mathcal{G} \simeq \gamma,$$

and form the fiber product

694

Uncertainty retained from cancelled passages.

140-3-s695-D001 Starting|Start

140-3-s695-D002 (17)

140-3-s695-D003 φ_i

[scan 696]

$$(19) \qquad g = \underset{\substack{i \in I \\ \text{(fiber product over } \gamma)}}{\prod_{\gamma} G_i} \supset \mathcal{G}^I$$

g is an extension of γ by $\mathcal{G}^I$

The datum of a transitive system of isomorphisms between the G_i, inducing the identity on the subgroups $\mathcal{G}$, and compatible with the system † transitive of isomorphisms already considered between the $\gamma_i = N_i/SN_i \simeq G_i/\mathcal{G}$, is then equivalent to the datum of a subgroup G of g, satisfying the conditions

$$(20) \qquad \begin{cases} \text{a)} & G \cap \mathcal{G}^I = \text{diagonal } \delta(\mathcal{G}) \text{ of } \mathcal{G}^I \\ \text{b)} & \text{The projection } G \longrightarrow \gamma \text{ is an epimorphism} \end{cases}$$

In other words, we have constructed, with the help of (13) and of the transitive system of isomorphisms between the γ_i, an extension g of their "common value" γ, by $\mathcal{G}^I$ — and the additional structure, corresponding to the datum (11), is the "restriction" of this extension to an extension of γ by the diagonal subgroup of $\mathcal{G}^I$. 695

Uncertainty retained from cancelled passages.

140-3-s696-D001 of isomorphisms|of the

Cote 140-4

[scan 1]

Pages|Page 650–787 + table of contents [scan 2]

N.B. Page 650 appears twice (numbering error) 1

50. Schematic envelopes of discrete groups: metabelian case 650 [scan 3]

1°) Unipotent sections same

2°) Attempt at a characterization of quasi-unipotent sections 652′

3°) Group scheme associated with certain extensions of discrete groups (abelian kernel) 655′

4°) Schematic envelope of a metabelian group 658′

5°) Universal characterizatio^{n} of the schematic envelope 662′

51. Example: the Fermat tower. 664

1°) The groups $\mathfrak{S} = \mathrm{Sl}(2,\mathbb{Z})/[\Pi_0,\Pi_0]$, and $\underline{\mathfrak{S}}$ same

2°) $\underline{Z}|Z_{\rho,\sigma}$: provisional version 665′

3°) The groups Φ_N^* and variants 673

4°) The sequence

$$1 \to \underline{N}^{0+}_{\rho,\sigma} \to \underline{N}_{\rho,\sigma} \to \Phi_{12}^* \to 1$$

, and relations with $\mathrm{Sl}(2,\mathbb{Z}/12\mathbb{Z})$ 677

5°) $\underline{M}_{\rho,\sigma}$, $\underline{M}^{\natural|\sharp}_{\rho,\sigma}$ via $N_{\rho,\sigma}$ 682

6°) † sequences

$$1 \to \mathbb{G}_a \to \underline{Z}_{\rho,\sigma} \to \widehat{\Phi}_{12} \to 1$$

, and

$$1 \to \underline{\mathfrak{S}}^{+} \to \underline{G}_{\rho,\sigma} \to \widehat{\Phi}_{12} \to 1$$

684

7°) $\underline{N}_{\rho}$, $\underline{N}^{\natural|\sharp}_{\rho}$ and so on 686

8°) $\underline{N}_{\sigma}$, $\underline{N}^{\natural|\sharp}_{\sigma}$ and so on 689

9°) Return to $\underline{M}_{\rho,\sigma}$, $\underline{M}^{\natural|\sharp}_{\rho,\sigma}$ and so on 690

10°) $\underline{M}_{\rho,\sigma}(0)|[0]$, $\underline{M}_{\rho,\sigma}[\frac{1}{2}]|(\frac{1}{2})$, $\underline{M}_{\rho,\sigma}(-\bar{j})$ and so on 691′

11°) The homomorphism $M \longrightarrow M_{\rho,\sigma} = \underline{M}_{\rho,\sigma}(\widehat{\mathbb{Z}})^{\wedge}$, and relations with the tower of Fermat curves: statement of the problem. 695

12°) Splittings of the extension $G_{\rho,\sigma}/\mu_G$ of $\widehat{\mathbb{Z}}^* \times \mathfrak{S}_3$ by $\widehat{M}$. 700

13°) †Characterization of the Fermat tower; †relations of compatibility, with $M \longrightarrow G_{\rho,\sigma}$ †707

14°) Relation with generalized Jacobians 720′

15°) Interpretation of $\Gamma_{\mathbb{Q}} \longrightarrow N_{\sigma}^{\natural|\sharp} \simeq \widehat{\mathbb{Z}}^*$. $\widehat{\mathbb{Z}}$, and digression on Kummer extensions 725′

16°) Reflections on Fermat's theorem 733 2

Uncertainty retained from cancelled passages.

140-4-s003-D001 $\underline{Z}_{\rho,\sigma}$

140-4-s003-D002 Digression on

140-4-s003-D004 conjectures

140-4-s003-D005 703

52. Lie approximations of Galois-Teichmüller systems. [scan 4]

1°) Notion of L- and ΠL-group: discrete case 737

2°) Profinite case 744

3°) †Lie approximations of systems of Galois-Teichmüller 748

4°) Lie approximation induced by a "geometric" Lie approximation 758

53. The special Teichmüller group $S\mathcal{T}_{1,1} \simeq \mathrm{Gl}(2,\mathbb{Z})^{\sim}$, and the group $SN_{1,1}$ 760

1°) Differential action and special Teichmüller group (provisional statement) same

2°) The groups $\mathrm{Gl}(2,\mathbb{R})^{\pm 1\sim}$, $\mathrm{Sl}(2,\mathbb{R})^{\sim}$ 761′

3°) The group $\mathrm{Gl}(2,\mathbb{Z})^{\sim}$ 763

4°) The isomorphism $\mathrm{Gl}(2,\mathbb{Z})^{\sim} \simeq S\mathcal{T}_{1,1}$ 766′|765′

5°) Digression on coverings of germs of spaces, and relations between topological germs and differentiable germs of surfaces 768′

6°) 3

Uncertainty retained from cancelled passages.

140-4-s004-D002 Discretification admissible homomorphisms

[scan 5]

[p. 650 bis]

50. Schematic envelopes of discrete groups.

†(Metabelian case)

1.† Let G be a group scheme locally of finite presentation over a base scheme S (the case important for us will be $S = \mathrm{Spec}(\mathbb{Z})$). A section g of G over S is said to be unipotent if on every geometric fiber, † $G_{\bar{s}}$, the element $g_{\bar{s}} \in G_{\bar{s}}(\bar{k}|k^a)$ $(k = k(\bar{s}))$ is unipotent, i.e. if there exists a group homomorphism Unipotent sections

$$(1) \qquad \mathbb{G}_{ak} \xrightarrow{\varphi_{\bar{s}}} G_k \qquad k = k(\bar{s})$$

such that

$$(2) \qquad g_{\bar{s}} = \varphi_{\bar{s}}(1)$$

We say that g is quasi-unipotent if for every $\bar{s}$, there exists $n \in \mathbb{N}^*$ such that $g_{\bar{s}}^n$ is unipotent. If the characteristic of $\bar{s}$ is $p > 0$, then this also means that $g_{\bar{s}}$ is of finite order. We see that if S is quasi-compact, then g is quasi-unipotent iff $\exists n \in \mathbb{N}^*$ such that g^n is unipotent.

†

† For g to be unipotent, it suffices that there exist a homomorphism

$$(3) \qquad \varphi : \mathbb{G}_a \longrightarrow G$$

4

Uncertainty retained from cancelled passages.

140-4-s005-D001 2

140-4-s005-D002 g

140-4-s005-D003 of

140-4-s005-D004 $g|G$ if|which unipotent|unipotent for a

140-4-s005-D005 ?

[scan 6]

such that we have

$$(4) \qquad \varphi(1) = g.$$

I wonder whether this is also necessary, in "good" cases. Let us first take the case where $S = \operatorname{Spec} K$. In characteristic 0, we have a uniqueness result for φ, which allows us to apply descent theory (valid over any base in characteristic 0): if after an extension $S' \longrightarrow S$ faithfully flat quasi-compact, there exists φ' having the desired properties, then φ exists (over S itself|=). So over a field of characteristic 0 it is OK. In characteristic $p > 0$ the difficulty comes from the fact that there are group homomorphisms $\mathbb{G}_a \longrightarrow \mathbb{G}_a$: (such as $x \mapsto x^p - x$) which are nontrivial, and which are trivial on 1. We shall get around this perhaps by applying descent with φ which are monomorphisms. This raises the question:|This leads to the question: Is a subgroup of G isomorphic to $\mathbb{G}_a$ known when we know an everywhere nontrivial section? Is this at least true over a field as base?

I do not feel like launching into these questions (certainly interesting!), and prefer 5

[scan 7]

to limit myself to a case closer to the one I am after, $S = \operatorname{Spec} \mathbb{Z}$, assuming S noetherian [p. 651]
integral, † with generic point η of characteristic 0. This already implies, by the result over a field of characteristic 0, that there exists a nonempty open U

$$(5) \qquad \underset{\substack{\text{open}\\ \text{nonempty}}}{U \subset S} \quad \text{and} \quad \varphi_U : \mathbb{G}_a|U \longrightarrow G|U$$

such that we have (4) on U – and (for U chosen) this φ_U is unique. By uniqueness, we thus see that there is a largest open $\mathcal{U}|U$ over which we can give such a $\varphi|\psi_{\mathcal{U}|U}$, and the question comes down to seeing whether we necessarily have $\mathcal{U}|U = S$.

If we had $U \neq S$, localizing if necessary at a maximal point of $S \setminus \mathcal{U}|S \setminus U$, we † reduce to examining the case where S is local, with closed point s – and to trying to|a little of prove that in this case, we can extend φ_U over all of S (whence a contradiction…). In the case which interests us, S is regular of dimension 1, so by localizing we find a trait. I shall confine myself to this case, crucial 6

Uncertainty retained from cancelled passages.

140-4-s007-D001 of characteristic

140-4-s007-D003 can

[scan 8]

,|. probably if we want to elucidate the general case, and which suffices for my purpose.

Let η be the generic point of S, s its closed point. Consider the subgroup $L_\eta = \varphi_\eta(\mathbb{G}_{a\eta}) \subset G_\eta$, let L be its schematic closure in G. which is a reduced subgroup scheme of G. Let † $L^0 = L \setminus \big(\underset{\substack{\text{components}\\ \text{of } L_s \text{ non-}\\ \text{identity}}}{\bigcup} L_{s,i} \big)$ be its identity component. We have won if we prove that $L^0 \simeq \mathbb{G}_{aS}$. In fact, we are not even|= sure that (L_s^0 † $(= (L_s)^0)$ is reduced! But for $\varphi : \mathbb{G}_{aS} \longrightarrow G$ extending φ_η to exist, it is probably not necessary that we have $L^0 \simeq \mathbb{G}_a$ – for it is possible that φ is not a monomorphism, i.e. that φ_s is not a monomorphism – it could even|= be the trivial homomorphism (if $g_s = 1$)! But even if $g_s \neq 1$, it is not clear that φ_s is a monomorphism.

Let us take the example where

$$(6) \qquad G = \underline{\operatorname{Aut}}_{\mathcal{O}_S}(M) \simeq \operatorname{Gl}(n)_S$$

M a † free module of rank n over S, then we shall have (since η of characteristic 0)

$$(7) \qquad \varphi_\eta(\lambda) = \sum \frac{X_\eta^i}{i!} \lambda^i$$

where

$$X_\eta | X \in \mathrm{End}_K(M_K) \qquad K = k(\eta)$$

7

Uncertainty retained from cancelled passages.
140-4-s008-D001 [illeg. 1]
140-4-s008-D002 is
140-4-s008-D004 a

[scan 9]
[p. 652]

is a nilpotent endomorphism of M_K. thus an element

$$(8) \qquad X \in \mathrm{End}_A(M) \otimes_A K, \qquad X^n = 0.$$

(A the affine ring of S, which is a † gauge) We shall have by hypothesis

$$g_\eta = \varphi_\eta(1) = \sum \frac{X^i}{i!} \in \begin{matrix} \mathrm{Aut}(M) \\ \cap \\ \mathrm{Aut}(M_K) \end{matrix}$$

i.e.

$$(9) \qquad \sum \frac{X^i}{i!}, \quad \sum (-1)^i \frac{X^i}{i!} \in \mathrm{End}_A(M)$$

and the question whether φ_η comes from a homomorphism of groups $\mathbb{G}_a \longrightarrow G$ comes down to that of knowing whether

$$(10) \qquad \frac{X^i}{i!} \in \mathrm{End}_A(M) \qquad \forall\, (1 \leq i \leq n-1) | 1 \leq i \leq n-1$$

(for $i = 0$ and $i \geq n | i = n$ this is trivial)

For $n = 2$ this is trivial, For $n = 3$ ditto|ditto (in (9) there is a single term with a denominator).

the first doubtful case is $n = 4$

$$\varphi_\eta(\lambda) = 1 + X\lambda + \frac{X^2}{2}\lambda^2 + \frac{X^3}{6}\lambda^3 \qquad \begin{matrix} X \in M_4(K) \\ X^4 = 0 \end{matrix}$$

we have by hypothesis $\varphi_\eta(1)$ and $\varphi_\eta(-1)$ in $M_4(A)$, i.e.

$$(11) \qquad \frac{X^2}{2} \pm \frac{X^3}{6} \in M_4(A)$$

we wonder whether it follows that

$$(12) \qquad \frac{X^2}{2}, \frac{X^3}{6} \in M_4(A) \qquad ?$$

If 2 invertible in A this is clear, for we deduce from (11) that $X^2 \in M_4(A)$ hence $\dfrac{X^2}{2} \in M_4(A)$, hence also $X^3/6 \in M_4(A)$, OK. Thus the difficulty

8

Uncertainty retained from cancelled passages.
140-4-s009-D002 [illeg. 1]

arises here in residual characteristic 2. † Suppose † for simplicity that 2 is a uniformizer π of A, we shall have, if $X \notin M_4(A)$ [scan 10]

$$X = \frac{1}{\dagger 2^{\nu}} X_0 \quad \text{with} \quad \begin{matrix} X_0 \in M_4(A) \ \dagger \\ X_0 \notin \dagger 2 M_4(A) \end{matrix}$$

hence † the datum of X is equivalent to that of

$$X_0 = \mid \in \dagger (a_{ij})_{1 \leq i,j \leq 4}, \quad \dagger \quad \text{the } a_{ij} \in A \text{ not all}$$

divisible by 2, with †.

$$(11\,\text{bis}) \qquad X_0^4 = 0, \qquad \frac{1}{2}\frac{X_0^2}{2^{2\nu}} \pm \frac{1}{6}\frac{X_0^3}{2^{3\nu}} \in M_4(A)$$

— and the question is whether it follows that we have|have

$$(12\,\text{bis}) \qquad \frac{1}{2}\frac{X_0^2}{2^{2\nu}}, \frac{1}{2}\frac{X_0^3}{2^{3\nu}} \in M_4(A) \qquad ?$$

Of† course, the † answer is trivially|totally yes, if $X_0^3 = 0$ i.e. $X^3 = 0$ — so we are interested in the case where $X_0^3 \neq 0$, i.e. † g_η is "principal unipotent.
m1391

Marginal note. m1391 Attempt at a characterization of quasi-†unipotent sections

2. To get to the bottom of it, one would need a counterexample in the preceding context (with a nilpotent matrix of order 4) — or a theorem ! But for my purpose, it seems that it is not essential to have a solution of this problem.

On the other hand, I shall try the

Theorem (!) Let G be a group scheme (separated) locally of finite presentation over a scheme S† which is quasi-compact, and let g be a section of G. For g to be quasi-unipotent, it 9

Uncertainty retained from cancelled passages.
140-4-s010-D001 Let
140-4-s010-D002 π
140-4-s010-D003 π
140-4-s010-D004 πM_4
140-4-s010-D005 π
140-4-s010-D006 and the
140-4-s010-D007 [*illeg.* 1]
140-4-s010-D008 $\in M_4(A)| \in A$
140-4-s010-D009 X_0^4
140-4-s010-D010 [illeg. 1]
140-4-s010-D011 q
140-4-s010-D012 $X_{0\eta}$
140-4-s010-D013 [illeg. 1]
140-4-s010-D014 [illeg. 1]

m1393

[scan 11]
[p. 653]

Marginal note. m1393

is necessary and sufficient that there exist an integer $n > 0$, and a homomorphism of group schemes $\varphi : \mathbb{G}_a \longrightarrow G$, such that we have

$$(13) \qquad \varphi(1) = g^n.$$

Sufficiency being trivial, it is a matter of examining necessity. For simplicity, we shall again suppose S noetherian and integral — once S is assumed noetherian, I think that the reduction to the integral case should be done by easy dévissages, which I do not feel like doing, for lack of essential need ! I shall therefore examine the case S noetherian integral, generic point η

a) Characteristic of η is > 0. Then † the hypothesis on g_η is that g_η is of finite order — hence $\exists n > 0$ such that $g_\eta^n = 1$, hence $g^n = 1$. We shall take in (13) this n, and φ the trivial homomorphism.

b) Case where η of characteristic zero. Let φ

$$(14) \qquad \varphi_{1\eta} : \mathbb{G}_{a\eta} \longrightarrow G_\eta$$

be defined by the condition

$$g_\eta = \varphi_{1\eta}(1)$$

let U_1 be its "open set of definition" in S.

Let

$$\varphi_{n\eta}(\lambda) = \lambda^n = \varphi_{1\eta}(n\lambda) : \mathbb{G}_{a\eta} \longrightarrow G_\eta$$

and let U_n be its open set of definition. We thus have

$$(15) \qquad \varphi_{n\eta}(1) = g^n \quad \text{on } U_n$$

10

Uncertainty retained from cancelled passages.

140-4-s011-D001 g_η

and [scan 12]

$$(16) \qquad n \mid n' \quad \Longrightarrow \quad U_n \subset U_{n'}.$$

We must prove that the union of the U_n is S. We are reduced to the case S local, and where φ_1 is already defined in

$$(17) \qquad U_1 = S \setminus \{s\}, \qquad s \text{ the closed point of } S$$

and this comes down to the

Lemma (!) Let S be noetherian integral, G a group scheme separated locally of finite presentation over S, U a nonempty open set of S, $\varphi_U : \mathbb{G}_{aU} \longrightarrow G_U$ a homomorphism of groups, we assume that $\varphi_U(1) = g_U$ extends to a section g of G over all of S (necessarily unipotent, by known general results of|of the SGA 3) Then there exists an integer > 0 such that $\varphi_U^n : \lambda \longmapsto \varphi_U(\lambda^n)$ extends to|over S a homomorphism

to

$$\varphi_n : \mathbb{G}_a \longrightarrow G.$$

I am not going to try to prove it in this general context of general group schemes, and shall restrict myself, as a decisive test (to carry conviction that it works) the case where $G = \mathrm{Gl}(n)_S$, as earlier, and that where S moreover is regular of dimension 1.

Thus we are reduced to the situation of a gauge A, and of $X \in M_n(K)$ (K the field of fractions) satisfying 11

[scan 13]
[p. 654]

$$(18) \qquad \begin{cases} X \in M_n(K) \\ X^n = 0, \quad \sum \frac{X^i}{i!} \in M_n(A), \quad \sum \frac{(-1)^i X^i}{i!} \in M_n(A) \end{cases}$$

and we want to show that there exists an integer $d > 0$ such that we have

$$(19) \qquad \frac{d^i X^i}{i!} \in M_n(A) \qquad (1 \leq i \leq n-1)$$

If A is of characteristic 0 i.e. if it is a $\mathbb{Q}$-algebra, this is equivalent moreover :

$$\frac{X^\dagger}{\dagger} \in M_n(A) \qquad \dagger$$

m1397

Marginal note. m1397



m1398

Marginal note. m1398 the case of characteristic zero is settled in the affirmative by means of the logarithm function — setting

$$X + \frac{X^2}{2!} + \cdots + \frac{X^{n-1}}{(n-1)!} = Y,$$

we shall have

$$\exp X = 1 + Y, \quad \text{hence}$$
$$X = \log(1+Y) = Y - \frac{Y^2}{2} + \frac{Y^3}{3} \ldots$$

Works over any base ring of characteristic zero…

b|0

i.e. It would have to be that in the case of characteristic 0, it is φ_U itself that can already extend to all of S. The first doubtful case in this regard seems to me $n|m = 5$

$$(20) \qquad \begin{cases} \dagger \quad X \in M_5(A|K), \ X^5 = 0 \\ \dfrac{X^2}{2} \pm \dfrac{X^3}{3!} + \dfrac{X^4}{\dagger 4!} \in M_5(A) \\ \overset{?}{\Longrightarrow} \quad \dagger \quad X \in M_5(A) \end{cases}$$

On the other hand, in the case considered a little earlier ($n = 4$, residual characteristic 2 — the only one that causes difficulties) it is clear that it suffices to take for d a sufficiently large power of 2.

This remark applies to the mixed characteristic cases. We should indeed have this

Corollary to the lemma dubious : if S is local, of residual characteristic $p > 0$, and if φ is defined on an open set which contains $S - V(p)$, (i.e. if the complement of U is of characteristic $p> \mid \geq 0$ — in fact unnecessary of course that S be local) then $\exists q = p^{r|n}$ such that φ^q extends. 12

Uncertainty retained from cancelled passages.

140-4-s013-D002 $i|0$

140-4-s013-D003 $i!$

140-4-s013-D004 [illeg. 1]

140-4-s013-D005 $X^5 = 0$

140-4-s013-D006 2

140-4-s013-D007 X^2, X^3, X^4

When G is affine, this should be trivial. We are going to try the [scan 14]

Proposition Let|Let S be a quasi-compact prescheme, n an integer $\geq 1| \geq 0$, $\dagger$ $S_0 = V(n) \subset S$ the closed subscheme defined by $n.1_S$ (thus if $x \in S$, we have $x \in S_0$ iff $\dagger$ $\dagger$ characteristic $x \mid n$), $U = S \setminus S_0$, G an affine group scheme of finite presentation|finite type over S, and

$$(21) \qquad \varphi_U : \mathbb{G}_a|U \longrightarrow G|U$$

a group homomorphism. We assume U schematically dense in S. Then there exists a power integer $m = n^\nu$ of n, and a homomorphism of group schemes

$$\psi_m : \mathbb{G}_a \longrightarrow G; |,$$

such that we have

$$(22) \qquad \psi_m|U = (\varphi_U^m : \lambda \longmapsto \varphi_U(m\lambda) = \varphi_U(\lambda)^m)$$

Let us indeed consider the endomorphism

$$u_m : \mathbb{G}_a \longrightarrow \mathbb{G}_a \qquad \lambda \longmapsto m\lambda$$

Then formula (22) means that we have commutativity

$$\begin{array}{ccc} \mathbb{G}_{aU} & \xrightarrow{(u_m)_U} & \mathbb{G}_{aU} \\ & \searrow^{\psi_U} & \downarrow{\scriptstyle \varphi_U} \\ & & G_U \end{array}$$

and the existence of the $\psi_i|\psi_m$ for large m, means that for large m the composite homomorphism

$$\varphi_U \circ u_m : \mathbb{G}_{aU} \longrightarrow |\text{—} G_U$$

extends to a group homomorphism $\mathbb{G}_a \longrightarrow G$. 13

Uncertainty retained from cancelled passages.
140-4-s014-D001 prime|prime
140-4-s014-D002 characteristic x is of
140-4-s014-D003 characteristic which divides n

[scan 15]

For reasons of schematic density and uniqueness, it suffices to prove that φ_U extends to a [p. 655]
morphism of schemes – this extension will automatically be a homomorphism of group schemes. † We may assume S Affine, let A be its ring, and let B be that of G. So the ring of U will be † $A_n \overset{\text{déf}}{=} A[1/n]$, that of G will be $B \otimes_A A_n = B_n$, by hypothesis we have a homomorphism (corresponding to $\varphi_U^*|\varphi_U$)

$$(23) \qquad B_n \xrightarrow{\varphi} A_n[T]$$

whereas the homomorphism † corresponding to ψ_{mU} will be the composite of the preceding one with

$$(24) \qquad \begin{array}{ccc} A_n[T] & \xrightarrow{\psi_m^*|u_m^*} & A_n[T] \\ T & \longmapsto & mT \qquad \text{thus} \qquad T^i \longmapsto m^i T^i \end{array}$$

Let † $x = a_0 + a_1 T + \cdots + a_N T^N \in A_n[T]$ of which|thus the $a_i \in A_n$ we shall therefore have

$$\psi_m^*(x) = \sum a_i m^i T^i = \sum (a_i n^{i\nu}) T^i$$

thus if $m = n^\nu$ is large enough, we shall have

$$\dagger$$

$$m^i a_i \in A \quad \text{i.e} \quad m^i a_i \in A|T^i[T] \qquad \forall i \underline{\underline{\geq | >}} 1$$

thus

$$\psi_m^*(x) \equiv a_0 \pmod{A[T]}$$

which shows

$$(25) \qquad \psi_m(x) \in A[T] \text{ for } m = n^\nu \text{ large} \iff a_0 \in A$$

14

Uncertainty retained from cancelled passages.
140-4-s015-D002 that of [illeg. 1]
140-4-s015-D003 =
140-4-s015-D004 composite
140-4-s015-D005 x
140-4-s015-D006 $\sim$ A

[scan 16]

Applying this to the images of elements of B which generate B as an A-algebra, we thus have

Corollary S † n, S_0, U being as in the proposition, and G being an affine scheme of finite type over S (there is no longer any question of a group structure), and $\varphi_U : \mathbb{E}^1_U \longrightarrow G_U$ a morphism of schemes, † g_U † $\overset{\text{déf}}{=} \varphi_U(0)$, For there to exist a † power $m = n^\nu$ of n such that $\varphi_U \circ \psi_m|u_m$ † (where u_m given by (24)) extends to S, it is necessary and sufficient that g_U extend to a section g of G over S.

Remark the preceding arguments prove in any case the dubious "theorem" on page 662|652' in the case where S is noetherian of dimension 1 (the case of interest to us being the one where the generic point of S is of characteristic 0).

m1403

Marginal note. m1403 Group schemes associated with extensions of discrete groups

3. Let $M_0|M$ now be a free $\mathbb{Z}$-module of finite type; we are going to construct a pro-group scheme over S, associated with the projective system of sub-$\mathbb{Z}$-modules of finite index in M. To carry out the construction, I am going to work out † a 15

Uncertainty retained from cancelled passages.
140-4-s016-D001 S_0
140-4-s016-D002 exist
140-4-s016-D003 =
140-4-s016-D004 [illeg. 1]
140-4-s016-D005 (where
140-4-s016-D007 thus

[scan 17]

m1405 [p. 656]

Marginal note. m1405 I am complicating my life over the construction of $\underline{\mathfrak{S}}$, it suffices to set $\underline{\mathfrak{S}} = \mathfrak{S}_{S_0} \times_{M_{S_0}} \underline{M}_{S_0}$.

preliminary exercise, relative to a situation

$$(26) \qquad 1 \longrightarrow M \longrightarrow \mathfrak{S} \longrightarrow H \longrightarrow 1$$

exact sequence of discrete groups, with M commutative – the case which will interest us later on being the one where M is a free $\mathbb{Z}$-module of finite type. In any case M is a $\mathbb{Z}$-module, and as such defines a Group over $\dagger S_0 = \operatorname{Spec}\mathbb{Z}$, namely

$$A \longmapsto M \otimes_{\mathbb{Z}} A$$

(A, commutative $\mathbb{Z}$-algebra) – which we shall denote by $\underline{M}$. In fact, $\underline{M}$ is representable, i.e. is a group scheme iff M is free of finite type. I feel like also defining Groups over S_0 (which will be group schemes if M is) $\underline{\mathfrak{S}}, \underline{H}$, and an exact sequence

$$(27) \qquad 1 \longrightarrow \underline{M} \longrightarrow \underline{\mathfrak{S}} \longrightarrow \underline{H} \longrightarrow 1$$

of Groups over S_0, and an isomorphism of exact sequences

$$(28) \qquad \begin{array}{ccccccccc} 1 & \longrightarrow & M & \longrightarrow & \mathfrak{S} & \longrightarrow & H & \longrightarrow & 1 \\ & & \downarrow\wr & & \downarrow\wr & & \downarrow\wr & & \\ 1 & \longrightarrow & \underline{M}(\mathbb{Z}) & \longrightarrow & \underline{\mathfrak{S}}(\mathbb{Z}) & \longrightarrow & \underline{H}(\mathbb{Z}) & \longrightarrow & 1 \end{array}$$

we shall take

$$(29) \qquad \underline{H} = H_{S_0} \qquad \text{constant Group with value } H$$

and the outermost vertical arrows of (28) 16

Uncertainty retained from cancelled passages.
140-4-s017-D001 $\mathbb{Z}$

[scan 18]

will be|are the canonical isomorphisms.

We can think of defining $\underline{\mathfrak{S}}$ as a functor (covariant) in $A/\mathbb{Z}$ or (contravariant) in S/S_0, by

$$(30) \qquad \underline{\mathfrak{S}}(A) = \mathfrak{S} \wedge_M \underbrace{M(A)}_{\dagger M \otimes_{\mathbb{Z}} A}$$

†. $\underline{\mathfrak{S}}(A)$ thus being defined as the extension of H by $M(A)$ deduced from the extension $\mathfrak{S}$ of H by M by the homomorphism

$$(31) \qquad M \longrightarrow M \otimes_{\mathbb{Z}} A \ ,$$

which we consider as a homomorphism of H-$\mathbb{Z}$-modules, by letting H operate on $M \otimes_{\mathbb{Z}} A$ via its operation on M.

The small difficulty is that the group|groups-functor thus defined by (30) is not itself representable – it gives the correct expression for $\underline{\mathfrak{S}}(A)$ only if $\operatorname{Spec} A$ is connected – it corresponds to the subfunctor of

$$S \longmapsto \underset{\sim}{:}\underline{\mathfrak{S}}(A) = \dagger\underline{\mathfrak{S}}(S) \qquad (S = \operatorname{Spec} A)$$

consisting of the $S \longrightarrow \underline{\mathfrak{S}}$ such that the composite

$$(32) \qquad S \longrightarrow \underline{H}_S$$

(which corresponds to a locally constant|constant function on S with values in H) is (entirely) constant. 17

Uncertainty retained from cancelled passages.

140-4-s018-D001 $A\otimes_{\mathbb{Z}}$

140-4-s018-D002 As we have

140-4-s018-D003 Spec |$\underline{\mathfrak{S}}$

[scan 19]

The correct expression will be obtained by starting from the [illeg. 1] "incorrect" functor [p. 657]

$$[\underline{\mathfrak{S}}] : A \longmapsto \mathfrak{S} \wedge_M (M \otimes_{\mathbb{Z}} A)$$

or better, from the contravariant functor|contrafunctor

$$S \longmapsto \mathfrak{S} \wedge_M \Gamma(S, M \otimes_{\mathbb{Z}} \underline{\mathcal{O}|O}_S) \ ,$$

and taking the associated Zariski sheaf – in fact we shall have. =

$$(33) \qquad \underline{\mathfrak{S}}(S) = \varinjlim_{\mathcal{U}=(U_i)_{i\in S|I}} \prod_{i\in I} [\mathfrak{S}](U_i)$$

$(U_i)_{i\in I}$ a partition of S into schemes open $\qquad S \simeq \coprod U_i$

Here is another way of presenting the construction of $\underline{\mathfrak{S}}$, using the general construction of §49 – in the sheaf-theoretic context. We have a system

$$(34) \qquad \underbrace{\underline{M}}_{\text{Group over } S_0} \ , \quad \underbrace{\underline{\mathfrak{S}}_{S_0} \supset M_{S_0}}_{\substack{\text{Groups (constant)} \\ \text{over } S_0,\ M_{S_0} \\ \text{normal}}} \ , \quad \text{Op.}^{\Theta|\theta} \text{ of } \mathfrak{S}_{S_0} \text{ on } \underline{M},$$

$$\text{homomorphism } \varphi : \dagger M_{S_0} \longrightarrow \underline{M}$$

(where the operation of $\mathfrak{S}_{S_0}$ on $\underline{M}$, which amounts to an operation of $\mathfrak{S}$ on $\underline{M}$, is the one deduced from the operation of $\mathfrak{S}$ on M, by functoriality) these data satisfying the compatibility conditions a) b) of loc. cit. By the construction of|of loc. cit., we construct 18

Uncertainty retained from cancelled passages.
140-4-s019-D001 𝔖

[scan 20]

with the aid by giving oneself a system

$$(35)\qquad \begin{array}{c}\underline{\mathcal{G}} \supset \underline{M}\\ \text{Group over } S_0,\\ \text{with } \underline{M} \text{ as}\\ \text{normal subgroup}\end{array} \qquad \text{, and } \Phi\colon \mathcal{G}_{S_0} \longrightarrow \underline{\mathcal{G}}$$

such that (34) is reconstituted from (35) by taking $\underline{M}$, $\mathcal{G}_{S_0}$ in (35), M_{S_0} †= the inverse image of $\underline{M} \subset \underline{\mathcal{G}}$ under Φ, φ induced by Φ, and Θ the operation deduced from Φ. Moreover, Φ induces an isomorphism.

$$\underbrace{\mathcal{G}_{S_0}/M_{S_0}}_{H_{S_0}} \xrightarrow{\sim} \underline{\mathcal{G}}/\underline{M}$$

– and we have here an "axiomatic" description of the system (35) in terms of (34).
The exact sequence of †Groups over $S|S_0$

$$(36)\qquad 1 \longrightarrow \underline{M} \longrightarrow \underline{\mathcal{G}} \longrightarrow \underline{\mathcal{G}}/\underline{M} \longrightarrow 1 \quad (\underline{\mathcal{G}}/\underline{M} \wr\!| \; H_{S_0}.)$$

in fact, over ∀ affine base scheme S (more generally, S such that $H^1(S, M \otimes_{\mathbb{Z}} \underline{\mathcal{O}}_S) = 0$) some exact sequences
m1409

Marginal note. m1409 The hypothesis made about H^1 being zero|zero is entirely useless, for the $\underline{M}$-torsors over S_0 defined by the inverse images of the $h(\in H)$ are obviously trivial

$$(37)\qquad \begin{array}{ccccccccc} 1 & \longrightarrow & \underline{M}(S) & \longrightarrow & \underline{\mathcal{G}}(A)|(/A) & \longrightarrow & H_{S_0}(S) & \longrightarrow & 1\\ & & \wr\!| & & & & \wr\!| & & \\ & & M \otimes_{\mathbb{Z}} A & & & & H \text{ if } S \text{ connected} & & \end{array}$$

if S affine with ring A

19

Uncertainty retained from cancelled passages.
140-4-s020-D001 [illeg. 1]
140-4-s020-D002 schemes

[scan 21]
[p. 658]

In fact, by construction we have a commutative diagram of exact sequences

$$(38)\qquad \begin{array}{ccccccccc} 1 & \longrightarrow & M_{S_0} & \longrightarrow & \mathcal{G}_{S_0} & \longrightarrow & H_{S_0} & \longrightarrow & 1\\ & & \downarrow & & \downarrow & & \Downarrow & & \\ 1 & \longrightarrow & \underline{M} & \longrightarrow & \underline{\mathcal{G}} & \longrightarrow & H_{S_0} & \longrightarrow & 1 \end{array}$$

which on passing to sections gives S affine (say) connected a commutative diagram

$$(39)\qquad \begin{array}{ccccccccc} 1 & \longrightarrow & M & \longrightarrow & \mathcal{G} & \longrightarrow & H & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \Downarrow & & \\ 1 & \longrightarrow & M\otimes_{\mathbb{Z}} A & \longrightarrow & \underline{\mathcal{G}}(A) & \longrightarrow & H & \longrightarrow & 1. \end{array}$$

For $S = S_0$ we recover the announced isomorphism (28).
m1411

Marginal note. m1411 N.B. As a sheaf (without S-group structure) $\underline{\mathcal{G}}$ is isomorphic to the †sheaf sum of the H copies of $\underline{M}$ – it is therefore representable if $\underline{M}$ is.

This construction of the S_0-group scheme $\underline{\mathcal{G}}$, from an exact sequence (27), is manifestly functorial with respect|with respect to the latter: a homomorphism of exact sequences

$$(40)\qquad \begin{array}{ccccccccc} 1 & \longrightarrow & M & \longrightarrow & \mathcal{G} & \longrightarrow & H & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & M' & \longrightarrow & \mathcal{G}' & \longrightarrow & H' & \longrightarrow & 1 \end{array}$$

gives rise to a commutative diagram of exact sequences 20

Uncertainty retained from cancelled passages.
140-4-s021-D003 scheme

[scan 22]

$$(41)\qquad \begin{array}{ccccccccccc} 1 & \longrightarrow & M_{S_0} & \longrightarrow & \mathcal{G}_{S_0} & \longrightarrow & H_{S_0} & & \longrightarrow & 1 & \\ & & \downarrow & & \downarrow & & \| & & & & \\ 1 & \longrightarrow & M'_{S_0} & \longrightarrow & \mathcal{G}'_{S_0} & \longrightarrow & H'_{S_0} & & \longrightarrow & 1 & \\ & & \downarrow & & \downarrow & & \downarrow & & & & \\ 1 & \longrightarrow & \underline{M} & \longrightarrow & \underline{\mathcal{G}} & \longrightarrow & H_{S_0} & & \longrightarrow & 1 & \dagger \\ & & \downarrow & & \downarrow & & \downarrow & & & & \\ 1 & \longrightarrow & \underline{M}' & \longrightarrow & \underline{\mathcal{G}}' & \longrightarrow & \underline{H}' & & \longrightarrow & 1 & \end{array}$$

m1413

Marginal note. m1413 Discrete schematic envelope of groups that are metabelian

4. Let us then take up again a free $\mathbb{Z}$-module of finite type M, and for $\forall$ sub-†$\mathbb{Z}$-module of finite index, let us consider the exact sequence

$$(41)\qquad 1 \longrightarrow M_i \longrightarrow M \longrightarrow M/M_i \longrightarrow 1.$$

More generally, if we start from an exact sequence (27), for $\forall$ subgroup M_i of finite index in M, normal in $\mathcal{G}$ i.e. stable under H, we have an exact sequence

$$(43) \qquad 1 \longrightarrow M_i \longrightarrow \mathcal{G} \longrightarrow \underbrace{\mathcal{G}/M_i}_{H_i} \longrightarrow 1$$

where now $H_i = \mathcal{G}/M_i$ is the|an extension of

$$(44) \qquad 1 \longrightarrow M/M_i \longrightarrow H_i \longrightarrow H \longrightarrow 1$$

deduced from (27) by

$$M \longrightarrow M/M_i.$$

21

Uncertainty retained from cancelled passages.
140-4-s022-D001 1
140-4-s022-D002 g

[scan 23]

We note that †, the $M_i \subset M$ of finite index and stable under H are cofinal in the set of all [p. 659]
the M_i of finite index in M. Let us therefore consider the associated exact sequence of group schemes take the nM !

$$(45) \qquad 1 \longrightarrow \underline{M_i} \longrightarrow \underline{\mathcal{G}}_i \longrightarrow H_{i\,S_0} \longrightarrow 1$$

For varying i, they form a projective system of extensions, hence an extension of progroups over $S_0 = \operatorname{Spec} \mathbb{Z}$.

†

We consider the pro-object $(\underline{\mathcal{G}}_i)$ as a pro-object of C the category of group schemes (smooth etc ...) over S_0. We note that the transition morphisms

$$\underline{\mathcal{G}}_j \longrightarrow \underline{\mathcal{G}}_i$$

are affine – the question being local on $\underline{\mathcal{G}}_i$, we are reduced to looking at it for the homomorphism

†

$$\begin{array}{ccc} E(M_i, M_j) & \longrightarrow & E(M_i, M_i) \\ & & \| \\ M_i \supset M_j & & \underline{M_i} \end{array}$$

extension of $(M_i/M_j)_{S_0}$ by $\underline{M_j}$

22

Uncertainty retained from cancelled passages.
140-4-s023-D001 having number generators
140-4-s023-D004 Remark|The pro-group $\underline{\mathcal{G}}$
140-4-s023-D005 image

[scan 24]

Now the first is affine, as a finite sum of affine schemes, the second is affine — so the morphism is affine.

This being so, the projective limit of the $\underline{\mathcal{G}}_i$ in the category of schemes exists (more precisely, the projective limit functor of the $\underline{\mathcal{G}}_i$ is representable) — it is perhaps not absurd to pass to the limit outright, and to work with this group scheme — not of finite type.

Remark The pro-group $(\underline{\mathcal{G}}_i)$ over $S_0 = \operatorname{Spec} \mathcal{G}|\mathbb{Z}$ depends only on the , discrete group $\mathcal{G}$, equipped with a class of subgroups † modulo commensurability — the hypothesis to be made being that among these subgroups, there are some which are $\mathbb{Z}$-modules of finite type (for taking them small enough, they are free of finite type). If M_0 is itself of finite index in $\mathcal{G}$ i.e. if one of the|$\forall$ of the H_i are finite, 23

Uncertainty retained from cancelled passages.
140-4-s024-D002 invariant

[scan 25]

then this pro-object depends only on $\mathcal{G}$ itself — subject to the condition that there exists a [p. 660]
subgroup of finite index in $\mathcal{G}$ which is a commutative group, of finite type as a $\mathbb{Z}$-module. (We then say that $\mathcal{G}$ is "metabelian".) 3

Thus, for $\forall$ discrete group $\mathcal{G}$ that is metabelian, we have constructed a pro-group scheme, projective system of affine group schemes, which I will denote by

$$\text{(46)} \qquad \operatorname{Env}(\mathcal{G}) = (\underline{\mathcal{G}}_i)$$

(the index set being made up of the normal subgroups of $\mathcal{G}$ which are commutative and torsion-free). This pro-group depends functorially on $\mathcal{G}$. It has properties of exactnesses|of exactness that are interesting and $\pm$ obvious, for example it is multiplicative

$$\text{(47)} \qquad \operatorname{Env}(\mathcal{G} \times \mathcal{G}') \xrightarrow{\sim} \operatorname{Env}(\mathcal{G}) \times_{S_0} \operatorname{Env}(\mathcal{G}').$$

If we have an exact sequence

$$\text{(48)} \qquad 1 \longrightarrow \mathcal{G}'' \longrightarrow \mathcal{G} \longrightarrow \mathcal{G}' \longrightarrow 1$$

of metabelian groups, we have an exact sequence for the envelopes

$$\text{(49)} \qquad 1 \longrightarrow \operatorname{Env}(\mathcal{G}'') \longrightarrow \operatorname{Env}(\mathcal{G}) \longrightarrow \operatorname{Env}(\mathcal{G}') \longrightarrow 1$$

24
[scan 26]

in a sense which should be made precise on occasion — for the associated group schemes (obtained by passing to the limit) — we do indeed have a faithfully flat homomorphism whose kernel is the right one …

The typical case, to which the general case reduces in a fairly immediate way, is the one where $\mathcal{G} = M = \mathbb{Z}$. We have for $\forall$ integer $n \geq 1$ the exact sequence

$$\text{(50)} \qquad 1 \longrightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z} \longrightarrow 1$$

I am going to † make explicit the construction of the corresponding extension of $\mathbb{Z}/n\mathbb{Z}$ by $\underline{\mathbb{Z}} = \mathbb{G}_a$, i.e. $\underline{\mathcal{G}}_n$.

It is an affine scheme, whose coordinate ring is isomorphic — in a not very canonical way — to the product of the n copies of that of $\underline{\mathbb{Z}} = \mathbb{G}_a$, namely $\mathbb{Z}[T]$ — we find such an isomorphism by choosing a set-theoretic section of $\mathbb{Z}$ over $\mathbb{Z}/n\mathbb{Z}$, for example by the set of integers $\mathbb{Z} \cap [0, n-1]$. The homomorphisms of 25

Uncertainty retained from cancelled passages.
140-4-s026-D001 construct

[scan 27]

transition between the $\underline{\mathcal{G}}_n$ have nothing flat about them of course. Passing to the projective [p. 661]
limit, we find an extension of

$$(\widehat{\mathbb{Z}})_{S_0} \overset{\text{dfn}}{=} \varprojlim_n (\mathbb{Z}/n\mathbb{Z})_{S_0}$$

— constant profinite group scheme over S_0 — by the projective limit of the schemes $\mathbb{G}_a$, mapping to itself by the homomorphisms seen † of 2°) — it is the spectrum of the inductive limit ring of the

$$\mathbb{Z}[T_n] \subset \mathbb{Q}[T]$$
$$T_n = \frac{1}{n}T$$

which is thus, strange to say, the † sub-$\mathbb{Z}$-algebra of $\mathbb{Q}[T]$ formed|formed by the polynomials for which the constant term is an integer.

We have an analogous description of the projective limit of the schematic envelope of $\mathbb{Z}^n$ — it is an extension of $(\widehat{\mathbb{Z}}^n)_{S_0}$ (beware of the abuse of notation) by an identity component, which is the spectrum of the subring of $\mathbb{Q}[T_1, \dots, T_n]$ formed by the polynomials whose constant term is an integer … 26

Uncertainty retained from cancelled passages.
140-4-s027-D001 of the §b
140-4-s027-D002 [illeg. 1]

[scan 28]

It is thus a group scheme over $S_0 = \operatorname{Spec}\mathbb{Z}$ all of whose fibers are reduced to the trivial group, except for the generic fiber, which is $\mathbb{G}_{a\,\mathbb{Q}}^n$.

† In the case of an arbitrary metabelian group $\mathcal{G}$, denoting|let us denote by $\widehat{\mathcal{G}}$ its profinite completion, by $\widehat{\mathcal{G}}_{S_0}$ the profinite group scheme over S_0.

$$(50) \qquad \widehat{\mathcal{G}}_{S_0} \overset{\text{dfn}}{=} \varprojlim_i (H_i)_{S_0}$$

we shall thereby have a canonical exact sequence

$$(51) \qquad 1 \longrightarrow \operatorname{Env}^0(\mathcal{G}) \xrightarrow{\dagger} \varprojlim \operatorname{Env}(\mathcal{G}) \longrightarrow (\widehat{\mathcal{G}})_{S_0} \longrightarrow 1$$

where $\operatorname{Env}^0(\mathcal{G})$ denotes the identity component of $\varprojlim \operatorname{Env}(\mathcal{G})$, i.e.

$$(52) \qquad \operatorname{Env}^0(\mathcal{G}) \overset{\text{dfn}}{=} \varprojlim_i \underline{M}_i$$
$$\simeq \varprojlim_n n.\underline{M}_0$$

where the M_i are the subgroups of finite index in $\mathcal{G}$ † which are free $\mathbb{Z}$-modules of finite type, where (as above) for such a $\mathbb{Z}$-module M,

$$(53) \qquad \boxed{\underline{M} = \check{V}(M) = \operatorname{Spec}\operatorname{Sym}_{\mathbb{Z}}(\check{M})}$$

denotes the vector bundle covariant|canonical in M which it defines, and 27

Uncertainty retained from cancelled passages.
140-4-s028-D001 And
140-4-s028-D002 dfn
140-4-s028-D003 such that|such that

[scan 29]

where M_0 is one of the M_i, giving rise to the system (cofinal in that of all the M_i) of the nM_0. The choice of a basis of M_0 over $\mathbb{Z}$ gives an isomorphism [p. 662]

$$\begin{aligned} \mathrm{Env}^0(\mathcal{G}) &\simeq \mathrm{Env}^0(M_0) \simeq \left(\mathrm{Env}^0(\mathbb{Z})^0\right)^n \\ &\simeq (\mathcal{E}^1)^n \end{aligned} \tag{54}$$

m1423

Marginal note. m1423 † where

$$(\mathbb{E}^1, mn|nm) \longrightarrow (\mathbb{E}^1, \underline{n})$$

is multiplication by m

$$\begin{aligned} \mathcal{E}^1 &\simeq \mathrm{Env}^0(\mathbb{Z}) \simeq \mathrm{Spec}(\mathbb{Z} + T\mathbb{Q}[T]) \\ &\simeq \varprojlim_n (\mathbb{E}^1, n)\,\dagger \end{aligned} \tag{55}$$

is the group scheme considered above, which represents the group-valued functor

$$A \longmapsto \mathcal{E}^1(A) = \dagger \left\{ (x_n) \in A^{\mathbb{N}^*} \;\middle|\; \begin{array}{l} x_n = m x_{mn} \\ \forall m, n \in \mathbb{N}^* \end{array} \right\}. \tag{56}$$

[If A is such that there exists $n \in \mathbb{N}^*$ with $nA = | \simeq 0$ — † which means that the residue characteristics of A divide n, and † there probably exists such an n as soon as the residue characteristics of A are all > 0 — then $\mathcal{E}^1(A) = \{1\}$ — if on the other hand A of characteristic 0, we have $\mathcal{E}^1(A) \simeq A$] 28

Uncertainty retained from cancelled passages.
140-4-s029-D001 $\varprojlim$
140-4-s029-D002 precisely
140-4-s029-D003 it is

[scan 30]

5. Universal characterization of the schematic envelope of a metabelian group.

Let G be a group scheme locally of finite presentation over a base S. We say that a homomorphism of group schemes

$$H \longrightarrow G$$

is † unipotent (resp. quasi-unipotent) if the image of ∀ element of H is a unipotent element (resp. quasi unipotent) of G — which is checked fiber by fiber.

A homomorphism from a discrete group

$$\mathcal{G} \longrightarrow \Gamma(G/S)$$

is said to be unipotent (resp. quasi-unipotent) if the homomorphism of group schemes † which corresponds to it

$$\mathcal{G}_S \longrightarrow G$$

is unipotent (resp. quasi-unipotent). 29

Uncertainty retained from cancelled passages.
140-4-s030-D001 quasi-
140-4-s030-D002 associated

[scan 31]

Note that by the properties of $\varprojlim$ of filtered projective systems (with affine transition morphisms) of schemes, if $\mathcal{G}$ is metabelian, whence a projective system of affine group schemes over S_0, hence over S, denoted $\mathrm{Env}_{\underline{S}|S}(\mathcal{G})$, and a projective limit [p. 663]

$$(57) \qquad \begin{aligned} L\,\mathrm{Env}_S(\mathcal{G}) &= L\,\mathrm{Env}(\mathcal{G}) \times_{S_0} S \\ &\simeq \varprojlim L\,\mathrm{Env}_S(\mathcal{G}), \end{aligned}$$

we have, as soon as G is locally of finite presentation
(58)

$$\begin{array}{ccc} \mathrm{Hom}_{\mathrm{progr}/S}(\mathrm{Env}_S(\mathcal{G}), G) & \xrightarrow{\sim} & \mathrm{Hom}(\mathrm{Env}_S(\mathcal{G}), G) \\ {\scriptstyle \mathrm{def}}\Vert & & \\ \varprojlim_i \mathrm{Hom}(\underline{\mathcal{G}}_{i\,S}, G) & & \end{array}$$

Note on the other hand that we have a † canonical homomorphism
(59)

$$\mathcal{G}_S \longrightarrow \dagger L\,\mathrm{Env}_S(\mathcal{G})$$

deduced by change of base from the homomorphism over the universal base S_0
(59 bis)

$$\mathcal{G}_{S_0} \longrightarrow L\,\mathrm{Env}(\mathcal{G})$$

coming from the system of canonical homomorphisms 30

Uncertainty retained from cancelled passages.
140-4-s031-D002 a
140-4-s031-D003 $\varprojlim$

[scan 32]

$$\underline{\mathcal{G}}_{S_0} \longrightarrow \underline{\mathcal{G}}_i$$

.
Thus, via (59) we find
(60)

$$\begin{array}{ccc} \mathrm{Hom}_{S\text{-gr}}(L\,\mathrm{Env}_S(\mathcal{G}), G) & \longrightarrow & \mathrm{Hom}_{S\text{-gr}}(\mathcal{G}_S, G) \\ & & \wr| \\ & & \mathrm{Hom}_{\mathrm{gr}}(\mathcal{G}, \Gamma(G/S)) \end{array}$$

1
Composing (58) and (60), we find

$$(61) \quad \begin{cases} \mathrm{Hom}_{\text{pro-}S\text{-gr}}\,\ell(\mathrm{Env}_S(\mathcal{G}), G) \simeq \mathrm{Hom}_{S\text{-gr}}(L\,\mathrm{Env}_S(\mathcal{G}), G) \\ \qquad\qquad \longrightarrow \mathrm{Hom}_{S\text{-gr}}(\mathcal{G}_S, G) \simeq \mathrm{Hom}_{\mathrm{gr}}(\mathcal{G}, G(S)) \end{cases}$$

This being so, we have the following result
Theorem † Under the preceding conditions
m1427

Marginal note. m1427

($\mathcal{G}$ a discrete metabelian group, G a group scheme locally of finite presentation over a base scheme S), the canonical map (61) is † injective, and has as image the set of homomorphisms quasi-unipotent from $\mathcal{G}$ into $G(S)$ — at least if we assume, to simplify life for ourselves, that S is noetherian and G affine over $S|S_0$. 31

Uncertainty retained from cancelled passages.
140-4-s032-D001 If
140-4-s032-D002 myself|oneself

[scan 33]
[p. 664]

51|0 Example : The Fermat "tower".
1. The groups $\mathcal{G}, \underline{\mathcal{G}}$.

I am interested in the quotient of $\mathcal{G}_0^{\dagger} = \mathrm{GL}(2, \mathbb{Z})$ obtained by taking the quotient by $[\Pi_0, \Pi_0]$ — we find an extension $\mathcal{G}$ of $\mathcal{G}_0/\Pi_0$ by the group
m1429

Marginal note. m1429 Π_0 generated by the $\lambda_i = -\ell_i$, $\ell_0 = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$ $\ell_1 = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$.

$$(1) \qquad M \overset{\text{dfn}}{=} \Pi_{0\,\text{ab}} \simeq \Big(\overset{\dagger}{\mathbb{Z}}(0) + \overset{\dagger}{\mathbb{Z}}(1) + \overset{\dagger}{\mathbb{Z}}(\infty)\Big)/\text{diagonal}$$

Thus we have an exact sequence of profinite groups
(2)

$$1 \longrightarrow \underset{\wr\,\Pi_{0\,\text{ab}},\ \text{also denoted}\ \mathcal{G}^\circ}{M} \longrightarrow \mathcal{G} \longrightarrow \overset{H}{\overbrace{\underset{\wr\,\text{GL}(2,\mathbb{Z}/4\mathbb{Z})/(\dot\ell_0,\dot\ell_1)}{\mathcal{G}_0/\Pi_0}}} \longrightarrow 1$$

In terms of generators and relations we have

$$(3) \quad \mathcal{G} = \left\{ \tau', \rho, \sigma \;\middle|\; \begin{array}{l} \rho^6 = \sigma^4 = \tau'^2 = 1 \\ \rho^3 = \sigma^2, \quad \tau'(\rho) = \rho^{-1}, \quad \tau'(\sigma) = \sigma^{-1} \\ \dagger[\lambda_0, \lambda_1] = 1, \quad \text{where} \end{array} \right.$$

$$\begin{cases} \lambda_0 = -\ell_0 = -\sigma\rho\sigma\rho = \sigma(\rho)\rho^\dagger \\ \lambda_1 = -\ell_1 = -\rho\sigma\rho\sigma = \rho\sigma(\rho) \end{cases}$$

of which|thus the commutation relation $[\lambda_0, \lambda_1] = 1$ is also written

$$(4) \qquad \sigma(\rho)\rho^2\sigma(\rho) = \rho\sigma(\rho^2)\rho \qquad \text{or again}$$

$$(5) \qquad -\sigma\,\underbrace{\rho\sigma\rho^{-1}\sigma\rho}\,\sigma = \underbrace{\rho\sigma\rho^{-1}\sigma\rho}$$

where the play of the $-$ signs can be made explicit in the formalism of generators and relations by removing this sign from the formula, and replacing in return one (or an odd numᵇer|number) of the σ in the formula by σ^{-1}. For example, the formula 32

Uncertainty retained from cancelled passages.
140-4-s033-D001 ∧
140-4-s033-D003 ∧
140-4-s033-D004 ∧
140-4-s033-D005 ∧
140-4-s033-D006 ρ
140-4-s033-D007 $-1|4$

[scan 34]

is written, introducing

N.B. $\lambda_\infty \overset{\text{dfn}}{=} -\ell_\infty$

$$(6) \qquad \lambda_\infty\dagger = \rho\sigma\rho^{-1}\sigma\rho = \rho(\sigma)\sigma\rho = -\rho\sigma(\rho)^{-1}\rho = \rho\sigma\rho^{-1}(\sigma)$$

in the form

$$(7) \qquad [\sigma, \lambda_\infty] = 1.$$

N.B. $\sigma = \sigma_\infty$

m1433

Marginal note. m1433 N.B. We find

$$\begin{cases} [\ell_0,\ell_1]=[\lambda_0,\lambda_1]=(\sigma(\rho)\rho^{-1})^3 \\ [\underbrace{\sigma_\dagger}_{\sigma},\ell_\infty]=[\underbrace{\sigma_\infty}_{\sigma},\lambda_\infty]=(\sigma(\rho^{-1})\rho)^3 \\ \qquad\qquad =\mathrm{int}(\sigma\rho^{-1})\cdot[\ell_0,\ell_1] \end{cases}$$

We are going to define a group scheme $\underline{\mathcal{G}}$ over $\mathbb{Z}$, whose $\widehat{\mathbb{Z}}|\mathbb{Z}$-points will be (up to an isomorphism say) the points of $\widehat{\mathcal{G}}$. We put

$$\underline{M}=\underline{\mathcal{G}}^\circ=\mathbb{G}_a|\mathbb{E}_a{}^{\{0,1,\infty\}}/\text{diagonal} \tag{8}$$

m1434

Marginal note. m1434 It is better to denote it by $\underline{M}$ — there are already too many $\mathcal{G}$'s!

We are going to have †

$$\mathcal{G}\dagger\simeq \mathrm{GL}(2,\mathbb{Z})/[\Pi_0,\Pi_0] \tag{\dagger(4)}$$

operate on $\underline{M}\dagger$, † through

$$\underbrace{\mathcal{G}_3\times\mu^\dagger_{\tau|2}}_{\simeq T_{0,3}|\Gamma_{0,3}}\simeq\mathcal{G}/M\dagger\simeq\mathcal{G}_{0,3}/\Pi. \tag{5}$$

Denoting by $\dot\lambda_0,\dot\lambda_1,\dot\lambda_\infty$ the images of the $\lambda_i=-\ell_i$ in $\dagger\mathcal{G}$, identified with elements of

$$M=\mathcal{G}^\circ=\underline{\mathcal{G}}^\circ(\widehat{\mathbb{Z}})$$

— in fact we have

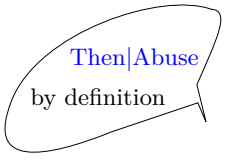
$$\begin{cases} \dot\lambda_0,|>\dot\lambda_1,\dot\lambda_\infty\in\dagger\underline{M}(\mathbb{Z})\subset\dagger\underline{M}(\widehat{\mathbb{Z}})=M\otimes_{\cdot|\mathbb{Z}}\mathbb{Z}|\widehat{\mathbb{Z}} \\ \text{thus,}\quad \dot\lambda_0+\dot\lambda_1+\dot\lambda_\infty=0 \end{cases} \tag{6}$$

m1435

Marginal note. m1435 (here we write the group law in $M=\underline{\mathcal{G}}^\circ$ additively)

m1436

Marginal note. m1436



$\mathcal{G}_3$ operates on $\underline{M}=\underline{\mathcal{G}}^\circ$ by its action on the canonical † generators $\dagger\dot\lambda_0,\dot\lambda_1,\dot\lambda_\infty$, thus

$$\begin{cases} \sigma(\dot\lambda_0)=\dot\lambda_1,\quad \sigma(\dot\lambda_1)=\dot\lambda_0\quad\text{hence}\quad \sigma(\dot\lambda_\infty)=\dot\lambda_\infty \\ \rho(\dot\lambda_i)=\dot\lambda_{i+1} \end{cases} \tag{7}$$

m1437

Marginal note. m1437 This is seen first for $i = 0, 1$ and, is deduced from it for $i = \infty$.

On the other hand we have by definition

$$(8) \qquad \tau(\dot\lambda_i) = -\dot\lambda_i$$

where

$$\begin{array}{ccccc} \tau & = & \sigma & \tau' \\ \| & & & \| \\ \tau_\infty & & & \tau'_\infty \end{array}$$

33

Uncertainty retained from cancelled passages.

140-4-s034-D001 $= -\ell_\infty$
140-4-s034-D002 ∞
140-4-s034-D003 $\mathcal{G}/M$
140-4-s034-D004 4
140-4-s034-D005 $= \mathrm{GL}(2, \mathbb{Z})$
140-4-s034-D006 $= \underline{\mathcal{G}}^\circ$
140-4-s034-D007 by describing in this way
140-4-s034-D008 2
140-4-s034-D009 $\widehat{\mathbb{Z}}$
140-4-s034-D011 M
140-4-s034-D012 $\underline{\mathcal{G}}^\circ$
140-4-s034-D013 $\underline{\mathcal{G}}^\circ$
140-4-s034-D014 bases
140-4-s034-D015 λ_0

[scan 35]

. The announced extension $\underline{\mathcal{G}}$ is of course that of § 50 3°), developed in the meantime [p. 665] (since the first lines of the present §, noting precisely that I was feeling cramped in making the construction explicit) :

(9)

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \underline{M} & \longrightarrow & \underline{\mathcal{G}} & \longrightarrow & H_{S_0} & \longrightarrow & 1 \\ & & \| & & & & & & \\ & & \check V(M) & & & & & & \end{array}$$

where $S_0 = \operatorname{Spec} \mathbb{Z}$.

For $\forall$ scheme S, we will put

$$\begin{array}{c} \underline{\mathcal{G}}^\Delta(S) \subset \underline{\mathcal{G}}(S) \\ \| \text{ dfn} \\ \text{Inverse image of the "diagonal" of} \\ \Gamma(H_{S|S_0}/S) \quad \text{(i.e. of the image of } H \longrightarrow \Gamma(H_{S|S_0}, S)) \\ \text{under the canonical homomorphism } \underline{\mathcal{G}|\mathcal{E}}(S) \longrightarrow H_S(S), \end{array} \qquad (10)$$

so that by the construction of loc. cit. we will have a canonical isomorphism

$$\underline{\mathcal{G}}^\Delta(S) \simeq \mathcal{G} \underset{M}{\wedge} \underbrace{\dagger\, \Gamma(S, \underline{\mathcal{O}}_S \otimes_{\mathbb{Z}} M)}_{\simeq \dagger M \otimes_{\mathbb{Z}} A \text{ if}} \qquad (11)$$

and a canonical exact sequence
(12)

$$1 \longrightarrow \underbrace{\Gamma(S, \underline{\mathcal{O}}_S \otimes_{\mathbb{Z}} M)}_{A \otimes_{\mathbb{Z}} M} \longrightarrow \underline{\mathcal{G}}^{\Delta}(S) \longrightarrow H \longrightarrow 1.$$

S affine with ring A

Taking $A = \widehat{\mathbb{Z}}$, we find in particular (and this is again a general fact about group schemes associated with extensions of discrete groups with abelian kernel)

$$\mathcal{G}^{\wedge} \simeq \underline{\mathcal{G}}^{\Delta}(\widehat{\mathbb{Z}}). \tag{13}$$

34

Uncertainty retained from cancelled passages.
140-4-s035-D001 *A*
140-4-s035-D002 *A*

[scan 36]

2. The group $\widetilde{Z}_{\rho,\sigma}$: provisional version.

. I am now going to make explicit for $\widehat{\mathcal{G}}$ and $\widehat{\mathcal{G}}_{0|\bullet} \longrightarrow \widehat{\mathcal{G}}$ the general constructions of § 49 VI, taking there †

$$\underline{\mathcal{G}}^{\natural} \overset{\text{dfn}}{=} \underline{M} = \underline{\mathcal{G}}^{\circ} \quad \text{(identity component of } \underline{\mathcal{G}}) \tag{14}$$

In a case like this one, where $\underline{\mathcal{G}}^{\natural}$ is precisely the closed subgroup of the (ρ,σ)-group $\mathcal{G}_{\rho,\sigma}$ (= $\mathcal{G}^{\wedge}$ here) generated by $\lambda_0, \lambda_1, \lambda_\infty$, we will dispense in the notation — where there is no ambiguity — with including the notation $\natural$. For $\forall\ A$ we will have to distinguish for example between $G_{\rho,\sigma}$ and $G^{\natural}_{\rho,\sigma}$ etc. We will however carry out the constructions in the scheme-theoretic setting, rather than the profinite one.
m1440

Marginal note. m1440 I think I will compare the two possible constructions at the end. The underlines will indicate the group schemes...

The essential step is to determine the group $\underline{Z}_{\rho,\sigma}$

$$\begin{array}{c} \underline{Z}_{\rho,\sigma} \subset \underline{\mathrm{Aut}}(\underline{\mathcal{G}}^{+}) \\ \| \text{ dfn} \\ \left\{ u \in \underline{\mathrm{Aut}}(\mathcal{G}^{+}) \;\middle|\; \begin{array}{ll} \dagger\ \text{a)}\ \exists \alpha \in \underline{\mathcal{G}}^{\natural}_{\tau} = \{1, \tau\}.\underline{M}, & u(\rho) = \alpha(\rho) \\ \text{b)}\ \exists \beta \in \text{------} & u(\sigma) = \beta(\sigma) \\ \text{c)}\ u(L_0^{\dagger \mathcal{G}}) = L_0^{\mathcal{G}} & \dagger \\ \qquad (\Longleftrightarrow u(L_0^{\Pi_0}) = L_0^{\Pi_0}) & \\ [\text{d) (automatic)} \quad u(\mathcal{G}^{\natural}) \subset \mathcal{G}^{\natural}] & \end{array} \right\} \end{array} \tag{15}$$

m1441

Marginal note. m1441 here, "$\exists\alpha$" "$\exists\beta$" of course means the local existence of α, β...

Let us note that u is determined by the values

$$\begin{cases} u(\rho) = \xi\rho \\ u(\sigma) = \eta\sigma \end{cases} \qquad \xi, \eta \in \Gamma\underline{\Pi} \tag{16}$$

m1442

Marginal note. m1442 where $\underline{\Pi} = \underline{M} \rtimes \dagger\mu_2$ ($\underline{M}$ subgroup|is a group of index 2 in $\underline{\Pi}$)

ξ, η satisfying the following conditions ; 35

Uncertainty retained from cancelled passages.

140-4-s036-D001 $\mathcal{G}$

140-4-s036-D002 1°

140-4-s036-D003 Π_0

140-4-s036-D004 $(\Longleftrightarrow L_0^{\Pi_0} = L_0^\natural)$

140-4-s036-D005 $\{1, \tau\}$

[scan 37]
[p. 666]

$$\begin{cases} \text{a) } \xi\rho \text{ (locally) conjugate to } \rho \text{ via } \underline{\mathcal{G}^\natural} = \underline{M} \\ \text{b) } \eta\sigma \text{ – – – – – – – – } \sigma \text{ " – – —} \\ \text{c) loop equation} \\ \quad \xi = \eta\ell_1^\nu \quad \text{where } \nu \in \Gamma\mathbb{G}_a \text{ suitable} \\ \qquad \text{(such that } \mu \overset{\text{dfn}}{=} 2\nu + 1 \in \Gamma\mathbb{G}_m) \end{cases} \tag{17}$$

.

Remark It is no longer possible here to denote by -1 the nontrivial element ω of the center $\mu|\mu_2$ of $\mathcal{G}^\wedge$, and to denote by $-x$ (for $x \in \mathcal{G}^\wedge$) the product of x with it, because of the confusion with the notation $-x$ (for $x \in M$) in the abelian group M for the inverse of x — but † $\omega x \neq -x$, since $\omega x \notin M$, $-x \in M$...

We of course put

$$\varepsilon_3(u) = \varepsilon_{\mathcal{G}}(\alpha), \qquad \varepsilon_2(u) = \varepsilon_{\mathcal{G}}(\beta) \tag{18}$$

(well defined, since ρ and ρ^{-1} are not conjugate on any fiber, idem for σ, σ^{-1}), how to make them explicit in terms of ξ, η ? We will have

†

$$\begin{cases} \xi = \alpha\rho(\alpha)^{-1} \begin{cases} \in \Gamma\underline{M} \quad \text{if } \alpha \in \Gamma\underline{M} = \Gamma\underline{\mathcal{G}^\natural} \text{ i.e. } \varepsilon_3(u) = 1 \\ = \underbrace{\alpha_0}_{\in\Gamma\underline{M}} \tau\rho(\alpha_0\tau)^{-1} = \alpha_0 \underbrace{(-\ell_1^{-1})}_{\lambda_1^{-1}\in\underline{M}} \rho(\alpha_0)^{-1} \in \Gamma\underline{M} \quad \text{if } \varepsilon_3(u) = -1 \end{cases} \\ \eta = \beta\sigma(\beta)^{-1} \begin{cases} \in \Gamma\underline{M} \quad \text{if } \beta \in \Gamma\underline{M} = \Gamma\underline{\mathcal{G}^\natural} \text{ i.e. } \varepsilon_2(u) = 1 \\ = (\underbrace{\beta_0}_{\in\Gamma\underline{M}} \tau)\sigma(\beta_0\tau)^{-1} = \omega\beta_0\sigma(\beta_0)^{-1} \in \omega\underline{M} \quad \text{if } \varepsilon_2(u) = -1 \end{cases} \end{cases} \tag{19}$$

thus in any case

$$\begin{cases} \xi \in \Gamma\underline{M} \\ \eta \in \Gamma(\omega^{\nu_2(u)}.\underline{M}) \end{cases} \tag{19 bis}$$

36

Uncertainty retained from cancelled passages.
140-4-s037-D002 the element
140-4-s037-D003 (17)

[scan 38]
[p. 676]

from which we get, by the loop equation (17)

$$\ell_1^\nu = (\omega\lambda_1)^\nu \in \Gamma(\omega^{\nu_2(u)}.\underline{M}) \quad \text{i.e.} \iff \omega^\nu \in \Gamma(\omega^{\nu_2(u)}.\underline{M}) \tag{20}$$

We can write it in the form

$$\underline{\nu} = 2\underline{\nu'} + \underline{\nu}_2 \tag{21}$$

where

$$\begin{cases} \underline{\nu}, \underline{\nu'} : Z_{\rho,\sigma} \longrightarrow \mathbb{G}_a \\ \underline{\nu}_2, \underline{\nu}_3 : Z_{\rho,\sigma} \rightrightarrows (\mathbb{Z}/2\mathbb{Z})_{S_0}^{\text{que|cste}} \end{cases} \tag{22}$$

† are well-defined morphisms of schemes, related to one another and to the group homomorphisms †

$$\underline{\mu} : Z_{\rho,\sigma} \longrightarrow \mathbb{G}_m, \qquad \underline{\varepsilon}_{1|2}, \underline{\varepsilon}_3 : Z_{\rho,\sigma} \longrightarrow \{\pm 1\}_{S_0}$$

by the formulas (21) and

$$\begin{cases} \underline{\mu} = 2\underline{\nu} + 1\dagger \quad (= 4\underline{\nu'} + 2\underline{\nu}_2 + 1) \\ \underline{\varepsilon}_2 = (-1)^{\underline{\nu}_2}, \qquad \underline{\varepsilon}_3 = (-1)^{\underline{\nu}_3}. \end{cases} \tag{23}$$

We will therefore have, by virtue of (24|21) † 37

Uncertainty retained from cancelled passages.
140-4-s038-D001 ε
140-4-s038-D002 μ
140-4-s038-D003 $+1$
140-4-s038-D004 to thus|will be able to $2\nu'$ $\nu_2 = 0|\nu_2 = 1$ $\varepsilon_2 = 1$

[scan 39]
[p. 667]

$$l_1^\nu = \lambda_1^\nu \omega^{\nu_2}. \tag{24}$$

Setting (cf. (19))

$$\begin{cases} \beta = \beta_0 \tau^{\nu_2} \qquad (\beta_0 | \beta \in \Gamma \underline{M}) \\ \eta_0 = \beta_0 \sigma(\beta_0)^{-1} \end{cases} \tag{25}$$

we shall thus have

$$\eta \overset{\text{dfn}}{=} \beta\sigma(\beta)^{-1} = \beta_0\sigma(\beta_0)\omega^{\nu_2} = \eta_0\omega^{\nu_2}$$

and the loop equation takes the form of an equation in $\underline{M}$

$$\zeta|\xi = \dagger\eta_0\lambda_1^\nu, \qquad \text{i.e. side of (21)} \tag{26}$$

$$\zeta|\xi = \tag{26bis}$$

where $\zeta|\xi, \eta_0$ are two sections of $\underline{M}$ which determine one another (for given ν). We have thus expressed in a convenient way the loop equation, i.e. the condition c)of (17), it remains to express a) and b), which are also formulated as

$$\begin{cases} \text{a) } \zeta|\xi \text{ locally of the form } \overset{(\nu_3=0)}{\alpha_0\rho(\alpha_0)^{-1}} \text{ or } \overset{(\nu_3=1)}{\alpha_0\lambda_1^{-1}\rho(\alpha_0)^{-1}} \\ \quad \text{with } \alpha_0 \in \Gamma\underline{M} \\ \text{b) } \eta_0 \text{ locally of the form } \beta_0\sigma(\beta_0)^{-1} \end{cases} \tag{27}$$

(Conditions whose formulation is independent of ν, ν_2 †, but depends apparently on ν_3.) To make these conditions explicit, we are going to write $\underline{M}$ additively, the relations read

We have thus also set $\alpha = \alpha_0 \tau^{\nu_3}$

$$\begin{cases} \zeta|\xi = \alpha_0 - \rho(\alpha_0) \quad \overset{\text{case}}{(\nu_3 = 0)} \quad \text{or} \quad \zeta|\xi = \alpha_0 - \rho(\alpha_0) - \lambda_1 \quad \overset{\text{case}}{(\nu_3 = 1)} \\ \eta_0 = \beta_0 - \sigma(\beta_0), \end{cases} \tag{28}$$

38

Uncertainty retained from cancelled passages.

140-4-s039-D001 β

140-4-s039-D002 for$|\nu_3$

[scan 40]

which we also express by

$$\begin{cases} \zeta|\xi \in \mathrm{Im}(\mathrm{Id}_{\underline{M}} - \rho_{\underline{M}}) \quad (\nu_3 = 0) \quad \text{or} \quad \zeta|\xi \in -\lambda_1 + \mathrm{Im}(\mathrm{Id}_{\underline{M}} - \rho_{\underline{M}}) \\ \eta_0 \in \mathrm{Im}(\mathrm{Id}_{\underline{M}} - \sigma_{\underline{M}}). \end{cases} \tag{29}$$

It is time to move on to explicit matrix calculations, by referring $\underline{M}$ to the basis λ_0, λ_1. In this basis, the operations of $\rho_{\underline{M}}, \sigma_{\underline{M}}, \tau_{\underline{M}}$ are given by

$$\begin{cases} \rho_{\underline{M}} = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix} & \mathrm{id}_{\underline{M}} - \rho_{\underline{M}} = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix} \\ \sigma_{\underline{M}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & \mathrm{id}_{\underline{M}} - \sigma_{\underline{M}} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \\ \tau_{\underline{M}} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}. \end{cases} \tag{30}$$

We have

$$\det(\mathrm{id}_{\underline{M}} - \rho_{\underline{M}}) = 3, \qquad \det(\mathrm{id}_{\underline{M}} - \sigma_{\underline{M}}) = 0$$

which shows that the endomorphisms $\mathrm{id}_{\underline{M}} - \rho_{\underline{M}}$ and $\mathrm{id}_{\underline{M}} - \sigma_{\underline{M}}$ of $\underline{M}$ are not quite automorphismsi.e. epimorphisms — It screws up in characteristic 3 for the first, everywhere †(in every characteristic) for the second. We shall have reason to take as "parameters" not $\zeta|\xi$ and η_0, but quite obviously α_0 and β_0 (in addition to the discrete "parameters" ν_2, ν_3 and ν'), and to make a group (if possible…) with the systems

39

Uncertainty retained from cancelled passages.

140-4-s040-D002 characteristic 2

[scan 41]

†

[p. 668]

$$\widetilde{\underline{Z}}_{\rho,\sigma} = \left\{ (\nu, \nu', \nu_2, \nu_3, \alpha_0, \beta_0) \;\middle|\; \begin{cases} \nu, \nu' \in \Gamma \mathbf{G}_a, \quad \nu_2, \nu_3 \in \Gamma\{0,1\}_{S_0}, \quad \alpha_0, \beta_0 \in \Gamma \underline{M} \\ \nu = 2\nu' - | + \nu_2 \qquad 2\nu + 1 \in \Gamma \mathbf{G}_m \\ \underbrace{\alpha_0 - \rho(\alpha_0) - | + \nu_3 \lambda_1}_{\zeta|\xi} = \underbrace{\beta_0 - \sigma(\beta_0)}_{\eta} + \nu \lambda_1 \end{cases} \right\} \tag{31}$$

N.B. ν' determines ν which is redundant

The loop equation, in α_0, β_0, thus takes the form

$$(\mathrm{id}|1 - \rho_{\underline{M}})\alpha_0 - | = (\mathrm{id} - \sigma_{\underline{M}})(\beta_0) = (\underbrace{2\nu' - | + \nu_2}_{\nu} + | - \nu_3)\lambda_1 \tag{32}$$

m1450

Marginal note. m1450 This must indeed come from the fact that the homomorphism

$$\underline{M}\times\underline{M}\longrightarrow\underline{M}$$

$$(\alpha_0,\beta_0)\longmapsto(\mathrm{id}-\rho_{\underline{M}})\alpha_0-(\dagger 1-\sigma_{\underline{M}})(\beta_0)$$

is|is indeed an epimorphism... But this is the case only in characteristic $\neq 3$!

Remark The scheme $\underline{Z}_{\rho,\sigma}$ itself, as it was defined in § 49 † $\overline{\mathrm{VI}}$, is not smooth it seems to me — it gets stuck|gets complicated in characteristics 2 and 3. On the other hand the finer group $\widetilde{\underline{Z}}_{\rho,\sigma}$, defined by bringing in the α,β right away (instead of $(\zeta|\xi,\eta)$) does seem smooth. This leads once again to modifying a little the presentation of the constructions of § 49, by denying myself|not using yet another group $\underline{Z}_{\rho,\sigma}$ (tied to the coordinates $\zeta|\xi,\eta$), and by replacing it with a group which will appear|appears at the same time|in a way as a normalizer of L_0 (or of $L_0^\sharp$) in $\mathcal{M}_{\rho,\sigma}$ (or $\mathcal{M}_{\rho,\sigma}^{\natural|\sharp}$). We shall give a presentation along these lines in the † § that follows, and we assume here that this construction is already in hand, 40

Uncertainty retained from cancelled passages.

140-4-s041-D001 (24)
140-4-s041-D002 id
140-4-s041-D003 § 6
140-4-s041-D004 sequel

[scan 42]

Whence the † (scheme $\widetilde{\underline{Z}}_{\rho,\sigma}$ will thus have a suitable group structure, and operates on $\underline{C}^+|\underline{G}^+$ via

$$\begin{cases}u(\rho)=(\alpha_0-\rho(\alpha_0)+\nu_3\lambda_1)\rho\\ u(\sigma)=(\beta_0-\sigma(\beta_0))\omega^{\nu_2}\sigma.\end{cases}\tag{33}$$

But we must all the same check that these formulas (33) really do determine an automorphism u of $\underline{C}^+|\underline{G}^+$. The essential check : make sure that the †relation (7)

$$[\sigma,\lambda_\infty]\tag{34}$$

is respected by u.

I am going to make explicit the action of u on $\underline{M}$, by making explicit its values on

$$\begin{cases}\lambda_0=\omega l_0=\omega\sigma\rho\sigma\rho\\ \lambda_1=\omega l_1=\omega\rho\sigma\rho\sigma\end{cases}$$

in terms of †

$$\begin{cases}\zeta|\xi=\alpha_0-\rho(\alpha_0)+\nu_3\lambda_1\\ \eta_0=\beta_0-\sigma(\beta_0)\end{cases}\qquad \alpha|\&\qquad\in\Gamma\underline{M}\tag{35}$$

Linked to u by

$$u\rho=\zeta|\xi\rho,\qquad u\sigma=\omega^{\nu_2}\eta_0\sigma.\tag{36}$$

We shall have by the loop relation

$$u(\varepsilon_0)=\dagger\varepsilon_0^{\dagger(\mu)}\qquad(\mu=2\nu+1=4\nu'+2\nu_2+1)$$

whence 41

Uncertainty retained from cancelled passages.

140-4-s042-D001 sphere|scheme

140-4-s042-D003 [7]

140-4-s042-D004 $\zeta|\xi$ intro

140-4-s042-D006 μ

140-4-s042-D007 μ

[scan 43]

[p. 669]

$$u(l_0) = u(\varepsilon_0^2) = l_0^{(\mu)}$$

and

$$u(\lambda_0) = u(\omega l_0) = \underbrace{u(\omega)}_{=\omega}\, u(l_0) = \omega l_0^{(\mu)} = \underbrace{\omega l_0}_{\lambda_0}(\underbrace{l_0^{(2\nu)}}_{\lambda_0^{2\nu}}) = \lambda_0^{(2\nu+1)} = \lambda_0^{(\mu)}$$

as it should. (To reassure myself, I have checked that the second formula (33) does indeed give

$$u(\sigma^2) = (\eta_0\sigma)(\eta_0\sigma) = (\underbrace{\eta_0\, \sigma(\eta_0)}_{1})\dagger\omega = \omega \quad \text{i.e.}$$

$$u(\omega) = \omega \tag{37}$$

We shall have

$$u(\lambda_1) = u(\rho(\lambda_0)) = (\zeta|\xi\rho u)(\lambda_0) = (\zeta|\xi\rho)u(\lambda_0) = (\zeta|\xi\rho)(\lambda_0^\mu) = \lambda_1^\mu$$

thus finally, as predicted

$$u(\lambda) = \mu.\lambda \qquad \text{for} \quad \lambda \in \Gamma\underline{M} \tag{38}$$

We shall indeed have that $u(\sigma) = (\beta_0 - \sigma(\beta_0))\omega^{\nu_2}.\sigma$ commutes with $u(\lambda_\infty) = \mu.\lambda_\infty$.

where $\mu = 2\nu + 1 = 4\nu' + 2\nu_2 + 1$

All these calculations are a bit murky, logically, for at the same time I want to prove the existence of a $\dagger$ u of $\underline{C^+}|\underline{G^+}$, and in the preceding calculations I already assume it (implicitly, in a "bigger" group, namely the smooth|free group scheme (heuristic) generated by $\rho, \sigma \ldots$. One way of making them rigorous would be to use the intrinsic description of $\underline{C^+}|\underline{G^+}$: $\dagger$

$$\underline{C^+}|\underline{G^+} = C^+_{S_0}|G^+_{S_0} \underset{M_{S_0}}{\wedge} \underline{M}_{S_0} \tag{39}$$

thus defining a homomorphism$u$ from $\underline{C^+}|\underline{G^+}$ to an S_0-group G (here $G = \underline{C^+}|\underline{G^+}$) amounts to 42

Uncertainty retained from cancelled passages.

140-4-s043-D001 σ^2

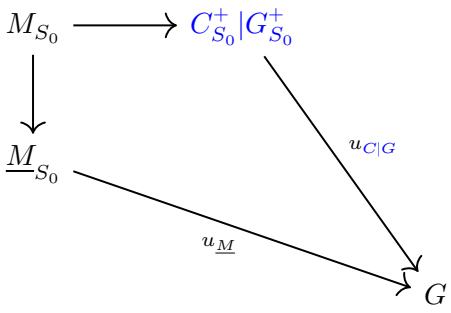
140-4-s043-D003 homomorphism

140-4-s043-D004 prove

define homomorphisms [scan 44]

$$\underline{M}_{S_0} \xrightarrow{u_{\underline{M}}} G^\sharp|G^\natural\ , \qquad C_{S_0}|G_{S_0} \xrightarrow{u_{C|G}} G^\sharp|G^\natural \quad \text{i.e.} \quad C|G \to G^\sharp(S_0)|G^\natural(S_0) \tag{40}$$

satisfying the following two conditions:
a) Commutativity of the diagram



, (41)

and b)

$$\begin{gathered} u_{\underline{M}} \text{ is compatible with the actions} \\ \text{of } C^+_{S_0}|G^+_{S_0} \text{ on } \underline{M}_{S_0} \text{ (via its action on } M_{S_0}) \\ \text{and on } G \text{ (via } u_{C|G}). \end{gathered} \tag{42}$$

In the present case, we will have $G = \underline{C}|\underline{G}$

$$u_{\underline{M}}(\lambda) = \dagger\, i(\mu.\lambda) \tag{43}$$

where

$$i : \quad \underline{M} \hookrightarrow \underline{C}^+|\underline{G}^+$$

is the inclusion, and

$$u_{C|G} : \; C^+_{S_0}|G^+_{S_0} \longrightarrow G \qquad \text{defined by (33)}, \tag{44}$$

We must therefore first check that (33), does define a homomorphism

$$C_{S_0}|G_{S_0} \to G$$

, or what amounts to the same,

$$C|G \to G(S_0)$$

. 43

Uncertainty retained from cancelled passages.
140-4-s044-D001 λ

For this we † use the presentation † of $C^+|G^+$ [scan 45] [p. 670]

†

$$C^+|G^+ = \left\{ \rho, \sigma \;\middle|\; \begin{array}{cc} \rho^6 = \sigma^4 = 1, & \rho^3 = \sigma^2 \\ [\sigma, \lambda_\infty] = 1 & \text{where} \quad \lambda_\infty = \dagger\rho\sigma\rho^{-1}\sigma\rho \end{array} \right\}, \tag{44}$$

thus setting

$$\begin{cases} \rho' = \zeta\rho|u(\rho), \quad \sigma' = \eta_0\omega^{\nu_2}\sigma & \text{cf. (33) and} \\ \zeta|\xi = \alpha_0 - \rho(\alpha_0) + \nu_3\lambda_1, \quad \eta|\eta_0 = \beta_0 - \sigma(\beta_0) \end{cases} \tag{45}$$

we must check the relations

$$\rho'^6 = \sigma'^4 = [\sigma', \lambda'_\infty] = 1, \quad \rho'^3 = \sigma'^2 \quad \text{where} \quad \lambda'_\infty = \dagger\rho'\sigma'\rho'^{-1}\sigma'\rho'. \tag{46}$$

We will begin by checking †

$$\rho'^3 = \sigma'^2 = \omega \tag{47}$$

which implies †

$$\rho'^6 = \sigma'^4 = 1, \quad \text{thus} \quad \exists \begin{array}{c} C_0^+|G_0^+ \xrightarrow{u_{C_0|G_0}} G(S_0) \\ \rho \longmapsto \rho', \quad \sigma \mapsto \sigma' \end{array} \tag{48}$$

Next, noting that we will have

$$\lambda'_\infty = u_{C_0|G_0}(\lambda_\infty),$$

it will suffice to † check

$$\lambda'_\infty = \mu\lambda_\infty \tag{49}$$

it will be obvious that λ'_∞ commutes with

$$\sigma' = u(\sigma) = \eta_0\sigma \qquad \dagger$$

since σ and $\eta_0 \in M$ commute with it. Thus to construct $u_{C|G}$, the essential checks are (47) and (49) — moreover once (47) is proved, the preceding calculations leading to (38) are justified (at least where they make sense, i.e., for $u_{C|G}$ and $\lambda \in M$), 44

Uncertainty retained from cancelled passages.

140-4-s045-D001 note that to the of index $C|G$
140-4-s045-D002 of
140-4-s045-D003 $C_0^+|G_0^+ \simeq \mathrm{SL}(2,\mathbb{Z})$
140-4-s045-D004 ρ
140-4-s045-D005 $\rho'(\lambda'_0)$
140-4-s045-D006 that|more
140-4-s045-D007 that
140-4-s045-D010 this
140-4-s045-D011 pui

so (49) will indeed be verified. [scan 46]

We have

$$\begin{aligned} \rho'^3 &= (\zeta|\xi\rho(\zeta|\xi)\rho^2(\zeta|\xi))\rho^3 \overset{\dagger}{=} \zeta|\xi\rho(\zeta|\xi)\rho^2(\zeta|\xi)\,\omega \\ \sigma'^2 &= (\eta_0\sigma(\eta_0))\sigma^2 = (\eta_0\sigma(\eta_0))\omega \end{aligned}$$

so formulas (47) reduce to:

$$\zeta|\xi\rho(\zeta|\xi)\rho^2(\zeta|\xi) = 1, \qquad \eta_0\sigma(\eta_0) = 1 \tag{50}$$

i.e., additively

$$\zeta|\xi + \rho(\zeta|\xi) + \rho^2(\zeta|\xi) = 0, \qquad \eta_0 + \sigma(\eta_0) = 0.$$

whose verification is trivial, given that

$$\rho_{\underline{M}}^3 = \sigma_{\underline{M}}^2 = \mathrm{id}_{\underline{M}}, \qquad (1 + \rho_{\underline{M}} + \rho_{\underline{M}}^2) = \dagger 0_{\underline{M}}.$$

We have thus indeed constructed a homomorphism †

$$u_{C_0^+|G_0^+}: \quad C_0^+|G_0^+ \longrightarrow \underline{C}_0^+(S_0)|\underline{G}_0^+(S_0) \ ,$$

and it remains to check (41) and (42).

For (41), we have already shown it at the level of the existence of a $u_{C_0|G_0}$ — it amounts to proving

$$(\quad u_{C_0|G_0}(\lambda_0) = \dagger\mu\lambda_0, \quad u_{C_0|G_0}(\lambda_1) = \mu\lambda_1$$

I will all the same redo the verification, to be sure.

$$\varepsilon_0' = u_{C_0|G_0}(\underbrace{\varepsilon_0}_{\sigma\rho}) = u_{C_0|G_0}(\sigma)u_{C_0|G_0}(\rho) = \eta_0\omega^{\nu_2}\dagger\sigma\zeta|\xi\rho = \eta_0\omega^{\nu_2}\sigma(\zeta|\xi)\sigma\rho$$
$$= \eta_0\sigma(\zeta|\xi)\omega^{\nu_2}\varepsilon_0$$

45

Uncertainty retained from cancelled passages.

140-4-s046-D002 $u(\rho) =$

140-4-s046-D004 $\mathrm{id}_{\underline{M}}$

140-4-s046-D005 to the

140-4-s046-D006 λ_0

140-4-s046-D007 $\zeta|\xi$

[scan 47]

I claim that [p. 671]

$$\varepsilon_0' = \varepsilon_0^{(\mu)} \quad \text{i.e.} \quad \eta_0\sigma(\zeta|\xi)\omega^{(\nu_2)} = \underbrace{\varepsilon_0^{(\mu-1)}}_{l_0^\nu} \quad \text{where } \mu - 1 = 2\nu\dagger, \quad \text{i.e.}$$
$$\eta_0\sigma(\zeta|\xi)\omega^{\nu_2} = l_0^{(\nu)} \quad \text{where } \nu = 2\nu' - |+\nu_2 \quad \text{i.e.}$$
$$\eta_0\sigma(\zeta|\xi) = \underbrace{l_0^{(2\nu')}}_{\lambda_0^{2\nu'}} \underbrace{(\omega l_0)^{(-\nu_2)}|(\omega l_0^{(-\nu_2)})}_{\lambda_0^{-\nu_2}} = \underbrace{\lambda_0^{(2\nu'-|+\nu_2)}}_{\lambda_0^\nu} \quad \text{i.e.}$$
$$\eta_0\sigma(\zeta|\xi) = \lambda_0^{(\nu)}$$

i.e., applying σ and noting that $\sigma(\eta_0) = \eta_0^{-1}$, $\sigma^2(\zeta|\xi) = \zeta|\xi$, $\sigma(\lambda_0) = \lambda_1$,

$$\eta_0^{-1}\zeta|\xi = \lambda_1(\nu) \quad \text{i.e. finally } \dagger\zeta|\xi = \eta_0\lambda_1(\nu)|\lambda_1^{(\nu)} \quad \text{or}$$
$$\zeta|\xi = \eta_0 + \nu\lambda_1 \quad \text{(written additively)}$$

which is true by hypothesis (cf. (31)).

We will then have († writing u instead of $u_{C_0|G_0}$ to simplify)

$$u_\dagger(\lambda_1) = u\rho(\lambda_0) = (\zeta|\xi\rho u)(\lambda_0) = \zeta|\xi\rho(u(\lambda_0)) = \zeta|\xi\rho(\lambda_0(\mu)) = \lambda_1(\mu)$$

OK.

It remains to check (42), which † amounts to the commutativity of $u_{\underline{M}} = \mu\,\mathrm{id}_{\underline{M}}$ with $(\rho_{\underline{M}}, \rho'_{\underline{M}})$, $(\sigma_{\underline{M}}, \sigma'_{\underline{M}})$ — this means also that $\rho_{\underline{M}}$ and $\rho'_{\underline{M}}$ on the one hand, $\sigma_{\underline{M}}$ and $\sigma'_{\underline{M}}$ on the other hand, †are equal, which is also obvious.

Thus, the group $\widetilde{Z}_{\rho,\sigma}$ is well and truly constructed — at least, as a scheme over $S_0 = \operatorname{Spec}\mathbb{Z}$, with morphisms of schemes

$$\begin{array}{l} \widetilde{\underline{Z}}_{\rho,\sigma} \xrightarrow{\nu'} G_a \\ \xrightarrow{\dagger} \dagger \\ \xrightarrow{\nu_2} \{0,1\}_{S_0} \\ \xrightarrow{\nu_3} \{0,1\}_{S_0} \end{array} \qquad \left| \qquad \begin{array}{l} \text{whence } \mu = 4\nu' + 2 + \nu_2: \quad \widetilde{Z}_{\rho,\sigma} \longrightarrow G_a \\ \qquad\qquad\qquad\qquad\qquad\quad \searrow \uparrow \\ \qquad\qquad\qquad\qquad\qquad\quad G_m \\ \varepsilon_2, \varepsilon_3: \quad \widetilde{Z}_{\rho,\sigma} \Longrightarrow \mu_{S_0} = \{\pm 1\}_{S_0} \end{array} \qquad (51)$$

Uncertainty retained from cancelled passages.

140-4-s047-D001 $+1| = 1$

140-4-s047-D002 $\zeta_0 = |\xi_0 =$

140-4-s047-D003 for $u = u_{C_0|G_0}|$for $u_{C_0|G_0}$

140-4-s047-D004 $C_0|G_0$

140-4-s047-D005 means

140-4-s047-D006 [illeg. 2]

140-4-s047-D007 $\nu|\varepsilon$

140-4-s047-D008 $\{\pm 1\}^2_{S_0}|\mu^2_{S_0}$

[scan 48]

and a morphism of † presheaves over S_0

$$\begin{array}{ccc} \underline{\widetilde{Z}}_{\rho,\sigma} & \longrightarrow & \underline{\mathrm{Aut}}_{\mathrm{gr}}(\underline{C}^+|\underline{G}^+), \\ \underline{u} & \longmapsto & u \end{array} \tag{52}$$

giving rise to a morphism

$$\begin{array}{ccc} \underline{\widetilde{Z}}_{\rho,\sigma} & \longrightarrow & \underline{\mathrm{Aut}}_{\mathrm{gr}}(\underline{C}^+|\underline{G}^+) \times \mathbf{G}_{\mathrm{m}} \times \mu_{S_0} \times \mu_{S_0} \\ \underline{u} & \longmapsto & u, \mu(\underline{u}), \varepsilon_2(\underline{u}), \varepsilon_3(\underline{u}) \end{array} \tag{53}$$

— calculations already done thirty-six times in various contexts should show that this morphism (which is monomorphic ? in characteristic $\neq 2, 3$) is a morphism of groups, for a suitable group|groups structure on $\underline{\widetilde{Z}}_{\rho,\sigma}$, uniquely determined (since it is so in characteristic $\neq 2, 3$, and we conclude by reason of schematic density). We will have †

$$1 \longrightarrow \underline{\widetilde{Z}}^{+0}_{\rho,\sigma} \longrightarrow \underline{\widetilde{Z}}_{\rho,\sigma} \longrightarrow \mathbf{G}_{\mathrm{m}} \times \mu_{S_0} \times \mu_{S_0} \longrightarrow 1 \tag{54}$$

m1459

Marginal note. m1459 a bit cavalier, since

$$\underline{Z}_{\rho,\sigma} \longrightarrow \mathbf{G}_{\mathrm{m}} \times \mu_{S_0} \times \mu_{S_0}$$

is not an epimorphism, because

$$\begin{array}{ccc} \underline{M} \times \underline{M} & \longrightarrow & \underline{M} \\ (\alpha_0, \beta_0) & \longmapsto & (1 - \rho_{\underline{M}})\alpha_0 - | + (1 - \sigma_{\underline{M}})\beta_0 \end{array}$$

is not epimorphic — where there is a swindle in characteristic 3 — cf. below…

making explicit

$$\begin{array}{ccc} \underline{\widetilde{Z}}^{+0}_{\rho,\sigma} & \underset{\text{subgroup scheme}}{\subset} & \underline{M} \times \underline{M} \times \mathbf{G}_{\mathrm{a}} \\ & \| & \\ \left\{ (\alpha_0, \beta_0, \nu') \,\middle|\, \begin{array}{c} \dagger 4\nu' = 0 \\ (1 + \rho_{\underline{M}})\alpha_0 = (1 - \sigma_{\underline{M}})\beta_0 + 2\nu'\lambda_1 \end{array} \right\} & & \end{array} \tag{55}$$

We will thus have

$$(56)\qquad 1 \longrightarrow \underline{\widetilde{Z}}^{+00}_{\rho,\sigma} \longrightarrow \underline{\widetilde{Z}}^{+0}_{\rho,\sigma} \longrightarrow \underline{{}_4\mathbf{G}_{\mathrm{a}}} \longrightarrow 1$$

47

Uncertainty retained from cancelled passages.
140-4-s048-D001 functors
140-4-s048-D002 this is not
140-4-s048-D003 α_0

[scan 49]
[p. 672]

where

$$(57)\qquad \begin{aligned} \underline{\widetilde{Z}}^{+00}_{\rho,\sigma} &\overset{\text{dfn}}{=} \left\{ \underline{u} \in \Gamma \underline{\widetilde{Z}}_{\rho,\sigma} \,\middle|\, \nu'(\underline{u}) = \nu_2(\underline{u}) = \nu_3(u|\underline{u}) = 0 \right\} \\ &\simeq \\ &\left\{ (\alpha_0, \beta_0) \in \Gamma(\underline{M} \times \underline{M}) \,\middle|\, (1-\rho_{\underline{M}})\alpha_0 = (1-\sigma_{\underline{M}})\beta_0 \right\}. \end{aligned}$$

m1461

Marginal note. m1461 in characteristic 3 its relative dimension jumps from 2 to 3 !

group scheme $\underline{Z}^{+00}_{\rho,\sigma}$ is smooth in characteristic $\neq 3$ of relative dimension 2 — it is even canonically a Module scheme, we will have

$$(58)\qquad \underline{\widetilde{Z}}^{+00}_{\rho,\sigma} \simeq \mathbf{V}|\mathbb{V}(\underline{P})$$

where P is the free $\mathbb{Z}$-module of rank 2 defined † by the exact sequence

$$(59)\qquad \begin{array}{ccccccccc} 0 & \longrightarrow & \dagger P & \longrightarrow & M \times M & \longrightarrow & M & \longrightarrow & 0 \\ & & & & (\alpha_0, \beta_0) & \longmapsto & (1-\rho_{\underline{M}})\alpha_0 - (1-\sigma_{\underline{M}})\beta_0 & & \end{array}$$

The homomorphism $M \times M \longrightarrow M$ is made explicit, setting

$$(60)\qquad \begin{cases} \alpha_0 = a_0\lambda_0 + a_1\lambda_1 & \text{hence} \quad \alpha_0 - \rho\alpha_0 = (a_0 + a_1)\lambda_0 + (2a_1|b_1 - a_0)\lambda_1 \\ \beta_0 = b_0\lambda_0 + b_1\lambda_1 & \text{hence} \quad \beta_0 - \sigma\beta_0 = (b_0 - b_1)\lambda_0 + (b_1 - b_0)\lambda_1 \end{cases}$$

as

$$(61)\qquad (\alpha_0, \beta_0) \longmapsto \underbrace{(a_0 + a_1 + (b_1 - | + b_0))}_{c_0}\lambda_0 + \underbrace{(2a_1 - a_0 - (b_1 - b_0))}_{c_1}\lambda_1$$

† This is a surjective homomorphism except in characteristic 3, † — the elements of the image are the

$$(62)\qquad \left\{ c_0\lambda_0 + c_1\lambda_1 \,\middle|\, \underbrace{c_0 + c_1}_{=3a_1} \equiv 0 \quad (3) \right\}$$

Remark If we take †a u defined over $\widehat{\mathbb{Z}}$ coming from an element of $M[0]$, then 48

Uncertainty retained from cancelled passages.
140-4-s049-D001 [illeg. 1]
140-4-s049-D002 M
140-4-s049-D003 We
140-4-s049-D005 we will set $a_1|c_1$ [illeg. 1]

140-4-s049-D006 a

[scan 50]

we will have

$$\mu \in \widehat{\mathbb{Z}}^*, \quad \mu \equiv \dagger(-1)^{\nu_3}(=1-2\nu_3) \mod (3)$$

and the condition on $(\mu, \varepsilon_2, \varepsilon_3)$ for being in the image of

$$\widetilde{\underline{Z}}_{\rho,\sigma} \longrightarrow \mathbf{G}_{\mathrm{m}} \times \mu_{S_0} \times \mu_{S_0} \ ,$$

which is written in general (by virtue of (62) & and of (32))

$$2\nu' - \nu_2 + \nu_3 \equiv 0 \quad (3)$$

becomes, on multiplying by 2

$$\underbrace{4\nu' - 2\nu_2}_{\mu-1} + 2\nu_3 \equiv 0 \quad (3)$$

i.e.

$$\mu \equiv 1 - 2\nu_3 \quad (3),$$

and it is automatically satisfied.

† m1463

Marginal note. m1463 by expressing|requiring $\mu|\nu$ are written

$$\left.\begin{array}{r} \mu \equiv 1 - 2\nu_2 \quad (4) \\ \mu \equiv 1 - 2\nu_3 \quad (6) \end{array}\right\rceil$$

$$\begin{array}{rl} \mu = 1 - |+ 2\nu_2 + 4\nu' & \text{with} \\ \mu = 1 - 2\nu_3 + 6\nu'' & \\ -\nu_2 + 2\nu' = -\nu_3 + 3\nu'' & \text{i.e.} \\ 2\nu' + \nu_3 - \nu_2 = 3\nu'' & \end{array}$$

† 49

Uncertainty retained from cancelled passages.

140-4-s050-D002 $\varepsilon_3|\varepsilon_2$

140-4-s050-D003 Z $\Gamma|\mu$ $\mathbf{G}_{\mathrm{a}}$ $-|+$ ν_2 1 $2\nu' - \nu_2 + \nu_3$

140-4-s050-D009 will|have ν_3 $-|+$ loop|loop $\underline{\lambda_1}|\lambda_1$

[scan 51]

3. The groups † $\mathbb{O}_N^*$. [p. 673]

We are going to introduce an auxiliary construction, by defining a group scheme which corresponds to the datum of quantities $\mu \in \Gamma\mathbf{G}_{\mathrm{m}}$, $\nu', \nu'' \in \Gamma\mathbf{G}_{\mathrm{a}}$, $\nu_2, \nu_3 \in \Gamma\{0,1\}_{S_0}$, satisfying

$$(63) \qquad \mu = 4\nu' - 2\nu_2 + 1 = 6\nu'' - 2\nu_3 + 1, \qquad 2\nu' - \nu_2 = 3\nu'' - \nu_3.$$

In general, let $N \in \mathbb{N}^*$ be a fixed natural integer (the case which interests us for what follows is $N = 12$), and let us consider the scheme in groups, $\mathbb{O}_N$ over $\mathbb{Z}$ associated to the extension of discrete groups

$$(64) \qquad 1 \longrightarrow \mathbb{Z} \xrightarrow{N} \mathbb{Z} \longrightarrow \mathbb{Z}/N\mathbb{Z} \longrightarrow 1$$

which fits into an exact sequence

$$(65) \qquad 1 \longrightarrow \mathbf{E}^1|\mathbb{E}^1 \xrightarrow{i} \mathbb{O}_N \longrightarrow (\mathbb{Z}/N\mathbb{Z})_{S_0} \longrightarrow 1.$$

Let us recall the † simple construction

$$\mathbb{O}_N = (\mathbb{Z})_{S_0} \underset{(N\mathbb{Z})}{\wedge} \underline{N\mathbb{Z}}. \tag{66}$$

I say that there is in|on $\mathbb{O}_N$ a unique Ring structure, whose addition is the abelian group structure already envisaged, and where multiplication is obtained as follows: A section of $\mathbb{O}_N$ is of the form † $n + N[\lambda]$, with the rules of calculation

$$\begin{cases} \text{a)} & (n + N[\lambda]) + (n' + N[\lambda']) = (n + n') + N[\lambda + \lambda'] \\ \text{b)} & n + N[\lambda] = n' + N[\lambda'] \iff \dagger\ n + N\lambda = n' + N\lambda' \text{ in } \Gamma\mathbf{G}_{\mathrm{a}}. \end{cases} \tag{67}$$

50

Uncertainty retained from cancelled passages.
140-4-s051-D001 $\widetilde{Z}_{\rho,\sigma}$ revised [illeg. 2]
140-4-s051-D003 defi
140-4-s051-D006 $k|x$
140-4-s051-D007 $\xi|n$

[scan 52]

We then multiply the sections by

$$(n + N[\lambda])(n' + N[\lambda']) = nn' + N[n\lambda' + n'\lambda + \lambda\lambda'], \tag{68}$$

we check that it is † compatible with the relations (67 b), that it is biadditive, commutative, associative, that $1 + N[0]$ is the unit… Another way of † seeing the ring structure. Let more generally A be a † $\mathbb{Z}$-algebra, † associative unital|unitary (say) † equipped with an ideal J two-sided which is a free $\mathbb{Z}$-module of finite type, let us consider † and the extension structure

$$0 \longrightarrow \dagger J \longrightarrow A \longrightarrow \overbrace{A/\dagger J}^{B} \longrightarrow 0 \tag{69}$$

hence a commutative group scheme $\underline{A}_{[N]}$, which is obtained as a quotient of † semidirect product $A_{S_0} \cdot \underline{J}$)

$$\begin{gathered} 0 \longrightarrow \underline{J} \longrightarrow A_{S_0} \times \underline{J} \longrightarrow \underline{A}_N \longrightarrow 0 \\ \lambda \longmapsto (\lambda, -\lambda) \end{gathered} \tag{70}$$

and we find that $A_{S_0} \cdot \underline{J}$ has a $\frac{1}{2}$-direct product Ring structure, and that 51

Uncertainty retained from cancelled passages.
140-4-s052-D001 biad
140-4-s052-D002 it|it con|en
140-4-s052-D005 $\mathbb{Z}|z$
140-4-s052-D006 com
140-4-s052-D007 principal
140-4-s052-D008 an
140-4-s052-D010 NA
140-4-s052-D011 $NA|N$
140-4-s052-D012 $S|S_0$ $NB|NA$ $\xi \to |p$

[scan 53]

the image of $\underline{J}$ in it is an ideal … [p. 674]

We do find a ring structure on $\underline{A}_{[N]}$, and an exact sequence †

$$1 \longrightarrow \underline{J} \longrightarrow \underline{A}_{[N]} \longrightarrow B_{S_0} \longrightarrow 1 \tag{71}$$

where $\underline{A}_{[N]} - B_{S_0}$ is a morphism of ring schemes. Passing to the multiplicative groups †, we find an exact sequence

$$\begin{array}{ccccccccc} 1 & \longrightarrow & (1+\underline{J})^* & \longrightarrow & A^*_{[N]} & \longrightarrow & (B_{S_0})^* & \longrightarrow & 1 \\ & & & & & & \| & & \\ & & & & & & (B^*)_{S_0} & & \end{array} \tag{72}$$

where for $A^*_{[N]}$, $(B_{S_0})^{*|'}$ and B^*, the $*$ denotes the passage to the multiplicative subgroup and where $(1+\underline{J})^*$ has the analogous meaning

$$(1+\underline{J})^* \subset \mid \subseteq 1+\underline{J}$$

$$\| \tag{73}$$

$$\{(1+\lambda) \mid 1+\lambda \text{ invertible in } \underline{A}\}$$

$$\wr$$

$$\{(\lambda,\lambda') \in \underline{J}\times\underline{J} \mid \lambda+\lambda'+\lambda\lambda' = 0 \text{ in } \underline{J}\}.$$

Thus we will have, in the special case $A=\mathbb{Z}$, $J=N\mathbb{Z}$

$$1 \longrightarrow (\underline{1+N\mathbb{Z}})^* \longrightarrow \mathbb{O}^*_N \longrightarrow (\mathbb{Z}/N\mathbb{Z})^*_{S_0} \longrightarrow 1 \tag{74}$$

where the kernel 52

Uncertainty retained from cancelled passages.

140-4-s053-D001 of

140-4-s053-D002 rings

† [scan 54]

$(1+N\mathbb{Z})^*$ is defined as

$$(1+N\mathbb{Z})^* \overset{\text{dfn}}{=} \{\lambda \in \dagger\underline{\mathbb{Z}} = \mathbf{G}_a \mid 1+N\lambda \text{ invertible}\} \tag{75}$$

†

The $\mathbb{O}_N$ for varying N form a projective system of Rings, the $\mathbb{O}^*_N$ form a projective system of commutative group schemes.

I am not going to look more closely at the structure of these projective systems, nor examine the projective limits — questions which arise more general|generally when one starts from an A as above. We are going to make explicit the canonical homomorphism

$$\mathbb{O}^*_N \longrightarrow \mathbb{O}^*_1 = \mathbb{O}^* = \mathbf{G}_m \tag{76}$$

†

which in general is neither an epimorphism, nor a monomorphism. Its kernel is † isomorphic to the subgroup|system of $+$

$$\{1+N[\lambda] \mid 1+N\lambda = 1 \text{ i.e. } N\lambda = 0\} \tag{77}$$

which as a scheme (without group structure) is isomorphic to ${}_N\mathbf{G}_a$, 53

Uncertainty retained from cancelled passages.
140-4-s054-D001 674
140-4-s054-D002 $\mathbf{G}_a$
140-4-s054-D003 homomorphism in part
140-4-s054-D004 φ
140-4-s054-D007 φ λ

[scan 55]

which is a group scheme (not flat if $N \neq 1$) which is zero over $\operatorname{Spec}\mathbb{Z}[1/N]$, and † the fibers [p. 675]
at the $p \mid N$ are $\simeq \mathbf{G}_a$.

I realize now that it is a clumsy move to leave the group $\mathbb{O}_N^*$, which is perfect of its kind, to refer to $\mathbf{G}_m$.

I am going to examine in particular

$$1 \longrightarrow (1+12\mathbb{Z})^* \longrightarrow \mathbb{O}_{12}^* \longrightarrow (\mathbb{Z}/12\mathbb{Z})^*_{S_0} \longrightarrow 1. \tag{78}$$

Let us note that

$$(\mathbb{Z}/12\mathbb{Z})^* \simeq (\mathbb{Z}/4\mathbb{Z})^* \times (\mathbb{Z}/6\mathbb{Z})^* \simeq \mu \times \mu.$$

Thus (78) takes the form †

$$1 \longrightarrow (1+12\mathbb{Z})^* \longrightarrow \mathbb{O}_{12}^* \xrightarrow{\varepsilon_2 \times \varepsilon_3} \mu \times \mu \longrightarrow 1 \tag{79}$$

The quantities $\nu_{2|3}$ ν_3 are also morphisms defined on $\mathbb{O}_{12}^*$

†

$$\mathbb{O}_{12}^* \xrightarrow{(\nu_2,\nu_3)} \{0,1\}_{S_0} \times \{0,1\}_{S_0} \tag{80}$$

they are the composites

$$\mathbb{O}_{12}^* \xrightarrow{(\varepsilon_2,\varepsilon_3)} \mu \times \mu \xrightarrow{\sim} \{0,1\}_{S_0} \times \{0,1\}_{S_0} \tag{81}$$

deduced from † $\begin{array}{ccc} 1 & \mapsto & 0 \\ -1 & \mapsto & 1 \end{array}$

54

Uncertainty retained from cancelled passages.
140-4-s055-D001 dimension|of dimension
140-4-s055-D003 the|the [illeg. 1] of|of the group|kernel over β
140-4-s055-D004 here
140-4-s055-D005 $(+1) \mapsto (0)$

[scan 56]

†

and they are group homomorphisms (for addition modulo 2 on $\{0,1\}_{S_0}$.

We also have $\varepsilon_2 = (-1)^{\nu_2} = 1 - 2\nu_2, \quad \varepsilon_3 = (-1)^{\nu_3} = 1 - 2\nu_3$

The "four cases of signs" in $(\mathbb{Z}/12\mathbb{Z})^*$ are given in the table

$\varepsilon_3 \backslash \varepsilon_2$	$+1$ i.e. $\nu_2 = 0$	-1 i.e. $\nu_2 = 1$
$+1$ i.e. $\nu_3 = 0$	1	$+5$
-1 i.e. $\nu_3 = 1$	-5	-1

(82)

thus we find a set-theoretic section of (79) by taking the sections of $\underline{\mathbb{O}}^*_{12}$ of the form

$$\underline{\mu}_0 = (-5)^{\nu_2}(+5)^{\nu_3} \begin{cases} \equiv (-1)^{\nu_3} & (6) \\ \equiv (-1)^{\nu_2} & (4) \end{cases} \tag{83}$$

Thus a general section of $\mathbb{O}^*_{12}$ is written

$$\underline{\mu} = \underbrace{(-5)^{\nu_2}(+5)^{\nu_3}}_{\mu_0} + 12[\dagger\nu_0|\nu_0] \tag{84}$$

where

$$\begin{cases} (\nu_2, \nu_3) \in \Gamma(\{0,1\}_{S_0} \times \{0,1\}_{S_0}) \\ \lambda_0 \in \Gamma(\mathbf{G}_a). \end{cases}$$

† Let us note that we have

$$5^{\nu_3} = 1 + 4\nu_3$$

and

$$(-5)^{\nu_2} = \left[\begin{array}{r} (-1)^{\nu_2}5^{\nu_2} = (1-2\nu_2)(1+4\nu_2) \\ = 1 + 2\nu_2 - 8\nu_2^2 \end{array}\right] = 1 - 6\nu_2$$

N.B. $\underline{\nu_2 = \nu_2^2}$

thus

$$\begin{aligned} \underline{\mu}_0 &= (1+4\nu_3)(1-6\nu_2)\dagger \\ &\dagger \\ &\dagger \\ &= 1 + 4\nu_3 - 6\nu_2 - 24\nu_2\nu_3 \end{aligned} \tag{85}$$

†

†

55

Uncertainty retained from cancelled passages.

140-4-s056-D001 that
140-4-s056-D002 λ_0
140-4-s056-D003 argue by|more modulo
140-4-s056-D005 -8
140-4-s056-D006 (-8)
140-4-s056-D007 $-\ \nu_2$
140-4-s056-D008 could
140-4-s056-D009 85′|85

[scan 57]
[p. 676]

If we have ((84) and (85))

$$\underline{\mu} = \underline{\mu}_0 + 12[\dagger\nu_0|\nu_0] = 1 + 4\nu_3 - 6\nu_2 + 12(\nu_0 - 2\nu_2\nu_3)$$

we conclude from this (by considering the congruences modulo 4)

$$\mu = 1 - 2\nu_2 + 4\underbrace{[\nu_3 - \nu_2 - |+3\nu_2\nu_3 + 3\nu_0]}_{\nu'} \tag{86}$$

and by|for congruence|congruences modulo 6

$$\mu = 1 - 2\nu_3 + 6\underbrace{[\nu_3 - \nu_2 - 2\nu_2\nu_3 + 2\nu_0]}_{\nu''}, \tag{87}$$

where the underline in $\underline{\mu}$ denotes the section in $\mathbb{O}_{12}^*$, the non-underlined μ † the corresponding|corresponding section of $\mathbf{G}_m = \mathbb{O}_1^*$. We will indeed have

$$\begin{aligned} 2\nu' - \nu_2 &= 2\nu_3 - 3\nu_2 - 6\nu_2\nu_3 + 6\nu_0 \qquad '' \\ 3\nu'' - \nu_3 &= 2\nu_3 - 3\nu_2 - 6\nu_2\nu_3 + 6\nu_0 \end{aligned}$$

In summary

Proposition Let us introduce on the smooth group|group scheme $\mathbb{O}_{12}^*$ the functions $\underline{\mu} \mapsto \nu_2|\xi_2, \nu_3$, † (with values in $\{0,1\}_{S_0} \subset \mathbf{G}_a = \mathbb{O}_1$), ν', ν'' † (with values in $\mathbb{O}_1|\mathbf{G}_a$) and μ (with values in ${\mathbf{G}_m}^* = \mathbb{O}_1^* \subset \mathbb{O}_1$) defined by the conditions

$$\left\{ \begin{array}{lll} \mu = & & \text{canonical image of } \underline{\mu} \text{ in } \mathbb{O}_1^* \text{ under} \\ & & \text{the homomorphism of Groups } \mathbb{O}_N^* \longrightarrow | \; -\mathbb{O}_1^* \\ (-1)^{\nu_2} & & \\ \quad \dagger \quad & = & \text{canonical image of } \underline{\mu} \text{ in } (\mathbb{Z}/4\mathbb{Z})_{S_0}^* \\ \quad \dagger & & \\ (-1)^{\nu_3} & = & \qquad\qquad\qquad\qquad\qquad\qquad (\mathbb{Z}/6\mathbb{Z})_{S_0}^* \end{array} \right. \tag{88}$$

whence

$$\dagger \quad \left\{ \begin{array}{l} \underline{\mu} = \dagger(-5)^{\nu_2}(+5)^{\nu_3} + 12[\nu_0|\lambda_0] \\ \quad = (1+4\nu_3)(1-6\nu_2) + 12\nu_0 \\ \quad = 1 + 4\nu_3 - 6\nu_2 + 12[\nu_0 - 2\nu_2\nu_3] \end{array} \right. \tag{89}$$

for a well-determined section ν_0 of $\mathbf{G}_a = \mathbb{O}_1$ 56

Uncertainty retained from cancelled passages.

140-4-s057-D001 λ_0
140-4-s057-D002 a
140-4-s057-D003 $\xi_3|\varepsilon_3$ and
140-4-s057-D004 $\mu_3|\nu_3$
140-4-s057-D005 the image
140-4-s057-D006 $\varepsilon_2|\xi_2$
140-4-s057-D007 $\left\{\right.$
140-4-s057-D008 μ_0

[scan 58]

Let us also put

$$\left\{ \begin{array}{l} \nu' = \nu_3 - \nu_2 - 3\nu_2\nu_3 + 3\nu_0 \\ \nu'' = \nu_3 - \nu_2\dagger - |\dagger + 2\nu_2 \cdot \nu_3 + 2\nu_0. \end{array} \right. \tag{90}$$

We then have the relations

$$\left\{ \begin{array}{ll} \mu = 1 - 2\nu_2 + 4\nu' & (= \varepsilon_2 + 4\nu') \\ \mu = 1 - 2\nu_3 + 6\nu'' & (= \varepsilon_3 + 6\nu'') \\ \quad . & \\ \underbrace{-\nu_2 + 2\nu'}_{\nu} = \underbrace{-\nu_3 + 3\nu''}_{\nu} & \text{i.e. } 2\nu' - \nu_2 + \nu_3 = 3\nu''. \end{array} \right. \tag{91}$$

m1475

Marginal note. m1475 where $\varepsilon_p = (-1)^{\nu_p}$
whence †
$\varepsilon_p - 1 = -2\nu_p$
$(p \in \{2, 3\})$

N.B. The condition $\underline{\mu} = 1$ is equivalent to the conditions

$$\nu_2 = \nu_3 = \nu_0 = 0 \tag{92}$$

and implies

$$\nu = \nu' = \nu'' = 0 \tag{93}$$

Uncertainty retained from cancelled passages.
140-4-s058-D001 +
140-4-s058-D002 —
140-4-s058-D003 $\varepsilon_p = 1 - 2\nu_p$

[scan 59]
[p. 677]

4. The groups|The group † $N_{\rho,\sigma}$, †and the exact sequence

$$1 \to N^{0+}_{\rho,\sigma} \to N_{\rho,\sigma} \to \mathbb{O}^*_{12} \text{ —— } 1.$$

Relations with $\mathrm{Gl}(2, \mathbb{Z}/12\mathbb{Z})$.
We define it as embedded in $\mathbb{O}^*_{12} \times \underline{M} \times \underline{M}$:

$$\widetilde{Z}_{\rho,\sigma} \subset \mathbb{O}^*_{12} \times \underline{M} \times \underline{M} \tag{94}$$
$$\{(\underline{\mu}, \alpha_0, \beta_0) \mid (1 - \rho_{\underline{M}})(\alpha_0) - (1 - \sigma_{\underline{M}})(\beta_0) = 3\nu''(\underline{\mu})\lambda_1\}$$

Note that we have

$$(1 - \rho_{\underline{M}})(\underbrace{\lambda_1 - \lambda_0}) = 3\lambda_1 \tag{95}$$

so the cocycle relation in (72|77) thus takes the form

$$(1 - \rho_{\underline{M}})(\alpha_0 - \nu''(\lambda_1 - \lambda_0)) - (1 - \sigma_{\underline{M}})(\beta_0) = 0 \tag{96}$$

I shall assume again that we have a natural group structure on $\widetilde{Z}_{\rho,\sigma}$, for which the projection $\widetilde{Z}_{\rho,\sigma} \to \mathbb{O}^*_{12}$ is a group homomorphism. We now find an exact sequence
(97)

$$1 \text{ ——— } \widetilde{Z}^{+0\dagger}_{\rho,\sigma} \text{ ———→ } \widetilde{Z}_{\rho,\sigma} \text{ ———→ } \mathbb{O}^*_{12} \text{ ——— } 1$$

where

$$\widetilde{Z}^{+0}_{\rho,\sigma} \overset{\text{dfn}}{=} \mathrm{Ker}(\widetilde{Z}_{\rho,\sigma} - \mathbb{O}^*_{12}) \tag{98}$$
$$\wr\!|$$
$$\{(\alpha_0, \beta_0) \mid (1 - \rho_{\underline{M}})(\alpha_0) - (1 - \sigma_{\underline{M}})(\beta_0) = 0\}$$

This group scheme is smooth in characteristic $\neq 3$, but not in characteristic 3, where its dimension goes

Uncertainty retained from cancelled passages.
140-4-s059-D001 and the sequence [illeg. 2]
140-4-s059-D002 exact
140-4-s059-D003 *

[scan 60]

from relative dimension 2 to relative dimension 3. This fact of an exceptional fiber of larger dimension cannot, it seems to me, be eliminated by a dodge, † by introducing some ad-hoc definitions for a "better" group $\widetilde{Z}_{\rho,\sigma}$: there are "more" automorphisms of $\underline{\mathcal{G}}_k$ (for k an algebraically closed field), having the properties envisaged, if k is of characteristic 3, than in the other cases of characteristic.

However, it seems to me that we can construct a subgroup scheme † of $\widetilde{Z}_{\rho,\sigma}$, which I feel like denoting $N_{\rho,\sigma}$, which is smooth over $S_0 = \mathrm{Spec}(\mathbb{Z})$ and coincides with $\widetilde{Z}_{\rho,\sigma}$ over † $S_{0,3|2} = \mathrm{Spec}\,\mathbb{Z}[1/3] = S_0 \setminus \{3\}$. To describe it, I note that in the loop equation(96), the left-hand side is automatically|automatic in †the submodule of $\underline{M}$ corresponding to the congruence

$$c_0 + c_1 \equiv 0 \ (3) \qquad (\text{cf } 62)$$

and if we introduce the corresponding sub-$\mathbb{Z}$-module of index

Uncertainty retained from cancelled passages.
140-4-s060-D001 of
140-4-s060-D003 of
140-4-s060-D004 $S_{0,3}$
140-4-s060-D006 which

[scan 61]
[p. 678]

3, M_0 of M

(99) $\quad M_0 \subset M$
'' dfn
$\{c_0\lambda_0 + c_1\lambda_1 \mid c_0 + c_1 \equiv 0 \ (3)\}$,

whence a vector bundle $\underline{M}_0$ over S_0 — we in fact have a morphism of schemes

(100)

$$\widetilde{Z}_{\rho,\sigma} \longrightarrow \underline{M}_0$$

$$(\underline{\mu}, \alpha_0, \beta_0) \longmapsto (1 - \rho_{\underline{M}})(\underbrace{\alpha_0 + \nu''(\lambda_1 - \lambda_0)}_{\widetilde{\alpha}_0}) + (1 - \sigma_{\underline{M}})(\beta_0)$$

which we can make explicit as follows: we have a homomorphism of $\underline{\mathbb{Z}}$-modules
(101)

$$M \times M \times \mathbb{Z} \longrightarrow M_0$$

$$(\alpha_0, \beta_0, \nu'') \longmapsto (1-\rho_M)(\alpha_0 + \nu''(\lambda_1 - \lambda_0)) + (1-\sigma_M)(\beta_0)$$

whence a homomorphism for the corresponding module schemes
(102)

$$\underline{M} \times \underline{M} \times \underset{\mathbb{G}_a}{\underline{\mathbb{Z}}} \longrightarrow \underline{M_0}$$

which we compose with
(103)

$$\widetilde{Z}_{\rho,\sigma} \longrightarrow \underline{M} \times \underline{M} \times \underline{\mathbb{Z}}$$

$$(\underline{\mu}, \alpha_0, \beta_0) \longmapsto (\alpha_0, \beta_0, \nu''(\mu))$$

We can then consider the cocycle equation (96) as an equation in $\underline{M_0}$ — it thus becomes stricter,

[scan 62]

whence a closed subscheme of $\widetilde{Z}_{\rho,\sigma}$,

(104) $N_{\rho,\sigma} \subset \widetilde{Z}_{\rho,\sigma}$
''
$\{(\underline{\mu}, \alpha_0, \beta_0) \in \widetilde{Z}_{\rho,\sigma} \mid$ the loop equation (96) valid in $\underline{M_0}\}$.

Let us then consider the morphism

$$N_{\rho,\sigma} \longrightarrow \mathbb{O}_{12}^*$$

m1476

Marginal note. m1476 N.B. The group $N_{\rho,\sigma}$ should also be denotable by $M|\widetilde{N}_{\rho,\sigma}[0]$, unless I am mistaken it Or|In rather it would be $(M^{\sharp|b}_{\rho,\sigma}/M_\rho)[0]$ † with the notations of § 49, †VI

It is clear that this morphism is still an epimorphism, the fiber over a $\underline{\mu} = (\nu_2, \nu_3, \nu_0)$ being made up of the $(\alpha_0, \beta_0) \in \Gamma\, \underline{M} \times \underline{M}$ satisfying (96) in M_0, † is identified|being identified with the scheme of $(\tilde{\alpha}_0, \beta_0)$ satisfying the linear equation

†

in $\tilde{\alpha}_0, \beta_0$

(105) $\qquad (1-\rho_{\underline{M}})(\tilde{\alpha}_0) - (1-\sigma_{\underline{M}})(\beta_0) = 0 \qquad$ in $\underline{M_0}$.

Now the homomorphism of Modules
(106)

$$\begin{cases} \underline{M} \times \underline{M} \longrightarrow \underline{M}_0 \\ (\tilde{\alpha}_0, \beta_0) \longmapsto (1-\rho_{\underline{M}})(\tilde{\alpha}_0) - (1-\sigma_{\underline{M}})(\beta_0) \end{cases}$$

is a smooth epimorphism — being associated to a corresponding homomorphism

$$M \times M \longrightarrow M_0$$

which is now|manifestly an epimorphism. Thus † $N_{\rho,\sigma}$ is smooth over S_0. † Being a closed subscheme of $\widetilde{Z}_{\rho,\sigma}$

Uncertainty retained from cancelled passages.
140-4-s062-D002 according to
140-4-s062-D003 2
140-4-s062-D004 take the composite homomorphism
140-4-s062-D005 $\mathbb{Z}$
140-4-s062-D008 $\widetilde{Z}$
140-4-s062-D009 This

[scan 63]
[p. 679]

which coincides with it over $S_0 \setminus \{3\}$, it follows that it is the schematic closure that|by

$$(107) \qquad N_{\rho,\sigma} = \widetilde{Z}_{\rho,\sigma} | (\mathrm{Spec}(\mathbb{Z}) \setminus \{3\}),$$
(schematic closure)

which implies that it is also a subgroup scheme.

Perhaps this is the moment to look a little more closely at the sub-$\mathbb{Z}$-module M_0 of M,

$$(108) \qquad \begin{aligned} M_0 &= \mathbb{Z}(\lambda_1 - \lambda_0) + \mathbb{Z}(3\lambda_1) \\ &= \mathbb{Z}\underbrace{(\lambda_1 - \lambda_0)}_{\delta} + \mathbb{Z}(1-\rho_M)\underbrace{(\lambda_1 - \lambda_0)}_{\delta} \\ &= \mathbb{Z}(\delta, |) \oplus \mathbb{Z}(1-\rho_M)(\delta) \\ &= \mathbb{Z}\delta + \mathbb{Z}\rho_M\delta. \end{aligned}$$

Note that M_0 is stable under ρ, σ, in general by construction, for $\forall\ x \in M$, we

$$(109) \qquad \begin{aligned} x &\equiv \rho_M(x) \quad (M_0) \quad (\text{i.e. } (1-\rho_M)(x) \in M_0) \\ x &\equiv \sigma_M(x) \quad (M_0) \quad (1-\sigma_M)(x) \in M_0. \end{aligned}$$

It is of course also stable under $\tau_M = -\mathrm{id}_M$.

Thus M_0 is normalized by $\mathcal{G}$, hence as a subgroup of $\mathcal{G}$) normal in $\mathcal{G}$. It therefore corresponds to a normal subgroup of $\mathcal{G}_0 = \mathrm{Gl}(2,\mathbb{Z})$,

[scan 64]

containing $[\Pi_0, \Pi_0]$, of index 3 in Π_0. I presume that it contains the congruence subgroup relative to 12 — i.e. that it corresponds to a normal subgroup of $\mathrm{Sl}(2, \mathbb{Z}/12\mathbb{Z})$. It is of index $3.12 = 36$ in $\mathrm{Sl}(2,\mathbb{Z})$, , † that

$$\mathrm{Sl}(2,\mathbb{Z}/12\mathbb{Z}) \simeq \underbrace{\mathrm{Sl}(2,\mathbb{Z}/4\mathbb{Z})}_{48} \times \underbrace{\mathrm{Sl}(2,\mathbb{Z}/3\mathbb{Z})}_{24}$$

— it would therefore be a subgroup of order 32 of $\mathrm{Sl}(2,\mathbb{Z}/12\mathbb{Z})$, it is obviously the normal subgroup generated by the elements † the image of ,

$$\begin{cases} (\sigma(\rho)\rho^{-1})^3 = [\lambda_0,\lambda_1], \text{ or } [\sigma_\infty,\lambda_\infty] \\ \text{cf. (7)} \\ \delta \overset{\text{dfn}}{=} \lambda_1\lambda_0^{-1} \ \dagger \\ \rho(\delta) \end{cases}$$

or again the normal subgroup generated by the single element

$$(110) \qquad \begin{cases} \delta = \lambda_1\lambda_0^{-1} = \ell_1\ell_0^{-1} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 1 & +2 \\ 0 & 1 \end{pmatrix} \\ \quad = \begin{pmatrix} 1 & +2 \\ 2 & 5 \end{pmatrix} \end{cases}$$

The question is therefore whether this normal subgroup of $\mathrm{Sl}(2,\mathbb{Z}/12\mathbb{Z})$ is well and truly of order 32, hence of index 36 — its order could be larger, its index smaller. To begin with, what is the order of δ?

Uncertainty retained from cancelled passages.

140-4-s064-D003 while

140-4-s064-D004 [illeg. 1]

140-4-s064-D006 $\in \Pi_0$

[scan 65]
[p. 680]

modulo 12? We have

$$\overset{\dagger}{\text{whence}} \quad \begin{cases} \delta^2 = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix} = \begin{pmatrix} 5 & 12 \\ 12 & 29 \end{pmatrix} \equiv \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix} \quad (12) \\ \delta^4 \equiv 1 \quad (12), \end{cases}$$

that is already a good point! We have even

$$(112) \qquad \begin{cases} \delta \equiv 1 \quad (2) & \delta \not\equiv 1 \ (4), \ \delta \not\equiv 1 \ (3) \\ \qquad\qquad \Downarrow & \\ \delta^2 \equiv 1 \quad (4) & \delta^2 \equiv -1 \quad (3) \end{cases}$$

Let us consider

$$(113) \qquad \mathrm{Sl}(2,\mathbb{Z}/12\mathbb{Z}) \simeq \underbrace{\mathrm{Sl}(2,\mathbb{Z}/4\mathbb{Z})}_{48} \times \underbrace{\mathrm{Sl}(2,\mathbb{Z}/3\mathbb{Z})}_{24} \overset{u\times v}{\to} \underbrace{\mathcal{G}_0^+/\Pi_0}_{12} \times \underbrace{\mathcal{G}_3^+}_{3}$$

where :

$$u : \mathrm{Sl}(2,\mathbb{Z}/4\mathbb{Z}) \longrightarrow \mathcal{G}_0^+/\Pi_0$$

is the canonical surjective homomorphism, whose kernel is isomorphic to $\dagger\mu \times \mu$ — the three nonzero elements of the kernel are just $\lambda_0, \lambda_1, \lambda_\infty$ permuted among themselves by ρ, and $\dot\lambda_1\dot\lambda_0^{-1} = \dot\lambda_1\dot\lambda_0 = \dot\lambda_\infty$ and ... i.e. † the $\dot\lambda_i$ are none other than $\dot\delta$ and its permuted images under ρ — and

$$v : \mathrm{Sl}(2,\mathbb{Z}/3\mathbb{Z}) \longrightarrow \mathcal{G}_3$$

is the composite homomorphism

$$\mathrm{Sl}(2,\mathbb{Z}/3\mathbb{Z}) \overset{\text{epimorphism}}{\longrightarrow} A_4|G_4 \overset{\text{epimorphism}}{\longrightarrow} \mathcal{G}_3^+$$
$$\cap$$
$$\mathcal{G}_{\mathbb{P}^1(\mathbb{Z}/3\mathbb{Z})}$$

whose kernel is of order 8. The kernel of $u \times v$ is of order $4.8 = 32$, and it obviously contains $\dot{\delta}$. This shows that it is indeed the normal subgroup generated by δ (whose order a priori was a multiple of 32), and

Uncertainty retained from cancelled passages.
140-4-s065-D001 (111)
140-4-s065-D003 $\mathbb{Z}$
140-4-s065-D004 the

[scan 66]

that † the subgroup under consideration of $\Pi_0 \subset \mathrm{Sl}(2,\mathbb{Z})$, of index 3 in Π_0, is the inverse image of this preceding subgroup of $\mathrm{Sl}(2,\mathbb{Z}/12\mathbb{Z})$.

Let us sum up the results obtained:

Proposition Let us consider in $\mathcal{G}_0^+ = \mathrm{Sl}(2,\mathbb{Z})$ the tower of invariant subgroups (even in $\mathcal{G}_0$)

(114)

$$\mathcal{G}_0^+ \xleftarrow{6} \Pi = \Pi[2] \xleftarrow{2} \Pi_0 \xleftarrow{3} \Pi_{00}\ (\supset [\Pi_0, \Pi_0])$$

$\Pi = \Pi[2]$: $\|_{\mathrm{dfn}}$ $\{u \in \mathcal{G}_0^+ \mid u \equiv 1\ (2)\}$ — † (i)|(1)

Π_0: $\|$ Subgroup generated by the $-\ell_i = \lambda_i$

Π_{00} $(\supset [\Pi_0, \Pi_0])$: $\|$ Subgroup invariant generated by $\lambda_1 \lambda_0^{-1} = \ell_1 \ell_0^{-1}$

Let us consider on the other hand the quotient group

(115)
$$\mathrm{Sl}(2,\mathbb{Z}/12\mathbb{Z}) \simeq \mathcal{G}_0^+ / \underbrace{\Pi[12]}_{\simeq\{u\in\mathcal{G}_0^+ \mid u\equiv 1\ (12)\}}$$
$$\simeq$$
$$\mathrm{Sl}(2,\mathbb{Z}/4\mathbb{Z}) \times \mathrm{Sl}(2,\mathbb{Z}/3\mathbb{Z}),$$

and the tower of the following subgroups

(116)
$$\left\{\begin{array}{ll} & H = \mathrm{Sl}(2,\mathbb{Z}/4\mathbb{Z}) \times \mathrm{Sl}(2,\mathbb{Z}/3\mathbb{Z}) \quad \text{order } 48\times 24 \\ 6 \cup & \qquad\qquad \cup \\ & \Pi^\bullet = \mathfrak{sl}(2,\mathbb{F}_2) \times \mathrm{Sl}(2,\mathbb{Z}/3\mathbb{Z}) \quad \text{order } 8\times 24 \\ 2 \cup & \qquad\qquad \cup \\ & \Pi_0^\bullet = \mathfrak{sl}(2,\mathbb{F}_2)' \times \mathrm{Sl}(2,\mathbb{Z}/3\mathbb{Z}) \quad \text{order } 4\times 24 \\ 3 \cup & \qquad\qquad \cup \\ & \Pi_{00}^\bullet = \mathfrak{sl}(2,\mathbb{F}_2)' \times \underbrace{\mathrm{Syl}_2}_{\simeq \dagger D_4} \quad \text{order } 4\times 8, \end{array}\right.$$

m1477

Marginal note. m1477 N.B. $(\mathcal{G}_0^+)_{\mathrm{ab}} \simeq \mathbb{Z}/12\mathbb{Z}$ is identified with a † quotient of $H/\Pi_{00}^\bullet$ by a subgroup (116) of order 3 — that is again the subgroup generated by the element $(-\dot{\rho}, 1)$ of

$$\underbrace{\mathrm{Sl}(2,\mathbb{Z}/4\mathbb{Z})/\mathfrak{sl}(2,\mathbb{F}_2)'}_{\mathcal{G}_0^+/\Pi_0} \times \mathrm{Sl}(2,\mathbb{Z}/3\mathbb{Z})/\mathrm{Syl}_2.$$

Where †

(117)
$$\begin{cases} \text{a)} & \mathfrak{sl}(2,\mathbb{F}_2) \simeq \\ & \operatorname{Ker}(\mathrm{Sl}(2,\mathbb{Z}/4\mathbb{Z}) \longrightarrow \mathrm{Sl}(2,\mathbb{Z}/2\mathbb{Z})) \\ & \qquad\qquad \text{normal subgroup } \simeq \\ & \qquad\qquad \dagger\ \mathbb{F}_2 \times \mathbb{F}_2 \text{ ; whose three elements } \neq 0 \\ & \qquad\qquad \text{are } \dot\lambda_0, \dot\lambda_1, \dot\lambda_\infty \text{ (images} \\ \text{b)} & \mathfrak{sl}(2,\mathbb{F}_2)' \subset \mathfrak{sl}(2,\mathbb{F}_2), \quad \text{in } \mathrm{Sl}(2,\mathbb{Z}/4\mathbb{Z})) \text{ — it is} \\ & \qquad\qquad \text{a complement to its} \\ & \qquad\qquad \text{Lie subalgebra } \mu \simeq \mathbb{F}_2 \text{ of matrices} \\ & \qquad\qquad \text{constant|scalar} \\ \text{c)} & \mathrm{Syl}_2 \subset \mathrm{Sl}(2,\mathbb{Z}/3\mathbb{Z}), \\ & \quad \| \\ & \operatorname{Ker}(\mathrm{Sl}(2,\mathbb{Z}/3\mathbb{Z}) \to \mathcal{G}_3^+) \\ & \text{Sylow subgroup, of order 8} \\ & \text{(which is normal)} \end{cases}$$

Uncertainty retained from cancelled passages.

140-4-s066-D001 M_0
140-4-s066-D004 $\mathfrak{sl}(2,\mathbb{F}_2)$
140-4-s066-D007 [*illeg.* 1]
140-4-s066-D008 (115)|(116) Sl(2)
140-4-s066-D009 we have
140-4-s066-D010 $\mathbb{F}_3|\mathbb{F}_2$

[scan 67]
[p. 681]

This being so, the tower of subgroups (114) is the inverse image of the tower (116).

† To satisfy my conscience, I am going to take up the preceding sketch|sketch of a proof again. We note that the homomorphism $\mathcal{G}_0^+ \longrightarrow H$ maps the groups appearing in (114) into those of (116) — for Π and Π_0, we have in fact known for a while that they are the inverse images of $\Pi^\bullet$, $\Pi_0^\bullet$! $*|\leftarrow$ (For this, it was enough|would be enough to examine $\mathrm{Sl}(2,\mathbb{Z}/4\mathbb{Z})$, instead of $\mathrm{Sl}(2,\mathbb{Z}/12\mathbb{Z})$ …).

Thus the inverse image of $\Pi_{00}^\bullet$ in $\mathcal{G}_0^+$ contains Π_{00}. As both are of index 36, these two groups are equal, q.e.d. (The same argument could be applied to the case of Π_0, $\Pi_0^\bullet$ …)

Corollary The canonical homomorphism

$$\mathcal{G}_0^* = \mathrm{Gl}(2,\mathbb{Z}) \to \mathrm{Gl}(2,\mathbb{Z}/12\mathbb{Z})/(\Pi_{00}^\bullet\dagger = \mathfrak{sl}(2,\mathbb{F}_2)' \times \mathrm{Syl}_2)$$

factors through

$$\mathcal{G} = \mathrm{Gl}(2,\mathbb{Z})/[\Pi_0,\Pi_0]$$

studied in the present § 51, and the inverse image of $\Pi_{00}^\bullet$ is none other than the subgroup M_0 of (99) (108).

m1478

Marginal note. m1478 N.B. But $\mathcal{G}_0^{+|*} \longrightarrow$ Sl|Gl$(2,\mathbb{Z}/12\mathbb{Z})$ does not factor through $\mathcal{G}^{+|*}$, for if this were so $\Pi_0^{\bullet|\prime}$ would be a sub-quotient of $\Pi_0/[\Pi_0,\Pi_0]$ and hence would be commutative — now all we see two|therefore which are both.

Uncertainty retained from cancelled passages.

140-4-s067-D001 Remark|Remark
140-4-s067-D002 $\simeq$

[scan 68]

The preceding considerations make fairly natural|natural the consideration of the covering of order 3 of $U_{0,\overline{\mathbb{Q}}}$ † corresponding to the subgroup Π_{00} of $\Pi_0 \simeq \Pi_1(U_{0,\overline{\mathbb{Q}}})$.

it is even a covering of $M_{1,1}$…

We check at once that it is a curve of genus 1 – if then we take it into our heads to work with $[\Pi_{0,0}, \Pi_{0,0}]$ instead of $[\Pi_0, \Pi_0]$, we find a group extension of $\Pi_0/[\Pi_0, \Pi_0] \simeq M$, by a group of rank 2 which is "elliptic", and on which M will operate in a highly nontrivial way – later on we would have to examine this action, and put it in relation with the action we have found, subordinate to the homomorphism

$$\mathrm{Sl}(2, \mathbb{Z})^\wedge \longrightarrow \mathrm{Sl}(2, \widehat{\mathbb{Z}}).$$

m1480

Marginal note. m1480 it is an elliptic curve of invariant 0 (for it is equipped with an automorphism of order 3)

Uncertainty retained from cancelled passages.
140-4-s068-D001 or of 1|0

[scan 69]
[p. 682]

5) The groups $\underline{M}_{\rho,\sigma}$, $\underline{M}^\sharp_{\rho,\sigma}$ (via $\underline{N}_{\rho,\sigma}$).

To every $\ell \in \Gamma\underline{M}_0$ there corresponds an inner automorphism $\mathrm{int}(\ell)$, which is the identity on $\underline{M}_0$. We will have, tautologically

$$(118) \qquad \begin{cases} \mathrm{int}(\ell)(\rho) = \alpha_{0|\cdot}(\rho) \\ \mathrm{int}(\ell)(\sigma) = \beta(\sigma) \end{cases} \qquad \alpha = \beta = \ell$$

But will we for all that have the loop relation (relative to $\underline{\mu} = 1$ hence $\nu'' = 0$)

$$(1 - \rho_M)(\underbrace{\alpha}_{\ell}) = (1 - \sigma_M)(\underbrace{\beta}_{\ell}).$$

$$\underset{(*)|\alpha_1}{\text{i.e.}} \qquad \rho_M(\ell) = \sigma_M(\ell) \qquad ?$$

Setting again

$$(119) \qquad \ell = c_0\lambda_0 + c_1\lambda_1$$

the relation $(*)$ is written

$$c_0\lambda_1 + c_1 \underbrace{\lambda_\infty}_{-\lambda_0-\lambda_1} = c_0\lambda_1 + c_1\lambda_0$$

$$\text{i.e.} \qquad -c_1\lambda_0 + (c_0 - c_1)\lambda_1 = c_1\lambda_0 + c_0\lambda_1$$

i.e. $c_1 = 0$ i.e. $\ell = c_0\lambda_0$. Thus it is subject to the condition|conditions that we restrict ourselves to considering $\ell \in \Gamma\underline{L}_0^{\underline{M}_0|\underline{M}}\Rightarrow$; corresponding|corresponding

$$(120) \quad \dagger \quad \ell = c_0\lambda_0 \in \underline{L}^{\underline{M}}_{0\dagger} \overset{\text{dfn}}{=} \mathrm{Im}\Big(\begin{array}{c} \underline{\mathbb{Z}} \longrightarrow \underline{M} \\ \simeq \\ \mathbb{G}_a \\ \ni \\ c_0 \mapsto c_0\lambda_0 \end{array}$$

We have a † diagram

$$(121)\quad \begin{array}{ccc} \mathbb{G}_a \simeq \underline{L}_0^{M_0|M} & \longrightarrow & \underline{N}_{\rho,\sigma} \qquad\qquad \dagger \\ \downarrow & & \\ \underline{\mathcal{G}}^+ & & \end{array}$$

Uncertainty retained from cancelled passages.

140-4-s069-D003 $\underline{L}_0$

140-4-s069-D004 $, \rho, \sigma$

140-4-s069-D005 dia

140-4-s069-D006 $\underline{L}_0^M$

[scan 70]

the image of

$$(127|122)\quad \underline{L}_0^{M_0|M} \longrightarrow \underline{N}_{\rho,\sigma}$$

is invariant – the homomorphism (121) being in fact compatible with the action of $N_{\rho,\sigma}$ – via $\underline{\mu}$ on $\underline{L}_0^{M_0} \simeq \mathbb{G}_a$ (i.e. induced by the operation of $\underline{N}_{\rho,\sigma}$ of on!| the $\underline{\mathcal{G}}^+ \supset \underline{M} \supset \underline{L}_0^M$), and the two operations of † $\underline{L}_0^M$ on $\underline{\mathcal{G}}^+$ deduced from (121) being identical|identical. Thus according to the general theory|theorem we find|will find a group scheme $\underline{M}_{\rho,\sigma}$ and a diagram of exact sequences

$$(123)\quad \begin{array}{ccccccccc} 1 & \longrightarrow & \underline{L}_0^M & \longrightarrow & \underline{N}_{\rho,\sigma} & \Longrightarrow & \overset{\text{dfn}}{=} \underline{N}_{\rho,\sigma}/\underline{L}_0^M & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \| & & \\ 1 & \longrightarrow & \underline{\mathcal{G}}^{+a|\alpha} & \longrightarrow & \underline{M}_{\rho,\sigma} & \Longrightarrow & \underline{\Gamma}_{\rho,\sigma} & \longrightarrow & 1 \end{array}$$

m1481

Marginal note. m1481 with the notations of § 49 VI, the group denoted here $\underline{M}_{\rho,\sigma}$ would probably be denoted $\underline{M}_{\rho,\sigma}/H_\rho|H_\Pi$, and $\underline{M}^\sharp_{\rho,\sigma}$ would be $\underline{M}^\sharp_{\rho,\sigma}/H_\rho$ (loc. cit. 8°)

Replacing $\underline{\mathcal{G}}^+$ by $\underline{M}$, we likewise find $\underline{M}^\sharp_{\rho,\sigma}$, and

(124)
$$\begin{array}{ccccccccc} 1 & \longrightarrow & \underline{L}_0^M & \longrightarrow & \underline{N}_{\rho,\sigma} & \longrightarrow & \underline{\Gamma}_{\rho,\sigma} & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \Downarrow & & \\ 1 & \longrightarrow & \dagger\underline{M} & \longrightarrow & \underline{M}^\sharp_{\rho,\sigma} & \longrightarrow & \underline{\Gamma}_{\rho,\sigma} & \longrightarrow & 1 \end{array}$$

,
moreover we have a homomorphism from (124) into (123) (compare § 49 VI 5°…), and

(125) $\qquad M^\sharp_{\rho,\sigma} \subset M_{\rho,\sigma} \qquad \ldots$

Uncertainty retained from cancelled passages.
140-4-s070-D002 $N_\rho | A_\rho$
140-4-s070-D003 $\underline{\mathcal{G}}^+$

[scan 71]
[p. 683]

We must specify the structure of $\Gamma_{\rho,\sigma}$, noting that $L_0^M \subset N^{0+}_{\rho,\sigma}$, hence

(126|125) $\quad 1 \longrightarrow \dagger\underline{\Gamma}^{0+}_{\rho,\sigma} \longrightarrow \underline{\Gamma}_{\rho,\sigma} \longrightarrow \underline{\mathbb{O}}^*_{12} \longrightarrow 1$

with

(127) $\quad 1 \longrightarrow \underline{L}_0^M \longrightarrow \underline{N}^{0+}_{\rho,\sigma} \longrightarrow \underline{\Gamma}^{0+}_{\rho,\sigma} \longrightarrow 1$

†

Let us consider thesurjective homomorphism of $\mathbb{Z}$-modules

$$(128)\quad N^{0+}_{0,\sigma}|N^{0+}_{\rho,\sigma} = \dagger\,\mathrm{Ker}(\quad M\times M \longrightarrow M_0)\dagger$$
$$\cap$$
$$M$$

$$\dagger \mapsto \dagger\,\dagger$$

$$(\ \underset{\|}{\alpha_0}\ ,\ \underset{\|}{\beta_0}\ ) \longrightarrow (1-\rho_M)(\alpha_0)-(1-\sigma_M)(\beta_0)$$
$$a_0\lambda_0+a_1\lambda_1 \qquad b_0\lambda_0+b_1\lambda_1$$

kernel consisting of the (α_0,β_0) such that

$$\begin{aligned} c_0 &\overset{\text{dfn}}{=} a_0+a_1+(b_1-b_0)=0 \\ c_1 &\overset{\text{dfn}}{=} 2a_1-a_0-(b_1-b_0)=0 \end{aligned} \quad \Big|\ \text{cf. (61)}$$

Let $\underline{N}^{0+}$ be the kernel, so that we have

$$(129)\qquad \underset{\rho,\sigma}{N^{0+}} \simeq \underline{N}^{0+}.$$

Let us consider the homomorphism

$$(130)\qquad \begin{cases} \mathbb{Z} \longrightarrow N^{0+} \\ n \longmapsto (n\lambda_0, n\lambda_0) \end{cases}$$

the image is interpreted as made up|made up of the systems (a_0,a_1,b_0,b_1) such that †

$$a_1=b_1=0, \qquad a_0=b_0, \qquad ''$$

† i.e. we have an exact sequence

$$\dagger \qquad 0 \longrightarrow \mathbb{Z} \longrightarrow N^{0+} \longrightarrow \mathbb{Z}^3$$

$$(a_0,a_1,b_0,b_1) \longmapsto (a_1,b_1,a_0-b_0)$$

hence † $N^{0+}/\mathbb{Z}$ is † a free $\mathbb{Z}$-module of rank 1,

---

Uncertainty retained from cancelled passages.

140-4-s071-D001 $N_{\rho,\sigma}$
140-4-s071-D002 it is C_f|it is G_f
140-4-s071-D005 $M\times M$
140-4-s071-D006 $\longrightarrow 0$
140-4-s071-D007 N
140-4-s071-D008 α_0,β_0
140-4-s071-D009 (α_0,β_0)
140-4-s071-D011 $a_0=b_0$

140-4-s071-D012 a [illeg. 2]
140-4-s071-D013 (131)
140-4-s071-D014 this
140-4-s071-D015 flat|free

[scan 72]

thus an exact sequence of $\mathbb{Z}$-modules

$$(131)\quad 1 \to \mathbb{Z} \longrightarrow N^{0+} \longrightarrow \Gamma^{0+} \longrightarrow 0$$
$$n \mapsto (n\lambda_0, n\lambda_0) \qquad N^{0+}/\operatorname{Im}\mathbb{Z} \overset{\wr}{=} \text{dfn}$$

with Γ^{0+} a free $\mathbb{Z}$-module of rank 1.

We conclude from this

$$(132)\qquad \Gamma^{0+}_{\rho,\sigma} \simeq \underline{\Gamma}^{0+} \qquad (\simeq \mathbb{G}_a)$$

† Thus, all the group schemes we have just constructed are smooth over $\operatorname{Spec}\mathbb{Z}$.

Can we define a more-or-less natural section of the extension ($N_{\rho,\sigma}$ of $\Gamma_{\rho,\sigma}$ by $L_0^{\underline{M}}$, hence of $M^\sharp_{\rho,\sigma}$ of $\underbrace{\Gamma_{\rho,\sigma}}_{\substack{\text{smooth of relative}\\ \text{dimension 2}}}$ by $\underline{M}$?

Taking into account the loop equation which describes $N_{\rho,\sigma}$, it seems that we find such a section as soon|some as we choose a splitting of

$$\begin{cases} 0 \to N^{0+} \to M \times M \to \overset{\text{rg 2}}{M_0} \to 0 \qquad \text{and of} \\ \\ (133)\quad 0 \to \underset{\text{rg 1}}{\mathbb{Z}} \longrightarrow N^{0+} \longrightarrow \underset{\text{rg 1}}{\Gamma^{0+}} \to 0 \end{cases}$$

(category of $\mathbb{Z}$-modules). But we must check that the "set-theoretic" sections of schemes that this allows us to define are multiplicative…

Uncertainty retained from cancelled passages.

140-4-s072-D001 Hence

[scan 73]

6°) $\underline{Z}_{\rho,\sigma}$ and $\underline{G}_{\rho,\sigma} = \mathbb{O}_{12}^* \cdot \underline{\mathcal{G}}^+$, $\quad \underline{G}^\sharp_{\rho,\sigma} \simeq \mathbb{O}_{12}^* \cdot \underline{\mathcal{G}}^\sharp = \mathbb{O}_{12}^* \cdot \underline{M}$. [p. 684]

Once $\mathbb{O}_{12}^*$ and $M_0 \subset M$ have been introduced, it seems to me that the group $\underline{Z}_{\rho,\sigma}$ (in place of $\widetilde{\underline{Z}}_{\rho,\sigma}$, in which we took account of the α, β and not only of ξ, η) again becomes nice:

$$\underline{Z}_{\rho,\sigma} \subset \underset{''}{\underline{\operatorname{Aut}}}(\underline{\mathcal{G}}^+) \times \mathbb{O}_{12}^*$$

$$(134)\quad \left\{ (u, \underline{\mu}) \;\middle|\; \begin{array}{ll} \text{a)} & u(\rho) = \xi\rho \quad \text{(locally) } \underline{\mathcal{G}}^\sharp_\tau\text{-conjugate to } \rho, \text{ by an} \\ & \alpha \in \underline{\mathcal{G}}^\sharp_\tau \text{ such that } \chi(\alpha) = \varepsilon_3(\underline{\mu}) \\ \text{b)} & u(\sigma) = \eta\sigma \quad \text{(locally) } \underline{\mathcal{G}}^\sharp_\tau\text{+|−conjugate to } \sigma, \text{ by a} \\ & \beta \in \underline{\mathcal{G}}^\sharp_\tau \text{ such that } \chi(\beta) = \varepsilon_2(\underline{\mu}) \\ \text{c)} & \text{loop equation} \\ & \xi_0 = \eta_0(\lambda_1^3)^{\nu''} \quad \text{i.e.} \quad \xi_0 = \eta_0 + 3\nu''\lambda_1 \\ & \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \text{in } \underline{M}_0 \end{array} \right.$$

where $\nu'' = \nu''(\underline{\mu}) \in \Gamma\mathbb{G}_a$ (cf. page verso page 686), and above[t])

$$(135) \qquad \begin{cases} \xi_0 = \xi - \nu_3\lambda_1 \in \Gamma\underline{M}_0 & \nu_3 = \nu_3(\underline{\mu}) \\ \eta_0 = \eta\omega^{\nu_2} \in \Gamma\underline{M}_0 & \nu_2 = \nu_2(\underline{\mu}) \end{cases}$$

We shall have moreover that

$$(136) \qquad \begin{cases} \xi_0 & \text{locally}|\text{locally}^{mt} \text{ in the image of } (1-\rho_{\underline{M}})(\underline{M}) \\ \eta_0 & \text{locally}|\text{locally}^{mt} \text{ in the image } (1-\sigma_{\underline{M}})(\underline{M}) \end{cases}$$

and we shall have

$$(137) \qquad \begin{aligned} \underline{Z}_{\rho,\sigma} \simeq \underline{\Sigma}^{\text{eff}}_{\rho,\sigma} \overset{\text{dfn}}{=} \{(\underline{\mu}, \xi_0, \eta_0) \in \Gamma\mathbb{O}^*_{12} \times \underline{M}_0 \times M_0 \mid \\ 1^\circ) \quad \xi_0 = \eta_0 + \dagger\nu''(3\lambda_1) \quad \text{in } \underline{M}_0 \\ \text{where } \nu'' = \nu''(\underline{\mu}) \\ 2^\circ) \quad \text{we have (136)} \quad \dagger \}. \end{aligned}$$

Now

$(1-\rho_{\underline{M}})$:

$$\underline{M} \longrightarrow \underline{M}_0$$

$(1-\sigma_{\underline{M}})$:

$$\underline{M} \longrightarrow \underline{\mathbb{Z}} = \mathbb{G}_a \longrightarrow \underline{M}_0$$

$$(b_0\lambda_0 + b_1\lambda_1) \longrightarrow \underset{\dagger}{(b_1 - b_0)}\,\dagger \longmapsto \lambda \underbrace{(\lambda_1 - \lambda_0)}_{\delta} = \lambda\delta$$

Uncertainty retained from cancelled passages.

140-4-s073-D003 ξ

140-4-s073-D004 }

140-4-s073-D005 $4|\lambda$

140-4-s073-D006 λ

[scan 74]

thus (136) † is a condition on η_0 alone, namely

(136bis) $\eta_0 \in \Gamma\,\mathrm{Im}($

$$\mathbb{G}_a \longrightarrow \underline{M}_0$$

$$\lambda \longmapsto \lambda\delta$$

$) = \Gamma\,\mathrm{Ker}($

$$M_0 \longrightarrow \mathbb{G}_a$$

$$c_0\lambda_0 + c_1\lambda_1 \longrightarrow c_0 + c_1$$

$)$

† Let us note that we have

$$(138) \qquad \begin{cases} M_0 = \mathbb{Z}\underbrace{\delta}_{\lambda_1-\lambda_0} \oplus \mathbb{Z}\,.\,\underbrace{(3\lambda_1)}_{(1-\rho)\delta} \\ \underline{M}_0 = \dagger\ \underline{\mathbb{Z}\delta} \times \underline{\mathbb{Z}\,.\,(3\lambda_1)}\ . \end{cases} \qquad \text{whence}$$

and we find

(139)

$$\underline{\Sigma}^{\mathrm{eff}}_{\rho,\sigma} \xrightarrow{\ \simeq\ } \mathbb{O}^*_{12} \times \dagger\ \dagger\ \underbrace{\underline{\mathbb{Z}\delta}}_{\in|\simeq\mathbb{G}_a}$$

$$(\underline{\mu}, \xi_0, \eta_0|\beta_0) \longmapsto (\underline{\mu}, \eta_0|\beta_0)$$

.

For $\xi_0|\zeta_0$ is recovered with $\underline{\mu}, \eta_0|\beta_0$ † the loop equation

†

N.B. If on the other hand we give ourselves ξ_0, we obtain from it by the loop equation and (138) the value of $\eta_0|\beta_0$, and that of ν'' – thus if we know in addition ν_2 and ν_3, we deduce the value of $2\nu_0$ hence (over a base where 2 is a regular section of $\mathcal{O}_S$) ν_0, i.e. finally $\underline{\mu}$. Thus over a

---

Uncertainty retained from cancelled passages.

140-4-s074-D001 is written
140-4-s074-D002 as
140-4-s074-D004 $\mathbb{G}_a$
140-4-s074-D006 $\underline{\mathbb{Z}}\delta$
140-4-s074-D007 $\mathbb{G}_a$
140-4-s074-D008 from
140-4-s074-D009 $\alpha_0|\xi_0 \quad \eta_0|\beta_0 \quad = \nu''\delta + (1-\rho_{\underline{M}})^{-1}\eta_0 \quad \xi_0|\eta_0$

[scan 75]
[p. 685]

base S where 2 is regular, we find
(140)

$$\underline{\Sigma}^{\mathrm{eff}}_{\rho,\sigma}(S) \hookrightarrow (\underline{M}_0\dagger \times \{0,1\}\dagger \times \{0,1\})(S)$$

$$(\underline{\mu}, \xi_0, \eta_0) \longmapsto (\xi_0, \varepsilon_2(\underline{\mu}), \varepsilon_3(\underline{\mu}))$$

the image being formed of the systems $(\xi_0, \varepsilon_2|\xi_2, \varepsilon_3|\xi_3)$ such that, setting

$$\dagger \qquad \xi_0 = \eta_0|\beta_0\delta + \nu''(3\lambda_1) \qquad \eta_0, \nu'' \in \dagger\mathbb{G}_a(S)$$

thanks to (138), we have

$$(141) \qquad \begin{cases} \nu'' + \nu_2 - \nu_3[+2\nu_2\nu_3] & \text{divisible by 2} \\ \underbrace{(-5)^{\nu_2}(+5)^{\nu_3} + 6(\nu'' + \nu_2 - \nu_3 + 2\nu_2\nu_3)}_{1-2\nu_3-2\nu_2\nu_3+6\nu''} & \text{inverse} \end{cases}$$

In any case we see that $\underline{\Sigma}^{\mathrm{eff}}_{\rho,\sigma}$ hence also $\underline{Z}_{\rho,\sigma}$ is smooth of relative dimension 2
(142)

$$1 \longrightarrow \underline{Z}^{+0}_{\rho,\sigma} \longrightarrow \underline{Z}_{\rho,\sigma} \longrightarrow \mathbb{O}^{*|\circ}_{12} \longrightarrow 1$$

$$\wr|$$

$$\mathbb{G}_a \simeq \underline{\mathbb{Z}\delta}$$

(the isomorphism

$$\begin{array}{ccccc} \mathbb{G}_a & \simeq & \underline{Z}^{+0}_{\rho,\sigma} & \overset{\simeq}{\nearrow} & \underline{\Sigma}^{\mathrm{eff}+0}_{\rho,\sigma} \\ & & \dagger & & \dagger \end{array}$$

being given by

$$\lambda \longmapsto (\underline{\mu} = 1, \xi_0\dagger = \lambda\delta, \eta_0 = \xi_0 = \lambda\delta)$$

)
We shall set

$$(143) \qquad \underline{L}^\sharp_0|L^\sharp_0 \overset{\mathrm{dfn}}{=} \{\lambda\,.\,\lambda_0 \mid \lambda \in \Gamma\mathbb{G}_a\} \subset \underline{M}$$

and we define
(144)

$$\underline{L}^\sharp_0 \longrightarrow \underline{Z}^{+0}_{\rho,\sigma}\Big(\overset{\simeq}{\longrightarrow} \underline{\Sigma}^{\mathrm{eff}+0}_{\rho,\sigma}\Big)$$

$$\begin{array}{ccccc} \ell & \longmapsto & (\mathrm{int}(\ell), 1) & \longmapsto & (\dagger 1, (1-\rho_{\underline{M}})(\ell), (1-\sigma_{\underline{M}})(\ell)) \\ {\scriptstyle \prime\prime\lambda.\lambda_0} & & & & = (1, -\lambda\delta, -\lambda\delta) \end{array}$$

Uncertainty retained from cancelled passages.

140-4-s075-D001 (S)
140-4-s075-D002 (S)
140-4-s075-D003 $141|140$
140-4-s075-D004 Γ
140-4-s075-D007 $\mathbb{G}_a$
140-4-s075-D008 $\overset{\simeq}{\leftarrow}$
140-4-s075-D009 $= \eta_0$
140-4-s075-D011 μ

[scan 76]

thus we have

$$\underline{L}_0^\sharp \simeq \underline{Z}_{\rho,\sigma}^{+0} \tag{145}$$

we shall have

$$\underline{\gamma}_{\rho,\sigma} \overset{\text{dfn}}{=} \underline{Z}_{\rho,\sigma}/\underline{L}_0^\sharp \simeq \underline{\mathbb{O}}_{12}^* \ . \tag{146}$$

If we now construct $G_{\rho,\sigma}$

†

as

$$\underline{G}_{\rho,\sigma} = (\underline{Z}_{\rho,\sigma} \cdot \underline{\mathcal{G}}^+)/\underline{L}_0^\sharp \tag{147}$$

we find

(148)

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \underline{L}_0^\sharp & \longrightarrow & \underline{Z}_{\rho,\sigma} & \longrightarrow & \overbrace{\underline{\gamma}_{\rho,\sigma}}^{\mathbb{O}_{12}^*} & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \wr & & \\ 1 & \longrightarrow & \underline{\mathcal{G}}^+ & \longrightarrow & G_{\rho,\sigma} & \longrightarrow & \gamma_{\rho,\sigma} & \longrightarrow & 1 \end{array}$$

.

Moreover we have a canonical splitting of $\underline{Z}_{\rho,\sigma}$ over $\gamma_{\rho,\sigma} = \mathbb{O}_{12}^*$

(149)

$$\begin{array}{ccccc} \mathbb{O}_{12}^{*|\wedge} & \longrightarrow & \underline{Z}_{\rho,\sigma} & \xrightarrow{\simeq} & \underline{\Sigma}_{\rho,\sigma}^{\text{eff}} \\ \underline{\mu} & \longmapsto & u_{\underline{\mu}}|\underline{\mu} & \longrightarrow & (\underline{\mu}, \xi_0 = \nu''(3\lambda_1), \eta_0 = 0) \end{array}$$

m1482

Marginal note. m1482 N.B. We set $G_{\rho,\sigma}^\sharp = \underbrace{\mathbb{O}_{12}^*}_{\gamma_{\rho,\sigma}} \cdot \underbrace{\underline{\mathcal{G}}^\sharp}_{\underline{M}} \subset G_{\rho,\sigma}$.

whence

$$\begin{cases} \underline{Z}_{\rho,\sigma} \simeq \mathbb{O}_{12}^* \cdot \underline{L}_0^\sharp \\ G_{\rho,\sigma} \simeq \mathbb{O}_{12}^* \cdot \underline{\mathcal{G}}^+ \end{cases} \tag{150}$$

The operation of $\mathbb{O}_{12}^*$ on $\underline{L}_0^\sharp$ is via

$$\mathbb{O}_{12}^* \text{ ——— } \mathbb{G}_m$$

and homotheties, that of $\mathbb{O}_{12}^*$ on $\underline{\mathcal{G}}^+$ is given by

$$(151)\qquad \begin{cases} & u_{\underline{\mu}}(\rho) = [\dagger 6\nu''|6\nu\lambda_1]\rho \quad (= \lambda_1^{6\nu''|6\nu}\rho \quad \text{in exponential notation}) \\ & u_{\underline{\mu}}(\sigma) = \omega^{\nu_2}\sigma \\ \dagger & u_{\underline{\mu}}(x) = \mu x \qquad \text{if } x \in \Gamma\underline{M}\,. \end{cases}$$

Uncertainty retained from cancelled passages.

140-4-s076-D001 147

140-4-s076-D002 $3\nu''|\nu_3$

140-4-s076-D004 $u_{\underline{\mu}}$

[scan 77]
[p. 686]
$\sigma\,\mathrm{D}\,\sigma$

7°) The groups $\underline{N}_\rho$, † etc, $\underline{N}_\rho^{\natural|\sharp}$ etc.

Let us study the subgroup

$$(152)\qquad \begin{array}{c} \underline{N}_\rho \subset \underline{G}_{\rho,\sigma}\,: \\ \| \\ \{\underline{u} \in \Gamma\underline{G}_{\rho,\sigma} \mid u(\rho) = \rho^{\varepsilon_3(\underline{u})}\}. \end{array}$$

m1484

Marginal note. m1484 Provisional notation, cf. below page 697

The composite homomorphism

$$\underline{N}_\rho \hookrightarrow \underline{G}_{\rho,\sigma} \xrightarrow{\underline{\chi}} \mathbb{O}_{12}^*, \qquad \underline{N}_\rho \xrightarrow{\underline{\chi}_\dagger} \mathbb{O}_{12}^*$$

is an epimorphism, let us check this to reassure ourselves.

Let $\underline{\mu}$ be a local section of $\mathbb{O}_{12}^*$, corresponding to $\dagger\nu_2, \nu_3, \dagger\nu_0, \mu, \nu', \nu'', \nu\dagger$ ... We will have

$$u_{\underline{\mu}}(\rho) = \lambda_1^{\nu}\rho = (\lambda_1^{3\nu''}\lambda_1^{-\nu_3})\rho = \mathrm{int}(\delta^{\nu''}\tau^{\nu_3})(\rho).$$

†

†

whence

$$(153)\qquad \mathrm{int}\{(\delta^{\nu''}\tau_1^{\nu_3})^{-1}\cdot u_{\underline{\mu}}\}(\rho) = \rho$$

i.e.

$$(154)\qquad \begin{cases} \underbrace{(\delta^{\nu''}\tau_1^{\nu_3})^{-1}u_{\underline{\mu}}}_{u} = \tau_1^{\nu_3}\delta^{-\nu''}u_{\underline{\mu}} \in \Gamma\underline{N}_\rho \\ \underline{\chi}(u) = \dagger\underline{\chi}((\delta^{\nu''}\tau_1^{\nu_3})^{-1})\cdot\underbrace{\underline{\chi}(u_{\underline{\mu}})}_{\underline{\mu}} \\ \underline{\chi}(\tau_1^{\nu_3}) = \underbrace{(-1)^{\nu_3}}_{\varepsilon_3} \end{cases}$$

m1485

Marginal note. m1485 N.B. We denote by $\underline{\chi}$ the canonical homomorphism

$$\underline{G}_{\rho,\sigma} \longrightarrow \mathbb{O}_{12}^*$$

Thus setting

$$v = (\omega\tau')^{\nu_3} u = \big(\underbrace{\omega\dagger\tau'\tau}_{\sigma}\big)^{\nu_3} \delta^{-\nu''} u_{\underline{\mu}}$$

we will have

$$(154) \qquad \underline{\chi}(v) = \underline{\mu}, \quad \text{with } v = \sigma^{\nu_3}\delta^{-\nu''} u_{\underline{\mu}} \in \Gamma \underline{N}_\rho$$
$$\left(\delta = \lambda_1\lambda_0^{-1}, \quad \begin{matrix}\nu_3 = \nu_3(\underline{\mu}),\\ \nu'' = \nu''(\underline{\mu})\end{matrix}\right).$$

76

Uncertainty retained from cancelled passages.

140-4-s077-D001 $\underline{N'}_\rho$
140-4-s077-D003 ρ
140-4-s077-D005 $\underline{\mu}|\nu_0$
140-4-s077-D006 ν_0
140-4-s077-D007 $, \nu_2, \nu_3$
140-4-s077-D010 if $\Leftrightarrow$
140-4-s077-D011 $(1-\rho_{\underline{M}})(\delta) \Leftarrow \lambda_1$ i.e.
140-4-s077-D013 μ
140-4-s077-D014 τ

[scan 78]

We will thus have an exact sequence

(155)

$$1 \longrightarrow \underline{Z|N^\dagger}_\rho \longrightarrow \underline{N}_\rho \longrightarrow \mathbb{O}_{12}^* \longrightarrow 1$$
$$\mathbb{O}_{12}^* \Vert \gamma_{\rho,\sigma}$$

and

$$\underline{Z|N^\dagger}_\rho = \underline{\mathrm{Centr}}_{\underline{\mathcal{G}|\widetilde{G}^+}}(\rho).$$

We have, if

$$(156) \qquad \begin{cases} H^+_{S_0} \overset{\text{dfn}}{=} \underline{\mathcal{G}|\widetilde{G}^+} \,/\, \underbrace{\dagger\underline{\mathcal{G}|\widetilde{G}^0}}_{\substack{\text{neutral}\\ \text{component,}\\ \simeq \underline{M}}} \simeq (H^+)_{S_0} \\ \text{i.e.} \quad H^+ = \mathcal{G}|\widetilde{G}^+/M, \end{cases}$$

a canonical homomorphism

(157)

$$\underline{Z}_\rho \longrightarrow \underline{\mathrm{Cent}}_{H^+_{S_0}}(\rho) \simeq \dagger\underline{L}_\rho \quad (\simeq (\mathbb{Z}/6\mathbb{Z})_{S_0})$$

† which is an epimorphism (since there is even a section, coming from the inclusion $\underline{L}_\rho \subset \underline{Z}_\rho$) and whose kernel is

$$\underline{Z}^0_\rho = \underline{Z}_\rho \cap \underline{M} = \{x \in \Gamma\underline{M} \mid \rho_{\underline{M}}\underline{x} = x \quad \text{i.e.} \quad (1-\rho_{\underline{M}}|1+\rho_{\underline{M}})x\dagger = 0\}. \tag{158}$$

This kernel is zero over $\underbrace{\operatorname{Spec}\mathbb{Z}}_{S_0}\setminus\{3\}$, but of dimension 1 in dimension 3. Yet we do feel that the "good" definition of $\underline{Z}^0_\rho$ must correspond to an equation

$$\ell|\varepsilon(1-\rho_{\underline{M}})x = 0 \qquad \text{in } \underline{\underline{M}}_0, \tag{159}$$

77

Uncertainty retained from cancelled passages.

140-4-s078-D001 0
140-4-s078-D002 0
140-4-s078-D003 $\underline{Z}$
140-4-s078-D004 $\underline{Z}$
140-4-s078-D005 surjective
140-4-s078-D006 ρ

[scan 79]
[p. 687]

which in fact amounts simply, to replacing $\underline{N}_\rho$ defined in (152), by the schematic closure of its restriction over $S_0 - \{3\}$; We thus find a better subgroup, which from now on I denote by $\underline{N}_\rho$, fitting into the exact sequence

(159)

$$1 \longrightarrow \underline{L}_\rho \quad \hookrightarrow | \leftarrow \quad \underline{N}_\rho \xrightarrow{\chi_\rho} \underline{\mathbb{O}}^*_{12} \longrightarrow 1$$

Let us set

$$\begin{cases} \underline{G}^{\natural|\sharp}_{\rho,\sigma} = \underbrace{\underline{\mathbb{O}}^*_{12}}_{\gamma_{\rho,\sigma}} \cdot \underbrace{\underline{\mathcal{G}|\widetilde{G}}^{\natural|\sharp}}_{\underline{M}} \subset \underline{\mathbb{O}}^*_{12} \cdot \underline{\mathcal{G}|\widetilde{G}}^{+} = \underline{G}_{\rho,\sigma}, \\ \underline{N}^{\natural|\sharp}_\rho = \underline{N}_\rho \cap \underline{G}^{\natural|\sharp}_{\rho,\sigma} \end{cases} \tag{160}$$

we then find

$$\underline{N}^{\natural|\sharp}_\rho \cap \underline{L}_\rho = \underline{M} \cap \underline{L}_\rho = \{1\} \tag{161}$$

thus

(161)

$$\underline{N}^{\natural|\sharp}_\rho \hookrightarrow \underbrace{\underline{\mathbb{O}}^*_{12}}_{\gamma_{\rho,\sigma}}$$

I say that the image is

$$\underline{\gamma}'_{\rho,\sigma} = \underline{\mathbb{O}}'^*_{12} \overset{\text{dfn}}{=} \left\{ \underline{\mu} \in \Gamma\underline{\mathbb{O}}^*_{12} \;\middle|\; \begin{array}{c} \varepsilon_3(\underline{\mu}) = 1 \\ \text{i.e. } \nu_3(\underline{\mu}) = 0 \end{array} \right\}. \tag{162}$$

m1488

Marginal note. m1488 thus we have $\underline{N}^{\natural|\sharp}_\rho \subset \underline{Z}_\rho$, it is better to denote it by $\underline{Z}^{\natural|\sharp}_\rho$.

That the image contains $\underline{\mathbb{O}}'^{*}_{12}$ follows from (154), let us show that it is contained.
Let

$$u = f^{-1}u_{\underline{\mu}}, \qquad f \in \Gamma\underline{\mathcal{G}|\widetilde{G}}^{\natural|\sharp} = \Gamma\underline{M}, \quad \underline{\mu} \in \Gamma\underline{\mathbb{O}}^{*}_{12}$$

be a local section of $\underline{G}^{\natural|\sharp}_{\rho,\sigma} = \mathbb{O}^{*}_{12}\underline{\mathcal{G}|\widetilde{G}}^{\natural|\sharp}$, let us write that it is in $\underline{N}_{\rho}$, whence 78

[scan 80]

$$\underbrace{\mathrm{int}(f^{-1})\, u_{\underline{\mu}}(\rho)}_{\lambda_1^{\nu}\rho} = \rho^{\varepsilon_3} \qquad \text{i.e.} \qquad \lambda_1^{\nu}\rho = \mathrm{int}(f)\rho^{\varepsilon_3}$$

reducing modulo $\underline{M}$ we find

$$\rho = \dagger\rho^{\varepsilon_3} \qquad \text{in } H^{+}$$

which implies $\varepsilon_3 = 1$, OK. Thus we have

(163)

$$\dagger\, \underbrace{\underline{N}^{\natural|\sharp}_{\rho,\dagger}}_{=\underline{Z}^{\natural|\sharp}_{\rho}} \xrightarrow{\sim} \underline{\gamma}'_{\rho,\sigma} = \underline{\mathbb{O}}'^{*}_{12}$$

— thus $\underline{Z}^{\natural|\sharp}_{\rho} = \underline{N}^{\natural|\sharp}_{\rho,\dagger}$ is a partial section, over $\underline{\mathbb{O}}'^{*\prime|*\prime}_{12}$, of the extension $N_{\rho,\sigma}$ of $\underline{\mathbb{O}}^{*}_{12}$ by L_{ρ}.
† This partial section does

not contain τ', which is a section of $\underline{N}_{\rho\dagger}$. Does τ' normalize $\underline{Z|N}^{\natural|\sharp}_{\rho}$ — since it normalizes $\underline{N}_{\rho}$, it would suffice for it to normalize $\dagger\underline{G}^{\natural|\sharp}_{\rho,\sigma}$, †

Since $\tau' = \tau\sigma$, and $\tau \in \Gamma\underline{G|\mathcal{G}}^{\natural|\sharp}_{\rho,\sigma}$ normalizes $\underline{G|\mathcal{G}}^{\natural|\sharp}_{\rho,\sigma}$, it amounts to the same thing to say that σ normalizes $\underline{G|\mathcal{G}}^{\natural|\sharp}_{\rho,\sigma} = \underline{\mathbb{O}}^{*}_{12} \cdot \underline{\mathcal{G}|\widetilde{G}}^{\natural|\sharp}_{\dagger}$. Since σ normalizes $\underline{\mathcal{G}|\widetilde{G}}^{\natural|\sharp} = \underline{\mathcal{G}|\widetilde{G}}^{0}$ which is normal, it suffices to look at the action of σ on $\underline{\mathbb{O}}^{*}_{12} \subset G_{\rho,\sigma}$, † i.e. on the $u_{\underline{\mu}}$ of (151). The question is whether the

$$\sigma u_{\underline{\mu}}\sigma^{-1}u_{\underline{\mu}}^{-1} \in \Gamma\underline{\mathcal{G}|\widetilde{G}}^{\natural|\sharp},$$

now 79

Uncertainty retained from cancelled passages.

140-4-s080-D001 int f
140-4-s080-D002 $\natural|\sharp$
140-4-s080-D003 σ
140-4-s080-D004 σ
140-4-s080-D005 Let us set
140-4-s080-D007 , σ
140-4-s080-D008 $\underline{\mathcal{G}}^{\natural}$
140-4-s080-D009 $\mathcal{G}|\widetilde{G}$ $\natural|\sharp$ $\mathcal{G}|\widetilde{G}$
140-4-s080-D011 ρ, σ
140-4-s080-D012 defined|defined

[scan 81]
[p. 688]

$$\sigma u_{\underline{\mu}}\sigma^{-1}u_{\underline{\mu}}^{-1} = \sigma u_{\underline{\mu}}(\sigma)^{-1} = \sigma(\omega^{\nu_2}\sigma)^{-1} = \omega^{\nu_2}$$

thus

Proposition. For a section x of $\underline{G}^{\natural|\sharp}_{\rho,\sigma}$, † $\sigma(x)$ is a section of $\underline{G}^{\natural|\sharp}_{\rho,\sigma} \cdot \underbrace{\{1,\omega\}_{S_0}}_{\mu_{S_0}}$ and its image in $\Gamma\{1,\omega\}$ is equal to $\nu_2\dagger(x)$ — so it is in $\underline{G}^{\natural|\sharp}_{\rho,\sigma}$ iff $\underline{\mu}(x)$ is a section of

$$\underline{\mathbb{O}}^{*\prime\prime}_{12} \overset{\text{def|dfn}}{=} \{\underline{\mu} \in \Gamma\underline{\mathbb{O}}^{*}_{12} \mid \varepsilon_2(\underline{\mu}) = 1\} \tag{164}$$

Corollary 1. $\underline{G}^{\natural|\sharp}_{\rho,\sigma}$ is not normalized by σ or τ', but

$$\underline{G}^{\prime\prime\natural|\sharp}_{\rho,\sigma} \overset{\text{dfn}}{=} \{x \in \Gamma\underline{G}^{\natural|\sharp}_{\rho,\sigma} \mid \varepsilon_2(x) = 1\} \tag{165}$$

is equal to

$$\underline{G}^{\natural|\sharp}_{\rho,\sigma} \cap \sigma(\underline{G}^{\natural|\sharp}_{\rho,\sigma}) = \underline{G}^{\natural|\sharp}_{\rho,\sigma} \cap \tau'(\underline{G}^{\natural|\sharp}_{\rho,\sigma})$$

and is normalized by σ and τ'.

Corollary 2. $\underline{Z|N}^{\natural|\sharp}_{\rho}$ is not normalized by σ or τ', but

$$\underline{Z|N}^{0\natural|\sharp}_{\rho} \overset{\text{dfn}}{=} \underline{Z|N}^{\natural|\sharp}_{\rho} \cap \underline{G}^{\prime\prime\natural|\sharp}_{\rho,\sigma}\dagger \subset \underline{Z|N}^{\natural|\sharp}_{\rho}\dagger \tag{166}$$

m1491

Marginal note. m1491 N.B. We have $N^{\prime\prime\natural|\sharp}_{\rho} = (N^{\natural|\sharp}_{\rho})^0$ (identity component)

is normalized by σ and τ' — †

(167)

$$\begin{array}{ccc} \underline{Z|N}^{0\natural|\sharp}_{\rho} & \xrightarrow[\sim]{\chi_\rho} & \underline{\mathbb{O}}^{*0}_{12} \quad (= \underline{\mathbb{O}}^{*\prime}_{12} \cap \underline{\mathbb{O}}^{*\prime\prime}_{12}) \\ \| & & \\ (\underline{Z|N}^{\natural|\sharp}_{\rho})^0 & & \end{array}$$

m1492

Marginal note. m1492 Do we have $\underline{Z}^{\prime\prime\natural|\sharp}_{\rho} = (\underline{N}_{\rho})^0$ (identity component) ? We should also † $\underline{G}^0_{\rho,\sigma}$, $\underline{G}_{\rho,\sigma}/\underline{G}^0_{\rho,\sigma}$... compare with the diagrams (38) page 622'.

Corollary 3. Let

$$\underline{N}^{\prime\prime\prime\dagger}_{\rho(-\bar{\jmath})} = \underbrace{\{1,\tau'\}_{S_0}}_{(\mu_{\tau'})_{S_0}} \underbrace{\underline{Z|N}^{0\natural|\sharp}_{\rho}}_{(\underline{Z|N}^{\natural|\sharp}_{\rho})^0} \subset \underline{N}^{\dagger}_{\rho}, \tag{168}$$

then we have

(169)

$$\underline{N}^{\prime\prime\prime\dagger}_{\rho(-\bar{\jmath})} \xrightarrow[\sim]{\chi_\rho} \underline{\mathbb{O}}^{*\prime\prime\prime}_{12} = \{\underline{\mu} \in \Gamma\underline{\mathbb{O}}^{*}_{12} \mid (\varepsilon_2\varepsilon_3)(\underline{\mu}) = 1\}$$

Uncertainty retained from cancelled passages.

140-4-s081-D001 σ
140-4-s081-D002 χ
140-4-s081-D003 $= | \subset$
140-4-s081-D004 $\underline{N}_\rho$
140-4-s081-D005 is identified
140-4-s081-D006 relations
140-4-s081-D007 $\natural|\sharp$
140-4-s081-D008 $\natural|\sharp$
140-4-s081-D009 $\natural|\sharp$

[scan 82]

Thus $N_\rho^{'''\natural|\sharp}$ is a partial section of N_ρ over $\mathbb{O}_{12}^{*'''}$. Finally, let us put

$$(170) \qquad \underline{Z}^!_\rho = \underline{Z}^{0\natural|\sharp}_\rho \times \mu_{S_0} \qquad (\mu = \{1, \omega\})$$

then $\underline{Z}^!_\rho$ is normalized by τ', and we can introduce

$$(171) \qquad \dagger \underline{N}^?_\rho \overset{\text{dfn}}{=} (\mu_{\tau'} | \mu\tau')_{S_0} \cdot \underline{Z}^!_\rho \subset \underline{N}_\rho$$

so that $\underline{Z}^{0\natural|\sharp}_\rho$ has index 4 in $\underline{N}^?_\rho$. We then have

Corollary 4. We have an exact sequence

$$(172) \quad 1 \longrightarrow \underset{\substack{\| \text{dfn} \\ \mu_\rho}}{\mu_{S_0}} \longrightarrow \underline{N}^?_\rho \longrightarrow \underbrace{\underline{\mathbb{O}}^*_{12}}_{\gamma_{\rho,\sigma}} \longrightarrow 1$$

m1494

Marginal note. m1494 N.B. There is no need to redesignate|distinguish $\mathbb{O}^*_{12}$ — there is no, $N_\rho | N^!_\rho$ etc, no point of risk of confusion of the groups [illeg. 1]

$\underline{N}^?_\rho$ is a bi-section of the extension $\underline{N}_\rho$ of $\underline{\mathbb{O}}^*_{12}$ by L_ρ.

It would be worthwhile to develop the diagram of notable subgroups of $\underline{N}_\rho$, modeled on the diagram (38) of pages $622^{1|'}$. 81

Uncertainty retained from cancelled passages.

140-4-s082-D001 Z

[scan 83]
[p. 689]

8°) The groups $N_\sigma, N_\sigma^{\natural|\sharp}$ ....

Let us put

$$(173)\qquad \begin{cases} \underline{N}_\sigma \subset \underline{G}_{\rho,\sigma} \\ \overset{\text{dfn}}{=} \left\{ \underline{u} \in \Gamma \underline{G}_{\rho,\sigma} \;\middle|\; \underline{u}(\sigma) = \sigma^{\varepsilon_2(\underline{u})} \left(= \omega^{\nu_2(\underline{u})}\sigma\right) \right\}. \end{cases}$$

$$(174)\qquad \underline{N}_\sigma^{\natural|\sharp} \overset{\text{dfn}}{=} \underline{N}_\sigma \cap \underline{G}_{\rho,\sigma}^{\natural|\sharp}$$

The composite homomorphism

$$\underline{N}_\sigma^{\natural|\sharp} \overset{\dagger}{\hookrightarrow} \dagger \underline{G}_{\rho,\sigma}^{\natural|*} \overset{\underline{\chi}}{\twoheadrightarrow} \mathbb{O}_{12}^*, \qquad \underline{N}_\sigma^{\natural|\sharp} \overset{\underline{\chi}_\sigma}{\twoheadrightarrow} \mathbb{O}_{12}^*$$

is an epimorphism. Indeed, the $u_{\underline{\mu}}$ are in $\dagger\, N_{\sigma|\rho}^{\natural|\sharp}$ — so we even have a splitting

$$\mathbb{O}_{12}^* \hookrightarrow \underline{N}_\sigma^{\natural|\sharp}$$

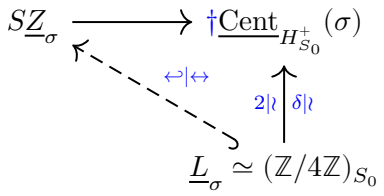
, hence decompositions into $\frac{1}{2}$ direct products

$$(175)\qquad \begin{cases} \underline{N}_\sigma^{\natural|\sharp} \simeq \mathbb{O}_{12}^* \cdot S\underline{Z}_\sigma^{\natural|\sharp} \\ \underline{N}_\sigma \simeq \mathbb{O}_{12}^* \cdot S\underline{Z}_\sigma \end{cases}$$

where we have put

$$(176)\qquad \begin{cases} \underline{Z}_\sigma = \mathrm{Ker}\left(\underline{N}_\sigma \overset{\varepsilon_2 \circ \underline{\chi}_\sigma}{\longrightarrow} \mu_{S_0} \right) = \underline{\mathrm{Cent}}_{\underline{G}_{\rho,\sigma}}(\sigma) \\ S\underline{Z}_\sigma = \underline{Z}_\sigma \cap \underline{\mathcal{G}|\widetilde{G}}^+ = \underline{\mathrm{Cent}}_{\underline{\mathcal{G}|\widetilde{G}}^+}(\sigma) \\ \underline{Z}_\sigma^{\natural|\sharp} = \underline{Z}_\sigma \cap \underline{G}_{\rho,\sigma}^{\natural|\sharp} \\ S\underline{Z}_\sigma^{\natural|\sharp} = S\underline{Z}_\sigma \cap \underbrace{\underline{\mathcal{G}|\widetilde{G}}^{\natural|\sharp}}_{=\underline{\mathcal{G}|\widetilde{G}}^0 = \underline{M}} \dagger \end{cases}$$

Moreover we have



which has a section, hence

$$S\underline{Z}_\sigma \simeq L_\sigma \times S\underline{Z}^{\natural|\sharp}_\sigma \tag{177}$$

82

Uncertainty retained from cancelled passages.

140-4-s083-D002 $\underline{\chi}_\sigma$
140-4-s083-D003 $\mathbb{O}$
140-4-s083-D004 G
140-4-s083-D006 Z $\mathcal{G}|\widetilde{G}$ $\natural|\sharp$ $\mathcal{G}|\widetilde{G}$
140-4-s083-D007 H

[scan 84]

Finally

$$\begin{array}{c} S\underline{Z}^{\natural|\sharp}_\sigma \subset \underline{M} \\ \| \\ \left\{x \in \Gamma\underline{M} \,\middle|\, (1-\sigma_{\underline{M}})(x) = 0\right\} = \{\dagger a_0\lambda_0 + a_1\lambda_1 \mid a_0 = a_1\} \\ = \{a(\lambda_0+\lambda_1) \mid a \in \Gamma\mathbb{G}_a\} \simeq \mathbb{G}_a, \end{array} \tag{178}$$

hence, taking 177 into account :

$$\begin{cases} S\underline{Z}_\sigma \simeq \underline{L}_\sigma \times \underbrace{\mathbb{G}_a}_{S\underline{Z}^{\natural|\sharp}_\sigma} \\ S\underline{Z}^{\natural|\sharp}_\sigma = (S\underline{Z}_\sigma)^0 \quad \text{(identity component)} \simeq \mathbb{G}_a \end{cases} \tag{179}$$

We thus have

$$1 \longrightarrow \underset{\wr \\ S\underline{Z}^{\natural\sharp}_{\sigma}}{\mathbb{G}_a} \longrightarrow \underline{N}^{\natural|\sharp}_{\sigma} \longrightarrow \underset{\| \\ \gamma_{\rho,\sigma}}{\mathbb{O}^*_{12}} \longrightarrow 1$$

, more precisely the isomorphisms (175)(177)(179) give

$$(180) \qquad \begin{array}{l} \underline{N}^{\natural|\sharp}_{\sigma} \simeq \mathbb{O}^*_{12} \cdot \mathbb{G}_a \\ \underline{N}_{\sigma} \simeq \mathbb{O}^*_{12} \cdot (\underline{L}_{\sigma} \times \mathbb{G}_a) \end{array}$$

In these formulas, $\mathbb{O}^*_{12}$ operates on $\mathbb{G}_a$ via

$$\chi : \mathbb{O}^*_{12} \longrightarrow \mathbb{G}_m$$

, and on L_σ via

$$\varepsilon_2 : \mathbb{O}^*_{12} \longrightarrow (\pm 1)_{S_0}$$

:

$$(181) \qquad \begin{cases} \underline{\mu}(\lambda) = \mu \cdot \lambda \\ \underline{\mu}(\sigma) = \sigma^{\varepsilon_2} \end{cases} \qquad \text{where } \mu = \chi(\underline{\mu}),\ \varepsilon_2 = \varepsilon_2(\underline{\mu})$$

83

Uncertainty retained from cancelled passages.

140-4-s084-D001 $x = a_0\lambda_0 + a_1\lambda_1$

[scan 85]
[p. 690]

9°) Return to $\underline{M}_{\rho,\sigma}$.

Drawing inspiration from the developments of § 49 (in particular VI, 5°)), we shall set

$$(182)\qquad \begin{cases} S_0|s_0\underline{M}_{\rho,\sigma} = \underline{Z}_{\rho,\sigma} \times_\gamma \underline{N}^?_\rho \times_\gamma \underline{N}^{\natural|\sharp}_\sigma \times_\gamma \underline{G}_{\rho,\sigma} \\ S_0|s_0\underline{M}^{\natural|\sharp}_{\rho,\sigma} = \underline{Z}_{\rho,\sigma} \times_\gamma \underline{N}^?_\rho \times_\gamma \underline{N}^{\natural|\sharp}_\sigma \times_\gamma \underline{G}^{\natural|\sharp}_{\rho,\sigma} \end{cases}$$

$$(183)\qquad \begin{cases} \dagger\underline{M}_{\rho,\sigma} = \underline{N}^?_\rho \times_\gamma \underline{N}^{\natural|\sharp}_\sigma \times_\gamma \underline{G}_{\rho,\sigma} \\ \underline{M}^{\natural|\sharp}_{\rho,\sigma} = \underline{N}^?_\rho \times_\gamma \underline{N}^{\natural|\sharp}_\sigma \times_\gamma \underline{G}^{\natural|\sharp}_{\rho,\sigma} \end{cases}$$

$$(184)\qquad \begin{cases} S_0|s_0\Gamma_{\rho,\sigma} = \underline{Z}_{\rho,\sigma} \times_\gamma \underline{N}^?_{\rho|1} \times_\gamma \underline{N}^{\natural|\sharp}_\sigma \\ \Gamma_{\rho,\sigma} = \underline{N}^?_\rho \times_\gamma \underline{N}^{\natural|\sharp}_\sigma \end{cases}$$

where †

$$\gamma = \gamma_{\rho,\sigma} = \mathbb{O}^*_{12}.$$

In fact, the groups $S_0|s_0\underline{M}_{\rho,\sigma}$, $S_0|s_0\underline{M}^{\natural|\sharp}_{\rho,\sigma}$, $S_0|s_0\Gamma_{\rho,\sigma}$ play rather an auxiliary technical role, they are extensions of the groups $\underline{M}_{\rho,\sigma}$, $\underline{M}^{\natural|\sharp}_{\rho,\sigma}$ or $\underline{\Gamma}_{\rho,\sigma}$ by

$$L_0^{\underline{M}} = \mathrm{Ker}\left(\underline{Z}_{\rho,\sigma} \longrightarrow \underset{\gamma}{\mathbb{O}^*_{12}}\right),$$

the first two deduced from the extension $S_0|s_0\underline{\Gamma}_{\rho,\sigma}$ of $\underline{\Gamma}_{\rho,\sigma}$ by $L_0^{\underline{M}} \simeq \mathbb{G}_a$, by inverse image via the epimorphisms $\underline{M}_{\rho,\sigma} \longrightarrow \underline{\Gamma}_{\rho,\sigma}$ and the induced epimorphism $\underline{M}^{\natural|\sharp}_{\rho,\sigma} \longrightarrow \underline{\Gamma}_{\rho,\sigma}$.

We shall turn our attention chiefly to the other groups (without $S_0|s_0$), linked by a diagram of exact sequences (compare page 639 (79)): 84

Uncertainty retained from cancelled passages.

140-4-s085-D001 S_0

140-4-s085-D002 $\gamma =$

[scan 86]

$$(190)\qquad \begin{array}{ccccccccc} & & \dagger & \dagger & \dagger & & & & \\ & & & & \dagger & & & & \\ & & & & \dagger & & & & \\ 1 & \longrightarrow & \underbrace{\underline{\mathcal{G}|\widetilde{G}}^{\natural|\sharp}}_{\underline{M}} & \longrightarrow & \underline{M}^{\natural|\sharp}_{\rho,\sigma} & \longrightarrow & \underline{\Gamma}_{\rho,\sigma} & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \downarrow 1 & & \\ 1 & \longrightarrow & \underline{\mathcal{G}|\widetilde{G}}^{+} & \longrightarrow & \underline{M}_{\rho,\sigma} & \longrightarrow & \underline{\Gamma}_{\rho,\sigma} & \longrightarrow & 1 \end{array}$$

and

$$(191)\qquad \begin{array}{ccccccccc} 1 & \longrightarrow & S\underline{N}^{?}_{\rho} \times S\dagger\underline{Z}^{\natural|\sharp}_{\dagger\sigma} \times \underbrace{\underline{\mathcal{G}|\widetilde{G}}^{\natural|\sharp}}_{\underline{M}} & \longrightarrow & \underline{M}^{\natural|\sharp}_{\rho,\sigma} & \longrightarrow & \underset{\dagger}{\mathbb{O}^{*}_{12}} & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \Downarrow \,|\!|\!| & & \\ 1 & \longrightarrow & S\underline{N}^{?}_{\rho} \times S\dagger\underline{Z}^{\natural|\sharp}_{\sigma} \times \underline{\mathcal{G}|\widetilde{G}}^{+} & \longrightarrow & \underline{M}_{\rho,\sigma} & \longrightarrow & \underbrace{\mathbb{O}^{*|\wedge}_{12}}_{\gamma_{\rho,\sigma}} & \longrightarrow & 1 \end{array}$$

,

$$(192)\qquad 1 \longrightarrow S\underline{N}^{?}_{\rho} \times S\dagger\underline{Z}^{\natural|\sharp}_{\sigma} \longrightarrow \underline{\Gamma}_{\rho,\sigma} \longrightarrow \underbrace{\mathbb{O}^{*|\wedge}_{12}}_{\gamma_{\rho,\sigma}} \longrightarrow 1$$

,
where we have

$$(193)\qquad \begin{cases} S\underline{N}^{?}_{\rho} = \mu_{\rho} \simeq \{\pm 1\}_{S_0} & (172) \\ SZ^{\natural|\sharp}_{\sigma} \simeq \mathbb{G}_a & (179) \end{cases} .$$

These exact sequences become more precise, thanks to the 1/2 direct decompositions:

$$\dagger,$$

$$(194)\qquad \begin{cases} \underline{Z}_{\rho,\sigma} \simeq \mathbb{O}^{*|\wedge}_{12} \cdot \mathbb{G}_a & (150, 151) \\ \underline{G}_{\rho,\sigma} \simeq \mathbb{O}^{*}_{12} \cdot \underline{\mathcal{G}|\widetilde{G}}^{+} & \mathrm{id} \\ \underline{G}^{\natural|\sharp}_{\rho,\sigma} \simeq \mathbb{O}^{*|\wedge}_{12} \cdot \underbrace{\dagger\underline{M}\dagger}_{\underline{\mathcal{G}|\widetilde{G}}^{\natural|\sharp}} & (160) \\ \underline{N}^{\natural|\sharp}_{\sigma} \simeq \mathbb{O}^{*}_{12} \cdot \mathbb{G}_a & (180) \\ \text{and finally} & \\ \underline{N}_{\rho}/\mu_{\rho} \simeq \mathbb{O}^{*}_{12} & (172) \end{cases}$$

85

Uncertainty retained from cancelled passages.

140-4-s086-D001 1
140-4-s086-D002 $\longrightarrow$
140-4-s086-D003 M $\natural|\sharp$
140-4-s086-D005 $\mathcal{G}|\widetilde{G}$
140-4-s086-D006 $\mathcal{G}|\widetilde{G}$ $\natural|\sharp$
140-4-s086-D008 N
140-4-s086-D009 ρ
140-4-s086-D010 γ
140-4-s086-D011 N
140-4-s086-D012 N
140-4-s086-D014 $Z|G$ $*|\wedge$ S $N|Z$ $\natural|\sharp$ $\simeq$
140-4-s086-D015 $\mathcal{G}|\widetilde{G}$
140-4-s086-D016 $\natural|\sharp$

[scan 87]
[p. 691]

We then find canonical isomorphisms

$$(195)\qquad \begin{cases} \underline{M}_{\rho,\sigma}/\mu_\rho & \simeq & \mathbb{O}^*_{12}\cdot(\mathbb{G}_a\times\underline{\mathcal{G}|\widetilde{G}^+}) \\ \uparrow & & \| \qquad\qquad \uparrow \\ \underline{M}^{\natural|\sharp}_{\rho,\sigma}/\mu_\rho & \simeq & \mathbb{O}^*_{12}\cdot(\underbrace{\mathbb{G}_a\times\underline{M}}_{\mathbb{G}_a\times\underline{\mathcal{G}|\widetilde{G}^+}}) \\ \\ \underline{\Gamma}_{\rho,\sigma} & \simeq & \mathbb{O}^*_{12}\cdot\mathbb{G}_a \end{cases}$$

m1500

Marginal note. m1500 operations of $\mathbb{O}^*_{12}$ on $\mathbb{G}_a\times\underline{\mathcal{G}|\widetilde{G}^+}$ given by (181) and (151). On $\mathbb{G}_a\times\underline{M}$ †, $\mathbb{O}^*_{12}$ operates simply by homothety.

$\underline{M}_{\rho,\sigma}$ and $\underline{M}^{\natural|\sharp}_{\rho,\sigma}$, as extensions, of $\underline{M}_{\rho,\sigma}/\mu_\rho$ and $\underline{M}^{\natural|\sharp}_{\rho,\sigma}/\mu_\rho$ by μ_ρ, are deduced from the extension

$$(196)\qquad 1\to\mu_\rho\to N^?_\rho\to\mathbb{O}^*_{12}\to 1$$

of $\mathbb{O}^*_{12}$ by μ_ρ, by inverse images under the canonical homomorphisms from the groups (195) onto the factor $\mathbb{O}^*_{12}$. We should come back to the extension (196), which is surely the inverse image of an extension of $(\mu\times\mu)_{S_0}$ by μ, via

$$\underline{\mathbb{O}}^*_{12}\xrightarrow{(\varepsilon_2,\varepsilon_3)}(\mu\times\mu)_{S_0}$$

(Compare § 49 V 8°).

The operation of $\mathbb{O}^*_{12}$ on $\mathbb{G}_a\times\underline{\mathcal{G}|\widetilde{G}^+}$ (the one on $\mathbb{G}_a\times\underline{\mathcal{G}|\widetilde{G}^{\natural|\sharp}}=\mathbb{G}_a\times\underline{M}$) takes place via operations on $\mathbb{G}_a$ and $\underline{\mathcal{G}|\widetilde{G}^+}$, which are made explicit by (181) and (151). 86

Uncertainty retained from cancelled passages.
140-4-s087-D001 it

[scan 88]

10°) The subgroups $\underline{M}_{\rho,\sigma}[0]$, $\underline{M}_{\rho,\sigma}(0)$, $\underline{M}_{\rho,\sigma}(\frac{1}{2})$, $\underline{M}'_{\rho,\sigma}(-\bar\jmath)$, $[M'''_{\rho,\sigma}(-\bar\jmath), M^?_{\rho,\sigma}(-\bar\jmath)]$.

We use the embedding|embedding

$$(195)\qquad \begin{cases} \underline{M}_{\rho,\sigma}=\underline{N}^?_\rho\times_\gamma\underline{N}^{\natural|\sharp}_\sigma\times_\gamma\underline{G}_{\rho,\sigma}\subset\underline{N}^?_\rho\times\underline{N}^{\natural|\sharp}_\sigma\times\underline{G}_{\rho,\sigma} \\ \| \\ \{(a,b,U)\,\{\in\Gamma\dagger\underline{N}^?_\rho\times\underline{N}_\sigma\times\underline{G}_{\rho,\sigma}\mid\underline{\chi}_\rho(a)\equiv|=\underline{\chi}_\sigma(b)\equiv|=\dagger\underline{\chi}_G(U)\} \end{cases}$$

We thus define

$$(196)\qquad \begin{array}{c} \underline{M}_{\rho,\sigma}[0]\subset\underline{M}^\dagger_{\rho,\sigma} \\ \| \\ \{(a,b,U)\in\Gamma\underline{M}_{\rho,\sigma}\mid U\in Z_{\rho,\sigma}\} \end{array}$$

m1502

Marginal note. m1502 N.B. we will even have $\underline{M}_{\rho,\sigma}[0] \subset \underline{M}^{\natural\sharp}_{\rho,\sigma}$.

thus

$$\underline{M}_{\rho,\sigma}[0] \simeq \underline{N}^{?}_{\rho} \times_{\gamma} \underline{N}^{\natural\sharp}_{\sigma} \times_{\gamma} Z_{\rho,\sigma} \overset{\sim}{\to} \dagger S_0|s_0\underline{\Gamma}_{\rho,\sigma}$$

Let

$$(197) \quad \underline{Z}_{\rho,\sigma}(0) \subset \underline{Z}_{\rho,\sigma}$$

be the $\frac{1}{2}$ direct factor $\mathbb{O}^*_{12}$ of (150) (151),

m1503

Marginal note. m1503 N.B. $\underline{Z}_{\rho,\sigma}(0) \subset \underline{Z}_{\rho,\sigma} \subset \underline{G}_{\rho,\sigma}$ is also the $\frac{1}{2}$ direct factor $\mathbb{O}^*_{12}$ of $\underline{G}_{\rho,\sigma}$ in (150), it could also be denoted $\underline{G}_{\rho,\sigma}(0)$.

We have then, let us put $\underline{M}_{\rho,\sigma}(0) \simeq \underline{N}^{?}_{\rho} \times_{\gamma} \underline{N}^{\natural\sharp}_{\sigma} \times_{\gamma} Z_{\rho,\sigma}(0)$

$$(198) \quad \left\{ \begin{array}{ccc} \underline{M}_{\rho,\sigma}(0) & \subset & \underline{M}_{\rho,\sigma}[0] \\ \downarrow \sim & & \downarrow \sim \\ S\underline{\Gamma}_{\rho,\sigma}(0) & \hookrightarrow & S_0|s_0\underline{\Gamma}_{\rho,\sigma} \\ \downarrow \sim & & \\ \underline{\Gamma}_{\rho,\sigma}(0) & & \end{array} \right.$$

$$(199) \quad \begin{aligned} \underline{M}_{\rho,\sigma}(0) &= \{(a,b,U) \in \Gamma\underline{M}_{\rho,\sigma} \mid U \in \Gamma\underline{Z}_{\rho,\sigma}(0)\} \\ &= \underline{\mu} \cdot \begin{pmatrix} \lambda & , & 0 \\ \cup & & \cup \\ \Gamma\mathbb{G}_a & & \Gamma\underline{\mathcal{G}|\widetilde{G}^{+}} \end{pmatrix} \end{aligned}$$

m1504

Marginal note. m1504 in the description (195) of $\underline{M}_{\rho,\sigma}$ as a $\frac{1}{2}$ direct product.

87

Uncertainty retained from cancelled passages.

140-4-s088-D001 $=$
140-4-s088-D002 χ
140-4-s088-D003 $\natural|\sharp$
140-4-s088-D004 $\underline{Z}$

[scan 89]
[p. 692]

m1506

Marginal note. m1506 N.B. $\dagger\ \mu_\rho \subset \underline{M}_{\rho,\sigma}(0) \subset \underline{M}_{\rho,\sigma}[0]$ i.e.

$$(200) \quad \underline{M}_{\rho,\sigma}(0)/\mu_\rho \simeq \mathbb{O}^*_{12} \cdot \mathbb{G}_a \hookrightarrow \underline{M}_{\rho,\sigma} \simeq \mathbb{O}^*_{12}(\mathbb{G}_a \times \underline{\mathcal{G}|\widetilde{G}^{+}})$$

(corresponding to the canonical inclusion † of $\mathbb{G}_a$ in $\mathbb{G}_a \times \underline{\mathcal{G}|\widetilde{G}^+}$).
On the other hand we will have

$$(201)\qquad \underline{M}_{\rho,\sigma}[0]/\mu_\rho \simeq \mathbb{O}_{12}^* \cdot (\mathbb{G}_a \times L_0^{\underline{M}}) \hookrightarrow \overbrace{\mathbb{O}_{12}^*(\mathbb{G}_a \times \underline{\mathcal{G}|\widetilde{G}^+})}^{M_{\rho,\sigma}\simeq}$$

On the other hand we define

$$(202)\quad \underline{M}_{\rho,\sigma}(\tfrac{1}{2}) = \{(a,b,U) \in \Gamma\underline{M}_{\rho,\sigma} \mid U = b\}$$

(thus we will have $\underline{M}_{\rho,\sigma}(\frac{1}{2}) \subset \underline{M}^{\natural|\sharp}_{\rho,\sigma}$)
Thus we have|the

$$(203)\quad \begin{cases} \underline{M}_{\rho,\sigma}(\frac{1}{2}) \xrightarrow{\sim} \underline{\Gamma}_{\rho,\sigma} \\ \underline{M}^{\natural|\sharp}_{\rho,\sigma} \simeq \underline{M}_{\rho,\sigma}(\frac{1}{2}) \cdot \underline{M} \\ \underline{M}_{\rho,\sigma} \simeq \underline{M}_{\rho,\sigma}(\frac{1}{2}) \cdot \underline{\mathcal{G}|\widetilde{G}^+}. \end{cases}$$

In the representation of $\underline{M}_{\rho,\sigma}/\mu_\rho$, $\underline{M}^{\natural|\sharp}_{\rho,\sigma}/\mu_\rho$ as $\frac{1}{2}$ direct products, we will have the description

i.e. $\mu_\rho \subset \underline{M}_{\rho,\sigma}(\frac{1}{2})$

$$(204)\quad \begin{cases} \underline{M}_{\rho,\sigma}(\frac{1}{2})/\mu_\rho = \left\{ \underline{\mu} \cdot \left(\underset{\Gamma\mathbb{G}_a}{\underset{\cup}{\lambda}} , \underbrace{\lambda(\lambda_1+\lambda_2)}_{\Gamma\underline{M}} \right) \;\middle|\; -\lambda|\lambda \in \Gamma\mathbb{G}_a \right\} \\ \quad \wr\| \\ \mathbb{O}_{12}^* \cdot \mathbb{G}_a \hookrightarrow \mathbb{O}_{12}^*(\mathbb{G}_a \times \underline{M}) \end{cases} \qquad \boxed{M^{\natural|\sharp}_{\rho,\sigma} \simeq}$$

associated to

$$\mathbb{G}_a \longrightarrow \mathbb{G}_a \times M$$

$$\lambda \longmapsto \lambda, \lambda(\lambda_1+\lambda_2)$$

Uncertainty retained from cancelled passages.
140-4-s089-D001 $\underline{M}_{\rho,\sigma}(0)$
140-4-s089-D003)

[scan 90]

Finally we put

$$(205)\qquad \begin{array}{c} \underline{M}^{\dagger|'}_{\rho,\sigma}(-\bar{\jmath}) \subset \underline{M}_{\rho,\sigma} \\ \| \\ \{(a,b,U) \in \Gamma\underline{M}_{\rho,\sigma} \mid U = a\} \end{array}$$

here we will not necessarily have $U \in \Gamma(\underline{G}^{\natural|\sharp}_{\rho,\sigma}\dagger)$ since $\dagger\, \underline{N}^?_\rho \not\subset \underline{G}^{\natural|\sharp}_{\rho,\sigma}$, thus

$$\underline{M}^{\dagger|'}_{\rho,\sigma}(-\bar{\jmath}) \not\subset \underline{M}^{\natural|\sharp}_{\rho,\sigma},$$

contrary to what happens for $\underline{M}_{\rho,\sigma}(0)$, $\underline{M}_{\rho,\sigma}[0]$, $\underline{M}_{\rho,\sigma}(\frac{1}{2})$. We will have a

$$(206)\qquad \begin{cases} \underline{M}^{\dagger|'}_{\rho,\sigma}(-\bar{\jmath}) \xrightarrow{\sim} \underline{\Gamma}_{\rho,\sigma} \\ \underline{M}_{\rho,\sigma} \simeq \underline{M}_{\rho,\sigma}(-\bar{\jmath}) \cdot \underline{\mathcal{G}|\widetilde{G}}^{+} \end{cases}$$

But we †have

$$\underline{\mu}_{\rho} \not\subset \underline{M}_{\rho,\sigma}(-\bar{\jmath})$$

more precisely

$$\underline{\mu}_{\rho} \cap \underline{M}_{\rho,\sigma}(-\bar{\jmath}) = \{1\}$$

thus $\underline{M}_{\rho,\sigma}(-\bar{\jmath})$ does not define (like $\underline{M}_{\rho,\sigma}(\frac{1}{2})$) a section of $\underline{M}_{\rho,\sigma}/\underline{\mu}_{\rho}$ †on top of †$\underline{\Gamma}_{\rho,\sigma}/\underline{\mu}_{\rho}$ ($\simeq \underline{N}^{\natural|\sharp}_{\sigma} \simeq \mathbb{O}^{*}_{12} \cdot \mathbb{G}_{a}$). We nevertheless have

$$\underline{M}_{\rho,\sigma}(-\bar{\jmath}) \cdot \underline{\mu}_{\rho} = \underline{M}_{\rho,\sigma}(-\bar{\jmath}) \cdot \underline{\mu}_{G}$$

and putting

$$(207)\qquad \begin{aligned} \underline{M}^{\sim}_{\rho,\sigma}(-\bar{\jmath}) &\overset{\text{déf}}{=} \underline{M}_{\rho,\sigma}(-\bar{\jmath}) \cdot \underline{\mu}_{\rho}/\underline{\mu}_{\rho} \\ &= \underset{\text{(isomorphic}}{\operatorname{Im}} \left(\underline{M}_{\rho,\sigma}(-\bar{\jmath}) \to \underline{M}_{\rho,\sigma}/\mu_{\rho} \right), \end{aligned}$$

$\underline{\Gamma}^{\sim}_{\rho,\sigma} = \underline{\Gamma}_{\rho,\sigma}/\underline{\mu}_{\rho} = (\underline{M}_{\rho,\sigma}/\underline{\mu}_{\rho})/\underline{\mathcal{G}|\widetilde{G}}^{+}$ 89

Uncertainty retained from cancelled passages.

140-4-s090-D001 ?
140-4-s090-D002 $\underline{M}$
140-4-s090-D003 [illeg. 2]
140-4-s090-D004 ?
140-4-s090-D005 ?
140-4-s090-D006 will not have
140-4-s090-D008 over
140-4-s090-D009 μ_2

[scan 91]
[p. 693]

m1510

Marginal note. m1510 N.B. We have

$$\underline{\Gamma}^{\sim}_{\rho,\sigma} \to \underline{N}_{\rho}/\underline{\mu}_{\rho}$$

epimorphism with kernel $\simeq$

$$\operatorname{Ker}\left(\underline{N}_{\sigma} \to \underset{\gamma_{\rho,\sigma}}{\mathbb{O}^{*}_{12}} \right) \simeq \mathbb{G}_{a},$$

and We have Cartesian diagram

$$\begin{array}{ccccc} \underline{\Gamma}^{\sim}_{\rho,\sigma} & \longrightarrow & \underline{N}_{\rho}/\underline{\mu}_{\rho} & \to & 1 \\ \uparrow & & \uparrow{\scriptstyle \text{can}} & & \\ \dagger\underline{\Gamma}^{\dagger}_{\rho,\dagger} & & & & \\ \simeq \underline{M}_{\rho,\sigma}(-\bar{\jmath}) & \longrightarrow & \underline{N}_{\rho} & \longrightarrow & 1 \\ \simeq \underline{M}^{\sim}_{\rho,\sigma}(-\bar{\jmath}) & & & & \end{array}$$

the diagram of subgroups of $\underline{N}_\rho$ of 7° defines a thus diagram of subgroups of $\underline{M}_{\rho,\sigma}(-\bar{\jmath})$…

we will have

$$(208)\qquad 1 \to \underline{\mu}_G \to \underline{M}^{\sim}_{\rho,\sigma}(-\bar{\jmath}|\dagger\bar{\jmath}) \to \underline{\Gamma}^{\sim}_{\rho,\sigma}\dagger \to 1$$

i.e. $\underline{M}^{\sim}_{\rho,\sigma}(-\frac{1}{2})$ is a bisection of $\underline{M}^{\sim}_{\rho,\sigma} \overset{\text{déf}}{=} \underline{M}_{\rho,\sigma}/\underline{\mu}_\rho$ over $\underline{\Gamma}^{\sim}_{\rho,\sigma}$. Putting

$$(209)\qquad \underline{M'}^{\sim}_{\rho,\sigma}(-\bar{\jmath}) \overset{\text{déf}}{=} \underline{M}^{\sim}_{\rho,\sigma}(-\bar{\jmath}|\dagger\bar{\jmath}) \cap \underbrace{\underline{M}^{\sim\natural|\sharp}_{\rho,\sigma}}_{\| \text{ déf}\ \underline{M}^{\natural|\sharp}_{\rho,\sigma}/\mu_\rho} \dagger$$

we find

m1511

Marginal note. m1511 $\underline{M'}^{\sim}_{\rho,\sigma}(-\bar{\jmath})$ of index 4 in $\underline{M}^{\sim}_{\rho,\sigma}(-\bar{\jmath})$

$$(209)\qquad \underline{M'}^{\sim}_{\rho,\sigma}(-\bar{\jmath}) \twoheadrightarrow \underbrace{\underline{\Gamma'}^{\sim}_{\rho,\sigma}}_{\| \text{ déf}\ \operatorname{Ker}\left(\underline{\Gamma}^{\sim}_{\rho,\sigma} \overset{\varepsilon_2}{\to} \underline{\mu}_{S_0}\right)} \overset{\dagger}{\subset} \underline{\Gamma}^{\sim}_{\rho,\sigma},$$

partial section of $\underline{M}^{\sim}_{\rho,\sigma}$ over $\underline{\Gamma}^{\sim}_{\rho,\sigma}$, with image p|pr $\Gamma'^{\sim}_{\rho,\sigma}$ of index 2 in $\underline{\Gamma}'_{\rho,\sigma}$. We will thus have

$$(210)\qquad \begin{cases} \underline{M}^{\sim\prime}_{\rho,\sigma} \overset{\text{déf}}{=} | \simeq \operatorname{Ker}\left(\underline{M}^{\sim}_{\rho,\sigma} \overset{\varepsilon_2}{\to} \underline{\mu}_{S_0}\right) \simeq M'^{\sim}_{\rho,\sigma}(-\bar{\jmath}) \cdot \underline{\mathcal{G}|\widetilde{G}}^{+} \\ \underline{M}^{\sim\prime\natural|\sharp}_{\rho,\sigma} \overset{\text{déf}}{=} \operatorname{Ker}\left(\underline{M}^{\sim\natural|\sharp}_{\rho,\sigma} \overset{\varepsilon_2}{\to} \underline{\mu}\right) \simeq \underline{M'}^{\sim}_{\rho,\sigma}(-\bar{\jmath}) \cdot \underline{M}. \end{cases}$$

Remark I stop here in the developments of interesting subgroups of $\underline{M}^{\sim}_{\rho,\sigma}(-\bar{\jmath})$ — I will come back to it once again, probably, in the context of the relations to the general groups $\underline{M}_{\rho,\sigma}$. It is becoming clear that it will be necessary to choose, 90

Uncertainty retained from cancelled passages.

140-4-s091-D001 $\underline{\mu}$
140-4-s091-D002 σ
140-4-s091-D003 $\sim$
140-4-s091-D004 $\frac{1}{2}$
140-4-s091-D005 $/\mu_\rho$
140-4-s091-D006 $\frac{1}{2}$
140-4-s091-D007 $/\mu_\rho$
140-4-s091-D008 déf

later on, between the respective possible notations for the groups denoted here $\underline{M}_{\rho,\sigma}$ and $\underline{M}_{\rho,\sigma}/\underline{\mu}_\rho \simeq \underline{M}^{\sim}_{\rho,\sigma}$ (and ditto with $\natural|\sharp$) — it is these latter, it is true, that seem to me the most interesting, for it is they* that link up with the universal group $M|\mathcal{M} = N|\mathcal{N}^{\sim}_{1,1}$ (extension of $\Gamma^{\sim}_{\mathbb{Q}}$ by $\mathcal{G}|G^{\wedge}_0 = (\mathrm{SL}(2,\mathbb{Z}))^{\wedge}$). [scan 92] m1513

Marginal note. m1513 rather, the group of their $\widehat{\mathbb{Z}}$-points respct…|respect…

It would seem that these groups would deserve the simplest notation, in this case $\underline{M}_{\rho,\sigma}$, and the groups denoted previously $\underline{M}_{\rho,\sigma}$ would take on their $\sim$, or an analogous sign. (The $\sim$ is

already used in a rather different sense in $\Gamma^{\sim}_{\mathbb{Q}}$, $M^{\sim}_{0,3}$ etc — on the other hand the superscripts $'$, †, $''$, $'''$, $^{\circ}$, +, !, ♮|♯ † are already used in different senses. The double use of $\sim$ moreover risks leading to conflicts of notation sooner or later, it is perhaps preferable to replace the $\sim$ in $\Gamma^{\sim}_{\mathbb{Q}}$ by letters such as $\Gamma^{\text{ét}}_{\mathbb{Q}}$ (suggesting "extended") or † "generalized" gen… † I will have to resolve these di- 91

Uncertainty retained from cancelled passages.

140-4-s092-D001 ♮|♯

140-4-s092-D002 $'$, $'$, $'$, $'$, +

140-4-s092-D003 gen

140-4-s092-D004 [illeg. 1]

lemmas of notation, for the time being I keep the notations introduced for the need|the needs in the present section, which will be revised|revised with care in a special § later on. [scan 93] [p. 694]

We should still describe the subgroups $\underline{M}^{\sim}_{\rho,\sigma}(-\bar{j})$, $\underline{M}'^{\sim}_{\rho,\sigma}(-\bar{j})$ in $\underline{M}^{\sim}_{\rho,\sigma}$ written as a $\frac{1}{2}$ direct product, we find

$$(211)\qquad \underline{M}^{\sim}_{\rho,\sigma}(-\bar{j})\dagger\dagger \subset \underline{M}^{\sim}_{\rho,\sigma} \simeq \mathbb{O}^{*}_{12}(\mathbb{G}_a \times \underline{\mathcal{G}|\widetilde{G}}^{+})$$
$$= \left\{ \dagger \begin{pmatrix} \underset{\mathbb{G}_a}{\lambda}, & \underset{\underline{\mathcal{G}|\widetilde{G}}^{+}}{x} \end{pmatrix} \cdot \underset{\mathbb{O}^{*}_{12}}{\underline{\mu}} \;\middle|\; x \cdot \dagger \underbrace{\underline{u_{\mu}}\dagger}_{\in \underline{G}_{\rho,\sigma}} \in N^{?}_{\rho} \right\}$$

where $u_{\underline{\mu}}$ is defined in (151), in particular

$$u_{\underline{\mu}}(\rho) = \lambda_1^{\nu}\{\rho\}|\rho\dagger \qquad \text{where} \quad \nu = \nu(\underline{\mu}),$$

$$\overset{\dagger}{\uparrow}$$
$$\delta^{\nu''}\tau^{\nu_3}$$

we have seen (154)

$$v = \underbrace{\sigma^{\nu_3}\delta^{-\nu''}u_{\underline{\mu}}}_{v|u} \in \Gamma\underline{N}_{\rho}, \qquad \underline{\chi}(v) = \underline{\mu}$$

and putting $\sigma = \omega\tau'\tau$, the first m[b]|side can also be written where $\delta = \lambda_1\lambda_0^{-1}$.

$$(212)\qquad v = \tau'^{\nu_3} \overset{\dagger}{\begin{pmatrix} \omega^{\nu_3} \cdot \underbrace{\tau^{\nu_3}\delta^{-\nu''}u_{\underline{\mu}}}_{\substack{\in \Gamma\dagger\underline{G}^{\sharp|\sharp}_{\rho,\sigma}, \text{ for the} \\ \text{three factors} \\ \text{are; hence} \\ \in \Gamma\underline{N}_{\rho} \cap \underline{G}^{\sharp|\sharp}_{\rho,\sigma} = \Gamma\underline{Z}^{\sharp|\sharp}_{\rho}}} \end{pmatrix}}$$

92

Uncertainty retained from cancelled passages.

140-4-s093-D001 $\simeq \underline{\mathbb{O}}^*_{12}(\mathbb{G}_a \times \underline{\mu}_G)$

140-4-s093-D002 $\underline{\mu}(\lambda, x)$

140-4-s093-D003 $\underline{\mu}$

140-4-s093-D004 $\underline{\mu}$

140-4-s093-D005 ρ

140-4-s093-D006 $=$

140-4-s093-D010 ν

140-4-s093-D011 $\omega\tau'\quad u$

140-4-s093-D012 N

and we know that $v \in \dagger N^?_\rho$, hence the general form $\dagger$ of the sections of $N^?_\rho$ over $\underline{\mu} = (\nu_2, \nu_3, \gamma_0|\nu_0) \in \Gamma\mathbb{O}^*_{12}$ [scan 94]

$$(212)\qquad (\underline{N^?_\rho})_{\underline{\mu}} = \left\{ \omega^i \sigma^{\nu_3} \delta^{-\nu''} u_{\underline{\mu}} \;\middle|\; i \in \Gamma\{0,1\}_{S_0} \right\}$$

with|where $\nu_3 = \nu_3(\underline{\mu})$, $\nu'' = \nu''(\underline{\mu})$, $\delta = \lambda_1\lambda_0^{-1}$. Thus we find

$$(213)\qquad \underline{M^\sim_{\rho,\sigma}}(-\bar{j}) = \left\{ \begin{pmatrix} \lambda, & x \\ \Downarrow & \Downarrow \\ \mathbb{G}_a & \underline{\mathcal{G}|\widetilde{G}^+} \end{pmatrix} \cdot \begin{matrix} \underline{\mu} \\ \Downarrow \\ \mathbb{O}^*_{12} \end{matrix} \;\middle|\; \begin{matrix} x = \omega^i \sigma^{\nu_3} \delta^{-\nu''} u_{\underline{\mu}}, \text{ where} \\ i \in \Gamma\{0,1\}_{S_0} \end{matrix} \right\}$$

$\delta = \lambda_1\lambda_0^{-1}$

We will therefore have

$$(214)\qquad \underline{M'^\sim_{\rho,\sigma}}(-\bar{j}) = \underline{M^\sim_{\rho,\sigma}}(-\bar{j}) \cap \underline{M^{\sharp|\sharp}_{\rho,\sigma}}$$
$$= \left\{ (\lambda, \delta^{-\nu''})\underline{\mu} \;\middle|\; \begin{matrix} \underline{\mu} \in \Gamma\mathbb{O}^{*\prime}_{12} \quad (\text{hence } \varepsilon_3(\underline{\mu}) = 0|1) \\ \lambda \in \Gamma\mathbb{G}_a \\ \nu'' = \nu''(\underline{\mu}) \end{matrix} \right\}$$

93

Uncertainty retained from cancelled passages.

140-4-s094-D001 $\underline{Z}$

140-4-s094-D002 elements

11°) The homomorphism $M \longrightarrow \underline{M}_{\rho,\sigma}(\widehat{\mathbb{Z}})^\Delta$, relation with the tower of Fermat curves. [scan 95]

Let us note that we have for every integer $N \in \mathbb{N}^*$ [p. 695]

$$(215)\qquad \mathbb{O}^\Delta_N(\widehat{\mathbb{Z}}) \simeq \widehat{\mathbb{Z}} \qquad \text{hence} \qquad \mathbb{O}^*_N(\widehat{\mathbb{Z}})^\Delta \simeq \widehat{\mathbb{Z}}^*.$$

m1519

Marginal note. m1519 The Δ indicates that we take the sections which are constant modulo the identity component.

We therefore have, taking into account the decompositions into semidirect products (150)

etc, isomorphisms

$$(216)\qquad \begin{cases} Z^{\ell|\wedge}_{\rho,\sigma} \overset{\text{dfn}}{=} \underline{Z}^{\dagger}_{\rho,\sigma}(\widehat{\mathbb{Z}})^{\Delta} = \widehat{\mathbb{Z}}^* \cdot \dagger \widehat{L}_0 \quad \text{where} \quad \widehat{L}_0 \simeq \widehat{\mathbb{Z}} \cdot \lambda_0 \simeq \widehat{\mathbb{Z}} \\ G_{\rho,\sigma} \overset{\text{dfn}}{=} \underline{G}^{\dagger}_{\rho,\sigma}(\widehat{\mathbb{Z}})^{\Delta} = \widehat{\mathbb{Z}}^* \cdot \underline{\mathcal{G}|\widetilde{G}}^{+\wedge} \\ G^{\natural|\sharp}_{\rho,\sigma} \overset{\text{dfn}}{=} \underline{G}^{\natural|\sharp}_{\rho,\sigma}(\widehat{\mathbb{Z}})^{\Delta} \simeq \widehat{\mathbb{Z}}^* \cdot \widehat{M} \\ N^{\sim\dagger}_{\rho\dagger} \overset{\text{df}}{=} \underline{N}^{\sim}_{\rho}(\widehat{\mathbb{Z}})^{\Delta} \simeq \mathbb{O}^*_{12}(\widehat{\mathbb{Z}})^{\Delta} \simeq \widehat{\mathbb{Z}}^* \\ N^{\natural|\sharp}_{\sigma} \overset{\text{df}}{=} \underline{N}^{\natural|\sharp}_{\sigma}(\widehat{\mathbb{Z}})^{\Delta} \simeq \widehat{\mathbb{Z}}^* \cdot \widehat{\mathbb{Z}} \\ M^{\sim}_{\dagger\ \rho,\sigma} \overset{\text{df}}{=} \underline{M}^{\dagger}_{\rho,\sigma}(\widehat{\mathbb{Z}})^{\Delta} \simeq \widehat{\mathbb{Z}}^* \cdot (\widehat{\mathbb{Z}} \times \underline{\mathcal{G}|\widetilde{G}}^{+\wedge}) \\ M^{\sim\natural|\sharp}_{\rho,\sigma} \overset{\text{df}}{=} \underline{M}^{\natural|\sharp}_{\rho,\sigma}(\widehat{\mathbb{Z}})^{\Delta} \simeq \widehat{\mathbb{Z}}^* \cdot (\widehat{\mathbb{Z}} \times \widehat{M}) \end{cases}$$

where the operations of $\widehat{\mathbb{Z}}^*$ on the $2^{\text{nd}}|2^{\text{nd}}$ $\frac{1}{2}$ direct factors are made explicit in the preceding sections. We will note that, despite the trappings|appearances of the notations and definitions, the profinite groups $Z_{\rho,\sigma}$ and preceding ones are also those which would be constructed † directly, in terms of $(\mathcal{G}|\widetilde{G}^{+|\natural\wedge}, \mathcal{G}|\widetilde{G}^{\natural|\sharp\wedge} = M^{\wedge})$ (equipped* with ρ, σ, τ), in § 49 VI. We therefore have a canonical homomorphism

$$(217)\qquad \begin{cases} M = N|\mathcal{N}^{\text{ét}}_{1,1} \longrightarrow M^{\sim}_{\rho,\sigma} \\ \cup \qquad\qquad\qquad \cup \\ M^{!} \longrightarrow M^{\sim\natural|\sharp}_{\rho,\sigma} \end{cases}$$

which as usual decomposes into 94

Uncertainty retained from cancelled passages.

140-4-s095-D001 $\natural|\sharp$
140-4-s095-D002 $\widehat{\mathbb{Z}}$
140-4-s095-D003 $\Delta|\natural$
140-4-s095-D004 $,\sigma$
140-4-s095-D005 $\natural$
140-4-s095-D007 ∞
140-4-s095-D008 $\natural|\sharp$
140-4-s095-D009 $\mathcal{G}$

[scan 96]

three, compatible with one and the same homomorphism

$$\mathcal{M}_{\dagger} \longrightarrow \dagger\gamma_{\rho,\sigma} \overset{\text{def}}{=} \underbrace{\underline{\gamma^*}_{\rho,\sigma}\ \underline{|\gamma^{\bullet}}_{\rho,\sigma}}_{\mathbb{O}^*_{12}}(\widehat{\mathbb{Z}})^{\Delta} = \dagger\widehat{\mathbb{Z}}^*$$

which is none other than the cyclotomic character χ. One† of these three,

$$\begin{array}{ccccc} \mathcal{M} & \longrightarrow & N^{\sim}_{\rho} & \twoheadrightarrow & \widehat{\mathbb{Z}}^*, \\ & \searrow \quad \nearrow & & & \\ & \Gamma^{\text{et}}_{\mathbb{Q}} & & & \end{array}$$

reduces here to the said character, and thus raises no interesting question. We have on the other hand
(218)

$$\begin{array}{ccc} \mathcal{M} & \longrightarrow & G_{\rho,\sigma} = \widehat{\mathbb{Z}}^* \cdot \mathfrak{S}^{+\wedge} \\ \downarrow & & \downarrow \\ \mathcal{M}^! & \longrightarrow & G_{\rho,\sigma}^{\natural|\sharp} = \widehat{\mathbb{Z}}^* \cdot M^{\wedge}, \end{array}$$

which will have to be examined more closely — as the cyclotomic character is (even on $\mathcal{M}^!$) surjective, we know in any case that these two homomorphisms are surjective. Finally, we have a homomorphism $\mathcal{M} \longrightarrow N_\sigma^{\natural|\sharp\dagger}$, which factors as
(219)

$$\Gamma_{\mathbb{Q}}^{\mathrm{et}} \longrightarrow N_\sigma^{\natural|\sharp\dagger} = \widehat{\mathbb{Z}}^* \cdot \widehat{\mathbb{Z}}$$

($\frac{1}{2}$ direct)

which will also have to be examined — in particular see whether it is surjective. Of course, I have a chance of doing so only by the "arithmetic route", by proving that the homomorphism induced on $\Gamma_{\mathbb{Q}}$ itself is an epimorphism. 95

Uncertainty retained from cancelled passages.
140-4-s096-D001 0
140-4-s096-D002 $\mathbb{O}|\Phi$
140-4-s096-D003 $\mathbb{O}_{12}^*$
140-4-s096-D004 the|of
140-4-s096-D005 ab
140-4-s096-D006 ab

[scan 97]
[p. 696]

I feel like putting these homomorphisms in relation with the "tower" of algebraic curves defined over $\mathbb{Q}$, indexed by $n \in \mathbb{N}^*|\mathbb{N}^\wedge$
(220)

$$\begin{array}{ll} X_n : & x_0^n + x_1^n + x_\infty^n = 0 \\ \cap & \\ \mathbb{P}^2_{\mathbb{Q}} & \end{array}$$

(Fermat curve of index n)

Let U_n be the open set of points where $x_0 x_1 x_\infty \neq 0$

On X_n the group scheme $\mu_n^{\{0,1,\infty\}}/\underbrace{\mu_n}_{\text{diag.}}$ operates ($\mu_n = {}_n\mathbb{G}_m$), $\dagger$ as does $\mathfrak{S}_3 = \mathfrak{S}_{\{0,1,\infty\}}$, hence also the semidirect product

$$(221) \qquad \underline{F}_n = \mathfrak{S}_3 \cdot \underbrace{\mu_n^{\{0,1,\infty\}}}_{\underline{F}_n^\circ}/\mu_n,$$

m1522

Marginal note. m1522 Notation F_n not very happy — it is possible that there will be need later for the letter F_n, F_∞ for the Fermat coverings U_n, U_∞ themselves… (compare corrections|rectifications page 719!)

we have

$$X_n/\underline{F}^{\circ}_n \simeq X_1 \simeq U_{0,3}, \tag{222}$$

the isomorphism compatible with the actions of $\mathfrak{S}_3$. More generally, we have, if $n = n'd$, a morphism

(223)

$$\begin{array}{ccc} X_{n=n'd} & \longrightarrow & X_{n'} \\ x_i & \longmapsto & x_i^d \end{array}$$

and in this way U_n becomes a principal covering (= torsor) over $U_{n'}$, with group $\mu_d^{\{0,1,\infty\}}/\mu_d$, More precisely still,

$$\begin{array}{ccc} \| & \xrightarrow{\quad\dagger\quad} & \| \end{array}$$

$\dagger$ 96

Uncertainty retained from cancelled passages.

140-4-s097-D001 we have

140-4-s097-D008 canonical

140-4-s097-D012 x_∞^d

[scan 98]

we have a canonical exact sequence

(224)

$$1 \longrightarrow \mu_d^{\{0,1,\infty\}} \longrightarrow \underset{\mathfrak{S}_{3\,S_0}\cdot\mu_n^{\{0,1,\infty\}}}{\underline{F}_n} \to \underset{\mathfrak{S}_{3\,S_0}\cdot\mu_{n'}^{\{0,1,\infty\}}}{\underline{F}_{n'}} \to 1$$

$$(x_0, x_1, x_\infty) \longmapsto (x_0, x_1, x_\infty)$$

$$(x_0, x_1, x_\infty) \longmapsto (x_0^d, x_1^d, x_\infty^d)$$

$$g\dagger \longmapsto g$$

$$\text{for } \begin{array}{l} x_0, x_1, x_\infty \in \Gamma\mu_n, \\ g \in \Gamma\mathfrak{S}_{3\,S_0} \end{array} \quad)$$

and the morphism|the isomorphism (223) is compatible with (224), and induces an isomorphism (compatible with actions of $F_{n'}$)

$$X_{n'} \simeq X_n/(\mu_d^{\{0,1,\infty\}}/\mu_d), \tag{225}$$

Moreover, the action of $\dagger$

$$\underline{F}^{\circ}_n = \mu_n^{\{0,1,\infty\}}/\mu_n$$

on U_n is free, and a fortiori that of $\mu_d^{\{0,1,\infty\}}/\mu_d$ is, so that U_n is indeed a torsor over $U_{n'}$ with group

$$\underline{F}^{\circ}_d = \mu_d^{\{0,1,\infty\}}/\mu_d$$

Passing to the limit, we find

$$(226)\qquad \begin{cases} X_\infty = \varprojlim_{n\in\mathbb{N}^*} X_n, \quad U_\infty = \varprojlim U_n \\ \underline{F}_\infty = \varprojlim \underline{F}_n = (\mathfrak{S}_3)_{S_0} \cdot \dagger \left(\widehat{\mathbb{Z}}^{\{0,1,\infty\}}/\widehat{\mathbb{Z}}\right) \otimes_{\widehat{\mathbb{Z}}} T_\bullet(\mathbb{G}_m)|T(\mathbb{G}_m) \end{cases}$$

97

Uncertainty retained from cancelled passages.

140-4-s098-D003 $\in \mathfrak{S}_3$

140-4-s098-D004 $\mu_n^{\{0,}$

140-4-s098-D008 $\widehat{\mathbb{Z}}^2$

[scan 99]

[p. 697]

with operation of F_∞ on U_∞

$$(227)\qquad F^\circ_\infty = \varprojlim_n F^\circ_n \simeq \widehat{\mathbb{Z}}^{\{0,1,\infty\}} \otimes_{\widehat{\mathbb{Z}}} T_\bullet | T_*(\mathbb{G}_m)^\dagger$$

operating freely on U_∞ — the quotient being $U_1 \simeq U_{0,3}$.

The pro-étale group $\underline{F}_\infty$ over $\mathbb{Q}$ corresponds profinite group on which $\Gamma_\mathbb{Q}$ operates, it is †

$$\longrightarrow$$

the group

$$(228)\qquad F^\circ_\infty \overset{\text{def}}{=} \underline{F}^\circ_\infty(\overline{\mathbb{Q}}) \simeq \dagger \cdot \overbrace{\left(\underbrace{\widehat{\mathbb{Z}}^{\dagger\{0,1,\infty\}}/\widehat{\mathbb{Z}}}_{\dagger}\right)}^{\widehat{M}|M^*} \otimes_{\widehat{\mathbb{Z}}} \underset{T_\bullet(\mathbb{G}_m)(\mathbb{C})}{\widetilde{\widehat{\mathbb{Z}}}}$$

on which $\Gamma_\mathbb{Q}$ operates † via (the † cyclotomic character operating as a factor †. We † make $\mathfrak{S}_3$ operate on the inverse image of this group over $U_{0,3\,\mathbb{Q}}$, in a way compatible with the action of $\mathfrak{S}_3$ on $U_{0,3\,\mathbb{Q}}$, † by permutation of the factors, and — this corresponds to an operation of † $\mathcal{M}_{0,3} = \Pi_1(U_{0,3\,\mathbb{Q}}, \mathfrak{S}_3; -\bar{j})$ † on F°_∞ — which is done via the canonical homomorphisms

$$(229)\qquad \mathcal{M} \longrightarrow \mathfrak{S}_3, \qquad \mathcal{M} \xrightarrow{\chi} \widehat{\mathbb{Z}}^*.$$

†

m1525

Marginal note. m1525 N.B. The action of $\mathcal{M}_{0,3}$ on $\widehat{M} \simeq \underline{M}(\widehat{\mathbb{Z}})$ is none other than that induced by

$$\mathcal{M} \to G_{\rho,\sigma}$$

, since $\mathcal{M}$ operates on the invariant subgroup $\widehat{M}$ of $G_{\rho,\sigma}$ — the action of ω being trivial, is factored into an action of $\dagger\mathcal{M}^{\text{et}}_{0,3}$.

$$(230)\qquad U_\infty(-\bar{j})$$

a torsor under F°_∞, on which $\mathcal{M}_{0,3} = \Pi_1(U_{0,3\,\mathbb{Q}}, \mathfrak{S}_3; -\bar{j})$ operates in a way compatible with its operations on F°_∞.

The datum of this profinite $(F^\circ_\infty, \mathcal{M}_{0,3})$-torsor is equivalent to that of the principal covering U_∞ of $\widehat{U}_{0,3\,\mathbb{Q}}$, $\underline{F}^\circ_\infty$, 98

Uncertainty retained from cancelled passages.

140-4-s099-D003 $\{0,1,\infty\}$
140-4-s099-D008 defined [illeg. 2] independent
140-4-s099-D011 $\mathfrak{S}_3$
140-4-s099-D012 $*$
140-4-s099-D013 F°_{∞}
140-4-s099-D014 via the trivial action
140-4-s099-D016 anti.
140-4-s099-D017 $\{0,1,\infty\}$
140-4-s099-D018 the profinite group [illeg. 1]
140-4-s099-D020 [illeg. 2] $\widehat{\mathbb{Z}}$ $U_{0,3}$ [illeg. 1]
140-4-s099-D021 $\mathcal{M}_{0,3}$
140-4-s099-D022 permits to the fiber [illeg. 4]
140-4-s099-D023 X_{∞} [illeg. 2]
140-4-s099-D025 $\mathcal{M}^{\mathrm{et}}_{0,3}$

[scan 100]

with the operations of $\mathfrak{S}_3$ on it compatible with the operations of $\mathfrak{S}_3$ on $\underline{F}^{\circ}_{\infty}$.
†
†
Let us consider on the other hand the exact sequence

$$(231)\qquad 1 \longrightarrow \underline{M} \to \underline{G}_{\rho,\sigma}/\underline{\mu}_G \,|\, \mu_G \xrightarrow{(\chi,\mathrm{can})} \mathbb{O}^*_{12} \times \underbrace{(\mathfrak{S}_3)_{S_0}}_{(H^+/\mu)_{S_0}\,|\,(H^+/\mu_{\mathfrak{S}})_{S_0}} \to 1$$

it gives, by passage to $(\ )(\widehat{\mathbb{Z}})^{\Delta}$, an exact sequence

$$(232)\qquad 1 \longrightarrow \widehat{M} \to G_{\rho,\sigma}/\mu_G \to \widehat{\mathbb{Z}}^* \times \mathfrak{S}_3 \longrightarrow 1,$$

which gives back the operation already considered of $\widehat{\mathbb{Z}}^* \times \mathfrak{S}_3$ on $\widehat{M}$,
It contains the exact sequence

$$(233)\qquad 1 \to \widehat{M} \to \overset{\mathfrak{S}^{+\wedge}/\mu_{\mathfrak{S}}}{\underset{\dagger}{\rightarrow}} \to \mathfrak{S}_3 \to 1$$

which † is a quotient of the homotopy exact sequence 99

Uncertainty retained from cancelled passages.

140-4-s100-D001 $\widehat{\mathbb{Z}}^{\{0,1,\infty\}}$ integral
140-4-s100-D003 Γ (which|(by
140-4-s100-D005 $G^{\wedge}_{\rho,\sigma}/\mu_G|\mathfrak{S}^{+\wedge}_{0,3}/\mu_{\mathfrak{S}}$
140-4-s100-D006 is

[scan 101]
[p. 698]

$$(234)\quad 1 \to \Pi_1(U_{0,3\,\overline{\mathbb{Q}}}, -\bar{\jmath}) \to \Pi_1(U_{0,3\,\overline{\mathbb{Q}}}, \mathfrak{S}_3; -\bar{\jmath}) \to \mathfrak{S}_3 \to 1$$

by the † commutator subgroup of the group $\Pi_1(U_{0,3\,\overline{\mathbb{Q}}}, -\bar{\jmath})$. This extension structure (233) of $\mathfrak{S}_3$ by $M^\wedge$ is not trivial, unless I am mistaken, although the induced extension of $\{1, \dot\rho, \dot\rho^2\} \simeq \mathbb{Z}/3\mathbb{Z}$ by $M^\wedge$, or of $\{1, \dot\sigma\}$ by $\widehat{M}$, is. A fortiori (232) is not trivial (although it has canonical splittings below $\widehat{\mathbb{Z}}$, above $\{1, \dot\rho, \dot\rho^2\}$ and above $\{1, \dot\sigma\}$). Thus (subject to checking these points presently...) the group $G_{\rho,\sigma}/\mu_G$ cannot be identified with a subgroup of the affine|linear group

yes, it is trivial, cf. below...

$$\boxed{\mathrm{Aff}_{\widehat{\mathbb{Z}}}(\widehat{M}) \simeq \mathrm{Aut}_{\widehat{\mathbb{Z}}}(\widehat{M}) \,.\, \widehat{M}} \tag{234}$$

corresponding to the subgroup $\widehat{\mathbb{Z}}^* \times \mathfrak{S}_3$ of $\mathrm{Aut}_{\widehat{\mathbb{Z}}}(\widehat{M})$.

† Let us consider the sub-group

$$\mathrm{Aff} \simeq (\widehat{\mathbb{Z}}^* \times \mathfrak{S}_3) \,.\, \widehat{M} \tag{235}$$

We have a diagram

$$\begin{array}{ccccccccc} (237) & 1 & — & M^\wedge & — & \mathrm{Aff} & \longrightarrow & \widehat{\mathbb{Z}}^* \times \mathfrak{S}_3 & — & 1 \\ & & & & & & \nearrow & & & \\ & & & & & \mathcal{M}_{0,3} & & & & \end{array}$$

with exact rows. The giving of the structure (230) is equivalent, up to isomorphism (nonunique), to the

100

Uncertainty retained from cancelled passages.

140-4-s101-D001 invariant

140-4-s101-D003 This|And

[scan 102]

giving of an $\widehat{M}$-conjugacy class of liftings of $\mathcal{M}_{0,3} \longrightarrow \mathfrak{S}_3 \times \widehat{\mathbb{Z}}^*$ to a

$$\mathcal{M}_{0,3} \longrightarrow \mathrm{Aff}\,. \tag{238}$$

m1530

Marginal note. m1530 N.B. This homomorphism induces on $\Pi^\wedge_{0,3}$ the canonical homomorphism

$$\Pi^\wedge_{0,3} \to (\Pi^\wedge_{0,3})_{\mathrm{ab}} \xrightarrow{\sim} \widehat{M}$$

an essential property of this homomorphism.

The choice of such a lifting depends on the choice of a point of $U_\infty(\overline{\mathbb{Q}})$ above the point $-\bar{\jmath}$ of $U_{0,3}(\overline{\mathbb{Q}})$ — the chg^t of this origin point is made by translation by an $x \in \widehat{M}$, and the lifting (238) is then changed by inner automorphism by this element x.

Thus the Fermat tower provides such a conjugacy class of liftings(238), knowledge of which is in principle equivalent to that of the Fermat tower up to isomorphism. We would like to put, if possible, these liftings in correspondence with the|the homomorphisms (217), and more particularly one of them (218), which provides a canonical homomorphism from $\mathcal{M}_{0,3}$, not into † Aff, but into † the analogous group $G_{\rho,\sigma}/\mu_G$ of (232), which lifts the same homomorphism $\mathcal{M}_{0,3} \longrightarrow \dagger\widehat{\mathbb{Z}}^* \times \mathfrak{S}_3$.

101

Uncertainty retained from cancelled passages.
140-4-s102-D001 [illeg. 1]
140-4-s102-D002 $G_{\rho,\sigma}/\mu_G$
140-4-s102-D003 $\mathbb{O}|\Phi$

[scan 103]
[p. 699]

$$(239)\quad 1 \to M^\wedge \longrightarrow G_{\rho,\sigma}/\mu_G \longrightarrow \widehat{\mathbb{Z}}^* \times \mathfrak{S}_3 \longrightarrow 1.$$
$$\wr\,|$$
$$\widehat{\mathbb{Z}}^* \,.\, \mathfrak{S}^{+\wedge}/\mu_{\mathfrak{S}} \qquad \mathcal{M}_{0,3} = \Pi_1(U_{0,3\,\mathbb{Q}}, \mathfrak{S}_3; -\bar{\jmath})$$

On $\mathfrak{S}^{+\wedge}_{0,3}, \simeq \Pi_1(U_{0,3\,\overline{\mathbb{Q}}}, \mathfrak{S}_3, -\bar{\jmath}) \subset \mathcal{M}_{0,3}$, this homomorphism induces the canonical homomorphism † from the quotient of $\mathfrak{S}^{+\wedge}_{0,3}$ obtained "by making $\Pi^\wedge_{0,3}$ abelian", to $G^{\natural|\sharp}_{\rho,\sigma}/\mu_G \simeq \mathfrak{S}^{+\wedge}/\mu_{\mathfrak{S}}$, an extension of $\mathfrak{S}_3 \subset \widehat{\mathbb{Z}}^* \times \mathfrak{S}_3$ by $M^\wedge$.

Let us take up again the homomorphism $\mathcal{M}_{0,3} \longrightarrow$ Aff (238), it fits into an exact sequence

$$(240)\quad \begin{array}{ccccccccc} 1 & \to & \Pi^\wedge_{0,3} & \to & \mathcal{M}_{0,3} & \to & \mathcal{N}_{\dagger 0,3} & \to & 1 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & \widehat{M} & \longrightarrow & \mathrm{Aff} & \to & \dagger\widehat{\mathbb{Z}}^* \times \mathfrak{S}_3 & \to & 1, \end{array} \qquad (\simeq \dagger\Gamma_{\mathbb{Q}} \times \mathfrak{S}_3)$$

where $\Gamma_{\mathbb{Q}} \times \mathfrak{S}_3 \longrightarrow \widehat{\mathbb{Z}}^* \times \mathfrak{S}_3$ is the homomorphism $\chi \times \mathrm{id}$ of (237), and where $\Pi^\wedge_{0,3} \longrightarrow \widehat{M}$ is the canonical homomorphism

$$(241)\quad \Pi^\wedge_{0,3} \to (\Pi^\wedge_{0,3})_{\mathrm{ab}} \simeq ((\Pi_{0,3})_{\mathrm{ab}})^\wedge \xrightarrow{\sim} \widehat{M}.$$

If we denote by † $\mathcal{M}_1$ the quotient of $\mathcal{M}_{0,3}$ by $[\Pi^\wedge_{0,3}, \Pi^\wedge_{0,3}]$, we find therefore 102

Uncertainty retained from cancelled passages.
140-4-s103-D001 quotient $\mathfrak{S}|G$
140-4-s103-D002 $\mathfrak{S}_3$
140-4-s103-D003 $\mathbb{Q}$
140-4-s103-D004 $\Gamma_{\mathbb{Q}}$
140-4-s103-D005 $\mathcal{M}_{0,2}|\mathcal{M}_{0,3}$

[scan 104]

a homomorphism of sequences that are exact

$$(242)\quad \begin{array}{ccccccccc} 1 & \to & M^\wedge & \to & \mathcal{M}_1 & \to & \mathcal{N}_{\dagger 0,3} & \to & 1 \\ & & \| & & \downarrow & & \downarrow & & \\ 1 & \to & M^\wedge & \to & \mathrm{Aff} & \to & \widehat{\mathbb{Z}}^* \times \mathfrak{S}_3 & \to & 1, \end{array}$$

(depending on the choice of a p^t of $U_\infty(\overline{\mathbb{Q}})$ above $-\bar{\jmath}$) which shows that the extension $\mathcal{M}_1$ of $\Gamma_{\mathbb{Q}}$ by $\widehat{M}$ is therefore with|: the inverse image of the trivial extension Aff of $\widehat{\mathbb{Z}}^* \times \mathfrak{S}_3$ by $M^\wedge$, hence is trivial. More precisely still, let us consider the quotient $\mathcal{M}_2$ of $\mathcal{M}_1$ by the kernel of $\mathcal{M}_1 \longrightarrow$ Aff, we find a diagram

$$\begin{array}{ccccccccc} 1 & \longrightarrow & M^\wedge & \to & \mathcal{M}_2 & \longrightarrow & N_2 & \longrightarrow & 1 \\ & & \| & & \| & & \| & & \\ (242\text{ bis}) \quad 1 & \longrightarrow & M^\wedge & \to & \mathrm{Aff} & \longrightarrow & \widehat{\mathbb{Z}}^* \times \mathfrak{S}_3 & \longrightarrow & 1 \end{array}$$

(where $N_2 \overset{\text{def}}{\simeq} \mathcal{M}_2/\operatorname{Im} M^\wedge$). We thus see that the extension of N_2 by $M^\wedge$ is trivial — we have an $M^\wedge$-conjugacy class of splittings of this extension — the splittings corresponding to the p^{ts} of † $U_\infty(\overline{\mathbb{Q}})$ above the p^t $-\bar{\jmath}$ of $U_{0,3}(\overline{\mathbb{Q}})$ 103

Uncertainty retained from cancelled passages.

140-4-s104-D002 $\mathbb{Q}$

140-4-s104-D004 $U_{0,2}|U_{0,3}$

[scan 105]

We would really like to show that this homomorphism $\mathcal{M}_{0,3} \longrightarrow \mathrm{Aff}$ is isomorphic to the homomorphism $\mathcal{M}_{0,3} \longrightarrow G_{\rho,\sigma}/\mu_G$ of (239), by means of a suitable isomorphism of extensions between Aff and $G_{\rho,\sigma}/\mu_G$. This would imply that this latter extension is split. The isomorphisms of this extension onto Aff correspond exactly to the splittings of this extension. [p. 700]

12°) Splittings of the extension $\underline{G}_{\rho,\sigma}/\mu_G$ of $\mathbb{O}_{12}^* \times \mathfrak{S}_3$ by $\underline{M}$.

Since we already have a lifting

$$\mathbb{O}_{12}^* \hookrightarrow \underline{G}_{\rho,\sigma}$$

(150|180, 151), I am going to look at the splittings of the extension (231) which extend this one — or what amounts to the same thing, the splittings of the induced extension on $\mathfrak{S}_3$: m1534

Marginal note. m1534 not quite — one would have to|has to restrict oneself to the splittings

$$(\mathfrak{S}_3)_{S_0} \to \underline{\mathfrak{S}}^+/\mu_{\mathfrak{S}}$$

which commute with $\underline{G}_{\rho,\sigma}(0) \simeq \mathbb{O}_{12}^*$

$$(243) \quad 1 \longrightarrow \underline{M} \to \underline{\mathfrak{S}}^+/\mu_{\mathfrak{S}} \to \underbrace{(\mathfrak{S}_3)_{S_0}}_{(H/\mu)_{S_0}} \to 1$$

m1535

Marginal note. m1535 We look at the † splittings over an arbitrary base S.

The splittings correspond † bijectively to the pairs 104

Uncertainty retained from cancelled passages.

140-4-s105-D001 compatible

140-4-s105-D002 is

[scan 106]

$$\begin{cases} (\rho', \sigma') = (\zeta\rho, \eta\rho) & \text{with } \zeta, \eta \in \Gamma\underline{M} \dagger \\ \text{satisfying} & \\ \rho'^3 = \sigma'^2 = 1, \qquad \sigma'(\rho') = \rho'^{-1}. & \end{cases} \tag{244}$$

We are going to interpret these conditions in terms of ζ, η. The condition $\rho'^3 = 1$ is written (since $\dot{\rho}|\rho^3 = 1$ in $(\mathfrak{S}_3)_{S_0}$)

$$\zeta\rho(\zeta)\rho^2(\zeta) = 1 \qquad \text{i.e. additively} \qquad (1 + \rho_{\underline{M}} + \rho^2_{\underline{M}})(\zeta) = 0 \tag{245}$$

this condition is always|always satisfied, for we have

$$1 + \rho_{\underline{M}} + \rho^2_{\underline{M}} = 0, \tag{245}$$

The condition $\sigma'^2 = 1$ is written

$$\eta\sigma(\eta) = 1 \qquad \text{i.e. additively} \qquad (1 + \sigma_{M|\underline{M}})(\eta) = 0 \tag{246}$$

and it means that we have

$$\eta = b\delta \qquad \begin{matrix}\text{with } b|\beta \in \Gamma\mathbb{G}_a. \\ \delta = \lambda_1 - \lambda_0\end{matrix} \tag{247}$$

We shall set

$$\zeta = a_0\lambda_0 + a_1\lambda_1 \qquad a_0, a_1 \in \Gamma\mathbb{G}_a, \tag{248}$$

and we are going to express in terms of a_0, a_1, b the condition $\sigma'(\rho') = \rho'^{-1}$.

We have

$$\sigma'(\rho') = \mathrm{int}(\eta)\sigma(\zeta\rho) = \mathrm{int}(\eta)\left[\sigma(\zeta)\underbrace{\sigma(\rho)}_{\rho^{-1}\lambda_1}\right]$$

the condition $\sigma'(\rho') = \underbrace{\rho'^{-1}}_{\rho^{-1}\zeta^{-1}}$ is written

$$\sigma(\zeta)\rho^{-1}\lambda_1 \overset{\dagger}{=} \eta^{-1}\rho^{-1}\zeta^{-1}\eta$$

i.e., multiplying on the left|left by ρ

$$\rho(\sigma(\zeta)) \,.\, \lambda_1 = \rho(\eta)^{-1}\zeta^{-1}\eta$$

or again, additively: 105

Uncertainty retained from cancelled passages.

140-4-s106-D001 $\widehat{M}$

140-4-s106-D002 $=$

[scan 107]
[p. 701]

$$\underbrace{\rho_M\sigma_M}_{\varepsilon_{1M}}(\zeta) + \lambda_1 = -\rho_M(\eta) - \zeta + \eta \qquad \text{or again}$$

$$\underbrace{(1 + (\varepsilon_1)_M)(\zeta)}_{(2a_1 - a_0)\lambda_1} + \lambda_1 = \underbrace{(1 - \rho_M)(\eta)}_{3b\lambda_1}, \tag{249}$$

or again, by an immediate calculation

$$2a_1 - a_0 + 1 = 3b \qquad \text{or} \qquad 2a_1 - a_0 - 3b = | \neq -1. \tag{250}$$

This shows that for $\zeta \in \Gamma\underline{M}$ given (i.e. for ρ' given), for there to exist an η (i.e. $\sigma'|b$) satisfying with ζ these conditions (244) — i.e. $\dagger$ such that $(\rho' = \zeta\rho, \sigma' = \eta\sigma)$ defines a splitting of (243) — it is necessary and sufficient that we have

$$2a_1 - a_0 + 1 \equiv 0 \qquad (3) \tag{251}$$

and then η is uniquely determined by (250), at least if 3 is a regular section of $\underline{\mathcal{O}}_S$ on the base S under consideration (which is the case if $S = \operatorname{Spec}\widehat{\mathbb{Z}}$).

Thus the splittings of (243) form a scheme Sc over S_0, which is given up to isomorphism by

$$\begin{aligned}\underline{\mathrm{Sc}} &\simeq \{(a_0, a_1, b) \in \Gamma\mathbb{E}^3 \mid 2a_1 - a_0 - 3b = -1\} \\ &\simeq \{(\zeta, \eta) \in \Gamma(M \times \underline{\mathbb{Z}\delta}) \mid (1+\varepsilon_M)(\zeta) - (1-\rho_M)(\eta) = -\lambda_1\}\end{aligned} \tag{252}$$

This scheme is naturally a 106

Uncertainty retained from cancelled passages.
140-4-s107-D001 of|and

[scan 108]

torsor under the module scheme

$$\left\{\begin{array}{c} \underline{\mathrm{Sc}}_0 \simeq \underline{K}, \quad \text{where } K \text{ is the kernel of} \\ \text{the exact sequence of } \mathbb{Z}\text{-modules} \\ 1 \longrightarrow K \longrightarrow \dagger\mathbb{Z}^3 \longrightarrow \mathbb{Z} \longrightarrow 0. \\ (\alpha_0, \alpha_1, \beta) \longmapsto 2\alpha_1 - \alpha_0 - 3\beta \end{array}\right. \tag{253}$$

$K \simeq \mathbb{Z}^2$

On the other hand we have an operation of $\underline{M}$ on $\underline{\mathrm{Sc}}$ by automorphisms that are inner|that are inner, which I am going to make explicit; To $\lambda \in \Gamma\underline{M}$ corresponds the operation

$$(\zeta, \eta) \longmapsto \lambda\zeta\rho(\lambda)^{-1}, \lambda\eta\sigma(\lambda)^{-1}$$

i.e., additively

$$(\zeta, \eta) \longmapsto (\zeta + (1-\rho_M)(\lambda), \eta + (1-\sigma_M)(\lambda)). \tag{254}$$

Thus, the operation in question comes from the $\underline{\mathrm{Sc}}_0$-torsor structure on $\underline{\mathrm{Sc}}$, and from a homomorphism of Groups|groups

$$\left\{\begin{array}{l} \underline{M} \longrightarrow \underline{\mathrm{Sc}}_0 \\ \lambda \longmapsto \big(\underbrace{(1-\rho_M)(\lambda)}_{(c_0+c_1)\lambda_0+(2c_1-c_0)\lambda_1}, \underbrace{(1-\sigma_M)(\lambda)}_{(c_1-c_0)\delta} \big) \quad \text{where } \lambda = c_0\lambda_0 + c_1\lambda_1 \end{array}\right. \tag{255}$$

which is in fact a homomorphism of M-dules, associated with the homomorphism of $\mathbb{Z}$-modules

$$\left\{\begin{array}{l} M \longrightarrow K \hookrightarrow \mathbb{Z}^3 \\ c_0\lambda_0 + c_1\lambda_1 \longmapsto (\alpha_0 = c_0 + c_1, \alpha_1 = 2c_1 - c_0, \beta = c_1 - c_0) \end{array}\right. \tag{256}$$

107

Uncertainty retained from cancelled passages.
140-4-s108-D001 $\mathbb{Z}|M$

[scan 109]

We see at once that this homomorphism is universally injective (since the composite $M \longrightarrow \mathbb{Z}^3$ [p. 702] is), hence an isomorphism for reasons of rank. We have thus found:

Proposition The extension $\underline{\mathfrak{S}^+/\mu_{\mathfrak{S}}}$ (243) of $(\mathfrak{S}_3)_{S_0}$ by $\underline{M}$ is split over S_0 — hence over every|every other base — the splittings over $S_0 = \operatorname{Spec}\mathbb{Z}$ correspond 1-1 to the splittings of $\mathfrak{S}^+/\mu_{\mathfrak{S}}$ as an extension of $\mathfrak{S}_3$ by M. Over an arbitrary base S, two splittings are conjugate by a unique section of $\underline{M}$ over S, so the scheme of splittings of (243|247) is † naturally a torsor under $\underline{M}$.

Let us note that it is immediate, over an arbitrary base S, that the splittings of (243) correspond one-to-one|one-to-one to the splittings of the extension of groups

$$1 \longrightarrow \underline{M}(S) \longrightarrow \underline{\mathfrak{S}^+/\mu_{\mathfrak{S}}}(S)^{\Delta} \longrightarrow \mathfrak{S}_3 \longrightarrow 1 \ , \tag{244}$$

which also identifies with the extension of 108

Uncertainty retained from cancelled passages.

140-4-s109-D003 a

[scan 110]

$\mathfrak{S}_3$ by $\underline{M}(S) = \Gamma(S, \underline{\mathcal{O}}_S) \otimes_{\mathbb{Z}} M$, deduced from the universal extension $\mathfrak{S}^+/\mu_{\mathfrak{S}}$ of $\mathfrak{S}_3$ by M, via the homomorphism

$$M \longrightarrow \underline{M}(S) = M|\widehat{M} \otimes_{\mathbb{Z}} \Gamma(S, \underline{\mathcal{O}}_S) \ .$$

Let us apply this to the case $S = \operatorname{Spec}(\widehat{\mathbb{Z}})$, we find the

Corollary The extension $\mathfrak{S}^{+\wedge}/\mu_{\mathfrak{S}}$ of $\mathfrak{S}_3$ by $M^\wedge$ is split, and its splittings form a torsor under $M^\wedge$ (by automorphisms that are inner|that are inner)

Let us now study the splittings of $\underline{G_{\rho,\sigma}/\mu_G}$ in its entirety i.e.|: by trying to determine, for a given splitting

$$(\rho' = \zeta\rho, \sigma' = \eta\sigma) \qquad \begin{cases} \zeta = a_0\lambda_0 + a_1\lambda_1 \\ \eta = b\delta = b(\lambda_1 - \lambda_0) \end{cases} \tag{257}$$

of $\underline{\mathfrak{S}^+/\mu_{\mathfrak{S}}}$ over $(\mathfrak{S}_3)_{S_0}$, the liftings

$$\mathbb{O}_{12}^* \longrightarrow \underline{G_{\rho,\sigma}/\mu_G}$$

compatible with the preceding splitting, i.e. which commute with ρ', σ'. For this 109

[scan 111]

I|I am going to determine the commutant of (ρ', σ') † in $\underline{G_{\rho,\sigma}/\mu_\rho|\mu_G}$, i.e. the commutant of the|of the partial section subgroup [p. 703]

$$(258) \qquad \underline{H}(\rho', \sigma') \subset \underline{G_{\rho,\sigma}/\mu_G}.$$

The image of this commutant in $(\mathfrak{S}_3)_{S_0} \simeq (\underline{G_{\rho,\sigma}/M})/\mathbb{O}_{12}^*$ is contained in the center of $(\mathfrak{S}_3)_{S_0}$, hence is trivial|trivial, whence we conclude

$$(259) \qquad \underline{\operatorname{Centr}}_{\underline{G_{\rho,\sigma}/\mu_G}}(\underline{H}(\rho', \sigma')) \subset \mathbb{O}_{12}^* \,.\, \underline{M} \simeq \underline{G}_{\rho,\sigma}^{\natural|\sharp} \dagger$$

$$\left\{ \lambda u_{\underline{\mu}} \;\middle|\; \begin{matrix} \lambda \overset{\|}{\in} \Gamma\underline{M} \\ \underline{\mu} \in \Gamma\mathbb{O}_{12}^* \end{matrix} \right\}$$

— where $u_{\underline{\mu}}$ is defined in (150,151). We are therefore going to determine the centralizer of $\underline{H}(\rho', \sigma')$ by expressing the condition for a $\lambda u_{\underline{\mu}}$ to lie in it, i.e. to commute with ρ', σ'.

We have:

$$(\lambda u_{\underline{\mu}})(\underbrace{\rho'}_{\breve{\zeta}\rho}) = \mathrm{int}(\lambda)\, u_{\underline{\mu}}(\zeta\rho) = \mathrm{int}(\lambda)\ \underbrace{u_{\underline{\mu}}(\zeta)}_{\zeta^\mu}\ \underbrace{u_{\underline{\mu}}(\rho)}_{\lambda_1^\nu\rho} = \mathrm{int}(\lambda)(\zeta^\mu\lambda_1^\nu\rho)$$
$$= \lambda\zeta^\mu\lambda_1^\nu\rho(\lambda^{-1})\rho$$

where $\nu = \nu(\underline{\mu})$, so the relation $\lambda u_{\underline{\mu}}(\rho') = \rho'$ can also be written

$$\lambda\zeta^\mu\lambda_1^\nu\rho(\lambda^{-1})\rho = \zeta\rho \qquad \text{i.e.} \qquad \lambda\zeta^\mu\lambda_1^\nu\rho(\lambda^{-1}) = \zeta,$$

or in additive notation

$$\lambda + \mu\zeta + \nu\lambda_1 - \rho(\lambda) = \zeta, \qquad \text{i.e.}$$

taking into account that $\mu - 1 = 2\nu$ 110

Uncertainty retained from cancelled passages.

140-4-s111-D001 of the|in

140-4-s111-D002 $/\mu_G$

[scan 112]

$$(260) \qquad (1-\rho)(\lambda) + 2\nu\zeta + \nu\lambda_1 = 0.$$

The relation $(\lambda u_{\underline{\mu}})(\sigma') = \sigma'$ becomes explicit, via

$$(\lambda u_{\underline{\mu}})(\sigma') = \mathrm{int}(\lambda) u_{\underline{\mu}}(\eta\sigma) = \mathrm{int}(\lambda) \begin{pmatrix} & \\ \underbrace{u_{\underline{\mu}}(\eta)}_{\eta^\mu} & \underbrace{u_{\underline{\mu}}(\sigma)}_{\substack{\omega^{\nu_2}\sigma=\sigma \\ \text{since } \omega=1 \\ \text{in } \underline{G}_{\rho,\sigma}/\mu_G}} \end{pmatrix} = \lambda\eta^\mu\sigma\lambda^{-1}$$
$$= \lambda\eta^\mu\sigma(\lambda^{-1})\sigma$$

so the relation $\lambda u_{\underline{\mu}}(\sigma') = \sigma'$ is written

$$\lambda\eta^\mu\sigma(\dagger\lambda^{-1})\sigma = \eta\sigma \qquad \text{i.e.} \qquad \lambda\eta^\mu\sigma(\lambda^{-1}) = \eta$$

or again, additively

$$\lambda + \mu\eta - \sigma(\lambda) = \eta$$
$$(261|264) \qquad (1-\sigma)\lambda + 2\nu\eta = 0.$$

In terms of a_0, a_1, b of (257), these equations are made explicit as conditions on

$$\lambda = x\lambda_0 + y\lambda_1$$

by the equations

$$(262|282) \qquad \left\{ \begin{array}{rcll} -x+y & = & -2\nu b & \text{(expresses (261))} \\ \left.\begin{array}{rcl} x+y & = & -2\nu a_0 \\ -x+2y & = & -\nu(2a_1+1) \end{array}\right\} & & & \text{(express (260))} \end{array} \right. ,$$

where the quantities a_0, a_1, b appearing on the right-hand sides (in|at addition to the multiplicative factor $\nu = \nu(\underline{\mu})$) satisfy (250)

$$(263) \qquad -a_0 + 2a_1 - 3b + 1 = 0.$$

The first and third equation (262|263) can be solved for x, y

$$(264) \qquad \left\{ \begin{array}{l} y = \nu| - \nu(-2a_1 + 2b - 1) \\ x = y + 2\nu b = \nu(-2a_1 + 4b - 1) \end{array} \right.$$

and the second equation (262|261) becomes, on 111

Uncertainty retained from cancelled passages.
140-4-s112-D001 ζ

[scan 113]
[p. 704]

substituting these values of x, y into it, a relation linking a_0, a_1, b, ν:

$$\nu(-4a_1 + 6b - 2) = -2\nu a_0 \qquad \text{i.e.}$$

$$2\nu(a_0 - 2a_1 + 3b - 1) = 0$$

which is a consequence of (263)! Thus we find:

Proposition For every splitting of the extension $\underline{\mathfrak{S}}^+/\mu_{\underline{\mathfrak{S}}}$ of $(\underline{\mathfrak{S}}_3)_{S_0}$ by $\underline{M}$ (over an arbitrary base), the commutant in $\underline{G}_{\rho,\sigma}/\underline{\mu}_G$ of the section subgroup is † a section subgroup of the projection the extension $\underline{G}^{\natural|\sharp}_{\rho,\sigma}$ of $\mathbb{O}^*_{12}$ by $\underline{M}$. [The lifting

$$(264) \qquad u_{\rho',\sigma'} : \qquad \mathbb{O}^*_{12} \longrightarrow \underline{G}^{\natural|\sharp}_{\rho,\sigma}$$

which is associated with ρ', σ' given by (257)

$$\rho' = \zeta\rho, \quad \sigma' = \eta\sigma \qquad \begin{matrix} \zeta = a_0\lambda_0 + a_1\lambda_1, \quad \eta = b\delta = b(\lambda_1 - \lambda_0) \\ a_0, a_1, b \in \Gamma\mathbb{G}_a \end{matrix} ,$$

is given by

$$(265) \qquad u_{\rho',\sigma'}(\underline{\mu}) = \lambda u_{\underline{\mu}}$$

where $u_{\underline{\mu}}$ is defined in (150,151), and where $\lambda = x\lambda_0 + y\lambda_1$, with $x, y \in \Gamma\mathbb{G}_a$ described by 264, where $\nu = \nu(\underline{\mu})$).

Corollary The canonical morphism

$$(266) \qquad \underline{\text{Scindages}}\left(\underline{G}_{\rho,\sigma}/\underline{\mu}_G \text{ †over } \mathbb{O}^*_{12} \times \underline{\mathfrak{S}}_3\right) \to \underline{\text{Scindages}}\left(\underline{\mathfrak{S}}^+ \text{ over } \underline{\mathfrak{S}}_3\right)$$

112

Uncertainty retained from cancelled passages.
140-4-s113-D002 again|a
140-4-s113-D004 ⟶ |over

[scan 114]

is an isomorphism of sheaves † $(\text{Sch})_{S_0}$, and hence (proposition page 702) the first is an $\underline{M}$ torsor via conjugation.

This statement is made explicit in a corresponding statement on the set-theoretic extensions, as in the Corollary to the preceding proposition. We find in particular, over $S = \text{Spec}(\widehat{\mathbb{Z}})$

Corollary The splittings of the extension $G_{\rho,\sigma}/\mu_G$ of $\widehat{\mathbb{Z}}^* \times \mathfrak{S}_3$ by $\widehat{M}$ form a torsor under $\widehat{M}$, isomorphic (by restriction) to the torsor of splittings of the extension $\mathfrak{S}^{+\wedge}/\mu_{\mathfrak{S}}$ of $\mathfrak{S}_3$ by $\widehat{M}$.

Variant of the preceding study for the extension $G_{\rho,\sigma}$ of $\mathbb{O}^*_{12} \times H^+_{S_0}$ by $\underline{M}$ (without division by $\underline{\mu}_G = \{1, \omega\}_{S_0}$). Let us first consider the question of the splittings of the subextension $\underline{\mathfrak{S}}^+$ of $H^+_{S_0}$ by $\underline{M}$ — they|they correspond to the pairs 113

Uncertainty retained from cancelled passages.
140-4-s114-D001 in

[scan 115]
[p. 705]

$$\begin{cases} \rho' = \zeta\rho,\ \ \sigma' = \eta\sigma \quad (\zeta, \eta \in \Gamma\underline{M}) \quad \text{such that we have} \\ \qquad\qquad \rho'^3 = \sigma'^2 = \omega, \qquad \sigma'(\rho') = \rho'^{-1}. \end{cases} \tag{267}$$

(We note|shall note that a splitting must transform ω, which operates trivially on $\underline{M}$ and is central in $\underline{H}^+$, into a central † section, now † see|here is why we have

$$\begin{cases} \underline{M} \cap \underline{\mathrm{Cent}}(\underline{\mathfrak{S}}^+) = \underline{M}^{\mathfrak{S}_3} = \dagger\,\{1\}|\{0\}, \quad \text{hence} \\ \qquad \underline{\mathrm{Cent}}(\underline{\mathfrak{S}}^+) \simeq \underline{\mu}_{\underline{\mathfrak{S}}} = \{1, \omega\}_{\underline{\mathfrak{S}}} \end{cases} \tag{268}$$

— which shows that a splitting must transform ω into ω, whence easily the characterization (267) of the ρ', σ' corresponding|corresponding to a splitting). We shall have from this

$$\begin{cases} \rho'^3 = (\zeta\rho)^3 = \zeta\rho(\zeta)\rho^2(\zeta)\,\underbrace{\rho^3}_{\omega} = \zeta\rho(\zeta)\rho^2(\zeta)\omega \\ \sigma'^2 = (\eta\sigma)^2 = \eta\sigma(\eta)\,\underbrace{\sigma^2}_{\omega} = \eta\sigma(\eta)\omega \end{cases}$$

thus the equations $\rho'^3 = \omega$ and $\sigma'^2 = \omega$ are equivalent respectively to

$$\begin{cases} \zeta\rho(\zeta)\rho^2(\zeta) = 1 \quad \text{i.e.} \quad (1 + \rho_M + \rho_M^2)(\zeta) = 0 \\ \qquad \eta\sigma(\eta) = 1 \qquad\quad \text{i.e.} \quad (1 + \sigma_M)(\eta) = 0 \end{cases}$$

i.e. this reduces to equations (245) (246), and we have seen that the condition on ζ is automatically satisfied, that on η amounts to:

$$(247) \qquad \eta = b\delta = b(\lambda_1 - \lambda_0), \quad \lambda|b \in \Gamma\mathbb{G}_a.$$

The condition $\sigma'(\rho') = \rho'^{-1}$ is made explicit on the other hand as on page 700′, we come back 114

Uncertainty retained from cancelled passages.
140-4-s115-D001 element
140-4-s115-D002 the inverse
140-4-s115-D003 {0}

[scan 116]

to equation (249), which is made explicit by (250). This shows that there exists a splitting even over $S_0 = \mathrm{Spec}\,\mathbb{Z}$ — which amounts to a splitting of the discrete group $\mathfrak{S}^+$ extension of $H^+ = \mathfrak{S}^+/M \simeq \mathfrak{S}_0/\Pi_0$ (by $M \simeq (\Pi_0)_{\mathrm{ab}}$) — and that the splittings in question form a torsor under $\underline{K}$, where $K \simeq \mathbb{Z}^2$ is defined in (253), and in fact † we have an isomorphism $M \simeq K$ (254), i.e. $\underline{M} \simeq \underline{K}$, corresponding to the operation of $\underline{M}$ on the scheme of splittings, by conjugation. Thus we find

Proposition (Over an arbitrary base S).

The extension $\underline{\mathfrak{S}}^+$ of $\underline{H}^+$ by $\underline{M}$ is split — two splittings are conjugate by a unique section of $\underline{M}$ — thus the map "passage to the quotient by $\mu_{\mathfrak{S}}$" defines a bijection from the set of splittings of $\underline{\mathfrak{S}}^+$ over $\underline{H}^+$, onto the set of splittings of $\underline{\mathfrak{S}}^+/\underline{\mu}_{\underline{\mathfrak{S}}}$ over $\underline{H}^+/\underline{\mu}_{\underline{H}^+} \simeq \underline{\mathfrak{S}}_3$. 115

Uncertainty retained from cancelled passages.
140-4-s116-D001 *K*

[scan 117]

Let us now study the centralizer in $\underline{G}_{\rho,\sigma}$ of a section subgroup over $\underline{H}^+$, say denoted $\underline{H}(\rho',\sigma')$. [p. 706] We must now have

$$\underline{\mathrm{Cent}}_{\underline{G}_{\rho,\sigma}}(\underline{H}(\rho',\sigma')) \subset \left\{ \lambda\omega^i u_{\underline{\mu}} \;\middle|\; \begin{array}{c} \mathbb{O}_{12}^* \cdot (\underline{M} \times \underbrace{\underline{\mu}_{\mathfrak{S}}}_{=\{1,\omega\}_{S_0}}) \\ \| \\ \lambda \in \Gamma \underline{M} \\ \underline{\mu} \in \Gamma \mathbb{O}_{12}^* \\ i \in \Gamma(\mathbb{Z}/2\mathbb{Z})_{S_0} \end{array} \right\} \tag{269}$$

Moreover $\lambda\omega^i u_{\underline{\mu}}$ centralizes $\underline{H}(\rho',\sigma')$ iff $\lambda u_{\underline{\mu}}$ centralizes. Commutation with ρ' is again made explicit by condition (260), the calculation on page 703 applying ne varietur. For the condition of commutation with σ', the calculation on page 703′ must be modified for we can no longer write $\omega^{\nu_2} = 1$, the condition is written

$$\omega^{\nu_2} \cdot \lambda\eta^{\mu}\sigma(\lambda^{-1})\sigma = \eta\sigma \quad \text{i.e.} \quad \omega^{\nu_2}\lambda\eta^{\mu}\sigma(\lambda^{-1}) = \eta$$

and is equivalent to the conjunction of the condition

$$\nu_2(\underline{\mu}) = 0 \quad \text{i.e.} \quad \underline{\mu} \in \Gamma\mathbb{O}_{12}^{*\prime\prime}, \tag{270}$$

and of relation (261). In the presence of $\nu_2(\underline{\mu}) = 0$, the calculations on pages 703′, 704 apply, finally we find:

Corollary For every splitting of $\underline{\mathfrak{S}}^+$ over $\underline{H}^+$, † let $\underline{T}$ be the centralizer † in $\underline{G}_{\rho,\sigma}$ of the section subgroup $\underline{H}(\rho',\sigma')$, 116

Uncertainty retained from cancelled passages.
140-4-s117-D002 [illeg. 1]
140-4-s117-D003 *T*

[scan 118]

†We then have an exact sequence

$$1 \longrightarrow \underline{\mu}_{\mathfrak{S}|G} \hookrightarrow \dagger \longrightarrow \underline{T} \longrightarrow \mathbb{O}_{12}^{*\prime\prime} \longrightarrow 1 \tag{271}$$

and for $\underline{T}^{\natural|\sharp} = \underline{T} \cap \underline{G}_{\rho,\sigma}^{\natural|\sharp}$, we have

$$\underline{T}^{\natural|\sharp} \simeq \mathbb{O}_{12}^{*\prime\prime}, \tag{272}$$

i.e. $\underline{T}^{\natural|\sharp}$ is a partial section of $\underline{G}_{\rho,\sigma}^{\natural|\sharp}$ over $\mathbb{O}_{12}^{*\prime\prime}$.

Corollary The extension $\underline{G}_{\rho,\sigma}$ of $\mathbb{O}_{12}^* \times \underline{H}^+$ by $\underline{M}$ is not split over any nonempty base S. The extension $\underline{G}''_{\rho,\sigma}$ of $\mathbb{O}_{12}^{*\prime\prime} \times \underline{H}^+$ by $\underline{M}$ is split over every base S, and two splittings are conjugate by a unique section of $\underline{M}$. The natural map gives a bijection of the set of splittings of $\underline{G}''_{\rho,\sigma}$ over $\mathbb{O}_{12}^{*\prime\prime} \times \underline{H}^+$, with the set of splittings of $\underline{\mathfrak{S}}^+$ over $\underline{H}^+$.

Corollary The extension $G_{\rho,\sigma}$ of $\widehat{\mathbb{Z}}^* \times H^+$ by $M^\wedge$ is not split. That of $G''_{\rho,\sigma}$ of $\widehat{\mathbb{Z}}^{*\prime\prime} \times H^+$ by $M^\wedge$ is split, and two splittings are conjugate by a unique element of $\widehat{M}$. The natural map defines a bijection of the set of splittings of $G''_{\rho,\sigma}$ over $\widehat{\mathbb{Z}}^* \times H^+$ with the set of splittings of † $\mathfrak{S}^\wedge$ over H^+. 117

Uncertainty retained from cancelled passages.
140-4-s118-D001 a section
140-4-s118-D002 G
140-4-s118-D003 G_ρ

[scan 119]

Let us make explicit a splitting of the extension $\mathfrak{S}^+$ of H^+ by M — corresponding to a splitting of $\underline{\mathfrak{S}}^+$ of $\underline{H}^+$ by $\underline{M}$ — , via a particular solution of the equation [p. 707]

$$a_0 - 2a_1 + 3b - 1 = 0 \tag{273}$$

taking

$$a_1 = b = 0 \qquad \text{hence} \qquad a_0 = 1$$

i.e.

$$\begin{cases} \zeta = \lambda_0, \ \eta = 0 \quad \text{(additive notation)} \\ \rho' = \lambda_0 \rho, \quad \sigma' = \sigma \end{cases} \tag{274}$$

the three formulas give a splitting of $\mathfrak{S}^+/\mu_{\mathfrak{S}}$ over $\mathfrak{S}_3 \simeq H^+/\mu_{H^+}$. Let us make explicit the corresponding liftings of $\mathbb{O}_{12}^*$ into $\underline{G}_{\rho,\sigma}/\mu_G$ resp. of $\mathbb{O}_{12}^{*\prime\prime}$ into $\underline{G}_{\rho,\sigma}$;,

$$\begin{cases} u_{\rho',\sigma'} : \mathbb{O}_{12}^* \longrightarrow \underline{G}_{\rho,\sigma}/\mu_G \quad \text{resp.} \quad \mathbb{O}_{12}^{*\prime\prime} \text{ — } \underline{G}_{\rho,\sigma} \\ \qquad \underline{\mu} \longmapsto \lambda_\infty^\nu u_{\underline{\mu}} \qquad (\nu = \nu(\underline{\mu})) \end{cases} \tag{275}$$

where $u_{\underline{\mu}}$ defined in (151). (N.B.§ (275) is contained in proposition page 704 (265) and (264), which give

$$u_{\rho',\sigma'} = \lambda u_{\underline{\mu}} \qquad \lambda = x\lambda_0 + y\lambda_1 \qquad x = y = -\nu$$
$$\text{hence} \qquad \lambda = -\nu(\lambda_0 + \lambda_1) = \nu\lambda_\infty.$$

We shall thus have, putting

$$u'_{\underline{\mu}} = u_{\rho',\sigma'}(\underline{\mu})$$

$$\left.\begin{cases} u'_{\underline{\mu}}(\rho') = \rho', \quad u'_{\underline{\mu}}(\sigma') = \sigma' \\ u'_{\underline{\mu}}(\lambda) = \mu\lambda \quad (\lambda \in \Gamma\underline{M}) \quad \text{where } (\mu = \chi(\underline{\mu})) \end{cases}\right\} \quad \begin{matrix} \text{(which suffices} \\ \text{to condition|characterize } u'_{\underline{\mu}}) \end{matrix} \tag{276}$$

118
[scan 120]

and we also deduce from this

$$\begin{cases} u'_{\underline{\mu}}(\rho) = \lambda_0^{-2\nu}\rho \\ u'_{\mu|\underline{\mu}}(\sigma) = \omega^{\nu_2}\sigma \qquad (= \sigma \text{ in } \underline{G}_{\rho,\sigma}/\underline{\mu}_G \ \ldots). \end{cases} \tag{277}$$

119
[scan 121]
[p. 708]

13°) Return to the Fermat tower.

Let us begin by bringing out a general topossic notion, which was a little muddled between the lines in 11°). Let Γ (here $\Gamma_{\mathbb{Q}} \times \mathfrak{S}_3\dagger = N_{0,3}$) be operating on a topos $\mathcal{X}$ (here $\dagger\mathcal{U}_{0,3\overline{\mathbb{Q}}} \simeq M'_{0,4\overline{\mathbb{Q}}}$) connected locally connected, $\dagger$ equipped with a "point" (geometric) x_0 (here: the point $-\bar{j}$ with values in $\overline{\mathbb{Q}}$), let m1553

Marginal note. m1553 Discrete context ($\mathcal{X}$ locally simply connected) or profinite, relatively[illeg. 1] reasonable…

$$\begin{cases} \Pi = \Pi_1(\mathcal{X}, x_0), \qquad \mathcal{M} = \Pi_1(\mathcal{X}, \Gamma; x_0) \\ M = \Pi_{\text{ab}} = \Pi/[\Pi,\Pi], \qquad \dagger\widetilde{\mathcal{M}} = \mathcal{M}/[\Pi,\Pi], \end{cases} \tag{278}$$

hence exact sequences and homomorphisms of exact sequences

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \Pi & \longrightarrow & \mathcal{M} & \longrightarrow & \Gamma & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & \text{epimorphism} & & \text{epimorphism} & & \sim\wr & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & M & \longrightarrow & \dagger\widetilde{\mathcal{M}} & \longrightarrow & \Gamma & \longrightarrow & 1 \end{array} \quad . \tag{279}$$

(Instead of taking $M = \Pi/(\Pi,\Pi)|[\Pi,\Pi] = \Pi_{ab}$, we would have|had been able to take a quotient of Π_{ab} invariant under the action of Γ, but no matter…). Let us consider "the universal abelian covering" $\mathcal{X}'_0$ of $\mathcal{X}$ at x_0, i.e. the quotient by $[\Pi,\Pi]$ of the universal covering of $\mathcal{X}$ at x_0 — it is thus a principal covering of $\mathcal{X}$, with group M. We 120

Uncertainty retained from cancelled passages.

140-4-s121-D001)|}

140-4-s121-D002 U

140-4-s121-D003 and connected

140-4-s121-D004 $\mathfrak{S}$

140-4-s121-D006 $\mathfrak{S}$

[scan 122]

proposes to examine the ways of lifting the action of Γ on $\mathcal{X}$ to an action on $\mathcal{X}'_0$, compatible with the action of Γ on the group of operators M †. More generally, if $\mathcal{X}'$ is an $M_{\mathcal{X}}$-torsor on $\mathcal{X}$ isomorphic to $\mathcal{X}'_0$, hence obtained †from $\mathcal{X}'_0$ by twisting with an M-torsor E ($\simeq$ | = fiber of $\mathcal{X}'_0$ at x_0), let us study the analogous liftings of the action of †Γ on $\mathcal{X}$ to an action of †Γ on $\mathcal{X}'$, compatible with the action of †Γ on $\mathcal{X}$.

† 121

Uncertainty retained from cancelled passages.

140-4-s122-D004 or what amounts

140-4-s122-D005 by a

140-4-s122-D007 $\mathfrak{S}$

140-4-s122-D008 $\mathfrak{S}$

140-4-s122-D009 $\mathfrak{S}$

140-4-s122-D010 while|and

[scan 123]

We thus want to make $\mathcal{X}'$, which is a covering of $\mathcal{X}$, given by its fiber E at x_0 and the action of $\Pi = \Pi_1(\mathcal{X}, x_0)$ on E (by translations, thanks to an M torsor structure on E) into a covering of $(\mathcal{X}', \dagger\Gamma)$ — i.e. extend the given action of Π on the|the fiber E, to an action of $\mathcal{M}$ on E — which moreover necessarily factors through $\dagger\widetilde{\mathcal{M}}$. I say that this action necessarily goes by way of the affine group of the torsor E — to be precise, Γ operates on M, let γ_M be the operation defined by $\gamma \in \Gamma$, and let $g \in \dagger\widetilde{\mathcal{M}}$ be over $\gamma \in \Gamma$, we will thus have, for $x \in E$, $\lambda \in M$ [p. 709]

$$\begin{array}{ccccc} g & (\lambda & \cdot & x &) = \gamma_M(\lambda)\,.\,gx \\ \cap & \cap & & \cap & \\ \dagger\widetilde{\mathcal{M}} & M & & E & \end{array} \tag{280}$$

which is trivial by calculation. †

$$g(\lambda.x) = (g\lambda)x = (g\lambda g^{-1}g)x = \underbrace{(g\lambda g^{-1})}_{\gamma_M(\lambda)}.(gx)$$

122

Uncertainty retained from cancelled passages.

140-4-s123-D002 $\mathfrak{S}$
140-4-s123-D003 $\mathfrak{S}$
140-4-s123-D004 $\mathfrak{S}$
140-4-s123-D005 $\mathfrak{S}$
140-4-s123-D006 for|is

[scan 124]

Thus if we introduce the group $\mathrm{Aff}(E)$, we will have a homomorphism of exact sequences

$$\begin{array}{ccccccccc} 1 & \longrightarrow & M & \longrightarrow & \dagger\widetilde{\mathcal{M}} & \longrightarrow & \Gamma & \longrightarrow & 1 \\ & & \Big\Vert{\scriptstyle \dagger \mathrm{id}_M} & & \Big\downarrow{\scriptstyle \Theta_{\dagger\widetilde{\mathcal{M}}}} & & \Big\downarrow{\scriptstyle \Theta_\Gamma} & & \\ 1 & \longrightarrow & M & \longrightarrow & \mathrm{Aff}(E) & \longrightarrow & \mathrm{Aut}(M) & \longrightarrow & 1 \end{array} \tag{281}$$

where the outer vertical arrows id_M and Θ_Γ are known, and where the middle arrow is to be determined. We can say that the category of the $\mathcal{X}'$ sought|sought (with the operation of Γ on them satisfying the desired conditions) is equivalent to that of pairs $(E, \Theta_{\dagger\widetilde{\mathcal{M}}})$, where E is an M-torsor, and $\Theta_{\dagger\widetilde{\mathcal{M}}}$ a homomorphism of groups $\dagger\widetilde{\mathcal{M}} \longrightarrow \mathrm{Aff}(E)$ † compatible with id_M and Θ_Γ i.e. which makes|makes (281) commutative. We can also say: once E is chosen, (and this choice is harmless up to nonunique isomorphism!!!|…) that this defines an extension $\mathrm{Aff}(E)$ of $\mathrm{Aut}(M)$ by M, whence thanks to $\Gamma \longrightarrow \mathrm{Aut}(M)$ an extension 123

Uncertainty retained from cancelled passages.

140-4-s124-D001 Θ
140-4-s124-D002 $\mathfrak{S}$
140-4-s124-D003 $\mathfrak{S}$
140-4-s124-D004 $\mathfrak{S}$
140-4-s124-D005 $\mathfrak{S}$
140-4-s124-D006 $\mathfrak{S}$
140-4-s124-D007 making

[scan 125]

m1558

[p. 700]

Marginal note. m1558 the splittings of $\mathrm{Aff}(E)$ determined by the points of E – shift of the origin inner splittings…

by M, which is moreover split. (That said, the liftings of the action of Γ on it correspond to the isomorphisms of this extension of Γ by M, with the extension $\mathfrak{S}$ itself. Thus it exists iff the extension of $\mathfrak{S}$ of Γ by M is split, † i.e. the expression

$$c^2(\mathfrak{S}) \in H^2(\Gamma, M) \quad \text{is zero.} \tag{282}$$

If such a lifting exists i.e. such an isomorphism, these form a torsor under|over the group of automorphisms of the extension (trivial) $\mathfrak{S}$ of Γ by M which is identified with the group

$$(\qquad Z^1(\Gamma, M)$$

of crossed homomorphisms from Γ to M. The M-conjugacy classes of such † isomorphisms therefore correspond to the elements of

$$H^1(\Gamma, M).$$

Another way of saying it, by taking $E =$ trivial M-torsor $= M$, so $\mathrm{Aff}(E)$ split, is that the chosen liftings correspond to the splittings of the extension $\mathfrak{S}$ of Γ by M, and the M-conjugacy classes of the liftings of the action of Γ on $\mathcal{X}'_0$ 124

Uncertainty retained from cancelled passages.
140-4-s125-D001 which
140-4-s125-D002 (or

[scan 126]

correspond to the M-conjugacy classes of the splittings of the said extension $\mathfrak{S}$ of Γ by M. This being well set out, let us return to the situation of

$$\left(\underbrace{U_{0,3\overline{\mathbb{Q}}}}_{\mathcal{X}}, \quad \Gamma = \underbrace{\Gamma_{\mathbb{Q}} \times \mathfrak{S}_3}_{N_{0,3}} \right),$$

the Γ-coverings of $\mathcal{X}$ correspond to the $\mathfrak{S}_3$-coverings of $U_{0,3\mathbb{Q}}$, i.e. to the coverings of $(U_{0,3\mathbb{Q}}, \mathfrak{S}_3)$.
m1560

Marginal note. m1560 N.B. The group we have just denoted M becomes here $\widehat{M}$, $\mathcal{M}$ becomes $\mathcal{M}_{0,3}$, π becomes $\widehat{\Pi}_{0,3}$.

Using the general constructions of § 49 VI, from the quotient

†

$$\mathfrak{S}^{+\wedge}/\mu_{\mathfrak{S}} \simeq \mathcal{M}^{+}_{0,3}/[\widehat{\Pi}_{0,3}, \widehat{\Pi}_{0,3}] \subset \widetilde{\mathcal{M}}$$

we could denote it $\widetilde{\mathcal{M}}'$!

(with the notations of the preceding pages), extension of $\mathfrak{S}_3 \subset \Gamma (= \Gamma_{\mathbb{Q}} \times \mathfrak{S}_3)$ by $M^{\wedge}$, we deduced a profinite group, † which was denoted $G_{\rho,\sigma}/\mu_G$ or $G_{\rho,\widetilde{\sigma}}$ in the preceding sections and appeared there as a $\frac{1}{2}$-direct product

$$G_{\rho,\widetilde{\sigma}} = {}^{\dagger}\widehat{\mathbb{Z}}^* \,.\, \widetilde{\mathcal{M}}, \qquad \text{extension of } \widehat{\mathbb{Z}}^* \times \mathfrak{S}_3 \quad \text{by } M^{\wedge}$$

and a canonical homomorphism

$$\widetilde{\mathcal{M}} = \mathcal{M}_{0,3}/(\widehat{\Pi}_{0,3}, \widehat{\Pi}_{0,3}) \longrightarrow G_{\rho,\widetilde{\sigma}}$$

fitting into a homomorphism of exact sequences 125

Uncertainty retained from cancelled passages.
140-4-s126-D002 $G_{\rho,\sigma}$ $\mathfrak{S}$ μ_G $\widehat{\Pi}_{0,3}$ $\widehat{\Pi}_{0,3}$
140-4-s126-D005 and
140-4-s126-D006 $\mathbb{O}|\Phi$

[scan 127]
[p. 711]

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \widehat{M} & \longrightarrow & \widetilde{\mathcal{M}} & \longrightarrow & \underbrace{\Gamma}_{N_{0,3}=\Gamma_{\mathbb{Q}}\times\mathfrak{S}_3} & \longrightarrow & 1 \\ & & \| & & \downarrow & & \downarrow{\scriptstyle \chi\times \mathrm{id}_{\mathfrak{S}_3}} & & \\ 1 & \longrightarrow & \widehat{M} & \longrightarrow & G_{\rho,\widetilde{\sigma}} & \longrightarrow & \widehat{\mathbb{Z}}^* \times \mathfrak{S}_3 & \longrightarrow & 1. \end{array} \tag{282}$$

Moreover against all|all expectation we found that the extension $G_{\rho,\widetilde{\sigma}}$ of $\widehat{\mathbb{Z}}^* \times \mathfrak{S}_3$ by $\widehat{M}$ was split, whence it follows that the extension $\widetilde{\mathcal{M}}$. of Γ by $\widehat{M}$ is so too († this last fact is also a consequence of the existence of the Fermat tower, by applying to it the general reflections of the preceding pages). Better still, we have seen that there is a unique $\widehat{M}$-conjugacy class of splittings of the extension $G_{\rho,\widetilde{\sigma}}$, which induces a canonical (but not unique!) $\widehat{M}$-conjugacy class of † splittings of $\widetilde{\mathcal{M}}$ over Γ. This therefore defines a canonical $\widehat{M}$-conjugacy class of liftings of the action of Γ on $\mathcal{X} = U_{0,3\overline{\mathbb{Q}}}$ to $\mathcal{X}'_0$ – or what amounts to the same thing, a torsor on $(U_{0,3\mathbb{Q}}, \mathfrak{S}_3)$, with group a profinite group

m1563

Marginal note. m1563 N.B. $H^1(\Gamma, \widehat{M})$ is a very big group! – But see below, cf. Theorem page 716 126

Uncertainty retained from cancelled passages.

140-4-s127-D002 precedes|preceded

140-4-s127-D003 $\widehat{M}$

[scan 128]

commutative over $(U_{0,3\mathbb{Q}}, \mathfrak{S}_3)$ which over † $U_{0,3\overline{\mathbb{Q}}}$ is isomorphic to the covering $\mathcal{X}'_0$ considered above.

Let us return for a moment to this general case at the beginning. Note that the datum of a diagram (281) defines a quotient $\widetilde{\widetilde{\mathcal{M}}}$ of $\widetilde{\mathcal{M}}$ – namely the coimage of $\Theta_{\widetilde{\mathcal{M}}}$ – † such that $M \hookrightarrow \widetilde{\widetilde{\mathcal{M}}}$ is injective, and thus giving rise to an exact sequence

$$\begin{array}{ccccccccc} 1 & \longrightarrow & M & \longrightarrow & \widetilde{\widetilde{\mathcal{M}}} & \longrightarrow & \tilde{\Gamma} & \longrightarrow & 1, \\ & & & & & & \| \text{déf} & & \\ & & & & & & \widetilde{\widetilde{\mathcal{M}}}/\operatorname{Im} M & & \end{array} \tag{283}$$

then|where we have

$$\tilde{\Gamma} \hookrightarrow \operatorname{Aut}(M), \tag{284}$$

$\tilde{\Gamma}$ identifies with the image of Γ in $\operatorname{Aut}(M)$ under the action of Γ on M coming from the extension structure $\widetilde{\mathcal{M}}$ of Γ by M.

This quotient $\widetilde{\widetilde{\mathcal{M}}}$ of $\mathcal{M}$, giving rise to an extension (283), is not arbitrary – apart from the fact that M maps into it injectively, and that $\tilde{\Gamma} = \widetilde{\widetilde{\mathcal{M}}}/\operatorname{Im} M$ 127

Uncertainty retained from cancelled passages.

140-4-s128-D001 $\overline{\mathbb{Q}}$ [illeg. 2]

140-4-s128-D003 giving

[scan 129]

operates † faithfully on M : in addition, the extension (283) is trivial, and the fact that it comes from a $\Theta_{\widetilde{\mathcal{M}}}$ implies an additional structure on this extension, namely the datum of an M-conjugacy class of splittings. [p. 712]

More precisely, we will have a canonical map

$$E \longrightarrow \mathrm{Scind}(\widetilde{\widetilde{\mathcal{M}}} \text{ as|conceived extension of } \tilde{\Gamma} \text{ by } M) \tag{285}$$

compatible with the operations of M on E (by translation) and on Scind (by composition with|or inner automorphisms|automorphisms). The M-conjugacy class of splittings under consideration is none other than the image of E.

For x and $y \in E$ to have the same image in Scind, it is necessary and sufficient that the element $u = y - x \in M$ be in $M^\Gamma = M^{\tilde{\Gamma}}$, which shows that

$$\text{the homomorphism } E \longrightarrow \mathrm{Scind} \text{ is injective} \tag{286}$$

$$\text{iff} \quad M^\Gamma = \{0\} \quad .$$

In the case where $M^\Gamma = \{0\}$, we thus see that E identifies exactly with an M-conjugacy class of the splittings of $\widetilde{\widetilde{\mathcal{M}}}$. 128

Uncertainty retained from cancelled passages.
140-4-s129-D001 injective|injectively

[scan 130]

Let us sum up these considerations, in a

Proposition Let $\mathcal{X}$ be a † connected locally connected topos equipped with a geometric point x_0, we place ourselves in the discrete context ($\mathcal{X}$ locally|locally simply|simply connected) or profinite (covering means profinite covering). m1567

Marginal note. m1567



We have a group Γ of automorphisms of $\mathcal{X}$ (Γ profinite in the profinite case — to do things properly we should spell out the reasonable hypotheses to be made to take account of the profinite topology of Γ…). We put

$$\Pi = \Pi_1(\mathcal{X}, x_0), \quad \mathcal{M} = \Pi_1(\mathcal{X}, \Gamma; x_0) \qquad \text{hence}$$

$$(*) \quad 1 \longrightarrow \Pi \longrightarrow \mathcal{M} \longrightarrow \Gamma \longrightarrow 1$$

we give ourselves a quotient M of Π, † by a normal subgroup in $\mathcal{M}$, or what amounts to the same thing a quotient of $\Pi_{\mathrm{ab}} = \Pi/[\Pi, \Pi]$ stable under the action of Γ, hence a quotient extension of $(*)$,

$$(287) \qquad 1 \longrightarrow M \longrightarrow \widetilde{\mathcal{M}} \longrightarrow \Gamma \longrightarrow 1$$

where M is commutative|commutative so Γ operates on M.

We are interested in the category of coverings † $\mathcal{X}'$ of $(\mathcal{X}, \Gamma)$, satisfying the conditions 129

Uncertainty retained from cancelled passages.

140-4-s130-D001 c

140-4-s130-D002 a

140-4-s130-D005 $M_{\mathcal{X}}$ over

[scan 131]

Following : after forgetting the operations of Γ, we find a connected Galois covering (and even abelian) of $\mathcal{X}$, corresponding to the quotient group M of $\Pi = \Pi_1(\mathcal{X}, x_0)$ — so that $\mathcal{X}'$ becomes in a natural way a torsor under $M_{\mathcal{X}}$. m1569 [p. 713]

Marginal note. m1569



The functor

$$\mathcal{X}' \longmapsto \mathcal{X}'(x_0) \qquad \text{(fiber at } x_0\text{)}$$

from the category of these $\mathcal{X}'$ to the category of sets, in fact goes to the sets E equipped with the following structures

a) E is an M-torsor, and as such gives rise to an extension

$$1 \longrightarrow M \longrightarrow \mathrm{Aff}(E) \longrightarrow \mathrm{Aut}(M) \longrightarrow 1$$

hence an induced subextension

$$1 \longrightarrow M \longrightarrow \mathrm{Aff}_\Gamma(E) \longrightarrow \tilde{\Gamma} \longrightarrow 1 \tag{288}$$

where

$$\tilde{\Gamma} = \mathrm{Im}(\Gamma \longrightarrow \mathrm{Aut}(M)).$$

b) We have a homomorphism of extensions

$$\begin{array}{ccccccccc} 1 & \longrightarrow & M & \longrightarrow & \widetilde{\mathcal{M}} & \longrightarrow & \Gamma & \longrightarrow & 1 \\ & & \Big\Vert{\scriptstyle \mathrm{id}} & & \Big\downarrow{\scriptstyle \Theta_{\widetilde{\mathcal{M}}}} & & \Big\downarrow{\scriptstyle \mathrm{can}} & & \\ 1 & \longrightarrow & M & \longrightarrow & \mathrm{Aff}_\Gamma(E) & \longrightarrow & \tilde{\Gamma} & \longrightarrow & 1 \end{array} \tag{289}$$

The category of the $\mathcal{X}'$ (with isomorphisms…) is equivalent to that of † pairs $(E, \Theta_{\widetilde{\mathcal{M}}})$, where E is an M-torsor, and $\Theta_{\widetilde{\mathcal{M}}}$ a homomorphism of extensions (289). 130

Uncertainty retained from cancelled passages.

140-4-s131-D001 *M*

[scan 132]

The automorphisms of an object $\mathcal{X}'$ correspond to "translations" by elements of $M^\Gamma = \mathrm{Cent}(\widetilde{\mathcal{M}})$, (in the terms of $E, \Theta_{\widetilde{\mathcal{M}}}$) to translations of E by † elements of M^Γ, †(in so far as elements of $\mathrm{Aff}_\Gamma(E)$ they are central, and by conjugating by these elements we find the same homomorphism $\Theta_{\widetilde{\mathcal{M}}}$…). m1571

Marginal note. m1571



The isomorphism classes of such elements †correspond one-to-one to the M-conjugacy classes of splittings of the extension (287) of Γ by $\widetilde{\mathcal{M}}$. In precise terms, to an $(E, \Theta_{\widetilde{\mathcal{M}}})$ is associated canonical|canonically a † homomorphism of M-sets

$$E \longrightarrow \text{Scind}(\widetilde{\mathcal{M}} \text{ extension of } \Gamma \text{ by } M) \tag{290}$$

whose image is the conjugacy class describing the isomorphism class under consideration. †The homomorphism (290) is|of factored through $M^\Gamma\backslash E$ as a monomorphism

$$M^\Gamma\backslash E \hookrightarrow \text{Scind}(\widetilde{\mathcal{M}} \text{ extension of } \Gamma \text{ by } M)\ , \tag{291}$$

and if $M^\Gamma = 0$, we see that E is canonically reconstructed from the class of 131

Uncertainty retained from cancelled passages.

140-4-s132-D001 elements|elements

140-4-s132-D002 so far

140-4-s132-D003 identify

140-4-s132-D004 M

140-4-s132-D005 The image

[scan 133]

conjugacy under consideration. The (affine) operation of $\widetilde{\mathcal{M}}$ on E (defined by $\Theta_{\widetilde{\mathcal{M}}}$) is none other than the operation (induced by the) conjugation on the set of splittings (which of course does not take one out of an M-conjugacy class, even when conjugating by arbitrary elements of $\widetilde{\mathcal{M}}$). m1573 [p. 714]

Marginal note. m1573



Corollary Suppose we have

$$\underbrace{H^0(\tilde{\Gamma}, M)}_{M^\Gamma} = H^1(\tilde{\Gamma}, M) = 0 \tag{292}$$

i.e. that † two splittings of a trivial extension of $\tilde{\Gamma}$ by M are conjugate by a unique element of M. This implies therefore (because of $M^\Gamma = \{0\}|0$) that the category of the $\mathcal{X}'$, or again that of the pairs $(E, \Theta_{\widetilde{\mathcal{M}}})$, is rigid, i.e. an object is known up to an isomorphism which is unique when we know its isomorphism class. This being so, the isomorphism classes in question (i.e. the classes of † conjugacy of the splittings of the extension $\widetilde{\mathcal{M}}$ of Γ by M) correspond one-to-one to the quotient groups $\widetilde{\widetilde{\mathcal{M}}}$ m1574

Marginal note. m1574



132

Uncertainty retained from cancelled passages.
140-4-s133-D002 all|all
140-4-s133-D005 conjugation|conjugation

[scan 134]

m1576

Marginal note. m1576



of $\mathcal{M}$ satisfying the following conditions
a) $M \longrightarrow \widetilde{\widetilde{\mathcal{M}}}$ (composite $M \longrightarrow \widetilde{\mathcal{M}} \longrightarrow \widetilde{\widetilde{\mathcal{M}}}$) is injective,
†
hence extension
(293)

$$1 \text{ — } M \text{ — } \widetilde{\widetilde{\mathcal{M}}} \text{ — } \dagger \underset{\overset{\text{dfn}}{=} \widetilde{\widetilde{\mathcal{M}}}/M}{\Gamma'} \text{ — } 1$$

b) † $\Gamma' \overset{\text{dfn}}{=} \widetilde{\widetilde{\mathcal{M}}}/M$ operates faithfully on M,
i.e. we have

$$\Gamma' \simeq \tilde{\Gamma} = \mathrm{Im}(\Gamma \longrightarrow \mathrm{Aut}(M)).$$

c) The extension $\widetilde{\widetilde{\mathcal{M}}}$ of $\Gamma' \simeq \tilde{\Gamma}$ by M is trivial (i.e. its class in $H^2(\tilde{\Gamma}, M)$ is zero).

For such a quotient $\widetilde{\widetilde{\mathcal{M}}}$ of $\widetilde{\mathcal{M}}$, we recover E as the set of all the splittings of the extension $\widetilde{\widetilde{\mathcal{M}}}$ of $\tilde{\Gamma}$ by M, † on which we have $\widetilde{\mathcal{M}}$ operate via $\widetilde{\widetilde{\mathcal{M}}}$ by conjugation – the restriction of this operation to $M \subset \widetilde{\mathcal{M}}$ makes E an E-torsor (by hypothesis (292)).

Let us return again to the particular case that interests us. In the preceding number^{o} (12°) we verified the † relations (292), showing at the same time that this famous extension $G_{\rho,\tilde{\sigma}}$ of $\tilde{\Gamma} \simeq \widehat{\mathbb{Z}}^* \times \mathfrak{S}_3$ by $M^\wedge$ (denoted M in the general considerations) 133

Uncertainty retained from cancelled passages.
140-4-s134-D001 Γ' $M|\Gamma$
140-4-s134-D002 $\widetilde{\widetilde{\mathcal{M}}}/M$
140-4-s134-D003 $\widetilde{\widetilde{\mathcal{M}}}/M$
140-4-s134-D004 by|which
140-4-s134-D005 tr

[scan 135]

was trivial, and that two splittings were conjugate by an element of M and only one (it is this second assertion of existence and uniqueness of conjugation that is the geometric content of (292)). Moreover, we have † indeed constructed, in number° 11, a surjective homomorphism [p. 715]

$$\mathcal{M} = \mathcal{M}_{0,3} \longrightarrow G_{\rho,\tilde{\sigma}}$$

thus making $G_{\rho,\tilde{\sigma}}$ a quotient of $\mathcal{M}$ and even of $\widetilde{\mathcal{M}}$, which has the properties a) b) c) of the preceding corollary. This therefore defines for us, up to unique isomorphism, an object of the category of $\mathcal{X}'$, or what amounts to the same, a pro-étale covering of $(U_{0,3\,\mathbb{Q}}, \mathfrak{S}_3) \simeq M'_{1,1\,\mathbb{Q}}$, which after forgetting the operations of $\mathfrak{S}_3$ and extension of the base from $\mathbb{Q}$ to $\overline{\mathbb{Q}}$, becomes the universal abelian covering of $U_{0,3\,\overline{\mathbb{Q}}}$.

But on the other hand the tower of Fermat coverings provides us with quite exactly such an object – and if 134

Uncertainty retained from cancelled passages.
140-4-s135-D001 thus

[scan 136]

I had not had this Fermat tower at my disposal, I certainly would not have suspected that † subextension $\mathfrak{S}^{+\wedge}/\mu_{\mathfrak{S}} = \widetilde{\mathfrak{S}}^{+\wedge}$ † of $\widetilde{H}^{+} \simeq \mathfrak{S}_3$ by $M^{\wedge}$ was split!
m1579

Marginal note. m1579



The conjecture that presents itself at once is that these two pro-étale coverings are isomorphic – the isomorphism between the two is then unique.

† This is the moment to determine the isomorphism classes of the classes of splittings of the extension $\widetilde{\mathcal{M}} \simeq \mathcal{M}_{0,3}/[\widehat{\Pi}_{0,3}, \widehat{\Pi}_{0,3}]$ of $\Gamma = \Gamma_{\mathbb{Q}} \times \mathfrak{S}_3$ by $M^{\wedge}$, we have by the exact sequence associated with the Hochschild spectral sequence

(294)

$$0 \to H^1(\Gamma_{\mathbb{Q}}, H^0(\dagger\mathfrak{S}_3, \widehat{M})) \to H^1(\Gamma, \widehat{M}) \to H^0(\Gamma_{\mathbb{Q}}, H^1(\mathfrak{S}_3, \widehat{M})) \to H^2(\Gamma_{\mathbb{Q}}, H^0(\mathfrak{S}_3, M))$$

Now we have seen that

$$(295) \qquad H^0(\mathfrak{S}_3, \widehat{M}) = 0 \qquad (\text{and } \overline{\text{m}}\ H^0(L_\rho | L_j, \widehat{M}|\overline{M}) = 0)$$

m1580

Marginal note. m1580 hence $H^1(\Gamma, \widehat{M}) \simeq H^0(\Gamma_{\mathbb{Q}}, H^1(\mathfrak{S}_3, \widehat{M}))$

† Let us compute on the other hand

$$H^1(\mathfrak{S}_3, \widehat{M}) = H^1(\mathfrak{S}_3, M) \otimes_{\mathbb{Z}} \widehat{\mathbb{Z}},$$

135

Uncertainty retained from cancelled passages.

140-4-s136-D001 $\widetilde{\widetilde{\mathcal{M}}}$ extension|even this extension
140-4-s136-D002 H
140-4-s136-D003 And
140-4-s136-D004 $\widetilde{\Gamma}$
140-4-s136-D005 And

[scan 137]

which returns us to computing [p. 716]

$$H^1(\mathfrak{S}_3, M) \quad ?$$

But the study made of the extension $\mathfrak{S}^+/\mu_{\mathfrak{S}}$ of $\mathfrak{S}_3$ by M – which has shown us that this extension was split, and that two splittings were conjugate by a unique element of M, gives us in fact

$$(296) \qquad \begin{cases} H^0(\mathfrak{S}_3, M) = H^1(\mathfrak{S}_3, M) = 0 \\ H^0(\mathfrak{S}_3, \widehat{M}|\overline{M}) = H^1(\mathfrak{S}_3, \widehat{M}|\overline{M}) = 0 \end{cases} \text{hence} \quad ,$$

hence finally, thanks to (295), the triumphant relation

$$(297) \qquad \boxed{H^1(\Gamma_{\mathbb{Q}} \times \mathfrak{S}_3, \widehat{M}) = 0} \qquad !$$

We thus find – once again against all expectations, I confess! – the

Theorem There exists a pro-étale covering m1582

Marginal note. m1582

† U_∞ of $(U_{0,3\,\mathbb{Q}}, \mathfrak{S}_3) = M'_{1,1\,\mathbb{Q}}$, having the following property: The inverse image $\overline{U}_\infty$ of U_∞ over $U_{0,3\,\overline{\mathbb{Q}}}$ (†forgetting operations of $\mathfrak{S}_3$) is "the" universal abelian covering of $U_{0,3\,\overline{\mathbb{Q}}}$. For two such cover- 136

Uncertainty retained from cancelled passages.

140-4-s137-D002 $X|Z$
140-4-s137-D003 without

[scan 138]

ings pro-étale over $(U_{0,3\,\mathbb{Q}}, \mathfrak{S}_3) = M'_{1,1\,\mathbb{Q}}$, there exists a unique isomorphism from one onto the other.

We have obtained existence in two ways: by the Fermat tower, and by the general theory of § 49 III, applied to the present case – which gives us a U_∞ and a U'_∞, each described up to unique isomorphism! The uniqueness theorem follows from the preceding little calculation – I was afraid of the "big" †

$$H^1(\Gamma_{\mathbb{Q}}, \widehat{M}) \simeq M \otimes_{\mathbb{Z}} \underbrace{H^1(\Gamma_{\mathbb{Q}}, T_{\infty|*}(G_m))}_{\widehat{\mathbb{Q}^*}},$$

but in the end it is the vanishing of $H^i(\mathfrak{S}_3, M)$ for $i = 0, 1$ that implies (without knowing anything about $\Gamma_{\mathbb{Q}}$!) that of $H^1(\Gamma_{\mathbb{Q}} \times \mathfrak{S}_3, \widehat{M})$, which ensures the uniqueness we needed. 137

Uncertainty retained from cancelled passages.

140-4-s138-D001 $H^1(\Gamma_{\mathbb{Q}}, \widehat{M}) \simeq H^1(\Gamma$

[scan 139]

The preceding theorem completely elucidates, it seems to me, the arithmetico-geometric [p. 717] meaning (in terms of the Fermat tower) of the homomorphism ((218) page 695')

$$\mathcal{M} = N_{1,1}^{\text{ét}} \longrightarrow G_{\rho,\sigma} \simeq \widehat{\mathbb{Z}}^* \cdot \mathfrak{S}_{\bullet|0}^{+\wedge}$$

obtained in section 11°) by purely algebraic means — or at least that of the slightly weaker homomorphism

$$\mathcal{M} \longrightarrow \overbrace{G_{\rho,\sigma/\mu_{\mathfrak{S}}}}^{G_{\rho,\widetilde{\sigma}} \overset{\text{dfn}}{=}} \simeq \widehat{\mathbb{Z}}^* \cdot \mathfrak{S}^{+\wedge}/\mu_{\mathfrak{S}}$$

which factors into a homomorphism

$$\underbrace{\widetilde{\mathcal{M}}}_{\text{denoted } \mathcal{M} \text{ in the preceding pages}} \overset{\text{dfn}}{=} \mathcal{M}/(\mu_{\mathcal{M}} = \mu_{\mathfrak{S}_{0|\bullet}}) \longrightarrow G_{\rho,\widetilde{\sigma}} \qquad \text{extension of } \tilde{\Gamma} = \widehat{\mathbb{Z}}^* \times \mathfrak{S}_3 \text{ by } M^\wedge.$$

The passage from $G^\dagger_{\rho,\sigma}$ to $G_{\rho,\widetilde{\sigma}}$, from the present point of view, had as its essential role (a posteriori!) to provide an extension by $M^\wedge$ (playing the role of an identity component) of a group operating faithfully on $M^\wedge$, which was not the case for the extension $G_{\rho,\sigma}$ of $\widehat{\mathbb{Z}}^* \times H^+$ by $M^\wedge$, since the † center $\mu_{H^+} = \{1, \omega_{H^+}\}$ † of H^+ operated trivially. 138

Uncertainty retained from cancelled passages.

140-4-s139-D003 $\sim$

140-4-s139-D004 center|element

140-4-s139-D005 H^+

[scan 140]

This, together with the relations

$$H^0(\widehat{\mathbb{Z}}^* \times \mathfrak{S}_3, \widehat{M}) = H^1(\widehat{\mathbb{Z}}^* \times \mathfrak{S}_3, \widehat{M}) = 0,$$

made it possible to interpret the group $G_{\rho,\widetilde{\sigma}}$ as a group of transformations on the set E of its own splittings, which made it possible to interpret the homomorphism $\mathcal{M} \to G_{\rho,\widetilde{\sigma}}$ (or $\widetilde{\mathcal{M}} \to G_{\rho,\widetilde{\sigma}}$) as an † operation of $\mathcal{M}$ † (or better, †of $\widetilde{\mathcal{M}} = \pi_1(U_{0,3\mathbb{Q}}, \mathfrak{S}_3) = \pi_1(M'_{1,1\mathbb{Q}})$ on E), thus corresponding to a pro-étale covering of $(U_{0,3\mathbb{Q}}, \mathfrak{S}_3) \simeq M'_{1,1\mathbb{Q}}$ — and this covering is none other than the Fermat pro-covering.

It remained to interpret in an analogous way the homomorphism $\boxed{\mathcal{M} = N_{1,1} \longrightarrow G_{\rho,\sigma}}$, fitting into the homomorphism of exact sequences

$$\begin{array}{ccccccccc} & & \mu_{\mathcal{M}} & & \mathcal{M} & & \widetilde{\mathcal{M}} & & \\ 1 & \to & \| & \to & \| & \to & \| & \to & 1 \\ & & (\{1,\omega\}) & & (N_{1,1}) & & (N'_{1,1} = \mathcal{M}_{0,3}) & & \\ & & \downarrow \simeq & & \downarrow & & \downarrow & & \\ & & \mu_G & & & & & & \\ 1 & \to & \| & \to & G_{\rho,\sigma} & \longrightarrow & G_{\rho,\tilde{\sigma}} & \longrightarrow & 1 \\ & & \{1,\omega\} & & & & & & \end{array} \quad , \tag{298}$$

which moreover makes it possible to interpret $\mathcal{M}$, as a central extension of $\widetilde{\mathcal{M}}$ 139

Uncertainty retained from cancelled passages.
140-4-s140-D001 ho
140-4-s140-D002 of H
140-4-s140-D003 $\widetilde{\mathcal{M}}$

[scan 141]

by μ, as the inverse image of the extension $G_{\rho,\sigma}$ of $G_{\rho,\tilde{\sigma}}$ by μ, under the preceding homomorphism $\widetilde{\mathcal{M}} \to G_{\rho,\tilde{\sigma}}$. There is a striking analogy with what had happened, moreover, when we had started from (in § 48, and in|of certain developments which preceded it) from the quotient group ("of Lie type") $\mathrm{SL}(2,\widehat{\mathbb{Z}})'$ of $\mathfrak{S}^{+\wedge}_{0,3}$, to construct a homomorphism [p. 718]

$$\dagger \qquad \mathcal{M}_{0,3} \longrightarrow \mathrm{GL}(2,\widehat{\mathbb{Z}})/\{\pm 1\}$$

from which it was possible for us to reconstruct † the extension $\mathcal{M} = N_{1,1}$ of $\mathcal{M}_{0,3}$ by μ precisely as the inverse image of the extension $\mathrm{GL}(2,\widehat{\mathbb{Z}})$ of $\mathrm{GL}(2,\widehat{\mathbb{Z}})/\{\pm 1\}$ by $\mu = \{\pm 1\}$. From the present point of view where I am trying to make arithmetic-geometric aspects as explicit as possible in terms of automorphisms of the profinite groups with loops $\Pi^{\wedge}_{0,3}$ — the remarkable fact is that the two 140

Uncertainty retained from cancelled passages.
140-4-s141-D001 $\mathcal{M}$
140-4-s141-D002 $\mathcal{M}$

[scan 142]

extensions of $\widetilde{\mathcal{M}}$ by μ, coming out of apparently rather different calculations, are canonically isomorphic.

I presume that a general explanation of this fact will not present|would not present any difficulty, in the present context — because in both cases, the bi-covering of GL|$\mathrm{SL}(2,\widehat{\mathbb{Z}})/\pm 1$ resp. of $G_{\rho,\tilde{\sigma}}$ came from the fact that I had started from a quotient group ($\mathrm{SL}(2,\widehat{\mathbb{Z}})$ or $\mathfrak{S}^{+\wedge}$) not of $\mathfrak{S}^{+\wedge}_{0,3} \simeq \mathrm{SL}(2,\mathbb{Z})^{\wedge}/\pm 1$, but of $\mathrm{SL}(2,\mathbb{Z})^{\wedge} \simeq N_{1,1}$ itself —

If we nevertheless look for an interpretation (of the "Fermat tower" kind) of the homomorphism $\mathcal{M} \to G_{\rho,\sigma}$, it would be necessary to interpret $G_{\rho,\sigma}$ (as was $G_{\rho,\tilde{\sigma}}$) as a group of automorphisms of a "typical fiber" "not too big" — the difficulty being to invent something on which that|it operates nontrivially. 141

[scan 143]
[p. 719]

Interpreting $\mathcal{M}_{0,3} = N'_{1,1}$ as $\pi_1(M'_{1,1})$, where $M'_{1,1}$ is the modular scheme/$\mathbb{Q}$ of elliptic curves up to symmetry, we see that the Fermat tower amounts to this: over every base scheme S (say of characteristic 0) and every|every quasi-elliptic curve $\dagger\widetilde{X}$ over S, let us consider the corresponding étale covering of degree three S' of S (of the "points of order 2 of $\widetilde{X}$"), let

$$\dagger \quad M_0(\widetilde{X}/S|B) \overset{\text{dfn}}{=} p_*(\mathbb{Z}_{S'|B'})/\mathbb{Z}_S \tag{299}$$

be the corresponding twisted constant $\mathbb{Z}$-module free of rank two, and let us consider

$$\dagger \quad M_1(\widetilde{X}/S) \overset{\text{dfn}}{=} M_0(\widetilde{X}/S) \otimes_{\mathbb{Z}} \underbrace{\widehat{\mathbb{Z}}(1)}_{\| \text{ dfn } T_\infty(\mathbb{G}_m)} , \tag{300}$$

which is a pro-étale group over S, a system of coefficients locally isomorphic (in the broad sense) to $\widehat{\mathbb{Z}}^2$. Of course, $M_1(\widetilde{X}/S)$ depends functorially on $\widetilde{X}/S$ for isomorphisms and inverse images (we have a cartesian section of an ad-hoc fibered category ...). This being so, 142

Uncertainty retained from cancelled passages.

140-4-s143-D001 E

140-4-s143-D003 M_2

140-4-s143-D005 M_2

[scan 144]

giving the "Fermat tower" is equivalent to giving, for $\forall\, \widetilde{X}/S$ as above, a torsor

$$F_\infty(\widetilde{X}/S) \quad \text{torsor over } S \text{ under } M_1(\widetilde{X}/S), \tag{301}$$

m1590

Marginal note. m1590 for a geometric construction cf. (320) page 724

compatible with isomorphisms and changes of base.

The preceding theorem says essentially that such an $F_\infty(\widetilde{X}/S)$ exists, and is unique up to unique isomorphism, requiring in addition that the natural homomorphism

$$\Pi^\wedge_{0,3} \longrightarrow M^\wedge$$

associated with such a datum F_∞ be the canonical homomorphism ....

On the other hand, let X be a relative elliptic curve over a base S of characteristic 0, and let $S' \subset X$ be the closed subscheme, étale of degree 3 over S, corresponding to the points of order 2. Let us consider the generalized Jacobian of X deduced from this multisection, let $\mathcal{J}^0$ be its part of degree 0, we thus have 143

[scan 145]

an exact sequence [p. 720]

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \mathcal{J}^0_0 & \longrightarrow & \mathcal{J}^0 & \longrightarrow & X & \longrightarrow & 1 \\ & & \wr\| & & & & & & \\ & & \mathbb{G}_m \otimes_{\mathbb{Z}} M_0 & & & & & & \end{array} \tag{302}$$

where $M_0 = M_0(X/S)$ is defined above (299). From this we deduce an exact sequence

$$\begin{array}{ccccccccc} 1 & \longrightarrow & M_1 & \longrightarrow & T_\infty(\mathcal{J}^0) & \longrightarrow & T_\infty(X) & \longrightarrow & 1 \\ & & \| & & & & & & \\ & & M_0 \otimes_{\mathbb{Z}} T_\infty(\mathbb{G}_m) & & & & & & \end{array} \tag{303}$$

where

$$T_\infty(G) = \varprojlim_n {}_n G \tag{304}$$

is the local system of the Π_1 of the smooth commutative group scheme with connected fibers over S. Thus giving a torsor under M_1 is equivalent to giving a torsor under $T_{\infty|\alpha}(\mathcal{J}^0)$, equipped with a section of the associated torsor of the|of group $T_\infty(X)$. Is there|Is there a remarkable description of the Fermat torsor $F_\infty(X/S)$, in terms of the geometry of the relative elliptic curve X/S? 144

[scan 146]

14°) Relation to generalized Jacobians.

Let us consider the "generalized Jacobian" of

$$U_{0,3} = U_{0,3\,\dagger\mathbb{Z}} = {}^\dagger\mathbb{P}^1_{\dagger\mathbb{Z}} - \{0,1,\infty\}_{\dagger\mathbb{Z}},$$

it is thus a group scheme $\mathcal{J}^*_{U_{0,3}} = \mathcal{J}^*$, augmented|mapping to $\mathbb{Z}$, fitting into the exact sequence

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \mathcal{J}^0 & \longrightarrow & \mathcal{J}^* & \xrightarrow{\deg} & \mathbb{Z}_{S_0\dagger} & \longrightarrow & 1 \\ & & \wr\| & & & & & & \\ & & \mathbb{G}_{mS_0} \otimes_{\mathbb{Z}} M_{0|\mathbb{Z}} & & & & & & \end{array} , \tag{305}$$

and equipped with a canonical morphism

N.B. $S_0 \overset{\text{df|dfn}}{=} \underline{\text{Spec}}\,\mathbb{Z}$

$$\varphi : U_{0,3\,\dagger\mathbb{Z}} \longrightarrow \underbrace{\mathcal{J}^1}_{\substack{\text{subscheme of } \mathcal{J}^* \\ \text{inverse image of} \\ \text{the section 1 of } \mathbb{Z}_{S_0}}} \tag{306}$$

The operations of $\mathfrak{S}_3$ on $U_{0,3}$ define by functoriality operations of $\mathfrak{S}_3$ on $\mathcal{J}^*$, compatible with the extension structure (305), and such that (306) commutes with the operations of $\mathfrak{S}_3$. Let us note that $\mathcal{J}^1$ is a $\mathcal{J}^0$-torsor and 145

Uncertainty retained from cancelled passages.

140-4-s146-D001 $\mathbb{Q}$
140-4-s146-D002 $\mathbb{A}_{\mathbb{Q}}$
140-4-s146-D003 $\mathbb{Q}$
140-4-s146-D004 $\mathbb{Q}$
140-4-s146-D005 $\mathbb{Q}$
140-4-s146-D007 $\mathbb{Q}$

[scan 147]

even an $(\mathfrak{S}_3, \mathcal{J}^0)$-torsor (a $\mathcal{J}^0$-torsor on which $\mathfrak{S}_3$ operates in a way compatible with its operations on $\mathcal{J}^0$), hence classified by an element of $H^1(S_0, \mathfrak{S}_3; \mathcal{J}^0)$. Now I presume that we have [p. 721]

$$H^0(S_0, \mathfrak{S}_3, \mathcal{J}^0) = H^1(S_0, \mathfrak{S}_3, \mathcal{J}^0) = 0 \tag{307}$$

(where

$$\left\{ \begin{array}{l} S_0 = \text{Spec}\,\mathbb{Z}, \quad \mathcal{J}^0 = \mathbb{G}_m \times_{\mathbb{Z}} M \simeq \mathbb{G}_m^{\{0,1,\infty\}}/\text{diag} \\ \quad \simeq \text{Ker} \left(\begin{array}{c} \mathbb{G}_m^{\{0,1,\infty\}} \longrightarrow \mathbb{G}_m^\dagger \\ \\ (x_0,x_1,x_\infty) \mapsto {}^{x_0\cdot x_1\cdot x_\infty}_{\dagger} \end{array} \right) \end{array} \right. \tag{308}$$

m1595

Marginal note. m1595 Careful, this last isomorphism is obtained via

$$(x_0, x_1, x_\infty) \longmapsto (x_1 x_\infty^{-1}, x_\infty x_0^{-1}, x_0 x_1^{-1})$$

, which commutes with $\mathfrak{S}_3^+$, but not|not with $\mathfrak{S}_3$ as a whole

We have indeed

$$\begin{aligned} H^0(S_0, \mathfrak{S}_3; \mathcal{J}^0) &\simeq \left\{ (x_0, x_1, x_\infty) \in \mathbb{Z}^{*\{0,1,\infty\}} \;\middle|\; \begin{array}{l} x_0 x_1 x_\infty = 1 \\ (x_0, x_1, x_\infty) \text{ invariant under } \mathfrak{S}_3 \end{array} \right\} \\ &= \{(x, x, x) \mid x \in \mathbb{Z}^{*\{0,1,\infty\}}, \quad x^3 = 1\} \\ &= 1 \end{aligned}$$

m1596

Marginal note. m1596 Calculation correct, for it rests on this $\mathfrak{S}_3^+$-isomorphism which is not an $\mathfrak{S}_3$-isomorphism, it seems here to be|there seems not to be need of $H^0(S, \mu_3) = 1$, whereas a calculation already made of the splittings of extensions (cf. 720) shows that S – over any base

$$\begin{array}{ccccc} H^0(S, \mathfrak{S}_3, \mathcal{J}^0) & \simeq & H^1(\dagger\mathfrak{S}_3; H^0(S|S_0, \mathcal{J}^0)) \simeq 0 & \Leftarrow \\ \| & & & \\ H^0(\mathfrak{S}_3, H^0(S, \mathcal{J}^0)) & & & \end{array}$$

(the only cube † root of 1 in $\mathbb{Z}$ (or even in $\mathbb{Q}$) is 1. This establishes the first relation (307). For the second, we use the spectral sequence 146

Uncertainty retained from cancelled passages.
140-4-s147-D001 $\{0, 1, \infty\}$
140-4-s147-D002 $(x_1 x_\infty^{-1}, x_\infty x_0^{-1}, x_0 x_1^{-1})$
140-4-s147-D003 S,
140-4-s147-D004 of

[scan 148]

$$H^{*|n}(S_0, \mathfrak{S}_3; \mathcal{J}^0) \Longleftarrow H^p(\mathfrak{S}_3, \underline{H}^q(S_0, \mathcal{J}^0)) \quad E_2^{pq}=$$

which gives the exact sequence

$$\begin{array}{ccccccc} 0 \to & H^1(\mathfrak{S}_3, H^0(S_0, \mathcal{J}^0)) & \xrightarrow{\dagger|\simeq} & H^1(S_0, \mathfrak{S}_3; \mathcal{J}^0) & \longrightarrow & H^0(\mathfrak{S}_3, H^1(S_0, \mathcal{J}^0)) \\ & & & & \searrow & \downarrow \\ & & & (H^{0|q}(\mathfrak{S}_3, M_0 \otimes_{\mathbb{Z}} H^1(S_0, \mathbb{G}_m))) & & H^2(\mathfrak{S}_3, H^0(S_0, \mathcal{J}^0)) \end{array}$$

Now we have

$$H^1(S_0, \mathcal{J}^0) \simeq H^1(S_0, \mathbb{G}_m^2) \simeq \mathrm{Pic}(S|S_0)^2 = 0$$

m1598

Marginal note. m1598 N.B. we do not need the relation $\mathrm{Pic}(S) = 0$

hence we get

$$H^1(S_0, \mathfrak{S}_3; \mathcal{J}^0) \simeq H^1(\mathfrak{S}_3, \underbrace{H^0(S_0, \mathcal{J}^0)})$$

But in fact, the calculations made elsewhere show that we have

$$H^0(\mathfrak{S}_3, M_{0|\alpha} \otimes_{\mathbb{Z}} I|I_0) = H^1(\mathfrak{S}_3, M_{0|\alpha} \otimes_{\mathbb{Z}} I) = 0 \tag{309}$$

for any $\mathbb{Z}$-module I (for example $I = J^0|\mathcal{J}^0(S)$, S an arbitrary|arbitrary scheme, or $I = H^1(S, \mathbb{G}_m) = \mathrm{Pic}(S)$), whence finally,

$$H^0(\dagger|S, \mathfrak{S}_3; \mathcal{J}^0) \simeq H^1(S, \mathfrak{S}_3, \mathcal{J}^0) = 0 \tag{310}$$

for $\forall$ scheme S. We thus find: 147

Uncertainty retained from cancelled passages.
140-4-s148-D001 $\simeq$
140-4-s148-D004 S_0

[scan 149]

Proposition Let $\dagger$ $\mathcal{J}^0 = \mathbb{G}_m^{\{0,1,\infty\}} / \underbrace{\mathbb{G}_m}_{\text{diag.}} \simeq \mathbb{G}_m \otimes_{\mathbb{Z}} M_0|M$ Then for $\forall$ $(\mathfrak{S}_3, \mathcal{J}^0)$-torsor over an [p. 722] arbitrary base S, there exists a unique trivialization of it, i.e. a unique section, invariant under the operations of $\mathfrak{S}_3$.

Corollary 1 Over any base S, there exists a unique morphism

$$\psi : U_{0,3\,S} \longrightarrow \mathcal{J}^0_S \tag{311}$$

which is compatible with the operations of $\mathfrak{S}_3$, and which (via the homomorphism $\mathcal{J}^*_S \longrightarrow \mathcal{J}^0_S$ which|that it|it is factored by, by the universal property of the [illeg. 1] generalized Jacobians) induces the identity homomorphism $\underset{\mathcal{J}^*_S}{\mathcal{J}^0_S} - \mathcal{J}^0_S$.

m1600

Marginal note. m1600 N.B. With the notation of 695, we have $U_{0,3\,S} \simeq (U_1)_S$, where $U_1 \subset \mathbb{G}_m^3/\mathbb{G}_m$ is defined by

$$U_1 = \{(x_0, x_1, x_\infty) \in \Gamma\mathbb{G}_m^{\{0,1,\infty\}} \mid x_0 + x_1 + x_\infty = 0\} \mod \mathrm{diag}_3 \mathbb{G}_m. \tag{312}$$

Corollary 3 Let Y be a scheme of 148

Uncertainty retained from cancelled passages.
140-4-s149-D001 $\mathcal{J}$

[scan 150]

Brauer-Severi over S, of relative dimension 1 (i.e. a form of $\mathbb{P}^1_S$), equipped with a closed subscheme $S' \subset Y$, étale of degree 3 over S, and let $\dagger$ $U = Y \setminus S'$, $\mathcal{J}^*_{U/S}$ the generalized Jacobian, so that we have

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \mathcal{J}^0_{U/S} & \longrightarrow & \mathcal{J}^*_{U/S} & \longrightarrow & \mathbb{Z}_S & \longrightarrow & 1 \\ & & \wr & & & & & & \\ & & \mathbb{G}_m \otimes_{\mathbb{Z}} M_0(S'/S) & & & & & & \end{array} \tag{313}$$

where $M_0(S'/S) \simeq p_*|\rho_*(\mathbb{Z}_{S'}) / \underbrace{\mathbb{Z}_S}_{\text{diag}}$ is the S-scheme of $\mathbb{Z}$-modules (locally isom. to $\mathbb{Z}^2$, deduced from $M_{0\,S}$ by twisting it (via the action of $\mathfrak{S}_3$) by the $\mathfrak{S}_3$-torsor over S associated with S'.

There exists then a unique morphism

$$(314) \qquad \psi : U \longrightarrow \mathcal{J}^0_{U/S}$$

commuting with the action of $\underline{\mathrm{Aut}}(S'/S) \xleftarrow{\sim} \underline{\mathrm{Aut}}_S(Y, S')$, and inducing (by the universal property) † the composite homomorphism

$$\mathcal{J}^0_{U/S} \hookrightarrow \mathcal{J}^*_{U/S} \xrightarrow{\widetilde{\psi}} \mathcal{J}^0_{U/S}$$

which is the identity. 149

Uncertainty retained from cancelled passages.

140-4-s150-D001 X

140-4-s150-D005 a

[scan 151]

Let us make explicit the|the morphism [p. 723]

$$(315) \qquad \begin{array}{ccc} U_{0,3} = \mathbb{P}^1 \setminus \{0,1,\infty\}_{S_0} & \xhookrightarrow{\ \psi\ } & \underbrace{\mathbb{G}_m^{\{0,1,\infty\}}/\mathbb{G}_m}_{\mathcal{J}^0} \\ \| & & \uparrow \\ \dagger & & \dagger U_1 = \{(x_0,x_1,x_\infty)^\dagger \in \Gamma\mathbb{G}_m^3/\mathbb{G}_m \mid \\ (\mathbb{G}_m \setminus \{1\})_{S_0} \simeq \{z \in \Gamma\mathbb{G}_m \mid z-1 \text{ inv}\} & & x_0+x_1+x_\infty = 0\} \end{array}$$

† † isomorphism of $U_{0,3}$ with U_1 which † corresponds: †

$$(316) \qquad \begin{cases} \overline{\psi} : \widehat{U}_{0,3}|\overline{U}_{0,3} \longrightarrow \widehat{U}_1 \subset \mathbb{P}^2_{S_0} \\ \quad \| \qquad\qquad\qquad\qquad \| \\ \quad \mathbb{P}^1_{S_0} \qquad\qquad \{(x_0,x_1,x_\infty) \in \widehat{\mathbb{P}}^2_{S_0}|\mathbb{P}^2_{S_0} \mid x_0+x_1+x_\infty = 0\} \\ \overline{\psi}(0) = (0,1,-1) \\ \overline{\psi}(1) = (-1,0,1) \\ \overline{\psi}(\infty) = (1,-1,0) \end{cases},$$

it suffices to take

$$(317) \qquad \psi(z) = (-z, z-1, 1) = (1, 1/z-1, -1/z) = \left(\frac{-z}{z-1}, 1, \frac{1}{z-1}\right).$$

Let us now return to the datum of a relative elliptic curve X over S modulo symmetry (corresponding to a morphism $S \longrightarrow \dagger\mathcal{M}'_{1,1}$). 150

Uncertainty retained from cancelled passages.

140-4-s151-D002 Γ

140-4-s151-D003 $\mathcal{J}^1$

140-4-s151-D004 $\mathbb{Z}$

140-4-s151-D005 it

140-4-s151-D006 it is

140-4-s151-D007 is

140-4-s151-D008 one must|extends

140-4-s151-D009 Spec

[scan 152]

We assume 2 invertible on S, so that the datum of X modulo ω_X corresponds, by the Legendre carpet $X \mapsto (X/\dagger\{1,\omega_X\})$, to the datum of a (Y,S') as above, with in addition a section of $U = Y - S'$.

In fact, the datum of (Y,S') amounts to the datum of S', i.e. of an $\mathfrak{S}_3$-torsor T over S — and Y is recovered from there|it as the twisted scheme $T \wedge_{\mathfrak{S}_3} \mathbb{P}^1_S$ † with

$$U = Y \setminus S' \simeq (T \wedge_{\mathfrak{S}_3} \mathbb{P}^1_S) \setminus \underbrace{T \wedge_{\mathfrak{S}_3} \{0,1,\infty\}_S}_{S'}$$

canonically embedded in

$$T \wedge_{\mathfrak{S}_3} \mathcal{J}^0_{(U_{0,3})_S} \simeq \mathcal{J}^0_{U/S}.$$

The embedding $U \subset \mathcal{J}^0_{U/S}$ can also be interpreted as a canonical morphism

$$X \longrightarrow \mathcal{J}^0_{U/S} \underbrace{= \mathcal{J}^0_{0\,X/S}|\mathcal{J}^0_{X/S}}_{\substack{\text{with|in notation}\\ \text{of page 720}}} \tag{318}$$

which should be easy to characterize axiomatically 151

Uncertainty retained from cancelled passages.

140-4-s152-D001 modulo symmetry

140-4-s152-D003 $\wedge_{\mathfrak{S}_3}$ and|or $\mathcal{J}^0$

[scan 153]

ment, † or to make explicit in terms of standard "elliptic functions". [p. 724]

Without dwelling on such an explicitation, and returning rather to the corresponding embedding

$$U \hookrightarrow \mathcal{J}^0_{U/S} \qquad (318)$$

and to the marked section

$$e \in U(S) = H^0(S, \mathcal{J}^0_{U/S}), \qquad (319)$$

we can use the tower of isogenies

$$1 \longrightarrow {}_n\mathcal{J}^0_{U/S} \hookrightarrow \mathcal{J}^0_{U/S} \xrightarrow{n.\mathrm{id}_{\mathcal{J}^0_{U/S}}} \mathcal{J}^0_{U/S} \longrightarrow 1 \qquad (321)$$

(for varying n), i.e. the universal isogeny deduced by passage to $\varprojlim$:

$$\begin{array}{ccccccccc} 1 & \longrightarrow & T_\infty(\mathcal{J}^0_{U/S}) & \longrightarrow & \widetilde{\mathcal{J}}^0_{U/S} & \xrightarrow{p} & \mathcal{J}^0_{U/S} & \longrightarrow & 1 \\ & & \wr| & & \| \mathrm{def} & & & & \\ & & T_\infty(\mathbb{G}_m)_S \otimes_{\mathbb{Z}} M_0(S'/S) & & \varprojlim_n {}_n(\mathcal{J}^0_{U/S}) & & & & \end{array} \qquad (322)$$

to deduce, by inverse image of e, a torsor under $T_\infty(\mathcal{J}^0_{U/S}) \simeq T_\infty(\mathbb{G}_m)_S \otimes_{\mathbb{Z}} M_0(S'/S)$. This being so, we will have (cf. (301) page 719')

$$F_\infty(\widetilde{X}/S) \simeq p^{-1}(e), \quad \text{the } T_\infty(\mathcal{J}^0_{U/S})\text{-torsor} \qquad (323)$$

deduced from the section e of $\mathcal{J}^0_{U/S}$ over S. 152

Uncertainty retained from cancelled passages.
140-4-s153-D001 [illeg. 1]
140-4-s153-D003 314

[scan 154]

Remark The isomorphism class of the $T_\infty(\mathcal{J}^0_{U/S})$-torsor obtained is an element of $H^1(S, T_\infty(\mathcal{J}^0_{U/S}))$, image of the element e of $H^0(S, \mathcal{J}^0_{U/S})$, under the canonical morphism

$$(324) \qquad H^0(S, \mathcal{J}^0) \longrightarrow H^1(S, T_\infty(\mathcal{J}^0)) \ ,$$

whose kernel consists of the subgroup of $H^0(S, \mathcal{J}^0)$ of "infinitely divisible" elements. In the case where S is reduced and is of finite type over † $\mathbb{Z}$ – for example reduced of finite type over $\mathbb{Q}$ – we know (and this is † "elementary") that this kernel is reduced to zero. This therefore means that the section e of U over S (i.e. the "elliptic parameter" under consideration) is then entirely determined, when one knows 153

Uncertainty retained from cancelled passages.
140-4-s154-D001 $\mathbb{Q}$
140-4-s154-D002 a theorem

[scan 155]

1°) the † associated étale triple covering S' of S, corresponding to the points of order 2 of [p. 725]
$\widetilde{X}/S$ – whence $\mathcal{J}^0 = \mathcal{J}^0_{U/S} \simeq \mathcal{J}^0_{0\dagger}(\widetilde{X}, S')$ – and

2°) The $T_\infty(\mathcal{J}^0_\dagger)$-torsor $\mathbb{F}_\infty | F_\infty(\widetilde{X}/S)$; 154

Uncertainty retained from cancelled passages.
140-4-s155-D001 [illeg. 1]
140-4-s155-D002 S
140-4-s155-D003 U/S

[scan 156]

15°) Interpretation of $\Gamma_{\mathbb{Q}} \longrightarrow N_\sigma^{\natural|\sharp}$.

In number 11° we introduced the canonical homomorphism

$$\begin{array}{ccc} \mathcal{M} \overset{\text{def}}{=} \mathcal{N}^{\text{ét}}_{1,1} & \longrightarrow & \mathcal{M}^{\sim}_{\rho,\sigma} \\ \cup & & \cup \\ \mathcal{M}^{!} & \longrightarrow & \mathcal{M}^{\sim\natural|\sharp}_{\rho,\sigma} \end{array} \qquad \text{cf. (217) page 695)}$$

which we have "decomposed" into two homomorphisms (in addition to a trivial homomorphism

$$\begin{array}{ccc} \mathcal{M} & \longrightarrow & N^{\sim}_\rho \simeq \widehat{\mathbb{Z}}^\times \\ & \searrow & \uparrow \chi \\ & & \Gamma^{\text{ét}}_{\mathbb{Q}} \end{array}$$

) into two homomorphisms

$$\begin{array}{ccc} \mathcal{M} & \longrightarrow & G_{\rho,\sigma} \simeq \widehat{\mathbb{Z}}^*.\mathfrak{S}^{+\wedge} \\ (325) \quad \cup & & \cup \qquad (\text{cf. (218)}) \\ \mathcal{M}^! & \longrightarrow & G^{\natural|\sharp}_{\rho,\sigma} \simeq \widehat{\mathbb{Z}}^*.M^{\wedge} \end{array}$$

and

$$(326) \qquad \Gamma^{\text{ét}}_{\mathbb{Q}} \longrightarrow N^{\natural|\sharp\dagger}_{\sigma} \simeq \widehat{\mathbb{Z}}^*.\widehat{\mathbb{Z}} \qquad (\text{cf. (219)}).$$

The first homomorphism (322) was characterized in arithmetico-†geometric terms (as regards its restriction to $\mathcal{N}_{1,1} \subset \mathcal{N}^{\text{ét}}_{1,1}$ itself, and reducing modulo μ_G to find a homomorphism

$$(327) \qquad \dagger \qquad \mathcal{M}_{0,3} \longrightarrow G^{\sim}_{\rho,\sigma} \overset{\text{def}}{\simeq} G_{\rho,\sigma}/\mu_G \ ,$$

in section 13°), via the tower of 155

Uncertainty retained from cancelled passages.
140-4-s156-D002 ab| ∼
140-4-s156-D003 algebraic
140-4-s156-D004 $\mathcal{M}_{0,3}$

[scan 157]

Fermat curves. From the present point of view, where we are trying to make explicit the algebraic geometry of essentially finite type over $\mathbb{Q}$ in terms of automorphisms of profinite groups, we can also say, conversely, that we have managed to "describe" the tower of Fermat curves, via the homomorphism (327) constructed here "by brute force". [p. 725|726]

It would now remain to find an arithmetico-geometric explicitation of the homomorphism † from $\Gamma_{\mathbb{Q}} \subset \Gamma^{\text{ét}}_{\mathbb{Q}}$ into $\widehat{\mathbb{Z}}^*.\widehat{\mathbb{Z}}$, induced by (326). But † using the embedding (

$$N^{\natural|\sharp\dagger}_{\sigma} \hookrightarrow G^{\natural|\sharp\dagger}_{\rho,\sigma} \simeq \widehat{\mathbb{Z}}^*.\underline{M} \ ,$$

the homomorphism (313|323) appears as the composite

$$\Gamma^{\text{ét}}_{\mathbb{Q}} \xhookrightarrow[\kappa_{1/2}]{} \mathcal{M}^! \xrightarrow[\text{i.e. "Fermat"}]{(325)} G^{\natural|\sharp\dagger}_{\rho,\sigma} \ ,$$

more precisely we have a commutative diagram: 156

Uncertainty retained from cancelled passages.
140-4-s157-D001 (322)
140-4-s157-D002 this the homomorphism
140-4-s157-D003 ab| ∼
140-4-s157-D004 ab| ∼
140-4-s157-D005 ab| ∼

[scan 158]

$$\begin{array}{ccc} [\Gamma_{\mathbb{Q}} \hookrightarrow]\Gamma^{\text{ét}}_{\mathbb{Q}} & \longrightarrow & N_\sigma^{\natural|\sharp} \simeq \widehat{\mathbb{Z}}^*.\widehat{\mathbb{Z}} \\ {\scriptstyle \kappa_{1/2}}\downarrow & & \downarrow \\ [\ \underbrace{\mathcal{M}^!_{0,3}}_{\Pi_1(\mathcal{U}_{0,3\mathbb{Q}})} \hookrightarrow]\mathcal{M}^{!\,\infty|\text{ét}} \simeq N|\mathcal{N}^{!\,\text{ét}}_{\rho\beta|\rho 3} & \xrightarrow[(325)]{\text{“Fermat”}} & G^{\natural|\sharp}_{\rho,\sigma} \simeq \widehat{\mathbb{Z}}^*.\underline{M}^\wedge. \end{array} \tag{328}$$

This therefore means that we find the homomorphism induced by (326)

$$\Gamma_{\mathbb{Q}} \longrightarrow N_\sigma^{\natural|\sharp} = \widehat{\mathbb{Z}}^*.\widehat{\mathbb{Z}} \tag{329}$$

by looking at the image in $N_\sigma^{\natural|\sharp} \subset \widehat{\mathbb{Z}}^*.M^\wedge$ of the homomorphism

$$\Gamma_{\mathbb{Q}} \longrightarrow \widehat{\mathbb{Z}}^*.\underline{M}^\wedge \subset \mathrm{Aff}(M^\wedge), \tag{330}$$

† deduced from the tower of fermat curves by looking at the fiber of the said tower at the point $1/2$, which gives a torsor under $T_\infty(\mathbb{G}_m)\otimes_{\widehat{\mathbb{Z}}}\widehat{M}$, being made explicit Galois-theoretically precisely by the homomorphism (326). Let us note that the †point $1/2$ † of $\mathcal{U}_{0,3\mathbb{Q}}$ corresponds, under the isomorphism ψ (317) of $\mathcal{U}_{0,3\mathbb{Q}}$ with $U_{1,\mathbb{Q}}$, to the point $(-\frac{1}{2},-\frac{1}{2},1)$ 157

Uncertainty retained from cancelled passages.

140-4-s158-D001 $\natural|\sharp$

140-4-s158-D002 fiber

140-4-s158-D003 canonically isomorphic

of † [scan 159] [p. 727]

$$s_{1/2} = (-\tfrac{1}{2},-\tfrac{1}{2},1) \in \dagger \begin{cases} U_{1\mathbb{Q}} \subset \mathbb{G}^3_{m\mathbb{Q}}/\mathbb{G}_{m\mathbb{Q}} \subset \mathbb{P}^2_{\mathbb{Q}} \\ \| \\ \{(x_0,x_1,x_\infty) \in \Gamma(\mathbb{G}^3_{m\mathbb{Q}}/\mathbb{G}_{m\mathbb{Q}}) \mid x_0+x_1+x_\infty = 0\} \end{cases} \tag{331}$$

The geometric|geometric fiber of U_n at the said point is thus identified with the systems of solutions of the equations below:

$$\begin{aligned} \dagger \quad U_n(s_{1/2}) &\simeq \{(x_0,x_1,x_\infty) \in \Gamma\mathbb{G}^3_m \mid x_0^n = -\tfrac{1}{2},\ x_1^n = -\tfrac{1}{2},\ x_\infty^n = 1|-1\} \\ &\quad\ \dagger \\ &\simeq \{(x_0,x_1,x_\infty) \in \Gamma(\mathbb{G}^3_m) \mid x_0 = -\tfrac{1}{2},\ x_1^n = -\tfrac{1}{2},\ x_\infty^n = 1|-1\} \\ &\simeq \{(x_0,x_{1|\infty}) \in \Gamma\mathbb{G}^2_m \mid x_0^n = -\tfrac{1}{2},\ x_{1|\infty}^n = -\tfrac{1}{2}\} \\ &\simeq \dagger T_n(-\tfrac{1}{2}) \times T_n(\tfrac{1}{2}|-\tfrac{1}{2}) \end{aligned} \tag{332}$$

where we have put, for $\forall\ \zeta|\xi \in \mathbb{G}_m(S)$ (S any scheme, here $S = \mathrm{Spec}\,\mathbb{Q}$)

$$\begin{gathered} T_n(\zeta|\xi) = \dagger \\ \mu_n\text{-torsor of } \dagger n^{\text{th}} \text{ roots of } \zeta|\xi, \end{gathered} \tag{333}$$

Passing to the limit we find an isomorphism

$$U_\infty(s_{1/2}) \simeq T_\infty(-\tfrac{1}{2}) \times T_\infty(-\tfrac{1}{2}), \tag{334}$$

where we put, as is only right

$$T_\infty(\zeta|\xi) = \varprojlim_n T_n(\zeta|\xi) \qquad (\text{a } T_\infty(\mathbb{G}_m)\text{-torsor}). \tag{335}$$

158

Uncertainty retained from cancelled passages.

140-4-s159-D001 U
140-4-s159-D002 $U_{1/2}$
140-4-s159-D003 $\bar{s}|s$
140-4-s159-D005 mod
140-4-s159-D006 $\mu_n\times$
140-4-s159-D007 image of $\zeta|\xi$
140-4-s159-D008 of

Thus, the homomorphism (329) should be interpretable in terms of the Galois homomorphism [scan 160]

$$\Gamma_{\mathbb{Q}} \longrightarrow \mathrm{Gal}(\widetilde{\mathbb{Q}}[\sqrt[\infty]{-\tfrac{1}{2}}]/\mathbb{Q}) \tag{336}$$

where $\widetilde{\mathbb{Q}}$ denotes the universal cyclotomic extension of $\mathbb{Q}$

$$\widetilde{\mathbb{Q}} \simeq \mathbb{Q}[\sqrt[\infty]{1}]. \tag{336}$$

To appreciate the "size" of the image of $\Gamma_{\mathbb{Q}}$ in $\widehat{\mathbb{Z}}^*.\widehat{\mathbb{Z}} \simeq N_\sigma^{\natural|\sharp}$, it is appropriate to make a digression of a general nature, on the Galois groups over $\mathbb{Q}$ of extensions of the form

$$\widetilde{\mathbb{Q}}[\sqrt[\infty]{\zeta_1|\xi_1}, \dots, \sqrt[\infty]{\zeta_N|\xi_N}].$$

For this, over a base $S/\mathbb{Q}$, we have for simplicity (we will be|restrict ourselves here reduced|restricted to the case $S = \mathrm{Spec}\,\mathbb{Q}$) let us give ourselves an abelian group A (in the case that interests us here we will take $A = \mathbb{Z}$), and 159

a homomorphism [scan 161]

$$A \xrightarrow{\zeta|\xi} \mathbb{G}_m(S) = H^0(S, \mathbb{G}_{mS}) \tag{337}$$ [p. 728]

(in the present case, it is given by $1 \mapsto -1/2$), or what amounts to the same thing, a homomorphism

$$A_S \xrightarrow{\zeta|\xi} \mathbb{G}_{mS}. \tag{338 †}$$

Let us consider the tower of Kummer isogenies

$$1 \longrightarrow \mu_n \longrightarrow \mathbb{G}_m \xrightarrow{\lambda\mapsto\lambda^n} \mathbb{G}_m \longrightarrow 1 \tag{33†9}$$

whence, by inverse image via (338), a tower of extensions of étale group schemes over S

$$1 \longrightarrow \mu_n \longrightarrow E_n(\zeta|\xi) \longrightarrow A_S \longrightarrow 1 \tag{339}$$

and by passage|passage to the $\varprojlim$

$$1 \longrightarrow T_\infty(\mathbb{G}_{mS}) \longrightarrow E_\infty(\zeta|\xi) \longrightarrow A_S \longrightarrow 1. \tag{340}$$

Let us suppose S is 0-connected, and take a † geometric point $\bar{s}$ of S, we will therefore have a canonical homomorphism

$$\Pi_1(S, \bar{s}) \longrightarrow \mathrm{Aut}_{\substack{\mathrm{gr}\\ \text{profinite}}}(E_\infty(\zeta|\xi)_{\bar{s}}), \tag{341}$$

160

Uncertainty retained from cancelled passages.
140-4-s161-D001 *bis*
140-4-s161-D002 8
140-4-s161-D003 fixed

where the fiber $E_\infty(\zeta|\xi)_{\bar s}$ is itself equipped with an abelian extension structure [scan 162]

$$1 \longrightarrow \underbrace{T_\infty(\bar k^*)}_{T} \longrightarrow E_\infty(\zeta|\xi)_{\bar s} \longrightarrow A \longrightarrow 1. \tag{342}$$

The homomorphism (341) is compatible with this extension structure, when one lets $\Pi_1(S,\bar s)$ act trivially on A, and on $T_\infty(\bar k^*)$ †(free $\widehat{\mathbb{Z}}$-module of rank 1) by multiplication by the cyclotomic character

$$\Pi_1(S,\bar s) \longrightarrow \Pi_1(\mathbb{Q},\bar s) \xrightarrow{\chi} \widehat{\mathbb{Z}}^*. \tag{343}$$

Let

$$G_{\zeta|\xi} \subset \mathrm{Aut}(E_\infty(\zeta|\xi)_{\bar s})$$
$$\| \tag{344}$$
$$\left\{ u \in \mathrm{Aut} \;\middle|\; \begin{array}{c} u \text{ respects the extension structure by} \\ \text{inducing } \mathrm{id}_A \text{ — and in } T_\infty(\bar k^*) \text{ a} \\ \text{homothety by an element} \\ \mu \in \widehat{\mathbb{Z}}^* \text{ uniquely determined} \end{array} \right\}$$

we thus have a canonical homomorphism$u \mapsto \mu$

$$G \longrightarrow \widehat{\mathbb{Z}}^*,$$

fitting into a diagram

$$1 \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(A, T_\infty(\bar k^*)) \longrightarrow G_{\zeta|\xi} \longrightarrow \widehat{\mathbb{Z}}^*, \tag{345}$$

161

Uncertainty retained from cancelled passages.
140-4-s162-D001 to the

[scan 163]

where the image of $G_{\zeta|\xi} \longrightarrow \widehat{\mathbb{Z}}^*$ consists of the $\mu \in \widehat{\mathbb{Z}}^*$ such that the homothety, μ_T does [p. 729] not change the class of the extension (342) of A by T. The homomorphism (341) can thus be interpreted as fitting into a diagram

$$\begin{array}{ccccccc} 1 & \longrightarrow & \mathrm{Hom}_{\mathbb{Z}}(A, T_\infty(\bar k^*)) & \longrightarrow & G_{\zeta|\xi} & \longrightarrow & \widehat{\mathbb{Z}}^* \\ & & & & \uparrow{\scriptstyle \gamma_{\zeta|\xi}} & \nearrow_{\text{commutative}} & \\ & & & & \Pi_1 & \text{cyclotomic character"} & \end{array} \tag{346}$$

N.B.

If S is essentially of finite type over $\mathbb{Q}$, then the image of $\Pi_1 \longrightarrow \widehat{\mathbb{Z}}^*$ is an open subgroup of $\widehat{\mathbb{Z}}^*$, hence the same is a fortiori true of the image of $G_{\zeta|\xi} \longrightarrow \widehat{\mathbb{Z}}^*$ — which will even be surjective if the cyclotomic character on Π_1 is surjective, i.e. if † the only roots of 1 on S are ± 1 (for example if $S = \mathrm{Spec}\,\mathbb{Q}\ldots$).

The diagram (346) has an an obvious functoriality property for variable A (say), i.e. for a 162

Uncertainty retained from cancelled passages.
140-4-s163-D002 one

[scan 164]

homomorphism of abelian groups

$$A' \xrightarrow{j} A \ , \tag{347}$$

giving a composite

$$A' \xrightarrow{j} A \xrightarrow{\zeta|\xi} \mathbb{G}_m(S) \quad , \qquad A' \xrightarrow{\zeta|\xi'} \mathbb{G}_m(S) \tag{348}$$

the extension $E_\infty(\zeta|\xi')_{\bar{s}}$ is identified with the inverse image of the extension $E_\infty(\zeta|\xi)_{\bar{s}}$ under (347), whence a homomorphism

$$G_{\zeta|\xi} \longrightarrow G_{\zeta|\xi'}$$

†

fitting into a commutative diagram

injective if j is surjective

†

$$\begin{array}{ccccccc}
1 & \longrightarrow & \mathrm{Hom}_{\mathbb{Z}}(A',T) & \longrightarrow & G_{\zeta|\xi'} & \longrightarrow & \widehat{\mathbb{Z}}^* \\
 & & \uparrow & & \uparrow & & \uparrow 1 \\
1 & \longrightarrow & \mathrm{Hom}_{\mathbb{Z}}(A,T) & \xrightarrow{\gamma_{\zeta|\xi'}} & G_{\zeta|\xi} & \longrightarrow & \widehat{\mathbb{Z}}^* \\
 & & & & \uparrow \gamma_{\zeta|\xi} & \nearrow & \text{cyclotomic character} \\
 & & & & \Pi_1 & &
\end{array} \tag{349}$$

where the arrow on the $\mathrm{Hom}_{\mathbb{Z}}$ is the transpose of (347). This shows that to study the structure of the image of $\gamma_{\zeta|\xi}$ (346), one is reduced, by functoriality, to its|to

163

Uncertainty retained from cancelled passages.
140-4-s164-D002 $G_{\zeta|\xi}|\mathbb{Q}_{\zeta|\xi}$
140-4-s164-D003 (349)

[scan 165]

[p. 730]

study in the case where $j|\zeta$ is injective

$$j|\zeta : \quad A \hookrightarrow \mathbb{G}_m(S) \ . \tag{350}$$

One can moreover, "divide" the diagram (346) a little further, by introducing

$$\Pi_1^+ = \Pi_1^+(S,\bar{s}) \overset{\text{def}}{=} \mathrm{Ker}(\Pi_1(S,\bar{s}) \xrightarrow{\text{cyclot.}} \widehat{\mathbb{Z}}^*), \tag{351}$$

and the homomorphism of exact sequences

$$\begin{array}{ccccccc}
1 & \longrightarrow & \Pi_1^+ & \longrightarrow & \Pi_1 & \xrightarrow{\chi_\Pi} & \widehat{\mathbb{Z}}^* \\
 & & \downarrow \gamma^+_{\zeta|\xi} & & \downarrow \gamma_{\zeta|\xi} & & \downarrow 1 \\
1 & \longrightarrow & \mathrm{Hom}_{\mathbb{Z}}(A,T) & \longrightarrow & \dagger G_{\zeta|\xi} & \longrightarrow & \widehat{\mathbb{Z}}^* .
\end{array} \tag{352}$$

One can consider that the study of the "size" of $\mathrm{Im}(\gamma_{\zeta|\xi} : \Pi_1 \longrightarrow G_{\zeta|\xi})$ is essentially reduced to that of the size of $\mathrm{Im}\,\chi_\Pi$ and of that of $\mathrm{Im}\,.\gamma^+_{\zeta|\xi}$; precisely, if we put

$$\mathrm{Im}\,\gamma_{\zeta|\xi} = \widetilde{\Pi}_1, \quad \mathrm{Im}\,\gamma^+_{\zeta|\xi} = \widetilde{\Pi}^+_1, \quad \mathrm{Im}\,\chi_\Pi = \dagger\Gamma \tag{353}$$

we will have an exact sequence

$$1 \longrightarrow \widetilde{\Pi}^+_1 \longrightarrow \widetilde{\Pi}_1 \longrightarrow \Gamma \longrightarrow 1\ . \tag{354}$$

I would like to try to give some 164

Uncertainty retained from cancelled passages.

140-4-s165-D002 $H|K$

140-4-s165-D003 $\widehat{\mathbb{Z}}^*|\widetilde{\Gamma}$

[scan 166]

indications on the size of $\widetilde{\Pi}^+_1 \underset{\zeta|\gamma}{\subset} \underbrace{\mathrm{Hom}(A,T)}_{\simeq \mathrm{Hom}(A,\widehat{\mathbb{Z}})}$ in the case where we assume

†noncanonical^t|noncanonical!

$$\begin{cases} 1°) & A \quad \mathbb{Z}\text{-module of finite type} \\ 2°) & S \quad \text{essentially}^{t|l} \text{ of finite type over } \mathbb{Q} \text{ and } \underline{\text{normal}} \\ 3°) & j|\zeta : A \longrightarrow \mathbb{G}_m(S) \text{ is injective } \dagger. \end{cases} \tag{355}$$

The hypothesis 2°) already implies as we have seen, that $\Gamma \subset \widehat{\mathbb{Z}}^*$ is an open subgroup. The hypothesis that S is normal implies moreover that the groups sought $\widetilde{\Pi}^+_1$ and Γ do not change on^h|on^t on replacing S by its generic point (we assume for simplicity that $\bar{s}$ is above the said generic point), so suppose that $S = \mathrm{Spec}\,K$, where K is a field extension of finite type of $\mathbb{Q}$. And one can also assume, if one cares to, that S is affine, and smooth over $\mathbb{Q}$....

Let us also note, putting

$$\check{A} = \mathrm{Hom}_{\mathbb{Z}}(A, \mathbb{Z}), \tag{356}$$

165

Uncertainty retained from cancelled passages.

140-4-s166-D001 [*illeg.* 1]

140-4-s166-D002 pure

[scan 167]

that $\check{A}$ is a free $\mathbb{Z}$-module of finite type (of rank n equal to $\mathrm{rg}_{\mathbb{Z}}\,A$), and that we have a [p. 731] canonical isomorphism

$$\mathrm{Hom}_{\mathbb{Z}}\left(A, \underbrace{T}_{\substack{\simeq\widehat{\mathbb{Z}} \\ (\text{noncanonical}^t|\text{noncanonical!})}}\right) \simeq \underbrace{\check{A} \otimes_{\mathbb{Z}} T}_{\check{A}(-1)} \quad \left(\begin{matrix} \simeq \widehat{\mathbb{Z}}^n \\ \text{noncanonical}^t|\text{noncanonical!} \end{matrix}\right) \tag{357}$$

I would be inclined to think that the image † $\widetilde{\Pi}^+_1$ of Π_1 in $\mathrm{Hom}_{\mathbb{Z}}(A,T) \simeq \check{A} \otimes_{\mathbb{Z}} T$ is $\underline{\text{open}}$. I even wonder, in the case S_1 normal 0-connected of finite type over $\mathbb{Z}$, dimension $2| \geq 2$ and $A = H^0(S_1, \mathbb{G}_m)$, putting $S = S_{1\mathbb{Q}}$, whether the homomorphism

$$\Pi_1 = \Pi_1(S, \bar{s}) \longrightarrow \mathrm{Hom}(A,T) \simeq \check{A}(-1)$$

is not necessarily surjective??

(More generally, this may be so if $\mathbb{G}_m(S)/\,\mathrm{Im}\,A$ is torsion-free, a case to which one should always be able to reduce, under the hypothesis (355), at the cost of enlarging A to $A' \subset H^0(S, \mathbb{G}_m)$, to|or an A' such that A'/A is finite...). 166

Uncertainty retained from cancelled passages.
140-4-s167-D001 of

[scan 168]

I will return to the case $S = \operatorname{Spec}\mathbb{Q}$, then †

$$\mathbb{G}_m(S) = \mathbb{G}_m(\mathbb{Q}) = \mathbb{Q}^* \simeq \{\pm 1\} \times \mathbb{Z}^{(\mathbb{P})}$$

where $\mathbb{P}$ is the set of prime numbers,

$$\Pi_1 = \operatorname{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) = \Gamma_{\mathbb{Q}}.$$

The question is thus to study the homomorphisms

$$\xi|\zeta_p : \Gamma_{\mathbb{Q}}^+ \longrightarrow \widehat{\mathbb{Z}}(-1) \tag{358}$$

associated to the $p \in \mathbb{P}$, and in particular to answer the questions

a) are the $\xi|\zeta_p$ surjective, or at least is the image an open subgroup?

b) are the $\xi|\zeta_p$ linearly independent over $\widehat{\mathbb{Z}}$? More precisely, if $p_1 \dots p_n$ are distinct prime numbers, is the homomorphism

$$\Gamma_{\mathbb{Q}}^+ \xrightarrow{(\xi|\zeta_{p_1},\dots,\xi|\zeta_{p_n})} \widehat{\mathbb{Z}}^n(-1|1)$$

surjective, or at least is the image of $\Gamma_{\mathbb{Q}}^+$ an open subgroup?

I will start by looking a little at question a) (the case at which we were in the context of Fermat was the one where $p = 2$). For this, we decompose $\xi|\zeta_p$ into its components

$$\xi|\zeta_{p,\dagger\ell} : \Gamma_{\mathbb{Q}}^+ \longrightarrow \widehat{\mathbb{Z}}_\ell(-1|1) \qquad (\ell \in \mathbb{P}) \tag{359}$$

167

Uncertainty retained from cancelled passages.
140-4-s168-D001 Gal
140-4-s168-D002 ℓ

[scan 169]

I would therefore like to prove (at least!) that these are surjective homomorphisms, or at least surjective except for a finite number of ℓ, and with open image for $\forall\ell$. To study $\xi|\zeta_{p,\ell}$ is to restrict oneself to the tower of isogenies [p. 732]

$$1 \longrightarrow \mu_n \longrightarrow \mathbb{G}_m \xrightarrow{\lambda\mapsto\lambda^n} \mathbb{G}_m \longrightarrow 1 \tag{339}$$

for n a power of ℓ

$$n = \ell^\nu, \tag{360}$$

and study †the tower of inverse images of $p \in \mathbb{G}_m(\mathbb{Q})$ under its isogenies. It is natural then to look at "the" decomposition group †

$$D_\ell = | \simeq \Pi_1(\mathbb{Q}_\ell, \overline{\mathbb{Q}}_\ell) \hookrightarrow \Gamma_{\mathbb{Q}} \tag{361}$$

and the cyclotomic ℓ-adic character induces

$$D_\ell \xrightarrow{\chi_\ell} \mathbb{Z}_\ell^* , \tag{362}$$

or better, the inertia subgroup

$$I_\ell \subset D_\ell \tag{363}$$

and the character induced by χ_ℓ on I_ℓ. Indeed we will have

$$I_\ell^+ \overset{\text{def}}{=} \operatorname{Ker}(\chi_\ell : I_\ell \longrightarrow \mathbb{Z}_\ell^*) \subset \Gamma_{\mathbb{Q}}^+ \tag{364}$$

for the $\chi_{\ell'}$ $(\ell' \neq \ell)$ are trivial on I_ℓ (they are not trivial however on D_ℓ!). 168

Uncertainty retained from cancelled passages.
140-4-s169-D001 the|the
140-4-s169-D002 (360)|(359)

[scan 170]

We are therefore going to look at the restriction

$$\xi|\zeta_{p,\ell} \mid I_\ell^+ : I_\ell^+ \longrightarrow \widehat{\mathbb{Z}}_\ell(-1) \tag{365}$$

and one can already ask whether this homomorphism is surjective, or whether at least its image is open|open?

= |–
169
[scan 171]
[p. 733]

16°) Reflections on Fermat's theorem.
"Fermat's theorem" can be stated by saying that for $n \geq 3$, the affine Fermat curve over $\mathbb{Q}$

$$\begin{aligned} U_n &\overset{\text{def}}{=} \left\{ (x_0, x_1, x_\infty)^{\cdot} \in \Gamma\mathbb{P}^2 \;\middle|\; \begin{array}{c} x_0, x_1, x_\infty \text{ inv.} \\ x_0^n + x_1^n + x_\infty^n = 0 \end{array} \right\} \\ &\simeq \left\{ (x_0, x_1|a_1, x_\infty|a_\infty)^{\cdot} \in \Gamma\mathbb{G}_m^{\dagger\{0,1,\infty\}}/\mathbb{G}_m \;\middle|\; x_0^n + x_1^n + x_\infty^n = 0 \right\} \end{aligned} \tag{366}$$

has no † rational point /$\mathbb{Q}$. We had introduced †the "tower" of Fermat curves in §11°) (page 696 and following), and in 14°) there appeared the description of U_n as the inverse image of

$$\begin{array}{c} U_1 \subset \overbrace{\mathbb{G}_{m_{\mathbb{Q}}} \otimes_{\mathbb{Z}} M}^{T} \simeq \mathbb{G}_{m_{\mathbb{Q}}}^{\{0,1,\infty\}}/\mathbb{G}_{m_{\mathbb{Q}}} \\ \| \\ \left\{ (x_0, x_1, x_\infty)^{\cdot} \in \Gamma\mathbb{G}_m^{\{0,1,\infty\}}/\mathbb{G}_m \;\middle|\; x_0 + x_1 + x_\infty = 0 \right\} \end{array} \tag{367}$$

under the Kummer isogeny of index n

$$1 \longrightarrow \mu_n \otimes_{\mathbb{Z}} M \longrightarrow T \xrightarrow{x \mapsto x^n} T \longrightarrow 1 \ . \tag{368}$$

Let us calculate the genus g_n of $\widehat{U}_n$, which is a covering of degree n^2 of $\widehat{U}_1 \underset{*|\neq}{\simeq} \widehat{\mathcal{U}|\mathcal{M}}\dagger_{0,3}$, with† ramificationof degree n at the points above $0, 1, \infty$, and étale elsewhere. The Hurwitz formula 170

Uncertainty retained from cancelled passages.
140-4-s171-D001 3
140-4-s171-D002 solution|solutions
140-4-s171-D003 this
140-4-s171-D004 $\mathbb{Q}|0, 3$
140-4-s171-D005 totally

[scan 172]

gives
†

$$\underbrace{\underbrace{(2-2g_n)}_{\text{EP of } \widehat{U}_n} - 3n}_{\text{EP}_! \text{ of } U_n} = n^2 \left(\underbrace{\underbrace{2}_{\text{EP of } \widehat{U}_1} \quad -3}_{\text{EP of } U_1} \right)$$

i.e.

$$2(g_n - 1) = n^2 - 3n \qquad \dagger \qquad \text{i.e.}$$

$$(369) \qquad g_n = \dagger 1 + \frac{1}{2}n(n-3)$$

So we have

$$(370) \qquad g_1 = g_2 = 0, \quad g_3 = 1, \quad g_n > 1 \quad \text{if } n \geq \dagger 4.$$

In any event, all the U_n are anabelian, and even the $\widehat{U}_n$ for $n \geq 4$.

If the "anabelian" yoga developed in the first part of these notes is valid, the † rational/$\mathbb{Q}$ points of a curve U_n † correspond to the splittings of the extension of profinite groups
(371)

$$1 \to \Pi_1(U_{n\bar{\mathbb{Q}}}) \to \underset{\substack{\cap \\ \Pi_1(U_1)\simeq M^!_{0,3}}}{\Pi_1(U_n)} \to \Gamma_{\mathbb{Q}} \to 1$$

which are distinct from the "splittings at infinity" (localized at the 3n points at infinity of U_n…), or again, the splittings such that the centralizer of $(\Gamma^+_{\mathbb{Q}} = \mathrm{Ker}(\Gamma_{\mathbb{Q}} \overset{\chi}{\to} \widehat{\mathbb{Z}}^*))$ in $\Pi_1(U_n)$ is reduced to $\{1\}$. So – if the "theorem

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Uncertainty retained from cancelled passages.

140-4-s172-D001 $g_n \geq 3$

140-4-s172-D002 $2 - 2g_n = -n^2$

140-4-s172-D003 2+

140-4-s172-D004 3

140-4-s172-D005 existence

140-4-s172-D007 are explained by|are expressed by

[scan 173]

of Fermat" is indeed valid (for U_n, say), it should be reflected|reflected in the nonexistence [p. 734] of such a splitting. Besides, if we succeeded in proving the nonexistence of splittings of (371) other than those "coming from the points at infinity", the theorem of Fermat would be proved for U_n, even without having established the "anabelian yoga" (a proof of which seems quite out of reach!). Now it is by no means excluded that a sufficiently thorough analysis of $M_{0,3}$ and $N_{1,1}$ in the present direction – also involving finer and finer "approximations" of $\Pi_1(U_n)$ and of the extension structure (371), might eventually make it possible to prove the nonexistence of "nontrivial" splittings of (371), for all $n \geq 3$. Each new refinement of the "Lie approximations" of $M_{0,3}$ should lead to a reexamination of the situation (371) and of the question of its splittings, with a view to a possible proof of the "theorems of Fermat".

[scan 174]

It is imperative to start from Lie approximations at least as fine as those|that of the present §, which appears as the "minima" approximation in which one can express "algebraically" the tower of Fermat curves, † as we saw in 13°), starting from $M_{0,3} \subset M^{\text{ét}}_{0,3}$.

Taking into account the operations of $\mathfrak{S}_3$ on U_n, the exact sequence (371) moreover embeds in a richer exact sequence

(372)

$$1 \to \Pi_1(U_{n\overline{\mathbb{Q}}}) \to \underset{\Pi_1(U_1,\mathfrak{S}_3)\simeq M_{0,3}}{\Pi_1(U_n,\mathfrak{S}_3)} \to \Gamma_{\mathbb{Q}} \times \mathfrak{S}_3 \to 1,$$

and the calculations made in the present § seem to suggest that this richer extension structure will play an important role.

From the point of view of the Lie approximation of $M_{0,3}$ developed in the present §, it is appropriate to replace the extension structure (372) by a less fine quotient structure, by dividing by an invariant subgroup of $\Pi_1(U_n, \mathfrak{S}_3)$ contained in

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Uncertainty retained from cancelled passages.

140-4-s174-D001 where

[scan 175]

$\Pi_1(U_n, \overline{\mathbb{Q}})$, namely the intersection in $M_{0,3}$ of this group with $[\widehat{\Pi}_{0,3}, \widehat{\Pi}_{0,3}]$. [p. 735]

We find a commutative diagram

(373)

$$\begin{array}{ccccccccc}
1 & \to & \widehat{M}[n] & \to & \overbrace{\Pi_1(U_n,\mathfrak{S}_3)^{\sim}}^{\mathcal{E}_n} & \to & \Gamma_{\mathbb{Q}} \times \mathfrak{S}_3 & \to & 1 \\
 & & \downarrow{\scriptstyle 1} & & \downarrow{\scriptstyle \dagger}\ \downarrow{\scriptstyle \text{epimorphism}} & & \downarrow{\scriptstyle \chi\times\mathrm{id}} & & \\
1 & \to & \widehat{M}[n] & \to & \widetilde{G}_{\rho,\sigma}[n] & \to & \widehat{\mathbb{Z}}^* \times \mathfrak{S}_3 & \to & 1 \\
 & & \downarrow & & \downarrow & & \downarrow{\scriptstyle 1} & & \\
1 & \to & \widehat{M} & \to & \widetilde{G}_{\rho,\sigma} & \to & \widehat{\mathbb{Z}}^* \times \mathfrak{S}_3 & \to & 1
\end{array}$$

where the image of $\widehat{M}[n] \to \widehat{M}$ is $n\widehat{M}$.

Here is † a way of constructing "algebraically" the diagrams (373) (for variable n) from the diagram
(374)

$$\begin{array}{ccccccccc}
1 & \to & \widehat{M} & \to & \overbrace{\Pi_1(U_1,\mathfrak{S}_3)^{\sim}}^{\mathcal{E}_1} & \to & \Gamma_{\mathbb{Q}}\times\mathfrak{S}_3 & \to & 1 \\
 & & \downarrow{\scriptstyle 1} & & \| & & \downarrow{\scriptstyle \chi\times\mathrm{id}} & & \\
 & & & & (M_{0,3}/[\widehat{\Pi}_{0,3},\widehat{\Pi}_{0,3}]) & & & & \\
 & & & & \downarrow & & & & \\
1 & \to & \widehat{M} & \to & \widetilde{G}_{\rho,\sigma} & \to & \widehat{\mathbb{Z}}^*\times\mathfrak{S}_3 & \to & 1. \\
 & & & & \simeq(\widehat{\mathbb{Z}}^*\times\mathfrak{S}_3).\widehat{M} & & & &
\end{array}$$

We choose a † splitting of the first i.e. a splitting of the second (two such splittings are conjugate by a unique element of $\widehat{M}$…). m1629

Marginal note. m1629 We have seen that these splittings correspond|correspond † bijectively to those of the extension $\mathfrak{S}^{+\wedge}/\mu$ of $\mathfrak{S}_3$ by $\widehat{M}$, and we have defined a canonical one † 12°).

† It can be interpreted as a geometric point of $U_\infty|U_n$ above the

Uncertainty retained from cancelled passages.
140-4-s175-D001 $\widetilde{G}_{\rho,\sigma}[n]$
140-4-s175-D002 [illeg. 1]
140-4-s175-D003 which
140-4-s175-D004 being
140-4-s175-D005 in
140-4-s175-D006 This is moreover

[scan 176]

base point $-\bar{j}$ of $U_{0,3\mathbb{Q}}\simeq U_{1\mathbb{Q}}$ – it defines a geometric point of each of the $U_{n\mathbb{Q}}$ above $-\bar{j}$. In terms of this splitting of $\mathcal{E}_1$, allowing us to write

$$(375)\qquad \mathcal{E}_1\simeq(\Gamma_{\mathbb{Q}}\times\mathfrak{S}_3).\widehat{M}\qquad(\tfrac{1}{2}\text{ direct product}),$$

the $\underset{\wr|}{\mathcal{E}_n}$ $\subset\mathcal{E}_1$ are identified with the semidirect products
$\Pi_1(U_n,\mathfrak{S}_n|\mathfrak{S}_3)/\ldots$

$$(376)\qquad \mathcal{E}_n=(\Gamma_{\mathbb{Q}}\times\mathfrak{S}_3).\underbrace{(n\widehat{M})}_{\overset{\text{definition}}{=}\widehat{M}[n]}$$

(the interpretations are at the level of generalities of the proposition-lemma on page 712′ ff, and would deserve to appear in the scholium|scholium-proposition in question…)

A points|point x_n rational|rational/$\mathbb{Q}$ of U_n defines a splitting of the extension (of $\Gamma_{\mathbb{Q}}$ by $n\widehat{M}=\widehat{M}[n]$)

$$(377)\qquad \underset{\varphi}{\mathcal{E}_n^!}\simeq\Gamma_{\mathbb{Q}}.(n\widehat{M})\qquad(\tfrac{1}{2}\text{ direct})$$

quotient of (371), knowledge of which

[scan 177]
[p. 736]

modulo † $\widehat{M}[n]$-conjugation implies that of the corresponding splitting of $\mathcal{E}_1$, hence implies knowledge of the corresponding point x_1 of $U_1(\mathbb{Q})$ †, which means that x_n is known modulo the operation of $\gamma_n = \mu_n(\mathbb{Q})^{\{0,1,\infty\}}/\mu_n(\mathbb{Q})$. Now we have

$$\gamma_n \overset{\text{dfn}}{=} \mu_n(\mathbb{Q})^{\{0,1,\infty\}}/\mu_n(\mathbb{Q}) = 1 \quad \text{if } n \text{ odd} \tag{378}$$

for $\mu_n(\mathbb{Q}) = 1$ if n odd – so in this case x_n is known when one knows the corresponding splitting, and even only its $\widehat{M}[n]$-conjugacy class.

This leads us to determine all these possible $\widehat{M}[n]$-conjugacy classes of such splittings, in order to examine (if it can be done) which ones can reasonably correspond to an $x_n \in U_n(\mathbb{Q})$. These classes correspond to the elements of

$$H^1(\Gamma_{\mathbb{Q}}, n\widehat{M}) \simeq H^1(\Gamma_{\mathbb{Q}}, \widehat{M}) \simeq \underbrace{H^1(\Gamma_{\mathbb{Q}}, \widehat{\mathbb{Z}}(-1))}_{(\mathbb{Q}^*)^\wedge|(\mathbb{Q}^*]^\wedge} \otimes_{\mathbb{Z}} M \simeq G(\mathbb{Q})^\wedge \tag{379}$$

Uncertainty retained from cancelled passages.

140-4-s177-D001 n

140-4-s177-D002 x_1

[scan 178]

where

$$G \overset{\text{dfn}}{=} \mathbb{G}_{m\,\mathbb{Q}}|G_{m\,\mathbb{Q}}{}^{\{0,1,\infty\}}/\mathbb{G}_m|G_m$$

is the torus of dimension 2 over $\mathbb{Q}$ in which the U_n are embedded. We see that everything happens as if we were studying simply the existence (!) and the cohomological locating of the $\mathbb{Q}$-rational points of G itself – without concerning ourselves with the equation $x_0^n + x_1^n + x_\infty^n = 1$ which defines U_n. This shows plainly that the approximation on which we are working is much too coarse – in fact, in practice we remain in the domain of a typically "abelian" analysis. It is clear that it is necessary to pass to finer approximations.

[scan 179]

52 Schematic envelopes of|of the L-groups. [p. 737]

1°) Notion of L-group, of LT-group (discrete case).

The notion of L-group is the natural generalization of that of "group with loops", which is called for if we want to be able to encompass mixed fundamental groups of orientable surfaces (of "finite character", i.e. homeomorphic to an $X \setminus S$, X a compact surface, S a finite subset) with a finite group of automorphisms.

We have seen that if Π is a group with loops (discrete), then every subgroup of finite index Π_0 inherits a loop structure, taking as "loop subgroups" of Π_0 the intersections with Π_0 of the loop subgroups of Π. [N.B. we assume that we are not in the cases (g, ν) † with $g = 0$, $\nu \in \{0, 1\}$, where the loop structure is not determined by the sole datum of a family of conjugacy classes of subgroups † ($\simeq \mathbb{Z}$) – the only case that would cause a problem here

Uncertainty retained from cancelled passages.

140-4-s179-D001 with

140-4-s179-D002 discrete

[scan 180]

is the case $(0, 1)$ of $\Pi \simeq \mathbb{Z}$ – in the case $(0, 0)$, we have $\Pi = \{1\}$ and $\Pi_0 = \Pi$ ...]. Conversely, the loop structure of Π is recovered from that of $\Pi_{0*}|\Pi_0$ (when it exists!), the loop subgroups of Π being the centralizers in Π of the loop-subgroups of Π_0.

Let Π be a discrete group, and let $\mathcal{L}_0(\Pi)$ be the set of group-with-loops structures on Π, of type (g, ν) †anabelian i.e. $\neq (0, 0|1), \dagger(0, 1), \dagger(1, 0)$. For the subgroups Π_i of Π of finite index, the $\mathcal{L}_0(\Pi_i)$ form an inductive system † of sets. If Π_i is normal,

†

we † set|take

$$(1) \qquad \mathcal{L}_0^{\natural|\sharp}(\Pi_i) = \text{set of loop structures on } \Pi_i \text{ invariant under } G_i = \Pi/\Pi_i.$$

† If there exists a Π_{i_0} of finite index such that $\mathcal{L}(\Pi_{i|i_0}) \neq \emptyset$, then Π_{i_0} hence also Π admits a finite generating set, and it follows that the set of normal subgroups of finite index is cofinal in the set of subgroups of finite index. We shall then set

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Uncertainty retained from cancelled passages.

140-4-s180-D004 with|≠

140-4-s180-D005 $(0, 2)$

140-4-s180-D006 $(1, 0)$

140-4-s180-D008 with a map|with maps

140-4-s180-D010 $\mathcal{L}(\Pi_i)|\mathcal{L}_0(\Pi_i)$

140-4-s180-D011 can

140-4-s180-D013 that hence a system

[scan 181]
[p. 738]

$$(2) \qquad \mathcal{L}(\Pi) = \varinjlim_{\substack{\Pi_i \text{ normal} \\ \text{of finite index} \\ \text{in } \Pi}} \mathcal{L}_0^{\natural|\sharp}(\Pi_i).$$

m1630

Marginal note. m1630 N.B. If Π_i is not normal in Π, we can define $\mathcal{L}_0^{\natural|\sharp}(\Pi_i)$ as the subset of $\mathcal{L}(\Pi_i)$ formed by the loop structures λ "invariant under Π", i.e. such that for $\forall g \in \Pi$, we have that λ (on Π_i) and $\mathrm{int}(g)|\mathrm{Int}(g)\lambda$ (on $\mathrm{int}(g)|\mathrm{Int}(g)(\Pi_i)$) coincide in $\Pi_i \cap \mathrm{Int}(g)|\mathrm{int}(g)(\Pi_i)$ (which is of finite index in Π ...)

the canonical morphisms (Π_i normal of finite index in Π)

$$(3) \qquad \mathcal{L}_0^{\natural|\sharp}(\Pi_i) \longrightarrow \mathcal{L}(\Pi)$$

are injective. If $\lambda \in \mathcal{L}(\Pi)$, we say that λ is an L-structure. on Π, (Π, λ) is called an L-group, (and if $\lambda \in \mathrm{Im}(\mathcal{L}_0^{\natural|\sharp}(\Pi_i) \to \mathcal{L}(\Pi))$), we say that Π_i is a subgroup of definition of the L-group Π – we regard it as a group with loops via the unique $\lambda_i \in \mathcal{L}_0^{\natural|\sharp}(\Pi_i)$ giving rise to λ.

Let Π be a discrete group, $\Pi'_0|\Pi_0$ a subgroup of Π of finite index. Then we have a canonical homomorphism for induction of L-structures

$$(4) \qquad \mathcal{L}|\mathcal{L}_0(\Pi) \hookrightarrow \mathcal{L}(\Pi')$$

which is injective, and whose image consists of the set of L-structures on Π' that are

[scan 182]

"invariant under Π" in an obvious sense. In particular, if we have an extension of a finite group G by a discrete group Π_0, then the L structures on Π correspond to the L-structures

on Π_0 invariant under Π/Π_0†$\longrightarrow$ Autext(Π_0), operating on the set of L-structures of Π in an obvious way. [N.B. $\Pi \mapsto \mathcal{L}(\Pi)$ is a functor in the outer groups Π, with outer isomorphisms as arrows ...]

Standard example: let X be an anabelian surface "of finite character" 0-connected, G a finite group operating on X,

$$\Pi = \Pi_1(X, G; \bullet) \tag{†5}$$

where a is a base point of X, so that we have an exact sequence

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \Pi_0 & \longrightarrow & \Pi & \longrightarrow & G & \longrightarrow & 1. \\ & & \| & & & & & & \\ & & \Pi_1(X, \bullet) & & & & & & \end{array} \tag{6}$$

We then have on Π_0 a loop structure

Uncertainty retained from cancelled passages.

140-4-s182-D001 $\simeq G$

140-4-s182-D002 4

[scan 183]

invariant under the outer operations of G, hence an L-structure on Π. It seems that $\forall$ L-group [p. 739] is isomorphic to a group obtained this way — but this is a result that seems deep (essentially equivalent to that of|of the the existence of fixed points of finite subgroups of Teichmüller groups ...), and which is not completely understood. We should bring out a general "good statement", linking the 2-category of L-groups, with that of surfaces X of finite character, † equipped with a|with a "signature" $\underline{d}$ with finite† support (with values in $\mathbb{N}^*$) punctured, by associating to such a surface its $\Pi_1(X, \underline{d}, a)$, relative to a point $a \in X - \operatorname{supp} \underline{d}$. Excluding a few singular cases "trivial", we should then find an L-group Π, isomorphic to a $\Pi_1(X; G, a)$,|; with G operating faithfully, which should be expressed

Uncertainty retained from cancelled passages.

140-4-s183-D001 by some

140-4-s183-D002 e

[scan 184]

in terms of Π by the fact that the centralizer of $\forall$ subgroup of finite index in Π is reduced to $\{e\}$. This|This should give a complete intrinsic description, in topological terms, of the L-groups having this additional property. There would then still be some work of fitting together to do, in terms of certain extensions of such L-groups by finite groups, to treat|find the general case. Moreover, we should make the connection with the associated "L-Galois topoi", paraphrasing the corresponding developments on Galois topoi etc. associated to groups with loops.

[scan 185]

It could be that the "right" general context for defining and studying general Teichmüller [p. 740] groups is that of L-groups, rather than that of groups with loops.

Taking for example the group †

$$\begin{aligned} \dagger\mathfrak{S}^+_{0,3} &\simeq \mathrm{Sl}(2, \mathbb{Z})/\{\pm 1\} \\ &\simeq \Pi_1(U_{0,3}(\mathbb{C}), \mathfrak{S}_3; -\bar{j}), \end{aligned} \tag{4}$$

this is equipped with an L-structure, which can be defined in terms of the unique conjugacy class of the subgroup L_0 generated by $\varepsilon_0 = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$, or of L_1 generated by $\varepsilon_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$. The L-group (discrete) has the remarkable property that every L-automorphism "direct" (inducing the identity on the orientation module) of it is inner. It could be that this property, added to the property that the center is trivial (or better, perhaps, that the centralizer

Uncertainty retained from cancelled passages.
140-4-s185-D001 $\mathfrak{M}|\mathfrak{S}$
140-4-s185-D002 $\mathfrak{M}^+_{0,3}|\mathfrak{S}^+_{0,3}$

[scan 186]

of every subgroup of finite index is reduced to $\{1\}$) characterizes $\mathfrak{S}^+_{0,3}$ up to (nonunique) isomorphism.

N.B. To prove that every direct L-automorphism|automorphism u of $\dagger\mathfrak{S}^+_{0,3}$ is inner, the essential point seems to me to be to prove that u preserves the subgroup $\Pi_{0,3}$, kernel of

$$1 \longrightarrow \Pi_{0,3} \longrightarrow \dagger\mathfrak{S}^+_{0,3} \longrightarrow \mathfrak{S}_3 \longrightarrow 1.$$

In fact, every automorphism of the group structure of $\mathfrak{S}^+_{0,3}$ (not necessarily an L-automorphism|automorphism) preserves the normal subgroup $\Pi_{0,3}$. This follows from the

Lemma Let $\Pi \subset \mathfrak{S}^+_{0,3}$ be a normal subgroup such that $\mathfrak{S}^+_{0,3}/\Pi \simeq \mathfrak{S}_3$, then $\Pi = \Pi_{0,3}$. More precisely, every surjective homomorphism $p : \mathfrak{S}^+_{0,3}\dagger \longrightarrow \mathfrak{S}_3$ is conjugate to the canonical homomorphism (hence its kernel is $\Pi_{0,3}$).

Proof We use the presentation of $\mathfrak{S}^+_{0,3}$ by $\rho^3 = \sigma^2 = 1$,

Uncertainty retained from cancelled passages.
140-4-s186-D001 $\mathfrak{M}$
140-4-s186-D002 $\mathfrak{M}$
140-4-s186-D003 Π

[scan 187]

which implies that a homomorphism p from $\mathfrak{S}^+_{0,3}$ to another group, whose image is not of order $1, 2$ or 3 i.e. of order ≥ 4, sends ρ and σ resp. to elements of order 3 and of order 2. So if the image is $\mathfrak{S}_3$, ρ is sent to $\dot\rho$ or $\dot\rho^{-1}$ (we may suppose, conjugating if necessary by $\dot\sigma_\infty$, that it is $\dot\rho$) and σ is sent to $\dagger\ \dot\sigma_0, \dot\sigma_1$ or $\dot\sigma_\infty$ (we may suppose, conjugating by $\dot\rho$ or $\dot\rho^2$ if necessary, that it is $\dot\sigma_\infty$), and we win. [p. 741]

We may call the Teichmüller group of an L-group Π, the group of its outer L-automorphisms — or perhaps, better, the group of outer L-automorphisms of the subgroup Π^+ (of index 1 or 2) of Π, consisting of the $g \in \Pi$ such that the automorphism $\mathrm{int}(g)$ is direct. The property envisaged for $\mathfrak{S}^+_{0,3}$ is that its "direct" Teichmüller group is reduced to 1. The property

Uncertainty retained from cancelled passages.
140-4-s187-D001 $\dot\sigma$

[scan 188]

corresponding to this for the L-group $\mathfrak{S}_{0,3} \simeq \mathrm{Gl}(2,\mathbb{Z})/\pm 1$ is that every L-automorphism of the subgroup $\mathfrak{S}^+_{0,3}$ is induced by a unique element of $\mathfrak{S}_{0,3}$. This property is perhaps characteristic of the discrete L-group $\mathfrak{S}_{0,3}$.

I feel like presenting in a slightly different form the notion of L-groupΠ, in terms of the datum of a $\mathbb{Z}$-module T free of rank 1 (the $\mathbb{Z}$-module of orientations) and of a$\dagger$ finite set $(k_i^{\natural|\sharp})_{i\in I}$ of germs of homomorphisms that are outer

$$k_i^{\natural|\sharp} : \quad \dagger T \longrightarrow \Pi \qquad (i \in I) \qquad (5)$$
(germ of outer homomorphism)

,

where the notion of "germ" is relative to that of subgroups of finite index of T. For such a datum,

Uncertainty retained from cancelled passages.
140-4-s188-D002 e
140-4-s188-D004 $\mathbb{Z}$

[scan 189]
[p. 742]

and any subgroup † Π_0 of Π of finite index, we find for $\forall i$ a set of germs of outer homomorphisms

$$T \longrightarrow \Pi_0$$

induced by the actual homomorphisms † $nT \longrightarrow \Pi$ of the class $k_i^{\natural|\sharp}$ — we find a finite number of germs of outer homomorphisms $T \longrightarrow \Pi_0$ for fixed i, hence also for i varying in I. This said, the only axiom of an L-structure, it is that we can find a subgroup Π_0 of finite index, such that the "induced structure" comes from a loop structure on Π_0.

We can generalize the notion of L-structure, taking inspiration from the preceding definition (for example), and dropping the requirement there that Π_0 be of finite index — but while

Uncertainty retained from cancelled passages.
140-4-s189-D001 to
140-4-s189-D002 outer

[scan 190]

taking care on the other hand that Π_0 be normal. We require that for $\forall i \in I$, the germs of homomorphisms $k_i : \dagger T \longrightarrow \Pi$ of the class $k_i^{\natural|\sharp}$ factor through Π_0 — which gives a set of germs of outer homomorphisms from T to Π_0 — m1631

Marginal note. m1631 N.B. it is not clear a priori that this set of germs of outer homomorphisms is finite

and we require that this structure in question on Π_0 come (for suitable Π_0) from a loop structure on Π_0, ξ|& obviously uniquely determined, once Π_0 has been chosen. As for Π_0, it is defined up to commensurability — i.e. if Π_0, Π_0' will do, then $\Pi_0 \cap \Pi_0'$ is of finite index in Π_0 and Π_0', and will do as well.

I feel like (for want of a better suggestion) calling the groups Π equipped with such a

Uncertainty retained from cancelled passages.
140-4-s190-D002 *n*

[scan 191]
[p. 743]

structure – linked to a class of discrete subgroups commensurable with one another – of *L-T*-groups. (where L is the initial of "loops", and T of "Teichmüller") – these are sorts of mixtures of Teichmüller groups and L-groups. To illustrate this point, let us start from an extension

$$(6) \qquad 1 \longrightarrow \Pi \longrightarrow \mathcal{G} \longrightarrow \mathcal{T} \longrightarrow 1$$

where Π is equipped with an L-structure invariant under $\mathcal{G}$, then $\mathcal{G}$ is an L-T-group admitting (if Π is itself a †group with loops) Π as a subgroup of definition. Starting from the group with loops Π, one can take for example

$$(7) \qquad \begin{cases} \mathcal{G} = \mathrm{Aut}_L(\Pi) \\ \mathcal{T} = \mathrm{Autext}_L(\Pi) \end{cases} \qquad \text{whence}\,,$$

Uncertainty retained from cancelled passages.
140-4-s191-D002 *L-*

[scan 192]

and thus the ambient group $\mathcal{G}$, an extension of the Teichmüller group $\mathcal{T}$ by the group with loops Π, appears as an LT-group.

†

Uncertainty retained from cancelled passages.
140-4-s192-D001 2

[scan 193]
[p. 744]

2°) Profinite case

The extension of the notion of an L-group or of an LT-group Π, in terms of germs of outer homomorphisms

$$(8) \qquad k_i^{\natural|\sharp} : T \longrightarrow \Pi \ ,$$

is essentially evident. Here is the most important example for us. †

Let first k be a field , and let M be an † † algebraic multiplicity over k, which is 0-connected, of finite type over k, separated/k and smooth in|of relative dimension 1. One should be able to prove then that M can be described by a smooth and quasi-projective algebraic curve U over k, a finite group † G which operates on U, as

$$(9) \qquad M = (U, G) = \begin{array}{l}\text{(small) Ringed topos of}\\ \text{étale sheaves on}\\ U \text{ where } G \text{ operates ...}\end{array}$$

Uncertainty retained from cancelled passages.
140-4-s193-D001 Let
140-4-s193-D003 0-connected
140-4-s193-D004 0-connected
140-4-s193-D006 *K*

[scan 194]

† – More intrinsically, it should be true that there exists a finite étale covering of M (which one can suppose Galois, with group G) † which is an|a scheme – which will necessarily be quasi-projective|quasi-projective, smooth, of relative dimension 1 over k. Let us suppose in any case that this is so, choose a geometric base point a of M, and consider

$$(10) \qquad \Pi = \Pi_1(M, a) \simeq \Pi_1(U|M, G; a) \, .$$

We shall henceforth suppose k of characteristic zero. Let us first suppose k algebraically closed. Using the exact sequence

$$(11) \qquad 1 \longrightarrow \underbrace{\Pi_1(U, a)}_{\Pi_0} \longrightarrow \underbrace{\Pi_1(U|M, G; a)}_{\Pi} \longrightarrow G \longrightarrow 1$$

and the natural † loop structure on Π_0, one finds an L-structure on $\Pi = \Pi_1(M, a)$, which manifestly does not

Uncertainty retained from cancelled passages.

140-4-s194-D001 or

140-4-s194-D002 and|such

140-4-s194-D004 L

[scan 195]

depend on the choice of the finite Galois étale covering U of M, and on the chosen lifting of the geometric point a of M to a geometric point of U. We shall always tacitly equip $\Pi = \Pi_1(M, a)$ with this L-structure. [p. 745]

If again k is an arbitrary field (the most interesting case being the one where k is an extension of finite type of $\mathbb{Q}$), we shall have an exact sequence

$$(12) \qquad 1 \longrightarrow \Pi_1(M_{\overline{k}}, a) \longrightarrow \Pi_1(M, a) \longrightarrow \Gamma_{\overline{k}/k} \longrightarrow 1$$

and $\Pi_1(M_{\overline{k}}, a|o)$ appears as equipped with an L-structure invariant under $\Pi_1(M, a|o) = \Pi$, so that Π appears as a TL-group.

I am particularly interested in the "universal" case of type (g, ν) – but the natural base is not then the

[scan 196]

spectrum of a field k, but the|the modular † multiplicity $M_{g,\nu}$. [To tell the truth, for the discrete L-group structure Π_0, playing the role of reference group, one should be able to introduce a corresponding modular † multiplicity over $\mathbb{Q}$, and even over $\mathbb{Z}$. The fact which risks causing conceptual difficulties for such a construction, it is that a "modular" multiplicity (over $\mathbb{Q}$, or over $\mathbb{Z}$…) describes in principle a fibered category — such as that of relative curve schemes of type (g, ν) — whereas the relative 1|L-multiplicities corresponding to a given "type" Π_0 form a 2-category (in groupoids) fibered over (Sch), and not just a fibered category…] [p. †]

Uncertainty retained from cancelled passages.

140-4-s196-D001 745

140-4-s196-D002 scheme

140-4-s196-D004 scheme

[scan 197]

We shall therefore have an exact sequence [p. 746]

$$(13) \quad 1 \to \widehat{\Pi}_{g,\nu} \to \underbrace{\Pi_1(U|M_{g,\nu_{\mathbb{Q}}})}_{\mathcal{M}_{g,\nu}} \to \underbrace{\Pi_1(M_{g,\nu_{\mathbb{Q}}})}_{\mathcal{N}_{g,\nu}} \to 1,$$

where $U_{g,\nu_{\mathbb{Q}}}$ is the "universal curve of|of the type g, ν" (in characteristic 0). Here it is $\Pi_1(U_{g,\nu}\dagger|\xi_{\mathbb{Q}})$ which appears as a profinite TL-group.

I would like to bring out a property common to this "universal" case (13), † † and to the case 1-multiplicity|of 1 multiplicity [illeg. 1] relative to a base field k an extension of finite type of $\mathbb{Q}$ (the first case reducing moreover essentially to the second, by taking a "generic point" of $M_{g,\nu}$…). For this, let us consider an LT-group $\mathcal{M}|\mathcal{A}$, and a homomorphism

$$k_i : T \longrightarrow \dagger\,\mathcal{M}$$

belonging to one of the $k_i^{\natural|\sharp}$. One can

Uncertainty retained from cancelled passages.
140-4-s197-D001 $\mathbb{Q}$
140-4-s197-D002 and
140-4-s197-D003 is reduced essential-|essentially
140-4-s197-D005 $\mathcal{G}|\Pi$

[scan 198]

for example choose a normal subgroup of definition Π in $\mathcal{M}$, which will therefore be a group with loops, † an exact sequence

$$(14) \quad 1 \to \Pi \to \mathcal{M} \to \mathcal{N} \to 1,$$

and a homomorphism

$$(15) \qquad k : T \hookrightarrow \Pi \ , \qquad k(T) \overset{\text{dfn}}{=} L$$

relative to the loop structure of Π. Let us consider the normalizer of $L|T$ in $\mathcal{M}$

$$(16) \qquad \mathrm{Norm}_{\mathcal{M}}(L) \overset{\text{dfn}}{=} N(L)$$

and consider the action of $N(L)/L \subset \mathcal{N}$ † on L. (N.B. One readily sees that the image of $N(L)/L$ in $\mathcal{N}$ is an open subgroup of $\mathcal{N}$…) This action is by a "cyclotomic character", induced by a character

$$(17) \qquad \mathcal{N} \xrightarrow{\chi_{\mathcal{N}}} \widehat{\mathbb{Z}}^* \ .$$

The essential hypothesis is that the image

Uncertainty retained from cancelled passages.
140-4-s198-D001 and one will have|and one has
140-4-s198-D002 in

[scan 199]
[p. 747]

of $\mathcal{N}$ under χ (or also of $N(L)/L$ under χ) is an open subgroup of $\widehat{\mathbb{Z}}^*$. This hypothesis has a character intrinsic to the LT-group $\mathcal{M}$ itself (independently of the choice of Π) — for the cyclotomic character

$$(18) \qquad \chi_{\mathcal{M}} : \mathcal{M} \longrightarrow \widehat{\mathbb{Z}}^*$$

has a character intrinsic to $\mathcal{M}$, and the hypothesis means that the image of the latter in $\widehat{\mathbb{Z}}^*$ is open.

Returning to L and $N(L)$, one sees that the hypothesis made|made implies this:

(19) Every open subgroup of $N(L)/L$ operates nontrivially on L.

One knows, by standard|standard arguments that this implies that for every field K † (in particular an ℓ-adic field $\mathbb{Q}_\ell$),

Uncertainty retained from cancelled passages.
140-4-s199-D003 0

[scan 200]

and every homomorphism from $\mathcal{M}$ into a group $G(K)$; where G is an affine group scheme of finite type over K; †

$$u : \mathcal{M} \longrightarrow G(K) \ , \tag{20}$$

the restriction of u to L is "quasi-unipotent", i.e.

(21) $\exists$ open subgroup L' of L, such that $u(L') \subset G(K)$ is made up of unipotent elements,

— which amounts to the same thing, if ℓ_i is a generator of $L = L_i$, that one has

(22) $u(\ell_i)$ is a quasi-unipotent element of $G(K)$, i.e. $\exists\ n \in \mathbb{N}^*$, such that ℓ_i^n is unipotent.

Uncertainty retained from cancelled passages.
140-4-s200-D001 the|u

[scan 201]
[p. 748]

③ Schematic envelope of a T-L-profinite group : *a)* The admissible homomorphisms.

Let $\mathcal{M}$ be a profinite T-L-group, whose structural character (18)

$$\mathcal{M} \xrightarrow{\chi_{\mathcal{M}}} \widehat{\mathbb{Z}}^*$$

has an open image, Π a normal subgroup of definition (it is therefore a profinite †group with loops)|], and $\Pi_0 \subset \Pi$ a discrete subgroup of Π, which is a "discretification" — it is therefore a discrete group with loops, etc.|and Π is identified with its profinite completion…

Let $G_{\mathbb{Q}}$ be a group scheme affine of finite type over $\mathbb{Q}$. Let $\mathcal{A}_{\mathbb{Q}}$ be the ring of adeles of $\mathbb{Q}$

$$\mathcal{A}_{\mathbb{Q}} = \widehat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q} \subset \prod_{\ell \in \mathbb{P}} \mathbb{Q}_\ell \tag{23}$$

and let us consider †|the group of adelic points

$$G(\mathcal{A}_{\mathbb{Q}}) \subset \prod_{\ell \in \mathbb{P}} G(\mathbb{Q}_\ell). \tag{24}$$

If $G_{\mathbb{Q}}$ comes from a group scheme|scheme

Uncertainty retained from cancelled passages.
140-4-s201-D001 *L*-
140-4-s201-D002 the

[scan 202]

affine of finite type G over $\mathbb{Z}$, then we have

$$G(\mathcal{A}_{\mathbb{Q}}) \simeq (|\{(x_\ell)_{\ell \in \mathbb{P}} \in \prod_{\ell} G(\mathbb{Q}_\ell) \mid x_\ell \in G(\mathbb{Z}_\ell) \text{ for almost tt}|\forall \ell\} \tag{25}$$

†

We give ourselves in addition a homomorphism

$$(26)\quad G_{\mathbb{Q}} \xrightarrow{\chi_G} \mathbb{G}_{m\mathbb{Q}}|G_{m\mathbb{Q}}.$$

We are interested in †some group homomorphisms

$$(27)\quad \mathcal{M} \longrightarrow G(\mathcal{A}_{\mathbb{Q}})$$

satisfying properties which we are going to enumerate.

1°) Continuity. This is equivalent to saying that a) the homomorphisms $\mathcal{M} \longrightarrow G(\mathbb{Q}_\ell)$ are continuous|continuous, and b) $\exists S$ a finite subset of $\mathbb{P}$, such that for $\ell \notin S$, one has † a factorization (continuous)

$$(28)\quad \mathcal{M} \longrightarrow G(\widehat{\mathbb{Z}}_\ell) \qquad (\ell \in \mathbb{P} \setminus S)$$

— condition which indeed does not depend on the chosen "model over $\mathbb{Z}$" G of $G_{\mathbb{Q}}$.

N.B. (28) is a homomorphism of profinite groups.

Uncertainty retained from cancelled passages.

140-4-s202-D001 a|one

140-4-s202-D002 a|these

140-4-s202-D003 [illeg. 1]

[scan 203]

ReN.B. Unless I am mistaken, property 1°) means that one can find an affine group scheme G of finite type over $\mathbb{Z}$, which defines|defines $G_{\mathbb{Q}}$, and such that (27) factors through $G(\widehat{\mathbb{Z}})$. [p. 749]

$$(29)\quad \dagger\quad \mathcal{M} \longrightarrow G(\widehat{\mathbb{Z}}).$$

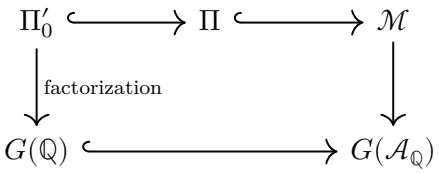
2°) Compatibility with χ :

(30)

$$\begin{array}{ccc} \mathcal{M} & \longrightarrow & G(\mathcal{A}_{\mathbb{Q}}) \\ {\scriptstyle \chi_{\mathcal{M}}}\searrow & & \swarrow{\scriptstyle \chi_G} \\ \widehat{\mathbb{Z}} & \hookrightarrow & \mathbb{G}_m|G_m(\mathcal{A}_{\mathbb{Q}}) \simeq \mathcal{A}_{\mathbb{Q}}^* \subset \prod_{\ell\in\mathbb{P}} \mathbb{Q}_\ell^* \end{array}$$

is commutative.

3°). Compatibility with the discretification Π_0.
(31)



, for a suitable subgroup of finite index Π_0' of Π_0.
N.B. Changing notation if need be we may assume|we may assume $\Pi_0' = \Pi_0$.
4°) Minimality (irredundancy|irredundancy) :
If $G'_{\mathbb{Q}}$ is a subgroup scheme of

Uncertainty retained from cancelled passages.
140-4-s203-D001 *G*

[scan 204]

$G_{\mathbb{Q}}$, such that † $\mathcal{M} \longrightarrow G(\mathcal{A}_{\mathbb{Q}})$ factors through $G'(\mathcal{A}_{\mathbb{Q}})$, then $G = G'$.
N.B. If this hypothesis is not satisfied, then among all the $G'_{\mathbb{Q}} \subset G_{\mathbb{Q}}$ which violate it, there is a (unique) smallest one, and on replacing χ_G by $\chi_{G'} = \chi_G \mid G'_{\mathbb{Q}}$ ..., the hypotheses 1°) to 4°) are satisfied for $G'_{\mathbb{Q}}, \chi'$, and $\mathcal{M} \longrightarrow G'(\mathcal{A}_{\mathbb{Q}})$.
Remark. Let $\mathcal{G}_0 \subset \mathcal{M}$ be the subgroup of $\mathcal{M}$ formed of the $g \in \mathcal{M}$ which are such that $g(\Pi_0)|g\ell(\Pi_0)$ and Π_0 are commensurable. That is|That is the g which normalize the "discretification germ" Π_0 of Π.
m1632

Marginal note. m1632 $\mathcal{G}_0/\Pi_0$ plays the role of a discrete Teichmüller group

It is possible that condition 3° will have to be strengthened, by requiring that we have

$$\mathcal{G}_0 \longrightarrow G(\mathbb{Q}).$$

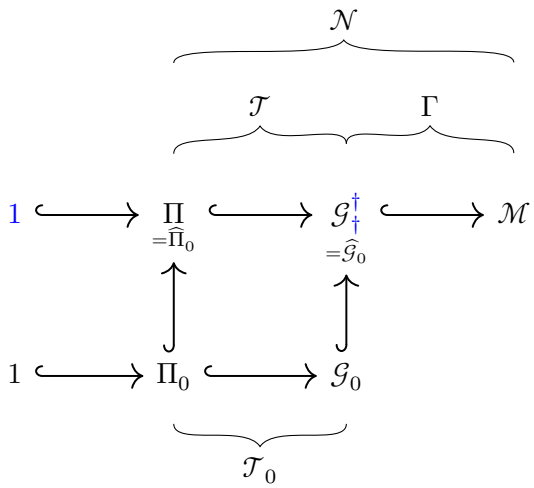
The future will decide it!

Uncertainty retained from cancelled passages.
140-4-s204-D001 *G*

[scan 205]
[p. 750]

Reflection done|done, I prefer to put things in a slightly different|different way, by introducing, in the profinite LT-group $\mathcal{M}$, a normal subgroup † Π with an L-structure (not necessari^{ly}|necessarily a loop structure — for example one can have $\Pi = \mathrm{SL}(2, \mathbb{Z})$) defining the LT-structure of $\mathcal{M}$, a discretification Π_0 of Π (in an obvious sense, for L-structures) a discrete subgroup $\mathcal{G}_0 \supset \Pi_0$ normalizing Π_0 (N.B. in important special cases will be $\mathcal{G}_0 = \Pi_0$), such that its closure $\mathcal{G}_0^{\dagger} \overset{\text{dfn}}{=} \mathcal{G}$ in $\mathcal{M}$ is normal in $\mathcal{M}$.

We thus have a diagram of groups
(32)



,

where the first row is formed of profinite groups, the second of discrete groups. We will note that $\mathcal{G}$ is identified with a quotient of the profinite completion of $\mathcal{G}_0$, but

---

Uncertainty retained from cancelled passages.

140-4-s205-D001 structure|structure

140-4-s205-D002 ∧|∗

140-4-s205-D003 ∧|∗

140-4-s205-D004 0

[scan 206]

I do not require for the moment that it be isomorphic to this completion itself. Thus idem|item for $\mathcal{T}$ and $\mathcal{T}_0$. The group $\mathcal{T}_0 = \mathcal{G}_0/\Pi_0$ plays the role of a discrete "direct" Teichmüller group, the group $\mathcal{T}$ that of a completion of a Teichmüller group, the group $\Gamma = \mathcal{M}/\mathcal{G}$ plays the role of a profinite Galois group. † I am interested mainly in two extreme situations

a) $\mathcal{G}_0 = \Pi_0$ whence $\mathcal{G} = \Pi$ — situation coming from a smooth relative 1-multiplicity over a field extension of finite type of $\mathbb{Q}$, or more generally over a scheme (or a multiplicity) integral of finite type over $\mathbb{Q}$, $\dagger\mathcal{N}\xrightarrow{\sim}| \simeq\Gamma$ is † the fundamental group of the said † scheme or of the said multiplicity,

b) $\mathcal{G}_0/\Pi_0 \simeq$ Teichmüller group of Π_0 or of finite index in this Teichmüller group — in this case, one is interested mainly in a "modular" situation, where

---

Uncertainty retained from cancelled passages.

140-4-s206-D001 situation

140-4-s206-D004 $\Gamma|k$ in fact

140-4-s206-D005 is

140-4-s206-D006 this

[scan 207]

$\mathcal{N}$ is interpreted as the fundamental group of their|the appropriate modular multiplicity, $\mathcal{T}$ as its "geometric" part, $\mathcal{T}_0$ as its "transcendental" part, and Γ as $\Gamma_{\overline{\mathbb{Q}}_0|\overline{\mathbb{Q}}/\mathbb{Q}}$ .... [p. 751]

We make on (32) the hypothesis that the cyclotomic character on $\mathcal{M}$ (operating on Π by automorphisms of L-structures) is trivial on $\mathcal{G}$, a fortiori on Π (thus

$$(33) \qquad \Pi = \Pi^{+}, \quad \mathcal{G} = \mathcal{G}^{+} \qquad),$$

i.e. that it factors as

$$(34) \qquad \Gamma \xrightarrow{\chi_\Gamma} \widehat{\mathbb{Z}}^* \ ,$$

whose image is an open subgroup.

If $(G_{\mathbb{Q}}, \chi_G)$ is as above (page 748), we are interested in the homomorphisms (27)

$$\mathcal{M} \longrightarrow G(\mathcal{A}_{\mathbb{Q}})$$

satisfying the conditions 1°) 2°) †3°) 4°) as above, but where 3°) is formulated in the slightly stronger form

(3° strong) (Compatibility with Teichmüller discretification $\mathcal{G}_0$) : We have a factoriza-

Uncertainty retained from cancelled passages.
140-4-s207-D001 2

[scan 208]

tion $\mathcal{G}_0 \longrightarrow G(\mathbb{Q})$, making thus the diagram commutative

$$(35) \qquad \begin{array}{ccc} \mathcal{G}_0 & \hookrightarrow & \mathcal{M} \\ \downarrow & & \downarrow \\ G(\mathbb{Q}) & \hookrightarrow & G(\mathcal{A}_{\mathbb{Q}}) \end{array} \ .$$

Structures of the form (32) probably deserve a name, I am going to call them — if only provisionally, Galois-Teichmüller structures. m1633

Marginal note. m1633 Leaving it open later to introduce more draconian conditions on this structure

If G is an affine algebraic group over $\mathbb{Q}$, a homomorphism $\mathcal{M} \longrightarrow |-G(\mathcal{A}_{\mathbb{Q}})$ satisfying the conditions 1°) 2°) 3°) (strong) 4°) is called a "Lie homomorphism" of the Galois-Teichmüller structure.

I will henceforth denote by $\underline{\mathcal{M}}_{\mathbb{Q}}$ (or simply $\underline{\mathcal{M}}$?) the group scheme $G_{\mathbb{Q}}$ over $\mathbb{Q}$. †

I denote by † $\underline{\mathcal{G}}_{\mathbb{Q}}$ the subgroup scheme of $\underline{\mathcal{M}}_{\mathbb{Q}}$ generated by the

Uncertainty retained from cancelled passages.
140-4-s208-D001 defined
140-4-s208-D002 $\mathcal{G}|\mathcal{M}$

[scan 209]
[p. 752]

image of $\mathcal{G}_0 \longrightarrow \dagger G(\mathbb{Q})$, by $\underline{\Pi}_{\mathbb{Q}}$ the one generated by the image of $\Pi \longrightarrow G(\mathbb{Q})$

We thus have † subgroup schemes

$$(36) \qquad \underline{\Pi}_{\mathbb{Q}} \subset \dagger\underline{\mathcal{G}}_{\mathbb{Q}} \subset \underline{\mathcal{M}}_{\mathbb{Q}}$$

normal in $\underline{\mathcal{M}}_{\mathbb{Q}}$, giving rise to quotient group schemes

$$(37) \qquad \begin{cases} \underline{\mathcal{T}}_{\mathbb{Q}} \simeq \underline{\mathcal{G}}_{\mathbb{Q}}/\underline{\Pi}_{\mathbb{Q}}, \quad \underline{\Gamma}_{\mathbb{Q}} \simeq \underline{\mathcal{M}}_{\mathbb{Q}}/\underline{\mathcal{G}}_{\mathbb{Q}}, \\ \underline{\mathcal{N}}_{\mathbb{Q}} \simeq \underline{\mathcal{M}}_{\mathbb{Q}}/\underline{\Pi}_{\mathbb{Q}} \end{cases}$$

— giving rise to four exact sequences of group schemes, the last of which is

$$(38) \qquad 1 \longrightarrow \underline{\mathcal{T}}_{\mathbb{Q}} \longrightarrow \underline{\mathcal{N}}_{\mathbb{Q}} \longrightarrow \underline{\Gamma}_{\mathbb{Q}} \longrightarrow 1 \ .$$

†Let us note that the character $\chi_G : G_{\mathbb{Q}} \longrightarrow \mathbb{G}_m|G_m$ is an epimorphism, and factors through $\underline{\Gamma}_{\mathbb{Q}}$ as

$$(39) \qquad \underline{\Gamma}_{\mathbb{Q}} \xrightarrow{\chi_{\underline{\Gamma}}} \mathbb{G}_{m\mathbb{Q}}|G_{m\mathbb{Q}} \ .$$

Uncertainty retained from cancelled passages.

140-4-s209-D001 $\mathcal{M}|G$
140-4-s209-D002 a
140-4-s209-D003 $\mathcal{M}$
140-4-s209-D004 am going to|see reflect|recall

The system of groups over $\mathbb{Q}$ (36, 37), equipped with the homomorphism from [scan 210]

$$\Pi \subset \mathcal{G} \subset M$$

into †

$$(40) \qquad \underline{\Pi}_{\mathbb{Q}}(\mathbb{A}'_{\mathbb{Q}}) \subset \underline{\mathcal{G}}_{\mathbb{Q}}(\mathbb{A}_{\mathbb{Q}}) \subset \underline{M}_{\mathbb{Q}}(\mathbb{A}_{\mathbb{Q}})$$

induced by $M \longrightarrow G(\mathbb{A}_{\mathbb{Q}})$ (satisfying conditions 1°) to 4°) above †, i.e. of a "Lie homomorphism") will also be called a "Lie approximation" of the Galois-Teichmüller system under consideration.

Consider a homomorphism

$$(41) \qquad k_i^{\dagger} \quad T_0 \longrightarrow \Pi_0$$

belonging|corresponding to the L-structure of Π, thus defines by composition with $\Pi_0 \longrightarrow \dagger\underline{\Pi}_{\dagger}(\mathbb{Q})$ a homomorphism

$$(42) \qquad T_0 \longrightarrow \dagger\underline{\Pi}(\mathbb{Q}).$$

m1634

Marginal note. m1634 N.B. for fixed i, k_i is determined modulo inner automorphisms of Π_0 and modulo composition with $n\,\mathrm{id}_{T_0}$, for $n \in \mathbb{N}^*$. As for this last point, there is often (let us fix Π_0) a "best choice", from which all the others are deduced by composition with $n\,\mathrm{id}_{T_0}$.

We see ((21) page 747′) that this homomorphism is quasi-unipotent, and that it therefore defines a homomorphism of † group schemes 209

Uncertainty retained from cancelled passages.

140-4-s210-D001 $\mathcal{G}$
140-4-s210-D003)
140-4-s210-D005 ♯
140-4-s210-D006 $\mathcal{G}$
140-4-s210-D007 $\mathbb{Q}$
140-4-s210-D009 $\mathcal{G}$
140-4-s210-D011 g

[scan 211]
[p. 753]

$$(43) \qquad \underline{k}_i : \underset{(\simeq G_{a\mathbb{Q}} \otimes_{\mathbb{Z}} T_0)}{\underline{T}_{0\mathbb{Q}}} \longrightarrow \underline{\Pi}_{\mathbb{Q}}$$

(not just a germ!) which defines, by passage to the $\mathbb{Q}$-rational points, the germ of the homomorphism (42). We thus find a finite set (indexed by the set of germs of loop subgroups of Π_0, or of Π which amounts to the same) of outer homomorphisms (43) of group schemes. I

feel like calling the Lie appro-†ximation nondegenerate if the preceding homomorphisms are all nontrivial, hence are monomorphisms.

$$(44) \qquad \underline{k}_i : \underline{T}_{0\mathbb{Q}} \simeq G_{a\mathbb{Q}} \otimes_{\mathbb{Z}} T_0 \hookrightarrow \underline{\Pi}_{\mathbb{Q}}.$$

Suppose that $\underline{M}_{\mathbb{Q}}$ comes from a group scheme (affine and of finite type) $\underline{M}$ over $\mathbb{Z}$, such that we have

$$(45) \qquad M \longrightarrow \underline{M}(\widehat{\mathbb{Z}})$$

(it seems to me that one can always find such an $\underline{M}$...). We then denote by 210

Uncertainty retained from cancelled passages.
140-4-s211-D002 representation

$\underline{\Pi}$, $\underline{\mathcal{G}}$ the schematic closures in $\underline{M}$ of $\underline{\Pi}_{\mathbb{Q}}$, $\underline{\mathcal{G}}_{\mathbb{Q}}$, whence a diagram of group schemes over $\mathbb{Z}$ [scan 212]

$$(46) \qquad \begin{array}{ccccc} & & \underline{\mathcal{T}} & & \underline{\Gamma} \\ \underline{\Pi} & \subset & \underline{\mathcal{G}} & \subset & \underline{M} \\ & & & & \underline{N} \end{array}$$

(subject to representability of the quotients $\underline{\mathcal{T}}$, $\underline{\Gamma}$, $\underline{N}$ — but I think that for Raynaud there is never any problem...). We † will have factorizations

$$(47) \qquad \begin{array}{ccccc} \Pi_0 & \hookrightarrow & \mathcal{G}_0 & \hookrightarrow & M \\ \downarrow & & \downarrow & & \downarrow \\ \underline{\Pi}(\mathbb{Z}) & \hookrightarrow & \underline{\mathcal{G}}(\mathbb{Z}) & \hookleftarrow & \underline{M}(\widehat{\mathbb{Z}}). \end{array}$$

We assume of course † that the homomorphism $\chi_{\underline{M}} : \dagger\underline{M}_{\mathbb{Q}} \longrightarrow G_{m\mathbb{Q}}$ extends to

$$(48) \qquad \chi_{\underline{M}} : \underline{M} \longrightarrow G_{m\mathbb{Z}},$$

which obviously factors through

$$(49) \qquad \chi_{\underline{\Gamma}} : \underline{\Gamma} \longrightarrow G_{m\mathbb{Z}}.$$

211

Uncertainty retained from cancelled passages.
140-4-s212-D001 are going to assume|assume
140-4-s212-D002 (is there
140-4-s212-D003 $\mathbb{Q}$

It would be interesting to bring out a theorem of existence, (or even of uniqueness, under still stronger conditions) of a "model" $\underline{M}$ smooth over $\mathbb{Z}$ of $\underline{M}_{\mathbb{Q}}$, satisfying the preceding conditions [scan 213] [p. 754]

$$\mathcal{T}$$

$$(M \longrightarrow \underline{M}(\mathbb{Z}), \quad \text{and } \underline{M} \xrightarrow{\chi_{\underline{M}}} G_{m\mathbb{Z}}).$$

Here it will no longer be true in general, I presume, that the outer homomorphisms $\underline{k}_i$ (43) extend as they stand to homomorphisms $\underline{T}_i \underset{\text{df}}{=} \underline{T}_{i\mathbb{Z}} \longrightarrow \underline{\Pi}$. But I am going to|have one consider

the centralizer of $k_i : T_0 \longrightarrow \Pi_0$ in Π_0, i.e. $\widetilde{L}_{i,0}$ which is a metabelian group: $\mathrm{Im}(T_0 \overset{k_i}{\longrightarrow} \Pi_0)$ is a normal subgroup of finite index in it. Consider the homomorphism

$$\begin{array}{ccc} & \widetilde{L}_{i,0} & \longrightarrow \underline{\Pi}(\mathbb{Z}), \\ (50) & \| & \\ & \mathrm{Centr}_{\Pi_0}(k_i : T_0 \longrightarrow \Pi_0) & \end{array}$$

which is entirely unipotent — it therefore factors through a homomorphism of group schemes

$$(51) \qquad \mathrm{Env}(\widetilde{L}_{i,0}) \longrightarrow \underline{\Pi}$$

(cf. § 50, 4°, page 658′ ff), factoring 212

through one of the smooth quotient|quotients group schemes $\underline{Z}_{i,n}$ of $\mathrm{Env}(\widetilde{L}_{i,0})$, of which the latter is the $\varprojlim$: [scan 214]

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \underline{nT_0} & \longrightarrow & \underline{Z}_{i,n} & \longrightarrow & (H_{i,n})_{S_0} & \longrightarrow & 1 \\ (52) & & & & & & \wr\| & & \\ & & & & & & (\widetilde{L}_{i,0}/nT_0) & & \end{array}$$

But of course, the restriction of the homomorphism obtained $S_0 = \mathrm{Spec}\,\mathbb{Z}$

$$(5|(s \qquad \underline{Z}_{i,n} \longrightarrow \underline{\Pi}$$

to

$$\underline{Z}^{\circ}_{i,n} \simeq \underline{nT_0}$$

is not in general a monomorphism — if it is for fixed n, then if $n \mid n'$, $n \neq n'$, it no longer is for n'. A particularly favorable case would be one where one can find n such that the induced homomorphism

$$(53) \qquad \underline{nT_0} \longrightarrow \underline{\Pi}$$

is a monomorphism. This implies that the schematic closure in $\underline{\Pi}$ of the "1-parameter additive subgroup" $\dagger\underline{L}_{i\mathbb{Q}} = \underline{k}_i(\underline{T_0})$ is a smooth subgroup scheme of $\underline{\Pi}$. this n is then unique. 213

Uncertainty retained from cancelled passages.

140-4-s214-D001 $\mathcal{G}$

[scan 215]

This condition should also be sufficient – at least if one requires that this schematic closure [p. 755] have an identity component which is $= \mid \simeq$ isomorphic to G_{aS_0}. We then find, in any case, a unique monomorphism (up to conjugation)

$$(54) \qquad \dagger\underline{\underline{k'_i}} : \qquad \underline{T_0} \hookrightarrow \underline{\Pi}$$

such that $\underline{\underline{k'_i}}_{\mathbb{Q}}$ is of the form $\underline{k_i}_{\mathbb{Q}} \circ (r_i \mathrm{id})$, with a suitable $\| \; |!!\; r_i \in \mathbb{Q}^{*+}$. I am not sure that under these conditions, † the homomorphism deduced from $\underline{k'_i}$ by passing to the $\mathbb{Z}$-points

$$T_0 \simeq \underline{T_0}(\mathbb{Z}) \longrightarrow \underline{\Pi}(\mathbb{Z})$$

factors into a homomorphism

$$(55) \qquad T_0 \overset{k'_i\varphi|?}{\longrightarrow} \Pi_0 \overset{\mathrm{can}}{\longrightarrow} \underline{\Pi}(\mathbb{Z}) \;,$$

where $T_0 \longrightarrow \Pi_0$ belongs to the "germ" $\underline{k_i}$, —

I will nevertheless take it as an additional hypothesis, when the $\underline{k'_i}$ exist.

I feel like stating a few other "good hypotheses", which we will try to realize 214

Uncertainty retained from cancelled passages.
140-4-s215-D001 k
140-4-s215-D003 can

[scan 216]

in working on "integral" Lie approximations i.e. over $\operatorname{Spec}\mathbb{Z}$.

Consider all possible homomorphisms $\underline{k_i'}$ (i variable, but also by conjugating the original $k_i : \dagger\, T_0 \longrightarrow \Pi_0$ by arbitrary elements of Π_0), and the corresponding homomorphisms on the Lie algebras

$$(56) \qquad \operatorname{Lie}(\underline{k_i'}) : \begin{array}{ccc} \dagger\, T_0 & \longrightarrow & \Gamma\underline{\operatorname{Lie}(\Pi)} \underset{\text{df}}{=} \operatorname{Lie}(\underline{\Pi}) \\ \wr & & \\ \mathbb{Z} & & \end{array}$$

(not canonical)

which are by hypothesis † universal monomorphisms, we would like $\Gamma\underline{\operatorname{Lie}}(\underline{\Pi})$ to be generated (as a $\mathbb{Z}$-module) by their images, or (that which amounts to the same) by the images of a generator of T_0. m1643

Marginal note. m1643 this hypothesis seems reasonable in a case like $\Pi_0 = \mathrm{SL}(2,\mathbb{Z})$, but not if $I = \emptyset$ for example…!

It|I amounts to the same to say that if one chooses for $\forall i$ one $\underline{k_i'}$, whence the image in $\Gamma\underline{\operatorname{Lie}}(\underline{\Pi})$ of a generating element of T_0, the I elements obtained in $\dagger\operatorname{Lie}(\underline{\Pi})$, and their conjugates by the elements of Π_0, generate the $\mathbb{Z}$-module $\operatorname{Lie}(\underline{\Pi})$. This implies that 215

Uncertainty retained from cancelled passages.
140-4-s216-D001 T_0
140-4-s216-D002 $\mathbb{Z}$
140-4-s216-D003 good|a
140-4-s216-D004 Γ

[scan 217]
[p. 756]

this group scheme $\underline{\Pi}^\circ$ over $S_0 = \operatorname{Spec}\mathbb{Z}$, as an étale sheaf on the "big étale topos" of S_0, is generated by a finite number of subgroups $k_i'(\underline{\mathrm{G}_a})$, (with possible repetition of the same conjugacy class), or again †that $\underline{\Pi}$ is † generated by the the $k_i'(\mathrm{G}_a)$ (their various conjugates|one of each say) and the discrete group $(\Pi_0)_{S_0}$. (And there arises the interesting question of making the relations explicit… is it enough to write the relations of Π_0, plus the commutativity of

$$(57) \qquad \begin{array}{ccc} T_0 & \xrightarrow{k_i'} & \Pi_0 \\ \downarrow{\scriptstyle \text{can}} & & \downarrow{\scriptstyle \text{can}} \\ \underline{T_0} & \xrightarrow{\underline{k_i'}} & \underline{\Pi} \end{array} \qquad ? \qquad)$$

Other conditions, concerning the group schemes

$$(58) \qquad \underline{\gamma}_{\underline{\Pi}} = \underline{\Pi}/\underline{\Pi}^\circ, \quad \underline{\gamma}_{\underline{\mathcal{G}}} = \underline{\mathcal{G}}/\underline{\mathcal{G}}^\circ, \quad \underline{\gamma}_{\underline{M}} = \underline{M}/\underline{M}^\circ$$

– we would like these to be constant group schemes, corresponding to the † finite discrete groups which describe the corresponding quotients of the groups over $\mathbb{Q}$, which discrete groups are quotients 216

Uncertainty retained from cancelled passages.
140-4-s217-D003 rather
140-4-s217-D004 arises infinitely|the interesting questi-
140-4-s217-D005 schemes

[scan 218]

$$(59)\qquad \begin{cases} \gamma_\Pi \text{ (resp. } \gamma_{\mathcal{G}}, \text{ resp. } \gamma_M) \quad \text{quotients of } \Pi, \mathcal{G}, M \\ \underline{\gamma}_{\underline{\Pi}} \simeq (\gamma_\Pi)_{S_0}, \quad \underline{\gamma}_{\underline{\mathcal{G}}} \simeq (\gamma_{\mathcal{G}})_{S_0}, \quad \underline{\gamma}_{\underline{M}} \simeq (\gamma_M)_{S_0} \end{cases}$$

which fit into a sequence of homomorphisms

$$(60)\qquad \gamma_\Pi \longrightarrow \gamma_{\mathcal{G}} \longrightarrow \gamma_M$$

and into exact sequences

$$(61)\qquad \begin{cases} \gamma_\Pi \longrightarrow \gamma_{\mathcal{G}} \longrightarrow \gamma_{\mathcal{T}} \longrightarrow 1 \\ \gamma_\Pi \longrightarrow \gamma_M \longrightarrow \gamma_N \longrightarrow 1 \\ \gamma_{\mathcal{G}} \longrightarrow \gamma_M \longrightarrow \gamma_{\dagger\Gamma} \longrightarrow 1 \\ \gamma_{\mathcal{T}} \longrightarrow \gamma_N \longrightarrow \gamma_\Gamma \longrightarrow 1 \end{cases} .$$

This hypothesis on $\underline{\Pi}, \underline{\mathcal{G}}, \underline{M}$ makes it possible to define groups $\underline{\Pi}^\Delta(\widehat{\mathbb{Z}})^\Delta, \underline{\mathcal{G}}^\Delta(\widehat{\mathbb{Z}})^\Delta, \underline{M}^\Delta(\widehat{\mathbb{Z}})$, and we shall necessarily have the first two factorizations of

$$(62)\qquad \begin{cases} \Pi \longrightarrow \underline{\Pi}^\Delta(\dagger\widehat{\mathbb{Z}}) \\ \mathcal{G} \longrightarrow \underline{\mathcal{G}}^\Delta(\widehat{\mathbb{Z}}) \\ M \longrightarrow \underline{M}^\Delta(\widehat{\mathbb{Z}}) \end{cases}$$

(for

$$\begin{array}{l} \Pi_0 \longrightarrow \underline{\Pi}(\mathbb{Z}) \subset \underline{\Pi}^\Delta(\widehat{\mathbb{Z}}) \\ \mathcal{G}_0 \longrightarrow \underline{\mathcal{G}}(\mathbb{Z}) \subset \underline{\mathcal{G}}^\Delta(\widehat{\mathbb{Z}}) \end{array}).$$

As for the third factorization, it should be included in the starting conditions for a Lie homomorphism of a Teichmüller - Galois structure - namely that $\underline{M}_{\mathbb{Q}}/\underline{M}^\circ_{\mathbb{Q}}$ is constant, which makes it possible

†
217

Uncertainty retained from cancelled passages.
140-4-s218-D002 $\mathcal{T}$
140-4-s218-D003 $\widehat{\mathbb{Q}}$
140-4-s218-D004 smooth

[scan 219]
[p. 757]

then to define $\underline{M}^{\Delta}_{\mathbb{Q}}(A|\mathbb{A}_{\mathbb{Q}})$, and we require that $M - \underline{M}_{\mathbb{Q}}(A|\mathbb{A}_{\mathbb{Q}})$ factor through

$$(63) \qquad M \longrightarrow \underline{M}^{\Delta}_{\mathbb{Q}}(A|\mathbb{A}_{\mathbb{Q}}) \quad .$$

If moreover we assume that the images of the three homomorphisms (62) are open subgroups, (a condition which considerably strengthens the index condition given 4°) (p. 749), provided we combine it with the surjectivity of the composites

$$(64) \qquad \begin{cases} \Pi \longrightarrow \underline{\Pi}^{\Delta}(\widehat{\mathbb{Z}}) \longrightarrow \Gamma\gamma_{\underline{\Pi}} \\ \mathcal{G} \longrightarrow \underline{\mathcal{G}}^{\Delta}(\widehat{\mathbb{Z}}) \longrightarrow \Gamma\gamma_{\underline{\mathcal{G}}} \\ M \longrightarrow \underline{M}^{\Delta}(\widehat{\mathbb{Z}}) \longrightarrow \Gamma\gamma_{\underline{M}} \end{cases},$$

we shall be particularly happy. We shall say that we have a Lie †approximation (over $\mathbb{Z}$, hence "integral") nondegenerate, if all the additional conditions considered previously are satisfied… 218

Uncertainty retained from cancelled passages.
140-4-s219-D001 homomorphism

Remark In the preceding presentation, the group Π_0 (or Π) itself seems to recede into the background, in favor of $\mathcal{G}_0$ (or $\mathcal{G}$) and its LT-structure. It seems to me indeed that the importance of the role of Π_0 fades behind that of the sub-group † of finite index Π_0' (if one exists) which would be a group with loops and of which the $k_i'(T_0)$ † would be † exactly the loop subgroups. It is not too clear, even if such a Π_0' exists, that it is necessarily unique (let us take|think of the case for example where $I = \emptyset$!). The case which had occupied me until now, $\Pi_0 = \mathcal{G}_0 = \mathcal{G}^+_{0,3} = |-\mathcal{G}^+_{1,1}|\mathcal{T}^+_{1,1}$, has been a little "misleading", for we then took $\Pi_0 := \mathcal{G}_0$ † precisely. If we had thought † in desire right at the beginning of my work, where by taking only $\Pi_0 = \Pi_{0,3}$ as the starting group, we would have seen (as was moreover actually the case) the role of $\Pi_{0,3}$ dwindle in favor of the smaller subgroups (type $\Pi^{\sharp}$, giving rise to the $\mathcal{G}^{\leftrightarrow}$ of § 49 VI…) [scan 220] 219

Uncertainty retained from cancelled passages.
140-4-s220-D001 created|side
140-4-s220-D003 and
140-4-s220-D004 Π_0'' closure
140-4-s220-D005 this
140-4-s220-D006 As we

4°) Lie approximation of $(\Pi_0, \mathcal{G}_0, M)$ induced by an approximation of $\mathcal{G}_0$. [scan 221]

Let us start with a system $(\mathcal{G}_0 \supset \Pi_0)$ of a discrete LT-group $\mathcal{G}_0$, and of an invariant L-subgroup Π_0 of $\mathcal{G}_0$. Suppose given group scheme $\underline{\mathcal{G}}_{\mathbb{Q}}$ over $\mathbb{Q}$, and a homomorphism [p. 758] m1650

Marginal note. m1650 We assume that the homomorphism $\mathcal{G}_0 \longrightarrow \mathbb{Z}^* = \{\pm 1\}$ defined by the action of $\mathcal{G}_0$ on the L-group Π_0 (multipliers) is trivial…

$$(65) \qquad \varphi : \mathcal{G}_0 \longrightarrow \underline{\mathcal{G}}(\mathbb{Q})$$

having the property that for ∀ homomorphism

$$(66)\quad k_i : T_0 \longrightarrow \mathcal{G}_0$$

which is part of the LT-structure, the composite homomorphism

$$T_0 \longrightarrow \mathcal{G}_0 \longrightarrow \underline{\mathcal{G}}(\mathbb{Q})$$

is quasi-unipotent. We assume moreover that $\underline{\mathcal{G}}_{\mathbb{Q}}$ is generated by the set of † $\mathbb{Q}$-rational points defined by the elements of $\mathcal{G}_0$ via (65).

We consider $\underline{\mathcal{G}}_{\mathbb{Q}}$ as equipped with the homomorphism (65) and with the family of outer homomorphisms

$$(67)\quad \underline{k}_i : \underline{T_0}_{\mathbb{Q}} \longrightarrow \underline{\mathcal{G}}_{\mathbb{Q}}$$

220

Uncertainty retained from cancelled passages.
140-4-s221-D002 its

giving rise to compatibilities [scan 222]

$$\begin{array}{ccc} T_0 & \xrightarrow{k_i} & \mathcal{G}_0 \\ {\scriptstyle \text{can}}\downarrow{\scriptstyle \dagger} & & \downarrow{\scriptstyle (65)} \\ \underline{T_0}_{\mathbb{Q}}(\mathbb{Q}) & \xrightarrow{\underline{k}_i(\mathbb{Q})} & \underline{\mathcal{G}}(\mathbb{Q}). \end{array} \qquad (68)$$

I then consider the group scheme over $\mathbb{Q}$ of pairs (u, μ) of an automorphisms|automorphism of $\underline{\mathcal{G}}_{\mathbb{Q}}$, and of a $\mu \in \Gamma G_{\mathrm{m}}$, such|such that for ∀ $i \in I$, there exists locally a section f_i of $\underline{\Pi}_{\mathbb{Q}}$ (generated by the image of Π_0 in $\underline{\mathcal{G}}(\mathbb{Q})$), such|such that we have

$$(69)\qquad u(\underline{k}_i'(\lambda)) = \mathrm{int}(f_i)^{-1}(\mu\lambda).$$

I denote by $\underline{M}^{\varphi}_{\mathbb{Q}}$ this group scheme over $\mathbb{Q}$, augmented to $G_{\mathrm{m}\mathbb{Q}}$ by $(u, \mu) \longmapsto \mu$, by a homomorphism which is trivial on $\mathrm{Im}(\underline{\mathcal{G}}_{\mathbb{Q}} \to \underline{M}^{\varphi}_{\mathbb{Q}})$, hence factors through a homomorphism

$$(70)\quad \underline{\Gamma}^{\varphi}_{\mathbb{Q}} \overset{\text{déf}}{=} \underline{M}^{\varphi}_{\mathbb{Q}} / \mathrm{Im}\, \underline{\mathcal{G}}_{\mathbb{Q}} \xrightarrow{\chi_{\underline{\Gamma}}} G_{\mathrm{m}\mathbb{Q}}.$$

If now $(\mathcal{G}_0, \Pi_0)$ is part of a Galois-Teichmüller system 221

Uncertainty retained from cancelled passages.
140-4-s222-D001 φ

[scan 223]
[p. 759]

$$\begin{array}{ccccc} (\Pi & \subset & \mathcal{G} & \subset & M), \\ \cup & & \cup & & \\ \Pi_0 & \subset & \mathcal{G}_0 & & \end{array}$$

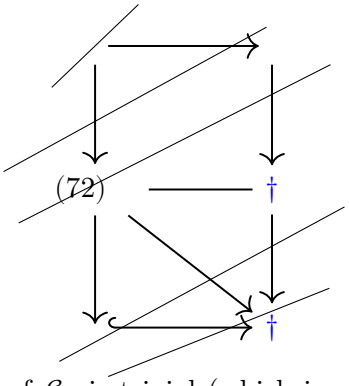
and if $\underline{\mathcal{G}_0} \overset{\varphi}{\to} \underline{\mathcal{G}}^{\dagger}_{\mathbb{Q}}$ comes from a Lie homomorphism of this system, $M \longrightarrow \underline{M}(A|\mathbb{A}_{\mathbb{Q}})$, by the method of the preceding section, then we find a natural homomorphism

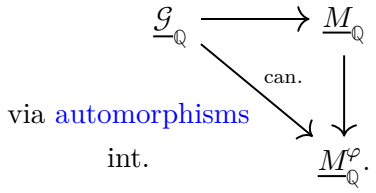
$$(71) \quad \underline{M}_{\mathbb{Q}} \longrightarrow \underline{M}^{\varphi}_{\mathbb{Q}},$$

m1653

Marginal note. m1653 via the operation of $\underline{M}_{\mathbb{Q}}$ on $\underline{\mathcal{G}}_{\mathbb{Q}}$, and $\chi_{\underline{M}}$

making the diagram commutative





via automorphisms
int.

If the center of $\underline{\mathcal{G}}_{\mathbb{Q}}$ is trivial (which is a "credible" hypothesis especially if the center of $\mathcal{G}_0$ is so…) then $\underline{\mathcal{G}}_{\mathbb{Q}} \longrightarrow \underline{M}^{\varphi}_{\mathbb{Q}}$ is injective, and $\underline{\mathcal{G}}_{\mathbb{Q}}$ is interpreted as "the $\underline{\mathcal{G}}_{\mathbb{Q}}$" associated to a Lie homomorphism

$$\begin{pmatrix} \pi|\Pi & \subset & \mathcal{G} & \subset & M \\ \cup & & \cup & & \\ \Pi_0 & \subset & \mathcal{G}_0 & & \end{pmatrix} \longrightarrow (\underline{\Pi} \subset \underline{\mathcal{G}}_{\mathbb{Q}} \subset \underline{M}^{\varphi}_{\mathbb{Q}}),$$

which has a certain universal character with respect to the given homomorphism (65) $\varphi : \mathcal{G}_0 \longrightarrow \underline{\mathcal{G}}(\mathbb{Q})$. 222

Uncertainty retained from cancelled passages.

140-4-s223-D001 φ

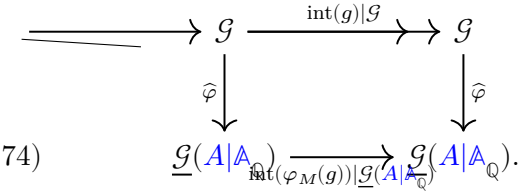
140-4-s223-D006 $A|\mathbb{A}$

140-4-s223-D008 $A|\mathbb{A}$

When we do not assume in advance the existence of a † Lie homomorphism in which $\underline{\mathcal{G}}_{\mathbb{Q}}$ fits, it [scan 224] is not automatic that we can find a homomorphism

$$(73) \quad \varphi_M : \ M \longrightarrow \underline{M}^{\varphi}_{\mathbb{Q}}(A|\mathbb{A}_{\mathbb{Q}})$$

which accounts for the operations of M on $\mathcal{G}_{\dagger}$, through the commutativity of a diagram, for $\forall\ g \in M$



(74)

We are mainly interested in those φ (65) for which (73) exists — at least after possibly restricting to an open subgroup|subgroup of M… 223

Uncertainty retained from cancelled passages.
140-4-s224-D001 $M|H$
140-4-s224-D002 $0|\mathbb{Q}$

[scan 225]

53 [p. 760]

1°) Differentiable|Differential action and special Teichmüller group.

Let X be a connected † differentiable, orientable surface (compact in the case of interest to us), I a discrete subset of X. For $\forall i \in I$, let $\mathbb{S}_i$ be the space (homeomorphic to the † circle) of homogeneous half-lines in the tangent space, and let $\tilde{\mathbb{S}}_i$ be a universal covering of $\mathbb{S}_i$.

Let † u be a differentiable automorphism of (X, S), it induces an automorphism of

$$(1) \qquad \mathbb{S} = \mathbb{S}_I \overset{\text{déf}}{=} \coprod_{i \in I} \mathbb{S}_i,$$

denoted $u_{\mathbb{S}}$. Let us consider the set of liftings $\tilde{u}$ of $u_{\mathbb{S}}$ to an automorphism of †

$$(2) \qquad \tilde{\mathbb{S}} = \coprod_{i \in I} \tilde{\mathbb{S}}_i.$$

Let us note, if T_0 is the orientation module of X, that T_0 operates on each $\tilde{\mathbb{S}}_i$, 224

Uncertainty retained from cancelled passages.
140-4-s225-D001 compact
140-4-s225-D002 c|d
140-4-s225-D003 Let $\tilde{g}$
140-4-s225-D004 $\mathbb{S}$ $\tilde{\mathbb{S}}$

[scan 226]

so that T_0 in fact operates on $\mathbb{S}$ and

$$(3) \qquad \tilde{\mathbb{S}}/T_0 \simeq \mathbb{S}.$$

This shows that (if u is as above) the group T_0^I operates on the left (for example) on the set of liftings $\tilde{u}$. which thus becomes a torsor under T_0^I.

Let G be a topological group, which operates differentiably on (X, S), with † one of the continuous actions on $\mathbb{S}$. We set

$$(4) \qquad \widetilde{G} = \left\{ (\dagger u, \tilde{u}) \;\middle|\; \begin{array}{l} u \in G, \quad \tilde{u} \text{ a continuous lifting} \\ \text{of } u_{\mathbb{S}} \text{ to } \tilde{\mathbb{S}} \end{array} \right\},$$

we find on $\widetilde{G}$ a canonical topology (convergence of u and uniform convergence of $\tilde{u}$), which is such that the homomorphism $G \longrightarrow \widetilde{G}$ makes|will make G a covering of $\widetilde{G}$, with group T_0^I:

$$(6) \qquad 1 \longrightarrow T_0^I \longrightarrow \widetilde{G} \longrightarrow G \longrightarrow 1 \ .$$

m1657

Marginal note. m1657 N.B. G operates on T_0^I via the character † of orientation

Proposition † Suppose that X is compact|compact, 225

Uncertainty retained from cancelled passages.
140-4-s226-D001 of|some
140-4-s226-D002 g
140-4-s226-D003 (52)|(ε)
140-4-s226-D004 If

[scan 227]
[p. 761]

and consider the canonical homomorphism

$$G \longrightarrow G/G^\circ \longrightarrow \mathcal{T}(X,S)$$

(Teichmüller group of (X,S)), we then have a canonical homomorphism of extensions

$$(7)\qquad \begin{array}{ccccccccc} 1 & \longrightarrow & T_0^I & \longrightarrow & \widetilde{G} & \longrightarrow & G & \longrightarrow & 1 \\ & & \downarrow{\scriptstyle 1} & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & T_0^I & \longrightarrow & \mathrm{Sp}\mathcal{T}(X,S) & \longrightarrow & \mathcal{T}(X,S) & \longrightarrow & 1 \end{array},$$

$\mathrm{Sp}\mathcal{T}(X,S)$ is the special Teichmüller group of (X,S).
m1659

Marginal note. m1659 For the proof cf. the developments of 5°) (pages 766′ ff, in particular pages 784 ff)

N.B. We shall note that if for $\forall i \in I$, we fix an origin e_i on $\mathbb{S}_i$, then it follows from the universal property of $\widetilde{\mathbb{S}}$ that giving a lifting $\tilde{u}$ of $u_{\mathbb{S}}$ to $\widetilde{\mathbb{S}}$ amounts to giving, for $\forall i \in I$, a lifting of $u_{\mathbb{S}}(e_i)$ to a point $\tilde{a}_i$ of $\widetilde{\mathbb{S}}_{u(e_i)|u(i)}$ (i.e., of $\widetilde{\mathbb{S}}_{u(i)}$). 226

[scan 228]

2°) The groups $\mathrm{GL}(2,\mathbb{R})^{\pm 1\sim}$|and †. We set

$$(8)\qquad \mathrm{GL}(2,\mathbb{R})^{\pm 1} = \{u \in \mathrm{GL}(2,\mathbb{R}) \mid \det u \in \{\pm 1\}\}.$$

This group operates differentiably on $\mathbb{C} \simeq \mathbb{R}^2$ leaving the origin fixed, we can therefore apply the construction of 1°), Here $\mathbb{S}_0$ is identified with the unit circle

$$(9)\qquad \begin{array}{ccc} \mathbb{U} \overset{\text{déf}}{=} \{z \in \mathbb{C} \mid |z| = 1\} & \longrightarrow & \mathbb{S}_0^{\dagger|*} \\ z & \longmapsto & \mathbb{R}^+ z \end{array}$$

and for $u \in \mathrm{GL}(2,\mathbb{R})^{\pm 1}$, its action on $\mathbb{S}_0^{\dagger|*}$ is identified with the action on $\mathbb{U}$ given by

$$(10)\qquad \begin{array}{ccc} z & \longmapsto & \dfrac{u(z)}{|z|} \\ \cup & & \\ \mathbb{U} & & \end{array}.$$

A lifting of this action $u_{\mathbb{S}_0^{\dagger|*}}$ to an action on $\widetilde{\mathbb{S}}_0^{\dagger|*}$ amounts to a lifting of $u(e_0 = 1)$ to an element of $\widetilde{\mathbb{S}}_0$. † We † shall construct|construct $\widetilde{\mathbb{S}}$ † by choosing a base point e_0 of $\mathbb{S}_0$, corresponding to the 227

Uncertainty retained from cancelled passages.

140-4-s228-D001 GL(2, ℤ)
140-4-s228-D002 *
140-4-s228-D003 *
140-4-s228-D004 *
140-4-s228-D005 *
140-4-s228-D006 and
140-4-s228-D007 shall choose|shall describe
140-4-s228-D008 $\tilde{\mathbb{S}}_0$

[scan 229]

point 1 of $\mathbb{U}$, so that the universal covering of $\mathbb{S}_0$ pointed over e_0 is identified with the [p. 762] exponential covering

$$\begin{array}{lll} \mathbb{R} \longrightarrow \mathbb{U} & & (\tilde{\to} \mathbb{S}_0) \\ t \longmapsto \exp 2i\pi t & & (\mapsto \mathbb{R}^+ \exp 2i\pi t. \end{array} \tag{11}$$

Thus, a lifting of $u_{\mathbb{S}^1}$ to $\tilde{u}$ amounts to the choice of $t \in \mathbb{R}$ such that

$$\exp 2i\pi t = \frac{u(1)}{|u(1)|} \qquad \dagger \tag{12}$$

i.e. to the choice of an argument t of $u(1)$.

We apply the construction of $\widetilde{G}$ of 1°) to the case of $G = \mathrm{Gl}(2,\mathbb{R})^{\pm 1}$, we find an extension structure

$$1 \longrightarrow \mathbb{Z} \longrightarrow \mathrm{Gl}(2,\mathbb{R})^{\pm 1\sim} \longrightarrow \mathrm{Gl}(2,\mathbb{R})^{\pm 1} \longrightarrow 1. \tag{13}$$

†which induces over the identity component $\mathrm{Sl}(2,\mathbb{R})$ of $\mathrm{Gl}(2,\mathbb{R})^{\pm 1}$ an extension structure

$$1 \longrightarrow \mathbb{Z} \longrightarrow \mathrm{Sl}(2,\mathbb{R})^{\sim} \longrightarrow \mathrm{Sl}(2,\mathbb{R}) \longrightarrow 1. \tag{14}$$

| Proposition The group $\mathrm{Sl}(2,\mathbb{R})^{\sim}$ is 1-connected, and even † homeomorphic to $\mathbb{R}^3$. 228

Uncertainty retained from cancelled passages.

140-4-s229-D001 is written
140-4-s229-D002 Propo
140-4-s229-D003 contra

[scan 230]

Proof Since the inclusion

$$\mathbb{U} \hookrightarrow \mathrm{Sl}(2,\mathbb{R}) \tag{15}$$

is a homotopy equivalence, † is identified topologically with the inclusion $\mathbb{U} \longrightarrow \mathbb{U} \times \mathbb{R}^2$, we are reduced to proving that † the inverse image $\mathbb{U}^{\sim}$ of $\mathbb{U}$ in $\mathrm{Sl}(2,\mathbb{R})^{\sim}$ is identified with the universal covering † $\mathbb{R}$ of $\mathbb{U}$ (pointed over 1 by 0 ...) Now if $u \in \mathbb{U}$, the † choice of a $\tilde{u}$ over u, i.e. of a lifting of $\dfrac{u(1)}{|u(1)|}$ to $\mathbb{R}$ by formula (12), is identified (since $\dfrac{u(1)}{|u(1)|} = u(1) = u$) with the choice of a $t \in \mathbb{R}$ such that $\exp 2i\pi t \in u$, O.K.

Corollary 1 The inverse subgroup in $\mathrm{Sl}(2,\mathbb{R})^{\sim}$ of a finite cyclic subgroup of order n is $\simeq \mathbb{Z}$ ($\exists!$ generator $\overline{\omega}$ of it, such that we have $\overline{\omega}^n = \omega_0'$, where ω_0' is the † generator of the $\mathrm{Ker}(\mathrm{Sl}(2,\mathbb{R})^{\sim} \to \mathrm{Sl}(2,\mathbb{R}))$).

m1663

Marginal note. m1663 Applying this to the center $\{\pm 1\}$ of $\mathrm{Sl}(2,\mathbb{R})$, we find a canonical element $\omega_0 \in \mathrm{Sl}(2,\mathbb{R})^\sim$ with $\omega_0^2 = \omega_0'$.

Corollary 2 The action of a $g \in \mathrm{Gl}(2,\mathbb{R})^{-1}$ on $\dagger\,\omega_0'$ (or on the center $\omega_0^{\mathbb{Z}}$ of $\mathrm{Sl}(2,\mathbb{R})^\sim$) is via $\omega_0' \mapsto \omega_0'^{-1}$ (resp. $\omega_0 \mapsto \omega_0^{-1}$) ... 229

Uncertainty retained from cancelled passages.

140-4-s230-D001 which
140-4-s230-D002 over
140-4-s230-D003 $\mathbb{R}$
140-4-s230-D004 last
140-4-s230-D005 canonical
140-4-s230-D006 $\overline{\omega}$

[scan 231]

3°) The group $\mathrm{Gl}(2,\mathbb{Z})^\sim$. [p. 763]

Let $\mathrm{Gl}(2,\mathbb{Z})^\sim$ be the inverse image of $\mathrm{Gl}(2,\mathbb{Z})$ in $\mathrm{Gl}(2,\mathbb{R})^{\pm 1\sim}$, so that we have an extension structure

$$\begin{array}{ccccccccc} 1 & \to & \mathbb{Z} & \to & \mathrm{Gl}(2,\mathbb{Z})^\sim & \to & \mathrm{Gl}(2,\mathbb{Z}) & \to & 1. \\ & & \uparrow & & & & & & \\ & & \dagger & \longmapsto & \dagger & & & & \end{array} \tag{16}$$

$$n \longmapsto \omega_0^n | \omega_0'^{\,n}$$

I would like to describe $\mathrm{Gl}(2,\mathbb{Z})^\sim$ by generators and relations. For this, I note this:

> Proposition The inverse image in $\mathrm{Gl}(2,\mathbb{Z})^\sim$ of the cyclic subgroups $L_\rho \simeq \mathbb{Z}/6\mathbb{Z}$ and $L_\sigma \simeq \mathbb{Z}/4\mathbb{Z}$ generated by ρ and σ are isomorphic to $\mathbb{Z}$.

This follows from the

Corollary (to the proposition p. 762): For every finite cyclic subgroup $\mathfrak{Z}|\mathfrak{z} \subset \mathrm{Sl}(2,\mathbb{R})$, the inverse image of $\mathfrak{Z}|\mathfrak{z}$ in $\mathrm{Sl}(2,\mathbb{R})^\sim$ is isomorphic to $\mathbb{Z}$.

Indeed, by conjugation we may assume that $\mathfrak{Z}|\mathfrak{z} \subset \mathbb{U}$, i.e. $\mathfrak{Z}|\mathfrak{z} = \mu_n(\mathbb{C})$, but then $\tilde{\mathfrak{Z}}|\tilde{\mathfrak{z}}$ is identified with the set of $t \in \mathbb{R}$ with $(\exp 2i\pi t)^n = 1$ i.e. $\exp 2i\pi(nt) = 1$ i.e. 230

Uncertainty retained from cancelled passages.

140-4-s231-D001 n
140-4-s231-D002 ω_0'

[scan 232]

$$t \in \frac{1}{n}\mathbb{Z},$$

the group of these t is $\simeq \mathbb{Z}$, q.e.d.

Corollary There exist $\tilde{\rho} \in \mathrm{Sl}(2,\mathbb{Z})^\sim$ and $\tilde{\sigma} \in \mathrm{Sl}(2,\mathbb{Z})^\sim$, over ρ and σ respectively, such that we have

$$\tilde{\rho}^3 = \tilde{\sigma}^{-2} \tag{17}$$

m1666

Marginal note. m1666 for a more pleasant statement, see (20) below ...

The set of these systems $(\tilde\rho, \tilde\sigma)$ is a torsor under $\mathbb{Z}$, by the operation

$$(\tilde\rho, \tilde\sigma) \longmapsto (\tilde\rho\, {\omega'_0}^{2n}, \tilde\sigma\, {\omega'_0}^{3n}), \tag{18}$$

where ω'_0 is the canonical generator of $\mathrm{Sl}(2,\mathbb{Z})^\sim \longrightarrow \mathrm{Sl}(2,\mathbb{Z})$.

We can also say that the inverse image of the element † of the center $\{\pm 1\}$ of $\mathrm{Sl}(2,\mathbb{Z})$ † † is a group isomorphic to $\mathbb{Z}$, with a unique generator ω_0 satisfying †

$$\omega_0^2 = \omega'_0 \quad (\text{the canonical generator of } \mathrm{Sl}(2,\mathbb{Z})^\sim \longrightarrow \mathrm{Sl}(2,\mathbb{Z}) \ ) \tag{19}$$

m1667

Marginal note. m1667 it is better to say: ∃! generator $\tilde\rho$ (resp. $\tilde\sigma$) of the inverse image $\widetilde{L}_\rho | L_\rho$ of L_ρ (resp. L_σ) in $\mathrm{Sl}(2,\mathbb{Z})^\sim$, over ρ (resp. σ), and we then have relations (20)

and this being so, it is possible to lift ρ to $\tilde\rho$, σ to $\tilde\sigma$ satisfying

$$\tilde\rho^{\,3} = \omega_0^{-1}, \qquad \tilde\sigma^{\,2} = \omega_0 \tag{20}$$

And this time, $\tilde\rho$ and $\tilde\sigma$ are uniquely deter- Do the verification 231

Uncertainty retained from cancelled passages.
140-4-s232-D001 = −id
140-4-s232-D002 the unique
140-4-s232-D003 distinct from|≠
140-4-s232-D005 of a

[scan 233]

mined. This follows † † at once from the preceding proposition. [p. 764]

i) Let us consider the inverse image of $\{1,\tau\} \subset \mathrm{Gl}(2,\mathbb{Z})$ in $\mathrm{Gl}(2,\mathbb{Z})^\sim$, † extension of $\{1,\tau\} \simeq \mathbb{Z}/2\mathbb{Z}$ by $\mathbb{Z}$ — I say that this extension is split.

m1670

Marginal note. m1670 N.B. $\tau = \tau_\infty = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ i.e. $\tau(x+iy) = -x+iy$ (reflection with respect to the y-axis)

Indeed, the action of τ on $\mathcal{S}_0$ lifts to the action of $\tilde\tau$ of order 2 on † $\tilde{\mathcal{S}}_0 \simeq \mathbb{R}$, which is given by

$$\tilde\tau t = \frac{1}{2} - t \tag{21}$$

for we have for $z \in \mathbb{U}$, $z = \exp 2i\pi t$

$$\begin{aligned} \tau z = -\bar z = -\exp 2i\pi t = \exp(\pi i - 2i\pi t) \\ = \exp\{\, 2i\pi \underbrace{\left(\frac{1}{2} - t\right)}_{\tilde\tau(t)} . \end{aligned} \tag{22}$$

It is perhaps more convenient to work with $\tau' = \tau'_\infty = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, which exchanges $e_0 = 1$ and $e_1 = i$, i.e. the reflection with respect|by to the main diagonal. This automorphism stabilizes $\mathbb{U}$, and fixes the point $\exp \frac{\pi i}{4|8} = \frac{\sqrt{2}}{2}(i+1)$), it lifts to

$$\widetilde{\tau'}(t) = 2i\pi \left(\frac{1}{8|4} - t\right). \tag{23}$$

Let us note that † the action of $\widetilde{\tau'}$ (like that 232

Uncertainty retained from cancelled passages.
140-4-s233-D001 in a way
140-4-s233-D002 from the
140-4-s233-D003 a group
140-4-s233-D005 $\mathbb{R}$
140-4-s233-D008 $\widetilde{\tau'}|\tau'$

[scan 234]

of $\tilde{\tau}$) on ω_0 is the symmetry $\omega_0 \mapsto \omega_0^{-1}$ (cf. end of the preceding §). I say that we have

$$(24) \quad \tilde{\tau}'(\tilde{\rho}) = \tilde{\rho}^{-1}, \qquad \tilde{\tau}'(\tilde{\sigma}) = \tilde{\sigma}^{-1}.$$

The formulas are also|thus written

$$\tilde{\tau}'(\tilde{\rho}^{-1}) = \tilde{\rho}, \qquad \tilde{\tau}'(\tilde{\sigma}^{-1}) = \tilde{\sigma}$$

they are valid † in $\mathrm{Sl}(2,\mathbb{Z})$, so to see that they are valid in $\mathrm{Sl}(2,\mathbb{Z})^{\sim}$, it suffices to show that the left-hand sides satisfy the characteristic relations (20), which is written

$$\tilde{\tau}'(\underbrace{\tilde{\rho}^{-3}}_{\omega_0^{\dagger}}) = \omega_0^{-1}, \qquad \tilde{\tau}'(\underbrace{\tilde{\sigma}^{-2}}_{\omega_0^{-1}}) = \omega_0$$

i.e. $\tilde{\tau}'(\omega_0) = \omega_0^{-1}$, OK.

> Theorem. The group $\mathrm{Sl}(2,\mathbb{Z})^{\sim}$ admits
> the presentation (with two generators $\tilde{\rho}, \tilde{\sigma}$)
> (25) $\boxed{\tilde{\rho}^3 \cdot \tilde{\sigma}^2 = 1}$

Proof Let $\Pi|\pi$ be the group † defined by this presentation, and let us consider the homomorphism

$$\begin{aligned} \Pi|\pi &\longrightarrow \mathrm{Sl}(2,\mathbb{Z})/\pm 1 = \{\rho,\sigma \mid \rho^3 = \sigma^2 = 1\} \\ (\tilde{\rho},\tilde{\sigma}) &\longmapsto (\rho,\sigma) \bmod\{\pm 1\} \end{aligned} \qquad (26)$$

233

Uncertainty retained from cancelled passages.
140-4-s234-D001 and
140-4-s234-D002 -1
140-4-s234-D003 generated

[scan 235]

the group $\mathrm{Sl}(2,\mathbb{Z})/\pm 1$ is identified with the quotient of $\pi|\Pi$ by the normal subgroup generated [p. 765]
by the element $\underline{\omega_0} \overset{\text{dfn}}{=} \tilde{\rho}^{-3} = \tilde{\sigma}^2$. But this element being central, the group generated by $\underline{\omega_0}$ is already normal.

thus we have an exact sequence

$$(27)\qquad \begin{array}{ccccccccc} n & \longmapsto & \underline{\omega_0}^n & & & & & & \\ \mathbb{Z} & \longrightarrow & \Pi|\pi & \longrightarrow & \mathrm{Sl}(2,\mathbb{Z})/\pm 1 & \longrightarrow & 1 & & \end{array}$$

which maps into the exact sequence

$$\dagger\qquad \begin{array}{ccccccccc} 1 & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathrm{Sl}(2,\mathbb{Z})^{\sim} & \longrightarrow & \mathrm{Sl}(2,\mathbb{Z})/\pm 1 & \longrightarrow & 1 \\ & & n & \longmapsto & \omega_0^n & & & & \end{array}$$

inducing identity isomorphisms on the end factors — thus by the lemma of the|of 5 $\pi|\Pi \longrightarrow \mathrm{Sl}(2,\mathbb{Z})^{\sim}$ is an isomorphism, q.e.d.

> Corollary 1 The group $\mathrm{Gl}(2,\mathbb{Z})^{\sim}$ admits
> the presentation (with three generators $\tilde{\rho},\tilde{\sigma},\tilde{\tau}'$)

$$(28)\qquad \boxed{\tilde{\tau}'^2 = \dagger\,\tilde{\rho}^3\cdot\tilde{\sigma}^2 = 1,\quad \tilde{\tau}'(\tilde{\rho}) = \tilde{\rho}^{-1},\quad \tilde{\tau}'(\tilde{\sigma}) = \tilde{\sigma}^{-1}}$$

I am going to try to present the † group $\mathrm{Gl}(2,\mathbb{Z})^{\sim}$ with|in three involutive generators, † $\tilde{r}_0,\tilde{r}_1,\tilde{r}_\infty$ over the elements $r_0, r_1|\sigma_0, r_\infty$ of $\mathrm{Gl}(2,\mathbb{Z})^{-}$, by drawing inspiration from the formulary|from the formularies of $\mathrm{Sl}(2,\mathbb{Z})$: 234

Uncertainty retained from cancelled passages.

140-4-s235-D001 (28)|(27)

140-4-s235-D002 $1, |\omega_0$

140-4-s235-D003 element

140-4-s235-D004 at the of

[scan 236]

$$(29)\qquad \begin{cases} r_\infty = \tau_\infty = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \\ r_0 = \tau'_\infty = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ r_1 = -\tau'_0 = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix} \end{cases}$$

m1674

Marginal note. m1674 $\left[\right.$ (30) $\begin{cases} r_0(\rho) = \rho^{-1},\quad r_0(\sigma) = \sigma^{-1} \\ r_1 = \rho^{-1}r_0 = r_0\rho \\ r_\infty = \sigma r_0 = r_0\sigma^{-1} \end{cases}$

Note that $(\tilde{\rho},\tilde{\tau}')$ generate a group isomorphic to D_∞, which I denote by $D_{\tilde{\rho}}$, likewise $(\tilde{\sigma},\tilde{\tau}')$ generates|generate a group $D_{\tilde{\sigma}}$, isomorphic to D_∞. Finally $(\omega_0,\tilde{\tau}')$ generate a group D_{ω_0}, again isomorphic to D_∞. We have a diagram

$$(31)\qquad \begin{array}{ccccc} & & D_{\omega_0} & & \\ & \overset{i}{\swarrow} & & \overset{j}{\searrow} & \\ D_{\tilde{\rho}} & & & & D_{\tilde{\sigma}|\tilde{\rho}} \end{array} \qquad \begin{cases} i(\omega_0) = \tilde{\rho}^{-3} \\ i(\tilde{\tau}') = \tilde{\tau}' \end{cases} \qquad \begin{cases} j(\omega_0) = \tilde{\sigma}^2 \\ j(\tilde{\tau}') = \tilde{\tau}' \end{cases}$$

and corollary 1 is written in the form

$$(32)\quad \mathrm{Gl}(2,\mathbb{Z})^{\sim} \simeq D_{\tilde\rho} *_{D_{\omega_0}} D_{\tilde\sigma}$$

(amalgamated sum of groups).

Now we can take in $D_{\tilde\rho}, D_{\tilde\sigma}$ the involutive generators

$$(33)\quad \begin{cases} \tilde r_0 = \tilde\tau' = \tilde\tau'_\infty & \text{and} \quad \tilde r_1 \overset{\text{dfn}}{=} \tilde\rho^{-1}\tilde r_0 = \tilde r_0\tilde\rho \quad \text{in } D_{\tilde\rho} \\ \tilde r_0 = \cdots\cdots & \text{and} \quad \tilde r_\infty \overset{\text{dfn}}{=} \tilde\sigma\,\tilde r_0 = \tilde r_0\tilde\sigma^{-1} \end{cases}$$

235

[scan 237]

and we then find

[p. 766]

Corollary 2 $\mathrm{Gl}(2,\mathbb{Z})^{\sim}$ admits a presentation (with three involutive generators $\tilde r_0, \tilde r_1, \tilde r_\infty$):

$$(34)\quad \boxed{\tilde r_0^2 = \tilde r_1^2 = \tilde r_\infty^2 = 1, \quad \Big(\underbrace{\tilde r_0^{\dagger}\tilde r_1}_{\tilde\rho}\Big)^3 \overset{\dagger}{\cdot} \Big(\underbrace{\tilde r_\infty \tilde r_0^{\dagger}}_{\tilde\sigma}\Big)^2 = 1}$$

$\dagger$

236

Uncertainty retained from cancelled passages.

140-4-s237-D001 -1

140-4-s237-D002 $=$

140-4-s237-D003 -1

140-4-s237-D004 ω_0

[scan 238]

4°) The isomorphism $\mathrm{Gl}(2,\mathbb{Z})^{\sim} \simeq S\mathcal{T}_{1,1}$ ✓

Put

$$(35)\quad \dagger \quad X_1 = \mathbb{R}^2/\mathbb{Z}^2 \simeq (\mathbb{R}/\mathbb{Z})^2 \simeq \mathbb{U}\times\mathbb{U}.$$

(model surface of genus 1), write its composition law additively, and let

$$(36)\quad a_{1,0} = 0 \in X_1, \quad X_1 \setminus \{a_{1,0}\} = X_1 \setminus \{0\} = U_{1,1},$$

model surface of type $(g,\nu) = (1,1)$.

The group $\mathrm{Gl}(2,\mathbb{Z})$ is identified with the group of automorphisms of the topological group X_1, and thus operates on X_1 leaving fixed the origin 0, hence it operates on $U_{1,0}$, whence a homomorphism

$$\mathrm{Gl}(2,\mathbb{Z}) \xrightarrow{\ \sim\ } \mathcal{T}_{1,1} = \mathcal{T}(U_{1,1})$$

$$(37)\qquad \begin{array}{c} \text{group of isotopy classes} \\ \text{of homeomorphisms} \\ \text{of } U_{1,1}, \end{array}$$

which we know to be an isomorphism. Moreover, $G = \mathrm{Gl}(2,\mathbb{Z})$ operates differentiably|differentially on $\dagger$ X_1, and the action on the tangent space is identified with the tautological action of $\mathrm{Gl}(2,\mathbb{Z}) \subset \mathrm{Gl}(2,\mathbb{R})$ on $\mathbb{R}^2$. Thus the group denoted by $\widetilde{G}$ in

237

Uncertainty retained from cancelled passages.
140-4-s238-D001 X_1
140-4-s238-D003 $\mathcal{T}|\tau$

[scan 239]

1°) e is canonically isomorphic, as an extension of G by $T_0 = \mathbb{Z}$ (N.B. $\mathbb{R}^2$ and its quotient X_1 [p. 767]
are canonically oriented) to the group $\mathrm{Gl}(2,\mathbb{Z})^\sim$ of 3°.

Thus we find † (cf. proposition p. 761) the

Theorem The inverse image under the isomorphism (37) of the extension $S\dagger\mathcal{T}_{1,1}$ of $\mathcal{T}_{1,1}$ by $\mathbb{Z}$ is canonically identified with $\mathrm{Gl}(2,\mathbb{Z})^\sim$, so it admits the two explicit presentations of 3°

$$\text{a)}\quad \tilde\rho^3 \underset{\dagger}{\cdot} \tilde\sigma^2 \underset{\dagger}{=} 1, \quad \tilde\tau'(\tilde\rho) = \tilde\rho^{-1}, \quad \tilde\tau'(\tilde\sigma) = \tilde\sigma^{-1}, \quad \tilde\tau'^{\,2} = 1$$

$$\text{b)}\quad \tilde r_0^2 = \tilde r_1^2 = \tilde r_\infty^2 = 1, \quad (\tilde r_0 \overset{\dagger}{\tilde r_1})^3 \underset{\dagger}{\cdot} (\tilde r_\infty \tilde r_0)^2 = 1.$$

Let us now interpret $\mathcal{T}_{1,1}$ and $S\mathcal{T}_{1,1}$ in terms of the fundamental group of $U_{1,1}$, which we shall compute by choosing (judiciously ?) a base point $a_{1,1}$ in $U_{1,1}$,

$$(38)\qquad \Pi_{1,1} \overset{\text{dfn}}{=} \pi_1(U_{1,1}, a_{1,1}),$$

$$(39)\qquad \mathcal{T}_{1,1} \simeq \mathrm{Aut}\,\mathrm{ext}_{\mathrm{lac}}(\Pi_{1,1})$$

238

Uncertainty retained from cancelled passages.
140-4-s239-D001 Theorem
140-4-s239-D002 $*$
140-4-s239-D003 $\omega\widetilde{\sigma}$
140-4-s239-D004 ω_0
140-4-s239-D005 ω_0
140-4-s239-D006 -1

[scan 240]

m1680

Marginal note. m1680 I use the notation ℓ'_0 instead of ℓ_0, since ℓ_0 already denotes a well-determined element of $\mathrm{Gl}(2,\mathbb{Z})$!

We have a presentation of $\Pi_{1,1}$ by

$$(40)\qquad \Pi_{1,1} = \{x, y, \ell'_0 \mid [x,y]\ell'_0 = 1\}$$

and denoting by L the subgroup-loops of $\Pi_{1,1}$ generated by ℓ'_0, we shall have

$$(41)\qquad S\mathcal{T}_{1,1} \simeq \mathrm{Norm}_{\mathrm{Aut}_{\mathrm{lac}}(\Pi_{1,1})}(L).$$

Thus we should make explicit the action †

$$S\mathcal{T}_{1,1} \simeq \mathrm{Gl}(2,\mathbb{Z})^\sim \longrightarrow \mathrm{Aut}_{\mathrm{lac}}(\Pi_{1,1}),$$

by the action of the generators $\tilde\rho, \tilde\sigma, \tilde\tau'$ or $\tilde r_0 = \tilde\tau', \tilde r_1, \tilde r_\infty$ — action which must normalize L, and more precisely, transform ℓ'_0 into ℓ'_0 (for $\tilde\rho, \tilde\sigma$) or into $\ell_0'^{-1}$ (for $\tilde\tau' = \tilde r_0, \tilde r_1, \tilde r_\infty$). Thus we must make explicit the action of the generators on x and on y. (N.B. $\Pi_{1,1}$ is the free group having as generators x and y !)

Let us consider the principal covering $U'^{a}_{1,1}$ with group $\mathbb{Z}^2$ of $U_{1,1} = X_1 - \{0\}$, induced by the principal covering $\mathbb{R}^2$ of $X_1 = \mathbb{R}^2/\mathbb{Z}^2$, — the inverse image of $U_{1,1}$ is $\mathbb{R}^2 \setminus \mathbb{Z}^2$. Let us choose a base point †

239

Uncertainty retained from cancelled passages.
140-4-s240-D001 of S
140-4-s240-D002 a

[scan 241]
[p. 768]

†

$$a \in \mathbb{R}^2 \setminus \mathbb{Z}^2,$$

up above $a_{1,1}$, we will thus have a canonical isomorphism

$$(42) \qquad \Pi_{1,1} \simeq \pi_1(\mathbb{R}^2 \setminus \mathbb{Z}^2, \mathbb{Z}^2; a)$$

whence an exact sequence

$$(43) \qquad 1 \to \pi_1(\mathbb{R}^2 \setminus \mathbb{Z}^2) \longrightarrow \Pi_{1,1} \longrightarrow \mathbb{Z}^2 \to 1.$$

$$\Pi_{1,1} \quad \| \quad \pi_1(U_{1,1} = X_1 \setminus \{a_{1,0}\}) \qquad \mathbb{Z}^2 \quad \wr \quad \pi_1(X_1) \quad \wr \quad (\Pi_{1,1})_{\mathrm{ab}}$$

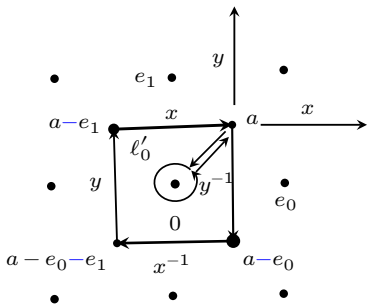
The elements of the mixed π_1 are made explicit by Jean Malgoire's calculation using the pairs

$$(44) \qquad (\gamma, \ell) \qquad \begin{pmatrix} \gamma \in \mathbb{Z}^2, \quad \ell \text{ class of paths in} \\ \mathbb{R}^2 \setminus \mathbb{Z}^2 \text{ from } a \text{ to } \dagger a + \gamma \end{pmatrix}$$

which we "compose like functions".

We will take, over the generators e_0, e_1 of $\mathbb{Z}^2$, the elements x, y defined by the straight-line paths from a to $a + e_0$, resp. $a + e_1$, starting from the origin

$$(45) \qquad a = \tilde{a}_{1,1} = \left(+\frac{1}{2}, +\frac{1}{2}\right)$$



Then the element $xyx^{-1}y^{-1}$ is represented by the path above, which is visibly opposite to the positively oriented "straight-line" loop from a around the origin 0. 240

Uncertainty retained from cancelled passages.
140-4-s241-D001 (44)|(42)
140-4-s241-D002 $\gamma|\tau$

[scan 242]

5°) Digression on the coverings of germs of spaces.
m1684

A) Coverings of germs

Marginal note. m1684 here we exclude † not the improper filter ($\emptyset \in \Phi$) corresponding to the empty germ of spaces

A "germ of spaces" is defined by a topological space S and a filter Φ on S, a morphism from (S, Φ) to (S', Φ') is an element of

$$(46) \qquad \varinjlim_{S_i \in \Phi} \mathrm{Hom}_{\Phi_{S_i|S'_i}, \Phi'}(S_i, S'),$$

where for $S_i \in \Phi$, Φ_{S_i} is the filter induced by Φ on S_i, and $\mathrm{Hom}_{\Phi_{S_i}, \Phi'}$ denotes the set of continuous maps f from S_i to S' compatible with Φ_{S_i}, Φ', i.e. such that $\forall S'_j \in \Phi'$, † we have $f^{-1}(S'_j) \in \Phi_{S_i}$. The composition of morphisms of germs is obvious.

A covering (étale) of a germ (S, Φ) of topological spaces is an object of the category

$$(47) \qquad \underline{\mathrm{Rev}}(S, \Phi) = \varinjlim_{S_i \in \Phi} \underline{\mathrm{Rev}}(S_i); \qquad *$$

these|the coverings form a category, which we could probably interpret (at least in "good cases") as the category of coverings of a suitable topos (namely the projective limit topos 241

Uncertainty retained from cancelled passages.
140-4-s242-D001 the
140-4-s242-D004 f

[scan 243]
m1686 [p. 769]

Marginal note. m1686 this has been done by Olivier Leroy?

of the (small) étale topoi of the $S_i \in \Phi$).

This category, "in the good cases", tends to be multi-Galois, or even Galois.

We say that a germ † of spaces is 1-connected, or simply connected, if the natural functor

$$(48) \qquad \begin{array}{ccc} (\mathrm{Ens}) & \longrightarrow & \underline{\mathrm{Rev}}(S, \Phi) \\ E & \longmapsto & {}^1E_{(S,\Phi)}|\widetilde{E}_{(S,\Phi)}, \text{ constant covering with} \\ & & \text{value } E|\Phi \end{array}$$

is an equivalence of categories. This means that

1°) This functor is fully faithful (we say that (S, Φ) is 0-connected), which means, if S is locally connected, that $\forall S_i \in \Phi$, $\exists S_j \in \Phi$, $S_j \subset S_i$, such that $S_j \neq \emptyset$ and $(\pi_0(S_j) \longrightarrow \pi_0(S_i))$ is a constant map

m1687

Marginal note. m1687 thus this implies that this germ of spaces is nonempty

2°) the functor is moreover essentially!|essentially^t surjective, which means that for every $S_i \in \Phi$, $\exists S_j \in \Phi$, $S_j \subset S_i$ such that for every connected component $S_{j,\alpha}$ of S_j, the outer homomorphism $\pi_1(S_{j,\alpha}) \longrightarrow \pi_1(S_i)$ is trivial.

Coverings of germs of spaces form a fibered category over that 242

Uncertainty retained from cancelled passages.
140-4-s243-D001 is

[scan 244]

of germs of spaces. The category of topological spaces † can be regarded as a full subcategory of that of germs of topological spaces by associating to every space S, the germ (S, Φ), where Φ is the trivial filter on S. This being so, the fibered category of coverings of germs of spaces induces|induced that of coverings of topological spaces.

Let $\dagger\mathcal{X}$ be a germ of spaces such that $\underline{\mathrm{Rev}}\mathcal{X}$ is a multi-Galois category, and let $\mathcal{S}$ be a 1-connected germ of spaces, $\mathcal{S} \longrightarrow \mathcal{X}$ a morphism. Then we have

$$\underline{\mathrm{Rev}}(\mathcal{X}) \longrightarrow \underline{\mathrm{Rev}}(\mathcal{S}) \xleftarrow{\sim} (\mathrm{Ens}) \tag{49}$$

fiber functor on $\underline{\mathrm{Rev}}(\mathcal{X})$, which makes it possible to define

$$\pi_1(\dagger\mathcal{X}, \mathcal{S}) \tag{50}$$

as the π_1 relative to this fiber functor. If $\mathcal{X}$ is 0-connected so $\underline{\mathrm{Rev}}(\mathcal{X})$ is Galois, we will thus have, if $\Pi \simeq \pi_1(\mathcal{X}, \mathcal{S})$, an equivalence of categories 243

Uncertainty retained from cancelled passages.
140-4-s244-D001 [illeg. 1]
140-4-s244-D002 $\mathcal{S}$
140-4-s244-D004 $\mathcal{S}$

[scan 245]
[p. 770]

$$\underline{\mathrm{Rev}}(\mathcal{X}) \simeq \mathrm{Ens}(\Pi). \tag{51}$$

The group $\pi_1(\mathcal{X}, \mathcal{S})$ depends functorially on the triple $(\mathcal{S}, \mathcal{X}, \mathcal{S} \longrightarrow \mathcal{X})$ i.e. on the arrow $\mathcal{S} \longrightarrow \mathcal{X}$ in the category of germs of spaces, where the source $\mathcal{S}$ is 1-connected.

Let|Let X be a topological space, Y a subset of X, we call the germ of X around Y the germ of spaces (X, Φ), where Φ is the filter of neighborhoods of Y in|in X, the residual germ of X around Y the germ of spaces $(X \setminus Y, \Phi_{X \setminus Y})$ induced on $X - Y$ (it is empty iff Y is open).
m1690

Marginal note. m1690 We have a diagram of canonical morphisms

$$\begin{array}{ccc} \mathrm{Germ}_Y(X) & \hookleftarrow & \mathrm{Gerres}_Y(X) \\ \downarrow & & \downarrow \\ X & \hookleftarrow & X - Y \end{array} \tag{52}$$

We will use this notion † in the case where $Y = \{x\}$, speaking also of the germ and residual germ of X at x. (N.B. the germ of X at x is always † 1-connected, but not the residual germ). Every germ of spaces (S, Φ) can be interpreted as the residual germ of a space † $X = S \cup \{\omega\}$ at a suitable point ω … 244

Uncertainty retained from cancelled passages.
140-4-s245-D002 especially
140-4-s245-D003 simp
140-4-s245-D004 S Φ

[scan 246]

. I am interested in the case where $x \in X$ has a neighborhood homeomorphic to $\mathbb{R}^2$ — i.e. X is a surface at x. I will denote by

$$(53) \qquad D^*_{X,x} \quad \text{or} \quad D^*_x$$

the residual germ of X at x (contained in the germ $D_{X,x}$ or D_x of X at x).

I am interested in the choice of a universal covering $D^{*\sim}_x$ of D^*_x — but for which|whom I have to make a digression on the abstract notion of universal covering of a germ of spaces.

Let (S, Φ) be a germ of spaces. A covering of (S, Φ) comes from a covering S'_i of an $S_i \in \Phi$, and defines a germ of spaces $(S'_i, \Phi_{S'_i})$, where $\Phi_{S'_i}$ denotes the inverse image of $\Phi_{\dagger S_i}$ in S'_i. This germ of spaces over $\mathcal{S} = (S, \Phi)$ (up to unique isomorphism of germs) does not depend on the choice of S_i, S'_i, but only on the element of Rev$(\mathcal{S})$. We thus find a canonical functor 245

Uncertainty retained from cancelled passages.
140-4-s246-D002 S'_i

[scan 247]
[p. 771]

$$(54) \qquad \text{Rev}(\mathcal{S}) \longrightarrow (\text{Germesp})/\mathcal{S}$$

which is fully faithful. A germ of spaces $\mathcal{S}'$ over $\mathcal{S}$ is called † an étale covering of $\mathcal{S}$, if it is in the essential image. Thus the category Rev$(\mathcal{S})$ of étale coverings of the germ of spaces $\mathcal{S}$ is equivalent to the full subcategory of the † category of germs of spaces, over the germ $\mathcal{S}$, formed by the étale coverings of the said germ $\mathcal{S}$.

Let us assume always|always that the category Rev$(\mathcal{S})$ is † multi-Galois. The 1-connected objects are called universal coverings of $\mathcal{S}$ — they form a groupoid, connected iff Rev$(\mathcal{S})$ is Galois i.e. iff the germ $\mathcal{S}$ is 0-connected. The universal coverings!|coverings of $\mathcal{S}$ can be interpreted as germs of spaces. Let $\tilde{\mathcal{S}}$ be such a universal covering, then $\tilde{\mathcal{S}}$ is indeed a 1-connected|1. connected germ of spaces, and $\tilde{\mathcal{S}} \longrightarrow \mathcal{S}$ makes it possible to define $\pi_1(\mathcal{S}, \tilde{\mathcal{S}})$. This being so, we have a canonical isomorphism

$$(55) \qquad \pi_1(\mathcal{S}, \tilde{\mathcal{S}}) \simeq \text{Aut}_{\text{Rev}(\mathcal{S})}(\tilde{\mathcal{S}})$$

m1693

Marginal note. m1693 Which expresses that $\pi_1(\mathcal{S}, \widetilde{\mathcal{S}})$ is none other than the π_1 of the "abstract" multi-Galois category Rev$(\mathcal{S})$, "calculated" at the 1-connected element $\widetilde{\mathcal{S}}$.

246

Uncertainty retained from cancelled passages.
140-4-s247-D001 the
140-4-s247-D002 g
140-4-s247-D005 g

†

[scan 248]

That said, let us return to the situation of (53), of a topological space X, and of an $x \in X$ such that X is a surface at x. Then the category †

B) Case of punctured surfaces

$$\text{Rev}(D^*_x)$$

is a Galois category, abelian, whose fundamental group is identified canonically with the orientation module $\overline{T}_x$ of X at x

$$(56) \qquad \pi_1(D_x^*) \overset{\text{def}}{=} \pi_1(\underline{\text{Rev}}(D_x^*)) \simeq \overline{T}_x.$$

This comes from the fact that the germ D_x^* † admits a "basis" formed of spaces $D_{x,i}^*$ † arising from the disks with center x, and the π_1 of the said disks form a constant projective system…

Thus the universal coverings $D_x^{*\sim}$ of D_x^* form a connected groupoid, with fundamental groupoid|group † $\overline{T}_x$.

Let us therefore choose such a $D_x^{*\sim}$, and consider the composite morphisms

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Uncertainty retained from cancelled passages.

140-4-s248-D001 In short.

140-4-s248-D002 coverings

140-4-s248-D003 is [illeg. 1]

140-4-s248-D004 isomorphic to

140-4-s248-D005 $\mathbb{Z}$

[scan 249]
[p. 772]

$$D_x^{*\sim} \longrightarrow D_x^* \hookrightarrow X - \{x\}$$

- more generally, let S be a subset of X, containing x as an isolated point (for example a discrete subset of X), so that we have

$$(57) \qquad D_x^* \hookrightarrow U \overset{\text{def}}{=} X \setminus S$$

and consider the composites

$$(58) \qquad D_x^{*\sim} \longrightarrow D_x^* \hookrightarrow U.$$

If $\underline{\text{Rev}}(U)$ is multi-Galois (for example U locally 1|0-connected), then we can therefore consider the corresponding fundamental group of U

$$(59) \qquad \pi_1(U, D_x^{*\sim}),$$

I am going to define a canonical homomorphism

$$(60) \qquad K_x|\kappa_x : \overline{T}_x \longrightarrow \underbrace{\pi_1(U, D_x^{*\sim})}_{\simeq \text{ Aut of the fiber functor } \underline{\text{Rev}}(U) \longrightarrow \underline{\text{Rev}}(D_x^{*\sim}) \text{ on } \underline{\text{Rev}}(U)},$$

we can describe it for example as

$$(61) \qquad \begin{array}{ccc} \pi_1(D_x^*, D_x^{*\sim}) & \longrightarrow & \pi_1(U, D_x^{*\sim}) \\ \wr & & \\ \overline{T}_x & & \end{array}$$

[scan 250]

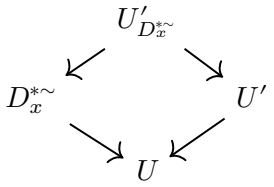
induced by the inclusion

$$D_x^* \hookrightarrow U$$

or again, † in terms of automorphisms of a fiber functor, as the †operation of $\overline{T}_x$ on

$$\Gamma(D_x^{*\sim}, \underbrace{U' \times_{\dagger U} D_x^{*\sim}}_{U'_{D_x^{*\sim}}})$$

coming from the operations of $\overline{T}_x$ on $D_x^{*\sim}$.

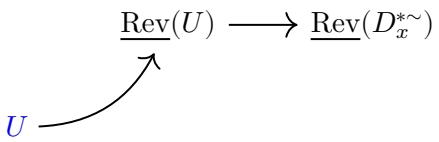


†

Suppose that we have already chosen separately a fundamental group $\pi_1(U, \widetilde{U}) = \pi$ of U, by the choice of a base point, or, more generally, of a universal covering $\widetilde{U}$ of U. To compare it with $\pi_1(U, \dagger D_x^{*\sim})$, we shall therefore choose a "path of domination" from $D_x^{*\sim}/U$ to|: $\widetilde{U}/U$, i.e. an isomorphism

$$(62) \qquad \widetilde{U} \xrightarrow{\sim} \mathrm{Revuni}(U, D_x^{*\sim})$$

(universal covering of $\dagger U$ defined by the fiber functor



m1697

Marginal note. m1697 N.B. if † $D_x^{*\sim}$ defined by a universal covering $D_{x,i}^{*\sim}$ of a punctured disk $D_{x,i}^*$, and if $\tilde{y} \in D_{x,i}^{*\sim}$, $\tilde{z} \in \widetilde{U}$, y, z the images in U, then

$$\dagger\mathrm{Revuni}(U, D_x^{*\sim}) \simeq \mathrm{Revuni}(U, y)$$

and $\widetilde{U} = \mathrm{Revuni}(U, z)$: the choice of (62) is equivalent to that of a path from y to z (U locally 1-connected).

Uncertainty retained from cancelled passages.

140-4-s250-D001 by
140-4-s250-D002 homomorphism
140-4-s250-D003 D
140-4-s250-D004 [illeg. 1] Each time
140-4-s250-D005 S
140-4-s250-D006 $\widetilde{U}$
140-4-s250-D007 [illeg. 1]
140-4-s250-D008 D

[scan 251]
[p. 773]

(the indeterminacy is modulo an † element † of π, or of $\pi_1(U, D_x^{*\sim})$) which induces an isomorphism

$$(63) \qquad \pi_z|\pi_s|\pi = \pi_1(U, \widetilde{U}) \xleftrightarrow{\sim} \pi_1(U, D_x^{*\sim})$$

(which is changed by an inner automorphism, when we change|compose (62) by an element of π).

Composing (63) and (60), we therefore find a homomorphism

$$(64) \qquad \overline{T}_x \longrightarrow \pi = \pi_1(U, \widetilde{U}),$$

which depends on the choice of (62) – and which, independently of this choice, defines a well-determined outer homomorphism – which is none other than the outer homomorphism

$$(65) \qquad \begin{array}{ccc} \pi_1(D_x^*) & \longrightarrow & \pi_1(U). \\ \wr & & \wr \\ \overline{T}_x & & \pi \end{array}$$

m1698

Marginal note. m1698 C) Relation with germs of arcs (playing the role † of "base points|point")

I am going to describe a convenient way of fixing a universal covering $D_x^{*\sim}$ of D_x^*, by a germ of an arc of a curve of X, issuing

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Uncertainty retained from cancelled passages.

140-4-s251-D001 automorphism

140-4-s251-D002 inner

140-4-s251-D003 origin

from x. In general, let $I = [0, 1]$, and $[I]$ be the residual germ of I at 0. This is obviously a [scan 252] 1-connected germ.

m1699

Marginal note. m1699 Rather than parametrize by germs of neighborhoods of 0 in $[0, 1]$, it would be more convenient for what follows to define a germ of an arc of X issuing from x as a germ of a subset of X at x, satisfying certain properties….

If X is a topological space and $x \in X$, we call a germ of an arc of a curve X issuing from x, the morphisms

$$(66) \qquad [I] \xrightarrow{\gamma} \text{germ rés}_x(X) \overset{\text{def}}{=} D_x^*$$

from the germ $[I]$ into the residual germ $D_{X,x}^*$ of X at x. Such a germ is therefore described by a continuous map from an interval $[0, \varepsilon]$ $(0 < \varepsilon \leq 1)$

$$(67) \qquad [0, \varepsilon] \xrightarrow{\varphi\,\dagger} X$$

such that we have

$$(68) \qquad \begin{cases} \varphi(0) = x \\ \varphi(t) \neq x \quad \text{if } t \in]0, \varepsilon] \end{cases} .$$

In the case we are interested in (X a surface in a neighborhood of x, hence $\underline{\mathrm{Rev}}(D_x^*)$ † Galois abelian with group $\overline{T}_x$), we can use this morphism to define a fiber functor on D_x^*

$$(69) \qquad \underline{\mathrm{Rev}}(D_x^*) \longrightarrow \underline{\mathrm{Rev}}([I]) \xleftarrow{\sim} (\mathrm{Ens})$$

Uncertainty retained from cancelled passages.

140-4-s252-D003 γ

140-4-s252-D004 Galois

[scan 253]

[p. 774]

hence a universal covering

$$D_x^{*\sim} \overset{\text{déf}}{=} \mathrm{Rev\,uni}(D_x^*, ([I], \varphi)). \tag{70}$$

Of course, we shall have a factorization

$$[I] \longrightarrow D_x^{*\sim} \longrightarrow D_x^* \ (\hookrightarrow U)$$
$$\varphi$$

so we have a canonical isomorphism

$$\mathrm{Rev\,uni}(U, D_x^{*\sim}) \simeq \mathrm{Rev\,uni}(U, [I]), \tag{71}$$

where I is regarded as a 1-connected germ over U by the composite morphism

$$[I] \longrightarrow D_x^* \longrightarrow U.$$

Thus we have

$$(72) \quad \pi_1(U, D_x^{*\sim}) = \pi_1(U, [I]) \xleftarrow{K_x} \overline{T}_x.$$

m1700

Marginal note. m1700 D) Relation with differential structure

When X is equipped with a differentiable structure in a neighborhood of x, we say that a germ of a curve in X issuing from x is differentiable, if the|the germ of a continuous map

$$I \longrightarrow X$$

which defines it is differentiable, †and moreover

Uncertainty retained from cancelled passages.

140-4-s253-D002 Let us take|For this

[scan 254]

the tangent map at the origin, †

$$(73)\quad \mathbb{R} \longrightarrow \mathrm{Tang}_x(X),$$

is injective. Then the image of $\mathbb{R}^+$ is a homogeneous half-line in the tangent space to X at x, which we call the half-tangent of the germ of an arc of a curve under consideration at x.

Returning for a moment to the purely topological case, we are going to consider the set Germarc|Courarc(X,x) of germs of arcs of X issuing from x, as the set of objects of a groupoid (the local fundamental groupoid of X at x), taking as morphisms from γ to γ' † the isomorphisms from $D^{*\sim}_{x,\gamma}$ to $D^{*\sim}_{x,\gamma'}$

$$\mathrm{Isom}(\gamma,\gamma') \overset{\text{déf}}{=} \mathrm{Isom}_{D^*_x}(D^{*\sim}_{x\gamma}, D^{*\sim}_{x,\gamma'}). \tag{74}$$

We thus find by construction an equivalence of connected groupoids

$$(75)\quad \underline{\text{Germarc|Courarc}}(X,x) \longrightarrow \underline{\text{Rev uni}}(D^*_x).$$

Uncertainty retained from cancelled passages.

140-4-s254-D001 [illeg. 2] leads to morphism

140-4-s254-D002 a|the

[scan 255]

[p. 775]

Let on the other hand, in the case where X is differentiable at x, †

$$\underline{\text{Germarc diff|Courarc diff}}(X,x) \tag{76}$$

be the full subcategory of † $\underline{\text{Germarc|Courarc}}(X,x)$ formed by the differentiable germs of arcs. Let $\mathbb{S}_x \simeq \mathbb{S}^1$ be the space of half-lines of the Tangent space $\mathrm{Tang}_x(X)$

$$\mathbb{S}_x = \begin{matrix}\text{space of}\\ \text{Half-lines}\\ \text{of } \mathrm{Tang}_x(X)\end{matrix} \simeq \mathbb{S}^1, \tag{77}$$

and let us consider the fundamental groupoid

$$(78)\quad \Pi_1(\mathbb{S}_x) \xrightarrow{\approx} \underline{\text{Rev uni}}(\mathbb{S}_x).$$

m1701

Marginal note. m1701



I am going to define an equivalence of categories

$$(79)\quad \begin{array}{ccc} \underline{\text{Germarc diff|Courarc diff}}(X,x) & \xrightarrow{\approx} & \Pi_1(\mathbb{S}_x) \\ \downarrow\wr & & \downarrow\wr \\ \underline{\text{Rev uni}}(D^*_x) & & \underline{\text{Rev uni}}(\mathbb{S}_x) \end{array}$$

which, on the sets of objects, is to be the map associating to ∀ differentiable germ of an arc at x its half-tangent. The question is thus, if γ, γ' are two such arcs, corresponding to half-tangents Δ, Δ', to define a canonical bijection

Uncertainty retained from cancelled passages.
140-4-s255-D001 Germ
140-4-s255-D002 $\mathcal{C}_x|\mathcal{G}_x$

[scan 256]

m1702

Marginal note. m1702

$$\begin{array}{ccc} (80)\quad \mathrm{Isom}(\gamma,\gamma') & \xrightarrow{\sim} & \mathrm{Isom}_{\Pi_1 \mathbb{S}_x}(\Delta,\Delta') \\ \| \text{ déf} & & \| \text{ déf} \\ \mathrm{Isom}(D^{*\sim}_{x,\gamma}, D^{*\sim}_{x,\gamma'}) & & \text{classes of paths in } \mathbb{S}^*_x \text{ from } \Delta \text{ to } \Delta' \end{array}$$

† Instead of defining the bijection (80) directly, I am going to define rather a functor (which will be an equivalence of categories)

$$(81)\quad \Theta : \Pi_1(\mathbb{S}_x) \xrightarrow{(\dagger\approx)} \underline{\mathrm{Rev\,uni}}(D^*_x),$$

Let

$$\left\{ \begin{array}{l} \mathcal{A}_x \text{ set of closed 1-connected subsets of } \mathbb{S}_x \\ \text{(i.e. arcs of a circle in } \mathbb{S}_x, \\ \quad \text{possibly reduced to points,} \\ \quad \text{but } \neq \mathbb{S}_x) \end{array} \right. \qquad (82)$$

this is an ordered set, and one|thus defines an ordered category $\underline{\mathcal{A}}_x$ (the objects are the $\in \mathcal{A}_x$, the arrows are the inclusions) — and we know that the enveloping groupoid $\underline{\mathrm{Gr}}(\underline{\mathcal{A}}_x)$ of this ordered category is canonically equivalent to $\Pi_1(\mathbb{S}_x)$,

Uncertainty retained from cancelled passages.
140-4-s256-D001 [illeg. 1]
140-4-s256-D002 $\to$ | $\sim$

[scan 257]

more precisely, let $\dagger\underline{\mathrm{Gr}}_0(\underline{\mathcal{A}}_x)$ be the full subcategory of $\underline{\mathrm{Gr}}(\underline{\mathcal{A}}_x)$ (whose set of objects is $\mathcal{A}_x$) whose objects are the one-point arcs, & we then have a canonical equivalence [p. 776]

$$(83)\quad \underline{\mathrm{Gr}}_0(\underline{\mathcal{A}}_x) \xrightarrow{\approx} \Pi_1(\mathbb{S}_x)$$

which on objects reduces to $\{a\} \longmapsto a$, and which, for any arc † A with endpoints a, b, whence inclusion isomorphisms in|in $\underline{\mathrm{Gr}}(\mathcal{A}_x)$

$$(84) \quad \left\{ \qquad a \xrightarrow{\sim} A \xleftarrow{\sim} b \right.$$

$$\text{whence} \quad a \xrightarrow{\sim} b \qquad \text{in } \underline{\mathrm{Gr}}(\mathcal{A}_x)$$

associates to this arrow $a \xrightarrow{\sim} b$ of $\underline{\mathrm{Gr}}_0(\mathcal{A}_x)$ the arrow of $\Pi_1(\mathbb{S}_x)$, defined by the arc A regarded as a path from a to b.

Thus, for $\forall$ groupoid $\mathcal{G}$, giving a functor

$$(85) \quad \Psi : \Pi_1(\mathbb{S}_x) \longrightarrow \mathcal{G}$$

is equivalent to giving a functor

Uncertainty retained from cancelled passages.

140-4-s257-D001 $\underline{\mathcal{A}}_{0x}|\underline{\mathcal{A}}_x$

140-4-s257-D002 $\mathbb{S}|\xi$

[scan 258]

$$\Psi : \underline{\mathcal{A}}_x \longrightarrow \mathcal{G} \tag{86}$$

and (85) is an equivalence of categories iff $\mathcal{G}$ is † connected and † its $\pi_1|\Pi_1$ is $\simeq \mathbb{Z}$, and if, for †a pair (a, b) of distinct points a, b of $\mathbb{S}_x$, defining two arcs A and B with endpoints a, b, considering the automorphism of a in $\underline{\mathrm{Gr}}_0(\mathcal{A}_x)$ i.e. in $\underline{\mathrm{Gr}}(\mathcal{A}_x)$, composed of

$$\{a\} \xrightarrow{\sim} A \xleftarrow{\sim} \{b\} \xrightarrow{\sim} B \xleftarrow{\sim} \{a\} \tag{86bis}$$

the corresponding automorphism of $\Psi(\{a\})$ in $\mathcal{G}$ is a generator of $\mathrm{Aut}_{\mathcal{G}}(\Psi(\{a\}))$.

We are thus going to define a functor

$$\underline{\mathcal{A}}_x \longrightarrow \underline{\mathrm{Revuni}}(D_x^*) \tag{87}$$

(satisfying the additional condition stated). To define such a functor, it suffices to define a functor

$$\underline{\mathcal{A}}_x \longrightarrow \begin{array}{c}\text{category of 1-connected germs} \\ \text{of spaces, over|over the} \\ \text{germ } D_x^*\end{array} \tag{88}$$

Uncertainty retained from cancelled passages.

140-4-s258-D001 connected

140-4-s258-D002 the

140-4-s258-D003 an x of

by composing this functor with the functor

[scan 259]

[p. 777]

$$\begin{matrix}\text{category of 1-connected germs}\\ \text{of spaces over|over}\\ \text{the germ } D_x^*\end{matrix} \longrightarrow \underline{\text{Revuni}}(D_x^*). \tag{89}$$

In fact, (88) will associate|we associate to $\forall\ A \in \dagger\mathcal{A}_x$ a † filter on $X - x$ tending to x, of which we will take the germ †.

† The elements A of $\mathcal{A}_x$ are obviously identified with the cones †closed C in $\text{Tang}_x(X)$, † distinct from $\{0\}$ and from the tangent^t|tangent space $\forall$ entire. † Let $C\mathcal{A}_x|C \in \mathcal{C}\mathcal{A}_x$ be one of these cones. Let us consider a local isomorphism †

$$f : (\text{Tang}_x(X), 0)\dagger \xrightarrow{\sim} (X, x), \tag{90}$$

whose map on tangent spaces is the identity. If C is a cone $\in \mathcal{C}\mathcal{A}_x$, its image under such an isomorphism depends on the choice of the local isomorphism in question. But let us consider the † filter base †of cones $C' \in \mathcal{C}\mathcal{A}_x$ containing † C strictly (i.e. corresponding to arcs A' which are neighborhoods of A, or again such that $C' - \{0\}$ is a neighborhood of $C \setminus \{0\}$), then the

Uncertainty retained from cancelled passages.

140-4-s259-D001 $\mathcal{E}$
140-4-s259-D002 subspace
140-4-s259-D003 residual
140-4-s259-D004 The half-lines
140-4-s259-D005 open
140-4-s259-D006 disti
140-4-s259-D007 Let C be a
140-4-s259-D008 a
140-4-s259-D009 &
140-4-s259-D012 family|set of
140-4-s259-D013 arcs
140-4-s259-D014 $\in$

the image under (90) † of the filter defined by the base of a filter on $\text{Tang}_x(X)$ formed by the $V \cap (C' \setminus \{0\})$, where|: V ranging over the neighborhoods of 0 in $\text{Tang}_x(X)$, is a filter on $X \setminus \{x\}$ which no longer depends on the choice of f, and which tends to x, hence defines a germ of spaces $\widetilde{A}$ or $\widetilde{C}$ over † D_x^*. This germ of spaces, † isomorphic to the germ of spaces defined by $\text{Tang}_x(X)$ endowed with the filter with base the $V \cap (C' - \{0\})$ — where one can take V convex closed symmetric|symmetric, hence $V \cap (C' - \{0\})$ contractible — is obviously 1-connected. For A i.e. C varying i.e. for $A_1 \subset A_2$ i.e. $C_1 \subset C_2$ we have the corresponding inclusion relation †

[scan 260]

$$\widetilde{C_1} \longrightarrow \widetilde{C_2}$$

on the germs of spaces in D_x^*. We thus do get (88), whence (87)and (81), and we check that the latter is indeed an equivalence of groupoids, via the verification|verification^t † made explicit|made explicit above ((86bis)).

Uncertainty retained from cancelled passages.

140-4-s260-D002 of this
140-4-s260-D003 $X_x|X$
140-4-s260-D004 is é|and it
140-4-s260-D005 A
140-4-s260-D007 by

Let us now consider, once again, a differentiable arc γ in X issuing from x, and its half-tangent

[scan 261]
[p. 778]

Δ. Then $\{\Delta\} \in \mathcal{A}_x$, and we can consider the germ of spaces $\{\Delta\}^\sim$ over D^*_x. We have the|some two canonical inclusion morphisms

$$\gamma \hookrightarrow \{\Delta\}^\sim \hookrightarrow D^*_x \tag{91}$$

and we deduce from this a canonical isomorphism

$$\begin{array}{ccc} \mathrm{Revuni}(D^*_x, \gamma) & \simeq & \mathrm{Revuni}(D^*_x, \{\Delta\}^\sim) \\ \| \text{déf} & & \| \text{déf} \\ D^{*\sim}_{x,\gamma} & & \Theta(\Delta) \end{array} \tag{92}$$

If γ' is another differentiable arc at x, corresponding to the half-tangent Δ', we thus have

$$D^{*\sim}_{x,\gamma} \simeq \Theta(\Delta), \qquad D^{*\sim}_{x,\gamma'} = \Theta(\Delta')$$

whence

$$\begin{array}{ccc} \mathrm{Isom}(D^{*\sim}_{x,\gamma}\}, D^{*\sim}_{x,\gamma'}) & \simeq & \mathrm{Isom}(\Theta(\Delta), \Theta(\Delta')) \\ \| \text{déf} & & \sim \uparrow \Theta \\ \mathrm{Isom}(\gamma, \gamma') & & \mathrm{Isom}_{\Pi_1(\mathbb{S}_x)}(\Delta, \Delta') \end{array}$$

which is the † bijection (80) sought.

In summary, we have an essentially commutative diagram of equivalences of groupoids

Uncertainty retained from cancelled passages.

140-4-s261-D001 right

[scan 262]

$$\begin{array}{ccc} \underline{\mathrm{germesdiff}}(X, x) & \xrightarrow[\approx]{\gamma \longmapsto \underline{\Delta}(\gamma)} & \Pi_1(\mathbb{S}_x) \\ {\scriptstyle \gamma \mapsto \mathrm{Revuni}(D^*_x, \gamma) = D^{*\sim}_{x,\gamma}} \downarrow \approx & \Theta \approx \swarrow & \downarrow \approx \quad {\scriptstyle a \mapsto \mathrm{Revuni}(\mathbb{S}_x, \cdot)} \\ \underline{\mathrm{Rev.uni.}}(D^*_x) & \xrightarrow{\approx} & \mathrm{Revuni.}(\mathbb{S}_x) \end{array} \tag{93}$$

m1703

Marginal note. m1703 Of course, the difficulty here came from the fact that we had no direct morphism between the germ D^*_x, and the space $\mathbb{S}_x$...

The only functor that has not yet made explicit is the functor

$$\begin{array}{ccc} \underline{\mathrm{Revuni}}(D^*_x) & \xrightarrow{\approx} & \mathrm{Revuni}(\mathbb{S}_x) \\ & {\scriptstyle \Theta^{-1}} \searrow \quad \nearrow {\scriptstyle \mathrm{can}} & \\ & \Pi_1(\mathbb{S}_x) & \end{array} \tag{94}$$

defined as the composite of Θ^{-1} with the canonical functor $\Pi_1(\mathbb{S}_x) \longrightarrow \mathrm{Rev.uni.}(\mathbb{S}_x)$.

Finally, in these considerations of differential topological geometry, the key construction is that of the functor Θ (81). We have taken some pains to define it in an invariant way. In the case where we have a local isomorphism

$$f : (\underbrace{\mathrm{Tang}_x(X)}_{V}, 0) \longrightarrow (X, x), \tag{90}$$

the construction of this|— Θ can be made explicit as a canonical construction in a vector space V. Every homogeneous half-line

[scan 263]

Δ in V defines a germ of arcs (differentiable) in V issuing from 0, whence a corresponding universal covering $D^{*\sim}_{V,0;\Delta}$ of $D^*_{V,0}$, hence a universal covering $D^{*\sim}_{x,\Delta}$ of D^*_x. We must see how this $D^{*\sim}_{x,\Delta}$ depends "continuously" on Δ, and what amounts to the same thing, how the universal covering $V^{*\sim}_\Delta$ of $V^* = V \setminus \{0\}$ depends continuously on Δ, which is quite standard....

[p. 779]

Now let $u|\varphi$ be an an automorphism of X, fixing x, and differentiable at x. We thus have † automorphisms induced

$$ {}^{u|\varphi}D^*_x \quad \text{automorphism of } D^*_x, \qquad {}^{u|\varphi}\mathbb{S}_x \quad \text{automorphism of } \mathbb{S}_x. \tag{95|94} $$

Consider on the other hand a universal covering $\tilde{\mathbb{S}}_x$ of $\mathbb{S}_x$, and let $D^{*\sim}_x$ be an associated universal covering of D^*_x, thanks to the equivalence (94). I say that we then have a

Uncertainty retained from cancelled passages.

140-4-s263-D002 some

[scan 264]

canonical bijection

$$ \begin{array}{c}\text{set of liftings}\\ \text{of } {}^uD^*_x \text{ to an}\\ \text{automorphism } {}^uD^{*\sim}_x \text{ of the}\\ \text{germ } D^{*\sim}_x\end{array} \quad \simeq \quad \begin{array}{c}\text{set of liftings}\\ \text{of } {}^u\mathbb{S}_x \text{ to an}\\ \text{automorphism } {}^u\mathbb{S}^\sim_x \text{ of }\\ \mathbb{S}^\sim_x\end{array} \text{ ,} \tag{96} $$

m1704

Marginal note. m1704 the construction (96) will be carried out in the following pages, leading to (122) page 783

bijection which is an isomorphism of $\overline{T}_x$-torsors. (where $\overline{T}_x$ is the $\mathbb{Z}$-orientation module of X at x, let us recall).

m1705

Marginal note. m1705 E) Axiomatization : the structure|structure of correlation between universal coverings $\mathbb{S}^\sim_x$ and $D^{*\sim}_x$.

I feel like giving an axiomatic description of (94) and (96).

1) Let $D_x = (X, x)$ be a germ of a differentiable surface, whence $V_{X,x} = V_x = \mathbb{V} = \mathrm{Tang}_x(X)$, and $\mathbb{S}_x = V^*/\mathbb{R}^{*+}$ (where $\mathbb{V}^* = V \setminus \{0\}$), as well as $D^*_{X,x} = D^*_x$. m1706

Marginal note. m1706 "germ" : we tacitly include the filter of neighborhoods of x in X

† We denote by † $\mathcal{A}_{X,x}$ or $\mathcal{A}_x$ the set of arcs of $\mathbb{S}_x$, in one-to-one correspondence with the set $\mathcal{C}_{X,x} = \mathcal{C}_x$ of † "closed cones" proper of V ($\neq V$ and $\{0\}$). For $A \in \mathcal{A}_x$ or $C \in \mathcal{C}_x$, we denote by † $\mathrm{Ep}|E_p(A)$ or $\mathrm{Ep}|E_p(C)$ the corresponding germ of spaces over D^*_x. If $\tilde{\mathbb{S}}$ is a universal covering of $\mathbb{S}$, $\widetilde{D}^*_x$ a universal covering of $\widetilde{D}^*_x$, we consider the sets

"thickening" † of A (or of C) in D^*_x

Uncertainty retained from cancelled passages.

140-4-s264-D001 resp. universal

140-4-s264-D002 $\mathcal{A}|\in$

140-4-s264-D003 angular sectors proper

140-4-s264-D004 $\widetilde{A}|f(A)$ or $\widetilde{C}|f(C)$

140-4-s264-D005 image|germ

[scan 265]
[p. 780]

$$\begin{array}{ccc} \Gamma(\dagger A, \mathbb{S}_x^\sim/\mathbb{S}_x) & , & \Gamma(\mathrm{Ep}|E_p(\widetilde{A}), D_x^{*\sim}/D_x^*) \\ \| & & \| \\ \mathrm{Hom}_{\mathbb{S}}(A, \mathbb{S}_x^\sim) & & \mathrm{Hom}_{D_x^*}(\widetilde{A}, D_x^{*\sim}) \\ \text{déf} \quad \| & & \text{déf} \quad \| \\ \mathbb{S}_x^\sim(A) & & D_x^{*\sim}(\mathrm{Ep}|E_p(A)) \end{array} \tag{97}$$

which are both Torsors under the same group $\overline{T}_{X,x} = \overline{T}_x$ (the $\mathbb{Z}$-module of orientations of X at the point x). Consider the set

$$H(A) = \mathrm{Isom}_{T_x|\overline{T}_x}(\mathbb{S}_x^\sim(A), \dagger) D_x^{*\sim}(\mathrm{Ep}|E_p(A)) \tag{98}$$

which is itself a $\overline{T}_x$-torsor. If we have $\dagger$ an inclusion

$$A \subset A'$$

we deduce from it, by restriction, isomorphisms of $\overline{T}_x$-torsors

$$\mathbb{S}_x^\sim(A') \xrightarrow{\sim} \mathbb{S}_x^\sim(A), \qquad D_x^{*\sim}(\mathrm{Ep}|E_p(A'))-| \longrightarrow D_x^{*\sim}(\mathrm{Ep}|E_p(A)) \tag{99}$$

whence a canonical isomorphism

$$H(A) \xrightarrow{\sim} H(A') \tag{100}$$

so that $\dagger$

$$A \longmapsto H(A), \quad \text{functor} \quad \underline{\mathcal{A}}_x \xrightarrow{H} \underline{\mathrm{Tors}}(\overline{T}_x), \tag{101}$$

where $\underline{\mathcal{A}}_x$ is the ordered category associated to the ordered set $\mathcal{A}_x$. We thus find by factorization a functor from the|from enveloping groupoid of $\underline{\mathcal{A}}_x$ to $\underline{\mathrm{Tors}}(\overline{T}_x)$;

$$\begin{array}{ccc} & \underline{\mathcal{A}}_x & \\ {\scriptstyle \rho|\gamma}\swarrow & & \searrow{\scriptstyle H} \\ \underline{\mathrm{Gr}}(\underline{\mathcal{A}}_x) & \xrightarrow{\underline{\mathrm{Gr}}(H)} & \underline{\mathrm{Tors}}(\overline{T}_x) \end{array} \quad . \tag{102}$$

Uncertainty retained from cancelled passages.

140-4-s265-D001 $\mathbb{S}|\widetilde{A}$

140-4-s265-D002 $D_x^{*\sim}(\widetilde{A})$

140-4-s265-D003 [illeg. 1]

140-4-s265-D004 $\longmapsto$ | $\longleftarrow$ defines

[scan 266]

We know that $\underline{\mathrm{Gr}}(\underline{\mathcal{A}}_x)$ is † a $\overline{T}_x$-groupoid, more precisely the functor

$$\begin{cases} \underline{\mathcal{A}}_x & \longrightarrow & \underline{\mathrm{Revuni}}(\mathbb{S}_x) \overset{\text{déf}}{=} \mathbb{S}^{\dagger\sim}_{x,A} \\ A & \longmapsto & \dagger\,\mathrm{Revuni}(\mathbb{S}, A) \end{cases} \tag{103}$$

which factors as a morphism of groupoids

$$\underline{\mathrm{Gr}}(\underline{\mathcal{A}}_x) \overset{(\simeq)}{\longrightarrow} \underline{\text{Rev. uni.}}(\mathbb{S}_x) \tag{104}$$

gives an equivalence of categories (104), hence

$$\mathrm{Hom}_{\underline{\mathrm{Gr}}(\underline{\mathcal{A}}_x)}(A, A') \xrightarrow{\sim} \mathrm{Hom}_{\dagger\mathbb{S}_x}(\mathbb{S}^{\sim}_{x,A}, \mathbb{S}^{\sim}_{x,A'}) \tag{105}$$

m1708

Marginal note. m1708 i.e. we have for $A, A' \in \mathcal{A}_x$, the canonical isomorphism

$$\mathrm{Hom}_{\underline{\mathrm{Gr}}(\underline{\mathcal{A}}_x)}(A, A') \simeq \mathrm{Isom}_{\underline{\mathrm{Revuni}}(\mathbb{S}_x)}(\mathbb{S}^{\sim}_{x,A}, \mathbb{S}^{\sim}_{x,A'}).$$

This † recalled, I say that the functor $\underline{\mathrm{Gr}}(H)$ of (102) is essentially constant i.e. it defines on the π_1 the zero homomorphism $\overline{T}_x \overset{0}{\to} \overline{T}_x$ — or what amounts to the same thing, there is between the $H(A)$, including the isomorphisms (100), a transitive system of isomorphisms. We then denote the "common value" of these $H(A)$, which depends only on the choice of $\mathbb{S}^{\sim}_x$ and of $D^{*\sim}_x$, by $H(\mathbb{S}^{\sim}_x, D^{*\sim}_x)$. †This set is endowed with the structure defined by isomorphisms of $\overline{T}_x$-torsors|functors

$$i_A|\tilde{\imath}_A : H(A) \xrightarrow{\sim} H(\mathbb{S}^{\sim}_x, D^{*\sim}_x) \tag{106}$$

satisfying the compatibility condition, express-

Uncertainty retained from cancelled passages.

140-4-s266-D002 isomorphic
140-4-s266-D004 *
140-4-s266-D005 $\mathbb{S}$
140-4-s266-D007 Revuni
140-4-s266-D008 done|being
140-4-s266-D010 We thus have

[scan 267]
[p. 781]

mant|me by the commutativity of the diagrams

$$(107) \qquad \begin{array}{ccc} H(A) & \overset{i_A}{\xrightarrow{\sim}} & \\ {\scriptstyle (100)}\downarrow{\scriptstyle \sim} & & H(\mathbb{S}^{\sim}_x, \mathbb{D}^{*\sim}_x) \\ H(A') & \overset{i_{A'}}{\xrightarrow{\sim}} & \end{array} .$$

To verify the fact that the functor $\underline{\mathrm{Gr}}(H)$ of (102) is essentially constant, I consider the two functors (independent of the choice of $\mathbb{S}^{\sim}_x, \mathbb{D}^{*\sim}_x$)

$$(108) \qquad \begin{cases} A \longmapsto \mathbb{S}^{\sim}_{xA}, & A \longmapsto \mathbb{D}^{*\sim}_{x,A} \\ \underline{\mathcal{A}}_x \to \underline{\mathrm{Revuni}}(\mathbb{S}_x) & \underline{\mathcal{A}}_x \to \underline{\mathrm{Revuni}}(\mathbb{D}^*_x) \end{cases} ,$$

which factor into equivalences

$$(109)\qquad \underline{\mathrm{Gr}}(\underline{\mathcal{A}}_x) \overset{F}{\underset{\approx}{\to}} \underline{\mathrm{Revuni}}(\mathbb{S}_x), \qquad \underline{\mathrm{Gr}}(\underline{\mathcal{A}}_x) \overset{G}{\underset{\approx}{\to}} \underline{\mathrm{Revuni}}(\mathbb{D}_x^*)$$

This said, using $\mathbb{S}_x^\sim, \mathbb{D}_x^{*\sim}$, we have for $A \in \mathrm{Ob}\,\underline{\mathcal{A}}_x = \mathcal{A}_x$ also, isomorphisms functorial in A

$$(110)\qquad \begin{cases} \mathbb{S}^\sim | H(A) \simeq \mathrm{Isom}_{\underline{\mathrm{Revuni}}(\mathbb{S}_x)} \dagger \left(\underbrace{F(A)}_{\mathbb{S}_{x,A}^\sim}, \mathbb{S}^\sim \right) \\ \mathbb{D}_x^{*\sim}(\mathrm{Ep}|E_p A) \simeq \mathrm{Isom}_{\underline{\mathrm{Revuni}}(\mathbb{D}_x^*)} \left(\underbrace{G(A)}_{\mathbb{D}_{x,A}^{*\sim}}, \mathbb{D}_x^{*\sim} \right), \end{cases}$$

and by the universal property of $\underline{\mathrm{Gr}}(\underline{\mathcal{A}}_x)$, these functorial isomorphisms extend over the category $\underline{\mathrm{Gr}}(\underline{\mathcal{A}}_x)$. Thus the situation is one where we have a commutative group $T|\overline{T}$ $(= T_x)$, three connected T-groupoids $\mathcal{G}, \mathcal{G}', \mathcal{G}''$ 266

Uncertainty retained from cancelled passages.

140-4-s267-D001 $\mathbb{S}$

[scan 268]

(namely $\underline{\mathrm{Gr}}(\underline{\mathcal{A}}_x)$, $\underline{\mathrm{Revuni}}(\mathbb{S}_x)$, $\underline{\mathrm{Revuni}}(\mathbb{D}_x^*)$), equivalences $\dagger$ of T-groupoids

$$(111)\qquad \mathcal{G} \overset{F}{\underset{\approx}{\longrightarrow}} \mathcal{G}', \qquad \mathcal{G} \overset{G}{\underset{\approx}{\longrightarrow}} \mathcal{G}''$$

and objects $x' \in \mathrm{Ob}\,\mathcal{G}'$, $x'' \in \mathrm{Ob}\,\mathcal{G}''$ (namely $\mathbb{S}^\sim$ and $\mathbb{D}_x^{*\sim}$), and if we consider the functors

$$(112)\qquad \begin{cases} \mathcal{G} \overset{f=f_{x'}}{\longrightarrow} \underline{\mathrm{Tors}}(T), & \mathcal{G} \overset{g=g_{x''}}{\longrightarrow} \underline{\mathrm{Tors}}(T) \\ x \mapsto \mathrm{Hom}_{\mathcal{G}'}(F(x), x'), & x \mapsto \mathrm{Hom}_{\mathcal{G}''}(G(x), x'') \end{cases}$$

and the corresponding functor

$$(113)\qquad \begin{cases} \mathcal{G} \longrightarrow \underline{\mathrm{Tors}}(T) \\ x \longmapsto \mathrm{Hom}_{\underline{\mathrm{Tors}}(T|G)}(f(x), g(x)) \end{cases},$$

to prove that this last functor is essentially constant. We have then, if

$$(114)\qquad F_1\dagger : \mathcal{G}' \overset{\approx}{\longrightarrow} \mathcal{G}, \qquad {}'G_1|G_1 : \mathcal{G}'' \overset{\approx}{\longrightarrow} \mathcal{G}$$

denote quasi-inverses of F, G, isomorphisms functorial in x

$$(115)\qquad \begin{cases} f(x) \overset{\mathrm{def}}{=} \mathrm{Hom}_{\mathcal{G}'}(F(x), x') \simeq \mathrm{Hom}_{\mathcal{G}}(x, F_1(x')) \\ g(x) \overset{\mathrm{def}}{=} \mathrm{Hom}_{\mathcal{G}''}(G(x), x'') \simeq \mathrm{Hom}_{\mathcal{G}}(x, F_2(x'')) \end{cases}$$

whence follows|follow isomorphisms of T-torsors

$$(116)\qquad \mathrm{Isom}(f(x), g(x)) \simeq \mathrm{Isom}(F_1(x'), F_2(x''))$$

whence the assertion. 267

Uncertainty retained from cancelled passages.
140-4-s268-D001 of
140-4-s268-D003 f

[scan 269]

Definition We say that we have a correlation between a universal covering $\mathbb{S}_x^{\sim}$ of $\mathbb{S}_x$, and another $\mathbb{D}_x^{*\sim}$ of $\mathbb{D}_x^*$, when we have given ourselves an element of $H(\mathbb{S}_x^{\sim}, \mathbb{D}_x^{*\sim})$. [p. 782]

Thus, if $A \in \mathcal{A}_x$, the datum of a correlation amounts to the datum of an element of $H(A)$, i.e. of an isomorphism † between the T_x-torsors $\mathbb{S}_x^{\sim}(A)$ and $\mathbb{D}_x^{*\sim}(\mathrm{Ep}|E_p(A))$. This identification is compatible with the inclusion relations between arcs $A \subset A'$. † The correlations between $\mathbb{S}_x^{*|\sim}$ and $\mathbb{D}_x^{*\sim}$ form a torsor under|over T_x. †

Let $\underline{\mathrm{Corr}}_x(\mathfrak{X}|X) = \underline{\mathrm{Corr}}_x(\dagger X)$ be the category of †triples $(\tilde{\mathbb{S}}_x, \widetilde{\mathbb{D}}_x^*, h)$ of correlated universal coverings of $\mathbb{S}_x$ and of $\mathbb{D}_x^*$, we have † functors

(117)
$$\begin{array}{ccccc} & & (\tilde{\mathbb{S}}_x, \mathbb{D}_x^{*\sim}, h) & & \\ & \swarrow & & \searrow & \\ \mathbb{S}_x^{*\sim|\sim} & & & & \mathbb{D}_x^{*\sim} \\ & \searrow^{\approx} & & \swarrow_{\approx} & \\ & & \underline{\mathrm{Corr}}_x(X) & & \\ & \swarrow & & \searrow & \\ \underline{\mathrm{Revuni}}(\mathbb{S}_x) & & & & \underline{\mathrm{Revuni}}(\dagger\mathbb{D}_x^*) \end{array} ,$$

which are equivalences of categories. 268

Uncertainty retained from cancelled passages.
140-4-s269-D001 of
140-4-s269-D002 D
140-4-s269-D003 $\mathbb{S}$
140-4-s269-D004 $\mathbb{S}$
140-4-s269-D005 pairs
140-4-s269-D006 equi|equivalences
140-4-s269-D007 $\mathbb{S}$

[scan 270]

All these constructions have been carried out intrinsically carefully enough to pass to isomorphisms of germs of differentiable surfaces $(X, x) \xrightarrow{u}_{\sim} (X', x')$. Suppose chosen, in addition to $(\mathbb{S}_x^{\sim}, \mathbb{D}_x^{*\sim}, h_x)$ relative|relative to (X, x), also $(\mathbb{S}_{x'}^{\sim}, a\,\mathbb{D}_{x'}^{*\sim}, h'_{x'})$ relative|relative to (X', x'). Now the isomorphism of the germs being given, it therefore induces

(118)
$$\begin{cases} u_{\mathbb{S}} : & \mathbb{S}_x \simeq \mathbb{S}_{x'} & \text{isomorphism of topological spaces} \\ u_{\mathbb{D}^*} : & \mathbb{D}_x^* \xrightarrow{\sim} \mathbb{D}_{x'}^* & \text{isomorphism of germs of spaces} \end{cases}$$

$$u_T : \quad T_x \simeq T_{x'} \qquad \text{isomorphism of the } \mathbb{Z}\text{-modules topological free of rank 1}$$

Let us then consider the two sets

(119)
$$\begin{cases} F(u) = \text{set of isomorphisms} & \mathbb{S}_x^{\sim} \simeq \mathbb{S}_{x'}^{\sim} \text{ over } u_{\mathbb{S}} \\ & \dagger \\ G(u) = \text{set of isomorphisms} & \mathbb{D}_x^{*\sim} \simeq \mathbb{D}_{x'}^{*\sim} \text{ over } u_{\mathbb{D}^*}, \end{cases}$$

they are both T_x-torsors. This being so, I say that we have a canonical isomorphism of T_x-torsors

$$(120) \qquad F(u) \xrightarrow{\sim} G(u)$$

— i.e. "it comes to the = to lift $u_{\mathbb{S}}$ to $u_{\tilde{\mathbb{S}}} : \mathbb{S}^{\sim}_x \simeq S^{\sim}_{x'}$, or to lift $u_{\mathbb{D}^*}$ to † $u_{\mathbb{D}^{*\sim}} : \mathbb{D}^{*\sim}_x \longrightarrow |-\mathbb{D}^{*\sim}_{x'}$". To see this, we note that by † transport of structure the 269

Uncertainty retained from cancelled passages.

140-4-s270-D001 $u_{\mathbb{S}}$

140-4-s270-D002 $u_{\tilde{\mathbb{S}}}$

140-4-s270-D003 constructions

[scan 271]

inverse images under $u_{\mathbb{S}|\mathcal{S}}$ and under $u_{\mathbb{D}|D^*}$ of $\mathbb{S}|\mathcal{S}^{\sim}_{x'}$ and of $\mathbb{D}|D^{*\sim}_{x'}$, that is|that is $\mathbb{S}|\mathcal{S}'^{\sim}_x$, $\mathbb{D}|D^{*\prime\sim}_x$, [p. 783] are equipped with a correlation, which I denote by h'_x. This posited, † canonical isomorphisms of T_x-torsors

$$(121) \quad \begin{aligned} F(u) &\simeq \mathrm{Isom}_{\mathbb{S}|\mathcal{S}_x}(\mathbb{S}|\mathcal{S}^{\sim}_x, \mathbb{S}|\mathcal{S}'^{\sim}_x) \\ G(u) &\simeq \mathrm{Isom}_{\mathbb{D}|D^*_x}(\mathbb{D}|D^{*\sim}_x, \mathbb{D}|D^{*\prime\sim}_x) \end{aligned}$$

Now by the diagram of equivalences (117) we do have isomorphisms of T_x-torsors

$$(122)$$

$$\begin{array}{ccc} & \mathrm{Isom}((\mathbb{S}|\mathcal{S}^{\sim}_x, \mathbb{D}|D^{*\sim}_x, h_x), & \\ & (\mathbb{S}|\mathcal{S}'^{\sim}_x, \mathbb{D}|D^{*\prime\sim}_x, h'_x)) & \\ \overset{\sim}{\swarrow} & & \overset{\sim}{\searrow} \\ F(u) = & & G(u) = \\ \mathrm{Isom}(\mathbb{S}|\mathcal{S}^{\sim}_x, \mathbb{S}|\mathcal{S}'^{\sim}_x]|) & & \mathrm{Isom}(\mathbb{D}|D^{*\sim}_x, \mathbb{D}|D^{*\prime\sim}_x) \end{array} ,$$

whence the canonical bijection sought (120) — † which gives in particular (96).

† The canonical bijection (120) has the following ± obvious property, which characterizes it:

(123) † Let $u_{\mathbb{S}|\mathcal{S}^{\sim}}$ and $u_{\mathbb{D}|D^{*\sim}}$ be †liftings of u †† $\mathbb{S}|\mathcal{S}^{\sim}_x \simeq \mathbb{S}|\mathcal{S}^{\sim}_{x'}$, $\mathbb{D}|D^{*\sim}_x \longrightarrow \mathbb{D}|D^{*\sim}_{x'}$, associated by (120), let $A \in \mathcal{A}_x$, $A' \in \mathcal{A}'_x$ its image under $u_{\mathbb{S}|\mathcal{S}}$, let α be a section of $\mathbb{S}|\mathcal{S}^{\sim}_x$ over A, β the section of $\mathbb{D}|D^{*\sim}_x$ over $\mathrm{Ep}|E_p(A)$ associated|associated by the correlation h_x, let α' be the section of $\mathbb{S}|\mathcal{S}^{\sim}_{x'}$ over A' deduced from α by applying $u_{\mathbb{S}|\mathcal{S}^{\sim}}$, β' the section of $\mathbb{D}|D^{*\sim}_{x'}$ over $\mathrm{Ep}(A')$ deduced from β by applying $u_{\mathbb{D}|D^{*\sim}}$, then α' and β' are associated by the correlation h'.

270

Uncertainty retained from cancelled passages.

140-4-s271-D001 the existence

140-4-s271-D002 of the

140-4-s271-D003 The

140-4-s271-D004 We

140-4-s271-D005 extensions

140-4-s271-D006 $u_{\mathbb{S}}$

140-4-s271-D007 associated

[scan 272]

This † (for A, α fixed) characterizes the map (120), because for $u_{\mathbb{S}|\mathcal{S}^{\sim}}$ fixed, it determines $u_{\mathbb{D}|D^{*\sim}}$ as follows: we have β by h_x, α' deduced|deduced from α by $u_{\mathbb{S}|\mathcal{S}}$, β' deduced|deduced from α' by $h_{x'}$, now β' must be equal to the section deduced from β by applying $u_{\mathbb{D}|D^{*\sim}} : \mathbb{D}|D^{*\sim}_x \longrightarrow |-\mathbb{D}|D^{*\sim}_{x'}$, and this indeed determines $u_{\mathbb{D}|D^{*\sim}}$ uniquely ...

N.B. We can use this construction, for A, α fixed, to define the map (120), we find that it is a T_x-morphism (hence an isomorphism) and, †at the same|= time, that it does not depend on the choice of α. But a priori it depends perhaps on the choice of A, and the point which requires verification is that this bijection in fact does not depend on the choice of A. We know indeed that for $A \subset A'$, it is the same|= for A and A'; |, whence at once the result, for the equivalence relation generated in $\mathcal{A}_x$ by the inclusion relation is the indiscrete relation 271

Uncertainty retained from cancelled passages.
140-4-s272-D001 characterizes
140-4-s272-D003 that it|since it

[scan 273]
[p. 784]

F) Application : differential interpretation of the special Teichmüller groups.

Let X be a topological space, equipped with a subset I that is discrete finite, such that X is a surface in a neighborhood of I — moreover we give ourselves a differentiable structure (C_1) in a neighborhood of the|of I. Let G † be a topological group, operating continuously on (X, I), and respecting the differentiable structure of X in a neighborhood of S. Let|Let

$$(124)\quad \begin{cases} \mathbb{S}|\mathcal{S}_{X,I} \text{ or}| \sim \mathbb{S}|\mathcal{S}_I = \coprod_{x\in I} \underbrace{\mathbb{S}|\mathcal{S}_{X,x}}_{\text{or } \mathbb{S}|\mathcal{S}_x} , \\ \mathbb{D}|D^*_{X,I} \text{ or}| \sim \mathbb{D}|D^*_I = \coprod_{x\in I} \underbrace{\mathbb{D}|D^*_{X,x}}_{\text{or } \mathbb{D}|D^*_x} \end{cases}$$

we assume that the operation of G on $\mathbb{S}|\mathcal{S}_S$ induced by the action of G is also continuous.

Let, for $\forall\ x \in I$, $\mathbb{D}|D^{*\sim}_x$ be a universal covering of $\mathbb{D}|D^*_x$, and put

$$(125)\quad \mathbb{D}|D^{*\sim}_{X,I} \text{ or}|- \mathbb{D}|D^{*\sim}_I = \coprod_{x\in S} \mathbb{D}|D^{*\sim}_x \ .$$

Let us for the moment forget the differentiable data, and consider the group

$$(126)\quad \widetilde{G} = \left\{ u, \tilde{u} \;\middle|\; \begin{array}{l} u \in G,\ \tilde{u} \text{ a lifting of} \\ u_{\mathbb{D}|D^*_I} \text{ to an automorphism } u_{\mathbb{D}|D^{*\sim}_I} \\ \text{of } \mathbb{D}|D^{*\sim}_I \end{array} \right\}$$

We have a homomorphism $(u, \tilde{u}) \longmapsto \tilde{u}$, $\widetilde{G} \longrightarrow G$, which is 272

Uncertainty retained from cancelled passages.
140-4-s273-D001 [illeg. 1]

[scan 274]

an epimorphism, and, putting

$$(127)\quad T_I = \prod_{x\in I} T_x \ ,$$

we have an exact sequence

$$(128)\quad 1 \to T_I \to \widetilde{G} \to G \to 1 \ .$$

I would like to introduce on $\widetilde{G}$ a topology such that (128) becomes an exact sequence of topological groups. Replacing G if necessary by the open subgroup G' of finite index kernel of the canonical homomorphism

$$(129)\quad G \longrightarrow \mathfrak{S}|\mathcal{S}_I \ ,$$

hence $\widetilde{G}$ by the inverse image $\widetilde{G'}$ of G', we are reduced to the case where G operates trivially on I, † Then $\widetilde{G}$ is expressed † † as the fiber product over G of the family, indexed by I, of the extensions

$$(130) \qquad 1 \to T_x \to \widetilde{G}_x \to G \to 1$$

corresponding to the cases $I_x \overset{\text{def}}{=} \{x\}$. Thus we are reduced to the case where $\operatorname{card} I|S = 1$. 273

Uncertainty retained from cancelled passages.

140-4-s274-D001 [illeg. 1]
140-4-s274-D002 subgroup [illeg. 1]
140-4-s274-D003 = that

[scan 275]
[p. 785]

Let †

$$(131) \qquad \{\Delta_{x,0}\dagger$$

be a|a † of neighborhoods|neighborhood of x in X which is|are † an open disks|open disk, † We can then find a universal covering $\Delta^{*\sim}_{x,0}$ of $\Delta^{*}_{x,0} = \Delta_{x,0} \setminus \{x\}$ which induces|induces $\mathbb{D}|D^{*\sim}_{x}$ on $\mathbb{D}|D^{*}_{x}$. By the continuity hypothesis, there exists a neighborhood U of 1 in G, and a neighborhood $\Delta_{x,1}$ of x, such that we have

$$(132) \qquad \begin{array}{ccc} U \times \Delta_{x,1} & \longrightarrow & \Delta_{x,0} \\ \cap & & \cap \\ G \times X & \longrightarrow & X \end{array},$$

and we can suppose that $\Delta_{x,1}$ is an open disk|open disks. This taken; for fixed $u \in U$, giving the lifting $\tilde{u}$ of $u_{\mathbb{D}|D_x}$ to $\mathbb{D}|D^{\sim}_{x}$ is equivalent to giving a lifting to †

$$(133) \qquad \widetilde{\Delta}_{x,1} \xrightarrow{\tilde{u}_{0,1}} \Delta^{\sim}_{x,0}$$

compatible with

$$u \mid \Delta^{*}_{x,1} : \qquad \Delta^{*}_{x,1|0} \longrightarrow \Delta^{*}_{x,0} \ .$$

That said, in the inverse image $\widetilde{U}$ in $\widetilde{G}$ of the neighborhood U of 1 in G, we are going to 274

Uncertainty retained from cancelled passages.

140-4-s275-D001 $\Delta_x|\mathbb{D}_x$
140-4-s275-D002 $\}_{\alpha\in\mathbf{N}}$
140-4-s275-D003 decreasing sequence
140-4-s275-D004 some
140-4-s275-D005 form basis
140-4-s275-D006 $\Delta|\mathbb{D}$

[scan 276]

introduce the topology the coarsest making continuous † the maps

$$(134) \qquad \begin{cases} p_1|p_i : \widetilde{U} \longrightarrow U & \text{induced|induced by } \widetilde{G} \longrightarrow G \\ q : \widetilde{U} \implies \Big|_{\dagger} \longrightarrow \operatorname{Cont}(\Delta^{*\sim}_{x,1}, \Delta^{*\sim}_{x,0}) & (u, \tilde{u}) \longmapsto \dagger\tilde{u}_{0,1} \\ & \dagger \\ q' = q \circ \sigma, \quad \text{where } \sigma(u,\tilde{u}) = (u,\tilde{u})^{-1} : & \widetilde{U} \longrightarrow \operatorname{Cont}(\Delta^{*\sim}_{x,1}, \Delta^{*\sim}_{x,0}) \end{cases}$$

We shall have|I shall have taken care to take $U|\widetilde{U}$ stable under $u \mapsto u^{-1}$, which implies that the same|= holds for $\widetilde{U}$.

We thus find, with the help of this topology on $\widetilde{U}$, † the filter of neighborhoods of $1_{\widetilde{G}}$ in $\widetilde{U}$, hence also its image filter in $\widetilde{G}$. We must check that the latter is indeed the filter of neighborhoods of 1 for a topological group topology, and that for this topology, $\widetilde{G} \to G$ is a covering of G. †I admit that everything works.

When X is a surface compact and orientable, connected, and if $T|\overline{T}$ denotes its module of 275

Uncertainty retained from cancelled passages.

140-4-s276-D001 compact convergence of the|of

140-4-s276-D002 $\longrightarrow$

140-4-s276-D003 $(\widetilde{u})_{0,1}$

140-4-s276-D004 $(u, \widetilde{u})$

140-4-s276-D007 a [illeg. 1] a filter

140-4-s276-D008 I suppose

[scan 277]

orientations, then all the T_x are canonically isomorphic to $T|\overline{T}$ [p. 786]

$$(135) \qquad T_x \simeq T|\overline{T} \qquad \forall x \in I,$$

and the exact sequence (128) takes the form

$$(136) \qquad 1 \longrightarrow T^I|\overline{T}^I \longrightarrow \widetilde{G} \longrightarrow G \longrightarrow 1$$

The "universal" case is the one where

$$(137) \qquad G = \mathcal{A} = \operatorname{Aut}\operatorname{top}(X, S)$$

(with the topology of uniform convergence of the u and of the u^{-1}), Then in the anabelian cases (the type of (X, S) being g, ν, and

$$(138) \qquad \begin{array}{l} 2g - 2 + \nu > 0 \quad \text{i.e.} \quad 2g + \nu \geq 3 \\ \text{i.e.} \quad (g, \nu) \neq (0,0), (0,1), (0,2), (1,0) \end{array}$$

)|, then we know that $\mathcal{A}^0$ is 1-connected (is|and even|= contractible) hence every|every †extension of $\mathcal{A}^0$ by a discrete group is canonically split, and the extension $\widetilde{\mathcal{A}}$ of G then comes from an extension $\widetilde{\mathcal{T}|\tau}$

$$(139) \qquad 1 \longrightarrow \mathbb{Z}^I|T^I \longrightarrow \widetilde{\mathcal{T}|\tau} \longrightarrow \mathcal{T}|\tau \longrightarrow 1$$

of the Teichmüller group

$$(140) \qquad \mathcal{T}|\tau = \mathcal{T}|\tau(X, S) = \mathcal{A}/\mathcal{A}^0 = \pi_0(\underbrace{\operatorname{Aut}\operatorname{top}(X, S)}_{A|\mathcal{A}})$$

276

Uncertainty retained from cancelled passages.

140-4-s277-D001 covering|covering of

[scan 278]

which|that thus takes the name of special Teichmüller group of (X, S). For arbitrary G the action of G on (X, S) amounts to giving a homomorphism of topological groups

$$G \longrightarrow \mathcal{A}|A$$

fitting into a diagram of topological groups

$$\begin{array}{ccc} G & \longrightarrow & \mathcal{A}|A \\ \downarrow & & \downarrow \\ \pi_0(G) & \longrightarrow & \mathcal{T}|\tau = \pi_0(\mathcal{A}|A) \end{array} , \tag{141}$$

and we then find, by functoriality, that the $\widetilde{G}$ constructed previously is the inverse image of the extension $\widetilde{\mathcal{T}|\tau} = \operatorname{Sp}\mathcal{T}|\tau\ a|\alpha$ of $\mathcal{T}|\tau$ by $\mathbb{Z}^I$, via $G \longrightarrow \mathcal{T}|\tau$.

These preliminaries being taken, let us return to the situation where X admits a differentiable structure C_1 in a neighborhood of $I|S$, and let for $\forall\, x \in I|S$, 277

[scan 279]

$\widetilde{\mathbb{S}|\mathcal{S}}_x$ be "the" universal covering of $\mathbb{S}|\mathcal{S}_x$ correlated with the universal covering $\mathbb{D}|D_x^{*\sim}$ of [p. 787] $\mathbb{D}|D_x^*$. Put

$$\widetilde{\mathbb{S}|\mathcal{S}}_I = \coprod_{x\in I} \widetilde{\mathbb{S}|\mathcal{S}}_x, \tag{142}$$

and note that we have, for $\forall\, u \in G$, a one-to-one correspondence between the set of liftings $\tilde{u}_{\mathbb{D}|D_I^*}$ of $u_{\mathbb{D}|D_I^*}$ to automorphisms of $\mathbb{D}|D_I^{*\sim}$, and the set of liftings $\tilde{u}_{\mathbb{S}|\mathcal{S}_I}$ of $u_{\mathbb{S}|\mathcal{S}}$ to automorphisms of $\widetilde{\mathbb{S}|\mathcal{S}}_I$, which is an isomorphism of T_I-torsors. This correspondence is compatible with compositions, and we thus find a canonical isomorphism

$$\widetilde{G} \simeq \text{group of the } \left\{ u, \tilde{u}_{\mathbb{S}|\mathcal{S}} \;\middle|\; \begin{array}{l} u \in G,\ \tilde{u}_{\mathbb{S}|\mathcal{S}_I} \text{ lifting} \\ \text{of } u_{\mathbb{S}|\mathcal{S}_I} \text{ to an} \\ \text{automorphism of } \mathbb{S}|\mathcal{S}_I^{\sim} \end{array} \right\} \tag{143}$$

We check that this is a topological isomorphism by putting on the second † the coarsest topology making continuous the maps

$$\begin{aligned} & p_1^a|p_1 : \widetilde{G} \longrightarrow G \\ & q_1 : \widetilde{G} \longrightarrow \operatorname{Aut}\operatorname{top}(\widetilde{\mathbb{S}|\mathcal{S}}_I) \qquad (u, \tilde{u}_{\mathbb{S}|\mathcal{S}}) \longmapsto \tilde{u}_{\mathbb{S}|\mathcal{S}} \\ & q_1' = q_1 \circ \sigma : \widetilde{G} \longrightarrow \operatorname{Aut}\operatorname{top}(\widetilde{\mathbb{S}|\mathcal{S}}_I) \end{aligned} \tag{144}$$

trans|tras

278

Uncertainty retained from cancelled passages.

140-4-s279-D001 met

[scan 280]

where $\operatorname{Aut}\operatorname{top}(\widetilde{\mathbb{S}|\mathcal{S}}_I)$ is equipped with compact convergence.

In the case where X is a surface compact and differentiable and orientable, we thus find an interpretation of the preceding groups, † even|way differential, in so far as an extension|as an extension of $G|\widetilde{G}$ by $T_I|\overline{T}_I$, as the inverse image of a special Teichmüller group $S\mathcal{T}|\operatorname{Sp}\mathcal{T} \longrightarrow \mathcal{T}|\tau$, by the canonical homomorphism $G \longrightarrow \mathcal{T}|\tau$ of (141). 279

Uncertainty retained from cancelled passages.

140-4-s280-D001 constructed|explicit by